vmware by Broadcom

VMware vSphere 9.1

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Release Notes

Release notes include product enhancements and notices, bug fixes, and resolved issues.

For vSphere 9.1 release notes, read VMware Cloud Foundation 9.1.1 release notes.

For single host management release notes, read VMware Host Client 9.1 release notes.

vSphere Express

Learn how to deploy, upgrade, and license a vSphere Express environment.

Components in vSphere Express

The full functionality is implemented by a set of mandatory and optional components.

Mandatory components:

- ESX
- vCenter
- VCF Operations
- License server

Optional components:

- vSphere Replication (part of protection and recovery)
- VMware Tools
- VMware Remote Console

Deploying vSphere Express

You can build a fully operational environment by deploying and configuring components manually.

Table 1: Deployment Order

#	Component	Documentation
1	ESX	Deploy ESX Hosts
2	vCenter	Deploy vCenter
3	VCF Operations	Deploy VCF Operations
4	License server	Deploy a License Server
5	(Optional) vSphere Replication	Deploy Protection and Recovery
6	(Optional) VMware Tools	Installing VMware Tools
7	(Optional) VMware Remote Console	Installing VMware Remote Console

Upgrading from vSphere 8 to vSphere Express

To upgrade your vSphere 8.x and Aria Operations 8.x virtual infrastructure to a fully functional and licensed vSphere Express 9.1.1 and later, comply with the source versions of your existing components and follow the order of upgrade and deployment procedures.

Minimum required version for components to upgrade:

- vCenter Server 8.0
- ESXi 8.0
- (Optional) Aria Operations 8.18.6
- (Optional) vSphere Replication 9.0.2.2

Table 2: Upgrade Order

#	Component	Documentation
1	vSphere Replication	Converge or Upgrade to Protection and Recovery 9.1
2	Aria Operations	Deploy VCF Operations 9.1 or Upgrade Aria Operations 8.18 and later
3	License server	Deploy a License Server
4	vCenter	Upgrade the vCenter Instance
5	ESX	Upgrade the ESX Hosts
6	vSphere Distributed Switch	Upgrade vSphere Distributed Switches
7	VMware Tools	Upgrade VMware Tools to 13.1
8	Virtual machine compatibility	Upgrade Virtual Machines

Converging vSphere Express to VCF

You can use your existing vSphere Express virtual infrastructure as building blocks to converge it to a full VCF deployment. See Converge a vCenter Instance, ESX Hosts, VMware Aria Operations.

Extending to vSphere Foundation

You can extend your existing vSphere Express virtual infrastructure to a vSphere Foundation environment by applying the respective license and optionally deploy VCF management services. See Deploy VCF Management Services for vSphere Foundation.

Licensing vSphere Express

vSphere Express uses the same licensing infrastructure as VMware Cloud Foundation and vSphere Foundation with some restrictions to license management functions.

You purchase a subscription, a default primary license appears in the VCF Business Services console, you add a license to VCF Operations and the license server, and you assign the license to assets by using VCF Operations. To license vSphere Express, you use this workflow with some restrictions on maximum license capacity, VCF Operations capabilities, and license management operations.

The following restrictions apply to vSphere Express.

License capacity maximum	A maximum of 128 cores per customer. The VCF Business Services console prevents any allocation that exceeds 128 cores per customer or site ID.
Default license only	You use only the default license to license your environment. You cannot split the default license into smaller licenses. You cannot share the default license across sites.
VCF Operations for licensing	With vSphere Express, you use VCF Operations only for licensing your environment.
You cannot use add-on licenses	No add-ons are available for vSphere Express.
Version 8 license keys are not available	vSphere Express is only available for version 9 or later.

For more information about the licensing model and how to manage and assign licenses, see the VCF Licensing guide.

vSphere Features in vSphere Express

vSphere Express entitles you to certain vSphere features. For more information on vSphere Express features, consult your sales representative.

ESX Installation and Setup

VMware ESX Installation and Setup describes how to install and configure VMware ESX™.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we have updated this guide to remove instances of non-inclusive language.

Intended Audience

VMware ESX Installation and Setup is intended for experienced administrators who want to install and configure ESX.

This information is written for experienced Windows or Linux system administrators who are familiar with virtual machine technology and data center operations. The information about using the Image Builder and VMware vSphere®Auto Deploy™ is written for administrators who have experience with Microsoft PowerShell and VMware Cloud Foundation® PowerCLI™.

In this guide, see:

- Installing and Setting Up ESX
- What is VMware vSphere Distributed Services Engine
- Customizing Installations with vSphere ESX Image Builder
- Installing ESX
- Setting Up ESX
- Troubleshooting ESX Booting
- Decommission an ESX Host

Installing and Setting Up ESX

See how install and set up ESX on your physical hardware so that it acts as a platform for virtual machines.

The two core components of vSphere are ESX and vCenter. ESX is the virtualization platform on which you can create and run virtual machines and virtual appliances. vCenter is a service that acts as a central administrator for ESX hosts connected in a network. vCenter lets you pool and manage the resources of multiple hosts. For detailed information about the vCenter deployment process, see vCenter Deployment and Setup.

In this guide, see how to install and set up ESX. Start with How to Install ESX and follow up with:

- Preparing for Installing ESX
- ESX Requirements
- Installing ESX
- Setting Up ESX
- After You Install and Set Up ESX

How to Install ESX

ESX is the virtualization platform on which you can create and run virtual machines and virtual appliances.

ESX 9.1 and later provides various options for installation and setup that define a corresponding sequence of tasks, but the basic sequence is as follows:

- Read the release notes for ESX 9.1 and later.
- Verify that your system meets the minimum hardware requirements. See ESX Requirements.
  Determine the ESX installation option to use: interactive, scripted or by using vSphere Auto Deploy.

  Attention:

  You can provision ESX hosts by using vSphere Auto Deploy, but Auto Deploy is installed together with vCenter, which is the service that acts as a central administrator for ESX hosts connected in a network. vCenter lets you pool and manage the resources of multiple hosts.

Determine where you want to locate and boot the ESX installer. See Media Options for Booting the ESX Installer If you are using PXE to boot the installer, verify that your network PXE infrastructure is properly set up. See Network Booting the ESX Installer.

Create a worksheet with the information you will need when you install ESX. See Required Information for ESX Installation.

Read and consider the General ESX Security Recommendations.

- Install ESX:
  - Install ESX Interactively
  - Scripted Installation
  - Installing ESX Using vSphere Auto Deploy
- Configure the ESX boot and network settings, the direct console, and other settings. See Setting Up ESX and After You Install and Set Up ESX.
- Consider setting up a syslog server for remote logging, to ensure sufficient disk storage for log files. Setting up logging on a remote host is especially important for hosts with limited local storage. See Required Free Space for System Logging and Configure Syslog on ESX Hosts.
- You deploy the vCenter appliance, a preconfigured virtual machine optimized for running vCenter and the vCenter components. You can deploy the vCenter appliance on ESX hosts or on vCenter instances. For detailed information about the vCenter installation process, see vCenter Installation and Setup.

About ESX Evaluation and Licensed Modes

Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. For more information on how to license your environment, see Licensing Overview in the *Licensing* guide.

The evaluation period is 90 days and begins when you turn on the ESX hosts, except for hosts that you provision by using Auto Deploy as stateless hosts. Such hosts have no evaluation period because booting ESX hosts by network with in-memory or cached images, preserving no state, was deprecated in ESX 9.0.

Preparing for Installing ESX

Before you install ESX, determine the installation option that is suitable for your environment and prepare for the installation process.

- To get started with ESX installation, you first need to read:
  - Download the ESX Installer
  - Required Information for ESX Installation
  - Media Options for Booting the ESX Installer

Download the ESX Installer

You can obtain the ESX installer software either from an OEM or from the Broadcom Support Portal.

Register for an account on the Broadcom Support Portal. For more information, see Broadcom knowledge base article 145581.

1. For product download instructions, see Broadcom knowledge base article 142814.
2. For download of offline bundle ZIP files for ESX patches and updates, see Broadcom knowledge base article 194106.
3. For more information, see Broadcom knowledge base article 278681.

Required Information for ESX Installation

Interactive installation prompts you for the required system information, but you must supply this information in the installation script.

In an interactive installation, the system prompts you for the required system information. In a scripted installation, you must supply this information in the installation script. For future use, note the values you use during the installation. These notes are useful if you must reinstall ESX and reenter the values that you originally selected.

Table 3: Required Information for ESX Installation

Information	Required or Optional	Default	Comments
Keyboard layout	Required	U.S. English	
VLAN ID	Optional	None	Range: 0 through 4095
IP address	Optional	DHCP	You can allow DHCP to configure the network during installation. After installation, you can change the network settings.
Subnet mask	Optional	Calculated based on the IP address	
Gateway	Optional	Based on the configured IP address and subnet mask	
Primary DNS	Optional	Based on the configured IP address and subnet mask	
Secondary DNS	Optional	None	
Host name	Required for static IP settings	None	The vSphere Client can use either the host name or the IP address to access the ESX host.
Install location	Required	None	Must be at least 8 GB if you install the components on a single disk.
Root password	Required	None	The root password must contain between 8 and 40 characters. For information about passwords see the *vSphere Security* documentation.

Media Options for Booting the ESX Installer

The installer must be accessible to the system on which you are installing ESX.

The following boot media are supported for the ESX installer:

- Boot from a CD/DVD. See Download and Burn the ESX Installer ISO Image to a CD or DVD.
- Boot from a USB flash drive. See Format a USB Flash Drive to Boot the ESX Installation or Upgrade.
- Boot from a network. See Network Booting the ESX Installer.
- Boot from a remote location using a remote management application. See Using Remote Management Applications.

Download and Burn the ESX Installer ISO Image to a CD or DVD

If you do not have an ESX installation CD/DVD, you can create one.

You can also create an installer ISO image that includes a custom installation script. See Create an Installer ISO Image with a Custom Installation or Upgrade Script.

1. Burn the ISO image to a CD or DVD.
2. Follow the procedure Preparing for Installing ESX.

Format a USB Flash Drive to Boot the ESX Installation or Upgrade

You can format a USB flash drive to boot the ESX installation or upgrade.

- Linux machine with superuser access to it.
- USB flash drive that can be detected by the Linux machine.
- The ESX ISO image, `VMware-VMvisor-Installer-`*`version_number-build_number`*`.x86_64.iso`, which includes the `isolinux.cfg` file.

The instructions in this procedure assume that the USB flash drive is detected as `/dev/sdb`.

Note:
The `ks.cfg` file that contains the installation script cannot be located on the same USB flash drive that you are using to boot the installation or upgrade. The kickstart file does not have any dependency on BIOS or UEFI boot.

1. Boot Linux, log in, and enter superuser mode by using a `su` or `sudo root` command.
2. If your USB flash drive is not detected as `/dev/sdb`, or you are not sure how your USB flash drive is detected, determine how it is detected.
   a) Plug in your USB flash drive.
   b) At the command line, run the command for displaying the current log messages.

      ```
      tail -f /var/log/messages
      ```

      You see several messages that identify the USB flash drive in a format similar to the following message.

      ```
      Oct 25 13:25:23 ubuntu kernel: [  712.447080] sd 3:0:0:0: [sdb] Attached SCSI
       removable disk
      ```

      In this example, `sdb` identifies the USB device. If your device is identified differently, use that identification in place of *`sdb`*.
3. Overwrite the entire USB drive with the ISO image. This operation overwrites the partition table and any previous content on the USB drive.

   ```
   dd bs=10M if=VMware-VMvisor-Installer-version_number-build_number.x86_64.iso of=/dev/
   sdb
   ```
4. Eject the USB drive.

   ```
   eject /dev/sdb
   ```

You can use the USB flash drive to boot the ESX installer.

Create a USB Flash Drive to Store the ESX Installation Script or Upgrade Script

You can use a USB flash drive to store the ESX installation script or upgrade script that is used during scripted installation or upgrade of ESX.

- Linux machine.
- ESX installation or upgrade script, the `ks.cfg` kickstart file.
- USB flash drive.

When multiple USB flash drives are present on the installation machine, the installation software searches for the installation or upgrade script on all attached USB flash drives.

The instructions in this procedure assume that the USB flash drive is detected as `/dev/sdb`.

Note:
Do not store the `ks` file containing the installation or upgrade script on the same USB flash drive that you are using to boot the installation or upgrade.

1. Attach the USB flash drive to a Linux machine that has access to the installation or upgrade script.
2. Create a partition table.

   ```
   /sbin/fdisk /dev/sdb
   ```
   a) Type `d` to delete partitions until they are all deleted.
   b) Type `n` to create primary partition 1 that extends over the entire disk.
   c) Type `t` to set the type to an appropriate setting for the FAT32 file system, such as `c`.
   d) Type `p` to print the partition table.

      The result should be similar to the following text:

      ```
      Disk /dev/sdb: 2004 MB, 2004877312 bytes
      255 heads, 63 sectors/track, 243 cylinders
      Units = cylinders of 16065 * 512 = 8225280 bytes
         Device Boot      Start         End      Blocks   Id  System
      ```

```
/dev/sdb1            1          243      1951866   c   W95 FAT32 (LBA)
```

e) Type `w` to write the partition table and quit.

3. Format the USB flash drive with the FAT32 file system.

   ```
   /sbin/mkfs.vfat -F 32 -n USB /dev/sdb1
   ```

4. Create a destination directory and mount the USB flash drive to it.

   ```
   mkdir -p /usbdisk
   mount /dev/sdb1 /usbdisk
   ```

5. Copy the ESX installation script to the USB flash drive.

   ```
   cp ks.cfg /usbdisk
   ```

6. Unmount the USB flash drive.

   ```
   umount /usbdisk
   ```

The USB flash drive contains the installation or upgrade script for ESX.

When you boot the ESX installer, point to the location of the USB flash drive for the installation or upgrade script. See Enter Boot Options to Run an Installation or Upgrade Script and PXELINUX Configuration Files.

Create an Installer ISO Image with a Custom Installation or Upgrade Script

You can customize the standard ESX installer ISO image with your own installation or upgrade script. This customization enables you to perform a scripted, unattended installation or upgrade when you boot the resulting installer ISO image.

- Linux machine.
- The ESX ISO image `VMware-VMvisor-Installer-x.x.x-XXXXXX.x86_64.iso`, where *x.x.x* is the version of ESX you are installing, and *XXXXXX* is the build number of the installer ISO image.
- Your custom installation or upgrade script, the `KS_CUST.CFG` kickstart file.

See also Installing ESX by Using a Script and About the boot.cfg File .

1. Download the ESX ISO image from the Broadcom Support Portal.
2. Mount the ISO image in a folder.

   ```
   mount -o loop VMware-VMvisor-Installer-x.x.x-XXXXXX.x86_64.iso /esx_cdrom_mount
   ```

   `XXXXXX` is the ESX build number for the version that you are installing or upgrading to.

3. Copy the contents of `esx_cdrom` to another folder.

   ```
   cp -r /esx_cdrom_mount/* /esx_cdrom
   ```

4. Copy the kickstart file to `/esx_cdrom`.

   ```
   cp KS_CUST.CFG /esx_cdrom
   ```

5. Modify the `boot.cfg` file in both `/esx_cdrom/efi/boot/boot.cfg` (for UEFI boot) and `/esx_cdrom/boot.cfg` (for legacy BIOS boot) to specify the location of the installation or upgrade script by using the `kernelopt` option.

   You must use uppercase characters to provide the path of the script, for example,

   ```
   kernelopt=runweasel ks=cdrom:/KS_CUST.CFG
   ```

   The installation or upgrade becomes completely automatic, without the need to specify the kickstart file during the installation or upgrade.

6. Recreate the ISO image using the `mkisofs` or the `genisoimage` command.

Command	Syntax
mkisofs	`mkisofs -relaxed-filenames -J -R -o custom_esx.iso -b ISOLINUX.BIN -c BOOT.CAT -no-emul-boot -boot-load-size 4 -boot-info-table -eltorito-alt-boot -eltorito-platform efi -b EFIBOOT.IMG -no-emul-boot /esx_cdrom`

Command	Syntax
genisoimage	`genisoimage -relaxed-filenames -J -R -o custom_esx.iso -b ISOLINUX.BIN -c BOOT.CAT -no-emul-boot -boot-load-size 4 -boot-info-table -eltorito-alt-boot -e EFIBOOT.IMG -no-emul-boot /esx_cdrom`

You can use this ISO installer image for regular boot or UEFI secure boot. However, the vSphere Lifecycle Manager cannot verify the checksum of such an ISO image and you cannot use it for upgrades by using vSphere Lifecycle Manager workflows.

The ISO image includes your custom installation or upgrade script.

Install ESX from the ISO image.

ESX Requirements

To install or upgrade ESX, your system must meet specific hardware and software requirements.

See the full list of ESX requirements:

- ESX System Storage Overview
- ESX Hardware Requirements
- Using Remote Management Applications
- Recommendations for Enhanced ESX Performance
- Incoming and Outgoing Firewall Ports for ESX Hosts
- Required Free Space for System Logging
- VMware Host Client System Requirements
- ESX Passwords and Account Lockout

ESX System Storage Overview

ESX 9.1 and later has a system storage layout that allows flexible management of partitions and third-party components, while facilitating debugging.

ESX System Storage

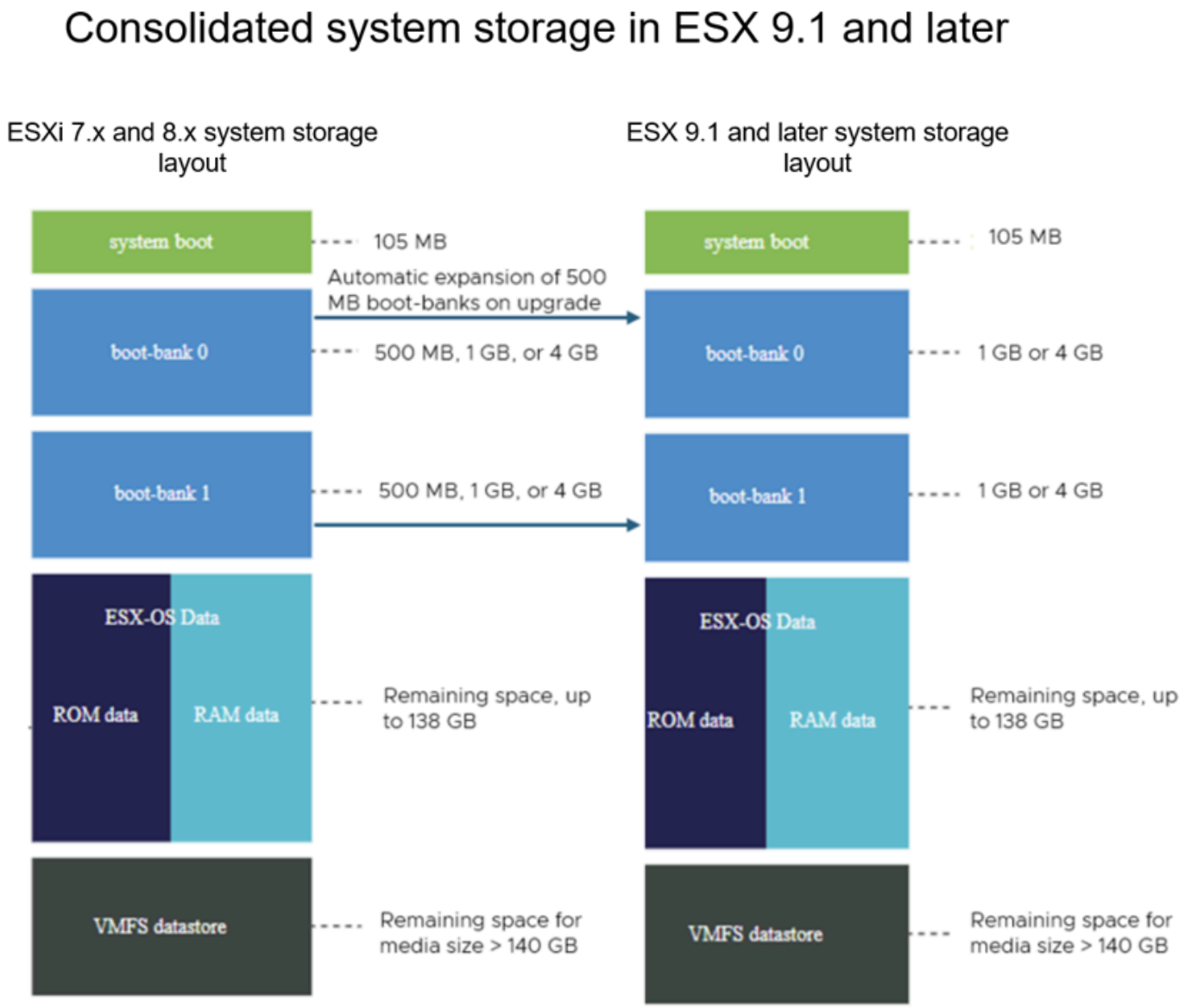
The ESX 9.1 and later system storage layout consists of four partitions:

Table 4: ESX system storage partitions:

Partition	Use	Type
System Boot	Stores boot loader and EFI modules.	FAT16
Boot-bank 0	System space to store ESX boot modules.	FAT16
Boot-bank 1	System space to store ESX boot modules.	FAT16
ESX-OSData	Acts as the unified location to store additional modules. Not used for booting and virtual machines. Consolidates the legacy `/scratch` partition, locker partition for VMware Tools, and core dump destinations. **CAUTION:** In case the installation media is a USB or an SD card device, best practice is to create ESX-OSData partitions on persistent storage device that is not shared between ESX hosts.	VMFS-L

The ESX-OSData volume is divided into two high-level categories of data, persistent and non-persistent data. Persistent data contains of data written infrequently, for example, VMware Tools ISOs, configurations, and core dumps.

Non-persistent data contains of frequently written data, for example, logs, VMFS global traces, vSAN Entry Persistence Daemon (EPD) data, vSAN traces, and real-time databases.



ESX System Storage Sizes

Partition sizes, except for the system boot partition, can vary depending on the size of the boot media used. If the boot media is a high-endurance one with capacity larger than 140 GB, a VMFS datastore is created automatically to store virtual machine data.

You can review the boot media capacity and the automatic sizing as configured by the ESX installer by using the vSphere Client and navigating to the **Partition Details** view. Alternatively, you can use ESXCLI, for example the `esxcli storage filesystem list` command.

Table 5: ESX System Storage Sizes, Depending on the Used Boot Media and Its Capacity.

Boot Media Size	8-32 GB	32-128 GB	>128 GB
System Boot	~ 105 MB	~ 105 MB	~ 105 MB
Boot-bank 0	~ 1 GB	~ 4 GB	~ 4 GB
Boot-bank 1	~ 1 GB	~ 4 GB	~ 4 GB
ESX-OSData	remaining space	remaining space	up to 138 GB
VMFS datastore			remaining space for media size > 140 GB

You can use the ESX installer boot option `systemMediaSize` to increase the size of system storage partitions on the boot media. The `systemMediaSize` parameter accepts the following values:

- default (138 GB)
- max (consume all available space, for multi-terabyte servers)

To set the boot option at install time, refer to Enter Boot Options to Run an Installation or Upgrade Script. For more information, see Broadcom knowledge base article 345195.

ESX System Storage Links

The sub-systems that require access to the ESX partitions, access these partitions by using the following symbolic links:

Table 6: ESX system storage symbolic links.

System Storage Volume	Symbolic Link
Boot-bank 0	`/bootbank`
Boot-bank 1	`/altbootbank`

Persistent data	`/productLocker`
	`/locker`
	`/var/core`
	`/usr/lib/vmware/isoimages`
	`/usr/lib/vmware/floppies`

Non-persistent data	`/var/run`
	`/var/log`
	`/var/vmware`
	`/var/tmp`
	`/scratch`

Storage Behavior

When you start ESX, the host enters an autoconfiguration phase during which system storage devices are configured with defaults.

When you reboot the ESX host after installing the ESX image, the host configures the system storage devices with default settings. Starting with ESX 7.0, you can activate the option autoPartition, which automatically formats all available empty devices with VMFS, except for legacy SD and USB devices. The default is `autoPartition=FALSE`, which formats with VMFS only boot devices with size larger than 140 GB. For more information, see Broadcom knowledge base article 313264.

> **CAUTION:**
> ESX overwrites any disks that appear to be blank. Disks are considered to be blank if they do not have a valid partition table or partitions. If you are using software that uses such disks, in particular if you are using logical

volume manager (LVM) instead of, or in addition to, conventional partitioning schemes, ESX might cause local LVM to be reformatted. Back up your system data before you power on ESX for the first time.

On the hard drive or USB device that the ESX host is booting from, the disk-formatting software retains existing diagnostic partitions that the hardware vendor creates. In the remaining space, the software creates the partitions described below.

Partitions Created by ESX on the Host Drive

For fresh installations, several new partitions are created for the system boot, boot banks, and ESX-OSData. Fresh ESX installations use GUID Partition Tables (GPT) instead of MSDOS-based partitioning. The installer creates boot banks of varying size depending on the size of the disk. For more information on the scratch partition see About the Scratch Partition.

The installer affects only the installation disk. The installer does not affect other disks of the server. When you install on a disk, the installer overwrites the entire disk. When the installer autoconfigures storage, the installer does not overwrite hardware vendor partitions.

To create the VMFS datastore, the ESX installer requires a minimum of 140 GB of free space on the installation disk.

You might want to override this default behavior if, for example, you use shared storage devices instead of local storage. To prevent automatic disk formatting, detach the local storage devices from the host under the following circumstances:

- Before you start the host for the first time.
- Before you start the host after you reset the host to the configuration defaults.

To override the VMFS formatting if automatic disk formatting already occurred, you can remove the datastore.

Upgrade on Systems with Existing 500 MB Boot-banks

Starting with ESX 9.0, boot-banks on the boot disk can only be 1 GB or 4 GB. For systems with existing 500 MB boot-banks, the upgrade process to ESX 9.1 or later automatically repartitions the boot device to create 1 GB boot-banks.

The following events occur during the repartitioning process:

- Existing 500 MB boot banks expand to ~1GB.
- The ESX-OSData volume is reduced by ~1 GB.
- Some non-critical system data might be lost.

Important:

Rollback from ESX 9.1 to 8.x is not possible in these cases due to the repartitioning process of the boot device. Rollback from ESX 9.1 to 8.x is possible as long as no changes to the bootbank partitions have been made and no corrupt partition is detected.

About the Scratch Partition

For new installations of ESX, during the autoconfiguration phase, a scratch partition is created as part of the ESX-OSData partition.

The scratch partition serves to store system logs, which you need when you create a support bundle. If the scratch partition is not present, system logs are stored in a ramdisk. If no scratch partition is created, you can configure one. You can also override the default configuration.

You can create the scratch partition on a remote SAN or NFS-mounted directory.

Set the Scratch Partition from the vSphere Client

If a scratch partition is not set up, you might want to configure one, especially if the host is low on memory. When a scratch partition is not present, system logs are stored in a ramdisk.

The directory to use for the scratch partition must exist on the host.

1. From the vSphere Client, connect to vCenter.
2. Select the ESX host in the inventory.
3. Click the **Configure** tab.
4. Select **System**.
5. Select **Advanced System Settings**.
   The setting **ScratchConfig.CurrentScratchLocation** shows the current location of the scratch partition.
6. In the **ScratchConfig.ConfiguredScratchLocation** text box, enter a directory path that is unique for this host. For example, `/vmfs/volumes/DatastoreUUID/DatastoreFolder`.
7. Reboot the host for the changes to take effect.

ESX Hardware Requirements

Make sure that the ESX host meets the minimum hardware configurations supported by ESX 9.1 and later.

Hardware and System Resources

Note:

Starting with VCF 9.1, support is deprecated and will be removed in the next major release for the following:

- USB and SD boot devices.
- ESX without a persistent system storage device for OSDATA partition.
- ESX with system storage device of less than 32 GB.
- APIs for reconfiguring the system storage locations.

To install or upgrade ESX, your hardware and system resources must meet the following requirements:

- Supported server platform. For a list of supported platforms, see the Broadcom Compatibility Guide.
- ESX 9.1 and later requires a host with at least two CPU cores.
- ESX 9.1 and later supports a broad range of multi-core 64-bit x86 processors. For a complete list of supported processors, see the Broadcom Compatibility Guide.
- ESX 9.1 and later requires the NX/XD bit to be enabled for the CPU in the BIOS.
- ESX 9.1 and later requires a minimum of 8 GB of physical RAM. Provide at least 12 GB of RAM to run virtual machines in typical production environments.
- To support 64-bit virtual machines, support for hardware virtualization (Intel VT-x or AMD RVI) must be enabled on x64 CPUs.
- One or more Gigabit or faster Ethernet controllers. For a list of supported network adapter models, see the Broadcom Compatibility Guide.
- ESX 9.1 and later requires a boot disk of at least 32 GB of persistent storage such as HDD, SSD, or NVMe. A boot device must not be shared between ESX hosts.
- SCSI disk or a local, non-network, RAID LUN with unpartitioned space for the virtual machines.
- For Serial ATA (SATA), a disk connected through supported SAS controllers or supported on-board SATA controllers. SATA disks are considered remote, not local. These disks are not used as a scratch partition by default because they are seen as remote.

Storage Systems

For a list of supported storage systems, see the Broadcom Compatibility Guide.

ESX Booting Requirements

- In ESX 9.1and later, support for legacy BIOS is limited and booting ESX hosts from the Unified Extensible Firmware Interface (UEFI) is recommended. For more information on Broadcom plans to deprecate support for legacy BIOS in server platforms, see Deprecation of legacy BIOS support in vSphere.

- With UEFI, you can boot systems from hard drives, CD-ROM drives, or USB media. vSphere Auto Deploy supports network booting and provisioning of ESX hosts with UEFI.
- If your system has supported Data Processing Units (DPU), you can only use UEFI to install and boot ESX on the DPUs.
- If the system firmware and the firmware on any add-in card that you are using support it, ESX can boot from a disk larger than 2 TB. See the vendor documentation.

Storage Requirements for ESX 9.1 Installation or Upgrade

For best performance of an ESX 9.1 installation, when deploying new servers, use a persistent storage device that is a minimum of 128 GB for boot devices. Upgrading servers with boot disk of size less than 8 GB to ESX 9.1 is blocked.

Note:

In ESX 9.1, the ESX-OSData volume is considered a unified partition and the separate components, such as `/scratch` and VMware Tools, are consolidated into a single persistent OSDATA partition.

Other options for best performance of an ESX 9.1 installation are the following:

- A persistent storage device that supports the minimum of 128 TB written (TBW).
- A device that delivers at least 100 MB/s of sequential write speed.
- To provide resiliency in case of device failure, a RAID 1 mirrored device is recommended.

Legacy SD and USB devices are supported with the following limitations:

- SD and USB devices are supported for boot bank partitions. The use of SD and USB devices for storing ESX-OSData partitions is being deprecated and best practice is to provide a separate persistent local device with a minimum of 128 GB to store the ESX-OSData volume. The persistent local boot device can be an industrial grade M.2 flash (SLC and MLC), SAS, SATA, HDD, SSD, or a NVMe device.
- If you do not provide persistent storage, you see an alarm such as `Secondary persistent device not found. Please move installation to persistent storage as support for SD-Card/USB only configuration is being deprecated`.
- You must use an SD flash device that is approved by the server vendor for the particular server model on which you want to install ESX on an SD flash storage device. You can find a list of validated devices in the Broadcom Compatibility Guide.
- See SD card/USB boot device revised guidance on updated guidance for SD card or USB-based environments.

If you use USB or SD devices to perform an upgrade, best practice is to allocate an ESX-OSData region on an available persistent disk or a SAN LUN. If persistent storage or a SAN LUN are not available, ESX-OSData is automatically created on a RAM disk. VMFS can also be used for ESX-OSData partition.

After upgrade, if ESX-OSData resides on a RAM disk and a new persistent device is found on subsequent boots, and this device has the setting `autoPartition=True`, ESX-OSData is automatically created on the new persistent device. ESX-OSData does not move between persistent storage automatically, but you can manually change the ESX-OSData location on a supported storage.

To reconfigure `/scratch`, see Set the Scratch Partition from the vSphere Client.

To configure the size of ESX system partitions, you can use the `systemMediaSize` option. For more information, see Boot option to configure the size of ESX system partitions.

In vSphere Auto Deploy installations, the installer attempts to allocate a scratch region on an available local disk or datastore. If no local disk or datastore is found, installation fails.

For environments that boot from a SAN or use vSphere Auto Deploy, the ESX-OSData volume for each ESX host must be set up on a separate SAN LUN.

Using Remote Management Applications

Remote management applications allow you to install ESX on servers that are in remote locations.

Remote management applications supported for installation include HP Integrated Lights-Out (iLO), Dell Remote Access Card (DRAC), IBM management module (MM), Fujitsu iRMC, and Remote Supervisor Adapter II (RSA II). For support on remote management applications, contact the vendor.

You can use remote management applications to do both interactive and scripted installations of ESX remotely.

If you use remote management applications to install ESX, the virtual CD might encounter corruption problems with systems or networks operating at peak capacity. If a remote installation from an ISO image fails, complete the installation from the physical CD media.

Supported Remote Management Server Models and Firmware Versions

You can use remote management applications to install or upgrade ESX, or to manage hosts remotely.

Table 7: Supported Remote Management Server Models and Minimum Firmware Versions

Remote Management Server Model	Firmware Version	Java
Dell DRAC 10	1.10.17.00	N/A
Dell DRAC 9	6.0.30.00	N/A
Dell DRAC 7	1.30.30 (Build 43)	1.7.0_60-b19
Dell DRAC 6	1.54 (Build 15), 1.70 (Build 21)	1.6.0_24
Dell DRAC 5	1.0, 1.45, 1.51	1.6.0_20,1.6.0_203
Dell DRAC 4	1.75	1.6.0_23
Fujitsu iRMC S5	1.10P	1.7.0_60-b19
Fujitsu iRMC S6	1.06S	N/A
HP ILO	1.81, 1.92	1.6.0_22, 1.6.0_23
HP ILO 2	1.8, 1.81	1.6.0_20, 1.6.0_23
HP ILO 3	1.28	1.7.0_60-b19
HP ILO 4	1.13	1.7.0_60-b19
HP ILO 5	2.72	N/A
HPE ILO 6	1.10	N/A
HPE ILO 7	1.20	N/A
IBM RSA 2	1.03, 1.2	1.6.0_22
Lenovo XClarity Controller 2 (XCC2)	8.10 (iyx346k-8.10)	N/A
Lenovo XClarity Controller 3 (XCC3)	9.11 (qgx3a2g-9.11)	N/A

Recommendations for Enhanced ESX Performance

To enhance performance, install or upgrade ESX on a robust system with more RAM than the minimum required and with multiple physical disks.

For ESX system requirements, see ESX Hardware Requirements.

Table 8: Recommendations for Enhanced Performance

System Element	Recommendation
RAM	ESX hosts require more RAM than typical servers. ESX 9.1 requires a minimum of 8 GB of physical RAM. Provide at least 12 GB of RAM to take full advantage of ESX features and run virtual machines in typical production environments. An ESX host must have sufficient RAM to run concurrent virtual machines. For possible memory savings from using variable overhead memory for each virtual machine, see *vSphere Resource Management*.
Dedicated Fast Ethernet adapters for virtual machines	Place the management network and virtual machine networks on different physical network cards. Dedicated Gigabit Ethernet cards for virtual machines, such as Intel PRO 1000 adapters, improve throughput to virtual machines with high network traffic.
Disk location	Place all data that your virtual machines use on physical disks allocated specifically to virtual machines. Performance is better when you do not place your virtual machines on the disk containing the ESX boot image. Use physical disks that are large enough to hold disk images that all the virtual machines use.
VMFS6 partitioning	The ESX installer creates the initial VMFS volumes on the first blank local disk found. To add disks or modify the original configuration, use the vSphere Client. This practice ensures that the starting sectors of partitions are 64K-aligned, which improves storage performance. **Note:** For SAS-only environments, the installer might not format the disks. For some SAS disks, it is not possible to identify whether the disks are local or remote. After the installation, you can use the vSphere Client to set up VMFS.
Processors	Faster processors improve ESX performance. For certain workloads, larger caches improve ESX performance.
Hardware compatibility	Use devices in your server that are supported by ESX drivers. See the Broadcom Compatibility Guide.

Required Free Space for System Logging

See the recommended minimum size and rotation configuration for hostd, vpxa, and fdm logs.

If you use Auto Deploy to install your ESX 9.1 host, or if you set up a log directory separate from the default location in a scratch directory on the VMFS volume, you might need to change your current log size and rotation settings to ensure that enough space is available for system logging. All vSphere components use this infrastructure. The default values for log capacity in this infrastructure vary, depending on the amount of storage available and on how you have configured system logging. ESX hosts that are deployed with Auto Deploy store logs on a RAM disk, which means that the amount of space available for logs is small.

If your ESX host is deployed with Auto Deploy, reconfigure your log storage in one of the following ways:

- Redirect logs over the network to a remote host, called syslog collector, such as VCF Operations Log Management (formerly
  VCF Operations for Logs, VMware vRealize Log Insight, and vCenter Log Insight), to collect syslog messages.
- Redirect logs to a NAS or NFS store.

If you redirect logs to non-default storage, such as a NAS or NFS store, you might also want to reconfigure log sizing and rotations for ESX hosts that are installed to disk.

You do not need to reconfigure log storage for ESX hosts that use the default configuration, which stores logs in a scratch directory on the VMFS volume. For these hosts, ESX 9.1 configures logs to best suit your installation, and provides enough space to accommodate log messages.

Table 9: Recommended Minimum Size and Rotation Configuration for hostd, vpxa, and fdm Logs

Log	Maximum Log File Size	Number of Log Files to Preserve	Minimum Disk Space Required
Management Agent (hostd)	10 MB	10	100 MB
VirtualCenter Agent (vpxa)	5 MB	10	50 MB
vSphere HA agent (Fault Domain Manager, fdm)	5 MB	10	50 MB

For information about setting up a remote log server, see Configure Syslog on ESX Hosts.

You can optionally install Log Management, which provides log aggregation and analytics. For more information, see Deploy Log Management.

VMware Host Client System Requirements

Make sure that your browser supports the VMware Host Client.

The following guest operating systems and Web browser versions are supported for the VMware Host Client.

Supported Browsers	Mac OS	Windows 32-bit and 64-bit	Linux
Google Chrome	120+	120+	120+
Mozilla Firefox	115+	115+	115+
Microsoft Edge	120+	120+	N/A

What is VMware vSphere Distributed Services Engine®

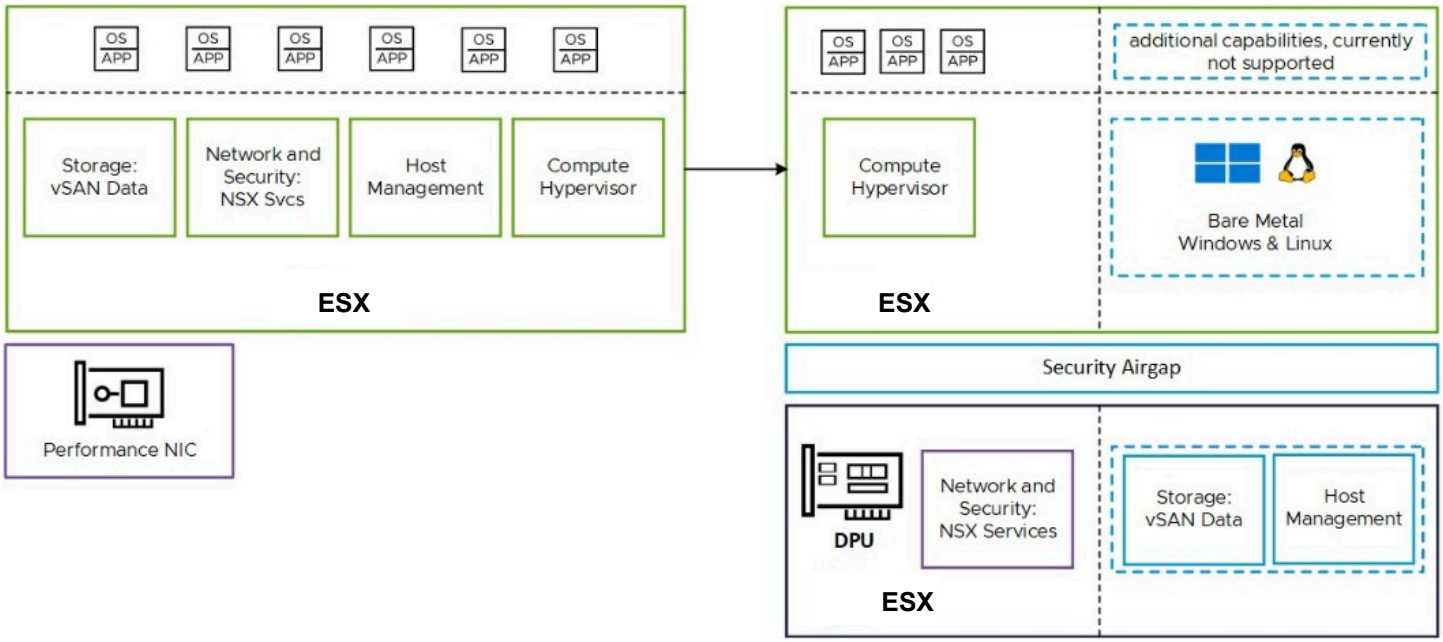
vSphere Distributed Services Engine® is a core vSphere capability that enables customers to use DPUs with vSphere and VMware Cloud Foundation.

ESX 8.0 and later enables breakthrough workload performance to meet ever-increasing throughput and latency needs. With vSphere Distributed Services Engine, infrastructure services are distributed across the different compute resources available on the ESX host, with networking functions offloaded to the DPU. Such a capability works well for modern applications, which are developed using a microservices architecture approach that seeks to break down the application into multiple independent but cooperating services. This increased complexity places new demand for the CPU. For example, processing storage requests or shuttling network traffic for these microservices leaves fewer CPU cycles for the actual workload. In this context, purpose-built accelerators such as DPUs can take on the new compute burden and help you improve the performance and efficiency of infrastructure.

With vSphere Distributed Services Engine, DPUs can accelerate the performance of your network and increase data throughput, while placing no operational burden of managing the lifecycle of DPUs, as the existing Day-0, Day-1, and Day-2 vSphere experience does not change. vSphere Distributed Services Engine is supported by DPUs from NVIDIA and AMD, and server designs from Dell, HPE, Lenovo, and Fujitsu. vSphere Distributed Services Engine is available on servers with pre-installed DPUs.

Starting with ESX 8.0, you can offload functionality that runs on the core CPU onto the DPU to significantly improve the network and security performance. As illustrated in the Evolving vSphere Architecture diagram, DPUs can also handle additional capabilities such as storage offload and bare metal management, but these additional capabilities are currently not supported.

Figure 1: Evolving vSphere Architecture



vSphere Distributed Services Engine offloads and accelerates infrastructure functions on the DPU by introducing a VMware vSphere Distributed Switch on the DPU and VMware NSX Networking and Observability, which allows you to proactively monitor, identify, and mitigate network infrastructure bottlenecks without complex network taps. The DPU becomes a new control point to scale infrastructure functions and enables security controls that are agentless and decoupled from the workload domain.

With vSphere Distributed Services Engine, you can:

- Install and update ESX images simultaneously on the x86 server and the attached supported DPU to reduce operational overhead of DPU lifecycle management with integrated vSphere workflows.
- Set alarms for DPU hardware alerts and monitor performance metrics on core, memory, and network throughput from the familiar vCenter interfaces, without the need of new tools.
- Accelerate vSphere Distributed Switch on the DPU to improve network performance and utilize available CPU cycles to achieve higher workload consolidation per ESX host.
- Get vSphere DRS and vSphere vMotion support for VMs running on ESX hosts with DPUs attached to get the benefits of passthrough without sacrificing on VM portability.
- Improve the security of infrastructure with zero-trust security.

vSphere Distributed Services Engine does not require a separate ESX license. An internal network that is isolated from other networks, connects the DPUs with ESX hosts. ESX 8.0 and later server builds are unified images, which contain both x86 and DPU content. In your environment, you see DPUs as new objects during installation and upgrade, and in networking, storage, and host profile workflows.

High Availability with VMware vSphere Distributed Services Engine

With ESX 8.0 Update 3 and later, you can opt for a VMware vSphere Distributed Services Engine installation with 2 data processing units (DPUs) to achieve high availability.

In vSphere systems with a single DPU, the device might become the single point of failure for workloads offloaded to the DPU, such as networking functions, and impact data and productivity. Starting with ESX 8.0 Update 3, vSphere Distributed Services Engine is also available on servers with 2 pre-installed DPUs, which provides hardware redundancy and resiliency.

You can utilize the two DPUs in Active/Standby mode to provide high availability. Such configuration provides redundancy in the event one of the DPUs fails. In the high availability configuration, both DPUs are assigned to the same NSX-backed vSphere Distributed Switch. For example, DPU-1 is attached to vmnic0 and vmnic1 of the vSphere Distributed Switch and DPU-2 is attached to vmnic2 and vmnic3 of the same vSphere Distributed Switch.

You can also utilize the two DPUs as independent devices to increase offload capacity per ESX host. Each DPU is attached to a separate vSphere Distributed Switch and you have no failover between DPUs in such configuration.

Dual-DPU systems can use NVIDIA or Pensando devices. In ESX 8.0 Update 3 and later, dual-DPU systems are supported by Lenovo server designs. The DPU devices on a dual DPU server must be identical in all aspects: same vendor, same hardware version and same firmware. For a list of current vendors and server designs for VMware vSphere Distributed Services Engine, see the Broadcom Compatibility Guide.

Installation of VMware vSphere Distributed Services Engine with 2 DPUs

vSphere Distributed Services Engine does not require a separate ESX license. ESX 8.0 Update 3 and later server builds are unified images, which contain both x86 and DPU content, and you cannot install x86 and DPU content separately. The installation procedure on both DPUs, either interactive or scripted, also happens in parallel and you see minimal performance loss as compared to a single-DPU system.

Starting with ESX 8.0 Update 3, you can get a pre-installed server configuration with 2 DPUs from Dell or Lenovo, or add a second DPU to a single DPU system on the supported dual DPU servers from Dell or Lenovo.

Note:

In any case, you need to run a complete fresh ESX 8.0 Update 3 or later installation on your system, not only on the newly added DPUs.

Error Handling, Failover, and Rollback for VMware vSphere Distributed Services Engine

Before installing VMware vSphere Distributed Services Engine, see the error handling, failover, and rollback options.

Error Handling

An installation failure of either x86 and DPU content on an ESX host marks the entire installation procedure as failed.

While the expectation is that the software state of DPUs remains identical at all times, in the unlikely case of an error during a lifecycle operation, such as installation or upgrade of a component, the operation might pass on one DPU but fail on the other. Since each lifecycle operation occurs within the boundaries of each DPU, errors do not affect the state of the other DPU, but the overall result of the installation is still marked as a failure.

During interactive install, in vSphere Lifecycle Manager workflows, and when you use ESXCLI, you receive information about the DPU on which the operation failed.

After a successful installation, in case of DPU errors, the recommended action is to restart the affected ESX host. If the DPU is still accessible from the ESX host, the general log bundle collection is sufficient for troubleshooting. If the DPU is

not accessible from the ESX host, logging in to the DPU from a BMC, iLO, or iDRAC interface can provide troubleshooting logs.

Failover

Failover support in ESX 8.0 Update 3 and later is limited to one of the DPUs becoming non-functional due to software errors within the DPU or a physical disconnect of one of the DPUs, such as cable disconnect. Failover due to Peripheral Component Interconnect (PCI) level errors is not supported.

Rollback

Rollback is a best effort mechanism to restore the system to a previous working state in case of a failure before the jumpstart phase of the ESX boot. Rollback on both x86 servers and the attached supported DPUs is automatic in case of an error during booting. You can also opt for a manual rollback by pressing `Shift+R` before the bootloader starts, to return to a previous good state.

Any failure after the jumpstart phase starts does not result in a rollback.

Table 10: Rollback scenarios for VMware vSphere Distributed Services Engine installation

Scenario	Number of reboots required
Both DPUs boot correctly. ESX does not boot correctly.	2
Both DPUs do not boot correctly. ESX boots correctly.	1
One of the DPUs boots with an earlier version than the other DPU and ESX .	2
One of the DPUs boots with an earlier version than the other DPU and ESX does not boot correctly.	2

Customizing Installations with vSphere ESX Image Builder

You can use VMware vSphere®ESX™ Image Builder CLI to create ESX installation images with a customized set of updates, patches, and drivers.

You can use vSphere ESX Image Builder with the vSphere Client or with VCF PowerCLI to create an ESX installation image with a customized set of ESX updates and patches. You can also include third-party network or storage drivers that are released between vSphere releases.

You can deploy an ESX image created with vSphere ESX Image Builder in either of the following ways:

- By burning it to an installation DVD.
- Through vCenter, using the Auto Deploy feature.

In this chapter, you can read about:

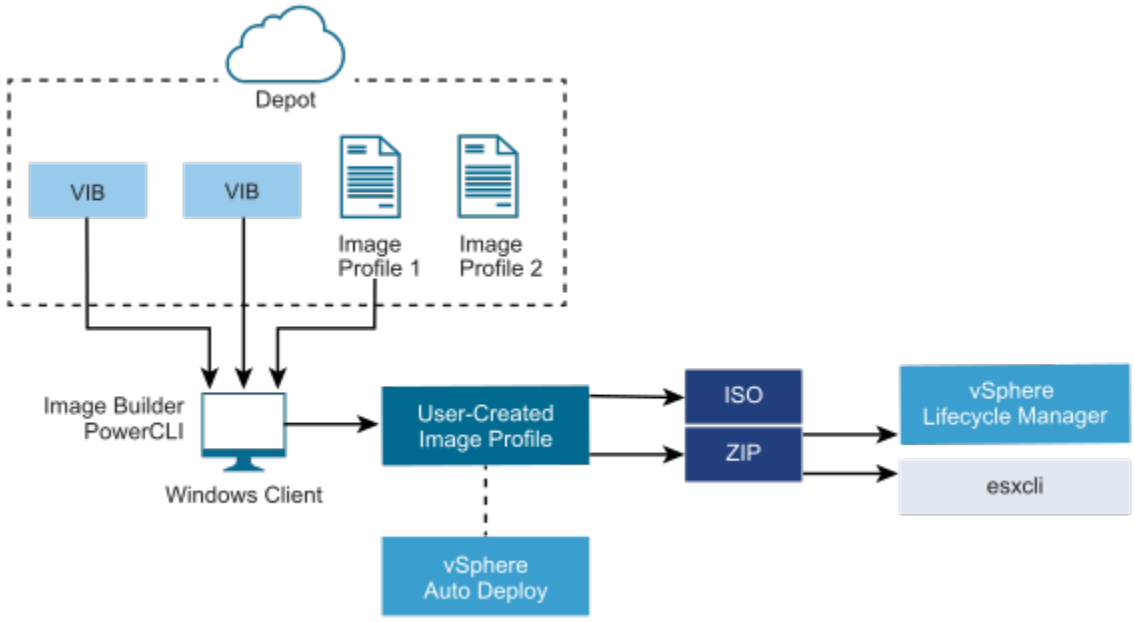
- How the vSphere ESX Image Builder Works
- Structure of ImageProfile, SoftwarePackage, and ImageProfileDiff Objects
- Configure vSphere ESX Image Builder
- Using VMware.Image Builder Cmdlets
- ESX Image Profile Tasks
- vSphere ESX Image Builder Workflows with VCF PowerCLI Cmdlets

How the vSphere ESX Image Builder Works

Create ESX image profiles for use by vSphere Auto Deploy, add custom third-party drivers to image profiles and export them, or perform upgrades.

With vSphere ESX Image Builder you can create ESX image profiles for use by vSphere Auto Deploy, add custom third-party drivers to existing image profiles and export to ISO or bundle, and perform upgrades. For basic concepts related to the way vSphere software is created, packaged, and distributed, see the Software Packaging Units That vSphere Lifecycle Manager Can Consume topic in the *Managing Host and Cluster Lifecycle* guide.

Figure 2: Image Builder Architecture



You use vSphere ESX Image Builder cmdlets for managing the software to deploy to your ESX hosts in several different situations.

Table 11: Cases Where You Can Use vSphere ESX Image Builder

Use Case for vSphere ESX Image Builder	Description
Create image profiles for use by vSphere Auto Deploy	Use vSphere ESX Image Builder to create an image profile that defines the VIBs that vSphere Auto Deploy uses to provision hosts.
Add custom third-party drivers to existing image profile and export to ISO or bundle	When you add a third-party driver or extension custom VIBs to your ESX hosts, use vSphere ESX Image Builder to clone the base image provided by Broadcom, add the custom VIBs, and export to ISO or to offline bundle ZIP file.
Perform upgrades	If you upgrade a system that includes custom extensions or drivers, you can use vSphere ESX Image Builder to create a custom image profile that includes vSphere 8.0 and later compatible VIBs for the custom extensions. Export the custom image profile to an ISO or to a ZIP to upgrade your system by using vSphere Lifecycle Manager baselines.

The vSphere ESX Image Builder cmdlets take image profiles and VIBs as input and produce various outputs.

Table 12: Input and Output to the vSphere ESX Image Builder Cmdlets

Parameter	Description
Input	Image profiles and VIBs that are located in a software depot are used as input to VCF PowerCLI cmdlets running on a Windows client.
Output	VCF PowerCLI cmdlets create custom image profiles that can be exported to an ISO image or an offline depot ZIP file. ISO images are used for installation. The ZIP depot can be used by vSphere Lifecycle Manager or by `esxcli software` commands to update or install images. Image profiles are also used in vSphere Auto Deploy rules to customize the software to provision ESX hosts with.

Image Profiles

Image profiles define the set of VIBs that an ESX installation or update process uses. Image profiles apply to ESX hosts provisioned with vSphere Auto Deploy. You define and manipulate image profiles with vSphere ESX Image Builder.

Image Profile Requirements

You can create a custom image profile from scratch or clone an existing profile and add or remove VIBs. A profile must meet the following requirements to be valid.

- Each image profile must have a unique name and vendor combination.
- Each image profile has an acceptance level. When you add a VIB to an image profile with an vSphere ESX Image Builder cmdlet, Image Builder checks that the VIB matches the acceptance level defined for the profile.
- You cannot remove VIBs that are required by other VIBs.
- You cannot include two versions of the same VIB in an image profile. When you add a new version of a VIB, the new version replaces the existing version of the VIB.

Image Profile Validation

An image profile and its VIBs must meet several criteria to be valid.

- Image profiles must contain at least one base VIB and one bootable kernel module.
- If any VIB in the image profile depends on another VIB, that other VIB must also be included in the image profile. VIB creators store that information in the SoftwarePackage object's Depends property.
- VIBs must not conflict with each other. VIB creators store conflict information in the SoftwarePackage object's Conflicts property.
- Two VIBs with the same name, but two different versions, cannot coexist. When you add a new version of a VIB, the new version replaces the existing version of the VIB.
- No acceptance level validation issues exist.

When you make a change to an image profile, vSphere ESX Image Builder checks that the change does not invalidate the profile.

Dependency Validation	When you add or remove a VIB, vSphere ESX Image Builder checks that package dependencies are met. Each SoftwarePackage object includes a Depends property that specifies a list of other VIBs that VIB depends on. See Structure of ImageProfile, SoftwarePackage, and ImageProfileDiff Objects .
Acceptance Level Validation	vSphere ESX Image Builder performs acceptance level validation each time an image profile is created or changed. vSphere ESX Image Builder checks the acceptance level of VIBs in the image profile against the minimum allowed acceptance level of the profile. The acceptance level of the VIB is also validated each time the signature of a VIB is validated.

VIB Validation During Export

When you export an image profile to an ISO, vSphere ESX Image Builder validates each VIB by performing the following actions.

- Checks that no conflicts exist by checking the Conflicts property of each SoftwarePackage object.
- Performs VIB signature validation. Signature validation prevents unauthorized modification of VIB packages. The signature is a cryptographic checksum that guarantees that a VIB was produced by its author. Signature validation also happens during installation of VIBs on an ESX host and when the vSphere Auto Deploy server uses VIBs.
- Checks that VIBs follow file path usage rules. VMware tests `VMwareCertified` and `VMwareAccepted` VIBs to guarantee those VIBs always follow file path usage rules.

Working with Acceptance Levels

Hosts, image profiles, and individual VIBs have acceptance levels. VIB acceptance levels show how the VIB was tested. Understanding what each acceptance level implies, how to change levels, and what a change implies is an important part of installation and update procedures.

Acceptance levels are set for hosts, image profiles, and individual VIBs. The default acceptance level for an ESX image or image profile is PartnerSupported.

Host acceptance levels The host acceptance level determines which VIBs you can install on a host. You can change a host's acceptance level with ESXCLI commands. By default, ESX hosts have an acceptance level of `PartnerSupported` to allow for easy updates with `PartnerSupported` VIBs.

> **Note:**
>
> VMware supports hosts at the `PartnerSupported` acceptance level. For problems with individual VIBs with `PartnerSupported` acceptance level, contact your partner's support organization.

Image profile acceptance levels The image profile acceptance level is set to the lowest VIB acceptance level in the image profile. If you want to add a VIB with a low acceptance level to an image profile, you can change the image profile acceptance level with the `Set-EsxImageProfile` cmdlet. See Set the Image Profile Acceptance Level.

The vSphere Lifecycle Manager does not display the actual acceptance level. Use vSphere ESX Image Builder cmdlets to retrieve the acceptance level information for VIBs and image profiles.

VIB acceptance levels A VIB's acceptance level is set when the VIB is created. Only the VIB creator can set the acceptance level.

If you attempt to provision a host with an image profile or VIB that has a lower acceptance level than the host, an error occurs. Change the acceptance level of the host to install the image profile or VIB. See Change the Host Acceptance Level. Changing the acceptance level of the host changes the support level for that host.

The acceptance level of a host, image profile, or VIB lets you determine who tested the VIB and who supports the VIB. VMware supports the following acceptance levels .

VMwareCertified The VMwareCertified acceptance level has the most stringent requirements. VIBs with this level go through thorough testing fully equivalent to VMware in-house Quality Assurance testing for the same technology. Today, only I/O Vendor Program (IOVP) program drivers are published at this level. VMware takes support calls for VIBs with this acceptance level.

VMwareAccepted VIBs with this acceptance level go through verification testing, but the tests do not fully test every function of the software. The partner runs the tests and VMware verifies the result. Today, CIM providers and PSA plug-ins are among the VIBs published at this level. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.

PartnerSupported VIBs with the PartnerSupported acceptance level are published by a partner that VMware trusts. The partner performs all testing. VMware does not verify the results. This level is used for a new or nonmainstream technology that partners want to enable for VMware systems. Today, driver VIB technologies such as Infiniband, ATAoE, and SSD are at this level with nonstandard hardware drivers. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.

CommunitySupported The CommunitySupported acceptance level is for VIBs created by individuals or companies outside of VMware partner programs. VIBs at this level have not gone through any VMware-approved testing program and are not supported by VMware Technical Support or by a VMware partner.

Change the Host Acceptance Level

You can lower the host acceptance level to match the acceptance level for a VIB or image profile you want to install.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

The acceptance level of each VIB on a host must be at least as high as the acceptance level of the host. For example, you cannot install a VIB with PartnerSupported acceptance level on a host with `VMwareAccepted` acceptance level. You must first lower the acceptance level of the host. For more information on acceptance levels, see Working with Acceptance Levels.

> **Warning:**
>
> Changing the host acceptance level to `CommunitySupported` affects the supportability of your host and might affect the security of your host.

1. Retrieve the acceptance level for the VIB or image profile.

Option	Description
View information for all VIBs	`esxcli --server=`*`server_name`* `software`

Option	Description
	`sources vib list --depot=depot_URL`
View information for a specified VIB	`esxcli --server=server_name software sources vib list --viburl=vib_URL`
View information for all image profiles	`esxcli --server=server_name software sources profile list --depot=depot_URL`
View information for a specified image profile	`esxcli --server=server_name software sources profile get --depot=depot_URL --profile=profile_name`

2. View the host acceptance level.
   `esxcli --server=server_name software acceptance get`
3. Change the acceptance level of the host.
   ```
   esxcli
       --server=server_name software acceptance set --level=acceptance_level
   ```
   The value for *acceptance_level* can be `VMwareCertified`, `VMwareAccepted`, `PartnerSupported`, or `CommunitySupported`. The values for *acceptance_level* are case-sensitive.

   Note:
   If the host has a higher acceptance level than the VIB or image profile you want to add, you can run commands in the `esxcli software vib` or `esxcli software profile` namespace with the --force option. When you use the --force option, a warning appears because you enforce a VIB or image profile with lower acceptance level than the acceptance level of the host and your setup is no longer consistent. The warning is repeated when you install VIBs, remove VIBs, or perform certain other operations on the host that has inconsistent acceptance levels.

Set the Image Profile Acceptance Level

If you want to add a VIB to an image profile, and the acceptance level of the VIB is lower than that of the image profile, you can clone the image profile with a lower acceptance level or change the image profile's acceptance level.

Install VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.

You can specify `VMwareCertified`, `VMwareAccepted`, `PartnerSupported`, or `CommunitySupported` as an acceptance level of an image profile. If you lower the acceptance level, the level of support for the image profile and hosts that you provision with it changes. For more information, see Working with Acceptance Levels.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.
2. Get the acceptance level for the image profile.
   `Get-EsxImageProfile -Name string`
3. Set the acceptance level of the image profile.
   `Set-EsxImageProfile -Name string -AcceptanceLevel level`

Structure of ImageProfile, SoftwarePackage, and ImageProfileDiff Objects

Knowing the structure of `ImageProfile`, `SoftwarePackage`, and `ImageProfileDiff` objects helps you manage deployment and upgrade processes.

ImageProfile Object Properties

The `ImageProfile` object, which is accessible with the `Get-EsxImageProfile` VCF PowerCLI cmdlet, has the following properties.

Name	Type	Description
`AcceptanceLevel`	`AcceptanceLevel`	Determines which VIBs you can add to the profile. Levels are VMwareCertified, VMwareAccepted, PartnerSupported, and CommunitySupported. See Working with Acceptance Levels.
`Author`	`System.String`	The person who created the profile. 60 characters or fewer
`CreationTime`	`System.DateTime`	The timestamp of creation time
`Description`	`System.String`	The full-text description of profile. No length limit.
`GUID`	`System.String`	Globally unique ID of the image profile.
`ModifiedTime`	`System.DateTime`	The timestamp of last modification time.
`Name`	`System.String`	The name of the image profile. 80 characters or fewer.
`ReadOnly`	`System.Boolean`	When set to `true`, the profile cannot be edited. Use `Set-EsxImageProfile -Readonly` to make your custom image profiles read only.
`Rules`	`ImageProfileRule[]`	Any OEM hardware requirements and restrictions that the image profile might have. If matching hardware is available, vSphere Auto Deploy verifies the value of this property when deploying an image profile and deploys the profile.
`Vendor`	`System.String`	The organization that publishes the profile. 40 characters or fewer.
`VibList`	`SoftwarePackage[]`	The list of VIB IDs the image contains.

SoftwarePackage Object Properties

When preparing an image profile, you can examine software packages to decide which packages are suitable for inclusion. The `SoftwarePackage` object has the following properties.

Name	Type	Description
`AcceptanceLevel`	`AcceptanceLevel`	The acceptance level of this VIB.
`Conflicts`	`SoftwareConstraint[]`	A list of VIBs that cannot be installed at the same time as this VIB. Each constraint uses the following format: `package-name[<<|<=|=|>=|<< version]`
`Depends`	`SoftwareConstraint[]`	A list of VIBs that must be installed at the same time as this VIB. Same constraint format as `Conflicts` property.
`Description`	`System.String`	The long description of the VIB.
`Guid`	`System.String`	The unique ID for the VIB.
`LiveInstallOk`	`System.Boolean`	True if live installations of this VIB are supported.
`LiveRemoveOk`	`System.Boolean`	True if live removals of this VIB are supported.
`MaintenanceMode`	`System.Boolean`	True if hosts must be in maintenance mode for installation of this VIB.

Name	Type	Description
Name	System.String	The name of the VIB. Uniquely describes the package on a running ESX system.
Provides	SoftwareProvides	The list of virtual packages or interfaces this VIB provides. See SoftwareProvide Object Properties.
ReferenceURLs	SupportReference[]	The list of SupportReference objects with in-depth support information. The SupportReference object has two properties, Title and URL, both of type System.String.
Replaces	SoftwareConstraint[]	The list of SoftwareConstraint objects that identify VIBs that replace this VIB or make it obsolete. VIBs automatically replace VIBs with the same name but lower versions.
ReleaseDate	System.DateTime	Date and time of VIB publication or release.
SourceUrls	System.String[]	The list of source URLs from which this VIB can be downloaded.
StatelessReady	System.Boolean	True if the package supports host profiles or other technologies that make it suitable for use with vSphere Auto Deploy.
Summary	System.String	A one-line summary of the VIB.
Tags	System.String[]	An array of string tags for this package, defined by the vendor or publisher. Tags can be used to identify characteristics of a package.
Vendor	System.String	The VIB vendor or publisher.
Version	System.String	The VIB version.
VersionObject	Software.Version	The VersionObject property is of type SoftwareVersion. The SoftwareVersion class implements a static Compare method to compare two versions of strings. See SoftwareVersion Object Properties.

ImageProfileDiff Object Properties

When you run the Compare-EsxImageProfile cmdlet, you pass in two parameters, first the reference profile, and then the comparison profile. The cmdlet returns an ImageProfileDiff object, which has the following properties.

Name	Type	Description
CompAcceptanceLevel	System.String	The acceptance level for the second profile that you passed to Compare-EsxImageProfile.
DowngradeFromRef	System.String[]	The list of VIBs in the second profile that are downgrades from VIBs in the first profile.
Equal	System.Boolean	True if the two image profiles have identical packages and acceptance levels.
OnlyInComp	System.String	The list of VIBs found only in the second profile that you passed to Compare-EsxImageProfile.
OnlyInRef	System.String[]	The list of VIBs found only in the first profile that you passed to Compare-EsxImageProfile.
PackagesEqual	System.Boolean	True if the image profiles have identical sets of VIB packages.

Name	Type	Description
RefAcceptanceLevel	System.String	The acceptance level for the first profile that you passed to Compare-EsxImageProfile.
UpgradeFromRef	System.String[]	The list of VIBs in the second profile that are upgrades from VIBs in the first profile.

SoftwareVersion Object Properties

The SoftwareVersion object lets you compare two version strings. The object includes a Compare static method that accepts two strings as input. If the first version string is a higher number than the second version stringand, it returns 1. Compare returns 0 if two versions strings are equal. Compare returns -1 if the second version string is a higher number than the first string. The object has the following properties.

Name	Type	Description
Version	System.String	The part of the version before the hyphen. This part indicates the primary version.
Release	System.String	The part of the version after the hyphen. This part indicates the release version.

SoftwareConstraint Object Properties

The SoftwareConstraint object implements a MatchesProvide method. The method accepts a SoftwareProvides or SoftwarePackage object as input. If the constraint matches the SoftwareProvide or the SoftwarePackage, it returns True or returns False otherwise.

The SoftwareConstraint object includes the following properties.

Name	Type	Description
Name	System.String	The name of the constraint. This name should match a corresponding SoftwareProvideName property.
Relation	System.String	An enum, or one of the following comparison indicators: <<, <=, = >=, >>. This property can be $null if the constraint does not have a Relation and Version property.
Version	System.String	The version to match the constraint against. This property can be $null if the constraint does not have a Relation and Version property.
VersionObject	SoftwareVersion	The version represented by a SoftwareVersion object.

SoftwareProvide Object Properties

The SoftwareProvide object includes the following properties.

Name	Type	Description
Name	System.String	The name of the provide.
Version	System.String	The version of the provide. Can be $null if the provider does not specify a version.
Release	System.String	The version of the provide as represented by a SoftwareVersion object. See Structure of ImageProfile, SoftwarePackage, and ImageProfileDiff Objects.

Configure vSphere ESX Image Builder

Before you can run vSphere ESX Image Builder cmdlets, you must install VCF PowerCLI and all prerequisite software.

Before you can run vSphere ESX Image Builder cmdlets, you must install VCF PowerCLI and all prerequisite software. The vSphere ESX Image Builder module is included with the VCF PowerCLI installation. If you want to manage vSphere ESX Image Builder with VCF PowerCLI cmdlets, verify that PowerShell 7.5 is installed. See the VCF PowerCLI User's Guide.

1. Open PowerShell on your workstation.
2. To install all VCF PowerCLI modules, run the command: `Install-Module VCF.PowerCLI -Scope CurrentUser`. Alternatively, you can install individual VCF PowerCLI modules by running the `Install-Module` cmdlet with the module name. If you see a warning that you are installing modules from an untrusted repository, press **y** and then press **Enter** to confirm the installation.

   You can verify that the VCF PowerCLI module is available by using the command:

   ```
   Get-Module -Name VMware.PowerCLI* -ListAvailable
   ```

Review Using VMware.Image Builder Cmdlets. If you are new to VCF PowerCLI, read the *VCF PowerCLI User's Guide*.

Use vSphere ESX Image Builder cmdlets and other VCF PowerCLI cmdlets and PowerShell cmdlets to manage image profiles and VIBs. Use `Get-Help cmdlet_name` at any time for command-line help.

Configure the vSphere ESX Image Builder

Before you can use vSphere ESX Image Builder with the vSphere Client, you must verify that the service is enabled and running.

Starting with ESX 9.0, due to the increase in the ESX image size, the default cache size of the vSphere Auto Deploy repository is set to 6 GB. The vSphere Auto Deploy server uses the repository to store data it needs, including the rules and rule sets you create and the VIBs and image profiles that you specify in your rules. Best practice is to allocate enough room for four image profiles and some extra space. Determine how much space to reserve for the vSphere Auto Deploy repository by considering how many image profiles you expect to use.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. Click **Enable Image Builder** to activate the service.

   The **Software Depots** tab appears.

- Add a Software Depot.
- Import a Software Depot.
- Clone an Image Profile.
- Create an Image Profile.
- Prepare Your System for vSphere Auto Deploy.
- You can change the default configuration properties of the **Image Builder Service**. For more information, see "Configuring vCenter" in the *vCenter and Host Management* documentation.

Using VMware.ImageBuilder Cmdlets

With VMware.ImageBuilder cmdlets, you can take advantage of all VCF PowerCLI features.

VMware.Image Builder cmdlets are implemented as Microsoft PowerShell cmdlets and included in VCF PowerCLI. You can take advantage of all VCF PowerCLI features by using VMware.ImageBuilder cmdlets. Experienced PowerShell users can use VMware.ImageBuilder cmdlets just like other PowerShell cmdlets. If you are new to PowerShell and VCF PowerCLI, follow these tips.

You can type cmdlets, parameters, and parameter values in thePowerShell console.

- Get help for any cmdlet by running `Get-Help cmdlet_name`.
- Remember that PowerShell is not case-sensitive.
- Use tab completion for cmdlet names and parameter names.
- Format any variable and cmdlet output by using `Format-List` or `Format-Table` or their short forms `fl` or `ft`. See `Get-Help Format-List`.

- Use wildcards for searching and filtering VIBs and image profiles. All wildcard expressions are supported.

Passing Parameters by Name

You can pass in parameters by name in most cases and surround parameter values that contain spaces or special characters with double quotes.

```
Add-EsxSoftwarePackage -ImageProfile profile42 -SoftwarePackage "partner package 35"
```

Passing Parameters as Objects

If you want to do scripting and automation, you can pass parameters as objects. You can use the technique with cmdlets that return multiple objects or with cmdlets that return a single object.

1. Bind the output of a cmdlet that returns multiple objects to a variable.
   ```
   $profs = Get-EsxImageProfile
   ```
2. When you run the cmdlet that needs the object as input, access the object by position, with the list starting with 0.
   ```
   Add-EsxSoftwarePackage -ImageProfile $profs[4] -SoftwarePackage partner-pkg
   ```

The example adds the specified software package to the fifth image profile in the list returned by `Get-EsxImageProfile`.

Most of the examples in the *vCenter Deployment and Setup* documentation pass in parameters by name. vSphere ESX Image Builder Workflows with VCF PowerCLI Cmdlets includes examples that pass parameters as objects.

VMware.ImageBuilder Cmdlets Overview

The VMware.ImageBuilder component of VCF PowerCLI provides cmdlets for managing VIBs, image profiles, and other content in software depots.

vSphere 7.0 and later introduce new ways of packaging VIBs along with legacy bulletins and patches. Software depots contain base images, vendor addons, and components, along with VIBs and image profiles. VCF.PowerCLI provide cmdlets that work with the new content in software depots.

VMware.ImageBuilder includes the following cmdlets.

> **Note:**
> When you run VMware.ImageBuilder cmdlets, provide all parameters on the command line when you invoke the cmdlet. Supplying parameters in interactive mode is not recommended.

Run `Get-Help` *`cmdlet_name`* at the VCF PowerCLI prompt for detailed reference information.

Table 13: VMware.ImageBuilder Cmdlets Used with Legacy Content in Software Depots

Cmdlet	Description
`Add-EsxSoftwareDepot`	Adds the software depot or ZIP file at the specified location to your current environment. Downloads metadata from the depot and analyzes VIBs for dependencies.
`Remove-EsxSoftwareDepot`	Disconnects from the specified software depot.
`Get-EsxSoftwareDepot`	Returns a list of software depots that are in the current environment. If you want to examine and manage image profiles and VIBs, you must first add the corresponding software depot to your environment.
`Get-EsxSoftwarePackage`	Returns a list of software package objects (VIBs). Use this cmdlet's options to filter the results.
`Get-EsxImageProfile`	Returns an array of `ImageProfile` objects from all currently added depots.
`New-EsxImageProfile`	Creates a new image profile. In most cases, creating a new profile by cloning an existing profile is recommended. See Clone an Image Profile with VCF PowerCLI Cmdlets.
`Set-EsxImageProfile`	Modifies a local `ImageProfile` object and performs validation tests on the modified profile. The cmdlet returns the modified object but does not persist it.
`Export-EsxImageProfile`	Exports an image profile as either an ESX ISO image for ESX installation, or as a ZIP file.

Cmdlet	Description
`Compare-EsxImageProfile`	Returns an `ImageProfileDiff` structure that shows whether the two profiles have the same VIB list and acceptance level. See Working with Acceptance Levels.
`Remove-EsxImageProfile`	Removes the image profile from the software depot.
`Add-EsxSoftwarePackage`	Adds one or more new packages (VIBs) to an existing image profile.
`Remove-EsxSoftwarePackage`	Removes one or more packages (VIBs) from an image profile.
`Set-ESXImageProfileAssociation`	Associates the specified image profile with the specified ESX system.

Table 14: VMware.ImageBuilder Cmdlets Used with New Content in Software Depots

Cmdlet	Description
`Get-DepotAddons`	Retrieves an array of objects that provide basic information about addons in a software depot.
`Get-DepotBaseImages`	Retrieves an array of objects that provide basic information about base images in a software depot.
`Get-DepotComponents`	Retrieves an array of objects that provide basic information about components in a software depot.
`Get-DepotInfo`	Retrieves basic information about the software depot located at the specified file path or URL address.
`Get-DepotVibs`	Retrieves an array of objects that provide basic information about VIBs in a software depot.
`New-IsoImage`	Generates an ISO image by using the specified software depot and software specification at the specified file path.
`New-PxeImage`	Generates a PXE image by using the specified software depot and software specification at the specified file path.
`New-OfflineBundle`	Generates a collection of metadata and VIBs in an offline bundle that you can use for ESX updates.

ESX Image Profile Tasks

Manipulate software depots, image profiles, and VIBs by using either the VMware.Image Builder component of VCF PowerCLI or the vSphere Client.

- Add a Software Depot
- Import a Software Depot
- Clone an Image Profile
- Create an Image Profile
- Edit an Image Profile
- Compare Image Profiles
- Move an Image Profile to a Different Software Depot
- Export an Image Profile to ISO or Offline Bundle ZIP
- Regenerate an Image Profile
- Preserve Image Profiles Across Sessions with VCF PowerCLI
- Compare VIBs with VCF PowerCLI Cmdlets

Add a Software Depot

Add one or more software depots to the vSphere ESX Image Builder inventory to customize image profiles.

Before you can work with software depots and customize image profiles, you must add one or more software depots to the vSphere ESX Image Builder inventory. You can add a software depot by using the vSphere Client.

Verify that the vSphere ESX Image Builder service is enabled and running. See Configure the vSphere ESX Image Builder.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.

2. On the **Software Depots** tab, click **New**.

   The **Add Software Depot** window appears.

   Note:

   The URLs of the official VMware online depot are now unique for each customer. To update the download sources URLs, see the following Broadcom knowledge base articles 390121 and 390123.

3. Select the type of depot that you want to create.

Option	Action
Online Depot	1. Enter a name of the depot in the inventory. 2. Enter the URL of the online depot.
Custom Depot	Enter the name of the depot in the inventory.

   The VMware online software depot is located at a url such as https://dl.broadcom.com/<Your Download Token>/PROD/COMP/ESX_HOST/main/vmw-depot-index.xml.

4. Click **Add**.
5. Optional: Click the **Software Packages** tab to view the contents of the selected depot and additional information about the packages.
6. Optional: If you added an **Online depot**, you can also:
   - **Check for Updates** to get the latest depot packages.
   - Click **More info** to get additional depot details.

The software depot is added to the list.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See Create a Deploy Rule or Clone a Deploy Rule.
- You can associate an image profile with an ESX host. See Add a Host to the vSphere Auto Deploy Inventory.
- Edit the Image Profile Association of a Host.
- **Remove** a custom software depot.

Import a Software Depot

If an offline depot is located on your local file system, import the ZIP file to the vSphere ESX Image Builder inventory.

If an offline depot is located on your local file system, you can import the ZIP file to the vSphere ESX Image Builder inventory by using the vSphere Client.

Verify that the vSphere ESX Image Builder service is enabled and running. see *Configure vSphere ESX Image Builder*.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.

2. On the **Software Depots** tab, click **Import** .
3. Enter the name of the software depot in the inventory.
4. Click **Browse** and select a ZIP file from the local system, that contains the software depot you want to import.
5. Click **Upload**.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See *Create a Deploy Rule* or *Clone a Deploy Rule*.
- You can associate an image profile with an ESX host. See *Add a Host to the vSphere Auto Deploy Inventory*.
- *Edit the Image Profile Association of a Host*.

Clone an Image Profile

Use the vSphere Client to clone image profiles.

You can use the vSphere Client to clone image profiles. You can clone an image profile when you want to make small changes to the VIB list in a profile. Another use case is when you want to use hosts from different vendors and plan to use the same basic profile, but also add vendor-specific VIBs.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.
- Verify that there is at least one custom depot in the vSphere ESX Image Builder inventory.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.
2. On the **Software Depots** tab, use the drop-down menu to select the software depot that contains the image profile that you want to work with.
3. From the list of image profiles in the depot, select the image profile that you want to clone and click **Clone**.
4. Enter an image profile name, vendor, and description.

   You must enter a unique image profile name.
5. From the **Software depot** drop-down menu, select in which custom depot to add the new image profile.
6. Click **Next**. The Select software packages page appears.
7. From the drop-down menu, select an acceptance level for the image profile.

   The acceptance level of the VIBs you add to the base image must be at least as high as the level of the base image. If you add a VIB with a lower acceptance level to the image profile, you must lower the image profile acceptance level. For more information, see *Working with Acceptance Levels*.
8. Select the VIBs that you want to add to the image profile and deselect the ones that you want to remove, and click **Next**.

   Note:
   The image profile must contain a bootable ESX image to be valid.

   vSphere ESX Image Builder verifies that the change does not invalidate the profile. Some VIBs depend on other VIBs and become invalid if you include them in an image profile separately. When you add or remove a VIB, vSphere ESX Image Builder checks whether the package dependencies are met.
9. On the **Ready to complete** page, review the summary information for the new image profile and click **Finish**.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See *Create a Deploy Rule* or *Clone a Deploy Rule*.
- You can associate an image profile with an ESX host. See *Add a Host to the vSphere Auto Deploy Inventory*.
- *Edit the Image Profile Association of a Host*.

Clone an Image Profile with VCF PowerCLI Cmdlets

Cloning a published profile is the easiest way to create a custom image profile. Cloning a profile is especially useful if you want to remove a few VIBs from a profile. You might also want to use hosts from different vendors and reuse the same basic profile, but add vendor-specific VIBs. VMware partners or large installations might consider creating a profile.

- Install the VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.
- Verify that you have access to the software depot that contains the image profile you want to clone.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.

Option	Action
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

The cmdlet returns one or more `SoftwareDepot` objects.

2. Optional: Run the `Get-EsxImageProfile` cmdlet to find the name of the profile that you want to clone.

   You can use filtering options with `Get-EsxImageProfile`.

3. Run the `New-EsxImageProfile` cmdlet to create the new profile and use the `-CloneProfile` parameter to specify the profile you want to clone.

   ```
   New-EsxImageProfile -CloneProfile My_Profile -Name "Test Profile 42" -Vendor <vendor-name>
   ```

   This example clones the profile named *My_Profile* and assigns it the name Test Profile 42. You must specify a unique combination of name and vendor for the cloned profile.

See Examine Depot Contents for some examples of filtering.

Customize the image profile by adding or removing VIBs. See Add VIBs to an Image Profile with VCF PowerCLI Cmdlets.

Create an Image Profile

You can create a new image profile by using the vSphere Client instead of cloning an existing one.

You might consider creating a new image profile if it differs significantly from the image profiles in your inventory.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.
- Verify that there is at least one custom depot in the vSphere ESX Image Builder inventory.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.

2. From the **Software depot** drop-down menu, select in which custom depot to add the new image profile.
3. On the Image Profiles tab, click **New Image Profile**.
4. Enter an image profile name, vendor, and description.

   You must enter a unique image profile name.

5. Click **Next**. The Select software packages page appears.
6. From the drop-down menu, select an acceptance level for the image profile.

   The acceptance level of the VIBs you add to the base image must be at least as high as the level of the base image. If you add a VIB with a lower acceptance level to the image profile, you must lower the image profile acceptance level. For more information, see *Working with Acceptance Levels*.

7. Select the VIBs that you want to add to the image profile and deselect the ones that you want to remove, and click **Next**.

   Note:
   The image profile must contain a bootable ESX image to be valid.

   vSphere ESX Image Builder verifies that the change does not invalidate the profile. Some VIBs depend on other VIBs and become invalid if you include them in an image profile separately. When you add or remove a VIB, vSphere ESX Image Builder checks whether the package dependencies are met.

8. On the **Ready to complete** page, review the summary information for the new image profile and click **Finish**.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See *Create a Deploy Rule* or *Clone a Deploy Rule*.

- You can associate an image profile with an ESX host. See *Add a Host to the vSphere Auto Deploy Inventory*.
- *Edit the Image Profile Association of a Host*.
- Select and **Delete** an Image profile.
- **View Software Packages** for the selected image profile.

Create a Custom ESX ISO Image with VCF PowerCLI Cmdlets

With ESX Image Builder, you can customize an ESX image profile, but not combine content from different depots to generate an ISO image. With VCF PowerCLI, you can customize ISO images by using content from multiple software depots and a custom software specification.

- Verify that your VCF PowerCLI version is 9.1 or later.
- Verify that you downloaded the following ZIP files.
  - ESX patch for the ESX version specified in the VMware Cloud Foundation Release Notes or in the list of supported async patches in Broadcom knowledge base article 88287. You can download patches from the Broadcom Support Portal.
  - OEM add-on for ESX from the Broadcom Support Portal. If the ESX version specified in the BOM is not available, contact your vendor to determine which OEM add-on version to use.

1. Gather the required information for the software spec that is used to create the custom ISO.
   a) In VCF PowerCLI, use the Get-DepotBaseImages cmdlet to get the base image version from the zip file for the ESX patch that you downloaded from the patches portal. For example: `Get-DepotBaseImages "c:\temp\VMware-ESX-9.1.x-xxxxxxx-depot.zip"`
   b) Use the Get-DepotAddons cmdlet to get the add-on name and version from the zip file for the OEM add-on for ESX that you downloaded from the Broadcom Support Portal. (if applicable). For example: `Get-DepotAddons "c:\temp\HPE-9.1.x.x.x-xxx-Synergy-Addon-depot.zip"`
2. Create the software spec using the information you gathered in step 1.The software spec is a JSON file that contains information about the ESX version and vendor add-on (if applicable). For example:

```
{
"add_on": {
   "name": "HPE-Custom-Syn-AddOn",
   "version": "9.1.x.x.x.x-xx"
},
"base_image": {
   "version": "9.1.x-x.xx.xxxxxxxx"
},
"components": null,
"hardware_support": null,
"solutions": null
}
```

3. In VCF PowerCLI, use the New-IsoImage cmdlet to generate a custom ISO. For example: `New-IsoImage -SoftwareSpec "c:\temp\HPE-70U1d-custom.JSON" -Depots "c:\temp\VMware-ESX-9.0-17551050-depot.zip" , "c:\temp\HPE-9.1.x.xx.x.x.xx-xxxxxxx-Synergy-Addon-depot.zip" -Destination "c:\temp\HPE-9.1-custom.iso"`

Provide the path to the software spec you created in step 2. The depot(s) include the path to the zip files for the supported ESX version and vendor add-on.The destination includes the path and file name for the custom ISO file.For more information about the New-IsoImage cmdlet, see New-IsoImage.

Create a Custom PXE Image with VCF PowerCLI Cmdlets

Starting with VCF PowerCLI, you can create a custom PXE image by using any software depot and a custom software specification.

- Install VCF PowerCLI.
- Verify that you have access to the software depot that contains the software specification you want to use.

1. Gather the required information for the software specification that you use to create a custom PXE image.
   1. Get the base image version for the required patch or upgrade by running the `Get-DepotBaseImages` cmdlet:
      ```
      PS C:\> Get-DepotBaseImages -Depot C:\VMware-ESX-9.xxxx-xxxxx-depot.zip
      ```
      The command output is:
      ```
      Version                                 Vendor                                  Release date
      -------                                 ------                                  ------------
      9.x.x.xxx.xxxxx                      VMware, Inc.                            04/29/20xx 00:00:00
      ```
   2. Get other packages, such as OEM addons, with cmdlets used with new metadata in software depots. For example:
      ```
      PS C:\> Get-DepotAddons -Depot C:\addon-depot.zip
      ```

   The command output is:
   ```
   Name                    Version                 ID                      Vendor                  Release
    date
   ----                    -------                 --                      ------
    ------------
   testaddonv1             1.0.0-1                 testaddonv1:1.0.0-1     ESXLifecycle QE         02/20/20xx
    18:28:23
   ```

   You can also list all components in a sofware depot with the `Get-DepotComponents` cmdlet:
   ```
   PS C:\> Get-DepotComponents -Depot C:\Intel-i40en_1.12.3.0-1OEM.700.1.0.15843807_18058526.zip
   ```

   The command output is:
   ```
   Name                            Version                         ID
            Vendor
   ----                            -------                         --
           ------
   Intel-i40en                     1.12.3.0-1OEM.xxxxx    Intel-i40en:1.12.3.0-1OEM.xxxxx     Intel
   ```

   You can use any number and combination of online and offline software depots.

2. Create a software specification. For example:
   ```
   {
       "base_image": {
           "version": "9.1.xxxxx"
       },
       "add_on": {
           "name": "testaddonv1",
           "version": "1.0.0-1"
       },
       "components": {
          "Intel-i40en": "1.12.3.0-1OEM.xxxxx"
      }
   }
   ```

   The software specification is a JSON file that contains information about the ESX base image and additional packages, such as a vendor add-on.

3. Generate a custom PXE image by running the `New-PxeImage` cmdlet with the parameters `Depots`,`SoftwareSpec` and `Destination`. For example:
   ```
   New-PxeImage -Depots “c:\temp\VMware-ESX-9.1xxxxx-xxxxx-depot.zip” , “c:\temp\HPE-9.1xxxxx-xxx-Synergy-
   Addon-depot.zip” -SoftwareSpec “c:\temp\HPE-xxx-custom.JSON” -Destination “C:\pxe-image”
   ```

   The depot(s) include the path to the zip files for the supported ESX version and vendor add-on. The destination include the path and file name for the custom PXE file.

   You can pass additional kernel options, create a live image, overwrite existing files, or check acceptance levels for individual VIBs used during the creation of the image. For more information about the `New-PxeImage` cmdlet, see https://developer.broadcom.com/powercli/latest/vmware.imagebuilder/commands/new-pxeimage.

You can use the PXE image in remediation workflows of PXE booted ESX hosts.

Edit an Image Profile

You can edit image profiles by using the vSphere Client. You can change the name, details, and VIB list of an image profile.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.
- Verify that there is at least one custom depot in the vSphere ESX Image Builder inventory.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.

2. On the **Software Depots** tab, use the drop-down menu to select the software depot that contains the image profile that you want to work with.
3. On the **Image Profiles** tab, select the image profile that you want to edit and click **Edit**.

   The **Edit Image Profile** wizard appears.
4. Optional: Change the vendor and description information of the image profile.
5. Click **Next**.

   The **Select software packages** page appears. DHCP server that you can configure for PXE boot
6. From the drop-down menu, select an acceptance level for the image profile.

   The acceptance level of the VIBs you add to the base image must be at least as high as the level of the base image. If you add a VIB with a lower acceptance level to the image profile, you must lower the image profile acceptance level. For more information, see *Working with Acceptance Levels*.
7. Select the VIBs that you want to add to the image profile and deselect the ones that you want to remove, and click **Next**.

   Note:
   The image profile must contain a bootable ESX image to be valid.

   vSphere ESX Image Builder verifies that the change does not invalidate the profile. Some VIBs depend on other VIBs and become invalid if you include them in an image profile separately. When you add or remove a VIB, vSphere ESX Image Builder checks whether the package dependencies are met.
8. On the **Ready to complete** page, review the summary information for the edited image profile and click **Finish**.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See *Create a Deploy Rule* or *Clone a Deploy Rule*.
- You can associate an image profile with an ESX host. See *Add a Host to the vSphere Auto Deploy Inventory*.
- *Edit the Image Profile Association of a Host*.

Add VIBs to an Image Profile with VCF PowerCLI Cmdlets

If an image profile is not set to read only, you can add one or more VIBs to that image profile. If the new VIB depends on other VIBs or conflicts with other VIBs in the profile, a message is displayed at the PowerShell prompt and the VIB is not added.

Install the VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.

You can add VIBs from VMware or from VMware partners to an image profile. If you add VMware VIBs, vSphere ESX Image Builder performs validation. If you add VIBs from two or more OEM partners simultaneously, no errors are reported but the resulting image profile might not work. Install VIBs from only one OEM vendor at a time.

If an error about acceptance level problems appears, change the acceptance level of the image profile and the acceptance level of the host. Consider carefully whether changing the host acceptance level is appropriate. VIB acceptance levels are set during VIB creation and cannot be changed.

You can add VIBs even if the resulting image profile is invalid.

Note:
VMware can support only environments and configurations that are proven to be stable and fully functional through rigorous and extensive testing. Use only those supported configurations. You can use custom VIBs if you lower your host acceptance level, and as a result, supportability. In that case, track the changes you made, so you can revert them if you want to remove custom VIBs and restore the host acceptance level to the default (PartnerSupported) later. See Working with Acceptance Levels.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Run the `Get-EsxImageProfile` cmdlet to list all image profiles in all currently visible depots.

   The cmdlet returns all available profiles. You can narrow your search by using the optional arguments to filter the output.

3. Clone the profile.

```
New-EsxImageProfile -CloneProfile My_Profile -Name "Test Profile 42" -Vendor "My Vendor"
```

   Image profiles published by VMware and its partners are read only. To make changes, you must clone the image profile. The `vendor` parameter is required.

4. Run the `Add-EsxSoftwarePackage` cmdlet to add a new package to one of the image profiles.

```
Add-EsxSoftwarePackage -ImageProfile My_Profile -SoftwarePackage partner-package
```

   The cmdlet runs the standard validation tests on the image profile. If validation succeeds, the cmdlet returns a modified, validated image profile. If the VIB that you want to add depends on a different VIB, the cmdlet displays that information and includes the VIB that can resolve the dependency. If the acceptance level of the VIB that you want to add is lower than the image profile acceptance level, an error occurs.

Compare Image Profiles

You can compare two image profiles by using the vSphere Client to check if they have the same VIB list, version, or acceptance level.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.

2. On the **Software Depots** tab, use the drop-down menu to select the software depot that contains the image profile that you want to work with.

3. On the **Image Profiles** tab, select an image profile and click **Compare To**.

   The **Compare Image Profile** wizard appears.

4. Click **Change** to select a second image profile.

   The **Select Image Profile** page appears.

5. Select a software depot from the drop-down menu and click on the second image profile.
6. In the **Compare Image Profile** page, select a comparison option from the **Software packages** drop-down menu.
   The left side of the list displays details of the VIBs that the first chosen image profile contains. The right part of the list provides information about the second image profile. The VIBs marked as `Same` are identical in both profiles. VIBs that are present in one of the image profiles are marked as `Missing` next to the image profile that they are not present in.

Compare Image Profiles with VCF PowerCLI Cmdlets

You can compare two image profiles by using the `Compare-EsxImageProfile` cmdlet, for example, to see if they have the same VIB list or acceptance level. Comparing image profiles or their properties is also possible by using the PowerShell comparison operators.

Install the VCF PowerCLI and all prerequisite software. See *Configure vSphere ESX Image Builder*.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.
2. Optional: Run the `Get-EsxImageProfile` cmdlet to view a list of all image profiles in all available depots.
   In the list, you can locate the names of the image profiles you want to compare.
3. Before comparing the image profiles, assign them to variables.
   For example, you can create variables `$imageProfile1` and `$imageProfile2` to hold the names of the compared images profiles.

```
$imageProfile1
    = Get-EsxImageProfile -Name "ImageProfile1"
$imageProfile2
    = Get-EsxImageProfile -Name "ImageProfile2"
```

4. Compare the two image profiles by using the `Compare-EsxImageProfile` cmdlet or the `-eq` comparison operator, which returns a Boolean value.
   - Compare the two image profiles to get a full description of the differences by using the `Compare-EsxImageProfile` cmdlet.

```
Compare-EsxImageProfile -ReferenceProfile
      $imageProfile1 -ComparisonProfile $imageProfile2
```

   - Compare the two image profiles by VIB list and acceptance level using the `-eq` comparison operator.

```
if ($imageProfile1 -eq $imageProfile2) {
   Write-host "Successfully verified that both image profiles are equal."
} else {
   Write-host "Failed to verify that the image profiles are equal."
}
```

   - Compare the two image profiles by a specific property using the `-eq` comparison operator.

```
if ($imageProfile1.vendor -eq $imageProfile2.vendor) {
   Write-host "Successfully verified that both image profiles are equal."
} else {
   Write-host "Failed to verify that the image profiles are equal."
}
```

Move an Image Profile to a Different Software Depot

You can move image profiles between custom depots by using the vSphere Client and modify it without affecting the source depot's configuration.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.
- Verify that there is at least one custom depot in the vSphere ESX Image Builder inventory.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.
2. On the **Software Depots** tab, use the drop-down menu to select the software depot that contains the image profile that you want to work with.
3. On the **Image Profiles** tab, select an image profile and click **Move to**.
4. From the drop-down menu, select the custom depot in which you want to move the image profile.
5. Click **OK**.

The image profile is in the new software depot.

Export an Image Profile to ISO or Offline Bundle ZIP

You can export an image profile to an ISO image or a ZIP file by using the vSphere Client.

You can export an image profile to an ISO image or a ZIP file by using the vSphere Client. You can use the ISO image as an ESX installer or to upgrade hosts with vSphere Lifecycle Manager. The ZIP file contains metadata and the VIBs of the image profile. You can use it for ESX upgrades or as an offline depot.

- Verify that the vSphere ESX Image Builder service is enabled and running. See *Configure the vSphere Image Builder*.
- Add or import a software depot to the vSphere ESX Image Builder inventory. See *Add a Software Depot* or *Import a Software Depot*.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere ESX Image Builder service.
2. On the **Software Depots** tab, use the drop-down menu to select the software depot that contains the image profile that you want to work with.
3. On the **Image Profiles** tab, select the image profile that you want to export and click **Export**.

   The **Export Image Profile** window appears.
4. Select the type of the exported file.

Option	Description
ISO	Exports the image profile to a bootable ISO image. If you want to create an ISO image that you can burn to a CD or DVD and use to start a stateless ESX instance, select the **Do not include an installer on the ISO** check box.
ZIP	Exports the image profile to a ZIP file.

5. Optional: If you want to bypass the acceptance level verification of the image profile, select **Skip acceptance level checking**.
6. Click **Ok**.

   The **Download** link starts generating in the "Download Image Profiles" column of the selected image profile.
7. When the image generates successfully, click **Download** to save the exported file.

- You can associate an image profile with a new vSphere Auto Deploy rule to provision ESX hosts. See *Create a Deploy Rule* or *Clone a Deploy Rule*.

- You can associate an image profile with an ESX host. See *Add a Host to the vSphere Auto Deploy Inventory*.
- *Edit the Image Profile Association of a Host*.

Export an Image Profile to an ISO or Offline Bundle ZIP with VCF PowerCLI Cmdlets

You can export an image profile to an ISO image or a ZIP file of component files and folders. You cannot create both by running the cmdlet once. You can use the ISO image as an ESX installer or upload the ISO into vSphere Lifecycle Manager for upgrades.

Install the VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Run `Export-EsxImageProfile` to export the image profile.

Export Format	Cmdlet
ISO images	`Export-EsxImageProfile` with the `-ExportToIso` parameter
Offline depot ZIP files	`Export-EsxImageProfile` with the `-ExportToBundle` parameter

For the ISO image, vSphere ESX Image Builder validates VIB signatures, adds VIB binaries to the image, and downloads the image to the specified location. For the ZIP file, vSphere ESX Image Builder validates VIB signatures and downloads the VIB binaries to the specified location.

Exporting an Image Profile

Follow these steps to export an image profile to an ISO image.

1. Add the software depot.
   ```
   Add-EsxSoftwareDepot -DepotUrl url_or_file
   ```
2. View all available image profiles to find the name of the image profile to export.
   ```
   Get-EsxImageProfile
   ```
3. Export the image profile.
   ```
   Export-EsxImageProfile -ImageProfile "myprofile" -ExportToIso -FilePath iso_name
   ```

Follow these steps to export an image profile to a ZIP file of component files and folders.

1. Add the software depot.
   ```
   Add-EsxSoftwareDepot -DepotUrl url_or_file
   ```
2. View all available image profiles to find the name of the image profile to export.
   ```
   Get-EsxImageProfile
   ```
3. Export the image profile.
   ```
   Export-EsxImageProfile -ImageProfile "myprofile" -ExportToBundle -FilePath C:\my_bundle.zip
   ```

Use the ISO image in an ESX installation or upload the ISO image into vSphere Lifecycle Manager to perform upgrades.

Use the ZIP file to upgrade an ESX installation.

- Import the ZIP file into vSphere Lifecycle Manager for use with patch baselines.
- Download the ZIP file to an ESX host or a datastore and run `esxcli software vib` commands to import the VIBs in the ZIP file.

See the *vSphere Upgrade* documentation.

Regenerate an Image Profile

If you use Auto Deploy to add stateful ESX hosts to a cluster that you manage by using an image, all hosts must have the same software specification.

When an Auto Deploy rule is created and the target cluster is managed by an image, a Preboot Execution Environment (PXE) image is created based on the image specification of the cluster. The generated PXE image is cached and is not automatically updated. As a result, if you modify the image specification in the vSphere Lifecycle Manager, you must update the PXE image manually.

For information how to create a rule to add hosts to a cluster managed by an image, see *Use Auto Deploy to Add a Host to a Cluster Managed by an Image* from the *vSphere Lifecycle Manager* documentation.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, select the desired rule.
   You must select a rule, which matches ESX hosts to a cluster that you manage by an image.
3. If the rule is active, first you must deactivate the rule.
   a) Click the **Activate/Deactive Rules** tab
   b) In the dialog box, select the rule.
   c) In the dialog box, select **Deactivate** and click **OK**.
4. Select **Reacreate Image Profile** and in the confirmation dialog box, click **Recreate**.
5. Optional: Activate the rule again.
   a) Click the **Activate/Deactive Rules** tab
   b) In the dialog box, select the rule.
   c) In the dialog box, select **Activate** and click **OK**.

The PXE image is running the latest image specification.

Preserve Image Profiles Across Sessions with VCF PowerCLI

You can export the image profile to a ZIP file software depot and add that depot in the next session.

When you create an image profile and exit the VCF PowerCLI session, the image profile is no longer available when you start a new session. You can export the image profile to a ZIP file software depot, and add that depot in the next session.

Install the VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.

1. In a VCF PowerCLI session, create an image profile, for example by cloning an existing image profile and adding a VIB.
2. Export the image profile to a ZIP file by calling `Export-EsxImageProfile` with the `ExportToBundle` parameter.
   ```
   Export-EsxImageProfile -ImageProfile "my_profile" -ExportToBundle -FilePath
       "C:\isos\temp-base-plus-vib25.zip"
   ```
3. Exit the VCF PowerCLI session.
4. When you start a new VCF PowerCLI session, add the depot that contains your image profile to access it.
   ```
   Add-EsxSoftwareDepot "C:\isos\temp-base-plus-vib25.zip"
   ```

Compare VIBs with VCF PowerCLI Cmdlets

You can compare two VIBs or their properties by using the PowerShell comparison operators.

Install the VCF PowerCLI and all prerequisite software. See Configure vSphere ESX Image Builder.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Optional: Run the `Get-EsxSoftwarePackage` cmdlet to view all available VIBs.
   In the list, you can locate the names of the VIBs you want to compare.
3. Before comparing the VIBs, assign them to variables.
   For example, you can create variables `$vib1` and `$vib2` to hold the names of the compared VIBs.

```
$vib1 = Get-EsxSoftwarePackage -Name "ReferenceVIB"
$vib2 = Get-EsxSoftwarePackage -Name "ComparisonVIB"
```

4. Use a comparison operator to compare the VIBs by contents and acceptance level or by a specific property.
   - Compare the two VIBs by their contents and acceptance level.

```
if ($vib1 -eq $vib2) {
   Write-host "Successfully verified that both VIBs are equal."
} else {
   Write-host "Failed to verify that the VIBs are equal."
}
```

   - Compare a specific property of the VIBs by using a comparison operator such as `-eq`, `-lt`, `-le`, `-gt`, or `-ge`.

```
if ($vib1.VersionObject -lt $vib2.VersionObject) {
   Write-host "Successfully verified that both the VIBs are equal."
} else {
   Write-host "Failed to verify that the VIBs are equal."
}
```

vSphere ESX Image Builder Workflows with VCF PowerCLI Cmdlets

vSphere ESX Image Builder workflows are examples for cmdlet use and do not represent actual tasks.

vSphere ESX Image Builder workflows are examples for cmdlet use. Workflows do not represent actual tasks, but illustrate how you might explore different ways of using a cmdlet. Administrators trying out the workflows benefit from some experience with VCF PowerCLI, Microsoft PowerShell, or both.

Examine Depot Contents

You can examine software depots and VIBs with vSphere ESX Image Builder cmdlets by using all kinds of wildcard expressions.

Verify that VCF PowerCLI and prerequisite software is installed. See Configure vSphere ESX Image Builder.

The workflow itself passes parameters by name. However, you can pass parameters as objects by accessing variables.

You can use filtering options and wildcard expressions to examine depot contents.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Retrieve image profiles.
   You can filter by vendor, name, and acceptance level.
   - `Get-EsxImageProfiles`

     Returns an array of `ImageProfile` objects from all depots you added to the session.
   - `Get-EsxImageProfile -Vendor "C*"`

     Returns all image profiles created by a vendor with a name that starts with the letter C.
3. Retrieve software packages by using the `Get-EsxSoftwarePackage` cmdlet.
   You can filter, for example by vendor or version, and you can use the standard PowerShell wildcard characters.
   - `Get-EsxSoftwarePackage -Vendor "V*"`

     Returns all software packages from a vendor with a name that starts with the letter V.
   - `Get-EsxSoftwarePackage -Vendor "V*" -Name "*scsi*"`

     Returns all software packages with a name that contains the string `scsi` in it from a vendor with a name that starts with the letter V.
   - `Get-EsxSoftwarePackage -Version "2.0*"`

     Returns all software packages with a version string that starts with 2.0.
4. Use `-Newest` to find the latest package.
   - `Get-EsxSoftwarePackage -Vendor "V*" -Newest`

     Returns the newest package for the vendors with a name that starts with the letter V, and displays the information as a table.
   - `Get-EsxSoftwarePackage -Vendor "V*" -Newest | format-list`

     Returns detailed information about each software package by using a pipeline to link the output of the request for software packages to the PowerShell `format-list` cmdlet.
5. View the list of VIBs in the image profile.
   ```
   (Get-EsxImageProfile -Name "Robin's Profile").VibList
   ```
   `VibList` is a property of the `ImageProfile` object.
6. Retrieve software packages released before or after a certain date by using the `CreatedBefore` or `CreatedAfter` parameter.
   ```
   Get-EsxSoftwarePackage -CreatedAfter 7/1/2020
   ```

   Depot Content Examination Using Variables

   This workflow example examines depot contents by passing in parameters as objects accessed by position in a variable, instead of passing in parameters by name. You can run the following commands in sequence from the VCF PowerCLI prompt. Replace names with names that are appropriate in your installation.

   ```
   Get-EsxSoftwarePackage -Vendor "V*"
   Get-EsxSoftwarePackage -Vendor "V*" -Name "r*"
   Get-EsxSoftwarePackage -Version "2.0*"
   $ip1 = Get-EsxImageProfile -name ESX-9.1.0-123456-full
   $ip1.VibList
   Get-EsxSoftwarePackage -CreatedAfter 7/1/2020
   ```

Create Image Profiles by Cloning Workflow

Use vSphere ESX Image Builder cmdlets to check available depots, add a depot, view image profile information, and to clone a new image profile.

You can use vSphere ESX Image Builder cmdlets to check which depots are available, to add a depot, to view image profile information, and to create a new image profile by cloning one of the available image profiles.

Verify that VCF PowerCLI and prerequisite software is installed. See Configure vSphere ESX Image Builder.

Published profiles are usually read only and cannot be modified. Even if a published profile is not read-only, cloning instead of modifying the profile is a best practice, because modifying the original profile erases the original. You cannot revert to the original, unmodified profile except by reconnecting to a depot.

A profile cloning workflow might include checking the current state of the system, adding a software depot, and cloning the profile.

1. In a PowerShell window, check whether any software depots are defined for the current session.

   ```
   $DefaultSoftwareDepots
   ```

   If you just started PowerShell, it returns the currently defined depots or nothing.

2. If the depot containing the profile that you want to clone does not appear in the results, add it to the current session.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl` *`depot_url`*.
ZIP file	1. Download the ZIP file to a local file path. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\file_path\`*`offline-bundle`*`.zip`

   PowerShell adds the specified depot to your current session and lists all current depots.

3. Optional: Check the `$DefaultSoftwareDepots` variable, which now returns the newly added depot.

4. View all available image profiles.

   ```
   Get-EsxImageProfile
   ```

5. To clone an image profile, enter its name, a new name for the new profile, and a name of the vendor.

   ```
   $ip = New-EsxImageProfile -CloneProfile base-tbd-v1 -Name "Test Profile 42" -Vendor "Vendor20"
   ```

6. Optional: View the newly created image profile, `$ip`.

   PowerShell returns the information about the image profile in tabular format.

   ```
   Name                Vendor          Last Modified          Acceptance Level
   ----                ------          -------------          ----------------
   Test Profile 42     Vendor20        9/15/2020 5:45:43...   PartnerSupported
   ```

   Creating Image Profile by Cloning Using Variables

   This workflow example repeats the steps of this workflow by passing in parameters as objects accessed by position in a variable, instead of passing in parameters by name. You can run the following cmdlets in sequence from the VCF PowerCLI prompt.

   ```
   $DefaultSoftwareDepots
   Add-EsxSoftwareDepot -DepotUrl depot_url
   $DefaultSoftwareDepots
   $profs = Get-EsxImageProfile
   $profs
   $ip = New-EsxImageProfile -CloneProfile $profs[2] -Name "new_profile_name" -Vendor "my_vendor"
   $ip
   ```

Create New Image Profiles Workflow

You can clone an existing image profile or create a new image profile, for which you must define dependencies and acceptance levels.

In most situations, you create an image profile by cloning an existing profile. Some VMware customers or partners might need to create a new image profile. Pay attention to dependencies and acceptance levels if you create an image profile from scratch.

- VCF PowerCLI and prerequisite software is installed. See Configure vSphere ESX Image Builder.

- You have access to a depot that includes a base image and one or more VIBs. VMware and VMware partners have public depots, accessible by a URL. VMware or VMware partners can create a ZIP file that you can unzip to your local environment and access by using a file path.

The system expects that the acceptance level of the VIBs you add to the base image is at least as high as the level of the base image. If you have to add a VIB with a lower acceptance level to the image profile, you must lower the image profile acceptance level. For more information, see Set the Image Profile Acceptance Level.

For string parameters, you can use the PowerShell prompting mechanism as an alternative to specify parameters on the command line. Prompting does not work for other parameters such as objects.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Run the `Get-EsxImageProfile` cmdlet to list all image profiles in all currently visible depots. You can narrow your search by using the optional arguments to filter the output.

```
Get-EsxSoftwarePackage -CreatedAfter 7/1/2020
```

3. Create a new profile, assign it a name and vendor, and add a base package.

```
$esxBase = Get-EsxSoftwarePackage 'esx-base'
$vsanHealth = Get-EsxSoftwarePackage 'vsanhealth'

New-EsxImageProfile -NewProfile -Name "Test #2" -Vendor "Vendor42" -SoftwarePackage
 esxBase[0],vsanHealth[0]
```

   The example uses the esx-base package. Names that contain spaces are surrounded by quotes.

   Starting with ESX 8.0 Update 2, removing some VIBs from their Components can cause runtime issues with ESX. For more information, see *Remove VIBs from a Host* topic in the *ESX Upgrade* guide.

4. Use a pipeline to pass the new image profile to `format-list` for detailed information about the new package.

```
(Get-EsxImageProfile -Name "Test #2").VibList | format-list
```

   Creating Image Profiles from Scratch Using Variables

   This command sequence repeats the steps of the workflow, but passes parameters as objects, accessed by position in a variable, instead of passing parameters by name. You can run the following commands in sequence at the VCF PowerCLI prompt.

```
Add-EsxSoftwareDepot depoturl
$pkgs = Get-EsxSoftwarePackage -CreatedAfter 7/1/2020
$ip2 = New-EsxImageProfile -NewProfile -Name "Test #2" -vendor "Vendor42" -SoftwarePackage $pkgs[0]
$ip2.VibList | format-list
```

Edit Image Profiles Workflow

You can create a custom image by cloning and editing an image profile by using VCF PowerCLI.

You can create a custom image by cloning and editing an image profile by using VCF PowerCLI. You can add or remove one or more VIBs in the existing profile. If adding or removing VIBs prevents the image profile from working correctly, an error occurs.

- VCF PowerCLI and prerequisite software is installed. See Configure vSphere ESX Image Builder.

- You have access to a depot that includes a base image and one or more VIBs. VMware and VMware partners make public depots, accessible by a URL, available. VMware or VMware partners can create a ZIP file that you can download to your local environment and access by using a file path.

1. In a PowerCLI session, run the `Add-EsxSoftwareDepot` cmdlet for each depot you want to work with.

Option	Action
Remote depot	Run `Add-EsxSoftwareDepot -DepotUrl <depot_url>`.
ZIP file	1. Download the ZIP file to a local file system. 2. Run `Add-EsxSoftwareDepot -DepotUrl C:\<file_path>\<offline-bundle>.zip`

   The cmdlet returns one or more `SoftwareDepot` objects.

2. Use a pipeline to pass the image profile you intend to edit to `format-list` to see detailed information.

   In this example, the image profile created in Create New Image Profiles Workflow contains only the base image. A newly created image profile is not added to the connected depot(s). It persists for the duration of the session and can be accessed in the list returned by `Get-EsxImageProfile`, by querying for it by name or binding it to a variable at the time it is created.

```
Get-EsxImageProfile "Test #2" | format-list
```

   PowerShell returns the information.

```
Name            : Test #2
Vendor        : Vendor42
...
VibList       : {esx-base 5.0.0.-...,}
```

3. Optional: If you are adding a VIB with a lower acceptance level than that of the image profile, change the acceptance level of the image profile.

```
Set-EsxImageProfile -ImageProfile "Test #2" -AcceptanceLevel VMwareAccepted
```

   PowerShell returns the information about the changed profile in tabular format.

```
Name           Vendor           Last Modified       Acceptance Level
----           ------           -------------       ----------------
Test #2        Vendor42         9/22/2010 12:05:... VMwareAccepted
```

4. Add a software package (VIB) to the image profile. You can add the package by name.

```
Add-EsxSoftwarePackage -ImageProfile "Test #2"
-SoftwarePackage NewPack3
```

   PowerShell returns the information about the image profile in tabular format.

```
Name           Vendor           Last Modified       Acceptance Level
----           ------           -------------       ----------------
Test #2        Vendor42         9/22/2010 12:05:... VMwareAccepted
```

   Note:

   If an error occurs when you add the software package, you might have a problem with acceptance levels, see Working with Acceptance Levels.

5. View the image profile again.

```
Get-EsxImageProfile "Test #2" | format-list
```

   The VIB list is updated to include the new software package and the information is displayed.

```
Name            : Test #2
Vendor        : Vendor42
...
```

```
VibList       : {esx-base 5.0.0.-..., NewPack3}
```

Editing Image Profiles by Using Variables

This cmdlet sequence repeats the steps of the workflow but passes parameters as objects, accessed by position in a variable, instead of passing parameters by name. You can run the following cmdlets in sequence from the VCF PowerCLI prompt.

```
Add-EsxSoftwareDepot -DepotUrl depot_url
$ip2 = Get-EsxImageProfile -name "Test #2"
$ip2 | format-list
Set-EsxImageProfile -ImageProfile $ip2 -AcceptanceLevel VMwareAccepted
Add-EsxImageSoftwarePackage -ImageProfile $ip2 -SoftwarePackage NewPack3
$ip2 | format-list
```

Installing ESX

You can install ESX interactively, with a scripted installation, or with vSphere Auto Deploy.

- Installing ESX Interactively
- Installing or Upgrading Hosts by Using a Script
- Overview of the Network Boot Installation Process
- Installing ESX Using vSphere Auto Deploy

Installing ESX Interactively

Use the interactive installation option for small deployments of fewer than five hosts.

In a typical interactive installation, you boot the ESX installer and respond to the installer prompts to install ESX to the local host disk. The installer reformats and partitions the target disk and installs the ESX boot image. If you have not installed ESX on the target disk before, all data on the drive is overwritten, including hardware vendor partitions, operating system partitions, and associated data.

> **Note:**
> To ensure that you do not lose any data, migrate the data to another machine before you install ESX.

If you are installing ESX on a disk that contains a previous installation of ESX, or a VMFS datastore, the installer provides you with options for upgrading. See the *vSphere Upgrade* documentation.

Interactive ESX Installation

You boot the server from an attached vSphere ESX CD or DVD, or from a bootable USB device, or by PXE booting the server from a location on the network. You follow the prompts in the installation wizard to install ESX to disk.

- Interactive ESX Installation
- Install ESX Interactively
- Install ESX on a Software iSCSI Disk

Install ESX Interactively

You use the ESX CD/DVD or a USB flash drive to install the ESX software onto a SAS, SATA, SCSI hard drive, or USB drive.

- You must have the ESX installer ISO in one of the following locations:
  - On CD or DVD. If you do not have the installation CD/DVD, you can create one. See Download and Burn the ESX Installer ISO Image to a CD or DVD
  - On a USB flash drive. See Format a USB Flash Drive to Boot the ESX Installation or Upgrade.

> **Note:**
> You can also PXE boot the ESX installer to run an interactive installation or a scripted installation. See Network Booting the ESX Installer.

- Verify that the server hardware clock is set to UTC. This setting is in the system BIOS or UEFI.
- Verify that a keyboard and monitor are attached to the machine on which the ESX software is installed. Alternatively, use a remote management application. See Using Remote Management Applications.
- Consider disconnecting your network storage. This action decreases the time that it takes the installer to search for available disk drives. When you disconnect network storage, any files on the disconnected disks are unavailable at installation.

  Do not disconnect a LUN that contains an existing ESX installation. Do not disconnect a VMFS datastore that contains the Service Console of an existing ESX installation. These actions can affect the outcome of the installation.
- Gather the information required by the ESX installation wizard. See *Required Information for ESX Installation*.
- Verify that ESX Embedded is not present on the host machine. ESX Installable and ESX Embedded cannot exist on the same host.

1. Insert the ESX installer CD/DVD into the CD/DVD-ROM drive, or attach the Installer USB flash drive and restart the machine.
2. Set the BIOS or UEFI to boot from the CD-ROM device or the USB flash drive.

   Note:

   If your system has data processing units (DPUs), you can only use UEFI to install and boot ESX on the DPUs.

   See your hardware vendor documentation for information on changing boot order.
3. On the welcome screen, press **Enter** to continue.
4. Accept the End User License Agreement by pressing **Enter**.

   Starting with ESX 8.0 Update 3, after the scanning for available devices completes, if your system has DPUs, you see them automatically listed with their respective PCI slots. You no longer select a slot. The DPU devices must be identical: same vendor, same hardware version, and same firmware.
5. On the **Select a Disk to Install or Upgrade** ESX screen, select the drive on which to install ESX and press **Enter**.

   Press F1 for information about the selected disk.

   Note:
   Do not rely on the disk order in the list to select a disk. The disk order is determined by the BIOS or UEFI and might be out of order. This might occur on systems where drives are continuously being added and removed.

   If you select a disk that contains data, the **Confirm Disk Selection** page appears.

   If you are installing on a disc with a previous ESX or ESX installation or VMFS datastore, the installer provides several choices.

   Important:
   If you are upgrading or migrating an existing ESX installation, see the *VMware ESX Upgrade* documentation.

   If you select a disk that is in vSAN disk group, the resulting installation depends on the type of disk and the group size:

   - If you select an SSD, the SSD and all underlying HDDs in the same disk group are wiped.
   - If you select an HDD, and the disk group size is greater than two, only the selected HDD is wiped.
   - If you select an HDD disk, and the disk group size is two or less, the SSD and the selected HDD is wiped.
6. Select the keyboard type for the host.

   You can change the keyboard type after installation in the direct console.
7. Enter the root password for the host.

   You can change the password after installation in the direct console.
8. In the **Confirm Install** screen, if your vSphere system has DPUs, you see each listed on a separate row. Press `F11` to confirm the start of the installation.

   Starting with 8.0 Update 3, if your systems has DPUs, you see a single progress bar for the ESX and DPU installation, with dynamic updates to the label showing what stage of the installer is being run.

   For more information about managing vSAN disk groups, see the *vSphere Storage* documentation.

If you select an SD or USB device, you see a warning that prompts you to select a persistent disk to store the ESX-OSData partition. In the **Select a Disk to store ESX OSData** screen, select a persistent storage device with minimum 32 GB of available space.

9. When the installation is complete, remove the installation CD, DVD, or USB flash drive.

   Starting with 8.0 Update 3, if the installation is not successful on any of the targets, you see the **Operation Failed** screen. Record the error log in the screen to help troubleshooting.

10. Press **Enter** to reboot the host.
11. Set the first boot device to be the drive on which you installed ESX in step 5.

    For information about changing boot order, see your hardware vendor documentation.

    Note:

    UEFI systems might require additional steps to set the boot device. See Host Fails to Boot After You Install ESX in UEFI Mode.

After the installation is complete, you can migrate existing VMFS data to the ESX host.

You can boot a single machine from each ESX image. Booting multiple devices from a single shared ESX image is not supported.

Set up basic administration and network configuration for ESX. See After You Install and Set Up ESX.

Install ESX on a Software iSCSI Disk

When you install ESX to a software iSCSI disk, you must configure the target iSCSI qualified name (IQN).

- Verify that the target IQN is configured in the iBFT BIOS target parameter setting. This setting is in the option ROM of the network interface card (NIC) to be used for the iSCSI LUN. See the vendor documentation for your system.
- Deactivate the iBFT adapter option to boot to the iSCSI target. This action is necessary to make sure that the ESX installer boots, rather than the iSCSI target. When you start your system, follow the prompt to log in to your iBFT adapter and deactivate the option to boot to the iSCSI target. See the vendor documentation for your system and iBFT adapter. After you finish the ESX installation, you can reenable the option to boot from the LUN you install ESX on.

During system boot, the system performs a Power-On Self Test (POST), and begins booting the adapters in the order specified in the system BIOS. When the boot order comes to the iSCSI Boot Firmware Table (iBFT) adapter, the adapter attempts to connect to the target, but does not boot from it. See Prerequisites.

If the connection to the iSCSI target is successful, the iSCSI boot firmware saves the iSCSI boot configuration in the iBFT. The next adapter to boot must be the ESX installation media, either a mounted ISO image or a physical CD-ROM.

1. Start an interactive installation from the ESX installation CD/DVD or mounted ISO image.
2. On the **Select a Disk** screen, select the iSCSI target you specified in the iBFT BIOS target parameter setting.

   If the target does not appear in this menu, make sure that the TCP/IP and initiator iSCSI IQN settings are correct. Check the network Access Control List (ACL) and confirm that the adapter has adequate permissions to access the target.

3. Follow the prompts to complete the installation.
4. Reboot the host.
5. In the host BIOS settings, enter the iBFT adapter BIOS configuration, and change the adapter parameter to boot from the iSCSI target.

   See the vendor documentation for your system.

On your iBFT adapter, reenable the option to boot to the iSCSI target, so the system will boot from the LUN you installed ESX on.

Installing ESX by Using a Script

The installation/upgrade script is a text file, for example `ks.cfg`, that contains supported commands.

The command section of the script contains the ESX installation options. This section is required and must appear first in the script.

- Scripted ESX Installation
- Installation and Upgrade Scripts Used for ESX Installation
- Configuring External Entropy Sources During Scripted Installation
- Install or Upgrade ESX from a CD or DVD by Using a Script
- Install or Upgrade ESX from a USB Stick by Using a Script
- Network Boot the ESX Installer for a Scripted Installation or Upgrade

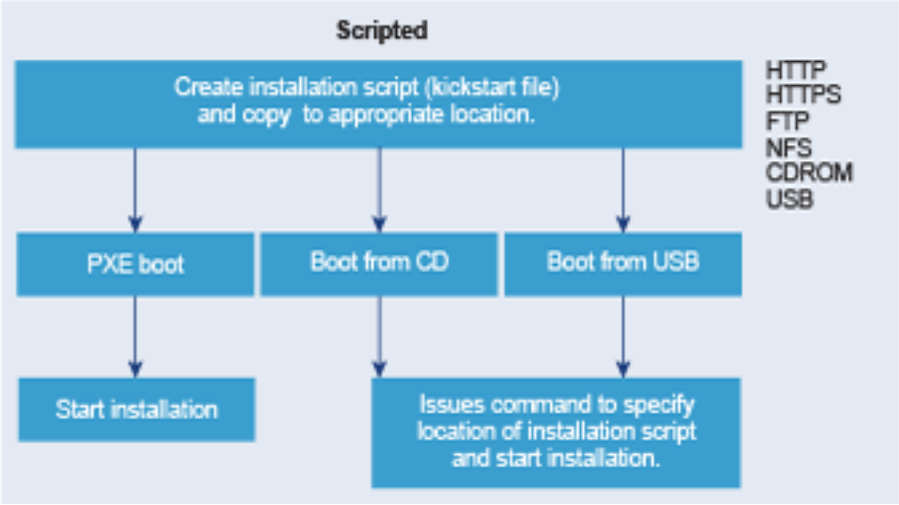
Scripted ESX Installation

Running a script is an efficient way to deploy multiple ESX hosts with an unattended installation.

The installation script contains the host configuration settings. You can use the script to configure multiple hosts with the same settings. See the chapter *Installation and Upgrade Scripts Used for ESX Installation* in the *ESX Upgrade* guide.

The installation script must be stored in a location that the host can access by HTTP, HTTPS, FTP, NFS, CDROM, or USB. You can PXE boot the ESX installer or boot it from a CD/DVD or USB drive.

Figure 3: Scripted Installation



Approaches for Scripted Installation

You can install ESX on multiple machines using a single script for all of them or a separate script for each machine.

For example, because disk names vary from machine to machine, one of the settings that you might want to configure in a script is the selection for the disk to install ESX on.

Table 15: Scripted Installation Choices

Option	Action
Always install on the first disk on multiple machines.	Create one script.
Install ESX on a different disk for each machine.	Create multiple scripts.

For information about the commands required to specify the disk to install on, see Installation and Upgrade Script Commands.

Enter Boot Options to Run an Installation or Upgrade Script

You can start an installation or upgrade script by typing boot options at the ESX installer boot command line.

At boot time you might need to specify options to access the kickstart file. You can enter boot options by pressing Shift+O in the boot loader. For a PXE boot installation, you can pass options through the `kernelopts` line of the `boot.cfg` file. See About the boot.cfg File and Network Booting the ESX Installer.

To specify the location of the installation script, set the `ks=filepath` option, where *`filepath`* indicates the location of your kickstart file. Otherwise, a scripted installation or upgrade cannot start. If `ks=filepath` is omitted, the text installer is run.

Supported boot options are listed in Boot Options .

1. Start the host.
2. When the ESX installer window appears, press `Shift+O` to edit the boot options.
3. At the `runweasel` command prompt, type `ks=`*`location of installation script plus boot command-line options`*.

   Boot Option

   You type the following boot options:

   ```
   ks=http://00.00.00.00/kickstart/ks-osdc-pdp101.cfg nameserver=00.00.0.0 ip=00.00.00.000
    netmask=255.255.255.0 gateway=00.00.00.000
   ```

Boot Options

When you perform a scripted installation, you might need to specify options at boot time to access the kickstart file.

Supported Boot Options

Table 16: Boot Options for ESX Installation

Boot Option	Description
`BOOTIF=`*`hwtype-MAC address`*	Similar to the `netdevice` option, except in the PXELINUX format as described in the IPAPPEND option under SYSLINUX at the `syslinux.org` site.
`gateway=`*`ip address`*	Sets this network gateway as the default gateway to be used for downloading the installation script and installation media.
`ip=`*`ip address`*	Sets up a static IP address to be used for downloading the installation script and the installation media. Note: the PXELINUX format for this option is also supported. See the IPAPPEND option under SYSLINUX at the `syslinux.org` site.
`ks=cdrom:/`*`path`*	Performs a scripted installation with the script at *`path`*, which resides on the CD in the CD-ROM drive. Each CDROM is mounted and checked until the file that matches the path is found. **Important:** If you have created an installer ISO image with a custom installation or upgrade script, you must use uppercase characters to provide the path of the script, for example, `ks=cdrom:/KS_CUST.CFG`.
`ks=file://`*`path`*	Performs a scripted installation with the script at *`path`*.
`ks=`*`protocol`*`://`*`serverpath`*	Performs a scripted installation with a script located on the network at the given URL. *`protocol`* can be `http`, `https`, `ftp`, or `nfs`. An example using NFS protocol is `ks=nfs://`*`host/porturl-path`*. The format of an NFS URL is specified in RFC 2224. **Important:** You must verify the URLs before using them, because the ESX installer does not validate SSL certificates for URLs.

Boot Option	Description
`ks=usb`	Performs a scripted installation, accessing the script from an attached USB drive. Searches for a file named `ks.cfg`. The file must be located in the root directory of the drive. If multiple USB flash drives are attached, they are searched until the `ks.cfg` file is found. Only FAT16 and FAT32 file systems are supported.
`ks=usb:/path`	Performs a scripted installation with the script file at the specified path, which resides on USB.
`ksdevice=device`	Tries to use a network adapter `device` when looking for an installation script and installation media. Specify as a MAC address, for example, `00:50:56:C0:00:01`. This location can also be a `vmnicNN` name. If not specified and files need to be retrieved over the network, the installer defaults to the first discovered network adapter that is plugged in.
`nameserver=ip address`	Specifies a domain name server to be used for downloading the installation script and installation media.
`netdevice=device`	Tries to use a network adapter `device` when looking for an installation script and installation media. Specify as a MAC address, for example, `00:50:56:C0:00:01`. This location can also be a `vmnicNN` name. If not specified and files need to be retrieved over the network, the installer defaults to the first discovered network adapter that is plugged in.
`netmask=subnet mask`	Specifies subnet mask for the network interface that downloads the installation script and the installation media.
`vlanid=vlanid`	Configure the network card to be on the specified VLAN.
`systemMediaSize`	You can use this option to increase the size of system storage partitions on the boot media. The `systemMediaSize` parameter accepts the following values: • default (138 GB) • max (consume all available space, for multi-terabyte servers)

For more information on ESX booting options post installation, see Broadcom knowledge base article 313264.

Installation and Upgrade Scripts Used for ESX Installation

You can use a default script to perform a standard ESX installation to the first detected disk.

During an ESX installation, you can use a default script to perform a standard installation to the first detected disk, and the boot loader configuration file to specify the kernel, the kernel options, and the boot modules that the mboot.c32 or mboot.efi boot loader uses for the installation.

- About the Default ks.cfg Installation Script
- About the boot.cfg File
- Locations Supported for Installation or Upgrade Scripts
- Path to the Installation or Upgrade Script
- Installation and Upgrade Script Commands
- Disk Device Names

About the Default ks.cfg Installation Script

The ESX installer includes a default installation script that performs a standard installation to the first detected disk.

The default `ks.cfg` installation script is located in the initial RAM disk at `/etc/vmware/weasel/ks.cfg`. You can specify the location of the default `ks.cfg` file with the `ks=file://etc/vmware/weasel/ks.cfg` boot option. See Enter Boot Options to Run an Installation or Upgrade Script.

When you install ESX using the `ks.cfg` script, the default root password is `myp@ssw0rd`.

You cannot modify the default script on the installation media. After the installation, you can use the vSphere Client to log in to the vCenter that manages the ESX host and modify the default settings.

With ESX 8.0 and later, if your system has data processing units (DPU), you use the `ks.cfg` script also to install ESX on DPUs.

The default script contains the following commands:

```
#
# Sample scripted installation file
#

# Accept the VMware End User License Agreement
vmaccepteula

# Set the root password for the DCUI and Tech Support Mode
rootpw myp@ssw0rd

# Install on the first local disk available on machine
install --firstdisk --overwritevmfs
 In case you system has DPUs, you also specify a PCI slot:
install --firstdisk --overwritevmfs --dpupcislots=<PCIeSlotID>

# Set the network to DHCP on the first network adapter
network --bootproto=dhcp --device=vmnic0

# A sample post-install script
%post --interpreter=python --ignorefailure=true
import time
stampFile = open('/finished.stamp', mode='w')
stampFile.write( time.asctime() )
```

About the boot.cfg File

The boot loader configuration file `boot.cfg` specifies the kernel, the kernel options, and the boot modules that the `mboot.c32` or `mboot.efi` boot loader uses in an ESX installation.

The `boot.cfg` file is provided in the ESX installer. You can modify the kernelopt line of the `boot.cfg` file to specify the location of an installation script or to pass other boot options.

The `boot.cfg` file has the following syntax:

```
# boot.cfg -- mboot configuration file
#
# Any line preceded with '#' is a comment.

title=STRING
prefix=DIRPATH
kernel=FILEPATH
kernelopt=STRING
modules=FILEPATH1---FILEPATH2... ---FILEPATHn

# Any other line must remain unchanged.
```

The commands in `boot.cfg` configure the boot loader.

Table 17: Commands in boot.cfg

Command	Description
`title=`*`STRING`*	Sets the boot loader title to *`STRING`*.
`prefix=`*`STRING`*	(Optional) Adds *`DIRPATH/`* in front of every *`FILEPATH`* in the `kernel=` and `modules=` commands that do not already start with / or with `http://`.

Command	Description
`kernel=FILEPATH`	Sets the kernel path to *FILEPATH*.
`kernelopt=STRING`	Appends *STRING* to the kernel boot options.
`modules=FILEPATH1 --- FILEPATH2... --- FILEPATHn`	Lists the modules to be loaded, separated by three hyphens (`---`).

See Create an Installer ISO Image with a Custom Installation or Upgrade Script and Network Booting the ESX Installer.

Locations Supported for Installation or Upgrade Scripts

In scripted installations and upgrades, the ESX installer can access the installation or upgrade script, also called the kickstart file, from several locations.

The following locations are supported for the installation or upgrade script:

- CD/DVD. See Create an Installer ISO Image with a Custom Installation or Upgrade Script.
- USB Flash drive. See Create a USB Flash Drive to Store the ESX Installation Script or Upgrade Script.
- A network location accessible through the following protocols: NFS, HTTP, HTTPS, FTP.

Path to the Installation or Upgrade Script

You can specify the path to an installation or upgrade script.

`ks=http://XXX.XXX.XXX.XXX/kickstart/KS.CFG` is the path to the ESX installation script, where*XXX.XXX.XXX.XXX*is the IP address of the machine where the script resides. See Installing ESX by Using a Script.

To start an installation script from an interactive installation, you enter the ks= option manually. See Enter Boot Options to Run an Installation or Upgrade Script.

Installation and Upgrade Script Commands

To modify the default installation or upgrade script or to create your own script, use supported commands. Use supported commands in the installation script, which you specify with a boot command when you boot the installer.

To determine which disk to install or upgrade ESX on, the installation script requires one of the following commands: `install`, `upgrade`, or `installorupgrade`. The `install` command creates the default partitions, including a VMFS datastore that occupies all available space after the other partitions are created.

With ESX 8.0 and later, if your system has supported data processing units (DPU), always consider the installation, re-installation or upgrade of ESX on the DPUs along with ESX on hosts. ESX update and upgrade on DPUs is not supported by the interactive or scripted method, you can only use vSphere Lifecycle Manager.

> **Note:**
> The use of SD and USB devices for storing ESX-OSData partitions is being deprecated. You can use SD and USB devices only to create boot bank partitions, `boot-bank 0` and `boot-bank 1`. Additionally, you can provide a persistent disk of minimum 32 GB on which to install the ESX-OSData partition. You define such disks by using the parameter `systemDisk` in the `install` command.

accepteula or vmaccepteula (Required)

Accepts the ESX license agreement.

> **Note:**
>
> Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. After you license the vCenter instance connected to the ESX hosts, they are licensed automatically. For more information, see Licensing Overview in the *Licensing* guide.

clearpart (Optional)

Clears any existing partitions on the disk. Requires the `install` command to be specified. Carefully edit the `clearpart` command in your existing scripts.

--drives=

Remove partitions on the specified drives.

--alldrives

Ignores the `--drives=` requirement and allows clearing of partitions on every drive.

--ignoredrives=

Removes partitions on all drives except those specified. Required unless the `--drives=` or `--alldrives` flag is specified.

--overwritevmfs

Allows overwriting of VMFS partitions on the specified drives. By default, overwriting VMFS partitions is not allowed.

--firstdisk=
disk-type1
[*disk-type2*,...]

Partitions the first eligible disk found. By default, the eligible disks are set to the following order:

1. Locally attached storage (`local`)
2. Network storage (`remote`)

You can change the order of the disks by using a comma-separated list appended to the argument. If you provide a filter list, the default settings are overridden. You can combine filters to specify a particular disk, including esx for the first disk with ESX installed on it, model and vendor information, or the name of the VMkernel device driver. For example, to prefer a disk with the model name ST3120814A and any disk that uses the mptsas driver rather than a normal local disk, the argument is --firstdisk=ST3120814A,mptsas,local. You can use `localesx` for local storage that contains ESX image or `remoteesx` for remote storage that contains ESX image.

dryrun (Optional)

Parses and checks the installation script. Does not perform the installation.

install

Specifies that this is a fresh installation. Either the `install`, `upgrade`, or `installorupgrade` command is required to determine which disk to install or upgrade ESX on.

--disk= or --drive=

Specifies the disk to partition. In the command `--disk=`*`diskname`*, the *`diskname`* can be a disk name or a full disk filesystem path in ESX, for example:

- Disk name: `--disk=naa.6d09466044143600247aee55ca2a6405` or
- Device path: `--disk=`*`/vmfs/devices/disks/mpx.vmhba1:C0:T0:L0`*

For accepted disk name formats, see Disk Device Names.

--firstdisk=
***disk-type1*,**
[*disk-type2*,...]

Partitions the first eligible disk found. By default, the eligible disks are set to the following order:

1. Locally attached storage (`local`)
2. Network storage (`remote`)

You can change the order of the disks by using a comma-separated list appended to the argument. If you provide a filter list, the default settings are overridden. You can combine filters to specify a particular disk, including esx for

the first disk with ESX installed on it, model and vendor information, or the name of the VMkernel device driver. For example, to prefer a disk with the model name ST3120814A and any disk that uses the mptsas driver rather than a normal local disk, the argument is --firstdisk=ST3120814A,mptsas,local. You can use `localesx` for local storage that contains ESX image or `remoteesx` for remote storage that contains ESX image.

--ignoressd

Excludes solid-state disks from eligibility for partitioning. This option can be used with the `install` command and the --firstdisk option. This option takes precedence over the --firstdisk option. This option is invalid with the --drive or --disk options and with the `upgrade` and `installorupgrade` commands. See the *vSphere Storage* documentation for more information about preventing SSD formatting during auto-partitioning.

--overwritevsan

You must use the --overwritevsan option when you install ESX on a disk, either SSD or HDD (magnetic), that is in a vSAN disk group. If you use this option and no vSAN partition is on the selected disk, the installation fails. When you install ESX on a disk that is in vSAN disk group, the result depends on the disk that you select:

- If you select an SSD, the SSD and all underlying HDDs in the same disk group is wiped.
- If you select an HDD, and the disk group size is greater than two, only the selected HDD is wiped.
- If you select an HDD disk, and the disk group size is two or less, the SSD and the selected HDD is wiped.

For more information about managing vSAN disk groups, see the *vSphere Storage* documentation.

--overwritevmfs

Required to overwrite an existing VMFS datastore on the disk before installation.

--preservevmfs

Preserves an existing VMFS datastore on the disk during installation.

--novmfsondisk

Prevents a VMFS partition from being created on this disk. Must be used with --overwritevmfs if a VMFS partition exists on the disk.

--systemdisk

If you use an USB or SD device, `systemDisk` specifies local persistent disk on which to install the ESX-OSData partition. For example, `install --firstdisk = usb --systemDisk=<diskID>`. As a result, boot bank partitions are placed on the USB device, while the OSData partition is on the disk specified in the `systemDisk` parameter.

--repartitionsystemdisk

If you use an USB or SD device and the local disk that you specify with the `systemDisk` parameter is not empty or contains a datastore, you can use `repartitionSystemDisk` to make sure that the persistent disk is repartitioned before use.

Note:
If a local persistent disk is not available or the disk size is less than 32GB, you see warning messages, but installation continues.

--forceunsupportedinstall

Blocks the installation of deprecated CPUs.

installorupgrade

Either the `install`, `upgrade`, or `installorupgrade` command is required to determine which disk to install or upgrade ESX on.

--disk= or --drive=

Specifies the disk to partition. In the command `--disk=`*`diskname`*, the *`diskname`* can be a disk name or a full disk filesystem path in ESX, for example:

- Disk name: `--disk=naa.6d09466044143600247aee55ca2a6405` or
- Device path: `--disk=`*`/vmfs/devices/disks/mpx.vmhba1:C0:T0:L0`*

For accepted disk name formats, see Disk Device Names.

--firstdisk=
disk-type1,
[*disk-type2*,...]

Partitions the first eligible disk found. By default, the eligible disks are set to the following order:

1. Locally attached storage (`local`)
2. Network storage (`remote`)

You can change the order of the disks by using a comma-separated list appended to the argument. If you provide a filter list, the default settings are overridden. You can combine filters to specify a particular disk, including esx for the first disk with ESX installed on it, model and vendor information, or the name of the VMkernel device driver. For example, to prefer a disk with the model name ST3120814A and any disk that uses the mptsas driver rather than a normal local disk, the argument is --firstdisk=ST3120814A,mptsas,local. You can use `localesx` for local storage that contains ESX image or `remoteesx` for remote storage that contains ESX image.

--overwritevsan

You must use the --overwritevsan option when you install ESX on a disk, either SSD or HDD (magnetic), that is in a vSAN disk group. If you use this option and no vSAN partition is on the selected disk, the installation fails. When you install ESX on a disk that is in a vSAN disk group, the result depends on the disk that you select:

- If you select an SSD, the SSD and all underlying HDDs in the same disk group is wiped.
- If you select an HDD, and the disk group size is greater than two, only the selected HDD is wiped.
- If you select an HDD disk, and the disk group size is two or less, the SSD and the selected HDD is wiped.

For more information about managing vSAN disk groups, see the *vSphere Storage* documentation.

--overwritevmfs

Install ESX if a VMFS partition exists on the disk, but no ESX installation exists. Unless this option is present, the installer fails if a VMFS partition exists on the disk, but anESX installation is missing.

keyboard (Optional)

Sets the keyboard type for the system.

keyboardType

Specifies the keyboard map for the selected keyboard type. *`keyboardType`* must be one of the following types.

- Belgian
- Brazilian
- Croatian
- Czechoslovakian
- Danish
- Estonian
- Finnish
- French
- German
- Greek
- Icelandic
- Italian
- Japanese
- Latin American
- Norwegian
- Polish
- Portuguese
- Russian
- Slovenian
- Spanish

- Swedish
- Swiss French
- Swiss German
- Turkish
- Ukrainian
- United Kingdom
- US Default
- US Dvorak

serialnum or vmserialnum (Optional)

The command is supported in ESX version 5.1 and later. Configures licensing. If not included, ESX installs in evaluation mode.

--esx=<license-key>

> **Note:**
>
> Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. After you license the vCenter instance connected to the ESX hosts, they are licensed automatically. For more information, see Licensing Overview in the *Licensing* guide.

network (Optional)

Specifies a network address for the system.

--bootproto=[dhcp|static]

Specifies whether to obtain the network settings from DHCP or set them manually.

--device=

Specifies either the MAC address of the network card or the device name, in the form `vmnicNN`, as in `vmnic0`. This option refers to the uplink device for the virtual switch.

--ip=

Sets an IP address for the machine to be installed, in the form `xxx.xxx.xxx.xxx`. Required with the --bootproto=static option and ignored otherwise.

--gateway=

Designates the default gateway as an IP address, in the form `xxx.xxx.xxx.xxx`. Used with the --bootproto=static option.

--nameserver=

Designates the primary name server as an IP address. Used with the `--bootproto=static` option. Omit this option if you do not intend to use DNS.

The `--nameserver` option can accept two IP addresses. For example: `--nameserver="10.126.87.104[,10.126.87.120]"`

--netmask=

Specifies the subnet mask for the installed system, in the form `255.xxx.xxx.xxx`. Used with the `--bootproto=static` option.

--hostname=

Specifies the host name for the installed system.

--vlanid= ***`vlanid`***

Specifies which VLAN the system is on. Used with either the --bootproto=dhcp or `--bootproto=static` option. Set to an integer from 1 to 4095.

--addvmportgroup=(0|1)

Specifies whether to add the VM Network port group, which is used by virtual machines. The default value is 1.

paranoid (Optional)

Causes warning messages to interrupt the installation. If you omit this command, warning messages are logged.

part or partition (Optional)

Creates an extra VMFS datastore on the system. Only one datastore per disk can be created. Cannot be used on the same disk as the `install` command. Only one partition can be specified per disk and it can only be a VMFS partition.

datastore name

Specifies where the partition is to be mounted.

--ondisk= or --ondrive=

Specifies the disk or drive where the partition is created.

--onfirstdisk=
disk-type1,
[*disk-type2*,...]

Partitions the first eligible disk found. By default, the eligible disks are set to the following order:

1. Locally attached storage (`local`)
2. Network storage (`remote`)

You can change the order of the disks by using a comma-separated list appended to the argument. If you provide a filter list, the default settings are overridden. You can combine filters to specify a particular disk, including esx for the first disk with ESX installed on it, model and vendor information, or the name of the VMkernel device driver. For example, to prefer a disk with the model name ST3120814A and any disk that uses the mptsas driver rather than a normal local disk, the argument is --onfirstdisk=ST3120814A,mptsas,local. You can use `localesx` for local storage that contains ESX image or `remoteesx` for remote storage that contains ESX image.

reboot (Optional)

Reboots the machine after the scripted installation is complete.

<--noeject>

The CD is not ejected after the installation.

rootpw (Required)

Sets the root password for the system.

--iscrypted

Specifies that the password is encrypted.

password

Specifies the password value.

upgrade

Either the `install`, `upgrade`, or `installorupgrade` command is required to determine which disk to install or upgrade ESX on.

--disk= or --drive=

Specifies the disk to partition. In the command `--disk=`*`diskname`*, the *`diskname`* can be a disk name or a full disk filesystem path in ESX, for example:

- Disk name: `--disk=naa.6d09466044143600247aee55ca2a6405` or
- Device path: `--disk=`*`/vmfs/devices/disks/mpx.vmhba1:C0:T0:L0`*

For accepted disk name formats, see Disk Device Names.

--firstdisk=
disk-type1,
[*disk-type2*,...]

Partitions the first eligible disk found. By default, the eligible disks are set to the following order:

1. Locally attached storage (`local`)
2. Network storage (`remote`)

You can change the order of the disks by using a comma-separated list appended to the argument. If you provide a filter list, the default settings are overridden. You can combine filters to specify a particular disk, including esx for the first disk with ESX installed on it, model and vendor information, or the name of the VMkernel device driver. For example, to prefer a disk with the model name ST3120814A and any disk that uses the mptsas driver rather than a normal local disk, the argument is --firstdisk=ST3120814A,mptsas,local. You can use `localesx` for local storage that contains ESX image or `remoteesx` for remote storage that contains ESX image.

%include or include (Optional)

Specifies another installation script to parse. This command is treated similarly to a multiline command, but takes only one argument.

filename

For example: `%include part.cfg`

%pre (Optional)

Specifies a script to run before the kickstart configuration is evaluated. For example, you can use it to generate files for the kickstart file to include.

--interpreter
=[python|busybox]

Specifies an interpreter to use. The default is busybox.

%post (Optional)

Runs the specified script after package installation is complete. If you specify multiple `%post` sections, they run in the order that they appear in the installation script.

--interpreter
=[python|busybox]

Specifies an interpreter to use. The default is busybox.

--timeout=secs

Specifies a timeout for running the script. If the script is not finished when the timeout expires, the script is forcefully stopped.

--ignorefailure
=[true|false]

If true, the installation is considered a success even if the `%post` script stops with an error.

%firstboot

Creates an `init` script that runs only during the first boot. The script has no effect on subsequent boots. If multiple `%firstboot` sections are specified, they run in the order that they appear in the kickstart file.

Note:
You cannot check the semantics of `%firstboot` scripts until the system is booting for the first time. A `%firstboot` script might contain potentially catastrophic errors that are not exposed until after the installation is complete.

Important:
The `%firstboot` script does not run, if secure boot is enabled on the ESX host.

--interpreter
=[python|busybox]

Specifies an interpreter to use. The default is busybox.

Note:
You cannot check the semantics of the `%firstboot` script until the system boots for the first time. If the script contains errors, they are not exposed until after the installation is complete.

Disk Device Names

The `install`, `upgrade`, and `installorupgrade` installation script commands require the use of disk device names.

Table 18: Disk Device Names

Format	Example	Description
NAA	naa.6d09466044143600247aee55ca2a6405	SCSI INQUIRY identifier
EUI	eui.3966623838646463	SCSI INQUIRY identifier
T10	t10.SanDisk00Cruzer_Blade000000004C5300011711 18101244	SCSI INQUIRY identifier
VML	vml.00025261	Legacy VMkernel identifier
MPX	mpx.vmhba0:C0:T0:L0	Path-based identifier

You can use the `esxcli storage core device list` command to display all device names in the vSphere CLI. The output is similar to the following example:

```
# esxcli storage core device list
naa.XXX
   Display Name: DGC Fibre Channel Disk(naa.XXX)
   ...
   Other UIDs: vml.000XXX
mpx.vmhba1:C0:T0:L0
   Display Name: Local VMware Disk (mpx.vmhba1:C0:T0:L0)
   ...
   Other UIDs: vml.0000000000XYZ
```

For more information on storage device names, see ESX Storage Device Names and Identifiers in the *vSphere Storage* documentation.

Install or Upgrade ESX from a CD or DVD by Using a Script

You can install or upgrade ESX from a CD-ROM or DVD-ROM drive by using a script that specifies the installation or upgrade options.

Before you run the scripted installation or upgrade, verify that the following prerequisites are met:

- The system on which you are installing or upgrading meets the hardware requirements. See ESX Hardware Requirements.
- You have the ESX installer ISO on an installation CD or DVD. See Download and Burn the ESX Installer ISO Image to a CD or DVD.
- The default installation or upgrade script (`ks.cfg`) or a custom installation or upgrade script is accessible to the system. See Installing ESX by Using a Script.
- You have selected a boot command to run the scripted installation or upgrade. See Enter Boot Options to Run an Installation or Upgrade Script. For a complete list of boot commands, see Boot Options .

You can start the installation or upgrade script by entering a boot option when you start the host. You can also create an installer ISO image that includes the installation script. With an installer ISO image, you can perform a scripted,

unattended installation when you boot the resulting installer ISO image. See Create an Installer ISO Image with a Custom Installation or Upgrade Script.

1. Boot the ESX installer from the local CD-ROM or DVD-ROM drive.
2. When the ESX installer window appears, press `Shift+O` to edit boot options.
3. Type a boot option that calls the default installation or upgrade script or an installation or upgrade script file that you created.

   The boot option has the form `ks=`.
4. Press Enter.

The installation, upgrade, or migration runs, using the options that you specified.

Install or Upgrade ESX from a USB Stick by Using a Script

You can install or upgrade ESX from a USB flash drive by using a script that specifies the installation or upgrade options.

Before running the scripted installation or upgrade, verify that the following prerequisites are met:

- The system that you are installing or upgrading to ESX meets the hardware requirements for the installation or upgrade. See ESX Hardware Requirements.
- You have the ESX installer ISO on a bootable USB flash drive. See Format a USB Flash Drive to Boot the ESX Installation or Upgrade.
- The default installation or upgrade script (`ks.cfg`) or a custom installation or upgrade script is accessible to the system. See Installing ESX by Using a Script.
- You have selected a boot option to run the scripted installation, upgrade, or migration. See Enter Boot Options to Run an Installation or Upgrade Script.

Supported boot options are listed in Boot Options .

1. Boot the ESX installer from the USB flash drive.
2. When the ESX installer window appears, press `Shift+O` to edit boot options.
3. Type a boot option that calls the default installation or upgrade script or an installation or upgrade script file that you created.

   The boot option has the form `ks=`.
4. Press Enter.

The installation, upgrade, or migration runs, using the options that you specified.

Network Boot the ESX Installer for a Scripted Installation or Upgrade

ESX 9.1 provides many options for booting the installer over a network and using an installation or upgrade script.

- For information about setting up a network infrastructure, see Network Booting the ESX Installer.
- For information about creating and locating an installation script, see Installing ESX by Using a Script.
- For specific procedures to network boot the ESX installer and use an installation script, see one of the following topics:
  - Boot the ESX Installer by Using Native UEFI HTTP
  - Boot the ESX Installer by Using iPXE and HTTP
  - Boot the ESX Installer by Using PXE and TFTP
- For information about using vSphere Auto Deploy to perform a scripted installation by using PXE to boot, see Installing ESX Using vSphere Auto Deploy.

Configuring External Entropy Sources During Scripted Installation

Starting with ESX 8.0 Update 1, you can configure external entropy sources in the kickstart file for scripted installation.

You can configure ESX in a highly secure environment to consume entropy from external entropy sources, such as a Hardware Security Module (HSM), and align with standards such as BSI Common criteria, EAL4, and NIST FIPS CMVP, by using the scripted installation method.

Starting with ESX 8.0 Update 1, a new entropy daemon, `entropyd`, is active and creates a vAPI endpoint to provide REST API to query and send entropy data to ESX hosts. The entropy daemon has several configurable parameters: `memory-cache size`, `storage-cache size`, `memory-low watermark`, and `entropy-lost-timeout`. The entropy daemon forwards entropy received from external devices to the VMkernel entropy pool.

If you do not need to select external entropy sources, you do not need to change anything in your existing scripts.

You can configure external entropy sources only for new installations. The entropy daemon uses entropy data passed during installation at first boot.

Note:

You cannot use any other method, such as interactive install, a cluster image, or Auto Deploy, to configure external entropy sources. You cannot enable the feature on an existing ESX host.

Before you update the kickstart file, you must:

- Get legitimate binary entropy data from an external source such as HSM.
- Save this data as a file of size between 512 KB to 10 MB, for example, `entropy_data.dat`.
- Encode the contents of the `entropy_data` file in a valid base64 format to create another file, for example `entropy_data.b64`.

In the kickstart file, you provide the following new parameters as part of the `entropy` command:

- `data` - the base64-encoded entropy data from an external source.
- `sources` - This bitmask value is set in the kernel settings during install time. The bitmask value is a 4-bits value:
  - binary(0000) = decimal(0) = default (all sources except entropyd if RDSEED is not supported, or FIPS-compliant sources if RDSEED is supported)
  - binary(0001) = decimal(1) = interrupts
  - binary(0010) = decimal(2) = RDRAND
  - binary(0100) = decimal(4) = RDSEED
  - binary(1000) = decimal(8) = entropyd (external entropy source)

  Source values are combined using bitwise OR operations: To enable multiple sources, sum the values. For example, if you combine RDSEED and entropyd, the calculation is 0100 | 1000 = 1100 or 4 + 8 = 12 in decimal, so you must use `entropySources=12`.

  If you decide to enable all entropy sources, then calculation is 0001 | 0010 | 0100 | 1000 = 1111 or 1+2+4+8 = 15 in decimal. The valid decimal range is 0-15.

  For EAL4 compliance, `8` is the only required value. Source values persist as a kernel setting during installation.

Note:

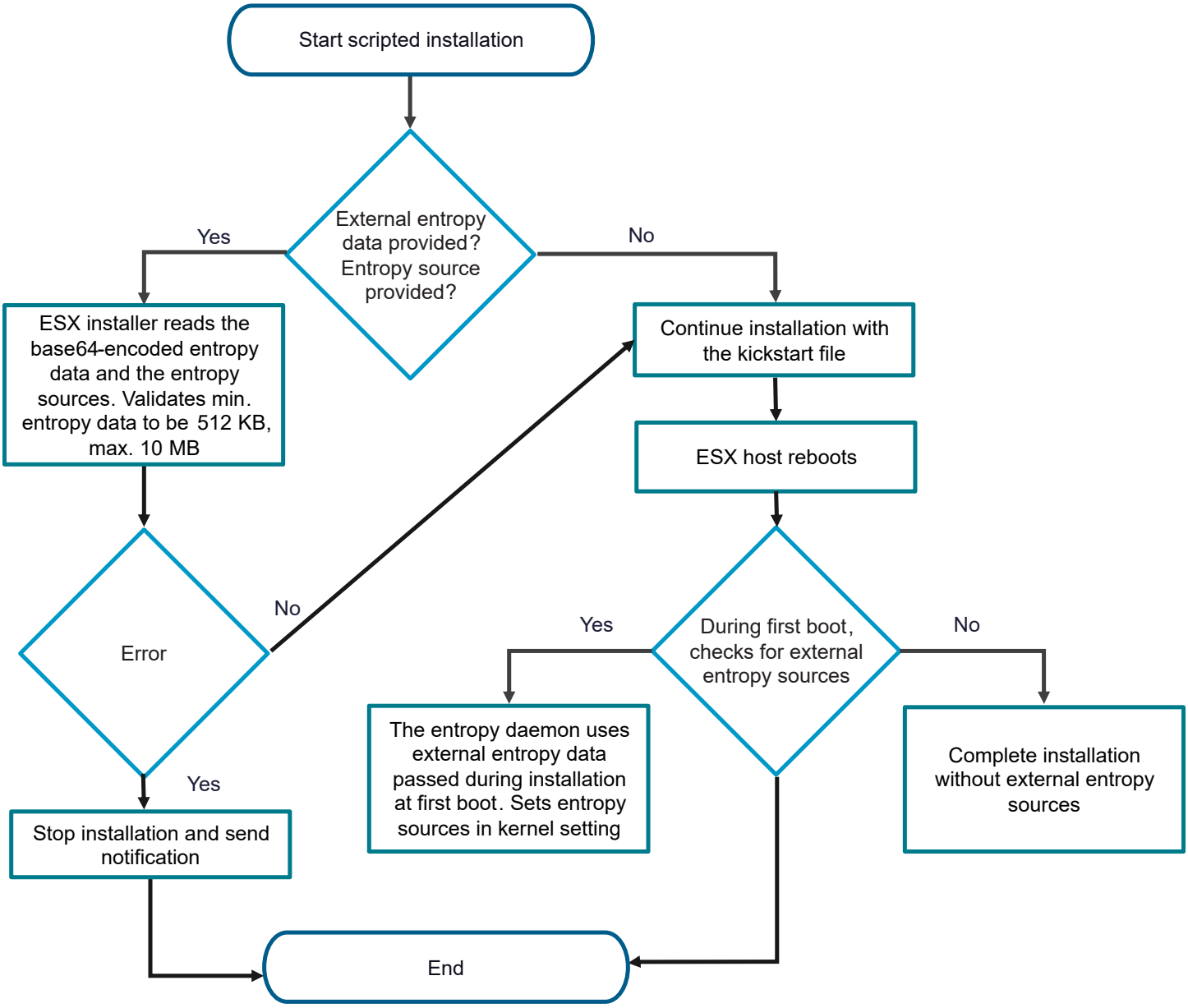
If you configure an ESX host with exteral entropy sources only, that is when the `sources` value is set to `8`, you must keep supplying external entropy to the host by using the entropy API. If external entropy is exhausted in the host, the host becomes unresponsive and it might require a hard reboot or re-installation to recover the host from such a situation.

If the ESX host is unresponsive even after a hard reset, you must boot the host with multiple entropy sources to recover.

A sample kickstart with the entropy parameters:

```
vmaccepteula
rootpw xxxxxxxx
entropy --sources=8 --data=xxxxxxx/xx/xxxxx/xx/xxxx...
install --firstdisk --overwritevmfs
network --bootproto=dhcp
```

Figure 4: Scripted installation workflow for adding external entropy sources



After installation completes, you can log in to the ESX host and define some parameters for the entropy daemon from the shell by using the following ESXCLI commands:

ESXCLI commands

1. Get Commands

##	esxcli system entropyd get	Command Description
1	`<no argument>`	Get currently configured and default values of all `entropyd` parameters
2	`--default-values`	Get default values

2. Set Commands

##	esxcli system entropyd set	Command Description
1	`--help`	Print details of "esxcli system entropyd set" command and it's arguments

2	`--reset=all`
	`--reset=memory-cache-size`
	`--reset=memory-cache-low-watermark`
	`--reset=storage-cache-size`
	`--reset=external-entropy-lost-timeout`
		Reset a parameter or all parameters to its default values.

3	`--memory-cache-size=` <value in KiB>	Set memory cache size in KiB.
4	`--memory-cache-low-watermark=` <value in %>	Set memory cache low water mark in percentage.
5	`--storage-cache-size=`<value in KiB>	Set storage cache size in KiB.
6	`--external-entropy-lost-timeout=` <value in seconds>	Set external entropy lost timeout in seconds.

Example:

```
$ esxcli system entropyd set --external-entropy-lost-timeout=70 --memory-cache-low-watermark=30 --memory-
cache-size=612 --storage-cache-size=5096
$ esxcli system entropyd get
External Entropy Lost Timeout Seconds: 70
Memory Cache Low Watermark Percentage: 30
Memory Cache Size Kibibytes: 612
Storage Cache Size Kibibytes: 5096

$ esxcli system entropyd get --default-values
External Entropy Lost Timeout Seconds: 60
Memory Cache Low Watermark Percentage: 20
Memory Cache Size Kibibytes: 512
Storage Cache Size Kibibytes: 4096
```

During installation, the ESX installer updates the entropy sources value in the kernel settings so that it persists in the ConfigStore after the installation. This change does not affect secure booting of ESX hosts.

Verification, monitoring, and troubleshooting

To check configured entropy sources, use the command `esxcli system settings kernel list -o entropySources`.

For troubleshooting, you can examine the following logs:

- Entropyd logs: `/var/run/log/entropyd.log`

- Installer logs: `/var/log/esxi_install.log`
- VMkernel logs for resource pool and vmkaccess violation: `/var/run/log/vmkernel.log`

You can also use VMkernel Observation (VOBs), which are kernel messages with unique identifier, to check for entropy alerts under **Monitor > Tasks and Events > Events** in the VMware Host Client. Alternatively, you can check entropy alerts in the ESX console by pressing `Alt+F11`.

For more information, see the vSphere Security and vSphere Automation SDKs Programming guides.

Network Booting the ESX Installer

If your host uses legacy BIOS or UEFI, you can use preboot execution environment (PXE) to boot an ESX host from a network device.

Alternatively, if your ESX host supports native UEFI HTTP, you can use hypertext transfer protocol (HTTP) to boot the host from a network device. ESX is distributed in an ISO format that is used to install to flash memory or to a local hard drive. You can extract the files and boot them over a network interface.

PXE uses Dynamic Host Configuration Protocol (DHCP) and Trivial File Transfer Protocol (TFTP) to boot an operating system over a network.

PXE booting requires some network infrastructure and a machine with a PXE-capable network adapter. Most machines that can run ESX have network adapters that can PXE boot.

Native UEFI HTTP uses DHCP and HTTP to boot over a network. UEFI HTTP boot requires a network infrastructure, UEFI firmware version on the ESX host that includes HTTP boot feature, and a network adapter that supports UEFI networking.

Booting by using HTTP is faster and more reliable than using TFTP. This is due to the capabilities of the TCP protocol that underlies the HTTP, such as built-in streaming and lost packet recovery. If your ESX hosts do not support native UEFI HTTP, you can use iPXE HTTP for the boot process.

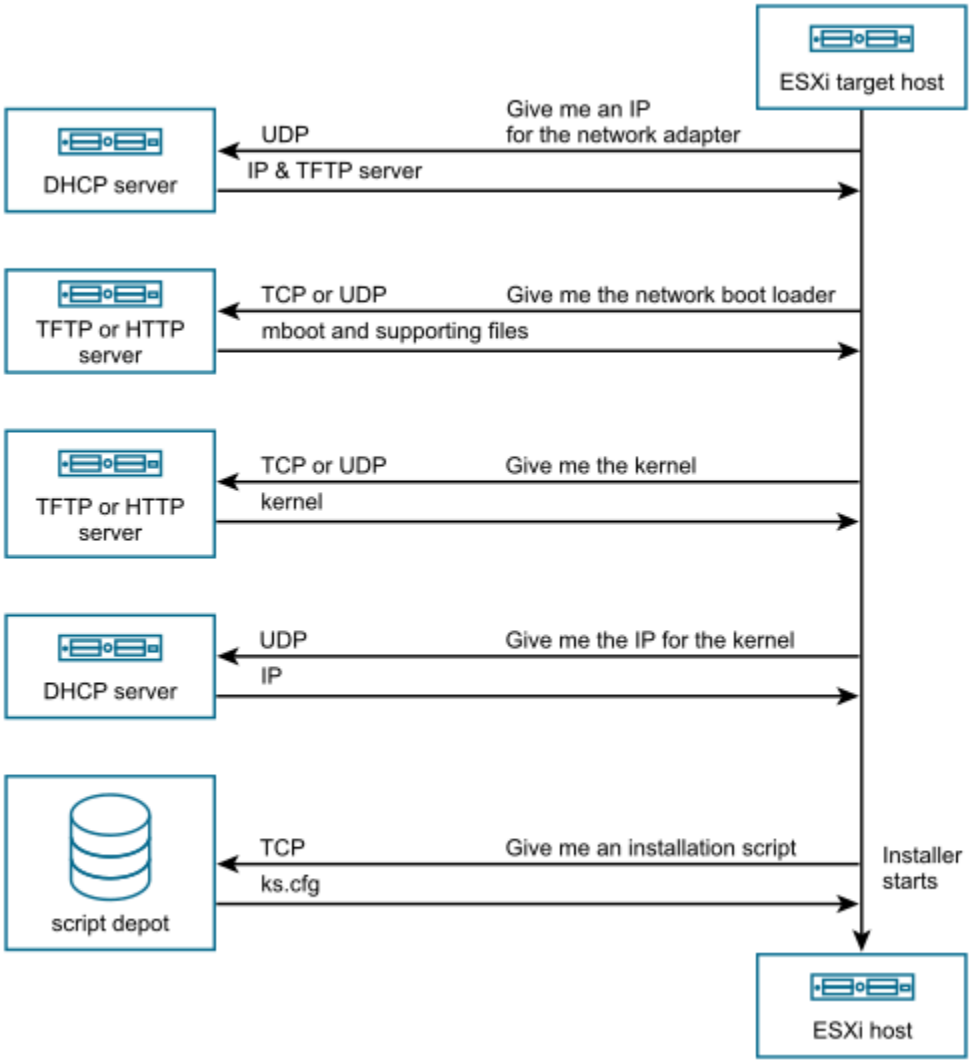
> **Note:**
> Network booting with legacy BIOS firmware is possible only over IPv4. Network booting with UEFI BIOS firmware is possible over IPv4 or IPv6.

Overview of the Network Boot Installation Process

You can boot an ESX host from a network interface. The network boot process varies depending on whether the target host is using legacy BIOS or UEFI firmware, and whether the boot process uses PXE TFTP, iPXE HTTP, or UEFI HTTP.

When you boot a target host, it interacts with the different servers in the environment to get a network adapter, boot loader, kernel, IP address for the kernel, and finally an installation script. When all components are in place, installation starts, as shown in the following illustration.

Figure 5: Overview of PXE Boot Installation Process



The interaction between the ESX host and other servers proceeds as follows:

1. The user boots the target ESX host.
2. The target ESX host makes a DHCP request.
3. The DHCP server responds with the IP information, the location of the TFTP or HTTP server, and the filename or URL of the initial network boot loader.
4. The ESX host contacts the TFTP or HTTP server and requests the filename or URL that the DHCP server specified.
5. The TFTP or HTTP server sends the network boot loader, and the ESX host runs it. The initial boot loader might load additional boot loader components from the server.
6. The boot loader searches for a configuration file on the TFTP or HTTP server, downloads the kernel and other ESX components as specified in the configuration file, and boots the kernel on the ESX host.
7. The installer runs interactively or using a kickstart script, as specified in the configuration file.

Network Boot Background Information

Understanding the network boot process can help you during troubleshooting.

TFTP Server

Trivial File Transfer Protocol (TFTP) is similar to the FTP service, and is typically used only for network booting systems or loading firmware on network devices such as routers. TFTP is available on Linux and Windows.

- Most Linux distributions include a copy of the tftp-hpa server. If you require a supported solution, purchase a supported TFTP server from your vendor of choice. You can also acquire a TFTP server from one of the packaged appliances on the VMware Marketplace.
- If your TFTP server runs on a Microsoft Windows host, use tftpd32 version 2.11 or later. See http://tftpd32.jounin.net/.

SYSLINUX and PXELINUX

If you are using PXE in a legacy BIOS environment, you must understand the different boot environments.

- SYSLINUX is an open-source boot environment for machines that run legacy BIOS firmware. The ESX boot loader for BIOS systems, `mboot.c32`, runs as a SYSLINUX plugin. You can configure SYSLINUX to boot from several types of media, including disk, ISO image, and network. You can find the SYSLINUX package at http://www.kernel.org/pub/linux/utils/boot/syslinux/.
- PXELINUX is a SYSXLINUX configuration for booting from a TFTP server according to the PXE standard. If you use PXELINUX to boot the ESX installer, the `pxelinux.0` binary file, `mboot.c32`, the configuration file, the kernel, and other files are transferred by TFTP.

  Note:
  VMware builds the `mboot.c32` plugin to work with SYSLINUX version 3.86 and tests PXE booting only with that version. Other versions might be incompatible. *The Open Source Disclosure Package for VMware vSphere Hypervisor* includes bug fixes for SYSLINUX version 3.86.

iPXE

iPXE is open-source software that provides an implementation of HTTP. You can use the software to perform an initial boot. For more information, see https://ipxe.org/.

VMware includes a build of iPXE as part of Auto Deploy. The source tree for this build is available in *The Open Source Disclosure Package for VMware vCenter*.

UEFI PXE and UEFI HTTP

Most UEFI firmware natively includes PXE support that allows booting from a TFTP server. The firmware can directly load the ESX boot loader for UEFI systems, `mboot.efi`. Additional software such as PXELINUX is not required.

Some UEFI firmware support native UEFI HTTP boot. The feature is introduced in version 2.5 of the UEFI specification. The firmware can load the ESX boot loader from an HTTP server, without additional software, such as iPXE.

Note:
Apple Macintosh products do not include PXE boot support. They include support for network booting through an Apple-specific protocol instead.

Alternative Approaches to Network Booting

Alternative approaches to network booting different software on different hosts are also possible, for example:

- Configuring the DHCP server to provide different initial boot loader filenames to different hosts depending on MAC address or other criteria. See your DCHP server's documentation.
- Approaches using iPXE as the initial bootloader with an iPXE configuration file that selects the next bootloader based on the MAC address or other criteria.

PXELINUX Configuration Files

You need a PXELINUX configuration file to boot the ESX installer on a legacy BIOS system. The configuration file defines the menu displayed to the target ESX host as it starts.

This section gives general information about PXELINUX configuration files.

For syntax details, see the SYSLINUX website at http://www.syslinux.org/.

Required Files

In the PXE configuration file, you must include paths to the following files:

- `mboot.c32` is the boot loader.
- `boot.cfg` is the boot loader configuration file.

See About the boot.cfg File

Filename for the PXE Configuration File

For the filename of the PXE configuration file, select one of the following options:

- `01-mac_address_of_target_ESX_host`. For example, `01-23-45-67-89-0a-bc`
- The target ESX host IP address in a hexadecimal notation.
- `default`

The initial boot file `pxelinux.0` tries to load a PXE configuration file in the following order:

1. It tries with the MAC address of the target ESX host, prefixed with its ARP type code, which is 01 for Ethernet.
2. If that attempt fails, it tries with the hexadecimal notation of target ESX system IP address.
3. Ultimately, it tries to load a file named `default`.

File Location for the PXE Configuration File

Save the file in `/tftpboot/pxelinux.cfg/` on the TFTP server.

For example, you might save the file on the TFTP server at `/tftpboot/pxelinux.cfg/01-00-21-5a-ce-40-f6`. The MAC address of the network adapter on the target ESX host is 00-21-5a-ce-40-f6.

Boot the ESX Installer by Using PXE and TFTP

You can use a TFTP server to PXE boot the ESX installer. The process differs slightly depending on whether you use UEFI or boot from a legacy BIOS.

Because most environments include ESX hosts that support UEFI boot and hosts that support only legacy BIOS, this topic discusses prerequisites and steps for both types of hosts.

Verify that your environment meets the following prerequisites.

- ESX installer ISO image, downloaded from the Broadcom Support Portal.
- Target host with a hardware configuration that is supported for your version of ESX. See the Broadcom Compatibility Guide.
- Network adapter with PXE support on the target ESX host.
- DHCP server that you can configure for PXE booting. See Sample DHCP Configurations.
- TFTP server.
- Network security policies to allow TFTP traffic (UDP port 69).
- For legacy BIOS, you can use only IPv4 networking. For UEFI PXE boot, you can use IPv4 or IPv6 networking.
- (Optional) Installation script (kickstart file).
- Use a native VLAN in most cases. If you want to specify the VLAN ID to be used with PXE booting, check that your NIC supports VLAN ID specification.

For legacy BIOS systems, obtain version 3.86 of the SYSLINUX package. For more information, see Network Boot Background Information.

- For legacy BIOS machines, the procedure supports booting multiple different versions of the ESX installer by using the same `pxelinux.0` initial boot loader for all target machines, but potentially different PXELINUX configuration files depending on the target machine's MAC address.
- For UEFI machines, the procedure supports booting multiple different versions of the ESX installer by using the same `mboot.efi` initial boot loader for all target machines, but potentially different `boot.cfg` files depending on the target machine's MAC address.

1. If your ESX host runs legacy BIOS firmware only, obtain and configure PXELINUX.
   a) Obtain SYSLINUX version 3.86, unpack it, and copy the `pxelinux.0` file to the top-level `/tftpboot` directory on your TFTP server.
   b) Create a PXELINUX configuration file using the following code model.

      ESX `-9.x.x-XXXXXX` is the name of the TFTP subdirectory that contains the ESX installer files.

      ```
      DEFAULT install
      NOHALT 1
      LABEL install
        KERNEL ESX-9.x.x-XXXXXX/mboot.c32
        APPEND -c ESX-9.x.x-XXXXXX/boot.cfg
        IPAPPEND 2
      ```

   c) Save the PXELINUX file in the `/tftpboot/pxelinux.cfg` directory on your TFTP server with a filename that will determine whether all hosts boot this installer by default:

Option	Description
Same installer	Name the file `default` if you want all host to boot this ESX installer by default.
Different installers	Name the file with the MAC address of the target host machine (01-*mac_address_of_target_ESX_host*) if you want only a specific host to boot with this file, for example, `01-23-45-67-89-0a-bc`.

2. If your ESX host runs UEFI firmware, copy the `efi/boot/bootx64.efi` and `efi/boot/crypto64.efi` files from the ESX installer ISO image to the `/tftpboot` folder on your TFTP server.
3. Rename the `efi/boot/bootx64.efi` file to `mboot.efi`.

   Note:
   Newer versions of `mboot.efi` can generally boot older versions of ESX, but older versions of `mboot.efi` might be unable to boot newer versions of ESX. If you plan to configure different hosts to boot different versions of the ESX installer, use the `mboot.efi` from the newest version.
4. Configure the DHCP server.
5. Create a subdirectory of your TFTP server's top-level `/tftpboot` directory and name it after the version of ESX it will hold, for example, `/tftpboot/ESX-9.x.x-xxxxx`.
6. Copy the contents of the ESX installer image to the newly created directory.
7. Modify the `boot.cfg` file
   a) Add the following line:

      ```
      prefix=ESX-9.x.x-xxxxxx
      ```

      Here, `ESX-9.x.x-xxxxxx` is the pathname of the installer files relative to the TFTP server's root directory.
   b) If the filenames in the `kernel=` and `modules=` lines begin with a forward slash (/) character, delete that character.
   c) If the `kernelopt=` line contains the string `cdromBoot`, remove the string only.
8. Optional: For a scripted installation, in the `boot.cfg` file, add the `kernelopt` option to the line after the kernel command, to specify the location of the installation script.

   Use the following code as a model, where `XXX.XXX.XXX.XXX` is the IP address of the server where the installation script resides, and `esxi_ksFiles` is the directory that contains the `ks.cfg` file.

   ```
   kernelopt=ks=http://XXX.XXX.XXX.XXX/esxi_ksFiles/ks.cfg
   ```

9. If your ESX host runs UEFI firmware, specify whether you want all UEFI hosts to boot the same installer.

Option	Description
Same installer	Copy or link the `boot.cfg` file to `/tftpboot/boot.cfg`
Different installers	1. Create a subdirectory of `/tftpboot` named after the MAC address of the target host machine

Option	Description
	(01-*mac_address_of_target_ESX_host*), for example, `01-23-45-67-89-0a-bc`. 2. Place a copy of (or a link to) the host's `boot.cfg` file in that directory, for example, `/tftpboot/01-23-45-67-89-0a-bc/boot.cfg`.

Boot the ESX Installer by Using iPXE and HTTP

You can use iPXE to boot the ESX installer from an HTTP server.

The prerequisites and steps depend on the support of UEFI boot or legacy BIOS only. You can use iPXE to boot the ESX installer from an HTTP server. The following topic discusses prerequisites and steps for ESX hosts that support UEFI boot and hosts that support legacy BIOS only.

Verify that your environment has the following components:

- ESX installer ISO image, downloaded from the Broadcom Support Portal.
- Target host with a hardware configuration that is supported for your version of ESX. See the *Broadcom Compatibility Guide*.
- Network adapter with PXE support on the target ESX host.
- DHCP server that you can configure for PXE booting. See Sample DHCP Configurations.
- TFTP server.
- Network security policies to allow TFTP traffic (UDP port 69).
- For legacy BIOS, you can use only IPv4 networking. For UEFI PXE boot, you can use IPv4 or IPv6 networking.
- (Optional) Installation script (kickstart file).
- Use a native VLAN in most cases. If you want to specify the VLAN ID to be used with PXE booting, check that your NIC supports VLAN ID specification.

Verify that your environment also meets the following prerequisites required for PXE boot using an HTTP Server:

- Verify that the HTTP server is accessible by your target ESX hosts.
- If your ESX host runs legacy BIOS firmware only, obtain version 3.86 of the SYSLINUX package. For more information, see Network Boot Background Information.
- For legacy BIOS machines, the procedure supports booting multiple different versions of the ESX installer by using the same `pxelinux.0` initial boot loader for all target machines, but potentially different PXELINUX configuration files depending on the target machine's MAC address.
- For UEFI machines, the procedure supports booting multiple different versions of the ESX installer by using the same `mboot.efi` initial boot loader for all target machines, but potentially different `boot.cfg` files depending on the target machine's MAC address.

1. Obtain and configure iPXE.
   a) Obtain the iPXE source code.
   b) On the iPXE download page, follow the build instructions, but run one of the following commands.
      - For ESX hosts that run legacy BIOS firmware only, run `make bin/undionly.kpxe`.
      - For ESX hosts that run UEFI firmware, run `make bin-x86_64-efi/snponly.efi`.
   c) Copy the `undionly.kpxe` or `snponly.efi` file to the `/tftpboot` directory on your TFTP server.
2. If your ESX host runs legacy BIOS firmware only, obtain and configure PXELINUX.
   a) Obtain SYSLINUX version 3.86, unpack it, and copy the `pxelinux.0` file to the `/tftpboot` directory on your TFTP server.
   b) Create a PXELINUX configuration file using the following code model.

      `ESX-9.x.x-XXXXXX` is the name of the TFTP subdirectory that contains the ESX installer files.

      ```
      DEFAULT install
      NOHALT 1
      LABEL install
        KERNEL ESX-9.x.x-XXXXXX/mboot.c32
      ```

```
APPEND -c ESX-9.x.x-XXXXXX/boot.cfg
IPAPPEND 2
```

c) Save the PXELINUX file in the `/tftpboot/pxelinux.cfg` directory on your TFTP server.
The filename determines whether all hosts boot this installer by default.

Option	Description
Same installer	Name the file `default` if you want all host to boot this ESX installer by default.
Different installers	Name the file with the MAC address of the target host machine (01-*mac_address_of_target_ESX_host*), if only a specific host must boot this file. For example, `01-23-45-67-89-0a-bc`.

3. If your ESX host runs UEFI firmware, copy the `efi/boot/bootx64.efi` file from the ESX installer ISO image to the `/tftpboot` folder on your TFTP server, and rename the file to `mboot.efi`.

   Note:
   Newer versions of `mboot.efi` can generally boot older versions of ESX, but older versions of `mboot.efi` might be unable to boot newer versions of ESX. If you plan to configure different hosts to boot different versions of the ESX installer, use the `mboot.efi` from the newest version.

4. Configure the DHCP server.
5. Create a directory on your HTTP server with the same name as the version of ESX it will hold. For example, `/var/www/html/ESX-9.x.x-XXXXXX`.
6. Copy the contents of the ESX installer image to the newly created directory.
7. Modify the `boot.cfg` file.
   a) Add the following line:
   ```
   prefix=http://XXX.XXX.XXX.XXX/ESX-9.x.x-XXXXXX
   ```
   where `http://XXX.XXX.XXX.XXX/ESX-9.x.x-XXXXXX` is the location of the installer files on the HTTP server.
   b) If the filenames in the `kernel=` and `modules=` lines begin with a forward slash (/) character, delete that character.
   c) If the `kernelopt=` line contains the string `cdromBoot`, remove the string only.
8. Optional: For a scripted installation, in the `boot.cfg` file, add the `kernelopt` option to the line after the kernel command, to specify the location of the installation script.
   Use the following code as a model, where `XXX.XXX.XXX.XXX` is the IP address of the server where the installation script resides, and `esxi_ksFiles` is the directory that contains the `ks.cfg` file.
   ```
   kernelopt=ks=http://XXX.XXX.XXX.XXX/esxi_ksFiles/ks.cfg
   ```
9. If your ESX host runs UEFI firmware, specify whether you want all UEFI hosts to boot the same installer.

Option	Description
Same installer	Copy or link the `boot.cfg` file to `/tftpboot/boot.cfg`
Different installers	1. Create a subdirectory of `/tftpboot` named after the MAC address of the target host machine (01-*mac_address_of_target_ESX_host*), for example, `01-23-45-67-89-0a-bc`. 2. Place a copy of (or a link to) the host's `boot.cfg` file in that directory, for example, `/tftpboot/01-23-45-67-89-0a-bc/boot.cfg`.

Boot the ESX Installer by Using Native UEFI HTTP

You can boot the ESX installer directly from an HTTP server, without additional software to support the process.

Verify that your environment has the following components:

- ESX host with UEFI firmware that supports the HTTP boot feature.

  - ESX installer ISO image, downloaded from the Broadcom Support Portal.
  - Target host with a hardware configuration that is supported for your version of ESX. See the Broadcom Compatibility Guide.
- Network adapter with UEFI networking support on the target ESX host.
- DHCP server that you can configure for UEFI HTTP booting. See Sample DHCP Configurations.
  - (Optional) Installation script (kickstart file)
  - Use a native VLAN in most cases. If you want to specify the VLAN ID to be used with HTTP(S) Booting, check that your NIC supports VLAN ID specification.

UEFI HTTP supports booting multiple versions of the ESX installer. You use the same `mboot.efi` initial boot loader for all target machines, but potentially different `boot.cfg` files depending on the MAC address os the target machine.

Note:
Do not mix IPv4 or IPv6 networking during the boot process. Use either IPv4 or IPv6 networking.

1. Copy the `efi/boot/bootx64.efi` file from the ESX installer ISO image to a directory on your HTTP server and rename the file to `mboot.efi`. For example, `http://www.example.com/esx/mboot.efi`.

   Note:
   Newer versions of `mboot.efi` can generally boot older versions of ESX, but older versions of `mboot.efi` might be unable to boot newer versions of ESX. If you plan to configure different hosts to boot different versions of the ESX installer, use the `mboot.efi` from the newest version.

2. Configure the DHCP server.
3. Create a directory on your HTTP server with the same name as the version of ESX it holds. For example, `http://www.example.com/esx/ESX-9.x.x-XXXXXX`.
4. Copy the contents of the ESX installer image to the newly created directory.
5. Modify the `boot.cfg` file.
   a) Add the following line with the URL of the newly created directory.

      ```
      prefix=http://www.example.com/esx/ESX-9.x.x-XXXXXX
      ```

   b) If the filenames in the `kernel=` and `modules=` lines begin with a forward slash (/) character, delete that character.
   c) If the `kernelopt=` line contains the string `cdromBoot`, remove the string only.
6. Optional: For a scripted installation, in the `boot.cfg` file, add the `kernelopt` option to the line after the kernel command, to specify the location of the installation script.

   For example, `kernelopt=ks=http://www.example.com/esx_ksFiles/ks.cfg`
7. Optional: You can use the virtual machine configuration parameters *`networkBootProtocol`* and *`networkBootUri`* to specify from where a virtual machines can boot. The setting *`networkBootProtocol`* specifies the boot protocol, IPv4 or IPv6. For example, `networkBootProtocol = httpv4`. The setting *`networkBootUri`* specifies the HTTP URL to the ESX bootloader (bootx64.efi). For example, `networkBootUri = http://xxx.xxx.xx.x/esx9.xxxx/efi/boot/bootx64.efi`.
8. Specify whether you want all UEFI hosts to boot the same installer.

Option	Description
Same installer	Add the `boot.cfg` file to the same directory as `mboot.efi`. For example, `http://www.example.com/esx/boot.cfg`
Different installers	1. Create a subdirectory of the directory that contains the `mboot.efi` file. Name the directory as the MAC address of the target host machine (01-*`mac_address_of_target_ESX_host`*), for example, `01-23-45-67-89-0a-bc`. 2. Add the custom `boot.cfg` file in the directory. For example, `http://www.example.com/esx/01-23-45-67-89-0a-bc/boot.cfg`.

You can use both installer types. ESX hosts without custom `boot.cfg` file on your HTTP server, boot from the default `boot.cfg` file.

Sample DHCP Configurations

The DHCP server must send the address of the TFTP or HTTP server and the filename of the initial boot loader to the ESX host.

When the target machine first boots, it broadcasts a packet across the network requesting information to boot itself. The DHCP server responds. The DHCP server must be able to determine whether the target machine is allowed to boot and the location of the initial boot loader binary. For PXE boot, the location is a file on a TFTP server. For UEFI HTTP boot, the location is a URL.

> **CAUTION:**
> Do not set up a second DHCP server if your network already has one. If multiple DHCP servers respond to DHCP requests, machines can obtain incorrect or conflicting IP addresses, or can fail to receive the proper boot information. Talk to a network administrator before setting up a DHCP server. For support on configuring DHCP, contact your DHCP server vendor.

There are many DHCP servers that you can use. The following examples are for an ISC DHCP server. If you are using a version of DHCP for Microsoft Windows, see the DHCP server documentation to determine how to pass the next-server and filename arguments to the target machine.

Example of Booting Using PXE and TFTP with IPv4

This example shows how to configure an ISC DHCP server to PXE boot ESX using a TFTP server at IPv4 address `xxx.xxx.xxx.xxx`.

```
#
# ISC DHCP server configuration file snippet.  This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
allow booting;
allow bootp;
option client-system-arch code 93 = unsigned integer 16;
class "pxeclients" {
   match if substring(option vendor-class-identifier, 0, 9) = "PXEClient";
   next-server xxx.xxx.xxx.xxx;
   if option client-system-arch = 00:07 or option client-system-arch = 00:09 {
      filename = "mboot.efi";
   } else {
      filename = "pxelinux.0";
   }
}
```

When a machine attempts to PXE boot, the DHCP server provides an IP address and the location of the `pxelinux.0` or `mboot.efi` binary file on the TFTP server.

Example of Booting Using PXE and TFTP with IPv6

This example shows how to configure an ISC DHCPv6 server to PXE boot ESX using a TFTP server at IPv6 address xxxx:xxxx:xxxx:xxxx::xxxx.

```
#
# ISC DHCPv6 server configuration file snippet.  This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
```

```
allow booting;
allow bootp;
option dhcp6.bootfile-url code 59 = string;
option dhcp6.bootfile-url "tftp://[xxxx:xxxx:xxxx:xxxx::xxxx]/mboot.efi";
```

When a machine attempts to PXE boot, the DHCP server provides an IP address and the location of the `mboot.efi` binary file on the TFTP server.

Example of Booting Using iPXE and HTTP with IPv4

This example shows how to configure an ISC DHCP server to boot ESX by loading iPXE from a TFTP server at IPv4 address `xxx.xxx.xxx.xxx`.

```
#
# ISC DHCP server configuration file snippet.  This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
allow booting;
allow bootp;
option client-system-arch code 93 = unsigned integer 16;
class "pxeclients" {
   match if substring(option vendor-class-identifier, 0, 9) = "PXEClient";
   next-server xxx.xxx.xxx.xxx;
   if option client-system-arch = 00:07 or option client-system-arch = 00:09 {
      if exists user-class and option user-class = "iPXE" {
         # Instruct iPXE to load mboot.efi as secondary bootloader
         filename = "mboot.efi";
      } else {
         # Load the snponly.efi configuration of iPXE as initial bootloader
         filename = "snponly.efi";
      }
   } else {
      if exists user-class and option user-class = "iPXE" {
         # Instruct iPXE to load pxelinux as secondary bootloader
         filename = "pxelinux.0";
      } else {
         # Load the undionly configuration of iPXE as initial bootloader
         filename = "undionly.kpxe";
   }
}
```

When a machine attempts to PXE boot, the DHCP server provides an IP address and the location of the `undionly.kpxe` or `snponly.efi` binary file on the TFTP server. In the legacy BIOS case, iPXE then asks the DHCP server for the next file to load, and the server returns `pxelinux.0` as the filename. In the UEFI case, iPXE then asks the DHCP server for the next file to load, and this time the server returns `mboot.efi` as the filename. In both cases, iPXE is resident and the system has HTTP capability. As a result, the system can load additional files from an HTTP server.

Example of Booting Using iPXE and HTTP with IPv6

This example shows how to configure an ISC DHCPv6 server to boot ESX by loading iPXE from a TFTP server at IPv6 address `xxxx:xxxx:xxxx:xxxx::xxxx`.

```
#
# ISC DHCPv6 server configuration file snippet. This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
allow booting;
```

```
allow bootp;

option dhcp6.bootfile-url code 59 = string;
if exists user-class and option user-class = "iPXE" {
   # Instruct iPXE to load mboot.efi as secondary bootloader
   option dhcp6.bootfile-url "tftp://[xxxx:xxxx:xxxx:xxxx::xxxx]/mboot.efi";
} else {
   # Load the snponly.efi configuration of iPXE as initial bootloader
   option dhcp6.bootfile-url "tftp://[xxxx:xxxx:xxxx:xxxx::xxxx]/snponly.efi";
}
```

When a machine attempts to PXE boot, the DHCP server provides an IP address and the location of the `snponly.efi` (iPXE) binary file on the TFTP server. iPXE then asks the DHCP server for the next file to load, and this time the server returns `mboot.efi` as the filename. iPXE is resident and the system has HTTP capability. As a result, the system can load additional files from an HTTP server.

Example of Booting Using UEFI HTTP with IPv4

This example shows how to configure an ISC DHCP server to boot ESX by using native UEFI HTTP over IPv4 from Web server `www.example.com`.

```
#
# ISC DHCP server configuration file snippet. This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
allow booting;
allow bootp;
option client-system-arch code 93 = unsigned integer 16;
class "httpclients" {
   match if substring(option vendor-class-identifier, 0, 10) = "HTTPClient";
   option vendor-class-identifier "HTTPClient";

   if option client-system-arch = 00:10 {
      # x86_64 UEFI HTTP client
      filename = http://www.example.com/esx/mboot.efi;
   }
}
```

Example of Booting Using UEFI HTTP with IPv6

This example shows how to configure an ISC DHCPv6 server to boot ESX by using native UEFI HTTP over IPv6 from Web server `www.example.com`.

```
#
# ISC DHCPv6 server configuration file snippet. This is not a complete
# configuration file; see the ISC server documentation for details on
# how to configure the DHCP server.
#
allow booting;
allow bootp;

option dhcp6.bootfile-url code 59 = string;
option dhcp6.user-class code 15 = { integer 16, string };
option dhcp6.vendor-class code 16 = { integer 32, integer 16, string };

if option dhcp6.client-arch-type = 00:10 {
      # x86_64 HTTP clients
      option dhcp6.vendor-class 0 10 "HTTPClient";
```

```
        option dhcp6.bootfile-url "http://www.example.com/esx/mboot.efi";
}
```

Installing ESX Using vSphere Auto Deploy

vSphere Auto Deploy lets you provision hundreds of physical hosts with ESX software.

Using Auto Deploy, experienced system administrators can manage large deployments efficiently. Hosts are network-booted from a central Auto Deploy server. Optionally, hosts are configured with a host profile of a reference host. The host profile can be set up to prompt the user for input. After boot up and configuration complete, the hosts are managed by vCenter just like other ESX hosts.

Important:

The vSphere Host Profiles capability is deprecated in ESX 9.0 and will be removed in a future release. Instead of Host profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level.

Auto Deploy can also be used for stateless caching or stateful installations.

Important:
Auto Deploy requires a secure separation between the production network and the management or deployment networks as discussed in vSphere Auto Deploy Security Considerations. Using Auto Deploy without this separation is insecure.

Stateless caching	By default, Auto Deploy does not store ESX configuration or state on the host disk. Instead, an image profile defines the image that the host is provisioned with, and other host attributes are managed through host profiles. A host that uses Auto Deploy for stateless caching still needs to connect to the Auto Deploy server and the vCenter.
Stateful installs	You can provision a host with Auto Deploy and set up the host to store the image to a disk. On subsequent boots, the host boots from the disk.

Understanding vSphere Auto Deploy

vSphere Auto Deploy can provision hundreds of physical hosts with ESX software.

You can specify the image to deploy and the hosts to provision with the image. Optionally, you can specify host profiles to apply to the hosts, a vCenter location (datacenter, folder, or cluster), and assign a script bundle for each host.

- Introduction to vSphere Auto Deploy
- Rules and Rule Sets

Introduction to vSphere Auto Deploy

vSphere Auto Deploy uses network boot infrastructure with host profiles, a desired image, or vSphere Configuration Profiles to provision ESX hosts.

State Information for ESX Hosts

Note:
You cannot use Auto Deploy on ESX hosts configured with DPUs as part of the vSphere Distributed Services Engine feature.

When you start a physical host that is set up for vSphere Auto Deploy, vSphere Auto Deploy uses PXE boot infrastructure in conjunction with vSphere Host Profiles, a vSphere Lifecycle Manager desired image, or vSphere Configuration Profiles to provision and customize that host. No state is stored on the host itself. Instead, the vSphere Auto Deploy server manages state information for each host. vSphere Auto Deploy stores the information for the ESX hosts to be provisioned in different locations. Information about the location of image profiles, host profiles, or clusters that you manage by either a single image or by a configuration on a cluster level is initially specified in the rules that map machines to image profiles and host profiles.

Important:

The vSphere Host Profiles capability is deprecated in ESX 9.0 and will be removed in a future release. Instead of Host profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level.

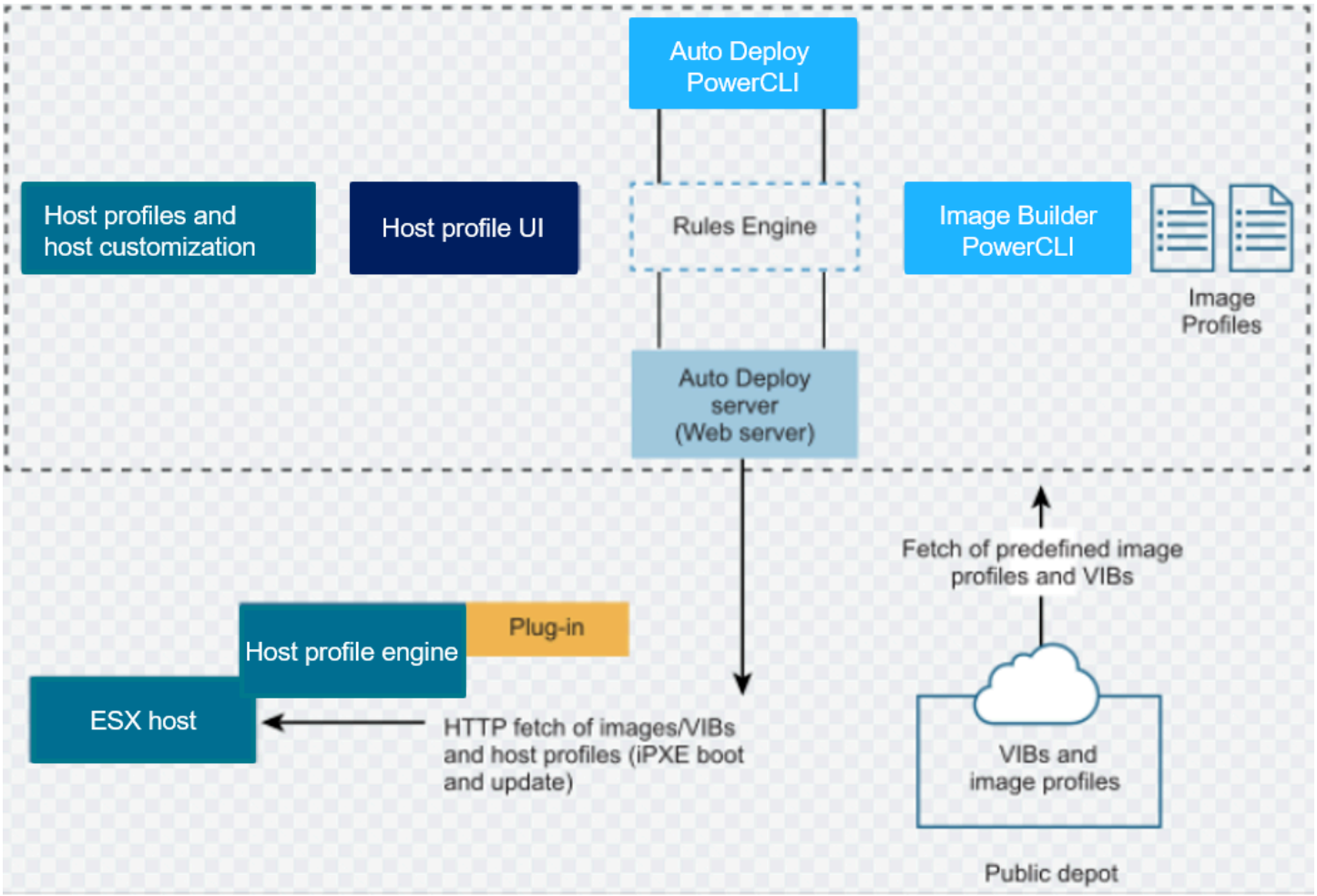
Table 19: vSphere Auto Deploy Stores Information for Deployment

Information Type	Description	Source of Information
Image state	The executable software to run on an ESX host.	Image profile, created with vSphere ESX Image Builder or a vSphere Lifecycle Manager image.
Configuration state	The configurable settings that determine how the host is configured, for example, virtual switches and their settings, driver settings, boot parameters, and so on.	Host profiles, created by using the vSphere Client, or a configuration that you create when setting up a cluster that manages all ESX host settings at a cluster level, such as vSphere Configuration Profiles.
Dynamic state	The runtime state that is generated by the running software, for example, generated private keys or runtime databases.	Host memory, lost during reboot.
Virtual machine state	The virtual machines stored on a host and virtual machine autostart information (subsequent boots only).	Virtual machine information sent by vCenter to vSphere Auto Deploy must be available to supply virtual machine information to vSphere Auto Deploy.
User input	State that is based on user input, for example, an IP address that the user provides when the system starts up, cannot automatically be included in the host profile.	Host customization information, stored by vCenter during first boot. You can create a host profile that requires user input for certain values. When vSphere Auto Deploy applies a host profile that requires user provided information, the host is placed in maintenance mode. In the vSphere Client, check the host profile compliance, and respond to the prompt to customize the host.

vSphere Auto Deploy Architecture

The vSphere Auto Deploy infrastructure consists of several components.

Figure 6: vSphere Auto Deploy Architecture



vSphere Auto Deploy server	Serves images and host profiles to ESX hosts.
vSphere Auto Deploy rules engine	Sends information to the vSphere Auto Deploy server which image profile and which host profile to serve to which host. Administrators use vSphere Auto Deploy to define the rules that assign image profiles and host profiles to hosts. For more information on vSphere Auto Deploy rules and rule sets, see Rules and Rule Sets. Apart from legacy image profiles that you create by using the VMware Image Builder and host profiles, you can also create vSphere Auto Deploy rules to deploy ESX by using a single vSphere Lifecycle Manager image or vSphere Configuration Profiles.
Image profiles	Define the set of VIBs with which to boot ESX hosts.

	• Broadcom and Broadcom partners make image profiles and VIBs available in public depots. Use vSphere ESX Image Builder to examine the depot and use the vSphere Auto Deploy rules engine to specify which image profile to assign to which host. • You use vSphere Lifecycle Manager images to apply software and firmware updates to the ESX hosts in a cluster. Using a single image to manage all hosts in a cluster ensures cluster-wide host image homogeneity. • Starting with ESX 8.0, you can also use vSphere Configuration Profiles to manage the configuration of all hosts in a cluster. • You can create a custom image profile based on the public image profiles and VIBs in the depot and apply that image profile to the host. See Customizing Installations with vSphere ESX Image Builder. •
Host profiles	Define machine-specific configuration such as networking or storage setup. Use the host profile UI to create host profiles. You can create a host profile for a reference host and apply that host profile to other hosts in your environment for a consistent configuration. For more information, see the *vSphere Host Profiles* documentation or the Setting Up a vSphere Auto Deploy Reference Host section.
Host customization	Stores information that the user provides when host profiles are applied to the host. Host customization might contain an IP address or other information that the user supplied for that host. For more information about host customizations, see the *vSphere Host Profiles* documentation. Host customization was called answer file in earlier releases of vSphere Auto Deploy.

Auto Deploy Certificates

By default, the Auto Deploy server provisions each host with certificates that are signed by the VMware Certificate Authority (VMware CA). For more information, see the *Managing Certificates for ESX Hosts* in the *vSphere Security* guide.

Alternatively, if your corporate policy requires that you use custom certificates, you can set up the Auto Deploy server to provision all hosts with custom certificates that are not signed by VMware CA. The Auto Deploy server becomes a subordinate certificate authority of your third-party CA. In the Custom Certificate Authority mode, you are responsible for managing the certificates. You cannot refresh and renew certificates from the vSphere Client. In this mode, you also cannot select only a set of hosts to provision with custom certificates, and you can manually sign custom certificates only for stateful hosts. For more information, see *Use Custom Certificates with Auto Deploy* in the *vSphere Security* guide.

Auto Deploy provides a third option that allows you to generate a certificate outside vSphere and become independent of the certificate management in vCenter. For example, you can generate a custom certificate by using a custom script or by using a provider of domain name registry services such as Verisign. You can use custom certificates for only a set of ESX hosts. You can provide custom certificates for stateless hosts as well. ESX hosts are identified by the MAC address of the NIC used for network booting, or the BIOS UUID of the ESX host. You update the VMware Endpoint Certificate Store (VECS) with the custom certificate by using VCF PowerCLI. For more information on the new VCF PowerCLI cmdlets, see How vSphere Auto Deploy Works with VCF PowerCLI. The VMware CA must trust the custom ESX certificates so you must add the CA public certificate for the custom certificates to the TRUSTED_ROOTS store in VECS. Auto Deploy also stores the custom certificates and when it recognizes a booting host with the respective MAC address of the NIC used for network booting, or the BIOS UUID of the ESX host, it automatically provides the custom certificate. You do not need to stop or restart Auto Deploy or vCenter when you add a custom certificate to VECS, only restart the host for which you upload a custom certificate. For more information, see *Use Custom Certificates with Auto Deploy* in the *vSphere Security* guide.

Rules and Rule Sets

You specify the behavior of the vSphere Auto Deploy server by using a set of rules. The vSphere Auto Deploy rules engine checks the rule set for matching host patterns to decide which items (image profile, host profile, vCenter location, or script object) to provision each host with.

The rules engine maps software and configuration settings to hosts based on the attributes of the host. For example, you can deploy image profiles or host profiles to two clusters of hosts by writing two rules, each matching on the network address of one cluster.

For hosts that have not yet been added to a vCenter system, the vSphere Auto Deploy server checks with the rules engine before serving image profiles, host profiles, and inventory location information to hosts. For hosts that are

managed by a vCenter system, the image profile, host profile, and inventory location that vCenter has stored in the host object is used. If you make changes to rules, you can use the vSphere Client or vSphere Auto Deploy cmdlets in a VCF PowerCLI session to test and repair rule compliance. When you repair rule compliance for a host, that host's image profile and host profile assignments are updated.

Note:

Auto Deploy can merge rules to complete a configuration. For example, if you have three rules, one specifying only an image, another only a cluster, and a third one with a configuration, if an ESX host matches the 3 rules, it can get the image from the first, the cluster from the second, and the configuration from the third rule. This is useful when you aim to write fewer rules for larger deployments, where all clusters in a RAC can get a configuration from several rules rather than creating complete rules for each cluster.

The rules engine includes rules and rule sets.

Rules

Rules can assign image profiles and host profiles to a set of hosts, or specify the location (folder or cluster) of a host on the target vCenter system. A rule can identify target hosts by boot MAC address, SMBIOS information, BIOS UUID, Vendor, Model, or fixed DHCP IP address. In most cases, rules apply to multiple hosts. You create rules by using the vSphere Client or vSphere Auto Deploy cmdlets in a VCF PowerCLI session. After you create a rule, you must add it to a rule set. Only two rule sets, the active rule set and the working rule set, are supported. A rule can belong to both sets, the default, or only to the working rule set. After you add a rule to a rule set, you can no longer change the rule. Instead, you copy the rule and replace items or patterns in the copy. If you are managing vSphere Auto Deploy with the vSphere Client, you can edit a rule if it is in inactive state.

You can specify the following parameters in a rule.

Parameter	Description
Name	Name of the rule, specified with the `-Name` parameter.
Item	One or more items, specified with the `-Item` parameter. An item can be an image profile, a host profile, a vCenter inventory location (datacenter, folder, cluster) for the target host, or a custom script. You can specify multiple items separated by commas.
Pattern	The pattern specifies the host or group of hosts to which the rule applies. **vendor** Machine vendor name. **model** Machine model name. **serial** Machine serial number. **hostname** Machine hostname. **domain** Domain name. **UUID** The Universally Unique Identifier. **ipv4** IPv4 address of the machine. **Gateway IPV4** A Gateway IPv4 address. **ipv6** IPv6 address of the machine. PXE booting with BIOS firmware is possible only with IPv4, PXE booting with UEFI firmware is possible with either IPv4 or IPv6. **mac** Boot NIC MAC address. **asset** Machine asset tag. **oemstring** OEM-specific strings in the SMBIOS. You can specify `-AllHosts` to apply the item or items to all hosts.

Active Rule Set

When a newly started host contacts the vSphere Auto Deploy server with a request for an image profile, the vSphere Auto Deploy server checks the active rule set for matching rules. The image profile, host profile, vCenter inventory location, and script object that are mapped by matching rules are then used to boot the host. If more than one item of the same type is mapped by the rules, the vSphere Auto Deploy server uses the item that is first in the rule set.

Working Rule Set

The working rule set allows you to test changes to rules before making the changes active. For example, you can use vSphere Auto Deploy cmdlets for testing compliance with the working rule set. The test verifies that hosts managed by a vCenter system are following the rules in the working rule set. By default, cmdlets

add the rule to the working rule set and activate the rules. Use the `NoActivate` parameter to add a rule only to the working rule set.

You use the following workflow with rules and rule sets.

1. Make changes to the working rule set.
2. Test the working rule set rules against a host to make sure that everything is working correctly.
3. Refine and retest the rules in the working rule set.
4. Activate the rules in the working rule set.
   If you add a rule in a VCF PowerCLI session and do not specify the `NoActivate` parameter, all rules that are currently in the working rule set are activated. You cannot activate individual rules.

See the VCF PowerCLI command-line help and How vSphere Auto Deploy Works with VCF PowerCLIfor more information on using vSphere Auto Deploy with VCF PowerCLI cmdlets. See Managing vSphere Auto Deploy with the vSphere Client for more information on using vSphere Auto Deploy with the vSphere Client.

Install and Configure vSphere Auto Deploy

Before you can start using vSphere Auto Deploy, you must prepare your environment in several steps.

You start with server setup and hardware preparation. You must configure the vSphere Auto Deploy service startup type in the vCenter system that you plan to use for managing the ESX hosts you provision, and install VCF PowerCLI.

- vSphere Auto Deploy Preinstallation Checklist
- Prepare Your System for vSphere Auto Deploy
- Using vSphere Auto Deploy Cmdlets

vSphere Auto Deploy Preinstallation Checklist

Before you can start the tasks in this vSphere Auto Deploy scenario, make sure that your environment meets the hardware and software requirements, and that you have the necessary permissions for the components included in the setup.

Table 20: Preinstallation Checklist

Required Software and Hardware	Details
vCenter	The vSphere Auto Deploy server is part of vCenter. You must enable and start the vSphere Auto Deploy service on the vCenter system. You can perform many of the setup tasks by logging in to vCenter. See Prepare Your System for vSphere Auto Deploy.
Storage	Storage for ESX datastores NFS, iSCSI, or Fibre Channel, with servers and storage arrays that are configured so the servers can detect the LUNs. • A list of target IP addresses for NFS or iSCSI. • A list of target volume information for NFS or iSCSI.
ESX hosts information	A list of target IP addresses for NFS or iSCSI. A list of target volume information for NFS or iSCSI. • Default route, net mask, and primary and secondary DNS server IP addresses. • IP address and net mask for the VMkernel primary management network. • IP address and net mask for other VMkernel networks such as storage, vSphere FT, or VMware vMotion. vSphere Auto Deploy does not overwrite existing partitions by default.
VCF PowerCLI	See *Install PowerCLI* topic in the *ESX Install* guide.
ESX software depot	The location of the ESX software depot on the Broadcom Support Portal. You use a URL to point to the image profile stored at that location, or you download a ZIP file to work with a local depot. Do not download the ESX image.
TFTP server	TFTP installer software such as WinAgents TFTP server.

Required Software and Hardware	Details
DHCP server	The DHCP server is included in the vSphere supported Windows Server versions.
DNS server	A working DNS server. You must add entries in both Forward (A Record) and Reverse (PTR Record) Zones for each target host.

You also need information about and administrator privileges to the core servers of the environment, including the Active Directory server, DNS server, DHCP server, NTP server, and so on.

You must have complete control of the broadcast domain of the subnet in which you deploy the setup. Ensure that no other DHCP, DNS, or TFTP server are on this subnet.

Prepare Your System for vSphere Auto Deploy

Before you can PXE boot an ESX host with vSphere Auto Deploy, you must install prerequisite software and set up the DHCP and TFTP servers that vSphere Auto Deploy interacts with.

- Verify that the hosts that you plan to provision with vSphere Auto Deploy meet the hardware requirements for ESX. See ESX Hardware Requirements.
- Verify that the ESX hosts have network connectivity to vCenter and that all port requirements are met. See the *vCenter Upgrade* guide.
- Verify that you have a TFTP server and a DHCP server in your environment to send files and assign network addresses to the ESX hosts that Auto Deploy provisions. See teh topics *Install the TFTP Server* and *Prepare the DHCP Server for vSphere Auto Deploy Provisioning* in the *ESX Installation* guide.
- Verify that the ESX hosts have network connectivity to DHCP, TFTP, and vSphere Auto Deploy servers.
- If you want to use VLANs in your vSphere Auto Deploy environment, you must set up the end to end networking properly. When the host is PXE booting, the firmware driver must be set up to tag the frames with proper VLAN IDs. You must do this set up manually by making the correct changes in the UEFI/BIOS interface. You must also correctly configure the ESX port groups with the correct VLAN IDs. Ask your network administrator how VLAN IDs are used in your environment.
- Verify that you have enough storage for the vSphere Auto Deploy repository. The vSphere Auto Deploy server uses the repository to store data it needs, including the rules and rule sets you create and the VIBs and image profiles that you specify in your rules.

  Best practice for ESX 9.0 and later is to allocate 6 GB to have enough room for four image profiles and some extra space. Each image profile requires approximately 400 MB. Determine how much space to reserve for the vSphere Auto Deploy repository by considering how many image profiles you expect to use.
- Obtain administrative privileges to the DHCP server that manages the network segment you want to boot from. You can use a DHCP server already in your environment, or install a DHCP server. For your vSphere Auto Deploy setup, replace the `gpxelinux.0` filename with `snponly64.efi.vmw-hardwired` for UEFI or `undionly.kpxe.vmw-hardwired` for BIOS. For more information on DHCP configurations, see Sample DHCP Configurations.
- Secure your network as for any other PXE-based deployment method. vSphere Auto Deploy transfers data over SSL to prevent casual interference and snooping. However, the authenticity of the client or the vSphere Auto Deploy server is not checked during a PXE boot.
- If you want to manage vSphere Auto Deploy with VCF PowerCLI cmdlets, verify that PowerShell 7.5 is installed. See the *vSphere PowerCLI User's Guide*.
- Set up a remote syslog server. See the *vCenter and Host Management* documentation for syslog server configuration information. Configure the first host you boot to use the remote syslog server and apply that host's host profile to all other target hosts. Optionally, install and use VMware Cloud Foundation Operations for logs, which provides log aggregation and analytics for VMware and non-VMware products, virtual and physical, with near real-time search and analytics of log events.
- Install ESX Dump Collector, set up your first host so that all core dumps are directed to ESX Dump Collector, and apply the host profile from that host to all other hosts. See Configure ESX Dump Collector with ESXCLI.

- If the hosts that you plan to provision with vSphere Auto Deploy are with legacy BIOS, verify that the vSphere Auto Deploy server has an IPv4 address. PXE booting with legacy BIOS firmware is possible only over IPv4. PXE booting with UEFI firmware is possible with either IPv4 or IPv6.

If you want to manage vSphere Auto Deploy with VCF PowerCLI cmdlets, see *Set Up vSphere Auto Deploy and Provision Hosts with vSphere PowerCLI*.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. Click **Enable Auto Deploy and Image Builder** to activate the service.

   If the **Image Builder** service is already active, select the **Configure** tab and click **Enable Auto Deploy Service**.

   The **Software Depot** page appears.
3. Configure the TFTP server.
   a) Click the **Configure** tab.
   b) Click **Download TFTP Boot Zip** to download the TFTP configuration file and unzip the file to the directory in which your TFTP server stores files.
   c) Optional: To use a proxy server, click **Add** on the *Auto Deploy Runtime Summary* pane and enter a proxy server URL in the text box.

      Using reverse proxy servers can offload the requests made to the vSphere Auto Deploy server.
4. Set up your DHCP server to point to the TFTP server on which the TFTP ZIP file is located.
   a) Specify the TFTP Server's IP address in DHCP option 66, frequently called next-server.
   b) Specify the boot filename, which is `snponly64.efi.vmw-hardwired` for UEFI or `undionly.kpxe.vmw-hardwired` for BIOS in the DHCP option 67, frequently called `boot-filename`.
5. Set each host you want to provision with vSphere Auto Deploy to network boot or PXE boot, following the manufacturer's instructions.
6. Optional: If you set up your environment to use Thumbprint mode, you can use your own Certificate Authority (CA) by replacing the OpenSSL certificate `rbd-ca.crt` and the OpenSSL private key `rbd-ca.key` with your own certificate and key file.

   The files are in `/etc/vmware-rbd/ssl/`.

   By default, vCenter uses VMware Certificate Authority (VMCA).

When you start an ESX host that is set up for vSphere Auto Deploy, the host contacts the DHCP server and is directed to the vSphere Auto Deploy server, which provisions the host with the image profile specified in the active rule set.

- You can change the default configuration properties of the **Auto Deploy Service**. For more information, see "Configuring vCenter" in the *vCenter and Host Management* documentation.
- You can change the default configuration properties of the **Image Builder Service**. For more information, see "Configuring vCenter" in the *vCenter and Host Management* documentation.
- Define a rule that assigns an image profile and optional host profile, host location, or script bundle to the host. For managing vSphere Auto Deploy with VCF PowerCLI cmdlets, see the How vSphere Auto Deploy Works with VCF PowerCLI section. For managing vSphere Auto Deploy with the vSphere Client, see the Managing vSphere Auto Deploy with the vSphere Client section.
- Configure the first host that you provision as a reference host. Use the storage, networking, and other settings you want for your target hosts to share. Create a host profile for the reference host and write a rule that assigns both the already tested image profile and the host profile to target hosts.
- If you want to have vSphere Auto Deploy overwrite existing partitions, set up a reference host to do auto partitioning and apply the host profile of the reference host to other hosts. See Configure a Reference Host for Auto-Partitioning .
- If you have to configure host-specific information, set up the host profile of the reference host to prompt for user input. For more information about host customizations, see the *vSphere Host Profiles* documentation.

Using vSphere Auto Deploy Cmdlets

vSphere Auto Deploy cmdlets are implemented as Microsoft PowerShell cmdlets and included in VCF PowerCLI. Users of vSphere Auto Deploy cmdlets can take advantage of all VCF PowerCLI features.

Experienced PowerShell users can use vSphere Auto Deploy cmdlets just like other PowerShell cmdlets. If you are new to PowerShell and VCF PowerCLI, the following tips might be helpful.

You can type cmdlets, parameters, and parameter values in the VCF PowerCLI shell.

- Get help for any cmdlet by running `Get-Help`*`cmdlet_name`*.
- Remember that PowerShell is not case sensitive.
- Use tab completion for cmdlet names and parameter names.
- Format any variable and cmdlet output by using `Format-List` or `Format-Table`, or their short forms `fl` or `ft`. For more information, run the `Get-Help Format-List` cmdlet.

Passing Parameters by Name

You can pass in parameters by name in most cases and surround parameter values that contain spaces or special characters with double quotes.

```
Copy-DeployRule -DeployRule testrule -ReplaceItem MyNewProfile
```

Most examples in the *vCenter Deployment and Setup* documentation pass in parameters by name.

Passing Parameters as Objects

You can pass parameters as objects if you want to perform scripting and automation. Passing in parameters as objects is useful with cmdlets that return multiple objects and with cmdlets that return a single object. Consider the following example.

1. Bind the object that encapsulates rule set compliance information for a host to a variable.
   ```
   $tr = Test-DeployRuleSetCompliance MyEsx42
   ```
2. View the `itemlist` property of the object to see the difference between what is in the rule set and what the host is currently using.
   ```
   $tr.itemlist
   ```
3. Remediate the host to use the revised rule set by using the `Repair-DeployRuleSetCompliance` cmdlet with the variable.
   ```
   Repair-DeployRuleSetCompliance $tr
   ```

The example remediates the host the next time you boot the host.

How vSphere Auto Deploy Works with VCF PowerCLI

You can manage vSphere Auto Deploy with VCF PowerCLI cmdlets to create rules that associate ESX hosts with various objects on the vCenter target.

You can manage vSphere Auto Deploy with VCF PowerCLI cmdlets to create rules that associate hosts with image profiles, host profiles, custom scripts and locations on the vCenter target. You can also update hosts by testing rule compliance and repairing compliance issues.

- Overview of the vSphere Auto Deploy Process by Using VCF PowerCLI
- Inventory of vSphere Auto Deploy VCF PowerCLI Cmdlets
- Assign an Image Profile to Hosts
- Write a Rule and Assign a Host Profile to Hosts
- Write a Rule and Assign a Host to a Folder or Cluster
- Configure a Stateless System by Running a Custom Script
- Test and Repair Rule Compliance
- Register a Caching Proxy Server Address with vSphere Auto Deploy

Overview of the vSphere Auto Deploy Process by Using VCF PowerCLI

You can write VCF PowerCLI scripts that provision hosts and boot hosts with an image that you specify, customizing the image profile, host profile, or vCenter location.

You can manage vSphere Auto Deploy with VCF PowerCLI cmdlets to create rules that associate hosts with image profiles, host profiles, custom scripts and locations on the vCenter target. You can also update hosts by testing rule compliance and repairing compliance issues.

Getting started with vSphere Auto Deploy requires that you learn how vSphere Auto Deploy works, install the vSphere Auto Deploy server, install VCF PowerCLI, write VCF PowerCLI rules that provision hosts, and power on your hosts to be booted with the image profile you specify. You can customize of the image profile, host profile, and vCenter location.

See vSphere PowerCLI Scenario for vSphere Auto Deploy for a step-by-step exercise that helps you set up your first vSphere Auto Deploy environment.

To provision the hosts in your environment with vSphere Auto Deploy successfully, you can follow these steps.

1. Deploy vCenter.
   The vSphere Auto Deploy server is included.
2. Configure the vSphere Auto Deploy service startup type.
   See Prepare Your System for vSphere Auto Deploy.
3. Install VCF PowerCLI, which includes vSphere Auto Deploy and vSphere ESX Image Builder cmdlets.
   See Configure vSphere ESX Image Builder, Using vSphere Auto Deploy Cmdlets, and Using VMware.ImageBuilder Cmdlets.
4. Find the image profile that includes the VIBs that you want to deploy to your hosts.
   - Usually, you add the depots containing the required software to your VCF PowerCLI session, and then select an image profile from one of those depots.
   - To create a custom image profile, use vSphere ESX Image Builder cmdlets to clone an existing image profile and add the custom VIBs to the clone.

   You must use vSphere ESX Image Builder for customization only if you have to add or remove VIBs. In most cases, you can add the depot where VMware hosts the image profiles to your VCF PowerCLI session as a URL.
5. Start a PowerShell session and connect to the vCenter system.
6. Use the `New-DeployRule`VCF PowerCLI cmdlet to write a rule that assigns the image profile to one host, to multiple hosts specified by a pattern, or to all hosts.
   ```
   New-DeployRule -Name "testrule" -Item image-profile -AllHosts
   ```
   See Assign an Image Profile to Hosts.

   Important:

   With ESX 9.0, support for stateless installation is deprecated. To continue using stateless hosts, you must use the `Set-ClusterNetworkBootMode` cmdlet that allows you to use network booting and provision stateless hosts in a selected cluster managed by vSphere Lifecycle Manager images.

   Note:
   vSphere Auto Deploy is optimized for provisioning hosts that have a fixed MAC address to IP address mapping in DHCP (sometimes called DHCP reservations). If you want to use static IP addresses, you must set up the host profile to prompt for host customization. For more information, see the *vSphere Host Profiles* documentation.
7. Power on the hosts that you want to provision.
8. If you do not use a vSphere Lifecycle Manager image or a vSphere Configuration Profile, set up a host you provision as a reference host for a host profile.
   You can specify the reference host syslog settings, firewall settings, storage, networking, and so on.
9. Set up the host you provisioned as a reference host for your host profile.
   You can specify the reference host syslog settings, firewall settings, storage, networking, and so on. See Setting Up a vSphere Auto Deploy Reference Host.
10. Create and export a host profile for the reference host.
    See the *Host Profiles* documentation.
11. To provision multiple hosts with the host profile, use the `Copy-DeployRule` cmdlet to edit the previously created rule.

You can revise the rule to assign not only an image profile but also a host profile, a vCenter location, and a custom script bundle.

```
Copy-DeployRule -DeployRule "testrule" -ReplaceItem my_host_profile_from_reference_host,my_target_cluster
                -ReplacePattern  "ipv4=192.XXX.1.10-192.XXX.1.20"
```

Where *my_host_profile_from_reference_host* is the name of the reference host profile, and *my_target_cluster* is the name of the target cluster.

12. Perform the test and repair compliance operations to remediate the hosts.
    See Test and Repair Rule Compliance .
13. Verify that the hosts you provisioned meet the following requirements.
    - Each host is connected to the vCenter system.
    - The hosts are not in maintenance mode.
    - The hosts have no compliance failures.
    - Each host with a host profile that requires user input has up-to-date host customization information.

    Remediate host associations and compliance problems and reboot hosts until all hosts meet the requirements.

Read *Understanding vSphere Auto Deploy* for an introduction to the boot process, differences between first and subsequent boots, and an overview of using host customization.

Inventory of vSphere Auto Deploy VCF PowerCLI Cmdlets

You specify the rules that assign image profiles and host profiles to hosts using a set of VCF PowerCLI cmdlets that are included in VCF PowerCLI.

If you are new to VCF PowerCLI, read the VCF PowerCLI documentation and review Using vSphere Auto Deploy Cmdlets. You can get help for any command at the PowerShell prompt.

- Basic help: `Get-Helpcmdlet_name`
- Detailed help: `Get-Help cmdlet_name-Detailed`

Note:
When you run vSphere Auto Deploy cmdlets, provide all parameters on the command line when you invoke the cmdlet. Supplying parameters in interactive mode is not recommended.

Important:

Starting with ESX 9.0, you cannot use the `New-DeployRule` cmdlet for images earlier than ESX 8.0.

Table 21: Rule Engine VCF PowerCLI Cmdlets

Command	Description
`Get-DeployCommand`	Returns a list of vSphere Auto Deploy cmdlets.
`Set-ClusterNetworkBootMode`	Configures the network boot options for ESX hosts in clusters managed by vSphere Lifecycle Manager images. When used with the `stateless` parameter, the cmdlet allows the use of UEFI PXE and UEFI HTTP(S) network booting to provision stateless hosts in a selected cluster. Equals the **Enable Auto Deploy Stateless** option in the vSphere Client. When used with the `bootstrap` parameter, the cmdlet allows the use of UEFI HTTP(S) network booting to provision stateful hosts in a selected cluster. Equals the **Enable Zero Touch Provisioning** option in the vSphere Client.
`Reset-ClusterNetworkBootMode`	Restores the default stateful provisioning of hosts in a cluster from the network boot mode.
`New-DeployRule`	Creates a new rule with the specified items and patterns.
`Set-DeployRule`	Updates an existing rule with the specified items and patterns. You cannot update a rule that is part of a rule set.
`Get-DeployRule`	Retrieves the rules with the specified names.

Command	Description
`Copy-DeployRule`	Clones and updates an existing rule.
`Add-DeployRule`	Adds one or more rules to the working rule set and, by default, also to the active rule set. Use the `NoActivate` parameter to add a rule only to the working rule set.
`Remove-DeployRule`	Removes one or more rules from the working rule set and from the active rule set. Run this command with the `-Delete` parameter to completely delete the rule.
`Set-DeployRuleset`	Explicitly sets the list of rules in the working rule set.
`Get-DeployRuleset`	Retrieves the current working rule set or the current active rule set.
`Switch-ActiveDeployRuleset`	Activates a rule set so that any new requests are evaluated through the rule set.
`Get-VMHostMatchingRules`	Retrieves rules matching a pattern. For example, you can retrieve all rules that apply to a host or hosts. Use this cmdlet primarily for debugging.
`Test-DeployRulesetCompliance`	Checks whether the items associated with a specified host are in compliance with the active rule set.
`Repair-DeployRulesetCompliance`	Given the output of `Test-DeployRulesetCompliance`, this cmdlet updates the image profile, host profile, and location for each host in the vCenter inventory. The cmdlet might apply image profiles, apply host profiles, or move hosts to prespecified folders or clusters on the vCenter system.
`Apply-EsxImageProfile`	Associates the specified image profile with the specified host.
`Get-VMHostImageProfile`	Retrieves the image profile in use by a specified host. This cmdlet differs from the `Get-EsxImageProfile` cmdlet in vSphere mage Builder.
`Repair-DeployImageCache`	Use this cmdlet only if the vSphere Auto Deploy image cache is accidentally deleted.
`Get-VMHostAttributes`	Retrieves the attributes for a host that are used when the vSphere Auto Deploy server evaluates the rules.
`Get-DeployMachineIdentity`	Returns a string value that vSphere Auto Deploy uses to logically link an ESX host in vCenter to a physical machine.
`Set-DeployMachineIdentity`	Logically links a host object in the vCenter database to a physical machine. Use this cmdlet to add hosts without specifying rules.
`Get-DeployOption`	Retrieves the vSphere Auto Deploy global configuration options. This cmdlet currently supports the vlan-id option, which specifies the default VLAN ID for the ESX Management Network of a host provisioned with vSphere Auto Deploy. vSphere Auto Deploy uses the value only if the host boots without a host profile.
`Set-DeployOption`	Sets the value of a global configuration option. Currently supports the vlan-id option for setting the default VLAN ID for the ESX Management Network.
`Add-ProxyServer`	Adds a proxy server to the vSphere Auto Deploy database. Run the command with the `-Address` parameter to specify the IPv4 or IPv6 address. The address can include a port number.
`List-ProxyServer`	Lists the proxy servers that are currently registered with vSphere Auto Deploy.
`Delete-ProxyServer`	Deletes one or more proxy servers from the list of proxy servers that are registered with vSphere Auto Deploy. You can run the command with the `-id` parameter from the list of proxy servers or with the`-Address` parameter by specifying the IPv4 or IPv6 address of the proxy server you want to delete.
`Add-ScriptBundle`	Adds one or more script bundles to the vSphere Auto Deploy server.
`Get-ScriptBundle`	Retrieves the list of script bundles available on the vSphere Auto Deploy server and the scripts they contain.
`Remove-ScriptBundle`	Removes a script bundle from vSphere Auto Deploy. Applicable for vSphere version 6.7 and later.

Command	Description
`Get-CustomCertificates`	Retrieves the custom host certificate uploaded into AutoDeploy. You must run the command with the `-HostId [MAC_Address | BIOS_UUID]` parameter. The first time you add custom certificates, you don't see any certificates returned by this cmdlet.
`List-CustomCertificates`	Retrieves information about all custom host certificates used by Auto Deploy. The list provides details for the name of the certificate, `Host ID`, and `Associated Host Name`, which reflects the name of the vCenter for the Auto Deploy server.
`Add-CustomCertificate`	Adds a custom certificate to the VMware Endpoint Certificate Store and associates it with an ESX host. The certificate becomes active upon host reboot. You can use the `Get-CustomCertificate` cmdlet to retrieve the custom host certificate key. You can run the command with the `-HostId [MAC_Address | BIOS_UUID]` parameter to associate the certificate to the host, specifying a `-Key [file:///path/to/key.key]` and `-Cert [file:///path/to/cert.crt]`. Using this cmdlet requires the **AutoDeploy.Rule.Create** privilege on the root folder of vCenter.
`Remove-CustomCertificates`	Removes a set of custom host certificates from Auto Deploy. The certificate entries are deleted from the database and the certificate files are removed from the filestore. Hosts that have already booted with a custom certificate must be rebooted to receive a new certificate. You must provide at least one of `-Cert` or `-HostId` parameters. Using this cmdlet requires the **AutoDeploy.Rule.Create** privilege on the root folder of vCenter.

Assign an Image Profile to Hosts

Before you can provision a host, you must create rules that assign an image profile to each host that you want to provision by using vSphere Auto Deploy.

vSphere Auto Deploy extensibility rules enforce that VIBs at the CommunitySupported level can only contain files from certain predefined locations, such as the ESXCLI plug-in path, jumpstart plug-in path, and so on. If you add a VIB that is in a different location to an image profile, a warning results. You can override the warning by using the `force` option.

If you call the `New-DeployRule` cmdlet on an image profile that includes VIBs at the CommunitySupported level which violate the rule, set `$DeployNoSignatureCheck = $true` before adding the image profile. With that setting, the system ignores signature validation and does not perform the extensibility rules check.

> **Note:**
> Image profiles that include VIBs at the CommunitySupported level are not supported on production systems.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Determine the location of a public software depot, or define a custom image profile by using vSphere ESX Image Builder.

3. Run `Add-EsxSoftwareDepot` to add the software depot that contains the image profile to the VCF PowerCLI session.

Depot Type	Cmdlet
Remote depot	Run `Add-EsxSoftwareDepot` *depot_url*.
ZIP file	1. Download the ZIP file to a local file path. 2. Run `Add-EsxSoftwareDepot C:\`*file_path*`\`*my_offline_depot*`.zip`.

4. In the depot, find the image profile that you want to use by running the `Get-EsxImageProfile` cmdlet.

   By default, the ESX depot includes one base image profile that includes VMware Tools and has the string `standard` in its name, and one base image profile that does not include VMware Tools.

5. Define a rule in which hosts with certain attributes, for example a range of IP addresses, are assigned to the image profile.

   ```
   New-DeployRule -Name "testrule" -Item "My Profile25" -
   Pattern "vendor=Acme,Zven", "ipv4=192.XXX.1.10-192.XXX.1.20"
   ```

   Double quotes are required if a name contains spaces, optional otherwise. Specify `-AllHosts` instead of a pattern to apply the item to all hosts.

   The cmdlet creates a rule named `testrule`. The rule assigns the image profile named My Profile25 to all hosts with a vendor of Acme or Zven that also have an IP address in the specified range.

6. Add the rule to the rule set.

   ```
   Add-DeployRule testrule
   ```

   By default, the rule is added to both the working rule set and the active rule set. If you use the `NoActivate` parameter, the working rule set does not become the active rule set.

When the host boots from iPXE, it reports attributes of the machine to the console. Use the same format of the attributes when writing deploy rules.

```
**********************************************************************
* Booting through VMware AutoDeploy...
*
* Machine attributes:
* . asset=No Asset Tag
* . domain=vmware.com
* . hostname=myhost.mycompany.com
* . ipv4=XX.XX.XXX.XXX
* . mac=XX:Xa:Xb:Xc:Xx:XX
* . model=MyVendorModel
* . oemstring=Product ID: XXXXXX-XXX
* . serial=XX XX XX XX XX XX...
* . uuid=XXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX
* . vendor=MyVendor
**********************************************************************
```

- For hosts already provisioned with vSphere Auto Deploy, perform the compliance testing and repair operations to provision them with the new image profile. See Test and Repair Rule Compliance .
- Turn on unprovisioned hosts to provision them with the new image profile.

Write a Rule and Assign a Host Profile to Hosts

vSphere Auto Deploy can assign a host profile to one or more ESX hosts.

The host profile might include information about storage configuration, network configuration, or other characteristics of the host. If you add a host to a cluster, that cluster's host profile is used.

- Install PowerCLI and all prerequisite software.
- Export the host profile that you want to use.

In many cases, you assign a host to a cluster instead of specifying a host profile explicitly. The host uses the host profile of the cluster.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Using the vSphere Client, set up a host with the settings you want to use and create a host profile from that host.
3. Find the name of the host profile by running `Get-VMhostProfile` VCF PowerCLI cmdlet, passing in the ESX host from which you create a host profile.
4. At the VCF PowerCLI prompt, define a rule in which host profiles are assigned to hosts with certain attributes, for example a range of IP addresses.

   ```
   New-DeployRule -Name "testrule2" -Item my_host_profile -Pattern "vendor=Acme,Zven",
    "ipv4=192.XXX.1.10-192.XXX.1.20"
   ```

   The specified item is assigned to all hosts with the specified attributes. This example specifies a rule named **testrule2**. The rule assigns the specified host profile *my_host_profile* to all hosts with an IP address inside the specified range and with a manufacturer of Acme or Zven.
5. Add the rule to the rule set.

   ```
   Add-DeployRule testrule2
   ```

   By default, the working rule set becomes the active rule set, and any changes to the rule set become active when you add a rule. If you use the `NoActivate` parameter, the working rule set does not become the active rule set.

- Assign a host already provisioned with vSphere Auto Deploy to the new host profile by performing compliance test and repair operations on those hosts. For more information, see Test and Repair Rule Compliance .
- Power on unprovisioned hosts to provision them with the host profile.

Write a Rule and Assign a Host to a Folder or Cluster

vSphere Auto Deploy can assign a host to a folder or cluster. When the host boots, vSphere Auto Deploy adds it to the specified location on the vCenter. Hosts assigned to a cluster inherit the cluster's host profile.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Verify that the folder you select is in a data center or in a cluster. You cannot assign the host to a standalone top-level folder.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.
2. Define a rule in which hosts with certain attributes, for example a range of IP addresses, are assigned to a folder or a cluster.

   ```
   New-DeployRule -Name testrule3 -Item "my folder"-Pattern "vendor=Acme,Zven",
    "ipv4=192.XXX.1.10-192.XXX.1.20"
   ```

   This example passes in the folder by name. You can instead pass in a folder, cluster, or data center object that you retrieve with the `Get-Folder`, `Get-Cluster`, or `Get-Datacenter` cmdlet.
3. Add the rule to the rule set.

   ```
   Add-DeployRule testrule3
   ```

   By default, the working rule set becomes the active rule set, and any changes to the rule set become active when you add a rule. If you use the `NoActivate` parameter, the working rule set does not become the active rule set.

- Assign a host already provisioned with vSphere Auto Deploy to the new folder or cluster location by performing test and repair compliance operation. See Test and Repair Rule Compliance .
- Power on unprovisioned hosts to add them to the specified vCenter location.

Configure a Stateless System by Running a Custom Script

You can use vSphere Auto Deploy to configure one or more hosts by associating custom scripts with a vSphere Auto Deploy rule.

- Verify that the script bundle you want to associate with a vSphere Auto Deploy rule is in `.tgz` format, with a maximum size of 10 MB, and written in Python or BusyBox ash scripting language.

The scripts run in alphabetical order after the initial ESX boot workflow of the host.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Run the `Add-ScriptBundle` cmdlet to add the script bundle that contains the necessary scripts to the vSphere Auto Deploy inventory.

   ```
   Add-ScriptBundle c:/temp/MyScriptBundle.tgz
   ```

   The name of the script bundle without the `.tgz` extension is the name identifier or object of the script bundle item. You can update an existing script bundle by using the `-Update` parameter with the `Add-ScriptBundle` cmdlet.

3. Optional: Run the `Get-ScriptBundle` cmdlet to verify that the script bundle is added to the vSphere Auto Deploy inventory.

4. Define a rule in which hosts with certain attributes, for example a range of IP addresses, are assigned to the script bundle.

   ```
   New-DeployRule -Name "testrule4" -Item "MyScriptBundle" -
   Pattern "vendor=Acme,Zven", "ipv4=192.XXX.1.10-192.XXX.1.20"
   ```

   Double quotes are required if a name contains spaces, optional otherwise. Specify `-AllHosts` instead of a pattern to apply the item to all hosts.

   You create a rule named *testrule4*. The rule assigns the script bundle named My Script Bundle to all hosts with a vendor of Acme or Zven that also have an IP address in the specified range. You can use the name identifier of the script bundle or the object returned by the `Get-ScriptBundle` cmdlet to identify the script bundle you want to associate with the rule.

5. Add the rule to the rule set.

   ```
   Add-DeployRule testrule4
   ```

   By default, the rule is added to both the working rule set and the active rule set. If you use the `NoActivate` parameter, the working rule set does not become the active rule set.

- For hosts already provisioned with vSphere Auto Deploy, perform the compliance testing and repair operations to provision them with the new scripts. See Test and Repair Rule Compliance .
- Turn on unprovisioned hosts to provision them with the new scripts.

Test and Repair Rule Compliance

Test new or modified rules for compliance and repair accordingly, as changes in the vSphere Auto Deploy rule set are not updated automatically.

When you add a rule to the vSphere Auto Deploy rule set or modify one or more rules, hosts are not updated automatically. vSphere Auto Deploy applies the new rules only when you test their rule compliance and perform remediation.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Verify that your infrastructure includes one or more ESX hosts provisioned with vSphere Auto Deploy, and that the host on which you installed VCF PowerCLI can access those ESX hosts.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Use VCF PowerCLI to check which vSphere Auto Deploy rules are currently available.

   ```
   Get-DeployRule
   ```

   The system returns the rules and the associated items and patterns.

3. Modify one of the available rules.

   For example, you can change the image profile and the name of the rule.

   ```
   Copy-DeployRule -DeployRule testrule -ReplaceItem MyNewProfile
   ```

   You cannot edit a rule already added to the active rule set. Instead, you can copy the rule and replace the item or pattern you want to change.

4. Verify that you can access the host for which you want to test rule set compliance.

   ```
   Get-VMHost -Name MyEsxi42
   ```

5. Run the cmdlet that tests rule set compliance for the host, and bind the return value to a variable for later use.

   ```
   $tr = Test-DeployRuleSetCompliance MyEsxi42
   ```

6. Examine the differences between the contents of the rule set and configuration of the host.

   ```
   $tr.itemlist
   ```

   If the host for which you want to test the new rule set compliance is compliant with the active rule set, the system returns a table of current and expected items.

   ```
   CurrentItem                              ExpectedItem
   -----------                              ------------
   My Profile 25MyNewProfile
   ```

7. Remediate the host to use the revised rule set the next time you boot the host.

   ```
   Repair-DeployRuleSetCompliance $tr
   ```

If the rule you changed specified the inventory location, the change takes effect when you repair compliance. For all other changes, reboot your host to have vSphere Auto Deploy apply the new rule and to achieve compliance between the rule set and the host.

Register a Caching Proxy Server Address with vSphere Auto Deploy

Simultaneously booting large number of stateless hosts places a significant load on the vSphere Auto Deploy server. You can load balance the requests between the vSphere Auto Deploy server and one or more proxy servers that you register with vSphere Auto Deploy.

See Prepare Your System for vSphere Auto Deploy.

1. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Register a caching proxy server addresses with vSphere Auto Deploy by running the `Add-ProxyServer` cmdlet.

   ```
   Add-ProxyServer -Address 'https://proxy_server_ip_address:port_number'
   ```

   You can run the cmdlet multiple times to register multiple proxy servers. The address can contain a port number.

3. Optional: Run the `List-ProxyServer` cmdlet to verify that the caching proxy server is registered with vSphere Auto Deploy.

vSphere Auto Deploy Tasks by Using the vSphere Client

- Managing vSphere Auto Deploy with the vSphere Client

- Overview of the vSphere Auto Deploy Process by Using the vSphere Client
- Create a Deploy Rule
- Clone a Deploy Rule
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- Activate, Deactivate, and Reorder Deploy Rules
- View Host Associations
- Edit the Image Profile Association of a Host
- Remediate a Non-compliant Host
- Add a Host to the vSphere Auto Deploy Inventory
- Add a Host to a Cluster That Uses a Single Image
- Add a Host to a Cluster Managed by vSphere Configuration Profiles
- Working with Script Bundles
- Download vSphere Auto Deploy Logs
- Start, Stop, or Restart the vSphere Auto Deploy Service

Managing vSphere Auto Deploy with the vSphere Client

You can add ESX hosts to the vSphere Auto Deploy inventory, create, monitor, and manage rules, and host associations by using the vSphere Client.

- Managing vSphere Auto Deploy with the vSphere Client
- Overview of the vSphere Auto Deploy Process by Using the vSphere Client
- Create a Deploy Rule
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- Download vSphere Auto Deploy Logs
- Start, Stop, or Restart the vSphere Auto Deploy Service

Overview of the vSphere Auto Deploy Process by Using the vSphere Client

In the vSphere Client, similar to VCF PowerCLI, you can create deploy rules that provision hosts and boot hosts with an image that you specify.

Getting started with vSphere Auto Deploy requires that you learn how vSphere Auto Deploy works, start the vSphere Auto Deploy and vSphere ESX Image Builder vCenter services, create deploy rules that provision hosts, and power on your hosts to be booted with the image or profile you specify.

> **Important:**
>
> The vSphere Host Profiles capability is deprecated in ESX 9.0 and will be removed in a future release.

The workflow for provisioning the hosts in your environment with vSphere Auto Deploy includes the following tasks:

1. Deploy vCenter.
   The vSphere Auto Deploy server is included.
2. Configure the vSphere Auto Deploy and vSphere ESX Image Builder service startup types.
   See Prepare Your System for vSphere Auto Deploy and Configure the vSphere ESX Image Builder.
3. Add or import a software depot to the vSphere Auto Deploy inventory.

See Add a Software Depot or Import a Software Depot.

4. (Optional) If you want to create a custom image profile, clone, or create an image profile by using the vSphere Client.
   See Clone an Image Profile or Create an Image Profile.

   Important:

   With ESX 9.0, support for stateless installation is deprecated. To continue using stateless ESX hosts, you must either have an empty cluster or create a new one in the vSphere Lifecycle Manager, and activate stateless support on that cluster in the Auto Deploy interface in the vSphere Client.

   In the vSphere Client, under **Auto Deploy > Clusters** you see a button **Activate Stateless Support**.

   You manage the lifecycle of hosts part of stateless clusters or standalone hosts only by using Auto Deploy. vSphere Lifecycle Manager operations relevant for stateful hosts such as single image association, image compliance, and remediation, are not supported for stateless clusters.
5. Create a deploy rule that assigns a host location or an image profile to one host, to multiple hosts specified by a pattern, or to all hosts.
   See Create a Deploy Rule.

   Note:
   vSphere Auto Deploy is optimized for provisioning hosts that have a fixed MAC address to IP address mapping in DHCP (sometimes called DHCP reservations). If you want to use static IP addresses, you must set up the host profile to prompt for host customization. For more information, see the *vSphere Host Profiles* documentation.
6. Power on the hosts that you want to provision.
7. If you do not use a vSphere Lifecycle Manager image or a vSphere Configuration Profile, set up a host you provision as a reference host for a host profile.
   You can specify the reference host syslog settings, firewall settings, storage, networking, and so on.
8. Extract a host profile from the reference host or use a specific location, such as a cluster managed by a vSphere Lifecycle Manager image or by a vSphere Configuration Profile.
9. To provision multiple hosts with the host profile, clone or edit the previously created rule by using the vSphere Client.
   See Clone a Deploy Rule or Edit a Deploy Rule.
10. Activate the new rule and deactivate the old one.
    See Activate, Deactivate, and Reorder Deploy Rules.
11. Remediate the host associations to apply the new rule to the host.
    See Remediate a Non-compliant Host.
12. Verify that the hosts you provisioned meet the following requirements.
    - Each host is connected to the vCenter system.
    - The hosts are not in maintenance mode.
    - The hosts have no compliance failures.
    - Each host with a host profile that requires user input has up-to-date host customization information.

    Remediate host associations and compliance problems and reboot hosts until all hosts meet the requirements.

Create a Deploy Rule

Before you provision ESX hosts with vSphere Auto Deploy, you must create rules that assign host locations, image and host profiles to the hosts.

Important:

With ESX 9.0, support for stateless installs is deprecated. To continue using stateless ESX hosts, you must either have an empty cluster or create a new one in the vSphere Lifecycle Manager, and activate stateless support on that cluster in the Auto Deploy interface in the vSphere Client.

You cannot switch an existing cluster from stateful to stateless cluster or the other way around.

You cannot switch a cluster with attached stateful host profile from stateful to stateless.

An ESX host can match more than one vSphere Auto Deploy rule criteria, when this is the case, the rule order is considered.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- If you want to include an image profile to the rule, verify that the software depot you need is added to the inventory. See *Add a Software Depot* or *Import a Software Depot*.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, click **New Deploy Rule**.
   The **New Deploy Rule** wizard appears.
3. On the **Name and hosts** page of the wizard, enter a name for the new rule.
4. Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
5. On the **Configuration** page of the wizard, you can optionally include items in the rule.
   Each enabled item adds a new page to the wizard.

Option	Action
Host Location	Add the hosts that match the criteria of the rule to a specific location, such as a cluster managed by a vSphere Lifecycle Manager image or by a vSphere Configuration Profile.
Image	Assign an image profile to the hosts that match the rule criteria.
Host Settings	Assign a host profile to the hosts that match the rule criteria.
Script Bundle	Assign a script bundle to the host that match the rule criteria.

6. Optional: On the **Select host location** page of the wizard, select a data center, folder, or cluster as host location for the hosts that match the rule. With ESX 9.0, support for stateless installs is deprecated and along with host location, you must define the type of hosts that the rule deploys. Select the Stateless (Network boot) option under **Applicability** if you want to create stateless hosts.
7. Optional: On the **Select Image** page of the wizard, use the drop-down menu to select a software depot and choose an image profile from the list.
   If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.
8. Optional: On the **Select host settings** page of the wizard, select a host profile from the list.
9. Optional: On the **Select script bundle** page of the wizard, select a script bundle from the list.
10. On the **Ready to complete** page, review the summary information for the new rule.

You can view the newly created rule listed on the **Deploy Rules** tab.

- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.
- Edit a vSphere Auto Deploy rule. See *Edit a Deploy Rule*.
- Clone a vSphere Auto Deploy rule. See *Clone a Deploy Rule*.
- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- Remediate non-compliant hosts. See *Remediate a Non-compliant Host*.
- Change the image profile association of a host. See *Edit the Image Profile Association of a Host*.

Clone a Deploy Rule

You can use a vSphere Auto Deploy rule as a template and modify only parts of the rule instead of creating a new one.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.

- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.
- If you want to include an image profile to the rule, verify that the software depot you need is added to the inventory. See *Add a Software Depot* or *Import a Software Depot*.

You can clone an existing vSphere Auto Deploy rule by using the Clone Deploy Rule wizard.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, select a rule from the list.
3. Click **Clone**.
   The **Clone Deploy Rule** wizard appears.
4. On the **Name and hosts** page of the wizard, enter a name for the new rule.
5. Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
6. On the **Configuration** page of the wizard, you can optionally include items in the rule.
   Each enabled item adds a new page to the wizard.

Option	Action
Host Location	Add the hosts that match the criteria of the rule to a specific location, such as a cluster managed by a vSphere Lifecycle Manager image or by a vSphere Configuration Profile.
Image	Assign an image profile to the hosts that match the rule criteria.
Host Settings	Assign a host profile to the hosts that match the rule criteria.
Script Bundle	Assign a script bundle to the host that match the rule criteria.

7. On the **Select host location** page of the wizard, select a location for the hosts that match the rule.

Option	Action
If you want to keep the host location used in the cloned rule	Select the **Same Host location** check box.
If you want to select a new location for the selected hosts	1. Select the **Browse for Host location** check box. 2. Select a data center, folder, or cluster as host location. 3. Click **Next**.

8. On the **Select Image** page of the wizard, select an image profile.

Option	Action
If you do not want to change the image profile	Select the **Same image profile** check box.
If you want to assign a new image profile to the selected hosts	1. Select the **Browse for Image Profile** check box. 2. Select a software depot from the drop-down menu. 3. Select an image profile from the list. 4. (Optional) If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.

9. On the **Select host settings** page of the wizard, select a host profile.

Option	Action
If you want to keep the host profile used in the cloned rule	Select the **Same Host profile** check box.
If you want to assign a new host profile to the selected hosts	1. Select the **Browse for Host Profile** check box. 2. Select a host profile from the list and click **Next**.

10. On the **Select script bundle** page of the wizard, select a script bundle from the list.
11. On the **Ready to complete** page, review the summary information for the new rule.

- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.
- Edit a vSphere Auto Deploy rule. See *Edit a Deploy Rule*.

Edit a Deploy Rule

You can edit the name of an inactive Auto Deploy rule, its matching hosts, the host location, the image profile, and the host profile.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, from the list of rules in the inventory select the rule and click **Edit**.
   The Edit Deploy Rule dialog box appears.
3. Optional: On the **Name and hosts** page of the wizard, enter a new name for the rule.
4. Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
5. On the **Configuration** page of the wizard, you can optionally include items in the rule.
   Each enabled item adds a new page to the wizard.

Option	Action
Host Location	Add the hosts that match the criteria of the rule to a specific location, such as a cluster managed by a vSphere Lifecycle Manager image or by a vSphere Configuration Profile.
Image	Assign an image profile to the hosts that match the rule criteria.
Host Settings	Assign a host profile to the hosts that match the rule criteria.
Script Bundle	Assign a script bundle to the host that match the rule criteria.

6. On the **Select host location** page of the wizard, select a location for the hosts that match the rule.

Option	Action
If you want to keep the host location used in the cloned rule	Select the **Same Host location** check box.
If you want to select a new location for the selected hosts	1. Select the **Browse for Host location** check box. 2. Select a data center, folder, or cluster as host location. 3. Click **Next**.

7. On the **Select Image** page of the wizard, select an image profile.

Option	Action
If you do not want to change the image profile	Select the **Same image profile** check box.
If you want to assign a new image profile to the selected hosts	1. Select the **Browse for Image Profile** check box. 2. Select a software depot from the drop-down menu. 3. Select an image profile from the list. 4. (Optional) If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.

8. On the **Select host settings** page of the wizard, select a host profile.

Option	Action
If you want to keep the host profile used in the cloned rule	Select the **Same Host profile** check box.
If you want to assign a new host profile to the selected hosts	1. Select the **Browse for Host Profile** check box. 2. Select a host profile from the list and click **Next**.

9. On the **Select script bundle** page of the wizard, select a script bundle from the list.
10. On the **Ready to complete** page, review the summary information for the new rule.

- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.
- Clone a vSphere Auto Deploy rule. See *Clone a Deploy Rule*.

Activate, Deactivate, and Reorder Deploy Rules

After you create a vSphere Auto Deploy rule, the rule is in inactive state. You must activate the rule for it to take effect.

You can use the Activate and Reorder wizard to activate, deactivate, and change the order of the rules.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.

The upper list on the **Activate and Reorder** page of the wizard displays the rules in the active rule set. The lower list displays the inactive rules.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, click **Activate/Deactivate rules**.

   The Activate and Reorder wizard appears.
3. Optional: If you want to deactivate an active rule, select the rule from the active rules list and click the **Deactivate** button.
4. From the list of inactive rules, select the rule that you want to activate and click the **Activate** button.
5. Optional: If you want to reorder the rules in the active rule list, select a rule that you want to move up or down in the list and click **Move up** or **Move down** above the list of active rules.

   The rules are listed by priority. For example, if two or more rules apply to the same host but are set to provision the host with different host locations, image profiles, and host profiles, the rule that is highest in the list takes effect on the host.
6. Optional: If you want to test an inactive rule before activation, click **Test rules before activation**.
   a) Select a host from the list and click **Check Compliance** to view the current status of the host and the changes that are expected after the activation of the rule.

      If the host is compliant with the rule, you do not need to remediate the host after you activate the rule.

b) Optional: If you want to remediate the selected hosts after the rule activation, enable the toggle button or select the **Remediate all host associations after rule activation** check box to remediate all hosts.

7. Review the list of active rules and click **OK**.

On the **Deploy Rules** tab, the rule is listed as active in the Status column.

- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- Remediate non-compliant hosts. See *Remediate a Non-compliant Host*.

View Host Associations

Some ESX hosts in the vSphere Auto Deploy inventory might not be compliant with the active deploy rules and you must check for compliance.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.
- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.

1. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.

2. Check the host associations compliance.

   The **Check Host Associations Compliance** window displays the status of the host and whether the host is compliant with the active rule set. You can view the currently assigned host location, image profile, host profile, script bundle, and the associations that will take effect after a remediation of the host.

Option	Steps
If you want to check the host associations compliance of a single host	1. On the **Deployed Hosts** tab, select an ESX host. 2. Click **Check Host Associations Compliance**. 3. Check if the host associations are compliant with the current active rule set. 4. If you want to remediate the host, click **Remediate**. 5. Close the **Check Host Associations Compliance** window.
If you want to check the host associations compliance of multiple hosts	1. On the **Deployed Hosts** tab, select multiple ESX hosts. 2. Click **Check Host Associations Compliance**. 3. Confirm that you want to check the compliance of all selected hosts. 4. Review the compliance status of the hosts in the left pane. 5. (Optional) Select a host to view the compliance status details. 6. (Optional) Select a host and click **Remediate** . 7. (Optional) Select the **Remediate all host associations after rule activation** check box to remediate all hosts. 8. **Close** the **Check Host Associations Compliance** window.

- Remediate non-compliant hosts. See *Remediate a Non-compliant Host*.
- Edit the image profile association of a host. See Edit the Image Profile Association of a Host.
- Edit a vSphere Auto Deploy rule. See *Edit a Deploy Rule*.

Edit the Image Profile Association of a Host

You can edit the image profile association of a single host if the host is not associated with a vSphere Auto Deploy rule.

Alternatively, you might not want to change the image profile association of multiple hosts by editing a rule.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.
- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deployed Hosts** tab, select an ESX host.
3. Click **Edit Image Profile Association**.
   The Edit Image Profile Association dialog box appears.
4. Edit the image profile association of the host.

Option	Action
If you do not want to change the image profile	Select the **Same image profile** check box.
If you want to assign a new image profile to the selected hosts	1. Select the **Browse for Image Profile** check box. 2. Select a software depot from the drop-down menu. 3. Select an image profile from the list. 4. (Optional) If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.

5. Click **OK**.

The new image profile is listed in the Associated Image Profile column after a refresh of the page.

- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- If the host is associated with a rule and you want to revert to the image profile defined in the rule, remediate the host. See Remediate a Non-compliant Host.

Remediate a Non-compliant Host

Remediate the ESX host associations when you add a rule to the vSphere Auto Deploy active rule set or make changes to one or more rules.

When you add a rule to the vSphere Auto Deploy active rule set or make changes to one or more rules, hosts are not updated automatically. You must remediate the host associations to apply the new rules to the host.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.
- Create a vSphere Auto Deploy rule. See *Create a Deploy Rule*.
- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.
- If the remediation of a host, results in a change in its location, the host must be placed in maintenance mode.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deployed Hosts** tab, select a single or multiple ESX hosts.
3. Click **Remediate Host Associations**.
   If you remediate a host that has an edited image profile association, the host reverts to the settings defined in the rule that it matches.
   You can monitor the progress of the remediation process in the Recent Tasks pane.

- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- Change the image profile association of a host. See *Edit the Image Profile Association of a Host*.

Add a Host to the vSphere Auto Deploy Inventory

You can view the hosts that do not match any vSphere Auto Deploy rule and manually add a host to the vSphere Auto Deploy inventory.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy*.

- To assign an image profile to the host, add the software depot that you need to the inventory. See Add a Software Depot or Import a Software Depot.

To add a host to the current vSphere Auto Deploy inventory of deployed hosts, you can create a new rule or edit an existing rule to include a host that is not deployed with vSphere Auto Deploy and associate it with a specific host location, image profile, host profile, and script bundle. Alternatively, you can manually add a host to the inventory by assigning it a host location, image profile, host profile, and script bundle.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Discovered Hosts** tab, select one or more hosts that you want to provision with a host location, image profile, and host profile.
3. Select **Add to Inventory**.
   Alternatively, click **Remove** to dismiss the selected hosts from the **Discovered Hosts** tab.
   The Add to Inventory wizard appears.
4. On the **Select host location** page of the wizard, select a data center, folder, or cluster as host location for the hosts that match the rule. With ESX 9.0, support for stateless installs is deprecated and along with host location, you must define the type of hosts that the rule deploys. Select the Stateless (Network boot) option under **Applicability** if you want to create stateless hosts.
5. On the **Select Image** page of the wizard, use the drop-down menu to select a software depot and choose an image profile from the list.
   If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.
6. On the **Select host settings** page of the wizard, select a host profile from the list.
7. On the **Select host profile** page of the wizard, use the **Filter** to search the host profiles list or select the **Do not include a host profile** check box to continue without adding a host profile.
8. On the **Select script bundle** page of the wizard, select a script bundle from the list.
9. On the Ready to complete page, review the selected host associations.

- Edit a vSphere Auto Deploy rule. See *Edit a Deploy Rule*.
- Clone a vSphere Auto Deploy rule. See *Clone a Deploy Rule*.
- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- Remediate non-compliant hosts. See *Remediate a Non-compliant Host*.

Add a Host to a Cluster That Uses a Single Image

Create a rule in Auto Deploy that assigns a cluster that you manage by a single image as the host location for newly added ESX hosts.

To add ESX hosts to a cluster that you manage by a single image, you create a rule in Auto Deploy that assigns such a cluster as the host location for newly added hosts. Each host inherits the same image, which enables hardware compatibility checking, cluster-wide remediation, and easier upgrades.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy* in the *VMware ESX Installation and Setup* documentation.
- Verify that each ESX host is version 8.0 or later.
- Verify that all hosts in the cluster are stateful and have a physical storage attached.
- Verify that solutions, which are not integrated with vSphere Lifecycle Manager are not enabled for the cluster.

By creating an Auto Deploy rule, where the host target location is a cluster managed by an image, you can transition stateful ESX hosts to the cluster. Based on host identification mechanisms, the rule adds the hosts to the target cluster.

Such Auto Deploy rule is not allowed to contain an Image Profile or a Host Profile, as the image specification and configuration of hosts in the target cluster are created automatically.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, click **New Deploy Rule**.
   The **New Deploy Rule** wizard appears.
3. On the **Name and hosts** page of the wizard, enter a name for the new rule.
4. Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
5. On the **Configuration** page of the wizard, select the **Host Location** check box, and click **Next**.
   You can optionally include a script bundle to the ESX hosts that match the rule criteria by selecting the **Script Bundle** check box.

   Note:
   The image profile and host profile attached to the target cluster are used.
6. On the **Select host location** page of the wizard, select a cluster that uses a single image.
7. On the **Select script bundle** page of the wizard, select a script bundle from the list.
8. On the **Ready to complete** page, review the summary information for the new rule.
9. Click **Finish**.

You can view the newly created rule listed on the **Deploy Rules** tab.

- Activate a vSphere Auto Deploy rule. See Activate, Deactivate, and Reorder Deploy Rules.
- For more information on stateless caching and stateful installations, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

Add a Host to a Cluster Managed by vSphere Configuration Profiles

Create an Auto Deploy rule that assigns newly added hosts to a cluster that manages ESX configuration collectively with vSphere Configuration Profiles.

To add ESX hosts to a cluster that manages ESX configuration at a cluster level, you create a rule in Auto Deploy that assigns such a cluster as the host location for newly added hosts, which inherit the same settings and do not require manual configuration.

- Prepare your system and install the Auto Deploy Server. For more information, see Install and Configure vSphere Auto Deploy.
- Verify that each ESX host is version 8.0 or later.

Note:

Starting with VCF 9.1, Auto Deploy supports provisioning bare-metal hosts with vSphere Distributed Switch (DVS) and NSX settings in vSphere Configuration Profile (VCP) clusters.

By creating an Auto Deploy rule to manage the configuration of all hosts collectively by using vSphere Configuration Profiles on a cluster level, you remove the need to use Host Profiles or manual configurations. At the same time, you keep the flexibility to define custom settings per host or override the cluster-level settings for a group of hosts. For more information on how to set up clusters that you manage with a configuration at a cluster level, see What is vSphere Configuration Profiles.

Before you start using Auto Deploy with vSphere Configuration Profiles, you must have a ready configuration, either in a `.json` file or imported from a reference host. The best practice is to:

- Create a VCP cluster.
- Manually configure the desired settings on an existing ESX host that you select as a reference host.
- Import the reference host configuration to create a draft, make changes, precheck, and apply it.

Successful provisioning of hosts in VCP clusters requires a valid, non-empty desired configuration document. If the cluster configuration lacks networking details, Auto Deploy rules might fail, because hosts cannot come online. Always validate the configuration before you create Auto Deploy rules and attempt to provision hosts.

Note:

Once you set up a cluster that manages ESX configuration at a cluster level, you cannot roll back to using Host Profiles or single images, and you cannot use the Quickstart option for such clusters. However, you can switch from a cluster that you manage with a single image to a cluster that you manage with a configuration at a cluster level just by selecting the **Set Up Host Settings** option under **Configure** > **Desired State** > **Host Settings**.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, click **New Deploy Rule**.
   The **New Deploy Rule** wizard appears.
3. On the **Name and hosts** page of the wizard, enter a name for the new rule.
4. Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
5. On the **Configuration** page of the wizard, select the **Host Location** check box, and click **Next**.
   You can ignore the **Image**, **Host Settings**, and **Script Bundle** check boxes. Image specification and configuration of hosts in the target cluster, and any post-installation settings, are created automatically. Even if you select any of the check boxes, the rule ignores the selections.
6. On the **Select host location** page of the wizard, select a cluster that manages ESX configuration at a cluster level.
7. On the **Ready to complete** page, review the summary information for the new rule.
8. Click **Finish**.

You can view the newly created rule listed on the **Deploy Rules** tab.

- Activate a vSphere Auto Deploy rule. See Activate, Deactivate, and Reorder Deploy Rules.
- For more information on stateless caching and stateful installs, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

Working with Script Bundles

You can add a custom script for additional post-deployment host configuration. The script runs after you provision an ESX host with Auto Deploy. For example, you can create a custom ESX firewall rule and other configurations not available with Host Profiles.

- Verify that you can run the script in the ESX Shell.

Since ESX 6.7 Update 1, you can add or remove a custom script by using the vSphere Client. A script bundle can include multiple scripts and must be delivered as a single compressed file with the `.tgz` extension. After uploaded to the vCenter, you can include the script bundle to an Auto Deploy rule.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. Select the **Script Bundles** tab.
3. Click **Upload**.
4. Browse to a script bundle file and select **Upload**.
   The script is present in the **Script Bundles** list.

5. Optional: Select a script bundle, click **Remove**, and confirm the selection.
   The script bundle is deleted from the list.

- Activate a vSphere Auto Deploy rule. See *Activate, Deactivate, and Reorder Deploy Rules*.
- Edit a vSphere Auto Deploy rule. See *Edit a Deploy Rule*.
- Clone a vSphere Auto Deploy rule. See *Clone a Deploy Rule*.
- View the host location, image profile, host profile, and added script bundles. See *View Host Associations*.
- Remediate non-compliant hosts. See *Remediate a Non-compliant Host*.
- Change the image profile association of a host. See *Edit the Image Profile Association of a Host*.

Download vSphere Auto Deploy Logs

You can use the vSphere Auto Deploy logging information from the vSphere Client to resolve problems that you encounter with vSphere Auto Deploy.

Use the vSphere Client to log in to the vCenter instance that vSphere Auto Deploy is registered with.

1. Navigate to **Home** > **Administration** and select **Deployment** > **System Configuration**.
2. Select one of the nodes for which you want to retrieve a support bundle. The support bundle holds the services logs.
3. Click **Export Support Bundle**.
4. Select only **VirtualAppliance** > **Auto Deploy**.
5. At **Select information to include**, select a custom support bundle.
6. Click **Export** to download the log files.

Start, Stop, or Restart the vSphere Auto Deploy Service

You can start, stop, or restart the Auto Deploy service in the vCenter Management Interface.

Verify that you have a root access to the vCenter Management Interface.

To start, stop, and restart services in vCenter, you use the vCenter Management Interface.

1. Log in to the vCenter Management Interface, https://*IP-address-or-FQDN*:5480.
2. Click **Services**.
   The **Services** pane displays a table of all installed services. You can sort them by name, startup type, health, and state.
3. Select the **Auto Deploy** service and select your action.
   The available actions depend on whether the Auto Deploy service is already running or not.
   - Click **Restart** to restart the service.
     Restarting the service requires confirmation and might lead to the Auto Deploy functionality becoming temporarily unavailable.
   - Click **Start** to start the service.
   - Click **Stop** to stop the service.
     Stopping the service requires confirmation.

Provision ESX Hosts with vSphere Auto Deploy

Use vSphere Auto Deploy to provision or reprovision hundreds of physical hosts with ESX software.

You can use vSphere Auto Deploy to provision hundreds of physical hosts with ESX software either for the first time (first boot), or reboot hosts, or reprovision hosts with a different image profile, host profile, custom script, or folder, or cluster location. You can also select to provision the hosts with an image profile that does not contain VMware Tools binaries.

- vSphere Auto Deploy Boot Process

- Reprovisioning Hosts
- Provision ESX Host with an Image Profile Without VMware Tools

vSphere Auto Deploy Boot Process

When you boot a host that you want to provision or reprovision with vSphere Auto Deploy, the vSphere Auto Deploy infrastructure supplies the image profile and, optionally, a host profile, a vCenter location, and script bundle for that host.

The boot process is different for hosts that have not yet been provisioned with vSphere Auto Deploy (first boot) and for hosts that have been provisioned with vSphere Auto Deploy and added to a vCenter system (subsequent boot).

First Boot Prerequisites

Before a first boot process, you must set up your system. Setup includes the following tasks, which are discussed in more detail in Install and Configure vSphere Auto Deploy.

- Set up a DHCP server that assigns an IP address to each host upon startup and that points the host to the TFTP server to download the iPXE boot loader from.
- If the hosts that you plan to provision with vSphere Auto Deploy are with legacy BIOS, verify that the vSphere Auto Deploy server has an IPv4 address. PXE booting with legacy BIOS firmware is possible only over IPv4. PXE booting with UEFI firmware is possible with either IPv4 or IPv6.
- Identify an image profile to be used in one of the following ways.
  - Choose an ESX image profile in a public depot.
  - Create a custom image profile by using vSphere ESX Image Builder, and place the image profile in a depot that the vSphere Auto Deploy server can access. The image profile must include a base ESX VIB.
- If you have a reference host in your environment, export the host profile of the reference host and define a rule that applies the host profile to one or more hosts. See Setting Up a vSphere Auto Deploy Reference Host.
- Specify rules for the deployment of the host and add the rules to the active rule set.

First Boot Overview

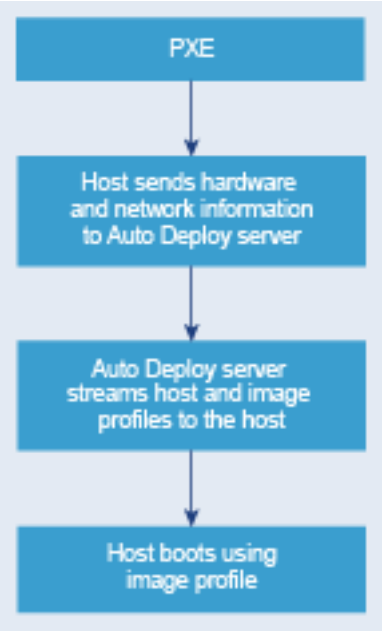
When a host that has not yet been provisioned with vSphere Auto Deploy boots (first boot), the host interacts with several vSphere Auto Deploy components.

1. When the administrator turns on a host, the host starts a PXE boot sequence.
   The DHCP Server assigns an IP address to the host and instructs the host to contact the TFTP server.
2. The host contacts the TFTP server and downloads the iPXE file (executable boot loader) and an iPXE configuration file.
3. iPXE starts executing.
   The configuration file instructs the host to make a HTTP boot request to the vSphere Auto Deploy server. The HTTP request includes hardware and network information.
4. In response, the vSphere Auto Deploy server performs these tasks:
   a. Queries the rules engine for information about the host.
   b. Streams the components specified in the image profile, the optional host profile, and optional vCenter location information.
5. The host boots using the image profile.
   If the vSphere Auto Deploy server provided a host profile, the host profile is applied to the host.
6. vSphere Auto Deploy adds the host to the vCenter system that vSphere Auto Deploy is registered with.
   a. If a rule specifies a target folder or cluster on the vCenter system, the host is placed in that folder or cluster. The target folder must be under a data center.
   b. If no rule exists that specifies a vCenter inventory location, vSphere Auto Deploy adds the host to the first datacenter displayed in the vSphere Client UI.
7. If the host profile requires the user to specify certain information, such as a static IP address, the host is placed in maintenance mode when the host is added to the vCenter system.
   You must reapply the host profile and update the host customization to have the host exit maintenance mode. When you update the host customization, answer any questions when prompted.

8. If the host is part of a DRS cluster, virtual machines from other hosts might be migrated to the host after the host has successfully been added to the vCenter system.

See Provision a Host (First Boot).

Figure 7: vSphere Auto Deploy Installation, First Boot



Subsequent Boots Without Updates

For hosts that are provisioned with vSphere Auto Deploy and managed by vCenter, subsequent boots can become completely automatic.

1. The administrator reboots the host.
2. As the host boots up, vSphere Auto Deploy provisions the host with its image profile and host profile.
3. Virtual machines are brought up or migrated to the host based on the settings of the host.
   - Standalone host. Virtual machines are powered on according to autostart rules defined on the host.
   - DRS cluster host. Virtual machines that were successfully migrated to other hosts stay there. Virtual machines for which no host had enough resources are registered to the rebooted host.

If vCenter is not responsive, the host contacts the vSphere Auto Deploy server and is provisioned with an image profile. The host continues to contact the vSphere Auto Deploy server until vSphere Auto Deploy reconnects to the vCenter.

vSphere Auto Deploy cannot set up vSphere distributed switches if vCenter is not responsive, and virtual machines are assigned to hosts only if they participate in an vSphere HA cluster. Until the host is reconnected to vCenter and the host profile is applied, the switch cannot be created. Because the host is in maintenance mode, virtual machines cannot start. See Reprovision Hosts with Simple Reboot Operations.

Any hosts that are set up to require user input are placed in maintenance mode. See Update the Host Customization in the vSphere Client.

Subsequent Boots With Updates

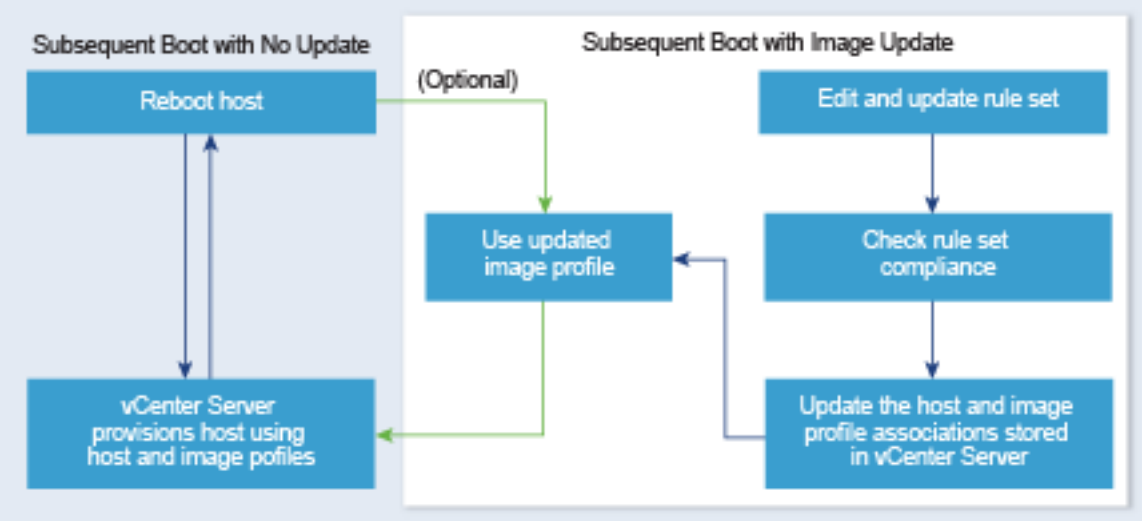
You can change the image profile, host profile, vCenter location, or script bundle for hosts. The process includes changing rules and testing and repairing the host's rule compliance.

1. The administrator uses the `Copy-DeployRule`VCF PowerCLI cmdlet to copy and edit one or more rules and updates the rule set. See Overview of the vSphere Auto Deploy Process by Using VCF PowerCLI for an example.
2. The administrator runs the `Test-DeployRulesetCompliance` cmdlet to check whether each host is using the information that the current rule set specifies.
3. The host returns a VCF PowerCLI object that encapsulates compliance information.

4. The administrator runs the `Repair-DeployRulesetCompliance` cmdlet to update the image profile, host profile, or vCenter location the vCenter system stores for each host.
5. When the host reboots, it uses the updated image profile, host profile, vCenter location, or script bundle for the host.
   If the host profile is set up to request user input, the host is placed in maintenance mode. Follow the steps in Update the Host Customization in the vSphere Client.

See Test and Repair Rule Compliance .

Figure 8: vSphere Auto Deploy Installation, Subsequent Boots



Provisioning of Systems that Have Distributed Switches

You can configure the host profile of a vSphere Auto Deploy reference host with a distributed switch.

When you configure the distributed switch, the boot configuration parameters policy is automatically set to match the network parameters required for host connectivity after a reboot.

When vSphere Auto Deploy provisions the ESX host with the host profile, the host goes through a two-step process.

1. The host creates a standard virtual switch with the properties specified in the boot configuration parameters field.
2. The host creates the VMkernel NICs. The VMkernel NICs allow the host to connect to vSphere Auto Deploy and to the vCenter system.

When the host is added to vCenter, vCenter removes the standard switch and reapplies the distributed switch to the host.

> **Note:**
> Do not change the boot configuration parameters to avoid problems with your distributed switch.

Provision a Host (First Boot)

Provisioning a host that has never been provisioned with vSphere Auto Deploy (first boot) differs from subsequent boot processes. You must prepare the host and fulfill all other prerequisites before you can provision the host. You can optionally define a custom image profile with vSphere ESX Image Builder by using the vSphere Client or VCF PowerCLI cmdlets.

- Make sure your host meets the hardware requirements for ESX hosts.
  See ESX Hardware Requirements.
- Prepare the system for vSphere Auto Deploy. See Install and Configure vSphere Auto Deploy.
- Write rules that assign an image profile to the host and optionally assign a host profile and a vCenter location to the host. See
  How vSphere Auto Deploy Works with VCF PowerCLI or Managing vSphere Auto Deploy with the vSphere Client.
  When the setup is complete, the vSphere Auto Deploy service is enabled, DHCP setup is complete, and rules for the host that you want to provision are in the active rule set.

1. Turn on the host.

   The host contacts the DHCP server and downloads iPXE from the location the server points it to. Next, the vSphere Auto Deploy server provisions the host with the image specified by the rule engine. The vSphere Auto Deploy server might also apply a host profile to the host if one is specified in the rule set. Finally, vSphere Auto Deploy adds the host to the vCenter system that is specified in the rule set.

2. Optional: If vSphere Auto Deploy applies a host profile that requires user input such as an IP address, the host is placed in maintenance mode. Reapply the host profile with the vSphere Client and provide the user input when prompted.

After the first boot process, the host is running and managed by a vCenter system. The vCenter stores the host's image profile, host profile, and location information.

You can now reboot the host as needed. Each time you reboot, the host is reprovisioned by the vCenter system.

Reprovision hosts as needed. See Reprovisioning Hosts.

If you want to change the image profile, host profile, custom script, or location of the host, update the rules and activate them by using the vSphere Client or perform a test and repair compliance operation in a VCF PowerCLI session. See Rules and Rule Sets or Test and Repair Rule Compliance .

Reprovisioning Hosts

Use vSphere Auto Deploy to reprovision ESX hosts with a different image profile or a different host profile.

vSphere Auto Deploy supports multiple reprovisioning options. You can perform a simple reboot or reprovision with a different image profile or a different host profile.

A first boot using vSphere Auto Deploy requires that you set up your environment and add rules to the rule set. See Install and Configure vSphere Auto Deploy.

The following reprovisioning operations are available.

- Simple reboot.
- Reboot of hosts for which the user answered questions during the boot operation.
- Reprovision with a different image profile.
- Reprovision with a different host profile.

Reprovision Hosts with Simple Reboot Operations

You can reprovision ESX hosts with the image profile, host profile, custom script, and vCenter location assigned during first boot.

A simple reboot of a host that is provisioned with vSphere Auto Deploy requires only that all prerequisites are still met. The process uses the previously assigned image profile, host profile, custom script, and vCenter location.

- Verify that the setup you performed during the first boot operation is in place. See Provision a Host (First Boot).
- Verify that all associated items like are available. An item can be an image profile, host profile, custom script or vCenter inventory location.
- Verify that the host has the identifying information (asset tag, IP address) it had during previous boot operations.

1. Place the host in maintenance mode.

Host Type	Action
Host is part of a DRS cluster	VMware DRS migrates virtual machines to appropriate hosts when you place the host in maintenance mode.
Host is not part of a DRS cluster	You must migrate all virtual machines to different hosts and place each host in maintenance mode.

2. Reboot the host.

The host shuts down. When the host reboots, it uses the image profile that the vSphere Auto Deploy server provides. The vSphere Auto Deploy server also applies the host profile stored on the vCenter system.

Use VCF PowerCLI To Reprovision a Host

You can use vSphere Auto Deploy to reprovision a host with a new image profile in a VCF PowerCLI session.

- Verify that the image profile you want to use to reprovision the host is available. Use vSphere ESX Image Builder in a VCF PowerCLI session. See Customizing Installations with vSphere ESX Image BuilderCustomizing Installations with vSphere ESX Image Builder.
- Verify that the setup you performed during the first boot operation is in place.

Several options for reprovisioning hosts exist.

- If the VIBs that you want to use support live update, you can use an `esxcli software vib update` command. In that case, you must also update the rule set to use an image profile that includes the new VIBs.
- During testing, you can apply an image profile to an individual host with the `Apply-EsxImageProfile` cmdlet and reboot the host so the change takes effect. The `Apply-EsxImageProfile` cmdlet updates the association between the host and the image profile but does not install VIBs on the host.
- In all other cases, use this procedure.

1. At the PowerShell prompt, run the `Connect-VIServer` VCF PowerCLI cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_or_ipv6_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate warnings result. In a development environment, you can ignore the warning.

2. Determine the location of a public software depot that contains the image profile that you want to use, or define a custom image profile with vSphere ESX Image Builder.

3. Run `Add-EsxSoftwareDepot` to add the software depot that contains the image profile to the VCF PowerCLI session.

Depot Type	Cmdlet
Remote depot	Run `Add-EsxSoftwareDepot` *`depot_url`*.
ZIP file	1. Download the ZIP file to a local file path or create a mount point local to the VCF PowerCLI machine. 2. Run `Add-EsxSoftwareDepot C:\`*`file_path`*`\`*`my_offline_depot`*`.zip`.

4. Run `Get-EsxImageProfile` to see a list of image profiles, and decide which profile you want to use.

5. Run `Copy-DeployRule` and specify the `ReplaceItem` parameter to change the rule that assigns an image profile to hosts.

   The following cmdlet replaces the current image profile that the rule assigns to the host with the *`my_new_imageprofile`* profile. After the cmdlet completes, `myrule` assigns the new image profile to hosts. The old version of `myrule` is renamed and hidden.

   ```
   Copy-DeployRule myrule -ReplaceItemmy_new_imageprofile
   ```

6. Test the rule compliance for each host that you want to deploy the image to.

   a) Verify that you can access the host for which you want to test rule set compliance.

   ```
   Get-VMHost-NameESX_hostname
   ```

   b) Run the cmdlet that tests rule set compliance for the host, and bind the return value to a variable for later use.

   ```
   $tr = Test-DeployRuleSetComplianceESX_hostname
   ```

   c) Examine the differences between the contents of the rule set and configuration of the host.

   ```
   $tr.itemlist
   ```

   The system returns a table of current and expected items if the host for which you want to test the new rule set compliance is compliant with the active rule set.

   ```
   CurrentItem                          ExpectedItem
   -----------                          ------------
    my_old_imageprofilemy_new_imageprofile
   ```

d) Remediate the host to use the revised rule set the next time you boot the host.

```
Repair-DeployRuleSetCompliance $tr
```

7. Reboot the host to provision it with the new image profile.

Reprovision a Host with a New Image Profile by Using the vSphere Client

You can use vSphere Auto Deploy to reprovision a host with a new image profile with the vSphere Client by changing the rule that the host corresponds to and activating the rule.

- Verify that the image profile you want to use to reprovision the host is available. See Create an Image Profile.
- Verify that the setup you performed during the first boot operation is in place.

1. Navigate to **Home** > **Auto Deploy**.
   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
2. On the **Deploy Rules** tab, from the list of rules in the inventory select the rule and click **Edit**.
   The Edit Deploy Rule dialog box appears.
3. Optional: On the **Name and hosts** page of the wizard, enter a name for the new rule.
4. Optional: Select to either apply the rule to all hosts in the inventory or only to hosts that match a specific pattern.
   You can select one or more patterns.
   For example, the rule can apply only to hosts in a vCenter Single Sign-On domain, with a specific host name, or that match a specific IPv4 range.
5. On the **Configuration** page of the wizard, you can optionally include items in the rule.
   Each enabled item adds a new page to the wizard.

Option	Action
Host Location	Add the hosts that match the criteria of the rule to a specific location, such as a cluster managed by a vSphere Lifecycle Manager image or by a vSphere Configuration Profile.
Image	Assign an image profile to the hosts that match the rule criteria.
Host Settings	Assign a host profile to the hosts that match the rule criteria.
Script Bundle	Assign a script bundle to the host that match the rule criteria.

6. Click **Next** to skip the Host Location selection.
7. On the **Select image profile** page of the wizard, assign an image profile to the hosts that match the rule criteria.

Option	Action
If you do not want to change the image profile	Select the **Same image profile** check box.
If you want to assign a new image profile to the selected hosts	1. Select the **Browse for Image Profile** check box. 2. Select a software depot from the drop-down menu. 3. Select an image profile from the list. 4. (Optional) If you want to bypass the acceptance level verification for the image profile, select the **Skip image profile signature check** check box.

8. Click **Next** to skip the Host profile selection.
9. On the **Ready to complete** page, review the summary information for the new image profile and click **Finish**.
10. Click **Activate/Deactivate rules**.
11. From the list of inactive rules, select the rule that you want to activate and click the **Activate** button.
12. Optional: If you want to reorder the rules in the active rule list, select a rule that you want to move up or down in the list and click **Move up** or **Move down** above the list of active rules.

    The rules are listed by priority. For example, if two or more rules apply to the same host but are set to provision the host with different host locations, image profiles, and host profiles, the rule that is highest in the list takes effect on the host.
13. Optional: If you want to test an inactive rule before activation, click **Test rules before activation**.
    a) Select a host from the list and click **Check Compliance** to view the current status of the host and the changes that are expected after the activation of the rule.

       If the host is compliant with the rule, you do not need to remediate the host after you activate the rule.
    b) Optional: If you want to remediate the selected hosts after the rule activation, enable the toggle button or select the **Remediate all host associations after rule activation** check box to remediate all hosts.
14. Review the list of active rules and click **OK**.
15. Reboot the host to provision it with the new image profile.

Update the Host Customization in the vSphere Client

If a host required user input during a previous boot, the answers are saved with the vCenter. If you want to prompt the user for new information, you must remediate the host.

Attach a host profile that prompts for user input to the host.

1. Migrate all virtual machines to different hosts, and place the host into maintenance mode.

Host Type	Action
Host is part of a DRS cluster	VMware DRS migrates virtual machines to appropriate hosts when you place the host in maintenance mode.
Host is not part of a DRS cluster	You must migrate all virtual machines to different hosts and place each host in maintenance mode.

2. Navigate to **Home** > **Auto Deploy**.

   By default, only the administrator role has privileges to use the vSphere Auto Deploy service.
3. On the **Deployed Hosts** tab, select an ESX host.
4. Click **Remediate Host Associations**.

   You can monitor the progress of the remediation process in the Recent Tasks pane.
5. When prompted, provide the user input.
6. Direct the host to exit maintenance mode.

The host customization is saved and takes effect the next time you boot the host.

Provision ESX Host with an Image Profile Without VMware Tools

You can select to provision ESX hosts with an image profile that does not contain VMware Tools binaries.

When you provision ESX hosts with vSphere Auto Deploy, you can select to provision the hosts with an image profile that does not contain VMware Tools binaries. This image profile is smaller, has a lower memory overhead, and boots faster in a PXE-boot environment.

Download the *xxxxx*`-no-tools` image profile from the Broadcom Support Portal.

If the network boot time is too slow when using the standard image, or if you want to save space on the hosts, you can use an image profile that does not include VMware Tools and place the VMware Tools binaries on a shared storage.

1. Boot an ESX host that was not provisioned with vSphere Auto Deploy.
2. Copy the `/productLocker` directory from the ESX host to a shared storage.

   You can connect to an ESX host using an SSH client.
3. Change the *`UserVars.ProductLockerLocation`* variable to point to the new `/productLocker` directory location.
   a) In the vSphere Client, select the reference host and click the **Configure** tab.
   b) Under **System**, click **Advanced System Settings**.
   c) Click **Edit**.
   d) Filter the settings for `uservars`, and select **UserVars.ProductLockerLocation**.
   e) Click the current value and edit the location so it points to the shared storage.
4. Create a host profile from the reference host.
5. Create a vSphere Auto Deploy rule that assigns the *`xxxxx`*`-no-tools` image profile and host profile from the reference host to all other hosts.
6. Boot your target hosts with the rule so they pick up the product locker location from the reference host.

Use vSphere Auto Deploy for Stateless Caching and Stateful Installs

With stateless caching, you can cache the image of an ESX host. With stateful installs, you can install hosts over the network.

The vSphere Auto Deploy stateless caching feature lets you cache the host's image. The vSphere Auto Deploy stateful installs feature lets you install hosts over the network. After the initial network boot, these hosts boot like other ESX hosts. The stateless caching solution is primarily intended for situations when several hosts boot simultaneously. The locally cached image helps prevent a bottleneck that results if several hundreds of hosts connect to the vSphere Auto Deploy server simultaneously. After the boot operation is complete, hosts connect to vSphere Auto Deploy to complete the setup.

The stateful installs feature lets you provision hosts with the image profile over the network without having to set up the PXE boot infrastructure.

Introduction to Stateless Caching and Stateful Installs

You can use the System Cache Configuration host profile to provision hosts with vSphere Auto Deploy stateless caching and stateful installs.

Examples of Stateless Caching and Stateful Installs

Hosts provisioned with vSphere Auto Deploy cache the image (stateless caching)	Set up and apply a host profile for stateless caching. You can cache the image on a local disk, a remote disk, or a USB drive. Continue provisioning this host with vSphere Auto Deploy. If the vSphere Auto Deploy server becomes unavailable, for example because hundreds of hosts attempt to access it simultaneously, the host boots from the cache. The host attempts to reach the vSphere Auto Deploy server after the boot operation to complete configuration.
Hosts provisioned with vSphere Auto Deploy become stateful hosts	Set up and apply a host profile for stateful installs. When you provision a host with vSphere Auto Deploy, the image is installed on the local disk, a remote disk, or a USB drive. For subsequent boots, you boot from the disk. The host no longer uses vSphere Auto Deploy.

Preparation

To successfully use stateless caching or stateful installs, decide how to configure the system and set the boot order.

Table 22: Preparation for Stateless Caching or Stateful Installs

Requirement or Decision	Description
Decide on VMFS partition overwrite	When you install ESX by using the interactive installer, you are prompted whether you want to overwrite an existing VMFS datastore. The System Cache Configuration host profile provides an option to overwrite existing VMFS partitions. The option is not available if you set up the host profile to use a USB drive.
Decide whether you need a highly available environment	If you use vSphere Auto Deploy with stateless caching, you can set up a highly available vSphere Auto Deploy environment to guarantee that virtual machines are migrated on newly provisioned hosts and that the environment supports vSphere Distributed Switch even if the vCenter system becomes temporarily unavailable.
Set the boot order	The boot order you specify for your hosts depends on the feature you want to use. • To set up vSphere Auto Deploy with stateless caching, configure your host to first attempt to boot from the network, and to then attempt to boot from disk. If the vSphere Auto Deploy server is not available, the host boots using the cache. • To set up vSphere Auto Deploy for stateful installs on hosts that do not currently have a bootable disk, configure your hosts to first attempt to boot from disk, and to then attempt to boot from the network. **Note:** If you currently have a bootable image on the disk, configure the hosts for one-time PXE boot, and provision the host with vSphere Auto Deploy to use a host profile that specifies stateful installs.

Stateless Caching and Loss of Connectivity

If the ESX hosts that run your virtual machines lose connectivity to the vSphere Auto Deploy server, the vCenter system, or both, some limitations apply the next time you reboot the host.

- If vCenter is available but the vSphere Auto Deploy server is unavailable, hosts do not connect to the vCenter system automatically. You can manually connect the hosts to the vCenter, or wait until the vSphere Auto Deploy server is available again.
- If both vCenter and vSphere Auto Deploy are unavailable, you can connect to each ESX host by using the VMware Host Client, and add virtual machines to each host.
- If vCenter is not available, vSphere DRS does not work. The vSphere Auto Deploy server cannot add hosts to the vCenter. You can connect to each ESX host by using the VMware Host Client, and add virtual machines to each host.
- If you make changes to your setup while connectivity is lost, the changes are lost when the connection to the vSphere Auto Deploy server is restored.

Understanding Stateless Caching and Stateful Installs

When you want to use vSphere Auto Deploy with stateless caching or stateful installs, you must set up a host profile, apply the host profile, and set the boot order.

When you apply a host profile that enables caching to a host, vSphere Auto Deploy partitions the specified disk. What happens next depends on how you set up the host profile and how you set the boot order on the host.

- vSphere Auto Deploy caches the image when you apply the host profile if **Enable stateless caching on the host** is selected in the System Cache Configuration host profile. No reboot is required. When you later reboot, the host

continues to use the vSphere Auto Deploy infrastructure to retrieve its image. If the vSphere Auto Deploy server is not available, the host uses the cached image.

- vSphere Auto Deploy installs the image if **Enable stateful installs on the host** is selected in the System Cache Configuration host profile. When you reboot, the host initially boots using vSphere Auto Deploy to complete the installation. A reboot is then issued automatically, after which the host boots from disk, similar to a host that was provisioned with the installer. vSphere Auto Deploy no longer provisions the host.

You can apply the host profile from the vSphere Client, or write a vSphere Auto Deploy rule in a VCF PowerCLI session that applies the host profile.

Using the vSphere Client to Set Up vSphere Auto Deploy for Stateless Caching or Stateful Installs

You can create a host profile on a reference host and apply that host profile to additional hosts or to a vCenter folder or cluster. The following workflow results.

1. You provision a host with vSphere Auto Deploy and edit that host's System Image Cache Configuration host profile.
2. You place one or more target hosts in maintenance mode, apply the host profile to each host, and instruct the host to exit maintenance mode.
3. What happens next depends on the host profile you selected.
   - If the host profile enabled stateless caching, the image is cached to disk. No reboot is required.
   - If the host profile enabled stateful installs, the image is installed. When you reboot, the host uses the installed image.

Using VCF PowerCLI to Set Up vSphere Auto Deploy for Stateless Caching or Stateful Installs

You can create a host profile for a reference host and write a vSphere Auto Deploy rule that applies that host profile to other target hosts in a VCF PowerCLI session. The following workflow results.

1. You provision a reference host with vSphere Auto Deploy and create a host profile to enable a form of caching.
2. You write a rule that provisions additional hosts with vSphere Auto Deploy and that applies the host profile of the reference host to those hosts.
3. vSphere Auto Deploy provisions each host with the image profile or by using the script bundle associated with the rule. The exact effect of applying the host profile depends on the host profile you selected.
   - For stateful installs, vSphere Auto Deploy proceeds as follows:
     - During first boot, vSphere Auto Deploy installs the image on the host.
     - During subsequent boots, the host boots from disk. The hosts do not need a connection to the vSphere Auto Deploy server.
   - For stateless caching, vSphere Auto Deploy proceeds as follows:
     - During first boot, vSphere Auto Deploy provisions the host and caches the image.
     - During subsequent boots, vSphere Auto Deploy provisions the host. If vSphere Auto Deploy is unavailable, the host boots from the cached image, however, setup can only be completed when the host can reach the vSphere Auto Deploy server.

Configure a Host Profile to Use Stateless Caching

If the vSphere Auto Deploy Server is not available, the host uses a cached image and to use stateless caching, you must configure a host profile.

When a host is set up to use stateless caching, if the vSphere Auto Deploy Server is not available, the host uses a cached image. To use stateless caching, you must configure a host profile. You can apply that host profile to other hosts that you want to set up for stateless caching.

> **Important:**
>
> The vSphere Host Profiles capability is deprecated in ESX 9.0 and will be removed in a future release.

- Decide which disk to use for caching and determine whether the caching process will overwrite an existing VMFS partition.

- In production environments, protect the vCenter system and the vSphere Auto Deploy server by including them in a highly available environment. Having the vCenter in a management cluster guarantees that VDS and virtual machine migration are available. If possible, also protect other elements of your infrastructure. See Set Up Highly Available vSphere Auto Deploy Infrastructure.
- Set up your environment for vSphere Auto Deploy. See *Install and Configure vSphere Auto Deploy*.
- Verify that a disk with at least 4GB of free space is available. If the disk is not yet partitioned, partitioning happens when you apply the host profile.
- Set up the host to attempt a network boot first and to boot from disk if network boot fails. See your hardware vendor's documentation.
- Create a host profile. See the *Host Profiles* documentation.

1. Navigate to **Home** > **Policies and Profiles** > **Host Profiles**.
2. Click the host profile you want to configure and select the **Configure** tab.
3. Click **Edit Host Profile**.
4. On the Edit host profile page of the wizard, select **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
5. In the **System Image Cache Profile Settings** drop-down menu, choose a policy option.

Option	Description
Enable stateless caching on the host	Caches the image to disk.
Enable stateless caching to a USB disk on the host	Caches the image to a USB disk attached to the host.

6. Depending on the policy option you select, you must do the following:
   a) If you select **Enable stateful installs on the host**:
      1. Specify the information about the disk to use.

Option	Description
Arguments for first disk	When configuring a System Image Install disk, you have multiple options to define the device you want ESX to be installed to and booted from. You can use the following arguments to define the disk for the installation: • `localesx` – The first disk detected containing a valid installation of ESX • `local` – The first local disk detected by ESX after boot • `remoteesx` - The first remote disk detected containing a valid installation of ESX • `sortedremoteesx`- The first remote disk sorted by the lowest LUN ID detected containing a valid installation of ESX • `remote` - The first remote disk detected by ESX after boot • `sortedremote` - The first remote disk sorted by the lowest LUN ID detected by ESX after boot • `device model` • `device vendor` • `vmkernel device driver name` You can get the values for the device model and vendor arguments by running the command `esxcli storage core device list` in a console to the ESX host, logging in as the root. You get the vmkernel device driver name argument by running the command `esxcli storage core adapter list`. You must then identify the storage adapter to which your boot device is connected. By default, the system attempts to replace an existing ESX installation, and then attempts to write to the local disk. You can use the **Arguments for first disk** field to specify a comma-separated list of disks to use, in order of preference. You can specify more than one disk. Use `localesx` for the first disk with ESX installed on it, model and vendor information, or specify the name of the vmkernel device driver. For example, to have the system first look for a disk with the model name ST3120814A, second for any disk that uses the mptsas driver, and third for the local disk, specify `ST3120814A,mptsas,local` as the value of this field. The first disk setting in the host profile specifies the search order for determining which disk to use for the cache. The search order is specified as a comma delimited list of values. The default setting `localesx,local` specifies that vSphere Auto Deploy should first look for an existing local cache disk. The cache disk is identified as a disk with an existing ESX software image. If vSphere Auto Deploy cannot find an existing cache disk, it searches for an available local disk device. When searching for an available disk vSphere Auto Deploy uses the first empty disk that does not have an existing VMFS partition. You can use the first disk argument only to specify the search order. You cannot explicitly specify a disk. For example, you cannot specify a specific LUN on a SAN.
Check to overwrite any VMFS volumes on the selected disk	If you select this check box, the system overwrites existing VMFS volumes if not enough space is available to store the image, image profile, and host profile.
Check to ignore any SSD devices connected to the host	If you select this check box, the system ignores any existing SSD devices and does not store image profiles and host profiles on them.

      2. In the **System Disk Configuration** drop-down menu, select **User must explicitly choose the policy option**.
   b) If you select **Enable stateful installs to a USB disk on the host**:
      1. In the **System Disk Configuration** drop-down menu, select **System disk specified by user in host customization**
      2. Under **Hosts and Clusters**, right-click the host and select **Host Profiles** > **Edit Host Customizations**. Define a disk with persistent storage in the **Value** field for the **System disk** property.

7. Click **Save** to complete the host profile configuration.

Apply the host profile to individual hosts by using the Host Profiles feature in the vSphere Client. See the *Host Profiles* documentation. Alternatively, you can create a rule to assign the host profile to hosts with the vSphere Client or by using VCF PowerCLI. See *Write a Rule and Assign a Host Profile to Hosts*.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Configure a Host Profile to Enable Stateful Installs

To set up an ESX host provisioned with vSphere Auto Deploy to boot from disk, you must configure a host profile.

- Decide which disk to use for storing the image, and determine whether the new image will overwrite an existing VMFS partition.
- Set up your environment for vSphere Auto Deploy. See *Install and Configure vSphere Auto Deploy*.
- Verify that a disk with at least 4GB of free space is available. If the disk is not yet partitioned, partitioning happens when you apply the host profile.
- Set up the host to boot from disk. See your hardware vendor's documentation.
- Create a host profile. See the *Host Profiles* documentation.

You can configure the host profile on a single host. You can also create a host profile on a reference host and apply that host profile to other hosts.

Important:

The vSphere Host Profiles capability is deprecated in ESX 9.0 and will be removed in a future release.

1. Navigate to **Home** > **Policies and Profiles** > **Host Profiles**.
2. Click the host profile you want to configure and select the **Configure** tab.
3. Click **Edit Host Profile**.
4. On the Edit host profile page of the wizard, select **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
5. In the **System Image Cache Profile Settings** drop-down menu, choose a policy option.

Option	Description
Enable stateful installs on the host	Caches the image to a disk.
Enable stateful installs to a USB disk on the host	Caches the image to a USB disk attached to the host.

6. Depending on the policy option you select, you must do the following:
   a) If you select **Enable stateful installs on the host**:
      1. Specify the information about the disk to use.

Option	Description
Arguments for first disk	When configuring a System Image Install disk, you have multiple options to define the device you want ESX to be installed to and booted from. You can use the following arguments to define the disk for the installation: • `localesx` – The first disk detected containing a valid installation of ESX • `local` – The first local disk detected by ESX after boot • `remoteesx` - The first remote disk detected containing a valid installation of ESX • `sortedremoteesx`- The first remote disk sorted by the lowest LUN ID detected containing a valid installation of ESX • `remote` - The first remote disk detected by ESX after boot • `sortedremote` - The first remote disk sorted by the lowest LUN ID detected by ESX after boot • `device model` • `device vendor` • `vmkernel device driver name` You can get the values for the device model and vendor arguments by running the command `esxcli storage core device list` in a console to the ESX host, logging in as the root. You get the vmkernel device driver name argument by running the command `esxcli storage core adapter list`. You must then identify the storage adapter to which your boot device is connected. By default, the system attempts to replace an existing ESX installation, and then attempts to write to the local disk. You can use the **Arguments for first disk** field to specify a comma-separated list of disks to use, in order of preference. You can specify more than one disk. Use `localesx` for the first disk with ESX installed on it, model and vendor information, or specify the name of the vmkernel device driver. For example, to have the system first look for a disk with the model name ST3120814A, second for any disk that uses the mptsas driver, and third for the local disk, specify `ST3120814A,mptsas,local` as the value of this field. The first disk setting in the host profile specifies the search order for determining which disk to use for the cache. The search order is specified as a comma delimited list of values. The default setting `localesx,local` specifies that vSphere Auto Deploy should first look for an existing local cache disk. The cache disk is identified as a disk with an existing ESX software image. If vSphere Auto Deploy cannot find an existing cache disk, it searches for an available local disk device. When searching for an available disk vSphere Auto Deploy uses the first empty disk that does not have an existing VMFS partition. You can use the first disk argument only to specify the search order. You cannot explicitly specify a disk. For example, you cannot specify a specific LUN on a SAN.
Check to overwrite any VMFS volumes on the selected disk	If you select this check box, the system overwrites existing VMFS volumes if not enough space is available to store the image, image profile, and host profile.
Check to ignore any SSD devices connected to the host	If you select this check box, the system ignores any existing SSD devices and does not store image profiles and host profiles on them.

      2. In the **System Disk Configuration** drop-down menu, select **User must explicitly choose the policy option**.
   b) If you select **Enable stateful installs to a USB disk on the host**:
      1. In the **System Disk Configuration** drop-down menu, select **System disk specified by user in host customization**
      2. Under **Hosts and Clusters**, right-click the host and select **Host Profiles** > **Edit Host Customizations**. Define a disk with persistent storage in the **Value** field for the **System disk** property.

7. Click **Save** to complete the host profile configuration.

Apply the host profile to individual hosts by using the Host Profiles feature in the vSphere Client. See the *Host Profiles* documentation. Alternatively, you can create a rule to assign the host profile to hosts with the vSphere Client or by using VCF PowerCLI. See *Write a Rule and Assign a Host Profile to Hosts*.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Setting Up a vSphere Auto Deploy Reference Host

In an environment where no state is stored on the ESX host, a reference host helps you set up multiple hosts with the same configuration.

You configure the reference host with the logging, coredump, and other settings that you want, save the host profile, and write a rule that applies the host profile to other hosts as needed. You can configure the storage, networking, and security settings on the reference host and set up services such as syslog and NTP.

Configure an ESX Dump Collector

You can configure an ESX Dump Collector to store core dumps either by using ESXCLI commands or by configuring a reference host.

Hosts provisioned with vSphere Auto Deploy do not have a local disk on which to store core dumps. You can configure an ESX Dump Collector to store core dumps either by using ESXCLI commands or by configuring a reference host to use the ESX Dump Collector by using the Host Profiles feature in the vSphere Client.

Configure ESX Dump Collector with ESXCLI

Hosts provisioned with vSphere Auto Deploy do not have a local disk to store core dumps on. You can configure ESX Dump Collector by using ESXCLI commands and keep core dumps on a network server for use during debugging.

Install ESXCLI if you want to configure the host to use ESX Dump Collector. In troubleshooting situations, you can use ESXCLI in the ESX Shell instead.

A core dump is the state of working memory if there is host failure. By default, a core dump is saved to the local disk. ESX Dump Collector is especially useful for vSphere Auto Deploy, but is supported for any ESX host. ESX Dump Collector supports other customization, including sending core dumps to the local disk and is included with the vCenter management node.

> **Note:**
> ESX Dump Collector is not supported to be configured on a VMkernel interface that is running on a NSX-T N-VDS switch.

If you intend to use IPv6, and if both the ESX host and ESX Dump Collector are on the same local link, both can use either local link scope IPv6 addresses or global scope IPv6 addresses.

If you intend to use IPv6, and if ESX and ESX Dump Collector are on different hosts, both require global scope IPv6 addresses. The traffic routes through the default IPv6 gateway.

1. Set up an ESX system to use ESX Dump Collector by running `esxcli system coredump` in the local ESX Shell or by using ESXCLI.

   ```
   esxcli system coredump network set --interface-name vmk0 --server-ip 10xx.xx.xx.xx --server-port 6500
   ```

   You must specify a VMkernel NIC and the IP address and optional port of the server to send the core dumps to. You can use an IPv4 address or an IPv6 address. If you configure an ESX system that is running on a virtual machine that is using a vSphere standard switch, you must select a VMkernel port that is in promiscuous mode.

2. Enable ESX Dump Collector.
   ```
   esxcli system coredump network set --enable true
   ```
3. Optional: Verify that ESX Dump Collector is configured correctly.
   ```
   esxcli system coredump network check
   ```

The host on which you have set up ESX Dump Collector is configured to send core dumps to the specified server by using the specified VMkernel NIC and optional port.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Configure ESX Dump Collector from the Host Profiles Feature in the vSphere Client

Hosts provisioned with vSphere Auto Deploy do not have a local disk to store core dumps on. You can configure a reference host to use ESX Dump Collector by using the Host Profiles feature in the vSphere Client.

- Verify that you have created the host profile on which you want to configure a coredump policy. For more information on how to create a host profile, see the *vSphere Host Profiles* documentation.
- Verify that at least one partition has sufficient storage capability for core dumps from multiple hosts provisioned with vSphere Auto Deploy.

Best practice is to set up hosts to use ESX Dump Collector with the `esxcli system coredump` command and save the host profile. For more information, see Configure ESX Dump Collector with ESXCLI.

1. Navigate to **Home** > **Policies and Profiles** > **Host Profiles**.
2. Click the host profile you want to configure and select the **Configure** tab.
3. Click **Edit Host Profile**.
4. Select **Networking Configuration** > **Network Coredump Settings**.
5. Select the **Enabled** check box.
6. Specify the host NIC to use, the Network Coredump Server IP, and the Network Coredump Server Port.
7. Click **Save** to complete the host profile configuration.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Configure Syslog from Host Profiles in the vSphere Client

Specify a remote syslog server by applying a host profile.

Hosts provisioned with vSphere Auto Deploy usually do not have sufficient local storage to save system logs. You can specify a remote syslog server for those hosts by setting up a reference host, saving the host profile, and applying that host profile to other hosts as needed.

- If you intend to use a remote syslog host, set up that host before you customize host profiles.
- Verify that you have access to the vSphere Client and the vCenter system.

Best practice is to set up the syslog server on the reference host with the vSphere Client or the `esxcli system syslog` command and to save the host profile. You can also set up syslog from the Host Profiles feature in the vSphere Client.

1. Navigate to **Home** > **Policies and Profiles** > **Host Profiles**.
2. Optional: If no reference host exists in your environment, click **Extract Profile from Host** to create a host profile.
3. Click the host profile you want to configure and select the **Configure** tab.
4. Click **Edit Host Profile**.
5. Select **Advanced Configuration Settings** > **Advanced Options** > **Advanced configuration options**.
   You can select specific sub-profiles and edit the syslog settings.
6. Optional: To create an advanced configuration option.
   a) Click the **Add sub-profile** icon.
   b) From the **Advanced option** drop-down list select **Configure a fixed option**.
   c) Specify *Syslog.global.loghost* as the name of the option, and your host as the value of the option.
7. Click **Save** to complete the host profile configuration.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Enable NTP Client on a Reference Host in the vSphere Client

When you collect logging information in large environments, you must ensure that log times are coordinated.

You can set up the reference host to use the NTP server in your environment, extract the host profile and create a vSphere Auto Deploy rule to apply it to other hosts.

1. Navigate to **Home** > **Hosts and Clusters**, and select an ESX host that you want to use as a reference host.
2. Select the **Configure** tab.
3. Under **System**, select **Time Configuration**.
4. Select the **Use Network Time Protocol** radio button.
   This option synchronizes the time and date of the host with an NTP server. The NTP service on the host periodically takes the time and date from the NTP server.
5. From the **NTP Service Startup Policy** drop-down menu, select **Start and stop with host**.
6. In the **NTP Servers** text box, enter the IP addresses or host names of the NTP servers that you want to use.
7. Click **OK**.

- Extract a host profile from the reference host. See the *Host Profiles* documentation.
- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.

Configure Networking for Your vSphere Auto Deploy Host in the vSphere Client

You can set up networking for your vSphere Auto Deploy reference host and apply the host profile to all other hosts to guarantee a fully functional networking environment.

Provision the host you want to use as your reference host with an ESX image by using vSphere Auto Deploy.

1. Navigate to **Home** > **Hosts and Clusters**, and select an ESX host that you want to use as a reference host.
2. Select the **Configure** tab and navigate to **Networking**.
3. Perform the networking setup.
   If you are using virtual switches and not vSphere Distributed Switch, do not add other VMkernel NICs to vSwitch0.

4. After the reference host is configured, reboot the system to verify that vmk0 is connected to the Management Network.
5. If no host profile exists for your reference host, create a host profile.

- Create a rule that applies the host profile to all hosts that you want to provision with the settings specified in the reference host. For writing a rule in a VCF PowerCLI session, see *Write a Rule and Assign a Host Profile to Hosts*.
- For hosts that are already provisioned with vSphere Auto Deploy, perform the test and repair compliance operations in a VCF PowerCLI session, see *Test and Repair Rule Compliance*.
- Power on unprovisioned hosts to provision them with the new host profile.

Configure a Reference Host for Auto-Partitioning

You can create a reference host to auto-partition all host you provision with vSphere Auto Deploy.

By default, vSphere Auto Deploy provisions hosts only if a partition is available on the host. The auto-partitioning option creates a VMFS datastore on your host's local storage. You can set up a reference host to auto-partition all hosts that you provision with vSphere Auto Deploy.

- Provision the host that you want to use as your reference host with an ESX image by using vSphere Auto Deploy.
- Verify that you have access to a vSphere Client that can connect to the vCenter system.

> **CAUTION:**
> If you change the default auto-partitioning behavior, vSphere Auto Deploy overwrites existing partitions regardless of their content. If you turn on this option, ensure that no unintended data loss results.
>
> To ensure that local SSDs remain unpartitioned during auto-partitioning, you must set the parameter `skipPartitioningSsds=TRUE` on the reference host.
>
> For more information about preventing SSD formatting during auto-partitioning, see the *vSphere Storage* documentation.

1. Navigate to **Home** > **Hosts and Clusters**, and select an ESX host that you want to use as a reference host.
2. Select the **Configure** tab.
3. Under **System**, select **Advanced System Settings** and click **Edit**.
4. Search for the `VMkernel.Boot.autoPartition` key and set the value to **true**.
5. Optional: If you want the local SSDs to remain unpartitioned, search for the `VMkernel.Boot.skipPartitioningSsds` key and set the value to **true**.
6. Click **OK**.
7. If no host profile exists for your reference host, create a host profile.

Auto-partitioning is performed when the hosts boot.

- Use vSphere Auto Deploy to create a rule that applies the host profile of your reference host to all hosts immediately when they boot. To create a rule with the vSphere Client, see Create a Deploy Rule. For writing a rule in a VCF PowerCLI session, see Write a Rule and Assign a Host Profile to Hosts.

Converting Stateless Hosts to Stateful Hosts

You can add physical storage to your stateless ESX, convert them to stateful ESX hosts, and add the hosts to a cluster that you manage by an image.

Your ESX hosts must have physical storage attached, because a cluster managed by an image does not support stateless ESX hosts, that use a Preboot Execution Environment (PXE) boot.

To ensure that image components are installed on a physical disk of an ESX host during a boot process, Auto Deploy verifies that a host profile is part of the PXE boot image. The host profile must contain a "System Image Cache Profile Settings" policy with configuration set to "Enable stateful installs on the host" or "Enable stateful installs to a USB disk on the host". If the attached host profile does not contain this policy or the policy's configuration differs, then the policy is

automatically configured to support a stateful installation. If a host profile is missing, a new host profile is attached to the cluster, containing only the "System Image Cache Profile Settings" policy with a configuration for a stateful installation.

Convert a Diskless ESX Host

To add diskless ESX hosts to a cluster managed by an image, add physical storage, convert them to stateful hosts and install an ESX 9.0 image.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy* in the *VMware ESX Installation and Setup* documentation.
- The ESX host is booting from an Auto Deploy rule that contains an Image Profile.
- The Auto Deploy service is running.
- You have an empty USB stick or other supported storage.

For information on stateless caching and stateful installs, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

1. From the home menu, click **Policies and Profiles**, and select **Host Profiles**.
2. Extract a host profile from a running ESX host or duplicate an existing host profile with a specified configuration and a host location.

   The existing host profile can be attached either to a host or to a cluster.
3. Right-click the new host profile, select **Edit Host Profile**, and browse to **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
4. In the **System Image Cache Profile Settings** drop-down menu, choose a policy option.

Option	Description
Enable stateless caching on the host	Caches the image to disk.
Enable stateless caching to a USB disk on the host	Caches the image to a USB disk attached to the host.

5. Depending on the policy option you select, you must do the following:
   a) If you select **Enable stateless caching on the host**:
      1. Enter arguments for the first disk, and select the check boxes if needed.
      2. In the **System Disk Configuration** drop-down menu, select **User must explicitly choose the policy option**.

   b) If you select **Enable stateless caching to a USB disk on the host**:
      1. In the **System Disk Configuration** drop-down menu, select **System disk specified by user in host customization**
      2. Under **Hosts and Clusters**, right-click the host and select **Host Profiles** > **Edit Host Customizations**. Define a disk with persistent storage in the **Value** field for the **System disk** property.
6. Click **Save** to complete the host profile configuration.
7. If you duplicated an existing host profile and the host profile was attached to a cluster, attach the new host profile to the cluster.
8. From the home menu, click **Auto Deploy**.
9. Deactivate the Auto Deploy rule that contains an Image Profile, and click **Edit**.

   The Edit Deploy Rule dialog box appears.
10. On the **Select image profile** page of the wizard, select an ESX 8.0 image profile.
11. On the **Select host profile** page of the wizard, select the new host profile.
12. Activate the rule and move the rule to the rule's initial position in the ordered list.
13. On the **Deployed Hosts** tab, select a single or multiple ESX hosts.
14. Click **Remediate Host Associations** for the ESX host.

    You can monitor the progress of the remediation process in the Recent Tasks pane.

15. Shut down the remediated ESX hosts.
16. After the ESX hosts are powered off, install the selected storage as local boot disks.
17. Power on each ESX host, enter the BIOS/UEFI setup, and change the boot order to boot first from the newly added storage and then from the network.

    Since the newly added storage is empty, each ESX host boots from the network and installs the ESX 9.1 image that you specified earlier on the storage. After the installation, each ESX host reboots and boots from the newly added storage.

The ESX hosts boot by default from the new storage and operate as if the ESX9.1 image is installed from a standard DVD.

Add the ESX hosts to a cluster that you manage by a single image. For more information, see *Add a Host* from the *vCenter and Host Management* documentation.

Convert a Stateless ESX Host with Enabled Stateless Caching

To add your stateless ESX hosts to a cluster that you manage by an image, convert them to stateful hosts by installing a standard 9.1 image.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy* in the *VMware ESX Installation and Setup* documentation.
- The ESX host is booting from an Auto Deploy rule that contains an Image Profile.
- The Auto Deploy service is running.

For information on stateless caching and stateful installs, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

1. From the home menu, click **Policies and Profiles**, and select **Host Profiles**.
2. Extract a host profile from a running ESX host or duplicate an existing host profile with a specified configuration and a host location.

   The existing host profile can be attached either to a host or to a cluster.
3. Right-click the new host profile, select **Edit Host Profile**, and browse to **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
4. Select **Enable stateful installs on the host** from the drop-down menu, enter `local` in the **Arguments for first disk** text box, and select the **Check to overwrite any VMFS volumes on the selected disk** check box.
5. If you duplicated an existing host profile and the host profile was attached to a cluster, attach the new host profile to the cluster.
6. From the home menu, click **Auto Deploy**.
7. Deactivate the Auto Deploy rule that contains an Image Profile, and click **Edit**.

   The Edit Deploy Rule dialog box appears.
8. On the **Select image profile** page of the wizard, select an ESX 8.0 image profile.
9. On the **Select host profile** page of the wizard, select the new host profile.
10. Activate the rule and move the rule to the rule's initial position in the ordered list.
11. On the **Deployed Hosts** tab, select a single or multiple ESX hosts.
12. Click **Remediate Host Associations** for the ESX host.

    You can monitor the progress of the remediation process in the Recent Tasks pane.
13. Restart the ESX hosts.

    If an ESX host has a legacy BIOS, you can change the boot order in the BIOS setup to first to boot from the local storage. For UEFI-based ESX hosts the boot order changes automatically during the reboot.

    Each ESX host boots from the network and installs the ESX9.1 image that you specified earlier on the storage used previously for caching. After the installation, each ESX host reboots again and boots from the local storage.

The ESX hosts boot by default from the new storage and operate as if the ESX9.1 image is installed from a standard DVD.

Add the ESX hosts to a cluster that you manage by a single image. For more information, see *Add a Host* from the *vCenter and Host Management* documentation.

Convert a Stateless ESX Host with a Single VMFS Partition on the Local Disk

To add your stateless ESX hosts to a cluster that you manage by an image, you must first convert the hosts to stateful hosts by repartitioning the VMFS partitions and installing a standard ESX 9.1 image.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy* in the *VMware ESX Installation and Setup* documentation.
- The ESX host is booting from an Auto Deploy rule that contains an Image Profile.
- The Auto Deploy service is running.

For information on stateless caching and stateful installs, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

1. From the home menu, click **Policies and Profiles**, and select **Host Profiles**.
2. Extract a host profile from a running ESX host or duplicate an existing host profile with a specified configuration and a host location.

   The existing host profile can be attached either to a host or to a cluster.
3. Right-click the new host profile, select **Edit Host Profile**, and browse to **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
4. Select **Enable stateful installs on the host** from the drop-down menu, enter `localesx` in the **Arguments for first disk** text box.
5. If you duplicated an existing host profile and the host profile was attached to a cluster, attach the new host profile to the cluster.
6. From the home menu, click **Auto Deploy**.
7. Deactivate the Auto Deploy rule that contains an Image Profile, and click **Edit**.

   The Edit Deploy Rule dialog box appears.
8. On the **Select image profile** page of the wizard, select an ESX 8.0 image profile.
9. On the **Select host profile** page of the wizard, select the new host profile.
10. Activate the rule and move the rule to the rule's initial position in the ordered list.
11. On the **Deployed Hosts** tab, select a single or multiple ESX hosts.
12. Click **Remediate Host Associations** for the ESX host.

    You can monitor the progress of the remediation process in the Recent Tasks pane.
13. Restart the ESX hosts.

    If an ESX host has a legacy BIOS, you can change the boot order in the BIOS setup to first to boot from the local storage. For UEFI-based ESX hosts the boot order changes automatically during the reboot.

    Each ESX host boots from the network, repartitions the VMFS partition to a standard ESX installation partition, and installs the ESX9.1 image that you specified earlier on the partition. After the installation, each ESX host reboots again and boots from the standard ESX partition.

The ESX hosts boot by default from the new partition and operate as if the ESX9.1 image is installed from a standard DVD.

Add the ESX hosts to a cluster that you manage by a single image. For more information, see *Add a Host* from the *vCenter and Host Management* documentation.

Convert a Stateless ESX Host with a Single VMFS Partition on a Remote Disk

To add your stateless ESX hosts to a cluster that you manage by an image, first you must convert the hosts to stateful hosts by repartitioning their remote VMFS partitions and installing a standard ESX9.1 image. For example, your ESX host can boot from Fibre Channel SAN or from iSCSI SAN.

- Prepare your system and install the Auto Deploy Server. For more information, see *Prepare Your System for vSphere Auto Deploy* in the *VMware ESX Installation and Setup* documentation.
- The ESX host is booting from an Auto Deploy rule that contains an Image Profile.
- The Auto Deploy service is running.

For information on stateless caching and stateful installs, see Use vSphere Auto Deploy for Stateless Caching and Stateful Installs.

For more information on booting from a remote disk, see the *Booting from iSCSI SAN* in the *vSphere Storage* documentation.

1. From the home menu, click **Policies and Profiles**, and select **Host Profiles**.
2. Extract a host profile from a running ESX host or duplicate an existing host profile with a specified configuration and a host location.

   The existing host profile can be attached either to a host or to a cluster.
3. Right-click the new host profile, select **Edit Host Profile**, and browse to **Advanced Configuration Settings** > **System Image Cache Configuration** > **System Image Cache Configuration**.
4. Select **Enable stateful installs on the host** from the drop-down menu, enter `remoteesx,remote` or `sortedremoteesx,sortedremote` in the **Arguments for first disk** text box.
5. If you duplicated an existing host profile and the host profile was attached to a cluster, attach the new host profile to the cluster.
6. From the home menu, click **Auto Deploy**.
7. Deactivate the Auto Deploy rule that contains an Image Profile, and click **Edit**.

   The Edit Deploy Rule dialog box appears.
8. On the **Select image profile** page of the wizard, select an ESX 8.0 image profile.
9. On the **Select host profile** page of the wizard, select the new host profile.
10. Activate the rule and move the rule to the rule's initial position in the ordered list.
11. On the **Deployed Hosts** tab, select a single or multiple ESX hosts.
12. Click **Remediate Host Associations** for the ESX host.

    You can monitor the progress of the remediation process in the Recent Tasks pane.
13. Restart the ESX hosts.

    Each ESX host boots from the network, repartitions the VMFS partition to a standard ESX installation partition, and installs the ESX9.1 image that you specified earlier on the partition. After the installation, each ESX host reboots again and boots from the remote standard ESX partition.

The ESX hosts boot by default from the new partition and operate as if the ESX9.1 image is installed from a standard DVD.

Add the ESX hosts to a cluster that you manage by a single image. For more information, see *Add a Host* from the *vCenter and Host Management* documentation.

Using Zero Touch Provisioning

With Zero Touch Provisioning, also called vSphere Elastic Provisioning, you can install ESX on hosts at scale from a single UEFI HTTPS boot URL or a static network configuration on bare metal.

In this chapter, see:

- What is Zero Touch Provisioning with Auto Deploy
- How to Choose and Switch Between Provisioning Methods in Auto Deploy
- Image ESX Hosts by Network with a UEFI HTTPS Boot URL Configured Directly in the UEFI Firmware
- How to Provision a Heterogeneous Cluster with Zero Touch Provisioning
- How to Provision ESX Hosts in Heterogeneous Clusters with the Auto Deploy - stateless Method
- How Zero Touch Provisioning Works with vSphere Configuration Profiles

What is Zero Touch Provisioning with Auto Deploy

Zero Touch Provisioning combines a vCenter-level UEFI HTTPS boot URL and a UEFI HTTPS boot facility to enable ESX provisioning at scale.

UEFI firmware and UEFI HTTPS boot allow you to avoid configuring PXE infrastructure and using network services such as TFTP, DHCP, and DHCP relay agents in complicated network configurations.

Starting with VCF 9.1, the management of UEFI HTTPS Boot moves from Auto Deploy to the Zero Touch Provisioning capability. The Boot Routing Service (BRS), which is part of Zero Touch Provisioning and is managed by the vSphere Lifecycle Manager, now serves the UEFI HTTPS boot URL. However, BRS does not have its own rule engine and depends on the Auto Deploy rule matching functionality, so Zero Touch Provisioning and Auto Deploy work together for UEFI HTTPS Boot scenarios.

Along wth BRS, Zero Touch Provisioning also adds a new first-stage bootloader, vSphere Boot Manager (vBM), replacing the iPXE bootloader used by Auto Deploy, and the Network Boot Service (NBS), which allows provisioning ESX hosts over the network.

Zero Touch Provisioning also introduces support for heterogeneous clusters that are managed by the vSphere Lifecycle Manager, provisioning multiple ESX images in a single cluster, including clusters using a vSphere Configuration Profile.

In the Zero Touch Provisioning scenarios, ESX hosts are provisioned directly from an HTTPS boot URL that is exposed by BRS and served by NBS.

You can use the Zero Touch Provisioning method to provision ESX hosts with all versions starting from ESXi 8.0 Update 3.

> **Note:**
>
> Network booting adds support to the common port 443 along with port 6501 that is used for iPXE boot in existing environments.

With Zero Touch Provisioning, you can:

- Use the UEFI HTTPS boot protocol to image ESX hosts at scale by using a single boot URL.
- Use a single boot URL on vCenter level for imaging devices from different hardware vendors to clusters with many alternative images based on hardware information such as SMBIOS data.
- Image ESX hosts over the network without DHCP and TFTP servers, using only static network configuration with UEFI HTTPS boot URL configured directly in the UEFI firmware.

Table 23: Comparison of the Auto Deploy and Zero Touch Provisioning operational components

Classic Auto Deploy (legacy)	Zero Touch Provisioning
• iPXE bootloader • TFTP server required • Per-host DHCP config (must have) • PXE infrastructure	• vSphere Boot Manager (vBM) • Single HTTPS URL for all hosts • Per-host DHCP config (optional) • BRS + NBS in vSphere Lifecycle Manager

Zero Touch Provisioning depends on the existence of HTTPS boot support in the underlying hardware.

You have the following options for setting the boot URL when performing an HTTPS boot, depending on the firmware capabilities:

- using a DHCP server to serve the boot URL as part of the network configuration
- using a management console such as iDRAC or inventory management tools
- interacting with the firmware setup/boot menu through the device console

For the HTTPS boot to work, ESX hosts must trust the vCenter certificate, which means that you must upload a vCenter certificate on the server. For more information, see Image ESX Hosts by Network with a UEFI HTTPS Boot URL Configured Directly in the UEFI Firmware.

Boot Routing Service

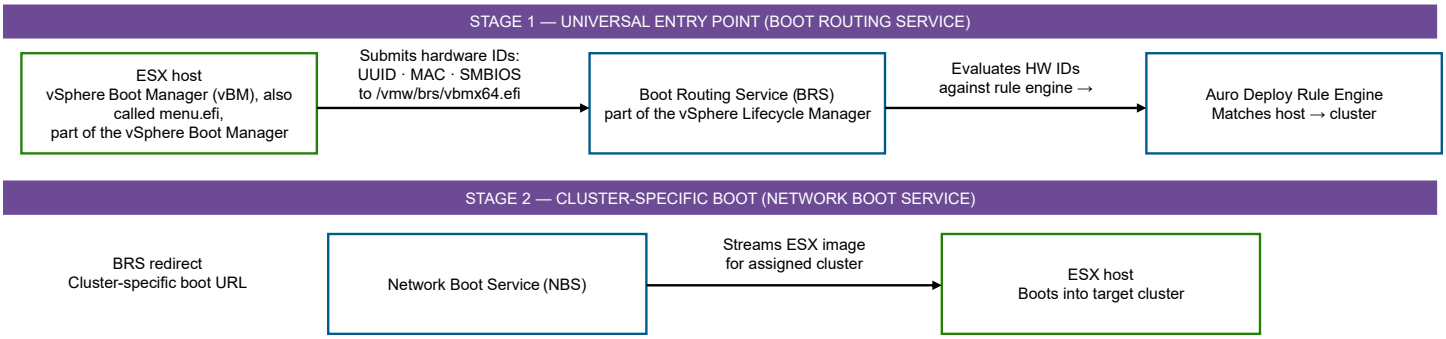
BRS exposes a single vCenter-level HTTPS boot URL, `https://<vcenter-ip/fqdn>/vmw/brs/vbmx64.efi`, that is configured once — either in a DHCP server, UEFI firmware, or management console — and applies to all physical machines managed by the vCenter instance. When an ESX host boots, vBM (menu.efi) submits hardware identifiers such as UUID, MAC address, and SMBIOS data to BRS. BRS evaluates these identifiers against the Auto Deploy Rule engine and issues a redirect to the appropriate cluster boot URL.

Network Boot Service

NBS serves cluster-specific Network Bootbanks (NBF) over HTTPS, eliminating the dependency on legacy PXE/TFTP infrastructure. ESX boot images and configurations are prepared in advance, stored on the vCenter filesystem, and exposed by cluster-scoped URLs (`/vmw/nbs/clusters/<cluster-id>/`).

BRS and NBS implement a two-stage network boot pipeline. BRS operates as the universal entry point: a single boot URL serves all devices managed by a vCenter instance, regardless of the intended destination in its inventory. NBS provides the cluster-specific boot. streaming ESX images for the assigned cluster. Both services are part of the vSphere Lifecycle Manager.

Figure 9: BRS + NBS Two-Stage Boot Pipeline



Upon receiving a boot request, BRS collects hardware attributes by vBM and consults the Auto Deploy Rule engine to determine the target cluster. BRS then redirects the booting host to the cluster-specific boot URL served by NBS. For stateless deployments, BRS redirects to an Auto Deploy-derived boot URL instead of an NBS cluster URL, preserving full backward compatibility with existing Auto Deploy rule sets.

In the Auto Deploy screen, when you select Zero Touch Provisioning as provisioning method, you see two tasks running in the **Recent Tasks** panel:

1. **Enable network boot**, which changes the network configuration of the cluster.
2. **Create network bootbank**, during which the vSphere Lifecycle Manager downloads VIBs from an online depot to a network bootbank, from which ESX hosts are booting.

For Zero Touch Provisioning, you only need the host location to identify the cluster to deploy the rule. The rule uses the image that has been created during the **Create network bootbank** task and the configuration of the cluster.

How to Choose and Switch Between Provisioning Methods in Auto Deploy

With ESX 9.1 and later, you can choose and switch between three methods to provision ESX hosts at scale by using Auto Deploy rules.

Starting with ESX 9.1, you can choose between Zero Touch Provisioning, Auto Deploy - stateful, and Auto Deploy - stateless options to provision ESX hosts at scale, depending on your needs. The **Clusters** tab under Auto Deploy in the vSphere Client adds a column **Provisioning Method**, which tracks your selections.

Note:

In the **Lifecycle Management Method** column, you might see the value **Baselines** for clusters that are not updated to VCF 9.0 and later, and neither of the **Provisioning Method** options works. If your environment has clusters that are managed by using legacy baselines, the **Provisioning Method** column is blank. You cannot create new such clusters, but the existing ones remain active and you can add Auto Deploy rules to them. You cannot change their provisioning method before you update to VCF 9.0 and later.

If an ESX host does not match any of the existing rules, you see it listed under the **Discovered Hosts** tab.

For a new empty cluster, you can select between:

- Auto Deploy - stateful, which defaults to provisioning ESX by using a vSphere Lifecycle Manager image stored on the device.
- Zero Touch Provisioning, which can use desired images, a vSphere Configuration Profile, or simply a cluster. Zero Touch Provisioning is not supported on folder level.
- Auto Deploy - stateless to select provisioning that does not cache the ESX image.

All images, except for stateless, which uses Auto Deploy images, are managed by the vSphere Lifecycle Manager. You can manage cluster configuration by either vSphere Configuration Profiles or vSphere Host Profiles, which are deprecated.

Table 24: Transitions between provisioning methods

Provisioning Method	Cluster Empty or Not	Transition to Auto Deploy - stateful	Transition to Auto Deploy - stateless	Transition to Zero Touch Provisioning
Auto Deploy - stateful	Empty cluster	N/A	Allowed	Allowed
Auto Deploy - stateful	Cluster with hosts	N/A	Not allowed	Allowed
Auto Deploy - stateless	Empty cluster	Allowed	N/A	Allowed
Auto Deploy - stateless	Cluster with hosts	Not allowed	N/A	Not allowed
Zero Touch Provisioning	Empty cluster	Allowed	Not allowed	N/A
Zero Touch Provisioning	Cluster with hosts	Allowed	Not allowed	N/A

If a stateful Host Profile is attached to the cluster or if a vSphere Configuration Profile is enabled on the cluster, you cannot transition clusters from stateful to stateless.

When creating a cluster in vCenter, it defaults to the provisioning method Auto Deploy - stateful. This means that when you create a rule for this cluster, Auto Deploy automatically picks up a vSphere Lifecycle Manager image for this cluster. ESX hosts added to such cluster are provisioned with that image by using the PXE infrastructure. Since the image is locally stored on the ESX hosts, this provisioning method is known as stateful installation. The Auto Deploy - stateful method does not support provisioning of heterogeneous, or composite, clusters.

From the default Auto Deploy - stateful method, you can switch to Zero Touch Provisioning by selecting **Enable Zero Touch Provisioning**, which is also a stateful installation method. Zero Touch Provisioning rules use the BRS and NBS services instead of the PXE infrastructure, as long as the firmware provides HTTP boot support. For servers that use BIOS, you cannot use the Zero Touch Provisioning method. With Zero Touch Provisioning, you can provision both homogeneous and heterogeneous clusters, because this provisioning method works for both.

Reverting the method from Zero Touch Provisioning to Auto Deploy - stateful excludes support for provisioning heterogeneous clusters and deletes the network bootbank that is created for this cluster. In such a case, all existing rules that are tied to that cluster deactivate and you need either to edit the existing or create new rules.

You can also switch an empty cluster to Auto Deploy - stateless by selecting **Enable AutoDeploy Stateless Support**, which means that no image is locally stored on ESX hosts in this cluster.

For empty clusters, you can revert from the Zero Touch Provisioning to the Auto Deploy - stateful or Auto Deploy - stateless method.

> **Note:**
>
> When you select Zero Touch Provisioning, you can track progress in the **Recent Tasks** panel, represented as tasks. The tasks are: Create Network Bootbank, Delete Network Bootbank, Deduplicate Network Bootbank, and Host Network Boot Progress. If the tasks are not complete and you create a rule, the rule restores the default stateful install option, because it cannot find a network bootbank.

After you create a rule for Zero Touch Provisioning, if you switch the rule to Auto Deploy - stateful, the rule automatically updates and changes the bootbank. Except when you are using a heterogeneous cluster, then the rule is deactivated.

When reverting the method from Zero Touch Provisioning to Auto Deploy - stateless, you manage the lifecycle of hosts part of stateless clusters or standalone hosts only by using Auto Deploy. vSphere Lifecycle Manager operations relevant to stateful hosts such as single image association, image compliance, and remediation, are not supported for stateless clusters.

Image ESX Hosts by Network with a UEFI HTTPS Boot URL Configured Directly in the UEFI Firmware

With UEFI HTTPS network boot support, you can provision ESX hosts by using a static network configuration with the boot URL configured directly in the firmware.

Even without a TFTP server, the DHCP server returns the IP and a boot URL required for network booting. You can use these parameters to set a static network configuration with the boot URL configured directly in the firmware, also called bare metal provisioning. You can manually modify the configuration parameters in the firmware to a UEFI HTTPS boot URL. You can set such a URL as a static value in the configuration of any server that supports UEFI HTTPS.

When you activate Zero Touch Provisioning on a cluster, in the **Zero Touch Provisioning** table at **Auto Deploy** > **Configure**, you can find the UEFI HTTPS Boot URL, such as `https://<vcenter-ip/fqdn>/vmw/brs/bootx64.efi`, which you can use to create a static network configuration directly in the UEFI firmware and not depend on a DHCP server.

If you use iDRAC to log in to a Dell server, under **Configuration** > **BIOS Settings** > **Network Settings**, you should disable all PXE devices and enable an HTTP device with the following settings:

- Deactivate DHCP,
- Set the static IP address or VLAN, subnet mask, gateway, and DNS servers,
- in the URI field, paste the HTTP(S) boot url that you copy from the **Zero Touch Provisioning** table at **Auto Deploy** > **Configure**.

> **Note:**
>
> Starting with VCF 9.1, you can use the common port 443 for network booting along with port 6501. Both IPv4 and IPv6 are supported.

Alternatively, you can paste the HTTPS boot URL in the DHCP configuration.

For the HTTPS boot to work, ESX hosts must trust the vCenter certificate, which means that you must upload a vCenter certificate on the server.

To export the security certificate from the vCenter instance, see Download and Install the vCenter Certificate. When you extract the zip package, open the `win` folder and see the certificate with extension `.0.crt`. To convert the certificate in the recommended Distinguished Encoding Rules (DER) format, use the command `openssl x509 -outform der -in <certificate file name>.crt -out <certificate file name>.der`.

You can use alternative ways to get vCenter certificates but use only the vCenter CA certificate.

When uploading the vCenter certificate, the server interface might differ per vendor. In Dell iDRAC, for example, you must upload the file that you export in DER format in **Root Certification Configuration**.

When setting the UEFI HTTPS boot URL as a static value in a server configuration, you must set the **TLS Authentication Mode** under **TLS Authentication Configuration** to One Way.

If the downgrade support on your firmware is active, you can use Zero Touch Provisioning with TLS 1.1, TLS 1.2, and TLS 1.3.

VCF 9.0 and later supports TLS 1.3, but if your servers do not support this version, you must explore other configuration options for your environment. For example, you can set up a secure server for TLS 1.3 only, and can use another server for ESX host provisioning that uses TLS 1.2.

How to Provision a Heterogeneous Cluster with Zero Touch Provisioning

With Zero Touch Provisioning, you can provision custom images of ESX to hosts in one cluster, depending on different criteria.

With Zero Touch Provisioning, you can boot different devices with custom images, which are based on criteria such as hardware information, in heterogeneous clusters.

The following steps cover a possible end-to-end scenario where you provision ESX hosts in a cluster with a default image and an OEM vendor add-on image. Provisioning with the add-on image depends on the underlying hardware.

Before you begin, make sure that you have a default ESX image and an OEM vendor add-on image in an online or offline depot. For more information, see Add a Software Depot or Import a Software Depot.

- In the vSphere Client, create a datacenter and a cluster with a heterogenous software specification that contains a default image and an OEM vendor add-on image:
  - In the **Basics** tab of the **New Cluster** workflow, select **Create a new image** and **Manage configuration at a cluster level** to allow the use of vSphere Configuration Profiles.
  - In the **Image** tab, select a default ESX version and a vendor add-on image.

    A heterogeneous cluster is created.
- In the cluster, click **Updates** > **Image** > **Edit** to add the vendor add-on image to the default one and check the **Select hosts by host's hardware criteria** option, specifying the vendor name.

  This selection defines that only servers from the respective OEM vendor get the custom add-on image. All other servers in the cluster get the default desired image for the cluster.
- In the **Auto Deploy** tab, select the **Enable Zero Touch Provisioning** option to activate UEFI HTTPS network boot support.

  In the **Recent Tasks** panel, you see two tasks appearing: **Enable network boot task on the cluster** and **Create network bootbank task**. Wait until both tasks are completed.

  When the **Create network bootbank** task completes, you see a boot URL configured in the **Target** field. The boot URL cannot work without the Auto Deploy rule engine, which redirects the booting host to the network boot folder corresponding to the target cluster.
- Create a Deploy Rule in Auto Deploy.

  All hosts that try to boot from the network boot URL are redirected to boot from the network bootbank generated for this cluster.
  - In **Name and hosts**, select **All hosts**.
  - In **Configuration**, select only **Host location**.
  - In **Select host location**, select the cluster.

    Any host on this cluster is going to use the network bootbank generated for this cluster.
  - Activate the rule.
- Monitor provisioning:

  Starting with VCF 9.1, you can use remote consoles for each host to monitor the provisioning. Imaging ESX hosts by Zero Touch Provisioning provides new output from the new boot logger service, vSphere Boot Manager (vBM).

  vBM extracts the SMBIOS data from virtual machines, sends it back to the boot routing service (BRS) and the boot routing service translates the SMBIOS data to Auto Deploy. Auto Deploy then matches the host against the rule that you have created. For example, hosts that are based on Dell servers can boot with an add-on from the vendor, while other hosts boot with the desired image for the cluster. The ESX installer downloads over the network. The installation completes automatically. Once the installation is complete, hosts reboot from the local disk, the desired configuration is applied, and they are added to the target cluster in vCenter.
- Confirm image compliance:
  - Go to the **Image Compliance** panel under the **Updates** tab and click **Check Compliance** to confirm that the correct desired images are selected for the correct virtual machines.

– Go to **Configure tab** > **Desired State** > **Configuration** and click **Check Compliance** to confirm the configuration.

How to Provision ESX Hosts in Heterogeneous Clusters with the Auto Deploy - stateless Method

You can provision ESX hosts in a single cluster with different images by using the Auto Deploy - stateless method and separate rules for each image.

Add or import at least two online or offline depots in the **Software Depots** tab of the Auto Deploy screen. See Add a Software Depot or Import a Software Depot.

Selecting the Auto Deploy - stateless method to provision ESX hosts limits the lifecycle management of hosts only to using Auto Deploy. But Auto Deploy - stateless allows you to provision hosts with custom images in a heterogeneous cluster.

You can create rules for more than one image that you want to use with the stateless cluster and provision ESX hosts in a single cluster with different images.

Starting with VCF 9.1, Auto Deploy - stateless works with both the Auto Deploy rule engine, which uses iPXE infrastructure, and the Boot Routing Service (BRS), which enables UEFI HTTPS booting.

1. Create a cluster in vCenter that you plan to use for stateless ESX deployments and select **Host Location** pointing to that cluster.

   The cluster defaults to the Auto Deploy - stateful provisioning method.

2. In the **Auto Deploy** tab, select the cluster and click **Enable Stateless Support**.

   The value in the Provisioning Method column changes to **Auto Deploy - stateless**.

3. Create a deploy rule and select the cluster in **Host Location**.

   This step can repeat as many times as the number of images you want to use in the heterogeneous stateless cluster.

4. Click **Deploy Rules**, select the cluster, and click **Edit** to add an image.

   When you create rules for a custom image that you want to use with the stateless cluster, for each rule, click **Hosts that match the following pattern** and then **Add Pattern** to select your custom criteria. For example, you can specify a server vendor. Select the same cluster in **Host Location** and add a custom image for a given OEM for example. This way, different rules can deploy different images to servers in the same cluster.

5. In the **Configuration** step of the wizard, select **Image** and **Host Location**.

6. In the **Select Image** step, select the image that you want to use for stateless provisioning.

7. Activate the rule.

   When you create multiple rules using different images on the same cluster, mind the order of the rules. Make sure that rules for a specific pattern are higher in the list of rules than rules that match all hosts. If a rule that matches all hosts precedes a rule for a specific pattern, all hosts match the image from the preceding rule.

   For example, if you have three rules:

   1. For a custom image in a stateless cluster.
   2. For a VMware image in a stateless cluster,
   3. For stateful installation.

   You can order the rules so that:

   1. The first rule is for the custom image,
   2. The second rule is for all other cases for stateless provisioning,
   3. The third rule covers stateful provisioning.

How Zero Touch Provisioning Works with vSphere Configuration Profiles

You can collectively manage the ESX configuration of a cluster with vSphere Configuration Profiles by using Zero Touch Provisioning.

With Zero Touch Provisioning, you can select between vSphere Lifecycle Manager desired images, vSphere Configuration Profiles on vCenter cluster level, or simply a cluster in a vCenter instance. When you create a cluster, the default provisioning method is Auto Deploy - stateful. You can switch to either Zero Touch Provisioning or Auto Deploy - stateless, but only Auto Deploy - stateful and Zero Touch Provisioning support vSphere Configuration Profiles.

Before you start using Zero Touch Provisioning with vSphere Configuration Profiles, you must have a ready configuration, either in a `.json` file or exported from a reference host. The best practice is to:

- Create a VCP cluster.
- Manually configure the desired settings, such as IP, VMK adapter, virtual switch, and firewall on an existing ESX host that you select as a reference host.
- Import the reference host configuration to create a draft, make changes, precheck, and apply it.

Successful provisioning of hosts in VCP clusters requires a valid, non-empty desired configuration document. If the cluster configuration lacks networking details, Auto Deploy rules might fail, because hosts cannot come online. Always validate the configuration before you create Auto Deploy rules and attempt to provision hosts.

For new clusters, when you complete the vSphere Configuration Profile configuration and create a rule in Auto Deploy, the provisioning method defaults to Auto Deploy - stateful. Select the cluster and click **Enable Zero Touch Provisioning**. Then you can start provisioning all other hosts in the cluster with the Sphere Configuration Profile.

If the imported ESX host is part of a cluster with an active Auto Deploy rule, the new cluster appears under the **Clusters** tab in Auto Deploy.

vSphere Auto Deploy Best Practices

Set up a highly available vSphere Auto Deploy infrastructure in large production environments or when using stateless caching.

Follow best practices when installing vSphere Auto Deploy and when using vSphere Auto Deploy with other vSphere components. Set up a highly available vSphere Auto Deploy infrastructure in large production environments or when using stateless caching. Follow all security guidelines that you would follow in a PXE boot environment, and consider the recommendations in this chapter.

vSphere Auto Deploy Best Practices

Follow vSphere Auto Deploy best practices to set up networking, configure vSphere HA, and optimize your environment for vSphere Auto Deploy.

For Auto Deploy

vSphere Auto Deploy and vSphere HA Best Practices

You can improve the availability of the virtual machines running on hosts provisioned with vSphere Auto Deploy by following best practices.

Some environments configure the hosts provisioned with vSphere Auto Deploy with a distributed switch or configure virtual machines running on the hosts with Auto Start Manager. In such environments, deploy the vCenter system so that its availability matches the availability of the vSphere Auto Deploy server. Several approaches are possible.

- Deploy vCenter. The vSphere Auto Deploy server is included.
- Run vCenter in a vSphere HA enabled cluster and configure the virtual machine with a vSphere HA restart priority of high. Include two or more hosts in the cluster that are not managed by vSphere Auto Deploy and pin the vCenter virtual machine to these hosts by using a rule (vSphere HA DRS required VM to host rule). You can set up the rule and then deactivate DRS if you do not want to use DRS in the cluster. The greater the number of hosts that are not managed by vSphere Auto Deploy, the greater your resilience to host failures.

  Note:
  This approach is not suitable if you use Auto Start Manager. Auto Start Manager is not supported in a cluster enabled for vSphere HA.

vSphere Auto Deploy Networking Best Practices

Prevent networking problems by following vSphere Auto Deploy networking best practices.

vSphere Auto Deploy and IPv6	Because vSphere Auto Deploy takes advantage of the iPXE infrastructure, if the hosts that you plan to provision with vSphere Auto Deploy are with legacy BIOS, the vSphere Auto Deploy server must have an IPv4 address. PXE booting with legacy BIOS firmware is possible only over IPv4. PXE booting with UEFI firmware is possible with either IPv4 or IPv6.
IP Address Allocation	Use DHCP reservations for address allocation. Fixed IP addresses are supported by the host customization mechanism, but providing input for each host is not recommended.
VLAN Considerations	Use vSphere Auto Deploy in environments that do not use VLANs. If you intend to use vSphere Auto Deploy in an environment that uses VLANs, make sure that the hosts that you want to provision can reach the DHCP server. How hosts are assigned to a VLAN depends on the setup at your site. The VLAN ID might be assigned by the switch or the router, or might be set in the host's BIOS or through the host profile. Contact your network administrator to determine the steps for allowing hosts to reach the DHCP server.

vSphere Auto Deploy and VMware Tools Best Practices

When you provision hosts with vSphere Auto Deploy, you can select an image profile that includes VMware Tools, or select the smaller image associated with the image profile that does not contain VMware Tools.

You can download two image profiles from the VMware download site.

- *xxxxx*-standard: An image profile that includes the VMware Tools binaries, required by the guest operating system running inside a virtual machine. The image is usually named esx-*version*-*xxxxx*-standard.
- *xxxxx*-no-tools: An image profile that does not include the VMware Tools binaries. This image profile is usually smaller has a lower memory overhead, and boots faster in a PXE-boot environment. This image is usually named esx-*version*-*xxxxx*-no-tools.

You can deploy ESX using either image profile.

- If the network boot time is of no concern, and your environment has sufficient extra memory and storage overhead, use the image that includes VMware Tools.
- If you find the network boot time too slow when using the standard image, or if you want to save some space on the hosts, you can use the image profile that does not include VMware Tools, and place the VMware Tools binaries on shared storage. See, Provision ESX Host with an Image Profile Without VMware Tools.

vSphere Auto Deploy Load Management Best Practices

Simultaneously booting large numbers of hosts places a significant load on the vSphere Auto Deploy server. Because vSphere Auto Deploy is a Web server at its core, you can use existing Web server scaling technologies to help distribute the load. For example, one or more caching reverse proxy servers can be used with vSphere Auto Deploy. The reverse proxies serve up the static files that make up the majority of an ESX boot image. Configure the reverse proxy to cache static content and pass all requests through to the vSphere Auto Deploy server.

Use multiple TFTP servers to point to different proxy servers. Use one TFTP server for each reverse proxy server. After that, set up the DHCP server to send different hosts to different TFTP servers.

When you boot the hosts, the DHCP server redirects them to different TFTP servers. Each TFTP server redirects hosts to a different server, either the vSphere Auto Deploy server or a reverse proxy server, significantly reducing the load on the vSphere Auto Deploy server.

After a massive power outage, bring up the hosts on a per-cluster basis. If you bring multiple clusters online simultaneously, the vSphere Auto Deploy server might experience CPU bottlenecks. All hosts might come up after a delay. The bottleneck is less severe if you set up the reverse proxy.

vSphere Auto Deploy Logging and Troubleshooting Best Practices

To resolve problems that you encounter with vSphere Auto Deploy, use the vSphere Auto Deploy logging information from the vSphere Client and set up your environment to send logging information and core dumps to remote hosts.

vSphere Auto Deploy Logs	Download the vSphere Auto Deploy logs by going to the vSphere Auto Deploy page in the vSphere Client. See, Download vSphere Auto Deploy Logs.
Setting Up Syslog	Set up a remote syslog server. See the *vCenter and Host Management* documentation for syslog server configuration information. Configure the first host you boot to use the remote syslog server and apply that host's host profile to all other target hosts. Optionally, install and use the vSphere Syslog Collector, a vCenter support tool that provides a unified architecture for system logging, enables network logging, and lets you combine logs from multiple hosts.
Setting Up ESX Dump Collector	Hosts provisioned with vSphere Auto Deploy do not have a local disk to store core dumps on. Install ESX Dump Collector and set up your first host so that all core dumps are directed to ESX Dump Collector, and apply the host profile from that host to all other hosts. See Configure ESX Dump Collector with ESXCLI.

Using vSphere Auto Deploy in a Production Environment

When you move from a proof of concept setup to a production environment, take care to make the environment resilient.

- Protect the vSphere Auto Deploy server. See vSphere Auto Deploy and vSphere HA Best Practices.
- Protect all other servers in your environment, including the DHCP server and the TFTP server.
- Follow VMware security guidelines, including those outlined in vSphere Auto Deploy Security Considerations.

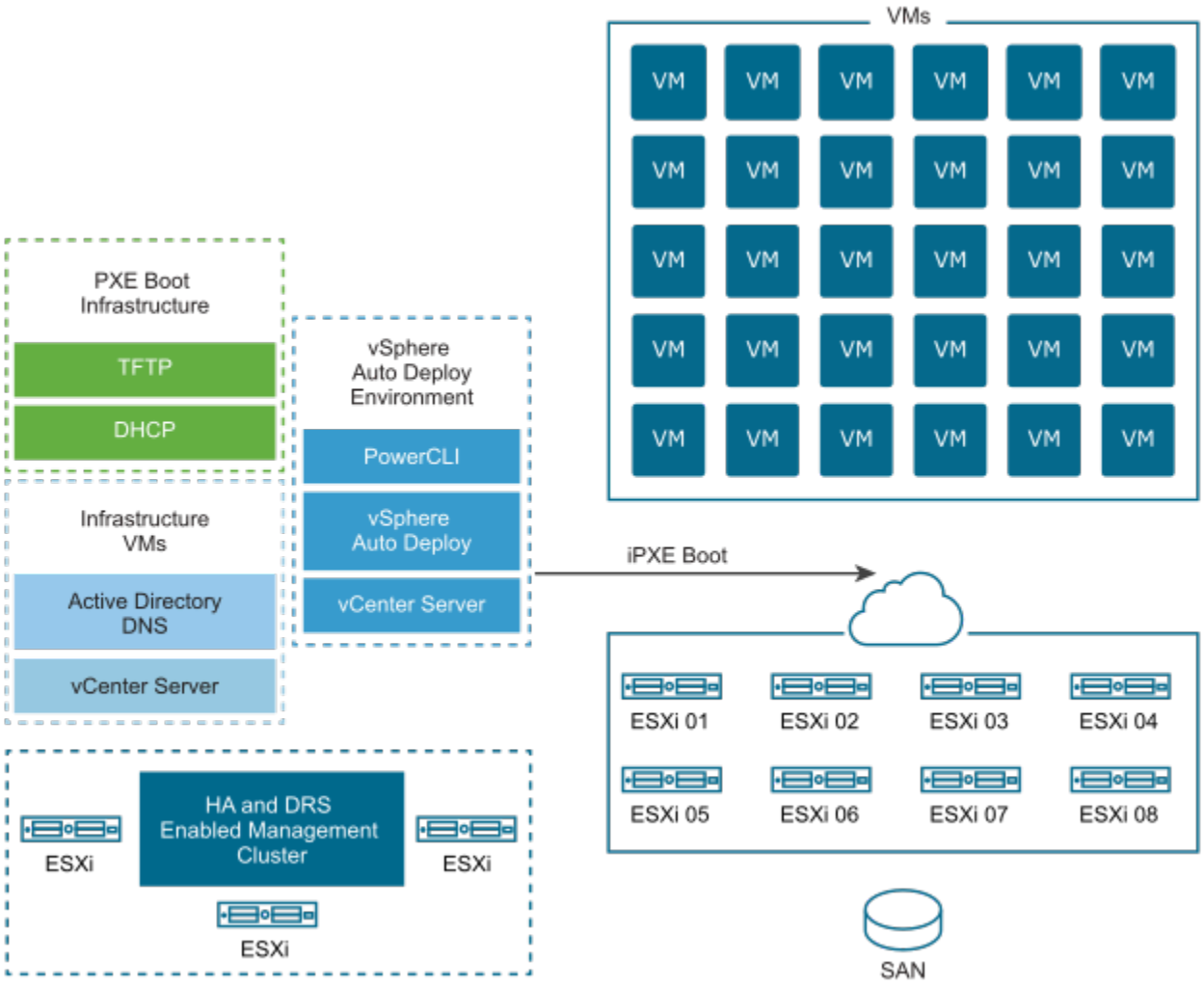
Set Up Highly Available vSphere Auto Deploy Infrastructure

Highly available vSphere Auto Deploy infrastructure prevents data loss and is a prerequisite for using vSphere Auto Deploy with stateless caching.

In many production situations, a highly available vSphere Auto Deploy infrastructure is required to prevent data loss. Such infrastructure is also a prerequisite for using vSphere Auto Deploy with stateless caching.

For the management cluster, install ESX on three hosts. Do not provision the management cluster hosts with vSphere Auto Deploy.

Figure 10: Highly Available vSphere Auto Deploy Infrastructure



1. Enable vSphere HA and vSphere DRS on the management cluster.
2. Set up the following virtual machines on the management cluster.

Infrastructure Component	Description
PXE boot infrastructure	TFTP and DHCP servers.
Infrastructure VM	Active Directory, DNS, vCenter.
vSphere Auto Deploy environment	VCF PowerCLI, vSphere Auto Deploy server, vCenter. Set up this environment on a single virtual machine or on three separate virtual machines in production systems.

   The vCenter on the infrastructure virtual machine differs from the vCenter in the vSphere Auto Deploy environment.
3. Set up vSphere Auto Deploy to provision other hosts as needed.

   Because the components on the management cluster are protected with vSphere HA, high availability is supported.

Device Alias Configuration

Device aliases, also called device names, are short names associated with I/O adapters in an I/O subsystem.

For example, network uplinks have aliases such as vmnic0, vmnic1, and so on. SCSI adapter objects in the storage subsystem and graphics device objects also have aliases. A hardware device can be presented as multiple I/O adapters

in the I/O subsystem. The I/O adapters can be of a different type from the underlying physical device. For example, an FCoE device is a storage I/O adapter that uses NIC hardware. Software iSCSI is a storage adapter using the network stack at the IP layer. Therefore, in the ESX native driver model, aliases formally refer only to I/O adapters, and not to physical devices such as a PCI NIC or a PCI HBA.

Device Alias Assignment

A stateless ESX deployment model is one where the ESX host is not installed on hard disks, and is typically booted by using PXE. A stateful ESX deployment model is one where the ESX host is installed on local hard disks. Device alias assignment occurs during a stateless ESX boot or a fresh installation of stateful ESX. The ESX host assigns aliases to I/O adapters in an order which is based on the underlying hardware enumeration order. The ESX host assigns aliases first to on-board devices and then to add-in cards based on slot order. The ESX host cannot assign aliases to absent devices or devices without supported drivers.

An uplink that uses a NIC that is built into the motherboard receives a vmnicN alias with lower number compared to an uplink of a PCI add-in card. The NIC driver might register more than one uplink. If one of the uplinks does not correspond to an enumerable hardware device, the ESX host assigns the next available alias to the uplink after the uplink is registered with the system.

Persistence of Device Alias Configuration

After the ESX host assigns aliases, alias configuration is persisted. The ESX host attempts to keep the alias of each device the same regardless of the ESX version updates, or hardware changes, such as adding or removing devices from slots.

The persistence of the alias configuration depends on the deployment model.

- In stateful systems, the alias configuration is persisted locally on the host.
- In stateless systems, if you do not manage the stateless system by using host profiles, the alias configuration is not persisted locally on the host.
- In stateful and stateless systems that you manage by using host profiles, the alias configuration is persisted in the host profile. If you apply a host profile to a stateful host, the host profile overrides any locally persisted alias configuration.

Changes in the Device Alias Configuration

The persistence of alias configuration is based on the bus addresses of devices. If the bus address of a device is altered, the persisted alias configuration becomes inapplicable and the aliases assigned to the device might change.

Changes in the device alias configuration might occur in the following cases:

- A driver upgrade might enumerate or present an I/O adapter differently to the system compared to how the I/O adapter is presented before the driver upgrade.
- A stack upgrade might result in changes to parts of a multi-module driver setup, or to the ESX I/O stack that supports a multi-module driver.
- BIOS or device firmware upgrades might lead to incomplete port or slot information.
- Changes in the slot position of a device.

Note:
If you remove a device from the system, the alias configuration of the I/O adapters of the device is removed. If you add the same devices back to the system later, the I/O adapters of the device might not receive their previous aliases.

Device Alias Configuration in ESX Clusters

Initial alias configuration is the same across a cluster of identical systems. However, even on a cluster that is considered homogeneous, small differences in hardware or firmware might result in differences in the alias configuration between hosts.

Differences in the processing order during driver binding can also result in differences in alias configuration. For example, a NIC driver registers two uplinks, uplink-1 and uplink-2, for two ports of the same PCI device, where one of the ports is not hardware enumerable by the system. Timing changes in the order of registration of the uplinks might result in

differences in how the ESX hosts assign aliases to the uplinks. One ESX host might assign the hardware-based alias to uplink-1, and another ESX host might assign the hardware-based alias to uplink-2.

To match alias configuration across homogeneous hosts, you can use host profiles. The Device Alias Configuration host profile applies alias configuration to an ESX host by mapping devices in the alias configuration to the ESX host devices. The mapping operation is based on the hardware information sources that are used as the basis for initial alias assignment. For more information about information sources used for alias assignment, see KB 2091560.

The Device Alias Configuration host profile also flags errors, for example, when a device is present in the host profile but not present on the host.

A heterogeneous cluster does not have the same default alias configuration across its hosts. Due to the differences between the devices, a host profile cannot be applied cleanly.

Using ESX Shell Commands to View Device Alias Information

On a running ESX system, you can view information about I/O adapter aliases by running commands in an ESX Shell.

Using ESX Shell Commands to View Device Alias Information

Command	Description
`device alias list`	Lists all current I/O adapter aliases.
`device alias get -n<alias>`	Displays the physical device that an I/O adapter alias maps to.
`network nic list`	Lists aliases and general information about network devices.
`storage core adapter list`	Lists all storage I/O adapters.

> **Note:**
> ESXCLI commands are supported commands. Using alternative sources for displaying alias information is not recommended.

Change Device Aliases on an ESX Host by Using Host Profiles

You can modify a device alias on ESX hosts by editing the device alias section of the host profile attached to the hosts.

A device alias change can occur, for example, when you apply a BIOS or device firmware update. For more information about changes in the device alias configuration, see Device Alias Configuration.

I/O adapters that are based on PCI hardware usually have a logical and a PCI alias entry. Both aliases must have the same value. Some I/O adapters usually have a logical alias entry only. PCI hardware devices without I/O adapters usually have a PCI alias entry only. Modify a PCI alias entry only when you need the alias for another device.

> **Note:**
> Two different I/O adapters must not have the same alias, except I/O adapters that are based on PCI hardware and have a logical and a PCI alias entry.

Procedure

For information about exporting, extracting, and editing a host profile, see the *vSphere Host Profiles* documentation.

1. Export the current host profile attached to the ESX host.
2. Extract a new host profile from the ESX host, but do not apply the host profile.
3. Remove all entries from the device alias section of the current host profile.
4. Transfer all entries from the device alias section of the new host profile to the same section of the current host profile. You can modify the aliases in the entries before adding them to the current host profile.
5. Reapply the current host profile to the ESX host.

Change Device Aliases on a Stateful ESX Host by Using ESXCLI Commands

You can modify a device alias on a stateful ESX host that does not have an attached host profile by running ESXCLI commands on the host.

A device alias change can occur, for example, when the slot position of a device changes. For more information about changes in the device alias configuration, see Device Alias Configuration.

Note:
The following ESXCLI commands might not be compatible with ESX version 6.7 and earlier.

1. To list the current assignment of aliases to device addresses, in the ESX Shell, run `localcli --plugin-dir /usr/lib/vmware/esxcli/int/ deviceInternal alias list`.

   For I/O adapters that are based on PCI hardware, you usually see a logical and a PCI alias entry.

   For some I/O adapters, you usually see a logical alias entry only.

   For PCI hardware devices without I/O adapters, you usually see a PCI alias entry only.

2. To change an alias, replace the `ALIAS`, `PCI_ADDRESS`, and `LOGICAL_ADDRESS` placeholder values with actual values, and follow one of the substeps.
   a) If an I/O adapter has a logical and a PCI alias, run `localcli --plugin-dir /usr/lib/vmware/esxcli/int/ deviceInternal alias store --bus-type logical --alias ALIAS --bus-address LOGICAL_ADDRESS` and `localcli --plugin-dir /usr/lib/vmware/esxcli/int/ deviceInternal alias store --bus-type pci --alias ALIAS --bus-address PCI_ADDRESS`.

      The logical and PCI aliases must have the same value.
   b) If an I/O adapter has a logical alias entry only, run `localcli --plugin-dir /usr/lib/vmware/esxcli/int/ deviceInternal alias store --bus-type logical --alias ALIAS --bus-address LOGICAL_ADDRESS`.
   c) If a PCI hardware device has a PCI alias entry only, run `localcli --plugin-dir /usr/lib/vmware/esxcli/int/ deviceInternal alias store --bus-type pci --alias ALIAS --bus-address PCI_ADDRESS`.

      Modify a PCI alias only when you need the alias for another device.

   Note:
   Two different I/O adapters must not have the same alias, except I/O adapters that are based on PCI hardware and have a logical and a PCI alias entry.

3. Reboot the system.

vSphere PowerCLI Scenario for vSphere Auto Deploy

Scenario to set up and configure a working vSphere Auto Deploy environment by using PowerCLI.

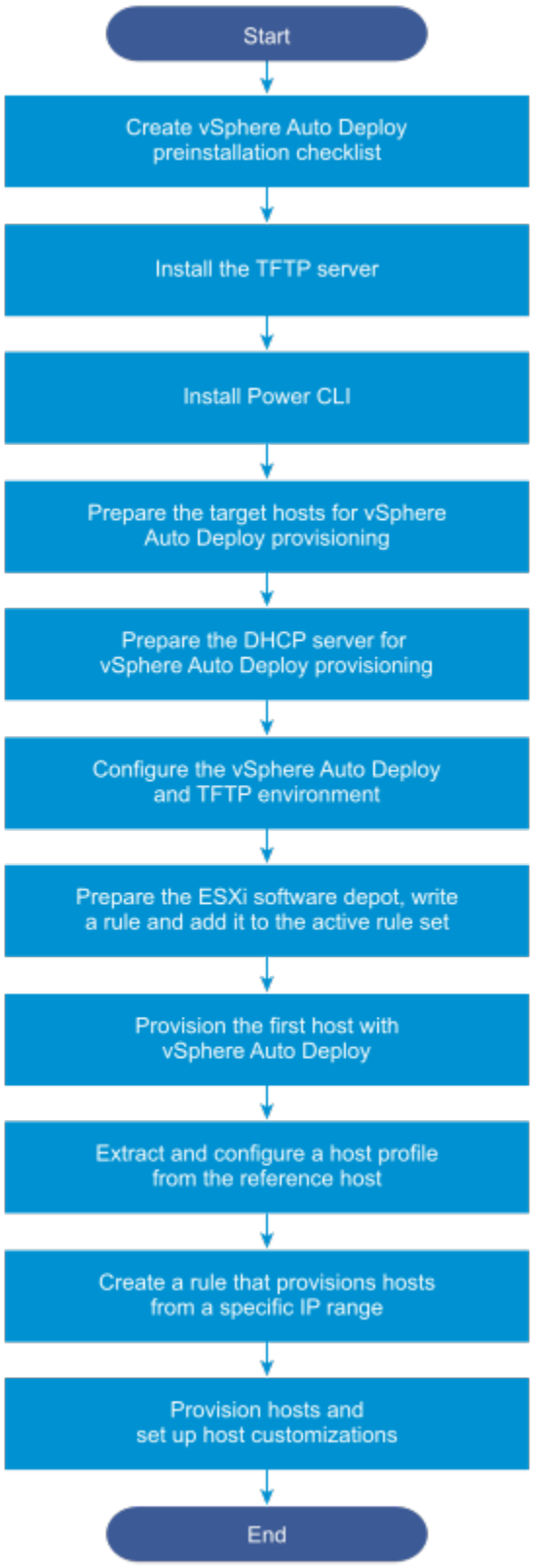
In this scenario, you are going to set up and configure a working vSphere Auto Deploy environment that includes four hosts. You will create rules and provision two of the hosts with an image profile and the other two with the same image profile and a host profile that is set up to prompt for user input. This scenario can provide you with the basis for a production environment. The task descriptions assume that you are using a flat network with no VLAN tagging between the physical hosts and the rest of your environment.

To perform the tasks in this scenario, you should have the following background knowledge and privileges.

- Experience with vSphere (vCenter and ESX).
- Basic knowledge of Microsoft PowerShell and VCF PowerCLI.
- Administrator rights to a Windows system and a vCenter system.

Follow the tasks in the order presented in this scenario. Some steps can be performed in a different order, but the order used here limits repeated manipulation of some components. For details on the preinstallation checklist and other prerequisites to configure vSphere Auto Deploy, see Install and Configure vSphere Auto Deploy.

Figure 11: vSphere Auto Deploy Setup and Hosts Provisioning Workflow



vSphere Auto Deploy takes advantage of the iPXE infrastructure and PXE booting with legacy BIOS firmware is possible only over IPv4. If the hosts that you want to provision with vSphere Auto Deploy are with legacy BIOS, the vSphere Auto Deploy server must have an IPv4 address. PXE booting with UEFI firmware is possible with either IPv4 or IPv6.

Install the TFTP Server

To set up a vSphere Auto Deploy infrastructure, you must install a TFTP server in your environment. vSphere Auto Deploy relies on a TFTP server for sending a boot image to the hosts that it provisions.

This task only installs the TFTP server. You later download a configuration file to the server. See Configure the vSphere Auto Deploy and TFTP Environment in the vSphere Client.

1. Download your preferred TFTP server to a location that has network access to your vCenter and install the server.
2. Configure the TFTP root directory, for example `D:\TFTP_Root\`.

Install VCF PowerCLI, to manage vSphere Auto Deploy with VCF PowerCLI cmdlets.

Install VCF PowerCLI

Before you can manage vSphere Auto Deploy with rules that you create with VCF PowerCLI cmdlets, you must install VCF PowerCLI.

Use the VMware PowerCLI User's Guide to learn about PowerShell basics, VCF PowerCLI concepts, and how to install and configure VCF PowerCLI.

Configure the settings of your target hosts to prepare them for provisioning with vSphere Auto Deploy.

Prepare the vSphere Auto Deploy Target Hosts

You must configure the BIOS settings of the four hosts and reconfirm the MAC address of the primary network device to prepare the target hosts for provisioning with vSphere Auto Deploy.

Hosts that you want to provision with vSphere Auto Deploy must meet the requirements for ESX.

See ESX Hardware Requirements.

1. Change the BIOS settings of each of the four physical hosts to force the hosts to boot from the primary network device.
2. Reconfirm the MAC address of the primary network device.

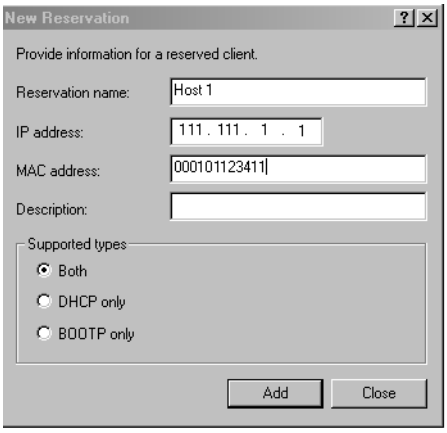
Set up the DHCP server to serve each target host with an iPXE binary.

Prepare the DHCP Server for vSphere Auto Deploy Provisioning

When you prepare the vSphere Auto Deploy target hosts, you must set up the DHCP server in this scenario to serve each target host with an iPXE binary.

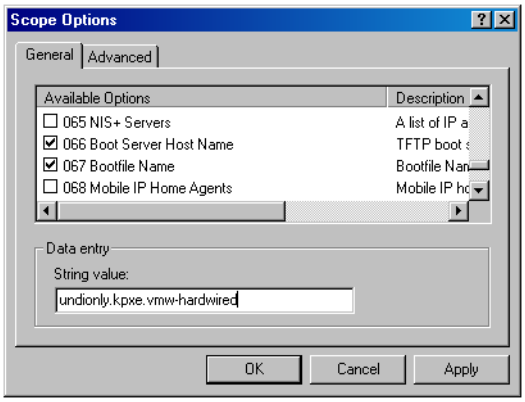
The environment in this scenario uses Active Directory with DNS and DHCP. The DHCP server is included in the vSphere supported Windows Server versions.

1. Log in to your DHCP Server with administrator privileges.
2. Create a DHCP scope for your IP address range.
   a) Click **Start > Settings > Control Panel > Administrative Tools** and click **DHCP**.
   b) Navigate to **DHCP** > ***hostname*** > **IPv4**.
   c) Right-click **IPv4** and select **New Scope**.
   d) On the Welcome screen, click **Next**, and specify a name and description for the scope.
   e) Specify an IP address range and click **Next**.
   f) Click **Next** until you reach the Configure DHCP Options screen and select **No, I will configure this option later**.
3. Create a DHCP reservation for each target ESX host.
   a) In the DHCP window, navigate to **DHCP** > ***hostname*** > **IPv4** > **Autodeploy Scope** > **Reservations**.
   b) Right-click **Reservations** and select **New Reservation**.
   c) In the New Reservation window, specify a name, IP address, and the MAC address for one of the hosts. Do not include the colon (:) in the MAC address.



d) Repeat the process for each of the other hosts.

4. Set up the DHCP Server to point the hosts to the TFTP Server.
   a) In the DHCP window, navigate to **DHCP** > ***hostname*** > **IPv4** > **Autodeploy Scope** > **Scope Options**.
   b) Right click **Scope Options** and choose **Configure Options**.
   c) In the Scope Options window, click the **General** tab.
   d) Click **066 Boot Server Host Name** and enter the address of the TFTP server that you installed in the String value field below the Available Options.



   e) Click **067 Bootfile Name** and enter `undionly.kpxe.vmw-hardwired`.

      The `undionly.kpxe.vmw-hardwired` iPXE binary will be used to boot the ESX hosts.
   f) Click **Apply** and click **OK** to close the window.

5. In the DHCP window, right-click **DHCP** > ***hostname*** > **IPv4** > **Scope** > **Activate** and click **Activate**.
6. Do not log out from the DHCP Server if you are using Active Directory for DHCP and DNS, or log out otherwise.

Start the vCenter service of vSphere Auto Deploy and configure the TFTP server.

Configure the vSphere Auto Deploy and TFTP Environment in the vSphere Client

After you prepare the DHCP server, you must start the vSphere Auto Deploy vCenter service and configure the TFTP server. You must download a TFTP ZIP file from your vSphere Auto Deploy server. The customized FTP server serves the boot images that vSphere Auto Deploy provides.

1. Use the vSphere Client to connect to the vCenter system that manages the vSphere Auto Deploy server.
2. Start the vSphere Auto Deploy service.
   a) Navigate to **Home** > **Auto Deploy**.
   b) On the **Auto Deploy** page, select your vCenter from the drop-down menu at the top.
   c) Click **Enable Auto Deploy and Image Builder** to activate the service.

      If the **Image Builder** service is already active, select the **Configure** tab and click **Enable Auto Deploy Service**.

3. In the Auto Deploy inventory, click the **Configure** tab.
4. Click the **Download TFTP Zip File**.
5. Save the file `deploy-tftp.zip` to the `TFTP_Root` directory that you created when you installed the TFTP Server, and unzip the file.

Add a software depot to your inventory and use an image profile from the depot to create a rule for host provisioning.

Prepare the ESX Software Depot and Write a Rule

After you configure the vSphere Auto Deploy infrastructure, you must add an ESX software depot, specify an image profile, write a rule, and add it to the active rule set.

Verify that you can access the ESX hosts that you want to provision from the system on which you run VCF PowerCLI.

vSphere Auto Deploy provisions hosts with image profiles that define the set of VIBs that an ESX installation process uses. Image profiles are stored in software depots. You must make sure that the correct image profile is available before you start provisioning hosts. When you add a software depot to a VCF PowerCLI session, it is available only during the current session. It does not persist across sessions.

The steps in this task instruct you to run VCF PowerCLI cmdlets. For additional information about the vSphere Auto Deploy cmdlets that you can run in a VCF PowerCLI session, see Inventory of vSphere Auto Deploy VCF PowerCLI Cmdlets.

1. Log in as an administrator to the console of the Windows system, either directly or by using RDP.

   This task assumes that you installed VCF PowerCLI on the system on which the vCenter system is running.
2. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate issues occur. In a development environment, you can ignore the warning.
3. Enter the vCenter credentials.
4. Run `Add-EsxSoftwareDepot` to add the online depot to the VCF PowerCLI session.

   ```
   Add-EsxSoftwareDepot https://dl.broadcom.com/<Download Token>/PROD/COMP/ESX_HOST/main/vmw-depot-index.xml
   ```

   Adding the software depot is required each time you start a new VCF PowerCLI session.

   Note:

   The URLs of the official VMware online depot are now unique for each customer. To update the download sources URLs, see the following KB articles: 390121 and 390123 .
5. Validate that you successfully added the software depot by checking the contents of the depot with the `Get-EsxImageProfile` cmdlet.

   The cmdlet returns information about all image profiles in the depot.
6. Create a new rule by running the `New-DeployRule` cmdlet.

   ```
   New-DeployRule -Name "InitialBootRule" -Item ESX-9.0.0-xxxxxxx-standard -AllHosts
   ```

   The cmdlet creates a rule that assigns the specified image profile to all hosts in the inventory.
7. Add the new rule to the active rule set to make the rule available to the vSphere Auto Deploy server.

   ```
   Add-DeployRule -DeployRule "InitialBootRule"
   ```

Provision your first host with vSphere Auto Deploy and verify its image provisioning.

Provision the First Host with vSphere Auto Deploy

After creating a rule and adding it to the active rule set, you can provision the first host and check its vCenter location to complete verification of the image provisioning of your setup.

1. Open a console session to the physical host that you want to use as the first ESX target host, boot the host, and look for messages that indicate a successful iPXE boot.

   During the boot process, DHCP assigns an IP address to the host. The IP address matches the name you specified earlier in the DNS server. The host contacts the vSphere Auto Deploy server and downloads the ESX binaries from the HTTP URL indicated in the iPXE tramp file that you downloaded earlier to the TFTP_Root directory. Each instance of vSphere Auto Deploy produces a custom set of files for the TFTP Server.
2. Use the vSphere Client to connect to the vCenter system that manages the vSphere Auto Deploy server.
3. On the vSphere Client Home page, click **Hosts and Clusters**.
4. Verify that the newly provisioned host is now in the vCenter inventory at the datacenter level.

   By default, vSphere Auto Deploy adds hosts at the datacenter level when the boot process completes.

Extract a host profile from the host and configure it to require user input.

Extract and Configure a Host Profile from the Reference Host

After provisioning the first host, you can extract and configure a host profile that can be used to apply the same configuration to other target hosts. Configuration that differs for different hosts, such as a static IP address, can be managed through the host customization mechanism.

vSphere Auto Deploy can provision each host with the same host profile. vSphere Auto Deploy can also use host customization that allows you to specify different information for different hosts. For example, if you set up a VMkernel port for vSphere vMotion or for storage, you can specify a static IP address for the port by using the host customization mechanism.

1. Use the vSphere Client to connect to the vCenter system that manages the vSphere Auto Deploy server.
2. Click **Policies and Profiles** and select **Host Profiles**.
3. Click **Extract Host Profile**.
4. On the **Select host** page of the wizard, select the reference host that you configured earlier and click **Next**.
5. On the **Name and Description** page of the wizard, enter a name and description for the new profile and click **Finish**.
6. Select the host profile that you want to edit and click the **Configure** tab.
7. Click **Edit Host Profile**.
8. Select **Security and Services** > **Security Settings** > **Security** > **User Configuration** > **root**.
9. From the **Password** drop-down menu, select **User Input Password Configuration**.
10. Click **Save** to configure the host profile settings.

Create a vSphere Auto Deploy rule to apply the host profile to other ESX hosts.

Create a Rule that Provisions Hosts from a Specific IP Range

After creating a host profile from a reference host, you can create a rule that applies the previously verified image profile and the host profile that you extracted to target hosts from a specific IP range.

1. Log in with administrator privileges to the console of the Windows system, either directly or by using RDP.
2. In a VCF PowerCLI session, run the `Connect-VIServer` cmdlet to connect to the vCenter system that vSphere Auto Deploy is registered with.

   ```
   Connect-VIServer ipv4_address
   ```

   The cmdlet might return a server certificate warning. In a production environment, make sure no server certificate issues occur. In a development environment, you can ignore the warning.
3. Run `Add-EsxSoftwareDepot` to add the online depot to the VCF PowerCLI session.

   ```
   Add-EsxSoftwareDepot https://dl.broadcom.com/<Download Token>/PROD/COMP/ESX_HOST/main/vmw-depot-index.xml
   ```

   Adding the software depot is required each time you start a new VCF PowerCLI session.

Note:

The URLs of the official VMware online depot are now unique for each customer. To update the download sources URLs, see the following KB articles: 390121 and 390123 .

4. Required: Display the rules in the active rule set by running the `Get-DeployRuleset` cmdlet.
5. Create a rule that instructs vSphere Auto Deploy to provision the set of hosts from a specified IP range with the image profile that you previously selected and the host profile that you created from the reference host.

   ```
   New-DeployRule -name "Production01Rule" -item "image_profile",ESXGold -Pattern "ipv4=IP_range"
   ```

6. Add the new rule to the active rule set.

   ```
   Add-DeployRule -DeployRule "Production01Rule"
   ```

7. Check the active rule set by running the `Get-DeployRuleset` command.

   VCF PowerCLI displays information similar to the following example.

   ```
   Name:              Production01Rule
   PatternList:       {ipv4=address_range}
   ItemList:          {ESX-version-XXXXXX-standard, Compute01, ESXGold}
   ```

Provision the hosts and set up the host customizations.

Provision Hosts and Set Up Host Customizations

With the rule in place that provisions hosts using an image profile and a host profile, you can provision specific target hosts. If any host profile items are set to prompt the user for input, the host comes up in maintenance mode. You apply the host profile or check host compliance to be prompted for the information. The system associates the host customization with the host.

1. Boot the remaining hosts you want to provision.

   vSphere Auto Deploy boots the hosts, applies the host profile, and adds the hosts to the vCenter inventory. The hosts remain in maintenance mode because the host profile from the reference host is set up to require user input for each host.

2. Use the vSphere Client to connect to the vCenter system that manages the vSphere Auto Deploy server.
3. Click **Policies and Profiles** and select **Host Profiles**.
4. Right-click the newly created host profile for Auto Deploy and click **Edit Host Customizations**.
5. Select the hosts, enter the required host customizations and click **Finish**.

   Alternatively, you can also **Import Host Customizations** file.

6. Apply the host profile to each of the hosts and get the hosts out of maintenance mode. Alternatively, you can reboot each host.

   When the reboot progress completes, all hosts are running with the image you specify and use the configuration in the reference host profile. The cluster shows that all hosts are fully compliant.

All hosts are now configured with the shared information through the reference host profile and with the host-specific information through the host customization mechanism. The next time you boot the hosts, they receive the complete Host Profile information, including the host-specific information, and boot up completely configured and out of Maintenance Mode.

Troubleshooting vSphere Auto Deploy

The vSphere Auto Deploy troubleshooting topics cover some situations when provisioning hosts with vSphere Auto Deploy does not work as expected.

- vSphere Auto Deploy TFTP Timeout Error at Boot Time
- vSphere Auto Deploy Host Boots with Wrong Configuration
- Host Is Not Redirected to vSphere Auto Deploy Server

- Package Warning Message When You Assign an Image Profile to a vSphere Auto Deploy Host
- vSphere Auto Deploy Host with a Built-In USB Flash Drive Does Not Send Coredumps to Local Disk
- vSphere Auto Deploy Host Reboots After Five Minutes
- vSphere Auto Deploy Host Cannot Contact TFTP Server
- vSphere Auto Deploy Host Cannot Retrieve ESX Image from vSphere Auto Deploy Server
- vSphere Auto Deploy Host Does Not Get a DHCP Assigned Address
- vSphere Auto Deploy Host Does Not Network Boot
- Recovering from Database Corruption on the vSphere Auto Deploy Server

vSphere Auto Deploy Rule Takes Long to Complete

After you run an Auto Deploy rule, it takes long to complete, and you see no progress for the tasks.

Auto Deploy automatically detects if a new version of the image that a cluster uses is available, or when the cache lifetime of files expires, and refreshes the cache by re-downloading the files from the latest version of the software depot. As a result, you might see a delay in the implementation of Auto Deploy rules.

When the cache lifetime of files cached by Auto Deploy expires, Auto Deploy automatically refreshes the cache from the software depot. Since a default software depot is usually more than 300 MB, depending on the network, the download might take long.

You can also see a delay in deploying Auto Deploy rules when converting a cluster that you manage with a single image to one that you manage with vSphere Configuration Profiles. If a host attempts to boot during the time Auto Deploy caches the configuration file, you might see a delay, because of the time Auto Deploy needs to create the cache.

A general vSphere infrastructure problem might also prevent Auto Deploy rules from running in a timely manner.

Regardless of any connectivity issues or breakage, Auto Deploy keeps active sessions and persists in booting hosts until it succeeds.

vSphere Auto Deploy TFTP Timeout Error at Boot Time

A TFTP Timeout error message appears when a host provisioned with vSphere Auto Deploy boots. The text of the message depends on the BIOS.

A TFTP Timeout error message appears when a host provisioned with vSphere Auto Deploy boots. The text of the message depends on the BIOS.

The TFTP server is down or unreachable.

- Ensure that your TFTP service is running and reachable by the host that you are trying to boot.
- To view the diagnostic logs for details on the present error, see your TFTP service documentation.

vSphere Auto Deploy Host Boots with Wrong Configuration

A host is booting with a different ESX image, host profile, or folder location than the one specified in the rules.

A host is booting with a different ESX image profile or configuration than the image profile or configuration that the rules specify. For example, you change the rules to assign a different image profile, but the host still uses the old image profile.

After the host has been added to a vCenter system, the boot configuration is determined by the vCenter system. The vCenter system associates an image profile, host profile, or folder location with the host.

Use the `Test-DeployRuleSetCompliance` and `Repair-DeployRuleSetCompliance` vSphere PowerCLI cmdlets to reevalute the rules and to associate the correct image profile, host profile, or folder location with the host.

Host Is Not Redirected to vSphere Auto Deploy Server

During boot, a host that you want to provision with vSphere Auto Deploy loads iPXE. The host is not redirected to the vSphere Auto Deploy server.

During boot, a host that you want to provision with vSphere Auto Deploy loads iPXE. The host is not redirected to the vSphere Auto Deploy server.

The `tramp` file that is included in the TFTP ZIP file has the wrong IP address for the vSphere Auto Deploy server.

Correct the IP address of the vSphere Auto Deploy server in the `tramp` file, as explained in the *vSphere Installation and Setup* documentation.

Package Warning Message When You Assign an Image Profile to a vSphere Auto Deploy Host

When you run a vSphere PowerCLI cmdlet that assigns an image profile that is not vSphere Auto Deploy ready, a warning message appears.

When you write or modify rules to assign an image profile to one or more hosts, the following error results:

```
Warning: Image Profile <name-here> contains one or more software packages that are not stateless-ready. You may experience problems when using this profile with Auto Deploy.
```

Each VIB in an image profile has a `stateless-ready` flag that indicates that the VIB is meant for use with vSphere Auto Deploy. You get the error if you attempt to write a vSphere Auto Deploy rule that uses an image profile in which one or more VIBs have that flag set to FALSE.

> **Note:**
> You can use hosts provisioned with vSphere Auto Deploy that include VIBs that are not stateless ready without problems. However booting with an image profile that includes VIBs that are not stateless ready is treated like a fresh install. Each time you boot the host, you lose any configuration data that would otherwise be available across reboots for hosts provisioned with vSphere Auto Deploy.

1. Use vSphere ESX Image Builder cmdlets in a vSphere PowerCLI session to view the VIBs in the image profile.
2. Remove any VIBs that are not stateless-ready.
3. Rerun the vSphere Auto Deploy cmdlet.

vSphere Auto Deploy Host with a Built-In USB Flash Drive Does Not Send Coredumps to Local Disk

If your vSphere Auto Deploy host has a built-in USB flash drive, and an error results in a coredump, the coredump is lost. Set up your system to use ESX Dump Collector to store coredumps on a networked host.

If your vSphere Auto Deploy host has a built-in USB Flash, and if it encounters an error that results in a coredump, the coredump is not sent to the local disk.

1. Install ESX Dump Collector on a system of your choice.

   ESX Dump Collector is included with the vCenter installer.
2. Use ESXCLI to configure the host to use ESX Dump Collector.

   ```
   esxcli conn_options system coredump network set IP-addr,port
   esxcli system coredump network set -e true
   ```
3. Use ESXCLI to deactivate local coredump partitions.

   ```
   esxcli conn_options system coredump partition set -e false
   ```

vSphere Auto Deploy Host Reboots After Five Minutes

A vSphere Auto Deploy host boots and displays iPXE information, but reboots after five minutes.

A host to be provisioned with vSphere Auto Deploy boots from iPXE and displays iPXE information on the console. However, after five minutes, the host displays the following message to the console and reboots.

```
This host is attempting to network-boot using VMware
AutoDeploy. However, there is no ESX  image associated with this host.
Details: No rules containing an Image Profile match this
host. You can create a rule with the New-DeployRule PowerCLI cmdlet
```

```
and add it to the rule set with Add-DeployRule or Set-DeployRuleSet.
The rule should have a pattern that matches one or more of the attributes
listed below.
```

The host might also display the following details:

```
Details: This host has been added to VC, but no Image Profile
is associated with it. You can use Apply-ESXImageProfile in the
PowerCLI to associate an Image Profile with this host.
Alternatively, you can reevaluate the rules for this host with the
Test-DeployRuleSetCompliance and Repair-DeployRuleSetCompliance cmdlets.
```

The console then displays the host's machine attributes including vendor, serial number, IP address, and so on.

No image profile is currently associated with this host.

You can assign an image profile to the host by running the `Apply-EsxImageProfile` cmdlet, or by creating the following rule:

1. Run the `New-DeployRule` cmdlet to create a rule that includes a pattern that matches the host with an image profile.
2. Run the `Add-DeployRule` cmdlet to add the rule to a ruleset.
3. Run the `Test-DeployRuleSetCompliance` cmdlet and use the output of that cmdlet as the input to the `Repair-DeployRuleSetCompliance` cmdlet.

vSphere Auto Deploy Host Cannot Contact TFTP Server

The host that you provision with vSphere Auto Deploy cannot contact the TFTP server.

When you attempt to boot a host provisioned with vSphere Auto Deploy, the host performs a network boot and is assigned a DHCP address by the DHCP server, but the host cannot contact the TFTP server.

The TFTP server might have stopped running, or a firewall might block the TFTP port.

- If you installed the WinAgents TFTP server, open the WinAgents TFTP management console and verify that the service is running. If the service is running, check the Windows firewall's inbound rules to make sure the TFTP port is not blocked. Turn off the firewall temporarily to see whether the firewall is the problem.
- For all other TFTP servers, see the server documentation for debugging procedures.

vSphere Auto Deploy Host Cannot Retrieve ESX Image from vSphere Auto Deploy Server

The host that you provision with vSphere Auto Deploy stops at the iPXE boot screen.

When you attempt to boot a host provisioned with vSphere Auto Deploy, the boot process stops at the iPXE boot screen and the status message indicates that the host is attempting to get the ESX image from the vSphere Auto Deploy server.

The vSphere Auto Deploy service might be stopped or the vSphere Auto Deploy server might be inaccessible.

1. Log in to the system on which you installed the vSphere Auto Deploy server.
2. Check that the vSphere Auto Deploy server is running.
   a) Click **Start** > **Settings** > **Control Panel** > **Administrative Tools**.
   b) Double-click **Services** to open the Services Management panel.
   c) In the Services field, look for the VMware vSphere Auto Deploy Waiter service and restart the service if it is not running.
3. Open a Web browser, enter the following URL, and check whether the vSphere Auto Deploy server is accessible.
   https://*Auto_Deploy_Server_IP_Address*:*Auto_Deploy_Server_Port*/vmw/rdb

   Note:
   Use this address only to check whether the server is accessible.

4. If the server is not accessible, a firewall problem is likely.
   a) Try setting up permissive TCP Inbound rules for the vSphere Auto Deploy server port.

      The port is 6501 unless you specified a different port during installation.
   b) As a last resort, deactivate the firewall temporarily and enable it again after you verified whether it blocked the traffic. Do not deactivate the firewall on production environments.

      To deactivate the firewall, run `netsh firewall set opmode disable`. To enable the firewall, run `netsh firewall set opmode enable`.

vSphere Auto Deploy Host Does Not Get a DHCP Assigned Address

The host you provision with vSphere Auto Deploy fails to get a DHCP Address.

When you attempt to boot a host provisioned with vSphere Auto Deploy, the host performs a network boot but is not assigned a DHCP address. The vSphere Auto Deploy server cannot provision the host with the image profile.

You might have a problem with the DHCP service or with the firewall setup.

1. Check that the DHCP server service is running on the Windows system on which the DHCP server is set up to provision hosts.
   a) Click **Start** > **Settings** > **Control Panel** > **Administrative Tools**.
   b) Double-click **Services** to open the Services Management panel.
   c) In the Services field, look for the DHCP server service and restart the service if it is not running.
2. If the DHCP server is running, recheck the DHCP scope and the DHCP reservations that you configured for your target hosts.

   If the DHCP scope and reservations are configured correctly, the problem most likely involves the firewall.
3. As a temporary workaround, turn off the firewall to see whether that resolves the problem.
   a) Open the command prompt by clicking **Start** > **Program** > **Accessories** > **Command prompt**.
   b) Type the following command to temporarily turn off the firewall. Do not turn off the firewall in a production environment.

      ```
      netsh firewall set opmode disable
      ```
   c) Attempt to provision the host with vSphere Auto Deploy.
   d) Type the following command to turn the firewall back on.

      ```
      netsh firewall set opmode enable
      ```
4. Set up rules to allow DHCP network traffic to the target hosts.

   See the firewall documentation for DHCP and for the Windows system on which the DHCP server is running for details.

vSphere Auto Deploy Host Does Not Network Boot

The host you provision with vSphere Auto Deploy comes up but does not network boot.

When you attempt to boot a host provisioned with vSphere Auto Deploy, the host does not start the network boot process.

You did not enable your host for network boot.

1. Reboot the host and follow the on-screen instructions to access the BIOS configuration.
2. In the BIOS configuration, enable Network Boot in the Boot Device configuration.

Recovering from Database Corruption on the vSphere Auto Deploy Server

In some situations, you might have a problem with the vSphere Auto Deploy database. The most efficient recovery option is to replace the existing database file with the most recent backup.

When you use vSphere Auto Deploy to provision the ESX hosts in your environment, you might encounter a problem with the vSphere Auto Deploy database.

Important:
This is a rare problem. Follow all other vSphere Auto Deploy troubleshooting strategies before you replace the current database file. Rules or associations that you created since the backup you choose are lost.

This problem happens only with hosts that are provisioned with vSphere Auto Deploy.

1. Stop the vSphere Auto Deploy server service.
2. Find the vSphere Auto Deploy log by going to the vSphere Auto Deploy page in the vSphere Client.
3. Check the logs for the following message:
   `DatabaseError: database disk image is malformed.`
   If you see the message, replace the existing database with the most recent backup.
4. Go to the vSphere Auto Deploy data directory at `/var/lib/rbd`.
   The directory contains a file named `db`, and backup files named `db-yyy-mm-dd`.
5. Rename the current `db` file.
   VMware Support might ask for that file if you call for assistance.
6. Rename the most recent backup to `db`.
7. Restart the vSphere Auto Deploy server service.
8. If the message still appears in the log, repeat the steps to use the next recent backup until vSphere Auto Deploy works without database errors.

Setting Up ESX

These topics provide information about using the direct console user interface and configuring defaults for ESX.

- ESX Autoconfiguration
- About the Direct Console ESX Interface
- Enable ESX Shell and SSH Access with the Direct Console User Interface
- Set the Password for the Administrator Account
- Configuring the BIOS Boot Settings
- Configuring Network Settings
- Configure System Logging

Initial ESX Configuration

- ESX Autoconfiguration
- Managing ESX Remotely
- About the Direct Console ESX Interface
- Configure the Keyboard Layout for the Direct Console
- Create a Security Banner for the Direct Console
- Redirecting the Direct Console to a Serial Port
- Redirect the Direct Console to a Serial Port by Setting the Boot Options Manually
- Redirect the Direct Console to a Serial Port from the vSphere Client
- Redirect the Direct Console to a Serial Port in a Host Deployed with Auto Deploy

ESX Autoconfiguration

When you turn on the ESX host for the first time or after resetting the configuration defaults, the host enters an autoconfiguration phase. This phase configures system network and storage devices with default settings.

By default, Dynamic Host Configuration Protocol (DHCP) configures IP, and all visible blank internal disks are formatted with the virtual machine file system (VMFS) so that virtual machines can be stored on the disks.

Managing ESX Remotely

You can use the VMware Host Client, the vSphere Client, and vCenter to manage your ESX hosts.

For instructions about downloading and installing vCenter and the vCenter components, see *vCenter Deployment and Setup*. For information about installing the VMware Host Client, see *vSphere Single Host Management*.

About the Direct Console ESX Interface

Use the direct console interface for initial ESX configuration and troubleshooting.

Connect a keyboard and monitor to the host to use the direct console. After the host completes the autoconfiguration phase, the direct console appears on the monitor. You can examine the default network configuration and change any settings that are not compatible with your network environment.

Key operations available to you in the direct console include:

- Configuring hosts
- Setting up administrative access
- Troubleshooting

You can also use vSphere Client to manage the host by using vCenter.

Table 25: Navigating in the Direct Console

Action	Key
View and change the configuration	F2
Change the user interface to high-contrast mode	F4
Shut down or restart the host	F12
View the VMkernel log	Alt+F12
Switch to the shell console	Alt+F1
Switch to the direct console user interface	Alt+F2
Move the selection between fields	Arrow keys
Select a menu item	Enter
Toggle a value	Spacebar
Confirm sensitive commands, such as resetting configuration defaults	F11
Save and exit	Enter
Exit without saving	Esc
Exit system logs	q

Configure the Keyboard Layout for the Direct Console

You can configure the layout for the keyboard that you use with the direct console.

1. From the direct console, select **Configure Keyboard** and press Enter.
2. Select the layout to use.
3. Press the spacebar to toggle selections on and off.
4. Press Enter.

Create a Security Banner for the Direct Console

A security banner is a message that is displayed on the direct console **Welcome** screen.

1. From the vSphere Client, connect to the vCenter instance.
2. Select the host in the inventory.
3. Click the **Configure** tab.
4. Under System, select **Advanced System Settings**.
5. Select `Annotations.WelcomeMessage`.
6. Click the **Edit** icon.
7. Enter a security message.

The message is displayed on the direct console **Welcome** screen.

Redirecting the Direct Console to a Serial Port

To manage your ESX host remotely from a serial console, you can redirect the direct console to a serial port.

ESX supports the VT100 terminal type and the PuTTy terminal emulator to view the direct console over the serial port.

You can redirect the direct console to a serial port in several ways.

Redirect the Direct Console to a Serial Port by Setting the Boot Options Manually

When you redirect the direct console to a serial port by setting the boot options, the change does not persist for subsequent boots.

Verify that the serial port is not in use for serial logging and debugging.

1. Start the host.
2. When the Loading VMware Hypervisor window appears, press Shift+O to edit boot options.
3. Deactivate the logPort and gdbPort on com1 and set tty2Port to com1 by entering the following boot options:

   ```
   "gdbPort=none logPort=none tty2Port=com1";
   ```

   To use com2 instead, replace `com1` with `com2`.

The direct console is redirected to the serial port until you reboot the host. To redirect the direct console for subsequent boots, see Redirect the Direct Console to a Serial Port from the vSphere Client

Redirect the Direct Console to a Serial Port from the vSphere Client

You can manage the ESX host remotely from a console that is connected to the serial port by redirecting the direct console to either of the serial ports com1 or com2. When you use the vSphere Client to redirect the direct console to a serial port, the boot option that you set persists after subsequent reboots.

- Verify that you can access the host from the vSphere Client.
- Verify that the serial port is not in use for serial logging and debugging, or for ESX Shell (tty1Port).

1. From the vSphere Client, connect to the vCenter instance.
2. Select the host in the inventory.
3. Click the **Configure** tab.
4. Under System, select **Advanced System Settings**.
5. Make sure that the **VMkernel.Boot.logPort** and **VMkernel.Boot.gdbPort** fields are not set to use the com port that you want to redirect the direct console to.
6. Set **VMkernel.Boot.tty2Port** to the serial port to redirect the direct console to: `com1` or `com2`.
7. Reboot the host.

You can now manage the ESX host remotely from a console that is connected to the serial port.

Redirect the Direct Console to a Serial Port in a Host Deployed with Auto Deploy

After you redirect the direct console to a serial port, you can make that setting part of the host profile that persists when you reprovision the host with Auto Deploy.

The serial port must not already be in use for serial logging and debugging.

1. From the vSphere Client, connect to the vCenter instance.
2. Select the host in the inventory.
3. Under System, select **Advanced System Settings**.
4. Make sure that the **VMkernel.Boot.logPort** and **VMkernel.Boot.gdbPort** fields are not set to use the com port that you want to redirect the direct console to.
5. Set **VMkernel.Boot.tty2Port** to the serial port to redirect the direct console to: `com1` or `com2`.
6. Click **OK**.
7. Save the host profile and attach the host to the profile. See the *vSphere Host Profiles* documentation.

The setting to redirect the direct console to a serial port is stored by vCenter and persists when you reprovision the host with Auto Deploy.

Enable ESX Shell and SSH Access with the Direct Console User Interface

Use the direct console user interface to enable the ESX Shell.

1. From the Direct Console User Interface, press F2 to access the System Customization menu.
2. Select **Troubleshooting Options** and press Enter.
3. From the Troubleshooting Mode Options menu, select a service to enable.
   - Enable ESX Shell
   - Enable SSH
4. Press Enter to enable the service.
5. Optional: Set the timeout for the ESX Shell.

   By default, timeouts for the ESX Shell is 0 (not active).

   The availability timeout setting is the number of minutes that can elapse before you must log in after the ESX Shell is enabled. After the timeout period, if you have not logged in, the shell is deactivated.

   Note:
   If you are logged in when the timeout period elapses, your session will persist. However, the ESX Shell is deactivated, preventing other users from logging in.

a) From the Troubleshooting Mode Options menu, select **Modify ESX Shell and SSH timeouts** and press Enter.
b) Enter the availability timeout in minutes.

   The availability timeout is the number of minutes that can elapse before you must log in after the ESX Shell is enabled.
c) Press Enter.
d) Enter the idle timeout.

   The idle timeout is the number of minutes that can elapse before the user is logged out of an idle interactive sessions. Changes to the idle timeout apply the next time a user logs in to the ESX Shell and do not affect existing sessions.

6. Press Esc until you return to the main menu of the Direct Console User Interface.

Set the Password for the Administrator Account

You can use the direct console to set the password for the administrator account (root).

The administrative user name for the ESX host is root. By default, the administrative password is not set.

1. From the direct console, select **Configure Password**.
2. Optional: If a password is already set up, type the password in the **Old Password** line and press Enter.
3. In the **New Password** line, type a new password and press Enter.
4. Retype the new password and press Enter.

Configuring the BIOS Boot Settings

If your server has multiple drives, you might need to configure the BIOS settings.

The BIOS boot configuration determines how your server boots. Generally, the CD-ROM device is listed first.

Note:

If you are using ESX Embedded, the BIOS boot configuration determines whether your server boots into the ESX boot device or another boot device. Generally, the USB flash device is listed first in the BIOS boot settings on the machine that hosts ESX.

When you install or upgrade ESX in UEFI mode, the installer creates a UEFI boot option named VMware ESX and makes it the default boot option, so you do not need to change the boot order.

You can change the boot setting by configuring the boot order in the BIOS during startup or by selecting a boot device from the boot device selection menu. When you change the boot order in the BIOS, the new setting affects all subsequent reboots. When you select a boot device from the boot device selection menu, the selection affects the current boot only.

Some servers do not have a boot device selection menu, in which case you must change the boot order in the BIOS even for one-time boots, and then change it back again during a subsequent reboot.

Change the BIOS Boot Setting for ESX

Configure the BIOS boot setting for ESX if you want the server to boot into ESX by default.

ESX Installable and ESX Embedded cannot exist on the same host.

1. While the ESX host is powering on, press the key required to enter your host's BIOS setup.

   Depending on your server hardware, the key might be a function key or Delete. The option to enter the BIOS setup might be different for your server.
2. Select the BIOS boot setting.

Option	Description
If you are using the installable version of ESX	Select the disk on which you installed the ESX software and move it to the first position in the list. The host boots into ESX .

Option	Description
If you are using ESX Embedded	Select the USB flash device and move it to the first position in the list. The host starts in ESX mode.

Configure the Boot Setting for Virtual Media

If you are using remote management software to set up ESX, you might need to configure the boot setting for virtual media.

ESX Installable and ESX Embedded cannot exist on the same host.

Virtual media is a method of connecting a remote storage media such as CD-ROM, USB mass storage, ISO image, and floppy disk to a target server that can be anywhere on the network. The target server has access to the remote media, and can read from and write to it as if it were physically connected to the server's USB port.

1. Connect the media to the virtual device.
   For example, if you are using a Dell server, log in to the Dell Remote Access Controller (DRAC) or a similar remote management interface and select a physical floppy or CD-ROM drive, or provide a path to a floppy image or CD-ROM image.
2. Reboot the server.
3. While the server is powering on, enter the device selection menu.
   Depending on your server hardware, the key might be a function key or Delete.
4. Follow the instructions to select the virtual device.

The server boots from the configured device once and goes back to the default boot order for subsequent boots. When you install or upgrade ESX in UEFI mode, you do not need to change the boot order, because the system default boot order is set to VMware ESX .

Configuring Network Settings

ESX requires one IP address for the management network. To configure basic network settings, use the vSphere Client or the direct console.

Use the vSphere Client if you are satisfied with the IP address assigned by the DHCP server.

Use the direct console for network configuration in the following cases:

- You are not satisfied with the IP address assigned by the DHCP server.
- You are not allowed to use the IP address assigned by the DHCP server.
- ESX does not have an IP address. This situation might occur if the autoconfiguration phase did not succeed in configuring DHCP.
- The wrong network adapter was selected during the autoconfiguration phase.

Use ESXCLI commands to configure your network settings. See ESXCLI Command Line Reference.

For security considerations, see Securing ESX Hosts.

Network Access to Your ESX Host

The default behavior is to configure the ESX management network using DHCP. You can override the default behavior and use static IP settings for the management network after the installation is completed.

Table 26: Network Configuration Scenarios Supported by ESX

Scenario	Approach
You want to accept the DHCP-configured IP settings.	In the ESX direct console, you can find the IP address assigned through DHCP to the ESX management interface. You can use that IP address to connect to the host from the vSphere Client and customize settings, including changing the management IP address.
One of the following is true: • You do not have a DHCP server. • The ESX host is not connected to a DHCP server. • Your connected DHCP server is not functioning properly.	During the autoconfiguration phase, the software assigns the link local IP address, which is in the subnet 169.254.x.x/16. The assigned IP address appears on the direct console. You can override the link local IP address by configuring a static IP address using the direct console.
The ESX host is connected to a functioning DHCP server, but you do not want to use the DHCP-configured IP address.	During the autoconfiguration phase, the software assigns a DHCP-configured IP address. You can make the initial connection by using the DHCP-configured IP address. Then you can configure a static IP address. If you have physical access to the ESX host, you can override the DHCP-configured IP address by configuring a static IP address using the direct console.
Your security deployment policies do not permit unconfigured hosts to be powered on the network.	Follow the setup procedure in Configure the Network Settings on a Host That Is Not Attached to the Network.

Choose Network Adapters for the Management Network

Traffic between an ESX host and any external management software is transmitted through an Ethernet network adapter on the host. You can use the direct console to choose the network adapters that are used by the management network.

Examples of external management software include the vCenter and SNMP client. Network adapters on the host are named vmnic*N*, where N is a unique number identifying the network adapter, for example, vmnic0, vmnic1, and so forth.

During the autoconfiguration phase, the ESX host chooses vmnic0 for management traffic. You can override the default choice by manually choosing the network adapter that carries management traffic for the host. In some cases, you might want to use a Gigabit Ethernet network adapter for your management traffic. Another way to help ensure availability is to select multiple network adapters. Using multiple network adapters enables load balancing and failover capabilities.

1. From the direct console, select **Configure Management Network** and press Enter.
2. Select **Network Adapters** and press Enter.
3. Select a network adapter and press Enter.

After the network is functional, you can use the vSphere Client to connect to the ESX host through vCenter.

Set the VLAN ID

You can set the virtual LAN (VLAN) ID number of the ESX host.

1. From the direct console, select **Configure Management Network** and press Enter.
2. Select **VLAN** and press Enter.
3. Enter a VLAN ID number from 1 through 4094.

Configuring IP Settings for ESX

By default, DHCP sets the IP address, subnet mask, and default gateway.

For future reference, write down the IP address.

For DHCP to work, your network environment must have a DHCP server. If DHCP is not available, the host assigns the link local IP address, which is in the subnet 169.254.x.x/16. The assigned IP address appears on the direct console. If

you do not have physical monitor access to the host, you can access the direct console using a remote management application. See Using Remote Management Applications.

When you have access to the direct console, you can optionally configure a static network address. The default subnet mask is 255.255.0.0.

Configure IP Settings from the Direct Console

If you have physical access to the host or remote access to the direct console, you can use the direct console to configure the IP address, subnet mask, and default gateway.

1. Select **Configure Management Network** and press Enter.
2. Select **IP Configuration** and press Enter.
3. Select **Set static IP address and network configuration**.
4. Enter the IP address, subnet mask, and default gateway and press Enter.

Configure IP Settings from the vSphere Client

If you do not have physical access to the host, you can use the vSphere Client to configure static IP settings.

1. Log in to the vCenter from the vSphere Client.
2. Select the host in the inventory.
3. On the **Configure** tab, expand **Networking**.
4. Select **VMkernel adapters**.
5. Select **vmk0 Management Network** and click the edit icon.
6. Select **IPv4 settings**.
7. Select **Use static IPv4 settings**.
8. Enter or change the static IPv4 address settings.
9. Optional: Set static IPv6 addresses.
   a) Select **IPv6 settings**.
   b) Select **Static IPv6 addresses**.
   c) Click the add icon.
   d) Type the IPv6 address and click **OK**.
10. Click **OK**.

Configuring DNS for ESX

You can select either manual or automatic DNS configuration of the ESX host.

The default is automatic. For automatic DNS to work, your network environment must have a DHCP server and a DNS server.

In network environments where automatic DNS is not available or not desirable, you can configure static DNS information, including a host name, a primary name server, a secondary name server, and DNS suffixes.

Configure DNS Settings from the Direct Console

If you have physical access to the host or remote access to the direct console, you can use the direct console to configure DNS information.

1. Select **Configure Management Network** and press Enter.
2. Select **DNS Configuration** and press Enter.
3. Select **Use the following DNS server addresses and hostname**.
4. Enter the primary server, an alternative server (optional), and the host name.

Configure DNS Suffixes

If you have physical access to the host, you can use the direct console to configure DNS information. By default, DHCP acquires the DNS suffixes.

1. From the direct console, select **Configure Management Network**.
2. Select **Custom DNS Suffixes** and press Enter.
3. Enter new DNS suffixes.

Configure the Network Settings on a Host That Is Not Attached to the Network

Some highly secure environments do not permit unconfigured hosts on the network to be powered on. You can configure the host before you attach the host to the network.

Verify that no network cables are connected to the host.

1. Power on the host.
2. Use the direct console user interface to configure the password for the administrator account (root).
3. Use the direct console user interface to configure a static IP address.
4. Connect a network cable to the host.
5. Optional: Use the vSphere Client to connect to a vCenter system.
6. Optional: Add the host to the vCenter inventory.

Test the Management Network

You can use the direct console to do simple network connectivity tests.

The direct console performs the following tests.

- Pings the default gateway
- Pings the primary DNS name server
- Pings the secondary DNS nameserver
- Resolves the configured host name

1. From the direct console, select **Test Management Network** and press Enter.
2. Press Enter to start the test.

Restart the Management Agents

The management agents synchronize VMware components and let you access the ESX host by using the vSphere Client and vCenter.

The vSphere Client and vCenter are installed together. You might need to restart the management agents if remote access is interrupted. Restarting the management agents restarts all management agents and services that are installed and running in `/etc/init.d` on the ESX host. Typically, these agents include hostd, ntpd, sfcbd, slpd, wsman, and vobd. The software also restarts the Fault Domain Manager (FDM) if installed.

Users accessing this host by using the vSphere Client and vCenter lose connectivity when you restart management agents.

1. From the direct console, select **Troubleshooting Options** and press Enter.
2. Select **Restart Management Agents** and press Enter.
3. Press F11 to confirm the restart.

The ESX host restarts the management agents and services.

Restart the Management Network

Restarting the management network interface might be required to restore networking or to renew a DHCP lease.

Restarting the management network will result in a brief network outage that might temporarily affect running virtual machines.
If a renewed DHCP lease results in a new network identity (IP address or host name), remote management software will be disconnected.

1. From the direct console, select **Restart Management Network** and press Enter.
2. Press F11 to confirm the restart.

Test Connectivity to Devices and Networks

You can use the direct console to perform some simple network connectivity tests, and specify other devices and networks.

1. From the direct console, select **Test Management Network** and press Enter.
2. Type addresses to ping or another DNS host name to resolve.
3. Press Enter to start the test.

Restoring the Standard Switch

A vSphere Distributed Switch functions as a single virtual switch across all associated hosts.

Verify that your management network is connected to a distributed switch.

Virtual machines can maintain a consistent network configuration as they migrate across multiple hosts. If you migrate an existing standard switch, or virtual adapter, to a Distributed Switch and the Distributed Switch becomes unnecessary or stops functioning, you can restore the standard switch to ensure that the host remains accessible.When you restore the standard switch, a new virtual adapter is created and the management network uplink that is currently connected to Distributed Switch is migrated to the new virtual switch.
You might need to restore the standard switch for the following reasons:

- The Distributed Switch is not needed or is not functioning.
- The Distributed Switch needs to be repaired to restore connectivity to vCenter and the hosts need to remain accessible.
- You do not want vCenter to manage the host. When the host is not connected to vCenter, most Distributed Switch features are unavailable to the host.

1. From the direct console, select **Restore Standard Switch** and press Enter.
   If the host is on a standard switch, this selection is dimmed, and you cannot select it.
2. Press F11 to confirm.

Configuring System Logging

ESX hosts run the syslog service (`vmsyslogd`) that writes messages from system components to log files and can forward messages to syslog collectors.

You can configure the amount and location of the logs. You can also create and apply log filters to modify the logging policy of an ESX host.

When configuring remote hosts for syslog, you also need to open some specified ports in the ESX host firewall to allow the transmission of log messages. For more information, see Broadcom knowledge base article 312032.

Configure Syslog on ESX Hosts

You can use the vSphere Client, the VMware Host Client, or the `esxcli system syslog` command to configure the syslog service.

The syslog service receives, categorizes, and stores log messages for analyses that help you take preventive action in your environment.

For information about using the `esxcli system syslog` command and other ESXCLI commands, see ESXCLI Command Line Reference. For details how to open the ESX firewall for the port specified in each remote host specification, see the vSphere Security guide.

Set ESX Syslog by using the vSphere Client

You can use the command to configure the syslog service globally and edit various advanced settings:

1. Browse to the host in the inventory.
2. Click **Configure**.
3. Under **System**, click **Advanced System Settings**.
4. Click **Edit**.
5. Filter for *syslog*.
6. To set up logging globally and configure various advanced settings, see ESX Syslog Options.
7. To overwrite the default log size and log rotation for any of the logs:
   a. Click the name of the log that you want to customize.
   b. Enter the number of rotations and the log size you want.
8. Click **OK**.

Syslog parameter settings that you define by using the vSphere Client or VMware Host Client are effective immediately. However, most settings you define by using ESXCLI require an additional command to take effect. For more details, see ESX Syslog Options.

Set ESX Syslog by using the VMware Host Client

You can use the VMware Host Client to configure and edit syslog service parameters on hosts.

1. In the VMware Host Client, under **Host**, click **Manage > System > Advanced settings**.
2. In the **Search** panel, type a syslog setting you want to define. See ESX Syslog Options.
3. Select the setting and click **Edit option**.
4. Set the value as described in the table of parameters in ESX Syslog Options.
5. Click **Save**.

Set ESX Syslog by using ESXCLI

You can configure the syslog service on hosts by using the ESXCLI command: `esxcli system syslog config set <syslog option>`.

For information about using the `esxcli system syslog` command and other ESXCLI commands, See the ESXCLI Command Line Reference on the Broadcom Developer Portal. For details how to open the firewall for the port specified in each remote host specification, see Configuring the ESX Firewall in the vSphere Security guide.

Using ESXCLI requires ESX to open SSH logins, which is a security risk, and is not recommended. If you chose to use ESXCLI, make sure you use the `esxcli system syslog reload` command after setting each parameter to make sure that it takes effect.

1. Use the ESXCLI command `esxcli system syslog config set <syslog option>` to set a syslog option that you decide to enable. For example, to set the `Syslog.global.logHost` option, use the command `esxcli system syslog config set --loghost=<str>`
2. After setting `Syslog.global.logHost`, ESX hosts open and maintain connections to the syslog collectors, and the transmission of messages begins immediately. When ESX generates a syslog message, it writes it to the appropriate log file on the ESX host and also forwards it to all configured syslog collectors.

Fine-tune Syslog on ESX Hosts

By using the right syslog settings, you can achieve proactive monitoring of your environment, reduce downtime, and take preventive action on servers.

While setting up syslog, you need to consider several parameters that affect log file retention, syslog transmission, transmission length, error handling, and the set up of SSL certificates for secure syslog message transmission. What follows are recommendations for fine-tuning your syslog parameters. You can see a description of all available parameters at ESX Syslog Options.

How to Specify Log File Retention

By default, log files cannot expand past a configured size. Once a log file reaches the configured size, logging is routed to a new log file and the oldest log file is deleted.

> **Note:**
> Best practice is to balance the rotate and size settings. Increasing the rotate setting ensures that syslog files are generated often enough to prevent any potential corruption or destruction from the other log files. Increasing the size setting reduces the time for switching to another log file. Optimal size settings are a multiple of 1024 KiB.

Use the Syslog.global.defaultSize setting to specify the log file maximum size in KiB, and Syslog.global.defaultRotate to set the maximum number of old log files to keep before rotating to a new log file. To change the log file retention parameters associated with a specific program, use the `Syslog.loggers.<progName>.rotate` and `Syslog.loggers.<progName>.size settings`, where `<progName>` is the name of the program whose parameters you want to adjust.

Manage Settings that Affect the Virtual Machine Log File

You can configure some settings that affect the virtual machine log file, `vmware.log`, either in the `vmx` file or in the `/etc/vmware/config` file. You must power off a virtual machine to edit the `vmx` file and edits take effect only on that virtual machine. If you use the `/etc/vmware/config` file, you must add the prefix *`"vmx"`* to the setting, for example `vmx.log.keepOld = "20"`, and edits affect all virtual machines on the ESX host.

Table 27: Configurable settings for the vmware.log file

Parameter	Description	Example	Notes
`logging`	Disables all virtual machine logging.	The default value is `logging = "TRUE"` To disable virtual machine logging: `logging = "FALSE"`	Do not use this setting, because disabling virtual machine logging makes it difficult or impossible to get support for virtual machine problems. If you need to use this setting for some reason, you can only place it in the `vmx` file of a virtual machine.
`log.throttleBytesPerSec`	Controls when a log file throttles. Log file throttling occurs when writes to the `vmware.log` exceed the specified rate for a significant amount of time. This occurs when code within the VMX process, which controls a virtual machine, creates excessive log messages. The default value for this setting is 1 KB/sec. In case of log thottling, you see `<<< Log Throttled >>>` in the `vmware.log` file.	`log.throttleBytesPerSec = "1500"` To disable log throttling, use `log.throttleBytesPerSec = "0xFFFFFFFF"`	Log file thottling might obscure information necessary to diagnose problems with the affected virtual machine. If you need to disable log throttling, place the line in the example in the `vmx` file of the affected virtual machine. Remove the line after the debugging session ends.

Parameter	Description	Example	Notes
log.keepOld	Controls the number of older vmware.log file to retain.	log.keepOld = "20"	Do not put the value of this setting below the default value (10). If virtual machines are frequently modified or moved, consider raising this setting to 20 or more.
log.rotateSize	Controls the maximum size of a vmware.log file in bytes.	log.rotateSize = "2500000" To disable limiting the maximum size of a vmware.log file, use log.rotateSize = "0"	A value of this setting below 100,000 can cause a loss of critical log messages and affect virtual machine performance. In ESX 7.x and earlier, the default value of this setting places no limit on the size of a vmware.log file. In ESX 8.x and later, the default value of this setting is 2,048,000.
log.fileName	Controls the name and location of virtual machine log files.	log.fileName = "myVMLog" This setting changes the name of the virtual machine log files from vmware.log to myVMlog. log.fileName = "/vmfs/volumes/vol1/myVM/myVM.log" This setting directs virtual machine log files to a directory on a different VMFS volume (vol1) by using myVM for a file name.	Do not place a log file outside the virtual machine directory to make sure that the collection of host support bundles picks up the log file, which can be critical to debug virtual machine problems.
log.fileLevel	Controls the minimum level at which messages are written to vmware.log. Every log message has a level associated with it. Levels below the specified setting are not added to a virtual machine log file. The virtual machine message log levels (from most to least restricted) are: • error • warning • notice • info (default) • trivia • debug • debug1 • debug2 • debug3 • debug4 • debug5 • debug6 • debug7 • debug8 • debug9 • debug10	log.fileLevel = "debug1"	Do not set a more restrictive level than "info" to avoid filtering out messages that are necessary for debugging virtual machine problems. Lower the level below "info" only upon request by licensed support. Restore the setting to "info" after debugging ends.

Parameter	Description	Example	Notes
`log.filter.minLogLevel.<groupName>`	Controls the output of specialized debugging messages.	`log.filter.minLogLevel.vsish = "debug5"`	Use this setting only upon request by licensed support, who should provide one or more `<groupName>` parameters. Remove the setting after debugging ends.
`log.sysogID`	Activates the sending of virtual machine log messages to the system logger of an ESX host, such as the syslog.	`log.syslogID = "vmx"`	Use `"vmx"` as value for this setting to allow the ESX syslog daemon, `vmsyslogd`, to send these messages to a separate log file.
`log.syslogLevel`	Controls the minimum level at which messages are output to the system logger of an ESX host, such as the syslog.	`log.syslogLevel = "debug"`	The levels and functioning of this setting are identical to those for the `log.fileLevel` setting.

How to Specify Message Transmission to Remote Hosts

Optionally, you can configure ESX to send syslog messages to one or more remote hosts, called syslog collectors, such as VCF Operations Log Management (formerly VCF Operations for Logs,VMware vRealize Log Insight, and vCenter Log Insight), to collect syslog messages.

Note:

Best practice is that you configure each ESX host to send syslog messages to at least one syslog collector. This helps ensure that the messages are preserved in case of a catastrophic system event and that you can process syslog messages in various ways, such as real-time categorization and analysis (for example, by type, time span, or machine), or archive messages.

Use the `Syslog.global.logHost` setting to define remote host specifications. Separate multiple remote host specifications with a comma (,). After setting `Syslog.global.logHost`, ESX hosts open and maintain connections to the syslog collectors, and the transmission of messages begins immediately. When ESX generates a syslog message, it writes it to the appropriate log file on the ESX host and also forwards it to all configured syslog collectors.

In addition to syslog messages, audit messages can also be transmitted to syslog collectors for security purposes. Audit records track security-related activity on the ESX host. For more information about audit records, see Log File Formats.

Note:
Consult with your company security response team if and how to set audit records. Certified configurations usually require audit records to be activated.

What follows is the syntax for `Syslog.global.logHost` remote host specifications:

`protocol://target[:port][?formatter=value[&framing=value]]`

Parameter	Description	Notes
`protocol`	Specifies the networking protocol. Valid values are `udp`, `tcp`, and `ssl`.	The `ssl` protocol specifies that transmission of syslog messages is encrypted. The `tcp` and `udp` protocols do not encrypt the transmission. **Note:** If capturing syslog messages or audit messages is critical to your system, avoid using the `udp` protocol because the networking infrastructure external to ESX might drop UDP messages.
`target`	Specifies the remote host. You can use either an IPV4 or IPV6 address, or a host name.	When you use an IPV6 address, you must embed it in square brackets `[xxx]`, where `xxx` is the IPV6 address.
`port`	(Optional) Specifies the remote host port to use. If you use UDP or TCP, the default port is 514. If you use SSL, the default port is 1514. If you choose to use different ports from 514 or 1514, you must adjust the ESX firewall to open the port.	For details how to open the ESX firewall for the port specified in each remote host specification, see the topic about *configuring the ESX firewall* in the *vSphere Security* guide.
`formatter`	Specifies how transmissions are formatted. The formatter must be RFC 3164 or RFC 5424.	RFC 3164 is the default.
`framing`	Specifies if transmissions are framed. Framing must be `non_transparent` or `octet_counting`.	The default is `non_transparent`. Transmissions in RFC 5424 format must specify `octet_counting` framing. For more information, see Protocols, Formats, and Framing of ESX Syslog Messages.

Examples of remote machine specifications:

Syslog.global.logHost string example	Notes
tcp://10.176.130.7:12345?formatter=RFC_3164	Transmits syslog messages to 10.176.130.7 using TCP/IP and port 12345. Transmission format is RFC 3164 with no framing.
tcp://10.176.130.7:12345?formatter=RFC_3164&framing=octet_counting	Transmits syslog messages to 10.176.130.7 using TCP/IP and port 12345. Transmission format is RFC 3164 and framing is `octet_counting`.
tcp://10.176.130.7:12345?formatter=RFC_5424&framing=octet_counting	Transmits syslog messages to 10.176.130.7 using TCP/IP and port 12345. Transmission format is RFC 5424 and framing is `octet_counting`.
tcp://[2001:db8:85a3:8d3:1319:8a2e:370:7348]	Transmits syslog messages to an IPV6 address using port 1514.
tcp://[2001:db8:85a3:8d3:1319:8a2e:370:7348]:5432	Transmit syslog messages to an IPV6 address using port 54321.
udp://company.com	Transmits syslog messages to `company.com` using UDP and port 514.
udp://company.com,tcp://10.20.30.40:1050	Transmits syslog messages to two remote hosts. The first remote host uses UDP to communicate with `company.com` using port 514. The second remote host uses TCP to communicate with the IPV4 address `10.20.30.40` using port 1050.
ssl://company.com	Transmits syslog messages to `company.com` using SSL (TLS) and port 514.

Maximum Message Transmission Length

If you use UDP, the maximum syslog message transmission length is 480 bytes for IPV4 and 1180 bytes for IPV6.

For TCP or SSL, the default maximum syslog message transmission length is 1 kibibyte (KiB). You can increase this length by using the `Syslog.global.remoteHost.maxMsgLen` parameter. The maximum value is 16 KiB. Messages longer than 16 KiB are truncated.

> **Note:**
> If increasing the maximum transmission length is necessary, best practice is to increase the length only as much as specifically necessary.

Increasing the maximum syslog message length can cause problems if the networking and syslog infrastructure external to ESX is unable to handle messages longer than 1 KiB.

> **Note:**
> Best practice is that you do not use UDP to transmit syslog messages due to the packet length constraints and the possibility that the external networking infrastructure might drop the messages.

Considerations for Certificates When Configuring SSL Transmissions to Remote Hosts

When configuring ESX to transmit syslog messages to remote hosts using SSL, you must add an SSL certificate for each remote host to the ESX host CA store. For more information, see the topic about managing certificates for ESX hosts in the *vSphere Security* guide.

> **Note:**
> Consult with your syslog collector documentation on how to configure the collector for secure receipt of syslog messages using SSL and a private key.

Additional SSL Transmission Parameters

An ESX system complying with security certification requirements might require activating X509 CRL checks.
You turn on the advanced settings `Syslog.global.certificate.strictX509Compliance` and `Syslog.global.certificate.checkCRL` by changing the default value of `false` to `true`. Due to implementation limitations, if you activate CRL checks by using the setting `Syslog.global.certificate.checkSSLCerts`, then all certificates in a certificate chain must provide a CRL link. By default, the setting is active. You can deactivate SSL certificate checks by changing the setting to `false`, but this is not a best practice. You might need to turn off SSL certificate checks when troubleshooting communications with a remote host, but do this only for a limited time.

Where to Find Syslog Daemon Error and Status Information

The ESX syslog daemon uses the log file `/var/run/log/vmsyslogd.log` to store status and error information, including dropped messages. If audit record transmission is active, the syslog daemon also emits audit records related to its operation, such as daemon start, stop, and error conditions, which allows you to verify that the syslog daemon runs properly.

How to Change the Default Syslog Log File Storage Area

The default syslog log file storage area is `/var/run/log`, local to each ESX host. Use the `Syslog.global.logDir` syslog configuration variable to change the default syslog log file storage area, as long as the location resides on persistent storage. If `Syslog.global.logDir` is configured to a persistent store shared by multiple ESX hosts to store their syslog log files, change the `Syslog.global.logDirUnique` setting to `true` to prevent mixing logs. The `Syslog.global.logDirUnique` setting makes sure that each ESX machine a unique name added to the `Syslog.global.logDir` path, separating the log files from other hosts.

Syslog Message Queueing for Remote Hosts and Message Drops

Once syslog emissions start, they never stop except for ESX reboots and failures, or a syslog reconfiguration to stop.

To avoid dropping messages, ESX uses an in-memory queue that allows the `vmsyslogd` service to handle the following conditions for a short time:

- ESX generates log messages at a faster rate than the `vmsyslogd` service can process and transmit
- Network connectivity between ESX and the remote host fails

If either of these conditions continues for a long period, the capacity of the in-memory queue might not be sufficient and `vmsyslog` might stop transmitting messages and audit logs to remote hosts. No data is lost, because dropped messages are written to the `/var/run/log/vmsyslogd-dropped.log` file.

To minimize the chance that message dropping occurs, place the `vmsyslogd` log files on the fastest available storage and configure `vmsyslogd` and its syslog collectors on a network with end-to-end bandwidth greater than 1 GigE.

> **Note:**
> Consider an end-to-end bandwidth of 2.5 GigE or more, preferably 10 GigE, to optimize performance and prevent message drops.

If you see excessive logs from an application, file a support service request for analysis and correction.

You can see statistics about message drops in the syslog daemon log file.

You can see dropped messages at `/var/run/log/vmsyslogd-dropped.log`. This log file has retention settings specific to it, similar to those for the program-specific retention parameters. The dropped messages log file retention parameters are: `Syslog.global.droppedMsgs.fileRotate` and `Syslog.global.droppedMsgs.fileSize`.

ESX Syslog Options

You can define the behavior of ESX syslog files and transmissions by using a set of syslog options.

Apart from the base settings, such as `Syslog.global.logHost`, starting from ESX 7.0 Update 1, a list of advanced options is available for customizations, and NIAP compliance.

> **Note:**
> Always configure persistent storage before you set any of the audit record parameters or the `Syslog.global.logDir` parameter.

> **Note:**
> All audit record settings, beginning with `Syslog.global.auditRecord`, take effect immediately. However, for other settings that you define by using ESXCLI, make sure to run the `esxcli system syslog reload` command to enable the changes.

Table 28: Legacy Syslog Options

Option	ESXCLI command	Description
`Syslog.global.logHost`	`esxcli system syslog config set --loghost=<str>`	Defines a comma-delimited list of remote hosts and specifications for message transmissions. If the `loghost=<str>` field is blank, no logs are forwarded. While no hard limit to the number of remote hosts to receive syslog messages exists, good practice is to keep the number of remote hosts to five or less. The format of a remote host specification is: `protocol://hostname|ipv4|'['ipv6']'[:port]`. The protocol must be one of TCP, UDP, or SSL. The value of a port can be any decimal number from 1 through 65535. If a port is not provided, SSL and TCP use 1514. UDP uses 514. For example: `ssl://hostName1:1514`.
`Syslog.global.defaultRotate`	`esxcli system syslog config set --default-rotate=<long>`	Maximum number of old log files to keep. You can set this number globally and for individual subloggers (see `Syslog.global.defaultSize`).

Option	ESXCLI command	Description
Syslog.global.defaultSize	esxcli system syslog config set --default-size=<long>	Default size of log files, in KiB. After a file reaches the default size, the syslog service creates a new file. You can set this number globally and for individual subloggers.
Syslog.global.logDir	esxcli system syslog config set --logdir=<str>	Directory where logs reside. The directory can be on mounted NFS or VMFS volumes. Only the /scratch directory on the local file system is persistent across reboots. Specify the directory as [*datastorename*] *path_to_file*, where the path is relative to the root of the volume backing the datastore. For example, the path [storage1] /systemlogs maps to the path /vmfs/volumes/storage1/systemlogs.
Syslog.global.logDirUnique	esxcli system syslog config set --logdir-unique=<bool>	Specifies the ESX host name to be concatenated to the value of Syslog.global.logDir. It is critical that you enable this setting when multiple ESX hosts log to a shared file system. Selecting this option creates a subdirectory with the name of the ESX host under the directory specified by **Syslog.global.LogDir**. A unique directory is useful if the same NFS directory is used by multiple ESX hosts.
Syslog.global.certificate.checkSSLCerts	esxcli system syslog config set --check-ssl-certs=<bool>	Enforces checking of SSL certificates when transmitting messages to remote hosts.

Table 29: Syslog Options Available Starting from ESX 7.0 Update 1 and later

Option	ESXCLI command	Description
Syslog.global.auditRecord.storageCapacity	esxcli system auditrecords local set --size=<long>	Specifies the capacity of the audit record storage directory located on the ESX host, in MiB. You cannot decrease the capacity of the audit record storage. You can increase the capacity before or after the audit record storage is enabled (see Syslog.global.auditRecord.storageEnable).
Syslog.global.auditRecord.remoteEnable	esxcli system auditrecords remote enable	Enables sending audit records to remote hosts. Remote hosts are specified by using the Syslog.global.logHost parameter.
Syslog.global.auditRecord.storageDirectory	esxcli system auditrecords local set --directory=<dir>	Creates an audit record storage directory and unless specified, sets /scratch/auditLog as the default location. You must not manually create an audit record storage directory and you cannot change the audit record storage directory while audit record storage is enabled (see Syslog.global.auditRecord.storageEnable).
Syslog.global.auditRecord.storageEnable	esxcli system auditrecords local enable	Enables the storage of audit records on an ESX host. If the audit record storage directory does not exist, it is created with the capacity specified by Syslog.global.auditRecord.storageCapacity.

Option	ESXCLI command	Description
`Syslog.global.certificate.checkCRL`	`esxcli system syslog config set --crl-check=<bool>`	Enables checking the revocation status of all the certificates in an SSL certificate chain. Enables verification of X.509 CRLs, which are not checked by default in compliance with industry conventions. A NIAP-validated configuration requires CRL checks. Due to implementation limitations, if CRL checks are enabled, then all certificates in a certificate chain must provide a CRL link. Do not enable the `crl-check` option for installations not related to certification, because of the difficulty in properly configuring an environment that uses CRL checks.
`Syslog.global.certificate.strictX509Compliance`	`esxcli system syslog config set --x509-strict=<bool>`	Enables strict compliance with X.509. Performs additional validity checks on CA root certificates during verification. These checks are generally not performed, as CA roots are inherently trusted, and might cause incompatibilities with existing, misconfigured CA roots. A NIAP-validated configuration requires even CA roots to pass validations. Do not enable the `x509-strict` option for installations not related to certification, because of the difficulty in properly configuring an environment that uses CRL checks.
`Syslog.global.droppedMsgs.fileRotate`	`esxcli system syslog config set --drop-log-rotate=<long>`	Specifies the number of old dropped message log files to keep.
`Syslog.global.droppedMsgs.fileSize`	`esxcli system syslog config set --drop-log-size=<long>`	Specifies the size of each dropped message log file before switching to a new one, in KiB.
`Syslog.global.logCheckSSLCerts`	`esxcli system syslog config set --check-ssl-certs=<bool>`	Enforces checking of SSL certificates when transmitting messages to remote hosts. **Note:** Deprecated. Use `Syslog.global.certificate.checkSSLCerts` in ESX 7.0 Update 1 and later.

Option	ESXCLI command	Description
`Syslog.global.logFilters`	`esxcli system syslog config logfilter [add | remove | set] ...`	Specifies one or more log filtering specifications. Each log filter must be separated by a double vertical bar "||". The format of a log filter is: `numLogs | ident | logRegexp`. `numLogs`sets the maximum number of log entries for the specified log messages. After reaching this number, the specified log messages are filtered and ignored. `ident` specifies one or more system components to apply the filter to the log messages that these components generate. `logRegexp` specifies a case-sensitive phrase with Python regular expression syntax to filter the log messages by their content.
`Syslog.global.logFiltersEnable`		Enables the use of log filters.
`Syslog.global.logLevel`	`esxcli system syslog config set --log-level=<str>`	Specifies the log filtering level. You must change this parameter only when troubleshooting an issue with the syslog daemon. You can use the values `debug` for the most detailed level, `info` for the default detail level, `warning` for only warnings or errors, or `error`, only for errors.
`Syslog.global.msgQueueDropMark`	`esxcli system syslog config --queue-drop-mark=<long>)`	Specifies the percent of the message queue capacity at which messages are dropped.
`Syslog.global.remoteHost.connectRetryDelay`	`esxcli system syslog config set --default-timeout=<long>`	Specifies the delay before retrying to connect to a remote host after a connection attempt fails, in seconds.
`Syslog.global.remoteHost.maxMsgLen`	`esxcli system syslog config set --remote-host-max-msg-len=<long>`	Specifies the maximum length of a syslog transmission before truncation occurs, in bytes. The default maximum length for remote host messages is 1 KiB for TCP, SSL, and UDP protocols. You can increase the maximum message length to up to 16 KiB. However, raising this value above 1 KiB does not ensure that long transmissions arrive untruncated to a syslog collector. For example, when the syslog infrastructure that issues a message is external to ESX. For versions earlier than ESXi 8.0 Update 3i, this parameter applies only to TCP and SSL protocols. UDP maximum message transmission length is limited at 480 bytes for IPV4 and 1180 bytes for IPV6.
`Syslog.global.vsanBacking`	`esxcli system syslog config set --vsan-backing=<bool>`	Allows log files and the audit record storage directory to be placed on a vSAN cluster. However, enabling this parameter might cause the ESX host to become unresponsive.

Protocols, Formats, and Framing of ESX Syslog Messages

Starting with ESX 8.0, the ESX syslog daemon, vmsyslogd, uses three parameters to define how to send messages and audit records - protocol, formatting, and framing.

The supported protocols are UDP, TCP, and TLS (SSL). Formatting of syslog messages is defined by either RFC 3164 or RFC 5424. Framing specifies how a message is encapsulated. Framing of encapsulated messages is defined as transparent, also called octet_counting, or non-transparent, if a message is not encapsulated. Transparent framing ensures that new lines embedded in a message do not confuse a syslog collector. Syslog messages sent by using the UDP protocol are considered transparently framed; a syslog collector is expected to understand this and accept the transmission as a single message.

RFC 3164 sets the maximum total length of a syslog message at 1024 bytes, while RFC 5424 extends this maximum to 2048 bytes.

The default maximum length for remote host messages in ESX is 1 KiB. You can increase the maximum message length to up to 16 KiB. However, raising this value above 1 KiB does not ensure that long transmissions arrive untruncated to a syslog collector. For example, when the syslog infrastructure that issues a message is external to ESX.

Syslog messages that the vmsyslogd transmits consist of structured data, a property list formatted in compliance with RFC 5424, and free format, or unstructured, data.

When a message is longer than the maximum length, ESX 8.0 and later mitigates the message, trying to preserve as much of the structured data as possible.

When a message is mitigated, three parameters are either added to existing structured data or structured data is created to contain these parameters: `msgModified`, `remoteHostMaxMsgLen`, and `originalLen` .

The `msgModified` parameter indicates how the mitigation impacts the message: only structured data, only unstructured data, or both.

The `remoteHostMaxMsgLen` parameter specifies the maximum message length that ESX can handle.

The `originalLen` parameter specifies the message length before it is mitigated.

Supported options for protocols, formatting, and framing of ESX syslog messages:

Formatting	Framing	UDP	TCP	SSL	Comments
Unspecified	Unspecified	Supported RFC 5426	Supported	Supported	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message. Framing defaults to non-transparent with TCP or SSL (TLS) and embedded newlines in structured data might corrupt messages. With UDP, packets are framed.
Unspecified	Non_transparent	Forbidden	Supported	Supported	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message. Framing defaults to non-transparent with TCP or SSL (TLS) and embedded newlines in structured data might corrupt messages.
Unspecified	Octet_counting	Forbidden	Supported RFC 6587	Supported RFC 6587	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message.

Formatting	Framing	UDP	TCP	SSL	Comments
RFC 5424	Unspecified	Supported RFC 5426	Supported RFC 5425	Supported RFC 5424	Formatting of messages complies to RFC 5424. Framing defaults to octet-counting with TCP or SSL (TLS). With UDP, framing might not be explicitly specified.
RFC 5424	Non_transparent	Forbidden	Not Supported	Not Supported	Not supported because embedded newlines in structured data might create corrupted messages.
RFC 5424	Octet_counting	Forbidden	Supported RFC 5425	Supported RFC 5425	Formatting of messages complies to RFC 5424.
RFC 3164	Unspecified	Supported RFC 5426	Supported	Supported	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message. Framing defaults to non-transparent with TCP or SSL (TLS) and embedded newlines in structured data might corrupt messages. With UDP, packets are framed.
RFC 3164	Non_transparent	Forbidden	Supported	Supported	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message. Framing defaults to non-transparent with TCP or SSL (TLS) and embedded newlines in structured data might corrupt messages.
RFC 3164	Octet_counting	Forbidden	Supported RFC 6587	Supported RFC 6587	Formatting of messages complies to RFC 3164, only timestamps are in RFC 3339 format. Structured data is prepended to each message.

Log File Formats

Starting with ESX 8.0, the format of log files is standardized and is expressed in Augmented Backus-Naur Form (ABNF).

In ESX 8.0 and later, log files are written either directly, from a single service such as VMX, or indirectly, when logs from a service are submitted to a syslog. For example, VMX always writes log messages in the `vmware.log` file of each virtual machine. To spare system resources, VMX does not submit log messages to the syslog. On the other hand, in some log files that `vmsyslogd` generates, you see messages from multiple programs, because the ESX syslog daemon creates and manages all log files, and messages to these files, from multiple services.

Format for direct log messages:

Parameter	Value
`LOG-MSG`	HEADER SP MSG
`HEADER`	TIMESTAMP SP SEVERITY SP THREAD-NAME SP OPID
`TIMESTAMP`	FULL-DATE T FULL-TIME (Complies to RFC 5424 with the requirement for UTC/GMT formatting and resolution in milliseconds, or more granular where possible.)
`FULL-DATE`	DATE-FULLYEAR - DATE-MONTH - DATE-MDAY
`DATE-FULLYEAR`	4DIGIT

DATE-MONTH	2DIGIT ; 01-12
DATE-MDAY	2DIGIT ; 01-28, 01-29, 01-30, 01-31 based on month/year
FULL-TIME	TIME-HOUR : TIME-MINUTE : TIME-SECOND[TIME-SECFRAC] Z
TIME-HOUR	2DIGIT ; 00-23
TIME-MINUTE	2DIGIT ; 00-59
TIME-SECOND	2DIGIT ; 00-59
TIME-SECFRAC	'.' 1*6DIGIT
SEVERITY	SEVERITY-STRING SEVERITY-VALUE [LINE-MARKER]
SEVERITY-STRING	Em/ Al / Cr / Er / Wa / No / In / Db (The 8 severity levels specified in RFC 5424 are abbreviated as follows: • Em - Emergency • Al - Alert • Cr - Critical • Er - Error • Wa - Warning • No - Notice • In - Informational • Db - Debug
SEVERITY-VALUE	(*DIGIT) (The SEVERITY-VALUE is an optional expression of the numeric value associated with the SEVERITY-STRING. This allows levels supported by a logger to be collapsed into the 8 required strings with no loss of information (e.g. Db(5) - debug, level 5).)
LINE-MARKER	+ (The LINE-MARKER is added to each subsequent line generated from a multi-line submission. It identifies multiline submissions and prevents a log injection security attack.)
NILVALUE	- (A single threaded program might not have a thread name and NILVALUE is acceptable.)
THREAD-NAME	NILVALUE / 1*32PRINTUSASCII (The component (APP-NAME) is implied when a single program writes the file and no component field is necessary, only the thread name.)
OPID	NILVALUE / 1*128UTF-8-STRING
STRUCTURED-DATA	1*SD-ELEMENT
SD-ELEMENT	[SD-ID *(SP SD-PARAM)]
SD-PARAM	PARAM-NAME %d34 PARAM-VALUE %d34
SD-ID	SD-NAME
PARAM-NAME	SD-NAME
PARAM-VALUE	UTF-8-STRING ; characters '', '\' and ']' MUST be escaped.
SD-NAME	1*32PRINTUSASCII ; except '', SP, ']', %d34 ()
MSG	[STRUCTURED-DATA SP] UTF-8-STRING

Format for log files that the vmsyslogd service manages:

Parameter	Value
LOG-MSG	HEADER SP MSG
HEADER	TIMESTAMP SP SEVERITY SP APP-NAME [PROC-IDENTIFIER] :
APP-NAME	1*32PRINTUSASCII
PROC-IDENTIFIER	[*DIGITS] ; the PID associated with APP-NAME
TIMESTAMP	FULL-DATE T FULL TIME (resolution in milliseconds or more granular where possible.)
FULL-DATE	DATE-FULLYEAR - DATE-MONTH - DATE-MDAY
DATE-FULLYEAR	4DIGIT

DATE-MONTH	2DIGIT ; 01-12
DATE-MDAY	2DIGIT ; 01-28, 01-29, 01-30, 01-31 based on month/year
FULL-TIME	TIME-HOUR : TIME-MINUTE : TIME-SECOND[TIME-SECFRAC] Z
TIME-HOUR	2DIGIT ; 00-23
TIME-MINUTE	2DIGIT ; 00-59
TIME-SECOND	2DIGIT ; 00-59
TIME-SECFRAC	'.' 1*6DIGIT
SEVERITY-STRING	Em/ Al / Cr / Er / Wa / No / In / Db (The 8 severity levels specified in RFC 5424 are abbreviated as follows: • Em - Emergency • Al - Alert • Cr - Critical • Er - Error • Wa - Warning • No - Notice • In - Informational • Db - Debug
SEVERITY	SEVERITY-STRING PRI-STRING [LINE-MARKER]
PRIVAL	1*3DIGIT ; range 0 .. 191 (the MSG PRI; contains facility and severity values, ORed together)
PRI-STRING	(PRIVAL) (The PRIVAL contains the bits from the message PRI. This allows one to see the Facility of the message, as well as the severity bits themselves._
LINE-MARKER	+ (The LINE-MARKER is added to each subsequent line generated from a multi-line submission. It identifies multiline submissions and prevents a log injection security attack.)
STRUCTURED-DATA	1*SD-ELEMENT
SD-ELEMENT	[SD-ID *(SP SD-PARAM)]
SD-PARAM	PARAM-NAME %d34 PARAM-VALUE %d34
SD-ID	SD-NAME
PARAM-NAME	SD-NAME
PARAM-VALUE	UTF-8-STRING ; characters '', '\' and ']' MUST be escaped.
SD-NAME	1*32PRINTUSASCII ; except '', SP, ']', %d34 ()
MSG	[STRUCTURED-DATA SP] UTF-8-STRING

Audit Records

ESX audit records, with facility code 13, are compliant to both RFC 3164 and 5424 formats and you find them in the structured data section. In the audit record, you also find event-based traceability information when such data is available. Audit records are stored in a special format, not in a regular log file. You can access audit records locally by using the viewAudit program and the Virtual Infrastructure Management functionality FetchAuditRecords. Do not read, use, or edit an audit record storage file directly. Locally stored audit records comply with RFC 5424 transmission format where the HOSTNAME and MSGID are always NILVALUE.

ESX Syslog Message Transmission Formats

ESX 8.0 and later formats syslog messages in compliance with either RFC 3164 or RFC 5424.

> **Note:**
> The timestamps associated with RFC 3164 messages are in RFC 3339 format, an exception to the RFC 3164 specification.

The definition of the ESX transmission formats for RFC 3164 and RFC 5424 is in Augmented Backus-Naur Form (ABNF).

RFC 3164 Transmission Message Format

Parameter	Value
SYSLOG-MSG	HEADER SP MSG
HEADER	PRI TIMESTAMP SP HOSTNAME SP APP-NAME [PROC-IDENTIFIER] ":"
PRI	"<" PRIVAL ">"
PRIVAL	1*3DIGIT ; range 0 .. 191 (the MSG PRI; contains facility and severity values, ORed together)
APP-NAME	1*32PRINTUSASCII
HOSTNAME	1*255PRINTUSASCII
PROC-IDENTIFIER	"[" *DIGITS "]" ; the PID associated with APP-NAME
TIMESTAMP	FULL-DATE "T" UTC-TIME (Never set a TIME-OFFSET)
FULL-DATE	DATE-FULLYEAR "-" DATE-MONTH "-" DATE-MDAY
DATE-FULLYEAR	4DIGIT
DATE-MONTH	2DIGIT ; 01-12
DATE-MDAY	2DIGIT ; 01-28, 01-29, 01-30, 01-31 based on month/year
UTC-TIME	TIME-HOUR ":" TIME-MINUTE ":" TIME-SECOND [TIME-SECFRAC] "Z"
TIME-HOUR	2DIGIT ; 00-23
TIME-MINUTE	2DIGIT ; 00-59
TIME-SECOND	2DIGIT ; 00-59
TIME-SECFRAC	"." 1*6DIGIT
STRUCTURED-DATA	1*SD-ELEMENT
SD-ELEMENT	"[" SD-ID *(SP SD-PARAM) "]"
SD-PARAM	PARAM-NAME "%d34 PARAM-VALUE %d34
SD-ID	SD-NAME
PARAM-NAME	SD-NAME
PARAM-VALUE	UTF-8-STRING ; characters "", '\' and ']' MUST be escaped.
SD-NAME	1*32PRINTUSASCII ; except ", SP, ']', %d34 (")
MSG	[STRUCTURED-DATA SP] UTF-8-STRING

RFC 5424 Transmission Message Format

Parameter	Value
SYSLOG-MSG	HEADER SP STRUCTURED-DATA [SP MSG]
HEADER	PRI VERSION SP TIMESTAMP SP HOSTNAME SP APP-NAME SP PROCID SP MSGID
PRI	"<" PRIVAL ">"
PRIVAL	1*3DIGIT ; range 0 .. 191; contains facility and severity data
VERSION	NONZERO-DIGIT 0*2DIGIT
HOSTNAME	1*255PRINTUSASCII
APP-NAME	1*48PRINTUSASCII
NILVALUE	'-'
PROCID	NILVALUE | *DIGITS ; the PID associated with APP-NAME
MSGID	NILVALUE
TIMESTAMP	FULL-DATE "T" UTC-TIME

FULL-DATE	DATE-FULLYEAR "-" DATE-MONTH "-" DATE-MDAY
DATE-FULLYEAR	4DIGIT
DATE-MONTH	2DIGIT ; 01-12
DATE-MDAY	2DIGIT ; 01-28, 01-29, 01-30, 01-31 based on month/year
UTC-TIME	TIME-HOUR ":" TIME-MINUTE ":" TIME-SECOND [TIME-SECFRAC] "Z"
TIME-HOUR	2DIGIT ; 00-23
TIME-MINUTE	2DIGIT ; 00-59
TIME-SECOND	2DIGIT ; 00-59
TIME-SECFRAC	"." 1*6DIGIT
STRUCTURED-DATA	NILVALUE / 1*SD-ELEMENT
SD-ELEMENT	"[" SD-ID *(SP SD-PARAM) "]"
SD-PARAM	PARAM-NAME " %d34 PARAM-VALUE %d34
SD-ID	SD-NAME
PARAM-NAME	SD-NAME
PARAM-VALUE	UTF-8-STRING ; characters '"', '\' and ']' MUST be escaped.
SD-NAME	1*32PRINTUSASCII ; except ", SP, ']', %d34 (")
MSG	MSG-UTF8
MSG-UTF8	BOM UTF-8-STRING
BOM	%xEF.BB.BF

Configure Log Filtering on ESX Hosts

The log filtering capability lets you modify the logging policy of the syslog service that is running on an ESX host.

You can create log filters to reduce the number of repetitive entries in the ESX logs and to denylist specific log events entirely.

Install ESXCLI. See *Getting Started with ESXCLI*. For troubleshooting, run `esxcli` commands in the ESX Shell.

You can add log filters and enable log filtering by using ESXCLI. A log filter, once established, remains in place until it is removed, even across ESX reboots.

Log filters affect all log events that are processed by the ESX host `vmsyslogd` service, whether they are recorded to a log directory or to a remote syslog server.

You must enable the log filtering capability and reload the syslog daemon to activate the log filters on the ESX host.

ESXCLI commands to configure log filters follow this pattern: `esxcli system syslog config logfilter {cmd} [cmd options]`.

For example, to get the list of available log filters, run the following command: `[root@xxx-xx-dhcp-xx-xx:~] esxcli system syslog config logfilter list`.

Use the `set` command to activate or deactivate log filtering: `[root@xxx-xx-dhcp-xx-xx:~] esxcli system syslog config logfilter set`.

Use the `add` command to add a log filter and the `remove` command to remove a log filter.

Use the `get` command to determine if log filtering is enabled.

A log filter is specified by three components and uses the following syntax: `numLogs | ident | logRegexp`.

Parameter	Description
`numLogs`	Specifies the number of matches of the logRegexp Python regular expression that will be allowed before filtering begins.
`ident`	The `ident` string is how an application identifies itself to the syslog facility. The `logRegexp` filter must be associated with the same application. You can find the `ident` string of an application by inspecting the log files in `/var/run/log`. The third field of each log file begins with the `ident` string and ends with `[`.
`logRegexp`	Python regular expression that identifies the messages which you want to filter out.

For example, to filter out all messages from the hostd daemon that contain the word "mark" after the tenth occurrence, use the following command: `esxcli system syslog config logfilter add --filter="10|Hostd|mark"`.

To remove the log filter, use the command `esxcli system syslog config logfilter remove --filter="10|Hostd|mark"`.

Set the Host Image Profile Acceptance Level

The Host Image Profile acceptance level determines which vSphere installation bundles (VIBs) are accepted for installation.

Required privileges: **Host** > **Configuration** > **SecurityProfile** and **Host** > **Configuration** > **Firewall**

VIB signatures are checked and accepted for installation based on a combination of the VIB acceptance level and the host image profile acceptance level. VIBs are tagged with an acceptance level that depends on their signature status.

See Working with Acceptance Levels.

1. From the vSphere Client, connect to the vCenter Server.
2. Select the host in the inventory.
3. Click the **Configure** tab.
4. Under System, select **Security Profile**.
5. Scroll down to Host Image Profile Acceptance Level, and click **Edit**.
6. Select the acceptance level and click **OK**.

Table 30: Host Image Profile Acceptance Levels

Host Image Profile Acceptance Level	Accepted Levels of VIBs
VMware Certified	VMware Certified
VMware Accepted	VMware Certified, VMware Accepted
Partner Supported	VMware Certified, VMware Accepted, Partner Supported
Community Supported	VMware Certified, VMware Accepted, Partner Supported, Community Supported

Remove All Custom Packages on ESX

After adding custom packages, you might decide to remove them.

Before you remove custom packages, shut down or migrate running virtual machines off of the ESX host.

1. Reboot the ESX host.
2. In the direct console, select **Remove Custom Extensions** and press F11 to confirm.
3. Reboot the host.

All custom packages are removed.

Modify ESX Configuration Files

You can modify ESX configuration files in the ESX Configuration Store (ConfigStore) by using the `/bin/configstorecli` tool.

The goal of the ESX Shell tool `configstorecli`, introduced in ESX 7.0 Update 1, is to manage all configurations for an ESX host centrally, instead of using different methods and a variety of configuration files.

For more information on how to manage ConfigStore, see Broadcom knowledge base article 312190.

Deactivate Support for Non-ASCII Characters on ESX

You can deactivate support for non-ASCII characters for virtual machine files and directory names by using two methods, depending on the ESX version.

By default, ESX supports the use of non-ASCII characters for virtual machine file and directory names.

If you need to deactivate support for non-ASCII characters, for ESX 7.0 Update 2 and later, see Broadcom knowledge base articles 393232 and 312190.

Prior to ESX 7.0 Update 2, you can deactivate this support by modifying the `/etc/vmware/hostd/config.xml` file with the following steps:

1. Using a text editor, open the `/etc/vmware/hostd/config.xml` file for the ESX host.
2. Within the `<config></config>` tag, add the following code.

   ```
   <g11nSupport>false</g11nSupport>
   ```
3. Save and close the file.
4. Reboot the host.

   After you deactivate this support, you can still enter non-ASCII characters for virtual machine names. vSphere user interfaces display the virtual machine names in the non-ASCII characters, but ESX converts the actual file and directory names to ASCII strings.

Reset the System Configuration

If you are having trouble determining the source of a problem with your ESX host, you can reset the system configuration.

Before resetting the configuration, back up your ESX configuration in case you want to restore your configuration.

Changes in the system configuration can be related to various problems, including problems with connectivity to the network and devices. Resetting the system configuration might solve such problems. If resetting the system configuration does not solve the problem, it can still rule out configuration changes made since the initial setup as the source of the problem.

When you reset the configuration, the software overrides all your configuration changes, deletes the password for the administrator account (root), and reboots the host. Configuration changes made by your hardware vendor, such as IP address settings and license configuration, might also be deleted.

Resetting the configuration does not remove virtual machines on the ESX host. After you reset the configuration defaults, the virtual machines are not visible, but you make them visible again by reconfiguring storage and reregistering the virtual machines.

> **CAUTION:**
> When you reset the configuration defaults, users accessing the host lose connectivity.

1. Back up the configuration by using the `Get-VMHostFirmware` VCF PowerCLI cmdlet.
2. From the direct console, select **Reset System Configuration** and press **Enter**.
3. Press **F11** to confirm.

The system reboots after all settings are reset to the default values.

After You Install and Set Up ESX

After you install and set up ESX, you can use manage hosts by various interfaces, license the hosts, and back up your configuration.

After ESX is installed and set up, you can manage the host by using the vSphere Client and vCenter, license the host, and back up your ESX configuration. You can also use the VMware Host Client to connect directly to the ESX host and to manage it. For information about installing and using the VMware Host Client, see *vSphere Single Host Management*.

> **Note:**
>
> When you install a standalone ESX host with TMP enabled that is not connected to a vCenter instance, create a backup of the ESX configuration recovery key. To get the recovery key, run the command `esxcli system settings encryption recovery list` on the ESX host and note it down. It is possible that the host might not be able to complete booting due to host configuration encryption related problems and in such cases, you can restore the host configuration by using the recovery key and running the configuration recovery process.
>
> For best practices and VMware recommendations, see the *vSphere Security* guide.

View System Logs

System logs provide detailed information about system operational events.

1. From the direct console, select **View System Logs**.
2. Press a corresponding number key to view a log.
   vCenter Agent (vpxa) logs appear if you add the host to vCenter.
3. Press Enter or the spacebar to scroll through the messages.
4. Perform a regular expression search.
   a) Press the slash key (/).
   b) Type the text to find.
   c) Press Enter.
   The found text is highlighted on the screen.
5. Press q to return to the direct console.

See also Configure Syslog on ESX Hosts.

Troubleshooting ESX Booting

The ESX booting troubleshooting topics provide solutions to problems that you might encounter during the ESX booting.

- Host Stops Unexpectedly at Bootup When Sharing a Boot Disk with Another Host
- Host Fails to Boot After You Install ESX in UEFI Mode

Host Stops Unexpectedly at Bootup When Sharing a Boot Disk with Another Host

When more than one host, either physical or virtual, boots from the same shared physical disk or LUN, they cannot use the same scratch partition.

The host stops at bootup when sharing a boot disk with another host.

More than one ESX host can share the same physical disk or LUN. When two such hosts also have the same scratch partition configured, either of the hosts can fail at bootup.

1. Set the hosts to boot sequentially, and boot the hosts.
   This setting lets you start the hosts so that you can change the scratch partition for one of them.
2. From the vSphere Client, connect to the vCenter.
3. Select the host in the inventory.
4. Click the **Configure** tab.
5. Under System, select **Advanced System Settings**.
6. Select **ScratchConfig**.
   The **ScratchConfig.CurrentScratchLocation** text box shows the current location of the scratch partition.
7. In the **ScratchConfig.ConfiguredScratchLocation** text box, enter a directory path that is unique for this host. For example, `/vmfs/volumes/`*`DatastoreUUID`*`/`*`DatastoreFolder`*.
8. Reboot the host for the changes to take effect.

Host Fails to Boot After You Install ESX in UEFI Mode

After you install ESX on a host machine in UEFI mode, the machine might fail to boot.

When you install or upgrade ESX, the installer tries to create aUEFI boot option named VMwareESX and make it the default boot option. When you reboot after installing ESX the reboot might fail. This problem is accompanied by an error message similar to `No boot device available.`

- When the installer creates the UEFI boot option, a problem occurs while writing to the NVRAM on the host motherboard.
- The host firmware does not recognize the attempt to set the UEFI boot option as the first boot option, or the firmware overrides the boot order.
- Either or both BIOS and firmware are outdated.
- The boot disk has an MBR or MSDOS partition table. Due to a technical limitation, the UEFI boot option is only created for a GUID Partition Table (GPT) partition table.

  Note:

  UEFI firmware attempts to load the boot image from the EFI system partition, which is FAT based, on the disk. Booting from the EFI system partition only works if the disk is laid out by using a GPT. If the boot disk has an MBR or MSDOS partition table, a UEFI boot fails. You cannot add a boot entry for MBR. If the disk is fully consumed by ESX, it cannot be converted to GPT, and you must boot in legacy BIOS mode.

1. While the error message is displayed on screen, open the boot options menu. Depending on your system, the boot options menu might open with a keyboard shortcut, in the BIOS menu, or in a BMC, iLO or iDRAC interface.
2. Check if a boot option VMwareESX exists and try to boot from it. If the boot is successful, change the boot order and set VMwareESX as the first boot option.
3. If the problem is not resolved, select an option similar to **Add boot option**.
   The wording and location of the option might vary, depending on your system.
4. Select the file `\EFI\BOOT\BOOTx64.EFI` on the disk that you installed ESX on.
5. Change the boot order so that the host boots from the option that you added.

Decommission an ESX Host

If you do not want your server to be an ESX host, you can decommission the ESX host machine.

1. Remove VMFS datastores on the internal disks so that the internal disks are no longer set up to store virtual machines.
2. Change the boot setting in the BIOS so that the host no longer boots into ESX.

   If you installed ESX in UEFI mode, delete the boot option VMwareESX or any other boot option created manually.
3. Install another operating system in its place.

ESX Upgrade

VMware ESX Upgrade describes how to upgrade VMware ESX™ to the current version.

Intended Audience

VMware ESX Upgrade is for anyone who needs to upgrade from earlier versions of ESX. These topics are for experienced Microsoft Windows or Linux system administrators who are familiar with virtual machine technology and data center operations.

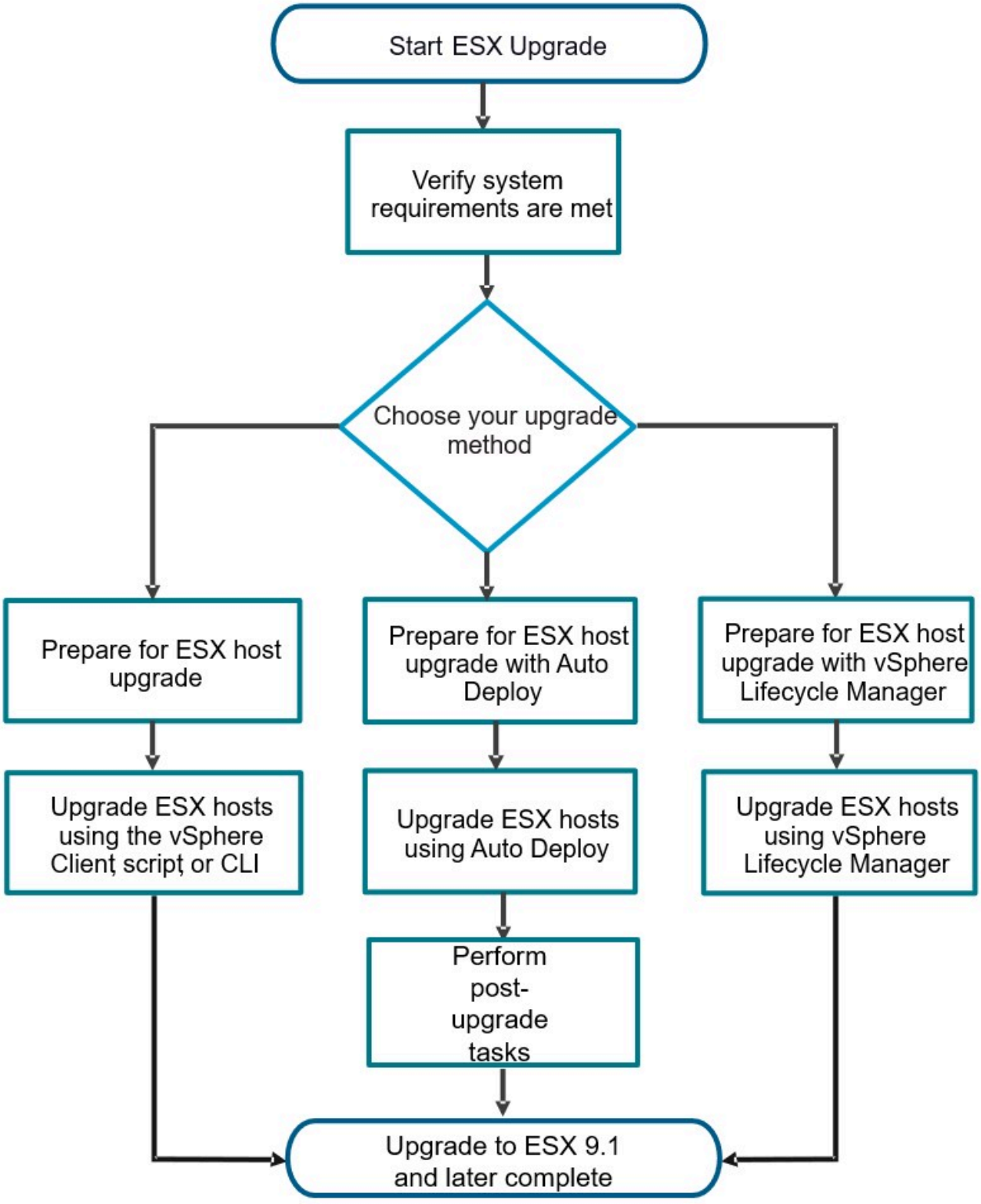
Overview of the ESX Host Upgrade Process

The details and level of support for an upgrade to ESX 9.1 depend on the ESX host to be upgraded and the upgrade method that you use.

Before you start an upgrade, verify that the upgrade path from your current version of ESX to the version to which you are upgrading, is supported. For more information, see the Broadcom Compatibility Guide.

You can upgrade an ESX host to version 9.1 by using an interactive upgrade from a CD, DVD, or USB, a scripted upgrade, ESXCLI, or vSphere Lifecycle Manager. When you upgrade an ESX host that has custom VIBs to version 9.1, all supported custom VIBs are migrated. For more information, see Upgrading Hosts That Have Third-Party Custom VIBs.

Figure 12: Overview of the ESX Host Upgrade Process



The following high-level steps are for upgrading ESX.

1. Verify that your system meets the upgrade requirements. See ESX Requirements.
2. Prepare your environment before upgrading. See Before Upgrading ESX Hosts.

3. Determine where you want to locate and boot the ESX installer. See Media Options for Booting the ESX Installer. If you are network booting the installer, verify that your network boot infrastructure is properly set up. See Network Booting the ESX Installer.
4. Upgrade ESX.
5. After upgrading ESX hosts, they automatically connect back to the vCenter and you must apply licenses. See After You Upgrade ESX Hosts.

CAUTION:

If you upgrade ESX hosts managed by vCenter, you must upgrade vCenter before you upgrade the ESX hosts. If you do not upgrade your environment in the correct order, you might lose data and lose access to servers.

The following methods are supported for a direct upgrade to ESX 9.1.

- Use the interactive graphical user interface (GUI) installer from a CD, DVD, or USB drive.
- Perform a scripted upgrade.
- Use ESXCLI.
- Use vSphere Auto Deploy. If the ESX host is deployed by using vSphere Auto Deploy, you can use vSphere Auto Deploy to reprovision the ESX host with a 9.1 image.
- Use the vSphere Lifecycle Manager.

Graphical User Interface (GUI) Installer	You can upgrade interactively by using an ESX installer ISO image on a CD, DVD, or USB flash drive or by network booting the installer. This method is appropriate for deployments with a few hosts. If during the installation process, you select a target disk that contains an ESX installation, the installer upgrades the host to ESX version 9.1. The installer also gives you the option to migrate some existing ESX host settings and configuration files and to preserve the existing VMFS datastore. See Upgrade Hosts Interactively.
Perform a Scripted Upgrade	To do a scripted upgrade, you can use the ESX 9.1 installer from a CD, DVD, or USB flash drive or by network booting the installer. This method is an efficient way to deploy multiple hosts. For more information, see Installing or Upgrading Hosts by Using a Script.
ESXCLI	You can use ESXCLI to upgrade hosts to ESX 9.1. ESX 9.1 introduces configuration files, components, base images, and add-ons as new software deliverables that you can use to update or patch ESX 9.1 hosts. For information about managing components, base images, and add-ons on ESX, see *ESXCLI Concepts and Examples*. To use ESXCLI commands, you must install Standalone ESXCLI. For more information about installing and using ESXCLI, see the following documents. • Getting Started with ESXCLI • ESXCLI Concepts and Examples See Upgrading Hosts by Using ESXCLI Commands .
vSphere Auto Deploy	If an ESX host is deployed with vSphere Auto Deploy, you can use vSphere Auto Deploy to reprovision the host and reboot it with a new image profile or a configuration that you manage on a cluster level. An image profile contains an ESX upgrade or patch, a host configuration profile, and optionally, third-party drivers or management agents that are provided by VMware partners. To add ESX hosts to a cluster that manages ESX configuration at a cluster level, you create a rule in Auto Deploy that assigns such a cluster as the host location for newly added hosts, which inherit the same settings and do not require manual configuration.You can build custom images by using vSphere ESX Image Builder CLI. For more information, see Using vSphere Auto Deploy to Reprovision Hosts .
vSphere Lifecycle Manager	vSphere Lifecycle Manager is a vCenter service for installing, upgrading, and updating ESX hosts. By using images and baselines, vSphere Lifecycle Manager enables centralized and simplified lifecycle management for multiple ESX hosts at a cluster level. For more information about performing orchestrated installations, upgrades, and updates, see the *Managing Host and Cluster Lifecycle* documentation.

For more information, see Upgrading to VMware Cloud Foundation 9.1.x and Upgrading to vSphere Foundation 9.1.x.

Upgrading Virtual Machines and VMware Tools

After you upgrade an ESX host, you can upgrade the virtual machines on the host to take advantage of new features.

You have the following tools for upgrading virtual machines.

vSphere Client	You can use the vSphere Client to upgrade a virtual machine step by step. For more information about upgrading virtual machines, see the *vSphere Virtual Machine Administration* documentation.

vSphere Lifecycle Manager	You can use the vSphere Lifecycle Manager to upgrade the virtual machine hardware and VMware Tools versions of the virtual machines in your environment. The vSphere Lifecycle Manager automates the upgrade process and verifies that the steps occur in the correct order. For more information, see the *Managing Host and Cluster Lifecycle* documentation.

Before Upgrading ESX Hosts

For a successful upgrade of your ESX hosts, understand and prepare for the changes that are involved.

For a successful ESX upgrade, follow these best practices:

1. Make sure that you understand the ESX upgrade process, the effect of that process on your existing deployment, and the preparation required for the upgrade.
   - If your vSphere system includes VMware solutions or plug-ins, make sure they are compatible with the vCenter Server version that you are upgrading to. See the Broadcom Product Interoperability Matrix.
   - Read Overview of the ESX Host Upgrade Process to understand the upgrade scenarios that are supported, and the options and tools that are available to perform the upgrade.
   - Read the vCenter and ESX Release Notes for known installation issues.
2. Prepare the system for the upgrade.
   - Make sure that the current ESX version is supported for the upgrade. See Overview of the ESX Host Upgrade Process.
   - Make sure that the system hardware complies with ESX requirements. See ESX Requirements and Broadcom Compatibility Guide. Check for system compatibility, I/O compatibility with network and host bus adapter (HBA) cards, storage compatibility, and backup software compatibility.
   - Make sure that sufficient disk space is available on the host for the upgrade.
   - If a SAN is connected to the host, detach the Fibre Channel system before continuing with the upgrade. Do not deactivate HBA cards in the BIOS.
3. Back up the host before performing an upgrade. If the upgrade fails, you can restore the host.
4. If you are using Auto Deploy to provision hosts, the user who is running the process must have local administrator privileges on the ESX host that is being provisioned. By default the installation process has these privileges and certificate provisioning happens as expected. However, if you are using another method than the installer, you must run it as a user who has the local administrator privileges.
5. Depending on the upgrade option you choose, you might need to migrate or power off all virtual machines on the host. See the instructions for your upgrade method.
   - For an interactive upgrade from CD, DVD, or USB drive: see Upgrade Hosts Interactively.
   - For a scripted upgrade: see Installation and Upgrade Scripts Used for ESX Installation.
   - For vSphere Auto Deploy: see Using vSphere Auto Deploy to Reprovision Hosts . If the ESX 8.0x or 9.0.x host was deployed by using vSphere Auto Deploy, you can use vSphere Auto Deploy to reprovision the host with a 9.1 image.
   - For the `esxcli` command method: see Upgrading Hosts by Using ESXCLI Commands .
6. Plan for the tasks that must be performed after the ESX host upgrade:
   - Test the system to ensure that the upgrade completed successfully.
   - Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. After you license the vCenter instance connected to the ESX hosts, they are licensed automatically. For more information, see Licensing Overview in the *Licensing* guide.
   - Consider setting up a syslog server for remote logging, to ensure sufficient disk storage for log files. Setting up logging on a remote host is especially important for hosts with limited local storage. You can optionally install Log Management, which provides log aggregation and analytics. See Required Free Space for System Logging.
7. If the upgrade was unsuccessful and you backed up the host, you can restore the host.

Upgrading Hosts That Have Third-Party Custom VIBs

A host can have custom vSphere Installation Bundles (VIBs) installed, for example, third-party drivers or management agents.

When you upgrade an ESX host to 9.1, all supported custom VIBs are migrated, regardless of whether the VIBs are included in the installer ISO. If the host or the installer ISO image contains a VIB that creates a conflict and prevents the

upgrade, an error message identifies the VIB that created the conflict. To upgrade the host, take one of the following actions:

- Remove the VIB that created the conflict from the ESX host and retry the upgrade. You can remove a VIB from the host by using `esxcli` commands. For more information, see Remove VIBs from a Host .
- Use the vSphere ESX Image Builder CLI to create a custom installer ISO image that resolves the conflict. See Customizing Installations with vSphere ESX Image Builder.

Upgrading ESX Hosts in an Environment With VMware NSX

In an NSX setup, before an upgrade to ESX 9.0 and later, make sure that the NSX kernel module is part of the image or upgrade baseline.

If your vSphere system includes NSX, before you start an upgrade of your ESX hosts, you must ensure that the NSX kernel module is part of the desired software specification or baseline that you use for the upgrade. When you upgrade an ESX host to 9.0 or later, all supported custom VIBs are migrated, regardless of whether the VIBs are included in the installer ISO. However, the NSX kernel module is not automatically migrated to the installer ISO image. Before you proceed to the upgrade operation, see Broadcom knowledge base article 312013.

Upgrade Hosts Interactively

To upgrade ESX hosts to ESX 9.1, you can boot the ESX installer from a CD, DVD, or USB flash drive.

- Verify that the ESX installer ISO in one of the following locations.
  - On CD or DVD. If you do not have the installation CD or DVD, you can create one. See Download and Burn the ESX Installer ISO Image to a CD or DVD.
  - On a USB flash drive. See Format a USB Flash Drive to Boot the ESX Installation or Upgrade.

    Note:

    You can also use PXE to boot the ESX installer to run an interactive installation or a scripted installation. See Overview of the Network Boot Installation Process.
- Verify that the server hardware clock is set to UTC. This setting is in the system BIOS or UEFI.
- If you are upgrading an ESX host, supported custom VIBs that are not included in the ESX installer ISO are migrated. See Upgrading Hosts That Have Third-Party Custom VIBs
- See your hardware vendor documentation for information about changing the boot order.

Before upgrading, consider disconnecting the network storage. This action decreases the time it takes the installer to search for available disk drives. When you disconnect network storage, any files on the disconnected disks are unavailable at installation. Do not disconnect a LUN that contains an existing ESX installation.

Note:

Upgrading an ESX host with a data processing unit (DPU) is not supported through the interactive or scripted installer. To upgrade such hosts, use vSphere Lifecycle Manager or ESXCLI.

1. Insert the ESX installer CD or DVD in the CD-ROM or DVD-ROM drive, or attach the Installer USB flash drive and restart the machine.
2. Set the BIOS or UEFI to boot from the CD-ROM device or the USB flash drive.
3. In the **Select a Disk** panel, select the drive on which to install or upgrade ESX and press Enter.

   Press F1 for information about the selected disk.

   Note:
   Do not rely on the disk order in the list to select a disk. The disk order is determined by the BIOS or UEFI. On systems where drives are continuously being added and removed, they might be out of order.
4. Upgrade or install ESX if the installer finds an existing ESX installation and VMFS datastore.

   If an existing VMFS datastore cannot be preserved, you can choose only to install ESX and overwrite the existing VMFS datastore, or to cancel the installation. If you choose to overwrite the existing VMFS datastore, back up the datastore first.

5. Press F11 to confirm and start the upgrade.
6. Remove the installation CD or DVD or USB flash drive when the upgrade is complete.
7. Press **Enter** to reboot the host.
8. Set the first boot device to be the drive which you selected previously when you upgraded ESX.

Installing or Upgrading Hosts by Using a Script

You can quickly deploy ESX hosts by using scripted, unattended installations or upgrades.

Scripted installations or upgrades provide an efficient way to deploy multiple hosts. The installation or upgrade script contains the installation settings for ESX. You can apply the script to all hosts that you want to have a similar configuration.

For a scripted installation or upgrade, you must use the supported commands to create a script. You can edit the script to change settings that are unique for each host.

> **Note:**
>
> Upgrading an ESX host with a data processing unit (DPU) is not supported through the interactive or scripted installer. To upgrade such hosts, use vSphere Lifecycle Manager or ESXCLI.

The installation or upgrade script can reside in one of the following locations:

- FTP server
- HTTP/HTTPS server
- NFS server
- USB flash drive
- CD-ROM drive

In the ESX Installation and Setup guide, see the following topics:

- Enter Boot Options to Run an Installation or Upgrade Script
- Boot Options
- About the boot.cfg File
- Locations Supported for Installation or Upgrade Scripts
- Path to the Installation or Upgrade Script
- Installation and Upgrade Script Commands
- Install or Upgrade ESX from a CD or DVD by Using a Script
- Install or Upgrade ESX from a USB Stick by Using a Script
- Network Boot the ESX Installer for a Scripted Installation or Upgrade

How to Upgrade Hosts by Using ESXCLI Commands

You can use ESXCLI commands to upgrade hosts.

In this chapter see:

- Upgrading Hosts by Using ESXCLI Commands
- See if ESX Host Update Requires Maintenance Mode or Reboot
- Place a Host in Maintenance Mode
- Update a Host with Individual VIBs
- Upgrade or Update a Host with Image Profiles
- Update ESX Hosts by Using Zip Files
- Remove VIBs from a Host
- Add Third-Party Extensions to Hosts with an ESXCLI Command
- Perform a Dry Run of an ESXCLI Installation or Upgrade
- List the Available VIBs and Profiles After Host Reboot
- Display the Image Profile and Acceptance Level of the Host

Upgrading Hosts by Using ESXCLI Commands

By using ESXCLI, you can upgrade to version 9.1 and update or patch ESX 9.1 hosts.

ESX 9.1 introduces configuration files, components, base images, and add-ons as new software deliverables that you can use to update or patch ESX9.1 hosts. For information about managing components, base images, and add-ons on ESX, see ESXCLI Concepts and Examples.

To use ESXCLI commands, you must install Standalone ESXCLI. For more information about installing and using ESXCLI, see the content under the Getting Started with ESXCLI chapter in VMware vSphere SDKs and Tools 9.1.

Note:

If you press Ctrl+C while an `esxcli` command is running, the command-line interface exits to a new prompt without displaying a message. However, the command continues to run to completion.

For ESX hosts deployed with vSphere Auto Deploy, the tools VIB must be part of the base booting image used for the initial Auto Deploy installation. The tools VIB cannot be added separately later.

VIBs, Image Profiles, and Software Depots

Upgrading ESX with esxcli commands requires an understanding of VIBs, image profiles, and software depots.

The following technical terms are used throughout the vSphere documentation set in discussions of installation and upgrade tasks.

VIB	A VIB is an ESXi software package. VMware and its partners package solutions, drivers, CIM providers, and applications that extend the ESXi platform as VIBs. VIBs are available in software depots. You can use VIBs to create and customize ISO images or to upgrade ESXi hosts by installing VIBs asynchronously onto the hosts.
Image Profile	An image profile defines an ESXi image and consists of VIBs. An image profile always includes a base VIB, and might include more VIBs. You examine and define an image profile by using vSphere ESXi Image Builder.
Software Depot	A software depot is a collection of VIBs and image profiles. The software depot is a hierarchy of files and folders and can be available through an HTTP URL (online depot) or a ZIP file (offline depot). VMware and VMware partners make depots available. Companies with large VMware installations might create internal depots to provision ESXi hosts with vSphere Auto Deploy, or to export an ISO for ESXi installation.

Understanding Acceptance Levels for VIBs and Hosts

Each VIB is released with an acceptance level that cannot be changed. The host acceptance level determines which VIBs can be installed to a host.

The acceptance level applies to individual VIBs installed by using the `esxcli software vib install` and `esxcli software vib update` commands, to VIBs installed using vSphere Lifecycle Manager, and to VIBs in image profiles.

The acceptance level of all VIBs on a host must be at least as high as the host acceptance level. For example, if the host acceptance level is VMwareAccepted, you can install VIBs with acceptance levels of VMwareCertified and VMwareAccepted, but you cannot install VIBs with acceptance levels of PartnerSupported or CommunitySupported. To install a VIB with a restrictive acceptance level that is less than the acceptance level of the host, you can change the setting of the host by using the vSphere Client or by running `esxcli software acceptance` commands.

Setting host acceptance levels is a best practice that allows you to specify which VIBs can be installed on a host and used with an image profile, and the level of support you can expect for a VIB. For example, you might set a more restrictive acceptance level for hosts in a production environment than for hosts in a testing environment.

VMware supports the following acceptance levels.

VMwareCertified	The VMwareCertified acceptance level has the most stringent requirements. VIBs with this level go through thorough testing fully equivalent to VMware in-house Quality Assurance testing for the same technology. Today, only I/O Vendor Program (IOVP) program drivers are published at this level. VMware takes support calls for VIBs with this acceptance level.

VMwareAccepted	VIBs with this acceptance level go through verification testing, but the tests do not fully test every function of the software. The partner runs the tests and VMware verifies the result. Today, CIM providers and PSA plug-ins are among the VIBs published at this level. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.
PartnerSupported	VIBs with the PartnerSupported acceptance level are published by a partner that VMware trusts. The partner performs all testing. VMware does not verify the results. This level is used for a new or nonmainstream technology that partners want to enable for VMware systems. Today, driver VIB technologies such as Infiniband, ATAoE, and SSD are at this level with nonstandard hardware drivers. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.
CommunitySupported	The CommunitySupported acceptance level is for VIBs created by individuals or companies outside of VMware partner programs. VIBs at this level have not gone through any VMware-approved testing program and are not supported by VMware Technical Support or by a VMware partner.

Table 31: VIB Acceptance Levels Required to Install on Hosts

Host Acceptance Level	VMwareCertified VIB	VMwareAccepted VIB	PartnerSupported VIB	CommunitySupported VIB
VMwareCertified	x			
VMwareAccepted	x	x		
PartnerSupported	x	x	x	
CommunitySupported	x	x	x	x

Match a Host Acceptance Level with an Update Acceptance Level

You can change the host acceptance level to match the acceptance level for a VIB or image profile that you want to install. The acceptance level of all VIBs on a host must be at least as high as the host acceptance level.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run esxcli commands in the ESX Shell.

Use this procedure to determine the acceptance levels of the host and the VIB or image profile to install, and to change the acceptance level of the host, if necessary for the update.

When you specify a target server by using `--server=<server_name>`, the server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see Getting Started with ESXCLI, or run `esxcli --help` at the ESXCLI command prompt.

1. Retrieve the acceptance level for the VIB or image profile.

Option	Description
List information for all VIBs	`esxcli --server=<server_name> software sources vib list --depot=<depot_URL>`
List information for a specified VIB	`esxcli --server=<server_name> software sources vib list --viburl=<vib_URL>`
List information for all image profiles	`esxcli --server=<server_name> software sources profile list --depot=<depot_URL>`
List information for a specified image profile	`esxcli --server=<server_name> software sources profile get --depot=<depot_URL> --profile=<profile_name>`

2. Retrieve the host acceptance level.

   ```
   esxcli --server=<server_name> software acceptance get
   ```

3. Optional: If the acceptance level of the VIB is more restrictive than the acceptance level of the host, change the acceptance level of the host.

   ```
   esxcli --server=<server_name> software acceptance set --level=<acceptance_level>
   ```

   The *acceptance_level* can be `VMwareCertified`, `VMwareAccepted`, `PartnerSupported`, or `CommunitySupported`. The values for *acceptance_level* are case-sensitive.

Note:
You can use the --force option for the `esxcli software vib` or `esxcli software profile` command to add a VIB or image profile with a lower acceptance level than the host. A warning appears. Because your setup is no longer consistent, the warning is repeated when you install VIBs, remove VIBs, and perform certain other operations on the host.

See if ESX Host Update Requires Maintenance Mode or Reboot

VIBs that you can install with a live installation do not require the host to be rebooted, but might require the host to be placed in maintenance mode.

Other VIBs and profiles might require the host to be rebooted after the installation or update.

Install vCLI or deploy the vSphere Management Assistant (vMA) virtual machine. See the chapter Getting Started with ESXCLI. For troubleshooting, run ESXCLI commands in the ESX Shell.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Check whether the VIB or image profile that you want to install requires the host to be placed in maintenance mode or to be rebooted after the installation or update.

   Run one of the following commands.

Option	Description
Check the VIB	`esxcli --server=<server_name> software sources vib get -v <vib_URL>`
Check the VIBs in a depot	`esxcli --server=<server_name> software sources vib get --depot=<depot_name>`
Check the image profile in a depot	`esxcli --server=<server_name> software sources profile get --depot=<depot_name>`

2. Review the return values.

   The return values, which are read from the VIB metadata, indicate whether the host must be in maintenance mode before installing the VIB or image profile, and whether installing the VIB or profile requires the host to be rebooted.

   Note:
   vSphere Lifecycle Manager relies on an internal ESX software scan API to determine whether maintenance mode is required or not. When you install a VIB on a live system, if the value for `Live-Install-Allowed` is set to false, the installation result instructs vSphere Lifecycle Manager to reboot the host. When you remove a VIB from a live system, if the value for `Live-Remove-Allowed` is set to false, the removal result instructs vSphere Lifecycle Manager to reboot the host. In either case, when the remediation starts, vSphere Lifecycle Manager automatically puts the host into maintenance mode.

If necessary, place the host in maintenance mode. See Place a Host in Maintenance Mode. If a reboot is required, and if the host belongs to a VMware HA cluster, remove the host from the cluster or disable HA on the cluster before the installation or update. Also, place the host in maintenance mode to minimize boot disk activity during the upgrade.

Place a Host in Maintenance Mode

Some installation and update operations that use live install require the host to be in maintenance mode.

Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Maintenance mode is required when an update operation requires a reboot. However, you only put the host in maintenance mode manually when you use esxcli commands for update and upgrade operations.

To determine whether an upgrade operation requires the host to be in maintenance mode, see See if ESX Host Update Requires Maintenance Mode or Reboot

Note:

If the host is a member of a vSAN cluster, and any virtual machine object on the host uses the "Number of failures to tolerate=0" setting in its storage policy, the host might experience unusual delays when entering maintenance mode. The delay occurs because vSAN has to evacuate this object from the host for the maintenance operation to complete successfully.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Check whether the host is in maintenance mode.

   ```
   esxcli --server=<server_name> system maintenanceMode get
   ```

2. Power off each virtual machine running on the ESX host.

   Note:
   You can list all running virtual machines and retrieve the World ID of each one by running the following command.

   ```
   esxcli --server=<server_name> vm process list
   ```

Option	Command
To shut down the guest operating system and then power off the virtual machine	`esxcli --server=<server_name> vm process kill --type soft --world-id <vm_ID>`
To power off the virtual machine immediately	`esxcli --server=<server_name> vm process kill --type hard --world-id <vm_ID>`
To force the power off operation	`esxcli --server=<server_name> vm process kill --type force --world-id <vm_ID>`

   Alternatively, to avoid powering off virtual machines, you can migrate them to another host. See the topic about migrating virtual machines in the *vCenter and Host Management* documentation.

3. Place the host in maintenance mode.

   ```
   esxcli --server=<server_name> system maintenanceMode set --enable true
   ```

4. Verify that the host is in maintenance mode.

   ```
   esxcli --server=<server_name> system maintenanceMode get
   ```

Update a Host with Individual VIBs

You can update a host with VIBs stored in a software depot that is accessible through a URL or in an offline ZIP depot.

- Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.
- Determine whether the update requires the host to be in maintenance mode or to be rebooted. If necessary, place the host in maintenance mode.
  See See if ESX Host Update Requires Maintenance Mode or Reboot. See Place a Host in Maintenance Mode.
- If the update requires a reboot, and if the host belongs to a vSphere HA cluster, remove the host from the cluster or deactivate HA on the cluster.

Important:

If you are updating ESX from a zip bundle in a VMware-supplied depot, either online from the Broadcom Support Portal or downloaded locally, VMware supports only the update method specified for VMware-supplied depots in the topic Upgrade or Update a Host with Image Profiles .

Note:

The `esxcli software vib update` and `esxcli software vib install` commands are not supported for upgrade operations. See Upgrade or Update a Host with Image Profiles .

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Determine which VIBs are installed on the host.

   ```
   esxcli --server=<server_name> software vib list
   ```

2. Find out which VIBs are available in the depot.

Option	Description
From a depot accessible by URL	`esxcli --server=<server_name> software sources vib list --depot=http://<web_server>/<depot_name>`
From a local depot ZIP file	`esxcli --server=<server_name> software sources vib list --depot=<absolute_path_to_depot_zip_file>`

   You can specify a proxy server by using the --proxy option.

3. Update the existing VIBs to include the VIBs in the depot or install new VIBs.

Option	Description
Update VIBs from a depot accessible by URL	`esxcli --server=<server_name> software vib update --depot=http://<web_server>/<depot_name>`
Update VIBs from a local depot ZIP file	`esxcli --server=<server_name> software vib update --depot=<absolute_path_to_depot_ZIP_file>`
Install all VIBs from a ZIP file on a specified offline depot (includes both VMware VIBs and partner-supplied VIBs)	`esxcli --server=<server_name> software vib install --depot <path_to_VMware_vib_ZIP_file> \<VMware_vib_ZIP_file> --depot <path_to_partner_vib_ZIP_file> \<partner_vib_ZIP_file>`

   Options for the `update` and `install` commands allow you to perform a dry run, to specify a specific VIB, to bypass the acceptance level verification, and so on. Do not bypass verification on production systems. See the *ESXCLI Reference*.

4. Verify that the VIBs are installed on your ESX host.

   ```
   esxcli --server=<server_name> software vib list
   ```

Upgrade or Update a Host with Image Profiles

You can upgrade or update a host with image profiles stored in a software depot that is accessible through a URL or in an offline ZIP depot.

- Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.
- Determine whether the update requires the host to be in maintenance mode or to be rebooted. If necessary, place the host in maintenance mode.
  See See if ESX Host Update Requires Maintenance Mode or Reboot. See Place a Host in Maintenance Mode.

  Important:
  When you use ESXCLI to update or upgrade the host, put the host manually in maintenance mode to ensure that the boot disk is not actively in use before the upgrade begins.
- If the update requires a reboot, and if the host belongs to a vSphere HA cluster, remove the host from the cluster or deactivate HA on the cluster.

You can use the `esxcli software profile update` or `esxcli software profile install` command to upgrade or update an ESX host.

When you upgrade or update a host, the `esxcli software profile update` or `esxcli software profile install` command applies a later version (major or minor) of a full image profile onto the host. After this operation and a reboot, the host can join a vCenter environment of the same or later version.

The `esxcli software profile update` command brings the entire content of the ESX host image to the same level as the corresponding upgrade method using an ISO installer. However, the ISO installer performs a pre-upgrade check for potential problems, such as insufficient memory or unsupported devices.

For more about the ESX upgrade process and methods, see Overview of the ESX Host Upgrade Process.

Important:

If you are upgrading or updating ESX from a VMware online depot or an offline .zip bundle downloaded locally from the Broadcom Support Portal, VMware supports only the update command `esxcli software profile update --depot=<depot_location> --profile=<profile_name>`. ESXCLI commands support 2 types of depot location: either a full path to a locally downloaded offline .zip bundle, such as `/vmfs/volumes/<datastore_name>/<depot_ name>.zip`, or an `http/https` link to the `index.xml` in an online depot, such as `https//<web_server>/<depot_name>/index.xml`. You can extract an offline .zip bundle and host it as an online depot on a private `http/https` server accessible to an ESX host.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

Note:

Options to the `update` and `install` commands allow you to perform a dry run, to bypass acceptance level verification, to ignore hardware compatibility check warnings, and so on. The option to bypass hardware compatibility check warnings is only available for ESX 6.7 Update 1 or later. Do not bypass verification on production systems.

For options help, type `esxcli software profile install --help` or `esxcli software profile update --help`. For the complete listing of available command-line options, see the ESXCLI Reference.

1. Determine which VIBs are installed on the host.

   ```
   esxcli --server=<server_name> software vib list
   ```

2. Determine which image profiles are available in the depot.

   ```
   esxcli --server=<server_name> software sources profile list --depot=http://<depot_location>/<depot_name>
   ```

   You can specify a proxy server by using the --proxy option.

3. Update the existing image profile to include the VIBs or install new VIBs.

Option	Description
Update the image profile from a VMware-supplied ZIP bundle, in a depot, accessible online from the Broadcom Support Portal or downloaded to a local depot	`esxcli software profile update --depot=<depot_location> --profile=<profile_name>` **Important:** This is the only update method that VMware supports for zip bundles supplied by VMware. VMware-supplied ZIP bundle names take the form: `VMware-ESX-<version_number>-<build_number>-depot.zip`. The profile name for VMware-supplied zip bundles takes one of the following forms. • `ESX-<version_number>-<build_number>-standard` • `ESX-<version_number>-<build_number>-notools` (does not include VMware Tools)
Update the image profile from a depot accessible by URL	`esxcli --server=<server_name> software profile update --depot=http://<depot_location>/<depot_name> --profile=<profile_name>`

Option	Description
Update the image profile from ZIP file stored locally on the target server	`esxcli --server=<server_name> software profile update --depot=file:///<path_to_profile_ZIP_file>/<profile_ZIP_file> --profile=<profile_name>`
Update the image profile from a ZIP file on the target server, copied into a datastore	`esxcli --server=<server_name> software profile update --depot=<datastore_name>/<profile_ZIP_file> --profile=<profile_name>`
Update the image profile from a ZIP file copied locally and applied on the target server	`esxcli --server=<server_name> software profile update --depot=/<root_dir>/<path_to_profile_ZIP_file>/<profile_ZIP_file> --profile=<profile_name>`
Install all new VIBs in a specified profile accessible by URL	`esxcli --server=<server_name> software profile install --depot=http://<depot_location>/<depot_name> --profile=<profile_name>`
Install all new VIBs in a specified profile from a ZIP file stored locally on the target.	`esxcli --server=<server_name> software profile install --depot=file:///<path_to_profile_ZIP_file>/<profile_ZIP_file> --profile=<profile_name>`
Install all new VIBs from a ZIP file on the target server, copied into a datastore	`esxcli --server=<server_name> software profile install --depot=<datastore_name>/<profile_ZIP_file> --profile=<profile_name>`
Install all new VIBs from a ZIP file copied locally and applied on the target server	`esxcli --server=<server_name> software profile install --depot=/<root_dir>/<path_to_profile_ZIP_file>/<profile_ZIP_file> --profile=<profile_name>`

Important:
The `software profile update` command updates existing VIBs with the corresponding VIBs from the specified profile, but does not affect other VIBs installed on the target server. The `software profile install` command installs the VIBs present in the depot image profile, and removes any other VIBs installed on the target server.

4. Reboot the ESX host immediately if in the ESXCLI output you see `Reboot Required: true`.

   Note:
   If you do not reboot the ESX host and continue, further modifications to the host, such as installation of a VIB, might fail or not take effect due to the pending reboot.

5. Verify that the VIBs are installed on your ESX host.

   ```
   esxcli --server=<server_name> software vib list
   ```

Update ESX Hosts by Using Zip Files

You can update hosts with VIBs or image profiles by downloading a ZIP file of a depot.

- Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.
- Download the ZIP file of a depot bundle from a third-party VMware partner.
- Determine whether the update requires the host to be in maintenance mode or to be rebooted. If necessary, place the host in maintenance mode.
  See See if ESX Host Update Requires Maintenance Mode or Reboot. See Place a Host in Maintenance Mode.
- If the update requires a reboot, and if the host belongs to a vSphere HA cluster, remove the host from the cluster or deactivate HA on the cluster.

VMware partners prepare third-party VIBs to provide management agents or asynchronously released drivers.

Important:
If you are upgrading or updating ESX from a zip bundle in a VMware-supplied depot, either online from the Broadcom Support Portal or downloaded locally, the `esxcli software vib update` and `esxcli software vib install` commands are not supported. VMware supports only the update and upgrade methods specified for VMware-supplied depots in the topic Upgrade or Update a Host with Image Profiles .If the depot is not supplied by VMware and does not have an image profile, you can use the `esxcli software vib update` and `esxcli software vib install` commands for individual or multiple VIB updates as described in Update a Host with Individual VIBs and the procedure in this topic.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

Install the ZIP file.

```
esxcli --server=<server_name> software vib update --depot=/<path_to_vib_ZIP>/<ZIP_file_name>.zip
```

Remove VIBs from a Host

You can uninstall third-party VIBs or VMware VIBs from your ESX host, unless the VIB is part of a Component that is required for the ESX lifecycle or a driver.

- If the removal requires a reboot, and if the host belongs to a vSphere HA cluster, deactivate HA for the host.
- Determine whether the update requires the host to be in maintenance mode or to be rebooted. If necessary, place the host in maintenance mode.
  See See if ESX Host Update Requires Maintenance Mode or Reboot. See Place a Host in Maintenance Mode.
- Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

  Note:

  To ensure that the boot disk is not actively in use when you use ESXCLI to update or upgrade the host, put the host manually in maintenance mode.

VMware partners prepare third-party VIBs to provide management agents or asynchronously released drivers.

Important:
Starting with ESX 8.0 Update 2, removing some VIBs from their Components can cause runtime issues with ESX.

Table 32: VIBs required for ESX image completeness.

Component	VIBs
ESX	bmcal
	bmcal-esxio
	clusterstore
	cpu-microcode
	crx
	drivervm-gpu
	esx-base
	esx-dvfilter-generic-fastpath
	esx-ui
	esx-xserver
	esxio
	esxio-base
	esxio-combiner

Component	VIBs
	esxio-combiner-esxio
	esxio-dvfilter-generic-fastpath
	gc
	gc-esxio
	native-misc-drivers
	infravisor
	native-misc-drivers-esxio
	pensandoatlas
	trx
	vdfs
	vsan
	vsanhealth
esx-update	esx-update
	loadesx
esxio-update	esxio-update
	loadesxio
Intel-ne1000	ne1000
Mellanox-nmlx5	nmlx5-core
	nmlx5-rdma
	nmlx5-rdma-esxio
	nmlx5-core-esxio
	nmlx5-cc
	nmlx5-cc-esxio
Mellanox-nmlxbf-gige	nmlxbf-gige-esxio
nipmi	nipmi
Pensando-ionic-en	ionic-en-esxio
	ionic-en
VMware-ahci	vmw-ahci
VMware-bfedac	bfedac-esxio
VMware-dwi2c	dwi2c
	dwi2c-esxio
VMware-iser	iser
VMware-mlnx-bfbootctl	mlnx-bfbootctl-esxio
VMware-mnet	mnet-esxio
VMware-nmlxbf-pmc	nmlxbf-pmc-esxio
VMware-NVMe-PCIe	nvme-pcie
	nvme-pcie-esxio
VMware-nvme-pcie-plugin	lsuv2-nvme-pcie-plugin
VMware-nvme-plugin	vmware-esx-esxcli-nvme-plugin-esxio
	vmware-esx-esxcli-nvme-plugin

Component	VIBs
VMware-NVMeoF-RDMA	nvmerdma
VMware-NVMeoF-TCP	nvmetcp
	nvmetcp-esxio
VMware-nvmxnet3	nvmxnet3
	nvmxnet3-esxio
VMware-nvmxnet3-ens	nvmxnet3-ens-esxio
	nvmxnet3-ens
VMware-penedac	penedac-esxio
VMware-pengpio	pengpio-esxio
VMware-penspi	penspi-esxio
VMware-pvscsi	pvscsi
	pvscsi-esxio
VMware-rd1173	rd1173-esxio
VMware-SDHCI-Driver	vmksdhci
	vmksdhci-esxio
VMware-spidev	spidev-esxio
VMware-vmkata	vmkata
VMware-vmkusb	vmkusb
	vmkusb-esxio

1. Power off each virtual machine running on the ESX host.

   Note:
   You can list all running virtual machines and retrieve the World ID of each one by running the following command.
   ```
   esxcli --server=<server_name> vm process list
   ```

Option	Command
To shut down the guest operating system and then power off the virtual machine	`esxcli --server=<server_name> vm process kill --type soft --world-id <vm_ID>`
To power off the virtual machine immediately	`esxcli --server=<server_name> vm process kill --type hard --world-id <vm_ID>`
To force the power off operation	`esxcli --server=<server_name> vm process kill --type force --world-id <vm_ID>`

   Alternatively, to avoid powering off virtual machines, you can migrate them to another host. See the topic about migrating virtual machines in the *vCenter and Host Management* documentation.

2. Place the host in maintenance mode.
   ```
   esxcli --server=<server_name> system maintenanceMode set --enable true
   ```
3. If necessary, shut down or migrate virtual machines.
4. Determine which VIBs are installed on the host.
   ```
   esxcli --server=<server_name> software vib list
   ```
5. Remove the VIB.
   ```
   esxcli --server=<server_name> software vib remove --vibname=<name>
   ```
   Specify one or more VIBs to remove in one of the following forms.

- `<name>`
- `<name>:<version>`
- `<vendor>:<name>`
- `<vendor>:<name>:<version>`

For example, the command to remove a VIB specified by vendor, name and version can take the following form.

```
esxcli --server myEsxiHost software vib remove --
vibname=PatchVendor:patch42:version3
```

Note:
The `remove` command supports several more options. See the *ESXCLI Reference*.

Add Third-Party Extensions to Hosts with an ESXCLI Command

You can use the `esxcli software vib` command to add a third-party extension released as a VIB package to the system.

When you use this command, the VIB system updates the firewall rule set and refreshes the host daemon after you reboot the system.

Otherwise, you can use a firewall configuration file to specify port rules for host services to enable for the extension. The *vSphere Security* documentation discusses how to add, apply, and refresh a firewall rule set and lists the `esxcli network firewall` commands.

Perform a Dry Run of an ESXCLI Installation or Upgrade

You can use the --dry-run option to preview the results of an installation or upgrade operation.

A dry run of the installation or update procedure does not make any changes, but reports the VIB-level operations that will be performed if you run the command without the --dry-run option.

Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Enter the installation or upgrade command, adding the --dry-run option.
   - `esxcli --server=<server_name> software vib install --dry-run`
   - `esxcli --server=<server_name> software vib update --dry-run`
   - `esxcli --server=<server_name> software profile install --dry-run`
   - `esxcli --server=<server_name> software profile update --dry-run`
2. Review the output that is returned.
   The output shows which VIBs will be installed or removed and whether the installation or update requires a reboot.

List the Available VIBs and Profiles After Host Reboot

You can use the --rebooting-image option to list the VIBs and profiles that are installed on the host and will be active after the next host reboot.

Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Enter one of the following commands.

Option	Description
For VIBs	`esxcli --server=<server_name> software vib list --rebooting-image`
For Profiles	`esxcli --server=<server_name> software profile get --rebooting-image`

2. Review the output that is returned.

   The output displays information for the ESX image that will become active after the next reboot. If the pending-reboot image has not been created, the output returns nothing.

Display the Image Profile and Acceptance Level of the Host

You can use the `software profile get` command to display the currently installed image profile and acceptance level for the specified host.

Install Standalone ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

This command also shows details of the installed image profile history, including profile modifications.

When you specify a target server by using `--server=<server_name>` in the procedure, the specified server prompts you for a user name and password. Other connection options, such as a configuration file or session file, are supported. For a list of connection options, see ESXCLI Syntax, or run `esxcli --help` at the vCLI command prompt.

1. Enter the following command.

   ```
   esxcli --server=<server_name> software profile get
   ```

2. Review the output.

Using vSphere Auto Deploy to Reprovision Hosts

You can reprovision ESXi hosts deployed by using vSphere Auto Deploy with a new image profile for a different version of ESXi.

> **Note:**
> If you upgrade the host to use an ESX 6.0 or later image, the vSphere Auto Deploy server provisions the ESX host with certificates that are signed by VMCA. If you are currently using custom certificates, you can set up the host to use the custom certificates after the upgrade. See *vSphere Security*.

If a host was deployed using vSphere Auto Deploy, you can use vSphere Auto Deploy to reprovision the host with a new image profile that contains a different version of ESX. You can use vSphere ESXi Image Builder to create and manage image profiles. The vSphere Auto Deploy server is automatically upgraded if you upgrade the corresponding vCenter system. Starting with version 6.0, the vSphere Auto Deploy server is always on the same management node as the vCenter system.

For more information, see Install and Configure vSphere Auto Deploy in the ESXi Installation guide:

- vSphere Auto Deploy Boot Process
- Reprovisioning Hosts
- Provision ESX Host with an Image Profile Without VMware Tools

After You Upgrade ESX Hosts

To complete a host upgrade, you ensure that the host is reconnected to its managing vCenter system and reconfigured if necessary. You also check that the host is licensed correctly.

After you upgrade an ESX host, take the following actions:

- View the upgrade logs. You can use the vSphere Client to export the log files.

- If a vCenter system manages the host, you must reconnect the host to vCenter by right-clicking the host in the vCenter inventory and selecting **Connect**.
- When the upgrade is complete, the ESX host is in evaluation mode. The evaluation period is 90 days. Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. After you license the vCenter instance connected to the ESX hosts, they are licensed automatically. For more information, see Licensing Overview in the *Licensing* guide.
- The host sdX devices might be renumbered after the upgrade. If necessary, update any scripts that reference sdX devices.
- Upgrade virtual machines on the host. See Upgrading Virtual Machines and VMware Tools.
- Set up the vSphere Authentication Proxy service. Earlier versions of the vSphere Authentication Proxy are not compatible with vSphere 9.1. See the *vSphere Security* documentation for details about configuring the vSphere Authentication Proxy service.

Licensing ESX Hosts After Upgrade

Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client.

After you license the vCenter instance connected to the ESX hosts, they are licensed automatically. For more information, see Licensing Overview in the *Licensing* guide.

Run the Secure Boot Validation Script After ESX Upgrade

After you upgrade an ESX host from a version that does not support UEFI secure boot, you must check if you can activate secure boot.

After you upgrade an ESX host from an older version of ESX that did not support UEFI secure boot, you might be able to activate secure boot. Whether you can activate secure boot depends on how you performed the upgrade and whether the upgrade replaced all the existing VIBs or left some VIBs unchanged. You can run a validation script after you perform the upgrade to determine whether the upgraded installation supports secure boot.

- Verify that the hardware supports UEFI secure boot.
- Verify that all VIBs are signed with an acceptance level of at least PartnerSupported. If you include VIBs at the CommunitySupported level, you cannot use secure boot.

For secure boot to succeed, the signature of every installed VIB must be available on the system. Older versions of ESX do not save the signatures when installing VIBs.

- If you upgrade using ESXCLI commands, the old version of ESX performs the installation of the new VIBs, so their signatures are not saved and secure boot is not possible.
- If you upgrade using the ISO, new VIBs do have their signatures saved. This is true also for vSphere Lifecycle Manager upgrades that use the ISO.
- If old VIBs remain on the system, the signatures of those VIBs are not available and secure boot is not possible.
  - If the system uses a third-party driver, and the VMware upgrade does not include a new version of the driver VIB, then the old VIB remains on the system after upgrade.
  - In rare cases, VMware might drop ongoing development of a specific VIB without providing a new VIB that replaces or obsoletes it, so the old VIB remains on the system after upgrade.

  Note:
  UEFI secure boot also requires an up-to-date bootloader. This script does not check for an up-to-date bootloader.

1. Upgrade the ESX and run the following command.

   ```
   /usr/lib/vmware/secureboot/bin/secureBoot.py -c
   ```

2. Check the output.

   The output either includes `Secure boot can be enabled` or `Secure boot CANNOT be enabled`.

Configure System Logging

ESX hosts run the syslog service (vmsyslogd) that writes messages from system components to log files and can forward messages to syslog collectors. You can configure the amount and location of the logs. You can also create and apply log filters to modify the logging policy of an ESX host.

For configuring the system logs on ESX hosts, see the following topics:

- Configure Syslog on ESX Hosts
- Fine-tune Syslog on ESX Hosts
- ESX Syslog Options
- Protocols, Formats, and Framing of ESX Syslog Messages
- Log File Formats
- ESX Syslog Message Transmission Formats
- Configure Log Filtering on ESX Hosts

When configuring remote hosts for syslog, you also need to open some specified ports in the ESX host firewall to allow the transmission of log messages. For more information, see Opening the firewall for syslog emission to remote hosts.

Collect Logs for Troubleshooting Failed ESX Upgrades

If an installation or upgrade fails, you can collect installation or upgrade log files for ESX that help you identify the source of the failure.

To collect files that help you in troubleshooting failed ESX upgrades, follow these steps:

- Enter the `vm-support` command in the ESX Shell or through SSH.
- Navigate to the `/var/tmp/directory`.
- Retrieve the log files from the `.tgz` file.

vCenter Deployment and Setup

vCenter Deployment and Setup describes how to deploy the VMware vCenter® appliance.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

vCenter Deployment and Setup is for anyone who must install and configure VMware vSphere®. These topics are for experienced Microsoft Windows or Linux system administrators who are familiar with virtual machine technology and data center operations.

Introduction to vSphere Deployment and Setup

The two core components of vSphere are ESX and vCenter. ESX is the virtualization platform on which you can create and run virtual machines and virtual appliances. vCenter is a service that acts as a central administrator for ESX hosts connected in a network. vCenter lets you pool and manage the resources of multiple hosts.

You deploy the vCenter appliance, as a preconfigured virtual machine optimized for running vCenter and associated services. You can deploy the vCenter appliance on ESX hosts or on vCenter instances.

For detailed information about the vCenter deployment process, see vCenter Deployment and Setup.

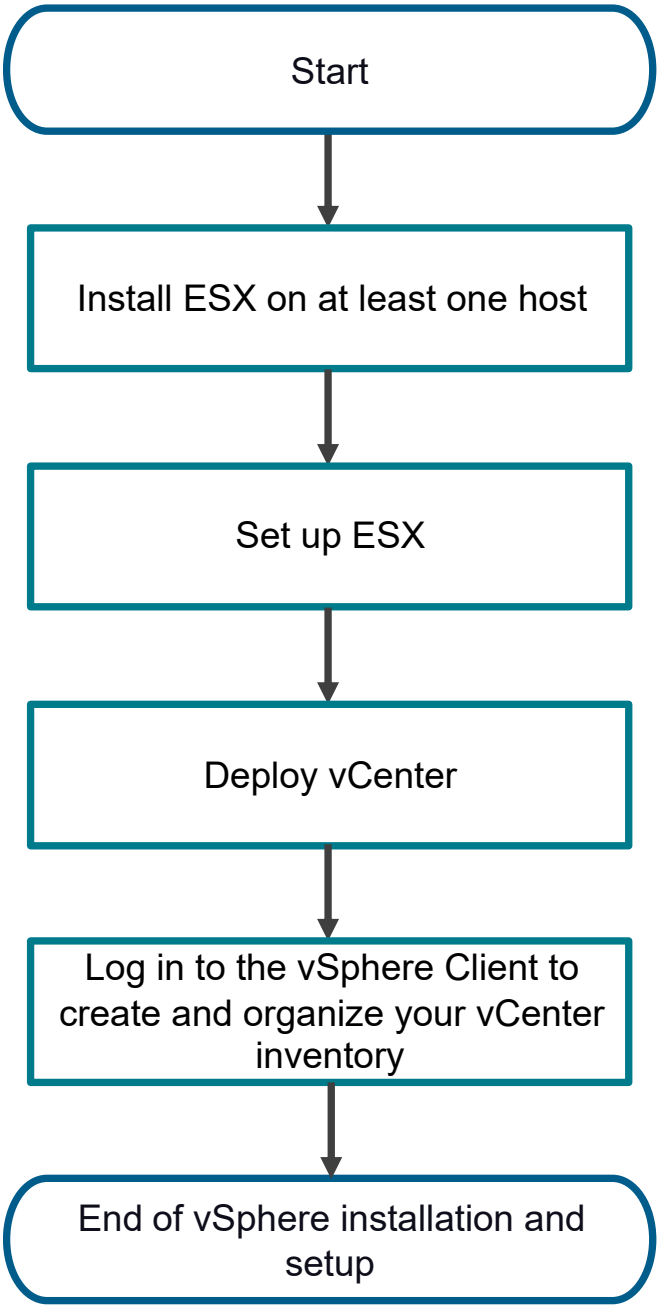
ESX Licensing

Starting with ESX 9.0, you no longer license ESX hosts by using the vSphere Client. For more information on how to license your environment, see Licensing Overview in the *Licensing* guide.

How to Deploy and Set Up vSphere

vSphere is a sophisticated product with multiple components to install and set up. To ensure a successful vSphere deployment, understand the sequence of tasks required.

Figure 13: vSphere Installation and Setup Workflow



Installing vSphere includes the following tasks:

1. Read the vSphere Release Notes.
2. Install ESX.
3. Use the direct console to configure ESX boot and network settings. See VMware ESX Installation and Setup for information.
4. Consider setting up a syslog server for remote logging, to ensure sufficient disk storage for log files. Setting up logging on a remote host is especially important for hosts with limited local storage. See VMware ESX Installation and Setup.
5. Deploy vCenter.
   You can deploy the vCenter appliance on an ESX host.
   1. Review the topics in System Requirements for the vCenter Appliance and verify that your system meets the hardware and software requirements for deploying the appliance.
   2. Determine the deployment method to use.

You can use the GUI method to deploy the appliance interactively. Alternatively, you can use the CLI method to perform a silent deployment of the appliance. See GUI Deployment of the vCenter Appliance and CLI Deployment of the vCenter Appliance.

3. Use the topic Required Information for Deploying a vCenter Appliance to create a worksheet with the information you need for the GUI deployment, or use the topic Prepare Your JSON Configuration File for CLI Deployment to create your JSON templates for the CLI deployment.
4. Deploy the appliance.

6. Connect to vCenter from the vSphere Client. See After You Deploy the vCenter Appliance.
7. Configure the vCenter instance. See vCenter Configuration.

vCenter Components and Services

vCenter provides a centralized platform for management, operation, resource provisioning, and performance evaluation of virtual machines and ESX hosts.

When you deploy the vCenter appliance, vCenter, its components, and the authentication services are deployed on the same system.

The following components are bundled with the vCenter appliance:

- The authentication services include vCenter Single Sign-On, License service, Lookup Service, and VMware Certificate Authority.
- The vCenter group of services includes vCenter, vSphere Client, vSphere Auto Deploy, and vSphere ESX Dump Collector. The vCenter appliance also contains the VMware vSphere Lifecycle Manager Extension service and the VMware vCenter Lifecycle Manager.

Authentication Services

vCenter Single Sign-On

The vCenter Single Sign-On authentication service provides secure authentication services to the vSphere software components. By using vCenter Single Sign-On, the vSphere components communicate with each other through a secure token exchange mechanism, instead of requiring each component to authenticate a user separately with a directory service like Active Directory.

vCenter Single Sign-On can authenticate users through:

- External identity provider federation
  You can configure vCenter for an external identity provider using federated authentication. In such a configuration, you replace vCenter as the identity provider. Currently, vSphere supports Active Directory Federation Services (AD FS) as the external identity provider. In this configuration, AD FS interacts with the identity sources on behalf of vCenter.
- vCenter built-in identity provider
  vCenter includes a built-in identity provider. By default, vCenter uses the vsphere.local domain as the identity source (but you can change it during deployment). You can configure the vCenter built-in identity provider to use Active Directory (AD) as its identity source using LDAP/S, OpenLDAP/S, and Integrated Windows Authentication (IWA). Such configurations allow customers to log in to vCenter using their AD accounts.

Authenticated users can then be assigned registered solution-based permissions or roles within a vSphere environment.

vCenter Single Sign-On is required with vCenter.

vSphere License Service

The vSphere License service provides common license inventory and management capabilities to all vCenter systems within the vCenter Single Sign-On domain.

VMware Certificate Authority

VMware Certificate Authority (VMCA) provisions each ESX host with a signed certificate that has VMCA as the root certificate authority, by default. Provisioning occurs when the ESX host is added to vCenter explicitly or as part of the ESX host installation process. All ESX certificates are stored locally on the host.

For information about all authentication services and capabilities, see vSphere Authentication.

What is the vCenter Appliance

The vCenter appliance is a preconfigured virtual machine that is optimized for running vCenter and the associated services.

The vCenter appliance includes the following software:

- Photon OS® 5.0
- vSphere authentication services
- PostgreSQL
- VMware vSphere Lifecycle Manager Extension
- VMware vCenter Lifecycle Manager

Version 9.1 of vCenter is deployed with virtual hardware version 10, which supports up to 64 virtual CPUs per virtual machine in ESX.

During the deployment, you can choose the vCenter appliance size for your vSphere environment size and the storage size for your database requirements.

vCenter supports high availability. For information about configuring vCenter in a vCenter High Availability cluster, see vSphere Availability.

vCenter supports file-based backup and restore. For information on backing up and restoring, see File-Based Backup and Restore of vCenter.

For information about the vCenter maximums, see VMware Configuration Maximums.

Understanding vSphere Domains and Domain Names

Each vCenter is associated with a vCenter Single Sign-On domain. The domain name defaults to vsphere.local, but you can change it during deployment. The domain determines the local authentication space.

vCenter Single Sign-On Domain

When you deploy a vCenter appliance, you are prompted to create a vCenter Single Sign-On domain or join an existing domain.

The domain name is used by the VMware Directory Service (vmdir) for all Lightweight Directory Access Protocol (LDAP) internal structuring.

You can give your domain a unique name. To prevent authentication conflicts, use a name that is not used by OpenLDAP, Microsoft Active Directory, and other directory services.

After you specify the name of your domain, you can add users and groups. You can add an Active Directory or LDAP identity source and allow the users and groups in that identity source to authenticate. You can also add vCenter instances to the domain.

vCenter Enhanced Linked Mode

vCenter Enhanced Linked Mode (ELM) allows you to log in to any single instance of vCenter and view and manage the inventories of all the vCenter systems in the group.

You can join up to 15 vCenter appliance deployments with vCenter Enhanced Linked Mode in a single vSphere Single Sign-On domain.

You can create a vCenter Enhanced Linked Mode group during the deployment of vCenter appliance.

You can also join a vCenter Enhanced Linked Mode group by moving, or repointing, a vCenter from one vSphere domain to another existing domain. See Repoint vCenter to Another vCenter in a Different Domain for information on repointing a vCenter node.

Note:

With vCenter 9.0, ELM is deprecated and will be removed in a future release. ELM is supported in vCenter 9.0 to allow smooth upgrade of existing infrastructure to VMware Cloud Foundation (VCF), but the recommended alternative is the grouping capability under VCF Operations. Group admins can enable alternatives for each ELM capability, such as vCenter Single Sign-On, consolidated licensing, vCenter roles and privileges sync, vCenter tags replication, and vCenter single API endpoint.

Before you can use new features in 9.0, such as VCF Single Sign-On and vCenter linking, you must deactivate ELM for all vCenter instances in your VCF Instance. For more information about deactivating ELM, see Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utility in the VCF documentation.

For more information on linking vCenter instances, see Linking vCenter Instances in VCF Operation.

vCenter Enhanced Linked Mode for vCenter Appliance

When you enable Enhanced Linked Mode for vCenter appliance deployments, you can connect multiple vCenter appliance deployments together to form a domain.

Other features include:

- A simplified backup and restore process. See File-Based Backup and Restore of vCenter for more information.
- A simplified HA process, removing the need for load balancers.
- Up to 15 vCenter appliance deployments can be linked together using enhanced linked mode and displayed in a single inventory view.
- For a vCenter High Availability (vCenter HA) cluster, three nodes are considered one logical vCenter node. See HA Architecture Overview in *vSphere Availability* for the vCenter HA architecture overview.

Enhanced Linked Mode with Read-Only Replication

If a vCenter High Availability (vCenter HA) instance is connected with another vCenter instance with enhanced linked mode and vCenter HA failover occurs to the passive node and is unable to communicate with its replication partner on the other vCenter node, the replica on the vCenter HA node enters read-only mode.

Joining a vCenter Enhanced Linked Mode Domain

You can join a vCenter appliance to another node during deployment of the vCenter appliance.

Note:

You can also join a vCenter Enhanced Linked Mode group by moving, or repointing, a vCenter from one vSphere domain to another existing domain. See Repoint vCenter to Another vCenter in a Different Domain for information on repointing a vCenter node.

For example, you can deploy two vCenter appliance systems and join the two nodes using vCenter Enhanced Linked Mode.

Before you deploy a new vCenter as part of a vCenter Enhanced Linked Mode, create an image-based backup of the existing nodes in your environment. You can use the backup as a precaution in case there is a failure during the deployment process. It is recommended that you shut down all linked vCenter instances and take offline snapshots to ensure that their databases are in sync during the backup process.

Enhanced Linked Mode requires vCenter instances to be of the same version for database replication to work across the different linked vCenter instances. During the upgrade process, if some of the vCenter instances are of different versions, the Enhanced Link Mode replication features might not work as expected.

If you are deploying the vCenter appliance nodes with the UI Installer:

1. For Appliance 1, deploy the vCenter appliance as an instance on ESX Host 1. Synchronize the time settings with ESX Host 1.

2. For Appliance 2, deploy the vCenter appliance as an instance on ESX Host 1 and configure the time settings so that Appliance 2 is synchronized with ESX Host 1. In stage 2 you select to join the vCenter Single Sign-On domain of the deployed appliance on Appliance 1. For specific instructions, see Stage 2 - Set up the Newly Deployed vCenter Appliance.

If you are deploying the vCenter appliance nodes with the CLI:

1. Configure the JSON configuration template `embedded_vCSA_on_VC.json` (or `embedded_vCSA_on_ESXi.json`) for Appliance 1 as an instance on ESX Host 1. See Prepare Your JSON Configuration File for CLI Deployment for specific instructions on preparing the JSON configuration file.
2. Deploy Appliance 1 by running the `vcsa-cli-installer` command. See Deploy a vCenter Appliance by Using the CLI for instructions.
3. Configure the JSON configuration template `embedded_vCSA_replication_on_VC.json` (or `embedded_vCSA_replication_on_ESXi.json`) for Appliance 2 as an instance on ESX Host 1. Enter the hostname of the first embedded node in the `replication_partner_hostname` field in the `sso` section.
4. Deploy Appliance 2 by running the `vcsa-cli-installer` command using the `embedded_vCSA_replication_on_VC.json` (or `embedded_vCSA_replication_on_ESXi.json`) file.

Deploying the vCenter Appliance

You can deploy the vCenter appliance to manage your vSphere environment.

You can deploy the vCenter appliance on an ESX host 9.0 or later, or on an ESX host or vSphere cluster from the inventory of a vCenter 9.0 instance or later.

For information about the software included in the vCenter appliance 9.1, see What is the vCenter Appliance.

For information about the software and hardware requirements for deploying the vCenter appliance, see System Requirements for the vCenter Appliance.

The vCenter installer contains executable files both for GUI and CLI deployments.

- The GUI deployment is a two stage process. The first stage is a deployment wizard that deploys the OVA file of the appliance on the target ESX host or vCenter instance. After the OVA deployment finishes, you are redirected to the second stage of the process that sets up and starts the services of the newly deployed appliance.
- The CLI deployment method involves running a CLI command against a JSON file that you previously prepared. The CLI installer parses the configuration parameters and their values from the JSON file and generates an OVF Tool command that automatically deploys and sets up the appliance. The CLI deployment automatically runs both stage 1 then stage 2, with no user interaction required.

The vCenter appliance has the following default user names:

User Name	Description
root	Use this user name to log in to the appliance operating system and the vCenter Management Interface. You set the password while deploying the virtual appliance.
administrator@*your_domain_name*	Use this user name for vCenter Single Sign-On login. You set the password while creating the vCenter Single Sign-On domain. You create a vCenter Single Sign-On domain during the deployment of a vCenter appliance in a new vCenter Single Sign-On domain. After you create a vCenter Single Sign-On domain, only the administrator@*your_domain_name* user has the privileges required to log in to vCenter Single Sign-On and vCenter. The administrator@*your_domain_name* user can proceed as follows: • Add an identity source in which additional users and groups are defined to vCenter Single Sign-On. • Give permissions to the users and groups. For information about adding identity sources and giving permissions to the users and groups, see *vSphere Authentication*.

For information about upgrading and patching the vCenter appliance, see vCenter Upgrade.

For information about configuring vCenter, see vCenter Configuration.

If you want to set up vCenter to use an IPv6 address version, use the fully qualified domain name (FQDN) or host name of the appliance. To set up an IPv4 address, the best practice is to use the FQDN or host name of the appliance, because the IP address can change if assigned by DHCP.

System Requirements for the vCenter Appliance

You can deploy the vCenter 9.1 appliance on an ESX 8.0 Update 1 or later hosts, or vCenter 8.0 or later instances. The upgrade of the appliance is a migration of the old version to the new version, which includes deploying a new appliance of version 9.1. Your system must also meet specific software and hardware requirements.

When you use fully qualified domain names (FQDN), verify that the client machine from which you are deploying the appliance and the network on which you are deploying the appliance uses the same DNS server.

Before you deploy the appliance, synchronize the clocks of the target ESX host and all vCenter instances on the vSphere network. Unsynchronized clocks might result in authentication problems and can cause the deployment to fail or prevent the appliance services from starting. See Synchronizing Clocks on the vSphere Network.

See the full list of vCenter system requirements:

- Hardware Requirements for the vCenter Appliance
- Storage Requirements for the vCenter Appliance
- Software Requirements for the vCenter Appliance
- DNS Requirements for the vCenter Appliance
- Required Ports for vCenter
- vSphere Client Software Requirements

Hardware Requirements for the vCenter Appliance

When you deploy or upgrade the vCenter appliance, you can select an appliance size suitable for your vSphere environment. The option that you select determines the number of CPUs and the amount of memory for the appliance.

The hardware requirements for a vCenter appliance depend on the size of your vSphere inventory.

	Number of vCPUs	Memory
Tiny environment (up to 10 hosts or 100 virtual machines)	2	14 GB
Small environment (up to 100 hosts or 1,000 virtual machines)	4	21 GB
Medium environment (up to 400 hosts or 4,000 virtual machine)	8	30 GB
Large environment (up to 1,000 hosts or 10,000 virtual machines)	16	39 GB
X-Large environment (up to 2,000 hosts or 35,000 virtual machines)	24	58 GB

Note:
If you want to add an ESX host with more than 512 LUNs and 2,048 paths to the vCenter inventory, you must deploy a vCenter appliance for a large or x-large environment.

Storage Requirements for the vCenter Appliance

When you deploy or upgrade the vCenter appliance, the ESX host or DRS cluster on which you deploy the appliance must meet minimum storage requirements. The required storage depends not only on the size of the vSphere environment and the storage size, but also on the disk provisioning mode.

The storage requirements are different for each vSphere environment size and depend on your database size requirements.

	Default Storage Size	Large Storage Size	X-Large Storage Size
Tiny environment • up to 10 hosts or 100 virtual machines	619 GB	2059 GB	4319 GB
Small environment • up to 100 hosts or 1,000 virtual machines	734 GB	2084 GB	4344 GB
Medium environment • up to 400 hosts or 4,000 virtual machine	933 GB	2233 GB	4493 GB
Large environment • up to 1,000 hosts or 10,000 virtual machines	1383 GB	2283 GB	4543 GB
X-Large environment • up to 2,000 hosts or 35,000 virtual machines with default and large storage • up to 2,500 hosts or 45,000 virtual machines with X-Large storage	2308 GB	2408 GB	4668 GB

For more information on disk provisioning, see Storage Provisioning and Space Reclamation in vSphere.

Note:
The storage requirements include the requirements for the vSphere Lifecycle Manager that runs as a service in the vCenter appliance.

Software Requirements for the vCenter Appliance

The vCenter 9.1 appliance can be deployed on ESX 8.0 Update 1 or later hosts, or on vCenter 8.0 or later instances or later.

You can deploy the vCenter 9.1 appliance using the GUI or CLI installer. You run the installer from a client machine that you use to connect to the target ESX host and deploy the appliance on the server. You can connect directly to an ESX 8.0 Update 1 or later host on which to deploy the appliance. You can also connect to a vCenter 8.0 or later instance to deploy the appliance on an ESX host or vCenter managed cluster that resides in the vCenter inventory.

For information about the requirements for network client machine, see System Requirements for the vCenter Installer.

DNS Requirements for the vCenter Appliance

When you deploy the vCenter appliance, similar to any network server, you can assign a fixed IP address and an FQDN that is resolvable by a DNS server so that clients can reliably access the service.

When you deploy the vCenter appliance with a static IP address, you ensure that in case of system restart, the IP address of the appliance remains the same.

Before you deploy the vCenter appliance with a static IP address, you must verify that this IP address has a valid internal domain name system (DNS) registration.

When you deploy the vCenter appliance, the configuration of the web server component that supports the vSphere Client fails if the installer cannot look up the fully qualified domain name (FQDN) for the appliance from its IP address. Reverse lookup is implemented using PTR records.

If you plan to use an FQDN for the appliance system name, you must verify that the FQDN is resolvable by a DNS server, by adding forward and reverse DNS A records.

After the vCenter upgrade, the appliance frees this static IP address and assumes the network settings of the old appliance.

Important:

To avoid upgrade or install failures in vSphere 9.1.x, your DNS server must comply with RFC 4074 behavior for negative `AAAA` responses. The DNS server must return `RCODE = 0` and an empty answer section for `AAAA` queries when there is no IPv6 record, rather than `NXDOMAIN` or other errors. See the Broadcom Knowledge Base article: 377682.

You can use the `nslookup` command to verify that the DNS reverse lookup service returns an FQDN when queried with the IP address and to verify that the FQDN is resolvable.

```
nslookup -nosearch -nodefname FQDN_or_IP_address
```

If you use DHCP instead of a static IP address for the vCenter appliance, verify that the appliance name is updated in the domain name service (DNS). If you can ping the appliance name, the name is updated in DNS.

Ensure that the ESX host management interface has a valid DNS resolution from the vCenter and all vSphere Client instances. Ensure that the vCenter has a valid DNS resolution from all ESX hosts and vSphere Client.

vSphere Client Software Requirements

Use of the vSphere Client requires a supported web browser.

VMware has tested and supports the following guest operating systems and browser versions for the vSphere Client.

Supported Guest Operating Systems	• Windows 32-bit and 64-bit • Mac OS • Linux
Supported Browser Versions	• Google Chrome 120 or later • Mozilla Firefox 115 or later • Microsoft Edge 120 or later

Note:
Later versions of these browsers are likely to work, but have not been tested.

Preparing for Deployment of the vCenter Appliance

Before you deploy the vCenter appliance, you must download the vCenter installer ISO file and mount it to a network virtual machine or physical server from which you want to perform the deployment.

The machine from which you deploy the appliance must run on a Windows, Linux, or Mac operating system that meets the operating system requirements. See System Requirements for the vCenter Installer.

- Synchronizing Clocks on the vSphere Network

System Requirements for the vCenter Installer

You can run the vCenter GUI or CLI installer from a client machine that is running on a Windows, Linux, or Mac operating system of a supported version.

To ensure optimal performance of the GUI and CLI installers, use a client machine that meets the minimum hardware requirements.

Table 33: System Requirements for the GUI and CLI Installers

Operating System	Supported Versions	Minimum Hardware Configuration for Optimal Performance
Windows	• Windows 11 • Windows 2022 • Windows 2025	4 GB RAM, 2 CPU having 4 cores with 2.3 GHz, 32 GB hard disk, 1 NIC
Linux	• SUSE 15.6 • Ubuntu 22.04, 24.04	4 GB RAM, 1 CPU having 2 cores with 2.3 GHz, 16 GB hard disk, 1 NIC **Note:** The CLI installer requires 64-bit OS.
Mac	• macOS 13, 14, 15	8 GB RAM, 1 CPU having 4 cores with 2.4 GHz, 150 GB hard disk, 1 NIC

Note:
For client machines that run on Mac 13 or later, concurrent GUI deployments of multiple appliances are unsupported. You must deploy the appliances in sequence.

Note:
Deploying the vCenter appliance with the GUI requires a minimum resolution of 1024x768 to properly display. Lower resolutions can truncate the UI elements.

Download and Mount the vCenter Installer

The vCenter appliance ISO image contains GUI and CLI installers for the vCenter appliance.

- Access to the Broadcom Support Portal.
- Verify that your client machine meets the system requirements for the vCenter installer. See System Requirements for the vCenter Installer.

With the GUI and CLI executable files that are included in the vCenter installer, you can:

- Deploy the vCenter appliance.
- Upgrade the vCenter appliance.
- Restore a vCenter appliance from a file-based backup.

1. Log in to the Broadcom Support Portal.
2. Download the vCenter appliance ISO image. See Broadcom knowledge base article 366685.
3. Confirm that the md5sum is correct by using an MD5 checksum tool.
4. Mount the ISO image to the client machine from which you want to deploy, upgrade, or restore the appliance.

   Note:
   ISO mounting software that does not allow more than eight directory levels, for example, MagicISO Maker on Windows, is unsupported.

   For Linux OS and Mac OS, Archive Manager is unsupported.

   For Mac OS, you can use DiskImageMounter.

   For Ubuntu 22.04 and 24.04, you can use Disk Image Mounter.

   For SUSE 15.6 OS, you can use the terminal.

   ```
   $ sudo mkdir mount_dir
   $ sudo mount -o loop VMware-vCSA-all-version_number-build_number.iso mount_dir
   ```

Open the `readme.txt` file and review the information about the other files and directories in the vCenter appliance ISO image.

Synchronizing the Client and Server System Clock

To establish a secure TLS connection to a vCenter appliance, the system where you are running the CLI installer (the client) must not have its system clock slower or faster than the server's system clock by an acceptable limit (tolerance).

See Table 34: Client Clock Tolerance for specific values for each deployment scenario.

Note:

The client clock values are applicable only for vCenter 9.0 and later.

Table 34: Client Clock Tolerance

Deployment Scenario	Clock Tolerance
Linking one vCenter with another vCenter	When deploying the second vCenter, the clock tolerance for the client and the first vCenter must not exceed 10 minutes.
Installing a vCenter appliance using a container vCenter with a `*._on_vc.json` template.	The maximum clock tolerance between the client and the container vCenter is 8 hours 20 minutes.

Prerequisites for Deploying the vCenter Appliance

To ensure a successful deployment of the vCenter appliance, you must perform some required tasks and pre-checks before running the installer.

General Prerequisites

- Download and Mount the vCenter Installer.

Target System Prerequisites

- Verify that your system meets the minimum software and hardware requirements. See System Requirements for the vCenter Appliance.
- If you want to deploy the appliance on an ESX host, verify that the ESX host is not in lockdown or maintenance mode and not part of a fully automated vSphere cluster.
- If you want to deploy the appliance on a vSphere cluster of the inventory of a vCenter instance, verify that the cluster contains at least one ESX host that is not in lockdown or maintenance mode.
- If you plan to use NTP servers for time synchronization, verify that the NTP servers are running and that the time between the NTP servers and the target server on which you want to deploy the appliance is synchronized.
- If you want to deploy the appliance on a vSAN ESA cluster with vSAN ESA encryption, you must enable vSAN ESA encryption before installing vCenter. vSAN Express Storage Architecture is a next-generation architecture designed to get the most out of high-performance storage devices, resulting in greater performance and efficiency. You can enable vSAN ESA encryption through vSAN SDK or vSAN API.

vCenter Enhanced Linked Mode Prerequisites

When deploying a new vCenter as part of an Enhanced Linked Mode deployment, create an image-based backup of the existing vCenter nodes in your environment. You can use the backup as a precaution in case there is a failure during the deployment process.

If the deployment fails, delete the newly deployed vCenter appliance, and restore the vCenter nodes from their respective image-based backups. You must restore all the nodes in the environment from their image-based backups. Failing to do so can cause the replication partners to be out of synchronization with the restored node.

- To learn more about creating vCenter Enhanced Linked Mode deployments, see vCenter Enhanced Linked Mode.
- To learn about image-based backs, see Image-Based Backup and Restore of a vCenter Environment.

Network Prerequisites

If you plan to assign a static IP address and an FQDN as a system name in the network settings of the appliance, verify that you have configured the forward and reverse DNS records for the IP address.

GUI Deployment of the vCenter Appliance

You can use the GUI installer to perform an interactive deployment of a vCenter appliance.

When you perform the GUI deployment, you download the vCenter installer on a client machine, run the deployment wizard from the client machine, and provide the inputs that are required for the appliance deployment and setup.

The GUI deployment process includes a series of two stages.

Figure 14: Stage 1 - OVA Deployment

The first stage walks you through the deployment wizard to choose the deployment type and appliance settings. This stage completes the deployment of the OVA file on the target ESX host with the deployment type and appliance settings that you provide.

As an alternative to performing the first stage of the deployment with the GUI installer, you can deploy the OVA file of the vCenter appliance by using the vSphere Client. After the OVA deployment, you must log in to the vCenter Management Interface of the newly deployed appliance to proceed with the second stage of the deployment process. See Deploy and Export OVF and OVA Templates in *vSphere Virtual Machine Administration* for information about deploying an OVA file using the vSphere Client.

Figure 15: Stage 2 - Appliance Setup

The second stage walks you through the setup wizard to configure the appliance time synchronization and vCenter Single Sign-On. This stage completes the initial setup and starts the services of the newly deployed appliance.

As an alternative to performing the second stage of the deployment with the GUI installer, you can log in to the vCenter Management Interface of the newly deployed appliance, https://`vcenter_IP_address_or_fqdn`:5480.

Required Information for Deploying a vCenter Appliance

When you use the GUI method to deploy a vCenter appliance, the wizard prompts you for deployment and setup information. It is a best practice to keep a record of the values that you enter in case you must reinstall the product.

You can use this worksheet to record the information that you need for deploying a vCenter appliance.

Table 35: Required Information During Stage 1 of the GUI Deployment Process

Required Information	Default	Your Entry
FQDN or IP address of the target server on which you want to deploy the appliance. The target server can be either an ESX host or a vCenter instance.	-	
HTTPS port of the target server	443	

Required Information	Default	Your Entry
User name with administrative privileges on the target server • If your target server is an ESX host, use root. • If your target server is a vCenter instance, use *user_name*@*your_domain_name*, for example, administrator@vsphere.local.	-	
Password of the user with administrative privileges on the target server	-	
Data center from the vCenter inventory on which you want to deploy the appliance. Target server must be a vCenter instance. Optionally you can provide a data center folder.	-	
ESX host or DRS cluster from the data center inventory on which you want to deploy the appliance.	-	
VM name for the appliance: • Must not contain a percent sign (%), backslash (\), or forward slash (/). • Must be no more than 80 characters in length.	vCenter	
Password for the root user of the appliance operating system • Must be at least 8 characters, but no more than 20 characters in length. • Must contain at least one uppercase letter. • Must contain at least one lowercase letter. • Must contain at least one number. • Must contain at least one special character, for example, a dollar sign ($), hash key (#), at sign (@), period (.), or exclamation mark (!).	-	
Deployment size of the vCenter appliance for your vSphere environment. See Hardware Requirements for the vCenter Appliance for information about the deployment sizes you can select. The option that you select determines the number of CPUs and the amount of memory for the appliance.	Tiny	
Storage size of the vCenter appliance for your vSphere environment. The required storage depends not only on the size of the vSphere environment, but also on the disk provisioning mode. See Storage Requirements for the vCenter Appliance. Increase the default storage size if you want larger volume for SEAT data (stats, events, alarms, and tasks).	Default	
Name of the datastore on which you want to store the configuration files and virtual disks of the appliance. **Note:** The installer displays a list of datastores that are accessible from your target server.	-	
Activate or deactivate Thin Disk Mode.	Deactivated	

Required Information	Default	Your Entry
Name of the network to which to connect the appliance **Note:** The installer displays a drop-down menu with networks that depend on the network settings of your target server. If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are not supported and are not displayed in the drop-down menu. The network must be accessible from the client machine from which you perform the deployment.	-	
IP version for the appliance address Can be either IPv4 or IPv6.	IPv4	
IP assignment for the appliance address Can be either static or DHCP.	static	
FQDN For a static IP assignment vCenter uses FQDN or IP address as the system name.	-	
IP address	-	
For IPv4 networks, you can use either a subnet mask or a network prefix. Subnet mask uses a dot decimal notation (for example, 255.255.255.0). An IPv4 network prefix is an integer between 0 and 32. For IPv6 networks, you must use a network prefix. An IPv6 network prefix is an integer between 0 and 128.	-	
Default gateway	-	
DNS servers separated by commas	-	
System name (FQDN) Only if you use DHCP with IPv4 and have a Dynamic DNS (DDNS) server available in your environment.	-	

Table 36: Required Information During Stage 2 of the GUI Deployment Process

Required Information	Default	Your Entry
Time synchronization settings You can synchronize the time of the appliance either with the time of the ESX host or with one or more NTP servers. If you want to use more than one NTP servers, you must provide the IP addresses or FQDNs of the NTP servers as a comma-separated list.	Synchronize time with NTP servers	
Activate or deactivate SSH access **Note:** vCenter High Availability requires remote SSH access to the appliance.	Deactivate	
Name for the new vCenter Single Sign-On domain For example, vsphere.local.	-	

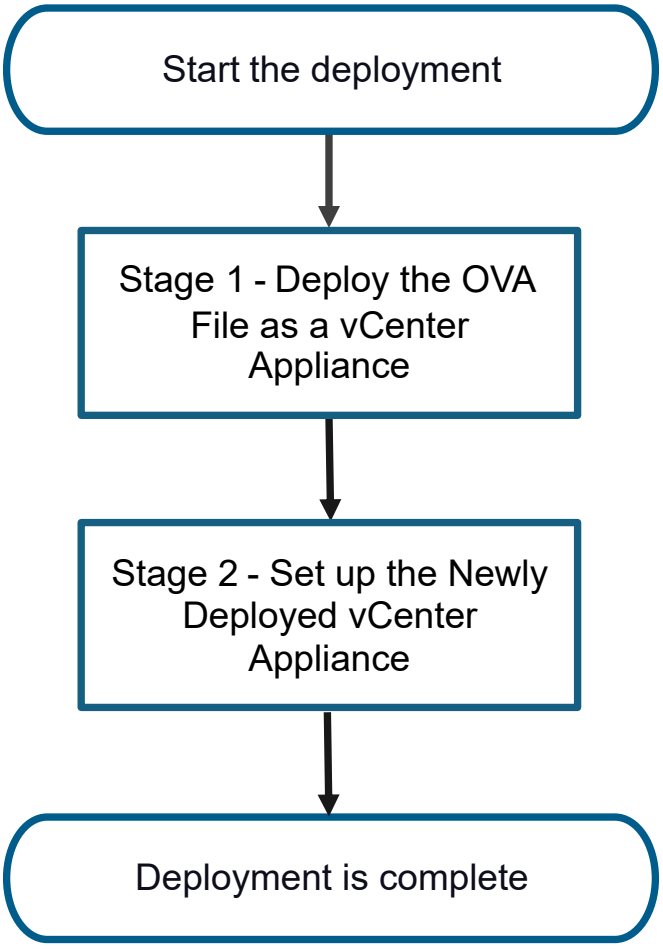
Required Information	Default	Your Entry
Password for the administrator account, administrator@*your_domain_name* • Must be at least 8 characters, but no more than 20 characters in length. • Must contain at least one uppercase letter. • Must contain at least one lowercase letter. • Must contain at least one number. • Must contain at least one special character, such as ampersand (&), hash key (#), and percent sign (%).	-	
Password of the vCenter Single Sign-On administrator user for the domain.	-	
Join or do not participate in the VMware Customer Experience Improvement Program (CEIP). For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.	Join the CEIP	

Deploy the vCenter Appliance by Using the GUI

You can use the GUI installer to perform an interactive deployment of a vCenter appliance. You must run the GUI deployment from a Windows, Linux, or Mac machine that is in the network on which you want to deploy the appliance.

- See Prerequisites for Deploying the vCenter Appliance.
- See Required Information for Deploying a vCenter Appliance.

Figure 16: Deployment Workflow of a vCenter Appliance



Stage 1 - Deploy the OVA File as a vCenter Appliance

With stage 1 of the deployment process, you deploy the OVA file, which is included in the vCenter installer ISO, as a vCenter appliance.

1. In the vCenter installer, navigate to the `vcsa-ui-installer` directory, go to the subdirectory for your operating system, and run the installer executable file.
   - For Windows OS, go to the `win32` subdirectory, and run the `installer.exe` file.
   - For Linux OS, go to the `lin64` subdirectory, and run the `installer` file.
   - For Mac OS, go to the `mac` subdirectory, and run the `Installer.app` file.
2. On the Home page, click **Install** to start the deployment wizard.
3. Review the Introduction page to understand the deployment process and click **Next**.
4. Read and accept the license agreement, and click **Next**.
5. Connect to the target server on which you want to deploy the vCenter appliance.

Option	Steps
You can connect to an ESX host on which to deploy the appliance.	1. Enter the FQDN or IP address of the ESX host. 2. Enter the HTTPS port of the ESX host. 3. Enter the user name and password of a user with administrative privileges on the ESX host, for example, the root user. 4. Click **Next**. 5. Verify that the certificate warning displays the SHA1 thumbprint of the SSL certificate that is installed on the target ESX host, and click **Yes** to accept the certificate thumbprint.
You can connect to a vCenter instance and browse the inventory to select an ESX host or DRS cluster on which to deploy the appliance.	1. Enter the FQDN or IP address of the vCenter instance. 2. Enter the HTTPS port of the vCenter instance. 3. Enter the user name and password of user with vCenter Single Sign-On administrative privileges on the vCenter instance, for example, the administrator@*your_domain_name* user. 4. Click **Next**. 5. Verify that the certificate warning displays the SHA1 thumbprint of the SSL certificate that is installed on the target vCenter instance, and click **Yes** to accept the certificate thumbprint. 6. Select the data center or data center folder that contains the ESX host or DRS cluster on which you want to deploy the appliance, and click **Next** **Note:** You must select a data center or data center folder that contains at least one ESX host that is not in lockdown or maintenance mode. 7. Select the ESX host or DRS cluster on which you want to deploy the appliance, and click **Next**.

6. On the Set up appliance VM page, enter a name for the vCenter appliance, set the password for the root user, and click **Next**.

   The appliance name must not contain a percent sign (%), backslash (\), or forward slash (/) and must be no more than 80 characters in length.

   The password must contain only lower ASCII characters without spaces, at least eight characters, a number, uppercase and lowercase letters, and a special character, for example, an exclamation mark (!), hash key (#), at sign (@), or brackets (()).

7. Select the deployment size for the vCenter appliance for your vSphere inventory.

   See Hardware Requirements for the vCenter Appliance for information about the deployment sizes you can select. The option that you select determines the number of CPUs and the amount of memory for the appliance.

8. Select the storage size for the vCenter appliance, and click **Next**.

   The required storage depends not only on the size of the vSphere environment, but also on the disk provisioning mode. See Storage Requirements for the vCenter Appliance.

9. Select the storage location for the vCenter appliance where all the virtual machine configuration files and virtual disks will be stored.

Option	Action
Install on an existing datastore accessible from the target host	Select a datastore from the list of compatible datastores.
Install on a new vSAN cluster containing the target host	Specify the required details to create a new vSAN cluster or a vSAN Express Storage Architecture (vSAN ESA) cluster to store the vCenter appliance.
Install on an existing vSAN datastore and claim additional disks	Specify the required details to create a cluster on the vSAN datastore. This option is displayed only if your environment contains a vSAN datastore.

   To enable thin provisioning, select **Enable Thin Disk Mode**. NFS datastores are thin provisioned by default.

10. Optional: If you selected vSAN as your storage location, you must claim disks for storage.
    - For vSAN, claim disks separately for cache tier and capacity tier.
    - For vSAN ESA, claim disks from the list of compatible disks.

11. On the Configure network settings page, set up the network settings.

    The IP address or the FQDN of the appliance is used as a system name. It is recommended to use an FQDN. However, if you want to use an IP address, use static IP address allocation for the appliance, because IP addresses allocated by DHCP might change.

Option	Action
Network	Select the network to which to connect the appliance. The networks displayed in the drop-down menu depend on the network settings of the target server. If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are not supported and are not displayed in the drop-down menu.
IP version	Select the version for the appliance IP address. You can select either IPv4 or IPv6.
IP assignment	Select how to allocate the IP address of the appliance. • **static** The wizard prompts you to enter the IP address and network settings. • **DHCP** A DHCP server is used to allocate the IP address. Select this option only if a DHCP server is available in your environment. If there is an enabled DDNS in your environment, you can enter a preferred fully qualified domain name (FQDN) for the appliance.
Common Ports	You can customize the HTTP and HTTPS ports (optional). If specifying a custom HTTP and HTTPS port number, ensure that you do not use a port number already in use by vCenter , or the default HTTP and HTTPS ports of 80 and 443.

Option	Action
	Note: 1. You can only customize the HTTP and HTTPS ports during installation. You cannot customize any other port. 2. You cannot change any port after installation. 3. Upgrade or migration procedures persist all port configuration. Port customizations that were previously supported in Windows OS deployments are persisted in a Windows to Linux migration.

12. On the Ready to complete stage 1 page, review the deployment settings for the vCenter appliance and click **Finish** to start the OVA deployment process.
13. Wait for the OVA deployment to finish, and click **Continue** to proceed with stage 2 of the deployment process to set up and start the services of the newly deployed appliance.

    Note:
    If you exit the wizard by clicking **Close**, you must log in to the vCenter Management Interface to set up and start the services.

The newly deployed vCenter appliance is running on the target server but the services are not started.

Stage 2 - Set up the Newly Deployed vCenter Appliance

When the OVA deployment finishes, you are redirected to stage 2 of the deployment process to set up and start the services of the newly deployed vCenter appliance.

1. Review the introduction to stage 2 of the deployment process and click **Next**.
2. Configure the time settings in the appliance, optionally enable remote SSH access to the appliance, and click **Next**.

Option	Description
Synchronize time with the ESX host	Enables periodic time synchronization, and VMware Tools sets the time of the guest operating system to be the same as the time of the ESX host.
Synchronize time with NTP servers	Uses a Network Time Protocol server for synchronizing the time. If you select this option, you must enter the names or IP addresses of the NTP servers separated by commas.

3. Create a new vCenter Single Sign-On domain or join an existing domain.

Option	Description
Create a new Single Sign-On domain	Creates a new vCenter Single Sign-On domain. 1. Enter the domain name, for example `vsphere.local`. **Note:** Ensure that the domain name does not contain any upper-case letters. 2. Set the password for the vCenter Single Sign-On administrator account. This is the password for the user administrator@*`your_domain_name`*. 3. Confirm the administrator password, and click **Next**.
Join an existing Single Sign-On domain	Joins a new vCenter Single Sign-On server to an existing vCenter Single Sign-On domain. You must provide the information about the vCenter Single Sign-On server to which you join the new vCenter Single Sign-On server. 1. Enter the fully qualified domain name (FQDN) or IP address of the vCenter Single Sign-On server to join.

Option	Description
	2. Enter the HTTPS port to use for communication with the vCenter Single Sign-On server. 3. Enter the domain name for the vCenter Single Sign-On you are joining, for example `vsphere.local`. 4. Enter the password of the vCenter Single Sign-On administrator account. 5. Click **Next**.

When you select to join an existing vCenter Single Sign-On domain, you enable the Enhanced Linked Mode feature. The infrastructure data is replicated with the joined vCenter Single Sign-On server.

4. Review the VMware Customer Experience Improvement Program (CEIP) page and choose if you want to join the program.

   For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.

5. On the Ready to complete page, review the configuration settings for the vCenter appliance, click **Finish**, and click **OK** to complete stage 2 of the deployment process and set up the appliance.

6. Optional: After the initial setup finishes, enter the URL from the browser with **https://*vcenter_IP_address_or_fqdn*/ui** to go to the vSphere Client and log in to the vCenter instance in the vCenter appliance, or click the **https://*vcenter_IP_address_or_fqdn*:443** to go the vCenter appliance Getting Started page.

7. Click **Close** to exit the wizard.

   You are redirected to the vCenter appliance Getting Started page.

You can configure high availability for the vCenter appliance. For information about providing vCenter appliance high availability, see *vSphere Availability*.

CLI Deployment of the vCenter Appliance

You can use the CLI installer to perform a silent deployment of a vCenter appliance on an ESX host or vCenter instance.

The CLI deployment process includes downloading the vCenter installer on a client machine or physical server from which you want to perform the deployment, preparing a JSON configuration file with the deployment information, and running the deployment command.

Prepare Your JSON Configuration File for CLI Deployment

Before you run the CLI installer to deploy a vCenter appliance, you must prepare a JSON file with configuration parameters and their values for your deployment specification.

- You must be familiar with the JSON syntax.
- Download and Mount the vCenter Installer.

The vCenter installer contains JSON templates for all deployment options. For information about the templates, see JSON Templates for CLI Deployment of the vCenter Appliance.

You can deploy an appliance with minimum configurations by setting values to the configuration parameters in the JSON template for your specification. You can edit the preset values, remove configuration parameters, and add configuration parameters for custom configurations.

For a complete list of the configuration parameters and their descriptions, navigate to the installer subdirectory for your operating system and run the `vcsa-deploy install --template-help` command or see Deployment Configuration Parameters.

1. In the vCenter installer ISO, navigate to the `vcsa-cli-installer` directory, and open the `templates` subfolder.
2. Copy the deployment templates from the `install` subfolder to your workspace.

   Important:
   The path to the JSON configuration files must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.
3. In a text editor, open the template file for your specification.

   To ensure the correct syntax of your JSON configuration file, use a JSON editor.
4. Fill in the values for the required configuration parameters and, optionally, enter additional parameters and their values.
   For example, if you want to use an IPv4 DHCP assignment for the network of the appliance, in the `network` subsection of the template, change the value of the `mode` parameter to `dhcp` and remove the default configuration parameters that are for a static assignment.

   ```
   "network": {
       "ip_family": "ipv4",
       "mode": "dhcp"
   },
   ```

   Important:

   The string values, including the passwords, must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

   To set a value that contains a backslash (\) or quotation mark (") character, you must precede the character with the backslash (\) character. For example, `"password":"my\"password"` sets the password my"password, `"image":"G:\\vcsa\\VMware-vCenter-Server-Appliance-9.1.1.XXXX-YYYYYYY_OVF10.ova"` sets the path `G:\vcsa\VMware-vCenter-Server-Appliance-9.1.1.XXXX-YYYYYYY_OVF10.ova`.

   The Boolean values must contain only lowercase characters, that is, a value can be either `true` or `false`. For example, `"ssh_enable":false`.
5. Optional: Use a JSON editor of your choice to validate the JSON file.
6. Save in UTF-8 format and close the file.

You can create and save additional templates if needed for your deployment specification.

JSON Templates for CLI Deployment of the vCenter Appliance

The vCenter installer ISO includes JSON templates with the minimum configuration parameters for all deployment options. The templates are located in the `vcsa-cli-installer/templates/install` directory.

For each deployment option, there is one template for deploying the appliance on an ESX host and another template for deploying the appliance on a vCenter instance.

Table 37: Deployment JSON Templates Included in the vCenter Installer

Template	Description
`embedded_vCSA_on_ESXi.json`	Contains the minimum configuration parameters that are required for the deployment of a vCenter appliance on an ESX host.
`vCSA_with_cluster_on_ESXi.json`	Contains the minimum configuration parameters that are required for the deployment of a vCenter appliance with a single node vSAN and a vLCM managed cluster on an ESX host.
`embedded_vCSA_on_VC.json`	Contains the minimum configuration parameters that are required for the deployment of a vCenter appliance on a vCenter instance.

Template	Description
embedded_vCSA_replication_on_ESXi.json	Contains the minimum configuration parameters that are required for the deployment of a vCenter appliance as a replication partner to another embedded vCenter on an ESX host.
embedded_vCSA_replication_on_VC.json	Contains the minimum configuration parameters that are required for the deployment of a vCenter appliance replication partner to another vCenter appliance on a vCenter instance.

Deployment Configuration Parameters

When you prepare your JSON configuration files for a CLI deployment, you must set parameters and values to provide input data for the deployment of a vCenter appliance.

Sections and Subsections of Configuration Parameters in the JSON Deployment Files

The configuration parameters in the JSON configuration files for a CLI upgrade are organized in sections and subsections.

Table 38: Sections and Subsections of Configuration Parameters in the JSON Deployment Files

Section	Subsection	Description
new_vcsa - describes the appliance that you want to deploy	esxi	Use only if you want to deploy the appliance directly on an ESX host. Contains the configuration parameters that describe the target ESX host. Depending on whether you want to configure a vSAN or a vSphere Lifecycle Manager managed cluster during the deployment, see the relevant section for a detailed information about the configuration parameters. • Table 39: Configuration Parameters in the new_vcsa Section, esxi Subsection for Deployments with a vSAN and a vSphere Lifecycle Manager Cluster • Table 40: Configuration Parameters in the new_vcsa Section, esxi Subsection for Deployments without a vSAN and a vSphere Lifecycle Manager Cluster **Note:** You must fill in either the esxi or the vc subsection.
	vc	Use only if you want to deploy the appliance on the inventory of a vCenter instance. Contains the configuration parameters that describe the target ESX host or DRS cluster from the vCenter inventory. See Table 41: Configuration Parameters in the new_vcsa Section, vc Subsection. **Note:** You must fill in either the vc or the esxi subsection.
	appliance	Contains the configuration parameters that describe the appliance. See Table 42: Configuration Parameters in the new_vcsa Section, appliance Subsection.
	install_parameters	Optional subsection for the prefix base MAC address allocation options. See Configuration Parameters in the new_vcsa Section, install_parameters Subsection.
	network	Contains the configuration parameters that describe the network settings for the appliance. See Table 44: Configuration Parameters in the new_vcsa Section, network Subsection.
	os	Contains the configuration parameters that describe the operating system settings for the appliance. See Table 46: Configuration Parameters in the new_vcsa Section, os Subsection.
	sso	Contains the configuration parameters that describe the vCenter Single Sign-On settings for the appliance. See Table 47: Configuration Parameters in the new_vcsa Section, sso Subsection.

Section	Subsection	Description
	`ovftool_arguments`	Optional subsection for adding arbitrary arguments and their values to the OVF Tool command that the installer generates. **Important:** The vCenter installer does not validate the configuration parameters in the `ovftool_arguments` subsection. If you set arguments that the OVF Tool does not recognize, the deployment might fail.
`ceip` - describes joining the VMware Customer Experience Improvement Program (CEIP)	`settings`	Contains only the `ceip_enabled` configuration parameter to join or not to join the VMware Customer Experience Improvement Program (CEIP). See Table 48: Configuration Parameters in the ceip Section, settings Subsection. **Note:** If set to `true`, you must run the CLI deployment command with the `--acknowledge-ceip` argument. For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.

Important:

The string values, including the passwords, must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

To set a value that contains a backslash (\) or a quotation mark (") character, you must precede the character with the backslash (\) character. For example, `"password":"my\"password"` sets the password my"password. Similarly, `"image":"G:\\vcsa\\VMware-vCenter-Server-Appliance-8.0.0.XXXX-YYYYYYY_OVF10.ova"` sets the path `G:\vcsa\VMware-vCenter-Server-Appliance-8.0.0.XXXX-YYYYYYY_OVF10.ova`.

Boolean values must contain only lowercase characters. It must be either `true` or `false`. For example, `"ssh_enable":false`.

Configuration Parameters in the `new_vcsa` Section

Table 39: Configuration Parameters in the `new_vcsa` Section, `esxi` Subsection for Deployments with a vSAN and a vSphere Lifecycle Manager Cluster

Name	Type	Description
`hostname`	string	The IP address or FQDN of the target ESX host on which you want to deploy the appliance.
`username`	string	A user name with administrative privileges on the target ESX host, for example, root.
`password`	string	The password of the user with administrative privileges on the target ESX host.
`deployment_network`	string	The name of the network to which to connect the appliance. **Note:** The network must be accessible from the target ESX host. Ignored if the target ESX host has only one network.
`datacenter`	string	A specific data center that you want to create.
`cluster`	string	The name of the vSAN or vSphere Lifecycle Manager managed cluster.
`compression_only`	Boolean	Set the value to `true` to enable compression on the vSAN cluster. If this parameter is set to `true`, the `deduplication_and_compression` parameter must be set to`false`.
`deduplication_and_compression`	Boolean	Set the value to `true` to enable compression and deduplication on the vSAN cluster. If this parameter is set to `true`, the `compression_only` parameter must be set to `false`.

Name	Type	Description
`cache_disk`		A list of UUIDs or canonical names of disks you want to use for cache. Specify only SSDs.
`capacity_disk`		A list of UUIDs or canonical names of disks you want to use for storage. Specify either SSDs or HDDs.
`enable_vlcm`	Boolean	Set to `true`. **CAUTION:** Do not modify the value to `false`. The default type of cluster created is a vSphere Lifecycle Manager managed cluster. Setting this field to `false` results in failure during cluster creation.
`enable_vsan_esa`	Boolean	Set to `true` to create a vSAN cluster with vSAN ESA enabled. vSAN Express Storage Architecture is a next-generation architecture that is designed to get the most out of high-performance storage devices, resulting in greater performance and efficiency.
`single_tier`	Array	A list of UUIDs or canonical names of disks you want to add to a vSAN storage pool. Required only if you set `enable_vsan_esa` to `true`.
`vsan_hcl_database_path`	String	Local vSAN HCL database path. If the vSAN HCL database present in this location is outdated, the installer will download and replace the older version with the latest version. Required only if you set `enable_vsan_esa` to `true`.
`datastore`	string	The name of the datastore where you want to store the configuration files and virtual disks of the appliance. **Note:** The datastore must be accessible from the ESX host. If you are using the thin disk mode, the datastore size should have a minimum of 25-GB space.
`port`	integer	The HTTPS reverse proxy port of the target ESX host. The default port is 443. Use only if the target ESX host uses a custom HTTPS reverse proxy port.

`ssl_certificate_verification`	string	The CLI verifies that a server's security certificate is signed by a Certificate Authority (CA), and establishes a secure connection. If the certificate is self-signed, the CLI stops the upgrade unless you specify the full certificate path.
		• Set `verification_mode` to `NONE`.
		`"verification_mode": "NONE"`
		• Specify the full path to the server certificate placed on the client machine that is running on Windows, Linux, or Mac with the following option:
		`"certificate_path": "C:\\server_cert.pem"`
		If you are connecting to a server with a self-signed certificate, and fail to enter the full certificate path, the CLI displays the server's self-signed certificate , and prompts you to accept or reject the certificate.

Table 40: Configuration Parameters in the `new_vcsa` Section, `esxi` Subsection for Deployments without a vSAN and a vSphere Lifecycle Manager Cluster

Name	Type	Description
`hostname`	string	The IP address or FQDN of the target ESX host on which you want to deploy the appliance.
`username`	string	A user name with administrative privileges on the target ESX host, for example, root.
`password`	string	The password of the user with administrative privileges on the target ESX host.

Name	Type	Description
deployment_network	string	The name of the network to which to connect the appliance. **Note:** The network must be accessible from the target ESX host. Ignored if the target ESX host has only one network.
datastore	string	The name of the datastore where you want to store the configuration files and virtual disks of the appliance. **Note:** The datastore must be accessible from the ESX host. If you are using the thin disk mode, the datastore size should have a minimum of 25-GB space.
port	integer	The HTTPS reverse proxy port of the target ESX host. The default port is 443. Use only if the target ESX host uses a custom HTTPS reverse proxy port.
ssl_certificate_verification	string	The CLI verifies that a server's security certificate is signed by a Certificate Authority (CA), and establishes a secure connection. If the certificate is self-signed, the CLI stops the upgrade unless you specify the full certificate path. • Set verification_mode to NONE. "verification_mode": "NONE" • Specify the full path to the server certificate placed on the client machine that is running on Windows, Linux, or Mac with the following option: "certificate_path": "C:\\server_cert.pem" If you are connecting to a server with a self-signed certificate, and fail to enter the full certificate path, the CLI displays the server's self-signed certificate thumbprint, and prompts you to accept or reject the certificate thumbprint.

Table 41: Configuration Parameters in the new_vcsa Section, vc Subsection

Name	Type	Description
hostname	string	The IP address or FQDN of the target vCenter instance on which you want to deploy the appliance.
username	string	vCenter Single Sign-On administrator user name on the target vCenter instance, for example, administrator@vsphere.local.
password	string	The password of the vCenter Single Sign-On administrator user on the target vCenter instance.
deployment_network	string	The name of the network to which to connect the appliance. **Note:** The network must be accessible from the target ESX host or DRS cluster on which you want to deploy the appliance. Ignored if the target ESX host or DRS cluster has only one network.
datacenter	array	The vCenter datacenter that contains the target ESX host or DRS cluster on which you want to deploy the appliance. If the datacenter is located in a folder or a structure of folders, specify the value as a comma-separated list of strings. For example, ["parent_folder", "child_folder", "datacenter_name"] **Note:** The value is case-sensitive.

Name	Type	Description
`datastore`	string	The name of the datastore that you want to store the configuration files and virtual disks of the appliance. **Note:** The datastore must be accessible from the target ESX host or DRS cluster. The datastore must have at least 25 GB of free space.
`port`	integer	The HTTPS reverse proxy port of the target vCenter instance. The default port is 443. Use only if the target vCenter instance uses a custom HTTPS reverse proxy port.

`ssl_certificate_verification`	string	The CLI verifies that a server's security certificate is signed by a Certificate Authority (CA), and establishes a secure connection. If the certificate is self-signed, the CLI stops the upgrade unless you specify the full certificate path.
		• Set `verification_mode` to `NONE`.
		`"verification_mode": "NONE"`
		• Specify the full path to the server certificate placed on the client machine that is running on Windows, Linux, or Mac with the following option:
		`"certificate_path": "C:\\server_cert.pem"`
		If you are connecting to a server with a self-signed certificate, and fail to enter the full certificate path, the CLI displays the server's self-signed certificate thumbprint, and prompts you to accept or reject the certificate thumbprint.

`target`	array	The target ESX host or DRS cluster on which you want to deploy the appliance.
		
		Important:
		You must provide the name that is displayed in the vCenter inventory. For example, if the name of the target ESX host is an IP address in the vCenter inventory, you cannot provide an FQDN.
		If the target ESX host or DRS cluster is located in a folder or a structure of folders, specify the value as a comma-separated list of strings. For example,
		`["parent_folder", "child_folder", "esxi-host.domain.com"]`
		If the target ESX host is part of a cluster, specify the path as a comma-separated list of strings. For example,
		`["cluster_name", "esxi-host.domain.com"]`
		
		Note:
		The value is case-sensitive.

`vm_folder`	string	Optional. The name of the VM folder where the appliance is deployed.

Table 42: Configuration Parameters in the `new_vcsa` Section, `appliance` Subsection

Name	Type	Description
`thin_disk_mode`	Boolean	Set to `true` to deploy the appliance with thin virtual disks.
`deployment_option`	string	The size of the appliance. • Set to `tiny` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the default storage size. Deploys an appliance with 2 CPUs, 12 GB of memory, and 315 GB of storage. • Set to `tiny-lstorage` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the large storage size. Deploys an appliance with 2 CPUs, 12 GB of memory, and 1390 GB of storage. • Set to `tiny-xlstorage` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the x-large storage size. Deploys an appliance with 2 CPUs, 12 GB of memory, and 3145 GB of storage. • Set to `small` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the default storage size. Deploys an appliance with 4 CPUs, 19 GB of memory, and 380 GB of storage. • Set to `small-lstorage` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the large storage size. Deploys an appliance with 4 CPUs, 19 GB of memory, and 1435 GB of storage. • Set to `small-xlstorage` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the x-large storage size. Deploys an appliance with 4 CPUs, 19 GB of memory, and 3195 GB of storage. • Set to `medium` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the default storage size. Deploys an appliance with 8 CPUs, 28 GB of memory, and 600 GB of storage. • Set to `medium-lstorage` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the large storage size. Deploys an appliance with 8 CPUs, 28 GB of memory, and 1600 GB of storage. • Set to `medium-xlstorage` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the x-large storage size. Deploys an appliance with 8 CPUs, 28 GB of memory, and 3360 GB of storage. • Set to `large` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the default storage size. Deploys an appliance with 16 CPUs, 37 GB of memory, and 965 GB of storage. • Set to `large-lstorage` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the large storage size. Deploys an appliance with 16 CPUs, 37 GB of memory, and 1665 GB of storage. • Set to `large-xlstorage` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the x-large storage size. Deploys an appliance with 16 CPUs, 37 GB of memory, and 3425 GB of storage. • Set to `xlarge` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the default storage size. Deploys an appliance with 24 CPUs, 56 GB of memory, and 1705 GB of storage. • Set to `xlarge-lstorage` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the large storage size. Deploys an appliance with 24 CPUs, 56 GB of memory, and 1805 GB of storage. • Set to `xlarge-xlstorage` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the x-large storage size. Deploys an appliance with 24 CPUs, 56 GB of memory, and 3565 GB of storage.
`image`	string	Optional. A local file path or URL to the vCenter appliance installation package. By default the installer uses the installation package that is included in the ISO file, in the `vcsa` folder.
`name`	string	The VM name for the appliance. Must contain only ASCII characters except a percent sign (%), backslash (\), or forward slash (/) and must be no more than 80 characters in length.

Name	Type	Description
ovftool_path	string	Optional. A local file path to the OVF Tool executable file. By default the installer uses the OVF Tool instance that is included in the ISO file, in the vcsa/ovftool folder.

Table 43: Configuration Parameters in the new_vcsa Section, install_parameters Subsection

Name	Type	Description
mac_allocation_prefix	string	The MAC prefix if a prefix-based MAC allocation scheme is used for the VM managed by the appliance.
mac_allocation_prefix_length	integer	The MAC prefix length if a prefix-based MAC allocation scheme is used for the VM managed by the appliance.

Table 44: Configuration Parameters in the new_vcsa Section, network Subsection

Name	Type	Description
ip_family	string	IP version for the network of the appliance. Set to ipv4 or ipv6.
mode	string	IP assignment for the network of the appliance. Set to static or dhcp.
system_name	string	Primary network identity. Can be an IP address or FQDN, preferably FQDN. It is used only when the mode parameter is set to static. You cannot change the value of this parameter after the deployment. The FQDN and dotted-decimal numbers must comply with the RFC 1123 guidelines. Mandatory if you provide network configuration settings in the v4 and/or v6 subsections and the mode parameter is set to static.
ip	string	IP address for the appliance. Required only if you use static assignment, that is, if you set the mode parameter to static. You must set an IPv4 or IPv6 address that corresponds to the network IP version, that is, to the value of the ip.family parameter. An IPv4 address must comply with the RFC 790 guidelines. An IPv6 address must comply with the RFC 2373 guidelines.
prefix	string	Network prefix length. Use only if the mode parameter is set to static. Remove if the mode parameter is set to dhcp. The network prefix length is the number of bits that are set in the subnet mask. For example, if the subnet mask is 255.255.255.0, there are 24 bits in the binary version of the prefix length, so the network prefix length is 24. For IPv4 version, the value must be between 0 and 32. For IPv6 version, the value must be between 0 and 128.
gateway	string	IP address of the default gateway. For IPv6 version, the value can be default.

Name	Type	Description

dns_servers	string or array	IP addresses of one or more DNS servers.
		To set more than one DNS server, use a comma-separated list of strings or a comma-separated list as a single string to provide the path. For example,
		`["x.y.z.a", "x.y.z.b"]`
		or
		`"x.y.z.a, x.y.z.b"`
		Optional when the mode parameter is set to static. Not supported when mode is set to DHCP.
		Mandatory if you provide network configuration settings in the v4 and/or v6 subsections and the mode parameter is set to static.

ports	string	Optional. Port numbers that the vCenter appliance uses for direct HTTP connections. By default, port 80 redirects requests to HTTPS port 443. You can customize the vCenter HTTP and HTTPS ports. If specifying a custom HTTP and HTTPS port number, ensure that you do not use a port already in use by vCenter, or the default HTTP and HTTPS ports of 80 and 443. The options to specify a custom port are: "rhttpproxy.ext.port1":"*port_number*" for the HTTP port, and "rhttpproxy.ext.port2:"*port_number*" for the HTTPS port. The following example specifies ports 81 and 444 for the HTTP and HTTPS ports: `ports: {"rhttpproxy.ext.port1":"81", "rhttpproxy.ext.port2":"444"}` For more information on ports in use by vCenter , see Required Ports for vCenter .
v4	string	Network configuration settings for IPv4 for dual IP stack. To use more than one network, dual IP stack (IPv4 and IPv6), with vCenter 9.1 you can set the v4 and v6 parameters in your JSON template. During vCenter firstboot, both IPv4 and IPv6 network configurations are applied. If the hostname parameter in the esxi subsection is an IPv4 address, the primary network is IPv4 and the secondary is IPv6.
v6	string	Network configuration settings for IPv6 for dual IP stack. To use more than one network, dual IP stack (IPv4 and IPv6), with vCenter 9.1 you can set the v4 and v6 parameters in your JSON template. During vCenter firstboot, both IPv4 and IPv6 network configurations are applied. If the hostname parameter in the esxi subsection is an IPv6 address, the primary network is IPv6 and the secondary is IPv4.

Table 45: Configuration Parameters in the `v4` and `v6` Subsections, `network` Subsection

Name	Type	Description
mode	string	IP assignment for the network of the appliance. Set to static or dhcp.
prefix	string	Network prefix length. Use only if the mode parameter is set to static. Remove if the mode parameter is set to dhcp. The network prefix length is the number of bits that are set in the subnet mask. For example, if the subnet mask is 255.255.255.0, there are 24 bits in the binary version of the prefix length, so the network prefix length is 24. For IPv4 version, the value must be between 0 and 32. For IPv6 version, the value must be between 0 and 128.
ip	string	IP address for the appliance. Required only if you use static assignment, that is, if you set the mode parameter to static. You must set an IPv4 or IPv6 address that corresponds to the network IP version, that is, to the value of the ip.family parameter. An IPv4 address must comply with the RFC 790 guidelines. An IPv6 address must comply with the RFC 2373 guidelines.

Name	Type	Description
gateway	string	IP address of the default gateway. For IPv6 version, the value can be `default`.

Table 46: Configuration Parameters in the `new_vcsa` Section, `os` Subsection

Name	Type	Description
password	string	The password for the root user of the appliance operating system. The password must contain between 8 and 20 characters, at least one uppercase letter, at least one lowercase letter, at least one number, and at least one special character, for example, a dollar sign ($), hash key (#), at sign (@), period (.), or exclamation mark (!). All characters must be lower ASCII characters without spaces.

ntp_servers	string or array	Optional. Host names or IP addresses of one or more NTP servers for time synchronization.
		To set more than one NTP server, use a comma-separated list of strings or a comma-separated list as a single string to provide the path. For example,
		`["x.y.z.a", "x.y.z.b"]`
		or
		`"x.y.z.a, x.y.z.b"`

ssh_enable	Boolean	Set to `true` to enable SSH administrator login to the appliance. **Note:** vCenter appliance high availability requires remote SSH access to the appliance.
time_tools_sync	Boolean	Optional. Set to `true` to deploy the appliance with the VMware Tools time synchronization. VMware Tools synchronizes the time of the appliance with the time of the ESX host. Ignored if you set NTP servers for time synchronization, that is, if you set the `ntp.servers` parameter.

Table 47: Configuration Parameters in the `new_vcsa` Section, `sso` Subsection

Name	Type	Description
password	string	Password of the vCenter Single Sign-On administrator user, administrator@*`your_domain_name`*. If you are deploying a vCenter appliance as the first instance in a new vCenter Single Sign-On domain, you must set the password for thevCenter Single Sign-On administrator user. The password must contain between 8 and 20 characters, at least one uppercase letter, at least one lowercase letter, at least one number, and at least one special character, for example, a dollar sign ($), hash key (#), at sign (@), period (.), or exclamation mark (!). All characters must be ASCII characters.
domain_name	string	vCenter Single Sign-On domain name, for example, vsphere.local. If you are deploying a vCenter appliance as the first instance in a new vCenter Single Sign-On domain, you must set the name for the new vCenter Single Sign-On domain.
replication_partner_hostname	string	The system name of the partner vCenter. Required only if you are deploying a replication partner in an existing vCenter Single Sign-On domain.
first_instance	string	Set `first_instance` to `true` if you do not want to set the replication partner for this node. If you want to set the replication partner for this node, set `first_instance` to `false` and provide the value of the `replication_partner_hostname` as the FQDN of a previously converged node which is currently in an embedded vCenter topology.

Name	Type	Description
sso_port	integer	The HTTPS reverse proxy port of the partner vCenter . The default port is 443. Use only if the partner uses a custom HTTPS reverse proxy port.

Configuration Parameters in the ceip Section

Table 48: Configuration Parameters in the ceip Section, settings Subsection

Name	Type	Description
ceip_enabled	Boolean	Set to true to join the CEIP for this appliance.

Syntax of the CLI Deployment Command

You can use command arguments to set the execution parameters of the deployment command.

You can add a space-separated list of arguments to the CLI deployment command.

```
vcsa-deploy install path_to_the_json_filelist_of_arguments
```

Argument	Description
--accept-eula	Accepts the end-user license agreement. Required for executing the deployment command.
--acknowledge-ceip	Confirms your acknowledgement of your VMware Customer Experience Improvement Program (CEIP) participation. Required if the ceip.enabled parameter is set to true in the JSON deployment template.
-v, --verbose	Adds debug information to the console output.
-t, --terse	Hides the console output. Displays only warning and error messages.
--log-dir*LOG_DIR*	Sets the location of the log and other output files.
--skip-ovftool-verification	Performs basic verification of the configuration parameters in the JSON file and deploys the appliance. Does not perform verification of the OVF Tool parameters.
--no-esx-ssl-verify	Skips the SSL verification for ESX connections. **Important:** Avoid using this option because it might cause problems during deployment or after deployment because of not validated identity of the target ESX host.
--no-ssl-certificate-verification	Skips security certificate verification for all server connections.
--operation-id*OPERATION_ID*	Provides an operation ID to track installation activities.
--pause-on-warnings	Pauses and waits for acknowledgment of warnings.
--verify-template-only	Performs basic template verification of the configuration parameters in the JSON file. Does not deploy the appliance.
--precheck-only	Performs only the basic template verification and OVF Tool parameter verification. Does not deploy the appliance.
-h, --help	Displays the help message for the vcsa-deploy install command.
--template-help	Displays the help message for the use of configuration parameters in the JSON deployment file.

Argument	Description
`--provide-passwd-in-stdin`	Allows you to send passwords through standard input to the CLI installer. Passwords provided through standard input are given precedence over the passwords provided through a JSON template. Passwords can be send to the CLI Installer in the following way: `echo '{"section1.subsection1.passwordkey1": "password1", "section2.subsection2.passwordkey2": "password2"}' | vcsa-deploy install <template.json> <arguments> --provide-passwd-in-stdin` The argument can be used for deployment and upgrade operations.

After the execution finishes, you can get the exit code of the command.

Exit Code	Description
0	Command ran successfully
1	Runtime error
2	Validation error
3	Template error

Deploy a vCenter Appliance by Using the CLI

You can use the CLI installer to perform an unattended deployment of a vCenter appliance. You must run the CLI deployment from a Windows, Linux, or Mac machine that is in the network on which you want to deploy the appliance.

- See Prerequisites for Deploying the vCenter Appliance.
- Prepare Your JSON Configuration File for CLI Deployment.
- Review Syntax of the CLI Deployment Command.
- Verify that the user name with which you are logged in to your client machine, the path to the vCenter installer, the path to your JSON configuration file, and the string values in your JSON configuration file contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

1. Navigate to the `vcsa-cli-installer` subdirectory for your operating system.
   - If you are running the deployment on Windows OS, navigate to the `vcsa-cli-installer\win32` directory.
   - If you are running the deployment on Linux OS, navigate to the `vcsa-cli-installer/lin64` directory.
   - If you are running the deployment on Mac OS, navigate to the `vcsa-cli-installer/mac` directory.
2. Optional: Run a pre-deployment check without deploying the appliance to verify that you prepared the deployment template correctly.

   ```
   vcsa-deploy install --accept-eula --precheck-only path_to_the_json_file
   ```

3. Run the deployment command.

   ```
   vcsa-deploy install --accept-eula --acknowledge-ceip optional_argumentspath_to_the_json_file
   ```

   Use *optional_arguments* to enter space-separated arguments to set additional execution parameters of the deployment command.

   For example, you can set the location of the log and other output files that the installer generates.

   ```
   vcsa-deploy install --accept-eula --acknowledge-ceip --log-dir=path_to_the_locationpath_to_the_json_file
   ```

Deploy Multiple vCenter Appliances Using the CLI

You can deploy multiple instances vCenter appliances concurrently (in batch mode) using the CLI installer.

To deploy multiple instances concurrently, create JSON templates for all the vCenter instances in your deployment. The CLI installer assesses the topology of the deployment using the JSON templates, and determines the order. For this reason, the JSON templates must use static IP addresses for all vCenter instances in the deployment that are dependant upon one another.

> **Important:**
> The JSON templates that you create for each appliance must use a static IP address to resolve the network addresses of other appliances in the deployment upon which they have a dependency.

To perform the batch deployment, place the JSON templates defining your deployment in a single directory. When invoked, the CLI installer deploys your existing deployment using the topology that is defined in the JSON templates.

1. In your workspace, create a folder to contain the JSON files for batch deployment. For example, *`MyWorkspace/BatchDeploy`*.
2. Prepare each JSON configuration file and copy the file to your batch deployment folder. See Prepare Your JSON Configuration File for CLI Deployment for instructions on configuring the JSON files.
3. Navigate to the `vcsa-cli-installer` subdirectory for your operating system.
   - If you are running the deployment on Windows OS, navigate to the `vcsa-cli-installer\win32` directory.
   - If you are running the deployment on Linux OS, navigate to the `vcsa-cli-installer/lin64` directory.
   - If you are running the deployment on Mac OS, navigate to the `vcsa-cli-installer/mac` directory.
4. Optional: Run a pre-deployment check without deploying the appliance to verify that you prepared the deployment template correctly. For example:

   ```
   vcsa-deploy install --precheck-only MyWorkspace/BatchDeploy
   ```

5. Run the deployment command. For example,

   ```
   vcsa-deploy install --accept-eula --acknowledge-ceip optional_argumentsMyWorkspace/BatchDeploy
   ```

   Use *`optional_arguments`* to enter space-separated arguments to set additional execution parameters of the deployment command.

   For example, you can set the location of the log and other output files that the installer generates.

   ```
   vcsa-deploy install --accept-eula --acknowledge-ceip --log-dir=path_to_the_locationMyWorkspace/BatchDeploy
   ```

File-Based Backup and Restore of vCenter

vCenter supports a file-based backup and restore mechanism that helps you to recover your environment after failures.

You can use the vCenter Management Interface to create a file-based backup of the vCenter. After you create the backup, you can restore it by using the GUI installer of the appliance.

You use the vCenter Management Interface to perform a file-based backup of the vCenter core configuration, inventory, and historical data of your choice. The backed-up data is streamed over FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB to a remote system. The backup is not stored on the vCenter.

You can perform a file-based restore only for a vCenter that you have previously backed up by using the vCenter Management Interface. You can perform such restore operation by using the GUI installer of the vCenter appliance. The process consists of deploying a new vCenter appliance and copying the data from the file-based backup to the new appliance.

> **Important:**
> If you back up a vCenter High Availability cluster, the backup operation only backs up the primary vCenter instance. Before restoring a vCenter High Availability cluster, you must power off the active, passive, and witness nodes. The restore operation restores the vCenter in non- vCenter High Availability mode. You must reconstruct the cluster after the restore operation completes successfully.

Considerations and Limitations for File-Based Backup and Restore

When you back up or restore a vCenter instance, consider these considerations and limitations.

Protocols

The following considerations apply to file-based backup and restore protocols:

- FTP and HTTP are not secure protocols.
- Backup servers must support minimum of 10 simultaneous connections for each vCenter.
- You can use an FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB server that is up and running with sufficient disk space to store the backup.
- You must have a complete backup location path created with write and read permissions.
- Only explicit mode is supported for FTPS.
- If you use HTTP or HTTPS, you must enable WebDAV on the backup web server.
- You can use only FTP, FTPS, HTTP, or HTTPS to transmit data through an HTTP proxy server.
- You can use IPv4 and IPv6 URLs in file-based backup and can restore of a vCenter. Mixed mode of IP versions between the backup server and the vCenter is unsupported.

Configuration

After a restore, the following configurations revert to the state when the backup was taken.

- Virtual machine resource settings
- Resource pool hierarchy and setting
- Cluster-host membership
- DRS configuration and rules

Storage DRS

If the configuration changes, the following might change after a restore.

- Datastore Cluster configuration
- Datastore Cluster membership
- Datastore I/O Resource Management (Storage I/O Control) settings
- Datastore-Datacenter membership
- Host-Datastore membership

Distributed Power Management

If you put an ESX host into standby mode after a backup, the vCenter might force the host to exit standby mode when you restore to the backup.

Distributed Virtual Switch

If you use a distributed virtual switch, you are advised to export separately the distributed virtual switch configuration before you restore from a backup. You can import the configuration after the restore. If you omit this consideration, you may lose the changes made to a distributed virtual switch after the backup. For detailed steps, see the Broadcom knowledge base article 338595.

However, when you restore the host-membership after a backup, you will not lose any host that you have added or removed from the distributed virtual switch after the backup.

Note:
You can only restore the distributed virtual switch-host membership if the host is part of the cluster.

Content Libraries

If you delete libraries or items after a backup, you cannot access or use these libraries or items after the restore. You can only delete such libraries or items. A warning message notifies you that there are missing files or folders in the storage backup.

If you create new items or item files after the backup, the Content Library Service has no record of the new items or files after the restore operation. A warning notifies you that extra folders or files were found on the storage backup.

If you create new libraries after the backup, the Content Library Service has no record of the new libraries after restore. The library content exists on the storage backing, but no warning is displayed. You must manually clean the new libraries.

Virtual Machine Life Cycle Operations

- Restoring vCenter from a backup that was taken during in-flight relocation operations in the vCenter instance.
  After you restore vCenter, the vCenter view of the virtual machines might be out of sync with the ESX view of the virtual machines. This is also true if you performed the backup during in-flight operations on vCenter. If virtual machines disappear after you restore vCenter, you can refer to the following cases.
  a. The missing virtual machine is located on the destination ESX host and is registered with the destination ESX host, but it is either an orphan or not in the vCenter inventory. You must manually add the virtual machine to the vCenter inventory.
  b. The missing virtual machine is located on the destination ESX host, but it is not registered with the destination ESX host and it is not in the vCenter inventory. You must manually register the virtual machine to the ESX host and add the virtual machine back to the vCenter inventory.
  c. The missing virtual machine is located on the destination ESX host, but it is not registered with the destination ESX host. In the vCenter instance, the missing virtual machine is marked as orphaned. You must remove the virtual machine from the vCenter inventory and add it again.
- Restoring vCenter from a backup that has an out-of-date linked clone virtual machine layout.
  If you create a linked clone virtual machine after the backup and you restore vCenter from the old backup, then after the restore, the vCenter does not know about the new linked clone virtual machine until vCenter discovers the new linked clone virtual machine. If you remove all existing virtual machines before the new linked clone virtual machine is discovered, then the removal of existing virtual machines corrupts the new linked clone due to missing disks. To avoid this, you must wait until all linked clone virtual machines are discovered by the vCenter before you remove virtual machines.
- Restoring vCenter from a backup that was taken during virtual machine registration.
  If you are registering a virtual machine during the backup and you restore vCenter from the old backup, then after the restore, the virtual machine is marked as orphaned in the vCenter instance. You must manually add the virtual machine to the vCenter inventory.

vCenter Enhanced Linked Mode

To restore a vCenter instance that is part of an Enhanced Linked Mode, you must have at least one running vCenter that has the VMware Directory Service database.

If the replication partners are temporarily unavailable or network-isolated, fix the replication partners before restoring the node. If all nodes in the federation have failed, you can ignore the warning and restore only the first node without replication partners. Restore the rest of the nodes according to the Enhanced Linked Mode topology.

vSphere High Availability

Restoring vCenter from a backup might cause it to rollback to older version for the vSphere HA cluster state (HostList, ClusterConfiguration, VM protection state) while the hosts in the cluster have the latest version for the cluster state. You need to make sure the vSphere HA cluster state stays the same during restore and backup operations. Otherwise, the following problems might occur.

- If hosts are added or removed to or from the vSphere HA cluster after backup and before vCenter restore, virtual machines could potentially failover to hosts not being managed by the vCenter but are still part of the HA cluster.

- Protection state for new virtual machines is not updated on the vSphere HA agents on the hosts that are part of the vSphere HA cluster. As a result, virtual machines are not protected or unprotected.
- New cluster configuration state is not updated on the vSphere HA agents on the hosts that are part of the vSphere HA cluster.

vCenter High Availability

Restoring vCenter requires vCenter HA to be reconfigured.

Storage Policy Based Management

Restoring vCenter from a backup can lead to the following inconsistencies related to storage policies, storage providers, and virtual machines.

- Registered storage providers after backup are lost.
- Unregistered storage providers after backup re-appear and might show a different provider status.
- Changes, such as create, delete, or update, performed on storage policies after backup are lost.
- Changes, such as create, delete, or update, performed on storage policy components after backup are lost.
- Default policy configuration changes for datastores performed after backup are lost.
- Changes in the storage policy association of the virtual machine and its disks, and in their policy compliance might occur.

Virtual Storage Area Network

Restoring vCenter from a backup might cause inconsistencies in the vSAN. For information on how to check vSAN health, see *Administering VMware vSAN*.

Patching

Restoring vCenter from a backup might result in missing security patches. You must apply them again after the restore is complete. For information on patching the vCenter, see Patching vCenter in *vSphere Upgrade*.

Schedule a File-Based Backup

You can schedule file-based backups for vSphere 8.0 and later. You can set up a schedule that is used to perform periodic backups.

- You must have an FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB server up and running with sufficient disk space to store the backup.

The schedule can be set up with information about the backup location, recurrence, and retention for the backups.

You can only set up one schedule at a time.

1. In a web browser, go to the vCenter Management Interface, https://`appliance-IP-address-or-FQDN`:5480.
2. Log in as root.
3. In the vCenter Management Interface, click **Backup**.
4. Click **Configure** to set up a backup schedule.
5. Enter the backup location details.

Option	Description
Backup location	Enter the backup location, including the protocol to use to connect to your backup server, the port, the server address, and the backup folder to store the backup files. Use one of the following protocols: FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB. For FTP, FTPS, HTTP, or HTTPS the path is relative to the home directory configured for the service.

Option	Description
Backup server credentials	Enter a user name and password of a user with**write** privileges on the backup server.

6. Set the schedule recurrence and time for the backup.

   The recurrence can be set daily, weekly, or you can customize the schedule to run the backup on a specific day or days of the week. You can specify the time of day to run the backup. The default time is 11:59pm.

7. Optional: Enter an Encryption Password if you want to encrypt your backup file.

   If you select to encrypt the backup data, you must use the encryption password for the restore procedure.

8. Select **Retain all backups** or enter the number of backups to retain.

   The retention information provides the number of backups to retain for a given vCenter.

   Note:
   This policy applies to both manual and scheduled backups.

9. Optional: Select **Stats, Events, and Tasks** to back up additional historical data from the database.

10. Optional: Select **Supervisors Control Plane** to activate the backup and restore of a Supervisor in the case of any disaster scenario. This allows you to record the state of the Supervisors Control Planes in a vCenter system.

    For more information, see Backing Up and Restoring vSphere Supervisor.

11. Click **Create**.

    The backup schedule information is populated in the Backup page.

The complete and in progress backups are listed under Activity.
You can perform an immediate backup with the existing schedule information by selecting **Use backup location and user name from backup schedule** from the backup schedule on the Backup Now dialog box.

Manually Back Up vCenter by Using the vCenter Management Interface

You can use the vCenter Management Interface to back up the vCenter instance. You can select whether to include historical data, such as stats, events, and tasks, in the backup file.

- You must have and FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB server up and running with sufficient disk space to store the backup.

  Note:

  The backup operation for a vCenter High Availability cluster, backs up only the active node.

1. In a web browser, go to the vCenter Management Interface, https://*vcenter-IP-address-or-FQDN*:5480.
2. Log in as root.
3. In the vCenter Management Interface, click **Backup**.

   The table under **Activity** displays the most current backup version that is taken of the vCenter.

4. Click **Back up now**.

   The **Back up Now** wizard opens.

5. Optional: Select **Use backup location and user name from backup schedule** to use the information from a scheduled backup.
6. Enter the backup location details.

Option	Description
Backup location	Enter the backup location, including the protocol to use to connect to your backup server, the port, the server address, and the backup folder to store the backup files. Use one of the following protocols: FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB.

Option	Description
	For FTP, FTPS, HTTP, or HTTPS the path is relative to the home directory configured for the service.
Backup server credentials	Enter a user name and password of a user with **write** privileges on the backup server. **Note:** Username and password should only contain ASCII characters.

7. Optional: Enter an Encryption Password if you want to encrypt your backup file.
   If you select to encrypt the backup data, you must use the encryption password for the restore procedure.
8. Optional: Select **Stats, Events, and Tasks** to back up additional historical data from the database.
9. Optional: In the **Description** text box, enter a description of the backup.
10. Click **Start** to begin the backup process.

The complete and in progress backups are listed under Activity.

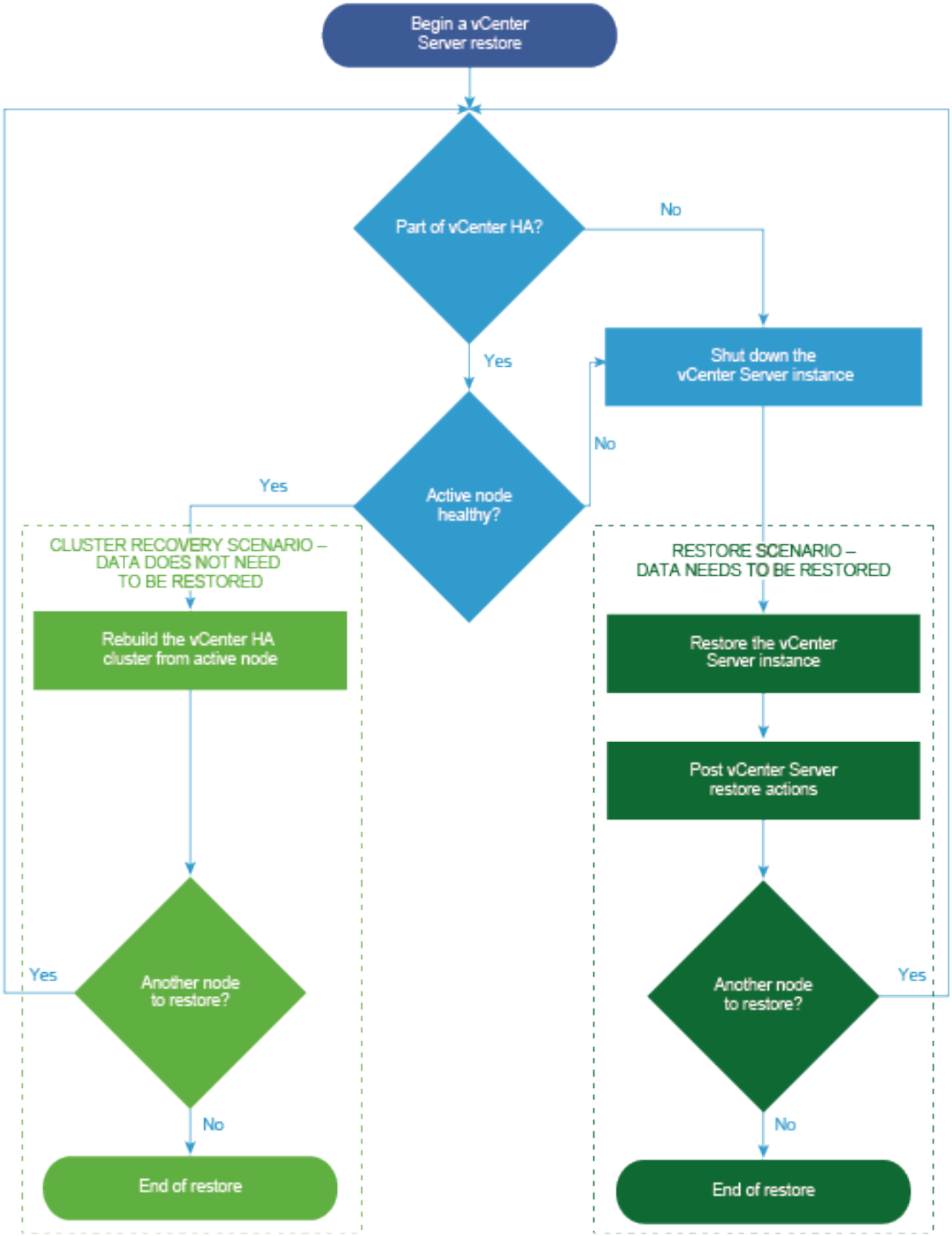
Restore vCenter from a File-Based Backup

You can use the vCenter appliance GUI installer to restore a vCenter to an ESX host or a vCenter instance. The restore procedure has two stages. The first stage deploys a new vCenter appliance. The second stage populates the newly deployed vCenter appliance with the data that is stored in the file-based backup.

- Verify that your system meets the minimum software and hardware requirements. See System Requirements for the vCenter Appliance.

- If the vCenter instance is part of a vCenter High Availability cluster, you must power off the active, passive, and witness nodes of the cluster before restoring the vCenter.

Figure 17: vCenter Restore Workflow



Stage 1 - Deploy a New Appliance

In stage 1 of the restore process, you deploy the OVA file, which is included in the vCenter GUI installer.

- Download and mount the vCenter installer. See Download and Mount the vCenter Installer.

  Note:

  If you are restoring a backup from a product that has a vCenter product patch applied, you must download the full ISO of that particular patch.
- If you plan to restore the vCenter on an ESX host, verify that the target ESX host is not in lockdown or maintenance mode or part of a fully automated DRS cluster.
- If you plan to restore the vCenter on a DRS cluster of a vCenter inventory, verify that the cluster contains at least one ESX host that is not in lockdown or maintenance mode.
- If you plan to assign a static IP address to the appliance, verify that you have configured the forward and reverse DNS records for the IP address.
- If you are attempting to restore a vCenter instance that is still running, power off the backed-up vCenter before you start the restore operation.

As an alternative to performing the first stage of the restore with the GUI installer, you can deploy the OVA file of the new vCenter appliance by using the vSphere Client. After the OVA deployment, you must log in to the vCenter Management Interface of the newly deployed appliance to proceed with the second stage of the restore process. See "Deploy an OVF or OVA Template" in *vSphere Virtual Machine Administration* vSphere Virtual Machine Administration for information about deploying an OVA file using the vSphere Client.

1. In the vCenter installer, navigate to the `vcsa-ui-installer` directory, go to the subdirectory for your operating system, and run the installer executable file.
   - For Windows OS, go to the `win32` subdirectory, and run the `installer.exe` file.
   - For Linux OS, go to the `lin64` subdirectory, and run the `installer` file.
   - For Mac OS, go to the `mac` subdirectory, and run the `Installer.app` file.
2. On the Home page, click **Restore**.
3. Review the Introduction page to understand the restore process and click **Next**.
4. Read and accept the license agreement, and click **Next**.
5. On the Enter backup details page, enter the details of the backup file that you want to restore, and click **Next**.

Option	Description
Backup location	Enter the server address and backup folder where the backup files are stored. You can also enter the IP address or hostname of the backup server and browse for the location of the backup folder. Specify the protocol to use to retrieve the backup from your backup server. You can select FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB. For example, `ftp://<server_IP_address>/tmp/vCenter/hostname_vcenter.com/<backup_filename>`
User name	Enter the user name of a user with **read** privileges on the backup server.
Password	Enter the password of the user with **read** privileges on the backup server.

6. Review the backup information, and click **Next**.
7. Connect to the ESX host or vCenter on which you want to deploy the vCenter appliance to use for the restore operation.

Option	Steps
You can connect to an ESX host on which to deploy the appliance to use for the restore operation.	1. Enter the FQDN or IP address of the ESX host. 2. Enter the HTTPS port of the ESX host. 3. Enter the user name and password of a user with administrative privileges on the ESX host, for example, the root user. 4. Click **Next**. 5. Verify that the certificate warning displays the SHA1 thumbprint of the SSL certificate that is installed on the target ESX host, and click **Yes** to accept the certificate thumbprint.
You can connect to a vCenter instance and browse the inventory to select an ESX host or DRS cluster on which to deploy the appliance to use for the restore operation.	1. Enter the FQDN or IP address of the vCenter instance. 2. Enter the HTTPS port of the vCenter instance. 3. Enter the user name and password of user with vCenter Single Sign-On administrative privileges on the vCenter instance, for example, the administrator@*your_domain_name* user. 4. Click **Next**. 5. Verify that the certificate warning displays the SHA1 thumbprint of the SSL certificate that is installed on the target vCenter instance, and click **Yes** to accept the certificate thumbprint. 6. Select the data center or data center folder that contains the ESX host or DRS cluster on which you want to deploy the appliance, and click **Next**. **Note:** You must select a data center or data center folder that contains at least one ESX host that is not in lockdown or maintenance mode. 7. Select the ESX host or DRS cluster on which you want to deploy the appliance, and click **Next**.

8. Accept the certificate warning.
9. Enter a name for the vCenter appliance, set up the password for the root user, and click **Next**.
10. Select the deployment size for the new vCenter appliance depending on the size of your vSphere inventory.

Deployment Profile	Description
Tiny	Deploys an appliance with 2 vCPUs and 14 GB of memory. Suitable for environments with up to 10 hosts or 100 virtual machines
Small	Deploys an appliance with 4 vCPUs and 21 GB of memory. Suitable for environments with up to 100 hosts or 1,000 virtual machines
Medium	Deploys an appliance with 8 vCPUs and 30 GB of memory. Suitable for environments with up to 400 hosts or 4,000 virtual machines
Large	Deploys an appliance with 16 vCPUs and 39 GB of memory. Suitable for environments with up to 1,000 hosts or 10,000 virtual machines
X-Large	Deploys an appliance with 24 vCPUs and 58 GB of memory.

Deployment Profile	Description
	Suitable for environments with up to 2,500 hosts or 30,000 virtual machines. Suitable for environments with up to 2,500 hosts or 45,000 virtual machines with X-Large Storage.

11. Select the storage size for the new vCenter appliance, and click **Next**.

    Important:
    You must consider the storage size of the appliance that you are restoring.

Storage Size Option	Tiny	Small	Medium	Large	X-Large
Default	Deploys an appliance with 619 GB of storage.	Deploys an appliance with 734 GB of storage.	Deploys an appliance with 933 GB of storage.	Deploys an appliance with 1383 GB of storage.	Deploys an appliance with 2308 GB of storage.
Large	Deploys an appliance with 2059 GB of storage.	Deploys an appliance with 2084 GB of storage.	Deploys an appliance with 2233 GB of storage.	Deploys an appliance with 2283 GB of storage.	Deploys an appliance with 2408 GB of storage.
X-Large	Deploys an appliance with 4319 GB of storage.	Deploys an appliance with 4344 GB of storage.	Deploys an appliance with 4493 GB of storage.	Deploys an appliance with 4543 GB of storage.	Deploys an appliance with 4668 GB of storage.

12. From the list of available datastores, select the location where all the virtual machine configuration files and virtual disks will be stored and, optionally, enable thin provisioning by selecting **Enable Thin Disk Mode**.
13. On the Configure network settings page review the settings populated from the backup file of the vCenter.
14. Optional: Edit the network configuration to match the current network environment where the vCenter is restored.
15. On the Ready to complete stage 1 page, review the deployment settings for the restored vCenter appliance and click **Finish** to start the OVA deployment process.
16. Wait for the OVA deployment to finish, and click **Continue** to proceed with stage 2 of the restore process to transfer the data to the newly deployed appliance.

    Note:
    If you exit the wizard by clicking **Close**, you must log in to the vCenter Management Interface to transfer the data.

The newly deployed vCenter appliance is running on the target server but the data is not copied from the backup location.

Stage 2 - Transfer Data to the Newly Deployed Appliance

After the OVA deployment finishes, you are redirected to stage 2 of the restore process in which the data from the backup location is copied to the newly deployed vCenter appliance.

1. Review the introduction to stage 2 of the restore process and click **Next**.
2. Review the backup details and click **Next**.
3. If you are restoring a node with enhanced linked mode support for vCenter appliance, you are asked to provide the Single Sign-On credentials. Enter the Single Sign-On user name and password, then click **Validate and Recover**.
4. On the Ready to complete page, review the details, click **Finish**, and click **OK** to complete stage 2 of the restore process.

   The restore process restarts the vCenter Management Service. You cannot access the vCenter Management API during the restart.

   Important:
   If a restore operation of a vCenter appliance VM results with a failure, you must power off and delete the partially restored VM. After that you can try to restore the VM again.

5. Optional: After the restore process finishes, click the **https://*vcenter_IP_or_fqdn*/ui** to go to the vSphere Client and log in to the vCenter instance , or click the **https://*vcenter_IP_or_fqdn*:443** to go to the vCenter Getting Started page.
6. Click **Close** to exit the wizard.

   You are redirected to the vCenter Getting Started page.
7. If the backed up vCenter node is part of a vCenter High Availability cluster, the last needs to be reconfigured after the restore operation completes successfully.

   For information about how to perform backup and restore operations, see *vSphere Availability*.

After successful completion of the restore operation, in the event of a total loss of all storage and servers where all the physical hardware or the data in the hardware is lost, follow the instructions in the Broadcom knowledge base article 332503 in order to recover the system to a pre-restore state at the time of the backup.

Image-Based Backup and Restore of a vCenter Environment

You can use vSphere APIs with a third-party product to back up and restore a virtual machine that contains vCenter.

You can perform a full image backup of a virtual machine that contains vCenter. The virtual machine must use a fully qualified domain name (FQDN) with correct DNS resolution, or the hostname must be configured to be an IP address.

Considerations and Limitations for Image-Based Backup and Restore

When you restore a vCenter environment, take into account these considerations and limitations.

> **Note:**
> Restoring a vCenter instance with DHCP network configuration results in changing its IP address. The changed IP address prevents some vCenter services from starting properly. To start all vCenter services successfully, after the restore, you must reconfigure the IP address of the restored vCenter instance to the IP address that the instance was set to when you performed the backup.

Synchronizing Clocks with NTP Time Synchronization Prior to Backup

Before creating a backup of your vCenter deployment, verify that all components on the vSphere network have their clocks synchronized using NTP time synchronization. See Synchronizing Clocks on the vSphere Network.

Configuration

After a restore, the following configurations revert to the state when the backup was taken.

- Virtual machine resource settings
- Resource pool hierarchy and setting
- Cluster-host membership
- DRS configuration and rules

Storage DRS

If the configuration changes, the following might change after a restore.

- Datastore Cluster configuration
- Datastore Cluster membership
- Datastore I/O Resource Management (Storage I/O Control) settings
- Datastore-Datacenter membership
- Host-Datastore membership

Distributed Power Management

If you put a host into standby mode after a backup, the vCenter might force the host to exit standby mode when you restore to the backup.

Distributed Virtual Switch

If you use a distributed virtual switch, you are advised to export separately the distributed virtual switch configuration before you restore to a backup. You can import the configuration after the restore. If you omit this consideration, you may lose the changes made to a distributed virtual switch after the backup. For detailed steps, see the Broadcom knowledge base article 338595.

Content Libraries

If you delete libraries or items after a backup, you cannot access or use these libraries or items after the restore. You can only delete such libraries or items. A warning message notifies you that there are missing files or folders in the storage backup.

If you create new items or item files after the backup, the Content Library Service has no record of the new items or files after the restore operation. A warning notifies you that extra folders or files were found on the storage backup.

If you create new libraries after the backup, the Content Library Service has no record of the new libraries after restore. The library content exists on the storage backing, but no warning is displayed. You must manually clean the new libraries.

Virtual Machine Life Cycle Operations

- Restoring vCenter from a backup that was taken while there are in-flight relocation operations within the vCenter instance.

  After you restore vCenter, the vCenter view of the virtual machines may be out of sync with the ESX view of the virtual machines. This is also true if you performed the backup while there were in-flight operations on vCenter. If virtual machines disappear after you restore vCenter, you can refer to the following cases.

  a. The missing virtual machine is located on the destination ESX host and is registered with the destination ESX host, but it is not in the vCenter inventory. You must manually add the virtual machine to the vCenter inventory.
  b. The missing virtual machine is located on the destination ESX host, but it is not registered with the destination ESX host and it is not in the vCenter inventory. You must manually register the virtual machine to the ESX and add the virtual machine back to the vCenter inventory.
  c. The missing virtual machine is located on the destination ESX host, but it is not registered with the destination ESX host. Within the vCenter instance, the missing virtual machine is marked as orphaned. You must remove the virtual machine from the vCenter inventory and add it again.
- Restoring vCenter from a backup that has an out of date linked clone virtual machine layout.

  If you create a linked clone virtual machine after the backup and you restore vCenter from the old backup, then after the restore, vCenter does not know about the new linked clone virtual machine until vCenter discovers the new linked clone virtual machine. If you remove all existing virtual machines before the new linked clone virtual machine is discovered, then the removal of existing virtual machines corrupts the new linked clone due to missing disks. To avoid this corruption, you must wait until all linked clone virtual machines get discovered by the vCenter before you remove virtual machines.

vSphere High Availability

Restoring vCenter from a backup may cause it to roll back to older version for the vSphere HA cluster state (HostList, ClusterConfiguration, VM protection state) while the hosts in the cluster have the latest version for the cluster state. Ensure that the vSphere HA cluster state stays the same during restore and backup operations. Otherwise, the following potential problems are present.

- If hosts are added or removed to/from the vSphere HA cluster after backup and before vCenter restore, virtual machines could potentially fail over to hosts not managed by the vCenter but are still part of the HA cluster.
- Protection states for new virtual machines are not updated on the vSphere HA agents on the hosts which are part of the vSphere HA cluster. As a result, virtual machines are not protected/unprotected.
- New cluster configuration state is not updated on the vSphere HA agents on the hosts which are part of the vSphere HA cluster.

vCenter High Availability

Restoring vCenter requires vCenter HA to be reconfigured.

Storage Policy Based Management

Restoring vCenter from a backup can lead to the following inconsistencies related to storage policies, storage providers, and virtual machines.

- Registered storage providers after backup are lost.
- Unregistered storage providers after backup reappear and might show different provider status.
- Changes, such as create, delete, or update, performed on storage policies after backup are lost.
- Changes, such as create, delete, or update, performed on storage policy components after backup are lost.
- Default policy configuration changes for datastores performed after backup are lost.
- Changes in the storage policy association of the virtual machine and its disks, and in their policy compliance might occur.

Virtual Storage Area Network

Restoring vCenter from a backup may cause inconsistencies in the vSAN. For information how to check vSAN health, see Monitoring vSAN Skyline Health.

Patching

Restoring vCenter from a backup might result in missing security patches. You must apply them again after the restore is complete. For information on patching the vCenter appliance, see vCenter Upgrade.

Restore a vCenter Image-based Environment

You can use a third-party product that uses vSphere APIs to restore a virtual machine that contains vCenter.

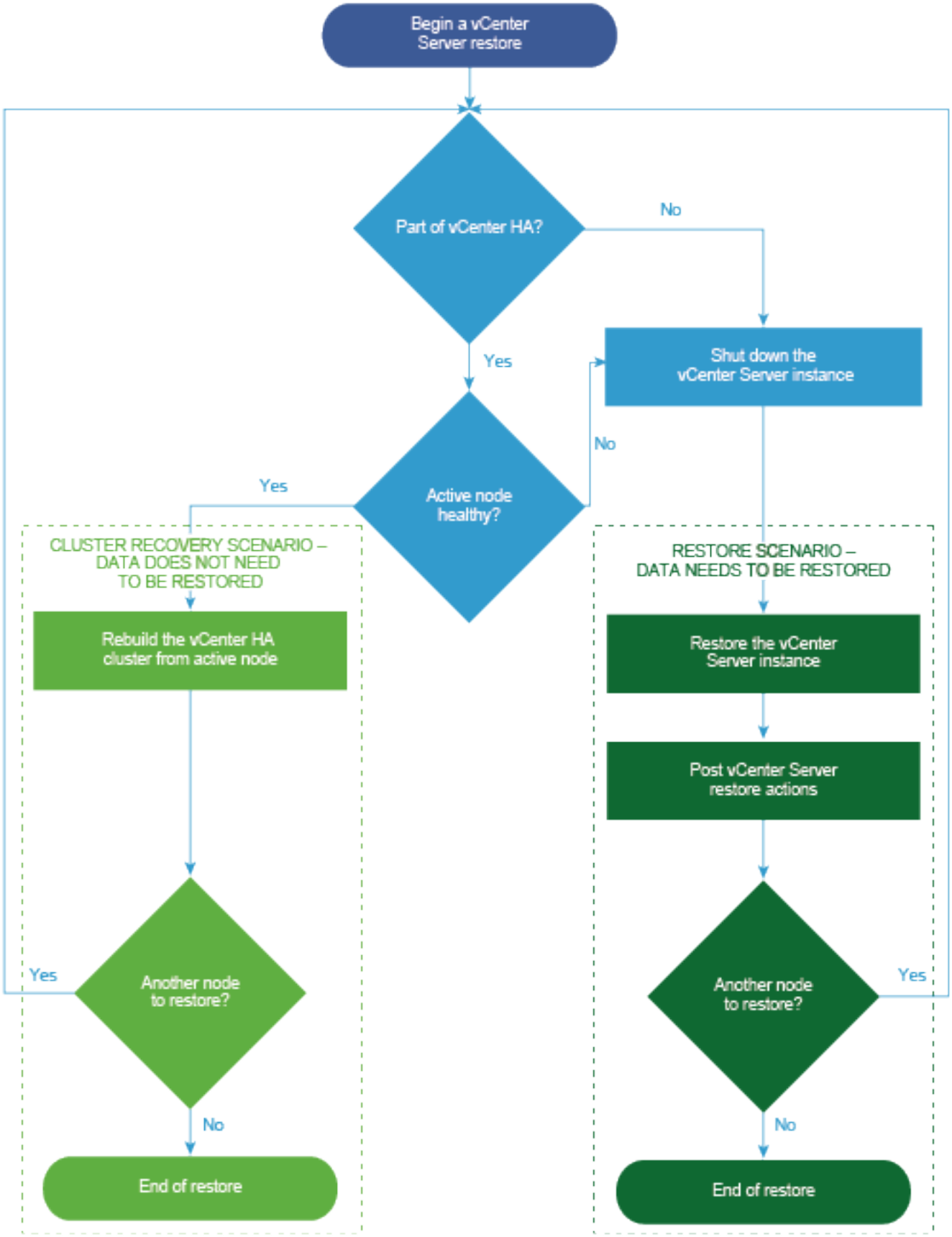
You can perform an image-based restore of a virtual machine that contains vCenter. The virtual machine must use a fully qualified domain name (FQDN) with correct DNS resolution, or the host name of the machine must be configured to be an IP address.

You can restore a virtual machine to the original location by either overwriting the backed up virtual machine or by creating a new virtual machine that contains the restored vCenter on the same ESX host. You can also restore the virtual machine on a new ESX host.

You can restore a virtual machine that contains vCenter directly on the ESX host that is running the third-party appliance when the vCenter service becomes unavailable or when you cannot access the third-party user interface by using the vSphere Client.

> **Important:**
> Restoring virtual machines that have snapshots or that are configured with Fault Tolerance is unsupported.

Figure 18: vCenter Restore Workflow



Restore a vCenter Instance

You can use a third-party product to restore a vCenter environment.

Important:
You can back up and restore only virtual machines that contain vCenter instances. You cannot back up and restore physical machines that are running vCenter by using a third-party product.

1. Restore the vCenter virtual machine onto the ESX host using a third-party solution.
2. Use the `service-control --status --all` command to verify that the services have started.
   Wait for all the vCenter services to start, which can take several minutes.
3. Launch the vCenter Management Interface, https://*appliance-IP-address-or-FQDN*:5480.
4. When you launch the vCenter Management Interface, based on the change in the vCenter appliance data you will see one of the following:
   a) If there is a change in the vCenter appliance data that needs to be synchronized, reconciliation window will be displayed. Follow the prompts to complete the reconcilation and then provide the Single Sign-On credentials to log into the vCenter Management Interface.
   b) If there is no change in the vCenter appliance data, the vCenter Management Interface login screen will be displayed.

Power on the restored virtual machine that contains the restored vCenter instance.

Restore a vCenter Enhanced Linked Mode Environment

You can restore a vCenter Enhanced Linked Mode environment using a third-party solution.

Important:
You can back up and restore only virtual machines that contain vCenter instances. You cannot back up and restore physical machines that are running vCenter by using a third-party product.

1. Restore the vCenter virtual machine onto the ESX host using a third-party solution.
   The services are masked and are not yet running.
2. Use the `systemctl status applmgmt` command to verify that the `systemd` instance of the `applmgmt` service has started.
   Wait for all the vCenter services to start, which can take several minutes.
3. Log in to the vCenter Management Interface, https://*appliance-IP-address-or-FQDN*:5480.

Power on the restored virtual machine that contains the restored vCenter instance.

After You Deploy the vCenter Appliance

After you deploy the appliance, consider these post installation options before adding inventory for the vCenter to manage.

If you plan to install Windows 11 as a guest operating system on a virtual machine, you must configure a key provider. Installing Windows 11 requires a Trusted Platform Module (TPM) 2.0. When installing Windows 11 as a guest operating system on a virtual machine, instead of using a physical TPM, you can use a virtual Trusted Platform Module (vTPM). A vTPM is a software-based representation of a physical TPM 2.0 chip. A vTPM depends on virtual machine encryption to secure vital TPM data, and so requires that you configure a key provider. For information about the key providers supported by vSphere, see the Virtual Machine Encryption chapter in the *vSphere Security* documentation. The easiest way is to configure a VMware vSphere®Native Key Provider™. vSphere Native Key Provider is included in all vSphere editions and does not require an external key server. For information about configuring a vSphere Native Key Provider, see Configuring and Managing vSphere Native Key Provider chapter in the *vSphere Security* documentation. As with all security solutions, consider the system design, implementation considerations, and tradeoffs of using vSphere Native Key Provider.

For information about configuring the vSphere Authentication Proxy service, see Start the vSphere Authentication Proxy Service in *vSphere Security*.

For information about upgrading vCenter, see the vCenter Upgrade documentation.

Log in to Your vCenter by Using the vSphere Client

Learn how to use the vSphere Client to log in to vCenter and manage your vSphere inventory.

The vSphere Client is automatically installed as part of the vCenter appliance deployment. This way, the vSphere Client always points to the same vCenter Single Sign-On instance.

1. Open a web browser and enter the URL for your vCenter instance: `https://vcenter_IP_address_or_fqdn`
   As an alternative, you can open a web browser and enter the URL for the vSphere Client: `https://vcenter_IP_address_or_fqdn/ui`.
2. If a warning message about a potential security risk appears, select to continue to the website.

Browser	Action
Microsoft Edge	1. Click **Details**. 2. Under the additional message that appears, click **Go on to the webpage**.
Mozilla Firefox	1. Click **Advanced**. 2. Under the additional message that appears, click **Accept the Risk and Continue**.
Google Chrome	1. Click **Advanced**. 2. Under the additional message that appears, click **Proceed to** *vcenter_IP_address_or_fqdn*.

3. If the warning message about a potential security risk appears again, repeat Step 2.
4. You can log in through vCenter Single Sign-On or your local user account.

The vSphere Client connects to all the vCenter systems on which the specified user has permissions, and you can view and manage the vSphere inventory.

Repoint vCenter to Another vCenter in a Different Domain

You can move a vCenter from one vCenter Single Sign-On domain to another vCenter Single Sign-On domain. Services such as tagging and licensing are retained and migrated to the new domain.

The following use cases are supported:

- You can migrate a vCenter from an existing domain to an another existing domain with or without replication. The migrated vCenter moves from its current vCenter Single Sign-On domain and joins the other existing domain as another vCenter connected via enhanced linked mode.
  - See Repoint a Single vCenter Node to an Existing vCenter Single Sign-On Domain without a Replication Partner for instructions on repointing a single embedded node from one domain to an existing domain without a replication partner.
  - See Repoint a vCenter Node to an Existing vCenter Single Sign-On Domain with a Replication Partner for instructions on repointing an embedded node from one domain to an existing domain using replication.
- You can migrate a vCenter from an existing domain to a newly created domain (where the migrated vCenter is the first instance). See Repoint a vCenter Node to a New vCenter Single Sign-On Domain for instructions of this type of repointing. In this case, there is no replication partner.

Repoint a Single vCenter Node to an Existing vCenter Single Sign-On Domain without a Replication Partner

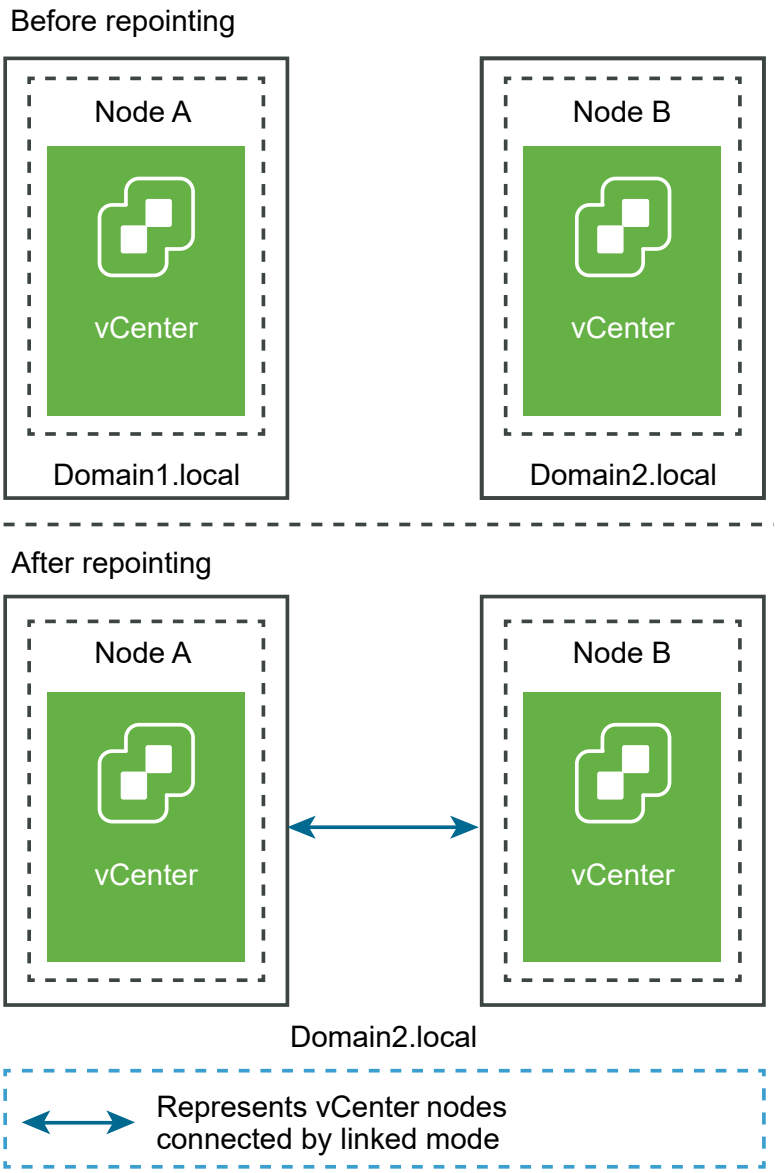
You can repoint a single vCenter from one vCenter Single Sign-On domain to an existing vCenter Single Sign-On domain without a replication partner. Each vCenter Single Sign-On domain contains a single vCenter.

- Repointing is only supported with vCenter 6.7 Update 1 and later.

- You must repoint to a vCenter that is of the same version and nodes that are of the same version and build number.
- To ensure no loss of data, take a file-based backup of each node before proceeding with repointing the vCenter.

See Figure: Repointing a Single from One Domain to an Existing Domain for an example of repointing a single vCenter from one domain to another existing domain. This is one of many ways to create an Enhanced Linked Mode node. In this case, there is no replication.

Figure 19: Repointing a Single vCenter from One vCenter Single Sign-On Domain to an Existing vCenter Single Sign-On Domain



1. Make sure that both vCenter nodes are powered on before beginning the repointing process.
2. Optional: Run the pre-check mode command. The pre-check mode fetches the tagging (tags and categories) and authorization (roles and privileges) data from the vCenter. Pre-check does not migrate any data, but checks for conflicts between the source and destination vCenter. For example, run the pre-check with the following CLI:

   ```
   cmsso-util domain-repoint -m pre-check --src-emb-admin Administrator --replication-partner-fqdn FQDN_of_destination_node --replication-partner-admin PSC_Admin_of_destination_node --dest-domain-name destination_PSC_domain
   ```

   Note:
   Pre-check is not required if a replication partner does not exist (repointing to a newly created domain).

See Syntax of the Domain Repoint Command for argument definitions for the `cmsso-util domain-repoint` command.

The pre-check writes the conflicts to the `/storage/domain-data` directory.

3. Optional: Review the conflicts and apply resolutions for all conflicts, or apply a separate resolution for each conflict.

   The conflict resolutions are:

   - Copy: Create a duplicate copy of the data in the target domain.
   - Skip: Skips copying the data in the target domain.
   - Merge: Merges the conflict without creating duplicates.

   Note:
   The default resolution mode for Tags and Authorization conflicts is Copy, unless overridden in the conflict files generated during pre-check.

4. Run the `execute` command. In execute mode, the data generated during the pre-check mode is read and imported to the target node. Then, the vCenter is repointed to the target domain. For example, repointing without a replication partner, run the `execute` command with the following:

   ```
   cmsso-util domain-repoint -m execute --src-emb-admin Administrator --replication-partner-
   fqdn FQDN_of_destination_node --replication-partner-admin PSC_Admin_of_destination_node --dest-domain-
   name destination_PSC_domain
   ```

   See Syntax of the Domain Repoint Command for argument definitions for the `cmsso-util domain-repoint` command.

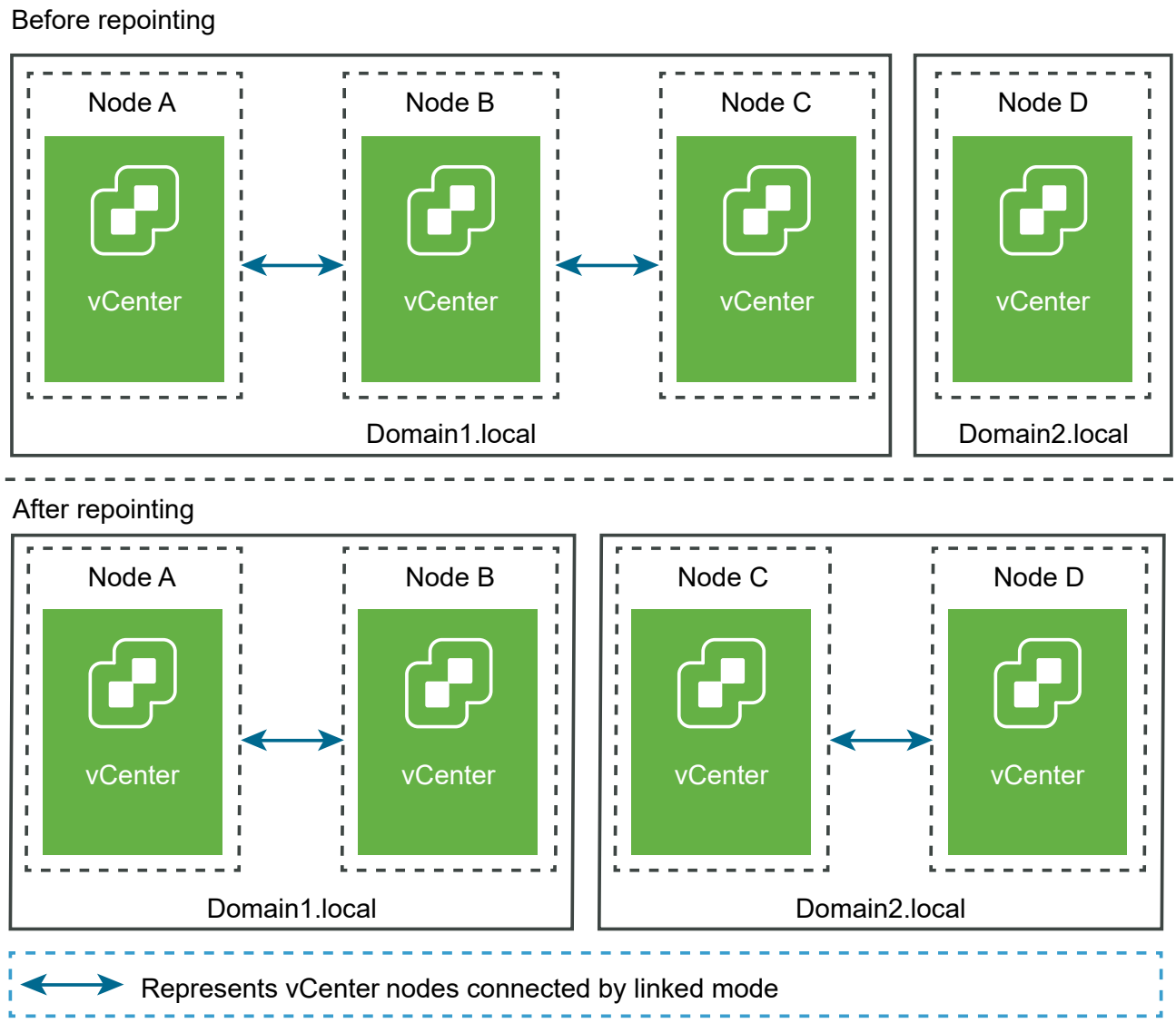
Repoint a vCenter Node to an Existing vCenter Single Sign-On Domain with a Replication Partner

You can repoint a vCenter from one vCenter Single Sign-On domain to an existing vCenter Single Sign-On domain using a replication partner.

- Repointing is only supported with vCenter 6.7 Update 1 and later.
- You must repoint to a vCenter that is of the same version and nodes that are of the same version and build number.
- To ensure no loss of data, take a file-based backup of each node before proceeding with repointing the vCenter.

See Figure 20: Repointing a vCenter from One vCenter Single Sign-On Domain to an Existing vCenter Single Sign-On Domain for an example of repointing to an existing domain. In this case, there is replication.

Figure 20: Repointing a vCenter from One vCenter Single Sign-On Domain to an Existing vCenter Single Sign-On Domain



1. Shut down the node (for example, Node C) that is being repointed (moved to a different domain).
2. Decommission the vCenter node that is being repointed. For example, to decommission Node C, log into Node B (on the original domain) and run the following command:

   ```
   cmsso-util unregister --node-pnid Node_C_FQDN --username Node_B_sso_administrator@sso_domain.com --
   passwd Node_B_sso_adminuser_password
   ```

   After unregistering Node C, services are restarted. References to Node C are deleted from Node B and any other nodes that were linked with Node C on the original domain.
3. Power on Node C to begin the repointing process.
4. Optional: Run the pre-check mode command. The pre-check mode fetches the tagging (tags and categories) and authorization (roles and privileges) data from the vCenter. Pre-check does not migrate any data, but checks the conflicts between the source and destination vCenter. For example, run the pre-check with the following CLI:

   ```
   cmsso-util domain-repoint -m pre-check --src-emb-admin Administrator --replication-partner-
   fqdn FQDN_of_destination_node --replication-partner-admin PSC_Admin_of_destination_node --dest-domain-
   name destination_PSC_domain
   ```

Note:
Pre-check is not required if a replication partner does not exist (repointing to a newly created domain).

See Syntax of the Domain Repoint Command for argument definitions for the `cmsso-util domain-repoint` command.

The pre-check writes the conflicts to the `/storage/domain-data` directory.

5. Optional: Check conflicts and apply resolutions for all conflicts or apply a separate resolution for each conflict.

   The conflict resolutions are:

   - Copy: Create a duplicate copy of the data in the target domain.
   - Skip: Skips copying the data in the target domain.
   - Merge: Merges the conflict without creating duplicates.

   Note:
   The default resolution mode for Tags and Authorization conflicts is Copy, unless overridden in the conflict files generated during pre-check.

6. Run the execute command. In execute mode, the data generated during the pre-check mode is read and imported to the target node. Then, the vCenter is repointed to the target domain. For example, run the execute command with the following:

   ```
   cmsso-util domain-repoint -m execute --src-emb-admin Administrator --replication-partner-fqdn FQDN
    _of_destination_node --replication-partner-admin destination_node_PSC_Admin_user_name --dest-domain-
   name destination_PSC_domain
   ```

   See Syntax of the Domain Repoint Command for argument definitions for the `cmsso-util domain-repoint` command.

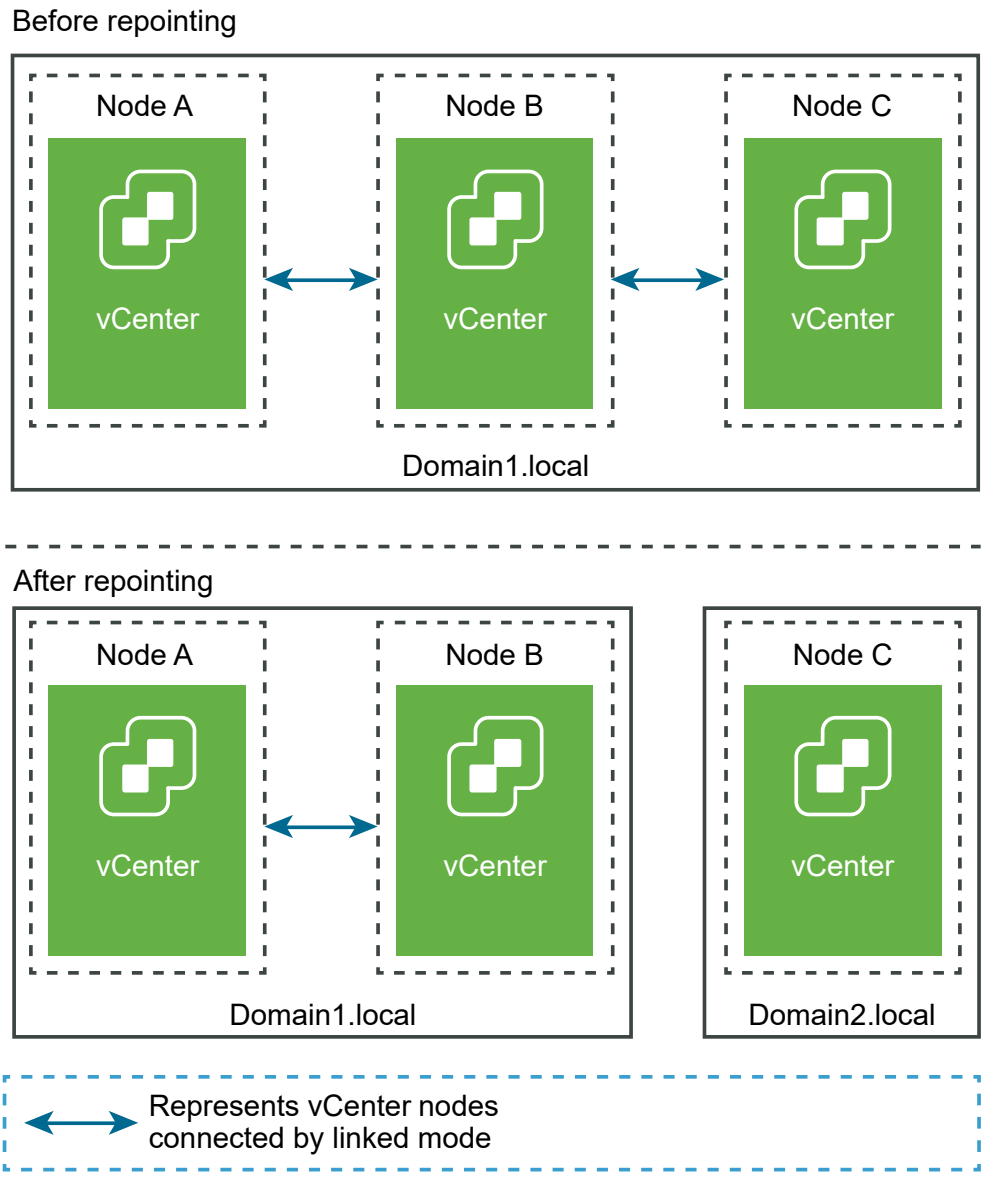
Repoint a vCenter Node to a New vCenter Single Sign-On Domain

You can repoint a vCenter from an existing vCenter Single Sign-On domain to a newly created vCenter Single Sign-On domain.

- Repointing is only supported with vCenter 6.7 Update 1 and later.
- You must repoint to a vCenter that is of the same version and nodes that are of the same version and build number.
- To ensure no loss of data, take a file-based backup and a snapshot in powered-off state of each node before proceeding with repointing the vCenter.

See Figure 21: Repointing a vCenter from One vCenter Single Sign-On Domain to a New vCenter Single Sign-On Domain for an example of repointing to a new domain. In this case, there is no replication partner.

Figure 21: Repointing a vCenter from One vCenter Single Sign-On Domain to a New vCenter Single Sign-On Domain



1. Shut down the node (for example, Node C) that is being repointed (moved to a different domain).

   Note:

   If the vCenter is part of a vCenter HA cluster, remove the vCenter HA configuration before proceeding with the domain repoint. For more information, see *vSphere Availability*.

2. Decommission the vCenter node that is being repointed. For example, to decommission Node C, log into Node B (on the original domain) and run the following command:

   ```
   cmsso-util unregister --node-pnid Node_C_FQDN --username Node_B_sso_administrator@sso_domain.com --
   passwd Node_B_sso_adminuser_password
   ```

   After unregistering Node C, services are restarted. References to Node C are deleted from Node B and any other nodes that were linked with Node C on the original domain.

3. Power on Node C to begin the repointing process.
4. Run the execute command. In execute mode, the data generated during the pre-check mode is read and imported to the target node. Then, the vCenter is repointed to the target domain. For example, repointing with no replication partner (repointing to a new domain), run the execute command with the following:

   ```
   cmsso-util domain-repoint -m execute --src-emb-admin Administrator --dest-domain-
   name destination_PSC_domain
   ```

   See Syntax of the Domain Repoint Command for argument definitions for the `cmsso-util domain-repoint` command.

Syntax of the Domain Repoint Command

You can use command arguments to set the execution parameters of the domain-repoint command.

The `cmsso-util domain-repoint` CLI repoints vCenter from one domain to another.

You can add a space-separated list of arguments to the CLI repoint command.

Use the following command to repoint a vCenter to another vCenter node:

```
cmsso-util domain-repoint -m execute --src-emb-admin Administrator --replication-partner-fqdn FQDN
 _of_destination_node --replication-partner-admin destination_node_PSC_Admin_user_name --dest-domain-
name destination_PSC_domain
```

Argument	Description
`-m`, `--mode`	*mode* can be `pre-check` or `execute`. The `pre-check` argument runs the command in pre-check mode. The `execute` argument runs the command in execute mode.
`-spa`, `--src-psc-admin`	SSO administrator user name for the source vCenter. Do not append the `@`*domain*.
`-dpf`, `--dest-psc-fqdn`	The FQDN of the vCenter to repoint.
`-dpa`, `--dest-psc-admin`	SSO administrator user name for the destination vCenter. Do not append `@`*domain*.
`-ddn`, `--dest-domain-name`	SSO domain name of the destination vCenter.
`-dpr`, `--dest-psc-rhttps`	(Optional) HTTPS port for the destination vCenter. If not set, the default `443` is used.
`-dvf`, `--dest-vc-fqdn`	The FQDN of the vCenter pointing to a destination vCenter. The vCenter is used to check for component data conflicts in the pre-check mode. If not provided, conflict checks are skipped and the default resolution (COPY) is applied for any conflicts found during the import process. **Note:** This argument is optional only if the destination domain does not have a vCenter. If a vCenter exists in the destination domain, this argument is mandatory.
`-sea`, `--src-emb-admin`	Administrator for the vCenter with embedded vCenter. Do not append `@domain` to the administrator id.
`-rpf`, `--replication-partner-fqdn`	(Optional) The FQDN of the replication partner node to which the vCenter is replicated.
`-rpr`, `--replication-partner-rhttps`	(Optional) The HTTPS port for the replication node. If not set, the default is `443`.
`-rpa`, `--replication-partner-admin`	(Optional) SSO administrator user name of the replication partner vCenter.
`-dvr`, `--dest-vc-rhttps`	(Optional) The HTTPS port for the vCenter pointing to the destination vCenter. If not set, the default `443` is used.
`--ignore-snapshot`	(Optional) Ignore snapshot warnings.
`--no-check-certs`	(Optional) Ignore certification validations.
`--debug`	(Optional) Retrieves command execution detail.
`-h`, `--help`	(Optional) Displays the help message for the `cmsso-util domain repoint` command.

Understanding Tagging and Authorization Conflicts

When you run the domain repoint command in pre-check mode, data from the vCenter is exported, examined, and conflicts are written to a file.

The following data is exported to the `/storage/domain-data/` or `ProgramData/VMWare/vCenterServerdata/domain-data` folder:

- `All_Privileges.json`
- `All_Roles.json`
- `All_TagCategories.json`
- `All_Tags.json`

These files contain the all the data (Authorization and Tagging) from the vCenter on which this command was run.

If a secondary vCenter is provided using the `-dvf` or `--dest-vc-fqdn` option, any conflicts are also exported to the same folder:

- `Conflicts_Roles.json`
- `Conflicts_TagCategories.json`
- `Conflicts_Tags.json`

The following is a sample conflicts file:

```
<---- Sample Conflict file code block --->
  {
  "global" : {
    "resolution" : "MERGE|SKIP|COPY",
    "description" : "Default resolution option used to resolve Role Conflicts is COPY. The
conflicts list describes the differences between Role entities on source and target vCenter instance. If
the source information represents an empty JSON array, it simply means that all the entity
attributes from source and target are identical. If the source lists few entries, it means
that only these entity attributes are missing from the target. If the target lists few entries,
it means that only these entity attributes are missing from the source. Though a global resolution
can be set, it can also be overridden at each conflict level by providing individual resolution
mode."
  },
  "conflicts-count" : 1,
  "conflicts-list" : {
    "NoCryptoAdmin" : {
      "source" : {
        "privileges" : "[]"
      },
      "target" : {
        "privileges" : "[Group-1.SamplePriv-1, Group-1.SamplePriv-4, Group-2.SamplePriv-10,
Group-2.SamplePriv-3, Group-2.SamplePriv-7, Group-3.SamplePriv-2, Group-3.SamplePriv-9]"
      },
      "resolution" : ""
    }
}
<----- End of code block --->
```

The parts of the sample conflict files are:

- `description`. Provides the details on how the respective conflicts file is read and understood.
- `source` and `target`. JSON objects that list only the differences between the source and target vCenter objects.
- `resolution`. User supplies one valid resolution. Valid resolutions are `MERGE`, `COPY`, and `SKIP`.

To specify the resolution for handling conflicts, you can provide a default resolution option all conflicts in the `"global": "resolution" = "MERGE|SKIP|COPY"` section. If you do not provide a valid global resolution type for `resolution` or leave it unedited, the system uses `COPY` as the default resolution option.

You can also provide a valid resolution option for each of the conflicts by editing the `resolution` property at each conflict level which overrides the global resolution option.

The types of conflicts listed in Table 49: Conflict Types.

Table 49: Conflict Types

Conflict	Properties used to compare Category Objects	Conflict Types	Conflicting Properties	Conflict Resolution Options
Role conflict	• `name`: Name of the category. • `privilegeId`: List of privileges for the role.	`RoleName` conflict occurs while importing roles and a role with the same name exists in the target vCenter but with different privileges.	Properties that can be conflicting for `RoleName` conflict type can be `Privileges`.	• `COPY`. A copy of the conflicting role is created in the target vCenter, with `--copy` appended to the role name. The new role is created with a new role ID with the same set of privilege IDs. The new role ID is updated in the `VPX_ACCESS` table. The new role ID is applicable for both role name conflict and role ID conflict. **Note:** The default resolution option to resolve Role conflicts is `COPY`. • `MERGE`.The `MERGE` option is resolved in the following sequence: a. If the source vCenter has a role with the same name and privilege list as a role in the target vCenter, but the role IDs are different, the role ID from the target vCenter is used and updated in the `VPX_ACCESS` table. b. If the source vCenter has a role with the same name as a role in the target vCenter, but with a different privilege list, then the privilege lists for both roles are merged. • `SKIP`. Do nothing. The specific role is skipped.
Tag Category conflict: A category name must be unique in a vCenter.	• `name`: Name of the category. • `cardinality`: Cardinality of Category, either Single or Multiple. • `associableEntityType`: List of vCenter object that can be associated with a tag from this category. A value of `All` indicates all vCenter objects.	Only one type of conflict can be seen while importing Tag Categories, `CategoryName` conflict. This conflict indicates that a category with the same name exists in the target vCenter but with different properties (`cardinality` or `associableEntityType`).	Properties that can be conflicting for conflict type `CategoryName` can be at least one of two types: `Cardinality` or `AssociableTypes`.	• `COPY`. A copy of the conflicting category is created in the target vCenter, with `--copy` appended to the category name. The new category is created with the same property name as in the source vCenter. All the tags that were present under this category is imported under the newly created `CategoryCopy`. **Note:** The default resolution option to resolve `CategoryName` conflicts is `COPY`. • `MERGE`. Conflicting properties are merged with the category that is already present in the SSO. Properties are merged as follows: 1. `Description`. The description that is already present is used. 2. `Cardinality`. Cardinality cannot shrink. If there is a cardinality conflict, the cardinality is set to `multiple`. It cannot be reduced to single.

Conflict	Properties used to compare Category Objects	Conflict Types	Conflicting Properties	Conflict Resolution Options
Tags Conflict: A `tag` object always belongs to a `category` Object. A tag Name must be unique only inside a category.	• `name` • `description`	Only one type of conflict can be seen while importing tags: `TagName` conflict. This conflict indicates that a Tag with the same name exists under the same category and in the target vCenter but with different properties.	Properties that can be conflicting for a conflict of type: `TagName` can be `Description`.	• `COPY`. A copy of the conflicting tag is created in the target vCenter, with `--copy` appended to the tag name. Take the `MoRef`(Internal tag ID) of the newly created tag and update the tag association if necessary. **Note:** The default resolution option to resolve `CategoryName` conflicts is `COPY`. • `MERGE`.Keep the existing description. Take the `MoRef`(Internal Tag ID) and update one or more Tag Associations if necessary. • `SKIP`. Do nothing. Do not create this tag. Clean up any Tag Associations.

vCenter Domain Repoint License Considerations

Domain repointing copies license keys to a new domain. Copying the license keys ensures that valid licensing of all assets is maintained after repointing.

For more information about cross-domain repointing, see "Repoint vCenter to Another vCenter in a Different Domain" in *vCenter Deployment and Setup*.

License Keys Not in Use in a Domain

If after completing repointing, a license key appears in more than one domain, but is not in use in some of those domains, you can remove the license key from any domain in which it is not in use.

License Keys in Use in Multiple Domains

If after completing repointing, a license key is in use (that is, assigned to assets) in more than one domain, to remove the license key from all but one domain, first a different license key must be assigned to each asset in domains from which the license key will be removed. Two common approaches:

- If you have other license keys available with sufficient unused capacity, you might use these other keys in place of a license key to be removed.
- You might divide the license keys used in more than one domain into separate license keys, one for each domain. To divide the license keys, see Broadcom knowledge base article 281796.

  Each of the resulting license keys can then be added to a different domain and assigned in vCenter to assets previously licensed with the original license key.

Troubleshooting vCenter Deployment

The vCenter deployment troubleshooting topics provide solutions to problems that you might encounter during the vCenter appliance deployment process.

Collecting Logs for Troubleshooting a vCenter Deployment or Upgrade

You can collect installation or upgrade log files for vCenter. If a deployment or upgrade fails, checking the log files can help you identify the source of the failure.

You can also collect deployment log files for vCenter.

Retrieve Installation Logs Manually

You can retrieve the installation log files manually for examination.

1. Navigate to the installation log file locations.
   - `%PROGRAMDATA%\VMware\vCenterServer\logs` directory, usually `C:\ProgramData\VMware\vCenterServer\logs`
   - `%TEMP%` directory, usually `C:\Users\`*`username`*`\AppData\Local\Temp`

   The files in the `%TEMP%` directory include `vc-install.txt`, `vminst.log`, `pkgmgr.log`, `pkgmgr-comp-msi.log`, and `vim-vcs-msi.log`.

   Note:
   - In Linux, the installation logs are in `/tmp/vcsaUiInstaller`
   - In MacOS, the installation logs are in `/var/folders/**/vcsaUiInstaller`
     The installer randomly picks a temporary folder to place the log files. Locate the `vcsaUiInstaller` folder within `/var/folders/` for the log files.
2. Open the installation log files in a text editor for examination.

Collect Deployment Log Files for the vCenter Appliance

If the vCenter appliance deployment fails, you can retrieve the log files and examine them for the reason of the failure.

The full path to the log files is displayed in the vCenter appliance deployment wizard.

In case of firstboot failure, you can download the support bundle on a Windows host machine and examine the log files to determine which firstboot script failed. See Export a vCenter Support Bundle for Troubleshooting.

1. On the Windows machine that you use for deploying the vCenter appliance, navigate to the log files folder.
   You must have administrator privilege to access the log files and the log files are available at:
   - Windows: `C:\Users\{user name}\AppData\Local\Temp\vcsaUiInstaller`
   - Linux: `/tmp/vcsaUiInstaller`
   - Mac : `/var/folders/**/**/**/vcsaUiInstaller`
2. Open the installation log files in a text editor for examination.

Export a vCenter Support Bundle for Troubleshooting

You can export the support bundle of the vCenter instance in the appliance for troubleshooting using the URL displayed on the DCUI home screen.

You can also collect the support bundle from the vCenter appliance Bash shell by running the `vc-support.sh` script.

The support bundle is exported in `.tgz` format.

1. Log in to the Windows host machine on which you want to download the bundle.
2. Open a Web browser and enter the URL to the support bundle displayed in the DCUI.
   https://*`appliance-fully-qualified-domain-name`*:443/appliance/support-bundle
3. Enter the user name and password of the root user.
4. Click **Enter**.
   The support bundle is downloaded as `.tgz` file on your Windows machine.
5. Optional: To determine which firstboot script failed, examine the `firstbootStatus.json` file.
   If you ran the `vc-support.sh` script in the vCenter appliance Bash shell, to examine the `firstbootStatus.json` file, run
   ```
   cat /var/log/firstboot/firstbootStatus.json
   ```

vCenter Upgrade

vCenter Upgrade describes how to upgrade VMware vCenter™ to the current version.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

vCenter Upgrade is for anyone who must upgrade from earlier versions of vSphere. These topics are for experienced Microsoft Windows or Linux system administrators who are familiar with virtual machine technology and data center operations.

vSphere Upgrade Options

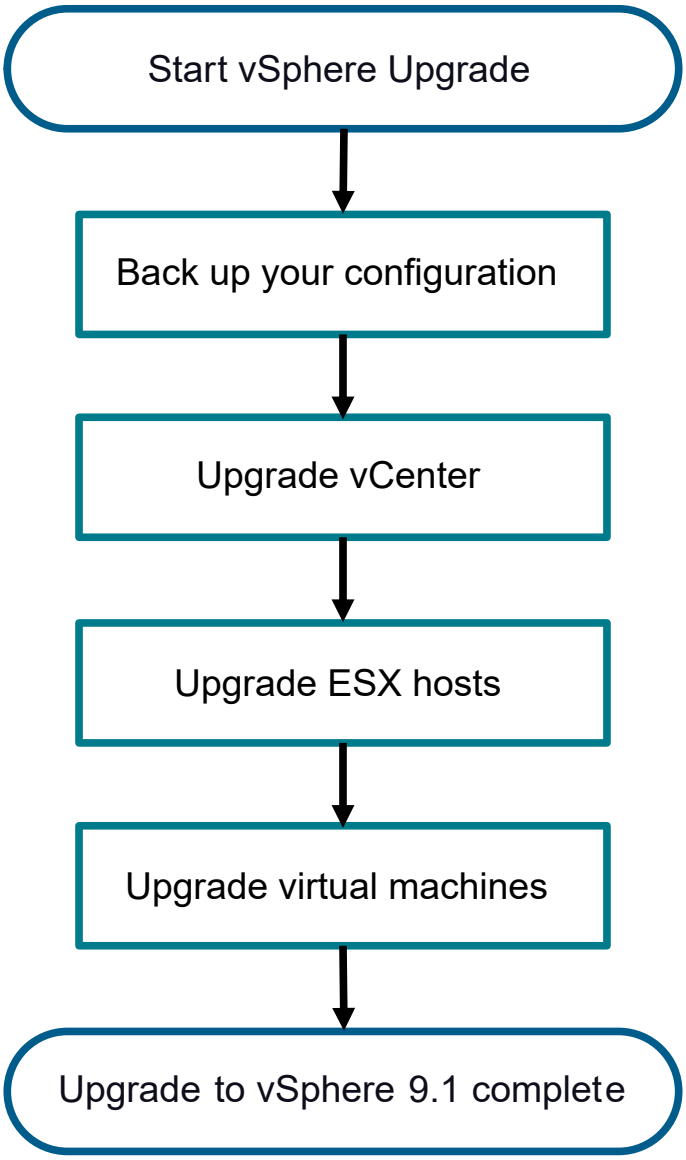
vSphere 9.1 provides many options for upgrading your vSphere deployment. For a successful vSphere upgrade, you must understand the upgrade options, the configuration details that impact the upgrade process, and the sequence of tasks.

The two core components of vSphere are VMware ESX™ and VMware vCenter™. ESX is the virtualization platform on which you can create and run virtual machines and virtual appliances. vCenter is a management plane that acts as a central administrator for ESX hosts connected in a network. You use the vCenter system to pool and manage the resources of multiple hosts.

How to Upgrade vSphere

vSphere is a sophisticated product with multiple components to upgrade. Understanding the required sequence of tasks is vital for a successful vSphere upgrade.

Figure 22: Overview of vSphere Upgrade Tasks



Upgrading vSphere includes the following tasks:

1. In the VMware Cloud Foundation Release Notes, read the information about VMware vCenter.
   See Release Notes.
2. Verify that you have backed up your configuration.
3. If your vSphere system includes VMware solutions or plug-ins, verify that they are compatible with the vCenter appliance version to which you are upgrading.
   See Product Interoperability Matrix.
4. Upgrade vCenter.
   See Overview of the vCenter Upgrade Process.
5. To ensure sufficient disk storage for log files, consider setting up a syslog server for remote logging. Setting up logging on a remote host is especially important for hosts with a limited amount of local storage.
   See Overview of the ESX Host Upgrade Process.
6. Optionally, upgrade your VM hardware version manually or by using vSphere Lifecycle Manager to perform an orchestrated upgrade.
   See Managing Host and Cluster Lifecycle.

When you upgrade vSphere, you must perform all procedures in the specified sequence to avoid possible data loss and to minimize downtime. You can perform the upgrade process for each component in only one direction. For example, after

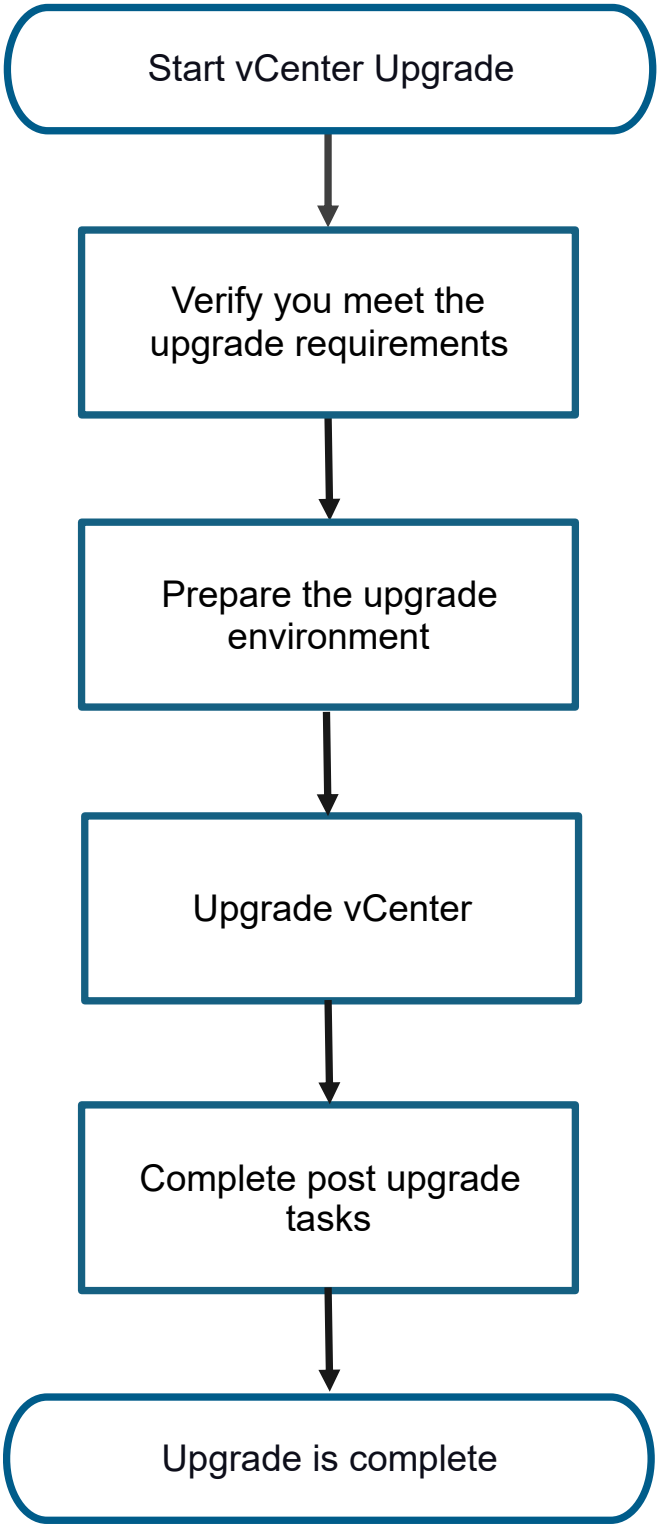
you upgrade to vCenter 9.1, you cannot revert to vCenter 9.0 or 7.0. With backups and some planning, however, you can restore your original vCenter environment.

Overview of the vCenter Upgrade Process

VMware provides many options to upgrade to vCenter 9.1 and later.

You can upgrade your vCenter version 8.0 and later installation to version 9.1 and later.

Figure 23: Overview of vCenter Upgrade Tasks



High-level steps for upgrading vCenter:

1. Verify that your system meets the hardware and software requirements. See .
2. Prepare your environment for the upgrade or migration.
3. Upgrade your vCenter appliance deployment.
4. Finish any required post-upgrade or post-migration tasks.

Concurrent upgrades are not supported and upgrade order matters.

vCenter Supported Upgrade Methods

Graphical User Interface (GUI) Installer	The GUI installer provides a two-step upgrade method using an OVA file that you deploy, and the vCenter Management Interface. The first step deploys a vCenter appliance using an OVA file. The second step uses the vCenter Management Interface to configure the new appliance using the source deployment data.
Command-Line Interface (CLI) Installer	The CLI installer provides advanced users with a CLI method for upgrading the vCenter appliance. You can upgrade to a vCenter appliance using customized CLI templates.
Deprecated vCenter Deployment Models	When upgrading from deprecated deployment models, you must first upgrade your deployment to a currently supported deployment model before attempting to upgrade it to a vCenter 9.1 deployment.
Patching and Updating vCenter	A patch or update brings the vCenter 9.1 software up to the current, minor version. You can use the patching process to make minor upgrades to your 9.1 deployment. See Differences Between vSphere Upgrades, Patches, and Updates and Patching and Updating vCenter 9.1 Deployments.

vCenter Upgrade Compatibility

Upgrading to vCenter 9.1 affects other software components of the data center.

Table 50: Upgrading vCenter and Related VMware Products and Components summarizes how upgrading vCenter can affect your data center components.

vCenter 9.1 and later can manage ESX version 9.0 hosts in the same cluster with ESX 9.1 hosts. vCenter 9.1 and later cannot manage ESX 7.0 or earlier hosts.

vSphere supports upgrades from vCenter 8.0 and later to vCenter 9.1 and later.

Table 50: Upgrading vCenter and Related VMware Products and Components

Product or Component	Compatibility
vCenter	Verify support for the upgrade path from your current version of vCenter to your planned upgrade version. See the Product Interoperability Matrix.
ESX hosts	Verify that your ESX host works with the vCenter version that you are upgrading to. vCenter 9.1 requires ESX host version 8.0 Update 1 or later. Upgrade if necessary. See the Product Interoperability Matrix.
VMware Host Profiles	Host Profiles is a vCenter tool for designing and deploying ESX hosts. Ensure that you are using Host Profiles version 6.0 or later. See Broadcom knowledge base article 318864. For more information on upgrade problems related to Host Profiles, see Upgrade Issues with vCenter Containing Host Profiles and the sections on Host Profiles upgrade workflows in the *vSphere Host Profiles* documentation.
VMFS volumes	You cannot upgrade a VMFS5 datastore to VMFS6. If you have a VMFS5 datastore in your environment, create a VMFS6 datastore and migrate virtual machines from the VMFS5 datastore to VMFS6. For more information on VMFS datastores, see vSphere VMFS Datastore Concepts and Operations in the *vSphere Storage* documentation.
Virtual machines	Upgrade options depend on your current version. For more information about upgrading virtual machines, see Upgrading Virtual Machines and VMware Tools in the *ESX Upgrade* documentation.
VMware Tools	Upgrade options depend on your current version. For more information, see Upgrading Virtual Machines and VMware Tools in the *ESX Upgrade* documentation.
Auto Deploy	To ensure compatibility and best performance, when you upgrade to vCenter 9.1, use Auto Deploy to upgrade ESX hosts to the same version.

Product or Component	Compatibility
vSphere Distributed Virtual Switch (DVS)	You must upgrade to DVS version 7.0 or later before you upgrade to vCenter 9.1.
vSphere Network I/O Control	DVS version 6.0 or later supports only Network I/O Control version 3. If you are using an earlier version of Network I/O Control you must upgrade to Network I/O Control version 3. For more information, see What is vSphere Network I/O Control in the *vSphere Networking* documentation.
vSAN	To avoid potential faults because of differences in the vSAN support in vCenter and ESX, synchronize the versions of vCenter and ESX. For the best integration between vSAN components on vCenter and ESX, deploy the latest version of these two vSphere components. For more information, see the ESX Installation and Setup, vSphere Deployment and Setup, and ESX Upgrade documentation. Specific vSAN configurations, such as a stretched cluster, require a witness host. Instead of using a dedicated physical ESX host as a witness host, you can deploy the vSAN witness appliance. For more information, see vSAN Witness Appliance.
vSAN disk version	vSAN has several different on-disk format versions available depending on the version and upgrade history of the cluster. Some on-disk format versions are transient while others are intended for long-term production. As certain vSAN features are tied to the on-disk format version, the format version must be accounted for when determining interoperability. See Broadcom knowledge base article 327034.

Differences Between vSphere Upgrades, Patches, and Updates

vSphere products distinguish between upgrades, which make major changes to the software, patches and updates, which make smaller changes to the software.

VMware product versions are numbered with two digits, for example, vSphere 9.1. A release that changes either digit, for example, 8.0 to 9.0, involves major changes in the software, and requires an upgrade from the previous version. A release that makes a smaller change, requiring only a patch or update, is indicated by an update number, for example, vSphere 8.0 Update 3 or vSphere 9.1.1 and vSphere 9.1.1.0100.

For information about upgrading vCenter installations, see Upgrading the vCenter Appliance.

For information about patching or updating vCenter, see Patching and Updating vCenter 9.1 Deployments.

When you upgrade an ESX host, some host configuration information is preserved in the upgraded version. After rebooting, the upgraded host can join a vCenter instance that has been upgraded to the same level. Because updates and patches do not involve major changes to the software, the host configuration is not affected. For more information, see the Overview of the ESX Host Upgrade Process article in the*ESX Upgrade* documentation.

Component Behavior Changes That Affect vCenter Upgrade

When upgrading to vCenter 9.1, it is important to understand changes in component behavior for version 9.1 that can affect the upgrade process.

Understanding changes from previous versions of vSphere can help in your upgrade planning. For a complete list of new features in vSphere 9.1, see the Release Notes for version 9.1 releases.

Upgrade Methods

There are multiple methods for upgrading vCenter to version 9.1.

Support for Graphical User Interface (GUI) Deployments of vCenter	You can upgrade an existing vCenter appliance version 9.0 deployment to vCenter 9.1 using the GUI. See GUI Upgrade of the vCenter Appliance.
Support for Command Line Interface (CLI) Deployments of vCenter appliance	You can upgrade an existing vCenter appliance version 9.0 deployment to vCenter 9.1 using the CLI. See CLI Upgrade of the vCenter Appliance .

Support for Reduced Downtime Upgrade through vSphere Client	You can use a migration-based approach to upgrade vCenter between major and minor versions that reduces downtime to several minutes. See Reduced Downtime Upgrade.

Changes in Supported Deployment Types

Changes from previous versions of vSphere can affect your deployment type.

Mixed IPv4 and IPv6 Upgrade

- Upgrade from vCenter 9.0 to 9.1 is supported for pure IPv4 or pure IPv6 management networks only.
- Using dual IP stacks (IPv4 and IPv6) is not supported.

Support for Upgrading a vCenter with Multi-Homing

When upgrading a vCenter appliance with two or more network interface cards (NICs), the upgrade transfers the IP addresses from each network interface from the source vCenter to the target vCenter.

During the upgrade process, the Upgrade installer performs a pre-check of the source vCenter and records the network settings. When this information is transferred to the target vCenter appliance, the network settings are transferred as part of the configuration. When the upgrade process finishes, you can log in to the vCenter appliance using the vSphere Client, and verify that the IP addresses have been successfully transferred to the newly upgraded vCenter appliance.

With two or more NICs per vCenter appliance, you can better manage network traffic in your environment. For example, with multiple network interfaces you can:

- Configure backup traffic to use a different network than management traffic.
- Keep ESX hosts on a physically different LAN segment from your management traffic. You can connect one network interface to the network with your ESX hosts, and connect the other to a network from which administrative clients connect to vCenter.

Support for Federal Information Processing Standard 140-2

vCenter 9.1 supports Federal Information Processing Standard (FIPS) 140-2.

FIPS 140-2 is a U.S. and Canadian government standard that specifies security requirements for cryptographic modules. By default, FIPS 140-2 is always activated after installation or upgrade of vCenter 9.1.

To learn more about support for FIPS 140-2 in VMware by Broadcom products, see https://www.vmware.com/security/certifications/fips.html.

To learn how to activate or deactivate FIPS 140-2 support, see Activate and Deactivate FIPS on the vCenter Appliance the *vSphere Security* documentation.

Support for Transport Security Layer 1.3

Starting in 8.0 Update 3, vSphere supports TLS 1.3 and 1.2 through the use of TLS profiles. By default, vSphere supports the Transport Security Layer (TLS) 1.3 encryption protocol.

Upgrading to vSphere 8.0 Update 3 activates the default TLS profile, named COMPATIBLE, on ESX and vCenter hosts. The COMPATIBLE profile supports TLS 1.3 and some TLS 1.2 connections.

To learn how to manage TLS configuration, see vSphere TLS Configuration in the *vSphere Security* guide.

Upgrading the vCenter Appliance

You can upgrade the vCenter appliance 8.0 or later to 9.1 only in the context of VMware vSphere Foundation or VMware Cloud Foundation. All the installation files that are necessary for the upgrade are included in the vCenter installer, which you can download from the Broadcom Support portal.

Note:

Before upgrading the vCenter appliance 8.0 or later to 9.1, you must upgrade your components in a specific sequence:

- For upgrading to VMware vSphere Foundation 9.1, you must first upgrade VMware Aria Operations 8.x to VMware Cloud Foundation Operations 9.1. See Upgrading to vSphere Foundation 9.1.
- For upgrading to VMware Cloud Foundation 9.1, see Upgrading to VMware Cloud Foundation 9.1.
- For upgrading vSphere 8.0 and later and Aria Operations 8.18 and later, see Upgrading vSphere 8 and Aria Operations 8 to 9.1.
- If the vSphere Lifecycle Manager database in your environment is reset, before upgrading vCenter see Broadcom knowledge base article 455737.

For information on the upgrade compatibility of vCenter, see the Broadcom Product Interoperability Matrix.

The upgrade of the vCenter appliance is a migration of the old version to the new version, which includes deploying a new vCenter appliance of version 9.1. You can deploy the new appliance on an ESX host 8.0 Update 1 or later, or on the inventory of a vCenter instance 8.0 or later. You assign a temporary IP address to the new appliance to facilitate the configuration and services data migration from the old appliance to the newly deployed appliance. After the migration, the IP address and host name of the old appliance are applied to the new upgraded appliance of version 9.1. At the end of the upgrade, the temporary IP address is released and the old appliance is powered off.

Version 9.1. of vCenter uses an embedded PostgreSQL database. During the upgrade, you must select a storage size for the new appliance that is suitable for the database size.

vCenter 9.1 uses an embedded vSphere Lifecycle Manager service that allows you to perform centralized and simplified lifecycle management of clusters with ESX 9.0 hosts. vSphere Lifecycle Manager in vSphere 9.1 includes the functionality that Update Manager provided in earlier vSphere releases for ESX host upgrade and patching operations, and upgrade of virtual machine hardware and VMware Tools.

For information about the software included in the vCenter 9.1 , see What is the vCenter Appliance in *vCenter Deployment and Setup*.

The vCenter installer contains executable files for both GUI and CLI upgrades which you can use alternatively.

- The GUI upgrade is a two stage process. The first stage is a deployment wizard that deploys the OVA file of the new appliance on the target ESX host or vCenter instance. After the OVA deployment finishes, you are redirected to the second stage of the process that sets up and transfers the services and configuration data from the old appliance to the newly deployed appliance.
- The CLI upgrade method involves running a CLI command against a JSON file that you previously prepared. The CLI installer parses the configuration parameters and their values from the JSON file and generates an OVF Tool command that deploys the new appliance. The OVF Tool command also transfers services and configuration data and from the old appliance to the new appliance.

For information about the vCenter appliance upgrade requirements, see System Requirements for the vCenter Appliance.

Important:
If the appliance that you are upgrading is configured in a mixed IPv4 and IPv6 environment, only the IPv4 settings are preserved. For information on the transfer of networking configuration settings for mixed mode IPv4 and IPv6 deployments, see Mixed IPv4 and IPv6 Upgrade.

If you are deploying the vCenter appliance directly on an ESX host, non-ephemeral distributed virtual port groups are not supported and are not shown. After the upgrade, you can manually connect the appliance to the original non-ephemeral distributed virtual port group. This is not a limitation when deploying the appliance through a vCenter, and you can deploy to ephemeral or non-ephemeral distributed virtual port groups.

For information on the upgrade compatibility of vCenter, see the Broadcom Compatibility Guide.

For information about deploying the vCenter, see Introduction to vSphere Deployment and Setup in *vCenter Deployment and Setup*.

For information about configuring the vCenter, see How to Configure vCenter in *vCenter Configuration*.

About the Upgrade Process of the vCenter appliance

You can upgrade the vCenter appliance from version 9.0 to version 9.1.

Attention:

If your current vCenter appliance is version 7.0 or earlier, you must upgrade to version 8.0 or later before upgrading to version 9.1.

Important:
In environments with multiple instances of vCenter appliance, you cannot perform parallel upgrades. Each vCenter instance must be upgraded separately. The reason for this is that VMware Directory Services (`vmdird`) encounters replication problems with single-sign on and certificate information.

When you run the GUI or CLI upgrade, the process includes:

1. Deploying a new vCenter 9.1 with a temporary network configuration. If you are upgrading a vCenter, you must select a deployment size for the new appliance that is suitable for your vSphere environment size. You must also select a storage size for the new appliance that is suitable for the vCenter database.
2. Exporting the services and configuration data from the source appliance of version 9.0 that you want to upgrade.
   You must select the data types that you want to transfer to the new appliance.
3. Transferring the exported data to the newly deployed appliance.
   If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are not supported and do not appear as options during the upgrade. After the upgrade, you can manually connect the appliance to the original non-ephemeral distributed virtual port group. This process is not a limitation when deploying the appliance through a vCenter, and you can deploy to ephemeral or non-ephemeral distributed virtual port groups.
4. If you are upgrading a vCenter appliance that uses an external Platform Services Controller, it is converted to a vCenter instance with those services embedded in the appliance.
   The new vCenter appliance contains all Platform Services Controller services, preserving the functionality and workflows, including authentication, certificate management, and licensing. It is no longer necessary nor possible to deploy and use an external Platform Services Controller. All Platform Services Controller services are consolidated into vCenter during the upgrade.
5. Powering off the source appliance. The new upgraded vCenter appliance appliance assumes the network configuration of the source appliance.

- For the new appliance requirements, see System Requirements for the vCenter Appliance.
- For the appliance upgrade preparation, see Preparing to Upgrade the vCenter appliance.
- For the appliance upgrade procedures, see Upgrading the vCenter Appliance.
- For the appliance post-upgrade procedures, see After Upgrading vCenter.

Preparing to Upgrade the vCenter appliance

Before you upgrade the vCenter appliance, you must download the vCenter installer ISO file and mount it to a network virtual machine or physical server from which you want to perform the upgrade.

The computer from which you upgrade the appliance must run on a Windows, Linux, or Mac operating system that meets the operating system requirements. See System Requirements for the vCenter Installer.

To avoid any issues due to expired VMware Certificate Authority (VMCA) root certificate and to minimize the downtime for their renewal, in vCenter 9.1 the VMCA root certificate and the solution user certificates are automatically renewed if they are set to expire in less than 2 years. The automatic certificate renewal is triggered as part of the vCenter upgrade process. After the vCenter system is successfully upgraded, the VMCA root certificate is valid for 10 years.

For more information about vCenter certificate management, see the vSphere Authentication documentation.

Before upgrading a vCenter appliance, you must prepare the ESX hosts in the inventory.

- Download and Mount the vCenter Installer
- Synchronizing Clocks on the vSphere Network
- Transferring Data from an Existing vCenter Appliance

- Prepare ESX Hosts for vCenter Appliance Upgrade

Transferring Data from an Existing vCenter Appliance

Data transfer happens as part of the upgrade downtime. You can choose how much data to transfer during the upgrade to control the length of the downtime and the storage the new appliance needs.

During the upgrade, you can transfer the minimum data required for the vCenter appliance to be operational, or historical event, task, and performance data, from the source appliance to the embedded PostgreSQL database of the new vCenter appliance. All transferred data is copied during the upgrade, and the original data on the source appliance remains unchanged.

You can migrate the following types of data to the embedded PostgreSQL database in use by version 9.1 of the vCenter appliance.

Configuration data	The minimum type of data you must transfer from your existing vCenter appliance to upgrade to a functioning vCenter appliance. This data can be transferred in a reasonably short amount of time, minimizing downtime during the upgrade to vCenter 9.1.
Configuration and historical data	In addition to the configuration data, you can choose to transfer historical data such as usage statistics, events, and tasks. The vCenter installer gives you the option to copy both the configuration and historical data at one time, or to copy only the configuration data during the upgrade process. You can then choose to copy the historical data in the background after you start your new vCenter 9.1. **Attention:** If you choose to copy all data at one time, you cannot start vCenter until all the data has been transferred to the embedded PostgreSQL database within vCenter 9.1.
Configuration, historical, and performance metrics data	The largest amount of data you can choose to transfer to vCenter. This option requires the longest transfer time and the largest storage footprint, and vCenter does not start until all the data has been transferred. To minimize the upgrade time and the storage required for the new appliance, select to transfer only the configuration data.

Prepare ESX Hosts for vCenter Appliance Upgrade

Before upgrading to vCenter 9.1, you must prepare your ESX hosts.

To upgrade vCenter appliance, your ESX hosts must meet the requirements for upgrade.

- ESX hosts must be at version 9.0 or later. For information on ESX compatibility, see the VMware by Broadcom Compatibility Guide.
- Your source and target ESX hosts must not be in lockdown or maintenance mode, and not part of fully automated DRS clusters.
- Review your environment's Certificate Authority (CA) signed SSL certificates. To learn about manually reviewing CA-signed SSL certificates in a vSphere environment, see Broadcom knowledge base article 321380.

1. If you have Custom or Thumbprint certificates, see ESX Host Upgrades and Certificates to determine your preparatory steps and certificate replacement behavior.

   If you use custom CA-signed certificates, ensure you review how to Change the ESX Certificate Mode to prevent VMCA from overwriting custom certificates.

2. If you have vSphere HA clusters, SSL certificate checking must be enabled.

   If certificate checking is not enabled when you upgrade ESX hosts, vSphere HA fails to configure on the hosts.

   a) Select the vCenter appliance instance in the vSphere Client inventory tree.
   b) Select the **Configure** tab, and then under **Settings** select **General**.
   c) Click **Edit**.
   d) Select **SSL settings**.

      Verify that **vCenter requires verified host SSL certificates** is selected.

Your ESX hosts are ready for the vCenter appliance upgrade.

Prerequisites for Upgrading the vCenter Appliance

To ensure a successful upgrade of the vCenter appliance, you must perform the following required tasks and pre-checks before beginning the upgrade process.

General Prerequisites

- Download and Mount the vCenter Installer.
- Verify that the clocks of all systems on the vSphere network are synchronized. See Synchronizing Clocks on the vSphere Network.

Target System Prerequisites

- Verify that your system meets the minimum software and hardware requirements. See System Requirements for the vCenter Appliance.
- If you plan to deploy the new appliance on an ESX host, verify that the target ESX host is not in lockdown or maintenance mode.
- If you plan to deploy the new appliance on an ESX host managed by vCenter, review the Distributed Resource Scheduler (DRS) settings on the cluster. Ensure that the DRS settings for the cluster on which the target ESX host resides are not configured for Full Automation. Set the Automation Level to Manual or Partially Automated. This Automation Level ensures that the target ESX host does not reboot during the upgrade process.
- If you plan to deploy the new appliance on a DRS cluster of the inventory of a vCenter instance, verify that the cluster contains at least one ESX host that is not in lockdown or maintenance mode.

Source System Prerequisites

- If the source appliance is on a DRS cluster of the inventory of a vCenter instance and during upgrade:
  - If you plan to use the ESX host where the appliance resides, review the Distributed Resource Scheduler (DRS) settings on the cluster. Ensure that the DRS settings of the cluster is set to manual or partially automated so that the VMs of the source system do not move during the upgrade.
  - If you plan to use the vCenter instance where the appliance resides, you need not modify the DRS settings of the cluster.
- Verify that port 22 is open on the vCenter appliance that you want to upgrade. The upgrade process establishes an inbound SSH connection to download the exported data from the source vCenter appliance.
- Verify that port 443 is open on the source ESX host on which the appliance that you want to upgrade resides. The upgrade process establishes an HTTPS connection to the source ESX host to verify that the source appliance is ready for upgrade and to set up an SSH connection between the new and the existing appliance.

  Note:
  When upgrading, the temporary vCenter instance requires the same access rights as the permanent vCenter instance to port 443. Ensure that any firewalls in your environment allow both the temporary and permanent vCenter instances to access port 443.
- Verify that you have sufficient free disk space on the appliance that you want to upgrade so that you can accommodate the data for the upgrade.
- Create an image-based backup (snapshot) of the vCenter appliance you are upgrading as a precaution in case there is a failure during the upgrade process.

  Important:

  To take a pre-upgrade image-based backup in an Enhanced Linked Mode environment, power off all the vCenter appliances in your environment, and back up each node. After you have taken backups of all the nodes, you can restart them and proceed with the upgrade procedure.

  If the upgrade fails, delete the newly deployed vCenter appliance, and restore the vCenter appliance from backup. You must restore all the nodes in the environment from their backups. Failing to do so will cause the replication partners to be out of synchronization with the restored node.

  To learn about image-based backup, see Image-Based Backup and Restore of a vCenter Environment in *vCenter Deployment and Setup*.

Network Prerequisites

- Verify that the new appliance can connect to the source ESX host and vCenter instance on which resides the appliance that you want to upgrade.
- If you plan to assign a static IP address and an FQDN as a system name in the temporary network settings of the appliance, verify that you have configured the forward and reverse DNS records for the IP address.
- If you plan to assign a DHCP IP address in the temporary network settings of the new vCenter appliance, verify that the ESX host on which you want to deploy the new appliance is in the same subnet as the ESX host on which the existing vCenter appliance runs.
- If you plan to assign a DHCP IPv4 address in the temporary network settings of the new vCenter appliance, verify that the ESX host on which you want to deploy the new appliance is connected to at least one network that is associated with a port group which accepts MAC address changes. Consider the default security policy of a distributed virtual switch, which is to reject MAC address changes. For information about how to configure the security policy for a switch or port group, see What is Security Policy in *vSphere Networking*.

GUI Upgrade of the vCenter Appliance

You can use the GUI installer to perform an interactive upgrade of a vCenter appliance.

To perform the GUI upgrade, download the vCenter appliance installer to a client machine, run the Upgrade Wizard from the client machine, and provide the inputs for the deployment and setup of the new upgraded appliance.

The GUI upgrade process includes a series of two stages.

Figure 24: Stage 1 - OVA Deployment

The first stage walks you through the deployment wizard to get the deployment type of the source appliance that you want to upgrade and configure the new appliance settings. During this stage, you deploy the new appliance with temporary network settings. This stage finishes the deployment of the OVA file on the target server with the same deployment type as the source appliance and the appliance settings that you provide.

As an alternative to performing the first stage of the upgrade with the GUI installer, you can deploy the OVA file of the new vCenter appliance using the vSphere Client or VMware Host Client. To deploy the OVA file on an ESX host or vCenter instance 9.0 or later, you can also use the vSphere Client. After the OVA deployment, you must log in to the vCenter Management Interface of the newly deployed appliance to proceed with the second stage of the upgrade process.

Figure 25: Stage 2 - Appliance Setup

The second stage walks you through the setup wizard to select the data types to transfer from the old to the new appliance. The new appliance uses the temporary network settings until the data transfer finishes. After the data transfer finishes, the new appliance assumes the network settings of the old appliance. This stage finishes the data transfer, starts the services of the new upgraded appliance, and powers off the old appliance.

As an alternative to performing the second stage of the upgrade with the GUI installer, you can log in to the vCenter Management Interface of the newly deployed appliance, https://`FQDN_or_IP_address`:5480.

Required Information for Upgrading a vCenter Appliance 8.0 and Later to Version 9.1

The GUI upgrade wizard prompts you for information about the vCenter appliance (version 8.0 or later) that you want to upgrade, deployment information for the new vCenter appliance 9.1. It is a best practice to keep a record of the values that you enter.

You can use this worksheet to record the information that you need for upgrading a vCenter appliance.

Table 51: Required Information During Stage 1 of the Upgrade

Required for Upgrade of	Required Information	Default
All deployment types	FQDN or IP address of the source appliance that you want to upgrade.	-
	HTTPS port of the source appliance.	443
	vCenter Single Sign-On administrator user name of the source appliance. **Important:** The user must be administrator@*your_domain_name*.	administrator@vsphere.local
	Password of the vCenter Single Sign-On administrator user.	-
	Password of the root user of the source appliance.	-
All deployment types	FQDN or IP address of the source server on which resides that appliance that you want to upgrade. The source server can be either an ESX host or a vCenter instance. **Note:** The source server cannot be the vCenter appliance that you want to upgrade. In such cases, use the source ESX host.	-
	HTTPS port of the source server.	443
	User name with administrative privileges on the source server. • If your source server is an ESX host, use root. • If your source server is a vCenter instance, use *user_name*@*your_domain_name*, for example, administrator@vsphere.local.	-
	Password of the user with administrative privileges on the source server.	-
All deployment types	FQDN or IP address of the target server on which you want to deploy the new appliance. The target server can be either an ESX host or a vCenter instance. **Note:** The target server cannot be the vCenter appliance that you want to upgrade. In such cases, use an ESX host as a target server.	-
	HTTPS port of the target server.	443
	User name with administrative privileges on the target server: • If your target server is an ESX host, use root. • If your target server is a vCenter instance, use *user_name*@*your_domain_name*, for example, administrator@vsphere.local.	-
	Password of the user with administrative privileges on the target server.	-

Required for Upgrade of	Required Information	Default
All deployment types Only if your target server is a vCenter instance.	Data center from the vCenter inventory on which you want to deploy the new appliance. Optionally you can provide a data center folder.	-
	ESX host or DRS cluster from the data center inventory on which you want to deploy the new appliance.	-
All deployment types	The virtual machine name for the new vCenter 9.1 appliance. • Must not contain a percent sign (%), backslash (\), or forward slash (/). • Must be no more than 80 characters in length.	vCenter Appliance
All deployment types	Password for the root user of the appliance operating system. • Must contain only the lower ASCII character set without spaces. • Must be at least 8 characters, but no more than 20 characters in length. • Must contain at least one uppercase letter. • Must contain at least one lowercase letter. • Must contain at least one number. • Must contain at least one special character, for example, a dollar sign ($), hash key (#), at sign (@), period (.), or exclamation mark (!).	-
vCenter appliance 8.0 and later	Deployment size of the new vCenter appliance 9.1 for your vSphere environment. • Tiny Deploys an appliance with 2 CPUs and 12 GB of memory. Suitable for environments with up to 10 hosts or 100 virtual machines. • Small Deploys an appliance with 4 CPUs and 19 GB of memory. Suitable for environments with up to 100 hosts or 1,000 virtual machines. • Medium Deploys an appliance with 8 CPUs and 28 GB of memory. Suitable for environments with up to 400 hosts or 4,000 virtual machines. • Large Deploys an appliance with 16 CPUs and 37 GB of memory. Suitable for environments with up to 1,000 hosts or 10,000 virtual machines. • X-Large Deploys an appliance with 24 CPUs and 56 GB of memory. Suitable for environments with up to 2,000 hosts or 35,000 virtual machines.	The size of the source vCenter appliance determines the default deployment size. Using this information, the upgrade installer calculates the best size to use for your environment, which might be the same as your existing deployment, or calculated to the next largest size.
All deployment types	Name of the datastore on which you want to store the configuration files and virtual disks of the new 9.1 . **Note:** The installer displays a list of datastores that are accessible from your target server.	-
	Enable or deactivate **Thin Disk Mode**.	Disabled

Required for Upgrade of	Required Information	Default
All deployment types	Name of the network to which to connect the new vCenter appliance 9.1. **Note:** The installer displays a drop-down menu with networks that depend on the network settings of your target server. If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are not supported and are not displayed in the drop-down menu. The network must be accessible from the source server on which resides that appliance that you want to upgrade. The network must be accessible from the physical client machine from which you perform the deployment.	-
	IP version for the appliance temporary address Can be either IPv4 or IPv6.	IPv4
	IP assignment for the appliance temporary address Can be either static or DHCP.	static
All deployment types Only if you use a static assignment for the temporary IP address.	Temporary system name (FQDN or IP address) The system name is used for managing the local system. The system name must be FQDN. If a DNS server is not available, provide a static IP address.	-
	Temporary IP address.	-
	For IPv4 version, a subnet mask as a dot decimal notation or a network prefix as an integer between 0 and 32. For IPv6 version, a network prefix as an integer between 0 and 128.	-
	Default gateway.	-
	DNS servers separated by commas.	-
All deployment types Only if you use a DHCP assignment with IPv4 version for the temporary IP address and a DDNS server is available in your environment.	Temporary system name (FQDN).	-

Table 52: Required Information During Stage 2 of the Upgrade

Required for	Required Information	Default
vCenter appliance 8.0 and later	Data types to transfer from the old appliance to the new vCenter appliance 9.1. In addition to the configuration data, you can transfer the events, tasks, and, performance metrics. **Note:** For minimum upgrade time and storage requirement for the new appliance, select to transfer only the configuration data.	-
vCenter appliance 8.0 and later	Join or do not participate in the VMware Customer Experience Improvement Program (CEIP). For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.	Join the CEIP.

Upgrade a vCenter Appliance 8.0 or later to Version 9.1 by Using the GUI

You can use the GUI installer to perform an interactive upgrade of a vCenter appliance 8.0 or later to vCenter appliance 9.1. You must run the GUI upgrade from a Windows, Linux, or Mac OS X machine that is in the same network as the appliance that you want to upgrade.

- See Prerequisites for Upgrading the vCenter Appliance.
- See Required Information for Upgrading a vCenter Appliance 8.0 and Later to Version 9.1.

Stage 1 - Deploy the OVA File of the New vCenter Appliance

With stage 1 of the upgrade process, you deploy the OVA file, which is included in the vCenter appliance installer, for the new vCenter appliance.

1. In the vCenter installer, navigate to the `vcsa-ui-installer` directory, go to the subdirectory for your operating system, and run the installer executable file.
   - For Windows OS, go to the `win32` subdirectory, and run the `installer.exe` file.
   - For Linux OS, go to the `lin64` subdirectory, and run the `installer` file.
   - For Mac OS, go to the `mac` subdirectory, and run the `Installer.app` file.
2. On the Home page, click **Upgrade**.
3. Review the Introduction page to understand the upgrade process and click **Next**.
4. Read and accept the license agreement, and click **Next**.
5. Connect to the source appliance that you want to upgrade.
   a) Enter the information about the source vCenter appliance that you want to upgrade, and click **Connect to Source**.

Option	Action
Appliance FQDN or IP address	Enter the IP address or FQDN of the vCenter appliance that you want to upgrade.
Appliance HTTPS port	If the source appliance uses a custom HTTPS port, change the default value to that of the custom port. The default port value is 443. Custom port values are supported beginning with vCenter appliance version 6.5 Update 2. If you are upgrading from earlier versions, you cannot specify a custom port.

   b) Enter the information about thevCenter Single Sign-On administrator and root user.

Option	Action
SSO user name	Enter the vCenter Single Sign-On administrator user name. **Important:** The user must be administrator@*your_domain_name*.
SSO password	Enter the password of the vCenter Single Sign-On administrator.
Appliance (OS) root password	Enter the password of the root user.

c) Enter the information about the source ESX host or vCenter instance on which resides the vCenter appliance that you want to upgrade and click **Next**.

Option	Description
Source server or host name	IP address or FQDN of the source ESX host or vCenter instance on which the vCenter appliance that you want to upgrade resides. **Note:** The source vCenter instance cannot be the vCenter appliance that you want to upgrade. In such cases, use the source ESX host.
HTTPS port	If the ESX host or vCenter instance uses a custom HTTPS port, change the default value. The default value is 443.
User name	User name of a user with administrative privileges on the ESX host or vCenter instance.
Password	Password of the user with administrative privileges on the ESX host or vCenter instance.

6. Verify that the certificate warning displays the SHA1 thumbprints of the SSL certificates that are installed on the source appliance and its source server, and click **Yes** to accept the certificate thumbprints.
7. Connect to the target server on which you want to deploy the new vCenter appliance.

Option	Steps
You can connect to an ESX host on which to deploy the new appliance.	1. Enter the FQDN or IP address of the ESX host. 2. Enter the HTTPS port of the ESX host. 3. Enter the user name and password of a user with administrative privileges on the ESX host, for example, the root user. 4. Click **Next**. 5. Accept the certificate warning, if any, by clicking **Yes**.
You can connect to a vCenter instance and browse the inventory to select an ESX host or DRS cluster on which to deploy the new appliance. **Note:** The target server cannot be the vCenter appliance that you want to upgrade. In such cases, use an ESX host as a target server.	1. Enter the FQDN or IP address of the vCenter instance. 2. Enter the HTTPS port of the vCenter instance. 3. Enter the user name and password of a vCenter Single Sign-On user with administrative privileges on the vCenter instance, for example, the administrator@*your_domain_name* user. 4. Click **Next**. 5. Accept the certificate warning, if any, by clicking **Yes**. 6. Select the data center or data center folder that contains the ESX host or DRS cluster on which you want to deploy the new appliance, and click **Next** **Note:** You must select a data center or data center folder that contains at least one ESX host that is not in lockdown or maintenance mode. 7. Select the ESX host or DRS cluster on which you want to deploy the new appliance, and click **Next**.

8. On the Set up target appliance VM page, enter a name for the new vCenter appliance, set the password for the root user, and click **Next**.

   The appliance name must not contain a percent sign (%), backslash (\), or forward slash (/) and must be no more than 80 characters in length.

   The password must contain only lower ASCII characters without spaces, at least eight characters, a number, uppercase and lowercase letters, and a special character, for example, an exclamation mark (!), hash key (#), at sign (@), or brackets (()).

Note:
The root password of the old appliance is not transferred to the new upgraded appliance.

9. Select the deployment size for the new vCenter appliance 9.1 for your vSphere inventory.

Deployment Profile	Description
Tiny	Deploys an appliance with 2 vCPUs and 14 GB of memory. Suitable for environments with up to 10 hosts or 100 virtual machines
Small	Deploys an appliance with 4 vCPUs and 21 GB of memory. Suitable for environments with up to 100 hosts or 1,000 virtual machines
Medium	Deploys an appliance with 8 vCPUs and 30 GB of memory. Suitable for environments with up to 400 hosts or 4,000 virtual machines
Large	Deploys an appliance with 16 vCPUs and 39 GB of memory. Suitable for environments with up to 1,000 hosts or 10,000 virtual machines
X-Large	Deploys an appliance with 24 vCPUs and 58 GB of memory. Suitable for environments with up to 2,500 hosts or 30,000 virtual machines. Suitable for environments with up to 2,500 hosts or 45,000 virtual machines with X-Large Storage.

10. Select the storage size for the new vCenter appliance 9.1 and click **Next**.

Important:
You must consider the storage size of the appliance that you are upgrading and the database size if external.

Storage Size Option	Tiny	Small	Medium	Large	X-Large
Default	Deploys an appliance with 619 GB of storage.	Deploys an appliance with 734 GB of storage.	Deploys an appliance with 933 GB of storage.	Deploys an appliance with 1383 GB of storage.	Deploys an appliance with 2308 GB of storage.
Large	Deploys an appliance with 2059 GB of storage.	Deploys an appliance with 2084 GB of storage.	Deploys an appliance with 2233 GB of storage.	Deploys an appliance with 2283 GB of storage.	Deploys an appliance with 2408 GB of storage.
X-Large	Deploys an appliance with 4319 GB of storage.	Deploys an appliance with 4344 GB of storage.	Deploys an appliance with 4493 GB of storage.	Deploys an appliance with 4543 GB of storage.	Deploys an appliance with 4668 GB of storage.

11. From the list of available datastores, select the location where all the virtual machine configuration files and virtual disks will be stored and, optionally, enable thin provisioning by selecting **Enable Thin Disk Mode**. NFS datastores are thin provisioned by default.
12. Configure the temporary network for communication between the vCenter appliance that you want to upgrade and the new vCenter appliance, and click **Next**.

Option	Action
Choose a network	Select the network to which to connect the new appliance temporarily. The networks displayed in the drop-down menu depend on the network settings of the target server. If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are unsupported and are not displayed in the drop-down menu. **Important:** If you want to assign a temporary IPv4 address with DHCP allocation, you must select a network that is associated with a port group which accepts MAC address changes.
IP Address family	Select the version for the temporary IP address of the new appliance. Can be either IPv4 or IPv6.
Network type	Select the allocation method for the temporary IP address of the appliance. • **Static** The wizard prompts you to enter the temporary IP address, subnet mask or prefix length, default gateway, and DNS servers. • **DHCP** A DHCP server is used to allocate the temporary IP address. Select this option only if a DHCP server is available in your environment. Optionally, you can provide a temporary system name (FQDN) if a DDNS server is available in your environment.

13. On the Ready to complete stage 1 page, review the deployment settings for the new vCenter appliance and click **Finish** to start the OVA deployment process.
14. Wait for the OVA deployment process to finish and click **Continue** to proceed with stage 2 of the upgrade process to transfer the data from the old appliance and start the services of the new appliance.

    Note:
    If you exit the wizard by clicking **Close**, you must log in to the vCenter Management Interface of the newly deployed vCenter appliance to transfer the data from the old appliance and set up the services.

The newly deployed vCenter appliance 9.1 is running on the target server but is not configured.

Important:
The data from the old appliance is not transferred and the services of the new appliance are not started.

Stage 2 - Transfer the Data and Set up the Newly Deployed vCenter Appliance

When the OVA deployment finishes, you are redirected to stage 2 of the upgrade process to transfer the data from the old appliance and start the services of the newly deployed vCenter appliance 9.1.

- Verify that stage 1 has completed successfully.

- Verify that you have the root and vCenter Single Sign-On credentials for the source vCenter appliance.
- Ensure that port 5480 is accessible on the newly deployed appliance if you are accessing stage 2 using the vCenter Management Interface.

1. Launch stage 2.

Option	Description
Recommended. Continue directly in the GUI Installer.	1. On the stage 1 completion screen in the desktop installer, click **Continue**. 2. If an untrusted certificate warning appears for the target appliance on port `5480`, review the SHA-256 fingerprint and click **Accept**. 3. The installer window loads the stage 2 interface automatically.
Launch stage 2 manually from a web browser. **Use this option if you closed the GUI installer after stage 1 or you are resuming the upgrade from a different client machine.**	1. Open a web browser and navigate to `https://<target_vCenter_appliance_IP_FQDN>:5480`. 2. Accept the browser security certificate warning. 3. On the vCenter Management Interface landing page, click **Upgrade**. 4. Log in using the `root` user name and the `root` password that is configured for the target vCenter appliance during stage 1.

2. Review the introduction to stage 2 of the upgrade process and click **Next**.
3. Run and wait for the pre-upgrade checks to finish and read the pre-upgrade checks results if any.
   - If the pre-upgrade check results contain error messages, read the messages and click **Export Logs** to export and download the support bundle for troubleshooting.

     You cannot proceed with the upgrade until you have corrected the errors.

     Important:

     If you have provided incorrect vCenter Single Sign-On user name and password of the source appliance during stage 1, the pre-upgrade check fails with an authentication error.
   - If the pre-upgrade check results contain warning messages, read the messages and click **Close**.

     After you have verified that your system meets the requirements from the warning message, you can proceed with the upgrade.
4. Connect to the source appliance that you want to upgrade.
   a) Enter the information about the source vCenter appliance that you want to upgrade, and click **Connect to Source**.

Option	Action
Appliance FQDN or IP address	Enter the IP address or FQDN of the vCenter appliance that you want to upgrade.
Appliance HTTPS port	If the source appliance uses a custom HTTPS port, change the default value to that of the custom port. The default port value is 443. Custom port values are supported beginning with vCenter appliance version 6.5 Update 2. If you are upgrading from earlier versions, you cannot specify a custom port.

b) Enter the information about thevCenter Single Sign-On administrator and root user.

Option	Action
SSO user name	Enter the vCenter Single Sign-On administrator user name. **Important:** The user must be administrator@*your_domain_name*.
SSO password	Enter the password of the vCenter Single Sign-On administrator.
Appliance (OS) root password	Enter the password of the root user.

c) Enter the information about the source ESX host or vCenter instance on which resides the vCenter appliance that you want to upgrade and click **Next**.

Option	Description
Source server or host name	IP address or FQDN of the source ESX host or vCenter instance on which the vCenter appliance that you want to upgrade resides. **Note:** The source vCenter instance cannot be the vCenter appliance that you want to upgrade. In such cases, use the source ESX host.
HTTPS port	If the ESX host or vCenter instance uses a custom HTTPS port, change the default value. The default value is 443.
User name	User name of a user with administrative privileges on the ESX host or vCenter instance.
Password	Password of the user with administrative privileges on the ESX host or vCenter instance.

5. On the **Select migration data** page, choose the types of data that you want to transfer from the old appliance to the new vCenter 9.1 appliance and click **Next**.

Data Option	Description
Configuration	Imports only core inventory and configuration settings. Provides the fastest downtime and lowest storage requirements.
Configuration and Events/Tasks	Imports core configuration plus historical event and task records.
Configuration, Events, Tasks, and Performance Metrics	Imports all historical data, including performance metrics. Requires the longest transfer time and largest storage footprint.

A large amount of data requires more time to be transferred to the new appliance. For the minimum upgrade time and storage requirements for the new appliance, select to transfer only the configuration data.

6. Review the VMware Customer Experience Improvement Program (CEIP) page and choose if you want to join the program.

   For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.

7. On the Ready to complete page, review the upgrade settings, accept the backup acknowledgment, and click **Finish**.
8. Read the shutdown warning message confirming that the source vCenter appliance will be powered off once data export complete, sand click **OK**.
9. Wait for the data export, source node shutdown, network identity reassignment, data import, and service startup to finish, and click **OK** to go to the vCenter Getting Started page.

The vCenter appliance is upgraded. The old vCenter appliance is powered off and the new appliance starts.

- Verify Your vCenter Upgrade Is Successful.
- If the old vCenter appliance uses a non-ephemeral distributed virtual port group, to preserve the port group setting, you can manually connect the new appliance to the original non-ephemeral distributed virtual port group. For information about configuring virtual machine networking on a vSphere distributed switch, see *vSphere Networking*.
- You can configure high availability for the vCenter appliance. For information about providing vCenter appliance high availability, see *vSphere Availability*.

Upgrading vCenter in High Availability Environments

You can use the GUI installer to perform an interactive upgrade of a vCenter appliance in a high availability (HA) environment.

Prerequisites for Upgrading vCenter High Availability Environments

To ensure a successful upgrade of vCenter in a high availability (HA) environment, your environment must meet certain prerequisites before running the upgrade.

General Prerequisites

Ensure that your environment meets the prerequisites for upgrading vCenter appliance. See Prerequisites for Upgrading the vCenter Appliance.

High Availability Prerequisites

In addition to the prerequisites for upgrading a standard vCenter appliance, the vCenter High Availability (vCenter HA) upgrade has the following prerequisites.

- A vCenter HA cluster consists of three vCenter appliances that act as Active, Passive, and Witness nodes. The Active node must be configured as the vCenter HA node.
- The Active node is part of the vCenter HA cluster.
- All nodes must be present in the cluster.
- The vCenter HA cluster must be in a healthy state.
- The vCenter HA cluster must be in enabled mode.
- The host that has the vCenter virtual machine (VM) must be managed by a container vCenter. It must not be standalone.
- The target vCenter must have the same placement as the source vCenter.
- vCenter HA cannot be set up on a vCenter that has disks on more than one datastore.

To learn more about vCenter HA configuration, see High Availability.

Upgrade a vCenter Appliance HA Cluster by Using the GUI

You can use the GUI installer to perform an interactive upgrade of a vCenter HA cluster of version 9.0 or later to vCenter appliance 9.1. You must run the GUI upgrade from a Windows, Linux, or Macintosh computer that is in the same network as the appliance that you want to upgrade.

- Ensure that your environment meets the prerequisites for upgrading a vCenter HA cluster. See Prerequisites for Upgrading vCenter High Availability Environments.
- See Required Information for Upgrading a vCenter Appliance 8.0 and Later to Version 9.1.

A vCenter HA cluster consists of three vCenter appliances that act as Active, Passive, and Witness nodes. The Active vCenter HA node is upgraded using a migration-based upgrade while preserving the existing configuration.

Stage 1 - Deploy the OVA File of the New vCenter High Availability Cluster

With stage 1 of the upgrade process, you deploy the OVA file of the new vCenter appliance 9.1.

Ensure that your environment meets the prerequisites for upgrading a vCenter HA cluster. See Prerequisites for Upgrading vCenter High Availability Environments.

1. In the vCenter installer, navigate to the `vcsa-ui-installer` directory, go to the subdirectory for your operating system, and run the installer executable file.
   - For Windows OS, go to the `win32` subdirectory, and run the `installer.exe` file.
   - For Linux OS, go to the `lin64` subdirectory, and run the `installer` file.
   - For Mac OS, go to the `mac` subdirectory, and run the `Installer.app` file.
2. On the Home page, click **Upgrade**.
3. Review the Introduction page to understand the upgrade process and click **Next**.
4. Read and accept the license agreement, and click **Next**.
5. Connect to the source appliance that you want to upgrade. This appliance is the Active vCenter HA node.
   a) Enter the information about the source vCenter appliance that you want to upgrade, and click **Connect to Source**.

Option	Action
Appliance FQDN or IP address	Enter the IP address or FQDN of the Active vCenter HA node that you want to upgrade.
Appliance HTTPS port	The default value (443) is displayed and cannot be edited.

   b) Enter the information about the vCenter Single Sign-On administrator and root user.

Option	Action
SSO user name	Enter the vCenter Single Sign-On administrator user name. **Important:** The user must be administrator@*your_domain_name*.
SSO password	Enter the password of the vCenter Single Sign-On administrator.
Appliance (OS) root password	Enter the password of the root user.

   c) Enter the information about the source vCenter instance on which resides the vCenter appliance that you want to upgrade and click **Next**.

Option	Description
Source server or host name	IP address or FQDN of the Active node. The Active node must be configured as the vCenter HA node.
HTTPS port	If the vCenter instance uses a custom HTTPS port, change the default value. The default value is 443.
User name	User name of a user with administrative privileges on the ESX host or vCenter instance.
Password	Password of the user with administrative privileges on the ESX host or vCenter instance.

6. Verify that the certificate warning displays the SHA1 thumbprints of the SSL certificates that are installed on the source appliance and its source server, and click **Yes** to accept the certificate thumbprints.
7. If vCenter HA is successfully detected, the target appliance is set to the manager of the source appliance. Click OK.
   The appliance deployment target information is populated.

8. Connect to the target server on which you want to deploy the new vCenter appliance.

Option	Steps
You can connect to an ESX host on which to deploy the new appliance.	1. Enter the FQDN or IP address of the ESX host. 2. Enter the HTTPS port of the ESX host. 3. Enter the user name and password of a user with administrative privileges on the ESX host, for example, the root user. 4. Click **Next**. 5. Accept the certificate warning, if any, by clicking **Yes**.
You can connect to a vCenter instance and browse the inventory to select an ESX host or DRS cluster on which to deploy the new appliance. **Note:** The target server cannot be the vCenter appliance that you want to upgrade. In such cases, use an ESX host as a target server.	1. Enter the FQDN or IP address of the vCenter instance. 2. Enter the HTTPS port of the vCenter instance. 3. Enter the user name and password of a vCenter Single Sign-On user with administrative privileges on the vCenter instance, for example, the administrator@*your_domain_name* user. 4. Click **Next**. 5. Accept the certificate warning, if any, by clicking **Yes**. 6. Select the data center or data center folder that contains the ESX host or DRS cluster on which you want to deploy the new appliance, and click **Next** **Note:** You must select a data center or data center folder that contains at least one ESX host that is not in lockdown or maintenance mode. 7. Select the ESX host or DRS cluster on which you want to deploy the new appliance, and click **Next**.

9. On the Set up target appliance VM page, enter a name for the new vCenter appliance, set the password for the root user, and click **Next**.

   The appliance name must not contain a percent sign (%), backslash (\), or forward slash (/) and must be no more than 80 characters in length.

   The password must contain only lower ASCII characters without spaces, at least eight characters, a number, uppercase and lowercase letters, and a special character, for example, an exclamation mark (!), hash key (#), at sign (@), or brackets (()).

   Note:
   The root password of the old appliance is not transferred to the new upgraded appliance.

10. Select the deployment size for the new vCenter appliance for your vSphere inventory.

Deployment Profile	Description
Tiny	Deploys an appliance with 2 vCPUs and 14 GB of memory. Suitable for environments with up to 10 hosts or 100 virtual machines
Small	Deploys an appliance with 4 vCPUs and 21 GB of memory. Suitable for environments with up to 100 hosts or 1,000 virtual machines
Medium	Deploys an appliance with 8 vCPUs and 30 GB of memory. Suitable for environments with up to 400 hosts or 4,000 virtual machines
Large	Deploys an appliance with 16 vCPUs and 39 GB of memory. Suitable for environments with up to 1,000 hosts or 10,000 virtual machines
X-Large	Deploys an appliance with 24 vCPUs and 58 GB of memory.

Deployment Profile	Description
	Suitable for environments with up to 2,500 hosts or 30,000 virtual machines. Suitable for environments with up to 2,500 hosts or 45,000 virtual machines with X-Large Storage.

11. Select the storage size for the new vCenter appliance, and click **Next**.

    Important:
    You must consider the storage size of the appliance that you are upgrading and the database size if external.

Storage Size Option	Tiny	Small	Medium	Large	X-Large
Default	Deploys an appliance with 619 GB of storage.	Deploys an appliance with 734 GB of storage.	Deploys an appliance with 933 GB of storage.	Deploys an appliance with 1383 GB of storage.	Deploys an appliance with 2308 GB of storage.
Large	Deploys an appliance with 2059 GB of storage.	Deploys an appliance with 2084 GB of storage.	Deploys an appliance with 2233 GB of storage.	Deploys an appliance with 2283 GB of storage.	Deploys an appliance with 2408 GB of storage.
X-Large	Deploys an appliance with 4319 GB of storage.	Deploys an appliance with 4344 GB of storage.	Deploys an appliance with 4493 GB of storage.	Deploys an appliance with 4543 GB of storage.	Deploys an appliance with 4668 GB of storage.

12. From the list of available datastores, select the location to store the virtual machine (VM) configuration files and virtual disks, and, optionally, enable thin provisioning by selecting **Enable Thin Disk Mode**. NFS datastores are thin provisioned by default.

    Note:

    vCenter HA cannot be set up on a vCenter that has disks on more than one datastore.

13. Configure the temporary network for communication between the vCenter appliance that you want to upgrade and the new vCenter appliance, and click **Next**.

Option	Action
Choose a network	Select the network to which to connect the new appliance temporarily. The networks displayed in the drop-down menu depend on the network settings of the target server. If you are deploying the appliance directly on an ESX host, non-ephemeral distributed virtual port groups are unsupported and are not displayed in the drop-down menu. **Important:** If you want to assign a temporary IPv4 address with DHCP allocation, you must select a network that is associated with a port group which accepts MAC address changes.
IP Address family	Select the version for the temporary IP address of the new appliance. Can be either IPv4 or IPv6.

Option	Action
Network type	Select the allocation method for the temporary IP address of the appliance. • **Static** The wizard prompts you to enter the temporary IP address, subnet mask or prefix length, default gateway, and DNS servers. • **DHCP** A DHCP server is used to allocate the temporary IP address. Select this option only if a DHCP server is available in your environment. Optionally, you can provide a temporary system name (FQDN) if a DDNS server is available in your environment.

14. On the Ready to complete stage 1 page, review the deployment settings for the new vCenter appliance and click **Finish** to start the OVA deployment process.
15. Wait for the OVA deployment process to finish and click **Continue** to proceed with stage 2 of the upgrade process to transfer the data from the old appliance and start the services of the new appliance.

    Note:
    If you exit the wizard by clicking **Close**, you must log in to the vCenter Management Interface of the newly deployed vCenter appliance to transfer the data from the old appliance and set up the services.

The newly deployed target vCenter appliance 9.1 is running on the target server but is not configured.

Important:
The data from the source vCenter is not transferred and the services of the target appliance are not started. Transfer the data from the old appliance and start the services of the newly deployed vCenter appliance. See Stage 2 - Transfer Data and Set-Up the Newly Deployed vCenter High Availability Cluster.

Stage 2 - Transfer Data and Set-Up the Newly Deployed vCenter High Availability Cluster

When the OVA deployment finishes, you are redirected to stage 2 of the upgrade process to transfer the data from the old appliance and start the services of the newly deployed vCenter appliance. When the deployment finishes, vCenter has high availability protection.

- Verify that stage 1 has completed successfully.
- Verify that you have the root user password and vCenter Single Sign-On administrator credentials for the source vCenter appliance.
- Verify that you have administrative credentials for the ESX host or vCenter on which the source vCenter HA Active node resides.
- Verify that port `5480` and port `443` are accessible between your deployment client machine, the source appliance, and the new target appliance.
- Verify that you have performed a full file-based backup or VM snapshot of the source vCenter appliance before proceeding with stage 2.

1. Launch stage 2.

Option	Description
Recommended. Continue directly in the GUI Installer.	1. On the stage 1 completion screen in the desktop installer, click **Continue**. 2. If an untrusted certificate warning appears for the target appliance on port `5480`, review the SHA-256 fingerprint and click **Accept**. 3. The installer window loads the stage 2 interface automatically.
Launch stage 2 manually from a web browser.	1. Open a web browser and navigate to `https://<target_vCenter_appliance_IP_FQDN>:5480`. 2. Accept the browser security certificate warning.

Option	Description
Use this option if you closed the GUI installer after stage 1 or you are resuming the upgrade from a different client machine.	3. On the vCenter Management Interface landing page, click **Upgrade**. 4. Log in using the `root` user name and the `root` password that is configured for the target vCenter appliance during stage 1.

2. Review the introduction to stage 2 of the upgrade process and click **Next**.
3. Run and wait for the pre-upgrade checks to finish and read the pre-upgrade checks results if any.
   - If the pre-upgrade check results contain error messages, read the messages and click **Export Logs** to export and download the support bundle for troubleshooting.

     You cannot proceed with the upgrade until you have corrected the errors.

     Important:

     If you have provided incorrect vCenter Single Sign-On user name and password of the source appliance during stage 1, the pre-upgrade check fails with an authentication error.
   - If the pre-upgrade check results contain warning messages, read the messages and click **Close**.

     After you have verified that your system meets the requirements from the warning message, you can proceed with the upgrade.
4. Enter the connection and credential details for the source vCenter HA Active node and its hosting ESX host or vCenter and click **Next**.

Field	Action/Description
Appliance FQDN or IP address	Enter the FQDN or temporary IP address of the source vCenter Active node.
Appliance HTTPS port	Enter the HTTPS port of the source appliance. Default value is `443`.
vCenter Single Sign-On user name	Enter the vCenter Single Sign-On administrator account (`administrator@vsphere.local`).
vCenter Single Sign-On password	Enter the password for the vCenter Single Sign-On administrator account.
Appliance (OS) root password	Enter the `root` user password for the source appliance.
Source host or server name	Enter the FQDN or IP address of the ESX host or vCenter hosting the source Active VM.
Source Hhst HTTPS port	Enter the HTTPS port of the hosting ESX host or vCenter. Default value is `443`.
Source host user name	Enter an administrative user account for the hosting server (`root` for ESX host).
Source host password	Enter the password for the administrative user account on the hosting server.

5. On the **Select migration data** page, choose the types of data that you want to transfer from the old appliance to the new vCenter 9.1 appliance and click **Next**.

Data Option	Description
Configuration	Imports only core inventory and configuration settings. Provides the fastest downtime and lowest storage requirements.
Configuration and Events/Tasks	Imports core configuration plus historical event and task records.
Configuration, Events, Tasks, and Performance Metrics	Imports all historical data, including performance metrics. Requires the longest transfer time and largest storage footprint.

A large amount of data requires more time to be transferred to the new appliance. For the minimum upgrade time and storage requirements for the new appliance, select to transfer only the configuration data.

6. Select whether to join the VMware Customer Experience Improvement Program (CEIP) and click **Next**.
7. On the Ready to complete page, review the upgrade settings, accept the backup acknowledgment, and click **Finish**.
8. Read the shutdown warning message confirming that the source vCenter appliance will be powered off once data export complete, sand click **OK**.
9. Wait for the data export, source node shutdown, network identity reassignment, data import, and service startup to finish, and click **OK** to go to the vCenter Getting Started page.

The vCenter appliance is upgraded. The old vCenter appliance is powered off and the new appliance starts.

After the Active node is upgraded, a new Passive and Witness node is created using a clone operation. These nodes are created automatically for Auto Deployment. For Manual Deployment, nodes are not created automatically. You must clone the Passive and Witness VMs and set the cluster mode to **enabled**.

When the deployment finishes, vCenter has high availability protection. You can click **Edit** to enter Maintenance Mode, Deactivate, or Remove vCenter HA. You can also initiate vCenter HA failover.

- Verify Your vCenter Upgrade Is Successful.
- If the old vCenter appliance uses a non-ephemeral distributed virtual port group, to preserve the port group setting, you can manually connect the new appliance to the original non-ephemeral distributed virtual port group. For information about configuring virtual machine networking on a vSphere distributed switch, see *vSphere Networking*.
- Upgrade all vCenter instances in the vCenter Single Sign-On domain.
  You can configure high availability for the vCenter appliance. For information about providing vCenter appliance high availability, see *vSphere Availability*.

CLI Upgrade of the vCenter Appliance

You can use the CLI installer to perform an unattended upgrade of a vCenter appliance on an ESX host or vCenter instance.

The CLI upgrade process includes downloading the vCenter appliance installer on a network virtual machine or physical server from which you want to perform the upgrade, preparing a JSON configuration file with the upgrade information, and running the upgrade command.

> **Important:**
> The user name that you use to log in to the machine from which you want to run the CLI upgrade, the path to the vCenter appliance ISO file, the path to your JSON configuration file, and the string values in your JSON configuration file, including the passwords, must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

The vCenter appliance ISO file contains templates of JSON files that contain the minimum configuration parameters that are required for upgrading a vCenter appliance. For information about preparing JSON templates for CLI upgrade of the vCenter appliance, see Prepare Your JSON Configuration File for CLI Upgrade.

Prepare Your JSON Configuration File for CLI Upgrade

Before you run the CLI command to upgrade a vCenter appliance, you must prepare a JSON file with configuration parameters and their values for your upgrade specification.

- You must be familiar with the JSON syntax.
- Download and Mount the vCenter Installer.

The vCenter installer contains JSON templates for all upgrade types. For information about the templates, see JSON Templates for CLI Upgrade of the vCenter Appliance.

You can upgrade an appliance with minimum configurations by setting values to the configuration parameters in the JSON template for your specification. You can edit the preset values, remove configuration parameters, and add configuration parameters for custom configurations.

For a complete list of the configuration parameters and their descriptions, navigate to the installer subdirectory for your operating system and run the `vcsa-deploy upgrade --template-help` command or see Upgrade Configuration Parameters for CLI Upgrade of vCenter Appliance .

1. In the vCenter appliance installer, navigate to the `vcsa-cli-installer` directory, and open the `templates` subfolder.
2. Copy the upgrade templates from the `upgrade` subfolder to your workspace.

   Important:
   The path to the JSON configuration files must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.
3. Open the template file for your use case in a text editor.

   To ensure the correct syntax of your JSON configuration file, use a JSON editor.
4. Fill in the values for the required configuration parameters and, optionally, enter additional parameters and their values.
   For example, if you want to use an IPv4 DHCP assignment for the temporary network of the new appliance, in the `temporary_network` subsection of the template, change the value of the `mode` parameter to `dhcp` and remove the default configuration parameters that are for a static assignment.

   ```
   "temporary_network": {
       "ip_family": "ipv4",
       "mode": "dhcp"
   },
   ```

   Important:
   The string values, including the passwords, must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

   To set a value that contains a backslash (\) or quotation mark (") character, you must precede the character with the backslash (\) character. For example, `"password":"my\"password"` sets the password my"password, `"image":"G:\\vcsa\\VMware-vCenter-Server-Appliance-9.1.X.XXXX-YYYYYYY_OVF10.ova"` sets the path `G:\vcsa\VMware-vCenter-Server-Appliance-9.1.x.XXXX-YYYYYYY_OVF10.ova`.

   The Boolean values must contain only lowercase characters, that is, a value can be either `true` or `false`. For example, `"ssh_enable":false`.
5. Optional: Use a JSON editor of your choice to validate the JSON file.
6. Save in UTF-8 format and close the file.

You can create and save additional templates if needed for your upgrade specification.

JSON Templates for CLI Upgrade of the vCenter Appliance

The vCenter appliance installer contains JSON templates that are located in the `vcsa-cli-installer/templates` directory. In the `upgrade` subfolder, you can find the JSON templates with the minimum configuration parameters for all upgrade types.

For each upgrade type, there is one template for deploying the new appliance on an ESX host and another template for deploying the new appliance on a vCenter instance.

Table 53: Upgrade JSON Templates Included in the vCenter Appliance Installer

Location	Template	Description
`vcsa-cli-installer\templates\upgrade\vcsa8.0`	`embedded_vCSA_on_ESXi.json`	Contains the minimum configuration parameters that are required for upgrade of a vCenter appliance 9.0 to vCenter appliance 9.1 on an ESX host.
	`embedded_vCSA_on_VC.json`	Contains the minimum configuration parameters that are required for upgrade of a vCenter appliance 9.0 to vCenter appliance 9.1 on a vCenter instance.

Upgrade Configuration Parameters for CLI Upgrade of vCenter Appliance

When you prepare your JSON configuration files for CLI upgrade, you must set parameters and values to provide input data for the upgrade of a vCenter appliance.

Sections and Subsections of Configuration Parameters in the JSON Upgrade Files

The configuration parameters in the JSON configuration files for CLI upgrade are organized in sections and subsections.

Table 54: Sections and Subsections of Configuration Parameters in the JSON Upgrade Files

Section	Subsection	Description
`new_vcsa` - describes the new appliance that you want to deploy.	`esxi`	Use only if you want to deploy the new appliance directly on an ESX host. Contains the configuration parameters that describe the target ESX host. See Table 55: Configuration Parameters in the new_vcsa Section, esxi Subsection. **Note:** You must fill in either this subsection or the `vc` subsection.
	`vc`	Use only if you want to deploy the new appliance on the inventory of a vCenter instance. Contains the configuration parameters that describe the target ESX host or DRS cluster from the vCenter inventory. See Table 56: Configuration Parameters in the new_vcsa Section, vc Subsection. **Note:** You must fill in either this subsection or the `esxi` subsection. The target vCenter instance cannot be the vCenter appliance that you want to upgrade. In such cases, use the `esxi` subsection.
	`appliance`	Contains the configuration parameters that describe the new appliance. See Table 57: Configuration Parameters in the new_vcsa Section, appliance Subsection
	`os`	Contains only the `ssh_enable` configuration parameter to set the SSH administrator login to the new appliance.
	`ovftool_arguments`	Optional. Use this subsection for adding arbitrary arguments and their values to the OVF Tool command that the installer generates. **Important:** The vCenter appliance installer does not validate the configuration parameters in the `ovftool_arguments` subsection. If you set arguments that the OVF Tool does not recognize, the deployment might fail.
	`temporary_network`	Contains the configuration parameters that describe the temporary network settings for the new appliance. See Table 58: Configuration Parameters in the new_vcsa Section, temporary_network Subsection
	`user_options`	Contains only the `vcdb_migrateSet` configuration parameter to set the types of data that you want to transfer from the old appliance to the new appliance. See Table 59: Configuration Parameters in the new_vcsa Section, user_options Subsection
`source_vc` - describes the existing appliance that you want to upgrade.	`managing_esxi_or_vc`	Contains the configuration parameters that describe the source ESX host or vCenter appliance instance on which resides the appliance that you want to upgrade. See Table 61: Configuration Parameters in the source_vc Section, managing_esxi_or_vc Subsection.
	`vc_vcsa`	Contains the configuration parameters that describe the source appliance that you want to upgrade. See Table 62: Configuration Parameters in the source_vc Section, vc_vcsa Subsection.

Section	Subsection	Description
`ceip` - describes joining the VMware Customer Experience Improvement Program (CEIP).	`settings`	Contains only the `ceip_enabled` configuration parameter to join or not to join the VMware Customer Experience Improvement Program (CEIP). See Table 63: Configuration Parameters in the ceip Section, settings Subsection. **Note:** If the `ceip_enabled` configuration parameter is set to `true`, you must run the CLI deployment command with the `--acknowledge-ceip` argument. For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.

Important:
The string values, including the passwords, must contain only ASCII characters. Extended ASCII and non-ASCII characters are unsupported.

To set a value that contains a backslash (\) or quotation mark (") character, you must precede the character with the backslash (\) character. For example, `"password":"my\"password"` sets the password my"password, `"image":"G:\\vcsa\\VMware-vCenter-Server-Appliance-9.1.x.XXXX-YYYYYYY_OVF10.ova"` sets the path `G:\vcsa\VMware-vCenter-Server-Appliance-9.1.x.XXXX-YYYYYYY_OVF10.ova`.

The Boolean values must contain only lowercase characters. Can be either `true` or `false`. For example, `"ssh_enable":false`.

Configuration Parameters in the `new_vcsa` Section

Table 55: Configuration Parameters in the `new_vcsa` Section, `esxi` Subsection

Name	Type	Description
`hostname`	string	The IP address or FQDN of the target ESX host on which you want to deploy the new appliance.
`username`	string	A user name with administrative privileges on the target ESX host, for example, root.
`password`	string	The password of the user with administrative privileges on the target ESX host.
`deployment_network`	string	The name of the network to which to connect the new appliance. The network must be part of the target ESX host or vCenter appliance instance (as identified by the `managing_esxi_or_vc` configuration parameter) network configuration. **Note:** The network must be accessible from the source ESX host or vCenter appliance instance (as identified by the `managing_esxi_or_vc` configuration parameter) on which resides the appliance that you want to upgrade. The network must be also accessible from the client machine from which you are performing the upgrade. Ignored if the target ESX host has only one network.
`datastore`	string	The name of the datastore on which to store the virtual machine configuration files and virtual disks of the new appliance. The datastore must be available to the target ESX host. **Note:** The datastore must have at least 25 GB of free space.
`port`	integer	The HTTPS reverse proxy port of the target ESX host. The default port is 443. Use only if the target ESX host uses a custom HTTPS reverse proxy port.

Name	Type	Description

ssl_certificate_verification	string	The CLI verifies that a server's security certificate is signed by a Certificate Authority (CA), and establishes a secure connection. If the certificate is self-signed, the CLI stops the upgrade unless you specify one of the following SSL certificate configuration options:
		Specify the Secure Hash Algorithm 1 (SHA-1) certificate thumbprint. A certificate thumbprint is a hexadecimal string that uniquely identifies a certificate. The thumbprint is calculated from the content of the certificate using a thumbprint algorithm.
		`"thumbprint": "certificate SHA-1 thumbprint"`
		Set verification_mode to NONE.
		`"verification_mode": "NONE"`
		If you are connecting to a server with a self-signed certificate, and fail to either specify the SHA-1 certificate thumbprint or set the verification mode to NONE, the CLI displays the server's self-signed certificate thumbprint, and prompts you to accept or reject the certificate thumbprint.
		You may also specify that the CLI ignore the self-signed using the vcsa-deploy upgrade command parameter --no-ssl-certificate-validation. See Syntax of the CLI Upgrade Command .

Table 56: Configuration Parameters in the new_vcsa Section, vc Subsection

Name	Type	Description
hostname	string	The IP address or FQDN of the target vCenter instance on which you want to deploy the new appliance.
username	string	vCenter Single Sign-On administrator user name on the target vCenter instance, for example, administrator@vsphere.local.
password	string	The password of the vCenter Single Sign-On administrator user on the target vCenter instance.
deployment_network	string	The name of the network to which to connect the new appliance. The network must part of the target ESX host or DRS cluster network configuration. **Note:** The network must be accessible from the source ESX host on which resides the appliance that you want to upgrade. The network must be also accessible from the client machine from which you are performing the upgrade. Ignored if the target ESX host or DRS cluster has only one network.

datacenter	array	The vCenter data center that contains the target ESX host or DRS cluster on which you want to deploy the new appliance.
		If the data center is located in a folder or a structure of folders, specify the as a comma-separated list of strings. For example,
		`["parent_folder", "child_folder", "datacenter_name"]`
		If there is no folder path on the data center, then use only the data center name. For example,
		`["datacenter_name"]`
		or
		`"datacenter_name"`
		Note:
		The value is case-sensitive.

datastore	string	The name of the datastore that you want to store all virtual machine configuration files and virtual disks of the new appliance. **Note:** The datastore must be available to the target ESX host or DRS cluster. The datastore must have at least 25 GB of free space.

Name	Type	Description
`port`	integer	The HTTPS reverse proxy port of the target vCenter instance. The default port is 443. Use only if the target vCenter instance uses a custom HTTPS reverse proxy port.

`target`	array	The target cluster, ESX host, or resource pool on which you want to deploy the new appliance. This is the target you specified with the `datacenter` parameter. This path must end with a cluster name, ESXi hostname, or resource pool name.
		
		Important:
		You must provide the name that is displayed in the vCenter inventory. For example, if the name of the target ESX host is an IP address in the vCenter inventory, you cannot provide an FQDN.
		
		Note:
		All values are case-sensitive.
		If you want the deployed appliance to be listed in a different location within the data center's hierarchy, use the `vm_folder` parameter described later in this section.
		If the target cluster, ESX host, or resource pool is located in a folder or a structure of folders, specify the value as a comma-separated list of strings. For example,
		`["parent_folder", "child_folder", "esxi-host.domain.com"]`
		If the target ESX host is part of a cluster, specify the path as a comma-separated list of strings. For example,
		`["cluster_name","esxi-host.domain.com"]`
		
		If you are deploying to a resource pool, include the label `Resources` before the resource pool name. For example:
		`["cluster_name", "Resources", "resource_pool_name"]`
		
		Note:
		Pre-checks verify only the memory of the resource pool.

`vm_folder`	string	Optional. The name of the virtual machine (VM) folder to which to add the new appliance.

Table 57: Configuration Parameters in the `new_vcsa` Section, `appliance` Subsection

Name	Type	Description
`thin_disk_mode`	Boolean	Set to `true` to deploy the new appliance with thin virtual disks.
`deployment_option`	string	The size for the new appliance. • Set to `tiny` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the default storage size. Deploys an appliance with 2 CPUs, 10 GB of memory, and 300 GB of storage. • Set to `tiny-lstorage` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the large storage size. Deploys an appliance with 2 CPUs, 10 GB of memory, and 825 GB of storage. • Set to `tiny-xlstorage` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the x-large storage size. Deploys an appliance with 2 CPUs, 10 GB of memory, and 1700 GB of storage. • Set to `small` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the default storage size. Deploys an appliance with 4 CPUs, 16 GB of memory, and 340 GB of storage. • Set to `small-lstorage` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the large storage size. Deploys an appliance with 4 CPUs, 16 GB of memory, and 870 GB of storage. • Set to `small-xlstorage` if you want to deploy a vCenter appliance for up to 100 hosts and 1,000 virtual machines with the x-large storage size. Deploys an appliance with 4 CPUs, 16 GB of memory, and 1750 GB of storage. • Set to `medium` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the default storage size. Deploys an appliance with 8 CPUs, 24 GB of memory, and 525 GB of storage. • Set to `medium-lstorage` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the large storage size. Deploys an appliance with 8 CPUs, 24 GB of memory, and 1025 GB of storage. • Set to `medium-xlstorage` if you want to deploy a vCenter appliance for up to 400 hosts and 4,000 virtual machines with the x-large storage size. Deploys an appliance with 8 CPUs, 24 GB of memory, and 1905 GB of storage. • Set to `large` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the default storage size. Deploys an appliance with 16 CPUs, 32 GB of memory, and 740 GB of storage. • Set to `large-lstorage` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the large storage size. Deploys an appliance with 16 CPUs, 32 GB of memory, and 1090 GB of storage. • Set to `large-xlstorage` if you want to deploy a vCenter appliance for up to 1,000 hosts and 10,000 virtual machines with the x-large storage size. Deploys an appliance with 16 CPUs, 32 GB of memory, and 1970 GB of storage. • Set to `xlarge` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the default storage size. Deploys an appliance with 24 CPUs, 48 GB of memory, and 1180 GB of storage. • Set to `xlarge-lstorage` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the large storage size. Deploys an appliance with 24 CPUs, 48 GB of memory, and 1230 GB of storage. • Set to `xlarge-xlstorage` if you want to deploy a vCenter appliance for up to 2,000 hosts and 35,000 virtual machines with the x-large storage size. Deploys an appliance with 24 CPUs, 48 GB of memory, and 2110 GB of storage. • Set to `management-tiny` if you want to deploy a vCenter appliance for up to 10 hosts and 100 virtual machines with the default storage size. Deploys an appliance with 2 CPUs, 10 GB of memory, and 300 GB of storage.
`image`	string	Optional. A local file path or URL to the vCenter appliance installation package. By default the installer uses the installation package that is included in the ISO file, in the `vcsa` folder.

Name	Type	Description
name	string	The VM name for the new appliance. Must contain only ASCII characters except a percent sign (%), backslash (\), or forward slash (/) and must be no more than 80 characters in length.
ovftool_path	string	Optional. A local file path to the OVF Tool executable file. By default the installer uses the OVF Tool instance that is included in the ISO file, in the vcsa/ovftool folder.

Table 58: Configuration Parameters in the `new_vcsa` Section, `temporary_network` Subsection

Name	Type	Description
ip_family	string	IP version for the temporary network of the new appliance. Set to ipv4 or ipv6.
mode	string	IP assignment for the temporary network of the new appliance. Set to static or dhcp.
ip	string	Temporary IP address for the new appliance. Required only if you use a static assignment, that is, if you set the mode parameter to static. You must set an IPv4 or IPv6 address that corresponds to the temporary network IP version, that is, to the value of the ip.family parameter. An IPv4 address must comply with the RFC 790 guidelines. An IPv6 address must comply with the RFC 2373 guidelines.

dns_servers	string or array	IP addresses of one or more DNS servers for the temporary network of the new appliance.
		To set more than one DNS server, use a comma-separated list of strings or a comma-separated list as a single string to provide the path. For example,
		`["x.y.z.a", "x.y.z.b"]`
		or
		`"x.y.z.a, x.y.z.b"`
		Required only if you use static network mode for the temporary IP address allocation, that is, if you set the mode parameter to static.

prefix	string	Network prefix length for the temporary network of the new appliance. Use only if the mode parameter is set to static. Remove if the mode parameter is set to dhcp. The network prefix length is the number of bits that are set in the subnet mask. For example, if the subnet mask is 255.255.255.0, there are 24 bits in the binary version of the prefix length, so the network prefix length is 24. For IPv4 version, the value must be between 0 and 32. For IPv6 version, the value must be between 0 and 128.
gateway	string	IP address of the default gateway for the temporary network of the new appliance. For IPv6 version, the value can be default.

Table 59: Configuration Parameters in the `new_vcsa` Section, `user_options` Subsection

Name	Type	Description
`vcdb_migrateSet`	string	Select the types of data to migrate from the old appliance to the new appliance. Data is copied from the source vCenter to the target server. The original source of the data remains unchanged. • Set to `core` if you want to transfer only the configuration data. This option provides the fastest data migration, keeping system downtime to a minimum. • Set to `core_events_tasks` if you want to transfer the configuration and historical data (events and tasks) immediately. vCenter will not start until all data has been migrated from the source vCenter appliance. • Set to `all` if you want to transfer the configuration, historical, and performance metrics data immediately. vCenter will not start until all data has been migrated from the source vCenter for Windows. This option transfers the largest amount of data, and requires more downtime than other data migration options. **Note:** To minimize the upgrade time and the amount of storage required for the new vCenter appliance, use the `core` value. For more information on the types of data you can transfer from your existing vCenter to the new, upgraded vCenter, see Transferring Data from an Existing vCenter Appliance.

Table 60: Configuration Parameters in the `new_vcsa` Section, `sso` Subsection

Name	Type	Description
`first_instance`	string	Set `first_instance` to `true` if this is the first converge upgrade operation in your vCenter SSO domain, or if you do not want to set the replication partner for this node. If you want to set the replication partner for this node, set `first_instance` to `false` and provide the value of the `replication_partner_hostname` as the FQDN of a previously converged node which is currently in an embedded vCenter topology.
`replication_partner_hostname`	string	Host name of the replication partner. Remove if it is the `first_instance`.

Configuration Parameters in the `source_vc` Section

Table 61: Configuration Parameters in the `source_vc` Section, `managing_esxi_or_vc` Subsection

Name	Type	Description
`hostname`	string	The IP address or FQDN of the source ESX or vCenter host on which resides the appliance that you want to upgrade.
`username`	string	A user name with administrative privileges on the source ESX host, for example, root.
`password`	string	The password of the user with administrative privileges on the source ESX host.
`port`	integer	The HTTPS reverse proxy port of the source ESX host. The default port is 443. Use only if the source ESX host uses a custom HTTPS reverse proxy port.

Table 62: Configuration Parameters in the `source_vc` Section, `vc_vcsa` Subsection

Name	Type	Description
`hostname`	string	The IP address or FQDN of the source appliance that you want to upgrade.
`username`	string	vCenter Single Sign-On administrator user on the source appliance, for example administrator@vsphere.local. **Important:** The user must be administrator@*`your_domain_name`*.
`password`	string	The password of the vCenter Single Sign-On administrator user on the source appliance.
`root_password`	string	The password for the root user of the operating system of the source appliance.
`export_dir`	string	Directory to export source configuration and data.

Configuration Parameters in the `ceip` Section

Table 63: Configuration Parameters in the `ceip` Section, `settings` Subsection

Name	Type	Description
`ceip_enabled`	Boolean	Set to `true` to join the CEIP for the new upgraded appliance.

Upgrade a vCenter Appliance by Using the CLI

You can use the CLI installer to perform an unattended upgrade of a vCenter appliance. You must run the CLI upgrade from a Windows, Linux, or Mac machine that is in the same network as the appliance that you want to upgrade.

- See Prerequisites for Upgrading the vCenter Appliance.
- Prepare Your JSON Configuration File for CLI Upgrade.
- Review the arguments for running the CLI upgrade. See Syntax of the CLI Upgrade Command.
- Verify that the user name with which you are logged in to your machine, the path to the vCenter appliance installer, the path to your JSON configuration file, and the string values in your JSON configuration file contain only ASCII characters. Extended ASCII and non-ASCII characters are not supported.

1. Navigate to the `vcsa-cli-installer` subdirectory for your operating system.
   - If you are running the upgrade on Windows, navigate to the `vcsa-cli-installer\win32` directory.
   - If you are running the upgrade on Linux, navigate to the `vcsa-cli-installer/lin64` directory.
   - If you are running the upgrade on Mac, navigate to the `vcsa-cli-installer/mac` directory.
2. Optional: Verify that you prepared the upgrade template correctly by running a basic template verification.

   ```
   vcsa-deploy upgrade --verify-template-only path_to_the_json_file
   ```
3. Optional: Gather and validate the upgrade requirements by running a pre-upgrade check.

   ```
   vcsa-deploy upgrade --precheck-only path_to_the_json_file
   ```

   The pre-upgrade check installs the Upgrade Runner on the source appliance that you want to upgrade without upgrading the appliance.

   The Upgrade Runner validates the configurations such as ESX, network settings, and NTP servers. The Upgrade Runner also checks if you have selected a suitable deployment size and storage size for the new appliance against the compute resources required for the upgrade.
4. Perform the upgrade by running the following command.

   ```
   vcsa-deploy upgrade --accept-eula optional_arguments path_to_the_json_file
   ```

   Use *`optional_arguments`* to enter space-separated arguments to set additional execution parameters of the upgrade command.

For example, you can set the location of the log and other output files that the installer generates. This example also confirms participation in the VMware Customer Experience Improvement Program (CEIP). If the `ceip_enabled` parameter is set to true in the JSON deployment template, you must include the argument `--acknowledge-ceip`.

```
vcsa-deploy upgrade --accept-eula --acknowledge-ceip --log-dir=path_to_the_locationpath_to_the_json_file
```

Verify Your vCenter Upgrade Is Successful.

Syntax of the CLI Upgrade Command

You can use command arguments to set the execution parameters of the upgrade command.

You can add a space-separated list of arguments to the CLI upgrade command.

```
vcsa-deploy upgrade path_to_the_json_file list_of_arguments
```

Argument	Description
`--accept-eula`	Accepts the end-user license agreement. Required for executing the deployment command.
`--acknowledge-ceip`	Confirms your acknowledgment of your VMware Customer Experience Improvement Program (CEIP) participation. Required if the `ceip_enabled` parameter is set to `true` in the JSON deployment template.
`-v`, `--verbose`	Adds debug information to the console output.
`-t`, `--terse`	Hides the console output. Displays only warning and error messages.
`--log-dir` *LOG_DIR*	Specifies the location of the log and other output files that the installer generates.
`--skip-ovftool-verification`	Performs basic verification of the configuration parameters and deploys the vCenter appliance, but does not validate the OVF Tool parameters in the `ovftool_arguments` subsection of the JSON template. If you set arguments that the OVF Tool does not recognize, the deployment might fail.
`--no-ssl-certificate-verification`	Prohibits SSL verification of for all server connections. The CLI verifies that a server's security certificate is signed by a Certificate Authority (CA), and establishes a secure connection. If the certificate is self-signed, the CLI stops the upgrade unless you specify that the CLI ignore the self-signed certificate using the `--no-ssl-certificate-verification` command parameter. If you are connecting to a server with a self-signed certificate, and fail to specify that the CLI accept it, the CLI displays the server's self-signed certificate thumbprint, and prompts you to accept or reject it. You can also specify that the CLI ignore self-signed certificates using the `ssl_certificate_verification` configuration parameter in the JSON template. See Upgrade Configuration Parameters for CLI Upgrade of vCenter Appliance . **Important:** Avoid using this option as it may cause problems during or after upgrade due to an un-validated identity of the target host.
`--operation-id`	Lets you provide an identifier to track the concurrent installation, migration, or upgrade of multiple vCenter instances. If do not provide an operation ID, the CLI generates a universally unique identifier (UUID) which you can use to identify the different instances of vCenter and their installation or upgrade status.
`--pause-on-warnings`	Pauses and waits for acknowledgment of warnings.

Argument	Description
`--verify-template-only`	Performs basic template verification without installing Upgrade Runner, running prechecks, and upgrading or migrating the vCenter appliance.
`--precheck-only`	Installs Upgrade Runner on the source appliance and runs a complete set of prechecks without performing the upgrade.
`--upgrade-framework`	Lets you provide the framework for the type of upgrade you want to perform. Specify `legacy` to run the regular upgrade process or `reduced_downtime_upgrade`.
`-h, --help`	Displays the help message for the `vcsa-deploy upgrade` command.
`--template-help`	Displays the help message for the use of configuration parameters in the JSON upgrade file.

After the execution finishes, you can get the exit code of the command.

Exit Code	Description
0	Command ran successfully
1	Runtime error
2	Validation error
3	Template error

Reduced Downtime Upgrade

Reduced downtime upgrade is a migration-based approach to upgrade vCenter between major and minor versions that reduces downtime to less than 5 minutes under ideal network, CPU, memory, and storage conditions.

You can perform a reduced downtime upgrade for single self-managed vCenter instances, vCenter instances managed by another vCenter, vCenter instances enabled with vCenter HA, and vCenter instances connected in Enhanced Linked Mode configuration.

Supported Upgrade Paths

Source Version	Target Version	Upgrade Interface and Capabilities
vCenter 9.1.0	vCenter 9.1.1	Full reduced downtime upgrade support by using the vSphere Client wizard and the `vcsa-deploy` CLI. vCenter HA and Enhanced Linked Mode supported.
vCenter 9.0.x	vCenter 9.1 / vCenter 9.1.1	Full reduced downtime upgrade support by using the vSphere Client wizard and the `vcsa-deploy` CLI. vCenter HA and Enhanced Linked Mode supported.
vCenter 8.0 Update 3	vCenter 9.1 / vCenter 9.1.1	Full reduced downtime upgrade support by using the vSphere Client wizard and the `vcsa-deploy` CLI. vCenter HA and Enhanced Linked Mode supported.
vCenter 8.0 Update 2	vCenter 9.1 / vCenter 9.1.1	Full reduced downtime upgrade support by using the vSphere Client wizard and the `vcsa-deploy` CLI. First version to support reduced downtime upgrade for vCenter HA clusters.

Source Version	Target Version	Upgrade Interface and Capabilities
vCenter 8.0 P02	vCenter 9.1 / vCenter 9.1.1	CLI reduced downtime upgrade support by using the `vcsa-deploy` CLI only. Minimum source patch level for reduced downtime upgrade. The vSphere Client wizard and vCenter HA reduced downtime upgrade are not available on this version.
vCenter 8.0 Update 1 vCenter 8.0	vCenter 9.1 / vCenter 9.1.1	**Not supported.** Use the standard two-stage migration upgrade, or update the source to vCenter 8.0 P02 or later to use reduced downtime upgrade.

Note:

Reduced downtime upgrade is not available for vCenter HA clusters of versions earlier than vCenter 8.0 Update 2.

Note:

vCenter 8.0 P02 is the minimum source patch level containing reduced downtime upgrade engine binaries. Source vCenter instances on 8.0 P02 can perform reduced downtime upgrade to 9.1.1 by using the `vcsa-deploy` CLI.

To use the vSphere Client wizard for reduced downtime upgrade or to upgrade a vCenter HA cluster, update the source instance to vCenter 8.0 Update 2 or later.

About the Reduced Downtime Upgrade Process

The reduced downtime upgrade process uses a migration-based approach. In this approach, a new vCenter appliance is deployed and the current vCenter data and configuration is copied to it. The reduced downtime upgrade does not replace the current vCenter upgrade process.

During the reduced downtime upgrade process, the source vCenter appliance and all resources remain online. The downtime occurs when the source vCenter appliance is stopped, the configuration is switched over to the target vCenter, and the services are started. The downtime is expected to take less than 5 minutes under ideal network, CPU, memory, and storage provisioning.

You can use either the vSphere Client or the CLI method to perform the reduced downtime upgrade.

Reduced Downtime Upgrade in High Availability Environments

Reduced downtime upgrade is integrated with vCenter HA deployment.

The reduced downtime upgrade is integrated with the vCenter HA automatic and manual configuration deployment, including self-managed vCenter and non-self-managed vCenter.

You can use the vSphere Client and API-based approaches to perform a reduced downtime upgrade in a high availability (HA) environment.

For more information, see Reduced Downtime Upgrade for vCenter HA.

Reduced Downtime GUI Upgrade

You can use the vSphere Client to perform a reduced downtime upgrade of a vCenter appliance.

The high-level steps for the reduced downtime upgrade are:

1. Configure the repository URL.

Note:

Perform this step when you are upgrading from versions earlier than vSphere 8.0 Update 2 to vSphere 8.0 Update 2 or later versions.

Go to Step 2 if you are upgrading from vSphere 8.0 Update 2 to later versions.

2. Select the target upgrade version for your vCenter choosing from the available version in the Broadcom online software depot or from the downloaded and mounted ISO files.
3. Verify that you have a backup of the source vCenter appliance.
4. Upgrade the vCenter Lifecycle Manager service plug-in and run prechecks.
5. Configure the target vCenter appliance.
6. Prepare the upgrade and and switch over to the target vCenter appliance.

Validations are run at every stage of the process. If you face an issue, resolve the issue and retry the upgrade steps. If the upgrade is canceled or an error occurs at any stage, the configuration reverts to the source vCenter instance. The source vCenter instance does not shut down until the upgrade process is complete and the target vCenter is up and running.

Note:

In the event of a failure, the upgrade process is cancelled automatically and the configuration reverts to the source vCenter instance. The vCenter Lifecycle Manager service plug-in does not revert.

Configure a Custom Repository URL

You can configure the vCenter appliance to use the default or a custom repository URL as the target version. By default the current repository for URL-based patching is the legacy public VMware repository URL.

Log in to the vCenter Appliance Management Interface (VAMI) as root.

There are two ways for configuring the repository URL:

- If vCenter is not connected to the Internet or if your security policy does not allow you to connect to the Internet, you can build and configure a custom repository. The custom repository runs on a local web server within your data center and replicates the data from the default repository.

  Note:

  You must perform this procedure when you are upgrading from versions earlier than vSphere 8.0 Update 2.
- If your vCenter system is connected to the Internet, you must replace the default download URL with the Broadcom online repository URL that requires a customer-specific token. See Download Updates from the Broadcom Online Repository.

1. If you want to configure a custom repository URL, build the repository on your local web server.
2. Log in to the Broadcom Support Portal.
3. Download the vCenter appliance ISO image. See Broadcom knowledge base article 366685.
4. Confirm that the md5sum is correct by using an MD5 checksum tool.
5. On your web server, create a repository directory under the root.

   For example, create the `vc_update_repo` directory.
6. In the vCenter Appliance Management Interface, click **Update**.
7. Click **Settings**.
8. Select the Repository settings.

Option	Description
Default	Uses the legacy VMware repository URL that is preset for the build profile of the appliance. This download URL is no longer active and must be replaced with the new Broadcom online repository. See Broadcom knowledge base article 390120.
Specified	Uses a custom repository download URL. You can enter the custom repository URL, for example, *`https://`*

Option	Description
	web_server_name.your_company.com/ vc_update_repo. The custom repository URL must use a secure protocol such as HTTPS or FTPS.

9. Optional: If the specified repository requires authentication, enter the user name and password.
10. If you do not want to perform a security certificate check, deselect the **Check Certificate** check box.
    If you trust the URL for the repository, you can choose to bypass the certificate check for the repository URL.
11. Click **SAVE**.

    In the **Current version details** pane, you can view the vCenter version and build number.

Download and Mount the ISO Image

VMware releases the vCenter appliance ISO image, which contains the installers for the vCenter appliance.

- Create an account at https://support.broadcom.com/.
- Verify that you have the following privileges:
  - **Virtual machine .Interaction.Configure CD media** on the virtual machine.
  - **Datastore.Browse datastore** on the datastore to which you upload the installation media ISO image.
  - **Datastore.Low level file operations** on the datastore to which you upload the installation media ISO image.
- Download the ISO image to the content library or datastore connected to the vCenter virtual machine or the Management vCenter.
- Mount the ISO image on the CD-ROM device of the vCenter virtual machine.

1. Log in to the Broadcom Support Portal.
2. Click the software divisions drop-down menu next to your user name, and select **VMware Cloud Foundation**.
3. On the left-hand side, click **My Downloads** and search for VMware vSphere Foundation (VVF) or VMware Cloud Foundation (VCF).

   After selecting the right version of one of the products, a list of its components is displayed.
4. Review and accept the Terms and Conditions.
5. Click the **View Group** button to view all vCenter download options.
6. Click the download button next to the vCenter appliance ISO image to download it.
7. In vSphere Client, right-click a virtual machine in the vCenter inventory and select **Edit Settings**.
8. Expand **CD/DVD drive**, and select **Datastore ISO File** from the drop-down menu.
   The **Select File** dialog box opens.
9. Browse to select the file that you downloaded and click **OK**.
10. Select **Connect At Power On** to connect the device when the virtual machine powers on.
11. Click **OK**.
12. Expand the **VM Hardware** panel and click the **Connected** icon next to the datastore ISO file to connect the device.

Download Updates from the Broadcom Online Repository

You can download the latest patches and updates directly from the Broadcom online software repository.

Log in to the vCenter Appliance Management Interface (VAMI) as root.

With vCenter 9.1, you can provision your custom URL to the Broadcom online repository and can download all latest updates and patches available. See Broadcom knowledge base article 390120.

In an air-gapped environment, you can build and configure a custom repository. See Configure a Custom Repository URL.

1. In the vCenter Appliance Management Interface, click **Update**.
2. Click **Settings**.
3. Select the Repository settings.

Option	Description
Default	Use the legacy VMware repository URL that is preset for the build profile of the appliance. These download URLs are no longer active and must be replaced with the new Broadcom online repository. See Broadcom knowledge base article 390120.
Specified	Use the Broadcom online repository URL. You can enter the Broadcom online repository URL, for example, `https://dl.broadcom.com/<download_token>/PROD/COMP/VCENTER/vmw/8d167796-34d5-4899-be0a-6daade4005a3/9.1/`. For more information about how to reconfigure the default online repository, see Broadcom knowledge base article 390120.

4. If you do not want to perform a security certificate check, deselect the **Check Certificate** check box.

   If you trust the URL for the repository, you can choose to bypass the certificate check for the repository URL.
5. Click **Save**.

   In the **Current version details** pane, you can view the vCenter version and build number.

Upgrade the vCenter Lifecycle Plug-in

After you download and mount the ISO image, upgrade the vCenter Lifecycle plug-in.

- Verify that you have downloaded and mounted the ISO image in an air-gapped environment. See Download and Mount the ISO Image.
- Verify that you have reconfigured the Broadcom online repository download URL. See Download Updates from the Broadcom Online Repository.
- Verify that you have a backup of the source vCenter appliance.
- Concurrent upgrades are not supported.
- Perform steps 3-5 if you are upgrading from a version earlier than vSphere 8.0 Update 2.
- If you are upgrading from version vSphere 8.0 Update 2 or later versions, you can mount the ISO file and can upgrade the plug-in.

With vCenter 9.1 and later, when the upgrade of the vCenter Lifecycle plug-in fails, your vCenter system is reverted to its original state. The automatic rollback is a configurable option which is enabled by default. See Configure the Automatic Rollback of the vCenter Lifecycle Plug-In.

1. In the vSphere Client, select the vCenter that you want to upgrade.
2. On the **Updates** tab, click **vCenter Server** > **Upgrade**.
3. Click **Target Version** and select one of the available options.

Upgrade Option	Description
Upgrade with online depot	If you reconfigured the download URL to use the Broadcom online repository, you can select from the latest available updates and patches for vCenter. See Download Updates from the Broadcom Online Repository.
Upgrade with ISO file	If you configured the vCenter appliance to use the custom repository URL, you can select from the downloaded and mounted ISO files. See Configure a Custom Repository URL.

4. From the **Target Version** field, click **Select version**.

   The **Select the available update** popup dialog opens.

5. Select the vCenter version and click **OK**.
6. Click **Next**.
7. In **Backup**, select the check box to indicate that you have backed up the vCenter.
   The backups are listed.
8. In **Upgrade Plug-in**, select **Upgrade plug-in** to update the vCenter Lifecycle service plug-in.
   After the plug-in upgrades, the UI reloads with the version from the ISO file.
9. Click **Next** to proceed with configuring the target vCenter appliance.

   Warning:
   Clicking **Next** before the upgrade completes might result in crash of the plug-in. Wait for the progress bar to reach 100% before you click **Next**.

10. Click **Run prechecks** to run source prechecks and click **Next**.
11. Optional: Click **Discard** to discard the upgrade workflow and return to the mount the ISO step.

    With vCenter 9.1 and later, the vCenter Lifecycle plug-in reverts to its previous state.

Configure the Automatic Rollback of the vCenter Lifecycle Plug-In

You can configure the upgrade behavior of the vCenter Lifecycle plug-in to automatically roll back to its previous state in case the upgrade fails.

1. Log in to the vCenter appliance as the root user.
2. Set the value of the `async_patch` > `enabled` field to `false` to deactivate the automatic rollback of the vCenter Lifecycle plug-in.

   By default the automatic rollback option is activated.

```
root@<vcenter_PNID> [ /etc/applmgmt/appliance ]# cat patching_backup_config.json
{
   "backup" :
   {
     "cap_registry": "/usr/lib/vmware-capengine/scripts/user_registry.sh",
     "cap_status_url": "http://127.0.0.1:15051/capengine/api/v1/workflow/instance/{0}",
     "cap_url": "http://127.0.0.1:15051/capengine/api/v1/workflow/execute",
     "enable_auto_backup": true,
     "partitions_to_exclude": "log_vg,lifecycle_vg",
     "patching_user": "patchingOrchestrator",
     "pwd_len": 16,
     "retry_count": 4,
     "retry_interval": 1,
     "snapshot_size": "10%",
     "snapshot_profile": "vg_root_0:100%,db_vg:30%,dblog_vg:30%",
     "status_retry_count": 1800,
     "include_boot_backup_restore": "yes"
   },
   "async_patch" :
   {
     "allowed_base_version" : "9.1.0.0",
     "enabled" : true
   }
 }
```

Configure the Target vCenter appliance

When the vCenter Lifecycle service plug-in update completes, configure the target vCenter appliance.

After the upgrade, the target vCenter has the same IP address and identity as the source vCenter.

1. In **Upgrade**, go to the **Target Appliance** section.
2. Click **Configure target appliance**.
   The **Target VM deployment** wizard appears.
3. Read and accept the license agreement, and click **Next**.
4. Review the VMware Customer Experience Improvement Program (CEIP) page and choose if you want to join the program.

   For information about the CEIP, see the Configuring Customer Experience Improvement Program section in *vCenter and Host Management*.
5. Click **Next**.
6. Enter location details if the source vCenter is managed by another vCenter.
   Enter the container location, HTTPS port, user name,and password of the source vCenter.
7. View and accept the certificate of the managing vCenter which manages the source vCenter and click **Next**.
8. Optional: In **HA Containers Locations**, enter the details of the managing vCenter which manages the Passive and Witness nodes if the vCenter HA is configured with the manual configuration,

   You can enter the details of the managing vCenter which manages the Passive and Witness nodes to create them automatically after the upgrade completes. If you do not enter the details, vCenter HA is removed during upgrade and you must configure HA manually after the upgrade completes.

   Enter the container location, HTTPS port, user name, and password of the vCenter that manages the Passive and Witness nodes.
9. Select where you want to deploy the target vCenter appliance and click **Next**.

   You can deploy the target vCenter appliance on the same location as the source appliance or choose a different location. If you select a different location, the target location must be a different vCenter or host and you must enter the connection details such as, host name or IP address, HTTPS port, user name, and password of the vCenter.

   Note:

   When vCenter HA (automatic or manual) is configured, you cannot deploy the target vCenter appliance on a different location.

   The **Deployment Type** page is displayed if you choose to deploy the target on the same location as the source vCenter appliance.
10. If you select a different location for the target vCenter, view and accept the certificate of the target deployment and click **Next**.
11. Select the deployment type of the target vCenter appliance and click **Next**.
    Select **Same configuration** to deploy the target vCenter appliance with same configuration as the source.

    Select **Detailed configuration** to enter all the configuration parameters for the target appliance.

    When vCenter HA (automatic or manual) is configured, you cannot provide detailed configuration parameters.

    a) Select the data center or virtual machine folder where you want to deploy the target vCenter appliance.
    b) Select the compute resource required for the target deployment.
    c) Click **Next**.
12. Configure the target VM appliance details and click **Next**.

    Enter the VM name and set and confirm the temporary root password. By default, the VM name is same as the source after the upgrade. You can retain it or enter a different VM name.

13. Select a deployment size and storage size for the target vCenter and click **Next**.
14. Select a datastore for the deployment and click **Next**.

    Optionally, enable thin provisioning.
15. In the **Network Settings** page, enter a temporary IP address of the target vCenter for static deployments.

    This IP address is used during the deployment of the target vCenter appliance, during upgrade and for switchover.

    If the temporary IP address mode is DHCP, you do not need to enter the IP address.

    If you select **Automatic**, the temporary IP address is automatically discovered from the local link layer IP range and you are not required to provide the IP address.
16. In the **Review** page, review the configuration and click **Finish**.

    The **Target Appliance** section under the **Upgrade** window appears.
17. To change the settings, click **Edit configuration** and make the changes required.
18. Click **Next** to proceed with the upgrade and switchover.
19. Click **Discard** to discard the upgrade workflow and return to the mount the ISO step. The vCenter Lifecycle plug-in does not revert.

    You cannot discard the configuration when the upgrade is progress.

Prepare Upgrade and Switch Over to the Target vCenter appliance

Once the target vCenter appliance is configured, you can upgrade and switch over the services to it.

The upgrade and switchover happen in two stages.

- **Preparation** - The first stage is the preparation stage where the upgrade is initialized and the target vCenter appliance is created with the configuration that you entered. During this stage, data from the source machine is replicated on the target machine. The duration of this process depends on your environment and the vCenter usage patterns.
- **Switchover** - The second stage is the switchover stage where the target vCenter appliance assumes the identity of the source appliance. During the switchover, there is a short downtime. You can choose a manual, automatic, or scheduled switchover.

The switchover process retains the source vCenter appliance. After the switchover completes, the source appliance remains powered off in the inventory, and the switchover process does not delete it. You must delete the source vCenter virtual machine manually after the switchover completes.

The switchover process shuts down the guest operating system of the source appliance. The timing of the shutdown depends on the source shutdown option that you select before starting the upgrade: during the upgrade or after the upgrade succeeds.

After a successful upgrade, the upgrade process appends an `-old` suffix to the source virtual machine name, and the target vCenter appliance takes the original name of the source virtual machine.

> **Important:**
>
> If the automated shutdown of the source vCenter appliance does not succeed, a warning message appears. Shut down the source appliance manually.

1. In **Update Planner**, go to the **Upgrade** section.
2. Click **Start upgrade**.
3. Select the type of switchover that you want to perform.
   - **Manual Switchover** - Select this option if want to manually switch over to the target vCenter appliance. You must click **Switchover** to trigger the switchover after the target appliance is created with the selected configuration.
   - **Automated Switchover** - Select this option to continue with an automatic switchover after the target appliance is created with the selected configuration.
   - **Scheduled Switchover** - Use this option to schedule the switchover for a time that is a minimum of 4 hours later than the current time.

4. If you select manual switchover, enter the switchover information.

   When the preparation stage completes and the switchover starts, you can see the status of the switchover stage.

   After the switchover completes, you can log in to the target vCenter appliance.

5. Click **Open vSphere Client** to log in.

   Note:

   If the upgrade fails, the configuration reverts to the source vCenter appliance.

Reduced Downtime CLI Upgrade

You can use the CLI installer to perform the reduced downtime upgrade of a vCenter appliance.

The CLI upgrade process includes downloading the vCenter appliance installer on a network virtual machine or physical server from which you want to perform the upgrade, preparing a JSON configuration file with the upgrade information, and running the upgrade command.

The vCenter appliance ISO file contains templates of JSON files that contain the minimum configuration parameters that are required for upgrading a vCenter appliance.

When you run the command to perform the reduced downtime upgrade, the installer patches the vSphere Lifecycle Manager, runs the CLI prechecks and the reduced downtime upgrade prechecks, and calls the Reduced Downtime Upgrade APIs.

CAUTION:

If NSX is not upgraded to a compatible version before the vCenter upgrade, the reduced downtime upgrade state will show as `Blocked`. The reduced downtime upgrade may remain in a blocked state even if the CLI session times out. Upgrade NSX to a compatible version to proceed the reduced downtime upgrade.

Prepare Your JSON Configuration File for the Reduced Downtime CLI Upgrade

Before you run the CLI command to perform the reduced downtime upgrade, you must prepare a JSON file with configuration parameters and their values for your upgrade specification.

- You must be familiar with the JSON syntax.
- Download and Mount the vCenter installer.

The vCenter installer contains JSON templates for all upgrade types. For information about the templates, see JSON Templates for CLI Upgrade of the Appliance.

For a complete list of the configuration parameters and their descriptions, navigate to the installer subdirectory for your operating system and run the `vcsa-deploy upgrade --template-help` command or see Upgrade Configuration Parameters for CLI Upgrade of Appliance.

1. In the the vCenter installer, navigate to the `vcsa-cli-installer` directory, and open the `templates` subfolder.
2. Copy the `upgrade` templates from the `upgrade` subfolder to your workspace.
3. Open the template file for your use case in a text editor.

   To ensure the correct syntax of your JSON configuration file, use a JSON editor.
4. Fill in the values for the required configuration parameters and, optionally, enter additional parameters and their values.
5. Optional: Use a JSON editor of your choice to validate the JSON file.
6. Save in UTF-8 format and close the file.

JSON Templates for the Reduced Downtime CLI Upgrade

The installer contains JSON templates that are located in the vcsa-cli-installer/templates directory. In the upgrade subfolder, you can find the JSON templates with the minimum configuration parameters for all upgrade types.

For each upgrade type, there is one template for deploying the new appliance on an ESX host and another template for deploying the new appliance on a vCenter instance.

For information about the templates, see JSON Templates for CLI Upgrade of the Appliance.For a complete set of parameters required for upgrade, see Upgrade Configuration Parameters for CLI Upgrade of Appliance.

Perform a Reduced Downtime Upgrade by Using the CLI

You can use the CLI installer to perform an reduced downtime upgrade of a vCenter appliance. You must run the upgrade from a Windows, Linux, or Mac machine that is in the same network as the appliance that you want to upgrade.

- Prepare Your JSON Configuration File for the Reduced Downtime CLI Upgrade.
- Review the arguments for running the CLI upgrade. See Syntax of the CLI Upgrade Command.
- Verify that the user name with which you are logged in to your machine, the path to the vCenter appliance installer, the path to your JSON configuration file, and the string values in your JSON configuration file contain only ASCII characters. Extended ASCII and non-ASCII characters are not supported.

1. Navigate to the `vcsa-cli-installer` subdirectory for your operating system.
   - If you are running the upgrade on Windows, navigate to the `vcsa-cli-installer\win32` directory.
   - If you are running the upgrade on Linux, navigate to the `vcsa-cli-installer/lin64` directory.
   - If you are running the upgrade on Mac, navigate to the `vcsa-cli-installer/mac` directory.
2. Optional: Verify that you prepared the upgrade template correctly by running a basic template verification.

   ```
   vcsa-deploy upgrade --verify-template-only path_to_the_json_file
   ```
3. Perform the reduced downtime upgrade by running the following command.

   ```
   vcsa-deploy upgrade --upgrade-framework reduced_downtime_upgrade path_to_the_json_file
   ```

   Or

   ```
   vcsa-deploy upgrade path_to_the_json_file
   ```

   To run the existing upgrade runner, run the following command

   ```
   vcsa-deploy upgrade --upgrade-framework legacy path_to_the_json_file
   ```

   The installer patches the vSphere Lifecycle Manager, runs the CLI prechecks and the reduced downtime upgrade prechecks, and calls the Reduced Downtime Upgrade APIs.

After Upgrading vCenter

After you upgrade vCenter 9.1, consider the post-upgrade options and requirements.

- Go to VMware Cloud Foundation Operations and assign your vCenter instance a license.
- Activate your vCenter instance for management by the VCF Operations cluster. See Broadcom knowledge base article 424630.
- Complete any component reconfigurations that might be required for changes during the upgrade or Reduced Downtime Upgrade cutover.
- Verify that you understand the authentication process and identify your identity sources. For detailed steps on verifying and reconfiguring Active Directory (LDAPS / IWA) or OIDC identity provider federation post-upgrade, see Identity Sources for vCenter with vCenter Single Sign-On.
- Upgrade any additional extension modules or third-party solution plugins that are linked to this instance of vCenter, such as VMware NSX, VMware Live Site Recovery, VMware Cloud Foundation Operations® (formerly VMware Aria Operations), and backup and monitoring applications.
- Optionally, upgrade the ESX hosts in the vCenter inventory to the same version as the vCenter instance using vSphere Lifecycle Manager cluster image management.
- If you are using smart card authentication, ensure to keep the smart card port open in the client environment. By default, the smart card port is open in vCenter. For details about the smart card port, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com

- If you plan to install Windows 11 as a guest operating system on a virtual machine, you must configure a key provider. Installing Windows 11 requires a Trusted Platform Module (TPM) 2.0. When installing Windows 11 as a guest operating system on a virtual machine, instead of using a physical TPM, you can use a virtual Trusted Platform Module (vTPM). A vTPM is a software-based representation of a physical TPM 2.0 chip. A vTPM depends on virtual machine encryption to secure vital TPM data, and so requires that you configure a key provider. For information about the key providers supported by vSphere, see the *Virtual Machine Encryption* chapter in the *vSphere Security* documentation. The easiest way is to configure a VMware vSphere®Native Key Provider™. vSphere Native Key Provider is included in all vSphere editions and does not require an external key server. For information about configuring a vSphere Native Key Provider, see *Configuring and Managing vSphere Native Key Provider* chapter in the *vSphere Security* documentation. As with all security solutions, consider the system design, implementation considerations, and tradeoffs of using vSphere Native Key Provider.
- If you had modified the DRS cluster automation level before upgrade, you can either continue to use the modified settings or revert the automation level to fully automated.
- Perform a baseline VAMI File-Based Backup (port `5480`) immediately following upgrade completion to establish an initial post-upgrade restore point.
- Retain the powered-off source vCenter appliance VM in the inventory for a 3 to 7 days observation period to verify stability, then permanently delete it from disk to reclaim datastore capacity. Do not power on the source VM on the production network post-upgrade.

Verify Your vCenter Upgrade Is Successful

You can verify the success of your vCenter upgrade.

You must be logged into the upgraded vCenter instance. If you created a reference of required information based or a CLI template, you can use it to validate the upgrade or migration success.

1. Verify that the IP address is correct.
2. Verify that the Active Directory registration has not changed.
3. Verify the Network registration is correct.
4. Verify the Domain is correct.
5. Verify the certificates are valid.
6. Verify the inventory data is correctly migrated.
   a) Review the events history.
   b) Review the performance charts.
   c) Review the users, permissions, and roles.

If the postupgrade configuration conforms to your required information or CLI template reference and expectations, the vCenter upgrade or migration is complete.

Use the vSphere Client to log in to the vCenter instance. See Log in to Your vCenter by Using the vSphere Client.

Reregister Plug-In Solution in vCenter After Upgrade or Migration

You reregister a previously registered plug-in solution and any third-party client plug-in packages with vCenter after the SSL certificate has been refreshed following an upgrade or migration.

Consult the vendor documentation for any solution-based vCenter extensions and client plug-ins for instructions to re-register after a vCenter upgrade or migration.

If the procedure provided by your plug-in solution vendor fails to reregister the plug-in, you can use the following procedure to remove the plug-in registration, and then register it again with vCenter. For information on registering plug-ins, see the *vCenter and Host Management* documentation. For information on removing or disabling unwanted plug-ins from vCenter, see Broadcom knowledge base article 344648.

1. In a web browser, navigate to the Managed Object Browser of your vCenter.
   `https://vcenter_server_ip_address_or_fqdn/mob/?moid=ExtensionManager`

2. Log in with your vCenter credentials.
3. On the `ManagedObjectReference:ExtensionManager` page, under Methods, click **UnregisterExtension**.
4. On the `void UnregisterExtension` page, in the text box inside the Value column, enter the value for the `key` property of the `Extension` data object of your vSphere Client extension.
5. To unregister the extension click **Invoke Method**.

Go to the solution registration page and register the plug-in.

Verify that your extension is registered successfully with vCenter by using one of the following approaches.

- In the vSphere Client, go to Administration and under Solutions, select **Client Plug-Ins** and click **Check for New Plug-Ins**.
- Log out and log in again to the vSphere Client. The vSphere Client checks for new plug-ins for each new user session.

Monitor and Manage Historical Data Migration

You can monitor and manage the background migration of historical data using the vCenter Management Interface.

- Verify that vCenter is successfully deployed and running.
- You must have selected to import historical data from the database in use by the previous version vCenter, to the embedded PostgreSQL database in vCenter 9.1. See Transferring Data from an Existing vCenter Appliance.

Using the vCenter Management Interface, you can perform the following data management tasks:

- Monitor the progress of the data migration.
- Pause the data migration.
- Cancel the data migration.

1. In a web browser, go to the vCenter Management Interface, https://*`appliance-IP-address-or-FQDN`*:5480.
2. Log in as root.

   The default root password is the password you set when deploying vCenter.
3. A status bar at the top of the vCenter Management Interface displays the percentage of data that has been copied from the source vCenter to the PostgreSQL database embedded in the target vCenter appliance.
4. To pause or cancel the data migration, click **Manage**.

Option	Description
Pause	During the data migration, vCenter performance might not be optimal. Pausing the migration lets vCenter run with better performance until such time that you can resume the process when it does not interfere with the needs of your business.
Resume	You can resume the data migration at a time that does not interfere with the business needs of your vCenter deployment.
Cancel	If you cancel the data import, the historical data will fail to import to the embedded PostgreSQL database. You cannot recover the data once you cancel the operation. If you cancel the import process, and want to import the historical data later, you must restart the upgrade or migration process from Stage 1 of the GUI installer.

When the data has been migrated, a success message displays in the vCenter Management Interface status bar.

Managing Software Updates, Upgrades, and Product Compatibility

You can view available vCenter updates and upgrades and produce interoperability reports about VMware products associated with vCenter. The reports help you plan vCenter updates and upgrades in your environment.

You can perform an interoperability check of VMware products within your environment against both source and target vCenter versions. You can also generate a pre-update report to ensure your environment meets the minimum software

and hardware requirements for a successful upgrade of vCenter. You can plan vCenter updates and upgrades based on the information in the reports you generate.

Note:

With vCenter 9.1 and later, you can perform interoperability check without joining the VMware Customer Experience Improvement Program (CEIP).

Note:

With vCenter 9.1 and later, you can perform interoperability checks in an air-gapped environment. First, download the compatibility data bundle and upload the file on your vCenter system. The compatibility data bundle holds all data from the Broadcom Compatibility Guide (BCG) combined with the Broadcom Product Interoperability Matrix.

For more information about how to download and use the compatibility data bundle in an air-gapped environment, see Broadcom knowledge base article 423296.

Interoperability report for the current vCenter	Lists VMware products in your environment and their compatibility against the current, or source, version of vCenter.
Interoperability reports for target versions of vCenter	Lists the VMware products in your environment and their compatibility against the target version of vCenter to which you want to upgrade.
Pre-update reports	With the pre-update report, you can ensure that your system meets the minimum software and hardware requirements for a successful upgrade of vCenter. The report provides information about problems that might prevent the completion of a software upgrade, and actions you can take to remedy those problems.

Monitor Interoperability for the Current vCenter Version

You can view an interoperability report that lists products in your environment associated with the current vCenter version and their compatibility.

Staring with vSphere 9.1, you can use Update Planner without joining the VMware Customer Experience Improvement Program (CEIP). For information about the CEIP, see the *Configuring Customer Experience Improvement Program* section in *vCenter and Host Management*.

The product interoperability report lists available products in your environment and their compatibility with the selected vCenter version.

Note:

In vSphere 9.1, you can perform interoperability checks in an air-gapped environment. First, download the compatibility data bundle and upload the file on your vCenter system. The compatibility data bundle holds all data from the Broadcom Compatibility Guide (BCG) combined with the Broadcom Product Interoperability Matrix.

For more information about how to download and use the compatibility data bundle in an air-gapped environment, see Broadcom knowledge base article 423296.

1. In the vSphere Client navigate to a vCenter for which you want to view an interoperability report.
2. On the **Monitor** tab, click **Interoperability**.

   The Product Interoperability report lists all available products in your environment associated with the selected vCenter.
3. (Optional) If there are VMware products in your environment that go undetected, you can manually add them to the list of products to check and regenerate the interoperability report.
   a) Click **Add Product** in the Product Interoperability pane.
   b) Select the VMware product and version you want to include in the interoperability report.

      Repeat this step for each VMware product you want to add to the report.

c) To add the selected products to the product inventory click **Done**.
d) Regenerate the report and verify that the products you added are included in the report.

4. (Optional) Click **Export** in the Product Interoperability pane to export and save a copy of the report as a comma-separated values (CSV) file, which allows data to be saved in a tabular format.

- Using the information in the report, you can apply patches and updates to your vCenter environment. See Patching and Updating vCenter 9.1 Deployments.
- You can generate a pre-upgrade report that lists actions that you must address to ensure a successful upgrade of vCenter in your environment. See Generate an Interoperability Report for Target vCenter Versions.

Generate an Interoperability Report for Target vCenter Versions

You can perform pre-checks and generate reports that provide product interoperability information for recommended update and upgrade versions of vCenter.

Staring with vSphere 9.1, you can use Update Planner without joining the VMware Customer Experience Improvement Program (CEIP). For information about the CEIP, see the *Configuring Customer Experience Improvement Program* section in *vCenter and Host Management*.

You can create a vCenter pre-update interoperability report to check the compatibility of VMware products in your environment against a pending version of vCenter. You can view the list of products in your environment that are associated with the current vCenter version, and whether those products are compatible with a suggested update version.

Note:

In vSphere 9.1, you can perform interoperability checks in an air-gapped environment. First, download the compatibility data bundle and upload the file on your vCenter system. The compatibility data bundle holds all data from the Broadcom Compatibility Guide (BCG) combined with the Broadcom Product Interoperability Matrix.

For more information about how to download and use the compatibility data bundle in an air-gapped environment, see Broadcom knowledge base article 423296.

1. In the vSphere Client, select a vCenter for which you want to view an interoperability report.
2. On the **Updates** tab, click **vCenter Server** > **Prechecks**.
3. Click **Select Version**.
4. Select the target version from the list of available updates and click **OK**.

   The list contains minor updates and major upgrade versions of vCenter.
5. Click **Product Interoperability** to view the report.
6. (Optional) If there are VMware products in your environment that go undetected, you can manually add them to the list of products to check and regenerate the interoperability report.
   a) Click **Modify Product List**.
   b) Select the VMware product and version that you want to include in the interoperability report.

      Repeat this step for each VMware product you want to add to the report.
   c) To add the selected products to the Update product inventory, click **Done**.
   d) Regenerate the report and verify that the products you added are included in the report.
7. (Optional) Click **Export** to save the report as a comma-separated values (CSV) file, which allows data to be saved in a tabular format.

- Using the information in the pre-upgrade report, you can apply patches and updates to your vCenter environment. See Patching and Updating vCenter 9.1 Deployments.
- You can generate a pre-upgrade report that lists actions that you must address to ensure a successful upgrade of vCenter in your environment. See Generate Pre-Update Reports.

Generate Pre-Update Reports

You can run pre-checks and generate reports that provide pre-update information. You use this information to make sure that your system meets the minimum software and hardware requirements to ensure a successful upgrade of vCenter.

When you generate a pre-update report, the report lists the actions you must address to ensure a successful upgrade of vCenter in your environment. The report also provides information about problems that might prevent the completion of a software upgrade or update.

1. In the vSphere Client, select a vCenter for which you want to view a pre-update report.
2. On the **Updates** tab, click **vCenter Server** > **Prechecks**.
3. Click **Select Version**.
4. Select the target version from the list of available updates and click **OK**.

   The list contains minor updates and major upgrade versions of vCenter.
5. Select **Source Prechecks** to view pre-update report.
6. (Optional) Click **Export** to save the report as a comma-separated values (CSV) file, which allows data to be saved in a tabular format.

- Using the information in the pre-upgrade report, you can apply patches and updates to your vCenter environment. See Patching and Updating vCenter 9.1 Deployments.
- You can generate an interoperability report that lists available VMware products in your environment, and the versions compatible with the vCenter version for which you are generating the report. See Generate an Interoperability Report for Target vCenter Versions.

Use the Latest Interoperability Data in an Air-Gapped Environment

You can generate update interoperability reports and perform system pre-checks for vCenter instances without Internet connectivity.

- Verify that you have a download token. See Broadcom knowledge base article 390098.

Starting with vCenter 9.1, you can use the VCF Download Tool to download binaries, metadata, and the compatibility data bundle from the Broadcom Support Portal. The compatibility data bundle combines the Broadcom Compatibility Guide (BCG) with the Broadcom Product Interoperability Matrix.

1. Download the VCF Download Tool from the Broadcom Support Portal.
   a) Log in to the Broadcom Support Portal and navigate to the **My Downloads** section.
   b) On the **Drivers & Tools** tab, search for the VCF Download Tool.
   c) Read and accept the *Terms and Conditions* and *Compliance Reporting Terms and Conditions*.
   d) Click the **HTTPS Download** icon next to the VCF Download Tool to initiate the download.
2. Navigate to the directory where you downloaded the VCF Download Tool and extract it.

   ```
   tar -xvzf vcf-download-tool-<vcf_product_version>.<build_number>.tar.gz
   ```
3. Run the VCF Download Tool on an Internet-connected system to download the latest compatibility metadata bundle.

   ```
   ./vcf-download-tool metadata download -d=<download_directory> --depot-download-token-
   file=<path_to_downloadtoken>.txt
   ```
4. Navigate to `<download_directory>/PROD/metadata/Compatibility`.

   The `/v1` and `/v2` directories contain the `VmwareCompatibilityData.json` files that provide the latest compatibility data.
5. Upload the downloaded compatibility data bundle to an offline vCenter system.
   a) Locate the compatibility data JSON files and compress them using the following command.

   ```
   cd <download_directory>/PROD/metadata/Compatibility
      gzip v1/VmwareCompatibilityData.json
   ```

b) Use Base64 encoding to convert the compressed binary data into a string format.

```
base64 v1/VmwareCompatibilityData.json.gz > compatibility_bundle_b64.txt
```

c) Log in to your vCenter system and copy the returned session ID.

```
curl -k -X POST -u 'username:password' https://<vcenter_ip_address_or_fqdn>/rest/com/vmware/cis/session
```

d) Send the compatibility data to your vCenter instance.

```
ENCODED_DATA=$(cat <encoded_compatibility_bundle_file_name>.txt)
curl --request PUT \
  --url 'https://<vcenter_ip_address_or_fqdn>/lcm/rest/vcenter/lcm/interop/bundle?vmw-task=true' \
  --header "vmware-api-session-id: <returned_session_ID>" \
  --header 'Content-Type: application/json' \
  --data "{\"data\": \"${ENCODED_DATA}\"}"
```

The API call returns the `task-id`, for example, `ea0a3939-2b97-4e8f-afee-cab46919a9da:com.vmware.vcenter.lcm.interop.interop_bundle`.

e) Use the `task-id` to view the result of the compatibility bundle upload task.

```
curl --request GET \
  --url https://<vcenter_ip_address_or_fqdn>/lcm/rest/cis/tasks/<returned_task_id> \
  --header "vmware-api-session-id: <returned_session_ID>"
```

The latest compatibility data bundle is uploaded to your vCenter instance in an air-gapped environment.

- You can generate an interoperability report that lists available VMware products in your environment, and the versions compatible with the vCenter version for which you are generating the report. See Generate an Interoperability Report for Target vCenter Versions.
- You can generate a pre-upgrade report that lists actions that you must address to ensure a successful upgrade of vCenter in your environment. See Generate Pre-Update Reports.

Patching and Updating vCenter 9.1 Deployments

VMware regularly releases patches for the vCenter. You can use the vCenter Appliance Management Interface (VAMI) or the appliance shell to apply patches to a vCenter.

VMware makes patches available on a monthly basis. These patches can only be applied in between major releases of vCenter. For example, patches released for the initial release of vCenter9.0, are not applicable to vCenter8.0 Update 1, as any patches previously made available will be included with the Update 1 release.

These patches can be for core product functionality, other packages in the vCenter such as Photon, or both.

Note:
You must use only the patches provided by VMware to update the packages in your vCenter. Updating these packages through any other means may impact the product functionality.

VMware distributes the available patches in two forms, one for ISO-based and one for URL-based models of patching.

- You can download the patch ISO images from the Broadcom Support Portal.
  VMware publishes a single type of ISO image that contains patches.

Download Filename	Description
`VMware-vCenter-Server-Appliance-`*`product_version-build_number`*`-patch-FP.iso`	Full product patch for the appliance, which contains the VMware software patches and the fixes related to security and third-party products (e.g. JRE and Photon OS components).

- You can configure the vCenter appliance to use a repository URL as a source of available patches.

Note:

The default online depot URLs are now unique for each customer. To update the default URLs of the online depot, see Broadcom knowledge base articles 390098 and 390120.

vCenter Quick Patches

You can apply express patches that carry security or critical bug fixes with minimal disruption to your workload management operations.

In vCenter 9.1, a mechanism is provided to apply a patch to your vCenter systems without disruption to your workload management operations. Such patches are called quick patches, and are delivered whenever there are some security or other critical bug fixes. Quick patches are not used to deliver major upgrades and updates.

Based on the services included in the patch payload, three groups of services are determined:

- Services with zero downtime such as `vlcm`, `vsphere-ui`, and `eam`. Upgrading services from this group does not impact the workload management operations.
- Services with near to zero downtime, up to 180 seconds such as `vmware-envoy` and `vmware-cis-license`. This group contains services that are critical for the workload management operations and must be restarted after an update.
- Services with downtime equivalent to the reduced downtime upgrade times, from 3 through 5 minutes. For example, `vmware-vpxd`, `vmca`, `lookupsvc`, and `vmware-vmon` have significant dependencies on other services and restarting them leads to cascading downtime for all dependant services. The downtime, that is required for a full recovery after a service from this group is restarted, is usually more than 3 minutes.

You can apply a quick patch to a vCenter system using the vCenter Management Interface and VCF API.

Patching vCenter Using the vCenter Management Interface

You can use the vCenter Management Interface to view the installed patches, check for new patches and install them, and configure automatic checks for available patches.

To perform ISO-based patching, you download an ISO image, attach the ISO image to the virtual CD/DVD drive of the appliance, check for available patches in the ISO image, and install the patches.

To perform URL-based patching, you check for available patches in a repository URL and install the patches. The vCenter is preset with a legacy VMware repository URL for the build profile of the appliance. You must reconfigure the appliance to use the Broadcom online repository repository URL or a custom repository URL, for example, a repository URL that you previously built on a local Web server running within your data center. See Download Updates from the Broadcom Online Repository and Configure a Custom Repository URL.

See Log In to the vCenter Appliance Management Interface.

Automatic Creation of LVM Snapshot

The vCenter patching orchestrator automatically creates a Logical Volume Manager (LVM) snapshot of the vCenter before performing a patch installation. If the patching fails, the orchestrator provides options to either continue patching after fixing the issue or roll back to the automatically created LVM snapshot, before starting the patching operation.

The rollback option is only available when you are patching vCenter 8.0 Update 2 and higher versions.

When you initiate the installation of a patch, prechecks are performed to verify that a snapshot can be taken and there is sufficient disk space for the LVM snapshot. If the precheck returns an error, the patching workflow exits.

After prechecks run successfully, the VMware Directory Services (`vmdird`) is stopped and the orchestrator takes an image-based backup (LVM snapshot). The snapshot is saved as snapshot disk on the vCenter appliance. If there is a failure while taking the snapshot, the `vmdird` is started in normal replication mode, an error is thrown and the patching workflow exits and the snapshot is cleaned up.

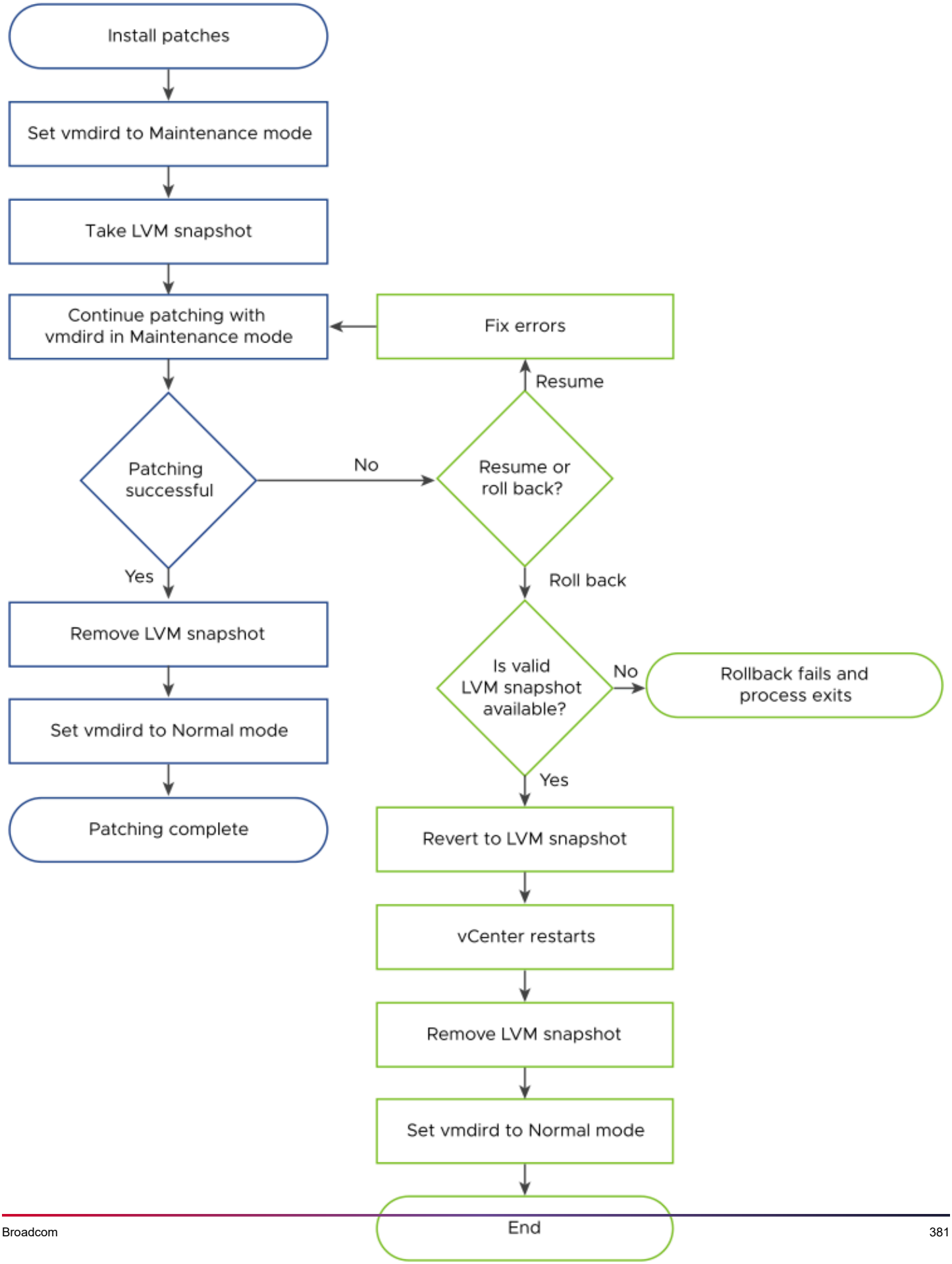
If the snapshot is taken successfully, the patching resumes with `vmdird` in standalone mode.

If an error occurs during patching, the vCenter Management Interface displays options to resume or roll back the process. You can select **Resume the vCenter installation** and correct the problem preventing the patching to finish staging the patches or select **Revert to the last vCenter backup version** to roll back.

If you choose to revert, a patching rollback API is called. The API checks if an LVM snapshot is available and reverts to the snapshot. After reverting, the snapshot is removed to reclaim the disk space and `vmdird` is set to normal replication

mode. The API also performs a cleanup of any changes made by the patching install workflow before creating the backup. If a valid snapshot is not available, the installation exits with an error.

Figure 26: Workflow for automatic creation of LVM snapshot and rollback



Check for and Stage Patches to the vCenter Appliance

Before you install available patches, you can stage the patches to the appliance. You can use the vCenter Management Interface to stage patches either from a local repository by attaching an ISO image to the appliance, or from a remote repository directly by using a repository URL.

- If you are staging patches from an ISO image that you previously downloaded from the Broadcom Support Portal, you must attach the ISO image to the CD/DVD drive of the vCenter. You can configure the ISO image as a datastore ISO file for the CD/DVD drive of the appliance using the vSphere Client. See *vSphere Virtual Machine Administration*.
- If you are staging patches from a remote repository or directly from the Broadcom online repository, verify that you have configured the repository settings and that the current repository URL is accessible. See Configure URL-Based Patching.

During the process of staging, the vCenter Management Interface validates that a patch is a VMware patch, that the staging area has enough free space, and that the patches are not altered. Only new patches or patches for existing packages that can be upgraded are staged.

If you are patching vCenter 8.0 Update 2 and higher versions, the patching orchestrator automatically creates a backup of the vCenter before performing a patch installation. For more information, see Automatic Creation of LVM Snapshot.

If a problem that prevents the successful staging of the patches occurs, vCenter suspends the staging process. Review the error messages, correct the problem, and, in many cases, you can resume staging the patches from the point at which vCenter encountered the problem.

1. Log in to the vCenter Management Interface as root.

   The default root password is the password you set when deploying vCenter.

2. Click **Update**.

3. Click **Check Updates** and select a source.

Option	Description
Check CD ROM + URL	Scans the attached ISO image as well as the configured repository URL for available patches.
Check CD ROM	Scans the ISO image that you attached to the CD/DVD drive of the appliance for available patches.

   In the Available Updates pane, you can view the details about the available patches in the source that you selected.

   Important:
   Some updates might require a reboot of the system. You can see information about these updates in the Available Updates pane.

   Note:

   Quick patches are marked with the **Quick patch** label. In the details view for an incoming quick patch, you can view the estimated downtime, the estimated patch install time, and the affected services.

4. You can run a pre-check of an update to verify that it is compatible with your current deployment.

5. Click the staging option that you would like to use.

Option	Description
Stage only	Stages the selected patches to the vCenter appliance for installation at a later time.
Stage and Install	Stages and installs the selected patches to the vCenter appliance. For more information on installing patches, see Install vCenter Patches.
Unstage	Unstages the selected patches.
Resume the vCenter installation	If vCenter encounters a problem staging the patches, the vCenter Management Interface displays the **Resume** button. Correct the problem preventing the patches from staging, and click **Resume** to finish staging the patches. The staging process

Option	Description
	resumes from the point at which vCenter encountered the problem.
Revert to last vCenter backup option	Select this option to roll back to the automatically created snapshot. If you choose to revert, a patching rollback API is called. The API checks if a snapshot is available and reverts to the snapshot. **Note:** The automatic creation of a backup is available only when you are patching vCenter 8.0 Update 2 and higher versions.

If you decided to stage the available patches for installation at a later time, you can now install. See Install vCenter Patches.

Configure the Repository for URL-Based Patching

For URL-based patching, by default the vCenter appliance is configured to use the default VMware repository URL that is preset for the build profile of the appliance. You can configure a custom repository URL as the current source of patches for your environment requirements.

Log in to the vCenter Appliance Management Interface as root.

By default the current repository for URL-based patching is the default VMware repository URL.

If vCenter is not connected to the Internet or if your security policy requires it, you can build and configure a custom repository. The custom patching repository runs on a local Web server within your data center and replicates the data from the default repository. Optionally, you can set up an authentication policy for accessing the Web server that hosts the custom patching repository.

1. If you want to configure a custom repository URL, build the repository on your local web server.
2. Log in to the Broadcom Support Portal.
3. Download the vCenter appliance ISO image. See Broadcom knowledge base article 366685.
4. Confirm that the md5sum is correct by using an MD5 checksum tool.
5. On your web server, create a repository directory under the root.

   For example, create the `vc_update_repo` directory.
6. Extract the ZIP file into the repository directory.

   The extracted files are in the `manifest` and `package-pool` subdirectories.
7. In the vCenter Management Interface, click **Update**.
8. Click **Settings**.
9. Select the Repository settings.

Option	Description
Default repository	Uses the default VMware repository URL that is preset for the build profile of the appliance.
Specified repository	Uses a custom repository. You must enter the repository URL, for example, *`https://web_server_name.your_company.com/`*`vc_update_repo`. The repository URL must use a secure protocol such as HTTPS or FTPS.

10. If the specified repository requires authentication, enter the user name and password.
11. (Optional) If you do not want to perform a security certificate check, deselect the **Check Certificate** check box.
    If you trust the URL for the repository, you can choose to bypass the certificate check for the repository URL.
12. Click **OK**.

Install vCenter Patches

Install vCenter Patches

You can check for and can install patches either from an ISO image, directly from a custom repository URL, or from the Broadcom online repository.

- Log in to the vCenter Appliance Management Interface as root.
- Before you can install available patches, you check for new patches and stage the patches to the vCenter appliance. See Check for and Stage Patches to the vCenter Appliance.
- If you are patching the appliance from an ISO image that you previously downloaded from Broadcom Support Portal Downloads, you must attach the ISO image to the CD/DVD drive of the vCenter appliance. You can configure the ISO image as a datastore ISO file for the CD/DVD drive of the appliance by using the vSphere Client. See *vSphere Virtual Machine Administration*.
- If you are patching the appliance from a custom repository URL or from the Broadcom online repository, verify that you have configured the repository settings and that the current repository URL is accessible. See Configure the Repository for URL-Based Patching.
- Create an image-based backup and take a powered-off snapshot of the vCenter Appliance you are patching as a precaution in case there is a failure during the patching process.

  Important:
  The services running in the vCenter appliance become unavailable during the installation of the patches. You must perform this procedure during a maintenance period. As a precaution if there is a failure, you can back up the vCenter. For information on backing up and restoring vCenter, see *vCenter Deployment and Setup*.

1. In the vCenter Management Interface, click **Update**.

   In the **Current version details** pane, you can view the vCenter version and build number.

   In the **Available updates** pane, you can view the available updates with type and release type.

   The **Version** indicates the available version and whether the patch is a quick patch.

   The **Type** indicates the type of the update. The values include:

   - **BUG FIX**
   - **SECURITY**
   - **BUG FIX AND SECURITY**
   - **ENHANCEMENT**
   - **INFORMATIONAL**

   The release type values include the following:

   - **UPDATE**
   - **PATCH**

   The **Reboot Required** indicates if the update requires a reboot of the vCenter system or not, and the values are **Yes** or **No**.

   The **Critical** indicates if the update is critical or not and the values are **Yes** or **No**.

2. Select the range of staged patches to apply and click **Install**.

   Important:
   Some updates might require a reboot of the system. You can see information about these updates in the Available Updates pane.

3. Read and accept the End-User License Agreement.
4. A system pre-check verifies that the patches can be successfully installed with the provided information.

   If the pre-check discovers missing or incorrect information, or other problems preventing a successful installation, you are prompted to correct the problem and resume the installation.
5. After the installation finishes, click **OK**.
6. If the patch installation requires the appliance to reboot, click **Summary**, and click **Reboot** to reset the appliance.

In the Available Updates pane, you can see the changed update status of the vCenter appliance.

Enable Automatic Checks for vCenter Patches

You can configure vCenter to perform automatic checks for available patches in the configured repository URL at a regular interval.

- Log in to the vCenter Appliance Management Interface as root.
- Verify that you have configured the repository settings and that the current repository URL is accessible. See Configure the Repository for URL-Based Patching.

1. In the vCenter Management Interface, click **Update**.
2. Click **Settings**.
3. Select **Check for updates automatically**, and select the day and time in UTC to perform automatic checks for available patches.
4. Click **OK**.

vCenter appliance performs regular checks for available patches in the configured repository URL. In the Available Updates pane, you can view information about the available patches. You can also view the vCenter health status for notifications about available patches. See View Health Status in *vCenter Configuration*.

Patching the vCenter Appliance by Using the Appliance Shell

You can use the `software-packages` utility in the appliance shell of a vCenter appliance to see the installed patches, stage new patches, and install new patches.

To perform ISO-based patching, you download an ISO image, mount the ISO image to the CD/DVD drive of the appliance, optionally stage the available patches from the ISO image to the appliance, and install the patches. For steps to mount the ISO image to the CD/DVD drive, see Configure a Datastore ISO File for the CD/DVD Drive section in the vSphere Virtual Machine Administration.

To perform URL-based patching, you optionally stage the available patches from a repository URL to the appliance and install the patches. The vCenter appliance is preset with a default VMware repository URL for the build profile of the appliance. You can use the `update.set` command to configure the appliance to use the default VMware repository URL or a custom repository URL, for example, a repository URL that you previously built on a local web server running within your data center. You can also use the `proxy.set` command to configure a proxy server for the connection between the vCenter appliance and the repository URL.

View a List of All Installed Patches in the vCenter Appliance

You can use the `software-packages` utility to see a list of the patches currently applied to the vCenter appliance. You can also view the list of the installed patches in chronological order and details about a specific patch.

1. Access the appliance shell and log in as a user who has a super administrator role.

   The default user with a super administrator role is root.
2. To view the full list of patches and software packages installed in the vCenter appliance, run the following command:

   ```
   software-packages list
   ```

3. To view all patches applied to the vCenter appliance in chronological order, run the following command:

   ```
   software-packages list --history
   ```

   You see the list in chronological order. A single patch in this list can be an update of multiple different packages.

4. To view details about a specific patch, run the following command:

   ```
   software-packages list --patch patch_name
   ```

   For example, if you want to view the details about the VMware-vCenter-Server-Appliance-Patch1 patch, run the following command:

   ```
   software-packages list --patch VMware-vCenter-Server-Appliance-Patch1
   ```

   You can see the complete list of details about the patch, such as vendor, description, and installation date.

Configure URL-Based Patching

For URL-based patching, the vCenter appliance is preset with a default VMware repository URL for the build profile of the appliance. You can use the `update.set` command to configure the appliance to use the default or a custom repository URL as the current source of patches and enable automatic checks for patches.

By default the current repository for URL-based patching is the default VMware repository URL.

> **Note:**
> You can use the `proxy.set` command to configure a proxy server for the connection between vCenter and the repository URL. For more information about the API commands in the appliance shell, see *vCenter Configuration*.

If vCenter is not connected to the Internet or if your security policy requires it, you can build and configure a custom repository. The custom patching repository runs on a local Web server within your data center and replicates the data from the default repository. Optionally, you can set up an authentication policy for accessing the Web server that hosts the custom patching repository.

1. Log in to the Broadcom Support Portal.
2. Download the vCenter appliance ISO image. See Broadcom knowledge base article 366685.
3. Confirm that the md5sum is correct by using an MD5 checksum tool.
4. On your web server, create a repository directory under the root.

   For example, create the vc_update_repo directory.

5. Extract the ZIP file into the repository directory.

   The extracted files are in the manifest and package-pool subdirectories.

6. Access the appliance shell and log in as a user who has a super administrator role.

   The default user with a super administrator role is root.

7. To see information about the current URL-based patching settings, run the `update.get` command.

   You can see information about the current repository URL, the default repository URL, the time at which the appliance last checked for patches, the time at which the appliance last installed patches, and the current configuration of automatic checks for patches.

8. Configure the current repository for URL-based patching.

   - To configure the appliance to use the default VMware repository URL, run the following command:

     ```
     update.set --currentURL default
     ```

   - To configure the appliance to use a custom repository URL, run the following command:

     ```
     update.set --currentURL https://web_server_name.your_company.com/vc_update_repo [--username username]
      [--password password]
     ```

     The square brackets [] enclose the command options.

     If the custom repository requires authentication, use the `--username` *`username`* and `--password` *`password`* options.

9. To activate automatic checks for vCenter appliance patches in the current repository URL at regular intervals, run the following command:

   ```
   update.set --CheckUpdates enabled [--day day] [--time HH:MM:SS]
   ```

   The square brackets [] enclose the command options.

   Use the `--day day` option to set the day for performing the regular checks for patches. You can set a particular day of the week, for example, `Monday`, or `Everyday`. The default value is `Everyday`.

   Use the `--time HH:MM:SS` option to set the time in UTC for performing the regular checks for patches. The default value is `00:00:00`.

   The appliance performs regular checks for available patches in the current repository URL.

10. To deactivate automatic checks for vCenter patches, run the following command:

    ```
    update.set --CheckUpdates disabled
    ```

If you configured the appliance to perform automatic checks for available patches, you can regularly view the vCenter appliance health status for notifications about available patches. See View vCenter Health Status in *vCenter Configuration*.

Stage Patches to the vCenter Appliance

Before you install available patches, you can stage the patches to the appliance. You can use the `software-packages` utility to stage patches either from a local repository by attaching an ISO image to the appliance, or from a remote repository directly by using a repository URL.

- If you are staging patches from an ISO image that you previously downloaded from the Broadcom Support Portal, you must attach the ISO image to the CD/DVD drive of the vCenter appliance. You can configure the ISO image as a datastore ISO file for the CD/DVD drive of the appliance by using the vSphere Client. See *vSphere Virtual Machine Administration*.
- If you are staging patches from a remote repository, verify that you have configured the repository settings and that the current repository URL is accessible. See Configure URL-Based Patching.

1. Access the appliance shell and log in as a user who has a super administrator role.

   The default user with a super administrator role is root.

2. Stage the patches.

   - To stage the patches included in the attached ISO image, run the following command:

     ```
     software-packages stage --iso
     ```

   - To stage the patches included in the current repository URL, run the following command:

     ```
     software-packages stage --url
     ```

     By default the current repository URL is the default VMware repository URL.

     If you want to stage only the third-party patches, use the `--thirdParty` option.

   - To stage the patches included in a repository URL that is not currently configured in the appliance, run the following command:

     ```
     software-packages stage --url URL_of_the_repository
     ```

     If you want to stage only the third-party patches, use the `--thirdParty` option.

   If you want to directly accept the End-User License Agreement, use the `--acceptEulas` option.

   For example, to stage only the third-party patches from the current repository URL with directly accepting the End-User License Agreement, run the following command:

   ```
   software-packages stage --url --thirdParty --acceptEulas
   ```

   In the process of staging, the command validates that a patch is a VMware patch, that the staging area has enough free space, and that the patches are not altered. Only completely new patches or patches for existing packages that can be upgraded are staged.

3. Optional: To see information about the staged patches, run the following command:

   ```
   software-packages list --staged
   ```

   Each patch includes a metadata file that contains information such as patch version, product name, whether a restart of the system is required, and so on.

4. Optional: To view a list of the staged patches, run the following command:

   ```
   software-packages list --staged --verbose
   ```

5. Optional: To unstage the staged patches, run the following command:

   ```
   software-packages unstage
   ```

   All directories and files generated by the staging process are removed.

Install the staged patches. See Install vCenter Patches.

Important:
If you staged the patches from an ISO image, keep the ISO image attached to the CD/DVD drive of the appliance. The ISO image must be attached to the CD/DVD drive of the appliance throughout the staging and installation processes.

Install vCenter Patches

You can use the `software-packages` utility to install the staged patches. You can also use the `software-packages` utility to install patches directly from an attached ISO image or repository URL without staging the patch payload.

- If you are installing staged patches, verify that you staged the correct patch payload. See Stage Patches to the vCenter Appliance.
- If you are installing patches that you previously staged from an ISO image, verify that the ISO image is attached to the CD/DVD drive of the vCenter appliance. See Stage Patches to the vCenter Appliance.
- If you are installing patches directly from an ISO image that you previously downloaded from the Broadcom Support Portal, you must attach the ISO image to the CD/DVD drive of the vCenter appliance. You can configure the ISO image as a datastore ISO file for the CD/DVD drive of the appliance by using the vSphere Client. See Change the CD/DVD Drive Configuration in *vSphere Virtual Machine Administration*.
- If you are installing patches directly from a repository, verify that you have configured the repository settings and that the current repository URL is accessible. See Configure URL-Based Patching.
- Create an image-based backup and take a powered-off snapshot of the vCenter appliance you are patching as a precaution in case there is a failure during the patching process.

Important:

The services running in the appliance become unavailable during the installation of the patches. You must perform this procedure during a maintenance period. As a precaution in case of failure, you can back up the vCenter appliance. For information about backing up and restoring vCenter, see File-Based Backup and Restore of vCenter and Image-Based Backup and Restore of a vCenter Environment in *vCenter Deployment and Setup*.

1. Access the appliance shell and log in as a user who has a super administrator role.

   The default user with a super administrator role is root.

2. Install the patches.

   - To install staged patches, run the following command:

     ```
     software-packages install --staged
     ```

   - To install patches directly from an attached ISO image, run the following command:

     ```
     software-packages install --iso
     ```

   - To install patches directly from the current repository URL, run the following command:

     ```
     software-packages install --url
     ```

     By default the current repository URL is the default VMware repository URL.

   - To install patches directly from a repository URL that is not currently configured, run the following command:

```
software-packages install --url URL_of_the_repository
```

If you want to directly accept the End-User License Agreement, use the `--acceptEulas` option.

For example, to install patches from the current repository URL without staging the patches with directly accepting the End-User License Agreement, run the following command:

```
software-packages install --url --acceptEulas
```

3. If the patch installation requires a reboot of the appliance, run the following command to reset the appliance.

```
shutdown now -r "patch reboot"
```

Patching a vCenter High Availability Environment

To patch a vCenter appliance configured in a vCenter High Availability (HA) cluster, you must remove the vCenter High Availability configuration and delete the Passive and Witness nodes. After patching the vCenter appliance, you must re-create your vCenter High Availability clusters.

This procedure describes how to remove the vCenter HA configuration.

1. In the vSphere Client, log in to the vCenter from which you want to remove the vCenter HA configuration.
2. Click the **Configure** tab, and select vCenter **HA**.
3. Select **Remove vCenter HA** .
   - The vCenter HA cluster's configuration is removed from the Active, Passive, and Witness nodes.
   - The Active node continues to run as a standalone vCenter appliance.
4. Delete the Passive and Witness nodes.

Patch the vCenter appliance as described in either Patching vCenter Using the vCenter Management Interface or Patching the vCenter Appliance by Using the Appliance Shell.

After patching the vCenter appliance, you can configure vCenter HA. For information about configuring vCenter HA, see You Can Minimize Downtime with vSphere in *vSphere Availability*.

Troubleshooting vCenter Upgrade

The installation and upgrade software enables you to identify problems on the client machine that can cause an installation, upgrade, or migration to fail.

For interactive installations, upgrades, and migrations, the errors or warnings are displayed on the final panel of the installer, where you are asked to confirm or cancel the installation or upgrade. For scripted installations, upgrades, or migrations, the errors or warnings are written to the installation log file. You can also consult the product release notes for known problems.

vSphere Update Manager provides custom messages for these errors or warnings. To see the original errors and warnings that are returned by the precheck script during an Update Manager host upgrade scan, review the Update Manager log file `vmware-vum-server-log4cpp.log`.

The *vSphere Upgrade* guide describes how to use VMware products and their features. If you encounter problems or error situations that are not described in this guide, you may find a solution in the Broadcom Support Portal. You can also use Broadcom Community Forums to find others with the same problem or ask for help, or you can open Support Request to get help from a VMware service professional.

- Export a vCenter Support Bundle for Troubleshooting

Errors and Warnings Returned by the Installation and Upgrade Precheck Script

The installation and upgrade precheck script runs tests to identify problems on the host machine that can cause an installation, upgrade, or migration to fail.

For interactive installations, upgrades, and migrations, the errors or warnings are displayed on the final screen of the GUI installer, where you are asked to confirm or cancel the installation or upgrade. For scripted installations, upgrades, or migrations, the errors and warnings are written to the installation log file.

vSphere Update Manager provides custom messages for these errors or warnings. To see the original errors and warnings returned by the precheck script during an Update Manager host upgrade scan, review the Update Manager log file `vmware-vum-server-log4cpp.log`.

Table 64: Error and Warning Codes That Are Returned by the Installation and Upgrade Precheck Script

Error or Warning	Description
64BIT_LONGMODESTATUS	The host processor must be 64-bit.
COS_NETWORKING	Warning. An IPv4 address was found on an enabled service console virtual NIC that has no corresponding address in the same subnet in the VMkernel. A separate warning appears for each such occurrence.
CPU_CORES	The host must have at least two cores.
HARDWARE_VIRTUALIZATION	Warning. If the host processor does not have hardware virtualization, or if hardware virtualization is not turned on in the host BIOS, host performance reduces. You can enable hardware virtualization in the host machine boot options panel. See your hardware vendor's documentation.
MD5_ROOT_PASSWORD	This test checks that the root password is encoded in MD5 format. If a password is not encoded in MD5 format, it might be significant only to eight characters. In this case, any characters after the first eight are no longer authenticated after the upgrade, which can create a security problem.
MEMORY_SIZE	The host requires the specified amount of memory to upgrade.
PACKAGE_COMPLIANCE	vSphere Update Manager upgrade to vSphere Lifecycle Manager only. This test checks the existing software on the host against the software contained on the upgrade ISO to determine whether the host has been successfully upgraded. If any of the packages are missing or are an older version than the package on the upgrade ISO, the test returns an error. The test results indicate which software was found on the host and which software was found on the upgrade ISO.
PARTITION_LAYOUT	You can upgrade or migrate software only if at most one VMFS partition on the disk is being upgraded. The VMFS partition must begin after sector 1843200.
POWERPATH	This test checks for the installation of EMC PowerPath software, consisting of a CIM module and a kernel module. If either of these components is found on the host, the test checks that matching components, such as CIM or the VMkernel and module, also exist in the upgrade. If they do not exist, the test returns a warning that indicates which PowerPath components were expected on the upgrade ISO and which, if any, were found.
PRECHECK_INITIALIZE	This test checks that the precheck script can be run.
SANE_ESX_CONF	The `/etc/vmware/esx.conf` file must exist on the host.
SPACE_AVAIL_ISO	vSphere Update Manager only. The host disk must have enough free space to store the contents of the installer CD or DVD.
SPACE_AVAIL_CONFIG	vSphere Update Manager upgrade to vSphere Lifecycle Manager only. The host disk must have enough free space to store the legacy configuration between reboots.
SUPPORTED_ESX_VERSION	You can upgrade to ESX9.0 only from version 8.0 ESX hosts.
TBOOT_REQUIRED	This message applies only to vSphere Update Manager upgrade to vSphere Lifecycle Manager only. The upgrade fails with this error when the host system is running in trusted boot mode (tboot), but the ESX upgrade ISO does not contain any tboot VIBs. This test prevents an upgrade that can make the host less secure.
UNSUPPORTED_DEVICES	Warning. This test checks for unsupported devices. Some PCI devices are not supported in ESX9.0.

Error or Warning	Description
UPDATE_PENDING	This test checks the host for VIB installations that require a reboot. This test fails if one or more such VIBs are installed, but the host has not yet been rebooted. In these conditions, the precheck script is unable to determine reliably which packages are currently installed on the host. If this test fails, it may not be safe to rely on the rest of the precheck tests to determine whether an upgrade is safe. If you encounter this error, restart the host and retry the upgrade.

Upgrade Issues with vCenter Containing Host Profiles

Most common issues that might occur during a vCenter upgrade to version 9.1, that contains host profiles.

- For issues occurring during a vCenter upgrade or ESX upgrade, see Troubleshooting vSphere Host Profiles in *vSphere Host Profiles*
- If an error occurs when you import a host profile to an empty vCenter inventory, see Reference Host is Unavailable in *vSphere Host Profiles* for Reference Host is Unavailable.
- If a host profile compliance check fails for NFS datastore, see vSphere Host Profile without an NFS Datastore in *vSphere Host Profiles*.

vCenter Configuration

vCenter Configuration provides information about configuring VMware vCenter®.

Intended Audience

This information is intended for anyone who wants to configure VMware vCenter®. The information is written for experienced system administrators who are familiar with virtual machine technology and data center operations.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

vSphere Client and vCenter Management Interface

Instructions in this guide reflect the vSphere Client, an HTML5-based GUI, and the vCenter Management Interface. Some additional functions can be performed using the vCenter appliance shell and the Direct Console User Interface.

How to Configure vCenter

You deploy the vCenter appliance as a preconfigured virtual machine optimized for running vCenter and associated services.

During the deployment of vCenter, you can create a vCenter Single Sign-On domain or join an existing domain. For information about the vCenter deployment, see Introduction to vSphere Deployment and Setupin the *vCenter Deployment and Setup* documentation.

The vCenter appliance package contains the following software:

- Photon OS® 5.0
- The vSphere authentication services
- PostgreSQL
- VMware vSphere Lifecycle Manager Extension
- VMware vCenter Lifecycle Manager

Customization of the preconfigured virtual machine is unsupported except for adding memory, CPU, and disk space.

vCenter has the following default user names:

- root user with the password that you set during the deployment of the appliance. You use the root user to log in to the vCenter Appliance Management Interface and to the preconfigured virtual machine operating system.

  Important:
  The password for the root account of vCenter expires after 90 days by default. For information about changing the root password and configuring the password expiration settings, see Change the Password and Password Expiration Settings of the Root User.
- administrator@*your_domain_name* which is the vCenter Single Sign-On user with the password and domain name that you set during the deployment of the appliance.

  When you install vCenter, you can change the vSphere domain. Do not use the same domain name as the domain name of your Microsoft Entra ID domain name.

  Initially, only the user administrator@*your_domain_name* has the privileges to log in to the vCenter system. By default, the administrator@*your_domain_name* user is a member of the **SystemConfiguration** > **Administrators** group. This user can add an identity source in which additional users and groups are defined to vCenter Single Sign-On or give permissions to the users and groups. For more information, see *vSphere Authentication*and *vSphere Security*.

You can configure vCenter settings in four ways:

- Use the vCenter Management Interface.

  You can edit the system settings such as access, network, time synchronization, and the root password settings. This is the preferred way for configuring vCenter.

- Use the vSphere Client.
  You can navigate to the system configuration settings of vCenter and configure the deployment to an external identity provider domain. You can manage the services that are running in vCenter and modify various settings such as access, network, and firewall settings.
- Use the Bash shell.
  You can use TTY1 to log in to the console or can use SSH and run configuration, monitoring, and troubleshooting commands in vCenter.
- Use the Direct Console User Interface.
  You can use TTY2 to log in to the vCenter Direct Console User Interface to change the password of the root user, configure the network settings, or enable access to the Bash shell or SSH.

Using the vCenter Management Interface to Configure vCenter

After you deploy vCenter, you can log in to the vCenter Management Interface and edit the settings.

For information about patching vCenter and enabling automatic checks for vCenter patches, see Patching and Updating vCenter 9.1 Deployments in the *vCenter Upgrade* documentation.

For information backing up and restoring vCenter, see File-Based Backup and Restore of vCenterin the *vCenter Deployment and Setup* documentation.

Log In to the vCenter Appliance Management Interface

Log in to the vCenter Appliance Management Interface to access the vCenter configuration settings.

Verify that the vCenter is successfully deployed and running.

> **Note:**
> The login session expires if you leave the vCenter Appliance Management Interface idle for 10 minutes.

1. In a web browser, go to the vCenter Appliance Management Interface, https://*`appliance-IP-address-or-FQDN`*:5480.
2. Log in as root.
   The default root password is the password that you set while deploying vCenter.

View vCenter Health Status

You can use the vCenter Management Interface to view the overall health status of vCenter and health messages.

Log in to the vCenter Appliance Management Interface as root.

The overall health status of vCenter is based on the status of the hardware components such as CPU, memory, database, storage, and swap. It is also based on the update component, which shows whether the software packages are up to date according to the last check for available patches.

> **Important:**
>
> If you do not perform regular checks for available patches, the health status of the update component might become out-of-date. For information about how to check for vCenter patches and enable automatic checks for vCenter patches, see Patching and Updating vCenter 9.1 Deployments in the *vCenter Upgrade* documentation.

1. In the vCenter Management Interface, click **Summary**.
2. In the Health Status pane, view the Overall Health badge.

Table 65: Health Status

Badge Icon	Description
	Good. All components are healthy.
	Warning. One or more components might become overloaded soon. View the details in the Health Messages pane.
	Alert. One or more components might be degraded. Nonsecurity patches might be available. View the details in the Health Messages pane.
	Critical. One or more components might be in an unusable status and vCenter might become unresponsive soon. Security patches might be available. View the details in the Health Messages pane.
	Unknown. No data is available.

Reboot or Shut Down vCenter

You can use the vCenter Appliance Management Interface to restart or power off the virtual machine running.

Log in to the vCenter Appliance Management Interface as root.

1. From the top menu pane, click the **Actions** drop-down menu.
2. Click **Reboot** or **Shutdown** to restart or power off the virtual machine.
3. In the confirmation dialog box, click **Yes** to confirm the operation.

Create a Support Bundle

You can create a support bundle that contains the log files for the vCenter instance running in the appliance. You can analyze the logs locally on your machine or can send the bundle to Broadcom Support.

Log in to the vCenter Appliance Management Interface as root.

1. Navigate to the **Create Support Bundle** option.
   a) From the header navigation, click the **Actions** drop-down menu.
   b) Click **Monitor** from the vertical navigation and select the **Logs** tab.
2. Click **Create Support Bundle**.
3. Select the components that you require the logs for.
4. Add a description and click **SAVE**.

   The created support bundle appears in the table on the **Monitor** > **Logs** tab.
5. Select the support bundle from the list and click **DOWNLOAD**.
6. Delete the bundle after you download it.

   Note:

   Generated support bundles are stored temporarily on the system. If you do not manually delete a support bundle after downloading it, the system automatically deletes it after 12 hours by default to preserve storage space. Ensure that you download the bundle before it expires.

The support bundle is downloaded as a `.tgz` file on your local machine.

Monitor CPU and Memory Use

You can use the vCenter Appliance Management Interface to monitor the overall CPU and memory use of vCenter.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Monitor**.
2. On the Monitor page, click the **CPU & Memory** tab.
3. From the **date range** drop-down menu, select the time period for which you want to generate a CPU utilization trending graph and a memory utilization trending graph.
4. Point to the graphs to see the CPU and memory use for a particular date and time.

Monitor Disk Use

You can use the vCenter Appliance Management Interface to monitor the disk use of vCenter.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Management Interface, click **Monitor**.
2. On the Monitor page, click the **Disks** tab.

The Monitor Disks pane shows a disk, sortable by name, partition, or utilization.

Monitor Network Use

You can use the vCenter Appliance Management Interface to monitor the network use of vCenter in the last day, week, month, or quarter.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Monitor**.
2. On the Monitor page, click the **Network** tab.
3. From the **date range** drop-down menu, select the time period for generating the network utilization graph.
4. From the table below the graph grid, select a packet or transmit byte rate to monitor.
   The options vary depending on your network settings.
   The network utilization graph refreshes to display the use of the item you select.
5. Point to the network utilization graph to see the network use data for a particular date and time.

Monitor Database Use

You can use the vCenter Appliance Management Interface to monitor the use of the embedded database of the vCenter by data type. You can also monitor space use trending graphs and filter any of the largest data types.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Monitor**.
2. On the Monitor page, click the **Database** tab to monitor the consumed and free space for the vCenter database.
3. From the **date range** drop-down menu, select the time period for which you want to generate the space utilization trending graphs.
4. At the base of the graph, click the title a particular database component to include or exclude that component from the graph.

Option	Description
Seat space utilization trend graph	Allows you to select and view alarm, event, task, and stat trendlines.
Overall space utilization trend graph	Allows you to select and view SEAT, DB Log, and core trendlines.

5. Point to the space utilization graph to see the database use value for a particular date and time.

Enable or Disable SSH and Bash Shell Access

You can use the vCenter Appliance Management Interface to edit the access settings for the appliance.

Log in to the vCenter Appliance Management Interface as root.

You can enable or disable an SSH administrator login to the appliance. You can also enable access to the vCenter Bash shell for a specific time interval.

1. In the vCenter Management Interface, click **Access**, and click **Edit**.
2. Edit the access settings for vCenter.

Option	Description
Activate SSH login	Activates SSH access to vCenter.
Activate DCUI	Activates DCUI access to vCenter.
Activate Console CLI	Activates console CLI access to vCenter.
Activate Bash Shell	Activates Bash shell access to vCenter for the number of minutes that you enter.

3. Click **OK** to save the settings.

Configure the DNS, IP Address, and Proxy Settings

You can set the combination of static and DHCP, IPv4 and IPv6 addresses, edit the DNS settings, and define the proxy settings for vCenter.

- To change the IP address of the appliance, verify that the system name of the appliance is an FQDN. The system name is used as a primary network identifier. If you set an IP address as a system name during the deployment of the appliance, you can later change the PNID to an FQDN.

  Note:
  You can set only IPv4 IP address as system name. The IPv4 IP address must be enabled before this setting.
- To restore a dual stack VC, the base VC after stage 1 deployment should be configured as:
  - If PNID of backed up VC resolves to IPv4 and IPv4 is configured static, the base VC during stage1 should have static or DHCP IPv4 configured.
  - If PNID of backed up VC resolves to IPv4 and IPv4 is configured DHCP, the base VC during stage1 should have DHCP IPv4 configured.
  - If PNID of backed up VC resolves to IPv6 and IPv6 is configured static, the base VC during stage1 should have static or DHCP IPv6 configured.
  - If PNID of backed up VC resolves to IPv6 and IPv6 is configured DHCP, the base VC during stage1 should have DHCP IPv6 configured.
- Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Networking**.
2. From the Network Settings page, click **Edit**.
3. Select **Network Adapter** and click **NEXT**.
4. Expand the Hostname and DNS section to configure the DNS settings.

   Note:
   Valid value for Hostname is either FQDN resolving to enabled IP address or IPv4 IP address.

Option	Description
Obtain DNS settings automatically	Obtains the DNS settings automatically from the network.
Enter DNS settings manually	Lets you set the DNS address settings manually. If you select this option, you must provide the following information: • The IP address of the preferred DNS server. • The IP address of the alternate DNS server.

5. Expand the NIC 0 section to configure the gateway settings.

   Note:
   PNID and network APIs support only NIC 0 as primary NIC.

   Recommended steps to configure network other than NIC 0, when vCenter has multiple NICs configured are:

   - NIC 1 is reserved for VCHA and should not be used in multi-NIC configurations.
   - As mentioned in the UI specifications, DGW should be set to NIC 0.
   - As access to vSphere Client is redirected to PNID, only access to NIC 0 is supported or recommended even in multi-NIC configurations.

6. Edit the IPv4 address settings.

Option	Description
Activate or Deactivate IPv4 settings	Activates or deactivates the IPv4 address based on the toggle switch option.
Obtain IPv4 settings automatically	Obtains the IPv4 address for the appliance automatically from the network.
Enter IPv4 settings manually	Uses an IPv4 address that you set manually. You must enter the IP address, subnet prefix length, and the default gateway. **Note:** For static IPv4 or IPv6 addresses, DNS server must be set manually.

   Note:
   Second and third party solutions need to re-register, when there is a change in the IPv4 and IPv6 IP addressess.

7. Edit the IPv6 address settings.

Option	Description
Enable or Disable IPv6 settings	Enables or disables the IPv6 address based on the toggle switch option.
Obtain IPv6 settings automatically through DHCP	Assigns IPv6 addresses to the appliance automatically from the network by using DHCP.
Obtain IPv6 settings automatically through router advertisement	Assigns IPv6 addresses to the appliance automatically from the network by using router advertisement.
Use static IPv6 addresses	Uses static IPv6 addresses that you set up manually. 1. Click the checkbox. 2. Enter the IPv6 address and the subnet prefix length. 3. Click **Add** to enter additional IPv6 addresses. 4. Click **Save**. **Note:** For static IPv4 or IPv6 addresses, DNS server must be set manually.

   You can configure the appliance to obtain the IPv6 settings automatically through both DHCP and router advertisement. You can assign a static IPv6 address at the same time.

   Note:
   Second and third-party solutions must re-register when there is a change in the IPv4 and IPv6 IP addresses.

8. In **SSO credentials**, enter the SSO administrator credentials for vCenter and Click **NEXT**.
9. Review and save the networking configuration.
10. To configure a proxy server, in the Proxy Settings section of the the Network Settings page, click **Edit**.
11. Select the proxy setting to activate.

Option	Description
FTP	Activate to configure the FTP proxy settings. **Note:** Ensure that the ICMP is enabled on the proxy server.
HTTPS	Activate to configure the HTTPS proxy settings.
HTTP	Activate to configure the HTTP proxy settings.

12. Optional: To add proxy bypass list entries, enter the IPv4 addresses, FQDNs, or domain names that you want to bypass as a comma separated list.
13. Enter the server hostname or IP address.
14. Enter the port.
15. Enter the username (optional).
16. Enter the password (optional).
17. Click **Save**.

Reconfigure the Primary Network Identifier

You can change the FQDN, IP, or PNID of the management network of vCenter.

Log in to the vCenter Appliance Management Interface as root.

The system name is used as a primary network identifier. If you set an IP address as a system name during the deployment of the appliance, you can later change the PNID to an FQDN. You cannot change an FQDN back to an IP address.

Note:

If vCenter High Availability (HA) is enabled, you must deactivate the vCenter HA setup before reconfiguring the PNID.

1. In the vCenter Appliance Management Interface, navigate to the **Networking** page and click **Edit**.
2. Select the NIC to be modified and click **Next**.
3. In the **Edit Settings** pane, change the host name and provide the new IP address. Click **Next**.
4. In the **SSO Credentials** pane, provide the administrator SSO credentials. You must use `administrator@<domain_name>` credentials.
5. In the **Ready to Complete** pane, review your new settings and check the backup acknowledgment box. Click **Finish**.

   A taskbar shows the status of the network update. To cancel the update, click **Cancel Network Update**. When the network reconfiguration is complete, the UI redirects to the new IP address.
6. To finish the reconfiguration process and restart services, log in using your administrator SSO credentials.
7. On the **Networking** page, verify the new host name and IP address.

- Re-register all deployed plug-ins.
- Regenerate all custom certificates.
- If vCenter HA was enabled, reconfigure vCenter HA.
- If Active Domain was enabled, reconfigure Active Domain.
- If Hybrid Link mode was enabled, reconfigure Hybrid Link with the Cloud vCenter.

Edit the Firewall Settings

After you deploy vCenter, you can edit its firewall settings and create firewall rules using the Management Interface.

Verify that the user who logs in to the vCenter instance is a member of the `SystemConfiguration.Administrators` group in vCenter Single Sign-On.

You can set up firewall rules to accept or block traffic between vCenter and specific servers, hosts, or virtual machines. You cannot block specific ports, you block all the traffic.

1. In the vCenter Appliance Management Interface, click **Firewall**.
2. Edit the firewall settings.

Command	Action
Add	1. To create a firewall rule, click **Add**. 2. Select a network interface of the virtual machine. 3. Enter the IP address of the network to apply this rule to. The IP address can be IPv4 and IPv6 address. 4. Enter a subnet prefix length. If you only block one IP address, use a 32-bit subnet mask. 5. From the **Action** drop-down menu, select whether to **Accept**, **Ignore**, **Reject**, or **Return** the connection between vCenter and the network that you entered. 6. Click **Save**.
Edit	1. Select a rule and click **Edit**. 2. Edit the settings of the rule. 3. Click **Save**.
Delete	1. Select a rule and click **Delete**. 2. At the prompt, click **Delete** again.
Reorder	1. Select a rule and click **Reorder**. 2. In the **Reorder** pane, select the rule to move. 3. Click **Move Up** or **Move Down**. 4. Click **Save**.

Configure the System Time Zone and Time Synchronization Settings

After you deploy vCenter, you can change the system time zone and time synchronization settings from the vCenter Appliance Management Interface.

Log in to the vCenter Appliance Management Interface as root.

When you deploy vCenter, you either use the time settings of the ESX host on which vCenter is running or you configure the time synchronization using an NTP server. If the time settings in your vSphere network change, you can edit the time zone and time synchronization settings in the appliance.

1. In the vCenter Appliance Management Interface, click **Time**.
2. Configure the system time zone settings.
   a) In the Time zone pane, click **Edit**.
   b) From the **Time zone** drop-down menu, select a location or time zone and click **Save**.
3. Configure the time synchronization settings.
   a) In the Time synchronization pane, click **Edit**.
   b) From the **Mode** drop-down menu, configure the time synchronization method.

Option	Description
Disabled	No time synchronization. Uses the system time zone settings.
Host	Enables VMware Tools time synchronization. Uses VMware Tools to synchronize the time of the appliance with the time of the ESX host.
NTP	Enables NTP synchronization. You must enter the IP address or FQDN of one or more NTP servers.

   c) Click **Save**.

Start, Stop, and Restart Services

You can use the vCenter Appliance Management Interface to view the status of vCenter components and to start, stop, and restart services.

Log in to the vCenter Appliance Management Interface as root.

You can use the `vimtop` utility plug-in to monitor vSphere services that run in the vCenter instance. See Using the vimtop Plug-In to Monitor the Resource Use of Services.

1. In the vCenter Appliance Management Interface, click **Services**.
   The Services page displays a table of the installed services that can be sorted by name, startup type, health, and state.
2. Select a service and click **SET STARTUP TYPE** to configure either Manual or Automatic start of a service.
3. Select a service and click **START** to start a service.
4. Select a service and click **STOP** to stop, or **RESTART** to restart a service, then click **OK**.

   Warning:
   Stopping or Restarting some services may lead to functionality becoming temporarily unavailable.

Configure Update Settings

You can use the vCenter Appliance Management Interface to configure your update settings and check for new updates.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Update**.
2. To configure your update settings, click **Settings**.
   a) To check for updates automatically, select the checkbox.
   b) Select to use a default or custom repository.
   c) If you selected a custom repository, enter the repository URL, username (optional), and password (optional). Click **Save**.
   For the URL, HTTPS and FTPS protocols are supported.
3. To manually check for updates, click the **Check Updates** drop-down menu.
   a) Select to check a **CD-ROM** or a **CD-ROM + URL** for updates.

The Available Updates table displays available updates sortable by version, type, release date, reboot requirement, and severity.

Change the Password and Password Expiration Settings of the Root User

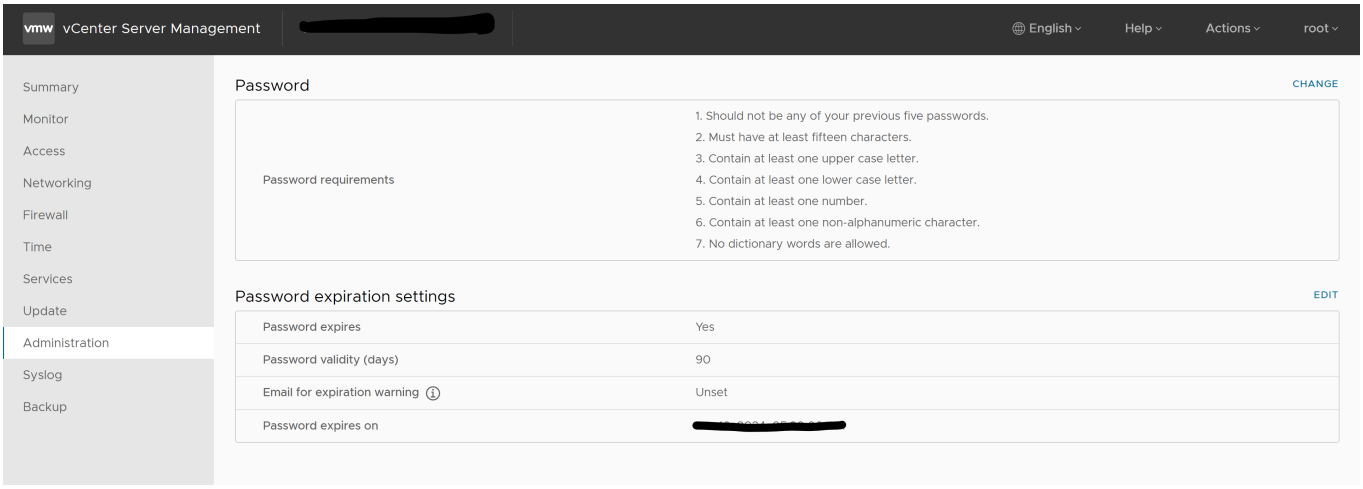
When you deploy vCenter, you set the initial password of the root user, which expires after 90 days by default. You can change the root password and the password expiration settings from the vCenter Appliance Management Interface.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Administration**.
2. In the Password section, click **Change**.
3. Enter the current password and the new password, then click **Save**.
4. Configure the password expiration settings for the root user.
   a) In the Password expiration settings section, click **Edit** and select the password expiration policy.

Option	Description
Yes	The password of the root user expires after a specified number of days. You must provide the following information: • **Root password validity (days)** The number of days after which the password expires. • **Email for expiration warning** The email address to which vCenter sends a warning message before the expiration date.
No	The password of the root user never expires.

   b) In the Password expiration settings pane, click **Save** to apply the new password expiration settings.



The Password expiration settings section displays the new expiration date.

Forward vCenter Log Files to Remote Syslog Server

You can forward the vCenter log files to a remote syslog server to conduct an analysis of your logs.

Log in to the vCenter Appliance Management Interface as root.

Note:

ESX can be configured to send log files to a vCenter rather than storing them to a local disk. The recommended maximum numbers of supported hosts to collect logs from is 30. See Broadcom knowledge base article 318939 for information on how to configure ESX log forwarding. This feature is intended for smaller environments with stateless ESX hosts. For all other cases, use a dedicated log server. Using vCenter to receive ESX log files might impact vCenter performance.

1. In the vCenter Appliance Management Interface, select **Syslog**.
2. In the Forwarding Configuration section, click **Configure** if you have not configured any remote syslog hosts. Click **Edit** if you already have configured hosts.
3. In the Create Forwarding Configuration pane, enter the server address of the destination host. The maximum number of supported destination hosts is three.
4. From the **Protocol** drop-down menu, select the protocol to use.

Menu Item	Description
TLS	Transport Layer Security
TCP	Transmission Control Protocol
RELP	Reliable Event Logging Protocol
UDP	User Datagram Protocol

5. In the **Port** text box, enter the port number to use for communication with the destination host.
6. In the Create Forwarding Configuration pane, click **Add** to enter another remote syslog server.
7. Click **Save**.
8. Verify that the remote syslog server is receiving messages.
9. In the Forwarding Configuration section, click **Send Test Message.**
10. Verify on the remote syslog server that the test message was received.
    The new configuration settings are shown in the Forwarding Configuration section.
11. Optional: To view the logs that are exported, log in to the vCenter Appliance Management Interface as root user and view the list of log files in `/etc/vmware-syslog/` directory.

Configure Log Insight Agent to Send Custom vCenter Logs to Log Management

You can configure the Log insight Agent service to collect and send custom vCenter logs to log management.

In vCenter 9.1, the Log insight Agent service is added as part of the vCenter appliance and is deactivated by default. When you configure the Log Insight Agent service, the service is started automatically.

If you installed and configured the Log Insight Agent on your vCenter system before upgrading to version 9.1, your configuration settings are not preserved and you must reconfigure the service.

1. On the vSphere Client home page, click **Developer Center** and select the **API Explorer** tab.
2. From the **Select Endpoint** drop-down menu, select a vCenter instance from your environment.
3. From the **Select API** drop-down menu, select `appliance`.
4. Locate the **logging/liagent/log_collection** API category and expand the `PUT` method.
5. Under **Try it out**, in the **request_body**, enter the following information:

```
{
    "configurations": [ {
        "enabled": true,
        "certificates": [ "-----BEGIN CERTIFICATE-----\n...\n-----END CERTIFICATE-----" ],
        "content": "<raw_string_li_agent_configuration>",
        "host": "<li_agent_server_host_name>",
        "loglevel": "<desired_log_level>",
        "port": 9543,
        "protocol": "cfapi",
        "unitid": "",
        "unittype": "VCENTER"
    } ]
}
```

The `ApplianceLoggingLiagentLogCollectionSet` is the data structure that is used by Log Insight Agent service to configure the collection of logs from a vCenter system or ESX hosts and send them to log management.

When building your request payload, populate the following fields:

Field	Data types	Description
`enabled`	boolean	**Required**. Specifies whether log collection is enabled for the component that is specified with `unittype` field. Possible values are `true` and `false`.
`unitid`	String	**Required**. A unique identifier for the log collection unit, typically aligned with the logical or infrastructure identity of the component. Allows component granularity in complex environments, for example, the hostname or FQDN.
`unittype`	String	**Required**. Defines the possible unit types for vCenter server. Possible values are `VCENTER` and `ESX`.
`content`	String	**Required**. Specify the configuration of the log collector agent as a raw string by using the INI format. For example, `"\n[server]\nhostname=<fqdn_vcf_ops_for_logs>\nproto=cfapi \nport=9543\n[update]\nauto_update=0\n[filelog|vsphere8_0-linux-applmgmt]\ndirectory=/var/log/vmware/applmgmt \ninclude=\"*.log*\";\"\"\nexclude=\"*.gz;*.stderr;*.stdout \"\ntags={\"vmw_product\": \"vcenter\"}"`
`host`	String	Specify the host name of the log management server.
`port`	integer	Enter the port on the log management server to which you want to send log traffic. The value of this field is used with the `host` field to fully identify the destination endpoint for the collected logs.
`protocol`	String	Define the protocol that is used to forward logs to the log managementserver. Currently, the supported values are `cfapi` and `syslog`.
`certificates`	Array<String>	Provide a list of PEM-encoded (Privacy-Enhanced Mail) certificates where each string represents a single certificate. These certificates are used to establish a secure connection between the component, for example, vCenter, and the log management server.
`loglevel`	String	**Optional**. Enter the desired overall log verbosity for the component. Supported log levels are: `info`, `debug`, `verbose`, `warning`, and `error`. If unset, the `loglevel` value is taken from the `content` payload. For the `VCENTER`component this field is not applicable.

6. Click **Execute** to configure the Log insight Agent.

Your configuration is preserved when you update or upgrade your vCenter system.

You must configure log collection in log management for your vCenter. See the *Configuring and Analyzing Logs* chapter in the *Infrastructure Operations* section of the *VMware Cloud Foundation* documentation.

Configure and Schedule Backups

You can use the vCenter Appliance Management Interface to set a backup location, create a backup schedule, and monitor backup activity.

Log in to the vCenter Appliance Management Interface as root.

1. In the vCenter Appliance Management Interface, click **Backup**.
2. To create a backup schedule, click **Configure**. To edit an existing backup schedule, click **Edit**.
   a) In the **Backup Schedule** pane, enter the backup location using the format `protocol://server-address<:port-number>/folder/subfolder`.
   Supported protocols for backup are FTPS, HTTPS, SFTP, FTP, NFS, SMB, and HTTP.

b) Enter the username and password for the backup server.
c) Enter the time and frequency for the backup to occur.
d) (Optional) Enter an encryption password for the backup.
e) Indicate the number of backups to retain.
f) Indicate the types of data to be backed up.

3. To initiate a manual backup, click **Backup Now**.

Information for scheduled and manual backups is displayed in the **Activity** table.

Using the vSphere Client to Configure vCenter

You can perform some configuration operations from the vSphere Client such as configuring vCenter Identity Provider Federation, networking, and other settings.

Configuring vCenter

You can configure vCenter from the vSphere Client and from the vCenter Appliance Management Interface.

What you can do depends on your deployment.

You can change many of the vCenter settings, including statistics collection, logging, and more.

Configuring Statistics Settings

To set up how statistical data is recorded, you configure collection intervals for statistics. You can access the stored statistical information through command-line monitoring utilities or by viewing performance charts in the vSphere Client.

Configure Statistics Collection Intervals in the vSphere Client

Statistic collection intervals determine the frequency at which statistic queries occur, the length of time statistical data is stored in the database, and the type of statistical data that is collected. You can view the collected statistics through the performance charts in the vSphere Client or through command-line monitoring utilities.

Required privilege: **Performance** > **ModifyIntervals**

> **Note:**
> Not all interval attributes are configurable.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. To enable or disable a statistics interval, check the box for that interval.
6. To change a statistics interval attribute value, select a value from the drop-down menu.
   a) In **Interval Duration**, select the time interval in which statistics data is collected.
   b) In **Save For**, select for how long the archived statistics are kept in the database.
   c) In **Statistics Level**, select a new level for collecting statistics.

      The lower the level is, the fewer number of statistic counters are used. Level 4 uses all statistics counters. Use it only for debugging purposes.

      The statistics level must be less than or equal to the statistics level that is set for the preceding statistics interval. This requirement is a vCenter dependency.
7. Optional: In Database Size, estimate the effect of the statistics settings on the database.
   a) Enter the number of **Physical Hosts**.
   b) Enter the number of **Virtual Machines**.

      The estimated space required and number of database rows required are calculated and displayed.

c) If necessary, make changes to your statistics collection settings.

8. Click **Save**.

Relationships Between the Default Settings for Statistics Intervals

- Samples that are collected every 5 minutes are stored for 1 day.
- Samples that are collected every 30 minutes are stored for 1 week.
- Samples that are collected every 2 hours are stored for 1 month.
- Samples that are collected on 1 day are stored for 1 year.

For all statistics intervals, the default level is 1. It uses the Cluster Services, CPU, Disk, Memory, Network, System, and Virtual Machine Operations counters.

Configure Runtime Settings for vCenter

You can change the vCenter ID, managed address, and name. You might need to make changes if you run multiple vCenter systems in the same environment.

Required privilege: **Global** > **Settings**

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. In the Edit vCenter Settings dialog box, select **Runtime Settings**.
6. In **vCenter unique ID**, enter a unique ID.

   You can change this value to a number from 0 through 63 to identify each vCenter system running in a common environment. By default, an ID value is generated randomly.
7. In **vCenter managed address**, enter the vCenter system address.

   The address can be IPv4, IPv6, a fully qualified domain name, an IP address, or another address format.
8. In **vCenter name**, enter the name of the vCenter system.

   If you change the DNS name of the vCenter, you can use this text box to modify the vCenter name to match.
9. Click **Save**.

If you made changes to the vCenter system unique ID, you must restart the vCenter system for these changes to take effect.

Configure User Directory Settings

You can configure some of the ways vCenter interacts with the user directory server that is configured as an identity source.

Required privilege: **Global** > **Settings**

These settings apply to vCenter Single Sign-On identity sources.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. In the Edit vCenter general settings window, select **User directory**.
6. In **User directory timeout**, type the timeout interval in seconds for connecting to the directory server.
7. Enable the **Query Limit** box to set a query limit size.
8. In **Query Limit Size**, enter the number of users and groups for which you can associate permissions on the child inventory objects of the vCenter system.

   Note:
   You can associate permissions with users and groups from the Add Permissions dialog box that displays when you click **Add permissions** in **Manage** > **Permissions** for a vSphere inventory object.
9. Click **SAVE**.

Configure Mail Sender Settings

You must configure the email address of the sender account to use vCenter operations, such as sending email notifications as alarm actions. You can send email alerts and alarms using either anonymous or authentication mode.

Required privilege: **Global** > **Settings**

SMTP authentication is available for:

- vSphere 7.0 Update 1 and later versions only.
- Office 365 mailbox users only.
- SMTP mail sender should meet the basic requirements for SMTP Authentication as mentioned in Microsoft's document Requirement for SMTP AUTH client submission.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. Select **Mail** and enter the settings that vCenter uses to send email alerts.
6. In **Mail server** text box, enter the SMTP server information.

   The SMTP server is the DNS name or IP address of the SMTP gateway to use for sending email messages.

   - For sending mails anonymously, you can enter any SMTP server information as the mail server name.
   - For SMTP authentication, enter the mail server name as `smtp.office365.com`, unless you have some customized configuration.

     Note:
     Do not use IP address as mail server as IP address is not supported for SMTP authentication.
7. In **Mail sender** text box, enter the sender account information.

   The sender account is the email address of the sender.

   For SMTP authentication, you must enter a valid SMTP account name in the **Mail sender** text box.

   Note:
   You must enter the full email address, including the domain name.

   For example, *`mail_server@example.com`*

8. Click **SAVE**.
9. This step is applicable only for SMTP authentication.
   You must configure the SMTP user settings as below:
   1. Select the **Configure** tab.
   2. Select **Advanced Settings**.
   3. Click **EDIT SETTINGS** and enter the following values for the configuration parameters:

Name	Value
`mail.smtp.username`	Valid SMTP account name.
`mail.smtp.password`	Valid SMTP account password. **Note:** Currently, the account password is not masked and is visible. You must use a dedicated SMTP mail user until the masking is available in upcoming releases.
`mail.smtp.port`	587

   4. Click **SAVE**.

You can perform the following steps to test the mail settings:

1. Create an alarm that is triggered by a user action.
   For example, the user action can be powering off a virtual machine.
2. Verify that you receive an email, when the alarm is triggered.

Configure SNMP Settings

You can configure up to four receivers to receive SNMP traps from vCenter. For each receiver, specify a host name, port, and community.

Required privilege: **Global** > **Settings**

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. Select **SNMP receivers**.
6. Select the **Enable receiver 1** box.
7. In **Primary Receiver URL**, enter the host name or IP address of the SNMP receiver.
8. In **Receiver port**, enter the port number of the receiver.
   The port number must be a value between 1 and 65535.

9. In **Community string**, enter the community identifier.
10. To send alarms to multiple receivers, select the additional **Enable receiver** boxes and enter the host name, port number, and community identifier for those receivers.
11. Click **Save**.

View Port Settings

You can view the ports used by the Web service to communicate with other applications. You cannot configure these port settings.

The Web service is installed as part of the vCenter installation. The Web service is a required component for third-party applications that use the VMware SDK application programming interface (API). For information about installing the Web service, see the Introduction to vSphere Deployment and Setupin the *vCenter Deployment and Setup* documentation.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. Select **Ports**.
   The ports used by the Web service are displayed.
6. Click **Save**.

Configure the Timeout Settings for the vSphere Client

By default, the vSphere Client sessions close after 120 minutes of inactivity. You can adjust this timeout setting within the vSphere Client.

Required privilege: **Global** > **Settings**

1. In the vSphere Client Home menu, select **Administration**.
2. Under **Deployment**, click **Client Configuration**.
3. On the **Client Configuration** page, click **Edit**.

   The **Edit client configuration** dialog opens.
4. Enter a new value in minutes in the **Session timeout** text field.
5. Click **Save**.
6. Re-login the vSphere Client for the changes to take effect.

Configure Logging Options

You can configure the amount of detail that vCenter collects in log files.

Required privilege: **Global** > **Settings**

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. Select **Logging settings**.
6. Select the logging options.

Option	Description
None	Turns off logging

Option	Description
Error	Displays only error log entries
Warning	Displays warning and error log entries
Info	Displays information, error, and warning log entries
Verbose	Displays information, error, warning, and verbose log entries
Trivia	Displays information, error, warning, verbose, and trivia log entries

7. Click **SAVE**.

Changes to the logging settings take effect immediately. You do not need to restart the vCenter system.

Configure Advanced Settings

In **Advanced Settings**, you can modify the vCenter configuration file, `vpxd.cfg`.

Required privilege: **Global** > **Settings**

You can use **Advanced Settings** to add entries to the `vpxd.cfg` file, but not to edit or delete them. Broadcom recommends that you change these settings only when instructed to do so by Broadcom Support or when you are following specific instructions in the documentation.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Select **Advanced Settings**.
4. Click **Edit Settings**.
5. In the **Name** field, type a name for the setting. Names must start with 'config.' For example: `config.log`.
6. In the **Value** field, type the value for the specified setting.
7. Click **Add**.
8. Click **Save**.

Newly added advanced settings have `config.` appended to the setting keys in the `vpxd.cfg` file. For example:

```
config.example.setting = exampleValue
```

Many advanced settings changes require you to restart the vCenter system. Consult VMware technical support to determine if your changes require a restart.

Send a Message to Other Logged In Users

Administrators can send messages to users who are currently logged in to a vCenter system. The message might announce maintenance or ask users to log out temporarily.

1. In the vSphere Client, navigate to the vCenter instance.
2. Click **Configure**.
3. Select **Settings** > **Message of the Day** and click **Edit**.
4. Enter a message and click **OK**.

The message appears at the top of the vSphere Client in each active user session.

Add a User to the SystemConfiguration.BashShellAdministrators Group

To enable access to the appliance Bash shell by using the vSphere Client, the user you use to log in must be a member of the SystemConfiguration.BashShellAdministrators group. By default, this group is empty and you must add a user to the group manually.

Verify that the user you use to log in to the vCenter instance is a member of the SystemConfiguration.Administrators group in the vCenter Single Sign-On domain.

1. Use the vSphere Client to log in as administrator@*your_domain_name* to the vCenter instance.
   The address is of the type http://*appliance-IP-address-or-FQDN*/ui.
2. From the vSphere Client menu, select **Administration**.
3. Select **Single Sign On** > **Users and Groups**.
4. Click on the **Groups** tab, select **SystemConfiguration.BashShellAdministrators** from the options available in the Group Name column.
5. Click **EDIT**.
6. In the **Edit Group** window, to add members, select the domain from the dropdown menu and then search for the required users.
7. Click **SAVE**.

Reboot a Node

In the vSphere Client, you can reboot a node in vCenter.

Verify that the user you use to log in to the vCenter instance is a member of the SystemConfiguration.Administrators group in the vCenter Single Sign-On domain.

1. Use the vSphere Client to log in as administrator@*your_domain_name* to the vCenter instance.
2. On the vSphere Client main page, click **Administration** > **Deployment** > **System Configuration**.
3. Under System Configuration, select a node from the list.
   You can view the health status of a node. See View the Health Status of Nodes.
4. Click **REBOOT NODE**.

Export a Support Bundle

You can export a support bundle containing the log files for a specific product included in vCenter.

Verify that the user who logs in to the vCenter instance is a member of the `SystemConfiguration.Administrators` group in vCenter Single Sign-On.

1. Use the vSphere Client to log in as administrator@*your_domain_name* to the vCenter instance.
   The address is of the type http://*appliance-IP-address-or-FQDN*/ui.
2. On the vSphere Client home page, click **Administration** > **Deployment** > **System Configuration**.
3. Select a node from the list and click **Export Support Bundle**.
4. In the **Export Support Bundle** window, expand the trees to view the services running in the appliance and deselect the services for which you do not want to export log files.
   All the services are selected by default. If you want to export the support bundle and send it to Broadcom Support, leave all check boxes selected. The services are separated in two categories: a Cloud infrastructure category, which contains the services of specific products in the appliance, and a Virtual appliance category, which contains the services specific for the appliance and the vCenter product.
5. Click **Export Support Bundle** and save the bundle on your local machine.

You saved the support bundle to your machine and can explore it.

Using the Appliance Shell to Configure vCenter

You can access all of the vCenter API commands and plug-ins that you can use for monitoring, troubleshooting, and configuring the appliance by using the appliance shell.

You can run all commands in the appliance shell with or without the `pi` keyword.

Access the Appliance Shell

To access the plug-ins included in the appliance shell and to be able to see and use the API commands, first access the appliance shell.

1. Access the appliance shell.
   - If you have direct access to the appliance console, press Alt+F1.
   - If you want to connect remotely, use SSH or another remote console connection to start a session to the appliance.
2. Enter a user name and password recognized by the appliance.

You are logged in to the appliance shell and can see the welcome message.

Enable and Access the Bash Shell from the Appliance Shell

If you log in to the appliance shell as a user who has a super administrator role, you can enable access to the Bash shell of the appliance for other users. The root user has access to the appliance Bash shell by default.

The appliance Bash shell is enabled by default for the root

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. If you want to enable the Bash shell access for other users, run the following command.
   ```
   shell.set --enabled true
   ```
3. To access the Bash shell run `shell` or `pi shell`.

Keyboard Shortcuts for Editing Commands

You can use various keyboard shortcuts to enter and edit commands in the appliance Bash shell.

Table 66: Keyboard Shortcuts and Function

Keyboard Shortcut	Details
Tab	Completes the current command. If you enter a part of the command name and press the Tab key, the system completes the command name. To view the commands that match a set of characters that you enter, type a character and press the Tab key.
Enter (at the command line)	Runs the command that you entered.
Enter (at the **More** prompt)	Displays the next page of output.
Delete or Backspace	Deletes the character that is on the left of the cursor.
Left arrow or Ctrl+B	Moves the cursor one character to the left. When you enter a command that extends beyond a single line, you can press the Left Arrow or Ctrl-B keys to go back to the beginning of the command.
Right arrow or Ctrl+F	Moves the cursor one character to the right.
Esc, B	Moves the cursor one word back.
Esc, F	Moves the cursor one word forward.
Ctrl+A	Moves the cursor to the beginning of the command line.
Ctrl+E	Moves the cursor to the end of the command line.
Ctrl+D	Deletes the character selected by the cursor.
Ctrl+W	Deletes the word next to the cursor.
Ctrl+K	Deletes the line forward. When you press Ctrl+K, everything that you entered starting from the cursor location to the end of the command line is deleted.

Keyboard Shortcut	Details
Ctrl+U or Ctrl+X	Deletes the line backward. When you press Ctrl+U, everything from the beginning of the command line to the cursor location is deleted.
Ctrl+T	Changes the places of the character to the left of the cursor with the character selected by the cursor.
Ctrl+R or Ctrl+L	Displays the system prompt and command line.
Ctrl+V or Esc, Q	Inserts a code to indicate to the system that the following keystroke must be treated as a command entry, not as an editing key.
Up arrow, or Ctrl+P	Recalls commands in the history buffer, beginning with the most recent command.
Down arrow or Ctrl+N	Returns to more recent commands in the history buffer after you use the Up arrow or Ctrl+P to recall commands.
Ctrl+Y	Recalls the most recent entry in the delete buffer. The delete buffer contains the last 10 items you have cut or deleted.
Esc, Y	Recalls the next entry in the delete buffer. The delete buffer contains the last 10 items you have cut or deleted. Press Ctrl+Y first to recall the most recent entry, and then press Esc, Y up to nine times to recall the remaining entries in the buffer.
Esc, C	Capitalizes the character selected by the cursor.
Esc, U	Changes the casing for all characters in the word selected by the cursor, up to the next space, to uppercase.
Esc, L	Changes the capitalized letters in a word from the character selected by the cursor to the end of the word to lowercase.

Get Help on APIs and Utilities

You can access the vCenter plug-ins and API commands from the appliance shell. You can use the plug-ins and commands for monitoring, troubleshooting, and configuring the appliance.

You can use the Tab key to auto complete API commands, plug-in names, and API parameters. Plug-in parameters do not support auto completion.

1. Access the appliance shell and log in.
2. To get help about the plug-ins, run the `help pi list` or the `? pi list` command.
   You receive a list with all the plug-ins in the appliance.
3. To get help about the API commands, run the `help api list` or the `? api list` command.
   You receive a list with all the API commands in the appliance.
4. To get help about a particular API command, run the `help api` *`api_name`* or the `? api` *`api_name`* command. For example, to receive help about the `com.vmware.appliance.version1.timesync.set` command, run `help api timesync.set` or `? api timesync.set`.

Plug-Ins in vCenter Shell

The plug-ins in vCenter provide you with access to various administrative tools. The plug-ins reside in the CLI itself. The plug-ins are standalone Linux or VMware utilities, which do not depend on any VMware service.

Table 67: Plug-Ins Available in the vCenter

Plug-In	Description
`com.vmware.clear`	A plug-in that you can use to clear the terminal screen.
`com.vmware.cmsso-util`	A plug-in that you use for orchestrating changes to PNID, Machine Certificate, unregistering a node from Component Manager, vCenter Single Sign-On, and reconfiguring vCenter.

Plug-In	Description
com.vmware.dcli	vAPI based CLI client.
com.vmware.nslookup	A plug-in that you can use to query the Domain Name System (DNS) to obtain domain name or IP address mapping or for any other specific DNS record.
com.vmware.pgrep	A plug-in that you can use to search for all named processes.
com.vmware.pgtop	A plug-in that you can use to monitor the PostgreSQL database.
com.vmware.ping	A plug-in that you can use to ping a remote host. Accepts the same arguments as bin/ping.
com.vmware.ping6	A plug-in that you can use to ping a remote host. Accepts the same arguments as bin/ping6.
com.vmware.portaccess	A plug-in that you can use to troubleshoot the port access of a host.
com.vmware.ps	A plug-in that you can use to see statistics on running processes.
com.vmware.rvc	Ruby vSphere Console
com.vmware.service-control	A plug-in that you can use to manage VMware services.
com.vmware.shell	A plug-in that allows access to the appliance Bash shell.
com.vmware.showlog	A plug-in that you can use to browse the log files.
com.vmware.shutdown	A plug-in that you can use to restart or power off the appliance.
com.vmware.software-packages	A plug-in that you can use to update the software packages in the appliance.
com.vmware.support-bundle	A plug-in that you can use to create a bundle on the local file system and export it to a remote Linux system. If you use the plug-in with the `stream` command, the support bundle is not created on the local file system, but is directly exported to the remote Linux system.
com.vmware.top	A plug-in that displays process information. Accepts the same arguments as /usr/bin/top/.
com.vmware.tracepath	A plug-in that traces the path to a network host. Accepts the same arguments as /sbin/tracepath.
com.vmware.tracepath6	A plug-in that traces the path to a network host. Accepts the same arguments as /sbin/tracepath6.
com.vmware.updatemgr-util	A plug-in that you can use to configure VMware Update Manager.
com.vmware.vcenter-restore	A plug-in that you can use to restore vCenter.
com.vmware.vimtop	A plug-in that you can use to view a list of vSphere services and their resource use.

Browse Log Files Using showlog Plug-In

You can browse the log files in vCenter to examine them for errors.

1. Access the appliance shell and log in.
2. Type the `showlog` command, add a space, and press the Tab key to view all the contents of the `/var/log` folder.
3. Run the command for viewing the firstboot log files.

```
showlog /var/log/firstboot/cloudvm.log
```

API Commands in Appliance Shell

The API commands in vCenter let you perform various administrative tasks. The API commands are provided by appliance management service. You can edit time synchronization settings, monitor processes and services, set up the SNMP settings, and so on.

Table 68: API Commands Available in vCenter

API Command	Description
com.vmware.appliance.health.applmgmt.get	Get the health of the applmgmt service.
com.vmware.appliance.health.databasestorage.get	Get the health of the database storage.
com.vmware.appliance.health.load.get	Get the CPU load health.
com.vmware.appliance.health.mem.get	Get the memory health.
com.vmware.appliance.health.softwarepackages.get	Get the health of the system update.
com.vmware.appliance.health.storage.get	Get the overall storage health.
com.vmware.appliance.health.swap.get	Get the swap health.
com.vmware.appliance.health.system.get	Get the system health.
com.vmware.appliance.health.system.lastcheck	Get the time of the last check of the health status.
com.vmware.appliance.monitoring.list	Get a list of monitored items.
com.vmware.appliance.monitoring.get	Get monitored item information.
com.vmware.appliance.monitoring.query	Query a range of values for the monitored items.
com.vmware.appliance.recovery.backup.job.cancel	Cancel a backup job by id.
com.vmware.appliance.recovery.backup.job.create	Start a backup job.
com.vmware.appliance.recovery.backup.job.get	Get a status of the backup job by id.
com.vmware.appliance.recovery.backup.job.list	Get a list of backup jobs.
com.vmware.appliance.recovery.backup.parts.list	Get a list of the vCenter components that can be included in a backup job.
com.vmware.appliance.recovery.backup.parts.get	Get detailed info for a backup part.
com.vmware.appliance.recovery.backup.validate	Validate parameters for a backup job without starting the job.
com.vmware.appliance.recovery.restore.job.cancel	Cancel a restore job.
com.vmware.appliance.recovery.restore.job.create	Start a restore job.
com.vmware.appliance.recovery.restore.job.get	Get status of the restore job.
com.vmware.appliance.recovery.restore.validate	Validate restore parameters of a restore job without starting the job.
com.vmware.appliance.system.uptime.get	Gets the system uptime.
com.vmware.appliance.version1.access.consolecli.get	Get information about the state of the console-based controlled CLI (TTY1).
com.vmware.appliance.version1.access.consolecli.set	Set enabled state of console-based controlled CLI (TTY1).
com.vmware.appliance.version1.access.dcui.get	Get information about the state of the Direct Console User Interface (DCUI TTY2).
com.vmware.appliance.version1.access.dcui.set	Set enabled state of the Direct Console User Interface (DCUI TTY2).
com.vmware.appliance.version1.access.shell.get	Get information about the state of Bash shell, that is, access to Bash shell from within the controlled CLI.

API Command	Description
com.vmware.appliance.version1.access.shell.set	Set enabled state of Bash shell, that is, access to Bash shell from within the controlled CLI.
com.vmware.appliance.version1.access.ssh.get	Get enabled state of the SSH-based controlled CLI.
com.vmware.appliance.version1.access.ssh.set	Set enabled state of the SSH-based controlled CLI.
com.vmware.appliance.version1.localaccounts.user.add	Create a new local user account.
com.vmware.appliance.version1.localaccounts.user.delete	Delete a local user account.
com.vmware.appliance.version1.localaccounts.user.get	Get the local user account information.
com.vmware.appliance.version1.localaccounts.user.list	List local user accounts.
com.vmware.appliance.version1.localaccounts.user.password.update	Update the password of a logged in user or of the user that you specify in the username parameter.
com.vmware.appliance.version1.localaccounts.user.set	Update local user account properties, such as role, full name, enabled status, and password.
com.vmware.appliance.version1.monitoring.snmp.disable	Stop an enabled SNMP agent.
com.vmware.appliance.version1.monitoring.snmp.enable	Start a disabled SNMP agent.
com.vmware.appliance.version1.monitoring.snmp.get	Return an SNMP agent configuration.
com.vmware.appliance.version1.monitoring.snmp.hash	Generate localized keys for secure SNMPv3 communications.
com.vmware.appliance.version1.monitoring.snmp.limits	Get SNMP limits information.
com.vmware.appliance.version1.monitoring.snmp.reset	Restore settings to factory defaults.
com.vmware.appliance.version1.monitoring.snmp.set	Set SNMP configuration.
com.vmware.appliance.version1.monitoring.snmp.stats	Generate diagnostics report for SNMP agent.
com.vmware.appliance.version1.networking.dns.domains.add	Add domains to DNS search domains.
com.vmware.appliance.version1.networking.dns.domains.list	Get a list of DNS search domains.
com.vmware.appliance.version1.networking.dns.domains.set	Set DNS search domains.
com.vmware.appliance.version1.networking.dns.hostname.get	Get the Fully Qualified Domain Name.
com.vmware.appliance.version1.networking.dns.hostname.set	Set the Fully Qualified Domain Name.
com.vmware.appliance.version1.networking.dns.servers.add	Add a DNS server. This method fails if you use DHCP.
com.vmware.appliance.version1.networking.dns.servers.get	Get DNS server configuration.
com.vmware.appliance.version1.networking.dns.servers.set	Set the DNS server configuration. If the host is configured to acquire DNS servers and host name by using DHCP, a DHCP refresh is forced.
com.vmware.appliance.version1.networking.firewall.addr.inbound.add	Add a firewall rule to allow or deny access from an incoming IP address.
com.vmware.appliance.version1.networking.firewall.addr.inbound.delete	Delete a specific rule at a given position or delete all rules.
com.vmware.appliance.version1.networking.firewall.addr.inbound.list	Get an ordered list of inbound IP addresses that are allowed or denied by a firewall rule.

API Command	Description
com.vmware.appliance.version1.networking.interfaces.get	Get information about a particular network interface.
com.vmware.appliance.version1.networking.interfaces.list	Get a list of available network interfaces, including interfaces that are not yet configured.
com.vmware.appliance.version1.networking.ipv4.get	Get IPv4 network configuration for interfaces.
com.vmware.appliance.version1.networking.ipv4.list	Get IPv4 network configuration for all configured interfaces.
com.vmware.appliance.version1.networking.ipv4.renew	Renew IPv4 network configuration on interfaces. If the interface is configured to use DHCP for IP address assignment, the lease of the interface is renewed.
com.vmware.appliance.version1.networking.ipv4.set	Set IPv4 network configuration for an interface.
com.vmware.appliance.version1.networking.ipv6.get	Get IPv6 network configuration for interfaces.
com.vmware.appliance.version1.networking.ipv6.list	Get IPv6 network configuration for all configured interfaces.
com.vmware.appliance.version1.networking.ipv6.set	Set IPv6 network configuration for an interface.
com.vmware.appliance.version1.networking.routes.add	Add static routing rules. A destination/prefix of the type 0.0.0.0/0 (for IPv4) or ::/0 (for IPv6) refers to the default gateway.
com.vmware.appliance.version1.networking.routes.delete	Delete static routing rules.
com.vmware.appliance.version1.networking.routes.list	Get routing table. A destination/prefix of the type 0.0.0.0/0 (for IPv4) or ::/0 (for IPv6) refers to the default gateway.
com.vmware.appliance.ntp.get	Get NTP configuration settings. If you run the `tymesync.get` command, you can retrieve the current time synchronization method (by using NTP or VMware Tools). The `ntp.get` command always returns the NTP server information, even when the time synchronization method is not set to NTP. If time synchronization method is not set by using NTP, the NTP status is displayed as down.
com.vmware.appliance.version1.ntp.server.add	Add NTP servers. This command adds NTP servers to the configuration. If the time synchronization is NTP-based, then NTP daemon is restarted to reload the new NTP servers. Otherwise, this command just adds servers to the NTP configuration.

API Command	Description
com.vmware.appliance.version1.ntp.server.delete	Delete NTP servers. This command deletes NTP servers from the configuration. If the time synchronization mode is NTP-based, the NTP daemon is restarted to reload the new NTP configuration. Otherwise, this command just deletes servers from the NTP configuration.
com.vmware.appliance.version1.ntp.server.set	Set NTP servers. This command deletes old NTP servers from the configuration and sets the input NTP servers in the configuration. If the time synchronization is set by using NTP, the NTP daemon is restarted to reload the new NTP configuration. Otherwise, this command just replaces the servers in NTP configuration with the NTP servers that you provide as input.
com.vmware.appliance.version1.resources.cpu.stats.get	Get CPU statistics.
com.vmware.appliance.version1.resources.load.health.get	Get load health.
com.vmware.appliance.version1.resources.load.stats.get	Get load averages (over 1, 5, and 15-minute intervals).
com.vmware.appliance.version1.resources.mem.health.get	Get memory health.
com.vmware.appliance.version1.resources.mem.stats.get	Get memory statistics.
com.vmware.appliance.version1.resources.net.stats.get	Get network statistics.
com.vmware.appliance.version1.resources.net.stats.list	Get network statistics for all interfaces that are up and running.
com.vmware.appliance.version1.resources.processes.stats.list	Get statistics on all processes.
com.vmware.appliance.version1.resources.softwarepackages.health.get	Get the health of the update component.
com.vmware.appliance.version1.resources.storage.health.get	Get storage health statistics.
com.vmware.appliance.version1.resources.storage.stats.list	Get storage statistics for each logical disk.
com.vmware.appliance.version1.resources.swap.health.get	Get swap health.
com.vmware.appliance.version1.resources.swap.stats.get	Get swap statistics.
com.vmware.appliance.version1.resources.system.health.get	Get the overall health of the system.
com.vmware.appliance.version1.resources.system.stats.get	Get the system status.
com.vmware.appliance.version1.services.list	Get the list of all known services.
com.vmware.appliance.version1.services.restart	Restart a service.
com.vmware.appliance.version1.services.status.get	Get the status of a service.
com.vmware.appliance.version1.services.stop	Stop a service.
com.vmware.appliance.version1.system.storage.list	Gets disk to partition mapping.
com.vmware.appliance.version1.system.storage.resize	Resizes all partitions to 100 percent of disk size.
com.vmware.appliance.version1.system.time.get	Gets system time.
com.vmware.appliance.version1.system.update.get	Get the URL-based patching configuration.
com.vmware.appliance.version1.system.update.set	Set the URL-based patching configuration.

API Command	Description
com.vmware.appliance.version1.system.version.get	Get the version of the appliance.
com.vmware.appliance.version1.timesync.get	Get the time synchronization configuration.
com.vmware.appliance.version1.timesync.set	Set the time synchronization configuration.

Configuring SNMP for vCenter

vCenter includes an SNMP agent that can send trap notifications and receive GET, GETBULK, and GETNEXT requests.

You can use the appliance shell API commands to enable and configure the vCenter SNMP agent. You configure the agent differently depending on whether you want to use SNMP v1/v2c or SNMP v3.

SNMP v3 informs are not supported. vCenter supports only notifications such as v1 and v2c traps, and v3 traps with all security levels.

Configure the SNMP Agent for Polling

If you configure the vCenter SNMP agent for polling, it can listen for and respond to requests from SNMP management client systems, such as GET, GETNEXT, and GETBULK requests.

By default, the embedded SNMP agent listens on UDP port 161 for polling requests from management systems. You can use the snmp.set --port command to configure an alternative port. To avoid conflicts between the port for the SNMP agent and the ports of other services, use a UDP port that is not defined in /etc/services.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Run the snmp.set --port command to configure the port.
   For example, run the following command:
   ```
   snmp.set --port port
   ```
   Here *port* is the port for the SNMP agent to use for listening for polling requests.

   Important:
   The port you specify must not be already in use by other services. Use IP addresses from the dynamic range, port 49152 and up.
3. Optional: If the SNMP agent is not enabled, enable it by running the snmp.enable command.

Configure vCenter for SNMP v1 and v2c

When you configure the vCenter SNMP agent for SNMP v1 and v2c, the agent supports sending notifications and receiving GET requests.

In SNMP v1 and v2c, community strings are namespaces that contain one or more managed objects. Namespaces can act as a form for authentication, but this does not secure the communication. To secure the communication, use SNMP v3.

Configure SNMP Communities

To enable the vCenter SNMP agent to send and receive SNMP v1 and v2c messages, you must configure at least one community for the agent.

An SNMP community defines a group of devices and management systems. Only devices and management systems that are members of the same community can exchange SNMP messages. A device or management system can be a member of multiple communities.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Run the snmp.set --communities command to configure an SNMP community.
   For example, to configure public, east, and west network operation center communities, run the following command:

```
snmp.set --communities public,eastnoc,westnoc
```

Each time you specify a community with this command, the settings you specify overwrite the previous configuration.

To specify multiple communities, separate the community names with a comma.

Configure SNMP Agent to Send v1 or v2c Notifications

You can use the vCenter SNMP agent to send virtual machine and environmental notifications to management systems.

To send SNMP v1 and v2c notifications with the SNMP agent, you must configure the target, that is the receiver, unicast address, community, and an optional port. If you do not specify a port, the SNMP agent sends notifications to UDP port 162 on the target management system by default.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.

   The default user with super administrator role is root.

2. Run the `snmp.set --targets` command:

   ```
   snmp.set --targets target_address@port/community
   ```

   Here *`target_address`*, *`port`*, and *`community`* are the address of the target system, the port number to send the notifications to, and the community name, respectively. The port value is optional. If you do not specify a port, the default port,161, is used.

   Each time you specify a target with this command, the settings you specify overwrite all previously specified settings. To specify multiple targets, separate them with a comma.

   For example, run the following command for configuring the targets 192.0.2.1@678/targetcommunity and 2001:db8::1/anothercom:

   ```
   snmp.set --targets 192.0.2.1@678/targetcommunity,2001:db8::1/anothercom
   ```

3. Optional: If the SNMP agent is not enabled, enable it by running the `snmp.enable` command.
4. Optional: To send a test trap to verify that the agent is configured correctly, run the `snmp.test` command.

   The agent sends a `warmStart` trap to the configured target.

Configure vCenter for SNMP v3

When you configure the SNMP agent for SNMP v3, the agent supports sending traps. SNMP v3 also provides stronger security than v1 or v2c, including cryptographic authentication and encryption.

SNMP v3 informs are not supported. vCenter supports only notifications such as v1/v2c traps and v3 traps with all security levels.

Configure SNMP Engine ID

Every SNMP v3 agent has an engine ID, which serves as a unique identifier for the agent. The engine ID is used with a hashing function to generate localized keys for authentication and encryption of SNMP v3 messages.

If you do not specify an engine ID before you enable the SNMP agent, when you enable the standalone SNMP agent, an engine ID is generated.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.

   The default user with super administrator role is root.

2. Run the `snmp.set --engineid` command to configure the target.
   For example, run the following command:

   ```
   snmp.set --engineid 80001adc040102030405060708091011
   ```

   where, the value can be broken into hexidecimal digits,

   ```
   80:00:1a:dc:04:24:17:e2:02:b8:61:3f:54:00:00:00:00
   ```

   and decoded as:

   80: -- RFC 3411 encoding applies as high bit '10000000' 0x80 and lower 7 bits 0 are part of enterprise id.

   00:1a:dc -- rest of enterprise id, the value parsed by vmware is, 1a:dc hexidecimal (6876 in decimal).

04: -- type of engineid as per RFC 3411, page 42. Using here '4' to say remaining octets can be tranlated to text.

0102030405060708091011 -- operator specified text encoded in hexidecmal characters.

Configure SNMP Authentication and Privacy Protocols

SNMP v3 optionally supports authentication and privacy protocols.

Authentication is used to ensure the identity of users. Privacy allows for encryption of SNMP v3 messages to ensure confidentiality of data. The privacy protocols provide a higher level of security than is available in SNMP v1 and v2c, which use community strings for security.

Both authentication and privacy are optional. However, you must enable authentication if you plan to enable privacy.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.

   The default user with super administrator role is root.

2. Optional: Run the `snmp.set --authentication` command to configure authentication.
   For example, run the following command:

   ```
   snmp.set --authentication protocol
   ```

   Here, *`protocol`* must be either `none`, for no authentication,`SHA1`, or `MD5`.

3. Optional: Run the `snmp.set --privacy` command to configure privacy protocol.
   For example, run the following command:

   ```
   snmp.set --privacy protocol
   ```

   Here, *`protocol`* must be either `none`, for no privacy, or `AES128`.

Configure SNMP Users

You can configure up to five users who can access SNMP v3 information. User names must be no more than 32 characters long.

- Verify that you have configured the authentication and privacy protocols before configuring users.
- Verify that you know the authentication and privacy passwords for each user that you plan to configure. Passwords must be at least eight characters long. Store these passwords in files on the host system.

While configuring a user, you generate authentication and privacy hash values based on the user's authentication and privacy passwords and on the SNMP agent's engine ID. After configuring users, if you change the engine ID, the authentication protocol, or the privacy protocol, the users are no longer valid and must be reconfigured.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.

   The default user with super administrator role is root.

2. If you are using authentication or privacy, get the authentication and privacy hash values for the user by the running `snmp.hash --auth_hash --priv_hash` command.
   For example, run the following command:

   ```
   snmp.hash  --auth_hashsecret1--priv_hashsecret2
   ```

   Here, *`secret1`* is the path to the file containing the user's authentication password and *`secret2`* is the path to the file containing the user's privacy password. Alternatively, you can specify the flag `--raw_secret` and set the boolean parameter to *`true`*.

   The authentication and privacy hash values are displayed.

3. Configure the user by running `snmp.set --user`.
   For example, run the following command:

   ```
   snmp.set --useruserid/authhash/privhash/security
   ```

   The parameters in the command are as follows.

Parameter	Description
`userid`	Replace with the user ID.
`authhash`	Replace with the authentication hash value.

Parameter	Description
privhash	Replace with the privacy hash value.
security	Replace with the level of security enabled for that user, which can be `auth`, for authentication only, `priv`, for authentication and privacy, or `none`, for no authentication or privacy.

Configure SNMP v3 Targets

Configure SNMP v3 targets to allow the SNMP agent to send SNMP v3 traps.

You can configure a maximum of three SNMP v3 targets, in addition to a maximum of three SNMP v1 or v2c targets.

To configure a target, you must specify a host name or IP address of the system that receives the traps, a user name, a security level, and whether to send traps. The security level can be either `none`, for no security, `auth`, for authentication only, or `priv`, for authentication and privacy.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Run the `snmp.set --v3targets` command to set up the SNMP v3 target.
   For example, run the following command:
   ```
   snmp.set --v3targetshostname@port/userid/secLevel/trap
   ```
   The parameters in the command are as follows.

Parameter	Description
hostname	Replace with the host name or IP address of the management system that receives the traps.
port	Replace with the port on the management system that receives the traps. If you do not specify a port, the default port, 161, is used.
userid	Replace with the user name.
secLevel	Replace with either `none`, `auth`, or `priv` to indicate the level of authentication and privacy you have configured. Use `auth` if you have configured authentication only, `priv` if you have configured both authentication and privacy, and `none` if you have configured neither.

3. Optional: If the SNMP agent is not enabled, enable it by running the `snmp.enable` command.
4. Optional: To send a test trap to verify that the agent is configured correctly, run the `snmp.test` command.
   The agent sends a `warmStart` trap to the configured target.

Configure SNMP Agent to Filter Notifications

You can configure the vCenter SNMP agent to filter out notifications if you do not want your SNMP management software to receive those notifications.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Run the `snmp.set --notraps` command to filter traps.
   - To filter specific traps, run the following command:
     ```
     snmp.set  --notraps oid_list
     ```
     Here, *oid_list* is a list of object IDs for the traps to filter, separated by commas. This list replaces any object IDs that were previously specified using this command.
   - To clear all trap filters, run the following command:
     ```
     snmp.set  --notraps reset
     ```
3. Optional: If the SNMP agent is not enabled, enable it by running the `snmp.enable` command.

The traps identified by the specified object IDs are filtered out of the output of the SNMP agent, and are not sent to SNMP management software.

Configure SNMP Management Client Software

After you have configured vCenter to send traps, you must configure your management client software to receive and interpret those traps.

Download the VMware MIB files. See Broadcom knowledge base article 313538.

To configure your management client software, specify the communities for the managed device, configure the port settings, and load the VMware MIB files. See the documentation for your management system for specific instructions for these steps.

1. In your management software, specify the vCenter instance as an SNMP-based managed device.
2. If you are using SNMP v1 or v2c, set up appropriate community names in the management software.
   These names must correspond to the communities set for the SNMP agent on vCenter.
3. If you are using SNMP v3, configure users and authentication and privacy protocols to match the protocols configured on vCenter.
4. If you configured the SNMP agent to send traps to a port on the management system other than the default UDP port 162, configure the management client software to listen on the port you configured.
5. Load the VMware MIBs into the management software to view the symbolic names for the vCenter variables.
   To prevent lookup errors, load these MIB files in the following order before loading other MIB files:
   1. `VMWARE-ROOT-MIB.mib`
   2. `VMWARE-TC-MIB.mib`
   3. `VMWARE-PRODUCTS-MIB.mib`

The management software can now receive and interpret traps from vCenter.

Reset SNMP Settings to Factory Defaults

You can reset SNMP settings to factory defaults. You can also reset the value of a specific argument to the factory default.

You can reset a specific argument, such as the communities or targets. You can also reset the SNMP configuration to the factory defaults.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. To reset specific arguments, run the command `snmp.set --arguments reset`.
   For example, to reset the communities that you configured, run the following command:
   ```
   snmp.set --communities reset
   ```
3. To reset the whole SNMP configuration to the factory defaults, run the command `snmp.reset`.

Configuring Time Synchronization Settings in vCenter

You can change the time synchronization settings in vCenter after deployment.

When you deploy vCenter, you can choose the time synchronization method to be either by using an NTP server or by using VMware Tools. In case the time settings in your vSphere network change, you can edit the vCenter and configure the time synchronization settings by using the commands in the appliance shell.

When you enable periodic time synchronization, VMware Tools sets the time of the guest operating system to be the same as the time of the ESXhost.

After time synchronization occurs, VMware Tools checks once every minute to determine whether the clocks on the guest operating system and the host still match. If not, the clock on the guest operating system is synchronized to match the clock on the host.

Native time synchronization software, such as Network Time Protocol (NTP), is typically more accurate than VMware Tools periodic time synchronization and is therefore preferred. You can use only one form of periodic time synchronization

in vCenter. If you decide to use native time synchronization software, vCenter VMware Tools periodic time synchronization is deactivated.

Use VMware Tools Time Synchronization

You can set up vCenter to use VMware Tools time synchronization.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Run the command to enable VMware Tools time synchronization.
   ```
   timesync.set --mode host
   ```
3. Optional: Run the command to verify that you successfully applied the VMware Tools time synchronization.
   ```
   timesync.get
   ```
   The command returns that the time synchronization is in host mode.

The time of the appliance is synchronized with the time of the ESX host.

Add or Replace NTP Servers in the vCenter Configuration

To set up the vCenter to use NTP-based time synchronization, you must add the NTP servers to the vCenter configuration.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.
2. Add NTP servers to the vCenter configuration by running the following `ntp.set` command.
   ```
   ntp.set --servers IP-addresses-or-host-names
   ```
   In this command, *`IP-addresses-or-host-names`* is a comma-separated list of IP addresses or host names of the NTP servers.
   This command removes the current NTP servers (if any) and adds the new NTP servers to the configuration. If the time synchronization is based on an NTP server, then the NTP daemon is restarted to reload the new NTP servers. Otherwise, this command replaces the current NTP servers in the NTP configuration with the new NTP servers you specify.
3. Optional: To verify that you successfully applied the new NTP configuration settings, run the following command.
   ```
   ntp.get
   ```
   The command returns a space-separated list of the servers configured for NTP synchronization. If the NTP synchronization is activated, the command returns that the NTP configuration is in Up status. If the NTP synchronization is deactivated, the command returns that the NTP configuration is in Down status.
4. Optional: To verify if the NTP server is reachable, run the following command.
   ```
   ntp.test --servers IP-addresses-or-host-names
   ```
   The command returns the status of the NTP servers.

If the NTP synchronization is deactivated, you can configure the time synchronization settings in the vCenter to be based on an NTP server. See Synchronize the Time in vCenter with an NTP Server.

Synchronize the Time in vCenter with an NTP Server

You can configure the time synchronization settings in the vCenter to be based on an NTP server.

Set up one or more Network Time Protocol (NTP) servers in the vCenter configuration. See Add or Replace NTP Servers in the vCenter Configuration.

1. Access the appliance shell and log in as a user who has the administrator or super administrator role.
   The default user with super administrator role is root.

2. Run the command to enable NTP-based time synchronization.
   ```
   timesync.set --mode NTP
   ```
3. Optional: Run the command to verify that you successfully applied the NTP synchronization.
   ```
   timesync.get
   ```
   The command returns that the time synchronization is in NTP mode.

Managing Appliance Local User Accounts in vCenter

If you log in to the appliance shell as a super administrator, you can manage the local user accounts in vCenter by running commands in the appliance shell. The default user with a super administrator role is root.

User Roles in vCenter

There are three main user roles in vCenter.

The local users of vCenter have the rights to perform various tasks. Three user roles are available in vCenter:

Operator	Local users with the operator user role can read vCenter configuration.
Administrator	Local users with the administrator user role can configure vCenter.
Super Administrator	Local users with the super administrator user role can configure vCenter, manage the local accounts, and use the Bash shell.

Get a List of the Appliance Local User Accounts in vCenter

You can see the list of the local user accounts so that you can decide which user account to manage from the appliance shell.

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. Run the `localaccounts.user.list` command.
   You can see a list of the local users. The information about a user includes the user name, status, role, status of the password, full name, and email.

   Note:
   The list of local users includes only the local users who have their default shell as appliance shell.

Create an Appliance Local User Account in vCenter

You can create a new appliance local user account.

- To create or modify appliance local accounts using the appliance shell, you must log in as `root` or as a local account assigned the `superAdmin` role.

For information about the user roles, see User Roles in vCenter.

Note:

Appliance Local User Accounts and vCenter SSO Users

- Appliance local user accounts
  The `localaccounts.user` commands manage local Linux OS users on the vCenter appliance, such as `root` or other local CLI accounts. You can also manage these accounts through the vCenter Management Interface (VAMI) at `https://<vcenter-ip>:5480`.
- vCenter SSO users (`vsphere.local`)
  Appliance shell `localaccounts` commands do not manage vCenter Single Sign-On users, such as `administrator@vsphere.local`. To create, modify, or reset passwords for SSO users, log in to the vSphere Client and navigate to **Administrator** > **Single Sign-On** > **Users and Groups**.

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. Run the `localaccounts.user.add --role <role> --username <user_name> --password <password>` command.
   For example, to add the local user account `test` with the `operator` user role, run the following command:
   ```
   localaccounts.user.add --role operator --username test --password testPass
   ```
   The role can be `operator`, `admin`, or `superAdmin`.
   You can also set up a new local user account and specify an email and the full name of the user. For example, to add the local user account `test1` with the `operator` user role, full name `TestName` and the email address `test1@mymail.com`, run the following command:
   ```
   localaccounts.user.add --role operator --username test1 --password --fullname TestName --email
    test1@mymail.com
   ```
   You cannot use spaces in full names.
3. Enter and confirm the password of the new local user when prompted.

You created a new local user in the appliance.

Update Password of an Appliance Local User in vCenter

You can update the password of a local user in vCenter for security reasons.

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. Run the `localaccounts.user.password.update --username user name --password` command.
   For example, to change the password of a user with user name test, run the following command:
   ```
   localaccounts.user.password.update --username test --password
   ```
3. Enter and confirm the new password when prompted.
   The password expiration period for local accounts is 90 days by default.

Update an Appliance Local User Account in vCenter

You can update an existing local user account in vCenter.

For information about the user roles, see User Roles in vCenter.

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. Run the `localaccounts.user.set --username` command to update an existing local user.
   - To update the role of the local user, run the following command:
     ```
     localaccounts.user.set --username user name --role new role
     ```
     Here, *`user name`* is the name of the user that you want to edit and *`new role`* is the new role. The role can be `operator`, `admin`, or `superAdmin`.
   - To update the email of the local user, run the following command:
     ```
     localaccounts.user.set --username user name --email new email address
     ```
     Here, *`user name`* is the name of the user that you want to edit and *`new email address`* is the new email address.
   - To update the full name of the local user, run the following command:
     ```
     localaccounts.user.set --username user name --fullname new full name
     ```
     Here, *`user name`* is the name of the user that you want to edit and *`new full name`* is the new full name of the user.

- To update the status of the local user, run the following command:
  `localaccounts.user.set --username user name --status new status`
  Here, *user name* is the name of the user that you want to edit and *status* is the new status of the local user. The status can be either `disabled` or `enabled`.

Delete an Appliance Local User Account in vCenter

You can delete a local user account in vCenter.

1. Access the appliance shell and log in as a user who has a super administrator role.
   The default user with a super administrator role is root.
2. Run the `localaccounts.user.delete --username` command.
   For example, to delete the user with user name test, run the following command:
   `localaccounts.user.delete --username test`
   The user is deleted.

Monitor Health Status and Statistics in vCenter

You can monitor the hardware health status of vCenter by using the API commands in the appliance shell. You can also monitor the health status of the update component for information about available patches.

You can view the status of the hardware components such as memory, CPU, storage, and network, as well as the update component that shows if the software packages are up to date according to the last check for available patches.
A particular health status can be green, yellow, orange, red, or gray. For more information, see View vCenter Health Status.

For a complete list of the API commands that you can use for monitoring statistics and health of the vCenter system, see API Commands in Appliance Shell.

1. Access the appliance shell and log in.
   The user name that you use to log in can be of a user with an operator, administrator, or super administrator user role.
2. View the health status of a particular component.
   - To view the health of the memory in vCenter, run the `mem.health.get` command.
   - To view the health of the storage in vCenter, run the `storage.health.get` command.
   - To view the health of the swap in vCenter, run the `swap.health.get` command.
   - To view the health of the update component in vCenter, run the `softwarepackages.health.get` command.

     Important:
     If you do not perform regular checks for available patches, the health status of the update component might become out-of-date. For information about checking for vCenter patches and enabling automatic checks for vCenter patches, see *vSphere Upgrade*.
   - To view the overall health of the vCenter system, run the `health.system.get` command.
3. To view statistics about a particular hardware component, run the respective command.
   For example, to view storage statistics for each logical disk, run the `storage.stats.list` command.

Using the Direct Console User Interface to Configure vCenter

After you deploy vCenter, you can reconfigure the network settings and enable access to the Bash shell for troubleshooting. To access the Direct Console User Interface, you must log in as root.

The home page of the Direct Console User Interface contains a link to the support bundle of the vCenter. The link to the support bundle is of the type https://*appliance-host-name*:443/appliance/support-bundle.

Log In to the Direct Console User Interface

The Direct Console User Interface lets you interact with vCenter locally by using text-based menus.

1. From the vSphere Client, navigate to the host and click **Configure** > **Services**. Confirm that the SSH and Direct Console User Interface services are running.
2. Open an SSH client and connect to the ESX host.
3. Log in with the root account.
4. Enter `DCUI` to start the Direct Console User Interface.
5. Click inside the console window and press F2 to customize the system.
6. Type the password for the root user and press Enter.

   Important:
   If you enter invalid credentials thrice, the root account is locked for five minutes.

You logged in to the Direct Console User Interface. You can change the password of the root user, edit the network settings, and enable access to the vCenter appliance Bash shell.

Change the Password of the Root User

To prevent unauthorized access to the vCenter Direct Console User Interface, you can change the password of the root user.

The default root password for the vCenter instance is the password you enter during deployment.

Important:
The password for the root account of vCenter expires after 90 days. You can change the expiry time for an account by logging as root to the vCenter Bash shell, and running `chage -M number_of_days -W warning_until_expirationuser_name`. To increase the expiration time of the root password to infinity, run the `chage -M -1 -E -1 root` command.

1. Log in to the Direct Console User Interface.
2. Select **Configure Password** and press Enter.
3. Type the old password of the root user, and press Enter.
4. Set up the new password and press Enter.
5. Press Esc until you return to the main menu of the Direct Console User Interface.

You changed the password of the root user of the appliance.

Configure the Management Network of vCenter

The vCenter instance can obtain networking settings from a DHCP server, or use static IP addresses. You can change the networking settings of vCenter from the Direct Console User Interface. You can change the IPv4, IPv6, and DNS configuration.

To change the IP address of the vCenter instance, verify that the system name is an FQDN. If, during deployment you set an IP address as a system name, you cannot change the IP address after the deployment. The system name is always used as a primary network identifier.

1. Log in to the Direct Console User Interface of vCenter.
2. Select **Configure Management Network** and press Enter.
3. Change the IPv4 settings from **IP Configuration**.

Option	Description
Use dynamic IP address and network configuration	Obtains networking settings from a DHCP server if one is available on your network
Set static IP address and network configuration	Sets static networking configuration

4. Change the IPv6 settings from **IPv6 Configuration**.

Option	Description
Enable IPv6	Enables or disables IPv6
Use DHCP stateful configuration	Uses a DHCP server to obtain IPv6 addresses and networking settings
Use ICMP stateless configuration	Uses a Stateless Address Autoconfiguration (SLAAC) to obtain IPv6 addresses and network settings

5. Change the DNS settings from **DNS Configuration**.

Option	Description
Obtain DNS server address and hostname automatically	Obtains the DNS server address and host name automatically. Use this option if the IP settings are obtained automatically from a DHCP server.
Use the following DNS server address and hostname	Sets the static IP address and host name for the DNS server.

6. Set custom DNS suffixes from **Custom DNS Suffixes**.
   If you do not specify any suffixes, a default suffix list is derived from the local domain name.
7. Press Esc until you return to the main menu of the Direct Console User Interface.

Restart the Management Network of vCenter

Restart the management network of vCenter to restore the network connection.

1. Log in to the Direct Console User Interface of vCenter.
2. Select **Restart Management Network** and press Enter.
3. Press F11.

Enable Access to the Bash Shell

You can use the Direct Console User Interface to enable local and remote access to the Bash shell. Bash shell access enabled through Direct Console User Interface remains enabled for 3600 seconds.

1. Log in to the Direct Console User Interface of vCenter.
2. Select **Troubleshooting Options** and press Enter.
3. From the Troubleshooting Mode Options menu, select to enable either Bash shell or SSH.
4. Press Enter to enable the service.
5. Press Esc until you return to the main menu of the Direct Console User Interface.

Access the vCenter Bash shell for troubleshooting.

Access the Bash Shell for Troubleshooting

Log in to the Bash shell for troubleshooting purposes only.

1. Access the shell using one of the following methods.
   - If you have direct access to the vCenter instance, press Alt+F1.
   - If you want to connect remotely, use SSH or another remote console connection to start a session.
2. Enter a user name and password.
3. In the shell, enter the command `pi shell` or `shell` to access the Bash shell.

vSphere Authentication

The *vSphere Authentication* documentation provides information to help you perform common tasks such as certificate management and vCenter Single Sign-On configuration.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

vSphere Authentication explains how you can manage certificates for vCenter and related services, and set up authentication with vCenter Single Sign-On.

Table 69: vSphere Authentication Highlights

Topics	Content Highlights
Getting Started with Authentication	• Managing authentication services. • Managing vCenter using the vCenter Management Interface.
vSphere Security Certificates	• Certificate model, and options for replacing certificates. • Replace certificates from the UI (simple cases). • Replace certificates using the Certificate Manager utility. • Replace certificates using the CLI (complex situations). • Certificate management CLI reference.
vSphere Authentication with vCenter Single Sign-On	• Architecture of the authentication process. • How to add identity sources so users in your domain can authenticate. • Two-factor authentication. • Managing users, groups, and policies. • vCenter Identity Provider Federation

Related Documentation

A companion document, *vSphere Security*, describes available security features and the measures that you can take to safeguard your environment from attack. That document also explains how you can set up permissions, and includes a reference to privileges.

In addition to these documents, VMware publishes the *vSphere Security Configuration Guide* (formerly known as the *Hardening Guide*) for each release of vSphere, accessible at https://core.vmware.com/security. The *vSphere Security Configuration Guide* contains guidelines on security settings that can or should be set by the customer, and security settings delivered by VMware that should be audited by the customer to ensure that they are still set to default.

Intended Audience

This information is intended for administrators who want to configure vCenter authentication and manage certificates. The information is written for experienced Linux system administrators who are familiar with virtual machine technology and data center operations.

Getting Started with vSphere Certificate Management and Authentication

vSphere provides common infrastructure services to manage certificates for both vCenter and ESX components, and to manage authentication with vCenter Single Sign-On.

How Do You Manage vSphere Certificates

By default, vSphere enables you to provision vCenter components and ESX hosts with VMware Certificate Authority (VMCA) certificates. You can also use custom certificates, which are stored in the VMware Endpoint Certificate Store (VECS). For more information, see What Options Do You Have to Manage vSphere Certificates.

What Is vCenter Single Sign-On

vCenter Single Sign-On allows vSphere components to communicate with each other through a secure token mechanism. vCenter Single Sign-On uses specific terms and definitions that are important to understand.

Table 70: vCenter Single Sign-On Glossary

Term	Definition
Principal	An entity that can be authenticated, such as a user.
Identity Provider	A service that manages identity sources and authenticates principals. Examples: Microsoft Active Directory Federation Services (AD FS) and vCenter Single Sign-On.
Identity Source (Directory Service)	Stores and manages principals. Principals consist of a collection of attributes about a user or a service account such as name, address, email, and group membership. Examples: Microsoft Active Directory and VMware Directory Service (vmdir).
Authentication	The means of determining whether someone or something is, in fact, who or what it declares itself to be. For example, users are authenticated when they provide their credentials, such as user name and correct password, and so on.
Authorization	The process of verifying what objects principals have access to.
Token	A signed collection of data comprising the identity information for a given principal. A token might include not only basic information about the principal such as email address and full name, but also, depending on the token type, the principal's groups and roles.
vmdir	VMware Directory Service. The internal (local) LDAP repository in vCenter that contains user identities, groups, and configuration data.
OAuth 2.0	An open authorization standard that enables the exchange of information among principals and web services without exposing principals' credentials.
OpenID Connect (OIDC)	Authentication protocol based on OAuth 2.0 that augments OAuth with user-identifying information. It is represented by the ID token that the authorization server returns together with the access token during OAuth authentication. vCenter uses OIDC capabilities when interacting with Active Directory Federation Services (AD FS), Okta, Microsoft Entra ID, and PingFederate.
System for Cross-domain Identity Management (SCIM)	The standard for automating the exchange of user identity information between identity domains or IT systems.
VMware Identity Services	Starting in version 8.0 Update 1, VMware Identity Services is a built-in container within vCenter that you can use for identity federation to external identity providers. It serves as an independent identity broker within vCenter and comes with its own set of APIs. Currently, VMware Identity Services support Okta, Microsoft Entra ID, and PingFederate as external identity providers.

Term	Definition
Tenant	A VMware Identity Services concept. A tenant provides a logical separation of data from other tenants' data in the same virtual environment.
JSON Web Token (JWT)	A token format defined by the OAuth 2.0 specification. A JWT token carries authentication and authorization information about a principal.
Relying party	A relying party "relies" on the authorization server, VMware Identity Services, or AD FS, for identity management. For example, through federation, vCenter establishes relying party trust to VMware Identity Services or AD FS.
Security Assertion Markup Language (SAML)	An XML-based open standard for exchanging authentication and authorization data between parties that is used by vCenter. Principals obtain a SAML token from vCenter Single Sign-On and then send it to the vSphere Automation API endpoint for a session identifier.

What Are the vCenter Single Sign-On Authentication Types

vCenter Single Sign-On uses different types of authentication, depending on whether the built-in vCenter identity provider or an external identity provider is involved.

Table 71: vCenter Single Sign-On Authentication Types

Authentication Type	What Acts as the Identity Provider?	Does vCenter Handle the Password?	Description
Token-Based Authentication	External identity provider. For example, AD FS.	No	vCenter contacts the external identity provider through a particular protocol and obtains a token, which represents a particular user identity.
Simple Authentication	vCenter	Yes	The user name and password are passed directly to vCenter, which validates the credentials through its identity sources.

vSphere Authentication Options

In vSphere 9.0 and later, external identity provider federation is the preferred authentication method for vCenter. With VCF 9.0 we are moving towards a modern authentication which leverages federation of vCenter with an external IDP provider. Native vCenter features for multi-factor authentication are deprecated in favor of leveraging the external identity providers capabilities or multi-factor authentication.

The vCenter authentication will transparently delegate the authentication call to the federated IDP which performs the authentication by all supported means including multi-factor authentication, and then return a token to vCenter which is used for authentication. All multi-factor authentication options including the token validity time can be configured at the external identity provider settings. For more information on VCF Single Sign-On, see Configuring VCF Single Sign-On.

vCenter supports the following external identity providers:

- **ADFS** (vSphere 7.0 and later)
- **OKTA** (vSphere 8.0 Update 1 and later)
- **Microsoft Entra ID**, formerly called Azure AD (vSphere 8.0 Update 2 and later)
- **PingFederate** (starting in vSphere 8.0 Update 3)

Consider the version and timelines when you plan your migration to a federated vCenter authentication.

Two-Factor Authentication Methods

Government agencies or large enterprises often require two-factor authentication. vSphere supports the following two-factor authentication method.

External Identity Provider Federation	With external identity provider federation, you can use the authentication mechanisms supported by the external identity provider, including multi-factor authentication.

Specifying a vCenter Non-default Authentication Method

You can set up a non-default authentication method from the vSphere Client, or by using the `sso-config` script.

Combining vCenter Authentication Methods

You can activate or deactivate each authentication method separately by using `sso-config`. Leave user name and password authentication enabled initially, while you are testing a two-factor authentication method, and set only one authentication method to enabled after testing.

Managing vCenter Certificates

You manage vCenter certificates from the vSphere Client, or by using an API, scripts, or CLIs.

The following table describes the interfaces that you can use to manage vCenter certificates.

Table 72: Interfaces for Managing vCenter Certificates

Interface	Description
vSphere Client	Web interface (HTML5-based client). See Managing Certificates Using the vSphere Client.
vSphere Automation API	See *VMware vSphere Automation SDKs Programming Guide*.
Certificate Management utility	Command-line tool that supports Certificate Signing Request (CSR) generation and certificate replacement. See Managing Certificates Using the vSphere Certificate Manager Utility.
CLIs for managing certificate and directory services	Set of commands for managing certificates, the VMware Endpoint Certificate Store (VECS), and VMware Directory Service (vmdir). See vSphere Certificates and Services CLI Command Reference.

Manage vCenter Certificates Using the vSphere Client

You can manage vCenter certificates from the vSphere Client.

1. Log in to a vCenter as a user with administrator privileges in the local vCenter Single Sign-On domain.
   The default domain is vsphere.local.
2. Select **Administration**.
3. Under **Certificates**, click **Certificate Management**.
   Certificate tabs for the different types of certificates appear.
4. Perform certificate tasks, such as viewing certificate details, renewing or refreshing a certificate, and adding a Trusted Root certificate.
   For more information, see Managing Certificates Using the vSphere Client.

Manage vCenter Certificates Using CLIs

vCenter includes CLIs for generating Certificate Signing Requests (CSRs), managing certificates, and managing services.

Enable SSH login to vCenter. You can use the **Access** tab in the vCenter Management Interface (`https://vcenter_server_ip:5480`) for SSH login activation and deactivation.

For example, you can use the `certool` command to generate CSRs and to replace certificates.

Use the CLIs for management tasks that the vSphere Client does not support, or to create custom scripts for your environment.

Table 73: CLIs for Managing vCenter Certificates and Associated Services

CLI	Description	Links
`certool`	Generate and manage certificates and keys. Part of the VMware Certificate Authority (VMCA).	certool Initialization Commands Reference
`vecs-cli`	Manage the contents of VMware Certificate Store instances. Part of VMware Authentication Framework Daemon (VMAFD).	vecs-cli Command Reference
`dir-cli`	Create and update certificates in VMware Directory Service. Part of VMAFD.	dir-cli Command Reference
`sso-config`	Update Security Token Service (STS) certificates.	Replace a vCenter STS Certificate Using the Command Line
`service-control`	Command for starting, stopping, and listing services.	Run this command to stop services before running other CLI commands.

1. Log in to the vCenter shell.

   Usually, you have to be the root user. See vSphere Certificates and Services CLI Command Reference for details.

2. Access a CLI at one of the following default locations.

   The required privileges depend on the task that you want to perform. Sometimes, you are prompted for the password twice to safeguard sensitive information.

   ```
   /usr/lib/vmware-vmafd/bin/vecs-cli
   /usr/lib/vmware-vmafd/bin/dir-cli
   /usr/lib/vmware-vmca/bin/certool
   /opt/vmware/bin/sso-config.sh
   ```

   The `service-control` command does not require that you enter the path.

   For more information, see Manual vSphere Certificate Replacement.

Managing vCenter Authentication Services

You manage authentication services from the vSphere Client, or by using the CLI. You can also manage the vCenter Identity Provider Federation configuration process by using an API.

You can manage vCenter authentication using different interfaces.

Table 74: Interfaces for Managing vCenter Authentication Services

Interface	Description
vSphere Client	Web interface (HTML5-based client).
API	Manage the vCenter Identity Provider Federation configuration process.
`sso-config`	Command-line utility for configuring the vCenter built-in identity provider.

Manage vCenter Authentication Services Using the vSphere Client

You can manage vCenter authentication services from the vSphere Client.

1. Log in to a vCenter as a user with administrator privileges in the local vCenter Single Sign-On domain.
   The default domain is vsphere.local.
2. Select **Administration**.
3. Under **Single Sign On**, click **Configuration** to manage identity providers and configure password and lockout policies.
   For more information, see vSphere Authentication with vCenter Single Sign-On.

Manage vCenter Authentication Services Using Scripts

vCenter includes a utility, `sso-config`, for managing authentication services.

Enable SSH login to vCenter. You can use the **Access Settings** tab in the vCenter Management Interface (`https://vcenter_server_ip:5480`) for SSH login activation and deactivation.

Use the `sso-config` utility for management tasks that the vSphere Client does not support, or to create custom scripts for your environment.

Table 75: CLIs for Managing Authentication and Associated Services

CLI	Description	Links
`sso-config`	Command-line utility for configuring the vCenter built-in identity provider.	Refer to the `sso-config` help by running `sso-config.sh -help`, or see the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=67304 for usage examples.
`service-control`	Command for starting, stopping, and listing services.	Run this command to stop services before running other CLI commands. The `service-control` command does not require that you specify the path.

1. Log in to the vCenter shell.
   Usually, you have to be the root or Administrator user. See vSphere Certificates and Services CLI Command Reference for details.
2. Access the `sso-config` utility at following default location.
   ```
   /opt/vmware/bin/sso-config.sh
   ```
   The required privileges depend on the task that you want to perform. Sometimes, you are prompted for the password twice to safeguard sensitive information.

Managing the vCenter

You can manage the vCenter from the vCenter Management Interface or from the vCenter shell.

For more information about managing vCenter, see *vCenter Configuration*.

Table 76: Interfaces for Managing the vCenter

Interface	Description
vCenter Management Interface	Use this interface to reconfigure the system settings. See Manage vCenter Using the Management Interface.
vCenter shell	Use this command-line interface to perform service management operations on VMCA, VECS, and VMDIR. See Managing Certificates Using the vSphere Certificate Manager Utility and vSphere Certificates and Services CLI Command Reference.

Manage vCenter Using the Management Interface

You can use the vCenter Management Interface to configure the system settings.

Settings in the vCenter Management Interface include time synchronization, network settings, and SSH login settings. You can also change the root password, join the appliance to an Active Directory domain, and leave an Active Directory domain.

> **Note:**
> Under the **Networking** pane, the virtual NIC 0 is reserved for management traffic. You cannot reassign the traffic from NIC 0 to another NIC. If you are using VCHA, this traffic uses NIC 1. You can add NICs to the vCenter appliance. See the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2147155.

1. In a browser, go to the web interface at `https://vcenter_server_ip:5480`.
2. If a warning message about an untrusted SSL certificate appears, resolve the issue based on your company security policy and the browser that you are using.
3. Log in as root.
   The default root password is the root password that you set when deploying the vCenter.

You see the Summary page of the vCenter Management Interface.

Manage vCenter Using the vCenter Shell

You can use service management utilities and CLIs from the vCenter shell. You can use TTY1 to log in to the console, or can use SSH to connect to the shell.

1. Activate SSH login if necessary.
   a) Log in to the vCenter Management Interface at https://*vcenter_server_ip*:5480.
   b) In the Navigator, select **Access** and click **Edit**.
   c) Toggle on **Activate SSH Login** and click **OK**.
   You can follow the same steps to activate the Bash shell for the vCenter.
2. Access the shell.
   - If you have direct access to the vCenter console, select **Log in**, and press Enter.
   - To connect remotely, use SSH or another remote console connection to start a session to the vCenter.
3. Log in as root with the password that you set when you initially deployed the vCenter.
   If you changed the root password, use the new password.

vSphere Security Certificates

vSphere provides security by using certificates to encrypt communications, authenticate services, and sign tokens.

How Does vSphere Use Certificates

vSphere uses certificates to:

- Encrypt communications between two nodes, such as a vCenter and an ESX host.
- Authenticate vSphere services.
- Perform internal actions such as signing tokens.

What is the VMware Certificate Authority

vSphere's internal certificate authority, VMware Certificate Authority (VMCA), provides all the certificates necessary for vCenter and ESX. VMCA is installed on every vCenter instance, immediately securing the solution without any other modification. Keeping this default configuration provides the lowest operational overhead for certificate management. vSphere provides a mechanism to renew these certificates in the event they expire. Starting with VCF 9.1, the VMCA and

solution certificates that are due to expire within two years are automatically renewed, and replaced when you upgrade your vCenter to major or minor version. A new VMCA root certificate is provisioned if the expiry threshold is met.

vSphere also provides a mechanism to replace certain certificates with your own certificates. However, replace only the SSL certificate that provides encryption between nodes, to keep your certificate management overhead low.

What Options Do You Have to Manage vSphere Certificates

The following options are recommended for managing certificates.

Table 77: Recommended Options for Managing vSphere Certificates

Mode	Description	Advantages
VMCA Default Certificates	VMCA provides all the certificates for vCenter and ESX hosts.	Simplest and lowest overhead. VMCA can manage the certificate life cycle for vCenter and ESX hosts.
VMCA Default Certificates with External SSL Certificates (Hybrid Mode)	You replace the vCenter SSL certificates, and allow VMCA to manage certificates for solution users and ESX hosts. Optionally, for high-security conscious deployments, you can replace the ESX host SSL certificates as well.	Simple and secure. VMCA manages internal certificates but you get the benefit of using your corporate-approved SSL certificates, and having those certificates trusted by your browsers.

What Tools Are Available to Replace vSphere Certificates

You can use the following options to replace the existing certificates.

Table 78: Different Approaches to vSphere Certificate Replacement

Option	See
Use the vSphere Client.	Managing Certificates Using the vSphere Client
Use the vSphere Automation API to manage the life cycle of certificates.	*VMware vSphere Automation SDKs Programming Guide*
Use the vSphere Certificate Manager utility from the command line.	Managing Certificates Using the vSphere Certificate Manager Utility
Use CLI commands for manual certificate replacement.	vSphere Certificates and Services CLI Command Reference

vSphere Certificate Requirements for Different Solution Paths

Certificate requirements depend on whether you use the VMware Certificate Authority (VMCA) as an intermediate certificate authority or you use custom certificates. Requirements are also different for machine certificates.

Before you begin to change certificates, ensure that all nodes in your vSphere environment are time synchronized.

> **Note:**
> vSphere deploys only RSA certificates for server authentication and does not support generating ECDSA certificates. vSphere verifies ECDSA certificates presented by other servers. For example, if vSphere connects to a syslog server and the syslog server has an ECDSA certificate, vSphere supports verifying that certificate.

Requirements for All Imported vSphere Certificates

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded). The vSphere Client and API still accept a key size up to 16384 bits when generating the Certificate Signing Request.

Note:
In vSphere 8.0, you can only generate CSRs with a minimum key length of 3072 bits when using the vSphere Client or the vSphere Certificate Manager. vCenter still does accept custom certificates bearing a key length of 2048 bits. In vSphere 8.0 Update 1 and later, you can use the vSphere Client to generate a CSR with a key length of 2048 bits.

Note:
vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

- PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When you add keys to VECS, they are converted to PKCS8.
- x509 version 3
- SubjectAltName must contain DNS Name=*machine_FQDN*
- CRT format
- Contains the following Key Usages: Digital Signature, Non-Repudiation, Key Encipherment.
- Contains the following Key Usages: Digital Signature, Key Encipherment.
- Exempting the vpxd-extension solution user certificate, Extended Key Usage can be either empty or can contain Server Authentication.

vSphere does not support the following certificates.

- Certificates with wildcards.
- The algorithms md2WithRSAEncryption, md5WithRSAEncryption, RSASSA-PSS, dsaWithSHA1, ecdsa_with_SHA1, and sha1WithRSAEncryption are not supported.
- When creating a custom machine SSL certificate for vCenter, Server Authentication and Client Authentication are not supported, and must be removed when using the Microsoft Certificate Authority (CA) templates. For more information, see the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112009.

vSphere Certificate Compliance to RFC 2253

The certificate must be in compliance with RFC 2253.

If you do not generate CSRs using vSphere Certificate Manager, ensure that the CSR includes the following fields.

String	X.500 AttributeType
CN	`commonName`
L	`localityName`
ST	`stateOrProvinceName`
O	`organizationName`
OU	`organizationalUnitName`
C	`countryName`
STREET	`streetAddress`
DC	`domainComponent`
UID	`userid`

If you generate CSRs using vSphere Certificate Manager, you are prompted for the following information, and vSphere Certificate Manager adds the corresponding fields to the CSR file.

- The password of the administrator@vsphere.local user, or for the administrator of the vCenter Single Sign-On domain that you are connecting to.
- Information that vSphere Certificate Manager stores in the `certool.cfg` file. For most fields, you can accept the default or provide site-specific values. The FQDN of the machine is required.
  - Password for administrator@vsphere.local
  - Two-letter country code
  - Company name

- Organization name
- Organization unit
- State
- Locality
- IP address (optional)
- Email (optional)
- Host name, that is, the fully qualified domain name of the machine for which you want to replace the certificate. If the host name does not match the FQDN, certificate replacement does not complete correctly and your environment might end up in an unstable state.
- IP address of the vCenter node on which you run vSphere Certificate Manager.
- OU (organizationalUnitName) (optional)

Certificate Requirements When Using VMCA as an Intermediate Certificate Authority

When you use VMCA as an intermediate CA, the certificates must meet the following requirements.

Certificate Type	Certificate Requirements

Root certificate	• You can use vSphere Certificate Manager to create the CSR. See Generate CSR Using the Certificate Manager and Prepare Root Certificate (Intermediate CA).
	• If you prefer to create the CSR manually, the certificate that you send to be signed must meet the following requirements.
	– Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
	– PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
	– x509 version 3
	– The CA extension must be set to true for root certificates, and cert sign must be in the list of requirements. For example:
	`basicConstraints = critical,CA:true`
	`keyUsage = critical,digitalSignature,keyCertSign`
	– CRL signing must be enabled.
	– Extended Key Usage can be either empty or contain Server Authentication.
	– No explicit limit to the length of the certificate chain. VMCA uses the OpenSSL default, which is 10 certificates.
	– Certificates with wildcards or with more than one DNS name are not supported.
	– You cannot create subsidiary CAs of VMCA.
	See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112009, Creating a Microsoft Certificate Authority Template for SSL certificate creation in vSphere 6.x, for an example using Microsoft Certificate Authority.

Machine SSL certificate	You can use vSphere Certificate Manager to create the CSR or create the CSR manually. If you create the CSR manually, it must meet the requirements that are listed previously under *Requirements for all Imported vSphere Certificates*. You also have to specify the FQDN for the host.
Solution user certificate	You can use vSphere Certificate Manager to create the CSR or create the CSR manually. **Note:** You must use a different value for Name for each solution user. If you generate the certificate manually, this might show up as **CN** under **Subject**, depending on the tool you use. If you use vSphere Certificate Manager, the tool prompts you for certificate information for each solution user. vSphere Certificate Manager stores the information in `certool.cfg`. For the vpxd-extension solution user, you can either leave Extended Key Usage empty or use "TLS WWW client authentication".

Requirements When Using Custom Certificates

When you want to use custom certificates, the certificates must meet the following requirements.

Certificate Type	Certificate Requirements
Machine SSL certificate	The machine SSL certificate on each node must have a separate certificate from your third-party or enterprise CA. • You can generate the CSR using the vSphere Client or the vSphere Certificate Manager, or create the CSR manually. The CSR must meet the requirements listed previously under *Requirements for all Imported vSphere Certificates*. • For most fields, you can accept the default or provide site-specific values. The FQDN of the machine is required.
Solution user certificate	Each solution user on each node must have a separate certificate from your third-party or enterprise CA. • You can generate the CSRs using vSphere Certificate Manager or prepare the CSR yourself. The CSR must meet the requirements listed previously under *Requirements for all Imported vSphere Certificates*. • If you use vSphere Certificate Manager, the utility prompts you for certificate information for each solution user. vSphere Certificate Manager stores the information in `certool.cfg`. **Note:** You must use a different value for Name for each solution user. A manually generated certificate might show up as **CN** under **Subject**, depending on the tool you use. When later you replace solution user certificates with custom certificates, provide the complete signing certificate chain of the third-party CA. For the vpxd-extension solution user, you can either leave Extended Key Usage empty or use "TLS WWW client authentication".

Note:

Solution users are deprecated in vCenter 9.0 and are being replaced by Service Accounts. For more information, see VCF 9.0 Release Notes.

vSphere Certificate Management

The work that is required for setting up or updating your vSphere certificate infrastructure depends on the requirements in your environment. You must consider whether you are performing a fresh installation or an upgrade, and whether you are considering ESX or vCenter.

Environments That Use VMware Certificate Authority Certificates

VMware Certificate Authority (VMCA) can handle all certificate management. VMCA provisions vCenter components and ESX hosts with certificates that use VMCA as the root certificate authority.

If you do not currently replace VMware certificates, your environment starts using VMCA-signed certificates instead of self-signed certificates.

Environments That Use Custom Certificates

If your company policy requires certificates that are signed by a third-party or enterprise certificate authority, or that require custom certificate information, you have several choices for a fresh installation.

- Have the VMCA root certificate signed by a third-party CA or enterprise CA. Replace the VMCA root certificate with that signed certificate. In this scenario, the VMCA certificate is an intermediate certificate. VMCA provisions vCenter components and ESX hosts with certificates that include the full certificate chain.
- If your company policy does not allow intermediate certificates in the chain, you can replace certificates explicitly. You can use the vSphere Client, vSphere Certificate Manager utility, or perform manual certificate replacement using the certificate management CLIs.

When upgrading an environment that uses custom certificates, you can retain some of the certificates.

- ESX hosts keep their custom certificates during upgrade. Make sure that the vCenter upgrade process adds all the relevant root certificates to the TRUSTED_ROOTS store in VMware Certificate Endpoint Store (VECS) on the vCenter. After the upgrade to vSphere 6.0 or later, you can set the certificate mode to **Custom**. If the certificate mode is VMCA (the default), and you perform a certificate refresh from the vSphere Client, the VMCA-signed certificates replace the custom certificates.
- For an upgrade of a simple vCenter installation to an embedded deployment, vCenter retains custom certificates. After the upgrade, your environment works as before. The existing vCenter and vCenter Single Sign-On certificates are retained. The certificates are used as machine SSL certificates. In addition, VMCA assigns a VMCA-signed certificate to each solution user (collection of vCenter services). The solution user uses this certificate only to authenticate to vCenter Single Sign-On. VMware does not recommend replacing solution user certificates.

vSphere Certificate Interfaces

For vCenter, you can view and replace certificates with the following tools and interfaces.

Table 79: Interfaces for Managing vCenter Certificates

Interface	Use
vSphere Client	Perform common certificate tasks with a graphical user interface.
vSphere Automation API	See *VMware vSphere Automation SDKs Programming Guide*.
vSphere Certificate Manager utility	Perform common certificate replacement tasks from the command line of the vCenter installation.
vSphere Certificate Management CLIs	Perform all certificate management tasks with `dir-cli`, `certool`, and `vecs-cli`.
`sso-config` utility	Perform STS certificate management from the command line of the vCenter installation.
PowerCLI 12.4 or later (also requires vSphere 7.0 or later)	Perform trusted certificate store management, manage vCenter Machine SSL certificates, and manage ESX Machine SSL certificates.

For ESX, you perform certificate management from the vSphere Client. VMCA provisions certificates and stores them locally on the ESX host. VMCA does not store ESX host certificates in VMDIR or in VECS. See the *vSphere Security* documentation.

Supported vCenter Certificates

For vCenter and related machines and services, the following certificates are supported:

- Certificates that are generated and signed by VMware Certificate Authority (VMCA).
- Custom certificates.
  - Enterprise certificates that are generated from your own internal PKI.
  - Third-party CA-signed certificates that are generated by an external PKI such as Verisign, GoDaddy, and so on.

Self-signed certificates that were created using OpenSSL in which no Root CA exists are not supported.

Replacing vSphere Certificates

You can perform different types of certificate replacement depending on your company policy and requirements for the system that you are configuring. You can perform certificate replacement from the vSphere Client, by using the vSphere Certificate Manager utility, or manually by using the CLIs included with your installation.

The VMware Certificate Authority (VMCA) is included in each vCenter deployment. VMCA provisions each node, each vCenter solution user, and each ESX host with a certificate that is signed by VMCA as the certificate authority.

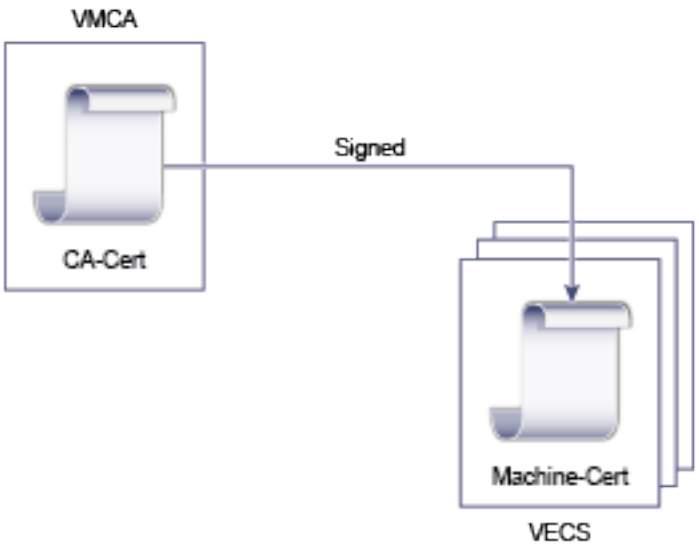
You can replace the default certificates. For vCenter components, you can use a set of command-line tools included in your installation. You have several options.

Note:
If your vCenter is linked to NSX Manager, and you replace vCenter certificates, you must update the thumbprint of the vCenter compute manager. See the topic titled "Add a Compute Manager" in *NSX Administrator Guide*.

Replacing Certificates with VMCA-Signed Certificates

If your VMCA certificate expires or you want to replace it for other reasons, you can use the certificate management CLIs to perform that process. By default, the VMCA root certificate expires after 10 years, and all certificates that VMCA signs expire when the root certificate expires, that is, after a maximum of 10 years. Starting with VCF 9.1, the VMCA and solution certificates that are due to expire within two years are automatically renewed and replaced when you upgrade your vCenter to major or minor version. A new VMCA root certificate is provisioned if the expiry threshold is met.

Figure 27: Certificates Signed by VMCA Are Stored in VECS



You can use the following vSphere Certificate Manager options:

- Replace Machine SSL Certificate with VMCA Certificate
- Replace Solution User Certificate with VMCA Certificate

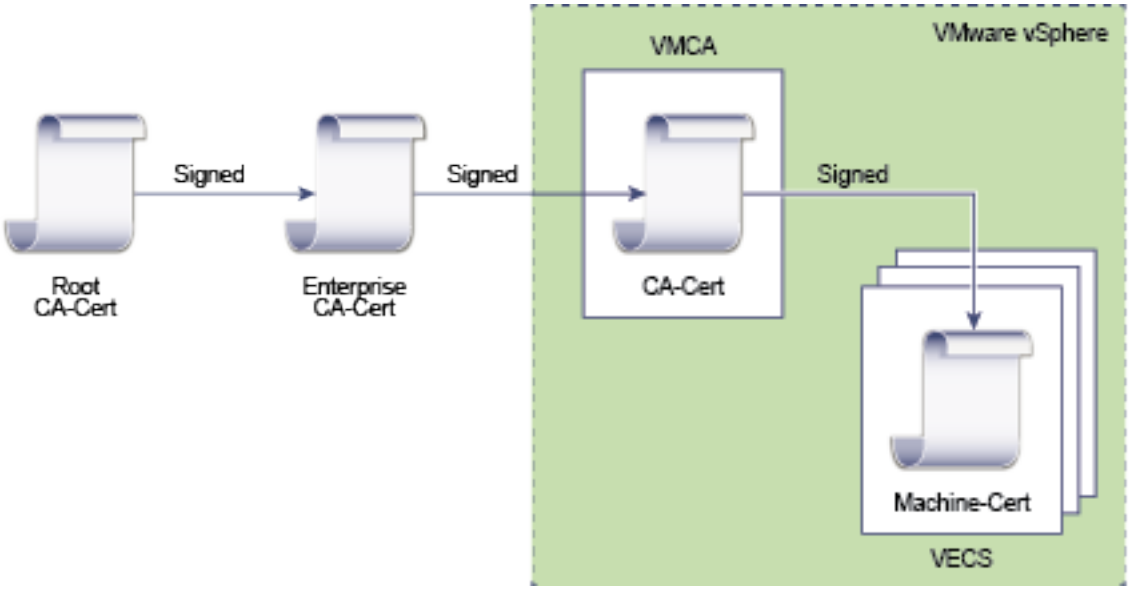
For manual certificate replacement, see Replace Existing VMCA-Signed Certificates with New VMCA-Signed Certificates Using the CLI.

Making VMCA an Intermediate Certificate Authority

You can replace the VMCA root certificate with a certificate that is signed by an enterprise certificate authority (CA), or a third-party CA. VMCA signs the custom root certificate each time it provisions certificates, making VMCA an intermediate CA.

Note:
If you perform a fresh install with a vCenter, replace the VMCA root certificate before you add ESX hosts. If you do, VMCA signs the whole chain, and you do not have to generate new certificates.

Figure 28: Certificates Signed by a Third-Party or Enterprise CA Use VMCA as an Intermediate CA



You can use the following vSphere Certificate Manager options:

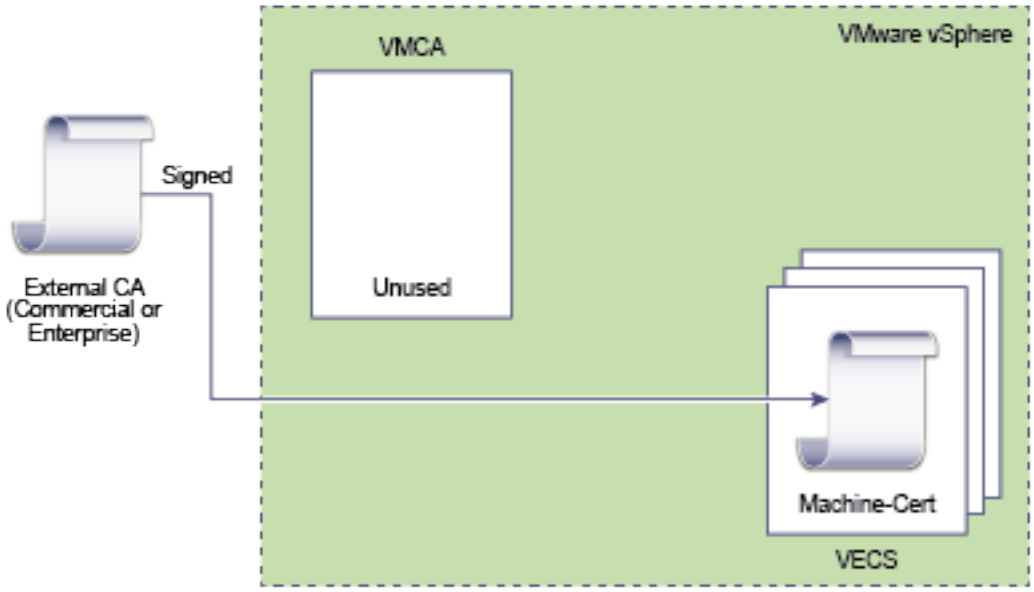
- Replace VMCA Root Certificate with Custom Signing Certificate and Replace All Certificates
- Replace Machine SSL Certificate with VMCA Certificate (multi-node enhanced linked mode deployment)
- Replace Solution User Certificate with VMCA Certificate (multi-node enhanced linked mode deployment)

For manual certificate replacement, see Make VMCA Into an Intermediate Certificate Authority Using the CLI.

Replacing VMCA-Signed Certificates with Custom Certificates

You can replace the existing VMCA-signed certificates with custom certificates. If you use that approach, you are responsible for all certificate provisioning and monitoring.

Figure 29: External Certificates Are Stored Directly in VECS



You can use the following vSphere Certificate Manager options:

- Replace Machine SSL Certificate with Custom Certificate
- Replace Solution User Certificates with Custom Certificates

For manual certificate replacement, see Replace Certificates with Custom Certificates Using the CLI.

You can also use the vSphere Client to generate a CSR for a machine SSL certificate (custom), and replace the certificate after the CA returns it. See Generate Certificate Signing Request for Machine SSL Certificate Using the vSphere Client (Custom Certificates).

Using the Hybrid Approach to Certificate Deployment

In the hybrid approach, you can have VMCA supply some of the certificates, but use custom certificates for other parts of your infrastructure. For example, because solution user certificates are used only to authenticate to vCenter Single Sign-On, consider having VMCA provision those certificates. Replace the machine SSL certificates with custom certificates to secure all SSL traffic.

Company policy often does not allow intermediate CAs. For those cases, hybrid deployment is a good solution. It minimizes the number of certificates to replace, and secures all traffic. The hybrid deployment leaves only internal traffic, that is, solution user traffic, to use the default VMCA-signed certificates.

For more information, see the blog post titled *New Product Walkthrough - Hybrid vSphere SSL Certificate Replacement* at https://blogs.vmware.com/vsphere/2017/01/walkthrough-hybrid-ssl-certificate-replacement.html.

ESX Certificate Replacement

For ESX hosts, you can change certificate provisioning behavior from the vSphere Client. See the *vSphere Security* documentation for details.

Table 80: ESX Certificate Replacement Options

Option	Description
VMware Certificate Authority mode (default)	When you renew certificates from the vSphere Client, VMCA issues the certificates for the hosts. If you changed the VMCA root certificate to include a certificate chain, the host certificates include the full chain.
Custom Certificate Authority mode	Allows you to update and use certificates manually that are not signed or issued by VMCA.
Thumbprint mode	Can be used to retain 5.5 certificates during refresh. Use this mode only temporarily in debugging situations.

Where vSphere Uses Certificates

The VMware Certificate Authority (VMCA) provisions your environment with certificates. Certificates include machine SSL certificates for secure connections, solution user certificates for authentication of services to vCenter Single Sign-On, and certificates for ESX hosts.

The following certificates are in use.

Table 81: Certificates in vSphere

Certificate	Provisioned	Comments
ESX certificates	VMCA (default)	Stored locally on ESX host.
Machine SSL certificates	VMCA (default)	Stored in VMware Endpoint Certificate Store (VECS).
Solution user certificates	VMCA (default)	Stored in VECS.
vCenter Single Sign-On SSL signing certificate	Provisioned during installation.	Manage this certificate from the command line. **Note:** Do not change this certificate in the filesystem or unpredictable behavior results.
VMware Directory Service (VMDIR) SSL certificate	Provisioned during installation.	In vSphere 6.5 and later, the machine SSL certificate is used as the vmdir certificate.

Certificate	Provisioned	Comments
SMS self-signed certificates	Provisioned during registration of the IOFilter Provider.	In vSphere 7.0 and later, SMS self-signed certificates are stored in `/etc/vmware/ssl/iofilter vp_castore.pem`. Before vSphere 7.0, SMS self-signed certificates are stored in `/etc/vm ware/ssl/castore.pem`. In addition, the SMS Store can also store VVOL VASA Provider's (version 4.0 and earlier) self-signed certificates when retainVasaProviderCertificate=True.

ESX Certificates

ESX certificates are stored locally on each host in the `/etc/vmware/ssl` directory. ESX certificates are provisioned by VMCA by default, but you can use custom certificates instead. ESX certificates are provisioned when the host is first added to vCenter and when the host reconnects. For more information, see the *vSphere Security* documentation.

Machine SSL Certificates

The machine SSL certificate for each node is used to create an SSL socket on the server side. SSL clients connect to the SSL socket. The certificate is used for server verification and for secure communication such as HTTPS or LDAPS.

Each vCenter node has its own machine SSL certificate. All services that are running on a vCenter node use the machine SSL certificate to expose their SSL endpoints.

The following services use the machine SSL certificate.

- The reverse proxy service. SSL connections to individual vCenter services always go to the reverse proxy. Traffic does not go to the services themselves.
- The vCenter service (vpxd).
- The VMware Directory Service (vmdir).

VMware products use standard X.509 version 3 (X.509v3) certificates to encrypt session information. Session information is sent over SSL between components.

Solution User Certificates

A solution user encapsulates one or more vCenter services. Each solution user must be authenticated to vCenter Single Sign-On. Solution users use certificates to authenticate to vCenter Single Sign-On through SAML token exchange.

A solution user presents the certificate to vCenter Single Sign-On when it first has to authenticate, after a reboot, and after a timeout has elapsed. The timeout (Holder-of-Key Timeout) can be set from the vSphere Client and defaults to 2592000 seconds (30 days).

For example, the vpxd solution user presents its certificate to vCenter Single Sign-On when it connects to vCenter Single Sign-On. The vpxd solution user receives a SAML token from vCenter Single Sign-On and can then use that token to authenticate to other solution users and services.

The following solution user certificate stores are included in VECS:

- `machine`: Used by the license server and the logging service.

  Note:
  The machine solution user certificate has nothing to do with the machine SSL certificate. The machine solution user certificate is used for the SAML token exchange. The machine SSL certificate is used for secure SSL connections for a machine.
- `vpxd`: vCenter service daemon (vpxd) store. vpxd uses the solution user certificate that is stored in this store to authenticate to vCenter Single Sign-On.
- `vpxd-extension`: vCenter extensions store. Includes the Auto Deploy service, inventory service, and other services that are not part of other solution users.
- `vsphere-webclient`: vSphere Client store. Also includes some additional services such as the performance chart service.

- `wcp`: VMware vSphere® with VMware Tanzu™ store. Also used for vSphere Cluster Services.

Internal Certificates

vCenter Single Sign-On certificates are not stored in VECS and are not managed with certificate management tools. As a rule, changes are not necessary, but in special situations, you can replace these certificates.

vCenter Single Sign-On Signing Certificate	The vCenter Single Sign-On service includes an identity provider service that issues SAML tokens that are used for authentication throughout vSphere. A SAML token represents the user's identity, and also contains group membership information. When vCenter Single Sign-On issues SAML tokens, it signs each token with its signing certificate so that clients of vCenter Single Sign-On can verify that the SAML token comes from a trusted source. You can replace this certificate from the CLI. See Replace a vCenter STS Certificate Using the Command Line.
VMware Directory Service SSL Certificate	In vSphere 6.5 and later, the machine SSL certificate is used as the VMware directory certificate.
vSphere Virtual Machine Encryption Certificates	The vSphere Virtual Machine Encryption solution connects with a key server. Depending on how the solution authenticates to the key server, it might generate certificates and store them in VECS. See the *vSphere Security* documentation.

VMware Certificate Authority and VMware Core Identity Services

Core identity services are part of every vCenter system. VMware Certificate Authority (VMCA) is part of every VMware core identity services group. Use the management CLIs and the vSphere Client to interact with these services.

VMware core identity services include several components.

Table 82: Core Identity Services

Service	Description
VMware Directory Service (vmdir)	Identity source that handles SAML certificate management for authentication with vCenter Single Sign-On.
VMware Certificate Authority (VMCA)	Issues certificates for VMware solution users, machine certificates for machines on which services are running, and ESX host certificates. VMCA can be used as is, or as an intermediary certificate authority. VMCA issues certificates only to clients that can authenticate to vCenter Single Sign-On in the same domain.
VMware Authentication Framework Daemon (VMAFD)	Includes the VMware Endpoint Certificate Store (VECS) and several other authentication services. VMware administrators interact with VECS. The other services are used internally.

VMware Endpoint Certificate Store

VMware Endpoint Certificate Store (VECS) serves as a local (client-side) repository for certificates, private keys, and other certificate information that can be stored in a keystore. You can decide not to use VMCA as your certificate authority and certificate signer, but you must use VECS to store all vCenter certificates, keys, and so on. ESX certificates are stored locally on each host and not in VECS.

VECS runs as part of the VMware Authentication Framework Daemon (VMAFD). VECS runs on every vCenter node, and holds the keystores that contain the certificates and keys.

VECS polls VMware Directory Service (vmdir) periodically for updates to the trusted root store. You can also explicitly manage certificates and keys in VECS using `vecs-cli` commands. See vecs-cli Command Reference.

VECS includes the following stores.

Table 83: Stores in VECS

Store	Description
Machine SSL store (MACHINE_SSL_CERT)	• Used by the reverse proxy service on every vSphere node. • Used by the VMware Directory Service (vmdir) on each vCenter node. All services in vSphere 6.0 and later communicate through a reverse proxy, which uses the machine SSL certificate. For backward compatibility, the 5.x services still use specific ports. As a result, some services such as vpxd still have their own port open.
Solution user stores • `machine` • `vpxd` • `vpxd-extension` • `vsphere-webclient` • `wcp`	VECS includes one store for each solution user. The subject of each solution user certificate must be unique, for example, the machine certificate cannot have the same subject as the vpxd certificate. Solution user certificates are used for authentication with vCenter Single Sign-On. vCenter Single Sign-On checks that the certificate is valid, but does not check other certificate attributes. The following solution user certificate stores are included in VECS: • `machine`: Used by the license server and the logging service. **Note:** The machine solution user certificate has nothing to do with the machine SSL certificate. The machine solution user certificate is used for the SAML token exchange. The machine SSL certificate is used for secure SSL connections for a machine. • `vpxd`: vCenter service daemon (vpxd) store. vpxd uses the solution user certificate that is stored in this store to authenticate to vCenter Single Sign-On. • `vpxd-extension`: vCenter extensions store. Includes the Auto Deploy service, inventory service, and other services that are not part of other solution users. • `vsphere-webclient`: vSphere Client store. Also includes some additional services such as the performance chart service. • `wcp`: VMware vSphere® with VMware Tanzu™ store. Also used for vSphere Cluster Services. Each vCenter node includes a `machine` certificate.
Trusted root store (TRUSTED_ROOTS)	Contains all trusted root certificates.
vSphere Certificate Manager Utility backup store (BACKUP_STORE)	Used by VMCA (VMware Certificate Manager) to support certificate revert. Only the most recent state is stored as a backup, you cannot go back more than one step.
Other stores	Other stores might be added by solutions. For example, the Virtual Volumes solution adds an SMS store. Do not modify the certificates in those stores unless Broadcom documentation or a Broadcom Knowledge Base article instructs you to do so. **Note:** Deleting the TRUSTED_ROOTS_CRLS store can damage your certificate infrastructure. Do not delete or modify the TRUSTED_ROOTS_CRLS store.

The vCenter Single Sign-On service stores the token signing certificate and its SSL certificate on disk. You can change the token signing certificate from the CLI.

Some certificates are stored on the file system, either temporarily during startup or permanently. Do not change the certificates on the file system.

Note:
Do not change any certificate files on disk unless instructed by VMware documentation or Knowledge Base Articles. Unpredictable behavior might result otherwise.

Managing vSphere Certificate Revocation

If you suspect that one of your certificates has been compromised, replace all existing certificates, including the VMCA root certificate.

vSphere supports replacing certificates but does not enforce certificate revocation for ESX hosts or for vCenter systems.

Remove revoked certificates from all nodes. If you do not remove revoked certificates, a man-in-the-middle attack might enable compromise through impersonation with the account's credentials.

Replacing vSphere Certificates in Large Deployments

When replacing certificates in deployments with large numbers of vCenter instances, you can use the vSphere Certificate Management utility or replace certificates manually using the CLIs. Some best practices guide the process you choose.

Replacement of Machine SSL Certificates in Environments with Multiple vCenter Instances

If your environment includes multiple vCenter instances, you can replace machine SSL certificates with the vSphere Client or the vSphere Certificate Manager utility, or manually with CLI commands.

Using the vSphere Certificate Manager to Replace Machine SSL Certificates on Multiple vCenter Instances	You run vSphere Certificate Manager on each instance. Depending on the task you perform, you are also prompted for certificate information. See the following topics for details: • Replace VMCA Root Certificate with Custom Signing Certificate and Replace All Certificates Using the Certificate Manager • Replace Machine SSL Certificate with VMCA Certificate (Intermediate CA) Using the Certificate Manager • Replace Solution User Certificates with VMCA Certificates (Intermediate CA) Using the Certificate Manager
Using the CLI to Manually Replace Machine SSL Certificates on Multiple vCenter Instances	For manual certificate replacement, you run the certificate replacement CLI commands on each instance. See the following topics for details: • Replace Machine SSL Certificates with VMCA-Signed Certificates Using the CLI • Replace Machine SSL Certificates (Intermediate CA) Using the CLI • Replace Machine SSL Certificates with Custom Certificates Using the CLI

Replacement of Solution User Certificates in Environments with Multiple vCenter Instances in Enhanced Linked Mode

If your environment includes multiple vCenter instances in enhanced linked mode, follow these steps for replacing solution user certificates.

Note:

ELM is not supported in 9.0.

Note:
When you list solution user certificates in large deployments, the output of `/usr/lib/vmware-vmafd/bin/dir-cli list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

Using the vSphere Certificate Manager to Replace Machine SSL	You run vSphere Certificate Manager on each instance. Depending on the task you perform, you are also prompted for certificate information. See Managing Certificates Using the vSphere Certificate Manager Utility.

Certificates on vCenter Instances in ELM

Using the CLI to Manually Replace Machine SSL Certificates on vCenter Instances in ELM

The high-level steps to manually replace machine SSL certificates on vCenter in ELM include:

1. Generating or requesting a certificate.
   You need the following certificates:
   - A certificate for the machine solution user on each vCenter.
   - A certificate for each of the following solution users on each node:
     - `vpxd` solution user
     - `vpxd-extension` solution user
     - `vsphere-webclient` solution user
     - `wcp` solution user
2. Using the CLI commands to replace the certificates on each node.
   The precise process depends on the type of certificate replacement that you are performing. See the following topics for details:
   - Replace Solution User Certificates with New VMCA-Signed Certificates Using the CLI
   - Replace Solution User Certificates (Intermediate CA) Using the CLI
   - Replace Solution User Certificates with Custom Certificates Using the Certificate Manager

Certificate Replacement in VMware Environments That Include External Solutions

Some solutions, such as VMware Live Recovery or VMware vSphere Replication, are always installed on a different machine than the vCenter instance. If you replace the default machine SSL certificate on the vCenter instance, a connection error results if the solution attempts to connect to the vCenter instance.

You can run the `ls_update_certs` script to resolve the issue. See the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2109074.

Managing Certificates Using the vSphere Client

You can view and manage certificates by using the vSphere Client.

The vSphere Client enables you to perform these management tasks.

- View the machine SSL, VMware Certificate Authority (VMCA) root, Trusted Root, and Security Token Service (STS) certificates.
- Add new Trusted Root certificates, and renew or replace existing machine SSL and STS certificates.
- Generate a custom Certificate Signing Request (CSR) for a machine SSL certificate and replace the certificate when the Certificate Authority returns it.

Most parts of the certificate replacement workflows are supported fully from the vSphere Client. Other certificate replacement workflows are supported by the vSphere Certificate Manager utility. See Managing Certificates Using the vSphere Certificate Manager Utility.

To understand more about options for replacing the default certificates, see Replacing vSphere Certificates.

> **Note:**
>
> If you use the VMCA as an intermediate CA, or use custom certificates, you might encounter significant complexity and the potential for a negative impact to your security, and an unnecessary increase in your operational risk. For more information about managing certificates within a vSphere environment, see the blog post titled *New Product Walkthrough - Hybrid vSphere SSL Certificate Replacement* at https://blogs.vmware.com/vsphere/2017/01/walkthrough-hybrid-ssl-certificate-replacement.html.

Explore Certificate Stores Using the vSphere Client

A VMware Endpoint Certificate Store (VECS) instance is included on each vCenter instance. You can explore the different stores inside the VMware Endpoint Certificate Store from the vSphere Client, including machine SSL, STS, and trusted root certificates.

For most management tasks, you must have the password for the administrator for the local domain account, administrator@vsphere.local or a different domain if you changed the domain during installation.

See VMware Endpoint Certificate Store for details on the different stores inside VECS.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Explore the certificates stored inside the VMware Endpoint Certificate Store (VECS).

   VMware Endpoint Certificate Store explains what is in the individual stores.
6. To view details for a certificate, select the appropriate certificate tab, select the certificate, and expand the certificate to view details.

Set the Threshold for vCenter Certificate Expiration Warnings Using the vSphere Client

vCenter monitors all certificates in the VMware Endpoint Certificate Store (VECS) and issues an alarm when a certificate is 30 days or less from its expiration. You can use the vSphere Client to change how soon you are warned with the vpxd.cert.threshold advanced option.

1. Log in to the vSphere Client.
2. Select the vCenter object and click **Configure**.
3. Click **Advanced Settings**.
4. Click **Edit Settings** and filter for `threshold`.
5. Change the setting of vpxd.cert.threshold to the desired value and click **Save**.

Renew VMCA Certificates with New VMCA-Signed Certificates Using the vSphere Client

You can replace all VMCA-signed certificates with new VMCA-signed certificates. This process is called renewing certificates. You can renew selected certificates or all certificates in your environment from the vSphere Client.

For certificate management, you have to supply the password of the administrator of the local domain (administrator@vsphere.local by default). If you are renewing certificates for a vCenter system, you also have to supply the vCenter Single Sign-On credentials for a user with administrator privileges on the vCenter system.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.

3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Renew the VMCA-signed machine SSL certificate for the local system.
   a) From the **Machine SSL** tab, select the desired certificate and click **Renew**.
   b) Specify the duration of the certificate in days.
   c) Click the checkbox to acknowledge that you have backed up vCenter and its databases.
   d) Click **Renew**.
      The system renews the certificate and displays a success message.
   e) When the certificate has been changed message appears, click **Refresh** to refresh your browser.

Replace Certificates with Custom Certificates Using the vSphere Client

You can use the vSphere Client to replace the default certificates with custom certificates.

You can use the vSphere Client to generate CSRs for each machine, and replace certificates when you receive them from your internal or third-party Certificate Authority (CA). When you submit the CSRs to your internal or third-party CA, the CA returns signed certificates and the root certificate. You can upload both the root certificate and the signed certificates from the vSphere Client.

Generate Certificate Signing Request for Machine SSL Certificate Using the vSphere Client (Custom Certificates)

The machine SSL certificate is used by the reverse proxy service on every vCenter node. Each machine must have a machine SSL certificate for secure communication with other services. You can use the vSphere Client to generate a Certificate Signing Request (CSR) for the machine SSL certificate and to replace the certificate once it is ready.

The certificate must meet the following requirements:

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded). The vSphere Client and API still accept a key size up to 16384 bits when generating the Certificate Signing Request.
- CRT format
- x509 version 3
- SubjectAltName must contain DNS Name=<machine_FQDN>.
- Contains the following Key Usages: Digital Signature, Key Encipherment

Note:
vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. Enter the credentials of your vCenter.
5. Generate the CSR.
   a) Under the **Machine SSL** tab, select the desired certificate and click **Generate Certificate Signing Request (CSR)**.
   b) Enter your certificate information and click **Next**.
      2048 (bits) is the default value for the key size. Change this value as required.

Note:
When you use vCenter to generate a CSR with a large key size, the generation takes a few minutes to complete because of the CPU-intensive nature of the operation.

c) Copy or download the CSR.
d) Click **Finish**.
e) Provide the CSR to your Certificate Authority.

When the Certificate Authority returns the certificate, replace the existing certificate in the certificate store. See Add Custom Certificates Using the vSphere Client.

Add a Trusted Root Certificate to the Certificate Store Using the vSphere Client

If you want to use third-party certificates in your environment, you must add a trusted root certificate to the certificate store. You can do so using the vSphere Client.

Obtain the custom root certificate from your third-party or in-house certificate authority (CA).

vSphere accepts only valid CA certificates for import. To be valid, a CA certificate must have the CA bit and the keyCertSign bit set in the basic constraint and the key usage X.509 v3 certificate extensions respectively. This implies that the certificate is a CA and its purpose is for certificate signing. See https://www.rfc-editor.org/rfc/rfc5280 for more information.

Ensure that the keyCertSign bit is set for all the certificates in the chain.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@`mydomain`.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **Trusted Root** tab, click **Add Trusted Root Certificate**.
6. Click **Browse** and select the location of the certificate chain.
   You can use a file of type CER, PEM, or CRT.
7. Click **Add**.
   The certificate is added to the store.

   Note:
   In vSphere 8.0 Update 2 and later, the **Start Root certificate push to vCenter Hosts** check box is removed. vCenter pushes the root certificates to all connected hosts in the inventory when a certificate is added. When a host with different root certificates from vCenter is connected, vCenter pushes the root certificates to correct this difference. In this case, vCenter root certificates overwrite the ones on the host, so administrators can ensure that any custom root certificates needed throughout the inventory are added to vCenter.

Add Custom Certificates Using the vSphere Client

You can use the vSphere Client to add custom Machine SSL certificates to the certificate store.

Generate certificate signing requests (CSRs) for each certificate that you want to replace. See Generate Certificate Signing Request for Machine SSL Certificate Using the vSphere Client (Custom Certificates). Place the certificate and private key in a location that the vCenter can access.

Usually, replacing the machine SSL certificate for each component is sufficient.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **Machine SSL** tab, select the certificate then click **Import and Replace Certificate**.
6. Click the appropriate certificate replacement option and click **Next**.

Option	Description
Replace with VMCA certificate	Creates a VMCA-generated CSR to replace the current certificate.
Replace with external CA certificate where CSR is generated from vCenter (private key embedded)	Use a certificate signed using a vCenter generated CSR to replace the current certificate.
Replace with external CA certificate (requires private key)	Use a certificate signed by an external CA to replace the current certificate.

7. Enter the CSR information, or upload the appropriate certificates.
8. Click the checkbox to acknowledge that you have backed up vCenter and its databases.
9. Review the information and click **Finish**.
   The system replaces the certificate and displays a success message.
10. When the certificate has been changed message appears, click **Refresh** to refresh your browser.

Generate a VMCA Leaf Certificate

You can generate a leaf certificate that is signed by the VMware Certificate Authority (VMCA) for use in your VMware infrastructure.

Generate a Certificate Signing Request (CSR) on the host in your VMware infrastructure where you want to install the leaf certificate.

In addition to VMware Certificate Authority (VMCA) handling all certificate management, it can generate leaf certificates. Leaf certificates are signed by VMCA and are used to identify other VMware resources. VMCA-generated leaf certificates are not stored in VECS. Also, vCenter does not track these leaf certificates for expiration.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **Trusted Root** tab, select the VMCA root certificate and click **Issue New Leaf Certificate**.
6. Browse for the CSR that you previously generated, specify a duration, then click **Next**.
7. Click **Download Certificates** to save the Leaf and Root certificates.

The generated Leaf and Root certificates are created and downloaded to the specified location.

Import the Leaf and Root certificates to the target host in your VMware infrastructure.

Managing Certificates Using the vSphere Certificate Manager Utility

The vSphere Certificate Manager utility allows you to perform most certificate management tasks interactively from the command line. vSphere Certificate Manager prompts you for the task to perform, for certificate locations and other information as needed, and then stops and starts services and replaces certificates for you.

To understand more about options for replacing the default certificates, see Replacing vSphere Certificates.

> **Note:**
> If you use the VMCA as an intermediate CA, or use custom certificates, you might encounter significant complexity and the potential for a negative impact to your security, and an unnecessary increase in your operational risk. For more information about managing certificates within a vSphere environment, see the blog post titled *New Product Walkthrough - Hybrid vSphere SSL Certificate Replacement* at http://vmware.com/go/hybridvmca.

If you use vSphere Certificate Manager, you are not responsible for placing the certificates in VECS (VMware Endpoint Certificate Store) and you are not responsible for starting and stopping services.

You run vSphere Certificate Manager options in sequence to complete a workflow. Several options, for example, generating CSRs, are used in different workflows. Before you run vSphere Certificate Manager, be sure that you understand the replacement process and procure the certificates that you want to use.

> **CAUTION:**
> vSphere Certificate Manager supports one level of revert. If you run vSphere Certificate Manager twice and notice that you unintentionally corrupted your environment, the tool cannot revert the first of the two runs.

vSphere Certificate Manager Utility Location

The vSphere Certificate Manager utility is located at:

```
/usr/lib/vmware-vmca/bin/certificate-manager
```

> **Note:**
>
> When running vSphere Certificate Manager, some options prompt you as follows:
>
> ```
> Enter proper value for VMCA 'Name':
> ```
>
> Respond to this prompt by entering the fully qualified domain name of the machine on which the certificate configuration is running.

Workflows in the vSphere Certificate Manager Utility

The following table presents an overview of the certificate replacement workflows you can accomplish by using the vSphere Certificate Manager utility.

Table 84: Workflows in the vSphere Certificate Management Utility

Workflow	Description	See
Replacing VMCA root certificate with custom signing certificate and replacing all certificates	To generate the VMCA root certificate, and replace all certificates, use Option 4, Regenerate a new VMCA Root certificate and replace all Certificates.	Regenerate a New VMCA Root Certificate and Replace All Certificates Using the Certificate Manager
Making VMCA an intermediate certificate authority	To make VMCA an intermediate CA, you must run the vSphere Certificate Manager utility several times and use multiple options. This workflow gives the complete set of steps for replacing both machine SSL certificates and solution user certificates.	Make VMCA an Intermediate Certificate Authority Using the Certificate Manager

Workflow	Description	See
Replacing all certificates with custom certificates	To replace all certificates with custom certificates, you must run the vSphere Certificate Manager utility several times and use multiple options. This workflow gives the complete set of steps for replacing both machine SSL certificates and solution user certificates.	Replace All Certificates with a Custom Certificate Using the Certificate Manager
Reverting the last performed operation	To revert the last performed certificate operation and return to the previous state, use Option 7, Revert last performed operation by re-publishing old certificates.	Revert Last Performed Operation by Republishing Old Certificates Using the Certificate Manager
Resetting all certificates	To replace all existing vCenter certificates with certificates that are signed by VMCA, use Option 8, Reset all Certificates.	Reset All Certificates Using the Certificate Manager

Regenerate a New VMCA Root Certificate and Replace All Certificates Using the Certificate Manager

You can use the vSphere Certificate Manager utility to regenerate the VMCA root certificate, and replace the local machine SSL certificate and the local solution user certificates with VMCA-signed certificates. When multiple vCenter instances are connected in Enhanced Linked Mode configuration, you must replace certificates on each vCenter.

You must know the following information when you run vSphere Certificate Manager with this option.

- Password for administrator@vsphere.local.
- The FQDN of the machine for which you want to generate a new VMCA-signed certificate. All other properties default to the predefined values but can be changed.

When you replace the existing machine SSL certificate with a new VMCA-signed certificate, vSphere Certificate Manager prompts you for information and enters all values, except for the password and the IP address of the vCenter, into the `certool.cfg` file.

- Password for administrator@vsphere.local
- Two-letter country code
- Company name
- Organization name
- Organization unit
- State
- Locality
- IP address (optional)
- Email (optional)
- Host name, that is, the fully qualified domain name of the machine for which you want to replace the certificate. If the host name does not match the FQDN, certificate replacement does not complete correctly and your environment might end up in an unstable state.
- IP address of vCenter
- VMCA name, that is, the fully qualified domain name of the machine on which the certificate configuration is running.
- OU (organizationalUnitName) (optional)

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select option 4, Regenerate a new VMCA Root Certificate and replace all certificates.
3. Enter the administrator user and password.
4. Respond to the prompts.

   vSphere Certificate Manager generates a new VMCA root certificate based on your input and replaces all certificates on the system where you are running vSphere Certificate Manager. The replacement process is complete after vSphere Certificate Manager has restarted the services.
5. To replace the machine SSL certificate, run vSphere Certificate Manager with option 3, Replace Machine SSL certificate with VMCA Certificate.
6. To replace the solution user certificates, run Certificate Manager with option 6, Replace Solution user certificates with VMCA certificates.

Make VMCA an Intermediate Certificate Authority Using the Certificate Manager

You can use the vSphere Certificate Manager utility to make VMCA an Intermediate CA. After you complete the process, VMCA signs all new certificates with the full chain. If you want, you can use vSphere Certificate Manager to replace all existing certificates with new VMCA-signed certificates.

To make VMCA an intermediate CA, you must run vSphere Certificate Manager several times. The high-level steps for replacing both machine SSL certificates and solution user certificates include:

1. Launching the vSphere Certificate Manager utility.
2. Generating a CSR by running Option 2, Replace VMCA Root certificate with Custom Signing Certificate and replace all Certificates. You might have to provide some information about the certificate next. When prompted for an option again, select Option 1, Generate Certificate Signing Request(s) and Key(s) for VMCA Root Signing certificate.
3. Submitting the CSR to your external or enterprise CA. You receive a signed certificate and a root certificate from the CA.
4. Combining the VMCA root certificate with the CA root certificate and saving the file.
5. Replacing certificates by running Option 2, Replace VMCA Root certificate with Custom Signing Certificate and replace all Certificates, and following the prompts. This process replaces all certificates on the local machine.
6. (Optional) Replacing certificates on each node when multiple vCenter instances are connected in Enhanced Linked Mode configuration by:
   a. First replacing the machine SSL certificate with the (new) VMCA certificate (Option 3, Replace Machine SSL certificate with VMCA Certificate).
   b. Then replacing the solution user certificates with the (new) VMCA certificate (Option 6, Replace Solution user certificates with VMCA certificates).

Generate CSR Using the Certificate Manager and Prepare Root Certificate (Intermediate CA)

You can use the vSphere Certificate Manager utility to generate Certificate Signing Requests (CSRs). Submit those CSRs to your enterprise CA or to an external certificate authority for signing. You can use the signed certificates with the different supported certificate replacement processes.

vSphere Certificate Manager prompts you for information. The prompts depend on your environment and on the type of certificate that you want to replace.

For any CSR generation, you are prompted for the password of the administrator@vsphere.local user, or for the administrator of the vCenter Single Sign-On domain that you are connecting to.

- You can use vSphere Certificate Manager to create the CSR.

  Note:
  In vSphere 8.0 and later, if you use the vSphere Certificate Manager to generate the CSR, the minimum key size is changed to 3072 bits from 2048. In vSphere 8.0 Update 1 and later, use the vSphere Client to generate a CSR with a key size of 2048 bits.

  Note:
  vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

- If you prefer to create the CSR manually, the certificate that you send to be signed must meet the following requirements.
  - Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
  - PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
  - x509 version 3
  - The CA extension must be set to true for root certificates, and cert sign must be in the list of requirements. For example:

    ```
    basicConstraints        = critical,CA:true
    keyUsage                = critical,digitalSignature,keyCertSign
    ```
  - CRL signing must be enabled.
  - Extended Key Usage can be either empty or contain Server Authentication.
  - No explicit limit to the length of the certificate chain. VMCA uses the OpenSSL default, which is 10 certificates.
  - Certificates with wildcards or with more than one DNS name are not supported.
  - You cannot create subsidiary CAs of VMCA.

    See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112009, Creating a Microsoft Certificate Authority Template for SSL certificate creation in vSphere 6.x, for an example using Microsoft Certificate Authority.

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```
2. Select Option 2, Replace VMCA Root certificate with Custom Signing Certificate and replace all Certificates.

   Initially, you use this option to generate the CSR, not to replace certificates.
3. Enter the administrator user and password.
4. Select Option 1, Generate Certificate Signing Request(s) and Key(s) for VMCA Root Signing certificate, to generate the CSR and answer the prompts.

   As part of the process, you have to provide a directory. vSphere Certificate Manager places the certificate to be signed (`*.csr` file) and the corresponding key file (`*.key` file) in the directory.
5. Name the certificate signing request (CSR) `root_signing_cert.csr`.
6. Send the CSR to your enterprise or external CA for signing and name the resulting signed certificate `root_signing_cert.cer`.
7. In a text editor, combine the certificates as follows.

   ```
   -----BEGIN CERTIFICATE-----
   Signed VMCA root certificate
   -----END CERTIFICATE-----
   -----BEGIN CERTIFICATE-----
   CA intermediate certificates
   -----END CERTIFICATE-----
   -----BEGIN CERTIFICATE-----
   Root certificate of enterprise or external CA
   -----END CERTIFICATE-----
   ```
8. Save the file as `root_signing_chain.cer`.

Replace the existing root certificate with the chained root certificate. See Replace VMCA Root Certificate with Custom Signing Certificate and Replace All Certificates Using the Certificate Manager.

Replace VMCA Root Certificate with Custom Signing Certificate and Replace All Certificates Using the Certificate Manager

You can use the vSphere Certificate Manager utility to generate a CSR and send the CSR to an enterprise or third-party CA for signing. You can then replace the VMCA root certificate with a custom signing certificate and replace all existing certificates with certificates that are signed by the custom CA.

- Generate the certificate chain.
  - You can use vSphere Certificate Manager to create the CSR or create the CSR manually.
  - After you receive the signed certificate from your third-party or enterprise CA, combine it with the initial VMCA root certificate to create the full chain.

    See Generate CSR Using the Certificate Manager and Prepare Root Certificate (Intermediate CA) for certificate requirements and the process of combining the certificates.
- Gather the information that you need.
  - Password for administrator@vsphere.local
  - Valid custom certificate for Root (`.crt` file)
  - Valid custom key for Root (`.key` file)

You run vSphere Certificate Manager on vCenter to replace the VMCA root certificate with a custom signing certificate.

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```
2. Select Option 2, Replace VMCA Root certificate with Custom Signing Certificate and replace all Certificates.
3. Enter the administrator user and password.
4. Select Option 2, Import custom certificate(s) and key(s) to replace existing VMCA Root Signing certificate, and respond to the prompts.
   a) Specify the full path to the root certificate when prompted.
   b) If you are replacing certificates for the first time, you are prompted for information to be used for the machine SSL certificate.

      This information includes the required FQDN of the machine and is stored in the `certool.cfg` file.

Replace Machine SSL Certificate with VMCA Certificate (Intermediate CA) Using the Certificate Manager

When you use VMCA as an intermediate CA, you can replace the machine SSL certificate explicitly using the vSphere Certificate Manager utility. First you replace the VMCA root certificate on the vCenter, then you can replace the machine SSL certificate, which will be signed by the new root of the VMCA. You can also use this option to replace machine SSL certificates that are corrupt or about to expire.

- You must know the following information to run vSphere Certificate Manager with this option.
  - Password for administrator@vsphere.local.
  - The FQDN of the machine for which you want to generate a new VMCA-signed certificate. All other properties default to the predefined values but can be changed.
  - Host name or IP address of the vCenter system.

When you replace the existing machine SSL certificate with a new VMCA-signed certificate, vSphere Certificate Manager prompts you for information and enters all values, except for the password and the IP address of the vCenter, into the `certool.cfg` file.

- Password for administrator@vsphere.local
- Two-letter country code
- Company name
- Organization name
- Organization unit
- State
- Locality
- IP address (optional)
- Email
- Host name, that is, the fully qualified domain name of the machine for which you want to replace the certificate. If the host name does not match the FQDN, certificate replacement does not complete correctly and your environment might end up in an unstable state.
- IP address of vCenter

- VMCA name, that is, the fully qualified domain name of the machine on which the certificate configuration is running.

  Note:
  The OU (organizationalUnitName) field is no longer mandatory.

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select Option 3, Replace Machine SSL certificate with VMCA Certificate.
3. Enter the administrator user and password.
4. Respond to the prompts.

   vSphere Certificate Manager stores the information in the `certool.cfg` file.

vSphere Certificate Manager replaces the machine SSL certificate.

Replace Solution User Certificates with VMCA Certificates (Intermediate CA) Using the Certificate Manager

When you use VMCA as an intermediate CA, you can replace the solution user certificate explicitly using the vSphere Certificate Manager utility. First you replace the VMCA root certificate on the vCenter, then you can replace the solution user certificate, which is signed by the new root of the VMCA. You can also use this option to replace solution certificates that are corrupt or about to expire.

- Restart all vCenter nodes explicitly if you replaced the VMCA root certificate in a deployment consisting of multiple instances of vCenter in Enhanced Linked Mode configuration.
- You must know the following information to run vSphere Certificate Manager with this option.
  - Password for administrator@vsphere.local
  - Host name or IP address of the vCenter system

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select option 6, Replace Solution user certificates with VMCA certificates.
3. Enter the administrator user and password.
4. Respond to the prompts.

   See the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112281 for more information.

vSphere Certificate Manager replaces all solution user certificates.

Replace All Certificates with a Custom Certificate Using the Certificate Manager

You can use the vSphere Certificate Manager utility to replace all certificates with custom certificates. Before you start the process, you must send CSRs to your certificate authority (CA). You can use Certificate Manager to generate the CSRs.

One option is to replace only the machine SSL certificate, and to use the solution user certificates that are provisioned by VMCA. Solution user certificates are used only for communication between vSphere components.

When you use custom certificates, you replace the VMCA-signed certificates with custom certificates. You can use the vSphere Client, the vSphere Certificate Manager utility, or CLIs for manual certificate replacement. Certificates are stored in VECS.

To replace all certificates with custom certificates, you must run the vSphere Certificate Manager utility several times. The high-level steps for replacing both machine SSL certificates and solution user certificates include:

1. Launching the vSphere Certificate Manager utility.
2. Generating certificate signing requests for the machine SSL certificate and the solution user certificates separately on each machine.

a. To generate CSRs for the machine SSL certificate, select Option 1, Replace Machine SSL certificate with Custom Certificate. When prompted for an option again, select Option 1, Generate Certificate Signing Request(s) and Key(s) for Machine SSL certificate.
b. If company policy does not allow a hybrid deployment, select Option 5, Replace Solution user certificates with Custom Certificate.

3. Submitting the CSR to your external or enterprise CA. You receive a signed certificate and a root certificate from the CA.
4. After receiving the signed certificates and the root certificate from your CA, replacing the machine SSL certificate on each machine by using Option 1, Replace Machine SSL certificate with Custom Certificate.
5. If you also want to replace the solution user certificates, select Option 5, Replace Solution user certificates with Custom Certificate.
6. Finally, when multiple vCenter instances are connected in Enhanced Linked Mode configuration, repeating the process on each node.

Generate Certificate Signing Requests Using the Certificate Manager (Custom Certificates)

You can use the vSphere Certificate Manager utility to generate Certificate Signing Requests (CSRs) that you can then use with your enterprise CA or send to an external certificate authority. You can use the certificates with the different supported certificate replacement processes.

vSphere Certificate Manager prompts you for information. The prompts depend on your environment and on the type of certificate you want to replace.

- For any CSR generation, you are prompted for the password of the administrator@vsphere.local user, or for the administrator of the vCenter Single Sign-On domain that you are connecting to.
- You are prompted for the host name or IP address of the vCenter.
- To generate a CSR for a machine SSL certificate, you are prompted for certificate properties, which are stored in the `certool.cfg` file. For most fields, you can accept the default or provide site-specific values. The FQDN of the machine is required.

  Note:
  In vSphere 8.0 and later, if you use the vSphere Certificate Manager to generate the CSR, the minimum key size is changed to 3072 bits from 2048. In vSphere 8.0 Update 1 and later, use the vSphere Client to generate a CSR with a key size of 2048 bits.

  Note:
  vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

1. Log in to each vCenter (the vCenter shell) in your environment and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Selection Option 1, Replace Machine SSL certificate with Custom Certificate.
3. Enter the administrator user and password.
4. Select Option 1, Generate Certificate Signing Request(s) and Key(s) for Machine SSL certificate, to generate the CSR, answer the prompts and exit vSphere Certificate Manager.

   As part of the process, you have to provide a directory. vSphere Certificate Manager places the certificate and key files in the directory.
5. If you also want to replace all solution user certificates, restart vSphere Certificate Manager and selection Option 5, Replace Solution user certificates with Custom Certificate.
6. Supply the password and the vCenter IP address or host name if prompted.
7. Select Option 1, Generate Certificate Signing Request(s) and Key(s) for Solution User Certificates, to generate the CSRs, answer the prompts and exit vSphere Certificate Manager.

   As part of the process, you have to provide a directory. Certificate Manager places the certificate and key files in the directory.

To perform certificate replacement, see Replace Machine SSL Certificate with Custom Certificate Using the Certificate Manager.

Replace Machine SSL Certificate with Custom Certificate Using the Certificate Manager

You can use the vSphere Certificate Manager utility to replace the machine SSL certificate on each node with a custom certificate. The machine SSL certificate is used by the reverse proxy service on every vCenter node. Each machine must have a machine SSL certificate for secure communication with other services.

Before you start, you need a CSR for each machine in your environment. You can generate the CSR using vSphere Certificate Manager or explicitly.

1. To generate the CSR using vSphere Certificate Manager, see Generate Certificate Signing Requests Using the Certificate Manager (Custom Certificates).
2. To generate the CSR explicitly, request a certificate for each machine from your third-party or enterprise CA. The certificate must meet the following requirements:
   – Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
   – CRT format
   – x509 version 3
   – SubjectAltName must contain DNS Name=<machine_FQDN>.
   – Contains the following Key Usages: Digital Signature, Key Encipherment

See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112014, Obtaining vSphere certificates from a Microsoft Certificate Authority.

1. Log in to vCenter and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Selection Option 1, Replace Machine SSL certificate with Custom Certificate.
3. Enter the administrator user and password.
4. Select Option 2, Import custom certificate(s) and key(s) to replace existing Machine SSL certificate, to start certificate replacement and respond to the prompts.

   vSphere Certificate Manager prompts you for the following information:

   - Password for administrator@vsphere.local
   - Valid Machine SSL custom certificate (`.crt` file)
   - Valid Machine SSL custom key (`.key` file)
   - Valid signing certificate for the custom machine SSL certificate (`.crt` file)
   - IP address of the vCenter

Replace Solution User Certificates with Custom Certificates Using the Certificate Manager

Many companies only require that you replace certificates of services that are accessible externally. However, the vSphere Certificate Manager utility also supports replacing solution user certificates. Solution users are collections of services, for example, all services that are associated with the vSphere Client.

Before you start, you need a CSR for each machine in your environment. You can generate the CSR using vSphere Certificate Manager or explicitly.

1. To generate the CSR using vSphere Certificate Manager, see Generate Certificate Signing Requests Using the Certificate Manager (Custom Certificates).
2. Request a certificate for each solution user on each node from your third-party or enterprise CA. You can generate the CSR using vSphere Certificate Manager or prepare it yourself. The CSR must meet the following requirements:
   – Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
   – CRT format
   – x509 version 3
   – SubjectAltName must contain DNS Name=<machine_FQDN>.
   – Each solution user certificate must have a different `Subject`. Consider, for example, including the solution user name (such as vpxd) or other unique identifier.
   – Contains the following Key Usages: Digital Signature, Key Encipherment

See also the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112014, Obtaining vSphere certificates from a Microsoft Certificate Authority.

When you are prompted for a solution user certificate, provide the complete signing certificate chain of the third-party CA.

The format looks similar to the following.

```
-----BEGIN CERTIFICATE-----
Signing certificate
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
CA intermediate certificates
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
Root certificate of enterprise or external CA
-----END CERTIFICATE-----
```

1. Log in to vCenter and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select Option 5, Replace Solution user certificates with Custom Certificate.
3. Enter the SSO user and password.
4. Select Option 2, Import custom certificate(s) and key(s) to replace existing Solution User Certificates, and respond to the prompts.

   vSphere Certificate Manager prompts you for the following information:

   - Password for administrator@vsphere.local
   - Certificate and key for machine solution user
   - The certificate and key (`vpxd.crt` and `vpxd.key`) for the machine solution user
   - The full set of certificates and keys (`vpxd.crt` and `vpxd.key`) for all solution users

Revert Last Performed Operation by Republishing Old Certificates Using the Certificate Manager

When you perform a certificate management operation by using the vSphere Certificate Manager utility, the current certificate state is stored in the BACKUP_STORE store in VECS before certificates are replaced. You can revert the last performed operation and return to the previous state.

> **Note:**
> The revert operation restores what is currently in the BACKUP_STORE. If you run vSphere Certificate Manager with two different options and you then attempt to revert, only the last operation is reverted.

1. Log in to the vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select Option 7, Revert last performed operation by re-publishing old certificates.
3. Enter the administrator user and password.
4. To continue, enter `Y`.

Reset All Certificates Using the Certificate Manager

You can use the vSphere Certificate Manager utility to replace all existing vCenter certificates with certificates that are signed by VMCA.

When you use this option, you overwrite all custom certificates that are currently in VMware Endpoint Certificate Store (VECS).
vSphere Certificate Manager can replace all certificates. Which certificates are replaced depends on which options you select.

1. Log in to vCenter shell and start the vSphere Certificate Manager.

   ```
   /usr/lib/vmware-vmca/bin/certificate-manager
   ```

2. Select Option 8, Reset all Certificates.
3. Enter the administrator user and password.
4. When prompted, enter your certificate information.

After certificates are replaced and services are restarted, verify your certificate information.

Manual vSphere Certificate Replacement

For some special certificate replacement cases, you cannot use the vSphere Certificate Manager utility. Instead, you can use the CLIs included with your installation for certificate replacement.

Guidelines for Stopping and Starting of vCenter Services

For certain parts of manual certificate replacement, you must stop all vCenter services and then start only the services that manage the certificate infrastructure. If you stop services only when needed, you can minimize downtime.

You have to stop and start services as part of the certificate replacement process. You can use the `service-control` command for starting and stopping services. You can start and stop all services or individual services. See the command-line help for more information.

Follow these guidelines.

- Do not stop services to generate new public/private key pairs or new certificates.
- If you are the only administrator, you do not have to stop services when you add a new root certificate. The old root certificate remains available, and all services can still authenticate with that certificate.
- Stop services right before you delete a machine SSL certificate in VMware Endpoint Certificate Store (VECS).

Replace Existing VMCA-Signed Certificates with New VMCA-Signed Certificates Using the CLI

If the VMware Certificate Authority (VMCA) root certificate expires in the near future, or if you want to replace it for other reasons, you can use the CLI to generate a new root certificate and add it to the VMware Directory Service. You can then generate new machine SSL certificates and solution user certificates using the new root certificate. Starting with VCF 9.1, the VMCA and solution certificates that are due to expire within two years are automatically renewed, and replaced when you upgrade your vCenter to major or minor version. A new VMCA root certificate is provisioned if the expiry threshold is met.

Only administrator@vsphere.local or other users in the CAAdmins group can perform certificate management tasks. See Add Members to a vCenter Single Sign-On Group.

Use the vSphere Certificate Manager utility to replace certificates for most cases.

If you need fine-grained control, this scenario gives detailed step-by-step instructions for replacing the complete set of certificates using CLI commands. You can instead replace only individual certificates using the procedure in the corresponding task.

Generate a New VMCA-Signed Root Certificate Using the CLI

You can generate new VMCA-signed certificates with the `certool` CLI and publish the certificates to vmdir.

1. On the vCenter, generate a new self-signed certificate and private key.

   ```
   certool --genselfcacert --outprivkey <key_file_path> --outcert <cert_file_path> --config <config_file>
   ```

2. Replace the existing root certificate with the new certificate.

   ```
   certool --rootca --cert <cert_file_path> --privkey <key_file_path>
   ```

   The command generates the certificate, adds it to vmdir, and adds it to VECS.

3. Stop all services and start the services that handle certificate creation, propagation, and storage.

   ```
   service-control --stop --all
   ```

```
service-control --start vmafdd
service-control --start vmdird
service-control --start vmcad
```

4. Optional: Publish the new root certificate to vmdir.

   ```
   dir-cli trustedcert publish --cert newRoot.crt
   ```

   The command updates all instances of vmdir immediately. If you do not run the command, propagation of the new certificate to all nodes might take a while.

5. Restart all services.

   ```
   service-control --start --all
   ```

Generate a New VMCA-Signed Root Certificate

The following example shows all the steps for verifying the current root CA information, and for regenerating the root certificate.

1. (Optional) On the vCenter, list the VMCA root certificate to make sure it is in the certificate store.

   ```
   /usr/lib/vmware-vmca/bin/certool --getrootca
   ```

   The output looks similar to this:

   ```
   output:
   Certificate:
       Data:
           Version: 3 (0x2)
           Serial Number:
               cf:2d:ff:49:88:50:e5:af
       ...
   ```

2. (Optional) List the VECS TRUSTED_ROOTS store and compare the certificate serial number there with the output from Step 1.

   ```
   /usr/lib/vmware-vmafd/bin/vecs-cli entry list --store TRUSTED_ROOTS --text
   ```

   In the simplest case with only one root certificate, the output looks like this:

   ```
   Number of entries in store :    1
   Alias : 960d43f31eb95211ba3a2487ac840645a02894bd
   Entry type :    Trusted Cert
   Certificate:
       Data:
           Version: 3 (0x2)
           Serial Number:
               cf:2d:ff:49:88:50:e5:af
   ```

3. Generate a new VMCA root certificate. The command adds the certificate to the TRUSTED_ROOTS store in VECS and in vmdir (VMware Directory Service).

   ```
   /usr/lib/vmware-vmca/bin/certool --selfca --config=/usr/lib/vmware-vmca/share/config/
   certool.cfg
   ```

Replace Machine SSL Certificates with VMCA-Signed Certificates Using the CLI

After you generate a new VMCA-signed root certificate, you can use the `vecs-cli` command to replace all machine SSL certificates in your environment.

Be prepared to stop all services and to start the services that handle certificate propagation and storage.

Each machine must have a machine SSL certificate for secure communication with other services. When multiple vCenter instances are connected in Enhanced Linked Mode configuration, you must run the Machine SSL certificate generation commands on each node.

1. Make one copy of `certool.cfg` for each machine that needs a new certificate.

   You can find the `certool.cfg` file in the `/usr/lib/vmware-vmca/share/config/` directory.

2. Edit the custom configuration file for each machine to include that machine's FQDN.

   Run `NSLookup` against the machine's IP address to see the DNS listing of the name, and use that name for the Hostname field in the file.

3. Generate a public/private key file pair and a certificate for each file, passing in the configuration file that you just customized.

   For example:

   ```
   certool --genkey --privkey=machine1.priv --pubkey=machine1.pub
   certool --gencert --privkey=machine1.priv --cert machine1.crt --Name=Machine1_Cert --config machine1.cfg
   ```

4. Stop all services and start the services that handle certificate creation, propagation, and storage.

   ```
   service-control --stop --all
   service-control --start vmafdd
   service-control --start vmdird
   service-control --start vmcad
   ```

5. Add the new certificate to VECS.

   All machines need the new certificate in the local certificate store to communicate over SSL. You first delete the existing entry, then add the new entry.

   ```
   vecs-cli entry delete --store MACHINE_SSL_CERT --alias __MACHINE_CERT
   vecs-cli entry create --store MACHINE_SSL_CERT --alias __MACHINE_CERT --cert machine1.crt --key
    machine1.priv
   ```

6. Restart all services.

   ```
   service-control --start --all
   ```

Replacing Machine Certificates with VMCA-Signed Certificates

1. Create a configuration file for the SSL certificate and save it as `ssl-config.cfg` in the current directory.

   ```
   Country = US
   Name = vmca-<FQDN-example>
   Organization = <my_company>
   OrgUnit = <my_company Engineering>
   State = <my_state>
   Locality = <mytown>
   Hostname = <FQDN>
   ```

2. Generate a key pair for the machine SSL certificate. In a deployment of multiple vCenter instances connected in Enhanced Linked Mode configuration, run this command on each vCenter node.

   ```
   /usr/lib/vmware-vmca/bin/certool --genkey --privkey=ssl-key.priv --pubkey=ssl-key.pub
   ```

   The `ssl-key.priv` and `ssl-key.pub` files are created in the current directory.

3. Generate the new machine SSL certificate. This certificate is signed by VMCA. If you replaced the VMCA root certificate with custom certificate, VMCA signs all certificates with the full chain.

   ```
   /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vmca-ssl.crt --privkey=ssl-key.priv --
   config=ssl-config.cfg
   ```

   The `new-vmca-ssl.crt` file is created in the current directory.

4. (Optional) List the content of VECS.

   ```
   /usr/lib/vmware-vmafd/bin/vecs-cli store list
   ```

   – Sample output on vCenter:

   ```
   output (on vCenter):
   MACHINE_SSL_CERT
   TRUSTED_ROOTS
   TRUSTED_ROOT_CRLS
   machine
   vsphere-webclient
   vpxd
   vpxd-extension
   hvc
   ```

```
data-encipherment
APPLMGMT_PASSWORD
SMS
wcp
KMS_ENCRYPTION
```

5. Replace the Machine SSL certificate in VECS with the new Machine SSL certificate. The `--store` and `--alias` values have to exactly match with the default names.
   – On each vCenter, run the following commands to update the Machine SSL certificate in the MACHINE_SSL_CERT store. You must update the certificate for each machine separately because each has a different FQDN.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store MACHINE_SSL_CERT --alias
 __MACHINE_CERT
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store MACHINE_SSL_CERT --alias
 __MACHINE_CERT --cert new-vmca-ssl.crt --key ssl-key.priv
```

You can also replace the certificates for your ESX hosts. See the *vSphere Security* publication.

Replace Solution User Certificates with New VMCA-Signed Certificates Using the CLI

After you replace the machine SSL certificates, you can use the `dir-cli` command to replace all solution user certificates. Solution user certificates must be valid, that is, not expired, but none of the other information in the certificate is used by the certificate infrastructure.

Be prepared to stop all services and to start the services that handle certificate propagation and storage.

Many VMware customers do not replace solution user certificates. They replace only the machine SSL certificates with custom certificates. This hybrid approach satisfies the requirements of their security teams.

- Certificates either sit behind a proxy, or they are custom certificates.
- No intermediate CAs are used.

You replace the machine solution user certificate and the solution user certificate on each vCenter system.

> **Note:**
> When you list solution user certificates in large deployments, the output of `/usr/lib/vmware-vmafd/bin/dir-cli list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

1. Make one copy of `certool.cfg`, remove the Name, IP address, DNS name, and email fields, and rename the file, for example, to `sol_usr.cfg`.

   You can name the certificates from the command line as part of generation. The other information is not needed for solution users. If you leave the default information, the certificates that are generated are potentially confusing.

2. Generate a public/private key file pair and a certificate for each solution user, passing in the configuration file that you just customized.

   For example:

```
/usr/lib/vmware-vmca/bin/certool --genkey --privkey=vpxd.priv --pubkey=vpxd.pub
/usr/lib/vmware-vmca/bin/certool --gencert --privkey=vpxd.priv --cert vpxd.crt --Name=VPXD_1 --config
 sol_usr.cfg
```

3. Find the name for each solution user.

```
/usr/lib/vmware-vmafd/bin/dir-cli service list
```

   You can use the unique ID that is returned when you replace the certificates. The input and output might look as follows.

```
/usr/lib/vmware-vmafd/bin/dir-cli service list
Enter password for administrator@vsphere.local:
1. machine-623bef28-0311-436e-b21f-6e0d39aa5179
```

2. vsphere-webclient-623bef28-0311-436e-b21f-6e0d39aa5179
3. vpxd-623bef28-0311-436e-b21f-6e0d39aa5179
4. vpxd-extension-623bef28-0311-436e-b21f-6e0d39aa5179
5. hvc-623bef28-0311-436e-b21f-6e0d39aa5179
6. wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e

In a deployment of multiple vCenter instances connected in Enhanced Linked Mode configuration, the output of `/usr/lib/vmware-vmafd/bin/dir-cli service list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

4. Stop all services and start the services that handle certificate creation, propagation, and storage.

```
service-control --stop --all
service-control --start vmafdd
service-control --start vmdird
service-control --start vmcad
```

5. For each solution user, replace the existing certificate in vmdir and then in VECS.

   The following example shows how to replace the certificates for the vpxd service.

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name <vpxd-xxxx-xxx-7c7b769cd9f4> --cert ./vpxd.crt
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vpxd --alias vpxd
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vpxd --alias vpxd --cert vpxd.crt --key vpxd.priv
```

   Note:
   Solution users cannot authenticate to vCenter Single Sign-On if you do not replace the certificate in vmdir.

6. Restart all services.

```
service-control --start --all
```

Using VMCA-Signed Solution User Certificates

1. Generate a public/private key pair for each solution user on each vCenter node in an Enhanced Linked Mode configuration. That includes a pair for the machine solution and a pair for each additional solution user (vpxd, vpxd-extension, vsphere-webclient, wcp).
   a. Generate a key pair for the machine solution user.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=machine-key.priv --pubkey=machine-key.pub
      ```
   b. Generate a key pair for the vpxd solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vpxd-key.priv --pubkey=vpxd-key.pub
      ```
   c. Generate a key pair for the vpxd-extension solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vpxd-extension-key.priv --pubkey=vpxd-extension-key.pub
      ```
   d. Generate a key pair for the vsphere-webclient solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vsphere-webclient-key.priv --pubkey=vsphere-webclient-key.pub
      ```
   e. Generate a key pair for the wcp solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=wcp-key.priv --pubkey=wcp-key.pub
      ```
2. Generate solution user certificates that are signed by the new VMCA root certificate for the machine solution user and for each additional solution user (vpxd, vpxd-extension, vsphere-webclient, wcp) on each vCenter node.

   Note:
   The `--Name` parameter has to be unique. Including the name of the solution user store name makes it easy to see which certificate maps to which solution user. The example includes the name, for example vpxd or vpxd-extension in each case.
   a. Make one copy of the `/usr/lib/vmware-vmca/share/config/certool.cfg` file, then modify or remove the Name, IP address, DNS name, and email fields as required, and rename the file, for example, to `sol_usr.cfg`.

b. Generate a certificate for the machine solution user on each node.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-machine.crt --privkey=machine-key.priv
 --Name=machine --config sol_usr.cfg
```

c. Generate a certificate for the vpxd solution user on each node.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vpxd.crt --privkey=vpxd-key.priv --
Name=vpxd --config sol_usr.cfg
```

d. Generate a certificate for the vpxd-extensions solution user on each node.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vpxd-extension.crt --privkey=vpxd-
extension-key.priv --Name=vpxd-extension --config sol_usr.cfg
```

e. Generate a certificate for the vsphere-webclient solution user on each node by running the following command.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vsphere-webclient.crt --
privkey=vsphere-webclient-key.priv --Name=vsphere-webclient --config sol_usr.cfg
```

f. Generate a certificate for the wcp solution user on each node by running the following command.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-wcp.crt --privkey=wcp-key.priv --
Name=wcp --config sol_usr.cfg
```

3. Replace the solution user certificates in VECS with the new solution user certificates.

Note:
The `--store` and `--alias` parameters have to exactly match the default names for services.

a. Replace the machine solution user certificate on each node:

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store machine --alias machine
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store machine --alias machine --cert new-
machine.crt --key machine-key.priv
```

b. Replace the vpxd solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vpxd --alias vpxd
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vpxd --alias vpxd --cert new-
vpxd.crt --key vpxd-key.priv
```

c. Replace the vpxd-extension solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vpxd-extension --alias vpxd-
extension
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vpxd-extension --alias vpxd-
extension --cert new-vpxd-extension.crt --key vpxd-extension-key.priv
```

d. Replace the vsphere-webclient solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vsphere-webclient --alias vsphere-
webclient
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vsphere-webclient --alias vsphere-
webclient --cert new-vsphere-webclient.crt --key vsphere-webclient-key.priv
```

e. Replace the wcp solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store wcp --alias wcp
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store wcp --alias wcp --cert new-wcp.crt
 --key wcp-key.priv
```

4. Update VMware Directory Service (vmdir) with the new solution user certificates. You are prompted for a vCenter Single Sign-On administrator password.

a. Run `/usr/lib/vmware-vmafd/bin/dir-cli service list` to get the unique service ID suffix for each solution user. You run this command on a vCenter system.

```
/usr/lib/vmware-vmafd/bin/dir-cli service list
output:
1. machine-623bef28-0311-436e-b21f-6e0d39aa5179
2. vsphere-webclient-623bef28-0311-436e-b21f-6e0d39aa5179
3. vpxd-623bef28-0311-436e-b21f-6e0d39aa5179
4. vpxd-extension-623bef28-0311-436e-b21f-6e0d39aa5179
5. hvc-623bef28-0311-436e-b21f-6e0d39aa5179
6. wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e
```

Note:
When you list solution user certificates in large deployments, the output of `/usr/lib/vmware-vmafd/bin/dir-cli list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

b. Replace the machine certificate in vmdir on each vCenter node. For example, if machine-6fd7f140-60a9-11e4-9e28-005056895a69 is the machine solution user on the vCenter, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name
 machine-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-machine.crt
```

c. Replace the vpxd solution user certificate in vmdir on each node. For example, if vpxd-6fd7f140-60a9-11e4-9e28-005056895a69 is the vpxd solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name
 vpxd-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vpxd.crt
```

d. Replace the vpxd-extension solution user certificate in vmdir on each node. For example, if vpxd-extension-6fd7f140-60a9-11e4-9e28-005056895a69 is the vpxd-extension solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name vpxd-
extension-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vpxd-extension.crt
```

e. Replace the vsphere-webclient solution user certificate on each node. For example, if vsphere-webclient-6fd7f140-60a9-11e4-9e28-005056895a69 is the vsphere-webclient solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name vsphere-
webclient-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vsphere-webclient.crt
```

f. Replace the wcp solution user certificate on each node. For example, if wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e is the wcp solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name wcp-1cbe0a40-e4ce-4378-
b5e7-9460e2b8200e --cert new-wcp.crt
```

Restart all services on each vCenter node.

Make VMCA Into an Intermediate Certificate Authority Using the CLI

You can use the CLI to replace the VMCA root certificate with a third-party CA-signed certificate that includes VMCA in the certificate chain. Going forward, all certificates that VMCA generates include the full chain. You can replace existing certificates with newly generated certificates.

If you use VMCA as an intermediate CA, or use custom certificates, you might encounter significant complexity and the potential for a negative impact to your security, and an unnecessary increase in your operational risk. For more information about managing certificates within a vSphere environment, see the blog post titled *New Product Walkthrough - Hybrid vSphere SSL Certificate Replacement* at https://blogs.vmware.com/vsphere/2017/01/walkthrough-hybrid-ssl-certificate-replacement.html.

Replace the Root Certificate (Intermediate CA) Using the CLI

The first step in replacing the VMCA certificates with custom certificates is generating a CSR, and sending the CSR to be signed. You then use the CLI to add the signed certificate to VMCA as a root certificate.

You can use the Certificate Manager utility or other tool to generate the CSR. The CSR must meet the following requirements:

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
- PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
- x509 version 3
- The CA extension must be set to true for root certificates, and cert sign must be in the list of requirements. For example:

```
basicConstraints        = critical,CA:true
keyUsage                = critical,digitalSignature,keyCertSign
```

- CRL signing must be enabled.
- Extended Key Usage can be either empty or contain Server Authentication.
- No explicit limit to the length of the certificate chain. VMCA uses the OpenSSL default, which is 10 certificates.
- Certificates with wildcards or with more than one DNS name are not supported.
- You cannot create subsidiary CAs of VMCA.

  See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2112009, Creating a Microsoft Certificate Authority Template for SSL certificate creation in vSphere 6.x, for an example using Microsoft Certificate Authority.

  Note:
  vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

VMCA validates the following certificate attributes when you replace the root certificate:

- Key size: 2048 bits (minimum) to 8192 bits (maximum).
- Key Usage: Cert Sign
- Basic Constraint: Subject Type CA

1. Generate a CSR and send it to your CA.

   Follow your CA's instructions.

2. Prepare a certificate file that includes the signed VMCA certificate and the full CA chain of your third-party CA or enterprise CA. Save the file, for example as `rootca1.crt`.

   You can accomplish this step by copying all CA certificates in PEM format into a single file. You start with the VMCA root certificate and end up with the root CA PEM certificate. For example:

   ```
   -----BEGIN CERTIFICATE-----
   <Certificate of VMCA>
   -----END CERTIFICATE-----
   -----BEGIN CERTIFICATE-----
   <Certificate of intermediary CA>
   -----END CERTIFICATE-----
   -----BEGIN CERTIFICATE-----
   <Certificate of Root CA>
   -----END CERTIFICATE-----
   ```

3. Stop all services and start the services that handle certificate creation, propagation, and storage.

   ```
   service-control --stop --all
   service-control --start vmafdd
   service-control --start vmdird
   service-control --start vmcad
   ```

4. Replace the existing VMCA root CA.

   ```
   certool --rootca --cert=rootca1.crt --privkey=root1.key
   ```

   When you run this command, it:

   - Adds the new custom root certificate to the certificate location in the file system.
   - Appends the custom root certificate to the TRUSTED_ROOTS store in VECS (after a delay).
   - Adds the custom root certificate to vmdir (after a delay).

5. Optional: To propagate the change to all instances of vmdir (VMware Directory Service), publish the new root certificate to vmdir, supplying the full file path for each file.

   For example, if the certificate has only one certificate in the chain:

   ```
   dir-cli trustedcert publish --cert rootca1.crt
   ```

   If the certificate has more than one certificate in the chain:

   ```
   dir-cli trustedcert publish --cert rootca1.crt --chain
   ```

Replication between vmdir nodes happens every 30 seconds. You do not have to add the root certificate to VECS explicitly because VECS polls vmdir for new root certificate files every 5 minutes.

6. Optional: If necessary, you can force a refresh of VECS.

```
vecs-cli force-refresh
```

7. Restart all services.

```
service-control --start --all
```

Replacing the Root Certificate

Replace the VMCA root certificate with the custom CA root certificate using the `certool` command with the `--rootca` option.

```
/usr/lib/vmware-vmca/bin/certool --rootca --cert=<path>/root.pem --privkey=<path>/root.key
```

When you run this command, it:

- Adds the new custom root certificate to the certificate location in the file system.
- Appends the custom root certificate to the TRUSTED_ROOTS store in VECS.
- Adds the custom root certificate to vmdir.

You can remove the original VMCA root certificate from the certificate store if your company policy requires it. If you do, you have to replace the vCenter Single Sign-On Signing certificate. See Replace a vCenter STS Certificate Using the Command Line.

Replace Machine SSL Certificates (Intermediate CA) Using the CLI

After you have received the signed certificate from the CA, you can use the CLI to make it the VMCA root certificate and replace all machine SSL certificates.

For each machine SSL certificate, the `SubjectAltName` must contain `DNS Name=<Machine FQDN>`.

These steps are essentially the same as the steps for replacing with a certificate that uses VMCA as the certificate authority. However, in this case, VMCA signs all certificates with the full chain.

Each machine must have a machine SSL certificate for secure communication with other services. When multiple vCenter instances are connected in Enhanced Linked Mode configuration, you must run the Machine SSL certificate generation commands on each node.

1. Make one copy of `certool.cfg` for each machine that needs a new certificate.

   The `certool.cfg` file is located in the `/usr/lib/vmware-vmca/share/config/` directory.

2. Edit the custom configuration file for each machine to include that machine's FQDN.

   Run `NSLookup` against the machine's IP address to see the DNS listing of the name, and use that name for the Hostname field in the file.

3. Generate a public/private key file pair and a certificate for each machine, passing in the configuration file that you just customized.

   For example:

```
certool --genkey --privkey=machine1.priv --pubkey=machine1.pub
certool --gencert --privkey=machine1.priv --cert machine42.crt --Name=Machine42_Cert --config machine1.cfg
```

4. Stop all services and start the services that handle certificate creation, propagation, and storage.

```
service-control --stop --all
service-control --start vmafdd
service-control --start vmdird
service-control --start vmcad
```

5. Add the new certificate to VECS.

   All machines need the new certificate in the local certificate store to communicate over SSL. You first delete the existing entry, then add the new entry.

```
vecs-cli entry delete --store MACHINE_SSL_CERT --alias __MACHINE_CERT
vecs-cli entry create --store MACHINE_SSL_CERT --alias __MACHINE_CERT --cert machine1.cert
--key machine1.priv
```

6. Restart all services.

```
service-control --start --all
```

Replacing Machine SSL Certificates (VMCA Is Intermediate CA)

1. Create a configuration file for the SSL certificate and save it as `ssl-config.cfg` in the current directory.

```
Country = US
Name = vmca-<FQDN-example>
Organization = VMware
OrgUnit = VMware Engineering
State = California
Locality = Palo Alto
Hostname = <FQDN>
```

2. Generate a key pair for the machine SSL certificate. In a deployment of multiple vCenter instances connected in Enhanced Linked Mode configuration, run this command on each vCenter node.

```
/usr/lib/vmware-vmca/bin/certool --genkey --privkey=ssl-key.priv --pubkey=ssl-key.pub
```

   The `ssl-key.priv` and `ssl-key.pub` files are created in the current directory.

3. Generate the new machine SSL certificate. This certificate is signed by VMCA. If you replaced the VMCA root certificate with custom certificate, VMCA signs all certificates with the full chain.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vmca-ssl.crt --privkey=ssl-key.priv --
config=ssl-config.cfg
```

   The `new-vmca-ssl.crt` file is created in the current directory.

4. (Optional) List the content of VECS.

```
/usr/lib/vmware-vmafd/bin/vecs-cli store list
```

   – Sample output on vCenter:

```
output (on vCenter):
MACHINE_SSL_CERT
TRUSTED_ROOTS
TRUSTED_ROOT_CRLS
machine
vsphere-webclient
vpxd
vpxd-extension
hvc
data-encipherment
APPLMGMT_PASSWORD
SMS
wcp
KMS_ENCRYPTION
```

5. Replace the Machine SSL certificate in VECS with the new Machine SSL certificate. The `--store` and `--alias` values have to exactly match with the default names.

   – On each vCenter, run the following commands to update the Machine SSL certificate in the MACHINE_SSL_CERT store. You must update the certificate for each machine separately because each has a different FQDN.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store MACHINE_SSL_CERT --alias
 __MACHINE_CERT
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store MACHINE_SSL_CERT --alias
 __MACHINE_CERT --cert new-vmca-ssl.crt --key ssl-key.priv
```

Replace Solution User Certificates (Intermediate CA) Using the CLI

After you replace the machine SSL certificates, you can use the CLI to replace the solution user certificates.

Each solution user certificate must have a different `Subject`. Consider, for example, including the solution user name (such as vpxd) or other unique identifier.

Many VMware customers do not replace solution user certificates. They replace only the machine SSL certificates with custom certificates. This hybrid approach satisfies the requirements of their security teams.

- Certificates either sit behind a proxy, or they are custom certificates.
- No intermediate CAs are used.

You replace the machine solution user certificate and the solution user certificate on each vCenter system.

> **Note:**
> When you list solution user certificates in large deployments, the output of `/usr/lib/vmware-vmafd/bin/dir-cli list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

1. Make one copy of `certool.cfg`, remove the Name, IP address, DNS name, and email fields, and rename the file, for example, to `sol_usr.cfg`.

   You can name the certificates from the command line as part of generation. The other information is not needed for solution users. If you leave the default information, the certificates that are generated are potentially confusing.

2. Generate a public/private key file pair and a certificate for each solution user, passing in the configuration file that you just customized.

   For example:

   ```
   certool --genkey --privkey=vpxd.priv --pubkey=vpxd.pub
   certool --gencert --privkey=vpxd.priv --cert vpxd.crt --Name=VPXD_1 --config sol_usr.cfg
   ```

3. Find the name for each solution user.

   ```
   /usr/lib/vmware-vmafd/bin/dir-cli service list
   ```

   You can use the unique ID that is returned when you replace the certificates. The input and output might look as follows.

   ```
   /usr/lib/vmware-vmafd/bin/dir-cli service list
   Enter password for administrator@vsphere.local:
   1. machine-623bef28-0311-436e-b21f-6e0d39aa5179
   2. vsphere-webclient-623bef28-0311-436e-b21f-6e0d39aa5179
   3. vpxd-623bef28-0311-436e-b21f-6e0d39aa5179
   4. vpxd-extension-623bef28-0311-436e-b21f-6e0d39aa5179
   5. hvc-623bef28-0311-436e-b21f-6e0d39aa5179
   6. wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e
   ```

   In a deployment of multiple vCenter instances connected in Enhanced Linked Mode configuration, the output of `/usr/lib/vmware-vmafd/bin/dir-cli service list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

4. Stop all services and start the services that handle certificate creation, propagation, and storage.

   ```
   service-control --stop --all
   service-control --start vmafdd
   service-control --start vmdird
   service-control --start vmcad
   ```

5. Replace the existing certificate in vmdir and then in VECS.

   For solution users, you must add the certificates in that order. For example:

   ```
   dir-cli service update --name <vpxd-xxxx-xxx-7c7b769cd9f4> --cert ./vpxd.crt
   vecs-cli entry delete --store vpxd --alias vpxd
   vecs-cli entry create --store vpxd --alias vpxd --cert vpxd.crt --key vpxd.priv
   ```

Note:

Solution users cannot log in to vCenter Single Sign-On if you do not replace the certificate in vmdir.

6. Restart all services.

```
service-control --start --all
```

Replacing Solution User Certificates (Intermediate CA)

1. Generate a public/private key pair for each solution user on each vCenter node in an Enhanced Linked Mode configuration. That includes a pair for the machine solution and a pair for each additional solution user (vpxd, vpxd-extension, vsphere-webclient, wcp).
   a. Generate a key pair for the machine solution user.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=machine-key.priv --pubkey=machine-key.pub
      ```
   b. Generate a key pair for the vpxd solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vpxd-key.priv --pubkey=vpxd-key.pub
      ```
   c. Generate a key pair for the vpxd-extension solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vpxd-extension-key.priv --pubkey=vpxd-extension-key.pub
      ```
   d. Generate a key pair for the vsphere-webclient solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=vsphere-webclient-key.priv --pubkey=vsphere-webclient-key.pub
      ```
   e. Generate a key pair for the wcp solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --genkey --privkey=wcp-key.priv --pubkey=wcp-key.pub
      ```
2. Generate solution user certificates that are signed by the new VMCA root certificate for the machine solution user and for each additional solution user (vpxd, vpxd-extension, vsphere-webclient, wcp) on each vCenter node.

   Note:
   The `--Name` parameter has to be unique. Including the name of the solution user store name makes it easy to see which certificate maps to which solution user. The example includes the name, for example vpxd or vpxd-extension in each case.
   a. Make one copy of the `/usr/lib/vmware-vmca/share/config/certool.cfg` file, then modify or remove the Name, IP address, DNS name, and email fields as required, and rename the file, for example, to `sol_usr.cfg`.
   b. Generate a certificate for the machine solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-machine.crt --privkey=machine-key.priv
       --Name=machine --config sol_usr.cfg
      ```
   c. Generate a certificate for the vpxd solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vpxd.crt --privkey=vpxd-key.priv --
      Name=vpxd --config sol_usr.cfg
      ```
   d. Generate a certificate for the vpxd-extensions solution user on each node.
      ```
      /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vpxd-extension.crt --privkey=vpxd-extension-key.priv --Name=vpxd-extension --config sol_usr.cfg
      ```
   e. Generate a certificate for the vsphere-webclient solution user on each node by running the following command.
      ```
      /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-vsphere-webclient.crt --
      privkey=vsphere-webclient-key.priv --Name=vsphere-webclient --config sol_usr.cfg
      ```
   f. Generate a certificate for the wcp solution user on each node by running the following command.
      ```
      /usr/lib/vmware-vmca/bin/certool --gencert --cert=new-wcp.crt --privkey=wcp-key.priv --
      Name=wcp --config sol_usr.cfg
      ```
3. Replace the solution user certificates in VECS with the new solution user certificates.

   Note:
   The `--store` and `--alias` parameters have to exactly match the default names for services.

a. Replace the machine solution user certificate on each node:

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store machine --alias machine
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store machine --alias machine --cert new-machine.crt --key machine-key.priv
```

b. Replace the vpxd solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vpxd --alias vpxd
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vpxd --alias vpxd --cert new-vpxd.crt --key vpxd-key.priv
```

c. Replace the vpxd-extension solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vpxd-extension --alias vpxd-extension
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vpxd-extension --alias vpxd-extension --cert new-vpxd-extension.crt --key vpxd-extension-key.priv
```

d. Replace the vsphere-webclient solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store vsphere-webclient --alias vsphere-webclient
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store vsphere-webclient --alias vsphere-webclient --cert new-vsphere-webclient.crt --key vsphere-webclient-key.priv
```

e. Replace the wcp solution user certificate on each node.

```
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store wcp --alias wcp
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store wcp --alias wcp --cert new-wcp.crt
 --key wcp-key.priv
```

4. Update VMware Directory Service (vmdir) with the new solution user certificates. You are prompted for a vCenter Single Sign-On administrator password.

a. Run `/usr/lib/vmware-vmafd/bin/dir-cli service list` to get the unique service ID suffix for each solution user. You run this command on a vCenter system.

```
/usr/lib/vmware-vmafd/bin/dir-cli service list
output:
1. machine-623bef28-0311-436e-b21f-6e0d39aa5179
2. vsphere-webclient-623bef28-0311-436e-b21f-6e0d39aa5179
3. vpxd-623bef28-0311-436e-b21f-6e0d39aa5179
4. vpxd-extension-623bef28-0311-436e-b21f-6e0d39aa5179
5. hvc-623bef28-0311-436e-b21f-6e0d39aa5179
6. wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e
```

Note:
When you list solution user certificates in large deployments, the output of `/usr/lib/vmware-vmafd/bin/dir-cli list` includes all solution users from all nodes. Run `/usr/lib/vmware-vmafd/bin/vmafd-cli get-machine-id --server-name localhost` to find the local machine ID for each host. Each solution user name includes the machine ID.

b. Replace the machine certificate in vmdir on each vCenter node. For example, if machine-6fd7f140-60a9-11e4-9e28-005056895a69 is the machine solution user on the vCenter, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name
 machine-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-machine.crt
```

c. Replace the vpxd solution user certificate in vmdir on each node. For example, if vpxd-6fd7f140-60a9-11e4-9e28-005056895a69 is the vpxd solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name
 vpxd-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vpxd.crt
```

d. Replace the vpxd-extension solution user certificate in vmdir on each node. For example, if vpxd-extension-6fd7f140-60a9-11e4-9e28-005056895a69 is the vpxd-extension solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name vpxd-
extension-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vpxd-extension.crt
```

e. Replace the vsphere-webclient solution user certificate on each node. For example, if vsphere-webclient-6fd7f140-60a9-11e4-9e28-005056895a69 is the vsphere-webclient solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name vsphere-
webclient-6fd7f140-60a9-11e4-9e28-005056895a69 --cert new-vsphere-webclient.crt
```

f. Replace the wcp solution user certificate on each node. For example, if wcp-1cbe0a40-e4ce-4378-b5e7-9460e2b8200e is the wcp solution user ID, run this command:

```
/usr/lib/vmware-vmafd/bin/dir-cli service update --name wcp-1cbe0a40-e4ce-4378-
b5e7-9460e2b8200e --cert new-wcp.crt
```

Replace Certificates with Custom Certificates Using the CLI

If your company policy requires it, you can use the CLI to replace some or all certificates used in vSphere with certificates that are signed by a third-party or enterprise CA. If you do that, VMCA is not in your certificate chain. You are responsible for storing all vCenter certificates in VECS.

You can replace all certificates or use a hybrid solution. For example, consider replacing all certificates that are used for network traffic but leaving VMCA-signed solution user certificates. Solution user certificates are used only for authentication to vCenter Single Sign-On. vCenter uses solution user certificates for internal communication only. Solution user certificates are not used for external communication.

Note:
If you do not want to use VMCA, you are responsible for replacing all certificates yourself, for provisioning new components with certificates, and for tracking certificate expiration.

Even if you decide to use custom certificates, you can still use the VMware Certificate Manager utility for certificate replacement. See Replace All Certificates with a Custom Certificate Using the Certificate Manager.

Request Certificates and Import a Custom Root Certificate Using the CLI

You can use custom certificates from an enterprise or third-party CA. The first step is requesting the certificates from the certificate authority then using the CLI to import the root certificates into VMware Endpoint Certificate Store (VECS).

The certificate must meet the following requirements:

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
- PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
- x509 version 3
- For root certificates, the CA extension must be set to true, and the cert sign must be in the list of requirements.
- SubjectAltName must contain DNS Name=<machine_FQDN>.
- CRT format
- Contains the following Key Usages: Digital Signature, Key Encipherment
- Start time of one day before the current time.
- CN (and SubjectAltName) set to the host name (or IP address) that the ESX host has in the vCenter inventory.

Note:
vSphere's FIPS certificate only validates RSA key sizes of 2048 bits and 3072 bits.

1. Send the Certificate Signing Requests (CSRs) for the following certificates to your enterprise or third-party certificate provider.
   - A machine SSL certificate for each machine. For the machine SSL certificate, the SubjectAltName field must contain the fully qualified domain name (DNS NAME=*machine_FQDN*).
   - Optionally, five solution user certificates for each node. Solution user certificates do not need to include IP address, host name, or email address. Each certificate must have a different certificate Subject.

   Typically, the result is a PEM file for the trusted chain, plus the signed SSL certificates for each vCenter node.

2. List the TRUSTED_ROOTS and the machine SSL stores.

   ```
   vecs-cli store list
   ```

   a) Ensure that the current root certificate and all machine SSL certificates are signed by VMCA.
   b) Note down the Serial number, issuer, and Subject CN fields.
   c) Optional: With a Web browser, open an HTTPS connection to a node where the certificate is to be replaced, view the certificate information, and ensure that it matches the machine SSL certificate.

3. Stop all services and start the services that handle certificate creation, propagation, and storage.

   ```
   service-control --stop --all
   service-control --start vmafdd
   service-control --start vmdird
   service-control --start vmcad
   ```

4. Publish the custom root certificate.

   ```
   dir-cli trustedcert publish --cert <my_custom_root>
   ```

   If you do not specify a user name and password on the command line, you are prompted.

5. Restart all services.

   ```
   service-control --start --all
   ```

You can remove the original VMCA root certificate from the certificate store if your company policy requires it. If you do, you have to refresh the vCenter Single Sign-On certificate. See Replace a vCenter STS Certificate Using the Command Line.

Replace Machine SSL Certificates with Custom Certificates Using the CLI

After you receive the custom certificates, you can use the CLI to replace each machine certificate.

You must have received a certificate for each machine from your third-party or enterprise CA.

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
- CRT format
- x509 version 3
- SubjectAltName must contain DNS Name=<machine_FQDN>.
- Contains the following Key Usages: Digital Signature, Key Encipherment

Perform the steps on each vCenter host.

You must have the following information before you can start replacing the certificates:

- Password for administrator@vsphere.local
- Valid Machine SSL custom certificate (`.crt` file)
- Valid Machine SSL custom key (`.key` file)
- Valid custom certificate for Root (`.crt` file)

1. Back up the current machine SSL certificate.

   ```
   /usr/lib/vmware-vmafd/bin/vecs-cli entry getcert --store MACHINE_SSL_CERT --alias __MACHINE_CERT >
    oldmachine.crt
   /usr/lib/vmware-vmafd/bin/vecs-cli entry getkey --store MACHINE_SSL_CERT --alias __MACHINE_CERT >
    oldmachinekey.key
   ```

2. Log in to each host and add the new machine certificates that you received from the certificate authority to VECS.

   All hosts need the new certificate in the local certificate store to communicate over SSL.

   a) Delete the existing certificate.

      ```
      /usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store MACHINE_SSL_CERT --alias __MACHINE_CERT
      ```

   b) Add the new certificate.

      ```
      /usr/lib/vmware-vmafd/bin/vecs-cli entry create --store MACHINE_SSL_CERT --alias __MACHINE_CERT --cert
       <cert-file-path> --key <key-file-path>
      ```

3. Extract the hash of the old certificate to be replaced.

   ```
   openssl x509 -in <path_to_old_machinessl_certificate> -noout -sha1 -fingerprint
   ```

   Output similar to the following appears:

   ```
   SHA1 Fingerprint=13:1E:60:93:E4:E6:59:31:55:EB:74:51:67:2A:99:F8:3F:04:83:88
   ```

4. Update the lookup service registration endpoint manually.

   ```
   /usr/lib/vmware-lookupsvc/tools/ls_update_certs.py --url https://<vCenterServer_FQDN>/lookupservice/sdk
    --certfile <cert-file-path> --user 'administrator@vsphere.local' --password '<password>' --fingerprint
    <SHA1_hash_of_the_old_certificate_to_replace>
   ```

   If you encounter problems running `ls_update_certs.py`, see thenowledge base article at http://knowledge.broadcom.com/external/article?legacyId=95982.

5. Restart all services.

   ```
   service-control --stop --all && service-control --start --all
   ```

vSphere Certificates and Services CLI Command Reference

You can manage VMCA (VMware Certificate Authority), VECS (VMware Endpoint Certificate Store), VMware Directory Service (vmdir), and Security Token Service (STS) certificates by using a set of CLIs. The vSphere Certificate Manager utility supports many related tasks as well, but the CLIs are required for manual certificate management and for managing other services.

You normally access the CLI tools for managing certificates and associated services by using SSH to connect to the appliance shell. See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2100508 for more information.

Manual vSphere Certificate Replacement gives examples for replacing certificates using CLI commands.

Table 85: vSphere CLI Tools for Managing Certificates and Associated Services

CLI	Description	See
`certool`	Generate and manage certificates and keys. Part of VMCAD, the VMware Certificate Management service.	certool Initialization Commands Reference
`vecs-cli`	Manage the contents of VMware Certificate Store instances. Part of VMware Authentication Framework Daemon (VMAFD).	vecs-cli Command Reference
`dir-cli`	Create and update certificates in VMware Directory Service. Part of VMAFD.	dir-cli Command Reference
`sso-config.sh`	Manage STS certificates.	Command-line help. Entering `sso-config.sh` with no options displays the command-line help.
`service-control`	Start or stop services, for example as part of a certificate replacement workflow.	Run this command to stop services before running other CLI commands.

vSphere CLI Locations

By default, you find the CLIs in the following locations.

```
/usr/lib/vmware-vmafd/bin/vecs-cli
/usr/lib/vmware-vmafd/bin/dir-cli
/usr/lib/vmware-vmca/bin/certool
/opt/vmware/bin/sso-config.sh
```

Note:
The `service-control` command does not require that you specify the path.

Required Privileges for Running vSphere CLIs

Required privileges depend on the CLI that you are using and on the command that you want to run. For example, for most certificate management operations, you have to be an administrator for the local vCenter Single Sign-On domain (vsphere.local by default). Some commands are available for all users.

dir-cli	You must be a member of the Administrators group in the local domain (vsphere.local by default) to run `dir-cli` commands. If you do not specify a user name and password, you are prompted for the password for the administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
vecs-cli	Initially, only the store owner and users with blanket access privileges have access to a store. Users in the Administrators group have blanket access privileges. The MACHINE_SSL_CERT and TRUSTED_ROOTS stores are special stores. Only the root user or the administrator user, depending on the type of installation, has complete access.
certool	Most of the `certool` commands require that the user is in the Administrators group. All users can run the following commands. • `genselfcacert` • `initscr` • `getdc` • `waitVMDIR` • `waitVMCA` • `genkey` • `viewcert`

Changing the certool Configuration Options

When you run `certool --gencert` or certain other certificate initialization or management commands, the command reads all the values from a configuration file. You can edit the existing file, override the default configuration file with the `--config=<file name>` option, or override values on the command line.

The configuration file, `certool.cfg`, is located in the `/usr/lib/vmware-vmca/share/config/` directory by default.

The file has several fields with the following default values:

```
Country = US
Name= Acme
Organization = AcmeOrg
OrgUnit = AcmeOrg Engineering
State = California
Locality = Palo Alto
IPAddress = 127.0.0.1
Email = email@acme.com
Hostname = server.acme.com
```

Note:
The OU (organizationalUnitName) field is no longer mandatory.

You can change the values by specifying a modified file on the command line, or by overriding individual values on the command line, as follows.

- Create a copy of the configuration file and edit the file. Use the --config command-line option to specify the file. Specify the full path to avoid path name issues.
- `/usr/lib/vmware-vmca/bin/certool --gencert --config /tmp/myconfig.cfg`
- Override individual values on the command line. For example, to override Locality, run this command:
  `/usr/lib/vmware-vmca/bin/certool --gencert --privkey=private.key --Locality="Mountain View"`

Specify `--Name` to replace the CN field of the Subject name of the certificate.

- For solution user certificates, the name is <sol_user name>@<domain> by convention, but you can change the name if a different convention is used in your environment.
- For machine SSL certificates, the FQDN of the machine is used.
  VMCA allows only one DNSName (in the `Hostname` field) and no other Alias options. If the IP address is specified by the user, it is stored in SubAltName as well.

Use the `--Hostname` parameter to specify the DNSName of a certificate's SubAltName.

certool Initialization Commands Reference

The `certool` initialization commands allow you to generate certificate signing requests, view and generate certificates and keys that are signed by VMware Certificate Authority (VMCA), import root certificates, and perform other certificate management operations.

In many cases, you pass a configuration file in to a `certool` command. See vSphere Certificates and Services CLI Command Reference. See Replace Existing VMCA-Signed Certificates with New VMCA-Signed Certificates Using the CLI for some usage examples. The command-line help provides details about the options.

certool --initcsr

Generates a Certificate Signing Request (CSR). The command generates a PKCS10 file and a private key.

Option	Description
--gencsr	Required for generating CSRs.
--privkey <key_file>	Name of the private key file.
--pubkey <key_file>	Name of the public key file.
--csrfile <csr_file>	File name for the CSR file to be sent to the CA provider.
--config <config_file>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.

Example:

```
certool --gencsr --privkey=<filename> --pubkey=<filename> --csrfile=<filename>
```

certool --selfca

Creates a self-signed certificate and provisions the VMCA server with a self-signed root CA. Using this option is one of the simplest ways to provision the VMCA server. You can instead provision the VMCA server with a third-party root certificate so that VMCA is an intermediate CA. See Make VMCA Into an Intermediate Certificate Authority Using the CLI.

This command generates a certificate that is predated by three days to avoid time zone conflicts.

Option	Description
--selfca	Required for generating a self-signed certificate.
--predate <number_of_minutes>	Allows you to set the Valid Not Before field of the root certificate to the specified number of minutes before the current time. This option can be helpful to account for potential time zone issues. The maximum is three days.
--config <config_file>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
machine-70-59:/usr/lib/vmware-vmca/bin # ./certool --predate=2280  --selfca --server= 192.0.2.24 --srp-
upn=administrator@vsphere.local
```

certool --rootca

Imports a root certificate. Adds the specified certificate and private key to VMCA. VMCA always uses the most recent root certificate for signing, but other root certificates remain trusted until you manually delete them. That means you can update your infrastructure one step at a time, and finally delete certificates that you no longer use.

Option	Description
--rootca	Required for importing a root CA.
--cert <certfile>	Name of the certificate file.
--privkey <key_file>	Name of the private key file. This file must be in PEM encoded format.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --rootca --cert=root.cert --privkey=privatekey.pem
```

certool --getdc

Returns the default domain name that is used by vmdir.

Option	Description
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.
--port <port_num>	Optional port number. Defaults to port 389.

Example:

```
certool --getdc
```

certool --waitVMDIR

Wait until the VMware Directory Service is running or until the timeout specified by --wait has elapsed. Use this option along with other options to schedule certain tasks, for example returning the default domain name.

Option	Description
--wait	Optional number of minutes to wait. Defaults to 3.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.
--port <port_num>	Optional port number. Defaults to port 389.

Example:

```
certool --waitVMDIR --wait 5
```

certool --waitVMCA

Wait until the VMCA service is running or until the specified timeout has elapsed. Use this option in conjunction with other options to schedule certain tasks, for example, generating a certificate.

Option	Description
--wait	Optional number of minutes to wait. Defaults to 3.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.
--port <port_num>	Optional port number. Defaults to port 389.

Example:

```
certool --waitVMCA --selfca
```

certool --publish-roots

Forces an update of root certificates. This command requires administrative privileges.

Option	Description
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --publish-roots
```

certool Management Commands Reference

The `certool` management commands allow you to view, generate, and revoke certificates and to view information about certificates.

certool --genkey

Generates a private and public key pair. Those files can then be used to generate a certificate that is signed by VMCA.

Option	Description
--genkey	Required for generating a private and public key.
--privkey <keyfile>	Name of the private key file.
--pubkey <keyfile>	Name of the public key file.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --genkey --privkey=<filename> --pubkey=<filename>
```

certool --gencert

Generates a certificate from the VMCA server. This command uses the information in `certool.cfg` or in the specified configuration file. You can use the certificate to provision machine certificates or solution user certificates.

Option	Description
--gencert	Required for generating a certificate.
--cert <certfile>	Name of the certificate file. This file must be in PEM encoded format.
--privkey <keyfile>	Name of the private key file. This file must be in PEM encoded format.
--config <config_file>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --gencert --privkey=<filename> --cert=<filename> --config=<config_file>
```

certool --getrootca

Prints the current root CA certificate in human-readable form. This output is not usable as a certificate, it is changed to be human readable.

Option	Description
--getrootca	Required for printing the root certificate.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --getrootca --server=remoteserver
```

certool --viewcert

Print all the fields in a certificate in human-readable form.

Option	Description
--viewcert	Required for viewing a certificate.
--cert <certfile>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.

Example:

```
certool --viewcert --cert=<filename>
```

certool --enumcert

List all certificates that the VMCA server knows about. The required filter option lets you list all certificates or only revoked, active, or expired certificates.

Option	Description
--enumcert	Required for listing all certificates.
--filter [all | active]	Required filter. Specify all or active. The revoked and expired options are not currently supported.

Example:

```
certool --enumcert --filter=active
```

certool --status

Sends a specified certificate to the VMCA server to check whether the certificate has been revoked. Prints `Certificate: REVOKED` if the certificate is revoked, and `Certificate: ACTIVE` otherwise.

Option	Description
--status	Required to check the status of a certificate.
--cert <certfile>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.
--server <server>	Optional name of the VMCA server. By default, the command uses localhost.

Example:

```
certool --status --cert=<filename>
```

certool --genselfcacert

Generates a self-signed certificate based on the values in the configuration file. This command generates a certificate that is predated by three days to avoid time zone conflicts.

Option	Description
--genselfcacert	Required for generating a self-signed certificate.
--outcert <cert_file>	Name of the certificate file. This file must be in PEM encoded format.
--outprivkey <key_file>	Name of the private key file. This file must be in PEM encoded format.
--config <config_file>	Name of the configuration file. A sample configuration file is located at `/usr/lib/vmware-vmca/share/config/certool.cfg`. As a best practice, make a copy of the default configuration file and replace the required fields.

Example:

```
certool --genselfcert --privkey=<filename> --cert=<filename> --config=<config_file>
```

vecs-cli Command Reference

The `vecs-cli` command set allows you to manage instances of VMware Certificate Store (VECS). Use these commands together with `dir-cli` and `certool` to manage your certificate infrastructure and authentication services.

vecs-cli store create

Creates a certificate store.

Option	Description
--name <name>	Name of the certificate store.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

Example:

```
vecs-cli store create --name <store>
```

vecs-cli store delete

Deletes a certificate store. You cannot delete the MACHINE_SSL_CERT, TRUSTED_ROOTS and TRUSTED_ROOT_CRLS system stores. Users with required privileges can delete solution user stores.

Option	Description
--name <name>	Name of the certificate store to delete.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

Example:

```
vecs-cli store delete --name <store>
```

vecs-cli store list

List certificate stores.

Option	Description
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

VECS includes the following stores.

Table 86: Stores in VECS

Store	Description
Machine SSL store (MACHINE_SSL_CERT)	• Used by the reverse proxy service on every vSphere node. • Used by the VMware Directory Service (vmdir) on each vCenter node. All services in vSphere 6.0 and later communicate through a reverse proxy, which uses the machine SSL certificate. For backward compatibility, the 5.x services still use specific ports. As a result, some services such as vpxd still have their own port open.
Solution user stores • `machine` • `vpxd` • `vpxd-extension` • `vsphere-webclient` • `wcp`	VECS includes one store for each solution user. The subject of each solution user certificate must be unique, for example, the machine certificate cannot have the same subject as the vpxd certificate. Solution user certificates are used for authentication with vCenter Single Sign-On. vCenter Single Sign-On checks that the certificate is valid, but does not check other certificate attributes. The following solution user certificate stores are included in VECS: • `machine`: Used by the license server and the logging service. **Note:** The machine solution user certificate has nothing to do with the machine SSL certificate. The machine solution user certificate is used for the SAML token exchange. The machine SSL certificate is used for secure SSL connections for a machine. • `vpxd`: vCenter service daemon (vpxd) store. vpxd uses the solution user certificate that is stored in this store to authenticate to vCenter Single Sign-On. • `vpxd-extension`: vCenter extensions store. Includes the Auto Deploy service, inventory service, and other services that are not part of other solution users. • `vsphere-webclient`: vSphere Client store. Also includes some additional services such as the performance chart service. • `wcp`: VMware vSphere® with VMware Tanzu™ store. Also used for vSphere Cluster Services. Each vCenter node includes a `machine` certificate.
Trusted root store (TRUSTED_ROOTS)	Contains all trusted root certificates.
vSphere Certificate Manager Utility backup store (BACKUP_STORE)	Used by VMCA (VMware Certificate Manager) to support certificate revert. Only the most recent state is stored as a backup, you cannot go back more than one step.

Store	Description
Other stores	Other stores might be added by solutions. For example, the Virtual Volumes solution adds an SMS store. Do not modify the certificates in those stores unless Broadcom documentation or a Broadcom Knowledge Base article instructs you to do so. **Note:** Deleting the TRUSTED_ROOTS_CRLS store can damage your certificate infrastructure. Do not delete or modify the TRUSTED_ROOTS_CRLS store.

Example:

```
vecs-cli store list
```

vecs-cli store permissions

Grants or revokes permissions to the store. Use either the --grant or the --revoke option.

The owner of the store can perform all operations, including granting and revoking permissions. The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default, has all privileges on all stores, including granting and revoking permissions.

You can use `vecs-cli get-permissions --name <store-name>` to retrieve the current settings for the store.

Option	Description
--name <name>	Name of the certificate store.
--user <username>	Unique name of the user who is granted permissions.
--grant [read|write]	Permission to grant, either read or write.
--revoke [read|write]	Permission to revoke, either read or write. Not currently supported.

vecs-cli store get-permissions

Retrieves the current permission settings for the store.

Option	Description
--name <name>	Name of the certificate store.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

vecs-cli entry create

Creates an entry in VECS. Use this command to add a private key or certificate to a store.

Note:
Do not use this command to add root certificates to the TRUSTED_ROOTS store. Instead, use the `dir-cli` command to publish root certificates.

Option	Description
--store <NameOfStore>	Name of the certificate store.
--alias <Alias>	Optional alias for the certificate. This option is ignored for the trusted root store.
--cert <certificate_file_path>	Full path of the certificate file.

Option	Description
--key <key-file-path>	Full path of the key that corresponds to the certificate. Optional.
--password <password>	Optional password for encrypting the private key.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

vecs-cli entry list

Lists all entries in a specified store.

Option	Description
--store <NameOfStore>	Name of the certificate store.

vecs-cli entry getcert

Retrieves a certificate from VECS. You can send the certificate to an output file or display it as human-readable text.

Option	Description
--store <NameOfStore>	Name of the certificate store.
--alias <Alias>	Alias of the certificate.
--output <output_file_path>	File to write the certificate to.
--text	Displays a human-readable version of the certificate.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

vecs-cli entry getkey

Retrieves a key that is stored in VECS. You can send the key to an output file or display it as human-readable text.

Option	Description
--store <NameOfStore>	Name of the certificate store.
--alias <Alias>	Alias for the key.
--output <output_file_path>	Output file to write the key to.
--text	Displays a human-readable version of the key.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

vecs-cli entry delete

Deletes an entry in a certificate store. If you delete an entry in VECS, you permanently remove it from VECS. The only exception is the current root certificate. VECS polls vmdir for a root certificate.

Option	Description
--store <NameOfStore>	Name of the certificate store.
--alias <Alias>	Alias for the entry you want to delete.
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.
-y	Suppresses the confirmation prompt. For advanced users only.

vecs-cli force-refresh

Forces a refresh of VECS. By default, VECS polls vmdir for new root certificate files every 5 minutes. Use this command for an immediate update of VECS from vmdir.

Option	Description
--server <server-name>	Used to specify a server name if you connect to a remote VECS instance.
--upn <user-name>	User Principle Name that is used to log in to the server instance specified by --server <server-name> . When you create a store, it is created in the context of the current user. Therefore, the owner of the store is the current user context and not always the root user.

dir-cli Command Reference

The `dir-cli` utility supports creation and updates to solution users, account management, and management of certificates and passwords in VMware Directory Service (vmdir). You can use `dir-cli` to manage and query the domain functional level of vCenter instances.

dir-cli nodes list

Lists all the enhanced linked mode connected vCenter systems.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.
--server <psc_ip_or_fqdn>	Use this option to connect to another vCenter to see its replication partners.

dir-cli computer password-reset

Enables you to reset the password of the machine account in the domain.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.
--live-dc-hostname <server name>	Current name of the vCenter instance.

dir-cli service create

Creates a solution user. Primarily used by third-party solutions.

Option	Description
--name <name>	Name of the solution user to create
--cert <cert file>	Path to the certificate file. This can be a certificate signed by VMCA or a third-party certificate.
--ssogroups <comma-separated-groupnames>	Makes the solution user a member of the specified groups.
--wstrustrole <ActAsUser>	Makes the solution user a member of the built-in administrators or users group. In other words, determines whether the solution user has administrative privileges.
--ssoadminrole <Administrator/User>	Makes the solution user a member of the ActAsUser group. The ActAsUser role enables users to act on behalf of other users.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli service list

List the solution users that `dir-cli` knows about.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli svaccount list

List the service account users that `dir-cli` knows about.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli service delete

Delete a solution user in vmdir. When you delete the solution user, all associated services become unavailable to all management nodes that use this instance of vmdir.

Option	Description
--name	Name of the solution user to delete.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli service update

Updates the certificate for a specified solution user, that is, collection of services. After running this command, update the solution user certificate entry in VECS by running the `vecs-cli entry create` command. See vecs-cli Command Reference.

Option	Description
--name <name>	Name of the solution user to update .
--cert <cert_file>	Name of the certificate to assign to the service.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli user create

Creates a regular user inside vmdir. This command can be used for human users who authenticate to vCenter Single Sign-On with a user name and password. Use this command only during prototyping.

Option	Description
--account <name>	Name of the vCenter Single Sign-On user to create.
--user-password <password>	Initial password for the user.
--first-name <name>	First name for the user.
--last-name <name>	Last name for the user.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli user modify

Modifies the specified user inside vmdir.

Option	Description
--account <name>	Name of the vCenter Single Sign-On user to modify.
--password-never-expires	Set this option to true if you are modifying a user account for automated tasks that have to authenticate to vCenter, and you want to ensure that the tasks do not stop running because of password expiration. Use this option with care.
--password-expires	Set this option to true if you want to revert the --password-never-expires option.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli user delete

Deletes the specified user inside vmdir.

Option	Description
--account <name>	Name of the vCenter Single Sign-On user to delete.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli user find-by-name

Finds a user inside vmdir by name. The information that this command returns depends on what you specify in the `--level` option.

Option	Description
--account <name>	Name of the vCenter Single Sign-On user to find.
--level <info level 0|1|2>	Returns the following information: • Level 0 - Account and UPN • Level 1 - level 0 info + First and last name • Level 2 : level 0 + Account deactivated flag, Account locked flag, Password never expires flag, password expired flag and password expiry flag. The default level is 0.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli group modify

Adds a user or group to an existing group.

Option	Description
--name <name>	Name of the group in vmdir.
--add <user_or_group_name>	Name of the user or group to add.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli group list

Lists a specified vmdir group.

Option	Description
--name <name>	Optional name of the group in vmdir. This option allows you to check whether a specific group exists.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli ssogroup create

Create a group inside the local domain (vsphere.local by default).

Use this command if you want to create groups to manage user permissions for the vCenter Single Sign-On domain. For example, if you create a group and then add it to the Administrators group of the vCenter Single Sign-On domain, then all users that you add to that group have administrator permissions for the domain.

It is also possible to give permissions to vCenter inventory objects to groups in the vCenter Single Sign-On domain. See the *vSphere Security* documentation.

Option	Description
--name <name>	Name of the group in vmdir. Maximum length is 487 characters.
--description <description>	Optional description for the group.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli trustedcert publish

Publishes a trusted root certificate to vmdir. After running this command, VECS picks up the certificate change after one minute, or you can run the `vecs-cli force-refresh` command to sync the certificate immediately.

Note:

Starting in vSphere 8.0 Update 3, use either the vSphere Client or the API to publish a trusted root certificate and avoid having to restart services.

Option	Description
--cert <file>	Path to certificate file.
--crl <file>	This option is not supported by VMCA.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.
--chain	Specify this option if you are publishing a chained certificate. No option value is needed.

dir-cli trustedcert unpublish

Unpublishes a trusted root certificate currently in vmdir. Use this command, for example, if you added a different root certificate to vmdir that is now the root certificate for all other certificates in your environment. Unpublishing certificates that are no longer in use is part of hardening your environment.

Note:

Starting in vSphere 8.0 Update 3, use the API to unpublish a trusted root certificate and avoid having to restart services.

Option	Description
--cert <file>	Path to the certificate file to unpublish
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli trustedcert list

Lists all trusted root certificates and their corresponding IDs. You need the certificate IDs to retrieve a certificate with `dir-cli trustedcert get`.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli trustedcert get

Retrieves a trusted root certificate from vmdir and writes it to a specified file.

Option	Description
--id <cert_ID>	ID of the certificate to retrieve. The `dir-cli trustedcert list` command shows the ID.
--outcert <path>	Path to write the certificate file to.
--outcrl <path>	Path to write the CRL file to. Not currently used.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli password create

Creates a random password that meets the password requirements. This command can be used by third-party solution users.

Option	Description
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli password reset

Allows an administrator to reset a user's password. If you are a non-administrator user who wants to reset a password, use `dir-cli password change` instead.

Option	Description
--account	Name of the account to assign a new password to.
--new	New password for the specified user.
--login <admin_user_id>	The administrator of the local vCenter Single Sign-On domain, administrator@vsphere.local by default.
--password <admin_password>	Password of the administrator user. If you do not specify the password, you are prompted.

dir-cli password change

Allows a user to change their password. You must be the user who owns the account to make this change. Administrators can use `dir-cli password reset` to reset any password.

Option	Description
--account	Account name.
--current	Current password of the user who owns the account.
--new	New password of the user who owns the account.

vSphere Authentication with vCenter Single Sign-On

vCenter Single Sign-On is an authentication broker and security token exchange infrastructure. vCenter Single Sign-On issues a token when a user authenticates. The user can use the token to authenticate to vCenter services. The user can then perform the actions that user has privileges for.

Because traffic is encrypted for all communications, and because only authenticated users can perform the actions that they have privileges for, your environment is secure.

Users and service accounts authenticate with a token, or a user name and password. Solution users authenticate with a certificate. For information on replacing solution user certificates, see vSphere Security Certificates. For information about replacing solution user certificates, see vSphere Security Certificates.

The next step is authorizing the users who can authenticate to perform certain tasks. Usually, you assign vCenter privileges, typically by assigning the user to a group that has a role. vSphere includes other permission models such as global permissions. See the *vSphere Security* documentation.

How vCenter Single Sign-On Protects Your Environment

vCenter Single Sign-On allows vSphere components to communicate with each other through a secure token mechanism.

vCenter Single Sign-On uses the following services.

- Authentication of users through either external identity provider federation or the vCenter built-in identity provider. The built-in identity provider supports local accounts, Active Directory or OpenLDAP, and miscellaneous authentication mechanisms.
- Authentication of solution users through certificates.
- Security Token Service (STS).
- SSL for secure traffic.

vCenter Built-in Identity Provider

vCenter includes a built-in identity provider. By default, vCenter uses the vsphere.local domain as the identity source (but you can change the domain during installation). You can configure the vCenter built-in identity provider to use Active Directory (AD) as its identity source using LDAP/S, or OpenLDAP/S. Such configurations allow customers to log in to vCenter using their AD accounts.

vCenter and an External Identity Provider

In vSphere 9.0 and later, external identity provider federation is the preferred authentication method for vCenter.

vSphere supports the following identity providers.

- vSphere 7.0 and later: Active Directory Federation Services (AD FS)
- vSphere 8.0 Update 1 and later: Okta
- vSphere 8.0 Update 2 and later: Microsoft Entra ID (formerly called Azure AD)
- Starting in vSphere 8.0 Update 3: PingFederate

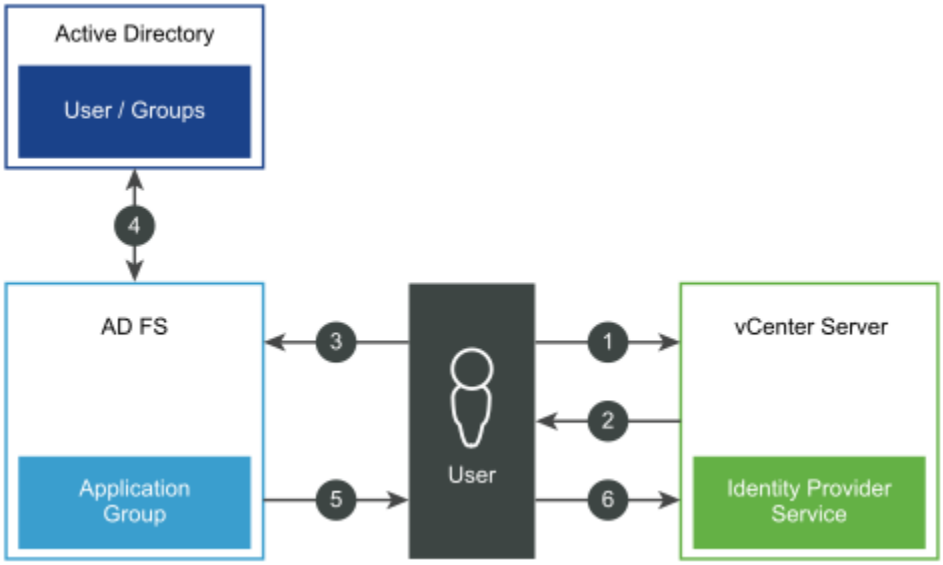
When you configure vSphere to use an external identity provider, the external identity provider interacts with the identity sources on behalf of vCenter.

User Login with vCenter Identity Provider Federated Authentication

When you use an external identity provider to authenticate with vCenter, vCenter redirects the login request to the external identity provider. The external identity provider authenticates the user with its directory service then issues a token for vCenter to use to log in the user.

For example, the following figure shows a detailed look at the user login flow for vCenter Identity Provider Federation using AD FS.

Figure 30: vCenter User Login Using AD FS Identity Provider Federation



vCenter, AD FS, and Active Directory interact as follows:

1. The user starts on the vCenter landing page by entering a user name.
2. If the user name is for a federated domain, vCenter redirects the authentication request to AD FS.
3. If needed, AD FS prompts the user to log in with Active Directory credentials.
4. AD FS authenticates the user with Active Directory.
5. AD FS issues a security token with group information from Active Directory.
6. vCenter uses the token to log in the user.

The user is now authenticated, and can view and modify any objects that the user's role has privileges for.

> **Note:**
> Initially, each user is assigned the No Access role. A vCenter administrator must assign the user at least to the Read Only role before the user can log in. See the *vSphere Security* documentation.

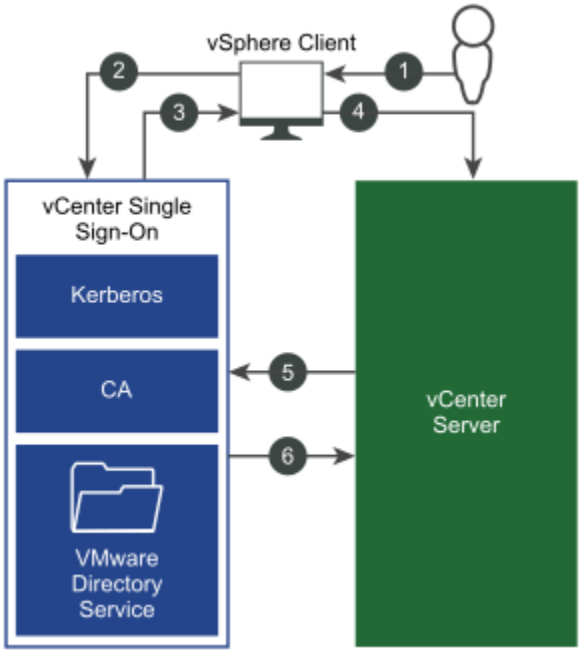
In case the external identity provider is unreachable, the login process falls back to the vCenter landing page, showing an appropriate information message. Users can still log in using their local accounts in the vsphere.local identity source.

The interaction between vCenter and Okta, Microsoft Entra ID, or PingFederate is similar to that of AD FS, with the exception that vCenter uses VMware Identity Services. See VMware Identity Services Authentication Process.

User Login with the vCenter Built-In Identity Provider

The following figure shows the user login flow when vCenter acts as the identity provider.

Figure 31: User Login with the vCenter Built-In Identity Provider



1. A user logs in to the vSphere Client with a user name and password to access the vCenter system or another vCenter service.
2. The vSphere Client passes the login information to the vCenter Single Sign-On service, which checks the SAML token of the vSphere Client. If the vSphere Client has a valid token, vCenter Single Sign-On then checks whether the user is in the configured identity source (for example, Active Directory).
   - If only the user name is used, vCenter Single Sign-On checks in the default domain.
   - If a domain name is included with the user name (*DOMAIN*\user1 or user1@*DOMAIN*), vCenter Single Sign-On checks that domain.
3. If the user can authenticate to the identity source, vCenter Single Sign-On returns a token that represents the user to the vSphere Client.
4. The vSphere Client passes the token to the vCenter system.
5. vCenter checks with the vCenter Single Sign-On server that the token is valid and not expired.
6. The vCenter Single Sign-On server returns the token to the vCenter system, using the vCenter Authorization Framework to allow user access.

The user is now authenticated, and can view and modify any objects that the user's role has privileges for.

> **Note:**
> Initially, each user is assigned the No Access role. A vCenter administrator must assign the user at least to the Read Only role before the user can log in. See the *vSphere Security* documentation.

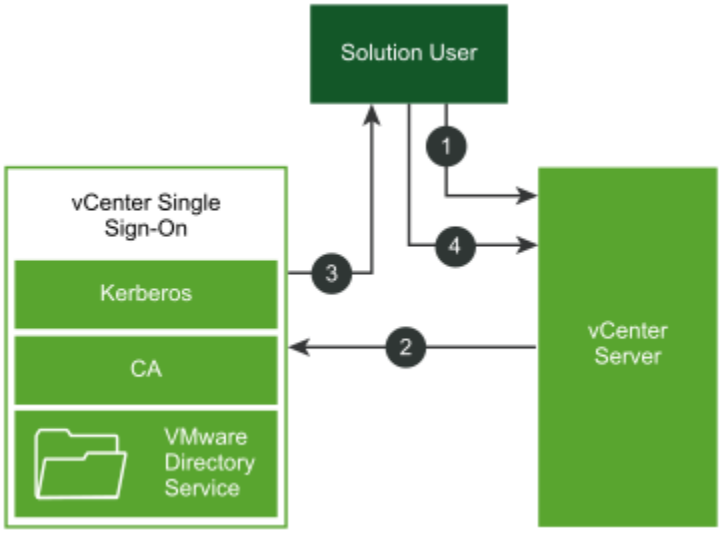
Login for Solution Users

Solution users are sets of services that are used in the vCenter infrastructure, for example, vCenter extensions. VMware extensions and potentially third-party extensions might also authenticate to vCenter Single Sign-On.

> **Note:**
> vCenter uses solution user certificates for internal communication only. Solution user certificates are not used for external communication.

The following figure shows the login flow for solution users.

Figure 32: Login for Solution Users



1. The solution user attempts to connect to a vCenter service.
2. The solution user is redirected to vCenter Single Sign-On. If the solution user is new to vCenter Single Sign-On, it has to present a valid certificate.
3. If the certificate is valid, vCenter Single Sign-On assigns a SAML token (bearer token) to the solution user. The token is signed by vCenter Single Sign-On.
4. The solution user is then redirected to vCenter Single Sign-On and can perform tasks based on its permissions. The next time the solution user has to authenticate, it can use the SAML token to log in to vCenter.

By default, this handshake is automatic because VMCA provisions solution users with certificates during startup. If your company policy requires third-party CA-signed certificates, you can replace the solution user certificates with third-party CA-signed certificates. If those certificates are valid, vCenter Single Sign-On assigns a SAML token to the solution user. See Replace Solution User Certificates with Custom Certificates Using the Certificate Manager.

Supported Encryption in vSphere

AES encryption, which is the highest level of encryption, is supported. The supported encryption affects security when vCenter Single Sign-On uses Active Directory as an identity source.

It also affects security anytime an ESX host or vCenter is joined to Active Directory.

vCenter Identity Provider Federation

In vSphere 9.0 and later, external identity provider federation is the preferred authentication method for vCenter.

To enable federated authentication to vCenter, you configure a connection to an external identity provider. The identity provider instance that you configure replaces vCenter as the identity provider. Currently, vCenter supports Active Directory Federation Services (AD FS), Okta, Microsoft Entra ID (formerly called Azure AD), and PingFederate as external identity providers. vCenter supports AD FS in vSphere 7.0 and later, Okta in vSphere 8.0 Update 1 and later, Microsoft Entra ID in vSphere 8.0 Update 2 and later, and PingFederate starting in vSphere 8.0 Update 3.

> **Note:**
> VMware encourages you to use federated authentication as vSphere moves towards token-based authentication. vCenter continues to have local accounts, for administrative access and error recovery.

How vCenter Identity Provider Federation Works

vCenter Identity Provider Federation enables you to configure an external identity provider for federated authentication. In this configuration, the external identity provider interacts with the identity source on behalf of vCenter. In vSphere 9.0 and later, external identity provider federation is the preferred authentication method for vCenter. With VCF 9.0 we are moving

towards a modern authentication which leverages federation of vCenter with an external IDP provider. Native vCenter features for multi-factor authentication is deprecated in favor of leveraging the external identity providers capabilities or multi-factor authentication. For more information on VCF single sign on, see VCF Single Sign-On.

vCenter Identity Provider Federation Basics

The vCenter authentication will transparently delegate the authentication call to the federated IDP which performs the authentication by all supported means including multi-factor authentication, and then return a token to VC which is used for authentication. All multi-factor authentication options including the token validity time can be configured at the external identity provider settings.

vCenter External Identity Provider Support

vCenter supports the following external identity providers:

- **ADFS** (vSphere 7.0 and later)
- **OKTA** (vSphere 8.0 Update 1 and later)
- **Microsoft Entra ID**, formerly called Azure AD (vSphere 8.0 Update 2 and later)
- **PingFederate** (starting in vSphere 8.0 Update 3)

Consider the version and timelines when you plan your migration to a federated VC authentication.

vCenter Identity Provider Federation Benefits

vCenter Identity Provider Federation provides the following benefits.

- You can use Single Sign-On with existing federated infrastructure and applications.
- You can improve data center security because vCenter never handles the user's credentials.
- You can use the authentication mechanisms, such as multi-factor authentication, supported by the external identity provider.

vCenter Identity Provider Federation Architecture

To establish a relying party trust between vCenter and an external identity provider, you must establish the identifying information and a shared secret between them. vCenter uses the OpenID Connect (OIDC) protocol to receive an identity token that authenticates the user with vCenter.

The high-level steps to configure an external identity provider with vCenter include:

1. Establishing a relying party trust between vCenter and the external identity provider by creating an OIDC configuration. For AD FS, you create an Application Group, or Application. For Okta, Microsoft Entra ID, and PingFederate, you create a native application with OpenID Connect as the sign-on method. The OIDC configuration consists of a Server application and a Web API. The two components specify the information that vCenter uses to trust and communicate with the external identity provider.
2. Creating a corresponding identity provider in vCenter.
3. Configuring group memberships in vCenter to authorize logins from users in the external identity provider domain.

The identity provider administrator must provide the following information to create the vCenter identity provider configuration:

- Client Identifier: The UUID string that is generated in AD FS when you create the Application Group (or Application) and that identifies the Application Group (or Application), or that is generated in Okta, Microsoft Entra ID, or PingFederate when you create the OpenID Connect application.
- Shared Secret: The secret that is generated in AD FS when you create the Application Group (or Application), or that is generated in Okta, Microsoft Entra ID, or PingFederate when you create the OpenID Connect application, and that is used to authenticate vCenter with the external identity provider.
- OpenID Address: The OpenID Provider Discovery endpoint URL of the external identity provider server, specifying a well-known address that is typically the issuer endpoint concatenated with the path `/.well-known/openid-configuration`. Here is an example OpenID address for an AD FS configuration:

  ```
  https://webserver.example.com/adfs/.well-known/openid-configuration
  ```

Likewise, here is an example OpenID address for an Okta configuration:

```
https://example.okta.com/oauth2/default/.well-known/openid-configuration
```

Here is an example OpenID address for a Microsoft Entra ID configuration:

```
https://login.microsoftonline.com/11111111-2222-3333-4444-555555555555/v2.0/.well-known/openid-configuration
```

Here is an example OpenID address for a PingFederate configuration:

```
https://pingfederate-fqdn-and-port/.well-known/openid-configuration
```

VMware Identity Services and Federated Authentication

In vSphere 8.0 Update 1 and later, VMware Identity Services provides integration with external identity providers as a federated identity provider. You can think of the VMware Identity Services as a "slimmed" down version of VMware Workspace ONE that is built-in to vSphere.

When you either install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Services are activated by default on vCenter. When you configure Okta, Microsoft Entra ID, or PingFederate as an external identity provider, vCenter uses VMware Identity Services to communicate with your Okta, Microsoft Entra ID, or PingFederate server.

vCenter supports Okta, Microsoft Entra ID, and PingFederate as an external identity provider in an Enhanced Linked Mode configuration. Even though, in an Enhanced Linked Mode configuration, multiple vCenter systems are running VMware Identity Services, only a single vCenter and, its VMware Identity Services communicate with your external identity provider server. For example, if you have an Enhanced Linked Mode configuration of three vCenter systems, A, B, and C, and you configure the Okta external identity provider on vCenter A, vCenter A is the only system to handle all the Okta logins. vCenter B and vCenter C do not directly communicate with the Okta server. To configure VMware Identity Services on other vCenter in the ELM configuration to interact with your external IDP server, see Activation Process for External Identity Providers in Enhanced Linked Mode Configurations.

Note:

When federated, vCenter supports only a single federation provider. No other external identity sources such as AD-over-LDAP are supported while vCenter is federated.

Note:
When configuring Okta as an external identity provider, all the vCenter systems in an Enhanced Linked Mode configuration must run at least vSphere 8.0 Update 1. For Microsoft Entra ID, the requirement is at least vSphere 8.0 Update 2. For PingFederate the requirement is at least vSphere 8.0 Update 3.

Warning:
When using an Enhanced Linked Mode configuration with Okta, Microsoft Entra ID, or PingFederate, you cannot remove the vCenter that runs the VMware Identity Services and that communicates with the identity provider from the ELM configuration.

VMware Identity Services Authentication Process

When you configure vCenter to use VMware Identity Services to communicate with your external identity provider, the following authentication process occurs:

1. A user logs in to vCenter with the vSphere Client.
2. vCenter Single Sign-On delegates the user authentication and redirects the user request to VMware Identity Services.
3. The VMware Identity Services process requests a token from the external identity provider to establish the user session.
4. The external identity provider authenticates the user (it can use Multifactor Authentication (MFA) or SSO credentials) and returns the token to VMware Identity Services.
   The token contains the user claims.
5. The VMware Identity Services process validates the identity provider token, generates a corresponding VMware Identity Services token, and sends the VMware Identity Services token to vCenter Single Sign-On.
6. vCenter Single Sign-On validates the token and grants the login request.

Note:
AD FS does not use VMware Identity Services for federated authentication.

How vCenter Interacts with Users and Groups Pushed by SCIM

When you configure your external identity provider, vCenter uses System for Cross-domain Identity Management (SCIM) for user and group management. SCIM is an open standard for automating the exchange of user identity information. A SCIM application that you create on your external IDP server manages the users and groups on external identity provider that you want to push to vCenter. vCenter also uses SCIM when searching for users and groups to assign permissions to vCenter objects.

Note:
An AD FS configuration searches the Active Directory by using LDAP. It does not use SCIM.

vCenter Identity Provider Federation Components

The following components comprise a vCenter Identity Provider Federation configuration:

- A vCenter
  - For AD FS: vCenter 7.0 or later
  - For Okta: vCenter 8.0 Update 1 or later
  - For Microsoft Entra ID: vCenter 8.0 Update 2 or later
  - For PingFederate: vCenter 8.0 Update 3
- An identity provider service configured on the vCenter
- An external identity provider (AD FS, Okta, Microsoft Entra ID, or PingFederate)
- An OpenID Connect (OIDC) configuration:
  - For AD FS: An Application Group (also called an Application)
  - For Okta, Microsoft Entra ID, or PingFederate: An OpenID Connect application
- A System for Cross-domain Identity Management (SCIM) application for user and group management (only for Okta, Microsoft Entra ID, or PingFederate)
- External identity provider groups and users that you map to vCenter groups and users
- VMware Identity Services enabled on vCenter (only for Okta, Microsoft Entra ID, or PingFederate)
- Optionally, for PingFederate, the SSL certificate, or certificate chain, of the PingFederate server, if this certificate was not issued by a well-known, public Certificate Authority. You import the PingFederate SSL certificate to the vCenter.

vCenter Identity Provider Federation Caveats and Interoperability

vCenter Identity Provider Federation can inter-operate with many other VMware features.

As you are planning your vCenter Identity Provider Federation strategy, consider possible interoperability limitations.

Authentication Mechanisms

In a vCenter Identity Provider Federation configuration, the external identity provider handles the authentication mechanisms (passwords, MFA, biometrics, and so on).

AD FS and Support for a Single Active Directory Domain

When you configure vCenter Identity Provider Federation for AD FS, the Configure Main Identity Provider wizard prompts you to enter LDAP information for the single AD domain that contains the users and groups you want to access vCenter. vCenter derives the AD domain to use for authorization and permissions from the User Base DN that you specify in the wizard. You can add permissions on vSphere objects only for users and groups from this AD domain. Users or groups from AD child domains or other domains in the AD forest are not supported by vCenter Identity Provider Federation.

Okta, Microsoft Entra ID, and PingFederate Support for Multiple Domains

When you configure vCenter Identity Provider Federation for Okta, Microsoft Entra ID, or PingFederate, the Configure Main Identity Provider wizard enables you to enter LDAP information for the multiple domains that contain the users and groups you want to access vCenter.

Password, Lockout, and Token Policies

When vCenter acts as the identity provider, you control vCenter password, lockout, and token policies for the default domain (vsphere.local, or the domain name you entered when installing vSphere). When using federated authentication with vCenter, the external identity provider controls the password, lockout, and token policies for the accounts stored in the identity source such as Active Directory.

Auditing and Compliance

When using vCenter Identity Provider Federation, vCenter continues to create log entries for successful user logins. However, the external identity provider is responsible for tracking and logging actions such as failed password-entry attempts and user account lockouts. vCenter does not log such events because they are no longer visible to vCenter. For example, when AD FS is the identity provider, AD FS tracks and logs errors for federated logins. When vCenter is the identify provider for local logins, vCenter tracks and logs errors for local logins. In a federated configuration, vCenter does continue to log user actions post-login.

Existing VMware Product Integration with External Identity Providers

VMware products that are integrated with vCenter (for example, VMware Cloud Foundation Operations (formerly Aria Operations), vSAN, NSX, and so on) continue to work as before.

Products That Integrate Post-Login

Products that integrate post-login (that is, they do not require a separate login) continue to work as before.

Simple Authentication for API, SDK, and CLI Access

Existing scripts, products, and other functionality that rely on API, SDK, or CLI commands that use Simple Authentication (that is, user name and password) continue to work as before. Internally, the authentication occurs by passing the user name and password. This passing of the user name and password compromises some of the benefits of using identity federation, because it exposes the password to vCenter (and your scripts). Consider migrating to token-based authentication where possible.

Accessing the vCenter Management Interface

If the user is a member of the vCenter Administrators group, accessing the vCenter Management Interface (formerly called vCenter Appliance Management Interface or VAMI) is supported.

Entering User Name Text on the AD FS Login Page

The AD FS login page does not support passing text to pre-populate the user name text box. As a result, during federated logins with AD FS, after entering your user name on the vCenter landing page and redirecting to the AD FS login page, you must reenter your user name on the AD FS login page. The user name that you enter on the vCenter landing page is necessary to redirect the login to the appropriate identity provider, and the user name on the AD FS login page is necessary to authenticate with AD FS. This inability to pass the user name to the AD FS login page is a limitation of AD FS. You cannot configure or change this behavior directly from vCenter.

Support for IPv6 Addresses

AD FS, Microsoft Entra ID, and Ping Federate support IPv6 addresses. Okta does not support IPv6 addresses.

VMware Identity Services Single Instance Configuration

By default, when you either install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Services are enabled on vCenter. When you configure Okta, Microsoft Entra ID, or PingFederate in an Enhanced Link Mode configuration, you use VMware Identity Services on a single vCenter system. For example, if you use Okta in an Enhanced Mode Link configuration that consists of three vCenter systems, only one vCenter its instance of VMware Identity Services is used to communicate with the Okta server.

> **Warning:**
>
> In an ELM configuration that uses VMware Identity Services, if the vCenter system that communicates with the external identity provider becomes unavailable, you can configure VMware Identity Services on other vCenter in the ELM configuration to interact with your external IDP server. See Activation Process for External Identity Providers in Enhanced Linked Mode Configurations.

Reconfiguring the Primary Network Identifier

Reconfiguring the Primary Network Identifier (PNID) of the vCenter requires that you update the external identity provider configuration as follows.

- AD FS: Add the new Redirect URIs to the AD FS server.
- Okta: Reconfigure Okta. See Configure vCenter Identity Provider Federation for Okta and follow the steps to create the identity provider on vCenter.
- Microsoft Entra ID: Reconfigure Entra ID. See Configure vCenter Identity Provider Federation for Microsoft Entra ID and follow the steps to create the identity provider on vCenter.
- PingFederate: Reconfigure PingFederate. See Configure vCenter Identity Provider Federation for PingFederate and follow the steps to create the identity provider on vCenter.

vCenter Identity Provider Federation Life Cycle

When managing the life cycle of vCenter Identity Provider Federation, there are some specific considerations.

You can manage your vCenter Identity Provider Federation life cycle in the following ways.

Migrating from Using Active Directory to an External Identity Provider

If you are using Active Directory as your identity source for vCenter, migrating to using an external identity provider is straight forward. If your Active Directory groups and roles match your identity provider groups and roles, you do not need to take any additional action. When the groups and roles do not match, then you must perform some additional work. If vCenter is a domain member, consider removing it from the domain as it is not needed or used for identity federation.

Cross-Domain Repointing and Migration

vCenter Identity Provider Federation supports cross-domain repointing, that is, moving a vCenter from one vSphere SSO domain to another. The repointed vCenter receives the replicated identity provider configuration from the vCenter system, or systems, to which it was pointed.

In general, you do not need to perform any additional identity provider reconfiguration for a cross-domain repoint, unless one of the following is true.

1. The identity provider configuration of the repointed vCenter differs from the identity provider configuration of the vCenter to which it was pointed.
2. This is the first time the repointed vCenter is receiving an identity provider configuration.

In these cases, some additional work is required. For example, for AD FS, you must add the vCenter system's Redirect URIs to the corresponding Application Group on the AD FS server. For example, if vCenter 1 with AD FS Application Group A (or no AD FS configuration) is repointed to vCenter 2 with AD FS Application Group B, you must add the Redirect URIs of vCenter 1 to Application Group B.

Users and Group Synchronization and vCenter Backup and Restore

Depending on when you synchronize your users and groups with vCenter and when you back up your vCenter, if you must restore your vCenter, you might need to re-synchronize your users and groups that were pushed by SCIM.

To restore a deleted user or group, you cannot simply push the user or group from the external identity provider to vCenter. You must update the SCIM 2.0 application on the external identity provider with the missing user or group. See Restore Deleted SCIM Users and Groups.

vCenter Identity Provider Federation and Enhanced Linked Mode

When you enable Identity Provider Federation in vCenter environments using enhanced linked mode, authentication and workflows continue to work as before.

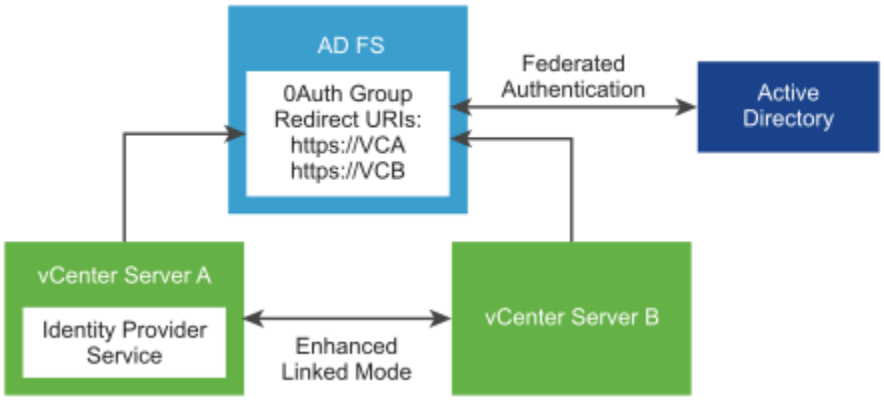
If you use Enhanced Linked Mode configuration, note the following when logging in to vCenter using federated authentication.

- Users continue to see the same inventory, and can perform the same actions, based on the vCenter permissions and roles model.
- vCenter hosts in enhanced linked mode are not required to have access to each other's identity providers. For example, consider two vCenter systems A and B, and that use enhanced linked mode. After vCenter A authorizes a user, then the user is authorized on vCenter B as well.

Enhanced Link Mode and AD FS

The following figure shows the authentication workflow when using AD FS with enhanced linked mode.

Figure 33: Enhanced Linked Mode and AD FS Identity Provider Federation



1. Two vCenter nodes are deployed in Enhanced Linked Mode configuration.
2. The AD FS setup has been configured on vCenter A using the Change Identity Provider wizard in the vSphere Client. Group memberships and permissions have also been established for AD FS users or groups.
3. vCenter A replicates the AD FS configuration to vCenter B.
4. All Redirect URIs for both vCenter nodes are added to the OAuth Application Group in AD FS. Only one OAuth Application Group is created.
5. When a user logs into and is authorized by vCenter A, the user is also authorized on vCenter B. If the user logs in to vCenter B first, the same holds true.

Enhanced Linked Mode Configuration Scenarios with AD FS

vCenter enhanced linked mode supports the following configuration scenarios for AD FS. In this section, the terms "AD FS settings" and "AD FS configuration" refer to the settings that you configure in the vSphere Client using the Change Identity Provider wizard, and any group memberships or permissions that you have established for AD FS users or groups.

Enable AD FS on an Existing Enhanced Linked Mode Configuration

High-level steps:

1. Deploy N vCenter nodes in Enhanced Linked Mode configuration.
2. Configure AD FS on one of the linked vCenter nodes.
3. The AD FS configuration is replicated to all other (N-1) vCenter nodes.
4. Add all Redirect URIs for all N vCenter nodes to the configured OAuth Application Group in AD FS.

Link a New vCenter to an Existing Enhanced Linked Mode AD FS Configuration

High-level steps:

1. (Prerequisite) Set up AD FS on a vCenter N-node Enhanced Linked Mode configuration.
2. Deploy a new independent vCenter node.
3. Repoint the new vCenter to the N-node AD FS enhanced linked mode domain, using one of the N nodes as its replication partner.
4. All AD FS settings in the existing Enhanced Linked Mode configuration are replicated to the new vCenter.
   The AD FS settings that are in the N-node AD FS enhanced linked mode domain overwrite any existing AD FS settings on the newly linked vCenter.
5. Add all Redirect URIs for the new vCenter to the existing configured OAuth Application Group in AD FS.

Unlink a vCenter from an Enhanced Linked Mode AD FS Configuration

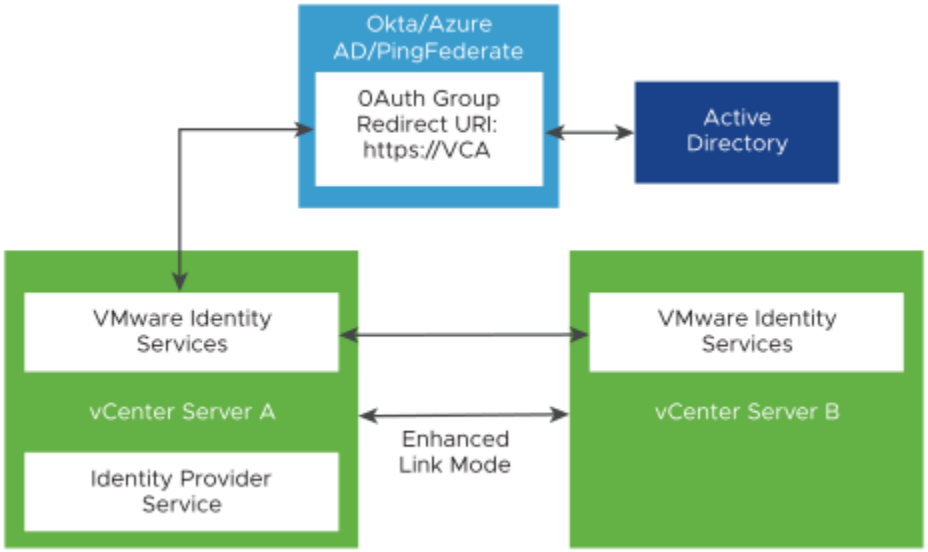
High-level steps:

1. (Prerequisite) Set up AD FS on an N-node vCenter Enhanced Linked Mode configuration.
2. Unregister one of the vCenter hosts in the N-node configuration and repoint it to a new domain to unlink it from the N-node configuration.
3. The domain repointing process does not preserve SSO settings, so all AD FS settings on the unlinked vCenter node are reverted and lost. To continue using AD FS on this vCenter unlinked node, you must reconfigure AD FS from the beginning or you must relink the vCenter to an Enhanced Linked Mode configuration where AD FS is already set up.

Enhanced Linked Mode and Okta, Microsoft Entra ID, or PingFederate Identity Provider Federation

The following figure shows the authentication workflow when using Okta, Microsoft Entra ID, or PingFederate with enhanced linked mode.

Figure 34: Enhanced Linked Mode and Okta, Microsoft Entra ID, or PingFederate Identity Provider Federation



Note:
When configuring Okta, Microsoft Entra ID, or PingFederate as an external identity provider, all the vCenter systems in an Enhanced Linked Mode configuration must run at least vSphere 8.0 Update 1 for Okta, vSphere 8.0 Update 2 for Microsoft Entra ID, and vSphere 8.0 Update 3 for PingFederate.

1. Two vCenter nodes are deployed in Enhanced Linked Mode configuration.
2. The Okta, Microsoft Entra ID, or PingFederate setup has been configured on vCenter A using the Change Identity Provider wizard in the vSphere Client. Also, group memberships and permissions have also been established for Okta, Microsoft Entra ID, or PingFederate users or groups.

   Note:
   Both vCenter A and B have VMware Identity Services enabled, but only the VMware Identity Services on vCenter A communicate with the identity provider server.
3. The VMware Identity Services running on vCenter A enable vCenter B to access its endpoint.
4. The Redirect URI for vCenter A is added to the OAuth Application in Okta, Microsoft Entra ID, or PingFederate. Only one OAuth Application is created.
5. When a user logs into and is authorized by vCenter A, the user is also authorized on vCenter B. If the user logs in to vCenter B first, the same holds true.

Enhanced Linked Mode Configuration Scenarios with Okta, Microsoft Entra ID, or PingFederate

vCenter enhanced linked mode supports the following configuration scenarios for Okta, Microsoft Entra ID, and PingFederate. In this section, the terms "Okta settings" and "Okta configuration," "Microsoft Entra ID settings" and "Microsoft Entra ID configuration," or "PingFederate settings" and "PingFederate configuration," refer to the settings that you configure in the vSphere Client using the Change Identity Provider wizard, and any group memberships or permissions that you have established for Okta, Microsoft Entra ID, or PingFederate users or groups.

Enable Okta, Microsoft Entra ID, or PingFederate on an Existing Enhanced Linked Mode Configuration

High-level steps:

1. Deploy N vCenter nodes in Enhanced Linked Mode configuration.
2. Configure Okta, Microsoft Entra ID, or PingFederate on one of the linked vCenter nodes.
3. The VMware Identity Services endpoint information is replicated to all other (N-1) vCenter nodes.
   The Okta, Microsoft Entra ID, or PingFederate configuration (shared client ID, and so on) information and user/group information is not replicated.

Link a New vCenter to an Existing Enhanced Linked Mode Okta, Microsoft Entra ID, or PingFederate Configuration

High-level steps:

1. (Prerequisite) Set up Okta, Microsoft Entra ID, or PingFederate on a vCenter N-node Enhanced Linked Mode configuration.
2. Deploy a new independent vCenter node.
3. Repoint the new vCenter to the N-node Okta, Microsoft Entra ID, or PingFederate enhanced linked mode domain, using one of the N nodes as its replication partner.
4. The VMware Identity Services endpoint information is replicated to all other (N-1) vCenter nodes.
   The Okta, Microsoft Entra ID, or PingFederate configuration (shared client ID, and so on) information and user/group information is not replicated.

   Note:
   You can add a vCenter node with an existing VMware Identity Services configuration. In this scenario, the existing VMware Identity Services configuration is replaced with the VMware Identity Services Enhanced Link Mode configuration that it is joining.

   You cannot add a vCenter node with an existing VMware Identity Services configuration to an ELM configuration that has not been configured with VMware Identity Services. In this scenario, first remove the existing VMware Identity Services configuration from the vCenter before adding it to the ELM configuration.

Unlink a vCenter from an Enhanced Linked Mode Okta, Microsoft Entra ID, or PingFederate Configuration

High-level steps:

1. (Prerequisite) Set up Okta, Microsoft Entra ID, or PingFederate on an N-node vCenter Enhanced Linked Mode configuration.
2. Unregister one of the vCenter hosts in the N-node configuration and repoint it to a new domain to unlink it from the N-node configuration.
3. The domain repointing process does not preserve SSO settings, so all Okta, Microsoft Entra ID, or PingFederate settings on the unlinked vCenter node are reverted and lost. To continue using Okta, Microsoft Entra ID, or PingFederate on this vCenter unlinked node, you must reconfigure Okta, Microsoft Entra ID, or PingFederate from the beginning or you must relink the vCenter to an Enhanced Linked Mode configuration where Okta, Microsoft Entra ID, or PingFederate is already set up.

   Note:
   You cannot unlink a vCenter with an active VMware Identity Services configuration.

Activation Process for External Identity Providers in Enhanced Linked Mode Configurations

Learn more about the availability considerations in Enhanced Linked Mode configurations with Okta, Microsoft Entra ID, or PingFederate.

- Two or more vCenter systems in an Enhanced Linked Mode configuration. For example, the systems are labeled as VC_1, VC_2, VC_3, through VC_N, where N is the number of vCenter systems in the Enhanced Linked Mode configuration.
- For Okta and Microsoft Entra ID, all vCenter systems must run vSphere 8.0 Update 2 or later. For PingFederate, all vCenter systems must run at least vSphere 8.0 Update 3.
- Okta, Microsoft Entra ID, or PingFederate is configured as an external identity provider on one of the vCenter systems. For example, the system is labeled as VC_1.
- The external identity provider is configured with all required OAuth2 and SCIM applications.

1. To activate a given vCenter VC_i where i is between 2 and N:
   a) Obtain local shell access to VC_i to run the activation script.

      Note:
      To perform the following steps, the vCenter user account with administrative privileges can be given either on the command line or in the console prompts.
   b) Execute `'status'` from the activation script to get the current activation state of the vCenter.

      ```
      python /usr/lib/vmware-trustmanagement/vmware_identity_services_activation.py status
      ```
   c) If the `'status'` command indicates that the vCenter is not activated, execute `'activate'` from the activation script:

      ```
      python /usr/lib/vmware-trustmanagement/vmware_identity_services_activation.py activate
      ```
   d) If the `'status'` command indicates that the vCenter is already activated, then execute the `'deactivate'` option then execute the `'activate'` option.

      ```
      python /usr/lib/vmware-trustmanagement/vmware_identity_services_activation.py deactivate
      ```

      - For example, execute the `'activate'` option.
      - Alternatively, you can specify the `'--force-replace'` option in the `'activate'` command.

2. Open a browser to the vCenter VC_i and log in as an administrator to vCenter.
   a) Navigate to **Home > Administration > Single Sign On > Configuration**.
   b) Under **User Provisioning**, verify that the **Tenant URL** contains the FQDN of VC_i.
   c) Copy the **Tenant URL** string and save this information for using it with the external identity provider.
   d) Under **Secret Token**, click the **Generate**, copy the generated token string, and save this information for using it with the external identity provider.
   e) Under **OpenID Connect**, verify that the **Redirect URI** contains the FQDN of VC_i.
   f) Copy the **Redirect URI** string and save this information for using it with the external identity provider.
3. Open a browser to the administration page of the external identity provider.

   Note:
   For more information, refer to the external identity provider specific details to perform the following steps.
   a) Find the OAuth2 registration that was setup when the external identity provider was originally configured in VC_1.
   b) Edit the OAuth2 registration and add the Redirect URI that was previously obtained for VC_i.
   c) If the external identity provider supports SCIM Push configurations with multiple destinations, then:
      - Find the SCIM Push configuration that was setup when the external identity provider was originally configured in VC_1.
      - Edit the SCIM Push configuration and add the **Tenant URL** and **Secret Token** that were previously obtained for VC_i.
   d) If the external identity provider supports SCIM Push configurations with only one destination:
      - Create a new SCIM Push configuration with the **Tenant URL** and **Secret Token** that were previously obtained for VC_i.
      - Ensure that the SCIM Push configuration is pushing the same user/group data as the SCIM Push configuration that was setup when the external identity provider was originally configured in VC_1.
   e) Initiate a SCIM push operation to ensure that VC_i is populated with the latest user or group data.

Configuring vCenter Identity Provider Federation

After you deploy vCenter initially, you can configure an external identity provider for federated authentication.

With VCF 9.0 we are moving towards a modern authentication which leverages federation of vCenter with an external IDP provider. Native vCenter features for multi-factor authentication is deprecated in favor of leveraging the external identity providers capabilities or multi-factor authentication.The vCenter authentication will transparently delegate the authentication call to the federated IDP which performs the authentication by all supported means including multi-factor authentication, and then return a token to VC which is used for authentication. All multi-factor authentication options including the token validity time can be configured at the external identity provider settings.

vCenter supports the following external identity providers:

- **ADFS** (vSphere 7.0 and later)
- **OKTA** (vSphere 8.0 Update 1 and later)
- **Microsoft Entra ID**, formerly called Azure AD (vSphere 8.0 Update 2 and later)
- **PingFederate** (starting in vSphere 8.0 Update 3)

You configure vCenter Identity Provider Federation from the vSphere Client or the API. You also must perform some configuration on your external identity provider. To configure vCenter Identity Provider Federation, you must have vCenter Single Sign-On administrator privileges. Having vCenter Single Sign-On administrator privileges is different from having the Administrator role on vCenter or ESX. In a new installation, only the vCenter Single Sign-On administrator (administrator@vsphere.local by default) can authenticate to vCenter Single Sign-On.

vCenter Identity Provider Federation Configuration Process Flow

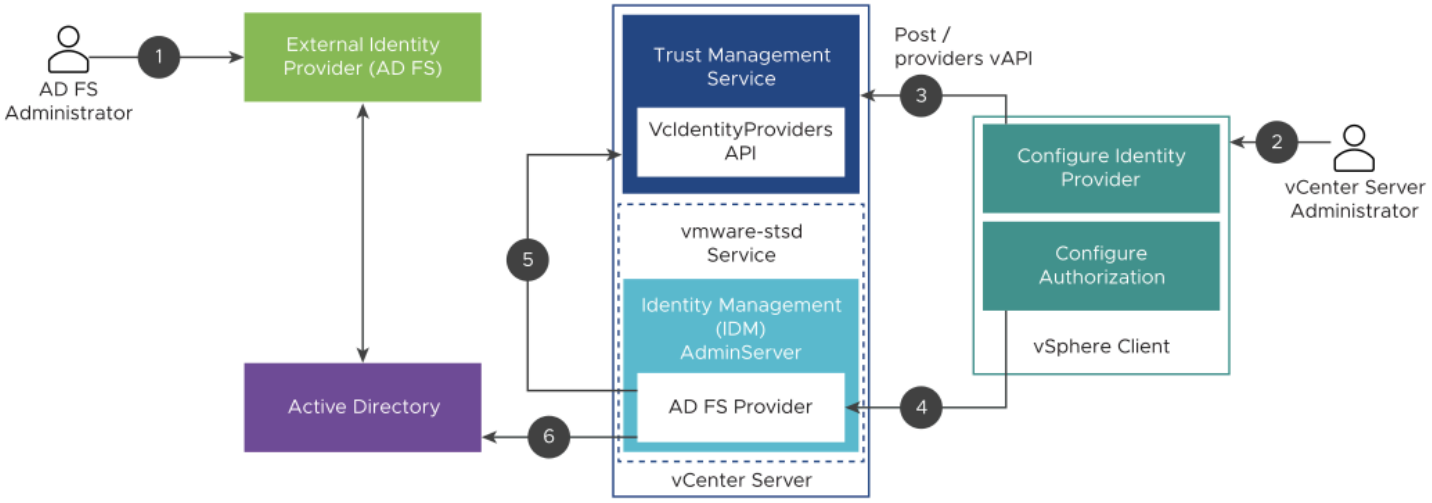
To configure vCenter Identity Provider Federation effectively, you must understand the communication flows that take place.

You can configure vCenter Identity Provider Federation for AD FS, Microsoft Entra ID (formerly called Azure AD), Okta, or PingFederate.

vCenter Identity Provider Federation Configuration Process Flow for AD FS

The following figure shows the process flow that occurs when configuring vCenter Identity Provider Federation for AD FS.

Figure 35: vCenter Identity Provider Federation Configuration Process Flow for AD FS



vCenter, AD FS, and Active Directory interact as follows.

1. The AD FS administrator configures an AD FS OIDC Application for vCenter.
2. The vCenter administrator logs in to the vCenter using the vSphere Client.
3. The vCenter administrator adds an AD FS identity provider to vCenter, and also enters information about the Active Directory domain.
   vCenter needs this information to make an LDAP connection to the Active Directory domain of the AD FS server. Using this connection, vCenter searches for users and groups, and adds them to vCenter local groups in the next step. See the section titled "Searching the Active Directory Domain" that follows for more information.
4. The vCenter administrator configures authorization permissions in vCenter for AD FS users.
5. The AD FS Provider queries the VcIdentityProviders API to get the LDAP connection information for the Active Directory source.
6. The AD FS Provider searches Active Directory for the queried users or groups to finish the Authorization configuration.

Searching the Active Directory Domain

You configure AD FS as the external identity provider in vCenter by using the Configure Main Identity Provider wizard in the vSphere Client. As part of the configuration process, you must enter information about your Active Directory domain, including user and group distinguished name information. Configuration of AD FS for authentication requires this Active Directory connection information. This connection is necessary to search for and map Active Directory user names and groups to roles and permissions in vCenter, whereas AD FS is used for the authentication of the user. This step of the Configure Main Identity Provider wizard does not create an Active Directory Over LDAP identity source. Rather, vCenter uses this information to establish a valid search-capable connection to your Active Directory domain to find users and groups there.

Consider an example using the following distinguished name entries:

- Base distinguished name for users: cn=Users,dc=corp,dc=local
- Base distinguished name for groups: dc=corp,dc=local
- User name: cn=Administrator,cn=Users,dc=corp,dc=local

If the AdfsUser@corp.local user is a member of the ADGroup@corp.local group, entering this information in the wizard allows a vCenter administrator to search for and find the ADGroup@corp.local group and add it to the vCenter Administrators@vsphere.local group. As a result, the AdfsUser@corp.local user is granted administrative privileges in vCenter upon logging in.

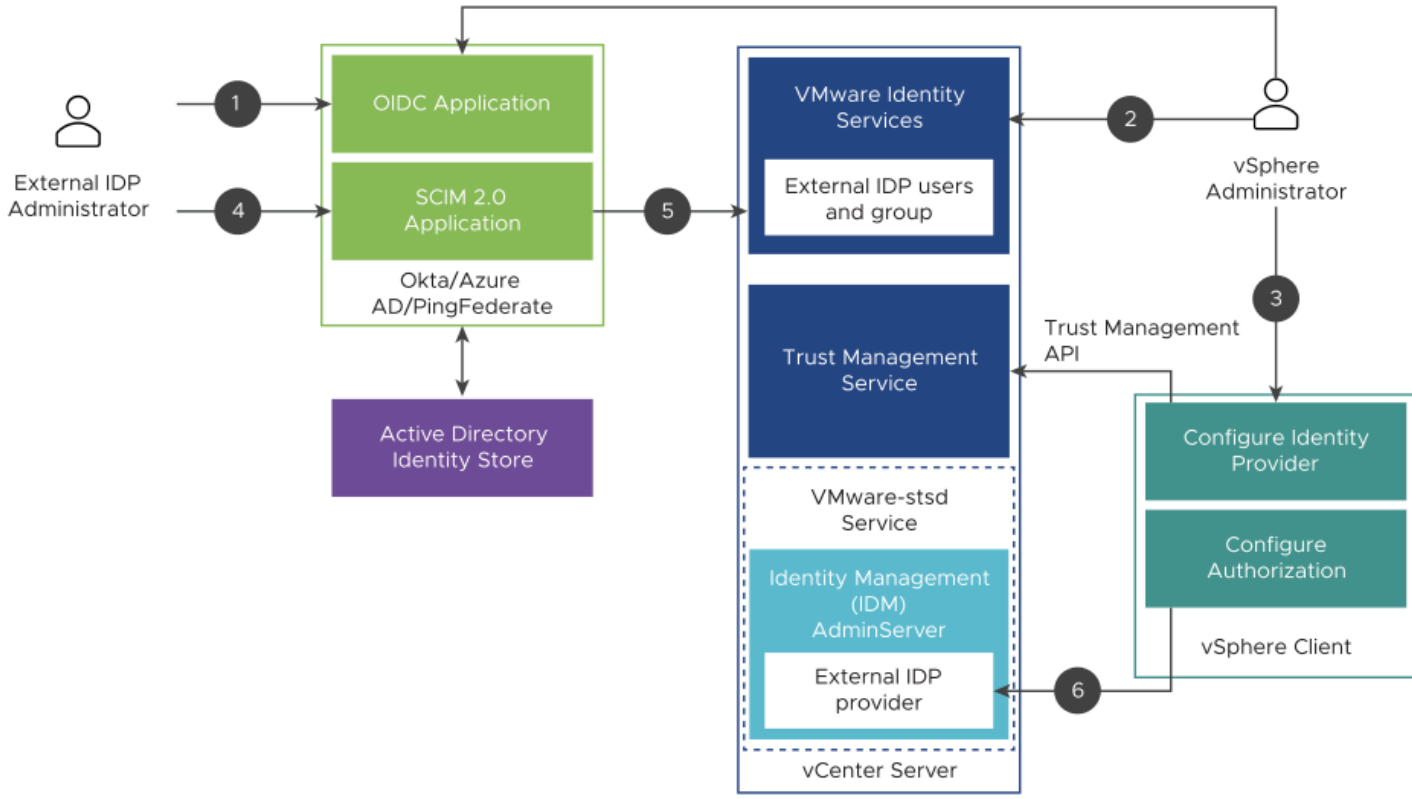
vCenter also uses this search process when you configure global permissions for Active Directory users and groups. In both cases, either configuring global permissions, or adding a user or group, you select the domain you entered for your

AD FS identify provider from the **Domain** drop-down menu to search for and select users and groups from your Active Directory domain.

vCenter Identity Provider Federation Configuration Process Flow Using VMware Identity Services

To configure Okta, Microsoft Entra ID, and PingFederate, you use VMware Identity Services. The following figure shows the process flow that occurs when configuring vCenter Identity Provider Federation using VMware Identity Services.

Figure 36: vCenter Identity Provider Federation Configuration Process Flow using VMware Identity



Services

vCenter, VMware Identity Services, and Active Directory interact as follows.

1. The external IDP administrator configures an OIDC Application for vCenter.
2. The vCenter administrator logs in to the vCenter using the vSphere Client, adds an identity provider to vCenter, and also enters domain information.
3. The vCenter administrator gives the Redirect URI (obtained from the identity provider configuration page in the vSphere Client) to the identity provider administrator to add to the OIDC Application created in step 2.
4. The external IDP administrator configures a SCIM 2.0 Application.
5. The external IDP administrator assigns the users and groups to the SCIM 2.0 Application, and pushes users and groups to vCenter.
6. The vCenter administrator configures authorization permissions in vCenter for external IDP users.

External IDP Users and Groups

Because an external identity provider uses System for Cross-domain Identity Management (SCIM) for users and groups, those users and groups reside in your vCenter. When you search for users and groups in your external identity provider, for example, to assign permissions, the search happens locally on the vCenter.

vCenter also uses this search process when you configure global permissions for external IDP users and groups. In both cases, either configuring global permissions, or adding a user or group, you select the domain you entered for your identity provider from the **Domain** drop-down menu to search for and select users and groups from your domain.

Use the Trusted Root Certificates Store Instead of the JRE truststore

If you imported a root CA certificate issued by your own internal Certificate Authority to the JRE truststore in vSphere 7.0, starting in vSphere 7.0 Update 1, you can register the certificate to the Trusted Root Certificates Store.

To configure vCenter Identity Provider Federation in vSphere 7.0 with a root CA certificate that was issued by your own internal Certificate Authority, you had to import it to the JRE truststore. Starting in vSphere 7.0 Update 1, you can register the certificate to the Trusted Root Certificates Store. This change means that you should add the root CA certificate that was issued by your own internal Certificate Authority to the Trusted Root Certificates Store (also called the VMware Endpoint Certificate Store, or VECS). Certificates in the JRE truststore continue to function, however, vCenter is standardizing on using the Trusted Root Certificates Store.

1. Log in with the vSphere Client to the vCenter.

   Note:
   To learn more, see Add a Trusted Root Certificate to the Certificate Store Using the vSphere Client.

2. Navigate to **Administration** > **Certificates** > **Certificate Management**.
3. Next to **Trusted Root Certificates**, click **Add**.
4. Browse for the AD FS root certificate and click **Add**.
   The certificate is added in a panel under **Trusted Root Certificates**.

Configure vCenter Identity Provider Federation for AD FS

After installing or upgrading to vSphere 7.0 or later, you can configure vCenter Identity Provider Federation for AD FS as an external identity provider.

Note:
This process of configuring an AD FS identity provider requires that you have administrative access to both your vCenter and your AD FS server. During the configuration process, you enter information first in your vCenter, then in your AD FS server, then in your vCenter.

Active Directory Federation Services requirements:

- AD FS for Windows Server 2016 or later must already be deployed.
- AD FS must be connected to Active Directory.
- An Application Group for vCenter must be created in AD FS as part of the configuration process. See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=78029.
- An AD FS server certificate (or a CA or intermediate certificate that signed the AD FS server certificate) that you add to the Trusted Root Certificates Store.
- You have created a vCenter administrators group in AD FS that contains the users you want to grant vCenter administrator privileges to.

For more information about configuring AD FS, see the Microsoft documentation.

vCenter and other requirements:

- vSphere 7.0 or later
- vCenter must be able to connect to the AD FS discovery endpoint, and the authorization, token, logout, JWKS, and any other endpoints advertised in the discovery endpoint metadata.
- You need the **VcIdentityProviders** > **Manage** privilege to create, update, or delete a vCenter Identity Provider that is required for federated authentication. To limit a user to view the Identity Provider configuration information only, assign the **VcIdentityProviders** > **Read** privilege.

Note:

These instructions are for vSphere 8.0 Update 1 and later. For vSphere 8.0, see the topic about configuring vCenter identity provider federation for AD FS in the *vSphere Authentication* documentation at https://techdocs.broadcom.com/us/en/vmware-cis/vsphere/vsphere/8-0.html

vCenter supports only one configured external identity provider (one source), and the vsphere.local identity source. You cannot use multiple external identity providers. vCenter Identity Provider Federation uses OpenID Connect (OIDC) for user login to vCenter.

This task describes how to add an AD FS group to the vSphere Administrators group as the way to control permissions. You can also configure privileges using AD FS Authorization through global or object permissions in vCenter . See the *vSphere Security* documentation for details about adding permissions.

CAUTION:

If you use an Active Directory identity source that you previously added to vCenter for your AD FS identity source, do not delete that existing identity source from vCenter. Doing so causes a regression with previously assigned roles and group memberships. Both the AD FS user with global permissions and users that were added to the Administrators group will not be able to log in.

Workaround: If you do not need the previously assigned roles and group memberships, and want to remove the previous Active Directory identity source, remove the identity source before creating the AD FS provider and configuring group memberships in vCenter.

1. Log in with the vSphere Client to the vCenter.
2. Add your AD FS server certificate (or a CA or intermediate certificate that signed the AD FS server certificate) to the Trusted Root Certificates Store.

   Note:
   To learn more, see Add a Trusted Root Certificate to the Certificate Store Using the vSphere Client.

   a) Navigate to **Administration** > **Certificates** > **Certificate Management**.
   b) Next to **Trusted Root Store**, click **Add**.
   c) Browse for the AD FS certificate and click **Add**.

   The certificate is added in a panel under **Trusted Root Certificates**.
3. Start to create the identity provider on vCenter.
   a) Use the vSphere Client to log in as an administrator to vCenter.
   b) Navigate to **Home** > **Administration** > **Single Sign On** > **Configuration**.
   c) Click **Change Provider** and select **ADFS**.

   The **Configure Main Identity Provider** wizard opens.
   d) In the **Prerequisites** panel, review the AD FS and the vCenter requirements.
   e) Click **Run Prechecks**.

   If the precheck finds errors, click **View Details** and take steps to resolve the errors as indicated.
   f) When the Precheck passes, click the confirmation check box then click **Next**.
   g) In the **Users and Groups** panel, enter user and group information for the Active Directory over LDAP connection to search for users and groups.

   vCenter derives the AD domain to use for authorization and permissions from the Base Distinguished Name for users. You can add permissions on vSphere objects only for users and groups from this AD domain. Users or groups from AD child domains or other domains in the AD forest are not supported by vCenter Identity Provider Federation.

Option	Description
Base distinguished name for users	Base Distinguished Name for users.
Base distinguished name for groups	The base Distinguished Name for groups.
Username	ID of a user in the domain who has a minimum of read-only access to Base DN for users and groups.
Password	ID of a user in the domain who has a minimum of read-only access to Base DN for users and groups.

Option	Description
Primary server URL	Primary domain controller LDAP server for the domain. Use the format `ldap://hostname:port` or `ldaps://hostname:port`. The port is typically 389 for LDAP connections and 636 for LDAPS connections. For Active Directory multi-domain controller deployments, the port is typically 3268 for LDAP and 3269 for LDAPS. A certificate that establishes trust for the LDAPS endpoint of the Active Directory server is required when you use `ldaps://` in the primary or secondary LDAP URL.
Secondary server URL	Address of a secondary domain controller LDAP server that is used for failover.
SSL certificates	If you want to use LDAPS with your Active Directory LDAP Server or OpenLDAP Server identity source, click `Browse` to select a certificate.

h) Click **Next**.

i) In the **OpenID Connect** panel, copy the Redirect URI and the Log-out Redirect URI.

Leave the other fields blank for now. You return to the **OpenID Connect** panel after creating the OpenID Connection configuration in the next step.

4. Create an OpenID Connect configuration in AD FS and configure it for vCenter.

To establish a relying party trust between vCenter and an identity provider, you must establish the identifying information and a shared secret between them. In AD FS, you do so by creating an OpenID Connect configuration known as an Application Group, which consists of a Server application and a Web API. The two components specify the information that vCenter uses to trust and communicate with the AD FS server. To enable OpenID Connect in AD FS, see the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=78029.

Note the following when you create the AD FS Application Group.

- You need the two vCenter Redirect URIs obtained in the preceding step.
- Copy the following information from the AD FS Application Group to a file or write it down for use when finishing the creation of the vCenter identity provider in the next step.
  - Client Identifier
  - Shared secret
  - OpenID address of the AD FS server

Note:
If necessary, obtain the OpenID address of your AD FS server by running the following PowerShell command as an AD FS administrator.

```
Get-AdfsEndpoint | Select FullUrl | Select-String openid-configuration
```

Copy the URL that is returned (select only the URL itself, not the closing bracket or the initial `"@{FullUrl="` part).

5. In the vCenter**OpenID Connect** panel:

a) Enter the following information, which you obtained from the previous step when creating the AD FS Application Group:

- Client Identifier
- Shared Secret
- OpenID Address

The Identity Provider Name is automatically filled in as Microsoft ADFS.

b) Click **Next**.

6. Review the information and click **Finish**.

vCenter creates the AD FS identity provider and displays the configuration information.

7. Configure group membership in vCenter for AD FS Authorization.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
   c) Click the **Groups** tab.
   d) Click the **Administrators** group and click **Add Members**.
   e) Select the domain from the drop-down menu.
   f) In the text box below the drop-down menu, enter the first few characters of AD FS group that you want to add then wait for the drop-down selection to appear.

      It might take several seconds for the selection to appear as vCenter establishes the connection to and searches Active Directory.
   g) Select the AD FS group and add it to the Administrators group.
   h) Click **Save**.
8. Verify logging in to vCenter with an Active Directory user.

Configure vCenter Identity Provider Federation for Okta

After installing or upgrading to vSphere 8.0 Update 1 or later, you can configure vCenter Identity Provider Federation for Okta as an external identity provider.

Okta requirements:

- You are using Okta and have a dedicated domain space, for example, https://*your-company*.okta.com.
- To perform OIDC logins and manage user and group permissions, you must create the following Okta applications.
  – An Okta native application with OpenID Connect as the sign-on method. The native application must include the grant types of authorization code, refresh token, and resource owner password.
  – A System for Cross-domain Identity Management (SCIM) 2.0 application with an OAuth 2.0 Bearer Token to perform user and group synchronization between the Okta server and the vCenter.

  See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=90835.
- You have identified the Okta users and groups that you want to share with vCenter. This sharing is a SCIM operation (not an OIDC operation).

Okta connectivity requirements:

- vCenter must be able to connect to the Okta discovery endpoint, and the authorization, token, JWKS, and any other endpoints advertised in the discovery endpoint metadata.
- Okta must also be able to connect with vCenter to send user and group data for the SCIM provisioning.

vCenter requirements:

- vSphere 8.0 Update 1 or later
- On the vCenter where you want to create the Okta identity source, verify that the VMware Identity Services are activated.

  Note:
  When you install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Servers are activated by default. You can use use the vCenter Management Interface to confirm the status of the VMware Identity Services. See Stop and Start the VMware Identity Services.

vSphere privileges requirements:

- You must have the **VcIdentityProviders** > **Manage** privilege to create, update, or delete a vCenter Identity Provider that is required for federated authentication. To limit a user to view the Identity Provider configuration information only, assign the **VcIdentityProviders** > **Read** privilege.

Enhanced Linked Mode requirements:

- You can configure vCenter Identity Provider Federation for Okta in an Enhanced Linked Mode configuration. When you configure Okta in an Enhanced Link Mode configuration, you configure the Okta identity provider to use VMware Identity Services on a single vCenter system. For example, if your Enhanced Mode Link configuration consists of two vCenter systems, only one vCenter and its instance of VMware Identity Services is used to communicate with the Okta

server. If this vCenter system becomes unavailable, you can configure VMware Identity Services on other vCenter in the ELM configuration to interact with your Okta server. For more information, see Activation Process for External Identity Providers in Enhanced Linked Mode Configurations.

- When configuring Okta as an external identity provider, all the vCenter systems in an Enhanced Linked Mode configuration must run at least vSphere 8.0 Update 1.

Networking requirements:

- If your network is not publicly available, you must create a network tunnel between your vCenter system and your Okta server, then use the appropriate publicly accessible URL as the Base Uri.

vCenter supports only one configured external identity provider (one source), and the vsphere.local identity source (local source). You cannot use multiple external identity providers. vCenter Identity Provider Federation uses OpenID Connect (OIDC) for user login to vCenter.

You can configure privileges using Okta groups and users through global or object permissions in vCenter. See the *vSphere Security* documentation for details about adding permissions.

1. Create an OpenID Connect application in Okta and assign groups and users to the OpenID Connect application.

   To create the OpenID Connect application and assign groups and users, see the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=90835. Follow the steps in the section titled "Create the OpenID Connect Application." After you create the Okta OpenID Connect application, copy the following information from the Okta OpenID Connect application to a file for use when configuring the vCenter identity provider in the next step.

   - Client Identifier
   - Client secret (shown as Shared secret in the vSphere Client)
   - Active Directory domain information, or Okta domain information if you are not running Active Directory

2. To create the identity provider on vCenter:
   a) Use the vSphere Client to log in as an administrator to vCenter.
   b) Navigate to **Home** > **Administration** > **Single Sign On** > **Configuration**.
   c) Click **Change Provider** and select **Okta**.

      The **Configure Main Identity Provider** wizard opens.
   d) In the **Prerequisites** panel, review the Okta and the vCenter requirements.
   e) Click **Run Prechecks**.

      If the precheck finds errors, click **View Details** and take steps to resolve the errors as indicated.
   f) When the Precheck passes, click the confirmation checkbox then click **Next**.
   g) In the **Directory Information** panel, enter the following information.

      - Directory Name: Name of the local directory to create on vCenter that stores the users and groups pushed from Okta. For example, `vcenter-okta-directory`.
      - Domain Name(s): Enter the Okta domain names that contain the Okta users and groups you want to synchronize with vCenter.

        After you enter your Okta domain name, click the Plus icon (+) to add it. If you enter multiple domain names, specify the default domain.

   h) Click **Next**.
   i) In the **OpenID Connect** panel, enter the following information.

      - Redirect UI: Filled in automatically. You give the redirect UI to your Okta administrator for use in creating the OpenID Connect application.
      - Identity Provider Name: Filled in automatically as Okta.
      - Client Identifier: Obtained when you created the OpenID Connect application in Okta in step 1. (Okta refers to Client Identifier as the Client ID.)
      - Shared Secret: Obtained when you created the OpenID Connect application in Okta in step 1. (Okta refers to Shared Secret as the Client Secret.)
      - OpenID Address: Takes the form https://*`Okta domain space`*/oauth2/default/.well-known/openid-configuration.

        For example, if your Okta domain space is example.okta.com, then the OpenID Address is: `https://example.okta.com/oauth2/default/.well-known/openid-configuration`

See https://developer.okta.com/docs/reference/api/oidc/#well-known-openid-configuration for more information.

j) Click **Next**.

k) Review the information and click **Finish**.

vCenter creates the Okta identity provider and displays the configuration information.

l) If necessary, scroll down and click the **Copy** icon for the Redirect URI and save it to a file.

You use the Redirect URI in the Okta OpenID Connection application.

m) Click the **Copy** icon for the Tenant URL and save it to a file.

Note:
If your network is not publicly available, you must create a network tunnel between your vCenter system and your Okta server. After creating the network tunnel, use the appropriate publicly accessible URL as the Base Uri.

n) Under **User Provisioning**, click **Generate** to create the secret token, select the token lifespan from the drop-down, then click **Copy to Clipboard**. Save the token to a secure location.

You use the Tenant URL and the token in the Okta SCIM 2.0 application. The Okta SCIM 2.0 application uses the token to synchronize the Okta users and groups into VMware Identity Services. This information is necessary to push Okta users and groups from Okta to vCenter.

3. Return to the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=90835 to update the Okta Redirect URI.

   Follow the steps in the section titled "Update the Okta Redirect URI."

4. To create the SCIM 2.0 application, remain in the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=90835.

   Follow the steps in the section titled "Create the SCIM 2.0 Application and Push Users and Groups to vCenter ."

   When done creating the SCIM 2.0 application as described in the knowledge base article, continue with the next step.

5. Configure vCenter for Okta Authorization.

   You can either assign Okta users to a vCenter group or assign inventory-level and global permissions to Okta users. The minimum permission required for logging in is Read-Only.

   To assign Okta users to a group, see Add Members to a vCenter Single Sign-On Group. To assign inventory-level and global permissions to Okta users, see the topic about managing permissions for vCenter components in the *vSphere Security* documentation.

6. Verify logging in to vCenter with an Okta user.

Configure vCenter Identity Provider Federation for Microsoft Entra ID

After installing or upgrading to vCenter 9.1, you can configure vCenter Identity Provider Federation for Microsoft Entra ID (formerly called Azure AD) as an external identity provider.

Microsoft Entra ID requirements:

- You are customer of Microsoft and have a Microsoft Entra ID account.

Microsoft Entra ID Connectivity requirements:

- You have created an enterprise (non-gallery) application with OpenID Connect as a sign-on method.
- Add authorization code, refresh token and resource owner password as grant types in the created application.
- For user and group sync, you need to configure VMware Identity Services Gallery Application for SCIM 2.0 provisioning in Microsoft Entra ID with OAuth 2.0 Bearer Token.

vCenter requirements:

- vCenter 9.1 or later, with the VMware Identity Services activated (they are activated by default).
- On the vCenter where you want to create the Microsoft Entra ID identity source, verify that the VMware Identity services are activated.
- The users and groups from the identity provider are provisioned in your vCenter.

vSphere privileges requirements:

- You must have the **VcIdentityProviders.Manage** privilege to create, update, or delete a vCenter Identity Provider that is required for federated authentication. To limit a user to view the Identity Provider configuration information only, assign the **VcIdentityProviders.Read** privilege.

Enhanced Linked Mode requirements:

- You can configure vCenter Identity Provider Federation for Microsoft Entra ID in an Enhanced Linked Mode configuration. When you configure Microsoft Entra ID in an Enhanced Link Mode configuration, you configure the Microsoft Entra ID identity provider to use VMware Identity Services on a single vCenter system. For example, if your Enhanced Mode Link configuration consists of two vCenter systems, only one vCenter and its instance of VMware Identity Services is used to communicate with the Microsoft Entra ID server. If this vCenter system becomes unavailable, you can configure VMware Identity Services on other vCenter systems in the ELM configuration to interact with your Microsoft Entra ID server. For more information, see Activation Process for External Identity Providers in Enhanced Linked Mode Configurations.
- When configuring Microsoft Entra ID as an external identity provider, all the vCenter systems in an Enhanced Linked Mode configuration must run at least vSphere 8.0 Update 2.

Networking requirements:

- If your network is not publicly available, you must create a network tunnel between your vCenter system and your Microsoft Entra ID server, then use the appropriate publicly accessible URL as the Base Uri.

vCenter supports only one configured external identity provider (one source), and the vsphere.local identity source (local source). You cannot use multiple external identity providers. vCenter Identity Provider Federation uses OpenID Connect (OIDC) for user login to vCenter.

You can configure privileges using Microsoft Entra ID groups and users through global or object permissions in vCenter. See the *vSphere Security* documentation for details about adding permissions.

For a walk-through of the configuration process, see the following video:

vCenter Authentication: AzureAD/Entra ID integration | vSphere 8 Update 2

1. Create an OpenID Connect application in Microsoft Entra ID and assign groups and users to the OpenID Connect application.

   To create the OpenID Connect application and assign groups and users, see the Broadcom knowledge base article: 322179. Follow the steps in the section titled "Create the OpenID Connect Application." After you create the OpenID Connect application, copy the following information from the Microsoft Entra ID OpenID Connect application to a file for use when configuring the vCenter identity provider in the next step.

   - Client Identifier
   - Client secret (shown as Shared secret in the vSphere Client).
   - Active Directory domain information, or Microsoft Entra ID domain information if you are not running Active Directory.

2. To create the identity provider on vCenter:
   a) Use the vSphere Client to log in as an administrator to vCenter.
   b) Navigate to **Home** > **Administration** > **Single Sign On** > **Configuration**.
   c) Click **Change Provider** and select **Microsoft Entra ID**.

      The **Configure Main Identity Provider** wizard opens.
   d) In the **Prerequisites** panel, review the Microsoft Entra ID and the vCenter requirements.
   e) Click **Run Prechecks**.

      If the precheck finds errors, click **View Details** and take steps to resolve the errors as indicated.
   f) When the Precheck passes, click the confirmation check box then click **Next**.
   g) In the **Directory Information** panel, enter the following information.

      - Directory Name: Name of the local directory to create on vCenter that stores the users and groups pushed from Microsoft Entra ID. For example, `vcenter-entraid-directory`.
      - Domain Names: Enter the Microsoft Entra ID domain names that contain the Microsoft Entra ID users and groups you want to synchronize with vCenter.

After you enter your Microsoft Entra ID domain name, click the Plus icon (+) to add it. If you enter multiple domain names, specify the default domain.

h) Click **Next**.

i) In the **OpenID Connect** panel, enter the following information.

- Redirect UI: Filled in automatically. You give the redirect UI to your Microsoft Entra ID administrator for use in creating the OpenID Connect application.
- Identity Provider Name: Filled in automatically as Microsoft Entra ID.
- Client Identifier: Obtained when you created the OpenID Connect application in Microsoft Entra ID in step 1. (Microsoft Entra ID refers to Client Identifier as the Client ID.)
- Shared Secret: Obtained when you created the OpenID Connect application in Microsoft Entra ID in step 1. (Microsoft Entra ID refers to Shared Secret as the Client Secret.)
- OpenID Address: Takes the form https://*`Microsoft Entra ID domain space`*/oauth2/default/.well-known/openid-configuration.

  For example, if your Microsoft Entra ID domain space is example.EntraID.com, then the OpenID Address is: `https://example.EntraID.com/oauth2/default/.well-known/openid-configuration`

j) Click **Next**.

k) Review the information and click **Finish**.

vCenter creates the Microsoft Entra ID identity provider and displays the configuration information.

l) If necessary, scroll down and click the **Copy** icon for the Redirect URI and save it to a file.

You use the Redirect URI in the Microsoft Entra ID OpenID Connection application.

m) Click the **Copy** icon for the Tenant URL and save it to a file.

Note:
If your network is not publicly available, you must create a network tunnel between your vCenter system and your Microsoft Entra ID server. After creating the network tunnel, use the appropriate publicly accessible URL as the Base URI.

n) Under **User Provisioning**, click **Generate** to create the secret token, select the token lifespan from the drop-down, then click **Copy to Clipboard**. Save the token to a secure location.

You use the Tenant URL and the token in the Microsoft Entra ID SCIM 2.0 application. The Microsoft Entra ID SCIM 2.0 application uses the token to synchronize the Microsoft Entra ID users and groups into VMware Identity Services. This information is necessary to push Microsoft Entra ID users and groups from Microsoft Entra ID to vCenter.

3. Return to the Broadcom knowledge base article: 322179 to update the Microsoft Entra ID Redirect URI.

   Follow the steps in the section titled "Update the Azure AD Redirect URI."

4. To create the SCIM 2.0 application, remain in the Broadcom knowledge base article: 322179.

   Follow the steps in the section titled "Create the SCIM 2.0 Application and Push Users and Groups to vCenter."

   When done creating the SCIM 2.0 application as described in the knowledge base article, continue with the next step.

5. Configure the group membership in vCenter for Microsoft Entra ID Authorization.

   You must configure group membership before Microsoft Entra ID users can log in to vCenter.

a) In the vSphere Client, while logged in as a local administrator, go to **Administration** > **Single Sign On** > **Users and Groups**.
b) Click the **Groups** tab.
c) Click the **Administrators** group and click **Add Members**.
d) Select the domain name of the Microsoft Entra ID group that you want to add from the drop-down menu.
e) In the text box below the drop-down menu, enter the first few characters of Microsoft Entra ID group that you want to add then wait for the drop-down selection to appear.
f) Select the Microsoft Entra ID group and add it to the Administrators group.
g) Click **Save**.

6. Verify logging in to vCenter with a Microsoft Entra ID user.
7. To assign inventory-level and global permissions to Microsoft Entra ID users, see the topic about managing permissions for vCenter components in the *vSphere Security* documentation.

Configuring vCenter Identity Provider for PingFederate

After installing or upgrading to vSphere 9.0, you can configure vCenter Identity Provider Federation for PingFederate as an external identity provider.

High-Level Steps to Configure vCenter Identity Provider for PingFederate

Configuring vCenter for PingFederate involves the following high-level steps:

1. On PingFederate, creating the vCenter/VMware Identity Services specific configuration, including the scopes and common configuration for PingFederate workflows.
2. On PingFederate, creating global items, including the Password Grant Flow configuration and Authorization Code Flow configuration.
3. On PingFederate, installing the SCIM Provisioner.
4. On vCenter, creating the PingFederate identity provider.
5. On PingFederate, creating the SCIM Application (SP Connection).
6. On vCenter, authorizing PingFederate users.

Note:
The instructions in this documentation create a typical setup for your PingFederate server. Your environment might differ, and thus you might make different selections.

Prerequisites to Configure vCenter Identity Provider for PingFederate

PingFederate requirements:

- You have installed an on-premises PingFederate server.
- From the vCenter on which you configure the PingFederate identity provider, you must obtain the trusted root certificates and import them into the PingFederate server.
- Optionally, you might need to import the PingFederate SSL certificate, or certificate chain, to the vCenter, if that certificate is self-signed (that is, not issued by a well-known, public Certificate Authority). If the PingFederate SSL certificate or one of the certificates in the chain was issued by a well-known Certificate Authority, vCenter automatically trusts that certificate, and you do not need to import it. If you are using one or more intermediate signing authorities for your PingFederate server SSL certificate, include the entire chain of certificates.

  To export the PingFederate SSL certificate, in the PingFederate Admin Console, go to **Security** > **SSL Server Certificates**, select the default certificate, and select **Export** from the **Select Action** drop-down.

  You import the PingFederate SSL certificate using the vSphere Client in the **OpenID Connect** panel as part of the configure identity provider workflow.
- To perform OIDC logins and manage user and group permissions, you must create the following PingFederate applications.
  – A PingFederate native application with OpenID Connect as the sign-on method. The native application must include the grant types of authorization code, refresh token, and resource owner password.

  - A System for Cross-domain Identity Management (SCIM) 2.0 application (in PingFederate, this is called the SP Connection), with an OAuth 2.0 Bearer Token to perform user and group synchronization between the PingFederate server and the vCenter.
- You have identified the PingFederate users and groups that you want to share with vCenter. This sharing is a SCIM operation (not an OIDC operation).

PingFederate connectivity requirements:

- vCenter must be able to connect to the PingFederate discovery endpoint, and the authorization, token, JWKS, and any other endpoints advertised in the discovery endpoint metadata.
- PingFederate must also be able to connect with vCenter to send user and group data for the SCIM provisioning.

vCenter requirements:

- vSphere 8.0 Update 3
- On the vCenter where you want to create the PingFederate identity source, verify that the VMware Identity Services are activated.

  Note:
  When you install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Servers are activated by default. You can use the vCenter Management Interface to confirm the status of the VMware Identity Services. See Stop and Start the VMware Identity Services.

vSphere privileges requirements:

- You must have the **VcIdentityProviders** > **Manage** privilege to create, update, or delete a vCenter Identity Provider that is required for federated authentication. To limit a user to view the Identity Provider configuration information only, assign the **VcIdentityProviders** > **Read** privilege.

Enhanced Linked Mode requirements:

- You can configure vCenter Identity Provider Federation for PingFederate in an Enhanced Linked Mode configuration. When you configure PingFederate in an Enhanced Link Mode configuration, you configure the PingFederate identity provider to use VMware Identity Services on a single vCenter system. For example, if your Enhanced Mode Link configuration consists of two vCenter systems, only one vCenter and its instance of VMware Identity Services is used to communicate with the PingFederate server. If this vCenter system becomes unavailable, you can configure VMware Identity Services on other vCenter in the ELM configuration to interact with your PingFederate server. For more information, see Activation Process for External Identity Providers in Enhanced Linked Mode Configurations.
- When configuring PingFederate as an external identity provider, all the vCenter systems in an Enhanced Linked Mode configuration must run at least vSphere 8.0 Update 3.

Create the Scopes

PingFederate supports the use of scopes to constrain and define access privileges.

Review the Prerequisites to Configure vCenter Identity Provider for PingFederate.

Log in to the PingFederate Admin console with an Administrator Account.

1. Go to **System** > **OAuth Settings** > **Scope Management**.
2. On the **Common Scopes** tab, add the following **Scope Values** along with a description. Click **Add** after entering each value and description.
   - **openid**
   - **profile**
   - **email**
3. Skip the **Exclusive Scopes** tab.
4. On the **Default Scope** tab, enter a description for **Default Scope**.
   A description is required. If **Default Scope Description** is empty, PingFederate logs the following error:

   The requested scope is invalid, unknown, malformed, or exceeds that which the client is permitted to request.

5. Click **Save**.

Continue with Create a Common Configuration for PingFederate Workflows.

Create a Common Configuration for PingFederate Workflows

Creating the common configuration for PingFederate includes creating the Access Token Manager, the objectID Attribute, the OpenID Connect Policy, and the OAuth Client Application.

Complete the following task:

- Create the Scopes

Log in to the PingFederate Admin console with an Administrator Account.

1. Create the Access Token Manager.
   a) Go to **Applications** > **OAuth** > **Access Token Management**.
   b) Click **Create New Instance**.
   c) On the **Type** tab:
      - **Instance Name**: Enter an instance name. For example, vIDB Access Token Manager.
      - **Instance ID**: Enter the instance ID. For example, vIDB.
      - **Type**: Select **JSON Web Tokens**.
      - **Parent Instance**: Leave the default, **None**.
   d) On the **Instance Configuration** tab:
      - **Use Centralized Signing Key**: Select the checkbox.
        Leaving this checkbox unselected causes PingFederate to expect "Active Signing Certificate Key ID" to be configured.
      - **JWS Algorithm**: Select an algorithm. For example, **RSA using SHA-256**.
      - At the bottom of the screen, click **Show Advanced Fields**.
        – **JWT ID Claim Length**: Add a number greater than zero (0). For example, 24. If you do not enter a value, the JTI claim is omitted in the access token.
   e) Click **Next**.
   f) On the **Access Token Attribute Contract** tab:
      - In the **Extend the Contract** text box, add the following claims to be generated in the Ping access token. Click **Add** after entering each claim.
        – **aud**
        – **iss**
        – **exp**
        – **iat**
        – **userName**
      - **Subject Attribute Name**: Select one claim to be used for auditing purposes. For example, **iss**.
   g) Click **Next** twice to skip the **Resource URIs** and **Access Control** tabs.
   h) Click **Save**.
2. Create the OpenID Connect Policy.
   a) Go to **Applications** > **OAuth** > **OpenID Connect Policy Management**.
   b) Click **Add Policy**.
   c) On the **Manage Policy** tab:
      - **Policy ID**: Enter a policy ID. For example, OIDC.
      - **Name**: Enter a policy name. For example, OIDC Policy.
      - **Access Token Manager**: Select the access token manager that you previously created. For example, vIDB Access Token Manager.

d) Click **Next**.
e) On the **Attribute Contract** tab:
   - Click **Delete** to remove all the attributes except **sub**. Otherwise, you must map the attributes to a value in the **Contract Fulfillment** tab later.
f) Click **Next**, then click **Next** again to skip the **Attributes Scope** tab.
g) On the **Attribute Sources & User Lookup** tab, click **Add Attribute Source**.
   After entering the information on each tab that follows, click **Next** to advance.
   - **Data Store**:
     – **Attribute Source ID**: Enter an attribute source ID. For example, vIDBLDAP.
     – **Attribute Source Description**: Enter a description. For example: vIDBLDAP.
     – **Active Data Store**: Select your Active Directory or OpenLDAP domain name from the drop-down.
   - **LDAP Directory Search**:
     – **Base DN**: Enter your base DN to find your users and groups.
     – **Search Scope**: Leave the default, **Subtree**.
     – **Attributes to return from search**: Select **<Show All Attributes>** and select **objectGUID**. Click **Add Attribute**.
   - **LDAP Binary Attribute Encoding Types**:
     – **ObjectGUID**: Select **Hex** for the **Attribute Encoding Type**.
   - **LDAP Filter**:
     – **Filter**: Enter a filter. For example, `userPrincipalName=${userName}.`
h) On the **Summary** page, click **Done**.
i) Click **Next** to advance, and on the **Contract Fulfillment** tab, map the **Attribute Contract** for the ID token:

Attribute Contract	Source	Value
sub	Select the Attribute Source ID previously created. In this documentation, the example used is vIDBLDAP.	**objectGUID**

j) Click **Next**, then click **Next** again to skip the **Insurance Criteria** tab.
k) Click **Save**.

3. Create the OAuth Client Application.
   a) Go to **Applications** > **OAuth** > **Clients**.
   b) Click **Add Client**.
   c) On the **Clients | Client** page:
      - **Client ID**: Enter the client ID. For example, vIDB.

      Note:
      Copy and save the client ID for use later when creating the vCenter identity provider for PingFederate.
      - **Name**: Enter a name. For example, vIDB.
      - **Client Authentication**: Select **Client Secret**.
        – **Client Secret**: You can input your own client secret or generate a secret. After you leave this page, you cannot view the secret again. You only have the option to change the secret.

      Note:
      Copy and save the secret for use later when creating the vCenter identity provider.
      - **Redirection URIs**: Enter the redirection URI(s) in the form: `https://`*`vCenter_Server_FQDN`*`:port/federation/t/CUSTOMER/auth/response/oauth2`.
        – Click **Add**.
      - **Allowed Grant Types**: Check **Authorization Code**, **Refresh Token**, **Client Credentials**, and **Resource Owner Password Credentials**.
      - **Default Access Token Manager**: Select the access token manager you previously created. For example, the one used in this documentation is vIDB Access Token Manager.

- **OpenID Connect**: For **Policy**, select the one you previously created. For example, the one used in this documentation is OIDC.

d) Click **Save**.

Continue with Create the Password Grant Flow Configuration.

Create the Password Grant Flow Configuration

For PingFederate to authenticate with vCenter, you set up the password grant flow.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows

Log in to the PingFederate Admin console with an Administrator Account.

1. Create the Password Credential Validator.
   a) Go to **System** > **Data & Credential Stores** > **Password Credential Validators**.
   b) Click **Create New Instance**.
   c) In the **Password Credential Validators | Create New Instance** page, enter the information as follows for each tab, then click **Next** to advance.
      - On the **Type** tab:
        – **Instance Name**: Enter the instance name. For example, vIDB Validator.
        – **Instance ID**: Enter the instance ID. For example, vIDB.
        – **Type**: Select **LDAP Username Password Credential Validator**.
      - On the **Instance Configuration** tab:
        – **LDAP Datastore**: Select the data store you are using.
        – **Search Base**: Enter your base DN to find your users and groups.
        – **Search Filter**: Enter a filter. For example, `userPrincipalName=${username}`.
        – **Scope of Search**: Select **Subtree**.
      - On the **Extended Contract** tab:
        – By default, the following are added:
          - **DN**
          - **email**
          - **givenName**
          - **username**
   d) Click **Next** then click **Save**.
2. Map the Validator in the Authorization Server Settings.
   a) Go to **System** > **OAuth Settings** > **Authorization Server Settings**.
   b) In **Password Credentials Validator**, select the one you previously created. For example, this documentation uses vIDB Validator.
   c) Click **Save**.
3. Create the Resource Owner Credentials Grant Mapping.
   a) Go to **Authentication** > **OAuth** > **Resource Owner Credentials Mapping**.
   b) In the **Resource Owner Credentials Grant Mapping** window:
      - **Source Password Validator Instance**: Select the one you previously created and click **Add Mapping**.
   c) In the **Resource Owner Credentials Grant Mapping | Resource Owner Credentials Mapping** page, click **Next** to skip the **Attribute Sources & User Lookup** tab.
   d) On the **Contract Fulfillment** tab:
      - For **USER_KEY**, select **Password Credentials Validator**, and for **Value**, select **username**.

e) Click **Next** to skip the **Insurance Criteria** tab then click **Save**.

4. Create the Access Token Mapping - Map the Password Credentials Validator to the Access Token Manager.

   This mapping is needed for the Password Grant workflow. If the mapping does not exist, PingFederate logs the following error:

   There are no access token managers available for the selected client and authentication context.

   a) Go to **Applications** > **Access Token Mappings**.
      - **Context**: Select the one you previously created. For example, this documentation uses vIDB Validator.
      - **Access Token Manager**: Select the one previously created. For example, this documentation uses vIDB Access Token Manager.
   b) Click **Add Mapping**.
   c) Click **Next** to skip the **Attribute Sources & User Lookup** tab.
   d) On the **Contract Fulfilment** tab, use the following table.

Contract	Source	Value
aud	**Context**	The client ID previously created. For example, the ID used in this documentation is vIDB.
exp	**No Mapping**	-
iat	**Expression**	Enter the following: `@org.jose4j.jwt.NumericDate@now().getValue()`

iss	**Expression**
	(If not visible, see the PingFederate documentation at https://docs.pingidentity.com/r/en-us/pingfederate-120/pf_enable_disable_express.)
		Enter the following:
		`#tmp=#this.get("context.HttpRequest").getOb`
		`#url=new java.net.URL(#tmp),`
		`#protocol=#url.getProtocol(),`
		`#host=#url.getHost(),#port=#url.getPort(),`
		`#result=(#port != -1) ?`
		`@java.lang.String@format("%s://%s:%d", #protocol, #host, #port) :`
		`@java.lang.String@format("%s://%s", #protocol, #host, #port)`

userName	**No Mapping**	- This contract is used later in the LDAP Filter for the OIDC Policy for Authorization Code workflow. For the PingFederate workflow, it is not needed.

e) Click **Next** to skip the **Insurance Criteria** tab then click **Save**.

Continue with Create the Authorization Code Flow Configuration.

Create the Authorization Code Flow Configuration

Creating the authorization code flow in PingFederate involves creating and configuring an IdP adapter.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows
- Create the Password Grant Flow Configuration

Log in to the PingFederate Admin console with an Administrator Account.

1. Create the IdP Adapter.
   a) Go to **Authentication** > **Integration** > **IdP Adapters**.
   b) Create **Create New Instance**.
   c) On the **Type** tab:
      - **Instance Name**: Enter a name, for example, HTML Form Auth Adapter.
      - **Instance ID**: Enter an ID, for example, HTMLFormAuthAdapter.
      - **Type**: Select HTML Form IdP Adapter.
      - **Parent Instance**: Select **None**.
   d) Click **Next**.
   e) On the **IdP Adapter** tab:

      Under **Password Credential Validator Instance**, click **Add a new row to 'Credential Validators.'** then select a validator (in this documentation, vIDB Validator is used), and click **Update**.
   f) On the **IdP Adapter** tab:
      - **Realm**: Select **USER_KEY**.
   g) Click **Next**.
   h) Click **Next** to skip the **Extended Contract** tab.
   i) On the **Adapter Attributes** tab:
      - **Unique User Key Attribute**: Select **username** and check **Pseudonym**.
   j) Click **Next** to skip the **Adapter Contract Mapping** tab then click **Save**.
2. Create IdP Adapter Grant Mapping.
   a) Go to **Authentication** > **OAuth** > **IdP Adapter Grant Mapping**.
   b) **Source Adapter Instance**: Select the Adapter instance you just created and click **Add Mapping**.
   c) On the **Attribute Sources & User Lookup** page, click **Add Attribute Source**.
   d) Enter the information as follows for each tab, then click **Next** to advance.
      - On the **Data Store** tab:
        – **Attribute Source ID**: Enter an ID with alphanumeric values.
        – **Attribute Source Description**. Enter a description.
        – **Active Data Store**: Select the active directory in use.
      - On the **LDAP Directory Search** tab:
        – **Base DN**: Enter your base DN to find your users and groups.
        – **Search Scope**: Use the default, **Subtree**.
        – **Attributes to return from search**: Select **<Show All Attributes>**, then, once loaded, from the attribute list, select **userPrincipalName**.
   e) Click **Add Attribute** then click **Next**.
   f) On the **LDAP Filter** tab:
      - **Filter**: Enter the filter. For example, `userPrincipalName=${username}`.
   g) Click **Next** then click **Save**.
   h) In the **IdP Adapter Grant Mapping | IdP Adapter Mapping** page, finish creating the IdP Grant Mapping.

      On the **Contract Fulfillment Lookup** tab, use the following table.

Contract	Source	Value
USER_KEY	Select the source previously created.	**Subject DN**
USER_NAME	Select the source previously created.	**userPrincipalName**

   i) Click **Next** then click **Save**.

      The created IdP Adapter Grant Mapping displays as '***Adapter Name***' **to Persistent Grant Contract**.

3. Map the IdP Adapter to Access Token Manager.
   a) Go to **Applications** > **OAuth** > **Access Token Mappings**.
      - **Context**: Select **IdP Adapter:*`Adapter Name`***.
      - **Access Token Manager**: Select the access token manager instance previously created. For example, in this documentation, it is vIDB Access Token Manager.
   b) Click **Add Mapping**.
      If you do not perform this mapping, PingFederate generates the following log file message:

      There are no mapped authentication sources to choose from. Please map an IdP Adapter or IdP connection first.
   c) Skip the **Attribute Sources & User Lookup** tab, and on the **Contract Fulfillment** tab, use the following table.

Contract	Source	Value
aud	**No Mapping**	-
exp	**No Mapping**	-
iat	**No Mapping**	-
iss	**No Mapping**	-
userName	**Adapter**	**username**

   d) Click **Next** to skip the **Insurance Criteria** tab then click **Save**.

Continue with Install the SCIM Provisioner.

Install the SCIM Provisioner

You create a System for Cross-domain Identity Management (SCIM) application that uses a token to synchronize the PingFederate users and groups into VMware Identity Services.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows
- Create the Password Grant Flow Configuration
- Create the Authorization Code Flow Configuration

On the PingFederate server, you must install the SCIM Provisioner to enable the provisioning of users and groups using SCIM.

> **Note:**
> If you are using an existing PingFederate environment, you might have already installed the SCIM Provisioner.

1. Download the SCIM Provisioner from https://marketplace.pingone.com/item/scim-provisioner.
   You must log in to the PingIdentity portal.
2. Copy the `pf-scim-quickconnection-1.4.jar` file to the folder that is mounted to the `/opt/out folder` of your PingFederate server.
   For example, place the file in the `/opt/out/instance/server/default/deploy` folder.
3. View the `/opt/out/instance/bin/run.properties` file and ensure that this setting is present: `pf.provisioner.mode=STANDALONE`
   According to the PingFederate documentation:

   STANDALONE - This server is a standalone instance that runs both the UI console and protocol engine (default).

4. If your PingFederate server instance is running as a container image, and you updated the `run.properties` file, you might need to restart the server. For example:
   a) Connect to the PingFederate server using SSH.
   b) Change to the `/root/ping` directory.
   c) Run the following commands:

```
docker-compose down
docker-compose up
```

The SCIM Connector displays as an option when configuring the user provisioning in Create the SCIM Application (SP Connection).

Continue with Configure vCenter Identity Provider Federation for PingFederate.

Configure vCenter Identity Provider Federation for PingFederate

After installing or upgrading to vSphere 8.0 Update 3, you can configure vCenter Identity Provider Federation for PingFederate as an external identity provider.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows
- Create the Password Grant Flow Configuration
- Create the Authorization Code Flow Configuration
- Install the SCIM Provisioner

Ensure that you have the following information from the PingFederate OpenID Connect application:

- Client Identifier
- Client secret (shown as Shared secret in the vSphere Client)
- Active Directory domain information, or PingFederate domain information if you are not running Active Directory

vCenter supports only one configured external identity provider (one source), and the vsphere.local identity source (local source). You cannot use multiple external identity providers. vCenter Identity Provider Federation uses OpenID Connect (OIDC) for user login to vCenter.

You can configure privileges using PingFederate groups and users through global or object permissions in vCenter. See the *vSphere Security* documentation for details about adding permissions.

1. To create the identity provider on vCenter:
   a) Use the vSphere Client to log in as an administrator to vCenter.
   b) Go to **Home** > **Administration** > **Single Sign On** > **Configuration**.
   c) Click **Change Provider** and select **PingFederate**.
      The **Configure Main Identity Provider** wizard opens.
   d) In the **Prerequisites** panel, review the PingFederate and the vCenter and other requirements.
   e) Click **Run Prechecks**.
      If the precheck finds errors, click **View Details** and take steps to resolve the errors as indicated.
   f) When the Precheck passes, click the confirmation checkbox then click **Next**.
   g) In the **Directory Information** panel, enter the following information.
      - Directory Name: Name of the local directory to create on vCenter that stores the users and groups pushed from PingFederate. For example, `vcenter-PingFederate-directory`.
      - Domain Name(s): Enter the PingFederate domain names that contain the PingFederate users and groups you want to synchronize with vCenter.
        After you enter your PingFederate domain name, click the Plus icon (+) to add it. If you enter multiple domain names, specify the default domain.

h) Click **Next**.
i) In the **OpenID Connect** panel, enter the following information.
   - Redirect UI: Filled in automatically. This redirect UI must match what you use in creating the OpenID Connect application in PingFederate.
   - Identity Provider Name: Filled in automatically as PingFederate.
   - Client Identifier: Obtained when you created the OpenID Connect application. (PingFederate refers to Client Identifier as the Client ID.)
   - Shared Secret: Obtained when you created the OpenID Connect application in PingFederate. (PingFederate refers to Shared Secret as the Client Secret.)
   - OpenID Address: Takes the form https://*PingFederate_domain_space*/idp/.well-known/openid-configuration.
     For example, if your PingFederate domain space is example.PingFederate.com, then the OpenID Address is: `https://example.PingFederate.com/idp/.well-known/openid-config`
   - SSL Certificate: Optionally, browse for the PingFederate SSL certificate, or certificate chain, if this certificate was not issued by a well-known, public Certificate Authority, to upload to vCenter. To export the PingFederate SSL certificate, in the Admin Console, go to **Security** > **SSL Server Certificates**, select the default certificate, and select **Export** from the **Select Action** drop-down. For more information, see the article, Exporting a certificate, at https://docs.pingidentity.com/r/en-us/pingfederate-111/nfv1585678806463. You can export the PingFederate SSL certificate without the private key, as it is not needed for the vCenter configuration.
j) Click **Next**.
k) Review the information and click **Finish**.
   vCenter creates the PingFederate identity provider and displays the configuration information.

2. Under **User Provisioning**, click **Generate** to create the secret token, select the token lifespan from the drop-down, then click **Copy to Clipboard**. Save the token to a secure location.
   When you create the PingFederate SP Connection (SCIM application), you use the token to synchronize the PingFederate users and groups into VMware Identity Services.

Continue with Create the SCIM Application (SP Connection).

Create the SCIM Application (SP Connection)

Creating a System for Cross-domain Identity Management (SCIM) 2.0 application is required, so that you can specify which PingFederate users and groups to push to vCenter.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows
- Create the Password Grant Flow Configuration
- Create the Authorization Code Flow Configuration
- Install the SCIM Provisioner
- Configure vCenter Identity Provider Federation for PingFederate

1. Add the vCenter Trusted Root Certificate to the PingFederate Server.
   Before you begin, export the trusted root certificate(s) from vCenter. You can obtain the certificate from the file system of the vCenter at `/var/lib/vmware/vmca/root.cer`. Or, see the Knowledge Base article at https://kb.vmware.com/s/article/2108294.
   a) Log in to the PingFederate Admin console with an Administrator Account.
   b) Go to **Security** > **Certificate & Key Management**.
   c) Select **Trusted CAs** then click **Import** to add the SSL certificate of the vCenter .
   d) If your PingFederate server instance is running as a container image, you might need to restart the server to add the certificate to the trust store. For example:
      1. Connect to the PingFederate server using SSH.
      2. Change to the `/root/ping` directory.
      3. Run the following commands:

```
docker-compose down
docker-compose up
```

2. Create the SP Connection.
   a) Log in to the PingFederate Admin console with an Administrator Account.
   b) Go to **Applications** > **Integration** > **SP Connections**.
   c) Click **Create Connection**.
   d) Select **Use a template for this connection** then select **SCIM Connector** from the drop-down.
      If the SCIM Connector option does not appear in the drop-down, check that you placed the SCIM Connector `.jar` file in the correct folder (the `/opt/out` folder of your PingFederate server).
   e) Click **Next**.
   f) Select only **Outbound Provisioning** then click **Next**.
   g) On the **General Info** tab:
      - **Partner's Entity ID (Connection ID)**: Update **SCIM Connector** to a name of your choice.
      - **Connection Name**: Enter a name.
      - **Base URL**: Enter the HTTPS address of the vCenter where you are configuring the PingFederate external identity provider, for example: `https://vcenter1.example.com`.
   h) Click **Next**.
   i) Click **Configure Provisioning**.
      On the **Target** tab:
      - **SCIM URL**: Enter the Usergroup endpoint.
        This is the Tenant URL obtained under **User Provisioning** on the **Configuration** page of the vCenter . For example: `https://vcenter1.example.com/usergroup/t/CUSTOMER/scim/v2`
      - **Authentication Method**: Select **OAuth 2 Bearer Token** from the drop-down.
      - **Access Token**: Paste the Secret Token that was generated from vCenter , and that you should have saved previously. See Step 2 in Configure vCenter Identity Provider Federation for PingFederate.
      - **Unique User Identifier**: Select **userName** from the drop-down.
      - **Filter Expression**: Copy the following expression into the text box: `externalId eq "%s"`
   j) Accept the rest of the default configuration setting values and click **Next**.
      - **Provisioning Options**: **User Create**, **User Update**, and **User Disable/ Delete** are checked.
      - **Remove User Action**: **Disable** is selected.

      Note:
      With **Disable** selected, when users are deleted from the Active Directory, they are not automatically shown as "disabled" in VMware Identity Services. This is expected behavior.
      - In Active Directory, instead of deleting the user on the next provisioning cycle, the user's property becomes "active"="false".
      - The user does not show as "disabled" in VMware Identity Services until another user is created or updated in Active Directory on the next provisioning cycle, and you follow the workaround at After deleting an AD user account, SaaS provisioner does not remove the user in the next provisioning cycle when Group DN is specified.
      - **Group Name Source**: **Common name** is selected.
   k) On the **Manage Channels** tab, click **Create**.
      - On the **Channel Info** tab:
        – **Channel Name**: Enter a name.
        – Accept the **Max threads** and **Timeout (Secs)** default values.
   l) Click **Next**.
      - On the **Source** tab:
        – **Active Data Store**: Choose your Active Directory domain.
   m) Click **Next**.
      - On the **Source Location** tab:

  - **Base DN**: Enter your base DN to find your users and groups.
  - **Users**: Customize to your environment. For example:
    - **Group DN:** Do not use.
    - **Filter**: Enter `(|(objectClass=person)(objectClass=organizationalPerson)(objectClass=user))`.
  - Groups: Customize to your environment. For example:
    - **Group DN**: Do not use.
    - **Filter**: Enter `(objectClass=group)`.

n) Click **Next**.
o) Accept the defaults on the **Attribute Mapping** tab.
p) Click **Next**.

On the **Activation & Summary** tab:

- **Channel Status**: Select **Active**.

q) Click **Done**.

The SP Connection is created and the SP Connections screen is displayed.

r) Click **Done**.
s) On the **Outbound Provisioning** tab, click **Next**.
t) Review the summary then click **Save**.
u) To make the connection active, toggle the **Enabled** slider.

PingFederate now pushes users and groups from the configured data store to vCenter. Allow some time for the push to occur. You can view the pushed users and groups in the vSphere Client. Go to **Administration** > **Single Sign On** > **Users and Groups**, and select the PingFederate domain.

Continue with Configure vCenter for PingFederate Authorization.

Configure vCenter for PingFederate Authorization

You can either assign PingFederate users to a vCenter group or assign inventory-level and global permissions to PingFederate users.

Complete the following tasks:

- Create the Scopes
- Create a Common Configuration for PingFederate Workflows
- Create the Password Grant Flow Configuration
- Create the Authorization Code Flow Configuration
- Install the SCIM Provisioner
- Configure vCenter Identity Provider Federation for PingFederate
- Create the SCIM Application (SP Connection)

The minimum permission required for a PingFederate user to log in is Read-Only.

1. To assign PingFederate users to a group, see Add Members to a vCenter Single Sign-On Group.
2. To assign inventory-level and global permissions to PingFederate users, see the topic about managing permissions for vCenter components in the *vSphere Security* documentation.
3. After assigning a PingFederate user permissions, verify that the user can log in.

Configure VMware Single Sign-On

After installing or upgrading to vSphere 8.0 Update 3, you can configure vCenter hosts for VMware Single Sign-On. When you configure VMware Single Sign-On, you use an external identity provider to sign into your vCenter hosts.

VMware Single Sign-On requirements:

- The vCenter on which you configure VMware Single Sign-On runs vSphere 8.0 Update 3.

- The vCenter hosts you want to connect to run at least vSphere 8.0 Update 1.
- You have configured one of the following external identity providers:
  - Microsoft Entra ID
  - Okta
  - PingFederate
- You must add the trusted root certificate from the vCenter host on which the external identity provider is configured to the vCenter host on which you configure VMware Single Sign-On.

VMware Single Sign-On enables you to connect vCenter hosts in a non-Enhanced Linked Mode configuration. That is, as long as you configure an external identity provider, you can leverage that configuration for single sign-on to other vCenter hosts. The vCenter host on which the external identity provider is configured acts as the identity provider for the other vCenter hosts.

You can configure multiple vCenter hosts to perform VMware Single Sign-On. To do so, you must configure each vCenter host to point to the vCenter host configured with an external identity provider.

After performing the VMware Single Sign-On configuration, you can still log in to your vCenter hosts with a local account.

Note:
VMware Single Sign-On does not share inventories between vCenter hosts as occurs in Enhanced Linked Mode.

1. Download the trusted root certificate from the vCenter host on which the external identity provider is configured. For example, see the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2108294.
2. Upload that trusted root certificate to the vCenter host on which you are configuring VMware SSO.
   See Add a Trusted Root Certificate to the Certificate Store Using the vSphere Client.
3. Use the vSphere Client to log in as an administrator to the vCenter host on which you are configuring VMware SSO.
4. Navigate to **Home** > **Administration** > **Single Sign On** > **Configuration**.
5. Click **Change Provider** and select **VMware SSO**.
   The **Configure Main Identity Provider** wizard opens.
6. In the **Prerequisites** panel, review the vCenter requirements.
7. Click **Run Prechecks**.
   If the precheck finds errors, click **View Details** and take steps to resolve the errors as indicated.
8. When the Precheck passes, click the confirmation checkbox then click **Next**.
9. In the **OpenID Connect** panel, enter the following information.
   - Identity Provider Name: Filled in with VMware SSO.
   - vCenter Server FQDN: Enter the FQDN of the vCenter host where the external identity provider is configured.
   - Port Number: Accept the default of 443, or change to the port you want to use.
   - User Name and Password: Enter the user name and password for an administrator account on this vCenter host where the external identity provider is configured.
10. Click **Next**.
11. Review the information and click **Finish**.
    vCenter creates the VMware SSO provider and displays the configuration information. This vCenter host now contains the same external identity provider configuration as the host on which the configuration was created. For example, when you compare the OpenID configurations between the two hosts, it is the same.
12. Configure this vCenter to use the external identity provider for authorization.
    You can either assign the external identity provider users to a vCenter group or assign inventory-level and global permissions to the users. The minimum permission required for logging in is Read-Only.
    To assign the external identity provider users to a group, see Add Members to a vCenter Single Sign-On Group. To assign inventory-level and global permissions to the users, see the topic about managing permissions for vCenter components in the *vSphere Security* documentation.

13. Verify logging in to this vCenter host with an external identity provider user.

    When you launch the vSphere Client, you see the Welcome to VMware vSphere screen, with the **Sign in with SSO** button. When you click this button, you are redirected to the external identity provider's sign-in screen.

Managing VMware Identity Services

You can stop and start VMware Identity Services, regenerate a SCIM token, and restore deleted SCIM users and groups.

Depending on the task, you use either the vSphere Client or the external identity provider's administration console.

Stop and Start the VMware Identity Services

To configure and run Okta, Microsoft Entra ID (formerly called Azure AD), or PingFederate as an external identity provider, VMware Identity Services must be started on vCenter. By default, when you either install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Services are started. You use the vCenter Management Interface to manage VMware Identity Services.

When you install or upgrade to vSphere 8.0 Update 1 or later, VMware Identity Services are automatically started. When you configure Okta, Microsoft Entra ID, or PingFederate as an external identity provider, you do not need to start VMware Identity Services, as they are already running. To start or stop VMware Identity Services, you must be root.

You configure the external identity provider on a single vCenter only. That vCenter, through its instance of VMware Identity Services, communicates to the identity provider. The other vCenter systems in the Enhanced Linked Mode configuration also have VMware Identity Services running, however, they do not communicate directly to the identity provider.

Starting in version 8.0 Update 1, vSphere includes the VMware Identity Services to support authenticating to Okta. Starting in version 8.0 Update 2, VMware Identity Services support authenticating to Microsoft Entra ID. Starting in version 8.0 Update 3, VMware Identity Services support authenticating to PingFederate.

1. In a Web browser, go to the vCenter Management Interface at https://*`vcenter-IP-address-or-FQDN`*:5480.
2. Log in as root.

   The default root password is the password that you set while deploying the vCenter.
3. Select **Services**.
4. View the status of VMware Identity Services.
5. To stop or start the service, select **VMware Identity Services**, then click either **Stop** or **Start**.

   After starting VMware Identity Services, no vCenter reboot is necessary.

Regenerate the SCIM Token in vCenter

In vCenter, you can regenerate a System for Cross-Domain Identity Management (SCIM) token for an external identity provider.

You must have created an external identity provider in vCenter.

If you generate another token, it becomes active immediately, and the previous token is revoked.

1. Log in as an administrator with the vSphere Client to the vCenter.
2. Navigate to the Configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Configuration**.
3. On the **Configuration** page, under **User Provisioning**/**Secret Token**, click **Regenerate** to regenerate the secret token, select the token lifespan from the drop-down, then click **Copy to Clipboard**. Save the token to a secure location.
4. The copied token is available for you to update your external identity provider configuration.

Restore Deleted SCIM Users and Groups

If the SCIM-pushed users and groups on your vCenter become out-of-sync with your external identity provider, you can take steps to fix the problem.

When you want to restore an SCIM-pushed user or group that you deleted from your vCenter, you cannot simply push the user or group from your identity provider. Because of the way vCenter uses System for Cross-domain Identity Management (SCIM) for user and group management, you must update the SCIM 2.0 application itself with the missing user or group.

1. Log in to the external IDP Admin console.
2. Navigate to the SCIM 2.0 application.
3. Assign the deleted or missing user or group.
4. Select the appropriate action to delete the pushed group or user to unlink the pushed group or user.
5. Select the appropriate action to push the group.
6. Verify on your vCenter that external IDP synchronized the group or user.

vCenter Single Sign-On

If you are not using an external identity provider, you must understand the underlying architecture of the built-in identity provider, vCenter Single Sign-On, and how it affects installation and upgrades.

vCenter Single Sign-On Components

vCenter Single Sign-On includes the Security Token Service (STS), an administration server, the vCenter Lookup Service, and the VMware Directory Service (vmdir). The VMware Directory Service is also used for certificate management.

During installation, the following components are deployed as part of a vCenter deployment.

STS (Security Token Service)	The STS service issues Security Assertion Markup Language (SAML) tokens. These security tokens represent the identity of a user in one of the identity source types supported by vCenter. The SAML tokens allow interactive, scripted, and service users (including solution users) who authenticate successfully to vCenter Single Sign-On to use any vCenter service that vCenter Single Sign-On supports without authenticating again to each service. The vCenter Single Sign-On service signs all tokens with a signing certificate, and stores the token signing certificate on disk. The certificate for the service itself is also stored on disk.
Administration server	The administration server allows users with administrator privileges to vCenter Single Sign-On to configure the vCenter Single Sign-On server and manage users and groups from the vSphere Client. Initially, only the user administrator@*`your_domain_name`* has these privileges. You can change the vSphere domain when you install vCenter. Do not name the domain name with your Microsoft Active Directory or OpenLDAP domain name.
VMware Directory Service (vmdir)	A VMware Directory Service (vmdir) is associated with the domain you specify during installation and is included in each vCenter deployment. This service is a multi-tenanted, peer-replicating directory service that makes an LDAP directory available on port 389. It also stores and manages vCenter Single Sign-On user accounts and passwords, which are secured by the SHA-512 hashing algorithm. If your environment includes multiple instances of vCenter configured in linked mode, an update of vmdir content in one vmdir instance is propagated to all other instances of vmdir.

	The VMware Directory Service stores not only vCenter Single Sign-On information but also certificate information.
Identity Management Service	Handles identity sources and STS authentication requests.

Using vCenter Single Sign-On with vSphere

When a user logs in to a vSphere component, or when a vCenter solution user accesses another vCenter service, vCenter Single Sign-On performs authentication. Users must be authenticated with vCenter Single Sign-On and have the necessary privileges for interacting with vSphere objects.

vCenter Single Sign-On authenticates both solution users and other users.

- Solution users represent a set of services in your vSphere environment. During installation, VMCA assigns a certificate to each solution user by default. The solution user uses that certificate to authenticate to vCenter Single Sign-On. vCenter Single Sign-On gives the solution user a SAML token, and the solution user can then interact with other services in the environment.
- When other users log in to the environment, for example, from the vSphere Client, vCenter Single Sign-On prompts for a user name and password. If vCenter Single Sign-On finds a user with those credentials in the corresponding identity source, it assigns the user a SAML token. The user can now access other services in the environment without being prompted to authenticate again.

  Which objects the user can view, and what a user can do, is usually determined by vCenter permission settings. vCenter administrators assign those permissions from the **Permissions** interface in the vSphere Client, not through vCenter Single Sign-On. See the *vSphere Security* documentation.

vCenter Single Sign-On and vCenter Users

Users authenticate to vCenter Single Sign-On by entering their credentials on the login page. After connecting to vCenter, authenticated users can view all vCenter instances or other vSphere objects for which their role gives them privileges. No further authentication is required.

After installation, the administrator of the vCenter Single Sign-On domain, administrator@vsphere.local by default, has administrator access to both vCenter Single Sign-On and vCenter. That user can then add identity sources, set the default identity source, and manage users and groups in the vCenter Single Sign-On domain.

After installation, the cloudadmin@vmc.local user vCenter has administrator access to both vCenter Single Sign-On and vCenter. That user can also add identity sources, set the default identity source, and set policies in the vmc.local domain. Certain management operations in the vmc.local domain are restricted to VMware Cloud on AWS operations staff.

All users that can authenticate to vCenter Single Sign-On can reset their password. See Change Your vCenter Single Sign-On Password . Only vCenter Single Sign-On administrators can reset the password for users who no longer have their password.

vCenter Single Sign-On Administrator Users

The vCenter Single Sign-On administrative interface is accessible from the vSphere Client.

To configure vCenter Single Sign-On and manage vCenter Single Sign-On users and groups, the user administrator@vsphere.local or a user in the vCenter Single Sign-On Administrators group must log in to the vSphere Client. Upon authentication, that user can access the vCenter Single Sign-On administration interface from the vSphere Client and manage identity sources and default domains, specify password policies, and perform other administrative tasks.

> **Note:**
> You cannot rename the vCenter Single Sign-On administrator user, which is administrator@vsphere.local by default or administrator@`mydomain` if you specified a different domain during installation. For improved security, consider creating additional named users in the vCenter Single Sign-On domain and assigning them administrative privileges. You can then stop using the administrator account.

Other User Accounts in vCenter

The following user accounts are created automatically within vCenter in the vsphere.local domain (or the default domain that you created at installation). These user accounts are shell accounts. The vCenter Single Sign-On password policy does not apply to these accounts.

Table 87: Other vCenter User Accounts

Account	Description
K/M	For Kerberos key management.
waiter-*random_string*	For Auto Deploy.

ESX Users

Standalone ESX hosts are not integrated with vCenter Single Sign-On. See *vSphere Security* for information on adding an ESX host to Active Directory.

If you create local ESX users for a managed ESX host with the VMware Host Client, ESXCLI, or PowerCLI, vCenter is not aware of those users. Creating local users can therefore result in confusion, especially if you use the same user names. Users who can authenticate to vCenter Single Sign-On can view and manage ESX hosts if they have the corresponding permissions on the ESX host object.

Note:
Manage permissions for ESX hosts through vCenter if possible.

How to Log In to vCenter Components

You can log in by connecting to the vSphere Client.

When a user logs in to a vCenter system from the vSphere Client, the login behavior depends on whether the user is in the domain that is set as the default identity source.

- Users who are in the default domain can log in with their user name and password.
- Users who are in a domain that has been added to vCenter Single Sign-On as an identity source but is not the default domain can log in to vCenter but must specify the domain in one of the following ways.
  - Including a domain name prefix, for example, MYDOMAIN\user1
  - Including the domain, for example, user1@mydomain.com
- Users who are in a domain that is not a vCenter Single Sign-On identity source cannot log in to vCenter. If the domain that you add to vCenter Single Sign-On is part of a domain hierarchy, Active Directory determines whether users of other domains in the hierarchy are authenticated or not.

If your environment includes an Active Directory hierarchy, see the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2064250 for details on supported and unsupported setups.

Groups in the vCenter Single Sign-On Domain

The vCenter Single Sign-On domain (vsphere.local by default) includes several predefined groups. Add users to one of those groups to permit them to perform the corresponding actions.

See Managing vCenter Single Sign-On Users and Groups.

For all objects in the vCenter hierarchy, you can assign permissions by pairing a user and a role with the object. For example, you can select a resource pool and give a group of users read privileges to that resource pool object by giving them the corresponding role.

For some services that are not managed by vCenter directly, membership in one of the vCenter Single Sign-On groups determines the privileges. For example, a user who is a member of the Administrators group can manage vCenter Single Sign-On. A user who is a member of the CAAdmins group can manage the VMware Certificate Authority, and a user who is in the LicenseService.Administrators group can manage licenses.

The following groups are predefined in vsphere.local. Many of these groups are internal to vsphere.local or give users high-level administrative privileges. Add users to any of these groups only after careful consideration of the risks.

CAUTION:
Do not delete any of the predefined groups in the vsphere.local domain. If you do, errors with authentication or certificate provisioning might result.

Table 88: Groups in the vsphere.local Domain

Privilege	Description
Users	Users in the vCenter Single Sign-On domain (vsphere.local by default).
SolutionUsers	Solution users group for vCenter services. Each solution user authenticates individually to vCenter Single Sign-On with a certificate. By default, VMCA provisions solution users with certificates. Do not add members to this group explicitly.
CAAdmins	Members of the CAAdmins group have administrator privileges for VMCA. Do not add members to this group unless you have compelling reasons.
DCAdmins	Members of the DCAdmins group can perform Domain Controller Administrator actions on VMware Directory Service. **Note:** Do not manage the domain controller directly. Instead, use the `vmdir` CLI or the vSphere Client to perform corresponding tasks.
SystemConfiguration.BashShellAdministrators	A user in this group has full access to all the Appliance Management APIs. By default, a user who connects to the vCenter with SSH can access only commands in the restricted shell, but users in this group have Bash Shell Access over SSH and gain full privileges similar to the root user.
ActAsUsers	Members of Act-As Users are allowed to get Act-As tokens from vCenter Single Sign-On.
ExternalIDPUsers	This internal group is not used by vSphere. VMware vCloud Air requires this group.
SystemConfiguration.Administrators	Members of the SystemConfiguration.Administrators group can view and manage the system configuration in the vCenter Management Interface running on port 5480. These users can view services, start and restart services, and troubleshoot services. These users can also access Appliance Management APIs except for those APIs that modify critical system configurations.
DCClients	This group is used internally to allow the management node access to data in VMware Directory Service. **Note:** Do not modify this group. Any changes might compromise your certificate infrastructure.
ComponentManager.Administrators	Members of the ComponentManager.Administrators group can invoke component manager APIs that register or unregister services, that is, modify services. Membership in this group is not necessary for read access on the services.
LicenseService.Administrators	Members of LicenseService.Administrators have full write access to all licensing-related data and can add, remove, assign, and unassign serial keys for all product assets registered in the licensing service.

Privilege	Description
Administrators	Administrators of the VMware Directory Service (vmdir). Members of this group can perform vCenter Single Sign-On administration tasks. Do not add members to this group unless you have compelling reasons and understand the consequences.
TrustedAdmins	Members of this group can perform VMware®vSphere Trust Authority™ configuration and administration tasks. By default, this group does not contain any members. You must add a member to this group so that you can perform vSphere Trust Authority tasks.
AutoUpdate	This group is used internally for vCenter Cloud Gateway.
SyncUsers	This group is used internally for vCenter Cloud Gateway.
vSphereClientSolutionUsers	This group is used internally for the vSphere Client.
ServiceProviderUsers	Members of this group can manage the vSphere with Tanzu and VMware Cloud on AWS infrastructure.
NsxAdministrators	This group is used for VMware NSX.
WorkloadStorage	Workload storage group.
RegistryAdministrators	Members of this group can manage the registry.
NsxAuditors	This group is used for VMware NSX.
NsxViAdministrators	This group is used for VMware NSX.
SystemConfiguration.SupportUsers	Members of the SystemConfiguration.SupportUsers group can access the support bundle API.
SystemConfiguration.ReadOnly	Members of this group can access vCenter appliance read-only operations under Appliance Management.
VCLSAdmin	Members of this group have administrative privileges for vSphere Cluster Services (vCLS).
AnalyticsService.Administrators	This group is used for the VMware Analytics Service APIs.
vStatsGroup	This group is used for vStats collection.

Configuring vCenter Single Sign-On Identity Sources

When a user logs in with just a user name, vCenter Single Sign-On checks in the default identity source whether that user can authenticate. When a user logs in and includes the domain name in the login screen, vCenter Single Sign-On checks the specified domain if that domain has been added as an identity source. You can add identity sources, remove identity sources, and change the default.

You configure vCenter Single Sign-On from the vSphere Client. To configure vCenter Single Sign-On, you must have vCenter Single Sign-On administrator privileges. Having vCenter Single Sign-On administrator privileges is different from having the Administrator role on vCenter or ESX. In a new installation, only the vCenter Single Sign-On administrator (administrator@vsphere.local by default) can authenticate to vCenter Single Sign-On.

You configure vCenter Single Sign-On from the vSphere Client.

Identity Sources for vCenter with vCenter Single Sign-On

You can use identity sources to attach one or more domains to vCenter Single Sign-On. A domain is a repository for users and groups that the vCenter Single Sign-On server can use for user authentication.

> **Note:**
> At any time, only one default domain exists. If a user from a non-default domain logs in, that user must add the domain name to authenticate successfully. The domain name is in the form:
> `DOMAIN\user`

The following identity sources are available.

- Active Directory over LDAP. vCenter Single Sign-On supports multiple Active Directory over LDAP identity sources.

- OpenLDAP versions 2.4 and later. vCenter Single Sign-On supports multiple OpenLDAP identity sources.

  Note:

  A Microsoft Windows update changed the default behavior of Active Directory to require strong authentication and encryption. This change impacts how vCenter authenticates to Active Directory. If you use Active Directory as your identity source for vCenter, you must enable LDAPS. For more information, see https://msrc.microsoft.com/update-guide/en-us/advisory/ADV190023.

Set the Default Domain for vCenter Single Sign-On

Each vCenter Single Sign-On identity source is associated with a domain. vCenter Single Sign-On uses the default domain to authenticate a user who logs in without a domain name. Users who belong to a domain that is not the default domain must include the domain name when they log in.

When a user logs in to a vCenter system from the vSphere Client, the login behavior depends on whether the user is in the domain that is set as the default identity source.

- Users who are in the default domain can log in with their user name and password.
- Users who are in a domain that has been added to vCenter Single Sign-On as an identity source but is not the default domain can log in to vCenter but must specify the domain in one of the following ways.
  - Including a domain name prefix, for example, MYDOMAIN\user1
  - Including the domain, for example, user1@mydomain.com
- Users who are in a domain that is not a vCenter Single Sign-On identity source cannot log in to vCenter. If the domain that you add to vCenter Single Sign-On is part of a domain hierarchy, Active Directory determines whether users of other domains in the hierarchy are authenticated or not.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Configuration**.
4. Under the **Identity Provider** tab, click **Identity Sources**, select an identity source, and click **Set as Default**.
5. Click **OK**.

   In the domain display, the default domain shows (default) in the Type column.

Managing the vCenter Security Token Service

The vCenter Security Token Service (STS) is a Web service that issues, validates, and renews security tokens.

As a token issuer, the Security Token Service (STS) uses a private key to sign the tokens and publishes the public certificates for services to verify the token signature. vCenter manages the STS signing certificates and stores them in the VMware Directory Service (vmdir). Tokens can have a significant lifetime, and historically might have been signed by any one of multiple keys.

Users present their primary credentials to the STS interface to acquire tokens. The primary credential depends on the type of user.

Table 89: STS Users and Credentials

Type of User	Primary Credentials
Solution user	Valid certificate.
Other users	User name and password available in a vCenter Single Sign-On identity source.

STS authenticates the user based on the primary credentials, and constructs a SAML token that contains user attributes.

By default, the VMware Certificate Authority (VMCA) generates the STS signing certificate. You can refresh the STS signing certificate with a new VMCA certificate. You can also import and replace the default STS signing certificate with a custom or third-party generated STS signing certificate. Do not replace the STS signing certificate unless the security policy of your company requires replacing all certificates.

You can use the vSphere Client to:

- Refresh STS certificates
- Import and replace custom and third-party generated STS certificates
- View STS certificate details, such as the expiration date

You can also use the command line to replace custom and third-party generated STS certificates.

STS Certificate Duration and Expiration

A fresh installation of vSphere 7.0 Update 1 and later creates an STS signing certificate with a duration of 10 years. When an STS signing certificate is close to expiring, an alarm warns you starting at 90 days once per week, and then daily when seven days away.

> **Note:**
> In certain circumstances, replacing your STS signing certificates can change the duration of the certificates. When performing certificate replacement, pay attention to the issuing and expiration dates.

STS Certificate Auto-Renewal

In vSphere 8.0 and later, vCenter Single Sign-On automatically renews a VMCA-generated STS signing certificate. The auto-renewal occurs before the STS signing certificate expires and before triggering the 90-day expiration alarm. If the auto-renewal fails, vCenter Single Sign-On creates an error message in the log file. If necessary, you can refresh the STS signing certificate manually.

> **Note:**
> vCenter Single Sign-On does not perform auto-renewal of custom generated or third-party STS signing certificates.

Refreshing and Importing and Replacing STS Certificates

In vSphere 8.0 and later, refreshing or importing and replacing the STS signing certificates does not require a vCenter restart and so avoids any downtime. Also, in a linked configuration, refreshing or importing and replacing the STS signing certificates on a single vCenter updates the STS certificates on all the linked vCenter systems.

> **Note:**
> In certain circumstances, a refresh or import and replace of STS signing certificates can require you to manually restart the vCenter systems.

Refresh a vCenter STS Certificate Using the vSphere Client

You can refresh your vCenter STS signing certificates using the vSphere Client. The VMware Certificate Authority (VMCA) issues a new certificate and replaces the current certificate.

For certificate management, you must supply the password of the administrator of the local domain (administrator@vsphere.local by default). If you are renewing certificates, you must also supply the vCenter Single Sign-On credentials for a user with administrator privileges on the vCenter system.

When you refresh STS signing certificates, the VMware Certificate Authority (VMCA) issues a new certificate and replaces the current certificate in the VMware Directory Service (vmdir). STS starts using the new certificate to issue new tokens. In an Enhanced Linked Mode configuration, vmdir uploads the new certificate from the issuing vCenter system to all linked vCenter systems. When you refresh STS signing certificates, you do not need to restart the vCenter system, nor any other vCenter system that is part of an Enhanced Linked Mode configuration.

If you are using a custom generated or third-party STS signing certificate, the refresh overwrites that certificate with a VMCA-issued certificate. To update custom generated or third-party STS signing certificates, use the import and replace option. See Import and Replace a vCenter STS Certificate Using the vSphere Client.

The VMCA-issued STS signing certificate is valid for 10 years and is not an external-facing certificate. Do not replace this certificate unless the security policy of your company requires it.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **STS Signing** tab, select the desired certificate and click **Refresh with vCenter Certificate**.

   If you are using a custom generated or third-party STS signing certificate, the refresh action overwrites that certificate with a VMCA-generated certificate.

   Note:
   If you were using third-party certificates for compliance reasons, the refresh might cause your vCenter systems to go out of compliance. Also, if you are using a custom generated or third-party STS signing certificate, the Security Token Service no longer uses that custom or third-party certificate for token signing.
6. Click **Refresh**.

   The VMCA refreshes the STS signing certificate on this vCenter system and on any linked vCenter systems.
7. Optional: If the **Force Refresh** button appears, vCenter Single Sign-On has detected a problem. Before clicking **Force Refresh**, consider the following potential results.
   - If all the impacted vCenter systems are not running at least vSphere 7.0 Update 3, they do not support the certificate refresh.
   - Selecting **Force Refresh** requires that you restart all vCenter systems and can render those systems inoperable until you do so.

   a) If you are unsure of the impact, click **Cancel** and research your environment.
   b) If you are sure of the impact, click **Force Refresh** to proceed with the refresh then manually restart your vCenter systems.

Import and Replace a vCenter STS Certificate Using the vSphere Client

You can import and replace the vCenter STS certificate with a custom generated or third-party certificate using the vSphere Client.

For certificate management, you must supply the password of the administrator of the local domain (administrator@vsphere.local by default). You also must supply the vCenter Single Sign-On credentials for a user with administrator privileges on the vCenter system.

To import and replace the default STS signing certificate, you must first generate a new certificate. When you import and replace STS signing certificates, the VMware Directory Service (vmdir) uploads the new certificate from the issuing vCenter system to all linked vCenter systems.

The STS certificate is not an external-facing certificate. Do not replace this certificate unless the security policy of your company requires it.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.

3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **STS Signing** tab, select the desired certificate and click **Import and Replace Certificate**.
6. Select the PEM file.
   The PEM file includes the signing certificate chain and the private key.
7. Click **Replace**.
   The STS signing certificate is replaced on this vCenter system and on any linked vCenter systems. Unless otherwise indicated, you do not need to restart the vCenter systems.

Replace a vCenter STS Certificate Using the Command Line

You can replace the vCenter STS certificate with a custom generated or third-party certificate using the CLI.

Enable SSH login to vCenter. See Manage vCenter Using the vCenter Shell.

To use a company required certificate or to refresh a certificate that is near expiration, you can replace the existing STS signing certificate. To replace the default STS signing certificate, you must first generate a new certificate.

The STS certificate is not an external-facing certificate. Do not replace this certificate unless the security policy of your company requires it.

> **CAUTION:**
> You must use the procedures described here. Do not replace the certificate directly in the file system.

1. Log in to the vCenter shell as root.
2. Create a certificate.
   a) Create a top-level directory to hold the new certificate and verify the location of the directory.

```
mkdir newsts
cd newsts
pwd
#resulting output: /root/newsts
```

   b) Copy the `certool.cfg` file into the new directory.

```
cp /usr/lib/vmware-vmca/share/config/certool.cfg /root/newsts
```

   c) Using a command-line editor such as Vim, open your copy of the `certool.cfg` file and edit it to use the local vCenter IP address and hostname. The country is required and has to be two characters, as shown in the following example.

```
#
# Template file for a CSR request
#

# Country is needed and has to be 2 characters
Country = US
Name = STS
Organization = ExampleInc
OrgUnit = ExampleInc Dev
State = Indiana
Locality = Indianapolis
IPAddress = 10.0.1.32
Email = chen@exampleinc.com
Hostname = homecenter.exampleinc.local
```

d) Generate the key.

```
/usr/lib/vmware-vmca/bin/certool --server localhost --genkey --privkey=/root/newsts/sts.key --pubkey=/root/newsts/sts.pub
```

e) Generate the certificate.

```
/usr/lib/vmware-vmca/bin/certool --gencert --cert=/root/newsts/newsts.cer --privkey=/root/newsts/sts.key --config=/root/newsts/certool.cfg
```

f) Create a PEM file with the certificate chain and private key.

```
cat newsts.cer /var/lib/vmware/vmca/root.cer sts.key > newsts.pem
```

3. Update the STS signing certificate, for example:

```
/opt/vmware/bin/sso-config.sh -set_signing_cert -t vsphere.local /root/newsts/newsts.pem
```

The VMCA refreshes the STS signing certificate on this vCenter system and on any linked vCenter systems.

View the Active vCenter STS Signing Certificate Chain Using the vSphere Client

You can use the vSphere Client to view the active vCenter STS signing certificate chain and the certificate information, such as the valid until date.

1. Log in with the vSphere Client to the vCenter.
2. Enter the user name and password for a user that has at least Read privileges.
3. Navigate to the Certificate Management UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Certificates**, click **Certificate Management**.
4. If the system prompts you, enter the credentials of your vCenter.
5. Under the **STS Signing** tab, select a certificate, then expand the certificate.
   Certificate and issue information is displayed, including:
   - Valid until date
   - A green check for a valid certificate, and an orange check warning of an expired certificate

Determine the Expiration Date of an LDAPS SSL Certificate Using the Command Line

When using Active Directory over LDAPS, you can upload an SSL certificate for the LDAP traffic. SSL certificates expire after a predefined lifespan. You can use the `sso-config.sh` command to view the certificate's expiration date so that you know to replace or renew the certificate before it expires.

Enable SSH login to vCenter. See Manage vCenter Using the vCenter Shell.

vCenter alerts you when an active LDAP SSL certificate is close to its expiration date.

You see certificate expiration information only if you use Active Directory over LDAP or an OpenLDAP identity source and specify an `ldaps://` URL for the server.

1. Log in as root to the vCenter.
2. Run the following command.

```
/opt/vmware/bin/sso-config.sh -get_identity_sources
```

Ignore the SLF4J messages.

3. To determine the expiration date, view the SSL certificate's details and verify the `NotAfter` field.

Managing vCenter Single Sign-On Policies

vCenter Single Sign-On policies enforce the security rules for local accounts and tokens in general. You can view and edit the default vCenter Single Sign-On password policy, lockout policy, and token policy.

Edit the vCenter Single Sign-On Password Policy

The vCenter Single Sign-On password policy determines the password format and password expiration. Password policy applies only to users in the vCenter Single Sign-On domain (vsphere.local).

By default, vCenter Single Sign-On built-in user account passwords expire after 90 days. The vSphere Client reminds you when your password is about to expire.

See Change Your vCenter Single Sign-On Password .

Note:
The administrator account (administrator@vsphere.local) does not get locked out nor does its password expire. Proper security practice is to audit logins from this account and rotate the password regularly.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Configuration**.
4. Click the **Local Accounts** tab.
5. Click **Edit** for the **Password Policy** row.
6. Edit the password policy.

Option	Description
Description	Password policy description.
Maximum lifetime	Maximum number of days that a password is valid before the user must change it. The maximum number of days you can enter is 999999999. A value of zero (0) means that the password never expires.
Restrict reuse	Number of previous passwords that cannot be reused. For example, if you enter 6, the user cannot reuse any of the last six passwords.
Maximum length	Maximum number of characters that are allowed in the password.
Minimum length	Minimum number of characters required in the password. The minimum length must be no less than the combined minimum of alphabetic, numeric, and special character requirements.
Character requirements	Minimum number of different character types that are required in the password. You can specify the number of each type of character, as follows: • Special: & # % • Alphabetic: A b c D • Uppercase: A B C • Lowercase: a b c • Numeric: 1 2 3 • Identical Adjacent: The number must be greater than 0. For example, if you enter 1, the following password is not allowed: p@$$word. The minimum number of alphabetic characters must be no less than the combined uppercase and lowercase characters. Non-ASCII characters are supported in passwords. In earlier versions of vCenter Single Sign-On, limitations on supported characters exist.

Note:
The password policy picks up the maximum length value only if the minimum length is greater than 20 characters. The behavior of the password policy is undefined or could result in failure of services when the minimum length value is greater than 20 characters and the maximum length is set to any value. To avoid a potential problem, leave the minimum length set to the default value of 8 characters, or no greater than 20 characters.

7. Click **Save**.

Edit the vCenter Single Sign-On Lockout Policy

If a user attempts to log in with incorrect credentials, a vCenter Single Sign-On lockout policy specifies when the user's vCenter Single Sign-On account is locked. Administrators can edit the lockout policy.

If a user logs in to vsphere.local multiple times with the wrong password, the user is locked out. The lockout policy allows administrators to specify the maximum number of failed login attempts, and set the time interval between failures. The policy also specifies how much time must elapse before the account is automatically unlocked.

Note:
The lockout policy applies only to user accounts, not to system accounts such as administrator@vsphere.local.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the Configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Configuration**.
4. Click the **Local Accounts** tab.
5. Click **Edit** for the **Lockout Policy** row.

   You might need to scroll down to see the **Lockout Policy** row.
6. Edit the parameters.

Option	Description
Description	Optional description of the lockout policy.
Maximum number of failed login attempts	Maximum number of failed login attempts that are allowed before the account is locked.
Time interval between failures	Time period in which failed login attempts must occur to trigger a lockout.
Unlock time	Amount of time that the account remains locked. If you enter 0, the administrator must unlock the account explicitly.

7. Click **Save**.

Edit the vCenter Single Sign-On Token Policy

The vCenter Single Sign-On token policy specifies token properties such as the clock tolerance and renewal count. You can edit the token policy to ensure that the token specification conforms to security standards in your corporation.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.

3. Navigate to the Configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Configuration**.
4. Click the **Local Accounts** tab.
5. Click **Edit** for the **Token Trustworthiness** row.
   You might need to scroll down to see the **Token Trustworthiness** row.
6. Edit the token policy configuration parameters.

Option	Description
Clock Tolerance	Time difference, in milliseconds, that vCenter Single Sign-On tolerates between a client clock and the domain controller clock. If the time difference is greater than the specified value, vCenter Single Sign-On declares the token invalid.
Maximum Token Renewal Count	Maximum number of times that a token can be renewed. After the maximum number of renewal attempts, a new security token is required.
Maximum Token Delegation Count	Holder-of-key tokens can be delegated to services in the vSphere environment. A service that uses a delegated token performs the service on behalf of the principal that provided the token. A token request specifies a DelegateTo identity. The DelegateTo value can either be a solution token or a reference to a solution token. This value specifies how many times a single holder-of-key token can be delegated.
Maximum Bearer Token Lifetime	Bearer tokens provide authentication based only on possession of the token. Bearer tokens are intended for short-term, single-operation use. A bearer token does not verify the identity of the user or entity that is sending the request. This value specifies the lifetime value of a bearer token before the token has to be reissued.
Maximum Holder-of-Key Token Lifetime	Holder-of-key tokens provide authentication based on security artifacts that are embedded in the token. Holder-of-key tokens can be used for delegation. A client can obtain a holder-of-key token and delegate that token to another entity. The token contains the claims to identify the originator and the delegate. In the vSphere environment, a vCenter system obtains delegated tokens on a user's behalf and uses those tokens to perform operations. This value determines the lifetime of a holder-of-key token before the token is marked invalid.

7. Click **Save**.

Managing vCenter Single Sign-On Users and Groups

A vCenter Single Sign-On administrator user can manage users and groups in the vsphere.local domain from the vSphere Client.

The vSphere Client presents a view of users and groups in your vSphere domain (vsphere.local by default). From this view, you can add, edit, and deactivate users. You can also add groups, and manage group membership.

Add vCenter Single Sign-On Users

Users listed on the **Users** tab in the vSphere Client are internal to vCenter Single Sign-On and belong to the vsphere.local domain. You add users to that domain from one of the vCenter Single Sign-On management interfaces.

You can select other domains and view information about the users in those domains, but you cannot add users to other domains from a vCenter Single Sign-On management interface.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. If vsphere.local is not the currently selected domain, select it from the drop-down menu.

   You cannot add users to other domains.
5. On the **Users** tab, click **Add**.
6. Enter a user name and password for the new user.

   The maximum number of characters allowed for the user name is 300.

   You cannot change the user name after you create a user. The password must meet the password policy requirements for the system.
7. Optional: Enter the first name and the last name of the new user.
8. Optional: Enter an email address and description for the user.
9. Click **Add**.

When you add a user, that user initially has no privileges to perform management operations.
Add the user to a group in the vsphere.local domain, for example, to the group of users who can administer VMCA (CAAdmins) or to the group of users who can administer vCenter Single Sign-On (Administrators). See Add Members to a vCenter Single Sign-On Group.

Deactivate and Activate vCenter Single Sign-On Users

When a vCenter Single Sign-On user account is deactivated, the user cannot log in to the vCenter Single Sign-On server until an administrator activates the account. You can deactivate and activate accounts from one of the vCenter Single Sign-On management interfaces.

You must be a member of the vCenter Single Sign-On Administrators group to deactivate and activate vCenter Single Sign-On users.

Deactivated user accounts remain available in the vCenter Single Sign-On system, but the user cannot log in or perform operations on the server. Users with administrator privileges can deactivate and activate accounts from the vCenter **Users and Groups** page.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.

3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Select a user name, click **More**, and click **Disable**.
5. Click **OK**.
6. To activate the user again, click **More**, click **Enable**, and click **OK**.

Delete a vCenter Single Sign-On User

You can delete users that are in the vsphere.local domain from a vCenter Single Sign-On management interface. You cannot delete local operating system users or users in another domain from a vCenter Single Sign-On management interface.

> **CAUTION:**
> If you delete the administrator user in the vsphere.local domain, you can no longer log in to vCenter Single Sign-On. Reinstall vCenter and its components.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Select **Users**, and select the vsphere.local domain from the drop-down menu.
5. In the list of users, select the user that you want to delete.
6. Click **Delete**.
   Proceed with caution. You cannot undo this action.
7. Click **Remove**.

Edit a vCenter Single Sign-On User

You can change the password or other details of a vCenter Single Sign-On user from a vCenter Single Sign-On management interface. You cannot rename users in the vsphere.local domain. That means you cannot rename administrator@vsphere.local.

You can create additional users with the same privileges as administrator@vsphere.local.

vCenter Single Sign-On users are stored in the vCenter Single Sign-On vsphere.local domain.

You can review the vCenter Single Sign-On password policies from the vSphere Client. Log in as administrator@vsphere.local and from the **Administration** menu, select **Configuration** > **Local Accounts** > **Password Policy**.

See also Edit the vCenter Single Sign-On Password Policy.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.

3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Click **Users**.
5. Select the user and click **Edit**.
6. Edit the user attributes.
   You cannot change the user name of the user.
   The password must meet the password policy requirements for the system.
7. Click **Save**.

Add a vCenter Single Sign-On Group

The vCenter Single Sign-On**Groups** tab shows groups in the local domain, vsphere.local by default. You add groups if you need a container for group members (principals).

You cannot add groups to other domains, for example, the Active Directory domain, from the vCenter Single Sign-On**Groups** tab.

If you do not add an identity source to vCenter Single Sign-On, creating groups and adding users can help you organize the local domain.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.
   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Select **Groups** and click **Create Group**.
5. Enter a name and description for the group.
   The maximum number of characters allowed for the group name is 300. You cannot change the group name after you create the group.
6. From the **Add Members** drop-down menu, select the identity source that contains the member to add to the group.
   If you have configured an external identity provider such as AD FS, the domain of that identity provider is available to select in the **Add Members** drop-down menu.
7. Enter a search term.
8. Select the member.
   You can add more than one member.
9. Click **Done**.

See Add Members to a vCenter Single Sign-On Group.

Add Members to a vCenter Single Sign-On Group

Members of a vCenter Single Sign-On group can be users or other groups from one or more identity sources. You can add new members from the vSphere Client.

See the knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=2095342 for the background information.

Groups listed on the **Groups** tab in the web interface are part of the vsphere.local domain. See Groups in the vCenter Single Sign-On Domain.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Click **Groups** and click the group (for example, Administrators).
5. Click **Edit**.
6. From the **Domain** drop-down menu, select the identity source that contains the member to add to the group.

   If you have configured an external identity provider, such as AD FS, the domain of that identity provider is available to select in the **Domain** drop-down menu.
7. Enter a search term.
8. Select the member.

   You can add more than one member.
9. Click **Save**.

Remove Members from a vCenter Single Sign-On Group

You can remove members from a vCenter Single Sign-On group by using the vSphere Client. When you remove a member (user or group) from a group, you do not delete the member from the system.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*mydomain*.
3. Navigate to the vCenter Single Sign-On user configuration UI.
   a) From the **Home** menu, select **Administration**.
   b) Under **Single Sign On**, click **Users and Groups**.
4. Click **Groups** and select a group.
5. Click **Edit**.
6. In the Current members list, click the user or group that you want to remove.
7. Click **Done**.

The user or group is removed from the group, but is still available in the system.

Change Your vCenter Single Sign-On Password

Users in the local domain, vsphere.local by default, can change their vCenter Single Sign-On passwords from the vSphere Client. Users in other domains change their passwords following the rules for that domain.

The vCenter Single Sign-On lockout policy determines when your password expires. By default, vCenter Single Sign-On passwords expire after 90 days, but administrator passwords such as the password for administrator@vsphere.local do not expire. vCenter Single Sign-On management interfaces show a warning when your password is about to expire.

Note:
You can change a password only if it is not expired.

If the password is expired, the administrator of the local domain, administrator@vsphere.local by default, can reset the password by using the `dir-cli password reset` command. Only members of the Administrator group for the vCenter Single Sign-On domain can reset passwords.

1. Log in with the vSphere Client to the vCenter.
2. Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group.

   If you specified a different domain during installation, log in as administrator@*`mydomain`*.
3. In the upper navigation pane, click your user name to pull down the menu and select **Change Password**.
4. Enter your current password.
5. Enter a new password and confirm it.

   The password must conform to the password policy.
6. Click **Confirm**.

   As an alternative, you can select **Single Sign On** > **Users and Groups**, select the user, and click **Edit**.

Managing the Login Message to the vSphere Client Login Page

You can create a message that is displayed on the vSphere Client login page.

You can set a message, disclaimer, or terms and conditions. Also, you can configure the message to require acknowledgment of the message before login.

Manage the Login Message to the vSphere Client Login Page

You can add a login message to the vSphere Client login page. You can also configure a custom login message and provide a check box for user consent.

1. Log in to the vSphere Client.

   Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group. If you specified a different domain during installation, log in as administrator@mydomain.
2. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Administration**.
3. Under **Single Sign On**, click **Configuration**.
4. Click the **Login Message** tab.
5. Click **Edit** and configure the login message.

Option	Description
Show login message	Toggle on **Show login message** to enable the login message. You cannot make changes to the login message unless you toggle on this switch.
Login message	Title of the message. By default, when **Consent checkbox** is toggled on, the login message text is `I agree to Terms and Conditions`. You must replace `Terms and Conditions` with your own text. If the **Consent checkbox** is toggled off, then `Login message` appears, over which you enter your message.
Consent checkbox	Toggle on **Consent checkbox** to require that the user clicks a check box before logging in. You can also display a message without a check box.
Details of login message	Message that the user sees when clicking the login message, for example, the text of the terms and conditions. You must enter some details in this text box.

6. Click **Save**.

vCenter Single Sign-On Security Best Practices

Follow vCenter Single Sign-On security best practices to protect your vSphere environment.

The vSphere authentication infrastructure enhances the security of your vSphere environment. To make sure that your infrastructure is not compromised, follow these vCenter Single Sign-On best practices.

Check Password Expiration

The default vCenter Single Sign-On password policy has a password lifetime of 90 days. After 90 days, the password expires and you can no longer log in. Check the expiration and refresh passwords in a timely fashion.

Configure Network Time Protocol

Use Network Time Protocol (NTP) to ensure that all systems use the same relative time source (including the relevant localization offset), and that the relative time source can be correlated to an agreed-upon time standard (such as Coordinated Universal Time—UTC). Synchronized systems are essential for vCenter Single Sign-On certificate validity, and for the validity of other vSphere certificates.

NTP also makes it easier to track an intruder in log files. Incorrect time settings can make it difficult to inspect and correlate log files to detect attacks, and can make auditing inaccurate.

See the *vSphere Security* documentation for instructions on configuring time synchronization using NTP.

Troubleshooting vCenter Authentication

The following topics provide a starting point for troubleshooting vCenter authentication problems. Search this documentation center and the Broadcom Knowledge Base system for additional pointers.

Determining the Cause of a Lookup Service Error

vCenter Single Sign-On installation displays an error referring to the vCenter or the vSphere Client.

vCenter and Web Client installers show the error `Could not contact Lookup Service. Please check VM_ssoreg.log....`

This problem has several causes, including unsynchronized clocks on the host machines, firewall blocking, and services that must be started.

1. Verify that the clocks on the host machines running vCenter Single Sign-On, vCenter , and the Web Client are synchronized.
2. View the specific log file found in the error message.

   In the message, system temporary folder refers to `%TEMP%`.
3. Within the log file, search for the following messages.

   The log file contains output from all installation attempts. Locate the last message that shows `Initializing registration provider...`

Message	Cause and solution
`java.net.ConnectException: Connection timed out: connect`	The IP address is incorrect, a firewall is blocking access to vCenter Single Sign-On, or vCenter Single Sign-On is overloaded. Ensure that a firewall is not blocking the vCenter Single Sign-On port (by default 7444). Ensure also that the machine on which vCenter Single Sign-On is installed has adequate free CPU, I/O, and RAM capacity.
`java.net.ConnectException: Connection refused: connect`	The IP address or FQDN is incorrect and the vCenter Single Sign-On service has not started or has started within the past minute.

Message	Cause and solution
	Verify that vCenter Single Sign-On is working by checking the status of vCenter Single Sign-On vmware-sso daemon. Restart the service. If restarting does not correct the problem, see the recovery section of the *vSphere Troubleshooting Guide*.
`Unexpected status code: 404. SSO Server failed during initialization`	Restart vCenter Single Sign-On. If restarting does not correct the problem, see the Recovery section of the *vSphere Troubleshooting Guide*.
The error shown in the UI begins with `Could not connect to vCenter Single Sign-On`	You also see the return code `SslHandshakeFailed`. This error indicates that the provided IP address or FQDN that resolves to vCenter Single Sign-On host was not the address used when you installed vCenter Single Sign-On . In the `VM_ssoreg.log`, find the line that contains the following message. `host name in certificate did not match: <install-configured FQDN or IP> != <A> or <B> or <C>` where A was the FQDN you entered during the vCenter Single Sign-On installation, and B and C are system-generated allowable alternatives. Correct the configuration to use the FQDN on the right of the `!=` sign in the log file. In most cases, use the FQDN that you specified during vCenter Single Sign-On installation. If none of the alternatives are possible in your network configuration, recover your vCenter Single Sign-On SSL configuration.

vCenter Login Fails Because the User Account Is Locked

When you log in to vCenter from the vSphere Client login page, an error indicates that the account is locked.

After several failed attempts, you cannot log in to the vSphere Client using vCenter Single Sign-On. You see the message that your account is locked.

You exceeded the maximum number of failed login attempts.

- If you attempted log in as a user from the system domain (vsphere.local by default), ask your vCenter Single Sign-On administrator to unlock your account. If the lock is set to expire in the lockout policy, you can wait until your account is unlocked. vCenter Single Sign-On administrators can use CLI commands to unlock your account.
- If you log in as a user from an Active Directory or LDAP domain, ask your Active Directory or LDAP administrator to unlock your account.

VMware Directory Service Replication Can Take a Long Time

If your environment includes multiple vCenter instances connected through enhanced linked mode, and if one of the vCenter instances becomes unavailable, your environment continues to function. When the vCenter becomes available again, user data and other information are usually replicated within 30 seconds with partners connected through enhanced linked mode. In certain circumstances, however, replication might take a long time.

In certain situations, for example, when your environment includes multiple vCenter instances in different locations, and you make significant changes while one vCenter is unavailable, you do not see replication across VMware Directory Service instances right away. For example, you do not see a new user that was added to the available vCenter instance in the other instance until replication is complete. Replication might take a long time, depending on your enhanced linked mode topology.

During normal operation, changes to a VMware Directory Service (vmdir) instance in one vCenter instance (node) show up in its direct replication partner within about 30 seconds. Depending on the replication topology, changes in one node might have to propagate through intermediate nodes before they arrive at each vmdir instance in each node. Information that is replicated includes user information, certificate information, license information for virtual machines that are created, cloned, or migrated with VMware vMotion, and more.

When the replication link is broken, for example, because of a network outage or because a node becomes unavailable, changes in the federation do not converge. After the unavailable node is restored, each node tries to catch up with all changes. Eventually, all vmdir instances converge to a consistent state but it might take a while to reach that consistent state if many changes occurred while one node was unavailable.

Your environment functions normally while replication happens. Do not attempt to solve the problem unless it persists for over an hour.

Export a vCenter Support Bundle

You can export a support bundle that contains the log files for the vCenter services from the vSphere Client, or by using an API. After the export, you can explore the logs locally or send the bundle to VMware Support.

Verify that the vCenter is successfully deployed and running.

For more information about the API, see *vCenter Management Programming Guide*.

1. From a Web browser, connect to the vCenter configuration management interface at https://*vcenter_server_ip*:5480.
2. Log in as the root user for the vCenter.
3. From the **Actions** menu, select **Create Support Bundle**.
4. Unless browser settings prevent an immediate download, the support bundle is saved to your local machine.

vCenter Authentication Services Logs Reference

The vCenter authentication services use syslog for logging. You can examine the log files to determine the reasons for failures.

Table 90: vCenter Authentication Services Logs

Service	Description
VMware Directory Service	By default, vmdir logging goes to `/var/log/messages` or `/var/log/vmware/vmdird/`. For issues at deployment time, `/var/log/vmware/vmdir/vmafdvmdirclient.log` might also contain useful troubleshooting data.
VMware Single Sign-On	vCenter Single Sign-On logging goes to `/var/log/vmware/sso/`.
VMWare Certificate Authority (VMCA)	VMCA service log is located in `/var/log/vmware/vmcad/vmcad-syslog.log`.
VMware Endpoint Certificate Store (VECS)	VECS service log is located in `/var/log/vmware/vmafdd/vmafdd-syslog.log`.
VMware Lookup Service	Lookup service log is located in `/var/log/vmware/sso/lookupServer.log`.

vCenter and Host Management

Learn how to use the VMware®vSphere Client components, configure and manage hosts, and migrate virtual machines in your vCenter environment.

vCenter and Host Management also provides brief introductions to the various tasks you can perform within the system, and it cross-references to the documentation that describes the tasks in detail.

vCenter and Host Management covers ESX and vCenter.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

vCenter and Host Management is intended for system administrators who are experienced Windows or Linux system administrators and who are familiar with virtual machine technology and data center operations.

What Are the Main vSphere Concepts and Features

VMware vSphere® uses the power of virtualization to transform data centers into simplified cloud computing infrastructures, enabling IT organizations to deliver flexible and reliable IT services.

The two core components of vSphere are VMware ESX™ and VMware vCenter®. ESX is the hypervisor on which you create and run virtual machines. vCenter is a service that acts as a central administrator for ESX hosts that are connected on a network. With vCenter, you can pool and manage the resources of multiple hosts. vCenter allows you to monitor and manage your physical and virtual infrastructure.

Some vSphere components are available as plug-ins that extend the functionality of the vSphere product.

What Are the Basics of Virtualization

A virtual machine is a software computer that, like a physical computer, runs an operating system and applications. ESX is the hypervisor in a vSphere environment. The hypervisor serves as a platform for running virtual machines and allows for the consolidation of computing resources.

Each virtual machine contains its own virtual, or software-based, hardware, including a virtual CPU, memory, hard disk, and network interface card.

ESX is installed on physical hardware in a virtualized data center, and acts as a platform for virtual machines. The hypervisor provides physical hardware resources dynamically to virtual machines to support the operation of the virtual machines. The hypervisor allows virtual machines to operate with a degree of independence from the underlying physical hardware. For example, a virtual machine can be moved from one physical host to another, or its virtual disks can be moved from one type of storage to another, without affecting the functioning of the virtual machine.

Because virtual machines are decoupled from the underlying physical hardware, virtualization allows you to consolidate physical computing resources such as CPUs, memory, storage, and networking into pools of resources. These resources can be dynamically and flexibly made available to virtual machines. With the vCenter management platform, you can increase the availability and security of your virtual infrastructure.

What Is the Physical Topology of a vSphere Data Center

A typical VMware vSphere data center consists of physical building blocks such as x86 virtualization servers, storage networks and arrays, IP networks, a management server, and desktop clients.

The vSphere data center includes the following components.

ESX Hosts	Industry standard x86 servers that run ESX on the bare metal. ESX software provides resources for and runs the virtual machines. You can group similarly configured x86 servers with connections to the same network and storage subsystems. This grouping creates an aggregate set of resources in the virtual environment, called a cluster.

Storage networks and arrays	VMware vSphere uses Fibre Channel SAN arrays, iSCSI SAN arrays, and NAS arrays to meet different data center storage needs. With storage area networks, you can connect and share storage arrays between groups of servers. This arrangement allows aggregation of the storage resources and provides more flexibility in provisioning them to virtual machines. For detailed information, see *vSphere Storage*.
IP networks	Each compute server can have multiple physical network adapters to provide high bandwidth and reliable networking to the entire VMware vSphere data center. For detailed information, see *vSphere Networking*.
vCenter	vCenter provides a single point of control to the data center. It provides essential data center services such as access control, performance monitoring, and configuration. It unifies the resources from the individual computing servers to be shared among virtual machines in the entire data center. It manages the assignment of virtual machines to the ESX hosts and the assignment of resources to the virtual machines within a given computing server. These assignments are based on the policies that the system administrator sets. Compute servers continue to function even if vCenter becomes unreachable (for example, if the network is severed). The ESX hosts can be managed separately and continue to run the virtual machines assigned to them based on the resource assignment that was last set. After connection to vCenter is restored, it can manage the data center as a whole again.
Management clients	VMware vSphere provides several interfaces for data center management and virtual machine access. These interfaces include vSphere Client for access through a web browser or vSphere Command-Line Interface (vSphere CLI).

What Are the Software Components of vSphere

VMware vSphere is a suite of software components for virtualization. These include ESX, vCenter, and other software components such as vCenter Single Sign-On, and vCenter plug-ins that fulfill several different functions in the vSphere environment.

vSphere includes the following software components:

ESX	The hypervisor runs virtual machines. Each virtual machine has a set of configuration and disk files that together perform all the functions of a physical machine. Through ESX, you run the virtual machines, install operating systems, run applications, and configure the virtual machines. Configuration includes identifying the virtual machine's resources, such as storage devices. The server provides bootstrapping, management, and other services that manage your virtual machines.
vCenter	A service that acts as a central administrator for VMware ESX hosts that are connected on a network. vCenter directs actions on the virtual machines and the ESX hosts. vCenter is installed to run automatically on a preconfigured virtual machine. The vCenter service runs continuously in the background. It performs its monitoring and managing activities even when no vSphere Clients are connected and when no one is logged on to the computer where it resides. It must have network access to all the hosts it manages. vCenter is deployed as a preconfigured virtual machine optimized for running vCenter and the vCenter components. You can deploy vCenter on ESX hosts 6.5 or later. All prerequisite services for running vCenter and the vCenter components are bundled in the vCenter installation. All vCenter services run as child processes of the VMware Service Library Lifecycle Manager service. See the *vCenter Installation and Setup* documentation for details about setting up this configuration.
vCenter Single Sign-On	A service that is part of the vCenter management infrastructure. The vCenter Single Sign-On authentication service makes the VMware Cloud infrastructure platform more secure by allowing the various vSphere software components to communicate with each other. The vCenter Single Sign-On authentication service uses a secure token exchange mechanism instead of requiring each component to authenticate a user separately with a directory service like Active Directory. For more information, see Configure VMware Single Sign-On.
vCenter plug-ins	Applications that provide additional features and functionality to vCenter. Typically, plug-ins consist of a server component and a client component. After the plug-in server is installed, it is registered with vCenter and the plug-in client is available to the vSphere Client for download. After a plug-in is installed on the vSphere Client, it might alter the interface by adding views, tabs, toolbar buttons, or menu items related to the added functionality.

	Plug-ins leverage core vCenter capabilities, such as authentication and permission management, but can have their own types of events, tasks, metadata, and privileges. Some vCenter features are implemented as plug-ins, and can be managed using the vSphere Client Plug-in Manager. These features include vCenter Storage Monitoring, vCenter Hardware Status, and vCenter Service Status.
vCenter database	Persistent storage for maintaining the status of each virtual machine, host, and user managed in the vCenter environment. The vCenter database can be remote or local to the vCenter system. The database is installed and configured during vCenter installation. If you are accessing your ESX host directly through the VMware Host Client, and not through a vCenter system and associated vSphere Client, you do not use a vCenter database.
tcServer	Many vCenter functions are implemented as web services that require the tcServer. The tcServer is installed on the vCenter machine as part of the vCenter installation. Features that require the tcServer to be running include: ICIM/Hardware Status tab, Performance charts, WebAccess, Storage Policy-Based services, and vCenter Service status.
vCenter agent	On each managed host, the software that collects, communicates, and runs the actions received from vCenter. The vCenter agent is installed the first time any host is added to the vCenter inventory.
Host agent	On each managed host, the software that collects, communicates, and runs the actions received through the vSphere Client. It is installed as part of the ESX installation.

What Are the Client Interfaces for vSphere

You can access vSphere components through the vSphere Client, the VMware Host Client, and the vSphere Command-Line Interfaces.

vSphere Client	The vSphere Client, introduced in vSphere 6.5, is an HTML5-based client and is included with vCenter. The vSphere Client is the primary interface for connecting to and managing vCenter instances.
VMware Host Client	The VMware Host Client is a Web-based application that you can use to manage individual ESX hosts that are not connected to a vCenter system. For more information about the VMware Host Client, see *vSphere Single Host Management - VMware Host Client*.
vSphere Command-Line Interfaces	vSphere supports multiple command-line interfaces for configuring virtual machines, ESX hosts, and vCenter.

What Are the Managed Inventory Objects in vSphere

In vSphere, the inventory is a collection of virtual and physical objects, such as data centers, networks, and datastores, on which you can place permissions, monitor tasks and events, and set alarms. You can rename inventory objects, and you can group most inventory objects by using folders to more easily manage them.

All inventory objects, with the exception of hosts, can be renamed to represent their purposes. For example, they can be named after company departments or locations or functions.

Note:
Managed object names cannot exceed 214 bytes (UTF-8 encoded).

vCenter monitors and manages the following inventory objects:

Data Centers	Unlike folders, which are used to organize specific object types, a data center is an aggregation of all the different types of objects used to work in virtual infrastructure. Within each data center, there are four separate hierarchies. • Virtual machines (and templates) • Hosts (and clusters) • Networks • Datastores The data center defines the namespace for networks and datastores. The names for these objects must be unique within a data center. You cannot have two datastores with the same name within a single

	data center, but you can have two datastores with the same name in two different data centers. Virtual machines, templates, and clusters need not be unique within the data center, but must be unique within their folder. Objects with the same name in two different data centers are not necessarily the same object. Because of this, moving objects between data centers can create unpredictable results. For example, a network named networkA in data_centerA might not be the same network as a network named networkA in data_centerB. Moving a virtual machine connected to networkA from data_centerA to data_centerB results in the virtual machine changing the network it is connected to.
Clusters	A collection of ESX hosts and associated virtual machines intended to work together as a unit. When you add a host to a cluster, the host's resources become part of the cluster's resources. vCenter manages the resources of all hosts in a cluster as one unit. If you enable VMware EVC on a cluster, you can ensure that migrations with vMotion do not fail because of CPU compatibility errors. If you enable vSphere DRS on a cluster, the resources of the hosts in the cluster are merged to allow resource balancing for the hosts in the cluster. If you enable vSphere HA on a cluster, the resources of the cluster are managed as a pool of capacity to allow rapid recovery from host hardware failures.
Datastores	A virtual representation of physical storage resources in the data center. A datastore is the storage location for virtual machine files. In an on-premises SDDC, these physical storage resources can come from the local SCSI disk of the ESX host, the Fibre Channel SAN disk arrays, the iSCSI SAN disk arrays, or Network Attached Storage (NAS) arrays. For both on-premises and cloud SDDCs, vSAN datastores hide the idiosyncrasies of the underlying physical storage and present a uniform model for the storage resources required by virtual machines.
Folders	Folders allow you to group objects of the same type so you can easily manage them. For example, you can use folders to set permissions across objects, to set alarms across objects, and to organize objects in a meaningful way. A folder can contain other folders, or a group of objects of the same type: data centers, clusters, datastores, networks, virtual machines, templates, or hosts. For example, one folder can contain hosts and a folder containing hosts, but it cannot contain hosts and a folder containing virtual machines. Data center folders form a hierarchy directly under the root vCenter and allow users to group their data centers. Within each data center is one hierarchy of folders with virtual machines and templates, one with hosts and clusters, one with datastores, and one with networks.
Hosts	The physical computer on which ESX is installed. All virtual machines run on hosts or clusters.
Networks	A set of virtual network interface cards (virtual NICs), distributed switches or vSphere Distributed Switches, and port groups or distributed port groups that connect virtual machines to each other or to the physical network outside of the virtual data center. All virtual machines that connect to the same port group belong to the same network in the virtual environment, even if they are on different physical servers. You can monitor networks and set permissions and alarms on port groups and distributed port groups.
Resource pools	Resource pools are used to compartmentalize the CPU and memory resources of a host or cluster. Virtual machines run in, and draw their resources from, resource pools. You can create multiple resource pools as direct children of a standalone host or cluster and then delegate control over each resource pool to other individuals or organizations. If DRS is enabled, provides various options for monitoring the status of the resources and adjusting or suggesting adjustments to the virtual machines using the resources. You can monitor resources and set alarms on them.
Templates	A template is a primary copy of a virtual machine that can be used to create and provision new virtual machines. Templates can have a guest operating system and application software installed. They can be customized during deployment to ensure that the new virtual machine has a unique name and network settings.
Virtual machines	A virtualized computer environment in which a guest operating system and associated application software can run. Multiple virtual machines can operate on the same managed host machine concurrently.
vApps	vSphere vApp is a format for packaging and managing applications. A vApp can contain multiple virtual machines.
Virtual Private Clouds (VPC) and subnets	Starting with VCF 9.0, VCP instances and subnets are available as inventory objects integrated from NSX. VPC is an industry standard for networking consumption. You can create a VPC instance and the underlying subnets, and attach virtual machines to the subnets directly from the vSphere Client, or through vCenter API or CLI. VPC instances that are created by using the NSX Manager are visible in the vCenter inventory tree. VPC instances that are created by using vCenter APIs or the vSphere Client are available in the NSX Manager.

Advanced vCenter Components

vCenter has several components that are packaged and installed with the base product.

vCenter components include:

VMware vSphere® vMotion®	Enables you to move running virtual machines from one ESX host to another ESX host without service interruption. It requires licensing on both the source and target host. vCenter centrally coordinates all vSphere vMotion activities.
vSphere® Storage vMotion®	Allows you to move the disks and configuration file of a running virtual machine from one datastore to another without service interruption. It requires licensing on the virtual machine's host.
VMware vSphere® High Availability	Enables a cluster with High Availability. If a host fails, all virtual machines that were running on the host are promptly restarted on different hosts in the same cluster. When you enable the cluster for vSphere HA, you specify the number of hosts you want to be able to recover. If you specify the number of host failures allowed as **1**, vSphere HA maintains enough capacity across the cluster to tolerate the failure of one host. All running virtual machines on that host can be restarted on remaining hosts. By default, you cannot turn on a virtual machine if doing so violates required failover capacity.
VMware vSphere® Distributed Resource Scheduler ™	Helps improve resource allocation and power consumption across all hosts and resource pools. vSphere DRS collects resource use information for all hosts and virtual machines in the cluster and gives recommendations (or migrates virtual machines) in one of two situations: • Initial placement – When you power on a virtual machine in the cluster for the first time, DRS either places the virtual machine or makes a recommendation. • Load balancing – DRS attempts to improve resource use across the cluster by performing automatic migrations of virtual machines (vSphere vMotion) or by providing a recommendation for virtual machine migrations. vSphere DRS includes VMware vSphere®Distributed Power Management ™ (DPM) capabilities. When DPM is enabled, the system compares cluster-level and host-level capacity to the demands of virtual machines that are running in the cluster. Based on the results of the comparison, DPM recommends (or implements) actions that can reduce the power consumption of the cluster.
VMware vSphere® Storage DRS™	Allows you to manage multiple datastores as a single resource, called a datastore cluster. A datastore cluster is an aggregation of multiple datastores into a single logical, load-balanced pool. You can treat the datastore cluster as a single flexible storage resource for resource management purposes. You can assign a virtual disk to a datastore cluster, and vSphere Storage DRS finds an appropriate datastore for it. The load balancer takes care of initial placement and future migrations based on workload measurements. Storage space balancing and I/O balancing minimize the risk of running out of space and the risk of I/O bottlenecks slowing the performance of virtual machines.
VMware vSphere® Fault Tolerance	vSphere Fault Tolerance provides continuous availability for virtual machines by creating and maintaining a Secondary VM that is identical to the Primary VM. This Secondary VM is continuously available to replace the Primary VM in a failover situation.

vCenter Plug-Ins

vCenter plug-ins extend the capabilities of vCenter by providing additional features and functions.

Some plug-ins are installed as part of the base vCenter product.

vCenter Storage Monitoring	Allows you to review information on storage use and to map relationships visually between all storage entities available in vCenter.
vCenter Hardware Status	Uses CIM monitoring to display the hardware status of hosts that vCenter manages.
vCenter Service Status	Displays the status of vCenter services.

Some plug-ins are packaged separately from the base product and require separate installation. You can update plug-ins and the base product independently of each other. VMware modules include:

vSphere Lifecycle Manager	Enables administrators to apply updates and patches across ESX hosts and all managed virtual machines. Administrators can create user-defined security baselines that represent a set of security standards. Security administrators can compare hosts and virtual machines against these baselines to identify and remediate systems that are not in compliance.

VMware Cloud Foundation Operations orchestrator	A workflow engine that enables you to create and run automated workflows in your vSphere environment. VMware Cloud Foundation Operations orchestrator coordinates workflow tasks across multiple VMware products and third-party management and administration solutions through its open plug-in architecture. VMware Cloud Foundation Operations orchestrator provides a library of workflows that are extensible. You can use any operation available in the vCenter API to customize VMware Cloud Foundation Operations orchestrator workflows.

Using the vSphere Client

Use the vSphere Client to connect to vCenter systems and manage vSphere inventory objects.

Use of the vSphere Client requires a supported web browser.

The home screen of the vSphere Client is a system dashboard that aggregates data from different sources in the environment together in a single, unified view.

VMware has tested and supports the following guest operating systems and browser versions for the vSphere Client.

Table 91: Supported Guest Operating Systems and Browser Versions for the vSphere Client.

Operating system	Browser
Windows 32-bit and 64-bit	Microsoft Edge: 120 and later. Mozilla Firefox: 115 and later. Google Chrome: 120 and later.
Mac OS	Microsoft Edge: 120 and later. Mozilla Firefox: 115 and later. Google Chrome: 120 and later.

Earlier versions of these browsers are likely to work, but have not been tested.

How to Use and Configure Your vSphere Client

Learn how to configure the vSphere Client timeout settings, edit the login banner, and refresh data in the vSphere Client.

Manage the Login Message to the vSphere Client Login Page

You can add a login message to the vSphere Client login page. You can also configure a custom login message and provide a check box for user consent.

1. Log in to the vSphere Client.

   Specify the user name and password for administrator@vsphere.local or another member of the vCenter Single Sign-On Administrators group. If you specified a different domain during installation, log in as administrator@mydomain.
2. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Administration**.
3. Under **Single Sign On**, click **Configuration**.
4. Click the **Login Message** tab.
5. Click **Edit** and configure the login message.

Option	Description
Show login message	Toggle on **Show login message** to enable the login message. You cannot make changes to the login message unless you toggle on this switch.
Login message	Title of the message. By default, when **Consent checkbox** is toggled on, the login message text is `I agree to Terms and Conditions`. You must replace `Terms`

Option	Description
	`and Conditions` with your own text. If the **Consent checkbox** is toggled off, then `Login message` appears, over which you enter your message.
Consent checkbox	Toggle on **Consent checkbox** to require that the user clicks a check box before logging in. You can also display a message without a check box.
Details of login message	Message that the user sees when clicking the login message, for example, the text of the terms and conditions. You must enter some details in this text box.

6. Click **Save**.

How to Configure Your vSphere Client Timeout Value

By default, the vSphere Client sessions close after 120 minutes of idle time, and require the user to log in again to resume using the client.

Change the Client Configuration in the vSphere Client

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Administration**.
2. Under **Deployment**, select **Client Configuration**.
3. In the **Client Configuration** pane, click **Edit**.
   The **Edit client configuration** dialog box appears.
4. Enter a new value in minutes in the session timeout text box.
5. Click **Save**.
6. Restart the VMware vSphere Client service.
   - Enables SSH access to vCenter, and use this Bash shell command `service-control --restart vsphere-ui`.
   - Use the Center Appliance Management Interface (VAMI).
     a. In a web browser, go to the vCenter Appliance Management Interface, https://**vcenter-IP-address-or-FQDN**:5480.
     b. Log in with the root user credentials.
     c. In the left navigation pane, click **Services**.
     d. From the list of services, select VMware vSphere Client, and click **Restart**.

How Do You Refresh Data in the vSphere Client

You must manually refresh the data in the vSphere Client to see changes made to objects by other users during your session.

For performance reasons, the vSphere Client does not continuously refresh data on all objects in the inventory. All changes that you make during your current session are immediately reflected in the client user interface. Changes made by other users or in other sessions are not reflected until you manually refresh the data.

To update all data in the current vSphere Client view, click the refresh icon ( ).

The client view is updated.

How to Use Your vSphere Client Inventory

Learn how to navigate, search, and sort the vSphere Client inventory.

How to Use Your Global Inventory Lists Navigator

You can use the vSphere **Global Inventory Lists** navigator to browse and select objects in the vSphere inventory as an alternative to the hierarchical inventory tree.

Unlike the inventory tree, which presents hierarchical arrangements of parent and child objects arranged in the **Hosts and Clusters**, **VMs and Templates**, **Storage**, and **Networking** inventory lists, the Global Inventory Lists navigator presents a list of all related objects in the inventory. You can navigate from an object to its related objects, regardless of their type.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Global Inventory Lists**.
2. In the left pane, select one of the object or resource categories to view objects of that type.
   For example, click **Hosts** to view the hosts in the vSphere inventory.
3. In the left navigation pane, click an object from the list once to display information about the object.
4. Optional: Click the object again to open it.

   Opening an object brings it to the top of the navigator and displays related object categories beneath it.

   For example, opening a host allows you to see the child resource pools, virtual machines, datastores, networks, and distributed switches associated with this host.
5. To access additional information or manage the selected object, click one of the tabs in the center pane.

Option	Description
Summary	You can view basic status and configuration for an object.
Monitor	You can view alarms, performance data, resource allocation, events, and other status information for an object.
Configure	Depending on the selected object, you can edit settings, alarm definitions, tags, permissions, and so on.
Permissions	You can view, add, change, and delete permissions. This tab is only available for logged in users with administration privileges.
Related Objects tabs	You can view and manage the objects related to the object that you selected. For example, if you select a host, the tabs that you see are **VMs**, **Resource Pools**, **Datastores**, and **Networks**.
Updates	Depending on the selected object, you can check the hardware compatibility of a host against the VMware Compatibility Guide, check the status of virtual machines, upgrade the VMware Tools version or the virtual hardware version of the virtual machines, and perform vSphere Lifecycle Manager operations. **Note:** The **Updates** tab is available only for certain types of inventory objects, for example, hosts, virtual machines, and clusters.

Inventory Keyboard Shortcuts in the vSphere Client

Discover some of the inventory keyboard shortcuts that you can use to quickly navigate to different inventories in the vSphere Client.

Table 92: Inventory Keyboard Shortcuts

Keyboard Combination on a Windows machine	Keyboard Combination on a Mac machine	Action
Ctrl+Alt+Home	Control+Option+Home	Home
Ctrl+Alt+1	Control+Option+1	Shortcuts
Ctrl+Alt+2	Control+Option+2	Hosts and Clusters

Keyboard Combination on a Windows machine	Keyboard Combination on a Mac machine	Action
Ctrl+Alt+3	Control+Option+3	VMs and Templates
Ctrl+Alt+4	Control+Option+4	Storage
Ctrl+Alt+5	Control+Option+5	Networking
Ctrl+Alt+6	Control+Option+6	Content Libraries
Ctrl+Alt+7	Control+Option+7	Supervisor Management
Ctrl+Alt+8	Control+Option+8	Global Inventory Lists
Ctrl+Alt+R	Control+Option+R	Refresh
Ctrl+Alt+S repeat twice	Control+Option+S repeat twice	Search

How to Sort Your vSphere Client Inventory Objects

You can use sortable columns and input filters to sort and find objects in your vSphere Client inventory that meet certain criteria.

You can sort list views by column. These list views can be selected in the **Global Inventory Lists** pane or found in search results.

For example, you can sort virtual machines by name, provisioned space, used space, and so on.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Global Inventory Lists**.
2. To open an object list view, select an object type from the list.
   If objects from the selected type are not present in the inventory, a blank page appears.
3. Optional: Click the column you want to sort objects by. You can click again on that column to reverse the sorting order.
4. Add or remove columns from the object list view:
   a) Click **Manage Columns** at the bottom left corner of the object list view.
      The **Show Columns** pop-up appears.
   b) Optional: To show a column in the object list view, select the check box for the column from the list.
   c) Optional: To hide a column from the object list view, deselect the check box for the column from the list.

How to Use Quick Filters to Find Objects in Your vSphere Client Inventory

From a list view, you can use the quick filter field to filter objects by entering specific values.

You can use input filters to sort and find objects in your vSphere Client inventory that meet certain criteria. You can filter inventory objects by name, state, status, and so on.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Global Inventory Lists**.
2. To open an object list view, select an object type from the list.
   If objects from the selected type are not present in the inventory, a blank page appears.

3. Click the drop-down menu on top of the inventory object list table, and select **Quick Filter**.
4. Enter your filtering parameter in the empty filter field. For example, in the virtual machine list view, to filter for all virtual machines that are in a powered on state, you must enter `powered on`.

   A list of inventory objects that meet your filtering criteria is displayed in your selected sorting order.

To clear the filtered list of inventory objects, delete the filtering parameter from the filter field.

How to Use Advanced Filters to Find Objects in Your vSphere Client Inventory

You can use the advanced filter options to create a set of filtering criteria and find specific objects in your vSphere Client.

1. In the vSphere Client, click the main navigation menu ( ), and select **Global Inventory Lists**.
2. To open an object list view, select an object type from the list.

   If objects from the selected type are not present in the inventory, a blank page appears.
3. Click the drop-down menu on top of the inventory object list table, and select **Advanced Filter**.
4. Click **Add New**.

   A pop-up window appears.
5. From the drop-down menu under Filter, select the filtering parameter.
6. Depending on the parameter you select, you must select values from a pre-defined list, or enter data in a filter operator.
   - From the Value list, select all values that you want to apply as filtering criteria.
   - From the filter operator, select how the filtering criteria must relate to each other and enter the filter value.

     Note:
     If you want to specify the filtering criteria further, you can click **Add Condition**, and enter the required details for the second filter operator.
7. Click **Apply**.

   A list of inventory objects that meet your filtering criteria is displayed in your selected sorting order. The filters that you apply are displayed on top of the inventory list.

   Note:
   You can apply multiple advanced filters to the inventory list. They are combined with the logical operator *and*.
8. Optional: To edit a filter that you applied, click the filter displayed on top of the inventory list, edit the filtering criteria, and click **Apply**.
9. Optional: To hide all filters applied to the inventory list, click **Hide All**.

   The filters continue to be applied.

To delete a filter that you applied, click the remove icon ( ) next to the filtering criterion. If the filter consists of more than one filtering criteria, you should remove each one separately.

To delete all filters applied to the inventory list, click **Clear All**.

How to Search Your vSphere Client Inventory

With the vSphere Client, you can search the inventory for objects that match specified criteria. You can search the inventories of all vCenter systems.

You can only view and search for inventory objects that you have permission to view.

Note:
If your permissions change while you are logged in, the search service might not immediately recognize these changes. To ensure that your search is performed with up-to-date permissions, log out of all your open sessions and log in again before you perform the search.

Perform a Quick Search

A quick search checks all types of objects for the specified search term within the name or other properties of the object.

1. Enter the search term in the search box at the top of the vSphere Client window.
   Multiple search terms in a quick or simple search are treated as if they are connected by ORs. For example, searching for `example machine` finds all objects with names containing either "example" or "machine".
   The search results appear below the search box as you type. The number of items displayed is limited to 10.
2. To run a saved search, click the saved search icon and select a saved search query.
3. To display an item in the inventory, click the item from the drop-down list of the search results.
4. To see more search results or more details about the search results, click the summary of the search results from the drop-down list of search results.
   a. To display an object in the inventory, click the object in the search results page.
   The search results are listed in tables arranged by object type. For example, if a search finds hosts and virtual machines, the following tables appear: **Hosts**, which shows only host results and **Virtual Machines**, which shows only virtual machine results.

Save a Search

You can save search queries in the vSphere Client so you can rerun, rename, or delete them later.

1. In a vSphere Client window, enter a query for a simple search in the search box.
2. Click the summary of the search results from the drop-down list with search results.
3. On the search results page, click **Save Search**.
4. Enter a name for the search query. Names must be lowercase with no spaces.
5. Click **Save**.
   The search query you entered is saved. You can reload the query later and repeat the search.

Manage a Saved Search

In the vSphere Client, you can rename, delete, or create a duplicate of a saved search query.

1. At the top of the vSphere Client window, click inside the search box.
2. Click the saved search icon and select a saved search query.
3. On the Search Results page, click **Actions** and select one of the options:

Option	Description
Save as	Creates a duplicate of the saved search query.
Rename	Renames the saved search query.
Delete	Deletes the saved search query.

4. Confirm the changes.

How to Export Your vSphere Inventory Lists

In the vSphere Client, you can export the contents of an inventory list view to a CSV file.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select**Global Inventory Lists**.
2. In the left pane, select one of the object or resource categories to view objects of that type.
   For example, click **Hosts** to view the hosts in the vSphere inventory.
3. Select whether you want all columns or your current selection of columns to be listed in the CSV file.
   a) Click **Manage Columns** at the bottom left corner of the object list view.
      The **Show Columns** pop-up appears.
   b) Optional: To show a column in the object list view, select the check box for the column from the list.
   c) Optional: To hide a column from the object list view, deselect the check box for the column from the list.
   d) Optional: To show all columns, click **SELECT ALL**.
4. Click **Export** at the bottom left corner of the list view.

The CSV file is generated and available for download.

How to Manage Your vCenter Client Plug-Ins

Learn how to monitor, add, or remove vCenter client plug-ins. You can monitor client plug-in activities including installation progress, failures, deployments, upgrades, and uninstallations.

From the vSphere Client, you can monitor vCenter client plug-ins using the **Recent Tasks** pane, the **Task Console**, or the **Administration** > **Client Plugins** view. In addition to monitoring installation progress, you can troubleshoot installation failures and plug-in incompatibilities. You can also add and remove client plug-ins.

For more information about how to develop client plug-ins, see the *Developing Remote Plug-ins with the vSphere Client SDK* documentation on https://developer.vmware.com.

View Details about a Client Plug-In

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Administration**.
2. Under **Solutions**, click **Client Plugins**.
   The **Client Plugins** main view shows a list of all deployed client plug-ins. You can view information about the name, version, deployment status, type, certification status, vendor, and plug-in ID of each client plug-in.
3. In the Name column, click a client plug-in name to open the client plug-in details view.
   You can use the client plug-in details view to view more information about the client plug-in instances, control the visibility of the client plug-ins on specific vCenter systems, and remove client plug-ins from the vSphere Client.

Hide and Show a Client Plug-In on a vCenter Instance

If your environment includes several vCenter systems joined in Enhanced Linked Mode, the vSphere Client can load views from client plug-ins added to any of the linked vCenter instances. You configure what is visible in a specific vCenter system, by using the **Hide** and **Show** options from the details view of a selected client plug-in.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Administration**.
2. Under **Solutions**, click **Client Plugins**.
3. From the Name column, click the client plug-in name.

The detailed client plug-in view opens.

4. Change the visibility of a client plug-in instance on a vCenter instance.
   a. Select the check box on the client plug-in instance.
      - To hide a client plug-in in the vSphere Client UI for a vCenter instance, select the check box on the instance version and click **Hide**.
      - To make a client plug-in instance visible in the vSphere Client UI for a vCenter instance, select the client plug-in version and click **Show**.
5. Confirm your choice in the pop-up dialog box that appears.

Add a Client Plug-In with the vSphere Client

Staring with vSphere 8.0, you can deploy remote client plug-ins using the vSphere Client. In an environment with several vCenter instances connected with Enhanced Linked Mode, you can select on which vCenter systems to deploy the client plug-in.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Administration**.
2. Under **Solutions**, click **Client Plugins**.
3. In the **Client Plugins** pane, click **Add**.
   The **Install new solution** wizard opens.
4. On the **Select an OVF template** page, specify the location of the client plug-in OVF descriptor file and click **Next**.

Option	Action
URL	Enter the URL to the OVF descriptor file located on the Internet. The only supported URL sources are HTTPS. **Note:** Make sure that all elements of the client plug-in are at the same location as the OVF descriptor file. Missing files might lead to failure in plug-in deployment.
Local file	Click **Upload files** and select all files associated with the client plug-in.

5. On the **Select a name and folder** page, enter a unique name for the virtual machine, select a deployment location, and click **Next**.
   The default name for the virtual machine is the same as the human-readable name of the plug-in defined in the OVF file. If you change the default name, choose a name that is unique within each vCenter virtual machine folder.
6. On the **Select a compute resource** page, select a resource where to run the deployed client plug-in VM, and click **Next**.
7. On the **Review details** page, verify the client plug-in details and click **Next**.
8. On the **License agreement** page, accept the end-user license agreements and click **Next**.
9. On the **Select storage** page, define where and how the files for the deployed client plug-in are stored.
   – Select a storage policy for the virtual machine.
     This option is available only if storage policies are enabled on the destination resource.
   – Select the **Disable Storage DRS for this virtual machine** check box to deactivate Storage DRS for the virtual machine.
   – Select a datastore to store the deployed client plug-in.
     The plug-in manifest file and virtual disk files are stored on the datastore. Select a datastore large enough to accommodate the virtual machine and all associated virtual disk files.
10. On the **Select networks** page, select a source network and map it to a destination network. Define the IP allocation settings of the client plug-in server. Click **Next**.
    The Source Network column lists all networks as defined in the OVF descriptor file.
11. On the **Customize template** page, view the read-only settings of the client plug-in and click **Next**.

12. On the **Associate vCenter** page, select the vCenter instances on which the client plug-in is deployed and click **Next**.
13. On the **Ready to complete** page, review your selections and click **Finish**.

As a result new tasks for downloading and installing the client plug-in appear in the **Recent Tasks** pane. After successful completion of the tasks, refresh your browser to see the newly added client plug-in.

Remove a Client Plug-In from a vCenter Instance

You can remove a client plug-in from the vSphere Client UI for a vCenter instance. In an environment with several vCenter instances connected with Enhanced Linked Mode, you can select from which vCenter systems to remove the client plug-in.

> **Note:**
>
> When you remove a client plug-in from the vSphere Client UI, the virtual machine on which the client plug-in instance is running is not removed from the vCenter instance. You must delete the virtual machine from the datastore to complete the client plug-in removal.

1. In the vSphere Client, click the main navigation menu ( ), and select **Administration**.
2. Under **Solutions**, click **Client Plugins**.
3. From the Name column in the **Client Plugins** pane, select the client plug-in.
   The detailed client plug-in view opens.
4. Select the check box on one of the client plug-in instances and click **Remove**.
5. In the confirmation dialog box, click **Yes** to remove the client plug-in instance.

As a result, the client plug-in is undeployed from the vCenter instance. Make sure that you refresh your browser to remove the client plug-in from the vSphere Client UI.

How to Start, Stop, and Restart Services in Your vSphere Client

In the vSphere Client, you can start, stop, and restart services that are running on vCenter. You can restart services upon a configuration change or for suspected functional or performance issues.

1. Log in to the vCenter Management Interface at https://<vcenter_fqdn>:5480.
   - Log in to the vCenter Management Interface directly.
   - In the vSphere Client home page, select **Administration** > **System Configuration**. Click a node from the list.
2. On the vCenter Management Interface home page, click **Services**.
3. Select a service from the service list.
4. From the top menu, click **Restart**, **Start**, or **Stop**.
   Unavailable options are dimmed.

   > **Note:**
   > Restarting the Content Library Service also restarts the Transfer Service and the OVF Service. The Content Library Service, the Transfer Service, and the OVF Service run on the same Tomcat server.

How to Provide Feedback in the vSphere Client

Use the feedback tool in the vSphere Client to provide timely feedback to our developers.

1. From the vSphere Client, click the feedback icon in the top-right corner.
2. Select the type of feedback that you want to give, and enter your feedback in the **Description** window.

   If you want to provide ideas for new features, use the VCF Feature Request Portal. For more information, see Use the VCF Feature Requests Portal to Submit vSphere Feature Requests.

3. Optional: Provide an email address and screenshots or other images.
4. Click **Send**.

Use the VCF Feature Requests Portal to Submit vSphere Feature Requests

You can provide suggestions for new vSphere features or share feature requests through the VCF Feature Requests Portal. All users with valid Broadcom Support Portal accounts can access the portal.

Verify that you have a valid Broadcom Support Portal account.

The VCF Feature Requests Portal is public. When you publish new ideas and feature requests, consider whether you want to share personal data.

You can access the VCF Feature Requests Portal at https://vcf.ideas.aha.io/portal_session/new by entering a valid Broadcom Support Portal user name and password.

1. On the VCF Feature Requests Portal login page, use your Broadcom Support Portal credentials to log in.
2. On the VCF Feature Requests Portal home page, click **Add a new idea**.
3. On the **VCF Feature Request Idea Portal Screen** page, in the **Your Idea** text field, enter a summary of your feature idea.
4. Enter a more detailed description of your idea.
5. From the **VCF Product Category** drop-down menu, select the respective vSphere category for which you want to submit an idea.
6. Click **Next**.
7. Optional: In the **Your Details** page, provide your company name and the customer impact of your idea.
8. Click **Next**.
9. Review the summary of your idea and click **Add Idea**.

Your idea appears in the **My Ideas** tab and in the list with submitted ideas. All users of the VCF Feature Requests Portal can view, subscribe, vote, and comment on your idea.

Configuring the Customer Experience Improvement Program

You can participate in the Customer Experience Improvement Program (CEIP) to provide anonymous feedback or information to VMware for quality, reliability, and functionality improvement of VMware products and services.

VMware Customer Experience Improvement Program

vCenter participates in VMware's Customer Experience Improvement Program (CEIP).

Details regarding the data collected through CEIP and the purposes for which it is used by VMware are set forth at the Trust & Assurance Center at https://www.broadcom.com/company/legal/privacy/data-usage-programs/ceip-products.

Join or Leave Customer Experience Improvement Program in vSphere Client

You can join your vCenter to the Customer Experience Improvement Program (CEIP), or decide to leave the CEIP at any time. To leave and rejoin your host to the CEIP, see *vSphere Single Host Management - VMware Host Client* documentation.

Obtain the user name and password of the administrator account.

1. Log in to the vSphere Client by using the credentials of the administrator account.
2. Click the main navigation menu
   ( ),
   and select **Administration**.
3. Under Deployment, click **Customer Experience Improvement Program**.
4. Click **Join** to join the CEIP or **Leave Program** to leave the Program.

Organizing Your vSphere Inventory

Plan how to set up your virtual inventory and consider how the virtual machines that it will support are going to be used and administered.

A large vSphere implementation might contain several virtual data centers with a complex arrangement of hosts, clusters, resource pools, and networks. It might involve multiple vCenter systems connected using vCenter linking. Smaller implementations might require a single virtual data center with a much less complex topology.

Here are the questions to answer as you create and organize an inventory of virtual objects:

- Will some virtual machines require dedicated resources?
- Will some virtual machines experience periodic spikes in workload?
- Will some virtual machines need to be administered as a group?
- Will some virtual objects require one set of system permissions, while other objects will require a different set of permissions?
- Do you want to use multiple vSphere Standard Switches, or you want to have a single vSphere Distributed Switch per data center?
- Do you want to use vMotion and Distributed Resource Management with certain virtual machines but not others?

The left pane of the vSphere Client displays your vSphere inventory. You can add and arrange objects in any way with the following considerations:

- The name of an inventory object must be unique with its parent.
- vApp names must be unique within the Virtual Machines and Templates view.
- System permissions are inherited and cascade. When you assign a system permission to an object in the inventory, the same permission propagates down the object hierarchy.

Tasks for Organizing Your Inventory

Populating and organizing your inventory involves the following activities:

> **Note:**
> vCenter supports an 80 character limit for all its entities. When you create a new inventory object, if the name exceeds this limit, the task fails and a warning appears.

- Creating data centers.
- Adding hosts to the data centers.
- Organizing inventory objects in folders.
- Setting up networking by using vSphere Standard Switches or vSphere Distributed Switches. To use services such as vMotion, TCP/IP storage, VMware vSAN™, and Fault Tolerance, setup VMkernel networking for these services. For more information, see *vSphere Networking*.
- Configuring storage systems and creating datastore inventory objects to provide logical containers for storage devices in your inventory. See *vSphere Storage*.
- Creating clusters to consolidate the resources of multiple hosts and virtual machines. You can enable vSphere HA and vSphere DRS for increased availability and more flexible resource management. See *vSphere Availability* for information about configuring vSphere HA, and *vSphere Resource Management* for information about configuring vSphere DRS.

- Creating resource pools to provide logical abstraction and flexible management of the resources in vSphere. Resource pools can be grouped into hierarchies and used to hierarchically partition available CPU and memory resources. See *vSphere Resource Management* for details.

How Do You Create a vSphere Data Center

A virtual data center is a container for all the inventory objects required to complete a fully functional environment for operating virtual machines. You can create multiple data centers to organize groups of environments to meet different user needs.

Required privileges:

- **Datacenter** > **Create datacenter**

For example, you can create a data center for each organizational unit in your enterprise or create some data centers for high-performance environments and other data centers for less demanding environments.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Right-click the vCenter object and select **New Datacenter**.
4. Optional: Enter a name for the data center and click **OK**.

Add hosts, clusters, resource pools, vApps, networking, datastores, and virtual machines to the data center.

How to Create a Folder in Your vSphere Inventory

In the vSphere Client,you can create folders and use them to group objects of the same type for easier management. For example, you can apply a common set of permissions to the folder and these permissions apply to all objects grouped in the folder.

A folder can contain other folders, or a group of objects of the same type. For example, one folder can contain both virtual machines and another folder that contains virtual machines, but it cannot contain both hosts and a folder that contains virtual machines.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select either a data center or another folder as a parent object for the folder that you want to create.
4. Right-click the parent object and click **New Folder**.
   - If the parent object is a folder, the new folder is of the same type as the parent folder - it can contain only objects of the same type that the parent folder contains.
   - If the parent object is a data center, you can create one of four types of folders: **Host and Cluster** folders, **Network** folders, **Storage** folders, and **VM and Template** folders.
5. Enter a name for the folder and click **OK**.

Move objects into the folder by right-clicking the object and selecting **Move To**. Select the folder as the destination. You can also move an object by dragging it to the destination folder.

How to Add a Host to Your vSphere Data Center or Folder

In the vSphere Client, you can add hosts under a data center object, a folder object, or a cluster object. If a host contains virtual machines, those virtual machines are also added to the inventory.

- Verify that a data center or a folder exists in the inventory.
- Obtain the user name and password of the root user account for the host.
- Verify that hosts behind a firewall are able to communicate with the vCenter system and all other hosts through port 902 or another custom-configured port.
- Verify that all NFS mounts on the host are active.
- Verify that you have the proper privileges. Different sets of privileges apply when you add multiple hosts to a cluster and a single host to a cluster or a data center. For more information, see Required vCenter Privileges for Common Tasks.
- If you want to add a host with more than 512 LUNs and 2,048 paths to the vCenter inventory, verify that the vCenter instance is suitable for a large or an x-large environment.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Navigate to a data center or folder within a data center.
4. Right-click the data center or folder and select **Add Host**.
5. On the **Name and location** page, enter the IP address or the name of the host and click **Next**.
6. On the **Connection settings** page, enter administrator credentials and click **Next**.
7. On the **Host summary** page, review the summary of the host details and click **Next**.
8. On the **Host lifecycle** page, the check box **Manage host with an image** check is selected by default.
   - If you want to manage the host with an image, leave the check box selected and click **Next**.

     For more information about using an image to manage the lifecycle of a standalone ESX host that is part of your vCenter inventory, see Managing the Lifecycle of Standalone ESX Hosts with vSphere Lifecycle Manager Images at Host Level in the *Managing Host and Cluster Lifecycle* documentation.
   - If you do not want to manage the host with an image, deselect the check box and click **Next**.
9. If you manage the host with an image, on the **Image** page, set up the desired image and click **Next**.
   a) Select an ESX version.
   b) Optional: Select a vendor add-on and the version of the add-on.
10. On the **Assign license** page, add a license to the host.
11. Optional: Select a lockdown mode option to deactivate the remote access for the administrator account after vCenter takes control of this host and click **Next**.
12. Optional: On the **VM location** page, select a location for the virtual machines that reside on the host and click **Next**.
13. Review the summary and click **Finish**.

A new task for adding the host appears in the Recent Tasks pane. It might take a few minutes for the task to complete.

How Do You Create and Configure Clusters in the vSphere Client

You can create clusters in the vSphere Client and then configure them manually or by using the Quickstart workflow.

A cluster is a group of hosts. When a host is added to a cluster, the resources of the host become part of the resources of the cluster. The cluster manages the resources of all hosts that it contains.

When you create clusters, you can enable vSphere High Availability (HA), vSphere Distributed Resource Scheduler (DRS), and the VMware vSAN features.

You can create a cluster that you manage with a single image. By using vSphere Lifecycle Manager images, you can easily update and upgrade the software and firmware on the hosts in the cluster. During cluster creation, you can select a reference host and use the image on that host as the image for the newly created cluster. For more information about using images to manage ESX hosts and clusters, see How to Use vSphere Lifecycle Manager Images in the *Managing Host and Cluster Lifecycle* documentation.

How Do You Create a vSphere Cluster with the vSphere Client

You can create a new vSphere cluster object by using the vSphere Client.

- Verify that a data center, or a folder within a data center, exists in the inventory.
- Verify that hosts have the same ESX version and patch level.
- Obtain the user name and password of the root user account for the host.
- If you want to extend a cluster with initially configured networking, verify that hosts do not have a manual vSAN configuration or a manual networking configuration.
- To create a cluster that you manage with a single image, review the requirements and limitations information in the*Managing Host and Cluster Lifecycle* documentation.

Required privileges:

- **Host** > **Inventory** > **Create cluster**

The clusters that you create can use vSphere Lifecycle Manager images for host updates and upgrades.

A vSphere Lifecycle Manager image is a combination of vSphere software, driver software, and desired firmware with regard to the underlying host hardware. The image that a cluster uses defines the full software set that you want to run on all ESX hosts in the cluster: the ESX version, additional VMware-provided software, and vendor software, such as firmware and drivers.

The image that you define during cluster creation is not immediately applied to the hosts. If you do not set up an image for the cluster, the cluster uses baselines and baseline groups. During cluster creation, you can select a reference host and use the image on that host as the image for the newly created cluster. For more information about using images and baselines to manage hosts in clusters, see the *Managing Host and Cluster Lifecycle* documentation.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a data center.
4. Right-click the data center and select **New Cluster**.
5. Enter a name for the cluster.
6. Select DRS, vSphere HA, or vSAN cluster features.

Option	Description
To use DRS with this cluster	1. Slide the switch to the right to enable the DRS service. 2. Click the info icon on the left to see the **Default Settings** for the DRS service. The default values are: – Automation Level: Fully Automated Migration – Threshold: 3
To use vSphere HA with this cluster	1. Slide the switch to the right to enable the vSphere HA service.

Option	Description
	2. Click the info icon on the left to see the **Default Settings** for the vSphere HA service. You are present with the following default values: Host Monitoring: Enabled Admission Control: Enabled VM Monitoring: Disabled
To use vSAN with this cluster	• Slide the switch to the right to enable the vSAN service. For more information on vSAN, see Preparing a New or Existing Cluster for vSAN in the *Planning and Configuring vSAN* chapter in the *VMware Cloud Foundation* documentation.

7. Select how to set up an image for the cluster. Click the info icon to the right of each option for more information.
8. Optional: If you want to manage the configuration at cluster level by using vSphere Configuration Profiles, select the check box **Manage configuration at a cluster level**. For more information about vSphere Configuration Profiles, see What is vSphere Configuration Profiles.
9. Click **Next**.
10. Select an image to assign to the cluster, and click **Next**.
11. Review the cluster details and click **Finish**.

    The cluster appears in the vCenter inventory. The **Quickstart** workflow appears under **Configure** > **Configuration**.

You can see the empty cluster in the vCenter inventory.

You can use the Quickstart workflow to easily configure and expand the cluster. You can also skip the Quickstart workflow and continue configuring the cluster and its hosts manually.

How to Extend and Configure Your vSphere Cluster by Using the Quickstart Workflow

You can extend and configure a cluster by using the simplified Quickstart workflow in the vSphere Client. Quickstart groups common tasks and offers configuration wizards that guide you through the process of configuring and extending a cluster.

After you provide the required information on each wizard, your cluster is configured based on your input. When you add hosts using the Quickstart workflow, hosts are automatically configured to match the cluster configuration.

After the cluster creation operation completes, you are redirected to the **Cluster quickstart** page located under **Configure** > **Configuration**. The Quickstart workflow consists of three cards for quickly extending and configuring your clusters.

Table 93: Quickstart workflow cards for extending and configuring a cluster

Cluster Quickstart Workflow Card	Description
Cluster basics	You can edit the cluster name, and activate or deactivate cluster services. The card lists the services that you enabled.
Add hosts	You can add new ESX hosts. After the hosts are added, the card shows the total number of hosts present in the cluster and displays health check validation for those hosts.
Configure cluster	You can configure network settings for vMotion and vSAN traffic, review and customize cluster services, and set up a vSAN datastore. After the cluster is configured, the card provides details on configuration mismatch and reports cluster health results through the vSAN Health service.

Characteristics of a Cluster Configured Using the Quickstart Workflows

A vSphere cluster that is configured using the Quickstart workflow has the following characteristics.

- Hosts are of version ESX 7.0 and later.

- Hosts end up having similar configuration, including network settings.
- Cluster configuration is based on the recommended default settings for networking and different services.
- Licenses cannot be assigned using the Quickstart workflow. You must manually assign the required licenses. See the VCF Licensing documentation.

Skipping Quickstart

The **Skip Quickstart** button prompts you to continue configuring the cluster and its hosts manually. To confirm exiting the simplified configuration workflow, click **Continue**. After you dismiss the **Cluster quickstart** workflow, you cannot restore it for the current cluster.

How to Add an ESX Host to Your vSphere Cluster by Using the Quickstart Workflow

You can add new and existing ESX hosts to a cluster in the vCenter inventory.

- Verify that hosts have the same ESX version and patch level.
- Obtain the user name and password of the root user account for the host.
- If you want to extend a cluster with initially configured networking, verify that hosts do not have a manual vSAN configuration or a manual networking configuration.
- Verify that you have the proper privileges. Different sets of privileges apply when you add multiple hosts to a cluster and a single host to a cluster or a data center.
- To add a host to a cluster that you manage with a single image, see Add Hosts to a Cluster that Uses an Image.

You can also add hosts to a DRS cluster. For more information, see the *vSphere Resource Management* documentation.

> **Note:**
> After you configure the cluster by using Quickstart, if you modify any cluster networking settings outside of Quickstart, you cannot use the Quickstart workflow to add and configure new hosts to the cluster.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Navigate to a cluster within a data center.
4. On the **Configure** tab, select **Configuration** > **Quickstart**.
5. Click **Add** in the **Add hosts** card.
6. On the **Add hosts** wizard, add new or existing hosts to the cluster.
   - Add hosts that are not part of the vCenter inventory.
     a. Click the **New hosts** tab.
     b. Populate the IP Address and credentials text boxes for those hosts.
     c. To add more new hosts, click the **Add Host** button.
     d. To reuse the credentials for all added hosts, select the **Use the same credentials for all hosts** check box.
   - Add hosts that are managed by your vCenter instance and are in the same data center as your cluster. The hosts must not be part of another cluster on the vCenter instance.
     a. Click the **Existing hosts** tab.
     b. From the list, select the hosts that you want to add to the cluster.
7. Click **Next**.

   The **Host summary** page lists all hosts that will be added to the cluster and related warnings.

Note:

If a host cannot be validated automatically by the system, you are prompted to manually validate its certificate and accept its thumbprint in the **Security Alert** pop-up.

8. On the **Host summary** page, review the details of the added hosts and click **Next**.
9. On the **Import Image** page, select the host whose image to use as the image for the cluster.

   The **Import Image** page appears when you add hosts to a cluster managed with a single image. For information about adding a host to a cluster that you manage with a single image, see Add Hosts to a Cluster that Uses an Image in the *Managing Host and Cluster Lifecycle* documentation.
10. On the **Review** page, review the IP addresses or FQDN of the added hosts and click **Finish**.

    Review the number of added hosts and the health check validation, performed by the vSAN Health service, in the **Add hosts**card.
11. Optional: Click **Re-validate** to trigger the validation of the hosts.

Note:

If an error occurs, it is visible in the **Recent Tasks** pane only.

All hosts are placed in maintenance mode and added to your cluster. You can manually exit the maintenance mode.

Configure your cluster default settings through the Quickstart workflow.

How to Configure Your vSphere Cluster by Using the Quickstart Workflow

Learn how to use the Quickstart workflow to quickly configure the settings of a vSphere cluster.

To configure the host networking settings on your host and to customize the cluster settings, start the **Configure cluster** wizard, part of the cluster Quickstart workflow. Learn how to use Quickstart to configure a vSAN cluster, see *Using Quickstart to Configure and Expand a vSAN Cluster* in the *vSAN Planning and Deployment* documentation.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a cluster.
4. On the **Configure** tab, select **Configuration** > **Quickstart**.

   The **Cluster Quickstart** page appears.

   Note:

   To configure your cluster host networking and services manually by referring to different parts of the vSphere software, click the **Skip quickstart** button. If you dismiss the **Cluster quickstart** workflow, you cannot restore it, and you have to configure manually any hosts that you add to this cluster in the future.
5. In the **Configure cluster** card, select **Configure**.

   The **Configure cluster** wizard appears.
6. On the **Distributed switches** page, configure the cluster networking.

   Alternatively, you can select the Configure networking settings later check box to hide all options that are related to cluster networking and to change the recommended host networking manually at a later time.

CAUTION:

After you select the Configure networking settings later check box, and complete the **Configure cluster** workflow, you cannot perform the cluster networking configuration in the future by using the **Configure cluster** wizard.

a) In the **Distributed switches** section, specify the number of distributed switches to create from the drop-down menu.

Note:
You can select up to three distributed switches.

The selected distributed switches are configured as part of this workflow and all hosts in the cluster connect to them.

b) Enter a unique name for each of the distributed switches you are about to create.

c) Optional: Click **Use Existing** to select an existing compatible distributed switch and an existing compatible distributed port group.

d) In the **Port groups** section, set up the vMotion network by selecting a distributed switch from the drop-down menu and assigning a new default port group to it.

e) In the **Physical adapters** section, for each physical network adapter (NIC), select the distributed switch name from the drop-down menu.

The new distributed switch must be assigned to at least one physical adapter.

Note:
If you are using an existing distributed switch, the physical adapter selection must match the current mapping of the distributed switch. Any variation results in an error.

This mapping of physical NICs to the distributed switches is applied to all hosts in this cluster.

f) Click **Next**.

7. If the vSphere DRS feature is enabled on the cluster, configure the networking options in the **vMotion traffic** page.

a) Optional: Select the **Use VLAN** check box and enter an ID for the vMotion distributed port group.

b) Optional: Select a protocol type from the drop-down menu.

c) Select an IP configuration type from the drop-down menu.

d) Optional: Populate the text boxes for each host in the cluster depending on the IP address type you need for setting up the networking.

If the IP address type is set to `DHCP`, DHCP is used to obtain the IP settings for the vMotion traffic.

e) Click **Next**.

The **Advanced options** page appears.

8. On the **Advanced options** page, customize the cluster settings depending on the services you enabled.

a) If you enabled the vSphere HA feature during cluster creation, use the options in the **vSphere HA** section to activate or deactivate host failure monitoring, virtual machine monitoring, and admission control.

If you enable admission control, you can specify the failover capacity by number of hosts.

b) If you enabled the vSphere DRS feature during cluster creation, the **Distributed Resource Scheduler** section is visible.

1. Set the Automation level to Fully Automated, Partially Automated or Manual.
2. Select one of the five migration settings from the Migration threshold drop-down menu.

c)

1. Select a lockdown mode from the drop-down menu.
2. Enter an NTP server address or FQDN.
   If you enter more than one NTP server addresses, separate them by using commas.
3. Select a host update option from the drop-down menu.

The settings are applied across all hosts in this cluster.

d) In the **Enhanced vMotion Compatibility** section, enable Enhanced vMotion™ Compatibility (EVC) and select the CPU model from the EVC mode drop-down menu.
e) Click **Next**.

The **Review** page appears.

9. Review the settings and select **Finish**.
The card closes, and the progress of the operation appears in the **Recent Tasks** pane.

You have created a fully configured cluster in the vCenter inventory.

Expand your cluster manually or by using the Quickstart workflow and the **Add hosts** card.

How to Extend Your vSphere Cluster with Host Networking Configuration

By using the Quickstart Workflow, you can extend a hyper-converged cluster by adding hosts and configuring their networking to match the cluster configuration.

- Verify that you have an existing cluster and hosts added to it.
- In the initial cluster configuration, you configured the host networking.
- Verify that hosts have the same ESX version and patch level.
- Obtain the user name and password of the root user account for the host.
- If you want to extend a cluster with initially configured networking, verify that hosts do not have a manual vSAN configuration or a manual networking configuration.
- To add a host to a cluster that you manage with a single image, see Add Hosts to a Cluster that Uses an Image in the *Managing Host and Cluster Lifecycle* documentation.

Note:
If you modify any cluster networking settings outside of Quickstart, you cannot use the Quickstart workflow to add and configure new hosts to the cluster.

1. In the vSphere Client, click the main navigation menu
( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a configured cluster.
3. On the **Add hosts** page, add new or existing hosts to the cluster.
   - Add hosts that are not part of the vCenter inventory.
     a. Click the **New hosts** tab.
     b. Populate the IP Address and credentials text boxes for those hosts.
     c. To add more new hosts, click the **Add Host** button.
     d. To reuse the credentials for all added hosts, select the **Use the same credentials for all hosts** check box.
   - Add hosts that are managed by your vCenter instance and are in the same data center as your cluster.
     a. Click the **Existing hosts** tab.
     b. From the list, select the hosts that you want to add to the cluster.
4. Click **Next**.
The **Host summary** page lists all hosts that will be added to the cluster and related warnings.

Note:
If a host cannot be validated automatically by the system, you are prompted to manually validate its certificate and accept its thumbprint in the **Security Alert** pop-up.

5. On the **Host summary** page, review the details of the added hosts and click **Next**.
6. Optional: On the **Import Image** page, select the host whose image to use as the image for the cluster.

   The **Import Image** page appears when you add hosts to a cluster managed with a single image. For information about adding a host to a cluster that you manage with a single image, see Add Hosts to a Cluster that Uses an Image in the *Managing Host and Cluster Lifecycle* documentation.
7. On the **Ready to complete** page, review the IP addresses or FQDN of the added hosts and click **Finish**.

   The **Extend Cluster Guide** page appears.
8. In the **Configure cluster** card, select **Configure**.

   The **Configure cluster** wizard appears.
9. If the vSphere DRS feature is enabled on the cluster, configure the networking options in the **vMotion traffic** page.
   a) Optional: Select the **Use VLAN** check box and enter an ID for the vMotion distributed port group.
   b) Optional: Select a protocol type from the drop-down menu.
   c) Select an IP type from the drop-down menu.
   d) Optional: Populate the text boxes for each host in the cluster depending on the IP address type you need for setting up the networking.

      If the IP address type is set to `DHCP`, DHCP is used to obtain the IP settings for the vMotion traffic.
10. Click **Next**.

    The **Review** page appears.
11. Review the settings and select **Finish**.

    The card closes, and the progress of the operation appears in the **Recent Tasks** pane.

After successful validation, your newly added hosts are configured as the existing hosts in your cluster and the **Configure** button in the **Configure hosts** card becomes inactive. You can only click **Re-validate** to verify the cluster configuration.

Add more hosts to the cluster.

How to Extend and Configure Your vSphere Cluster Manually

You can extend a cluster by manually adding hosts to that cluster. If you previously configured the cluster without setting up the host networking, the configuration of the existing hosts in the cluster is applied to the newly added hosts.

- Verify that you have an existing cluster and hosts added to it.
- Verify that hosts have the same ESX version and patch level.
- Obtain the user name and password of the root user account for the host.
- During the initial cluster configuration, select the Configure networking settings later check box. For more information, see How to Configure Your vSphere Cluster by Using the Quickstart Workflow.
- To add a host to a cluster that you manage with a single image, see Add Hosts to a Cluster that Uses an Image in the *Managing Host and Cluster Lifecycle* documentation.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a configured cluster.
3. On the **Add hosts** page, add new or existing hosts to the cluster.
   - Add hosts that are not part of the vCenter inventory.
     a. Click the **New hosts** tab.
     b. Populate the IP Address and credentials text boxes for those hosts.
     c. To add more new hosts, click the **Add Host** button.

d. To reuse the credentials for all added hosts, select the **Use the same credentials for all hosts** check box.

- Add hosts that are managed by your vCenter instance and are in the same data center as your cluster. The hosts must not be part of another cluster on the vCenter instance.

  a. Click the **Existing hosts** tab.
  b. From the list, select the hosts that you want to add to the cluster.

4. Click **Next**.

   The **Host summary** page lists all hosts that will be added to the cluster and related warnings.

   Note:

   If a host cannot be validated automatically by the system, you are prompted to manually validate its certificate and accept its thumbprint in the **Security Alert** pop-up.

5. On the **Host summary** page, review the details of the added hosts and click **Next**.
6. Optional: On the **Import Image** page, select the host from which an image must be imported to the cluster.

   The **Import Image** page appears when you add hosts to a cluster managed with a single image. For information about adding a host to a cluster managed with a single image, see Add Hosts to a Cluster that Uses an Image in the *Managing Host and Cluster Lifecycle* documentation.

7. On the **Review** page, click **Finish**.

   The **Cluster Quickstart** page appears.

8. Click **Skip Quickstart** to continue configuring the cluster manually.

   CAUTION:

   After you dismiss the Quickstart workflow, you cannot enable it again on the cluster.

   Note:
   You must manually configure all hosts added to the cluster after dismissing the Quickstart workflow.

Manually configure all cluster settings and add more hosts to the cluster.

vSphere Tags and Custom Attributes

Tags and attributes allow you to attach metadata to objects in the vSphere inventory to make it easier to sort and search for these objects.

What Are vSphere Tags and How to Use Them

You can create, edit, and delete tag categories and vSphere tags. You can also assign vSphere tags to objects in the vCenter inventory and manage the user privileges for working with tags and categories.

A tag is a label that you can apply to objects in the vSphere inventory. When you create a tag, you assign that tag to a category. Categories allow you to group related tags together. When you define a category, you can specify the object types for its tags, and whether more than one tag in the category can be applied to an object.

For example, if you wanted to tag your virtual machines by guest operating system type, you can create a category called operating system. You can specify that it applies to virtual machines only and that only a single tag can be applied to a virtual machine at any time. The tags in this category might be Windows, Linux, and Mac OS.

Tags and categories can span multiple vCenter instances:

- If multiple vCenter instances are configured to use vCenter linking, tags and tag categories are replicated across all these vCenter instances.
- To manage tags at scale, across vCenter instances managed by the same VCF Operations instance, you can also use the tag management capabilities in VCF Operations. For more information, see Tags and Categories Overview.

Tags can also be used when you configure compute or storage policies. For example, if you have critical workloads that need to be run on hosts with high network performance, you can create a specific tag and assign the tag to the virtual

machines with the respective workloads. You can configure a compute policy and specify that virtual machines with that specific tag must be placed in a host that has network performance.

How Do You Create and Edit a vSphere Tag Category

You use categories in your vSphere environment to group tags together and define how tags can be applied to objects. You create and edit a tag category from the vSphere Client.

The required privilege depends on the task that you want to perform.

Task	Privilege
Create a tag category	**vSphere Tagging** > **Create vSphere Tag Category**
Edit a tag category	**vSphere Tagging** > **Edit vSphere Tag Category**

You can create a tag category explicitly, as explained here, or as part of the tag creation process. Each tag has to belong to at least one tag category.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Tags & Custom Attributes**.
2. Click the **Tags** tab and click **Categories**
3. Start the task that you want to perform.

Task	Steps to perform
Create a tag category	Click **New**.
Edit a tag category	Select a category and click **Edit**.

4. Edit the category settings.

Setting	Description
Category Name	The category name must be unique to the currently selected vCenter system. Enter a category name that describes common characteristics, for example license requirements, or virtual machines that you plan to tag and include in the category.
Description	You can provide text in the description to describe the purpose or use of the category.
Tags Per Object	• If you select **One Tag**, you can apply only one tag from this category to an object. Use this option for categories whose tags are mutually exclusive. For example, if you have a category called Priority with tags High, Medium, and Low, then each object should have only one tag because an object can have only one priority. • If you select **Many tags**, you can apply more than one tag from the category to an object. Use this option for categories whose tags are not mutually exclusive. After you have set Tags Per Object, you can change from **One Tag** to **Many Tags**, but not from **Many Tags** to **One Tag**.
Associable Object Types	Select whether tags in this category can be assigned to all objects or only to a specific type of object, such as a virtual machine or a datastore. Changes to the associable object type are limited. • If you initially selected a single object type, you can later change the category to work for all object types.

Setting	Description
	• If you initially selected All Objects, you cannot restrict the category later.

5. Click **Create** or **Save**.

How Do You Delete a vSphere Tag Category

Learn how to remove a vSphere tag category from your vSphere environment.

Required privilege: InventoryService.Tagging.DeleteCategory.

Deleting a category also deletes all tags associated with the category.

1. In the vSphere Client, click the main navigation menu

   and select **Tags & Custom Attributes**.
2. Click the **Tags** tab, and click **Categories**.
3. Select a category from the list and click **Delete**.
   The **Delete Category** dialog opens.
4. To confirm the operation, click **Delete**.

The category and all associated tags are deleted.

How Do You Create and Edit a vSphere Tag

Read how to create or edit a vSphere tag. After you create a tag, you can use it to add metadata to inventory objects.

The required privilege depends on the task that you want to perform.

Task	Privilege
Create a tag	**vSphere Tagging** > **Create vSphere Tag** on any vCenter inventory object.
Edit a tag	**vSphere Tagging** > **Edit vSphere Tag** on any vCenter inventory object.

You can use tags to add metadata to inventory objects. You can also record information about your inventory objects in tags, and you can use the tags in searches. For information about creating and editing a tag category, see How Do You Create and Edit a vSphere Tag Category.

1. In the vSphere Client, click the main navigation menu

   and select **Tags & Custom Attributes**.
2. Select the **Tags** tab, and click **Tags**.
3. Start the task that you want to perform.

Task	Steps to perform
Create a tag	Click **New**.
Edit a tag	Select a tag and click **Edit**.

4. In the dialog box, enter or edit the name for the tag.
   You can use up to 256 characters for the tag name.

5. Enter or edit the description of the tag.
6. If you create a tag, specify the tag category.
   - Select an existing category from the **Category** drop-down menu.
   - Click **Create New Category** and select the newly created category.
7. Click **Create** or **Save**.

How Do You Delete a vSphere Tag

You can delete an existing vSphere tag when you no longer need it by using the vSphere Client.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Tags & Custom Attributes**.
2. Select the **Tags** tab and click **Tags**.
3. Select one or more tags from the list and click **Delete**.
4. To confirm the operation, click **Delete**.

The selected tags are deleted.

Use Advanced Filters to Filter Tags in the vSphere Client

Use the advanced filter options to create a set of filtering criteria and find specific tags and categories in your vSphere Client.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Tags & Custom Attributes**.
2. Select the **Tags** tab.
3. If you want to filter tags, click **Tags**.
4. If you want to filter categories, click **Categories**.
5. Click the drop-down menu in the left upper corner of the tags or categories table, and select **Advanced Filter**.
6. Click **Add New**.

   A pop-up window appears.
7. From the drop-down menu under Filter, select the filtering parameter.
8. From the drop-down menu under Operator, select how the filtering criteria must relate to each other.
9. Enter the filter value.
10. If you want to specify the filtering criteria further, click **Add Condition**, and enter the required details for the second filter operator.
11. Click **Apply**.

    A list of tags or categories that meet your filtering criteria is displayed in your selected sorting order. The filters that you apply are displayed on top of the inventory list.

    Note:

    You can apply multiple advanced filters to the inventory list. They are combined with the logical operator *and*.

12. Optional: Optional: To edit a filter that you applied, click the filter displayed on top of the inventory list, edit the filtering criteria, and click **Apply**.
13. Optional: Optional: To hide all filters applied to the inventory list, click **Hide All**.

    The filters continue to be applied.

To delete a filter that you applied, click the remove icon (

×

) next to the filtering criterion. If the filter consists of more than one filtering criteria, you should remove each one separately.

To delete all filters applied to the inventory list, click **Clear All**.

How Do You Assign or Remove a vSphere Tag

After you create vSphere tags, you can apply or remove them as metadata to objects in the vCenter inventory.

Required privilege: InventoryService.Tagging.AttachTag.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to an inventory tree.
3. Select an object from the list and click **Actions**.
4. From the drop-down menu, select **Tags and Custom Attributes**.
5. Select the required action.
   - To assign a tag, select **Assign Tag**.
   - To remove a tag, select **Remove tag**.
6. Select a tag from the list and click **Assign**.

   In the **Assign Tag** dialog box, you can also create new tags with **Add Tag**. For information about creating a tag, see How Do You Create and Edit a vSphere Tag.

How Do You Add Permissions for vSphere Tags and Tag Categories

You can manage the user privileges for working with tags and categories. The procedure for assigning permissions to tags is the same as the procedure for assigning permissions to tag categories.

When you create a tag, you can specify which users and groups can operate with that tag. For example, you can grant administrative rights only to administrators and set read-only permissions for all other users or groups. You must have vSphere administrator credentials to set and manage permissions for tags.

Permissions for tags work similar to permissions for vCenter inventory objects. For more information, see the *vSphere Security* documentation.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Tags & Custom Attributes**.
2. On the **Tags** tab, click the **Tags** or **Categories** button.

   Depending on the button that you click, you see the list of tags or the list of tag categories.

3. Select an item from the list and click **Add Permission**.
4. In the **Add Permission** dialog box, select a domain from the drop-down menu.
5. Search for a user or a group to add.
6. Select a role to add from the drop-down menu.
7. To enable permission inheritance, select the **Propagate to children** check box.
8. Click **OK**.

vSphere Tagging Best Practices

Incorrect tagging can lead to replication errors. To avoid these errors, diligently follow best practices when tagging objects in your vSphere environment.

When working with tags in multiple node situations, expect replication delays between the nodes (generally 30 seconds to 2 minutes depending on your setup).

Follow these best practices to avoid replication errors:

- After creating a tag, if you immediately assign that tag to a local object, assign it from the management node where you created the tag.
- After creating a tag, if you immediately assign that tag to a remote object, assign it from the management node to which the object is local. Depending on your environment setup, allow for replication time to propagate the new tag before you use the tag.
- Avoid simultaneously creating categories and tags from different management nodes before categories and tags across nodes can finish the replication process. If duplicate categories or tags are created from different nodes at the same time, the duplicates might not be detected and will appear. If you see these results, manually delete duplicates from one management node.
- Before you remove an object from the vCenter inventory that is not a virtual machine or a host, first remove the tags from the object.

For more information about tagging performance best practices, see the blog post VMware vCenter 8.0 U3 Tagging Performance Best Practices.

What Are Custom Attributes in the vSphere Client

You can use custom attributes in the vSphere Client to assign user-specific values for each object of the custom attribute type.

After you create the attributes, set the value for the attribute on each virtual machine or managed host, as appropriate. This value is stored with vCenter and not with the virtual machine or managed host. Use the new attribute to filter information about your virtual machines and managed hosts. If you no longer need the custom attribute, remove it. A custom attribute is always a string.

For example, suppose that you have a set of products and you want to sort them by sales representative. Create a custom attribute for the sales person's name, Name. Add the custom attribute, Name, column to one of the list views. Add the appropriate name to each product entry. Click the column title Name to sort alphabetically.

The custom attributes feature is available only when you are connected to a vCenter system.

How Do You Add and Edit Custom Attributes in the vSphere Client

You can create custom attributes in the vSphere Client and associate these custom keys with an object, such as a host, virtual machine, cluster, or network. You can then assign a user-specific value using the custom attributes to specific objects in your vCenter environment.

Required privileges: Global.ManageCustomFields

After you create the attributes, set an appropriate value for the attribute on each inventory object. The attribute value is stored with vCenter and not with the inventory object. Use the new attribute to filter your inventory objects. If you no longer need the custom attribute, remove it. A custom attribute is always a string.

For example, suppose that you have lots of virtual machines in your inventory and you want to sort them by owner.

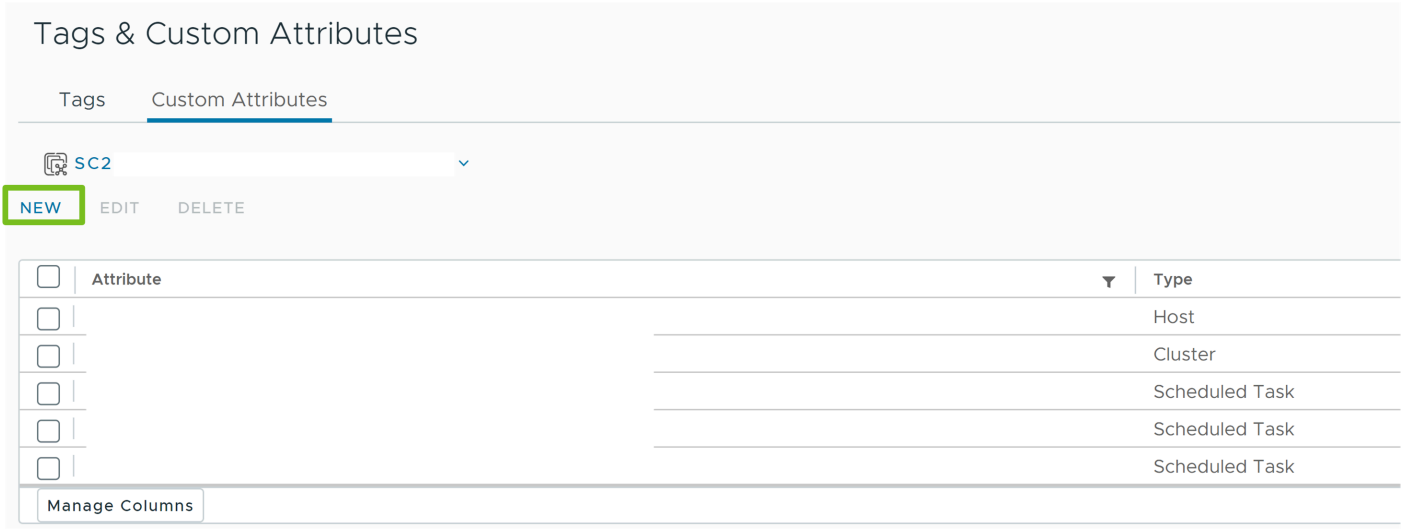
1. Create a `VM-owner` custom attribute for the virtual machine type of inventory objects.
2. Add the name of the owner to each VM entry you want and use the **Global Inventory Lists** pane to add the `VM-owner` custom attribute column to one of the list views.
3. You can now click the **VM-owner** column to sort alphabetically by owner.

   Tip:
   If you cannot use the custom attribute column to sort the list of inventory objects, you can export the content of the list view and then use the sorting options on the saved CSV file.

   Note:
   Tags and tag categories support a finer-grained mechanism for tagging your object. Consider using tags and tag categories instead of custom attributes.

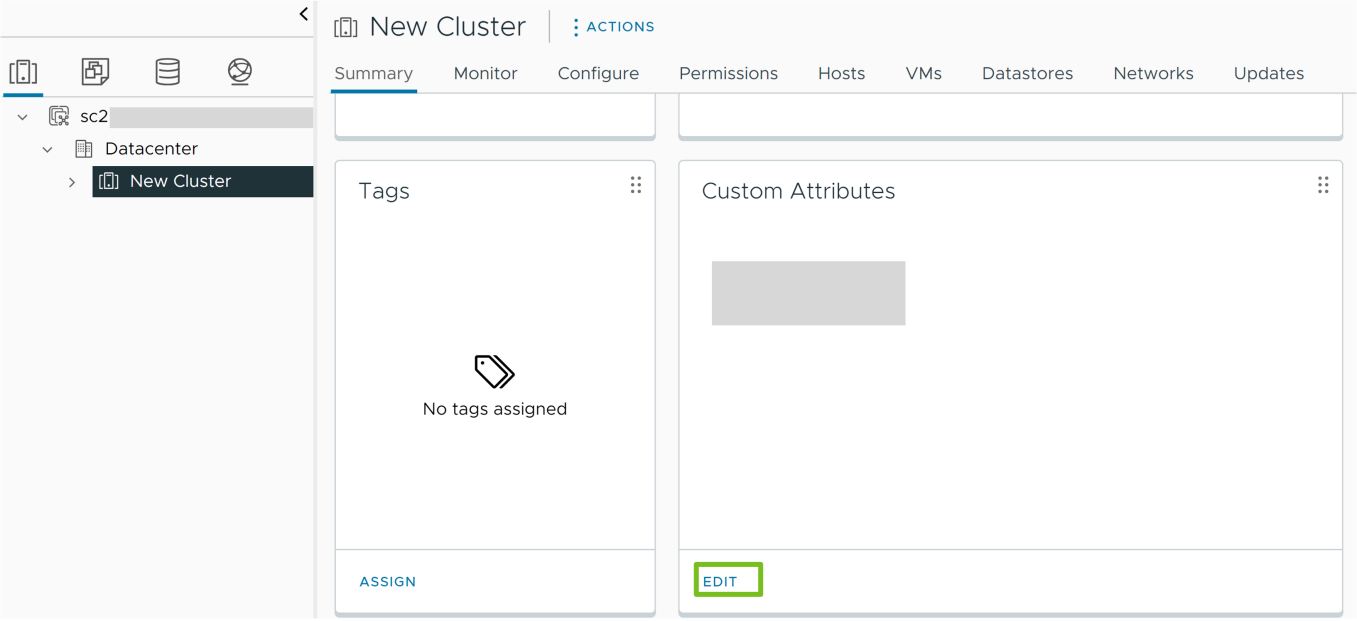
1. In the vSphere Client, click the main navigation menu ( ), and select **Tags & Custom Attributes**.
2. Select the **Custom Attributes** tab.

   All currently defined custom attributes for vCenter are displayed.
3. Click **New**.



4. Enter the values for the custom attribute.
   a) Enter the name of the attributes in the **Attribute** text box.
   b) Select the attribute type from the **Type** drop-down menu.
   c) Click **OK**.

   After you have defined an attribute on an object, it is available to all objects of that type in the inventory. However, the value you specify is applied only to the currently selected object.
5. You can later add a value to the custom attribute for each inventory object that you want.
   a) Navigate to an inventory object in your vCenter inventory.
   b) On the **Summary** tab, navigate to the **Custom Attributes** card.
   c) Click **Edit**.



- If you want to add a value to an existing custom attribute, enter or edit the attribute value in the Value column.
- If you want to create a new custom attribute, enter a name and value to the attribute and select the attribute type from the **Type** drop-down menu. Click **Add**.

d) Click **Save**.

vSphere Compute Policies

You can create policies that govern the placement of workload virtual machines in your vSphere inventory.

Creating and Managing Compute Policies

By using compute policies, you can specify how the vSphere Distributed Resource Scheduler (DRS) places virtual machines on hosts in a vSphere cluster. For more information about DRS, see Create a vSphere DRS Cluster.

Compute policies establish persistent placement constraints that are enforced not only at initial VM placement, but throughout the VM lifecycle — including during DRS load balancing, maintenance mode evacuation, and HA failover operations.

You can create, edit, or delete a compute policy. Policy conflicts are not detected when you create a policy. Policies in the system might be in conflict with each other. For example, a group of virtual machines can be part of both a VM-VM affinity policy and a VM-VM anti-affinity policy. In such cases, DRS might not be able to place the virtual machines in a way that complies with both policies.

Starting with vSphere 9.1.1, some policies are created automatically during VM creation in response to placement policy constraints specified in a virtual machine configuration specification. System-created policies are visible in the vSphere Client alongside user-created policies. In vSphere 9.1.1, you can identify system-created policies by the name prefix `AutoGenPolicy-` that is followed by the tag category name and a SHA-256 hash string. The system created policies have a description field that summarizes the full set of policy parameters in plain language.

For information on how to request system-created policies by using the bulk policy creation API, see VM-to-VM Affinity and Anti-Affinity Policies.

Note:

Affinity policies are not the same as the vSphere DRS affinity rules.

Compute Policies and vSphere Tags

Tags are the mechanism by which compute policies identify which virtual machines and hosts to govern. Before you create a VM-VM or VM-Host policy, you must have tag categories and tags in place, or you can allow the system to create

them automatically when you specify placement policies in a VM configuration specification. For more information about vSphere tags, see What Are vSphere Tags and How to Use Them.

Impact of Compute Policies During Host Replacement and Maintenance

If you configure a compute policy based on host tags, you must review the potential impact if a tagged host is placed in maintenance mode or fails and is replaced. Available resources are reduced for the duration of the replacement or maintenance operation. As a best practice, allocate a sufficient number of tagged hosts to support the workloads that are placed by compute policy when a tagged host is temporarily unavailable.

Compute Policy Types

Compute Policy	Description
VM-Host affinity	Places virtual machines in a specified category on hosts in a specified category. For more information, see Create a VM-Host Affinity Policy.
VM-Host anti-affinity	Keeps virtual machines in a specified category away from hosts in a specified category. For more information, see Create a VM-Host Anti-Affinity Policy.
VM-VM affinity	Places virtual machines sharing a specified tag on the same host, cluster, or zone. Your vSphere version must be 9.1.1 or later. For more information, see Create a VM-VM Affinity Policy.
VM-VM anti-affinity	Keeps virtual machines sharing a specified tag on different hosts, clusters, or zones. Your vSphere version must be 9.1.1 or later. For more information, see Create a VM-VM Anti-Affinity Policy.
Limit VM placement span plus one host for maintenance	Limits the number of hosts across which a group of VMs is spread, reserving capacity for maintenance. For more information, see Create a Limit VM Placement Span Plus One Host For Maintenance Policy.
VM evacuation by best effort restart on best host	Shuts down guest operating systems when a host enters maintenance mode and restarts VMs on the best available host. For more information, see Create a VM Evacuation by Best Effort Restart on Best Host Policy.

Monitoring Compute Policy Compliance

- Monitoring virtual machine compliance with a particular compute policy. In the vSphere Client, in the Summary page of each virtual machine, you can view the compute policies that apply to the virtual machine. You can also view the compliance status of the virtual machine with each policy.
- Monitoring overall compute policy compliance. In the vSphere Client, navigate to**Policies and Profiles** > **Compute Policies**. You can check the compliance status of the policy - the accumulated compliance status of all the virtual machines that are subject to that policy.

Create a VM-Host Affinity Policy

A VM-Host affinity policy establishes a relationship between a category of virtual machines and a category of ESX hosts.

- Verify that you have administrator privileges.
- Verify that DRS is activated.

By using compute policies, you can specify how the vSphere Distributed Resource Scheduler (DRS) places virtual machines on hosts in a resource pool. For more information about DRS, see Create a vSphere DRS Cluster.

You can use VM-Host affinity policies when virtual machines with workload-specific configurations require placement on hosts that have certain characteristics. You can also use VM-Host affinity policies when virtual machines that run certain applications must be placed on hosts that are licensed to run those applications.

To prevent a VM-Host affinity policy from blocking the upgrade of a host or cluster, VM-Host affinity policies cannot prevent some host-related and virtual machine-related operations.

- A policy cannot prevent a host from entering maintenance mode. If the host must be placed in maintenance mode, virtual machines with an affinity for the host are automatically migrated to another host in the cluster. The virtual machines are migrated back to the initial host when the maintenance completes and the host becomes available.

- A policy cannot prevent a host configured for HA from executing a failover. Virtual machines with an affinity for the failed host can be migrated to any available host in the cluster.
- A policy cannot prevent a virtual machine from powering-on. If a virtual machine subject to a VM-Host affinity policy specifies a resource reservation that no host can meet, it is powered on on any available host. When a host that can match the resource reservation becomes available, the virtual machine is migrated to it.

1. Log in to the vSphere Client.
2. Create a tag category and a tag for the virtual machines that you want to include in the VM-Host affinity policy.

   You can use existing tags and categories or create new ones. For more information about creating and using tags, see vSphere Tags and Custom Attributes.

   Enter a category name that describes common characteristics, such as license requirements, or virtual machines that you plan to tag and include in the category.
3. Create a tag category and a tag for the ESX hosts that you want to include in the VM-Host affinity policy.

   You can use existing tags and categories or create new ones.
4. Assign the tags to the virtual machines and hosts that you want to include in a VM-Host affinity policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.
5. Create a VM-Host affinity policy.
   a) In the vSphere Client, click the main navigation menu
   ( ).
   b) Select **Policies and Profiles** > **Compute Policies**.
   c) Click **Add**.

      The **New Compute Policy** wizard appears.
   d) Enter a name for the policy.

      The policy name must be unique within your inventory.
   e) From the **Policy type** drop-down menu, select **VM-Host affinity**.
   f) Enter a description of the policy.
   g) From the **VM tag** and **Host Tag** drop-down menu, select a category and tag to which the policy applies.

      If only one tag is associated with a category, it is selected by default after you select the tag category.

      Note:

      In a stretched cluster, each host is automatically tagged with an **Availability zones** tag. The same tag is assigned to all hosts in an availability zone (AZ). You can use these tags to establish an affinity between a virtual machine and any host in the cluster that has a specific availability zone tag.

      Compute policy does not affect the location of virtual machine storage. Ensure that the virtual machine storage policies locate a copy of virtual machine storage in the AZ where the virtual machine is deployed. Compute policy does not affect network paths to and from the virtual machine, which might transit a different AZ than the one where the virtual machine is deployed.
   h) From the **Policy requirement** drop-down menu, select how the policy is applied.
      - To allow placement with violations if necessary, click **Preferred**.
      - To prevent powering on the virtual machine if the requirement in the policy cannot be met, select **Mandatory**.

      Note:

      Policy requirements apply only to initial virtual machine placement.

      Maintenance operations might override the policy requirements.

i) Click **Create**.

Create a VM-Host Anti-Affinity Policy

A VM-Host anti-affinity policy describes a relationship between a category of VMs and a category of hosts.

- Verify that you have administrator privileges.
- Verify that DRS is activated.

By using compute policies, you can specify how the vSphere Distributed Resource Scheduler (DRS) places virtual machines on hosts in a resource pool. For more information about DRS, see Create a vSphere DRS Cluster.

You can use a VM-Host anti-affinity policy when you want to avoid placing virtual machines that have specific host requirements such as a GPU or other devices, or capabilities such as IOPS control, on hosts that do not support those requirements.

To prevent a VM-Host affinity policy from blocking the upgrade of a host or cluster, VM-Host affinity policies cannot prevent some host-related and virtual machine-related operations.

- A policy cannot prevent a host from entering maintenance mode.
- A policy cannot prevent a host configured for HA from executing a failover. VMs with an anti-affinity for the failed host can be migrated to any available host in the cluster.
- A policy cannot prevent a VM from powering-on. If a VM subject to a VM-Host anti-affinity policy specifies a resource reservation that no host can meet, it is powered on on any available host.

1. Log in to the vSphere Client.
2. Create a tag category and a tag for the virtual machines that you want to include in the VM-Host anti-affinity policy.

   You can use existing tags and categories or create new ones. For more information about creating and using tags, see vSphere Tags and Custom Attributes.

   Enter a category name that describes common characteristics, such as license requirements, or virtual machines that you plan to tag and include in the category.
3. Create a tag category and a tag for the ESX hosts that you want to include in the VM-Host affinity policy.

   You can use existing tags and categories or create new ones.
4. Assign the tags to the virtual machines and hosts that you want to include in a VM-Host affinity policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.
5. Create a VM-Host anti-affinity policy.
   a) In the vSphere Client, click the main navigation menu

      ( ).
   b) Select **Policies and Profiles** > **Compute Policies**.
   c) Click **Add**.

      The **New Compute Policy** wizard appears.
   d) Enter a name for the policy.

      The policy name must be unique within your inventory.
   e) From the **Policy type** drop-down menu, select **VM-Host anti-affinity**.
   f) Enter a description of the policy.
   g) From the **VM tag** and **Host Tag** drop-down menu, select a category and tag to which the policy applies.

      If only one tag is associated with a category, it is selected by default after you select the tag category.

      Note:

      In a stretched cluster, each host is automatically tagged with an **Availability zones** tag. The same tag is assigned to all hosts in an availability zone (AZ). You can use these tags to establish an affinity between a virtual machine and any host in the cluster that has a specific availability zone tag.

h) From the **Policy requirement** drop-down menu, select how the policy is applied.

- To allow placement with violations if necessary, click **Preferred**.
- To prevent powering on the virtual machine if the requirement in the policy cannot be met, select **Mandatory**.

Note:

Policy requirements apply only to initial virtual machine placement.

Maintenance operations might override the policy requirements.

i) Click **Create**.

Create a Limit VM Placement Span Plus One Host For Maintenance Policy

A policy of type limit VM placement span plus one host for maintenance describes a relationship between members of a category of virtual machines and a specified number of hosts.

- Verify that you have administrator privileges.
- Verify that DRS is activated.

By using compute policies, you can specify how the vSphere Distributed Resource Scheduler (DRS) places virtual machines on hosts in a resource pool. For more information about DRS, see Create a vSphere DRS Cluster.

A policy of type limit VM placement span plus one host for maintenance establishes a relationship between virtual machines in a given category and specified number of hosts. After the policy is created, DRS attempts to place the virtual machines from the category on a number of hosts that you specify. You can select from 1 through 96 hosts. When one of the hosts is entering maintenance-mode, it is not counted in the total number that you specified.

1. Log in to the vSphere Client.
2. Create a tag category and a tag for the virtual machines that you want to include in the policy.

   You can use existing tags and categories or create new ones. For more information about creating and using tags, see vSphere Tags and Custom Attributes.

   Enter a category name that describes common characteristics, such as license requirements, or virtual machines that you plan to tag and include in the category.
3. Assign the tags to the virtual machines that you want to include in a policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.
4. Create a limit VM placement span plus one host for maintenance policy.
   a) In the vSphere Client, click the main navigation menu

   ( ), and select **Policies and Profiles** > **Compute Policies**.
   b) Click **Add**.

   The **New Compute Policy** wizard appears.
   c) Enter a name for the policy.

   The policy name must be unique within your inventory.
   d) From the **Policy type** drop-down menu, select **limit VM placement span plus one host for maintenance**.
   e) Enter a description of the policy.
   f) From the **VM tag** drop-down menu, select a category and tag to which the policy applies.

   If only one tag is associated with a category, it is selected by default after you select the tag category.
   g) Enter the number of hosts.
   h) Click **Create**.

Create a VM Evacuation by Best Effort Restart on Best Host Policy

When you apply a policy *VM evacuation by best effort restart on best host* to a virtual machine, when the host enters maintenance mode, the guest operating systems for the virtual machine shuts down.

- Verify that you have administrator privileges.
- Verify that DRS is activated.

If the guest operating system shut down is unavailable for a virtual machine that is subject to the *VM evacuation by best effort restart on best host* policy, or the guest operating system shut down cannot complete in time, the virtual machine powers off. When a virtual machine powers off as a result of the policy application, the virtual machine attempts to power on every few minutes and might power on a different host.

1. Log in to the vSphere Client.
2. Create a tag category and a tag for the virtual machines that you want to include in the policy.

   You can use existing tags and categories or create new ones. For more information about creating and using tags, see vSphere Tags and Custom Attributes.

   Enter a category name that describes common characteristics, such as license requirements, or virtual machines that you plan to tag and include in the category.
3. Assign the tags to the virtual machines that you want to include in the policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.
4. Create a *VM evacuation by best effort restart on best host* policy.
   a) In the vSphere Client, click the main navigation menu

      (**),**

      and select **Policies and Profiles** > **Compute Policies**.
   b) Click **Add**.

      The **New Compute Policy** wizard appears.
   c) From the **Policy type** drop-down menu, select *VM evacuation by best effort restart on best host*.
   d) Enter a name for the policy.

      The policy name must be unique within your inventory.
   e) Enter a description of the policy.
   f) From the **VM tag** drop-down menu, select a category and tag to which the policy applies.

      If only one tag is associated with a category, it is selected by default after you select the tag category.
   g) Click **Create**.

Create a VM-VM Affinity Policy

Starting with vSphere 9.1.1, you can create a VM-VM affinity policy to place virtual machines that share a specified tag on the same host.

- Verify that you have administrator privileges.
- Verify that DRS is activated.
- Create a tag category and a tag for the virtual machines that you want to include in the VM-VM affinity policy. For more information about creating and using tags, see vSphere Tags and Custom Attributes.
- Assign the tags to the virtual machines that you want to include in a VM-VM affinity policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.

The VM-VM affinity policy is important for workloads with colocation requirements, such as distributed application tiers that must not share a single point of failure.

VM-VM affinity is enforced by DRS continuously, at initial placement, during load balancing, maintenance mode, and HA failover operations.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Policies and Profiles** > **Compute Policies**.
2. Click **Add**.

   The **New Compute Policy** wizard appears.
3. From the **Policy type** drop-down menu, select **VM-VM affinity**.
4. Enter a name for the policy.

   The policy name must be unique within your inventory.
5. (Optional): Enter a description of the policy.
6. From the **VM tag** drop-down menu, select a category and tag to which the policy applies.

   If only one tag is associated with a category, after you select the tag category, the tag is selected by default.
7. Click **Create**.

The compute policy is available in the list of compute policies that are created in the vCenter instance.

To view details about a compute policy, click the compute policy name from the list.

To delete a compute policy, select it from the list, and click **Delete**.

Create a VM-VM Anti-Affinity Policy

Starting with vSphere 9.1.1, you can create a VM-VM anti-affinity policy to place virtual machines that share a specified tag on different hosts.

- Verify that you have administrator privileges.
- Verify that DRS is activated.
- Create a tag category and a tag for the virtual machines that you want to include in the VM-VM anti-affinity policy. For more information about creating and using tags, see vSphere Tags and Custom Attributes.
- Assign the tags to the virtual machines that you want to include in a VM-VM anti-affinity policy. For more information about assigning a tag to an object, see How Do You Assign or Remove a vSphere Tag.

The VM-VM anti-affinity policy is important for workloads with isolation requirements, such as distributed application tiers that must not share a single point of failure.

DRS enforces VM-VM anti-affinity policies continuously — not only at initial placement but also during load balancing, maintenance mode, and HA failover operations.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Policies and Profiles** > **Compute Policies**.
2. Click **Add**.

   The **New Compute Policy** wizard appears.
3. From the **Policy type** drop-down menu, select **VM-VM anti-ffinity**.
4. Enter a name for the policy.

   The policy name must be unique within your inventory.

5. (Optional): Enter a description of the policy.
6. From the **VM tag** drop-down menu, select a category and tag to which the policy applies.

   If only one tag is associated with a category, after you select the tag category, the tag is selected by default.
7. Click **Create**.

The compute policy is available in the list of compute policies that are created in the vCenter instance.

To view details about a compute policy, click the compute policy name from the list.

To delete a compute policy, select it from the list, and click **Delete**.

Working with vSphere Tasks

vSphere tasks are activities and actions that occur on an object within the vSphere inventory.

vSphere tasks represent system activities that do not complete immediately, such as migrating a virtual machine. For example, powering off a virtual machine is a task. You can perform this task manually every evening, or you can set up a scheduled task to power off the virtual machine every evening.

How Do You View vSphere Tasks

Learn how to view vSphere tasks that are associated with a single object or all objects in a vSphere environment.

By default, the task list for an object also includes tasks performed on its child objects. You can filter the task list by entering the object that you want in the **Target** column filter.

If you are logged in to a vCenter instance that is part of a vCenter linking group, the **Server** column in the task list displays the name of the vCenter system on which the task is performed.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to an object in the inventory.
3. Click the **Monitor** tab.
4. Under **Tasks and Events**, select **Tasks**.

   The task list contains tasks performed on the object and detailed information, such as target, task status, initiator, and start/completion time of the task.

   You can limit the number of tasks visible in the **Tasks** pane by selecting an option from the **Tasks per page** drop-down menu.
5. Optional: Click the filter ( ) icon on a column header and filter the task list by entering your filtering criteria.
6. Optional: Click the expand row (^) icon to view related events for a task. You can view task information about multiple tasks at the same time.
7. Optional: Click **Open in new tab** to open only the **Task Console** in a separate tab on your browser.

How Do You Cancel a vSphere Task

Canceling a vSphere task stops a running task from occurring. Canceling a scheduled task does not cancel subsequent runs. To cancel a scheduled task that has not run, reschedule it.

Required privileges:

- Manual tasks:**Tasks** > **Update Task**
- Scheduled tasks: **Scheduled task** > **Remove task**

1. In the vSphere Client, navigate to the **Recent Tasks** pane, and locate the task.
   By default, the **Recent Tasks** pane is displayed at the bottom of the vSphere Client.
2. Click the cancel button (⊗) next to the progress bar in the Status column for the task you want.
   If the cancel option is unavailable, the selected task cannot be canceled.
3. In the **Cancel Task** pop-up dialog box, click **OK**.

The vCenter system or ESX host stops the progress of the task and returns the object to its previous state. The vSphere Client displays the task with a status*The task was canceled by a user.*

Cancelled tasks and the corresponding events are retained in the vCenter database. For information about the retention of tasks and events in the vCenter database, and how to configure the database settings, see the *vSphere Monitoring and Performance* documentation.

How Do You Filter vSphere Tasks

To lower the number of vSphere tasks displayed for a single object or all objects in your environment, filter the task history in the database using several criteria, such as status, initiator, and time range.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Inventory**.
2. Navigate to an object in the inventory.
3. Click the **Monitor** tab.
4. Under **Tasks and Events**, select **Tasks**.
5. Click **Filter**.
   The **Filter Tasks** dialog opens.
6. Filter the tasks you want to view.
   a) Select the respective check boxes in the **Status** options list to filter the tasks by their status.
   b) From the **Type** drop-down menu, select the task type.
      - **User** - Displays the tasks initiated by all users, such as `vsphere.local` users, Active Directory domain users, and system users.
      - **System** - Displays the tasks initiated only by system users. System users are user accounts associated with an application or service.
   c) To view the tasks initiated by all users or by specific users, select the **All users** or **Specific users** radio button.
      If you select to view the tasks performed by specific users, create a list of users.
      1. Select the domain for the users from the drop-down menu.
      2. Select the users who initiated the tasks.
         a. Enter a name in the Search box.
            The system searches from the users in the selected domain.
         b. Select a user and click **Add**.
   d) From the **Select time set** drop-down menu, select the time when the task was started, completed, or queued.
   e) From the **Select time range** options, select the time range when the tasks were performed.
      - **Relative** - If you want to filter the tasks for a relative time range, specify the relative time range from the **Last** and **Range** drop-down menus.
      - **Date range** - If you want to filter the tasks for a specific time period, specify the date and time from the drop-down menus.

For example, to filter the tasks that got queued for the last 2 weeks, from the **Select time set** drop down menu, select **Queued time**, and select the **Relative** radio button. Then, from the **Last** drop-down menu, select **2** and from the **Range** drop-down menu, select **Weeks**.

7. Click **Filter**.

The **Tasks** page displays the tasks that match your criteria. If you want to clear the task filter, click **Clear all filters** in the upper-right corner.

How Do You Export vSphere Tasks

You can export the detailed information for selected vSphere tasks or all tasks listed in the **Tasks** page to a CSV file.

To export the information about specific tasks, you can filter the task list and then select only the tasks that you want. You can also export the information about all tasks listed for an object or all objects in the vSphere environment.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to an object in the inventory.
3. Click the **Monitor** tab.
4. Under **Tasks and Events**, select **Tasks**.
5. Optional: To export one or more tasks, click the check boxes in front of the respective tasks.
6. Click **Export** at the top-left corner of the **Tasks** pane.

   Note:
   By default, if you did not select any tasks, the detailed information about all displayed tasks is exported.

   a) Optional: If you selected one or more tasks but you want to export the information about all displayed tasks, click **All Rows**.
   b) Optional: If you want to export only the information for the selected tasks, click **Selected Rows**.

   The CSV file is generated and available for download.

How Do You Export vSphere Tasks with Advanced Export

You can use the advanced export option to export vSphere tasks that meet your specific criteria, such as status, initiator, and time range.

With the advanced export option for tasks, you can export up to 100 tasks that meet your specific criteria. You can use the exported information for monitoring and troubleshooting purposes. For example, you can export the tasks performed by specific users during the time range when some suspicious activity was detected.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to an object in the inventory.
3. Click the **Monitor** tab.
4. Under **Tasks and Events**, select **Tasks**.
5. From the **Export** drop-down menu, select **Advanced export**.

   The **Advanced Export Tasks** dialog opens.

6. Filter the tasks for which you want to export information.
   a) To export only tasks with specific status, select the respective check boxes in the **Status** options list.
   b) To export tasks based on the initiator of the tasks, select one of the options from the **Type** drop-down menu.
      - **User** - Exports tasks initiated by all users, such as `vsphere.local` users, Active Directory domain users, and system users.
      - **System** - Exports tasks initiated only by system users. System users are user accounts associated with an application or service.
   c) From the **Users** radio buttons, select whether to export the tasks initiated by specific users or by all users.

      If you select the **Specify users** radio button, you must create a list of users.

      1. From the **Domain** drop-down menu, select the domain for the users.
      2. Create a list of users who initiated the tasks.
         a. Enter a name in the Search text box.
            The system searches from the users in the selected domain.
         b. Select a user from the list and click **Add**.
   d) From the **Select time set** drop-down menu, select a time set that defines the time range of the exported tasks.

      You can select between start time, completion time, and queued time.
   e) From the **Select time range** options, select the time range for the exported tasks.
      - **Relative** - If you want to export the tasks for a relative time range, specify the relative time range from the **Last** and **Range** drop-down menus.
      - **Data range** - If you want to export the tasks for a specific time period, specify the date and time from the drop-down menus.

      For example, to export the tasks that got queued between March 13, 2022 10:30 AM and June 13, 2022 10:30 PM from the **Select time set** drop-down menu, select **Queued time**. Then select the **Date range** radio button and specify the time period using the respective text fields.
   f) From the **Include columns** radio buttons, select whether you want to export the information from all columns or from specific columns.
7. Click **Export**.

A `.csv` file is generated and available for download.

Scheduling vSphere Tasks

You can schedule vSphere tasks to run once in the future or multiple times, at a recurring interval.

The vSphere tasks that you can schedule are listed in the following table.

Table 94: Scheduled Tasks

Scheduled Task	Description
Add a host	Adds the host to the specified data center or cluster.
Change the power state of a virtual machine	Powers on, powers off, suspends, or resets the state of the virtual machine.
Change cluster power settings	Activate or deactivate DPM for hosts in a cluster.
Change resource settings of a resource pool or virtual machine	Changes the following resource settings: • CPU – Shares, Reservation, Limit. • Memory – Shares, Reservation, Limit.
Check compliance of a profile	Checks that a host's configuration matches the configuration specified in a host profile.
Clone a virtual machine	Makes a clone of the virtual machine and places it on the specified host or cluster.
Create a virtual machine	Creates a new virtual machine on the specified host.
Deploy a virtual machine	Creates a new virtual machine from a template on the specified host or cluster.

Scheduled Task	Description
Migrate a virtual machine	Migrate a virtual machine to the specified host or datastore by using migration or migration with vMotion.
Make a snapshot of a virtual machine	Captures the entire state of the virtual machine at the time the snapshot is taken.
Scan for Updates	Scans templates, virtual machines, and hosts for available updates. This task is available only when vSphere Lifecycle Manager is installed.
Remediate	Installs missing patches from the baselines selected for remediation on the hosts discovered during the scan operation and applies the newly configured settings. This task is available only when vSphere Lifecycle Manager is installed.

You create scheduled tasks by using the **Scheduled Task** wizard. For some scheduled tasks, this wizard opens the wizard used specifically for that task. For example, if you create a scheduled task that migrates a virtual machine, the **Scheduled Task** wizard opens the **Migrate Virtual Machine** wizard, which you use to set up the migration details.

Scheduling one task to run on multiple objects is not possible. For example, you cannot create one scheduled task on a host that powers on all virtual machines on that host. You must create a separate scheduled task for each virtual machine.

After a scheduled task runs, you can reschedule it to run again at another time.

> **Note:**
> At the time when a scheduled task is about to run, the user who created or last edited the task must be authenticated and authorized to run the task. If, for example, the user permissions to create or edit a scheduled task are removed, or the user account is deleted, the scheduled task fails with an error message. An administrator or a user with the required privileges can reschedule the task for subsequent runs.

How Do You Create a Scheduled vSphere Task

You can create scheduled tasks for operations that you want to run automatically once or at a recurring interval.

Required privilege: **Schedule Task** > **Create tasks**

> **Note:**
> At the time when a scheduled task is about to run, the user who created the task must still have the **Schedule Task** > **Create tasks** privileges. If the user is no longer authenticated or authorized to run the task, for example, their permissions to create a scheduled task are removed, or the user account is deleted, the scheduled task fails with an error message. An administrator or another user with the required privileges can reschedule the task for subsequent runs.

If the task to schedule is not available in the vSphere Client, use the vSphere API. See the *vSphere SDK Programming Guide*.

> **CAUTION:**
> Do not schedule multiple tasks simultaneously on the same object. The results are unpredictable.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to the object in the inventory for which you want to schedule a task.
3. Select **Configure** > **Scheduled Tasks** > **New Scheduled Task**.
4. From the **New Scheduled Task** drop-down menu, select the task to schedule.
   The **Schedule New Task** wizard opens for the task with the task name appended next to the name of the wizard. The wizard contains a page where you configure the scheduling options for the task.

5. Enter the required scheduling options for the task.
   a) Enter a name and description of the task.
   b) Choose a frequency for the task.

Option	Description
Once	Runs the scheduled task at the time selected.
After vCenter startup	Runs the task a specified number of minutes after vCenter startup.
Hourly	1. Enter the repeat frequency. 2. Enter the start date and time. 3. Enter the end date and time. For example, to start a task at the half-hour mark of every fifth hour, enter `5` hours and `30` minutes.
Daily	1. Enter the repeat frequency. 2. Enter the start date and time. 3. Enter the end date and time. For example, to run the task at 2:30 pm every four days, enter `4` and `14:30`.
Weekly	1. Enter the repeat frequency. 2. Select the day of the week. 3. Enter the start date and time. 4. Enter the end date and time. For example, to run the task at 6 am every Tuesday and Thursday, enter **1** week, **6 am**, and select **Tuesday** and **Thursday**.
Monthly	1. Enter the repeat frequency. 2. Select the days by using one of the following methods. – Enter a specific day of the month and the number of months. For example, the tenth day every five months. – Select **first**, **second**, **third**, **fourth**, or **last**, and select the day of the week and the number of months. **last** runs the task on the last week in the month that the day occurs. For example, if you select the last Monday of the month and the month ends on a Sunday, the task runs six days before the end of the month.

   c) Optional: Enter one or more email addresses to receive an email notification when the task is complete.

   To schedule some tasks, you must enter additional details for the task. For example, to schedule taking a virtual machine snapshot, the **Schedule New Task (Take Snapshot)** wizard opens. In the **Scheduling Options** page, you set the scheduling options for the task, and in the **Snapshot Settings** page, you enter the properties for the snapshot.

6. Click **Schedule the task**.

How Do You Change or Reschedule a Scheduled vSphere Task

After a scheduled vSphere task is created, you can change the schedule, frequency, and other attributes of the task. You can edit and reschedule tasks before or after they run.

Required privilege:**Schedule Task** > **Modify**

> **Note:**
> At the time when a scheduled task is about to run, the user who last edited the scheduled task must still have the **Schedule Task** > **Modify** privileges. If the user is no longer authenticated or authorized to run the task, for example, their permissions to edit a scheduled task are removed, or the user account is deleted, the scheduled task fails with a warning. An administrator or another user with the required privileges can reschedule the task for subsequent runs.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to the object in the inventory for which you want to edit a scheduled task.

   To view all scheduled tasks for a vCenter instance, navigate to that vCenter instance.
3. Select **Configure**, and select **Scheduled Tasks**.
4. Select a task from the list on the left and click **Edit**.
5. Right-click the task and select **Edit**.
6. Change the task attributes as necessary.
7. Click **Save**.

How Do You Remove a Scheduled vSphere Task

Removing a scheduled task removes all future occurrences of the task. The history associated with all completed occurrences of the task remains in the vCenter database.

Required privilege:**Scheduled task** > **Remove**

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**..
2. Navigate to the object for which you want to remove a scheduled task.

   To view all scheduled tasks for a vCenter instance, navigate to that vCenter instance.
3. Select **Configure**, and select **Scheduled Tasks**.
4. Select a task from the list on the left and click **Remove**.

vSphere Host Configuration

Before you set up your virtual environment and consider how the virtual machines that it will support are going to be used and administered, you should configure ESX hosts in vCenter.

The configuration of ESX hosts involves several tasks. Learn how to configure the boot device of the ESX host, the agent VM settings, advanced host attributes, and time configuration settings.

For information on configuring network and storage access, and security settings, see the configuration information for the specific vSphere component in the *vSphere Security* documentation, the *vSphere Storage* documentation, and the *vSphere Networking* documentation.

How to Configure the Boot Device on Your ESX Host

On servers running ESX, you can select the device that the server boots from.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Under Hardware, select **Overview** and click the **Boot Options** button.
5. Select a boot device from the drop-down menu.
6. Optional: To reboot immediately from the device you selected, select **Apply and Reboot on OK**.
   If you do not select **Apply and Reboot on OK**, the new setting takes effect when the host reboots next.
7. Click **OK**.

How to Configure Your Agent VM Settings in the vSphere Client

In your vSphere environment, you can configure the datastore and network settings for the ESX agent virtual machines that you deploy on a host.

An ESX agent is a virtual machine, or a virtual machine and a vSphere Installation Bundle (VIB), that extend the functions of an ESX host to provide additional services that a vSphere solution requires.

For example, a solution might require a particular network filter or firewall configuration to function. A solution can use an ESX agent to connect to the vSphere Hypervisor and extend the host with functions specific to that solution. For example, the ESX agent can filter network traffic, act as a firewall, or gather other information about the virtual machines on the host.

When you configure the datastore and network settings for ESX agents on a host, all of the ESX agents that you deploy on the host use that datastore and network configuration.

> **Important:**
> ESX agents are deployed only if you configure the network and datastore settings.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Under **Virtual Machines**, select **Agent VM Settings**.
   The current settings for the ESX agents on the host, if any, appear.
5. Click **Edit**.
6. From the **Datastore** drop-down menu, select a datastore in which to deploy the ESX agent virtual machines.
7. From the **Network** drop-down menu, select a network to connect the ESX agents.
8. Click **OK**.

For information about ESX agents and ESX Agent Manager, see *Developing and Deploying vSphere Solutions, vServices, and ESX Agents*.

Set Advanced Host Attributes

You can set advanced attributes for a host.

CAUTION:
Changing advanced options is considered unsupported. Typically, the default settings produce the optimum result. Change the advanced options only when you get specific instructions from VMware technical support or a knowledge base article.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Under **System**, click **Advanced System Settings**.
5. Click the **Edit** button.
6. Find the appropriate item and change the value.
7. Click **OK**.

Advanced Memory Attributes

You can use the advanced memory attributes to customize memory resource usage.

Table 95: Advanced Memory Attributes

Attribute	Description	Default
Mem.ShareForceSalting	Mem.ShareForceSalting 0: Inter-virtual machine Transparent Page Sharing (TPS) behavior is still retained. The value of VMX option `sched.mem.pshare.salt` is ignored even if present. Mem.ShareForceSalting 1: By default the salt value is taken from `sched.mem.pshare.salt`. If not specified, it falls back to old TPS (inter-VM) behavior by considering salt values for the virtual machine as 0. Mem.ShareForceSalting 2: By default the salt value is taken from `sched.mem.pshare.salt` if present, or `vc.uuid`. If it does not exist, then the page sharing algorithm generates random and unique value for salting per virtual machine, which is not configurable by users.	2
Mem.SamplePeriod	Specifies the periodic time interval, measured in seconds of the virtual machine's execution time, over which memory activity is monitored to estimate working set sizes.	60
Mem.BalancePeriod	Specifies the periodic time interval, in seconds, for automatic memory reallocations. Significant changes in the amount of free memory also trigger reallocations.	15
Mem.IdleTax	Specifies the idle memory tax rate, as a percentage. This tax effectively charges virtual machines more for idle memory than for memory they are actively using. A tax rate of 0 percent defines an allocation policy that ignores working sets and allocates memory strictly based on shares. A high tax rate results in an allocation policy that allows idle memory to be reallocated away from virtual machines that are unproductively hoarding it.	75
Mem.ShareScanGHz	Specifies the maximum amount of memory pages to scan (per second) for page sharing opportunities for each GHz of available host CPU resource. For example, defaults to 4 MB/sec per 1 GHz.	4

Attribute	Description	Default
Mem.ShareScanTime	Specifies the time, in minutes, within which an entire virtual machine is scanned for page sharing opportunities. Defaults to 60 minutes.	60
Mem.CtlMaxPercent	Limits the maximum amount of memory reclaimed from any virtual machine using the memory balloon driver (`vmmemctl`), based on a percentage of its configured memory size. Specify 0 to deactivate reclamation for all virtual machines.	65
Mem.AllocGuestLargePage	Activates backing of guest large pages with host large pages. Reduces TLB misses and improves performance in server workloads that use guest large pages. 0=deactivate.	1
Mem.AllocUsePSharePool and Mem.AllocUseGuestPool	Reduces memory fragmentation by improving the probability of backing guest large pages with host large pages. If host memory is fragmented, the availability of host large pages is reduced. 0 = deactivate.	15
Mem.MemZipEnable	Activates memory compression for the host. 0 = deactivate.	1
Mem.MemZipMaxPct	Specifies the maximum size of the compression cache in terms of the maximum percentage of each virtual machine's memory that can be stored as compressed memory.	10
LPage.LPageDefragEnable	Activates large page defragmentation. 0 = deactivate.	1
LPage.LPageDefragRateVM	Maximum number of large page defragmentation attempts per second per virtual machine. Accepted values range from 1 to 1024.	32
LPage.LPageDefragRateTotal	Maximum number of large page defragmentation attempts per second. Accepted values range from 1 to 10240.	256
LPage.LPageAlwaysTryForNPT	Try to allocate large pages for nested page tables (called 'RVI' by AMD or 'EPT' by Intel). If you activate this option, all guest memory is backed with large pages in machines that use nested page tables (for example, AMD Barcelona). If NPT is not available, only some portion of guest memory is backed with large pages. 0= deactivate.	1

Advanced NUMA Attributes

You can use the advanced NUMA attributes to customize NUMA usage.

Table 96: Advanced NUMA Attributes

Attribute	Description	Default
Numa.RebalancePeriod	Controls the frequency of rebalance periods, specified in milliseconds. More frequent rebalancing can increase CPU overheads, particularly on machines with a large number of running virtual machines. More frequent rebalancing can also improve fairness.	2000
Numa.MigImbalanceThreshold	The NUMA rebalancer computes the CPU imbalance between nodes, accounting for the difference between each virtual machine’s CPU time entitlement and its actual consumption. This option controls the minimum load imbalance between nodes needed to trigger a virtual machine migration, in percent.	10
Numa.RebalanceEnable	Activate NUMA rebalancing and scheduling. Set this option to 0 to deactivate all NUMA rebalancing and initial placement of virtual machines, effectively deactivating the NUMA scheduling system.	1
Numa.RebalanceCoresTotal	Specifies the minimum number of total processor cores on the host required to activate the NUMA rebalancer.	4

Attribute	Description	Default
Numa.RebalanceCoresNode	Specifies the minimum number of processor cores per node required to activate the NUMA rebalancer. This option and Numa.RebalanceCoresTotal are useful when deactivating NUMA rebalancing on small NUMA configurations (for example, two-way Opteron hosts), where the small number of total or per-node processors can compromise scheduling fairness when you activate NUMA rebalancing.	2
Numa.AutoMemAffinity	Automatically set memory affinity for virtual machines that have CPU affinity set.	1
Numa.PageMigEnable	Automatically migrate pages between NUMA nodes to improve memory locality. Page migration rates set manually are still in effect.	1

Configuring the Time on Your ESX Host

Time synchronization on ESX hosts is the process of aligning the system clock of each host with a reliable external time source such as Network Time Protocol (NTP) or Precision Time Protocol (PTP). Accurate time on ESX hosts ensures that time-sensitive operations, including SSL certificate validation and SAML token authentication, function correctly across the vSphere network.

Unsynchronized clocks on ESX hosts can cause authentication failures, prevent the vmware-vpxd service from starting, and cause installations, migrations, or upgrades to fail. Time inconsistencies can also cause the first boot of a component in your environment to fail, depending on where in the environment the time is inaccurate and when synchronization occurs. Problems most commonly occur when the target host for a destination is not synchronized with NTP or PTP. Issues can also arise if a virtual machine migrates to a host that is set to a different time due to fully automated DRS.

vSphere supports three mechanisms for clock synchronization. To achieve optimal results, you must select the protocol that is best suited to your specific environment. For more information, see Synchronizing Clocks on the vSphere Network.

To ensure precise timestamping of events and synchronization of the time between an ESX host and the other components in the vSphere network, configure the time settings of the ESX host manually or synchronize the time and date of the host with an NTP or PTP server. For more information, see Configure the Date and Time on Your ESX Host Manually, Use PTP for Time and Date Synchronization of Your ESX Host, Use NTP Servers for Time and Date Synchronization of Your ESX Host.

Exact timestamping shows the precise sequence of events that occur in the vSphere network. Time synchronization between the components of the vSphere network can prevent authentication problems, backup issues, and incorrect logs. To ensure that all components in your environment have the correct time, as a best practice you must use the same time synchronization mechanism across the board.

Configure the Date and Time on Your ESX Host Manually

Learn how to configure the date and time of the ESX host manually in the vSphere Client. You can use the manual time configuration only after you stop synchronizing the time and date with NTP or PTP.

- Verify that the NTP and the PTP configurations are not used on the host.

If you notice some significant clock skew between a host in your environment and the rest of the vSphere components, you might need to first manually set the time and date on the host before synchronizing it to an NTP server or PTP.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Select **System** > **Time Configuration**.
5. Click **Manual Set-Up**.

   The **Manual time configuration** dialog box appears.
6. Enter a date and time and click **OK**.

   Note:

   ESX hosts use UTC (Coordinated Universal Time) and do not support changing time zones. In the vSphere Client, you see your local time as the current host time.

Use NTP Servers for Time and Date Synchronization of Your ESX Host

To avoid time synchronization problems between an ESX host and other components in the vSphere network, you can synchronize the time and date of the host to an NTP server.

- Verify that the PTP service is not running. For information about how to change its status, see *Manage the NTP and PTP Service Status on Your ESX Host.*

The NTP and the PTP services cannot run simultaneously. Deactivate the PTP service and then enable the NTP server synchronization. Additionally, when you enable the NTP service, the manual time configuration becomes inactive.

Note:

You can set a startup policy to control the start and stop of the NTP service. You can also change the NTP service status manually. For more information about services, see *Manage the NTP and PTP Service Status on Your ESX Host*.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Select **System** > **Time Configuration**.
5. Click **Add Service** and select **Network Time Protocol** from the drop-down menu.
6. In the **Network Time Protocol** dialog box, edit the network time protocol settings.
   a) To monitor all events in the vSphere environment, select **Enable monitoring events**.
   b) In the **NTP Servers** text box, enter the IP addresses or host names of the NTP servers that you want to use. A best practice is to use at least three NTP servers for the time synchronization of the host.
   c) Click **OK**.

   The NTP configuration is triggered immediately.

Use PTP for Time and Date Synchronization of Your ESX Host

To ensure that the time of an ESX host is synchronized with the time of other components of the vSphere network, you can synchronize the time and date of the host to PTP.

- Verify that the NTP service is not running. For information about how to change its status, see *Manage the NTP and PTP Service Status on Your ESX Host*.
- For a hardware-based timestamping, verify that you enabled PCI passthrough for a network device on the host. See the *vSphere Networking* documentation.

The Precision Time Protocol allows both software- and hardware-based timestamping on an ESX hosts and provides a highly accurate time synchronization. PTP provisions hardware-based timestamping for the virtual machines and the hosts within a network.

The PTP and the NTP services cannot run simultaneously. Stop the NTP service and then enable the PTP service. Additionally, when you enable the PTP service, the manual time configuration becomes inactive.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. On the **Configure** tab, select **System** > **Time Configuration**.
5. Click **Add Service** and select **Precision Time Protocol** from the drop-down menu.
6. In the **Precision Time Protocol** dialog box, edit the PTP settings.
   a) Select the PTP-enabled network device for time synchronization from the **Network adapter type** drop-down menu.

PCI passthrough	For configuring a hardware timestamping with a PCI passthrough network device, follow these steps: 1. From the **Device name** drop-down menu, select the PCI passthrough network device. For vSphere 7.0 U3, the supported network adapters are IntelX710 and E810. 2. Enter the IP configuration settings for the network device. – To use DHCP to obtain the IP address configuration of the network device, select **Obtain IPv4 settings automatically**. – To set a static IP configuration, select the **Use static IPv4 settings** option and enter the IPv4 configuration settings. – To use the IEEE 802.3 network transport, select **No IP configuration (use IEEE 802.3 transport)**.
VMkernel Adapter	For configuring software timestamping PTP, from the **Device name** drop-down menu, select the VMkernel adapter configured to use PTP.

7. Optional: Create a fallback mechanism in case the PTP synchronization fails.
   a) To monitor the events in vSphere, select the **Enable monitoring events** check box.
   b) To enable a fallback mechanism in case PTP become inaccessible, click **Enable fallback**.

      You can create a fallback mechanism only if you enabled tracking the events in vSphere.
   c) In the **NTP Servers** text box, enter the IP addresses or host names of the NTP servers that you want to use. A best practice is to use at least three NTP servers for the time synchronization of the host.
8. Click **OK**.

   Finishing the configuration immediately triggers the PTP synchronization.
9. Optional: To test whether the service used for time synchronization of the host is running correctly, click **Test Services**.

   The **Time Synchronization Services test** dialog box appears. You can see information about how the activated time configuration works by collecting information from various systems.

Manage the NTP and PTP Service Status on Your ESX Host

In the vSphere Client, you can configure a startup policy for the NTP or PTP service that runs on the ESX host. You can also manually start, stop, or restart the respective service and in this way override the configured startup policy.

1. In the vSphere Client, click the main navigation menu

   (**),** and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Select **System** > **Services**.
5. Change the status of the NTP or PTP service manually.

Option	Description
Option	Action
Change the NTP service status	1. Select **NTP Daemon**. 2. Click **Start**, **Stop**, or **Restart**.
Change the PTP service status	1. Select **PTP Daemon**. 2. Click **Start**, **Stop**, or **Restart**.

6. Optional: Select the NTP or PTP service and click **Edit Startup Policy**.

Option	Description
Start and stop with port usage	Starts or stops the service when a client port is activated or deactivated for access to the security profile of the host.
Start and stop with host	Starts and stops the service when the host powers on and shuts down.
Start and stop manually	You must manually control the status of the service.

Delete a Time Synchronization Service on Your ESX Host

Read how to delete a time synchronization service that is no longer needed by using the vSphere Client.

Verify that the service you want to delete is not used to synchronize the time of the host.

1. In the vSphere Client, click the main navigation menu

   (**),** and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select a host in the inventory and click the **Configure** tab.
4. Select **System** > **Time Configuration**.
5. Select the **Precision Time Protocol** or **Network Time Protocol** check box and click **Delete**.
6. Click **OK**.

Managing Hosts in the vSphere Client

To monitor all hosts in your vSphere environment from one place and to simplify host configuration, add the hosts to a vCenter instance.

The Managing Hosts in the vCenter section of the vCenter and Host Management publication explains the lifecycle and administrative tasks that you can perform on the ESX hosts that a vCenter instance manages.

You can disconnect and reconnect a managed host to temporarily suspend monitoring without removing the host or its virtual machines from the inventory, or remove a host to delete it and its associated virtual machines entirely.

The section also describes how to relocate an ESX host between vCenter instances, reboot or shut down a host, verify SSL certificates for legacy hosts, and enable SGX remote attestation on multi-socket hosts.

For information about the configuration management of ESX hosts, also see the vSphere Networking, vSphere Storage, and vSphere Security documentation.

Disconnecting and Reconnecting an ESX Host

You can disconnect and reconnect a host that a vCenter system manages. Disconnecting a managed host does not remove it from vCenter, but temporarily suspends all monitoring activities that vCenter performs.

The managed host and its associated virtual machines remain in the vCenter inventory. By contrast, removing a managed host from vCenter deletes the managed host and all its associated virtual machines from the vCenter inventory.

If an ESX host becomes disconnected due to a loss of network connectivity to vCenter, the ESX host reconnects automatically to vCenter after network connectivity is restored. If you manually disconnect an ESX host, see How to Reconnect Your Managed ESX Host for more information.

Reconnecting an ESX host automatically or manually does not impact the running virtual machines, unless the host is part of a cluster and there are resource pool privileges configured. For more information about creating and managing resource pools, see the *vSphere Resource Management* documentation.

How to Disconnect Your Managed ESX Host

Use the vSphere Client to disconnect a managed host from vCenter.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a host.
3. Right-click the host and select **Connection** > **Disconnect** from the pop-up menu.
4. In the confirmation dialog box that appears, click **Ok**.
   If the managed host is disconnected, the word “disconnected” is appended to the object name in parentheses, and the object is dimmed. All associated virtual machines are similarly dimmed and labeled.

How to Reconnect Your Managed ESX Host

Use the vSphere Client to reconnect a managed host to a vCenter system.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a host.
3. Right-click the host and select **Connection** > **Connect** from the pop-up menu.
   When the managed host’s connection status to vCenter is changed, the statuses of the virtual machines on that managed host are updated to reflect the change.

How Do You Reconnect ESX Hosts After Changes to the vCenter SSL Certificate

vCenter uses an SSL certificate to encrypt and decrypt host passwords stored in the vCenter database. If the certificate is replaced or changed, vCenter cannot decrypt host passwords, and therefore cannot connect to managed hosts. If vCenter fails to decrypt a host password, the host is disconnected from vCenter.

You must reconnect the host and supply the login credentials, which are encrypted and stored in the database using the new certificate.

Rename an ESX Host FQDN

Starting with vSphere 9.0, you can change the FQDN of an ESX host at any time. You do not need to disconnect the host, or put it in maintenance mode.

- Verify that the ESX host, and the vCenter instance are version 9.0 or later.
- Verify that you have the Host.Config.Settings privilege.
- Verify that the current ESX host FQDN remains valid until the rename task is complete.
- Verify that the new ESX host FQDN is registered with your DNS provider.
- If you manage your certificates, verify that the current and the new host FQDNs are present in the host certificate as Subject Alternative Name (SAN).

For example, you can change the host FQDN if you need to change the network configuration.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Right-click the host, and select **Rename**.
4. On the **Rename Host** page, select the check boxes to verify that you meet the prerequisites.
5. In the New host FQDN field, enter the name of the host, and click **Rename Host**.

You can check the task progress in the **Recent Tasks** pane. When the task is complete, the new host name appears in the inventory list.

How to Relocate Your ESX Host

You can move a host to another location within the vSphere inventory by dragging the host to the new location. The new location can be a folder, a cluster, or you can place the host as a standalone object in the data center.

Power off all virtual machines that are running on the host, or migrate the virtual machines to a new host by using vMotion.

When a host is moved from a cluster to another destination in the vSphere inventory, the resources which the host provides are deducted from the total cluster resources. You can either keep the virtual machines in the same cluster and migrate them to other hosts, or keep them on the host and remove them from the cluster. For information about removing a host from a cluster, see Removing a Host from a Cluster in the *vSphere Resource Management* documentation.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a host.
3. If the host is part of a cluster, put it in maintenance mode.
   a) Right-click the host and select **Maintenance Mode** > **Enter Maintenance Mode**.
   b) Optional: If the host is part of a DRS cluster, evacuate the powered off or suspended virtual machines to other hosts within the cluster by selecting the check box **Move powered-off and suspended virtual machines to other hosts in the cluster**.
   c) In the confirmation dialog box, click **OK**.
      The host enters maintenance mode.
4. Select the host in the vSphere inventory panel and drag it to the new location within the inventory.
5. Right-click the host and select **Maintenance Mode** > **Exit Maintenance Mode**.
6. Optional: Power on the virtual machines that you powered off before you put the host in maintenance mode.

How to Remove an ESX Host from Your vCenter Instance

Remove a managed host from vCenter to stop vCenter from monitoring and managing that host.

- Make sure that NFS mounts are active. If NFS mounts are unresponsive, the operation fails.
- If the host you want to remove from the cluster is connected to a distributed switch, remove the host from the switch. For more information, see Remove Hosts from a vSphere Distributed Switch in the *vSphere Networking* documentation.

If possible, remove managed hosts while they are connected. Removing a disconnected host does not remove the vCenter agent from the managed host.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a host.
3. Optional: If the host is part of a cluster, put it in maintenance mode.
   a) Right-click the host and select **Maintenance Mode** > **Enter Maintenance Mode** from the pop-up menu.
      If not all virtual machines on the host are powered off, the host does not enter maintenance mode.

      If the host is part of a DRS cluster, when the host enters maintenance mode, DRS attempts to evacuate powered on virtual machines from the host by using vMotion.
   b) In the confirmation dialog box, click **Ok**.
      If the host is part of a DRS cluster, you can evacuate powered off or suspended virtual machines to other hosts within the cluster. Select the check box **Move powered-off and suspended virtual machines to other hosts in the cluster**.

      The host icon changes and the term Maintenance Mode is added to the name in parentheses.
4. Right-click the host you want to remove in the inventory pane, and select **Remove from Inventory** from the pop-up menu.
5. In the confirmation dialog box, click **Yes** to remove the host.
   vCenter removes the host and the associated virtual machines from the vCenter instance. vCenter then returns the status of all associated processor and migration licenses to available.

How to Reboot or Shut Down Your ESX Host

You can power off or restart any ESX host using the vSphere Client. Powering off a managed host disconnects it from vCenter, but does not remove it from the inventory.

- Power off all virtual machines that are running on the host, or migrate the virtual machines to a new host by using vMotion.
- Put the host in maintenance mode.

To shut down or reboot an ESX host, you can use the ESXCLI command set. See the *Stopping and Rebooting Hosts with ESXCLI* topic in the *ESXCLI Concepts and Examples* documentation at https://developer.broadcom.com/tools/esxcli/latest.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select a host.
3. From the **Actions** menu, select **Power**.
4. Select the operation.
   - To power off and restart the ESX host, click **Reboot**.
   - To power off the ESX host, click **Shut Down**.
5. Provide a reason for the operation and click **OK**.
   This information is added to the log file.

How Do You Verify SSL Certificates for Legacy ESX Hosts

You can configure vCenter to check the SSL certificates of hosts to which it connects. If you configure this setting, vCenter and the vSphere Client check for valid SSL certificates before connecting to a host for operations such as adding a host or making a remote console connection to a virtual machine.

SSL certificates are signed by VMware Certificate Authority by default. You can instead use certificates from a third-party CA. Thumbprint mode is supported only for legacy hosts.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and navigate to the vCenter instance.
3. Select the **Configure** tab.
4. Under **Settings**, select **General**.
5. Click **Edit**.
6. Select **SSL settings**.
7. Determine the host thumbprint for each legacy host that requires validation.
   a) Log in to the direct console.
   b) Select **View Support Information** on the **System Customization** menu.
      The thumbprint is displayed in the column on the right.
8. Compare the thumbprint you obtained from the host with the thumbprint listed in the vCenter SSL settings dialog box.
9. If the thumbprints match, select the check box for the host.
   Hosts that are not selected will be disconnected after you click **Save**.

10. Click **Save**.

Enabling SGX Remote Attestation on Multi-Socket ESX Hosts

Intel® Software Guard Extensions (Intel® SGX) is a hardware-based security solution that allows you to isolate specific application code and data in private memory regions, called enclaves. Use the vSphere Client to register your SGX-capable hosts with multiple CPU sockets with the Intel Registration Server and use remote attestation for applications running inside your vSGX-enabled virtual machines.

You can enable the Virtual Intel® Software Guard Extensions (vSGX) on virtual machines and provide additional security to your workloads. See Securing Virtual Machines with Intel Software Guard Extensions in the *vSphere Security* documentation. Furthermore, you can use remote attestation for the vSGX-enabled virtual machines. Intel SGX remote attestation is a security mechanism that allows you to establish an authenticated and secure communication channel with a trusted remote entity. To use remote attestation for virtual machines using SGX enclaves, hosts with a single CPU socket do not require Intel registration.

Starting with vSphere 8.0, to enable the remote attestation on a virtual machine running on a host with multiple CPU sockets, you must first register the host with the Intel Registration Server. If an SGX-capable host with multiple CPU sockets is not registered with the Intel Registration Server, you can only power on vSGX-enabled virtual machines that do not require remote attestation.

When you add a host with SGX-capable CPUs, vCenter accesses the Unified Extensible Firmware Interface (UEFI) variables provided by the BIOS and reads the current registration status of the host. To enable vCenter to retrieve information about the SGX status of a host, you must set the firmware boot mode of the host to UEFI mode. See How to View the SGX Registration Status of Your ESX Host.

You can change the current SGX registration status of the host by using the registration options in vSphere Client or by rebooting the ESX host after microcode updates and adding or replacing a CPU package. After each host reboot, you can view the updated registration status of the host by using the vSphere Client.

SGX Registration Statuses of a Host

You can view the current status of an SGX-capable hosts using the vSphere Client and perform the necessary steps to register the hosts with the Intel Registration Server.

SGX Registration Status	Description
Not Applicable	SGX-capable hosts with a single CPU socket do not require registration with the Intel Registration Server to enable remote attestation.
Incomplete	The registration status is incomplete in one of the following use cases: • When you add a new host to a vCenter instance and the host is not registered yet. • After a host firmware update that performs an Intel SGX Trusted Computing Base (TCB) recovery. • For hosts which have multiple CPU packages, when a CPU package is added or replaced, you must manually perform an SGX factory reset on BIOS setup. Then the host must be registered as if a newly added host. • When you manually perform an SGX factory reset on the BIOS setup, you must register the host again.
Complete	The host is successfully registered with the Intel Registration Server.

How to View the SGX Registration Status of Your ESX Host

You can view the current SGX registration status of an ESX host by using the vSphere Client.

- Make sure that the host is installed on an Intel CPU with SGX capabilities and SGX is enabled.
- Set the firmware boot mode of the host to UEFI.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and select an SGX-capable host from the inventory.
3. On the **Summary** tab, navigate to the **Hardware** card.
4. Expand the **SGX** node to view the value of the Registration Status property.

   For more information about the different registration statuses, see Enabling SGX Remote Attestation on Multi-Socket ESX Hosts.

To use the remote attestation feature for vSGX-enabled virtual machines, you must register the host with the Intel Registration Server if the host registration is incomplete and the host has multiple CPU sockets. See How to Register Your Multi-Socket ESX Host with the Intel SGX Registration Server.

How to Register Your Multi-Socket ESX Host with the Intel SGX Registration Server

To use the SGX remote attestation feature for a multi-socket host, register the ESX host with the Intel Registration Server using the vSphere Client.

- Make sure that the host is installed on an Intel CPU with SGX capabilities and SGX is enabled.
- Set the firmware boot mode of the host to UEFI.

The Intel SGX attestation mechanism ensures the trust between the vSGX enclave and an external entity. To use this feature on a multi-socket host with enabled SGX capabilities, you must register the host with the Intel SGX Registration Server.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Select an SGX-capable host from the inventory, and click the **Configure** tab.
4. Under Hardware, select **SGX** and click **Register**.

Upon successful completion of the registration operation, the registration status of the host changes to Completed.

Enable remote attestation for a vSGX-enabled virtual machine. See Securing Virtual Machines with Intel Software Guard Extensions in the *vSphere Security* documentation.

vSphere Virtual Machines Migration

With vSphere vMotion, you can move virtual machines from one compute resource or storage location to another by using cold or hot migration.

For example, with vSphere vMotion you can move powered on virtual machines away from a host to perform maintenance, to balance loads, to collocate virtual machines that communicate with each other, to move virtual machines apart to minimize fault domain, to migrate to new server hardware, and so on.

Moving a virtual machine from one inventory folder to another folder or resource pool in the same data center is not a form of migration. Unlike migration, cloning a virtual machine or copying its virtual disks and configuration file in the same vCenter system are procedures that create a new virtual machine. Cloning and copying a virtual machine in the same vCenter system are also not forms of migration.

By using migration, you can change the compute resource that the virtual machine runs on. For example, you can move a virtual machine from one host to another host or cluster.

To migrate virtual machines with disks larger than 2 TB, the source and destination ESX hosts must be version 6.0 and later.

Depending on the power state of the virtual machine that you migrate, migration can be cold or hot.

Cold Migration	Moving a powered off or suspended virtual machine to a new host. Optionally, you can relocate configuration and disk files for powered off or suspended virtual machines to new storage locations. You can also use cold migration to move virtual machines from one virtual switch to another, and from one data center to another. You can perform cold migration manually or you can schedule a task.
Hot Migration	Moving a powered on virtual machine to a new host. Optionally, you can also move the virtual machine disks or folder to a different datastore. Hot migration is also called live migration or vSphere vMotion. With vSphere vMotion, you migrate the virtual machine without any interruption in its availability.

Depending on the virtual machine resource type, you can perform three types of migration.

Change compute resource only	Moving a virtual machine, but not its storage, to another compute resource, such as a host, cluster, resource pool, or vApp. You can move the virtual machine to another compute resource by using cold or hot migration. If you change the compute resource of a powered on virtual machine, you use vSphere vMotion.
Change storage only	Moving a virtual machine and its storage, including virtual disks, configuration files, or a combination of these, to a new datastore on the same host. You can change the datastore of a virtual machine by using cold or hot migration. If you move a powered on virtual machine and its storage to a new datastore, you use Storage vMotion.
Change both compute resource and storage	Moving a virtual machine to another host and at the same time moving its disk or virtual machine folder to another datastore. You can change the host and datastore simultaneously by using cold or hot migration.

You can move virtual machines between vSphere sites by using migration between the following types of objects.

Migrate to another virtual switch	Moving the network of a virtual machine to a virtual switch of a different type. You can migrate virtual machines without reconfiguring the physical and virtual network. By using cold or hot migration, you can move the virtual machine from a standard to a standard or distributed switch, and from a distributed switch to another distributed switch. When you move a virtual machine network between distributed switches, the network configuration and policies that are associated with the network adapters of the virtual machine are transferred to the target switch.
Migrate to another data center	Moving a virtual machine to a different data center. You can change the data center of a virtual machine by using cold or hot migration. For networking in the target data center, you can select a dedicated port group on a distributed switch.
Migrate to another vCenter system	Moving a virtual machine to a different vCenter instance. You can move a virtual machine to a vCenter instance that is connected to the source vCenter instance through vCenter linking. You can also move virtual machines between vCenter instances that are located across a long distance from each other. Starting with vSphere 7.0 Update 1c, you can migrate workloads across vCenter systems by using Advanced Cross vCenter vMotion. You can initiate migration of workloads both from on-premise environments and from cloud environments. Advanced Cross vCenter vMotion does not depend on vCenter linking and you can migrate virtual machines across vCenter systems in different vCenter Single Sign-On domains. Starting with vSphere 7.0 Update 3, you can use the Advanced Cross vCenter vMotion feature to clone virtual machines across vCenter systems.

Starting with vSphere 9.0, you can migrate the virtual machines to the vSphere Supervisor. For more information, see Import VMs into vSphere Supervisor.

Cold Migration in Your vSphere Environment

Cold migration is the migration of powered off or suspended virtual machines between hosts across clusters, data centers, and vCenter instances. By using cold migration, you can also move associated disks from one datastore to another.

You can use cold migration to have the target host checked against fewer requirements than when you use vMotion. For example, if you use cold migration when a virtual machine contains a complex application setup, the compatibility checks during vMotion might prevent the virtual machine from moving to another host.

You must power off or suspend the virtual machines before you begin the cold migration process. Migrating a suspended virtual machine is considered a cold migration because although the virtual machine is powered on, it is not running.

Migrating a virtual machine across its network subnets might result in the loss of network connectivity.

CPU Compatibility Check During Cold Migration

If you attempt to migrate a powered off virtual machine that is configured with a 64-bit operating system to a host that does not support 64-bit operating systems, vCenter generates a warning. Otherwise, CPU compatibility checks do not apply when you migrate powered off virtual machines with cold migration.

When you migrate a suspended virtual machine, the new host for the virtual machine must meet CPU compatibility requirements. This requirement allows the virtual machine to resume execution on the new host.

Operations During Cold Migration

A cold migration consists of the following operations:

1. If you select the option to move to a different datastore, the configuration files, including the NVRAM file (BIOS settings), log files, and the suspend file, are moved from the source host to the destination host's associated storage area. You can choose to move the virtual machine's disks as well.
2. The virtual machine is registered with the new host.
3. After the migration is completed, the old version of the virtual machine is deleted from the source host and datastore if you selected the option to move to a different datastore.

Network Traffic for Cold Migration

By default, data for VM cold migration, cloning, and snapshots is transferred through the management network. This traffic is called provisioning traffic. It is not encrypted but uses run-length encoding of data.

On a host, you can dedicate a separate VMkernel network adapter to the provisioning traffic, for example, to isolate this traffic on another VLAN. On a host, you can assign no more than one VMkernel adapter for provisioning traffic. For information about enabling provisioning traffic on a separate VMkernel adapter, see Setting Up VMkernel Networkingin the *vSphere Networking* documentation.

If you plan to transfer high volumes of virtual machine data that the management network cannot accommodate, redirect the cold migration traffic on a host to the TCP/IP stack that is dedicated to cold migration and cloning of powered off virtual machines. You can also redirect if you want to isolate cold migration traffic in a subnet different from the management network, for example, for migration over a long distance. See How to Place Traffic for Cold Migration, Cloning, and Snapshots on the Provisioning TCP/IP Stack.

How to Migrate a Powered Off or Suspended Virtual Machine

You can use cold migration to move a virtual machine and its associated disks from one datastore to another in your vSphere environment. The virtual machines are not required to be on shared storage.

- Make sure that you are familiar with the requirements for cold migration. See Cold Migration in Your vSphere Environment.
- Required privilege:**Resource** > **Migrate powered off virtual machine**

1. In the vSphere Client, click the main navigation menu

   (**),** and select **Inventory**.
2. Navigate to **Hosts and Clusters**, and navigate to the virtual machine.
3. Power off or suspend the virtual machine.
4. Right-click the virtual machine and select **Migrate**.
   a) To locate a virtual machine, select a data center, folder, cluster, resource pool, host, or vApp.
   b) Click the **Virtual Machines** tab.
5. Select the migration type and click **Next**.

Option	Description
Change compute resource only	Move the virtual machine to another host.
Change storage only	Move the virtual machine's configuration file and virtual disks.
Change both compute resource and storage	Move the virtual machine to another host and move its configuration file and virtual disks.

6. If you change the compute resource of the virtual machine, select the destination compute resource for this virtual machine migration and click **Next**.

   Any compatibility problem appears in the Compatibility panel. Fix the problem, or select another host or cluster.

   Possible targets include hosts and DRS clusters with any level of automation. If a cluster has no DRS enabled, select a specific host in the cluster rather than selecting the cluster.

   Important:

   If the virtual machine that you migrate has an NVDIMM device and uses PMem storage, the destination host or cluster must have available PMem resources. Otherwise, the compatibility check fails and you cannot proceed further with the migration.

   If the virtual machine that you migrate does not have an NVDIMM device but it uses PMem storage, you must select a host or cluster with available PMem resources, so that all PMem hard disks remain stored on a PMem datastore. Otherwise, all the hard disks use the storage policy and datastore selected for the configuration files of the virtual machine.

   Important:

   Migrating a virtual machine that has an NVDIMM device or a vPMem disk to a host that does not have the proper license fails and leaves the virtual machine in an unmanageable state for 90 seconds. You can afterwards retry the migration and select a destination host that is licensed to use PMem devices.
7. If you change the storage of the virtual machine, enter the required details in the **Select Storage** page.
   a) Select the storage type for the virtual machine configuration files and all the hard disks.
      - If you select the **Standard** mode, all virtual disks are stored on a standard datastore.
      - If you select the **PMem** mode, all virtual disks are stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine.
      - If you select the **Hybrid** mode, all PMem virtual disks remain stored on a PMem datastore. Non-PMem disks are affected by your choice of a VM storage policy and datastore or datastore cluster.

   Selecting the type of storage is possible only if PMem or Hybrid storage types are available in the data center.

b) Select the format for the virtual machine's disks.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation. Instead, it is zeroed out on demand on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it.

c) Select a virtual machine storage policy from the **VM Storage Policy** drop-down menu.

Storage policies specify storage requirements for applications that run on the virtual machine. You can also select the default policy for vSAN or Virtual Volumes datastores.

Important:

If the virtual machine hard disks use different storage policies, the new policy that you select only applies to non-PMem hard disks. PMem hard disks are migrated to the host-local PMem datastore of the destination host.

d) Select the datastore location where you want to store the virtual machine files.

Option	Action
Store all virtual machine files in the same location on a datastore.	Select a datastore from the list and click **Next**.
Store all virtual machine files in the same Storage DRS cluster.	1. Select a Storage DRS cluster. 2. To deactivate Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 3. Click **Next**.
Store virtual machine configuration files and disks in separate locations.	1. Click **Configure per disk**. **Note:** You can use the **Configure per disk** option to downgrade from or upgrade to PMem storage. 2. For the virtual machine configuration file and for each virtual disk, select **Browse**, and select a datastore or Storage DRS cluster. **Note:** Configuration files cannot be stored on a PMem datastore. 3. If you selected a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 4. Click **Next**.

8. If you change the compute resource of the virtual machine, select destination networks for the virtual machine migration.

   You can migrate a virtual machine network to another distributed switch in the same or to another data center or vCenter.

Option	Action
Select a destination network for all VM network adapters connected to a valid source network.	1. Click the arrow in the **Destination Network** column and select **Browse**. 2. Select a destination network and click **OK**. 3. Click **Next**.
Select a new destination network for each VM network adapter connected to a valid source network.	1. Click **Advanced**. 2. Click the arrow in the **Destination Network** column and select **Browse**. 3. Select a destination network and click **OK**. 4. Click **Next**.

9. On the **Ready to complete** page, review the details and click **Finish**.

vCenter moves the virtual machine to the new host or storage location.

Event messages appear in the **Events** tab. The data displayed on the **Summary** tab shows the status and state throughout the migration. If errors occur during migration, the virtual machines revert to their original states and locations.

Migrating Virtual Machines with vSphere vMotion

If you must take a host offline for maintenance, you can move the virtual machine to another host. Migration with vSphere vMotion allows virtual machine processes to continue working throughout a migration.

When you migrate a virtual machine with vSphere vMotion, the new host for the virtual machine must meet compatibility requirements so that the migration can proceed.

For more information about the vMotion migration types, see vSphere Virtual Machines Migration.

Transferred State Information

The state information includes the current memory content and all the information that defines and identifies the virtual machine. The memory content includes transaction data and the bits of the operating system and applications that are in the memory. The defining and identification information stored in the state includes all the data that maps to the virtual machine hardware elements. This information includes BIOS, devices, CPU, MAC addresses for the Ethernet cards, chipset states, registers, and so forth.

Stages in vSphere vMotion

Migration with vSphere vMotion occurs in three stages:

1. When you initiate a migration with vSphere vMotion, vCenter verifies that the existing virtual machine is in a stable state within its current host.
2. The virtual machine state information (memory, registers, and network connections) is copied to the target host.
3. The virtual machine resumes its activities on the new host.

If errors occur during migration, the virtual machine reverts to its original state and location.

Host Configuration for vSphere vMotion

Before using vSphere vMotion, you must configure your hosts correctly.

- Each host must be licensed.

- Each host must meet shared storage requirements for vSphere vMotion. See vSphere vMotion Shared Storage Requirements.
- Each host must meet the networking requirements for vSphere vMotion. See What Are the vSphere vMotion Networking Requirements.

vSphere vMotion Across Long Distances

You can perform reliable migrations between hosts and sites that are separated by high network round-trip latency times. No user configuration is necessary.

For long-distance migration, verify the network latency between the hosts and your license.

- The round-trip time between the hosts must be up to 150 milliseconds.
- You must place the traffic related to transfer of virtual machine files to the destination host on the provisioning TCP/IP stack. See How to Place Traffic for Cold Migration, Cloning, and Snapshots on the Provisioning TCP/IP Stack.

vSphere vMotion Shared Storage Requirements

Configure your hosts for vSphere vMotion with shared storage to ensure that virtual machines are accessible to both source and target hosts.

During a migration with vMotion, the migrating virtual machine must be on storage accessible to both the source and target hosts. Ensure that the hosts configured for vMotion use shared storage. Shared storage can be on a Fibre Channel storage area network (SAN), or can be implemented using iSCSI and NAS.

If you use vMotion to migrate virtual machines with raw device mapping (RDM) files, make sure to maintain consistent LUN IDs for RDMs across all participating hosts.

For information about SANs and RDMs, see the *vSphere Storage* documentation.

What Are the vSphere vMotion Networking Requirements

Migration with vMotion requires correctly configured network interfaces on source and target hosts.

Configure each host with at least one network interface for vMotion traffic. To ensure secure data transfer, the vMotion network must be a secure network, accessible only to trusted parties. Additional bandwidth significantly improves vMotion performance. When you migrate a virtual machine with vMotion without using shared storage, the contents of the virtual disk is transferred over the network as well.

Network traffic with vMotion can be encrypted. Encrypted vMotion depends on host configuration, or on compatibility between the source and destination hosts.

Requirements for Concurrent vMotion Migrations

You must ensure that the vMotion network has at least 250 Mbps of dedicated bandwidth per concurrent vMotion session. Greater bandwidth lets migrations complete more quickly. Gains in throughput resulting from WAN optimization techniques do not count towards the 250-Mbps limit.

To determine the maximum number of concurrent vMotion operations possible, see vCenter Limits on Simultaneous Migrations. These limits vary with a host link speed to the vMotion network.

Round-Trip Time for Long-Distance vMotion Migration

You can perform reliable migrations between hosts that are separated by high network round-trip latency times. The maximum supported network round-trip time for vMotion migrations is 150 milliseconds. This round-trip time lets you migrate virtual machines to another geographical location at a longer distance.

Multiple-NIC vMotion

You can configure multiple NICs for vMotion by adding two or more NICs to the required standard or distributed switch. For more information, see Broadcom Knowledge Base article 318899.

Network Configuration

Configure the virtual networks on vMotion enabled hosts as follows:

- On each host, configure a VMkernel port group for vMotion.
  To have the vMotion traffic routed across IP subnets, enable the vMotion TCP/IP stack on the host. See How to Place vSphere vMotion Traffic on the vMotion TCP/IP Stack of Your ESX Host.
- If you are using standard switches for networking, ensure that the network labels used for the virtual machine port groups are consistent across hosts. During a migration with vMotion, vCenter assigns virtual machines to port groups based on matching network labels.

  Note:

  By default, you cannot use vMotion to migrate a virtual machine that is attached to a standard switch with no physical uplinks configured, even if the destination host also has a no-uplink standard switch with the same label.

  To override the default behavior, set the `config.migrate.test.CompatibleNetworks.VMOnVirtualIntranet` advanced settings of vCenter to `false`. The change takes effect immediately. For information about configuring advanced settings of vCenter, see *vCenter Configuration*.
- If you use Intrusion Detection Systems (IDS) and firewalls to protect your environment, make sure that you configure them to allow connections to the ports used for vMotion on the ESX hosts. For the list of currently supported ports for vMotion, see the VMware Ports and Protocols Tool™ at https://ports.esp.vmware.com/home/vSphere.

For information about configuring the vMotion network resources, see Networking Best Practices for vSphere vMotion .

For more information about vMotion networking requirements, see Broadcom Knowledge Base article 342209.

Networking Best Practices for vSphere vMotion

Consider certain best practices for configuring the network resources for vMotion on an ESX host.

- Provide the required bandwidth in one of the following ways:

Physical Adapter Configuration	Best Practices
Dedicate at least one adapter for vMotion.	Use at least one 10 GbE adapter for workloads that have a small number of memory operations, or if you migrate workloads that have many memory operations. If only two Ethernet adapters are available, for best availability, combine both adapters into a team, and use VLANs to divide traffic into networks: one or more for virtual machine traffic and one for vMotion.
Direct vMotion traffic to one or more physical NICs that have high-bandwidth capacity and are shared between other types of traffic as well	• To distribute and allocate more bandwidth to vMotion traffic across several physical NICs, use multiple-NIC vMotion. • On a vSphere Distributed Switch 5.1 and later, use vSphere Network I/O Control shares to guarantee bandwidth to outgoing vMotion traffic. Defining shares also prevents from contention as a result from excessive vMotion or other traffic. • To avoid saturation of the physical NIC link as a result of intense incoming vMotion traffic, use traffic shaping in egress direction on the vMotion port group on the destination host. By using traffic shaping you can limit the average and peak bandwidth available to vMotion traffic, and reserve resources for other traffic types. • In vSphere 7.0 Update 1 or earlier, vMotion saturates 1 GbE and 10 GbE physical NICs with a single vMotion VMkernel NIC. Starting with vSphere 7.0 Update 2, vMotion saturates high speed links such as 25 GbE, 40 GbE and 100 GbE with a single vMotion VMkernel NIC. If you do not have dedicated uplinks for vMotion, you can use Network I/O Control to limit vMotion bandwidth use.

- Use jumbo frames for best vMotion performance.
  Ensure that jumbo frames are enabled on all network devices that are on the vMotion path including physical NICs, physical switches, and virtual switches.
- Place vMotion traffic on the vMotion TCP/IP stack for migration across IP subnets that have a dedicated default gateway that is different from the gateway on the management network. See How to Place vSphere vMotion Traffic on the vMotion TCP/IP Stack of Your ESX Host.

Virtual Machine Conditions and Limitations for vSphere vMotion

To migrate virtual machines with vMotion, the virtual machine must meet certain network, disk, CPU, USB, and other device requirements.

The following virtual machine conditions and limitations apply when you use vSphere vMotion:

- The source and destination management network IP address families must match. You cannot migrate a virtual machine from a host that is registered to vCenter with an IPv4 address to a host that is registered with an IPv6 address.
- Using 1 GbE network adapters for the vSphere vMotion network might result in migration failure, if you migrate virtual machines with large vGPU profiles. Use 10 GbE network adapters for the vSphere vMotion network.
- If virtual CPU performance counters are enabled, you can migrate virtual machines only to hosts that have compatible CPU performance counters.
- You can migrate virtual machines that have 3D graphics enabled. If the 3D Renderer is set to Automatic, virtual machines use the graphics renderer that is present on the destination host. The renderer can be the host CPU or a GPU graphics card. To migrate virtual machines with the 3D Renderer set to Hardware, the destination host must have a GPU graphics card.
- Starting with vSphere 6.7 Update 1 and later, vSphere vMotion supports virtual machines with vGPU.
- vSphere DRS supports initial placement of vGPU virtual machines running vSphere 6.7 Update 1 or later without load balancing support.
- You can migrate virtual machines with USB devices that are connected to a physical USB device on the host. You must enable the devices for vSphere vMotion.
- You cannot use migration with vSphere vMotion to migrate a virtual machine that uses a virtual device backed by a device that is not accessible on the destination host. For example, you cannot migrate a virtual machine with a CD drive backed by the physical CD drive on the source host. Disconnect these devices before you migrate the virtual machine.
- You cannot use migration with vSphere vMotion to migrate a virtual machine that uses a virtual device backed by a device on the client computer. Disconnect these devices before you migrate the virtual machine.

Using vSphere vMotion to Migrate vGPU Virtual Machines

You can use vMotion to perform a live migration of NVIDIA vGPU-powered virtual machines without causing data loss. Starting with vSphere 8.0 Update 3, vMotion for vGPU virtual machines is activate by default.

Stun time is the time when the virtual machine is inaccessible to users during vMotion. During the stun time, you are unable to access the VM, desktop, or application. After the migration is complete, access to the virtual machine resumes and all applications continue from their previous state. For information on frame buffer size in vGPU profiles, refer to the NVIDIA Virtual GPU documentation.

The expected VM stun times and the estimated worst-case stun times are listed in the following tables. The expected stun times were tested over a 10Gb network with NVIDIA Tesla V100 PCIe 32 GB GPUs:

Table 97: Expected Stun Times for vMotion of vGPU VMs

Used vGPU Frame Buffer (GB)	VM Stun Time (sec)
1	2
2	4

Used vGPU Frame Buffer (GB)	VM Stun Time (sec)
4	6
8	12
16	22
32	39

Table 98: Estimated Worst-Case Stun Times (sec)

vGPU Memory	VM Memory 4 GB	VM Memory 8 GB	VM Memory 16 GB	VM Memory 32 GB
1 GB	5	6	8	12
2 GB	7	9	11	15
4 GB	13	14	16	21
8 GB	24	25	28	32
16 GB	47	48	50	54
32 GB	91	92	95	99

Note:
When you consider the expected and estimated worst-case stun times, keep in mind the following points:

- The configured vGPU profile represents an upper bound to the used vGPU frame buffer. In many use cases, the amount of vGPU frame buffer memory used by the VM at any given time is below the assigned vGPU memory in the profile.
- Both expected and estimated worst-case stun times are only valid when migrating a single virtual machine. If you are concurrently migrating multiple virtual machines, that is, for a vSphere manual remediation process, the stun times will have adverse effects.
- The above estimates assume sufficient CPU, memory, PCIe, and network capacity to achieve 10 Gbps migration throughput.

DRS supports initial placement of vGPU VMs running vSphere 6.7 Update 1 and later without load balancing support.

VMware vSphere vMotion is supported only with and between compatible NVIDIA GPU device models and NVIDIA GRID host driver versions as defined and supported by NVIDIA. For compatibility information, refer to the NVIDIA Virtual GPU User Guide.

To check compatibility between NVIDIA vGPU host drivers, vSphere, and Horizon, refer to the Broadcom Compatibility Guide.

Related tasks

- How to Migrate Your Virtual Machine to a New Compute Resource
- How to Migrate Your Virtual Machine to a New Compute Resource and Storage
- How to Migrate Your Virtual Machine with Storage vMotion

How to Set a Stun Time Limit for Your vGPU Virtual Machines

Learn how to set a stun time limit per virtual machine when you migrate NVIDIA vGPU-powered virtual machines with vSphere vMotion. Setting a stun time limit might prevent vCenter from powering on the virtual machine or migrating it to a host and network whose estimated maximum stun time exceeds that limit.

- Verify that the vCenter instance is version 8.0 Update 2 or later.

- Verify that the source and destination ESX hosts are version 8.0 Update 2 or later.
- Verify that vMotion is enabled for vGPU virtual machines. The `vgpu.hotmigrate.enabled` advanced setting must be set to `true`. For more information about how to configure the vCenter advanced settings, see *Configure Advanced Settings* in the *vCenter Configuration* documentation.

Virtual machine stun time is the time when the virtual machine is inaccessible to users during vMotion. Starting with vSphere 8.0 Update 2, you can set a vMotion stun time limit for vGPU virtual machines.

The limit that you set must be higher than the estimated maximum stun time for the current device configuration. In case of multiple PCI devices, the stun time limit that you set must be higher than the sum of the contributions of all PCI devices. Setting a stun time limit that is lower than the estimated maximum stun time might prevent the virtual machine from powering on.

Note:
Maximum stun time estimates are only available for C-Series and Q-Series vGPU types.

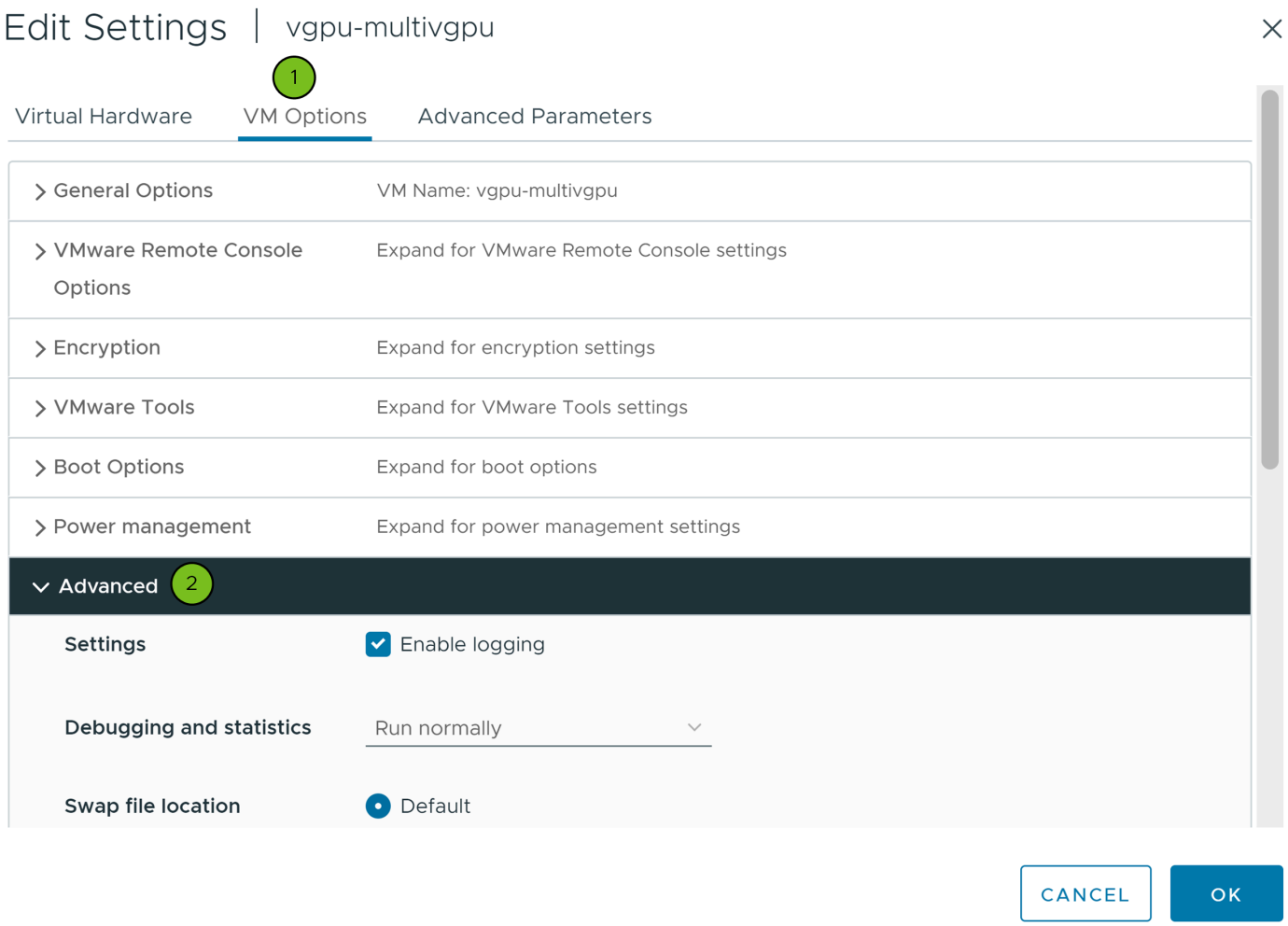
The maximum stun time is calculated on the basis of the bandwidth of the host on which the VM runs currently. The calculations might change at the time of migration if the destination host has lower bandwidth. For example, if a virtual machine is running on a host with a 25 Gbps vMotion NIC, but the destination host has a 10 Gbps vMotion NIC. At the time of migration the maximum stun time calculation is based on the 10 Gbps vMotion NIC.

Note:

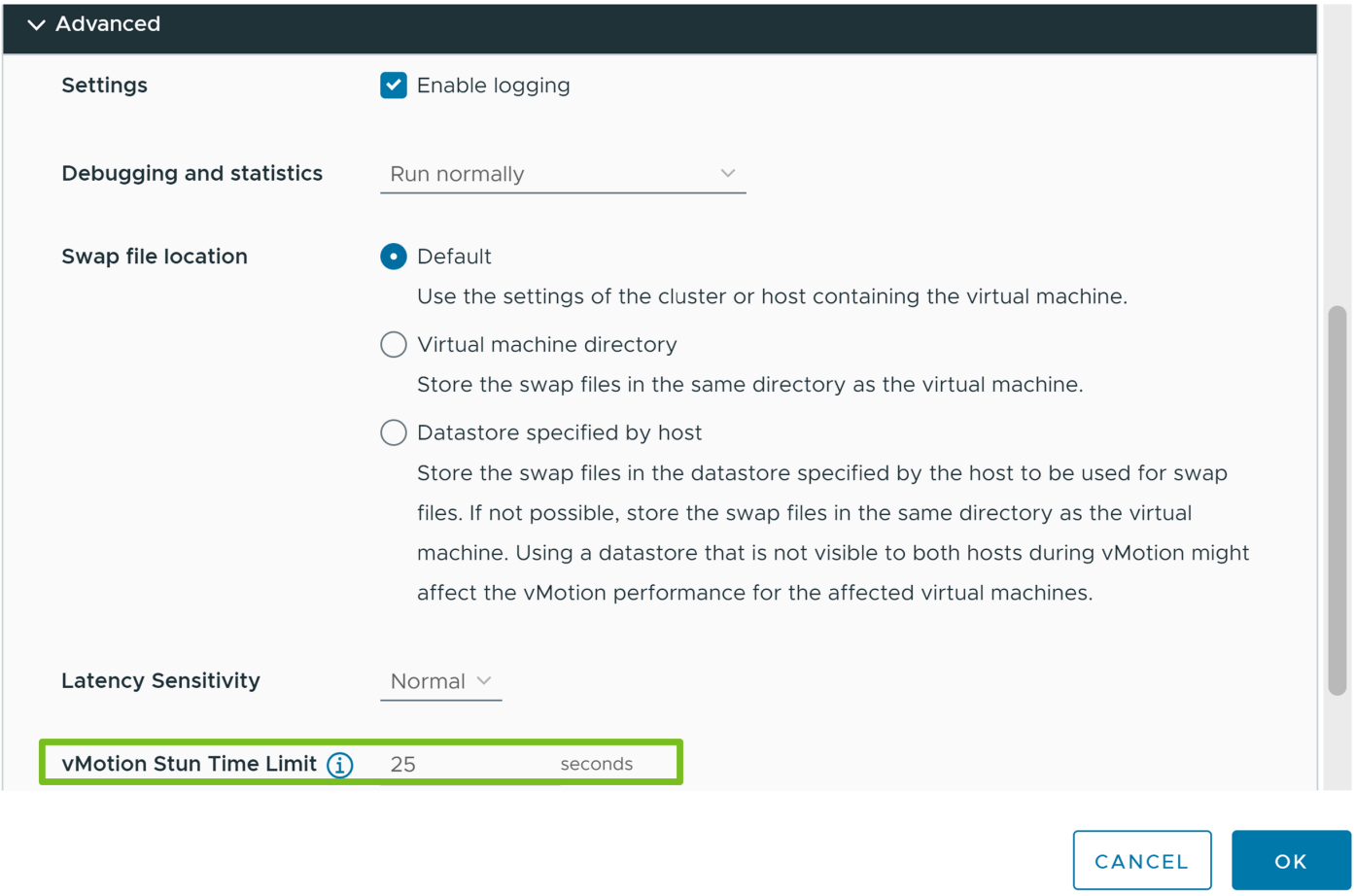
Starting with vSphere 8.0 Update 3, vMotion for vGPU virtual machines is activated by default. If you want to deactivate it, you must set the `vgpu.hotmigrate.enabled` advanced setting to `false`. For more information about how to configure the vCenter advanced settings, see *Configure Advanced Settings* in the *vCenter Configuration* documentation.

1. In the vSphere Client, click the main navigation menu

   (**),**
   and select **Inventory**.
2. Navigate to **Hosts and Clusters**.
3. Navigate to a vGPU virtual machine.
4. Right-click the virtual machine and select **Edit Settings**.
   The **Edit Settings** dialog box appears.
5. Click the **VM Options** tab and expand **Advanced**.



6. In the **vMotion Stun Time Limit** input field, enter the stun time limit in seconds.



If you set a stun time limit that is lower than the estimated maximum stun time for the PCI device configuration, a warning message appears. A warning message is also displayed on the **Virtual Hardware** tab.

7. Click **OK**.

Set a Stun Time Limit for a vMotion Migration of a Virtual Machine with a PCI Device

You can set a stun time limit per virtual machine when you migrate virtual machines with a PCI device with vSphere vMotion. Setting a stun time limit might prevent vCenter from powering on the virtual machine or migrating it to a host and network whose estimated maximum stun time exceeds that limit.

Verify that the vMotion network is configured through the Quickstart Workflow. For information about configuring the networking options for the vMotion traffic, see How to Add an ESX Host to Your vSphere Cluster by Using the Quickstart Workflow.

Virtual machine stun time is the time when the virtual machine is inaccessible to users during vMotion. You can set a vMotion stun time limit for the virtual machines with PCI devices. The limit that you set must be higher than the estimated maximum stun time for the current device configuration. If there are multiple PCI devices, the stun time limit that you set must be higher than the sum of the contributions of all PCI devices. Setting a stun time limit that is lower than the estimated maximum stun time might prevent the virtual machine from powering on.

For information on how to view the stun time for a PCI device, see View Stun Time Estimates During a vMotion Migration of a Virtual Machine with a PCI Device.

The maximum stun time is calculated based on the vMotion bandwidth of the host on which the virtual machine runs currently. If the destination host has lower bandwidth, the calculations might change at the time of migration. For example,

if a virtual machine is running on a host with a 25 Gbps vMotion NIC, but the destination host has a 10 Gbps vMotion NIC, at the time of migration the maximum stun time calculation is based on the 10 Gbps vMotion NIC.

Note:

You cannot set a stun time limit higher than 100 seconds.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Navigate to the virtual machine with PCI device for which you want to set a stun time limit.
4. Right-click the virtual machine and select **Edit Settings**.

   The **Edit Settings** dialog box appears.
5. Click the **VM Options** tab and expand **Advanced**.
6. In the **vMotion Stun Time Limit** input field, enter the stun time limit in seconds.

   If you set a stun time limit that is lower than the estimated maximum stun time for the PCI device configuration, a warning message appears. A warning message is also displayed on the **Virtual Hardware** tab.
7. Click **OK**.

View Stun Time Estimates During a vMotion Migration of a Virtual Machine with a PCI Device

You can view the expected PCI device stun times during a migration of a virtual machine with vMotion and the estimated worst-case stun times.

Stun time is the time when the virtual machine is inaccessible to users during vMotion. During the stun time, you are unable to access the virtual machine, desktop, or application. After the migration is completed, access to the virtual machine resumes and all applications continue from their previous state.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Navigate to the virtual machine with PCI device for which you want to view the stun time limit.
4. Right-click the virtual machine and select **Edit Settings**.

   The **Edit Settings** dialog box appears.
5. Click the **Virtual Hardware** tab, and expand the menu next to the PCI device for which you want to view the estimated stun times.
6. Next to **Estimated Max Stun Time**, you can see the estimated maximum stun time for the PCI device.
7. If you want to view the estimated stun times depending on the different bandwidth, click **View Reference Table**.

   When you consider the expected and estimated worst-case stun times, keep in mind the following points:

   - The estimated stun times are only valid when migrating a single virtual machine. If you are concurrently migrating multiple virtual machines, that is, for a vSphere manual remediation process, the stun times differ.
   - The estimates assume sufficient CPU, memory, PCIe, and network capacity to achieve 10-Gbps migration throughput.

Virtual Machines Swap File Location and vSphere vMotion Compatibility

Virtual machine swap file location affects vMotion compatibility in different ways depending on the version of ESX running on the virtual machine's host.

You can configure ESX 7.0 or later hosts to store virtual machine swap files with the virtual machine configuration file, or on a local swap file datastore specified for that host.

The location of the virtual machine swap file affects vMotion compatibility as follows:

- For migrations between hosts running ESX 7.0 and later, vMotion and migrations of suspended and powered-off virtual machines are allowed.
- During a migration with vMotion, if the swap file location on the destination host differs from the swap file location on the source host, the swap file is copied to the new location. This activity can result in slower migrations with vMotion. If the destination host cannot access the specified swap file location, it stores the swap file with the virtual machine configuration file.

For information about using swap files, see Using Swap Files with vSphere in the *vSphere Resource Management* documentation.

vSphere vMotion Notifications for Latency Sensitive Applications

Starting with vSphere 8.0, you can notify an application running inside the guest OS of a virtual machine when a vSphere vMotion event starts and finishes. This notification mechanism allows latency sensitive applications to prepare or delay a vSphere vMotion operation.

In cases of latency sensitive applications, such as VoIP applications and high frequency trading applications, vSphere vMotion and vSphere DRS are usually deactivated. vSphere 8.0 introduces a notification mechanism which allows you to notify an application that a vSphere vMotion is about to happen so that the application takes the necessary steps to prepare. vSphere vMotion pauses after generating the start event and waits for the application to acknowledge the start notification before it proceeds.

> **Note:**
>
> Enabling the notification mechanism for vSphere vMotion operations might result in an increase of the overall vSphere vMotion time.

How to Configure a Virtual Machine for vSphere vMotion Notifications

By default, the notification mechanism is deactivated. To enable the notification mechanism on a virtual machine, the virtual machine must be of hardware version 20 or above and you must configure the following advanced virtual machine configuration options.

You can use vSphere Web Services API to enable sending notifications to applications running inside the virtual machine, and to specify the maximum period of time in seconds that an application has to prepare for the vMotion operation.

- To enable sending notifications to applications running inside the virtual machine, set the `vmx.vmOpNotificationToApp.enabled` virtual machine property to `true`
- To specify the maximum period of time in seconds that an application has to prepare for the vMotion operation, use the `vmx.vmOpNotificationToApp.timeout` virtual machine property. When a vSphere vMotion operation generates a start event, the vMotion operation pauses and waits for an acknowledgment from the application to proceed. vSphere vMotion waits for the notification timeout that you specify.

  This property is optional and is not set by default. You can use it to configure a more restrictive timeout for a specific virtual machine. If left unset, the host notification configuration is used.

For more information, see the virtual machine data object properties in the vSphere Web Services API documentation. For information on how to use the vSphere Web Services SDK, see the vSphere Management SDK guide.

How to Configure a Host for vSphere vMotion Notifications

You can use the `VmOpNotificationToApp.Timeout` advanced host configuration setting, to specify a notification timeout that applies to all virtual machines on a host. Use the `ConfigManager` APIs to set a value to this configuration

property. The default notification timeout is `0`. In this case, application notifications are generated but the vSphere vMotion operations are not delayed.

If you set a notification timeout on a host and a virtual machine running on the host at the same time, the smaller value is used.

For information on how to use the vSphere Web Services SDK, see the vSphere Management SDK guide.

How to Register an Application for vSphere vMotion Notifications

To enable an application to receive notifications for vSphere vMotion events, you must have VMware Tools installed on the virtual machine hosting the application. For more information about installing and configuring VMware Tools, see the *VMware Tools* documentation.

With vSphere 8.0, you can register only one application per virtual machine. An application that wants to receive vSphere vMotion notifications, can use the following guest RPC calls and check for new vMotion events periodically, for example, every 1 or 2 seconds, using the `vm-operation-notification.check-for-event` guest RPC call.

For information on how to use guest RPC calls, see the vSphere Guest SDK guide.

Command	Description
`vm-operation-notification.register`	Registers an application to start receiving notifications for vSphere vMotion events.
`vm-operation-notification.unregister`	Unregisters an application from receiving notifications for vSphere vMotion events.
`vm-operation-notification.list`	Retrieves information about the registered application running on a virtual machine on the host.
`vm-operation-notification.check-for-event`	Retrieves information about the vSphere vMotion event registered at the time of the call.
`vm-operation-notification.ack-event`	Acknowledges a vSphere vMotion start event.

How to Migrate Your Virtual Machine to a New Compute Resource

Learn how to use vSphere vMotion and the **Migration** wizard to migrate a powered-on virtual machine from one compute resource to another. To relocate only the disks of a powered-on virtual machine, migrate the virtual machine to a new datastore by using Storage vMotion.

Verify that your hosts and virtual machines meet the requirements for migration with vMotion with shared storage.

- Verify that your hosts and virtual machines meet the requirements for migration with vMotion. See Host Configuration for vSphere vMotion and Virtual Machine Conditions and Limitations for vSphere vMotion.
- Verify that the storage that contains the virtual machine disks is shared between the source and target hosts. See vSphere vMotion Shared Storage Requirements.
- For migration across vCenter instances, verify whether your system meets additional requirements. See What Are the Requirements for vMotion Between vCenter Instances.
- For migration of a virtual machine with NVIDIA vGPU, verify that the target ESX host has a free vGPU slot. Also, verify that the `vgpu.hotmigrate.enabled` advanced setting is set to `true`. For more information about configuring vCenter advanced settings, see *Configure Advanced Settings* in the *vCenter Configuration* documentation.
- Required privilege: **Resource** > **Migrate powered on virtual machine**

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Select **Hosts and Clusters**, and navigate to the virtual machine.
3. Right-click the virtual machine and select **Migrate**.
   a) To locate a virtual machine, select a data center, folder, cluster, resource pool, host, or vApp.
   b) Click the **Virtual Machines** tab.
4. Click **Change compute resource only** and click **Next**.
5. Select a host, cluster, resource pool, or vApp to run the virtual machine, and click **Next**.
   Any compatibility problem appears in the Compatibility panel. Fix the problem, or select another host or cluster.

   Possible targets include hosts and fully automated DRS clusters in the same or another vCenter system. If your target is a non-automated cluster, select a host within the non-automated cluster.

   Important:

   If the virtual machine that you migrate has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resources. Otherwise, the compatibility check fails and you cannot proceed further with the migration.

   If the virtual machine that you migrate does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resources, so that all PMem hard disks remain stored on a PMem datastore. Otherwise, all the hard disks use the storage policy and datastore selected for the configuration files of the virtual machine.

   Important:
   Migrating a virtual machine that has an NVDIMM device or a vPMem disk to a host that does not have the proper license fails and leaves the virtual machine in an unmanageable state for 90 seconds. You can afterwards retry the migration and select a destination host that is licensed to use PMem devices.
6. Select a destination network for all VM network adapters connected to a valid source network and click **Next**.

   You can click **Advanced** to select a new destination network for each VM network adapter connected to a valid source network.

   You can migrate a virtual machine network to another distributed switch in the same or to another data center or vCenter.
7. Select the migration priority level and click **Next**.

Option	Description
Schedule vMotion with high priority	vCenter attempts to reserve resources on both the source and destination hosts to be shared among all concurrent migrations with vMotion. vCenter grants a larger share of host CPU resources. If sufficient CPU resources are not immediately available, vMotion is not initiated.
Schedule normal vMotion	vCenter reserves resources on both the source and destination hosts to be shared among all concurrent migrations with vMotion. vCenter grants a smaller share of host CPU resources. If there is a lack of CPU resources, the duration of vMotion can be extended.

8. Review the page and click **Finish**.

vCenter moves the virtual machine to the new host or storage location.

Event messages appear in the **Events** tab. The data displayed on the **Summary** tab shows the status and state throughout the migration. If errors occur during migration, the virtual machines revert to their original states and locations.

What is Migration with Storage vMotion

With Storage vMotion, you can migrate a virtual machine and its disk files from one datastore to another while the virtual machine is running. You can also move virtual machines off of arrays for maintenance or to upgrade.

With Storage vMotion you also have the flexibility to optimize disks for performance, or to transform disk types, which you can use to reclaim space.

You can choose to place the virtual machine and all its disks in a single location, or you can select separate locations for the virtual machine configuration file and each virtual disk. The virtual machine does not change execution host during a migration with Storage vMotion.

During a migration with Storage vMotion, you can change the disk provisioning type.

Migration with Storage vMotion changes virtual machine files on the destination datastore to match the inventory name of the virtual machine. The migration renames all virtual disk, configuration, snapshot, and `.nvram` files. If the new names exceed the maximum filename length, the migration does not succeed.

Storage vMotion has several uses in administering virtual infrastructure, including the following examples of use.

- Storage maintenance and reconfiguration. You can use Storage vMotion to move virtual machines off a storage device to allow maintenance or reconfiguration of the storage device without virtual machine downtime.
- Redistributing storage load. You can use Storage vMotion to redistribute virtual machines or virtual disks to different storage volumes to balance capacity or improve performance.

Storage vMotion Requirements and Limitations

A virtual machine and its host must meet resource and configuration requirements for the virtual machine disks to be migrated with Storage vMotion.

Storage vMotion is subject to the following requirements and limitations:

- Virtual machine disks must be in persistent mode or be raw device mappings (RDMs). For virtual compatibility mode RDMs, you can migrate the mapping file or convert to thick-provisioned or thin-provisioned disks during migration if the destination is not an NFS datastore. If you convert the mapping file, a new virtual disk is created and the contents of the mapped LUN are copied to this disk. For physical compatibility mode RDMs, you can migrate the mapping file only.
- Migration of virtual machines during VMware Tools installation is not supported.
- Because VMFS3 datastores do not support large capacity virtual disks, you cannot move virtual disks greater than 2 TB from a VMFS5 datastore to a VMFS3 datastore.
- ESX 4.0 and later hosts do not require vMotion configuration to perform migration with Storage vMotion.
- The host on which the virtual machine is running must have access to both the source and target datastores.
- For limits on the number of simultaneous migrations with vMotion and Storage vMotion, see vCenter Limits on Simultaneous Migrations.

How to Migrate Your Virtual Machine with Storage vMotion

Learn how to use Storage vMotion to migrate the configuration file of a virtual machine and its virtual disks to a new storage. You can migrate the virtual machine while it is powered on.

- Verify that your system satisfies the requirements for Storage vMotion. See Storage vMotion Requirements and Limitations.
- For migration of a virtual machine with NVIDIA vGPU, verify that the ESX host on which the virtual machine runs has a free vGPU slot when the host is with version 7.0 Update 2 and earlier. Starting with vSphere 7.0 Update 3, the source hosts are not required to have a free vGPU slot.
- Verify that the `vgpu.hotmigrate.enabled` advanced setting is set to `true`. For more information about configuring vCenter advanced settings, see Configure Advanced Settings in the *vCenter Configuration* documentation.
- Required privilege: **Resource** > **Migrate powered on virtual machine**

1. In the vSphere Client, click the main navigation menu

   ( ),

   and select **Inventory**.
2. Select **Hosts and Clusters**, and navigate to the virtual machine.
3. Right-click the virtual machine and select **Migrate**.
   a) To locate a virtual machine, select a data center, folder, cluster, resource pool, host, or vApp.
   b) Click the **Virtual Machines** tab.
4. Click **Change storage only** and click **Next**.
5. Select the format for the virtual machine's disks.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation. Instead, it is zeroed out on demand on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it.

6. Select a virtual machine storage policy from the **VM Storage Policy** drop-down menu.

   Storage policies specify storage requirements for applications that run on the virtual machine.

   Important:

   If the virtual machine hard disks use different storage policies, the new policy that you select only applies to non-PMem hard disks. PMem hard disks are migrated to the host-local PMem datastore of the destination host.
7. Select the datastore location where you want to store the virtual machine files.

Option	Action
Store all virtual machine files in the same location on a datastore.	Select a datastore and click **Next**.
Store all virtual machine files in the same Storage DRS cluster.	1. Select a Storage DRS cluster. 2. To deactivate Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 3. Click **Next**.
Store virtual machine configuration files and disks in separate locations.	1. Click **Advanced**. **Note:** You can use the **Advanced** option to downgrade from or upgrade to PMem storage. 2. For the virtual machine configuration file and for each virtual disk, select **Browse**, and select a datastore or Storage DRS cluster.

Option	Action
	Note: Configuration files cannot be stored on a PMem datastore. 3. If you selected a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 4. Click **Next**.

8. On the Ready to complete page, review the details and click **Finish**.

vCenter moves the virtual machine to the new storage location. Names of migrated virtual machine files on the destination datastore match the inventory name of the virtual machine.

Event messages appear in the **Events** tab. The data displayed on the **Summary** tab shows the status and state throughout the migration. If errors occur during migration, the virtual machines revert to their original states and locations.

How to Migrate Your Virtual Machine to a New Compute Resource and Storage

Learn how to move a virtual machine from your vSphere inventory to another compute resource and move its disks or virtual machine folder to another datastore. With vMotion, you can migrate a virtual machine and its disks and files while the virtual machine is powered on.

- Verify that your hosts and virtual machines meet the requirements for live migration. See Requirements and Limitations for vMotion Without Shared Storage.
- For migration across vCenter instances, verify whether your system meets additional requirements. See What Are the Requirements for vMotion Between vCenter Instances.
- For migration of a virtual machine with NVIDIA vGPU, verify that the target ESX host has a free vGPU slot. Also, verify that the `vgpu.hotmigrate.enabled` advanced setting is set to `true`. For more information about configuring vCenter advanced settings, see Configure Advanced Settings in *vCenter Configuration*.
- Required privilege: **Resource** > **Migrate powered on virtual machine**

Simultaneous migration to a new compute resource and datastore provides greater mobility for virtual machines by eliminating the vCenter boundary. Virtual machine disks or contents of the virtual machine folder are transferred over the vMotion network to reach the destination host and datastores.

To make disk format changes and preserve them, you must select a different datastore for the virtual machine files and disks. You cannot select the same datastore on which the virtual machine disks currently reside.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Select **Hosts and Clusters**, and navigate to the virtual machine.
3. Right-click the virtual machine and select **Migrate**.
   a) To locate a virtual machine, select a data center, folder, cluster, resource pool, host, or vApp.
   b) Click the **Virtual Machines** tab.
4. Select **Change both compute resource and storage** and click **Next**.
5. Select a destination resource for the virtual machine, and click **Next**.

   Any compatibility problems appear in the Compatibility panel. Fix the problem, or select another host or cluster.

   Possible targets include hosts and fully automated DRS clusters. If your target is a non-automated cluster, select a host within the non-automated cluster.

If your environment has more than one vCenter instances, you can move virtual machines from one vCenter inventory to another.

Important:

If the virtual machine that you migrate has an NVDIMM device and uses PMem storage, the destination host or cluster must have available PMem resources. Otherwise, the compatibility check fails and you cannot proceed further with the migration.

If the virtual machine that you migrate does not have an NVDIMM device but it uses PMem storage, you must select a host or cluster with available PMem resources, so that all PMem hard disks remain stored on a PMem datastore. Otherwise, all the hard disks use the storage policy and datastore selected for the configuration files of the virtual machine.

Important:

Migrating a virtual machine that has an NVDIMM device or a vPMem disk to a host that does not have the proper license fails and leaves the virtual machine in an unmanageable state for 90 seconds. You can afterwards retry the migration and select a destination host that is licensed to use PMem devices.

6. Select the format for the virtual machine's disks.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation. Instead, it is zeroed out on demand on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it.

7. Select a virtual machine storage policy from the **VM Storage Policy** drop-down menu.

   Storage policies specify storage requirements for applications that run on the virtual machine.

   Important:

   If the virtual machine hard disks use different storage policies, the new policy that you select only applies to non-PMem hard disks. PMem hard disks are migrated to the host-local PMem datastore of the destination host.

8. Select the datastore location where you want to store the virtual machine files.

Option	Action
Store all virtual machine files in the same location on a datastore.	Select a datastore and click **Next**.
Store all virtual machine files in the same Storage DRS cluster.	1. Select a Storage DRS cluster. 2. To deactivate Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 3. Click **Next**.
Store virtual machine configuration files and disks in separate locations.	1. Click **Advanced**.

Option	Action
	Note: You can use the **Advanced** option to downgrade from or upgrade to PMem storage. 2. For the virtual machine configuration file and for each virtual disk, select **Browse**, and select a datastore or Storage DRS cluster. **Note:** Configuration files cannot be stored on a PMem datastore. 3. If you selected a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 4. Click **Next**.

9. Select a destination network for all VM network adapters connected to a valid source network and click **Next**.

   You can click **Advanced** to select a new destination network for each VM network adapter connected to a valid source network.

   You can migrate a virtual machine network to another distributed switch in the same or to another data center or vCenter.

10. Select the migration priority level and click **Next**.

Option	Description
Schedule vMotion with high priority	vCenter attempts to reserve resources on both the source and destination hosts to be shared among all concurrent migrations with vMotion. vCenter grants a larger share of host CPU resources. If sufficient CPU resources are not immediately available, vMotion is not initiated.
Schedule normal vMotion	vCenter reserves resources on both the source and destination hosts to be shared among all concurrent migrations with vMotion. vCenter grants a smaller share of host CPU resources. If there is a lack of CPU resources, the duration of vMotion can be extended.

11. On the Ready to complete page, review the details and click **Finish**.

vCenter moves the virtual machine to the new host or storage location.

Event messages appear in the **Events** tab. The data displayed on the **Summary** tab shows the status and state throughout the migration. If errors occur during migration, the virtual machines revert to their original states and locations.

vSphere vMotion in Environments Without Shared Storage

You can use vSphere vMotion to migrate virtual machines to a different compute resource and storage simultaneously. Unlike vSphere Storage vMotion, which requires a single host to have access to both the source and destination datastore, you can migrate virtual machines across storage accessibility boundaries.

vSphere vMotion does not require environments with shared storage. This is useful for performing cross-cluster migrations, when the target cluster machines might not have access to the storage of the source cluster. Processes that are working on the virtual machine continue to run during the migration with vSphere vMotion.

You can use vSphere vMotion to migrate virtual machines across vCenter instances.

You can place the virtual machine and all its disks in a single location or select separate locations for the virtual machine configuration file and each virtual disk. In addition, you can change virtual disks from thick-provisioned to thin-provisioned or from thin-provisioned to thick-provisioned. For virtual compatibility mode RDMs, you can migrate the mapping file or convert from RDM to VMDK.

vSphere vMotion without shared storage is useful for virtual infrastructure administration tasks similar to vSphere vMotion with shared storage or vSphere Storage vMotion tasks.

- Host maintenance. You can move virtual machines from a host to allow maintenance of the host.
- Storage maintenance and reconfiguration. You can move virtual machines from a storage device to allow maintenance or reconfiguration of the storage device without virtual machine downtime.
- Storage load redistribution. You can manually redistribute virtual machines or virtual disks to different storage volumes to balance capacity or improve performance.

Requirements and Limitations for vMotion Without Shared Storage

If you want to migrate a virtual machine with vMotion in the absence of shared storage, the virtual machine and its host must meet resource and configuration requirements for the virtual machine files and disks.

vMotion in an environment without shared storage is subject to the following requirements and limitations:

- The hosts must meet the networking requirement for vMotion. See What Are the vSphere vMotion Networking Requirements.
- The virtual machines must be properly configured for vMotion. See Virtual Machine Conditions and Limitations for vSphere vMotion.
- Virtual machine disks must be in persistent mode or be raw device mappings (RDMs). See Storage vMotion Requirements and Limitations.
- The destination host must have access to the destination storage.
- When you move a virtual machine with RDMs and do not convert those RDMs to VMDKs, the destination host must have access to the RDM LUNs.
- Consider the limits for simultaneous migrations when you perform a vMotion migration without shared storage. This type of vMotion counts against the limits for both vMotion and Storage vMotion, so it consumes both a network resource and 16 datastore resources. See vCenter Limits on Simultaneous Migrations.

When you migrate a powered-on virtual machine with vMotion in the absence of shared storage, the migration goes over the network using the vMotion vmknic, and the data is transferred directly between the two ESX hosts.

Migrating Virtual Machines Between vCenter Systems

You can migrate powered-on and powered-off virtual machines between vCenter instances. With Advanced Cross vCenter vMotion, you can also move or clone workloads across vCenter systems both between on-premises environments and between cloud and on-premises environments.

Cold and Hot migration of virtual machines across vCenter systems is helpful in certain VM provisioning cases.

- Balance workloads across clusters and vCenter instances.
- Elastically expand or shrink capacity across resources in different vCenter instances in the same site or in another geographical area.
- Move virtual machines between environments that have different purposes, for example, from a development to production.
- Move virtual machines to meet different Service Level Agreements (SLAs) regarding storage space, performance, and so on.

> **Note:**
> During the migration of a virtual machine to another vCenter system, the performance data that has been collected about the virtual machine is lost.

What Are the Requirements for vMotion Between vCenter Instances

You can use vSphere vMotion across vCenter instances if your system meets certain requirements.

The following list sums the requirements that your system must meet so that you can use migration across vCenter instances:

- When you migrate virtual machines across vCenter instances without using Advanced Cross vCenter vMotion, verify that you meet the following requirements.
  - Both vCenter instances must be in the same vCenter Single Sign-On domain.
  - The source and destination vCenter instances and ESX hosts must be 6.7 or later.
  - Both vCenter instances must be time-synchronized with each other for correct vCenter Single Sign-On token verification.
  - For migration of compute resources only, both vCenter instances must be connected to the shared virtual machine storage.
- When you migrate virtual machines across vCenter instances in different vCenter Single Sign-On domains with Advanced Cross vCenter vMotion, verify that you meet the following requirements.
  - The vCenter instance from which you initiate the import or export of virtual machines must be version 7.0 Update 1c or later.

Network Compatibility Checks During vMotion Between vCenter Instances

Migration of virtual machines between vCenter instances moves virtual machines to new networks. The migration process performs checks to verify that the source and destination networks are similar.

vCenter performs network compatibility checks to prevent the following configuration problems:

- vMotion between distributed switches of different versions
- vMotion from an external network to an internal network
- vMotion to a distributed switch that is not working properly

vCenter does not perform checks for and notify you about the following problems:

- If the source and destination distributed switches are not in the same broadcast domain, virtual machines lose network connectivity after migration.
- If the source and destination distributed switches do not have the same services configured, virtual machines might lose network connectivity after migration.

MAC Address Management During Migration Between vCenter Systems

When you move a virtual machine between vCenter instances, the environment specifically handles MAC address migration to avoid address duplication and loss of data in the network.

In an environment with multiple vCenter instances, when a virtual machine is migrated, its MAC addresses are transferred to the target vCenter. The source vCenter adds the MAC addresses to a denylist so that it does not assign them to newly created virtual machines.

To reclaim unused MAC addresses from the denylist, contact VMware Technical Support for assistance.

Export or Clone a Virtual Machine with Advanced Cross vCenter vMotion

Learn how to export or clone workloads to a target vCenter system by using Advanced Cross vCenter vMotion. You can initiate migration of workloads both from on-premise environments and from cloud environments.

- Obtain the credentials for the administrator accounts of the vCenter instances to which you want to migrate virtual machines.
- Verify that the target vCenter instances to which you export or clone virtual machines are version 6.7 or later.
- Verify that the source vCenter instance from which you export virtual machines is version 7.0 Update 1c or later if you want to export virtual machines to other vCenter instances.
- Verify that the source vCenter instance from which you clone virtual machines is version 7.0 Update 3 or later.
- Verify that the destination ESX hosts are version 6.7 or later.
- If the virtual machine that you migrate has an NVDIMM device and uses PMem storage, verify that the destination host or cluster have available PMem resources.
- If you migrate a virtual machine that has an NVDIMM device or a vPMem disk, verify that the destination host has the proper license.

- Required privilege: **Resource.Query vMotion**
- When you migrate powered on virtual machines, verify that you meet the following requirements.
  - Required privilege: **Resource.Migrate powered on virtual machine**
  - Verify that your hosts and virtual machines meet the requirements for migration with vMotion. See Host Configuration for vSphere vMotion and Virtual Machine Conditions and Limitations for vSphere vMotion.
  - Verify that your hosts and virtual machines meet the requirements for live migration. See Requirements and Limitations for vMotion Without Shared Storage.
- When you migrate powered off virtual machines, verify that you meet the following requirements.
  - Required privilege: **Resource.Migrate powered off virtual machine**
  - Make sure that you are familiar with the requirements for cold migration. See Cold Migration in Your vSphere Environment.
- Required ports:
  - Port 8000 on the ESX host for vMotion.
  - Port 902 on the ESX host for NFC.
  - Port 443 between the two vCenter instances.

  For the list of all supported ports and protocols in VMware products, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/view.

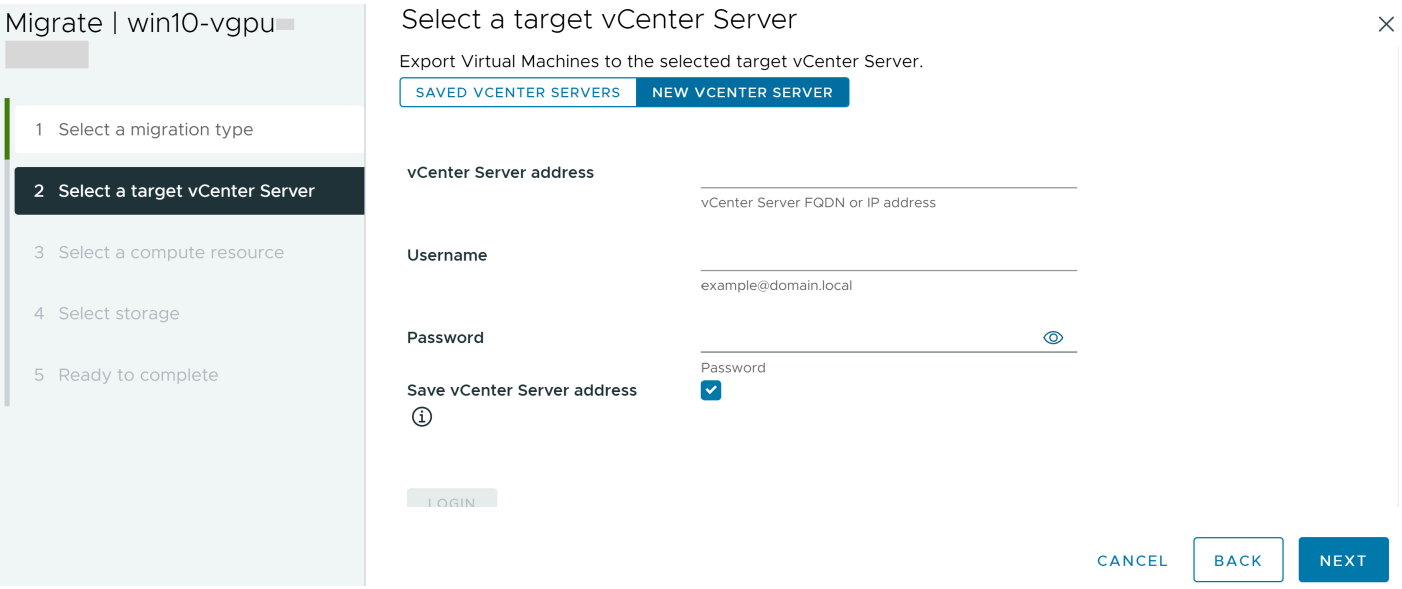
Advanced Cross vCenter vMotion does not depend on vCenter linking and you can migrate virtual machines across vCenter systems in different vCenter Single Sign-On domains. For example, when you deploy a new vCenter instance, you can migrate virtual machines from the earlier version of your vCenter instance to the newly provisioned one.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Navigate to**Hosts and Clusters**.
3. From the inventory tree, select the host or cluster that contains the virtual machines you want to migrate.
4. On the **VMs** tab, click the **Virtual Machines** button.
5. From the virtual machines list, select the virtual machines that you want to migrate.
6. Right-click the selection and click **Migrate**.
7. If a confirmation dialog box appears, click **Yes**.

   The migration wizard appears.
8. Select **Cross vCenter export** as the migration type.
   a) Optional: To create a clone of the virtual machine on the target vCenter, select the **Keep VMs on the source vCenter Server (performs a VM clone operation).** check box.
   b) Click **Next** to continue.
9. Select the target vCenter instance to which you want to export or clone virtual machines.



Option	Action
Saved vCenter Instances	From the drop-down menu, select the IP address or FQDN of a saved vCenter.
New vCenter Instance	1. Enter the IP address or FQDN, user name, and password of a vCenter instance. **Note:** The check box **Save vCenter address** is selected by default. If you do not want to save the vCenter address, deselect the check box. 2. Click **Login**.

The connection to a saved vCenter instance is valid only for the current user session. You must enter the credentials of the target vCenter instances for each login session.

10. If a **Security Alert** dialog box appears, click **Yes**.
11. Click **Next** to continue.
12. Select the destination compute resource for this virtual machine migration and click **Next**.

    Possible targets include hosts and DRS clusters with any level of automation. If DRS is not enabled for a cluster, select a specific host in the cluster rather than selecting the cluster.

    Important:

    If the virtual machine that you migrate does not have an NVDIMM device but it uses PMem storage, you must select a host or cluster with available PMem resources, so that all PMem hard disks remain stored on a PMem datastore. Otherwise, all the hard disks use the storage policy and datastore selected for the configuration files of the virtual machine.

    Important:
    When you try to migrate a virtual machine that has an NVDIMM device or a vPMem disk to a host that does not have the proper license, the operation fails and the virtual machine is in an unmanageable state for 90 seconds. After 90 seconds, you can retry the migration, and select a destination host that is licensed to use PMem devices.

13. If a compatibility problem appears in the Compatibility pane, fix the problem or select another host or cluster for the migration.
14. Select the destination storage for the virtual machine migration.
    a) Select the storage type for the virtual machine configuration files and all the hard disks.

Mode	Description
Standard	All virtual disks are stored on a standard datastore.
PMem	All virtual disks are stored on the local PMem datastore of the host. Configuration files cannot be stored on a PMem datastore. You must additionally select a regular datastore for the configuration files of the virtual machine.
Hybrid	All PMem virtual disks remain stored on a PMem datastore. Your choice of a VM storage policy and your choice of a datastore affects non-PMem disks.

    You can select the type of storage only if PMem or Hybrid storage types are available in the data center.
    b) Select the format for the virtual machine's disks.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation. Instead, it is zeroed out on demand on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it.

    c) Select a virtual machine storage policy from the **VM Storage Policy** drop-down menu.

    Storage policies specify storage requirements for applications that run on the virtual machine. You can also select the default policy for vSAN or Virtual Volumes datastores.

    Important:
    If the virtual machine hard disks use different storage policies, the new policy that you select only applies to non-PMem hard disks. PMem hard disks are migrated to the host-local PMem datastore of the destination host.
    d) Select the datastore location for the virtual machine files.

Option	Action
Store all virtual machine files in the same location on a datastore.	Select a datastore from the list and click **Next**.
Store all virtual machine files in the same Storage DRS cluster.	1. Select a Storage DRS cluster. 2. If you want to migrate the virtual machine to a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 3. Click **Next**.

Option	Action
Store virtual machine configuration files and disks in separate locations.	1. Click **Configure per disk**. **Note:** You can use the **Configure per disk** option to downgrade from or upgrade to PMem storage. 2. For the virtual machine configuration file and for each virtual disk, select **Browse**, and select a datastore or a Storage DRS cluster. **Note:** Configuration files cannot be stored on a PMem datastore. 3. If you want to migrate the virtual machine to a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 4. Click **Next**.

15. If a compatibility problem appears in the Compatibility pane, fix the problem or select another destination storage for the migration.
16. Select the destination folder for the virtual machine migration and click **Next**.
17. Select the destination networks for the virtual machine migration.

Option	Action
Select a destination network for all VM network adapters connected to a valid source network.	1. Click the arrow in the **Destination Network** column and select **Browse**. 2. Select a destination network and click **OK**. 3. Click **Next**.
Select a different destination network for each VM network adapter connected to a valid source network.	1. Click **Advanced**. 2. Click the arrow in the **Destination Network** column and select **Browse**. 3. Select a destination network and click **OK**. 4. Click **Next**.

18. On the **Ready to complete** page, review the details and click **Finish**.

The virtual machines are moved or cloned to the destination folder in the new vCenter instance. You can monitor the migration process in the **Recent tasks** pane. If errors occur during the migration, the virtual machines revert to their original states and locations.

Import or Clone a Virtual Machine with Advanced Cross vCenter vMotion

Learn how to use Advanced Cross vCenter vMotion to import or clone workloads from a source vCenter system. You can initiate migration of workloads both from on-premise environments and from cloud environments.

- Obtain the credentials for the administrator account of the vCenter instance from which you want to import or clone virtual machines.
- Verify that the source vCenter instances from which you import or clone virtual machines are version 6.7 or later.
- Verify that the target vCenter instance to which you import virtual machines is version 7.0 Update 1c or later.
- Verify that the target vCenter instance to which you clone virtual machines is version 7.0 Update 1c or later.
- Verify that the destination ESX hosts are version 6.7 or later.
- If the virtual machine that you migrate has an NVDIMM device and uses PMem storage, verify that the destination host or cluster have available PMem resources.
- If you migrate a virtual machine that has an NVDIMM device or a vPMem disk, verify that the destination host has the proper license.
- Required privilege: **Resource.Query vMotion**

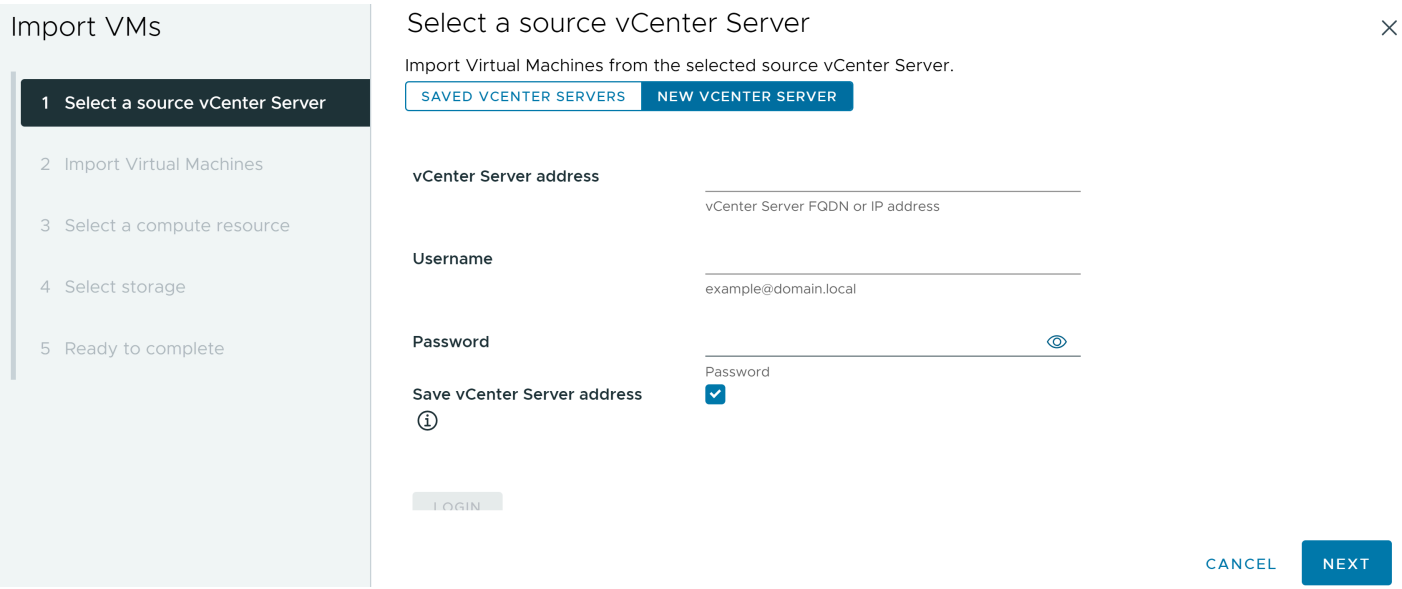
- When you migrate powered on virtual machines, verify that you meet the following requirements.
  - Required privilege: **Resource.Migrate powered on virtual machine**
  - Verify that your hosts and virtual machines meet the requirements for migration with vMotion. See Host Configuration for vSphere vMotion and Virtual Machine Conditions and Limitations for vSphere vMotion.
  - Verify that your hosts and virtual machines meet the requirements for live migration. See Requirements and Limitations for vMotion Without Shared Storage.
- When you migrate powered off virtual machines, verify that you meet the following requirements.
  - Required privilege: **Resource.Migrate powered off virtual machine**
  - Make sure that you are familiar with the requirements for cold migration. See Cold Migration in Your vSphere Environment.
- Required ports:
  - Port 8000 on the ESX host for vMotion.
  - Port 902 on the ESX host for NFC.
  - Port 443 between the two vCenter instances.

  For the list of all supported ports and protocols in VMware products, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/.

Advanced Cross vCenter vMotion does not depend on vCenter linking and you can migrate virtual machines across vCenter systems in different vCenter Single Sign-On domains. For example, when you deploy a new vCenter instance, you can migrate virtual machines from the earlier version of your vCenter instance to the newly provisioned one.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Select **Hosts and Clusters**.
3. From the inventory tree, right-click the host or cluster to which you want to import virtual machines.
4. Click **Import VMs**.

   The **Import VMs** wizard appears.
5. Select the source vCenter instance from which you want to import virtual machines.



Option	Action
Saved vCenter Instances	From the drop-down menu, select the IP address or FQDN of a saved vCenter instance.

Option	Action
New vCenter Instance	1. Enter the IP address or FQDN, user name, and password of a vCenter instance. **Note:** The check box **Save vCenter Server address** is selected by default. If you do not want to save the vCenter address, deselect the check box. 2. Click **Login**.

The connection to a saved vCenter instances is valid only for the current user session. You must enter the credentials of the source vCenter instances for each login session.

6. If a **Security Alert** dialog box appears, click **Yes**.
7. Click **Next**.
8. Select the virtual machines to import or clone.
   a) If you want to migrate more than one virtual machine, the selected virtual machines must be in the same power state.
   b) To keep a clone of the virtual machine on the source vCenter, select the **Keep VMs on the source vCenter (performs a VM clone operation).** check box.
   c) Click **Next** to continue.

   If you want to migrate more than one virtual machine, the selected virtual machines must be in the same power state.
9. Select the destination compute resource for this virtual machine migration and click **Next**.

   Possible targets include hosts and DRS clusters with any level of automation. If DRS is not enabled for a cluster, select a specific host in the cluster rather than selecting the cluster.

   Important:

   If the virtual machine that you migrate does not have an NVDIMM device but it uses PMem storage, you must select a host or cluster with available PMem resources, so that all PMem hard disks remain stored on a PMem datastore. Otherwise, all the hard disks use the storage policy and datastore selected for the configuration files of the virtual machine.

   Important:
   When you try to migrate a virtual machine that has an NVDIMM device or a vPMem disk to a host that does not have the proper license, the operation fails and the virtual machine is in an unmanageable state for 90 seconds. After 90 seconds, you can retry the migration, and select a destination host that is licensed to use PMem devices.
10. If a compatibility problem appears in the Compatibility pane, fix the problem or select another host or cluster for the migration.
11. Select the destination storage for the virtual machine migration.
    a) Select the storage type for the virtual machine configuration files and all the hard disks.

Mode	Description
Standard	All virtual disks are stored on a standard datastore.
PMem	All virtual disks are stored on the local PMem datastore of the host. Configuration files cannot be stored on a PMem datastore. You must additionally select a regular datastore for the configuration files of the virtual machine.
Hybrid	All PMem virtual disks remain stored on a PMem datastore. Your choice of a VM storage policy and your choice of a datastore affects non-PMem disks.

You can select the type of storage only if PMem or Hybrid storage types are available in the data center.

b) Select the format for the virtual machine's disks.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation. Instead, it is zeroed out on demand on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it.

c) Select a virtual machine storage policy from the **VM Storage Policy** drop-down menu.

Storage policies specify storage requirements for applications that run on the virtual machine.

Important:
If the virtual machine hard disks use different storage policies, the new policy that you select only applies to non-PMem hard disks. PMem hard disks are migrated to the host-local PMem datastore of the destination host.

d) Select the datastore location where you want to store the virtual machine files.

Option	Action
Store all virtual machine files in the same location on a datastore.	Select a datastore from the list and click **Next**.
Store all virtual machine files in the same Storage DRS cluster.	1. Select a Storage DRS cluster. 2. If you want to migrate the virtual machine to a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 3. Click **Next**.
Store virtual machine configuration files and disks in separate locations.	1. Click **Configure per disk**. **Note:** You can use the **Configure per disk** option to downgrade from or upgrade to PMem storage. 2. For the virtual machine configuration file and for each virtual disk, select **Browse**, and select a datastore or Storage DRS cluster. **Note:** Configuration files cannot be stored on a PMem datastore. 3. If you want to migrate the virtual machine to a Storage DRS cluster and do not want to use Storage DRS with this virtual machine, select **Disable Storage DRS for this virtual machine** and select a datastore within the Storage DRS cluster. 4. Click **Next**.

12. If a compatibility problem appears in the Compatibility pane, fix the problem or select another destination storage for the migration.
13. Select the destination folder for the virtual machines and click **Next**.
14. Select the destination networks for the virtual machine migration.

Option	Action
Select a destination network for all VM network adapters connected to a valid source network.	1. Click the arrow in the **Destination Network** column and select **Browse**. 2. Select a destination network and click **OK**. 3. Click **Next**.
Select a different destination network for each VM network adapter connected to a valid source network.	1. Click **Advanced**. 2. Click the arrow in the **Destination Network** column and select **Browse**. 3. Select a destination network and click **OK**. 4. Click **Next**.

15. On the **Ready to complete** page, review the details and click **Finish**.

The virtual machines are moved or cloned to the destination folder in the current vCenter instance. You can monitor the migration process in the **Recent tasks** pane. If errors occur during the migration, the virtual machines revert to their original states and locations.

Migrating Virtual Machines Between Servers with DPU Devices

vSphere 8.0 introduces the VMware vSphere®Distributed Services Engine™ feature which allow offloading of some of your networking operations from an x86 host to a data processing unit (DPU). You can migrate virtual machines in a vSphere 8.0 environment with hosts that have data processing units (DPUs).

In vSphere 8.0, if you enable network offloads to a DPU device, you can use vSphere vMotion for migrating virtual machines when both hosts have DPU devices. To enable vMotion between hosts with DPU devices, you must perform multiple steps on the virtual machines, on the vCenter system, and on VMware NSX.

1. Create a vSphere Distributed Switch on a data center and enable the network offloads compatibility on the vSphere Distributed Switch. For more information about how to enable the network offloads compatibility on a host with a DPU device, see What is Network Offloads Compatibility in the *vSphere Networking* documentation.
2. Add to the switch only hosts with DPUs from the same vendor and of the same model.
3. Deploy NSX and configure an NSX transport node for the vSphere Distributed Switch. See the *NSX Installation Guide* documentation.
4. Enable the UPT support on the virtual machine that you want to migrate to and from a host with DPU devices. See *Add a Network Adapter to a Virtual Machine* in the *vSphere Virtual Machine Administration* documentation.

You can also use a DPU device on a host as a standard NIC. In this case, no additional configuration is required. You can use vSphere vMotion to migrate virtual machines to and from hosts with and without DPU devices.

What Is Encrypted vSphere vMotion

Encrypted vSphere vMotion secures confidentiality, integrity, and authenticity of data that is transferred with vSphere vMotion. vSphere supports encrypted vMotion of unencrypted and encrypted virtual machines across vCenter instances.

vSphere vMotion always uses encryption when migrating encrypted virtual machines. For virtual machines that are not encrypted, you can select one of the encrypted vSphere vMotion options.

What Is Encrypted in Encrypted vSphere vMotion

For encrypted disks, the data is transmitted encrypted in all cases. For unencrypted disks, the following applies:

- If disk data is transferred within a host, that is without changing the host, you change only the datastore, the transfer is unencrypted.
- If disk data is transferred between hosts and encrypted vMotion is used, the transfer is encrypted. If encrypted vMotion is not used the transfer is unencrypted.

For virtual machines that are encrypted, migration with vSphere vMotion always uses encrypted vSphere vMotion. You cannot turn off encrypted vSphere vMotion for encrypted virtual machines.

Encrypted vSphere vMotion States for Unencrypted Virtual Machines

For virtual machines that are not encrypted, you can set encrypted vSphere vMotion to one of the following states. The default is Opportunistic.

Disabled	Do not use encrypted vSphere vMotion.
Opportunistic	Use encrypted vSphere vMotion if the source and the destination hosts support it. Only ESX hosts of version 6.5 and later support encrypted vSphere vMotion.
Required	Allow only encrypted vSphere vMotion. If the source or the destination host does not support encrypted vSphere vMotion, migration with vSphere vMotion is not allowed.

When you encrypt a virtual machine, the virtual machine keeps a record of the current encrypted vSphere vMotion setting. If you later deactivate encryption for the virtual machine, the encrypted vMotion setting remains at Required until you change the setting explicitly. You can change the settings using **Edit Settings**.

> **Note:**
>
> Currently, you must use the vSphere APIs to migrate or clone encrypted virtual machines across vCenter instances. See *vSphere Web Services SDK Programming Guide* and *vSphere Web Services API Reference* documentation.

Migrating or Cloning Encrypted Virtual Machines Across vCenter Instances

vSphere vMotion supports migrating and cloning encrypted virtual machines across vCenter instances.

When migrating or cloning encrypted virtual machines across vCenter instances, the source and destination vCenter instances must be configured to share the key provider that was used to encrypt the virtual machine. In addition, the key provider name must be the same on both the source and destination vCenter instances, and must have the following characteristics:

- Standard key provider: The same key server (or key servers) must be in the key provider.
- vSphere Native Key Provider: Must have the same KDK.

  > **Note:**
  > You cannot clone or migrate an encrypted virtual machine using vSphere Native Key Provider to a standalone host, no matter that the source host resides in a cluster.

The destination vCenter ensures the destination ESX host has encryption mode set, ensuring the host is cryptographically "safe."

The following privileges are required when using vSphere vMotion to migrate or clone an encrypted virtual machine across vCenter instances.

- Migrating: **Cryptographic operations** > **Migrate** on the virtual machine
- Cloning: **Cryptographic operations** > **Clone** on the virtual machine

Also, the destination vCenter must have the **Cryptographic operations** > **EncryptNew** privilege. If the destination ESX host is not in "safe" mode, the **Cryptographic operations** > **RegisterHost** privilege must also be on the destination vCenter.

Certain tasks are not allowed when migrating virtual machines (non-encrypted or encrypted), either on the same vCenter or across vCenter instances.

- You cannot change the VM Storage Policy.
- You cannot perform a key change.

  > **Note:**
  > You can change the VM Storage Policy while cloning virtual machines.

vSphere Native Key Provider vMotion and Cross-vCenter vMotion

vSphere Native Key Provider supports vMotion and Encrypted vMotion across ESX hosts. Cross-vCenter vMotion is supported if vSphere Native Key Provider is configured on the destination host.

Encrypted vMotion and Intel QuickAssist Technology (QAT) Hardware Accelerator

The vMotion encryption and decryption tasks, which use CPU resources, are now offloaded to the hardware devices. Some processors, for example, 4th Gen Intel Xeon Scalable Processors and 5th Gen Intel Xeon Scalable Processors, have built-in QAT accelerators. Starting with vSphere 9.1, if your processor has QAT accelerator and the device is available for vMotion, by default the crypto operation for ecrypted vMotion is offloaded to the QAT accelerator. This reduces the CPU consumption for the vMotion task. If the QAT accelerator is needed for a virtual machine operation at the same time as an encrypted vMotion task, the vMotion task does not interrupt the virtual machine operation.

The feature is activated by default during new deployment and after an upgrade to vSphere 9.1. You can see the following message in the vmkernel logs:

```
DAS Crypto AEAD session handle created successfully. Crypto workload will be offloaded.
```

vSphere is compliant with the Federal Information Processing Standards (FIPS). If you want to make your end-to-end solution compliant with FIPS, you must choose FIPS-compliant hardware. If your environment requires FIPS compliance, you must validate that third-party hardware solutions, including Intel QAT, meet the necessary standards.

How to Activate Encrypted vSphere vMotion on Your Virtual Machine

You can activate encrypted vSphere vMotion during virtual machine creation. You can later change the encrypted vMotion state from the virtual machine settings.

Encrypted vMotion is supported only in vSphere 6.5 and later.

You can change the encrypted vMotion state only for virtual machines that are not encrypted.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Inventory**.
2. Select **Hosts and Clusters**, and navigate to the virtual machine.
3. Right-click the virtual machine and select **Edit Settings**.
4. Select **VM Options**.
5. Click **Encryption**, and select an option from the **Encrypted VMotion** drop-down menu.

Disabled	Do not use encrypted vMotion.
Opportunistic	Use encrypted vMotion if source and destination hosts support it. Only ESX hosts of version 6.5 and later support encrypted vMotion.
Required	Allow only encrypted vMotion. If the source or destination host does not support encrypted vMotion, migration with vMotion fails.

How to Isolate Traffic for Migrating Your Virtual Machines

You can use the vSphere vMotion TCP/IP stack to isolate traffic for vMotion and the provisioning TCP/IP stack to isolate traffic for cold migration, VM clones, and snapshots.

How to Place vSphere vMotion Traffic on the vMotion TCP/IP Stack of Your ESX Host

Learn how to use the vMotion TCP/IP stack to isolate traffic for vMotion and to assign a dedicated default gateway, routing table, and DNS configuration for this traffic. To enable the vMotion TCP/IP stack, assign a new VMkernel adapter to it.

Verify that the host is running ESXi 6.0 or later

By using a separate TCP/IP stack, you can handle vMotion traffic according to the topology of the network and as required for your organization:

- Route the traffic for migration of powered on virtual machines by using a default gateway. The gateway might be different from the gateway assigned to the default stack on the host.
- Assign a separate set of buffers and sockets.
- Avoid routing table conflicts that might otherwise appear when many features are using a common TCP/IP stack.
- Isolate traffic to improve security.

1. In the vSphere Client, click the main navigation menu ( ), and select **Inventory**.
2. Select **Hosts and Clusters**, and select the host.
3. Click the **Configure** tab.
4. Select **Networking**, and click **VMkernel adapters**.
5. Click **Add networking**.
6. On the Select connection type page, select **VMkernel Network Adapter** and click **Next**.
7. On the **Select target device** page, select the switch for the VMkernel adapter, and click **Next**.

Option	Description
Select an existing network	Use the physical adapter configuration of an existing distributed port group to send data from the VMkernel adapter to the external network.
Select an existing standard switch	Use the physical adapter configuration for the VMkernel adapter of an existing standard switch.
New vSphere standard switch	Assign a new physical adapter configuration for the VMkernel adapter on a new standard switch.

8. On the Port properties page, select **vMotion** from the **TCP/IP stack** drop-down menu.

   The vMotion traffic becomes the only service that is enabled. You cannot use this VMkernel adapter for traffic types other than vMotion.
9. Set the label, VLAN ID, and IP mode of the VMkernel adapter, and click **Next**.
10. Optional: On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter** check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

11. Optional: On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click **Add IPv6 address** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**. 3. To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

12. Review your settings selections on the Ready to complete page and click **Finish**.

After you create a VMkernel adapter on the vMotion TCP/IP stack, you can use only this stack for vMotion on this host. The VMkernel adapters on the default TCP/IP stack are deactivated for the vMotion service. If a live migration uses the default TCP/IP stack while you are configuring VMkernel adapters with the vMotion TCP/IP stack, the migration completes successfully. However, the involved VMkernel adapters on the default TCP/IP stack are deactivated for future vMotion sessions.

Assign a default gateway, and configure the DNS settings, congestion control, and maximum number of connections for the vMotion TCP/IP stack.

For more information on how to change the configuration of a TCP/IP stack on a host, see Change the Configuration of an TCP/IP Stack on a Host in the *vSphere Networking* documentation.

How to Place Traffic for Cold Migration, Cloning, and Snapshots on the Provisioning TCP/IP Stack

Learn how to use the provisioning TCP/IP stack to isolate traffic for cold migration, VM clones, snapshots, and cold data in hot migration, and to assign a dedicated default gateway, routing table, and DNS configuration for this traffic. To enable the Provisioning TCP/IP stack, assign a new VMkernel adapter to it.

Verify that the host is running ESXi 6.0 or later

By using a separate TCP/IP stack, you can handle vMotion traffic according to the topology of the network and as required for your organization:

- Route the traffic for migration of powered on virtual machines by using a default gateway. The gateway might be different from the gateway assigned to the default stack on the host.
- Assign a separate set of buffers and sockets.
- Avoid routing table conflicts that might otherwise appear when many features are using a common TCP/IP stack.
- Isolate traffic to improve security.

1. In the vSphere Client, navigate to the host.
2. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
3. Select **Hosts and Clusters, and select the host**.
4. Click the **Configure** tab.
5. Select **Networking**, and click **VMkernel adapters**.
6. Click **Add networking**.
7. On the Select connection type page, select **VMkernel Network Adapter** and click **Next**.
8. On the **Select target device** page, select the switch for the VMkernel adapter, and click **Next**.

Option	Description
Select an existing network	Use the physical adapter configuration of an existing distributed port group to send data from the VMkernel adapter to the external network.
Select an existing standard switch	Use the physical adapter configuration for the VMkernel adapter of an existing standard switch.
New vSphere standard switch	Assign a new physical adapter configuration for the VMkernel adapter on a new standard switch.

9. On the Port properties page, select **Provisioning** from the **TCP/IP stack** drop-down menu.

   The provisioning traffic becomes the only service that is enabled. You cannot use this VMkernel adapter for traffic types other than provisioning.
10. Set the label, VLAN ID, and IP mode of the VMkernel adapter, and click **Next**.
11. Optional: On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter** check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

12. Optional: On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click **Add IPv6 address** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**.

Option	Description
	3. To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

13. Review your settings selections on the Ready to complete page and click **Finish**.

After you create a VMkernel adapter on the provisioning TCP/IP stack, you can use only this stack for cold migration, cloning, and snapshots on this host. The VMkernel adapters on the default TCP/IP stack are deactivated for the provisioning service. If a live migration uses the default TCP/IP stack while you configure VMkernel adapters with the provisioning TCP/IP stack, the data transfer completes successfully. However, the involved VMkernel adapters on the default TCP/IP stack are deactivated for future cold migration, cross-host cloning, and snapshot sessions.

vCenter Limits on Simultaneous Migrations

vCenter places limits on the number of simultaneous virtual machine migration and provisioning operations that can occur on each host, network, and datastore.

Each operation, such as a migration with vMotion or cloning a virtual machine, is assigned a resource cost. Each host, datastore, or network resource, has a maximum cost that it can support at any one time. Any new migration or provisioning operation that causes a resource to exceed its maximum cost does not proceed immediately, but is queued until other operations complete and release resources. Each of the network, datastore, and host limits must be satisfied for the operation to proceed.

vMotion without shared storage, migrating virtual machines to a different host and datastore simultaneously, is a combination of vMotion and Storage vMotion. This migration inherits the network, host, and datastore costs associated with those operations. vMotion without shared storage is equivalent to a Storage vMotion with a network cost of 1.

Network Limits

Network limits apply only to migrations with vMotion. Network limits depend on the version of ESX and the network type. All migrations with vMotion have a network resource cost of 1.

Table 99: Network Limits for Migration with vMotion

Operation	ESX Version	Network Type	Maximum Cost
vMotion	7.0, 8.0, 9.0, 9.1	1GigE	4
vMotion	7.0, 8.0, 9.0, 9.1	10GigE	8

Datastore Limits

Datastore limits apply to migrations with vMotion and with Storage vMotion. A migration with vMotion has a resource cost of 1 against the shared virtual machine's datastore. A migration with Storage vMotion has a resource cost of 16 against the source datastore and 16 against the destination datastore.

Table 100: Datastore Limits and Resource Costs for vMotion and Storage vMotion

Operation	ESX Version	Maximum Cost Per Datastore	Datastore Resource Cost
vMotion	7.0, 8.0, 9.0, 9.1	128	1
Storage vMotion	7.0, 8.0, 9.0, 9.1	128	16

Host Limits

Host limits apply to migrations with vMotion, Storage vMotion, and other provisioning operations such as cloning, deployment, and cold migration. All hosts have a maximum cost per host of 8. For example, on an ESX 7.0 host, you can perform 2 Storage vMotion operations, or 1 Storage vMotion and 4 vMotion operations.

Table 101: Host Migration Limits and Resource Costs for vMotion, Storage vMotion, and Provisioning Operations

Operation	ESX Version	Derived Limit Per Host	Host Resource Cost
vMotion	7.0, 8.0, 9.0, 9.1	8	1
Storage vMotion	7.0, 8.0, 9.0, 9.1	2	4
vMotion Without Shared Storage	7.0, 8.0, 9.0, 9.1	2	4
Other provisioning operations	7.0, 8.0, 9.0, 9.1	8	1

vMotion Migration Compatibility Checks

During migration, use the **Migrate Virtual Machine** wizard to view information about the compatibility checks carried out using various criteria between the destination host or cluster and the migrating virtual machine.

When you select a host or a cluster, the Compatibility panel at the bottom of the **Migrate Virtual Machine** wizard displays information about the compatibility of the selected host or cluster with the virtual machine's configuration.

Compatibility Check Results

If the virtual machine is compatible, the panel displays the message `Compatibility checks succeeded`. If the virtual machine is not compatible with either the host's or cluster's configured networks or datastores, the compatibility window might display both warnings and errors:

- Warning messages do not stop migration. Often the migration is justified and you can continue with the migration despite the warnings.
- Errors might stop migration if no error-free destination hosts are available among the selected destination hosts. In this case, if you click **Next**, the wizard displays the compatibility errors again, and you cannot proceed to the next step.

Compatibility Checks During Migration Setup

When you attempt to move only the compute resource, the **Migrate Virtual Machine** wizard examines the source and destination hosts, the target resource pool, the datastore, and the network. When you attempt to move only the storage, the wizard checks the compatibility of the same objects except for the network.

When you move compute resources and storage together, the **Migrate Virtual Machine** wizard runs fewer compatibility checks. For example, if you move the compute resource, you select the target host or cluster under a vCenter instance. The wizard performs all necessary validation only against the selected host, and does not check the datastores available on the destination host. When you attempt to move the virtual machine to a cluster, the **Migrate Virtual Machine** wizard examines the compatibility against the host recommendation from vSphere DRS. The wizard directly validates the compatibility of the target datastore when you select it later.

Another compatibility check is whether vMotion is enabled on the source and target hosts.

Compatibility Checks for Virtual Hardware

Effects of a specific host CPU feature on compatibility depend on whether ESX exposes or hides them from virtual machines.

- Features that are exposed to virtual machines are not compatible when they do not match on the source and target hosts.
- Features that are not exposed to virtual machines are considered as compatible regardless of whether they match on the hosts.

Specific items of virtual machine hardware can also cause compatibility problems. For example, a virtual machine using an Enhanced VMXNET virtual NIC cannot be migrated to a host running a version of ESX that does not support Enhanced VMXNET.

CPU Compatibility and vSphere Enhanced vMotion Compatibility

vCenter performs compatibility checks before it allows migration of running or suspended virtual machines to another host. Compatibility checks ensure that the virtual machine is compatible with the target host CPU.

When you initiate a migration with vMotion or a migration of a suspended virtual machine, the **Migrate Virtual Machine** wizard checks the destination host for compatibility. If compatibility problems prevent migration, the wizard displays an error message.

The CPU instruction set available to the operating system and to applications running in a virtual machine is determined at the time that a virtual machine is powered on. This CPU feature set is based on the following items:

- Host CPU family and model.
- Settings in the BIOS that might deactivate CPU features.
- ESX version running on the host.
- The virtual hardware compatibility setting of the virtual machine.
- The guest operating system of the virtual machine.

CPUs have new instructions or features added in each new CPU generation. While a virtual machine is powered-on, the set of features available to the virtual machine remains constant regardless of what host CPU the virtual machine runs on.

When you migrate a powered-on or suspended virtual machine, the set of CPU features available on the target host must include all of the features available to the virtual machine when it was powered-on prior to the migration. A compatibility check is performed to ensure that after you migrate a virtual machine, the target host CPU will support all of the instructions and features that the powered-on virtual machine requires. If the target host CPU does not supports all of them, you cannot proceed with the migration.

When you migrate a virtual machine by using vMotion, the processors of the target host must provide the same instructions to the virtual machine after migration that the processors of the source host provided before migration. Clock speed, cache size, and number of cores can differ between source and target processors.

To improve CPU compatibility between hosts of varying CPU feature sets, some host CPU features can be hidden from the virtual machine by placing the host in an Enhanced vMotion Compatibility (EVC) cluster. If the hosts in a cluster have processors from the same CPU vendor but different CPU generations, you can use Enhanced vMotion Compatibility to make all host CPUs migration compatible. When you configure EVC on a cluster, all the hosts in the cluster become migration compatible, even if they are a mixture of CPU generations. The host CPUs in that cluster are treated as if they are of the same generation and provide the same set of instructions and features.

For example, if a cluster contains hosts with Intel CPUs of different generations: Intel Cascade Lake, Intel Ice Lake, and Intel Sapphire Rapids, after you configure EVC, all hosts in the cluster can appear as Intel Cascade Lake during virtual machine power-on and migration.

Pre-Defined EVC Mode

The EVC mode of a cluster can be chosen from a list of pre-defined modes that closely match the CPU generations from Intel or AMD, such as Intel Cascade Lake and AMD Zen 2. After you select a pre-defined EVC mode and apply it to the cluster, you can only add ESX hosts with CPUs that have the same or greater features than that of the EVC mode of the cluster. After you add a host to a cluster with pre-defined EVC mode, the host CPU features that are available to a virtual machine are restricted to the features of the EVC mode. You can also select a pre-defined vSGA EVC mode and apply it to all hosts in the cluster. For more information about vSGA, see Virtual Shared Graphics Acceleration.

Note:

Pre-defined EVC modes closely match CPU generations. However, CPUs from the same generation might support a close but not identical set of instructions or features. Pre-defined EVC modes per generation match the highest set of instructions and features for the specified CPU generation. Hosts with CPUs from the same generation but with a less rich feature set might not be allowed to the cluster with a pre-defined EVC mode for this generation. In such cases you can use custom EVC mode.

Custom EVC Mode

Starting with vSphere 9.0, you can also use a custom EVC type. With the custom EVC type, you can select a set of hosts or clusters that are used to create a custom EVC mode based on the highest set of CPU and vSGA features common to all selected hosts and hosts included in the selected clusters. By using a custom EVC mode, you can select specific hosts or clusters, and their host CPUs are used to construct the new custom EVC mode.

A custom EVC mode can be created from a set of hosts or clusters within the vCenter instance, and from hosts or clusters managed by other vCenter instances if the vCenter instances operate in linked mode. However, the created custom EVC mode can only be applied to clusters managed by the current vCenter instance.

You can apply pre-defined or custom EVC modes to clusters and virtual machines.

CPU Families and Feature Sets Compatibility with vSphere vMotion

Processors are grouped into families. Processors within a given family generally have similar feature sets.

Processor vendors define processor families. You can distinguish different processor versions within the same family by comparing the processors' model, stepping level, and extended features. Each vendor's CPU generations, identifiable by family, model and stepping, add new instructions and features.

By default, vCenter does compatibility checks to ensure that the CPU features of a host are sufficient:

- To power-on a virtual machine on that host.
- For the host to be the target host of a vMotion migration.
- For the host to be the target host of migration of a suspended virtual machine.

For more information on identifying Intel processors and their features, see *How to Identify My Intel Processor*, available from Intel. For more information on identifying AMD processors and their features, see *Obtaining Processor Information Via the CPUID Instruction*, available from AMD.

Enhanced vMotion Compatibility

You can use the Enhanced vMotion Compatibility (EVC) feature to help ensure vMotion compatibility between all hosts in a cluster. EVC ensures that all hosts in a cluster present the same CPU feature set to virtual machines, even if the actual CPUs on the hosts differ. Using EVC prevents migrations with vMotion from failing because of incompatible CPUs.

When you configure EVC on a cluster, you configure all host processors in the cluster to present the same feature set. The EVC mode must be equivalent to, or a subset of, the feature set of the host with the smallest feature set in the cluster.

Activating EVC only affects processor features that affect vMotion compatibility. Activating EVC does not prevent a virtual machine from taking advantage of faster processor speeds or increased numbers of CPU cores that might be available on newer hosts.

EVC cannot prevent virtual machines from accessing hidden CPU features in all circumstances. Applications that do not follow CPU vendor recommended methods of feature detection might behave unexpectedly in an EVC environment. VMware EVC cannot be supported with ill-behaved applications that do not follow the CPU vendor recommendations. For more information about creating well-behaved applications, see *What is an ill-behaved application, and why does it affect EVC?*, in Broadcom Knowledge Base article 313545.

The EVC feature for Virtual Shared Graphics Acceleration (vSGA) allows multiple virtual machines to share GPUs installed on ESX hosts and leverage the 3D graphics acceleration capabilities. vSGA enables efficient use of hardware acceleration by grouping virtual machines with different performance requirements on one GPU. For more information about vSGA, see Virtual Shared Graphics Acceleration.

Pre-defined and Custom EVC Modes

Starting with vSphere 9.0, in addition to pre-defined EVC modes, you can also use a custom EVC mode. With the pre-defined EVC mode, you can choose from a list of EVC and vSGA modes which closely match vendors' CPU generations. With the custom EVC mode, you can select a set of hosts or clusters from which to construct a custom EVC mode based on the highest set of CPU and vSGA features common to all selected hosts. You cannot modify the feature sets of the EVC types.

Enhanced vMotion Compatibility Requirements for Hosts

To improve CPU compatibility between hosts that have varying CPU feature sets, you can place the host in a cluster with activated Enhanced vMotion Compatibility (EVC). Hosts in an EVC cluster and hosts that you add to an existing EVC cluster must meet EVC requirements.

- To activate or change the EVC mode of a cluster, power off all virtual machines in the cluster that require a feature set greater than the EVC mode that you intend to enable. You can also migrate these virtual machines out of the cluster.
- All hosts in the cluster must meet the following requirements:

Requirements	Description
Supported ESX version	ESX 7.0 or later.
vCenter	The host must be connected to a vCenter system.
CPUs	A single vendor, either AMD or Intel.
Advanced CPU features enabled	Enable these CPU features in the BIOS if they are available: • Hardware virtualization support (AMD-V or Intel VT) • AMD No eXecute(NX) • Intel eXecute Disable (XD) **Note:** Hardware vendors sometimes deactivate particular CPU features in the BIOS by default. You might have problems enabling EVC because the EVC compatibility checks detect the absence of features that are expected to be present for a particular CPU. If you cannot enable EVC on a system with a compatible processor, ensure that all features are enabled in the BIOS.
Supported CPUs for the EVC mode that you want to enable	To check EVC support for a specific processor or server model, see the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
Configured for vMotion	See Host Configuration for vSphere vMotion.

- Custom EVC is available in vCenter instances of version 9.0 or later.

How to Configure the EVC Mode of an Existing Cluster

Learn how to activate or change the vSphere Enhanced vMotion Compatibility (EVC) mode on an existing cluster to ensure vMotion CPU compatibility between the hosts in the cluster.

- Verify that the hosts in the cluster meet the requirements listed in Enhanced vMotion Compatibility Requirements for Hosts.
- Verify that all hosts in the cluster are connected and registered on vCenter. The cluster cannot contain a disconnected host.
- Verify that all hosts in the cluster have supported CPUs for the EVC mode you want to enable. For a list of supported CPUs, see the Broadcom Compatibility Guide at https://compatibilityguide.broadcom.com/.
- Power off all virtual machines with no EVC mode or with an EVC mode that is greater than the EVC mode that you plan to enable on the cluster. To determine the EVC mode of all virtual machines on the hosts in the cluster, see How to Determine the EVC Mode of Your Virtual Machine.

- When you change the EVC mode on a cluster, virtual machines must be in the following power states, depending on whether you raise or lower the EVC mode.

EVC Mode	Virtual Machine Power Action
Raise the EVC mode to a CPU baseline with more features.	Running virtual machines can remain powered on. New EVC mode features are not available to the virtual machines until they are powered off and powered back on again. A full power cycling is required. Rebooting the guest operating system or suspending and resuming the virtual machine is not sufficient. **Tip:** You can configure a virtual machine to power cycle when its guest OS is rebooted, by setting the `vmx.reboot.powerCycle` VM parameter to `TRUE`. When the virtual machine is powered back on, this configuration parameter is removed. To edit the virtual machine configuration settings, you can use the VMware Host Client. See *Edit the Configuration File Parameters in the VMware Host Client* topic in the *vSphere Single Host Management - VMware Host Client* guide. For information on how to set this parameter to a group of virtual machines using VMware PowerCLI, see the VMware vSphere Blog post.
Lower the EVC mode to a CPU baseline with fewer features.	Power off virtual machines if they are powered on and running at a higher EVC mode than the one you intend to enable.

- If you want to use custom EVC mode, verify that the vCenter version is 9.0 or later.

Several EVC approaches are available to ensure CPU compatibility:

- You can raise the EVC mode to expose more CPU features.
- You can lower the EVC mode to hide CPU features and increase compatibility.
- If all the hosts in a cluster are compatible with a newer EVC CPU or Graphics (vSGA) mode, you can change the EVC CPU or Graphics (vSGA) mode of an existing EVC cluster.

For more information about vSGA, see Virtual Shared Graphics Acceleration.

Note:

When you take a snapshot of a virtual machine in an EVC cluster, keep in mind the following use cases:

- If you take a snapshot of a running virtual machine before an EVC downgrade, an attempt to revert to that snapshot after the EVC downgrade fails and the virtual machine is left in a suspended state.
- If you take a snapshot of a virtual machine before an EVC upgrade, you can revert to that snapshot before or after the EVC upgrade.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Select a cluster in the vSphere inventory.
4. Power off all the virtual machines on the hosts with feature sets greater than the EVC mode.
5. Click the **Configure** tab.
6. Under Configuration, select VMware EVC, and click **Edit**.
7. If you want to select a pre-defined EVC type, select **Pre-defined** from the **EVC type**drop-down menu.
   a) From the **Host type**drop-down menu, select whether you want to activate the EVC mode for AMD hosts or Intel hosts.
   b) From the **CPU Mode** drop-down menu, select a CPU feature set.
   c) From the **Graphics Mode (vSGA)** drop-down menu, select a graphics feature set.

Option	Description
Baseline Graphics	Applies the Baseline Graphics feature set that includes features through Direct3D 10.1/OpenGL 3.3. **Note:** To configure the vSGA mode to apply the Baseline Graphics set that includes features through Direct3D 10.1/OpenGL 3.3, the virtual machine must be compatible with ESX 6.5 or later.
D3D 11.0 class features	Applies the baseline graphics feature set that includes features through Direct3D 11.0/OpenGL 4.3. **Note:** When you create a cluster and you enable EVC graphics mode Direct3D 11.0/OpenGL 4.3 class features, you can add only ESX 8.0 hosts that support Direct3D 11.0/OpenGL 4.3 to the cluster. When you try to add an ESX 8.0 host that does not support Direct3D 11.0/OpenGL 4.3 to the cluster, the operation fails with an error message.

8. If you want to configure a custom EVC mode, select Custom from the EVC type drop down menu.
   a) From the list of inventory objects, select the hosts or clusters to be used to construct the custom EVC mode feature set.

      Note:

      If your vCenter instances are part of a vCenter linking group, you can select hosts or clusters from any vCenter instance part of the vCenter linking group.

      If some of your clusters have EVC modes applied, you cannot select specific hosts from that cluster but only the whole cluster.
9. Click **Next**.

   A validation process starts.

   If the EVC CPU mode or the vSGA mode that you selected cannot be applied, the **Compatibility** pane displays the reason, and the relevant hosts for each reason.

10. Review the EVC mode details and click **Finish**.

How to Configure the EVC Mode of a Virtual Machine

Per virtual machine EVC is deactivated by default. You can activate, deactivate, and change the EVC mode of a virtual machine to ensure its seamless migration across clusters, vCenter systems, and data centers that have different processors.

- Power off the virtual machine.
- Verify that the virtual hardware version is 14 or later.

The per virtual machine EVC feature facilitates the migration of the virtual machine beyond the cluster and across vCenter systems and data centers that have different CPU generations. The EVC mode of a virtual machine is independent from the EVC mode defined at the cluster level. The cluster-based EVC mode limits the CPU features a host exposes to virtual machines. The per virtual machine EVC mode determines the set of host CPU features that a virtual machine requires to power on and migrate. By default, when you power on a newly created virtual machine, it inherits the feature set of its parent EVC cluster or host. However, you can change the EVC mode for each virtual machine separately. You can raise or lower the EVC mode of a virtual machine. Lowering the EVC mode increases the CPU compatibility of the virtual machine. You can also use the API calls to customize the EVC mode further.

There are several differences between the way the EVC feature works at the host cluster level and at the virtual machine level.

- Unlike cluster-based EVC, you can change the per-VM EVC mode only when the virtual machine is powered off.
- With cluster-based EVC, when you migrate a virtual machine out of the EVC cluster, a power cycle resets the EVC mode that the virtual machine has. With Per-VM EVC, the EVC mode becomes an attribute of the virtual machine. A power cycle does not affect the compatibility of the virtual machine with different processors.
- When you configure EVC at the virtual machine level, the per virtual machine EVC mode overrides cluster-based EVC. If you do not configure per virtual machine, when you power on the virtual machine, it inherits the EVC mode of its parent EVC cluster or host.
- If a virtual machine is in an EVC cluster and the per virtual machine EVC is also enabled, the EVC mode of the virtual machine cannot exceed the EVC mode of the EVC cluster in which the virtual machine runs. The baseline feature set that you configure for the virtual machine cannot contain more CPU features than the baseline feature set applied to the hosts in the EVC cluster. For example, if you configure a cluster with the Intel "Merom" Generation EVC mode, you should not configure a virtual machine with any other Intel baseline feature set. All other sets contain more CPU features than the Intel "Merom" Generation feature set and as a result of such configuration, the virtual machine fails to power on.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Navigate to a virtual machine in the inventory, and click the **Configure** tab.
4. Under Configuration, select **VMware EVC**.

   The pane shows details about the EVC mode of the virtual machine and CPUID details.

   Important:
   For newly created virtual machines, the EVC mode that shows in the **VMware EVC** pane is deactivated.

   For powered off virtual machines, the **VMware EVC** pane always shows the EVC status defined at the virtual machine level.

   For powered on virtual machines with per-VM EVC activated, the VMware EVC pane shows the EVC status defined at the virtual machine level.

   For powered on virtual machines with per-VM EVC deactivated, the VMware EVC pane shows the EVC mode that the virtual machine inherits from its parent EVC cluster or host.

5. Click the **Edit** button.

   The **Configure EVC mode** dialog box appears.

6. If you want to select a pre-defined EVC type, select **Pre-defined** from the **EVC type**drop-down menu.
   a) From the **Host type**drop-down menu, select whether you want to activate the EVC mode for AMD hosts or Intel hosts.
   b) From the **CPU Mode** drop-down menu, select the baseline CPU feature set that you want to enable for the cluster.

      Important:

      If the virtual machine is in an EVC cluster and the per-VM EVC mode exceeds the EVC mode for the cluster, the virtual machine fails to power on. The baseline CPU feature set for the virtual machine must not contain more CPU features than the baseline CPU feature set of the cluster.

   c) From the **Graphics Mode (vSGA)** drop-down menu, select a baseline graphics feature set.

Option	Description
Baseline Graphics	Applies the Baseline Graphics feature set that includes features through Direct3D 10.1/OpenGL 3.3. **Note:** To configure the vSGA mode to apply the Baseline Graphics set that includes features through Direct3D 10.1/OpenGL 3.3, the virtual machine must be compatible with ESX 7.0 Update 1 or earlier.
D3D 11.0 class features	Applies the baseline graphics feature set that includes features through Direct3D 11.0/OpenGL 4.3 **Note:** To configure the vSGA mode to apply the Baseline Graphics set that includes features through Direct3D 11.0/OpenGL 4.3, verify that the virtual machine is compatible with ESX 8.0 or later and has installed VMware Tools 12.0.0 or later. To power on a virtual machine, configured with Direct3D 11.0/OpenGL 4.3, verify that the ESX host graphics hardware is available.

7. If you want to configure a custom EVC type, select **Custom** from the **EVC type** drop-down menu.
   a) From the list of inventory objects, select the hosts or clusters to be used to construct the EVC mode feature set.

      Note:

      If your vCenter instances are part of a vCenter linking group group, you can select hosts or clusters from any vCenter instance part of the vCenter linking group group.

      If some of your clusters have EVC modes applied, you cannot select specific hosts from that cluster but only the whole cluster.

8. Click **Next**.

   A validation process starts.

   If the EVC CPU or the vSGA mode that you selected cannot be applied, the **Compatibility** pane displays the reason, and the relevant hosts for each reason.

9. Review the EVC mode details and click **Finish**.

How to Determine the EVC Mode of Your Virtual Machine

The EVC mode of a virtual machine determines the CPU and graphics features that a host must have in order for the virtual machine to migrate to that host and power on. The EVC mode of a virtual machine is independent from the EVC mode that you configure for the cluster in which the virtual machine runs.

The EVC mode of a virtual machine is determined when the virtual machine powers on. At power-on, the virtual machine also determines the EVC mode of the cluster in which it runs. If the EVC mode of a running virtual machine or the entire EVC cluster is raised, the virtual machine does not change its EVC mode until it is powered off and powered on again. This means that the virtual machine does not use any CPU features exposed by the new EVC mode until the virtual machine is powered off and powered on again.

For example, you create an EVC cluster that contains hosts with Intel processors and you set the EVC mode to Intel "Merom" Generation (Xeon Core 2). When you power on a virtual machine in this cluster, it runs in the Intel Merom Generation (Xeon Core 2) EVC mode. If you raise the EVC mode of the cluster to Intel "Penryn" Generation (Xeon 45 nm Core 2), the virtual machine retains the lower Intel "Merom" Generation (Xeon Core 2) EVC mode. To use the feature set of the higher EVC mode, such as SSE4.1, the virtual machine must be powered off and powered on again.

1. In the vSphere Client, click the main navigation menu

   (**),**

   and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Navigate to a cluster or a host in the inventory.
4. Click the **VMs** > **Virtual Machines** tab.

   A list of all virtual machines in the selected cluster or on the selected host appears.
5. To verify the status of the CPU mode, check the **EVC CPU Mode** column.
   a) If the column is not visible, click **Manage Columns** at the bottom left corner of the **Virtual Machines** table.

      The **Show Columns** pop-up appears.
   b) To show the **EVC CPU Mode** column, locate and select the respective check box from the list.

   The **EVC CPU Mode** column shows the CPU modes of all virtual machines in the cluster or on the host.

   Important:
   For each virtual machine, the **EVC CPU Mode** column displays the EVC mode defined at the virtual machine level.

   However, if you do not configure per-VM EVC for a virtual machine, the virtual machine inherits the EVC mode of its parent cluster or host. As a result, for all virtual machines that do not have per-VM EVC configured, the **EVC CPU Mode** column displays the inherited EVC mode of the parent host or cluster.

   If the virtual machine is in an EVC cluster, the EVC mode that you see in the **EVC CPU Mode** column is defined in the following manner.

   - When the virtual machine is powered on, the **EVC CPU Mode** column displays either the per-VM EVC mode, or the cluster-level EVC mode.

Per-VM EVC	Cluster-Level EVC	EVC Mode for the Virtual Machine
Activated	Activated	Enabled. The **EVC CPU Mode** column displays the EVC mode of the virtual machine.
Deactivated	Activated	Enabled. The **EVC CPU Mode** column displays the EVC mode of the EVC cluster.

- When the virtual machine is powered off, the **EVC CPU Mode** column displays the per-VM EVC mode. If per-VM EVC is deactivated, the **EVC CPU Mode** column for the virtual machine is empty.

When the virtual machine is not in an EVC cluster and per-VM EVC is not configured, the EVC mode that you see in the **EVC CPU Mode** column is defined in the following manner.

- When the virtual machine is powered on, the **EVC CPU Mode** column displays the EVC mode of the parent host.
- When the virtual machine is powered off, the **EVC CPU Mode** column is empty.

6. To verify the status of the graphics mode, check the **EVC Graphics Mode (vSGA)** column.
   a) If the column is not visible, click **Manage Columns** at the bottom left corner of the **Virtual Machines** table.
      The **Show Columns** pop-up appears.
   b) To show the **EVC Graphics Mode (vSGA)** column, locate and select the respective check box from the list.

   The **EVC Graphics Mode (vSGA)** column displays the baseline graphics features set. To view the baseline graphics, you must enable **3D graphics** in the virtual machine.

   For information about configuring 3D graphics in a virtual machine, see *Configure 3D Graphics and Video Cards* in the *vSphere Virtual Machine Administration* documentation.

How to View Feature Capabilities Details for Your EVC Cluster

The feature set that is exposed by an EVC cluster corresponds to the feature set of a particular type of processor. Processor feature sets are described by a set of feature flags, most of which correspond to flags returned by the CPUID instruction.

You can view the feature flags currently exposed by the hosts in an EVC cluster.

1. In the vSphere Client, click the main navigation menu
   (**),**
   and select **Inventory**.
2. Select **Hosts and Clusters**.
3. Select a cluster in the inventory.
4. Click the **Configure** tab.
5. Select **VMware EVC** and expand **Current Feature Capabilities Details**.

This VMware EVC panel displays the CPUID feature flags that EVC enforces for the hosts in this cluster. For information about CPUID feature flags, see the Intel and AMD websites.

What Is vSphere Developer Center

The Developer Center is a single point of entry for developers that provides tools to manage API structure and to capture user actions to translate them into executable code.

The vSphere Client Developer Center provides tools for automation experts, devops engineers, and developers to find the resources to manage API structures and to capture vSphere Client actions to translate them into PowerCLI.

What Is vSphere API Explorer

The API Explorer allows you to browse and invoke vSphere REST APIs supported by the system and provides information and context around the API calls.

With the API Explorer, you can choose an API endpoint from your environment and retrieve a list of vSphere REST APIs. You can review details like available parameters, expected responses, and response status codes, and you can invoke the APIs against the live environment. The available APIs depend on the role of the selected endpoint.

How Do You Retrieve APIs Using vSphere API Explorer

You can use the API Explorer to retrieve available vSphere REST APIs from a selected endpoint and receive information and context around API calls.

1. In the vSphere Client, click the main navigation menu
   ( ),
   and select **Developer Center**.
2. Click the **API Explorer** tab.
3. From the **Select Endpoint** drop-down menu, select an endpoint from the environment.
4. From the **Select API** drop-down menu, select an API. The listed APIs are the ones publicly provided by the API explorer in vCenter.
5. Optional: You can use the filter text box to filter your results. For example, enter `health` to view a list of methods related to monitoring the health of the selected API.
6. Select an API category from the list.
7. Select a method from the list.

   You can review deprecated APIs by using the toggle button next to each method from the list. Avoid using deprecated APIs. Deprecated APIs might become unresponsive in the future and cause unexpected failures in your automation scripts.

   Detailed information about the method appears.
8. If a section about parameter details appears for the selected method, enter a method parameter value in the **Value** text box.
9. Optional: To invoke the method against the live environment, click **Execute**.
   a) If a warning dialog box appears, click **Yes**.

   The result for the invoked method appears in the response box.
10. Optional: To copy the result for the invoked method to your clipboard, click **Copy Response**.
11. Optional: To download the result for the invoked method, click **Download**.

What Is vSphere Code Capture

Code Capture records user actions and translates them into executable code.

Code Capture gives you the ability to record actions taken in the vSphere Client and output them as usable PowerCLI code. You can then copy the code or download it as a script and use it in a PowerShell session to execute the task.

> **Note:**
>
> Only vCenter calls are recorded. Calls made on operations regarding roles, privileges, tags, content libraries, and storage policies are not recorded.

How Do You Record Actions by Using vSphere Code Capture

After you enable Code Capture, you can use it to record a session.

To use Code Capture to record a session, you must first enable Code Capture.

You can use Code Capture to record actions taken in the vSphere Client to produce a PowerCLI code output.

> **Note:**
> Calls made on operations regarding roles, privileges, tags, content libraries, and storage policies are not recorded. Sensitive data such as passwords are not recorded.

1. In the vSphere Client, click the main navigation menu ( ), and select **Developer Center**.
2. Click the **Code Capture** tab.
3. Optional: If Code Capture is not enabled, click the toggle to enable Code Capture.
4. To start a recording, navigate to your desired pane and click the red record button in the top pane. To start recording immediately, click **Start Recording**.

   While a recording is in progress, the red record button in the top pane blinks.
5. Optional: To clear the code captured in a previous session and start a new session, click **Clear and Start Another**.
6. To stop a recording, click the red record button in the top pane, or navigate to the **Code Capture** tab in the Developer Center and click **Stop Recording**.

   The recorded code appears in the code pane.
7. Optional: Click **Copy** to copy the code or **Download** to download it as a PowerCLI script.
8. To clear the current code and start another recording, click **Clear and Start Another** or navigate to your desired pane and click the red record button in the top pane.

The recorded code appears in the code pane. You can copy the code, download it, or clear the code to start another recording.

Automating Management Tasks Using VMware Cloud Foundation Operations orchestrator

VMware Cloud Foundation Operations orchestrator is a platform that provides a library of extensible workflows. By using the workflow library, you can automate and configure processes to manage the vSphere infrastructure, other VMware technologies, and third-party technologies.

VMware Cloud Foundation Operations orchestrator exposes every operation in the vCenter API so that you can integrate all these operations into your own automated processes.

What Are VMware Cloud Foundation Operations orchestrator Workflows

A workflow is a series of actions and decisions that are automated to run sequentially after you run the workflow. VMware Cloud Foundation Operations orchestrator (VCF Operations orchestrator) provides a library of workflows that perform common management tasks.

Basics of Workflows

Workflows consist of a schema, variables, and input and output parameters. The workflow schema is the main component of a workflow as it defines all the workflow elements and the logical task flow of the workflow. The workflow variables and parameters are used by workflows to transfer data. VMware Cloud Foundation Operations orchestrator saves a workflow token every time a workflow runs, recording the details of that specific run of the workflow. This token contains all parameters related to the workflow run. For example, if you run a workflow three times, three workflow tokens are saved.

With the vSphere Client, you can run and schedule workflows on selected objects from the vSphere inventory. You cannot create, delete, edit, and manage workflows in the vSphere Client. You develop and manage workflows in the VMware Cloud Foundation Operations orchestrator client.

Input Workflow Parameters

To run, most workflows require a certain set of input parameters. The workflow processes input parameters that the user, an application, another workflow, or an action passes to it.

For example, if a workflow resets a virtual machine, the workflow requires the name of the virtual machine as an input parameter.

Output Workflow Parameters

The workflow output parameters represent the result from the workflow run. Some workflows and workflow elements can change the output parameters of the workflow when they run. While they run, workflows can receive the output parameters of other workflows as input parameters.

For example, if a workflow creates a snapshot of a virtual machine, the output parameter for the workflow is the resulting snapshot.

Workflow Input Form

When you run a workflow in the vSphere Client, the client loads the workflow presentation. You provide the input parameters of the workflow in the workflow input form.

Waiting for Input

Some workflows require user input during their run and the run is suspended either until the user provides the required information, or until the workflow run times out.

What Administration Tasks Can You Perform on vSphere Objects by Using VCF Operations orchestrator

By using VCF Operations orchestrator view in the vSphere Client, you can perform administration tasks such as running and scheduling workflows, and viewing the list of available workflows.

From the VCF Operations orchestrator view in the vSphere Client, you can work with workflows. Working with workflows includes the following tasks:

- Associating workflows with specific vSphere inventory objects such as virtual machines, ESX hosts, clusters, resource pools, and folders.
- Exporting and importing existing associations of workflows with vSphere inventory objects for backup purposes or to import them to another vCenter instance.
- Editing associations of workflows with vSphere inventory objects such as virtual machines, ESX hosts, clusters, resource pools, folders, and others.
- Viewing information about workflow runs and about workflows waiting for user intervention.
- Running and scheduling workflows on vSphere objects.

Managing Context Actions of Workflows on Your vSphere Inventory Objects

By using VCF Operations orchestrator, you can associate workflows with the different vSphere object types to see more workflows displayed in the context menu when you right-click a vSphere inventory object. You can also run these workflows on more object types.

You can add and edit context actions, and export and import configuration files containing the context actions of workflows on vSphere objects.

Workflows associated with inventory object types are listed in the context menu that appears when you right-click the inventory objects and in the **Actions** menu.

You can assign Automation Orchestrator groups. The users in the Automation Orchestrator groups and users from the VCF Operations orchestrator Administrator group in the vSphere Client can use the defined context actions on the vSphere inventory objects.

Assign an Automation Orchestrator Group in the vSphere Client

With VCF Operations orchestrator, you can assign an Automation Orchestrator group. The users in the group can use the defined context actions on the vSphere inventory objects.

- Verify that groups are created in the VMware Cloud Foundation Operations orchestrator client. For more information about VCF Operations orchestrator groups and how to create them, see the *Workload Orchestration* documentation.
- Log in to the vSphere Client as a member of the VMware Cloud Foundation Operations orchestrator Administrators group. For more information about VMware Cloud Foundation Operations orchestrator roles and groups, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click the **Group Assignments** tab.

   A list of all groups which can use context actions is displayed.
3. Click **Assign**.
4. Select a group from the list and click **Assign**.

   The group appears in the list of assigned groups.

Associate Your Workflows with vSphere Inventory Object Types

You can associate VMware Cloud Foundation Operations orchestrator workflows with a vSphere object type, such as host, to run the workflows directly on the inventory objects of that type.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Log in to the vSphere Client as a member of the VMware Cloud Foundation Operations orchestrator Administrators group. For more information about VMware Cloud Foundation Operations orchestrator roles and groups, see the *Workload Orchestration* documentation.

Workflows associated with inventory object types are listed in the context menu that appears when you right-click an inventory object, and in the **Actions** menu.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click the **Context Actions** tab.
3. Click **Add** to add a workflow.
4. Navigate through the workflow library to find the workflow to add, and select the workflow.
5. Under Type of Object, select the vSphere object types with which you want to associate the workflow.
6. Optional: Use the add or remove icons to add or remove object types.

   To remove an object type, more than one object types must be selected.
7. Optional: Select the **Allow multiple selection** check box.

   With multi-selection enabled, you can select multiple vSphere objects of the same type when you run the workflow.
8. Click **Assign**.

   The workflow appears in the list of workflows.

Edit the Context Actions of Your Workflows on vSphere Objects

After you associate a VMware Cloud Foundation Operations orchestrator workflow with objects from the vSphere inventory, you can edit the context actions of workflows on the objects from the vSphere inventory.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

- Log in to the vSphere Client as a member of the VMware Cloud Foundation Operations orchestrator Administrators group. For more information about VMware Cloud Foundation Operations orchestrator roles and groups, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click the **Context Actions** tab.
3. Click the workflow that you want to edit.

   A dialog box appears.
4. Change the context action properties.
5. Click **Save**.

Export the Context Actions of Your Workflows on vSphere Objects

You can transfer the context actions of VMware Cloud Foundation Operations orchestrator workflows on objects in the vSphere inventory by using a JSON file.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Log in to the vSphere Client as a member of the VMware Cloud Foundation Operations orchestrator Administrators group. For more information about VMware Cloud Foundation Operations orchestrator roles and groups, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click the **Context Actions** tab.
3. Click **MORE** > **Export**.
4. Select a location where you want to save the JSON file, and click **Save**.

Import Context Actions of Your Workflows on vSphere Objects

You can import a JSON file that contains the context actions of VMware Cloud Foundation Operations orchestrator workflows on objects in the vSphere inventory.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Log in to the vSphere Client as a member of the VMware Cloud Foundation Operations orchestrator Administrators group. For more information about VMware Cloud Foundation Operations orchestrator roles and groups, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click the **Context Actions** tab.
3. Click **MORE** > **Import**.
4. Browse to select the JSON file to import, and click **Open**.

VMware Cloud Foundation Operations orchestrator compares the two sets of workflow context actions and imports the missing workflow context actions.

Working with VCF Operations orchestrator Workflows in the vSphere Client

In the vSphere Client, you can view information about VCF Operations orchestrator workflows, run, and schedule workflows.

You can perform some scheduling and running tasks on the VCF Operations orchestrator workflows from the VCF Operations orchestrator view in the vSphere Client.

You can schedule a workflow to run at a specified time or start a workflow directly, by right-clicking a vSphere inventory object and selecting **VCF Operations orchestrator**.

Workflow tasks include:

- Running workflows on vSphere inventory objects, such as virtual machines, ESX hosts, clusters, resource pools, and folders.
- Viewing information about workflow runs.
- Viewing information about workflows waiting for user interaction.
- Searching for a specific workflow from the list of available workflows.
- Scheduling workflows.

Run Your Workflows on vSphere Inventory Objects

You can automate management tasks in vSphere by running VCF Operations orchestrator workflows directly on objects from the vSphere inventory.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Verify that you have workflows associated with the vSphere inventory objects. See Associate Your Workflows with vSphere Inventory Object Types.

1. In the vSphere Client home page, select **Global Inventory Lists** .
2. From the left navigation pane, select an inventory category.
3. Right-click the object that you want to run the workflow on, and click **VCF Operations orchestrator** > **Run Workflow**.
   A dialog box appears.
4. From the drop-down menu, select the workflow that you want to run.

   Note:
   If you cannot find the expected workflows, you might have to associate them to the specified vSphere inventory object.
5. Provide the required workflow parameters.
6. Click **Run**.

Schedule Your Workflows on vSphere Inventory Objects

You can automate management tasks in vSphere by scheduling VCF Operations orchestrator workflows directly on objects from the vSphere inventory.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Verify that you have workflows associated with the vSphere inventory objects. See Associate Your Workflows with vSphere Inventory Object Types.

1. In the vSphere Client home page, select **Global Inventory Lists** .
2. From the left navigation pane, select an inventory category.
3. Right-click the object that you want to run the workflow on, and click **VCF Operations orchestrator** > **Schedule Workflow**.

   A dialog box appears.
4. In the **Task name** text box, enter the name of the scheduled task.
5. Optional: In the **Description** text box, enter a description of the scheduled task.
6. Schedule the date and time of the workflow run.
7. Optional: Specify the end date.
8. Optional: From the **Schedule** drop-down menu, select a recurrence option.
9. Click **Next**.
10. From the drop-down menu, select the workflow that you want to run.

    Note:
    If you cannot find the expected workflows, you might have to associate them to the specified vSphere inventory object.
11. Provide the required workflow parameters.
12. Click **Finish**.

Scheduling Workflows from the VCF Operations orchestrator view in the vSphere Client

From the VCF Operations orchestrator view in the vSphere Client, you can create tasks to schedule workflows, edit scheduled tasks, suspend scheduled tasks, and resume suspended scheduled tasks.

Schedule a VMware Cloud Foundation Operations orchestrator Workflow

In the vSphere Client, you can schedule a VMware Cloud Foundation Operations orchestrator workflow to run at a specified time. You can also set the recurrence for a scheduled workflow.

Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestratorplug-in for the vSphere Client, see the Workload Orchestration documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Workflows**.
3. Select the workflow that you want to schedule and click **Schedule**.
4. In the **Task name** text box, enter the name of the scheduled task.
5. Optional: In the **Description** text box, enter a description of the scheduled task.
6. Schedule the date and time of the workflow run.
7. Optional: Specify the end date.
8. Optional: From the **Schedule** drop-down menu, select a recurrence option.
9. Click **Next**.
10. Provide the required workflow parameters.
11. Click **Finish**.

Edit the Schedule of Your VMware Cloud Foundation Operations orchestrator Workflow

In the vSphere Client, you can modify the schedule of a VMware Cloud Foundation Operations orchestrator workflow and set it to run at an earlier or later time.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click **Scheduled workflows**.

   A list of the scheduled workflows appears.
3. Select the workflow whose schedule you want to edit and click **Edit**.
4. Modify the parameters of the scheduled workflow and click **Finish**.

Suspend a Scheduled VMware Cloud Foundation Operations orchestrator Workflow Run

In the vSphere Client, you can suspend a VMware Cloud Foundation Operations orchestrator scheduled task for a workflow run. You can also resume suspended scheduled tasks.

1. Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.
2. Click **Scheduled workflows**.

   A list of the scheduled workflows appears.
3. Select a workflow and click **Pause**.

   The workflow schedule is suspended.

The state of the scheduled task changes to Suspended.

Resume a Suspended Scheduled VMware Cloud Foundation Operations orchestrator Workflow Run

In the vSphere Client, you can resume a scheduled VMware Cloud Foundation Operations orchestrator workflow run that has been suspended.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Scheduled workflows**.

   A list of the scheduled workflows appears.

3. Select a workflow and click **Resume**.

   The workflow schedule is resumed from suspension.

The state of the scheduled task changes to Pending.

View Information About Your VCF Operations orchestrator Workflow Runs

In the vSphere Client, you can view information about the workflow runs for the connected VCF Operations orchestrator server. The available information includes the workflow name, start and end date, state of the workflow, and the user who started the workflow.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Run a VCF Operations orchestrator workflow.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Workflow Runs**.

   The list of workflow runs appears.

You can review the list of workflow runs, cancel a running workflow, delete a workflow run, or respond to a workflow that requires interaction.

View Information About the Runs of a Specific Workflow

In the vSphere Client, you can view information about the runs of a single VMware Cloud Foundation Operations orchestrator workflow such as start and end date, state of the workflow, and user who has started the workflow.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.
- Run the specific VCF Operations orchestrator workflow at least once.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Workflows**.

   A list of the available workflows appears.

3. Click the name of a workflow, and click the **All Runs** tab.

   A list of workflow runs appears.

You can review the list of workflow runs, cancel a running workflow, or respond to a workflow that requires interaction.

View Workflows That Are Waiting for User Input

In the vSphere Client, you can view the VMware Cloud Foundation Operations orchestrator workflows that are waiting for user input.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Waiting for input**.

A list of workflows that are waiting for user input appears.
You can provide values for the required parameters of workflows that are waiting for user input. For example, the user might have to provide an approval or further details about the workflow.

Find a VCF Operations orchestrator Workflow

You can browse for workflows in the inventory of the VCF Operations orchestrator server or filter the available workflows by a search keyword to find a particular workflow. In vSphere environments with many available workflows, filtering provides a quick way to navigate through the list of workflows.

- Verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

1. In the object navigator, click **VCF Operations Orchestrator**.

   Depending on your environment and the integration between vCenter and VMware Cloud Foundation Operations orchestrator, the name of the interface control might be **VCF Operations Orchestrator**.

2. Click **Workflows**.
3. In the filter text box, enter a search term or the name of the workflow that you want to find.

   You can see a list with the workflows that contain the search term in the workflow name or description.

Workflows for Managing Your vSphere Inventory Objects

The vCenter plug-in workflow library contains VCF Operations orchestrator workflows that you can use to run automated processes related to the vCenter and host management.

To access workflows in the vSphere Client, verify that you have configured the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client. For information about configuring the VMware Cloud Foundation Operations orchestrator plug-in for the vSphere Client, see the *Workload Orchestration* documentation.

The default workflows for managing vSphere inventory objects are the workflows included in the vCenter plug-in workflow library.

> **Note:**
> Only a predefined set of vCenter workflows are available by default in the context menu. You can associate additional workflows with each vSphere object. See Associate Your Workflows with vSphere Inventory Object Types.

For more information about the workflows available in the vCenter plug-in workflow library, see *vCenter Plug-In Workflow Library* in the *Using VMware Cloud Foundation Operations orchestrator Plug-Ins* documentation.

Working with Headless Systems

ESX supports the detection and configuration of headless systems.

A headless system is a system that can be operated without a monitor, keyboard, or mouse. Network Appliance boxes do not have Video Graphics Array (VGA), the primary interface is a single serial port. You can set up your existing headless systems to use ESX. You can add ESX appliances to a data center where virtual machines are managed with vCenter. All existing ESX features can be used with a headless system that is configured with either embedded flash or minimal local storage. ESX allows for dynamic switching between different serial modes, which is useful for diagnosing and debugging problems. You can switch between modes to view or modify system parameters.

How to Detect a Headless System

ESX automatically detects headless systems.

ESX automatically redirects the DCUI over a serial port connection to improve headless detection. When ESX automatically detects a headless system, ESX will set up the serial port as COM1, 115200 baud, and redirects the DCUI over this serial port. The specific settings of com port and baud rate are read from the Serial Port Console Redirection (SPCR) table, if it exists. This behavior can be deactivated using new boot parameters if the default settings are not acceptable. You can set the `headless` flag in the ACPI FADT table to mark a system as headless.

Serial Mode Dynamic Switching

ESX supports dynamic switching between four different serial port modes.

ESX supports serial mode dynamic switching to provide maximum platform flexibility, and to allow debugging and supportability in the text box. ESX examines the input characters for any serial port mode and switches the modes based on the input key sequence. DCUI, Shell, GDB, and Logging modes are supported. If you have two serial ports, only one of the four modes is allowed on each port. Two serial ports cannot be in the same mode. If you attempt a dynamic switch to a mode in use by the other serial port, the request is ignored. Dynamic switching eliminates the need to interrupt the boot process manually or to create a custom image to redirect to a serial port. It also addresses supportability issues regarding headless systems that only have one serial port, by making it possible to switch the serial port between different modes of operation.

What Serial Port Modes Does ESX Support

ESX supports four serial port modes — logging mode, GDB mode, Shell mode, and DCUI mode.

There are four serial port modes in ESX:

Logging mode – Logging mode is the default mode in a debug build. Logging mode sends the `vmkernel.log` over the serial port.

GDB mode – Use GDB mode for dedicated debugging.

Shell mode – Shell mode is the shell port access, which is similar to SSH.

DCUI mode – DCUI mode is a Direct Console User Interface. This is the user interface that is displayed when you boot ESX using a monitor.

> **Note:**
>
> Only COM1 and COM2 ports are supported. USB serial or PCI serial cards are not supported.

ESX Keystrokes for Dynamic Serial Mode Switching

ESX includes a unique keystroke sequence that allows dynamic serial mode switching.

Dynamic Switching Keystrokes

After the correct keystroke sequence is entered, the system switches the serial port to the desired mode.

Logging mode: `Ctrl+G, Ctrl+B, 1`

Shell mode: `Ctrl+G, Ctrl+B, 2`

DCUI mode: `Ctrl+G, Ctrl+B, 3`

GDB mode: `Ctrl+G, Ctrl+B, ?`

> **Note:**
> Once in GDB mode, you cannot switch modes again using a key sequence. You must use the CLI to switch modes.

How Do You Switch Serial Modes Through the ESX CLI

You can switch serial modes using the CLI.

Dynamic Switching Using the CLI

Use esxcfg-advcfg to set the current mode to **none**. Then set the new desired mode using the CLI.

Logging mode: `esxcfg-advcfg -s com1 /Misc/LogPort`

Shell mode: `esxcfg-advcfg -s com1 /Misc/ShellPort`

DCUI mode: `esxcfg-advcfg -s com1 /Misc/ConsolePort`

GDB mode: `esxcfg-advcfg -s com1 /Misc/GDBPort`

Example

If the serial mode is set to logging mode, enter these two commands to switch it to DCUI mode.

```
§. > esxcfg-advcfg –s none /Misc/LogPort
§. > esxcfg-advcfg –s com1 /Misc/ConsolePort
```

What Are the Keystrokes for Controlling the Serial DCUI

You can use alternate keystrokes to control the DCUI over a serial port. These alternate are useful when F2 or other function keys cannot be used.

Controlling the Serial DCUI

Alternate keystroke mappings for DCUI mode:

- Esc + 1 -> F1
- Esc + 2 -> F2
- Esc + 3 -> F3
- Esc + 4 -> F4
- Esc + 5 -> F5
- Esc + 6 -> F6
- Esc + 7 -> F7
- Esc + 8 -> F8
- Esc + 9 -> F9
- Esc + 0 -> F10
- Esc + ! -> F11
- Esc + @ -> F12

Troubleshooting Overview

The Troubleshooting section in the vCenter and Host Management documentation is a collection of common troubleshooting scenarios and solutions for problems that you might encounter when you manage vCenter and ESX hosts in a vSphere environment.

The section introduces a systematic methodology for effective troubleshooting, including how to gather diagnostic information, identify symptoms, and define the problem space before you apply fixes.

The section covers three main areas:

- **Troubleshooting with vCenter logs** — explains how to troubleshoot by using vCenter logs, which are located in the /var/log/vmware/ directory and record the activity of services and agents in your implementation. For more information, see Troubleshooting with vCenter Logs.
- **Troubleshooting vCenter and ESX host certificates** — addresses certificate-related problems, including the scenario where vCenter cannot connect to managed hosts after you replace default certificates and restart the system. For more information, see Troubleshooting vCenter and ESX Host Certificates.
- **Troubleshooting ESX hosts** — covers host-specific problems, such as resolving vSphere HA host states that indicate an error condition and can prevent vSphere HA from fully protecting the virtual machines on the host. For more information, see Troubleshooting ESX Hosts .

Guidelines for Troubleshooting Your Implementation of vSphere

To troubleshoot your implementation of vSphere, identify the symptoms of the problem, determine which of the components are affected, and test possible solutions.

Identifying Symptoms	A number of potential causes might lead to the under-performance or nonperformance of your implementation. The first step in efficient troubleshooting is to identify exactly what is going wrong.
Defining the Problem Space	After you have isolated the symptoms of the problem, you must define the problem space. Identify the software or hardware components that are affected and might be causing the problem and those components that are not involved.
Testing Possible Solutions	When you know what the symptoms of the problem are and which components are involved, test the solutions systematically until the problem is resolved.

Identifying Symptoms

Before you attempt to resolve a problem in your implementation, you must identify precisely how it is failing.

The first step in the troubleshooting process is to gather information that defines the specific symptoms of what is happening. You might ask these questions when gathering this information:

- What is the task or expected behavior that is not occurring?
- Can the affected task be divided into subtasks that you can evaluate separately?
- Is the task ending in an error? Is an error message associated with it?
- Is the task completing but in an unacceptably long time?
- Is the failure consistent or sporadic?
- What has changed recently in the software or hardware that might be related to the failure?

Defining the Problem Space

After you identify the symptoms of the problem, determine which components in your setup are affected, which components might be causing the problem, and which components are not involved.

To define the problem space in an implementation of vSphere, be aware of the components present. In addition to VMware software, consider third-party software in use and which hardware is being used with the VMware virtual hardware.

Recognizing the characteristics of the software and hardware elements and how they can impact the problem, you can explore general problems that might be causing the symptoms.

- Misconfiguration of software settings
- Failure of physical hardware
- Incompatibility of components

Break down the process and consider each piece and the likelihood of its involvement separately. For example, a case that is related to a virtual disk on local storage is probably unrelated to third-party router configuration. However, a local disk controller setting might be contributing to the problem. If a component is unrelated to the specific symptoms, you can probably eliminate it as a candidate for solution testing.

Think about what changed in the configuration recently before the problems started. Look for what is common in the problem. If several problems started at the same time, you can probably trace all the problems to the same cause.

Testing Possible Solutions

After you know the problem's symptoms and which software or hardware components are most likely involved, you can systematically test solutions until you resolve the problem.

With the information that you have gained about the symptoms and affected components, you can design tests for pinpointing and resolving the problem. These tips might make this process more effective.

- Generate ideas for as many potential solutions as you can.
- Verify that each solution determines unequivocally whether the problem is fixed. Test each potential solution but move on promptly if the fix does not resolve the problem.
- Develop and pursue a hierarchy of potential solutions based on likelihood. Systematically eliminate each potential problem from the most likely to the least likely until the symptoms disappear.
- When testing potential solutions, change only one thing at a time. If your setup works after many things are changed at once, you might not be able to discern which of those things made a difference.
- If the changes that you made for a solution do not help resolve the problem, return the implementation to its previous status. If you do not return the implementation to its previous status, new errors might be introduced.
- Find a similar implementation that is working and test it in parallel with the implementation that is not working properly. Make changes on both systems at the same time until few differences or only one difference remains between them.

Troubleshooting with vCenter Logs

You can often obtain valuable troubleshooting information by looking at the logs provided by the various services and agents that your implementation is using.

Most logs are located in `/var/log/vmware/<service_name>` for vCenter deployments.

Common Logs

The following logs are common to all vCenter deployments.

Table 102: Common Log Directories

Log Directory	Description
`../firstboot`	Stores first boot logs
`applmgmt` and `applmgmt-audit`	Stores logs related to the VMware Appliance Management Service
`cloudvm`	Stores logs for allotment and distribution of resources between services
`rhttpproxy`	Stores logs for the VMware HTTP Reverse Proxy service
`vapi`	Stores logs for the VMware vAPI Endpoint service
`vmafdd`	Stores logs for the VMware Authentication Framework - LDAP service
`vmdird`	Stores logs for the VMware Directory Service - LDAP service
`vmon`	Stores logs for the VMware Service Lifecycle Manager service

Management Node Logs

The following logs are available if a management node deployment is selected.

Table 103: Management Node Log Directories

Log Directory	Service
`rbd`	VMware vSphere Auto Deploy
`content-library`	VMware Content Library Service
`eam`	VMware ESX Agent Manager

Log Directory	Service
netdumper	VMware vSphere ESX Dump Collector
perfcharts	VMware Performance Charts Service
vmcam	VMware vSphere Authentication Proxy
vmdird	VMware Directory Service - LDAP
vmware-sps	VMware vSphere Profile-Driven Storage Service
vpxd	VMware vCenter
vpostgres	VMware Postgres service
vcha	VMware vCenter High Availability service

Troubleshooting vCenter and ESX Host Certificates

Certificates are automatically generated when you install vCenter. These default certificates are not signed by a commercial certificate authority (CA) and might not provide strong security. You can replace default vCenter certificates with certificates signed by a commercial CA. When you replace vCenter and ESX certificates, you might encounter errors.

New vCenter Certificate Does Not Appear to Load

After you replace default vCenter certificates, the new certificates might not appear to load.

When you install new vCenter certificates, you might not see the new certificate.

Existing open connections to vCenter are not forcibly closed and might still use the old certificate.

To force all connections to use the new certificate, use one of the following methods.

- Restart the network stack or network interfaces on the server.
- Restart the vCenter service.

vCenter Cannot Connect to Managed Hosts

After you replace default vCenter certificates and restart the system, vCenter might not be able to connect to managed hosts.

vCenter cannot connect to managed hosts after server certificates are replaced and the system is restarted.

Log into the host as the root user and reconnect the host to vCenter.

Troubleshooting ESX Hosts

The host troubleshooting topics provide solutions to potential problems that you might encounter when using your vCenter and ESX hosts.

Troubleshooting vSphere HA Host States

vCenter reports vSphere HA host states that indicate an error condition on the host. Such errors can prevent vSphere HA from fully protecting the virtual machines on the host and can impede vSphere HA's ability to restart virtual machines after a failure. Errors can occur when vSphere HA is being configured or unconfigured on a host or, more rarely, during normal operation. When this happens, you should determine how to resolve the error, so that vSphere HA is fully operational.

vSphere HA Agent Is in the Agent Unreachable State

The vSphere HA agent on a host is in the Agent Unreachable state for a minute or more. User intervention might be required to resolve this situation.

vSphere HA reports that an agent is in the Agent Unreachable state when the agent for the host cannot be contacted by the primary host or by vCenter. Consequently, vSphere HA is not able to monitor the virtual machines on the host and might not restart them after a failure.

A vSphere HA agent can be in the Agent Unreachable state for several reasons. This condition most often indicates that a networking problem is preventing vCenter or the primary host from contacting the agent on the host, or that all hosts in the cluster have failed. This condition can also indicate the unlikely situation that vSphere HA was deactivated and then reactivated on the cluster while vCenter could not communicate with the vSphere HA agent on the host, or that the ESX host agent on the host has failed, and the watchdog process was unable to restart it. In any of these cases, a failover event is not triggered when a host goes into the Unreachable state.

Determine if vCenter is reporting the host as not responding. If so, there is a networking problem, an ESX host agent failure, or a total cluster failure. After the condition is resolved, vSphere HA should work correctly. If not, reconfigure vSphere HA on the host. Similarly, if vCenter reports the hosts are responding but a host's state is Agent Unreachable, reconfigure vSphere HA on that host.

vSphere HA Agent is in the Uninitialized State

The vSphere HA agent on a host is in the Uninitialized state for a minute or more. User intervention might be required to resolve this situation.

vSphere HA reports that an agent is in the Uninitialized state when the agent for the host is unable to enter the run state and become the primary host or to connect to the primary host. Consequently, vSphere HA is not able to monitor the virtual machines on the host and might not restart them after a failure.

A vSphere HA agent can be in the Uninitialized state for one or more reasons. This condition most often indicates that the host does not have access to any datastores. Less frequently, this condition indicates that the host does not have access to its local datastore on which vSphere HA caches state information, the agent on the host is inaccessible, or the vSphere HA agent is unable to open required firewall ports. It is also possible that the ESX host agent has stopped.

Search the list of the host's events for recent occurrences of the event `vSphere HA Agent for the host has an error`. This event indicates the reason for the host being in the uninitialized state. If the condition exists because of a datastore problem, resolve whatever is preventing the host from accessing the affected datastores. If the ESX host agent has stopped, you must restart it. After the problem has been resolved, if the agent does not return to an operational state, reconfigure vSphere HA on the host.

> **Note:**
> If the condition exists because of a firewall problem, check if there is another service on the host that is using port 8182. If so, shut down that service, and reconfigure vSphere HA.

vSphere HA Agent is in the Initialization Error State

The vSphere HA agent on a host is in the Initialization Error state for a minute or more. User intervention is required to resolve this situation.

vSphere HA reports that an agent is in the Initialization Error state when the last attempt to configure vSphere HA for the host failed. vSphere HA does not monitor the virtual machines on such a host and might not restart them after a failure.

This condition most often indicates that vCenter was unable to connect to the host while the vSphere HA agent was being installed or configured on the host. This condition might also indicate that the installation and configuration completed, but the agent did not become a primary host or a secondary host within a timeout period. Less frequently, the condition is an indication that there is insufficient disk space on the host's local datastore to install the agent, or that there are insufficient unreserved memory resources on the host for the agent resource pool. Finally, for ESX 5.x hosts, the configuration fails if a previous installation of another component required a host reboot, but the reboot has not yet occurred.

When a Configure HA task fails, a reason for the failure is reported.

Reason for Failure	Action
Host communication errors	Resolve any communication problems with the host and retry the configuration operation.
Timeout errors	Possible causes include that the host crashed during the configuration task, the agent failed to start after being installed, or the agent was unable to initialize itself after starting up. Verify that vCenter is able to communicate with the host. If so, see vSphere HA Agent Is in the Agent Unreachable State or vSphere HA Agent is in the Uninitialized State for possible solutions.

Reason for Failure	Action
Lack of resources	Free up approximately 75MB of disk space. If the failure is due to insufficient unreserved memory, free up memory on the host by either relocating virtual machines to another host or reducing their reservations. In either case, retry the vSphere HA configuration task after resolving the problem.
Reboot pending	If an installation for a 5.0 or later host fails because a reboot is pending, reboot the host and retry the vSphere HA configuration task.

vSphere HA Agent is in the Uninitialization Error State

The vSphere HA agent on a host is in the Uninitialization Error state. User intervention is required to resolve this situation.

vSphere HA reports that an agent is in the Uninitialization Error state when vCenter is unable to unconfigure the agent on the host during the Unconfigure HA task. An agent left in this state can interfere with the operation of the cluster. For example, the agent on the host might elect itself as primary host and lock a datastore. Locking a datastore prevents the valid cluster primary host from managing the virtual machines with configuration files on that datastore.

This condition usually indicates that vCenter lost the connection to the host while the agent was being unconfigured.

Add the host back to a vCenter instance (version 5.0 or later). The host can be added as a stand-alone host or added to any cluster.

vSphere HA Agent is in the Host Failed State

The vSphere HA agent on a host is in the Host Failed state. User intervention is required to resolve the situation.

Usually, such reports indicate that a host has failed, but failure reports can sometimes be incorrect. A failed host reduces the available capacity in the cluster and, in the case of an incorrect report, prevents vSphere HA from protecting the virtual machines running on the host.

This host state is reported when the vSphere HA primary host to which vCenter is connected is unable to communicate with the host and with the heartbeat datastores that are in use for the host. Any storage failure that makes the datastores inaccessible to hosts can cause this condition if accompanied by a network failure.

Check for the noted failure conditions and resolve any that are found.

vSphere HA Agent is in the Network Partitioned State

The vSphere HA agent on a host is in the Network Partitioned state. User intervention might be required to resolve this situation.

While the virtual machines running on the host continue to be monitored by the primary hosts that are responsible for them, vSphere HA's ability to restart the virtual machines after a failure is affected. First, each primary host has access to a subset of the hosts, so less failover capacity is available to each host. Second, vSphere HA might be unable to restart an FT Secondary VM after a failure. See also *vSphere Availability* troubleshooting.

A host is reported as partitioned if both of the following conditions are met:

- The vSphere HA primary host to which vCenter is connected is unable to communicate with the host by using the management (or VMware vSAN™) network, but is able to communicate with that host by using the heartbeat datastores that have been selected for it.
- The host is not isolated.

A network partition can occur for a number of reasons including incorrect VLAN tagging, the failure of a physical NIC or switch, configuring a cluster with some hosts that use only IPv4 and others that use only IPv6, or the management networks for some hosts were moved to a different virtual switch without first putting the host into maintenance mode.

Resolve the networking problem that prevents the hosts from communicating by using the management networks.

vSphere HA Agent is in the Network Isolated State

The vSphere HA agent on a host is in the Network Isolated state. User intervention is required to resolve this situation.

When a host is in the Network Isolated state, there are two things to consider -- the isolated host and the vSphere HA agent that holds the primary role.

- On the isolated host, the vSphere HA agent applies the configured isolation response to the running VMs, determining if they should be shut down or powered off. It does this after checking whether a primary agent is able to take responsibility for each VM (by locking the VM's home datastore.) If not, the agent defers applying the isolation response for the VM and rechecks the datastore state after a short delay.
- If the vSphere HA primary agent can access one or more of the datastores, it monitors the VMs that were running on the host when it became isolated and attempts to restart any that were powered off or shut down.

A host is network isolated if both of the following conditions are met:

- Isolation addresses have been configured and the host is unable to ping them.
- The vSphere HA agent on the host is unable to access any of the agents running on the other cluster hosts.

> **Note:**
> If your vSphere HA cluster has vSAN enabled, a host is determined to be isolated if it cannot communicate with the other vSphere HA agents in the cluster and cannot reach the configured isolation addresses. Although the vSphere HA agents use the vSAN network for inter-agent communication, the default isolation address is still the gateway of the host. Hence, in the default configuration, both networks must fail for a host be declared isolated.

Resolve the networking problem that is preventing the host from pinging its isolation addresses and communicating with other hosts.

Configuration of vSphere HA on Hosts Times Out

The configuration of a vSphere HA cluster might time out on some of the hosts added to it.

When you enable vSphere HA on an existing cluster with a large number of hosts and virtual machines, the setup of vSphere HA on some of the hosts might fail.

This failure is the result of a time out occurring before the installation of vSphere HA on the host(s) completes.

Set the vCenter advanced option config.vpxd.das.electionWaitTimeSec to value=240. After this change is made, the time outs do not occur.

Authentication Token Manipulation Error

When you try to create a password that does not meet the authentication requirements of the ESX host, an error occurs.

When you create a password on the host, the following fault message appears: `A general system error occurred: passwd: Authentication token manipulation error`.

The following message is included: `Failed to set the password. It is possible that your password does not meet the complexity criteria set by the system.`

The host checks for password compliance using the default authentication plug-in, `pam_passwdqc.so`. If the password is not compliant, the error appears.

When you create a password, include a mix of characters from four character classes: lowercase letters, uppercase letters, numbers, and special characters such as an underscore or dash.

> **Note:**
> An uppercase character that begins a password does not count toward the number of character classes used. A number that ends a password does not count toward the number of character classes used.

Your user password must meet the following length requirements.

- Passwords containing characters from three character classes must be at least eight characters long.
- Passwords containing characters from all four character classes must be at least seven characters long.

When the PAM module determines if the password consists of dictionary words, the module might treat some numbers as letters and might reject a password which seems to meet the complexity requirements. For example, a password such as *P4$$w0rd* might be rejected because it is based on a dictionary word.

If the module rejects your password, you can transfer the password to the /bin/pwqcheck application on the ESX Shell, change some numbers or letters, and check if it is accepted.

For more information, see the *vSphere Security* documentation.

Managing Host and Cluster Lifecycle

Managing Host and Cluster Lifecycle provides information about configuring and using VMware®vSphere Lifecycle Manager to manage the ESX hosts and clusters in your environment.

Managing Host and Cluster Lifecycle provides instructions for configuring vSphere Lifecycle Manager, working with the vSphere Lifecycle Manager depot, and using baselines and images to install, update, or upgrade the software and firmware running on your ESX hosts.

Managing Host and Cluster Lifecycle also provides detailed guidelines about using vSphere Lifecycle Manager recommended images and performing hardware compatibility checks on single hosts or clusters. It also describes how you can configure and use the Update Manager Download Service (UMDS) to download software updates in deployments with no access to the Internet.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

This information is intended for experienced system administrators who are familiar with data center operations and virtual machine technology.

Client Interface

The instructions in this guide reflect the HTML5-based vSphere Client.

What is vSphere Lifecycle Manager

VMware vSphere Lifecycle Manager enables centralized and simplified lifecycle management for VMware ESX hosts by using images.

Lifecycle Management in vSphere

Lifecycle management refers to the process of installing software, maintaining it through updates and upgrades, and decommissioning it.

In the context of maintaining a vSphere environment, your clusters and hosts in particular, lifecycle management refers to tasks such as installing ESX and firmware on new hosts, and updating or upgrading the ESX version and firmware when required.

vSphere Lifecycle Manager Overview

vSphere Lifecycle Manager is a service that runs in vCenter. Upon deploying the vCenter appliance, the vSphere Lifecycle Manager user interface becomes automatically enabled in vSphere Client.

vSphere Lifecycle Manager encompasses the functionality that Update Manager provides in earlier vSphere releases and enhances it by adding new features and options for ESX lifecycle management at either a cluster or a standalone host at vCenter level.

In vSphere releases earlier than 7.0, Update Manager provides you with the ability to use baselines and baseline groups for host patching and host upgrade operations. Starting with vSphere 7.0, vSphere Lifecycle Manager introduces the option of using vSphere Lifecycle Manager images as the preferred way to manage the lifecycle of the hosts and clusters in your environment. You can also upgrade the virtual machine hardware and VMware Tools versions of the virtual machines in your environment.

Starting in VCF 9.1 and vSphere Foundation 9.1, vSphere Lifecycle Manager relies on the Software depot component that is part of the VCF management services. The VCF software depot centralizes the download of VMware ESX base images, vendor add-ons, components, and ESX patches and updates from Broadcom.

In air-gapped or secured networks without direct Internet access, vSphere Lifecycle Manager works with VCF software depot offline mode, the VCF download tool, or manually imported offline ZIP bundles. Direct configuration or editing of public Broadcom depot URLs in vCenter is no longer supported in vSphere 9.1 and is automatically managed by VCF software depot.

vSphere Lifecycle Manager Operations

The basic vSphere Lifecycle Manager operations are related to maintaining an environment that is up-to-date and ensuring smooth and successful updates and upgrades of the ESX hosts.

Operation	Description
Compliance Check	An operation of scanning ESX hosts to determine their level of compliance with a baseline attached to the cluster or with the image that the cluster uses. The compliance check does not alter the object.
Remediation Pre-Check	An operation that you perform before remediation to ensure that the health of a cluster is good and that no issues occur during the remediation process.
Remediation	An operation of applying software updates to the ESX hosts in a cluster. During remediation, you install software on the hosts. Remediation makes a non-compliant host compliant with the baselines attached to the cluster or with the image for cluster.
Staging	An operation that reduces the time ESX hosts spend in maintenance mode. When you stage an image or baseline to an ESX host, vSphere Lifecycle Manager downloads the respective bulletins or components from the vSphere Lifecycle Manager depot to the host without applying them immediately. Staging makes the components, patches, and extensions available locally on the hosts. You can choose to remediate the hosts at a later time, not immediately after staging.

The vSphere Lifecycle Manager Depot

Several components make up vSphere Lifecycle Manager and work together to deliver the vSphere Lifecycle Manager functionality and coordinate the major lifecycle management operations that it provides for. The vSphere Lifecycle Manager depot is an important component in the vSphere Lifecycle Manager architecture, because it contains all software updates that you use to create vSphere Lifecycle Manager images and baselines. In VCF 9.1, the vSphere Lifecycle Manager depot is populated and synchronized automatically through the VCF software depot. In addition to the VCF software depot, the vSphere Lifecycle Manager depot can consume third-party partner depots (for example, Nvidia, Mellanox), OEM partner depots (Dell, HPE, Lenovo, Cisco), and local offline ZIP bundles containing ESX patches and base images.

You can use vSphere Lifecycle Manager only if the vSphere Lifecycle Manager depot is populated with components, add-ons, base images, and legacy bulletins and patches.

For more information about software updates and how they are distributed, see Software Packaging Units That vSphere Lifecycle Manager Can Consume.

For more information about the vSphere Lifecycle Manager depot, see The vSphere Lifecycle Manager Depot.

Secure Hashing and Signature Verification in vSphere Lifecycle Manager

vCenter performs an automatic SHA-256 hash check on all software retrieved through the VCF software depot whether synchronized online, loaded from an offline VCF download tool patch store index, or imported directly as an offline ZIP bundle. The hash check verifies the SHA-256 checksum of the downloaded software to ensure its integrity. During remediation, before vSphere Lifecycle Manager installs any software on a host, the ESX host checks the signature of the installable units to verify that they are not corrupted or altered during the download.

When you import an ISO image into the vSphere Lifecycle Manager depot, vCenter performs an MD5 hash check on the ISO image to validate its MD5 checksum. During remediation, before the ISO image is installed, the ESX host verifies the signature inside the image.

If an ESX host is configured with UEFI Secure Boot, the ESX host performs full signature verification of each package that is installed on the host every time the host boots. For more information, see the *vSphere Security* documentation.

vSphere Lifecycle Manager Scalability

For information about the scalability that vSphere Lifecycle Manager supports, visit the VMware Configuration Maximums Matrix at https://configmax.broadcom.com/home.

The vSphere Lifecycle Manager User Interface in the vSphere Client

After you deploy the vCenter appliance, vSphere Lifecycle Manager becomes immediately visible in the vSphere Client.

The vSphere Lifecycle Manager user interface has two main views, which for convenience this guide calls the home view and the compliance view.

vSphere Lifecycle Manager Home View

The vSphere Lifecycle Manager home view is where you configure and administer the vSphere Lifecycle Manager instance that runs on your vCenter system. You go to the vSphere Lifecycle Manager home view to configure how vSphere Lifecycle Manager images and baselines work and to view all images used in a vCenter system.

You do not need any special privilege to access the vSphere Lifecycle Manager home view.

To access the vSphere Lifecycle Manager home view in the vSphere Client, select **Menu** > **Lifecycle Manager**.

In the vSphere Lifecycle Manager home view, you specify the vSphere Lifecycle Manager instance that you want to administer by selecting a vCenter system from the drop-down menu at the top of the **Lifecycle Manager** pane.

In the **Lifecycle Manager** pane, you have the following top-level tabs:

- If your vCenter instance is of version vSphere 9.0 and above, you have **Image Library**, **Software Depot**, **Updates**, **Baselines**, and **Settings**.
- If your vCenter instance is of version vSphere 8.0 Update 3 and earlier, you have **Software Depot**, **Updates**, **Imported ISOs**, **Baselines**, and **Settings**.

You use the **Image Library** and **Software Depot** tabs when you work with vSphere Lifecycle Manager images. You use the **Updates** and **Baselines** tabs when you work with vSphere Lifecycle Manager baselines. For more information about the **Software Depot** and **Updates** tabs, see Browsing the vSphere Lifecycle Manager Depot.

The **Settings** tab is where you configure all vSphere Lifecycle Manager remediation settings and download sources. You use the **Settings** tab with both vSphere Lifecycle Manager images and baselines. For more information about how to configure the vSphere Lifecycle Manager settings, see vSphere Lifecycle Manager Remediation Settings.

In the vSphere Lifecycle Manager home view, you can perform the following tasks:

- Browse the vSphere Lifecycle Manager depot.
- Create, import, and store vSphere Lifecycle Manager images that can be assigned to clusters and standalone hosts in the vCenter instance.
- Trigger the synchronization of updates with the configured online depots.
- Trigger the synchronization of hardware compatibility data.
- Import offline depots manually.
- Create and manage baselines and baseline groups.
- Configure the default vSphere Lifecycle Manager download source.
- Add a URL to an online depot to the list of download sources.
- Allow or disallow downloading from a download source.
- Configure host remediation settings.
- Configure virtual machine rollback settings.

vSphere Lifecycle Manager Compliance View

The vSphere Lifecycle Manager compliance view is where you perform the major vSphere Lifecycle Manager operations - checking the compliance of ESX hosts against an image or a baseline, staging, remediation pre-checks, remediation, and so on.

You go to the vSphere Lifecycle Manager compliance view to actually use the vSphere Lifecycle Manager images and baselines on your clusters and hosts.

To access the vSphere Lifecycle Manager compliance view in the vSphere Client, you must have the **View Compliance Status** privilege.

The vSphere Lifecycle Manager compliance view is on the **Updates** tab for a selected object.

Depending on the selected object, its product version, and whether you use images or baselines to manage the object, you access the vSphere Lifecycle Manager compliance view in two different ways. You create new clusters or add standalone hosts to a data center or folder only by assigning an image to these objects.

- To access the vSphere Lifecycle Manager compliance view for a cluster or a standalone host that you manage with an image, go to the **Updates** tab for the object and select **Image**.
  - If you select a vCenter instance, a data center, or a folder, in the **Images** card of the vSphere Lifecycle Manager compliance view, you can perform the following tasks:
    - Assign an image to one or many clusters or standalone hosts.
    - Check the compliance status of the ESX hosts in the cluster or the standalone host against the assigned image.
    - Run a remediation pre-check with the image associated with the cluster or the standalone host to ensure successful remediation. You can run remediation pre-checks on a single or many clusters or standalone hosts in the selected inventory object.
    - Stage an image to one or many clusters or standalone hosts to speedup the remediation process.
    - Remediate one or many clusters or standalone hosts with the image associated with them.
  - If you select a cluster or a standalone host, in the **Image** card of the vSphere Lifecycle Manager compliance view, you can perform the following tasks:
    - Export, import, and edit the image that the cluster or the standalone host are associated with.
    - Assign an image to the cluster or standalone host from the images currently available in the Image Library.
    - Upgrade the firmware of the ESX hosts in the cluster or the standalone host in case you have a hardware support manager configured.
    - Check for and view recommended images for the cluster or the standalone host.
    - Check the hardware compatibility for a selected ESX version against vSAN HCL.
    - Check the compliance status of the ESX hosts in the cluster or the standalone host against the image.
    - Run a remediation pre-check to ensure successful remediation.
    - Remediate the ESX hosts against the image that the cluster or the standalone host use.
- To access the vSphere Lifecycle Manager compliance view for a host or a cluster of ESX 8.x hosts that you manage with baselines, go to the **Updates** tab for the object and select **Baselines**.

  In the **Baselines** card of the vSphere Lifecycle Manager compliance view, you can perform the following tasks:
  - Check the compliance status of ESX hosts and clusters against baselines or baseline groups.
  - Attach and detach baselines and baseline groups to hosts and clusters.
  - Generate a remediation pre-check report that lists recommended actions to ensure successful remediation.
  - Stage patches or extensions to hosts.
  - Remediate hosts against baselines and baseline groups.
  - Remediate hosts that are part of a vSAN cluster against system-managed baselines.

On the **Updates** tab, you can also perform other tasks.

- Under **Hosts**, select **Hardware Compatibility** to check the hardware compatibility of a host against the Broadcom Compatibility Guide.
- Under **Hosts**, select **VMware Tools** or **VM Hardware** to check the status of virtual machines and upgrade the VMware Tools version or the virtual hardware version of the virtual machines.

vSphere Lifecycle Manager and vCenter Single Sign-On Domain

If your vCenter system is part of a vCenter Single Sign-On domain, you can access the settings for each vSphere Lifecycle Manager on a connected vCenter instance from a single page. Go to the vSphere Lifecycle Manager home view and from the **Lifecycle Manager** drop-down menu select a vCenter instance. The individual vSphere Lifecycle Manager instances do not share settings or objects and must be configured independently.

Using vSphere Lifecycle Manager with VMware vSphere®Distributed Services Engine™

vSphere Distributed Services Engine provides you with the ability to offload some of the network operations from your server CPU to a data processing unit, or the so-called DPU device on the server. You can use vSphere Lifecycle Manager images and all vSphere Lifecycle Manager operations to manage the life cycle of the software and firmware that runs on the DPU devices of your servers.

Using DPU devices for network acceleration frees up the CPU capacity for business-critical workloads. Besides accelerating networking performance, using DPU devices provides security and compression acceleration.

What Is a DPU Device?

A DPU device is a high-performance network interface card, with added embedded CPU cores, memory, and a hypervisor running on the device independently from the ESX hypervisor installed on the server. DPU devices are like limited-resource servers with multiple general-purpose compute cores.

The ESX hypervisor that runs on the DPU device is a fully functional hypervisor, but it can only run on ARM CPU architectures. You do not use ESX on the DPU to run and provision virtual machines and workloads. The hypervisor that runs on DPU devices is a trimmed version of the ESX hypervisor that runs on your servers. This trimmed ESX version is optimized for I/O activities, such as packet offloads, external management, and so on.

From vSphere perspective, the DPU device is a pre-configured device that you can start using without any further configuration or customizations.

You cannot manage DPU devices separately from managing your ESX hosts. All lifecycle operations that you use to manage the lifecycle of the software and firmware on your hosts are used to also manage the software and firmware lifecycle of DPU devices in a DPU-based environment.

The DPU devices that VCF 9.1 supports are from NVIDIA and AMD, and server designs from Dell, HPE, Lenovo, and Fujitsu.

How Do I Start Using a Server with a DPU Device?

To take advantage of the vSphere Distributed Services Engine, your server must include a supported DPU device. You cannot purchase a DPU device separately and insert it into a legacy non-DPU server.

To start utilizing the DPU device on a server to offload network services, you must perform a fresh ESX installation on the server. During installation, ESX is installed both on the server and on the DPU device. The ESX image contains VIBs for both the ESX version to be installed on a host and the ESX version to be installed on the DPU device on that host.

Managing the Lifecycle of the ESX and Firmware on a DPU Device

You use vSphere Lifecycle Manager to upgrade the ESX version and firmware on a DPU device. The VIBs that are needed to upgrade the DPU device are included in the base ESX images that VMware distributes and that you have in the vSphere Lifecycle Manager depot. You do not need to take care of the software and firmware on a DPU device separately. For example, during remediation vSphere Lifecycle Manager updates ESX both on the host and on the DPU device.

You can have a vSphere Distributed Services Engine installation with 2 DPUs, see High Availability with VMware vSphere Distributed Services Engine. vSphere Lifecycle Manager operations happen in parallel on both DPU devices for hosts with dual DPU setup.

vSphere Lifecycle Manager operations are compatible with DPU-based environments. However, keep the following operations in mind:

- Recommendation generation

- Hardware compatibility checks work only if the cluster does not contain a mix of DPU and non-DPU enabled hosts.

Solutions Interoperability with vSphere Distributed Services Engine

NSX is the only solution that is allowed to push VIBs to ESX on the server, on the DPU, or both. Currently, no other solutions can work together with the vSphere Distributed Services Engine.

System Requirements for Using vSphere Distributed Services Engine

- ESX 8.0 or later
- vCenter 8.0 or later

Limitations to Using vSphere Lifecycle Manager for a DPU-Backed Cluster

- The vSphere Distributed Services Engine works for clusters and standalone hosts that you manage with a vSphere Lifecycle Manager image. You cannot add a host with a DPU device to a cluster that you manage with baselines.
- A host can have one or two DPU devices. If a host has two DPU devices, they must run the same firmware and hardware revisions.
- Each host in the DPU-based cluster must have a DPU device.
- All hosts must have DPU devices from the same vendor and of the same model. The DPU generation might vary.

Software Packaging Units That vSphere Lifecycle Manager Can Consume

VMware and other software vendors create and package their software updates into specific units that vSphere Lifecycle Manager can consume. The smallest packaging construct is the component. Components can be further grouped into larger software units, such as add-ons and base images.

vSphere Lifecycle Manager uses software from VMware, original equipment manufacturers (OEMs), and third-party software providers.

- OEMs are VMware partners, such as, Dell, HPE, Lenovo.
- Third-party software providers are providers of I/O filters, device drivers, CIM modules, and so on.

What Is a vSphere Installation Bundle (VIB)?

The VIB is the basic building block for the creation of installation packages for ESX hosts. A VIB is a software package that contains metadata and a binary payload, which represents the actual piece of software to be installed on an ESX host. The VIB does not represent an entire feature, but just a single module of the feature. So, the VIB is the smallest installable software unit that VMware and other software vendors ship.

A VIB can install software on either the ESX running on the host or the ESX running on a DPU device on the host. Alternatively, a single VIB can carry updates for both the ESX versions on the host and on the DPU device.

Table 104: VIB Terminology

Term	Definition
VIB Metadata	An XML file (`descriptor.xml`) that describes the contents of the VIB. It also contains dependency information, textual descriptions, system requirements, and information about bulletins.
Standalone VIB	A VIB that is not included in a bulletin or component.

vSphere Lifecycle Manager does not consume and work with individual VIBs. VIBs must be further packaged into a higher-level construct, such as a base image, an add-on, and so on.

What Is a Bulletin?

The bulletin is a grouping of one or more VIBs. Bulletins are defined within the metadata of the VIB. You use bulletins, and not individual VIBs, to create vSphere Lifecycle Manager baselines, which you attach to inventory objects and use to update and upgrade ESX hosts.

Table 105: Bulletin Terminology

Term	Definition
Patch	A small software update that provides bug fixes or enhancements to the current version of the software. A patch groups one or more VIBs together to address a particular issue or to provide enhancements to the current version of the software.
Roll-up Bulletin	A collection of patches that are grouped to facilitate downloads and deployment.
Extension	A bulletin that defines a group of VIBs for adding an optional component to an ESX host. An extension is usually provided by a third party. The third-party provider is also responsible for providing patches and updates for the extension.

What Are Software Components?

The component is the basic packaging construct for VIBs. VMware, OEMs, and third-party software providers deliver software in the form of components. The component is a bulletin with additional metadata specifying the name and the version of the component. Unlike the bulletin, the component is a logical grouping of VIBs that provides you with a complete and visible feature upon installation. Components can contain VIBs for the ESX version on the server, ESX version the DPU device of the server, if the server is DPU-backed, or for both of them.

VMware and OEMs do not deliver components independently. VMware bundles components together into fully functional and bootable ESX base images. OEMs bundle components together into vendor add-ons. Third-party software vendors create and ship software, for example, drivers or adapters, as independent components.

What Is a Base Image?

The base image is an ESX image that VMware provides with every release of ESX. Base images are not provided by the original equipment manufacturers (OEMs).

The base image is a collection of components that is complete and cannot be modified. The base image is the immutable base building block of the vSphere Lifecycle Manager image. It can boot up a server.

Base images have a user-readable name and a unique version that is updated with every major or minor release of ESX.

The version of a base image corresponds to an ESX release and uses the following naming format:

Release Type	Base Image Naming Example
General Availablity release	ESX 9.0.0.0, ESX 9.1.0.0, ESX 8.0
Update release	ESX 9.0.1.0, ESX 9.0.2.0, ESX 8.0 U1, ESX 8.0 U2, and so on
Patch release	ESX 9.0.1.0100, ESX 9.0.1.0200, ESX 8.0a, ESX 8.0sa, and so on

Base images can contain components and VIBs applicable to the ESX version running on a DPU device. So, a single base image can contain software updates for both ESX on the server and ESX on the DPU device.

Base images are hosted and available in the VMware online depot. Additionally, you can download an ESX installer ISO file and an offline bundle (ZIP file) that contains the ESX version from the Broadcom Support Portal.

What Is a Vendor Add-On?

OEMs pack one or multiple components into a unit called an add-on. You use vendor add-ons to customize an ESX image with OEM content and drivers.

The add-on is a collection of components that does not represent a complete, bootable image. You cannot use vendor add-ons on their own. To customize an ESX release, you must add the vendor add-on to an ESX base image. The combination of a vendor add-on and an ESX base image is practically identical to the custom image that OEMs provide.

When combined with an ESX base image, the add-on can add, update, or remove components that are part of the ESX base image. In vSphere Client, for each add-on available in the depot, you can view the list of components that it adds to an ESX base image. Similarly, you can find information about the components that it removes from a base image.

In addition to custom ISO images used only for baselines and offline bundles, OEMs release ZIP files that contain only the vendor add-on - the delta between the custom image and the ESX base image. OEMs can release such add-on ZIP files at their discretion. The introduction of the concept of add-ons decouples the release cycle of OEMs from the release cycle of VMware. As a result, you can update vendor add-ons independently of updating the ESX version of your hosts. Also, the vendor add-on decouples the OEM customization from the VMware stock image. As a result, you can combine software components more freely.

Vendor add-ons support the ESX software that runs on DPU devices.

Components and Bulletins in the vSphere Lifecycle Manager Depot

vSphere Lifecycle Manager can consume both components and bulletins.

If you use vSphere Lifecycle Manager images to manage hosts and clusters, you can only work with components and the related notions of add-ons and base images. You can find the list of the components, add-ons, and ESX base images on the **Software Depot** tab in the vSphere Lifecycle Manager home view.

If you use baselines and baseline groups to update hosts and clusters, vSphere Lifecycle Manager reads and lists the software updates that are available in the vSphere Lifecycle Manager depot as bulletins. You can find the list of available bulletins on the **Updates** tab in the vSphere Lifecycle Manager home view.

vSphere Lifecycle Manager Images and Baselines

vSphere Lifecycle Manager enables you to manage ESX hosts and clusters with images.

vSphere Lifecycle Manager images and vSphere Lifecycle Manager baselines are different in their essence, the way they work, and the features they support.

> **Important:**
>
> Starting with vSphere 9.0, upgrading to the latest version of ESX is only possible using vSphere Lifecycle Manager images. Baseline support was deprecated in vSphere 8.0 and removed in 9.0. For more information, see KB 379388.

You use vSphere Lifecycle Manager images to perform the following tasks.

- Install a desired ESX version on all hosts in a cluster or on a standalone host.
- Install and update third-party software on all ESX hosts in a cluster or on a standalone host.
- Update and upgrade the ESX version on all hosts in a cluster or on a standalone host.
- Update the firmware of all ESX hosts in a cluster or of a standalone host.
- Generate recommendations and use a recommended image for your cluster or host.
- Check the hardware compatibility of hosts and clusters against the VMware Compatibility Guide and the vSAN Hardware Compatibility List.

In vSphere 9.x, you use vSphere Lifecycle Manager baselines and baseline groups to perform only the following tasks.

- Update and patch ESX hosts only of version 8.x.
- Update third-party software on ESX hosts.

vSphere Lifecycle Manager Images

You use vSphere Lifecycle Manager images to apply software and firmware updates to the ESX hosts in a cluster. Using an image to manage all hosts in a cluster ensures a consistent, declarative state.

You can use various methods and tools to deploy ESX hosts and maintain their software lifecycle. For example, you can upgrade hosts by using vSphere Image Builder, ESXCLI, vSphere Auto Deploy. The different deployment and upgrade choices involve different workflows and require you to use different ESX image formats. When you use vSphere Lifecycle Manager images, you follow one workflow and use the same ESX image format for all software lifecycle-related operations: install, upgrade, update, and patching, which significantly simplifies the lifecycle management process.

Understanding vSphere Lifecycle Manager Images

A vSphere Lifecycle Manager image represents a desired software specification to be applied to the hosts in a cluster. When you set up a vSphere Lifecycle Manager image, you can define the full software stack that you want to run on the hosts in a cluster: the ESX version, additional VMware software, vendor and third-party software, for example, firmware and drivers.

A vSphere Lifecycle Manager image can consist of the following four elements:

- ESX base image
  The base image contains an image of VMware ESX Server and additional components, such as drivers and adapters that are necessary to boot a server. The base image is the only mandatory element in a vSphere Lifecycle Manager image. All other elements are optional.
- Vendor add-on
  The vendor add-on is a collection of software components that OEMs create and distribute. The vendor add-on can contain drivers, patches, and solutions.
- Firmware and drivers add-on
  The firmware and drivers add-on is a special type of vendor add-on designed to assist in the firmware update process. The firmware and drivers add-on contains firmware for a specific server type and corresponding drivers. To add a firmware and drivers add-on to your image, you must install the hardware support manager plug-in provided by the hardware vendor for the hosts in the respective cluster.
- Independent components
  The component is the smallest discrete unit in an image. The independent components that you add to an image contain third-party software, for example drivers or adapters.

You can set up a composite image on a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or many alternative images. All images in a composite image contain the same base image and solutions but may require different vendor or firmware and driver add-ons.

You must set up or assign a vSphere Lifecycle Manager image for a cluster during the creation of the cluster or the addition of the standalone host to the data center or folder.

> **Note:**
>
> If you switch to using images, you cannot revert to using baselines for that cluster or standalone host. If you move a host out of a cluster managed with an image, the lifecycle of the standalone host is now managed by the image which is derived from the cluster's image. You cannot add this host to a cluster managed with baselines.

The Desired State Model

The concept of images that vSphere Lifecycle Manager introduces is based on the Desired State model for managing ESX hosts and clusters.

The desired state of an ESX host represents both the target software and target configuration for the host as opposed to the software and configuration that it currently runs. The Desired State model is the idea of managing hosts and clusters by defining and applying a desired state instead of listing and following steps to change the current state.

vSphere Lifecycle Manager Baselines and Baseline Groups

You can use baselines and baseline groups only to update or patch the ESX hosts of version 8.x in your environment. You must use images to update and upgrade the hosts of version vSphere 8.0 and later in your environment.

Note:

Upgrading to the latest version of ESX is only possible by using vSphere Lifecycle Manager images. Baseline support was deprecated in vSphere 8.0 and removed in 9.0 and later. For more information, see Broadcom knowledge base article 379388.

If you still have clusters or standalone hosts that use baselines, you must transition them to clusters and standalone hosts that use images to be able to upgrade to ESX version 9.0 and later. See Transition a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses Images.

Baselines

A baseline is a grouping of multiple bulletins. You can attach a baseline to an ESX host and check the compliance of the host against the associated baseline.

Baselines can be classified according to different criteria.

- Depending on the type of content, baselines are patch baselines and extension baselines. Patch and extension baselines contain bulletins of the respective kind.
- Depending on how the update content is selected, baselines are fixed and dynamic.
- Depending on how they are created and managed, baselines are predefined, recommendation, or custom baselines.

Baseline Groups

A baseline group is a collection of non-conflicting baselines. You can attach the entire baseline group to an inventory object to check the compliance status of the object against all the baselines in the group as a whole.

You can combine custom baselines with any of the predefined baselines to create baseline groups.

Host baseline groups can contain various patch and extension baselines.

To update ESX hosts by using baselines or baseline groups, you must first attach the baselines or baselines group to an inventory object.

Although you can attach baselines and baseline groups to individual objects, a more efficient method is to attach them to container objects, such as folders, vApps, clusters, and data centers. Individual vSphere objects inherit baselines attached to the parent container object. Removing an object from a container removes the inherited baselines from the object.

For more information about creating and managing baselines and baseline groups, see Types of vSphere Lifecycle Manager Baselines and Baseline Groups.

What Is the Difference Between vSphere Lifecycle Manager Images and Baselines

A vSphere Lifecycle Manager image is a combination of components, vendor add-on, and an ESX base image. A vSphere Lifecycle Manager baseline is a collection of bulletins. Some differences exist between the operations that you can perform on clusters that you manage with a vSphere Lifecycle Manager image and the operations on clusters that you manage with vSphere Lifecycle Manager baselines.

Operation	Images	Baselines
Distribution	Base image, vendor add-ons, and components are distributed through online depots and as offline bundles.	Bulletins are distributed through online depots and as offline bundles.
Validation	Supported. You can validate a vSphere Lifecycle Manager image to check if it is applicable to all hosts in the cluster. You can also perform a remediation pre-check.	Not supported. You do not validate a baseline before applying the updates to the hosts. You can only perform a remediation pre-check.

Operation	Images	Baselines
Import/Export	You can export an image and can use it to manage other clusters in the same or in a different vCenter instance. Images are portable across vCenter instances. You can export an image as an ISO, ZIP, or JSON file, but you can only import images that are in a JSON format when you create an image or add an image to Image Library.	You can create a custom baseline and can attach it to different objects in the same vCenter instance. You cannot export baselines and cannot distribute them across vCenter instances.
Compliance checks	With vSphere Lifecycle Manager images, you can check the compliance of the hosts against an image. To check the compliance against another image, you can use an image from the Image Library or set up a new image.	With baselines, you can check the compliance of an object against a single or against multiple baselines.
Staging	Supported. You can stage an image to a cluster or a host in the cluster before remediation. Staging reduces ESX downtime.	Supported. You can stage updates to the hosts before actually installing them.
Remediation	With vSphere Lifecycle Manager images, you can add, remove, or modify the components in the image that you use for a cluster. When you remediate the hosts against the new image, all modified components are applied to the host. So, you can upgrade and patch a host with a single remediation operation. vSphere Lifecycle Manager images define the precise image to be applied to the hosts after remediation. No deviation from the defined image is possible after remediation. vSphere Lifecycle Manager does not allow solutions to push VIBs to the hosts.	With vSphere Lifecycle Manager baselines, you can remediate an object against a single baseline or against multiple baselines. So, with a single operation, you can patch and update a host. However, vSphere Lifecycle Manager baselines list the updates to be applied to hosts, but the remediation result is not always predictable, because ESX image on the hosts might change after remediation.
Firmware updates	Supported. With vSphere Lifecycle Manager images, firmware updates are carried out through firmware and drivers add-ons, which you add to the image that you use to manage a cluster. Updating firmware with images requires an OEM-provided hardware support manager (HSM) plug-in, which integrates with vSphere Lifecycle Manager. **Note:** In VCF 9.1, vSphere Lifecycle Manager validates the currently installed firmware version for a given device against the vSAN HCL when you do not configure a hardware support manager. This check ensures that the current firmware remains compatible after the upgrade since the system does not update firmware during this process. Both vCenter and ESX must run version 9.1 or later to generate this hardware compatibility report for homogeneous or heterogeneous clusters.	Not supported.
Hardware compatibility checks	Supported You can check the hardware compatibility of the hosts in a cluster against the Broadcom Compatibility Guide (BCG). You can also check the compatibility of all hosts in a vSAN-enabled cluster against the vSAN Hardware Compatibility List (vSAN HCL).	Not supported.
Software recommendations	Supported. Based on the hardware of the hosts in the cluster, you get recommendations about available and applicable ESX updates or upgrades.	Limited support. Software recommendations are only available for vSAN clusters in the form of recommendation baselines.
vCenter /Data center-level operations	With vSphere Lifecycle Manager images, you can trigger all main image operations at a vCenter or data center level.	With vSphere Lifecycle Manager baselines, you can trigger any of the main operations at the vCenter or data center level.

Operation	Images	Baselines
Virtual machine management	You can upgrade the VMware Tools and virtual hardware versions of the virtual machines in a cluster that you manage with vSphere Lifecycle Manager images.	Supported.
VCF Download Tool	Supported.	Not supported.
Remote Office/Branch Offices (ROBO) support	Supported. With vSphere Lifecycle Manager images, use VCF software depot to manage and distribute software depots centrally across remote locations. If a specific remote cluster requires an alternate local HTTP mirror outside of VCF software depot, you can optionally configure a depot override. For more information, see Manage Depot Overrides for a Cluster or a Standalone Host.	Not supported. Although no specific optimization exists for ROBO deployments, you can still use baselines and baseline groups with ROBO clusters.
REST APIs	Available.	Not available.
Support for DPU devices	Supported. To manage a cluster of DPU-backed hosts, you can only use vSphere Lifecycle Manager images. All vSphere Lifecycle Manager operations that work for updating the ESX version on the host also deliver updates to the ESX version on the DPU device on the host.	Not supported. You cannot use baselines for clusters that contain hosts with DPU devices.
ESX Live Patching	Supported. To deploy patches with resolved security and functional issues to all ESX hosts in a cluster managed with an image with minimal disruption to your workloads. See Configuring for Live Patches.	Not supported.

What Are the Requirements for Using vSphere Lifecycle Manager

You must comply with a set of requirements to use images for software lifecycle management. To achieve your goals, you must also know the specifics in behavior and limitations that vSphere Lifecycle Manager has.

Important:

Upgrading to the latest version of ESX is only possible using vSphere Lifecycle Manager images. Baseline support was deprecated in vSphere 8.0 and removed in 9.0. For more information, see KB 379388.

Table 106: Requirements for Using vSphere Lifecycle Manager

Scenario	Requirements
Using an image to manage a cluster.	• All ESX hosts in the cluster must be of version 8.0 and later. • All ESX hosts in the cluster must be stateful. A stateful installation is one in which the host boots from a disk or from a vCenter-level UEFI HTTP(s) boot URL by using Zero Touch Provisioning. **Note:** In a vCenter instance of version 9.0 and later, you can assign vSphere Lifecycle Manager images to clusters with stateless hosts and standalone stateless hosts but only in specific cases. See Managing Stateless Hosts with an Image. • ESX hosts in the cluster can be from different vendors or from the same vendor but different generations or models. Different generations and models of servers may need different software drivers. When a vendor requires different VIBs for different models, you must set up a composite image to manage each generation or model. In vSphere 9.0 and later, you can use a composite vSphere Lifecycle Manager image to manage a heterogeneous cluster that has hosts from different vendors. You can remediate a cluster with a single or composite image depending on the homogeneity of the hosts in the cluster. vSphere Lifecycle Manager does not detect the hardware differences between the hosts in the cluster so you need to match the alternative images with the hosts on which you want them to be applied. • If you want to use DPU devices for network acceleration, all hosts in the cluster must have a DPU device from the same vendor and of the same model. You cannot add a host without a DPU device to a cluster of DPU-backed hosts. • The cluster must include only integrated components. For example: – VMware vSAN™ – VMware vSphere® High Availability (HA) – vSphere IaaS Control Plane – NSX
Using an image to manage a standalone host.	• The standalone ESX host must be of version 8.0 and later. • The standalone ESX host must be stateful. A stateful installation is one in which the host boots from a disk or a vCenter-level UEFI HTTP(s) boot URL (using Zero Touch Provisioning). vSphere Lifecycle Manager images can be assigned to standalone stateless hosts of version 9.0 and later but only in very specific cases. See Managing Stateless Hosts with an Image. • You manage the lifecycle of a standalone host only using a single image. • The host must include only integrated solutions. For example, the witness host for a vSAN stretch cluster must be a standalone host. For more information, see the *vSAN Planning and Deployment* documentation and vSAN Clusters and vSphere Lifecycle Manager.

Scenario	Requirements
Using baselines and baseline groups to update a cluster or a host.	• To use baselines for ESX host patching operations, vSphere Lifecycle Manager works only with ESX hosts of version 8.x. • vCenter 9.0 and later does not support baselines for ESX host upgrade operations.
Switching from using baselines to using an image to manage a cluster or a host.	• The cluster or the standalone host must meet the requirements for using an image. • The cluster or the standalone host must be eligible for the transition. For more information about the **Check cluster's eligibility to be managed with an image** and **Check host's eligibility to be managed with an image** tasks, see Cluster or Standalone Host Eligibility to Use vSphere Lifecycle Manager Images.
Upgrading virtual machine hardware and VMware Tools	For VMware Tools and virtual machine hardware upgrade operations, vSphere Lifecycle Manager works with ESX 9.x.

Required Privileges for Using vSphere Lifecycle Manager

To configure vSphere Lifecycle Manager settings and to use successfully vSphere Lifecycle Manager images and baselines, you must have the proper privileges.

You can assign vSphere Lifecycle Manager privileges to different roles in vSphere Client.

vSphere Lifecycle Manager Privileges For Using Images

When you use vSphere Lifecycle Manager images, you need a different set of privileges for each task.

Table 107: vSphere Lifecycle Manager Privileges For Using Images

Task	Privilege in vSphere Client	Privilege in the API
Set Up an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write` • `VcIntegrity.lifecycleSettings.Read`
Import an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read** • **VMware vSphere Lifecycle Manager > Upload File > Upload upgrade images and offline bundles**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write` • `VcIntegrity.lifecycleSettings.Read` • `VcIntegrity.FileUpload.com.vmware.vcIntegrity.ImportFile`
Export an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read`

Task	Privilege in vSphere Client	Privilege in the API
Edit an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write` • `VcIntegrity.lifecycleSettings.Read`
Work with Recommendations	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write` • `VcIntegrity.lifecycleSettings.Read`
Work with Depots	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Write**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write`
Manage Depot Overrides	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Write**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write`
Check Compliance	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read`
Run Remediation Pre-Check	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: General Privileges > Read** • **VMware vSphere Lifecycle Manager > ESXi Health Perspectives > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Remediation Privileges > Read**	• `VcIntegrity.lifecycleGeneral.Read` • `VcIntegrity.lifecycleHealth.Read` • `VcIntegrity.lifecycleSoftwareRemediation.Read`
Remediate Against an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: General Privileges > Read** • **VMware vSphere Lifecycle Manager > ESXi Health Perspectives > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Remediation Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Remediation Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read**	• `VcIntegrity.lifecycleGeneral.Read` • `VcIntegrity.lifecycleHealth.Read` • `VcIntegrity.lifecycleSoftwareRemediation.Read` • `VcIntegrity.lifecycleSoftwareRemediation.Write` • `VcIntegrity.lifecycleSettings.Read`

Task	Privilege in vSphere Client	Privilege in the API
Edit Remediation Settings	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Write**	• `VcIntegrity.lifecycleSettings.Read` • `VcIntegrity.lifecycleSettings.Write`
Update Firmware	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Read** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Privileges > Write** • **VMware vSphere Lifecycle Manager > Lifecycle Manager: Settings Privileges > Read**	• `VcIntegrity.lifecycleSoftwareSpecification.Read` • `VcIntegrity.lifecycleSoftwareSpecification.Write` • `VcIntegrity.lifecycleSettings.Write`
View Hardware Compatibility	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Hardware Compatibility Privileges > Access Hardware Compatibility**	• `VcIntegrity.HardwareCompatibility.Read`
Stage an Image	• **VMware vSphere Lifecycle Manager > Lifecycle Manager: Image Remediation Privileges > Write**	• `VcIntegrity.lifecycleSoftwareRemediation.Write`

For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation.

vSphere Lifecycle Manager Privileges For Using Baselines

Each of the vSphere Lifecycle Manager privileges that you need to use baselines and baseline groups covers a distinct functionality.

Table 108: VMware vSphere Lifecycle Manager Privileges For Using Baselines

Task	Privilege in vSphere Client	Privilege in the API	Description
Configure	**VMware vSphere Lifecycle Manager > Configure > Configure Service**	`VcIntegrity.General.com.vmware.vcIntegrity.Configure`	Configure the vSphere Lifecycle Manager service and the scheduled patch download task.
Manage Baseline	**VMware vSphere Lifecycle Manager > Manage Baselines > Attach Baseline**	`VcIntegrity.Baseline.com.vmware.vcIntegrity.AssignBaselines`	Attach baselines and baseline groups to objects in the vSphere inventory.
	VMware vSphere Lifecycle Manager > Manage Baselines > Manage Baselines	`VcIntegrity.Baseline.com.vmware.vcIntegrity.ManageBaselines`	Create, edit, or delete baselines and baseline groups.
Manage Patches and Upgrades	**Manage Patches and Upgrades > Remediate to Apply Patches, Extensions, and Upgrades**	`VcIntegrity.Updates.com.vmware.vcIntegrity.Remediate`	Remediate virtual machines and hosts to apply patches, extensions, or upgrades. In addition, this privilege allows you to view the compliance status of objects.

Task	Privilege in vSphere Client	Privilege in the API	Description
	VMware vSphere Lifecycle Manager > **Manage Patches and Upgrades** > **Scan for Applicable Patches, Extensions, and Upgrades**	`VcIntegrity.Updates.com.vmware.vcIntegrity.Scan`	Scan virtual machines and hosts to search for applicable patches, extensions, or upgrades.
	VMware vSphere Lifecycle Manager > **Manage Patches and Upgrades** > **Stage Patches and Extensions**	`VcIntegrity.Updates.com.vmware.vcIntegrity.Stage`	Stage patches or extensions to hosts. In addition, this privilege allows you to view the compliance status of hosts.
	VMware vSphere Lifecycle Manager > **Manage Patches and Upgrades** > **View Compliance Status**	`VcIntegrity.Updates.com.vmware.vcIntegrity.ViewStatus`	View baseline compliance information for an object in the vSphere inventory.
Upload File	**VMware vSphere Lifecycle Manager** > **Upload File** > **Upload upgrade images and offline bundles**	`VcIntegrity.FileUpload.com.vmware.vcIntegrity.ImportFile`	Upload upgrade images and offline patch bundles.

For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation.

The vSphere Lifecycle Manager Depot

The vSphere Lifecycle Manager depot is the source of software updates for vSphere Lifecycle Manager. Starting in VMware Cloud Foundation 9.1 and VMware vSphere Foundation 9.1, vCenter populates the depot through the integrated VCF software depot, replacing direct depot URL configuration.

The vSphere Lifecycle Manager depot is a local depot on the vCenter appliance. This local depot contains software updates that vSphere Lifecycle Manager downloads through the VCF software depot. You can also manually import updates into the vSphere Lifecycle Manager depot.

vSphere Lifecycle Manager manages software updates exclusively for VMware ESX hosts, including ESX base images, vendor add-ons, I/O device drivers (VIBs), async VMware Tools, and host patches. You can work with vSphere Lifecycle Manager only if the vSphere Lifecycle Manager depot contains software packages.

On March 31, 2025, Broadcom transitioned to a unified software depot (`dl.broadcom.com`) for all VCF and vSphere Foundation binaries. Access to this depot is managed centrally by the VCF software depot rather than by customer-specific token URLs configured directly in vSphere Lifecycle Manager.

Starting in VCF 9.1 and vSphere Foundation 9.1, the vSphere Lifecycle Manager depot authentication and synchronization are delegated to the VCF software depot, which runs in VCF Operations. Direct customer token URLs configured in vCenter are deprecated and replaced by centralized activation codes (see Broadcom knowledge base article 443647) in VCF Operations. UMDS is officially deprecated in 9.1 and replaced by the VCF software depot for online synchronization and the VCF download tool (VCFDT) for offline patch store generation.

In environments without access to the Internet, you can use the VCF download tool for downloading software from the Broadcom online depot and configure an offline depot (web server) for storing the downloads. Alternatively, you can use offline depots (offline bundles) to import updates into the vSphere Lifecycle Manager depot manually.

How Do Software Updates Reach the vSphere Lifecycle Manager Depot

In vSphere 9.1, vSphere Lifecycle Manager no longer connects directly to the Broadcom online depot. Instead, vCenter populates the depot through the integrated VCF software depot. The VCF software depot handles authorization with `dl.broadcom.com` by using activation codes that are configured in VCF Operations, and retrieves ESX metadata, VIBs, and compatibility bundles into the vSphere Lifecycle Manager depot.

When VCF software depot is registered, vSphere Lifecycle Manager automatically configures the service as the depot source and removes legacy direct Broadcom URLs. Administrators manage depot activation codes in VCF Operations,

not in vSphere Lifecycle Manager. Manually editing Broadcom depot URLs in vSphere Lifecycle Manager is not supported and can disrupt the VCF software depot integration.

For information about activation codes, see Broadcom knowledge base article 443647.

VCF Software Depot

In VCF 9.1 and later, the VCF software depot operates alongside SDDC Manager to coordinate depot downloads. In vSphere Foundation 9.1 and later, VCF software depot runs directly within VCF Operations as a standalone component without SDDC Manager.

VCF software depot supports three operating modes for populating the vSphere Lifecycle Manager depot, depending on the network connectivity of the environment:

- For environments with Internet access, VCF software depot communicates directly with `dl.broadcom.com`, appending the required authentication headers based on the activation code configured in VCF Operations.
- For environments without direct Internet access, VCF software depot downloads updates from an offline depot server, which an administrator prepares by using the VCF download tool.
- For fully isolated environments, VCF software depot delivers software metadata and packages from a local depot store that an administrator populates by using the VCF download tool.

vSphere Lifecycle Manager interacts with VCF software depot seamlessly regardless of the Internet access.

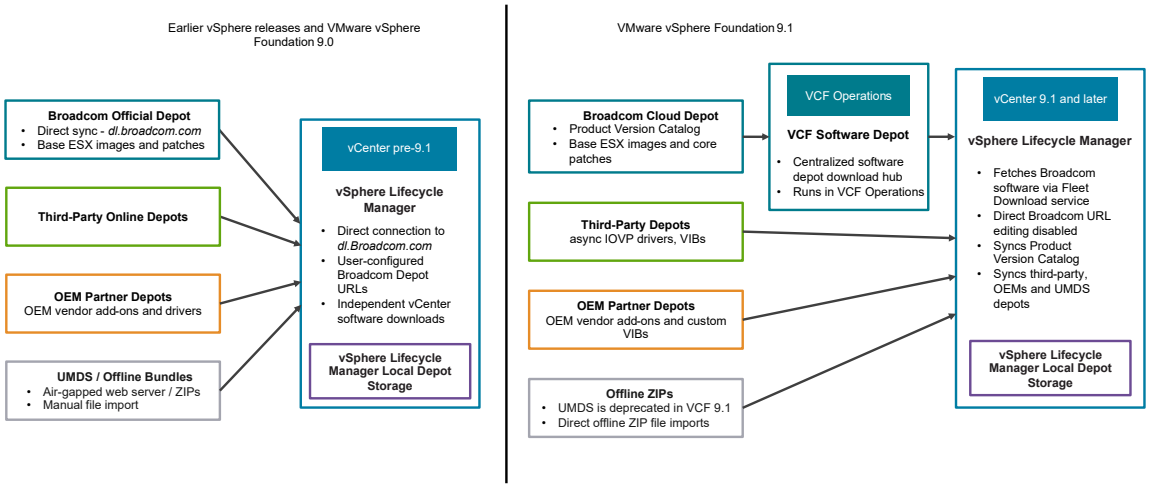
What Is an Online Depot and What Is the Default Broadcom Online Depot

VMware, OEMs, and third-party software providers create and ship their software updates in a well-defined XML file structure, which is called a software depot. Depending on the distribution format of the updates, software depots can be online and offline.

An online depot is the hosted version of the software updates that VMware, OEMs, and third-party software providers ship. In vSphere 9.1, vSphere Lifecycle Manager accesses the online depot through the VCF software depot, which handles authentication automatically by using the configured activation codes.

The Broadcom Online Depot

The unified software depot that Broadcom provides hosts ESX base images, vendor add-ons, ESX-compatible I/O device drivers certified by VMware, and async VMware Tools releases. All software that you need for installing, updating, or customizing your hosts is now centralized in the unified Broadcom software depot. You do not need to work with separate online depots to access OEM and third-party software updates.



Note:
Firmware updates are not hosted in the Broadcom depot. To perform firmware updates, you must install the hardware support manager plug-in that your hardware vendor provides. The plug-in gives you access to depots that contain the necessary firmware and related drivers updates.

You can configure vSphere Lifecycle Manager to also access and use other online depots besides the Broadcom unified depot, for example, third-party depots that contain additional components such as CIM modules. However, working with

additional third-party depots and independent components is rarely necessary. In most cases, the vendor add-ons that are available in the official Broadcom depot provide full OEM customization for ESX.

Synchronization Synchronization is the process during which vSphere Lifecycle Manager downloads the contents of the online depots that you configure it to use to the local vSphere Lifecycle Manager depot. During synchronization, only software metadata is downloaded. The actual payloads are downloaded when they are needed, for example, during staging or remediation. When you deploy vCenter, vSphere Lifecycle Manager synchronizes with the official VMware online depot automatically. After the initial synchronization, you can schedule a download task to run at regular intervals or you can initiate a download task manually.

What Is an Offline Depot

An offline depot, also called an offline bundle, is a ZIP file that you download from the Internet or copy from a media drive and you save on a local or a shared network drive. You then can import the offline bundle to the vSphere Lifecycle Manager depot.

All software updates hosted in the unified Broadcom depot are also available as offline bundles, which you can download from the Broadcom Support Portal Downloads site. You can also download offline bundles from the websites of third-party vendors.

In addition to distributing an `offline.zip` file, or offline bundle, and a custom ISO image, OEMs distribute an `Add-on.zip` file that contains the delta between the OEM custom image and the respective base image that VMware provides. For more information about OEM add-ons, see Software Packaging Units That vSphere Lifecycle Manager Can Consume.

Import Import is the operation through which you upload the contents of an offline bundle into the vSphere Lifecycle Manager depot. During an import operation, both the software metadata and the actual payloads are downloaded into the vSphere Lifecycle Manager depot.

In vCenter 9.0 and 9.1, importing ISO images to the vSphere Lifecycle Manager depot is no longer supported, and the **Imported ISOs** tab has been removed. Upgrades to ESX 9.x can only be performed using vSphere Lifecycle Manager images. Baselines in 9.1 cannot perform host upgrades and are restricted exclusively to minor patching of legacy ESX 8.x hosts using offline ZIP bundles.

You cannot use an ISO image for clusters that you manage with a vSphere Lifecycle Manager image.

Software Deliverables and Their Corresponding Distribution Formats

Software Vendor	Software Deliverable	Software Distribution Formats
VMware / Broadcom	ESX base images and host patches	• Unified Broadcom Online Depot (VCF software depot) • VCF download tool patch store • Offline bundles (ZIP files) imported manually
OEMs (Dell, HPE, Lenovo, etc.)	Vendor Add-ons and Custom Drivers	• Unified Broadcom Online Depot / OEM Online Depots • Offline bundles (ZIP files / Add-on ZIPs)
Third-Party Vendors	Components (I/O Drivers, VIBs)	For device drivers that are certified by VMware: • Offline bundles (ZIP files) For other third-party software that is verified and certified by OEMs, for example I/O filters, CIM module: • Third-Party Online Depots • Offline bundles (ZIP files)

vSphere Lifecycle Manager Download Sources

The vSphere Lifecycle Manager download source is the place from which vSphere Lifecycle Manager downloads software updates.

In vSphere Foundation 9.1, the default download source for vSphere Lifecycle Manager is VCF software depot. The VCF software depot manages the connection to the Broadcom online depot by using the configured activation codes. You do not need to manually configure Broadcom depot URLs in vSphere Lifecycle Manager.

VCF Download Tool Shared Repository

For environments without access to the Internet, you can configure vSphere Lifecycle Manager to use a VCF Download Tool shared repository as its download source. When you configure vSphere Lifecycle Manager to use a VCF Download Tool repository, the synchronization of update metadata is not triggered immediately. The metadata is downloaded according to the configured download schedule or when you initiate the download. When you use a VCF Download Tool repository as a download source for vSphere Lifecycle Manager, only the metadata of the updates is downloaded and stored in the vSphere Lifecycle Manager depot. The actual payload is downloaded during staging or remediation.

You can configure the VCF Download Tool repository to download and store the compatibility data bundle. See Recommended Images.

Cluster Depot Overrides (ROBO Deployments)

In Remote Office / Branch Office (ROBO) environments, use the VCF software depot to centrally manage and distribute software depots to remote sites. If a remote cluster must download updates from a local web server instead of VCF software depot, you can configure a depot override for that cluster. A depot override configures vSphere Lifecycle Manager to use a local depot with updates for a particular cluster instead of the depot that all clusters in that vCenter instance use by default. For more information, see Manage Depot Overrides for a Cluster or a Standalone Host.

Browsing the vSphere Lifecycle Manager Depot

You can use vSphere Client to view and browse the contents of the vSphere Lifecycle Manager depot.

You can view the vSphere Lifecycle Manager depot in the vSphere Lifecycle Manager home view. The contents of the vSphere Lifecycle Manager depot are displayed on two different tabs: **Software Depot** and **Updates**.

Software Depot

On the **Software Depot** tab, you can view all VMware base images, vendor add-ons, and components that are available in the vSphere Lifecycle Manager depot.

You can use the **ESXi Versions**, **Vendor Addons**, and **Component** links at the top of the tab for easier navigation through the lists.

The **ESXi Versions** list contains all base images available in the depot together with information about the version, release date, and category for each image. When you select an image from the list, an information panel appears on the right. The panel displays a list of all components that the base image applies to a host upon remediation.

The **Vendor Addons** list contains all vendor add-ons available in the depot together with information about the version, release date, and category for each add-on. When you select an add-on from the list, an information panel appears on the right. The panel displays information about the components that the add-on applies to the host and the components that the add-on removes from a host upon remediation.

The **Component** list contains all components that are available in the depot together with information about the version, release date, and category for each component. When you select a component from the list, an information panel appears on the right. The panel displays information about the VIBs that the component contains.

You can filter the **Component** list so that it displays only independent components or all components available in the vSphere Lifecycle Manager depot. Independent components are components that are not part of a vendor add-on.

You use the ESX images, vendor add-ons, and components visible on the **Software Depot** tab to set up images that you can use to manage hosts in clusters collectively. You store the setup images in the vSphere Lifecycle Manager Image Library. See Working with Image Library.

Updates

On the **Updates** tab, you can see all components available in the vSphere Lifecycle Manager depot as bulletins. You can use the **Filter by Baseline** drop-down menu to view only the bulletins that are part of a particular baseline.

When you select a bulletin from the list, additional information appears below the bulletins list. In the bottom pane, you see information about the baselines that include the selected bulletin.

You use the bulletins visible on the **Updates** tab to create baselines and baseline groups.

Because the official Broadcom depot hosts certified partner content in addition to VMware content, the **Updates** tab displays a broader set of OEM bulletins, for example vendor add-ons and VMware-certified device drivers. Some of these bulletins might have dependencies that must be pulled into the baselines that you create, so that the remediation against those baselines is successful. As a best practice, always consult the KB article for an individual bulletin to find information about its deployment specifics and required dependencies before including the bulletin in your baselines.

Changes are also introduced in the way VMware content is packaged. As a result, you might see additional bulletins on the **Updates** tab at patch and update releases. Those bulletins are usually of the Enhancement or BugFix category. When you include those bulletins in a baseline, you might need to also include a base ESX bulletin in that baseline. As a best practice, to ensure successful application of patches and updates, always include the appropriate rollup bulletin into your baselines. You can use the **Show only rollup updates** toggle switch that is on the **Updates** tab to filter the list of bulletins.

Working With the vSphere Lifecycle Manager Depot

The vSphere Lifecycle Manager synchronizes to its download source regularly and automatically. But you can change the contents of the vSphere Lifecycle Manager depot by manually importing or deleting updates or ISO images or by triggering synchronization at a convenient time.

Import Updates to the vSphere Lifecycle Manager Depot

You can use an offline bundle in ZIP format and import updates to the vSphere Lifecycle Manager depot manually. When you import offline bundles, you add both the update metadata and actual payload to the vSphere Lifecycle Manager depot.

- Verify that the updates that you import are in ZIP format.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Upload File** > **Upload File**.

You use the import option to populate the vSphere Lifecycle Manager depot with updates from an offline bundle. Apart from the legacy patches and extensions, an offline bundle can also contain an ESX base image, a vendor add-on, or third-party software, for example, asynchronous drivers specific to the OEM hardware requirements. For more information about base images, vendor add-ons, and components, see Software Packaging Units That vSphere Lifecycle Manager Can Consume.

If you want to use vSphere Lifecycle Manager baselines, you can import offline bundles that contain patches and extensions for hosts that run ESX 8.0 and later. In that case, you can use the contents of the offline bundle only for host patching operations. If you import an OEM offline bundle that contains an ESX image of a version earlier than 8.0, you cannot use the image for upgrade operations.

If you want to use vSphere Lifecycle Manager images, you can import offline bundles that contain software for hosts that run ESX 8.0 and later. In that case, you can use the contents of the offline bundle to set up vSphere Lifecycle Manager images, which you can use to upgrade a standalone host or multiple ESX hosts collectively.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. Select **Actions** > **Import Updates** at the top of the vSphere Lifecycle Manager home view.

   The **Import Update Resources** dialog box opens.

3. Enter a URL or browse to an offline bundle in ZIP format on your local machine.

   If the upload fails, check whether the structure of the ZIP file is correct and whether the vSphere Lifecycle Manager network settings are set up correctly.

4. Click **Import**.

   The Import updates task appears in the **Recent Tasks** pane.

You imported updates to the vSphere Lifecycle Manager depot. vSphere Lifecycle Manager automatically generates new image recommendations for the clusters or hosts that already have generated recommended images. However, if the imported updates are solution components only, vSphere Lifecycle Manager does not generate new recommendations automatically.

You can view the imported patches, extension, and base image on the **Updates** and **Software Depot** tabs in the vSphere Lifecycle Manager home view.

You can view the imported ESX images, vendor add-ons, and additional components on the **Software Depot** tab in the vSphere Lifecycle Manager home view.

Synchronize the vSphere Lifecycle Manager Depot

Instead of waiting for the predefined download task to run as scheduled, you can update your local vSphere Lifecycle Manager depot immediately.

At regular configurable intervals, vSphere Lifecycle Manager downloads updates through the VCF software depot. VCF software depot retrieves content from the Broadcom online depot in online mode, or from a VCF download tool shared repository in offline mode.

Regardless of the download schedule, you can initiate synchronization between the vSphere Lifecycle Manager depot and the configured download sources. Similar to scheduled synchronization, when you initiate synchronization manually, vSphere Lifecycle Manager downloads software from the download sources that VCF software depot manages. For more information about configuring the vSphere Lifecycle Manager download sources, see Configuring the vSphere Lifecycle Manager Download Sources.

During synchronization, vSphere Lifecycle Manager downloads only the update metadata, the actual payloads are downloaded during staging or remediation.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.

2. Select **Actions** > **Sync Updates** at the top of the vSphere Lifecycle Manager home view.

   The Sync updates task appears in the **Recent Tasks** pane.

You downloaded updates to the vSphere Lifecycle Manager depot. vSphere Lifecycle Manager automatically generates new image recommendations for the clusters that already have generated recommended images. However, if the updates are related to downloading solution components only, vSphere Lifecycle Manager does not generate new recommendations automatically.

You can view the downloaded patches and extension on the **Updates** tab in the vSphere Lifecycle Manager home view.

You can view the downloaded ESX images, vendor add-ons, and components on the **Software Depot** tab in the vSphere Lifecycle Manager home view.

Configuring the vSphere Lifecycle Manager Download Sources

Configure the download source for vSphere Lifecycle Manager. In vSphere 9.1, the VCF software depot is the default download source. For environments without Internet access, configure a VCF download tool shared repository.

In vSphere 9.1, the default download source for vSphere Lifecycle Manager is the VCF software depot. The VCF software depot manages the connection to the Broadcom online depot by using the configured activation codes. You do not need to manually configure Broadcom depot URLs in vSphere Lifecycle Manager. For information about activation codes, see Broadcom knowledge base article 443647.

For environments without access to the Internet, configure a VCF download tool shared repository as the download source. An administrator populates the repository by using the VCF download tool, and the VCF software depot retrieves content from it in offline mode.

vSphere Lifecycle Manager downloads only the metadata and not the actual binary payload of the updates. Downloading the metadata saves disk space and network bandwidth. The availability of regularly updated metadata in the vSphere Lifecycle Manager depot lets you perform compliance checks on hosts at any time.

Whatever the download source, vSphere Lifecycle Manager downloads the following types of information:

- Metadata about all ESX 8.x updates regardless of whether you have hosts of such versions in your environment.

  Note:

  In vSphere 9.x, the vSphere Lifecycle Manager depot contains also the metadata about the I/O Vendor Program (IOVP) program drivers for vSphere 7.x releases.
- Patch recalls for ESX 8x hosts.

  vSphere Lifecycle Manager supports the recall of patches for hosts that are running ESX 8.0 or later. A patch is recalled when it has problems or potential issues. After you scan the hosts in your environment, vSphere Lifecycle Manager alerts you if the recalled patch has been installed on any host. Recalled patches cannot be installed on hosts with vSphere Lifecycle Manager. vSphere Lifecycle Manager deletes all the recalled patches from the vSphere Lifecycle Manager depot. After a patch that fixes the problem is released, vSphere Lifecycle Manager downloads the new patch to its depot.

  If you have already installed the problematic patch, vSphere Lifecycle Manager notifies you that a fix is available and prompts you to apply the new patch.

Downloading host patches from the Broadcom online depot through the VCF software depot is a secure process.

- Patches are cryptographically signed with the VMware private keys. Before you try to install a patch on a host, the host verifies the signature.
- VCF software depot downloads the patch metadata and patch binaries over SSL connections. vSphere Lifecycle Manager verifies both the validity of the SSL certificates and the common name in the certificates. The common name in the certificates must match the names of the servers from which vSphere Lifecycle Manager downloads the patches. vSphere Lifecycle Manager downloads the patch metadata and binaries only after successful verification of the SSL certificates.

Download Sources

If your environment has access to the Internet, VCF software depot uses the default connection settings to download updates to the vSphere Lifecycle Manager depot. VCF software depot manages the connection to the Broadcom online depot automatically. You do not need to manually configure download URLs.

In environments that lack Internet connectivity, you can point vSphere Lifecycle Manager to a shared repository that an administrator creates and maintains by using the VCF download tool. VCF software depot retrieves content from this repository in offline mode.

For more information about VCF download tool, see Setting Up and Using the VCF Download Tool.

The default configuration is for the vSphere Lifecycle Manager to download information directly from the Internet. However, you can change the download source at any time. Changing the download source from a shared repository to the Internet and the reverse is a change in the vSphere Lifecycle Manager configuration. The two options are mutually exclusive. You cannot download updates from the Internet and a shared repository at the same time.

By default, vSphere Lifecycle Manager is configured to use the VCF software depot as the download source. VCF software depot manages the connection to the unified Broadcom online depot. When you deploy vCenter, synchronization through VCF software depot is triggered automatically. When you change the default download source, synchronization to the new download source is not triggered automatically. The synchronization task runs as per its schedule. To download

new data, you must run the **VMware vSphere Lifecycle Manager Update Download** task or trigger synchronization manually.

The **VMware vSphere Lifecycle Manager Update Download** task is a scheduled task that runs at regular intervals. You can change the schedule, and you can also trigger the **VMware vSphere Lifecycle Manager Update Download** task independently of its schedule.

If the **VMware vSphere Lifecycle Manager Update Download** task is running when you apply the new configuration settings, the task continues to use the old settings until it finishes. The next time the download task starts, vSphere Lifecycle Manager uses the new settings.

Using a Proxy Server

vSphere Lifecycle Manager does not manage the proxy configuration for depot downloads independently. VCF software depot uses the proxy settings of the VCF Operations instance or SDDC Manager where it runs.

Configure vSphere Lifecycle Manager to Use the Internet as a Download Source

If your environment has access to the Internet, vSphere Lifecycle Manager downloads ESX images, vendor add-ons, and other components from the Broadcom online depot to the local vSphere Lifecycle Manager depot through the VCF software depot. The VCF software depot manages the connection automatically by using the configured activation codes.

Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**

The Internet is the default download source for vSphere Lifecycle Manager. The VCF software depot operates in online mode and communicates directly with the Broadcom online depot. Downloading takes place at configurable regular intervals. To initiate downloading of updates regardless of the download schedule, see Synchronize the vSphere Lifecycle Manager Depot .

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Administration** > **Download Sources**.
3. Click the **Change Download Source** button.

   The **Change Download Source Type** dialog box opens.
4. Select the **Download directly from the internet** option and click **Save**.

Configure vSphere Lifecycle Manager to Use a Shared Repository as a Download Source

When direct Internet access is unavailable, configure vSphere Lifecycle Manager to use a VCF download tool shared repository as the download source for ESX images, vendor add-ons, and additional components. VCF software depot retrieves content from the repository in offline mode.

- Create a shared repository by using VCF download tool and host the repository on a web server or a local disk.
- Verify that VCF download tool is of version compatible with the version of vSphere Lifecycle Manager that you are using. For more information about compatibility, see Setting Up and Using the VCF Download Tool.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**.

You cannot use folders on a network drive as a shared repository. vSphere Lifecycle Manager does not download updates from folders on a network share in the Microsoft Windows Uniform Naming Convention form (such as `\\<server_name>\shared`), or on a mapped network drive (for example, `Z:\`). Repositories must be hosted on an HTTP/HTTPS web server or accessible through a valid Linux file system path on the vCenter appliance.

The downloading of updates takes place at configurable regular intervals. To initiate downloading of updates regardless of the download schedule, see Synchronize the vSphere Lifecycle Manager Depot .

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Administration** > **Download Sources**.
3. Click the **Change Download Source** button.

   The **Change Download Source Type** dialog box opens.
4. Select the **Download from a UMDS shared repository** option and enter the path or URL address to the shared repository.
   - Web server (HTTP/HTTPS) - Enter the URL pointing directly to the ESX host patch location in the VCF download tool directory structure.

     Example: `https://<web-server-address>/depotStore/PROD/COMP/ESX_HOST/patch-store/`
   - Local appliance disk/mount - Enter the absolute file system path on the vCenter appliance where the download files or mounted media reside.

     Example: `/storage/updatemgr/depotStore/` or `/mnt/usb/`
5. Click **Save**.

   vCenter validates the URL or path. Upon successful validation, vSphere Lifecycle Manager sets the shared repository as its active download source. If the validation fails, vSphere Lifecycle Manager reports a reason for the failure.

   Important:

   If the updates in the folder that you specify are downloaded with a VCF download tool version that is not compatible with the vCenter version that you use, the validation fails and you receive an error message.

The shared repository is used as the main source for downloading software updates.

Using a Folder or a Web Server as a Shared Repository

You can use a folder or a web server as a shared repository.

- When you use a folder as a shared repository, *`depotStore`* is the path to the top-level directory that stores the patches and notifications copied from the downloaded binaries with the VCF download tool.

  For example, use VCF download tool to download the patches and notifications to a folder `/depotStore/`, then manually copy only the host binaries from the `/depotStore/PROD/COMP/ESX_HOST/patch-store/` to a plugged-in USB device on the physical machine where VCF download tool is installed. Then, plug in the USB device to the physical machine where vSphere Lifecycle Manager runs. The device is mapped as `/mnt/usb` and the folder to configure as a shared repository for vSphere Lifecycle Manager is `/mnt/usb`.
- When you use a web server as a shared repository, *`depotStore`* is the path to the top-level directory on the web server that stores the ESX patches downloaded with the VCF download tool.

  For example, copy the ESX patches and notifications from VCF download tool to `/depotStore`. If the folder is configured on a web server and is accessible from other physical machines at the URL `http://example.com/depotStore/`, the URL to configure as a shared repository in vSphere Lifecycle Manager is `http://example.com/depotStore/PROD/COMP/ESX_HOST/patch-store/`. This URL contains only the ESX patches downloaded with the VCF download tool.

After changing the download source to a shared repository, initiate a manual synchronization task (**Actions** > **Sync Updates**) or run the scheduled **VMware vSphere Lifecycle Manager Update Download** task to populate the local vSphere Lifecycle Manager depot with the new metadata.

Run the VMware vSphere Lifecycle Manager Update Download Task

If you change the download source, you must run the VMware vSphere Lifecycle Manager Update Download task to download any new updates.

1. In the vSphere Client, navigate to a vCenter instance.
2. On the **Configure** tab, select **Scheduled Tasks**.
3. In the **Scheduled Tasks** pane, select the **VMware vSphere Lifecycle Manager Update Download** task and click **Run**.

You can see the running task that is listed in the **Recent Tasks** pane.

Add a New Download Source

You can add a new download source for vSphere Lifecycle Manager by providing a URL to the online depot or the VCF download tool repository that you want to use. Additionally, you can edit or delete a download source. You can also temporarily stop vSphere Lifecycle Manager downloading updates from a particular download source.

Add a New Download Source

If you use the Internet as a download source for updates, you can add URL addresses to third-party online depots. vSphere Lifecycle Manager downloads software updates from all the online depots that you configured it to use. Update metadata are downloaded from the online depots to the local vSphere Lifecycle Manager depot.

- Verify that vSphere Lifecycle Manager is configured to use the Internet or VCF software depot as its download source.
- Required privileges: **VMware vSphere Lifecycle ManagerConfigure**

The default download source for official ESX base images, patches, vendor add-ons, and VMware-certified drivers is automatically managed through the VCF software depot using central activation codes. See Broadcom knowledge base article 443647. You cannot manually add, edit, or delete official Broadcom depot URLs (`dl.broadcom.com`) in vSphere Lifecycle Manager.

Legacy customer token URLs, see Broadcom knowledge base article 390121, are deprecated and automatically cleaned up upon VCF software depot registration.

Downloading updates takes place at configurable regular intervals. To initiate the downloading of updates regardless of the download schedule, see Synchronize the vSphere Lifecycle Manager Depot .

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Administration** > **Download Sources**.
3. On the **Internet Download Sources** pane, click **New**.

   The **New Download Source** dialog box opens.
4. Enter a URL address to a new third-party download source..

   vSphere Lifecycle Manager supports both HTTP and HTTPS URL addresses. Use HTTPS URL addresses to download data securely. The URL address that you add must be complete and must point to a valid online depot containing an `index.xml` file that defines the vendor and component index.

   Note:

   Upstream proxy settings configured in vCenter or VCF Operations apply automatically when vSphere Lifecycle Manager connects to custom third-party download sources.

5. Optional: Enter a short description for the download source.
6. Click **Save**.

The new URL location is added to the list of download sources and downloading from it is enabled by default. Metadata is downloaded during the next scheduled download task, or immediately when you initiate a manual **Sync Updates** action.

Modify a Download Source

You can edit or delete a download source from the list of vSphere Lifecycle Manager download sources. You can also allow or disallow vSphere Lifecycle Manager to download updates from a particular download source.

Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Administration** > **Download Sources**.
3. In the **Internet Download Sources** pane, select a URL address from the list of download sources and select your task.
   - Click **Edit** to edit the source URL or the description for the selected download source.
   - Click **Enable** or **Disable** to allow or disallow downloading from the selected download source.
   - Click **Delete** to delete the selected download source.

     Note:
     You cannot edit or delete the default VMware download source for ESX updates. You can only allow or disallow vSphere Lifecycle Manager to use it for downloading update metadata from it.

Configure the vSphere Lifecycle Manager Automatic Download Task

Downloading host updates and related metadata is a predefined automatic process that you can modify. The automatic download task is enabled by default and starts immediately after you deploy vCenter. After the initial download, the task runs according to its schedule.

- Verify that the VCF software depot providing software updates to vSphere Lifecycle Manager has access to the Internet or VCF download tool is configured with an offline shared repository.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**.

The default schedule settings ensure frequent checks, but you can change the schedule if your environment requires you to adjust the frequency of the checks.

If you need the latest software updates, you might want to reduce the time interval between the checks for updates. By contrast, if you are not concerned about the latest updates, if you want to reduce the network traffic, or if you cannot access the update servers, you might want to increase the time interval between the checks for updates.

The automatic download of update metadata is enabled by default and the default task name is VMware vSphere Lifecycle Manager Update Download. You can change the configuration of the task.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.

2. On the **Settings** tab, select **Administration** > **Automatic Download**.
3. In the **Automatic Download Settings** pane, click the **Edit** button.
   The **Edit Automatic Download Settings** dialog box opens.
4. Select the **Enable Automatic Download** check box and configure the schedule and settings for the download.
   To receive notification emails after the download task finishes, you must configure mail settings for the vSphere Client. For more information, see the *vCenter and Host Management* documentation.
5. Click **Save** to save your changes and close the dialog box.

The automatic download task runs according to the schedule you configured.

Setting Up and Using the VCF Download Tool

In VCF 9.1, the VCF download tool is the command-line utility that you can use to download and manage software binaries and metadata that are required from vSphere Lifecycle Manager.

In VCF 9.1, vCenter no longer connects to the Internet for software depot access. Instead, VCF Operations acts as the egress endpoint, retrieving content from the Broadcom online depot or from a VCF download tool shared repository, based on the configured VCF software depot operating mode.

For security reasons and deployment restrictions, vSphere, including vSphere Lifecycle Manager, might be deployed in a secured or air-gapped network without direct access to the Internet. vSphere Lifecycle Manager requires access to software update metadata and patch information to function properly. In such an environment, you can install VCF download tool on a computer that has Internet access to download upgrades, patch binaries, and patch metadata. You can then export the downloads to a portable media drive on an internal web server so they become accessible as a shared repository.

> **Important:**
>
> The URL of the unified Broadcom online depot is `dl.broadcom.com`. You must include the download activation code, see Broadcom knowledge base article 443647, with your CLI requests when downloading software using the VCF download tool.

If the server on which vCenter is installed has restricted Internet access, but is connected to a server that has Internet access, you can automate the transfer of files from VCF download tool to the vSphere Lifecycle Manager depot by hosting the downloads on a local web server.

VCF download tool is available for installation on a Windows or Linux-based operating systems. In VCF 9.1, legacy Update Manager Download Service (UMDS) commands are replaced with the new `vcf-download-tool esx` commands for downloading ESX images and updates. The machine on which you install VCF download tool must have Internet access.

Compatibility Between the VCF Download Tool and vSphere Lifecycle Manager

Like the vSphere Update Manger Download Service (UMDS), the VCF download tool (VCFDT) must be of the same version as vSphere Lifecycle Manager. For example, vSphere Lifecycle Manager 9.1 is compatible and can work only with VCF download tool 9.1. If you are using an Update release version of vSphere Lifecycle Manager, VCF download tool must be of the same Update release version. To ensure the best possible experience, make sure that you run the latest version of VCFDT whenever possible.

Installing the VCF Download Tool

The VCF download tool 9.1 can be downloaded from the Broadcom Support Portal and installed on a Linux or Windows machine.

Install VCF Download Tool on a Linux OS

In environments where the VCF software depot providing software updates to vSphere Lifecycle Manager does not have direct Internet access, you can install VCF download tool (VCFDT) on a Linux-based operating system with Internet connectivity to download update binaries and metadata.

- Verify that you have administrative privileges on the Linux system where you install the VCF download tool.
- Verify that you have registered your software depot and generated an activation code. See Broadcom knowledge base article 443647.

1. Log into the Broadcom Support Portal and navigate to the **My Downloads** section.
2. Search for and select `VMware Cloud Foundation`.
3. On the **Products** tab, click **VMware Cloud Foundation 9** and select the latest release version from the **Release** column.
4. On the **Drivers & Tools** tab, search for the VCF download tool.
5. Read and accept the *Terms and Conditions* and *Compliance Reporting Terms and Conditions*.
6. Click the **HTTPS Download** icon next to the VCF download tool to initiate the download.

   You can create a directory for downloading the tool on your Linux machine, for example, `/vcfdt`.
7. Navigate to the directory where you downloaded the VCF download tool and extract the files from the `vcf-download-tool-9.1.0.0.<build_number>.tar.gz` archive.

   ```
   tar -xvzf vcf-download-tool-9.1.1.0.<build_number>.tar.gz
   ```
8. Optional: Navigate to the `./bin` directory and run the `--help` option of the `esx` command to validate the installation and see all available subcommands.

   ```
   ./bin/vcf-download-tool esx -h
   ```
9. Generate a software depot ID for the Linux machine on which the VCF download tool is installed.

   ```
   ./bin/vcf-download-tool configuration generate --software-depot-id
   ```

   Tip:

   To retrieve the software depot ID after generation, run the command that is provided below.

   ```
   ./bin/vcf-download-tool configuration get --software-depot-id
   ```

   A registration URL is generated.
10. Copy the registration URL into your browser or log in to the VCF Business Services console and navigate to **Software Depot Registration**.
11. Complete the registration of your VCF download tool instance in the **Register Software Depot** pane.

    An activation code is provided.
12. Save the activation code into a single-line `.txt` file.

    ```
    echo "<activation_code>" > vcfdt-activation-code.txt
    ```

    Include the activation code with every VCF download tool command to maintain a continuous, authenticated connection to the unified Broadcom software depot.

You set up and authenticated the VCF download tool on the Linux machine to the unified Broadcom software depot.

You can now download ESX images, vendor add-ons, and components and transfer them to an offline depot (web server) to be consumed by vSphere Lifecycle Manager in an air-gapped environment.

Install VCF Download Tool on Windows

In environments where the VCF software depot providing software updates to vSphere Lifecycle Manager does not have direct Internet access, you can install VCF download tool (VCFDT) on a Windows operating system with Internet connectivity to download update binaries and metadata.

- Verify that you have administrative privileges on the Windows system where you install the VCF download tool.
- Verify that you have registered your software depot and generated an activation code. See Broadcom knowledge base article 443647.

1. Log into Broadcom Support Portal and navigate to the **My Downloads** section.
2. Search for and select `VMware Cloud Foundation`.
3. On the **Products** tab, click **VMware Cloud Foundation 9** and select the latest release version from the **Release** column.
4. On the **Drivers & Tools** tab, search for the VCF download tool.
5. Read and accept the *Terms and Conditions* and *Compliance Reporting Terms and Conditions*.
6. Click the **HTTPS Download** icon next to the VCF download tool to initiate the download.

   You can create a directory for downloading the tool on your Windows machine, for example, `C:\Users\Administrator\Downloads\vcfdt`.
7. Navigate to the directory where you downloaded the VCF download tool, open PowerShell or Command Prompt (cmd) as an Administrator, and extract the files from the `vcf-download-tool-9.1.0.0.<build_number>.tar.gz` archive.

   ```
   cd C:\Users\Administrator\Downloads\vcfdt
   tar -xvzf vcf-download-tool-9.1.1.0.<build_number>.tar.gz
   ```
8. Optional: Navigate to the `\bin` folder and run the `--help` option of the `esx` command to validate the installation and see all available subcommands.

   ```
   C:\Users\Administrator\Downloads\vcfdt\bin\vcf-download-tool.bat esx -h
   ```
9. Generate a software depot ID for the Windows machine on which the VCF download tool is installed.

   ```
   C:\Users\Administrator\Downloads\vcfdt\bin\vcf-download-tool.bat configuration generate --software-depot-id
   ```

   Tip:

   To retrieve the software depot ID after generation, run the command that is provided below.

   ```
   C:\Users\Administrator\Downloads\vcfdt\bin\vcf-download-tool.bat configuration get --software-depot-id
   ```

   A registration URL is generated.
10. Copy the registration URL into your browser or log in to the VCF Business Services console and navigate to **Software Depot Registration**.
11. Complete the registration of your VCF download tool instance in the **Register Software Depot** pane.

    An activation code is provided.
12. Save the activation code into a single-line `.txt` file.
    - Command for Command Prompt (cmd):
      ```
      echo <activation_code_string> > vcfdt-activation-code.txt
      ```
    - Command for PowerShell:
      ```
      Set-Content -Path "vcfdt-activation-code.txt" -Value "<activation_code_string>" -NoNewline
      ```

    Include the activation code with every VCF download tool command to maintain a continuous, authenticated connection to the unified Broadcom software depot.

You set up and authenticated the VCF download tool on the Windows machine to the unified Broadcom software depot.

You can now download ESX images, vendor add-ons, and components and transfer them to an offline depot (web server) to be used by vCenter instances in an air-gapped environment.

Setting Up the VCF Download Tool

You can set up VCF download tool to download patches for ESX hosts. You can also set up VCF download tool to download ESX 9.x patch binaries, patch metadata, notifications from third-party portals and the compatibility data bundle.

You can set up VCF download tool to download ESX 9.x patches from the unified Broadcom depot and third-party portals.

Set Up the Data to Download with VCF Download Tool

By default, the VCF download tool downloads patch binaries, patch metadata, notifications for ESX hosts, and the compatibility data bundle. You can specify which patch binaries and patch metadata to download with VCF download tool.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. View the host vetions currently selected for download before activating or deactivating content.

   Tip:

   Always run `./bin/vcf-download-tool esx configuration --get` to inspect active patch selections before mutating configuration. Deactivating downloads for decommissioned major ESX versions significantly reduces download time and storage utilization in offline repositories.

   ```
   ./bin/vcf-download-tool esx configuration --get
   ```

   Example valid output showing enabled host platforms and log location:

   ```
   Version: 9.1.1.0.25713929
   Provided host platform(s) are enabled for download/export of ESX content
   Log file: C:\Users\Administrator\Downloads\log\vdt.log

   Host Versions for which patch content will be downloaded:
   esxio-9.1-INTL
   embeddedEsx-9.1-INTL
   embeddedEsx-9.0-INTL
   esxio-9.0-INTL
   esxio-8.0-INTL
   embeddedEsx-8.0-INTL
   embeddedEsx-7.0-INTL
   ```

4. Deactivate host update downloads for major product versions that are no longer present in your environment, for example, ESX 8.0 or 7.0.

   ```
   ./bin/vcf-download-tool esx configuration --disable-host embeddedEsx-8.0-INTL
   ```

   Example actual response when deactivating content:

   ```
   Version: 9.1.1.0.25713929
   Provided host platform(s) are disabled for download/export of ESX content.

   Log file: C:\Users\Administrator\Downloads\log\vdt.log
   ```

5. Optional: Activate patch downloads for specific versions.

   To re-activate a host version that was previously deactivated or to activate a new platform, for example, re-activate `embeddedEsx-8.0-INTL`), run:

   ```
   ./bin/vcf-download-tool esx configuration --enable-host embeddedEsx-8.0-INTL
   ```

   Example actual response when activating content:

   ```
   Version: 9.1.1.0.25713929
   Provided host platform(s) are enabled for download/export of ESX content.

   Log file: C:\Users\Administrator\Downloads\log\vdt.log
   ```

Proceed to run the download command to fetch the selected ESX patch binaries and metadata into your patch repository store.

Configure the VCF Download Tool Patch Repository Location

VCF download tool downloads ESX upgrades, patch binaries, patch metadata, notifications, and the compatibility data bundle to a folder that you can specify during the download.

The VCF download tool downloads patch binaries and patch metadata on a Linux or Windows system to a directory that you specify with the `--depot-store` option of the `download` and `metadata` command.

You can configure the folder in which VCF download tool downloads data after you install VCF download tool. The folder in which VCF download tool downloads patch binaries and patch metadata must be on the system on which VCF download tool is installed.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. Configure the patch repository directory when running the download command:

   ```
   ./bin/vcf-download-tool esx download --depot-store=myDepotStore --depot-download-activation-code-file=vcfdt-activation-code.txt
   ```

   In this example, `myDepotStore` is the directory in which you want to download the patch binaries and patch metadata, and `vcfdt-activation-code.txt` contains your authenticated activation code that you saved during the tool setup. See Setting Up and Using the VCF Download Tool and Broadcom knowledge base article 443647.

   Note:

   The first time you run the download command, the tool prompts you to activate the Broadcom Customer Experience Improvement Program (CEIP), which collects product usage data to improve customer experience.

   Do not leave the terminal unattended before responding to this prompt, or the download process will remain stalled waiting for input.

   VCF download tool automatically generates a standardized depot store structure for all downloaded artifacts that is easy to access and manage. For example, the depot structure may be as follows:

   ```
   /myDepotStore/
   └── PROD
       └── COMP
           └── ESX_HOST
               └── patch-store
                   ├── hostupdate
                   │   ├── DSP
                   │   │   ├── hostupdate20-consolidated-metadata-index__.xml
                   │   │   ├── metadata-583517650.zip
                   │   │   └── vib20
                   │   │       ├── my-vib-1
                   │   │       │   └── VMware_bootbank-vib-1_1.0-1.vib
                   │   │       └── my-vib-2
                   │   │           ├── VMware_bootbank-vib-2_1.0-1.vib
                   │   │           └── VMware_bootbank-vib-2_2.0-1.vib
                   │   └── hostupdate20-consolidated-index__.xml
                   ├── symlink-hostupdate -> /depotStore/PROD/COMP/ESX_HOST/patch-store/hostupdate
                   └── version.txt
   ```

   After displaying *Depots download started*, the tool may appear to hang or pause, especially during the initial run when downloading all artifacts: ESX base images, vendor add-ons, and patch packages.
4. Optional: Periodically check the disk size of your configured `--depot-store` directory in a separate terminal or file explorer to confirm that it is steadily growing.

You successfully configured the directory in which VCF download tool stores patch data.

Sync the binaries and metadata in the local depot with the data from the unified Broadcom software depot by using the VCF download tool.

Once the download completes successfully, host or export the generated `myDepotStore` folder on an internal HTTP/HTTPS web server or local mount so that vSphere Lifecycle Manager can consume it in an offline or air-gapped environment.

Configure the VCF Download Tool for Downloading from Third-Party Vendors

You can configure VCF download tool to connect to the websites of third-party vendors to download host patches and notifications.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. Configure VCF download tool to download data from the new URL address.

   To add a new URL address for downloading patches and notifications for ESX hosts, run the following command:

   ```
   ./bin/vcf-download-tool esx configuration --add-url https://<host_URL>/index.xml
   ```

4. Optional: Remove a URL address, so that VCF download tool does not download data from it anymore.

   Downloaded data is retained and can be exported. Use the following command:

   ```
   ./bin/vcf-download-tool esx configuration --remove-url https://<host_URL>/index.xml
   ```

You configured VCF download tool to download host patches and notifications from specific URL addresses.

Download the patches and notifications by using VCF download tool.

Using the VCF Download Tool

You use VCF download tool in air-gap scenarios, where vSphere Lifecycle Manager has restricted access to the Internet. You first use VCF download tool to download software and notifications and then you export the data, so that it becomes available to vSphere Lifecycle Manager.

To use VCF download tool, the system on which you install it must have Internet access. In more secure environments, the VCF download tool can connect to the Internet through an intermediary server. Setting up a proxy server allows you to filter and monitor all Internet-bound traffic, ensuring that only authorized requests, such as the software downloads from `dl.broadcom.com` are permitted.

After you download the upgrades, patch binaries, patch metadata, you can export the data to a web server or a portable media drive, such as CD or USB flash drive. Then, you must set up vSphere Lifecycle Manager to use the folder on the web server or the media drive (mounted as a local disk) as a download source.

Download Upgrade Binaries Using the VCF Download Tool

After you set up the VCF download tool, you can download upgrades, patches, notifications, and the compatibility data bundle to the system on which VCF download tool is installed.

Administrator-level access is a requirement for downloading data with VCF download tool that runs on Linux and Windows.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. Download the selected updates. See Set Up the Data to Download with VCF Download Tool.

   ```
   ./bin/vcf-download-tool esx download --depot-download-activation-code-file=vcfdt-activation-code.txt
    --depot-store=myDepotStore
   ```

   This command downloads all the upgrades, patches, notifications, and the compatibility data bundle from the configured sources for the first time. Before the download starts, VCF download tool checks whether the patch store

has sufficient storage space to complete the operation. Next, it downloads all new patches and notifications that are released after the previous VCF download tool download. VCF download tool downloads the compatibility data bundle, regardless of whether it has changed since the last download operation.

Export the downloaded upgrades, patches, notifications, and the compatibility data bundle.

Download All Metadata for ESX Hosts of Version 9.1 and 9.0 Through the VCF Download Tool

You can download only the metadata of all upgrades and patches for all 8.x and 9.x versions of ESX.

Administrator-level access is a requirement for downloading data with VCF download tool that runs on Linux or Windows.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. Download all metadata for ESX hosts of version 8.x and 9.x to the configured directory in which VCF download tool stores patch data.

   ```
   ./bin/vcf-download-tool esx metadata --depot-store=myDepotStore --depot-download-activation-code-file=vcfdt-activation-code.txt
   ```

You downloaded all metadata for the updates and patched for ESX hosts of version 8.x and 9.x to the configured patch folder.

Import an Offline Bundle and Merge It to the VCF Download Tool Patch Store

You can import an offline bundle and the compatibility data bundle to VCF download tool and can merge its content into the VCFDT patch store making it available to all vCenter instances.

1. Connect to a console or shell on the system where VCF download tool is installed.
2. Navigate to the directory where VCF download tool is installed.
3. Import the offline bundle or the compatibility data bundle to a specific patch store location of the VCF download tool.

   ```
   ./bin/vcf-download-tool esx import --depot-store=myDepotStore --offline-bundle=<offline_bundle_name>.zip
   ```

The offline bundle or the compatibility data bundle is downloaded and merged with the VCF download tool patch store.

vSphere Lifecycle Manager Remediation Settings

Whether you manage the ESX hosts in your environment with images or with baselines, you can configure the behavior of vSphere Lifecycle Manager during remediation.

The vSphere Lifecycle Manager remediation settings for hosts and clusters that use vSphere Lifecycle Manager images differ from the remediation settings for hosts and clusters that you manage with baselines. For example, allowing the installation of software on PXE booted hosts and the removal of media devices before maintenance mode are settings that you can configure only for hosts and clusters that use baselines. Virtual machine migration settings, maintenance mode settings, and Quick Boot are examples of remediation settings that you can configure for both hosts and clusters that use images or baselines.

You can modify the default vSphere Lifecycle Manager settings only if you have the appropriate privileges. The permission must be assigned to the vCenter instance where vSphere Lifecycle Manager runs. For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation. For a list of the vSphere Lifecycle Manager privileges and their descriptions, see Required Privileges for Using vSphere Lifecycle Manager.

If your vCenter system is connected to other vCenter systems by a common vCenter Single Sign-On domain, you can configure the remediation settings for each vSphere Lifecycle Manager instance. The configuration properties that you modify are applied only to the vSphere Lifecycle Manager instance that you specify, and are not propagated to the other instances in the domain.

How Do Cluster Settings Affect Remediation?

When you remediate ESX hosts that are in a cluster, certain cluster settings might cause remediation failure. You must configure the cluster settings in such a way as to ensure successful remediation.

Distributed Resource Scheduler (DRS)	Updates often require a host to enter maintenance mode during remediation. Virtual machines cannot run when a host is in maintenance mode. To ensure that the virtual machines can remain powered on while the hosts they currently reside on are patched, you can activate DRS for the cluster and you can configure it for vSphere vMotion. In this case, before the host is put in maintenance mode, vCenter migrates the virtual machines to another ESX host within the cluster. If you are unable to vMotion virtual machines between ESX hosts in the cluster due to incompatible hardware, you may consider enabling Enhanced vMotion Compatibility (EVC). For more information about EVC and the requirements that the hosts in an EVC cluster must meet, see KB 313545.
Distributed Power Management (DPM)	If a host has no running virtual machines, DPM might put the host in standby mode, which might interrupt a vSphere Lifecycle Manager operation. So, to make sure that all vSphere Lifecycle Manager operations finish successfully, you must deactivate DPM during these operations. For successful remediation, you must configure vSphere Lifecycle Manager to deactivate DPM. After the remediation task finishes, vSphere Lifecycle Manager restores DPM. If DPM has already put a host in standby mode, vSphere Lifecycle Manager powers on the host before compliance checks, remediation, and staging. After the respective task finishes, vSphere Lifecycle Manager turns on DPM and lets DPM put the host into standby mode, if needed. vSphere Lifecycle Manager does not remediate powered off hosts. If a host is put in standby mode and DPM is manually deactivated for a reason, vSphere Lifecycle Manager does not remediate or power on the host.
HA Admission Control	Within a cluster, you must deactivate HA admission control temporarily to let vSphere vMotion proceed. This action prevents downtime for the machines on the hosts that you remediate. You can configure vSphere Lifecycle Manager to deactivate HA admission control during remediation. After the remediation of the entire cluster is complete, vSphere Lifecycle Manager restores the HA admission control settings. vSphere Lifecycle Manager deactivates HA admission control before remediation, but not before compliance checks. Additionally, for clusters that you manage with baselines, vSphere Lifecycle Manager deactivates HA admission control before staging. You can also use the slot policy of the vSphere HA addmission control. See Slot Policy Admission Control . **Note:** Deactivating HA admission control before you remediate a two-node cluster that uses a vSphere Lifecycle Manager image causes the cluster to practically lose all its high availability guarantees. The reason is that when one of the two hosts enters maintenance mode, vCenter cannot failover virtual machines to that host and HA failovers are never successful. For more information about HA admission control, see the *vSphere Availability* documentation.
Fault Tolerance (FT)	If FT is turned on for any of the virtual machines on a host within a cluster, you must temporarily turn off FT before performing any vSphere Lifecycle Manager operation on the cluster. If FT is turned on for any of the virtual machines on a host, vSphere Lifecycle Manager does not remediate that host. You must remediate all hosts in a cluster with the same updates, so that FT can be reactivated after remediation. A primary virtual machine and a secondary virtual machine cannot reside on hosts of different ESX versions and patch levels. For more information about Fault Tolerance, see How Fault Tolerance Works in the *vSphere Availability* documentation.

Configuring vSphere Lifecycle Manager for Fast Upgrades

During cluster or host remediation, you can preserve the state of the virtual machines in the host memory and restore them from memory after the remediation finishes. Suspending virtual machines to memory and using the Quick Boot functionality significantly reduces the time for remediation, minimizes system boot time, and reduces the downtime of system and services.

Preparing a host for remediation, through vSphere Lifecycle Manager , and performing live migration or vSphere vMotion of all its virtual machines to other hosts, can take some time, depending on their CPU, memory, and network

characteristics. You may partially mitigate these evaluation times by choosing to configure vSphere Lifecycle Manager to suspend certain virtual machines to memory instead of migrating them, powering them off, or suspending them to disk.

You can use the suspend to memory functionality only for patching operations, for example, when you remediate a cluster or a standalone host to apply to it a hot patch, express patch, and so on. You cannot use the suspend to memory option for upgrade operations, for example, when you upgrade your ESX hosts from version 8.0 to version 9.0.

Suspend Virtual Machines to Memory

Suspend to memory is an option that you can use only for clusters or hosts that you manage with vSphere Lifecycle Manager images. The functionality works together with the Quick Boot setting to optimize the remediation process and minimize virtual machine downtime.

You enable vSphere Lifecycle Manager to suspend virtual machines to memory when you configure the vSphere Lifecycle Manager host remediation settings. During remediation pre-check and remediation, vSphere Lifecycle Manager verifies that the suspend to memory option is indeed applicable to the host or cluster under remediation. If suspend to memory is inapplicable, vSphere Lifecycle Manager reports an error and prevents remediation from proceeding.

During a suspend to memory operation, virtual machines remain in a suspended state for some time. The impact is similar to the impact that the suspend to disk operation might have on virtual machines and workloads.

> **CAUTION:**
>
> Suspending virtual machines to memory will impact the availability of workloads running on those virtual machines. Once you enable the suspend to memory option and the remediation operation starts, the virtual machines cannot be unsuspended from memory until the maintenance mode operation is completed or successfully canceled.

> **CAUTION:**
> As a best practice, consider taking snapshots of the virtual machines with critical workloads before you start remediation when the suspend to memory option is activated.

vSphere Lifecycle Manager might not suspend to memory all virtual machines on the host even if you activated the feature for the entire cluster or the standalone host. In some cases, vSphere Lifecycle Manager is still able to proceed with the remediation of the host, even if some virtual machines cannot be suspended to memory.

- vSphere ESX Agent Manager (EAM) virtual machines
  vSphere Lifecycle Manager powers off the EAM virtual machines after all other virtual machines are suspended. Similarly, vSphere Lifecycle Manager powers on the EAM virtual machines before any other virtual machines are resumed from memory. None of the suspended virtual machines is resumed until the EAM virtual machines are powered on.
- vSphere Cluster Services virtual machines
  vSphere Lifecycle Manager first migrates the vSphere Cluster Services virtual machines to another host, and then suspends to memory the rest of the virtual machines on the host.

Similarly, vSphere Lifecycle Manager does not suspend to memory the management virtual appliances for some VMware products and solutions. However, if a virtual machine for any of the following products or solutions runs on a host, the suspend to memory pre-check fails and vSphere Lifecycle Manager does not proceed with the remediation of the respective host:

- vCenter
- vSAN witness virtual machine
- vSphere IaaS Control Plane
- NSX
- VMware Cloud Foundation Operations HCX
- VMware vSphere Replication
- VMware Live Site Recovery
- VMware Cloud Foundation Operations and VMware Cloud Foundation Automation components

Note:
Third-party virtual machines do get suspended during remediation, if the Suspend to memory option is activated.

Quick Boot

Quick Boot is a setting that you can use with clusters or standalone hosts that you manage with vSphere Lifecycle Manager images and baselines. Using Quick Boot optimizes the host patching and upgrade operations. Quick Boot lets vSphere Lifecycle Manager reduce the remediation time for hosts that undergo patch and upgrade operations. Patch and upgrade operations do not affect the hardware of a host. If the Quick Boot feature is activated, vSphere Lifecycle Manager skips the hardware reboot (the BIOS or UEFI firmware reboot). As a result, the time an ESX host spends in maintenance mode is reduced and the risk of failures during remediation is minimized.

To configure vSphere Lifecycle Manager to suspend virtual machines to the host memory, you must activate Quick Boot. However, you can activate Quick Boot even if you decide not to use the Suspend to memory option.

Requirements for Using Suspend to Memory

Several factors might hinder the applicability of the suspended to memory option. If suspend to memory is inapplicable, vSphere Lifecycle Manager reports an error and prevents remediation from proceeding. Suspend to memory works under the following conditions:

- The host supports the suspend to memory functionality.
- Quick Boot is activated for the cluster or the standalone host and the host under remediation supports Quick Boot.
- The remediation does not involve host upgrades or firmware upgrade.
- The host and the virtual machines meet certain requirements.

Host Requirements	Virtual Machine Requirements
• The host has enough free memory. • The host has sufficient free low memory. • The host has enough free memory per NUMA node to start after a reboot. • The host has enough reservation available • The host does not use swapped or compressed pages of virtual machines.	• The virtual machines do not have any passthrough devices. • The virtual machines do not have latency sensitivity set to high. • The virtual machines are not fault tolerant. • The virtual machines are not encrypted. • The virtual machines do not use persistent memory. • The virtual machines do not have virtual SGX or SEV devices. • The virtual machines do not have the suspend feature deactivated. • The virtual machines are not frozen source virtual machines during an Instant Clone operation.

Suspend to Memory and vSphere High Availability (HA)

When you configure vSphere Lifecycle Manager to suspend virtual machines to memory during remediation, vSphere HA provides protection for the suspended virtual machines in cases of failure at the virtual machine or host level. By modifying the vSphere HA advanced options, you can set a timeout value for suspended to memory virtual machines. If a suspended to memory virtual machine is not responsive for the specified time, vSphere HA powers on the virtual machine on the original host or on another host.

- If you deactivate or reconfigure vSphere HA for the cluster during remediation, vSphere HA can no longer protect the suspended virtual machines. Before you change the vSphere HA configuration, make sure that no hosts in the cluster are in maintenance mode and the suspended virtual machines are powered on.
- If you modify the `das.failoverDelayForSuspendToMemoryVmsSecs` advanced option for vSphere HA after you configure vSphere Lifecycle Manager to use the suspend to memory option, the newly specified timeout value might not apply to the virtual machines. If you need to modify the default value of the `das.failoverDelayForSuspendToMemoryVmsSecs` option, ensure that you modify it before you start remediation to ensure that the new value is in effect.
- If the suspend to memory operation fails, vSphere HA determines the most appropriate failover host after the specified timeout value expires. The failover host might be the original host or another one.

- You must synchronize the server time for all ESX hosts in the cluster. If the hosts are not synchronized, vSphere HA might not respect the specified timeout period and initiate failover earlier or later.

For more information about using and configuring vSphere HA, see the *vSphere Availability* documentation.

Configuring vSphere Lifecycle Manager for Live Patches

With the Live Patch setting you apply security patches and urgent bug fixes to all hosts in a cluster that you manage with a vSphere Lifecycle Manager image with minimal disruption to your workloads.

You can use the Live Patch setting to deploy patches with resolved security and functional issues to the ESX hosts in a cluster managed with an image. When Live Patch is activated, cluster remediation has the following specifics:

- The live-patch eligible hosts are not placed into maintenance mode.
- The live-patch eligible hosts are not rebooted.
- The virtual machines on the live-patch eligible hosts are not migrated to other hosts in the cluster.

You can select a global and a cluster-specific Live Patch mode depending on the live-patch eligibility of the hosts in your clusters. By default, the automatic live patch mode is selected on a global level. All hosts in the clusters are remediated with the live-patch image, whether they are eligible or not.

When you run a remediation precheck, vSphere Lifecycle Manager identifies the live patch impact for each host within the cluster.

In vCenter version 9.0 and later, you can use the Live Patch functionality for applying VMkernel security patches, User Space daemons patches, and NSX transport node patches.

In vCenter version 9.1 and later, Live Patch is activated by default and vSphere Lifecycle Manager will perform live patch operations on the hosts in a cluster that is managed with an image when possible. See Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images.

Prerequisites and System Requirements for Using Live Patch

The following mandatory conditions must be met forvSphere Lifecycle Manager to execute live patch operations on a cluster:

- vCenter must be of version 9.0 and later.
- All ESX hosts in the cluster managed with a vSphere Lifecycle Manager image must be of version 8.0 Update 3 and later.
- You use the Live Patch functionality only to upgrade hosts in a cluster that you manage with a vSphere Lifecycle Manager image.
- vSphere DRS must be enabled and set to the **Fully Automated** automation level on the cluster that is managed with a single image before you try to install a live patch on all hosts.
- You must deactivate the parallel remediation setting before you try to install a live patch on the hosts in the cluster. If you have both settings enabled, the remediation pre-check reports this as an issue for a successful remediation and provides you with possible solutions. Live patches can only be installed on the hosts in a cluster managed with an image in sequence.
- vSphere Lifecycle Manager can install a live patch only to a host that is eligible to be updated with a specific live patch release of the VMware base image. vSphere Lifecycle Manager software depot provides information about all the possible versions of the ESX host image which you can update with a specific Live Patch release. For more information about how to browse the depot, see Browsing the vSphere Lifecycle Manager Depot.

Live Patch Feature Capabilities and Compatibility Limitations

The following guidelines outline supported features, operating modes, and hardware or virtual machine compatibility characteristics when using Live Patch:

- **TPM Compatibility.** Live Patch is compatible with systems that are configured with TPM devices.
- **DPU Incompatibility.** Live Patch is not compatible with systems that are configured with DPUs using vSphere Distributed Services Engine.
- **ESXCLI Restriction.** Live Patch is not supported when using ESXCLI commands to patch ESX hosts.

- **Live Patch Modes.**
  - By default, Live Patch is turned on and the **Auto** mode is selected. Only the hosts that are eligible for live patching are live-patched. The rest of the hosts in the cluster are put into maintenance mode or rebooted.
  - All hosts in the cluster managed wwith an image must support the Live Patch functionality when you select the **Enforce** mode. If you activate Live Patch for a vCenter instance or a cluster using the **Enforce** mode and you try to install a live patch on the cluster where the remediation requires a reboot, maintenance mode, or a non-Live Patch installation, remediation is blocked. You can view details about the compliance of the cluster against the selected live patch image for further information.

    This limitation does not apply to the asynchronous installation or upgrade of VMware Tools as these operations can happen even when Live Patch is activated.
- **Virtual Machine FSR Compatibility.** Virtual machines that do not support the Fast Suspend and Resume (FSR) mechanism must be migrated or power-cycled to apply hardware changes with Live Patch. FSR typically allows for the non-disruptive addition or removal of virtual hardware to powered-on virtual machines, but it is incompatible with:
  - Fault Tolerance (FT) configured VMs.
  - DirectPath I/O (Passthrough) VMs
  - vSphere pods
  - Shared-Disk Clustering, for example, Microsoft SQL Server VMs participating in FCI.

  While these VMs trigger a compliance check warning, they do not block Live Patch operations.
- **VMware Tools Upgrades.** The asynchronous installation or upgrade of VMware Tools is not restricted by Live Patch and can happen even when Live Patch is activated.

Live Patch Modes

By default, Live Patch is activated and vSphere Lifecycle Manager remediates all hosts in a cluster regardless their live-patch eligibility.

You can select between two Live Patch modes, the **Auto** and **Enforce** mode. The **Auto** live-patch mode allows vSphere Lifecycle Manager to remediate all hosts in the cluster with the selected live-patch image. First, vSphere Lifecycle Manager remediates the live-patch eligible hosts sequentially without placing them into maintenance mode, and then vSphere Lifecycle Manager remediates the remaining hosts that require maintenance mode or a reboot. If there is even one host in the cluster that requires a maintenance mode or a reboot, the **Enforce** live-patch mode forces vSphere Lifecycle Manager to stop the cluster remediation.

How to Configure the vSphere Lifecycle Manager Remediation Settings

In the vSphere Client, you configure separately the vSphere Lifecycle Manager remediation settings for the hosts and clusters that you manage with an image and with baselines.

The global remediation settings are valid for all hosts and clusters that you manage with images or baselines respectively. For clusters or standalone hosts that you manage with an image, you can override the global remediation settings and can use specific remediation settings for a particular cluster or host.

Hosts that are in a vSAN cluster can enter maintenance mode only one at a time. This behavior is a peculiarity of the vSAN cluster. For more information about the vSphere Lifecycle Manager behavior during the remediation of hosts in a vSAN cluster, see Remediation Specifics of vSAN Clusters .

Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images

You can configure how ESX hosts and VMs behave before and during the remediation of a cluster or a standalone host that you manage with an image and has vSphere Configuration Profiles enabled.

- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.
- To use Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.

For information about automatically triggered hardware compatibility checks, which is a functionality that is also applicable only to vSAN clusters, see When Does the Hardware Compatibility Check Task Run.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Remediation Settings** > **Clusters**.
3. On the **Clusters Remediation Settings** pane, click the **Edit** button.

   The **Edit Cluster Remediation Settings** dialog box opens.
4. Configure the remediation settings for clusters that use images.

Setting	Description
Common Settings	
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is turned on.
VM power state	The **VM power state**option lets you control the behavior of the virtual machines that run on the ESX host. You can select from the following options. • **Do not change power state** • **Suspend to disk** • **Suspend to memory** To select the **Suspend to memory** option, you must turn on Quick Boot. Otherwise, the **Suspend to memory** option is dimmed. Together with Quick Boot, the **Suspend to memory** option provides faster host upgrades. vSphere Lifecycle Managersuspends to the host memory and not to the disk the powered on virtual machines on the host. After the Quick Boot, the suspended virtual machines are resumed from memory. • **Power off** The default selection is **Do not change power state**.
VM migration	You can configure vSphere Lifecycle Manager to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster. By default, the **Migrate powered off and suspended VMs to other hosts in the cluster** check box is selected. If you have virtual machines with passthrough devices on a host in a DRS cluster, make sure that you automate the migration of such virtual machines. For more information, see the VMware knowledge base article at KB 318908.
Retry policy	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode or if staging or remediation for that host fails. You can configure vSphere Lifecycle Manager to wait for a specified retry delay period and to retry to put the host into maintenance mode as many times as you indicate in the **Number of retries** text box. By default, vSphere Lifecycle Manager reties 3 times to put the host into maintenance mode over a period of 5 minutes.
DPM	VMware Distributed Power Management (DPM) monitors the resources consumed by the running virtual machines in the cluster. If sufficient excess capacity exists, VMware DPM recommends moving virtual machines to other hosts in the cluster and placing the original host into standby mode to conserve power. If the capacity is insufficient, VMware DPM might recommend returning standby hosts to a powered-on state. • If you select the **Disable DPM on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables DPM after remediation is complete. • If you deselect the **Disable DPM on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which DPM is enabled. By default, the **Disable DPM on the cluster** check box is selected.

Setting	Description
HA admission control	Admission control is a policy that vSphere HA uses to ensure failover capacity within a cluster. If vSphere HA admission control is enabled during remediation, vMotion might be unable to migrate the virtual machines within the cluster. Deactivating admission control allows a virtual machine to be powered on even if it causes insufficient failover capacity. When this happens, no warnings are presented, and the cluster does not turn red. If a cluster has insufficient failover capacity, vSphere HA can still perform failovers, and uses the VM Restart Priority setting to determine which virtual machines to power on first. **Note:** Deactivating HA admission control before you remediate a two-node cluster causes the cluster to practically lose all its high availability guarantees. • If you select the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables HA admission control after remediation is complete. • If you deselect the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which HA admission control is enabled. By default, the **Disable HA admission control on the cluster** check box is deselected.
Image-Specific Settings	
Live Patch	Live Patch allows you to perform low-disruption updates of the ESX hosts in a cluster managed with an image. When you turn on Live Patch, you can install the latest security patches and bug fixes without placing the all hosts in a cluster into maintenance mode or evacuating virtual machines from the hosts. See Configuring vSphere Lifecycle Manager for Live Patches. By default, Live Patch is turned on and the **Auto** mode is selected. Only the hosts that are eligible for live patching will be live patched. The rest of the hosts in the cluster will be put into maintenance mode or rebooted. You can also select the **Enforce** live patch mode. The cluster is remediated only if all hosts are eligible for live patching. If any of the hosts in the cluster require maintenance mode, the live patch remediation is stopped.
Hardware compatibility issues	vSphere Lifecycle Manager performs a hardware compatibility check as part of the remediation pre-check and the remediation tasks for vSAN clusters. You can configure vSphere Lifecycle Manager to prevent remediation when hardware compatibility issues exist for the cluster. • If you select the **Prevent remediation if hardware compatibility issues are found** option, vSphere Lifecycle Manager reports hardware compatibility issues as an error, which prevents remediation. • If you deselect the **Prevent remediation if hardware compatibility issues are found** option, vSphere Lifecycle Manager reports hardware compatibility issues as a warning, which does not prevent remediation. If the cluster is not vSAN-enabled, vSphere Lifecycle Manager does not perform a hardware compatibility check as part of the remediation pre-check or the remediation tasks.
Parallel remediation	Enabling parallel remediation allows you to remediate multiple ESX hosts simultaneously. By selecting the **Parallel remediation** check box, you enable vSphere Lifecycle Manager to remediate all hosts that are in maintenance mode in parallel instead of in sequence. You can also specify the maximum number of concurrent remediations manually. By default, parallel remediation is deselected. **Note:** If the hosts have NSX virtual distributed switches that are ready to be migrated to vSphere Distributed Switches, you must manually set the maximum number of parallel remediations to no more than 4. In cases when host switch migration is needed, if more than 4 hosts are remediated in parallel, the remediation might fail, because the host switch migration takes more time than the time vSphere Lifecycle Manager needs to complete the parallel remediation.

5. Click **Save**.

These settings become the default remediation settings for clusters that you manage with vSphere Lifecycle Manager. For any cluster you can change the default settings when you configure individual remediation tasks.

Configure the Global vSphere Lifecycle Manager Remediation Settings for Standalone Hosts That You Manage with an Image

You can configure how ESX hosts and VMs behave before and during the remediation of a standalone host that you manage with an image.

- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.
- To use Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.

For information about automatically triggered hardware compatibility checks, which is a functionality that is also applicable only to vSAN clusters, see When Does the Hardware Compatibility Check Task Run.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Remediation Settings** > **Standalone Hosts**.
3. On the **Standalone Hosts Remediation Settings** pane, click the **Edit** button.

   The **Edit Standalone Host Remediation Settings** dialog box opens.
4. Configure the remediation settings for the standalone hosts that use images.

Setting	Description
Common Settings	
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is turned on.
VM power state	The **VM power state**option lets you control the behavior of the virtual machines that run on the ESX host. You can select from the following options. • **Do not change power state** • **Suspend to disk** • **Suspend to memory** To select the **Suspend to memory** option, you must turn on Quick Boot. Otherwise, the **Suspend to memory** option is dimmed. Together with Quick Boot, the **Suspend to memory** option provides faster host upgrades. vSphere Lifecycle Managersuspends to the host memory and not to the disk the powered on virtual machines on the host. After the Quick Boot, the suspended virtual machines are resumed from memory. • **Power off** The default selection is **Do not change power state**.
Retry policy	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode or if staging or remediation for that host fails. You can configure vSphere Lifecycle Manager to wait for a specified retry delay period and to retry to put the host into maintenance mode as many times as you indicate in the **Number of retries** text box. By default, vSphere Lifecycle Manager reties 3 times to put the host into maintenance mode over a period of 5 minutes.

5. Click **Save**.

These settings become the default remediation settings for standalone hosts that you manage with vSphere Lifecycle Manager images. For any host you can change the default settings when you configure individual remediation tasks.

Override the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images

You can customize the remediation settings for a particular cluster or a standalone host while the global remediation settings remain intact and applicable to all other clusters or hosts that you manage with vSphere Lifecycle Manager images and that are in the same vCenter instance.

- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.
- To enable Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.
- To use Live Patch, verify that the ESX host is compatible with the feature. For more information, see Configuring for Live Patches.

When you override the global remediation settings for a single cluster or host, the overrides are used during the remediation of that specific cluster or host. For all other clusters or hosts, the global remediation settings apply.

1. In vSphere Client, navigate to a cluster or host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the horizontal ellipsis icon and select **Edit remediation settings**.
4. In the **Edit Cluster Remediation Settings** or **Edit Standalone Host Remediation Settings** dialog box, configure the remediation settings of the target cluster or host and click **Save**.

Setting	Description
Common Settings	
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is turned on. You can configure this setting for a cluster or a standalone host.
VM power state	The **VM power state**option lets you control the behavior of the virtual machines that run on the ESX host. You can select from the following options. • **Do not change power state** • **Suspend to disk** • **Suspend to memory** To select the **Suspend to memory** option, you must turn on Quick Boot. Otherwise, the **Suspend to memory** option is dimmed. Together with Quick Boot, the **Suspend to memory** option provides faster host upgrades. vSphere Lifecycle Managersuspends to the host memory and not to the disk the powered on virtual machines on the host. After the Quick Boot, the suspended virtual machines are resumed from memory. • **Power off** The default selection is **Do not change power state**. You can configure these options for a cluster or a standalone host.
VM migration	You can configure vSphere Lifecycle Manager to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster. By default, the **Migrate powered off and suspended VMs to other hosts in the cluster** check box is selected. If you have virtual machines with passthrough devices on a host in a DRS cluster, make sure that you automate the migration of such virtual machines. For more information, see the VMware knowledge base article at KB 318908. You can configure this setting only for a cluster that you manage with an image.

Setting	Description
Retry policy	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode or if staging or remediation for that host fails. You can configure vSphere Lifecycle Manager to wait for a specified retry delay period and to retry to put the host into maintenance mode as many times as you indicate in the **Number of retries** text box. By default, vSphere Lifecycle Manager reties 3 times to put the host into maintenance mode over a period of 5 minutes. You can configure this setting for a cluster or a standalone host.
DPM	VMware Distributed Power Management (DPM) monitors the resources consumed by the running virtual machines in the cluster. If sufficient excess capacity exists, VMware DPM recommends moving virtual machines to other hosts in the cluster and placing the original host into standby mode to conserve power. If the capacity is insufficient, VMware DPM might recommend returning standby hosts to a powered-on state. • If you select the **Disable DPM on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables DPM after remediation is complete. • If you deselect the **Disable DPM on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which DPM is enabled. By default, the **Disable DPM on the cluster** check box is selected. You can configure this setting only for a cluster that you manage with an image.
HA admission control	Admission control is a policy that vSphere HA uses to ensure failover capacity within a cluster. If vSphere HA admission control is enabled during remediation, vMotion might be unable to migrate the virtual machines within the cluster. Deactivating admission control allows a virtual machine to be powered on even if it causes insufficient failover capacity. When this happens, no warnings are presented, and the cluster does not turn red. If a cluster has insufficient failover capacity, vSphere HA can still perform failovers, and uses the VM Restart Priority setting to determine which virtual machines to power on first. **Note:** Deactivating HA admission control before you remediate a two-node cluster causes the cluster to practically lose all its high availability guarantees. • If you select the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables HA admission control after remediation is complete. • If you deselect the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which HA admission control is enabled. By default, the **Disable HA admission control on the cluster** check box is deselected. You can configure this setting only for a cluster that you manage with an image.
Image-Specific Settings	
Live Patch	Live Patch allows you to perform low-disruption updates of the ESX hosts in a cluster managed with an image. When you turn on Live Patch, you can install the latest security patches and bug fixes without placing the all hosts in a cluster into maintenance mode or evacuating virtual machines from the hosts. See Configuring vSphere Lifecycle Manager for Live Patches. By default, Live Patch is turned on and the **Auto** mode is selected. Only the hosts that are eligible for live patching will be live patched. The rest of the hosts in the cluster will be put into maintenance mode or rebooted. You can also select the **Enforce** live patch mode. The cluster is remediated only if all hosts are eligible for live patching. If any of the hosts in the cluster require maintenance mode, the live patch remediation is stopped. You can configure this setting only for a cluster that you manage with an image.

Setting	Description
Hardware compatibility issues	vSphere Lifecycle Manager performs a hardware compatibility check as part of the remediation pre-check and the remediation tasks for vSAN clusters. You can configure vSphere Lifecycle Manager to prevent remediation when hardware compatibility issues exist for the cluster. • If you select the **Prevent remediation if hardware compatibility issues are found** option, vSphere Lifecycle Manager reports hardware compatibility issues as an error, which prevents remediation. • If you deselect the **Prevent remediation if hardware compatibility issues are found** option, vSphere Lifecycle Manager reports hardware compatibility issues as a warning, which does not prevent remediation. If the cluster is not vSAN-enabled, vSphere Lifecycle Manager does not perform a hardware compatibility check as part of the remediation pre-check or the remediation tasks. You can configure this setting only for a cluster that you manage with an image.
Parallel remediation	Enabling parallel remediation allows you to remediate multiple ESX hosts simultaneously. By selecting the **Parallel remediation** check box, you enable vSphere Lifecycle Manager to remediate all hosts that are in maintenance mode in parallel instead of in sequence. You can also specify the maximum number of concurrent remediations manually. By default, parallel remediation is deselected. **Note:** If the hosts have NSX virtual distributed switches that are ready to be migrated to vSphere Distributed Switches, you must manually set the maximum number of parallel remediations to no more than 4. In cases when host switch migration is needed, if more than 4 hosts are remediated in parallel, the remediation might fail, because the host switch migration takes more time than the time vSphere Lifecycle Manager needs to complete the parallel remediation. You can configure this setting only for a cluster that you manage with an image.

These settings become the remediation settings for the selected cluster or standalone host. vSphere Lifecycle Manager uses those settings for that cluster for all future remediation tasks. The global remediation settings remain unchanged and are applied to all other clusters.

In the **Image Compliance** card, vSphere Lifecycle Manager displays a message that the global remediation settings are overridden. Also, an option to reset the values appears in the card.

Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Hosts That You Manage with Baselines

You can configure how vSphere Lifecycle Manager behaves before and during remediation against a baseline or a baseline group. The remediation settings help ensure that vSphere Lifecycle Manager puts ESX hosts in maintenance mode before remediation.

- Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**
- To enable Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.

When you use vSphere Lifecycle Manager baselines, you can configure vSphere Lifecycle Manager to let other software initiate the remediation of PXE booted ESX hosts. The remediation installs software modules on the hosts, but typically those host updates are lost after a reboot. To retain updates on stateless hosts after a reboot, use a PXE boot image that contains the updates. You can update the PXE boot image before applying the updates with vSphere Lifecycle Manager, so that the updates are not lost because of a reboot. vSphere Lifecycle Manager itself does not reboot the hosts, because it does not install updates requiring a reboot on PXE booted ESX hosts.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.

2. On the **Settings** tab, select **Remediation Settings** > **Baselines**.
3. Click the **Edit** button.
   The **Edit Settings for Host Remediation** dialog box opens.
4. Configure the baselines remediation settings and click **Save**.

Option	Description
VM power State	You can configure vSphere Lifecycle Manager to power off or suspend all running virtual machines before host remediation. Alternatively, you can configure vSphere Lifecycle Manager not to change the power state of the virtual machines.
Maintenance mode failures	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode before remediation. If you enable vSphere Lifecycle Manager to retry to put the host in maintenance mode, you must specify the number of retries and the retry delay. vSphere Lifecycle Manager waits for as much time as you configure for the **Retry Delay** option, and retries to put the host in maintenance mode as many times as you indicate in the **Number of retries** text box.
PXE booted hosts	You can allow the installation of software for solutions on the PXE booted ESX hosts in the vSphere inventory that you manage with vSphere Lifecycle Manager baselines.
VM migration	If you enable virtual machine migration by selecting the respective option, vSphere Lifecycle Manager migrates the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster.
Removable media devices	vSphere Lifecycle Manager does not remediate hosts on which virtual machines have connected CD/DVD or floppy drives. All removable media drives that are connected to the virtual machines on a host might prevent the host from entering maintenance mode and interrupt remediation. So, you can disconnect all removable media devices to ensure that the respective host enters maintenance mode successfully. After remediation, vSphere Lifecycle Manager reconnects the removable media devices if they are still available.
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is deactivated.
Parallel remediation	Enabling parallel remediation allows you to remediate multiple ESX hosts simultaneously. By default, parallel remediation is deselected. By selecting the **Parallel remediation** option, you enable vSphere Lifecycle Manager to remediate all hosts that are in maintenance mode in parallel instead of in sequence. Alternatively, you can specify the maximum number of concurrent remediations manually. **Note:** If the hosts have NSX virtual distributed switches that are ready to be migrated to vSphere Distributed Switches, you must manually set the maximum number of parallel remediations to no more than 4. In cases when host switch migration is needed, if more than 4 hosts are remediated in parallel, the remediation might fail, because the host switch

Option	Description
	migration takes more time than the time vSphere Lifecycle Manager needs to complete the parallel remediation.

These settings become the default failure response settings with vSphere Lifecycle Manager baselines. You can specify different settings when you configure individual remediation tasks.

How to Use the Image Library in a vCenter Instance

The Image Library on each vCenter instance contains vSphere Lifecycle Manager images that are created through various operations. These operations include creating clusters, adding a standalone host to a data center or folder, transitioning a cluster or a standalone host from managed with baselines to a cluster or standalone host managed with images, setting up an image manually, or importing an image into the Image Library.

Understanding the Image Library in vCenter 9.1

Starting with vCenter 9.0, Image Library becomes a centralized image repository that stores desired image specification for clusters and standalone hosts in a single vCenter 9.x instance. Image Library contains only the desired image specifications while the vSphere Lifecycle Manager depot contains the software updates that you set up in an image.

You can assign the images in Image Library to multiple clusters and standalone hosts in a vCenter instance on a vCenter, data center, and folder level. You can also assign an image from Image Library directly to a single cluster or a standalone host. See Assign an Image from Image Library to a Cluster or a Standalone Host.

You can create custom images, edit an existing image, or import an image from a JSON file. When you create a cluster or add a standalone host to your inventory, Image Library stores the desired image specification that is associated with the cluster or the standalone host on the vCenter instance.

Each vCenter instance has its own Image Library. Images from one Image Library can be associated only with clusters and standalone hosts in the vCenter instance where the Image Library resides. You can reuse an image from one Image Library by exporting the image as JSON file and importing it into the Image Library on another vCenter instance.

Browsing the Image Library Tab

On the **Image Library** tab, you can view all image specifications available in a vCenter instance. You can create a new image specification, assign an image to a cluster or a standalone host, edit an unassigned image, or import or export an image.

Create an Image in the Image Library on a vCenter 9.1 Instance

You can manually create a new image in Image Library or import a JSON file that contains the desired image specification. Moreover, vSphere Lifecycle Manager automatically extracts and stores the images that are associated with all clusters and standalone hosts in your vCenter instance during the time of creation, addition, or transition.

Edit an Image in the Image Library on a vCenter 9.1 Instance

The ability to edit an image depends on whether the image is assigned to a cluster or a standalone host.

- The image is associated with a cluster or a standalone host.
  To maintain compliance, you cannot directly modify the software stack of an active image. You have three options:
  - Basic edits - You can only change the image name.
  - Clone and modify - Create a copy of the image to create a new image version. You can then add and remove drivers and components. To apply these changes, you must manually associate the new image with the clusters or the standalone hosts, you want.
  - Detach and edit - Remove the image from all associated clusters and standalone hosts. Once detached, you can edit it by adding or removing drivers and components, or adding or removing alternative images.
- The image is not associated with a cluster or a standalone host.
  If an image is not associated with any inventory objects, you have full control.

- Modify its components by adding, removing, or updating any image element.
- Add or replace alternative images to accommodate any requirements from different vendors or firmware and driver add-ons.

Export or Import an Image in the Image Library on a vCenter 9.1 Instance

You can reuse an image that is stored in one Image Library by exporting the image as a JSON file and then importing it into the Image Library of another vCenter instance. Depending on your goal, the image can be exported, distributed, and consumed in three different file formats: JSON, ISO, and ZIP. Composite images can be exported only in a JSON file. See Export an Image from Image Library.

You can import an image into Image Library only in the JSON file format. See Import an Image in Image Library.

Removing an Image from the Image Library on a vCenter 9.1 Instance

Removing an image from Image Library is only possible in case where the image is not assigned to any cluster or host in your vCenter instance.

Create an Image in Image Library Manually

You create an image in Image Library to be able to associate this image with a single or multiple clusters or standalone hosts in your vCenter instance or in another vCenter instance.

- Verify that you have the right privileges. See Required Privileges for Using vSphere Lifecycle Manager.
- Verify that the vCenter version is 9.0 and later.
- Verify that you have an ESX base image, vendor add-ons, and components available in the local vSphere Lifecycle Manager depot.

  Important:

  In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

  Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.

Each vCenter instance has its own Image Library that stores persistently the images that are used to manage the lifecycle of clusters and standalone hosts running on it. You manually create an image in Image Library to be able to assign it to a single or multiple clusters or standalone hosts.

When you have a fresh vCenter instance, Image Library is empty. Make sure to create or import an image to Image Library so that you can use it when creating a cluster or adding a standalone host.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.

2. On the **Image Library** tab, select **Create Image**.
3. On the **Create Image** pane, set up the desired image.
   a) Enter a unique image name.
   b) Select an ESX version.
   c) Optionally, select a vendor add-on and the version of the add-on.
   d) Optionally, add firmware and drivers add-on to the image.
   e) Optionally, manage the additional components of the image.
4. Optional: Add an alternative image that is applied on a specific group of hosts in the cluster.

   You create a composite image for clusters that may have hosts from different vendors or from the same vendor but different generations, family, or model.

   a) Click the **Add image** button in the upper right corner of the **Create Image** pane.

      The **Composite Image Name** text field appears above the **Default Image** tab.

      The **New Image** tab appears next to the **Default Image** tab.
   b) In the **Composite Image Name** text field, enter a unique name for the composite image.
   c) In the **Image Name** text field on the **New Image** tab, enter a unique name for the alternative image.
   d) Optionally, select a vendor add-on and the version of the add-on.
   e) Optionally, add firmware and drivers add-on to the image.
   f) Optionally, manage the additional components of the image.
   g) In the **Selection Criteria** section, specify the rules that are used to filter the hosts to which the alternative image is assigned.
      - Filter hosts by using their UUID - The UUID of a host is the `UUID` (Universal Unique Identifier) field in the SMBIOS (System Management BIOS) Type 1 (System Information) structure that is mapped to the offset 08h (8 bytes). Click the **Show details** drop-down menu to add the UUIDs of the hosts on which you want to apply the alternative image and click **Add**.
      - Filter hosts by using other SMBIOS fields like `Manufacturer` (Type 1 structure), `Product Name` (Type 1 structure), `Family` (Type 1 structure), or `OEM Strings` (Type 11 structure). Click the **Show details** drop-down menu to add values to any of these filter criteria and then click **Add**.
5. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.
6. Click **Save**.

The image is saved in Image Library and now you can assign it to clusters and standalone hosts in your vCenter instance.

Edit an Image in Image Library

You can edit an image in Image Library that is not currently assigned to any cluster or standalone host to add, remove, or update an image element. You can also duplicate an image that is associated with a cluster or standalone host and then edit the image elements and save is as a new image in Image Library.

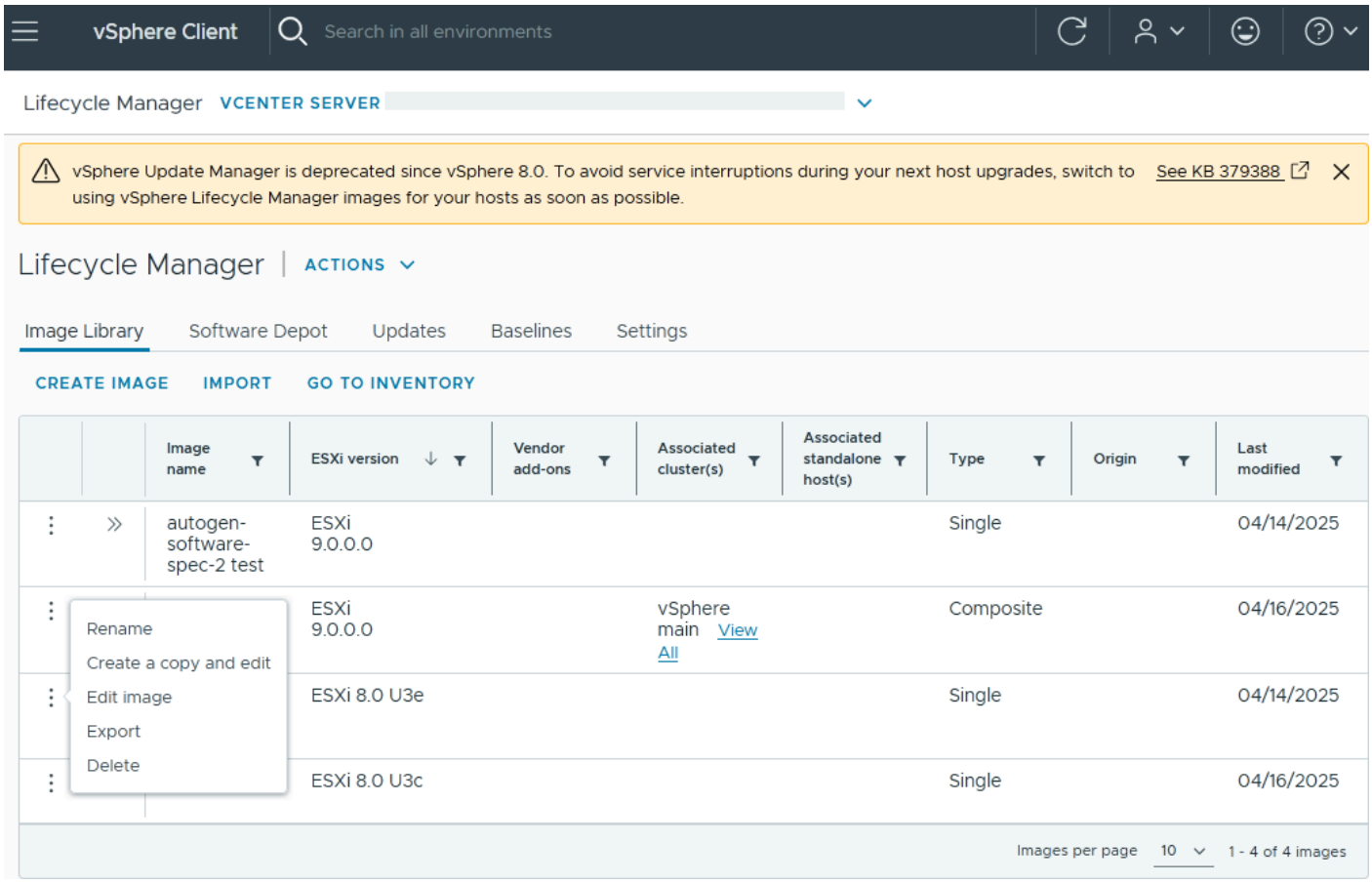
- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.
- Verify that the image is not assigned to a cluster or standalone host.
- Verify that the vCenter version is 9.0 and later.

When you start editing an image in Image Library, vSphere Lifecycle Manager saves a working copy of the image draft. If you edit an image but, for some reason, do not save the new setup, the next time you start editing, you can either use the saved draft as a starting point or discard the previous changes entirely.

You can validate an image draft before you save it in Image Library. During validation, vSphere Lifecycle Manager checks for missing dependencies and conflicting components. You must resolve all issues before you save the image in Image Library.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Image Library** tab, navigate the image that you want to edit and from the **Row actions** (three dots icon) menu, select one of the following options.



Option	Description
Rename an image	You can change the unique name of the image in Image Library. This name is unique for all vCenter instances. When you change the image name in Image Library, if this image is assigned to a cluster or standalone host, you see the changes reflected in the respective Inventory view under the **Updates** tab. **Note:** vSphere Lifecycle Manager does not allow two images to have the same name but different settings, or two images with the same settings to have different names in one vCenter instance.
Create a copy of an image and edit its settings	1. From the **Row actions** (three dots icon) menu, click **Create a copy and edit**.

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td></td><td>2. In the Edit Image copy pane, modify the image setup.
<table>
<tr><th>Image Element</th><th>Possible Modifications</th></tr>
<tr><td>ESX Version</td><td>From the ESXi Version drop-down menu, select a new ESX base image.</td></tr>
<tr><td>Vendor Addon</td><td>• To add a vendor add-on to the image, click Select and select a vendor add-on.
• To change the version of the vendor add-on in the image or to select a new vendor add-on, click the pencil icon and make your changes.
• To remove the vendor add-on from the image altogether, click the trash icon.</td></tr>
<tr><td>Firmware and Drivers Addon</td><td>• To add a firmware add-on to the image, click Select. In the Select Firmware and Drivers Addon dialog box, specify a hardware support manager and select a firmware add-on to add to the image.
• To select a new firmware add-on, click the pencil icon and make your changes.
• To remove the firmware add-on element from the image altogether, click the trash icon.
Selecting a firmware add-on for a family of vendor servers is possible only if the respective vendor-provided hardware support manager is registered as an extension to the vCenter where vSphere Lifecycle Manager runs.
Note:
In VCF 9.1, vSphere Lifecycle Manager validates the currently installed firmware version for a given device against the vSAN HCL when you do not configure a hardware support manage. This check ensures that the current firmware remains compatible after the upgrade since the system does not update firmware during this process. Both vCenter and ESX must run version 9.1 or later to generate this hardware compatibility report for homogeneous or heterogeneous clusters.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>Image Element</th><th>Possible Modifications</th></tr>
<tr><td>Components</td><td>Click Show details and view the list of additional components in the image.
To add components to the image, click Add Components and select the components and their corresponding versions to add to the image.
CAUTION:
If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can select and retain a lower version for an additional component that is part of the ESX base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster.
To delete a component from the image, click the trash icon in the table with components.
CAUTION:
If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can remove the VMware Host Client and / or ESX VM Tools components from the base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster. No other components can be removed from the base image.
To delete a manually added component that overrides a component in the selected vendor add-on or base image, click the undo icon in the table with components.
This action reverts the override.</td></tr>
</table>
3. If you edit a composite image for a cluster, add one or more alternative images.
You create a composite image for clusters that have hosts from different vendors or from the same vendor but different generations, family, or model.
a. Click the Add image button at the top right corner of the Step 1: Define Image card.
The Composite Image Name text field appears above the Default Image tab.
The New Image tab appears next to the Default Image tab.
b. In the Composite Image Name text field, enter the unique name for the composite image.
c. In the New Image tab, enter a unique name for the alternative image.
d. Modify the image setup to select a vendor add-on and the version of the add-on, add firmware and drivers add-on to the image, or manage the additional components of the image.
Note that you cannot change the base image of an alternative image directly. To modify the base image, you must do so from the Default Image tab of the composite image, and this change is reflected in all alternative images.
e. In the Selection Criteria section, specify the rules that are used to filter the hosts to which the alternative image is assigned.
• You can select hosts manually.
a. Click the Edit hosts button above the table.
The Select Hosts dialog box opens.
b. On the All Hosts tab, select the hosts to which you want to assign the alternative image.
c. Optionally, you can view all selected hosts on the Selected Hosts tab.
d. Click Save.
Selected hosts are added to the table in the Selection Criteria section.
</td></tr>
</table>

Option	Description
	• Filter hosts by using SMBIOS fields like `Manufacturer` (Type 1 structure), `Product Name` (Type 1 structure), `Family` (Type 1 structure), or `OEM Strings` (Type 11 structure). Click the **Show details** drop-down menu to add values to any of these filtering criteria and click **Add**. 4. If you edit a composite image, remove one or more alternative images.
Edit an unassigned image	If an image is not associated with a cluster or a host, you can edit its elements. 1. From the **Row actions** (three dots icon) menu, click **Edit image**. The **Edit Image** pane appears. 2. Edit the image settings in the same way as described above.

3. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.

4. Click **Save**.

The image is edited in Image Library and now you can assign it to clusters and standalone hosts in your vCenter instance.

Export an Image from Image Library

Export an image from Image Library when you want to use the same image for clusters or standalone hosts in the same or in a different vCenter instance.

- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.
- Verify that the vCenter version is 9.0 and later.

Depending on your goals and the type of image, you can export an image as a JSON file, as an installable ISO image, or as an offline bundle that contains all software packages included in the image. The export format depends on your needs and goals. Composite images can only be exported in a JSON file.

If you export an image as an ISO file, you cannot import back the installable ISO image on a vCenter instances of version 9.0 and later. On a vCenter instances of version 9.0 or later, you can only import offline bundles to the vSphere Lifecycle Manager depot or JSON files to the Image Library, a cluster, or a standalone host. See Import an Image in Image Library, Import an Image to a Cluster or Standalone Host, and Import Updates to the Depot.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Image Library** tab, navigate the image that you want to export and from the **Row actions** (three dots icon) menu, click **Export**.

   The **Export Image** dialog box opens.
3. In the **Export Image** dialog box, select a file format and click **Export**.

   For a composite image, only a JSON file is exported.

   You can export a single image in one file format at a time.

The exported file is saved on your local machine.

Import the image to the Image Library in the same or in a different vCenter instance. You can also import the image to a single cluster or standalone host in the same or in a different vCenter instance.

Import an Image in Image Library

You can import an image of a cluster or a standalone host as a JSON file in Image Library. The imported image can be assigned to a single or multiple clusters or standalone hosts in the vCenter instance.

- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.
- Verify that the vCenter version is 9.0 and later.

You can import an image only if it is in a JSON file format. The JSON file contains only the image metadata, but not the actual software payload. To successfully import an image to Image Library and apply the software specification to clusters or standalone hosts in the vCenter instance, all the components specified in the image must be available in the vSphere Lifecycle Manager depot.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Image Library** tab, click **Import**.
3. In the **Import Image** dialog box, select a JSON file and click **Next**.
   - Enter a URL address to the JSON file that you want to import.
   - Browse to a JSON on your local machine.
4. In the **Edit Image** pane, modify the image setup.
5. If the image contains conflicting components and unresolved dependencies, resolve the issues and retry the procedure.
6. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.
7. Click **Save**.

The JSON file is imported in Image Library and now you can assign it to a single or numerous clusters or standalone hosts.

Verifying Image Integrity with SHA-256

You can verify the integrity of the vSphere Lifecycle Manager images that you create, customize, import, and export. Using the SHA-256 checksum of the image, you validate that the image you use across your vCenter systems is valid and intact.

Starting with vSphere 9.1, you can view and copy the generated SHA-256 checksum of each image in your environment.

In the **Image Library** tab on the vSphere Lifecycle Manager home view, you can select to display the **Checksum** column.

From the **Image Library** tab in the vSphere Lifecycle Manager home view in the vSphere Client, open the **Image Details** card and copy the checksum for an image by using the **Copy to clipboard** button. You can then verify that the checksum is the same as the one of an imported image.

Example Use Case for Verifying Image Integrity

User A is a system administrator who creates a vSphere Lifecycle Manager image that does not have the VMware Host Client component and exports the customized image from vCenter A. The image is uploaded to a remote server and the URL is shared with user B.

User B is another system administrator who wants to import the image from vCenter A and then remediate a cluster on vCenter B. To validate the image integrity, the user B must compare the SHA-256 checksum that is generated when the image was created on vCenter A, with the checksum of the imported image.

After importing the image in the Image Library on vCenter B, the user B takes the SHA-256 checksum obtained from the image in vCenter A, and uses it to filter the images by the **Checksum** column in the Image Library on vCenter B. If the verification is successful, the table must list only the image that is imported from vCenter A. User B can then proceed to assign the image to the chosen clusters and remediate them.

Managing the Lifecycle of Clusters with vSphere Lifecycle Manager Images at Cluster Level

A vSphere Lifecycle Manager cluster is a cluster of ESX hosts that you manage with images. When you create a cluster, you must select the image that is associated with that cluster.

Creating a vSphere Lifecycle Manager Cluster

When you create a cluster, you must select the image for that cluster from the available options. During cluster creation, you can choose from the following options:

- Select an image from Image Library.
- Set up an image yourself.
- Use the image of a host within or outside the current vCenter instance.

In all cases, you can also enable vSphere Configuration Profiles for the cluster that you create.

The default selections in the **New Cluster** wizard are based on the presumption that you are creating a cluster that uses an image from the Image Library.

Adding Hosts to a vSphere Lifecycle Manager Cluster

You can add hosts of ESX version 8.0 or later to a cluster that you manage with images. You can add a host to a cluster and at the same time use the image on that host as an image for the entire cluster.

You can add hosts only of version 8.x to a cluster that you manage with baselines.

Removing Hosts from a vSphere Lifecycle Manager Cluster

Removing a host from a cluster is a straightforward procedure. If you remove a host from a cluster that uses an image, the host retains the software and firmware installed during the last remediation against the image for the cluster.

> **Note:**
> When you remove a host from a vSAN cluster that you manage with an image, vSphere Lifecycle Manager invalidates the results from the last hardware compatibility check for the cluster. To obtain valid hardware compatibility information about the cluster, you must re-run a hardware compatibility check. For instructions on how to check the hardware compatibility for a cluster, see Check the Hardware Compatibility of a Cluster.

All cluster-related operations are described in full detail in the *vCenter and Host Management* documentation.

- For information about using Auto Deploy to deploy and provision ESX hosts, see the *VMware ESX Installation and Setup* documentation.
- For information about how to use Zero Touch Provisioning to provision ESX hosts over the network without DHCP and TFTP servers, see the Using Zero Touch Provisioning chapter in the *VMware ESX Installation and Setup* documentation.
- For information about how to use images for managing stateless hosts, see How to Use Images for Managing Stateless Hosts.

Create a Cluster that Uses an Image from Image Library

You can create a cluster that uses an image by selecting a desired image from Image Library. Make sure that Image Library has at least one image that meets your requirements available before you initiate the cluster creation workflow.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that the hosts are of ESX version 8.0 or later.
- Verify that a data center exists in the vCenter inventory.
- Verify that you have the required ESX base images, vendor add-ons, and components available in the vSphere Lifecycle Manager depot.

  Important:

  In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

  Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.
- Verify that you have an image specification available in Image Library. See Working with Image Library.
- If you want to enable vSphere Configuration Profiles on the cluster, verify that the following requirements are met:
  - The hosts in the cluster are of ESX version 8.0 or later.
  - vCenter is version 9.0 or later.

To create a cluster in a vCenter instance, you must select an image to manage the lifecycle of the hosts in that cluster.

1. In the vSphere Client, navigate to the **Hosts and Clusters** inventory.
2. Right-click a data center and select **New Cluster**.

   The **New Cluster** wizard opens.
3. Enter a name for the cluster.
4. Select DRS, vSphere HA, or vSAN cluster features.

   By default, all options are deselected.

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td>To use DRS with this cluster</td><td>1. Slide the switch to the right to enable the DRS service.
2. Click the info icon on the left to see the Default Settings for the DRS service. The default values are:
– Automation Level: Fully Automated Migration
– Threshold: 3</td></tr>
<tr><td>To use vSphere HA with this cluster</td><td>1. Slide the switch to the right to enable the vSphere HA service.
2. Click the info icon on the left to see the Default Settings for the vSphere HA service. You are present with the following default values:
<table><tr><td>Host Monitoring:</td><td>Enabled</td></tr><tr><td>Admission Control:</td><td>Enabled</td></tr><tr><td>VM Monitoring:</td><td>Disabled</td></tr></table></td></tr>
<tr><td>To use vSAN with this cluster</td><td>• Slide the switch to the right to enable the vSAN service.
For more information on vSAN, see the Software-Defined Storage with vSAN section in the VMware Cloud Foundation 9.x documentation.</td></tr>
</table>

5. Leave the **Select an image from the Image Library** radio button selected.

   This option is selected by default.

6. Optional: To enable vSphere Configuration Profiles on the cluster, select the **Manage configuration at a cluster level**.
7. Click **Next**.
8. On the **Image** page, select the desired image from the table and click **Next**.

   The table lists all image specifications currently available in Image Library.
9. On the **Review** page, review your selections and the image setup.
10. If you are satisfied with the configuration, click **Finish** to finish the creation of the cluster.

A cluster that uses an image appears in the vCenter inventory. You can view and customize the cluster image on the **Updates** tab for the cluster.

If the cluster has vSphere Configuration Profiles enabled, you can configure the settings of the hosts in the cluster collectively by going to the **Configure** tab for the cluster and clicking **Desired State** > **Configuration**. See Edit the Host Settings of a Draft Configuration in vSphere Client.

Add hosts to the cluster.

Create a Cluster That Uses an Image by Composing a New Image

You can create a vSphere Lifecycle Manager cluster that uses an image by setting up the desired image during the creation of the cluster. When you choose to set up the image manually instead of importing it from a host, you must have the necessary software available in the vSphere Lifecycle Manager depot.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that the hosts are of ESX version 8.0 or later.
- Verify that a data center exists in the vCenter inventory.
- Verify that you have the required ESX base images, vendor add-ons, and components available in the vSphere Lifecycle Manager depot.

  Important:

  In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

  Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.
- If you want to enable vSphere Configuration Profiles on the cluster, verify that the following requirements are met:
  - The hosts in the cluster are of ESX version 8.0 or later.
  - vCenter is version 9.0 or later.

When you set up an image manually, you must specify the ESX version for the image and, optionally, a vendor add-on. After the cluster creation is complete, you can further customize the image. For example, you can add components to the image. You can also configure a hardware support manager and add a firmware and drivers add-on to the image. For information about editing an image, see Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.

If you decide to enable vSphere Configuration Profiles on the cluster, you must set up the host configuration manually after you complete the cluster creation.

1. In the vSphere Client, navigate to the **Hosts and Clusters** inventory.
2. Right-click a data center and select **New Cluster**.

   The **New Cluster** wizard opens.
3. Enter a name for the cluster.
4. Select DRS, vSphere HA, or vSAN cluster features.

   By default, all options are deselected.

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td>To use DRS with this cluster</td><td>1. Slide the switch to the right to enable the DRS service.
2. Click the info icon on the left to see the Default Settings for the DRS service. The default values are:
– Automation Level: Fully Automated Migration
– Threshold: 3</td></tr>
<tr><td>To use vSphere HA with this cluster</td><td>1. Slide the switch to the right to enable the vSphere HA service.
2. Click the info icon on the left to see the Default Settings for the vSphere HA service. You are present with the following default values:
<table><tr><td>Host Monitoring:</td><td>Enabled</td></tr><tr><td>Admission Control:</td><td>Enabled</td></tr><tr><td>VM Monitoring:</td><td>Disabled</td></tr></table></td></tr>
<tr><td>To use vSAN with this cluster</td><td>• Slide the switch to the right to enable the vSAN service.
For more information on vSAN, see the Software-Defined Storage with vSAN section in the VMware Cloud Foundation 9.0 documentation.</td></tr>
</table>

5. On the **Basics** page, enter a name for the cluster and enable vSphere DRS, vSphere HA, or vSAN.
6. Select the **Create a new image** radio button.
7. Optional: To enable vSphere Configuration Profiles on the cluster, select the **Manage configuration at a cluster level**.
8. Click **Next**.
9. On the **Image** page, set up the desired image.
   a) Select an ESX version.
   b) Optional: Select a vendor add-on and the version of the add-on.
10. Click **Next**.
11. On the **Review** page, review your selections and the image setup.
12. If you are satisfied with the configuration, click **Finish** to finish the creation of the cluster.

A cluster that uses an image appears in the vCenter inventory. You can view and customize the cluster image on the **Updates** tab for the cluster.

If the cluster has vSphere Configuration Profiles enabled, you can configure the settings of the hosts in the cluster collectively by going to the **Configure** tab for the cluster and clicking **Desired State** > **Configuration**. See Edit the Host Settings of a Draft Configuration in vSphere Client.

Add one or more hosts to the new cluster.

Create a Cluster That Uses an Image by Importing an Image from a Host

During cluster creation, instead of composing a new image or using an image from Image Library, you can import the desired software specification from a reference host.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that the vCenter version is 9.0 and later.
- Verify that a data center exists in the vCenter inventory.
- Verify that the reference host is version ESX 8.0 or later.
- Verify that the ESX-OSDATA volume on the reference host is backed by a dedicated NVMe, SSD, or HDD device.
- Obtain the user name and password of the root user account for the reference host if it is not in your vCenter instance.

Note:

If you attempt to use an image from a stateless host without a password set and use this image to manage the lifecycle of the cluster, the operation fails. Verify that you set a password for the root user account for the stateless host.

- If you want to enable vSphere Configuration Profiles on the cluster, verify that the following requirements are met:
  - The hosts in the cluster are of ESX version 8.0 or later.
  - vCenter is version 9.0 or later.

If you choose to import an image, vSphere Lifecycle Manager extracts the software specification from the reference host, adds it to Image Library if such image specification does not exist there, and uses it for the newly created cluster. Importing an image saves you the time and effort of ensuring that you have all necessary components and images available in the vSphere Lifecycle Manager depot. Also, because you use a ready-made image, you do not need to spend time validating the new image.

During image import, along with extracting the software specification from the reference host, vSphere Lifecycle Manager also extracts the software depot associated with the image, and imports the software components to the vSphere Lifecycle Manager depot in the vCenter instance where you create the cluster. As a result, in air-gap scenarios, you only need one reference host to obtain the necessary ESX image and components in the local depot and to create a software specification for your clusters.

You can import an image from an ESX host that is in the same or a different vCenter instance. You can also import an image from an ESX host that is not managed by vCenter. Along with importing the image, you can also choose to move the reference host to the cluster. As a result, the newly created cluster uses the same image as the image on the reference host, which is now part of that cluster. If the reference host is in another vCenter instance, you can import the image from that host, but you cannot move it to the cluster.

Note:
When you import an image from a host, vSphere Lifecycle Manager retrieves the ESX version, vendor add-on, and other components from the host. vSphere Lifecycle Manager does not extract the components from solutions, such as vmware-fdm or nsx-lcp,and firmware updates installed on the reference host. Therefore, the image for the new cluster does not contain solution components or a firmware and drivers add-on. To obtain firmware updates in the depot and update the firmware and drivers add-on on your cluster, you must configure a hardware support manager for the cluster after the cluster is created. For more information about firmware updates, see Firmware Updates with vSphere Lifecycle Manager.

You can use the image of a reference host that is in a DPU-based cluster and import it into a non-DPU cluster. Similarly, you can select a non-DPU host as the reference host for a DPU-based cluster.

1. In the vSphere Client, navigate to the **Hosts and Clusters** inventory.
2. Right-click a data center and select **New Cluster**.
   The **New Cluster** wizard opens.
3. Enter a name for the cluster.
4. On the **Basics** page, enter a name for the cluster and enable vSphere DRS, vSphere HA, or vSAN.
5. Choose the import method of creating an image for the cluster.
   - To import an image from a host that is in the same vCenter inventory, select the **Extract an image from a host in this vCenter instance** radio button.
   - To import an image from a host that is in a different vCenter instance or a standalone host that is not added to your vCenter, select the **Extract an image from a new host** radio button.

6. Optional: To enable vSphere Configuration Profiles on the cluster, select the **Manage configuration at a cluster level** check box.
7. Click **Next**.
8. Select the host image you want to import and use for the cluster.

Import Option	Steps
Import an image from an existing host in the vCenter inventory	1. On the **Image** page, select the reference host which you want to extract the image from. 2. Click **Extract Image** under the **Image Specification** section. **Note:** Detailed information about the image on the selected host appears at the bottom of the page. You can view the ESX version and all additional components. 3. Click **Next**.
Import an image from a new host	1. On the **Image** page, enter the host details and click the **Find Host** button. 2. To start extracting the image specification on the host, click **Extract image**. 3. If a **Security Alert** dialog box opens, click **Yes** to confirm that you want to connect to the host. 4. To move a standalone host to the cluster, select the **Also move selected host to cluster** check box and click **Next**.

9. On the **Review** page, review your selections and verify that the selected reference host and its image are what you need.
10. Click **Finish**.

A cluster that uses an image appears in the vCenter inventory. Depending on your selections, the reference host might be in the newly created cluster. The image for that cluster is identical to the image on the selected reference host. You can view and customize the cluster image on the **Updates** tab for the cluster.

If the cluster has vSphere Configuration Profiles enabled, you can configure the settings of the hosts in the cluster collectively by going to the **Configure** tab for the cluster and clicking **Desired State** > **Configuration**. See Edit the Host Settings of a Draft Configuration in vSphere Client.

Add other hosts to the cluster.

Add Hosts to a Cluster that Uses an Image

When you add hosts to a cluster, you can appoint one of those hosts as a reference host. vSphere Lifecycle Manager extracts and uses the image on the reference host as the new image for the cluster. The option to import a host's image to the cluster facilitates and simplifies the cluster upgrade operation by removing the need for you to manually import components to the vSphere Lifecycle Manager depot, set up and validate a new image, check the compliance of the hosts against the image, and then remediate the cluster against the image to apply the new software specification to all hosts.

- Verify that the vCenter version is 9.0 or later.
- Verify that the hosts to add are of version ESX 8.0 or later.
- Verify that the hosts that you add to the cluster are of the same main and patch version as the rest of the hosts.
- Obtain the user name and password of the root user account for the hosts that are not in your vCenter instance.
- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- To add a host and assign its image to the cluster, verify that the following requirements are met.
  - The vCenter instance is of version 9.0 or later.
  - The ESX version on the reference host is 8.0 or later.
- To add a host to a cluster that has vSphere Configuration Profiles enabled, verify that the following requirements are met.
  - The vCenter instance is of version 9.0 or later.

- The ESX host is of version 8.0 or later.
- The host can have NSX or vSphere Distributed Switch configured.
- No host profile is attached to the host.

Along with extracting the software specification from the appointed reference host, vSphere Lifecycle Manager also extracts the software depot associated with the image, and imports the components to the vSphere Lifecycle Manager depot in the vCenter instance where the target cluster is.

Note:
When you import an image from a host, vSphere Lifecycle Manager retrieves the ESX version, vendor add-on, and user-added components from the host. vSphere Lifecycle Manager does not extract the components from solutions and firmware updates installed on the reference host. Therefore, the new image for the cluster does not contain solution components or a firmware and drivers add-on. To obtain firmware updates in the depot and add a firmware and drivers add-on to your cluster image, you must configure a hardware support manager. For more information about firmware updates, see Firmware Updates with vSphere Lifecycle Manager.

Adding a host to a cluster and importing its image to the target cluster changes the compliance state of the other hosts in the cluster. After adding the host and setting its image as the new cluster image, you can run a compliance check. The newly added host is compliant against the new cluster image. The rest of the hosts become non-compliant. To apply the new cluster image to all the hosts in the cluster and make them compliant, you must remediate the cluster.

Note:
You cannot downgrade the software that is actually installed on the hosts in the cluster. If the image on the reference host contains software components of lower version, you can still import and use that image for the cluster. However, the hosts in the cluster become incompatible with the new image, and you cannot proceed and remediate the cluster against that image.

1. In vSphere Client, navigate to the **Hosts and Clusters** inventory.
2. Right-click cluster and select **Add Hosts**.
   The **Add Hosts** wizard opens.
3. On the **Add hosts**, specify the hosts that you want to add to the cluster and click **Next**.
   - Add a host that is not in the same vCenter inventory.
     a. Click the **New Hosts** tab.
     b. Enter the required information about the host in the text boxes.
     c. To add more new hosts, click the **Add Host** button and enter the required information.
     d. Optionally, if all hosts use the same credentials, select the **Use the same credentials for all hosts** check box.
   - Add a host that is in the same vCenter inventory.
     a. Click the **Existing hosts** tab.
     b. Select one or multiple hosts from the list.

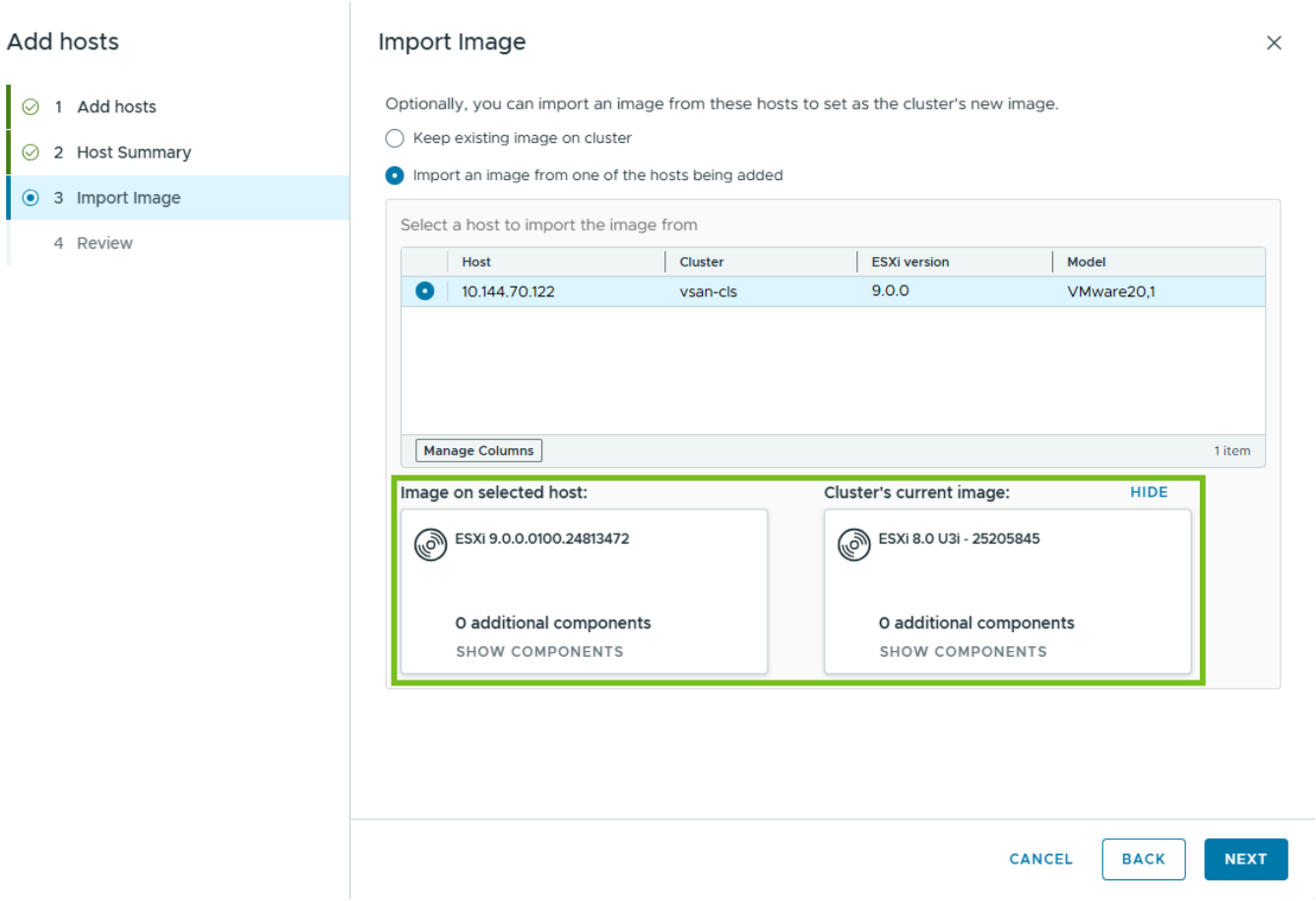
   Note:
   If you want to add multiple hosts, you do not need to specify only new hosts or only existing hosts. You can specify new hosts and select from the existing hosts at the same time.
4. On the **Host Summary** page, review the information about the hosts and click **Next**.
5. On the **Import Image** page, select the host whose image you want to use for the cluster.
   - To add the specified hosts to the cluster without changing the current image for that cluster, select the **Don't import an image** radio button.
   - To use any of the specified hosts as a reference host and use its image as the new image for that cluster, select the **Select which host to import the image from** radio button and select a host from the list.

     Note:
     If the image on the host that you select is of version earlier than ESX 8.0, you cannot proceed with importing the image from that host.

When you select a host to import an image from, a card with information about the image appears. You can view the ESX version, the vendor add-on, and the additional components that the image on the reference host contains.

6. If you select the **Select which host to import the image from** radio button, select one of the hosts from the table that appears.
   a) Click **Show Cluster's Current Image** at the bottom of the page.

      A card with information about the current image for the cluster appears. Before you proceed, you can review and compare the current cluster image with the image on the selected host before you proceed. In this way, you can ensure that you selected the right reference host.



7. Click **Next**.
8. On the **Ready to Complete** page, review the information about the selected hosts and the new cluster image.
9. Click **Finish** to add the hosts to the cluster.

The specified hosts are added to the cluster.

If you chose to import an image from a host during the host addition procedure, the image for the cluster changes. The new software specification for the cluster is identical to the one on the selected and added reference host.

If you chose to import an image from a host during the host addition procedure, remediate the cluster to apply the new image to all the hosts.

Managing the Lifecycle of Standalone ESX Hosts with vSphere Lifecycle Manager Images at Host Level

You can use an image to manage the lifecycle of any standalone ESX host that is part of your vCenter inventory.

A standalone host is a host that is connected to a vCenter instance but is not part of any cluster. In your inventory, standalone hosts are the hosts directly under a data center object or in a host folder.

The permissions and privileges that you need to manage a standalone host with an image are the same as the vSphere Lifecycle Manager permissions and privileges that you need for the respective operations at a cluster level.

Using vSphere Lifecycle Manager images does not reduce the number of standalone hosts a vCenter instance can manage.

You can perform on standalone hosts all vSphere Lifecycle Manager operations that you can perform at a cluster level by using vSphere Client.

When you add a standalone host to a data center or a folder, the option to use an image from Image Library is selected by default. You must make sure that Image Library contains at least one single image specification that you can use. During the addition of the standalone host, you can either set up an image manually, use the existing image on the host, or import an image from a reference host in the vCenter inventory. For information, see Set Up a New Image to a Host When Adding the Host to a Data Center or Folder , Import an Image from a Reference Host When Adding a Host to a Data Center or Folder , and Use the Current Image of a Host When Adding the Host to a Data Center or Folder.

After the standalone host is added to the data center or folder, you can further customize the image. For more information, see Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.

You can start using images on a standalone host that you still manage with baselines. When you take a host out of a cluster that you manage with vSphere Lifecycle Manager image, the host retains the image. You can move a standalone host that you manage with an image to a cluster that you manage with an image.

For information about how to transition an existing standalone host to a host that is managed with a vSphere Lifecycle Manager image and all available workflows for managing a standalone host with images, see How to Use vSphere Lifecycle Manager Images .

Select an Image from Image Library When Adding the Host to a Data Center or Folder

When you add a standalone host to a data center or folder in vCenter, you can select a single image from Image Library to manage the host lifecycle.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that a data center or a folder exists in the inventory.
- Verify that the vCenter instance is version 9.0 or later.
- Verify that the host is of ESX version 8.0 or later.
- Verify that an image specification exists in Image Library.
- Obtain the user name and password of the root user account for the host.
- Verify that you have the required ESX base images, vendor add-ons, and components available in the vSphere Lifecycle Manager depot.

  Important:

  In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

  Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.
- Verify that you have the proper privileges. For more information, see vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to a data center or folder within a data center.
2. Right-click the data center or folder and select **Add Host**.

   The **Add Host** wizard opens.

3. On the **Name and location** page, enter the IP address or the name of the host and click **Next**.
4. On the **Connection settings** page, enter administrator credentials and click **Next**.
5. On the **Host summary** page, review the host summary and click **Next**.
6. On the **Host lifecycle** page, leave the **Select an image from the Image Library** radio button selected.

   This option is selected by default.
7. On the **Image** page, select a single image from the listed images currently available in Image Library and click **Next**.
8. On the **Assign license** page, add a license to the host and click **Next**.
9. On the **Lockdown mode** page, select a lock-down mode option to deactivate the remote access for the administrator account after vCenter takes control of this host and click **Next**.
10. On the **VM location** page, select a location for the virtual machines that reside on the host and click **Next**.
11. On the **Ready to complete** page, review your selections and the image setup and click **Finish**.

A standalone host that uses a single image from Image Library appears in the vCenter inventory.

View and customize the host image on the **Updates** tab for the standalone host.

Import an Image from a Reference Host When Adding a Host to a Data Center or Folder

When you add a standalone host to a data center or folder, you can import an image from another host in vCenter to manage the host lifecycle.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that a data center or a folder exists in the inventory.
- Verify that the vCenter instance is version 8.0 Update 3 or later.
- Verify that the reference host is of ESX version 8.0 or later.
- Verify that the ESX-OSDATA volume on the reference host is backed by a dedicated NVMe, SSD, or HDD device.
- Obtain the user name and password of the root user account for the host.

  Note:

  If you attempt to import an image from a stateless host without a password set and use this image to manage the lifecycle of the standalone host, the operation fails. Verify that you set a password for the root user account for the stateless host.
- Verify that you have the proper privileges. For more information, see vSphere Lifecycle Manager Privileges For Using Images.

During the addition of a standalone host to a data center or a folder, you can choose to import an image from a host in the inventory and use this image to manage the lifecycle of the host. After the standalone host is added to the data center or folder, you can further customize the image. For example, you can customize the additional components or configure a hardware support manager and add firmware and drivers add-ons to the image. For information about editing an image, see Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.

1. In the vSphere Client, navigate to a data center or folder within a data center.
2. Right-click the data center or folder and select **Add Host**.

   The **Add Host** wizard opens.

3. On the **Name and location** page, enter the IP address or the name of the host and click **Next**.
4. On the **Connection settings** page, enter root credentials and click **Next**.
5. On the **Host summary** page, review the host summary and click **Next**.
6. On the **Host lifecycle** page, select the **Extract an image from a host in this vCenter instance** radio button and click **Next**.
7. On the **Image** page, select the host which image you want to assign to the host and click **Extract image**.

   You can view the ESX version and all additional components.

   After the image specification is extracted, you can view detailed information about the image in the **Image specification** section.
8. Click **Next**.
9. On the **Assign license** page, add a license to the host and click **Next**.
10. Optional: On the **Lockdown mode** page, select a lock-down mode option to deactivate the remote access for the administrator account after vCenter takes control of this host and click **Next**.
11. On the **VM location** page, select a location for the virtual machines that reside on the host and click **Next**.
12. On the **Ready to complete** page, review your selections and the image setup and click **Finish**.

A standalone host that uses an image from another host appears in the vCenter inventory.

View and customize the host image on the **Updates** tab for the standalone host.

Use the Current Image of a Host When Adding the Host to a Data Center or Folder

When you add a standalone host to a data center or folder in vCenter, you can use the current image on the host to manage its lifecycle.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that a data center or a folder exists in the inventory.
- Verify that the vCenter instance is version 9.0 or later.
- Verify that the host is of ESX version 8.0 or later.
- Obtain the user name and password of the root user account for the host.

  Note:

  If you attempt to add a stateless host without a password set and use its image, the operation fails. Verify that you set a password for the root user account for the stateless host.
- Verify that you have an ESX image available in the vSphere Lifecycle Manager depot.
- Verify that you have the proper privileges. For more information, see vSphere Lifecycle Manager Privileges For Using Images.

When you add a standalone host to a data center or a folder, you can use the image that is currently present on the host. After the standalone host is added to the data center or folder, you can further customize this image. For example, you can add additional components or configure a hardware support manager and add firmware and drivers add-ons to the image. For information about editing an image, see Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.

1. In the vSphere Client, navigate to a data center or folder within a data center.
2. Right-click the data center or folder and select **Add Host**.

   The **Add Host** wizard opens.

3. On the **Name and location** page, enter the IP address or the name of the host and click **Next**.
4. On the **Connection settings** page, enter administrator credentials and click **Next**.

   Note:

   If a host cannot be validated automatically by the system, you are prompted to manually validate its certificate and accept its thumbprint in the **Security Alert** pop-up if you recognize the certificate.
5. On the **Host summary** page, review the host summary and click **Next**.
6. On the **Host lifecycle** page, select the **Extract the image on the host** radio button and click **Next**.
7. On the **Image** page, view the detailed information about the image that appears in the **Image Specification** section and click **Next**.

   You can view the ESX version and all additional components.
8. On the **Assign license** page, add a license to the host and click **Next**.
9. Optional: On the **Lockdown mode** page, select a lock-down mode option to deactivate the remote access for the administrator account after vCenter takes control of this host and click **Next**.
10. On the **VM location** page, select a location for the virtual machines that reside on the host and click **Next**.
11. On the **Ready to complete** page, review your selections and the image setup and click **Finish**.

A standalone host that uses a vSphere Lifecycle Manager image appears in the vCenter inventory.

View and customize the host image on the **Updates** tab for the standalone host.

Set Up a New Image to a Host When Adding the Host to a Data Center or Folder

You can add a standalone host to a data center or folder and set up a new vSphere Lifecycle Manager image to manage the lifecycle of the host.

- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that a data center or a folder exists in the inventory.
- Verify that the host is of ESX version 8.0 or later.
- Obtain the user name and password of the root user account for the host.
- Verify that you have the required ESX base images, vendor add-ons, and components available in the vSphere Lifecycle Manager depot.

  Important:

  In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

  Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.
- Verify that you have the proper privileges. For more information, see vSphere Lifecycle Manager Privileges For Using Images.

When you add a standalone host to a data center or a folder, you must specify the ESX version for the image and, optionally, a vendor add-on. After the standalone host is added to the data center or folder, you can further customize the image. For example, you can add components to the image. You can also configure a hardware support manager and add a firmware and drivers add-on to the image. For information about editing an image, see Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.

1. In the vSphere Client, navigate to a data center or folder within a data center.
2. Right-click the data center or folder and select **Add Host**.

   The **Add Host** wizard opens.

3. On the **Name and location** page, enter the IP address or the name of the host and click **Next**.
4. On the **Connection settings** page, enter administrator credentials and click **Next**.
5. On the **Host summary** page, review the host summary and click **Next**.
6. On the **Host lifecycle** page, select the **Create a new image** radio button and click **Next**.
7. On the **Image** page, set up the desired image and click **Next**.
   a) Select an ESX version.
   b) Optional: Select a vendor add-on and the version of the add-on.
8. On the **Assign license** page, add a license to the host and click **Next**.
9. Optional: On the **Lockdown mode** page, select a lock-down mode option to deactivate the remote access for the administrator account after vCenter takes control of this host and click **Next**.
10. On the **VM location** page, select a location for the virtual machines that reside on the host and click **Next**.
11. On the **Ready to complete** page, review your selections and the image setup and click **Finish**.

A standalone host that uses an image appears in the vCenter inventory.

You can view and customize the host image on the **Updates** tab for the standalone host.

Lifecycle Management of Clusters and Standalone Hosts at Scale

Image Library stores vSphere Lifecycle Manager images that can be assigned to clusters and standalone hosts in your vCenter instance at a vCenter, data center, and folder level. After running compliance and remediation precheck, you can apply the image simultaneously on multiple clusters and standalone hosts.

All fundamental lifecycle management tasks that you perform on a single cluster or standalone host levels, can also be performed on multiple clusters or standalone hosts at a vCenter, data center, or folder level. Lifecycle management on a global level includes the following tasks:

- Assign an image from Image Library to multiple clusters or standalone hosts.
- Check the compliance of the clusters or standalone hosts against the assigned image.
- Run a remediation precheck to ensure the software and hardware compatibility with the assigned image.
- Stage the assigned image to multiple clusters or standalone hosts.
- Remediate multiple clusters and standalone hosts in a vCenter instance at scale.

The overall workflow for managing the lifecycle of multiple clusters and standalone hosts at scale is the following:

1. Create an image or import an image as JSON file in Image Library. See Working with Image Library.
2. Assign the image to multiple clusters or standalone hosts. See Assign an Image to Clusters or Standalone Hosts at a vCenter Data Center or Folder Level.
3. Check the compliance and run a remediation precheck of all selected clusters and standalone hosts.
4. Stage the assigned image to multiple clusters and standalone hosts.
5. Remediate multiple clusters and standalone hosts against the assigned image. See

Assign an Image to Clusters or Standalone Hosts at a vCenter, Data Center, or Folder Level

You can attach a desired image specification to multiple clusters and standalone hosts that are already managed with images and remediate them against this image.

- Verify that vCenter is of version 9.0 and later.
- Verify that the hosts you want to assign an image to are of ESX version 8.0 or later.
- Review the requirements for using an image in What Are the Requirements for Using vSphere Lifecycle Manager.
- Verify that you have clusters and hosts managed with images.
- Verify that a data center exists in the vCenter inventory and the data center is not empty.
- Verify that you have a folder created in the vCenter inventory and the folder is not empty
- Verify that you have at least one image available in Image Library.

- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.

1. In the vSphere Client, navigate to a vCenter instance, a data center, or a folder.
2. Click on the **Updates** tab and select **Hosts** > **Images**.
3. In the **Images** pane, click the **Clusters** or **Standalone Hosts** tab depending on the objects you want to select.
4. Select one or more check boxes next to the objects to which you want to assign an image and click **Assign Image**.

   The **Assign image** dialog box opens.
5. On the **Select image** page, select the image that you want to assign to the selected clusters or standalone hosts and click **Next**.

   You choose from the images currently available in Image Library. See Working with Image Library.

   Note:

   You manage the lifecycle of standalone hosts only with single images.
6. On the **Review** page, review the information about the image and the objects you want to assign the image to, and click **Assign**.

   The **Assign image to cluster(s) and standalone host(s)** tasks starts.

vSphere Lifecycle Manager starts the **Check hardware compatibility of cluster's host with image** task even for clusters without vSAN and performs compliance checks of all selected clusters and standalone hosts against the assigned image.

Remediate one or more of the clusters or standalone hosts to which you assigned an image from Image Library. Optionally, before you remediate any of the objects with the assigned image, you can check their compliance against the image, run remediation prechecks, and stage the image to the respective objects.

Check the Compliance of Multiple Clusters and Standalone Hosts Against an Image

When you check the compliance of multiple clusters and standalone hosts against an assigned image, vSphere Lifecycle Manager compares the software on each host in all selected clusters or standalone hosts with the software specified in the image.

- Verify that vCenter is of version 9.0 and later.
- Verify that you have clusters and hosts managed with images.
- Verify that you have a folder created in the vCenter inventory and the folder is not empty.
- Verify that a data center exists in the vCenter inventory and the data center is not empty.
- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.

You can select a group of clusters or standalone hosts at a vCenter, data center, or folder level to run compliance checks and retrieve information about the impact of the remediation operations on each host. For more information about the possible compliance states, see Check the Compliance Against an Image.

1. In the vSphere Client, navigate to a vCenter instance, a data center, or a folder.
2. Click on the **Updates** tab and select **Hosts** > **Images**.
3. In the **Images** pane, click the **Clusters** or **Standalone Hosts** tab depending on the objects you want to select.
4. Select one or more checkboxes next to the objects for which you want to run a compliance check and click **Check Compliance**.

   The `Check compliance of given entities against their respective images` task starts.

The compliance statuses are visible in the **Compliance** column on the **Clusters** and **Standalone Hosts** tabs.

Remediate the non-compliant clusters or standalone hosts. See Remediate Multiple Clusters and Standalone Hosts at Scale.

Run Remediation Prechecks for Multiple Clusters and Standalone Hosts

To ensure that the health of a group of clusters or standalone hosts is optimal and there are no issues during remediation, you may choose to perform remediation checks against some of all of these clusters or standalone hosts in parallel.

- Verify that vCenter is of version 9.0 and later.
- Verify that you have clusters and hosts managed with images.
- Verify that you have a folder created in the vCenter inventory and the folder is not empty.
- Verify that a data center exists in the vCenter inventory and the data center is not empty.
- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.

You can select a group of clusters or standalone hosts at a vCenter, data center, or folder level to run remediation prechecks.

For vSAN clusters, the remediation precheck operation includes a hardware compatibility check. Depending on how you configure the vSphere Lifecycle Manager remediation settings, vSphere Lifecycle Manager might prevent remediation if hardware compatibility issues exist.

1. In the vSphere Client, navigate to a vCenter instance, a data center, or a folder.
2. Click on the **Updates** tab and select **Hosts** > **Images**.
3. In the **Images** pane, click the **Clusters** or **Standalone Hosts** tab depending on the objects you want to select.
4. Select one or more checkboxes next to the objects for which you want to run a remediation precheck and click **Precheck**.
   - If you want to use the image that is currently assigned on the cluster or the standalone host, click **Precheck with cluster's image** or **Precheck with host's image**.
   - Optionally, you can use another image from Image Library to run the remediation prechecks.
     a. From the **Precheck** drop-down menu, click **Precheck with another image** .
        The **Run Precheck with another image** dialog box opens.
     b. Select an image from Image Library and click **Run precheck**.
     c. On the **Precheck results** page, view the progress and results of the remediation precheck operation.
     d. Click **Export**, to download the report if any issues are found during the precheck.
     e. Review the results from the precheck and click **Done**.

Fix all issues that vSphere Lifecycle Manager identifies during the remediation precheck and remediate the clusters or standalone hosts that you want.

Stage Images to Multiple Clusters and Standalone Hosts

You can download the image components from the vSphere Lifecycle Manager depot to a group of clusters or standalone hosts without applying the software and firmware updates immediately. Staging reduces the time that hosts spend in maintenance mode.

- Verify that vCenter is of version 9.0 and later.
- Verify that you have clusters and hosts managed with images.
- Verify that you have a folder created in the vCenter inventory and the folder is not empty.
- Verify that a data center exists in the vCenter inventory and the data center is not empty.
- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.

After the staging operation completes successfully, you see the statuses of the clusters and standalone hosts change to `Staged` in the **Staging** column on the **Clusters** and **Standalone Hosts** tabs. For more information about staging an image, see Staging Images to Hosts in a Cluster or Standalone Hosts.

1. In the vSphere Client, navigate to a vCenter instance, a data center, or a folder.
2. Click on the **Updates** tab and select **Hosts** > **Images**.
3. In the **Images** pane, click the **Clusters** or **Standalone Hosts** tab depending on the objects you want to select.
4. Select one or more checkboxes next to the objects to which you want to stage an image and click **Stage**.

After staging completes successfully, the status of the selected objects changes to `Staged`.

Remediate the clusters and standalone hosts to install the staged software and firmware to all hosts.

Remediate a Group of Clusters or Standalone Hosts

When you assign an image from Image Library to a group of clusters or standalone hosts, the software specified in the image is not immediately installed on the hosts in the clusters or the standalone hosts.

- Verify that vCenter is of version 9.0 and later.
- Verify that you have clusters and hosts managed with images.
- Verify that you have a folder created in the vCenter inventory and the folder is not empty.
- Verify that a data center exists in the vCenter inventory and the data center is not empty.
- Verify that you have the proper privileges. See Required Privileges for Using vSphere Lifecycle Manager.

Remediating a group of clusters or standalone hosts at a vCenter, data center, or folder level follows the same rules as running the remediation one cluster or standalone host at a time. See Remediating a Cluster or a Standalone Host Against an Image.

1. In the vSphere Client, navigate to a vCenter instance, a data center, or a folder.
2. Click on the **Updates** tab and select **Hosts** > **Images**.
3. In the **Images** pane, click the **Clusters** or **Standalone Hosts** tab depending on the objects you want to select.
4. Select one or more checkboxes next to the objects which you want to remediate and click **Remediate**.

   The **Review Remediation Impact** dialog box opens. The dialog box contains the **Foundation Agreement**.
5. Review the **Foundation Agreement** and, if satisfied, accept it by selecting the respective checkbox.

   The checkbox is selected by default.
6. Click the **Start remediation** button.

   The `Remediation of cluster(s) and standalone host(s)` task followed by separate remediation tasks for all selected clusters or standalone hosts appear in the **Recent Tasks** pane.

All clusters or standalone hosts are remediated successfully and their statuses change from `Unknown` to `OK` in the **Remediation** column on the **Clusters** and **Standalone Hosts** tabs.

Managing Hosts Provisioned with vSphere Auto Deploy or Zero Touch Provisioning

You can add stateless standalone hosts to a data center or folder or to a cluster managed with an image. With vSphere Foundation 9.1, host provisioning with Auto Deploy is enhanced by Zero Touch Provisioning of bare-metal hosts.

In addition to legacy PXE infrastructure, With Zero Touch Provisioning in vSphere Foundation 9.1, you can provision ESX hosts at scale by using the UEFI HTTP(s) boot protocol and the Auto Deploy rules. This allows hosts to boot directly from a URL served by vCenter, reducing reliance on complex DHCP and TFTP infrastructure. For maximum security, HTTPS boot is recommended. For more information about how to use Zero Touch Provisioning, see Using Zero Touch Provisioning in the *VMware ESX Installation and Setup* guide.

Requirements and Limitations for Managing Stateless Hosts in vCenter 9.0 and Later

In a vCenter instance of version 9.0 and later, you add a host to a data center or a folder, or create a cluster, by setting up an vSphere Lifecycle Manager image to manage its lifecycle. You can also create an Auto Deploy rule and use Image Builder-generated images when adding a host to a data center or a folder, or creating a cluster.

When managing stateless hosts in a vCenter instance, consider these requirements and limitations:

- If you want to add a stateless host to a cluster managed with an image, the cluster must be empty and you must first activate the stateless support for this cluster. You can activate the stateless host support from the **Clusters** tab on the **Auto Deploy** page. For more information about how to use vSphere Auto Deploy, see Installing ESX Using vSphere Auto Deploy chapter in the *ESX Installation and Setup* documentation.
- The vCenter instance to which you want to add a stateless host must be of version 9.0 and later.
- The stateless hosts which you want to manage in your vCenter instance must be of versions 8.0 and later.
- If you want to group stateless hosts in a cluster that is managed with an image that cluster must contain only stateless hosts.
- The lifecycle of standalone stateless hosts and stateless hosts in a cluster that is managed with an image is managed only from Auto Deploy.
- In vSphere Foundation 9.1, you can create Auto Deploy rules that install different Image Builder images to the stateless hosts in a cluster thus creating a heterogeneous cluster. The cluster must have the **Auto Deploy - stateless** provisioning method activated.

  Note:

  You must create the Auto Deploy rules for provisioning the stateless hosts in the cluster by selecting an Image Builder images for the different host vendors and not a composite one.
- The vSphere Configuration Profiles cannot be enabled to manage the host configuration of standalone stateless hosts and clusters with stateless hosts.

Requirements and Limitations for Managing Stateful Hosts Provisioned with Auto Deploy in vCenter 9.1 and Later

In vCenter 9.1, you can manage the lifecycle and configuration of stateful hosts in a cluster that is provisioned with vSphere Auto Deploy by using two methods:

- **Auto Deploy - stateful**
  - The provisioned vSphere Lifecycle Manager image is stored on the hosts in the cluster or the standalone host.
  - All newly created clusters default to this provisioning method for the hosts that you plan to add.
  - This provisioning method can be changed to **Auto Deploy - stateless** only when the cluster is still empty.
  - You cannot use a composite vSphere Lifecycle Manager image for provisioning hosts in such clusters.
  - You can use vSphere Configuration Profiles to manage the configuration of all stateful hosts in the cluster.
- **Zero Touch Provisioning**
  - Hosts can be bootstrapped directly from a single HTTP(s) boot URL served by the vCenter instance on port `443`.
  - Hosts can be provisioned with a single or composite vSphere Lifecycle Manager image. Create a rule by selecting a composite image to remediate a heterogeneous cluster.
  - You can use all functionality that is provided by vSphere Configuration Profiles to manage the configuration of the stateful hosts in the cluster.
  - This provisioning method can be changed to **Auto Deploy - stateful** when the cluster is empty or the cluster already has some stateful hosts added.
  - To use Zero Touch Provisioning, both your vCenter instance and the ESX hosts must be of version 9.1 and later.

How to Use vSphere Lifecycle Manager Images

Using vSphere Lifecycle Manager images provides a simplified and unified workflow for patching and upgrade of ESX hosts. You can also use vSphere Lifecycle Manager images for bootstrapping purposes and firmware updates.

An image defines the exact software stack to run on all ESX hosts in a cluster or on a standalone host.

Important:

Starting with vSphere 9.0, upgrading to the latest version of ESX is only possible by using vSphere Lifecycle Manager images. Baseline support for host upgrades was deprecated in vSphere 8.0 and removed in 9.0 and later.

In vCenter 9.x, baselines retain only minor update and patching support for legacy ESX 8.x hosts. You cannot use baselines to upgrade hosts or to manage ESX 9.x hosts. For more information, see Broadcom knowledge base article 379388.

General Workflow

Using images to apply software and firmware updates to ESX hosts is a multi-stage process.

1. Software updates must become available in the vSphere Lifecycle Manager depot.
   To set up and use an image, you use the software updates that are available in the vSphere Lifecycle Manager depot. The depot contains base images, vendor add-ons, and additional components.
   Updates get into the vSphere Lifecycle Manager local depot through synchronization with configurable download sources. In vSphere Foundation 9.1, vSphere Lifecycle Manager is configured to synchronize with the official Broadcom depot through the central VCF software depot, using central activation code. See Broadcom knowledge base article 443647. Direct editing of customer token URLs in vCenter (legacy KB 390121) is replaced by the VCF software depot topology in 9.1. You can also import updates into the depot manually or with the VCF download tool.
   You can see the contents of the vSphere Lifecycle Manager depot in the vSphere Lifecycle Manager home view.
   For more information, see The vSphere Lifecycle Manager Depot.
2. Start using vSphere Lifecycle Manager images or assign an image from Image Library.
   vSphere Lifecycle Manager provides you with the option to start using images during the procedure for creating a cluster or adding a host to a data center or folder. If after you upgrade your vCenter instance to version 9.0 and later, you still have clusters or standalone hosts that are managed with baselines, you must switch from using vSphere Lifecycle Manager baselines to using vSphere Lifecycle Manager images as soon as possible.
   Even when you save the image, no software is installed on the ESX hosts during image setup.
   You can also assign an image from Image Library after you create a cluster or add a standalone host to a data center or folder.
   For more information, see , Managing the Lifecycle of Clusters with vSphere Lifecycle Manager Images at Cluster Level, Managing the Lifecycle of Standalone ESX Hosts with vSphere Lifecycle Manager Images at Host Level, and Transition a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images.
3. Check the compliance of the ESX hosts in the cluster or the standalone host against the image specification.
   The compliance check compares the current image on the ESX hosts in the cluster or the standalone host against the desired image that you specified during the setup process, and defines compatibility status of the hosts.
   For more information, see Check the Compliance of a Cluster or Standalone Host Against a vSphere Lifecycle Manager Image and View Host Compliance Information .
4. Review the compliance statuses of the hosts in the cluster or the standalone host.
5. You can run a remediation pre-check on an ESX host to ensure software and hardware compatibility with the image.
   Running a remediation pre-check is optional. The remediation pre-check ensures that all requirements for successful remediation are met. For more information, see Run a Remediation Pre-Check for a Cluster.
6. Remediate the non-compliant ESX hosts in the cluster or the non-compliant standalone host.
   Remediation is the process through which the software specification defined by the vSphere Lifecycle Manager image that you use for a cluster or a host is actually applied to the hosts in the cluster or to the standalone host.
   For more information about remediating hosts against an image, see Remediating a Cluster or a Standalone Host Against an Image.

Limitations

- When you set up and save, or assign an image to a cluster or a standalone host, the image is not applied to the hosts in the cluster or to the standalone host unless you remediate the hosts. The mere action of changing the management method does not alter the hosts in the cluster or the standalone host.

- After you set up an image for the cluster or the standalone host and remediate the hosts in the cluster or the standalone host against the image, standalone VIBs are deleted from the hosts.
- After you set up an image for the cluster or the standalone host and remediate the hosts in the cluster or the standalone host against the image, non-integrated solution agents are deleted from the hosts.

Transition a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images

You can manage a cluster or a standalone host with either images or baselines. You use baselines only to update a cluster or a standalone host. If you still have clusters or standalone hosts that use baselines in a vCenter instance version 9.0 and later, you must switch them to using images.

To transition a cluster or a standalone host to vSphere Lifecycle Manager images, you can use an image from Image Library, set up a new image, or use an image from a reference host. Before you proceed with setting up the image, vCenter checks and reports if the cluster or host is eligible for using images. For more information about cluster or host eligibility, see Cluster or Standalone Host Eligibility to Use vSphere Lifecycle Manager Images.

> **Important:**
>
> With vSphere 9.0, using baselines to upgrade the clusters and standalone hosts in your vCenter instances of version 9.0 and later is deprecated. For more information, see KB 379388.

System Requirements

To switch to using images, the cluster or host must meet multiple requirements.

- All ESX hosts in the cluster or the standalone host must be of version 8.0 and later.
- All ESX hosts in the cluster or the standalone host must be stateful.
  A stateful installation is one in which the host boots from a disk or from a vCenter-level UEFI HTTP(s) boot URL by using Zero Touch Provisioning.

  > **Note:**
  >
  > In a vCenter instance of version 9.0 and later, vSphere Lifecycle Manager images can be assigned to clusters with stateless hosts and standalone stateless hosts but only in very specific cases. See Managing Stateless Hosts with an Image.
- No host in the cluster or standalone host can contain any unknown components.
- Some of the hosts in the cluster can be from different vendors or from the same vendor but different generations, family, or model. Such clusters must be upgraded to ESX 9.0 first, then transitioned to images, and managed with a composite image.

If a host is of version earlier than ESX 8.0, you must first upgrade the host and then you can successfully switch to using images. For more information about using baselines for host patching and update operations, see Using vSphere Lifecycle Manager Baselines and Baseline Groups .

For more information about transitioning a stateless host into a stateful host, find information about Auto Deploy and Zero Touch Provisioning in the *VMware ESX Installation and Setup* documentation.

Specifics of the Transitioning Workflow

Several behavioral specifics exist when you switch to using vSphere Lifecycle Manager images.

- If you switch to using images for a cluster or a standalone host, you cannot revert to using baselines for the cluster or the standalone host.
- When you set up and save an image for a cluster or host, the image is not applied to the hosts in the cluster or to the standalone host unless you remediate the hosts. The mere action of changing the management method does not alter the hosts in the cluster or the standalone host.
- After you set up an image for the cluster or the standalone host and remediate the hosts in the cluster or the standalone host against the image, standalone VIBs are deleted from the hosts.
- After you set up an image for the cluster or host and remediate the hosts in the cluster or the standalone host against the image, non-integrated solution agents are deleted from the hosts.

- If you take a host out of a cluster that you manage with a vSphere Lifecycle Manager image, the host becomes a standalone host managed with an image.
- If you take a host out of a cluster that you manage with baselines, the hosts becomes a standalone host managed with an image. The image assigned to the standalone host is the image currently running on the host. If an image cannot be extracted and assigned to the host, the movement operation fails and the host stays in the cluster managed with baselines.
- The reverse action, moving a standalone host that you manage with an image to a cluster that you manage with an image is supported. After you move the standalone host managed with a vSphere Lifecycle Manager image, you must check the compliance of the hosts in the cluster against the specified cluster image.

Cluster or Standalone Host Eligibility to Use vSphere Lifecycle Manager Images

Switching from baselines to images requires that you set up or import a vSphere Lifecycle Manager image to manage the cluster or host with. As part of the transition, before you set up the image, vCenter triggers an automatic task that checks whether the cluster or host is eligible to use vSphere Lifecycle Manager images.

The **Check cluster's eligibility to be managed with an image** and **Check host's eligibility to be managed with an image** tasks ensure that the cluster or host is not undergoing remediation against a baseline and check whether all requirements for using vSphere Lifecycle Manager images are met.

Also, the tasks check for standalone VIBs and ensure that no unintegrated solutions are enabled for the cluster or host. You might not be able to switch to using vSphere Lifecycle Manager images if unintegrated solutions are enabled on the cluster or host.

> **Note:**
>
> If you use any third-party products or solutions, you must confirm with your third-party software vendor whether the respective solution works with vSphere Lifecycle Manager.

The tasks return three types of notifications: error, warning, and info.

Errors

The **Check cluster's eligibility to be managed with an image** and **Check host's eligibility to be managed with an image** tasks report an error in case one of the following conditions occurs:

- The cluster contains at least one host that is not stateful or the standalone host is not stateful.
- The cluster contains at least one host that is not of a compatible ESX version or the standalone host is of version different than 8.0 and later.
- The cluster or the standalone host contain VIBs of unintegrated solutions. In that case, you must deactivate the unintegrated solution and retry the transition.

Warnings

The **Check cluster's eligibility to be managed with an image** and **Check host's eligibility to be managed with an image** tasks issue a warning if the cluster contains at least one host with or a standalone host contains a standalone VIB or an unknown VIB. Warnings do not block the transition to using vSphere Lifecycle Manager images, but they require special attention or a user action. For example, a warning is returned in one of the following cases:

- If a host in the cluster contains a standalone VIB, for example a driver, for which a component is available in the vSphere Lifecycle Manager depot. If you want to keep the VIB, you must add the respective component to the vSphere Lifecycle Manager image. Otherwise, the standalone VIB is deleted upon remediation.
- If a host in the cluster or a standalone host contain an unknown VIB. Unknown VIBs are standalone VIBs for which no component is available in the vSphere Lifecycle Manager depot. If vSphere Lifecycle Manager detects an unknown VIB, you must import a component that contains the VIB into the vSphere Lifecycle Manager depot and restart the transition. Otherwise, the unknown VIB is deleted upon remediation.

Info

The **Check cluster's eligibility to be managed with an image** and **Check host's eligibility to be managed with an image** tasks give an info notification if the cluster contains at least one host with a standalone VIB or the standalone host

contains a standalone VIB, but you can still proceed to setting up a vSphere Lifecycle Manager image for the cluster or host without any additional actions. For example, an info notificaiton is returned in the following case:

- If the cluster is enabled for an integrated solution, for example vSphere HA or vSAN.

Option 1: Use an Image from a Host in the Cluster

To start managing an existing cluster or host that uses baselines with an vSphere Lifecycle Manager image, you can select and use one of the images installed on a host in the cluster or the current image on the standalone host.

- Verify that all ESX hosts in the cluster or the standalone host are of version 8.0 and later.
- Verify that your vCenter instance is of version 9.0 and later.
- Verify that all ESX hosts in the cluster or the standalone host are stateful. A stateful install is one in which the host boots from a disk.
- If you use a single image, verify that all ESX hosts in the cluster are from the same hardware vendor.

  Note:

  You can have hosts in the cluster from different vendors or from the same vendor but different generations, family, or model only if you use a composite image.
- Verify that no unintegrated solution is enabled for the cluster or the host.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

If you switch to using images, you cannot revert to using baselines for the cluster. You can move the hosts to another cluster, which uses baselines, but you cannot change the cluster that already uses an image.

If you switch to using images for a standalone host, you cannot revert to using baselines for the host.

For conceptual information about vSphere Lifecycle Manager images, see vSphere Lifecycle Manager Images.

For information about how to use vSphere Lifecycle Manager images to manage hosts and clusters, see How to Use vSphere Lifecycle Manager Images .

Instead of importing or manually setting up an image, you can use an image from a host that is already in the cluster or the image currently running on the standalone host.

When you initiate the transitioning workflow for a cluster that uses baselines, vSphere Lifecycle Manager triggers the **Get Installed Host Images** task after the eligibility checks finish successfully. During this task, vSphere Lifecycle Manager retrieves information about all images currently installed on the hosts in the cluster. Then vSphere Lifecycle Manager calculates which are the most complete, valid, and suitable images that you can use to remediate the cluster. The list of suitable images is generated based on two criteria: the ESX version and the software running on the hosts in the cluster. Based on these criteria, vSphere Lifecycle Manager provides the following image options to choose from:

- The image that contains the latest ESX version by build number.
- The image that has the most widely used software on most of the hosts in the cluster.
- The images that have a different software setup than the most widely used setup on the hosts in the cluster.

After you select an image from a host in the cluster, you can make changes to the image or save it without editing.

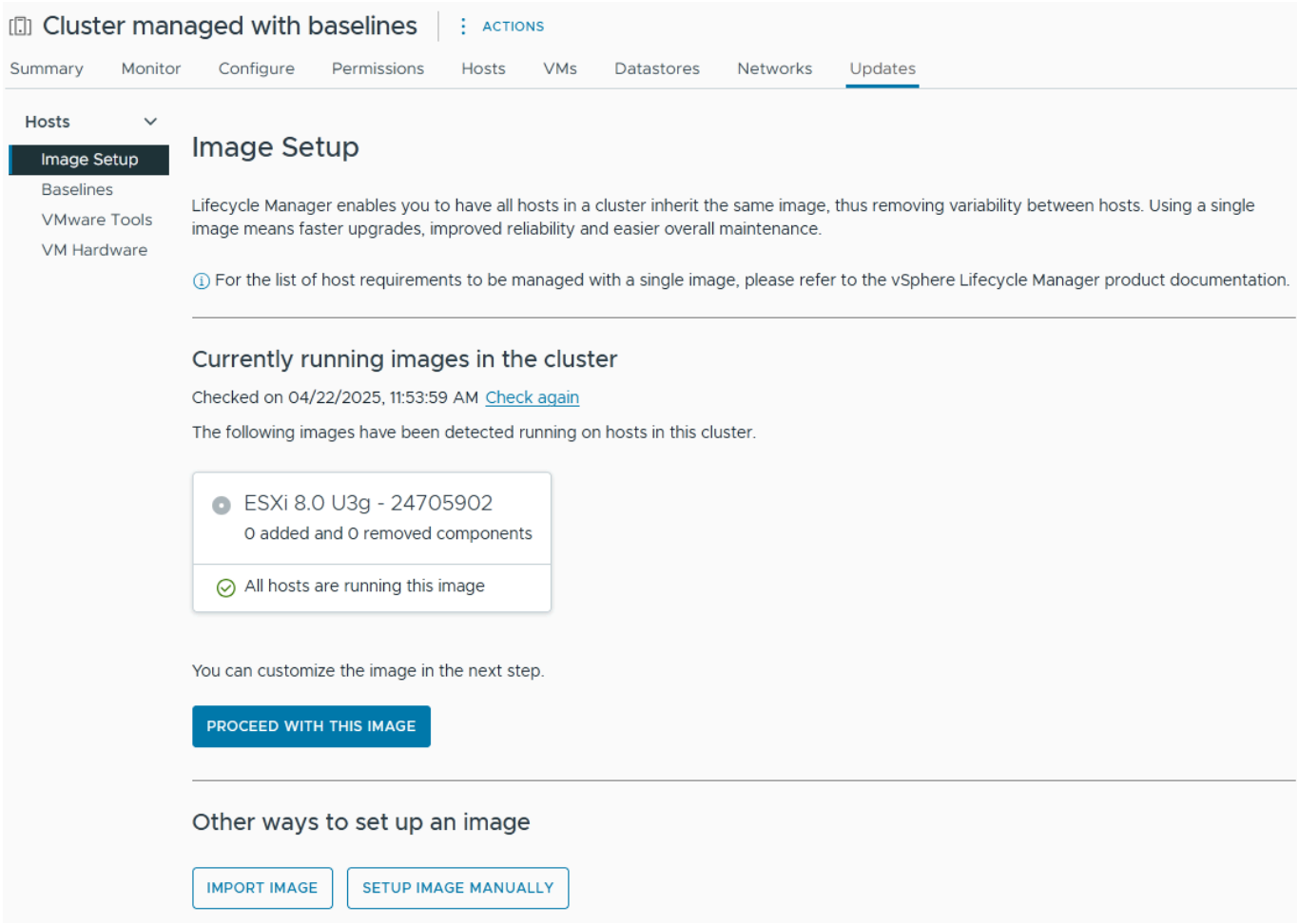
When you decide to add a host to or remove a host from a cluster that already has some generated list of suitable images, this list is invalidated. vSphere Lifecycle Manager automatically triggers a new **Get Installed Host Images** task. During this task, vSphere Lifecycle Manager retrieves information about the new set of images installed on the hosts in the cluster and computes a new list of suitable images based on the cluster membership changes. You can also regenerate the list of images at any time.

When you start transitioning a standalone host that uses baselines to a host that is managed with an image, you can view and use the image currently installed on the host. vSphere Lifecycle Manager automatically retrieves information about the current image on the standalone host. You can select and edit the current image on the standalone host or use it without changes.

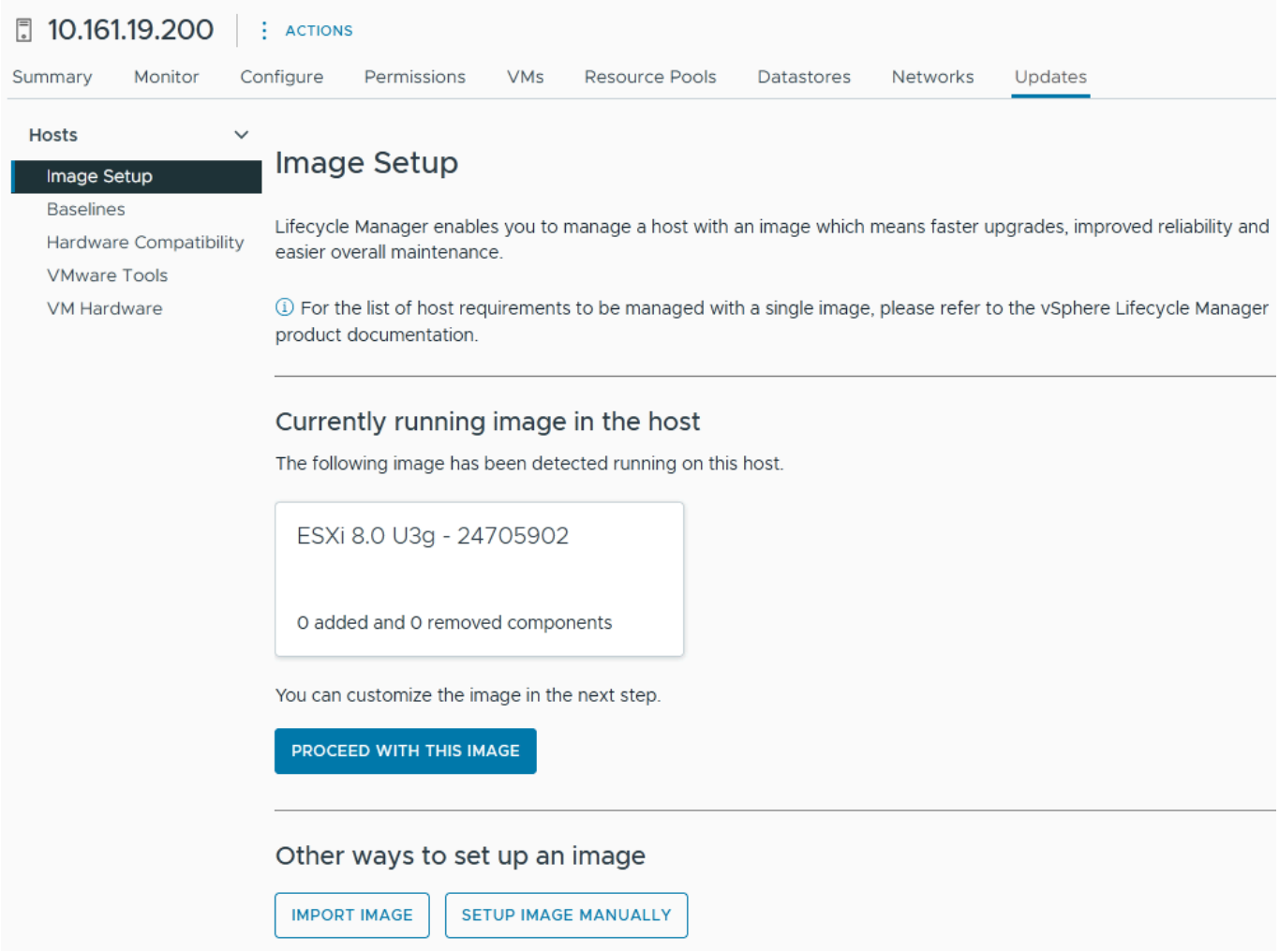
1. In the vSphere Client, navigate to a cluster or host that you manage with baselines and baseline groups.
2. On the **Updates** tab, click **Image Setup**.

   vSphere Lifecycle Manager starts checking if the cluster or host is eligible for using images. If no problems are reported, vSphere Lifecycle Manager retrieves information about the images currently installed on the hosts in the cluster or the image running on the standalone host.

   - If you selected a cluster, a list of the most valid and suitable images for the cluster is displayed in the **Currently running images in the cluster** section on the **Image Setup** card.



   - If you selected a standalone host, you can view the image card with details for the current host image in the **Currently running image in the host** section on the **Image Setup** card.



3. If you selected a cluster, select a host image from the provided list of images and click **Proceed with this image** to confirm your selection.

   The **Reading and extracting the selected image** progress bar is showing you the progress of the extract image operation.

   The **Step 1: Define Image** and **Step 2: Check Image Compliance** cards appear on the **Image Setup** card.

   The elements of the selected image appear in the **Step 1: Define Image** card.

4. If you selected a standalone host, click **Use image on host**.

   The **Reading and extracting the selected image** progress bar is showing you the progress of the extract image operation.

   The **Step 1: Define Image** and **Step 2: Check Image Compliance** cards appear on the **Image Setup** card.

   The elements of the selected image appear in the **Step 1: Define Image** card.

5. If you want to further customize the host image, edit any of the image elements.

6. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.

7. Click **Save**.

   Saving the image triggers an automatic compliance check. All hosts in the cluster or the standalone host are checked against the image.

8. In the **Step 2: Check Image Compliance** card, click **Check compliance** to run a compliance check of the cluster or standalone host against the image.

9. In the **Image Setup** card, finish the image setup.

   a) Click the **Finish image setup** button.

      The **Finish image setup** dialog box opens.

   b) In the **Finish image setup** dialog box, click **Yes, finish image setup**.

      CAUTION:

      Finalizing an image setup on a cluster where vSphere HA is enabled automatically initiates a vSphere HA reconfiguration task across all hosts in the cluster. When vSphere HA and VM Monitoring are both enabled, the temporary reset of HA agents during reconfiguration can lead to unexpected virtual machine restarts. Before you click **Yes, finish image setup**, temporarily set VM Monitoring to Disabled or perform the image conversion during a planned maintenance window. After the HA reconfiguration task completes, re-enable VM Monitoring. See Activate VM Monitoring in *vSphere Availability*.

You now manage all hosts in the cluster collectively or the standalone host with an image. Upon remediation, the image is installed on all hosts in the cluster or on the standalone host.

To apply the image to all hosts in the cluster or the standalone host, remediate the cluster or host against the image.

Option 2: Import an Image from a JSON File

To start managing an existing cluster or host that already uses baselines with a vSphere Lifecycle Manager image, you can import an image from another cluster or host instead of setting up a new image manually.

- Verify that all ESX hosts in the cluster or the standalone host are of version 8.0 and later.
- Verify that all ESX hosts in the cluster or the standalone host are stateful.
  A stateful installation is one in which the host boots from a disk.

  Note:

  In a vCenter instance of version 9.0 and later, you can assign vSphere Lifecycle Manager images to clusters with stateless hosts and standalone stateless hosts but only in very specific cases. See Managing Stateless Hosts with an Image.
- If you use a single image, verify that all ESX hosts in the cluster are from the same hardware vendor.

  Note:

  You can have hosts in the cluster from different vendors or from the same vendor but from different generations, family, or model only if you use a composite image.
- Verify that no unintegrated solution is enabled for the cluster or the host.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

If you switch to using images, you cannot revert to using baselines for the cluster. You can move the hosts to another cluster, which uses baselines, but you cannot change the cluster that already uses an image.

If you switch to using images for a standalone host, you cannot revert to using baselines for the host.

For conceptual information about vSphere Lifecycle Manager images, see vSphere Lifecycle Manager Images.

For information about how to use vSphere Lifecycle Manager images to manage hosts and clusters, see How to Use vSphere Lifecycle Manager Images .

1. In the vSphere Client, navigate to a cluster or host that you manage with baselines and baseline groups.
2. On the **Updates** tab, click **Image Setup**.
3. On the **Image Setup** card, click the **Import Image** button.
   The **Import Image** dialog box opens.
4. Select a JSON file to import and click **Next**.
   - Click the **Browse** button and select a JSON file on your local machine.
   - Enter a valid URL to a JSON file on a remote web server.

   vSphere Lifecycle Manager starts checking if the cluster or host is eligible for using images. If no problems are reported, the **Step 1: Define Image** and **Step 2: Check Image Compliance** cards appear on the **Image Setup** card.

   The elements of the selected image appear in the **Step 1: Define Image** card.
5. Optional: In the **Step 1: Define Image** card, customize the imported image by changing any of its elements.
6. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.
7. Click **Save**.
   Saving the image triggers an automatic compliance check. All hosts in the cluster or the standalone host are checked against the image.
8. In the **Step 2: Check Image Compliance** card, click **Check compliance** to run a compliance check of the cluster or standalone host against the image.
9. In the **Image Setup** card, finish the image setup.
   a) Click the **Finish image setup** button.

      The **Finish image setup** dialog box opens.
   b) In the **Finish image setup** dialog box, click **Yes, finish image setup**.

      CAUTION:

      Finalizing an image setup on a cluster where vSphere HA is enabled automatically initiates a vSphere HA reconfiguration task across all hosts in the cluster. When vSphere HA and VM Monitoring are both enabled, the temporary reset of HA agents during reconfiguration can lead to unexpected virtual machine restarts. Before you click **Yes, finish image setup**, temporarily set VM Monitoring to Disabled or perform the image conversion during a planned maintenance window. After the HA reconfiguration task completes, re-enable VM Monitoring. See Activate VM Monitoring in *vSphere Availability*.

You now manage all hosts in the cluster or the standalone host with an image. Upon remediation, the image is installed on all hosts in the cluster or on the standalone host.
To apply the image to all hosts in the cluster or the standalone host, remediate the cluster or host against the image.

Option 3: Set Up a New Image Manually

You can start using vSphere Lifecycle Manager images for an existing cluster or host that uses baselines by setting up an image manually.

- Verify that all ESX hosts in the cluster or the standalone host are of version 8.0 and later.
- Verify that all ESX hosts in the cluster or the standalone host are stateful.
  A stateful installation is one in which the host boots from a disk.

Note:

In a vCenter instance of version 9.0 and later, you can assign vSphere Lifecycle Manager images to clusters with stateless hosts and standalone stateless hosts but only in very specific cases. See Managing Stateless Hosts with an Image.

- If you use a single image, verify that all ESX hosts in the cluster are from the same hardware vendor.

  Note:

  You can have hosts in the cluster from different vendors or from the same vendor but from different generations, family, or model only if you use a composite image.
- Verify that no unintegrated solution is enabled for the cluster or the host.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

If you switch to using images, you cannot revert to using baselines for the cluster. You can move the hosts to another cluster, which uses baselines, but you cannot change the cluster that already uses an image.

If you switch to using images for a standalone host, you cannot revert to using baselines for the host.

For conceptual information about vSphere Lifecycle Manager images, see vSphere Lifecycle Manager Images.

For information about how to use vSphere Lifecycle Manager images to manage hosts and clusters, see How to Use vSphere Lifecycle Manager Images .

1. In the vSphere Client, navigate to a cluster or host that you manage with baselines and baseline groups.
2. On the **Updates** tab, click **Image Setup**.
3. Click the **Set up image manually** button.

   vSphere Lifecycle Manager starts checking if the cluster or host is eligible for using images. If no problems are reported,

   the **Step 1: Define Image** and **Step 2: Check Image Compliance** cards appear on the **Image Setup** card.

   The elements of the selected image appear in the **Step 1: Define Image** card.
4. If the **Check cluster's eligibility to be managed with a single image** or **Check host's eligibility to be managed with a single image** tasks report an error or a warning that requires an action, resolve the issue and restart the procedure.
5. In the **Step 1: Define Image**, select an ESX image from the **ESXi Version** drop-down menu.
6. Optional: Add a vendor add-on to the image.
   a) Click **Select**.

      The **Select Vendor Addon** appears.
   b) Select an add-on from the list.

      An information panel appears on the right. The information panel shows information about the components that the add-on adds to the ESX image and the components that it removes from the image.
   c) From the **Version** column use the drop-down menu for the selected add-on and select the add-on version.
   d) Click **Select**.
7. Optional: Add a firmware and drivers add-on to the image.

   Note:

   To manage the firmware of the ESX hosts in the cluster or the standalone host, you must first install the hardware support manager plug-in provided by the hardware vendor of the hosts.

   a) Click **Select**.

      The **Select Firmware and Drivers Addon** appears.
   b) Select a hardware support manager from the respective drop-down menu.

      A list of firmware and drivers add-ons appears.

c) Select an add-on from the list.

An information panel appears on the right. The information panel shows information about the supported ESX versions and whether the add-on contains a driver or not.

d) From the **Version** column use the drop-down menu for the selected add-on and select the add-on version.

e) Click **Select**.

8. Optional: Manage the additional components of the image.

a) Click **Show details**.

b) Click **Add components**.

The **Add Components** dialog box opens.

c) Optional: Use the **Show** drop-down menu to sift out the components that are not part of the selected vendor add-on.

d) Select one or multiple components from the list.

An information panel appears on the right. The information panel shows information about the component that you selected first.

e) From the **Version** column use the drop-down menu for the selected component to select the component version.

CAUTION:
If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can select and retain a lower version for an additional component that is part of the ESX base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster.

f) Click **Select**.

The selected components appear in the list of components that the image contains. You can use the **Show** drop-down menu to sift out the additional components. For example, you can list only the customized components.

g) Optional: From the **Notes** column, click the trash icon to remove a component.

CAUTION:
If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can remove the VMware Host Client and ESX VM Tools components from the base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster.

h) Optional: Click **Hide details** to hide the list of components.

9. Optional: In vSphere Foundation 9.x, if you have hosts in the cluster from different vendors or from the same vendor but different generations, family, or model, add one or more alternative images to manage their lifecycle.

a) Click the **Add image** button at the top right corner of the **Step 1: Define Image** card.

The **Composite Image Name** text field appears above the **Default Image** tab.

The **New Image** tab appears next to the **Default Image** tab.

b) In the **Composite Image Name** text field, enter the unique name for the composite image.

c) In the **New Image** tab, enter a unique name for the alternative image.

d) Modify the image setup to select a vendor add-on and the version of the add-on, add firmware and drivers add-on to the image, or manage the additional components of the image.

Note that you cannot change the base image of an alternative image directly. To modify the base image, you must do so from the **Default Image** tab of the composite image, and this change is reflected in all alternative images.

e) In the **Selection Criteria** section, specify the rules that are used to filter the hosts to which the alternative image is assigned.

- You can select hosts manually.
  a. Click the **Edit hosts** button above the table.
     The **Select Hosts** dialog box opens.
  b. On the **All Hosts** tab, select the hosts to which you want to assign the alternative image.
  c. Optionally, you can view all selected hosts on the **Selected Hosts** tab.
  d. Click **Save**.

Selected hosts are added to the table in the **Selection Criteria** section.

- Filter hosts by using SMBIOS fields like `Manufacturer` (Type 1 structure), `Product Name` (Type 1 structure), `Family` (Type 1 structure), or `OEM Strings` (Type 11 structure). Click the **Show details** drop-down menu to add values to any of these filtering criteria and click **Add**.

10. Optional: To validate the image, click the **Validate** button.

    You validate an image to check for missing dependencies and component conflicts.

    The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.

11. Click **Save**.

    Saving the image triggers an automatic compliance check. All hosts in the cluster or the standalone host are checked against the image.

12. In the **Step 2: Check Image Compliance** card, click **Check compliance** to run a compliance check of the cluster or standalone host against the image.

13. In the **Image Setup** card, finish the image setup.
    a) Click the **Finish image setup** button.

       The **Finish image setup** dialog box opens.
    b) In the **Finish image setup** dialog box, click **Yes, finish image setup**.

       CAUTION:

       Finalizing an image setup on a cluster where vSphere HA is enabled automatically initiates a vSphere HA reconfiguration task across all hosts in the cluster. When vSphere HA and VM Monitoring are both enabled, the temporary reset of HA agents during reconfiguration can lead to unexpected virtual machine restarts. Before you click **Yes, finish image setup**, temporarily set VM Monitoring to Disabled or perform the image conversion during a planned maintenance window. After the HA reconfiguration task completes, re-enable VM Monitoring. See Activate VM Monitoring in *vSphere Availability*.

You set up an image for the cluster or standalone host. You now manage all hosts in the cluster collectively or the standalone host with a vSphere Lifecycle Manager image for the cluster or host. Upon remediation, the image is installed on all hosts in the cluster or the standalone host.

To apply the image to all hosts in the cluster or the standalone host, remediate the cluster or host against the image.

Working with Images at a Cluster or Standalone Host Level

When you manage a cluster or host with an image, you can change the image at any time. You can edit the image by changing the software that it includes, for example, you can add or remove components, and you can also change the version of the included components. You can assign a new image from Image Library. You can also reuse an image by exporting it and importing it into a different cluster or a standalone host.

Using vSphere Lifecycle Manager images starts with setting up an image for a cluster or a host. After you start managing a cluster or a host with an image, you can edit the image at any time. You can validate the image before saving it to verify that it includes no conflicting components or missing dependencies.

Setting Up an Image for a Cluster or a Host

You can set up an image during the creation of the cluster or when you add a standalone host to a data center or a folder or later. During the creation of a cluster or the addition of a standalone host to a data center or a folder, you can only define the ESX version and, optionally, a vendor add-on to be included in the image for the cluster or host. You can later edit the image to include additional components or a firmware add-on. For detailed information about creating a cluster and adding hosts to it, and about adding a standalone host to a data center or folder, see the *vCenter and Host Management* documentation.

Note:

When you set up an image, you select an ESX version and a vendor add-on from the vSphere Lifecycle Manager depot. If no ESX base images and vendor add-ons are available in the vSphere Lifecycle Manager depot, you must populate the depot with software updates by synchronizing the depot or uploading updates to the depot manually. For detail information about the corresponding procedures, see Synchronize the vSphere Lifecycle Manager Depot and Import Updates to the vSphere Lifecycle Manager Depot.

If you do not set up an image during the creation of the cluster or the addition of the standalone host to a data center or folder, the cluster or the host use baselines, but you can switch to images at any time. When you set up an image during the transition workflow, you can define the full stack of software to run on the hosts in the cluster or on the standalone host. For more information about switching from baselines to images, see Transition a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images.

After the creation of the cluster or the addition of a standalone host to a data center or folder, you can choose to assign a different image from the image specifications that exist in Image Library. See Assign an Image from Image Library to a Cluster or a Standalone Host.

Viewing Details About the Image of a Cluster or Host

You can view the software elements that the image for a cluster or host contains on the **Updates** tab for that cluster or host.

The **Image** pane consists of two cards.

The **Image** card contains information about the image that the cluster or host uses. In that card, you perform all image-related operations. You edit the image, you export the image, you validate your selections, and so on. You can also check and view the recommendations that VMware provides.

The **Image Compliance** card contains compliance information about the hosts in the cluster or the standalone host. In that card, you perform host-related operations. You check the compliance of the hosts in the cluster or the standalone host, you run remediation pre-checks, you remediate the hosts in the cluster or the standalone host, and so on.

In the **Image Compliance** card, you edit remediation settings for that cluster or host and manage depot overrides.

Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level

For a cluster or host that you manage with an image, you can edit the image at any time to add, remove, or update an image element. For example, you can edit the image to update the vendor add-on version that it includes, to add or remove a driver, to upgrade the ESX version in the image, to add one or many alternative images, and so on.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

Working with Image Drafts	When you edit an image, vSphere Lifecycle Manager saves the working copy of the image as a draft. The draft is an edited but unsaved version of an image. If you edit an image, but neglect to save it, the edited image is automatically saved as a draft. When you return to the workflow later, you may choose to utilize the saved draft version as a starting point or discard the draft.
Validation	You can validate an image draft before you save it. Validation checks whether the image is correct and complete. During validation, vSphere Lifecycle Manager checks for missing dependencies and conflicting components. In case of issues, vSphere Lifecycle Manager returns messages with information about the existing issues. You must resolve all issues before you can save the image.

Editing an image and validating a draft before saving it are supported operations for clusters with hosts or standalone hosts that have a DPU device. During validation, vSphere Lifecycle Manager validates both the VIBs that are applicable to the ESX version on the host and to the ESX version on a DPU device.

1. In the vSphere Client, navigate to a cluster or host that you manage with a vSphere Lifecycle Manager image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** pane, click the **Edit** button.
4. In the **Edit Image** pane, modify the image setup.

Image Element	Possible Modifications
ESX Version	From the **ESXi Version** drop-down menu, select a new ESX base image.
Vendor Addon	• To add a vendor add-on to the image, click **Select** and select a vendor add-on. • To change the version of the vendor add-on in the image or to select a new vendor add-on, click the pencil icon and make your changes. • To remove the vendor add-on from the image altogether, click the trash icon.
Firmware and Drivers Addon	• To add a firmware add-on to the image, click **Select**. In the **Select Firmware and Drivers Addon** dialog box, specify a hardware support manager and select a firmware add-on to add to the image. • To select a new firmware add-on, click the pencil icon and make your changes. • To remove the firmware add-on element from the image altogether, click the trash icon. Selecting a firmware add-on for a family of vendor servers is possible only if the respective vendor-provided hardware support manager is registered as an extension to the vCenter where vSphere Lifecycle Manager runs.
Components	Click **Show details** and view the list of additional components in the image. • To add components to the image, click **Add Components** and select the components and their corresponding versions to add to the image. **CAUTION:** If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can select and retain a lower version for an additional component that is part of the ESX base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster. • To delete a component from the image, click the trash icon in the table with components. **CAUTION:** If your vCenter instance and all ESX hosts in the cluster are of version 8.0 Update 3 or later, you can remove the VMware Host Client and ESX VM Tools components from the base image. To avoid critical errors on any of the hosts in the cluster, make sure you perform thorough testing of the customized image before remediating the cluster. • To delete a manually added component that overrides a component in the selected vendor add-on or base image, click the undo icon in the table with components. This action reverts the override.

5. Optional: If you edit a composite image for a cluster, add one or more alternative images to manage their lifecycle.

   You create a composite image for clusters that have hosts from different vendors or from the same vendor but different generations, family, or model.

   a) Click the **Add image** button at the top right corner of the **Step 1: Define Image** card.

      The **Composite Image Name** text field appears above the **Default Image** tab.

      The **New Image** tab appears next to the **Default Image** tab.
   b) In the **Composite Image Name** text field, enter the unique name for the composite image.
   c) In the **New Image** tab, enter a unique name for the alternative image.
   d) Modify the image setup to select a vendor add-on and the version of the add-on, add firmware and drivers add-on to the image, or manage the additional components of the image.

      Note that you cannot change the base image of an alternative image directly. To modify the base image, you must do so from the **Default Image** tab of the composite image, and this change is reflected in all alternative images.
   e) In the **Selection Criteria** section, specify the rules that are used to filter the hosts to which the alternative image is assigned.

      - You can select hosts manually.
        a. Click the **Edit hosts** button above the table.
           The **Select Hosts** dialog box opens.
        b. On the **All Hosts** tab, select the hosts to which you want to assign the alternative image.
        c. Optionally, you can view all selected hosts on the **Selected Hosts** tab.
        d. Click **Save**.
           Selected hosts are added to the table in the **Selection Criteria** section.
      - Filter hosts by using SMBIOS fields like `Manufacturer` (Type 1 structure), `Product Name` (Type 1 structure), `Family` (Type 1 structure), or `OEM Strings` (Type 11 structure). Click the **Show details** drop-down menu to add values to any of these filtering criteria and click **Add**.

6. Optional: If you edit a composite image, remove one or more alternative images.
7. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.
8. Click **Save**.

   The save operation triggers validation. If the image is valid, vSphere Lifecycle Manager saves it and runs a compliance check against the new image. You can view compliance information in the **Image Compliance** pane.

   If the image is invalid, saving the image fails and vSphere Lifecycle Manager returns an error.

The new image is validated and displayed in the **Image** card. vSphere Lifecycle Manager uses the compatibility data bundle to perform an automatic hardware compatibility check against the new image.

If there are recommended images generated for the cluster or host, those recommendations become invalidated and updated. vSphere Lifecycle Manager automatically generates a new recommendation based on the new image.

Assign an Image from Image Library to a Cluster or a Standalone Host

You can change the image associated with a cluster or a standalone host, by assigning a new image from Image Library.

After you create a cluster or add a standalone host to a data center or a folder, you can change the image associated with these inventory objects.

1. In the vSphere Client, navigate to a cluster or host that you manage with a vSphere Lifecycle Manager image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** card, click the **Assign image** button.

   The **Assign image** dialog box opens.
4. Select an image from the images currently available in Image Library and click **Next**.

   You assign composite images to clusters that have hosts from different vendors or from the same vendor but different generations, family, or model. Standalone hosts are associated only with single images.
5. On the **Review** page, review the details of the selected image and click **Assign**.

The image from Image Library is assigned to the selected cluster or standalone host.

Check the compliance of the cluster or standalone host against the newly assigned image, run remediation precheck and remediate the cluster or standalone host.

Export or Import a vSphere Lifecycle Manager Image for a Cluster or Standalone Host

You can reuse an image that you already set up for a cluster or a standalone host by exporting it and importing it to another cluster or standlone host in the same or different vCenter instance. Depending on your goal, the image can be exported, distributed, and consumed in three different formats.

You can export a single image of a cluster or a standalone host as an ISO image, ZIP file, or a JSON file. You can export a composite image only as a JSON file.

You can import an image to a cluster or a standalone host that use vSphere Lifecycle Manager images only in JSON format.

The export and import operations work for clusters with hosts that have DPU devices.

ISO Image

Distributing an image created with vSphere Lifecycle Manager in an ISO format is useful when you need the image to perform clean installations of ESX and for bootstrapping purposes, for example, the kickstart workflow.

You cannot use an image exported as an ISO file to update another cluster or standalone host managed with a vSphere Lifecycle Manager image.

ZIP File

Distributing an image created with vSphere Lifecycle Manager as an offline bundle is useful when you want to import the components that the image contains into the depot of the target vSphere Lifecycle Manager instance.

You cannot use the ZIP file to create or update a vSphere Lifecycle Manager image for a cluster or a standalone host.

JSON File

Distributing an image created with vSphere Lifecycle Manager as a JSON file is useful when you want to reuse the same image for other clusters or standalone hosts that use images for lifecycle management.

When you distribute the JSON file to clusters or standalone hosts in a different vCenter instance, you must make sure that the depot of the target vSphere Lifecycle Manager instance contains all components that the JSON file contains.

The JSON file contains only metadata and not the actual software payloads.

Which Distribution Format to Choose?

To reuse an existing image for a cluster or standalone host in the same vCenter system, you must export the image as a JSON file and then import the JSON file to the target cluster or standalone host.

However, when you want to use an existing image for a cluster or a standalone host in another vCenter instance, exporting the image as a JSON file might not be enough. You might also need to export the image as a ZIP file. At the

target location, you must import the JSON file as an image to the target cluster or standalone host. But you might also need to import the ZIP file to the target vSphere Lifecycle Manager depot to make sure that all components included in the image are available to the target vSphere Lifecycle Manager instance.

Export an Image from a Cluster or Standalone Host

You export an image when you want to use the same image for another cluster or standalone host in the same or in a different vCenter instance.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

Depending on your goals and the type of image, you can export an image as a JSON file, as an installable ISO image, or as an offline bundle that contains all software packages included in the image. The export format depends on your needs and goals. Composite images can only be exported in a JSON file.

For example, if you intend to use the image for a cluster or host in another vCenter, you must export it as a JSON file and as a ZIP file. Afterwards, you must import both the JSON file and the ZIP file to the target vCenter system. For information about importing updates to the vSphere Lifecycle Manager depot, see Import Updates to the vSphere Lifecycle Manager Depot.

1. In the vSphere Client, navigate to a cluster or host that you manage with a vSphere Lifecycle Manager image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** card, click the horizontal ellipsis icon and select **Export**.
4. In the **Export Image** dialog box, select a file format, and click **Export**.

   If the image is composite, only a JSON file is exported.

   You can export the image in one file format at a time.

The exported file is saved to the directory you specify.

Import the image to a target cluster or standalone host in the same or in a different vCenter instance. You can also import the image to Image Library in the same or in a different vCenter instance.

Import an Image to a Cluster or Standalone Host

Instead of setting up a new image manually, you can reuse an existing image by importing it to a cluster or a standalone host. Upon remediation, the imported image is applied to all hosts in the cluster or to the standalone host.

- Verify that the vSphere Lifecycle Manager depot contains all components included in the image that you import.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

You can import an image only if it is in a JSON format. The JSON file contains only the image metadata, but not the actual software payload. To successfully import an image to a cluster or host and apply the software specification to the hosts in the cluster or to the standalone host, all the components specified in the image must be available in the vSphere Lifecycle Manager depot.

You can import images into Image Library that are exported from a single cluster or standalone host. See Import an Image in Image Library as a JSON File.

If you want to distribute and reuse an image across vCenter instances, importing the JSON file might not be enough if the components from the image are not available in the target vSphere Lifecycle Manager depot. In such cases, before you import the JSON file to the target cluster or host, you must first import an offline bundle that contains all components included in the image to the target vSphere Lifecycle Manager depot. If you try to import a JSON file to a cluster or host but the target vSphere Lifecycle Manager depot does not contain the corresponding components, the import operation fails due to validation errors.

For information about importing updates to the vSphere Lifecycle Manager depot, see Import Updates to the vSphere Lifecycle Manager Depot.

1. In the vSphere Client, navigate to a cluster or host that you manage with a vSphere Lifecycle Manager image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. Click the horizontal ellipsis icon and select **Import**.
4. In the **Import Image** dialog box, select a JSON file and click **Next**.
   - Enter a URL address to the JSON file that you want to import.
   - Browse to a JSON on your local machine.
5. Optional: In the **Edit Image** card, modify the image set-up.

Image Element	Possible Modifications
ESX Version	From the **ESXi Version** drop-down menu, select a new ESX base image.
Vendor Addon	• To add a vendor add-on to the image, click **Select**. • To change the version of the vendor add-on in the image or to select a new vendor add-on, click the pencil icon. • To remove the vendor add-on element from the image altogether, click the trash icon .
Firmware and Drivers Addon	• To add a firmware add-on to the image, click **Select**. • To select a new firmware add-on, click the pencil icon. • To remove the firmware add-on element from the image altogether, click the trash icon. Selecting a firmware add-on for a family of vendor servers is possible only if the respective vendor-provided hardware support manager is registered as an extension to the vCenter where vSphere Lifecycle Manager runs.
Components	Click **Show details** and view the list of additional components in the image. • To add components to the image, click **Add Components** and select components to add to the image. • To delete a component from the image, click the trash icon in the table with components. • To delete a manually added component that overrides a component in the selected vendor add-on, click the undo icon in the table with components. This action reverts the override.
Alternative Image	If you edit the image on a heterogeneous cluster, add one or more alternative images to the default image of the cluster. **Note:** You create a composite image only for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. 1. Click the **Add image** button at the top right corner of the pane. The **New Image** tab appears next to the **Default Image** tab. 2. In the **New Image** tab, enter a unique name for the composite image and for the alternative image. 3. Modify the image setup to select a vendor add-on and the version of the add-on, add firmware and drivers add-on to the image, or manage the additional components of the image. Note that you cannot change the base image of an alternative image directly. To modify the base image, you must do so from the **Default Image** tab of the composite image, and this change is reflected in all alternative images.

Image Element	Possible Modifications
	In the **Selection Criteria** section, specify the rules that are used to filter the hosts to which the alternative image is assigned. • You can select hosts manually. a. Click the **Edit hosts** button above the table. The **Select Hosts** dialog box opens. b. On the **All Hosts** tab, select the hosts to which you want to assign the alternative image. c. Optionally, you can view all selected hosts on the **Selected Hosts** tab. d. Click **Save**. Selected hosts are added to the table in the **Selection Criteria** section. • Filter hosts by using SMBIOS fields like `Manufacturer` (Type 1 structure), `Product Name` (Type 1 structure), `Family` (Type 1 structure), or `OEM Strings` (Type 11 structure). Click the **Show details** drop-down menu to add values to any of these filtering criteria and click **Add**.

6. If the image contains conflicting components or unresolved dependencies, resolve the issues and retry the procedure.
7. Optional: To validate the image, click the **Validate** button.

   You validate an image to check for missing dependencies and component conflicts.

   The compatibility data bundle is used to ensure that the target ESX version is compatible with the current version of the vCenter instance.
8. Click **Save**.

   A compliance check task is automatically triggered. You can view compliance information in the **Image Compliance** card.

The imported JSON file is imported and set as your new image for the target cluster or host. At that stage, nothing is installed on the hosts in the cluster or on the standalone host. The installation of software on the hosts happens during remediation.

Remediate the hosts in the cluster or the standalone host against the new image. See Run a Remediation Pre-Check for a Cluster and Remediate a Cluster or Selected Hosts in a Cluster Against an Image.

Check the Compliance Against an Image

When you check the compliance of a cluster or host against an image, vSphere Lifecycle Manager compares the software on each host in the cluster or on the standalone host with the software specified in the image. If the image contains a firmware and drivers add-on, the compliance check also calculates the firmware compliance of the hosts with the image. If some of the hosts in the cluster or the standalone host contain DPU devices, vSphere Lifecycle Manager compares the software and firmware on the DPU devices on the hosts with the software and firmware specified in the cluster or host image.

In addition to calculating the compliance state for each host in the cluster or the standalone host, the compliance check gives you information about the impact that the remediate operation will have on a single host, for example if remediation will cause host reboot or if maintenance mode is needed for the host.

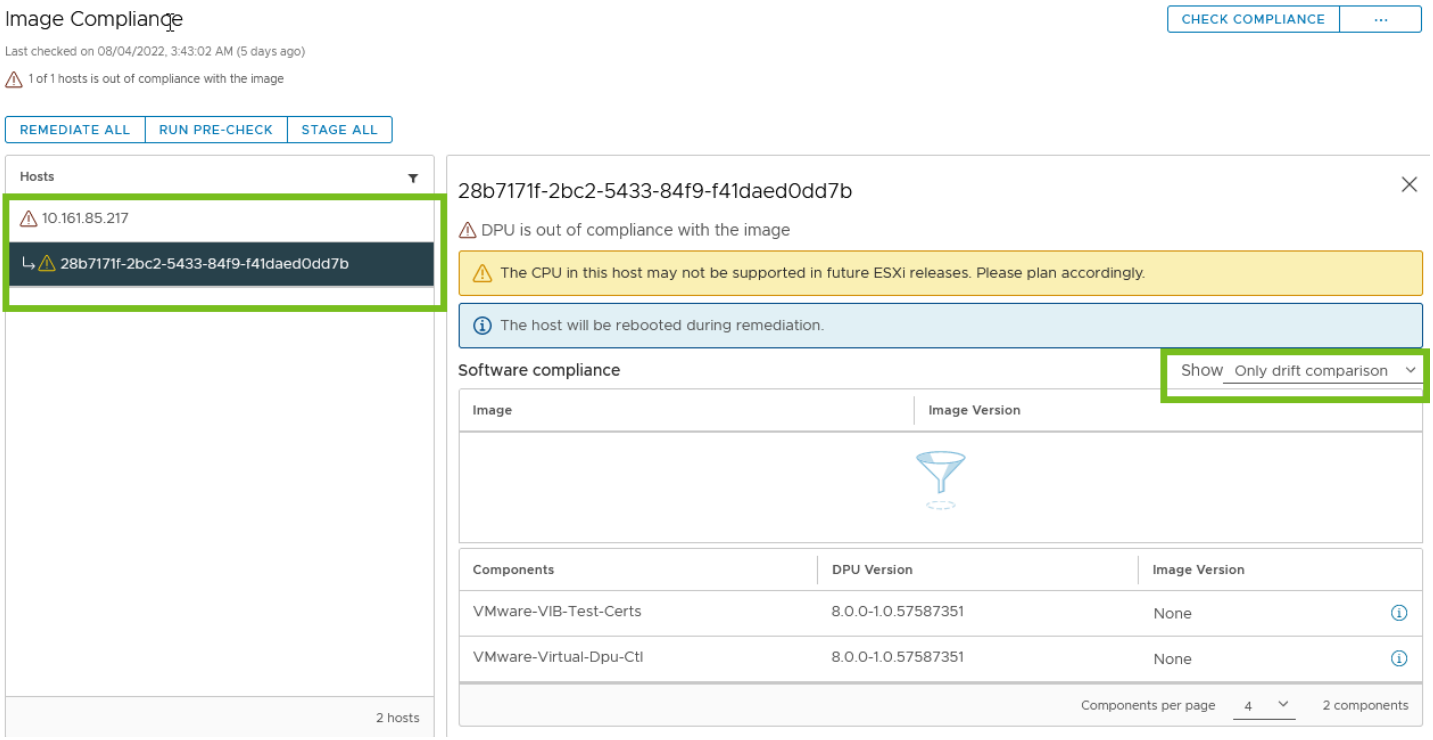
A host can have any of the four compliance states: compliant, non-compliant, incompatible, and unknown. If a host has a DPU device, the compliance state of the DPU device is merged into the overall host compliance result.

Compliant	A host is compliant if the image on the host matches the image that you set for the cluster or host. A DPU-backed host is compliant when the software and firmware on the server and on the DPU device are the same as the software and firmware specified in the image for the cluster or host.
Non-Compliant	A host is non-compliant if the image on the host does not match the image that you set for the cluster or host. A compliant host becomes non-compliant when you set a new image for the cluster or host or

	manually add or remove components on the host. Once you successfully remediate a non-compliant host, the host becomes compliant. For example, a host is non-compliant in the following cases. • The ESX version on the host is earlier than the ESX version included in the image for the cluster or host. • The firmware on the host is different from the firmware add-on in the image for the cluster or host. • The host has a component that is not included in the image for the cluster or host. • The host contains a standalone VIB. A DPU-backed host is non-compliant in the following cases: • ESX on the server, ESX on the DPU device, or both have software components of earlier versions than the software and firmware specified in the image for the cluster or host. • You edit the image for the cluster or host and add a component that contains one or multiple VIBs that are applicable to ESX on the server, ESX on the DPU device, or both platforms. • The remediation of a cluster or a host fails and as a result, the software and firmware components on the host are updated, but the DPU is not updated or the reverse.
Incompatible	A host is incompatible when the image for the cluster or host cannot be applied to the host. For example, a host is incompatible in the following cases. • The ESX version on the host is later than the ESX version included in the image for the cluster or host. • The host does not have sufficient resources, for example RAM. • The hardware of the host is incompatible with the vSphere Lifecycle Manager image for the cluster or host. A DPU-backed host has the incompatible compliance state when the ESX version on the DPU is later than the ESX version included in the image for the cluster or host.
Unknown	The unknown compliance state indicates that there is no compliance information about the host. For example, the compliance state of a host is unknown in the following cases. • You add a new host to the cluster. The compliance state of the newly added hosts is unknown until you perform a compliance check operation on the cluster. • You add a new standalone host to the data center or folder. The compliance state of the standalone host is unknown until you perform a compliance check operation on the host. • You edit the image for the cluster or host and save the modifications. The compliance state of all hosts in the cluster or the standalone host is unknown until you check the compliance of the cluster or host against the new image.

Compliance information about the hosts in a cluster or the standalone hosts is displayed on the **Updates** tab for that cluster or host, in the **Image Compliance** card. If you select a cluster, the **Image Compliance** card displays a list of all hosts in the cluster that are out of compliance with the image for the cluster. When you select a host from the cluster, the compliance information about the host appears on the right. If you select a standalone host, the **Image Compliance** card displays the compliance information about the host.

If a host has a DPU device, it is listed directly below the host. The compliance information about the DPU device appears on the right and is presented in an identical manner as the overall compliance information about the host. You can see full image comparison or drift comparison.



Check the Compliance of a Cluster or Standalone Host Against a vSphere Lifecycle Manager Image

You check the cluster compliance against an image to understand how each of the hosts in the cluster compares to the specified image.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

When you initiate the compliance check operation on a data center, folder, or vCenter instance, vSphere Lifecycle Manager performs compliance checks on all clusters or standalone hosts that are part of the selected inventory object.

1. In vSphere Client, navigate to a cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Check Compliance** button.

The **Image Compliance** card displays information about the overall number of non-compliant and incompatible hosts in the cluster or the standalone host. You can view detailed compliance information for every host. If the host has a DPU device, you can also view compliance information about the DPU device only.

Remediate the cluster or the standalone host to make the non-compliant hosts compliant. See Run a Remediation Pre-Check for a Cluster and Remediate a Cluster or Selected Hosts in a Cluster Against an Image.

View Host Compliance Information

You can view detailed compliance information for every non-compliant host in a cluster or for every standalone host that you manage with an image. You can also view compliance information about the DPU device on the host, if the host has a DPU device. As a result, you can easily find what causes the host to become out of compliance with the cluster or host image.

- Run a compliance check.
- Verify that no host is added to the cluster after your last compliance check.

Detailed compliance information is displayed only for hosts that are out of compliance with the image in the cluster or the standalone host. vSphere Lifecycle Manager displays no compliance details for compliant hosts.

For hosts that have the incompatible compliance state, vSphere Lifecycle Manager shows in a signpost information about what causes the compatibility issues.

1. In vSphere Client, navigate to a cluster or host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. View compliance information for the hosts in a cluster or for a standalone host.

Option	Action
View the compliance information for a host in the cluster.	1. In the **Image Compliance** card, select a host from the **Hosts** list. **Note:** Only non-compliant hosts are listed. 2. To view the full comparison between the image on the host and the image for the cluster, select **Full image comparison** from the drop-down menu for the **Software compliance** table. 3. To view only the image elements that make the host non-compliant with the image for the cluster, select **Only drift comparison** from the drop-down menu for the **Software compliance** table.
View the compliance information for a standalone host.	You can view the compliance information for a standalone host in the **Image Compliance** card. • To view the full comparison between the software that runs on the host and the image for the host, select **Full image comparison** from the drop-down menu for the **Software compliance** table. • To view only the image elements that make the host non-compliant with the image, select **Only drift comparison** from the drop-down menu for the **Software compliance** table.

4. If you want to view compliance details about a DPU device on a host, select the DPU device listed for the host.

   DPU devices are listed directly under the host they belong to.

   - To view the full comparison between the image on the DPU device and the image for the cluster or the standalone host, select **Full image comparison** from the drop-down menu for the **Software compliance** table.
   - To view only the image elements that make the DPU device non-compliant with the image for the cluster or the standalone host, select **Only drift comparison** from the drop-down menu for the **Software compliance** table.

   An information panel appears on the right. In the **Software compliance** table, you can see what software runs on the selected host and what is the software specification in the image for the cluster or the standalone host.

Staging vSphere Lifecycle Manager Images to ESX Hosts in a Cluster or Standalone Hosts

Staging is the process of downloading depot components from the vSphere Lifecycle Manager depot to the ESX hosts without applying the software and firmware updates immediately. Staging reduces the time that ESX hosts spend in maintenance mode.

Staging exists as a separate operation that you can initiate. Staging is also automatically triggered when you remediate a cluster, a host in the cluster, or a standalone host. By default, vSphere Lifecycle Manager performs the staging operation onto all hosts in the cluster in parallel.

If you manage a cluster with an image, you can stage the image to all hosts in the cluster or to selected host from that cluster. Staging is also supported for clusters with hosts and standalone hosts that have DPU devices. During staging, vSphere Lifecycle Manager downloads to the hosts all components that are applicable to the DPU device.

After staging is successful, a staged icon
( )
appears for any host to which the image is staged. The icon shows that both the software and firmware components

from the image are successfully staged to the respective host. If vSphere Lifecycle Manager can't stage the firmware components to the host, staging does not fail, but you don't see the staged icon for that host.

Staging as Part of Remediation

The remediation process happens in two steps, staging and then the actual remediation. vSphere Lifecycle Manager puts the ESX hosts in maintenance mode only after staging is complete. The result is reduced ESX downtime and disruption in the virtual workloads running on the virtual machines that reside on the respective hosts.

If staging is successful but the subsequent remediation process fails, the compliance status of the cluster or the standalone host changes from non-compliant to staged.

What Happens With the Staged Content?

The staged content is downloaded to the `/scratch` partition on vCenter if there is enough space available for the staged image. As a result, the staged content is persistent across power-cycles and reboots. If you delete the staged content, vSphere Lifecycle Manager re-stages the components needed for applying the image to the hosts in the cluster or the standalone host as part of the remediation process.

During staging, vSphere Lifecycle Manager validates the checksum and signature of the VIBs and downloads them to the appointed folder only after the validation is successfully completed. When the VIBs become available locally on the hosts, only a root user can modify them.

Staging Firmware Components

If an image contains a drivers and firmware add-on, during staging, the firmware VIBs are staged to the host or hosts along with the other software components defined in the image. If the firmware vendor does not support firmware staging, you get a notification in the **Image Compliance** card. You won't see the staged icon for that host even if staging the rest of the software is successful. In such a case, vSphere Lifecycle Manager installs the firmware components on the hosts during remediation.

Stage an Image to a Cluster

Staging an image to a cluster before remediating the cluster reduces the time that ESX hosts spend in maintenance mode. After successful staging, you can immediately proceed with remediation, or you can remediate the cluster at a later time.

- Verify that the hosts in the cluster are non-compliant with the image. See Check the Compliance Against an Image.
- Verify that the hosts in the cluster are connected to vCenter.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

When you stage an image to a cluster, vSphere Lifecycle Manager downloads all software and firmware components defined in the image from the vSphere Lifecycle Manager depot to all or only some of the hosts in the cluster. Before you can stage an image to a cluster or some hosts, you must run a compliance check for the cluster and ensure that hosts to which you stage the image are non-compliant with that image. Staging content to compliant hosts doesn't alter the hosts in any way. Staging the image on hosts whose compliance status is unknown triggers an automatic compliance check.

1. In the vSphere Client, navigate to a cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Stage (All)** button to stage all hosts in the cluster.
4. To stage the image to only some of the hosts in the cluster, select the hosts in the **Image Compliance** card and click the **Stage** button.
5. Optional: Click **Skip Remaining Hosts** if you want to stop the staging process while it is ongoing.
   The image is not staged to the skipped hosts.

After staging finishes successfully, an icon
( )
appears for all staged host in the cluster. The icon shows that all software and firmware from the image is successfully

staged to the selected hosts in the cluster. If vSphere Lifecycle Manager can't stage the firmware components to a host, staging is still successful, but you don't see the staged icon for that host.

Remediate the cluster to install the staged software and firmware to the hosts in the cluster.

Stage an Image to a Standalone Host

Staging is an operation that you can perform for a standalone host to reduce the time that the host spends into maintenance mode during remediation. Learn how to stage an image to a standalone host.

- Verify that the standalone host is non-compliant with the image. See Check the Compliance Against an Image.
- Verify that the standalone host is connected to vCenter.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to the standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Stage** button.

After staging finishes successfully, an icon ( ) appears for the host. The icon shows that all software and firmware components from the image are successfully staged to the host. If vSphere Lifecycle Manager can't stage the firmware components to the host, staging does not fail, but you don't see the staged icon for the host.

Remediate the host to install the staged software and firmware components to the host.

Remediation Precheck for a Cluster or a Standalone Host

To ensure that the health of a cluster or standalone host is optimal and that no problems occur during remediation, you can perform a remediation pre-check.

The remediation pre-check operation includes a series of checks for the cluster and for each host in the cluster, and for the standalone host. These checks include extensive health checks to determine whether the cluster or host is in a stable state and to ensure successful remediation. Also, the remediation pre-check triggers a compliance check for the cluster or for the standalone host. As a result, after the remediation pre-check, you can view compliance information for each host and whether host reboot or maintenance mode are necessary for successful remediation.

Remediation pre-checks leverage the compatibility data bundle to ensure that the currently installed ESX version can be upgraded to the target ESX version. See Recommended Images.

For vSAN clusters, the remediation pre-check operation includes a hardware compatibility check. Depending on how you configure the vSphere Lifecycle Manager remediation settings, vSphere Lifecycle Manager might prevent remediation if hardware compatibility issues exist. For information about configuring the global vSphere Lifecycle Manager remediation settings, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images. For information about configuring the remediation settings for a particular cluster, seeOverride the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images.

Run a Remediation Pre-Check for a Cluster

You can perform a remediation pre-check for all hosts in a cluster or only for selected hosts.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

Instead of generating a remediation pre-check report for the entire cluster, you can run a remediation pre-check task for one or more hosts in the cluster.

1. In vSphere Client, navigate to a cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Run pre-check (All)** button.

   You can observe the progress of the pre-check task. When the task finishes, vSphere Lifecycle Manager displays information about the issues found during the pre-check.
4. To run a precheck on some of the hosts in the cluster, select the hosts in the **Image Compliance** card and click the **Run pre-check** button.

If vSphere Lifecycle Manager reports no issues, remediate the cluster. See Remediate a Cluster or Selected Hosts in a Cluster Against an Image.

If issues are reported, resolve the issues before you remediate the cluster.

Run a Remediation Pre-Check for a Standalone Host

You can also run a remediation pre-check for a standalone host in your inventory.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

The remediation pre-check task ensures that the host can be successfully remediated.

1. In the vSphere Client, navigate to a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance card**, click the **Run pre-check** button.

   The **Running pre-check** dialog appears inside the **Image Compliance** card.

   You can observe the progress of the task.

When the pre-check finishes, vSphere Lifecycle Manager displays information about the issues found during the pre-check.

If vSphere Lifecycle Manager reports no issues, you can remediate the host. See Remediate a Standalone Host Against an Image.

If vSphere Lifecycle Manager reports any issues, resolve the issues before you remediate the host.

Remediating a Cluster or a Standalone Host Against an Image

When you set up or import a vSphere Lifecycle Manager image to use with a cluster or host, the software specified in the image is not immediately installed on the hosts in that cluster or on the standalone host. To apply the software specification from the image to the hosts, you must remediate the cluster or the standalone host against that image.

To initiate remediation of a cluster or host, you must have the required privileges. For a list of all vSphere Lifecycle Manager privileges and their descriptions, see vSphere Lifecycle Manager Privileges For Using Images. For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation

During remediation, the image that you set up for the cluster is installed on all ESX hosts in the cluster.

If DRS is not set to automated mode, remediating of a cluster of standalone ESX host containing powered on VMs will require manual actions.

For vSAN clusters, the remediation operation includes a hardware compatibility check. Depending on how you configure the vSphere Lifecycle Manager remediation settings, vSphere Lifecycle Manager might not proceed with the remediation task if hardware compatibility issues exist. For information about configuring the global vSphere Lifecycle Manager remediation settings, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images. For information about configuring the remediation settings for a particular cluster or standalone cluster, see Override the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images.

Maintenance Mode

If the update requires it, hosts are put into maintenance mode before remediation. Virtual machines cannot run when a host is in maintenance mode. To ensure a consistent user experience, vCenter migrates the virtual machines to other hosts within the cluster before a host is put into maintenance mode. vCenter can automatically evacuate these virtual machines to another host in the cluster if the following conditions are met:

- The destination hosts are compatible and have sufficient resources to run these virtual machines.
- DRS is configured for automated mode.
- vMotion is correctly configured.

You can configure vSphere Lifecycle Manager to deactivate HA admission control for the cluster before remediation. Though possible this option is not recommended, because deactivating HA admission control before you remediate a two-node cluster that uses a vSphere Lifecycle Manager image causes the cluster to practically lose all its high availability guarantees. The reason is that when one of the two hosts enters maintenance mode, vCenter cannot failover virtual machines to that host and HA failovers are never successful. For more information about HA admission control, see the *vSphere Availability* documentation.

Parallel Remediation

During the remediation of a cluster against a vSphere Lifecycle Manager image, the ESX hosts in the cluster are remediated sequentially by default. So, if the remediation for a single host in the cluster fails, the remediation of the entire cluster stops. However, you can configure vSphere Lifecycle Manager to remediate in parallel the hosts within a cluster that uses images. See Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images.

Parallel remediation reduces the overall remediation time, but is only possible if more than one cluster has sufficient free resources for multiple hosts to be offline at once without impacting virtual machine availability. Parallel remediation requires that you manually place multiple hosts into maintenance mode before patching can commence and exit maintenance mode after patching is complete. If you activate parallel remediation, vSphere Lifecycle Manager does not remediate the ESX hosts that are not in maintenance mode.

When you configure vSphere Lifecycle Manager to remediate hosts in parallel, you can set the maximum number of hosts to be remediated in a single remediation task. Alternatively, you can let vSphere Lifecycle Manager calculate the optimal number of hosts to remediate in parallel.

When you remediate hosts in parallel, if the remediation of a single host fails, the remediation task for the entire cluster does not stop and the rest of the hosts are remediated successfully. After remediation finishes, vSphere Lifecycle Manager reports an error for the respective host.

Parallel remediation is deactivated by default, but you can activate it during remediation or in the vSphere Lifecycle Manager general remediation settings. Parallel remediation and all other remediation settings apply to remediation tasks that you start in vCenter. Solutions such as NSX, for example, might have separate parallel remediation settings.

You cannot remediate in parallel the witness host and the hosts in the associated vSAN cluster. Parallel remediation is also not possible for clusters that are enabled for vSphere IaaS Control Plane or NSX.

Remediating a DPU-Backed Cluster or Standalone Host

During remediation, vSphere Lifecycle Manager applies VIBs to the ESX version on the DPU, if the host has a DPU device and the image for the cluster or host contains components with VIBs applicable to the DPU.

Remediate a Cluster or Selected Hosts in a Cluster Against an Image

By remediating a cluster against an image, you apply the software specified of the image to all hosts in the cluster or only to selected hosts in the cluster. So, by remediating a cluster, you make the non-compliant hosts compliant with the image that you set for the cluster.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

During remediation, the hosts in the cluster are remediated in sequence by default. You can configure vSphere Lifecycle Manager to remediate hosts in parallel.

The hosts that have the incompatible compliance state are not remediated.

If a vCenter HA failover is initiated during the remediation of a cluster, the remediation task is canceled. After the failover finishes, you must restart the remediation task on the new node.

1. In vSphere Client, navigate to a cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Remediate (All)** button.

   The **Review Remediation Impact** dialog box opens. The dialog box contains detailed information about all changes that remediation will enforce on the hosts in the cluster.
4. Optional: To remediate only some of the hosts in the cluster, select the hosts in the **Image Compliance** card, click the **Remediate** button.
5. In the **Review Remediation Impact** dialog box, review the impact summary, the applicable remediation settings, and the Foundation Agreement.
6. Optional: To save and review the impact details later, click **Export Impact Details**.
7. Accept the Foundation Agreement by selecting respective check box.

   The check box is selected by default.
8. Click the **Start remediation** button.

   The **Remediate Cluster** task appears in the **Recent Tasks** pane. You can also observe the progress of the remediation task in the **Image Compliance** card. If remediation fails, vSphere Lifecycle Manager gives information about the reasons for the failure.

Remediate a Standalone Host Against an Image

When you remediate a standalone host against the image assigned to the host, vSphere Lifecycle Manager applies the image to the host. Remediation is the operation that makes a non-compliant standalone host compliant with the image that you use for the host.

- Verify that the host is non-compliant with the image for the standalone host.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the **Remediate** button.

   The **Review Remediation Impact** dialog box opens. The dialog box contains detailed information about all changes that remediation will enforce on the host.
4. Review the impact summary, the applicable remediation settings, and the Foundation Agreement.
5. Accept the Foundation Agreement by selecting respective check box.

   The check box is selected by default.
6. Click the **Start remediation** button.

   The **Remediate Host** task appear in the **Recent Tasks** pane. You can also observe the progress of the remediation task in the **Image Compliance** card. If remediation fails, vSphere Lifecycle Manager gives information about the reasons for the failure.

View Last Remediation or Remediation Pre-Check Results for a Cluster or a Standalone Host that Uses an Image

You can view the remediation results from the last remediation or remediation pre-check that vSphere Lifecycle Manager performed.

Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to a cluster or a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the horizontal ellipsis icon and select your task.
   - To view the results from the last remediation pre-check performed on the cluster or the standalone host, select **Last pre-check results**.
   - To view the results from the last remediation of the cluster or the standalone host, select **Last remediation results**.

The **Image Compliance** displays detailed information about the last remediation or remediation pre-check task that ran on the cluster or the standalone host.

Manage Depot Overrides for a Cluster or a Standalone Host

In Remote Office and Branch Office (ROBO) scenarios, use VCF software depot to manage and distribute software updates centrally across remote locations. If a specific remote cluster requires an alternate local HTTP mirror outside VCF software depot, you can configure a depot override for that cluster or standalone host.

- Set up an online or offline web server depot accessible locally by the ESX hosts in the target cluster or standalone host, or use a VCF download tool patch repository structure.
- Export an offline bundle containing the required ESX base images, vendor add-ons, and components from the vSphere Lifecycle Manager image, and populate the local depot server.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

A ROBO cluster or a ROBO standalone host may have limited or no access to the Internet or limited connectivity to vCenter. As a result, clusters and standalone hosts in ROBO deployments might experience delays during compliance checks, remediation pre-checks, staging, and remediation operations when fetching packages over WAN links.

With vSphere Lifecycle Manager images, you can configure one or more local depots for a specific cluster or standalone host. When a depot override is configured, vSphere Lifecycle Manager evaluates metadata compatibility using the image specification defined in vCenter during compliance checks and pre-checks. During staging and remediation, ESX hosts download software payload binaries directly from the configured local override depot URL rather than transferring data across the WAN from the vSphere Lifecycle Manager local depot on the vCenter appliance. Local depot overrides save time and network bandwidth for Remote Office and Branch Office (ROBO) deployments by eliminating redundant WAN traffic.

If all depot overrides are removed or deactivated for a cluster or standalone host, vSphere Lifecycle Manager automatically reverts to the default vSphere Lifecycle Manager download source through VCF software depot.

You can manage local depot overrides for ROBO clusters and standalone hosts in the vSphere Client or programmatically by using the vSphere Automation API.

For each cluster or standalone host that you manage with an image, you can add and use multiple local depots instead of the default vSphere Lifecycle Manager depot. You can also delete the depot overrides that you configure. If depot overrides are not active for a cluster or a standalone host, the cluster or the host uses the general vSphere Lifecycle Manager depot in vCenter.

1. In vSphere Client, navigate to a cluster or a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image Compliance** card, click the horizontal ellipsis icon and select **Manage depot overrides**.

   The **Manage depot overrides** dialog box opens.
4. Select your task.

Option	Description
Add depot overrides	Enter a URL or a file path to a local depot and click **Add**. The depot is added to the **Depot override URL** list.
Delete a depot override	Click the horizontal ellipsis icon for a depot override from the list and click **Delete**. The depot is removed from the **Depot override URL** list.

5. Click **Close**.

In the **Image Compliance** pane, you see a notification if depot overrides are active for the cluster or the standalone host.

vSphere Lifecycle Manager Recommended Images

For the clusters that you manage with images, vSphere Lifecycle Manager can generate and provide you with recommendations in the form of pre-validated images that are compatible with the hardware and software of the hosts in a cluster. Recommended images are valid images that are based on the latest ESX version in a series or the latest and greatest ESX version. You can easily replace the current image of a cluster with the recommended image for that cluster.

Up until now, vSphere Lifecycle Manager recommendations for pre-validated images were generated based on the Broadcom Compatibility Guide (BCG) and vSAN Hardware Compatibility List (vSAN HCL). In vSphere 9.0, the vSphere Lifecycle Manager leverages the VMware Validation Service which is used to publish a compatibility data bundle. The compatibility data bundle holds all data from the Broadcom Compatibility Guide (BCG) combined with the Broadcom Product Interoperability Matrix. This bundle is provided in a compressed archived format that you can add to the vSphere Lifecycle Manager download sources and download an updated version of the compatibility data bundle when you synchronize the vSphere Lifecycle Manager depot through VCF software depot. See Import Updates to the Depot and Sync Compatibility Data. In case you use the VCF download tool, the compatibility data bundle is automatically downloaded along with the ESX host updates to the vSphere Lifecycle Manager depot.

When you set up or edit an image, you manually combine the ESX image version with the other image elements (vendor add-on, firmware add-on, and additional components) in such a way as to define the full software stack to run on all hosts in the cluster. vSphere Lifecycle Manager generates recommendations automatically under certain conditions and upon certain events. For example, when you validate an image against a cluster or a standalone host or run a remediation precheck, vSphere Lifecycle Manager uses the data from the compatibility data bundle to estimate whether the image is complete and valid, and suitable to your environment. You can also trigger recommendation generation manually to view the vSphere Lifecycle Manager recommendations. This saves you the effort of exploring the possible and applicable combinations of image elements on your own.

Recommended images are validated through a series of checks that ensure that a recommended image has no missing dependencies or conflicting components. For vSAN clusters, the validation also runs a hardware compatibility check against the vSAN Hardware Compatibility List (vSAN HCL). The extensive validation checks ensure that if you decide to use a recommended image for a cluster, the remediation against the recommended image is successful.

To generate recommendations, vSphere Lifecycle Manager checks what software is available in the vSphere Lifecycle Manager depot and if you selected a hardware support manager - what firmware is available in the depot. Based on the interoperability matrix, available software, firmware, and, for vSAN clusters, the hardware compatibility checks, for each cluster or standalone host that you manage with an image, vSphere Lifecycle Manager provides you with the latest in the current version series and with the latest version of the ESX base image that is appropriate for your environment.

> **Important:**
>
> Recommendation generation is not supported for clusters with composite images.

You can manually cancel the recommendation generation task from the **Recent Tasks** pane.

What Does the Latest in Current Series Mean

The recommendation that vSphere Lifecycle Manager generates is based on the major ESX version in the image that the cluster uses. vSphere Lifecycle Manager recommends the most recent ESX version that causes no hardware compatibility issues or regressions. The ESX version in the recommended image is always within the major release series of the ESX version of the image for the cluster.

The latest image in a series contains the latest minor ESX version or patch for that version. For example, if the current image for a cluster contains a base ESX image of version 8.0 and base images of version 8.0 Update 1, 8.0 Update 2, 8.0 Update 3, and 9.x are available in the vSphere Lifecycle Manager depot, the latest image in the current series recommendation contains ESX version 8.0 Update 3.

In some cases, the ESX version in a recommended image might be the same as the ESX version in the current image for a cluster, but the recommended image might contain a later version of any other image element, for example the vendor add-on, a component, or the firmware add-on.

What Does Latest and Greatest Mean

In case the vSphere Lifecycle Manager depot contains an ESX image with a higher major version that the version running on the hosts in the cluster, a recommendation is generated based on the Broadcom Interoperability Matrix. You are provided with the best upgrade path for the hosts in a cluster based on the interoperability between the ESX image currently running on the hosts and the base images available in the vSphere Lifecycle Manager depot.

Why Are Some ESX Versions not Recommended

The latest ESX major version available in the depot might not be recommended, because it might be in conflict with upgrade path restrictions, it causes hardware compatibility issues, or it is not interoperable with the current vCenter version. Images that are not recommended contain ESX versions that, if used for a cluster, might lead to regressions.

For example, the current image for a cluster contains a base ESX image of version 8.0 Update 1 and the vSphere Lifecycle Manager depot contains ESX images of version 8.0 Update 2, a hot patch for 8.0 Update 2, 8.0 Update 3, 9.0, and 9.1. Upgrading to the ESX 8.0 Update 2 with the hot patch is an allowed upgrade path. Upgrading further to ESX 9.1 causes regression, because the recommended image that contains ESX 9.1 does not contain the hot patch. So, vSphere Lifecycle Manager lists ESX 8.0 Update 2 with the hot patch as the latest in current series recommendation. ESX 9.1 is listed as a not recommended ESX version. When you upgrade the hosts to version 8.0 Update 2 with the hot patch, vSphere Lifecycle Manager changes the recommendation to version 8.0 Update 3 as latest in series and version 9.1 as latest and greatest. You can now safely upgrade to ESX 9.1.

In some cases, vSphere Lifecycle Manager might list as not recommended an ESX version within the current release series. The reason is possible hardware compatibility issues.

In vSphere Client, you can view a list of all ESX versions that are not recommended for use together with an explanation about the reason why they are not recommended.

Automatically Triggered Recommendation Generation

vSphere Lifecycle Manager generates a new image recommendation automatically in the following cases.

- The vSphere Lifecycle Manager depot gets updated.
  By default, the depot updates every 24 hours. Also, the content of the depot changes when you import an offline bundle to the depot or you manually trigger synchronization to the configurable download sources.

  Note:

  If the depot gets updated with solution components only, vSphere Lifecycle Manager does not generate new recommendations. Similarly, if the depot gets updated by adding or removing base images that are of a major ESX version as compared to the current image of a cluster, vSphere Lifecycle Manager does not start a recommendation generation task.
- You edit the image that you use for a cluster and save the new image setup.

  Note:

  If you edit an image by only adding solution components to the image, vSphere Lifecycle Manager generates no new recommendation.

The automated recommendation generation is available only for clusters that use a single image. When vSphere Lifecycle Manager starts the generation of a new recommendation automatically, the `Compute Image recommendations for cluster` task appears in the **Recent tasks** pane. You can observe the progress of the task or cancel it. vCenter issues an event when a recommendation generation task starts or ends. If the task fails, vCenter issues an alarm of the warning type. In cases of failure, you must check for recommended images for the cluster manually. The recommendation generation task cannot run simultaneously with other vSphere Lifecycle Manager operations, for example, remediation and compliance checks. If you must start another operation immediately, you can cancel the Compute Image recommendations for cluster task at any time.

In ROBO deployments, automatically triggered recommendation generation relies on software synchronization between the local depot and the central vSphere Lifecycle Manager depot. When using the central VCF software depot, synchronization across sites is managed automatically. However, when using manual depot overrides (local web servers), administrators must manually ensure that the local override depot is synchronized with the central vSphere Lifecycle Manager depot for recommended image generation to function accurately.

Add the Compatibility Data Bundle to the vSphere Lifecycle Manager Depot

To use the combined data from the Broadcom Compatibility Guide (BCG) and the Broadcom Product Interoperability Matrix, you must add the compatibility data bundle to the vSphere Lifecycle Manager depot.

Information from the compatibility data bundle is used by vSphere Lifecycle Manager each time you validate an image, run a remediation precheck, or trigger a recommendation generation for a cluster or standalone host. The compatibility data bundle is available on the VMware Verification Service site and is updated for each product release.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. Select **Actions** > **Import Updates** at the top of the vSphere Lifecycle Manager home view.

   The **Import Update Resources** dialog box opens.
3. Browse to the compatibility data bundle on your local machine.

   If the upload fails, check whether the structure of the compressed archive file is correct and whether the vSphere Lifecycle Manager network settings are set up correctly.

   You download the compatibility data bundle automatically when you use VCF download tool to download software and notification in air-gap scenarios. See Download the Specified Data Using VCF Download Tool.
4. Click **Import**.

   The `Import updates` task appears in the **Recent Tasks** pane.

Sync Compatibility Data

To initiate a hardware compatibility check for a host or generate recommendations for pre-validated images, the compatibility data bundle that combines the Broadcom Compatibility Guide (BCG) with the Broadcom Product Interoperability Matrix must become available to vSphere Lifecycle Manager.

Synchronizing compatibility ensures that the compatibility information from BCG and the Broadcom Product Interoperability Matrix becomes available to vSphere Lifecycle Manager. The synchronization task is automated when you synchronize the online depots or the VCFDT-created shared repository. See Synchronize the Depot.

When no compatibility data is available for use to vSphere Lifecycle Manager, you must trigger the compatibility data synchronization manually.

vSAN HCL data is updated through synchronization. If you want to check the hardware compatibility of a host that is in a vSAN cluster, you must first verify that vSAN HCL data is up to date. For more information about updating vSAN HCL data, see the vSAN documentation.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. Select **Actions** > **Compatibility Data** > **Sync**.

   The `Update compatibility data` task appears in the **Recent Taks** pane.

After the `Update compatibility data` task finishes, the compatibility data from the BCG and the Broadcom Product Interoperability Matrix becomes available to vSphere Lifecycle Manager.

Check for recommended images for the cluster or host. See Check for Recommended Images. Check the hardware compatibility of your hosts against BCG before you update or upgrade them to a later ESX version. See Hardware Compatibility Checks for Clusters and Hosts.

Check for Recommended Images

By using recommended images for your clusters or standalone hosts, you ensure that your environment runs the latest verified software. Although vSphere Lifecycle Manager generates image recommendation automatically when the cluster or the host changes or when new software is available in the vSphere Lifecycle Manager depot, you can manually check for recommendations at any time.

- Verify that remediation is not running for the cluster or the standalone host.
- Verify that you have synced the vSphere Lifecycle Manager depot to have the latest compatibility data bundle available.
- Verify that you have connection to the Internet.
- Verify that the cluster is manages with a single image. Generating recommendations is not supported for clusters managed with composite images.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

A recommended image contains updates for your cluster or host. The recommendations is based on the ESX versions available in the vSphere Lifecycle Manager depot. When you trigger the **Compute image recommendations** task, vSphere Lifecycle Manager first determines the recommended ESX version for the cluster or host. After that, vSphere Lifecycle Manager checks sequentially for newer versions of the vendor add-on, additional components, and firmware add-on that are compatible with the recommended ESX version and the hardware of the hosts in the cluster or the standalone host. So, sometimes, a recommended image might contain the same ESX version as the ESX version in the current image for the cluster or host but combined with an updated vendor add-on, component, or firmware add-on.

The **Compute image recommendations** task is non-cancellable. You must rerun the task periodically to ensure that the recommendations are valid and still suitable to the cluster or host.

Before you check for recommended images, you must ensure that the cluster or host is not being remediated. Recommendations generation and remediation are mutually exclusive operations. They cannot run simultaneously.

1. In vSphere Client, navigate to a cluster or a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** card, click the horizontal ellipsis icon and select **Check for recommended images**.
   When the task finishes, a blue badge appears in the **Image** card.
4. To view the recommended images, click the horizontal ellipsis icon and select **View recommended images**.

vSphere Lifecycle Manager suggests recommended images for the cluster or for the standalone host or reports that no recommended images are available for the cluster or the host. Images are recommended based on the latest ESX version in current series and based on the highest ESX version available in the depot and interoperable with the vCenter environment. When no recommended images are available, vSphere Lifecycle Manager displays detailed information about why no recommendations are available or why certain ESX versions are not recommended.

View the recommendations. You can import a recommended image to the cluster or host and replace the current image that the cluster or the host uses. See Use a vSphere Lifecycle Manager Recommended Image as the New Image for a Cluster or a Standalone Host.

View the Recommended or Not Recommended Images for a Cluster or a Standalone Host

You can view the pre-validated images that vSphere Lifecycle Manager recommends for a cluster or a standalone host and also the ESX versions that vSphere Lifecycle Manager does not recommend for that cluster or host because of possible hardware compatibility issues or regression.

- Verify that the **Compute image recommendations** task is performed for the cluster or the host or check for recommended images for the cluster or the host manually. See Check for Recommended Images.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to a cluster or a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. If no recommendations are available for the cluster or host at all, click **View Details** in the banner at the top of the card to see details.
4. If recommendations are available, in the **Image** card, click the horizontal ellipsis icon and select **View recommended images**.
   - To view the images that vSphere Lifecycle Manager recommends for the cluster or the host, check the **Latest in current series** or the **Latest and greatest** columns in the **Recommended Images** dialog box.
   - To view the ESX versions that vSphere Lifecycle Manager doesn't recommend for the cluster or the host, in the **Recommended Images** dialog box, view the table that lists the images not recommended for use, along with explanations of the reasons.

   If the **View recommended images** option is dimmed, no recommended images are available for this cluster or host. If recommendations are available, in the **Image** card, you see a label indicating the number of recommendations.

   The **Recommended Images** dialog box opens.

You can decide to use one of the recommended images and set it as the new image for the cluster or the host.

Use a vSphere Lifecycle Manager Recommended Image as the New Image for a Cluster or a Standalone Host

For any cluster or host that you manage with an image, you can view the images that vSphere Lifecycle Manager recommends and you can replace the current image for the cluster or the host with one of the recommended images. Using recommended images saves you the time and effort of identifying valid images that are applicable to all hosts in a cluster or to a standalone host.

- Check for recommended images for a cluster or a host. See Check for Recommended Images.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In vSphere Client, navigate to a cluster or to a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** card, click the horizontal ellipsis icon and select **View recommended images**.

   If the **View recommended images** option is dimmed, no recommended images are available for this cluster or host.

   The **Recommended Images** dialog box opens.
4. In the **Recommended Images** dialog box, select one of the recommended images by clicking the respective radio button and click **Continue**.

   The recommended image is based on the latest ESX version within the current release series or the latest and greatest major ESX version available in the vSphere Lifecycle Manager depot. For example, if the ESX version in your current image is 8.0, the first option might include ESX version 8.0 Update 3 and a related vendor add-on. The second recommended image can contain ESX 9.0 or any other minor release version within the 9.0 release series.

   The selected image is imported to the cluster or the standalone host as a draft. The **Image** card appears.
5. Optional: Edit the image and validate the new image setup.
6. Click **Save**.

   If you do not save the image, it is saved as a draft. The next time you start editing the image for that cluster or host, you can use the draft as a starting point.

The recommended image is saved for that cluster or host. If a draft exists for the cluster or the host, the draft is overridden by the recommended image. No software is installed on the hosts in the cluster or on the standalone host at this stage.

To apply the software specification that the image defines, remediate the cluster or the host against the new image. See Run a Remediation Pre-Check for a Cluster and Remediating a Cluster or a Standalone Host Against an Image.

Firmware Updates with vSphere Lifecycle Manager

You can use vSphere Lifecycle Manager images to perform firmware updates on the ESX hosts in a cluster or on the standalone hosts. With a single operation, you can update both the software and the firmware on the host.

In earlier vSphere releases, you could perform firmware updates on vSAN clusters by using system-managed baselines. For non- vSAN clusters, firmware updates had to be manual.

vSphere Lifecycle Manager enables you to easily update the firmware in any cluster or host that you manage with an image. Firmware updates are not available for clusters or hosts that you manage with baselines.

To apply firmware updates to the hosts in a cluster or to the standalone host that you manage with an image, you must include a special type of add-on, the firmware and drivers add-on, to the image and then remediate the cluster or the host. The firmware and drivers add-on is a vendor-provided component that holds firmware update packages and / or drivers.

Unlike vendor add-ons, firmware and drivers add-ons are not distributed through the official VMware online depot or as offline bundles available at the Broadcom Support Portal Downloads site. For a given hardware vendor, firmware updates are available in a special vendor depot, whose content you access through a software module called a hardware support manager. The hardware support manager is a plug-in that registers itself as a vCenter extension. Each hardware vendor provides and manages a separate hardware support manager that integrates with vSphere. For each cluster or host that you manage with an image, you select the hardware support manager that provides the firmware updates for the cluster or the host. After you determine the hardware support manager that you want to use for a cluster or a standalone host, the hardware support manager provides you with a list of the available firmware updates. When you include a firmware add-on in an image, that add-on might modify the specified image by adding or removing components. The firmware add-on also defines the firmware versions to be installed on the hosts. During remediation, vSphere Lifecycle Manager applies the image to the hosts and requests the selected hardware support manager to update the firmware on the hosts in accordance with the firmware add-on specified in the image.

By selecting a hardware support manager and including a firmware add-on in your image, vSphere Lifecycle Manager can also determine the firmware compliance of the cluster or standalone host during a compliance check. So, you can easily detect and remediate any unwanted drifts. The hardware support manager is also responsible for retrieving the firmware versions on the host hardware, and, in some cases, determining the appropriate drivers for the updated firmware version.

In VCF 9.1, if you do not configure a hardware support manager for a cluster or a standalone host, vSphere Lifecycle Manager assumes that you do not want to upgrade their firmware. vSphere Lifecycle Manager treats the currently running firmware on such clusters or standalone hosts as the target firmware during a hardware compatibility check. To generate a hardware compatibility report for a cluster without a hardware support manager, both vCenter and ESX must be of version 9.1 and later and the cluster can be either homogeneous or heterogeneous.

For vSAN clusters, the hardware support manager inspects the hosts in the cluster to determine their current I/O device controllers and firmware. During a hardware compatibility check for the cluster, vSphere Lifecycle Manager checks whether the firmware in the image is compatible with the hardware in the cluster as per vSAN Hardware Compatibility List (vSAN HCL). When vSphere Lifecycle Manager remediates the cluster, the hardware compatibility check ensures that the firmware and drivers on the hosts are certified for use with vSAN.

Firmware Updates on DPU Devices

In a DPU-based environment, only if the server vendor has an integrated hardware support manager, you can update the firmware on the DPU devices of the hosts through vSphere Lifecycle Manager. For servers that do not provide a hardware support manager, firmware updates are manual.

Deploying Hardware Support Managers

The deployment method, entitlement, licensing, and management of a hardware support manager plug-in are determined by the respective OEM.

Several of the major OEMs develop and supply hardware support managers. For example:

- Cisco
  The hardware support manager that Cisco provides is integrated with Cisco Intersight Infrastructure Service, which is part of Cisco Intersight and you activate the hardware support manager from within the Cisco Intersight SaaS-based management platform. No additional appliances are required on your vCenter instance.

- Dell
  The hardware support manager that Dell provides is part of their host management solution, OpenManage Integration for VMware vCenter (OMIVV), which you deploy as an appliance.
- Hitachi
  The hardware support manager that Hitachi provides, Hitachi Unified Compute Platform Advisor, is infrastructure automation and management software for all Hitachi converged, hyperconverged, and integrated systems, which you deploy as an appliance.
- HPE
  The hardware support managers that HPE provides are part of their management tools, HPE Compute Ops Management plug-in for VMware vCenter and HPE OneView for VMware vCenter.
- Lenovo
  The hardware support manager that Lenovo provides is part of their server management solution, Lenovo XClarity Integrator for VMware vCenter, which you deploy as an appliance.

You can find the full list of all VMware-certified hardware support managers in the Broadcom Compatibility Guide at Hardware Support Managers.

Deploying and Configuring Hardware Support Managers

Regardless of the hardware vendor, you must deploy the hardware support manager appliance on a host with sufficient memory, storage, and processing resources. Typically, hardware support manager appliances are distributed as OVF or OVA templates. You can deploy them on any host in any vCenter instance.

> **Note:**
>
> If vCenter is configured with a proxy for Internet access, the proxy must be able to reach any hardware support manager registered with that vCenter instance. You must select between the following options:
>
> - Assign a private IP address within the `10.x.x.x` range to the hardware support manager, which is automatically exempt from proxy use.
> - Enable a direct access to the registered hardware support manager by configuring the proxy settings with an exception for its IP address.

After you deploy the appliance, you must power on the appliance virtual machine and register the appliance as a vCenter extension. You might need to log in to the appliance as an administrator. Each hardware support manager might register with only one or multiple vCenter systems.

A vCenter plug-in user interface might become available in the vSphere Client after you deploy a hardware support manager appliance, but the hardware support manager might also have a separate user interface of its own. For example, OMIVV, HPE OneView for VMware vCenter, and Lenovo XClarity Integrator for vCenter all have a vCenter plug-in user interface, which helps you configure and work with the respective hardware support manager.

Each hardware support manager has its own mechanism of managing the actual firmware packages and making firmware add-ons available for you to choose.

The successful integration between the hardware support manager and vSphere Lifecycle Manager might require a specific configuration of the hardware support manager.

vSphere Lifecycle Manager and vSAN integrate to support proactive hardware management. You can deploy and register a hardware support manager that has proactive hardware support management functionality. All hosts in a vSAN cluster associated with such hardware support manager, start receiving proactive notifications about predictive disk failures. You can use these notifications to take early remediation steps before the actual disk failure occurs. Each vendor has a predefined set of health statuses that trigger the respective vCenter event. Upon receiving an event regarding disk failure, vSAN raises a health alarm and provides some troubleshooting guidance from the vendor. For more information about proactive hardware management, see the *Managing Proactive Hardware* section in Monitoring and Troubleshooting vSAN VCF documentation. For more information about how to replace failed disks, see the *Replace a Caching Device on a Host in vSAN Cluster* and *Replace a Capacity Device in vSAN OSA Cluster* procedures in the Monitoring and Troubleshooting vSAN VCF documentation.

For detailed information about deploying, configuring, and managing hardware support managers, refer to the respective OEM-provided documentation.

Use an Image for Firmware Updates

vSphere Lifecycle Manager allows you to manage the firmware lifecycle on ESX hosts that are part of a cluster or on a standalone host that you manage with an image.

- Deploy the vendor-provided hardware support manager and register it as a vCenter extension. For more information about deploying and managing a hardware support manager, see the respective OEM documentation.
- If you use the hardware support manager provided by Dell, create a cluster profile and associate it with the cluster. For more information, review the OpenManage Integration for VMware vCenter (OMIVV) documentation.
- If both vCenter and ESX are of version 9.1 and later and no hardware support manager is configured, you can generate a hardware compatibility report before cluster remediation.
- Verify that all hosts in the cluster are from the same vendor.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to a cluster or a standalone host that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Image**.
3. In the **Image** card, click the **Edit** button.
4. In the **Edit Image** card, for the **Firmware and Drivers Addon**, click **Select**.

   The **Firmware and Drivers Addon** dialog box opens.
5. In the **Firmware and Drivers Addon** dialog box, select a hardware support manager from the drop-down menu.

   The selected hardware support manager must be from the same hardware vendor as the hosts in the cluster or as the standalone host. Otherwise, during a compliance check, the hardware support manager reports the selected firmware and drivers add-on to be incompatible with the host or hosts that are from a different vendor. Firmware remediation fails.

   A list of all available firmware add-ons appears.
6. Select a firmware add-on from the list.

   An information panel appears on the right. The panel contains information about the supported ESX versions and whether the selected add-on contains the necessary drivers.
7. Click **Select**.

   The selected firmware and drivers add-on is included in the image.
8. In the **Image** card, validate and save the image.

   After the image is saved, a compliance check against the new image is triggered for the cluster or host.
9. In the **Image Compliance** card, review the compliance check results for the cluster and for the host.
10. If any host in the cluster or the standalone host has firmware that is non-compliant with the new image firmware, remediate the respective host or the cluster.
    a) Optional: In the **Image Compliance** card, run a remediation pre-check to ensure that remediation finishes successfully.
       - To run a pre-check for all hosts in the cluster, click the **Run Pre-check** button.
       - To run a pre-check for a single host in the cluster, click the vertical ellipsis icon for the host and select **Run Pre-check**.
       - To run a pre-check for a standalone host, click the **Run Pre-check**.

    b) In the **Image Compliance** card, initiate remediation.
       - To remediate all hosts in the cluster, click the **Remediate All** button.

         During cluster remediation, if the remediation of a single host fails, the remediation for the cluster ends prematurely.
       - To remediate a single host, click the vertical ellipsis icon for the host and select **Remediate**.
       - To remediate a standalone host, click the **Remediate** button.

    You are not obliged to start remediation immediately after setting up an image for a cluster or a standalone host. However, nothing is installed on the hosts unless you remediate them against the image for the cluster or the host.

The firmware on the hosts is actually updated only after successful remediation. You can remediate the objects in your environment at any time that is convenient for you.

The firmware on the hosts in the cluster or the standalone host is updated to the firmware version specified in the firmware add-on for the image.

vSphere Lifecycle Manager Hardware Compatibility Checks for Clusters and Hosts

vSphere Lifecycle Manager automates the process of validating the hardware compliance of hosts and clusters against a selected ESX version. Hardware compatibility checks ensure that the host or cluster hardware is compliant with the Broadcom Compatibility Guide (BCG) and vSAN Hardware Compatibility List (vSAN HCL).

Hardware Compatibility Lists

Hardware compatibility lists are lists of hardware that is certified for use with various vSphere releases. The BCG contains information about server models and I/O devices that are certified for use with particular vSphere releases. Besides BCG, vSAN maintains a separate hardware compatibility list that lists all I/O and networking device controller hardware and the respective firmware versions that are certified for use with vSAN. The vSAN HCL also contains information about the disk drives that a specific vSphere release supports and the earliest disk drive firmware version that is certified for use with vSAN.

With vSphere Lifecycle Manager, you can perform the following tasks.

- Check the hardware compatibility of a single host.
- Check the hardware compatibility of a vSAN cluster.

In general, hardware incompatibilities do not prevent remediation and are not resolved upon remediation. However, you can configure vSphere Lifecycle Manager to prevent remediation when hardware compatibility issues exist for a cluster. For information about configuring the global vSphere Lifecycle Manager remediation settings, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images. For information about configuring the remediation settings for a particular cluster, see Override the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images.

Cluster-Level Hardware Compatibility Checks

Running a hardware compatibility check on your clusters before remediating them helps you ensure good vSAN cluster health and avoid entering into unsupported and unwanted configurations after remediation.

Cluster-level hardware compatibility checks are available only for vSAN clusters that you manage with an image. If a vSAN cluster uses baselines, hardware compatibility checks are not available. Also, if a cluster uses a vSphere Lifecycle Manager image but vSAN is not enabled for that cluster, hardware compatibility checks for that cluster are not available. Hardware compatibility checks for vSAN clusters are performed against the vSAN HCL.

To perform a hardware compatibility check for a vSAN cluster, the vSAN HCL data that is available to vSphere Lifecycle Manager must be up to date. vSAN HCL data is synchronized automatically or, in environments without a connection to the Internet, manually. For more information about keeping the vSAN HCL data up to date, see the vSAN documentation.

When you initiate a hardware compatibility check for a cluster, vSphere Lifecycle Manager scans the image and verifies that all elements of the image are compatible with the hardware of all hosts within the cluster. vSphere Lifecycle Manager validates only those hardware devices that vSAN uses. Because cluster-level hardware compatibility checks validate the compatibility between the cluster hardware and the cluster image, the compatibility results might not be accurate unless you remediate the cluster successfully and apply the image to all hosts in the cluster.

> **Note:**
> vSphere Lifecycle Manager performs the full driver and firmware verification of PCI devices and disk drives only if you configure vSphere Lifecycle Manager with a hardware support manager and add a firmware add-on to the vSphere Lifecycle Manager image. Without using a hardware support manager, vSphere Lifecycle Manager only validates the PCI and network interface card (NIC) devices and their respective driver versions, and the disk drive version.

Note:

In VCF 9.1, if no hardware support manage is configured, vSphere Lifecycle Manager validates the currently installed firmware version for specific PCI devices (vSAN storage and network controllers) against the vSAN HCL. To generate such hardware compatibility report, both vCenter and ESX must be of version 9.1 and later and the cluster can be homogeneous or heterogeneous.

Hardware compatibility issues are reported as warnings, and as such, they do not prevent you from remediating the hosts in the cluster against the image unless you change that behavior by editing the remediation settings for the cluster. For information about configuring remediation settings for a cluster, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images and Override the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images.

Depending on the type of device, during a hardware compatibility check for a cluster, vSphere Lifecycle Manager performs different validations and verifications during a hardware compatibility check.

Device Type	vSphere Lifecycle Manager Validation
Storage I/O controllers	For storage I/O controllers, vSphere Lifecycle Manager performs the following verifications: • Verifies that all storage device controllers are certified for use with the ESX version specified in the image. • Verifies that the image contains the correct storage device driver and firmware versions as per the vSAN HCL. • Suggests a compatible storage device driver and firmware version combination as per the vSAN HCL.
Network controllers	For NIC devices, vSphere Lifecycle Manager performs the following verifications: • Verifies that the physical Remote Direct Memory Access (RDMA)-enabled NICs on the hosts in the cluster are certified for use with the ESX version specified in the image as per the vSAN HCL. • Verifies that the image that the cluster uses contains the correct NIC driver and firmware versions as per the vSAN HCL. • Suggests a compatible NIC driver and firmware version combination for the cluster as per the vSAN HCL. • Verifies that the device is certified for all active features that are enabled on that device. For example, if vSAN uses a RDMA-capable NIC, vSphere Lifecycle Manager verifies that the NIC is certified for RDMA.
Disk drives	For disk drives, vSphere Lifecycle Manager performs the following verifications: • Verifies that all disk drives in the cluster are certified for use with the ESX version specified in the image as per vSAN HCL. • Verifies that the disk drive firmware version specified in the image for the cluster is equal to or higher than the earliest supported firmware version for the device as per vSAN HCL. **Note:** vSphere Lifecycle Manager performs those verifications not only on directly connected disk drives, but also on physical disk drives behind RAID-0 logical volumes and behind a VMD controller.

What Are the Currently Active Features for a Device

The list of active features for a device is a list of the software features that are enabled on the device and that the device can support. The active features that a device has enabled are the result of enabling or using a solution or a service. In

the vSphere Client, information about the currently active features on a device is only available for PCI devices - storage controllers and network controllers. A list of the currently active features is not displayed for disk drives.

When a solution or service, for example vSAN or NSX, enables a set of features on a device, those features impose certain hardware compatibility constraints upon the device. The device must be certified for those active features that vSAN uses. During a hardware compatibility check, in addition to validating the driver and firmware combination for a PCI device, vSphere Lifecycle Manager also validates the constraints that the active features impose on that device. Therefore, by running a hardware compatibility check, you ensure that only certified features are enabled on the devices in the vSAN cluster. Deactivated features do not raise compatibility issues.

When the set of active features for a device changes, you must re-run the hardware compatibility check to get correct compliance results.

Hardware Compatibility Check Results

After a hardware compatibility check, vSphere Lifecycle Manager lists a device as compliant or non-compliant. In some cases, the compatibility status is unavailable.

Compliant	A device is compliant when it is compatible with the ESX, driver, and firmware versions defined for the image. For PCI devices, the driver-firmware version combination must also be certified for all active features enabled on the respective device.
Non-compliant	A device is non-compliant when it is incompatible with the ESX, driver, or firmware version defined for the image. Additionally, PCI devices are non-compliant when the driver-firmware version combination is not certified for all active features enabled on the respective device. If a device is not present in the vSAN HCL at all, vSphere Lifecycle Manager marks the device as non-compliant.
Unavailable	The hardware compatibility status for a NIC device is unavailable in the following cases: • Compatibility information for PCI devices is unavailable if the vSAN HCL data is not updated and does not contain information about the constraints imposed upon a device by the active features that are enabled on that device. • The cluster has no hosts or some hosts in the cluster are inaccessible. • No hardware support manager is registered in vCenter or the image for the cluster does not contain a firmware and drivers add-on.

When Does the Hardware Compatibility Check Task Run

vSphere Lifecycle Manager performs regular hardware compatibility checks for the vSAN clusters that you manage with an image. In addition, certain vSphere Lifecycle Manager operations also trigger an automatic hardware compatibility check. Automated hardware compatibility checks are available for vSAN clusters that use an image.

For information about hardware compatibility checks and instructions how to manually perform a hardware compatibility check for a cluster or for a single host, see vSphere Lifecycle Manager Hardware Compatibility Checks for Clusters and Hosts.

The Hardware Compatibility Check Task

The vSAN Hardware Compatibility List (vSAN HCL) database changes regularly. For example, when VMware certifies new OEM devices, drivers, or firmware, those become part of the vSAN HCL database. Similarly, devices, drivers, or firmware that are not supported anymore are removed from the vSAN HCL database.

Changes in the vSAN HCL database might make your hardware compatibility results invalid and outdated. To provide you with valid hardware compatibility information, vSphere Lifecycle Manager runs a periodic hardware compatibility check against the latest vSAN HCL data.

The periodic hardware compatibility check is a preconfigured scheduled task that you can edit and force to run at any time. By default, the task runs every 24 hours. The scheduled task is configured at a vCenter level. If a vCenter system contains no vSAN clusters that you manage with an image, vSphere Lifecycle Manager skips the scheduled hardware compatibility check. This periodic task runs only for vSAN clusters that you manage with an image.

vSphere Lifecycle Manager Operations That Trigger a Hardware Compatibility Check

vSphere Lifecycle Manager performs an automatic hardware compatibility check for any vSAN cluster that you manage with an image in the following cases.

- You assign an image from Image Library to one or more clusters at a cluster, vCenter, data center, or folder level.
  When you assign an image to one or more clusters, vSphere Lifecycle Manager starts the automatic **Check hardware compatibility of cluster's host with image** task even for clusters without vSAN.
- You edit the image for the cluster and save the image.
  When you edit and save an image, vSphere Lifecycle Manager starts the Check hardware compatibility of cluster's host with image task even for clusters without vSAN. In such case, vSphere Lifecycle Manager only returns a warning that image hardware compatibility is not verified in non- vSAN clusters.
  If the automatically triggered hardware compatibility task fails, you can still save the new image for the cluster.
- You initiate a remediation pre-check or remediation.
  The hardware compatibility check is a part of the remediation pre-check and remediation task for vSAN clusters. If a cluster is not vSAN-enabled, vSphere Lifecycle Manager does not perform a hardware compatibility check when you initiate a remediation pre-check or remediation.
  You can configure how vSphere Lifecycle Manager behaves in case of hardware compatibility issues by editing the remediation settings for the cluster. For information about configuring remediation settings for a cluster, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images and Override the Global vSphere Lifecycle Manager Remediation Settings for Clusters or Standalone Hosts That You Manage with Images.
- You add or remove a host to and from the cluster.
  When you add or remove a host to and from the cluster, vSphere Lifecycle Manager invalidates the hardware compatibility check results for the cluster and issues a warning. You must rerun a hardware compatibility check to obtain valid information about potential hardware compatibility issues. Alternatively, you can remediate the cluster or run a remediation pre-check, which both automatically trigger a hardware compatibility check.

Network Interface Card Validation

Hardware compatibility checks for vSAN clusters are enhanced to incorporate Network Interface Card (NIC) validation against the vSAN HCL. Determining the compatibility between the NICs in a vSAN cluster and the software defined in the image that the cluster uses is crucial to a successful cluster upgrade.

During a hardware compatibility check, for each NIC device, vSphere Lifecycle Manager verifies that the driver and firmware version combination defined in the image for the cluster is certified for use with all active features enabled on the device.

For NIC devices, vSphere Lifecycle Manager checks the exact firmware version during the hardware compatibility check.

In a vSAN cluster, during a hardware compatibility check, vSphere Lifecycle Manager validates only the RDMA-capable NICs that vSAN uses. That is, if a host has a RDMA-capable NIC but the NIC is not in use, vSphere Lifecycle Manager does not calculate the hardware compatibility of that device. vSphere Lifecycle Manager does not validate non-RDMA NICs. Non-RDMA NICs in use by vSAN need no certification and therefore vSphere Lifecycle Manager does not validate them during a hardware compatibility check.

System Requirements for Network Interface Card (NIC) Validation

- vCenter 8.0 and later
- ESX 8.0 and later

Disk Drive Validation

During a cluster-level hardware compatibility check, vSphere Lifecycle Manager verifies that the disk drives that vSAN uses are supported and certified as per the vSAN Hardware Compatibility List (vSAN HCL). vSphere Lifecycle Manager also ensures that the disk drive firmware version specified in the cluster image is compatible with the cluster hardware.

The disk drives in a vSAN cluster and the firmware installed on the drives are of paramount importance for the overall vSAN cluster health. For example, a faulty disk drive firmware can cause issues ranging from performance degradation to

data loss. You can use vSphere Lifecycle Manager hardware support managers to perform disk drive firmware upgrades. Before you upgrade the disk drive firmware, however, you must ensure that the target firmware version is supported as per the vSAN HCL.

Note:
For SAS and SATA disk drives, the vSAN HCL lists the earliest supported firmware version. All firmware versions later than the specified in the vSAN HCL are supported. For NVMe devices behind a Volume Management Device (VMD) controller, the vSAN HCL lists the exact driver-firmware combination certified for use with a given ESX base image version. Only that specific driver-firmware version combination listed in the vSAN HCL is supported.

Supported Disk Drives Types

vSphere Lifecycle Manager validates the following types of disk drives and storage device configurations:

- HDD (SAS/SATA)
- SSD (SAS/SATA)
- SAS/SATA disk drives behind single-disk RAID-0 logical volumes
- NVMe devices behind a VMD controller

Note:
vSphere Lifecycle Manager treats NVMe devices that are not VMD-enabled as PCI storage controllers. In the vSphere Client, hardware compatibility information about NVMe devices that are not VMD-enabled is available on the **PCI Devices** tab and not on the **Disks** tab.

System Requirements for Disk Drive Validation

- vCenter 8.0 and later
- ESX 8.0 and later
- For validation of NVMe devices behind a VMD controller, vCenter 8.0 and later and ESX 8.0 and later

Important:
Hardware compatibility checks do not validate NVMe devices behind a VMD controller, if the ESX version of the hosts is earlier than 8.0. If you want to check the hardware compatibility of NVMe devices behind a VMD controller for hosts running an earlier version of ESX, for example 8.0 Update 3, you have the following workarounds:

- Override manually the compliance status of the NVMe device behind a VMD controller.
- Upgrade your hosts to ESX 8.0 or later.

RAID-0 Logical Volumes

vSphere Lifecycle Manager can validate the physical SAS/SATA disk drives behind single-disk RAID-0 logical volumes. The following requirements exist:

- The RAID controller is in a RAID or mixed mode.
  For more information about the RAID and mixed mode, see the Broadcom knowledge base article KB 53573.
- vCenter 8.0 Update 3 and later
- ESX 8.0 and later
- The hardware support manager must be upgraded and certified to work with vSphere 8.0 Update 3.
  If the you do not use an upgraded version of the hardware support manager, the compliance status of the physical drives behind RAID-0 logical volumes is unknown. In this case, you must manually validate the disk drives and the target firmware version and override the compliance status for those disks.

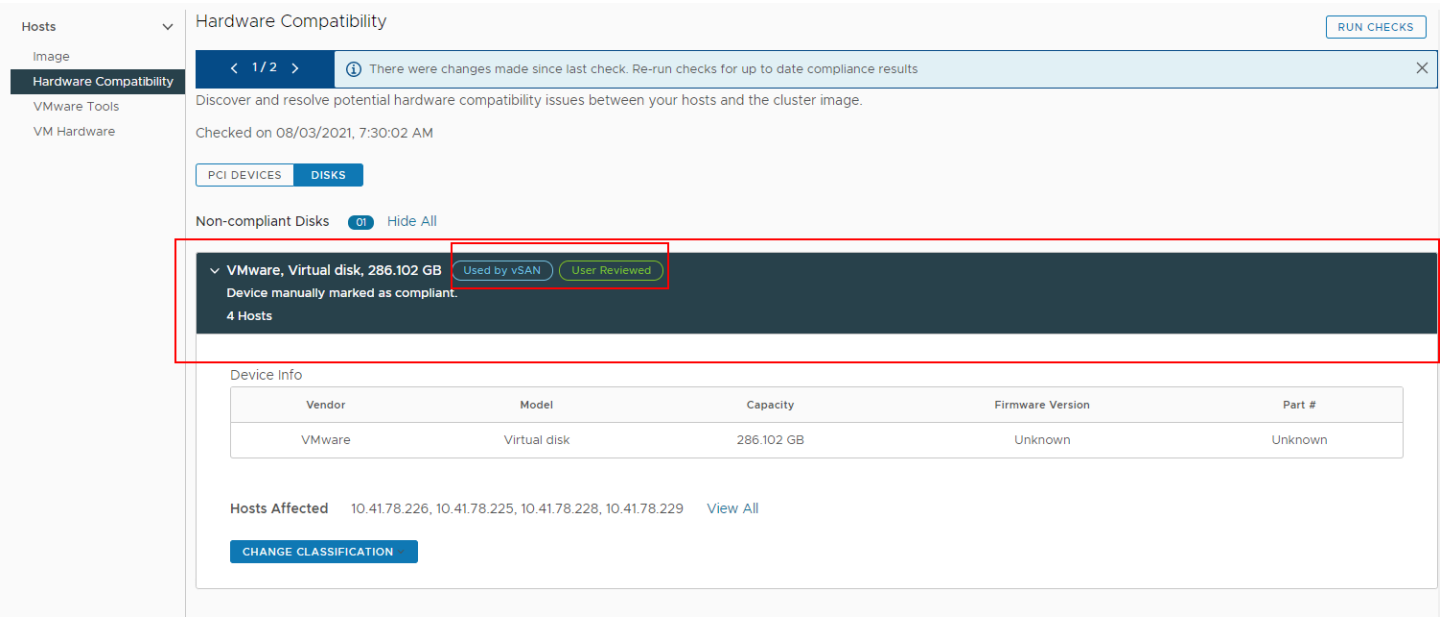
Disk Drive Validation Results

vSphere Lifecycle Manager does not display a disk drive compatibility status and compatibility information for every single disk in the vSAN disk group. vSphere Lifecycle Manager groups the disk drives that vSAN uses by vendor, model, target firmware version, capacity, and part number. That is, all disk drives by the same vendor, the same model, and with the same target firmware version form one entry in the list of disk devices.

Disk drives can be compliant or non-compliant. In the cases when vSphere Lifecycle Manager cannot find a unique match for a disk device in the vSAN HCL, vSphere Lifecycle Manager prompts you to manually specify the exact device that you want to validate. vSphere Lifecycle Manager then calculates the compliance status based on your selection.

When vSphere Lifecycle Manager is unable to determine the disk drive compliance, the respective devices are listed as non-compliant. You can manually validate those devices and set the compliance status to compliant or non-compliant. For more information, see Change the Compliance Status of a Disk Device Manually.

For each entry in the disk device list, you can view summarized information about the disk, the compliance status, the number of affected hosts, and a label that shows whether the compliance status is manually set or whether the device is certified. The **Used by vSAN** label is attached to all disk devices that vSAN uses.



If you expand the entry, you can view detailed compliance information about the respective disk device and the affected hosts.

When a new disk is added to a vSAN cluster, you must manually re-run the check to obtain the new compliance information about the cluster. Similarly, if you remove a disk from the vSAN disk group, you must re-run the hardware compatibility check to obtain an updated compliance information about the cluster.

Check the Hardware Compatibility of a Cluster

For a vSAN cluster that you manage with an image, you can check the compliance between the image components and the hardware in the cluster. The check is performed against the vSAN Hardware Compatibility List (vSAN HCL) and ensures that if the image is applied to the hosts, the result after remediation is in accordance with the vSAN HCL.

- Verify that vSAN is enabled for the cluster.
- Verify that the cluster uses an image.
- To validate the compatibility between the PCI device and disk device hardware and target firmware version, verify that the image for the cluster includes a firmware add-on.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In vSphere Client, navigate to a vSAN cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Hardware Compatibility**.
   In the **Hardware Compatibility** pane, you see the results from the previous compatibility check.
3. In the **Hardware Compatibility** pane, click the **Run Checks** button.

vSphere Lifecycle Manager displays all compatibility information and issues in the **Hardware Compatibility** pane. You can see detailed compatibility information for each PCI device or disk drive.

Review the result from the hardware compatibility check.

Resolve any issues before you remediate the cluster.

Change the Compliance Status of a Disk Device Manually

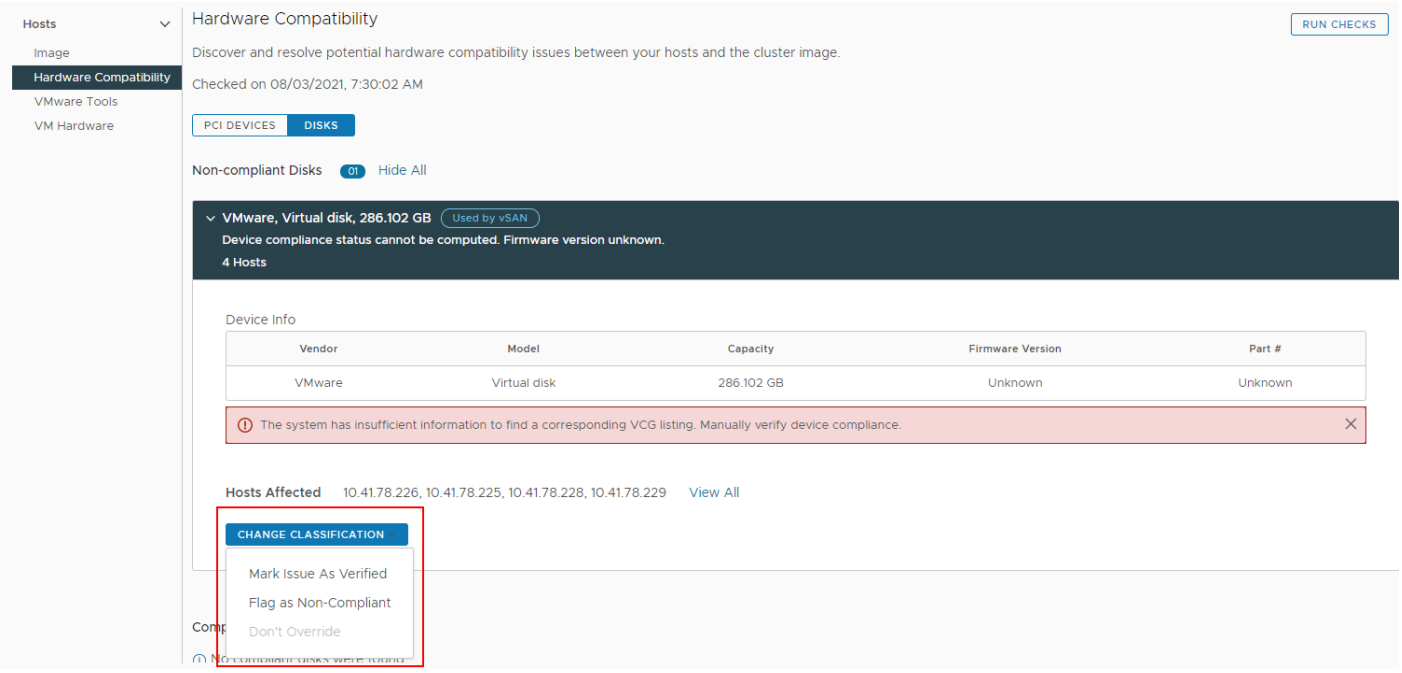
You can manually change the compliance status of a disk device and mark it as compliant or non-compliant.

- vCenter 8.0 or later.
- Verify that vSAN is enabled for the cluster.
- Verify that the cluster uses an image.
- Verify that all hosts in the cluster are from the same vendor.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

In cases when the compliance status of a disk drive is unavailable, you must perform the hardware compatibility check manually and mark the device as compliant or non-compliant.

1. In the vSphere Client, navigate to a vSAN cluster that you manage with an image.
2. On the **Updates** tab, select **Hosts** > **Hardware Compatibility**.
   In the **Hardware Compatibility** pane, you see the results from the previous compatibility check.
3. Click the **Disks** button.
   You see a list of all non-compliant and compliant disk devices.
4. Click the disk device whose compliance status you want to override.
5. Click the **Change Classification** button.

   A drop-down menu appears.



6. Select the compliance status to apply to the disk device.
   - To mark the issue as compliant, select **Mark Issue As Verified**.
   - To mark the issue as non-compliant, select **Flag as Non-Compliant**.

   The **Mark Issue As Verified** option is not available for compliant devices. The **Flag as Non-Compliant** option is unavailable if the disk device is non-compliant

7. Optional: To undo the override selection, select **Don't Override** from the **Change Classification** drop-down menu.

You changed the compliance status of a disk device. However, the disk device remains in its original list until you run a new hardware compatibility check.

If you marked the device as verified, a **User Reviewed** label appears for the disk group.

If you marked the device as non-compliant, a **Flagged** label appears for the disk group.

Run a new hardware compatibility check so that the new compliance status for a disk is saved.

Host-Level Hardware Compatibility Checks

You can run a hardware compatibility check for any host to determine which ESX version the host hardware is compatible with. The hardware compatibility check ensures that the host hardware, that is server model and I/O devices, is certified for use with a selected ESX version. Host-level hardware compatibility checks are performed against the Broadcom Compatibility Guide (BCG), even if the host is in a vSAN cluster.

You can check the hardware compatibility of any host, whether it is in a cluster that uses an image or baselines. You can also check the hardware compatibility of a standalone host. Host-level hardware compatibility checks do not validate the firmware on the host.

After the hardware compatibility check, vSphere Lifecycle Manager shows the compliance status for the server. These checks complete in one of three states: compatible, incompatible, and unknown. For more information about compatibility statuses, see Hardware Compatibility Report for a Host.

If the server status is incompatible, vSphere Lifecycle Manager does not proceed with checking the compatibility for the hardware devices.

Check the Hardware Compatibility of a Host

You can check the hardware compatibility of a host to determine whether the host hardware is certified for a specified ESX version. The hardware compatibility check is performed against the Broadcom Compatibility Guide (BCG) or, if the host is in a vSAN cluster, against the vSAN Hardware Compatibility List (HCL)

- If needed, synchronize hardware compatibility data. See Sync Compatibility Data.
- Verify that the Customer Experience Improvement Program is enabled.
- Verify that vCenter is connected to the Internet.
- Verify that the host is not part of an VxRail environment.
- Verify that you have the proper privileges. See vSphere Lifecycle Manager Privileges For Using Images.

1. In the vSphere Client, navigate to a standalone host or a host in a cluster.
2. On the **Updates** tab, select **Hosts** > **Hardware Compatibility**.
3. In the **Hardware Compatibility** pane, select your task.
   - To run a hardware compatibility check for the host for the first time, select a target ESX from the drop-down menu and click **Apply**.
   - To check the hardware compatibility between the host and the already selected target ESX version, click **Re-run Checks**.
   - To choose a new target ESX version for the hardware compatibility check, click **Edit** and select a new target ESX version.
   - To export the hardware compatibility report in a CSV format, click the **Export** button.

vSphere Lifecycle Manager displays the result from the compatibility check. You can see a list of the compatible, incompatible, and unknown devices. For each device, you can see full details by clicking the expand button.

Hardware Compatibility Report for a Host

The hardware compatibility report gives you information whether for a selected server model and hardware devices, vSphere Lifecycle Manager finds records for a target ESX version in the Broadcom Compatibility Guide (BCG).

Server Hardware Compatibility

Host Model is Not Compatible	This compatibility status indicates that there are no records for the selected ESX version in the BCG. If the host is not compatible with the selected ESX version, vSphere Lifecycle Manager does not proceed to checking the compatibility of the devices. In the **Host Model Compatibility** card, you can see details about the host: server model name, CPU model, and the BIOS version running on the host. At the bottom of the card, you see a list of all certified CPU series for the target ESX version.
Host Model is Compatible	This compatibility status indicates that the host is certified for use with the selected ESX version as per BCG. When the host is compatible, vSphere Lifecycle Manager proceeds with the device validation. In the **Host Model Compatibility** card, you can see details about the host: server model name, CPU model, and the BIOS version running on the host. Because in BCG the information about CPUs is based on CPU series, and not specific models, you might need to manually check if the CPU of the host is part of the supported CPU series. You might also need to manually check if the BIOS version on the host matches any of the compatible BIOS versions for the CPU series as per BCG.
Hardware Compatibility Checks Not Supported for the Host Vendor Model	When the server model is not part of the list of certified OEMs, vSphere Lifecycle Manager does not perform a hardware compatibility check and you do not see a hardware compatibility report for the selected host.

Device Hardware Compatibility

The compatibility statuses for devices are: compatible, incompatible, and unknown.

Unknown	Unknown devices are devices for which no records exist in BCG. When you click the expand button for the device, you see the following device information: device IDs, driver and firmware currently running on the device. No compatibility data is generated and displayed. The unknown status might also indicate that multiple matches exist in the BCG for the respective device. In such cases, use the device ID to manually check if the hardware device matches any of the supported devices for the target ESX version in the BCG.
Incompatible	The incompatible status indicates that no records exist in the BCG for the selected ESX version. When you click the expand button for the device, you see information about the ESX versions that are compatible with the device as per BCG.
Compatible	The compatible status indicates that the device is compatible with the selected ESX version as per BCG. When you click the expand button for the device, you see the following device information: device IDs, driver and firmware currently running on the device. For compatible devices, you might need to manually confirm that the driver-firmware combination running on the device is supported as per BCG.

Using vSphere Lifecycle Manager Baselines and Baseline Groups

Using baselines and baseline groups to manage the updates and patches of ESX 8.x hosts is a multi-stage process.

1. Populate the vSphere Lifecycle Manager repository with patches, extensions, and updates.
   The vSphere Lifecycle Manager repository contains software updates that you can use with both vSphere Lifecycle Manager baselines and vSphere Lifecycle Manager images. On the **Updates** tab of the vSphere Lifecycle Manager home view, you see all software updates available in the vSphere Lifecycle Manager depot as bulletins.
   Updates get into the vSphere Lifecycle Manager local depot through synchronization with configurable download sources. By default, vSphere Lifecycle Manager is configured to synchronize updates from the official VMware depot.

Note:

In vSphere Foundation 9.1, software binaries and metadata in vSphere Lifecycle Manager depot are synchronized automatically from the unified Broadcom software depot (`dl.broadcom.com`) through the VCF software depot, using central activation code. See Broadcom knowledge base article 443647.

Manual editing of Broadcom depot URLs in vCenter is unsupported, and legacy customer token URLs (KB 390121) are deprecated. In air-gapped environments, software packages are populated using the VCF Download Tool or manually imported offline ZIP bundles.

Important:

You can no longer upgrade hosts through baselines or import ISO images to the vSphere Lifecycle Manager depot.

For more information about working with the vSphere Lifecycle Manager depot, see The vSphere Lifecycle Manager Depot.

2. Create baselines by combining bulletins from the depot.
   You can also combine several non-conflicting baselines to create a baseline group. Baseline groups can contain different types of baselines.
   For more information about creating baselines and baseline groups, see Types of vSphere Lifecycle Manager Baselines and Baseline Groups.
3. Attach the baselines to individual ESX hosts or container objects for ESX hosts.
   For more information, see Attach Baselines and Baseline Groups to Objects.
4. Check the compliance of ESX hosts against a selected baseline or baseline group.
   You can run a compliance check on an individual ESX host or a container object.
   For more information about compliance checks against baselines and baseline groups, see Checking Compliance Against vSphere Lifecycle Manager Baselines and Baseline Groups.
5. Review the compliance status of the scanned object.
   For more information about compliance states, see Viewing Compliance Information About ESX Hosts and Updates.
6. Optionally, you can stage the patches and extensions to ESX hosts before remediation. Staging is not a mandatory, but recommended to reduce remediation time.
   For more information about staging updates before remediation, see Stage Patches and Extensions to ESX Hosts.
7. Remediate the non-compliant objects. After remediation, you can review the compliance status again to make sure that the updates are installed.
   For more information about remediating objects against baselines and baseline groups, see Remediating ESX Hosts Against vSphere Lifecycle Manager Baselines and Baseline Groups .

Types of vSphere Lifecycle Manager Baselines and Baseline Groups

You use baselines and baseline groups only to update the ESX hosts of version 8.x in your vSphere inventory. The vSphere Lifecycle Manager baselines are three types: predefined baselines, recommendation baselines, or custom baselines, which you create manually. Depending on their content, baselines can be patch and extension baselines.

When you initiate a compliance check for an ESX host, you evaluate it against baselines and baseline groups to determine its level of compliance to those baselines or baseline groups.

If your vCenter system is connected to other vCenter systems by a common vCenter Single Sign-On domain, the baselines and baseline groups that you create and manage are applicable only to the inventory objects managed by the vCenter system where the selected vSphere Lifecycle Manager instance runs.

In the vSphere Client, the baselines and baseline groups are displayed on the **Baselines** tab of the vSphere Lifecycle Manager home view.

Predefined, Recommendation, and Custom Baselines

Predefined baselines

Predefined baselines cannot be edited or deleted, you can only attach or detach them to inventory objects.

Predefined baselines can be of the following type:

- Host Security Patches

The Host Security Patches baseline checks ESX hosts for compliance with all security patches.

- Critical Host Patches
  The Critical Host Patches baseline checks ESX hosts for compliance with all critical patches.
- Non-Critical Host Patches
  The Non-Critical Host Patches baseline checks ESX hosts for compliance with all optional patches.

The Host Security Patches, and Critical Host Patches predefined baselines are attached by default to the vCenter instance where vSphere Lifecycle Manager runs.

Recommendation Baselines

Recommendation baselines are predefined baselines that vSAN generates.

You use recommendation baselines to update your vSAN clusters with recommended critical patches, drivers, or updates.

These baselines appear by default when you use vSAN clusters with ESX hosts of version 8.0 and later in your vSphere inventory. If your vSphere environment does not contain any vSAN clusters, no recommendation baselines are created.

Recommendation baselines update their content periodically, which requires vSphere Lifecycle Manager to have constant access to the Internet. The vSAN recommendation baselines are typically refreshed every 24 hours.

Recommendation baselines cannot be edited or deleted. You do not attach recommendation baselines to inventory objects in your vSphere environment. You can create a baseline group by combining multiple recommendation baselines, but you cannot add any other type of baseline to that group. Similarly, you cannot add a recommendation baseline to a baseline group that contains patch and extension baselines.

Custom Baselines

Custom baselines are the baselines that you create. You can create custom patch and extension baselines to meet the needs of your specific deployment.

Patch and Extension Baselines

Patch Baselines

Patch baselines define a number of patches that must be applied to a given host. Patch baselines can be either dynamic or fixed.

Baseline	Description
Dynamic Patch Baseline	A dynamic baseline is a set of patches that meet certain criteria. You specify the criteria for patch inclusion in the baseline. Only the patches that meet the criteria are included in the baseline. As the set of available patches in the vSphere Lifecycle Manager depot changes, dynamic baselines are updated as well. You can manually include or exclude patches from the baseline.
Fixed Patch Baseline	A fixed baseline is a set of patches that does not change as patch availability in the depot changes. You manually select the patches from the total set of patches available in the vSphere Lifecycle Manager depot.

Extension Baselines

Extension baselines contain additional software modules for ESX hosts, for example device drivers. This additional software might be VMware software or third-party software. All third-party software for ESX hosts is classified as host extension, but extensions are not restricted to just third-party software.You can install additional modules by using extension baselines, and update the installed modules by using patch baselines.

Extensions deliver additional host features, updated drivers for hardware, Common Information Model (CIM) providers for managing third-party modules on the host, improvements to the performance or usability of the existing host features, and so on.

The host extension baselines that you create are always fixed. You must carefully select the appropriate extensions for the ESX hosts in your environment.

You use extension baselines to install extensions on the ESX hosts in your environment. After an extension is installed on a host, you can update the extension module through either patch, or extension baselines.

Note:

When you use extension baselines, you must be aware of the functional implications that the installation of new modules on the host might have. Extension modules might alter the behavior of ESX hosts. During the installation of extensions, vSphere Lifecycle Manager only performs the checks and verifications expressed at the package level.

Baseline Groups

You create a baseline group by assembling existing and non-conflicting baselines. Baseline groups allow you to scan and remediate objects against multiple baselines at the same time.

The following are valid combinations of baselines that can make up a baseline group:

- Multiple host patch and extension baselines.

To create, edit, or delete baselines and baseline groups, you must have the **Manage Baseline** privilege. To attach baselines and baseline groups to target inventory objects, you must have the **Attach Baseline** privilege. The privileges must be assigned on the vCenter system where vSphere Lifecycle Manager runs.

For more information about managing users, groups, roles, and permissions, see the *vSphere SecurityvSphere Security* documentation.

For a list of all vSphere Lifecycle Manager privileges and their descriptions, see vSphere Lifecycle Manager Privileges For Using Baselines.

Creating Baselines in vCenter 9.x for ESX 8.x

Because in vSphere 9.0 and later releases the unified Broadcom online depot hosts certified partner content in addition to VMware content, and vSphere Lifecycle Manager retrieves this content through VCF software depot, a broader set of OEM bulletins are available in the vSphere Lifecycle Manager depot. As a result, in the **Create Baseline** and **Edit Baseline** wizards, you also see a broader set of OEM bulletins. Some of these bulletins might have dependencies that must be pulled into the baselines that you create, so that the remediation against those baselines is successful. Always consult the knowledge base article for an individual bulletin before you include it in a baseline. The knowledge base article contains information about the bulletin deployment specifics and required dependencies. You must include in the baseline, only bulletins compatible with the hardware on which the host is running. Otherwise, remediation might fail.

Changes are also introduced in the way VMware content is packaged. As a result, at patch and update releases, you might see additional bulletins on the patch selection page of the **Create Baseline** and **Edit Baseline** wizards. Those bulletins are usually of the Enhancement or BugFix category. When you include those bulletins in a baseline, you might need to also include base ESX bulletins in that baseline. To ensure successful application of VMware patches and updates, always include the appropriate roll-up bulletin into your baselines. Otherwise, remediation might fail.

Create, Edit, or Delete vSphere Lifecycle Manager Baselines and Baseline Groups

You can create update baselines for ESX hosts of version 8.x , which you must first import to the vSphere Lifecycle Manager depot.

- Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines**.

1. In the vSphere Client, click the bars icon and select **Lifecycle Manager**.
2. Optional: Select a vCenter system from the **Lifecycle Manager** drop-down menu.

   The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.

3. On the **Baselines** tab, select **New** > **Baseline**.
4. On the **Name and Description** page, enter information about the baseline, select the baseline type and click **Next**.
5. Follow the prompts to finish creating the baseline.

Baseline Type	Steps
Patch Baseline	To create a fixed patch baseline, perform the following steps: 1. On the **Select Patches Automatically** page, stop the automatic updates by deselecting the respective check box and click **Next**. 2. On the **Select Patches Manually** page, select the patches that you want to include in the baseline and click **Next**. – To view only the rollup bulletins in the list, turn on the **Show only rollup updates** toggle switch. – To filter the patches that are available in the vSphere Lifecycle Manager depot and find specific patches to include in the baseline, use the filter icon next to each column header. If you use several criteria to filter the patches, the relationship between those filter criteria is defined by the Boolean operator AND. 3. On the **Summary** page, review your selections and click **Finish**. To create a dynamic patch baseline, perform the following steps: 1. On the **Select Patches Automatically** page, set the criteria for adding patches to the baseline. a. Enable the automatic update of the baseline by selecting the respective check box. b. On the **Criteria** tab, specify the criteria that a patch must meet to be added to the baseline and click **Next**. **Note:** The relationship between the selected criteria is defined by the Boolean operator AND. For example, when you select a product and severity option, the patches are restricted to the ones that are applicable for the selected product and are of the specified severity level. • **Patch Vendor** specifies which patch vendor to use. • **Product** restricts the set of patches to the selected products or operating systems. **Note:** The asterisk at the end of a product name is a wildcard character for any version number. • **Severity** specifies the severity of patches to include. • **Category** specifies the category of patches to include. • **Release Date** specifies the range for the release dates of the patches. c. On the **Matched** tab, deselect patches from the ones that matched your criteria to exclude them permanently from the baseline. d. On the **Excluded** and **Selected** tabs, view the patches that are excluded from the baseline and the ones that are included in the baseline. You can use the filter icon next to each column header on the **Matched**, **Excluded**, and **Selected** tabs to filter the patches that are available in the vSphere Lifecycle Manager depot. This way, you can easily find specific patches to exclude from or include in the baseline. If you use several criteria to filter the patches, the relationship

Baseline Type	Steps
	between those filter criteria is defined by the Boolean operator AND. 2. On the **Select Patches Manually** page, select individual patches to include in the baseline and click **Next**. The patches that are displayed on this page are patches that do not meet the criteria you set on the **Select Patches Automatically** page. You can use the filter icon next to each column header to filter the patches that are available in the vSphere Lifecycle Manager depot and find specific patches to include in the baseline. If you use several criteria to filter the patches, the relationship between those filter criteria is defined by the Boolean operator AND. The patches that you add manually to the dynamic baseline stay in the baseline regardless of the automatically downloaded patches. 3. On the **Summary** page, review your selections and click **Finish**.
Extension Baseline	1. On the **Select Extensions** page, select individual extensions to include in the baseline and click **Next**. You can use the filter icon next to each column header to filter the extensions that are available in the vSphere Lifecycle Manager depot and find specific extensions to include in the baseline. If you use several criteria to filter the patches, the relationship between those filter criteria is defined by the Boolean operator AND. 2. On the **Summary** page, review your selections and click **Finish**.

The new baseline appears in the baselines list on the **Baselines** tab.You can attach the baseline to a data center, a cluster, or a host.

Create a Host Baseline Group

You can combine multiple baselines of different types into a baseline group. For example, you can combine multiple patch and extension baselines.

Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines**

A baseline group might contain a combination of host patch and host extension baselines.
You can create a baseline group and add baselines to it later.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.
      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Baselines** tab, select **New** > **Baseline Group**.
   The **Create Baseline Group** wizard opens.

3. On the **Name and Description** page, enter a unique name and, optionally, a description for the baseline group, and click **Next**.
4. Optional: On the **Patch Baselines** page, select patch baselines to include in the baseline group and click **Next**.
5. Optional: On the **Extension Baselines** page, select extension baselines to include in the baseline group and click **Next**.
6. On the **Summary** page, review your selections and click **Finish**.

The new host baseline group appears in the baselines list on the **Baselines** tab. You can attach the baseline group to a data center, a cluster, or a host.

Edit a vSphere Lifecycle Manager Baseline or Baseline Group

You can edit existing patch or extension baselines.

Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines**

1. In the vSphere Client, click the bars icon and select **Lifecycle Manager**.
2. Optional: Select a vCenter system from the **Lifecycle Manager** drop-down menu.
   The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
3. On the **Baselines** tab, select a baseline or a baseline group from the list and click **Edit**.
   The **Edit Baseline** or **Edit Baseline Group** wizard appears.
4. Follow the prompts to finish editing the selected baseline or baseline group.

Attach the edited baseline or baseline group to a data center, a cluster, or a host.

Add or Remove a Single Update from a Custom Baseline

You can edit the content of a custom baseline by adding or removing individual patches or extension from the baseline.

Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines**

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.
      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. Click the **Updates** tab.
   You see a list of all updates in the vSphere Lifecycle Manager depot.
3. Select a patch or extension from the list, and click **Add/Remove Baselines**.
   The **Add/Remove baselines** dialog box opens.
4. Select your task.
   - To add the patch to a baseline, select that baseline in the **Custom Patch Baselines** list.
   - To remove the patch from a baseline, deselect that baseline in the **Custom Patch Baselines** list.
5. Click **OK**.

Delete Baselines and Baseline Groups

You can delete the baselines and baseline groups that you no longer need. Deleting a baseline detaches it from all the objects to which it is attached. You cannot delete predefined and system-managed baselines.

Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines**

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Baselines** tab, select a baseline or a baseline group from the list and click **Delete**.

   The **Delete Baseline** dialog box opens.
3. Click **Yes** to confirm the deletion of the selected baseline or baseline group.

Attach Baselines and Baseline Groups to Objects

To check the compliance status of the objects in your inventory against selected baselines and baseline groups, you must first attach the respective baselines and baseline groups to the objects.

- If you want to attach a baseline or a baseline group to a cluster, verify that the cluster is not configured to use an image.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines** > **Attach Baseline**.

You attach baselines and baseline groups to individual hosts or objects that contain hosts, such as clusters, data centers, and vCenter instances. In the vSphere infrastructure hierarchy, the baseline and baseline groups that you attach to container objects are also attached to the child objects. For example, if you attach a baseline or baseline group to a folder, the baseline or the baseline groups is inherited by all the objects in the folder, including sub-folders.

You cannot use vSphere Lifecycle Manager to update the hosts in a cluster that uses a vSphere Lifecycle Manager image. For more information about using vSphere Lifecycle Manager images to manage hosts in clusters collectively, see How to Use vSphere Lifecycle Manager Images .

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.
3. In the **Attached Baselines** pane, click **Attach** > **Attach Baseline or Baseline Group**.

   The **Attach** dialog box opens.
4. Select one or more baselines or baseline groups and click **Attach**.

   If you select a baseline group, all the baselines in it are attached to the object.

   The selected baselines or baseline groups are attached to the object. They appear on the list in the **Attached Baselines** pane. If the selected object is a container object, the selected baselines or baseline groups are attached to all the child objects.

Scan the selected object against the attached baselines.

Detach Baselines and Baseline Groups from Objects

You can detach baselines and baseline groups from the objects to which the baselines or baseline groups are directly attached.

- Verify that the cluster is not configured to manage all its hosts collectively.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Baselines** > **Attach Baseline**.

vSphere inventory objects might have inherited properties, so instead of detaching baselines and baseline groups directly from an object, you might need to select its container object and detach the baselines or baseline groups from the container object. For example, if you want to detach a baseline or a baseline group from a host that is a part of a cluster, you must select the cluster and not the host.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.
3. In the **Attached Baselines** pane, select one or more baselines or baseline groups and click **Detach**.
   The **Detach** dialog box opens.
4. Select the object to detach the baseline or baseline group from and click **Detach**.
   The baseline is removed from the **Attached Baselines** list.

Checking Compliance Against vSphere Lifecycle Manager Baselines and Baseline Groups

Before you update an ESX host or a container object with vSphere Lifecycle Manager baselines, you must first check its compliance status.

You use vSphere Lifecycle Manager to check the compliance status of ESX hosts against the baselines and baseline groups that you attach to the hosts or to a parent container object. You perform a compliance check on hosts to determine whether they have the latest patches or extensions that conform to your security policies. During the compliance check, attributes of the host are evaluated against all patches and extensions from an attached baseline or baseline group.

You can check the compliance status of a single ESX host or a valid container object. Supported groups of ESX hosts include virtual infrastructure container objects such as folders, clusters, and data centers. When you initiate a compliance check for a container object, vSphere Lifecycle Manager scans all the ESX hosts in that container object.

> **Important:**
>
> In a vCenter instance of version 9.0 and later, vSphere Lifecycle Manager does not perform compliance checks for ESX hosts of version 9.0 and later that are still managed with baselines.
>
> **Note:**
> If you initiate a compliance check for an inventory object, for example data center, that contains clusters that use vSphere Lifecycle Manager images, the compliance check is not performed for those clusters.

To generate compliance information, you can initiate compliance checks manually or you can schedule the compliance checks to run at regular periods. Schedule compliance checks at a data center or vCenter system level to make sure that the objects in your inventory are up-to-date.

You check the compliance status of vSphere objects from the vSphere Lifecycle Manager compliance view.

To initiate or schedule compliance checks, you must have the **Scan for Applicable Patches, Extensions, and Upgrades** privilege.

For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation.

For a list of all vSphere Lifecycle Manager privileges and their descriptions, see vSphere Lifecycle Manager Privileges For Using Baselines.

Initiate a Compliance Check for ESX Hosts Manually

Before remediation, you must check the compliance of the vSphere objects against the attached baselines and baseline groups. To check the compliance status of hosts in the vSphere inventory immediately, initiate a compliance check manually.

- If you want to check the compliance status of a cluster, verify that the cluster is not configured to use an image.
- Verify that all ESX hosts in the cluster or the standalone host manager with baselines are not of version 9.0 and later.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.

   The **Baselines** card shows three panels. In those panels, you obtain host information about the selected object, host compliance information, and remediation information.
3. In the compliance information panel, click **Check Compliance**.

The selected inventory object and all child objects are scanned against all attached patch and extension baselines. The larger the virtual infrastructure and the higher up in the object hierarchy that you initiate the scan, the longer the scan takes.

Schedule Regular Compliance Checks for ESX Hosts Managed with Baselines

You can configure vSphere Lifecycle Manager to check the compliance status of ESX hosts that you manage with baselines at specific times or at intervals that are convenient for you.

- If you want to check the compliance status of a cluster, verify that the cluster is not configured to use an image.
- Verify that all ESX hosts in the cluster or the standalone host manager with baselines are of version 8.x.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.

   The **Baselines** card shows three panels. In those panels, you obtain host information about the selected object, host compliance information, and remediation information.
3. In the compliance information panel, click **Schedule**.

   The **Automatic compliance check** dialog box opens.
4. Configure the compliance check schedule.
   a) Set the frequency and the starting point of the compliance check.
   b) Enter a unique name, and optionally, a description for the scan task.
   c) Optional: Specify one or more email addresses to receive notification after the scan task is complete.

      You must configure mail settings for the vCenter system to enable this option.
5. Click **Save** to exit the **Automatic compliance check** dialog box.

View Information About the Patches and Extensions in a Baseline

You can view information about the patches and extensions included in a baseline or a baseline group.

- Verify that the host for which you want to view compliance information uses baselines and is not managed with a vSphere Lifecycle Manager image.
- Review the The vSphere Lifecycle Manager Compliance View topic.

For information about the different compliance statuses that an update might have, see Compliance Statuses of Updates.

1. In the vSphere Client, navigate to a single ESX host, cluster, or a valid container object.
2. On the **Updates** tab, select **Hosts** > **Baselines**.
3. In the **Attached Baselines and Baseline Groups** card, select a baseline.

   A new card appears below the **Attached Baselines and Baseline Groups** pane. Depending on the selected object, the bottom card might contain information about the updates and ESX images in the baseline that you select. If the

selected object is a container for ESX hosts, the bottom card shows the compliance of each ESX in the container object against the selected baseline.

Baseline Type	Available Information
Patch	The bottom card contains a table that lists all patches in the baseline. For each update, you can see the following information. • **Update Name** • **Update ID** The update ID is a vendor-assigned identification code of the patch. • **Status** The **Status** column shows the compliance status of the update. • **Severity** • **Category** • **Impact** The **Impact** column displays the actions that you must take to install the update. For example, rebooting the system or putting the host in maintenance mode. • **ESXi Version**
Extension	• **Update Name** • **Update ID** The update ID is a vendor-assigned identification code of the extension. • **Status** The **Status** column shows the compliance status of the update. • **Severity** • **Category** • **Impact** The **Impact** column displays the actions that you must take to install the update. For example, rebooting the system or putting the host in maintenance mode. • **ESXi Version**
Baseline Group	To view information about the patches and extension in a baseline group, select the respective tab in the bottom pane. • Click **Baselines** for information about the baselines that a baseline group contains. • Click **Updates** for information about the patches and extensions that the baseline group contains.

Viewing Compliance Information About ESX Hosts and Updates

vSphere Lifecycle Manager scans objects to determine how they comply with the baselines and baseline groups that you attach to those objects. You can view compliance information about a single ESX host or a group of hosts in a container object.

Supported groups of ESX hosts include virtual infrastructure container objects such as folders, clusters, and data centers.

The host or the container object must have an attached baseline or baseline group to be examined for compliance information. Compliance with baselines and baseline groups is assessed at the time of viewing.

Important:

In a vCenter instance of version 9.0 and later, vSphere Lifecycle Manager does not perform compliance checks for ESX hosts of version 9.0 and later that are still managed with baselines.

The overall compliance status of an ESX hosts depends on the compliance statuses of all baselines and baseline groups that are attached to the object. For information about the different compliance statuses that an object, a baseline or a baseline group might have, see Compliance Statuses of ESX Hosts, Baselines, and Baseline Groups .

The compliance status of a baseline depends on the compliance statuses of all updates in the baseline. For information about the compliance statuses that updates might have, see Compliance Statuses of Updates.

The ability to view the compliance status of vSphere objects depends on the privileges that you have. To view the compliance status of an inventory object, you must have the **View Compliance Status** privilege. Users that have privileges to remediate against patches and extensions and to stage patches and extensions on a particular inventory object, can view the compliance status of the same object even if they do not have the **View Compliance Status** privilege.

- Users with the privilege to view a container, but not all the contents of the container, can view the aggregate compliance status of all objects in the container.
- If a user does not have the permission to view an object, its contents, or a particular virtual machine, the results of those scans are not displayed.

For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation.

For a list of all vSphere Lifecycle Manager privileges and their descriptions, see vSphere Lifecycle Manager Privileges For Using Baselines.

For information about checking the compliance of hosts against an image, see Check the Compliance of a Cluster or Standalone Host Against a vSphere Lifecycle Manager Image.

The vSphere Lifecycle Manager Compliance View

Compliance information about inventory objects is displayed on the **Updates** tab for the object, in the so-called vSphere Lifecycle Manager compliance view.

The vSphere Lifecycle Manager compliance view for objects that you manage with baselines and baseline groups consists of three panes.

> **Important:**
>
> In a vCenter instance of version 9.0 and later, vSphere Lifecycle Manager does not perform compliance checks for ESX hosts of version 9.0 and later that are still managed with baselines.

Table 109: vSphere Lifecycle Manager Compliance View

Card	Description
Baselines	The **Baselines** card has three information panels. • Host Information panel For individual ESX hosts, this panel shows information about the ESX version installed on the host. You can also view all updates that are installed on the host. For container objects, this panel shows information about the ESX versions of all hosts in the container object. • Compliance Information panel For individual ESX hosts, this panel shows the overall compliance status of the host against all attached baselines and baseline groups. You can also view compliance information about the baselines and baseline groups attached to the host. For container objects, this panel shows the overall number of compliant and non-compliant hosts. This panel also shows the last time a compliance check was completed. • Remediation Information panel This panel shows the result of the remediation pre-check and indicates whether the selected object is ready for remediation. The panel also contains information about the issues that require user attention or action. This panel also shows the last time a remediation pre-check was completed. The information in the **Baselines** card changes dynamically depending on the inventory object, baselines, and baseline groups that you select.
Attached Baselines and Baseline Groups	Displays the baselines and baseline groups attached to the selected object.
Bottom card	The bottom card appears when you select a baseline or a baseline group from the **Attached Baselines and Baseline Groups** pane. The information in this card depends on the type of inventory object that you select. For individual hosts, the bottom card shows information about all updates in the baseline or baseline group that you select from the **Attached Baselines and Baseline Groups** pane. • If you select a patch baseline or extension baseline, the bottom card shows a list of all updates that the selected baseline contains. • If you select a baseline group, the bottom card shows all baselines included in the group along with their compliance statuses. You can also view all the updates that the baseline group contains. If the baseline group contains an ESX image, information about it is also displayed in the bottom pane. For container objects, the bottom card displays compliance information about the ESX hosts that have the selected baseline or baseline group. When you select a baseline or a baseline group from the **Attached Baselines and Baseline Groups** pane, the bottom card appears and displays all the hosts to which the selected baseline or baseline group is attached along with the individual compliance statuses of those hosts.

Compliance Statuses of Updates

When you work with vSphere Lifecycle Manager baselines, an update stands for all patches and extensions that you can apply with vSphere Lifecycle Manager baselines. The compliance status of the updates in the baselines and baseline groups that you attach to objects in your inventory is calculated after you check the compliance of the target object.

The compliance statuses of the updates in a baseline define the overall compliance status of that baseline. For more information about baseline compliance statuses, see Compliance Statuses of ESX Hosts, Baselines, and Baseline Groups .

Conflict	The update conflicts with either an existing update on the host or another update in the vSphere Lifecycle Manager depot. vSphere Lifecycle Manager reports the type of conflict. A conflict does not indicate any

	problem on the target object. It just means that the current baseline selection is in conflict. You can perform compliance checks, remediation, and staging operations.
Conflicting New Module	The host update is a new module that provides software for the first time, but it is in conflict with either an existing update on the host or another update in the vSphere Lifecycle Manager depot. vSphere Lifecycle Manager reports the type of conflict. A conflict does not indicate any problem on the target object. It just means that the current baseline selection is in conflict. You can perform scan, remediation, and staging operations. In most cases, you must resolve the conflict.
Incompatible Hardware	The hardware of the selected object is incompatible or has insufficient resources to support the update. For example, when you perform a host upgrade scan against a 32-bit host or if a host has insufficient RAM.
Installed	The update is installed on the target object and no further user action is required.
Missing	The update is applicable to the target object, but it is not yet installed. You must perform a remediation on the target object with this update, so that the update becomes compliant.
Missing Package	The metadata for the update is in the depot, but the corresponding binary payload is missing. The reasons can be that the product might not have an update for a given locale; the vSphere Lifecycle Manager depot is corrupt, and vSphere Lifecycle Manager no longer has Internet access to download updates; or you have manually deleted an update package from the vSphere Lifecycle Manager depot.
New Module	The update is a new module. An update with this compliance status cannot be installed when it is part of a host patch baseline. When it is part of a host extension baseline, the new module status indicates that the module is missing on the host and can be provisioned by remediation. The compliance status of the baseline depends on the type of baseline containing the update with the New Module status. If the baseline is a host patch baseline, the overall status of the baseline is compliant. If the baseline is a host extension baseline, the overall status of the baseline is non-compliant.
Not Applicable	The update is not applicable to the target object. A patch might have the not applicable compliance status for one of the following reasons: • There are other patches in the vSphere Lifecycle Manager depot that obsolete this patch. • The update does not apply to the target object.
Not Installable	The update cannot be installed. The compliance check might succeed, but the remediation of the target object cannot be performed.
Obsoleted By Host	This compliance status is mainly applicable to patches. The target object has a newer version of the patch. For example, if a patch has multiple versions, after you apply the latest version to the host, the earlier versions of the patch have the Obsoleted By Host compliance status.
Staged	This compliance status applies to host patches and host extensions. It indicates that the update is copied from the vSphere Lifecycle Manager depot to the host, but it is not yet installed. Staged compliance status might occur only when you check the compliance status of hosts running ESX 8.0 and later.
Unknown	A patch is in the unknown state for a target object until vSphere Lifecycle Manager successfully scans the object. A scan might not succeed if the target object is of an unsupported version, if vSphere Lifecycle Manager lacks metadata, or if the patch metadata is corrupt.

Compliance Statuses of ESX Hosts, Baselines, and Baseline Groups

Compliance statuses are computed after you initiate a compliance check for an inventory object against the attached baselines or baseline groups. The compliance statuses that baselines and baseline groups might have depend on the applicability of the patches and extensions contained in the baseline or baseline group attached to an object. The compliance status of a single host depends on the compliance statuses of all baselines and baseline groups attached to the host.

Compliant

The compliant status indicates that a vSphere object is compliant with all baselines in an attached baseline group or with all patches and extensions in an attached baseline. The compliant state requires no further action. If a baseline contains patches that are not relevant to the target object, the individual updates, and baselines or baseline groups that contain them, are treated as not applicable, and represented as compliant. Compliant are also hosts with attached patch baselines containing extensions or patches with the obsoleted by host status.

The compliant status occurs under the following conditions:

- Target objects are compliant with the baselines and baseline groups when all updates in the baseline or baseline group are either installed on the target object, obsoleted by host, or are not applicable to the target object.
- The updates in a baseline are compliant when they are installed on the target object, or are not applicable to the object.

Non-Compliant

The non-compliant status indicates that one or more baselines in a baseline group, or one or more patches and extensions in a baseline are applicable to the target object, but are not installed (missing) on the target. You must remediate the target object to make it compliant.

When a baseline contains a non-compliant update, the overall status of the baseline is non-compliant. When a baseline group contains a non-compliant baseline, the overall status of the baseline group is non-compliant. The non-compliant status takes precedence over the incompatible, unknown, and compliant states.

Unknown

When you attach a baseline or a baseline group to a vSphere object and you do not initiate a compliance check for the object, the status of the vSphere object against the baseline or baseline group is unknown. This status indicates that a compliance check is required, that the compliance check has failed, or that you initiated a compliance check on an unsupported platform.

When a baseline contains updates in the compliant and unknown states, the overall status of the baseline is unknown. When a baseline group contains unknown baselines and compliant baselines, the overall status of the baseline group is unknown. The unknown compliance status takes precedence over the compliant status.

Incompatible

The incompatible status requires attention and further action. You must determine the reason for incompatibility by probing further. You can remediate the objects that have this status, but the operation might not be successful. In most cases, vSphere Lifecycle Manager provides sufficient details for the incompatibility.

When a baseline contains updates in the incompatible, compliant, and unknown states, the overall status of the baseline is incompatible. When a baseline group contains incompatible, unknown, and compliant baselines, the overall status of the baseline group is incompatible. The incompatible compliance status takes precedence over the compliant and unknown compliance statuses.

View Compliance Information About ESX Hosts

You can check how the ESX hosts in your environment comply with the baselines and baseline groups that you attach to those hosts. You can check and view the compliance status of an individual host or a container object.

- Verify that the host for which you want to view compliance information uses baselines and is not managed with a vSphere Lifecycle Manager image.
- Verify that all ESX hosts in the cluster or the standalone host manager with baselines are not of version 9.0 and later.
- Review the The vSphere Lifecycle Manager Compliance View topic.

Compliance checks provide information about the degree of compliance of an object with the attached baselines and baseline groups.

In the compliance view for an object, you can view information about the compliance of the object with the attached baselines and baselines groups. You can also view the individual compliance statuses of the attached baselines and baseline groups. The compliance view changes dynamically and depends on the object that you want to view compliance information about. For a full description of the compliance information that you can obtain about an object, seeThe vSphere Lifecycle Manager Compliance View.

For information about the different compliance statuses that an object might have, see Compliance Statuses of ESX Hosts, Baselines, and Baseline Groups .

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines** and review the compliance information in the compliance view.

You see complete compliance information about the selected object.

Stage Patches and Extensions to ESX Hosts

Staging is the process during which vSphere Lifecycle Manager downloads patches and extensions on the ESX hosts. During staging, the patches and extensions are not installed on the host. Staging reduces the time that the host spends in maintenance mode during remediation.

- Attach a patch or extension baseline or a baseline group containing patches and extensions to the host.
- Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Patches and Upgrades** > **Stage Patches and Extensions**.

To stage patches or extensions to hosts, first attach a patch or extension baseline or a baseline group containing patches and extensions to the host. Staging patches and extensions does not require that hosts enter maintenance mode.

With the vSphere Client, you can stage a single baseline, multiple baselines, or baseline groups to a single host or a group of hosts in a container object.

Some limitations exist depending on the compliance status of the patches or extensions that you want to stage.

Patches cannot be staged if they are obsoleted by other patches in the baselines or baseline groups for the same stage operation. vSphere Lifecycle Manager stages only the patches that it can install in a subsequent remediation process, based on the current compliance status of the host. If a patch is obsoleted by patches in the same selected patch set, the obsoleted patch is not staged.

If a patch is in conflict with the patches in the vSphere Lifecycle Manager depot and is not in conflict with a host, after a compliance check, vSphere Lifecycle Manager reports this patch as a conflicting one. You can still stage the patch to the host and after the stage operation, vSphere Lifecycle Manager reports this patch as staged.

During the stage operation, vSphere Lifecycle Manager performs pre-scan and post-scan operations and updates the compliance status of the baseline.

For more information about the different compliance statuses that an update might have, see Compliance Statuses of Updates.

After you stage patches or extensions to hosts, you must remediate the hosts against all staged patches or extensions.

After remediation finishes, the host deletes all staged patches or extensions from its cache regardless of whether they were applied during the remediation. The compliance status of the patches or extensions that were staged but not applied to the hosts reverts from Staged to its previous value.

> **Important:**
>
> Staging patches and extensions is supported for hosts that are running ESX 8.x. You can stage patches to PXE booted ESX hosts, but if the host is restarted before remediation, the staged patches are lost and you must stage them again.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.
3. In the **Attached Baselines** pane, select one or more baselines.
4. Click **Stage**.
   The **Stage Patches** dialog box opens.

5. Select hosts on which to stage patches and extensions.
   The number of selected hosts is on the top of the list.
6. To view the patches or extensions that will download to the selected hosts, expand the **Stage** list.
7. Click **Stage**.

The staging operation starts. You can monitor the progress of the task in the **Recent Tasks** pane.

Remediate the host or hosts.

After remediation, all staged patches and extensions, whether installed or not during the remediation, are deleted from the host.

Remediating ESX Hosts Against vSphere Lifecycle Manager Baselines and Baseline Groups

Remediation is the process during which vSphere Lifecycle Manager applies patches and extensions to ESX hosts. Remediation renders the selected ESX hosts compliant with the attached baselines and baseline groups.

General Considerations

- vSphere Lifecycle Manager supports the remediation of ESX hosts against patch and extension baselines.
- You can initiate remediation manually or schedule a regular remediation task to run at a time that is convenient for you.
- You can remediate a single ESX host or multiple hosts in a container object. You can initiate remediation at a folder, a cluster, a data center, and even vCenter level.

  Note:
  If you initiate a remediation task against a baseline for an object that contains clusters that use a vSphere Lifecycle Manager image, remediation is not performed on those clusters.
- By default, the remediation process runs sequentially. That is, vSphere Lifecycle Manager remediates the hosts in a cluster or another container object one by one. However, you can configure vSphere Lifecycle Manager to remediate multiple hosts in parallel.
- If a vCenter instance is connected to other vCenter systems by a common vCenter Single Sign-On domain, you can remediate only the inventory objects managed by the vCenter system where the vSphere Lifecycle Manager instance that you use runs.
- To remediate vSphere objects against baselines or baseline groups, you must have the **Remediate to Apply Patches and Extensions** privilege. For more information about managing users, groups, roles, and permissions, see the *vSphere Security* documentation.
  For a list of all vSphere Lifecycle Manager privileges and their descriptions, see vSphere Lifecycle Manager Privileges For Using Baselines.
- If a vCenter HA failover is initiated during the remediation of a cluster, the remediation task is canceled. After the failover finishes, you must restart the remediation task on the new node. See How Do Cluster Settings Affect Remediation?.

Maintenance Mode

If the update requires it, vSphere Lifecycle Manager puts hosts into maintenance mode during remediation. Virtual machines cannot run when a host is in maintenance mode. To ensure a consistent user experience, vCenter migrates the virtual machines to other hosts within the cluster before the host is put in maintenance mode. vCenter can migrate the virtual machines if the cluster is configured for vMotion and if VMware Distributed Resource Scheduler (DRS) and VMware Enhanced vMotion Compatibility (EVC) are enabled. However, EVC is not a prerequisite for vMotion. For container objects or individual hosts that are not in a cluster, migration with vMotion cannot be performed. You must manually shut down all virtual machines. After remediation, hosts exit maintenance mode. In case of failure during remediation, hosts might be unable to exit maintenance mode.

Parallel Remediation

You can enable vSphere Lifecycle Manager to remediate in parallel the hosts within a cluster that uses baselines. Parallel remediation reduces the time needed for patching the hosts in your environment. You can remediate in parallel only

ESX hosts that are already in maintenance mode. During parallel remediation, hosts do not enter maintenance mode automatically. Similarly, after remediation finishes, the hosts do not exit maintenance mode automatically. To remediate hosts in parallel, you must manually enter and exit maintenance mode. If you enable parallel remediation, vSphere Lifecycle Manager does not remediate the ESX hosts that are not in maintenance mode.

When you configure vSphere Lifecycle Manager to remediate hosts in parallel, you can set the maximum number of hosts to be remediated in a single remediation task. Alternatively, you can let vSphere Lifecycle Manager to remediate all hosts in maintenance mode in parallel.

When you remediate hosts in parallel, if the remediation of a single host fails, the remediation task for the entire cluster does not stop and the rest of the hosts are remediated successfully. After remediation finishes, vSphere Lifecycle Manager reports an error for the respective host.

Parallel remediation is deactivated by default, but you can enable it during remediation or in the vSphere Lifecycle Manager general remediation settings.

You cannot remediate in parallel hosts in a vSAN cluster.

Remediation Pre-Check

Before you remediate an object, you can perform a remediation pre-check on the object. During that check, vSphere Lifecycle Manager identifies possible issues that might prevent successful remediation and takes or suggests actions to fix the issues.

For more information about the possible issues that might prevent successful remediation, see Remediation Pre-Check Report.

Remediation of PXE Booted ESX Hosts

vSphere Lifecycle Manager lets you remediate PXE booted ESX hosts. vSphere Lifecycle Manager does not apply the patches that require a reboot to PXE booted ESX hosts.

If there is any additional software installed on the PXE booted ESX host, the software might be lost if the host restarts. Update your image profile with the additional software so that it will be present after the reboot.

To patch PXE booted ESX hosts, you must enable the respective setting in the **Edit Settings for Host Remediation** dialog box, which you open from the **Settings** tab in the vSphere Lifecycle Manager home view.

Understanding the Remediation Operation

For ESX hosts, updates are all-inclusive. The most recent update contains the patches from all previous releases.

The ESX image on the host maintains two copies. The first copy is in the active boot and the second one is in the standby boot. When you patch an ESX host, vSphere Lifecycle Manager creates an image based on the content of the active boot and the content of the patch. The new ESX image is then located in the standby boot and vSphere Lifecycle Manager designates the active boot as the standby boot and reboots the host. When the ESX host reboots, the active boot contains the patched image and the standby boot contains the previous version of the ESX host image.

When you update an ESX host, vSphere Lifecycle Manager replaces the backup image of the host with the new image and replaces the active boot and the standby boot. During the update, the layout of the disk that hosts the boots changes. The total disk space for an ESX host remains 1GB, but the disk partition layout within that 1GB disk space changes to accommodate the new size of the boots where the ESX image is stored.

For rollback purposes, the term update refers to all ESX patches and extensions. Each time you update an ESX host, a copy of the previous ESX build is saved on your host.

If an update fails and the ESX host cannot boot from the new build, the host reverts to booting from the original boot build. ESX permits only one level of rollback. Only one previous build can be saved at a time. In effect, each ESX host stores up to two builds, one boot build and one standby build.

From the vSphere Lifecycle Manager settings, you can configure the host remediation process to skip a host reboot during host patch operations. This configuration setting is called Quick Boot. For more information about configuring the vSphere Lifecycle Manager remediation settings, see vSphere Lifecycle Manager Remediation Settings.

Types of Host Remediation

Host remediation runs in different ways depending on the types of baselines that you attach to an object and whether the remediated host is in a cluster or not.

Host Patch Remediation

Patching is the process of remediating ESX hosts against patch baselines.

vSphere Lifecycle Manager handles host patches in the following ways:

- If a patch in a patch baseline requires the installation of another patch, vSphere Lifecycle Manager detects the prerequisite in the depot and installs it together with the selected patch.
- If a patch is in a conflict with other patches that are installed on the host, the conflicting patch might not be staged or installed. However, if another patch in the baseline resolves the conflicts, the conflicting patch is installed. For example, consider a baseline that contains patch A and patch C, and patch A conflicts with patch B, which is already installed on the host. If patch C obsoletes patch B, and patch C is not in a conflict with patch A, the remediation process installs patches A and C.
- If a patch is in a conflict with the patches in the vSphere Lifecycle Manager depot and is not in a conflict with the host, after a compliance check, vSphere Lifecycle Manager reports this patch as a conflicting one. You can stage and apply the patch to the host.
- When multiple versions of the same patch are selected, vSphere Lifecycle Manager installs the latest version and skips installing the earlier versions.

During patch remediation, vSphere Lifecycle Manager automatically installs the prerequisites of the patches.

With vSphere Lifecycle Manager9.1, you can remediate hosts of version ESX 8.x managed with baselines against patches from offline bundles, which you import to the vSphere Lifecycle Manager depot manually. Hosts running ESX 9.0 and later cannot be managed or remediated using baselines. ESX 9.x hosts must use vSphere Lifecycle Manager images.

> **Important:**
>
> With vSphere Lifecycle Manager 9.1, you cannot remediate hosts of version ESX 9.0 and later that use baselines against patches with imported ISOs.

You can stage patches before remediation to reduce host downtime.

Host Extension Remediation

During extension remediation, vSphere Lifecycle Manager does not automatically install the prerequisites of the extension. Missing extension prerequisites cause some remediation operations to fail. If the missing prerequisite is a patch, you can add it to a patch baseline. If the missing prerequisite is an extension, you can add it to the same or another extension baseline. You can then remediate the host against the baseline or baselines that contain the prerequisite and the original extension baseline.

Remediating Hosts in a Cluster Managed with Baselines

For ESX hosts in a cluster, the remediation process is sequential by default.

When you remediate a cluster of hosts sequentially and one of the hosts fails to enter maintenance mode, vSphere Lifecycle Manager reports an error and the remediation process stops and fails. The hosts in the cluster that are remediated stay at the updated level. The ones that are not remediated after one host fails remain unupdated.

If you initiate remediation at a data center level, the remediation processes for the clusters run in parallel. Clusters that you manage with a vSphere Lifecycle Manager image are not remediated against the attached baselines or baseline groups. If the remediation process fails for one of the clusters within a data center, the remaining clusters are still remediated.

Before you start remediation, you can generate a report that shows which cluster, host, or virtual machine has the cluster features enabled. For more information, see Remediation Pre-Check Report.

Remediation and Cluster Settings

If a host in a DRS-enabled cluster runs a virtual machine on which vCenter is installed, DRS first attempts to migrate the virtual machine running vCenter to another host, so that the remediation succeeds. If the virtual machine cannot be migrated to another host, the remediation fails for the host, but the remediation process for the cluster does not stop. vSphere Lifecycle Manager proceeds to remediate the next host in the cluster.

In some cases, such as two-node clusters, remediation of hosts in a cluster managed with baselines requires that you temporarily deactivate cluster features such as VMware DPM and HA admission control. Also, you must turn off Fault Tolerance if it is enabled on any of the virtual machines on a host, and disconnect the removable devices connected to the virtual machines on a host, so that they can be migrated with vMotion. For more information about configuring the vSphere Lifecycle Manager remediation settings, see vSphere Lifecycle Manager Remediation Settings.

If a vCenter HA failover is initiated during the remediation of a cluster, the remediation task is canceled. After the failover finishes, you must restart the remediation task on the new node.

When you perform remediation on a cluster managed with baselines that consists of not more than two hosts, deactivating HA admission control might not be enough to ensure successful remediation. You might need to temporarily deactivate vSphere High Availability (HA) for the cluster. If you keep HA enabled, the remediation attempts on hosts in the cluster fail, because HA cannot provide recommendation to vSphere Lifecycle Manager to place any of the hosts into maintenance mode. The reason is that if one of the two hosts is placed into maintenance mode there is no failover host left available in the cluster. To ensure successful remediation on a two-node cluster, you must temporarily deactivate HA for the cluster or place the hosts in maintenance mode manually and then remediate the two hosts in the cluster. Remember to re-enable HA as soon as both hosts in the cluster have been remediated.

vSAN Clusters

vSphere Lifecycle Manager remediates hosts that are part of a vSAN cluster sequentially. The reason is that by design only one host from a vSAN cluster can be in a maintenance mode at any time. For more information about using vSphere Lifecycle Manager with vSAN clusters, see vSAN Clusters and vSphere Lifecycle Manager.

Generate a Pre-Remediation Check Report

When you generate a pre-remediation check report, vSphere Lifecycle Manager generates a list with actions that you must perform to ensure successful remediation of the hosts in your cluster.

The remediation pre-check report contains information about issues at the cluster, host, and VM level that might prevent the completion of remediation.

For information about the possible issues that might prevent successful remediation, see Remediation Pre-Check Report.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Host** > **Baselines**.
3. In the **Baselines** pane, click **Pre-Check Remediation**.
   The **Remediation Pre-check** dialog box opens.
4. Review the results from the pre-check and click **Done**.
   In the bottom pane of the **Remediation Pre-check** dialog box, you see a list of issues at the host and virtual machine level.

The **Remediation Pre-check** dialog box lists the issues with cluster, hosts, and virtual machines that might prevent successful remediation of the selected object.

In the upper pane of the **Remediation Pre-check** dialog box, you see a list of issues at a cluster level.

In the bottom pane of the **Remediation Pre-check** dialog box, you see a list of issues at the host and virtual machine level.

Fix all issues that vSphere Lifecycle Manager identifies during the pre-remediation check and remediate the selected object.

Remediation Pre-Check Report

The remediation pre-check report shows the results of a check that is performed on a cluster or a host before remediation. During that check, vSphere Lifecycle Manager identifies issues that might prevent successful remediation. Depending on the type of issue, vSphere Lifecycle Manager suggests actions that you must take to fix the issue or resolves the issue automatically.

You can generate a pre-check remediation report in the vSphere Lifecycle Manager compliance view for an object.

Table 110: Cluster Issues

Current Configuration/Issue	Recommended Action	Details
DRS is deactivated on the cluster.	Enable DRS on the cluster.	DRS enables vCenter to place and migrate virtual machines automatically on hosts to attain the best use of cluster resources.
vSAN health check fails during the pre-check.	Navigate to the vSAN Health page and address any health issues before proceeding with remediation.	The vSAN health check performs a series of tests on the hosts in the vSAN cluster. The vSAN health check must succeed to ensure the hosts are successfully remediated. If you start a remediation task in a vSAN cluster that failed the vSAN health check during the remediation pre-check, the hosts enter maintenance mode, get upgraded, but might fail to exit maintenance mode. The remediation eventually fails.
Insufficient licenses for one or multiple ESX hosts in the cluster.	Ensure that you have multiple licenses for the ESX hosts that have more than 32 cores per CPU.	One CPU license covers up to 32 physical cores. If a CPU has more than 32 cores, you must assign additional CPU licenses to the respective ESX host.
DPM is enabled on the cluster.	None. vSphere Lifecycle Manager deactivates DPM automatically.	If a host has no running virtual machines, DPM might put the host in standby mode before or during remediation and vSphere Lifecycle Manager cannot remediate them.
HA admission control is enabled on the cluster.	None. vSphere Lifecycle Manager deactivates HA admission control automatically.	HA admission control prevents the migration of virtual machines with vSphere vMotion and the hosts cannot enter maintenance mode.
EVC is deactivated on the cluster.	None. vSphere Lifecycle Manager enables EVC automatically, although no notification or message appears in the vSphere Client.	If EVC is deactivated for a cluster that is composed of hosts with heterogeneous CPU modes, the live migration of certain virtual machines with vSphere vMotion is blocked. The result is downtime of the machines on the hosts that you remediate with vSphere Lifecycle Manager.

Table 111: Host Issues

Current Configuration/Issue	Recommended Action	Details
A CD/DVD drive is attached to a virtual machine on the ESX host.	Disconnect the CD/DVD drive from the Guest OS.	Any CD/DVD drives or removable devices that are connected to the virtual machines on a host, including the virtual CD device that is attached when a VMware Tools installation is initiated but not completed, might prevent the host from entering maintenance mode. When you start a remediation operation, the hosts with virtual machines to which removable devices are connected are not remediated.
A floppy drive is attached to a virtual machine on the ESX host.	Disconnect the floppy drive from the Guest OS	Any floppy drives or removable devices that are connected to the virtual machines on a host might prevent the host from entering maintenance mode. When you start a remediation operation, the hosts with virtual machines to which removable devices are connected are not remediated.
Fault Tolerance (FT) is enabled for a virtual machine on the ESX host.	Migrate the FT- backed VM to another host	If FT is enabled for any of the virtual machines on a host, vSphere Lifecycle Manager cannot remediate that host.
A powered-on virtual machine is configured to use Virtual Flash Read Cache.	Deactivate Virtual Flash Read Cache before proceeding with the upgrade.	Virtual Flash Read Cache is not supported. During an upgrade operation, vSphere Lifecycle Manager removes Virtual Flash Read Cache for all virtual machines on the host. Before remediation, consult KB 310737.
vCenter is installed on a virtual machine on the ESX host and DRS is deactivated on the cluster.	Enable DRS on the cluster and ensure that virtual machines can be migrated with vSphere vMotion.	One of the virtual machines in the cluster runs the vCenter instance that you currently use. If you enable DRS on the cluster, vSphere vMotion can migrate the virtual machine where vCenter runs to ensure that the remediation of the hosts is successful.

Remediate ESX Hosts Against a Single Baseline or Multiple Baselines

Remediation makes the remediated ESX host compliant with the attached baselines and baseline groups. You can remediate a host against a single or multiple baselines, or against a baseline group. Baseline groups might contain multiple patch and extension baselines. You can remediate ESX hosts against a single attached update baseline at a time.

- Required privileges: **VMware vSphere Lifecycle Manager** > **Manage Patches and Upgrades** > **Remediate to Apply Patches, Extensions, and Upgrades**.
- Attach a patch or extension baseline or a baseline group containing patches and extensions to the host.
- Resolve any issues that occur during Remediation Pre-check.
- To enable Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.

You can update all hosts in your vSphere inventory by using a single patch or extension baseline. You can remediate a single ESX host or a group of ESX hosts in a container object, such as a folder, a cluster, or a data center. You can also initiate remediation at a vCenter level.

Because the unified Broadcom online depot hosts certified partner content in addition to VMware content, and vSphere Lifecycle Manager retrieves this content through VCF software depot, a broader set of OEM bulletins are available in the vSphere Lifecycle Manager depot. As a result, a broader set of OEM bulletins are included in the vSphere Lifecycle Manager predefined bulletins.

During remediation, always inspect the contents of those baselines to exclude the bulletins that you do not need in the baseline.

For the bulletins that you do need, consult the corresponding knowledge base articles for information about deployment specifics and dependencies. Verify that dependent bulletins are also included in the baselines that you use for remediation.

1. In vSphere Client, navigate to the vSphere Lifecycle Manager compliance view for an individual host or a container object.
   a) Navigate to a host, cluster, or a container object.
   b) Click the **Updates** tab.
2. Select **Hosts** > **Baselines**.
3. In the **Attached Baselines and Baseline Groups** pane, select the baselines and baseline groups to use for remediation.

   You can select a single baseline or baseline group. You can also select multiple baselines and baseline groups. Your selection must contain no more than one update baseline.
4. Click **Remediate**.

   The **Remediate** dialog box opens.
5. Expand the list of pre-check issues and review the actions that vSphere Lifecycle Manager must perform to ensure successful remediation.
6. Optional: To generate a full remediation pre-check report, click **Show Full Remediation Pre-Check Report**.

   If you select this option, the **Remediate** dialog box closes and vSphere Lifecycle Manager does not proceed with the remediation process. Instead, the **Remediation Pre-Check** dialog box opens. After you review the results from the remediation pre-check, you must initiate remediation again.
7. Expand the list of hosts to be remediated and deselect any host that you do not want to remediate.

   The list contains all the hosts to which the selected baselines and baseline groups are attached. Even if you navigated to a single host before initiating remediation, the list might still display multiple hosts to be remediated. All hosts in the list are selected by default. Deselecting hosts from the list changes the overall number of hosts to be remediated.
8. Optional: To view information about the updates that will be installed during the remediation, expand the list of updates.
9. Optional: To schedule the remediation task for a later time, expand **Scheduling Options** and configure a scheduled remediation task.

   By default, the remediation task starts immediately after closing the **Remediate** dialog box.
10. Expand **Remediation settings** and review and edit the remediation settings.
    - To turn on or turn off Quick Boot, select or deselect the respective check box in the **Remediation settings** table.
    - To allow or disallow health checks after remediation, select or deselect the respective check box in the **Remediation settings** table.
    - To ignore warnings about unsupported hardware devices, select the respective check box in the **Remediation settings** table.
    - To configure parallel remediation for the selected hosts, expand **Parallel remediation**, select the respective check box and configure the maximum number of concurrent remediations.

      Note:
      vSphere Lifecycle Manager remediates in parallel only the ESX hosts that are in maintenance mode. Hosts that are not in maintenance mode are not remediated. If you do not set the maximum number of concurrent remediations, vSphere Lifecycle Manager remediates all hosts that are in maintenance mode.

      If the hosts have NSX virtual distributed switches that are ready to be migrated to vSphere Distributed Switches, you must set the maximum number of parallel remediations to no more than 4. In cases when host switch migration is needed, if more than 4 hosts are remediated in parallel, the remediation might fail, because the host switch migration takes more time than the time vSphere Lifecycle Manager needs to complete the parallel remediation.

- To change any other of the remediation settings, click the **Close Dialog And Go To Settings** link above the table.

  If you select this option, the **Remediate** dialog box closes and vSphere Lifecycle Manager does not proceed with the remediation process. Instead, you are redirected to the **Baselines Remediation Settings** pane on the **Settings** tab of the vSphere Lifecycle Manager home view. To change any of the remediation settings, click the **Edit** button. Remediation does not resume automatically. After you make the desired changes, you must initiate remediation again.

11. Click **Remediate**.

Depending on the remediation schedule you configure, the remediation task starts immediately or runs later.

Upgrading Virtual Machines with vSphere Lifecycle Manager

You can use vSphere Lifecycle Manager to upgrade virtual machine hardware and the VMware Tools version of a virtual machine.

Whether you perform an upgrade of the virtual machine hardware version or the VMware Tools version, the upgrade is a multi-stage process.

1. You check the status of individual virtual machines or a container object.
   vSphere Lifecycle Manager checks the status of a virtual machine against the latest virtual machine hardware version supported by the host on which the virtual machine runs. Similarly, vSphere Lifecycle Manager checks the status of the virtual machine against the latest VMware Tools version supported by the host on which the virtual machine runs.
   For more information about checking virtual machine status, see Checking the Status of Virtual Machines.
2. You review the status of the scanned virtual machines.
3. You upgrade the virtual machine to match the host where it resides.
   With vSphere Lifecycle Manager, you can upgrade the virtual machine hardware version and the VMware Tools version that a virtual machine has. You can use vSphere Lifecycle Manager to upgrade the virtual machine hardware version to the latest hardware version, and to the latest VMware Tools version on the hosts.
   For more information about upgrading virtual machines, see Upgrading Virtual Machines.

Configure Virtual Machine Rollback Settings

By default, vSphere Lifecycle Manager takes snapshots of virtual machines before upgrading them. If the upgrade fails, you can use the snapshot to return a virtual machine to its state before the upgrade.

Required privileges: **VMware vSphere Lifecycle Manager** > **Configure**

You can configure vSphere Lifecycle Manager to keep snapshots for an indefinite or fixed period of time. Use the following guidelines when managing snapshots.

- Keeping snapshots indefinitely consumes a large amount of disk space and may degrade virtual machine performance.
- Keeping no snapshots saves space and does not impact virtual machine performance, but, in the event of a disaster, may impact mean time to recovery (MTTR). However, keeping no snapshots limits the availability of a rollback.
- Keeping a fixed number of snapshots for a fixed period of time, a middle-ground solution, allows you to balance space usage and performance with a short-term rollback option.

vSphere Lifecycle Manager does not take snapshots of fault tolerant virtual machines. If you decide to take snapshots of such virtual machines, the upgrade might fail.

If you configure vSphere Lifecycle Manager to automatically upgrade VMware Tools on power cycle for selected virtual machines, vSphere Lifecycle Manager does not take snapshots of the virtual machines before upgrading them and you cannot roll back.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.
      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Cluster lifecycle** > **VMs**.
3. Click the **Edit** button.
   The **Edit Default Settings for VM Rollback** dialog box opens.
4. Select the **Take snapshot of VMs** check box.
5. Select the time period for keeping the snapshots.
   - Do not delete snapshots
   - Keep snapshots for a configurable, fixed period of time
6. Click **Save** to save your changes and close the **Edit Default Settings for VM Rollback** dialog box.

These settings become the default rollback option settings for virtual machines. You can specify different settings when you configure individual remediation tasks.

Checking the Status of Virtual Machines

You use vSphere Lifecycle Manager to check the status of virtual machines before you upgrade them. The status check shows if the virtual machine is up-to-date or can be upgraded.

With vSphere Lifecycle Manager, you can check the status of a single virtual machine or a group of virtual machines in a parent container object.

Supported groups of virtual machines or ESX hosts include virtual infrastructure container objects such as folders, vApps, clusters, and data centers.

vSphere Lifecycle Manager checks the status of virtual machines in two aspects.

- You can use vSphere Lifecycle Manager to check the status of the virtual machines in respect with the VMware Tools version that they have installed.
  The status check is performed against the latest VMware Tools version that the parent host supports.
- vSphere Lifecycle Manager checks the status of the virtual machines in respect with their VM hardware compatibility.
  vSphere Lifecycle Manager compares the hardware compatibility of the virtual machines with the default VM hardware compatibility configured for the host.

Check the Status of an Individual Virtual Machine

You check the status of virtual machines to see whether the VMware Tools version they have is up to date and whether their hardware compatibility matches the default VM hardware compatibility for the host.

1. In the vSphere Client, navigate to a virtual machine.
2. On the **Updates** tab, click **Check Status**.
   The **Scan entity** task appears in the **Recent Tasks** pane. After the task finishes, status information appears in the **VMware Tools** and **VM Hardware Compatibility** panels.

The virtual machines are scanned for VMware Tools and VM hardware compliance.

Check the Status of the Virtual Machines in a Container Object

You check the status of virtual machines to see whether the VMware Tools version they have is up-to-date and whether their hardware compatibility matches the default VM hardware compatibility for the host where they reside.

When you perform a status check for a container object, vSphere Lifecycle Manager checks the VMware Tools and VM Hardware Compatibility statuses for all child virtual machines. The larger the virtual infrastructure and the higher up in the object hierarchy you initiate the status check, the longer the task takes.

1. In the vSphere Client, navigate to a virtual machine container object, such as a virtual machine folder, host, cluster, and so on.
2. Click the **Updates** tab.
3. Select your task.

Option	Action
Check the VMware Tools status of the virtual machines in the container object.	1. Select **Hosts** > **VMware Tools** > . 2. Click **Check Status**. The information about the VMware Tools status appears in the **Tools Status** column in the table that lists all virtual machines in the selected container object. If the container object is a data center or a vCenter instance, you must first specify the cluster that you want to see results for.
Check the VM Hardware compatibility status of the virtual machines in the container object.	1. Select **Hosts** > **VM Hardware**. 2. Click **Check Status**. The information about the VM Hardware Compatibility status appears in the **Status** column in the table that lists all virtual machines in the selected container object. For each virtual machine in the object, you can also see the VM hardware compatibility and the host compatibility. If the container object is a data center or a vCenter instance, you must first specify the cluster that you want to see results for.

The VMware Tools Status

Check the VMware Tools status for information whether the current version of VMware Tools is installed, supported, or whether upgrades are available.

Table 112: VMware Tools Status

VMware Tools Status	Description
Up to Date	VMware Tools is installed, supported, and the version is compliant.
	VMware Tools is installed, supported, and the version is newer than the version available on the ESX host.
Upgrade Available	VMware Tools is installed, but the version is old.
	VMware Tools is installed and supported, but a newer version is available on the ESX host.
Version Unsupported	VMware Tools is installed, but the version is old.
	VMware Tools is installed, but the version has a known issue and must be immediately upgraded.
	VMware Tools is installed, but the version is too new to work correctly with this virtual machine.
Not Installed	VMware Tools is not installed on this virtual machine.
Guest Managed	vSphere does not manage VMware Tools.
Unknown	The status of the virtual machine is not checked.

Upgrading Virtual Machines

With vSphere Lifecycle Manager, you can upgrade the VMware Tools version and the hardware version of a virtual machine. You can also upgrade multiple virtual machines simultaneously if the virtual machines are in a container object, such as a folder or vApp. You can also upgrade simultaneously all virtual machines that run on a host, in a cluster, or in a data center.

vSphere Lifecycle Manager supports upgrading powered on, suspended, and powered off virtual machines.

During the upgrade of VMware Tools, the virtual machines must be powered on. If a virtual machine is in the powered off or suspended state before remediation, vSphere Lifecycle Manager powers it on. After the upgrade completes, vSphere Lifecycle Manager restarts the machine and restores the original power state of the virtual machine.

During the virtual hardware upgrade, the virtual machines must be powered off. If a virtual machine is powered on, vSphere Lifecycle Manager powers the machine off, upgrades the virtual hardware, and then powers the virtual machine on.

You can also upgrade VMware Tools and the hardware version of a virtual machine template. A template is a copy of a virtual machine that you can use to create and provision new virtual machines.

You can set up automatic upgrades of VMware Tools on power cycle. For more information, see Automatically Upgrade VMware Tools on Reboot.

You can configure vSphere Lifecycle Manager to take snapshots of virtual machines and to keep the snapshots indefinitely or for a specific period of time. By using snapshots, you can roll back a virtual machine to its previous state if upgrading the virtual machine with vSphere Lifecycle Manager fails. After the upgrade finishes, you can delete the snapshots if you do not need them. For more information about configuring virtual machine rollback settings, see Configure Virtual Machine Rollback Settings.

You can upgrade virtual machines immediately or schedule an upgrade operation to run at a convenient time.

If a host is connected to vCenter by using an IPv6 address, you cannot scan and remediate virtual machines that run on the host.

Upgrade the VM Hardware Compatibility of Virtual Machines

You can upgrade the hardware of virtual machines to the latest hardware version that the host supports. You can upgrade immediately or schedule an upgrade at a time that is convenient for you.

With vSphere Lifecycle Manager, you can upgrade the hardware compatibility version of a single virtual machine or multiple virtual machines simultaneously. Supported container objects for virtual machines in the vSphere inventory are folders, vApps, data centers.

1. In the vSphere Client, navigate to a single virtual machine or a container object.

   You can also initiate upgrade at the level of any inventory object where virtual machines run. For example, you can start the upgrade operation at a host or cluster level.

2. Open the **Upgrade VM Hardware to Match Host** dialog box.

Inventory Object	Steps
Virtual Machine	1. Click the **Updates** tab. 2. In the **VM Hardware Compatibility** panel, click **Upgrade to Match Host**.
Container Object, Host, Cluster, Data Center, or vCenter Instance	1. Click the **Updates** tab. 2. Select **Hosts** > **VM Hardware**. 3. If the selected inventory object is a data center or a vCenter instance, select a cluster from the list. A list of all virtual machines in the cluster appears in the bottom pane. 4. Select the virtual machines to upgrade. 5. Click **Upgrade to Match Host**.

   A list of the virtual machines selected for upgrading is visible in the **Upgrade VM Hardware to Match Host** dialog box.

3. Optional: To change the selection of the virtual machines to upgrade, select or deselect virtual machines from the list.
4. Optional: To schedule the upgrade for a specific date and time, expand **Scheduling Options** and configure the scheduled task.
   a) Enter a name and, optionally, a description for the scheduled upgrade task.
   b) Use the **Powered On VMs**, **Powered Off VMs**, and **Suspended VMs** drop-down menus to configure the upgrade to run immediately or at a specific date and time.
5. Optional: To configure the use of snapshots, expand **Rollback Options** and change the default settings.
   a) To allow or disallow taking of snapshots of virtual machines before upgrading them, select or deselect the **Take snapshot of VMs** check box.

      The option to take snapshots is selected by default.
   b) Select a period for keeping the snapshots.
      - Keep the snapshots indefinitely.
      - Keep the snapshots for a fixed period.
   c) Enter a snapshot name and, optionally, a description for the snapshot.
   d) Include the virtual machine memory in the snapshot by selecting the respective check box.
6. Review your selections and click the **Upgrade to Match Host** button.

The hardware versions of the selected virtual machines are upgraded and the virtual machine status changes to Up to Date.

Upgrade the VMware Tools Version of Virtual Machines

You can upgrade the VMware Tools version of virtual machines to the latest version supported by the ESX host or vSphere environment. You can perform the upgrade immediately or schedule an upgrade task at a time convenient for your maintenance window.

- Before you run the **Upgrade to Match Host** operation, ensure that the host images or the VMware Tools repository location are provisioned with VMware Tools packages. Starting with ESX 8.0 Update 3 and vSphere 9.x, ESX host images might not include VMware Tools ISO packages that are bundled in the base ESX image.

Using vSphere Lifecycle Manager, you can upgrade the VMware Tools version of a single virtual machine or multiple virtual machines simultaneously. Supported container objects for virtual machines in the vSphere inventory are folders, vApps, clusters, data centers, and vCenter instances.

Important:

- Upgrading VMware Tools might trigger a guest operating system reboot or require a manual guest restart for updated device drivers, such as VMXNET3 network adapters or PVSCSI storage controllers, to take effect. Plan upgrades during approved maintenance windows.
- Modern Linux operating systems, such as RHEL, Ubuntu, SUSE, Debian, Rocky Linux, and Photon OS, manage `open-vm-tools` through the guest package manager (`yum`, `apt`, `zypper`). Do not use vSphere Lifecycle Manager to upgrade VMware Tools on Linux virtual machines running `open-vm-tools`. Instead, update `open-vm-tools` by using native Linux package management tools.

1. In the vSphere Client, navigate to a single virtual machine or a container object.

   You can initiate the upgrade at the level of any inventory object where virtual machines run. For example, you can start the upgrade operation at a host or cluster level.
2. Open the **Upgrade VMware Tools to Match Host** dialog box.

Inventory Object	Steps
Virtual Machine	1. Click the **Updates** tab. 2. In the **VMware Tools** panel, click **Upgrade to Match Host**.
Container Object **(Host, Cluster, Folder, Data Center, or vCenter Instance)**	1. Click the **Updates** tab. 2. Select **Hosts** > **VMware Tools**.

Inventory Object	Steps
	3. If the selected inventory object is a data center or a vCenter instance, select a cluster from the list. A list of all virtual machines in the cluster appears in the bottom pane. 4. In the **VMs in Cluster** pane, select the virtual machines to upgrade. 5. Click **Upgrade to Match Host**.

A list of the virtual machines selected for upgrading is visible in the **Upgrade VMware Tools to Match Host** dialog box.

3. Optional: In the **Upgrade VMware Tools to Match Host** dialog box, review the list of selected virtual machines. Adjust selection by selecting or deselecting individual virtual machines from the list.
4. Optional: To schedule the upgrade for a specific date and time, expand **Scheduling Options** and configure the scheduled task.
   a) Enter a name and, optionally, a description for the scheduled upgrade task.
   b) Use the **Powered On VMs**, **Powered Off VMs**, and **Suspended VMs** drop-down menus to configure the upgrade to run immediately or at a specific date and time.
5. Optional: To configure the use of snapshots, expand **Rollback Options** and change the default settings.
   a) To allow or disallow taking of snapshots of virtual machines before upgrading them, select or deselect the **Take snapshot of VMs** check box.
      The option to take snapshots is selected by default.
   b) Select a period for keeping the snapshots.
      - Keep the snapshots indefinitely.
      - Keep the snapshots for a fixed period.
   c) Enter a snapshot name and, optionally, a description for the snapshot.
   d) Include the virtual machine memory in the snapshot by selecting the respective check box.
6. Review your selections and click the **Upgrade to Match Host** button.

vSphere Lifecycle Manager upgrades the VMware Tools version on the selected virtual machines. Upon completion, and guest reboot if necessary, the VMware Tools status for each VM updates to *Up to Date* in the vSphere Client.

Automatically Upgrade VMware Tools on Reboot

You can automate the process of upgrading VMware Tools for the virtual machines in your inventory.

Verify that you have the**Manage Patches and Upgrades** > **Remediate to Apply Patches, Extensions, and Upgrades** privilege.

You can set up vSphere Lifecycle Manager to check the VMware Tools version of a virtual machine when the virtual machine is rebooted. If necessary, vSphere Lifecycle Manager upgrades VMware Tools to the latest version supported by the host on which the virtual machine runs.

Note:

When you perform a VMware Tools upgrade on power cycle, vSphere Lifecycle Manager does not take a snapshot of the virtual machine and you cannot roll back to the previous version of the virtual machine.

1. In the vSphere Client, navigate to a single virtual machine or an inventory object that contains virtual machines.
2. Configure vSphere Lifecycle Manager to automatically upgrade VMware Tools on reboot.

Inventory Object	Steps
Virtual machine	1. Click the **Updates** tab. 2. In the **VMware Tools** panel, click **Turn On**.

Inventory Object	Steps
Container Object, Host, cluster, data center, or vCenter instance	1. Click the **Updates** tab. 2. Select **Hosts** > **VMware Tools**. 3. If the selected inventory object is a data center or a vCenter instance, select a cluster from the list. A list of all virtual machines in the cluster appears in the bottom pane. 4. In the **VMs in cluster** pane, select the virtual machines for which you want to enable the automatic upgrade of VMware Tools. 5. Click **Set Auto Update** and select **On**. The new status is visible in the **Auto Update** column.

The next time you power on or restart a virtual machine, vSphere Lifecycle Manager checks the version of VMware Tools installed on the virtual machines and performs an upgrade, if necessary.

vSphere Lifecycle Manager Images and Other VMware Components and Solutions

You can use a vSphere Lifecycle Manager image to manage a cluster if the cluster only contains solutions integrated to work with vSphere Lifecycle Manager. If an unitegrated solution is enabled on a cluster, you cannot use vSphere Lifecycle Manager images to manage that cluster, but you can still use baselines and baseline groups.

A solution is a VMware component that integrates with vCenter and adds some new functionality to the ESX hosts in the inventory.

When you enable a solution for a cluster that uses a vSphere Lifecycle Manager image, the solution automatically uploads an offline bundle with components into the vSphere Lifecycle Manager depot and adds its component to all hosts in the cluster. You cannot control the lifecycle of solution components. For example, if you export the image, solution components are not part of the exported image.

Integrated Solutions

You can manage a cluster with an image if the cluster has any of the following solutions enabled.

- vSphere High Availability
- vSAN
  For more information about the integration between vSAN and vSphere Lifecycle Manager, see vSAN Clusters and vSphere Lifecycle Manager and the *vSAN* chapter in the *VMware Cloud Foundation 9.1.x* documentation.
- vSAN File Services
- VMware vSphere Supervisor
  For detailed information about the integration between vSphere Supervisor and vSphere Lifecycle Manager, see the *vSphere Supervisor Platform*
  chapter in the *VMware Cloud Foundation 9.1.x* documentation.
- NSX
  For more information about the integration between NSX and vSphere Lifecycle Manager, see the *NSX* chapter in the *VMware Cloud Foundation 9.1.x* documentation..
- VMware Cloud Foundation
  vSphere Lifecycle Manager is available as an option in the VMware Cloud Foundation Workload Domain. For more information, see the *Lifecycle Management* chapter in the *VMware Cloud Foundation 9.1.x* documentation..
- Protection and Recovery
  For more information about the integration between vSphere Lifecycle Manager and Protection and recovery, see the *Protection and Recovery* documentation.

You can also use baselines for clusters that have those solutions enabled.

vSAN Clusters and vSphere Lifecycle Manager

You can manage a vSAN cluster by using vSphere Lifecycle Manager image for that cluster. Working with vSAN clusters has its specifics regardless of whether you manage the cluster with an image or with baselines.

Managing a vSAN Cluster by Using Recommendation Baseline Groups

You can update and upgrade the hosts in a vSAN cluster by using automatically generated system-managed baselines groups. Those system-managed baseline groups are called recommendation baseline groups. Recommendation baseline groups do not contain firmware and driver updates. Recommendation baselines contain only patch or upgrade baselines.

If you want to switch to using images for a vSAN cluster that contains ESX hosts of versions earlier than 8.0, you must first use an upgrade baseline to upgrade the hosts. Then, you can switch to using a vSphere Lifecycle Manager image for the cluster.

For more information about recommendation baselines, see About Recommendation Baseline Groups.

For more information about using baselines to manage hosts and clusters, see Using vSphere Lifecycle Manager Baselines and Baseline Groups .

Managing a vSAN Cluster with an Image

The image that you use for a cluster defines the full software stack to run on the hosts in that cluster: ESX version, vendor customization, drivers, and firmware. When you manage a vSAN cluster with an image, you can take advantage of the functionalities that vSphere Lifecycle Manager images provide.

- You can update the firmware on all hosts in the vSAN cluster.
  You perform firmware update by setting up an image that contains a firmware add-on and remediating the vSAN cluster against that image. For more information about performing firmware updates by using vSphere Lifecycle Manager images, see Firmware Updates with vSphere Lifecycle Manager.
- You can run a hardware compatibility check for the cluster.
  The hardware compatibility check task verifies that the image for the cluster can be successfully applied to all hosts and that it is compliant with the vSAN Hardware Compatibility List (HCL). For more information about hardware compatibility checks, see vSphere Lifecycle Manager Hardware Compatibility Checks for Clusters and Hosts.
- You can check firmware compliance with the image.
  When you perform a compliance check against the image for a cluster, firmware compliance is also checked. As a result, you can easily notice if a driver or firmware in your cluster becomes non-compliant. For more information about checking the compliance of a cluster against an image, see Check the Compliance Against an Image.
- You can use vSphere Lifecycle Manager recommended images.
  When you manage a vSAN cluster with vSphere Lifecycle Manager images, the vSAN recommendation engine does not generate vSAN health alarms or recommendation baselines for that cluster. However, vSphere Lifecycle Manager generates pre-validated images include a recommended firmware version for the hosts in your vSAN cluster. For more information about vSphere Lifecycle Manager recommended images, see vSphere Lifecycle Manager Recommended Images.

Managing a vSAN Witness Host with an Image

You can manage a standalone vSAN witness host with an image. The witness host can be dedicated to a vSAN stretched cluster or support up-to 64 two-node vSAN clusters. You upgrade a witness host that you manage with a vSphere Lifecycle Manager image in the same way you upgrade any other standalone host. For more information, see How to Use vSphere Lifecycle Manager Images and vSphere Lifecycle Manager and the vSAN Witness Hosts.

Remediation Specifics of vSAN Clusters

Whether you manage a vSAN cluster with baselines or with an image, remediating the hosts that are part of a vSAN cluster has its specifics.

When you remediate hosts that are part of a vSAN cluster, take note of the following behavior:

- vSphere Lifecycle Manager can only put one host at a time in maintenance mode.

- vSphere Lifecycle Manager can only remediate hosts that are part of a vSAN cluster sequentially.
- Because vSphere Lifecycle Manager handles the remediation of the hosts sequentially, the host remediation process might take an extensive amount of time to finish.
- vSphere Lifecycle Manager remediates vSAN clusters with configured fault domains by upgrading all hosts from one fault domain first and then upgrading the hosts in the next fault domain.
- For a vSAN stretched cluster, vSphere Lifecycle Manager first remediates the hosts from the preferred site and then proceeds with remediating the hosts in the secondary site.

Host Maintenance Mode and vSAN Clusters

You can remediate a host that is in a vSAN cluster in two ways, depending on how you want to handle the virtual machines on the host:

- You can put the host in maintenance mode manually and remediate the host by using vSphere Lifecycle Manager.
- You can have the host enter maintenance mode during the vSphere Lifecycle Manager remediation process.

In the vSphere Client, when you put a host from a vSAN cluster into maintenance mode, you can choose between multiple options: Ensure accessibility, Full data evacuation, and No data evacuation. The Ensure accessibility option is the default option, and means that when you put a host in maintenance mode, vSAN ensures that all accessible virtual machines on the host remain accessible. To learn more about each of the options, see the "Place a Member of a vSAN Cluster in Maintenance Mode" topic in the *vSphere Storage* documentation.

During remediation, vSphere Lifecycle Manager, puts the hosts from the vSAN cluster in maintenance mode and handles the virtual machines on the host in the manner of the default Ensure accessibility option.

If a host is a part of a vSAN cluster, and any virtual machine on the host uses a VM storage policy with the setting for "Number of failures to tolerate=0", the host might experience unusual delays when it enters maintenance mode. The delay occurs because vSAN has to migrate the virtual machine data from one disk on the vSAN datastore cluster to another. Delays might take up to hours. You can work around this by setting the "Number of failures to tolerate=1" for the VM storage policy, which results in creating two copies of the virtual machine files on the vSAN datastore.

vSAN Health Check

vSphere Lifecycle Manager performs a remediation pre-check of vSAN clusters to ensure successful remediation. The vSAN health check is part of the remediation pre-check.

The vSAN health check gives you information about the cluster state and whether you must take extra actions to ensure successful remediation. Even if you do not take the recommended actions, you can still remediate the vSAN cluster or a host from the cluster. vSphere Lifecycle Manager successfully puts the host in maintenance mode and applies software updates on the host successfully. However, the host might fail to exit maintenance mode, and the remediation process might fail. As a result, the host from the vSAN cluster is upgraded, but you must take manual steps to take the host out of maintenance mode.

Using vSphere Lifecycle Manager Images to Remediate vSAN Stretched Clusters

When you manage a vSAN stretched cluster or a two-node Remote Office/Branch Office (ROBO) cluster with vSphere Lifecycle Manager, you can manage the hosts in the cluster with an image that is different from the image used to upgrade the dedicated witness host. You upgrade the vSAN witness host in the same way as you upgrade a standalone host.

What Is a Stretched Cluster

A stretched cluster is a deployment model in which two or more hosts are part of the same logical cluster but are located in separate geographical locations. Every vSAN stretched cluster or two-node ROBO cluster has a witness host, which is a standalone host that is not a member of the respective cluster but is associated with it. The witness host of a vSAN cluster is managed by the same vCenter where the respective stretched or ROBO cluster resides.

vSphere Lifecycle Manager and the vSAN Witness Hosts

The vSAN witness host is a physical or virtual ESX host that contains the witness components of virtual machine objects stored in the vSAN cluster. The witness host does not support workloads and is not a data node. A single stretched or two-node ROBO cluster can have only one witness host.

You can use vSphere Lifecycle Manager images to manage a vSAN stretched cluster and its witness host. You define separate images for the vSAN cluster and for the witness host. See Upgrading vSAN Stretched Clusters by Using a vSphere Lifecycle Manager Image. The following requirements exist:

- vCenter must be version 8.0 Update 2 and later.
- The witness host must be ESX version 8.0 and later.
- The witness host can be a virtual server or a physical server.
- The witness host can be a dedicated witness host or a shared witness host.
- The witness host must be upgraded before the hosts in the associated vSAN stretched or two-node cluster.
- The witness host and the associated vSAN clusters must not be upgraded in parallel.
- You cannot run virtual machines on a witness host. If vSphere Lifecycle Manager detects any stale virtual machines running on a witness host, during the remediation of the standalone host vSphere Lifecycle Manager sets the **VM power state** remediation setting to **Do not change power state**. For more information, see Configure the Global vSphere Lifecycle Manager Remediation Settings for Clusters That You Manage with Images.

You start using vSphere Lifecycle Manager images to manage the witness host by performing any of the following tasks:

- You switch from using vSphere Lifecycle Manager baselines to using vSphere Lifecycle Manager images for an existing vSAN stretched or two-node ROBO cluster and for the dedicated standalone host.
- You transition an existing vSAN cluster that uses an image into a stretched cluster with a virtual witness host.
- You upgrade to version 8.0 Update 2 and later for vCenter and version 8.0 or later for the witness host.

  Note:

  The witness host must be of the same major and minor version as the rest of the hosts in the cluster.

You stop using vSphere Lifecycle Manager images to manage the witness host in the following cases:

- You transition an existing vSAN stretched cluster that uses images into a regular vSAN cluster.
- You deactivate vSAN for an existing vSAN stretched cluster that you manage with an image.

Important:
You can use vSphere Lifecycle Manager images to manage standalone hosts in your vCenter inventory. You can apply a separate vSphere Lifecycle Manager image to the witness host of a vSAN cluster. You can start managing the witness host with a vSphere Lifecycle Manager image at the time of adding the host to the inventory or you can transition an existing standalone host that uses an image to a witness host.

Upgrading vSAN Stretched Clusters by Using a vSphere Lifecycle Manager Image

You upgrade a stretched vSAN cluster along with the associated witness host by using a vSphere Lifecycle Manager image, if the following requirements exist:

- vCenter must be version 8.0 and later.
- The witness host must be ESX version 8.0 and later.
- The witness host is an ESX host running inside a VM.
- The witness host is a standalone host dedicated to the stretched vSAN cluster.

Once you upgrade the vCenter instance, the virtual dedicated witness host is no longer managed with the vSphere Lifecycle Manager image that you define for the vSAN cluster. You can use a full vSphere Lifecycle Manager image to upgrade a witness host in the same way you upgrade a standalone host. The desired image you apply on a witness host can contain a base ESX image, and any user components, solution components, or vendor add-ons.

For stretched vSAN clusters, you must first upgrade the witness host with the separate vSphere Lifecycle Manager image you configured and then proceeds to remediating the hosts in the preferred site and the secondary site. If all hosts in the preferred site are in a compliant state, then vSphere Lifecycle Manager skips the preferred site and starts remediating the hosts from the secondary site. If any host in the entire cluster is in an incompatible state after the remediation started,

remediation stops. For more information about fault domain-aware remediation and the order in which vSphere Lifecycle Manager remediates the hosts in a vSAN cluster, see Using vSphere Lifecycle Manager Images to Remediate vSAN Clusters with Configured Fault Domains .

In VMware Cloud Foundation 9.0, by default a stretched vSAN cluster is upgraded automatically along with the dedicated witness host. If you remove components from the base image assigned to a stretched vSAN cluster and remediate the cluster, the removed components are removed from the data nodes and not from the dedicated witness host. The automatic upgrade of virtual dedicated witness nodes in a VCF 9.x environment applies only the base image from the desired state of the stretched vSAN cluster to the witness host. For more information about how to upgrade your management domain to VCF 9.1, see the Upgrade to ESX 9.1 article in the *VMware Cloud Foundation 9.1* documentation.

To remediate the witness host against a vSphere Lifecycle Manager image, the following requirements exist:

- vCenter must be version 8.0 Update 2 and later.
- The witness host must be ESX version 8.0 and later.
- The witness host can be a virtual server and a physical server.
- The witness host can be a dedicated witness host and a shared witness host.
- Depending on the storage architecture of the stretched vSAN cluster, OSA or ESA, there are different virtual appliances to the witness host.

Using vSphere Lifecycle Manager Images to Remediate vSAN Clusters with Configured Fault Domains

In vSAN clusters with configured fault domains, vSphere Lifecycle Manager remediates the hosts in an order that vSphere Lifecycle Manager calculates by factoring in the defined fault domains.

What Is a Fault Domain?

A fault domain consists of one or more vSAN hosts grouped according to their physical location in the data center. When configured, fault domains enable vSAN to tolerate failures of entire physical racks that are defined by the end user, as well as failures of a single host, capacity device, network link, or a network switch dedicated to a fault domain. You can configure fault domains for non-stretched and stretched vSAN clusters. For more information about configuring fault domains, see the *Administering VMware vSAN* documentation.

Upgrading vSAN Clusters Configured with Multiple Fault Domains

vSphere Lifecycle Manager remediates vSAN clusters with configured fault domains by remediating all hosts in one fault domain at a time. To define the order of fault domains, vSphere Lifecycle Manager calculates and assigns priority to each fault domain for the vSAN cluster.

Remediation starts with the fault domain that has the highest priority. The priority of a fault domain is determined by the number of non-compliant hosts in that fault domain. The fewer non-compliant hosts in a fault domain, the higher the priority of that fault domain. However, if multiple fault domains have the same priority, vSphere Lifecycle Manager selects the first fault domain from the list of fault domains.

After vSphere Lifecycle Manager selects a fault domain, vSphere Lifecycle Manager uses DRS recommendations to select the optimal host within that domain to be remediated.

For fault domain-aware remediation of vSAN clusters, the following requirements exist:

- vCenter must be version 8.0 and later
- The ESX hosts must be version 8.0 and later

Upgrading vSAN Clusters Enabled with NSX or vSphere Supervisor

You can remediate a vSAN cluster against a vSphere Lifecycle Manager image that contains the same ESX version as the ESX version currently on the hosts, but the latest versions of NSX and vSphere Supervisor components. In that case, vSphere Lifecycle Manager upgrades only those components, without upgrading the ESX version. Even in those cases, vSphere Lifecycle Manager still recognizes the configured fault domains for the vSAN cluster and performs the solution upgrade in accordance with the fault domain configuration.

For fault domain-aware remediation of vSAN clusters with enabled NSX or vSphere IaaS Control Plane, the following requirements exist:

- vCenter must be version 8.0 and later
- The ESX hosts version 8.0 and later

Updating Firmware in vSAN Clusters

You can use vSphere vSphere Lifecycle Manager images to upgrade the firmware of the servers that run in your vSAN clusters.

In a vSAN cluster, the SCSI controller firmware and the physical drive firmware are handling most of the data communication. To ensure your vSAN cluster health, you must perform controller firmware updates, when necessary.

Because firmware updates affect the hardware layer in your vSphere environment, they usually are rare events. Firmware updates occur during initial ESX host setup or during major updates of vSphere or vSAN.

In earlier vSphere releases, firmware updates are delivered as baselines in the vSAN-managed baseline group. You must use a special vendor-provided tool that vSAN uses to detect, download, and install firmware updates. However, the recommendation baseline group that vSAN generates contains only patch updates and driver updates. It no longer contains firmware updates. As a result, you cannot use baselines to update the firmware in your vSAN clusters if the ESX hosts are of version 8.0 and later. To perform firmware updates on hosts that are of ESX version 8.0 and later and that are in a vSAN cluster, you must satisfy the following requirements:

- The cluster must be managed with an image
- You must deploy an OEM-provided hardware support manager and register it as a vCenter extension.

The hardware support manager inspects the hardware of the hosts in the cluster and lists available and compatible firmware versions, which you can add to the image for the cluster. The actual firmware update happens upon remediating the cluster against an image that contains a firmware add-on.

For more information about the requirements for using images, see What Are the Requirements for Using vSphere Lifecycle Manager.

For more information about performing firmware updates by using images, see Firmware Updates with vSphere Lifecycle Manager.

About Recommendation Baseline Groups

vSAN creates system-managed baseline groups called recommendation baseline groups. You use recommendation baseline groups to upgrade the hosts in a vSAN cluster to the latest supported ESX version, to patch the hosts with critical patches, or to update drivers on the host.

vSAN generates recommendation baseline groups automatically. If your vSphere environment does not contain any vSAN clusters, no recommendation baseline groups are generated. For each vSAN cluster in the vSphere inventory, vSphere Lifecycle Manager displays a single recommendation baseline group. You cannot edit or delete a recommendation baseline group and you cannot add it to custom baseline groups.

Recommendation baseline groups can contain any of the following software updates:

- An upgrade baseline that contains an ESX upgrade image by a certified vendor with the latest tested and recommended version for the vSAN cluster.
- One or multiple patch baselines that contain recommended critical patches for the ESX version of the hosts in the vSAN cluster.
- Recommended drivers for the ESX hosts in the vSAN cluster.

> **Note:**
> Recommendation baseline groups no longer contain firmware updates. To update the firmware on your hosts, you must transition to using an image for the vSAN cluster.

How Does vSphere Lifecycle Manager Generate Recommendation Baselines?

A vSAN recommendation engine regularly checks the current state of the software installed on the hosts in the vSAN cluster against the vSAN Hardware Compatibility List (HCL). If update recommendations are detected, the engine downloads all new critical patches and upgrade images and generates a vSAN cluster-level baseline. All available baselines are packed together in a recommendation baseline group and made available for use by vSphere Lifecycle Manager.

Every 24 hours, vSphere Lifecycle Manager runs an automatic check for a recommendation baseline group with build recommendations coming from vSAN. If a new recommendation baseline group is detected, vSphere Lifecycle Manager automatically attaches the vSAN recommendation baseline group to the vSAN cluster.

After refreshing the vSAN recommendation baseline group, vSphere Lifecycle Manager automatically performs a compliance check operation on the vSAN clusters against the updated recommendation baseline group. Operations such as adding and removing hosts from an existing vSAN cluster also trigger refresh of the attached recommendation baseline group, followed by a compliance check.

System Requirements for Using vSAN Recommendation Baseline Groups

- vCenter 8.0 or later.
  vSphere Lifecycle Manager runs as a service in vCenter 8.0 and later.
- vSAN cluster that contains hosts of ESX version 8.0 and later.
- Constant access of the vSphere Lifecycle Manager host machine to the Internet or the latest version of the compatibility data bundle uploaded to the vSphere Lifecycle Manager depot. See Add the Compatibility Data Bundle to the vSphere Lifecycle Manager Depot.

vSphere Lifecycle Manager and vSphere Supervisor

You can enable vSphere Supervisor on vSphere clusters that you manage with vSphere Lifecycle Manager images. In vSphere 9.0, you can transition a vSphere cluster managed with baselines that has vSphere Supervisor enabled into a vSphere cluster managed with images.

vSphere Lifecycle Manager and vSphere Supervisor with vSphere Networking

You start using vSphere Supervisor with vSphere networking on a cluster that uses a vSphere Lifecycle Manager image by enabling the cluster for **Supervisor Management**. A cluster enabled for **Supervisor Management** is called a vSphere Supervisor. You enable a cluster for **Supervisor Management** from the **Supervisor Management** user interface in the vSphere Client.

Requirements

- Verify that all ESX hosts in the vSphere Supervisor are version 8.0 and later.
- Verify that vCenter is version 9.0 and later.
- **Note:**

  You can manage the lifecycle of a vSphere Supervisor with either vSphere Lifecycle Manager images or vSphere Lifecycle Manager baselines. You can transition a vSphere Supervisor that uses vSphere Lifecycle Manager baselines to a vSphere Supervisor that uses vSphere Lifecycle Manager images.
- Review the configuration requirements and additional information in *Working with vSphere Lifecycle Manager* chapter in the *Updating vSphere Supervisor* documentation.

Supported Lifecycle Workflows

The following workflows are supported for any vSphere Supervisor that uses vSphere Lifecycle Manager images and is configured to use the vSphere networking stack.

- You can upgrade a vSphere Supervisor to the latest version of VMware vSphere Supervisor. You can also upgrade the ESX version of the hosts in the vSphere Supervisor.

You perform the upgrade of the vSphere Supervisor from the **Workload Management** user interface in the vSphere Client.

You upgrade the ESX version of the hosts in the vSphere Supervisor by remediating the cluster from the vSphere Lifecycle Manager user interface in the vSphere Client.

Note:

You cannot combine an ESX upgrade with a vSphere Supervisor upgrade to reduce the upgrade window.

- You can deactivate vSphere Supervisor from the **Workload Management** user interface in the vSphere Client. You can deactivate vSphere Supervisor on a cluster that uses a vSphere Lifecycle Manager image.

  Warning:

  If you deactivate vSphere Supervisor on a cluster managed with an image, your managed namespaces and control plane VMs are also deleted.
- You can transition a vSphere Supervisor that uses vSphere Lifecycle Manager baselines to a vSphere Supervisor that uses vSphere Lifecycle Manager images like any other cluster in a vCenter instance of version 9.0 and later.
- You can add and remove hosts to and from a cluster that has both vSphere Supervisor and vSphere Lifecycle Manager enabled.

  For more information about adding and removing hosts to and from a cluster, see the *vCenter and Host Management* documentation.

For detailed information about working with a vSphere Supervisor that uses an image, see the *Working with vSphere Lifecycle Manager* section in the *Maintaining vSphere Supervisor* documentation.

Upgrading the vSphere Supervisor Components in a vSAN Cluster

To upgrade the vSphere Supervisor components installed on the hosts in a cluster, you must remediate the cluster against a vSphere Lifecycle Manager image that contains the latest version of those vSphere Supervisor components.

In a vSAN cluster with configured fault domains, vSphere Lifecycle Manager recognizes the configured fault domains for the cluster and performs the solution upgrade in accordance with the fault domain configuration. If the vSAN cluster is a stretched cluster, you must upgrade the witness host separately, after vSphere Lifecycle Manager finishes remediating all fault domains. For more information about remediating vSAN stretched clusters and vSAN clusters configured with fault domains, see Using vSphere Lifecycle Manager Images to Remediate vSAN Clusters with Configured Fault Domains . For more information about stretched clusters, see the *vSAN Planning and Deployment* documentation.

Scalability

For information about the scalability that vSphere Lifecycle Manager supports, visit the VMware Configuration Maximums Matrix at https://configmax.broadcom.com/.

vSphere Lifecycle Manager and vSphere Supervisor with NSX Networking

You can start using vSphere Supervisor with NSX networking on a cluster that uses a vSphere Lifecycle Manager image by enabling the cluster for **Workload Management**. A cluster enabled with **Workload Management** is called a vSphere Supervisor. You enable the cluster for **Workload Management** from the **Workload Management** user interface in vSphere Client. Upon enabling **Workload Management** on a vSphere Supervisor configured with the NSX networking stack, vSphere Lifecycle Manager installs the Spherelet VIB on every ESX host in the cluster.

Requirements

- Verify that all ESX hosts in the vSphere Supervisor are version 8.0 or later.
- Verify that vCenter is version 9.0 or later.
- **Note:**

  You can manage the lifecycle of a vSphere Supervisor with either vSphere Lifecycle Manager images or vSphere Lifecycle Manager baselines. You can transition a vSphere Supervisor that uses vSphere Lifecycle Manager baselines to a vSphere Supervisor that uses vSphere Lifecycle Manager images like any other cluster in a vCenter instance of version 9.0 and later.

- Review the configuration requirements and additional information in the *Working with vSphere Lifecycle Manager* chapter in the *Maintaining vSphere Supervisor* documentation.

Supported Lifecycle Workflows

The following workflows are supported for any vSphere Supervisor that uses vSphere Lifecycle Manager images and is configured to use the NSX networking stack.

- You can upgrade a vSphere Supervisor to the latest version. You can also upgrade the ESX version of the hosts in the vSphere Supervisor.
  You perform the upgrade of the vSphere Supervisor from the **Workload Management** user interface in the vSphere Client. During upgrade, vSphere Lifecycle Manager upgrades the Spherelet VIB on the hosts to make it compatible with the new version of vSphere Supervisor or the new version of ESX.
  You upgrade the ESX version of the hosts in the vSphere Supervisor by remediating the cluster from the vSphere Lifecycle Manager user interface in the vSphere Client.

  Note:

  You cannot perform a simultaneous upgrade of both vSphere Supervisor and ESX.
- You can deactivate vSphere Supervisor from the **Workload Management** user interface in the vSphere Client. You can deactivate vSphere Supervisor on a cluster that uses a vSphere Lifecycle Manager image.
  When you deactivate vSphere Supervisor, you can use the cluster for traditional virtual machine workloads.

  Warning:

  If you deactivate vSphere Supervisor on a cluster managed with an image, your managed namespaces and control plane VMs are also deleted.
- You can transition a vSphere Supervisor that uses vSphere Lifecycle Manager baselines to a vSphere Supervisor that uses vSphere Lifecycle Manager images like any other cluster in a vCenter instance of version 9.0 and later.
- You can add and remove hosts to and from a vSphere Supervisor that uses vSphere Lifecycle Manager images.
  When you add a host to a vSphere Supervisor that you manage with a vSphere Lifecycle Manager image, vSphere Lifecycle Manager automatically installs the Spherelet VIB on the newly added host.
  When you remove a host from a vSphere Supervisor that you manage with a vSphere Lifecycle Manager image, vSphere Lifecycle Manager removes the Spherelet VIB from the host. vSphere Lifecycle Manager also deletes the Spherelet VIB from a host that you move to another vSphere Supervisor.
  For more information about adding and removing hosts to and from a cluster, see the *vCenter and Host Management* documentation.

For detailed information about working with a vSphere Supervisor that uses an image, see the *Working with vSphere Lifecycle Manager* chapter in the *Maintaining vSphere Supervisor* documentation.

Upgrading the vSphere Supervisor Components in a vSAN Cluster

To upgrade the vSphere Supervisor components installed on the hosts in a cluster, you must remediate the cluster against a vSphere Lifecycle Manager image that contains the latest version of those vSphere Supervisor components.

In a vSAN cluster with configured fault domains, vSphere Lifecycle Manager recognizes the configured fault domains for the cluster and performs the solution upgrade in accordance with the fault domain configuration. If the vSAN cluster is a stretched cluster, you must upgrade the witness host separately, after vSphere Lifecycle Manager finishes remediating all fault domains. For more information about remediating vSAN stretched clusters and vSAN clusters configured with fault domains, see Using vSphere Lifecycle Manager Images to Remediate vSAN Clusters with Configured Fault Domains . For more information about stretched clusters, see the *vSAN Planning and Deployment* documentation.

Scalability

For information about the scalability that vSphere Lifecycle Manager supports, visit the Broadcom Configuration Maximums Matrix at https://configmax.broadcom.com.

vSphere Lifecycle Manager and VMware NSX

You can use vSphere Lifecycle Manager baselines for update operations in environments with NSX 3.0. With vSphere 9.0 and NSX 3.1 and later, you must use vSphere Lifecycle Manager images to manage clusters that have NSX enabled.

Using vSphere Lifecycle Manager Images in an Environment With NSX 3.1 and Later

You can use NSX on a cluster that you manage with a vSphere Lifecycle Manager image. As a result, you can use NSX Manager, to install, upgrade, or remove NSX components on the ESX hosts in a cluster that you manage with an image.

Requirements

- Verify that all ESX hosts in the cluster are version 8.0 or later.
- Verify that vCenter is version 9.0 or later.
- Verify that NSX is version 3.1 or later.
- Verify that a vSphere Distributed Switch (VDS) is configured to manage the NSX traffic.

Supported Lifecycle Workflows

The following workflows are supported for clusters that are enabled for both vSphere Lifecycle Manager images and NSX.

- You can enable NSX on a cluster that you manage with a vSphere Lifecycle Manager image.
  You perform the operation by configuring a transport node profile (TNP) for the cluster in the NSX Manager . In thec you can either manually add a TNP to the cluster, or automatically generate one in the **Getting Started** wizard. You can continue leveraging individual transport node configurations, but you must always use a TNP for the clusters that you manage with a vSphere Lifecycle Manager image. You cannot enable NSX on a cluster that uses a vSphere Lifecycle Manager image if you choose to only use individual transport node configurations for the hosts in the cluster.
- You can add hosts to a cluster that you manage with a vSphere Lifecycle Manager image and that is enabled with NSX. You can also remove hosts from such a cluster.
  You perform the add and remove host operations in the vSphere Client. When you add a host to the cluster, vSphere Lifecycle Manager automatically installs the NSX component to the newly added host. To add a host to a cluster that you manage with a vSphere Lifecycle Manager image, the host must be added to the VDS associated with the TNP. Otherwise, the host cannot fully work with NSX.
  When you move a host from one cluster that uses a vSphere Lifecycle Manager image to another, vSphere Lifecycle Manager applies the target cluster's image together with the target NSX component to the newly added host. If a host is deleted from the vCenter inventory, the NSX component is uninstalled from the host.
  For more information about adding and removing hosts to and from a cluster, see the *vCenter and Host Management* documentation.
- You can upgrade NSX 3.1 to a later version in a cluster that you manage with a vSphere Lifecycle Manager image.
  You perform the operation from the NSX Manager.
- You can upgrade both NSX and ESX in a vSphere Lifecycle Manager remediation task. The workflow is supported only if you upgrade from NSX version 3.1.
  In the NSX Manager, you stage the NSX upgrade as part of the image that the cluster uses. From the vSphere Lifecycle Manager user interface in the vSphere Client, you can further edit the image and you initiate remediation of the cluster. During remediation, vSphere Lifecycle Manager applies both the NSX and ESX upgrades to the hosts in the cluster. For more information, see the *Upgrading NSX Guide* documentation.
- You can switch from using vSphere Lifecycle Manager baselines to using a vSphere Lifecycle Manager image for a cluster that is enabled with NSX.
- You can uninstall NSX from a cluster that you manage with a vSphere Lifecycle Manager image.
- You can check the compliance, generate a remediation pre-check report, and remediate a cluster that you manage with a vSphere Lifecycle Manager image and that is enabled with NSX.
  You perform the check compliance, generate a remediation pre-check, and remediation operations in the vSphere Client. Whenever you change the NSX configuration in the NSX Manager, the compliance state of the cluster that you see on the **Updates** tab for the cluster in the vSphere Client changes to non-compliant. You can remediate non-compliant hosts and clusters in the vSphere Client or you can solve the issues that cause non-compliance in the NSX Manager.

- You can back up and restore NSX.
- You can export the vSphere Lifecycle Manager image of a cluster that is enabled with NSX and import this image to another cluster that has both vSphere Lifecycle Manager images and NSX enabled.

For detailed information about all workflows that you perform in the NSX Manager, see the *Software-Defined Networking with NSX* documentation.

Upgrading the NSX Components in a vSAN Cluster

To upgrade the NSX components installed on the hosts in a cluster, you must remediate the cluster against a vSphere Lifecycle Manager image that contains the latest version of those NSX components.

In a vSAN cluster with configured fault domains, vSphere Lifecycle Manager recognizes the configured fault domains for the cluster and performs the solution upgrade in accordance with the fault domain configuration. If the vSAN cluster is a stretched cluster, you must upgrade the witness host separately, after vSphere Lifecycle Manager finishes remediating all fault domains. For more information about remediating vSAN stretched clusters and vSAN clusters configured with fault domains, see Using vSphere Lifecycle Manager Images to Remediate vSAN Clusters with Configured Fault Domains . For more information about stretched clusters, see the *vSAN Planning and Deployment* documentation.

Scalability

For information about the scalability that vSphere Lifecycle Manager supports, visit the VMware Configuration Maximums Matrix at https://configmax.broadcom.com.

Using vSphere Lifecycle Manager Images for Standalone Hosts with NSX 4.2 and later

You can use NSX on a standalone host that you manage with a vSphere Lifecycle Manager image. As a result, vSphere Lifecycle Manager becomes responsible for installing, upgrading, and removing NSX components on the host.

Requirements

- Verify that the standalone ESX host and the vCenter instance are of version 8.0 Update 3 and later.
- Verify that NSX is of version 4.2 and later.

Supported Lifecycle Workflows

- You can enable NSX on a standalone host that uses vSphere Lifecycle Manager images.
  You perform the operation by configuring NSX on the individual host in the NSX Manager UI. For more information about how to configure individual hosts as transport nodes, see the *Prepare ESXi Individual Hosts as Transport Nodes* procedure in the *NSX Installation Guide*.
- You can uninstall NSX from a standalone host that you manage with a vSphere Lifecycle Manager image.

  Warning:

  This operation may break network connectivity for VMs.
- You can switch from using vSphere Lifecycle Manager baselines to using vSphere Lifecycle Manager images for an existing standalone host that is prepared as a transport node.
  During the eligibility check on the transport node, vSphere Lifecycle Manager performs several additional checks and allows the operation to continue in case all of the following conditions are met:
  - The vCenter instance is of version 9.0 and later.
  - The ESX host is of version 8.0 Update 3 and later.
  - The NSX version on the standalone host is 4.2 and later.
  - The **Multi NSX** mode is not enabled on the vCenter instance. See *Multiple NSX Managers Managing a Single VMware vCenter* in the *NSX Installation Guide*.
- You can upgrade both NSX and ESX on the standalone host by using a vSphere Lifecycle Manager remediation task. You run the remediation operation in the vSphere Client.
- You can add a standalone host that has NSX already installed to a data center or a folder and enable vSphere Lifecycle Manager on the host during this process.

During the eligibility check on the host, vSphere Lifecycle Manager performs several additional checks and allows the operation to continue in case all of the following conditions are met:

- The vCenter instance is of version 9.0 and later.
- The ESX host is of version 8.0 Update 3 and later.
- The NSX version on the standalone host is 4.2 and later.
- The **Multi NSX** mode is not enabled on the vCenter instance. See the *Software-Defined Networking with NSX* documentation.

- If you take a host out of a cluster that is prepared with a transport node profile (TNP) and is managed with a vSphere Lifecycle Manager image, the host becomes standalone. The host can retain the image of the cluster in case the following criteria are met:
  – The vCenter instance is of version 9.0 and later.
  – The ESX host is of version 8.0 Update 3 and later.
  – NSX is of version 4.2 or later.
  – During the move operation the host stays connected or reachable.

  Otherwise, the image is removed from the host and the host is reverted back to using baselines.
- You can add a standalone host that used baselines to a cluster that you manage with a vSphere Lifecycle Manager image and that is enabled with NSX.

  When you add the standalone host to the cluster, vSphere Lifecycle Manager automatically installs the NSX component to the newly added host. To add a host to a cluster that you manage with a vSphere Lifecycle Manager image, the host must be added to the VDS associated with the TNP. Otherwise, the host cannot fully work with NSX.

Backup and Restore Scenarios When Using vSphere Lifecycle Manager

Restoring a vCenter instance from a backup might impact the clusters in your environment in a seemingly unexpected way. Whether you use images or baselines to manage your clusters, vSphere Lifecycle Manager behaves in a specific manner during backup and restore operations.

When you back up a vCenter instance, you create a backup copy of all clusters in that vCenter instance.

Restoring vCenter After Switching from Baselines to Images for Cluster Lifecycle Management

Cluster A is a cluster that you manage by using baselines. You back up the vCenter instance where the cluster is. After the backup, you switch from using baselines to using images to manage cluster A and you remediate the cluster to apply the image to the hosts in the cluster. You now manage the lifecycle of cluster A by using a cluster image.

If you must restore the vCenter instance from the backup copy you created, the restored vCenter instance contains cluster A. Because cluster A was managed through baselines at the time when you backed up the vCenter system, the restored vCenter instance contains cluster A, but you must again use baselines to manage it.

Restoring vCenter After Remediating a Cluster Managed by an Image

After remediation, cluster A uses image X with components Y to manage all hosts in the cluster collectively. At a point in time T, you back up the vCenter system. Later, you remediate the cluster against a new image X+1 with new components Y+1. Now all hosts in the cluster use image X+1 with components Y+1.

If you must restore the vCenter system from the backup copy that you created at time T, the restored vCenter instance contains cluster A, but the compliance check lists the hosts in the cluster as incompatible with the image that cluster A uses. The reason for the incompatibility is that after the restore operation, cluster A reverts back to using image X with components Y, while the hosts in the cluster still run image X+1 with components Y+1. Because you cannot downgrade ESX, to make the hosts compliant with the cluster image, you must upgrade the cluster to image X+1 with components Y +1.

vSphere Configuration Profiles

Learn how to manage the configuration of all hosts in a cluster collectively in vSphere Client.

Intended Audience

This information is intended for experienced system administrators who are familiar with data center operations and virtual machine technology.

What is vSphere Configuration Profiles

With vSphere Configuration Profiles, you can manage the configuration of all hosts in a cluster collectively. You set up a desired configuration that you can apply to all hosts in a single operation. You can also use a reference host and make its configuration the desired configuration for an entire cluster. Using vSphere Configuration Profiles ensures consistency in host configuration.

vSphere Configuration Profiles uses a declarative, desired-state model to manage host configuration at the cluster level. Once you define the target cluster configuration, vSphere Configuration Profiles ensures all hosts in the cluster remain compliant with the desired state.

vSphere Configuration Profiles Capabilities

With vSphere Configuration Profiles, you can perform the following tasks:

- Set a desired host configuration at a cluster level. The configuration is created and managed in the form of a JSON file. You can edit the desired cluster configuration settings in the vSphere Client.
- Check the host compliance against the configuration for the cluster.
- Remediate the cluster to make non-compliant hosts compliant with the configuration set at the cluster level.

You can enable vSphere Configuration Profiles on a cluster that you manage with an image or baselines. For clusters that you manage with baselines, you can also use vSphere Host Profiles, see the *vSphere Host Profiles* documentation. vSphere Host Profiles is announced deprecated in VMware vSphere Foundation 9.0 and will be removed in a future release.

You start using vSphere Configuration Profiles by enabling the feature during cluster creation or by enabling vSphere Configuration Profiles on an existing cluster. You can switch from using host profiles to enabling and using vSphere Configuration Profiles at any time. The change is irreversible.

For information about how to enable vSphere Configuration Profiles during cluster creation and set a desired configuration for the cluster, see Enable vSphere Configuration Profiles During Cluster Creation.

For information about how to switch from using legacy configuration management tools, such as host profiles, to using vSphere Configuration Profiles, see Enable vSphere Configuration Profiles on an Existing Cluster.

System Requirements for Using vSphere Configuration Profiles

- The vCenter instance must be of version 9.0 and later
- All ESX hosts in a cluster that is managed with an image must be of version 8.0 Update 1 and later.
- All ESX hosts in a cluster that is managed with baselines must be of version 8.0 Update 3 and later.
- Required privileges. To perform all tasks that are mentioned before, you need the proper privileges, see Required Privileges for Using vSphere Configuration Profiles.

Limitations to Using vSphere Configuration Profiles

- You cannot deactivate vSphere Configuration Profiles.

- You can manage the vSphere Distributed Switch configuration of the hosts in a cluster with enabled vSphere Configuration Profiles. You can also use the workflows for managing and configuring the vSphere Distributed Switches that are described in the *vSphere Networking* documentation.
- You cannot enable vSphere Configuration Profiles on a cluster with DPU-backed hosts.
- You can utilize vSphere Configuration Profiles if NSX is enabled for the cluster. vSphere Configuration Profiles can manage configuration properties that are related to NSX and you can view any drifts in the cluster configuration when they occur.

  Note:

  If your ESX hosts are running version 8.x, you cannot enable vSphere Configuration Profiles on a cluster that also has NSX enabled. To use NSX with vSphere Configuration Profiles, vCenter, ESX, and NSX must be of version 9.0 or later.
- In vSphere Foundation 9.1, you can enable vSphere Configuration Profiles if vSAN is enabled. vSphere Configuration Profiles can manage configuration properties that are related to vSAN and you can view any drifts in the cluster configuration when they occur. See Using vSphere Configuration Profiles to Manage vSAN Cluster in the VMware Cloud Foundation 9.1 documentation.

The following table provides a comprehensive overview of the vSphere Configuration Profiles capabilities from vSphere 8.0 U1 to 9.1, detailing the progression from basic cluster management to full vSphere Distributed Switches, NSX, vSAN, and standalone host support.

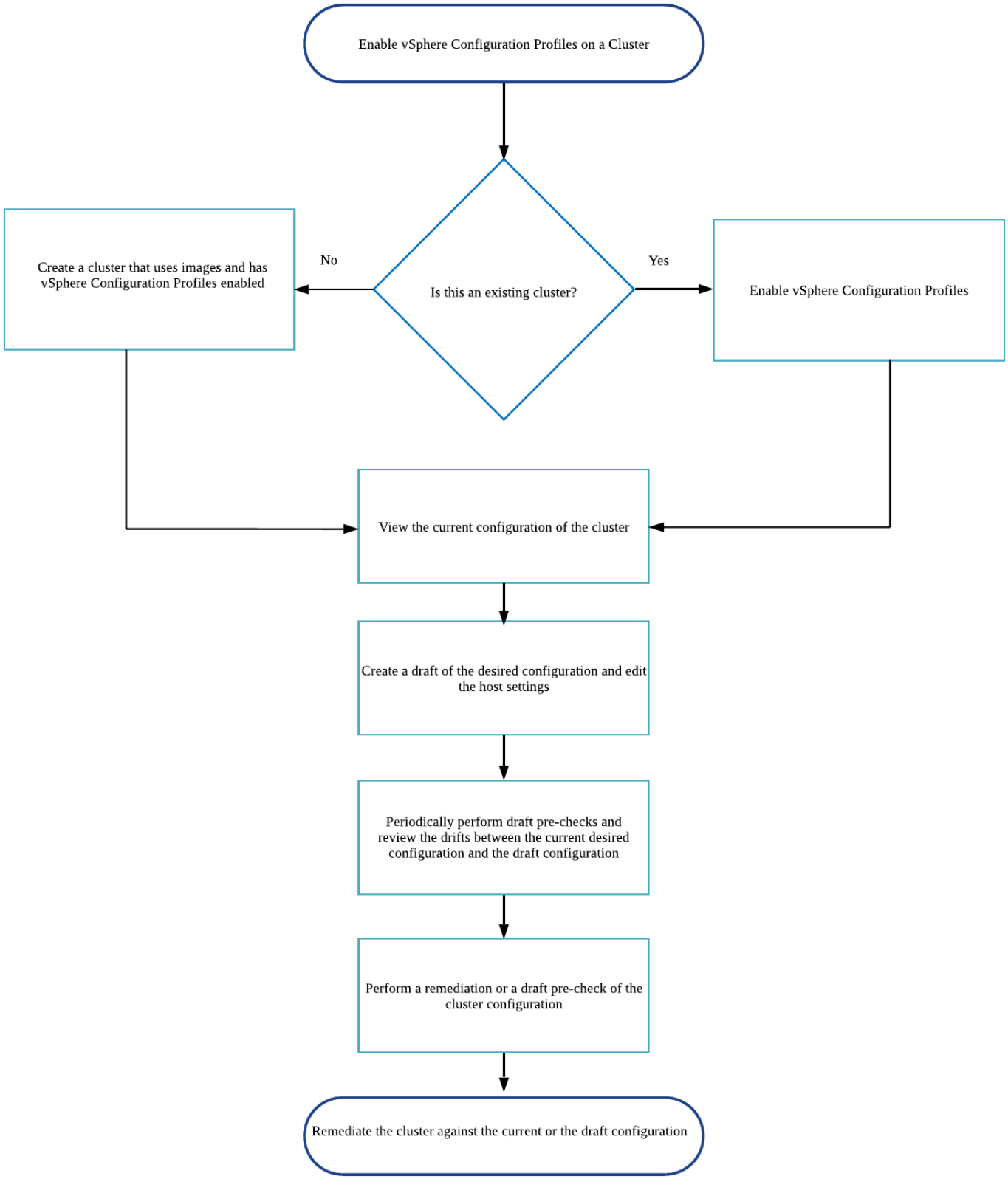
Feature or capability	vSphere 8.0 U1	vSphere 8.0 U3	vSphere 9.0	vSphere 9.1
Enable vSphere Configuration Profiles on a cluster	Supported.	Supported.	Supported.	Supported.
View vSphere Distributed Switches configuration	Supported. View only - (vCenter and ESX 8.0 U1+)	Supported. View	Supported. View	Supported. View
Manage vSphere Distributed Switches configuration	Unsupported. View only (Read-only compliance drift detection; vSphere Configuration Profiles cannot apply or remediate vSphere Distributed Switches settings)	Supported. Full management (vCenter + ESX 8.0 U3+)	Supported. Full management	Supported. Full management
NSX enabled on the cluster	Unsupported. Cannot enable vSphere Configuration Profiles if NSX is enabled.	Unsupported. Cannot enable vSphere Configuration Profiles if NSX is enabled.	Supported. vCenter, ESX, and NSX must be 9.0 or later. vSphere Configuration Profiles manages NSX - related config properties and shows drift. If ESX hosts run 8.x, you cannot enable vSphere Configuration Profiles on a cluster with NSX.	Supported. Same version 9.0+ requirement across vCenter, ESX, and NSX. vSphere Configuration Profiles manages NSX-related config properties and shows drift.
vSAN enabled on the cluster	Unsupported. Not managed by vSphere Configuration Profiles	Unsupported. Not managed by vSphere Configuration Profiles	Partial Support. Co-existence supported vSphere Configuration Profiles does not manage vSAN configuration. Use the existing vSAN APIs to configure vSAN.	Supported. Full Management vSphere Configuration Profiles actively manages vSAN- related config properties and reports drift.
Enable vSphere Configuration Profiles on Standalone Hosts	Unsupported. Cluster level only	Unsupported. Cluster level only	Supported.	Supported.

Feature or capability	vSphere 8.0 U1	vSphere 8.0 U3	vSphere 9.0	vSphere 9.1
DPU-backed hosts in the cluster	Unsupported.	Unsupported.	Unsupported.	Unsupported.
Deactivate vSphere Configuration Profiles	Unsupported. Cannot be deactivated once enabled	Unsupported. Cannot be deactivated once enabled	Unsupported. Cannot be deactivated once enabled	Unsupported. Cannot be deactivated once enabled
vSphere Auto Deploy and vSphere Distributed Switches (Stateless Hosts)	Workaround Required. Not directly supported Stateless boot mode blocks vSphere Configuration Profiles. Hosts must boot with vSphere Standard Switches first or must use stateful caching before migrating to vSphere Distributed Switches.	Workaround Required. Not directly supported Stateless boot mode blocks vSphere Configuration Profiles. Hosts must boot with vSphere Standard Switches first or must use stateful caching before migrating to vSphere Distributed Switches.	Workaround Required. Not directly supported Stateless boot mode blocks vSphere Configuration Profiles. Hosts must boot with vSphere Standard Switches first or must use stateful caching before migrating to vSphere Distributed Switches.	Workaround Required. Not directly supported Stateless boot mode blocks vSphere Configuration Profiles. Hosts must boot with vSphere Standard Switches first or must use stateful caching before migrating to vSphere Distributed Switches.

How to Start Using vSphere Configuration Profiles for Clusters That You Manage with an Image

You can enable vSphere Configuration Profiles during the cluster creation or you can enable vSphere Configuration Profiles on an existing cluster that is managed with a vSphere Lifecycle Manager image or baselines.

Figure 37: Enable vSphere Configuration Profiles on a Cluster Managed with an Image



When you enable vSphere Configuration Profiles during the cluster creation, the cluster is created with a default configuration and you must set up the desired cluster configuration by editing its settings. For more information, see Enable vSphere Configuration Profiles During Cluster Creation.

When you enable vSphere Configuration Profiles on an existing cluster, you set up a desired cluster configuration and as a result of this operation, all hosts in the cluster are compliant with the desired configuration. For more information, see Enable vSphere Configuration Profiles on an Existing Cluster.

You can create a draft configuration and can edit the configuration settings directly in the vSphere Client. For more information, see Create a Draft Configuration for a Cluster in vSphere Client.

How to Start Using vSphere Configuration Profiles on Clusters Managed with Baselines

You can enable vSphere Configuration Profiles during the cluster creation or you can enable vSphere Configuration Profiles on an existing cluster that is managed with baselines.

When you enable vSphere Configuration Profiles during the cluster creation operation, the cluster is created with an empty configuration. To create a desired cluster configuration, you must add at least one host to the cluster that you manage with baselines. Then you must select a reference host from which the configuration schema and the configuration settings are extracted and imported to the cluster as the desired cluster configuration. The reference host must be part of the cluster that is managed with baselines and must be of version 8.0 Update 3 and later. For more information, see Enable vSphere Configuration Profiles During Cluster Creation.

When you enable vSphere Configuration Profiles on an existing cluster that you manage with baselines, you use a host from the cluster as a reference host to set up the desired cluster configuration. When the enablement operation completes, all hosts in the cluster are compliant with the desired configuration. For more information, see Enable vSphere Configuration Profiles on an Existing Cluster.

Developers Support

You can use all vSphere Configuration Profiles features through several programming interfaces:

- VCF Automation API. For more information about the vSphere Configuration Profiles API, see the API Reference documentation on developer.broadcom.com.
- VCF PowerCLI. For more information about how to use scripts to automate the vSphere Configuration Profiles features, see the VCF PowerCLI documentation on developer.broadcom.com.

Required Privileges for Using vSphere Configuration Profiles

vSphere Configuration Profiles support fine-grained permissions for various lifecycle tasks.

Table 113: Required Privileges For Using vSphere Configuration Profiles

Task	Privilege in the vSphere Client	Privilege in the API	Description
Create Cluster with vSphere Configuration Profiles enabled	**Host** > **Inventory** > **CreateCluster**	`Host.Inventory.CreateCluster`	Create a cluster along with its initial configuration.
View Host Settings in the Cluster Configuration	**VMware vSphere Lifecycle Manager** > **Desired Configuration Management Privileges** > **Read-only access to desired configuration management platform**	`VcIntegrity.ClusterConfiguration.View`	View the draft configuration document, current configuration settings, compliance reports, and pre-check reports.
Check Cluster Compliance			
View Compliance Results			
Run Remediation Pre-Check			
View Remediation Pre-Check Results			
View Draft Pre-Check Results			

Task	Privilege in the vSphere Client	Privilege in the API	Description
Remediate a Cluster Against the Desired Cluster Configuration	**VMware vSphere Lifecycle Manager > Desired Configuration Management Privileges > Remediate cluster to the desired configuration.**	`VcIntegrity.ClusterConfiguration.Remediate`	Remediate a cluster against the draft or desired configuration.
Enable vSphere Configuration Profiles on a cluster			
Export a Configuration or Configuration Schema	**VMware vSphere Lifecycle Manager > Desired Configuration Management Privileges > Export desired cluster configuration**	`VcIntegrity.ClusterConfiguration.Export`	Export the desired cluster configuration document.
Extract a Configuration Document from a Reference Host	**VMware vSphere Lifecycle Manager > Desired Configuration Management Privileges > Modify desired cluster configuration.**	`VcIntegrity.ClusterConfiguration.Modify`	Modify the desired cluster configuration by crating a draft configuration, extracting a configuration document from a reference host, or importing a configuration.
Create a Draft Configuration for a Cluster			
Import a Configuration			

How to Configure Remediation Settings for Clusters and Standalone Hosts that Have vSphere Configuration Profiles Enabled

For clusters and standalone hosts managed with an image with vSphere Configuration Profiles enabled, you can customize host and VM behavior before and during remediation.

The global remediation settings are valid for all hosts and clusters that you manage with images or baselines respectively. For clusters or standalone hosts that you manage with an image, you can override the global remediation settings and can use specific remediation settings for a particular cluster or host.

Configure the Global Remediation Settings for Clusters That Have vSphere Configuration Profiles Enabled

You can configure how ESX hosts and VMs behave before and during the remediation of a cluster that you manage with an image and has vSphere Configuration Profiles enabled.

- Verify that you have the proper privileges. See Required Privileges for Using vSphere Configuration Profiles.
- To enable Quick Boot, verify that the ESX host is compatible with the feature. For more information, see Quick Boot.

1. Navigate to the vSphere Lifecycle Manager home view.
   a) In vSphere Client, select **Menu** > **Lifecycle Manager**.
   b) Select a vCenter system from the **Lifecycle Manager** drop-down menu.

      The drop-down menu is available only when multiple vCenter systems are connected by a common vCenter Single Sign-On domain. By selecting a vCenter system, you specify which vSphere Lifecycle Manager instance you want to administer.
2. On the **Settings** tab, select **Remediation Settings** > **Clusters**.
3. In the **Clusters Remediation Settings** pane, click the **Edit** button.

   The **Edit Cluster Remediation Settings** dialog box opens.

4. Configure the remediation settings for clusters that have vSphere Configuration Profiles enabled and click **Save**.

Setting	Description
Common Settings	
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is turned on. You can configure this setting for a cluster or a standalone host.
VM power state	The **VM power state**option lets you control the behavior of the virtual machines that run on the ESX host. You can select from the following options. • **Do not change power state** • **Suspend to disk** • **Suspend to memory** To select the **Suspend to memory** option, you must turn on Quick Boot. Otherwise, the **Suspend to memory** option is dimmed. Together with Quick Boot, the **Suspend to memory** option provides faster host upgrades. vSphere Lifecycle Managersuspends to the host memory and not to the disk the powered on virtual machines on the host. After the Quick Boot, the suspended virtual machines are resumed from memory. • **Power off** The default selection is **Do not change power state**. You can configure these options for a cluster or a standalone host.
VM migration	You can configure vSphere Lifecycle Manager to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster. By default, the **Migrate powered off and suspended VMs to other hosts in the cluster** check box is selected. If you have virtual machines with passthrough devices on a host in a DRS cluster, make sure that you automate the migration of such virtual machines. For more information, see the VMware knowledge base article at KB 318908. You can configure this setting only for a cluster that you manage with an image.
Retry policy	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode or if staging or remediation for that host fails. You can configure vSphere Lifecycle Manager to wait for a specified retry delay period and to retry to put the host into maintenance mode as many times as you indicate in the **Number of retries** text box. By default, vSphere Lifecycle Manager reties 3 times to put the host into maintenance mode over a period of 5 minutes. You can configure this setting for a cluster or a standalone host.
DPM	VMware Distributed Power Management (DPM) monitors the resources consumed by the running virtual machines in the cluster. If sufficient excess capacity exists, VMware DPM recommends moving virtual machines to other hosts in the cluster and placing the original host into standby mode to conserve power. If the capacity is insufficient, VMware DPM might recommend returning standby hosts to a powered-on state. • If you select the **Disable DPM on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables DPM after remediation is complete. • If you deselect the **Disable DPM on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which DPM is enabled. By default, the **Disable DPM on the cluster** check box is selected. You can configure this setting only for a cluster that you manage with an image.

Setting	Description
HA admission control	Admission control is a policy that vSphere HA uses to ensure failover capacity within a cluster. If vSphere HA admission control is enabled during remediation, vMotion might be unable to migrate the virtual machines within the cluster. Deactivating admission control allows a virtual machine to be powered on even if it causes insufficient failover capacity. When this happens, no warnings are presented, and the cluster does not turn red. If a cluster has insufficient failover capacity, vSphere HA can still perform failovers, and uses the VM Restart Priority setting to determine which virtual machines to power on first. **Note:** Deactivating HA admission control before you remediate a two-node cluster causes the cluster to practically lose all its high availability guarantees. • If you select the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables HA admission control after remediation is complete. • If you deselect the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which HA admission control is enabled. By default, the **Disable HA admission control on the cluster** check box is deselected. You can configure this setting only for a cluster that you manage with an image.
Configuration Profile-Specific Settings	
Serial Remediation	Enabling serial remediation allows you to remediate all ESX hosts in a cluster against the desired configuration sequentially. For example, during a Live Patch update, hosts that do not require reboot or to be put into maintenance mode, are remediated first, followed by hosts that require reboot or to be put into maintenance mode. All hosts are remediated one at a time. You can configure this setting only for a cluster that you manage with an image and vSphere Configuration Profiles is enabled.
Auto-remediate newly added hosts	You can configure how vSphere Configuration Profiles behaves when a new host is added to a cluster that has vSphere Configuration Profiles enabled. If auto-remediation is activated, any newly added hosts are automatically remediated according to the cluster configuration. Whenever the settings of a host differ from the cluster configuration, these differences are stored in the cluster configuration as host overrides. When a host is added to a cluster with the auto-remediation setting activated, vSphere Configuration Profiles pauses for one minute to see whether additional hosts will be added to that cluster before starting the remediation. **Note:** If any hosts are in maintenance mode when added to a cluster, after a successful auto-remediation, the hosts are put into maintenance mode again. By default, the auto-remediation option is deactivated.

These settings become the default remediation settings for clusters that have vSphere Configuration Profiles enabled. For any cluster, you can change the default settings when you configure individual remediation tasks.

Override the Global Remediation Settings for Clusters That Have vSphere Configuration Profiles Enabled

You can customize the remediation settings for a particular cluster while the global remediation settings remain intact and applicable to all other clusters that havevSphere Configuration Profiles enabled and that are in the same vCenter instance.

- Verify that you have the proper privileges. See Required Privileges for Using vSphere Configuration Profiles.

When you override the global remediation settings for a single cluster, the overrides are used during the remediation of that specific cluster. All other clusters in the vCenter instance continue to use the global remediation settings.

1. In vSphere Client, navigate to a cluster that has vSphere Configuration Profiles enabled.
2. On the **Configure** tab, select **Desired State** > **Configuration**.
3. In the **Configuration** card, click the **Compliance** tab and click **Edit remediation settings**.
4. In the **Edit Cluster Remediation Settings** dialog box, configure the remediation settings of the target cluster under the **Configuration Profile specific settings** section.

Setting	Description
Common Settings	
Quick Boot	Quick Boot reduces the host reboot time during remediation. By default, Quick Boot is turned on. You can configure this setting for a cluster or a standalone host.
VM power state	The **VM power state**option lets you control the behavior of the virtual machines that run on the ESX host. You can select from the following options. • **Do not change power state** • **Suspend to disk** • **Suspend to memory** To select the **Suspend to memory** option, you must turn on Quick Boot. Otherwise, the **Suspend to memory** option is dimmed. Together with Quick Boot, the **Suspend to memory** option provides faster host upgrades. vSphere Lifecycle Managersuspends to the host memory and not to the disk the powered on virtual machines on the host. After the Quick Boot, the suspended virtual machines are resumed from memory. • **Power off** The default selection is **Do not change power state**. You can configure these options for a cluster or a standalone host.
VM migration	You can configure vSphere Lifecycle Manager to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster. By default, the **Migrate powered off and suspended VMs to other hosts in the cluster** check box is selected. If you have virtual machines with passthrough devices on a host in a DRS cluster, make sure that you automate the migration of such virtual machines. For more information, see the VMware knowledge base article at KB 318908. You can configure this setting only for a cluster that you manage with an image.
Retry policy	You can configure how vSphere Lifecycle Manager behaves if a host fails to enter maintenance mode or if staging or remediation for that host fails. You can configure vSphere Lifecycle Manager to wait for a specified retry delay period and to retry to put the host into maintenance mode as many times as you indicate in the **Number of retries** text box. By default, vSphere Lifecycle Manager reties 3 times to put the host into maintenance mode over a period of 5 minutes. You can configure this setting for a cluster or a standalone host.
DPM	VMware Distributed Power Management (DPM) monitors the resources consumed by the running virtual machines in the cluster. If sufficient excess capacity exists, VMware DPM recommends moving virtual machines to other hosts in the cluster and placing the original host into standby mode to conserve power. If the capacity is insufficient, VMware DPM might recommend returning standby hosts to a powered-on state. • If you select the **Disable DPM on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables DPM after remediation is complete. • If you deselect the **Disable DPM on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which DPM is enabled. By default, the **Disable DPM on the cluster** check box is selected. You can configure this setting only for a cluster that you manage with an image.

Setting	Description
HA admission control	Admission control is a policy that vSphere HA uses to ensure failover capacity within a cluster. If vSphere HA admission control is enabled during remediation, vMotion might be unable to migrate the virtual machines within the cluster. Deactivating admission control allows a virtual machine to be powered on even if it causes insufficient failover capacity. When this happens, no warnings are presented, and the cluster does not turn red. If a cluster has insufficient failover capacity, vSphere HA can still perform failovers, and uses the VM Restart Priority setting to determine which virtual machines to power on first. **Note:** Deactivating HA admission control before you remediate a two-node cluster causes the cluster to practically lose all its high availability guarantees. • If you select the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager remediates the hosts in the cluster and re-enables HA admission control after remediation is complete. • If you deselect the **Disable HA admission control on the cluster** option, vSphere Lifecycle Manager skips remediating the clusters on which HA admission control is enabled. By default, the **Disable HA admission control on the cluster** check box is deselected. You can configure this setting only for a cluster that you manage with an image.
Configuration Profile-Specific Settings	
Serial Remediation	Enabling serial remediation allows you to remediate all ESX hosts in a cluster against the desired configuration sequentially. For example, during a Live Patch update, hosts that do not require reboot or to be put into maintenance mode, are remediated first, followed by hosts that require reboot or to be put into maintenance mode. All hosts are remediated one at a time. You can configure this setting only for a cluster that you manage with an image and vSphere Configuration Profiles is enabled.
Auto-remediate newly added hosts	You can configure how vSphere Configuration Profiles behaves when a new host is added to a cluster that has vSphere Configuration Profiles enabled. If auto-remediation is activated, any newly added hosts are automatically remediated according to the cluster configuration. Whenever the settings of a host differ from the cluster configuration, these differences are stored in the cluster configuration as host overrides. When a host is added to a cluster with the auto-remediation setting activated, vSphere Configuration Profiles pauses for one minute to see whether additional hosts will be added to that cluster before starting the remediation. **Note:** If any hosts are in maintenance mode when added to a cluster, after a successful auto-remediation, the hosts are put into maintenance mode again. By default, the auto-remediation option is deactivated.

5. Click **Save**.

These settings become the remediation settings for the selected cluster. vSphere Configuration Profiles uses those settings for that cluster for all future remediation tasks. The global remediation settings remain unchanged and are applied to all other clusters.

In the **Configuration** card, on the **Compliance** tab vSphere Configuration Profiles displays a message that the global remediation settings are overridden. Also, an option to reset the values appears on the tab.

How to Enable vSphere Configuration Profiles on a Cluster

You can enable vSphere Configuration Profiles during the process of creating a cluster or after the cluster is created.

Enable vSphere Configuration Profiles During Cluster Creation

You can enable vSphere Configuration Profiles during the process of creating a cluster. However, in this case, there are some steps which you must perform to create a desired cluster configuration.

When you create a cluster managed with an image, it is created with a default configuration. You must define the desired configuration for the cluster by editing the default settings after the cluster is successfully created. Adding hosts beforehand or using a reference host is optional.

The overall workflow for enabling vSphere Configuration Profiles on a cluster that you manage with an image and setting up a desired configuration is the following:

1. Create a cluster. For more information, see Managing the Lifecycle of Clusters with vSphere Lifecycle Manager Images at Cluster Level.
2. Set up the desired configuration for the cluster.
   You can use vSphere Client to create a draft configuration and edit the host configuration settings for a cluster in the user interface.
   - Option 1: Create a draft configuration by using one of the following options:
     - Use the settings of the default configuration for the cluster.
     - Reuse the configuration from an existing cluster by exporting its configuration document and importing it into the created draft configuration. See Import a Configuration Document or a Configuration from a Reference Host Into an Existing Draft Configuration
     - Use one of the hosts in the cluster as a reference host to create the desired cluster configuration. The desired cluster configuration is compliant with all hosts in the cluster.

     After creating the draft configuration, edit the configuration settings to achieve the desired configuration for the cluster. For more information, see Create a Draft Configuration for a Cluster in vSphere Client and Edit the Host Settings of a Draft Configuration in vSphere Client.
   - Option 2: Export the configuration schema for the newly created cluster and use it to create your own JSON configuration document, which you can import into the cluster.
     The exported configuration schema document follows the JSON Schema format. You use it to create and edit your own JSON configuration document in an external JSON editor tool.
   - Option 3: Export the default or the draft configuration document of the cluster, edit the JSON file manually, and import it back into the cluster. You can also export and use the default or the draft configuration schema for the cluster to facilitate yourself in editing the desired configuration document.
   - Option 4: Extract the settings from a host in the vCenter inventory and import them into the newly created cluster to use them as the common desired configuration.

The overall workflow for enabling vSphere Configuration Profiles on a cluster that you manage with baselines and setting up a desired configuration is the following:

1. Create a cluster and add at least one host to the cluster. For more information about creating a cluster, adding and removing hosts to and from a cluster, see the *vCenter and Host Management* documentation.
2. Set up the desired configuration for the cluster.
   - Option 1: Create a draft configuration by using one of the hosts in the cluster as a reference host. If any of the hosts in the cluster have settings that differ from the reference host, those settings are added as host overrides in the generated draft configuration.
     After creating the draft configuration, edit the configuration settings to achieve the desired configuration for the cluster. For more information, see Create a Draft Configuration for a Cluster in vSphere Client and Edit the Host Settings of a Draft Configuration in vSphere Client.
   - Option 2: Export the configuration schema for the newly created cluster and use it to create your own JSON configuration document, which you can import into the cluster.
     The exported configuration schema document follows the JSON Schema format. You use it to create and edit your own JSON configuration document in an external JSON editor tool.
   - Option 3: Export the draft configuration document of the cluster, edit the JSON file manually, and import it back into the cluster. You can also export and use the default or the draft configuration schema for the cluster to facilitate yourself in editing the desired configuration document.

Enable vSphere Configuration Profiles on an Existing Cluster

For any cluster that you manage with an image or with baselines, you can start using vSphere Configuration Profiles. The workflow allows you to enable vSphere Configuration Profiles on an existing cluster and set up the desired configuration for the cluster.

- Verify that vCenter is of version 9.0 and later.
- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that if the cluster uses vSphere Distributed Switch and you want to manage the vSphere Distributed Switch configuration with vSphere Configuration Profiles, ESX hosts are of version 8.0 Update 3 and later.
- Verify that both vCenter and ESX are of version 9.0 and later if you want to use NSX with vSphere Configuration Profiles.
- Verify that all hosts in the cluster are compliant with the image for the cluster if the cluster is managed with an image.
- Verify that the ESX hosts in the cluster have the required license.
- Verify that you have the required privileges for the operation. See Required Privileges for Using vSphere Configuration Profiles.

The vSphere Configuration Profiles enablement workflow for a cluster that you manage with an image starts with eligibility checks on the cluster and hosts. If the cluster is not eligible, the vSphere Configuration Profiles enablement workflow cannot proceed. If the cluster is eligible, user can import the cluster configuration using a host in the cluster as a reference host or using a JSON file. The configuration is then validated. You can edit the configuration at this stage by using the export and import options. vSphere Configuration Profiles runs a pre-check to ensure that the configuration can be applied to all hosts. After you confirm the enablement of vSphere Configuration Profiles, eligibility checks are re-run and the cluster is remediated against the configuration that you set up.

If the cluster that you manage with an image is empty at the time you enable vSphere Configuration Profiles, a predefined configuration is created. You can then add hosts to the cluster and edit the hosts configuration settings.

You can enable vSphere Configuration Profiles on a cluster that you manage with baselines. The vSphere Configuration Profiles enablement workflow starts with selecting a reference host whose configuration schema is imported and used as a desired cluster configuration schema.

You can enable vSphere Configuration Profiles on a cluster that uses a vSphere Distributed Switch.

> **Note:**
> vSphere Configuration Profiles can manage the vSphere Distributed Switch configuration of the hosts in a cluster that is managed with an image.

To manage any drifts related to the vSphere Distributed Switch configuration, you can use both options:

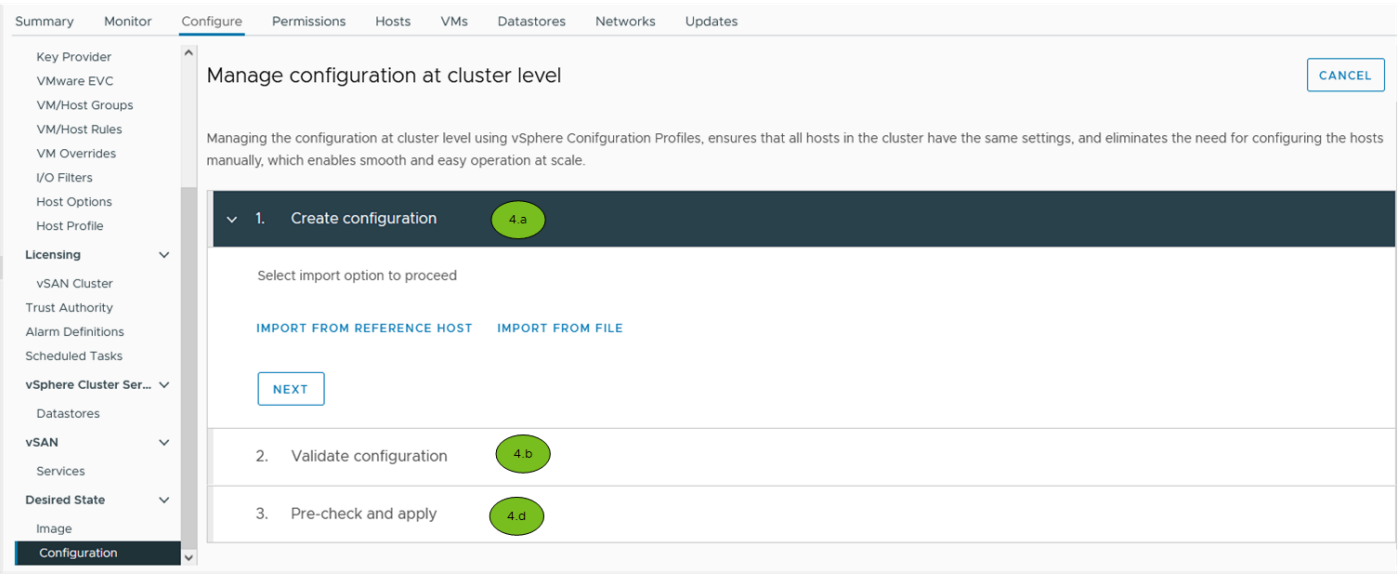
- The vSphere Distributed Switch workflows described in the *vSphere Networking* documentation.
- The vSphere Configuration Profiles workflows described in this chapter.

vSphere Configuration Profiles can be enabled on a cluster that has NSX or vSAN enabled. You can use vSphere Configuration Profiles to view and manage the NSX configuration properties. vSphere Configuration Profiles does not manage the vSAN configuration. Use the existing vSAN APIs to configure the vSAN settings.

1. In the vSphere Client, navigate to a cluster that you manage with an image or a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. Click **Create configuration**.

   The **Check Cluster Eligibility for Transition** task starts to ensure that the cluster is eligible for enabling the vSphere Configuration Profiles.

4. In the **Manage configuration at cluster level** pane, follow the prompts to set up a desired configuration for the cluster and apply it to all hosts.



a) Under **Create configuration**, select an import option and click **Next**.

- To use the configuration from a reference a host within the cluster, click **Import from reference host**, select the reference host, and confirm the import.
  The configuration of the selected host is extracted and compared to the configurations of all hosts in the cluster. If any differences are detected, they are recorded as host overrides in the generated cluster configuration.

Note:
This option is available for a cluster that you manage with baselines and for a cluster that you manage with an image.

- To import the configuration from a JSON file that you created or edited, click **Import from file**, locate the JSON file with the configuration that you want to import and confirm the import.

Note:
This option is only available for a cluster that you manage with an image.

The **Validate Configuration for Transition** task starts to ensure that the imported configuration is valid and can be used as the desired configuration for the cluster.

b) Optional: Under **Validate configuration**, view the validation messages and edit the imported configuration.

- Click **Export configuration** to download the current configuration document and edit its settings.
- Click **Export configuration schema** to download the JSON file that represents the configuration schema.
- Click **Import configuration** to import the edited configuration document back into the cluster.

c) Click **Next**.

The **Precheck Configuration for Transition** task starts to ensure that the configuration can be applied to all hosts successfully.

d) Under **Pre-check and apply**, view pre-check details and remediation impact and click **Finish and apply**.

e) In the **Finish and apply** dialog box, confirm that you want to apply the configuration to all hosts in the cluster.

vSphere Configuration Profiles is enabled for the cluster and the cluster is remediated against the configuration you set up during this operation.

f) Optional: Click **Go to configuration** to view and edit the cluster configuration settings.

You can cancel the operation at any stage of the workflow. Later, if you did not discard the changes that you made during the vSphere Configuration Profiles enablement workflow, you can resume the operation and use the configuration imported before canceling the operation. If you want to start the whole process anew, discard all your changes. When you discard your changes, the imported configuration is also deleted.

The cluster now uses vSphere Configuration Profiles to manage the configuration of its hosts. All hosts in the cluster are compliant with the desired configuration that you set up during the vSphere Configuration Profiles enablement operation.

View the current desired cluster configuration. For more information, see View Host Settings in the Desired Configuration. You can also create a draft configuration and edit the host settings in the vSphere Client. For more information, see Create a Draft Configuration for a Cluster in vSphere Client.

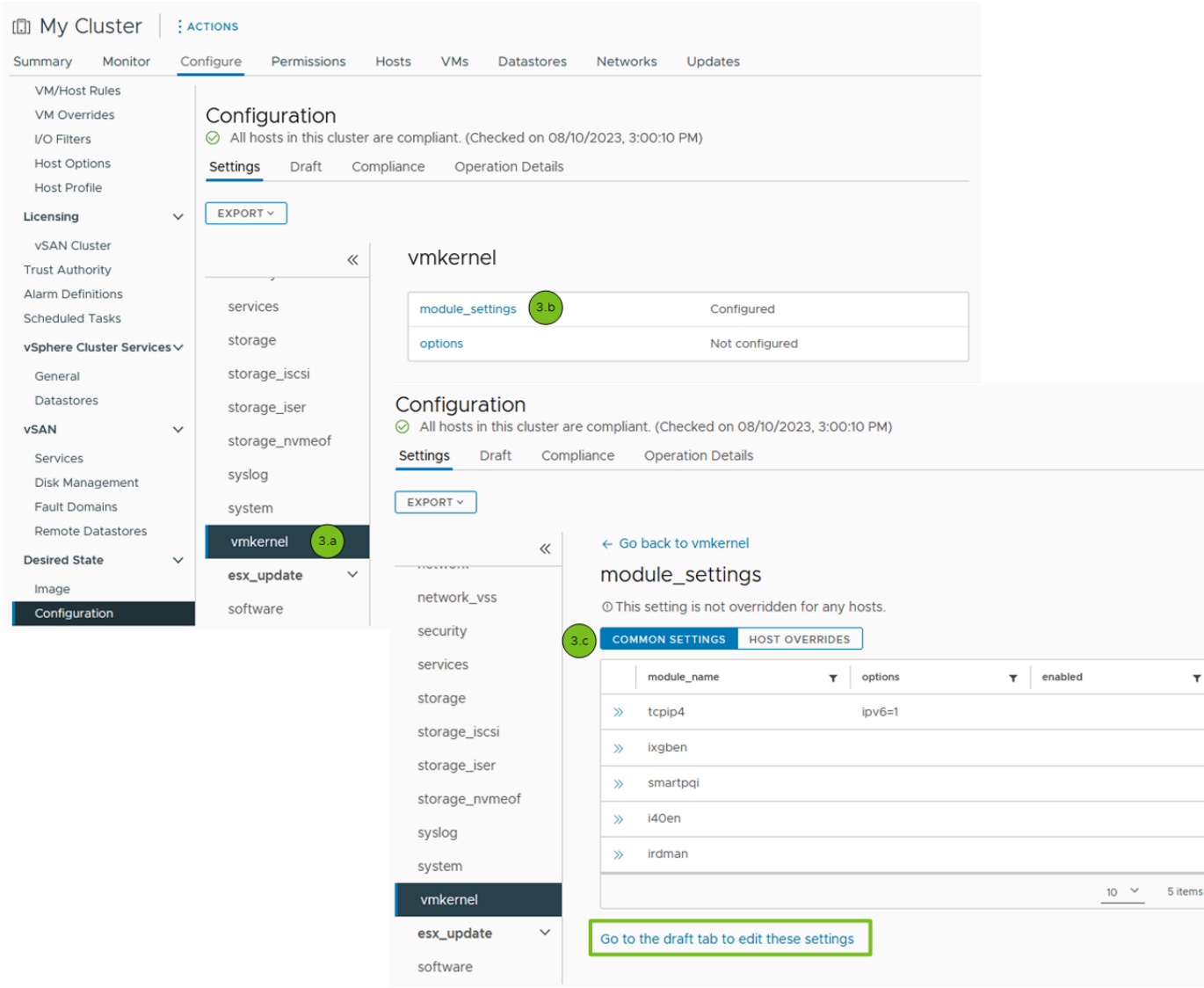
View Host Settings in the Desired Configuration

You can view information about how each host setting is configured in the current desired configuration for the cluster. For each host setting defined in the configuration document, you can see the values as defined for all hosts in the cluster and also any host overrides.

- Verify that vCenter is of version 9.0 and later.
- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you have the required privileges for viewing host settings in the desired configuration. See Required Privileges for Using vSphere Configuration Profiles.

The common settings, the host-specific settings, and the host overrides can be configured manually in the configuration document or in the vSphere Client. In the vSphere Client, you can view both, the settings of the currently applied desired configuration and the editable settings of the draft configuration of a cluster.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Settings** tab, view information about each setting defined in the current desired configuration



document.

a) Select a group of settings.
b) In the panel on the right, select a setting that is configured.
c) To view the common configuration for the selected setting, click **Common configuration**.
d) To view host overrides for the selected setting, select **Host overrides** and select the host for which you want to see the overrides.

Create a draft cluster configuration and edit the host settings by clicking the **Go to the draft tab to edit these settings** link. For more information, see Create a Draft Configuration for a Cluster in vSphere Client.

How Host Profile Configuration Settings Map to the vSphere Configuration Profiles Settings for ESX Hosts

The following table presents the mapping between the host configuration settings that are managed with vSphere Host Profile and their corresponding settings in vSphere Configuration Profiles.

Host profiles are available under **Menu** > **Policies and Profiles** > **Host Profiles**. For more information about how to use host profiles, see the *vSphere Host Profiles* documentation.

Note:

Each setting that is configurable through vSphere Configuration Profiles has a human readable path that is shown in the UI when needed.

For example, if you changed the welcome message for a cluster, after you run a compliance check for the cluster, you can see the `/profile/esx/system/welcome_message` setting location listed in the table showing the drifts.

vSphere Host Profile Setting				vSphere Configuration Profiles Setting		
Level 1	**Level 2**	**Level 3**	**Level 4**	**Component**	**Group**	**Key**
Advanced Configuration Settings	Advanced Option	Advanced Configuration Option	`Config.Etc.motd`	esx	system	`etc_motd`
			`Config.HostAgent.level[Vmsvc].logLevel`		services	`hostd`
			`Misc.HeartbeatTimeout`		advanced_options	`misc`
			`Misc.LogPort`			`misc`
			`OSData.configuredLocation`		storage	Not supported.
			`RdmFilter.HbalsShared`		services	`hostd` Enable the **Show advanced settings** option and navigate to **plugins** > **solo** > **rdmfilter** > **hba_is_shared** setting.
			`ScratchConfig.ConfiguredScratchLocation`		storage	`scratch_location`
			`Security.AccountLockFailures`		security	`settings` > `account_lock_failures`
			• `Syslog.global.logDir` • `Syslog.global.logHost`		syslog	• `global_settings` > `log_dir` • `global_settings` > `remote_logging` > `log_host`
			`Syslog.loggers.*.rotate` `Syslog.loggers.*.size`			`logger_settings`
	AgentVM Configuration			This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	Configuration Files	vCenter agent (vpxa) configurations		esx	services	`vpxa`

vSphere Host Profile Setting				vSphere Configuration Profiles Setting		
Level 1	**Level 2**	**Level 3**	**Level 4**	**Component**	**Group**	**Key**
		Feature State Switch Profile		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
		VMTools Configuration Settings				
		/etc/vmware/config		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	DirectPath IO Configuration			esx	hardware	`pci_devices`
	Graphics Configuration			esx	graphics	• `devices` • `policy`
	HostProfiles Log Configuration			This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	Hosts File Configuration	/etc/hosts file settings		esx	network	`etc_hosts`
	Power System Configuration			esx	advanced_options	`power`
		System Image Cache Configuration		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
General System Settings	Console Configuraiton	DCUI Keyboard		esx	system	`keyboard`
	Core Dump Configuration	Coredump File		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	Date and Time Configuration	Date and Time Configuration		esx	system	`system_time`
	Device Alias Configuration	Device Alias Configuration				Not supported
	Host Cache Configuration	Host Cache Settings				Not supported
	HostSpec Configuration	HostSpec Profile		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	Kernel Module Configuration	Kernel Module Configuration		esx	vmkernel	`module_settings`
	Management Agent Configuration	Management Agent Configuration			system	`snmp`
	Software Device Configuration	Software Device Configuration				`software_devices`
	System Resource Pool Configuration	Resource Pool		This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
	System Swap Configuration	System Swap Settings		esx	system	`system_swap`

vSphere Host Profile Setting				vSphere Configuration Profiles Setting		
Level 1	**Level 2**	**Level 3**	**Level 4**	**Component**	**Group**	**Key**
	vFlash Host Swap Cache Configuration	vFlash Host Swap Cache Settings				`system_swap > host_cache`
	Networking Configuration	Standard Switch			network_vss	`switches`
		Virtual Machine Portgroup			network_vss	`switches > vSwitch0 > port_groups > VM Network`
		Host Portgroup			network_vss	`switches > vSwitch0 > port_groups > Management Network`
		Physical Nic Configuration			network	`nics`
		vSphere Distributed Switch			network_vds	`switches`
		Host virtual nic			network	`vmknics`
		NSX Host vNIC		Not supported		
		NSX-T OpaqueSwitch		Not supported		
		Netstack Instance		esx	network	`net_stacks`
		Network Coredump Settings			system	`netdump`
Other	ESXInfo			This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
Security and Services	Firewall Configuration	Firewall Configuration	Ruleset Settings	esx	network	• `firewall` • `firewall_rule_sets`
	Security Settings	Authentication Configuration	Active Directory configuration		identity	• `active_directory` • `smart_card_authentication`
		Host Acceptance Level Configuration		esx_update	software	`acceptance_level`
		Domain Settings		esx	security	`domain_settings`
		Security	Role		authorization	`permissions`
			User Configuration		authentication	`user_accounts`
			Active Directory Permission	This is not considered a user-configurable setting. Therefore, vSphere Configuration Profiles does not support it.		
			Lockdown Mode	esx	authorization	`lockdown_settings`
	Service Configuration	Service Configuration	DCUI	Not supported		
			TSM	Not supported in this release		

vSphere Host Profile Setting				vSphere Configuration Profiles Setting		
Level 1	**Level 2**	**Level 3**	**Level 4**	**Component**	**Group**	**Key**
			TSM-SSH	Not supported in this release		
			API Forwarder	Not supported		
			DPD	Not supported		
			entropyd	Not supported		
			GPU Manager	Not supported		
			hbrsrv	Not supported		
			infravisor	Not supported		
			lbtd	Not supported		
			ntpd	Not supported		
			nvmf-discoveryd	Not supported		
			pcscd	Not supported		
			ptpd	Not supported		
			sfcbd-watchdog	Not supported		
			slpd	Not supported		
			vdtc	Not supported		
			vltd	Not supported		
			vmsyslogd	Not supported		
			xorg	Not supported		
Storage Configuration	NFS Storage Configuration			esx	storage	• `nfs_v3_datastores` • `nfs_v41_datastores`
	iSCSI Initiator Configuration	Independent Hardware iSCSI Adapters			storage_iscsi	`hardware_adapters`
		Software iSCSI Initiator				`software_adapter`
		Dependent Hardware iSCSI Adapters				`hardware_adapters`
	NFS UserHost Configuration				storage	`nfs_krb_credentials`
	Pluggable Storage Architecture (PSA) Configuration	PSA device sharing				`block_devices`
		PSA claimrule number				• `psa_filter_claim_rules` • `psa_mpp_claim_rules` • `psa_vaai_claim_rules`
		Host boot device configuration				block_devices
		PSA device setting				`block_devices`

vSphere Host Profile Setting				vSphere Configuration Profiles Setting		
Level 1	**Level 2**	**Level 3**	**Level 4**	**Component**	**Group**	**Key**
		PSA device configuration				`block_devices`
		PSA device inquiry cache				`block_devices`
	Native Muti-Pathing (NMP)	Storage Array Type Plugin (SATP) configuration	SATP Claimrule			`nmp_claim_rules`
			SATP default PSP configuration			`nmp_claim_rules`
		PSP and SATP configuration for NMP devices				`satp_default_psps`
		Path Selection Policy (PSP) configuration				`satp_default_psps`
	Virtual Volumes (VVOL)	VASA Provider configuration		Not supported		
		Storage Container configuration		Not supported		
		Vvol VASA Context configuration		Not supported		
	vSAN Configuration	VMkernel Network Adapter Configuration		Not supported		

Common ESX Configuration Settings

Learn how to use vSphere Configuration Profiles to manage some of the most common host configuration settings at cluster level.

To edit the configuration settings of the hosts in a cluster with vSphere Configuration Profiles, you must have a draft configuration. On the **Configuration** page, select the **Draft** tab and navigate to the settings you want to edit by choosing the component, then group and key from the page.

ESX Security Settings

You use the ESX security settings to configure all hosts in a cluster to comply with the security policies of your organization. For example, you can configure some of the following options:

- Lock an account after a specific number of failed log-in attempts.
- Configure password history setting to restrict the reuse of passwords on the ESX hosts.
- Configure an appropriate maximum password age of the ESX hosts.
- Change the restrictions on passwords and pass phrases.

To edit the values of the ESX security settings, select **esx** > **security** > **settings** and click **Configure settings**.

Security Setting	Description	Default Value
account_lock_failures	The maximum number of failed login attempts before the user account is locked. If you set `0` as the value to this setting, you deactivate account locking.	`5`
account_unlock_time	The number of seconds that a user is locked out.	`900` (15 minutes)
password_quality_control	The restrictions on passwords or pass phrases which ESX enforces for access from the Direct Console User Interface (DCUI), the ESX Shell, SSH, or the VMware Host Client. The format of the password follows the pam_passwdqc password strength checking module for PAM.	`random=0 retry=3 min=disabled,disabled,disabled,7,7` By default, you must include a mix of at least three of the four character classes: lowercase letters, uppercase letters, numbers, and special characters such as underscore or dash. By default, password length must be at least 7 characters and pass phrases are disallowed. By default, a user is prompted 3 times to provide a sufficiently strong password and to enter it twice the first time.
password_history	The number of passwords to remember for each user. If you set `0` as the value to this setting, you deactivate password history.	`5`
password_max_days	The maximum days before a password must be changed.	`99999`
ssh_session_limit	The maximum number of SSH sessions for all users except for the 'root' user.	`50`
default_shell_access	The property that you can use to configure the default access to ESX Shell that a newly created user has.	`true` By default, newly created users have access to the ESX Shell.

Starting with vSphere 9.1, you can configure whether SSH can be enabled or not on a cluster level by configuring the **ssh_allowed_policy** setting. To edit the values of the SSH policy setting, select**esx** > **system** > **ssh_server** and click **Configure settings**.

Security Setting	Description	Default Value
ssh_allowed_policy	By default, the setting is not configured which allows SSH to be enabled by using DCUI, VMware Host Client, or vSphere Client. When set to `false`, you can enable SSH on a host only through DCUI.	

See Securing ESX Hosts in the vSphere Security documentation for more information about how to configure the ESX passwords, account lockouts, and SSH access.

ESX Firewall Settings

By default, each ESX host has a firewall that is enabled after installation. You use the ESX firewall settings to configure the firewall and firewall rules across all hosts in a cluster. You can configure the following options:

- Configure the firewall to block all network traffic by default.
- Configure the ESX firewall to only allow traffic from authorized networks.

To edit the firewall settings of all hosts in a cluster, select **esx** > **network** > **firewall** and **esx** > **network** > **firewall_rule_sets** and click **Configure settings**.

ESX Firewall Setting		Description	Default Value
esx > **network** > **firewall**	**enabled**	Activate or deactivate the ESX firewall.	`true`
	default_action	Set the default behavior of the ESX firewall to either allow or drop the traffic that does not match a specific ruleset.	`DROP`
esx > **network** > **firewall_rule_sets**	**name**	Set the name of the ruleset.	
	enabled	Activate or deactivate the ruleset.	
	allow_all_ip	Allow all IP addresses to access a specific firewall ruleset. This option reduces security significantly, so consider using it for troubleshooting only and for a short time.	`true`
	allowed_ips > **address**	Restrict access to specific IP addresses or a range of IP addresses.	

For more information about how to configure the ESX firewall settings for a single host, see the vSphere Security documentation.

Syslog Settings

Configure the syslog service (`vmsyslogd`) that runs on each ESX host in a cluster. You can configure the following options:

- Enable audit record logging.
- Configure the time for storing the audit records.
- Configure a persistent log location for all locally stored audit records on the ESX host.
- Enable log filtering to reduce the repetitive entries in the ESX logs.
- Transmit audit records to a remote host.
- Log sufficient information for the events.
- Configure a persistent log location for all locally stored logs on the ESX host.
- Configure a remote log server for the ESX host.
- Enable verifying the certificates for remote logging endpoints.
- Configure the ESX host to use strict x509 verification for TLS-enabled remote logging endpoints.

To edit the syslog settings of all hosts in a cluster, select **esx** > **syslog** > **audit_record_settings**, **esx** > **syslog** > **global_settings**, **esx** > **syslog** > **logfilters**, and **esx** > **syslog** > **logger_settings**, and click **Configure settings**.

Note:

A blank or omitted default value indicates that the property key is absent. The key is optional and non-required, so it is not assigned a default value or documented with one in the JSON schema.

Syslog Setting		Description	Default Value
audit_record_settings	**enable_local_audit_record_storage**	Activate or deactivate the audit record storage.	`false`
	prevent_record_dropping	Prevent dropping audit records when the connection with the remote syslog server is lost or when some local storage issues occur.	`false`
	local_storage_dir	Indicate the directory where local copies of the audit logs are stored. The directory must be within a persistent file system. Only the `/scratch` directory on the local file system is persistent across reboots.	`/scratch/auditLog`

Syslog Setting			Description	Default Value
	local_storage_dir_capacity		Set a limit on the local storage to ensure sufficient capacity in case the remote syslog is unavailable. Set a value from 4 through 100 MiB.	`4`
global_settings	**vsan_backing**		Store log files and the audit records on a vSAN cluster.	`true`
	log_dir_unique		Specify that the ESX host name is concatenated to the value of the **log_dir** setting (by default, `/scratch/log`).	`false`
	rotations		Set the maximum number of local log files kept on the ESX host before the oldest one is deleted.	`8`
	rotation_size_in_kb		Specify the maximum size of the log file in KB, before it is rotated.	`1024`
	drop_log_rotate		Set the maximum number of older dropped log files to keep.	`10`
	drop_log_size		Specify the maximum size, in KiB, of each dropped log file before switching to a new one.	`100`
	queue_drop_mark		Specify the log entries queue capacity as a percentage after which logs are dropped.	`90`
	log_dir		Set a directory on a datastore of the local or remote ESX host.	`/scratch/log`
	log_level		Specify the general log level. Change this parameter only when troubleshooting an issue.	`INFO`
	remote_logging	**log_host**	Specify the remote host to which syslog messages are forwarded and the port on which the remote host receives the messages. Select between the UDP, TCP, and SSL protocols. You can use the default 514 port for UDP/TCP and 1514 for SSL.	`NOT_SPECIFIED` (not used by default) No hard limit for the number of remote hosts is set but it is recommended to configure five or less hosts.
		crl_check	Enable the Certificate Revocation List (CRL) checking to verify whether the remote host certificate is still valid.	`false`
		x509_strict	Enable strict compliance with X.509.	`false`
		check_ssl_certs	Enforce checking of the SSL certificates when you log in to a remote host.	`true`
		retry_timeout	Set the delay in seconds before retying to connect to a remote host after a connection attempt fails.	`180`
	remote_host_max_msg_len		Specify the maximum length of a syslog transmission in bytes. Set a value between 1024 through 16384 bytes.	`1024`
logfilters	**enable_logfilters**		Activate or deactivate the use of log filtering.	`false`

Syslog Setting		Description	Default Value
	filters	Specify one or more log filtering specifications. Use the following format to enter log filters: `numLogs | ident | logRegexp`. • `numLogs` - The maximum number of log entries before filtering and excluding from the system logs. • `ident` - The system components to apply the filter to. For example, you can set `vodb`, `hostd`, `vmkernel`, or `*` for all components. • `logRegexp` - A log expression conforming to the Python regular expression syntax that must be excluded.	
logger_settings	**logger**	Set the ID of the logger that you configure.	
	rotation_size_in_kb	Specify the maximum size of logs, in KiB, before rotation for the specified logger. Set a value between `1` and `102400` KiB.	
	rotations	Set the maximum number of log files that the specified logger must keep before deleting the oldest log file. Set a value between `0` and `100`.	

For more information about how to configure the ESX syslog settings for a single host, see the ESX Installation and Setup documentation.

ESX Lockdown Mode Settings

Configure the lockdown mode behavior of all hosts in a cluster and the users who have access privileges in lockdown mode. You can configure the following options:

- Add users to the DCUI access list.
- Specify the list of lockdown mode exception users.
- Enable normal or strict lockdown mode on all hosts.

To edit the lockdown mode settings of all hosts in a cluster, select **esx** > **authorization** > **dcui_access** and **esx** > **authorization** > **lockdown_settings** and click **Configure settings**.

Authorization and Lockdown Setting		Description	Default Value
dcui_access	**users**	Specify the uses who can access the hosts in a cluster when they are not accessible from vCenter and can change their lockdown mode settings.	
lockdown_settings	**mode**	Set `normal` or `strict` lockdown mode to apply the lockdown mode behavior to all hosts in a cluster. For more information about the lockdown mode options, see Configuring and Managing Lockdown Mode on ESX Hosts in the vSphere Security documentation.	

Authorization and Lockdown Setting		Description	Default Value
	exception_users	Specify the list of users who do not lose their access when the hosts enter lockdown mode. For more information about how to specify lockdown exception users, see Configuring and Managing Lockdown Mode on ESX Hosts in the vSphere Security documentation.	

For more information about lockdown mode, see the vSphere Security documentation.

ESX Advanced Settings

Configure some of the advanced system settings for all hosts in a cluster. You can configure the following options:

- Allow warnings about unmitigated hyperthreading vulnerabilities.
- Set a timeout to automatically terminate idle DCUI sessions on the ESX host.
- End idle host client sessions automatically.
- Deactivate virtual hardware management network interfaces.
- Enable volatile key destruction.
- Enable the Bridge Protocol Data Unit (BPDU) filter on the ESX host.
- Restrict the use of the dvFilter network APIs on the ESX host.
- Configure the inactivity timeout to automatically terminate idle ESX host shells.
- Set a timeout to limit how long the ESX Shell and SSH services are allowed to run.
- Allow warnings that the ESX shell is enabled.
- Enable the highest version of TLS supported.

To edit the advanced options settings for all hosts in a cluster, , select **esx** > **advanced_options** > **net**, **esx** > **advanced_options** > **mem**, or **esx** > **advanced_options** > **user_vars**, and click **Configure settings**.

ESX SSH Daemon Settings

Configure the SSH daemon (`sshd`) to minimize vulnerabilities and reduce attack surface when allowing the secure remote access to the ESX Shell on all hosts in a cluster. You can configure the following options:

- Use the FIPS 140-2/140-3 validated cryptographic modules.
- Use only the FIPS 140-2/140-3 validated ciphers.
- Prevent gateway ports on a ESX SSH daemon.
- Deactivate host-based authentication.
- Set a timeout count on the idle sessions.
- Set a timeout interval on the idle sessions.
- Configure the system login banner to be displayed before granting access to the system.
- Deactivate the use of `.rhosts` files.
- Deactivate the streaming local forwarding on the ESX SSH daemon.
- Deactivate the TCP forwarding.
- Prevent establishing tunnels.
- Configure the ESX SSH daemon to prohibit user environment settings.

To edit the ESX SSH daemon settings of all hosts in a cluster, select **esx** > **system** > **ssh_server** and click **Configure settings**.

ESX SSH Daemon Setting	Description	Default Value
macs	Configure the Message Authentication Code (MAC) algorithms that must be used in SSH connections to ensure authenticity and integrity of data exchange during SSH connections.	`hmac-sha2-256,hmac-sha2-512`
banner	Set the location to the banner file that holds the text that is displayed before authentication during an SSH login.	`/etc/issue`
ciphers	Configure the ciphers for the ESX SSH daemon that are used for secure communication.	`aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes192-ctr,aes128-ctr`
use_pam	Use the Pluggable Authentication Modules (PAM) with the ESX SSH daemon.	`true`
log_level	Set the logging level of the ESX SSH daemon.	`INFO`
print_motd	Configure ESX SSH daemon to show the message of the day (motd) after a successful SSH login.	`true`
rekey_limit	Set the maximum amount of data that may be transmitted and the maximum time that may pass before the session key is renegotiated.	`1G, 1H`
max_startups	Set the maximum number of concurrent unauthenticated connections that are allowed from the ESX SSH daemon.	`10:30:100`
gateway_ports	Disallow remote hosts to connect to forwarded ports on the ESX SSH daemon.	`NO`
ignore_rhosts	Configure the ESX SSH daemon to ignore `.rhosts` files to prevent security compromises.	`YES`
permit_tunnel	Disallow SSH tunnelling via the ESX SSH daemon to significantly decrease the attack surface of a host.	`NO`
kex_algorithms	Configure the Key Exchange (KEX) algorithms for the ESX SSH daemon to enable strong algorithms and deactivate deprecated or weak ones.	`ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256`
tcp_keep_alive	Configure the ESX SSH daemon to prevent inactive connections from being terminated due to inactivity timeout, for example.	`true`
syslog_facility	Specify the syslog facilities of the messages that are logged by the ESX SSH daemon to a remote syslog server.	`AUTH`
permit_root_login	Enable root login over the ESX SSH daemon.	`YES`
host_key_algorithms	Configure a range of host key algorithms which must be used to verify the identity of the host to the connecting SSH client.	`ecdsa-sha2-nistp256,ecdsa-sha2-nistp384,ecdsa-sha2-nistp521,rsa-sha2-256,rsa-sha2-512`

ESX SSH Daemon Setting	Description	Default Value
allow_tcp_forwarding	Activate the TCP forwarding for the ESX SSH daemon. By default, this setting is deactivated.	`NO`
client_alive_interval	Specify the period in seconds after which the ESX SSH daemon sends a message to check whether the SSH client is alive. The value to this setting can be between 0 and 65535	`200`
client_alive_count_max	Set the maximum number of client alive messages that can be sent without the ESX SSH daemon receiving any response from the SSH client. Set a value between 0 and 655355.	`3`
permit_user_environment	Prevent users from presenting environment options to the ESX SSH daemon which can potentially bypass access restrictions.	`false`
host_based_authentication	Activate host-based authentication that is based on a trust relationship established with a public key for the ESX SSH daemon.	`false`
allow_stream_local_forwarding	Configure the stream local forwarding in the ESX SSH daemon.	`NO`
challenge_response_authentication	Enable the challenge-response authentication with the ESX SSH daemon. The SSH server requests from the SSH client to respond to a challenge before granting authentication.	`true`

vSphere Standard Switch Settings

You can protect virtual traffic against impersonation and interception Layer 2 attacks by configuring the vSphere Standard Switch settings of all hosts in a cluster by using vSphere Configuration Profiles. Some of the options that you can configure are:

- Reject forged transmits for all vSphere Standard Switches or Standard Port Groups in a cluster.
- Reject guest MAC address changes for all vSphere Standard Switches or Standard Port Groups in a cluster.
- Reject guest promiscuous mode requests for all vSphere Standard Switches or Standard Port Groups in a cluster.
- Restrict the use of Virtual Guest Tagging (VGT) on the vSphere Standard Switches.

For more information about see the Securing vSphere Standard Switches article in the vSphere Security documentation.

To edit the vSphere Standard Switches settings of all hosts in a cluster, select **esx** > **network_vss** > **switches** and click **Configure settings**.

How to Manage a Desired Configuration for a Cluster

To change the configuration on the hosts in a cluster, you must first change the cluster desired configuration and then you must remediate the cluster using the created draft configuration.

Create a Draft Configuration for a Cluster in vSphere Client

When you want to change the desired configuration of a cluster that uses vSphere Configuration Profiles, you can create a draft configuration for that cluster in the vSphere Client.

- Verify that vCenter is of version 9.0 or later

- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you are using vSphere Configuration Profiles for the cluster.
- Verify that you have the required privileges for the operation. See Required Privileges for Using vSphere Configuration Profiles.

You can create only one draft configuration at a time for a cluster. When you finish changing the host settings, you must run a remediation pre-check. Then you may apply the changes to all hosts in the cluster thus making your draft configuration the new desired cluster configuration.

Note:

Multiple users with the required privileges can create draft configurations for a cluster simultaneously. However, only the first saved draft configuration becomes the new desired cluster configuration. If a user remediated the cluster with that desired cluster configuration, any other users editing the configuration for the same cluster receive a conflict error.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Draft** tab, choose the method of creating a draft configuration for the cluster.

Option	Steps
Create a copy of the current cluster configuration	To create a working copy from the current cluster configuration, click **Create draft**. • If you enabled vSphere Configuration Profiles during cluster creation, the draft configuration contains the default settings of the cluster that you manage with an image retrieved from the desired cluster image. • If you enable vSphere Configuration Profiles on a cluster that you manage with an image or a cluster that you manage with baselines, the draft configuration is created from the desired cluster configuration.
Import a configuration document into a cluster and edit it as a draft configuration	This option for creating a draft configuration is available only for clusters that you manage with an image. 1. On the **Draft** tab, select **Import from file**. The **Import Configuration from File** dialog box opens. 2. Click **Browse** to locate the file that you want to import and click **Import**. 3. Click **Close**.
Use the configuration from one of the hosts in the cluster as a reference configuration	Use this option to create the cluster configuration by using one of the hosts in the cluster as a reference host. **Note:** Importing host settings from a reference host deletes the current desired state configuration draft. **Tip:** If you enabled vSphere Configuration Profiles on a cluster that is managed with baselines, no default draft configuration is available for that cluster. You must instead import a configuration from a reference host within the cluster. 1. On the **Draft** tab, select **Import from host**.

Option	Steps
	The **Import from host** dialog box opens. 2. Choose how to handle host settings overrides. – Use the reference host settings as common settings and keep only the already existing host overrides from the current cluster configuration. During the import operation, the configuration of the reference host is extracted and is compared to the current cluster configuration. Only the existing host overrides in the current cluster configuration are preserved and the reference host settings become the common configuration settings for the cluster. You can view the host overrides for a specific host setting on the **Host overrides** tab for that setting or under the `host-override` section of the draft configuration document. – Use the reference host settings as common settings and add to the newly created cluster configuration all deviating host settings as overrides. During the import operation, the configuration of the reference host is extracted and is compared to all host configurations in the cluster. If a host in the cluster has configuration settings that differ from the settings of the reference host, these settings are marked in the draft configuration as host overrides. You can view the host overrides for a specific host setting on the **Host overrides** tab for that setting or under the `host-override` section of the draft configuration document. 3. Select one of the hosts that are listed for the cluster and click **Import**.

You created a draft configuration of the cluster which you can use to edit the host settings in the vSphere Client.

You can now edit the host settings within the draft configuration and can remediate the cluster against the draft configuration in a single operation. For more information, see Edit the Host Settings of a Draft Configuration in vSphere Client.

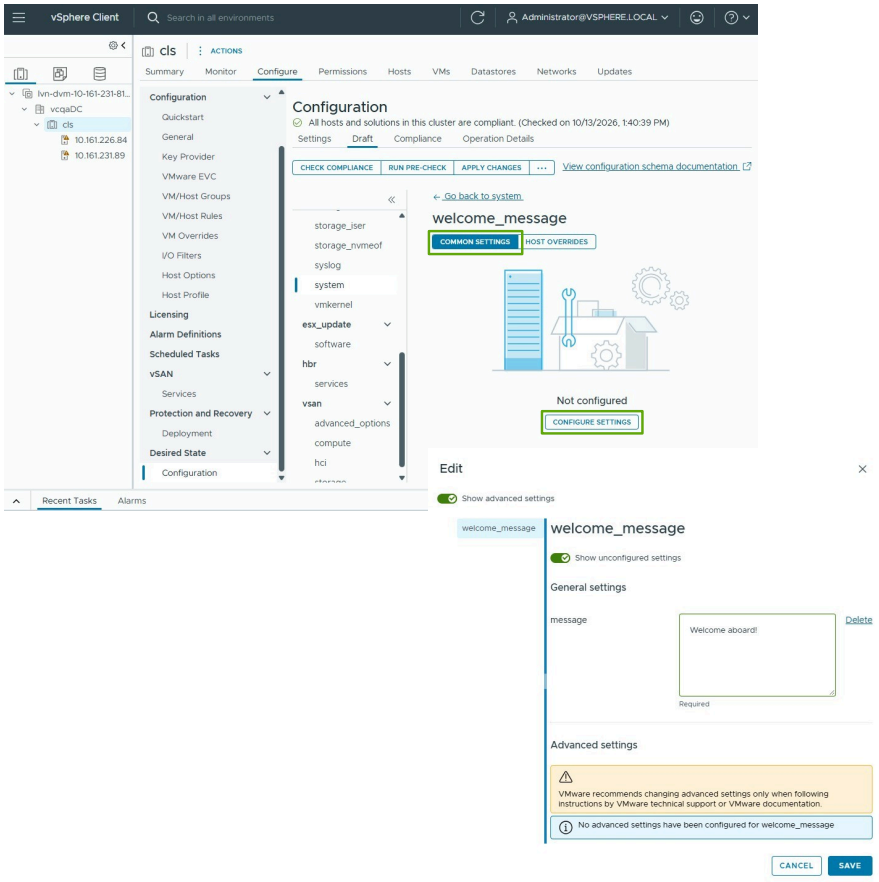
Edit the Host Settings of a Draft Configuration in vSphere Client

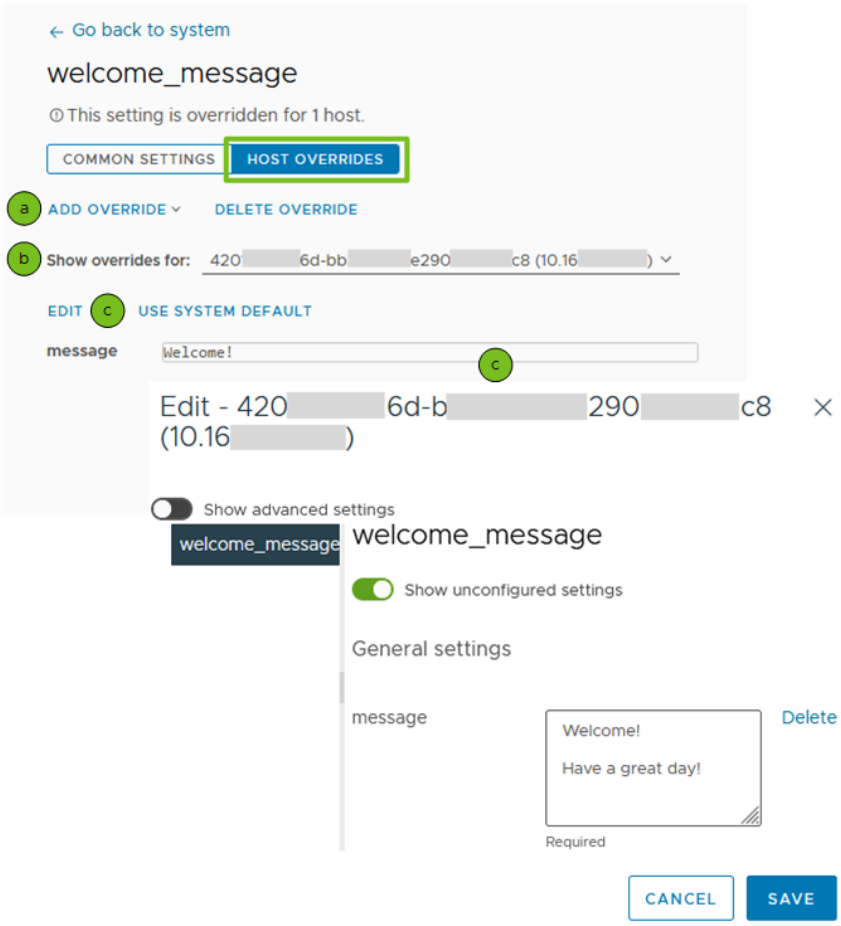
After you create a draft configuration for a cluster, you can edit the host settings in the vSphere Client.

- Verify that vCenter is of version 9.0 or later
- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you have a draft configuration that is created for the cluster.
- Verify that you have the required privileges for performing remediation. See Required Privileges for Using vSphere Configuration Profiles.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Draft** tab, navigate to the host setting that you want to edit.
4. Edit the host setting of the draft configuration.

Option	Steps
Configure a common setting that has no existing configuration	1. On the **Common settings** tab, click **Configure settings**. The **Edit** dialog box opens.

Option	Steps
	Figure 38: Configure a common setting  2. Select the common setting from the left sidebar and configure its value. The settings of the selected item appear on the right of the **Edit** dialog box. 3. (Optional) Activate the **Show advanced settings** toggle, to view and edit the advanced settings. If there is at least one advanced setting configured, the **Show advanced settings** toggle is automatically activated. 4. (Optional) Deactivate the **Show unconfigured settings** toggle, to view and edit only the configured settings. By default, the toggle is activated. 5. Click **Save**. The configured settings appear under the **Common settings** tab.
Edit a common setting that is already configured	1. On the **Common settings** tab, navigate to the current configuration for the selected setting and select it from the tree on the left. 2. Click **Edit**. The **Edit** dialog box opens. 3. Navigate to the configured setting by clicking the **View settings** links and edit the value of the setting. 4. Click **Save**.
Delete a common setting configuration	1. On the **Common settings** tab, select the setting that you want to delete from the list of configured settings and click **Delete**. 2. Click **Delete setting** to confirm the deletion of the setting configuration.
Add a host override for a configured common setting	1. On the **Host overrides** tab, click the **Add override** drop-down menu and select an option. – To add a host override by filtering the available hosts in the cluster using their host names, click **Add override by host name**. The **Add override for <setting_name>** dialog box opens. Select a host and click **Add override**.

Option	Steps
	– To add a host override by filtering the hosts using their BIOS UUID, click **Add override by BIOS UUID**. The **Add override for <setting_name>** dialog box opens. Enter the BIOS UUID of the host in the respective field and click **Add override**. You can use this option to add a host override for a host that is not currently part of the cluster. 2. From the **Show overrides for** drop-down menu, select the host override that you created. The configured common setting with its value appear under the drop-down menu. 3. Select from the options for editing. – To edit the current configuration of the common setting, click **Edit**. The **Edit - <BIOS UUID (<host_name>)>** dialog box opens. **Figure 39: Edit a host override for a configured common setting**  Navigate to the configured setting and change its value. – To apply the system default configuration which can be different from the common configuration of the setting, click **Use system default** and confirm your choice. 4. Click **Save**.

The settings of the draft configuration are configured.
You can view the changes between the current draft configuration and the desired configuration of the cluster by clicking **Show changes** on the **Draft** tab. You can run a draft pre-check to validate the draft configuration and ensure that it can be applied on all hosts in the cluster. For more information, see Run a Remediation Pre-Check or a Draft Pre-Check. You can also remediate the cluster against the draft configuration and make it the new desired cluster configuration. For more information, see Remediate a Cluster Against the Desired or Draft Configuration.

Working with Configuration Documents and Configuration Schemas

You can manually edit the configuration settings for a cluster by using the configuration document and the configuration schema available for the current configuration and for the draft configuration of the cluster.

What Are the Configuration Schema and the Configuration Document

You set up and manage the desired host configuration for an entire cluster by working with the configuration document for the respective cluster. The configuration document is a JSON file that you can download to your local machine and edit with a JSON editor tool.

You do not need to download the configuration document to edit it. Instead, you can directly edit the current cluster configuration in the vSphere Client. You create a working copy of the cluster configuration, called a draft, edit the configuration settings, and apply your changes to the cluster in a single operation. For more information, see Edit the Host Settings of a Draft Configuration in vSphere Client.

A valid configuration document contains a `profile` section and, optionally, a `host-specific` and `host-override` sections.

- The `profile` section contains common configuration that is applicable to all hosts in the cluster.
- The `host-specific` section represents configuration that can only be specified per host.
- The `host-override` section represents configuration that is overridden for a specific host in the cluster.

The configuration document is backed by a JSON schema, which is not editable. The configuration schema is a JSON file that represents the complete ESX configuration. The schema follows the JSON Schema format and contains the default values for all host properties. The configuration schema is generated from the desired software specification that is defined in the image for the cluster. The configuration schema changes when you change the software specification for the cluster.

Export the Cluster Configuration or Cluster Configuration Schema

You can export the current cluster configuration in the form of a JSON file, which you can manually edit. You can also export the configuration schema for the cluster and use it to help you create your own configuration document from scratch or edit an exported configuration document.

- Verify that vCenter is of version 9.0 or later.
- Verify that ESX hosts are of version 8.0 Update 1 or later.
- Verify that you have the required license.
- Verify that you have the required privileges for the export operation. See Required Privileges for Using vSphere Configuration Profiles.

1. In the vSphere Client, navigate to a cluster that you manage with an image or a baseline.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Settings** tab, select your task.
   - To download the current configuration for the cluster, from the **Export** drop-down menu, click **Cluster configuration**.
   - To download the configuration schema for the cluster, from the **Export** drop-down menu, click **Cluster configuration schema**.

   A dialog box appears that prompts you to download the JSON file that contains the current cluster configuration or the configuration schema.
4. Click **Download**.

The current configuration document or the configuration schema is saved locally on your machine.

Edit the configuration document or use the configuration schema to create a new JSON file with the desired host configuration for the cluster. Then, import the newly created or edited configuration document into the cluster.

Export the Draft Cluster Configuration or Draft Cluster Configuration Schema

You can export the draft cluster configuration in the form of a JSON file and manually edit it. You can also export the draft configuration schema for the cluster and use it to help you create your own configuration document from scratch or edit an exported configuration document.

- Verify that vCenter is of version 9.0 or later.
- Verify that ESX hosts are of version 8.0 Update 1 or later.
- Verify that you have the required license.
- Verify that you have a working copy of the configuration that is created for the cluster.
- Verify that you have the required privileges for the export operation. See Required Privileges for Using vSphere Configuration Profiles.

1. In the vSphere Client, navigate to a cluster that you manage with an image or a baseline.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Draft** tab, select your task.
   - To download the current draft configuration document for the cluster, click the horizontal ellipses icon and select **Export**.
   - To download the draft configuration schema for the cluster, click the horizontal ellipses icon and select **Export configuration schema**.

   A dialog box appears that prompts you to download the JSON file that contains the current draft cluster configuration or the draft configuration schema.
4. Click **Download**.

The current draft configuration document or the draft configuration schema is saved locally on your machine.

Edit the draft configuration document or use the draft configuration schema to create a new JSON file with the desired host configuration for the cluster. Then, import the newly created or edited draft configuration document into the cluster.

Extract The Configuration from a Reference Host

You can extract the configuration of any host in the vCenter inventory. The configuration is saved on your local machine as a JSON file, which you can edit. You can import the extracted configuration into any other cluster that has vSphere Configuration Profiles enabled.

- Verify that vCenter is of version 9.0 or later.
- Verify that ESX hosts are of version 8.0 Update 1 or later.
- Verify that you have the required license.
- Verify that you have the required privileges for the extract operation. See Required Privileges for Using vSphere Configuration Profiles.

1. In the vSphere Client, navigate to a cluster that you manage with an image or a baseline.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Settings** tab, click the **Export** drop-down menu and select **Reference host configuration**.

   The **Export Configuration** wizard opens.
4. Follow the prompts to download the configuration of the selected reference host.

   You can activate the **Show hosts from existing cluster** toggle to narrow down the list of hosts available in your vCenter to only the hosts available in the selected cluster.

The configuration of a selected reference host is exported and downloaded to your local machine as a JSON file.

Import the downloaded JSON configuration into another cluster to reuse the configuration. Alternatively, you can also edit the downloaded file first and then import it into the same or different cluster.

Import a Configuration Document or a Configuration from a Reference Host Into an Existing Draft Configuration

For any cluster that uses vSphere Configuration Profiles and has a draft configuration created, you can override and replace the configuration settings by importing a valid configuration document or using the configuration settings of a reference host in the cluster.

- Verify that vCenter is of version 9.0 or later
- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you have a draft configuration created for the cluster.
- Verify that you have the required privileges for the import operation. See Required Privileges for Using vSphere Configuration Profiles.

You can import the configuration document of another cluster and reuse the configuration defined in that other cluster. You can also import a configuration document that you manually edited or created from scratch. You can also import the settings of one of the hosts available in the cluster and use them as common settings for all hosts in the cluster.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Draft** tab, select your task.

Import Option	Steps
Import a JSON file	1. To import a configuration document, click the horizontal ellipses and select **Import from file**. The **Import Configuration from File** dialog box opens. 2. Click **Browse** to locate the file that you want to import and click **Import and overwrite draft**.
Use the configuration of a reference host in the cluster	1. To create the cluster configuration by using the configuration of one of the hosts in the cluster as reference, click the ellipses and select **Import from host**. The **Select reference host** dialog box opens. 2. Click the radio button for the respective host which configuration settings you want to use and click **Import and overwrite draft**.

4. Click **Close**.

The imported configuration settings become the new draft configuration for the entire cluster.

When you replace the settings in the draft configuration with the settings of a single host from the cluster, these settings become the common settings for all hosts in the cluster. If a host in the cluster has settings that differ from the new common settings for the cluster, the original host settings are saved in the new draft configuration as host overrides.

You can view the changes that are applied to the edited draft configuration of the cluster by clicking **Show changes**. You can also start a draft pre-check to ensure that the configuration is valid and to compute the remediation impact on all hosts in the cluster. Then you can remediate the cluster to make all non-compliant hosts compliant.

Delete a Draft Configuration for a Cluster

You can discard all changes that you made to a draft configuration by deleting it in the vSphere Client. The current desired state of the cluster remains unaffected.

- Verify that vCenter is of version 9.0 or later.
- Verify that ESX hosts are of version 8.0 Update 1 or later.
- Verify that you have a draft configuration that is created for a cluster.
- Verify that you have the required license.

- Verify that you have the required privileges for viewing host settings in the desired configuration. See Required Privileges for Using vSphere Configuration Profiles.

When you create and edit a draft configuration, your changes are stored on vCenter. When you delete your draft configuration, the changes you made will be permanently lost. If another user creates a draft configuration for the same cluster, that draft configuration remains unaffected when you delete your draft configuration. You can preserve your current cluster configuration and continue editing the draft configuration settings only if you export the draft configuration document. For more information, see Export the Draft Cluster Configuration or Draft Cluster Configuration Schema.

1. In the vSphere Client, navigate to a cluster that you manage with an image.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Draft** tab, click the horizontal ellipses icon and select **Discard draft**.
   The **Discard draft** dialog box opens.
4. Click **Discard draft**.

The draft configuration is permanently removed and you can no longer edit its settings.

Create a new draft configuration if you want to edit the cluster configuration in the vSphere Client. For more information, see Create a Draft Configuration for a Cluster in vSphere Client.

Check Host Compliance Against the Desired Configuration for the Cluster

You can check the compliance of each host in a cluster only against the desired configuration set for the entire cluster. If a host is out of compliance with the desired configuration, you can see detailed information about the settings that make the host non-compliant.

- Verify that vCenter is of version 9.0 or later.
- Verify that all ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that all ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you have the required privileges for checking compliance. See Required Privileges for Using vSphere Configuration Profiles.

The **Check Cluster Configuration Compliance** task runs automatically when you add a host to the cluster. You can manually run a compliance check only on the current desired configuration of the cluster.

You can enable vSphere Configuration Profiles on a cluster that contains hosts connected to a vSphere Distributed Switch. If you add a host connected to a vSphere Standard Switch to a cluster that contains hosts connected to a vSphere Distributed Switch, vSphere Configuration Profiles shows the added host as non-compliant. You must use the vSphere Distributed Switch workflows to remediate the host and make the host compliant with the desired cluster configuration. For more information about how to add a host to a vSphere Distributed Switch, see the *vSphere Networking* documentation.

> **Note:**
>
> vSphere Configuration Profiles can manage the vSphere Distributed Switch configuration of the hosts in a cluster managed with a desired configuration. You can also use the workflows for managing and configuring the vSphere Distributed Switches described in the *vSphere Networking* documentation.

vSphere Configuration Profiles can be enabled on a cluster that has NSX or vSAN enabled. You cannot use vSphere Configuration Profiles to manage the NSX or vSAN configuration settings, you can run a compliance check and view the drifts between the desired and the current cluster configuration.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Compliance** tab, click **Check compliance**.

   The **Check Cluster Configuration Compliance** task starts. After the task finishes, a list of all non-compliant hosts shows on the **Compliance** tab.
4. View the compliance results for each non-compliant host in the cluster.
   a) On the **Compliance** tab, select a host from the **Hosts** list.

      An information panel appears on the right. You see the host settings and the respective desired and current values that cause the noncompliance.

Remediate the cluster to make all non-compliant hosts compliant.

Set Up Alarms for Monitoring Configuration Drifts

You can create alarms that notify you when vSphere Configuration Profiles detects drifts from the desired cluster configuration for a cluster or a host. Receiving such notifications allows you to take the necessary actions and remediate the clusters which are out of compliance with the desired cluster configuration.

- Verify that the clusters and hosts you select have the vSphere Configuration Profiles enabled.
- Required privileges for creating alarms: **Alarms** > **Create alarm**.
- Verify that you have the required license.

vSphere Configuration Profiles generates events when a change in the configuration of a cluster or a host occurs. You can create alarms that activate in response of these events and notify you about the configuration drifts by sending emails or SNMP traps.

For detailed information about how to manage alarms in the vSphere Client, see the *vSphere Monitoring and Performance* documentation.

1. In the vSphere Client, navigate to a vCenter instance or an inventory object.
2. On the **Configure** tab, click **Alarm Definitions**.

   Depending on the inventory object you select, the location of the **Alarm Definitions** node varies.
3. Click **Add** to create a new alarm definition.

   The **New Alarm Definition** wizard opens.

4. On the **Name and Targets** page of the wizard, enter a name and use the **Target type** drop-down menu to select whether the alarm is triggered for clusters or hosts.
5. On the **Alarm rule 1** page of the wizard, select the event trigger and how you want to be notified about changes to the cluster or host configuration.
   a) From the drop-down menu select the event trigger. You can choose from `com.vmware.vcIntegrity.ClusterConfigurationOutOfCompliance` or `com.vmware.vcIntegrity.HostConfigurationOutOfCompliance`.
   a) From the severity drop-down menu, select the severity of the alarm.
   b) To send email notifications when alarms are triggered, activate the **Send email notifications** toggle and configure the recipient addresses and the subject of the email notifications.
   c) To send SNMP traps when alarms are triggered, activate the **Send SNMP traps** toggle.
   d) Click **Next**.
6. On the **Reset Rule 1** page of the wizard, specify the alarm reset rule if needed and click **Next**
7. On the **Review** page of the wizard, review the alarm definition and activate the **Enable this alarm** toggle.
8. Click **Create**.

When vSphere Configuration Profiles detects that all hosts in a cluster or only some of the hosts in the cluster are out of compliance with the desired cluster configuration, an alarm is triggered. You receive email notifications or SNMP traps and can take the necessary actions to remediate the cluster and make all non-compliant hosts compliant again.

Run a Remediation Pre-Check or a Draft Pre-Check

You run a remediation pre-check to ensure that the desired configuration is valid and compatible with the current state of the hosts in the cluster, and that the cluster is in good health. You run a draft pre-check to ensure that the draft configuration is valid and can be applied on all hosts in the cluster. A pre-check that finds no issues means that you can successfully remediate the cluster and apply the desired or draft configuration to all hosts.

- Verify that vCenter is of version 9.0 or later.
- Verify that you have a draft configuration created, if you want to run a draft pre-check.
- Verify that ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image
- Verify that ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that you have the required privileges for running a remediation or a draft pre-check task. See Required Privileges for Using vSphere Configuration Profiles.

You can create a draft of the cluster configuration in the vSphere Client and edit the configuration settings directly in the user interface. Before applying the draft configuration on a cluster, you can run a draft pre-check.

During a remediation or a draft pre-check, vSphere Configuration Profiles runs various checks on each host and the entire cluster to estimate and validate the impact of the desired or the draft configuration on the current state of the hosts in the cluster. The remediation and draft pre-checks also compute the remediation impact on the hosts.

You can enable vSphere Configuration Profiles on a cluster that has hosts connected to a vSphere Distributed Switch. If you add a host that is connected to a vSphere Standard Switch to such cluster, the remediation pre-check returns an error and you cannot remediate the cluster before you manually add this host to the vSphere Distributed Switch. For more information about how to add a host to a vSphere Distributed Switch, see the *vSphere Networking* documentation.

> **Note:**
> vSphere Configuration Profiles can manage the vSphere Distributed Switch configuration of the hosts in a cluster managed with a desired configuration. You can also use the workflows for managing and configuring the vSphere Distributed Switches described in the *vSphere Networking* documentation.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. Start a pre-check by selecting one of the following options.
   - To run a remediation pre-check of the current desired configuration, on the **Compliance** tab, click **Run pre-check**.

     The **Cluster Configuration Remediation Pre-Check** task starts.
   - To run a draft pre-check of the edited cluster configuration, on the **Draft** tab, click **Run pre-check**.

     The **Cluster Configuration Draft Pre-check** task starts.
4. Optional: Click the **Operation Details** tab to view details about the remediation or draft pre-check operations.

   The **Operation Details** tab gives summarized information or detailed information about the remediation operation, the remediation pre-check, or the draft pre-check operations. Any issues during those operations are also listed on the **Operation Details** tab.

Fix all issues that are reported during the remediation or the draft pre-check and remediate the cluster.

Remediate a Cluster Against the Desired or Draft Configuration

You remediate a cluster against the desired or draft configuration to make all non-compliant hosts compliant with the configuration for the cluster.

- Verify that vCenter is of version 9.0 or later.
- Verify that ESX hosts are of version 8.0 Update 1 or later for using vSphere Configuration Profiles on a cluster that you manage with an image.
- Verify that ESX hosts are of version 8.0 Update 3 or later for using vSphere Configuration Profiles on a cluster that you manage with baselines.
- Verify that you have the required license.
- Verify that at least one of the hosts in the cluster is out of compliance with the desired configuration.
- Verify that you have the required privileges for performing remediation. See Required Privileges for Using vSphere Configuration Profiles.

You can create a draft of the cluster configuration in the vSphere Client and can edit the configuration settings directly in the user interface. To make all hosts in the cluster compliant with the draft configuration, you must apply it to the cluster. The apply operation first sets the draft configuration as the new desired cluster configuration and then remediates all hosts in the cluster against the new desired cluster configuration.

Before the cluster is remediated, an automatic remediation or draft pre-check is triggered. As a result, the remediation impact is calculated for each host in the cluster. The remediation might require a reboot or the host to enter maintenance mode. You can manually trigger a remediation or a draft pre-check. See

You can enable vSphere Configuration Profiles on a cluster that contains hosts connected to a vSphere Distributed Switch. You cannot use vSphere Configuration Profiles to manage the vSphere Distributed Switch configuration of the hosts in the cluster. You must manually fix any drifts in the vSphere Distributed Switch networking configuration of each non-compliant host in the cluster before you start the remediation task.

> **Note:**
>
> vSphere Configuration Profiles can manage the vSphere Distributed Switch configuration of the hosts in a cluster managed with a desired configuration. You can also use the workflows for managing and configuring the vSphere Distributed Switches described in the *vSphere Networking* documentation.

1. In the vSphere Client, navigate to a cluster that you manage with an image or to a cluster that you manage with baselines.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. Start the remediation operation.
   - To remediate the cluster against the current desired configuration, on the **Compliance** tab, click **Remediate**.

- To remediate the cluster against the draft configuration, on the **Draft** tab, click **Apply changes**.

  The **Remediate** wizard opens. A remediation pre-check task starts automatically to ensure that remediation can run without issues.

4. On the **Pre-check** page, wait for the remediation pre-check to finish, view the pre-check results, and click **Next**.
5. On the **Remediation Settings** page, review the remediation settings that are configured for this operation and click **Next**.
6. On the **Review Impact** page, review the summary of the remediation impact on the cluster and on each host individually.
7. Click **Remediate** to close the **Remediate** wizard.

   The **Remediate Cluster Configuration** task starts.
8. Optional: Click the **Operation Details** tab to view details about the remediation, the remediation pre-check, or the draft pre-check operations.

   The **Operation Details** tab gives summarized information or detailed information about the remediation operation or the remediation and draft pre-check operations. Any issues during those operations are listed on the **Operation Details** tab.

All hosts in the cluster are compliant with the desired configuration.

> **Note:**
>
> More than one user with the required privileges can create a draft configuration for a cluster at the same time. If one of the users remediates the cluster with their draft configuration, the draft configuration becomes the new desired configuration for the cluster. In case another user attempts to remediate the cluster with their draft configuration at the same time, the remediation operation fails with an error. That user can either discard their draft configuration and start anew, or export their current draft and import it later.

Troubleshooting Clusters Managed with vSphere Configuration Profiles

Specific SDDC Manager operations and workflows related to clusters that have vSphere Configuration Profiles enabled can lead to drifts between the desired cluster configuration and the current settings on the hosts in the cluster. Follow the steps described in this article to resolve any issues that may occur.

When Do You See Drifts in the Cluster Configuration

Several SDDC Manager operations can cause a cluster to become non-compliant with the desired cluster configuration. Depending on the expected cluster configuration, you can resolve these drifts either by remediating the cluster with the desired configuration or by editing the desired cluster configuration and then remediating the cluster so that all hosts become compliant.

Compliance drifts may occur in the following scenarios:

- Adding a host to a cluster that has the vSphere Configuration Profiles enabled using the VCF Operations console or SDDC Manager .
- Stretching a vSAN cluster.
- Managing passwords, configuring VMware Customer Experience Improvement Program (CEIP), updating VCF DNS and NTP servers, or converting or importing existing vSphere environments into VCF using the VCF Operations console or the SDDC Manager UI.

> **Tip:**
>
> With VCF 9.1, this article is updated and moved to the vSphere Configuration Profiles publication.

Resolving Drifts in the Desired Cluster Configuration

Following steps can help you prevent potential problems when you have a cluster with vSphere Configuration Profiles enabled and you run any of the operations that are mentioned earlier.

1. Check host compliance against the desired cluster configuration. See Check Host Compliance Against the Desired Configuration for the Cluster.
2. If a host in the cluster has an unknown compliance status, create a new draft configuration by importing the configuration from a reference host or creating a working copy of the existing cluster configuration. See Import a Configuration Document or a Configuration from a Reference Host Into an Existing Draft Con… and Create a Draft Configuration in .
3. Optionally, you can edit the host settings of the draft configuration. See Edit the Host Settings of a Draft Configuration in .
4. If you changed the desired cluster configuration or created a draft configuration, run another compliance check against the new desired cluster configuration. See Check Host Compliance Against the Desired Configuration for the Cluster.
5. If you want to apply the current desired configuration on the hosts that are out of compliance, remediate the cluster against the desired cluster configuration. See Remediate a Cluster Against the Desired or Draft Configuration.
6. If you want to apply the re-created cluster configuration, remediate the hosts in the cluster against the draft cluster configuration. See Remediate a Cluster Against the Desired or Draft Configuration.
7. If you stretched a vSAN cluster, after remediating the hosts in the cluster, reapply the VM storage policy on the cluster. See the Reapply Virtual Machine Storage Policy topic in the *vSphere Storage* documentation.

As a result, all hosts in the cluster are compliant with the desired or draft cluster configuration and exit maintenance mode.

vSphere Host Profiles

Learn how to manage and configure Host Profiles in the vSphere Client.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

The *vSphere Host Profiles* documentation is intended for administrators who are familiar with vSphere host configuration.

What Are vSphere Host Profiles

vSphere Host Profiles are profiles that encapsulate the host configuration and help to manage the host configuration, especially in environments where an administrator manages multiple hosts or clusters in vCenter.

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

Host Profiles provide an automated and centrally managed mechanism for host configuration and configuration compliance. Host Profiles can improve efficiency by reducing reliance upon repetitive, manual tasks. Host Profiles capture the configuration of a pre-configured and validated reference host, store the configuration as a managed object and use the catalog of parameters contained within to configure networking, storage, security, and other host-level parameters. Host Profiles can be applied to individual hosts, a cluster, or all the hosts and clusters associated to a host profile. Applying a Host Profile to a cluster affects all hosts in the cluster and results in a consistent configuration across the applied hosts.

Host Profiles can be used to validate the configuration of a host by checking compliance of a host or cluster against the Host Profile that is associated with that host or cluster.

Reference Host Independence

A dedicated reference host is not required to be available to perform host profile tasks.

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

When you create a host profile, you extract the configuration information from a specified ESX reference host. In previous releases, vSphere required that the reference host was available for certain Host Profiles tasks, such as editing, importing, and exporting. From vSphere 6.0 and later, a dedicated reference host is no longer required to be available to perform these tasks. For host profile tasks that require a reference host, an ESX host that is compatible to the host profile is assigned as a reference host.

Sometimes, a compatible host is not available to validate the host profile during these tasks. If you make small changes to the host profile, validation is skipped and you see a warning indicating that no valid reference host is associated with the profile. You can then proceed and finish the task.

Due to the introduction of this feature, users can no longer edit or change the reference host from the vSphere Client. The reference host selection occurs at runtime, without notifying users, in the vCenter for ongoing tasks.

Reference Host Selection

To select a suitable reference host, vCenter performs complex version and compatibility checks on the target host.

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

A major-version vCenter instance can manage hosts of all ESX Update release versions within the respective major version. For example, a vCenter 8.0 Update 3 instance can manage ESX hosts and hosts profiles of version 8.0 GA, 8.0 Update 1, 8.0 Update 2, and 8.0 Update 3. This change affects the reference host selection for particular host profile operations.

Version Check

To select a reference host, vCenter checks whether the target host is of the same ESX Update release version as the host profile. For some operations, the target host can run a later ESX version than the ESX version of the host profile. For example, if you want to edit a 8.0 Update 3 host profile, the reference host that validates the host profile cannot be running ESX 8.0 GA, 8.0 Update 1, or 8.0 Update 2.

Compatibility Check

In addition to checking the version on the target host, vCenter performs an extensive compatibility check that includes a subprofile compatibility check in the existing releases. Starting with vSphere 7.0 Update 3, to appoint a reference host, vCenter checks the advanced options and the default roles in the target host. A reference host must support the advanced options of the host profile it validates. Also, the default roles in the reference host must match the default roles in the host profile.

Host Profile Operations and Required Reference Host Versions

A reference host is needed to validate the host profile during different host profile operations and workflows. Starting with vSphere 7.0 Update 3, depending on the operation, the reference host must be of exactly the same or later Update release version as the host profile it validates. The following table summarizes the compatible reference host versions for different host profile operations if your environment runs a vCenter instance 7.0 Update 3 or later.

Host Profile Operation	Compatible Reference Host Version
Edit	The same Update release version as the host profile.
Check Compliance	The same or later Update release version as the host profile.
Remediation	The same or later Update release version as the host profile.
Attach	The same or later version as the host profile.
Import/Export	The exact same version as the host profile.

How Do You Manage vSphere Host Profiles in the vSphere Client

By using the vSphere Client, you can perform basic tasks for host profiles, perform and schedule compliance checks of hosts and clusters, remediate hosts which are not compliant with the attached host profiles.

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

Host Profiles Usage Model

The Host Profiles workflow starts with the concept of a reference host. The configuration of the reference host, which is extracted as a host profile, serves as a configuration template for configuring other hosts. The reference host does not have to be related or associated with the host profile extracted from it.

Before you begin, ensure that you have an existing vSphere environment installation with at least one properly configured ESX host.

The sequence required to create a host profile from a reference host, apply the host profile to a host or cluster and check compliance against the host profile is as follows:

1. Set up and configure the reference host.
2. Create a host profile from the reference host.
3. Attach hosts or clusters to the host profile.
4. Check the compliance to the host profile. If all hosts are compliant with the reference host, they are correctly configured.
5. Apply (remediate).

If you want the Host Profile to use directory services for authentication, configure the reference host to use a directory service. See the *vSphere Security* documentation.

Hosts Provisioned with vSphere Auto Deploy

For hosts provisioned with vSphere Auto Deploy, the vSphere Client owns the entire host configuration, which is captured in a Host Profile. Usually, the Host Profile information is sufficient to store all configuration information. Sometimes the user is prompted for input when the host provisioned with Auto Deploy starts. See the *VMware ESX Installation and Setup* documentation for more information on Auto Deploy.

> **CAUTION:**
>
> Booting ESX hosts by network with in-memory or cached images, and preserving no state, is deprecated in ESX 9.0. For more information, see *Installing ESX Using vSphere Auto Deploy* in the *ESX Installation and Setup* guide.

How Do You Get Started with vSphere Host Profiles

Learn how to create a host profile, attach or detach hosts and clusters to a host profile, import or export a host profile, duplicate a host profile, and copy settings from a reference host to a host profile or from one host profile to another.

> **Note:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

> **Note:**
> When vSphere Configuration Profiles is enabled on a cluster, you cannot attach or edit host profiles in the cluster or in the hosts in that cluster. For more information on vSphere Configuration Profiles, see the *Managing Host and Cluster Lifecycle* guide.

How Do You Create a vSphere Host Profile?

You create a host profile by extracting the configuration from a designated reference host.

Verify that you have a working vSphere installation and at least one completely and properly configured host.

Host profiles do not capture offline or non-present devices. Any changes made to offline devices after extracting a host profile do not make a difference to the compliance check results.

> **Note:**
> When vSphere Configuration Profiles is enabled on a cluster, you cannot attach or edit host profiles in the cluster or in the hosts in that cluster. For more information on vSphere Configuration Profiles, see the *Managing Host and Cluster Lifecycle* guide.

> **Note:**
>
> Host profiles are not supported for ESX hosts with a Data Processing Unit (DPU).

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Click **Extract Host Profile**.
4. On the **Select host** page, select the vCenter instance where to create the host profile.
5. Select the host that acts as the reference host and click **Next**.
   The selected host must be a valid host.
6. Enter a name and, optionally, a description for the new profile, and click **Finish**.

The new profile appears in the **Host Profiles** pane.

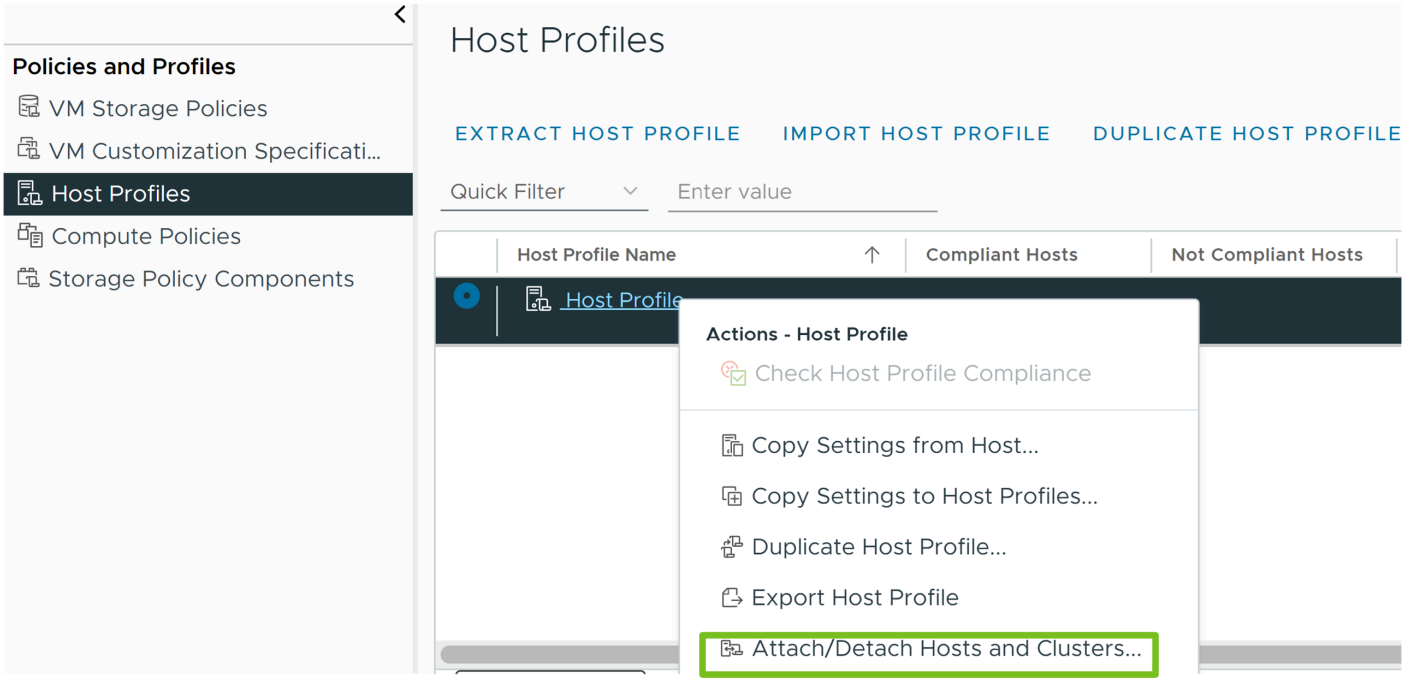
How Do You Attach or Detach Hosts and Clusters to a vSphere Host Profile?

After you create a host profile from a reference host, you must attach a host or a cluster to the host profile to associate that host profile with the selected host or cluster. To disassociate a configuration from an ESX host or an entire cluster, detach the host or cluster from the host profile.

When a host profile is attached to a cluster, the host or hosts within that cluster are also attached to the host profile. However, when the host profile is detached from the entire cluster, there is no association between the host or hosts within the cluster and that host profile. Detaching a host profile from an ESX host or a cluster does not delete that host profile. You can delete the host profile after detaching it from all the entities it is associated with.

> **Note:**
> When vSphere Configuration Profiles is enabled on a cluster, you cannot attach or edit host profiles in the cluster or in the hosts in that cluster. For more information on vSphere Configuration Profiles, see the *Managing Host and Cluster Lifecycle* guide.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile and select **Attach/Detach Hosts and Clusters** .



4. Select or deselect the hosts or clusters from the list, and click **Save**.
   You can use the **Filter** text box to search the hosts and clusters list.

The hosts or clusters are added or removed from the selected host profile.

How Do You Duplicate a vSphere Host Profile?

A host profile duplicate is a copy of an existing host profile. When you create the duplicate, you enter a new name and description for it and the duplicate appears in the list of host profiles.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile you want to duplicate and select **Duplicate Host Profile**.
4. Enter a different name and description for the duplicate host profile, and click **OK**.

A clone of the host profile appears in the **Host Profiles** list.

How Do You Copy Settings from a Reference Host to a vSphere Host Profile?

If the configuration of the reference host changes, you can update the host profile so that it matches the reference host's new configuration. You can make changes to the reference host and then copy the changes to the host profile. Alternatively, you can edit the host profile to match the new reference host configuration.

After you create a host profile, you can make incremental updates to the profile. When making changes to a host profile, consider the benefits and limitations of the two methods:

- Make the configuration changes to a host and copy that host's settings to the profile. The settings within the existing profile are updated to match those of the host. This method allows you to validate the configuration on a single host before rolling it to the other hosts that are attached to the profile.
- Update the profile directly by editing the host profile. This provides the ability to do more comprehensive and immediate remediation of those changes.

  Note:
  Fixed user password, system image cache and some of the host customized settings are not present in the newly updated host profile. Edit the host profile to update these settings.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile and click **Copy Settings from Host**.
4. Select the host from which you want to copy the configuration settings and click **OK**.

How Do You Import or Export a vSphere Host Profile?

You can import a host profile from a file that is in the VMware profile format (`.vpf`). You can also export a host profile to a file that is in the VMware profile format (`.vpf`).

When you export a host profile, administrator and user profile passwords are not exported. This is a security measure and stops passwords from being exported in plain text when the profile is exported. You are prompted to re-enter the values for the password after the profile is imported and the password is applied to a host.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. If you want to import a host profile, click **Import Host Profile**.
   a) To locate the VMware Profile Format file (`.vpf`) that you want to import, click **Choose File**.
   b) Enter the **Name** and **Description** for the imported Host Profile, and click **OK**.
   The imported host profile appears in the **Host Profiles** list.

4. If you want to export a host profile, right-click the profile, select **Export Host Profile**, and click **Save**.

How Do You Copy Settings from a vSphere Host Profile?

After you make changes to a host profile, you can propagate those changes to other host profiles in the inventory.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile from which you want to copy settings and select **Copy Settings to Host Profiles**.
4. Select the settings you want to copy to other host profiles, and click **Next**.
5. Select the target host profile that will be overwritten with the selected settings, and click **Next**.
   The differences between the host profile settings are displayed on the **Ready to complete** page.
6. Click **Finish**.

How Do You Manage Compliance Checks of Hosts and Clusters to a vSphere Host Profile

You can check the compliance of hosts and clusters to a vSphere host profile to verify if the host configuration matches the configuration of the host profile. Learn how to check compliance manually or schedule regular compliance checks of hosts and clusters, and how to remediate hosts that are not compliant with the host profile configuration.

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

How Do You Check Compliance of Hosts or Clusters to a vSphere Host Profile?

Learn how to check the compliance of a host or cluster to the host profile attached to them. By using the compliance check, you can see if any configuration parameters on a host are different from those specified in the host profile.

As a result of the compliance check, the host status is labeled as Compliant, Unknown, or Non-compliant.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click a host profile and click **Check Host Profile Compliance**.

> **Note:**
> Host profiles do not capture offline or unpresented devices. Any changes made to offline devices after extracting a host profile do not make a difference to the compliance check results.

The compliance status is updated as Compliant, Unknown, or Non-compliant.

A non-compliant status indicates a discovered and specific inconsistency between the profile and the host. To resolve this, you should remediate the host. Any unknown status indicates that the compliance of the host cannot be verified. To resolve the issue, remediate the host through the host profile. Often the compliance check fails because the host is disconnected.

During a compliance check of an ESX host to the host profile, two groups of advanced option parameters are not managed by the host profile and their values remain unchanged for the host:

- advanced option parameters which are disabled in the host profile.
- advanced option parameters which exist in the ESX host you want to remediate, and do not exist in the host profile.

To see more details on compliance failures, select a host profile from the **Host Profiles** main view for which the last compliance check displayed one or more failures. To see which parameters differ between the host that failed compliance

and the host profile, click the **Monitor** tab and select the **Compliance** view. Then select the failing host. The differing parameters are displayed in the Compliance window, below the host list.

How Do You Schedule a Compliance Check of Hosts or Clusters to a vSphere Host Profile?

You can schedule a regular compliance check of a host or cluster, using a standard vSphere Client scheduled tasks workflow. This automation determines whether any configuration parameters are different from those specified in the host profile.

Verify that the mail server is configured to send notifications.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Select the desired host profile and navigate to the **Monitor** tab.
4. From the **Monitor** tab, select **Scheduled Tasks** -> **New Scheduled Task** -> **Check Compliance**.
   The **Schedule New Tasks (Check Compliance)** dialog box appears.
5. Optional: Enter a new task name.
   Additional details about the task can be entered in the **Description** text box.
6. Choose a frequency and how often to run the task.
   You can postpone when the task is run for the first time and schedule when it ends.
7. Optional: To receive a notification after the task is finished, enter an email.
8. Click **Schedule The Task**.

The newly created task is visible on the **Scheduled Tasks** list. You can **Edit**, **Run** or **Remove** the task.

To understand more about compliance status, checks or failures, see **Check Compliance**.

How Do You Remediate a Host to Comply with a vSphere Host Profile Configuration?

When you check the compliance of a host to a host profile and a host is non-compliant, you can use the remediate option to apply the host profile settings to the host. During remediation, all host profile parameters are updated to the values defined in the host profile that is attached to the host.

Verify that a host profile is attached to the host.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile and select **Remediate**.

   Note:
   Certain Host Profile policy configurations require that the host is rebooted after remediation. In such cases, you are prompted to place the host into maintenance mode. You might be required to place hosts into maintenance mode before remediation. Hosts that are in a fully automated DRS cluster are placed into maintenance mode during remediation. Alternatively, if the host is not placed into maintenance mode when required, the remediation process stops.
4. Select the hosts you want to remediate against the host profile.

   Note:

   When you remediate an ESX host to comply with a host profile, two groups of advanced option parameters are not managed by the host profile and their values remain unchanged for the host:

   - advanced option parameters which are disabled in the host profile.
   - advanced option parameters which exist in the ESX host you want to remediate, and do not exist in the host profile.

The remediation pre-check results do not contain tasks related to these two groups of advanced option parameters.

5. Optional: To specify host properties, enter your host customizations or import a host customization file.

   You can modify the user input parameters for the host profiles policies by customizing the host and clicking **Next**.

6. To check if the selected hosts can be remediated, click **Pre-check Remediation**.

   The pre-check generates a list of tasks that will be performed on the host.

7. If needed, select the check box to reboot the hosts.

   Alternatively you can reboot the hosts manually after the remediation process.

8. Review the tasks that are necessary to remediate the host profile and click **Remediate**.

The compliance status is updated.

How Do You Configure vSphere Host Profiles

Learn how to edit host profile policies and components, and configure security and networking host profiles.

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

How Do You Manage Host Profile Policies and Policy Components

In the vSphere Client, you can view and edit host profiles policies, policy options, and parameters. You can change policy names or policy descriptions. If you do not want a host profile component to be considered during a compliance check, you can deactivate that specific component.

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

Edit a Host Profile

You can view and edit Host Profiles policies, select a policy to be checked for compliance, and change the policy name or description.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile that you want to edit and click **Edit Host Profile**.
4. Optional: Click the **Name and Description** tab to change the profile name and description.
5. In the **Settings** tab, expand each category to view or edit particular policies or settings.

   Note:

   See How Do You Edit a vSphere Host Profile Policy? for detailed instructions for editing a Host Profile policy. See How Do You Deactivate a Host Profile Component or Subprofile? for detailed instructions on enabling or disabling a policy from compliance check or remediation.

6. View **All** host profile configurations or only **Favorites** configurations.

   You can mark a configuration as a favorite with the star icon. Those are then added to **Favorites** configurations.

7. Optional: In the search field, filter the configuration names and values you want to view.
   For example, enter `SNMP`. All configurations that relate to **SNMP** are displayed.

8. Optional: Customize the hosts.
   Make any changes to the available configuration values for this profile and click **Save**.

   Note:
   The host customization settings page only appears if you changed any settings that require host customizations.

The changes are made when the "Update Host Profile" task is completed in the Recent Tasks status. If you attempt to remediate the profile before the task is complete, the profile configuration does not contain the change.

How Do You Edit a vSphere Host Profile Policy?

A host profile policy describes how a specific configuration setting is applied. You can edit policies belonging to a specific host profile.

To see the sub-profile components that comprise the host profile, you can expand the host profile's configuration hierarchy when you edit the host profile. These components are categorized by functional group or resource class to make it easier to find a particular parameter. Each subprofile component contains one or more attributes and parameters, with the policies and compliance checks.

Each policy consists of one or more options that contain one or more parameters. Each parameter consists of a key and a value. The value can be one of a few basic types, for example, integer, string, string array, or integer array.

Note:
Currently, there is no way to remove or replace policy options, policies, or sub-profiles that are deprecated in this release. To allow old host profiles to continue working, metadata is added to the deprecated policies. New host profiles are extracted with only non-deprecated parts of a host profile.

Table 114: Subset of Host Profile Subprofile Configurations

Component Categories	Configuration Settings	Notes and Examples
Advanced Configuration Settings	Advanced Options, Agent VM, DirectPath, Hosts file, Power System, System Image Cache, Configuration Files, Graphics Config, Host Profile Log config.	• Host profiles do not check advanced settings if they are the same as the default settings. vCenter copies only the advanced configuration settings that have changed and that differ from the default values. In addition, compliance checks are limited to the settings that are copied. • Host profiles do not support the configuration of PCI devices for virtual machine passthrough on the ESX host.
General System Settings	Console, Core Dump, Device Alias, Host Cache, Kernel Module, Management Agent, System Resource Pool, System Swap, vFlash Host Swap Cache, CIM-XML Indication Subscriptions, Host Spe, Software Design config.	For date and time configuration: • For the time zone, enter a UTC string. For example, "America/Los_Angeles" for United States Pacific time zone. • The default time zone is set to the local time and location of the vSphere Client machine. **Note:** Enter the host cache size as a whole GB in MB. For example, 1024MB.
Networking	vSwitch, Port groups, Physical NIC speed, security and NIC teaming policies, vSphere Distributed Switch, and vSphere Distributed Switch uplink port.	When DHCPv6 is enabled in the networking subprofile, manually turn on the corresponding rule set in the firewall subprofile.
Security	Firewall, Security Settings, Service.	

Component Categories	Configuration Settings	Notes and Examples
Storage	Configure storage options, including Native Multi-Pathing (NMP), Pluggable Storage Architecture (PSA), iSCSI adapters, vSAN, vVols and NFS storage.	• Use the vSphere CLI to configure or modify the NMP and PSA policies on a reference host, and then extract the Host Profile from that host. If you use the Profile Editor to edit the policies, to avoid compliance failures, make sure that you understand interrelationships between the NMP and PSA policies and the consequences of changing individual policies. For information about the NMP and PSA, see the *vSphere Storage* documentation. • Add the rules that change device attributes before extracting the Host Profile from the reference host. After attaching a host to the Host Profile, if you edit the profile and change the device attributes (for example, mask device paths or adding SATP rules to mark the device as SSD) you are prompted to reboot the host to make the changes. However, after rebooting, compliance failures occur because the attributes changed. Because Host Profiles extract device attributes before rebooting, if any changes occur after the reboot, it evaluates and finds those changes, and reports it as noncompliant. • To configure or modify the SatpDeviceProfile policy after extracting the host profile, use the vSphere Client. For compliance purposes, the policy option strings must be in the following format: – For an ALUA supported array, for example, SATP_ALUA, the policy options must be separated by a semicolon (;). For example: `implicit_support=<on/off>; explicit_support=<on/off>; action_onRetryErrors=<on/off>` – For an ALUA supported array with CX, for example, SATP_ALUA_CX, the policy options must be separated by a semicolon (;). For example: `navireg=<on/off>; implicit_support=<on/off>; action_onRetryErrors=<on/off>` – For a CX array, for example, SATP_CX or SATP_INV, the policy options must be separated by a space. For example: `navireg=<on/off> ipfilter=<on/off> action_onRetryErrors=<on/off>` **Note:** The policy configuration options that are marked with `off` are not present in the configuration string.

Other profile configuration categories include: user group, authentication, kernel module, DCUI keyboard, host cache settings, SFCB, resource pools, login banner, SNMP agent, power system, and CIM indication subscriptions.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Select the host profile that you want to edit and click the **Configure** tab.
4. Click **Edit Host Profile**.
5. Expand a subprofile until you reach the policy to edit.
6. Select the policy.
   The policy options and parameters appear on the right side of the **Edit Host Profile** dialog box.
7. Edit the policy options or parameters.

How Do You Deactivate a Host Profile Component or Subprofile?

You can decide whether a host profile component or a subprofile is applied or considered during a compliance check. This action allows administrators to eliminate non-critical attributes from consideration during the compliance check or ignore values that, while part of the host profile, are likely to vary between hosts.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Select the host profile that you want to edit and click the **Configure** tab.
4. Click **Edit Host Profile**.
5. Expand the Host Profile Component hierarchy until you reach the desired component or component element.
6. Deselect the check box next to a component.
   The check box is selected by default. If you deselect the check box, the other subprofiles that are enabled are still applied and selected.

During a host profile compliance check or host remediation, the component or component element is not considered.

> **Note:**
> Sometimes, deselecting the check box might remove the component or component element from the host. This action is displayed in the task list after the pre-check remediation.

How Do You Configure a Security vSphere Host Profile?

Learn how to manage role profiles, user account profiles, lockdown mode profiles, and active directory permission profiles that are grouped as part of the security host profile. Starting with vSphere 8.0, users with access to the ESX Shell and the Administrator role can remove or grant ESX Shell access to user accounts.

Make sure that you have the SecurityConfigProfile plugin available to validate the role, user account, and active directory permission profiles as there are dependencies between them.

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

You can configure the host profile options, part of the security profile.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Select the host profile that you want to edit and click the **Configure** tab.
4. Click **Edit Host Profile**.
5. Unfold the **Security and Services** > **Security Settings** profile category and open the **Security** folder.
   The following profiles are displayed:

Role	This profile allows you to view default roles and add custom roles within the ESX system.
User Configuration	This profile allows you to create and manage user accounts. By default, newly created user accounts have access to the ESX Shell. Starting with vSphere 8.0, administrators can use the advanced option Security.DefaultShellAccess to change this default behavior. Here are some of the operations that you can perform for user accounts: • Create a user account. • Configure the password for a user account. • Configure the password for the root user. • Activate or deactivate user account ESX Shell access. **Note:** Do not modify the shell access property for the root user account. • Configure the role for any user that is not the default one. • Assign a default or a custom role (configure permissions) for a local account. • Configure the SSH key for any user.
Active Directory Permission	This profile allows you to manage permissions for active directory users or groups. For example, you can create permissions that associate an active directory user or a group with a role. When an ESX host joins the active directory domain, an Admin permission is created for the DOMAIN group **ESX Admins**. Also, when an active directory user or group is given some permissions on the ESX host, a corresponding permission is created on that host. The Active Directory Permission profile captures that permission.
Lockdown Mode	This profile allows you to increase the security of your ESX hosts by restricting user permissions and privileges. You can configure the following lockdown mode settings: • Normal lockdown mode: An ESX host can be accessed from a local console and vCenter. DCUI service is not stopped. • Strict lockdown mode: An ESX host can be only accessed from vCenter. The DCUI service is stopped. • Exception users: The users that still have their permissions regardless of the lockdown mode state. • Disabled: You can disable the lockdown mode.

For more information on the security profile, see the *vSphere Security* documentation.

How Do You Configure a Networking vSphere Host Profile?

By default, hosts provisioned with vSphere Auto Deploy are assigned DHCP addresses by a DHCP server. You can use the vSphere Auto Deploy host customization mechanism to assign static IP addresses to hosts.

- Set up your vSphere Auto Deploy environment.

- Boot the host using vSphere Auto Deploy.
- Extract a host profile from the host.

Note:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Select the host profile that you want to edit and click the **Configure** tab.
4. Click **Edit Host Profile**.
5. Change the default IP address settings by locating **Networking configuration** > **Host virtual nic** > **Management Network** > **IP address settings**.
6. From the **IPv4 address** drop-down menu, select **User specified IPV4 address to be used while applying the configuration**.
7. If the host is in a different subnet than the vCenter system, select **Networking Configuration** > **NetStack Instance** > **defaultTcpipStack** > **IP route configuration** and enter the default route in the **Default IPv4 gateway** text box.
8. Select **Networking Configuration** > **NetStack Instance** > **defaultTcpipStack** > **DNS configuration**.
9. Make sure the **Flag indicating if DHCP should be used** check box is deselected, and click **Save**.
10. Right-click the host and select **Host Profiles** > **Attach Host Profile**.
11. Select the profile to attach, and click **Next**.
12. Provide the IP address and net mask, and click **Finish**.
13. Reboot the ESX host.

The IP address is saved as a host customization and applied to the host.

How Do You Customize Hosts by Using vSphere Host Profiles

To customize hosts with shared attributes, you can create a host profile in a reference host. To customize individual hosts, you can set up some fields in the host profile to prompt the user for input for each host. You can also edit and export host customizations.

Host Customization

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

Host profiles allow you to prespecify information, for example, the storage setup or Syslog setup in a reference host to and apply the host profile to a set of target hosts that share the same settings. You can also use host profiles to specify that certain settings are host dependent. If you do so, the host comes up in maintenance mode when you provision it with Auto Deploy. Remediate the host or reset the host customization to be prompted for input. The system stores your input and uses it the next time the host boots.

When the host profile is set to prompt for user input, you must specify a value in the dialog box that appears when you reset the host customization. An error results if you do not specify a value.

Table 115: Host Profile Options that Prompt for iSCSI User Input

Information to Request User Input For	Setting the Host Profile Option
When you apply a host profile on a system that includes a profile for iSCSI, you are prompted for several properties. For many of the properties, a system default is available. For some properties, you must specify a value or an error results. **IQN name** If the iSCSI setup uses an IQN name, you are prompted when you apply the host profile. You cannot continue until you provide the name. **CHAP information** If you set up iSCSI to require CHAP authentication, you are prompted for CHAP information including the user name and the secret when you apply the host profile. You cannot continue until you provide the name.	1. Select **Edit Host Profile**, click **Storage configuration**, and click **iSCSI Initiator Configuration**. 2. Select the folder for an already enabled initiator and set up the initiator. 3. Set up the initiator. For many fields, the user is prompted as part of host customization.

Table 116: Host Profile Options that Prompt for Storage User Input

Information to Request User Input For	Setting the Host Profile Option
You are setting up the Fixed PSP configuration and want to prompt for the adapter and target IDs for the storage arrays that should use the Fixed PSP.	You can set the option only if the adapter is set up to use the Fixed PSP. 1. Select **Edit Host Profile**, click **Storage configuration**. 2. Click **Native Multipathing (NMP)**. 3. Click **Path Selection Policy (PSP) configuration**. 4. In the Preferred Path window, select **Prompt the user for adapter and target IDs on the host**.

Table 117: Host Profile Options that Prompt for Security User Input

Information to Request User Input For	Setting the Host Profile Option
Administrator password for ESX host when the host boots for the first time.	1. Select **Edit Host Profile**, and click **Security and Services**. 2. click **Security Settings** and click **Security configuration**. 3. In the right panel, select **User Input Password to be Used to Configure Administrator Password** from the **Administrator password** drop-down menu.
Preconfigures a user for the ESX host but prompts for the password for that user on each host when the host boots for the first time.	You can perform this task only if a user configuration already exists. Configure the user by selecting one of the options. • **Assigned fixed user configurations** is available for compatibility with ESX/ESXi 4.1 system, this option displays the password in the clear. • **Assign advanced fixed user configurations** is for users of ESXi 5.0 and later systems. • **Specify the user configuration in the profile but prompt for password during host configuration** allows you to specify the information about the user but prompt for a password on each host.

Information to Request User Input For	Setting the Host Profile Option
Prompt the user for credentials when the host joins the Active Directory domain.	1. Set the Authentication configuration profile to use a fixed domain. a. Select **Edit Host Profile**, click **Security and Services**. b. Click **Security Settings**, and click **Authentication configuration**. c. Click **Active Directory configuration**. d. In the Domain Name drop-down menu, select **Configure a fixed domain name**. 2. Set the method for joining the domain to prompt the user. a. Select **Edit Host Profile**, click **Security and Services** and click **Authentication configuration**. b. Click **Active Directory configuration**. c. In the Join Domain Method drop-down menu, select **Use user specified AD credentials to join the host to domain**.

Table 118: Host Profile Options that Prompt for Networking User Input

Information to Request User Input For	Setting the Host Profile Option
Prompt the user for the MAC address for a port group. You can have the system prompt the user in all cases (User specified MAC address...) or prompt the user only if no default is available.	1. Select **Edit Host Profile,** click **Networking configuration**, and click **Host port group**. 2. Click **Management Network**. 3. In the **Determine how MAC address for vmknic should be decided** field, select how the system manages the MAC address. – **User specified MAC Address to be used while applying the configuration** – **Prompt the user for the MAC Address if no default is available**
Prompt the user for the IPv4 address for each ESX host to which the profile is applied. You can have the system prompt the user in all cases (User specified IPv4 address...) or prompt the user only if no default is available.	1. Select **Edit Host Profile,** click **Networking configuration**, and click **Host port group**. 2. Click **Management Network** and click **IP address settings**. 3. In the **IPv4 address** field, select how the system manages the IPv4 address. – **User specified IPv4 Address to be used while applying the configuration** – **Prompt the user for the IPv4 Address if no default is available**
Prompt the user for the IPv6 address for each ESX host to which the profile is applied. You can have the system prompt the user in all cases (User specified IPv6 address...) or prompt the user only if no default is available.	1. Select **Edit Host Profile,** click **Networking configuration**, and click **Host port group**. 2. Click **Management Network** and click **IP address settings**. 3. In the **Static IPv6 address** field, select how the system manages the IPv6 address. – **User specified IPv6 Address to be used while applying the configuration** – **Prompt the user for the IPv6 Address if no default is available**

Information to Request User Input For	Setting the Host Profile Option
Prompt the user for the DNS name of the host. You can have the system prompt the user in all cases (User specified host name...) or prompt the user only if no default is available.	1. Select **Edit Host Profile,** , and click **Networking configuration** > **Netstack instance** > **defaultTcpipStack** > **DNS configuration**. 2. In the Host name field, select how the system manages the DNS configuration. — **Prompt the user for host name if default is not available** — **User specified host name to be used while applying the configuration**
Prompt the user for the MAC address for a distributed switch, its port group, or one of its services. Right-click the Host virtual NIC folder icon and click the **Add sub-profile** icon to determine the component to which the setting is applied. You can decide to prompt the user in all cases or only if no default is available.	1. Open **Networking configuration**. 2. Click **Host virtual NIC**. 3. In the **Determine how MAC address for vmknic should be decided** field, select how the system manages the MAC address for the distributed switch. — **User specified MAC address to be used while applying the configuration** — **Prompt the user for the MAC address if no default is available**
Prompt the user for the IPv4 address for a distributed switch, its port group, or one of its services. Right-click the Host virtual NIC folder icon and click the **Add sub-profile** icon to determine the component to which the setting is applied. You can decide to prompt the user only if no default is available or in all cases.	1. Open **Networking configuration**. 2. Click **Host virtual NIC**. 3. Click **IP address settings**. 4. In the IPv4 address field, select how the system handles the IPv4 address for the distributed switch. — **User specified IPv4 address to be used while applying the configuration** — **Prompt the user for IPv4 address if no default is available**
Prompt the user for the IPv6 address for a distributed switch, its port group, or one of its services. Right-click the Host virtual NIC folder icon and click the **Add sub-profile** icon to determine the component to which the setting is applied. You can decide to prompt the user only if no default is available or in all cases.	1. Open **Networking configuration**. 2. Open **Host virtual NIC**. 3. Open **IP address settings**. 4. In the **Static IPv6 address** field, select how the system manages the IPv6 address for the distributed switch. — **User specified IPv6 address to be used while applying the configuration** — **Prompt the user for IPv6 address if no default is available**

How Do You Export Host Customizations?

Learn how to export the customized attributes of a vSphere host profile as a `.CSV` file to your desktop.

For security reasons, sensitive data such as passwords are not exported.

1. Navigate to **Hosts and Clusters** and select an ESX host with an attached host profile.
2. Right-click the ESX host and select **Host Profiles** > **Export Host Customizations**.
3. Click **Yes**.

   Note:
   Only an English version of the `.csv` file is supported.

   The file is saved as a `.csv` file.

After the file is saved to your desktop, you can manually edit the file and save it to apply the customizations later.

How Do You Edit Host Customizations?

Read how to edit host customizations for specific hosts attached to a vSphere host profile or a cluster attached to a host profile.

1. In the vSphere Client, select **Menu** > **Policies and Profiles**.
2. Under **Policies and Profiles**, click **Host Profiles**.
3. Right-click the host profile and select **Edit Host Customizations**.
4. Select the host or hosts for which you want to edit the customization, and click **Next**.
5. Optional: To import host customization from your desktop, you can select **Import Host Customization** . `csv` in the **Customize hosts** page.

   If you want to edit host customizations by using a `.csv` file, you can export the host customizations in `.csv` file format, edit the values in the answer file fields, and then import the file. For more information on how to export host customizations, see How Do You Export Host Customizations?.

   Note:
   User input in the **Customize hosts** pane is overwritten by the import process and the values from the `.csv` file.

   After importing the `.csv` file, the text boxes are updated with the information from the file.
6. Edit the host configuration values and click **Finish**.

Answer File Field and vSphere Host Profile Extraction

When a new profile is extracted from an ESX host or post upgrade, host profiles policy options and host customization values are changed. You can get more familiar with the changes by referring to the Answer File Field and Host Profile Extraction table.

After you extract a new host profile from an ESX host, the values for some of the policy options change. To keep these values, you have to save them manually before extracting the host profile. The *Answer File Field and Host Profile Extraction* table lists the policy options with changed values and the options that keep their values the same in the newly extracted host profile.

To locate the policy options, navigate to **Host Profiles** main view, right-click the extracted host profile, and select **Edit Host Customizations.**

Table 119: Answer File Field and Host Profile Extraction

Policy option path	Policy option	Variations in the policy options value in the newly extracted profile
Security and Services -> Security Settings -> Security	UserInputPasswordConfigOption	For default users, it leaves the password unchanged (root). For non-default users, it always extracts user input.
	UserInputAdminPasswordOption	Changes to fixed. Does not keep user input value.
Storage configuration -> iSCSI Initiator Configuration	AdvancedUserInputUserConfigOption	Changes to fixed. Does not keep user input value.
	UserInputIqn	Keeps user input value.
	UserInputAlias	Keeps user input value.
	UserInputChapName	Changes to fixed. Does not keep user input value.
	UserInputChapSecret	Changes to fixed. Does not keep user input value.
	UserInputFirstBurstLength	Changes to fixed. Does not keep user input value.
	UserInputMaxBurstLength	Changes to fixed. Does not keep user input value.
General System Settings -> Management Agent Configuration -> SNMP Agent Configuration	UserInputMaxReceiveSegmentLength	Changes to fixed. Does not keep user input value.
	UserInputUsers	Changes to fixed. Does not keep user input value.
	UserInputV3Targets	Changes to fixed. Does not keep user input value.
	UserInputEngineId	Changes to fixed. Does not keep user input value.
	UserInputContact	Changes to fixed. Does not keep user input value.
	UserInputLocation	Changes to fixed. Does not keep user input value.

Policy option path	Policy option	Variations in the policy options value in the newly extracted profile
Security and Services -> Security Settings -> Security -> Active Directory Permission	UserInputUserName	Keeps user input value.
	UserInputPassword	Keeps user input value.
Networking Configuration	UserInputMacAddress	Conditional, prompt user if no default value is available.
	UserInputHostName	Conditional, prompt user if no default value is available.
	UserInputIPAddress	Conditional, prompt user if no default value is available.
Advanced Configuration Settings -> Advanced Options	UserInputAdvancedConfigOption	Changes to fixed. Does not keep user input value.
Advanced Configuration Settings -> System Image Cache Configuration	Enable Stateless Caching	You must explicitly select the policy option.
	Enable Stateful Install	You must explicitly select the policy option.

Note:
This table does not list all policy options. You must be aware of the policy options you modified before extracting a new host profile.

Recommended Upgrade Workflows for vSphere Host Profiles

View the recommended upgrade path while upgrading your 8.0 environment. To ensure the best possible performance, when you upgrade an ESX host, also upgrade the host profile. The host profile version must match the ESX host version.

Upgrade Support Matrix

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

When you upgrade your vCenter from version 8.0 to version 9.0, you can keep your current ESX host version and host profiles version.

Note:
vCenter instances and ESX hosts of versions 7.0 or earlier are not supported in vCenter 9.0. You can use the *Recommended Host Profiles Upgrade Workflows* in the *vSphere Host Profiles 8.0* guide as intermediate upgrade steps.

Table 120: Upgrade Support Matrix

Upgrade Workflow	vCenter	ESX host	Host Profiles
8.0 → 9.0	Requires an upgrade	Optional upgrade	Optional upgrade

For more information about vCenter and ESX Version Interoperability, see the Broadcom Compatibility Guide at https://compatibilityguide.broadcom.com/.

How Do You Upgrade vSphere Host Profiles and vCenter from 8.0 to 9.0

Use the upgrade workflows to resolve version compliance errors when upgrading vCenter with version 8.0 to version 9.0.

CAUTION:

The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

How Do You Upgrade Host Profiles in a vCenter Instance with Stateful ESX9.0 Hosts Only?

You use the present workflow to resolve version compliance errors when upgrading vCenter with version 9.0 to version 9.1, if your cluster contains stateful ESX9.0 hosts.

Prerequisites

- Verify that your cluster contains ESX9.0 hosts only.
- Verify that a host profile with version 9.0 is attached to the cluster.
- Verify that your vCenter is with version 9.0.

Procedure

1. Upgrade your vCenter instance to version 9.1.
   The vCenter configuration is not changed during the upgrade.
   For more information, see the *vCenter Upgrade* documentation.
2. Upgrade all ESX hosts in the cluster to version 9.1.
   Edit host customization and host profile edit operations are not available, while compliance check, attach host profile, and remediate hosts operations are available.
3. Remediate the cluster against the host profile with version 9.0.
   During remediation, all host profile settings are applied.
4. Extract a new host profile from an ESX9.1 host.
   During host profile extraction, some parameters of the configured host profile policy change. For more information, see Answer File Field and vSphere Host Profile Extraction.
5. Attach the host profile with version 9.1 to the cluster.
   The host customization data updates automatically.
   All host profile operations become available.

How Do You Upgrade Host Profiles in a vCenter Instance With Stateless ESX9.0 Hosts Only?

You use the present workflow to resolve version compliance errors when upgrading vCenter with version 9.0 to version 9.1, if your cluster contains stateless ESX9.0 hosts.

CAUTION:

Booting ESX hosts by network with in-memory or cached images, and preserving no state, is deprecated in ESX 9.0. For more information, see *Installing ESX Using vSphere Auto Deploy* in the *ESX Installation and Setup* guide.

Prerequisites

- Verify that your cluster contains ESX9.0 hosts only.
- Verify that a host profile with version 9.0 is attached to the cluster.
- Verify that your vCenter is with version 9.0.

Procedure

1. Upgrade your vCenter instance to version 9.1.
   The vCenter configuration is not changed during the upgrade.
   For more information, see the *vCenter Upgrade* documentation.
2. Create a new rule or edit the existing rule with an image profile version 9.1 and select the cluster.

3. Remediate all ESX hosts in the cluster by clicking **Remediate Host Associations.**
4. Reboot all ESX hosts in the cluster.
   All hosts are compliant with the host profile version 9.0, but the host profile cannot be edited. Remediation and check compliance operations are available.

What to do next

To be able to customize your host profile, follow these steps.

1. Extract a new host profile from an ESX9.1 host.
   During host profile extraction, some parameters of the configured host profile policy change. For more information, see Answer File Field and vSphere Host Profile Extraction.
2. Attach the host profile with version 9.1 to the cluster.
   The host customization data updates automatically.
   All host profile operations become available.

Using vSphere Host Profiles to Migrate an NSX Virtual Distributed Switch to a vSphere Distributed Switch

For clusters enabled with VMware NSX, you can use the regular ESX upgrade workflow to migrate the NSX-managed Virtual Distributed Switches of the hosts to vCenter-managed vSphere Distributed Switches. By using host profiles, you can perform the upgrade and the host switch migration in an environment where all hosts are provisioned with Auto Deploy.

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

Requirements

- Verify that the ESX hosts and the vCenter instance are of version 9.0.
- Verify that VMware NSX is of version 3.1.1 or later.

Workflow

1. Upload a software depot that contains an ESX 9.0 image to vCenter.
   For more information, see the *VMware ESX Installation and Setup* documentation.
2. In NSX Manager, run the migration readiness pre-check, address any configuration issues, review the recommended topology, and create a vSphere Distributed Switch.

   Note:
   At this point, do not make any configuration changes to VMware NSX.

   For more information about the steps you need to perform in NSX Manager, see the *NSX Administration* documentation.
3. Create a new image rule or update the existing one with the uploaded ESX 9.0 image.
   For more information, see the *VMware ESX Installation and Setup* documentation.
4. Reboot the reference host.
   The reboot triggers the migration of the host switches from NSX Virtual Distributed Switches to vSphere Distributed Switches.
   After the reboot, the ESX version on the reference host is upgraded to 9.0. A new TNP with vSphere Distributed Switch is attached to the cluster.
5. Update the host profile by copying the settings from the reference host.
   The updated host profile is configured with vSphere Distributed Switch instead of NSX Virtual Distributed Switch.
   For more information, see How Do You Copy Settings from a Reference Host to a vSphere Host Profile?

6. Reboot all ESX hosts in the cluster.
   After the reboot, all hosts in the cluster use the new host profile, which is configured with vSphere Distributed Switch instead of NSX Virtual Distributed Switch.

Troubleshooting vSphere Host Profiles

The Host Profiles troubleshooting topics provide solutions to problems that you might encounter when performing tasks for vSphere Host Profiles in vCenter .

> **CAUTION:**
>
> The vSphere Host Profiles capability is deprecated in vSphere 9.0 and will be removed in a future release. Instead of Host Profiles, you can use vSphere Configuration Profiles to manage configurations at cluster level. For more details, see this blog.

If upgrading your vCenter to version 9.1 results with a failure due to an existing host profile in the inventory, see *vCenter Upgrade* for Upgrade Issues with vCenter Containing Host Profiles.

Host Customization Data Is Missing

Deploying a stateless host results with a failure in exiting maintenance mode.

When you are in the process of deploying a stateless host, the host fails in exiting maintenance mode.

> **CAUTION:**
>
> Booting ESX hosts by network with in-memory or cached images, and preserving no state, is deprecated in ESX 9.0. For more information, see *Installing ESX Using vSphere Auto Deploy* in the *ESX Installation and Setup* guide.

In an environment where the following takes place for a host profile:

- The host profile is attached to a stateless host.
- The host profile has host-specific settings.

When the stateless host is rebooting, applying the host profile fails because the host does not exit from a maintenance mode.

- Provide a host customization as described in How Do You Edit Host Customizations?.
- Remediate the host profile to the host again.
- Reboot the host, if needed.
- Refer to the host profiles command-line utility in host profiles CLI for customizing stateless hosts.

Reference Host is Unavailable

Creating a vSphere Host Profile in the vCenter inventory results with a failure.

When you perform Host Profiles operations in the vCenter inventory such as:

- Edit a Host Profile.
- Import a Host Profile.
- Export a Host Profile.

The process of creating a host profile fails with an error.

```
There is no suitable host in the inventory as reference host for the profile Host Profile. The profile does
 not have any associated reference host.
```

The vCenter inventory does not have a compatible host that acts as a reference host for the host profile you are trying to create.

Add a host that acts as a reference host in the vCenter inventory with the same version as the host profile you are trying to create.

Compliance Failures with vSphere Storage Host Profiles

When you use storage devices, that are not shared across a cluster, but the vSphere storage stack cannot detect them as local (for example, some SAS devices), applying a host profile might result with compliance failures.

To resolve the compliance failures caused by using local storage devices, use the upgraded Pluggable Storage Architecture (PSA) host profile policies.

To find out whether a device is detected as local or not, you can check the **Is Local** setting for a device by running the command `esxcli storage core device list -d naa.xxxx` in the ESX shell.

> **Note:**
> ESX diagnostic data that you obtain by running the `vm-support` command contains host profiles information which includes storage host profiles, PSA, NMP, and Virtual Volumes data. No sensitive information, such as passwords, is collected.

Compliance Failures with Serial Attached SCSI (SAS) Devices

This procedure resolves compliance failures for SAS devices that are not detected as local by the vSphere storage stack.

Extract a host profile from a reference host. See How Do You Create a vSphere Host Profile? for instructions.

1. Navigate to **Host Profiles** main view and select the host profile that you want to edit.
2. Click the **Configure** tab, and click **Edit Host Profile**.
3. Select **Storage configuration** > **Pluggable Storage Architecture configuration** > **PSA device sharing** > ***name of device***
4. For each device not shared across the cluster, deactivate **Device is shared clusterwide**.

   > **Note:**
   > By default, the **Device is shared clusterwide** setting is deactivated for devices detected as local and enabled for non-local devices. This setting allows storage host profiles to ignore local devices during compliance checks.

   The **Device is Shared Clusterwide** setting for PSA devices helps you determine which devices in the cluster should be configured by a host profile. Correctly setting this value for devices in the cluster eliminates compliance errors due to non-shared devices.
5. Remediate the host profile to the reference host for the changes to take effect.

Compliance failures with SAN Boot LUN Devices

This procedure resolves compliance failures for SAN boot LUN devices shared across the cluster, but logically local to the ESXhost.

Extract a host profile from a reference host. See How Do You Create a vSphere Host Profile? for instructions.

To avoid this compliance failure from occurring, set the **Device is shared clusterwide** value to **False** before extracting the host profile from the reference host. You can deactivate the value by using the command `esxcli storage core device setconfig -d naa.xxxx --shared-clusterwide=false`, where *naa.xxxx* is the unique device identifier generated in naa format.

1. Navigate to **Host Profiles** main view and select the host profile that you want to edit.
2. Click the **Configure** tab, and click **Edit Host Profile**.
3. Select **Storage configuration** > **Pluggable Storage Architecture configuration** > **PSA device sharing** > ***name of device***
4. Deactivate the **Device is shared clusterwide** in the host profile by deselecting the check box.

   When applying the host profile to the target host, the boot device settings for the remote boot LUN device are copied from the reference host into the target host.
5. Optional: Select **Storage configuration** > **Pluggable Storage Architecture configuration** > **Host boot device configuration** and verify that the boot LUN ID is correctly captured.
6. Remediate the host profile to the reference host for the changes to take effect.

   If you must re-extract the host profile (for example, if you attach more shared SAN boot LUNs to your cluster), you do not need to reconfigure the **Device is Shared Clusterwide** setting for devices that you previously configured through ESXCLI.

vSphere Host Profile without an NFS Datastore

A host profile compliance check fails for NFS datastore.

When you confirm the compliance of your ESX host to its attached Host Profile, a non-compliant status is present for the NFS datastore. The compliance check detects the mounted NFS datastore to the ESX host as an additional datastore.

The remediate or compliance check failure occurs when the following two conditions take place:

- The extracted Host Profile does not have NFS storage (NasStorageProfile).
- The ESX host, attached to the Host Profile, has an already mounted NFS storage.

1. Create a dummy NFS datastore in the Host Profile.
2. Deactivate the dummy NFS datastore so that the NFS storage profile is fully deactivated.

Upgraded Host is Not Compliant with a Legacy Host Profile

An upgraded ESX host with attached legacy vSphere host profile results in a non-compliant status after reboot.

When you upgrade a host to the latest ESX version and remediate the attached legacy host profile to that host, the compliance status might be updated to non-compliant after the host reboots.

You are using a legacy host profile, which is attached to an ESX host version 6.7 or earlier and contains the following sub-profiles:

- `Misc.LogPort`
- `host/vim/vmvisor/plugins/vmware_*`

- Make the ESX host compliant with the legacy host profile.
  a. From the **Host Profiles** main view, right-click the host profile that you want to edit and select **Edit Host Profile**.
  b. In the **Edit host profile** page, deactivate the sub-profile(s).
  c. **Check Host Profile Compliance**.
- Upgrade the legacy host profile to the same version as the ESX host.
  a. Right-click the legacy host profile and select **Copy Settings from Host**.
  b. Select the upgraded ESX host from which you want to copy the configuration settings and click **OK**.

vSphere Virtual Machine Administration

vSphere Virtual Machine Administration describes how to create, configure, and manage virtual machines in the VMware vSphere® environment.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we have updated this guide to remove instances of non-inclusive language.

This guide provides introductions to the tasks that you can perform within the system and also cross-references to the documentation that describes the tasks in detail.

This information focuses on managing virtual machines and includes the following information.

- Creating and deploying virtual machines, templates, and clones
- Deploying OVF templates
- Using content libraries to manage templates and other library items
- Configuring virtual machine hardware and virtual machine options
- Managing multi-tiered applications with VMware vSphere vApp
- Monitoring solutions with the vCenter Solutions Manager
- Managing virtual machines, including using snapshots
- Upgrading virtual machines
- Troubleshooting virtual machines

vSphere Virtual Machine Administration covers VMware ESX™ and VMware vCenter®.

Intended Audience

This information is written for experienced Windows or Linux system administrators who are familiar with virtualization.

Introduction to VMware vSphere Virtual Machines

A virtual machine is a software computer that, like a physical computer, runs an operating system and applications. The virtual machine consists of a set of specification and configuration files and is backed by the physical resources of a host. Every virtual machine has virtual devices that provide the same functionality as physical hardware but are more portable, more secure, and easier to manage.

Before you start creating and managing virtual machines, you benefit from some background information, for example, the virtual machine files, life cycle, components, and so on.

Virtual Machine Files

A virtual machine consists of several files that are stored on a storage device. The key files are the configuration file, virtual disk file, NVRAM setting file, and log file. You configure virtual machine settings through the vSphere Client, ESXCLI, or the vSphere Web Services SDK.

> **CAUTION:**
> Do not change, move, or delete virtual machine files without instructions from a VMware Technical Support representative.

Table 121: Virtual Machine Files

File	Usage	Description
`.vmx`	*vmname*`.vmx`	Virtual machine configuration file
`.vmxf`	*vmname*`.vmxf`	Additional virtual machine configuration files
`.vmdk`	*vmname*`.vmdk`	Virtual disk characteristics
`-flat.vmdk`	*vmname*`-flat.vmdk`	Virtual machine data disk

File	Usage	Description
.nvram	*vmname*.nvram or nvram	Virtual machine BIOS or EFI configuration
.vmem	*vmname*.vmem	Virtual machine paging backup file
.vmsd	*vmname*.vmsd	Virtual machine snapshots information (metadata) file
.vmsn	*vmname*.vmsn	Virtual machine memory snapshot file
.vswp	*vmname*.vswp	Virtual machine swap file
.vmss	*vmname*.vmss	Virtual machine suspend file
.log	vmware.log	Current virtual machine log file
-#.log	vmware-#.log (where # is a number starting with 1)	Old virtual machine log files
vmname.scoreboard	vm-1.scoreboard	Stores performance information for a virtual machine

Additional files are created when you perform certain tasks with the virtual machine.

- A .hlog file is a log file that is used by vCenter to keep track of virtual machine files that must be removed after a certain operation completes.
- A .vmtx file is created when you convert a virtual machine to a template. The .vmtx file replaces the virtual machine configuration file (.vmx file).
- A .vmfd files contain metadata for vmdk disks.

Virtual Machines and the Virtual Infrastructure

The infrastructure that supports virtual machines consists of at least two software layers, virtualization and management. In vSphere, ESX provides the virtualization capabilities that aggregate and present the host hardware to virtual machines as a normalized set of resources. Virtual machines run on ESX hosts that vCenter manages.

vCenter can pool the resources of multiple hosts and lets you effectively monitor and manage your data center infrastructure. You can manage resources for virtual machines, provision virtual machines, schedule tasks, collect statistics logs, create templates, and more. vCenter also provides vSphere vMotion™, vSphere Storage vMotion, vSphere Distributed Resource Scheduler (DRS), vSphere High Availability (HA), and vSphere Fault Tolerance. These services enable efficient and automated resource management and high availability for virtual machines.

The vSphere Client is the primary interface for managing vCenter, ESX hosts, and virtual machines. The vSphere Client also provides console access to virtual machines.

For information about running virtual machines on an isolated ESX host, see the *vSphere Single Host Management* documentation.

The vSphere Client presents the organizational hierarchy of managed objects in inventory views. Inventories are the hierarchical structure used by vCenter or the host to organize managed objects. This hierarchy includes the monitored objects in vCenter.

In the vCenter hierarchy that you see in the vSphere Client, a data center is the top-level container of ESX hosts, folders, clusters, resource pools, vSphere vApps, virtual machines, and so on.

Datastores are virtual representations of underlying physical storage resources. Datastores hide the idiosyncrasies of the underlying physical storage and present a uniform model for the storage resources required by virtual machines. A datastore is the storage location (for example, a physical disk or LUN on a RAID, or a SAN) for virtual machine files.

Virtual Machine Lifecycle

You have many options for creating and deploying virtual machines. You can create a single virtual machine and install a guest operating system and VMware Tools on it. You can clone an existing virtual machine or convert it to a template. You can also deploy OVF or OVA templates.

The vSphere Client **New Virtual Machine** wizard and the **Edit Settings** dialog box let you add, configure, or remove most of the virtual machine's hardware, options, and resources. You monitor CPU, memory, disk, network, and storage metrics through the performance charts in the vSphere Client. Snapshots let you capture the state of the virtual machine,

including the virtual machine memory, settings, and virtual disks. You can roll back to the previous virtual machine state when needed.

With vSphere vApps, you can manage multi-tiered applications. You use vSphere Lifecycle Manager to perform orchestrated upgrades to upgrade the virtual hardware and VMware Tools of virtual machines in the inventory at the same time.

When a virtual machine is no longer needed, you can remove it from the inventory without deleting it from the datastore, or you can delete the virtual machine and all its files.

Virtual Machine Components

Virtual machines typically have an operating system, VMware Tools, and virtual resources and hardware. You manage these components like you manage the components of a physical computer.

Operating System	You install a guest operating system on a virtual machine the same way that you install an operating system on a physical computer. You must have a CD/DVD-ROM or ISO image containing the installation files from an operating system vendor. After installation, you are responsible for securing and patching the operating system.
VMware Tools	VMware Tools is a suite of utilities that enhances the performance of the virtual machine guest operating system and improves management of the virtual machine. It includes device drivers and other software that is essential for your VM. With VMware Tools, you have more control over the virtual machine interface.
Compatibility Setting	In the vSphere Client, you assign each virtual machine to a compatible ESX host version, cluster, or data center by applying a compatibility setting. The compatibility setting determines which ESX host versions the virtual machine can run on and the hardware features available to the virtual machine.
Hardware Devices	Each virtual hardware device performs the same function for the virtual machine as hardware on a physical computer does. Every virtual machine has CPU, memory, and disk resources. CPU virtualization emphasizes performance and runs directly on the processor whenever possible. The underlying physical resources are used whenever possible. The virtualization layer runs instructions only as needed to make virtual machines operate as if they were running directly on a physical machine. All recent operating systems provide support for virtual memory, allowing software to use more memory than the machine physically has. Similarly, the ESX hypervisor provides support for overcommitting virtual machine memory, where the amount of guest memory configured for all virtual machines might be larger than the amount of the host's physical memory. You access the hardware devices in the **Edit Settings** dialog box. Not all devices are configurable. Some hardware devices are part of the virtual motherboard and appear in the expanded device list of the **Edit Settings** dialog box, but you cannot modify or remove them. For a list of hardware devices and their functions, see Virtual Machine Hardware Available to vSphere Virtual Machines. In the **Edit Settings** dialog box, you can also add virtual hardware devices to the virtual machine. You can use the memory or CPU hotplug options to add memory or CPU resources to a virtual machine while the virtual machine is running. You can deactivate Memory or CPU hotplug to avoid adding memory or CPUs while the virtual machine is running. Memory hotplug is supported on all 64-bit operating systems, but to use the added memory, the guest operating system must also support this feature. For more information, see the Broadcom Compatibility Guide.

A vSphere administrator or other privileged user can determine who can access or modify a virtual machine by setting permissions on the virtual machine. See the *vSphere Security* documentation.

Virtual Machine Hardware Available to vSphere Virtual Machines

VMware provides devices, resources, profiles, and vServices that you can configure or add to your virtual machine.

Not all hardware devices are available to every virtual machine. The host that the virtual machine runs on and the guest operating system must support devices that you add or configurations that you make. To verify support for a device in your environment, see the Broadcom Compatibility Guide or the Guest Operating System Installation Guide.

The PCI and SIO virtual hardware devices are part of the virtual motherboard, but cannot be configured or removed.

Table 122: Virtual Machine Hardware and Descriptions

Hardware Device	Description
CPU	You can configure a virtual machine that runs on an ESX host to have one or more virtual processors. A virtual machine cannot have more virtual CPUs than the actual number of logical CPUs on the host. You can change the number of CPUs allocated to a virtual machine and configure advanced CPU features, such as the CPU Identification Mask and hyperthreaded core sharing.
Chipset	The motherboard uses VMware proprietary devices based on the following chips: • Intel 440BX AGPset 82443BX Host Bridge/Controller • Intel 82371AB (PIIX4) PCI ISA IDE Xcelerator • National Semiconductor PC87338 ACPI 1.0 and PC98/99 Compliant SuperI/O • Intel 82093AA I/O Advanced Programmable Interrupt Controller
DVD/CD-ROM Drive	Installed by default when you create a virtual machine. You can configure DVD/CD-ROM devices to connect to client devices, host devices, or datastore ISO files. You can add, remove, or configure DVD/CD-ROM devices.
Hard Disk	Stores the operating system of a virtual machine, program files, and other data associated with its activities. A virtual disk is a large physical file, or a set of files, that can be copied, moved, archived, and backed up as easily as any other file.
IDE 0, IDE 1	By default, two Integrated Drive Electronics (IDE) interfaces are presented to the virtual machine. The IDE interface (controller) is a standard way for storage devices (floppy drives, hard drives, and CD-ROM drives) to connect to the virtual machine.
Keyboard	Provides keyboard input from any virtual machine consoles.
Memory	The virtual hardware memory size determines how much memory applications that are running inside the virtual machine have available to them. A virtual machine cannot benefit from more memory resources than its configured virtual hardware memory size.
Network Adapter	ESX networking features provide communication between virtual machines on the same host, between virtual machines on different hosts, and between other virtual and physical machines. When you configure a virtual machine, you can add network adapters (NICs) and specify the adapter type.
Parallel port	Interface for connecting peripherals to the virtual machine. The virtual parallel port can connect to a file. You can add, remove, or configure virtual parallel ports.
PCI controller	Bus on the virtual machine motherboard that communicates with components such as hard disks and other devices. One PCI controller is presented to the virtual machine. You cannot configure or remove this device.
PCI Device	You can add up to 16 PCI vSphere DirectPath devices to a virtual machine. The devices must be reserved for PCI passthrough on the host on which the virtual machine runs. Snapshots are not supported with DirectPath I/O passthrough devices.
Pointing device	Mirrors the pointing device that is connected to the virtual machine console when you first connect to the console.
Serial Port	Interface for connecting peripherals to the virtual machine. The virtual serial port can connect to a physical serial port, to a file on the host computer, or over the network. You can also use it to establish a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. You can configure a virtual machine with up to 32 serial ports. You can add, remove, or configure virtual serial ports.
SATA controller	Provides access to virtual disks and DVD/CD-ROM devices. The SATA virtual controller appears to a virtual machine as an AHCI SATA Controller.
SCSI controller	Provides access to virtual disks. The SCSI virtual controller appears to a virtual machine as different types of controllers, including LSI Logic Parallel, LSI Logic SAS, and VMware Paravirtual. You can change the SCSI controller type, allocate bus sharing for a virtual machine, or add a paravirtualized SCSI controller.
SIO controller	Provides serial and parallel ports, floppy devices, and performs system management activities. One SIO controller is available to the virtual machine. You cannot configure or remove this device.

Hardware Device	Description
USB controller	The USB hardware chip that provides USB 1.x and USB 2.0 function to the USB ports that it manages. The virtual USB Controller is the software virtualization of the USB 1.x and USB 2.0 host controller function in the virtual machine.
USB xHCI controller	The USB hardware chip that provides USB 3 function to the USB ports it manages. The virtual USB xHCI controller is the software virtualization of the USB 3 host controller function in the virtual machine.
USB device	You can add multiple USB devices to a virtual machine, such as security dongles and mass storage devices. The USB devices can be connected to an ESX host or a client computer.
VMCI	Virtual Machine Communication Interface device. Provides a high-speed communication channel between a virtual machine and the hypervisor. You cannot add or remove VMCI devices.
NVMe controller	NVM Express controller. NVMe is a logical device interface specification for accessing nonvolatile storage media attached through a PCI Express (PCIe) bus in real and virtual hardware.
NVDIMM controller	Provides access to the non-volatile memory resources of the host.
NVDIMM device	Non-Volatile Dual In-Line Memory Module. NVDIMM modules are memory devices that sit on an ordinary memory channel, but contain non-volatile memory. You can add up to 64 virtual NVDIMM devices to a virtual machine.
TPM device	Trusted Platform Module. When you add a virtual TPM 2.0 device to a virtual machine, the guest OS uses the device to store sensitive information, perform cryptographic tasks, or attest the integrity of the guest platform.
Virtual Precision Clock device	A virtual clock device that provides a virtual machine with access to the system time of the primary ESX host.
Virtual Watchdog Timer device	To ensure self-reliance related to the system performance within a virtual machine. If the guest operating system stops responding and cannot recover on its own due to software glitches or errors, the watchdog timer waits for a predefined period of time and then restarts the system.
vSGX device	Virtual Intel® Software Guard Extensions (vSGX) provides a virtual machine additional security to your workloads. Intel SGX is a processor-specific technology that defines private regions of memory, called enclaves. Intel SGX protects the enclave contents from disclosure and modification in such a way that code running outside the enclave cannot access them.
RDM disk	You can use a raw device mapping (RDM) to store virtual machine data directly on a SAN LUN, instead of storing it in a virtual disk file.
PS2 controller	PS2 controller provides access to the virtual keyboard and point to the PS2 interface.
Video card	A virtual graphics card that provides graphics acceleration and display capabilities for virtual machine consoles.

Virtual Machine Options

Use the available virtual machine options to fine-tune the settings and behavior of your virtual machine and to ensure maximum performance.

A virtual machine might be running in any of several locations, such as ESX hosts, data centers, clusters, or resource pools. Many of the options and resources that you configure have dependencies on and relationships with these objects.

VMware virtual machines have the following options.

General Options	View or modify the virtual machine name, and check the location of the configuration file and the working location of the virtual machine.
VMware Remote Console Options	Enhance the virtual machine access control and security by locking the guest OS screen when the console is closed, and limit the number of users that can connect simultaneously to the VM console.
Encryption Options	Enable or deactivate encryption for the virtual machine if the vCenter instance is in a trusted relationship with a KMS server.

	Activate the Fault Tolerance Encryption to encrypt Fault Tolerance log traffic. For more information, see the *vSphere Security* documentation. You can also enable or deactivate encrypted vMotion for virtual machines that are not encrypted. You can set encrypted vMotion to the deactivated, opportunistic, or required state. You can enable encrypted vMotion during virtual machine creation. Alternatively, you can change the encrypted vMotion state later. For more information, see the *vCenter and Host Management* documentation.
Confidential Compute	Enable hardware-based isolation and encryption of the virtual machine memory.
Power Management	Manage guest power options. Suspend the virtual machine or leave the virtual machine powered on when you put the guest operating system into standby.
VMware Tools	Manage the power controls for the virtual machine, run VMware Tools scripts, and upgrade VMware Tools during power cycling. Automatically synchronize the time between guest and host operating systems, and periodically synchronize the guest time with the host if the virtual machine guest operating system does not have a native time synchronization software.
Virtualization Based Security (VBS)	Enable VBS to provide an additional level of protection to the virtual machine. VBS is available on the latest Windows OS versions. For more information, see the *vSphere Security* documentation.
Boot Options	Set the boot delay when powering on virtual machines or to force BIOS setup and configure failed boot recovery.
Advanced Options	Enable logging, configure debugging and statistics, and change the swap file location. You can also change the latency sensitivity, add configuration parameters, and set the vMotion Stun Time limit.
Fibre Channel NPIV	Control virtual machine access to LUNs on a per-virtual machine basis. N-port ID virtualization (NPIV) provides the ability to share a single physical Fibre Channel HBA port among multiple virtual ports, each with unique identifiers.
vApp Options	Enable or deactivate the vApp functionality in a virtual machine. When you enable vApp options, you can view and edit vApp properties, vApp Deployment options, and vApp Authoring options. For example, you can configure an IP allocation policy or a network protocol profile for the vApp. A vApp option that is specified at the level of a virtual machine overrides the settings specified at the level of the vApp.
CPU Topology	Adjust Cores per socket and NUMA nodes, and enable CPU hot-add.

The vSphere Client

All administrative functions are available through the vSphere Client.

The vSphere Client is a cross-platform application that can connect only to vCenter. It has a full range of administrative functionality and an extensible plug-in-based architecture. Typical users are virtual infrastructure administrators, help desk, network operations center operators, and virtual machine owners.

Users can use the vSphere Client to access vCenter through a Web browser. vSphere Client uses the VMware API to mediate the communication between the browser and the vCenter.

The vSphere Client introduces some productivity enhancements and usability improvements. Watch the following video to learn about the usability improvements to working with virtual machine hard disks.

Where to Go From Here

You must create, provision, and deploy your virtual machines before you can manage them.

To begin provisioning virtual machines, determine whether to create a single virtual machine and install an operating system and VMware tools, work with templates and clones, or deploy virtual machines, virtual appliances, or vApps stored in Open Virtual Machine Format (OVF).

After you provision and deploy virtual machines into the vSphere infrastructure, you can configure and manage them. You can configure existing virtual machines by modifying or adding hardware or install or upgrade VMware Tools. You might need to manage multitiered applications with VMware vApps or change virtual machine startup and shutdown settings, use virtual machine snapshots, work with virtual disks, or add, remove, or delete virtual machines from the inventory.

Deploying Virtual Machines

VMware supports several methods to provision vSphere virtual machines. What works best in your environment depends on factors such as the size and type of your infrastructure and the goals that you want to achieve.

vSphere 9.1 provides a flexible mechanism for organizing virtual machines in your environment based on custom criteria by assigning vSphere tags during the following virtual machine and template creation operations:

- Creating a virtual machine.
- Cloning an existing virtual machine.
- Deploying a virtual machine from a template.
- Cloning a virtual machine to a template.
- Converting a template to a virtual machine.
- Cloning a template to another template.

Create a single virtual machine if no other virtual machines in your environment have the requirements you are looking for, such as a particular operating system or hardware configuration. You can also create a single virtual machine and install an operating system on it, and then use that virtual machine as a template from which to clone other virtual machines. For more information, see Create a Virtual Machine with the New Virtual Machine Wizard.

To use a preconfigured virtual machine, deploy and export virtual machines, virtual appliances, and vApps stored in Open Virtual Machine Format (OVF). A virtual appliance is a virtual machine that typically has an operating system and other software installed. You can deploy virtual machines from local file systems and from shared network drives. For more information, see Deploy and Export OVF and OVA Templates.

Create a template and deploy multiple virtual machines from it. A template is a primary copy of a virtual machine that you can use to create and provision virtual machines. Use templates to save time. If you frequently clone a virtual machine, consider making it a template. For more information, see Deploy a Virtual Machine from a Template .

If you are deploying many similar virtual machines, cloning a virtual machine can save time. You can create, configure, and install software on a single virtual machine. You can clone it multiple times, rather than creating and configuring each virtual machine individually. For more information, see Clone an Existing Virtual Machine.

Create a Virtual Machine with the New Virtual Machine Wizard

If no virtual machines in your environment meet your needs, you can create a single virtual machine, for example, of a particular operating system or hardware configuration.

Verify that you have the following privileges:

- **Virtual machine** > **Edit inventory** > **Create new** on the destination folder or data center.
- **Virtual machine** > **Change Configuration** > **Add new disk** on the destination folder or data center, if you are adding a new disk.
- **Virtual machine** > **Change Configuration** > **Add existing disk** on the destination folder or data center, if you are adding an existing disk.
- **Virtual machine** > **Configuration** > **Configure Raw device** on the destination folder or data center, if you are using an RDM or SCSI pass-through device.
- **Virtual machine** > **Change Configuration** > **Configure Host USB device** on the destination folder or data center, if you are attaching a virtual USB device backed by a host USB device.
- **Virtual machine** > **Change Configuration** > **Advanced configuration** on the destination folder or data center, if you are configuring advanced virtual machine settings.
- **Virtual machine** > **Configuration** > **Change Swapfile placement** on the destination folder or data center, if you are configuring swap file placement.
- **Virtual machine** > **Change Configuration** > **Toggle disk change tracking** on the destination folder or data center, if you are enabling change tracking on the virtual machine's disks.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **Datastore** > **Allocate space** on the destination datastore or datastore folder.
- **Network** > **Assign network** on the network that the virtual machine will be assigned to.
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

To assign vSphere tags to a virtual machine, verify that you have the following privileges:

- **vSphere Tagging** > **Assign or Unassign vSphere Tag** and **vSphere Tagging** > **Assign or Unassign vSphere Tag on Object**

To verify the privileges assigned to your role, click **Menu** > **Administration** > **Roles** and select the role.

If you want to create a virtual machine that uses persistent memory, choose a host or a cluster with an available PMem resource.

When you create a virtual machine without a template or clone, you can configure the virtual hardware, including processors, hard disks, and memory. You open the New Virtual Machine wizard from any object in the inventory that is a valid parent object of a virtual machine.

During the creation process, a default disk is configured for the virtual machine. You can remove this disk and add a new hard disk, select an existing disk, or add an RDM disk on the Virtual Hardware page of the wizard.

1. Right-click an inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **New Virtual Machine**.
2. On the **Select a creation type** page, select **Create a new virtual machine** and click **Next**.
3. On the **Select a name and folder** page, enter a unique name for the virtual machine and select a deployment location.
4. Click **Next**.
5. On the **Select a compute resource** page, select the host, cluster, resource pool, or vApp where the virtual machine will run and click **Next**.

   If creating the virtual machine at the selected location causes compatibility problems, an alarm appears in the **Compatibility** pane.
6. On the **Select storage** page, choose the storage type, the storage policy, and a datastore or datastore cluster where to store the virtual machine files, and click **Next**.

Option	Description
Create a virtual machine on a host that has PMem resource	1. Choose the type of storage by selecting the **Standard** or the **PMem** radio button. With the **PMem** storage option, every virtual machine disk file is stored on the host-local PMem datastore by default. You can change the datastore at a later time. The virtual machine home location must be on a non-PMem datastore. For more information about persistent memory and PMem storage, refer to the vSphere Resource Management guide. 2. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 3. To encrypt the virtual machine, select the **Encrypt this virtual machine** check box. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Select a datastore or a datastore cluster.
Create a virtual machine on a host that does not have PMem resource	1. Select a VM storage policy or leave the default one. 2. To encrypt the virtual machine, select the **Encrypt this virtual machine** check box. 3. Select a datastore or a datastore cluster. 4. If you do not want to use storage DRS with the virtual machine, select the **Disable Storage DRS for this virtual machine** check box.

   For information about creating an encrypted virtual machine, see *vSphere Security*.
7. On the **Select compatibility** page, select the virtual machine compatibility with ESX host versions and click **Next**.

   To have access to the latest hardware features, select the latest ESX host version.

8. On the **Select a guest OS** page, select the guest OS family and version.
   a) (Optional) Select the **Enable Windows Virtualization Based Security** check box.

      When you enable this option, hardware virtualization, IOMMU, EFI, and secure boot become available to the guest operating system. You must also enable **Virtualization Based Security** within the guest operating system of this virtual machine.

      The **Enable Windows Virtualization Based Security** option is available only for the latest Windows OS versions, for example, Windows 10 and Windows Server 2016. For more information about VBS, see the *vSphere Security* documentation.

   When you select a guest operating system, BIOS or Extensible Firmware Interface (EFI) is selected by default, depending on the firmware supported by the operating system. Mac OS X Server guest operating systems support only EFI. If the operating system supports BIOS and EFI, you can change the default by editing the virtual machine after you create it and before you install the guest operating system. If you select EFI, you cannot boot an operating system that supports only BIOS, and the reverse.

   Important:
   Do not change the firmware after the guest operating system is installed. The guest operating system installer partitions the disk in a particular format, depending on which firmware the installer was booted from. If you change the firmware, you will not be able to boot the guest.

9. On the **Customize hardware** page, configure the virtual machine hardware and options and click **Next**.

   You can leave the defaults and configure the virtual machine hardware and options later. For more information, see Configuring Virtual Machine Hardware and Configuring Virtual Machine Options.

   Important:
   If you chose to use PMem storage for the virtual machine, its default hard disk, the new hard disks that you configure, and the NVDIMM devices that you add to the virtual machine all share the same PMem resources. So, you must adjust the size of the newly added devices in accordance with the amount of the PMem available to the host. If any part of the configuration requires attention, the wizard alerts you.

10. On the **Assign Tag** page, select the vSphere Tags that you want to assign to the virtual machine and click **Next**.
    a) To filter the tags list, select **Advanced Filter** and click **Add New**. Use the **Advanced Filter** dialog box to filter by **Tag Name**, **Category**, or **Description**. You can then specify a logical operator, enter a comparison value, and click **Apply**. Alternatively, you can use the **Quick Filter** to find specific tags.
    a) Click the **Selected** tab to verify the tags you want to assign.
    b) To remove tags, on the **Selected** tab, select the tags that you want to remove, and click **Remove**.
11. On the **Ready to complete** page, review the details and click **Finish**.

The virtual machine appears in the vSphere Client inventory.

Clone a Virtual Machine to a Template

After you create a virtual machine, you can clone it to a template. Templates are primary copies of virtual machines that let you create ready-for-use virtual machines.

If a load generator is running in the virtual machine, stop it before you perform the clone operation.

Verify that you have the following privileges:

- **Virtual machine** > **Provisioning** > **Create template from virtual machine** on the source virtual machine.
- **Virtual machine** > **Edit inventory** > **Create from existing** on virtual machine folder where the template is created.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **Datastore** > **Allocate space** on all datastores where the template is created.
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

To clone a virtual machine and its assigned tags to a template, verify that you have the following privileges:

- **vSphere Tagging** > **Assign or Unassign vSphere Tag** and **vSphere Tagging** > **Assign or Unassign vSphere Tag on Object**

In 9.1 and later, when you clone a virtual machine to a template, vSphere tag information is stored within the template. You can replicate the tags when you convert the template to a virtual machine.

You can make changes to the template, such as installing additional software in the guest operating system, while preserving the original virtual machine.

You cannot modify templates after you create them. To alter an existing template, you must convert it to a virtual machine, make the required changes, and convert the virtual machine back to a template. To preserve the original state of a template, clone the template to a template.

1. Start the **Clone Virtual Machine To Template** wizard.

Option	Description
From a valid parent object of a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **New Virtual Machine**. 2. On the Select a creation type page, select **Clone virtual machine to template** and click **Next**. 3. On the Select a virtual machine page, select the virtual machine that you want to clone.
From a virtual machine	Right-click the virtual machine and select **Clone** > **Clone to Template**.

2. On the **Select a name and folder** page, enter a name for the template and select a data center or folder in which to deploy it, and click **Next**.

   The template name determines the name of the files and folder on the disk. For example, if you name the template win8tmp, the template files are named win8tmp.vmdk, win8tmp.nvram, and so on. If you change the template name, the names of the files on the datastore do not change.

   Folders provide a way to store virtual machines and templates for different groups in an organization and you can set permissions on them. If you prefer a flatter hierarchy, you can put all virtual machines and templates in a datacenter and organize them a different way.

3. On the **Select a compute resource** page, select a host or cluster resource for the template, and click **Next**.

   The **Compatibility** pane shows the result from the compatibility checks.

   Important:

   If the virtual machine that you clone has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, you cannot proceed with the task.

   If the virtual machine that you clone does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, all the hard disks of the template will use the storage policy and datastore selected for the configuration files of the source virtual machine.

4. On the Select storage page, select the datastore or datastore cluster in which to store the template configuration files and all of the virtual disks. Click **Next**.

Option	Description
Clone a virtual machine that has vPMem hard disks	1. Choose the type of storage for the template by selecting the **Standard**, the **PMem**, or the **Hybrid** radio button. If you select the **Standard** mode, all virtual disks are stored on a standard datastore. If you select the **PMem** mode, all virtual disks are stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. If you select the **Hybrid** mode, all PMem virtual disks remain stored on a PMem datastore. Non-PMem disks are affected by your choice of a VM storage policy and datastore or datastore cluster. For more information about persistent memory and PMem storage, see the *vSphere Resource Management* guide.

Option	Description
	2. From the **Select virtual disk format** drop-down menu, select a new virtual disk format for the template or keep the same format as the source virtual machine. 3. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 4. Select a datastore or a datastore cluster. 5. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 6. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.
Clone a virtual machine that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. **Same format as source** uses the same disk format as the source virtual machine. The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine. **Thick Provision Eager Zeroed** is a type of thick virtual disk that supports clustering features such as Fault tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat file format (`-flat.vmdk`), the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks. The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to the maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.

Important:
You cannot change the storage policy if you clone an encrypted virtual machine. For information about cloning an encrypted virtual machine, see *vSphere Security*.

5. On the Ready to complete page, review the template settings and click **Finish**.

   The progress of the clone task appears in the **Recent Tasks** pane. When the task completes, the template appears in the inventory.

Deploy a Virtual Machine from a Template

Deploying a virtual machine from a template creates a virtual machine that is a copy of the template. The new virtual machine has the virtual hardware, installed software, and other properties that are configured for the template.

You must have the following privileges to deploy a virtual machine from a template:

- **Datastore** > **Allocate space** on the destination datastore.
- **Network** > **Assign network** on the network to which the virtual machine is assigned. Required only if you customize the original hardware by adding a network card.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **vApp** > **Import**

- **Virtual machine** > **Change Configuration** > **Change Settings** to configure the virtual machine hardware.
- **Virtual machine** > **Change Configuration** > **Add new disk** on the data center or virtual machine folder. Required only if you customize the original hardware by adding a new virtual disk.
- **Virtual machine** > **Edit inventory** > **Create new** on the destination folder or data center.
- **Virtual machine** > **Interaction** > **PowerOn** to power on the virtual machine.
- **Virtual machine** > **Edit inventory** > **Create from existing** on the data center or virtual machine folder.
- **Virtual machine** > **Provisioning** > **Deploy template** on the source template.
- **Virtual machine** > **Provisioning** > **Customize guest** on the template or template folder if you are customizing the guest operating system.
- **Virtual machine** > **Provisioning** > **Read customization specifications** on the root vCenter if you are customizing the guest operating system.
- **Virtual machine** > **Provisioning** > **Modify customization specification**
- **vSphere Tagging** > **Assign or Unassign vSphere Tag** and **vSphere Tagging** > **Assign or Unassign vSphere Tag on Object**
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

1. Start the **Deploy From Template** wizard.

Option	Description
From a valid parent object of a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **New Virtual Machine**. 2. On the Select a creation type page, select **Deploy from template** and click **Next**. 3. On the **Select a template** page, select the template that you want to use.
From a template	Right-click a template and select **New VM from This Template**.

2. On the **Select a name and folder** page, enter a unique name for the virtual machine and select a deployment location, and click **Next**.
3. On the **Select a compute resource** page, select the host, cluster, resource pool, or vApp where the virtual machine will run and click **Next**.

   The virtual machine will have access to the resources of the selected object.

   Important:

   If the template that you deploy has an has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, you cannot proceed with the task.

   If the template that you deploy does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, all the hard disks of the virtual machine will use the storage policy and datastore selected for the configuration files of the source template.

   If creating the virtual machine at the selected location causes compatibility problems, an alarm appears in the **Compatibility** pane.
4. On the **Select storage** page, select the datastore or datastore cluster in which to store the virtual machine configuration files and all of the virtual disks. Click **Next**.

Option	Description
Deploy a virtual machine from a template that has vPMem hard disks	1. Choose the type of storage for the template by selecting the **Standard**, the **PMem**, or the **Hybrid** radio button. If you select the **Standard** mode, all virtual disks will be stored on a standard datastore. If you select the **PMem** mode, all virtual disks will be stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. If you select the **Hybrid** mode, all PMem virtual disks will remain stored on a PMem datastore. Non-PMem disks are affected by your choice of a VM storage policy and datastore or datastore cluster. For more information about persistent memory and PMem storage, see the *vSphere Resource Management* guide.

Option	Description
	2. From the **Select virtual disk format** drop-down menu, select a new virtual disk format for the template or keep the same format as the source virtual machine. 3. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 4. Select a datastore or a datastore cluster. 5. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 6. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.
Deploy a virtual machine from a template that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. **Same format as source** uses the same disk format as the source virtual machine. The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine. **Thick Provision Eager Zeroed** is a type of thick virtual disk that supports clustering features such as Fault tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types o disks. The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can grow to the maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.

5. On the **Select clone options** page, select additional customization options for the new virtual machine, and click **Next**.

   You can choose to customize the guest operating system or the virtual machine hardware. You can also power on the virtual machine after it is created or assign tags to the virtual machine.

   A cloned virtual machine automatically retains any tags assigned to the source virtual machine.

6. Optional: On the **Customize guest OS** page, apply a customization specification to the virtual machine.

Option	Description
Select an existing specification	Select a customization specification from the list.
Override	To change the guest customization specification for this deployment only, click **Override**, complete the steps in the **Override VM Customization Specification** wizard, and click **OK**.

   Customizing the guest OS prevents conflicts that might occur if you deploy virtual machines with identical settings, such as duplicate computer names.

Note:

To access customization options for Windows guest operating systems, Microsoft Sysprep tools must be available on the Windows guest operating system. The Sysprep Tool is built into Windows Vista and Windows 2008 and later operating systems. For details about this and other customization requirements, see Guest Operating System Customization Requirements.

7. Optional: On the **Customize hardware** page, configure the virtual machine hardware and options and click **Next**.

   You can leave the defaults and configure the virtual machine hardware and options later. For more information, see Configuring Virtual Machine Hardware and Configuring Virtual Machine Options.

   Important:
   If you chose to use PMem storage for the virtual machine, its default hard disk, the new hard disks that you configure, and the NVDIMM devices that you add to the virtual machine all share the same PMem resources. So, you must adjust the size of the newly added devices in accordance with the amount of the PMem available to the host. If any part of the configuration requires attention, the wizard alerts you.

8. Optional: On the **Assign Tag** page, select the vSphere Tags you want to assign to the virtual machine and click **Next**.
   a) To filter the tags list, select **Advanced Filter** and click **Add New**. Use the **Advanced Filter** dialog box to filter by **Tag Name**, **Category**, or **Description**. You can then specify a logical operator, enter a comparison value, and click **Apply**. Alternatively, you can use the **Quick Filter** to find specific tags.
   a) Click the **Selected** tab to verify the tags you want to assign.
   b) To remove tags, on the **Selected** tab, select the tags that you want to remove, and click **Remove**.

9. On the Ready to complete page, review the information and click **Finish**.

Clone an Existing Virtual Machine

Cloning a virtual machine creates a virtual machine that is a copy of the original. The new virtual machine is configured with the same virtual hardware, installed software, and other properties that were configured for the original virtual machine.

If a load generator is running in the virtual machine, before you perform the clone operation, you must stop the load generator.

You must have the following privileges to clone a virtual machine:

- **Virtual machine** > **Edit inventory** > **Create from existing** on the datacenter or virtual machine folder.
- **Virtual machine** > **Provisioning** > **Clone virtual machine** on the virtual machine you are cloning.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **Datastore** > **Allocate space** on the destination datastore or datastore folder.
- **Virtual machine** > **Provisioning** > **Read customization specifications** on the root vCenter if you are customizing the guest operating system.
- **Virtual machine** > **Provisioning** > **Customize guest** on the virtual machine or virtual machine folder if you are customizing the guest operating system.
- **Virtual machine** > **Provisioning** > **Modify Customization specification**
- **Network** > **Assign network** on the network to which you assign the virtual machine.
- **Virtual machine** > **Change Configuration** > **Add new disk** on the datacenter or virtual machine folder.
- **Virtual machine** > **Change Configuration** > **Change Settings** to configure the virtual machine hardware.
- **vSphere Tagging** > **Assign or Unassign vSphere Tag** and **vSphere Tagging** > **Assign or Unassign vSphere Tag on Object**
- **Virtual machine** > **Interact** > **PowerOn** to power on the virtual machine.
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.
- If the virtual machine that you clone has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have an available PMem resource. Otherwise, you cannot proceed with the task
- If the virtual machine that you clone does not have an NVDIMM device, but has virtual PMem hard disks, the destination host or cluster must have an available PMem resource. Otherwise, all hard disks of the destination virtual machine will use the storage policy and datastore selected for the configuration files of the source virtual machine.

- To access customization options for Windows guest operating systems, Microsoft Sysprep tools must be installed on the vCenter system. Sysprep Tool is built into Windows Vista and Windows 2008 and later operating systems. For details about this and other customization requirements, see Guest Operating System Customization Requirements.

For information about persistent memory and PMem storage, see the *vSphere Resource Management* guide.

For information on how to configure the virtual machine hardware options, see Configuring Virtual Machine Hardware and Configuring Virtual Machine Options

Note:
When heavily loaded applications, such as load generators, are running in the guest operating system during a clone operation, the virtual machine quiesce operation might fail. VMware Tools might be denied CPU resources and time out. You can quiesce the virtual machines running lower I/O disk operation.

Important:
If you clone an encrypted virtual machine, you cannot change the storage policy. For information about cloning an encrypted virtual machine, see *vSphere Security*.

1. Start the **Clone Existing Virtual Machine** wizard.

Option	Action
From a valid parent object of a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, cluster, vApp, resource pool, or host, and select **New Virtual Machine**. 2. On the **Select a creation type** page, select **Clone an existing virtual machine**, and click **Next**. 3. On the **Select a virtual machine** page, select the virtual machine that you want to clone.
From a virtual machine	Right-click a virtual machine and select **Clone** > **Clone to Virtual Machine**.

2. On the **Select a name and folder** page, enter a unique name for the new virtual machine, select a deployment location, and click **Next**.

   The template name determines the name of the files and folder on the disk. For example, if you name the template `win8tmp`, the template files are named `win8tmp.vmdk`, `win8tmp.nvram`, and so on. If you change the template name later, the names of the files on the datastore do not change.

   Folders provide a way to store virtual machines and templates for different groups in an organization and you can set permissions on them. If you prefer a flatter hierarchy, you can put all virtual machines and templates in a data center and organize them in a different way.

3. On the **Select a compute resource** page, select the host, cluster, resource pool, or vApp where the virtual machine will run and click **Next**.

   The **Compatibility** pane shows the result from the compatibility checks.

4. On the **Select storage** page, select the datastore or datastore cluster in which to store the template configuration files and all virtual disks.

Option	Action
Clone a virtual machine that has vPMem hard disks	1. Select the type of storage for the template by clicking the **Standard**, the **PMem**, or the **Hybrid** radio button. – If you select the **Standard** mode, all virtual disks are stored on a standard datastore. – If you select the **PMem** mode, all virtual disks are stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. – If you select the **Hybrid** mode, all PMem virtual disks remain stored on a PMem datastore. Your choice of a VM storage policy and datastore or datastore cluster affects the non-PMem disks. 2. From the **Select virtual disk format** drop-down menu, select a new virtual disk format for the template or keep the same format as the source virtual machine.

Option	Action
	3. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 4. Select a datastore or a datastore cluster. 5. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 6. Enable the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.
Clone a virtual machine that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. – The **Same format as source** option uses the same disk format as the source virtual machine. – The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. The space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine. – **Thick Provision Eager Zeroed** is a type of thick virtual disk that supports clustering features such as Fault tolerance. The space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks. – The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can grow to the maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Enable the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.

5. Click **Next**.
6. On the **Select clone options** page, select additional customization options for the new virtual machine and click **Next**.

   You can choose to customize the guest operating system or the virtual machine hardware. You can also power on the virtual machine after it is created or assign tags to the virtual machine.

   A cloned virtual machine automatically retains any tags that are assigned to the source virtual machine.
7. Optional: On the **Customize guest OS** page, apply a customization specification to the virtual machine and click **Next**.

   Customizing the guest operating system prevents from conflicts that might occur if you or other users clone virtual machines with identical settings, such as duplicate computer names.

Option	Action
Select an existing specification	Select a customization specification from the list.
Override	To change the guest customization specification for this deployment only, click **Override**, complete the steps in the **Override VM Customization Specification** wizard, and click **OK**.

8. Optional: On the **User settings** page, specify the required settings for the virtual machine.

   This page of the wizard appears only if the selected specification requires additional customization.

9. Optional: On the **Customize hardware** page, configure the virtual machine hardware and options and click **Next**.
   You can leave the defaults and configure the virtual machine hardware and options later.

   Important:
   If you chose to use PMem storage for the virtual machine, its default hard disk, the new hard disks that you configure, and the NVDIMM devices that you add to the virtual machine all share the same PMem resources. You must adjust the size of the newly added devices in accordance with the amount of the PMem available to the host. If any part of the configuration requires attention, the wizard alerts you.
10. Optional: On the **Assign Tag** page, select the vSphere Tags you want to assign to the virtual machine and click **Next**.
    a) To filter the tags list, select **Advanced Filter** and click **Add New**. Use the **Advanced Filter** dialog box to filter by **Tag Name**, **Category**, or **Description**. You can then specify a logical operator, enter a comparison value, and click **Apply**. Alternatively, you can use the **Quick Filter** to find specific tags.
    a) Click the **Selected** tab to verify the tags you want to assign.
    b) To remove tags, on the **Selected** tab, select the tags that you want to remove, and click **Remove**.
11. On the **Ready to complete** page, review the virtual machine settings and click **Finish**.

The new virtual machine appears in the inventory.

Cloning a Virtual Machine with Instant Clone

With the Instant Clone technology, you can create powered on virtual machines from the running state of another powered on virtual machine. The result is a new virtual machine that is identical to the source virtual machine.

With Instant Clone you can create new virtual machines from a controlled point in time. Instant cloning is convenient for large-scale application deployments because it ensures memory efficiency and allows for creating numerous virtual machines on a single host.

The result of an Instant Clone operation is a virtual machine that is called a destination virtual machine. The processor state, virtual device state, memory state, and disk state of the destination virtual machine are identical to those of the source virtual machine. To avoid network conflicts, you can customize the virtual hardware of the destination virtual machine during an Instant Clone operation. For example, you can customize the MAC addresses of the virtual NICs or the serial port configurations of the destination virtual machine. vSphere 9.0 and later does not support customization of the guest OS of the destination virtual machine. For information about manual guest OS customization, see the *vSphere Web Services SDK Programming Guide*.

During an Instant Clone operation, the source virtual machine is stunned for a short period of time, less than 1 second. While the source virtual machine is stunned, a new writable delta disk is generated for each virtual disk and a checkpoint is taken and transferred to the destination virtual machine. The destination virtual machine then powers on by using the source's checkpoint. After the destination virtual machine is fully powered on, the source virtual machine also resumes running.

Instant Cloned virtual machines are fully independent vCenter inventory objects. You can manage Instant Cloned virtual machines like regular virtual machines without any restrictions.

Starting with vSphere 9.0, you can Instant Clone a virtual machine only through the API calls.

For information about Instant Clone, see the *vSphere Web Services SDK Programming Guide*.

Clone a Template to a Template

Templates are primary copies of virtual machines that let you create ready-for-use virtual machines. You can change the template, for example, by installing additional software in the guest operating system, while preserving the state of the primary template.

Verify that you have the following privileges:

- **Virtual machine** > **Provisioning** > **Clone template** on the source template.
- **Virtual machine** > **Edit inventory** > **Create from existing** on the folder where the template is created.
- **Datastore** > **Allocate space** on all datastores where the template is created.
- **Resource** > **Assign virtual machine to resource pool**

- **vSphere Tagging** > **Assign or Unassign vSphere Tag** and **vSphere Tagging** > **Assign or Unassign vSphere Tag on Object** if the template has a tag associated with it.
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

In 9.1, when you convert a template to a virtual machine, vSphere tag information is stored within the template. You can replicate the tags when you convert the template to a virtual machine.

1. Start the **Clone Template to Template** wizard.

Option	Description
From a valid parent object of a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine and select **New Virtual Machine**. 2. Select **Clone template to template** and click **Next**. 3. On the **Select a template to clone** page, browse to the template that you want to clone or accept the default one.
From a template	Right-click a template and select **Clone to Template**.

2. On the **Select a name and folder** page, enter a unique name for the template and select the data center or folder in which to deploy it.

   The template name determines the name of the files and folder on the disk. For example, if you name the template `win10tmp`, the template files are named `win10tmp.vmdk`, `win10tmp.nvram`, and so on. If you change the template name, the names of the files on the datastore do not change.

   Folders provide a way to store virtual machines and templates for different groups in an organization and you can set permissions on them. If you prefer a flatter hierarchy, you can put all virtual machines and templates in a data center and organize them in a different way.

3. Click **Next**.
4. On the **Select a compute resource** page, select a host or cluster resource for the template, and click **Next**.

   The **Compatibility** pane shows the result of the compatibility checks.

   The template must be registered with an ESX host. The host handles all requests for the template and must be running when you create a virtual machine from the template.

   Important:

   If the template that you clone has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, you cannot proceed with the task.

   If the template that you clone does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, all the hard disks of the template use the storage policy and datastore selected for the configuration files of the source template.

5. On the **Select storage** page, select the datastore or datastore cluster in which to store the virtual machine configuration files and all of the virtual disks.

Option	Description
Clone a virtual machine that has vPMem hard disks	1. Choose the type of storage for the template by selecting the **Standard**, the **PMem**, or the **Hybrid** radio button. If you select the **Standard** mode, all virtual disks are stored on a standard datastore. If you select the **PMem** mode, all virtual disks are stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. If you select the **Hybrid** mode, all PMem virtual disks remain stored on a PMem datastore. Non-PMem disks are affected by your choice of a VM storage policy and datastore or datastore cluster. For more information about persistent memory and PMem storage, see the *vSphere Resource Management* guide. 2. From the **Select virtual disk format** drop-down menu, select a new virtual disk format for the template or keep the same format as the source virtual machine. 3. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one.

Option	Description
	4. Select a datastore or a datastore cluster. 5. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 6. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance issues. You can also convert a standard hard disk to a PMem hard disk.
Clone a virtual machine that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. **Same format as source** uses the same disk format as the source virtual machine. The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine. **Thick Provision Eager Zeroed** is a type of thick virtual disk that supports clustering features such as Fault tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types o disks. The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can expand to its maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance issues. You can also convert a standard hard disk to a PMem hard disk.

Important:
If you clone an encrypted virtual machine, you cannot change the storage policy. For information about cloning an encrypted virtual machine, see *vSphere Security*.

6. Click **Next**.
7. On the **Ready to complete** page, review the template settings and click **Finish**.

The progress of the clone task appears in the **Recent Tasks** pane. When the task completes, the template appears in the inventory.

Convert a Template to a Virtual Machine

Converting a template to a virtual machine (VM) does not make a copy. You convert a template to a VM to edit the template, or if you do not need to preserve the VM as a golden image for deploying VMs.

Verity that you have the following privileges:

- **Virtual machine** > **Provisioning** > **Mark as virtual machine** on the source template.
- **Resource** > **Assign virtual machine to resource pool** on the resource pool where the virtual machine runs.

In 9.1, when you convert a template to a virtual machine, vSphere tag information is stored within the template. You can replicate the tags when you convert the template to a virtual machine.

1. Start the **Convert Template to Virtual Machine** wizard.

Option	Description
From a valid parent object of a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine and select **New Virtual Machine**. 2. On the **Select a creation type** page, select **Convert template to virtual machine**, and click **Next**. 3. On the **Select a template** page of the wizard, select a template to deploy from the list.
From a template	Right-click the template and select **Convert to Virtual Machine**.

2. On the **Select a compute resource** page, select the host, cluster, vApp, or resource pool for the virtual machine to run in.

   Important:

   If the template that you convert has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, you cannot proceed with the task.

   If the template that you convert does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, all the hard disks of the virtual machine use the storage policy and datastore selected for the configuration files of the source template.

   The **Compatibility** pane shows the result from the compatibility checks.

3. Click **Next**.
4. On the **Ready to complete** page, review the settings, and click **Finish**.

The virtual machine appears in the inventory.

Deploy and Export OVF and OVA Templates

You can export virtual machines, virtual appliances, and vApps in Open Virtual Format (OVF) and Open Virtual Appliance (OVA). You can then deploy the OVF or OVA template in the same environment or in a different environment.

Note:
In vSphere 8.0 and later, you cannot export OVA templates, OVF templates is the only option.

Alternatively, you can use the VMware OVF Tool to deploy and export OVF templates.

In previous versions of vSphere, you needed to install the Client Integration Plug-in to deploy and export OVF or OVA templates. vSphere 6.5 no longer requires that you install the Client Integration Plug-in to export OVF templates or to deploy OVF and OVA templates.

OVF and OVA File Formats and Templates

OVF is an open standard which contains multiple files as a package. For example, `.ovf`, `.vmdk`, `.nvram`, and so on. OVF supports exchange of virtual appliances across products and platforms. OVA is a single-file distribution of the OVF file package. When you export a virtual machine as an OVF file, you download an OVF file and the virtual disk files.

The OVF and OVA formats offer the following advantages:

- OVF and OVA files are compressed, allowing for faster downloads.
- The vSphere Client validates an OVF or OVA file before importing it, and ensures that it is compatible with the intended destination server. If the appliance is incompatible with the selected host, it cannot be imported and an error message appears.
- OVF and OVA can encapsulate multi-tiered applications and more than one virtual machine.

Exporting OVF templates allows you to create virtual appliances that other users can import later. You can use the export function to distribute pre-installed software as a virtual appliance, or to distributing template virtual machines to users. You can make the OVF or OVA file available to users who cannot access your vCenter inventory.

Deploying an OVF or OVA template allows you to add pre-configured virtual machines or vApps to your vCenter or ESX inventory. Deploying an OVF or OVA template is similar to deploying a virtual machine from a template. However, you can deploy an OVF or OVA template from any local file system accessible from the vSphere Web Client, or from a remote web server. The local file systems can include local disks (such as `C:`), removable media (such as CDs or USB keychain drives), and shared network drives.

Deploy an OVF or OVA Template

You can deploy an OVF or OVA template from a local file system or from a URL.

Prerequisites

Verify that you have the following privileges:

- **Datastore** > **Allocate space** on the destination datastore.
- **Network** > **Assign network** on the network to which the virtual machine is assigned. Required only if you customize the original hardware by adding a network card.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **vApp** > **Export** on the vApp.
- **vApp** > **Import** on the destination host, cluster, or resource pool.
- **Virtual machine** > **Change Configuration** > **Add new disk** on the data center or virtual machine folder. Required only if you customize the original hardware by adding a new virtual disk.
- **Virtual machine** > **Change Configuration** > **Change Settings** to configure the virtual machine hardware.
- **Virtual machine** > **Edit inventory** > **Create new** on the destination folder or data center.
- **Virtual machine** > **Interaction** > **PowerOn** to power on the virtual machine.
- **Virtual machine** > **Provisioning** > **Customize guest** on the virtual machine or virtual machine folder, if you are customizing the guest operating system.
- **Virtual machine** > **Provisioning** > **Modify customization specification**
- **Virtual machine** > **Provisioning** > **Read customization specifications** on the root vCenter, if you are customizing the guest operating system.
- **VM storage policies** > **View VM storage policies** on the vCenter.

Some of the pages in the **Deploy OVF Template** wizard only appear if the OVF template that you deploy requires additional customization, contains deployment options, or has one or multiple vService dependencies.

1. Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **Deploy OVF Template**.
   The **Deploy OVF Template** wizard opens.
2. On the **Select an OVF template** page, specify the location of the source OVF or OVA template and click **Next**.

Option	Action
URL	Type a URL to an OVF or OVA template located on the Internet. Supported URL sources are HTTP and HTTPS. **Note:** Verify that all files from the OVF package are at the same location of the OVF descriptor file. Example: `http://broadcom.com/VMTN/appliance.ovf`.
Local file	Click **Browse** and select all the files associated with an OVF template or OVA file. This includes files such as `.ovf`, `.vmdk`, etc. If you do not select all the required files, a warning message displays.

3. On the **Select a name and folder** page, enter a unique name for the virtual machine or vApp, select a deployment location, and click **Next**.
   The default name for the virtual machine is the same as the name of the selected OVF or OVA template. If you change the default name, choose a name that is unique within each vCenter virtual machine folder.
   The default deployment location for the virtual machine is the inventory object where you started the wizard.
4. On the **Select a compute resource** page, select a resource where to run the deployed VM template,.
   a. To automatically power on the virtual machine that you deploy on the host, select the **Automatically power on deployed VM** check box.

b. Click **Next**.

5. On the **Review details** page, verify the OVF or OVA template details and click **Next**.

Option	Description
Publisher	Publisher of the OVF or OVA template, if a certificate included in the OVF or OVA template file specifies a publisher.
Download size	Size of the OVF or OVA file.
Size on disk	Size on disk after you deploy the OVF or OVA template.

6. (Optional) On the **Configuration** page, depending on the OVA template, select a deployment configuration and click **Next**.
7. On the **Select storage** page, define where and how to store the files for the deployed OVF or OVA template, and click **Next**.
   a. Select the disk format for the virtual machine virtual disks.

Format	Description
Thick Provision Lazy Zeroed	Creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine.
Thick Provision Eager Zeroed	A type of thick virtual disk that supports clustering features such as Fault tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat file format (`-flat.vmdk`), the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks.
Thin Provision	Use this format to save storage space. For the thin disk, you provision as much datastore space as the disk requires based on the value that you enter for the disk size. However, the thin disk starts small and at first, uses only as much datastore space as the disk needs for its initial operations.

   b. Select a VM Storage Policy.
      This option is available only if storage policies are enabled on the destination resource.
   c. Select the **Show datastores from Storage DRS clusters** check box if you do not want to use storage DRS with the virtual machine.
   d. Select a datastore to store the deployed OVF or OVA template.
      The configuration file and virtual disk files are stored on the datastore. Select a datastore large enough to accommodate the virtual machine or vApp and all associated virtual disk files.

      Note:

      If you want to use the API calls to deploy an OVF template that contains vPMem hard disks and that has been exported from a content library, see Broadcom knowledge base article 336060.

8. On the **Select networks** page, select a source network and map it to a destination network. Click **Next**.
   The Source Network column lists all networks that are defined in the OVF or OVA template.
9. On the **Customize template** page, customize the deployment properties of the OVF template and click **Next**.
10. (Optional) On the **vService bindings** page, depending on the OVA template, select a binding service provider and click **Next**.
11. On the **Ready to complete** page, review the page and click **Finish**.

As a result, a new task for creating the virtual machine appears in the Recent Task pane. After the task is complete, the new virtual machine is created on the selected resource.

Export an OVF Template

An OVF template captures the state of a virtual machine or vApp into a self-contained package. The disk files are stored in a compressed, sparse format.

Note:
You cannot export encrypted virtual machines.

Prerequisites

- Power off the virtual machine or vApp.
- Required privilege:**vApp** > **Export**

Procedure

1. Navigate to a virtual machine or vApp and from the **Actions** menu, select **Template** > **Export OVF Template**.
2. In the **Name** folder, enter the name of the template.
   Example: For example, enter `MyVm`.

   Note:
   When you export an OVF template with a name that contains asterisk (*) characters, those characters turn into underscore (_) characters.
3. In the **Annotation** field, enter a description.
4. Select the **Enable advanced options** check box if you want to include additional information or configurations in the exported template.
   The advanced settings include information about the BIOS UUID, MAC, and configuration settings used by other applications. These options limit portability.
5. To save each file associated with the template (`.ovf`, `.vmdk`, `.mf`), click **OK** and respond to the prompts.

Export an OVF Template with a vTPM Placeholder

With vSphere 8.0, you can manually export an OVF template of a virtual machine that contains a vTPM device. Because vTPMs are security devices with unique identities, a vTPM placeholder is added to the OVF template instead of a vTPM device and replaced with a new vTPM device upon instantiation. To deploy or export an OVF template of a virtual machine with a vTPM device, you can use the VMware OVF Tool. For more information about the VMware OVF Tool, see the OVF Tool User Guide.

Prerequisites

- Power off the virtual machine.
- Deactivate any applications in the guest operating system of the virtual machine that use the vTPM device, such as BitLocker.
- Verify that you have the required privilege: **Cryptographic operations** > **Decrypt**.

Procedure

1. Remove the vTPM device from the virtual machine.
   a. Navigate to the virtual machine and click **Actions**.
   b. Click **Edit Settings**.
   c. On the **Virtual Hardware** tab, locate the Trusted Platform Module entry.
   d. Move your pointer over the device and click the **Remove** icon.
      This icon appears only for the virtual hardware that you can safely remove.
   e. Click **Delete** to confirm you want to remove the device.
      The vTPM device is marked for removal.
   f. Click **OK**.
      Verify that the Virtual Trusted Platform Module entry no longer appears in the virtual machine **Summary** tab in the **VM Hardware** pane.
2. Decrypt the virtual machine if it is encrypted.
   For more information on how to decrypt a virtual machine, see Decrypt an Encrypted Virtual Machine or Virtual Disk.
3. Export the virtual machine to an OVF template.
4. Open the OVF (`.ovf`) descriptor file and manually add the vTPM device placeholder:

```
<Item ovf:required="false">
 <rasd:AutomaticAllocation>false</rasd:AutomaticAllocation>
 <rasd:ElementName>Virtual TPM</rasd:ElementName>
```

```
    <rasd:InstanceID>13</rasd:InstanceID>
    <rasd:ResourceSubType>vmware.vtpm</rasd:ResourceSubType>
    <rasd:ResourceType>1</rasd:ResourceType>
  </Item>
```

Note:
The OVF descriptor contains a placeholder for a vTPM device which does not have any secrets. The virtual machine encryption keys are generated when you deploy a virtual machine from this OVF template.

5. Open the OVF manifest file (`.mf`) and change the SHA-256 checksum for the `.ovf` file.
6. Resign the `.ovf` file.
7. Deploy the manually exported OVF template.

Export an OVF Template of a Virtual Machine without a vTPM Placeholder

To deploy an OVF template of a virtual machine with a vTPM device, you can use the VMware OVF Tool. For more information about the VMware OVF Tool, see the OVF Tool User Guide.

Prerequisites

- Power off the virtual machine.
- Deactivate any applications in the guest operating system of the virtual machine that use the vTPM device, such as BitLocker.

Procedure

1. Remove the vTPM device from the virtual machine.
   a. Navigate to the virtual machine and click **Actions**.
   b. Click **Edit Settings**.
   c. On the **Virtual Hardware** tab, locate the Trusted Platform Module entry.
   d. Move your pointer over the device and click the **Remove** icon.
      This icon appears only for the virtual hardware that you can safely remove.
   e. Click **Delete** to confirm you want to remove the device.
      The vTPM device is marked for removal.
   f. Click **OK**.
      Verify that the Virtual Trusted Platform Module entry no longer appears in the virtual machine **Summary** tab in the **VM Hardware** pane.
2. Decrypt the virtual machine if it is encrypted.
   For more information on how to decrypt a virtual machine, see Decrypt an Encrypted Virtual Machine or Virtual Disk.
3. Export the virtual machine to an OVF template.
4. Deploy the manually exported OVF template and add a vTPM device to the virtual machine.
   For more information on how to add a vTPM device to a virtual machine, see Add Virtual Trusted Platform Module to an Existing Virtual Machine.

Using Content Libraries

Content libraries are container objects for VM and vApp templates and other types of files, such as ISO images, text files, and so on.

To deploy virtual machines and vApps in the vSphere inventory, you can use the templates in the library. You can also use content libraries to share content across vCenter instances in the same or different locations. Sharing templates and files results in consistency, compliance, efficiency, and automation in deploying workloads at scale.

A content library stores and manages content in the form of library items. A single library item can consist of one file or multiple files. For example, the OVF template is a set of files (`.ovf`, `.vmdk`, and `.mf`). When you upload an OVF template to the library, you upload the entire set of files, but the result is a single library item of the OVF Template type.

Starting with vSphere 7.0 Update 3, you can protect the OVF items by applying the default OVF security policy to a content library. The OVF security policy enforces strict validation on OVF items when you deploy or update the item,

import items, or synchronize OVF and OVA templates. To make sure that the OVF and OVA templates are signed by a trusted certificate, you can add the OVF signing certificate from a trusted CA.

In previous versions of vSphere, content libraries supported only OVF templates. As a result, VM and vApp templates were both converted to OVF files when you uploaded them to a content library. Starting with vSphere 7.0 Update 1, content libraries also support VM templates. So, templates in the content library can be either of the OVF Template type, or the VM Template type. vApp templates are still converted to OVF files when you upload them to a content library.

You create and manage a content library from a single vCenter instance, but you can distribute the content to other vCenter instances if HTTP(S) traffic is allowed between the two systems. The distribution of VM templates additionally requires that the respective vCenter instances are in Enhanced Linked Mode or Hybrid Linked Mode and that the respective hosts are connected through a network.

> **CAUTION:**
>
> With vCenter 9.0, ELM, which allows unified view and administration of multiple vCenter instances, is deprecated and will be removed in a future release. ELM is supported in VCF 9.0 to allow smooth upgrade of existing infrastructure to VCF, but the recommended alternative is the grouping capability under VCF Operations. Group admins can enable alternatives for each ELM capability, such as VMware Single Sign-On, consolidated licensing, vCenter roles and privileges sync, vCenter tags replication, and vCenter single API endpoint.

You can create two types of content libraries.

- You can create a local content library to store and manage content in a single vCenter instance. If you want to share the contents of that library, you can enable publishing. When you enable publishing, other users can subscribe to the library and use its content. Alternatively, you can create subscriptions for the library, which gives you control over the distribution of content. For more information about managing a local library that has publishing enabled, see Create and Delete a Subscription for a Local Library.
- You can create a subscribed content library to subscribe to a published library and use its contents. You cannot upload or import items into a subscribed library. Subscribers only use the content in the published library, but it is the administrator of the published library who manages the templates. For more information about managing a subscribed library, see Synchronize the Contents of a Subscribed Library.

Content Library Administrator Role, Permissions Hierarchy, and Inheritance

vCenter provides a sample role that allows you to give users or groups privileges to manage content libraries.

Content Library Administrator Role

vCenter provides a sample role that allows you to give users or groups privileges to manage content libraries.

The Content Library Administrator role is a predefined role that gives a user privileges to monitor and manage a library and its contents.

You can modify the role or use it as an example to create custom roles for specific tasks you want to allow other users to perform.

If a user has this role on a library, that user can perform the following tasks on that library.

- Create, edit, and delete local or subscribed libraries.
- Create and delete subscriptions to a local library with publishing enabled.
- Publish a library or a library item to a subscription.
- Synchronize a subscribed library and synchronize items in a subscribed library.
- View the item types supported by the library.
- Configure the global settings for the library.
- Import items to a library.
- Export library items.
- Migrate a content library to another datastore.

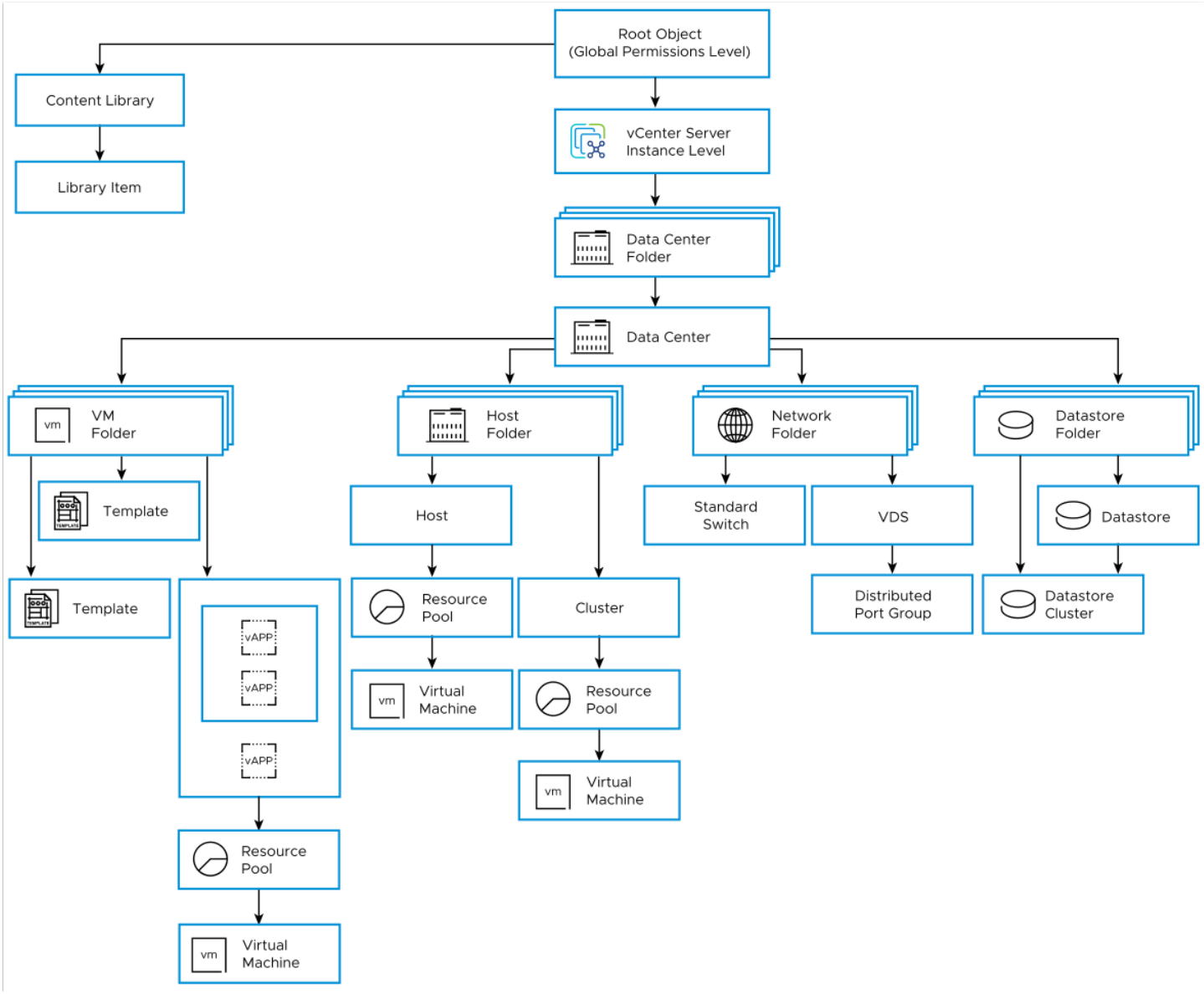
Content Library Permissions Hierarchy and Inheritance

vSphere objects inherit permissions from a parent object in the hierarchy. Content libraries work in the context of a single vCenter instance. However, content libraries are not direct children of a vCenter system from an inventory perspective.

The direct parent for content libraries is the global root. This means that if you set a permission at a vCenter level and propagate it to the children objects, the permission applies to data centers, folders, clusters, hosts, virtual machines, and so on, but does not apply to the content libraries that you see and operate with in this vCenter instance. To assign a permission on a content library, an Administrator must grant the permission to the user as a global permission. Global permissions support assigning privileges across solutions from a global root object.

The figure illustrates the inventory hierarchy and the paths by which permissions can propagate.

Figure 40: vSphere Inventory Hierarchy



To let a user manage a content library and its items, an Administrator can assign the Content Library Administrator role to that user as a global permission. The Content Library Administrator role is a sample role in the vSphere Client.

Users who are Administrators can also manage libraries and their contents. If a user is an Administrator at a vCenter level, they have sufficient privileges to manage the libraries that belong to this vCenter instance, but cannot see the libraries unless they have a Read-Only role as a global permission.

For example, if you have an Administrator role that is defined at a vCenter level and you navigate to Content Libraries in the object navigator, you see 0 libraries despite there are existing libraries in the vSphere inventory of that vCenter instance. To see the libraries, you need a Read-Only role that is assigned as a global permission.

Administrators whose role is defined as a global permission can see and manage the libraries in all vCenter instances that belong to the global root.

Because content libraries and their children items inherit permissions only from the global root object, when you navigate to a library or a library item and click **Configure** tab, you can see there is no **Permissions** tab. An Administrator cannot assign individual permissions on different libraries or different items within a library.

Create and Edit a Content Library

In the vSphere Client, you can create a local or a subscribed content library and change its configuration settings. By using content libraries, you can store and manage content in one vCenter instance. Alternatively, you can distribute content across vCenter instances to increase consistency and facilitate the deployment workloads at scale.

By using the OVF security policy, you can protect the items of your content library.

Create a Content Library

You can create a local content library and populate it with templates and other types of files. You can then use the content library templates to deploy virtual machines or vApps in your virtual environment. You can also share the contents of your local library with users from other vCenter instances.

Required privileges:

- **Content library** > **Create local library** or **Content library** > **Create subscribed library** on the vCenter instance where you want to create the library.
- **Datastore** > **Allocate space** on the destination datastore.

> **Note:**
> VM Templates cannot be stored in a library that uses NFS or SMB storage. OVF templates can only be stored on the datastore that is associated with the content library.

You can also create a subscribed library to use at will the contents of an already existing published local library.

1. Navigate to **Menu** > **Content Libraries**.
2. Click **Create**.
   The **New Content Library** wizard opens.
3. On the **Name and location** page, enter a name, select a vCenter instance for the content library, and click **Next**.
4. On the **Configure content library** page, select the type of content library that you want to create.

Option	Description
Local content library	A local content library is accessible only in the vCenter instance where you create it by default. 1. To make the content of the library available to other vCenter instances, select **Enable publishing** . 2. If you want to require a password for accessing the content library, select **Enable authentication** and set a password. 3. Click **Next**.
Subscribed content library	A subscribed content library originates from a published content library. Use this option to take advantage of already existing content libraries.

Option	Description
	Note: If the published content library does not have a security policy enabled, you cannot apply the OVF default security policy to the subscribed content library. If you create a subscribed library, depending on the **Subscription URL** that you provide, the system performs a check to determine whether the provided URL can have a security policy enabled or not. To see up-to-date content, you can synchronize the subscribed library with the published library, but you cannot add or remove content from the subscribed library. Only an administrator of the published library can add, modify, and remove contents from the published library. To subscribe to a library, provide the following information: 1. In the **Subscription URL** text box, enter the URL address of the published library. **Note:** The system performs a check to determine whether the **Subscription URL** that you provide can have a security policy enabled or not. 2. If authentication is enabled on the published library, select **Enable authentication** and enter the publisher password. 3. Select a download method for the contents of the subscribed library. – To download a local copy of all the items in the published library immediately after subscribing to it, select **immediately**. – To save storage space, select **when needed**. You download only the metadata for the items in the published library. If you must use an item, synchronize the item or the entire library to download its content. 4. If prompted, accept the SSL certificate thumbprint. The SSL certificate thumbprint is stored on your system until you delete the subscribed content library from the inventory. 5. Click **Next**.

5. Optional: On the **Apply security policy** page, select **Apply Security Policy** and select **OVF default policy**.

   Note:
   Currently, only the OVF default policy is supported.

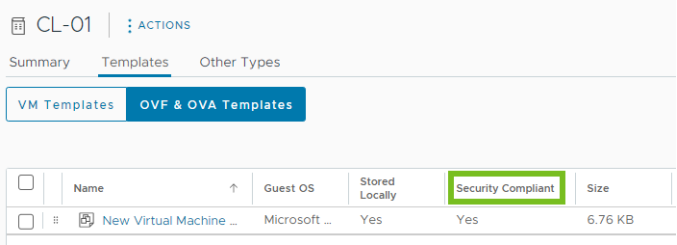
   When you apply the OVF default policy to a content library:

   - All OVF items are validated by signing the OVF package with a valid certificate.
   - All OVF or OVA items in the content library will have a `Yes` or **No Security Compliant** status based on the certificate validation.
   - All items that are not OVF or OVA items will have a **Yes Security Compliant** status.

   If the OVF default policy is not applied to a content library, all items in the content library will have a **Yes Security Compliant** status.

6. Click **Next**.
7. On the **Add storage** page, select a datastore as a storage location for the content library contents and click **Next**.
8. On the **Ready to complete** page, review the details and click **Finish**.

   You can view the **Security Compliant** status under the **Templates** tab of a content library.



Edit a Content Library

In the vSphere Client, you can edit a local library by changing its name, description, or tags. You can also change the configuration settings of a local or subscribed content library.

The privileges that you need depend on the task that you want to perform.

Task	Privilege
Edit Local Content Library Settings	**Content library** > **Update library** and **Content library** > **Update local library** on the library.
Edit Subscribed Content Library Settings	**Content library** > **Update subscribed library** and **Content library** > **Probe subscription information** on the subscribed library.
Delete Content Library	**Content library** > **Delete subscribed library** or **Content library** > **Delete local library** depending on the type of library that you want to delete.
Assign or unassign a tag	**VSphere Tagging** > **Assign or Unassign vSphere Tag**

To share the contents of a local library across multiple vCenter instances, you must enable publishing for the library. From the **Edit Setting** dialog box, you can obtain the URL of your library and send it to other users to subscribe.

If a library is already published, you can change its password for authentication. To keep access to the published library, users who are subscribed to your library must update the password.

For subscribed libraries, you can change the download method or enable automatic synchronization with the published library.

1. Navigate to **Menu** > **Content Libraries**.
2. Right-click a content library and select the action that you want to perform.
   - To edit the content library description, select **Edit notes**.
   - To change the name of the content library, select **Rename**.
   - To assign a tag to the content library, select **Tags** > **Assign Tag**.
   - To remove a tag from the content library, select **Tags** > **Remove Tag**.

     For information about tags and tag categories, see the *vCenter and Host Management* documentation.
   - To delete the content library, select **Delete**.
3. To edit the configuration settings of a content library, right-click the content library and click **Edit Settings**.

   The changes that you can make depend on the type of content library that you edit.

Content Library Type	Action
Local content library that is unpublished	To share the contents of a local library with other users, you can enable the publishing of the local library. 1. To publish the local library and share its contents with other users, select the **Enable publishing** check box. 2. To obtain the URL of your library and distribute it, click the **Copy Link** button.

Content Library Type	Action
	3. Select **Enable user authentication for access to this content library** and set a password for the library. If you protect the library with a password, you must provide both the URL and the password to users who want to subscribe to your library. 4. To activate or deactivate the security policy, select or deselect the **Apply Security Policy**. When you deactivate the security policy of a content library, you cannot reuse the existing OVF items. 5. Click **OK**.
Local content library that is published	You can change the following settings of a local library that is published. • You can copy the subscription URL to your library and send it to other users to subscribe. • You can unpublish the library by deselecting the **Enable publishing** check box. Users who are currently subscribed to this library can no longer synchronize to the library, but they can still use the previously synchronized content. • You can activate or deactivate authentication for the library. • In the vSphere Client, you can change the password for authentication if the library is published and password-protected. a. Enter the current password and the new password. Confirm the new password. b. Click **OK**. • To activate or deactivate the security policy, select or deselect the **Apply Security Policy**. When you deactivate the security policy of a content library, you cannot reuse the existing OVF items.
Subscribed content library	You can change the following settings of a subscribed library: • Activate or deactivate the automatic synchronization with the published library. • Modify the subscription URL. • Update the password for authentication to the published library. • Select a download method. You can either download all library content immediately or download the library content only when needed. If you switch from the option to download content only when needed to the option to download all library content immediately, a synchronization task starts and content starts downloading. The number and size of items in the published library determine the amount of time and network bandwidth that the task requires. • To activate or deactivate the security policy, select or deselect the **Apply Security Policy**. When you deactivate the security policy of a content library, you cannot reuse the existing OVF items.

Configure Advanced Content Library Settings

You can monitor and configure the content library service settings across different vCenter instances. This advanced configuration ensures the consistency in your environment.

Verify that you are a member of the **CAAdmins** and **SystemConfiguration.Administrators** group in the vCenter Single Sign-On domain.

1. Navigate to **Menu** > **Content Libraries**.
2. From the **Content Libraries** pane, select a library and click **Advanced**.
   The **Advanced Configuration** dialog box opens.
3. Configure the parameters.

Configuration Parameter	Description
Library Auto Sync Enabled	Enables automatic synchronization of subscribed content libraries.
Library Auto Sync Refresh Interval (minutes)	Interval between two consequent automatic synchronizations of the subscribed content library. Measured in minutes.
Library Auto Sync Setting Refresh Interval (seconds)	Interval after which the refresh interval for the automatic synchronization settings of the subscribed library will be updated

Configuration Parameter	Description
	if it has been changed. Measured in seconds. If you change the refresh interval, you must restart the vCenter system.
Library Auto Sync Start Hour	The time of the day when the automatic synchronization of a subscribed content library starts.
Library Auto Sync Stop Hour	The time of the day when the automatic synchronization of a subscribed content library stops. Automatic synchronization stops until the start hour.
Library Maximum Concurrent Sync Items	Maximum number of concurrently synchronizing library items for each subscribed library.
Max concurrent NFC transfers per ESX host	Maximum concurrent NFC transfers limit per ESX host.
Maximum Bandwidth Consumption	Bandwidth usage threshold across all transfers, measured in Mbps where 0 means unlimited bandwidth.
Maximum Number of Concurrent Priority Transfers	Concurrent transfer limit for priority files. If exceeded, transfers are queued. This threadpool is used only to transfer priority objects. For example, OVF. If you change the concurrent transfer limit for priority files, you must restart the vCenter system.
Maximum Number of Concurrent Transfers	Concurrent transfer limit. If exceeded, transfers are queued. If you change the concurrent transfer limit, you must restart the vCenter system.

4. Click **Save**.
5. From the **vCenter** drop-down menu, select the vCenter instance whose settings you want to configure.

   Note:

   The **vCenter** drop-down menu appears only if your environment has more than one vCenter instance.
6. Edit the configuration parameters and click **Save**.
7. To apply the changes that require a content library service restart, click **Restart from VAMI**.

   The **Restart Service** window opens, and you can log in with your credentials.

Create and Delete a Subscription for a Local Library

To share the contents of a local library with users from other vCenter instances, enable publishing for the library. This way, other users can subscribe to the published library and use its contents.

As an administrator, you can also create subscriptions for the library to gain control over the distribution of content.

Note:
You cannot create subscriptions to a local library that does not have publishing enabled. You also cannot deactivate publishing for a library that already has subscriptions. To deactivate publishing for a local library, you must first delete all of its subscriptions.

To avoid name collisions and failures when you publish content from the publisher library, the publisher and subscriber libraries must have unique folders dedicated to them.

Local Libraries

You use a local library to store and manage items in a single vCenter instance. When you enable publishing for the library, users from other vCenter instances can subscribe to it and gain access to the library items. In this case, only the subscriber initiates and controls the synchronization of content between the published library and the subscribed library.

If publishing is enabled, you can also create subscriptions to the local library. Creating a subscription creates a new subscribed library or links the publisher library to an existing subscriber. Unlike regular subscribed libraries, subscriptions give the administrator of the local library control over the distribution of content library items.

A publisher library is a local library with subscriptions.

Using Subscriptions

When you create a subscription for a local library, the result is a subscribed library. A publisher library is aware of its subscriptions. Subscriptions enable the administrator of the publisher library to control the content distribution. With subscriptions, content is distributed either when the subscriber initiates synchronization, or when the administrator of the local library publishes the library items to one or more of the existing subscriptions.

When you use subscriptions, you have the flexibility to decide how much of the library content you want to share with the subscribers. For example, you can publish some or all library items. You can also publish content to selected subscribers or to all subscribers.

The use of subscriptions allows content distribution between a publisher and a subscriber in the following scenarios.

- The publisher and subscriber are in the same vCenter instance.
- The publisher and subscriber are in vCenter instances that are in Enhanced Linked Mode.

  CAUTION:

  With vCenter 9.0, ELM, which allows a unified view and administration of multiple vCenter instances, is deprecated and will be removed in a future release. ELM is supported in VCF 9.0 to allow a smooth upgrade of existing infrastructure to VCF, but the recommended alternative is the grouping capability under VCF Operations. Group admins can enable alternatives for each ELM capability, such as VMware Single Sign-On, consolidated licensing, vCenter roles and privileges sync, vCenter tags replication, and vCenter single API endpoint.
- The publisher and subscriber are in vCenter instances that are in Hybrid Linked Mode. For more information about Hybrid Linked Mode, see the *VMware Cloud AWS*documentation.

  Note:
  Publishing content is possible only from an on-premises publisher to a cloud subscriber, and not in the reverse scenario.

Limitations in Content Distribution

Content is distributed when a publisher library publishes content to its subscribers or when a subscriber synchronizes the content in the respective subscribed library with the published library. You can publish and synchronize a single content library item or an entire library. The following limitations exist in the distribution of content between a published and a subscribed library.

- You can publish only VM templates. If you publish an entire library that contains both VM templates and OVF templates, only the VM templates are replicated to the subscriber. To synchronize OVF templates and other types of files, the subscriber must initiate the synchronization.
- You can synchronize only OVF templates. If a subscriber initiates synchronization with a published library that contains both VM templates and OVF templates, only the OVF templates are synchronized in the subscribed library. VM templates are synchronized when a publisher library publishes them to its subscribers.

Create a Subscription for a Local Library

Subscriptions enable you to publish library items to a subscriber whenever you want. Create a subscription for a publisher library to control the distribution of templates to the subscriber.

Required privileges:

- **Datastore** > **Allocate space**
- **Network** > **Assign network**
- **Resource** > **Assign virtual machine to resource pool**
- **Virtual machine** > **Edit inventory** > **Create from existing**
- **Virtual machine** > **Change Configuration** > **Set annotation**
- **Content library** > **Create subscribed library**
- **Content library** > **Create a subscription for a published library** on the content library for which you create a subscription.

When you create a subscription, you can link the publisher library to an existing subscribed library or create a new subscribed library. Creating a subscription to a new subscribed library triggers automatic synchronization. Creating a

subscription to an existing library does not trigger automatic synchronization. If you create a subscription to an existing library, the synchronization happens when you publish an item or the entire library.

When you create a subscription to a new subscribed library and you select the option to download library content only when needed, only the metadata for the publisher library contents is downloaded to the associated storage. When the subscriber needs to use a library item, you either publish the item to the subscriber or the subscriber synchronizes the item to download its content to their local storage. For more information about synchronizing an entire library or a library item, see Synchronize the Contents of a Subscribed Library.

1. Select **Menu** > **Content Libraries**.
2. Right-click a local library and select **New Subscription**.
   The **Create Subscription** wizard starts.
3. Select your task.

Task	Steps
Create a new subscription to a new subscribed library	1. On the **Select subscription type** page, select the **Create a new subscription to a new Subscriber library** radio button and click **Next**. 2. On the **Configure subscription** page, configure the subscription and click **Next**. – Select a vCenter instance to create the subscribed library in. – Enter a name and description for the new subscribed library. – Select the download method for the new subscribed library. 3. On the **Select folder** page, select a location for the library contents and click **Next**. 4. On the **Select compute resource** page, select the compute resource for the subscribed library and click **Next**. 5. On the **Select storage** page, select a storage location for the subscribed library items and click **Next**. 6. On the **Select network** page, select a network for the synchronization of the subscribed library items and click **Next**. 7. On the **Review** page, review the configuration and click **Finish**.
Create a new subscription to an existing subscribed library	1. On the **Select subscription type** page, select the **Create a new subscription to an existing Subscriber library** radio button and click **Next**. 2. On the **Configure subscription** page, configure the subscription and click **Next**. a. From the **vCenter** drop-down menu, select the vCenter instance to create the subscription in. A list of all libraries that are subscribed to the publisher and that are in the selected vCenter instance appears. b. Select an existing subscribed library from the list. 3. On the **Select folder** page, select a location for the subscription and click **Next**. 4. On the **Select compute resource** page, select the compute resource for the subscriber library and click **Next**. 5. On the **Select storage** page, select where to store the contents of the subscriber and click **Next**. 6. On the **Select network** page, select a network for the synchronization of the subscribed library items use and click **Next**. 7. On the **Review** page, review the configuration and click **Finish**.

A new subscription is created. When you go to the **Subscriptions** tab of the local library, you can view a list of all existing subscriptions for the library.

Delete a Subscription from a Local Library

Delete a subscription if you no longer want to share the contents of a publisher library with a subscriber. You might also need to delete a subscription if the corresponding subscribed library has been deleted.

Required privilege: **Content library** > **Delete subscription of a published library** on the library.

If you do not delete a subscription to a deleted subscribed library and you publish content to that subscriber, the task triggers an error. To avoid errors, always delete subscriptions that link to deleted subscribed libraries.

Deleting is a useful option when you must change a subscription. Because editing is currently not supported, your only choice is to delete the subscription that you want to modify and create a new one.

When you delete a subscription, the respective subscribed library and its contents are not deleted. Deleting a subscription means that you can no longer publish templates to the subscriber. The subscriber can still initiate the synchronization of OVF templates, if any. The subscriber can also continue using the previously synchronized content.

1. Navigate to the **Content Libraries** list.
2. Open a local library by clicking its name.
3. On the **Subscriptions** tab, select one or multiple subscriber libraries from the list.
4. Click the **Delete subscription** button and click **Delete** to delete the selected subscriber libraries.

The subscription is deleted and you cannot use the publish function to synchronize the content in the respective subscribed library .

Publish the Library Contents and a Single Template to a Subscriber

Publish a library to replicate all VM templates stored in the publisher library to one or multiple subscribers. If you want to replicate a single VM template from a publisher library to a subscriber, you can publish the item, and not the entire library.

Required privileges on the content library:

- **Content library** > **Publish a library to its subscribers**
- **Content library** > **Sync library item**

If the publisher library contains OVF templates and other types of files, publishing the entire library to a subscription only replicates or updates the VM templates.

1. Navigate to the **Content Libraries** list.
2. Open a local library by clicking its name.
3. Publish a content library contents or a VM template to a subscriber.

Action	Description
Publish a content library contents to a subscriber	1. On the **Subscriptions** tab, select the subscriber libraries to update. 2. Click the **Publish** button. 3. To confirm the process, in the **Publish Library** pop-up window, click **Publish**. A publish task appears in the **Recent Tasks** pane. All VM templates from the publisher library are published to the selected subscribers.
Publish a single template to a subscriber	1. On the **Templates** tab, right-click a template of the VM template type and select **Publish**. 2. In the **Publish Template** dialog box, select the subscribers to which to publish the selected template. 3. Click **OK**.

Action	Description
	After the publishing finishes, the item content and metadata are downloaded to the storage of the subscribed library. On the **Templates** tab for the subscription, the value for the item in the **Stored Content Locally** column changes to Yes.

Synchronize the Contents of a Subscribed Library

You create a subscribed library to subscribe to a published library. You cannot add content to a subscribed library. You can only synchronize the content of the subscribed library with the content of the published library.

You can create the subscribed library in the same vCenter instance where the published library is, or in a different vCenter system. When you create a subscribed library, you can download all the contents of the published library immediately after the subscribed library is created. Alternatively, you can download only the metadata for the items from the published library and later download the full content of only those items that you need.

To ensure that the contents of a subscribed library are up-to-date, the subscribed library automatically synchronizes to the source published library at regular intervals. You can also manually synchronize a single item or an entire subscribed library.

When you update VM templates on the publisher library through the check-in and check-out operations, the VM templates are available in the subscriber library after you publish the subscriber library from the **Subscriptions** tab of the publisher.

The vertical timeline view is not available in the subscriber library. You can monitor only the latest version of the VM template.

For information about the VM templates management and the vertical timeline view, see Manage VM Templates in a Content Library.

Download Methods for Synchronization

When you create a subscribed library, you can use the option to download content from the source published library immediately or only when needed to manage your storage space.

- When you synchronize a subscribed library that is configured to download all the contents of the published library immediately, the process synchronizes both the item metadata and the item contents. During synchronization, the library items that are new for the subscribed library are fully downloaded to the subscribed library storage. If some items are deleted from the published library, their contents remain at the storage location of your subscribed library, and you have to manually delete them.
- When you synchronize a subscribed library that is configured to download contents only when needed, the process synchronizes only the metadata for the library items from the published library, and does not download the contents of the items, which saves storage space. If you must use a library item, you need to synchronize that item. Synchronizing a library item downloads the full content of that item to your storage. When you no longer need the item, you can delete the item contents to free storage space.

You can take advantage of optimized transfer speed for synchronization between a published and a subscribed library under certain circumstances.

- If a published and a subscribed library belong to vCenter systems that are in the same vCenter Single Sign-On domain, and both libraries use datastores as backing storage, transfer speed for synchronization is faster. The transfer speed optimization is made possible if the libraries can store their contents to datastores managed by ESX hosts that are directly connected to each other. Therefore, the synchronization between the libraries is handled by a direct ESX host to ESX host transfer.
- If the datastores have VMware vSphere Storage APIs - Array Integration (VAAI) enabled, the library content synchronization between the published and the subscribed library is further optimized. In this case, the contents are synchronized by a direct datastore to datastore transfer.

The supported download methods depend on the source library. For more information, see the following table.

Table 123: Source Objects to Which You Can Subscribe by Creating a Subscribed Library in the vSphere Client

Source Object	Download library content immediately	Download library content when needed
A library running in a vCenter 6.x instance	Supported	Supported
A catalog running in a vCloud Director 5.5 and later instance	Supported	Not supported
A third-party library	Supported for third-party libraries that require authentication, if the username of the third-party library is `vcsp`. If the username of the source third-party library is different from `vcsp`, you can subscribe to it by using the VMware vCloud Suite API.	Supported for third-party libraries that require authentication, if the username of the third-party library is `vcsp`. If the username of the source third-party library is different from `vcsp`, you can subscribe to it by using the VMware vCloud Suite API.

Synchronize a Subscribed Content Library

To ensure that your subscribed library displays the latest content of the published library, you can manually initiate a synchronization task.

Required privilege: **Content library** > **Sync subscribed library** on the library.

You can also have subscribed libraries automatically synchronize with the content of the published library. Automatic synchronization requires a lot of storage spaces, because you download full copies of all the items in a published library.

To enable automatic synchronization, you must change the library settings. For information about changing content library settings, see Edit a Content Library.

1. Navigate to the **Content Libraries** list.
2. Right-click a subscribed library and select **Synchronize**.

A new task for synchronizing the subscribed library appears in the **Recent Tasks** pane. After the task is complete, you can see the updated list with library items on the **Templates** and **Other Types** tabs.

Synchronize a Library Item in a Subscribed Library

To update or download the content of a library item in a subscribed library, you can synchronize the library item.

Required privilege: **Content library** > **Sync library item** on the library item.

When you create a subscribed library, only the metadata for the library contents is downloaded to the associated storage if you selected the option to download library content only when needed. When you must use a library item, you synchronize it to download its content to your local storage.

When you no longer need the item, you can delete the content of the item to free storage space. You continue to see the item in your subscribed library, but it no longer takes up space on your storage because only the metadata of items remains on the storage. For information about deleting an item, see Delete a Content Library Item.

1. Navigate to the **Content Libraries** list.
2. Select a subscribed library from the list.
3. Synchronize the item that you must use.
   - On the **Templates** tab, right-click a template, and select **Synchronize Item**.

     Note:
     Synchronization is not available for VM templates. You can only synchronize OVF templates.
   - On the **Other Types** tab, right-click an item, and select **Synchronize Item**.

After the synchronization finishes, the item content and metadata are downloaded to the storage of the subscribed library. On the **Templates** tab for the subscribed library, the value for the item in the **Stored Content Locally** column changes to Yes.

How to Populate Libraries with Content

You can populate a content library with VM templates and OVF templates that you can use to provision new virtual machines. You can also add other files to a content library, such as ISO images, scripts, and text files.

You can populate a library with items in several ways.

Import Items to a Content Library

You can add items to a local content library by importing files from your local machine or from a web server. You can import OVF and OVA templates and other types of files, such as ISO images, certificates, and so on. You can keep the items in the library and share them with other users across multiple vCenter instances. You can also use the templates in the content library to deploy new virtual machines and vApps.

Required privilege: **Content library** > **Add library item** and **Content library** > **Update files** on the library.

1. Navigate to the **Content Libraries** list.
2. Right-click a local content library and select **Import Item**.
   The **Import Library Item** dialog box opens.
3. In the **Source** section, choose the source of the item.

Option	Description
Import from a URL	Enter the path to the web server where the item is. **Note:** You can import either an `.ovf` or `.ova` file. The resulting content library item is of the OVF Template type.
Import from a Local File	Click **Browse** to navigate to the file that you want to import from your local system. You can use the drop-down menu to filter files in your local system. **Note:** You can import either an `.ovf` or `.ova` file. When you import an OVF template, first select the OVF descriptor file (`.ovf`). Next, you are prompted to select the other files in the OVF template, for example the `.vmdk` file. The resulting content library item is of the OVF Template type.

   vCenter reads and validates the manifest and certificate files in the OVF package during importing. A warning is displayed in the **Import Library Item** wizard, if certificate issues exist, for example, if vCenter detects an expired certificate.

   Note:
   vCenter does not read signed content, if the OVF package is imported from an `.ovf` file from your local machine.
4. In the **Destination** section, enter a name and a description for the item.
5. Click **Import**.

In the **Recent Tasks** pane you see two tasks, one about creating a new item in the library, and the second about uploading the contents of the item to the library. After the task is complete, the item appears on the **Templates** tab or on the **Other Types** tab.

Clone a vApp to a Template in a Content Library

You can clone existing vApps to vApp templates in a content library. You can use the vApp templates later to provision new vApps on a cluster or a host in your vSphere inventory. The vApp is exported to a content library in the OVF format.

Verify that you have the following privileges:

- **Content library** > **Add library item** on the library.
- **Content library** > **Update files** on the library.
- **vApp** > **Export** on the vApp.

1. In the vSphere Client, select **Menu** > **VMs and Templates**.
2. Right-click a vApp and select **Clone** > **Clone to Template in Library**.
   The **Clone to Template in Content Library** wizard opens.
3. On the **Basic information** page, configure the content library template and click **Next**.
   a) Select the **New template** radio button.
   b) Enter a name and, optionally, notes for the template.
   c) Optional: Select the **Preserve MAC-addresses on network adapters** check box to preserve the MAC addresses of the network adapters.
   d) Optional: Select **Include extra configuration** to include vApp-related configuration in the template that you clone.
4. On the **Location** page, select a content library to clone the vApp to.
5. On the **Ready to complete** page, review the configuration details and click **Finish**

A new task for cloning to OVF package appears in the Recent Tasks pane. After the task finishes, the vApp template appears on the **Templates** tab for the content library.

Use the template to provision vApps on a host or a cluster in your vSphere inventory. See Create New vApp From a Template in a Content Library .

Clone a Virtual Machine or a Virtual Machine Template to a Template in a Content Library

You can add new templates to a content library by cloning virtual machines or virtual machine templates from your vCenter inventory to templates in the content library. You can use the content library items later to provision virtual machines on a cluster or a host. You can also update an existing template in the content library by cloning a virtual machine or virtual machine template from the vCenter inventory.

Verify that you have the following privileges:

- **Content library** > **Add library item** on the library.
- **Content library** > **Update files** on the library.
- **Datastore** > **Allicate space** on the destination datastore.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **vApp** > **Export** on the vApp, if you create an OVF Template in the content library.
- **Virtual machine** > **Edit inventory** > **Create from existing** on the data center or virtual machine folder.
- **Virtual machine** > **Provisioning** > **Clone template** on the source template, if cloning a virtual machine template.
- **Virtual machine** > **Provisioning** > **Create template from virtual machine** on the source virtual machine, if cloning a virtual machine.
- **VM storage policies** > **View VM storage policies** on the vCenter instance.

Templates are primary copies of virtual machines that you can use to create virtual machines that are ready for use. You can change the template, such as installing additional software in the guest operating system, while preserving the state of the original template.

When you clone a virtual machine from the vCenter inventory as a template to the content library, you can choose what type of content library item to create. You can choose to create a library item of either the VM Template type or OVF Template type.

> **Important:**
> If you choose to create a VM Template library item, an identical VM template is created in the vCenter inventory.

For more information, see Templates in Content Libraries.

1. Navigate to the virtual machine or template that you want to clone.
2. Select your task.

Option	Description
Clone a virtual machine	1. Right-click the virtual machine and select **Clone** > **Clone as Template to Library**. The **Clone Virtual Machine To Template** wizard opens.

Option	Description
	2. On the **Basic information** page, enter a name and notes for the template, select the template type, and select an inventory folder for the template. You can create an OVF Template or VM Template in the content library. 3. On the **Location** page, select a local content library in which you want to add the template. 4. On the **Select a compute resource** page, select the compute resource for the template. 5. On the **Select storage** page, select the storage for the template disk and configuration files. 6. On the **Ready to complete** page, review the details and click **Finish** to complete the cloning task.
Clone a virtual machine template	1. Right-click the virtual machine template and select **Clone to Library**. The **Clone to Template in Library** dialog box opens. 2. Select the **Clone as** option. You can create a template or you can choose an existing template to update. 3. From the content libraries list, select the library in which you want to add the template. 4. Enter a name and notes for the template. 5. Select the configuration data that you want to include in the template. You can select to preserve the MAC-addresses on the network adapters and include extra configuration. 6. Click **OK**.

A new task for cloning appears in the **Recent Tasks** pane. After the task is complete, the template appears in the **Templates** tab for the content library. You can view the type of template in the **Type** column.

Use the template to create virtual machines on hosts or clusters in the vSphere inventory.

Manage Content Library Items

You can perform various tasks with the items in a content library. For example, you can synchronize an item from a subscribed library to download all its contents and use the item to deploy a virtual machine. You can delete items you no longer need to use, and so on.

Each template or other type of file in a content library is a library item. An item can contain a single file or multiple files. For example, when you add an OVF template to the library, you actually upload all the files that are associated with the template, but in the vSphere Client you only see one library item of the OVF Template type.

Update a Content Library Item

Managing and keeping your virtual environment up-to-date might require you to update the content of a library item. For example, you can directly update a template when you want to add a patch to it, instead of deleting the existing template and creating a new one.

Verify that you have the Content Library Administrator role and the following privileges:

- **Content library** > **Update library item** on the library.
- **Content library** > **Update files** on the library.

Note:
You can update only templates from the **OVF and OVA Templates** type. Updating templates from the **VM Templates** type is not supported.

You cannot update the contents of a subscribed library. In local and published libraries, you can update only templates of the OVF Template type.

1. Navigate to the **Content Libraries** list.
2. Click a content library and select the file that you want to update.
   - From the **Templates** tab, click **OVF & OVA Templates**, right-click a template from the list, and select **Update Item**.
   - From the **Other Types** tab, right-click a file from the library that is not a template, and select **Update Item**.

   The **Update Library Item** dialog box opens.
3. In the **Source** section, select a file to overwrite the item in your library with.

Option	Description
URL	Enter the URL to a web server where the item is stored.
Browse	Navigate to an item that is stored on your local system.

4. Optional: In the **Destination** section, change the name of the item, the notes of the item, or both. Click **Update**.

The content of the item is updated. On the **Summary** tab for the item, you can view the time of the last update of the item.

Export an Item from a Content Library to Your Local Computer

You might need to export an item from a content library to your local system.

Required privilege: **Content library** > **Download files** on the library.

Note:
You can export only items from the **OVF and OVA Templates** type. Exporting items from the **VM Templates** type is not supported.

1. Navigate to the **Content Libraries** list.
2. Select a content library.
3. Select the type of file that you want to export.
   - From the **Templates** tab, click **OVF & OVA Templates**, right-click a template from the list, and select **Export Item**.
   - From the **Other Types** tab, right-click a file from the library that is not a template, and select **Export Item**.
4. In the **Export Library Item** dialog box click **OK**.
5. If you are exporting an OVF template, you are prompted to save each file that is associated with the template to the browser download location (for example, `.vmdk` and `.mf` files).

Clone Library Items from One Library to Another Library

You can clone a template from one content library to another in the same vCenter instance. The cloned template is an exact copy of the original template.

Verify that you have the following privileges:

- **Content library** > **Add library item** on the destination library.
- **Content library** > **Update files** on the destination library.

When cloning a template between libraries, you can select the source library to also be a destination library in the clone wizard.

A subscribed library can be the source of an item you want to clone, but you cannot clone items to a subscribed library. The subscribed libraries are filtered out from the list with destination libraries in the Clone Library Item dialog box. When the source library of an item you want to clone is a subscribed library with the setting to download items only when needed, the item is first downloaded to the source subscribed library and then cloned to the destination library.

1. Navigate to the **Content Libraries** list.
2. Click a content library and click the **Templates** tab.
3. Right-click a VM template and select **Clone Item**.
   The **Clone Library Item** dialog box opens.
4. Optional: Change the name and notes for the item you clone.
5. From the list of content libraries, select the library in which you want to clone the template and click **OK**.
   You can select the destination library to be the same as the source library if you want to have an identical copy of the template in the same library.

A new task for cloning the template appears in the Recent Tasks pane. After the task is complete, a clone of the template appears on the **Templates** tab of the destination content library.
Deploy a virtual machine from a template on a host or a cluster in your vSphere inventory.

Edit a Content Library Item

Edit a library item to change its name, description, or tag properties.

Required privileges on the library:

- **Content library** > **Update library item**
- **Content library** > **Update local library**

You can edit items only in a local library, regardless of whether it is published or not. Library items in subscribed libraries cannot be modified.

You can edit both VM templates and OVF templates.

1. Navigate to the **Content Libraries** list.
2. Open a local library by clicking its name.
3. Navigate the library item to edit.
   - To edit a template, click the **Templates** tab.
   - To edit another type of file, click the **Other Types** tab.
4. Right-click the item and select your task from the context menu.
   - To edit the description of the item, select **Edit Notes**.
   - To rename the item, select **Rename**.
   - To assign a tag to the item, select **Tags** > **Assign Tag**.
   - To remove a tag from the item, select **Tags** > **Remove Tag**.
   - **vSphere Tagging** > **Assign or Unassign vSphere Tag**

Delete a Content Library Item

If you use a subscribed library, and you synchronize it, you can later delete the library from storage, but keep the metadata. You can also completely delete a library item such as a template.

Required privileges

Task	Required Privileges
Delete the contents of a library item	**Content library** > **Evict library item**
Delete a library item	**Content library** > **Delete library item**

If a subscribed library is created with the option to download library content only when needed, only metadata for the library items is stored in the associated with the library storage. When you want to use a library item, for example, use a

VM template to deploy a virtual machine, you have to synchronize the item. Synchronization downloads the entire content to the associated storage.

In the vSphere Client, you can delete an item altogether.

1. Select **Menu** > **Content Libraries**.
2. Click a content library, select the type of item, and select the task that you want to perform with the item.
3. From the **Templates** tab, right-click a template from the library, and select **Delete**.
4. From the **Other Types** tab, right-click a file from the library that is not a template, and select **Delete**.

Convert a Subscribed Content Library to a Published Local Library

To prevent data loss if the original published local library becomes unavailable, you can convert a subscribed content library into a published local library.

Before you convert a subscribed content library to a published library, make sure that:

- Verify that your environment contains both a published and a subscribed content library. These libraries can reside on separate vCenter instances or within a single vCenter instance.
- The subscribed content library does not contain VM templates.
- The subscribed content library is fully synchronized and there are no ongoing synchronizations. For more information about the content library synchronization, see Synchronize a Subscribed Content Library.

  Note:

  For subscribed libraries configured to download content on demand, during conversion, the system prompts you to update the library to "Immediate" download to ensure all items are present locally.

The convert a subscribed library to a published local library feature, allows a subscribed library on another vCenter to become a published library and distribute content, maintaining availability. For example, if you decommission a vCenter instance that hosts a published library serving subscribed libraries.

After the conversion process, other existing subscribed libraries do not automatically detect the new publisher library. You must manually point each subscriber to the publisher's new subscription URL by editing the content library in the vSphere Client or through the content library API.

Note:

When you point a subscribed library to the new converted publisher, the system identifies the identical items already present in the library. Subscribed libraries do not require re-synchronization or download items that are already successfully synchronized from the original publisher.

1. Navigate to **Menu** > **Content Libraries**.
2. Right-click a subscribed content library from the **Content Libraries** pane, and select **Convert to Published Library**.
3. In the **Convert Subscribed Library to Published Library** dialog box, click **Convert**.
4. Click **Sync Library**.

Migrating Content Libraries and Library Items to Another Datastore

A content library uses a datastore for storage backing. When the storage hardware requires maintenance or you need to evacuate mapped storage, you can migrate the content library to another vSphere datastore, such as vSAN, VMFS, vVol, or NFS, within a single vCenter instance. The migration can be either homogeneous, from vSAN to vSAN, or heterogeneous, from vSAN to VMFS.

In earlier versions, when a datastore is at full capacity, you must migrate the content library to a new storage location with sufficient space. This process requires manually recreating the content library on a different datastore and reassociating subscribed items.

You can now seamlessly migrate a content library to another datastore without recreating the content library or manually reconfiguring the associated components.

When you migrate a content library to another datastore, you migrate the content library items, such as OVF and OVA templates, ISO images, and other types of files.

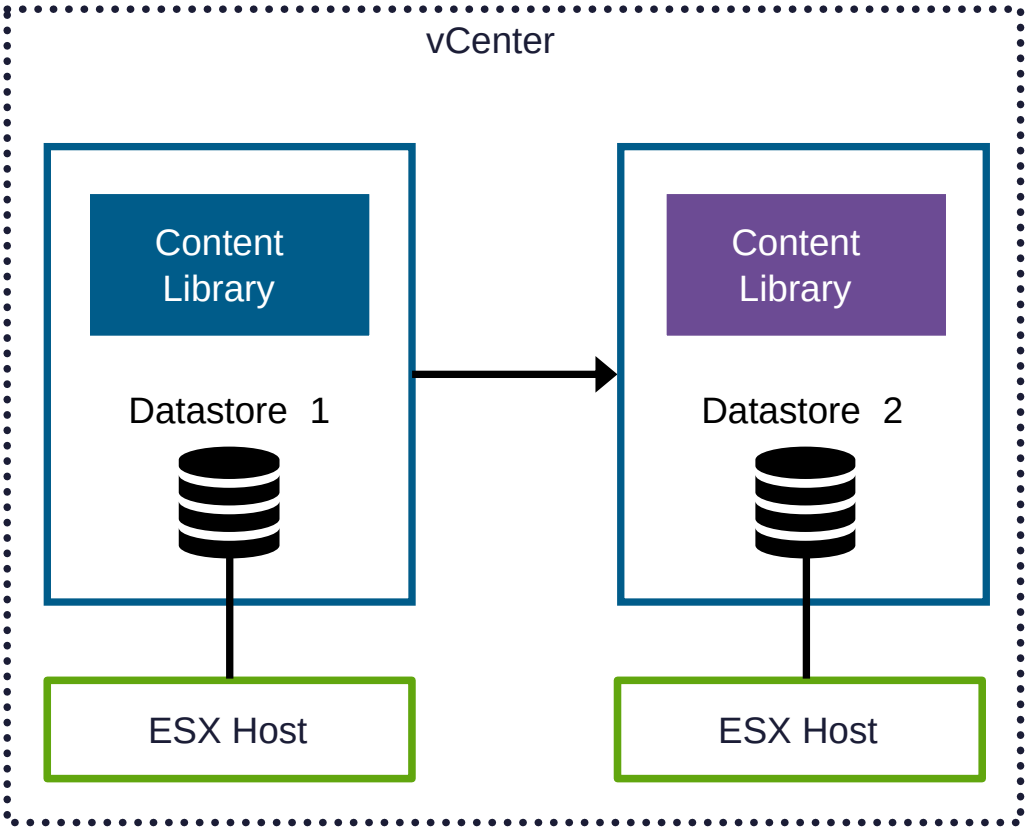
You can move a publisher library from one datastore to another without changing the subscription URL for the subscribed libraries. After the migration, the subscriber libraries do not require content synchronization.

Note:

A storage cannot enter maintenance mode until all content libraries are migrated. For information on storage decommissioning, see the *vSphere Storage* documentation.

Figure 41:

Content library migration from one datastore to another.



Subscriber library migration preserves all previously synchronized content on disk. After migration, you can continue synchronizing with the publisher without having to download existing content again.

Migrate a Content Library to Another Datastore

When you migrate a content library to another datastore, the system makes all library content temporarily inaccessible, deactivates any changes, and pauses synchronization with subscriber libraries.

To migrate a content library to another datastore, make sure that:

- The datastore to which you want to move the content library has enough space. The UI provides an alert if there is insufficient space.
- There are no operations in progress that the content library migration can impact.
- There is no ISO file mounted on a powered-on VM.

Verify that you have the following privileges:

- **Content library** > **Update local library** on the library, if migrating a local library.

- **Content library** > **Update subscribed library** on the library, if migrating a subscribed library.
- **Datastore** > **Allocate space** on the destination datastore.

Note:

Migrating a content library to another datastore does not support migrating VM templates. VM templates can reside in a different datastore. For more information on how to migrate VM templates, see Manage VM Templates in a Content Library.

1. Navigate to the **vSphere Menu** > **Content Libraries**.
2. From the **Content Libraries** pane, select a content library in the list.
3. Right-click the content library and select **Migrate**.
4. From the **Content to migrate** window, click **Next**.
5. From the **Select storage** window, select the datastore to which you want to migrate the content library and click **Next**.
6. On the **Review** page, review the details and click **Finish**.

From the **Recent Tasks** pane, you can monitor migration events for successes or failures.

If a failure occurs during migration, the content library rolls back to the previous storage backing, and any remaining copies are cleaned up.

Creating Virtual Machines and vApps from Templates in a Content Library

You can deploy virtual machines and vApps from the VM or OVF templates that are stored in a content library.

The library can be a local library to the vCenter instance where you want to deploy the VM or the vApp template, or can be a subscribed library to that vCenter instance.

The VM Template type is only supported in the vSphere Client. You can deploy virtual machines from VM Templates in a content library only in the vSphere Client.

Note:
You can also use the API calls to create and manage VM templates in a content library.

The use of templates results in consistency, compliance, and efficiency when you deploy virtual machines and vApps in your data center.

Deploy a Virtual Machine from an OVF Template in a Content Library

In content libraries, you can use the OVF template, which is either the template of a virtual machine or a vApp, to deploy a virtual machine to a host or a cluster in your vSphere inventory.

Verify that you have the following privileges:

- **Virtual machine** > **Edit Inventory** > **Create from existing**
- **Virtual machine** > **Edit Inventory** > **Create new**
- **Virtual machine** > **Change Configuration** > **Add new disk**
- **Virtual machine** > **Change Configuration** > **Advanced configuration**
- **Datastore** > **Allocate space**
- **Network** > **Assign network**
- **Virtual machine** > **Provisioning** > **Deploy template**
- **Resource** > **Assign virtual machine to resource pool**
- **vApp** > **Import**
- **vApp** > **Create**
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

1. Navigate to **Menu** > **Content Libraries**.
2. Select a content library and click the **Templates** tab.
3. Right-click an OVF template and select **New VM from This Template**.
   The **New Virtual Machine from Content Library** wizard opens.
4. On the **Select a name and folder** page, enter a name, select a location for the virtual machine, and select additional customization options for the new virtual machine.
   a) To customize the guest operating system of the virtual machine, select the **Customize the operating system** check box. A **Customize guest OS** page appears in the next window.
   b) To change the hardware settings of the virtual machine, select the **Customize this virtual machine's hardware** check box. A **Customize hardware** page appears if you want to customize the hardware of the virtual machine.
5. Click **Next**.
6. On the **Select a compute resource** page, select a host, a cluster, a resource pool, or a vApp where to run the deployed template, and click **Next**.

   Important:

   If the template that you deploy has an NVDIMM device and virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, you cannot proceed with the task.

   If the template that you deploy does not have an NVDIMM device, but it has virtual PMem hard disks, the destination host or cluster must have available PMem resource. Otherwise, all the hard disks of the virtual machine use the storage policy and datastore selected for the configuration files of the source template.
7. On the **Review details** page, verify the template details and click **Next**.
8. On the **Select storage** page, select the datastore or datastore cluster in which to store the virtual machine configuration files and all of the virtual disks. Click **Next**.

Option	Description
Deploy a virtual machine from a template that has vPMem hard disks	1. Choose the type of storage for the template by selecting the **Standard**, the **PMem**, or the **Hybrid** radio button. If you select the **Standard** mode, all virtual disks will be stored on a standard datastore. If you select the **PMem** mode, all virtual disks will be stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. If you select the **Hybrid** mode, all PMem virtual disks will remain stored on a PMem datastore. Non-PMem disks are affected by your choice of a VM storage policy and datastore or datastore cluster. For more information about persistent memory and PMem storage, see the *vSphere Resource Management* guide. 2. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.
Deploy a virtual machine from a template that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. **Same format as source** uses the same disk format as the source virtual machine. The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, on first write from the virtual machine.

Option	Description
	Thick Provision Eager Zeroed is a type of thick virtual disk that supports clustering features such as Fault tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks. The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can grow to the maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Turn on the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.

Note:

If you want to use the API calls to deploy an OVF template that contains vPMem hard disks and that has been exported from a content library, see Broadcom knowledge base article 336060.

9. On the **Select networks** page, select a network for each network adapter in the template and click **Next**.
10. On the **Customize hardware** page, configure the hardware options of the virtual machine and click **Next**.

    Note:
    You cannot change the existing virtual hardware disks, CD/DVD drives settings, and you cannot add new virtual disks or CD/DVD drives. You can change these settings after you deploy the virtual machine.
11. On the **Ready to complete** page, review the page and click **Finish**.

A new task for creating the virtual machine appears in the Recent Tasks pane. After the task is complete, the new virtual machine is created on the selected resource.

Deploy a Virtual Machine from a VM Template in a Content Library

In the vSphere Client, you can use a content library item of the VM Template type to deploy a virtual machine to a host or cluster in your vSphere environment.

Verify that you have the following privileges:

- **Datastore** > **Allocate space**
- **Virtual machine** > **Edit Inventory** > **Create new**
- **Virtual machine** > **Inventory** > **Create from existing**
- **Virtual machine** > **Change** > **Add new disk**
- **Virtual machine** > **Provisioning** > **Deploy template**
- **Virtual machine** > **Provisioning** > **Customize guest** on the template or template folder, if you customize the guest operating system.
- **Virtual machine** > **Provisioning** > **Read customization specifications** on the vCenter instance, if you customize the guest operating system.
  Resource > **Assign virtual machine to resource pool**
- **vApp** > **Import**
- **vApp** > **Create**
- **VM storage policies** > **View VM storage policies** to view and use defined storage capabilities and storage profiles.

- To access customization options for Windows guest operating systems, Microsoft Sysprep tools must be installed on the vCenter system. The Sysprep Tool is built into Windows Vista and Windows 2008 and later operating systems. For details about this and other customization requirements, see Guest Operating System Customization Requirements.

  Important:
  If the template that you deploy has an NVDIMM device and virtual PMem disks, the destination host or cluster must have an available PMem resourse. Otherwise, you cannot proceed with the task.

If the template that you deploy does not have an NVDIMM device, but has virtual PMem hard disks, the destination host or cluster must have an available PMem resource. Otherwise, all hard disks of the virtual machine will use the storage policy and datastore selected for the configuration files of the source template.

For information about persistent memory and PMem storage, see the *vSphere Resource Management* guide.

For more information about configuring the virtual machine hardware options, see Configuring Virtual Machine Hardware and Configuring Virtual Machine Options

Note:

If you want to use the API calls to deploy an OVF template that contains vPMem hard disks and that has been exported from a content library, see Broadcom knowledge base article 336060.

1. Navigate to **Menu** > **Content Libraries**.
2. To open a content library, click its name.
3. On the **Templates** tab, right-click a VM Template and select **New VM from This Template**.
   The **Deploy From VM Template** wizard opens.
4. On the **Select a name and folder** page, enter a name and select a location for the virtual machine.
5. On the **Select a compute resource** page, select a host, a cluster, a resource pool, or a vApp where to run the deployed VM template, and click **Next**.
6. On the **Select storage** page, select the datastore or datastore cluster in which to store the virtual machine configuration files and all virtual disks.

Option	Action
Deploy a virtual machine from a template that has vPMem hard disks	1. Select the type of storage for the template by clicking the **Standard**, the **PMem**, or the **Hybrid** radio button. – If you select the **Standard** mode, all virtual disks are stored on a standard datastore. – If you select the **PMem** mode, all virtual disks are stored on the host-local PMem datastore. Configuration files cannot be stored on a PMem datastore and you must additionally select a regular datastore for the configuration files of the virtual machine. – If you select the **Hybrid** mode, all PMem virtual disks remain stored on a PMem datastore. Your choice of a VM storage policy and datastore or datastore cluster affects the Non-PMem disks. 2. From the **VM Storage Policy** drop-down menu, select a virtual machine storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. To select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk, enable the **Configure per disk** option. **Note:** To convert a PMem hard disk to a regular one, you can use the **Configure per disk** option , but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.
Deploy a virtual machine from a template that does not have vPMem hard disks	1. Select the disk format for the virtual machine virtual disks. – The **Same format as source** option uses the same disk format as the source virtual machine.

Option	Action
	– The **Thick Provision Lazy Zeroed** format creates a virtual disk in a default thick format. The space required for the virtual disk is allocated when the virtual disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out later, on demand, or on first write from the virtual machine. – The **Thick Provision Eager Zeroed** format is a type of thick virtual disk that supports clustering features such as Fault Tolerance. The space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks. – The **Thin Provision** format saves storage space. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can extend to the maximum capacity allocated to it. 2. Select a VM storage policy or leave the default one. 3. Select a datastore or a datastore cluster. 4. Select the **Disable Storage DRS for this virtual machine** check box if you do not want to use storage DRS with the virtual machine. 5. Enable the **Configure per disk** option to select a separate datastore or a datastore cluster for the template configuration file and for each virtual disk. **Note:** You can use the **Configure per disk** option to convert a PMem hard disk to a regular one, but that change might cause performance problems. You can also convert a standard hard disk to a PMem hard disk.

7. Click **Next**.
8. On the **Select deploy options** page, apply a customization specification to the virtual machine and click **Next**.

   You can choose to customize the guest operating system or the virtual machine hardware. You can also choose to power on the virtual machine after its creation.
9. Optional: On the **Customize guest OS** page, select a customization specification to apply to the virtual machine.

   Customizing the guest operating system helps prevent conflicts that can result if you or other users deploy virtual machines with identical settings, such as duplicate computer names.

Option	Action
Select an existing specification	Select a customization specification from the list.
Override	To change the guest customization specification for this deployment only, click **Override**, complete the steps in the **Override VM Customization Specification** wizard, and click **OK**.

10. Optional: On the **User settings** page, specify the required settings for the virtual machine.

    This page of the wizard appears only if the selected specification requires additional customization.
11. Optional: On the **Customize hardware** page, configure the virtual machine hardware and options and click **Next**.

    You can leave the defaults and configure the virtual machine hardware and options later.

    Important:
    If you chose to use PMem storage for the virtual machine, its default hard disk, the new hard disks that you configure, and the NVDIMM devices that you add to the virtual machine all share the same PMem resources. You must adjust the size of the newly added devices in accordance with the amount of the PMem available to the host. If any part of the configuration requires attention, the wizard alerts you.
12. On the **Ready to complete** page, review the information and click **Finish**.

A new task for creating the virtual machine appears in the **Recent Tasks** pane. After the task finishes, the new virtual machine is created on the selected resource.

Create New vApp From a Template in a Content Library

You can use an OVF template to create a new vApp on a host or a cluster in your vSphere inventory.

Verify that you have the following privileges:

- **Datastore** > **Allocate space** on the destination datastore.
- **Network** > **Assign network** on the destination network.
- **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
- **vApp** > **Create** on the destination folder or data center.
- **vApp** > **Import** on the destination folder or data center.
- **Virtual machine** > **Edit inventory** > **Create from existing** on the destination folder or data center.
- **Virtual machine** > **Change Configuration** > **Add new disk** on the virtual machine.

1. Navigate to **Menu** > **Content Libraries**.
2. Open a content library by clicking its name, and click the **Templates** tab.
3. Right-click a template of a vApp and select **New vApp from This Template**.
   The **New vApp from Content Library** wizard opens.
4. On the **Select a name and folder** page, enter a name and select a location for the vApp, and click **Next**.
5. On the **Select a compute resource** page, select a host, a cluster, a resource pool, or a vApp to deploy the vApp to and click **Next**.
6. On the **Review detail**s page, verify the template details and click **Next**.
7. On the **Select storage** page, select disk format and a storage resource for the vApp.
8. On the **Select networks** page, select a destination network for each source network.
9. On the **Ready to complete** page, review the configurations you made for the vApp, and click **Finish**.

A new task for creating the vApp appears in the **Recent Tasks** pane. After the task is complete, the new vApp is created.

Manage VM Templates in a Content Library

In vSphere 8.0 and later, you can edit the contents of the VM templates by checking them out, making the necessary changes, and checking them in.

You can track the history of changes over time by using the vertical timeline view. The vertical timeline view provides you with detailed information about the different VM template versions, the updates that privileged users have made, and when the last change was made. By using the vertical timeline, you can revert VM templates back to their previous state or can delete the previous version of a VM template.

In addition, you can deploy a virtual machine from the latest version of the VM template without any disruptions while it is checked out for update. You can update the virtual machine and check it back in into the same VM template.

You can migrate VM Templates from one datastore to another using the following workflow:

1. Check out a virtual machine from a Template. For more information, see Check Out a Virtual Machine from a Template.
2. Migrate the storage with Storage vMotion. For more information, see Migrating Virtual Machines with vSphere vMotion.
3. Check in a VM to a Template. For more information, see Check In a Virtual Machine to a Template.

This is a supported workflow available in vSphere environments with Storage vMotion capabilities.

Templates in Content Libraries

Templates are primary copies of virtual machines that you can use to deploy virtual machines that are customized and ready for use. Templates promote consistency throughout your vSphere environment. You can use the content library to store and manage templates of virtual machines and vApps. You can use VM templates and vApp templates to deploy virtual machines and vApps to a destination object, such as a host or a cluster.

Content libraries support two types of templates, the OVF Template type and the VM Template type.

In a content library, you can store and manage virtual machine templates such as OVF templates or VM templates. vApps are always converted to OVF templates in the content library.

VM Templates in Content Libraries and in the vCenter Inventory

A VM template is a template of a virtual machine. You create a VM template by cloning a virtual machine into a template.

A VM template can be managed by vCenter or by a content library.

In previous releases of vSphere, you can manage VM templates only through the vCenter inventory list. When you cloned a virtual machine or a VM template to a content library template, the resulting content library item was in an OVF format. Starting with vSphere 7.0 Update 1, local content libraries support both OVF templates and VM templates. You choose the type of template when you clone the virtual machine into the content library.

When you create a VM template in a content library, the library item is backed by a VM template in the vCenter inventory. The content library item and the corresponding inventory object are related in the following ways.

- If you convert the VM template in the vCenter inventory to a virtual machine, the corresponding VM template library item is also deleted.
- If you rename the VM template in the vCenter, the corresponding VM template library item is also renamed.
- If you rename the VM template library item the associated VM template in the vCenter inventory is also renamed.
- If you delete the VM template in the vCenter inventory, the corresponding VM template library item is also deleted.
- If you delete the VM template library item, the associated VM template in the vCenter inventory is also deleted.

OVF Templates in Content Libraries

In a content library, an OVF template is either a template of a virtual machine, or a template of a vApp. When you clone a virtual machine into a template in a content library, you choose whether to create an OVF template or a VM template. However, if you clone a vApp into a template in a content library, the resulting content library item is always an OVF template. Because the OVF format is actually a set of files, if you export the template, all the files in the OVF template library item (`.ovf`, `.vmdk`, `.mf`) are saved to your local system.

Differences Between VM Templates and OVF Templates in Content Libraries

You can use both VM templates and OVF templates to deploy new virtual machines in your vSphere environment. However, the two types of templates have different properties and support different deployment options.

See the following table for a detailed list of the differences between VM templates and OVF templates in a content library.

Table 124: VM Templates and OVF Templates Properties

Property	VM Templates in Content Library	OVF Templates in Content Library
Datastore	VM templates can be stored on any datastore that you have privileges to. **Note:** VM templates cannot be stored in a library that uses NFS or SMB storage.	OVF templates can only be stored on the datastore that is associated with the content library.
Footprint	The default one.	Compressed or Thin.
Host/Datastore Maintenance Mode	When the host becomes inaccessible, VM templates are automatically migrated to another host.	When either the host or the datastore becomes inaccessible, you must manually migrate the OVF templates to another host or datastore.
Associated with a Host	Yes.	No.
Storage DRS	Supported.	Not supported.
Cross-vendor Compatibility	Not supported.	Supported.

Property	VM Templates in Content Library	OVF Templates in Content Library
Software License Agreement	Not supported.	Supported.
Encryption	Supported. You can create encrypted VM templates.	Not supported. While OVF templates cannot be encrypted themselves, you can still deploy an encrypted virtual machine from an OVF template.
Deployment Options	During the deployment of a VM template, hardware customization and guest OS customization are both supported.	During the deployment of an OVF template, only guest OS customization is supported. Hardware customization is not supported.

The supported operations on a content library template are different depending on the template type. You can edit the settings for both OVF and VM templates. However, you can update, export, and clone a template only if it is an OVF template.

Check Out a Virtual Machine from a Template

In the vSphere Client, you can edit the VM templates and monitor the changes that have been made by other privileged users. You can perform the checkout operation to update a virtual machine from the VM template. During this process, the VM template is not available for checkout from other users, but they can deploy a virtual machine from the VM template without any disruptions.

Verify that you have the following privileges:

- **Content library** > **Check out a template**
- **Resource** > **Assign virtual machine to resource pool**
- **Datastore** > **Allocate space**
- **Virtual machine** > **Edit inventory** > **Create from existing**
- **Virtual machine** > **Change Configuration** > **Set annotation**
- If you want to power on the checked out virtual machine, verify that you have the **Virtual machine** > **Interaction** > **Power On** privilege.

When you check out a VM template, you cannot convert the virtual machine to a template or migrate the virtual machine to a different vCenter inventory.

1. To check out a VM template

Option	Action
From a content library	1. Navigate to **Menu** > **Content Libraries**. 2. To open a local library, click its name. 3. On the **Templates** tab, select a VM template and click the **Check out VM from this template** button.
From the vSphere Client inventory	1. Navigate to **Menu** > **VMs and Templates** and click the VM template. 2. Click the **Versioning** tab and in the vertical timeline view, click **Check out VM from this template**.

   The **Check out VM from VM Template** dialog box opens.

2. On the **Name and location** page, enter a virtual machine name, select the virtual machine location, and click **Next**.
3. On the **Select compute resource** page, select the compute resource for the checked out virtual machine and click **Next**.
4. On the **Review** page, review the configuration.
5. Choose whether to power on the virtual machine after checkout by selecting the **Power on VM after checkout** check box.
6. Click **Finish**.

The checked out virtual machine appears in the selected location marked with a blue circle icon. You can perform the necessary configuration changes.

After you complete the virtual machine updates, you can check in the virtual machine back to the template.

Check In a Virtual Machine to a Template

After you check out a virtual machine from a template and update the virtual machine, you must check the virtual machine back into the VM template. When you check in the virtual machine to a template, you create a new version of the VM template containing the updated state of the virtual machine.

Verify that the virtual machine is powered off or suspended. You cannot check in a powered-on virtual machine to a VM template.

Required privileges:

- **Content library** > **Check in a template**

When you check in the virtual machine to the VM template, you allow the deployment of the last changes that you make to the virtual machine.

1. To check in a virtual machine to a template:

Option	Action
From a content library	1. Navigate to **Menu** > **Content Libraries**. 2. To open a content library, click its name. 3. On the **Templates** tab, select a VM template and click **Check in VM to template**.
From the vSphere Client inventory	1. Navigate to **Menu** > **VMs and Templates** and click the VM template. 2. Click the **Versioning** tab and in the vertical timeline view, click **Check in VM to template**.

   The **Check in VM** dialog box opens.
2. To describe the change, enter a comment in **Check in notes**.
3. Click **Check in**.

The updated version of the VM template appears in the vertical timeline. You can see the check-in comment, the name of the user who made the changes, and the date of the change.

Discard a Checked Out Virtual Machine

If you check out a VM template and make no updates to the virtual machine or perform an update that you do not want to keep, you can discard the checked out virtual machine. Each time you check in the virtual machine back to the template, you create a new version of the VM template. You can discard the checked out virtual machine to avoid creating new versions or to prevent other users from using a faulty version.

Verify that you have the following privilege:

- **Virtual machine** > **Edit inventory** > **Remove** on the checked out virtual machine.

1. To discard a checked out virtual machine:

Option	Action
From a content library	1. Navigate to **Menu** > **Content Libraries**. 2. To open a local library, click its name. 3. On the **Templates** tab, select a VM template. 4. From the vertical timeline, click the horizontal ellipsis icon (•••) that appears in the checked out VM template box and select **Discard Checked Out VM**.
From the vSphere Client inventory	1. Navigate to **Menu** > **VMs and Templates** and click the VM template. 2. Click the **Versioning** tab in the vertical timeline. 3. Click the horizontal ellipsis icon (•••) that appears in the checked out VM template box, and select **Discard Checked Out VM**.

   The **Discard Checked Out VM** dialog box opens.

2. To delete the checked out virtual machine and discard all changes, click **Discard**.

You deleted the virtual machine from the inventory and discarded all changes.

Revert to a Previous Version of a Template

If the latest VM template contains changes that you no longer want to keep or you made a mistake during your last checkin, you can revert the VM template to the previous version.

Required privileges:

- **Content library** > **Check in a template**

1. To revert to a previous version of a template:

Option	Action
From a content library	1. Navigate to **Menu** > **Content Libraries**. 2. To open a local library, click its name. 3. On the **Templates** tab, select a VM template.
From the vSphere Client inventory	1. Navigate to **Menu** > **VMs and Templates** and click the VM template. 2. Click the **Versioning** tab.

2. From the vertical timeline, navigate to the previous state of the VM template, click the horizontal ellipsis icon (•••), and select **Revert to This Version**.

   The **Revert to Version** dialog box opens.

3. Enter a reason for the reverted operation and click **Revert**.

The VM template that you revert to becomes the current VM template.

Delete a Previous Version of a VM Template

Delete a previous version of a VM template if you no longer want to allow the use of the template. Deleting a VM template removes the template and its content from the inventory.

Required privileges:

- **Content Library** > **Delete library item**

1. To delete a previous version of a template:

Option	Action
From a content library	1. Navigate to **Menu** > **Content Libraries**. 2. To open a local library, click its name. 3. On the **Templates** tab, select a VM template.
From the vSphere Client inventory	1. Navigate to **Menu** > **VMs and Templates** and click the VM template. 2. Click the **Versioning** tab.

2. From the vertical timeline, navigate to the previous state of the VM template, click the horizontal ellipsis icon (•••), and select **Delete Version**.

   The **Confirm Delete** dialog box opens.

3. To delete permanently the VM template and its contents, click **Yes**.

Configuring Virtual Machine Hardware

You can add or configure most virtual machine hardware settings during virtual machine creation or after you create the virtual machine and install the guest operating system.

When you configure the virtual machine hardware, you can view the existing hardware configuration and add or remove hardware. You can change nearly every setting that was selected during virtual machine creation.

Not all hardware devices are available to every virtual machine. The host that the virtual machine runs on and the guest operating system must support devices that you add or configurations that you make.

Virtual Machine Compatibility

When you create a virtual machine or upgrade an existing virtual machine, you use the virtual machine compatibility setting to select the ESX host versions that the virtual machine can run on.

The compatibility setting determines the virtual hardware available to the virtual machine, which corresponds to the physical hardware available on the host. Virtual hardware includes BIOS and EFI, available virtual PCI slots, the maximum number of CPUs, the maximum memory configuration, and other characteristics. New virtual hardware capabilities are typically released once a year with major or minor releases of vSphere.

Table 125: Virtual Machine Compatibility Options

Compatibility	Description
ESX 9.1 and later	This virtual machine (hardware version 22) is compatible with ESX 9.1 and later.
ESX 9.0 and later	This virtual machine (hardware version 22) is compatible with ESX 9.0, ESX 9.1 and later.
ESX 8.0 Update 2 and ESX 8.0 Update 3	This virtual machine (hardware version 21) is compatible with ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later.
ESX 8.0 and ESX 8.0 Update 1	This virtual machine (hardware version 20) is compatible with ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later..
ESX 7.0 Update 3 and later	This virtual machine (hardware version 19) is compatible with ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later..
ESX 7.0 Update 2 and later	This virtual machine (hardware version 19) is compatible with ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later.
ESX 7.0 Update 1 and later	This virtual machine (hardware version 18) is compatible with ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later.
ESX 7.0 and later	This virtual machine (hardware version 17) is compatible with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, ESX 9.1 and later.

The compatibility setting that appears in the **Compatible with** drop-down menu is the default for the virtual machine that you are creating. The following factors determine the default virtual machine compatibility:

- The ESX host version on which the virtual machine is created.
- The inventory object that the default virtual machine compatibility is set on, including a host, cluster, or data center.

You can accept the default compatibility or select a different setting. It is not always necessary to select the latest ESX host version. Selecting an earlier version can provide greater flexibility and is useful in the following situations:

- To standardize testing and deployment in your virtual environment.
- If you do not need the capabilities of the latest host version.
- To maintain compatibility with older hosts.

When you create a virtual machine, consider the environment that the virtual machine runs in and weighs the benefits of different compatibility strategies. Consider your options for these scenarios, which demonstrate the flexibility inherent with each virtual machine compatibility selection.

Objects in the Environment	Compatibility	Results
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 7.0 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 7.0 Update 1 and later. A virtual machine with such compatibility cannot run on ESX 6.7 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 7.0 Update 1 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 7.0 Update 2 and later. A virtual machine with such compatibility cannot run on ESX 7.0 and earlier.
Cluster with 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 7.0 Update 2 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 7.0 Update 3 and later. A virtual machine with such compatibility cannot run on ESX 7.0 Update 1 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 7.0 Update 3 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 8.0 and later. A virtual machine with such compatibility cannot run on ESX 7.0 Update 2 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 8.0 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 8.0 Update 1 and later. A virtual machine with such compatibility cannot run on ESX 7.0 Update 3 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 8.0 Update 1 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 8.0 Update 2 and later. A virtual machine with such compatibility cannot run on ESX 8.0 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 8.0 Update 2 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 8.0 Update 3 and later. A virtual machine with such compatibility cannot run on ESX 8.0 Update 1 and earlier.

Objects in the Environment	Compatibility	Results
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 8.0 Update 3 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 9.0 and later. A virtual machine with such compatibility cannot run on ESX 8.0 Update 2 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 9.0 and later	This virtual machine does not have all the capabilities available to virtual machines that run on ESX 9.1 and later. A virtual machine with such compatibility cannot run on ESX 8.0 Update 3 and earlier.
Cluster with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2,ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 hosts	ESX 9.1 and later	This virtual machine provides access to the latest virtual hardware features and ensures the best performance. However, a virtual machine with such compatibility cannot run on ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1,ESX 8.0 Update 2, ESX 8.0 Update 3, or ESX 9.0.

Set the Default Compatibility for Virtual Machine Creation

To ensure that virtual machines are compatible with the ESX hosts that reside in the vSphere environment, you can set the default compatibility for virtual machine creation on the host, cluster, or data center.

Required privileges:

- On the host or cluster:**Host** > **Inventory** > **Modify cluster**
- On the data center:**Datacenter** > **Reconfigure datacenter**

The following conditions apply:

- To set the default compatibility on the cluster, the cluster must contain hosts that are connected and not in maintenance mode.
- A default compatibility setting on the host overrides a default cluster or data center setting.
- A default compatibility setting on the cluster overrides a default data center setting.

Select a host, cluster, or data center in the inventory.

Option	Action
Host	1. Click the **Configure** tab, and under **Virtual Machines**, select **Default VM Compatibility**. 2. Click **Edit**. The **Edit Default VM Compatibility** dialog box opens. 3. From the **Compatible with** drop-down menu, select the compatibility and click **OK**. **Note:** You can set the compatibility only on hosts that are not part of a cluster.
Cluster	1. Click the **Configure** tab and under **Configuration**, select **General**. 2. In the **Default VM Compatibility** section, click **Edit**. 3. From the **Compatible with** drop-down menu, select the compatibility and click **OK**. When you change the compatibility for a cluster, the compatibility for all hosts in the cluster changes as well.
Data Center	1. Right-click the data center and select **Edit Default VM Compatibility**. 2. From the **Compatible with** drop-down menu, select the compatibility and click **OK**.

When you create a virtual machine on one of these objects, the default compatibility setting is used.

Schedule a Compatibility Upgrade for a Single Virtual Machine

To make the virtual machine compatible with the latest version of the host, you can schedule a compatibility upgrade. The compatibility level determines the virtual hardware available to a virtual machine, which corresponds to the physical hardware on the host.

- Required privilege: **Virtual machine** > **Change Configuration** > **Upgrade virtual machine compatibility** to create a backup or snapshot of the virtual machines.
- Upgrade to the latest version of VMware Tools. On Microsoft Windows virtual machines, if you upgrade the compatibility level before you upgrade VMware Tools, the virtual machine might lose its network settings.
- Verify that all `.vmdk` files are available to the ESX host on a VMFS5, or NFS datastore.
- Verify that the virtual machine is stored on VMFS5 or NFS datastores.

To schedule an upgrade for multiple virtual machines, see Schedule a Compatibility Upgrade for a Virtual Machine.

1. Right-click a virtual machine and select **Compatibility** > **Schedule VM Compatibility Upgrade**.
2. In the **Schedule VM Compatibility Upgrade** dialog box, confirm that you want to schedule a compatibility upgrade by clicking **Yes**.
3. From the **Compatible with** drop-down menu, select the compatibility to upgrade to.

   The virtual machine compatibility is upgraded the next time you restart the virtual machine.
4. Optional: To upgrade the compatibility when you do regularly scheduled guest maintenance, select **Only upgrade after normal guest OS shutdown**.

The virtual machine compatibility is upgraded and the new version appears on the virtual machine Summary tab.

Change the Default Virtual Machine Compatibility Setting

You can change the compatibility of a virtual machine by upgrading its compatibility or scheduling a compatibility upgrade.

- Create a backup or snapshot of the virtual machines. See Manage Virtual Machines With Snapshots.
- Upgrade to the latest version of VMware Tools. If you upgrade the compatibility before you upgrade VMware Tools, the virtual machine might lose its network settings.
- Verify that all `.vmdk` files are available to the ESX host on a VMFS3, VMFS5, or NFS datastore.
- Verify that the virtual machines are stored on VMFS3, VMFS5, or NFS datastores.
- Verify that the compatibility settings for the virtual machines are not the latest supported version.
- Determine the ESX versions that you want the virtual machines to be compatible with. See Virtual Machine Compatibility.

You can also change the default compatibility setting for the host, cluster, or data center.

The virtual machine compatibility determines the virtual hardware available to the virtual machine.

1. Optional: To determine the compatibility setting of a virtual machine, select the virtual machine in the inventory and click the **Summary** tab.
2. Change the default compatibility setting of a virtual machine.
   a) Right-click a virtual machine and click **Compatibility** > **Upgrade VM Compatibility**.
   b) Right-click a virtual machine and click **Compatibility** > **Schedule VM Compatibility Upgrade**.
   c) Change the default compatibility setting of a host or a cluster. For more information, see Set the Default Compatibility for Virtual Machine Creation.

Hardware Features Available with Virtual Machine Compatibility Settings

The virtual machine compatibility setting determines the virtual hardware available to the virtual machine, which corresponds to the physical hardware available on the host. To determine whether to upgrade the virtual machines in your environment, you can review and compare the hardware available for different compatibility levels.

For more information about the virtual machine compatibility and the supported hardware versions, see Broadcom knowledge base article 312100.

For more information about the UEFI version used in virtual machines, see Broadcom knowledge base article 313364.

Table 126: Supported Features for Virtual Machine Compatibility

Feature	Feature Availability by Virtual Hardware Version (HWV)		Recent Hardware Versions				
		22	21	20	19	18	17
ESX compatibility	See Broadcom knowledge base article 312100.	ESX 9.0 and later	ESX 8.0 Update 2 and later	ESX 8.0 and later	ESX 7.0 Update 2 and later	ESX 7.0 Update 1 and later	ESX 7.0 and later
BIOS firmware (1)	• HWV 4 or later	Y	Y	Y	Y	Y	Y
UEFI firmware (1)	• HWV 8 or later	Y	Y	Y	Y	Y	Y
Secure Boot	• HWV 13 or later	Y	Y	Y	Y	Y	Y
Maximum number of logical processors (2)	• 960 for HWV 22 or later • 768 for HWV 18 - 21 • 256 for HWV 15 - 17 • 128 for HWV 11 - 14 • 64 for HWV 9 - 10 • 32 for HWV 8 • 8 for HWV 7 • 4 for HWV 4	960	768	768	768	768	256
Maximum number of cores (virtual CPUs) per socket (3)	• 960 for HWV 22 or later • 256 for HWV 20 - 21 • 64 for HWV 9 - 19 • 32 for HWV 8 • 8 for HWV 7 • 1 for HWV 4	960	256	256	64	64	64
CPU Hot Add support	• HWV 11 or later	Y	Y	Y	Y	Y	Y
Maximum memory (GB) (4)	• 24560 for HWV 18 or later • 6128 for HWV 13 - 17 • 4080 for HWV 11 • 1011 for HWV 8 - 10 • 256 for HWV 7 • 64 for HWV 4	24560	24560	24560	24560	24560	6128
Memory Hot Add Support	• HWV 7 or later	Y	Y	Y	Y	Y	Y
Virtual NUMA Topology	• HWV 20 or later	Y	Y	Y	N	N	N
Synthetic Timers (18)	• HWV 20 or later	Y	Y	Y	N	N	N
Maximum PCI passthrough devices	• 128 for HWV 21 or later • HWV 20: – Max 32 for ESX 8.0 – Max 64 for ESX 8.0 Update 1 • 16 for HWV 13 - 19	128	128	32 for ESX 8.0 64 for ESX 8.0 Update 1	16	16	16

Feature	Feature Availability by Virtual Hardware Version (HWV)		Recent Hardware Versions				
Enable passthrough device peer to peer	• HWV 22 or later	Y	N	N	N	N	N
PCI Hot Add support	• HWV 13 or later	Y	Y	Y	Y	Y	Y
Virtual PMC support	• HWV 9 or later	Y	Y	Y	Y	Y	Y
Dynamic DirectPath	• HWV 17 or later	Y	Y	Y	Y	Y	Y
Enhanced DirectPath	• HWV 20 or later	Y	Y	Y	N	N	N
Vendor Device Groups	• HWV 20 or later	Y	Y	Y	N	N	N
Intel Advanced Matrix eXtensions (AMX)	• HWV 20 or later	Y	Y	Y	N	N	N
AMD SEV-ES	• HWV 18 or later	Y	Y	Y	Y	Y	N
AMD SEV-SNP	• HWV 22 or later	Y	N	N	N	N	N
Intel TDX	• HWV 22 or later	Y	N	N	N	N	N
E2E 4KN support - Support for 4KN VMDK and 4KN local drives	• HWV 22 or later	Y	N	N	N	N	N
Virtual Hyperthreading	• HWV 20 or later	Y	Y	Y	N	N	N
Virtual I/O MMU for Intel hosts	• HWV 14 or later	Y	Y	Y	Y	Y	Y
Virtual I/O MMU for AMD hosts	• HWV 18 or later	Y	Y	Y	Y	Y	N
Virtual I/O MMU for Enhanced DirectPath	• HWV 22 or later	Y	N	N	N	N	N
Virtual SGX device	• HWV 17 or later	Y	Y	Y	Y	Y	Y
Nested virtualization (5)	• HWV 9 or later	Y	Y	Y	Y	Y	Y
Microsoft VBS for Intel hosts	• HWV 14 or later	Y	Y	Y	Y	Y	Y
Microsoft VBS for AMD hosts	• HWV 19 or later	Y	Y	Y	Y	N	N
Communication Interfaces							
Maximum Serial ports	• 32 for HWV 11 or later • 4 for HWV 4 - 10	32	32	32	32	32	32
Maximum Parallel ports	3 for all HWV	3	3	3	3	3	3
USB 2.0 (13)	• HWV 7 or later	Y	Y	Y	Y	Y	Y
USB 3.1 SuperSpeed (14)	• HWV 8 or later	Y	Y	Y	Y	Y	Y
USB 3.1 SuperSpeedPlus (14)	• HWV 17 or later	Y	Y	Y	Y	Y	Y
USB 3.2 (14)	• HWV 21 or later	Y	Y	N	N	N	N

Feature	Feature Availability by Virtual Hardware Version (HWV)		Recent Hardware Versions				
Maximum USB devices	• HWV 7 or later	20	20	20	20	20	20
VMCI	• HWV 7 or later	Y	Y	Y	Y	Y	Y
DataSets Service	• HWV 20 or later	Y	Y	Y	N	N	N
vMotion App Notification	• HWV 20 or later	Y	Y	Y	N	N	N
Networking							
Maximum NICs	• 10 for HWV 7 or later • 4 for HWV 4	10	10	10	10	10	10
VMXNet3 version	• 9 for HWV 22 or later • 8 for HWV 20, 21 • 7 for HWV 19 • 6 for HWV 18 • 5 for HWV 15 and 17 • 4 for HWV 14 • 3 for HWV 13 • 2 for HWV 11 • 1 for HWV 7 - 10	9	8	8	7	6	5
VMXNet2 version	• HWV 4 or later	Y	Y	Y	Y	Y	Y
VMXNet version	• HWV 4 or later	Y	Y	Y	Y	Y	Y
PCNet32 (6)	• HWV 4 or later	Y	Y	Y	Y	Y	Y
E1000e	• HWV 8 or later	Y	Y	Y	Y	Y	Y
E1000	• HWV 4 or later	Y	Y	Y	Y	Y	Y
PVRDMA RoCE versions	• RoCEv2 for HWV 14 or later • RoCEv1 for HWV 13-19	RoCEv2	RoCEv2	RoCEv2	RoCEv1, RoCEv2	RoCEv1, RoCEv2	RoCEv1, RoCEv2
Maximum PVRDMA NICs	• 10 for HWV 13 or later	10	10	10	10	1	1
PVRDMA native endpoints (7)	• HWV 18 or later	Y	Y	Y	Y	Y (7)	N
ATA Storage							
Maximum Floppy Devices	• 2 for all HWV	2	2	2	2	2	2
Maximum IDE controllers	• HWV 4 or later	1	1	1	1	1	1
Maximum Virtual IDE disks and Virtual IDE CD-ROMs in any combination	• HWV 4 or later	4	4	4	4	4	4
Maximum Virtual IDE CD-ROMs	• HWV 4 or later	Y	Y	Y	Y	Y	Y
Virtual IDE Disks	• HWV 7 or later	Y	Y	Y	Y	Y	Y
IDE hot add support	• Not supported for all HWV	N	N	N	N	N	N
Maximum disks per Virtual SATA controller	• HWV 10 or later	30	30	30	30	30	30

Feature	Feature Availability by Virtual Hardware Version (HWV)		Recent Hardware Versions				
Maximum SATA controllers	• 4 for HWV 10 or later	4	4	4	4	4	4
SCSI Storage							
Maximum Virtual SCSI disks	• HWV 4 or later	256	256	256	256	256	256
Maximum SCSI controllers	• 4 for HWV 4 or later (12)	4	4	4	4	4	4
Maximum disks per VMware Paravirtual SCSI controllers	• HWV 7 or later	64	64	64	64	64	64
Maximum disks per LSI Logic SAS SCSI controllers	• HWV 7 or later	15	15	15	15	15	15
Maximum disks per LSI Logic SCSI controllers	• HWV 4 or later	15	15	15	15	15	15
Maximum disks per Bus Logic SCSI controllers	• HWV 4 or later	15	15	15	15	15	15
SCSI passthrough support	• HWV 4 or later	Y	Y	Y	Y	Y	Y
SCSI hot add support	• HWV 4 or later	Y	Y	Y	Y	Y	Y
SCSI Read only support	• HWV 21 or later	Y	Y	N	N	N	N
NVMe Storage							
Maximum Virtual NVMe disks	• 256 for HWV 21 or later • 60 for HWV 13 - 20	256	256	60	60	60	60
Maximum NVMe controllers	• 4 for HWV 13 or later	4	4	4	4	4	4
Maximum NVMe disks per controller	• 64 for HWV 21 or later • 15 for HWV 13 - 20	64	64	15	15	15	15
NVMe version (8)	• 1.4 for HWV 22 or later • 1.3c for HWV 19 - 21 • 1.0e for HWV 13 - 18 • See notes (10) and (11)	1.4 (19)	1.3c (11)	1.3c (10)	1.3c (10)	1.0e	1.0e
Virtual Disk							
Maximum Virtual disk size	• HWV 13 or later	62 TB	62 TB	62 TB	62 TB	62 TB	62 TB
Maximum redo log depth	• all HWV	255	255	255	255	255	255
Snapshots							
Maximum snapshot levels	• all HWV	32 (16)	32 (16)	32 (16)	32 (16)	32 (16)	32 (16)
Maximum total snapshots	• all HWV	optional (17)	optional (17)	optional (17)	optional (17)	optional (17)	optional (17)

Feature	Feature Availability by Virtual Hardware Version (HWV)		Recent Hardware Versions				
Persistent Memory							
Virtual PMEM disks support	• HWV 14 or later	Y	Y	Y	Y	Y	Y
Maximum NVDIMM controllers	• 1 for HWV 14 or later	1	1	1	1	1	1
Maximum NVDIMM devices	• 64 for HWV 14 or later	64	64	64	64	64	64
Graphics							
Maximum vGPUs	• 16 for HWV 21 • 8 for HWV 20 • 4 for HWV 13 - 19	16	16	8	4	4	4
Maximum SVGA displays	• 10 for HWV 7 or later • 1 for HWV 4	10	10	10	10	10	10
Maximum video memory (MB)	• 256 for HWV 18 or later • 128 for HWV 11 - 17 • 512 for HWV 9 - 10 • 128 for HWV 4 - 8	256	256	256	256	256	128
Maximum 3D graphics memory (GB)	• 8 for HWV 18 or later • 4 for HWV 17 • 2 for HWV 11 - 15	8	8	8	8	8	4
SVGA 3D hardware acceleration	• HWV 8 or later	Y	Y	Y	Y	Y	Y
WDDM version	• 1.2 for HWV 20 or later • 1.1 for HWV 17 - 19 • 1.0 for HWV 7 - 15	1.2	1.2	1.2	1.1	1.1	1.1
Direct3D version (9)	• 11.0 for HWV 19 or later • 10.1 for HWV 17 - 18 • 10.0 for HWV 13 - 15 • 9.x for HWV 7 - 12	11.0 (15)	11.0 (15)	11.0 (15)	11.0 (15)	10.1	10.1
OpenGL version	• 4.3 for HWV 20 or later • 4.0 for HWV 19 • 3.3 for HWV 13 - 18	4.3 (15)	4.3 (15)	4.3 (15)	4.0 (15)	3.3	3.3
Other Virtual Devices							
Memory ballooning	• HWV 4 or later	Y	Y	Y	Y	Y	Y
Virtual TPM 2.0	• HWV 14 or later	Y	Y	Y	Y	Y	Y
Virtual Precision Clock device	• HWV 17 or later	Y	Y	Y	Y	Y	Y
Virtual Watchdog Timer device	• HWV 17 or later	Y	Y	Y	Y	Y	Y

- (1) - By default, for each combination of hardware version and guest operating system, vSphere will use either UEFI or BIOS. Some guest operating systems require either UEFI or BIOS firmware, or support both firmware types. If the guest operating system supports BIOS and UEFI, change the default firmware type of a virtual machine before you install the guest operating system. Otherwise, you must reinstall the guest operating system. For more information about the UEFI version used in VMware virtual machines, see Broadcom knowledge base article 313364.

- (2) - UEFI firmware is required for more than 128 logical processors.
- (3) - UEFI firmware is required for more than 64 virtual CPUs per sockets.
- (4) - UEFI firmware is required for more than 6 TB of virtual memory.
- (5) - For more information about the nested virtualization support limits, see Broadcom knowledge base article 313547.
- (6) - 32-bit virtual machines only.
- (7) - PVRDMA native endpoints for virtual machines of hardware version 18 do not support vSphere vMotion.
- (8) - By default, Windows virtual machines of hardware version 20 use NVMe version 1.0e.
- (9) - Direct3D requires a VMware WDDM driver in the Windows guest operating system.
- (10) - NVMe version 1.0 is the default NVMe controller for Windows virtual machines of hardware versions 19 and 20.
- (11) - NVMe 1.3c is the default NVMe controller for virtual machines of hardware version 21 with Windows Server 2022 or later and Windows 11 or later guest operating systems.
- (12) - The maximum number of SCSI controllers is 4 for any combination of PVSCI, LSI, or BusLogic controllers.
- (13) - When you select USB 2.0 from the vSphere user interface, two controllers are added to the virtual machine - UHCI for USB 1.x and EHCI for USB 2.0.
- (14) - Virtual machines can have only one USB 3.x controller and an additional USB 2.0 controller.
- (15) - To power on a virtual machine that is configured with Direct3D 11.0/OpenGL 4.0 and later, verify that the ESX host graphics hardware is available. Virtual machines that are configured with Direct3D 10.1/OpenGL 3.3 do not require an ESX host graphics adapter.
- (16) - Depends on the number of virtual disks of a virtual machine:
  - 32 if the number is either less than or equal to 184.
  - 17 if the number is more than 184 but either less than or equal to 256.
  - 3 if the number is more than 256.
- (17) - Maximum total number of snapshots for a virtual machine that can be set with the `snapshot.maxSnapshots` configuration file option. The default value is `-1` or `unenforced`.
- (18) - Synthetic Timers are only available for Windows 10 and later guest operating systems.
- (19) - Hardware version 22 or later supports NVMe 1.4.

Virtual CPU Configuration and Limitations

You can set most of the CPU parameters when you create virtual machines or after you install the guest operating system. Some actions require that you power off the virtual machine before you change the settings.

VMware uses the following terminology. Understanding these terms can help you plan your strategy for CPU resource allocation.

CPU	The CPU, or processor, is the component of a computer system that performs the tasks required for computer applications to run. The CPU is the primary element that performs the computer functions. CPUs contain cores.
CPU Socket	A CPU socket is a physical connector on a computer motherboard that connects to a single physical CPU. Some motherboards have multiple sockets and can connect multiple multicore processors (CPUs).
Core	A core contains a unit containing an L1 cache and functional units needed to run applications. Cores can independently run applications or threads. One or more cores can exist on a single CPU.
Resource sharing	Shares specify the relative priority or importance of a virtual machine or resource pool. If a virtual machine has twice as many shares of a resource as another virtual machine, it is entitled to consume twice as much of that resource when the two virtual machines are competing for resources.
Resource allocation	You can change CPU resource allocation settings, such as shares, reservation, and limit, when available resource capacity does not meet demands. For example, if at year end, the workload on accounting increases, you can increase the accounting resource pool reserve.
vSphere Virtual Symmetric Multiprocessing (Virtual SMP)	Virtual SMP or vSphere Virtual Symmetric Multiprocessing is a feature that enables a single virtual machine to have multiple processors.

Virtual CPU Limitations

The maximum number of virtual CPUs that you can assign to a virtual machine is 960. The number of virtual CPUs depends on the number of logical CPUs on the host and the type of guest operating system that is installed on the virtual machine.

Be aware of the following limitations:

- A virtual machine cannot have more virtual CPUs than the number of logical cores on the host. The number of logical cores is equal to the number of physical cores if hyperthreading is deactivated or two times that number if hyperthreading is activated.
- If a running virtual machine has 128 virtual CPUs or less, you cannot use hot adding to further increase the number of virtual CPUs. To change the number of virtual CPUs beyond that limit, you must first power off the virtual machine. By contrast, if a running virtual machine already has more than 128 virtual CPUs, you can use hot adding to further increase the number of virtual CPUs to up to 960.
- The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.
- Not every guest operating system supports Virtual SMP, and guest operating systems that support this functionality might support fewer processors than are available on the host. For information about Virtual SMP support, see the Broadcom Compatibility Guide.

Configuring Multicore Virtual CPUs

VMware multicore virtual CPU support lets you control the number of cores per virtual socket in a virtual machine. This capability lets operating systems with socket restrictions use more of the host CPU cores, which increases overall performance.

> **Important:**
> When you configure your virtual machine for multicore virtual CPU settings, you must ensure that your configuration complies with the requirements of the guest operating system EULA.

Using multicore virtual CPUs can be useful when you run operating systems or applications that can take advantage of only a limited number of CPU sockets.

You can configure a virtual machine with ESX 9.1 and later compatibility to have up to 960 virtual CPUs. A virtual machine cannot have more virtual CPUs than the actual number of logical CPUs on the host. The number of logical CPUs means the number of physical processor cores or two times that number if hyperthreading is activated. For example, if a host has 128 logical CPUs, you can configure the virtual machine for 128 virtual CPUs.

You configure how the virtual CPUs are assigned in terms of cores and cores per socket. Determine how many CPU cores you want in the virtual machine, then select the number of cores you want in each socket, depending on whether you want a single-core CPU, dual-core CPU, tri-core CPU, and so on. Your selection determines the number of sockets that the virtual machine has.

The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.

For more information about multicore CPUs, see the *vSphere Resource Management* documentation.

Configure Virtual Machine CPU Resources

To improve the virtual machine performance, you can add, change, or configure CPU resources.

Enable CPU Hot Add

By default, you cannot add CPU resources to a virtual machine when the virtual machine is powered on. The CPU hot add option lets you add CPU resources to a running virtual machine.

- Verify that the virtual machine is configured as follows.
  - Latest version of VMware Tools installed.

  - Guest operating system that supports CPU hot add.
  - Virtual machine compatibility is ESX 7.x or later.
  - Virtual machine is powered off.
- Required privileges: **Virtual Machine** > **Change Configuration** > **Change CPU Count**

The following conditions apply.

- For best results, use virtual machines that are compatible with ESX 7.0 or later.
- Hot adding multicore virtual CPUs is supported only with virtual machines that are compatible with ESX 7.0 or later.
- If a virtual machine has 128 virtual CPUs or less, you cannot use hot adding to further increase the number of virtual CPUs to be over 128. To change the number of virtual CPUs beyond that limit, you must first power off the virtual machine. By contrast, if a virtual machine already has more than 128 virtual CPUs, you can use hot adding to further increase the number of virtual CPUs to up to 960.
- Not all guest operating systems support CPU hot add. You can deactivate these settings if the guest operating system is not supported.
- To use the CPU hot add feature with virtual machines that are compatible with ESX 7.x and later, set the **Number of cores per socket** to 1.
- Adding CPU resources to a running virtual machine with CPU hot add enabled disconnects and reconnects all USB passthrough devices that are connected to that virtual machine.

  Note:
  If the ESX host is of version 7.0 Update 2 and earlier, hot-adding virtual CPUs to a virtual machine with NVIDIA vGPU requires that the ESX host have a free vGPU slot. Starting with vSphere 7.0 Update 3, the source host does not require to have a free vGPU slot.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **VM Options** tab, expand **CPU Topology**, and select **Enable CPU Hot Add**.
3. Click **OK**.

You can now add CPUs even if the virtual machine is powered on.

Change the Number of Virtual CPUs

A virtual machine with ESX 9.1 and later compatibility can have up to 960 virtual CPUs. You can change the number of virtual CPUs while your virtual machine is powered off. If virtual CPU hot add is enabled, you can increase the number of virtual CPUs while the virtual machine is running.

- If CPU hot add is not enabled, power off the virtual machine before adding virtual CPUs.
- To hot add multicore CPUs, verify that the virtual machine is compatible with ESX 7.0 and later.
- Verify that you have the following privileges:
  - **Virtual machine** > **Change Configuration** > **Change CPU Count**
  - **Virtual machine** > **Change Configuration** > **Change resource**
  - **Virtual machine** > **Change Configuration** > **Change Settings**

Virtual CPU hot add is supported for virtual machines with multicore CPU support and ESX 6.5 and later compatibility. When the virtual machine is powered on and CPU hot add is enabled, you can hot add virtual CPUs to the running virtual machine. You can add only multiples of the number of cores per socket.

If a virtual machine has 128 virtual CPUs or less, you cannot use hot adding to further increase the number of virtual CPUs. To change the number of virtual CPUs beyond that limit, you must first power off the virtual machine. By contrast, if a virtual machine already has more than 128 virtual CPUs, you can use hot adding to further increase the number of virtual CPUs to up to 960.

The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.

Important:
When you configure your virtual machine for multicore virtual CPU settings, you must ensure that your configuration complies with the requirements of the guest operating system EULA.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **CPU**.
4. From the **CPU** drop-down menu, select the number of cores and click **OK**.

Expose VMware Hardware Assisted Virtualization

You can expose full CPU virtualization to the guest operating system so that applications that require hardware virtualization can run on virtual machines without binary translation or paravirtualization.

Note:
VMware does not support running third-party hypervisors on ESX with the exception of specifically defined and documented functionality. Currently, this is limited to Microsoft Hyper-V, strictly for VBS only and not for virtualizing multiple VMs.

For more information, see Broadcom knowledge base article 313547.

- Verify that the virtual machine compatibility is ESX 7.0 and later.
- Intel Nehalem Generation (Xeon Core i7) or later processors or AMD Opteron Generation 3 (Greyhound) or later processors.
- Verify that Intel VT-x or AMD-V is enabled in the BIOS so that hardware-assisted virtualization is possible.
- Required Privileges: **Virtual machine** > **Change Configuration** > **Change Settings** set on the vCenter system.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **CPU**, and select **Expose hardware assisted virtualization to the guest OS**.
3. Click **OK**.
   The **Configure** tab refreshes, and the Nested Hypervisor CPU option shows **Enabled.**

Activate Virtual CPU Performance Counters

You can use performance tuning tools in the guest operating system for software profiling. You can identify and improve processor performance problems. This capability is useful for software developers who optimize or debug software that runs in the virtual machine.

- Verify that the virtual machine compatibility is ESX 7.0 and later.
- Verify that the virtual machine is turned off.
- Verify that Intel Nehalem Generation (Xeon Core i7) or later processors or AMD Opteron Generation 3 ("Greyhound") or later processors are installed.
- Verify that Intel VT-x or AMD-V is enabled in the BIOS so that hardware-assisted virtualization is possible.
- Required Privileges:**Virtual machine** > **Change Configuration** > **Change Settings** is set on the vCenter system.

The following conditions apply:

- If virtual CPU performance counters are enabled, you can migrate the virtual machine only to hosts that have compatible CPU performance counters.
- If an ESX host's BIOS uses a performance counter, or if Fault Tolerance is enabled, some virtual performance counters might not be available for the virtual machine to use.

Note:
If a virtual machine resides on an ESX host in an EVC cluster, CPU counters are not supported for virtual machine creation or editing. You must deactivate CPU performance counters.

For a list of virtualized Model-Specific Registers (MSRs), see Broadcom knowledge base article 344161.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **CPU** and select the **Enable virtualized CPU performance counters** check-box.
3. Click **OK**.

Configure Processor Scheduling Affinity

The **Scheduling Affinity** option gives you detailed control over how virtual machine CPUs are distributed across the host's physical cores. The option supports hyperthreading if hyperthreading is enabled. ESX generally manages processor scheduling well, even when hyperthreading is enabled. These settings are useful only for fine-tuning critical virtual machines.

- Verify that the virtual machine is powered off.
- Verify that the virtual machine does not reside in a DRS cluster.
- Verify that the host has more than one physical processor core.
- Privileges: **Virtual machine** > **Change Configuration** > **Change resource** and **Virtual machine** > **Change Configuration** > **Change Settings**

Using CPU affinity, you can assign a virtual machine to a specific processor. This assignment allows you to restrict the assignment of virtual machines to a specific available processor in multiprocessor systems.

This setting does not appear for virtual machines in a DRS cluster or when the host has only one processor core and no hyperthreading.

For potential issues with CPU affinity, see the *vSphere Resource Management* documentation.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **CPU**, and enter a comma-separated list of hyphenated processor ranges in the **Scheduling Affinity** text box.
   For example, "0,4-7" would indicate affinity with CPUs 0, 4, 5, 6, and 7. Selecting all processors is identical to selecting no affinity. You must provide at least as many processor affinities as you have virtual CPUs.
3. Click **OK**.

Activate IO/MMU Virtualization Settings

ESX can determine whether a virtual machine needs hardware support for virtualization. ESX makes this determination based on the processor type and the virtual machine. Overriding the automatic selection can provide better performance for some use cases.

> **Important:**
> Modern x86 processors can fully support virtualized workloads without software assistance. So, the CPU/MMU Virtualization setting has been deprecated in ESX 6.7 and later. The CPU/MMU Virtualization setting is available only for virtual machines with ESX 6.5 and earlier compatibility.

You can use software MMU when your virtual machine runs heavy workloads, such as Translation Lookaside Buffers (TLBs) intensive workloads that have significant impact on the overall system performance. However, software MMU has a higher overhead memory requirement than hardware MMU. So, to support software MMU, the maximum overhead supported for virtual machine limit in the VMkernel must be increased.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **CPU**, and select the **Enabled** check box to activate the I/O MMU options for a virtual machine.

   > **Note:**
   > To take advantage of all features that virtual hardware version 13 provides, use the default hardware MMU setting.

3. Click **OK**.

Virtual Memory Configuration

You can add, change, or configure virtual machine memory resources or options to improve virtual machine performance. You can set most of the memory parameters during virtual machine creation or after the guest operating system is installed.

Some actions require that you power off the virtual machine before changing the settings.

The memory resource settings for a virtual machine determine how much of the host's memory is allocated to the virtual machine. The virtual hardware memory size determines how much memory is available to applications that run in the virtual machine. A virtual machine cannot benefit from more memory resources than its configured virtual hardware memory size. ESX hosts limit the memory resource use to the maximum amount useful for the virtual machine, so that you can accept the default of Unlimited memory resources.

Change the Memory Configuration

You can reconfigure the amount of memory allocated to a virtual machine to improve performance.

Verify that you have the following privileges on the virtual machine:

- **Virtual machine** > **Change Configuration** > **Change Memory**
- **Virtual machine** > **Change Configuration** > **Change resource**

Minimum memory size is 4 MB for virtual machines that use BIOS firmware. Virtual machines that use EFI firmware require at least 96 MB of RAM or they cannot power on.

Maximum memory size for virtual machines that use BIOS firmware is 24560 GB. You must use EFI firmware for virtual machines with memory size greater than 6128 GB.

Maximum memory size for a virtual machine depends on the physical memory of the ESX host and the virtual machine compatibility settings.

If the virtual machine memory is greater than the host memory size, swapping occurs, which can have a severe effect on virtual machine performance. The maximum for best performance represents the threshold above which the physical memory of the ESX host is insufficient to run the virtual machine at full speed. This value fluctuates as conditions on the host change, for example, as virtual machines are powered on or off.

The memory size must be a multiple of 4 MB.

Table 127: Maximum Virtual Machine Memory

Introduced in Host Version	Virtual Machine Compatibility	Maximum Memory Size
ESX 9.0	ESX 9.0 and later	24560 GB
ESX 8.0 Update 3	ESX 8.0 Update 3 and later	24560 GB
ESX 8.0 Update 2	ESX 8.0 Update 2 and later	24560 GB
ESX 8.0 Update 1	ESX 8.0 Update 1 and later	24560 GB
ESX 8.0	ESX 8.0 and later	24560 GB
ESX 7.0 Update 3	ESX 7.0 Update 3 and later	24560 GB
ESX 7.0 Update 2	ESX 7.0 Update 2 and later	24560 GB
ESX 7.0 Update 1	ESX 7.0 Update 1 and later	24560 GB
ESX 7.0	ESX 7.0 and later	6128 GB
ESX 6.7 Update 2	ESX 6.7 Update 2 and later	6128 GB
ESX 6.7	ESX 6.7 and later	6128 GB
ESX 6.5	ESX 6.5 and later	6128 GB

The ESX host version indicates when support began for the increased memory size. For example, the memory size of a virtual machine with ESX 6.0 and later compatibility running on ESX 6.5 is restricted to 4080 GB.

1. Click **Virtual Machines** in the inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Memory** drop-down menu.
   a) From the **Memory** drop-down menu, select the amount of RAM to assign to the virtual machine.
   b) Select whether the memory is specified in MB, GB, or TB.
4. Click **OK**.

Allocate Memory Resources to a Virtual Machine

You can change the amount of memory resources that are allocated to a virtual machine by using the shares, reservations, and limits settings. The host determines the appropriate amount of physical RAM to allocate to virtual machines based on these settings. You can assign a high or low shares value to a virtual machine, depending on its load and status.

Power off the virtual machine.

The following user-defined settings affect the memory resource allocation of a virtual machine.

Limit	Places a limit on the consumption of memory for a virtual machine. This value is expressed in megabytes.
Reservation	Specifies the guaranteed minimum allocation for a virtual machine. The reservation is expressed in megabytes. If the reservation cannot be met, the virtual machine will not power on.
Shares	Each virtual machine is granted a number of memory shares. The more shares a virtual machine has, the greater share of host memory it receives. Shares represent a relative metric for allocating memory capacity. For more information about share values, see the *vSphere Resource Management* documentation.

You cannot assign a reservation to a virtual machine that is larger than the virtual machine configured memory. If you give a virtual machine a large reservation and reduce the virtual machine configured memory size, the reservation is reduced to match the new configured memory size.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand Memory and allocate the memory capacity for the virtual machine.

Option	Description
Option	Description
Reservation	Guaranteed memory allocation for this virtual machine.
Limit	Upper limit for this virtual machine’s memory allocation. Select Unlimited to specify no upper limit.
Shares	The values Low, Normal, High, and Custom are compared to the sum of all shares of all virtual machines on the server.

3. Click **OK**.

Manage Persistent Memory

ESX 6.7 and later provide support for the latest computer memory technology, which is called non-volatile memory (NVM) or persistent memory (PMem).

PMem combines the high data transfer rate of volatile computer memory with the persistence and resiliency of traditional storage. PMem devices have low access latency and can retain stored data through reboots or power outages.

Modes of Consumption of the Persistent Memory Resources of the Host

When you add a physical PMem device to a host, ESX detects the PMem resource and exposes it as a host-local PMem datastore to the virtual machines that run on the host. Depending on the guest operating system, virtual machines can have direct access to the PMem resources.

Each host can have only one local PMem datastore that pools and represents all PMem resources of the host.

Persistent memory combines the properties of both memory and storage. So, virtual machines can consume the PMem resources of the ESX host as memory (through virtual NVDIMM devices) or as storage (through virtual PMem hard disks).

The host-local PMem datastore stores all direct-accessed NVDIMM devices and virtual PMem hard disks.

Virtual PMem (vPMem)

In this mode, if the guest operating system is PMem-aware, the virtual machine can have direct access to the physical PMem resources of the host and use them as standard, byte-addressable memory.

Virtual machines use virtual non-volatile dual in-line memory modules (NVDIMMs) for direct access to PMem. The NVDIMM is a memory device that sits on an ordinary memory channel, but contains non-volatile memory. In vSphere 9.0, the virtual NVDIMM is a new type of device that represents the physical PMem regions of the host. A single virtual machine can have up to 64 virtual NVDIMM devices. Each NVDIMM device is stored on the host-local PMem datastore.

> **Note:**
> To add an NVDIMM device to a virtual machine, the virtual machine must be of hardware version 14 and the guest operating system must support persistent memory. If the guest operating system is not PMem-aware, you can still use PMem, but you cannot add an NVDIMM device to the virtual machine.

Virtual PMem Disks (vPMemDisk)

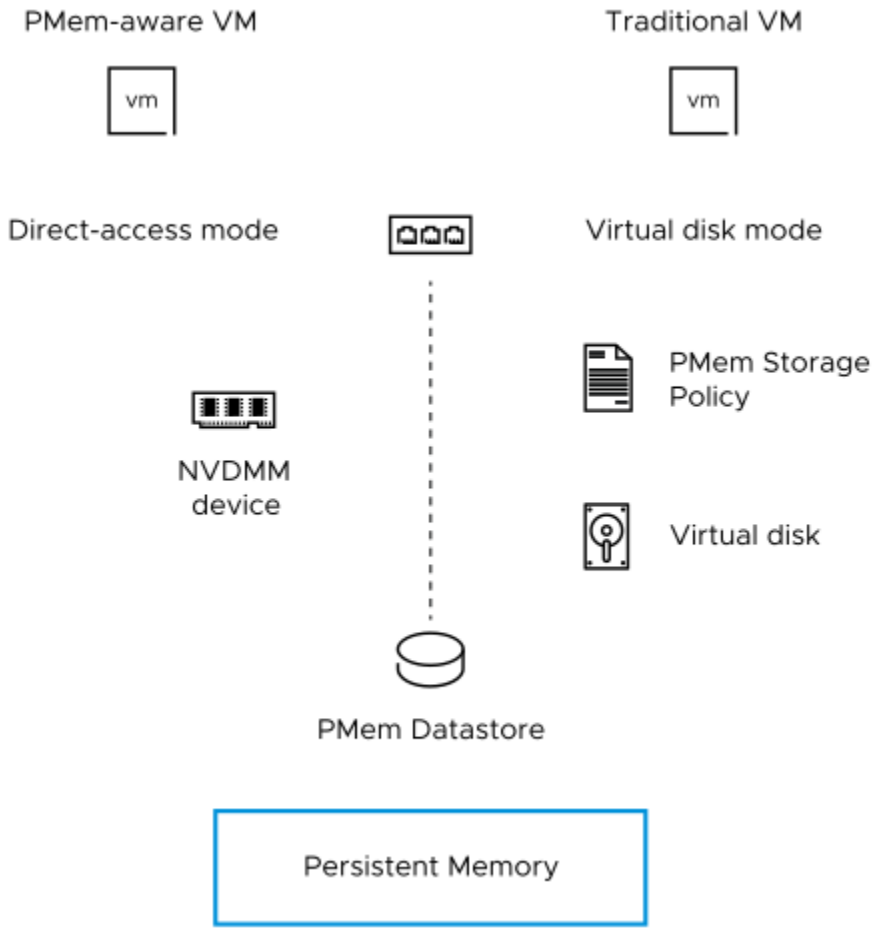
In this mode, the virtual machine does not have direct access to the PMem resources of the host.

You must add a virtual PMem hard disk to the virtual machine. A virtual PMem hard disk is a traditional SCSI disk to which the PMem Storage Policy is applied. The policy automatically places the hard disk on the host-local PMem datastore.

In this mode of usage, there are no requirements for the hardware version of the virtual machine and the guest operating system.

> **Note:**
> If the guest operating system is not PMem-aware, virtual machines can use PMem only through vPMemDisks.

The following diagram illustrates how the persistent memory components interact.



For information about how to configure and manage VMs with NVDIMMs or virtual persistent memory disks, see the *vSphere Resource Management* documentation.

Add an NVDIMM Device to a Virtual Machine

Add a virtual NVDIMM device to a virtual machine to enable it to use non-volatile, or persistent, computer memory. Non-volatile memory (NVM), or persistent memory (PMem), combines the high data transfer rates of the volatile memory with the persistence and resiliency of traditional storage. The virtual NVDIMM device is a virtual NVM device that can retain stored data through reboots or power source failures.

- Power off the virtual machine.
- Verify that the guest operating system of the virtual machine supports PMem.
- Verify that you have the **Datastore** > **Allocate space** privilege on the virtual machine.
- Verify that the host or the cluster on which the virtual machine resides has available PMem resources.
- To add an NVDIMM device to a virtual machine, verify that the virtual machine hardware is of version 14 or higher.
- To configure vSphere HA for PMem virtual machines, verify that the virtual machine is of version 19 or later, and vSphere HA is enabled on the cluster.

If an ESX host fails or the datastore is no longer accessible, when you add an NVDIMM device to a virtual machine, you can configure vSphere HA to failover all PMem virtual machines on another host.

> **Note:**
> If a host fails, NVDIMM PMem data cannot be restored. vSphere HA will restart the virtual machine on another host with a new and empty NVDIMM of the same size.

When you add an NVDIMM device to a virtual machine, you can take a snapshot of the virtual machine by configuring the NVDIMM snapshot mode of the virtual machine in the independent erase on revert mode or in the independent persistent mode.

NVDIMM Snapshot Conditions and Limitations	• You cannot configure the NVDIMM snapshot mode when the virtual machine has existing snapshots. • You cannot add another NVDIMM device to the virtual machine if the NVDIMM snapshot mode is independent persistent or independent erase on revert, and the virtual machine has existing snapshots. • You cannot add an NVDIMM device to a virtual machine, remove an NVDIMM device from a virtual machine, or change the NVDIMM device parameters if the virtual machine has snapshots. To allow these operations, remove all virtual machine snapshots. • You can use vMotion to migrate virtual machines that have snapshots containing NVDIMM devices and revert these snapshots.

For more information, the *vSphere Resource Management* documentation.

1. Add an NVDIMM device to a virtual machine when you deploy a virtual machine or when you edit an existing virtual machine.

Option	Action
Create a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine and select **New Virtual Machine**. 2. On the **Select a creation type page**, select **Create a new virtual machine**, and click **Next**. 3. Navigate through the pages of the wizard. 4. On the **Customize hardware** page, click the **Virtual Hardware** tab. 5. On the **Virtual Hardware** tab, click the **Add New Device** button. 6. From the drop-down menu, select **NVDIMM**.
Edit a virtual machine	1. Right-click a virtual machine in the inventory and select **Edit Settings**. 2. Click the **Virtual Hardware** tab. 3. On the **Virtual Hardware** tab, click the **Add New Device** button. 4. From the drop-down menu, select **NVDIMM**.

   The NVDIMM device appears in the Virtual Hardware devices list together with the virtual NVDIMM controller. Each virtual machine can have a maximum of one virtual NVDIMM controller and each NVDIMM controller can have up to 64 virtual NVDIMM devices.

   Note:
   You can change the size of the NVDIMM device at a later time. The virtual machine must be powered off.

2. In the **New NVDIMM** text box, enter the size of the NVDIMM device and select the units from the drop-down menu.

   Note:
   Adjust the size of the newly added devices in accordance with the amount of the PMem available to the host. If any part of the configuration requires attention, the wizard alerts you.

3. Expand the **New NVDIMM device** section and select the **Allow failover on another host for all NVDIMM devices** check box.

   Note:
   When you allow the failover process, if you add another NVDIMM device to the virtual machine, the NVDIMM device will have **PMem HA** enabled. If you want to preserve the NVDIMM content of the virtual machine during a host failure, make sure to deselect the **Allow failover on another host for all NVDIMM devices** check box.

4. From the **Snapshot Mode**, configure the NVDIMM snapshot mode.

Option	Description
Persistent	Persistent mode is the default NVDIMM snapshot mode. Taking snapshots of virtual machines with an NVDIMM device in persistent mode is not supported.
Independent - Persistent	Taking snapshots of virtual machines with an NVDIMM device in independent-persistent mode is supported. When you revert

Option	Description
	to the previous snapshot, the current contents of the NVDIMM device is preserved.
Independent - Erase on revert	Taking snapshots of virtual machines with an NVDIMM device in independent-erase on revert mode is supported. When you revert to the previous snapshot, the content of the NVDIMM device is erased.

5. If you deploy a virtual machine, click **Next**.
6. If you edit an existing virtual machine, click **OK**.

When you power on the virtual machine, you can view the vSphere HA Protection status of the virtual machine in the **vSphere HA** panel on the **Summary** tab.

Change Memory Hot Add Settings

Memory hot add lets you add memory resources for a virtual machine while that virtual machine is powered on.

- Power off the virtual machine.
- Verify that the virtual machine has a guest operating system that supports memory hot add functionality.
- Verify that the virtual machine compatibility is ESX 6.x and later.
- Verify that VMware Tools is installed.

Enabling memory hot add produces some memory overhead on the ESX host for the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **Memory**, and select the **Memory Hot Plug** check box to activate adding memory to the virtual machine while it is powered on.
4. Click **OK**.

You can now add memory to a virtual machine, even if the virtual machine is powered on.

Virtual Disk Configuration

You can add large-capacity virtual disks to virtual machines and add more space to existing disks, even when the virtual machine is running.

You can set most of the virtual disk parameters during virtual machine creation or after you install the guest operating system.

You can store virtual machine data in a new virtual disk, an existing virtual disk, or a mapped SAN LUN. A virtual disk appears as a single hard disk to the guest operating system. The virtual disk is composed of one or more files on the host file system. You can copy or move virtual disks on the same hosts or between hosts.

For virtual machines running on an ESX host, you can store virtual machine data directly on a SAN LUN instead of using a virtual disk file. This option is useful if in your virtual machines you run applications that must detect the physical characteristics of the storage device. Mapping a SAN LUN also allows you to use existing SAN commands to manage storage for the disk.

When you map a LUN to a VMFS volume, vCenter or the ESX host creates a raw device mapping (RDM) file that points to the raw LUN. Encapsulating disk information in a file allows vCenter or the ESX host to lock the LUN so that only one virtual machine can write to it. This file has a `.vmdk` extension, but the file contains only disk information that describes the mapping to the LUN on the ESX system. The actual data is stored on the LUN. You cannot deploy a virtual machine from a template and store its data on a LUN. You can store its data only in a virtual disk file.

The amount of free space in the datastore is always changing. Ensure that you leave sufficient space for virtual machine creation and other virtual machine operations, such as growth of sparse files, snapshots, and so on. To review space utilization for the datastore by file type, see the *vSphere Monitoring and Performance* documentation.

Thin provisioning lets you create sparse files with blocks that are allocated upon first access, which allows the datastore to be over-provisioned. The sparse files can continue growing and fill the datastore. If the datastore runs out of disk space while the virtual machine is running, it can cause the virtual machine to stop functioning.

Large Capacity Virtual Disk Conditions and Limitations

Virtual machines with large capacity virtual hard disks, or disks greater than 2 TB, must meet resource and configuration requirements for optimal virtual machine performance.

The maximum value for large capacity hard disks is 62 TB. When you add or configure virtual disks, always leave a small amount of overhead. Some virtual machine tasks can quickly consume large amounts of disk space, which can prevent successful completion of the task if the maximum disk space is assigned to the disk. Such events might include taking snapshots or using linked clones. These operations cannot finish when the maximum amount of disk space is allocated. Also, operations such as snapshot quiesce, cloning, Storage vMotion, or vMotion in environments without shared storage, can take significantly longer to finish.

Virtual machines with large capacity disks have the following conditions and limitations:

- The guest operating system must support large capacity virtual hard disks.
- You can move or clone disks that are greater than 2 TB to ESX 8.0 or later hosts or to clusters that have such hosts available.
- The datastore format must be one of the following:
  - VMFS5 or later
  - An NFS volume on a Network Attached Storage (NAS) server
  - vSAN
- Fault Tolerance is not supported.
- BusLogic Parallel controllers are not supported.

Change the Virtual Disk Configuration

If you run out of disk space, you can increase the size of the disk. You can change the virtual device node and the persistence mode for virtual disk configuration for a virtual machine.

If you change the disk mode to **Independent - Nonpersistent** or from **Independent - Nonpersistent**, power off the virtual machine.

Verify that you have the following privileges:

- **Virtual machine** > **Change Configuration** > **Modify device settings** on the virtual machine.
- **Virtual machine** > **Change Configuration** > **Extend virtual disk** on the virtual machine.
- **Datastore** > **Allocate space** on the datastore.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Hard disk** to view or change the disk settings, and click **OK**.

Option	Description
Maximum Size	Shows the maximum size of this hard disk on this VM. **Note:** Extending the size of a virtual hard disk causes stun time for the virtual machine. The stun time is longer if the virtual disk is of the Eager Zeroed Thick type.
VM storage policy	Select one of the available storage policies. For more information, see the *vSphere Storage* documentation. Alternatively, to change the VM storage policy: • Right-click the virtual machine and select **VM Policies** > **Edit VM Storage Policies**. • From the **Hard disk 1** drop-down menu, select a new policy and click **OK**.

Option	Description
	Note: You cannot change the VM storage policy of an existing PMem hard disk. You also cannot change the storage policy of an existing non-PMem disk to Host-local PMem Default Storage Policy.
Type	Shows the storage type. You cannot change this setting for an existing hard disk. You choose the storage type of a hard disk when you add the hard disk to the virtual machine. For more information about storage types and available disk formats, see the *vSphere Storage* documentation.
Sector Size	The sector size of the hard disk.
Sharing	Specifies sharing information.
Disk File	Lists disk files on the datastore.
Disk Mode	Disk mode determines how a virtual disk is affected by snapshots. You have the following options: • **Dependent**: Dependent mode is the default disk mode. When you take a snapshot of a virtual machine, dependent disks are included in snapshots. When you revert to the previous snapshot, all data are reverted to the point of taking a snapshot. • **Independent - Persistent**: Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode are written permanently to the disk even if you revert a snapshot. When you power off or reset a virtual machine, the disk and all its snapshots are preserved. • **Independent - Nonpersistent**: Disks in nonpersistent mode behave like read-only disks. Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset a virtual machine, or you delete a snapshot.
Virtual Device Node	Displays the virtual device node.

Use Disk Shares to Prioritize Virtual Machines

If multiple virtual machines access the same VMFS datastore and the same logical unit number (LUN), use disk shares to prioritize the disk accesses from the virtual machines. Disk shares distinguish high-priority from low-priority virtual machines.

You can allocate the host disk's I/O bandwidth to the virtual hard disks of a virtual machine. Disk I/O is a host-centric resource so you cannot pool it across a cluster.

Shares is a value that represents the relative metric for controlling disk bandwidth to all virtual machines. The values are compared to the sum of all shares of all virtual machines on the server.

Disk shares are relevant only within a given host. The shares assigned to virtual machines on one host have no effect on virtual machines on other hosts.

You can select an IOP limit, which sets an upper bound for storage resources that are allocated to a virtual machine. IOPs are the number of I/O operations per second.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Hard disk** to view the disk options.
3. From the **Shares** drop-down menu, select a value for the shares to allocate to the virtual machine.
4. Click **OK**.

Determine the Virtual Disk Format and Convert a Virtual Disk from Thin Provision to a Thick Provision Format

When the disk space is exhausted and a thin-provisioned disk cannot expand, the virtual machine cannot boot. If you created a virtual disk in the thin provision format, you can convert it to the thick provision format.

The thin provisioned disk starts small and at first, uses just as much storage space as it needs for its initial operations. After you convert the disk, it grows to its full capacity and occupies the entire datastore space provisioned to it during the disk's creation.

For more information about thin provisioning and available disk formats, see the vSphere Storage documentation.

> **Note:**
>
> You cannot create thick provisioned disks on NFS datastores. For more information, see Broadcom knowledge base article 340272.

1. Verify that the disk format of a virtual hard disk is Thin Provision.
   a) Right-click a virtual machine and click **Edit Settings**.
   b) On the **Virtual Hardware** tab, expand **Hard disk** and check the **Type** field.
   c) To exit the wizard, click **Cancel**.
2. To open the datastore management panel, click the **Datastores** tab, and click a datastore from the list.
   The datastore that stores the virtual machine files is listed.
3. Click the **Files** tab, and open the virtual machine folder.
4. Browse to the virtual disk file that you want to convert.
   The file has the `.vmdk` extension.
5. To convert the virtual disk to a thick provision format, select the virtual disk file and click the **Inflate**.
   The **Inflate** window appears.
6. Click **OK**.

The inflated virtual disk occupies the entire datastore space originally provisioned to it.

> **Note:**
> The **Inflate** option might not be available if the virtual disk is thick or when the virtual machine is powered on.

How do I Add a Hard Disk to a Virtual Machine

When you create a virtual machine, a default virtual hard disk is added. You can add another hard disk if you run out of disk space, if you want to add a boot disk, or for other file management purposes. When you add a hard disk to a virtual machine, you can create a virtual disk, add an existing virtual disk, or add a mapped SAN LUN.

When you add a hard disk to a virtual machine, you can create a virtual disk, add an existing virtual disk, or add a mapped SAN LUN.

You can add a virtual hard disk to a virtual machine before or after you add a SCSI or SATA storage controller. The new disk is assigned to the first available virtual device node on the default controller, for example (0:1). Only device nodes for the default controller are available unless you add additional controllers.

The following ways to add disks can help you plan your disk configuration. These approaches show how you can optimize controller and virtual device nodes for different disks. For storage controller limitations, maximums, and virtual device node behavior, see SCSI, SATA, and NVMe Storage Controller Conditions, Limitations, and Compatibility.

Add an existing hard disk that is configured as a boot disk during virtual machine creation.	To ensure that the virtual machine can boot, remove the existing disk before you add the boot disk. After you add a new hard disk to the virtual machine, you might need to go into the BIOS setup to ensure that the disk you were using to boot the virtual machine is still selected as the boot disk. You can avoid this problem by not mixing adapter types, and by using device node 0 on the first adapter as the boot disk.
Keep the default boot disk and add a new disk during virtual machine creation.	The new disk is assigned to the next available virtual device node, for example (0:1) You can add a new controller and assign the disk to a virtual device node on that controller, for example (1:0) or (1:1).
Add multiple hard disks to an existing virtual machine.	If you add multiple hard disks to a virtual machine, you can assign them to several SCSI or SATA controllers to improve performance. The controller must be available before you can select a virtual device node. For example, if you add controllers 1, 2, and 3, and add four hard disks, you might assign the fourth disk to virtual device node (3:1).

Add a New Hard Disk to a Virtual Machine

You can add a virtual hard disk to an existing virtual machine, or you can add a hard disk when you customize the virtual machine hardware during the virtual machine creation process. For example, you might need to provide additional disk space for an existing virtual machine with a heavy work load. During virtual machine creation, you might want to add a hard disk that is preconfigured as a boot disk.

- Ensure that you are familiar with the configuration options and caveats for adding virtual hard disks. See Virtual Disk Configuration.
- Before you add disks greater than 2 TB to a virtual machine, see Large Capacity Virtual Disk Conditions and Limitations.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add new disk** privilege on the destination folder or datastore.
- Verify that you have the **Datastore** > **Allocate space** privilege.

During virtual machine creation, a hard disk and a SCSI or SATA controller are added to the virtual machine by default, based on the guest operating system that you select. If this disk does not meet your needs, you can remove it and add a new hard disk at the end of the creation process.

If you add multiple hard disks to a virtual machine, you can assign them to several controllers to improve performance. For controller and bus node behavior, see SCSI, SATA, and NVMe Storage Controller Conditions, Limitations, and Compatibility.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **Hard Disk** from the drop-down menu.

   The hard disk appears in the Virtual Hardware devices list.

   Note:
   If the host where the virtual machine resides has available PMem resources, you can place the new hard drive on the host-local PMem datastore.

4. Expand **New hard disk** and customize the settings of the new hard disk.
   a) Enter a size for the hard disk and select the unit from the drop-down menu.
   b) From the **VM storage policy**, select a storage policy or leave the default one.
   c) From the **Location** drop-down menu, select the datastore location where you want to store virtual machine files.
   d) From the **Disk Provisioning** drop-down menu, select the format for the hard disk.

Option	Action
Same format as source	Use the same format as the source virtual machine.
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation, but is zeroed out on demand at a later time on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out during creation. It might take longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin provisioned format. At first, a thin provisioned disk uses only as much datastore space as the disk initially needs. If the thin disk needs more space later, it can grow to the maximum capacity allocated to it.

   e) From the **Sector Size** drop-down menu, select the virtual disk sector size.
   f) From the **Sharing** drop-down menu, select a value for the shares to allocate to the virtual disk.

      Shares is a value that represents the relative metric for controlling disk bandwidth. The values Low, Normal, High, and Custom are compared to the sum of all shares of all virtual machines on the host.
   g) From the **Disk Mode** drop-down menu, select a disk mode.

Option	Description
Dependent	Dependent disks are included in snapshots.
Independent - Persistent	Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode are written permanently to the disk even if you revert a snapshot. When you power off or reset a virtual machine, the disk and all its snapshots are preserved.
Independent - Nonpersistent	Disks in nonpersistent mode behave like read-only disks. Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset the virtual machine, or when you delete a snapshot.

   h) From the **Virtual Device Node**, select a virtual device node or leave the default one.

      In most cases, you can accept the default device node. For a hard disk, a nondefault device node is useful to control the boot order or to have different SCSI controller types. For example, you might want to boot from an LSI Logic controller and share a data disk with another virtual machine that is using a BusLogic controller with bus sharing turned on.

Add an Existing Hard Disk to a Virtual Machine

You can add an existing virtual hard disk to a virtual machine when you customize the virtual machine hardware during the virtual machine creation process or after the virtual machine is created. For example, you might want to add an existing hard disk that is preconfigured as a boot disk.

- Make sure that you are familiar with controller and virtual device node behavior for different virtual hard disk configurations. See How do I Add a Hard Disk to a Virtual Machine.
- Before you add disks greater than 2 TB to a virtual machine, see Large Capacity Virtual Disk Conditions and Limitations.
- Verify that you have the following privileges:
  - **Virtual machine** > **Change Configuration** > **Add existing disk** privilege on the destination folder or datastore.
  - **Virtual machine** > **Change Configuration** > **Add or remove device**
  - **Datastore** > **Browse datastore**

During virtual machine creation, a hard disk and a SCSI or SATA controller are added to the virtual machine by default, based on the guest operating system that you select. If this disk does not meet your needs, you can remove it and add an existing hard disk at the end of the creation process.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Optional: To delete the existing hard disk, move your pointer over the disk and click the **Remove** icon.
   The disk is removed from the virtual machine. If other virtual machines share the disk, the disk files are not deleted.
3. On the **Virtual Hardware** tab, click the **Add New Device** button.
4. Select **Existing Hard Disk** from the drop-down menu.
   The **Select File** dialog box opens.
5. In the **Select File**, expand a datastore, select a virtual machine folder, and select the disk to add.
6. Click **OK**.
   The disk file appears in the **Contents** column. The **File Type** drop-down menu shows the compatibility file types for this disk.
7. Optional: Expand **New Hard disk** and make further customizations for the hard disk.
8. Click **OK**.

Add an RDM Disk to a Virtual Machine

You can use a raw device mapping (RDM) to store virtual machine data directly on a SAN LUN, instead of storing it in a virtual disk file. You can add an RDM disk to an existing virtual machine, or you can add the disk when you customize the virtual machine hardware during the virtual machine creation process.

- Ensure that you are familiar with SCSI controller and virtual device node behavior for different virtual hard disk configurations. See How do I Add a Hard Disk to a Virtual Machine.
- Before you add disks greater than 2 TB to a virtual machine, see Large Capacity Virtual Disk Conditions and Limitations.
- Required privilege:**Virtual machine** > **Configuration** > **Configure Raw device** and **Virtual machine** > **Change Configuration** > **Add new disk** on the virtual machine.

When you give a virtual machine direct access to an RDM disk, you create a mapping file that resides on a VMFS datastore and points to the LUN. Although the mapping file has the same `.vmdk` extension as a regular virtual disk file, the mapping file contains only mapping information. The virtual disk data is stored directly on the LUN.

During virtual machine creation, a hard disk and a SCSI or SATA controller are added to the virtual machine by default, based on the guest operating system that you select. If this disk does not meet your needs, you can remove it and add an RDM disk at the end of the creation process.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button and select **RDM Disk** from the drop-down menu.
   The **Select Target LUN** dialog box opens.
3. In the **Select Target LUN** dialog box, select the target LUN for the raw device mapping and click **OK**.
   The disk appears in the virtual device list.

4. Select the location for the mapping file.
   - To store the mapping file with the virtual machine configuration file, select **Store with the virtual machine**.
   - To select a location for the mapping file, select **Browse** and select the datastore location for the disk.
5. Select a compatibility mode.

Option	Description
Physical	Allows the guest operating system to access the hardware directly. Physical compatibility is useful if you are using SAN-aware applications on the virtual machine. However, a virtual machine with a physical compatibility RDM cannot be cloned, made into a template, or migrated if the migration involves copying the disk.
Virtual	Allows the RDM to behave as if it were a virtual disk, so that you can use such features as taking snapshots, cloning, and so on. When you clone the disk or make a template out of it, the contents of the LUN are copied into a `.vmdk` virtual disk file. When you migrate a virtual compatibility mode RDM, you can migrate the mapping file or copy the contents of the LUN into a virtual disk.

6. Accept the default or select a different virtual device node.

   In most cases, you can accept the default device node. For a hard disk, a nondefault device node is useful to control the boot order or to have different SCSI controller types. For example, you might want to boot from an LSI Logic controller and share a data disk with another virtual machine using a BusLogic controller with bus sharing turned on.
7. Optional: If you selected virtual compatibility mode, select a disk mode to change the way that disks are affected by snapshots.

   Disk modes are not available for RDM disks using physical compatibility mode.

Option	Description
Dependent	Dependent disks are included in snapshots.
Independent - Persistent	Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode are written permanently to the disk even if you revert a snapshot. When you power off or reset a virtual machine, the disk and all its snapshots are preserved.
Independent - Nonpersistent	Disks in nonpersistent mode behave like read-only disks. Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset the virtual machine, or when you delete a snapshot.

8. Click **OK**.

SCSI, SATA, and NVMe Storage Controller Conditions, Limitations, and Compatibility

To access virtual disks, CD/DVD-ROM, and SCSI devices, a virtual machine uses storage controllers, which are added by default when you create the virtual machine. You can add controllers or change the controller type after virtual machine creation or while you are in the creation wizard.

If you know about node behavior, controller limitations, and compatibility of different types of controllers before you change or add a controller, you can avoid potential boot problems.

How Storage Controller Technology Works

Storage controllers appear to a virtual machine as different types of SCSI controllers, including BusLogic Parallel, LSI Logic Parallel, LSI Logic SAS, and VMware Paravirtual SCSI. AHCI, SATA, and NVM Express (NVMe) controllers are also available.

NVMe is a standardized protocol designed specifically for high-performance multi-queue communication with NVM devices. ESX supports the NVMe protocol to connect to local and networked storage devices. For more information, see the vSphere Storage documentation.

When you create a virtual machine, the default controller is optimized for the best performance. The controller type depends on the guest operating system, the device type, and sometimes, the virtual machine's compatibility. For example,

when you create virtual machines with Apple Mac OS X guests and ESX 5.5 and later compatibility, the default controller type for both the hard disk and the CD/DVD drive is SATA. When you create virtual machines with Windows Vista and later guests, a SCSI controller is the default for the hard disk and a SATA controller is the default for the CD/DVD drive.

In high performance storage environments you can benefit from using VMware Paravirtual SCSI controllers. The VMware Paravirtual SCSI controller ensures greater throughput and lower CPU use, which boosts performance as compared to the other SCSI controller options. For platform support for VMware Paravirtual SCSI controllers, see the Broadcom Compatibility Guide.

Each virtual machine can have a maximum of four SCSI controllers and four SATA controllers. The default SCSI or SATA controller is 0. When you create a virtual machine, the default hard disk is assigned to the default controller 0 at bus node (0:0).

When you add storage controllers, they are numbered sequentially 1, 2, and 3. If you add a hard disk, SCSI, or CD/DVD-ROM device to a virtual machine after virtual machine creation, the device is assigned to the first available virtual device node on the default controller, for example (0:1).

If you add a SCSI controller, you can reassign an existing or new hard disk or device to that controller. For example, you can assign the device to (1:z), where 1 is SCSI controller 1 and z is a virtual device node from 0 to 64. For SCSI controllers, z cannot be 7. By default, the virtual SCSI controller is assigned to virtual device node (z:7), so that device node is unavailable for hard disks or other devices.

If you add a SATA controller, you can reassign an existing or new hard disk or device to that controller. For example, you can assign the device to (1:z), where 1 is SATA controller 1 and z is a virtual device node from 0 to 29. For SATA controllers, you can use device nodes from 0 through 29, including 0:7.

Alternatively, each virtual machine can have a maximum of four NVMe controllers. You can reassign an existing or new hard disk or device to that controller. For example, you can assign the hard disk to (x:z), where x is NVMe controller and z is a virtual device node. x has values from 0 to 3, and z has values from 0 to 63.

Storage Controller Limitations

Storage controllers have the following requirements and limitations:

- LSI Logic SAS and VMware Paravirtual SCSI are available for virtual machines with ESX 4.x and later compatibility.
- AHCI SATA is available only for virtual machines with ESX 5.5 and later compatibility.
- NVMe is available only for virtual machines with ESX 6.5 and later compatibility.
- BusLogic Parallel controllers do not support virtual machines with disks larger than 2 TB.
- Disks on VMware Paravirtual SCSI controllers might not experience optimal performance if they have snapshots or if the hosts memory is overcommitted.

> **CAUTION:**
> Changing the controller type after the guest operating system is installed will make the disk and any other devices connected to the adapter inaccessible. Before you change the controller type or add a new controller, make sure that the guest operating system installation media contains the necessary drivers. On Windows guest operating systems, the driver must be installed and configured as the boot driver.

Storage Controller Compatibility

Adding different types of storage controllers to virtual machines that use BIOS firmware can cause operating system boot problems. In the following cases, the virtual machine might fail to boot correctly and you might have to enter the BIOS setup and select the correct boot device:

- If the virtual machine boots from LSI Logic SAS or VMware Paravirtual SCSI, and you add a disk that uses BusLogic, LSI Logic, or AHCI SATA controllers.
- If the virtual machine boots from AHCI SATA, and you add BusLogic Parallel or LSI Logic controllers.

Adding additional disks to virtual machines that use EFI firmware does not cause boot problems.

Table 128: VMware Storage Controller Compatibility

Existing Controller	Added Controller						
	BusLogic Parallel	**LSI Logic**	**LSI Logic SAS**	**VMware Paravirtual SCSI**	**AHCI SATA**	**IDE**	**NVMe**
BusLogic Parallel	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LSI Logic	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LSI Logic SAS	Requires BIOS setup	Requires BIOS setup	Usually Works	Usually Works	Requires BIOS setup	Yes	Usually Works
VMware Paravirtual SCSI	Requires BIOS setup	Requires BIOS setup	Usually Works	Usually Works	Requires BIOS setup	Yes	Usually Works
AHCI SATA	Requires BIOS setup	Requires BIOS setup	Yes	Yes	Yes	Yes	Yes
IDE	Yes	Yes	Yes	Yes	Yes	N/A	Yes
NVMe	Requires BIOS setup	Requires BIOS setup	Usually Works	Usually Works	Requires BIOS setup	Yes	Usually Works

Add a SATA Controller

If a virtual machine has multiple hard disks or CD/DVD-ROM devices, you can add up to three additional SATA controllers to assign the devices to. When you spread the devices among several controllers, you can improve performance and avoid data traffic congestion. You can also add additional controllers if you exceed the thirty-device limit for a single controller.

- Verify that the virtual machine compatibility is ESX 5.5 and later.
- Verify that you are familiar with storage controller behavior and limitations.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add or remove device** privilege on the virtual machine.

You can boot virtual machines from SATA controllers and use them for large-capacity virtual hard disks.

Not all guest operating systems support AHCI SATA controllers. Typically, when you create virtual machines with ESX 5.5 and later compatibility and Mac OS X guest operating systems, a SATA controller is added by default for the virtual hard disk and CD/DVD-ROM devices. Most guest operating systems, including Windows Vista and later have a default SATA controller for CD/DVD-ROM devices. To verify support, see the Broadcom Compatibility Guide.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **SATA Controller** from the drop-down menu.
   The controller appears in the Virtual Hardware devices list.
4. Click **OK**.

You can add a hard disk or CD/DVD drive to the virtual machine and assign it to the new controller.

Add a SCSI Controller to a Virtual Machine

Many virtual machines have a SCSI controller by default, depending on the guest operating system. If you have a heavily loaded virtual machine with multiple hard disks, you can add up to three additional SCSI controllers to assign the disks to. When you spread the disks among several controllers, you can improve performance and avoid data traffic congestion. You can also add additional controllers if you exceed the 15-device limit for a single controller.

Verify that you have the **Virtual machine** > **Change Configuration** > **Add or remove device** privilege on the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button. add a new SCSI controller.
3. Select **SCSI Controller** from the drop-down menu.

   The controller appears in the Virtual Hardware device list.
4. From the **Change Type** drop-down menu, select the controller type.

   Do not select a BusLogic Parallel controller for a virtual machines with disks larger than 2 TB. This controller does not support large capacity hard disks.
5. Expand **New SCSI Controller**, and select the type of sharing from the **SCSI Bus Sharing** drop-down menu.

Option	Description
Option	Description
None	Virtual disks cannot be shared by other virtual machines.
Virtual	Virtual machines can share virtual disks on the same ESX host. Select **Thick provision eager zeroed** when you create the disk.
Physical	Virtual machines can share virtual disks on any ESX host. Select **Thick provision eager zeroed** when you create the disk.

6. Click **OK**.

You can now add a hard disk or other SCSI device to the virtual machine and assign it to the new SCSI controller.

How to Activate and Deactivate the Automatic Replacement of the LSI SAS Controller

vSphere 8.0 can automatically and safely replace the LSI SAS controller with the native VMware PVSCSI controller for existing virtual machines.

- Verify that the guest operating system of the virtual machines is Windows 10 and later or Windows Server 2016 and later.
- Verify that you have VMware Tools version 12.1.0 or later.
- The automatic replacement is possible for virtual machines that have the VMware Paravirtual SCSI (PVSCSI) driver installed with VMware Tools.
- Verify that the VMware Tools Device Helper policy is enabled.

Starting with vSphere 8.0, you can activate and deactivate the automatic replacement of the LSI SAS controller by configuring the VMware Tools Device Helper policy, which is `unconfigured` by default. The default unconfigured policy setting deactivates the automatic replacement of the LSI SAS controller.

You can configure the VMware Tools policy settings by using the Local Group Policy Editor (`gpedit.msc`). You can also place the VMware Tools policy files to a domain controller and assign the policy to all virtual machines in the domain.

To enable the VMware Tools policy and activate the replacement of the LSI SAS controller with the VMware PVSCSI controller, perform the following steps.

1. Open a command-line window with the **Run as administrator** setting.
   For example, click **Start**, type `cmd`, right-click the `cmd.exe` icon, and select **Run as administrator**.

2. Enter `xcopy "c:\ProgramData\Vmware\Vmware Tools\deviceHelperPolicy.admx" "c:\Windows\PolicyDefinitions"`.
3. Enter `xcopy "c:\ProgramData\Vmware\Vmware Tools\en-US\deviceHelperPolicy.adml" "c:\Windows\PolicyDefinitions\en-US"`.
4. Start the Local Group Policy Editor (`gpedit.msc`) and navigate to **Computer Configuration** > **Administrative Templates** > **VMware Tools Configuration**.
5. Select the **Device Helper Enable** setting and select **Enabled**.
6. Click **OK** and exit the Local Group Policy Editor.
7. Stop and start the VMTools service.
   a) To stop the VMTools service, enter `sc stop vmtools`.
   b) To start the VMTools service, enter `sc start vmtools`.

   When you restart the VMTools service, the configuration is read and the Device Helper installs the PVSCSI driver for the next system restart.
8. Restart the virtual machine.

   The LSI SAS controller is replaced with the VMware PVSCSI controller.

You can view the changes in the virtual machine settings or in the Windows virtual machine by using the Device Manager utility.

You can deactivate the automatic replacement of the LSI SAS controller by using the Local Group Policy Editor.

Option	Action
Deactivate by using the Local Group Policy Editor	In the Windows virtual machine, set the VMware Tools policy to `Disabled` or `Not configured`.

Add a Paravirtualized SCSI Adapter

You can add a VMware Paravirtual SCSI high performance storage controller to a virtual machine to provide greater throughput and lower CPU use.

- Verify that the virtual machine has a guest operating system with VMware Tools installed.
- Verify that the virtual machine compatibility is ESX 7.x and later.
- Ensure that you are familiar with the VMware Paravirtual SCSI controller type.
- To access boot disk devices attached to a VMware Paravirtual SCSI controller, verify that the virtual machine has a Windows 2003 or Windows 2008 guest operating system.
- In some operating systems, before you change the controller type, create a virtual machine with an LSI Logic controller, install VMware Tools, and then change to paravirtual mode.
- Verify that you have the **Virtual machine** > **Configuration** > **Add or remove device** privilege on the virtual machine.

VMware Paravirtual SCSI controllers are best suited for environments, especially SAN environments, running I/O-intensive applications.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **SCSI Controller** from the drop-down menu.
4. Expand **New SCSI controller** and from the **Change Type** menu, select **VMware Paravirtual**.

   The controller appears at the bottom of the Virtual Hardware device list.

5. Click **OK**.

Add an NVMe Controller

If a virtual machine has multiple hard disks, you can add up to four virtual NVMe controllers to which to assign the virtual disks. Using an NVMe controller significantly reduces the software overhead for processing guest OS I/O, as compared to AHCI SATA or SCSI controllers.

- Verify that the virtual machine has a guest operating system that supports NVMe.
- Verify that the virtual machine compatibility is ESX 8.0 or later.
- Verify that you are familiar with storage controller behavior and limitations.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add new disk** privilege on the virtual machine.

NVMe controllers perform best with virtual disks on an all-flash disk array, local NVMe SSD, vSAN, and PMem storage.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **NVMe Controller** from the drop-down menu.
   The controller appears in the Virtual Hardware devices list.
4. Click **OK**.

You can add a hard disk to the virtual machine and assign it to the NVMe controller.

Change the SCSI Controller Configuration

You can specify the SCSI controller type and you can set the type of SCSI bus sharing for a virtual machine.

- Verify that you are familiar with the limitations and conditions for configuring SCSI controllers.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Modify device settings** privilege on the virtual machine.

The choice of a SCSI controller type does not affect whether your virtual disk is an IDE or SCSI disk. The IDE adapter is always ATAPI. The default for your guest operating system is already selected.

The choice of a SCSI bus sharing option determines whether virtual machines on different hosts can access the same virtual disk.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **SCSI controller**, and select a SCSI controller type from the **Change Type** drop-down menu.

   CAUTION:
   Changing the SCSI controller type might result in a virtual machine boot failure.

   Do not select a BusLogic Parallel controller for virtual machines with disks larger than 2 TB. This controller does not support large capacity hard disks.

   The vSphere Client displays information about what happens if you change the SCSI controller type. If you select a controller type that is not recommended for the virtual machine's guest operating system, a warning is displayed.
3. Expand **SCSI controller** and select the type of sharing in the **SCSI Bus Sharing** drop-down menu.

Option	Description
None	Virtual disks cannot be shared by other virtual machines.
Physical	Virtual disks can be shared by virtual machines on any ESX host.
Virtual	Virtual disks can be shared by virtual machines on the same ESX host.

   For virtual or physical bus sharing, select **Thick provision eager zeroed** when you create the disk.

4. Click **OK**.

Virtual Machine Network Configuration

vSphere networking features provide communication between virtual machines on the same host, between virtual machines on different hosts, and between other virtual and physical machines.

When you configure networking for a virtual machine, you select or change an adapter type, a network connection, and whether to connect the network when the virtual machine powers on.

Network Adapter Basics

When you configure a virtual machine, you can add network adapters (NICs) and specify the adapter type.

Network Adapter Types

The type of network adapters that are available depend on the following factors:

- The virtual machine compatibility, which depends on the host that created or most recently updated it.
- Whether the virtual machine compatibility has been updated to the latest version for the current host.
- The guest operating system.

Supported NICs currently differ between an on-premises environment and VMware Cloud on AWS. The following NIC types are supported in an on-premises deployment:

E1000E	Emulated version of the Intel 82574 Gigabit Ethernet NIC. E1000E is the default adapter for Windows 8 and Windows Server 2012.
VMXNET 3	A paravirtualized NIC designed for performance. VMXNET 3 offers all the features available in VMXNET 2 and adds several new features, such as multiqueue support (also known as Receive Side Scaling in Windows), IPv6 offloads, and MSI/MSI-X interrupt delivery. VMXNET 3 is not related to VMXNET or VMXNET 2.
SR-IOV passthrough	Representation of a virtual function (VF) on a physical NIC with SR-IOV support. The virtual machine and the physical adapter exchange data without using the VMkernel as an intermediary. This adapter type is suitable for virtual machines where latency might cause failure or that require more CPU resources. SR-IOV passthrough is available in ESX 6.0 and later for guest operating systems Red Hat Enterprise Linux 6 and later, and Windows Server 2008 R2 with SP2. An operating system release might contain a default VF driver for certain NICs, while for others, you must download and install it from a location provided by the vendor of the NIC or of the host. For information about assigning an SR-IOV passthrough network adapter to a virtual machine, see the *vSphere Networking* documentation.

For network adapter compatibility considerations, see the Broadcom Compatibility Guide.

Legacy Network Adapters and ESX Virtual Hardware Versions

The default network adapter types for all legacy virtual machines depend on the adapters available and compatible to the guest operating system and the version of virtual hardware on which the virtual machine was created.

If you do not upgrade a virtual machine to use a virtual hardware version, your adapter settings remain unchanged. If you upgrade your virtual machine to take advantage of newer virtual hardware, your default adapter settings will likely change to be compatible with the guest operating system and upgraded host hardware.

To verify the network adapters that are available to your supported guest operating system for a particular version of vSphere ESX, see the Broadcom Compatibility Guide.

Network Adapters and Legacy Virtual Machines

Legacy virtual machines are virtual machines that are supported by the product in use, but are not current for that product.

The default network adapter types for all legacy virtual machines depend on the adapters available and compatible to the guest operating system and the version of virtual hardware on which the virtual machine was created.

If you do not upgrade a virtual machine to correspond with an upgrade to a newer version of an ESX host, your adapter settings remain unchanged. If you upgrade your virtual machine to take advantage of newer virtual hardware, your default adapter settings will likely change to be compatible with the guest operating system and upgraded host hardware.

To verify the network adapters that are available to your supported guest operating system for a particular version of vSphere ESX, see the Broadcom Compatibility Guide.

Change the Virtual Machine Network Adapter Configuration

You can change the virtual machine network configuration, including its power on behavior and resource allocation.

Verify that you have the following privileges:

- **Virtual machine** > **Change Configuration** > **Modify device settings** on the virtual machine.
- **Network** > **Assign network** on a network if you are changing the network that the virtual machine connects to.
- To activate the UPT support, make sure that the virtual machine is of hardware version 20 or later.

Starting with vSphere 8.0, you can enable the Uniform Passthrough (UPT) compatibility on a VMXNET3 adapter.

For details about configuring the networking for virtual machine network adapters, see the vSphere Security documentation.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Network adapter**, and select the port group to connect to from the drop-down menu.

   The menu lists all standard and distributed port groups that are available for virtual machine use on the host.

   If you want to provision bandwidth to the network adapter from a reserved quota by using vSphere Network I/O Control version 3, select a port group that is associated with the network resource pool that provides the quota.
3. Optional: Change the **Status** settings.

Option	Description
Connected	To connect or disconnect the virtual network adapter, select or deselect this option while the virtual machine is running. This check box is not available when the virtual machine is powered off.
Connect at power on	To connect to the network when the virtual machine powers on, select this option for the virtual network adapter. If you do not select this option, you must manually connect the adapter in order for the virtual machine to access the network.

4. Select the network adapter type to use from the **Adapter Type** drop-down menu.
5. Optional: Select how to assign the **MAC address** from the drop-down menu.
   - Select **Automatic** to assign a MAC address automatically.
   - Select **Manual** to enter manually the MAC address that you want.
6. If you have selected SR-IOV passthrough as the adapter type, to allow changes in the maximum transmissions unit (MTU) of packets from the guest operating system, select an option form the **Allow Guest MTU Change** drop-down menu.
7. If you have selected SR-IOV passthrough as the adapter type, to enable the UPT, select the **Use UPT Support** check box.

   Note:
   When you activate the UPT support, the full memory size of the virtual machine is automatically reserved.
8. If the network adapter is connected to a distributed port group of a distributed switch that has vSphere Network I/O Control version 3 enabled, allocate bandwidth to the adapter.

   Note:
   You cannot allocate bandwidth to **SR-IOV passthrough** network adapters.

a) From the **Shares** drop-down menu, set the relative priority of the traffic from this virtual machine as shares from the capacity of the connected physical adapter.
b) In the **Reservation** text box, reserve a minimum bandwidth that must be available to the VM network adapter when the virtual machine is powered on.
c) In the **Limit** text box, set a limit on the bandwidth that the VM network adapter can consume.

9. Click **OK**.

How to Add a Network Interface Controller to a Virtual Machine

You can add a network interface controller (NIC), or network adapter to a virtual machine to connect to a network, to enhance communications, or to replace an older adapter.

Verify that you have the following privileges:

- **Network** > **Assign network** on a network.
- **Virtual machine** > **Change Configuration** > **Add or remove device**
- To add an SR-IOV Passthrough adapter, ensure that the virtual machine is of hardware version 10 and later and power off the virtual machine.
- To activate the UPT support, verify that:
  - The ESX host is installed on a DPU and the ESX supports UPT.
  - The adapter type is VMXNET3.
  - The virtual machine is of hardware version 20 or later.

When you add a NIC to a virtual machine, you select the adapter type, network connection, whether the device must connect when the virtual machine is powered on, and the bandwidth allocation.

Starting with vSphere 8.0, you can enable the Uniform Passthrough (UPT) compatibility on a VMXNET3 adapter.

For details about configuring the networking for virtual machine network adapters, see the vSphere Networking documentation.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click **Add New Device** and select **Network Adapter** from the drop-down menu.

   The new network adapter appears at the bottom of the network adapters list.
3. Expand **New Network** and select the standard or distributed port group to connect to.

   The menu lists all standard and distributed port groups that are available for virtual machine use on the host.

   If you want to provision bandwidth to the network adapter from a reserved quota by using vSphere Network I/O Control version 3, select a port group that is associated with the network resource pool that provides the quota.
4. Optional: Review and optionally change the **Status** settings.

Option	Description
Connected	To connect or disconnect the virtual network adapter, select this option while the virtual machine is running. This check box is not available when the virtual machine is powered off.
Connect at power on	To connect to the network when the virtual machine powers on, select this option for the virtual network adapter. If you do not select this option, you must manually connect the adapter for the virtual machine to access the network.

5. From the **Adapter Type** drop-down menu, select the network adapter type.
6. If you have selected SR-IOV passthrough as the adapter type, to allow changes in the maximum transmissions unit (MTU) of packets from the guest operating system, select an option form the **Allow Guest MTU Change** drop-down menu.
7. To enable the UPT, select the **Use UPT Support** check box.

   Note:
   When you activate the UPT support, the full memory size of the virtual machine is automatically reserved.

8. If the network adapter is connected to a distributed port group of a distributed switch that has vSphere Network I/O Control version 3 enabled, allocate bandwidth to the adapter.

   Note:
   You cannot allocate bandwidth to **SR-IOV passthrough** network adapters.

   a) From the **Shares** drop-down menu, set the relative priority of the traffic from this virtual machine as shares from the capacity of the connected physical adapter.
   b) In the **Reservation** text box, reserve a minimum bandwidth that must be available to the VM network adapter when the virtual machine is powered on.
   c) In the **Limit** text box, set a limit on the bandwidth that the VM network adapter can consume.

9. (Optional) Select how to assign the **Mac address** from the drop-down menu.
   - Select **Automatic** to assign a MAC address automatically.
   - Select **Manual** to enter manually the MAC address that you want.

10. Click **OK**.

Virtual Machine Parallel and Serial Port Configuration

Parallel and serial ports are interfaces for connecting peripherals to the virtual machine. The virtual serial port can connect to a physical serial port or to a file on the host computer.

You can also use it to establish a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. You can add parallel and serial ports and change the parallel and serial port configuration. Hardware version 11 and later versions allow you to configure virtual machines in such a way that serial and parallel ports are absent from the virtual chipset altogether.

Starting with vSphere 9.1, you cannot add, remove, and configure parallel ports. For information, see Broadcom knowledge base article 335056.

Using Serial Ports with vSphere Virtual Machines

You can set up virtual serial port connections for vSphere virtual machines in several ways. The connection method that you select depends on the task that you must accomplish.

You can set up virtual serial ports to send data in the following ways.

Physical serial port on the host	Sets the virtual machine to use a physical serial port on the host computer. This method lets you use an external modem or a hand-held device in a virtual machine.
Output to file	Sends output from the virtual serial port to a file on the host computer. This method lets you capture the data that a program running in the virtual machine sends to the virtual serial port.
Connect to a named pipe	Sets a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. With this method, two virtual machines or a virtual machine and a process on the host can communicate as if they were physical machines that are connected by a serial cable. For example, use this option for remote debugging of a virtual machine.
Connect over the network	Enables a serial connection to and from a virtual machine's serial port over the network. The Virtual Serial Port Concentrator (vSPC) aggregates traffic from multiple serial ports onto one management console. vSPC behavior is similar to physical serial port concentrators. Using a vSPC also allows network connections to a virtual machine's serial ports to migrate seamlessly when you use vMotion to migrate the virtual machine. For requirements and steps to configure the Avocent ACS v6000 virtual serial port concentrator, see Broadcom knowledge base article 345281.

Server and Client Connections for Named Pipe and Network Serial Ports

You can select a client or server connection for serial ports. Your selection determines whether the system waits for a connection or initiates it. Typically, to control a virtual machine over a serial port, you select a server connection. This selection lets you control the connections, which is useful if you connect to the virtual machine only occasionally. To use a serial port for logging, select a client connection. This selection lets the virtual machine connect to the logging server when the virtual machine starts and to disconnect when it stops.

Supported Serial Ports

When you use a physical serial port for serial port passthrough from an ESX host to a virtual machine, serial ports that are integrated into the motherboard are supported. A virtual machine can use up to 32 serial ports.

Unsupported Serial Ports

When you use a physical serial port for serial port passthrough from an ESX host to a virtual machine, the serial ports connected through USB are not supported for serial port passthrough. They might be supported by USB passthrough from an ESX host to a virtual machine. See USB Configuration from an ESX Host to a Virtual Machine.

In addition, you cannot use Migration with VMotion when you use a physical serial port for serial passthrough.

Adding a Firewall Rule Set for Serial Port Network Connections

If you add or configure a serial port that is backed by a remote network connection, ESX firewall settings can prevent transmissions.

Before you connect network-backed virtual serial ports, you must add one of the following firewall rule sets to prevent the firewall from blocking communication:

- **VM serial port connected to vSPC**. Use to connect the serial port output through a network with the **Use virtual serial port concentrator** option enabled to allow only outgoing communication from the host.
- **VM serial port connected over network**. Use to connect the serial port output through a network without the virtual serial port concentrator.

  Important:
  Do not change the allowed IP list for either rule set. Updates to the IP list can affect other network services that might be blocked by the firewall.

For details about allowing access to an ESX service through the firewall, see the *vSphere Security* documentation.

Configure Virtual Machine Communication Interface Firewall

You can configure the virtual machine Communication Interface firewall (VMCI) to restrict virtual machines accessing the hypervisor-based services and VMCI-based services.

You can restrict VMCI usage to a subset of VMCI-based services on each virtual machine. For example, you can allow certain virtual machines to access VMCI services and deny access to others for security reasons.

Currently, VMCI devices support guest to host communication. A virtual machine can communicate with VMCI services through the following means:

- ESX hypervisor
- Services installed on the host operating system in the form of a vmkernel module
- Applications installed by a verified vSphere Installation Bundle

Change the Serial Port Configuration

You can connect the virtual serial port to a physical serial port or to a file on the host computer. You can also use a host-side named pipe to set up a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. In addition, you can use a port or vSPC URI to connect a serial port over the network. You can add up to 32 serial ports to a virtual machine.

- Check that you know the correct media types for the port to access, vSPC connections, and any conditions that might apply. See Using Serial Ports with vSphere Virtual Machines.
- To connect a serial port over a network, add a Firewall rule set. See Adding a Firewall Rule Set for Serial Port Network Connections.
- To use authentication parameters with network serial port connections, see Authentication Parameters for Virtual Serial Port Network Connections.

- Required privileges:
  - **Virtual machine** > **Change Configuration** > **Modify device settings** on the virtual machine.
  - **Virtual machine** > **Interaction** > **Device connection** on the virtual machine to change the device connection status.

Virtual machines can be in a powered-on state during configuration.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Serial port**, and select a connection type.

Option	Action
Use physical serial port	Select this option to have the virtual machine use a physical serial port on the host computer. Select the serial port from the drop-down menu.
Use output file	Select this option to send output from the virtual serial port to a file on the host computer. Browse to select an output file to connect the serial port to.
Use named pipe	Select this option to set a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. 1. Type a name for the pipe in the **Pipe Name** field. 2. Select the **Near end** and **Far end** of the pipe from the drop-down menus.
Use Network	Select **Use network** to connect through a remote network. 1. Select the network backing. – Select **Server** to have the virtual machine monitor incoming connections from other hosts. – Select **Client** to have the virtual machine initiate a connection to another host. 2. Enter a **Port URI**. The URI is the remote end of the serial port to which the virtual machine's serial port should connect. 3. If vSPC is used as an intermediate step to access all virtual machines through a single IP address, select **Use Virtual Serial Port Concentrator** and enter the vSPC URI location.
Printer	Select **Printer** to connect to a remote printer.

3. Optional: Select **Yield CPU on poll**.

   Select this option only for guest operating systems that use serial ports in polled mode. This option prevents the guest from consuming excessive CPUs.
4. Optional: Select **Connect at power on** to connect the serial port when the virtual machine powers on.
5. Click **OK**.

Establishing Serial Port Network Connections to a Client or Server Without Authentication Parameters

If you do not use vSPC and you configure your virtual machine with a serial port connected as a server with a `telnet://:12345` URI, you can connect to your virtual machine's serial port from your Linux or Windows operating system.

```
telnetyourESXiServerIPAddress12345
```

Similarly, if you run the Telnet Server on your Linux system on port 23 (`telnet://yourLinuxBox:23`), you configure the virtual machine as a client URI.

```
telnet://yourLinuxBox:23
```

The virtual machine initiates the connection to your Linux system on port 23.

Authentication Parameters for Virtual Serial Port Network Connections

When you establish serial port connections over the network, you can use authentication parameters to secure the network. These parameters can support an encrypted connection with a remote system using SSL over Telnet or Telnets, or an encrypted connection with a concentrator using SSL over Telnet or Telnets.

URI Forms

If you do not use virtual serial port network connection (vSPC) and you configure your virtual machine with a serial port connected as a server with a `telnet://:12345` URI, you can connect to your virtual machine's serial port from your Linux or Windows operating system. You use one of the following formats:

- Telnet over TCP.
  `telnet://host:port`
  The virtual machine and remote system can negotiate and use SSL if the remote system supports the Telnet authentication option. If not, the connection uses unencrypted text (plain text).
- Telnets over SSL over TCP.
  `telnets://host:port`
  SSL negotiation begins immediately, and you cannot use the Telnet authentication option.

Authentication Parameters

For an encrypted connection, the URI includes a set of authentication parameters. Enter the parameters as key words or key/value pairs. You can enter authentication parameters for secure Telnet (`telnets`), or for Telnet (`telnet`) as shown in the following syntax:

```
telnet://host:port#key[=value] [&key[=value] ...]
```

The first parameter must have a number sign (#) prefix. Additional parameters must have an ampersand (&) prefix. The following parameters are supported.

thumbprint=*value*

Specifies a certificate thumbprint against which the peer certificate thumbprint is compared. When you specify a thumbprint, certificate verification is enabled.

peerName=*value*

Specifies the peer name that is used to validate the peer certificate. When you specify a peer name, certificate verification is enabled.

verify

Forces certificate verification. The virtual machine will verify that the peer certificate subject matches the specified peerName and that it was signed by a certificate authority known to the ESX host. Verification is enabled if you specify a thumbprint or peerName

cipherList=*value*

Specifies a list of SSL ciphers. The ciphers are specified as a list separated by colons, spaces, or commas.

Establishing Serial Port Network Connections to a Client or Server

Simple Server Connection	To connect to a virtual machine's serial port from a Linux or Windows operating system if you do not use vSPC, configure the virtual machine with a serial port connected as a server with a `telnet://:12345` URI. To access a virtual serial port from a client, use `telnet yourESXServerIPAddress 12345`.
Secure Server Connection	To enforce an encrypted connection to the virtual machine's serial port from a Linux operating system, you can configure Telnet to enforce encryption by configuring the virtual machine with a serial port connected as a server with a `telnet://:12345#verify` URI. To access a virtual serial port from a client, use `telnet-ssl yourESXServerName 12345`. This connection will fail if the Telnet program you are using does not support SSL encryption.
Simple Client Connection	If you are running a Telnet server on your system and you want the virtual machine to automatically connect to it, you can configure the virtual machine as a client using `telnet://yourLinuxBox:23`. The Virtual machine keeps initiating the Telnet connection to port 23 on *yourLinuxBox*.
Secure Client Connection	Additional URI options allow you to enforce a specific server certificate and restrict the ciphers being used. Virtual machines with a serial port configured as a client with `telnet://ipOfYourLinuxBox:23#cipherList=DHE-RSA-AES256-SHA256:DHE-RSA-AES256-SHA&peerName=myLinuxBoxName.withDomain`

will connect to *ipOfYourLinuxBox* only if the system supports one of two listed ciphers, and if it presents a trusted certificate issued to *myLinuxBoxName.withDomain*. Replace *.withDomain* with the full domain name, for example, example.org.

Add a Serial Port to a Virtual Machine

You can connect the virtual serial port to a physical serial port or to a file on the host computer. You can also use a host-side named pipe to set up a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. In addition, you can use a port or vSPC URI to connect a serial port over the network. A virtual machine can use up to 32 serial ports.

- Verify that the virtual machine is powered off.
- Check that you know the correct media types for the port to access, vSPC connections, and any conditions that might apply. See Using Serial Ports with vSphere Virtual Machines.
- To connect a serial port over a network, add a Firewall rule set. See Adding a Firewall Rule Set for Serial Port Network Connections.
- To use authentication parameter with network serial port connections, see Authentication Parameters for Virtual Serial Port Network Connections.
- Required privilege: **Virtual Machine** > **Change Configuration** > **Add or Remove Device**

  Important:
  With virtual hardware version 11 and later, if you configure a virtual machine without serial ports, they are entirely removed from the virtual chipset and they are not visible to the virtual machine OS.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **Serial Port** from the drop-down menu.
   The new serial port appears at the bottom of the device list.
4. From the **New Serial port** drop-down menu, select a connection type.

Option	Action
Use output file	Select this option to send output from the virtual serial port to a file on the host computer. To select an output file to connect to the serial port, click **Browse** .
Use physical serial port	Select this option to have the virtual machine use a physical serial port on the host computer. Select the serial port from the drop-down menu.
Use named pipe	Select this option to set a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. 1. Enter a name for the pipe in the **Pipe Name** field. 2. Select the **Near end** and **Far end** of the pipe from the drop-down menus.
Use Network	To connect through a remote network, select **Use network**. 1. Select the network backing. – To have the virtual machine monitor incoming connections from other hosts, select **Server**. – To have the virtual machine initiate a connection to another host, select **Client**. 2. Enter a **Port URI**. The URI is the remote end of the serial port to which the virtual machine's serial port should connect. 3. If vSPC is used as an intermediate step to access all virtual machines through a single IP address, select **Use Virtual Serial Port Concentrator** and enter the **vSPC URI** location.

5. Optional: Select **Yield CPU on poll**.
   Select this option only for guest operating systems that use serial ports in polled mode. This option prevents the guest from consuming excessive CPUs.

6. Click **OK**.

Establishing Serial Port Network Connections to a Client or Server Without Authentication Parameters

If you do not use vSPC and you configure your virtual machine with a serial port connected as a server with a `telnet://:12345` URI, you can connect to your virtual machine's serial port from your Linux or Windows operating system.

`telnet`*`yourESXiServerIPAddress`*`12345`

Similarly, if you run the Telnet Server on your Linux system on port 23 (`telnet://yourLinuxBox:23`), you configure the virtual machine as a client URI.

`telnet://`*`yourLinuxBox`*`:23`

The virtual machine initiates the connection to your Linux system on port 23.

Other Virtual Machine Device Configuration

You can add and configure virtual hardware, such as DVD/CD-ROM drives to virtual machines.

Not all devices are available to add and configure. For example, you cannot add a video card, but you can configure available video cards and PCI devices.

For information about how to add, remove, and configure floppy drives or SCSI devices, see Broadcom knowledge base article 335056.

Change the CD/DVD Drive Configuration

You can configure DVD or CD devices to connect to client devices, host devices, datastore ISO files, or content library ISO files.

Configure a Datastore ISO File for the CD/DVD Drive

To install a guest operating system and its applications on a new virtual machine, you can connect the CD/DVD device to an ISO file that is stored on a datastore accessible to the host.

Verify that you have the following privileges:

- **Virtual machine** > **Interaction** > **Configure CD media** on the virtual machine.
- **Virtual machine** > **Interaction** > **Power On**
- **Datastore** > **Browse datastore** on the datastore to which you upload the installation media ISO image.
- **Datastore** > **Low level file operations** on the datastore to which you upload the installation media ISO image.

If an ISO image is not available on a local or shared datastore, upload the file to a datastore from your local system by using the datastore file browser. See Upload ISO Image Installation Media for a Guest Operating System.

To avoid performance issues and possible conflicts between virtual machines that might try to simultaneously access the ISO image, unmount and disconnect the ISO file when the installation finishes.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Expand **CD/DVD drive**, and select **Datastore ISO File** from the drop-down menu.
   The **Select File** dialog box opens
3. Browse to select the file and click **OK**.
4. From the **Virtual Device Node** drop-down menu, select the node that the drive uses in the virtual machine.
5. Optional: Select **Connect At Power On** to connect the device when the virtual machine powers on.

   Note:
   If you install a guest operating system, select **Connect At Power On**.

6. Click **OK**.
7. Power on the virtual machine and click the **Summary** tab.
8. Expand the **VM Hardware** panel and click the **Connected** icon next to the datastore ISO file to connect the device

Configure a Content Library ISO File for the CD/DVD Drive

To install a guest operating system and its applications on a new virtual machine, you can connect the CD/DVD device to an ISO file that is stored in a content library.

Verify that you have the following privileges on the virtual machine:

- **Virtual machine** > **Interaction** > **Configure CD media**
- **Virtual machine** > **Interaction** > **Power On**

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Expand **CD/DVD drive** and select **Content Library ISO File** from the drop-down menu.
   The **Choose an ISO image to mount dialog** box opens.
3. Select the ISO file and click **OK**.
4. Optional: Select **Connect At Power On** to connect the device when the virtual machine powers on.
5. Click **OK**.
6. Power on the virtual machine and click the **Summary tab**.
7. Expand the **VM Hardware** panel and from the drop-down menu next to the **Connected** icon, select to connect the CD/DVD drive to a content library ISO file.

Configure a Host Device Type for the CD/DVD Drive

You can configure the CD/DVD drive of a virtual machine to connect to a physical CD or DVD device on the host so that you can install a guest operating system, VMware Tools, or other applications.

- Verify that the virtual machine is powered off.
- You cannot use vMotion to migrate virtual machines that have CD drives that are backed by the physical CD drive on the host. Disconnect these devices before you migrate the virtual machine.
- Verify that you have the **Virtual machine** > **Interaction** > **Configure CD media** and **Virtual machine** > **Interaction** > **Power On**privileges on the virtual machine.

When you create a virtual machine, a controller is added by default and the CD/DVD drive is attached to that controller. The controller and driver type depend on the guest operating system. Typically, virtual machines with newer guest operating systems have a SATA controller and CD/DVD drive. Other guests use an IDE controller and CD/DVD drive.

If you connect to media that does not require you to power off the virtual machine, you can select the media to connect to from the CD/DVD drive connection icon on the virtual machine **Summary** tab.

When you add a CD/DVD drive that is backed by a USB CD/DVD drive on the host, you must add the drive as a SCSI device.

For more information on how to add, remove, and configure SCSI devices, see Broadcom knowledge base article 335056.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **CD/DVD** and select **Host Device** from the drop-down menu.
3. Optional: Select **Connect At Power On** to connect the device when the virtual machine powers on.
4. If more than one type of CD/DVD media is available on the host, select the media.
5. In the **Virtual Device Node** drop-down menu, select the node that the drive uses in the virtual machine.

   The first available node is selected by default. Changing the default is typically not required.

6. Click **OK**.
7. Power on the virtual machine and click the **Summary** tab.

The connected CD/DVD device appears in the **VM Hardware** list.

Configure a Client Device Type for the CD/DVD Drive

To install a guest operating system and its applications or other media on a virtual machine, you can connect the CD/DVD device to a physical DVD or CD device on the system from which you access the vSphere Client.

Verify that the virtual machine is powered on.

By default, passthrough IDE mode is used for remote client device access. You can write or burn a remote CD only through passthrough mode access.

1. Navigate to a virtual machine in the inventory and click the **Summary** tab.
2. In the **VM Hardware** panel, click the **CD/DVD drive** connection icon, select an available drive to connect to, and browse for the CD/DVD media.

   An Access Control dialog box opens. Click **allow** to proceed. To change your selection, click the connection icon, select **Disconnect**, and select a different option.

How do I Add or Modify a Virtual Machine CD or DVD Drive

CD/DVD drives are necessary for installing a guest operating system and VMware Tools. You can use a physical drive on a client or host or you can use an ISO image to add a CD/DVD drive to a virtual machine.You can use an ISO image to add a CD/DVD drive to a virtual machine.

- Verify that the virtual machine is powered off.
- If an ISO image file is not available on a local or shared datastore, upload an ISO image to a datastore from your local system by using the datastore file browser. See Upload ISO Image Installation Media for a Guest Operating System.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add or remove device** privilege on the virtual machine.
- Verify that you have the **Datastore** > **Browse datastore** on the datastore.
- **Virtual machine** > **Change Configuration** > **Change Settings** to configure the virtual machine hardware.

When you power on the virtual machine, you can select the media to connect to from the **VM Hardware** panel on the virtual machine **Summary** tab.

The following conditions exist.

- If you add a CD/DVD drive that is backed by a USB CD/DVD drive on the host, you must add the drive as a SCSI device.
- You must disconnect virtual machines that have CD drives that are backed by the physical CD drive on the host, before you migrate the virtual machine.
- You access the host CD-ROM device through emulation mode. Passthrough mode is not functional for local host CD-ROM access. You can write or burn a remote CD only through passthrough mode access, but in emulation mode you can only read a CD-ROM from a host CD-ROM device.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Select your task.

Option	Description
Add a CD/DVD drive	On the **Virtual Hardware** tab, click the **Add New Device** button and select **CD/DVD Drive**.
Modify CD/DVD settings	On the **Virtual Hardware** tab, expand **CD/DVD drive** and change the configuration settings.

3. To change CD/DVD settings, select the device type from the **CD/DVD drive** drop-down menu.

Option	Action
Client Device	Select this option to connect the CD/DVD device to a physical DVD or CD device on the system from which you access the vSphere Client.

Option	Action
	From the **Device Mode** drop-down menu, select **Passthrough CD-ROM**.
Host Device	1. Select to connect the CD/DVD device to a physical DVD or CD device on the host. 2. From the **CD/DVD Media** drop-down menu, select the media to connect to. 3. From the **Device Mode** drop-down menu, select **Emulate CD-ROM**.
Datastore ISO File	Select this option to connect the CD/DVD device to an ISO file that is stored on a datastore accessible to the host. The **Select File** dialog box opens. 1. In the **Select File** dialog box, browse to the file containing the ISO image to connect to. 2. Click **OK**.
Content Library ISO File	Select this option to connect the CD/DVD device to an ISO file that is stored in a content library. The **Choose an ISO image to mount** dialog box opens 1. In the **Choose an ISO image to mount**, select the ISO image to connect to. 2. Click **OK**.

4. Optional: Specify additional settings for the CD/DVD drive.

Option	Description
Connect At Power On	Select this option to connect to the device when the virtual machine turns on.
Device Mode	Select **Passthrough CD-ROM** for a CD/DVD drive that is connected to the physical client machine. Select **Emulate CD-ROM** otherwise.
Virtual Device Node	Specify the location of the ISO that you are mounting. To change the device node from the default, select a new mode from the **Virtual Device Node** drop-down menu. For example, if you select SATA(0:2), the guest operating system detects the drive as ID 2 on controller 0.

5. Power on the virtual machine and click the **Summary** tab.
6. Expand the **VM Hardware** panel and click **Connected** next to select to.

You can now install the guest operating system or other applications.

How do I Add a PCI Device to a Virtual Machine

vSphere allows you to easily assign multiple PCI passthrough devices to a virtual machine without specifying an exact physical device on a particular ESX host.

Verify that you have the privileges that you need for the task that you plan to perform.

- If you plan to add a PCI device when you edit a virtual machine, verify that you have the **Virtual machine** > **Change Configuration** > **Add or remove device** privilege.
- If you plan to increase the memory reservation when you edit a virtual machine, verify that you have the**Virtual machine** > **Change Configuration** > **Change resource** privilege.
- If you plan to reduce the virtual machine memory when you edit a virtual machine, verify that you have the **Virtual machine** > **Change Configuration** > **Change Memory** privilege.
- Power off the virtual machine.
- To use Dynamic DirectPath, verify that the virtual machine is compatible with ESX 7.0 or later.
- To use Fixed DirectPath, verify that Intel Virtualization Technology for Directed I/O (VT-d) or AMD I/O Virtualization Technology (IOMMU) is enabled in the host BIOS.
- For Dynamic DirectPath and Fixed DirectPath, verify that the PCI devices are connected to the host and marked as available for passthrough. If your ESX host is configured to boot from a USB device, or if the active coredump partition is configured to be on a USB device or SD cards connected through USB channels, deactivate the USB controller for passthrough. VMware does not support USB controller passthrough for ESX hosts that boot from USB devices or SD

cards that are connected through USB channels. A configuration in which the active coredump partition is configured to be on a USB device or SD card that is connected through USB channels is also not supported. For more information, see Broadcom knowledge base article 315272.

- To use Enhanced DirectPath, verify that:
  - The virtual machine is compatible with ESX 8.0 or later.
  - You download and install special drivers provided by the hardware vendor.
- To use NVIDIA GRID vGPU graphic devices:
  - Verify that an NVIDIA GRID vGPU graphic device with an appropriate driver is installed on the host. See the *VMware ESX Upgrade* documentation.
  - Verify that the virtual machine is compatible with ESX 6.0 and later.
- To add multiple NVIDIA GRID vGPUs to a virtual machine:
  - Verify that the virtual machine is compatible with ESX 6.7 Update 2 and later.
  - Only Q-series and C-series vGPU types are supported.
- To use Vendor Device Groups, verify that:
  - The virtual machine is compatible with ESX 8.0 or later.

vSphere 9.1 extends the existing NVSwitch technology support to include the NVIDIA Blackwell GPUs. The NVIDIA Blackwell architecture supports up to 16 GPUs per host, providing a unified, high-performance environment for AI and HPC workloads.

For NVIDIA Hopper and Blackwell architectures, vSphere automatically organizes these interconnected GPUs into device groups. vSphere DRS recognizes a device group as a single unit, ensuring that a virtual machine is always placed on a host that meets the full hardware requirements of the group.

You can connect to the guest operating system of a virtual machine all PCI devices that are configured on an ESX host and made available for passthrough.

You can also take advantage of the vSphere Distributed Resource Scheduler (DRS) feature. vSphere DRS can move a virtual machine to a compatible ESX host when the virtual machine powers on.

Fixed DirectPath devices	Fixed DirectPath devices allow you to specify the physical location of the devices that you want to add to a virtual machine. Fixed DirectPath allows a virtual machine to access directly the physical PCI and PCIe devices that are connected to a specific host. This way you can directly access devices, such as high-performance graphics or sound cards. You can connect each virtual machine to up to 16 PCI devices. You configure PCI devices on an ESX host to make them available for passthrough to a virtual machine. See the *vSphere Networking* documentation. However, you must not enable PCI passthrough for ESX hosts that are configured to boot from USB devices. When Fixed DirectPath devices are made available to a virtual machine, you cannot perform certain operations on the virtual machine. These operations include suspending, migration with vMotion, and taking or restoring snapshots of the virtual machine.
Enhanced DirectPath devices	Enhanced DirectPath devices are an improvement that allows hardware device vendors to extend the capabilities of their devices when used in virtual machines. For example, if implemented by the hardware vendor, an Enhanced DirectPath device might support a virtual machine migration with vMotion or suspend and resume operations. Enhanced DirectPath devices might also have configuration parameters that are associated with them and you can view and change these configuration parameters when you add such PCI devices to a virtual machine. The configuration parameters, if implemented by the vendor, can help you further specify and select particular hardware devices in accordance with the amount of the resources available on the device or their type. Enhanced DirectPath devices allow vSphere DRS to identify a host within the cluster that has an available device with the required resources.
Dynamic DirectPath devices	Dynamic DirectPath devices allow you to specify the device type that you want to add to a virtual machine. Dynamic DirectPath provides you with the ability to assign multiple PCI passthrough devices to a virtual machine. Dynamic DirectPath allows vSphere DRS to identify a host within the cluster that has an available device with the same vendor and model name.
Vendor Device Groups	Vendor Device Groups is a collection of two or more hardware devices that you can allocate together to a virtual machine. By using Vendor Device Groups, other properties can be considered when allocating devices to a virtual machine. For example, one could ensure that a pair of devices is always allocated

on the same PCI switch. Without Vendor Device Groups, each device is allocated to a virtual machine independently of one another.

Note:
You can only add and remove the entire Vendor Device Groups entity and not an individual device of that group.

If you select a Vendor Device Groups that has a NIC, you must add an appropriate network for the NIC in that Vendor Device Groups. The network that you add is one of the networks that you configure for the SR-IOV NICs to be attached to.

NVIDIA GRID GPU devices

If an ESX host has an NVIDIA GRID GPU graphics device, you can configure a virtual machine to use the NVIDIA GRID virtual GPU (vGPU) technology.

NVIDIA GRID vGPU devices optimize complex graphics operations and make them run at high performance without overloading the CPU. By using NVIDIA GRID vGPU, you can share a single physical GPU among multiple virtual machines as separate vGPU-enabled passthrough devices.

Starting with vSphere 7.0 Update 2, you can configure a virtual machine to use the NVIDIA Multi-Instance GPU (MIG) feature. By using NVIDIA MIG, you can securely partition applicable GPUs into separate GPU instances. Each GPU instance has dedicated resources, such as memory, memory caches, and compute cores. If a GPU is in MIG mode, you can assign unique vGPU profile names to a virtual machine. VMware creates GPU and compute instances automatically, so you should not create them manually.

Note:
When you add a PCI device to a virtual machine, the full memory size of the virtual machine is automatically reserved.

1. Add a PCI device to a virtual machine when you deploy a virtual machine or you edit an existing virtual machine.

Option	Action
Create a new virtual machine	1. Select any inventory object that is a valid parent object of a virtual machine and click **Actions**. 2. Click **New Virtual Machine**. 3. On the **Select a creation type** page, select **Create a new virtual machine**, and click **Next**. 4. Navigate through the pages of the wizard. 5. On the **Customize hardware** page, click the **Virtual Hardware** tab.
Edit a virtual machine	1. Select any inventory object that is a valid parent object of a virtual machine and click **Actions**. 2. Click **Edit Settings**. 3. Click the **Virtual Hardware** tab.

2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. From the drop-down menu, under **Other Devices**, select **PCI Device**.
   The **Device Selection** window that contains a list of devices appears.
4. Select the PCI device to connect to the virtual machine and click **Select**.

Option	Action
Fixed DirectPath **Dynamic DirectPath** **Enhanced DirectPath**	1. Select a PCI device from the list and click **Select**. The PCI device appears in the Virtual Hardware device list. 2. Expand the **New PCI device** section and if present, enter the configuration parameters for the PCI device. 3. To remove the PCI device, click **Remove**. The device is removed from the list. 4. To add another PCI device, click the **Add Device** button or undo the operation by clicking **Undo**.
NVIDIA GRID vGPU	1. Select a PCI device from the list and click **Select**. The PCI device appears in the Virtual Hardware device list. 2. Expand the **New PCI device** section.

Option	Action
	3. From the **vGPU Profile** drop-down menu, select the NVIDIA GRID vGPU passthrough device to connect to the virtual machine. 4. To remove the NVIDIA GRID vGPU passthrough device, click **Remove**. The device is removed from the list. 5. To add another NVIDIA GRID vGPU passthrough device, click the **Add Device** button or undo the operation by clicking **Undo**.
Group	1. Select a Vendor Device Groups entry from the list and click **Select**. The Vendor Device Group devices appear in the Virtual Hardware device list. 2. Expand the **New Vendor Device Group** section. 3. If you add a Vendor Device Group that has a NIC, add an appropriate network. 4. From the **Network** drop-down menu click **Browse**, and select an appropriate network. 5. To remove the Vendor Device Group, click the vertical ellipsis icon and click **Remove device**. The Vendor Device Group is removed from the list. 6. To undo the operation, click **Undo**.

5. Click **Next** and complete the wizard.
6. Power on the virtual machine.
   The connected PCI devices appear:
   - On the **Virtual Hardware** tab of the **Edit Settings** wizard.
   - On the **Summary** tab in the **PCI Devices** panel.

   You can clone the virtual machine with the PCI passthrough devices.

How do I Configure 3D Graphics

When you create or edit a virtual machine, you can configure 3D graphics to take advantage of Windows AERO, CAD, Google Earth, and other 3D design, modeling, and multimedia applications.

Before you enable 3D graphics, become familiar with the available options and requirements.

How Enabling 3D Graphics Affects the Virtual Machine

You can use vMotion to migrate virtual machines that have 3D graphics enabled. If the 3D Renderer is set to Automatic, virtual machines use either the GPU on the destination host or a software renderer, depending on GPU availability. To migrate virtual machines with the 3D Renderer set to Hardware, the destination host must have a GPU.

You can set a group of virtual machines to use only Hardware rendering. For example, if you have virtual machines that run CAD applications or have other complex engineering capabilities, you might require that those virtual machines have persistent high-quality 3D capability present. When you migrate such virtual machines, the destination host must also have GPU capability. If the host does not have GPU, the migration cannot proceed. To migrate such virtual machines, you must turn them off and change the renderer setting to Automatic.

You can enable 3D on virtual machines that have Windows desktop or Linux guest operating systems. Not all guests support 3D graphics. To verify 3D support for a guest operating system, see the Broadcom Compatibility Guide.

Prerequisites

VMware supports only Intel Graphics Cards for 3D.

For more information about the graphics cards, see the Broadcom Compatibility Guide.

To use the graphics card or GPU hardware, download the appropriate VMware graphics driver from the Broadcom Support Portal.

Linux distributions must have a 3.2 or later kernel. If 3D is not available on a Linux guest, verify that the driver is available in the Linux kernel. If it is not available, upgrade to a more recent Linux distribution. The location of the kernel depends on whether the distribution is based on `deb` or `rpm`.

Table 129: Linux Driver Location

VMware Linux Guest Kernel Drivers	Debian Format	RPM Format
`vmwgfx.ko`	`dpkg -S vmwgfx.ko`	`rpm -qf vmwgfx.ko`
`vmwgfx_dri.so`	`dpkg -S vmwgfx_dri`	`rpm -qf vmwgfx_dri`
`vmware_drv.so`	`dpkg -S vmware_drv`	`rpm -qf vmware_drv`
`libxatracker.so.1`	`dpkg -S libxatracker`	`rpm -qf libxatracker`

3D Rendering Options

You can select the 3D rendering options for each virtual machine to be Hardware, Software, or Automatic.

Table 130: 3D Rendering Options

Rendering Option	Description
Hardware	The virtual machine must have access to a physical GPU. If the GPU is not available, the virtual machine cannot power on.
Software	The virtual machine's virtual device uses a software renderer and will not attempt to use a GPU, even if one is present.
Automatic	The default setting. The virtual device selects whether to use a physical GPU or software-based rendering. If a GPU is available on the system and has the resources required by the virtual machine, the virtual machine uses the GPU. Otherwise software rendering is used.

Configure 3D Graphics and Video Cards

When you enable 3D graphics, you can select a hardware or software graphics renderer and optimize the graphics memory allocated to the virtual machine. You can increase the number of displays in multi-monitor configurations and change the video card settings to meet your graphics requirements.

- Verify that the virtual machine is powered off.
- Verify that the virtual machine compatibility is ESX 5.0 and later.
- To enable 3D graphics in virtual machines with Windows 8 guest operating systems, the virtual machine compatibility must be ESX 5.1 or later.
- To use a Hardware 3D renderer, ensure that graphics hardware is available. See How do I Configure 3D Graphics.
- If you update the virtual machine compatibility from ESX 5.1 and later to ESX 5.5 and later, reinstall VMware Tools to get the latest SVGA virtual graphics driver and Windows Display Driver Model driver.
- Verify that you have the**Virtual machine** > **Change Configuration** > **Modify device settings** privilege on the virtual machine.

The default setting for total video RAM is adequate for minimal desktop resolution. For more complex situations, you can change the default memory. Typically, 3D applications require a video memory of 256 MB – 1 GB.

Fault Tolerance is not supported for virtual machines that have 3D graphics enabled.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Video Card**.
3. Select custom or automatic settings for your displays from the drop-down menu.

Option	Description
Auto-detect settings	Applies common video settings to the guest operating system.
Specify custom settings	Lets you select the number of displays and the total video memory.

4. Select the number of displays from the drop-down menu.
   You can set the number of displays and extend the screen across them.
5. Enter the required video memory.
6. Optional: Select **Enable 3D support**.
   This check box is active only for guest operating systems on which VMware supports 3D.
7. Optional: Select a 3D Renderer.

Option	Description
Automatic	Selects the appropriate option (software or hardware) for this virtual machine.
Software	Uses normal CPU processing for 3D calculations.
Hardware	Requires graphics hardware (GPU) for faster 3D calculations. **Note:** The virtual machine will not power on if graphics hardware is not available.

8. Click **OK**.

Sufficient memory allocation is set for this virtual machine's graphics.

Reduce Memory Overhead for Virtual Machines with 3D Graphics Option

Virtual machines with the 3D graphics option enabled can have higher memory consumption than other virtual machines. You can reduce the memory overhead by editing the configuration file (`.vmx` file) of your virtual machines and disabling certain memory-related settings. Reducing the memory overhead of virtual machines can help you increase the number of virtual machines per host.

Verify that your virtual machines are using hardware version 10 or later.

1. Shut down the virtual machine on which the 3D graphics option is enabled.
2. Deactivate the **Accelerate 3D Graphics** option.
3. Upgrade your ESX host to use the features available in hardware version 10 or later.
4. Set the maximum size of your display to the size you need.
5. Locate the configuration file (`.vmx`) of your virtual machine.
6. Open the virtual machine configuration file in a text editor and add the line, `svga.vgaOnly=TRUE`.
   This option removes all graphics and SVGA functionality from your SVGA device, but does not remove the settings that allow BIOS to enter VGA mode.
7. Save the changes and exit the text editor.
8. Power on your virtual machine and check the display console.
9. Verify the memory reservation settings in the `vmware.log` file.

How do I Add a Virtual Watchdog Timer Device to a Virtual Machine

To ensure self-reliance that is related to the system performance within a virtual machine, you can add a virtual Watchdog Timer (VWDT) device.

If the guest operating system stops responding and cannot recover on its own due to software glitches or errors, the VWDT waits for a predefined period of time and then restarts the system.

You can enable the VWDT to start either by the guest operating system, or by the BIOS or EFI firmware. If you chose the VWDT to start with the BIOS or EFI firmware, it starts before the guest operating system boots.

The VWDT has an important role in guest-based clustering solutions where each virtual machine in the cluster can recover on its own if it fails.

Add a Virtual Watchdog Timer Device to a Virtual Machine

To prevent the virtual machine from a guest operating system failure for an extended period of time, you can add a VWDT device to the virtual machine.

- Power off the virtual machine.
- Verify that the virtual machine hardware is of version 17 or later.
- Verify that the guest operating system of the virtual machine supports a watchdog timer:

  Note:
  The guest operating system may require you to explicitly enable the watchdog services. Failure to do so might cause the watchdog timer to power off or restart the virtual machine.
  - The VWDT device implements the Watchdog Resource Table (WDRT) and Watchdog Action Table (WDAT) specifications. Windows Server 2003 supports devices compatible with WDRT and Windows Server 2008 and later supports devices compatible with WDAT.
  - The Linux distributions based on 4.9 or later kernel, such as Ubuntu 18.04 and Red Hat Enterprise Linux 7.6, support devices compatible with WDAT if the `wdat_wdt.ko` driver is available.
  - The watchdog timer is not supported for other guest operating systems, such as FreeBSD and Mac OS X.
- Required privileges:
  - **Virtual Machine** > **Change Configuration** > **Add or remove device**
  - **Virtual machine** > **Change Configuration** > **Modify device settings**

1. Right-click a virtual machine from the vSphere inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click **Add New Device** and select **Watchdog Timer** from the drop-down menu.

   The New Watchdog timer device appears in the **Virtual Hardware** devices list.
3. To start the virtual watchdog timer with the BIOS or EFI firmware, select **Start with BIOS/EFI boot**.

   The virtual watchdog timer starts before the guest operating system boots.

   If the guest operating system takes too long to boot or it does not support the VWDT device, a warning message appears, and the VWDT device might constantly restart the virtual machine.
4. Click **OK**.

You can view the status of the VWDT device in the **VM Hardware** panel on the **Summary** tab.

How do I Add a Precision Clock Device to a Virtual Machine

A Precision Clock device is a virtual clock device that provides a virtual machine with access to the system time of the primary ESX host.

- To ensure that the Precision Clock device provides the guest operating system of a virtual machine with accurate time, synchronize the primary ESX host to use Network Time Protocol (NTP) or Precision Time Protocol (PTP). For more information on how to configure the host time synchronization, see the *vSphere Single Host Management - VMware Host Client* documentation.
- Power off the virtual machine.
- Verify that the virtual machine hardware version is 17 or later.
- Required privileges:
  - **Virtual Machine** > **Change Configuration** > **Add or remove device**

– **Virtual machine** > **Change Configuration** > **Modify device settings**

To synchronize the guest operating system of a virtual machine with the host in an efficient manner, add a Precision Clock device to the virtual machine. For information on how to use the Precision Clock device as a reference clock for time synchronization of the supported guest operating system, see the *vCenter and Host Management* documentation.

1. Right-click a virtual machine from the vSphere Client inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click **Add New Device**, and from the drop-down menu select **Precision Clock**.
   The Precision Clock device appears in the **Virtual Hardware** devices list.
3. Select the time synchronization protocol and click **OK**.

Option	Description
Host System Time (Any)	The ESX host time synchronization type has no limitations.
Host System Time (NTP)	You can power on the virtual machine on an ESX host that is configured with NTP time synchronization.
Host System Time (PTP)	You can power on the virtual machine on an ESX host that is configured with PTP time synchronization.

Configure Linux Guest Operating System for Precision Time Synchronization

After you add the Precision Clock device, configure the Linux guest operating system to use it as a PTP reference clock.

The `ptp_vmw` kernel module allows the Linux guest OS to communicate with the Precision Clock device. The module is introduced in Linux kernel version 5.7. If the version of your kernel is earlier than 5.7, you must compile the driver from source.

1. Verify or load the `ptp_vmw` driver.
   a) To verify or load the `ptp_vmw` driver, run the following command

   ```
   # Check whether the ptp_vmw module is loaded.
   lsmod | grep ptp_vmw
   ```

   b) Review the output when the driver is loaded. If the driver is loaded successfully, the system returns the following output:

   ```
   esxcli memtier device list

   ptp_vmw 12288 0
   ```

   c) Load the kernel manually, if the module is not loaded.

   ```
   # Manually load the ptp_vmw kernel module.
   modprobe ptp_vmw
   ```

2. Configure `chrony` to use PTP as its reference clock.
   a) Open the `chrony` configuration file, typically located at `/etc/chrnoy.conf` or `/etc/chrony/chrony.conf`. Verify the correct path for your Linux distribution before editing.
   b) To set the VMware PTP hardware clock (PHC) as the reference clock, append the following line to the configuration file:

   ```
   echo "refclock PHC /dev/ptp_vmware poll 2 dpoll -1" >> /etc/chrony/chrony.conf
   ```

   c) To apply the updated condition, restart `chrony`.

   ```
   bash # Add the VMware PTP hardware clock as a reference clock for chrony.
   systemctl restart chrony
   ```

3. To verify that `chrony` uses the Precision Clock, run the command `chronyc sourcestats -v`.
4. In the output, verify that the first entry is `PHC0` and that its offset converges to a small value in nanoseconds.

   Example of the correctly synchronized state

Name/IP Address	NP	NR	Span	Frequency	Freq Skew	Offset	Std Dev
PHC0	31	14	119	+0.000	0.000	+24 ns	2 ns
pool.ntp.org	2	0	65	-0.000	2000.000	+119 us	4000 ms

   The PHC0 entry corresponds to the Precision Clock device you configured. Depending on your grandmaster clock, the offset gradually converges and stabilizes at a small value (typically in nanoseconds).

   The following table describes each column in the `chronyc sourcestats` output:

Column	Description
Name/IP Address	Name or IP address of the time source. `PHC0` is the Precision Clock device.
NP	Number of sample points in the measurement set.
NR	The number of residual runs with the same sign. A higher value indicates a more stable source.
Span	Length of the measurement in seconds.
Frequency	Estimated clock frequency error, in parts per million (ppm). A value near zero indicates accurate frequency tracking.
Freq Skew	Estimated error in the frequency value, in ppm. A lower value indicates a more reliable frequency estimate.
Offset	Estimated offset of the source clock from the reference. For a correctly synchronized PTP source, this value is typically in nanoseconds.
Std Dev	Standard deviation of the offset measurements. A lower value indicates a more stable source.

Configure Windows Guest Operating System for Precision Time Synchronization

The VMware Time Provider is required to use the Precision Clock device on Windows guest operating systems. It is installed automatically as part of the VMware Tools installation.

After you install VMware Tools, verify that the VMware precision clock is active by running the following command:

```
w32tm /query /source
```

The expected output when the VMware Time Provider is active is `vmwTimerProvider`.

If the output does not show `vmwTimerProvider`, perform the following troubleshooting steps.

1. Verify that VMware Tools is installed correctly and that the VMware Time Provider service is running.
2. Verify that the Windows Time service is running: `sc query w32tm`.
   a. If the **State** field does not show **Running**, start the Windows Time service: `net start w32tm`.
3. If the Windows Time service is running, but `w32tm /query /source` still does not return `vmwTimerProvider`, verify that PTP is correctly configured on the ESX host.

Troubleshooting Precision Time Synchronization

ESX supports PTP with hardware timestamping for a specific set of network devices. The implementation passes a raw PCI network device to a system CRX (Container Runtime for ESX) instance that runs a PTP time synchronization agent (`linuxptp`). The CRX instance is fully integrated into the ESX host date and time services, and it supports APIs, graphical interfaces, and command-line interfaces. The CRX instance also supplies time to the virtual Precision Clock system for virtual machines to consume.

Note:

Hardware-based PTP requires a supported PCI network device that is passthrough-enabled and connected to a PTP network. For a complete list of supported devices, see `/usr/lib/vmware/trx/etc/device.list` on the ESX host.

For more information, see Troubleshooting Precision Time Protocol Setup in vSphere 9.0 and later.

If PTP does not work correctly, isolate the error along the path from the hardware PTP network interface card (NIC) to the PTP time synchronization agents inside the CRX container.

Check the status of the Precision Time Protocol service

1. In the vSphere Client inventory, select an ESX host.
2. On the **Configure** tab, select **System** > **Time Configuration**.
3. Select **Precision Time Protocol** and click **Test Services**.
4. Review the error messages in the **Time Synchronization Services** dialog box.

Check the log files for errors

On the ESX host, the log file resides on one of the following paths:

- `/var/log/trx.log`
- `/var/run/log/trx.log`

Verify hardware PTP prerequisites

Verify that all prerequisites are met for the hardware-based PTP service to operate correctly.

- The PCI device is passthrough-enabled.
- The passthrough PCI device is connected to a PTP network.
- The PCI device is on the supported device list at `/usr/lib/vmware/trx/etc/device.list` on the ESX host.

Virtual Shared Graphics Acceleration

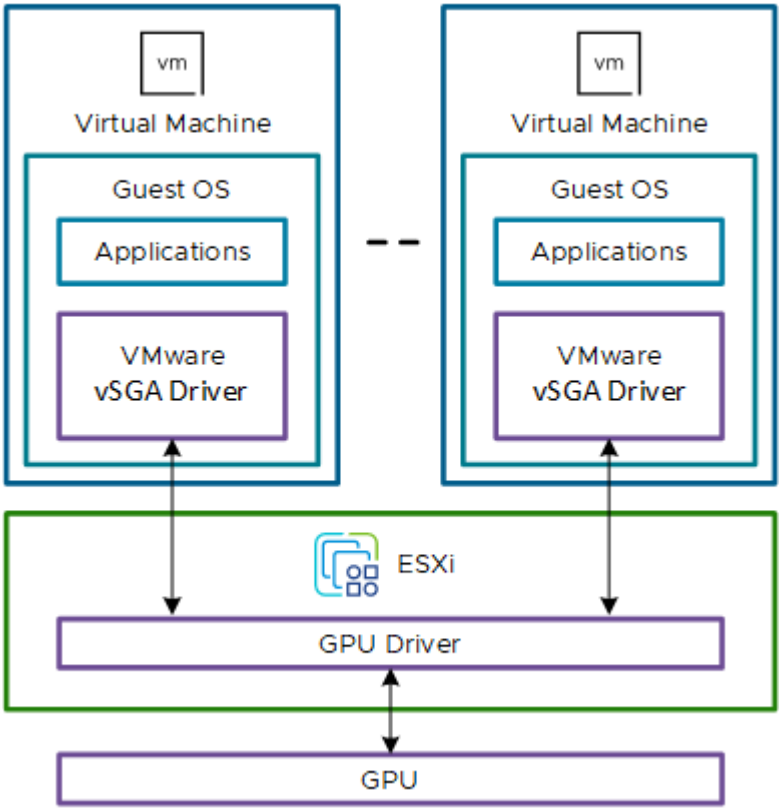
Virtual Shared Graphics Acceleration (vSGA) enables sharing a physical GPU across multiple virtual desktops.

To provide GPU hardware acceleration to VMs, you can use one of the following graphics accelerations:

- Virtual Shared Graphics Acceleration (vSGA).
- Virtual Shared Graphics Passthrough Graphics Acceleration (vGPU/DVX).
- Virtual Dedicated Graphics Acceleration (vDGA).

With vSGA, the physical GPUs in the ESX host are virtualized and shared across multiple VMs.

Figure 42: High-level architecture of vSGA



Graphics Acceleration Types Comparison

The following table compares the features of the three types of graphics acceleration.

Table 131: Types of graphics acceleration

	vSGA	vGPU/DVX	vDGA
Consolidation	High (limited by video memory)	Up to 1:32	None (1:1)
Performance Level	Lightweight to mid-level	Mid-level or Workstation	Workstation
Application Compatibility	Limited	All supported versions	All supported versions
Maximum DirectX Level	• vSphere 7.0 Update 2 or later • DirectX 11.0 / SM5.0	All supported versions	All supported versions
Maximum OpenGL version	• vSphere 8.0 or later • OpenGL 4.3	All supported versions	All supported versions
Video encoding and decoding	Software	Hardware	Hardware
OpenCL or CUDA compute	No	Yes	Yes
vSphere vMotion Support	Yes	Limited	No
Microsoft Virtualization-based Security (VBS)	Yes	No	No

Use Case for vSGA Graphics

vSGA is designed to address the knowledge worker use case where the graphics load is usually low. The knowledge workers use various non-specialized end-user applications that do not demand complex models to render.

Advantages of vSGA

The use of vSGA results in enhanced manageability, streamlined consolidation, and efficiency.

Manageability

vSGA supports the following features which allow an Administrator to easily manage their infrastructure at scale.

- Suspend and resume a VM.
- VM migration with vMotion.
- Guest operating system and ESX host driver compatibility across vSphere versions.
- vSphere DRS for initial placement of VMs and load-balancing.

Consolidation

vSGA allows efficient use of hardware acceleration by leveraging VMs with different performance requirements on a single GPU.

The elasticity of vSGA allows a higher density of VMs per graphics card compared to vGPU and DVX solutions.

Hardware Requirements for Hardware Accelerated Graphics

For more information about the hardware requirements for graphics acceleration solutions, see the following table.

Table 132: Hardware requirements for graphics acceleration

Component	Description
Physical Space for Graphics Cards	Many high-end graphics cards are full-height, full-length, and double-width. Most of them take two slots on the motherboard but use a single PCIe x16 slot. To accommodate the selected GPU card in the appropriate PCIe slot, verify that the ESX host has enough disk space.
Host Power Supply Units (PSU)	Check the power requirements of the GPU and cooling requirements of the server to make sure that the PSU is powerful enough and contains the proper power cables to power the GPU.
Two Display Adapters	If the ESX host does not have an extra graphics adapter, you can install an additional low-end display adapter to act as a primary display adapter. If the GPU is set as a primary display for the ESX host, vSGA cannot use the GPU for rendering. If two GPUs are installed, the server BIOS might have the option to select which GPU is primary and which is secondary.

Set up Your Environment for vSGA

To use vSGA, you must configure the ESX host and set up the VM.

Configure the ESX host

To enable vSGA feature, configure the ESX host by performing the following high-level steps.

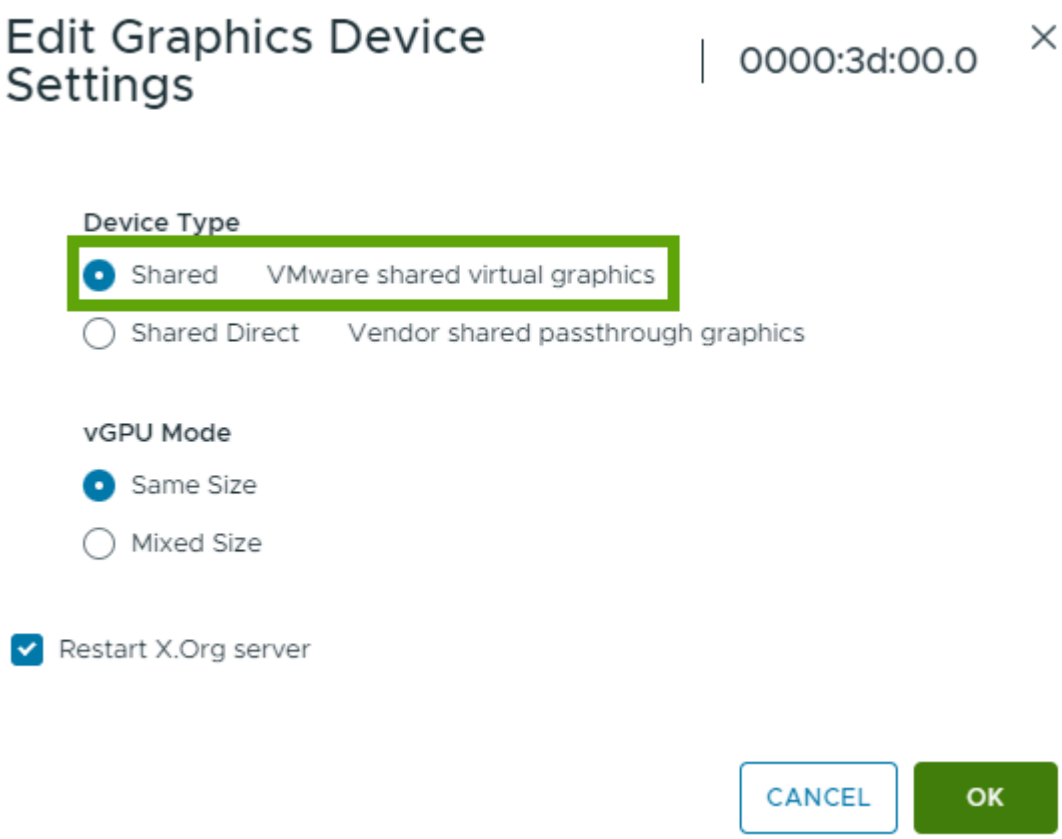
Prerequisites for vSGA

Verify that the ESX host is compatible with ESX 8.0 Update 2 or later.

Procedure

1. Install the graphics card on the ESX host.
2. Place the host in maintenance mode. For more information, see Place a Host in Maintenance Mode.
3. Install the associated GPU graphics card VIB files on the host. For example, run the following ESXCLI command.
   ```
   esxcli software vib install -v $PWD/VendorName-vsga-driver_x.y.z-1OEM.a.b.c.vib
   ```
4. In the vSphere Client, verify that the GPU graphics card is in **Shared** mode.

a. Select the host and navigate to **Configure** > **Hardware** > **Graphics**.
b. Click the **Graphics Device** tab and check the **Configured Type** column.
c. To change the **vGPU Mode**, select the graphics card and click **Edit**.
The **Edit Graphics Device Settings** window appears.



d. Select **Shared** and click **OK**.
5. Reboot the host and exit the maintenance mode.
6. To verify that at least one driver VM starts, from the SSH terminal, run the `crx-cli list` command.

Set up the Virtual Machine General Settings

After you configure the ESX host, you must setup the VM general settings.

Prerequisites for Virtual Machine General Settings

- Verify that the VM is powered off.
- Verify that the VM is not suspended.

Procedure

1. Right-click a VM in the inventory and select **Edit Settings**.

Table 133: General Settings of a VM

Setting	Description
Hardware Version	Make sure that the VM is of the latest hardware version that all hosts support. • Right-click the VM and select **Compatibility**. • If **Upgrade VM Compatibility** is not active, the VM is compatible with the latest hardware version. The minimum hardware version is 11. For DirectX 11.0 and OpenGL 4.3, the hardware version must be compatible with ESX 8.0 and later.
CPU	The required number of CPUs depend on the usage and is determined by the actual workload. To obtain optimal performance, allocate 2 virtual CPUs to the VM.
Memory	The virtual hardware memory size depends on the usage and the number of applications that are running inside the VM. Set the memory size of the VM to • 4 GB if the VM runs 3D applications. • 2 GB if the VM runs basic productivity applications.
Network Adapter	Select the VMXNET 3 adapter type.
SCSI Controller	Select the LSI Logic SAS controller type. For the highest workloads using local flash-based storage, use VMware Paravirtual.
Other Devices	Remove all devices that are not used, such as COM, LTP, DVD, or floppy.

Configure a VM for Enabling vSGA

To enable vSGA, configure the VM by performing the following high-level steps.

Procedure

1. To enable 3D graphics, select the **Enable 3D Support** check box. For more information, see How do I Configure 3D Graphics.
2. Set **3D Renderer** to **Automatic** or **Hardware**.
3. Select the size of the graphics **3D Memory**.
   – 512 MB if the VM runs basic productivity applications.
   – 2048 MB if the VM runs 3D applications. For maximum consolidation, set the **3D Memory** size to the minimum required by the guest OS applications. Guest operating systems typically cannot use more graphics memory than half of the available system memory. Therefore, to run larger graphics workloads, you must increase the Guest OS memory.

Configure a Windows VM for vSGA

After you configure the general settings for a VM, download and install the corresponding VMware Tools version in the VM.

Procedure

1. Install VMware Tools. For more information, see Installing VMware Tools.
2. Restart the VM and run `dxdiag.exe`.
3. Click the **Display** tab. You should see the following parameters.

Table 134: Device panel and Driver panel parameters

Device Panel	Drivers Panel
• Name: VMware SVGA 3D • Device Type: Full Display Device	• Direct3D DDI: 11.1 • Feature levels: 11_0, 10_1, 10_0, 9_3, 9_2, 9_1 • Driver Model: WDDM 1.2

Table 135: VMware Tools version support for WDDM, DirectX, and OpenGL

VMware Tools Version	Max WDDM Version	Max DirectX Version	Max OpenGL Version
VMware Tools 10.0	WDDM 1.0	DirectX 10.0	OpenGL 3.3
VMware Tools 10.3.2	WDDM 1.0	DirectX 10.1	OpenGL 3.3
VMware Tools 11.1.5	WDDM 1.1	DirectX 11.0	OpenGL 4.1
VMware Tools 12.0	WDDM 1.2	DirectX 11.0	OpenGL 4.3

Configure a Linux VM for vSGA

After you configure the general settings for a VM, install Open VM Tools (open-vm-tools) in the Linux VM.

Procedure

1. Install Open VM Tools. For more information, see the *VMware Tools* documentation Install Open VM Tools .
2. Verify that the VMware graphics driver is installed.

3. To make sure that OpenGL applications work in the VM, install Mesa in the VM and verify that the OpenGL version is 3.3 or later.
   For example, run the following command `glxinfo | grep "OpenGL version"`.
4. Verify the vmwgfx drm (direct rendering manager) driver version by using the drm_info tool.
   For example, run the following script.

```
drm_info | head -n 2
Node: /dev/dri/card0
——Driver: vmwgfx (Linux drm driver for VMware graphics devices) version 2.20.0 (20211206)
```

Table 136: Linux DRM, Mesa, and OpenGL versions

Linux DRM Version	Mesa Version	OpenGL Version
DRM 2.9	Mesa 11.1.0	OpenGL 3.3
DRM 2.18	Mesa 20.2.0	OpenGL 4.1
DRM 2.20	Mesa 22.0.0	OpenGL 4.3

USB Configuration from an ESX Host to a Virtual Machine

You can add multiple USB devices to a virtual machine when the physical devices are connected to an ESX host. USB passthrough technology supports adding USB devices to virtual machines that reside on the host to which the devices are connected, such as security dongles and mass storage devices.

How USB Device Passthrough Technology Works

When you attach a USB device to a physical host, the device is available only to virtual machines that reside on that host. The device cannot connect to virtual machines that reside on another host in the data center.

A USB device is available to only one virtual machine at a time. When you connect a device to a powered on virtual machine, the device is not available to connect to other virtual machines that run on the host. When you remove the active connection of a USB device from a virtual machine, it becomes available to the other virtual machines that run on the host.

To connect a USB passthrough device to a virtual machine that runs on the ESX host where the device is physically attached, you require an arbitrator, a controller, and a physical USB device or device hub.

USB Arbitrator	Manages connection requests and routes the USB device traffic. The arbitrator is installed and enabled by default on ESX hosts. It scans the host for USB devices and manages the device connection among virtual machines that reside on the host. It routes the device traffic to the correct virtual machine for delivery to the guest operating system. The arbitrator monitors the USB device and prevents other virtual machines from using it until you release it from the virtual machine it is connected to.
USB Controller	The USB hardware chip that provides a USB function to the USB ports that it manages. The virtual USB controller is the software virtualization of the USB host controller function in the virtual machine. USB controller hardware and modules that support USB devices, such as USB 3.2 Gen 2x2, USB 3.1 SuperSpeedPlus, USB 3.1 SuperSpeed, and USB 2.0 must exist on the host. A controller must be present before you can add a USB device to the virtual machine. The USB arbitrator can monitor a maximum of 15 USB controllers. Devices connected to controllers numbered 16 or greater are not available to the virtual machine.
USB Devices	You can add up to 20 USB devices to a virtual machine, which is the maximum number of devices supported for a simultaneous connection to one virtual machine. The maximum number of USB devices supported on a single ESX host for a simultaneous connection to one or more virtual machines is also 20. For more information, see Broadcom knowledge base article 315272.

USB 3.1 SuperSpeed Device Requirements

Starting with vSphere 5.5 Patch 3, USB 3.1 SuperSpeed devices are available for passthrough not only from a client computer to a virtual machine, but also from an ESX host to a virtual machine. USB 3.1 SuperSpeed devices still have the following virtual machine configuration requirement:

- The virtual machine must have an enabled xHCI controller, Windows 8 or later, Windows Server 2012 and later, or a Linux guest operating system with a 2.6.35 or later kernel.

USB 3.1 SuperSpeedPlus Device Requirements

Starting with vSphere 7.0, USB 3.1 SuperSpeedPlus devices are available for passthrough at their maximum speed (SuperSpeedPlus), not only from a client computer to a virtual machine, but also from an ESX host to a virtual machine. To operate their maximum transfer speed, USB 3.1 SuperSpeedPlus devices have the following virtual machine configuration requirements:

- The virtual machine must have an enabled xHCI controller, Windows 10 or later, Windows Server 2016 and later, or a Linux guest operating system with a 4.6 or later kernel.
- The virtual machine must be of hardware version 17 or later.
- For requirements and steps how to enable USB 3.1 SuperSpeedPlus, see Broadcom knowledge base article 324281.

USB 3.2 Gen 2x2 Device Requirements

Starting with vSphere 8.0 Update 2, USB 3.2 Gen 2x2 devices are available for passthrough at their maximum speed of 20 Gbps, not only from a client computer to a virtual machine but also from an ESX host to a virtual machine. To operate at their maximum data transfer speed, USB 3.2 Gen 2x2 devices have the following virtual machine configuration requirements:

- The virtual machine must have an enabled xHCI controller, Windows 10 or later, Windows Server 2016 or later, or a Linux guest operating system with a 4.6 or later kernel.
- The virtual machine must be of hardware version 21 or later.

How the USB Autoconnect Feature Works

When you add a USB device connection from an ESX host to a virtual machine, the autoconnect feature is activated for the device connection. It is not deactivated until you remove the device connection from the virtual machine.

With autoconnect enabled, the device reconnects in the following cases:

- The virtual machine is cycling through power operations, such as Power Off/Power On, Reset, Pause/Resume.
- The device is unplugged from the host, and then plugged back in to the same USB port.
- The device is power cycled but has not changed its physical connection path.
- The device is mutating identity while it is in use.
- A new virtual USB device is added.

The USB passthrough autoconnect feature identifies the device by using the USB path of the device on the host. It uses the physical topology and port location instead of the device identity.

If you plug the same device back into a different USB port on the host, it cannot reestablish the connection with the virtual machine. If you unplug the device from the host and plug in a different device to the same USB path, the new device appears. It is connected to the virtual machine by the autoconnect feature that you enabled in the previous device connection.

Autoconnect is useful when devices mutate during usage. For example, for iPhone and other similar devices, the device VID/PID changes during software or firmware upgrades. The upgrade process disconnects and reconnects the devices to the USB port.

The USB port is speed-specific. If you change a USB device with another USB device that works with different speed, the autoconnect feature might not work. For example, you might connect a USB 2.0 high-speed device to a port and connect that device to the virtual machine. If you unplug the device from the host and plug in a USB 3.2 Gen 2x2, USB 3.1 SuperSpeed, or 3.1 SuperSpeedPlus device to the same port, the device might not connect to the virtual machine.

For a list of supported USB devices for passthrough from an ESX host to a virtual machine, see Broadcom knowledge base article 315272.

vSphere Features Available with USB Passthrough

Migrations with vMotion and DRS are supported with USB device passthrough from an ESX host to a virtual machine.

Table 137: vSphere Features Available for USB Passthrough from an ESX Host to a Virtual Machine

Feature	Supported with USB Device Passthrough
vSphere Distributed Power Management (DPM)	No
vSphere Distributed Resource Scheduler (DRS)	Yes
vSphere Fault Tolerance	No
vSphere vMotion	Yes

For details about migration with vMotion, see Conditions and Limitations When Configuring USB Devices for vMotion.

If a host with connected USB devices resides in a DRS cluster with activated DPM, you must deactivate DPM for that host. Otherwise DPM might power off the host with the device, which disconnects the device from the virtual machine.

Conditions and Limitations When Configuring USB Devices for vMotion

With USB passthrough from a host to a virtual machine, you can migrate a virtual machine to another ESX host in the same datacenter and maintain the USB passthrough device connections to the original host.

If a virtual machine has USB devices attached that pass through to an ESX host, you can migrate that virtual machine with the devices attached.

For a successful migration, review the following conditions:

- You must configure all USB passthrough devices connected to a virtual machine for vMotion. If one or more devices is not configured for vMotion, the migration cannot proceed. For troubleshooting details, see the Troubleshooting USB Passthrough Devices documentation.
- When you migrate a virtual machine with attached USB devices away from the host to which the devices are connected, the devices remain connected to the virtual machine. However, if you suspend or power off the virtual machine, the USB devices are disconnected and cannot reconnect when the virtual machine is resumed. The device connections can be restored only if you move the virtual machine back to the host to which the devices are attached.
- If you resume a suspended virtual machine that has a Linux guest operating system, the resume process might mount the USB devices at a different location on the file system.
- If a host with attached USB devices resides in a DRS cluster with activated distributed power management (DPM), deactivate DPM for that host. Otherwise DPM might turn off the host with the attached device. This action disconnects the device from the virtual machine because the virtual machine migrated to another host.
- Remote USB devices require that the hosts be able to communicate over the management network following migration with vMotion, so the source and destination management network IP address families must match. You cannot migrate a virtual machine from a host that is registered to vCenter with an IPv4 address to a host that is registered with an IPv6 address.

Avoiding Data Loss with USB Devices

When a virtual machine connects to a physical USB device on an ESX host, virtual machine functions can affect USB device behavior and connections.

- Before you hot add memory, CPU, or PCI devices, you must remove any USB devices. Hot adding these resources disconnects USB devices, which might result in data loss.
- Before you suspend a virtual machine, make sure that a data transfer is not in progress. During the suspend or resume process, USB devices behave as if they have been disconnected, then reconnected. For information about suspend and resume behavior after migration with vMotion, see Conditions and Limitations When Configuring USB Devices for vMotion.
- Before you change the state of the arbitrator, make sure that USB devices residing on the host are not attached to a virtual machine. If USB devices become unavailable to a virtual machine, a host administrator might have deactivated the arbitrator. When an administrator stops or disconnects the arbitrator for troubleshooting or other purposes, USB devices attached to that host become unavailable to the virtual machine. If a data transfer is taking place at this time, you might lose the data. To reestablish the arbitrator, you must restart the host or restart the `usbarbitrator` and `hostd` services. Restarting the services requires that you power off and then power on the virtual machine.

Connecting USB Devices to an ESX Host

You can connect and chain multiple USB hubs and devices to an ESX host. Careful planning and knowledge of hub behavior and limitations can help ensure that your devices work optimally.

USB physical bus topology defines how USB devices connect to the host. Support for USB device passthrough to a virtual machine is available if the physical bus topology of the device on the host does not exceed tier seven. The first tier is the USB host controller and root hub. The last tier is the target USB device. You can cascade up to five tiers of external or internal hubs between the root hub and the target USB device. An internal USB hub attached to the root hub or built into a compound device counts as one tier.

The quality of the physical cables, hubs, devices, and power conditions can affect USB device performance. To ensure the best results, keep the host USB bus topology as simple as possible for the target USB device, and use caution when you deploy new hubs and cables into the topology. The following conditions can affect USB behavior:

- Communication delay between the host and virtual machine increases as the number of cascading hubs increases.
- Connecting or chaining multiple external USB hubs increases device enumeration and response time, which can make the power support to the connected USB devices uncertain.
- Chaining hubs together also increases the chance of port and hub error, which can cause the device to lose connection to a virtual machine.
- Certain hubs can cause USB device connections to be unreliable, so use care when you add a new hub to an existing setup. Connecting certain USB devices directly to the host rather than to a hub or extension cable might resolve their connection or performance issues.

Note:
To prevent additional problems, be aware of the physical constraints of long-term deployment in a machine room environment. Small devices are easily damaged by being stepped on or knocked loose.

In some cases, you must hard reset the device and hub to restore the device to a working state.

For a list of supported USB devices for passthrough from an ESX host to a virtual machine, see Broadcom knowledge base article 315272.

USB Compound Devices

For compound devices, the virtualization process filters out the USB hub so that it is not visible to the virtual machine. The remaining USB devices in the compound appear to the virtual machine as separate devices. You can add each device to the same virtual machine or to different virtual machines if they run on the same host.

For example, the Aladdin HASP HL Drive USB dongle package contains three devices (0529:0001 HASP dongle, 13fe:1a00 Hub, 13fe:1d00 Kingston Drive). The virtualization process filters out the USB hub. The remaining Aladdin HASP HL Drive USB dongle devices (one Aladdin HASP dongle and one Kingston Drive) appear to the virtual machine as individual devices. You must add each device separately to make it accessible to the virtual machine.

Add USB Devices to an ESX Host

You can connect multiple USB devices to an ESX host so that the virtual machines that run on the same host can access the devices. The number of devices that you can connect depends on the device type and how the devices and hubs chain together.

- If a host has attached USB devices and resides in a DRS cluster with activated DPM, deactivate DPM for that host. For instructions about overriding the default DPM setting for a single host, see *vSphere Resource Management*.
- To verify that the virtual machine meets the requirements for the USB device connection, see Connecting USB Devices to an ESX Host.
- To add eight xHCI controllers to the ESX host, verify that the current version of your ESX host is 6.0 or later.

Each ESX host has several USB ports. The number of ports on each host depends on the physical setup of the host. When you calculate the depth of hub chaining, remember that on a typical server the front ports connect to an internal hub.

The USB arbitrator can monitor a maximum of 15 USB controllers. If your system includes more than 15 controllers and you connect USB devices to them, the devices are not available to the virtual machine.

The host treats USB CD/DVD-ROM devices as SCSI devices.

- To add a USB device to an ESX host, connect the device to an available port or hub.

You can add the device to the virtual machine. See Add USB Devices from an ESX Host to a Virtual Machine.

Add a USB Controller to a Virtual Machine

To support USB passthrough from an ESX host or from a client computer to a virtual machine, you can add a USB controller to the virtual machine.

- Verify that the ESX host has USB controller hardware and modules that support USB 3.2, 2.0, and 3.1 devices.
- Verify that the client computers have USB controller hardware and modules that support USB 3.2, 2.0, and 3.1 devices present.
- To use the xHCI controller on a Linux guest operating system, verify that the Linux kernel version is 2.6.35 or later.
- Verify that the virtual machine is powered on.
- Required Privilege (ESX host passthrough): **Virtual Machine** > **Change Configuration** > **Add or Remove Device**.

In the vSphere Client, you can add one xHCI controller and one EHCI+UHCI controller.

- From hardware version 11 to hardware version 16, the supported number of root hub ports per xHCI controller is eight (four local USB 3.1 SuperSpeed ports and four logical USB 2.0 ports).

- From hardware version 17 to hardware version 20, the supported number of root hub ports per xHCI controller is eight (four local USB 3.1 SuperSpeedPlus ports and four logical USB 2.0 ports).
- With hardware version 21, the supported number of root hub ports per xHCI controller is eight (four local USB 3.2 Gen 2x2 ports and four logical USB 2.0 ports).

The conditions for adding a controller vary, depending on the device version, the type of passthrough (host or client computer), and the guest operating system.

Table 138: USB Controller Support

Controller Type	Supported USB Device Version	Supported for Passthrough from ESX host to a VM	Supported for Passthrough from Client Computer to a VM
EHCI+UHCI	2.0	Yes	Yes
xHCI	3.2, 3.1, 2.0	Yes USB 3.2, 3.1, and 2.0	Yes Windows 8 or later, Windows Server 2012 and later, or a Linux guest operating system with a 2.6.35 or later kernel.

For Mac OS X systems, the EHCI+UHCI controller is enabled by default and is required for access to a USB mouse and keyboard.

For virtual machines with Windows or Linux guest operating systems, you can add one or two controllers of different types. You cannot add two controllers of the same type.

For USB passthrough from an ESX host to a virtual machine, the USB arbitrator can monitor a maximum of 15 USB controllers. If your system includes more than 15 controllers and you connect USB devices to them, the devices are not available to the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and from the drop-down menu select **USB Controller**.
   The controller appears in the **Virtual Hardware** devices list.
4. Select the USB cotroller type from the **New USB Controller** drop-down menu.
   If compatibility errors appear, you must fix them before you can add the controller.
5. Click **OK**.

Add one or more USB devices to the virtual machine.

Add USB Devices from an ESX Host to a Virtual Machine

You can add one or more USB passthrough devices from an ESX host to a virtual machine if the physical devices are connected to the host on which the virtual machine runs.

- Verify that the virtual machine is compatible with ESX/ESX 7.0 and later.
- Verify that a USB controller is present. See Add a USB Controller to a Virtual Machine.
- To use vMotion to migrate a virtual machine with multiple USB devices, enable all attached USB devices for vMotion. You cannot migrate individual USB devices. For vMotion limitations, see Conditions and Limitations When Configuring USB Devices for vMotion.
- When you add a CD/DVD-ROM drive that is backed by a USB CD/DVD drive on the host, add the drive as a SCSI device.
- Verify that you know the virtual machine requirements for USB devices. See USB Configuration from an ESX Host to a Virtual Machine.
- Required privileges:
  – **Virtual Machine** > **Change Configuration** > **Configure Host USB device**

– **Virtual Machine** > **Change Configuration** > **Add or remove device**

If a USB device is connected to another virtual machine, you cannot add it until that machine releases it.

Note:

If you have the Apple Frontpanel Controller device in your environment, you can safely add it to a virtual machine. However, this device has no documented function and no known use. ESX hosts do not use it and do not provide Xserver functionality for USB passthrough.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, click the **Add New Device** button.
3. Select **Host USB Device** from the drop-down menu.
   The new USB device appears at the bottom of the Virtual Hardware device list.
4. Expand **New USB Device**, and select the device to add.
   You can add multiple USB devices, but only one device at a time.
5. If you do not plan to migrate a virtual machine with USB devices attached, deselect the **Support vMotion** option.
   This action reduces migration complexity, which results in better performance and stability.
6. Click **OK**.

Remove USB Devices That Are Connected Through an ESX Host

When you remove USB devices from a virtual machine, devices that use the passthrough technology from a host become available to other virtual machines that run on that host.

- Verify that the devices are not in use.
- To minimize the risk of data loss, follow the instructions to safely unmount or eject hardware for your operating system. Safely removing hardware allows accumulated data to be transmitted to a file. Windows operating systems typically include a Remove Hardware icon located in the System Tray. Linux operating systems use the `umount` command.

  Note:
  You might need to use the `sync` command instead of or in addition to the `umount` command, for example, after you issue a `dd` command on Linux or other UNIX operating systems.

1. Unmount or eject the USB device from the guest operating system.
2. Right-click a virtual machine in the inventory and select **Edit Settings**.
3. To remove the device, move the pointer over the device and click the **Remove device** icon.
4. Click **OK** to save your changes.

Remove USB Devices from an ESX Host

You can remove USB devices from the host if you must shut down the host for maintenance or if you do not want those devices to be available to virtual machines that run on the host. When you detach a USB device from the host, the device disconnects from the virtual machine.

Verify that the USB devices are not in use.

CAUTION:
If data transfer is taking place when you remove USB devices from a host, you can lose data.

- Follow the device manufacturers instructions to safely remove the device.
  When you remove the device from the host, it is no longer available to the virtual machines that run on the host.

USB Configuration from a Client Computer to a Virtual Machine

You can add multiple USB devices to a virtual machine when the physical devices are connected to the client computer on which you run the vSphere Client. The vSphere Client must be logged in to a vCenter instance that manages the ESX

host where the virtual machine resides. USB passthrough technology supports adding multiple USB devices, such as security dongles, mass storage devices, and smartcard readers to virtual machines.

How USB Device Passthrough Technology Works

The USB controller is the USB hardware chip that provides a USB function to the USB ports that it manages. The USB controller hardware and modules that support USB 3.2 Gen 2x2, USB 3.1 SuperSpeedPlus, USB 3.1 SuperSpeed, and USB 2.0 devices must be available on the virtual machine. Two USB controllers are available for each virtual machine. The controllers support multiple USB devices, such as USB 3.2 Gen 2x2, USB 3.1 SuperSpeedPlus, USB 3.1 SuperSpeed, and USB 2.0. The controller must be present before you can add USB devices to the virtual machine.

You can add up to 20 USB devices to a virtual machine, which is the maximum number of devices supported for a simultaneous connection to one virtual machine. You can add the devices only one at a time.

The virtual machine retains its connection to the device while it is in an S1 standby state. USB device connections are preserved when you migrate virtual machines to another host in the data center.

A USB device is available to only one powered on virtual machine at a time. When a virtual machine connects to a device, that device is no longer available to other virtual machines or to the client computer. When you disconnect the device from the virtual machine or shut down the virtual machine, the device returns to the client computer and becomes available to other virtual machines that the client computer manages.

For example, when you connect a USB mass storage device to a virtual machine, it is removed from the client computer and does not appear as a drive with a removable device. When you disconnect the device from the virtual machine, it reconnects to the client computer's operating system and is listed as a removable device.

Avoiding Data Loss

Before you connect a device to a virtual machine, make sure that the device is not in use on the client computer.

If the vSphere Client disconnects from the vCenter system or host, or if you restart or shut down the client computer, the device connection breaks. It is best to have a dedicated client computer for USB device use or to reserve USB devices connected to a client computer for short-term use, such as updating software or adding patches to virtual machines. To maintain USB device connections to a virtual machine for an extended time, use USB passthrough from an ESX host to the virtual machine.

USB 3.1 SuperSpeed Device Requirements

Starting with vSphere 5.5 Patch 3, USB 3.1 SuperSpeed devices are available for passthrough not only from a client computer to a virtual machine, but also from an ESX host to a virtual machine. USB 3.1 SuperSpeed devices still have the following virtual machine configuration requirement:

- The virtual machine must have an enabled xHCI controller, Windows 8 or later, Windows Server 2012 and later, or a Linux guest operating system with a 2.6.35 or later kernel.

USB 3.1 SuperSpeedPlus Device Requirements

Starting with vSphere 7.0, USB 3.1 SuperSpeedPlus devices are available for passthrough at their maximum speed (SuperSpeedPlus), not only from a client computer to a virtual machine, but also from an ESX host to a virtual machine. To operate their maximum transfer speed, USB 3.1 SuperSpeedPlus devices have the following virtual machine configuration requirements:

- The virtual machine must have enabled xHCI controller, Windows 10 or later, Windows Server 2016 or later, or a Linux guest operating system with a 4.6 or later kernel.
- Verify that the virtual machine hardware is of version 17 or later.
- For requirements and steps how to enable the USB 3.1 SuperSpeedPlus, see Broadcom knowledge base article 324281.

USB 3.2 Gen 2x2 Device Requirements

Starting with vSphere 8.0 Update 2, USB 3.2 Gen 2x2 devices are available for passthrough at their maximum speed of 20 Gbps, not only from a client computer to a virtual machine but also from an ESX host to a virtual machine. To

operate at their maximum data transfer speed, USB 3.2 Gen 2x2 devices have the following virtual machine configuration requirements:

- The virtual machine must have an enabled xHCI controller, Windows 10 or later, Windows Server 2016 or later, or a Linux guest operating system with a 4.6 or later kernel.
- The virtual machine must be of hardware version 21 or later.

How to Connect USB Devices to a Client Computer

You can connect multiple USB devices to a client computer so that virtual machines can access the devices. The number of devices that you can add depends on several factors, such as how the devices and hubs chain together and the device type.

USB physical bus topology defines how USB devices connect to the client computer. Support for USB device passthrough to a virtual machine is available if the physical bus topology of the device on the client computer does not exceed tier seven. The first tier is the USB host controller and root hub. The last tier is the target USB device. You can cascade up to five tiers of external or internal hubs between the root hub and the target USB device. An internal USB hub attached to the root hub or built into a compound device counts as one tier.

The quality of the physical cables, hubs, devices, and power conditions can affect USB device performance. To ensure the best results, keep the client computer USB bus topology as simple as possible for the target USB device, and use caution when you deploy new hubs and cables into the topology. The following conditions can affect USB behavior:

- Connecting or chaining multiple external USB hubs increases device enumeration and response time, which can make the power support to the connected USB devices uncertain.
- Chaining hubs together increases the chance of port and hub error, which can cause the device to lose connection to a virtual machine.
- Certain hubs can cause USB device connections to be unreliable, so use care when you add a new hub to an existing setup. Connecting certain USB devices directly to the client computer rather than to a hub or extension cable might resolve their connection or performance issues. In some cases, you must remove and reattach the device and hub to restore the device to a working state.

The USB arbitrator can monitor a maximum of 15 USB controllers. If your system includes controllers that exceed the 15-controller limit and you connect USB devices to them, the devices are not available to the virtual machine.

USB Compound Devices

For compound devices, the virtualization process filters out the USB hub so that it is not visible to the virtual machine. The remaining USB devices in the compound appear to the virtual machine as separate devices. You can add each device to the same virtual machine or to different virtual machines if they run on the same host.

For example, the Aladdin HASP HL Drive USB dongle package contains three devices (0529:0001 HASP dongle, 13fe:1a00 Hub, 13fe:1d00 Kingston Drive). The virtualization process filters out the USB hub. The remaining Aladdin HASP HL Drive USB dongle devices (one Aladdin HASP dongle and one Kingston Drive) appear to the virtual machine as individual devices. You must add each device separately to make it accessible to the virtual machine.

Procedure

To add a USB device to a client computer, connect the device to an available port or hub.

What to do next:

You can now add the USB device to the virtual machine.

For more information on how to add a USB controller to a virtual machine, see Add a USB Controller to a Virtual Machine.

Add USB Devices from a Client Computer to a Virtual Machine

You can add one or more USB passthrough devices from a client computer to a virtual machine in the vSphere Client. The devices must be connected to a client computer that connects to the ESX host on which the virtual machine resides.

- Verify that a USB device is connected to the client computer.
- Verify that the virtual machine is powered on.
- Verify that a USB controller is present.
- Verify that the vSphere Client has access to the ESX host on which the virtual machines are running.
- Verify that you have the following privileges:
  - **Virtual machine** > **Change Configuration** > **Add or remove device**
  - **Virtual machine** > **Interaction** > **Device connection**

Note:
If you connect to a USB device on a Mac OS X client computer, you can add only one device to the virtual machine at a time.

The devices maintain their virtual machine connections in S1 standby, if the vSphere Client is running and connected. After you add the USB device to the virtual machine, a message on the client computer states that the device is disconnected. The device remains disconnected from the client computer until you disconnect it from the virtual machine.

Fault Tolerance is not supported with USB passthrough from a client computer to a virtual machine.

1. In the vSphere Client, navigate to a virtual machine.
2. Launch the VMware Remote Console application.

   Note:
   You cannot connect a USB device to a virtual machine if you use the HTML5 console in the vSphere Client.
3. In the VMware Remote Console toolbar, click **VMRC** > **Removable Devices** and find the USB device.
4. Click **Connect (Disconnect from menu)**.

The USB device is connected to the virtual machine.

Remove USB Devices That Are Connected Through a Client Computer

You can remove USB devices from a virtual machine if the devices are no longer needed. When you disconnect a USB device from a virtual machine, the device is released from the virtual machine and is given back to the client computer, which starts using it.

- Verify that the virtual machine is powered on.
- To minimize the risk of data loss, follow the instructions to safely unmount or eject hardware for your operating system. Safely removing hardware allows accumulated data to be transmitted to a file. Windows operating systems typically include a Remove Hardware icon located in the System Tray. Linux operating systems use the `umount` command.

  Note:
  You might need to use the `sync` command instead of or in addition to the `umount` command, for example after you run a `dd` command on Linux or other UNIX operating systems.
- Required privileges: **Virtual machine** > **Change Configuration** > **Add or remove device** and **Virtual machine** > **Interaction** > **Connect devices** to disconnect from the user interface.

1. Unmount or eject the USB device from the guest operating system.
2. On the virtual machine **Summary** tab, click the disconnect icon on the right side of the USB device entry.
3. Select a device to disconnect from the drop-down menu.

   A **Disconnecting** label and a spinner appear, indicating that a disconnection is in progress. When the device is disconnected, after a slight delay, the **Summary** tab refreshes and the device is removed from the virtual machine configuration.

The device reconnects to the client computer and is available to add to another virtual machine. In some cases, Windows Explorer detects the device and opens a dialog box on the client computer. You can close this dialog box.

Remove a USB Controller from a Virtual Machine

You can remove a USB controller from the virtual machine if you do not want to connect to USB devices.

> **Note:**
> The ESX host supports hot-removal of a USB controller. You can remove a USB controller from a virtual machine when both the ESX host and the guest operating system of the virtual machine support the hot-removal functionality.
>
> To make sure that the guest operating system of the virtual machine supports the hot-removal functionality, refer to the vendors documentation.

- Verify that the USB controller is not in use.
- Verify that all USB devices are disconnected from the virtual machine.
- Required Privilege: **Virtual Machine** > **Change Configuration** > **Add or Remove Device**

1. Navigate to a data center, folder, cluster, resource pool, host, or vApp, click the **VMs** tab and click **Virtual Machines**.
2. Right-click a virtual machine and click **Edit Settings**.
3. On the **Virtual Hardware** tab, move the pointer over the USB controller and click the **Remove device** icon.
4. Click **OK** to confirm the deletion and close the dialog box.

The controller is no longer connected to the virtual machine, but you can add it later.

Add a Shared Smart Card Reader to Virtual Machines

You can configure multiple virtual machines to use a virtual shared smart card reader for smart card authentication. The smart card reader must be connected to a client computer on which the vSphere Client runs. All smart card readers are treated as USB devices.

- Verify that the smart card reader is connected to the client computer.
- Verify that the virtual machine is powered on.
- Verify that a USB controller is present.
- Required Privilege: **Virtual machine** > **Change Configuration** > **Add or remove device**

A license is required for the shared smart card feature. See *vCenter and Host Management*.

When you log out of Windows XP guest operating systems, to log back in, you must remove the smart card from the smart card reader and re-add it. You can also disconnect the shared smart card reader and reconnect it.

If the vSphere Client disconnects from the vCenter or host, or if the client computer is restarted or shut down, the smart card connection breaks. For this reason, it is best to have a dedicated client computer for smart card use.

To connect a USB smart card reader that is not shared, see USB Configuration from a Client Computer to a Virtual Machine.

1. Navigate to a datacenter, folder, cluster, resource pool, host, or vApp, and click the **Related Options** tab and click **Virtual Machines**.
2. Select a virtual machine, click it again, and click the **Summary** tab.
3. Click the USB icon on the right side of **USB Devices** under **VM Hardware**, and select an available shared smart card reader from the drop-down menu.

   Select a device that appears as **Shared** ***`the model name of your smart card reader`*** followed by a number.

   A **Connecting** label and a spinner appear showing that a connection is in progress. When the device has successfully connected and the Summary tab refreshes, the device is connected and the device name appears next to **USB Devices**.

You can now use smart card authentication to log in to virtual machines in the vSphere Client inventory.

Configuring Virtual Machine Options

You can set or change virtual machine options to run VMware Tools scripts, control user access to the remote console, configure startup behavior, and more.

The virtual machine options define a range of virtual machine properties, such as the virtual machine name and the virtual machine behavior with the guest operating system and VMware Tools.

You can view or change virtual machine settings from the vSphere Client. Not all options are available to every virtual machine and some options rarely must change from their defaults.

The host that the virtual machine runs on and the guest operating system must support any configurations that you make.

You can view and change virtual machine settings on the **VM Options** tab of the **Edit Settings** wizard.

You can select one of the following options.

Table 139: Virtual Machine Options in the vSphere Client

Options	Description
General Options	In this section, you can view or change the following settings. • Virtual machine name • Virtual machine configuration file location • Virtual machine working location • Guest operating system and OS version Currently, you can only edit the virtual machine name. The information about the other settings is read only. To change the operating system for a VM, you have to reinstall the OS - or consider deploying a new VM with your operating system of choice.
VMware Remote Console Options	In this section, you can change the locking behavior of a virtual machine and the settings for simultaneous connections.
Encryption	In this section, you can change the encryption settings of a virtual machine.
Confidential Compute	Enable hardware-based isolation and encryption of the virtual machine memory.
VMware Tools	In this section, you can change the behavior of VMware Tools scripts. You can also customize the automatic VMware Tools upgrades, automatically synchronize the guest time of the virtual machine on startup or resume with the host, and periodically synchronize the guest time with the host.
Virtualization Based Security	Activate or deactivate VBS for the virtual machine.
Boot Options	In this section, you can change the virtual machine boot options. For example, add a delay before booting, force entry into the BIOS or EFI setup screen, or set reboot options.
Power Management	In this section, you can change virtual machine suspend behavior.
Advanced	In this section, you can change the following advanced virtual machine options. • Acceleration and logging settings • Debugging and statistics • Swap file location • Latency sensitivity
CPU Topology	Adjust cores per socket and NUMA nodes, and enable CPU hot-add.

General Virtual Machine Options

View or change general virtual machine settings, such as the name and location of the virtual machine, configuration file location, and operating system.

Change the Virtual Machine Name

A virtual machine must have a name that is unique within the folder where the virtual machine is located. If you move a virtual machine to a different datastore folder or host that has an existing virtual machine of the same name, you must change the virtual machine's name to keep it unique.

Verify that you have the following privilege:

- **Virtual machine** > **Change Configuration** > **Rename** on the virtual machine.

When you change the name of a virtual machine, you change the name used to identify the virtual machine in the vCenter inventory. This action does not change the name used as the computer name by the guest operating system.

The virtual machine name also determines the name of the virtual machine files and folder on the disk. For example, if you name the virtual machine win8, the virtual machine files are named win8.vmx, win8.vmdk, win8.nvram, and so on. If you change the virtual machine name, the names of the files on the datastore do not change.

> **Note:**
> Migration with Storage vMotion changes the virtual machine filenames on the destination datastore to match the inventory name of the virtual machine. The migration renames all virtual disk, configuration, snapshot, and .nvram files. If the new names exceed the maximum filename length, the migration does not succeed.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **General Options**.
3. Delete the existing name and enter a new name for the virtual machine in the **VM Name** text box.
4. Click **OK**.

View the Virtual Machine Configuration and Working File Location

You can view the location of the virtual machine configuration and working files. You can use this information when you configure backup systems.

Verify that the virtual machine is powered off and that you have the following privileges:

Virtual machine > **Change Configuration** > **Change Settings** on the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click **VM Options** tab and expand **General Options**.

   The path to the location of the virtual machine configuration file appears in the **VM Config File** text box. The path to the virtual machine working location appears in the **VM Working Location** text box.

Change the Configured Guest Operating System

When you change the guest operating system type in the virtual machine settings, you change the setting for the guest operating system in the virtual machine's configuration file. To change the guest operating system itself, you must install the new operating system in the virtual machine.

Power off the virtual machine.

You might change the guest operating system, for example, if you are upgrading the guest operating system installed in the virtual machine.

When you set the guest operating system type for a new virtual machine, vCenter chooses configuration defaults based on the guest type. Changing the guest operating system type after the virtual machine is created does not retroactively change those settings. It affects the recommendations and setting ranges offered after the change.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **General Options**.
3. From the **Guest OS** drop-down menu, select the guest operating system family.
4. From the **Guest OS Version** drop-down menu, select the guest operating system version.
5. Click **OK**.

VMware Remote Console Options

Change the VMware Remote Console options to control the access to the virtual machine.

Change the Virtual Machine Console Options for Remote Users

You can limit the number of simultaneous connections to a virtual machine and lock the guest operating system when the last remote user disconnects from the virtual machine console.

- Verify that VMware Tools is installed and running.
- To use the **Guest OS lock** option, verify that you have a Windows XP or later guest operating system.
- Verify that you have the following privilege: **Virtual machine** > **Change Configuration** > **Display connection settings** on the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab, and expand **VMware Remote Console Options**.
3. Optional: Select the **Guest OS lock** check box to lock the guest operating system when the last remote user disconnects.
4. Optional: In the **Maximum number of sessions** text box, specify the number of simultaneous connections to the virtual machine.
5. Click **OK**.

Configure Virtual Machine Encryption Options

Starting with vSphere 9.0, you can take advantage of virtual machine encryption. Encryption protects not only your virtual machine but also virtual machine disks and other files.

You set up a trusted connection between vCenter and a key management server (KMS). vCenter can then retrieve keys from the KMS as needed.

For detailed information about virtual machine encryption, see the *vSphere Security* documentation.

Clone an Encrypted Virtual Machine

When you clone an encrypted virtual machine, the clone is encrypted with the same keys unless you change them. To change keys, you can use the vSphere Client, the PowerCLI, or the API.

- A key provider must be configured and enabled.
- Create an encryption storage policy, or use the bundled sample, VM Encryption Policy.
- Required privileges:
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Decrypt**
  - **Cryptographic operations** > **Encrypt**
  - **Cryptographic operations** > **Recrypt**
  - If the host encryption mode is not `Enabled`, you also must have **Cryptographic operations** > **Register host** privilege.
  - **Datastore** > **Allocate space** on the destination datastore.
  - **Resource** > **Assign virtual machine to resource pool** on the destination host, cluster, or resource pool.
  - **Virtual machine** > **Inventory** > **Create from existing** on the destination folder or data center.
    Virtual machine > **Provisioning** > **Clone virtual machine** on the source virtual machine.
  - **VM storage policies** > **View VM storage policies**

If you use the PowerCLI or the API, you can clone the encrypted virtual machine and change keys in one step. For more information, see *vSphere Web Services SDK Programming Guide*.

You can perform the following operations during clone.

- Create an encrypted virtual machine from an unencrypted virtual machine or virtual machine template.
- Create a decrypted virtual machine from an encrypted virtual machine or virtual machine template.
- Recrypt the destination virtual machine with different keys from that of source virtual machine.
- Starting in vSphere 8.0, selecting the **Replace** option for a virtual machine with a vTPM device starts with a new, blank vTPM, which gets its own secrets and identity.

Note:
vSphere 8.0 includes the vpxd.clone.tpmProvisionPolicy advanced setting to make the default clone behavior for vTPMs to be "replace."

You can create an instant clone virtual machine from an encrypted virtual machine with the caveat that the instant clone shares the same key with the source virtual machine. You cannot recrypt keys on either the source or the instant clone virtual machine. See *vSphere Web Services SDK Programming Guide*.

1. Navigate to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and select **Clone** > **Clone to Virtual Machine**.
3. Navigate through the pages of the wizard.
   a) On the **Select a name and folder** page, enter a name and select a data center or folder in which to deploy it.
   b) On the **Select a compute resource**, select an object for which you have privileges to create encrypted virtual machines. For information about prerequisites and required privileges for encryption tasks, see the *vSphere Security* documentation.
   c) Change the keys for the cloned vTPM.

      Cloning a virtual machine duplicates the entire virtual machine, including the vTPM and its secrets, which can be used to determine a system's identity. To change secrets on a vTPM, select **Replace** for **TPM Provision Policy**.

      Note:

      When you replace the secrets of a vTPM, all keys, including workload-related keys, are replaced. As a best practice, ensure that your workloads no longer use a vTPM before you replace the keys. Otherwise, the workloads in the cloned virtual machine might not function correctly.
   d) On the **Select storage** page, select the datastore or datastore cluster in which to store the template configuration files and all virtual disks. You can change the storage policy as part of the clone operation. For example, changing from using an encryption to non-encryption policy decrypts the disks.
   e) On the **Select clone options**, select additional customization options.
   f) On the **Ready to complete** page, review the information and click **Finish**.
4. Optional: Change the keys for the cloned virtual machine.

   By default, the cloned virtual machine is created with the same keys as its parent. Best practice is to change the cloned virtual machine keys to ensure that multiple virtual machines do not have the same keys.
   a) Decide upon a shallow or deep recrypt.

      To use a different DEK and KEK, perform a deep recrypt of the cloned virtual machine. To use a different KEK, perform a shallow recrypt of the cloned virtual machine. For a deep recrypt, you must power off the virtual machine. You can perform a shallow recrypt operation while the virtual machine is powered on, and if the virtual machine has snapshots present. Shallow recrypt of an encrypted virtual machine with snapshots is permitted only on a single snapshot branch (disk chain). Multiple snapshot branches are not supported. If the shallow recrypt fails before updating all links in the chain with the new KEK, you can still access the encrypted virtual machine if you have the old and new KEKs.
   b) Perform a recrypt of the clone using the API.

Virtual Machine Power Management Options

Configure virtual machine Power management options to define how the virtual machine responds when the guest OS is placed on standby.

Manage Power Management Settings for a Virtual Machine

If the guest operating system is placed on stand by, the VM can either remain powered on or be suspended. You can use the Power Management settings to control this behavior. Some desktop-based guests, such as Windows 7, have stand by enabled by default, so that the guest goes into standby after a predetermined time.

The following conditions apply:

- Power Management options are not available on every guest operating system.

- **Wake on LAN** supports only Windows guest operating systems and is not available on Vlance NICs or when a Flexible NIC is operating in Vlance mode. That is, the current VMware Tools are not installed on the guest operating system.
- **Wake on LAN** can resume virtual machines that are in an S1 sleep state only. It cannot resume suspended, hibernated, or powered off virtual machines.
- NICs that support **Wake on LAN** include Flexible (VMware Tools required), vmxnet, Enhanced vmxnet, and vmxnet 3.

  Note:

  To avoid having the guest operating system go into standby mode unintentionally, verify the settings before you deploy the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click **VM Options** and expand **Power Management**.
3. In the **Standby response** section, select the standby response of the virtual machine.
   - The **Suspend the virtual machine** option stops all processes, which saves resources, and copies the contents of the virtual machine's memory to the virtual machine's `.vmss` file. Writing the memory to the `.vmss` file is useful if you must copy the file to help with a troubleshooting scenario.
   - The **Put the guest OS in standby mode and leave the virtual machine powered on** option stops all processes, but leaves the virtual devices connected to the virtual machine.
4. In the **Wake on LAN** section, select one or multiple virtual network adapters to which to apply the Wake on LAN option.
5. To save your changes, click **OK**.

Configuring VMware Tools Options

Configure the VMware Tools options to define the power operations for the virtual machine and decide when to run VMware Tools Scripts. Through VMware Tools configuration, you can automatically synchronize the virtual machine guest operating system time with the host.

Configure the Virtual Machine Power States

Changing virtual machine power states is useful when you do maintenance on the host. You can use the system default settings for the virtual machine power controls, or you can configure the controls to interact with the guest operating system. For example, you can configure the **Power off** control to power off the virtual machine or shut down the guest operating system.

- Verify that you have privileges to perform the intended power operation on the virtual machine.
- To set optional power functions, install VMware Tools in the virtual machine.
- Power off the virtual machine before editing the VMware Tools options.

You can modify many virtual machine configurations while the virtual machine is running, but you might need to change the virtual machine power state for some configurations.

You cannot configure a **Power on**

( )

action. This action powers on a virtual machine when a virtual machine is stopped, or resumes the virtual machine and runs a script when it is suspended and VMware Tools is installed and available. If VMware Tools is not installed, it resumes the virtual machine and does not run a script.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **VMware Tools**.
3. Select an option for the virtual machine **Power Off**

   ( )

   control from the drop-down menu.

Option	Description
Power Off	Immediately stops the virtual machine. The power off action shuts down the guest operating system or powers off the virtual machine. A message indicates that the guest operating system might not shut down properly. Use this power off option only when necessary.
Shut Down Guest (Default)	Follows system settings. The current value of the system settings appears in parentheses. Uses VMware Tools to initiate an orderly system shut down of the virtual machine. Soft power operations are possible only if the tools are installed in the guest operating system.

4. Select an option for the **Suspend**

   ( )

   control from the drop-down menu.

Option	Description
Suspend (Default)	Suspends the virtual machine and leaves it connected to the network.
Suspend Guest	Follows system settings. The current value of the system setting appears in parentheses. Pauses all virtual machine activity. When VMware Tools is installed and available, the suspend action runs a script in the guest operating system and suspends the virtual machine. If VMware tools is not installed, the suspend action suspends the virtual machine without running a script.

5. Select an option for the **Reset**

   ( )

   control from the drop-down menu.

Option	Description
Reset	Shuts down and restarts the guest operating system without powering off the virtual machine. If VMware Tools is not installed, the reset action resets the virtual machine.
Default (Restart Guest)	Follows system settings. The current value of the system setting appears in parentheses. Uses VMware Tools to initiate an orderly restart. Soft power operations are possible only if the tools are installed in the guest operating system.

6. To save your changes, click **OK**.

Synchronize the Time of a Virtual Machine Guest Operating System with the Host

You can configure VMware Tools options to set the time of a virtual machine guest operating system to be the same as the time of the host. Starting with vSphere 7.0 Update 1, VMware Tools provides accurate and synchronized time between guest and host operating systems.

To synchronize the virtual machine guest operating system time with that on the host, install VMware Tools on the virtual machine.

You can synchronize the time between guest and host operating systems while the virtual machine is running. This operation does not depend on the hardware version of the virtual machine.

For information about the timekeeping best practices for Windows guest operating systems, see Broadcom knowledge base article 315346.

For information about the timekeeping best practices for Linux guest operating systems, see Broadcom knowledge base article 310053.

1. To synchronize the time of the virtual machine guest operating system with the host time, open VMware Tools options for your virtual machine while creating, editing, cloning, or deploying a virtual machine.

Option	Action
Create a virtual machine	1. Right-click any inventory object that is a valid parent object of a virtual machine and select **New Virtual Machine**. 2. On the **Select a creation type** page, select **Create a new virtual machine** and click **Next**. 3. Navigate through the pages of the wizard. 4. On the **Customize hardware** page, click the **VM Options** tab and expand **VMware Tools**.
Edit a virtual machine	1. Right-click a virtual machine in the inventory and select **Edit Settings**. 2. Click the **VM Options** tab and expand **VMware Tools**.
Clone an existing virtual machine	1. Right-click a virtual machine in the inventory and select **Clone** > **Clone to Virtual Machine**. 2. Navigate through the pages of the wizard. 3. On the **Select clone options** page, select **Customize this virtual machine's hardware** and click **Next**. 4. On the **Customize hardware** page, click the **VM Options** tab and expand **VMware Tools**.
Deploy a virtual machine from a template	1. Right-click a template in the inventory and select **New VM from This Template**. 2. Navigate through the pages of the wizard. 3. On the **Select clone options** page, select **Customize this virtual machine's hardware** and click **Next**. 4. On the **Customize hardware** page, click the **VM Options** tab and expand **VMware Tools**.

2. Select the time synchronization options.

Option	Action
Synchronize at startup and resume (recommended)	This option is selected by default to ensure the best time synchronization between guest and host operating systems after performing certain operations, for example: • When you resume a virtual machine from a suspended operation. • When you migrate a virtual machine with vMotion. • When you revert a snapshot.
Synchronize time periodically	Periodically synchronize the time of the guest operating system of a virtual machine with the host. **Note:** Select this option if the guest operating system of a virtual machine does not have a native time synchronization software.

3. To save your changes, click **OK**.

Virtualization Based Security

Microsoft virtualization-based security (VBS), a feature of Windows 10, Windows Server 2016, and Windows Server 2019 operating systems, uses hardware and software virtualization to enhance system security by creating an isolated, hypervisor-restricted, specialized subsystem.

For more detailed information about VBS, see the *vSphere Security* documentation.

Configure Virtual Machine Boot Options

Edit Boot Options to activate or deactivate UEFI Secure Boot and configure the boot behavior of the virtual machine.

For more information about activating or deactivating UEFI Secure Boot, see the *vSphere Security* documentation.

Delay the Boot Sequence

Delaying the boot operation is useful when you change BIOS or EFI settings such as the boot order. For example, you can change the BIOS or EFI settings to force a virtual machine to boot from a CD-ROM.

- Verify that vSphere Client is logged in to a vCenter system.
- Verify that you have access to at least one virtual machine in the inventory.
- Verify that you have privileges to edit boot options for the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click **VM Options** tab and expand **Boot Options**.
3. In the **Boot Delay** section, select the time in milliseconds to delay the boot operation.
4. Optional: In the **Force setup** section, select whether to force entry into the BIOS or EFI setup screen the next time the virtual machine boots.
5. Optional: In the **Failed Boot Recovery** section, select whether the virtual machine should reboot after a boot failure and enter the time in seconds.
6. Click **OK**.

Configuring Virtual Machine Advanced Options

You can edit the Advanced virtual machines settings when you must solve issues that are caused by an application or when you need log files and debugging information for troubleshooting purposes.

You can also add or change configuration parameters and change the latency sensitivity of a virtual machine.

Enable Virtual Machine Logging

To troubleshoot problems with your virtual machine, you can use logging to collect log files.

Required privilege: **Virtual machine** > **Change Configuration** > **Change Settings**

ESX hosts store virtual machine log files in the same directory as the virtual machine's configuration files. By default, the log file name is `vmware.log`. Archived log files are stored as `vmware-`*`n`*`.log`, where *`n`* is a number in sequential order beginning with 1.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **Advanced**.
3. In the Settings row, select **Enable logging** and click **OK**.

You can view and compare log files in the same storage location as the virtual machine configuration files.

Configure Virtual Machine Debugging and Statistics

You can run a virtual machine so that it collects additional debugging information that is helpful to VMware technical support in resolving issues.

Verify that you have the following privilege: **Virtual machine** > **Change Configuration** > **Change Settings** on the virtual machine.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **Advanced**.
3. Select a debugging and statistics option from the drop-down menu.
   - **Run normally**
   - **Record Debugging Information**
   - **Record Statistics**

The number of debugging and statistics options available depends on the host software type and version. On some hosts, some options are not available.

4. Click **OK**.

Change the Swap File Location

When a virtual machine is powered on, the system creates a VMkernel swap file to serve as a backing store for the virtual machine RAM contents.

- Power off the virtual machine.
- Verify that you have the following privilege: **Virtual machine** > **Change Configuration** > **Change Swapfile placement** on the virtual machine.

You can accept the default swap file location or save the file to a different location. By default, the swap file is stored in the same location as the virtual machine configuration file.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **VM Options** tab and expand **Advanced**.
3. Select a swap file location option.

Option	Description
Default	Stores the virtual machine swap file at the default location defined by the host or cluster swap file settings.
Virtual machine directory	Stores the virtual machine swap file in the same folder as the virtual machine configuration file.
Datastore specified by host	If the host or cluster settings define a location for the swap file, this location is used. Otherwise, the swap file is stored with the virtual machine.

4. Click **OK**.

Configure the Virtual Machine Latency Sensitivity

To improve the latency sensitive workloads, you can configure the latency sensitivity of a virtual machine.

- Verify that the virtual machine is of hardware version 20 or later.
- Verify that you have the following privilege: **Virtual machine** > **Change Configuration** > **Change Settings** on the virtual machine.

With vSphere 8.0 and later, you can use the Virtual Hyperthreading (vHT) feature to configure low latency VMs and expose vHT to the guest operating systems. When you enable vHT, each guest vCPU is treated as a single hyperthread of a virtual core.

For more information about the vHT feature, see the *vSphere Resource Management* documentation.

1. Navigate to a virtual machine in the inventory and click **Actions**.
2. Click **Edit Settings**.
3. On the **VM Options** tab, click **Advanced**.
4. From the **Latency Sensitivity** drop-down menu, select a setting.

Option	Description
Normal	The default setting. The CPU scheduler for a VM with normal latency sensitivity can move the virtual CPUs of the VM across any physical CPU of the host, depending on the system load.
High	High Latency Sensitivity requires you to set CPU and 100% memory reservation for the VM. Scheduling of the VM is adjusted for low-latency workload requirements. Each virtual CPU is granted exclusive access to a physical core.

Option	Description
High with Hyperthreading	High Latency Sensitivity requires you to set CPU and 100% memory reservation for the VM. Each virtual CPU is granted exclusive access to a hyperthread of a physical core. Each consecutive pair of virtual machines vCPUs is assigned to the pair of hyperthreads on a physical core.

For more information on how to calculate the full CPU reservation for vHT, see the vHT Full CPU Reservation section in the *vSphere Resource Management* documentation.

5. Click **OK**.

Configure Guest Variable Access Control

Guest variable access control allows you to control access to the guest variables by unprivileged guest users. You can configure the write or both read and write access to specific guest variables to secure virtual machines and ESX hosts. This measure ensures that only privileged users can access or modify guest variables.

To manage the guest variable access control, make sure that you have the following privileges:

- **Virtual machine** > **Change Configuration** > **Advanced configuration**

You can configure the guest variable access control by using the user interface of the vCenter system.

You can configure access control by applying the specification to:

- A single guest variable
- Groups of guest variables that share a common prefix
- All guest variables
- An existing guest variable or a guest variable that does not exist yet
- VMs that are powered on or powered off

Considerations and Limitations

- To limit the VMX configuration file size, guest variable access control supports only 256 configurations.
- When `guest_rpc.rpci.auth.cmd.<info-set/info-get>` is set to `TRUE`, it takes absolute precedence over all other guest variable access control configurations.
- Guest variable access control configuration overrides the system default access control configuration. This precedence does not apply to write access for "Read-Only" guest variables, which always prohibit write access for all guest users.
- The guest variable access control configuration takes precedence over the system default configurations for all `cloud-init` GuestInfo variables and the `guest_rpc.auth.cloud-init.set` setting is ignored.

To configure the guest variable access control, perform the following steps.

1. Right-click a virtual machine and select **Edit Settings**.
2. Click the **Advanced Parameters** tab.
3. In the **Attributes** text box, enter one of the following options.

Option	Description
Option	Description
`guestvar.deny.unpriv.name.<guest-var-key>`	Access is denied to unprivileged users for the guest variable named `<guest-var-key>`.
`guestvar.deny.unpriv.prefix.<guest-var-key>`	Access is denied to unprivileged guest users for the guest variable that is named `<guest-var-key>` and for guest variable names starting with `<guest-var-key>` with `.` as a separator, exept for those that match a more specific prefix or name setting.
`guestvar.deny.unpriv`	Access is denied to unprivileged guest users for all guest variables, except for those matching a prefix or name setting.

Note:

`<guest-var-key>` is a guest variable name or prefix (with `.` as a separator) without `guestinfo.` prefix.

4. In the **Value** text box, enter **`<access>`** where `<access>` is the level of access to be denied to unprivileged guest users for the guest variable named `<guest-var-key>` and click **Add**. Valid values are:
   - `read-write` - denies both read and write access.
   - `write` - denies only write access.
   - `none` - both read and write are allowed for unprivileged users.
5. Click **OK** to save the changes.

Configure the CPU Topology of a Virtual Machine

Different virtual CPU topologies impact scheduling and placement from within the guest operating system of a virtual machine.

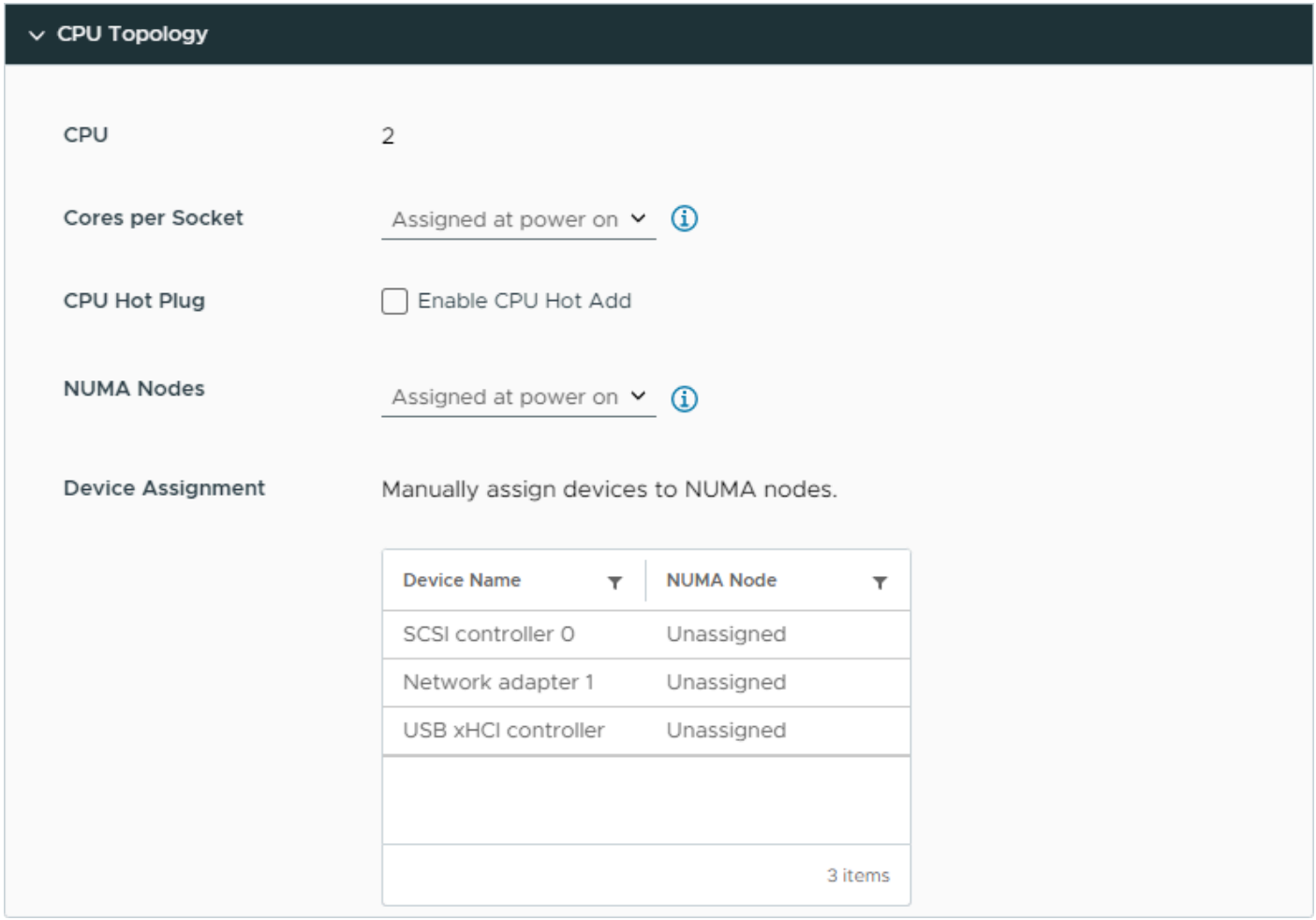
- Virtual machine requirements:
  - Hardware version 20 or later.
  - Verify that the virtual machine is powered off.

By default, the ESX host automatically selects a virtual topology that aligns with the underlying hardware. With vSphere 8.0, you can manually configure the virtual CPU topology. For example, you can change the number of virtual CPUs per NUMA node, assign a virtual device to a virtual NUMA node or leave it unassigned.

The configuration is preserved when you edit a virtual machine or migrate a virtual machine with vMotion.

For more information about the virtual CPU topology and virtual NUMA, see the *vSphere Resource Management* documentation.

1. Navigate to a virtual machine and from the **Actions** menu, click **Edit Settings**.
2. On the **VM Options** tab, click **CPU Topology**.
3. Configure the CPU topology for the virtual machine.



Option	Description
Cores per Socket	Select whether you want the system to automatically assign the cores per socket or manually specify the numbers of cores per socket.
CPU Hot Plug	When enabled, the VM hot adds virtual CPUs while it is running. When you activate **CPU Hot Plug**, you deactivate the virtual NUMA and the hot added CPUs run on a single NUMA node. To enable virtual NUMA and allow hot added CPUs to appear on new virtual NUMA nodes, on the **Advanced Parameters** tab, enter the `numa.allowHotadd` parameter in the **Attribute** field and set its value to `TRUE`. For more information, see Broadcom knowledge base article 313375.
NUMA Nodes	Configure the number of NUMA nodes automatically at power on or manually.

Option	Description
Device Assignment	If you select to configure the NUMA nodes manually, to assign devices to NUMA nodes, click the vertical ellipsis icon (⋮) that appears in front of each device and select a NUMA node.

4. Click **OK**.

From the **Summary** tab of the virtual machine, you can monitor the CPU topology exposed by the ESX host - either Assigned at power on or Manual.

Configure Virtual Machine Advanced File Parameters

You can change or add virtual machine configuration parameters when instructed by a Broadcom technical support representative, or if you see VMware documentation that instructs you to add or change a parameter to fix a problem with your system.

Verify that you have the **Virtual machine** > **Change Configuration** > **Advanced configuration** privilege.

> **Important:**
> Changing or adding parameters when a system does not have problems might lead to decreased system performance and instability.

The following conditions apply:

- To change a parameter, you change the existing value for the keyword/value pair. For example, if you start with the keyword/value pair, keyword/value, and change it to keyword/value2, the result is keyword=value2.
- You cannot delete a configuration parameter entry.

> **CAUTION:**
> You must assign a value to configuration parameter keywords. If you do not assign a value, the keyword can return a value of 0 or false, which can result in a virtual machine that cannot power on.

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. Click the **Advanced Parameters** tab.
3. Optional: To add a parameter, enter an attribute name and value for the parameter and click **Add**.
4. Optional: To change a parameter, enter a new value in the **Value** text box for that parameter and click ✓ to confirm changes.
5. Click **OK**.

Managing Multi-Tiered Applications with vSphere vApp

With vSphere vApp, you can package multiple interoperating virtual machines and software applications into a single unit that you can manage and distribute in OVF format.

A vApp can contain one or more virtual machines. Any operation carried out with the vApp, such as clone or power off, affects all virtual machines in the vApp container.

In the vSphere Client, you can navigate to the vApp **Summary** tab, where you can view the current status of the vApp, and you can manage the vApp.

> **Note:**
> Because the vApp metadata resides in the vCenter database, a vApp can be distributed across multiple ESX hosts. The metadata information might be lost if the vCenter database is cleared or if the standalone ESX host that contains the vApp is removed from the vCenter. Back up your vApps to an OVF package to avoid losing metadata.
>
> vApp metadata for the virtual machines within the vApp does not follow the snapshot semantics for virtual machine configuration. The vApp properties that you delete, modify, or define after you take a snapshot of a

virtual machine remain respectively deleted, modified, or defined, if the virtual machine reverts to that snapshot or any prior snapshots.

Create a vApp

A vApp allows you to perform resource management and certain other management activities such as power operations for multiple virtual machines at the same time.

Verify that one of those objects is available in your data center.

- A standalone host that is running ESX 7.0 or later
- A DRS cluster

You can think of the vApp as the container for the virtual machines, and you can perform the operations on the container.

When you create a vApp, you can add it to a folder, standalone host, resource pool, DRS cluster, or another vApp.

1. In the vSphere Client, right-click an object that supports vApp creation and click **New vApp**.
   The **New vApp** wizard starts.
2. On the **Select creation type** page, select **Create a new vApp** and click **Next**.
3. On the **Select a name and location** page, type a name and select a location for the vApp, and click **Next**.
   - If you start the creation process from a folder or vApp, you are prompted for a host, cluster, or resource pool.
   - If you start the creation process from a resource pool, host, or cluster, you are prompted for a folder or a data center.
4. On the Resource allocation page, allocate CPU and memory resources to the vApp.

Option	Description
Shares	Defines the CPU or memory shares for this vApp with respect to the parent's total. Sibling vApps share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values respectively in a 1:2:4 ratio. Select **Custom** to give each vApp a specific number of shares that expresses a proportional weight.
Reservation	Defines the guaranteed CPU or memory allocation for this vApp.
Reservation Type	Defines whether the reservation is expandable. Select the **Expandable** check box to make the reservation expandable. When the vApp is powered on, if the combined reservations of its virtual machines are larger than the reservation of the vApp, the vApp can use resources from its parent or ancestors.
Limit	Defines the upper limit for this vApp's CPU or memory allocation. Select **Unlimited** to specify no upper limit.

5. Click **Next**.
6. On the **Review and finish** page, review the vApp settings and click **Finish**.

Perform vApp Power Operations

One of the advantages of a vApp is that you can perform power operations on all virtual machines it contains at the same time.

Prerequisites depend on the task that you want to perform.

Task	Required Privileges
Power on a vApp	**vApp** > **Power On** on the vApp.
Power off a vApp	**vApp** > **Power Off** on the vApp.
Suspend a vApp	**vApp** > **Suspend**

When powering on a vApp within a DRS cluster in manual mode, no DRS recommendations are generated for virtual machine placements. The power-on operation performs as if DRS is run in a semiautomatic or automatic mode for the

initial placements of the virtual machines. This does not affect vMotion recommendations. Recommendations for individual powering on and powering off of virtual machines are also generated for vApps that are running.

1. Navigate to a vApp in the inventory.
2. Select one of the power operation options.

Task	Action
Power on	Right-click the vApp and select **Power** > **Power On**. You can power on a vApp to power on all its virtual machines and child vApps. Virtual machines are powered on according to the startup order configuration. If a delay is set in the startup settings of a virtual machine in the vApp, the vApp waits for the set length of time before powering on that virtual machine.
Power off	Right-click the vApp and select **Power** > **Power Off** You can power off a vApp to power off all its virtual machines and child vApps. Virtual machines are powered off in reverse startup order. If a delay is set in the shutdown settings of a virtual machine in the vApp, the vApp waits for the set length of time before powering off that virtual machine.
Suspend	Right-click the vApp and select **Power** > **Suspend**. You can suspend a vApp to suspend all its virtual machines and child vApps. Virtual machines are suspended in the reverse order of the specified startup order. All virtual machines are suspended regardless of the Suspend behavior you specified in the Power Management VM Option for the virtual machine.
Resume	Right-click a vApp that is powered off or suspended and select **Power On**. Virtual machines are resumed according to their startup order configuration.

On the **Summary** tab, the **Status** indicates the vApp status.

Create or Add an Object to a vApp

You can populate a vApp with objects by creating a virtual machine, resource pool, or child vApp within the vApp. You can also add an existing object from the inventory, such as a virtual machine or another vApp, to the vApp.

Select your task.

- Create an object inside a vApp.
  - Right-click a vApp in the inventory, and select **New Virtual Machine** to create a new virtual machine within the vApp.
  - Right-click a vApp in the inventory, and select **New Child vApp** to create a child vApp inside the vApp.
  - Right-click a vApp in the inventory, and select **New Resource Pool** to create a new resource pool within the vApp.
  - Right-click a vApp in the inventory, and select **Deploy OVF Template** to deploy an OVF template and add the corresponding virtual machine to the selected vApp.
- Add an existing object to a vApp.
  a. Navigate to an object in the inventory.
  b. Drag the object to the destination vApp.
  c. Release the mouse button.
     If the move is not permitted, the object is not added to the vApp.

The new object is now a part of the vApp in the vApp inventory.

Clone a vApp

Cloning a vApp is similar to cloning a virtual machine. When you clone a vApp, you clone all virtual machines and vApps within the vApp.

When you clone a vApp, you can add the clone to a folder, standalone host, resource pool, cluster enabled for DRS, or another vApp.

Verify that one of those objects is available in your datacenter.

- A standalone host that runs ESX 8.0 or later
- A DRS cluster

Make sure that the VMs in the vApp are powered off.

1. Start the cloning wizard.
   - Start the cloning wizard from a DRS cluster by right-clicking the cluster and selecting **New vApp** > **New vApp**.
   - Start the cloning wizard from an existing vApp by right-clicking the vApp and selecting **Clone** > **Clone**.
2. On the Select creation type page, select **Clone an existing vApp** and click **Next**.
3. On the Select source vApp page, select an existing vApp to clone, and click **Next**.
   The Select source vApp page appears only if you start the wizard from a DRS cluster.
4. On the Select destination page, select a valid host, vApp, or resource pool in which to run the vApp, and click **Next**.
5. On the Select a name and location page, type a name for the vApp, select location, and click **Next**.
6. On the Select storage page, select the virtual disk format and the target datastore and click **Next**.
7. On the Map networks page, configure the network mappings for the networks that the virtual machines in the vApp use.
8. On the Resource allocation page, allocate CPU and memory resources for the vApp and click **Next**.
9. On the Review and finish page, review the vApp settings, and click **Finish**.

Edit vApp Notes

You can add or edit notes for a particular vApp.

1. Right-click a vApp in the inventory and select **Edit Notes**.
2. Type your comments in the **Edit Notes** window.
3. Click **OK**.

Your comments appear on the **Summary** tab for the vApp.

Configure vApp Properties

Define and use custom properties to present custom information to all the virtual machines in the vApp. You can later assign values and edit those properties. If you deploy the vApp from an OVF file with properties predefined in that OVF, you can edit those properties as well.

- Power off the vApp.
- Required privilege:**vApp** > **vApp application configuration** on the vApp and **vApp** > **vApp instance configuration** to activate the set a value for the property.

The **Properties** pane contains a list of all the properties that are defined for a vApp. You can use the filters to browse through the list more easily.

1. Navigate to a vApp in the inventory.
2. On the **Configure** tab, select **Settings** > **vApp properties**.
   The **Properties** pane shows the list of properties and the permissible actions.
3. Select your task by clicking the respective button.

Option	Description
Add	Creates a new property. **Important:** If the virtual machine is connected to a distributed switch and has the vApp option enabled, you cannot select any of the following **Dynamic property** options: IP address, Subnet, Netmask,

Option	Description
	Gateway, Domain name, HTTP Proxy, Host prefix, DNS servers, DNS search path, Network name.
Edit	Edits the property. You can change the general information for the property, such as the property label, category, and description. You can also edit the type parameters. **Important:** If the virtual machine is connected to a distributed switch and has the vApp option enabled, you cannot select any of the following **Dynamic property** options: IP address, Subnet, Netmask, Gateway, Domain name, HTTP Proxy, Host prefix, DNS servers, DNS search path, Network name.
Set Value	Sets a value for the property. That value is different from the default value that you define when you create a property.
Delete	Removes a property from the list.

Edit vApp Settings

You can edit and configure several vApp settings, including startup order, resources, and custom properties.

1. Right-click a vApp in the inventory and click **Edit Settings**.
2. On the **Resources** tab, edit CPU and memory settings, such as shares, reservation, and limit.
3. On the **Start Order** tab, set and edit the start order of the virtual machines.
4. On the **IP Allocation** tab, specify IP protocol and choose an IP allocation scheme.
5. On the **Details** tab, view product information, such as name, vendor, product URL, and vendor URL.

Configure vApp CPU and Memory Resources

You can configure the CPU and memory resource allocation for the vApp.

Verify that you have the required privilege **vApp** > **vApp resource configuration** on the vApp.

Reservations on vApps and all their child resource pools, child vApps, and child virtual machines count against the parent resources only if those objects are powered on.

1. Right-click a vApp in the inventory and click **Edit Settings**.
2. To allocate CPU resources to the vApp, click the **Resources** tab and expand **CPU**.

Option	Description
Shares	CPU shares for this vApp with respect to the parent's total. Sibling vApps share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values respectively in a 1:2:4 ratio. Select **Custom** to give each vApp a specific number of shares, which express a proportional weight.
Reservation	Guaranteed CPU allocation for this vApp.
Reservation Type	To make the reservation expandable, select the **Expandable** check box. When the vApp is powered on, if the combined reservations of its virtual machines are larger than the reservation of the vApp, the vApp can use resources from its parent or ancestors.
Limit	Upper limit for this vApp's CPU allocation. Select **Unlimited** to specify no upper limit.

3. To allocate memory resources to the vApp, click the **Resources** tab and expand **Memory**.

Option	Description
Shares	Memory shares for this vApp with respect to the parent's total. Sibling vApps share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values respectively in a 1:2:4 ratio. Select **Custom** to give each vApp a specific number of shares, which express a proportional weight.
Reservation	Guaranteed memory allocation for this vApp.

Option	Description
Reservation Type	To make the reservation expandable, select the **Expandable** check box to make the reservation expandable. When the vApp is powered on, if the combined reservations of its virtual machines are larger than the reservation of the vApp, the vApp can use resources from its parent or ancestors.
Limit	Upper limit for this vApp's memory allocation. Select **Unlimited** to specify no upper limit.

4. Click **OK**.

Configure vApp IP Allocation Policy

If your vApp is set up to allow it, and if you have the required privileges, you can edit how IP addresses are allocated for the vApp.

Required privileges:

- **vApp** > **vApp instance configuration**
- **vApp** > **vApp application configuration**

You cannot configure the IP allocation policy during the vApp creation process.

Before you configure the IP allocation policy, you must specify the IP protocol and the IP allocation scheme that the vApp supports.

If you deploy a vApp from an OVF template, the IP allocation policy might still be editable.

1. Right-click a vApp in the inventory and click **Edit Settings**.
2. In the **Edit vApp** dialogue box, click the **IP Allocation** tab.
   This tab is only available in the vSphere Client.
3. In the Authoring section, define the IP protocol and the IP allocation scheme that the vApp supports.

   The IP protocol and the IP allocation scheme determine what options for IP allocation are available.

   A vApp can obtain its network configuration through the OVF environment or through a DHCP server. If you do not select any of these options, the IP addresses are manually allocated.

   The IP protocols that a vApp can support are IPv4, IPv6, or both.
4. In the Deployment section, select an IP allocation policy from the **IP allocation** drop-down menu.

Option	Description
Static - Manual	IP addresses are manually configured. No automatic allocation is performed.
Transient - IP Pool	IP addresses are automatically allocated using IP pools from a specified range when the vApp is powered on. The IP addresses are released when the appliance is powered off.
DHCP	A DHCP server is used to allocate the IP addresses. The addresses assigned by the DHCP server are visible in the OVF environments of virtual machines started in the vApp.
Static - IP Pool	IP addresses are automatically allocated from the managed IP network range of vCenter Server at power-on, and remain allocated at power-off.

   Static - IP Pool and Transient - IP Pool have in common that IP allocation is done through the range managed by the vSphere platform as specified by the IP pool range in a network protocol profile. The difference is that for a static IP Pool, the IP addresses are allocated at first power-on and remain allocated, while for a transient IP Pool, the IP addresses are allocated when needed, typically at power-on, but released during power-off.
5. Click **OK**.

Configure vApp Startup and Shutdown Options

You can change the order in which virtual machines and nested vApps within a vApp start up and shut down. You can also specify the delays and actions performed at startup and shutdown.

Required privilege:**vApp** > **vApp application configuration** on the vApp.

1. Right-click a vApp in the inventory and click **Edit Settings**.
2. Select a virtual machine and select its order group.

   Virtual machines and vApps in the same group are started before the objects in the next group. The first group of virtual machines to power on is Group 1, followed by Group 2, Group 3, and so forth. The reverse order is used for shutdown.
3. Click the **Start Order** tab and select a virtual machine from the list.
4. From the **Group** drop-down menu, select a group for the virtual machine.
5. Optional: Select the startup action for the virtual machine.

   The default is **Power On**. Select **None** to power on the virtual machine manually.
6. Optional: Specify when the startup action is to happen.
   - Enter a time delay in seconds for the startup action.
   - To perform the startup action when VMware Tools has started, select **Continue when VMware Tools are ready**.
7. Optional: Select the shutdown action for the virtual machine.

   The default shutdown action is **Power Off**. You can also select **Guest Shutdown** to shut down the guest OS and leave the virtual machine running, **Suspend**, or **None**.
8. Optional: Enter a time delay in seconds for the shutdown action.
9. Click **OK**.

Configure vApp Product Properties

You can configure product and vendor information for a vApp.

Required privilege:**vApp** > **vApp application configuration** on the vApp.

1. Right-click a vApp in the inventory and click **Edit Settings**.
2. To enter product and vendor information, click the **Details** tab.

vApp Setting	Description
Name	Product name.
Product URL	If you enter a product URL, a user can click the product name on the virtual machine summary page and go to the product's web page.
Vendor	Vendor name.
Vendor URL	If you enter a vendor URL, a user can click the vendor name from the **Summary** page of the virtual machine and go to the vendor's web page.

3. Click **OK**.

View vApp License Agreements

You can view the license agreement for the vApp that you edit.

- Required privilege:**vApp** > **vApp application configuration** on the vApp.
- Verify that the vApp is imported from an OVF template that includes one or more license agreements.

1. Navigate to a vApp in the inventory.
2. On the **Configure** tab, expand **Settings** and click **License agreements**.

Virtual Machine vApp Options

You can enable and configure vApp functionality for virtual machines that are not within a vApp. When a virtual machine vApp options are enabled, you can configure OVF properties, use the OVF environment, and specify IP allocation and product information for the virtual machine.

You can configure and modify the authoring vApp options of a virtual machine to specify custom information that is preserved and used when you export the virtual machine as an OVF template. If you later deploy that OVF template, the information that you specified is available for editing in the deployment vApp options of the virtual machine.

For a virtual machine with vApp options enabled, the authoring vApp options are preserved when you export the virtual machine as an OVF template and those options are used when you deploy a new virtual machine from that template. Deployment vApp options are available for virtual machines that are deployed from OVF templates.

Use a Network Protocol Profile to Allocate IP Addresses to a Virtual Machine or vApp

You can associate a network protocol profile with a port group of a standard switch or a distributed switch and use the profile to dynamically allocate IP addresses to a virtual machine that is within a vApp.

Verify that the virtual machine is connected to a port group that is associated with the network protocol profile. For more information, see Associate a Port Group with a Network Protocol Profile.

Select your task.

Option	Description
Use a Network Protocol Profile to Allocate IP Addresses to a Virtual Machine	1. Navigate to a virtual machine in the vCenter inventory. 2. On the **Configure tab**, expand **Settings** and select **vApp Options**. 3. Click the **Edit** button. The **Edit vApp options** dialog box opens. 4. If vApp options are not enabled, select the **Enable vApp options** check box. 5. Click the **IP Allocation** tab. 6. In the Authoring section, select **OVF environment** as an IP allocation scheme. 7. In the Deployment section, set the **IP allocation** to **Transient - IP Pool** or **Static - IP Pool**. 8. Click **OK**.
Use a Network Protocol Profile to Allocate IP Addresses to a vApp	1. Navigate to a vApp in the vCenter inventory 2. Right-click the vApp and select **Edit Settings**. The **Edit vApp** dialog box opens. 3. Click the **IP Allocation** tab. 4. In the Authoring section, select **OVF environment** as an IP allocation scheme. 5. In the Deployment section, set the **IP allocation** to **Transient - IP Pool** or **Static - IP Pool**. 6. Click **OK**.

Both the **Static - IP Pool** and **Transient - IP Pool** options allocate an IP address from the range defined in the network protocol profile that is associated with the port group. If you select **Static - IP Pool**, the IP address is assigned the first time the virtual machine or vApp is powered on. The assigned IP address persists across restarts. If you select **Transient - IP Pool**, an IP address is assigned every the virtual machine or vApp is powered on.

When the virtual machine is powered on, the adapters connected to the port group receive IP addresses from the range in the protocol profile. When the virtual machine is powered off, the IP addresses are released.

Activate vApp Options for a Virtual Machine

To activate the vApp functionality for a virtual machine that is not part of a vApp, you must enable vApp options at the virtual machine level. Virtual machine vApp options are saved when you export the virtual machine as an OVF template. Those vApp options are later used when you deploy the OVF template.

If you activate vApp options and export a virtual machine to OVF, the virtual machine receives an OVF Environment XML descriptor at boot time. The OVF descriptor might include values for custom properties, including network configuration and IP addresses.

The OVF environment can be transported to the guest in two ways:

- As a CD-ROM that contains the XML document. The CD-ROM is mounted on the CD-ROM drive.
- Through VMware Tools. The guest OS environment variable *guestinfo.ovfEnv* contains the XML document.

For information on how to create a vApp, see Create a vApp.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. Click the **Edit** button.
   The **Edit vApp Options** dialog box opens.
4. If vApp options are deactivated, select the **Enable vApp options** check box and click **OK**.

Edit Application Properties and OVF Deployment Options for a Virtual Machine

If a virtual machine is a deployed OVF template, you can view the application properties and the OVF deployment options that are defined in the OVF. The deployment options include information about unrecognized OVF sections and the IP allocation policy.

Verify that the virtual machine has the vApp options enabled. See Activate vApp Options for a Virtual Machine.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. Click the **Edit** button.
   The **Edit vApp Options** dialog box opens.
4. If the OVF template included editable IP allocation options, click the **IP Allocation** tab and modify the IP allocation options in the **Deployment** section.

Option	Description
Static - Manual	IP addresses are manually configured. No automatic allocation is performed.
Transient - IP Pool	IP addresses are automatically allocated using IP pools from a specified range when the vApp is powered on. The IP addresses are released when the appliance is powered off.
DHCP	A DHCP server is used to allocate the IP addresses. The addresses assigned by the DHCP server are visible in the OVF environments of virtual machines started in the vApp.
Static - IP Pool	IP addresses are automatically allocated from the managed IP network range of vCenter Server at power-on, and remain allocated at power-off.

   With the **Static - IP Pool** and **Transient - IP Pool** options, the allocation of IP addresses is done through an IP pool range defined in a network protocol profile. The difference between the two options is that with a static IP pool, the IP addresses are allocated at the first power on of the virtual machine and remain allocated, while with a transient IP pool, the IP addresses are allocated when needed, typically at power-on, and released during power-off.

OVF Authoring Options for a Virtual Machine

You can use the OVF authoring options that are included in a virtual machine vApp options to specify custom information that is preserved when you export the virtual machine as an OVF template.

vApp properties are a central concept to vApp deployment and self-configuration. They can turn a general OVF package into a running vApp instance with a custom configuration.

The set of properties assigned to a running vApp is determined by the OVF package from which the vApp is deployed.

- When an OVF package is created the author adds the set of properties necessary for the vApp to function in an unknown environment. For example, properties that contain network configuration, a property that contains the email address of the system administrator, or a property that contains the number of expected vApp users.
- Some property values are entered by the user when the vApp is deployed, while other property values are configured by vCenter when the vApp is powered on. How properties are handled depends on the property type and the vCenter configuration.

When vCenter powers on a vApp, it creates an XML document that contains all properties and their values. This document is made available to each virtual machine in the vApp, and allows the virtual machines to apply the properties to their own environment.

Edit vApp Product Information for a Virtual Machine

If you want to export a virtual machine as an OVF template, you can specify product information that becomes available when you deploy a new virtual machine from the OVF template.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. Click the **Edit** button.
   The **Edit vApp Options** dialog box opens.
4. To enter product and vendor information, click the **Details** tab.

vApp Setting	Description
Name	Product name.
Product URL	If you enter a product URL, a user can click the product name on the virtual machine summary page and go to the product's web page.
Vendor	Vendor name.
Vendor URL	If you enter a vendor URL, a user can click the vendor name from the **Summary** page of the virtual machine and go to the vendor's web page.

5. Click **OK**.

Manage vApp Custom Properties for a Virtual Machine

You can define and manage custom properties that are stored in the OVF template when you export a virtual machine or vApp and are used by vCenter when you deploy the OVF template. OVF templates support static properties, which are often configured by the user, and dynamic properties, which are always set by the vCenter.

To customize your virtual machine or vApp with properties, perform the following steps.

1. Define the OVF properties, for example a DNS address or gateway, in the virtual machine or vApp.
2. If you plan to export the virtual machine or the vApp to an OVF template:
   a. Set up the OVF environment transport to carry the settings into the virtual machine. See Edit OVF Details for a Virtual Machine.
   b. Write some glue-code to access and apply the information to the virtual machine.

See the VMware vApp Developer blog topic *Self-Configuration and the OVF Environment* for a discussion, sample code, and a video.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. You can manage custom properties from the **Properties** panel.

Option	Description
Add	To create a property, click **Add**.
Edit	To edit an existing property, select the property and click **Edit**.
Set Value	To set a new value to the existing property, select the property and click **Set Value**.
Delete	To delete an existing property, select the property and click **Delete**.

Edit vApp IP Allocation Policy for a Virtual Machine

You can set or edit the IP allocation policy that the virtual machine uses when you export it to an OVF template and deploy the OVF template.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. Click the **Edit** button.
   The **Edit vApp Options** dialog box opens.
4. If vApp options are not enabled, select the **Enable vApp options** check box.
5. Select the **IP protocol** and an **IP allocation scheme**.

   The supported protocols are IPv4, IPv6, or both.

   Consult the following table to learn more about the IP allocation schemes.

Option	Description
OVF environment	IP allocation is determined by the environment in which you deploy the OVF template.
DHCP	The IP addresses are allocated through a DHCP server when the virtual machine is powered on.

   The information you specify in the Authoring section will be used if you export the virtual machine to OVF and deploy the OVF at a later time.
6. Click **OK**.

Edit OVF Details for a Virtual Machine

A virtual machine's OVF settings allow you to customize the OVF environment, OVF transport, and boot behavior after OVF deployment. You can edit and configure settings that affect the OVF environment in the Virtual Machine Properties dialog box.

vApp options must be enabled in order to access these options.

1. Navigate to a virtual machine from the vCenter inventory.
2. On the **Configure** tab, expand **Settings** and select **vApp options**.
3. Click the **Edit** button.
   The **Edit vApp Options** dialog box opens.
4. If vApp options are not enabled, select the **Enable vApp options** check box.
5. To customize the OVF settings for the virtual machine, click the **OVF Details** tab.

Option	Description
OVF environment transport	• If you select **ISO image**, an ISO image that contains the OVF template information is mounted in the CD-ROM drive. • If you select **VMware Tools**, the VMware Tools `guestInfo.ovfEnv` variable is initialized with the OVF environment document.
Installation boot	If you click **Enable**, the virtual machine reboots after the OVF deployment completes. You can specify the delay time in seconds before the virtual machine starts the reboot operation.

6. Click **OK** to save your changes.
   In the **vApp Options are enabled** pane, you can view the **OVF Settings** panel with information about the OVF settings.
7. Optional: To view information about the OVF environment settings, power on the virtual machine and click the **View OVF Environment** button in the **vApp Options are enabled** pane.
   The information is displayed in XML format.

Managing Virtual Machines

You can manage individual virtual machines or a group of virtual machines that belongs to a host or cluster.

From the virtual machine's console, you can change the guest operating system settings, use applications, browse the file system, monitor system performance, and so on. Use snapshots to capture the state of the virtual machine at the time you take the snapshot.

To migrate virtual machines using cold or hot migration, including vMotion, vMotion in environments without shared storage, and Storage vMotion, see the vCenter and Host Management documentation.

Install a Guest Operating System

A virtual machine is not complete until you install the guest operating system and VMware Tools. Installing a guest operating system in your virtual machine is essentially the same as installing it in a physical computer.

The basic steps for a typical operating system are described in this section. For more information, see the Guest Operating System Installation Guide.

Using PXE with Virtual Machines

You can start a virtual machine from a network device and remotely install a guest operating system using a Preboot Execution Environment (PXE). You do not need the operating system installation media. When you turn on the virtual machine, the virtual machine detects the PXE server.

PXE booting is supported for Guest Operating Systems that are listed in the VMware Guest Operating System Compatibility list and whose operating system vendor supports PXE booting of the operating system.

The virtual machine must meet the following requirements:

- Have a virtual disk without operating system software and with enough free disk space to store the intended system software.
- Have a network adapter connected to the network where the PXE server resides.

For details about guest operating system installation, see the Guest Operating System Installation Guide.

Install a Guest Operating System from Media

You can install a guest operating system from a CD-ROM or from an ISO image. Installing from an ISO image is typically faster and more convenient than a CD-ROM installation.

- Verify that the installation ISO image is present on a datastore accessible to the ESX host, such as VMFS, vSAN, or network file system (NFS) volume.
  Alternatively, verify that an ISO image is present in a content library.
- Verify that you have the installation instructions that the operating system vendor provides.

1. Log in to the vCenter system or host on which the virtual machine resides.
2. Select an installation method and click **OK**.

Option	Action
Client Device	Insert the installation CD-ROM for your guest operating system into the CD-ROM drive of your ESX host.
ISO image	1. Right-click the virtual machine and select **Edit Settings**. The virtual machine Edit Settings dialog box opens. If the **Virtual Hardware** tab is not preselected, select it. 2. Select **Datastore ISO File** from the CD/DVD drop-down menu, and browse for the ISO image for your guest operating system.
ISO image from a Content Library	1. Right-click the virtual machine and select **Edit Settings**. The virtual machine Edit Settings dialog box opens. If the **Virtual Hardware** tab is not preselected, select it. 2. Select **Content Library ISO File** from the CD/DVD drop-down menu, and select an ISO image from the content library items.

3. Right-click the virtual machine and select **Power On**.
   A green right arrow appears next to the virtual machine icon in the inventory list.

4. Follow the installation instructions that the operating system vendor provides.

Install VMware Tools. VMware highly recommends running the latest version of VMware Tools on your guest operating systems. Although the guest operating system can run without VMware Tools, you lose important functionality and convenience without them. See How do I Upgrade Virtual Machines for instructions on installing and upgrading VMware Tools.See Introduction to VMware Tools for instructions on installing and upgrading VMware Tools.

Upload ISO Image Installation Media for a Guest Operating System

You can upload an ISO image file to a datastore from your local computer. You can do this when a virtual machine, host, or cluster does not have access to a datastore or to a shared datastore that has the guest operating system installation media that you require.

Required privileges:

- **Datastore** > **Browse datastore** on the datastore.
- **Datastore** > **Low level file operations** on the datastore.
- **Host** > **Configuration** > **System Management**

1. In the vSphere Client, select **Menu** > **Storage**.
2. Select the datastore from the inventory to which you upload the file.
3. Optional: On the **Files** tab, click the **New Folder** icon to create a folder.
4. Select an existing folder or the folder that you created, and click the **Upload Files** icon.
5. On the local computer, find the file and upload it.

   ISO upload times vary, depending on file size and network upload speed.
6. Refresh the datastore file browser to see the uploaded file in the list.

After you upload the ISO image installation media, you can configure the virtual machine CD-ROM drive to access the file.

Customizing Guest Operating Systems

When you clone a virtual machine or deploy a virtual machine from a template, you can customize the guest operating system of the virtual machine. You can change the computer name, network settings, and license settings.

Customizing guest operating systems helps prevent conflicts that can occur when virtual machines with identical settings are deployed, such as duplicate computer names. You can apply customization during virtual machine deployment or later.

- During the cloning or deployment process, you can specify customization settings or select an existing customization specification.
- You can create a customization specification from the **Policies and Profiles** menu and apply it to a virtual machine.

Guest Operating System Customization Requirements

To customize the guest operating system, you must configure the virtual machine and guest operating system to meet VMware Tools and virtual disk requirements.

Other requirements apply, depending on the guest operating system type.

VMware Tools Requirements

The latest version of VMware Tools must be installed on the virtual machine or template to customize the guest operating system during cloning or deployment. For information about VMware Tools support matrix, see the Broadcom Product Interoperability Matrix.

Virtual Disk Requirements

The guest operating system being customized must be installed on a disk attached as SCSI node 0:0 in the virtual machine configuration.

Windows Requirements

Customization of Windows guest operating systems requires the virtual machine to be running on an ESX host of version 6.5 or later.

Linux Requirements

Customization of Linux guest operating systems requires that Perl is installed in the Linux guest operating system.

> **Note:**
> Certain Linux OS versions do not include the D-Bus package in the minimal Linux installation option. Therefore, running commands such as `dbus-uuidgen` might fail during customization.

Verifying Customization Support for a Guest Operating System

To verify customization support for Windows operating systems or Linux distributions, and compatible ESX hosts, see the Broadcom Compatibility Guide. You can use this online tool to search for the guest operating system and ESX version. After the tool generates your list, click the guest operating system to see whether guest customization is supported.

Create a vCenter Application to Generate Computer Names and IP Addresses

Instead of entering computer names and IP addresses for virtual NICs when you customize guest operating systems, you can create a custom application and configure it in such a way that vCenter generates the names and addresses.

The application can be an arbitrary executable binary or script file appropriate for the corresponding operating system in which vCenter is running. After you configure an application and make it available to vCenter, every time you initiate a guest operating system customization for a virtual machine, vCenter runs the application.

The application must comply with the reference XML file in Broadcom knowledge base article 312203.

1. Create the application and save it on the vCenter system's local disk.
2. Select a vCenter instance in the inventory.
3. Click the **Configure** tab, click **Settings**, and click **Advanced Settings**.
4. Click **Edit Settings** and enter the configuration parameters for the script.
   a) In the **Name** text box, enter `config.guestcust.name-ip-generator.arg1`.
   b) In the **Value** text box, enter `c:\sample-generate-name-ip.pl` and click **Add**.
   c) In the **Name** text box, enter `config.guestcust.name-ip-generator.arg2`.
   d) In the **Value** text box, enter the path to the script file on the vCenter system and click **Add**. For example, enter `c:\sample-generate-name-ip.pl`.
   e) In the **Name** text box, enter `config.guestcust.name-ip-generator.program`.
   f) In the **Value** text box, enter `c:\perl\bin\perl.exe` and click **Add**.
5. Click **Save**.

You can select the option to use an application to generate computer names or IP addresses during guest operating system customization.

Customize Windows During Cloning or Deployment

You can customize the Windows guest operating systems of the virtual machines when you deploy a new virtual machine from a template or clone an existing virtual machine. Customizing the guest operating system helps prevent conflicts that might result if you deploy virtual machines with identical settings, such as duplicate computer names.

- Verify that all requirements for customization are met. See Guest Operating System Customization Requirements.
- Verify that there are customization specifications available for use. For information about creating a guest customization specification, see Create a Customization Specification for Windows.

You can prevent new virtual machines and templates from receiving the same Security IDs (SIDs) as the source virtual machine. Duplicate SIDs can cause Kerberos and New Technology LAN Manager (NTLM) authentication failures

regardless of whether the computers are part of a domain or a workgroup. Windows 11 (versions 24H2 and 25H2) and Windows Server 2025 enforce SID uniqueness checks, and duplicate SIDs on these systems can result in inaccessible shared network folders, failed Remote Desktop Protocol (RDP) connections, and Failover Clustering errors. To avoid these issues, always use Sysprep when cloning or duplicating a Windows installation to ensure that each virtual machine receives a unique SID. For more information, see Kerberos and NTLM authentication failures due to duplicate SIDs and the documentation for your Microsoft Windows operating system.

Important:
After customization, the default administrator password is not preserved for Windows Server 2008. During customization, the Windows Sysprep utility deletes and recreates the administrator account on Windows Server 2008. You must reset the administrator password when the virtual machine starts the first time after customization.

1. Right-click any vSphere inventory object that is a valid parent object of a virtual machine, such as data center, cluster, vApp, resource pool, or host, and select **New Virtual Machine**.
2. On the **Select a creation page**, select **Clone an existing virtual machine** or **Deploy from template**.
3. Click **Next**.
4. Follow the prompts until you reach the **Select clone options** page.
5. On the **Select clone options** page, select the **Customize the operating system** check box and click **Next.**
6. On the **Customize guest OS** page, apply a customization specification to the virtual machine, and click **Next**.

Option	Action
Select an existing specification	Select a customization specification from the list.
Override	To change the guest customization specification for this deployment only, click **Override**, complete the steps in the **Override VM Customization Specification** wizard, and click **OK**. **Note:** With vSphere 9.1, you can deactivate IPv4 on the network interface, allowing the guest operating system to proceed directly with IPv6 networking. For more information, see Create and Manage Customization Specifications for Windows and Linux.

7. On the **User settings** page, specify the required settings for the virtual machine.
   This page of the wizard appears only if the selected specification requires additional customization.
8. On the **Ready to complete** page, review the details and click **Finish**.

When the new virtual machine starts for the first time, the guest operating system runs finalization scripts to complete the customization process. The virtual machine might restart several times during this process.

If the guest operating system pauses when the new virtual machine starts, it might be waiting for you to correct errors, such as an incorrect product key or an invalid user name. To determine whether the system is waiting for information, open the virtual machine console.

After you deploy certain Windows operating systems that are not volume licensed, you might need to reactivate your operating system on the new virtual machine.

If the new virtual machine encounters customization errors while it is starting, the errors are logged to `%WINDIR%\temp\vmware-imc`. To view the error log file, from the Windows **Start** menu navigate to **Programs** > **Administrative Tools** > **Event Viewer**.

Customize Linux During Cloning or Deployment

When you deploy a virtual machine from a template or clone an existing virtual machine, you can customize the Linux guest operating systems of the virtual machine.

- Verify that all requirements for customization are met. See Guest Operating System Customization Requirements.
- Verify that there are customization specifications available for use. For information about creating a guest customization specification, see Create a Customization Specification for Linux.

1. Right-click any vSphere Client inventory object that is a valid parent object of a virtual machine, such as data center, cluster, vApp, resource pool, or host, and select **New Virtual Machine**.
2. On the **Select a creation page**, select **Clone an existing virtual machine** or **Deploy from template**.
3. Click **Next**.
4. Follow the prompts until you reach the **Select clone options** page.
5. On the **Select clone options** page, select the **Customize the operating system** check box and click **Next.**
6. On the **Customize guest OS** page, apply a customization specification to the virtual machine, and click **Next**.

Option	Action
Select an existing specification	Select a customization specification from the list.
Override	To change the guest customization specification for this deployment only, click **Override**, complete the steps in the **Override VM Customization Specification** wizard, and click **OK**. **Note:** With vSphere 9.1, you can deactivate IPv4 on the network interface, allowing the guest operating system to proceed directly with IPv6 networking. For more information, see Create and Manage Customization Specifications for Windows and Linux.

7. On the **User settings** page, specify the required settings for the virtual machine.
   This page of the wizard appears only if the selected specification requires additional customization.
8. On the **Ready to complete** page, review the details and click **Finish**.

When the new virtual machine starts for the first time, the guest operating system runs finalization scripts to complete the customization process. The virtual machine might restart several times during this process.

If the guest operating system pauses when the new virtual machine starts, it might be waiting for you to correct errors, such as an incorrect product key or an invalid user name. To determine whether the system is waiting for information, open the virtual machine console.

If the new virtual machine encounters customization errors during startup, the errors are reported through the guest's system logging mechanism. View the errors in the `/var/log/vmware-imc/toolsDeployPkg.log` file.

Apply a Customization Specification to an Existing Virtual Machine

To prevent conflicts that can occur if you deploy virtual machines with identical settings, such as duplicate computer names, you can apply a customization specification to an existing virtual machine.

- Verify the guest operating system is installed.
- Verify that VMware Tools is installed and running.
- Power off the virtual machine.

When you clone an existing virtual machine, or deploy a virtual machine from a VM template in a folder, you can customize the guest operating system of the resulting virtual machine during the clone or the deployment tasks.

When you deploy a virtual machine from a template in a content library, you can customize the guest operating system only after the deployment task is complete.

1. Right-click a virtual machine in the vSphere inventory, and select **Guest OS** > **Customize Guest OS**.
   The **Customize Guest OS** dialog box opens.
2. Select a customization specification from the list and click **OK**.
   If the specification requires you to configure additional settings, a new dialog box opens and you are prompted to enter information about the required settings.

Create and Manage Customization Specifications for Windows and Linux

You can create and manage customization specifications for Windows and Linux guest operating systems. Customization specifications are XML files that contain guest operating system settings for virtual machines.

When you apply a specification to the guest operating system during virtual machine cloning or deployment, you prevent conflicts that might occur if you deploy virtual machines with identical settings, such as duplicate computer names.

vCenter saves the customized configuration parameters in the vCenter database. If the customization settings are saved, the administrator and domain administrator passwords are stored in encrypted format in the database. The system relies on a unique certificate for password encryption. If you reinstall the vCenter instance or attach a new vCenter instance to the database, the encrypted passwords become invalid.

Because the certificate used to encrypt the passwords is unique to each vCenter system, if you reinstall vCenter or attach a new instance of the server to the database, the encrypted passwords become invalid. You must re-enter the passwords before you can use them.

You can deactivate IPv4 on the network interface to allow the guest OS to use an IPv6-only network directly, simplifying network configuration. Deactivating IPv4 also reduces dual-stack overhead, limits unnecessary IPv4 address allocation, and can improve startup performance by avoiding delays that are associated with the IPv4 DHCP timeout.

IPv4 deactivation guest operating system support

The following guest operating systems are supported for IPv4 deactivation:

- Windows Server 2008 R2 or Windows 7 SP1 and later
- SUSE Linux Enterprise Server 10 SP4 or later
- Red Hat Enterprise Linux 5.10 or later
- Oracle Linux 5.x or later
- CentOS 5.x or later
- Rocky Linux 8.x or later
- AlmaLinux OS 8.x or later
- Miracle Linux 8.x or later
- Ubuntu 18.04 or later

Create a Customization Specification for Linux

Save the system settings for a Linux guest operating system in a customization specification, which you can apply when cloning virtual machines or deploying virtual machines from templates. You can use the cloud-init utility as a standardized way to initialize a virtual machine when you deploy it in a cloud environment.

- Verify that all requirements for customization are met. See Guest Operating System Customization Requirements.
- To run the customization script:
  - Verify that VMware Tools version 10.1.0 or later is installed. The customization fails if VMware Tools is version earlier than 10.1.0 and you attempt to run the customization script.
  - In the VMware Tools configuration, the `enable-custom-scripts` option is deactivated by default for security reasons. When you attempt to run the customization script with a deactivated `enable-custom-scripts` option, the customization fails with a customization error.

    For example, to enable the `enable-custom-scripts` option, you must run `vmware-toolbox-cmd` as a root user with the `config` command:

    ```
    vmware-toolbox-cmd config set deployPkg enable-custom-scripts true
    cat /etc/vmware-tools/tools.conf
    [deployPkg]
    enable-custom-scripts = true
    ```

    To verify that you set the option correctly, you can run the following command:

    ```
    vmware-toolbox-cmd config get deployPkg enable-custom-scripts
    [deployPkg] enable-custom-scripts = true
    ```

For more information, see the *VMware Tools User Guide*.

- To customize the guest operating system of a virtual machine with cloud-init metadata and user data:
  - Verify that VMware Tools version 11.3.0 or later is installed.
  - Verify that cloud-init version 21.1 or later is installed.

1. Click the Main menu icon (☰) and click **Policies and Profiles**.
2. Click **VM Customization Specifications**.
3. To create a specification, from the **VM Customization Specifications** pane, click **New**.
   The **New VM Customization Specification** wizard opens.
4. On the **Name and target OS** page, enter a name and a description for the customization specification, and select **Linux** as a target guest OS.
5. Optional: To initiate customization with raw cloud-init data:
   a) Select the **Use cloud-init config data** check box and click **Next**.
   b) On the **Cloud-init metadata** page, apply a cloud-init metadata to the guest operating system of the VM.

      For more information about the metadata and user data format, see Broadcom knowledge base article 311895.

Option	Action
Upload a cloud-init metadata file	The cloud-init metadata file is a plain-text in YAML or JSON file format. The maximum file size is 512 KB. • Click **Upload** and navigate to the file on your local machine. The contents of the file appear in the text box.
Enter a cloud-init metadata script	• Enter the metadata script directly into the text box.

   c) Click **Next**.
   d) Optional: On the **Cloud-init user data**, apply a cloud-init user configuration to the guest operating system of the VM. For example, you can add a user to the guest operating system.

      For more information about the cloud-init user data, see the Cloud-init documentation.

      Create and manage

Upload a cloud-init user data file	The cloud-init user data file is a plain-text in raw cloud-init format. The maximum file size is 512 KB. • Click **Upload** and navigate to the file on your local machine. The contents of the file appear in the text box.
Enter a cloud-init user data	• Enter the cloud-init user data script directly into the text box.

e) Click **Next**.
f) On the **Ready to complete** page, review the details and click **Finish** to save your changes.

6. Optional: To initialize traditional customization:
   a) On the **Computer name** page, enter a computer name for the guest operating system and a domain name.
      The guest operating system uses the computer name to identify itself on the network. On Linux system, it is called the host name.

Option	Action
Use the virtual machine name	Select this option to use the virtual machine name. The computer name that vCenter creates is identical to the name of the virtual machine on which the guest operating system is running. If the name exceeds 63 characters, it is truncated.
Enter a name in the Clone/Deploy wizard	Select this option to be prompted to enter a name during cloning or deployment.
Enter a name	• Enter a name. The name can contain alphanumeric characters and a hyphen (-). It cannot contain a period (.), blank spaces, or special characters, and cannot contain digits only. Names are not case-sensitive. • To ensure that the name is unique, select the **Append a numeric value** check box. This action appends a hyphen followed by a numeric value to the virtual machine name. The name is truncated if it exceeds 63 characters when combined with the numeric value.
Generate a name using the custom application configured with vCenter	Enter a parameter that can be passed to the custom application.

   b) Enter the **Domain Name** for the computer and click **Next**.
   c) On the **Time zone** page, select the time zone for the virtual machine and click **Next**.
   d) On the **Customization script** page, apply a customization script to the guest operating system of the VM and click **Next**.

Option	Action
Upload a file containing the customization script	• Click **Browse** and navigate to the file on your local machine. The contents of the script appear in the **Script** text box.
Enter the customization script	Enter the customization script directly into the **Script** text box.

The customization script cannot exceed 1500 characters.

Note:

The default timeout period for the guest customization to complete is set to 100 seconds and includes the time for the script to run when you use a `"precustomization"` command-line parameter. If you run scripts that take a time exceeding the timeout, the guest customization fails.

When you add a customization script with the `"precustomization"` command-line parameter, it is called before the guest customization begins. As a result, the virtual NIC is disconnected and you cannot access the network.

When you add a customization script with the `"postcustomization"` command-line parameter, it is called after the guest customization finishes. As a result, the script is scheduled in the initialization process after the virtual machine powers on, the NIC is connected, and you can access the network. The time for the script to run is not included in the default timeout period and you avoid a guest customization failure.

Customization Script Example

```
#!/bin/sh
```

```
if [ x$1 == x"precustomization" ]; then
echo Do Precustomization tasks
elif [ x$1 == x"postcustomization" ]; then
echo Do Postcustomization tasks
fi
```

e) On the **Network** page, select the type of network settings to apply to the guest operating system and click **Next**.

Option	Action
Use standard network settings	• Select **Use standard network settings** so that vCenter configures all network interfaces from a DHCP server by using the default settings.
Manually select custom settings	• Select a network adapter from the list or add a new one. • For the selected NIC, click **Edit**. The **Edit Network** dialog box opens. • To configure the virtual machine to use an IPv4 network, click the **IPv4** tab. If you select the **Prompt the user for an IPv4 address when the specification is used** option, vCenter prompts for an IP address when you select to apply the customization specification during cloning or deployment. You are also prompted to configure the gateways during cloning and deployment. • To explicitly deactivate IPv4, select the **Disable IPv4** check box. • To configure the virtual machine to use an IPv6 network, click the **IPv6** tab. If you select the **Prompt the user for an address when the specification is used** option, vCenter prompts for an IP address when you select to apply the customization specification during cloning or deployment. You are also prompted to configure the gateways during cloning and deployment. • Click **OK**

f) On the **DNS settings** page, enter DNS server and domain settings.
The **Primary DNS**, **Secondary DNS**, and **Tertiary DNS** text boxes accept both IPv4 and IPv6 addresses.

g) On the **Ready to complete** page, review the details and click **Finish** to save your changes.

The customization specification that you created is listed in the Customization Specification Manager. You can use the specification to customize virtual machine guest operating systems.

Create a Customization Specification for Windows

Save specific Windows guest operating system settings in a customization specification, which you can apply when cloning virtual machines or deploying from templates.

Ensure that all requirements for customization are met. See Guest Operating System Customization Requirements.

Note:
The default administrator password is not preserved for Windows Server 2008 after customization. During customization, the Windows Sysprep utility deletes and recreates the administrator account on Windows Server 2008. You must reset the administrator password when the virtual machine starts the first time after customization.

Starting with vSphere 8.0 Update 2, you can specify an organizational unit (OU) for your remote desktops. An OU is a subdivision in Active Directory that contains users, groups, computers, or other OUs.

1. Select **Menu** > **Policies and Profiles**, and under Policies and Profiles, click **VM Customization Specifications**.
2. Click the **New** button.
   The **New VM Customization Specification** wizard opens.

3. On the **Name and target OS** page, enter a name and description for the customization specification and select **Windows** as a target guest OS.
4. Optional: Select the **Generate a new security identity (SID)** option and click **Next**.

   A Windows Security ID (SID) is used in some Windows operating systems to uniquely identify systems and users. If you do not select this option, the new virtual machine has the same SID as the virtual machine or template from which it was cloned or deployed.

   Duplicate SIDs do not cause problems when the computers are part of a domain and only domain user accounts are used. However, if the computers are part of a Workgroup or local user accounts are used, duplicate SIDs can compromise file access controls. For more information, see the documentation for your Microsoft Windows operating system.
5. On the **Registration Information** page, enter the virtual machine owner's name and organization, and click **Next**.
6. On the **Computer name** page, enter a computer name for the guest operating system and a domain name.

   The operating system uses the computer name to identify itself on the network. On Linux systems, it is called the host name.

Option	Action
Use the virtual machine name	Select this option to use the virtual machine name. The computer name that vCenter creates is identical to the name of the virtual machine on which the guest operating system is running. If the name exceeds 63 characters, it is truncated.
Enter a name in the Clone/Deploy wizard	Select this option to be prompted to enter a name during cloning or deployment.
Enter a name	1. Enter a name. The name can contain alphanumeric characters and a hyphen (-). It cannot contain a period (.), blank spaces, or special characters, and cannot contain digits only. Names are not case-sensitive. 2. To ensure that the name is unique, select the **Append a numeric value** check box. This action appends a hyphen followed by a numeric value to the virtual machine name. The name is truncated if it exceeds 63 characters when combined with the numeric value.
Generate a name using the custom application configured with vCenter	Enter a parameter that can be passed to the custom application.

7. On the **Windows license** page, provide licensing information for the Windows operating system and click **Next**.

Option	Action
For nonserver operating systems	Type the Windows product key for the new guest operating system.
For server operating systems	1. Type the Windows product key for the new guest operating system. 2. Select **Include Server License Information**. 3. Select either **Per seat** or **Per server**. 4. If you select **Per server**, enter the maximum number of simultaneous connections for the server to accept.

8. On the **Administrator Password** page, configure the administrator password for the virtual machine and click **Next**.
   a) Enter a password for the administrator and confirm the password by entering it again.
   b) Select the **Automatically logon as Administrator** check box to log users in to the guest operating system as Administrator, and select the number of times to log in automatically.
9. On the **Time zone** page, select the time zone for the virtual machine and click **Next**.
10. On the **Commands to run once** page, specify commands to run the first time a user logs in to the guest operating system and click **Next**.

    See the Microsoft Sysprep documentation for more information about `RunOnce` commands.

11. On the **Network** page, select the type of network settings to apply to the guest operating system and click **Next**.
    - Select **Use standard network settings** so that vCenter configures all network interfaces from a DHCP server by using the default settings.
    - Select **Manually select custom settings** and configure each network interface yourself.
        a. Select a network adapter from the list or add a new one.
        b. For the selected NIC, click **Edit**.
           The **Edit Network** dialog box opens.
        c. Click the **IPv4** tab to configure the virtual machine to use IPv4 network.
           You can configure all the settings at that stage, or you can select the **Prompt the user for an IPv4 address when the specification is used** option. In that case, vCenter prompts for an IP address when you select to apply that customization specification during cloning or deployment. With that option, you can also configure subnet and gateways during cloning or deployment.
           To explicitly deactivate IPv4, select the **Disable IPv4** check box. You can deactivate IPv4 on the network interface, allowing the guest OS to proceed directly with IPv6-only networking.
        d. Click the **IPv6** tab to configure the virtual machine to use an IPv6 network.
           You can configure all the settings at that stage, or you can select the **Prompt the user for an address when the specification is used** option. In that case, vCenter prompts for an IP address when you apply that customization specification during cloning or deployment. With that option, you can also configure the gateways during cloning or deployment.
        e. Click the **DNS** tab to specify DNS server details.
        f. Click **WINS** to specify primary and secondary WINS server information.
        g. Click **OK** to apply changes and click **Next**.
12. On the **Workgroup or Domain** page, select how the virtual machine participates in the network and click **Next**.

Option	Action
Workgroup	Enter a workgroup name. For example, `MSHOME`.
Windows Server Domain	1. Enter the domain name. 2. To add a computer to the specified domain, enter the user name and password for a user account that has permission. 3. Enter the OU path. For example, `OU=MyOU,DC=MyDom,DC=MyCompany,DC=com`

13. On the **Ready to complete** page, review the details and click **Finish** to save your changes.

The customization specification that you created is listed in the Customization Specification Manager. You can use the specification to customize virtual machine guest operating systems.

Create a Customization Specification for Windows Using a Custom Sysprep Answer File

A custom sysprep answer file is a file that stores various customization settings such as computer name, licensing information, and workgroup or domain settings. You can supply a custom sysprep answer file as an alternative to specifying many of the settings in the Guest Customization wizard.

Ensure that all requirements for customization are met. See Guest Operating System Customization Requirements.

Windows Server 2003 and Windows XP use a text file called `sysprep.inf`. Windows Server 2008, Windows Vista, and Windows 7 use an XML file called `sysprep.xml`. You can create these files using a text editor, or use the Microsoft Setup Manager utility to generate them. For more information about how to create a custom sysprep answer file, see the documentation for the relevant operating system.

> **Important:**
>
> If you use a custom sysprep answer file to deploy a virtual machine with an operating system Windows Vista or later, you must specify the network customization specifications in the sysprep file. The custom network settings that you configure in the **New VM Guest Customization Specification** wizard are not applied. For more information, see Broadcom knowledge base article 313515.

You can prevent Windows from assigning new virtual machines or templates with the same Security IDs (SIDs) as the original virtual machine. Duplicate SIDs do not cause problems when the computers are part of a domain and only domain user accounts are used. However, if the computers are part of a Workgroup or local user accounts are used, duplicate SIDs can compromise file access controls. For more information, see the documentation for your Microsoft Windows operating system.

1. Select **Menu** > **Policies and Profiles**, and under Policies and Profiles, click **VM Customization Specifications**.
2. Click the **Create a new specification** icon.
   The **New VM Customization Specification** wizard opens.
3. On the Name and target OS page, enter a name and description for the customization specification and select **Windows** as a target guest OS.
4. Optional: Select the **Generate a new security identity (SID)** option.
   A Windows Security ID (SID) is used in some Windows operating systems to uniquely identify systems and users. If you do not select this option, the new virtual machine has the same SID as the virtual machine or template from which it was cloned or deployed.

   Duplicate SIDs do not cause problems when the computers are part of a domain and only domain user accounts are used. However, if the computers are part of a Workgroup or local user accounts are used, duplicate SIDs can compromise file access controls. For more information, see the documentation for your Microsoft Windows operating system.
5. Select **Use Custom Sysprep Answer File** and click **Next**.
6. On the Custom sysprep file, select the option to import or create a sysprep answer file and click **Next**.

Option	Description
Import a Sysprep answer file	Click **Upload** and browse to the file.
Create a Sysprep answer file	Type the contents of the file in the text box.

7. On the **Network** page, select the type of network settings to apply to the guest operating system and click **OK**.
   - Select **Use standard network settings** so that vCenter configures all network interfaces from a DHCP server by using the default settings.
   - Select **Manually select custom settings** and configure each network interface yourself.
     a. Select a network adapter from the list or add a new one.
     b. For the selected NIC, click **Edit**.
        The **Edit Network** dialog box opens.
     c. Click the **IPv4** tab to configure the virtual machine to use an IPv4 network.
        You can configure all the settings at that stage, or you can select the **Prompt the user for an IPv4 address when the specification is used** option. In that case, vCenter prompts for an IP address when you apply that customization specification during cloning or deployment. With that option, you can also configure the subnet and gateways during cloning or deployment.
        To explicitly deactivate IPv4, select the **Disable IPv4** check box.
     d. Click the **IPv6** tab to configure the virtual machine to use an IPv6 network.
        You can configure all the settings at that stage, or you can select the **Prompt the user for an address when the specification is used** option. In that case, vCenter prompts for an IP address when you apply that customization specification during cloning or deployment. With that option, you can also configure the gateways during cloning or deployment.
     e. Click the **DNS** tab to specify the DNS server details.
     f. Click **WINS** to specify primary and secondary WINS server information.
     g. Click **OK** to apply the changes and click **Next**.
8. On the Ready to complete page, review the details and click **Finish** to save your changes.

The customization specification that you created is listed in the Customization Specification Manager. You can use the specification to customize virtual machine guest operating systems.

Manage Customization Specifications

You can edit, duplicate, export, or delete existing specifications.

1. In the vSphere Client, select **Menu** > **Policies and Profiles** and click **VM Customization Specifications**.
2. Select a customization specification and select your task.

Option	Description
Edit customization specification	You can change the customization spec, such as changing the networking configuration. Click **Edit** and make the necessary changes.
Duplicate customization specification	If you need a customization specification that is only slightly different from an existing specification, you can use the Customization Specification Manager to create a copy of the existing specification and modify it. For example, you must change the IP address or the administrator password.
Export customization specification	You can export customization specifications and can save them as `.xml` files.
Delete customization specification	You can remove customization specifications to free up storage.

Import a Customization Specification

You can import an existing specification and can use the specification to customize the guest operating system of a virtual machine.

Before you begin, you must have at least one customization specification that is saved as an xml file located on a file system accessible from the vSphere Client.

1. In the vSphere Client, select **Menu** > **Policies and Profiles** and click **VM Customization Specifications**.
2. Click the **Import** button.
3. Browse to the `.xml` file to import, specify a name and optional description, and click **OK**.

The imported specification is added to the list of customization specifications.

Configure User Mappings on Guest Operating Systems

As a vSphere administrator, you can enable guest OS access on certain Single Sign-On (SSO) accounts.

Enabling SSO accounts to log in to a guest OS provides users with additional capabilities to perform administrative tasks on guest virtual machines, such as installing or upgrading the VMware Tools or configuring apps.

Functionality to allow vSphere administrators to configure a guest operating system to use VGAuth authentication. The vSphere administrator must know the guest administrator password for the enrollment process and the virtual machine must be powered off.

To enroll SSO users to a guest user account, you must enroll SSO users to accounts in guest operating systems. The enrollment process maps a vSphere user to a particular account in the guest by using SSO certificates. Subsequent guest management requests use an SSO SAML token to log in to the guest.

You must configure VMs to accept X.509 certificates. X.509 certificates allow the vSphere administrators in your data center to use SAML tokens that are issued by a single sign-on service to access guest OSs.

View Existing SSO User Mappings

You can view the existing guest user mappings for guest operating systems on the selected virtual machine. You must authenticate your credentials to view guest mappings.

1. Navigate to the virtual machine and click the **Configure** tab.
2. Click the **Guest User Mappings** tab.
3. To log in to your guest OS account, enter your user name and password, and click **Log In**.

   The existing in-guest user mappings are displayed.

Add SSO Users to Guest Operating Systems

You can map a new SSO user to a guest user account by creating a user map. Mapping can be established for any type of SSO users, for example, solution and users.

Power on the virtual machine.

1. Navigate to the virtual machine and click **Configure** tab.
2. Click **Guest User Mappings**.
3. Enter your user name and password and click **Log In**.
4. In the **Guest User Mappings** pane, click the **Add** button.
   The **Add New User Mapping** dialog box opens.
5. From the list of SSO users, select the SSO user that you want to map to a guest account.
6. Specify a guest OS user name and click **Add**.
   The SSO user is mapped to a guest user account. A new guest user account is added to the list of **Guest User Mappings**.

Remove SSO Users from Guest Operating Systems

You can remove an existing SSO account from guest user mappings.

Power on your virtual machine.

1. Navigate to a virtual machine and click the **Configure** tab.
2. Click **Guest User Mappings**, enter you user name and password, and click **Log In**.
3. In the **Guest User Mappings** pane, select the SSO user from the list that you want to remove.
4. Click the **Remove** button.
5. Click **Yes** to confirm.
   The mapping between the selected SSO user account and guest OS account is removed.

Managing the Power States of a Virtual Machine

The basic power operations for a virtual machine include powering on, powering off, suspending, reset, and hard stop. These power options are analogous to the power operations on a physical computer.

Verify that you have the following privileges:

- **VirtualMachine** > **Interaction** > **PowerOn**
- **VirtualMachine** > **Interaction** > **PowerOff**
- **VirtualMachine** > **Interaction** > **Suspend**
- **VirtualMachine** > **Interaction** > **Reset**

For information about how to change the virtual machine power states, see Configure the Virtual Machine Power States.

1. Navigate to a virtual machine in the inventory.
2. Right-click the virtual machine or click **Actions**, and select **Power**.
3. Select a power operation.

Option	Description
Power On	Powers on a virtual machine when the virtual machine is stopped.
Power Off	Powers off a virtual machine. Powering off a virtual machine might cause loss of data within the guest operating system.
Suspend	Suspends a running virtual machine and leaves it connected to the network. When you resume a suspended virtual machine,

Option	Description
	the virtual machine continues operating at the same point the virtual machine was at when it was suspended.
Reset	Restarts the guest operating system. By using this operation, the virtual machine powers off and then powers on. By resetting a virtual machine, you might lose all unsaved information within the guest operating system.
Hard stop	You can use this operation only when you cannot power off a virtual machine or if a virtual machine becomes unresponsive. When you hard stop a virtual machine, all processes are ended and the virtual machine is powered off. By using this operation, you might lose all unsaved information. If you use the Hard stop operation, vSphere HA restarts the virtual machine.

Edit Virtual Machine Startup and Shutdown Settings

You can configure virtual machines running on an ESX host to start up and shut down with the host or after a delay. You can also set the default timing and startup order for virtual machines. This way, the operating system has enough time to save data when the host enters maintenance mode or is being powered off for another reason.

The Virtual Machine Startup and Shutdown (automatic startup) setting is deactivated for all virtual machines residing on hosts that are in a vSphere HA cluster. Automatic startup is not supported with vSphere HA.

> **Note:**
> You can also create a scheduled task to change the power settings for a virtual machine. See *vCenter and Host Management*.

1. In the vSphere Client, navigate to and select the host where the virtual machine is located.
2. Click the **Configure** tab.
3. Under **Virtual Machines**, select **VM Startup/Shutdown** and click **Edit**.
   The **Edit VM Startup/Shutdown Configuration** dialog box opens.
4. Select **Automatically start and stop the virtual machines with the system**.
5. Optional: In the **Default VM Settings** pane, configure the default startup and shutdown behavior for all virtual machines on the host.

Setting	Description
Startup Delay	After you start the ESX host, it starts powering on the virtual machines that are configured for automatic startup. After the ESX host powers on the first virtual machine, the host waits for the specified delay time and then powers on the next virtual machine. The virtual machines are powered on in the startup order specified under the **Default VM Settings** pane.
Continue if VMware Tools is started	Shortens the startup delay of the virtual machine. If VMware Tools starts before the specified delay time passes, the ESX host powers on the next virtual machine without waiting for the delay time to pass.
Shutdown delay	Shutdown delay is the maximum time the ESX host waits for a shutdown command to complete. When you power off the ESX host, the autostart manager initiates the automatic shutdown on the first virtual machine and waits within the specific delay time for the virtual machine to complete the power action. The power action can be Power Off, Guest Shutdown, or Suspended. The order in which virtual machines are shut down is the reverse of their startup order. After the ESX host shuts down the first virtual machine within the time that you specify, the host shuts down the next virtual machine. If a virtual machine does not shut down within the specified delay time, the host runs a power off command and then starts shutting down

Setting	Description
	the next virtual machine. The ESX host shuts down only after all virtual machines are shut down.
Shutdown action	Select a shutdown action that is applicable to the virtual machines on the host when the host shuts down. • **Guest Shutdown** • **Power Off** • **Suspend** • **None**

6. Optional: You can also configure the startup order and behavior for individual virtual machines.

   Use this option when you need the delay of the virtual machine to be different from the default delay for all machines. The settings that you configure for individual virtual machines override their default settings.

   a) To change the startup order of virtual machines, select a virtual machine from the **Manual Startup** category and use the up arrow to move it up to the **Automatic** or **Automatic Ordered** categories.

      Use the up and down arrows to change the startup order for virtual machines in the **Automatic** and **Manual Startup** categories. During the shutdown, the virtual machines shut down in the reverse order.

   b) To edit the startup and shutdown behavior of a virtual machine, select a virtual machine, use the up arrow to move it, and click the **Edit** icon.

      The **Virtual Machine Startup/Shutdown setting** dialog box opens.

   c) In the **Startup Settings** pane, configure the startup behavior of the virtual machine.

      You can decide to use the default startup delay or you can specify a new one. If you select **Continue immediately if VMware Tools starts**, the ESX host powers on the next virtual machine without waiting for the delay to pass.

   d) In the **Shutdown Settings** pane, configure the shutdown behavior of the virtual machine.

      You can use the default shutdown delay or specify a new one and select the shutdown action.

   e) Click **OK**.

7. Click **OK**.

Using a Virtual Machine Console

You can access a virtual machine console by displaying it in a separate web browser, or from the VMware Remote Console (VMRC).

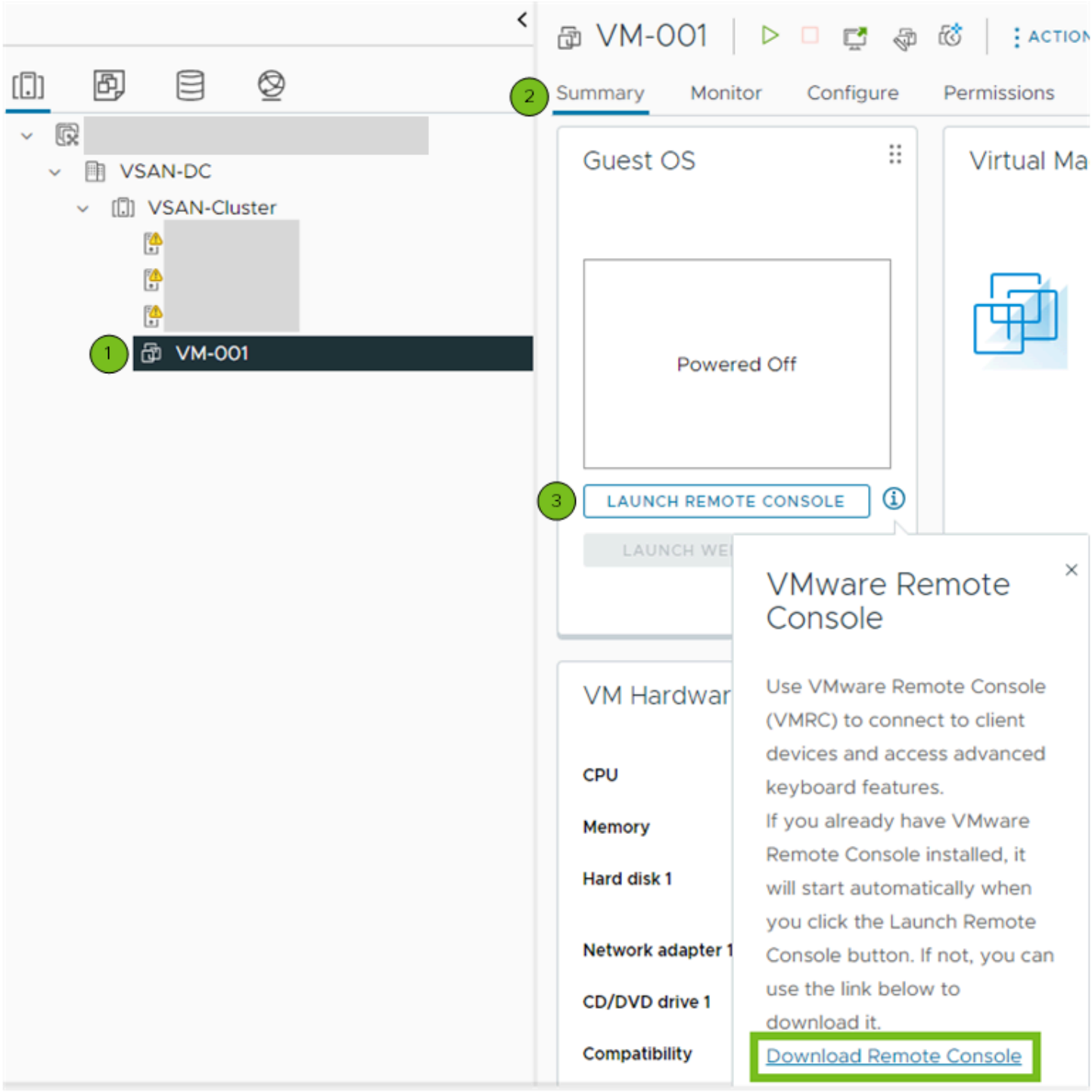
From the virtual machine remote console, you can perform tasks in the virtual machine such as installing an operating system, configuring the operating system settings, running applications, monitoring performance, and so on. The vSphere Client offers these choices:

- Launch the Web console to display the VM console in a separate browser tab.
- Download the VMware Remote Console (VMRC) standalone application, which opens in a separate window. The VMware Remote Console standalone application enables you to connect to client devices and launch virtual machine consoles on remote hosts.

Install the VMware Remote Console Application

The VMware Remote Console (VMRC) is a standalone console application. VMRC enables you to connect to client devices and open virtual machine consoles on remote hosts.

Download the VMRC if you do not already have the VMRC installed. For more information, see Broadcom knowledge base article 368995.


VM-001
ACTION
2
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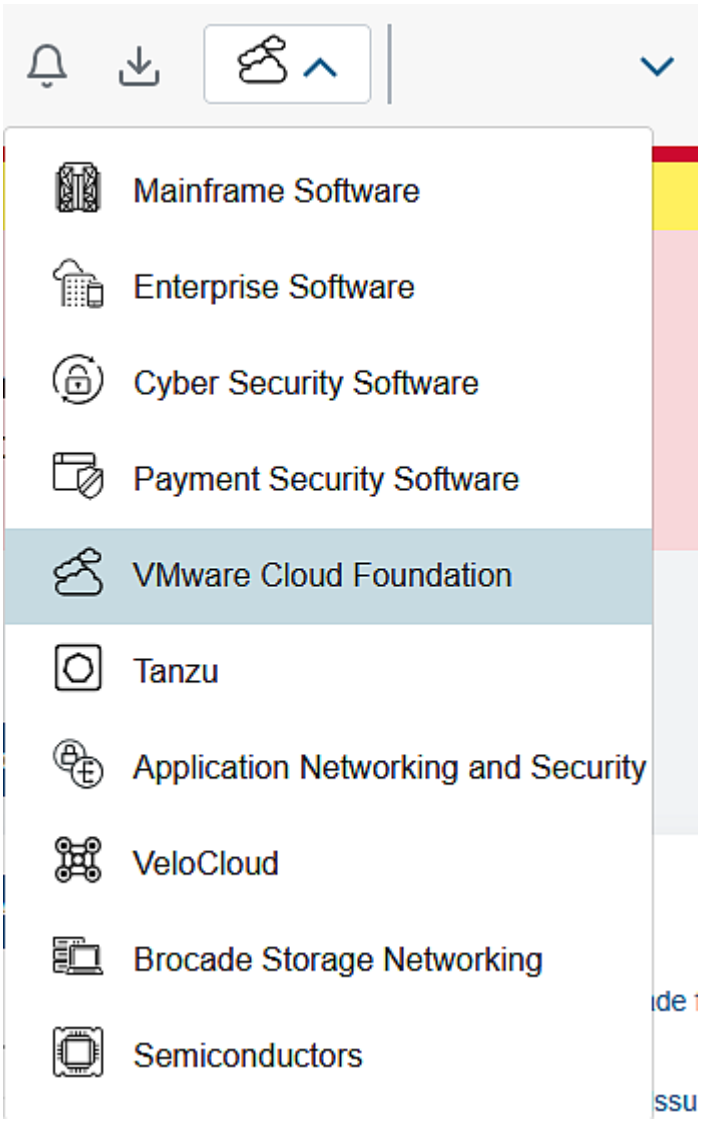
Monitor
Configure
Permissions
Guest OS
Virtual Ma
VSAN-DC
VSAN-Cluster
1
VM-001
Powered Off
3
LAUNCH REMOTE CONSOLE
VM Hardwar
CPU
Memory
Hard disk 1
Network adapter 1
CD/DVD drive 1
Compatibility
VMware Remote Console
Use VMware Remote Console (VMRC) to connect to client devices and access advanced keyboard features.
If you already have VMware Remote Console installed, it will start automatically when you click the Launch Remote Console button. If not, you can use the link below to download it.
Download Remote Console

1. In the vSphere Client, navigate to a virtual machine in the inventory.
2. Click the **Summary** tab.
3. From the **Launch Remote Console** link, click the information icon ( ) and click **Download Remote Console**.
4. Download the VMRC installer from the Broadcom Support Portal.

   Note:

   To download the VMRC installer, you must have a profile at the Broadcom Support Portal.

   a) Log in to the Broadcom Support Portal with your credentials.
   b) From the drop-down menu, select **VMware Cloud Foundation**.



   c) From the side navigation menu, click **My Downloads** and enter `VMware vSphere` in the **Search Product Name** bar.
   d) Click **Show Results** and click **VMware vSphere**.
   e) Click the **Products** tab and expand **VMware vSphere - Enterprise** from the list.
   f) Click the release version and click the **Drivers & Tools** tab.
   g) In the **Search** bar, enter `VMware Remote Console`.

   The list with VMRC versions appears.

h)
Expand the version that you want to download and click the download ( ) icon.

Start the VMware Remote Console Application

You can use the standalone VMRC application to connect to client devices.

- Verify that VMRC is installed on your local system. You can download the VMRC installer from the Broadcom Support Portal.
- Verify that you have the **Virtual machine** > **Interaction** > **Console interaction** privilege.

With VMRC, you can access the mouse and keyboard connected to remote virtual machines. To perform administrative tasks, you must log in to the VMRC as an administrator.

1. In the vSphere Client, navigate to a virtual machine in the inventory.
2. On the **Summary** tab, click **Launch Remote Console**.
   A dialog box opens that requires you to confirm that you want to open the remote console.
3. In the **Open VMware Remote Console** dialog box, confirm that you want to open VMRC.
   The VMRC opens as a standalone application for the selected virtual machine. You can also run more than one console to access remotely several virtual machines at the same time.

Open the Web Console

You can access a virtual machine desktop from the vSphere Client by opening the web console and performing various tasks in the virtual machine. For example, you can install an operating system, configure the operating system settings, run applications, monitor performance, and so on.

- Verify that the virtual machine has a guest operating system and that VMware Tools is installed.
- Verify that the virtual machine is powered on.
- Verify that you have the **Virtual machine** > **Interaction** > **Console interaction** privilege.

1. In the vSphere Client, navigate to a virtual machine in the inventory.
2. In the **Summary** tab, select **Launch Web Console**.
   The console opens in a new browser tab.
3. Click anywhere inside the console window to start using your mouse, keyboard, and other input devices in the console.

   Note:

   For information about supported international keyboards, refer to the VMware HTML Console SDK Release Notes.
4. Optional: Click **Send Ctrl-Alt-Delete** to send the Ctrl+Alt+Delete keystroke combination to the guest operating system.
5. Optional: Press Ctrl+Alt to release the pointer from the console window and work outside the console window.
6. Optional: Click **Full Screen** to view the console in full screen mode.
7. Optional: Press Ctrl+Alt+Enter to enter or exit full screen mode.

Manage the VMware Remote Console Proxy Configuration

VMware Remote Console proxy for vSphere (VMRC proxy) is a service in the vCenter system which transmits the network traffic between VMRC and the ESX hosts. When you use the VMRC proxy, VMRC does not require a direct network connection to the ESX host.

You can activate or deactivate the VMRC proxy. You also can change the VMRC proxy settings to reduce the workload on the vCenter system.

Activate the VMware Remote Console Proxy

The VMRC proxy is deactivated by default. You can activate the VMRC proxy service through **Advanced Settings** of your vCenter system.

Verify that you have the **Global.Settings** privilege.

1. In the vSphere Client, navigate to and select the vCenter instance.
2. On the **Configure** tab, select **Advanced Settings**.
3. Click **Edit Settings**.
   The **Edit Advanced vCenter Settings** dialog box opens.
4. In the **Name** text box, enter the name of the service - `config.mksdevproxy.enable`.
5. In the **Value** text box, enter `true` and click **Add**.
6. Click **Save**.
   The proxy setting appears in the list containing all configuration parameters.

Deactivate the VMware Remote Console Proxy

To simplify your vCenter configuration, you can deactivate the VMRC proxy.

Verify that you have the **Global.Settings** privilege.

1. In the vSphere Client, navigate to and select the vCenter instance.
2. On the **Configure** tab, select **Advanced Settings**.
3. Click **Edit Settings**.
   The **Edit Advanced vCenter Settings** dialog box opens.
4. Click the **Filter** icon in the **Name** column.
5. To view the VMRC proxy parameter, in the text box, enter `config.mksdevproxy.enable` and close the dialog box.
6. In the **Value** text box, enter `false`, and click **Save**.

Manage the VMware Remote Console Proxy Settings

To ensure that the vCenter system performs efficiently, you can change the VMRC proxy settings. You can configure the number of concurrent VMRC proxy connections and the maximum bandwidth for each concurrent VMRC proxy connection.

Configure a Concurrent VMRC Proxy Connection

The VMRC network traffic might impact the work of the vCenter instance and you can limit the number of the concurrent VMRC proxy connections.

Verify that you have the **Global.Settings** privilege.

To ensure that the vCenter performance is optimum, configure the number of the concurrent VMRC proxy connections to be between 1 and 1024. To meet your configuration needs, you can change the maximum concurrent VMRC proxy connections. By default, the maximum number of concurrent VMRC proxy connections is 32.

1. In the vSphere Client, navigate to and select the vCenter instance.
2. On the **Configure** tab, select **Advanced Settings**.
3. Click **Edit Settings**.
4. Configure the number of the concurrent VMRC proxy connections.

Option	Action
Configure a concurrent VMRC proxy connection	1. In the **Name** text box, enter `config.mksdevproxy.connLimit`.

Option	Action
	2. In the **Value** text box, enter a value for the maximum number of concurrent connections allowed. 3. Click **Add**.
Limit the number of the VMRC proxy connections	1. Click the **Filter** icon in the **Name** column. 2. To view the VMRC proxy parameter, in the text box enter `config.mksdevproxy.connLimit` and close the dialog box. 3. In the **Value** text box, change the number of the concurrent connections.

5. Click **Save**.

Configure the Maximum Bandwidth for a Concurrent VMRC Proxy Connection

You can limit the number of the available network bandwidth.

Verify that you have the **Global.Settings** privilege.

Each concurrent connection has a maximum of 300 KBps network bandwidth. VMRC requires minimum 50 KBps for the basic mouse, keyboard, and screen transfer.

1. In the vSphere Client, navigate to and select the vCenter instance.
2. On the **Configure** tab, select **Advanced Settings**.
3. Click **Edit Settings**.
4. Configure the maximum bandwidth for each concurrent VMRC proxy connection.

Option	Action
Configure the VMRC inbound traffic to the vCenter system	1. In the **Name** text box, enter `config.mksdevproxy.readthrottler`. 2. In the **Value** text box, enter the value of the inbound traffic in KBps. 3. Click **Add**.
Configure the vCenter outbound traffic to VMRC	1. In the **Name** text box, enter `config.mksdevproxy.writethrottler`. 2. In the **Value** text box, enter the value of the outbound traffic in KBps. 3. Click **Add**.
Limit the VMRC inbound traffic to the vCenter system	1. Click the **Filter** icon in the **Name** column. 2. To display the VMRC proxy parameter, in the text box, enter `config.mksdevproxy.readthrottler` and close the dialog box. 3. In the **Value** text box, enter the inbound bandwidth limit in KBps.
Limit the vCenter outbound traffic to VMRC	1. Click the **Filter** icon in the **Name** column. 2. To display the VMRC proxy parameter, in the text box, enter `config.mksdevproxy.writethrottler` and close the dialog box. 3. In the **Value** text box, enter the outbound bandwidth limit in KBps.

5. Click **Save**.

Answer Virtual Machine Questions

Virtual machine questions are messages that are generated by vCenter. Virtual machine questions appear whenever the virtual machine needs user intervention to continue its operation. In most cases, virtual machine questions appear when you power on a virtual machine .

Verify that the virtual machine hardware version is 11 or later.

To save time and ensure the consistency of your virtual environment, you can apply the same answer to multiple or to all virtual machines that have the same pending question.

1. Navigate to a virtual machine with a question.
2. Right-click the virtual machine and select **Guest OS** > **Answer Question**.
   The **Answer Question** wizard opens.
3. In the **Answer Question** dialog box, select your answer.
4. Optional: Apply the selected answer to other virtual machines that have the same pending question.
   a) Click the **Select other virtual machines** hyperlink.
      A list of all virtual machines with the same pending question appears.
   b) Select the virtual machines to which to apply the answer.
5. Click **OK**.

Remove and Reregister VMs and VM Templates

You can remove VMs and VM templates from the vCenter inventory or delete them from a disk. If you only removed the VM from the inventory, you can add it back from the datastore.

Adding Existing Virtual Machines to vCenter

When you add a host to vCenter, it discovers all the virtual machines on that managed host and adds them to the vCenter inventory.

If a managed host is disconnected, the already discovered virtual machines continue to be listed in the inventory.

If a managed host is disconnected and reconnected, any changes to the virtual machines on that managed host are identified, and the vSphere Client updates the list of virtual machines. For example, if node3 is removed and node4 is added, the new list of virtual machines adds node4 and shows node3 as orphaned.

Remove VMs or VM Templates from vCenter or from the Datastore

You can temporarily remove a virtual machine or a VM template from vCenter or you can permanently delete it from the datastore.

Power off the virtual machine.

The process is the same for a virtual machine or a VM template:

- When you remove a virtual machine from the inventory, you unregister it from the host and vCenter, you do not delete it from the datastore. Virtual machine files remain at the same storage location and you can later re-register the virtual machine by using the datastore browser. This helps if you want to edit the virtual machine configuration file. It is also useful to temporarily remove a virtual machine when you have reached the maximum number of virtual machines that your license or hardware allows.
- If you no longer need a virtual machine and want to free up space on the datastore, you can remove the virtual machine from vCenter and delete all virtual machine files from the datastore, including the configuration file and virtual disk files.

Log in to the vSphere Client and perform the task:

Option	Description
Temporarily remove the virtual machine or VM template	1. Right-click the virtual machine. 2. Select **Remove From Inventory** and click **Yes**.
Permanently delete the virtual machine or VM template	1. Right-click the virtual machine. 2. Select **Delete from Disk** and click **Yes**.

Register a VM or VM Template with vCenter

If you removed a VM or VM template from vCenter but did not delete it from disk, you can return it to the vCenter inventory by registering it with the vCenter.

1. In the vSphere Client inventory, right-click the datastore on which the virtual machine configuration file is stored and select **Register VM**.
2. On the Select a name and folder page, use the existing name or type a new name, select a datacenter or folder location and click **Next**.
3. On the **Select a compute resource** page, select a host, cluster, resource pool, or vApp on which to run the new virtual machine.

Option	Description
Option	Action
Run the virtual machine on a standalone host.	Select the host and click **Next**.
Run the virtual machine in a cluster with automatic DRS placement.	Select the cluster and click **Next**.
Run the virtual machine in a cluster without DRS automatic placement.	1. Select the cluster and click **Next**. 2. Select a host within a cluster and click **Next**.

4. On the **Ready to complete** page, review your selections and click **Finish**.

Manage Virtual Machine Templates

After you clone a virtual machine to a template, you can perform different operations with the template. You can rename the template, delete it from the vCenter inventory, or delete it from the disk. You can also register the template with vCenter.

Change the Template Name

If you move a template to another host or a data center folder, you can change the template name to make it unique in that folder.

1. Right-click the template and select **Rename**.
2. Enter a new name and click **OK**.

Delete and Remove Templates

You can delete a template by removing it from the inventory or deleting the template from the disk. If you remove the template from the inventory, it remains on the disk and can be reregistered with vCenter to restore it to the inventory.

By removing a templete, you unregister it from the vCenter inventory, but it remains on the disk at the same storage location and you can use the datastore browser to reregister the template at a later time. For example, if you decide to update the template rather than create one.

> **Note:**
> Templates that you delete are permanently removed from the system and you cannot recover such templates.

Action	Description
Remove a template from the inventory	1. Right-click the template and select **Remove from Inventory**. 2. To confirm removing the template from the vCenter inventory, click **Yes**. The template is unregistered from the vCenter inventory.
Delete a template from the disk	1. Right-click the template and select **Delete from Disk**. 2. To confirm removing the template from the datastore, click **Yes**.

Reregister Templates

You unregister templates when you remove them from the vCenter inventory or when the hosts with which they are associated are removed from the vCenter inventory.

You can reregister templates to the vCenter inventory.

1. In the vCenter inventory, navigate to the datastore that contains the template.
2. Select the datastore and click the **Files** tab.
3. Locate the template folder and click it to display the template files.
4. Select the `.vmtx` file and click the **Register VM** icon.**Result:** The **Register VM Template** wizard opens.
5. On the **Select a name and folder** page, specify a name and location for the template and click **Next**.
6. On the **Select a compute resource** page, select a host or cluster on which to store the template and click **Next**.
7. On the **Ready to complete** page, review your selections and click **Finish**.
8. (Optional) To verify that the template is registered, check the host or cluster inventory.

Inventory Object	Action
Host	1. Navigate to the host. 2. Click the **VMs** tab and select **VM Templates**.
Cluster	1. Navigate to the cluster. 2. Click the **VMs** tab and select **VM Templates**.

Result: The template is registered to the host and you can view the template when you navigate to the host and click **VM Templates**.

Manage Virtual Machines With Snapshots

Snapshots preserve the state and data of a virtual machine at the time that you take the snapshot. When you take a snapshot of a virtual machine, an image of the virtual machine in a given state is copied and stored.

Snapshots are useful when you want to revert repeatedly to a virtual machine state, but you do not want to create multiple virtual machines.

You can take multiple snapshots of a virtual machine to create restoration positions in a linear process. With multiple snapshots, you can save many positions to accommodate many kinds of work processes. Snapshots operate on individual virtual machines. Taking snapshots of multiple virtual machines, for example, taking a snapshot of a VM for each member of a team, requires that you take a separate snapshot of each team member's virtual machine.

Snapshots are useful as a short-term solution for testing software with unknown or potentially harmful effects. For example, you can use a snapshot as a restoration point during a linear or iterative process, such as installing update packages, or during a branching process, such as installing different versions of a program. Using snapshots ensures that each installation begins from an identical baseline.

With snapshots, you can preserve a baseline before you change a virtual machine.

Several operations for creating and managing virtual machine snapshots and snapshot trees are available in the vSphere Client. These operations enable you to create snapshots, revert any snapshot in the snapshot hierarchy, delete snapshots, and more. You can create snapshot trees where you save the virtual machine state at any specific time so that you can revert that virtual machine state later. Each branch in a snapshot tree can have up to 32 snapshots.

A snapshot preserves the following information:

- Virtual machine settings. The virtual machine directory, which includes the disks added or changed after you take the snapshot.
- Power state. The virtual machine can be powered on, powered off, or suspended.
- Disk state. State of all the virtual machine's virtual disks.
- (Optional) Memory state. The contents of the virtual machine's memory.

The Snapshot Hierarchy

The vSphere Client presents the snapshot hierarchy as a tree with one or more branches. Snapshots in the hierarchy have parent to child relationships. In linear processes, each snapshot has one parent snapshot and one child snapshot, except for the last snapshot, which has no child snapshot. Each parent snapshot can have more than one child. You can revert to the current parent snapshot or to any parent or child snapshot in the snapshot tree and create more snapshots from that snapshot. Each time you revert a snapshot and take another snapshot, a branch (child snapshot) is created.

Parent Snapshots The first virtual machine snapshot that you create is the base parent snapshot. The parent snapshot is the most recently saved version of the current state of the virtual machine. Taking a snapshot creates a delta disk file for each disk attached to the virtual machine and optionally, a memory file. The delta disk files and memory file are stored with the base `.vmdk` file. The parent snapshot is always the snapshot that appears immediately above the **You are here** icon in the Snapshot Manager. If you revert a snapshot, that snapshot becomes the parent of the **You are here** current state.

> **Note:**
> The parent snapshot is not always the snapshot that you took most recently.

Child Snapshots A snapshot of a virtual machine taken after the parent snapshot. Each child snapshot contains delta files for each attached virtual disk, and optionally a memory file that points from the present state of the virtual disk (You are here). Each child snapshot's delta files merge with each previous child snapshot until reaching the parent disks. A child disk can later be a parent disk for future child disks.

The relationship of parent and child snapshots can change if you have multiple branches in the snapshot tree. A parent snapshot can have more than one child. Many snapshots have no children.

> **CAUTION:**
> Do not manually manipulate individual child disks or any of the snapshot configuration files because doing so can compromise the snapshot tree and result in data loss. This restriction includes disk resizing and making modifications to the base parent disk by using the `vmkfstools` command.

Snapshot Behavior

Taking a snapshot preserves the disk state at a specific time by creating a series of delta disks for each attached virtual disk or virtual RDM and optionally preserves the memory and power state by creating a memory file. Taking a snapshot creates a snapshot object in the Snapshot Manager that represents the virtual machine state and settings.

Each snapshot creates an additional delta `.vmdk` disk file. When you take a snapshot, the snapshot mechanism prevents the guest operating system from writing to the base `.vmdk` file and instead directs all writes to the delta disk file. The delta disk represents the difference between the current state of the virtual disk and the state that existed at the time that you took the previous snapshot. If more than one snapshot exists, delta disks can represent the difference between each snapshot. Delta disk files can expand quickly and become as large as the entire virtual disk if the guest operating system writes to every block of the virtual disk.

Snapshot Files and Limitations

When you take a snapshot, you capture the state of the virtual machine settings and the virtual disk. If you are taking a memory snapshot, you also capture the memory state of the virtual machine. These states are saved to files that reside with the base files of the virtual machines.

Snapshot Files

A snapshot consists of files that are stored on a supported storage device. A Take Snapshot operation creates `.vmdk`, `-delta.vmdk` or `-sesparse.vmdk`, `.vmsd`, and `.vmsn` files. By default, the first and all delta disks are stored with the base `.vmdk` file. The `.vmsd` and `.vmsn` files are stored in the virtual machine directory.

SEsparse is a default format for all delta disks on the VMFS6 datastores.

Delta disk files A `.vmdk` file to which the guest operating system can write. The delta disk represents the difference between the current state of the virtual disk and the state that existed at the time that the previous snapshot was taken. When you take a snapshot, the state of the virtual disk is preserved, the guest operating system stops writing to it, and a delta or child disk is created.

	A delta disk has two files. One is a small descriptor file that contains information about the virtual disk, such as geometry and child-parent relationship information. The other one is a corresponding file that contains the raw data. The files that make up the delta disk are called child disks or redo logs.
Flat file	A `-flat.vmdk` file that is one of two files that comprises the base disk. The flat disk contains the raw data for the base disk. This file does not appear as a separate file in the Datastore Browser.
Database file	A `.vmsd` file that contains the virtual machine's snapshot information and is the primary source of information for the Snapshot Manager. This file contains line entries, which define the relationships between snapshots and between child disks for each snapshot.
Memory file	A `.vmsn` file that includes the active state of the virtual machine. Capturing the memory state of the virtual machine lets you revert to a turned on virtual machine state. With nonmemory snapshots, you can only revert to a turned off virtual machine state. Memory snapshots take longer to create than nonmemory snapshots. The time the ESX host takes to write the memory onto the disk depends on the amount of memory the virtual machine is configured to use.

A **Take Snapshot** operation creates `.vmdk`, `-delta.vmdk`, or `-sesparse.vmdk`, and `vmsn` files.

File	Description
vmname-number`.vmdk` and *vmname-number*`-delta.vmdk`	A snapshot file that represents the difference between the current state of the virtual disk and the state that existed at the time the previous snapshot was taken. The filename uses the following syntax, `S1vm-000001.vmdk` where `S1vm` is the name of the virtual machine and the six-digit number, `000001`, is based on the files that already exist in the directory. The number does not consider the number of disks that are attached to the virtual machine.
vmname`.vmsd`	Database of the virtual machine's snapshot information and the primary source of information for the Snapshot Manager.
vmname`.Snapshot`*number*`.vmsn`	Memory state of the virtual machine at the time you take the snapshot. The filename uses the following syntax, `S1vm.snapshot1.vmsn`, where `S1vm` is the virtual machine name, and `snapshot1` is the first snapshot. **Note:** A `.vmsn` file is created each time you take a snapshot, regardless of the memory selection. A `.vmsn` file without memory is much smaller than one with memory.

Snapshot Limitations

Snapshots can affect the virtual machine performance and do not support some disk types or virtual machines configured with bus sharing. Snapshots are useful as short-term solutions for capturing point-in-time virtual machine states and are not appropriate for long-term virtual machine backups.

- VMware does not support snapshots of raw disks, RDM physical mode disks, or guest operating systems that use an iSCSI initiator in the guest.
- Virtual machines with independent disks must be powered off before you take a snapshot.
- You can take a memory snapshot of a virtual machine with an independent disk only to analyze the guest operating system behavior of a virtual machine. You cannot use such snapshots for virtual machine backups because restoring this type of snapshots is unsupported.
- Quiesced snapshots require VMware Tools installation and guest operating system support.
- Snapshots are not supported with PCI vSphere DirectPath I/O devices.
- VMware does not support snapshots of virtual machines configured for bus sharing. If you require bus sharing, consider running backup software in your guest operating system as an alternative solution. If your virtual machine currently has snapshots that prevent you from configuring bus sharing, delete (consolidate) the snapshots.
- Snapshots provide a point-in-time image of the disk that backup solutions can use, but Snapshots are not intended for a robust method of backup and recovery. If the files containing a virtual machine are lost, its snapshot files are also

lost. Also, large numbers of snapshots are difficult to manage, consume large amounts of disk space, and are not protected if there is hardware failure.
- Snapshots can negatively affect the performance of a virtual machine. Performance degradation is based on how long the snapshot or snapshot tree is in place, the depth of the tree, and how much the virtual machine and its guest operating system have changed from the time you took the snapshot. Also, you might see a delay in the amount of time it takes the virtual machine to power on. Do not run production virtual machines from snapshots on a permanent basis.
- If a virtual machine has virtual hard disks larger than 2 TB, snapshot operations can take much longer to finish.

Manage Snapshots

You can view and manage all snapshots for an active virtual machine. You can review the snapshots information, revert to the latest snapshot, change the name and description, or delete a snapshot.

You can manage the snapshots when you select a virtual machine in the vSphere Client inventory and click the **Snapshots** tab.

The snapshot tree displays all snapshots of the virtual machine and the power state of the virtual machine when a snapshot was taken. The detailed information region contains the snapshot name and description, time of creation, and the disk space. You can also see whether you took a snapshot of the virtual machine memory and if you quiesced the guest file system.

The **You are here** pin represents the current and active state of the virtual machine and it is always visible.

Take Snapshots of a Virtual Machine

You can take one or more snapshots of a virtual machine to capture the settings state, disk state, and memory state at specific times. When you take a snapshot, you can also quiesce the virtual machine files and exclude the virtual machine disks from snapshots.

When you take a snapshot, other activity that is occurring in the virtual machine might affect the snapshot process when you revert to that snapshot. The best time to take a snapshot from a storage perspective is when you are not incurring a large I/O load. The best time to take a snapshot from a service perspective is when no applications in the virtual machine are communicating with other computers. The potential for problems is greatest if the virtual machine is communicating with another computer, especially in a production environment. For example, if you take a snapshot while the virtual machine is downloading a file from a server on the network, the virtual machine continues downloading the file and communicating its progress to the server. If you revert to the snapshot, communications between the virtual machine and the server are confused and the file transfer fails. Depending on the task that you are performing, you can create a memory snapshot or you can quiesce the file system in the virtual machine.

Memory Snapshots	The default selection for taking snapshots. When you capture the virtual machine's memory state, the snapshot retains the live state of the virtual machine. Memory snapshots create a snapshot at a precise time, for example, to upgrade software that is still working. If you take a memory snapshot and the upgrade does not complete as expected, or the software does not meet your expectations, you can revert the virtual machine to its previous state. When you capture the memory state, the virtual machine's files do not require quiescing. If you do not capture the memory state, the snapshot does not save the live state of the virtual machine and the disks are crash consistent unless you quiesce them.
Quiesced Snapshots	When you quiesce a virtual machine, VMware Tools quiesces the file system of the virtual machine. A quiesce operation ensures that a snapshot disk represents a consistent state of the guest file systems. Quiesced snapshots are appropriate for automated or periodic backups. For example, if you are unaware of the virtual machine's activity, but want several recent backups to revert to, you can quiesce the files. If the virtual machine is powered off or VMware Tools is not available, the `Quiesce` parameter is not available. You cannot quiesce virtual machines that have large capacity disks.

Important:
Do not use snapshots as your only backup solution or as a long-term backup solution.

Change Disk Mode to Exclude Virtual Disks from Snapshots

You can set a virtual disk to independent mode to exclude the disk from any snapshots taken of its virtual machine.

Prerequisites

Power off the virtual machine and delete any existing snapshots before you change the disk mode. Deleting a snapshot involves committing the existing data on the snapshot disk to the parent disk.

Required privileges:

- **Virtual machine** > **Change Configuration** > **Modify device settings**

Note:
An independent disk does not participate in virtual machine snapshots. That is, the disk state is independent of the snapshot state and creating, consolidating, or reverting to snapshots does not have effect on the disk.

You can take a memory snapshot of a virtual machine with an independent disk, but the snapshot cannot be restored. The main purpose of these snapshots is to capture the state of a virtual machine for further analysis through the Vmss2core or similar tools.

Procedure

1. Right-click a virtual machine in the inventory and select **Edit Settings**.
2. On the **Virtual Hardware** tab, expand **Hard disk**, and select an independent disk mode option.

Option	Description
Dependent	Dependent mode is the default disk mode. When you take a snapshot of a virtual machine, dependent disks are included in the snapshot. When you revert to the previous snapshot, all data are reverted to the point of taking a snapshot.
Independent - Persistent	Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode are written permanently to the disk even if you revert a snapshot. When you power off or reset a virtual machine, the disk and all its snapshots are preserved.
Independent - Nonpersistent	Disks in nonpersistent mode behave like read-only disks. Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset the virtual machine, or when you delete a snapshot.

3. Click **OK**.

Take a Snapshot of a Virtual Machine

You can take a snapshot when a virtual machine is powered on, powered off, or suspended. If you are suspending a virtual machine, wait until the suspend operation finishes before you take a snapshot.

When you create a memory snapshot, the snapshot captures the state of the virtual machine memory and the virtual machine power settings. When you capture the virtual machine memory state, the snapshot operation takes longer to complete. You might also see a momentary lapse in response over the network.

When you quiesce a virtual machine, VMware Tools quiesces the file system in the virtual machine. The quiesce operation pauses or alters the state of running processes on the virtual machine, especially processes that might modify information stored on the disk during a revert operation.

Application-consistent quiescing is not supported for virtual machines with IDE or SATA disks.

Note:
If you take a snapshot of a Dynamic Disk (a Microsoft-specific disk type), the snapshot technology preserves the quiesce state of the file system, but does not preserve the quiesce state of the application.

Prerequisites

- If you are taking a memory snapshot of a virtual machine that has multiple disks in different disk modes, verify that the virtual machine is powered off. For example, if you have a special purpose configuration that requires you to use an independent disk, you must power off the virtual machine before taking a snapshot.
- To capture the memory state of the virtual machine, verify that the virtual machine is powered on.
- To quiesce the virtual machine files, verify that the virtual machine is powered on and that VMware Tools is installed.
- Verify that you have the **Virtual machine** > **Snapshot management** > **Create snapshot** privilege on the virtual machine.

Procedure

1. In the vSphere Client, navigate to a virtual machine and click the **Snapshots** tab.
2. Click **Take Snapshot**.
   Result: The **Take snapshot** dialog box opens.
3. Enter a name for the snapshot.
4. (Optional) Enter a description for the snapshot.
5. (Optional) To capture the memory of the virtual machine, select the **Include virtual machine's memory** check box.
6. (Optional) To pause running processes on the guest operating system so that file system contents are in a known consistent state when you take a snapshot, select the **Quiesce guest file system (requires VMware Tools)** check box.

   Note:
   You can quiesce the virtual machine files only when the virtual machine is powered on and the **Snapshot the virtual machine's memory** check box is deselected.

   For more information about the virutal machine behavior when you quiesce a virtual machine, see Broadcom knowledge base article 343375.
7. Click **Create**.

Revert a Virtual Machine Snapshot

To return a virtual machine to its original state, or to return to another snapshot in the snapshot hierarchy, you can use the revert options.

Verify that you have the **Virtual machine** > **Snapshot management** > **Revert to snapshot** privilege on the virtual machine.

When you revert a snapshot, you return the virtual machine's memory, settings, and the state of the virtual machine disks to the state they were in when you took the snapshot. You can revert any snapshot in the snapshot tree and make that snapshot the parent snapshot of the current state of the virtual machine. Subsequent snapshots from this point create a new branch of the snapshot tree.

Restoring snapshots has the following effects:

- The current disk and memory states are discarded, and the virtual machine reverts to the disk and memory states of the parent snapshot.
- Existing snapshots are not removed. You can revert those snapshots at any time.
- If the snapshot includes the memory state, the virtual machine will be in the same power state as when you created the snapshot.

Table 140: Virtual Machine Power State After Restoring a Snapshot

Virtual Machine State When Parent Snapshot Is Taken	Virtual Machine State After Restoration
Powered on (includes memory)	Reverts to the parent snapshot, and the virtual machine is powered on and running.
Powered on (does not include memory)	Reverts to the parent snapshot and the virtual machine is powered off.
Powered off (does not include memory)	Reverts to the parent snapshot and the virtual machine is powered off.

Virtual machines running certain kinds of workloads can take several minutes to resume responsiveness after reverting from a snapshot.

> **Note:**
> vApp metadata for virtual machines in vApps does not follow the snapshot semantics for the virtual machine configuration. vApp properties that are deleted, modified, or defined after a snapshot is taken remain intact (deleted, modified, or defined) after the virtual machine reverts to that snapshot or any previous snapshots.

When you revert to a snapshot, disks that you added or changed after the snapshot was taken are reverted to the snapshot point. For example, when you take a snapshot of a virtual machine, add a disk, and revert the snapshot, the added disk is removed.

Independent disks are also removed when you revert to a snapshot that was taken before the disk was added. If the latest snapshot includes an independent disk, its contents do not change when you revert to that snapshot.

1. To revert a snapshot, navigate to a virtual machine in the vSphere Client inventory and click the **Snapshots** tab.
2. Navigate to a snapshot in the snapshot tree, click **Revert**, and click the **Revert** button.

Delete a Snapshot

Deleting a snapshot permanently removes the snapshot from the snapshot tree. The snapshot files are consolidated and written to the parent snapshot disk and merge with the virtual machine base disk. You can delete a single snapshot or all snapshots in a snapshot tree.

- Familiarize yourself with the delete and delete all actions and how they affect virtual machine performance.
- Required Privilege: **Virtual machine** > **Snapshot management** > **Remove Snapshot** on the virtual machine.

Deleting a snapshot does not change the virtual machine or other snapshots. Deleting a snapshot consolidates the changes between snapshots and previous disk states. Then it writes all the data from the delta disk that contains the information about the deleted snapshot to the parent disk. When you delete the base parent snapshot, all changes merge with the base virtual machine disk.

To delete a snapshot, a large amount of information must be read and written to a disk. This process can reduce the virtual machine performance until the consolidation is complete. Consolidating snapshots removes redundant disks, which improves the virtual machine performance and saves storage space. The time to delete snapshots and consolidate the snapshot files depends on the amount of data that the guest operating system writes to the virtual disks after you take the last snapshot. If the virtual machine is powered on, the required time is proportional to the amount of data the virtual machine is writing during consolidation.

Failure of disk consolidation can reduce the performance of virtual machines. You can check whether any virtual machines require separate consolidation operations by viewing a list. For information about locating and viewing the consolidation state of multiple virtual machines and running a separate consolidation operation, see Consolidate Snapshots.

Delete	Use the **Delete** option to remove a single parent or child snapshot from the snapshot tree. This option writes disk changes that occur between the state of the snapshot and the previous disk state to the parent snapshot. **Note:** Deleting a single snapshot preserves the current state of the virtual machine and does not affect any other snapshot. You can also use the **Delete** option to remove a corrupt snapshot and its files from an abandoned branch of the snapshot tree without merging them with the parent snapshot.
Delete All	Use the **Delete All** option to delete all snapshots from the snapshot tree. The **Delete all** option consolidates and writes the changes that occur between snapshots and the previous delta disk states to the base parent disk. It then merges them with the base virtual machine disk.

To prevent snapshot files from merging with the parent snapshot if, for example, an update or installation fails, first use the **Revert** button to revert to a previous snapshot. This action invalidates the snapshot delta disks and deletes the memory file. You can then use the **Delete** option to remove the snapshot and any associated files.

CAUTION:
Use care when you delete snapshots. You cannot revert a deleted snapshot. For example, you might want to install several browsers, a, b, and c, and capture the virtual machine state after you install each browser. The first, or base snapshot, captures the virtual machine with browser a and the second snapshot captures browser b. If you revert the base snapshot that includes browser a and take a third snapshot to capture browser c, and delete the snapshot that contains browser b, you cannot return to the virtual machine state that includes browser b.

To delete snapshots from a snapshot tree, navigate to a virtual machine in the vSphere Client inventory and click the **Snapshots** tab.

Option	Action
Delete a single snapshot	1. Navigate to and select a snapshot in the snapshots tree. 2. Click **Delete** and click the **Delete** button. The snapshot data is consolidated to the parent snapshot and the selected snapshot is removed from the snapshot tree.
Delete all snapshots	1. Click **Delete All**. 2. To delete all snapshots, select the **Delete all snapshots** and click **Delete All**. All immediate snapshots before the **You are here** current state are consolidated to the base parent disk. All existing snapshots are removed from the snapshot tree and the virtual machine. 3. To automatically delete snapshots once they reach a specified age, select **Delete snapshots older than __ days** check box and enter the number of days.

Consolidate Snapshots

The presence of redundant delta disks can adversely affect the virtual machine performance. You can combine such disks without violating a data dependency. After consolidation, redundant disks are removed, which improves the virtual machine performance and saves storage space.

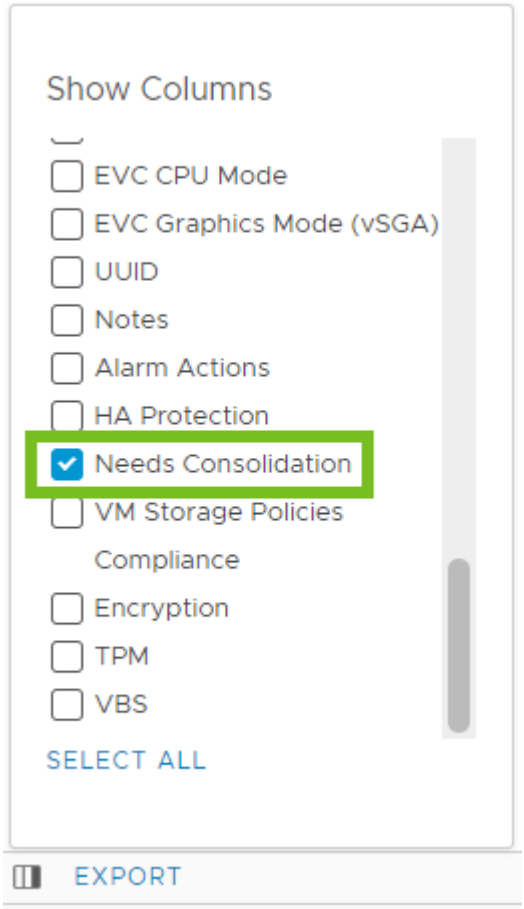
Required privilege: **Virtual machine** > **Snapshot management** > **Remove Snapshot**

Snapshot consolidation is useful when snapshot disks fail to compress after a **Revert**, **Delete**, or **Delete all** operation. This might happen, for example, if you delete a snapshot but its associated disk does not commit back to the base disk.

Resumable online and offline snapshot consolidations are supported. If you cancel a snapshot consolidation operation, it can be resumed from where it stopped.

1. Navigate to a virtual machine in the vSphere Client inventory and click the **Snapshots** tab.
2. Perform the necessary snapshot operations.
   If the virtual machine snapshot files must be consolidated, the **Consolidation is required** message appears.
3. Click the **Consolidate** button.
   The **Consolidate** dialog box appears.
4. Click **OK**.
5. To verify that the consolidation is successful, check the **Needs Consolidation** column.
   a) Navigate to an inventory object that contains a list of virtual machines, for example, a vCenter instance, a host, or a cluster.
   b) Click the **VMs** tab and click **Virtual Machines**.
   c) Click at the bottom-left corner of the VMs list panel.
   The **Show Columns** window appears.

d) Select **Needs Consolidation**.



A `Yes` status indicates that the snapshot files for the virtual machine must be consolidated. A `Not Required` status indicates that the files are consolidated.

How do I Schedule Tasks for Snapshot Deletion

By scheduling tasks for snapshot deletion, you can significantly reduce resource consumption in your environment. This is achieved by decreasing the number of VM snapshots and configuring the period in which they are preserved before deletion.

Required privileges:

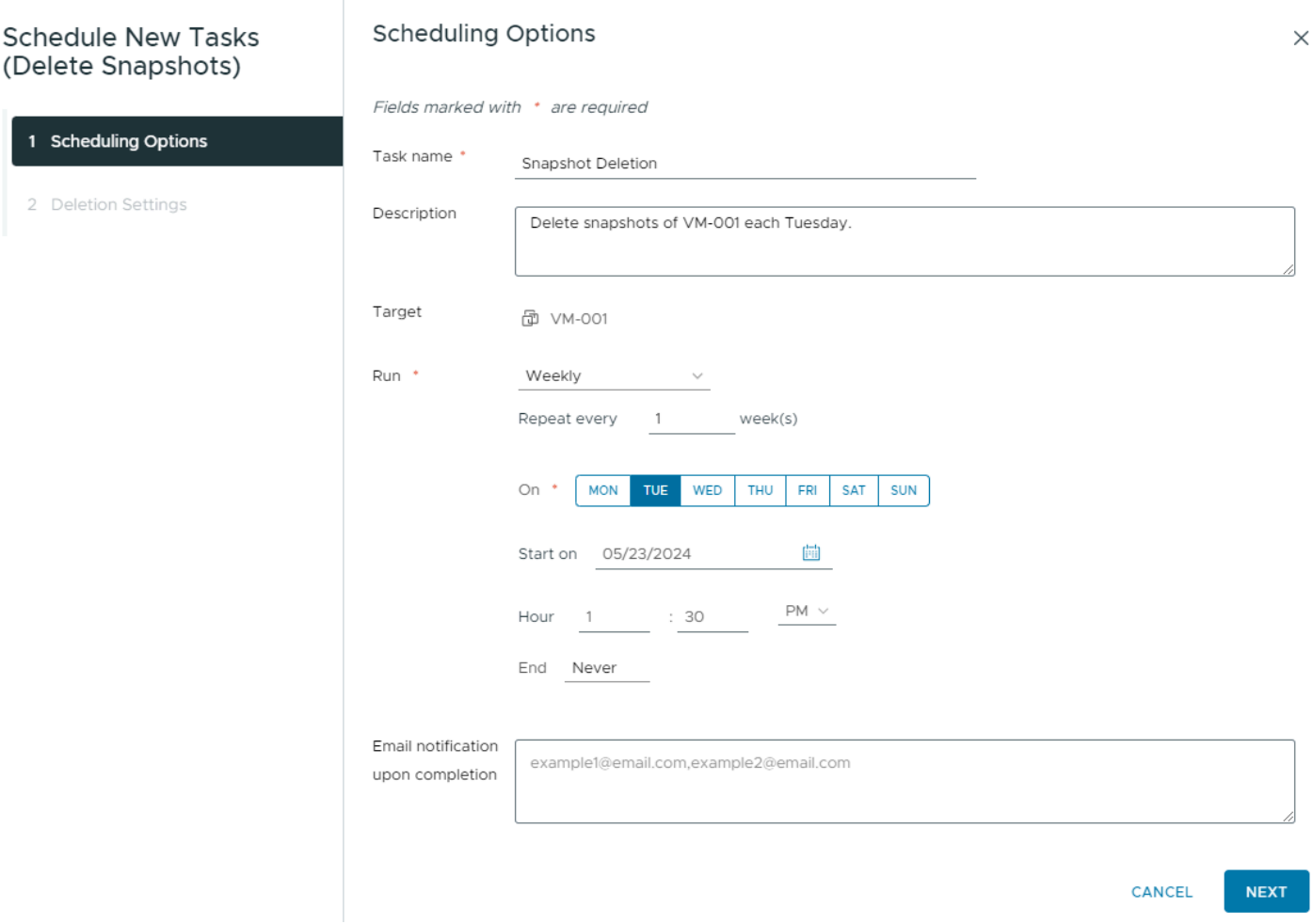
- **Virtual machine** > **Snapshot management** > **Remove Snapshot** on the virtual machine.
- **Scheduled task** > **Create task**

You can set these tasks to run once in the future or at a recurring interval.

The deletion process runs for 3 hours by default. You can set the exact time when it should stop. Ending the deletion process will not halt it for VM snapshots that have already started but will only prevent initiating a deletion task for new snapshots.

1. Navigate to a virtual machine in the inventory and click the **Configure** tab.
2. Select **Scheduled Tasks** and from the **New Scheduled Task** drop-down menu select **Delete Snapshots**.
   The **Schedule New Task** window opens.

3. On the **Scheduling Options** page set the scheduling options.



Option	Description
Task name	Enter a task name.
Description	Enter a brief summary of the criteria and schedule for the deletion process.

Option	Description
Run	Set when the deletion process should occur, the specific day and time, and set the specific day and time when the process should stop. • Once • After vCenter startup • Hourly • Daily • Weekly • Monthly
Email notification upon completion	Enter the email addresses to receive a notification when the deletion process is complete.

4. Click **Next**.
5. On the **Deletion Settings** page, specify the number of days the snapshots older than this will be removed.
6. Click **Schedule The Task**.

How do I Schedule Tasks for Snapshot Deletion for a Group of Virtual Machines

With vSphere 8.0 Update 3, you have the flexibility to schedule a task for snapshot deletion for multiple virtual machines so that you can optimize your work.

Required privileges:

- **Virtual machine** > **Snapshot management** > **Remove Snapshot** on the virtual machine.
- **Scheduled task** > **Create tasks**

To schedule a task for snapshot deletion of multiple virtual machines, perform the following steps.

1. Navigate to and select an ESX host in the inventory.
2. Click the **VMs** tab and select the virtual machines that you want to schedule a snapshot deletion.
3. Right-click the selection and select **Snapshots** > **Schedule Snapshots Deletion**.
4. To schedule the task, complete the steps in the wizard.

After scheduling snapshot deletion tasks, you can view detailed information about each task, such as the planned period and the exact time the following deletion process will run.

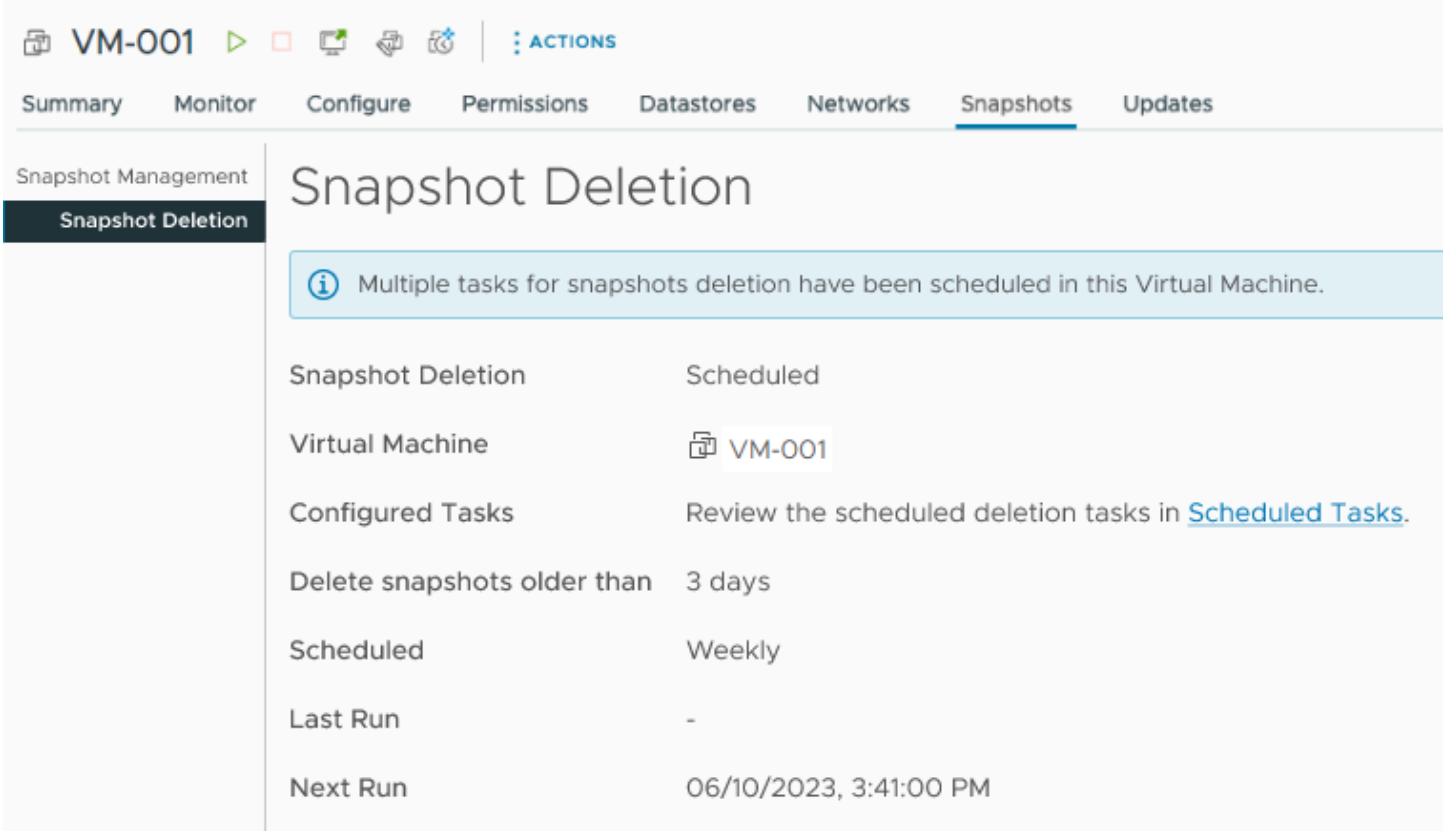
Monitor the Scheduled Tasks for Snapshot Deletion

After you schedule tasks for snapshot deletion on a virtual machine, you can monitor all planned tasks for better visibility of the process.

To observe the scheduled tasks for snapshot deletion, perform the following steps.

1. Navigate to and select a virtual machine in the inventory.
2. Click the **Snapshots** tab and under **Snapshot Management** select **Snapshot Deletion**.

You can view the information for the scheduled tasks.



Virtual Machine Storage DRS Rules

The Storage DRS rules that you define at the virtual machine level function in the same way as the affinity and anti-affinity rules that you define at the datastore cluster level. Virtual machine Storage DRS rules define whether virtual machine hard disks are placed and kept on the same datastore or on different datastores within a datastore cluster.

In the vSphere Client, you can create, edit, and delete Storage DRS rules. You can also create Storage DRS rules that place and keep all virtual disks of certain virtual machines on different datastores within a datastore cluster.

VMDK Affinity Rules

By default, all virtual machine hard disks are kept together on the same datastore within a datastore cluster that has Storage DRS enabled. That is, VMDK affinity rules are enabled by default for all virtual machines that are in a datastore cluster. You can override that rule for the datastore cluster or for individual virtual machines.

Storage DRS Anti-Affinity Rules

You can also create anti-affinity rules to place certain virtual hard disks or virtual machines on different datastores and keep them separated.

- VMDK anti-affinity rules ensure that two or more virtual hard disks of a single virtual machine are placed and kept on different datastores within the datastore cluster.
- VM anti-affinity rules ensure that all the virtual hard disks of two or more virtual machines are placed and kept on different datastores within the datastore cluster.

For more information about Storage DRS, see the *vSphere Resource Management* documentation.

Add a VMDK Affinity Rule

Create a VMDK affinity rule to place and keep all virtual disks of a virtual machine on the same datastore within a datastore cluster.

1. Navigate to a virtual machine in the inventory.
2. On the **Configure** tab, expand **Settings** and click **VM SDRS Rules**.
3. Click the **Add** button.
   The **Add SDRS rule** dialog box opens.
4. From the **Type** drop-down menu, select **VMDK affinity**.
5. From the **Datastore cluster** drop-down menu, select the datastore cluster where you want to keep the virtual machine disks.
   A datastore cluster is visible in the list only if the virtual machine configuration file or at least one of the virtual hard disks is placed on a datastore within the datastore cluster.
6. Optional: Deselect the **Keep VMDKs together** to create a rule that places and keeps the virtual hard disks on different datastores.
   If you leave the check box selected, the rule that you create is the same as the default Storage DRS rule that operates at the datastore cluster level.
   Deselecting the check box creates a Storage DRS rule that overrides the default VMDK affinity rule for the datastore cluster.
7. Click **OK**.

An Intra-VM Affinity rule is created that keeps VMDKs together. That is, all virtual hard disks of the selected virtual machine are placed and kept together on the same datastore within the datastore cluster.

Add a VMDK Anti-Affinity Rule

Create a VMDK anti-affinity rule to place and keep certain virtual hard disks of a virtual machine on different datastores within the datastore cluster.

When you create an anti-affinity rule, it applies to those of the virtual machine's hard disks that are on datastores within the selected datastore cluster. Anti-affinity Storage DRS rules operate during migrations that Storage DRS initiates or recommends, but they do not operate when the user initiates a migration.

1. Navigate to a virtual machine in the inventory.
2. On the **Configure** tab, expand **Settings** and click **VM SDRS Rules**.
3. Click the **Add** button.
   The **Add SDRS rule** dialog box opens.
4. From the **Type** drop-down menu, select **VMDK anti-affinity**.
5. In the **Rule name** text box, enter a name for the rule.
6. From the **Datastore cluster** drop-down menu, select the datastore cluster where the anti-affinity rule will operate.
   All virtual hard disks that are placed on datastores within the selected datastore cluster appear at the bottom of the dialog box.
7. Select the virtual hard disks to which the anti-affinity rule applies.
8. Optional: Deselect the **Enable the rule** check box.
   Deselecting the **Enable the rule** check box deactivates the rule. You can still create the rule, but after creation the rule is not applied to the selected virtual hard disks.
9. Click **OK**.

A VMDK anti-affinity rule is created. If the rule is enabled, all selected virtual hard disks are placed and kept on different datastores within the datastore cluster.

Add a VM Anti-Affinity Rule

Create a VM anti-affinity rule to place and keep all virtual hard disks of selected virtual machines on different datastores within the datastore cluster.

1. Navigate to a virtual machine in the inventory.
2. On the **Configure** tab, expand **Settings** and click **VM SDRS Rules**.
3. Click the **Add** button.
   The **Add SDRS rule** dialog box opens.
4. From the **Type** drop-down menu, select **VM anti-affinity**.
5. In the **Rule name** text box, enter a name for the rule.
6. From the **Datastore cluster** drop-down menu, select the datastore cluster where the anti-affinity rule will operate.
7. From the list of virtual machines at the bottom of the dialog box, select the virtual machines to which the anti-affinity rule applies.
   You can add or remove virtual machines to the list.
8. Optional: Deselect the **Enable the rule** check box.
   The rule is enabled by default.
   Deselecting the **Enable the rule** check box deactivates the rule. If an existing rule is deactivated, it is not applied to the virtual machine.
9. Click **OK**.

A VM anti-affinity rule is created. If the rule is enabled, all virtual hard disks of the selected virtual machines are placed and kept on different datastores within the datastore cluster.

Distributing Content with GuestStore

The GuestStore feature provides an easy and flexible mechanism to maintain and distribute different content types across multiple virtual machines on multiple ESX hosts at the same time.

By using the GuestStore framework, you can make sure that the distributed content is always consistent and you can improve the content management in your environment.

As a vSphere administrator, after you configure GuestStore on an ESX host, the virtual machines on the host can start accessing the GuestStore content immediately.

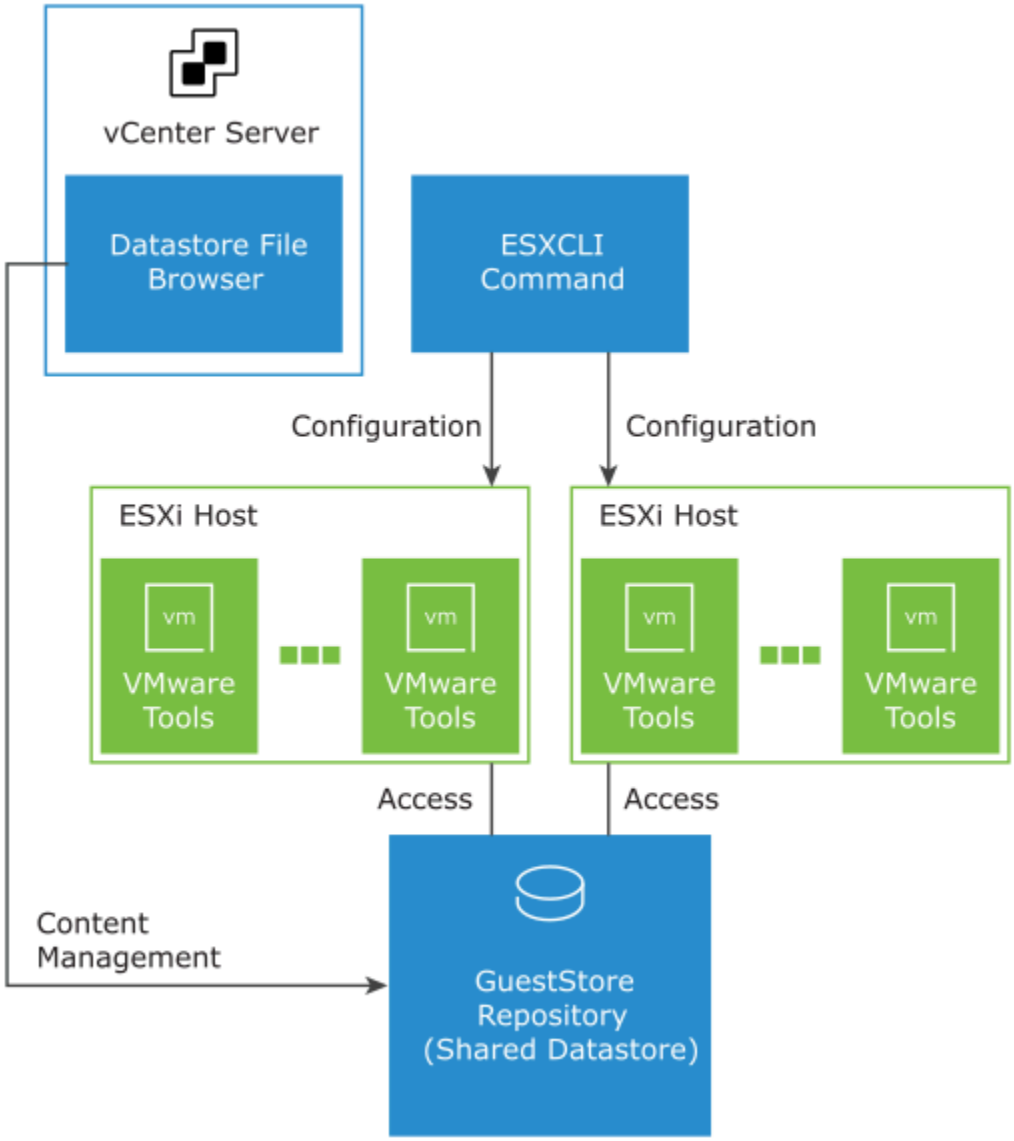
The GuestStore content consists of:

- Binary files that contain updates for VMware Tools and VMware agents.
- Scripts that are either provided by VMware or are custom scripts.
- Configuration files of VMware Tools and VMware agents.

With GuestStore you can:

- Maintain the content on a datastore, that is called a GuestStore repository.
- Fetch and distribute VMware Tools and VMware agent updates.
- Distribute configurations for VMware Tools and different VMware agents.
- Distribute a custom content - custom scripts, agents, and configuraton files.

For example, by importing VMware Tools in the GuestStore repository, you can schedule an automated VMware Tools upgrade at the next reboot of the virtual machine. From the GuestStore repository, you can also perform VMware Tools upgrade for particular virtual machines when necessary.



By using vSphere Client, you can configure a GuestStore repository on a shared datastore of an ESX host. You perform the configuration by using ESXCLI. You manage the contents of your datastore by using the datastore file browser, or, if you use an NFS datastore, you can mount the datastore to any NFS client machine.

By using a PowerCLI script, you can configure multiple hosts or a cluster managed by a vCenter system.

To access the GuestStore contents, the virtual machine guest operating system uses a resource path. The path to the datastore contents must be the same as the resource path of the guest operating system. For example, to access the `/example/myrepository/bar` file in the GuestStore repository, the resource path in the guest operating system must be the same, that means `/example/myrepository/bar`.

Starting with vSphere 7.0 Update 2, you can download and extract the VMware Tools content under the GuestStore repository path. The GuestStore framework allows you, as a vSphere Client administrator, to distribute a configuration file or VMware agent to set of various guest operating systems. For information about configuring GuestStore for VMware Tools, see the *VMware Tools* documentation.

Requirements for GuestStore

To use GuestStore, your vSphere environment must meet the following requirements:

- Virtual machines with Windows guest operating systems must be running on ESX 7.0 Update 2 and later and VMware Tools 11.2.5 and later versions.

- Virtual machines with Linux guest operating systems must be running on ESX 7.0 Update 3 and later and VMware Tools 11.3.0 and later versions.
- The file that is distributed through GuestStore must be 512 MB or smaller.

Set the GuestStore Repository with ESXCLI

You can use ESXCLI commands to set the URL to the GuestStore repository and verify the configuration by retrieving the currently set URL.

1. Set the GuestStore repository URL.

   Note:
   The URL must point to a datastore path where GuestStore content is stored. The datastore path must be accessible to the ESX host. If you want to set up a common GuestStore repository across multiple ESX hosts, you should use a shared datastore path.

   ```
   esxcli system settings gueststore repository set --url "<datastore_path>"
   ```

   The following example contains a possible syntax for the datastore path.

   ```
   esxcli system settings gueststore repository set --url "ds:///vmfs/volumes/<datastore_uuid>/GuestStore"
   ```

2. Retrieve the GuestStore repository URL.

   ```
   esxcli system settings gueststore repository get
   ```

   You receive the currently set URL in the output.

   ```
   URL: <datastore_path>
   ```

Clear the GuestStore Repository Setting with ESXCLI

You can use ESXCLI commands to clear the GuestStore repository URL setting and verify that the URL is not set.

1. Clear the GuestStore repository URL setting.

   ```
   esxcli system settings gueststore repository set --url ""
   ```

2. Retrieve the GuestStore repository URL.

   ```
   esxcli system settings gueststore repository get
   ```

   You receive the URL information in the output.

   ```
   URL: <not set>
   ```

Sharing Data Between the vSphere Client and a Virtual Machine Guest Operating System with DataSets

The DataSets feature provides a way to share data between the vSphere Client and a virtual machine guest operating system. By using DataSets, you can exchange data which can be used to configure the guest operating system and the software applications and agents running inside the guest operating system.

The DataSets feature is intended for information that does not change frequently. You must organize the intended information in DataSets entries. Persistent data in DataSets may be any key-value data, which then the software applications and agents can use.

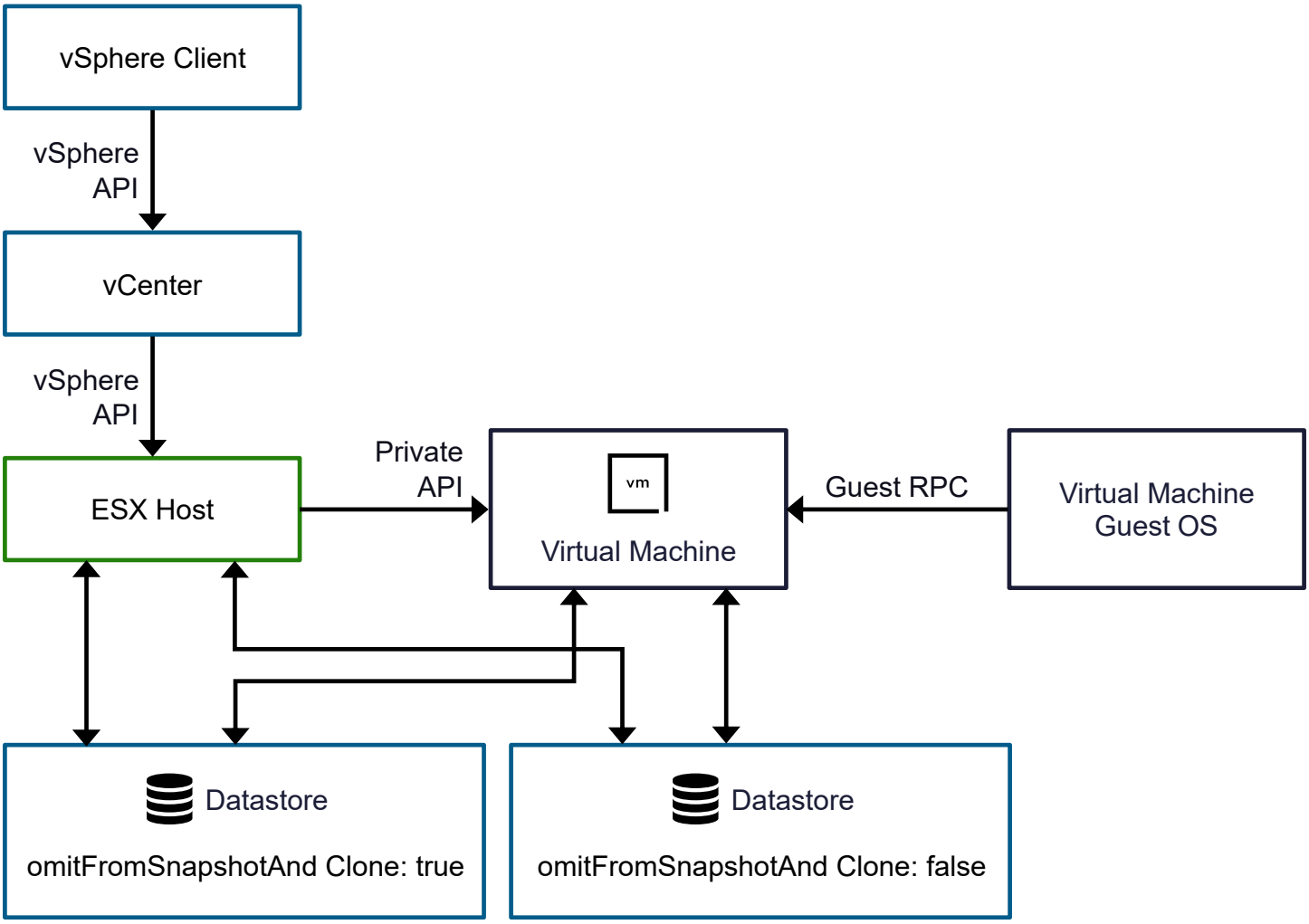
You can create key-value pairs within DataSets, and thereafter list the keys, update their values, or delete key-value pairs.

You can modify DataSets by using software applications regardless of the power state of the virtual machine. To avoid conflict with other applications, each application using DataSets must store its data in at least one unique DataSets entry. You can use the DataSets feature to implement:

- Guest deployment status - when you perform guest provisioning that includes running deployment scripts. DataSets can present configuration data and deployment scripts to the guest operating system. The guest operating system then reports back status including possible errors.

- Guest agent configuration - when a management application needs to configure an in-guest agent. Through the management side you can create and populate the DataSets with configuration data, and the guest agent reads and responds to that data.
- Guest related metadata or other items.

DataSets Architecture



The DataSets feature supports VM operations such as migration of a virtual machine with vMotion, cloning a virtual machine, and taking snapshots of a virtual machine. You can use the `omitFromSnapshotAndClone` parameter to preserve the DataSets information when you clone a virtual machine, create, or revert a snapshot.

For example, if you set `omitFromSnapshotAndClone` to `true`, the DataSets information is not preserved when you revert a snapshot.

If you set `omitFromSnapshotAndClone` to `false` and take a snapshot of a virtual machine, the DataSets information is included in snapshots. When you revert to the previous snapshot, the DataSets information is reverted to the point of taking a snapshot.

Requirements for DataSets

To use DataSets, your vSphere Client environment must meet the following requirements:

- vCenter 8.0 or later.
- The ESX host running in your environment must be at ESX 8.0 or later.
- Virtual machines must be of hardware version 20 or later.
- A virtual machine supports maximum 1024 DataSets and each DataSet can have maximum 1024 entries.
- Each virtual machine can have a maximum of 100 MB DataSets data.
- Keys must be no larger than 4 KB and values no larger than 1 MB.

For more information about Data Sets, see the *VMware Guest SDK Programming Guide*.

How do I Upgrade Virtual Machines

You can upgrade virtual machines to a higher level of compatibility and a higher version of VMware tools. After the upgrade, your VMs can take advantage of new hardware options and new features.

For a list of hardware features available to virtual machines with each ESX hardware compatibility setting, see Hardware Features Available with Virtual Machine Compatibility Settings.

To determine whether your virtual machines are compatible with a new version of ESX, see Virtual Machine Compatibility.

VMware Tools Upgrade

The first step in upgrading virtual machines is to upgrade VMware Tools. Installing VMware Tools is part of the process of creating a new virtual machine. If you are installing VMware Tools in multiple virtual machines with Windows guest operating systems, you can automate its installation and specify options for the components to include or exclude. For information about installing, upgrading, and configuring VMware Tools, see the *VMware Tools User Guide*.

If the virtual machines do not have VMware Tools installed, you can use the VMware Tools upgrade procedure to install VMware Tools. After you install or upgrade VMware Tools, upgrade the virtual machine compatibility.

Virtual Machine Compatibility Upgrade

VMware offers the following tools for upgrading virtual machines:

vSphere Client	Requires that you perform the virtual machine upgrade one step at a time, but does not require vSphere Lifecycle Manager. In the vSphere Client, you can upgrade virtual machines manually, or schedule upgrades. **Manual Upgrade** — Use this procedure to upgrade one or more virtual machines to the latest supported virtual hardware version immediately. **Schedule VM Upgrades** — Use this procedure to schedule an upgrade of one or more virtual machines at the next reboot of the virtual machine, and choose from all supported compatibility level upgrades.
vSphere Lifecycle Manager	Automates the process of upgrading and patching virtual machines, ensuring that the steps occur in the correct order. You can use vSphere Lifecycle Manager to directly upgrade virtual machine hardware, VMware Tools, and virtual appliances. You can also patch and update third-party software running on the virtual machines and virtual appliances. See the *vSphere Lifecycle Manager* documentation.

Note:
Do not use `vmware-vmupgrade.exe` to upgrade virtual machines.

Note:
Upgrading virtual machine hardware is a heavyweight operation that might cause some applications or the operating system to stop working properly.

Downtime for Upgrading Virtual Machines

During a virtual machine compatibility upgrade, you must shut down the virtual machine for all guest operating systems. For VMware Tools upgrade, downtime is not required for many Linux operating systems.

Table 141: Virtual Machine Downtime by Guest Operating System

Guest Operating System	Upgrade VMware Tools	Upgrade Virtual Machine Compatibility
Microsoft Windows	Downtime to restart the guest operating system.	Downtime to shut down and power on the virtual machine.
Linux	Downtime to restart the guest operating system is required to load drivers.	Downtime to shut down and power on the virtual machine.
NetWare	No downtime.	Downtime to shut down and power on the virtual machine.
Solaris	No downtime.	Downtime to shut down and power on the virtual machine.
FreeBSD	No downtime.	Downtime to shut down and power on the virtual machine.
Mac OS X	No downtime.	Downtime to shut down and power on the virtual machine.

Note:
For Linux guest operating systems, VMXNET3 and PVSCSI drivers are embedded in the Linux kernel. Linux virtual machines do not use VMware Tools to load VMXNET3 and PVSCSI drivers.

To load the new versions of the VMXNET driver, you must restart the virtual machine or manually reload the driver. Manual restart is not required for the Linux guest operating system using kernel version 3.10.

To verify that the drivers are configured in the Linux kernel and that the virtual hardware is available, see Broadcom knowledge base article 343109.

Planning Downtime for Virtual Machines

You can stagger virtual machine downtimes to accommodate a schedule convenient to you and your customers.

For example:

- If your virtual machine users are located in diverse time zones, you can prepare by migrating virtual machines to specific hosts to serve a given time zone. This way you can arrange host upgrades so that virtual machine downtime occurs transparently outside business hours for that time zone.
- If your virtual machine users operate around the clock, you can delay downtime for their virtual machines to normally scheduled maintenance periods. You do not need to upgrade any stage within a certain time period. You can take as long as needed at any stage.

Upgrade the Compatibility of a Virtual Machine Manually

The virtual machine compatibility determines the virtual hardware available to the virtual machine, which corresponds to the physical hardware available on the host machine. You can upgrade the compatibility level to make additional hardware available to the virtual machine.

- Create a backup or snapshot of the virtual machines. See Manage Virtual Machines With Snapshots.
- Upgrade VMware Tools. On Microsoft Windows VMs, the virtual machine might lose its network settings if you upgrade the compatibility before you upgrade VMware Tools.
- Verify that all virtual machines and their `.vmdk` files are stored on storage connected to the ESX host or the client machine.
- Determine the ESX versions that you want the virtual machines to be compatible with. For more information, see Virtual Machine Compatibility.

- Check whether the guest operating systems of the virtual machines that you upgrade require a power-off. For example, some Linux operating systems do not require a power-off before a virtual machine compatibility upgrade. For more information, see Virtual Machine Compatibility.

  Important:
  Upgrading virtual machine hardware might cause some applications or the operating system to stop working properly. Perform a hardware version upgrade only if you need a feature that comes with the newer hardware version.

1. In the vSphere Client, navigate to the virtual machine.
2. Optional: Right-click the virtual machine and select **Power** > **Power Off**.
3. Select **Actions** > **Compatibility** > **Upgrade VM Compatibility**.
4. Click **Yes** to confirm the upgrade.
5. Select a compatibility and click **OK**.

Schedule a Compatibility Upgrade for a Virtual Machine

The virtual machine compatibility determines the virtual hardware available to the virtual machine, which corresponds to the physical hardware available on the host. You can schedule a compatibility upgrade to make a virtual machine compatible with newer versions of ESX.

- Power off the virtual machine.
- Create a backup or snapshot of the virtual machine. See Manage Virtual Machines With Snapshots.
- Upgrade to the latest version of VMware Tools. If you upgrade the compatibility before you upgrade VMware Tools, the virtual machine might lose its network settings.
- Verify that all `.vmdk` files are available to the ESX host on a VMFS5 or NFS datastore.
- Verify that the virtual machine is stored on VMFS5 or NFS datastores.
- Verify that the compatibility settings for the virtual machine are not the latest supported version.
- Determine the ESX version that you want the virtual machine to be compatible with. See Virtual Machine Compatibility.

Use this procedure to schedule an upgrade for one virtual machine at the next reboot of the virtual machine, and choose from all supported compatibility level upgrades. To upgrade virtual machines immediately to the latest supported compatibility, see Upgrade the Compatibility of a Virtual Machine Manually.

You can use this procedure to schedule an upgrade for multiple virtual machines.

For information about virtual machine hardware versions and compatibility, see Virtual Machine Compatibility.

1. Navigate to a virtual machine in the inventory.
2. Right-click the virtual machine and select **Compatibility** > **Schedule VM Compatibility Upgrade**.
3. In the **Schedule VM Compatibility Upgrade** dialog box, confirm that you want to schedule a compatibility upgrade by clicking **Yes**.
4. From the **Compatible with** drop-down menu, select the compatibility to upgrade to.

   The virtual machine compatibility is upgraded the next time you restart the virtual machine.
5. Optional: To upgrade the compatibility when you do regularly scheduled guest maintenance, select **Only upgrade after normal guest OS shutdown**.

   This prevents the scheduled upgrade from occurring unless the guest operating system of the virtual machine is shut down or restarted normally.

Each of the selected virtual machines is upgraded to the compatibility that you chose at the next reboot of the virtual machine, and the Compatibility setting is updated in the Summary tab of the virtual machine.

Monitor the Guest Operating System Reboot Status

The guest OS reboot requirement is triggered when you install, upgrade, or uninstall VMware Tools. Some of the VMware Tools operations may also require a guest OS restart. For example, when you update drivers, user-mode components, VMware Tools services, or the Microsoft VC++ Redistributable package.

For large-scale environments, tracking the guest OS reboot status provides you with visibility into the reboot lifecycle across vCenter and the ESX host. The ESX host monitors the guest OS reboot status by using specific guest information that is reported by VMware Tools. The ESX host user interface and command-line utility use the guest information to display the VMware Tools installation status, running status, and VMware Tools version.

The ESX host preserves the guest OS reboot status during:

- Suspend and resume operations.
- Migration of virtual machines with vMotion.
- Snapshot operations.

System Requirements for Guest Operating System Reboot Status

To use the guest operating system reboot status, your environment must meet these requirements:

- vCenter 9.1 and later.
- ESX 9.1 and later.

  Note:

  ESX on Windows Arm 64-bit is not supported.
- All supported Windows guest operating system versions report the guest OS reboot status with VMware Tools 13.1.0 and later.

Procedure

To monitor the guest OS reboot status and identify which components require a restart after you install VMware Tools, perform the following steps.

1. Select and power on a virtual machine.
2. Click the **Summary** tab.

   Note:

   The **Issues and Alarms** card appears displaying details about the guest OS system reboot.
3. In the **Virtual Machine Details** card, click the information icon of **VMware Tools**.
4. You can track the running status, VMware Tools version, and the status of VMware Tools, and the installation type.
5. To view all guest OS system reboot requirements, click the **Monitor** tab and select **Issues and Alarms** > **All Issues**.

The **Summary** and **Monitor** tabs display the event message "Guest Operating System reboot was requested by VMware Tools installer (**list of one or more components**) on **timestamp**, where the list of components might consist of the following categories:

- **Drivers**: reported system drivers include vmci, vsock, pvscsi, vmxnet3, vmhgfs, hgfs, vmrawdsk, vmmemctl, vFileIntrospection, vnetwfp, vNetworkIntrospection, vsepamsi, vmmouse, efifw, VM3DService, and svga_wddm. For more information, see VMware Tools Device Drivers.
- **Packages**: Software dependencies reported as vcredist for Microsoft Visual C++ Redistributable.
- **User-Mode Services**: Core utilities reported as VMTools for VMware Tools Service, VGAuth for VGAuth Service, and VGAuthSecPkg for the VGAuth LSA plug-in.
- **Miscellaneous**: System-level changes or operations, reported as DiskTimeout for disk service, PerfSetting for performance settings, or MSI for the Microsoft installer, or MSIRepair for the repair installation.

Troubleshooting Virtual Machines

The virtual machine troubleshooting topics provide solutions to potential problems that you might encounter when using your virtual machines.

Troubleshooting USB Passthrough Devices

Information about feature behavior can help you troubleshoot or avoid potential problems when USB devices are connected to a virtual machine.

Error Message When You Try to Migrate Virtual Machine with USB Devices Attached

Migration with vMotion cannot proceed and issues a confusing error message when you connect multiple USB devices from an ESX host to a virtual machine and one or more devices are not enabled for vMotion.

The Migrate Virtual Machine wizard runs a compatibility check before a migration operation begins. If unsupported USB devices are detected, the compatibility check fails and an error message similar to the following appears: `Currently connected device 'USB 1' uses backing 'path:1/7/1', which is not accessible.`

To successfully pass vMotion compatibility checks, you must enable all USB devices that are connected to the virtual machine from a host for vMotion. If one or more devices are not enabled for vMotion, migration will fail.

1. Make sure that the devices are not in the process of transferring data before removing them.
2. Re-add and enable vMotion for each affected USB device.

Cannot Copy Data From an ESX Host to a USB Device That Is Connected to the Host

You can connect a USB device to an ESX host and copy data to the device from the host. For example, you might want to gather the vm-support bundle from the host after the host loses network connectivity. To perform this task, you must stop the USB arbitrator.

If the USB arbitrator is being used for USB passthrough from an ESX host to a virtual machine the USB device appears under `lsusb` but does not mount correctly.

This problem occurs because the nonbootable USB device is reserved for the virtual machine by default. It does not appear on the host's file system, even though `lsusb` can see the device.

1. Stop the `usbarbitrator` service:`/etc/init.d/usbarbitrator stop`
2. Physically disconnect and reconnect the USB device.
   By default, the device location is `/vmfs/devices/disks/mpx.vmhbaXX:C0:T0:L0`.
3. After you reconnect the device, restart the `usbarbitrator` service:`/etc/init.d/usbarbitrator start`
4. Restart `hostd` and any running virtual machines to restore access to the passthrough devices in the virtual machine.

Recover Orphaned Virtual Machines

Virtual machines appear with `(orphaned)` appended to their names.

Virtual machines that reside on an ESX host that vCenter manages might become orphaned in rare cases. Such virtual machines exist in the vCenter database, but the ESX host no longer recognizes them.

Virtual machines can become orphaned if a host failover is unsuccessful, or when the virtual machine is unregistered directly on the host. If this situation occurs, move the orphaned virtual machine to another host in the data center on which the virtual machine files are stored.

1. Determine the datastore where the virtual machine configuration (`.vmx`) file is located.
   a) Select the virtual machine in the inventory, and click the **Datastores** tab.
      The datastore or datastores where the virtual machine files are stored are displayed.
   b) If more than one datastore is displayed, select each datastore and click the **Files** tab to browse for the `.vmx` file.
   c) To verify the location of the `.vmx` file, select the virtual machine from **Datastore**.
2. Return to the virtual machine in the inventory, right-click it, and select **Remove from Inventory**.
3. Click **Yes** to confirm the removal of the virtual machine.
4. Reregister the virtual machine with vCenter.
   a) Right-click the datastore where the virtual machine file is located and select **Register VM**.
   b) Browse to the `.vmx` file and click **OK**.
   c) Select the location for the virtual machine and click **Next**.
   d) Select the host on which to run the virtual machine and click **Next**.
   e) Click **Finish**.

vSphere Networking

vSphere Networking provides information about configuring networking for VMware vSphere®, including how to create vSphere distributed switches and vSphere standard switches.

vSphere Networking also provides information on monitoring networks, managing network resources, and networking best practices.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

The information presented is written for experienced Windows or Linux system administrators who are familiar with network configuration and virtual machine technology.

Introduction to vSphere Networking

Get to know the basic concepts of the vSphere networking and learn to set up and configure a network in a vSphere environment.

Networking Concepts

A few concepts are essential for a thorough understanding of virtual networking. If you are new to vSphere, it is helpful to review these concepts.

Networking Concept	Description
Physical Network	A network of physical machines that are connected so that they can send data to and receive data from each other. VMware ESX runs on a physical machine.
Virtual Network	A network of virtual machines running on a physical machine that are connected logically to each other so that they can send data to and receive data from each other. Virtual machines can be connected to the virtual networks that you create when you add a network.
Physical Ethernet Switch	A physical Ethernet switch manages network traffic between machines on the physical network. A switch has multiple ports, each of which can be connected to a single machine or another switch on the network. Each port can be configured to behave in certain ways depending on the needs of the machine connected to it. The switch learns which hosts are connected to which of its ports and uses that information to forward traffic to the correct physical machines. Switches are the core of a physical network. Multiple switches can be connected together to form larger networks.
vSphere Standard Switch	It works much like a physical Ethernet switch. It detects which virtual machines are logically connected to each of its virtual ports and uses that information to forward traffic to the correct virtual machines. A vSphere standard switch can be connected to physical switches by using physical Ethernet adapters, also referred to as uplink adapters, to join virtual networks with physical networks. This type of connection is similar to connecting physical switches together to create a larger network. Even though a vSphere standard switch works much like a physical switch, it does not have some of the advanced functionality of a physical switch.
vSphere Distributed Switch	A vSphere distributed switch acts as a single switch across all associated hosts in a data center to provide centralized provisioning, administration, and monitoring of virtual networks. You configure a vSphere distributed switch on the vCenter system and the configuration is propagated to all hosts that are associated with the switch. This lets virtual machines maintain consistent network configuration as they migrate across multiple hosts. A vSphere distributed switch can be modified to allow running real-time workload and act as a base for industrial switch.

Networking Concept	Description
Host Proxy Switch	A hidden standard switch that resides on every host that is associated with a vSphere distributed switch. The host proxy switch replicates the networking configuration set on the vSphere distributed switch to the particular host.
Standard Port Group	Network services connect to standard switches through port groups. Port groups define how a connection is made through the switch to the network. Typically, a single standard switch is associated with one or more port groups. A port group specifies port configuration options such as bandwidth limitations and VLAN tagging policies for each member port.
Distributed Port	A port on a vSphere distributed switch that connects to a host's VMkernel or to a virtual machine's network adapter.
Distributed Port Group	A port group associated with a vSphere distributed switch that specifies port configuration options for each member port. Distributed port groups define how a connection is made through the vSphere distributed switch to the network.
NSX Distributed Port Group	A port group associated with a vSphere distributed switch that specifies port configuration options for each member port. To distinguish between vSphere distributed port groups and NSX port groups, in the vSphere Client the NSX virtual distributed switch, and its associated port group, is identified with the icon. NSX appears as an opaque network in vCenter, and you cannot configure NSX settings in vCenter. The NSX settings displayed are read only. You configure NSX distributed port groups using VMware NSX® Manager or the VMware NSX API management tools. Those NSX distributed port groups are created as segments or logical switch from VMware NSX. To learn about configuring NSX, see the *NSX Data Center for vSphere* documentation.
NSX Manager Appliance Plug-in	NSX Manager Appliance can be installed from the vSphere Web Client. You do not need to perform any installation operations from the NSX Manager UI. After the NSX Manager is installed, NSX appears as a plug-in in the vCenter that is ready to install VMware NSX for virtual networking or security use cases. For more information, see the *VMware NSX* documentation.
NIC Teaming	NIC teaming occurs when multiple uplink adapters are associated with a single switch to form a team. A team can either share the load of traffic between physical and virtual networks among some or all of its members, or provide passive failover if there is a hardware failure or a network outage.
VLAN	VLAN enable a single physical LAN segment to be further segmented so that groups of ports are isolated from one another as if they were on physically different segments. The standard is 802.1Q.
VMkernel TCP/IP Networking Layer	The VMkernel networking layer provides connectivity to hosts and handles the standard infrastructure traffic of vSphere vMotion, IP storage, Fault Tolerance, and vSAN.
IP Storage	Any form of storage that uses TCP/IP network communication as its foundation. iSCSI and NFS can be used as virtual machine datastores and for direct mounting of `.ISO` files, which are presented as CD-ROMs to virtual machines.
TCP Segmentation Offload	TCP Segmentation Offload, TSO, allows a TCP/IP stack to emit large frames (up to 64 KB) even though the maximum transmission unit (MTU) of the interface is smaller. The network adapter then separates the large frame into MTU-sized frames and depends an adjusted copy of the initial TCP/IP headers.

Network Services

A virtual network provides several services to the host and virtual machines. You can enable two types of network services in ESX:

- Connecting virtual machines to the physical network and to each other.
- Connecting VMkernel services (such as NFS, iSCSI, vSAN, or vMotion) to the physical network.

VMware ESX Dump Collector Support

The ESX Dump Collector sends the state of the VMkernel memory, that is, a core dump to a network server when the system encounters a critical failure. The ESX Dump Collector in ESX supports both vSphere standard and distributed Switches. The ESX Dump Collector can also use any active uplink adapter from the team of the port group that handles the VMkernel adapter for the collector.

Changes to the IP address for the ESX Dump Collector interface are automatically updated if the IP addresses for the configured VMkernel adapter changes. The ESX Dump Collector also adjusts its default gateway if the gateway configuration of the VMkernel adapter changes.

If you try to delete the VMkernel network adapter used by the ESX Dump Collector, the operation fails and a warning message appears. To delete the VMkernel network adapter, deactivate the dump collection and delete the adapter.

There is no authentication or encryption in the file transfer session from a crashed host to the ESX Dump Collector. You can configure the ESX Dump Collector on a separate VLAN when possible to isolate the ESX core dump from regular network traffic.

For information about installing and configuring the ESX Dump Collector, see the *vCenter Deployment and Setup* documentation.

Setting Up Networking with vSphere Standard Switches

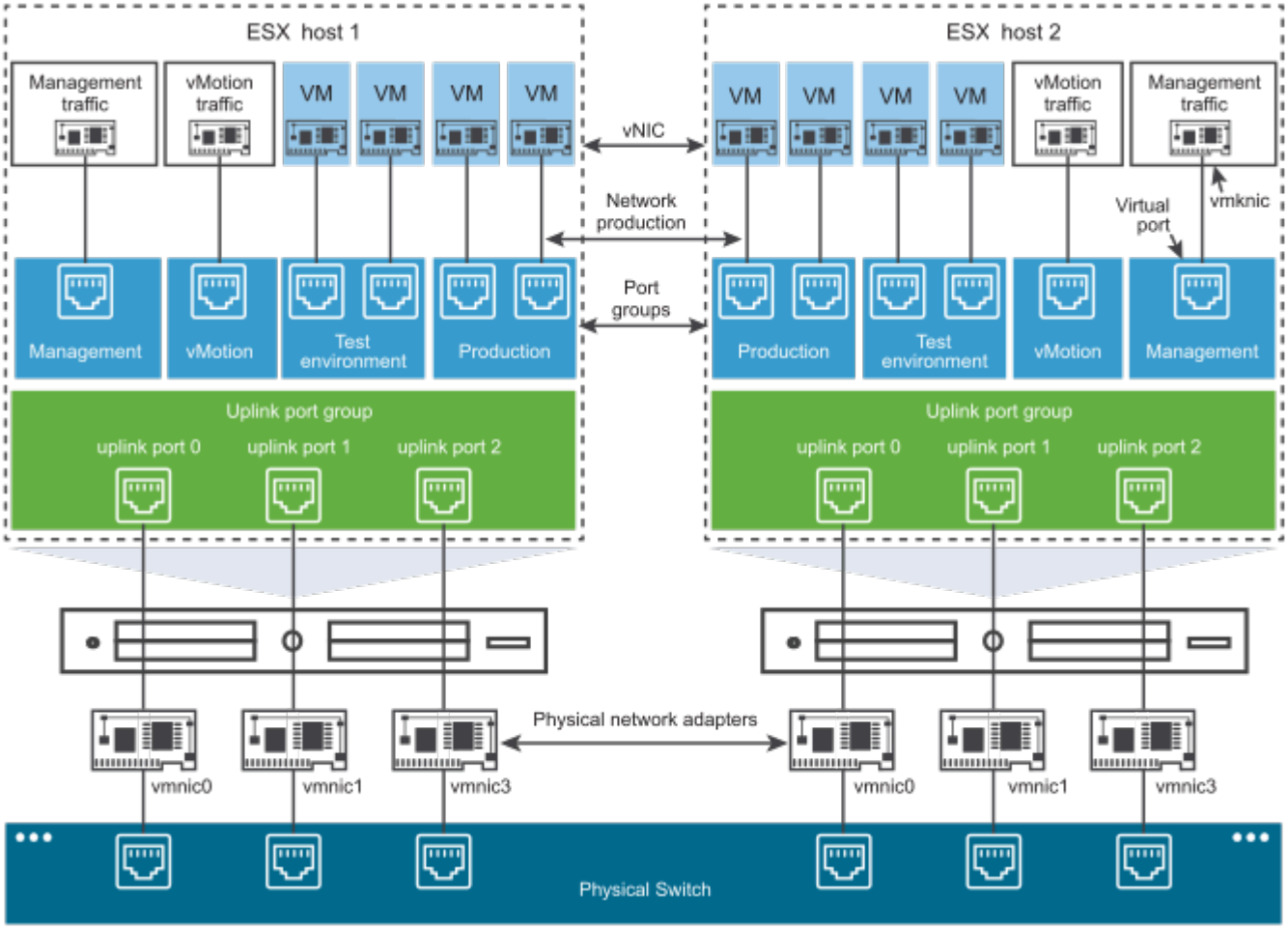
vSphere Standard Switches handle network traffic at the host level in a vSphere deployment. You can create abstracted network devices called vSphere Standard Switches. Learn how to use standard switches to provide network connectivity to hosts and virtual machines. A standard switch can bridge traffic internally between virtual machines in the same VLAN and link to external networks.

Standard Switch Overview

To provide network connectivity to hosts and virtual machines, you connect the physical NICs of the hosts to uplink ports on the standard switch. Virtual machines have network adapters (vNICs) that you connect to port groups on the standard switch. Every port group can use one or more physical NICs to handle their network traffic. If a port group does not have a physical NIC connected to it, virtual machines on the same port group can only communicate with each other and not with the external network. Each host has to be configured for networking separately. Settings are set only for that particular host.

Figure 43: vSphere Standard Switch architecture

vSphere Standard Switch architecture



A vSphere Standard Switch is similar to a physical Ethernet switch. Virtual machine network adapters and physical NICs on the host use the logical ports on the switch as each adapter uses one port. Each logical port on the standard switch is a member of a single port group. Settings that are changed at the virtual switch level are inherited by each standard port group. For information about maximum allowed ports and port groups, see the *Configuration Maximums* documentation.

Standard Port Groups

Each port group on a standard switch is identified by a network label, which must be unique to the current host. You can use network labels to make the networking configuration of virtual machines portable across hosts. You should give the same label to the port groups in a data center that use physical NICs connected to one broadcast domain on the physical network. Conversely, if two port groups are connected to physical NICs on different broadcast domains, the port groups should have distinct labels.

For example, you can create `Production` and `Test environment` port groups as virtual machine networks on the hosts that share broadcast domain on the physical network.

A VLAN ID, which restricts port group traffic to a logical Ethernet segment within the physical network, is optional. For port groups to receive the traffic that the same host sees, but from more than one VLAN, the VLAN ID must be set to VGT (VLAN 4095). Settings that are different from the virtual switch can be made at the port group level. For settings like, teaming policy or security settings, like promiscuous mode.

Number of Standard Ports

To ensure efficient use of host resources on ESX hosts, the number of ports of standard switches are dynamically scaled up and down. A standard switch on such a host can expand up to the maximum number of ports supported on the host.

Create a vSphere Standard Switch

Create a vSphere Standard Switch to provide network connectivity for hosts, virtual machines, and to handle VMkernel traffic. Depending on the connection type that you want to create, you can create a new vSphere Standard Switch with a VMkernel adapter, only connect physical network adapters to the new switch, or create the switch with a virtual machine port group.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual switches**.
3. Click **Add networking**.
4. Select a connection type for which you want to use the new standard switch and click **Next**.

Option	Description
VMkernel Network Adapter	Create a new VMkernel adapter to handle host management traffic, vMotion, network storage, fault tolerance, or vSAN traffic.
Physical Network Adapter	Add physical network adapters to an existing or a new standard switch.
Virtual Machine Port Group for a Standard Switch	Create a new port group for virtual machine networking.

5. Select **New standard switch** and click **Next**.
6. Add physical network adapters to the new standard switch.
   a) Under Assigned adapters, click **Add adapters**.
   b) Select one or more physical network adapters from the list and click **OK**.

      For higher throughput and to provide redundancy, configure at least two physical network adapters in the Active list.
   c) Optional: Use the **Move up** and **Move down** arrows in the **Assigned adapters** list to change the position of the adapter.
   d) Click **Next**.
7. If you create the new standard switch with a VMkernel adapter or virtual machine port group, enter connection settings for the adapter or the port group.

Option	Description
VMkernel adapter	1. Enter a label that indicates the traffic type for the VMkernel adapter, for example, `vMotion`. 2. Set a VLAN ID to identify the VLAN that the network traffic of the VMkernel adapter uses. 3. Select IPv4, IPv6 or both. 4. Select an option from the drop-down menu to set the MTU size. If you select Custom, enter a value for the MTU size. You can enable jumbo frames by setting an MTU value greater than 1500. You cannot set an MTU size greater than 9000 bytes. 5. Select an TCP/IP stack. After you set an TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you are able to use only this stack to handle vMotion or Provisioning traffic on the host. 6. If you use the default TCP/IP stack, select from the available services. 7. Configure IPv4 and IPv6 settings.
Virtual machine port group	1. Enter a network Label or the port group, or accept the generated label. 2. Set the VLAN ID to configure VLAN handling in the port group.

8. On the Ready to Complete page, click **Finish**.

- You might need to change the teaming and failover policy of the new standard switch. For example, if the host is connected to an Ether channel on the physical switch, you must configure the vSphere Standard Switch with Route based on IP hash as a load balancing algorithm. See What is Teaming and Failover Policy for more information.
- If you create the new standard switch with a port group for virtual machine networking, connect virtual machines to the port group.

Port Group Configuration for Virtual Machines

Learn how to add or modify a virtual machine port group to set up traffic management on a set of virtual machines.

The **Add Networking** wizard in the vSphere Client guides you through the process to create a virtual network to which virtual machines can connect, including creating a vSphere Standard Switch and configuring settings for a network label.

When you set up virtual machine networks, consider whether you want to migrate the virtual machines in the network between hosts. If so, ensure that the same broadcast domain is accessible by both hosts—that is, the same Layer 2 subnet.

ESX does not support virtual machine migration between hosts in different broadcast domains because the migrated virtual machine might require systems and resources that it would no longer have access to in the new network. Even if your network configuration is set up as a high-availability environment or includes intelligent switches that can resolve the virtual machine's needs across different networks, you might experience lag times as the Address Resolution Protocol (ARP) table updates and resumes network traffic for the virtual machines.

Virtual machines reach physical networks through uplink adapters. A vSphere Standard Switch can transfer data to external networks only when one or more network adapters are attached to it. When two or more adapters are attached to a single standard switch, they are transparently teamed.

Add a Virtual Machine Port Group

Learn how to add VM port groups on a vSphere Standard Switch to provide connectivity and common network configuration for virtual machines.

1. In the vSphere Client, navigate to the host.
2. Right-click the host and select **Add Networking**.
3. In **Select connection type**, select **Virtual Machine Port Group for a Standard Switch** and click **Next**.
4. In **Select target device**, select an existing standard switch or create a new standard switch.
5. If the new port group is for an existing standard switch, navigate to the switch.
   a) Click **Browse**.
   b) Select a standard switch from the list and click **OK**.
   c) Click **Next** and go to Step 7.
6. Optional: If you choose to create a new standard switch, enter a value for the MTU size and click **Next**.

   You can create a standard switch with or without adapters.

   If you create a standard switch without physical network adapters, all traffic on that switch is confined to that switch. No other hosts on the physical network or virtual machines on other standard switches can send or receive traffic over this standard switch. You might create a standard switch without physical network adapters if you want a group of virtual machines to be able to communicate with each other, but not with other hosts or with virtual machines outside the group.

a) Click **Add adapters**.
b) Select an adapter from the **Network Adapters** list and click **OK**.
c) Optional: Use the up and down arrows in the **Assigned adapters** list to change the position of the adapter if needed.
d) Click **Next**.

7. On the Connection settings page, identify traffic through the ports of the group.
   a) Type a **Network label** for the port group, or accept the generated label.

      Note:
      Port group names must not contain a colon character (:).
   b) Set the **VLAN ID** to configure VLAN handling in the port group.

      The VLAN ID also reflects the VLAN tagging mode in the port group.

VLAN Tagging Mode	VLAN ID	Description
External Switch Tagging (EST)	0	The virtual switch does not pass traffic associated with a VLAN.
Virtual Guest Tagging (VGT)	4095	Virtual machines handle VLANs. The virtual switch passes traffic from any VLAN.

   c) Click **Next**.
8. Review the port group settings in the Ready to complete page, and click **Finish**.

   Click **Back** if you want to change any settings.

Edit a Standard Switch Port Group

Learn how to edit the name and VLAN ID of a standard switch port group, and override networking policies at the port group level.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select a standard switch from the list.

   The topology diagram of the switch appears.
4. In the topology diagram of the switch, click the name of the port group.
5. Next to the topology diagram title, click the horizontal ellipsis icon and select **Edit settings**.
6. On the Properties page, rename the port group in the **Network label** text field.
7. Configure VLAN tagging in the **VLAN ID** drop-down menu.

VLAN Tagging Mode	VLAN ID	Description
External Switch Tagging (EST)	0	The virtual switch does not pass traffic associated with a VLAN.
Virtual Guest Tagging (VGT)	4095	Virtual machines handle VLANs. The virtual switch passes traffic from any VLAN.

8. On the Security page, override the switch settings for protection against MAC address changes, which are forged transmits and for running virtual machines in promiscuous mode.
9. On the Traffic shaping page, override at the port group level the size of average and peak bandwidth and of bursts.
10. On the Teaming and failover page, override the teaming and failover settings that are inherited from the standard switch.

    You can configure traffic distribution and rerouting between the physical adapters that are associated with the port group. You can also change the order in which host physical adapters are used upon failure.

11. Click **OK**.

Remove a Port Group from a vSphere Standard Switch

Learn how to remove port groups from vSphere Standard Switches in case you no longer need the associated labeled networks.

Verify that there are no powered-on virtual machines that are connected to the port group that you want to remove.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the standard switch.
4. From the topology diagram of the switch, select the port group that you want to remove by clicking its label.
5. From the toolbar in the switch topology, click the **Remove** action icon.

vSphere Standard Switch Properties

vSphere Standard Switch settings control switch-wide defaults for ports, which can be overridden by port group settings for each standard switch. Learn how to edit standard switch properties, such as the uplink configuration and the number of available ports.

Number of Ports on ESX Hosts

To ensure efficient use of host resources on ESX hosts, the ports of virtual switches are dynamically scaled up and down. A switch on such a host can expand up to the maximum number of ports supported on the host. The port limit is determined based on the maximum number of virtual machines that the host can handle.

Change the Size of the MTU on a vSphere Standard Switch

Learn how to change the size of the maximum transmission unit (MTU) on a vSphere Standard Switch to improve the networking efficiency by increasing the amount of payload data transmitted with a single packet, that is, enabling jumbo frames.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select a standard switch from the table and click **Edit settings**.
4. Change the **MTU (Bytes)** value for the standard switch.

   You can enable jumbo frames by setting an MTU value greater than 1500. You cannot set an MTU size greater than 9000 bytes.
5. Click **OK**.

Change the Speed of a Physical Adapter

Learn how to change the connection speed and duplex of a physical adapter to transfer data in compliance with the traffic rate.

If the physical adapter supports SR-IOV, you can enable it and can configure the number of virtual functions to use for virtual machine networking.

1. In the vSphere Client, navigate to a host.
2. On the **Configure** tab, expand **Networking** and select **Physical adapters**.

   The physical network adapters of the host appear in a table that contains details for each physical network adapter.

3. Select the physical network adapter from the list and click the **Edit adapter settings** icon.
4. Select the speed and duplex mode of the physical network adapter from the drop-down menu.

   The speed and duplex settings you select must match those configured on the physical switch. Speed is the rate of the interface, listed in megabits per second (Mbps). Duplex refers to how data flows on the interface.
5. Click **OK**.

Assign Physical Adapters to a vSphere Standard Switch

Learn how to assign a physical adapter to a standard switch to provide connectivity to virtual machines and VMkernel adapters on the host. You can form a team of NICs to distribute traffic load and to configure failover.

NIC teaming combines multiple network connections to increase throughput and provide redundancy should a link fail. To create a team, you associate multiple physical adapters to a single vSphere Standard Switch.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the standard switch that you want to add a physical adapter to.
4. Click **Manage Physical Adapters**.
5. Add one or more available physical network adapters to the switch.
   a) Click **Add adapters**, select one or more network adapters from the list and click **OK**.

      The selected adapters appear in the failover group list under the Assigned Adapters list.
   b) Optional: Use the up and down arrows to change the position of an adapter in the failover groups.

      The failover group determines the role of the adapter for exchanging data with the external network, that is, active, standby or unused. By default, the adapters are added as active to the standard switch.
6. Click **OK** to apply the physical adapter configuration.

Topology of a vSphere Standard Switch

View the structure and components of a vSphere Standard Switch by using its topology diagram.

The topology diagram of a standard switch provides a visual representation of the adapters and port groups that are connected to the switch.

From the diagram you can edit the settings of a selected port group and of a selected adapter.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the standard switch from the list.

The diagram appears under the list of virtual switches on the host.

Diagram of a Standard Switch That Connects the VMkernel and Virtual Machines to the Network

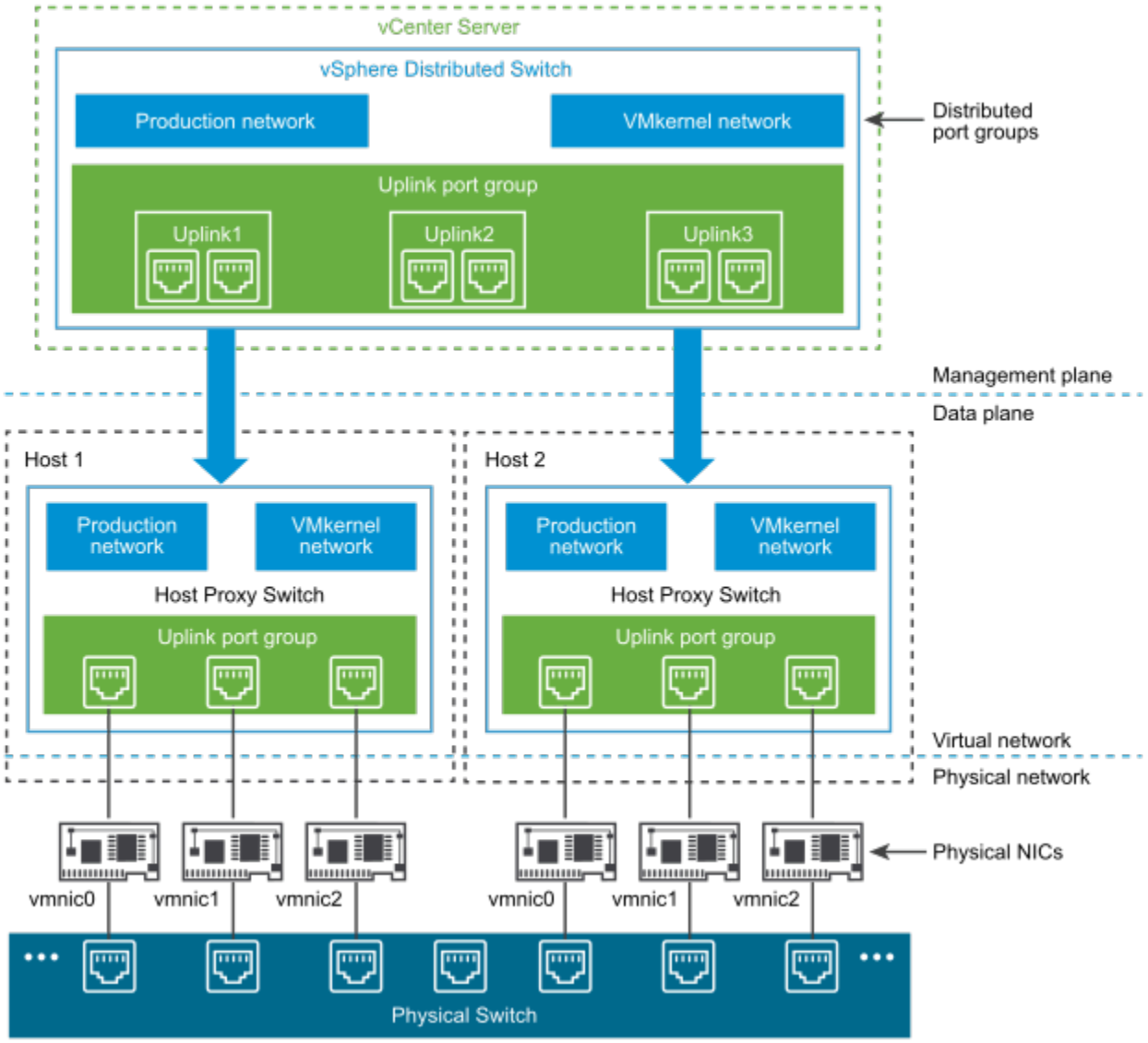
In your virtual environment, a vSphere Standard Switch handles VMkernel adapters for vSphere vMotion and for the management network, and virtual machines grouped. You can use the central topology diagram to examine whether a virtual machine or VMkernel adapter is connected to the external network and to identify the physical adapter that carries the data.

Figure 44: Topology Diagram of a vSphere Standard Switch

Setting Up Networking with vSphere Distributed Switches

With vSphere distributed switches you can set up and can configure networking in a vSphere environment.

Figure 45: vSphere Distributed Switch Architecture



A network switch in vSphere consists of two logical sections that are the data plane and the management plane. The data plane implements the packet switching, filtering, tagging, and so on. The management plane is the control structure that

you use to configure the data plane functionality. A vSphere Standard Switch contains both data and management planes, and you configure and maintain each standard switch individually.

A vSphere Distributed Switch separates the data plane and the management plane. The management functionality of the distributed switch resides on the vCenter system that lets you administer the networking configuration of your environment on a data center level. The data plane remains locally on every host that is associated with the distributed switch. The data plane section of the distributed switch is called as a host proxy switch. The networking configuration that you create on vCenter (the management plane) is automatically pushed down to all host proxy switches (the data plane).

The vSphere Distributed Switch introduces two abstractions that you use to create consistent networking configuration for physical NICs, virtual machines, and VMkernel services.

Uplink port group	An uplink port group or dvuplink port group is defined during the creation of the distributed switch and can have one or more uplinks. An uplink is a template that you use to configure physical connections of hosts and failover and load-balancing policies. You map physical NICs of hosts to uplinks on the distributed switch. At the host level, each physical NIC is connected to an uplink port with a particular ID. You set failover and load-balancing policies over uplinks and the policies are automatically propagated to the host proxy switches, or the data plane. In this way you can apply consistent failover and load-balancing configuration for the physical NICs of all hosts that are associated with the distributed switch.
Distributed port group	Distributed port groups provide network connectivity to virtual machines and accommodate VMkernel traffic. You identify each distributed port group by using a network label, which must be unique to the current data center. You configure NIC teaming, failover, load balancing, VLAN, security, traffic shaping, and other policies on distributed port groups. The virtual ports that are connected to a distributed port group share properties that are configured to the distributed port group. As with uplink port groups, the configuration that you set on distributed port groups on vCenter (the management plane) is automatically propagated to all hosts on the distributed switch through their host proxy switches (the data plane). In this way you can configure a group of virtual machines to share networking configuration by associating the virtual machines to the same distributed port group.

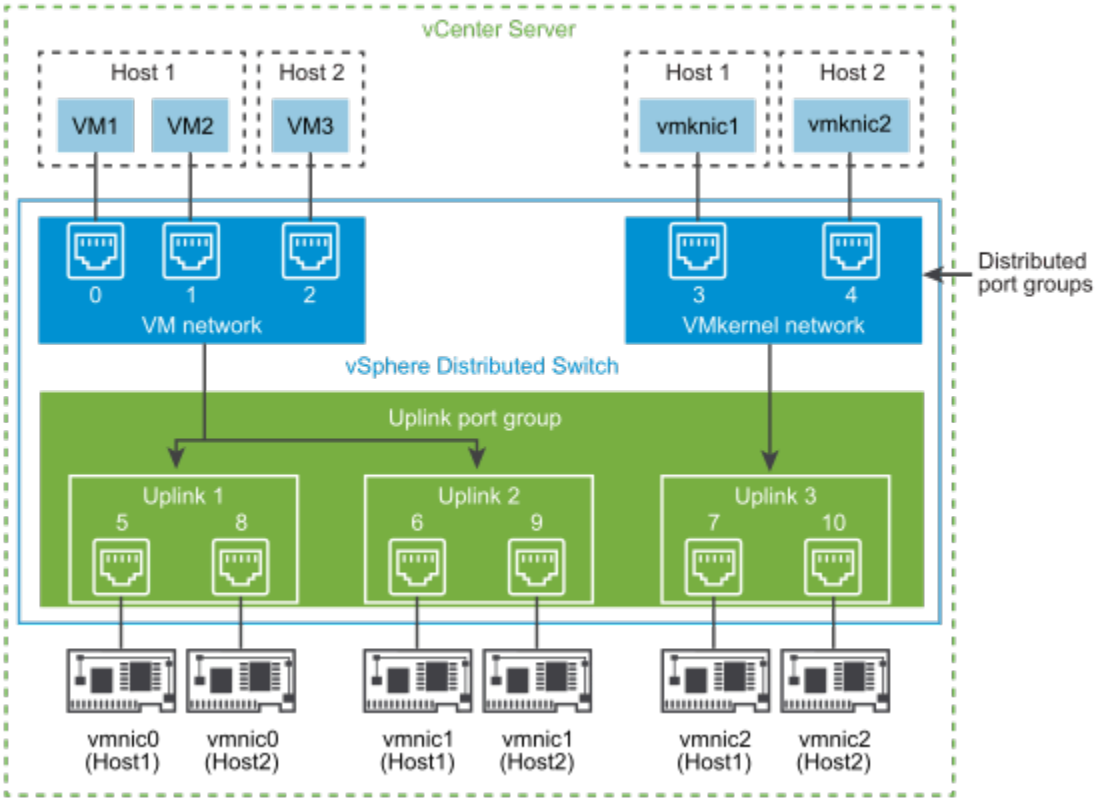
For example, suppose that you create a vSphere Distributed Switch on your data center and associate two hosts with it. You configure three uplinks to the uplink port group and connect a physical NIC from each host to an uplink. Each uplink has two physical NICs, one from each host which is mapped to it. For example, Uplink 1 is configured with vmnic0 from Host 1 and Host 2. Next you create the Production and the VMkernel network distributed port groups for virtual machine networking and VMkernel services. Respectively, a representation of the Production and the VMkernel network port groups is also created on Host 1 and Host 2. All policies that you set to the Production and the VMkernel network port groups are propagated to their representations on Host 1 and Host 2.

To ensure efficient use of host resources, the number of distributed ports of proxy switches is dynamically scaled up and down. A proxy switch on such a host can expand up to the maximum number of ports supported on the host. The port limit is determined based on the maximum number of virtual machines that the host can handle.

vSphere Distributed Switch Data Flow

The data flow from the virtual machines and VMkernel adapters down to the physical network depends on the NIC teaming and load balancing policies that are set to the distributed port groups. The data flow also depends on the port allocation on the distributed switch.

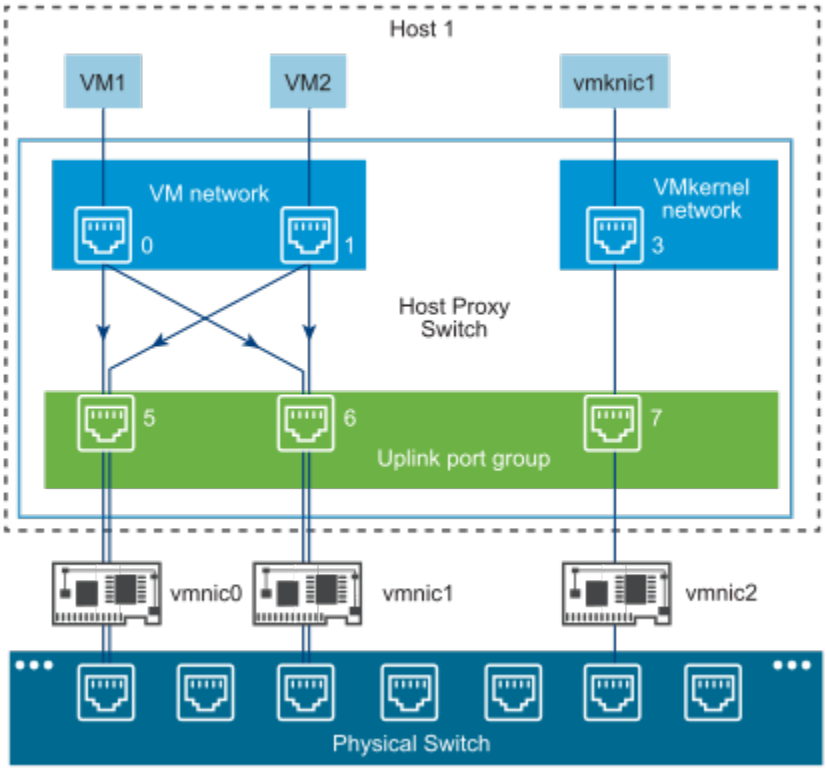
Figure 46: NIC Teaming and Port Allocation on a vSphere Distributed Switch



For example, suppose that you create the VM network and the VMkernel network distributed port groups, respectively with 3 and 2 distributed ports. The distributed switch allocates ports with IDs from 0 to 4 in the order that you create the distributed port groups. Next, you associate Host 1 and Host 2 with the distributed switch. The distributed switch allocates ports for every physical NIC on the hosts, as the numbering of the ports continues from 5 in the order that you add the hosts. To provide network connectivity on each host, you map vmnic0 to Uplink 1, vmnic1 to Uplink 2, and vmnic2 to Uplink 3.

To provide connectivity to virtual machines and to accommodate VMkernel traffic, you configure teaming and failover to the VM network and to the VMkernel network port groups. Uplink 1 and Uplink 2 handle the traffic for the VM network port group, and Uplink 3 handles the traffic for the VMkernel network port group.

Figure 47: Packet Flow on the Host Proxy Switch



On the host side, the packet flow from virtual machines and VMkernel services passes through particular ports to reach the physical network. For example, a packet sent from VM1 on Host 1 first reaches port 0 on the VM network distributed port group. Because Uplink 1 and Uplink 2 handle the traffic for the VM network port group, the packet can continue from uplink port 5 or uplink port 6 . If the packet goes through uplink port 5, it continues to vmnic0, and if the packet goes to uplink port 6, it continues to vmnic1.

What is Network Offloads Compatibility

In vSphere 8.0, vSphere Distributed Services Engine (vDSE) introduces the network offloading capability that uses data processing units (DPUs) to offload infrastructure functions from host or server CPUs.

Starting with VCF 9.1, the network offloading capabilities have expanded beyond vDSE to include advanced NIC passthrough for high-performance network adapters.

The network offloading compatibility enables you to shift networking operations from the host x86 CPUs to specialized hardware - a DPU or a high-performance NIC. As a result, system efficiency is improved by maximizing packets-per-second and also CPU overhead is reduced. A vSphere Distributed Switch configured for offloading supports the following operational states:

- Non-offloading mode
  Before NSX is enabled, the DPU or Performance NIC operates as a traditional NIC.
- Offloading mode
  After NSX is enabled, traffic forwarding logic is offloaded from the ESX host to the hardware pipeline of the DPU or Performance NIC.

The Network Offloads Compatibility setting is defined during the creation of the vSphere Distributed Switch and cannot be modified after hosts are associated. For DPU-based offloads (using vDSE), you must select the specific DPU vendor, for example, NVIDIA BlueFiled or Pensando. For Performance NIC offloads (UPTv2 or DVX), you must select Advanced NIC Passthrough.

Features Supported by Network Offload Switches:

- Creation and deletion of the vSphere distributed switch.

- Configuration management.
- vSphere distributed switch health check.
- Link aggregation control protocol (LACP).
- Port mirroring.
- Private LANs.
- Link layer discovery protocol.

Note:
Features not supported by vSphere Distributed Switch backed by DPU.

- Network I/O control.
- Traffic shaping policies.

Dual DPU and Performance NIC Availability

The deployment architecture for network offloading differs depending on whether you use vDSE with DPUs or Advanced NIC Passthrough with Performance NICs.

Dual DPU Support

vDSE supports the use of two DPUs for enhanced availability. For more information on dual DPUs, see What is VMware vSphere Distributed Services Engine.

Dual DPUs can be consumed in the following modes:

- High Availability (HA) mode
  Each DPU is managed by a single offloaded vSphere Distributed Virtual Switch (vDS). One DPU acts as the active processor, while the second DPU serves as a stand-by. The stand-by DPU is designated as a backup DPU. If the active DPU fails, then the active networking offload is switched to the stand-by DPU. This provides high availability for the DPU. This switchover minimizes the risk of failure on the active workloads.
  If ESX detects an active DPU failure, it triggers a failover to the shadow switch and ports on the stand-by DPU to enable packet processing.
- Non-HA mode
  Both DPUs act as active processors, but each is consumed by a separate offloaded VDS. This allows for active networking datapath offload across both devices without redundancy between them.

Perfromacnce NIC Availability (Advanced NIC Passhtrough)

For environments that use Performance NICs, such as ConnectX-7 or BlueField-3 in NIC mode, through Advanced NIC Passthrough, the following configuration requirements and limitations apply in VCF 9.1,:

- Only a single physical NIC per a vSphere Distributed Switch is supported for these offload modes. The NIC can have multiple ports configured.
- Dual physical adapters are not supported for a single offloaded datapath. However, High Availability is achieved through the use of multiple ports on the single supported NIC.
- Dual Performance NIC configuration for a single offloaded VDS are not supported.

Performance NIC Offloads

The network offloads support in offloading infrastructure operations from the host to a physical DPU or SmartNic for better performance. It frees up resources and improves the performance of the system. A performance NIC similar to a DPU handles demanding workloads that are offloaded to a NIC while offering high performance. These performance NICs support Uniform Pass Through UPT and DVX. Starting with VCF 9.1, UPTv2 for Advanced NIC allows you to efficiently use hardware on VMs that are equipped with VMXNET3. You can use the feature without compromising vMotion, High Availability, vSphere DRS, Hot Add for CPU, and so on.

To enable this feature, you must create a DVS with Advanced Nic in vCenter and must add a host to VDS which is equipped with an Advanced NIC. vCenter also performs a check if the host is attached to an Advanced Nic. You must also set Advanced NIC Passthrough in NSX-T Manager.

Enable Network Offloads

To enable the network offloads, you have to perform multiple steps in the vCenter and in VMware NSX®.

Step	Solution
Create a vSphere Distributed Switch	Create a vSphere Distributed Switch
Associate hosts to vSphere Distributed Switch	Add Hosts to a vSphere Distributed Switch
Configure network offloads with vSphere Distributed Services Engine	Configure Network Offloads with vSphere Distributed Services Engine
View the topology of the vSphere Distributed Switch with Network Offloads	View the Topology of Network Offloads Switch

What is an Industrial vSwitch

An Industrial vSwitch (IvS) is a networking component that helps VMs communicate between virtual and physical networks in a manufacturing industry, to support the industry transform their operations, and modernize their factory floors. Unlike standard switches, industrial switches provide network connectivity in industrial settings and factories. IvS is only supported with a vSphere Distributed Switch.

Starting with vSphere 9.0, a collaborative solution between the VCF networking team and data plane is introduced to support real-time industrial automation and control use cases while maintaining a clear difference between real-time and non-real-time functions. The Industrial vSwitch aims to meet the sub-millisecond latency that is required between industrial controller VMs and physical IO devices and support such deployment at scale.

PROFINET (Process Field Network) is an industrial Ethernet protocol that is used for communication and fast data exchange between controllers and devices. It helps achieve real-time communication in an industrial setting.

Parallel Redundancy Protocol (PRP) is a standard for Ethernet networks that are defined in IEC 62439-3, that provides uninterrupted data transmission and provides zero-time recovery if there is a single failure.

Under PRP, each node is connected to two separated, parallel Local Area Networks (LANs). For example, LAN A and LAN B with PRP sends two copies of each packet, one over each network. When the destination network receives a packet, it accepts the first copy and it discards the duplicate copy.

The two networks are assumed to be fail-independent. The destination node always receives at least one packet as long as either one of the two networks is operational.

You can use the vSphere Client to create, delete, reconfigure Industrial vSwitch assign an uplink port group, and tag the uplink to be a part of LAN A or LAN B if you plan to leverage PRP. PRP configuration is optional and should only be configured if you have two distinct LAN networks available. The Industrial vSwitch supports up to a maximum of four uplinks, two for each LAN network when PRP is enabled.

You can configure support for PROFINET and optionally, PRP in the NSX Manager. These are configured on transport node level or segment level. They are functions that can be activated or deactivated independently. The transport node level configurations are set as vSphere Distributed Switch properties, the segment level configurations are set as switch port properties on the ESX host. For more information about Industrial vSwitches and how to configure an Industrial vSwitch in NSX Manager, see Configure an Industrial Switch and Understanding Industrial vSwitch Segment Profile.

Create a vSphere Distributed Switch

Learn how to create a vSphere Distributed Switch on a data center to handle the networking configuration of multiple ESX hosts at a time from a central place.

1. In the vSphere Client, right-click a data center from the inventory tree.
2. Select **Distributed Switch** > **New Distributed Switch**.
3. On the **Name and location** page, enter a name for the new distributed switch, or accept the generated name, and click **Next**.
4. On the **Select version** page, select a distributed switch version and click **Next**.

Option	Description
Distributed Switch: 9.1.0	Compatible with ESX 9.1.0 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 9.0.0	Compatible with ESX 9.0.0 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 8.0.3	Compatible with ESX 8.0.3 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 8.0.0	Compatible with ESX 8.0 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 7.0.3	Compatible with ESX 7.0.3 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 7.0.2	Compatible with ESX 7.0.2 and later. Features that are released with later vSphere Distributed Switch versions are not supported.
Distributed Switch: 7.0.0	Compatible with ESX 7.0 and later. Features that are released with later vSphere Distributed Switch versions are not supported.

5. On the **Configure settings** page, configure the distributed switch settings.
   a) In the **Number of uplinks** field, enter a number or use the arrow buttons to set the number of uplink ports.

      Uplink ports connect the distributed switch to physical NICs on associated hosts. The number of uplink ports is the maximum number of allowed physical connections to the distributed switch per host.
   b) (Optional) Select the check box to enable the Industrial vSwitch.

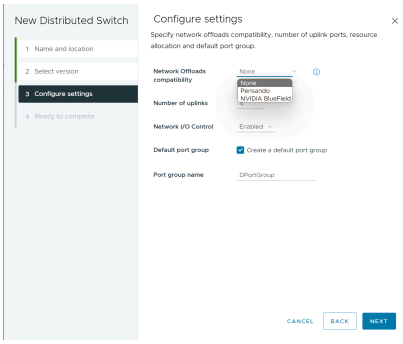
      Industrial vSwitch is an NSX EDP-based switch that facilitates real-time Profinet communication between Profinet devices across layer 2 networks. It enables the forwarding of Profinet traffic between virtual Programmable Logic Controllers (vPLCs) deployed in vSphere and IO devices outside the vSphere cluster. The Industrial vSwitch also supports Parallel Redundancy Protocol (PRP) which ensures seamless redundancy, high availability and provides zero-time recovery if there is a single failure. For more information, see What is an Industrial vSwitch.

      Note:

      - Industrial vSwitch is supported on vSphere Distributed Switch 9.0.0 and later.
      - The Industrial vSwitch can only be enabled when you create a vSphere Distributed Switch.
      - Network Offloads Compatibility, Network I/O control, Default Port Group, Port Group Name, Distributed Virtual Port Group, Port Mirroring, UPT, Passthrough, and vSphere Distributed Switch health checks are disabled if the **Industrial vSwitch** is enabled.

   c) Use the drop-down menu to select the type of **Network Offloads Compatibility**.

      Figure 48: Network Offloads Compatibility



By using network offloads compatibility, you can shift networking and security functions from the host CPU to a Data Processing Unit (DPU) or a Performance NIC. A DPU is an adapter with embedded compute capability, while Performance NICs use hardware pipelines to accelerate traffic via UPTv2 or Enhanced DirectPath I/O. Both technologies significantly improve performance by freeing up host CPU resources.

- **None**: If you select **None**, Network Offloads Compatibility is not supported.
- **Pensando**: If you select **Pensando**, Network Offloads Compatibility is supported. **Network I/O Control** is not supported.
- **NVIDIA BlueField**: If you select **NVIDIA BlueField**, Network Offloads Compatibility is supported. **Network I/O Control** is not supported.
- **Advanced NIC Passthrough**: If you select **Advanced NIC Passthrough**, Network Offloads Compatibility is supported. **Network I/O Control** is not supported.

Note:

You can configure **Advanced NIC Passthrough** when you use vSphere Distributed Switch 9.1.0 and later.

d) Use the drop-down menu to enable or disable **Network I/O Control**.

By using Network I/O Control, you can prioritize the access to network resources for certain types of infrastructure and workload traffic according to the requirements of your deployment. Network I/O Control continuously monitors the I/O load over the network and dynamically allocates available resources.

e) Optional: Select the **Create a default port group** check box to create a new distributed port group with default settings for this switch. Enter a **Port group name**, or accept the generated name.

If your system has custom port group requirements, create distributed port groups that meet those requirements after you add the distributed switch.

6. In the **DPU Failover Configuration** page, specify how the uplinks in a team are used when a failover occurs by configuring the failover order list. If you want to use some uplinks but reserve others for emergencies in case the uplinks in use fail, use the **Move Up** and **Move Down** to move uplinks into different groups.

   In the **DPU Failover Configuration** page, you can specify how DPU mapped uplinks can be consumed. When multiple DPUs are present on the host they can be attached to same distributed switch in high availability mode, that is one DPU will run the active traffic and other DPU will be available as standby.

   a) Select the type of DPU from the **Preset** drop-down menu.

   - **Single DPU**: If you select **Single DPU**, then only one DPU connects to the switch at a time as an active DPU.
   - **Dual DPU**: If you select **Dual DPU**, then both the DPUs connect to the switch at the same time. Only one of them is active.
   - **Custom**: If you select **Custom**, then you can specify the uplinks.

Option	Description
Active uplinks	Continue to use the uplink if the DPU adapter connectivity is up and active.
Standby uplinks	Use this uplink if the active DPU adapter is down.

7. On the **Ready to complete** page, review the settings that you selected and click **Finish**.

   Use the **Back** button to edit any settings.

A distributed switch is created in the data center. You can view the features that are supported on the distributed switch and other details by navigating to the new distributed switch and clicking the **Summary** tab.

Add hosts to the distributed switch and configure their network adapters on the switch.

Upgrade a vSphere Distributed Switch to a Later Version

Learn how to upgrade the vSphere Distributed Switch version to a later version. The upgrade lets the distributed switch take advantage of features that are available only in the later version.

- Upgrade vCenter to version 9.1.
- Upgrade all hosts connected to the distributed switch to ESX 9.1.

  CAUTION:

  Before you can upgrade ESX and vCenter to version 9.1, you must first upgrade your VDS to version 7.0 or later.

To be able to restore the connectivity of the virtual machines and VMkernel adapters if the upgrade fails, back up the configuration of the distributed switch.

If the upgrade is not successful, to recreate the switch with its port groups and connected hosts, you can import the switch configuration file. See Export vSphere Distributed Switch Configurations and Import a vSphere Distributed Switch Configuration.

Important:

Before upgrading the distributed switch, set DRS to Manual mode for all clusters that share the vDS to prevent unexpected vMotion operations during the upgrade. After the upgrade completes, return DRS to its original automation level. For the full list of best practices, see Broadcom knowledge base article 407905.

1. In vSphere Client, select **Inventory** from the main menu

   ( ).
2. Click **Networking**

   ( )

   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Upgrade** > **Upgrade Distributed Switch**.
4. Select the vSphere Distributed Switch version that you want to upgrade the switch to and click **Next**.

Option	Description
Distributed Switch: 9.1.0	Compatible with ESX 9.1.0 and later. New features and enhancements • Advanced NIC Passthrough. See What is Network Offloads Compatibility.
Distributed Switch: 9.0.0	Compatible with ESX 9.0.0 and later. New features and enhancements • Industrial vSwitch. See What is an Industrial vSwitch.
Distributed Switch: 8.0.3	Compatible with ESX 8.0.3 and later. New features and enhancements • Dual DPU Support. See What is Network Offloads Compatibility.
Distributed Switch: 8.0.0	Compatible with ESX 8.0.0 and later. New features and enhancements. See What is Network Offloads Compatibility.

Option	Description
Distributed Switch: 7.0.3	Compatible with ESX 7.0.3 and later. New features and enhancements • NVMe over TCP. See Create a VMkernel Adapter on a Host Associated with a vSphere Distributed Switch.
Distributed Switch: 7.0.2	Compatible with ESX 7.0.2 and later. New features and enhancements. • LACP fast mode. See LACP Support on a vSphere Distributed Switch.
Distributed Switch: 7.0.0	Compatible with ESX 7.0 and later. New features and enhancements. • NSX distributed port group. See the "Networking Concepts" section in Introduction to vSphere Networking.

5. Review the host compatibility and click **Next**.

   Some ESX instances that are connected to the distributed switch might be incompatible with the selected target version. Upgrade or remove the incompatible hosts, or select another upgrade version for the distributed switch.

6. Complete the upgrade configuration and click **Finish**.

   CAUTION:
   After you upgrade the vSphere Distributed Switch, you cannot revert it to an earlier version. You also cannot add ESX hosts that are running an earlier version than the new version of the switch.

Edit General and Advanced vSphere Distributed Switch Settings

Learn how to use basic settings for a vSphere Distributed Switch with setting the switch name and number of uplinks. You can also learn how to use advanced settings for a distributed switch including Cisco Discovery Protocol and the maximum MTU for the switch.

1. In vSphere Client, select **Inventory** from the main menu

   ( ).

2. Click **Networking**

   ( )

   and navigate to the distributed switch.

3. On the **Configure** tab, expand **Settings** and select **Properties**.
4. Click **Edit**.
5. Click **General** to edit the vSphere Distributed Switch settings.

Option	Description
Name	Enter the name for the distributed switch.
Industrial vSwitch	Select **Enable Industrial vSwitch** to enable this feature. **Note:** • The Industrial vSwitch can only be enabled only when you create a vSphere distributed switch. • **Network Offloads Compatibility**, **Network I/O control**, **Default Port Group**, **Port Group Name, Distributed Virtual Port Group, Port Mirroring, UPT, passthru, and vSphere distributed switch**

Option	Description
	health checks are disabled if the **Industrial vSwitch** is enabled.
Network I/O Control	To activate or deactivate **Network I/O control**, use the drop-down menu. **Note:** This menu option is deactivated if the **Network Offloads compatibility** is activated.
Network Offloads compatibility	To select the type of network offloads compatibility, use the drop-down menu. • **None**: If you select **None**, network offloads compatibility is not activated. • **Pensando**: If you select **Pensando** the **Network I/O control** is deactivated. • **NVIDIA BlueField**: If you select **NVIDIA BlueField** the **Network I/O control** is deactivated. • **Advanced NIC Passthrough**: If you select **Advanced NIC Passthrough** the **Network I/O control** is deactivated.
Description	Add or modify a description of the distributed switch settings.

6. Click **Advanced** to edit the vSphere Distributed Switch settings.

Option	Description
MTU (Bytes)	Maximum MTU size for the vSphere Distributed Switch. To enable jumbo frames, set a value greater than 1500 bytes.
Multicast filtering mode	• **Basic**. The distributed switch forwards traffic that is related to a multicast group based on a MAC address generated from the last 23 bits of the IPv4 address of the group. • **IGMP/MLD snooping**. The distributed switch forwards multicast traffic to virtual machines according to the IPv4 and IPv6 addresses of subscribed multicast groups by using membership messages defined by the Internet Group Management Protocol (IGMP) and Multicast Listener Discovery protocol.
Discovery Protocol	1. Select Cisco Discovery Protocol, Link Layer Discovery Protocol, or (disabled) from the **Type** drop-down menu. 2. Set **Operation** to Listen, Advertise, or Both. For information about the Discovery Protocol, see Switch Discovery Protocol.
Administrator Contact	Enter the name and other details of the administrator for the distributed switch.

7. Click **Uplinks** to edit the vSphere Distributed Switch settings.

Option	Description
Number of uplinks	Select **Add** to add uplink ports for the distributed switch. Edit uplink names to modify the names of the uplinks.

8. Click **OK**.

Manage Networking on Multiple Hosts on a vSphere Distributed Switch

Learn how to create and manage virtual networks on a vSphere Distributed Switch by adding hosts to the switch and connecting their network adapters to the switch. To create uniform networking configuration throughout multiple hosts on the distributed switch, you can use a host as a template and can apply its configuration to other hosts.

Tasks for Managing Host Networking on a vSphere Distributed Switch

You can add new hosts to a vSphere Distributed Switch, connect network adapters to the switch, and remove hosts from the switch. In a production environment, you might need to keep the network connectivity up for virtual machines and VMkernel services while you manage host networking on the distributed switch.

Adding Hosts to a vSphere Distributed Switch

Consider preparing your environment before you add new hosts to a distributed switch.

- Create distributed port groups for virtual machine networking.
- Create distributed port groups for VMkernel services. For example, create distributed port groups for management network, vMotion, and Fault Tolerance.
- Configure enough uplinks on the distributed switch for all physical NICs that you want to connect to the switch. For example, if the hosts that you want to connect to the distributed switch have eight physical NICs each, configure eight uplinks on the distributed switch.
- Make sure that the configuration of the distributed switch is prepared for services with specific networking requirements. For example, iSCSI has specific requirements for the teaming and failover configuration of the distributed port group where you connect the iSCSI VMkernel adapter.

You can use the **Add and Manage Hosts** wizard to add multiple hosts at a time.

Managing Network Adapters on a vSphere Distributed Switch

After you add hosts to a distributed switch, you can connect physical NICs to uplinks on the switch, configure virtual machine network adapters, and manage VMkernel networking.

If some hosts on a distributed switch are associated to other switches in your data center, you can migrate network adapters to or from the distributed switch.

If you migrate virtual machine network adapters or VMkernel adapters, make sure that the destination distributed port groups have at least one active uplink, and the uplink is connected to a physical NIC on the hosts. Another approach is to migrate physical NICs, virtual network adapters, and VMkernel adapters simultaneously.

If you migrate physical NICs, leave at least one active NIC that handles the traffic of port groups. For example, if *vmnic0* and *vmnic1* handle the traffic of the *VM Network* port group, migrate *vmnic0* and leave *vmnic1* connected to the group.

Removing Hosts from a vSphere Distributed Switch

Before you remove hosts from a distributed switch, you must migrate the network adapters that are in use to a different switch.

- To add hosts to a different distributed switch, you can use the **Add and Manage Hosts** wizard to migrate the network adapters on the hosts to the new switch all together. You can then remove the hosts safely from their current distributed switch.
- To migrate host networking to standard switches, you must migrate the network adapters in stages. For example, remove physical NICs on the hosts from the distributed switch by leaving one physical NIC on every host connected to the switch to keep the network connectivity up. Next, attach the physical NICs to the standard switches and migrate VMkernel adapters and virtual machine network adapters to the switches. Lastly, migrate the physical NIC that you left connected to the distributed switch to the standard switches.

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Add Hosts to a vSphere Distributed Switch

Learn to manage the networking of your vSphere environment by using a vSphere Distributed Switch, you must associate hosts with the switch. You connect the physical NICs, VMkernel adapters, and virtual machine network adapters of the hosts to the distributed switch.

- Verify that enough uplinks are available on the distributed switch to assign to the physical NICs that you want to connect to the switch.
- Verify that there is at least one distributed port group on the distributed switch.
- Verify that the distributed port group have active uplinks that are configured in its teaming and failover policy.

If you migrate or create VMkernel adapters for iSCSI, verify that the teaming and failover policy of the target distributed port group meets the requirements for iSCSI:

- Verify that only one uplink is active, the standby list is empty, and the rest of the uplinks are unused.
- Verify that only one physical NIC per host is assigned to the active uplink.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Add and Manage Hosts**.
4. On the Select task page, select **Add hosts**, and click **Next**.
5. On the Select hosts page,
   a) Select one or more hosts from the list of available hosts under **All hosts**.
   b) To view the selected hosts, click **Selected**.
      The selected hosts are displayed.
   c) To filter the hosts based on their compatibility, click **Compatibility**.

      Note:

      When you add hosts to a vSphere Distributed Switch with Network Offloads activated, you can only select adapters that are backed by a compatible DPU or a supported Performance NIC. vCenter automatically verifies that the hardware is capable of either DSE or Advanced NIC Passthrough.
   d) To select all the available hosts, click **Select All**.
6. Click **Next**.
7. On the **Manage physical adapters page**, you can add or remove network adapters to the distributed switch by assigning or unassigning an uplink.
8. To manage adapters on all hosts that have the same physical network adapter, select **Adapters on all hosts**.
   a) Click **Select All** to select all hosts.
   b) To assign an uplink to the host, select an uplink from the drop-down menu.

      If the distributed switch with Network Offloads is set to Advanced NIC passthrough, you can only add compatible adapters that are available for Advanced NIC passthrough.
   c) To unassign an uplink from the hosts, select **None** from the drop-down menu.
   d) To see more details about the hosts, expand the network adapter that is listed under **Physical network adapters**.
   e) You can view the switches that use this VMkernel adapter in **In use by switch**.

   For instance, if you assign *uplink1* to *vmnic1*, it is assigned to all the hosts that have *vmnic1* as its physical network adapter.
9. To manage adapters per host, select **Adapters per host**.
   a) Select the individual host from the list.
   b) To assign an uplink to the host, select an uplink from the drop-down menu.
   c) To unassign an uplink from the host, select **None** from the drop-down menu.

   If you select physical NICs that are assigned to other standard or distributed switches, the NICs are migrated to the current distributed switch.

   For consistent network configuration, you can connect the same physical NIC on every host to the same uplink on the distributed switch.

   For instance, if you are adding two hosts connect *vmnic1* on each host to *Uplink1* on the distributed switch.
10. Click **Next**.

    Note:
    If a host does not have an assigned physical network adapter, then a warning appears.

11. On the **Manage VMkernel adapters page**, you can manage VMkernel adapters to the distributed switch.
12. To manage VMkernel adapters on all hosts that have the same VMkernel adapter, select **Adapters on all hosts**.
    a) To select all the hosts, click **Select All**.
    b) Click **Assign Port Group**.
       You can see all the available port groups.
    c) To assign a port group, click **Assign**.
    d) To unassign a port group, click **Unassign**.
    e) You can view the switches that use this VMkernel adapter in **In use by switch**.
    f) To see more details about the hosts, expand the VMkernel adapter that is listed under **Name**.

    For instance, if you assign *DPortGroup1* to *vmk0*, the port group is assigned to all the hosts that has *vmk0* as its VMkernel network adapter.
13. To manage VMkernel adapter per host, select **Adapters per host**.
    a) Select the individual host from the list.
    b) Click **Assign Port Group**.
       You can see all the available port groups.
    c) To assign a port group, click **Assign**.
    d) To unassign a port group, click **Unassign**.
14. Click **Next**.
15. On the **Migrate VM networking** page, select the check box **Migrate virtual machine networking** to migrate virtual machines to a distributed switch.
16. To configure per network adapter, click **Assign port group**.
    a) To assign a port group, click **Assign**.
       For instance, the port group is assigned to all the virtual machines that have the same network adapter.
    b) To unassign port groups, click **Unassign**.
17. To configure per virtual machine, click **Assign port group**.
    a) To assign a port group, click **Assign**.
    b) To unassign a port group, click **Unassign**.
18. Click **Next**.
19. On the **Ready to Complete** page of the **Add and Manage Hosts** wizard, review the settings for the virtual machine.
20. Click **Finish**.
    You have now successfully added a host to the vSphere distributed switch.

Having hosts that are associated with the distributed switch, you can manage physical network adapters, VMkernel adapters, and virtual machine network adapters.

NSX-Managed Network Folders in vCenter

If the **Organize NSX Portgroups in vCenter folders** option is activated in NSX Manager during project creation, the NSX port goups that represent projects and VPC subnets get automatically organized into hierarchical folders in the vCenter inventory.

Upon activating the **Organize NSX Portgroups in vCenter folders** option, NSX creates an NSX-managed folder called `NSX Managed Folders` under the **Datacenter** object in the vCenter inventory. The hierarchically structured project and VPC folders are placed inside the`NSX Managed Folders` folder and contain all NSX port groups that represent networks and VPCs. The following limitations exist for you as the VI admin.

- The folders are strictly managed by the NSX manager. You, as a vSphere administrator, cannot edit, rename, or delete those NSX-managed folders. However, you can delete the VPC object itself. For more information about VPCs in vCenter, see Managing Virtual Private Clouds in vCenter.
- You, as a vSphere administrator, can set tags and permissions to these folders and objects.
- The NSX-managed folders contain only NSX port groups, but not vSphere-native distributed or standard port groups.

- Whenever changes occur in the network topology, the NSX port groups are moved to the corresponding folders automatically by NSX.

For more information about how NSX port groups are organized into vCenter folders, see Example: Organizing NSX Port Groups in VMware vCenter Folders.

Configure Physical Network Adapters on a vSphere Distributed Switch

Learn how to assign physical NICs to uplinks on the switch for the hosts that are associated with a distributed switch. You can configure physical NICs on the distributed switch for multiple hosts at a time.

For consistent networking configuration throughout all hosts, you can assign the same physical NIC on every host to the same uplink on the distributed switch. For example, you can assign *vmnic1* from hosts ESX *A* and ESX *B* to *Uplink 1*.

1. In vSphere Client, select **Inventory** from the main menu

   ( ).
2. Click **Networking**

   ( )

   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Add and Manage Hosts**.
4. On the **Select task** page, select **Manage host networking** and click **Next**.
5. On the **Select hosts** page, click the **Member Hosts** tab and select a host from the list and click **Next**.
6. On the **Manage physical adapters** page, you can add or remove network adapters to the distributed switch by assigning or unassigning them to an uplink.

   While adding hosts to the vSphere Distributed Switch with network offloads compatibility, you can only add a compatible adapter that is supported by a compatible DPU.

   a) To manage adapters on all hosts that have the same physical network adapter, click the **Adapters on all hosts** tab.

      - To assign an uplink to a NIC, select an uplink from the drop-down menu in the **Assign uplink** column.

      Note:

      In the uplinks drop-down menu, the `--` option indicates that not all the hosts with the same network adapter can be assigned to the same uplink. For example, if *vmnic0* is assigned to *uplink1* in ESX *A*, and to *uplink2* in ESX *B*, then the `--` option is selected in the drop-down menu. Selecting this option and moving to the next operation does not change any configuration.
      - To unassign an uplink from the NIC, select **None** from the drop-down menu in the **Assign uplink** column.
      - To see more details about a NIC, click the expand icon for the respective network adapter.

   b) To manage adapters per host, click the **Adapters per host** tab and click the name of a host from the list.

      - To assign an uplink to a NIC, select an uplink from the drop-down menu in the **Assign uplink** column.
      - To unassign an uplink from the NIC, select **None** from the drop-down menu in the **Assign uplink** column.
      - To see more details about a NIC, click the expand icon for the respective network adapter.

      For consistent network configuration, you can connect the same physical NIC on every host to the same uplink on the distributed switch. For example, if you are adding two hosts, connect *vmnic1* on each host to *Uplink1* on the distributed switch.

      If you select physical NICs that are already assigned to other standard or distributed switches, the NICs are migrated to the current distributed switch.

c) Click **Next**.

Note:

If a host does not have an assigned physical network adapter, then a warning appears.

7. On the **Manage VMkernel adapters** page, you can add VMkernel adapters to the distributed switch.
   a) To manage adapters on all hosts that have the same VMkernel adapter, click the **Adapters on all hosts** tab.
      - To assign a port group to a VMkernel adapter, click **Assign port group** in the **Destination port group** column and click **Assign** on the **Assign port group** tab..
      - To unassign a port group from a VMkernel adapter, click the X icon in the **Destination port group** column.
      - To see more details about a VMkernel adapter, click the expand icon for the respective network adapter and click the **Related hosts** tab.
   b) To manage adapters per host, click the **Adapters per host** tab and click the name of a host from the list.
      - To assign a port group to a VMkernel adapter, click **Assign port group** in the **Destination port group** column and click **Assign** on the **Assign port group** tab.
      - To unassign a port group from a VMkernel adapter, click the X icon in the **Destination port group** column.
      - To see more details about a VMkernel adapter, click the expand icon for the respective network adapter and click the **Settings** tab.
   c) Click **Next**.
8. On the **Migrate VM networking** page, select the **Migrate virtual machine networking** check-box to migrate virtual machines to a distributed switch and click **Next**.
9. On the **Ready to omplete** page, review the settings for the virtual machine and click **Finish**.

Migrate VMkernel Adapters to a vSphere Distributed Switch

Learn how to migrate VMkernel adapters to a distributed switch if you want to handle the traffic for VMkernel services by using only this switch when you no longer need the adapters on other standard or distributed switches.

1. In vSphere Client, select **Inventory** from the main menu

   ( ).
2. Click **Networking**

   ( )

   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Add and Manage Hosts**.
4. On the Select task page, select **Manage host networking** and click **Next**.
5. On the **Select hosts** page, under **Member Hosts**, select from the hosts that are associated with the distributed switch.
6. Click **Next**.
7. On the **Manage physical adapters page**, you can add or remove network adapters to the distributed switch by assigning or unassigning an uplink.
8. Click **Next**.

   Note:
   If a host does not have an assigned physical network adapter, then a warning appears.

9. On the **Manage VMkernel adapters page**, you can manage VMkernel adapters to the distributed switch.
10. To manage VMkernel adapters on all hosts that have the same VMkernel adapter, select **Adapters on all hosts**.
    a) Click **Assign Port Group**
       You can see all the available port groups.
    b) To assign a port group, click **Assign**.
    c) To unassign a port group, click **Unassign**.

    For instance, if you assign *DPortGroup1* to *vmk0*, the port group is assigned to all the hosts that has *vmk0* as its VMkernel network adapter.
11. To see more details about the hosts, expand the VMkernel adapter that is listed under **Name**.
12. You can view the switches that use this VMkernel adapter in **In use by switch**.
13. To manager VMkernel adapter per host, select **Adapters per host**.
    a) Select the individual host from the list.
    b) Click **Assign Port Group**.
       You can see all the available port groups.
    c) To assign a port group, click **Assign** .
    d) To unassign a port group, click **Unassign**.
14. Click **Next**.
15. On the **Migrate VM networking** page, select the check box **Migrate virtual machine networking** to migrate virtual machines to a distributed switch.
16. Click **Next**.
17. On the **Ready to Complete** page of the **Add and Manage Hosts** wizard, review the settings for the virtual machine.
18. Click **Finish**.

Create a VMkernel Adapter on a vSphere Distributed Switch

Learn how to create a VMkernel adapter on hosts associated with a distributed switch to provide network connectivity to the hosts and to handle the traffic for vSphere vMotion, IP storage, Fault Tolerance logging, and vSAN.

You should dedicate one distributed port group for each VMkernel adapter. One VMkernel adapter should handle only one traffic type.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. From the **Actions** menu, select **Add VMkernel Adapters**.
4. On the Select hosts page, click **Attached hosts**, select from the hosts that are associated with the distributed switch and click **OK**.
5. Click **Next**.
6. On the Configure VMkernel adapter page, configure the settings for the VMkernel adapter.

Option	Description
Network label	The network label is inherited from the label of the distributed port group.

Option	Description
IP settings	Select IPv4, IPv6, or both. **Note:** The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	Choose whether to get MTU for the network adapter from the switch or to set a custom size. You cannot set the MTU size to a value greater than 9000 bytes.
TCP/IP stack	Select a TCP/IP stack from the list. Once you set a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you will be able to use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration.
Available services	You can enable services for the default TCP/IP stack on the host. Select from the available services: • **vMotion**. Enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. The migration with vMotion to the selected host is not possible if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or there are no adapters using the vMotion TCP/IP stack. • **Provisioning**. Handles the data transferred for virtual machine cold migration, cloning, and snapshot migration. • **Fault Tolerance logging**. Enables Fault Tolerance logging on the host. You can use only one VMkernel adapter for FT traffic per host. • **Management**. Enables the management traffic for the host and vCenter. Typically, hosts have such a VMkernel adapter created when the ESX software is installed. You can create another VMkernel adapter for management traffic on the host to provide redundancy. • **vSphere Replication**. Handles the outgoing replication data that is sent from the source ESX host to the vSphere Replication server. • **vSphere Replication NFC**. Handles the incoming replication data on the target replication site. • **vSAN**. Enables the vSAN traffic on the host. Every host that is part of a vSAN cluster must have such a VMkernel adapter. • **vSphere Backup NFC**. VMkernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMkernel Adapter when vSphereBackup NFC service is enabled. • **NVMe over TCP**. VMkernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMkernel Adapter when NVMe over TCP adapter is enabled. For more information, see *vSphere Storage Guide*. • **NVMe over RDMA**. VMkernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMkernel Adapter when NVMe over RDMA adapter is enabled. For more information, see *vSphere Storage Guide*.

7. On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. To change the VMkernel default gateway, select **Configure on VMkernel adapters** or **Configure on TCP/IP stack and enter a gateway address**.

8. On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Enter the IPv6 address and subnet prefix length. 2. To change the VMkernel default gateway, select **Configure on VMkernel adapters** or **Configure on TCP/IP stack and enter a gateway address**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

9. Review your settings selections on the Ready to complete page and click **Finish**.

Migrate Virtual Machine Networking to the vSphere Distributed Switch

Learn how to manage virtual machine networking by using a distributed switch, migrate virtual machine network adapters to labeled networks on the switch.

Verify that at least one distributed port group that is intended for virtual machine networking exists on the distributed switch.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Add and Manage Hosts**.
4. On the **Select task** page, select **Manage host networking** and click **Next**.
5. On the **Select hosts** page, under **Member Hosts**, select from the hosts that are associated with the distributed switch.
6. Click **Next**.
7. On the **Manage physical adapters page**, you can add or remove network adapters to the distributed switch by assigning or unassigning an uplink.
8. Click **Next**.

   Note:
   If a host does not have an assigned physical network adapter, then a warning appears.
9. On the **Manage VMkernel adapters page**, you can add VMkernel adapters to the distributed switch.
10. On the **Migrate VM networking** page, select the check box **Migrate virtual machine networking** to migrate virtual machines to a distributed switch.
11. To configure per network adapter, click **Assign port group**.
    a) To assign a port group, click **Assign**.
       For instance, the port group is assigned to all the virtual machines that have the same network adapter.
    b) To unassign a port group, click **Unassign**.
12. To configure per virtual machine, click **Assign port group**.
    a) To assign a port group, click **Assign**.
    b) To unassign a port group, click **Unassign**.
13. Click **Next**.
14. On the **Ready to Complete** page of the **Add and Manage Hosts** wizard, review the settings for the virtual machine.
15. Click **Finish**.

Remove Hosts from a vSphere Distributed Switch

Learn how to remove hosts from a vSphere Distributed Switch if you have configured a different switch for the hosts.

- Verify that physical NICs on the target hosts are migrated to a different switch.
- Verify that VMkernel adapters on the hosts are migrated to a different switch.
- Verify that virtual machine network adapters are migrated to a different switch.

For details about migrating network adapters to different switches, see Tasks for Managing Host Networking on a vSphere Distributed Switch.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Add and Manage Hosts**.
4. On the **Select task** page, select **Remove hosts** and click **Next**.
5. On the **Select hosts** page, select the host or hosts that you want to remove and click **Next**.
6. On the **Ready to complete** page, review selections and click **Finish**.

Managing Networking on Host Proxy Switches

Learn how to change the configuration of the proxy switch on every host that is associated with a vSphere Distributed Switch. You can manage physical NICs, VMkernel adapters, and virtual machine network adapters.

For details about setting up VMkernel networking on host proxy switches, see Create a VMkernel Adapter on a vSphere Distributed Switch.

Migrate Network Adapters on a Host to a vSphere Distributed Switch

Learn how to migrate network adapters from a standard switch to the distributed switch for hosts that are associated with a distributed switch. You can migrate physical NICs, VMkernel adapters, and virtual machine network adapters at the same time.

To migrate virtual machine network adapters or VMkernel adapters, make sure that the destination distributed port groups have at least one active uplink, and the uplink is connected to a physical NIC on this host. Alternatively, migrate physical NICs, virtual network adapters, and VMkernel adapters at once.

To migrate physical NICs, make sure that the source port groups on the standard switch have at least one physical NIC to handle their traffic. For example, if you migrate a physical NIC that is assigned to a port group for virtual machine networking, make sure that the port group is connected to at least one physical NIC. Otherwise the virtual machines on the same VLAN on the standard switch has connectivity between each other but not to the external network.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the destination distributed switch and click the horizontal ellipsis icon next to **Manage physical adapters**.
4. Select **Migrate Networking**.
5. Configure physical NICs.
   a) From the **On other switches/unclaimed** list, select a physical NIC and click **Assign uplink**.
   b) Select an uplink and click **OK**.
   c) Click **Next**.
6. Configure VMkernel adapters.
   a) Select an adapter and click **Assign port group**.
   b) Select a distributed port group and click **OK**.
   You should connect one VMkernel adapter to one distributed port group at a time.

c) Click **Next**.

7. Configure virtual machine network adapters.
   a) Select the check box **Migrate virtual machine networking**.
   b) Select a virtual machine or a virtual machine network adapter and click **Assign port group**.
      If you select a virtual machine, you migrate all network adapters on the virtual machine. If you select a network adapter, you migrate only this network adapter.
   c) Select a distributed port group from the list and click **OK**.
   d) Click **Next**.
8. On the Ready to complete page, review the new networking configuration and click **Finish**.

Migrate a VMkernel Adapter on a Host to a vSphere Standard Switch

Learn how to migrate VMkernel adapters from the distributed to a standard switch if a host is associated with a distributed switch.

Verify that the destination standard switch has at least one physical NIC.

For details about creating VMkernel adapters on a vSphere Distributed Switch, see Create a VMkernel Adapter on a vSphere Distributed Switch.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the destination standard switch from the list.
4. Click **Migrate VMkernel Adapter**.
5. On the Select VMkernel adapter page, select the virtual network adapter to migrate to the standard switch from the list.
6. On the Configure settings page, edit the **Network label** and **VLAN ID** for the network adapter.
7. On the Ready to complete page, review the migration details and click **Finish**.
   Click **Back** to edit settings.

Assign a Physical NIC of a Host to a vSphere Distributed Switch

Learn how to assign physical NICs of a host that is associated with a distributed switch to the uplink port on the host proxy switch.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select a distributed switch from the list.
4. Click **Manage Physical Adapters**.
5. Select a free uplink from the list and click **Add adapters**.
6. Select a physical NIC and click **OK**.

Remove a Physical NIC from a vSphere Distributed Switch

Learn how to remove a physical NIC of a host from an uplink on a vSphere Distributed Switch.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the distributed switch.
4. Click **Manage Physical Adapters**.
5. Select an uplink a click **Remove selected**.
6. Click **OK**.

When you remove physical NICs from active virtual machines, you might see the NICs you removed reported in the . See Removing NICs from Active Virtual Machines.

Removing NICs from Active Virtual Machines

Learn how to remove NICs from active virtual machines, and see the NICs you have removed in the vSphere Client.

Removing NICs from an Active Virtual Machine without a Guest Operating System Installed

You cannot remove NICs from an active virtual machine on which no operating system is installed.

The vSphere Client might report that the NIC has been removed, but you continue to see it attached to the virtual machine.

Removing NICs from an Active Virtual Machine with a Guest Operating System Installed

You can remove a NIC from an active virtual machine, but it might not be reported to the vSphere Client for some time. If you click **Edit Settings** for the virtual machine, you might see the removed NIC listed even after the task is complete. The Edit Settings dialog box for the virtual machine does not immediately display the removed NIC.

You might also see the NIC attached to the virtual machine if the guest operating system of the virtual machine does not support hot removal of NICs.

Distributed Port Groups

Learn how a distributed port group specifies port configuration options for each member port on a vSphere Distributed Switch. Distributed port groups define how a connection is made to a network.

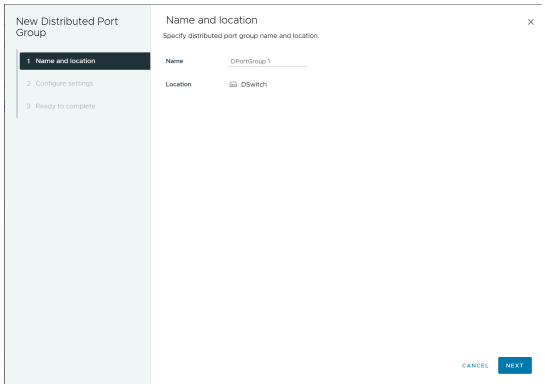
Add a Distributed Port Group

Learn how to create a distributed switch network for your virtual machines, and to associate VMkernel adapters, and add a distributed port group to a vSphere Distributed Switch.

Related to adding a port group, is applying VLAN tagging globally on all distributed ports. Using the VLAN options that you can select VLAN tags. To learn more, see Configure VLAN Tagging on a Distributed Port Group or Distributed Port

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Distributed port group** > **New distributed port group**.
4. On the **Name and location** page, enter the name of the new distributed port group, or accept the generated name, and click **Next**.

   Figure 49: New Distributed Port Group - Name and Location

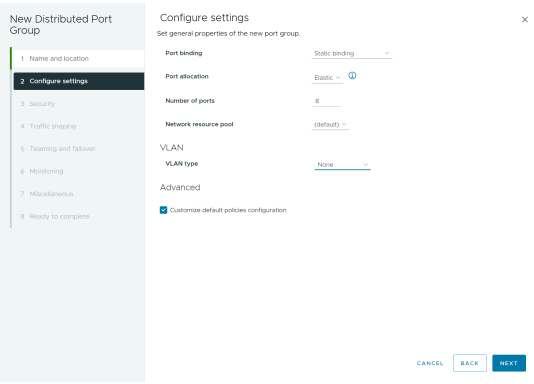


5. On the **Configure settings** page, set the general properties for the new distributed port group.

Setting	Description
Port binding	Select the ports that are assigned to virtual machines connected to this distributed port group. • **Static binding**: Assign a port to a virtual machine when the virtual machine connects to the distributed port group. • **Ephemeral - no binding**: No port binding. You can assign a virtual machine to a distributed port group with ephemeral port binding also when connected to the host.
Port allocation	• **Elastic**: The default number of ports is eight. When all the ports are assigned, a new set of eight ports is created. • **Fixed**: The default number of ports is set to eight. When all the ports are assigned, no additional ports are created.
Number of ports	Enter the number of ports on the distributed port group.
Network resource pool	To assign the new distributed port group to a user-defined network resource pool, use the drop-down menu If you have not created a network resource pool, this menu is empty. **Note:** You cannot assign Network Resource Pool if Network Offloads is enabled.
VLAN	Use the **VLAN type** drop-down menu to specify the type of VLAN traffic filtering and marking: • **None**: Do not use VLAN. Select **None** if you are using External Switch Tagging. • **VLAN**: In the **VLAN ID** text box, enter a number between 1 and 4094 for Virtual Switch Tagging. • **VLAN trunking**: Enter a VLAN trunk range.

Setting	Description
	Pass VLAN traffic with an ID to the guest OS. You can set multiple ranges and individual VLANs by using a comma-separated list. For example: `1702-1705, 1848-1849` Use this option for Virtual Guest Tagging. • **Private VLAN**: Associate the traffic with a private VLAN created on the distributed switch. If you did not create any private VLANs, this menu is empty.
Advanced	To customize the policy configurations for the new distributed port group, select this check box.

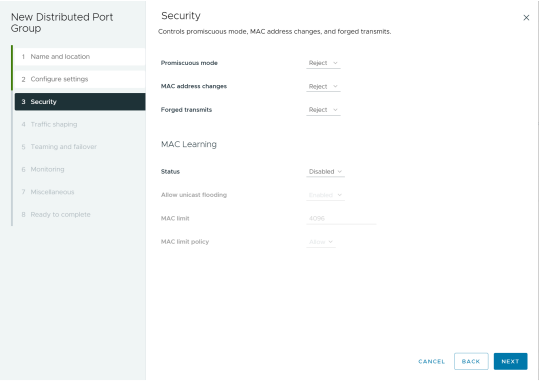
Figure 50: New Distributed Port Group - Configure settings



6. Click **Next**.
7. Optional: On the Security page, edit the security exceptions and click **Next**.

Setting	Description
Promiscuous mode	• **Reject**: Placing an adapter in promiscuous mode from the guest operating system does not result in receiving frames for other virtual machines. • **Accept**: If an adapter is placed in promiscuous mode from the guest operating system, the switch allows the guest adapter to receive all frames passed on the switch in compliance with the active VLAN policy for the port where the adapter is connected. Firewalls, port scanners, intrusion detection systems, and so on, must run in promiscuous mode.
MAC address changes	The MAC address change feature allows a VM to change its MAC address. A VM connected to a port can run an administrative command to change the MAC address of its vNIC and still send and receive traffic on that vNIC. • **Reject**: If the option is set to **Reject** and the guest OS changes the MAC address of the adapter to a value different from the address in the `.vmx` configuration file, then the switch drops all inbound frames to the virtual machine adapter. If the guest OS changes the MAC address back, the virtual machine receives frames again. • **Accept**: If the guest OS changes the MAC address of a network adapter, the adapter receives frames to its new address.
Forged Transmits	• **Reject**: The switch drops any outbound frame with a source MAC address that is different from the one in the `.vmx` configuration file. • **Accept**: The switch does not perform filtering and permits all outbound frames.

Figure 51: New Distributed Port Group - Security



8. Optional: On the Security page, edit the MAC Learning policy and click **Next**.

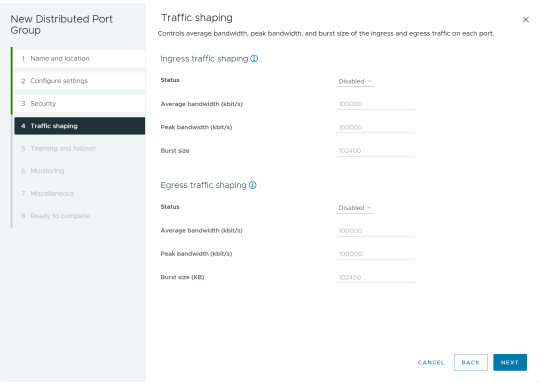
Setting	Description
Status	Enable or disable the MAC learning feature. The default is disabled.
Allow unicast flooding	When a packet that is received by a port has an unknown destination MAC address, the packet is dropped. With unknown unicast flooding enabled, the port floods unknown unicast traffic to every port on the switch that has MAC learning and unknown unicast flooding enabled. This property is enabled by default, if MAC learning is enabled.
MAC Limit	The number of MAC addresses that can be learned is configurable. The maximum value is 4096 per port, which is the default.
MAC Limit Policy	The policy for when the MAC limit is reached. The options are: • Drop - Packets from an unknown source MAC address are dropped. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled. • Allow - Packets from an unknown source MAC address are forwarded although the address will not be learned. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.

9. Optional: On the Traffic shaping page, enable or disable Ingress or Egress traffic shaping and click **Next**.

Setting	Description
Status	If you enable either **Ingress traffic shaping** or **Egress traffic shaping**, you are setting limits on the amount of networking bandwidth that is allocated for each virtual adapter that is associated with this particular port group. If you disable the policy, services have a free, clear connection to the physical network by default. **Note:** You cannot assign traffic shaping policies if Network Offloads Compatibility enabled.
Average bandwidth	This feature establishes the number of bits per second to allow across a port, averaged over time. It is the allowed average load.
Peak bandwidth	The maximum number of bits per second to allow across a port when it is sending and receiving a burst of traffic. It tops the

Setting	Description
	bandwidth that is used by a port whenever it is using its burst bonus.
Burst size	The maximum number of bytes to allow in a burst. If this parameter is set, a port can gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than specified by **Average bandwidth**, it can temporarily transmit data at a faster speed if a burst bonus is available. This parameter tops the number of bytes that can be accumulated in the burst bonus and as a result is transferred at a faster speed.

Figure 52: New Distributed Port Group - Traffic shaping

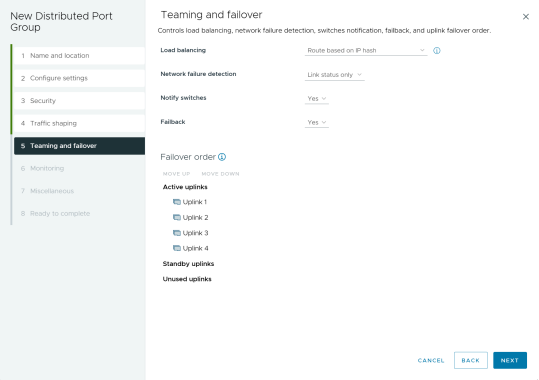


10. Optional: On the Teaming and failover page, edit the settings and click **Next**.

Setting	Description
Load balancing	Specify the way that an uplink is selected. • **Route based on originating virtual port**: Select an uplink based on the virtual port where the traffic entered the distributed switch. • **Route based on IP hash**: Select an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, whatever is at those offsets is used to compute the hash. • **Route based on source MAC hash**: Select an uplink based on a hash of the source Ethernet. • **Route based on physical NIC load**: Select an uplink based on the current loads of physical NICs. • **Use explicit failover order**: Always use the highest order uplink from the list of Active adapters which passes failover detection criteria. **Note:** IP-based teaming requires that the physical switch is configured with EtherChannel. For all other options, disable EtherChannel.
Network failure detection	Specify the method to use for failover detection. • **Link status only**: Relies solely on the link status that the network adapter provides. This option detects failures, such as cable pulls and physical switch power failures, but not configuration errors, such as a physical switch port being blocked by spanning tree or that is misconfigured to the wrong VLAN or cable pulls on the other side of a physical switch. • **Beacon probing**: Sends out and listens for beacon probes on all NICs in the team and uses this information, in addition

Setting	Description
	to link status, to determine link failure. This detects many of the failures previously mentioned that are not detected by link status alone. **Note:** Do not use beacon probing with IP-hash load-balancing.
Notify switches	Select **Yes** or **No** to notify switches in case of failover. If you select **Yes**, whenever a virtual NIC is connected to the distributed switch or whenever that virtual NIC's traffic can be routed over a different physical NIC in the team because of a failover event, a notification is sent out over the network to update the lookup tables on physical switches. In almost all cases, this process is desirable for the lowest latency of failover occurrences and migrations with vMotion. If the **Notify switches** is set to **Yes**, then all the connected ports, port groups, and distributed switches are connected back to the host when vCenter reconnects with ESX hosts. **Note:** Do not use this option when the virtual machines using the port group are using Microsoft Network Load Balancing in unicast mode. No such issue exists with NLB running in multicast mode.
Failback	Select **Yes** or **No** to disable or enable failback. This option determines how a physical adapter is returned to active duty after recovering from a failure. If failback is set to **Yes** (default), the adapter is returned to active duty immediately upon recovery, displacing the standby adapter that took over its slot, if any. If failback is set to **No**, a failed adapter is left inactive even after recovery until another currently active adapter fails, requiring its replacement.
Failover order	Specify how to distribute the workload for uplinks. To use some uplinks but reserve others for emergencies if the uplinks in use fail, set this condition by moving them into different groups: • **Active uplinks**: Continue to use the uplink when the network adapter connectivity is up and active. • **Standby uplinks** : Use this uplink if one of the active adapters' connectivity is down. • **Unused uplinks** : Do not use this uplink. **Note:** When using IP-hash load-balancing, do not configure standby uplinks.

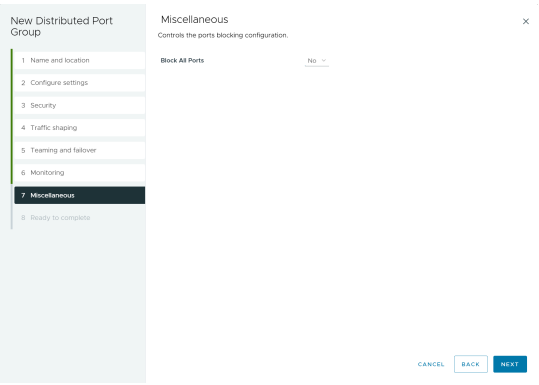
Figure 53: New Distributed Port Group - Teaming and failover



11. Optional: On the Miscellaneous page, select **Yes** or **No** and click **Next**.

    Selecting **Yes** shuts down all ports in the port group. This action can disrupt the normal network operations of the hosts or virtual machines using the ports.

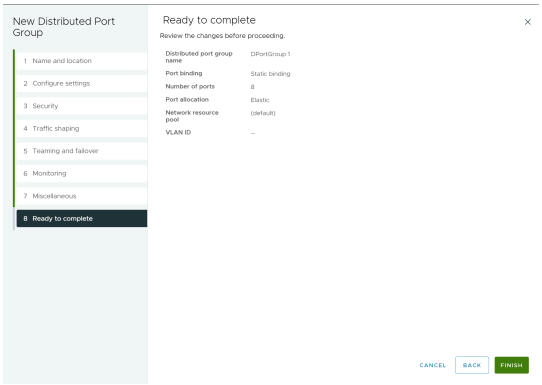
    Figure 54: New Distributed Port Group - Miscellaneous



12. On the Ready to complete page, review your settings and click **Finish**.

    To change any settings, click the **Back** button.

    Figure 55: New Distributed Port Group - Ready to complete



Edit General Distributed Port Group Settings

Learn to edit general distributed port group settings such as the distributed port group name, port settings, and network resource pool.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed port group.
3. Right-click the distributed port group and select **Edit Settings**.
4. Select **General** to edit the following distributed port group settings.

Option	Description
Name	The name of distributed port group. You can edit the name in the text field.
Port binding	Choose when ports are assigned to virtual machines connected to this distributed port group. • **Static binding**: Assign a port to a virtual machine when the virtual machine connects to the distributed port group. • **Ephemeral**: No port binding. You can also assign a virtual machine to a distributed port group with ephemeral port binding when connected to the host.
Port allocation	• **Elastic:** The default number of ports is set to eight. When all ports are assigned, a new set of eight ports is created. This is the default. • **Fixed**: The default number of ports is set to eight. No additional ports are created when all ports are assigned.
Number of ports	Enter the number of ports on the distributed port group.
Network resource pool	Use the drop-down menu to assign the new distributed port group to a user-defined network resource pool. If you have not created a network resource pool, this menu is empty.
Description	Enter any information about the distributed port group in the description field.

5. Click **OK**.

Remove a Distributed Port Group

Learn how to remove a distributed port group when you no longer need the corresponding labeled network to provide connectivity and configure connection settings for virtual machines or VMkernel networking.

- Verify that all virtual machines that are connected to the corresponding labeled network are migrated to a different labeled network.
- Verify that all VMkernel adapters connected to the distributed port group are migrated to a different port group, or are deleted.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed port group.
3. Right-click the distributed port group and select **Delete**.

Working with Distributed Ports

A distributed port is a port on a vSphere Distributed Switch that connects to the VMkernel or to a virtual machine's network adapter.

Default distributed port configuration is determined by the distributed port group settings, but some settings for individual distributed ports can be overridden.

Monitor the State of Distributed Ports in the vSphere Client

Learn how vSphere monitors distributed ports and provides information about the current state and runtime statistics of each port. Starting vSphere 8.0 Update 3, all the ports are displayed, you can filter all the columns in the ports list.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to a distributed port group.
3. Click the **Ports** tab and select a port from the list.
   All the ports are displayed on the grid.
   The ports table for the distributed switch displays runtime statistics for each distributed port.
4. Click the funnel icon next to **Port ID** to filter ports by Port ID.
5. Click the funnel icon next to **Name** to filter ports by the name of the port.
6. Click the funnel icon next to **Connectee** column to filter based on the conectee for each distributed port.

Option	Description
Physical Adapter	Search is based on the connected physical adapter.
VMkernel Adapter	Search is based on the connected host VmkVnic.
Virtual Machine	Search is based on the connected VM.
Host	Search is based on the connected host.
--	Search based on the connectee of this distributed port is unavailable.

7. Click the funnel icon next to **Runtime MAC Address** to filter ports by the MAC address of distributed port.
8. Click the funnel icon next to **Port Group** to filter ports by the name of the port group.
9. Click the funnel icon next to **State** column to filter based on the uplink connection for each distributed port.
   The **State** column displays the current state for each distributed port.

Option	Description
Link Up	The link for this distributed port is up.
Link Down	The link for this distributed port is down.
Blocked	This distributed port is blocked.
--	The state of this distributed port is unavailable.

10. Click the funnel icon next to **VLAN ID** column to filter based on the conectee for each distributed port.

Option	Description
VLAN	Search results are based on the VLAN ID.
Private VLAN	Search results are based on the Private VLAN that is identified by its primary VLAN ID
VLAN Trunk	Search results are based on the VLAN traffic with ID within the VLAN trunk range
--	Search is based on the VLAN ID of this distributed port that is unavailable.

11. Click the funnel icon next to **VIF ID** to filter ports by the virtual interface identifier.

Configure Distributed Port Settings in vSphere Client

Learn how to configure general distributed port settings such as the port name and description.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to a distributed port group.
3. Click the **Ports** tab, and select a distributed port from the table.
   Information about the distributed port appears at the bottom of the screen.
4. Click the vertical ellipsis icont and click **Edit**.
5. On the Properties page and policy pages, edit information about the distributed port and click **OK**.

   If overrides are not allowed, the policy options are disabled.

   You can allow overrides at the port level by changing the **Advanced** settings of the distributed port group. See Configure Overriding Networking Policies on Port Level.

Configure Virtual Machine Networking on a vSphere Distributed Switch

Connect virtual machines to a vSphere Distributed Switch either by configuring an individual virtual machine NIC or migrating groups of virtual machines from the vSphere Distributed Switch itself.

Connect virtual machines to vSphere Distributed Switches by connecting their associated virtual network adapters to distributed port groups. You can do this either for an individual virtual machine by modifying the virtual machine's network adapter configuration, or for a group of virtual machines by migrating virtual machines from an existing virtual network to a vSphere Distributed Switch.

Migrate Virtual Machines to or from a vSphere Distributed Switch

In addition to connecting virtual machines to a distributed switch at the individual virtual machine level, learn to migrate a group of virtual machines between a vSphere Distributed Switch network and a vSphere Standard Switch network.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and clusters** and navigate to a data center.
3. Right-click the data center and select **Migrate VMs to Another Network**.
4. Select a source network.
   - Select **Specific network** and use the **Browse** button to select a specific source network.
   - Select **No network** to migrate all virtual machine network adapters that are not connected to any other network.
5. Use **Browse** to select a destination network and click **Next**.
6. Select virtual machines from the list to migrate from the source network to the destination network and click **Next**.
7. Review your selections and click **Finish**.
   Click **Back** to edit any selections.

Connect an Individual Virtual Machine to a Distributed Port Group

Learn how to connect an individual virtual machine to a vSphere Distributed Switch by modifying the NIC configuration of the virtual machine.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and clusters** and navigate to a data center, folder, cluster, resource pool, or host, and click the **VMs** tab.
3. Click **Virtual Machines** and click the virtual machine from the list.
4. From the Actions menu, select Edit Settings.
5. Expand the **Network adapter** section and select **Browse** from the **Network adapter** drop-down menu.
6. In the Select Network dialog box, select a distributed port group and click **OK**.
7. Click **OK**.

Topology of a vSphere Distributed Switch

The topology diagrams of a vSphere Distributed Switch in the vSphere Client show the structure of virtual machine adapters, VMkernel adapters, and physical adapters in the switch.

You can examine the components, arranged in port groups, whose traffic is handled by the switch, and the connections between them. The diagram displays information about the physical adapter that connects the virtual adapters to the external network.

You can view the components that are running on the entire distributed switch and on each host participating in it.

Central Topology Diagram

You can use the central topology diagram of the switch to locate and edit the settings for distributed port groups and uplink groups that are associated with multiple hosts. You can initiate migration of virtual machine adapters from a port group to a destination on the same or different switch. You can also reorganize the hosts and their networking on the switch by using the **Add and Manage Hosts** wizard.

Topology Diagram of a Host Proxy Switch

The topology diagram of a host proxy switch shows the adapters that are attached to the switch ports on the host. You can edit the settings of the VMkernel and physical adapters.

Topology Diagram of a Network Offloads Switch

The topology diagram of a network offloads switch shows the adapters that are attached to the switch ports on the host. You can edit the settings of the VMkernel and physical adapters.

Diagram Filters

You can use diagram filters to limit the information displayed in topology diagrams. The default filter limits the topology diagram to display 32 port groups, 32 hosts, and 1024 virtual machines.

You can change the scope of the diagram by using no filters or by applying custom filters. By using a custom filter, you can view information only about a set of virtual machines, a set of port groups on certain hosts, or a port. You can create filters from the central topology diagram of the distributed switch.

View the Topology of a vSphere Distributed Switch

Examine the organization of components that are connected to the distributed switch across the hosts in a vCenter.

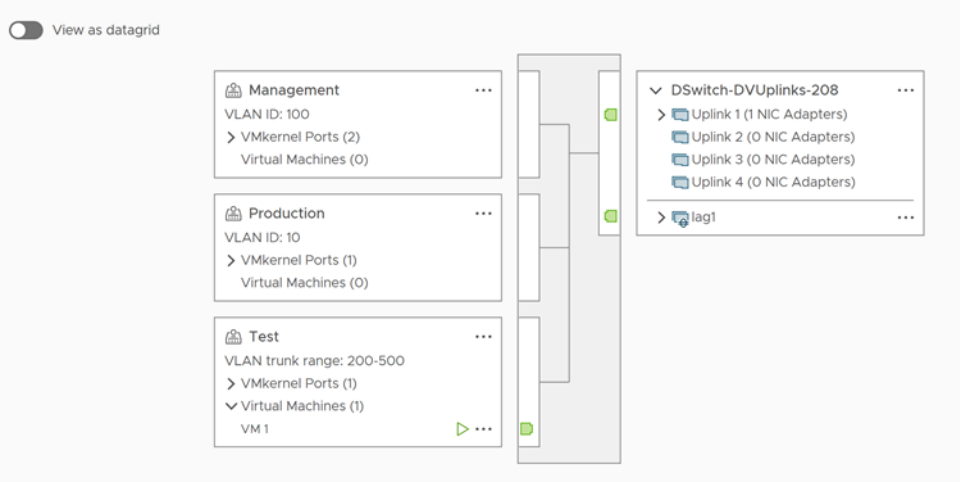
1. Navigate to the vSphere distributed switch in the vSphere Client.
2. On the **Configure** tab, expand **Settings** and select **Topology**.

By default the diagram shows up to 32 distributed port groups, 32 hosts, and 1024 virtual machines.

Diagram of a Distributed Switch That Connects the VMkernel and Virtual Machines to the Network

In your virtual environment, a vSphere Distributed Switch handles VMkernel adapters for vSphere vMotion and for the management network, and virtual machines grouped. You can use the central topology diagram to examine whether a virtual machine or VMkernel adapter is connected to the external network and to identify the physical adapter that carries the data.

Figure 56: Topology Diagram of a Distributed Switch That Handles VMkernel and Virtual Machine Networking



You can perform the following common tasks in the topology of the distributed switch:

- Use filters to view the networking components only for selected port groups on certain hosts, for selected virtual machines, or for a port.
- Locate, configure, and migrate virtual machine networking components across host and port groups by using the **Migrate Virtual Machine Networking** wizard.
- Detect the virtual machine adapters that have no network that is assigned and move them to the selected port group by using the **Migrate Virtual Machine Networking** wizard.
- Handle networking components on multiple hosts by using the **Add and Manage Hosts** wizard.
- View the physical NIC or NIC team that carries the traffic that is related to a selected virtual machine adapter or VMkernel adapter.

In this way you can also view the host on which a selected VMkernel adapter resides. Select the adapter, trace the route to the associated physical NIC, and view the IP address or domain name next to the NIC.
- Determine the VLAN mode and ID for a port group. For information about VLAN modes, see VLAN Configuration.

View the Topology of a Host Proxy Switch

Examine and reorganize the networking of the VMkernel and virtual machines that the vSphere Distributed Switch handles on a host.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select the distributed switch from the list.

The topology of the host proxy switch appears under the list.

View the Topology of Network Offloads Switch

Examine the organization of the distributed switch with network offloads.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. The

   icon indicates that network offloads are supported for this distributed switch.

Setting Up VMkernel Networking

Learn how to set up VMkernel adapters to provide network connectivity to hosts and to accommodate system traffic of vMotion, IP storage, Fault Tolerance logging, vSAN, and so on.

VMkernel Networking Layer

The VMkernel networking layer provides connectivity to hosts and handles the standard system traffic of vSphere vMotion, IP storage, Fault Tolerance, vSAN, and others. You can also create VMkernel adapters on the source and target vSphere Replication hosts to isolate the replication data traffic.

Multihoming in the VMkernel networking is defined as multiple VMkernel adapters in an TCP/IP stack.

Note:
If there is a single vmnic with multiple TCP/IP stacks, it cannot be defined as multihoming.

Multihoming is supported if the vmknics are in different IP subnets. The admin must ensure that the route entries reflect the intent. The ESX does not run any routing protocol and it does not support or add dynamic routes. For any specific route entry, it must be added statically from the vCenter or the ESX host. Configurations with more than one vmknic interface on the same IP subnet in a single network stack instance is not supported. Deploying such a configuration might lead to unexpected results like connectivity issues, low throughput and asymmetric routing. However, there are a few exceptions to this rule. For example, iSCSI, Multi-NIC vMotion, and NSX VTEPS. For more information, see Broadcom knowledge base article 318546.

TCP/IP Stacks at the VMkernel Level

Default TCP/IP stack	Provides networking support for the management traffic between vCenter and ESX hosts, and for system traffic such as vMotion, IP storage, Fault Tolerance, and so on.
vMotion TCP/IP stack	Supports the traffic for live migration of virtual machines. Use the vMotion TCP/IP to provide better isolation for the vMotion traffic. After you create a VMkernel adapter on the vMotion TCP/IP stack, you can use only this stack for vMotion on this host. The VMkernel adapters on the default TCP/IP stack are disabled for the vMotion service. If a live migration uses the default TCP/IP stack while you configure VMkernel adapters with the vMotion TCP/IP stack, the migration completes successfully. However, the involved VMkernel adapters on the default TCP/IP stack are disabled for future vMotion sessions.

Provisioning TCP/IP stack	Supports the traffic for virtual machine cold migration, cloning, and snapshot migration. You can use the provisioning TCP/IP to handle Network File Copy (NFC) traffic during long-distance vMotion. NFC provides a file-specific FTP service for vSphere. ESX uses NFC for copying and moving data between datastores. VMkernel adapters that are configured with the provisioning TCP/IP stack handle the traffic from cloning the virtual disks of the migrated virtual machines in long-distance vMotion. By using the provisioning TCP/IP stack, you can isolate the traffic from the cloning operations on a separate gateway. After you configure a VMkernel adapter with the provisioning TCP/IP stack, all adapters on the default TCP/IP stack are disabled for the Provisioning traffic.
Custom TCP/IP stacks	You can add custom TCP/IP stacks at the VMkernel level to handle networking traffic of custom applications.
Mirror TCP/IP stack	You can create a vmknic on mirror TCP/IP stack when you select the mirror stack for ERSPAN.

Securing System Traffic

Take appropriate security measures to prevent unauthorized access to the management and system traffic in your vSphere environment. For example, isolate the vMotion traffic in a separate network that includes only the ESX hosts that participate in the migration. Isolate the management traffic in a network that only network and security administrators can access. For more information, see the *vSphere Security* documentation and the *ESX Installation and Setup* documentation.

System Traffic Types

Dedicate a separate VMkernel adapter for every traffic type. For distributed switches, dedicate a separate distributed port group for each VMkernel adapter.

Management traffic	Carries the configuration and management communication for ESX hosts, vCenter, and host-to-host High Availability traffic. By default, when you install the ESX software, a vSphere Standard Switch is created on the host together with a VMkernel adapter for management traffic. To provide redundancy, you can connect two or more physical NICs to a VMkernel adapter for management traffic.
vMotion traffic	Accommodates vMotion. A VMkernel adapter for vMotion is required both on the source and the target hosts. Configure The VMkernel adapters for vMotion to handle only the vMotion traffic. For better performance, you can configure multiple NIC vMotion. To have multi-NIC vMotion, you can dedicate two or more port groups to the vMotion traffic, respectively every port group must have a vMotion VMkernel adapter that is associated with it. Then you can connect one or more physical NICs to every port group. In this way, multiple physical NICs are used for vMotion, which results in greater bandwidth. **Note:** vMotion network traffic is not encrypted. You should provision secure private networks for use by vMotion only.
Provisioning traffic	Handles the data that is transferred for virtual machine cold migration, cloning, and snapshot migration.
IP storage traffic and discovery	Handles the connection for storage types that use standard TCP/IP networks and depend on the VMkernel networking. Such storage types are software iSCSI, dependent hardware iSCSI, and NFS. If you have two or more physical NICs for iSCSI, you can configure iSCSI multipathing. ESX hosts support NFS 3 and 4.1.
Fault Tolerance traffic	Handles the data that the primary fault tolerant virtual machine sends to the secondary fault tolerant virtual machine over the VMkernel networking layer. A separate VMkernel adapter for Fault Tolerance logging is required on every host that is part of a vSphere HA cluster.
vSphere Replication traffic	Handles the outgoing replication data that the source ESX host transfers to the vSphere Replication server. Dedicate a VMkernel adapter on the source site to isolate the outgoing replication traffic.
vSphere Replication NFC traffic	Handles the incoming replication data on the target replication site.
vSAN traffic	Every host that participates in a vSAN cluster must have a VMkernel adapter to handle the vSAN traffic.
vSphere Backup NFC	VMkernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMkernel Adapter when vSphere Backup NFC service is enabled.
NVMe over TCP	VMkernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMkernel Adapter when NVMe over TCP adapter is enabled. For more information, see the *vSphere Storage* documentation.

NVMEe over RDMA	VMkernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMkernel Adapter when NVMe over RDMA adapter is enabled. For more information, see the *vSphere Storage* documentation.

View Information About VMkernel Adapters on a Host

You can view each VMkernel adapter's assigned services, associated switch, port settings, IP settings, TCP/IP stack, VLAN ID, and policies.

1. In the vSphere Client, navigate to the host.
2. Click the **Configure** tab and expand the **Networking** menu.
3. To view information about all VMkernel adapters on the host, select **VMkernel adapters**.
4. Select an adapter from the VMkernel adapters list to view its settings.

Tab	Description
All	Displays all configuration information about the VMkernel adapter. This information includes port and NIC settings, IPv4 and IPv6 settings, traffic shaping, teaming and failover, and security policies.
Properties	Displays the port properties and NIC settings of the VMkernel adapter. The port properties include the port group (network label) to which the adapter is associated, the VLAN ID, and the enabled services. The NIC settings include the MAC address and the configured MTU size.
IP Settings	Displays all IPv4 and IPv6 settings for the VMkernel adapter. IPv6 information is not displayed if IPv6 has not been enabled on the host.
Policies	Displays the configured traffic shaping, teaming and failover, and security policies that apply for the port group to which the VMkernel adapter is connected.

Create a VMkernel Adapter on a vSphere Standard Switch

Learn how to create a VMkernel network adapter on a vSphere Standard Switch to provide network connectivity for hosts and to handle the system traffic for vSphere vMotion, IP storage, Fault Tolerance logging, vSAN, and so on. You can also create VMkernel adapters on the source and target vSphere Replication hosts to isolate the replication data traffic. Dedicate a VMkernel adapter to only one traffic type.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Click **Add networking**.
4. On the Select connection type page, select **VMkernel Network Adapter** and click **Next**.
5. On the Select target device page, select either an existing standard switch or select **New standard switch**.
6. Optional: On the Create a Standard Switch page, assign physical NICs to the switch.

   You can create the standard switch without physical NICs and can configure them later. While physical NICs are not attached to the host, the host does not have network connectivity to the other hosts on the physical network. The virtual machines on the host are able to communicate with each other.

   a) Click **Add adapters** and select as many physical NICs as you need.
   b) Use the up and down arrows to configure the active and standby NICs.
7. On the Port properties page, configure the settings for the VMkernel adapter.

Option	Description
Network label	The network label is inherited from the label of the distributed port group.
IP settings	Select IPv4, IPv6, or both.

Option	Description
	Note: The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	Choose whether to get MTU for the network adapter from the switch or to set a custom size. You cannot set the MTU size to a value greater than 9000 bytes.
TCP/IP stack	Select a TCP/IP stack from the list. Once you set a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you will be able to use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration.
Available services	You can enable services for the default TCP/IP stack on the host. Select from the available services: • **vMotion**. Enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. The migration with vMotion to the selected host is not possible if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or there are no adapters using the vMotion TCP/IP stack. • **Provisioning**. Handles the data transferred for virtual machine cold migration, cloning, and snapshot migration. • **Fault Tolerance logging**. Enables Fault Tolerance logging on the host. You can use only one VMkernel adapter for FT traffic per host. • **Management**. Enables the management traffic for the host and vCenter. Typically, hosts have such a VMkernel adapter created when the ESX software is installed. You can create another VMkernel adapter for management traffic on the host to provide redundancy. • **vSphere Replication**. Handles the outgoing replication data that is sent from the source ESX host to the vSphere Replication server. • **vSphere Replication NFC**. Handles the incoming replication data on the target replication site. • **vSAN**. Enables the vSAN traffic on the host. Every host that is part of a vSAN cluster must have such a VMkernel adapter. • **vSphere Backup NFC**. VMkernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMkernel Adapter when vSphereBackup NFC service is enabled. • **NVMe over TCP**. VMkernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMkernel Adapter when NVMe over TCP adapter is enabled. For more information, see *vSphere Storage Guide*. • **NVMe over RDMA**. VMkernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMkernel Adapter when NVMe over RDMA adapter is enabled. For more information, see *vSphere Storage Guide*.

8. Optional: On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter** check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

9. Optional: On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click **Add IPv6 address** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**. 3. To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

10. Review your settings selections on the Ready to complete page and click **Finish**.

Create a VMkernel Adapter on a Host Associated with a vSphere Distributed Switch

Learn how to create a VMkernel adapter on a host that is associated with a distributed switch to provide network connectivity to the host and to handle the traffic for vSphere vMotion, IP storage, Fault Tolerance logging, vSAN, and others. You can set up VMkernel adapters for the standard system traffic on vSphere Standard Switches and on vSphere Distributed Switches.

You should dedicate a single distributed port group per VMkernel adapter. For better isolation, you should configure one VMkernel adapter with one traffic type.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Click **Add networking**.
4. On the Select connection type page, select **VMkernel Network Adapter** and click **Next**.
5. From the **Select an existing network** option, select a distributed port group and click **Next**.
6. On the **Port properties** page, configure the settings for the VMkernel adapter.

Option	Description
Network label	The network label is inherited from the label of the distributed port group.
IP settings	Select IPv4, IPv6, or both. **Note:** The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	Choose whether to get MTU for the network adapter from the switch or to set a custom size. You cannot set the MTU size to a value greater than 9000 bytes.
TCP/IP stack	Select an TCP/IP stack from the list. Once you set an TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you are able to use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration. If you select the vNetworking stack, no other service can run on the same VMkernel adapter and a dedicated VMkernel adapter for TEP Traffic is available.
Available services	You can enable services for the default TCP/IP stack on the host. Select from the available services: • **vMotion**. Enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. The migration with vMotion to the selected host is not possible if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or there are no adapters using the vMotion TCP/IP stack. • **Provisioning**. Handles the data that is transferred for virtual machine cold migration, cloning, and snapshot migration. • **Fault Tolerance logging**. Enables Fault Tolerance logging on the host. You can use only one VMkernel adapter for FT traffic per host. • **Management**. Enables the management traffic for the host and vCenter. Typically, hosts have such a VMkernel adapter that is created when the ESX software is installed. You can create another VMkernel adapter for management traffic on the host to provide redundancy. • **vSphere Replication**. Handles the outgoing replication data that is sent from the source ESX host to the vSphere Replication server. • **vSphere Replication NFC**. Handles the incoming replication data on the target replication site. • **vSAN**. Enables the vSAN traffic on the host. Every host that is part of a vSAN cluster must have such a VMkernel adapter. • **vSphere Backup NFC**. VMkernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMkernel Adapter when vSphereBackup NFC service is enabled. • **NVMe over TCP**. VMkernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMkernel Adapter when NVMe over TCP adapter is enabled. For more information, see vSphere Storage Guide. • **NVMe over RDMA**. VMkernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMkernel Adapter when NVMe over RDMA adapter is enabled. For more information, see vSphere Storage Guide. • **vNetworking**. To integrate NSX with other VCF components in vCenter, a VMkernel setting for dedicated vNetworking is added to support the overlay configuration in vCenter. For more information, see Building Cloud Infrastructure.

7. Optional: On the IPv4 settings page, select an option for obtaining IP addresses.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter** check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

8. Optional: On the IPv6 settings page, select an option for obtaining IPv6 addresses.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click **Add IPv6 address** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**. 3. To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

9. Review your settings selections on the Ready to complete page and click **Finish**.

Edit a VMkernel Adapter Configuration

Learn how to modify the supported traffic type for a VMkernel adapter, or the way IPv4 or IPv6 addresses are obtained.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Select the VMkernel adapter that resides on the target distributed or standard switch and click **Edit**.
4. On the Port properties page, edit the settings for the VMkernel adapter.

Option	Description
MTU	Choose whether to get MTU for the network adapter from the switch or to set a custom size. You cannot set the MTU size to a value greater than 9000 bytes.
TCP/IP stack	Select a TCP/IP stack from the list. Once you set a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you will be able to use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are disabled for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are disabled for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration.

Option	Description
Available services	You can enable services for the default TCP/IP stack on the host. Select from the available services: • **vMotion**. Enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. The migration with vMotion to the selected host is not possible if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or there are no adapters using the vMotion TCP/IP stack. • **Provisioning**. Handles the data transferred for virtual machine cold migration, cloning, and snapshot migration. • **Fault Tolerance logging**. Enables Fault Tolerance logging on the host. You can use only one VMkernel adapter for FT traffic per host. • **Management**. Enables the management traffic for the host and vCenter. Typically, hosts have such a VMkernel adapter created when the ESX software is installed. You can create another VMkernel adapter for management traffic on the host to provide redundancy. • **vSphere Replication**. Handles the outgoing replication data that is sent from the source ESX host to the vSphere Replication server. • **vSphere Replication NFC**. Handles the incoming replication data on the target replication site. • **vSAN**. Enables thevSAN traffic on the host. Every host that is part of a vSAN cluster must have such a VMkernel adapter. • **vSphere Backup NFC**. VMKernel port setting for dedicated backup NFC traffic. NFC traffic goes through the VMKernel Adapter when vSphereBackup NFC service is enabled. • **NVMe over TCP**. VMKernel port setting for dedicated NVMe over TCP storage traffic. NVMe over TCP storage traffic goes through the VMKernel Adapter when NVMe over TCP adapter is enabled. For more information, see *vSphere Storage Guide*. • **NVMe over RDMA**. VMKernel port setting for dedicated NVMe over RDMA storage traffic. NVMe over RDMA storage traffic goes through the VMKernel Adapter when NVMe over RDMA adapter is enabled. For more information, see *vSphere Storage Guide*.

5. Optional: On the IPv4 settings page, select the method by which IP addresses are obtained.

Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter** check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

6. Optional: On the IPv6 settings page, select an option for obtaining IPv6 addresses.

   Note:
   The IPv6 option does not appear on hosts that do not have IPv6 enabled.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses. In ESX 6.5 and later router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	1. Click **Add IPv6 address** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**. 3. To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

7. Click **OK**.

Overriding the Default Gateway of a VMkernel Adapter

Learn how to override the default gateway for a VMkernel adapter to provide a different gateway for vSphere vMotion.

Each TCP/IP stack on a host can have only one system default gateway. This default gateway is part of the routing table and all services that operate on the TCP/IP stack use it. However, you can override the default gateway for specific VMkernel adapters to provide alternative Layer 3 connectivity for services like vMotion.

For example, you can configure VMkernel adapters vmk0 and vmk1 on a host with default gateway 10.162.10.1.

- vmk0 is used for management traffic on the 10.162.10.0/24 subnet.
- vmk1 is used for vMotion traffic on the 172.16.1.0/24 subnet. You override the stack's default gateway and manually assign 172.16.1.1 as the gateway for this specific VMkernel adapter.

If you configure vmk1 to use 172.16.1.1 as its gateway, vMotion uses vmk1 as its egress interface with the gateway 172.16.1.1. The 172.16.1.1 gateway is a part of the vmk1 configuration and is not in the routing table. The override is a static property of the VMkernel adapter, and not a dynamic route for the host. Only the services that specify vmk1 as an egress interface use this gateway. This provides another Layer 3 connectivity options for services that need multiple gateways.

You can use the vSphere Client or an ESXCLI command to configure the default gateway of a VMkernel adapter.

See Create a VMkernel Adapter on a vSphere Standard Switch, Create a VMkernel Adapter on a Host Associated with a vSphere Distributed Switch, and Configure the VMkernel Adapter Gateway by Using esxcli Commands.

Configure the VMkernel Adapter Gateway by Using esxcli Commands

Learn how to override the default gateway of a VMkernel adapter to provide a different gateway for vSphere vMotion by using esxcli commands.

1. Open an SSH connection to the host.
2. Log in as the root user.
3. Run the following command.

Option	Description
IPv4	`esxcli network ip interface ipv4 set -i vmknic -t static -g IPv4 gateway -I IPv4 address -N mask`
IPv6	**Important:** You must turn off DHCPv6 or Router Advertisement before you can set the IPv6 vmknic gateway.

Option	Description
	esxcli network ip interface ipv6 set -i *vmknic* -d off -r off To add a static IPv6 address: esxcli network ip interface ipv6 address add -i *vmknic* -I *IPv6 address* To set the IPv6 vmknic gateway: esxcli network ip interface ipv6 set -i *vmknic* -g *IPv6 gateway*

Where *vmknic* is the name of the VMkernel adapter, *gateway* is the IP address of the gateway, *IP address* is the address of the VMkernel adapter, and *mask* is the network mask.

Configure the resolv.conf File by Using esxcli Commands

The `resolv.conf` file is used to configure centrally managed DNS servers. You can configure entries in the `/etc/resolv.conf` file using `esxcli` commands, which allows the modification to remain when rebooting the ESX host. The property can be set explicitly if DHCP is not enabled.

1. Open an SSH connection to the host.
2. Log in as the root user.
3. Run the following command.

Option	Description

Add DNS server	To add a new DNS server to the end of the list of DNS servers to use for this ESX host.
	`esxcli network ip dns server add`
	Command Options
	`-N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.
	`-s|--server=<str>`
	The IPV4 or the IPV6 address of the DNS server that you want to add to the DNS server list.
	Note:
	This command is required.
	`For example: esxcli network ip dns server add -N vmotion -s xx.xx.xx.xx`

Add DNS search	To add a search domain to the list of domains to be searched when trying to resolve a host name on the ESX host.
	`esxcli network ip dns search add`
	Command Options
	`-d|--domain=<str>`
	The string name of a domain that you want to add to the list of search domains.
	Note:
	This command is required.
	`-N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.

Remove DNS server	To remove a DNS server from the list of DNS servers to use for this ESX host. `esxcli network ip dns server remove`*IP address* Command Options `-a|--all`

Option	Description

	`-N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.
	`-s|--server=<str>`

Remove DNS search	To remove a search domain to the list of domains to be searched when trying to resolve an host name on the ESX host.
	`esxcli network ip dns search remove`
	Command Options
	`-d|--domain=<str>`
	The string name of a domain that you want to remove from the list of search domains.
	Note:
	This command is required.
	`-N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.

List DNS search	To print a list the DNS server currently configured on the system in the order in which they will be used.
	`esxcli network ip dns server list`
	Command Options
	`-N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.
	`For example: esxcli network ip dns server list -N vmotion`

List DNS search domain	To list the search domains currently configured on the ESX host in the order in which they will be used while searching.
	`esxcli network ip dns search list -N`
	Command Options
	`---N|--netstack=<str>`
	The network stack instance. If it is not specified, then use the default netstack instance.
	`For example: esxcli network ip dns search list -N vmotion`

Configure the DNS Hosts File by Using ESXCLI Commands

The DNS hosts file is used for mapping hostnames or domain names to IP addresses. You can configure entries in the `/etc/hosts` file using `esxcli` commands, which allows the modifications to remain unchanged while rebooting the ESX host.

1. Open an SSH connection to the host.
2. Log in as the root user.
3. Run the following command.

Option	Description

add	To add the hostname and IP address mapping, run the following command.
	`esxcli network ip hosts add --ip --hostname`
	`For example: esxcli network ip hosts add --hostname www.samplehostname.com --ip xx.xx.xx.xx`

remove	To remove a hostname mapping from the configuration, run the following command. `esxcli network ip hosts remove --hostname`

Option	Description
	For example: `esxcli network ip hosts remove --hostname www.samplehostname.com --ip xx.xx.xx.xx`
list	To list the IP addresses and their associated DNS entries, run the following command. `esxcli network ip hosts list`

Where *`IP address`* is the address of the VMkernel adapter, *`hostname`* is the DNS entry you want to associate with the IP address, *`alaises`* is any aliases you want to associate, and *`comment`* is a line about this entry.

Note:
ESXCLI network IP hosts add values only to the other hosts. The ESXCLI network IP host does not add values to the current host. The information about the current host is managed through the host name under the **Network Settings**. If you try to change the current hostname you might see an error similar to the following:

```
Error adding etc host item: User can not set item for management IP: X.X.X.X
```

For information on changing the host name of the current host, see Change the Configuration of an TCP/IP Stack on a Host.

View TCP/IP Stack Configuration on a Host

You can view the DNS and routing configuration of a TCP/IP stack on a host. You can also view the IPv4 and IPv6 routing tables, the congestion control algorithm, and the maximum number of allowed connections.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **TCP/IP configuration**.
3. Select a stack from the TCP/IP Stacks table.

   If no custom TCP/IP stacks are configured on the host, you view the default, vMotion, and Provisioning TCP/IP stacks on the host.

DNS and routing details about the selected TCP/IP stack appear below the TCP/IP Stacks table. You can view the IPv4 and IPv6 routing tables, and the DNS and routing configuration for the stack.

Note:
The IPv6 routing table is only visible if IPv6 is enabled on the host.

The **Advanced** tab contains information about the configured congestion control algorithm and the maximum number of allowed connections to the stack.

Change the Configuration of an TCP/IP Stack on a Host

Learn how to modify the DNS and default gateway configuration of an TCP/IP stack on a host. You can also change the congestion control algorithm, the maximum number of connections, and the name of custom TCP/IP stacks.

Note:
You can change the DNS and default gateway configuration of the default TCP/IP stack only. Multiple DNS and gateways configurations are supported when using separate TCP/IP stacks.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **TCP/IP configuration**.
3. Select a stack from the table, click **Edit** and make the appropriate changes.

Page	Option
DNS Configuration	Select a method of obtaining the DNS server. • Select **Obtain settings automatically from a VMkernel network adapter** and select a network adapter from the **VMKernel network adapter** drop-down menu • Select **Enter settings manually** and edit the DNS configuration settings. a. Edit the Host name. b. Edit the Domain name. c. Type a preferred DNS server IP address. d. Type an alternate DNS server IP address. e. Use the **Search domains** text box to specify DNS suffixes to use in DNS search when resolving unqualified domain names.
Routing	Edit the VMkernel gateway information. **Note:** Removing the default gateway might cause the client to lose connectivity with the host.
Name	Change the name of a custom TCP/IP stack
Advanced	Edit the maximum number of connections and the congestion control algorithm of the stack

4. Click **OK** to apply your changes.

You can add static routes to additional gateways by using CLI commands. For more information, see Broadcom knowledge base article 308786 .

Explicit Congestion Notification

Explicit Congestion Notification (ECN) allows a TCP sender to reduce the transmission rate to avoid packet drops. ECN is specified in RFC 3168. vSphere 7.0 and later supports ECN and it is enabled by default.

You can get the ECN status of any netstack using esxcli command.

1. In the ESX Shell on the host, use the following command.

   ```
   esxcli network ip netstack set -N <NetStack-Name> --ecn=<str>
   ```

2. You can set the status of ECN. The setting has the following possible values in ESX.

   ```
   ...
     --ecn=<str>           The status of ECN (Explicit Congestion Notification).
                               disabled: Completely disable ECN functionality
                               echo-only: Only echoing ECN, won't initiate
                               enabled: Fully enable ECN functionality
   ```

   The default value of this parameter is enabled. ESX is likely to use ECN if the value of this parameter is enabled. If a router or a network appliance in the environment cannot handle IP packets with ECN bits accurately, then the ECN capability can be set to disabled.

Create a Custom TCP/IP Stack

Learn how to create a custom TCP/IP stack on a host to forward networking traffic through a custom application.

Once you set a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select the vMotion or the Provisioning TCP/IP stack, you can use only these stacks to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are disabled for future vMotion sessions. If you set the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are disabled for operations that include Provisioning traffic, such as virtual machine cold migration, cloning, and snapshot migration.
If you must change the TCP/IP stack configuration, delete the existing VMkernel adapter and create a new one. You can then create a TCP/IP stack for that adapter.

1. Open an SSH connection to the host.
2. Log in as the root user.
3. Run the ESXCLI command.

```
esxcli network ip netstack add -N="stack_name"
```

The custom TCP/IP stack is created on the host. You can assign VMkernel adapters to the stack.

Remove a VMkernel Adapter

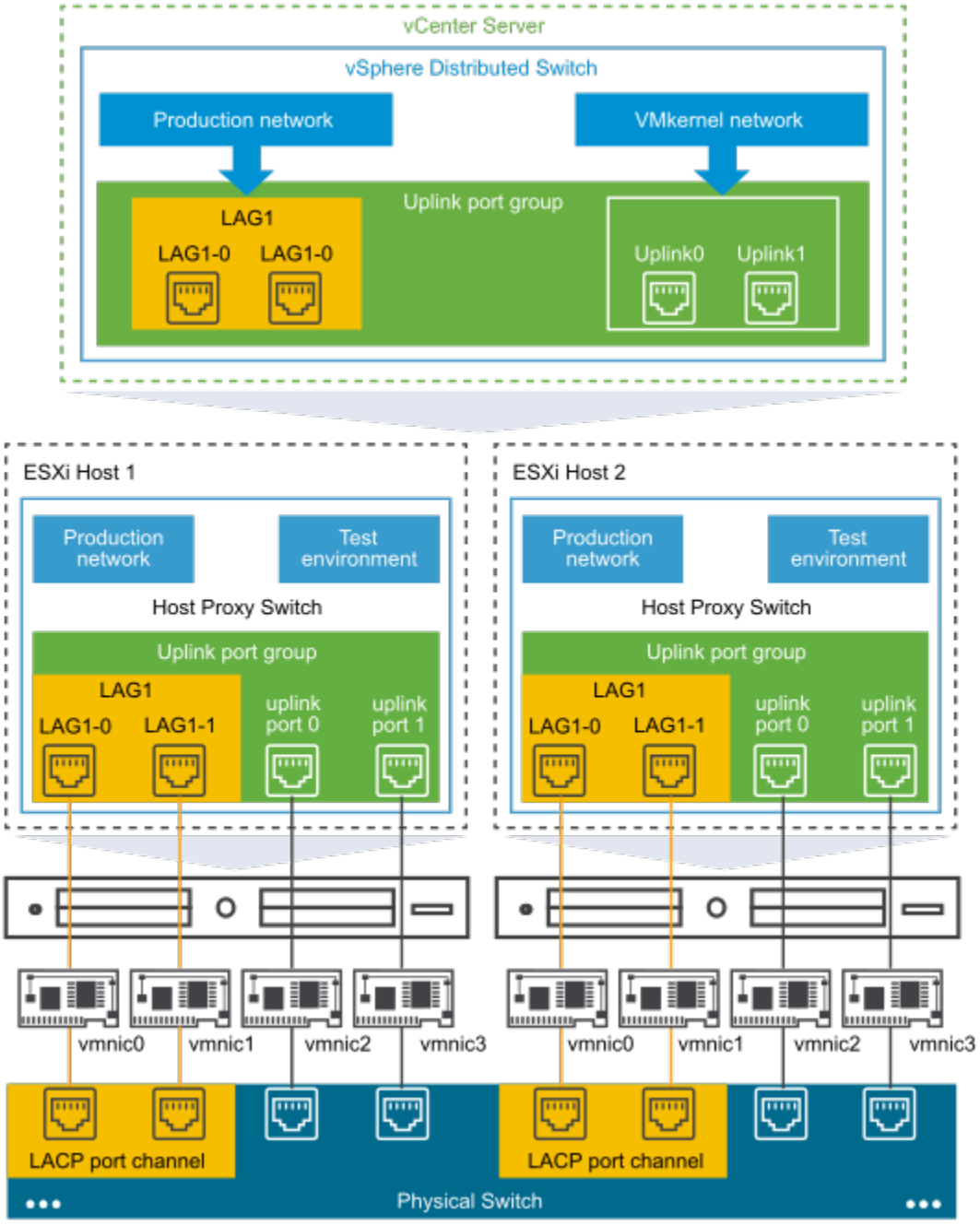
Learn how to remove a VMkernel adapter from a vSphere distributed or a standard switch when you no longer need the adapter. Make sure that you leave at least one VMkernel adapter for management traffic on the host to keep the network connectivity up.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Select a VMkernel adapter from the list, and click the **Remove selected network adapter** icon.
4. Click **Remove**.

LACP Support on a vSphere Distributed Switch

With Link Aggregation Control Protocol (LACP) support on a vSphere Distributed Switch, you can connect ESX hosts to physical switches by using dynamic link aggregation. You can create multiple link aggregation groups (LAGs) on a distributed switch to aggregate the bandwidth of physical NICs on ESX hosts that are connected to LACP port channels.

Figure 57: Enhanced LACP Support on a vSphere Distributed Switch



LACP Configuration on the Distributed Switch

You configure a LAG with two or more ports and connect physical NICs to the ports. LAG ports are teamed within the LAG, and the network traffic is load balanced between the ports through an LACP hashing algorithm. You can use a LAG to handle the traffic of distributed port groups to provide increased network bandwidth, redundancy, and load balancing to the port groups.

When you create a LAG on a distributed switch, a LAG object is also created on the proxy switch of every host that is connected to the distributed switch. For example, if you create LAG1 with two ports, LAG1 with the same number of ports is created on every host that is connected to the distributed switch.

On a host proxy switch, you can connect one physical NIC to only one LAG port. On the distributed switch, one LAG port can have multiple physical NICs from different hosts that are connected to it. The physical NICs on a host that you connect to the LAG ports must be connected to links that participate in an LACP port channel on the physical switch.

You can create up to 64 LAGs on a distributed switch. A host can support up to 32 LAGs. However, the number of LAGs that you can actually use depends on the capabilities of the underlying physical environment and the topology of the virtual network. For example, if the physical switch supports up to four ports in an LACP port channel, you can connect up to four physical NICs per host to a LAG.

You can configure the LACP timeout value. LACP functions by sending frames down all the links that have the protocol enabled. If it finds a device on the other end of the link that also has LACP enabled, it sends frames independently along the same links, enabling the two units to detect multiple links between themselves and then combine them into a single logical link. The timeout value is the amount of time that a LAG interface waits for a Protocol Data Unit (PDU) from the remote system before bringing the LACP session down. Periodic transmissions of LACP PDUs occur at either a slow or fast transmission rate.

Port Channel Configuration on the Physical Switch

For each host on which you want to use LACP, you must create a separate LACP port channel on the physical switch. You must consider the following requirements when configuring LACP on the physical switch:

- The number of ports in the LACP port channel must be equal to the number of physical NICs that you want to group on the host. For example, if you want to aggregate the bandwidth of two physical NICs on a host, you must create an LACP port channel with two ports on the physical switch. The LAG on the distributed switch must be configured with at least two ports.
- The hashing algorithm of the LACP port channel on the physical switch must be the same as the hashing algorithm that is configured to the LAG on the distributed switch.
- All physical NICs that you want to connect to the LACP port channel must be configured with the same speed and duplex.

Configure LACP Teaming and Failover for Distributed Port Groups

To handle the network traffic of distributed port groups by using a LAG, you assign physical NICs to the LAG ports and set the LAG as active in the teaming and failover order of distributed port groups.

Table 142: LACP Teaming and failover configuration of distributed port groups

Failover Order	Uplinks	Description
Active	A single LAG	You can only use one active LAG or multiple standalone uplinks to handle the traffic of distributed port groups. You cannot configure multiple active LAGs or mix active LAGs and standalone uplinks.
Standby	Empty	Having an active LAG and standby uplinks and the reverse is not supported. Having a LAG and another standby LAG is not supported.
Unused	All standalone uplinks and other LAGs if any	Because only one LAG must be active and the Standby list must be empty, you must set all standalone uplinks and other LAGs to unused.

Configure a Link Aggregation Group to Handle the Traffic for Distributed Port Groups

To aggregate the bandwidth of multiple physical NICs on hosts, you can create a link aggregation group (LAG) on the distributed switch and use it to handle the traffic of distributed port groups.

- Verify that for every host where you want to use LACP, a separate LACP port channel exists on the physical switch. See LACP Support on a vSphere Distributed Switch.
- Verify that the vSphere Distributed Switch where you configure the LAG is version 6.5 or later.
- Verify that enhanced LACP is supported on the distributed switch.

Newly created LAGs do not have physical NICs assigned to their ports and are unused in the teaming and failover order of distributed port groups. To handle the network traffic of distributed port groups by using a LAG, you must migrate the traffic from standalone uplinks to the LAG.

Create a Link Aggregation Group

To migrate the network traffic of distributed port groups to a link aggregation group (LAG), you create a LAG on the distributed switch.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. On the **Configure** tab, expand **Settings** and select **LACP**.
4. Click **New**.
5. In the **New Link Aggregation Group** dialog box, anter a name for the new LAG.
6. Set the number of ports to the LAG.

   Set the same number of ports for the LAG as the number of ports in the LACP port channel on the physical switch. A LAG port has the same function as an uplink on the distributed switch. All LAG ports form a NIC team in the context of the LAG.
7. Select the LACP negotiating mode of the LAG.

Option	Description
Active	All LAG ports are in an Active negotiating mode. The LAG ports initiate negotiations with the LACP port channel on the physical switch by sending LACP packets.
Passive	The LAG ports are in Passive negotiating mode. The LAG ports respond to LACP packets they receive but do not initiate LACP negotiation.

   If the LACP-enabled ports on the physical switch are in Active negotiating mode, you can set the LAG ports in Passive mode and the reverse.
8. Select a load-balancing mode from the hashing algorithms that LACP defines.

   Note:
   The hashing algorithm must be the same as the hashing algorithm set to the LACP port channel on the physical switch.
9. Select the link aggregation timeout mode.

   Periodic transmissions of LACP PDUs occur at either a slow or fast transmission rate, depending upon the LACP timeout preference you select. In a fast timeout, PDUs are sent every second, and, in a slow timeout, PDUs are sent every 30 seconds. A slow timeout is the default setting.

   Note:
   A fast timeout is supported only on distributed switch version 7.0.2 or later.
10. Set the VLAN policy for the LAG.

    This option is active when overriding the VLAN policy per individual uplink ports is enabled on the uplink port group. If you set the VLAN policy to the LAG, they override the policies set on the uplink port group level.
11. Click **OK**.

The new LAG is unused in the teaming and failover order of distributed port groups. No physical NICs are assigned to the LAG ports.

As with standalone uplinks, the LAG has a representation on every host that is associated with the distributed switch. For example, if you create LAG1 with two ports on the distributed switch, a LAG1 with two ports is created on every host that is associated with the distributed switch.

Set the LAG as standby in the teaming and failover configuration of distributed port groups. In this way, you create an intermediate configuration that lets you migrate the network traffic to the LAG without losing network connectivity.

Set a LAG as Standby in the Teaming and Failover Order

The new link aggregation group (LAG) by default is unused in the teaming and failover order of distributed port groups. Because only one LAG or only standalone uplinks can be active for distributed port groups, you must create an intermediate teaming and failover configuration, where the LAG is standby. This configuration lets you migrate physical NICs to the LAG ports by keeping the network connectivity up.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to a distributed switch.
3. Right-click the distributed switch and select **Distributed Port Group** > **Manage Distributed Port Groups**.
4. On the **Select port group policies** page, select **Teaming and failover** and click **Next**.
5. On the **Select port groups** page, select the port groups where you want to use the LAG.
6. On the **Teaming and failover** page, configure the failover order by selecting a LAG and using the **Move up** and **Move down** buttons.
7. Click **Next**, review the message that informs you about the usage of the intermediate teaming and failover configuration, and click **OK**.
8. On the Ready to complete page, click **Finish**.

Migrate physical NICs from standalone uplinks to the LAG ports.

Assign Physical NICs to the Ports of the Link Aggregation Group

Learn how to set the new link aggregation group (LAG) as standby in the teaming and failover order of distributed port groups. Having the LAG as standby lets you safely migrate the physical NICs from standalone uplinks to the LAG ports without losing network connectivity.

- Verify that either all LAG ports or the corresponding LACP-enabled ports on the physical switch are in active LACP negotiating mode.
- Verify that the physical NICs that you want to assign to the LAG ports have the same speed and are configured at full duplex.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch where the LAG resides.
3. From the **Actions** menu, select **Add and Manage Hosts**.
4. Select **Manage host networking**.
5. Select the host whose physical NICs you want to assign to the LAG ports and click **Next**.
6. On the Select network adapter tasks page, select **Manage physical adapters** and click **Next**.
7. On the Manage physical adapters page, select a NIC and click **Assign an uplink**.
8. Select a LAG port and click **OK**.
9. Repeat Step 6 and Step 7 for all physical NICs that you want to assign to the LAG ports.
10. Complete the wizard.

    Configure Two Physical NICs to a LAG in the Add and Manage Hosts Wizard

    For example, if you have a LAG with two ports, you configure a physical NIC to each LAG port in the **Add and Manage Hosts** wizard.

Set the LAG as active and all standalone uplinks to unused in the teaming and failover order of distributed port groups.

Set the Link Aggregation Group as Active in the Teaming and Failover Order of the Distributed Port Group

Learn how to set the LAG as active and move all standalone uplinks as unused in the teaming and failover order of the distributed port groups, after you have migrated physical NICs to the ports of the link aggregation group (LAG.

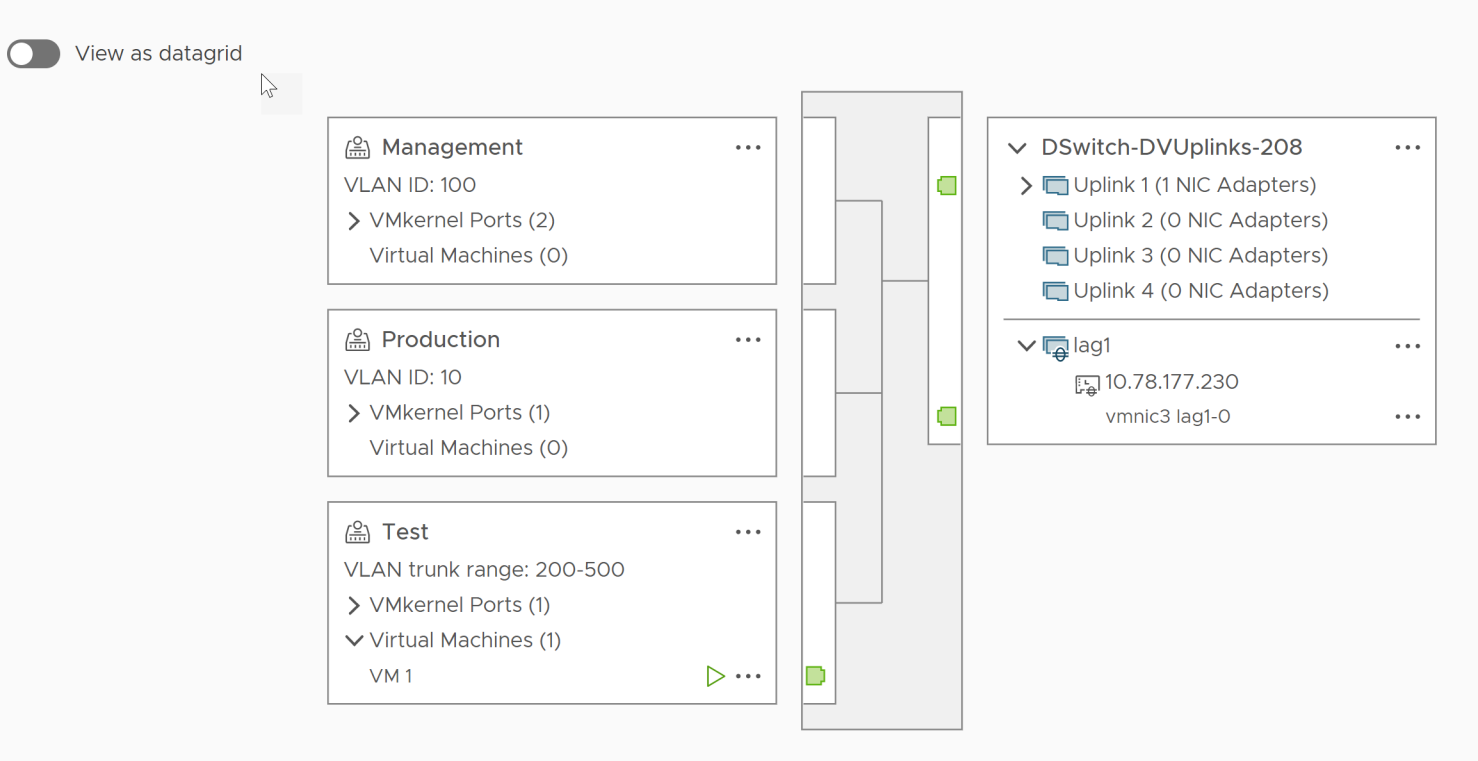
1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. From the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**.
4. Select **Teaming and failover** and click **Next**.
5. Select the port groups where you set the LAG as standby and click **Next**.
6. In Failover order, use the up and down arrows to move the LAG in the Active list, all standalone uplinks in the Unused list, and leave the Standby list empty.
7. Click **Next** and click **Finish**.

You safely migrated network traffic from standalone uplinks to a LAG for distributed port groups and created a valid LACP teaming and failover configuration for the groups.

Topology of a Distributed Switch that Uses a LAG

If you configure a LAG with two ports to handle the traffic of a distributed port group, you can check the topology of the distributed switch to view how it changed as a result of the new configuration.

Figure 58: Distributed Switch Topology with a LAG



Edit a Link Aggregation Group

Learn how to modify the settings of a link aggregation group (LAG) if you need to add more ports to the group or change the LACP negotiating mode, the load balancing algorithm, or the VLAN policy.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. On the **Configure** tab, expand **Settings** and select **LACP**.
4. Click the **New Link Aggregation Group** icon.
5. In the **Name** text box, type a new name for the LAG.
6. Change the number of ports for the LAG if you want to add more physical NICs to it.
   The new NICs must be connected to ports that are part of an LACP port channel on the physical switch.
7. Change the LACP negotiating mode of the LAG.
   If all ports on the physical LACP port channel are in Active LACP mode, you can change the LACP mode of the LAG to Passive and the reverse.

8. Change the load balancing mode of the LAG.

   You can select from the load balancing algorithms that LACP defines.

9. Select the link aggregation timeout mode.

   Periodic transmissions of LACP PDUs occur at either a slow or fast transmission rate, depending upon the LACP timeout preference you select. In a fast timeout, PDUs are sent every second, and, in a slow timeout, PDUs are sent every 30 seconds. A slow timeout is the default setting.

   Note:
   You can set a fast LACP timeout by running the following ESXCLI command.

   ```
   esxcli network vswitch dvs vmware lacp timeout set --vds DVS-name --lag-id <integer> --timeout 1
   ```

10. Change the VLAN policy.

    This option is active when the option for overriding the VLAN policy for individual ports is enabled on the uplink port group. If you change the VLAN policy for the LAG, they override the policies set at the uplink port group level.

11. Click **OK**.

Limitations of the LACP Support on a vSphere Distributed Switch

The LACP support on a vSphere Distributed Switch lets network devices negotiate automatic bundling of links by sending LACP packets to a peer. However, the LACP support on a vSphere Distributed Switch has limitations.

- The LACP is not supported with software iSCSI port binding. iSCSI multipathing over LAG is supported, if port binding is not used.
- The LACP support settings are not available in host profiles.
- The LACP support is not possible between nested ESX hosts.
- The LACP support does not work with the ESX dump collector.
- The LACP control packets (LACP PDU) do not get mirrored when port mirroring is enabled.
- The teaming and failover health check does not work for LAG ports. LACP checks the connectivity of the LAG ports.
- The enhanced LACP support works correctly when only one LAG handles the traffic per distributed port or port group.

Back Up and Restore Network Configurations

vSphere enables you to backup and restore the configuration of a vSphere Distributed Switch , distributed and uplink port groups in cases of invalid changes or a transfer to another deployment.

Back Up and Restore a vSphere Distributed Switch Configuration

vCenter provides the ability to backup and restore the configuration of a vSphere Distributed Switch. Learn how to restore the virtual network configuration in cases of database or upgrade failure. You can also use a saved switch configuration as a template to create a copy of the switch in the same or a new vSphere environment.

You can import or export a configuration of a distributed switch including its port groups. For information about exporting, importing, and restoring a port group configuration, see Export, Import, and Restore vSphere Distributed Port Group Configurations.

Note:
You can use a saved configuration file to restore policies and hosts associations on the distributed switch. You cannot restore the connection of physical NICs to uplink ports or ports of link aggregation groups.

Export vSphere Distributed Switch Configurations

Learn how to export vSphere Distributed Switch and distributed port group configurations to a file. The file preserves valid network configurations, enabling transfer of these configurations to other environments.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Settings** > **Export Configuration**.
4. Choose to export the distributed switch configuration, or export the distributed switch configuration and all port groups.
5. Optional: Enter notes about this configuration in the **Descriptions** field.
6. Click **OK.**
7. Click **Yes** to save the configuration file to your local system.

Use the exported configuration file to do the following tasks:

- Create a copy of the exported distributed switch in a vSphere environment. See Import a vSphere Distributed Switch Configuration.
- Overwrite the settings on an existing distributed switch. See Restore a vSphere Distributed Switch Configuration.

You can also export, import, and restore only port group configurations. See Export, Import, and Restore vSphere Distributed Port Group Configurations.

Import a vSphere Distributed Switch Configuration

Learn how to import a stored configuration file to create a new vSphere Distributed Switch or to restore a switch that has been deleted earlier.

The configuration file contains the networking settings of the switch. By using it you can also replicate the switch in other virtual environments.

> **Note:**
> You can use a saved configuration file to replicate the switch instance, its host associations, and policies. You cannot replicate the connection of physical NICs to uplink ports or ports on link aggregation groups.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and clusters** and navigate to a data center.
3. Right-click the data center and select **Distributed Switch** > **Import Distributed Switch**.
4. Browse to the location of the configuration file.
5. To assign the keys from the configuration file to the switch and its port groups, select the **Preserve original distributed switch and port group identifiers** check box and click **Next**.

   You can use the **Preserve original distributed switch and port group identifiers** option in the following cases:

   - Recreate a deleted switch.
   - Restore a switch whose upgrade has failed.

   All port groups are recreated and the hosts that have been connected to the switch are added again.

6. Review the settings for the switch and click **Finish**.

A new distributed switch is created with settings from the configuration file. If you have included distributed port group information in the configuration file, the port groups are also created.

Restore a vSphere Distributed Switch Configuration

Learn how to use the restore option to reset the configuration of an existing distributed switch to the settings in the configuration file. Restoring a distributed switch changes the settings on the selected switch back to the settings saved in the configuration file.

> **Note:**
> You can use a saved configuration file to restore policies and hosts associations on the distributed switch. You cannot restore the connection of physical NICs to uplink ports or ports of link aggregation groups.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch in the navigator and select **Settings** > **Restore Configuration**.
4. Browse for the configuration backup file to use.
5. Select **Restore distributed switch and all port groups** or **Restore distributed switch only** and click **Next**
6. Review the summary information for the restore.
   Restoring a distributed switch will overwrite the current settings of the distributed switch and its port groups. It will not delete existing port groups that are not part of the configuration file.
7. Click **Finish**.
   The distributed switch configuration has been restored to the settings in the configuration file.

Export, Import, and Restore vSphere Distributed Port Group Configurations

Learn to export vSphere distributed port group configurations to a file. The configuration file allows you to preserve valid port group configurations, enabling distribution of these configurations to other deployments.

You can export port group information at the same time you export distributed switch configurations. See Back Up and Restore a vSphere Distributed Switch Configuration.

Export vSphere Distributed Port Group Configurations

Learn how to export a distributed port group configurations to a file. The configuration preserves valid network configurations, enabling distribution of these configurations to other deployments.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to a distributed port group.
3. Right-click the distributed port group and select **Export Configuration**.
4. Optional: In the **Descriptions** field, type notes about this configuration.
5. Click **OK**.
   Click **Yes** to save the configuration file to your local system.

You now have a configuration file that contains all the settings for the selected distributed port group. You can use this file to create multiple copies of this configuration on an existing deployment, or overwrite settings of existing distributed port groups to conform to the selected settings.

You can use the exported configuration file to do the following tasks:

- To create a copy of the exported distributed port group, see Import a vSphere Distributed Port Group Configuration.
- To overwrite settings on an existing distributed port group, see Restore a vSphere Distributed Port Group Configuration.

Import a vSphere Distributed Port Group Configuration

Learn how to import to create a distributed port group from a configuration file.

If an existing port group has the same name as the imported port group, the new port group name has a number appended in parentheses. The settings from the imported configuration are applied to the new port group and the settings of the original port group remain unchanged.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to the distributed switch.
3. Right-click the distributed switch and select **Distributed Port Group** > **Import Distributed Port Group**.
4. Browse to the location of your saved configuration file and click **Next**.
5. Review the import settings before completing the import.
6. Click **Finish.**

Restore a vSphere Distributed Port Group Configuration

Learn how to use the restore option to reset the configuration of an existing distributed port group to the settings in a configuration file.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Networking**
   ( )
   and navigate to a distributed port group.
3. Right-click the distributed port group and select **Restore Configuration**.
4. Select one of the following options and click **Next**:
   - **Restore to previous configuration** to roll your port group configuration back one step. You cannot restore the port group configuration completely if you have performed more than one step.
   - **Restore configuration from a file** lets you restore the port group configuration from an exported backup file. You can also use a distributed switch backup file as long as it contains configuration information for the port group.
5. Review the summary information for the restore.
   The restore operation overwrites the current settings of the distributed port group with the settings from the backup. If you are restoring the port group configuration from a switch backup file, the restore operation does not delete existing port groups that are not a part of the file.
6. Click **Finish**.

ESX Configuration Manager Integration

You can manage core vSphere networking configurations in the vSphere Lifecycle Manager cluster level configuration manager. The vSphere networking configurations include *vmknics*, *netstacks*, *vdsvswitches*, *pnics*, *netdump.firewall. ipsec*, */etc/hosts*, */etc/resolv.conf* .

Starting vSphere 8.0 Update 3, vSphere Distributed Switch is integrated with vSphere Configuration Profiles (VCP). You can manage host distributed switch configuration in the vSphere Configuration Profiles (VCP) cluster.

Note:
Global distributed switch configuration such as MTU is out of the scope of VCP.

You can manage vSphere Distributed Switch in addition to all the host configurations at the cluster level by using a single configuration file in JSON format. You can check if all the hosts in the cluster are compliant with the distributed and the host configurations. You can migrate the ESX host networking from standard switch to a distributed switch using a remediation work flow at the cluster level. VMkernel network adapter on distributed switch is now fully supported by the configuration manager.

Using the configuration manager, you can configure VCPs that are attached to the VCP cluster which includes the host settings such as, the list of distributed switches that the hosts can join, and the PNIC to VDS uplink mapping.

Export Host Configuration

Learn how desired state configuration helps you to seamlessly manage your vSphere Networking configurations across geographies and different domains.

- You must have a vSphere Life Cycle Manager enabled cluster.
- Ensure that you have hosts in the cluster.

In this task, we are exporting a JSON file holding the desired state specification for a vmknic adapter.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and Clusters** and navigate to a cluster.
3. On the **Configure** tab, click **Desired State > Host Settings > Extract from reference host**. The Extract **Settings** dialog box appears.
4. Choose the host from the existing cluster.
5. Click **Next**
6. Click **Download Settings** to download the file.

The JSON file is downloaded.

You can then reuse the JSON file to apply the desired state that it contains to another cluster in the same or in a different vCenter instance.

> **Note:**
> Solution configuration is not included. Cached information is not included. Only host specific property is included.
>
> **Note:**
> If you have vSphere Life Cycle Manager enabled cluster, you can not join the host in the cluster to a vSphere Distributed Switch.

Import Host Configuration

Learn how to import a desired state as a JSON file.

- You must have a vSphere Life Cycle Manager enabled cluster.
- Ensure you have hosts in the cluster.
- Ensure you have a valid host configuration JSON file.

Learn how to import a JSON file with host configuration. Before applying a desired state on a cluster, you can scan all hosts in the cluster against the desired state and check the cluster compliance against the desired state. You can run compliance checks to detect any drifts instantly.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and Clusters** and navigate to a cluster.
3. On the **Configure** tab, click **Desired State > Host Settings > Import Settings**. The Import **Host** Settings dialog box appears.
4. Click **Browse** to select file. The compliance is validated.
   If there are hosts that are not compliant, an error message is displayed.
5. Click **Remediate** to remediate the host compliance issue. The **Remediate Settings** dialog box appears. The **Pre-check** is in-progress.
6. The **Pre-Check** is completed. The host level remediation details are displayed under **Review Impact**.
7. Click **Remediate** to remediate the configuration settings.

You can view the imported settings on the host.

Rollback and Recovery of the Management Network

Learn how to prevent and recover from misconfiguration of the management network by using the rollback and recovery support of the vSphere Distributed Switch and vSphere Standard Switch.

Rollback is available for use on both standard and distributed switches. To fix invalid configuration of the management network, you can connect directly to a host to fix the issues through the DCUI.

vSphere Networking Rollback

Learn how vSphere protects hosts from losing connection to vCenter as a result of misconfiguration of the management network by rolling back configuration changes.

In vSphere networking rollback is enabled by default. However, you can activate or deactivate rollbacks at the vCenter level.

Host Networking Rollbacks

Host networking rollbacks occur when an invalid change is made to the networking configuration for the connection with vCenter. Every network change that disconnects a host also triggers a rollback. The following examples of changes to the host networking configuration might trigger a rollback:

- Updating the speed or duplex of a physical NIC.
- Updating DNS and routing settings.
- Updating teaming and failover policies or traffic shaping policies of a standard port group that contains the management VMkernel network adapter.
- Updating the VLAN of a standard port group that contains the management VMkernel network adapter.
- Increasing the MTU of management VMkernel network adapter and its switch to values not supported by the physical infrastructure.
- Changing the IP settings of management VMkernel network adapters.
- Removing the management VMkernel network adapter from a standard or distributed switch.
- Removing a physical NIC of a standard or distributed switch containing the management VMkernel network adapter.
- Migrating the management VMkernel adapter from vSphere standard to distributed switch.

If a network disconnects for any of these reasons, the task fails and the host reverts to the last valid configuration.

vSphere Distributed Switch Rollbacks

Distributed switch rollbacks occur when invalid updates are made to distributed switches, distributed port groups, or distributed ports. The following changes to the distributed switch configuration trigger a rollback:

- Changing the MTU of a distributed switch.
- Changing the following settings in the distributed port group of the management VMkernel network adapter:
  - Teaming and failover
  - VLAN
  - Traffic shaping
- Blocking all ports in the distributed port group containing the management VMkernel network adapter.
- Overriding the policies on at the level of the distributed port for the management VMkernel network adapter.

If a configuration becomes invalid because of any of the changes, one or more hosts might become out of synchronization with the distributed switch.

If you know where the conflicting configuration setting is located, you can manually correct the setting. For example, if you have migrated a management VMkernel network adapter to a new VLAN, the VLAN might not be actually trunked on the physical switch. When you correct the physical switch configuration, the next distributed switch-to-host synchronization will resolve the configuration problem.

If you are not sure where the problem exists, you can restore the state of the distributed switch or distributed port group to an earlier configuration. See Restore a vSphere Distributed Port Group Configuration.

Disable Network Rollback

Rollback is enabled by default in vSphere. You can disable rollback in vCenter by using the vSphere Client.

1. In vSphere Client, select **Inventory** from the main menu
   ( ).
2. Click **Hosts and Clusters** and navigate to a vCenter instance.
3. On the **Configure** tab, expand **Settings** and select **Advanced Settings**.
4. Click **Edit Settings**.
5. Select the `config.vpxd.network.rollback` key, and change the value to false.
   If the key is not present, you can add it and set the value to false.
6. Click **OK**.
7. Restart vCenter to apply the changes.

Disable Network Rollback by Using the vCenter Configuration File

Rollback is enabled by default in vSphere. You can disable rollback by editing the `vpxd.cfg` configuration file of vCenter directly.

1. On the host machine of vCenter, navigate to the directory `/etc/vmware-vpx`.
2. Open the `vpxd.cfg` file for editing.
3. In the `<network>` element, set the `<rollback>` element to `false`:

```
<config>
 <vpxd>
  <network>
    <rollback>false</rollback>
  </network>
 </vpxd>
</config>
```

4. Save and close the file.
5. Restart the vCenter system.

Resolve Errors in the Management Network Configuration on a vSphere Distributed Switch

Learn how to use the Direct Console User Interface (DCUI) to restore the connection between vCenter and a host that accesses the management network through a distributed switch.

Verify that the management network is configured on a port group on the distributed switch.

If networking rollback is disabled, misconfiguring the port group for the management network on the distributed switch leads to loss of connection between vCenter and the hosts that are added to the switch. You have to use the DCUI to connect each host individually.

If the uplinks that you use to restore the management network are also used by VMkernel adapters that handle other types of traffic (vMotion, Fault Tolerance, and so on), the adapters loose network connectivity after the restore.

For more information about accessing and using the DCUI, see the *vSphere Security* documentation.

> **Note:**
>
> Recovery of the management connection on a distributed switch is not supported on stateless ESX instances.

1. Connect to the DCUI of the host.
2. From the **Network Restore Options** menu, select **Restore vDS**.
3. Configure the uplinks and optionally the VLAN for the management network.
4. Apply the configuration.

The DCUI creates a local ephemeral port and applies the values you provided for the VLAN and uplinks. The DCUI moves the VMkernel adapter for the management network to the new local port to restore connectivity to vCenter.

After the connection of the host to vCenter is restored, correct the configuration of the distributed port group and re-add the VMkernel adapter to the group.

vSphere Networking Policies

Policies set at the standard switch or distributed port group level apply to all of the port groups on the standard switch or to ports in the distributed port group. The exceptions are the configuration options that are overridden at the standard port group or distributed port level.

Apply Networking Policies on a vSphere Standard or Distributed Switch

You apply networking policies differently on vSphere Standard Switches and vSphere Distributed Switches. Not all policies that are available for a vSphere Distributed Switch are also available for a vSphere Standard Switch.

Table 143: Virtual Switch Objects Where Policies Apply

Virtual Switch	Virtual Switch Object	Description
vSphere Standard Switch	Entire switch	When you apply policies on the entire standard switch, the policies are propagated to all standard port groups on the switch.
	Standard port group	You can apply different policies on individual port groups by overriding the policies that are inherited from the switch.
vSphere Distributed Switch	Distributed port group	When you apply policies on a distributed port group, the policies are propagated to all ports in the group.
	Distributed port	You can apply different policies on individual distributed ports by overriding the policies that are inherited from the distributed port group.
	Uplink port group	You can apply policies at uplink port group level, and the are policies are propagated to all ports in the group.
	Uplink port	You can apply different policies on individual uplink ports by overriding the policies that are inherited from the uplink port group.

Table 144: Policies Available for a vSphere Standard Switch and vSphere Distributed Switch

Policy	Standard Switch	Distributed Switch	Description
Teaming and failover	Yes	Yes	Lets you configure the physical NICs that handle the network traffic for a standard switch, standard port group, distributed port group, or distributed port. You arrange the physical NICs in a failover order and apply different load balancing policies over them.
Security	Yes	Yes	Provides protection of traffic against MAC address impersonation and unwanted port scanning. The networking security policy is implemented in Layer 2 of the networking protocol stack.

Policy	Standard Switch	Distributed Switch	Description
Traffic shaping	Yes	Yes	Lets you restrict the network bandwidth that is available to ports, but also to allow bursts of traffic to flow through at higher speeds. ESX shapes outbound network traffic on standard switches and inbound and outbound traffic on distributed switches.
VLAN	Yes	Yes	Lets you configure the VLAN tagging for a standard or distributed switch. You can configure External Switch Tagging(EST), Virtual Switch Tagging (VST), and Virtual Guest Tagging (VGT).
Traffic filtering and marking	No	Yes	Lest you protect the virtual network from unwanted traffic and security attacks or apply a QoS tag to a certain traffic type.
Resources allocation	No	Yes	Lets you associate a distributed port or port group with a user-defined network resource pool. In this way, you can better control the bandwidth that is available to the port or port group. You can use the resource allocation policy with vSphere Network I/O Control version 2 and 3.
Port blocking	No	Yes	Lets you selectively block ports from sending and receiving data.

Configure Overriding Networking Policies on Port Level

Learn how to apply different policies for distributed ports, you configure the per-port overriding of the policies that are set at the port group level. You can also enable the reset of any configuration that is set on per-port level when a distributed port disconnects from a virtual machine.

1. Locate a distributed port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups**.
2. Right-click the distributed port group and select **Edit Settings**.
3. Select the **Advanced** page.

Option	Description
Configure reset at disconnect	From the drop-down menu, enable or disable reset at disconnect. When a distributed port is disconnected from a virtual machine, the configuration of the distributed port is reset to the distributed port group setting. Any per-port overrides are discarded.
Override port policies	Select the distributed port group policies to be overridden on a per-port level.

4. Optional: Use the policy pages to set overrides for each port policy.
5. Click **OK**.

What is Teaming and Failover Policy

NIC teaming lets you increase the network capacity of a virtual switch by including two or more physical NICs in a team. To determine how the traffic is rerouted in case of adapter failure, you include physical NICs in a failover order. To determine how the virtual switch distributes the network traffic between the physical NICs in a team, you select load balancing algorithms depending on the needs and capabilities of your environment.

NIC Teaming Policy

You can use NIC teaming to connect a virtual switch to multiple physical NICs on a host to increase the network bandwidth of the switch and to provide redundancy. A NIC team can distribute the traffic between its members and provide passive failover in case of adapter failure or network outage. You set NIC teaming policies at virtual switch or port group level for a vSphere Standard Switch and at a port group or port level for a vSphere Distributed Switch.

> **Note:**
> All ports on the physical switch in the same team must be in the same Layer 2 broadcast domain.

Load Balancing Policy

The Load Balancing policy determines how network traffic is distributed between the network adapters in a NIC team. vSphere virtual switches load balance only the outgoing traffic. Incoming traffic is controlled by the load balancing policy on the physical switch.

For more information about each load balancing algorithm, see Load Balancing Algorithms Available for Virtual Switches.

Network Failure Detection Policy

You can specify one of the following methods that a virtual switch uses for failover detection.

Link status only	Relies only on the link status that the network adapter provides. Detects failures, such as removed cables and physical switch power failures. However, link status does not detect the following configuration errors: • Physical switch port that is blocked by spanning tree or is misconfigured to the wrong VLAN. • Pulled cable that connects a physical switch to another networking devices, for example, an upstream switch.
Beacon probing	Sends out and listens for Ethernet broadcast frames, or beacon probes, that physical NICs send to detect link failure in all physical NICs in a team. ESX hosts send beacon packets every second. Beacon probing is most useful to detect failures in the closest physical switch to the ESX host, where the failure does not cause a link-down event for the host. Use beacon probing with three or more NICs in a team because ESX can detect failures of a single adapter. If only two NICs are assigned and one of them loses connectivity, the switch cannot determine which NIC needs to be taken out of service because both do not receive beacons and as a result all packets sent to both uplinks. Using at least three NICs in such a team allows for n-2 failures where n is the number of NICs in the team before reaching an ambiguous situation.

Failback Policy

By default, a failback policy is enabled on a NIC team. If a failed physical NIC returns online, the virtual switch sets the NIC back to active by replacing the standby NIC that took over its slot.

If the physical NIC that stands first in the failover order experiences intermittent failures, the failback policy might lead to frequent changes in the NIC that is used. The physical switch sees frequent changes in MAC addresses, and the physical switch port might not accept traffic immediately when an adapter becomes online. To minimize such delays, you might consider changing the following settings on the physical switch:

- Disable Spanning Tree Protocol (STP) on physical NICs that are connected to ESX hosts .
- For Cisco based networks, enable PortFast mode for access interfaces or PortfFast trunk mode for trunk interfaces. This might save about 30 seconds during the initialization of the physical switch port.
- Deactivate the trunking negotiation.

Notify Switches Policy

By using the notify switches policy, you can determine how the ESX host communicates failover events. When a physical NIC connects to the virtual switch or when traffic is rerouted to a different physical NIC in the team, the virtual switch sends notifications over the network to update the lookup tables on physical switches. Notifying the physical switch offers lowest latency when a failover or a migration with vSphere vMotion occurs.

Load Balancing Algorithms Available for Virtual Switches

Learn to configure various load balancing algorithms on a virtual switch to determine how network traffic is distributed between the physical NICs in a team.

Route Based on Originating Virtual Port

The virtual switch selects uplinks based on the virtual machine port IDs on the vSphere Standard Switch or vSphere Distributed Switch.

Route Based on Originating Virtual Port is the default load balancing method on the vSphere Standard Switch and vSphere Distributed Switch.

Each virtual machine running on an ESX host has an associated virtual port ID on the virtual switch. To calculate an uplink for a virtual machine, the virtual switch uses the virtual machine port ID and the number of uplinks in the NIC team. After the virtual switch selects an uplink for a virtual machine, it always forwards traffic through the same uplink for this virtual machine as long as the machine runs on the same port. The virtual switch calculates uplinks for virtual machines only once, unless uplinks are added or removed from the NIC team.

The port ID of a virtual machine is fixed while the virtual machine runs on the same host. If you migrate, power off, or delete the virtual machine, its port ID on the virtual switch becomes free. The virtual switch stops sending traffic to this port, which reduces the overall traffic for its associated uplink. If a virtual machine is powered on or migrated, it might appear on a different port and use the uplink, which is associated with the new port.

Table 145: Considerations on Using Route Based on Originating Virtual Port

Considerations	Description
Advantages	• An even distribution of traffic if the number virtual NICs is greater than the number of physical NICs in the team. • Low resource consumption, because in most cases the virtual switch calculates uplinks for virtual machines only once. • No changes on the physical switch are required.
Disadvantages	• The virtual switch is not aware of the traffic load on the uplinks and it does not load balance the traffic to uplinks that are less used. • The bandwidth that is available to a virtual machine is limited to the speed of the uplink that is associated with the relevant port ID, unless the virtual machine has more than one virtual NIC.

Route Based on Source MAC Hash

The virtual switch selects an uplink for a virtual machine based on the virtual machine MAC address. To calculate an uplink for a virtual machine, the virtual switch uses the virtual machine MAC address and the number of uplinks in the NIC team.

Table 146: Considerations on Using Route Based on Source MAC Hash

Considerations	Description
Advantages	• A more even distribution of the traffic than Route Based on Originating Virtual Port, because the virtual switch calculates an uplink for every packet. • Virtual machines use the same uplink because the MAC address is static. Powering a virtual machine on or off does not change the uplink that the virtual machine uses. • No changes on the physical switch are required.
Disadvantages	• The bandwidth that is available to a virtual machine is limited to the speed of the uplink that is associated with the relevant port ID, unless the virtual machine uses multiple source MAC addresses. • Higher resource consumption than Route Based on Originating Virtual Port, because the virtual switch calculates an uplink for every packet. • The virtual switch is not aware of the load of the uplinks, so uplinks might become overloaded.

Route Based on IP Hash

The virtual switch selects uplinks for virtual machines based on the source and destination IP address of each packet.

To calculate an uplink for a virtual machine, the virtual switch takes the last octet of both source and destination IP addresses in the packet, puts them through a XOR operation, and then runs the result through another calculation based on the number of uplinks in the NIC team. The result is a number between 0 and the number of uplinks in the team minus one. For example if a NIC team has four uplinks, the result is a number between 0 and 3 as each number is associated with a NIC in the team. For non-IP packets, the virtual switch takes two 32-bit binary values from the frame or packet from where the IP address would be located.

Any virtual machine can use any uplink in the NIC team depending on the source and destination IP address. In this way, each virtual machine can use the bandwidth of any uplink in the team. If a virtual machine runs in an environment with a large number of independent virtual machines, the IP hash algorithm can provide an even spread of the traffic between the NICs in the team. When a virtual machine communicates with multiple destination IP addresses, the virtual switch can generate a different hash for each destination IP. In this way, packets can use different uplinks on the virtual switch that results in higher potential throughput.

However, if your environment has a small number of IP addresses, the virtual switch might consistently pass the traffic through one uplink in the team. For example, if you have a database server that is accessed by one application server, the virtual switch always calculates the same uplink, because only one source-destination pair exists.

Physical Switch Configuration

To ensure that IP hash load balancing works correctly, you must have an Etherchannel configured on the physical switch. An Etherchannel bonds multiple network adapters into a single logical link. When ports are bound into an Etherchannel, every time the physical switch receives a packet from the same virtual machine MAC address on different ports, the switch updates its content addressable memory (CAM) table correctly.

For example, if the physical switch receives packets on ports 01 and 02 from MAC address A, the switch makes a 01-A and a 02-A entry in its CAM table. As a result, the physical switch distributes the incoming traffic to the correct ports. Without an Etherchannel, the physical switch first makes a record that a packet from MAC address A is received on port 01, then updates the same record that a packet from MAC address A is received on port 02. Hence, the physical switch forwards incoming traffic only on port 02, and might result in packets not reaching their destination and overloading the corresponding uplink.

Limitations and Configuration Requirements

- ESX hosts support IP hash teaming on a single physical switch or stacked switches.
- ESX hosts support only 802.3ad link aggregation in Static mode . You can only use a static Etherchannel with vSphere Standard Switches. LACP is not supported. If you enable IP hash load balancing without 802.3ad link aggregation and the reverse, you might experience networking disruptions.
- You must use Link Status Only as network failure detection with IP hash load balancing.
- You must set all uplinks from the team in the Active failover list . The Standby and Unused lists must be empty.
- The number of ports in the Etherchannel must be same as the number of uplinks in the team.

Considerations on Using Route Based on IP Hash

Considerations	Description
Advantages	• A more even distribution of the load compared to Route Based on Originating Virtual Port and Route Based on Source MAC Hash, as the virtual switch calculates the uplink for every packet. • A potentially higher throughput for virtual machines that communicate with multiple IP addresses.
Disadvantages	• Highest resource consumption compared to the other load balancing algorithms. • The virtual switch is not aware of the actual load of the uplinks. • Requires changes on the physical network. • Complex to troubleshoot.

Route Based on Physical NIC Load

Route Based on Physical NIC Load is based on Route Based on Originating Virtual Port, where the virtual switch checks the actual load of the uplinks and takes steps to reduce it on overloaded uplinks. Available only for vSphere Distributed Switch.

The distributed switch calculates uplinks for virtual machines by taking their port ID and the number of uplinks in the NIC team. The distributed switch tests the uplinks every 30 seconds, and if their load exceeds 75 percent of usage, the port ID of the virtual machine or vmkernel port with the highest I/O is moved to a different uplink.

Table 147: Considerations on Using Route Based on Physical NIC Load

Considerations	Description
Advantages	• Low resource consumption because the distributed switch calculates uplinks for virtual machines only once and checking the of uplinks has minimal impact. • The distributed switch is aware of the load of uplinks and takes care to reduce it if needed. • No changes on the physical switch are required.
Disadvantages	• The bandwidth that is available to virtual machines is limited to the uplinks that are connected to the distributed switch.

Use Explicit Failover Order

No actual load balancing is available with this policy. The virtual switch always uses the uplink that stands first in the list of Active adapters from the failover order and that passes failover detection criteria. If no uplinks in the Active list are available, the virtual switch uses the uplinks from the Standby list.

Configure NIC Teaming, Failover, and Load Balancing on a vSphere Standard Switch or Standard Port Group

Learn how to configure failover order to determine how network traffic is rerouted in case of adapter failure. Include two or more physical NICs in a team to increase the network capacity of a vSphere Standard Switch or standard port group. Select a load balancing algorithm to determine how the standard switch distributes the traffic between the physical NICs in a team.

Configure NIC teaming, failover, and load balancing depending on the network configuration on the physical switch and the topology of the standard switch. See What is Teaming and Failover Policy and Load Balancing Algorithms Available for Virtual Switches for more information.

If you configure the teaming and failover policy on a standard switch, the policy is propagated to all port groups in the switch. If you configure the policy on a standard port group, it overrides the policy inherited from the switch.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Navigate to the Teaming and Failover policy for the standard switch, or standard port group.

Option	Action
Standard Switch	1. Select the switch from the list. 2. Click **Edit settings** and select **Teaming and failover**.
Standard port group	1. Select the switch where the port group resides. 2. From the switch topology diagram, select the standard port group and click **Edit settings**. 3. Select **Teaming and failover**. 4. Select **Override** next to the policies that you want to override.

4. From the **Load balancing** drop-down menu, specify how the virtual switch load balances the outgoing traffic between the physical NICs in a team.

Option	Description
Route based on the originating virtual port	Select an uplink based on the virtual port IDs on the switch. After the virtual switch selects an uplink for a virtual machine or a VMkernel adapter, it always forwards traffic through the same uplink for this virtual machine or VMkernel adapter.
Route based on IP hash	Select an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, the switch uses the data at those fields to compute the hash .

Option	Description
	IP-based teaming requires that the physical switch is configured with EtherChannel.
Route based on source MAC hash	Select an uplink based on a hash of the source Ethernet.
Use explicit failover order	From the list of active adapters, always use the highest order uplink that passes failover detection criteria. No actual load balancing is performed with this option.

5. From the **Network failure detection** drop-down menu, select the method that the virtual switch uses for failover detection.

Option	Description
Link status only	Relies only on the link status that the network adapter provides. This option detects failures such as removed cables and physical switch power failures.
Beacon probing	Sends out and listens for beacon probes on all NICs in the team, and uses this information, in addition to link status, to determine link failure. ESX sends beacon packets every second. The NICs must be in an active/active or active/standby configuration because the NICs in an unused state do not participate in beacon probing.

6. From the **Notify switches** drop-down menu, select whether the standard or distributed switch notifies the physical switch in case of a failover.

   Note:

   If the **Notify switches** is set to **Yes**, then all the connected ports, port groups, and distributed switches are connected back to the host when vCenter reconnects with ESX hosts.

   Note:
   Set this option to **No** if a connected virtual machine is using Microsoft Network Load Balancing in unicast mode. No issues exist with Network Load Balancing running in multicast mode.

7. From the **Failback** drop-down menu, select whether a physical adapter is returned to active status after recovering from a failure.

   If failback is set to **Yes**, the default selection, the adapter is returned to active duty immediately upon recovery, displacing the standby adapter that took over its slot, if any.

   If failback is set to **No** for a standard port, a failed adapter is left inactive after recovery until another currently active adapter fails and must be replaced.

8. Specify how the uplinks in a team are used when a failover occurs by configuring the Failover Order list.

   If you want to use some uplinks but reserve others for emergencies in case the uplinks in use fail, use the up and down arrow keys to move uplinks into different groups.

Option	Description
Active adapters	Continue to use the uplink if the network adapter connectivity is up and active.
Standby adapters	Use this uplink if one of the active physical adapter is down.
Unused adapters	Do not use this uplink.

9. Click **OK**.

Configure NIC Teaming, Failover, and Load Balancing on a Distributed Port Group or Distributed Port

Learn how to configure failover order to determine how network traffic is rerouted in case of adapter failure. Include two or more physical NICs in a team to increase the network capacity of a distributed port group or port. Select a load balancing algorithm to determine how the distributed switch load balances the traffic between the physical NICs in a team.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

Configure NIC teaming, failover, and load balancing according with the network configuration on the physical switch and the topology of the distributed switch. See What is Teaming and Failover Policy and Load Balancing Algorithms Available for Virtual Switches for more information.

If you configure the teaming and failover policy for a distributed port group, the policy is propagated to all ports in the group. If you configure the policy for a distributed port, it overrides the policy inherited from the group.

Note:
Setting failback option is not supported with **Route based on physical NIC load** teaming policy.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Navigate the Teaming and Failover policy on the distributed port group or port.

Option	Action
Distributed port group	1. From the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**. 2. Select the port group and click **Next**. 3. Select **Teaming and failover**.
Distributed port	1. On the **Networks** tab, click **Distributed Port Groups** and double-click a distributed port group. 2. On the **Ports** tab, select a port and click **Edit distributed port settings**. 3. Select **Teaming and failover**. 4. Select **Override** next to the properties that you want to override.

3. From the **Load balancing** drop-down menu, specify how the virtual switch load balances the outgoing traffic between the physical NICs in a team.

Option	Description
Route based on the originating virtual port	Select an uplink based on the virtual port IDs on the switch. After the virtual switch selects an uplink for a virtual machine or a VMkernel adapter, it always forwards traffic through the same uplink for this virtual machine or VMkernel adapter.
Route based on IP hash	Select an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, the switch uses the data at those fields to compute the hash . IP-based teaming requires that the physical switch is configured with EtherChannel.
Route based on source MAC hash	Select an uplink based on a hash of the source Ethernet.
Route based on physical NIC load	Available for distributed port groups or distributed ports. Select an uplink based on the current load of the physical network adapters connected to the port group or port. If an uplink remains busy at 75 percent or higher for 30 seconds, the host proxy switch moves a part of the virtual machine traffic to a physical adapter that has free capacity.

Option	Description
	Note: Selecting **Route based on physical NIC load** prevents you from setting a failback option for a distributed port group.
Use explicit failover order	From the list of active adapters, always use the highest order uplink that passes failover detection criteria. No actual load balancing is performed with this option.

4. From the **Network failure detection** drop-down menu, select the method that the virtual switch uses for failover detection.

Option	Description
Link status only	Relies only on the link status that the network adapter provides. This option detects failures such as removed cables and physical switch power failures.
Beacon probing	Sends out and listens for beacon probes on all NICs in the team, and uses this information, in addition to link status, to determine link failure. ESX sends beacon packets every second. The NICs must be in an active/active or active/standby configuration because the NICs in an unused state do not participate in beacon probing.

5. From the **Notify switches** drop-down menu, select whether the standard or distributed switch notifies the physical switch in case of a failover.

   Note:

   If the **Notify switches** is set to **Yes**, then all the connected ports, port groups, and distributed switches are connected back to the host when vCenter reconnects with ESX hosts.

   Note:
   Set this option to **No** if a connected virtual machine is using Microsoft Network Load Balancing in unicast mode. No issues exist with Network Load Balancing running in multicast mode.

6. From the **Failback** drop-down menu, select whether a physical adapter is returned to active status after recovering from a failure.

   If failback is set to **Yes**, the default selection, the adapter is returned to active duty immediately upon recovery, displacing the standby adapter that took over its slot, if any.

   If failback is set to **No** for a distributed port, a failed adapter is left inactive after recovery only if the associated virtual machine is running. When the **Failback** option is **No** and a virtual machine is powered off, if all active physical adapters fail and then one of them recovers, the virtual NIC is connected to the recovered adapter instead of to a standby one after the virtual machine is powered on. Powering a virtual machine off and then on leads to reconnecting the virtual NIC to a distributed port. The distributed switch considers the port as newly added, and assigns it the default uplink port, that is, the active uplink adapter.

7. Specify how the uplinks in a team are used when a failover occurs by configuring the Failover Order list.

   If you want to use some uplinks but reserve others for emergencies in case the uplinks in use fail, use the up and down arrow keys to move uplinks into different groups.

Option	Description
Active adapters	Continue to use the uplink if the network adapter connectivity is up and active.
Standby adapters	Use this uplink if one of the active physical adapter is down.
Unused adapters	Do not use this uplink.

8. Review your settings and apply the configuration.

What is VLAN Policy

VLAN policies determine how VLANs function across your network environment.

A virtual local area network (VLAN) is a group of hosts with a common set of requirements, which communicate as if they were attached to the same broadcast domain, regardless of their physical location. A VLAN has the same attributes as a physical local area network (LAN), but it allows for end stations to be grouped together even if not on the same network switch.

The scope of VLAN policies can be distributed port groups and ports, and uplink port groups and ports.

Configure VLAN Tagging on a Distributed Port Group or Distributed Port

To apply VLAN tagging globally on all distributed ports, you must set the VLAN policy on a distributed port group. To integrate the virtual traffic on the port with physical VLANs in a different way from the parent distributed port group, you must use the VLAN policy on a distributed port.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Navigate to the VLAN policy on the distributed port group or distributed port.

Option	Action
Distributed port group	1. From the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**. 2. Select **VLAN** and click **Next**. 3. Select the port group and click **Next**.
Distributed port	1. On the **Networks** tab, click **Distributed Port Groups** and double-click a distributed port group. 2. On the **Ports** tab, select a port and click the **Edit distributed port settings** icon. 3. Select **VLAN**. 4. Select **Override** next to the properties to override.

3. From the **VLAN type** drop-down menu, select the type of VLAN traffic filtering and marking, and click **Next**.

Option	Description
None	Do not use VLAN. Use this option in case of External Switch Tagging.
VLAN	Tag traffic with the ID from the **VLAN ID** field. Type a number between 1 and 4094 for Virtual Switch Tagging.
VLAN Trunking	Pass VLAN traffic with ID within the **VLAN trunk range** to guest operating system. You can set multiple ranges and individual VLANs by using a comma-separated list. For example: `1702-1705, 1848-1849`. Use this option for Virtual Guest Tagging.
Private VLAN	Associate the traffic with a private VLAN created on the distributed switch.

4. Review your settings and apply the configuration.

Configure VLAN Tagging on an Uplink Port Group or Uplink Port

To configure VLAN traffic processing generally for all member uplinks, you must set the VLAN policy on an uplink port. To handle VLAN traffic through the port in a different way than for the parent uplink port group, you must set the VLAN policy on an uplink .

To override the VLAN policy at the port level, enable the port-level overrides. See Configure Overriding Networking Policies on Port Level.

Use the VLAN policy at the uplink port level to propagate a trunk range of VLAN IDs to the physical network adapters for traffic filtering. The physical network adapters drop the packets from the other VLANs if the adapters support filtering by VLAN. Setting a trunk range improves networking performance because physical network adapters filter traffic instead of the uplink ports in the group.

If you have a physical network adapter that does not support VLAN filtering, the VLANs still might not be blocked. In this case, configure VLAN filtering on a distributed port group or on a distributed port.

For information about VLAN filtering support, see the technical documentation from the adapter vendors.

1. In the vSphere Client, navigate to a distributed switch.
2. On the **Networks** tab, click **Uplink Port Groups**.
3. Navigate to the VLAN policy is on the uplink port group or port.

Option	Action
Uplink port group	1. Right-click an uplink port group in the list and select **Edit Settings**. 2. Click **VLAN**.
Uplink port	1. Double-click an uplink port group. 2. On the **Ports** tab, select a port and click the **Edit distributed port settings** tab. 3. Click **VLAN** and select **Override**.

4. Enter a **VLAN trunk range** value to propagate to the physical network adapters.
   For trunking of several ranges and individual VLANs, separate the entries with commas.
5. Click **OK**.

What is Security Policy

Networking security policy provides protection of traffic against MAC address impersonation and unwanted port scanning

The security policy of a standard or distributed switch is implemented in Layer 2 (Data Link Layer) of the network protocol stack. The three elements of the security policy are promiscuous mode, MAC address changes, and forged transmits. See the *vSphere Security* documentation for information about potential networking threats.

Configure the Security Policy for a vSphere Standard Switch or Standard Port Group

For a vSphere Standard Switch, you can configure the security policy to reject MAC address and promiscuous mode changes in the guest operating system of a virtual machine. You can override the security policy that is inherited from the standard switch on individual port groups.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Navigate to the Security policy for the standard switch or port group.

Option	Action
vSphere Standard Switch	1. Select a standard switch from the list. 2. Click **Edit settings**. 3. Select **Security**.
Standard port group	1. Select the standard switch where the port group resides. 2. In the topology diagram, select a standard port group. 3. Click **Edit settings**. 4. Select **Security** and select **Override** next to the options to override.

4. Reject or accept promiscuous mode activation or MAC address changes in the guest operating system of the virtual machines attached to the standard switch or port group.

Option	Description
Promiscuous mode	• **Reject**. The VM network adapter receives only frames that are addressed to the virtual machine. • **Accept**.The virtual switch forwards all frames to the virtual machine in compliance with the active VLAN policy for the port to which the VM network adapter is connected. **Note:** Promiscuous mode is insecure mode of operation. Firewalls, port scanners, intrusion detection systems, must run in promiscuous mode.
MAC address changes	• **Reject**. If the guest OS changes the effective MAC address of the virtual machine to a value that is different from the MAC address of the VM network adapter, the switch drops all inbound frames to the adapter. If the guest OS changes the effective MAC address of the virtual machine back to the MAC address of the VM network adapter, the virtual machine receives frames again. • **Accept**. If the guest OS changes the effective MAC address of the virtual machine to a value that is different from the MAC address of the VM network adapter, the switch allows frames to the new address to pass.
Forged transmits	• **Reject**. The switch drops any outbound frame from a virtual machine adapter with a source MAC address that is different from the one in the `.vmx` configuration file. • **Accept**. The switch does not perform filtering, and permits all outbound frames.
Status	Enable or disable the MAC learning feature. The default is disabled.
Allow unicast flooding	When a packet that is received by a port has an unknown destination MAC address, the packet is dropped. With unknown unicast flooding enabled, the port floods unknown unicast traffic to every port on the switch that has MAC learning and unknown unicast flooding enabled. This property is enabled by default, if MAC learning is enabled.
MAC Limit	The number of MAC addresses that can be learned is configurable. The maximum value is 4096 per port, which is the default.
MAC Limit Policy	The policy for when the MAC limit is reached. The options are: • Drop - Packets from an unknown source MAC address are dropped. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled. • Allow - Packets from an unknown source MAC address are forwarded although the address will not be learned. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.

5. Click **OK**.

Configure the Security Policy for a Distributed Port Group or Distributed Port

Learn how to set a security policy on a distributed port group to allow or reject promiscuous mode and MAC address changes from the guest operating system of the virtual machines associated with the port group. You can override the security policy inherited from the distributed port groups on individual ports.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Navigate to the Security policy for the distributed port group or port.

Option	Action
Distributed port group	1. From the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**. 2. Select **Security** and click **Next**. 3. Select the port group and click **Next**.
Distributed port	1. On the **Networks** tab, click **Distributed Port Groups** and double-click a distributed port group . 2. On the **Ports** tab, select a port and click the **Edit settings** icon. 3. Select **Security**. 4. Select **Override** next to the properties to override.

3. Reject or accept promiscuous mode activation or MAC address changes in the guest operating system of the virtual machines attached to the distributed port group or port.

Option	Description
Promiscuous mode	• **Reject**. The VM network adapter receives only frames that are addressed to the virtual machine. • **Accept**.The virtual switch forwards all frames to the virtual machine in compliance with the active VLAN policy for the port to which the VM network adapter is connected. **Note:** Promiscuous mode is insecure mode of operation. Firewalls, port scanners, intrusion detection systems, must run in promiscuous mode.
MAC address changes	• **Reject**. If the guest OS changes the effective MAC address of the virtual machine to a value that is different from the MAC address of the VM network adapter, the switch drops all inbound frames to the adapter. If the guest OS changes the effective MAC address of the virtual machine back to the MAC address of the VM network adapter, the virtual machine receives frames again. • **Accept**. If the guest OS changes the effective MAC address of the virtual machine to a value that is different from the MAC address of the VM network adapter, the switch allows frames to the new address to pass.
Forged transmits	• **Reject**. The switch drops any outbound frame from a virtual machine adapter with a source MAC address that is different from the one in the `.vmx` configuration file. • **Accept**. The switch does not perform filtering, and permits all outbound frames.
Status	Enable or disable the MAC learning feature. The default is disabled.
Allow unicast flooding	When a packet that is received by a port has an unknown destination MAC address, the packet is dropped. With unknown

Option	Description
	unicast flooding enabled, the port floods unknown unicast traffic to every port on the switch that has MAC learning and unknown unicast flooding enabled. This property is enabled by default, if MAC learning is enabled.
MAC Limit	The number of MAC addresses that can be learned is configurable. The maximum value is 4096 per port, which is the default.
MAC Limit Policy	The policy for when the MAC limit is reached. The options are: • Drop - Packets from an unknown source MAC address are dropped. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled. • Allow - Packets from an unknown source MAC address are forwarded although the address will not be learned. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.

4. Review your settings and apply the configuration.

What is Traffic Shaping Policy

A traffic shaping policy is defined by average bandwidth, peak bandwidth, and burst size. You can establish a traffic shaping policy for each port group and each distributed port or distributed port group.

ESX shapes outbound network traffic on standard switches and inbound and outbound traffic on distributed switches. Traffic shaping restricts the network bandwidth available on a port, but can also be configured to allow bursts of traffic to flow through at higher speeds.

Average Bandwidth	Establishes the number of bits per second to allow across a port, averaged over time. This number is the allowed average load.
Peak Bandwidth	Maximum number of bits per second to allow across a port when it is sending or receiving a burst of traffic. This number limits the bandwidth that a port uses when it is using its burst bonus.
Burst Size	Maximum number of bytes to allow in a burst. If this parameter is set, a port might gain a burst bonus if it does not use all its allocated bandwidth. When the port needs more bandwidth than specified by the average bandwidth, it might be allowed to temporarily transmit data at a higher speed if a burst bonus is available. This parameter limits the number of bytes that have accumulated in the burst bonus and transfers traffic at a higher speed.

Configure Traffic Shaping for a vSphere Standard Switch or Standard Port Group

ESX lets you shape outbound traffic on standard switches or port groups. The traffic shaper restricts the network bandwidth available to any port, but you can also configure it to temporarily allow bursts of traffic to flow through a port at higher speeds.

The traffic shaping policies that you set at switch or port group level are applied at each individual port that participates in the switch or port group. For example, if you set an average bandwidth of 100000 Kbps on a standard port group, 100000 Kbps averaged over time can pass through each port that is associated with the standard port group.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Navigate to the traffic shaping policy on the standard switch or port group.

Option	Action
vSphere Standard Switch	1. Select a standard switch from the list. 2. Click **Edit settings**. 3. Select **Traffic shaping**.
Standard port group	1. Select the standard switch where the port group resides. 2. In the topology diagram, select a standard port group.

Option	Action
	3. Click **Edit settings**. 4. Select **Traffic shaping** and select **Override** next to the options to override.

4. Configure traffic shaping policies.

Option	Description
Status	Enables setting limits on the amount of networking bandwidth allocated for each port that is associated with the standard switch or port group.
Average Bandwidth	Establishes the number of bits per second to allow across a port, averaged over time (the allowed average load).
Peak Bandwidth	The maximum number of bits per second to allow across a port when it is sending a burst of traffic. This setting tops the bandwidth used by a port whenever it is using its burst bonus. This parameter can never be smaller than the average bandwidth.
Burst Size	The maximum number of bytes to allow in a burst. If this parameter is set, a port might gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than the average bandwidth specifies, the port can temporarily transmit data at a higher speed if a burst bonus is available. This parameter tops the number of bytes that can accumulate in the burst bonus and can be transferred at a higher speed.

5. For each traffic shaping policy (**Average Bandwidth**, **Peak Bandwidth**, and **Burst Size**), enter a bandwidth value.
6. Click **OK**.

Edit the Traffic Shaping Policy on a Distributed Port Group or Distributed Port

You can shape both inbound and outbound traffic on vSphere distributed port groups or distributed ports. The traffic shaper restricts the network bandwidth for any port in the group, but might also be configured to temporarily allow “bursts” of traffic to flow through a port at higher speeds.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

The traffic shaping policies that you set at distributed port group level are applied on each individual port that participates in the port group. For example, if you set an average bandwidth of 100000 Kbps on a distributed port group, 100000 Kbps averaged over time can pass through each port that is associated with the distributed port group.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Navigate to the Traffic Shaping policy for the distributed port group or port.

Option	Action
Distributed port group	1. From the **Actions** menu, select **Distributed Port Group** > **Manage Distributed Port Groups**. 2. Select **Traffic shaping** and click **Next**. 3. Select the port group and click **Next**.
Distributed port	1. On the **Networks** tab, click **Distributed Port Groups** and double-click a distributed port group . 2. On the **Ports** tab, select a port and click the **Edit distributed port settings** icon. 3. Select **Traffic shaping**. 4. Select **Override** next to the properties to override.

3. Configure traffic shaping policies.

   Note:
   The traffic is classified to ingress and egress according to the traffic direction in the switch, not in the host.

Option	Description
Status	Enable either **Ingress traffic shaping** or **Egress traffic shaping** by using the **Status** drop-down menus.
Average Bandwidth	Establishes the number of bits per second to allow across a port, averaged over time, that is, the allowed average load.
Peak Bandwidth	The maximum number of bits per second to allow across a port when it is sending/sending or receiving a burst of traffic. This parameter tops the bandwidth used by a port whenever it is using its burst bonus.
Burst Size	The maximum number of bytes to allow in a burst. If this parameter is set, a port might gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than the average bandwidth specifies, the port can temporarily transmit data at a higher speed if a burst bonus is available. This parameter tops the number of bytes that might accumulate in the burst bonus and be transferred at a higher speed.

4. Review your settings and apply the configuration.

What is Resource Allocation Policy

The Resource Allocation policy allows you to associate a distributed port or port group with a user-created network resource pool. This policy provides you with greater control over the bandwidth given to the port or port group.

For information about creating and configuring network resource pools, see vSphere Network I/O Control.

Edit the Resource Allocation Policy on a Distributed Port Group

Learn how to associate a distributed port group with a network resource pool to give you greater control over the bandwidth that is given to the distributed port group.

- Enable Network I/O Control on the distributed switch, see Enable Network I/O Control on a vSphere Distributed Switch.
- Create and configure network resource pools, see Create a Network Resource Pool.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Right-click the distributed switch in the navigator and select **Distributed Port Groups** > **Manage Distributed Port Groups**.
3. Select the **Resource allocation** check box and click **Next**.
4. Select the distributed port group to configure and click **Next**.
5. Add or remove the distributed port group from the network resource pool and click **Next**.
   - To add the distributed port group, select a user-defined resource pool from the **Network resource pool** drop-down menu.
   - To remove the distributed port group, select **default** from the **Network resource pool** drop-down menu.
6. Review your settings in the **Ready to complete** section and click **Finish**.
   Use the **Back** button to change any settings.

What is Traffic Filtering and Marking Policy

In a vSphere distributed switch, by using the traffic filtering and marking policy, you can protect the virtual network from unwanted traffic and security attacks or apply a QoS tag to a certain type of traffic.

The traffic filtering and marking policy represents an ordered set of network traffic rules for security and for QoS tagging of the data flow through the ports of a distributed switch. In general, a rule consists of a qualifier for traffic, and of an action for restricting or prioritizing the matching traffic.

The vSphere distributed switch applies rules on traffic at different places in the data stream. The distributed switch applies traffic filter rules on the data path between the virtual machine network adapter and distributed port, or between the uplink port and physical network adapter for rules on uplinks. The traffic filtering and marking policy does not support vSphere Distributed Switch configured with Network Offloads Compatibility.

Traffic Filtering and Marking on a Distributed Port Group or Uplink Port Group

Set traffic rules at the level of distributed port groups or of uplink port groups to introduce filtering and priority tagging for traffic access over virtual machines, VMkernel adapters, or physical adapters.

Enable Traffic Filtering and Marking on a Distributed Port Group or Uplink Port Group

Enable the traffic filtering and marking policy on a port group if you want to configure traffic security and marking on all virtual machine network adapters or uplink adapters that are participating in the group.

> **Note:**
> You can disable the traffic filtering and marking policy on a particular port to avoid processing the traffic flowing through the port. See Disable Traffic Filtering and Marking on a Distributed Port or Uplink Port
>
> **Note:**
> The traffic filtering and marking policy does not support vSphere Distributed Switch configured with Network Offloads Capability.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. Click the **Enable and reorder** button.
5. Click **Enable all traffic rules**.
6. Click **OK**.

Set up traffic marking or filtering on the data that is flowing through the ports of the distributed port group or through the uplink port group. See Mark Traffic on a Distributed Port Group or Uplink Port Group and Filter Traffic on a Distributed Port Group or Uplink Port Group.

Mark Traffic on a Distributed Port Group or Uplink Port Group

Learn to assign priority tags to traffic, such as VoIP and streaming video, that has higher networking requirements for bandwidth, low latency, and so on. You can mark the traffic with a CoS tag in Layer 2 of the network protocol stack or with a DSCP tag in Layer 3.

Priority tagging is a mechanism to mark traffic that has higher QoS demands. In this way, the network can recognize different classes of traffic. The network devices can handle the traffic from each class according to its priority and requirements.

You can also re-tag traffic to either raise or lower the importance of the flow. By using a low QoS tag, you can restrict data tagged in a guest operating system.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. If traffic filtering and marking is disabled, click **Enable and reorder** > **Enable all traffic rules** > **OK**.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.
6. In the network traffic rule dialog box, select the **Tag** option from the **Action** drop-down menu.
7. Set the priority tag for the traffic within the scope of the rule.

Option	Description
CoS value	Mark the traffic matching the rule with a CoS priority tag in network Layer 2. Select the check box and type a value from 0 to 7.
DSCP value	Mark the traffic associated with the rule with a DSCP tag in network Layer 3. Select the check box and type a value from 0 to 63.

8. Specify the kind of traffic that the rule is applicable to.

   To determine if a data flow is in the scope of a rule for marking or filtering, the vSphere distributed switch examines the direction of the traffic, and properties like source and destination, VLAN, next level protocol, infrastructure traffic type, and so on.

   a) From the **Traffic direction** drop-down menu, select whether the traffic must be ingress, egress, or both so that the rule recognizes it as matching.

      The direction also influences how you are going to identify the traffic source and destination.

   b) By using qualifiers for system data type, Layer 2 packet attributes, and Layer 3 packet attributes set the properties that packets must have to match the rule.

      A qualifier represents a set of matching criteria related to a networking layer. You can match traffic to system data type, Layer 2 traffic properties, and Layer 3 traffic properties. You can use the qualifier for a specific networking layer or can combine qualifiers to match packets more precisely.

      - Use the system traffic qualifier to match packets to the type of virtual infrastructure data that is flowing through the ports of the group . For example, you can select NFS for data transfers to network storage.
      - Use the MAC traffic qualifier to match packets by MAC address, VLAN ID, and next level protocol.
        Locating traffic with a VLAN ID on a distributed port group works with Virtual Guest Tagging (VGT). To match traffic to VLAN ID if Virtual Switch Tagging (VST) is active, use a rule on an uplink port group or uplink port.
      - Use the IP traffic qualifier to match packets by IP version, IP address, and next level protocol and port.

9. In the rule dialog box, click **OK** to save the rule.

Voice over IP Traffic Marking

Voice over IP (VoIP) flows have special requirements for QoS in terms of low loss and delay. The traffic related to the Session Initiation Protocol (SIP) for VoIP usually has a DSCP tag equal to 26, which stands for Assured Forwarding Class 3 with Low Drop Probability (AF31).

For example, to mark outgoing SIP UDP packets to a subnet 192.168.2.0/24, you can use the following rule:

Rule Parameter	Parameter Value
Action	Tag
DSCP value	26
Traffic direction	Egress

Rule Parameter	Parameter Value
Traffic qualifiers	IP Qualifier
Protocol	UDP
Destination port	5060
Source address	IP address matches 192.168.2.0 with prefix length 24

Filter Traffic on a Distributed Port Group or Uplink Port Group

Learn how to allow or stop traffic for securing the data that flows through the ports of a distributed port group or uplink port group.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. If traffic filtering and marking is disabled, click **Enable and reorder** > **Enable all traffic rules** > **OK**.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.
6. In the network traffic rule dialog box, use the Action options to let traffic pass through the ports of the distributed port group or uplink port group, or to restrict it.
7. Specify the kind of traffic that the rule is applicable to.

   To determine if a data flow is in the scope of a rule for marking or filtering, the vSphere distributed switch examines the direction of the traffic, and properties like source and destination, VLAN, next level protocol, infrastructure traffic type, and so on.
   a) From the **Traffic direction** drop-down menu, select whether the traffic must be ingress, egress, or both so that the rule recognizes it as matching.

      The direction also influences how you are going to identify the traffic source and destination.
   b) By using qualifiers for system data type, Layer 2 packet attributes, and Layer 3 packet attributes set the properties that packets must have to match the rule.

      A qualifier represents a set of matching criteria related to a networking layer. You can match traffic to system data type, Layer 2 traffic properties, and Layer 3 traffic properties. You can use the qualifier for a specific networking layer or can combine qualifiers to match packets more precisely.

      - Use the system traffic qualifier to match packets to the type of virtual infrastructure data that is flowing through the ports of the group . For example, you can select NFS for data transfers to network storage.
      - Use the MAC traffic qualifier to match packets by MAC address, VLAN ID, and next level protocol.
        Locating traffic with a VLAN ID on a distributed port group works with Virtual Guest Tagging (VGT). To match traffic to VLAN ID if Virtual Switch Tagging (VST) is active, use a rule on an uplink port group or uplink port.
      - Use the IP traffic qualifier to match packets by IP version, IP address, and next level protocol and port.
8. In the rule dialog box, click **OK** to save the rule.

Network Traffic Rules on a Distributed Port Group or Uplink Port Group

Learn how to define traffic rules in a distributed port group or uplink port group to introduce a policy for processing traffic related to virtual machines or to physical adapters. You can filter specific traffic or describe its QoS demands.

> **Note:**
> You can override the rules of the policy for traffic filtering and marking at port level. See Network Traffic Rules on a Distributed Port or Uplink Port.

View Traffic Rules on a Distributed Port Group or Uplink Group

View the traffic rules that form the traffic filtering and marking policy of a distributed port group or uplink port group.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. If traffic filtering and marking is disabled, click **Enable and reorder** > **Enable all traffic rules** > **OK**.
5. Examine **Action** to see if the rule filters traffic (Allow or Drop) or marks traffic (Tag) with special QoS demands.
6. Select the rule for which you want to view the criteria for locating traffic.

   The traffic qualifying parameters of the rule appear in the Traffic Qualifiers list.

Edit a Traffic Rule on a Distributed Port Group or Uplink Port Group

Learn to create or edit traffic rules, and use their parameters to configure a policy for filtering or marking the traffic on a distributed port group or uplink port group.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. If traffic filtering and marking is disabled, click **Enable and reorder** > **Enable all traffic rules** > **OK**.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.

Name the network traffic rule, and deny, allow, or tag the target traffic.

Modify Rule Priorities on a Distributed Port Group or Uplink Port Group

Learn how to reorder the rules that form the traffic filtering and marking policy of a distributed port group or uplink port group to change the sequence of actions for processing traffic.

The vSphere distributed switch applies network traffic rules in a strict order. If a packet already satisfies a rule, the packet might not be passed to the next rule in the policy.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. Click the **Enable and reorder** button.
5. If traffic filtering and marking is disabled, click **Enable all traffic rules** to enable it.
6. Select a rule and use the **Move up** or **Move down** button to change its priority.
7. Click **OK** to apply the changes.

Delete a Traffic Rule on a Distributed Port Group or Uplink Port Group

Learn how to delete a traffic rule on a distributed port group or uplink port group to stop processing packets flowing to virtual machines or physical adapters in a specific way.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. If traffic filtering and marking is disabled, click **Enable and reorder** > **Enable all traffic rules** > **OK**.
5. Select the rule and click the**Delete** button.
6. Click **OK**.

Disable Traffic Filtering and Marking on a Distributed Port Group or Uplink Port Group

Learn how to let traffic flow to virtual machines or physical adapters without additional control related to security or QoS by disabling the traffic filtering and marking policy.

> **Note:**
> You can enable and set up the traffic filtering and marking policy on a particular port. See Enable Traffic Filtering and Marking on a Distributed Port or Uplink Port.

1. Locate a distributed port group or an uplink port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups** to see the list of distributed port groups, or click **Uplink Port Groups** to see the list of uplink port groups.
2. Click a distributed port group or an uplink port group and select the **Configure** tab.
3. Under Settings, select **Traffic Filtering And Marking**.
4. Click the **Enable and reorder** button.
5. Use the toggle button to disable all traffic rules.
6. Click **OK**.

Traffic Filtering and Marking on a Distributed Port or Uplink Port

Learn how to filter traffic or describe its QoS demands for an individual virtual machine, VMkernel adapter, or physical adapter by configuring the traffic filtering and marking policy on a distributed port or uplink port.

Enable Traffic Filtering and Marking on a Distributed Port or Uplink Port

Enable the traffic filtering and marking policy on a port to configure traffic security and marking on a virtual machine network adapter, VMkernel adapter, or uplink adapter. The traffic filtering and marking policy does not support vSphere Distributed Switch configured with Network Offloads Compatibility.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.

2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. Click the **Enable and reorder** button.
5. Use the toggle button to override the default settings.
6. Optional: Click **Enable all traffic rules**.

   If traffic rules are enabled on group level, after you override the default settings for the port, the traffic rules are automatically enabled.
7. Click **OK**.

Set up traffic filtering or marking for the data flowing through the distributed port or through the uplink port. See Mark Traffic on a Distributed Port or Uplink Port and Filter Traffic on a Distributed Port or Uplink Port.

Mark Traffic on a Distributed Port or Uplink Port

Assign priority tags in a rule for traffic that needs special treatment such as VoIP and streaming video. You can mark the traffic for a virtual machine, VMkernel adapter, or physical adapter with a CoS tag in Layer 2 of the network protocol stack or with a DSCP tag in Layer 3.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

Priority tagging is a mechanism to mark traffic that has higher QoS demands. In this way, the network can recognize different classes of traffic. The network devices can handle the traffic from each class according to its priority and requirements.

You can also re-tag traffic to either raise or lower the importance of the flow. By using a low QoS tag, you can restrict data tagged in a guest operating system.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. If traffic filtering and marking is not enabled at the port level, click the Enable and reorder button, override the default settings and click **Enable all traffic rules**.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.

   You can change a rule inherited from the distributed port group or uplink port group. In this way, the rule becomes unique within the scope of the port.
6. In the network traffic rule dialog box, select the **Tag** option from the **Action** drop-down menu.
7. Set the priority tag for the traffic within the scope of the rule.

Option	Description
CoS value	Mark the traffic matching the rule with a CoS priority tag in network Layer 2. Select the check box and type a value from 0 to 7.
DSCP value	Mark the traffic associated with the rule with a DSCP tag in network Layer 3. Select the check box and type a value from 0 to 63.

8. Specify the kind of traffic that the rule is applicable to.

   To determine if a data flow is in the scope of a rule for marking or filtering, the vSphere distributed switch examines the direction of the traffic, and properties like source and destination, VLAN, next level protocol, infrastructure traffic type, and so on.

   a) From the **Traffic direction** drop-down menu, select whether the traffic must be ingress, egress, or both so that the rule recognizes it as matching.

      The direction also influences how you are going to identify the traffic source and destination.

   b) By using qualifiers for system data type, Layer 2 packet attributes, and Layer 3 packet attributes set the properties that packets must have to match the rule.

      A qualifier represents a set of matching criteria related to a networking layer. You can match traffic to system data type, Layer 2 traffic properties, and Layer 3 traffic properties. You can use the qualifier for a specific networking layer or can combine qualifiers to match packets more precisely.

      - Use the system traffic qualifier to match packets to the type of virtual infrastructure data that is flowing through the ports of the group . For example, you can select NFS for data transfers to network storage.
      - Use the MAC traffic qualifier to match packets by MAC address, VLAN ID, and next level protocol.
        Locating traffic with a VLAN ID on a distributed port group works with Virtual Guest Tagging (VGT). To match traffic to VLAN ID if Virtual Switch Tagging (VST) is active, use a rule on an uplink port group or uplink port.
      - Use the IP traffic qualifier to match packets by IP version, IP address, and next level protocol and port.

9. In the rule dialog box, click **OK** to save the rule.

Filter Traffic on a Distributed Port or Uplink Port

By using a rule, permit or stop traffic for securing data flows through a virtual machine, a VMkernel adapter, or a physical adapter.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. If traffic filtering and marking is not enabled at the port level, click the Enable and reorder button, override the default settings and click **Enable all traffic rules**.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.

   You can change a rule inherited from the distributed port group or uplink port group. In this way, the rule becomes unique within the scope of the port.
6. In the network traffic rule dialog box, select the **Allow** action to let traffic pass through the distributed port or uplink port, or the **Drop** action to restrict it.
7. Specify the kind of traffic that the rule is applicable to.

   To determine if a data flow is in the scope of a rule for marking or filtering, the vSphere distributed switch examines the direction of the traffic, and properties like source and destination, VLAN, next level protocol, infrastructure traffic type, and so on.

   a) From the **Traffic direction** drop-down menu, select whether the traffic must be ingress, egress, or both so that the rule recognizes it as matching.

      The direction also influences how you are going to identify the traffic source and destination.

b) By using qualifiers for system data type, Layer 2 packet attributes, and Layer 3 packet attributes set the properties that packets must have to match the rule.

A qualifier represents a set of matching criteria related to a networking layer. You can match traffic to system data type, Layer 2 traffic properties, and Layer 3 traffic properties. You can use the qualifier for a specific networking layer or can combine qualifiers to match packets more precisely.

- Use the system traffic qualifier to match packets to the type of virtual infrastructure data that is flowing through the ports of the group . For example, you can select NFS for data transfers to network storage.
- Use the MAC traffic qualifier to match packets by MAC address, VLAN ID, and next level protocol.
  Locating traffic with a VLAN ID on a distributed port group works with Virtual Guest Tagging (VGT). To match traffic to VLAN ID if Virtual Switch Tagging (VST) is active, use a rule on an uplink port group or uplink port.
- Use the IP traffic qualifier to match packets by IP version, IP address, and next level protocol and port.

8. In the rule dialog box, click **OK** to save the rule.

Network Traffic Rules on a Distributed Port or Uplink Port

Learn how to define traffic rules in a distributed port or uplink port group to introduce a policy for processing traffic related to a virtual machine or to a physical adapter. You can filter specific traffic or describe its QoS demands.

View Traffic Rules on a Distributed Port or Uplink Port

Review the traffic rules that form the traffic filtering and marking policy of a distributed port or uplink port.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. If traffic filtering and marking is not enabled at the port level, click the Enable and reorder button, override the default settings and click **Enable all traffic rules**.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
5. Examine **Action** to see if the rule filters traffic (Allow or Drop) or marks traffic (Tag) with special QoS demands.
6. Select the rule for which you want to view the criteria for locating traffic.

   The traffic qualifying parameters of the rule appear in the Traffic Qualifiers list.

Edit a Traffic Rule on a Distributed Port or Uplink Port

Learn how to create or edit traffic rules, and use their parameters to configure a policy for filtering or marking the traffic on a distributed port or uplink port.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.

2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. If traffic filtering and marking is not enabled at the port level, click the Enable and reorder button, override the default settings and click **Enable all traffic rules**.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
5. Click **Add** to create a new rule, or select a rule and click **Edit** to edit it.

   You can change a rule inherited from the distributed port group or uplink port group. In this way, the rule becomes unique within the scope of the port.

Name the network traffic rule, and deny, allow, or tag the target traffic.

Modify Rule Priorities on a Distributed Port or Uplink Port

Learn how to reorder the rules that form the traffic filtering and marking policy of a distributed port or uplink port to change the sequence of actions for analyzing traffic for security and QoS.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

The vSphere distributed switch applies network traffic rules in a strict order. If a packet already satisfies a rule, the packet might not be passed to the next rule in the policy.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. Click the **Enable and reorder** button.
5. If traffic filtering and marking is not enabled at the port level, override the default settings and click Enable all traffic rules.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
6. Select a rule and use the **Move up** or **Move down** button to change its priority.
7. Click **OK** to apply the changes.

Delete a Traffic Rule on a Distributed Port or Uplink Port

Delete a traffic rule on a distributed port or uplink port to stop filtering or marking certain type of packets that are flowing to a virtual machine or a physical adapter.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.

2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. If traffic filtering and marking is not enabled at the port level, click the Enable and reorder button, override the default settings and click **Enable all traffic rules**.

   If traffic rules are enabled at group level, after you override the default settings for the port, the traffic rules are automatically enabled.
5. Select the rule and click the **Delete** button.
6. Click **OK**.

Disable Traffic Filtering and Marking on a Distributed Port or Uplink Port

Disable the traffic filtering and marking policy on a port to let traffic flow to a virtual machine or a physical adapter without filtering for security or marking for QoS.

To override a policy on distributed port level, enable the port-level override option for this policy. See Configure Overriding Networking Policies on Port Level.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Select the **Traffic Filtering and Marking** tab.
4. Click the **Enable and reorder** button.
5. Use the toggle buttons to override the default settings at the port level and disable all traffic rules.
6. Click **OK**.

Qualifying Traffic for Filtering and Marking

The traffic that you want to filter or want to mark with QoS tags can be matched to the type of carried infrastructure data, such as data for storage, vCenter management, and so on, and to Layer 2 and Layer 3 properties.

To match the traffic in the scope of the rule more precisely, you can combine criteria for system data type, Layer 2 header, and Layer 3 header.

System Traffic Qualifier

By using the system traffic qualifier in a rule for port group or port, you can determine whether certain system data traffic must be marked with a QoS tag, allowed, or dropped.

System Traffic Type

You can select the type of traffic through the ports of the group that carries system data, that is, traffic for management from vCenter, storage, VMware vSphere®vMotion®, and vSphere Fault Tolerance. You can mark or filter only a specific traffic type, or for all system data traffic except for a infrastructure feature. For example, you can mark with a QoS value or filter the traffic for management from vCenter , storage and vMotion, but not the traffic carrying the Fault Tolerance data.

MAC Traffic Qualifier

By using the MAC traffic qualifier in a rule, you can define matching criteria for the Layer 2 (Data Link Layer) properties of packets such as MAC address, VLAN ID, and next level protocol that consumes the frame payload.

Protocol Type

The **Protocol type** attribute of the MAC traffic qualifier corresponds to the EtherType field in Ethernet frames. EtherType represents the type of next level protocol that is going to consume the payload of the frame.

You can select a protocol from the drop-down menu or type its hexadecimal number. For example, to capture traffic for the Link Layer Discovery Protocol (LLDP) protocol, type `88CC`.

VLAN ID

You can use the VLAN ID attribute of the MAC traffic qualifier to mark or filter traffic in a particular VLAN.

> **Note:**
>
> The VLAN ID qualifier on a distributed port group works with Virtual Guest Tagging (VGT).
>
> If a flow is tagged with a VLAN ID through Virtual Switch Tagging (VST), it cannot be located by using this ID in a rule on a distributed port group or distributed port. The reason is that the distributed switch checks the rule conditions, including the VLAN ID, after the switch has already untagged the traffic. In this case, to match traffic by VLAN ID successfully, you must use a rule on an uplink port group or uplink port.

Source Address

By using the Source Address group of attributes, you can match packets by the source MAC address or network.

You can use a comparison operator to mark or filter packets that have or do not have the specified source address or network.

You can match the traffic source in several ways.

Table 148: Patterns for Filtering or Marking Traffic by MAC Source Address

Parameters to Match Traffic Source Address	Comparison Operator	Networking Argument Format
MAC address	**is** or **is not**	Type the MAC address for matching. Use colons to separate the octets in it.
MAC network	**matches** or **does not match**	Type the lowest address in the network and a mask. Set ones at the positions of the network bits, and zeroes for the host part.

For example, for a MAC network with prefix 05:50:56 that is 23 bits long, set the address as `00:50:56:00:00:00` and mask as `ff:ff:fe:00:00:00`.

Destination Address

By using the Destination Address group of attributes, you can match packets to their destination address. The MAC destination address options have the same format as those for the source address.

Comparison Operators

To match traffic in a MAC qualifier more closely to your needs, you can use affirmative comparison or negation. You can use operators such that all packets except the ones with certain attributes fall in the scope of a rule.

IP Traffic Qualifier

By using the IP traffic qualifier in a rule, you can define criteria for matching traffic to the Layer 3 (Network Layer) properties such as IP version, IP address, next level protocol, and port.

Protocol

The **Protocol** attribute of the IP traffic qualifier represents the next level protocol consuming the payload of the packet. You can select a protocol from the drop-down menu or type its decimal number according to RFC 1700.

For the TCP and UDP protocols, you can also match traffic by source and destination ports.

Source Port

By using the Source port attribute, you can match TCP or UDP packets by the source port. Consider the traffic direction when matching traffic to a source port.

Destination Port

By using the Destination port attribute, you can match TCP or UDP packets by the destination port. Consider the traffic direction when matching traffic to a destination port.

Source Address

By using the Source Address attribute, you can match packets by source address or subnet. Consider the traffic direction when matching traffic to a source address or network.

You can match traffic source in several ways.

Table 149: Patterns for Filtering or Marking Traffic by IP Source Address

Parameters to Match Traffic Source Address	Comparison Operator	Networking Argument Format
IP version	**any**	Select the IP version from the drop-down menu.
IP address	**is** or **is not**	Type the IP address that you want to match.
IP subnet	**matches** or **does not match**	Type the lowest address in the subnet and the bit length of the subnet prefix.

Destination Address

Use the Destination Address to match packets by IP address, subnet, or IP version. The destination address has the same format as the one for the source.

Comparison Operators

To match traffic in an IP qualifier more closely to your needs, you can use affirmative comparison or negation. You can define that all packets fall in the scope of a rule except packets with certain attributes.

Manage Policies for Multiple Port Groups on a vSphere Distributed Switch

Learn how to modify networking policies for multiple port groups on a vSphere Distributed Switch.

Create a vSphere Distributed Switch with one or more port groups.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Right-click the distributed switch in the object navigator and select **Distributed Port Group** > **Manage Distributed Port Groups**.
3. On the Select port group policies page, select the check box next to the policy categories to modify and click **Next**.

Option	Description
Security	Set MAC address changes, forged transmits, and promiscuous mode for the selected port groups.

Option	Description
Traffic shaping	Set the average bandwidth, peak bandwidth, and burst size for inbound and outbound traffic on the selected port groups.
VLAN	Configure how the selected port groups connect to physical VLANs.
Teaming and failover	Set load balancing, failover detection, switch notification, and failover order for the selected port groups.
Resource allocation	Set network resource pool association for the selected port groups.
Miscellaneous	Enable or disable port blocking on the selected port groups.

4. On the Select port groups page, select the distributed port group(s) to edit and click **Next**.
5. Optional: On the Security page, use the drop-down menus to edit the security exceptions and click **Next**.

Option	Description
Promiscuous mode	• **Reject**. Placing a guest adapter in promiscuous mode has no effect on which frames are received by the adapter. • **Accept**. Placing a guest adapter in promiscuous mode causes it to detect all frames passed on the vSphere Distributed Switch that are allowed under the VLAN policy for the port group that the adapter is connected to.
MAC address changes	• **Reject**. If set to **Reject** and the guest operating system changes the MAC address of the adapter to anything other than what is in the `.vmx` configuration file, all inbound frames are dropped. If the Guest OS changes the MAC address back to match the MAC address in the `.vmx` configuration file, inbound frames are passed again. • **Accept**. Changing the MAC address from the Guest OS has the intended effect. Frames to the new MAC address are received.
Forged transmits	• **Reject**. Any outbound frame with a source MAC address that is different from the one currently set on the adapter are dropped. • **Accept**. No filtering is performed and all outbound frames are passed.

6. Optional: On the VLAN page, use the drop-down menus to edit the VLAN policy and click **Next**.

Option	Description
None	Do not use VLAN.
VLAN	In the **VLAN ID** field, enter a number between 1 and 4094.
VLAN trunking	Enter a **VLAN trunk range**.
Private VLAN	Select an available private VLAN to use.

7. Optional: On the Traffic shaping page, use the drop-down menus to enable or disable Ingress or Egress traffic shaping and click **Next**.

Option	Description
Status	If you enable either **Ingress traffic shaping** or **Egress traffic shaping**, you are setting limits on the amount of networking bandwidth allocated for each VMkernel adapter or virtual network adapter associated with this port group. If you disable the policy, services have a free, clear connection to the physical network by default.
Average bandwidth	Establishes the number of bits per second to allow across a port, averaged over time, that is, the allowed average load.
Peak bandwidth	The maximum number of bits per second to allow across a port when it is sending or receiving a burst of traffic. This maximum

Option	Description
	number tops the bandwidth used by a port whenever it is using its burst bonus.
Burst size	The maximum number of bytes to allow in a burst. If this parameter is set, a port might gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than specified by **Average bandwidth**, it might be allowed to transmit data at a higher speed if a burst bonus is available. This parameter tops the number of bytes that can be accumulated in the burst bonus and transferred at a higher speed.

8. Optional: On the Teaming and failover page, use the drop-down menus to edit the settings and click **Next**.

Option	Description
Load balancing	IP-based teaming requires that the physical switch be configured with ether channel. For all other options, ether channel should be disabled. Select how to choose an uplink. • **Route based on the originating virtual port**. Choose an uplink based on the virtual port where the traffic entered the distributed switch. • **Route based on IP hash**. Choose an uplink based on a hash of the source and destination IP addresses of each packet. For non-IP packets, whatever is at those offsets is used to compute the hash. • **Route based on source MAC hash**. Choose an uplink based on a hash of the source Ethernet. • **Route based on physical NIC load**. Choose an uplink based on the current loads of physical NICs. • **Use explicit failover order**. Always use the highest order uplink, from the list of Active adapters, which passes failover detection criteria.
Network failure detection	Select the method to use for failover detection. • **Link status only**. Relies solely on the link status that the network adapter provides. This option detects failures, such as cable pulls and physical switch power failures, but not configuration errors, such as a physical switch port being blocked by spanning tree or that is misconfigured to the wrong VLAN or cable pulls on the other side of a physical switch. • **Beacon probing**. Sends out and listens for beacon probes on all NICs in the team and uses this information, in addition to link status, to determine link failure. Do not use beacon probing with IP-hash load balancing.
Notify switches	Select **Yes** or **No** to notify switches in the case of failover. Do not use this option when the virtual machines using the port group are using Microsoft Network Load Balancing in unicast mode. If you select **Yes**, whenever a virtual NIC is connected to the distributed switch or whenever that virtual NIC's traffic is routed over a different physical NIC in the team because of a failover event, a notification is sent out over the network to update the lookup tables on physical switches. Use this process for the lowest latency of failover occurrences and migrations with vMotion. If the **Notify switches** is set to **Yes**, then all the connected ports, port groups, and distributed switches are connected back to the host when vCenter reconnects with ESX hosts.

Option	Description
Failback	Select **Yes** or **No** to disable or enable failback. This option determines how a physical adapter is returned to active duty after recovering from a failure. • **Yes** (default). The adapter is returned to active duty immediately upon recovery, displacing the standby adapter that took over its slot, if any. • **No**. A failed adapter is left inactive even after recovery until another currently active adapter fails, requiring its replacement.
Failover order	Select how to distribute the work load for uplinks. To use some uplinks but reserve others in case the uplinks in use fail, set this condition by moving them into different groups. • **Active uplinks**. Continue to use the uplink when the network adapter connectivity is up and active. • **Standby uplinks** . Use this uplink if one of the active adapter's connectivity is down. When using IP-hash load balancing, do not configure standby uplinks. • **Unused uplinks** . Do not use this uplink.

9. Optional: On the Resource allocation page, use the **Network resource pool** drop-down menu to add or remove resource allocations and click **Next**.
10. Optional: On the Miscellaneous page, select **Yes** or **No** from the drop-down menu and click **Next**.

    Select**Yes** to shut down all ports in the port group. This shutdown might disrupt the normal network operations of the hosts or virtual machines using the ports.
11. Review your settings on the Ready to complete page and click **Finish**.

    Use the **Back** button to change any settings.

Port Blocking Policies

Port blocking policies allow you to selectively block ports from sending or receiving data.

Edit the Port Blocking Policy for a Distributed Port Group

You can block all ports in a distributed port group.

Blocking the ports of a distributed port group might disrupt the normal network operations of the hosts or virtual machines using the ports.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Right-click the distributed switch in the object navigator and select **Distributed Port Group** > **Manage Distributed Port Groups**.
3. Select the **Miscellaneous** check box and click **Next**.
4. Select one or more distributed port group to configure and click **Next**.
5. From the **Block all ports** drop-down menu, enable or disable port blocking, and click **Next**.
6. Review your settings and click **Finish**.

Edit the Blocking Policy for a Distributed Port or Uplink Port

You can block an individual distributed port or uplink port.

Enable the port-level overrides. See Configure Overriding Networking Policies on Port Level

Blocking the flow through a port might disrupt the normal network operations on the host or virtual machine using the port.

1. Navigate to a distributed switch and then navigate to a distributed port or an uplink port.
   - To navigate to the distributed ports of the switch, click **Networks** > **Distributed Port Groups**, click a distributed port group from the list, and click the **Ports** tab.
   - To navigate to the uplink ports of an uplink port group, click **Networks** > **Uplink Port Groups**, click an uplink port group from the list, and click the **Ports** tab.
2. Select a port from the list.
3. Click **Edit distributed port settings**.
4. In the **Miscellaneous** section, select the **Override** check box, and from the drop-down menu enable or disable port blocking.
5. Click **OK**.

What is the MAC Learning Policy

MAC learning provides network connectivity to deployments where multiple MAC addresses are used from one vNIC.

For example, in a nested hypervisor deployment where an ESX VM runs on an ESX host and multiple VMs run inside the ESX VM. Without MAC learning, when the ESX VM's vNIC connects to a switch port, it only contains a static MAC address. VMs running inside the ESX VM do not have network connectivity because their packets have different source MAC addresses. With MAC learning, the vSwitch inspects the source MAC address of every packet coming from the vNIC, learns the MAC address in its MAC table, and allows the packet to pass. If a MAC address that is learned is not used for a certain period of time, it is removed.

MAC learning also supports unknown unicast flooding. Normally, when a packet that is received by a port has an unknown destination MAC address, the packet is dropped. With unknown unicast flooding enabled, the port floods unknown unicast traffic to every port on the switch that has MAC learning and unknown unicast flooding enabled. This property is enabled by default, but only if MAC learning is enabled.

The number of MAC addresses that can be learned is configurable. The maximum value is 4096 per port, which is the default. You can also set the policy for when the limit is reached. The options are:

- Drop - Packets from an unknown source MAC address are dropped. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.
- Allow - Packets from an unknown source MAC address are forwarded although the address will not be learned. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.

In vSphere 6.7 and later, MAC Learning can be enabled on a Distributed Virtual Port group using the vSphere API. You can configure the MAC Learning Policy on vSphere Distributed Switch, Distributed Virtual Port Group, and Distributed Virtual Port. If a MAC learning policy is not set on Distributed Virtual Port Group, it is inherited from vSphere Distributed Switch and if it is not enabled on the DVport, it is inherited from Distributed Virtual Port Group. See *vSphere Web Services API Reference* for more information.

How to Use VLANs to Isolate Network Traffic

VLANs let you segment a network into multiple logical broadcast domains at Layer 2 of the network protocol stack.

VLAN Configuration

Virtual LANs (VLANs) enable a single physical LAN segment to be further isolated so that groups of ports are isolated from one another as if they were on physically different segments.

Benefits of Using VLANs in vSphere

The VLAN configuration in a vSphere environment provides certain benefits.

- Integrates ESX hosts into a pre-existing VLAN topology.
- Isolates and secures network traffic.
- Reduces congestion of network traffic.

VLAN Tagging Modes

vSphere supports three modes of VLAN tagging in ESX: External Switch Tagging (EST), Virtual Switch Tagging (VST), and Virtual Guest Tagging (VGT).

Tagging Mode	VLAN ID on switch port groups	Description
EST	0	The physical switch performs the VLAN tagging. The host network adapters are connected to access ports on the physical switch.
VST	Between 1 and 4094	The virtual switch performs the VLAN tagging before the packets leave the host. The host network adapters must be connected to trunk ports on the physical switch.
VGT	• 4095 for standard switch • Range of and individual VLANs for distributed switch	The virtual machine performs the VLAN tagging. The virtual switch preserves the VLAN tags when it forwards the packets between the virtual machine networking stack and the external switch. The host network adapters must be connected to trunk ports on the physical switch. The vSphere Distributed Switch supports a modification of VGT. For security reasons, you can configure a distributed switch to pass only packets that belong to particular VLANs. **Note:** For VGT you must have an 802.1Q VLAN trunking driver that is installed on the guest operating system of the virtual machine.

Private VLANs

Private VLANs are used to solve VLAN ID limitations by adding a further segmentation of the logical broadcast domain into multiple smaller broadcast subdomains.

A private VLAN is identified by its primary VLAN ID. A primary VLAN ID can have multiple secondary VLAN IDs associated with it. Primary VLANs are **Promiscuous**, so that ports on a private VLAN can communicate with ports configured as the primary VLAN. Ports on a secondary VLAN can be either **Isolated**, communicating only with promiscuous ports, or **Community**, communicating with both promiscuous ports and other ports on the same secondary VLAN.

To use private VLANs between a host and the rest of the physical network, the physical switch connected to the host needs to be private VLAN-capable and configured with the VLAN IDs being used by ESX for the private VLAN functionality. For physical switches using dynamic MAC+VLAN ID based learning, all corresponding private VLAN IDs must be first entered into the switch's VLAN database.

Create a Private VLAN

Create the necessary private VLANs on the vSphere Distributed Switch to be able to assign distributed ports to participate to a private VLAN.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and select **Private VLAN** .
3. Click **Edit**.
4. To add a primary VLAN, click **Add**.
   The primary private VLAN also appears under Secondary Private VLAN ID.
5. To add a secondary VLAN, in the right pane click **Add**.
6. From the drop-down menu in the **Secondary VLAN type** column, select either **Isolated** or **Community**.
7. Click **OK**.

Configure a distributed port group or port to associate traffic with the private VLAN. See Configure VLAN Tagging on a Distributed Port Group or Distributed Port.

Remove a Primary Private VLAN

Remove unused primary VLANs from the configuration of a vSphere Distributed Switch.

Verify that no port groups are configured to use the primary VLAN and its associated secondary VLANs.

When you remove a primary private VLAN, you also remove the associated secondary private VLANs.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and select **Private VLAN**.
3. Click **Edit**.
4. Select the primary private VLAN to remove.
5. Click `Remove` above the Primary VLAN ID list.
6. Click **OK.**

Remove a Secondary Private VLAN

Remove unused secondary private VLANs from the configuration of a vSphere Distributed Switch.

Verify that no port groups are configured to use the secondary VLAN.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and select **Private VLAN**.
3. Click **Edit**.
4. Select a primary private VLAN.
   The secondary private VLANs associated with it appear on the right.
5. Select the secondary private VLAN to remove.
6. Above the secondary VLAN ID list, click the **times sign (x)**button and click **OK**.

Manage Network Resources

vSphere provides several different methods to help you manage your network resources.

DirectPath I/O

DirectPath I/O allows virtual machine access to physical PCI functions on platforms with an I/O Memory Management Unit.

The following features are unavailable for virtual machines that are configured with DirectPath:

- Hot adding and removing of virtual devices

- Suspend and resume
- Record and replay
- Fault tolerance
- High availability
- DRS (limited availability. The virtual machine can be part of a cluster, but cannot migrate across hosts)
- Snapshots

Enable Passthrough for a Network Device on a Host

Passthrough devices provide the means to use resources efficiently and improve the performance of your environment. You can enable DirectPath for a network device on a host.

> **CAUTION:**
> If your ESX host is configured to boot from a USB device or an SD card attached to a USB channel, make sure that you do not enable DirectPath for the USB controller. Passing through a USB controller on an ESX host that boots from a USB device or SD card might put the host in a state where its configuration cannot be persisted.

1. Browse to a host in the vSphere Client navigator.
2. On the **Configure** tab, expand **Hardware** and click **PCI Devices**.
3. Locate and select the device on the host, click **Toggle Passthrough**.

   A list of available passthrough devices appears.

Icon	Description
green icon	The name of the device.
orange icon	The identifier of the device.

4. Select the network device to be used for passthrough and click **Ok**.

   The selected PCI device appears in the table. Device information is displayed at the bottom of the screen.

Configure DirectPath Mode

DirectPath allows virtual machines to connect directly to the physical devices on the same host through the devices that use the Passthrough method.

The DirectPath view integrates configuration for all the devices that can be used by virtual workloads in a single interface.

1. Browse to a host in the vSphere Client navigator.
2. On the **Configure** tab, expand **Hardware** and click **DirectPath**.

   A list of all PCI devices consumable through DirectPath is displayed. You can view the name of the device, identifier of the device, driver name, driver version, and the number of VMs using these devices. If the device has memory, and device memory usage is integrated with vSphere, memory metrics provide information on how device memory and reservation are used under the columns **Total Memory, Used Memory, Reserved Memory, and Used Reserved Memory.**
3. Select the PCI device and click **Configure DirectPath Mode**. The selected PCI device appears in the table. Device information is displayed at the bottom of the screen.

   You can only configure one device at a time. If multiple devices are selected, **Configure DirectPath Mode** is not available.

   When one or more PCI devices are selected, a detailed view appears below the PCI device list.

   - Show VM Associations: Displays all the VMs running on the selected devices

- Show VM Class Associations: Displays a list of VM classes from which a VM was incorporated and is using the selected devices. The reservations that are used in a VM class which has a reserved slot on a PCI device is also displayed.
- Show Profile Associations: Displays the list of all DirectPath profiles that are used by a VM running on a PCI device. To view the DirectPath Profiles Utilization, click and select a profile name or navigate to the **Host > Monitor > Hardware Health > DirectPath Profiles Utilization**. The DirectPath Profiles Utilization view provides the number of configurations that are in use and the number of configurations that remain free.

4. Click **Configure**.

Configure a PCI Device on a Virtual Machine

Passthrough devices provide the means to more efficiently use resources and improve performance in your environment. You can configure a passthrough PCI device on a virtual machine in the vSphere Client.

Verify that a passthrough networking device is configured on the host of the virtual machine. See Enable Passthrough for a Network Device on a Host.

When using passthrough devices with a Linux kernel version 2.6.20 or earlier, avoid MSI and MSI-X modes because these modes impact performance significantly.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Power off the virtual machine.
3. From the Actions menu, select Edit Settings.
4. Select the **Virtual Hardware** tab in the dialog box displaying the settings.
5. Expand the **Memory** section, and set the **Limit** to **Unlimited**.
6. Click the **Add new device** button and under Other devices, select **PCI Device**.
   The **New PCI device** drop-down menu is added to the list in the **Virtual Hardware** tab.
7. From the **New PCI device** drop-down menu, select the passthrough device to use, and click **OK**.
8. Power on the virtual machine.

Adding a DirectPath I/O device to a virtual machine sets memory reservation to the memory size of the virtual machine.

Hot-add and Hot-remove support for VM DirectPath Devices

Passthrough devices provide the means to use resources efficiently and improve the performance of your environment.

In vSphere 8.0 and earlier, passthrough devices were added or removed only when the VM was powered off. The current versions of vSphere offers hot-add and hot-remove support for VM DirectPath devices. Hot-add refers to the procedure of adding a passthrough device when the VM is powered on.

The following features are available for hot-add and hot-remove support for VM DirectPath devices.

- Hot-add of a passthrough enabled device to a VM by vSphere API. You can perform the operation in one of the following ways:
  a. You enable the passthrough for an existing device that is discovered at ESX boot time and hot-add the device to the VM.
  b. Alternatively, you can physically hot-add a new device to an empty PCIe hot-plug slot of the host, enable passthrough for the newly added device, and then hot-add the device to the VM.

  Note:
  This feature requires PCIe Native hot plug certification for the server.
- Hot-remove of a passthrough device from a VM by vSphere API.
- Surprise hot-remove of passthrough device which is in use by a VM, can be physically pulled out of the device from the host.

Note:
This feature requires PCIe Native surprise hot plug certification for the server.

Note:
Hot-add and hot-remove of VM DirectPath device is supported for only NVMe devices. Support for hot-add and hot-remove of VM DirectPath is not available in the vSphere Client.

The following features are not supported for VMs with hot plug of passthrough enabled.

- Input–output Memory Management Unit (IOMMU) exposed to the guest operating system.
- CPU hot plug.
- Memory hot plug.
- Dynamic Direct path I/O.
- Support for VMs with the number of vCPUs > 128.
- Virtual interrupts also known as interrupt posting might be disabled for VMs with the hot plug of the passthrough enabled.
- Graceful physical hot-removal of passthrough device from the host is not supported.
- UI support for hot-add and hot-remove.
- All features that are unavailable with VM Direct Path Devices.

Requirements for Platforms, Devices, and Guest Operating Systems

Server and Device Requirements

- Ensure that the hot-add and hot-remove of VMDirectPath I/O devices feature is supported by your server Original equipment manufacturer (OEM).
- The server platform firmware must be Unified Extensible Firmware Interface (UEFI).
- Ensure that the server platform and device conforms to the requirements mentioned in Broadcom knowledge base article 312208.
- The server should be certified for the VM Direct Path I/O.
- For surprise hot-remove of a passthrough device from a VM the server model should be certified for PCIe Native surprise hot plug.

Note:
For more information on server certification, see Broadcom Compatibility Guide.

Guest Operating Systems Requirements

- Ensure that the Guest Operating System has support of NVMe hot plug and NVMe surprise hot-removal.
- Use the latest and stable GOS distribution that contains all the fixes of NVMe driver I/O stack.

The vCenter and ESX Requirements

The ESX and the vCenter versions must be 8.0 update 1 or later.

Limits

- Maximum number of passthrough devices that can be hot-added in a single `ReconfigureVM` API call is 1.
- Maximum number of passthrough devices that can be hot-removed in a single `ReconfigureVM` API call is 1.
- Other VM reconfiguration changes cannot be requested in the same `ReconfigureVM` API call when hot-add or hot-remove a passthrough device is requested.
- The maximum number of NVMe passthrough devices that a VM can support is 32.

Enable Hot-Add and Hot-Remove of VM DirectPath Devices

You can enable hot-add and hot-remove of VMDirectPath I/O devices by preparing your vCenter and the ESX host.

Prepare the vCenter and the ESX host.

- Deploy the vCenter vCenter and stateful install ESX with the build combination.
- Set the VMKernel boot parameter.
  ```
  set -s maxIntrCookies -v 4096
  ```
- Restart the ESX host.
- Deploy the data center and add the host.

1. Browse to a host in the vSphere Client.
2. On the **Configure** tab, expand **Hardware** and click **PCI Devices**.
3. Select the **NVMe PCI** device and press **Toggle Passthrough**.

Prepare and configure the VM.

Prepare and configure VM

Passthrough devices provide the means to use resources efficiently and improve the performance of your environment. You can prepare and configure your virtual machine.

Deploy and configure the VM (UEFI boot).

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and select **Edit Settings**.
3. Expand the **Memory** section and set the **Reserve all guest memory (All locked)**.
4. Click **Save**.
5. Set **fixedPassthruHotPlugEnabled** to **TRUE** for the VM using vSphere API `ReconfigVM_Task()` For more information, see https://developer.broadcom.com.

   Note:
   Search for the term *Hotplug* using the search bar in https://developer.broadcom.com.

   a) On vCenter , **Right click on VM** → **Compatibility** → **Upgrade VM Compatibility** to **vmx-20** and click save.
   b) Set `motherboardLayout` to `ACPI` using the vSphere API. For more information, see https://developer.broadcom.com.

   Note:
   Search for the term *Hotplug* using the search bar in https://developer.broadcom.com.
6. (Optional) Add the passthrough enabled PCI devices to the VM.
7. Power on the VM.

   Note:
   Verify if the ongoing hot-plug operation has succeeded before proceeding with the next hot-plug operation.
8. Perform hot-add and hot-remove of passthrough devices using the vSphere API. For more information, see https://developer.broadcom.com.

   Note:
   Search for the term *Hotplug* using the search bar in https://developer.broadcom.com/.

Verify if the hot plug operation has succeeded.

Verify If the Hot Plug Operation is Successful using vSphere Client

To check if your hot-add, hot-remove, and surprise hot-remove operation has succeeded, you can verify using the vSphere Client.

1. Locate the virtual machine in the vSphere Client.
2. Select a data center, folder, cluster, resource pool, or host and click the VMs tab.
3. Click **Virtual Machines** and click the virtual machine from the list.
4. In the **PCI Devices** table, verify the following.

Option	Description
Verify if the hot-add of a passthrough -enabled device to a VM using vSphere API is successful.	The hot-added passthrough device is displayed.
Verify if the hot-remove of a passthrough device from a VM using vSphere API is successful.	The hot-removed passthrough device is not displayed.
Verify if the surprise hot-remove (physical hot-remove) of a passthrough device of a VM from the ESX host is successful.	• Surprise hot-removed device is not displayed. • In the Host View, a. Browse to a host in the vSphere Client. b. On the **Configure** tab, expand **Hardware**, and click **PCI Devices**. c. Verify that the surprise hot-removed device is not displayed.

Verify If the Hot Plug Operation is Successful using VM

To check if your hot-add, hot-remove, and surprise hot-remove operation has succeeded, you can verify using the virtual machine.

1. In the Virtual Machine, open an SSH connection to the Guest Operating System.
2. Log in as the root user.
3. Run the command `lspci`.
4. Verify the results.

Option	Description
Verify if the hot-add of a passthrough -enabled device to a VM using vSphere API is successful.	The hot-added passthrough device is displayed. **Note:** The passthrough device might have a different Spotfire Binary Data File (SBDF) address than the one on the host.
Verify if the hot-remove of a passthrough device from a VM by using vSphere API is successful.	The hot-removed passthrough device is not displayed.
Verify if the surprise hot-remove (physical hot-remove) of a passthrough device of a VM from the ESX host is successful.	The surprise hot-removed device is not displayed.

What is Single Root I/O Virtualization (SR-IOV)

vSphere supports Single Root I/O Virtualization (SR-IOV). You can use SR-IOV for networking of virtual machines that are latency sensitive or require more CPU resources.

Overview of SR-IOV

SR-IOV is a specification that allows a single Peripheral Component Interconnect Express (PCIe) physical device under a single root port to appear as multiple separate physical devices to the hypervisor or the guest operating system.

SR-IOV uses physical functions (PFs) and virtual functions (VFs) to manage global functions for the SR-IOV devices. PFs are full PCIe functions that are capable of configuring and managing the SR-IOV functionality. It is possible to configure or control PCIe devices using PFs, and the PF has full ability to move data in and out of the device. VFs are lightweight PCIe functions that support data flowing but have a restricted set of configuration resources.

The number of virtual functions provided to the hypervisor or the guest operating system depends on the device. SR-IOV enabled PCIe devices require appropriate BIOS and hardware support, as well as SR-IOV support in the guest operating system driver or hypervisor instance. See SR-IOV Support.

Using SR-IOV in vSphere

In vSphere, a virtual machine can use an SR-IOV virtual function for networking. The virtual machine and the physical adapter exchange data directly without using the VMkernel as an intermediary. Bypassing the VMkernel for networking reduces latency and improves CPU efficiency.

In vSphere, though a virtual switch (standard switch or distributed switch) does not handle the network traffic of an SR-IOV enabled virtual machine connected to the switch, you can control the assigned virtual functions by using switch configuration policies at port group or port level.

SR-IOV Support

vSphere supports SR-IOV in an environment with specific configuration only. Some features of vSphere are not functional when SR-IOV is enabled.

Supported Configurations

To use SR-IOV in vSphere, your environment must meet several configuration requirements.

Table 150: Supported Configurations for Using SR-IOV

Component	Requirements
Physical host	• Must be compatible with the ESX release. • Must have an Intel or AMD processor. • Must support I/O memory management unit (IOMMU), and must have IOMMU enabled in the BIOS. • Must support SR-IOV, and must have SR-IOV enabled in the BIOS. Contact the server vendor to determine whether the host supports SR-IOV.
Physical NIC	• Must be compatible with the ESX release. • Must be supported for use with the host and SR-IOV according to the technical documentation from the server vendor. • Must have SR-IOV enabled in the firmware. • Must use MSI-X interrupts.
PF driver in ESX for the physical NIC	• Must be certified by VMware. • Must be installed on the ESX host. The ESX release provides a default driver for certain NICs, while for others you must download and manually install it.
Guest OS	Must be supported by the NIC on the installed ESX release according to the technical documentation from the NIC vendor.
VF driver in the guest OS	• Must be compatible with the NIC. • Must be supported on the guest OS release according to the technical documentation from the NIC vendor. • Must be Microsoft WLK or WHCK certified for Windows virtual machines. • Must be installed on the operating system. The operating system release contains a default driver for certain NICs, while for others you must download and install it from a location provided by the vendor of the NIC or the host.

To verify that your physical hosts and NICs are compatible with ESX releases, see the *Broadcom Compatibility Guide*.

Availability of Features

The following features are not available for virtual machines configured with SR-IOV:

- vSphere vMotion
- Storage vMotion
- vShield
- VXLAN Virtual Wire
- vSphere High Availability
- vSphere Fault Tolerance
- vSphere DRS
- vSphere DPM
- Virtual machine suspend and resume
- Virtual machine snapshots
- MAC-based VLAN for passthrough virtual functions
- Hot addition and removal of virtual devices, memory, and vCPU
- Participation in a cluster environment
- Network statistics for a virtual machine NIC using SR-IOV passthrough

> **Note:**
> Attempts to enable or configure unsupported features with SR-IOV in the result in unexpected behavior in your environment.

Supported NICs

All NICs must have drivers and firmware that support SR-IOV. Some NICs might require SR-IOV to be enabled on the firmware. To learn what NICs are supported for virtual machines configured with SR-IOV, see the Broadcom Compatibility Guide.

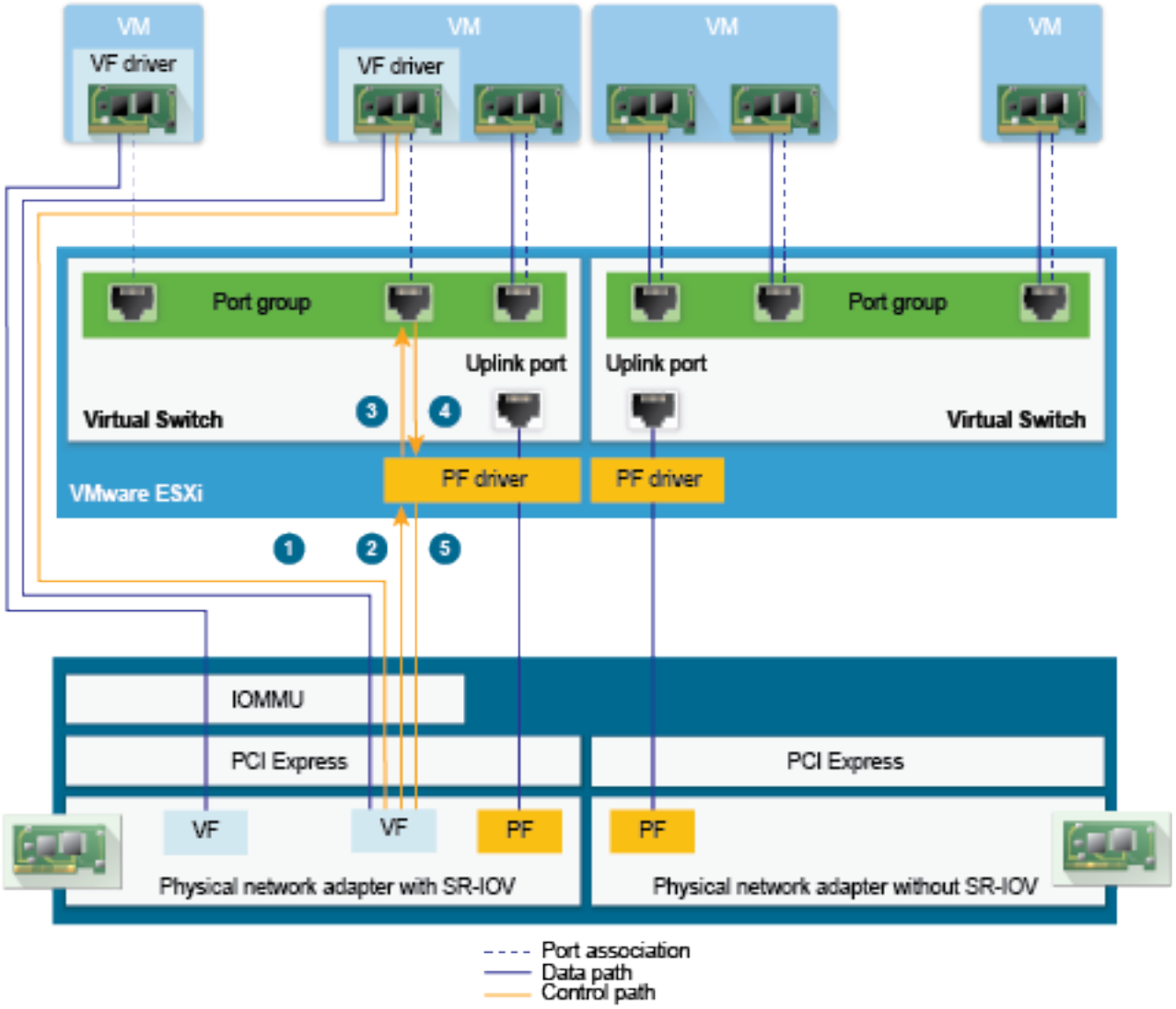
SR-IOV Component Architecture and Interaction

vSphere SR-IOV support relies on the interaction between the virtual functions (VFs) and the physical function (PF) of the NIC port for better performance, and interaction between the driver of the PF and the host switch for traffic control.

In a host that runs virtual machine traffic on top of SR-IOV physical adapters, virtual machine adapters directly contact the virtual functions to communicate data. However, the ability to configure networks is based on the active policies for the port holding the virtual machines.

On an ESX host without SR-IOV, the virtual switch sends external network traffic through its ports on the host from or to the physical adapter for the relevant port group. The virtual switch also applies the networking policies on managed packets.

Figure 59: Data and Configuration Paths in the SR-IOV Support of vSphere



Data Path in SR-IOV

After the virtual machine network adapter is assigned to a virtual function, the VF driver in the guest operating system uses the I/O memory management unit (IOMMU) technology to access the virtual function that must receive or send the data over the network. The VMkernel, that is, the virtual switch in particular, does not process the data flow, which reduces the overall latency of SR-IOV enabled workloads.

Configuration Path in SR-IOV

If the guest operating system attempts to change the configuration of a virtual machine adapter mapped to a VF, the change takes place if it is allowed by the policy on the port associated with the virtual machine adapter.

The configuration workflow consists of the following operations:

1. The guest operating system requests a configuration change on the VF.
2. The VF forwards the request to the PF through a mailbox mechanism.
3. The PF driver checks the configuration request with the virtual switch (standard switch or host proxy switch of a distributed switch).
4. The virtual switch verifies the configuration request against the policy on the port with which the VF enabled virtual machine adapter is associated.

5. The PF driver configures the VF if the new settings are in compliance with the port policy of the virtual machine adapter.

   For example, if the VF driver tries to modify the MAC address, the address remains the same if MAC address change is not allowed in the security policy for the port group or port. The guest operating system might show that the change is successful but a log message indicates that the operation has failed. As a result, the guest operating system and the virtual device save different MAC addresses. The network interface in the guest operating system might not be able to acquire an IP address and communicate. In this case, you have to reset the interface in the guest operating system to get the latest MAC address from the virtual device and acquire an IP address.

vSphere and Virtual Function Interaction

Virtual functions (VFs) are lightweight PCIe functions that contain all the resources necessary for data exchange but have a minimized set of configuration resources. Interaction between vSphere and VFs is limited.

- The physical NIC must use MSI-X interrupts.
- VFs do not implement rate control in vSphere. Every VF can potentially use the entire bandwidth of a physical link.
- When a VF device is configured as a passthrough device on a virtual machine, the standby and hibernate functions for the virtual machine are not supported.
- The maximum number of VFs that you can create and the maximum number of VFs that you can use for passthrough are different. The maximum number of VFs that you can instantiate depends on the NIC capability and on the hardware configuration of the host. However, due to the limited number of interrupt vectors available for passthrough devices, only a limited number of all instantiated VFs can be used on an ESX host.

  The total number of interrupt vectors on each ESX host can scale up to 4096 in the case of 32 CPUs. When the host boots, devices on the host such as storage controllers, physical network adapters, and USB controllers consume a subset of the 4096 vectors. If these devices require more than 1024 vectors, the maximum number of potentially supported VFs is reduced.
- The number of VFs that is supported on an Intel NIC might be different from the number that is supported on an Emulex NIC. See the technical documentation from the NIC vendor.
- If you have Intel and Emulex NICs present with SR-IOV enabled, the number of VFs available for the Intel NICs depends on how many VFs are configured for the Emulex NIC, and the reverse. You can use the following formula to estimate the maximum number of VFs for use if all 3072 interrupt vectors are available for passthrough:

  ```
  3X + 2Y < 3072
  ```

  where `X` is the number of Intel VFs, and `Y` is the number of Emulex VFs.

  This number might be smaller if other types of devices on the host use more than 1024 interrupt vectors from the total of 4096 vectors on the host.
- vSphere SR-IOV supports up to 1024 VFs on supported Intel and Emulex NICs.
- vSphere SR-IOV supports up to 64 VFs on a supported Intel or Emulex NIC.
- If a supported Intel NIC loses connection, all VFs from the physical NIC stop communication completely, including that between VFs.
- If a supported Emulex NIC loses connection, all VFs stop communication with the external environment, but communication between VFs still works
- VF drivers offer many different features, such as IPv6 support, TSO, and LRO checksum. See the technical documentation of the NIC vendor for more details.

DirectPath I/O vs SR-IOV

SR-IOV offers performance benefits and tradeoffs similar to those of DirectPath I/O. DirectPath I/O and SR-IOV have similar functionality but you use them to accomplish different things.

SR-IOV is beneficial in workloads with very high packet rates or very low latency requirements. Like DirectPath I/O, SR-IOV is not compatible with certain core virtualization features, such as vMotion. SR-IOV does, however, allow for a single physical device to be shared amongst multiple guests.

With DirectPath I/O you can map only one physical function to one virtual machine. SR-IOV lets you share a single physical device, allowing multiple virtual machines to connect directly to the physical function.

Configure a Virtual Machine to Use SR-IOV

To use the capabilities of SR-IOV, you must enable the SR-IOV virtual functions on the host and connect a virtual machine to the functions.

Verify that the configuration of your environment supports SR-IOV, see SR-IOV Support.

The traffic passes from an SR-IOV passthrough adapter to the physical adapter in compliance with the active policy on the associated port on the standard or distributed switch.

To examine which virtual function is assigned to an SR-IOV passthrough network adapter, on the **Summary** tab for the virtual machine expand the **VM Hardware** panel and check the properties of the adapter.

The topology diagram of the switch marks virtual machine adapters that use virtual functions with the

icon.

Set up the traffic passing through the virtual functions attached to the virtual machine by using the networking policies on the switch, port group, and port. See Networking Options for the Traffic Related to an SR-IOV Enabled Virtual Machine.

Enable SR-IOV on a Host Physical Adapter

Before you can connect virtual machines to virtual functions, use the vSphere Client to enable SR-IOV and set the number of virtual functions on your host.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Physical adapters**.
   You can look at the SR-IOV property to see whether a physical adapter supports SR-IOV.
3. Select the physical adapter and click **Edit adapter settings**.
4. Under SR-IOV, select **Enabled** from the **Status** drop-down menu.
5. In the **Number of virtual functions** text box, type the number of virtual functions that you want to configure for the adapter.
   A value of 0 means SR-IOV is not enabled for that physical function.
6. Click **OK**.
7. Restart the host.

The virtual functions become active on the NIC port represented by the physical adapter entry. They appear in the PCI Devices list in the **Settings** tab for the host.

You can use the `esxcli network sriovnic` vCLI commands to examine the configuration of virtual functions on the host.

Associate a virtual machine with a virtual function through an SR-IOV passthrough network adapter.

Assign a Virtual Function as SR-IOV Passthrough Adapter to a Virtual Machine

To ensure that a virtual machine and a physical NIC can exchange data, you must associate the virtual machine with one or more virtual functions as SR-IOV passthrough network adapters.

- Verify that the virtual functions exist on the host.
- Verify that the passthrough networking devices for the virtual functions are active in the PCI Devices list on the **Settings** tab for the host.
- Verify that the virtual machine compatibility is ESX 5.5 and later.
- Verify that Red Hat Enterprise Linux 6 and later or Windows has been selected as the guest operating system when the virtual machine was created.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Power off the virtual machine.
3. From the Actions menu, select Edit Settings.
4. Select the **Virtual Hardware** tab in the dialog box displaying the settings.
5. From the **Add new device** drop-down menu, select **Network Adapter**.
   The New Network section is added to the list in the **Virtual Hardware** tab.
6. Expand the New Network section and connect the virtual machine to a port group.

   The virtual NIC does not use this port group for data traffic. The port group is used to extract the networking properties, for example, VLAN tagging, to apply on the data traffic.

   SR-IOV allows a VM to communicate directly through a virtual function of the physical NIC. While this function improves performance, it bypasses most of the virtual switch controls. If spoofing is not restricted, it can create security risks. For more information on how to prevent MAC Spoofing, see Prevent MAC spoofing for VMs over SR-IOV.
7. From the **Adapter type** drop-down menu, select **PCI Device passthrough**.
8. From the **Physical function** drop-down menu, select the physical adapter to back the passthrough virtual machine adapter.
9. To allow changes in the MTU of packets from the guest operating system, use the **Guest OS MTU Change** drop-down menu.
10. Expand the Memory section, select **Reserve all guest memory (All locked)** and click **OK**.
    I/O memory management unit (IOMMU) must reach all virtual machine memory so that the passthrough device can access the memory by using direct memory access (DMA).
11. Power on the virtual machine.

When you power on the virtual machine, the ESX host selects a free virtual function from the physical adapter and maps it to the SR-IOV passthrough adapter. The host validates all properties of the virtual machine adapter and the underlying virtual function against the settings of the port group to which the virtual machine belongs.

Prevent MAC spoofing for VMs over SR-IOV

MAC spoofing is the ability for a virtual machine to send traffic using a MAC address different from the one that was assigned originally. Security policies usually isolate VMs by rejecting forged or unauthorized traffic. This policy ensures protection against IP/MAC impersonation, ARP spoofing, and rogue network activity.

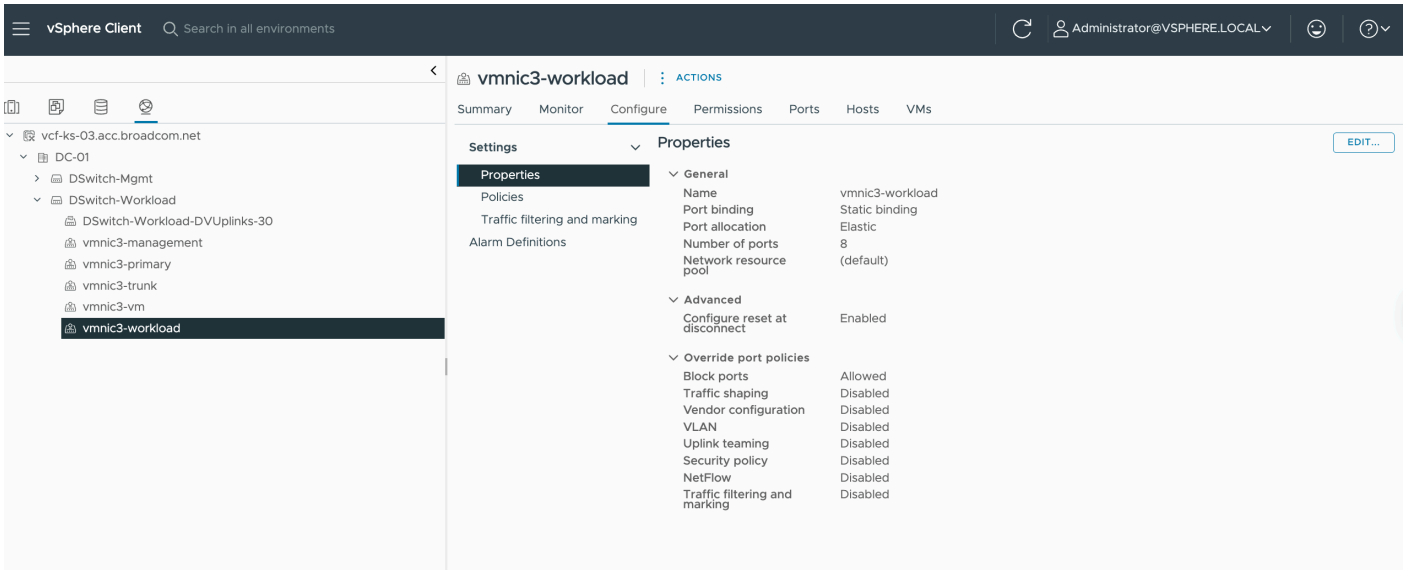
SR-IOV allows a VM to communicate directly through a VF of the physical NIC. This function improves performance but also bypasses many of the virtual switch controls. If spoofing is not restricted, it can create security risks.

By default **MAC Address Changes** is set to **Accept** when SR-IOV port groups are configured, allowing users to modify MAC address.

To control MAC spoofing for VMs over SR-IOV, **MAC Address Changes** should be set to **Reject** in the PortGroup configuration.

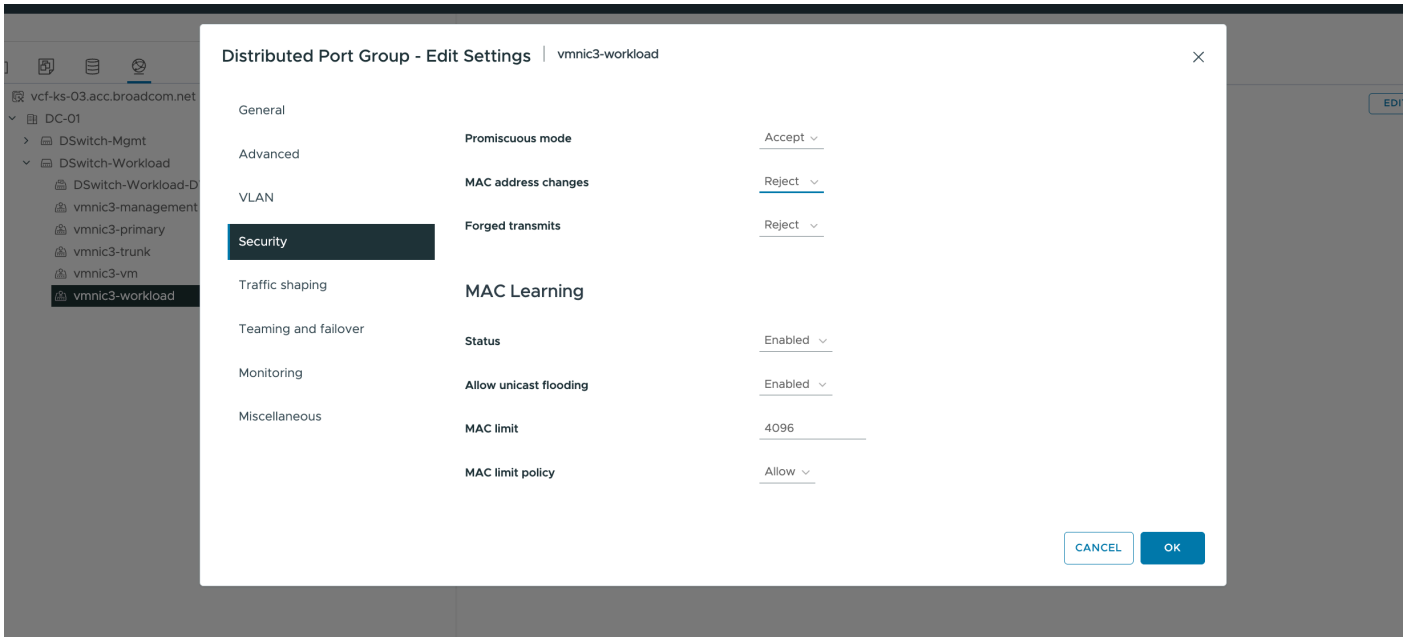
1. In the vSphere Client, navigate to **Networking > Port Groups**.
2. Find and select your **SR-IOV** port group and click **Edit.**

   SR-IOV Port Group


vSphere Client
Search in all environments
Administrator@VSPHERE.LOCAL
vcf-ks-03.acc.broadcom.net
DC-01
DSwitch-Mgmt
DSwitch-Workload
DSwitch-Workload-DVUplinks-30
vmnic3-management
vmnic3-primary
vmnic3-trunk
vmnic3-vm
vmnic3-workload
vmnic3-workload
ACTIONS
Summary
Monitor
Configure
Permissions
Ports
Hosts
VMs
Settings
Properties
Policies
Traffic filtering and marking
Alarm Definitions
Properties
EDIT...
General
Name vmnic3-workload
Port binding Static binding
Port allocation Elastic
Number of ports 8
Network resource pool (default)
Advanced
Configure reset at disconnect Enabled
Override port policies
Block ports Allowed
Traffic shaping Disabled
Vendor configuration Disabled
VLAN Disabled
Uplink teaming Disabled
Security policy Disabled
NetFlow Disabled
Traffic filtering and marking Disabled

3. Click **Security** and set **MAC Address Changes** to **Reject.**

Figure 60: Set MAC address Changes

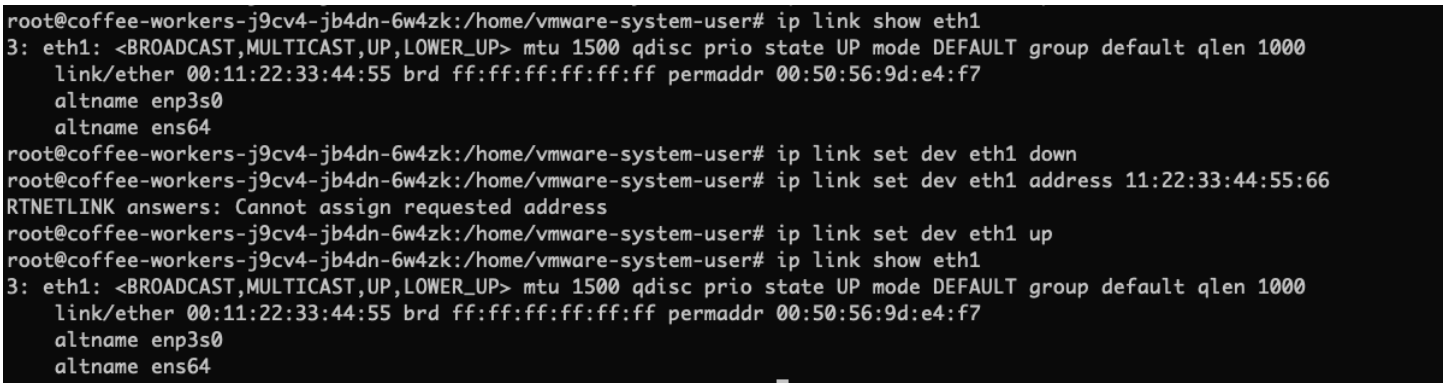


4. Click **Ok**.

Validate if MAC spoof is prevented based on the configuration When you attempt to update the MAC address again, you might get an error message.

```
"RTNETLINK answers: Cannot assign requested address"
```

Figure 61: Validate MAC Spoofing



Networking Options for the Traffic Related to an SR-IOV Enabled Virtual Machine

In vSphere you can configure certain networking features on a virtual machine adapter that is associated virtual function (VF). Use settings for the switch, for the port group, or for a port depending on the type of the virtual switch (standard or distributed) that handles the traffic.

Table 151: Networking Options for a Virtual Machine Adapter That Uses a VF

Networking Option	Description
MTU size	Change the size of the MTU, for example, to enable jumbo frames.
Security policy for VF traffic	• If the guest operating system changes the initially set MAC address of a virtual machine network adapter that uses a VF, accept or drop incoming frames for the new address by setting the **MAC address changes** option. • Enable global promiscuous mode for virtual machine network adapters, including adapters that use VFs.
VLAN tagging mode	Configure VLAN tagging in the standard or distributed switch, and let the tagged traffic reach the virtual machines that are associated with VFs, that is, enable Virtual Guest Tagging (VGT).

Using an SR-IOV Physical Adapter to Handle Virtual Machine Traffic

In vSphere both the physical function (PF) and virtual functions (VFs) of an SR-IOV capable physical adapter can be configured to handle virtual machine traffic.

The PF of an SR-IOV physical adapter controls the VFs that virtual machines use, and can carry the traffic flowing through the standard or distributed switch that handles the networking of these SR-IOV enabled virtual machines.

The SR-IOV physical adapter works in different modes depending on whether it backs the traffic of the switch.

Mixed Mode

The physical adapter provides virtual functions to virtual machines attached to the switch and directly handles traffic from non SR-IOV virtual machines on the switch.

You can check whether an SR-IOV physical adapter is in mixed mode in the topology diagram of the switch. An SR-IOV physical adapter in mixed mode appears with the

icon in the list of physical adapters for a standard switch or in the list of uplink group adapters for a distributed switch.

SR-IOV Only Mode

The physical adapter provides virtual functions to virtual machines connected to a virtual switch, but does not back traffic from non SR-IOV virtual machines on the switch.

To verify whether the physical adapter is in SR-IOV only mode, examine the topology diagram of the switch. In this mode, the physical adapter is in a separate list called External SR-IOV Adapters and appears with the

icon.

Non SR-IOV Mode

The physical adapter is not used for traffic related to VF aware virtual machines. It handles traffic from non SR-IOV virtual machines only.

Enable SR-IOV by Using Host Profiles or an ESXCLI Command

You can configure the virtual functions on an ESX host by using an ESXCLI command, or by using a host profile to set up multiple hosts simultaneously or to set up stateless hosts.

Enable SR-IOV in a Host Profile

For multiple hosts or a stateless host, you can configure the virtual functions of the physical NIC by using a host profile and apply the profile on a host by using Auto Deploy.

- Verify that the configuration of your environment supports SR-IOV. See SR-IOV Support.
- Create a host profile based on the SR-IOV capable host. See the *vSphere Host Profiles* documentation.

For information about running ESX by using Auto Deploy with host profiles, see the *vCenter Deployment and Setup* documentation.

You can also enable SR-IOV virtual functions on the host by using the `esxcli system module parameters set` vCLI command on the NIC driver parameter for virtual functions in accordance with the driver documentation. For more information about using ESXCLI commands, see *ESXCLI Concepts and Examples* documentation.

1. From the Home page, click **Host Profiles**.
2. Select the host profile from the list and click the **Configure** tab.
3. Click **Edit Host Profile** and expand the **General System Settings** node.
4. Expand **Kernel Module Parameter** and select the parameter of the physical function driver for creating virtual functions.
   For example, the parameter for the physical function driver of an Intel physical NIC is `max_vfs`.
5. In the **Value** text box, type a comma-separated list of valid virtual function numbers.

   Each list entry indicates the number of virtual functions that you want to configure for each physical function. A value of 0 ensures that SR-IOV is not enabled for that physical function.

   For example, if you have a dual port, set the value to *x*`,`*y* where *x* or *y* is the number of virtual functions you want to enable for a single port.

   If the target number of virtual functions on a single host is 30, you might have two dual port cards set to `0,10,10,10`.

   Note:
   The number of virtual functions supported and available for configuration depends on your system configuration.

6. Click **Finish**.
7. Remediate the host profile to the host as required.

The virtual functions appear in the PCI Devices list on the **Settings** tab for the host.

Associate a virtual function with a virtual machine adapter by using the SR-IOV passthrough network adapter type. See Assign a Virtual Function as SR-IOV Passthrough Adapter to a Virtual Machine.

Enable SR-IOV on a Host Physical Adapter by Using an ESXCLI Command

Learn how to run a console command on ESX to create SR-IOV virtual functions on a physical adapter to troubleshoot or to configure hosts directly.

Install the vCLI package, deploy the vSphere Management Assistant (vMA) virtual machine, or use the ESX Shell. See *Getting Started with ESXCLI*.

You can create SR-IOV virtual functions on the host by manipulating the NIC driver parameter for virtual functions in accordance with the driver documentation.

1. To create virtual functions by setting the parameter for virtual functions of the NIC driver, run the `esxcli system module parameters set` command at the command prompt.

   ```
   esxcli system module parameters set -m driver -p vf_param=w,x,y,z
   ```

   Where *`driver`* is the name of the NIC driver, and *`vf_param`* is the driver-specific parameter for creating the virtual function.

   You can use a comma-separated list to set values for the *`vf_param`* parameter, where each entry indicates the number of virtual functions for a port. A value of 0 ensures that SR-IOV is not enabled for that physical function.

   If you have two dual port NICs, you can set the value to *`w`*`,`*`x`*`,`*`y`*`,`*`z`*, where *`w`*,*`x`*,*`y`*, and *`z`* is the number of virtual functions you want to enable for a single port. For example, to create 30 virtual functions distributed on two dual port Intel cards by using the ixgbe driver, run the following command for the ixgbe driver and the `max_vfs` parameter:

   ```
   esxcli system module parameters set -m ixgbe -p max_vfs=0,10,10,10
   ```

2. Restart the host to create the virtual functions.

Associate a virtual function with a virtual machine adapter by using the SR-IOV passthrough network adapter type. See Assign a Virtual Function as SR-IOV Passthrough Adapter to a Virtual Machine.

What is Remote Direct Memory Access for Virtual Machines

vSphere 6.5 and later releases support remote direct memory access (RDMA) communication between virtual machines that have paravirtualized RDMA (PVRDMA) network adapters .

Overview of RDMA

RDMA allows direct memory access from the memory of one computer to the memory of another computer without involving the operating system or CPU. The transfer of memory is offloaded to the RDMA-capable Host Channel Adapters (HCA) . A PVRDMA network adapter provides remote direct memory access in a virtual environment.

Using RDMA in vSphere

In vSphere, a virtual machine can use a PVRDMA network adapter to communicate with other virtual machines that have PVRDMA devices. The virtual machines must be connected to the same vSphere Distributed Switch.

The PVRDMA device automatically selects the method of communication between the virtual machines . For virtual machines that run on the same ESX host with or without a physical RDMA device, the data transfer is a memcpy between the two virtual machines. The physical RDMA hardware is not used in this case.

For virtual machines that reside on different ESX hosts and that have a physical RDMA connection, the physical RDMA devices must be uplinks on the distributed switch. In this case, the communication between the virtual machines by way of PVRDMA uses the underlying physical RDMA devices.

For two virtual machines that run on different ESX hosts, when at least one of the hosts does not have a physical RDMA device, the communication falls back to a TCP-based channel and the performance is reduced.

PVRDMA Support

vSphere 6.5 and later supports PVRDMA only in environments with a specific configuration.

Supported Configurations

To use PVRDMA in vSphere 6.5 or later, your environment must meet several configuration requirements.

Table 152: Supported Configurations for Using PVRDMA

Component	Requirements
vSphere	• ESX host 6.5 or later. • vCenter 6.5 or later. • vSphere Distributed Switch.
Physical host	• Must be compatible with the ESX release.
Host Channel Adapter (HCA)	• Must be compatible with the ESX release. **Note:** Virtual machines that reside on different ESX hosts require HCA to use RDMA. You must assign the HCA as an uplink for the vSphere Distributed Switch. PVRDMA does not support NIC teaming. The HCA must be the only uplink on the vSphere Distributed Switch. For virtual machines on the same ESX hosts or virtual machines using the TCP-based fallback, the HCA is not required.
Virtual machine	• Virtual hardware version 13 or later.
Guest OS	• Linux (64-bit)

To verify that your physical hosts and HCAs are compatible with ESX releases, see the *Broadcom Compatibility Guide*.

Note:
Attempts to enable or configure unsupported features with PVRDMA might result in unexpected behavior in your environment.

Support for PVRDMA Namespaces

In releases previous to vSphere 7.0, PVRDMA virtualized public resource identifiers in the underlying hardware to guarantee that a physical resource can be allocated with the same public identifier when a virtual machine resumed operation following the use of vMotion to move it from one physical host server to another. To do this, PVRDMA distributed virtual to physical resource identifier translations to peers when creating a resource. This resulted in additional overhead that can be significant when creating large numbers of resources.

PVRDMA namespaces prevents these additional overheads by letting multiple virtual machines coexist without coordinating the assignment of identifiers. Each virtual machine is assigned an isolated identifier namespace on the RDMA hardware, such that any virtual machine can select its identifiers within the same range without conflicting with other virtual machines. The physical resource identifier no longer changes even after vMotion, so virtual to physical resource identifier translations are no longer necessary.

PVRDMA namespaces are enabled automatically on vSphere 7.0 and later with virtual machine hardware version 17 or later. The underlying hardware must also support PVRDMA namespaces. To learn how to enable PVRDMA namespaces on your environment's hardware, refer to the RDMA vendor documentation.

Support for PVRDMA Native Endpoints

PVRDMA native endpoints are supported in virtual machine hardware version 18 and later beginning with vSphere 7.0 Update 1 and later releases. PVRDMA native endpoints allow PVRDMA to communicate with non-PVRDMA endpoints. To use PVRDMA native endpoints, you must enable PVRDMA namespaces. To learn how to enable PVRDMA namespaces on your environment's specific hardware, refer to the RDMA vendor documentation.

You must configure the virtual machine to use PVRDMA native endpoints. See Configure a Virtual Machine to Use PVRDMA Native Endpoints.

Configure an ESX Host for PVRDMA

Configure the VMkernel adapter and firewall rule of an ESX host for PVRDMA communication.

Verify that your ESX host meets the requirements for PVRDMA. See PVRDMA Support.

Tag a VMkernel Adapter for PVRDMA

Learn how to select a VMkernel adapter and enable it for PVRDMA communication.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Click the **Edit** button.
5. Locate `Net.PVRDMAVmknic`by using the filter text field.
6. Click the value field and enter the value of the VMkernel adapter that you want to use, for example vmk0.
7. Click **OK** .

Enable the Firewall Rule for PVRDMA

Learn how to enable the firewall rule for PVRDMA in the security profile of the ESX host.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Firewall**.
4. Click the **Edit** button.
5. Locate the pvrdma rule by using the filter text field.
6. Select the check box next to the pvrdma rule and click **OK**.

Assign a PVRDMA Adapter to a Virtual Machine

To enable a virtual machine to exchange data by using RDMA, you must associate the virtual machine with a PVRDMA network adapter.

- Verify that the host on which the virtual machine is running is configured for RDMA. See Configure an ESX Host for PVRDMA.
- Verify that the host is connected to a vSphere Distributed Switch.
- Verify that the virtual machine uses virtual hardware version 13 and above.
- Verify that the guest operating system is a Linux 64-bit distribution.

You can add up to ten PVRDMA network adapters to a virtual machine when using vSphere 7.0.2 and later.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Power off the virtual machine.
3. From the Actions menu, select Edit Settings.
4. Select the **Virtual Hardware** tab in the dialog box displaying the settings.
5. From the **Add new device** drop-down menu, select **Network Adapter**.
   The New Network section is added to the list in the **Virtual Hardware** tab.
6. Expand the New Network section and connect the virtual machine to a distributed port group.
7. From the **Adapter type** drop-down menu, select PVRDMA.
8. Expand the **Memory** section, select **Reserve all guest memory (All locked)**, and click **OK**.
9. Power on the virtual machine.

Configure a Virtual Machine to Use PVRDMA Native Endpoints

PVRDMA native endpoints are available as an advanced virtual machine configuration.

Verify that your environment supports PVRDMA. See PVRDMA Support.

PVRDMA native endpoints are supported in virtual machine hardware version 18 and later beginning with vSphere 7.0 Update 1 and later releases. To use PVRDMA native endpoints, you must enable PVRDMA namespaces. To learn how to enable PVRDMA namespaces on your environment's specific hardware, refer to the vendor documentation.

You can use the vSphere Client to configure native endpoints or edit the virtual machine's VMX file. If editing the VMX file directly, add the parameter `vrdma`*`x`*`.nativeEndpointSupport = "TRUE"`, where *x* is the index of the PVRDMA adapter. The following procedure uses the vSphere Client to configure native endpoints.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click **VM Options**.
4. Expand **Advanced**.
5. Under Configuration Parameters, click the **Edit Configuration** button.
6. In the dialog box that appears, click **Add Row** to enter a new parameter and its value.
7. Enter the parameter `vrdma`*`x`*`.nativeEndpointSupport`, where *x* is the index of the PVRDMA adapter, and set the value to `TRUE`.
   The index *x* is the number of the PVRDMA adapter minus 1. For example, if the PVRDMA adapter you want to enable native endpoints with is labelled "Network Adapter 2," then the index is 1.

Configure a Virtual Machine to use PVRDMA Asynchronous Mode

Learn how to configure a virtual machine to use PVRDMA asynchronous mode. It is available as an advanced virtual machine configuration.

Verify that your environment supports PVRDMA. See PVRDMA Support.

PVRDMA asynchronous mode is available in virtual machines running on vSphere 8.0 and later. Asynchronous mode might improve the throughput and latency for RDMA workloads running in the virtual machine. When asynchronous mode is enabled, an increased CPU use might be observed in the host. When asynchronous mode is in use, it is recommended that the virtual machine is configured for high latency sensitivity.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click **VM Options**.
4. Expand **Advanced**.
5. Under Configuration Parameters, click the **Edit Configuration** button.
6. In the dialog box that appears, click **Add Row** to enter a new parameter and its value.
7. Enter the parameter `vrdma.asyncMode`, and set the value to `TRUE`.

Network Requirements for RDMA over Converged Ethernet

RDMA over Converged Ethernet ensures low-latency, light-weight, and high-throughput RDMA communication over an Ethernet network. RoCE requires a network that is configured for lossless traffic of information at layer 2 alone or at both layer 2 and layer 3.

RDMA over Converged Ethernet (RoCE) is a network protocol that uses RDMA to provide faster data transfer for network-intensive applications. RoCE allows direct memory transfer between hosts without involving the hosts' CPUs.

There are two versions of the RoCE protocol. RoCE v1 operates at the link network layer (layer 2). RoCE v2 operates at the Internet network layer (layer 3) . Both RoCE v1 and RoCE v2 require a lossless network configuration. RoCE v1 requires a lossless layer 2 network, and RoCE v2 requires that both layer 2 and layer 3 are configured for lossless operation.

Lossless Layer 2 Network

To ensure lossless layer 2 environment, you must be able to control the traffic flows. Flow control is achieved by enabling global pause across the network or by using the Priority Flow Control (PFC) protocol defined by Data Center Bridging group (DCB). PFC is a layer 2 protocol that uses the class of services field of the 802.1Q VLAN tag to set individual traffic priorities. It puts on pause the transfer of packets towards a receiver in accordance with the individual class of service priorities. This way, a single link carries both lossless RoCE traffic and other lossy, best-effort traffic. With traffic flow congestion, important lossy traffic can be affected. To isolate different flows from one another, use RoCE in a PFC priority-enabled VLAN.

Lossless Layer 3 Network

RoCE v2 requires that lossless data transfer is preserved at layer 3 routing devices. To enable the transfer of layer 2 PFC lossless priorities across layer 3 routers, configure the router to map the received priority setting of a packet to the corresponding Differentiated Serviced Code Point (DSCP) QoS setting that operates at layer 3. The transferred RDMA packets are marked with layer 3 DSCP, layer 2 Priority Code Points (PCP) or with both. To extract priority information from the packet routers use either DSCP or PCP. In case PCP is used, the packet must be VLAN-tagged and the router must copy the PCP bits of the tag and forward them to the next network. If the packet is marked with DSCP, the router must keep the DSCP bits unchanged.

Like RoCE v1, RoCE v2 must run on a PFC priority-enabled VLAN.

> **Note:**
> Do not team RoCE NICs, if you intend to use RDMA on those NICs.

For vendor-specific configuration information, refer to the official documentation of the respective device or the switch vendor.

Configure Remote Direct Memory Access Network Adapters

Learn how to install a remote direct memory access (RDMA) network adapter on your ESX hosts. Once installed, you can use the vSphere Client to view the RDMA adapter and its corresponding network adapter, and configure its VMkernel binding.

Install an RDMA-capable adapter on your ESX host. For example, Mellanox Technologies MT27700 Family ConnectX-4.

RDMA provides direct memory access from the memory of one host to the memory of another host without involving the remote operating system and CPU. This boosts network and host performance with lower latency, lower CPU load, and faster bandwidth.

Starting with VCF 9.1, you can use the `pktcap-uw --capture RDMASniffer` command for packet analysis and the `rdmaperf-uw` utility to validate network readiness for RDMA-based applications, such as vSAN over RDMA or NVMe over RDMA. With the RDMA Sniffer capability for native packet capture and the built-in `rdmaperf` utility, you can troubleshoot RoCEv2 connections, verify end-to-end connectivity between nodes, and evaluate throughput and latency metrics directly within ESX. For more information, see Use rdmaperf to Validate and Evaluate RDMA Connectivity Between ESX Hosts.

View RDMA Capable Network Adapter

ESX supports RDMA capable network adapters. After you install such an adapter on your ESX host, the vSphere Client displays its two components, the RDMA adapter, and a physical network adapter.

Install a RDMA-capable adapter that supports RDMA (RoCE v2) on your ESX host. For example, Mellanox Technologies MT27700 Family ConnectX-4.

You can use the vSphere Client to view the RDMA adapter and its corresponding network adapter.

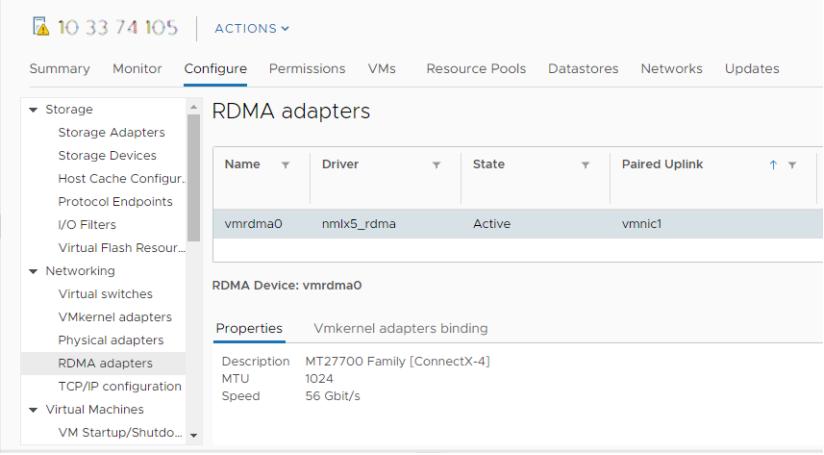
1. On your ESX host, install a RDMA-capable adapter that supports RDMA (RoCE v2).

   The host discovers the adapter and the vSphere Client displays its two components, an RDMA adapter, and a physical network adapter.

2. Navigate to the host.
3. Under **Networking**, click **RDMA adapters**.

   In this example, the RDMA adapter appears on the list as `vmrdma0`. The **Paired Uplink** column displays the network component as the `vmnic1` physical network adapter.

   Figure 62: RDMA adapters installed on an ESX host in a vSphere environment.



4. To verify the description of the adapter, select the RDMA adapter from the list, and click the **Properties** tab.

Configure Remote Direct Memory Access Network Adapters

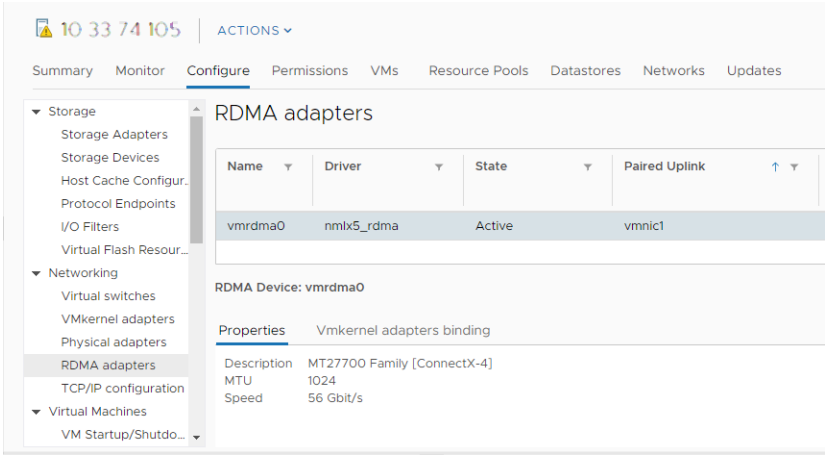
You can install a remote direct memory access (RDMA) network adapter and configure its VMkernel binding.

1. Install an RDMA-capable adapter that supports RDMA (RoCE v2) on your ESX host.

   The host discovers the adapter and the vSphere Client displays its two components, an RDMA adapter and a physical network adapter.

2. In the vSphere Client, verify that the RDMA adapter is discovered by your host.
   a) Navigate to the host.
   b) Click the **Configure** tab.
   c) Under **Networking**, click **RDMA adapters**.

      In this example, the RDMA adapter appears on the list as `vmrdma0`. The **Paired Uplink** column displays the network component as the `vmnic1` physical network adapter.



   d) To verify the description of the adapter, select the RDMA adapter from the list, and click the **Properties** tab.

3. Configure VMkernel binding for the RDMA adapter.

   In the configuration, you can use a vSphere standard switch or a vSphere Distributed Switch. The following steps use the standard switch as an example.

   a) Create a vSphere standard switch and add the network component to the switch.

      Note:
      Make sure to select the physical network adapter that corresponds to the RDMA adapter. In this example, it is the `vmnic1` adapter.

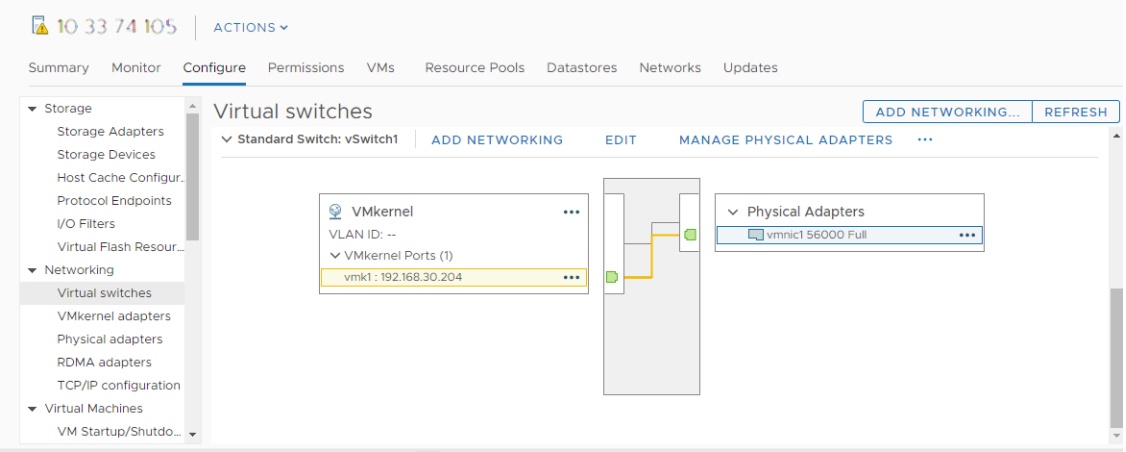
      For information about creating the switch, see Create a vSphere Standard Switch or Create a vSphere Distributed Switch.

   b) Add a VMkernel adapter to the vSphere standard switch that you created.

      Assign an appropriate static IPv4 or IPv6 address to the VMkernel adapter, so that your RDMA adapter can discover the NVMe over RDMA target.

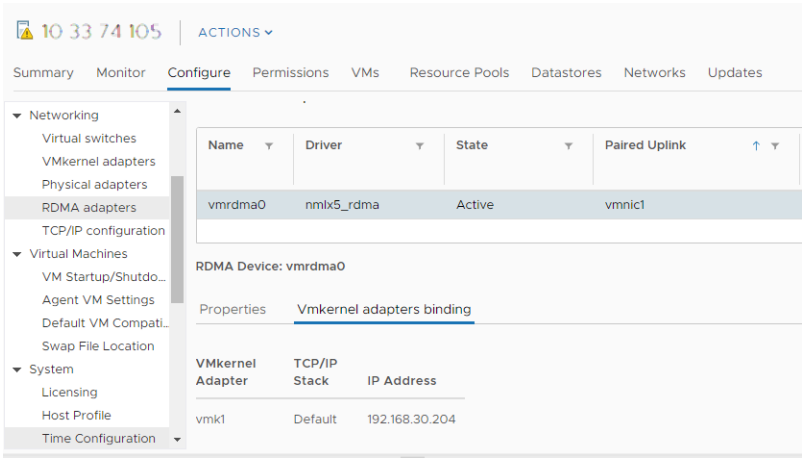
      For information about adding the VMkernel adapter, see Setting Up VMkernel Networking.

   The illustration shows that the physical network adapter and the VMkernel adapter are connected to the vSphere standard switch. Through this connection, the RDMA adapter is bound to the VMkernel adapter.



4. Verify the VMkernel binding configuration for the RDMA adapter.
   a) Navigate to the RDMA adapter.
   b) Click the **VMkernel adapters binding** tab and verify that the associated VMkernel adapter appears on the page.

      In this example, the `vmrdma0` RDMA adapter is paired to the `vmnic1` network adapter and is connected to the `vmk1` VMkernel adapter.



You can use the RDMA network component of the adapter for such storage configurations as iSER or NVMe over RDMA. To learn more, see the *vSphere Storage* documentation.

Capture RDMA Packets on an ESX Host

Starting with vSphere 9.1, ESX supports RDMA packet capture directly on the host through the RDMA Sniffer feature in the RDMA driver. Use the `esxcli` and `pktcap-uw` utilities to activate the sniffer and capture traffic. When your capture session is complete, you can deactivate the sniffer.

- Verify that the ESX host runs version 9.1 or later.
- Verify that the host has an RDMA-capable NIC installed.
- Verify that a VMkernal adapter is configured and bound to the RDMA adapter, see Configure Remote Direct Memory Access Network Adapters.

In earlier releases, capturing RDMA traffic on an ESX host required configuring a mirror port on the physical switch, because RDMA workloads bypass the ESX kernel network stack. The RDMA Sniffer feature eliminates that dependency. When activated on a supported adapter, `pktcap-uw` can capture RDMA packets directly through the paired uplink.

> **Important:**
>
> Deactivate the sniffer immediately after the capture session. Leaving the sniffer activated affects adapter performance.

1. Select a host and run the following command to verify that the RDMA addapter supports the RDMA Sniffer feature: `esxcli rdma device sniffer get`.

   You see the Sniffer status of the device in the output.

Status	Meaning
N/A	The adapter does not support RDMA Sniffer. Select a different adapter.
off	The adapter supports RDMA Sniffer; the sniffer is currently deactivated. Proceed to Step 2.
on	The adapter supports RDMA Sniffer and the sniffer is already activated.

   To proceed with the following steps, the adapter must support RDMA Sniffer, even if it is deactivated.

2. Identify the physical uplink adapter paired with the RDMA device by running the following command.
   a) Run the following command: `esxcli rdma device list`.
   b) In the output, locate the target RDMA device and note the value in the corresponding `Paired Uplink` column.
3. Activate the RDMA Sniffer on the target RDMA device.
   a) Run the following command: `esxcli rdma device sniffer set -e 1 -d <rdma_device>`.

      Replace *<rdma_device>* with a device name from the list in Step 2.
   b) Verify that the sniffer is activated by running the following command: `esxcli rdma device sniffer get -d <rdma_device>`.

      Replace *<rdma_device>* with a device name from the list in Step 2.
4. Start an RDMA workload on the ESX host, for example vSAN over RDMA, NVMe oer RDMA, iSER, or a test using the `rdmaperf-uw` utility.
5. Use `pktcap-uw` to capture RDMA packets:

```
pktcap-uw --capture RDMASniffer \
  --uplink <paired_uplink> \
  -c 1000 \
  -o <output_file> \
  --dstudpport 4791
```

   Replace *<paired_uplink>* with the uplink name from Step 2 and *<output_file>* with the target capture file path (for example, `/tmp/sniffer.pcap`)
6. After the capture session is complete, deactivate the RDMA Sniffer on the RDMA device.
   a) Run the following command: `esxcli rdma device sniffer set -e 0 -d <rdma_device>`.
   b) Run the following command to confirm that the Sniffer status for the device is off: `esxcli rdma device sniffer get`.

The captured RDMA packets are saved as a PCAP file at the path specified in Step 5. The RDMA Sniffer is deactivated and no longer affects adapter performance.

Analyze the PCAP file to diagnose RDMA connectivity issues or inspect RoCEv2 frame contents. To validate RDMA connectivity and measure performance between two hosts, see Use rdmaperf to Validate and Evaluate RDMA Connectivity Between ESX Hosts.

Use rdmaperf to Validate and Evaluate RDMA Connectivity Between ESX Hosts

Use the bult-in `rdmaperf` utility to confirm that two ESX hosts can form a direct RDMA connection over RoCEv2. You can also use the rdmaperf utility to measure throughput and latency, and also to collect CPU cost data, so that you can meaningfully compare RDMA and traditional TCP/IP networking.

- Verify that the ESX hosts run version 9.1.

- For two selected ESX hosts, create VMkernel adapters bound to the RDMA adapters, see Configure Remote Direct Memory Access Network Adapters.
- Verify that you have root access to both hosts through SSh or the ESX Shell.

The `rdmaperf` utility is a built-in kernel module. The user-space binary is located at `/usr/lib/vmware/rdma/bin/rdmaperf-uw`.

1. On the server host (host1), load the `rdmaperf` kernel module: `vmkload_mod rdmaperf`.
2. Start the `rdmaperf-uw` server process: `/usr/lib/vmware/rdma/bin/rdmaperf-uw -s -l <vmkernel_ip_server>`.

   Replace *<vmkernel_ip_server>* with he VMkernel adapter IP address of host1.

   The server process listens on port 18888 and blocks until the client connects.
3. On the client host (host2), load the `rdmaperf` kernel module: `vmkload_mod rdmaperf`.
4. On host2, run a write-bandwidth connectivity test against host1:

   ```
   /usr/lib/vmware/rdma/bin/rdmaperf-uw \
     -c -r <vmkernel_ip_server> \
     -l <vmkernel_ip_client> \
     -o write_bw -d 10
   ```

   Replace *<vmkernel_ip_server>* with he VMkernel adapter IP address of host1 and *<vmkernel_ip_client>* with he VMkernel adapter IP address of host2.

   When the connection is established, the utility displays a configuration summary followed by test results:

   ```
   ----------------------------------------------------
   Operation type:           WRITE_BW
   Queue pair type:          RC
   Local end-point address:  192.168.122.2
   Remote end-point address: 192.168.122.1
   Port number:              18888
   Is client:                YES
   Use RDMA CM:              YES
   Queue size:               64
   Post list size:           4
   Number of connections:    1
   Test duration:            10 seconds
   Message size:             4096
   ----------------------------------------------------
   --------------------------------
   Test results for conn id: 1
   Message size:         4096 bytes
   Iterations:           4284337
   Avg:                  6693.86 MB/s
   --------------------------------
   Total avg BW of this endpoint is: 8693.86 MB/s
   ----------------------------------------------------
   ```

   A non-zero `Avg` bandwidth value and a completed iteration count confirm a successful RDMA connection between the two hosts.

   If the test exits without producing a result block, verify NIC compatibility, VMkernel binding configuration, and IP reachability between the two VMkernel adapter addresses.
5. (Optional) Repeat the test with other operation types to collect full performance data:

   ```
   /usr/lib/vmware/rdma/bin/rdmaperf-uw \
     -c -r <vmkernel_ip_server> \
     -l <vmkernel_ip_client> \
   ```

```
-o <operation_type> \
-d <duration_seconds>
```

Replace *<operation_type>* with any value from the table:

-o value	Measures
write_bw	Write bandwidth (throughput)
read_bw	Read bandwidth (throughput)
write_lat	Write latency
read_lat	Read latency
send_bw	Send bandwidth
send_lat	Send latency

Also, adjust *<duration_seconds>* as needed.

Important:

Record the `Avg` and `Total avg BW` values for each operation type. Run the server process on host1 before each client test.

6. (Optional) Run an equivalent test over TCP/IP and compare the results.

   Use a standard TCP/IP throughput or latency tool, for example, `iperf3` between the same two hosts over a non-RDMA VMkernel path. Compare throughput (MB/s) and latency (µs) values to quantify the CPU load reduction and bandwidth improvement that RDMA ensures.

7. After all tests are complete, unload the `rdmaperf` module on both hosts: `vmkload_mod -u rdmaperf`.

Both hosts established a RoCEv2 RDMA connection, confirmed by a non-zero average bandwidth result. If you also do performance evaluation, the output provides per-connection and aggregate throughput (MB/s) or latency (µs) metrics depending on the operation type. Use these results to establish an RDMA performance baseline, identify bottlenecks, and validate hardware readiness before deploying RDMA-dependent workloads, such as NVMe over RDMA.

The `rdmaperf` module is unloaded on both hosts.

What are Jumbo Frames

Jumbo frames let ESX hosts send larger frames out onto the physical network. The network must support jumbo frames end-to-end that includes physical network adapters, physical switches, and storage devices.

Before enabling jumbo frames, check with your hardware vendor to ensure that your physical network adapter supports jumbo frames.

You can enable jumbo frames on a vSphere distributed switch or vSphere standard switch by changing the maximum transmission unit (MTU) to a value greater than 1280 bytes. With vCenter 7.0 Update 3, you can set the size of the maximum transmission unit (MTU) on a vSphere Distributed Switch to up to 9190 bytes to support switches with larger packet sizes.

Enable Jumbo Frames on a vSphere Distributed Switch

Enable jumbo frames for the entire traffic that passes through a vSphere Distributed Switch.

Important:
When you change the MTU size of a vSphere Distributed Switch, the physical NICs that are assigned as uplinks are brought down and up again. This causes a short network outage of between 5 to 10 milliseconds for virtual machines or services that are using the uplinks.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and select **Properties**.
3. Click **Edit**.
4. Click **Advanced** and set the **MTU** property to a value greater than 1500 bytes.
   You cannot set the MTU size to a value greater than 9190 bytes.
5. Click **OK**.

Enable Jumbo Frames on a vSphere Standard Switch

Enable jumbo frames for all traffic through a vSphere Standard Switch on a host.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **Virtual Switches**.
3. Select a standard switch from the virtual switch table and click **Edit settings**.
4. In the **Properties** section, set the **MTU** property to a value greater than 1500 bytes.
   You can increase the MTU size up to 9000 bytes.
5. Click **OK**.

Enable Jumbo Frames for a VMkernel Adapter

Jumbo frames reduce the CPU load caused by transferring data. Enable jumbo frames on a VMkernel adapter by changing the maximum transmission units (MTU) of the adapter.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Select a VMkernel adapter from the adapter table.
   The properties of the adapter appear.
4. Click **Edit**.
5. On the Port properties page, set the **MTU** property to a value greater than 1500.
   You can increase the MTU size up to 9000 bytes.
6. Click **OK**.

Enable Jumbo Frame Support on a Virtual Machine

Enabling jumbo frame support on a virtual machine requires an enhanced VMXNET adapter for that virtual machine.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. From the Actions menu, select Edit Settings.
3. Select the **Virtual Hardware** tab in the dialog box displaying the settings.
4. Expand the **Network adapter** section. Record the network settings and MAC address that the network adapter is using.
5. Click the times-circle button to remove the network adapter from the virtual machine.
6. From the **Add new device** drop-down menu, select **Network Adapter**.
   The New Network section is added to the list in the Virtual Hardware tab.

7. Expand the New Network section.
8. From the **Adapter Type** drop-down menu, select **VMXNET 2 (Enhanced)** or **VMXNET 3**.
9. Set the network settings to the ones recorded for the old network adapter.
10. Set the **MAC Address** to **Manual**, and type the MAC address that the old network adapter was using.
11. Click **OK**.

- Check that the enhanced VMXNET adapter is connected to a standard switch or to a distributed switch with jumbo frames enabled.
- Inside the guest operating system, configure the network adapter to allow jumbo frames. See the documentation of your guest operating system.
- Configure all physical switches and any physical or virtual machines to which this virtual machine connects to support jumbo frames.

What is TCP Segmentation Offload

Learn how to use TCP Segmentation Offload (TSO) in VMkernel network adapters and virtual machines to improve the network performance in workloads that have severe latency requirements.

TSO on the transmission path of physical network adapters, and VMkernel and virtual machine network adapters improves the performance of ESX hosts by reducing the overhead of the CPU for TCP/IP network operations. When TSO is enabled, the network adapter divides larger data chunks into TCP segments instead of the CPU. The VMkernel and the guest operating system can use more CPU cycles to run applications.

To benefit from the performance improvement that TSO provides, enable TSO along the data path on an ESX host including physical network adapters, VMkernel and guest operating system. By default, TSO is enabled in the VMkernel of the ESX host , and in the VMXNET 2 and VMXNET 3 virtual machine adapters.

For information about the location of TCP packet segmentation in the data path, see Broadcom knowledge base article 318877.

Manage Software TSO in the VMkernel

If a physical network adapter experiences problems with TSO, you can temporarily enable the software simulation of TSO in the VMkernel until you troubleshoot the problems.

Run these `esxcli network nic software set` console commands to activate or deactivate the software simulation of TSO in the VMkernel.

- Activate the software simulation of TSO in the VMkernel.

  `esxcli network nic software set --ipv4tso=1 -n vmnic`*X*`esxcli network nic software set --ipv6tso=1 -n vmnic`*X*

- Deactivate the software simulation of TSO in the VMkernel.

  `esxcli network nic software set --ipv4tso=0 -n vmnic`*X*`esxcli network nic software set --ipv6tso=0 -n vmnic`*X*

where *X* in vmnic*X* represents the number of the NIC ports on the host.

The configuration change persists across host reboots.

How to determine whether TSO is Supported on the Physical Network Adapters on an ESX Host

Examine whether a physical network adapter offloads TCP/IP packet segmentation when you estimate the networking performance on a host that runs latency-sensitive workloads. If a physical network adapter supports TSO, then TSO is enabled by default.

Run the following console command to determine whether TSO is enabled on the physical network adapters on a host.

```
esxcli network nic tso get
```

Manage TSO on an ESX Host

Activate TCP Segmentation Offload (TSO) on the transmission path to have the NIC divide larger data chunks into TCP segments. Deactivate TSO to have CPU perform TCP segmentation.

By default, a host uses hardware TSO if its physical adapters support it.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Edit the value of the `Net.UseHwTSO` parameter for IPv4 and of `Net.UseHwTSO6` for IPv6.
   - To activate TSO, set `Net.UseHwTSO` and `Net.UseHwTSO6` to `1`.
   - To deactivate TSO, set `Net.UseHwTSO` and `Net.UseHwTSO6` to `0`.
5. Click **OK** to apply the changes.
6. To reload the driver module of the physical adapter, run the `esxcli system module set` console command in the ESX Shell on the host.
   a) To deactivate the driver, run the `esxcli system module set` command with the --enabled false option.

```
esxcli system module set
--enabled false
--module
nic_driver_module
```

   b) To activate the driver, run the `esxcli system module set` command with the --enabled true option.

```
esxcli system module set
--enabled true
--module
nic_driver_module
```

If a physical adapter does not support hardware TSO, the VMkernel segments large TCP packets coming from the guest operating system and sends them to the adapter.

How to find if TSO Is Enabled on an ESX Host

Examine whether hardware TSO is enabled in the VMkernel when you estimate the networking performance on a host that runs latency-sensitive workloads. By default, hardware TSO is enabled on an ESX host.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Examine the value of the `Net.UseHwTSO` and `Net.UseHwTSO6` parameters.
   `Net.UseHwTSO` shows the TSO state for IPv4, and `Net.UseHwTSO6` for IPv6. TSO is enabled if the property is set to 1.

Manage TSO on a Linux Virtual Machine

Activate TSO support on the network adapter of a Linux virtual machine so that the guest operating system redirects TCP packets that need segmentation to the VMkernel.

- Verify that ESX supports the Linux guest operating system.
  See the *VMware Compatibility Guide* documentation.

- Verify that the network adapter on the Linux virtual machine is VMXNET2 or VMXNET3.

In a terminal window on the Linux guest operating system, to activate or deactivate TSO, run the `ethtool` command with the -K and tso options.

- To activate TSO, run the following command:
  `ethtool-K ethYtsoon`
- To deactivate TSO, run the following command:
  `ethtool-K ethYtsooff`

where *Y* in eth*Y* is the sequence number of the NIC in the virtual machine.

Manage TSO on a Windows Virtual Machine

By default, TSO is activated on a Windows virtual machine on VMXNET2 and VXMNET3 network adapters. For performance reasons, you might want to deactivate TSO.

- Verify that ESX supports the Windows guest operating system. See the *VMware Compatibility Guide* documentation.
- Verify that the network adapter on the Windows virtual machine is VMXNET2 or VMXNET3.

1. In the Network and Sharing Center on the Windows control panel, click the name of the network adapter.
2. Click its name.
   A dialog box displays the status of the adapter.
3. Click **Properties**, and beneath the network adapter type, click **Configure**.
4. On the **Advanced** tab, set the **Large Send Offload V2 (IPv4)** and **Large Send Offload V2 (IPv6)** properties to **Enabled** or **Disabled**.
5. Click **OK**.
6. Restart the virtual machine.

What is Large Receive Offload

Learn how to use Large Receive Offload (LRO) to reduce the CPU overhead for processing packets that arrive from the network at a high rate.

LRO reassembles incoming network packets into larger buffers and transfers the resulting larger but fewer packets to the network stack of the host or virtual machine. The CPU has to process fewer packets than when LRO is deactivated, which reduces its utilization for networking especially in the case of connections that have high bandwidth.

To benefit from the performance improvement of LRO, enable LRO along the data path on an ESX host including VMkernel and guest operating system. By default, LRO is activated in the VMkernel and in the VMXNET3 virtual machine adapters.

For information about the location of TCP packet aggregation in the data path, see Broadcom knowledge base article 318877.

Manage Hardware LRO for All VMXNET3 Adapters on an ESX Host

Activate the hardware capabilities of host physical adapters to aggregate incoming TCP packets for VXMNET3 VM adapters by using the LRO technology instead of consuming resources for assembling in the guest operating system.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Edit the value of the `Net.Vmxnet3HwLRO` parameter.
   - To activate hardware LRO, set `Net.Vmxnet3HwLRO` to `1`.
   - To deactivate hardware LRO, set `Net.Vmxnet3HwLRO` to `0`.

5. Click **OK** to apply the changes.

Manage Software LRO for All VMXNET3 Adapters on an ESX Host

Use software LRO in the VMkernel backend of VMXNET3 adapters to improve networking performance of virtual machines if the host physical adapters do not support hardware LRO.

vSphere supports software LRO for both IPv4 and IPv6 packets.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Edit the value of the `Net.Vmxnet3SwLRO` parameter for VMXNET3 adapters.
   - To activate software LRO, set `Net.Vmxnet3SwLRO` to 1.
   - To deactivate software LRO, set `Net.Vmxnet3SwLRO` to 0.
5. Click **OK** to apply the changes.

Determine Whether LRO Is Enabled for VMXNET3 Adapters on an ESX Host

Examine the status of LRO on an ESX when you estimate the networking performance on a host that runs latency-sensitive workloads.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Examine the value of the LRO parameters for VMXNET2 and VMXNET3.
   - For hardware LRO, examine the `Net.Vmxnet3HwLRO` parameter. If it is equal to 1, hardware LRO is enabled.
   - For software LRO, examine the `Net.Vmxnet3SwLRO` parameter. If it is equal to 1, software LRO is enabled.

Change the Size of the LRO Buffer for VMXNET 3 Adapters

You can change the size of the buffer for packet aggregation for virtual machine connections through VMXNET 3 network adapters. Increase the buffer size to reduce the number of TCP acknowledgments and improve efficiency in workloads.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Enter a value between 1 and 65535 for the `Net.VmxnetLROMaxLength` parameter to set the LRO buffer size in bytes.

   By default the size of the LRO buffer is equal to 32000 bytes.

Activate or Deactivate LRO for All VMkernel Adapters on an ESX Host

Use LRO in the VMkernel network adapters on an ESX host to improve the networking performance for incoming infrastructure traffic.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Edit the value of the `Net.TcpipDefLROEnabled` parameter.
   - To activate LRO for the VMkernel network adapters on the host, set `Net.TcpipDefLROEnabled` to `1`.
   - To deactivate software LRO for the VMkernel network adapters on the host, set `Net.TcpipDefLROEnabled` to `0`.
5. Click **OK** to apply the changes.

Change the Size of the LRO Buffer for VMkernel Adapters

You can change the size of the buffer for packet aggregation for VMkernel connections. Increase the buffer size to reduce the number of TCP acknowledgments and improve efficiency in the VMkernel.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System**.
3. Click **Advanced System Settings**.
4. Enter a value between 1 and 65535 for the `Net.TcpipDefLROMaxLength` parameter to set the LRO buffer size in bytes.

   By default the size of the LRO buffer is equal to 32768 bytes.

Manage LRO on a VMXNET3 Adapter on a Linux Virtual Machine

If LRO is enabled for VMXNET3 adapters on the host, activate LRO support on a network adapter on a Linux virtual machine to ensure that the guest operating system does not spend resources to aggregate incoming packets into larger buffers.

Verify that the Linux kernel is 2.6.24 and later.

In a terminal window on the Linux guest operating system, run the `ethtool` command with the -K and lro options.

- To activate LRO, run the following command:

  ```
  ethtool-K ethYlroon
  ```

  where *Y* in eth*Y* is the sequence number of the NIC in the virtual machine.
- To deactivate LRO, run the following command:

  ```
  ethtool-K ethYlrooff
  ```

  where *Y* in eth*Y* is the sequence number of the NIC in the virtual machine.

Manage LRO on a VMXNET3 Adapter on a Windows Virtual Machine

If LRO is enabled for VMXNET3 adapters on the host, activate LRO support on a network adapter on a Windows virtual machine to ensure that the guest operating system does not spend resources to aggregate incoming packets into larger buffers.

- Verify that the virtual machine runs Windows Server 2012 and later or Windows 8 and later.
- Verify that the virtual machine compatibility is ESX 6.0 and later.
- Verify that the version of the VMXNET3 driver installed on the guest operating system is 1.6.6.0 and later.
- Verify that LRO is enabled globally on a virtual machine that runs Windows Server 2012 and later or Windows 8 and later. See Manage LRO Globally on a Windows Virtual Machine.

On Windows, the LRO technology is also referred to as Receive Side Coalescing (RSC).

1. In the **Network and Sharing Center** of the guest operating system's Control Panel, click the name of the network adapter.

   A dialog box displays the status of the adapter.

2. Click **Properties**, and under the VMXNET3 network adapter type, click **Configure**.
3. On the **Advanced** tab, set both **Recv Segment Coalescing (IPv4)** and **Recv Segment Coalescing (IPv6)** to **Enabled** or **Disabled**.
4. Click **OK**.

Manage LRO Globally on a Windows Virtual Machine

To use Large Receive Offload (LRO) on a VMXNET3 adapter on a virtual machine that runs Windows 8 and later or Windows Server 2012 and later, you must enable LRO globally on the guest operating system. On Windows, the LRO technology is also referred to as Receive Side Coalescing (RSC).

1. To verify whether LRO is deactivated globally on a Windows 8 and later or Windows Server 2012 guest OS, run the `netsh int tcp show global` command at the command prompt.

   ```
   netsh int tcp show global
   ```

   The command displays the status of the global Transmission Control Protocol (TCP) parameters that are set on the Windows 8.x OS.

   ```
   TCP Global Parameters
   ----------------------------------------------
   Receive-Side Scaling State          : enabled
   Chimney Offload State               : disabled
   NetDMA State                        : disabled
   Direct Cache Access (DCA)           : disabled
   Receive Window Auto-Tuning Level    : normal
   Add-On Congestion Control Provider  : none
   ECN Capability                      : disabled
   RFC 1323 Timestamps                 : disabled
   Initial RTO                         : 3000
   Receive Segment Coalescing State    : disabled
   ```

   If LRO is globally deactivated on the Windows 8 and later or Windows Server 2012 machine, the Receive Segment Coalescing State property appears as disabled.

2. To enable LRO globally on the Windows OS, run the `netsh int tcp set global` command at the command prompt:

   ```
   netsh int tcp set global rsc=enabled
   ```

Enable LRO for the VMXNET3 adapter on the Windows 8 and later or Windows Server 2012 virtual machine. See Manage LRO on a VMXNET3 Adapter on a Windows Virtual Machine.

NetQueue and Networking Performance

NetQueue takes advantage of the ability of some network adapters to deliver network traffic to the system in multiple receive queues that can be processed separately, allowing processing to be scaled to multiple CPUs, improving receive-side networking performance.

To use Rx queues effectively in the physical NICs by managing the vNIC and the VMkernel adapter filters, the NetQueue balancer in ESX uses load balancing algorithms.

You can activate or deactivate different types of Rx queues. For more information, see the `esxcli network nic queue loadbalancer set` command in the *ESXCLI Reference* documentation.

Activate NetQueue on a Host

NetQueue is enabled by default. To use NetQueue after it has been deactivated, you must activate it again.

1. In the ESX Shell on the host, use the following command:

   ```
   esxcli system settings kernel set --setting="netNetqueueEnabled" --value="TRUE"
   ```

2. To configure the NIC driver to use NetQueue, use the `esxcli module parameters set` command .
   To configure the driver with eight receive queues.on a dual-port Emulex NIC, run this ESXCLI command.

   ```
   esxcli system module parameters set -m tg3 -p force_netq=8,8
   ```

3. Reboot the host.

Deactivate NetQueue on a Host

NetQueue is activated by default.

Familiarize yourself with the information on configuring NIC drivers in *Getting Started with ESXCLI*.

1. In the ESXCLI, use the following command depending on the host version:

   ```
   esxcli system settings kernel set --setting="netNetqueueEnabled" --value="FALSE"
   ```

2. To deactivate NetQueue on the NIC driver, use the `esxcli module parameters set` command.
   To configure the driver with one receive queues, run the ESXCLI commands on a dual-port Emulex NIC.

   ```
   esxcli system module parameters set -m tg3 -p force_netq=1,1
   ```

3. Reboot the host.

vSphere Network I/O Control

Use vSphere Network I/O Control to allocate network bandwidth to business-critical applications and to resolve situations where several types of traffic compete for common resources.

What is vSphere Network I/O Control

vSphere Network I/O Control version 3 introduces a mechanism to reserve bandwidth for system traffic based on the capacity of the physical adapters on a host. It enables fine-grained resource control at the VM network adapter level similar to the model that you use for allocating CPU and memory resources..

Version 3 of the Network I/O Control feature offers improved network resource reservation and allocation across the entire switch.

Models for Bandwidth Resource Reservation

Network I/O Control version 3 supports separate models for resource management of system traffic related to infrastructure services, such as vSphere Fault Tolerance, and of virtual machines.

The two traffic categories have different nature. System traffic is strictly associated with an ESX host. The network traffic routes change when you migrate a virtual machine across the environment. To provide network resources to a virtual machine regardless of its host, in Network I/O Control you can configure resource allocation for virtual machines that is valid in the scope of the entire distributed switch.

Bandwidth Guarantee to Virtual Machines

Network I/O Control version 3 provisions bandwidth to the network adapters of virtual machines by using constructs of shares, reservation and limit. Based on these constructs, to receive sufficient bandwidth, virtualized workloads can rely on admission control in vSphere Distributed Switch, vSphere DRS and vSphere HA. See Admission Control for Virtual Machine Bandwidth.

Availability of Features

SR-IOV is not available for virtual machines configured to use Network I/O Control version 3.

Enable Network I/O Control on a vSphere Distributed Switch

Enable network resource management on a vSphere Distributed Switch to guarantee minimum bandwidth to system traffic for vSphere features and to virtual machine traffic.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. From the **Actions** menu, select **Settings** > **Edit Settings**.
3. From the **Network I/O Control** drop-down menu, select **Enable**.

   Note:
   When Network Offloads compatibility is enabled, the **Network I/O Control** is disabled. When **Network Offloads** is set to **None**, **Network I/O Control** is supported.
4. Click **OK**.

When enabled, the model that Network I/O Control uses to handle bandwidth allocation for system traffic and virtual machine traffic is based on the Network I/O Control version that is active on the distributed switch. See What is vSphere Network I/O Control .

Allocate Bandwidth for System Traffic

You can configure Network I/O Control to allocate certain amount of bandwidth for traffic generated by vSphere Fault Tolerance, vSphere vMotion, and so on.

You can use Network I/O Control on a distributed switch to configure bandwidth allocation for the traffic that is related to the main vSphere features:

- Management
- Fault Tolerance
- NFS
- vSAN
- vMotion
- vSphere Replication
- vSphere Data Protection Backup
- Virtual machine
- NVMe over TCP

vCenter propagates the allocation from the distributed switch to each physical adapter on the hosts that are connected to the switch.

Network I/O Control for Provisioning Traffic

Provisioning traffic handles the data that is transferred for virtual machine cold migration, cloning, and snapshot migration. Provisioning traffic consumed high network bandwidth. It severely impacts the network bandwidth consumed by management traffic. In addition, *vSphere vMotion Unified Data Transport* speeds up the provisioning traffic and it results in higher network congestion on the management network. Starting vSphere 9.0, the network I/O control is introduced for provisioning traffic to add control over its network capacity and bandwidth during network congestion. It also eliminates network congestion by isolating it from the management traffic.

Bandwidth Allocation Parameters for System Traffic

By using several configuration parameters Network I/O Control allocates bandwidth to traffic from basic vSphere system features.

Table 153: Allocation Parameters for System Traffic

Parameter for Bandwidth Allocation	Description
Shares	Shares, from 1 to 100, reflect the relative priority of a system traffic type against the other system traffic types that are active on the same physical adapter. The amount of bandwidth available to a system traffic type is determined by its relative shares and by the amount of data that the other system features are transmitting.
Reservation	The minimum bandwidth, in Mbps, that must be guaranteed on a single physical adapter. The total bandwidth reserved among all system traffic types cannot exceed 75 percent of the bandwidth that the physical network adapter with the lowest capacity can provide. Reserved bandwidth that is unused becomes available to other types of system traffic. However, Network I/O Control does not redistribute the capacity that system traffic does not use to virtual machine placement.
Limit	The maximum bandwidth, in Mbps or Gbps, that a system traffic type can consume on a single physical adapter.

Example of Bandwidth Reservation for System Traffic

The capacity of the physical adapters determines the bandwidth that you guarantee. According to this capacity, you can guarantee minimum bandwidth to a system feature for its optimal operation.

For example, on a distributed switch that is connected to ESX hosts with 10 GbE network adapters, you might configure reservation to guarantee 1 Gbps for management through vCenter, 1 Gbps for vSphere Fault Tolerance, 1 Gbps for vSphere vMotion traffic, and 0.5 Gbps for virtual machine traffic. Network I/O Control allocates the requested bandwidth on each physical network adapter. You can reserve no more than 75 percent of the bandwidth of a physical network adapter, that is, no more than 7.5 Gbps.

You might leave more capacity unreserved to let the host allocate bandwidth dynamically according to shares, limits, and use, and to reserve only bandwidth that is enough for the operation of a system feature.

How to Allocate Bandwidth for System Traffic

Assign bandwidth for host management, virtual machines, NFS storage, vSphere vMotion, vSphere Fault Tolerance, vSAN, and vSphere Replication on the physical adapters that are connected to a vSphere Distributed Switch.

- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.

To enable bandwidth allocation for virtual machines by using Network I/O Control, configure the virtual machine system traffic. The bandwidth reservation for virtual machine traffic is also used in admission control. When you power on a virtual machine, admission control verifies that enough bandwidth is available.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Resource Allocation**.
3. Click **System Traffic**.
   You see the bandwidth allocation for the types of system traffic.
4. Select the traffic type according to the vSphere feature that you want to provision and click **Edit**.
   The network resource settings for the traffic type appear.
5. From the **Shares** drop-down menu, edit the share of the traffic in the overall flow through a physical adapter.

   Network I/O Control applies the configured shares when a physical adapter is saturated.

You can select an option to set a pre-defined value, or select **Custom** and type a number from 1 to 100 to set another share.

6. In the **Reservation** text box, enter a value for the minimum bandwidth that must be available for the traffic type.

   The total reservation for system traffic must not exceed 75% of the bandwidth supported by the physical adapter with the lowest capacity of all adapters connected to the distributed switch.

7. In the **Limit** text box, set the maximum bandwidth that system traffic of the selected type can use.
8. Click **OK** to apply the allocation settings.

vCenter propagates the allocation from the distributed switch to the host physical adapters that are connected to the switch.

Allocate Bandwidth for Virtual Machine Traffic

Version 3 of Network I/O Control lets you configure bandwidth requirements for individual virtual machines. You can also use network resource pools where you can assign a bandwidth quota from the aggregated reservation for the virtual machine traffic and then allocate bandwidth from the pool to individual virtual machines.

How to Allocate Bandwidth for Virtual Machines

Network I/O Control allocates bandwidth for virtual machines by using two models: allocation across the entire vSphere Distributed Switch based on network resource pools and allocation on the physical adapter that carries the traffic of a virtual machine.

Network Resource Pools

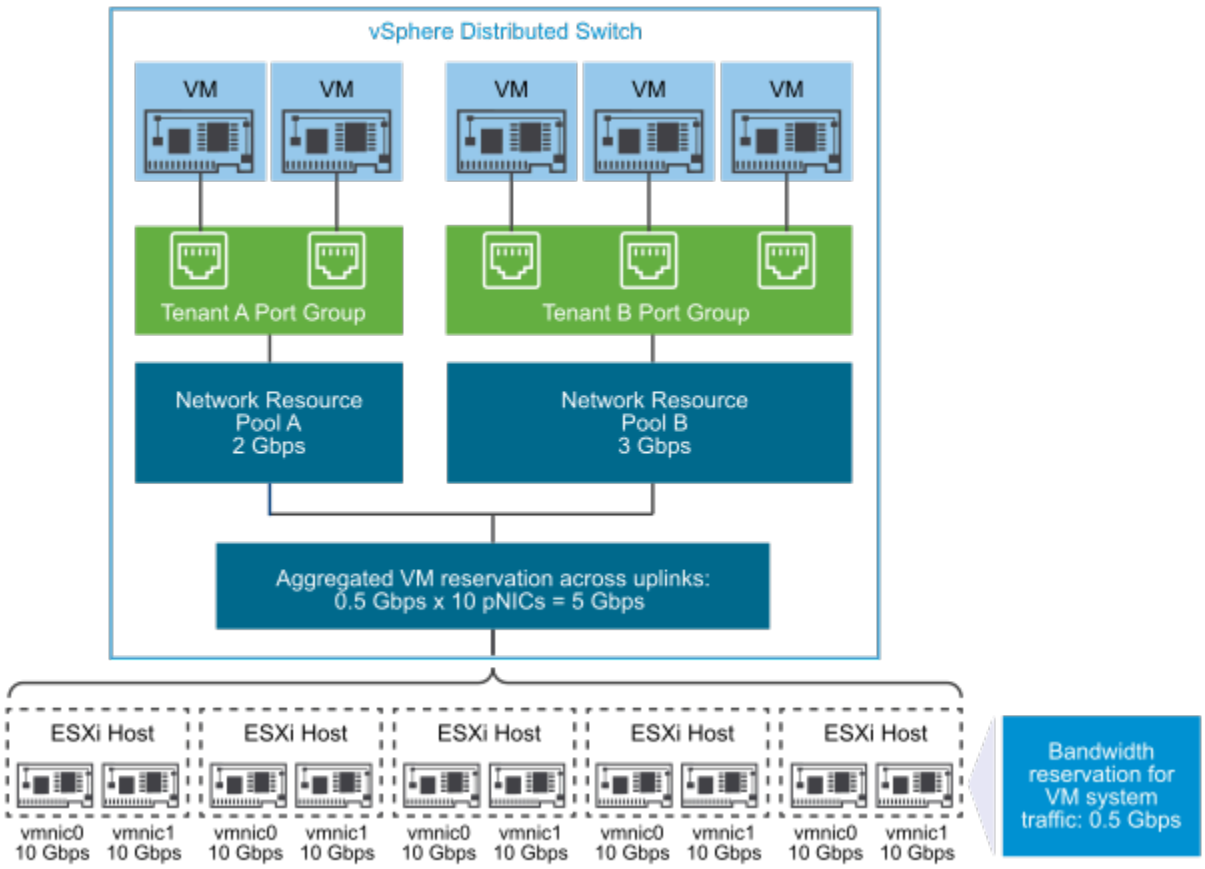
A network resource pool represents a part of the aggregated bandwidth that is reserved for the virtual machine system traffic on all physical adapters connected to the distributed switch.

For example, if the virtual machine system traffic has 0.5 Gbps reserved on each 10 GbE uplink on a distributed switch that has 10 uplinks, then the total aggregated bandwidth available for VM reservation on this switch is 5 Gbps. Each network resource pool can reserve a quota of this 5 Gbps capacity.

The bandwidth quota that is dedicated to a network resource pool is shared among the distributed port groups associated with the pool. A virtual machine receives bandwidth from the pool through the distributed port group the VM is connected to.

By default, distributed port groups on the switch are assigned to a network resource pool, called default, whose quota is not configured.

Figure 63: Bandwidth Aggregation for Network Resource Pools Across the Uplinks of a vSphere Distributed Switch



Defining Bandwidth Requirements for a Virtual Machine

You allocate bandwidth to an individual virtual machine similarly to allocating CPU and memory resources. Network I/O Control version 3 provisions bandwidth to a virtual machine according to shares, reservation, and limits that are defined for a network adapter in the VM hardware settings. The reservation represents a guarantee that the traffic from the virtual machine can consume at least the specified bandwidth. If a physical adapter has more capacity, the virtual machine may use additional bandwidth according to the specified shares and limit.

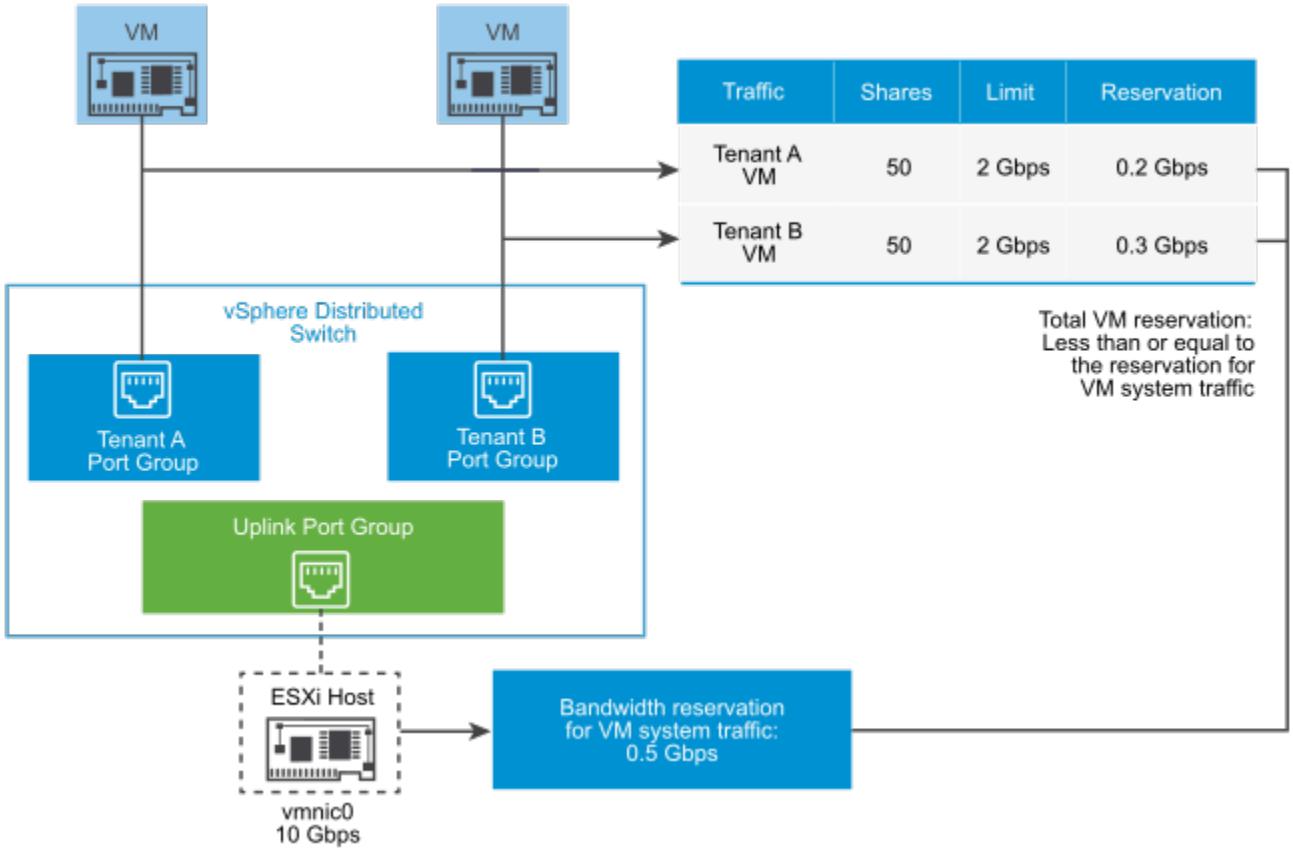
Bandwidth Provisioning to a Virtual Machine on the Host

To guarantee bandwidth, Network I/O Control implements a traffic placement engine that becomes active if a virtual machine has bandwidth reservation configured. The distributed switch attempts to place the traffic from a VM network adapter to the physical adapter that can supply the required bandwidth and is in the scope of the active teaming policy.

The total bandwidth reservation of the virtual machines on a host cannot exceed the reserved bandwidth that is configured for the virtual machine system traffic.

The actual limit and reservation also depends on the traffic shaping policy for the distributed port group the adapter is connected to. For example, if a VM network adapter requires a limit of 200 Mbps and the average bandwidth configured in the traffic shaping policy is 100 Mbps, then the effective limit becomes 100Mbps.

Figure 64: Configuration for Bandwidth Allocation for Individual Virtual Machines



The reservation of bandwidth for a VM in Network I/O Control is guaranteed against the reservation of VM system traffic on the physical adapter that sends the traffic from the VM.

Bandwidth Allocation Parameters for Virtual Machine Traffic

Network I/O Control version 3 allocates bandwidth to individual virtual machines based on configured shares, reservation, and limit for the network adapters in the VM hardware settings.

Table 154: Bandwidth Allocation Parameters for a VM Network Adapter

Parameter for Bandwidth Allocation	Description
Shares	The relative priority, from 1 to 100, of the traffic through this VM network adapter against the capacity of the physical adapter that is carrying the VM traffic to the network.
Reservation	The minimum bandwidth, in Mbps, that the VM network adapter must receive on the physical adapter.
Limit	The maximum bandwidth on the VM network adapter for traffic to other virtual machines on the same or on another host.

Admission Control for Virtual Machine Bandwidth

To guarantee that sufficient bandwidth is available to a virtual machine, vSphere implements admission control at host and cluster levels based on bandwidth reservation and teaming policy.

Bandwidth Admission Control in vSphere Distributed Switch

When you power on a virtual machine, the Network I/O Control feature on a distributed switch verifies that these conditions are satisfied on the host.

- A physical adapter on the host can supply the minimum bandwidth to the VM network adapters in accordance with the teaming policy and reservation.
- The reservation for a VM network adapter is less than the free quota in the network resource pool.

If you change the reservation for a network adapter of a running virtual machine, Network I/O Control verifies again whether the associated network resource pool can accommodate the new reservation. If the pool does not have enough unclaimed quota, the change is not applied.

To use admission control in vSphere Distributed Switch, perform the following tasks:

- Configure bandwidth allocation for the virtual machine system traffic on the distributed switch.
- Configure a network resource pool with a reservation quota from the bandwidth configured for virtual machine system traffic.
- Associate the network resource pool with the distributed port group that connects the virtual machines to the switch.
- Configure the bandwidth requirements of a virtual machine connected to the port group.

Bandwidth Admission Control in vSphere DRS

If you power on a virtual machine that is in a cluster, vSphere DRS places the virtual machine on a host that has the capacity to guarantee the bandwidth reserved for the virtual machine according to the active teaming policy.

vSphere DRS migrates a virtual machine to another host to satisfy the bandwidth reservation of the virtual machine in these situations:

- The reservation is changed to a value that the initial host can no longer satisfy.
- A physical adapter that carries traffic from the virtual machine is offline.

To use admission control in vSphere DRS, perform the following tasks:

- Configure bandwidth allocation for the virtual machine system traffic on the distributed switch.
- Configure the bandwidth requirements of a virtual machine that is connected to the distributed switch.

For more information about resource management according to bandwidth demands of virtual machines, see the *vSphere Resource Management* documentation.

Bandwidth Admission Control in vSphere HA

When a host fails or is isolated, vSphere HA powers on a virtual machine on another host in the cluster according to the bandwidth reservation and teaming policy.

To use admission control in vSphere HA, perform the following tasks:

- Allocate bandwidth for the virtual machine system traffic.
- Configure the bandwidth requirements of a virtual machine that is connected to the distributed switch.

For more information about vSphere HA provides failover based on the bandwidth demands of virtual machines, see the *vSphere Availability* documentation.

Create a Network Resource Pool

Create network resource pools on a vSphere Distributed Switch to reserve bandwidth for a set of virtual machines.

- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.

A network resource pool provides a reservation quota to virtual machines. The quota represents a portion of the bandwidth that is reserved for virtual machine system traffic on the physical adapters connected to the distributed switch. You can set aside bandwidth from the quota for the virtual machines that are associated with the pool. The reservation

from the network adapters of powered on VMs that are associated with the pool must not exceed the quota of the pool. See How to Allocate Bandwidth for Virtual Machines.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Resource Allocation**.
3. Click **Network resource pools**.
4. Click the **Add** icon.
5. Optional: Type a name and a description for the network resource pool.
6. Enter a value for **Reservation quota**, in Mbps, from the free bandwidth that is reserved for the virtual machine system traffic.

   The maximum quota that you can assign to the pool is determined according to the following formula:

   ```
   max reservation quota = aggregated reservation for vm system traffic - quotas of the other resource pools
   ```

   where

   - `aggregated reservation for vm system traffic` = configured bandwidth reservation for the virtual machine system traffic on each pNIC * number of pNICs connected to the distributed switch
   - `quotas of the other pools` = the sum of the reservation quotas of the other network resource pools
7. Click **OK**.

Add one or more distributed port groups to the network resource pool so that you can allocate bandwidth to individual virtual machines from the quota of the pool. See Add a Distributed Port Group to a Network Resource Pool.

Add a Distributed Port Group to a Network Resource Pool

Add a distributed port group to a network resource pool so that you can allocate bandwidth to the virtual machines that are connected to the port group.

Verify that Network I/O Control is enabled. See Enable Network I/O Control on a vSphere Distributed Switch.

To assign a network resource pool to several distributed port groups at once, you can use the Resource allocation policy in the **Manage Distributed Port Groups** wizard. See Manage Policies for Multiple Port Groups on a vSphere Distributed Switch.

Network I/O Control allocates bandwidth to the virtual machines associated with the distributed port group according to the model implemented in the Network I/O Control version that is active on the distributed switch. See What is vSphere Network I/O Control .

1. Locate a distributed port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups**.
2. Right-click the distributed port group and select **Edit settings**.
3. In the Edit Settings dialog box, click **General**.
4. From the **Network resource pool** drop-down menu, select the network resource pool and click **OK**.

   If the distributed switch does not contain network resource pools, you see only the **(default)** option in the drop-down menu.

Configure Bandwidth Allocation for a Virtual Machine

You can configure bandwidth allocation to individual virtual machines that are connected to a distributed port group. You can use shares, reservation, and limit settings for bandwidth.

- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. From the Actions menu, select Edit Settings.
3. Expand the Network adapter section of the VM network adapter.
4. If you want to configure bandwidth allocation for a new VM network adapter, from the **Add new device** drop-down menu select **Network adapter**.

   A New Network section displays options for bandwidth allocation and other network adapter settings.
5. If the VM network adapter is not connected to the distributed port group, select the port group from the drop-down menu next to the Network adapter or New Network label.
6. From the **Shares** drop-down menu, set the relative priority of the traffic from this virtual machine as shares from the capacity of the connected physical adapter.

   Network I/O Control applies the configured shares when a physical adapter is saturated.

   You can select an option to set a pre-defined value, or select **Custom** and type a number from 1 to 100 to set another share.
7. In the **Reservation** text box, reserve a minimum bandwidth that must be available to the VM network adapter when the virtual machine is powered on.

   If you provision bandwidth by using a network resource pool, the reservation from the network adapters of powered on VMs that are associated with the pool must not exceed the quota of the pool.

   If vSphere DRS is enabled, to power on the virtual machine, make sure that the reservation from all VM network adapters on the host does not exceed the bandwidth reserved for virtual machine system traffic on the host physical adapters.
8. In the **Limit** text box, set a limit on the bandwidth that the VM network adapter can consume.
9. Click **OK**.

Network
I/O Control allocates the bandwidth that you reserved for the network adapter of the virtual machine out of the reservation quota of the network resource pool.

Configure Bandwidth Allocation on Multiple Virtual Machines

With a single operation, configure bandwidth allocation on multiple virtual machines that are connected to a specific network resource pool, for example, after you upgrade Network I/O Control to version 3.

- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.
- Verify that the virtual machines are associated with a specific network resource pool through the connected distributed port groups. See Add a Distributed Port Group to a Network Resource Pool.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Resource Allocation**.
3. Click **Network resource pools**.
4. Select a network resource pool.
5. Click **Virtual Machines**.

   A list of the VM network adapters that are connected to the selected network resource pool appears.

6. Select the VM network adapters whose settings you want to configure and click **Edit**.
7. From the **Shares** drop-down menu, set the relative priority of traffic from these virtual machines in the scope of the physical adapters that carry the traffic.

   Network I/O Control applies the configured shares when a physical adapter is saturated.
8. In the **Reservation** text box, reserve a minimum bandwidth that must be available to each VM network adapter when the virtual machines are powered on.

   If you provision bandwidth by using a network resource pool, the reservation from the network adapters of powered on VMs that are associated with the pool must not exceed the quota of the pool.
9. In the **Limit** text box, set a limit on the bandwidth that each VM network adapter can consume.
10. Click **OK**.

Modify the Quota of a Network Resource Pool

Learn how to modify the quota of bandwidth that you can reserve for virtual machines connected to a set of distributed port groups.

- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Resource Allocation**.
3. Click **Network resource pools**.
4. Select a network resource pool from the list and click **Edit**.
5. In the **Reservation quota** text box, enter the bandwidth quota for virtual machines from the aggregation of free bandwidth that is reserved for virtual machine system traffic on all physical adapters on the switch.
6. Click **OK**.

Remove a Distributed Port Group from a Network Resource Pool

To stop allocating bandwidth to the virtual machines from the reservation quota of a network resource pool, remove the association between the port group to which the virtual machines are connected and the pool.

1. Locate a distributed port group in the vSphere Client.
   a) Select a distributed switch and click the **Networks** tab.
   b) Click **Distributed Port Groups**.
2. Right-click the distributed port group and select **Edit Settings**.
3. In the Edit Settings dialog box for the port group, click **General**.
4. From the **Network resource pool** drop-down menu, select **(default)** and click **OK**.

The distributed port group becomes associated with the default VM network resource pool.

Delete a Network Resource Pool

Delete a network resource pool that is no longer in use.

Uncouple the network resource pool from all associated distributed port groups. See Remove a Distributed Port Group from a Network Resource Pool.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Resource Allocation**.
3. Click **Network resource pools**.
4. Select a network resource pool and click **Remove**.
5. Click **OK** to delete the resource pool.

Move a Physical Adapter Out of Network I/O Control

Under certain conditions you might need to exclude physical adapters with low capacity from the bandwidth allocation model of Network I/O Control version 3.

- Verify that the host is running ESX 6.5 and later.
- Verify that vSphere Distributed Switch is version 6.5.0 and later.
- Verify that Network I/O Control on the switch is version 3.

For example, if the bandwidth allocation on a vSphere Distributed Switch is tailored on top of 10 GbE NICs, you might not be able to add a 1GbE NIC to the switch because it cannot meet the higher allocation requirements configured on the 10GbE NICs.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System** and select **Advanced System Settings** .
3. Set the physical adapters that you need to function outside the scope of Network I/O Control as a comma-separated list to the `Net.IOControlPnicOptOut` parameter.
   For example: `vmnic0,vmnic3`
4. Click **OK** to apply the changes.

MAC Address Management

MAC addresses are used in the Layer 2 (Data Link Layer) of the network protocol stack to transmit frames to a recipient. In vSphere, vCenter generates MAC addresses for virtual machine adapters and VMkernel adapters, or you can assign addresses manually.

Each network adapter manufacturer is assigned a unique three-byte prefix called an Organizationally Unique Identifier (OUI), which it can use to generate unique MAC addresses.

VMware supports several address allocation mechanisms, each of them with a separate OUI:

- Generated MAC addresses
  - Assigned by vCenter
  - Assigned by the ESX host
- Manually set MAC addresses
- Generated for legacy virtual machines, but no longer used with ESX

If you reconfigure the network adapter of a powered off virtual machine, for example by changing the automatic MAC address allocation type, or setting a static MAC address, vCenter resolves any MAC address conflict before the adapter reconfiguration takes effect.

MAC Address Assignment from vCenter

vSphere provides several schemes for automatic allocation of MAC addresses in vCenter. You can select the scheme that best suits your requirements for MAC address duplication, OUI requirements for locally administered or universally administered addresses, and so on.

The following schemes of MAC address generation are available in vCenter:

- VMware OUI allocation, default allocation
- Prefix-based allocation
- Range-based allocation

After the MAC address is generated, it does not change unless the virtual machine's MAC address conflicts with that of another registered virtual machine. The MAC address is saved in the configuration file of the virtual machine.

Note:
If you use invalid prefix- or range-based allocation values, an error is logged in the `vpxd.log` file. vCenter does not allocate MAC addresses when provisioning a virtual machine.

Preventing MAC Address Conflicts

The MAC address of a powered off virtual machine is not checked against the addresses of running or suspended virtual machines.

When a virtual machine is powered on again, it might acquire a different MAC address. The change might be caused by an address conflict with another virtual machine. While this virtual machine has been powered off, its MAC address has been assigned to another virtual machine that has been powered on.

If you reconfigure the network adapter of a powered off virtual machine, for example, by changing the automatic MAC address allocation type or setting a static MAC address, vCenter resolves MAC address conflicts before the adapter reconfiguration takes effect.

For information about resolving MAC address conflicts, see the *vSphere Troubleshooting* documentation.

What is VMware OUI Allocation

VMware Organizationally Unique Identifier (OUI) allocation assigns MAC addresses based on the default VMware OUI `00:50:56` and the vCenter ID.

VMware OUI allocation is the default MAC address assignment model for virtual machines. The allocation works with up to 64 vCenter instances, and each vCenter can assign up to 64000 unique MAC addresses. The VMware OUI allocation scheme is suitable for small scale deployments.

MAC Address Format

According to the VMware OUI allocation scheme, a MAC address has the format `00:50:56:`*`XX`*`:`*`YY`*`:`*`ZZ`* where `00:50:56` represents the VMware OUI, *`XX`* is calculated as (128+ vCenter ID), and *`YY`* and *`ZZ`* are random two-digit hexadecimal numbers.

The addresses created through the VMware OUI allocation are in the range `00:50:56:80:`*`YY`*`:`*`ZZ`* - `00:50:56:BF:`*`YY`*`:`*`ZZ`*.

What is Prefix-Based MAC Address Allocation

You can use prefix-based allocation to specify an OUI other than the default one `00:50:56` by VMware, or to introduce Locally Administered MAC Addresses (LAA) for a larger address space.

Prefix-based MAC address allocation overcomes the limits of the default VMware allocation to provide unique addresses in larger scale deployments. Introducing an LAA prefix leads to a very large MAC address space (2 to the power of 46) instead of an universally unique address OUI which can give only 16 million MAC addresses.

Verify that the prefixes that you provide for different vCenter instances in the same network are unique. vCenter relies on the prefixes to avoid MAC address duplication issues, see Troubleshoot MAC Address Allocation.

What is Range-Based MAC Address Allocation

You can use range-based allocation to include or exclude ranges of Locally Administered Addresses (LAA).

You specify one or more ranges using a starting and ending MAC addresses, for example, (`02:50:68:00:00:02`, `02:50:68:00:00:FF`). MAC addresses are generated only from within the specified range.

You can specify multiple ranges of LAA, and vCenter tracks the number of used addresses for each range. vCenter allocates MAC addresses from the first range that still has addresses available. vCenter checks for MAC address conflicts within its ranges.

When using range-based allocation, you must provide different instances of vCenter with ranges that do not overlap. vCenter does not detect ranges that might be in conflict with other vCenter instances. See the Troubleshoot MAC Address Allocation for more information about resolving issues with duplicate MAC addresses.

Note:
Range-based MAC address allocation settings are lost when you upgrade to a new version of vCenter. You must manually recreate the range-based MAC address allocation settings after the upgrade.

Assigning a MAC Address

Use the vSphere Client to enable prefix-based or range-based MAC address allocation and to adjust the allocation parameters.

If you are changing from one type of allocation to another, for example changing from the VMware OUI allocation to a range-based allocation, use the vSphere Client. However, when a schema is prefix-based or range-based and you want to change to a different allocation schema, you must edit the `vpxd.cfg` file manually and restart vCenter.

Change to or Adjust Range- or Prefixed-Based Allocations

By switching from the default VMware OUI to range- or prefixed-based MAC address allocation through thevSphere Client, you can avoid and resolve MAC address duplication conflicts in vSphere deployments.

Change the allocation scheme from the default VMware OUI to range- or to prefixed-based allocation by using the **Advanced Settings** available for the vCenter instance in thevSphere Client.

To switch from range- or prefixed-based allocation back to VMware OUI allocation, or between range- and prefixed-based allocation, edit the `vpxd.cfg` file manually. See Set or Change Allocation Type.

1. In thevSphere Client, navigate to avCenter instance.
2. On the **Configure** tab, expand **Settings** and select **Advanced Settings**.
3. Click **Edit Settings.**
4. Add or edit parameters for the target allocation type.
   Use only one allocation type.
   - Change to prefix-based allocation.

Key	Example Value
`config.vpxd.macAllocScheme.prefixScheme.prefix`	005026
`config.vpxd.macAllocScheme.prefixScheme.prefixLength`	23

`prefix` and `prefixLength` determine the range of MAC address prefixes that newly added vNICs have. `prefix` is the starting OUI of MAC addresses related to the vCenter instance, and `prefixLength` determines the length of the prefix in bits.

For example, the settings from the table result in VM NIC MAC addresses starting with 00:50:26 or 00:50:27.

- Change to range-based allocation.

Key	Example Value
`config.vpxd.macAllocScheme.rangeScheme.range[X].begin`	005067000000
`config.vpxd.macAllocScheme.rangeScheme.range[X].end`	005067ffffff

X in `range[X]` stands for the range sequence number. For example, `0` in `range[0]` represents the allocation settings of the first range for MAC address allocation.

5. Click **Save**.

Set or Change Allocation Type

If you are changing from range- or prefixed-based allocation to the VMware OUI allocation, you must set the allocation type in the `vpxd.cfg` file and restart the vCenter.

Decide on an allocation type before changing the `vpxd.cfg` file. For information on allocation types, see MAC Address Assignment from vCenter

1. On the host machine of vCenter, navigate to the directory `/etc/vmware-vpx`.
2. Open the `vpxd.cfg` file.
3. Decide on an allocation type to use and enter the corresponding XML code in the file to configure the allocation type.
   The following are examples of XML code to use.

   Note:
   Use only one allocation type.

   - VMware OUI allocation

     ```
     <vpxd>
        <macAllocScheme>
           <VMwareOUI>true</VMwareOUI>
        </macAllocScheme>
     </vpxd>
     ```

   - Prefix-based allocation

     ```
      <vpxd>
         <macAllocScheme>
           <prefixScheme>
              <prefix>005026</prefix>
              <prefixLength>23</prefixLength>
           </prefixScheme>
        </macAllocScheme>
     </vpxd>
     ```

   - Range-based allocation

     ```
     <vpxd>
        <macAllocScheme>
            <rangeScheme>
               <range id="0">
                  <begin>005067000001</begin>
                  <end>005067000001</end>
               </range>
            </rangeScheme>
        </macAllocScheme>
     </vpxd>
     ```

4. Save the `vpxd.cfg`.
5. Restart the vCenter host.

MAC Address Generation on ESX Hosts

An ESX host generates the MAC address for a virtual machine adapter when the host is not connected to vCenter. Such addresses have a separate VMware OUI to avoid conflicts.

The ESX host generates the MAC address for a virtual machine adapter in one of the following cases:

- The host is not connected to vCenter.
- The virtual machine configuration file does not contain the MAC address and information about the MAC address allocation type.

MAC Address Format

The host generates MAC addresses that consists of the VMware OUI `00:0C:29` and the last three octets in hexadecimal format of the virtual machine UUID. The virtual machine UUID is based on a hash calculated by using the UUID of the ESX physical machine and the path to the configuration file (`.vmx`) of the virtual machine.

Preventing MAC Address Conflicts

All MAC addresses that have been assigned to network adapters of running and suspended virtual machines on a given physical machine are tracked for conflicts.

If you import a virtual machine with a host-generated MAC address from one vCenter to another, select the **I Copied It** option when you power on the virtual machine to regenerate the address and avoid potential conflicts in the target vCenter or between the vCenter systems.

How to set a Static MAC Address to a Virtual Machine

In most network deployments, generated MAC addresses are a good approach. However, you might need to set a static MAC address for a virtual machine adapter with unique value.

The following cases show when you might set a static MAC address:

- Virtual machine adapters on different physical hosts share the same subnet and are assigned the same MAC address, causing a conflict.
- Ensure that a virtual machine adapter always has the same MAC address.

By default, VMware uses the Organizationally Unique Identifier (OUI) `00:50:56` for manually generated addresses, but all unique manually generated addresses are supported.

> **Note:**
> Make sure that no other non-VMware devices use addresses assigned to VMware components. For example, you might have physical servers in the same subnet, which use `11:11:11:11:11:11`, `22:22:22:22:22:22` as static MAC addresses. The physical servers do not belong to the vCenter inventory, and vCenter is not able to check for address collision.

VMware OUI in Static MAC Addresses

By default, static MAC addresses have the VMware Organizationally Unique Identifier (OUI) as the prefix. However, the range of free address provided by the VMware OUI is restricted.

If you decide to use the VMware OUI, part of the range is reserved for use by vCenter, host physical NICs, virtual NICs, and for future use.

You can set a static MAC address that contains the VMware OUI prefix in compliance with the following format:

```
00:50:56:XX:YY:ZZ
```

where *XX* is a valid hexadecimal number between 00 and 3F, and *YY* and *ZZ* are valid hexadecimal numbers between 00 and FF. To avoid conflict with MAC addresses that are generated by vCenter or are assigned to VMkernel adapters for infrastructure traffic, the value for *XX* must not be greater than 3F.

The maximum value for a manually generated MAC address is as follows.

```
00:50:56:3F:FF:FF
```

To avoid conflicts between the generated MAC addresses and the manually assigned ones, select a unique value for *XX*:*YY*:*ZZ* from your hard-coded addresses.

Assign a Static MAC Address

You can assign static MAC addresses to the virtual NIC of a powered off virtual machine by using the vSphere Client.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Power off the virtual machine.
3. From the Actions menu, select Edit Settings.
4. Select the **Virtual Hardware** tab in the dialog box displaying the settings.
5. Expand the network adapter section.
6. Under MAC Address, select **Manual** from the drop-down menu.
7. Type the static MAC address, and click **OK**.
8. Power on the virtual machine.

Assign a Static MAC Address in the Virtual Machine Configuration File

To set a static MAC address for a virtual machine, you can edit the configuration file of the virtual machine by using the vSphere Client.

1. Locate the virtual machine in the vSphere Client.
   a) Select a data center, folder, cluster, resource pool, or host and click the **VMs** tab.
   b) Click **Virtual Machines** and click the virtual machine from the list.
2. Power off the virtual machine.
3. From the **Actions** drop-down menu, select **Edit Settings**.
4. Select the **VM Options** tab and expand **Advanced**.
5. Click **Edit Configuration**.
6. To assign a static MAC address, add or edit parameters as required.

Parameter	Value
`ethernet` *`X`* `.addressType`	`static`
`ethernet` *`X`* `.address`	*`MAC_address_of_the_virtual_NIC`*

   `X` next to `ethernet` stands for the sequence number of the virtual NIC in the virtual machine.

   For example, `0` in `ethernet0` represents the settings of the first virtual NIC device added to the virtual machine.
7. Click **OK**.
8. Power on the virtual machine.

Configure vSphere for IPv6

Configure ESX hosts and vCenter for operation in a pure IPv6 environment for larger address space and improved address assignment.

IPv6 is designated by the Internet Engineering Task Force (IETF) as the successor to IPv4 providing the following benefits:

- Increased address length. The increased address space resolves the problem of address exhaustion and eliminates the need for network address translation. IPv6 uses 128-bit addresses compared with the 32-bit addresses used by IPv4.
- Ability for improved address autoconfiguration of nodes.

vSphere IPv6 Connectivity

In an environment that is based on vSphere 6.0 and later, nodes and features can communicate over IPv6 transparently supporting static and automatic address configuration.

IPv6 in the Communication between vSphere Nodes

The nodes in a vSphere deployment can communicate using IPv6 and accept assigned addresses according to the network configuration.

Table 155: IPv6 Support of the Nodes in a vSphere Environment

Connection Type	IPv6 Support	Address Configuration of vSphere nodes
ESX to ESX	Yes	• Static • Automatic: AUTOCONF/DHCPv6
vCenter machine to ESX	Yes	• Static • Automatic: AUTOCONF/DHCPv6
vCenter machine to machine	Yes	• Static • Automatic: AUTOCONF/DHCPv6
ESX to vSphere Client machine	Yes	• Static • Automatic: AUTOCONF/DHCPv6
Virtual machine to virtual machine	Yes	• Static • Automatic: AUTOCONF/DHCPv6
ESX to iSCSI Storage	Yes	• Static • Automatic: AUTOCONF/DHCPv6
ESX to NFS Storage	Yes	• Static • Automatic: AUTOCONF/DHCPv6
ESX to Active Directory	No Use LDAP through vCenter to connect ESX to the Active Directory database	-
vCenter to Active Directory	No Use LDAP to connect vCenter to the Active Directory database	-

IPv6 Connectivity of vSphere Features

Certain vSphere features do not support IPv6:

- vSphere DPM over Intelligent Platform Management Interface (IPMI) and Hewlett-Packard Integrated Lights-Out (iLO). vSphere 6.5 and later supports only Wake-On-LAN (WOL) to bring a host out of standby mode.
- Authentication Proxy
- vSphere Management Assistant and ESXCLI connected to Active Directory.
  Use LDAP to connect the vSphere Management Assistant or the ESXCLI to the Active Directory database.

IPv6 Connectivity of Virtual Machines

Virtual machines can exchange data in the network over IPv6. vSphere supports both static and automatic assignment of IPv6 addresses for virtual machines.

Configuring one or more IPv6 addresses is also possible when you customize the guest operating system of a virtual machine.

vSAN Connectivity

vSAN supports IPv6. Use NFS 4.1 with AUTH_SYS.

FQDNs and IPv6 Addresses

In vSphere, you should use fully qualified domain names (FQDNs) that are mapped to IPv6 addresses on the DNS server. You can use IPv6 addresses if they have a valid FQDN on the DNS server for reverse lookup.

To deploy vCenter in a pure IPv6 environment, you must use FQDNs only.

Deploy vSphere on IPv6

Run vSphere in a pure IPv6 environment to use an extended address space and flexible address assignment.

If you plan to deploy vCenter and ESX hosts in an IPv6 network, you must perform additional steps.

Enable IPv6 on a vSphere Installation

If you have a greenfield deployment of vSphere in an IPv6 network, configure ESX and vCenter for pure IPv6 management connection by configuring IPv6 on the deployment nodes and connecting them.

- Verify that the IPv6 addresses for vCenter, the ESX hosts and an external database, if used, are mapped to fully qualified domain names (FQDNs) on the DNS server.
- Verify that the network infrastructure provides IPv6 connectivity for the ESX hosts, vCenter and external database if used.
- Verify that vCenter is version 6.5 or later and is installed with FQDN that is mapped to an IPv6 address. See the *vCenter Deployment and Setup* documentation.
- Verify that the ESX hosts are version 6.5 or later. See the *vCenter Deployment and Setup* documentation.

1. In the Direct Console User Interface (DCUI), configure each ESX host as a pure IPv6 node.
   a) In the DCUI, press F2 and log in to the host.
   b) From the **Configure Management Network** menu, select **IPv6 Configuration** and press Enter.
   c) Assign an IPv6 address to the host.

Address Assignment Option	Description
Automatic address assignment using DHCPv6	1. Select the **Use dynamic IPv6 address and network configuration** option and select **Use DHCPv6**. 2. Press Enter to save the changes.
Static address assignment	1. Select the **Set static IPv6 address and network configuration** option and enter the IPv6 address of the host and the default gateway. 2. Press Enter to save the changes.

   d) From the **Configure Management Network** menu, select **IPv4 Configuration** and press Enter.
   e) Select **Disable IPv4 configuration for management network** and press Enter.
2. In the vSphere Client, add the hosts to the inventory.

Enable IPv6 on an Upgraded vSphere Environment

On an IPv4 deployment of vSphere that consists of an installed or upgraded vCenter and upgraded ESX, configure ESX and vCenter for pure IPv6 management connection by enabling IPv6 on the deployed nodes and reconnecting them.

- Verify that the network infrastructure provides IPv6 connectivity for the ESX hosts, vCenter and external database if used.
- Verify that the IPv6 addresses for vCenter, the ESX hosts and an external database, if used, are mapped to fully qualified domain names (FQDNs) on the DNS server.
- Verify that you have version 6.x or later of vCenter installed or upgraded. See the *vCenter Deployment and Setup* and *vCenter Upgrade* documentation.

- Verify that all ESX hosts are upgraded to version 6.x or later. See the *VMware ESX Upgrade* documentation.

1. In the vSphere Client, disconnect the hosts from vCenter.
2. Configure each ESX host as a pure IPv6 node.
   a) Open an SSH connection and log in to the ESX host.
   b) Run the following command:
      ```
      esxcli network ip interface ipv6 set -i vmk0 -e true
      ```
   c) Assign an IPv6 address to the management network.

Address Assignment Option	Description

Static address assignment	1. Open an SSH connection and log in to the ESX host.
	2. Set a static IPv6 address for the management network vmk0 by running the following command:
	`esxcli network ip interface ipv6 address add -I *IPv6_address* -i vmk0`
	3. Set the default gateway for the management network vmk0 by running the following command:
	`esxcli network ip interface ipv6 set -i vmk0 -g *default_gateway_IPv6_address*`
	4. Add a DNS server by running the following command:
	`esxcli network ip dns server add -s *DNS_server_IPv6_address*`

Automatic address assignment using DHCPv6	1. Open an SSH connection and log in to the ESX ESX host.
	2. Enable DHCPv6 for the management network vmk0 by running the following command:
	`esxcli network ip interface ipv6 -i vmk0 -enable-dhcpv6 = true`
	3. Enable IPv6 Router Advertised for the management network vmk0 by running the following command:
	`esxcli network ip interface ipv6 set -i vmk0 -enable-router-adv =true`
	4. Add a DNS server or use the DNS setting published by DHCPv6 by running one the following commands:
	`esxcli network ip dns server add -s *DNS_server_IPv6_address*`
	`esxcli network ip interface ipv6 set -i vmk0 --peer-dns=true`

3. Deactivate the IPv4 configuration for the management network.
   a) Open an SSH connection and log in to the ESX host.
   b) Run the following command:
      ```
      esxcli network ip interface ipv4 set -i vmk0 --type=none
      ```

4. If vCenter uses an external database, configure the database as an IPv6 node.
5. Configure vCenter as a pure IPv6 node and restart it.
6. Deactivate the IPv4 on the database server.
7. In the vSphere Client, add the hosts to the inventory.
8. Deactivate the IPv4 in the network infrastructure.

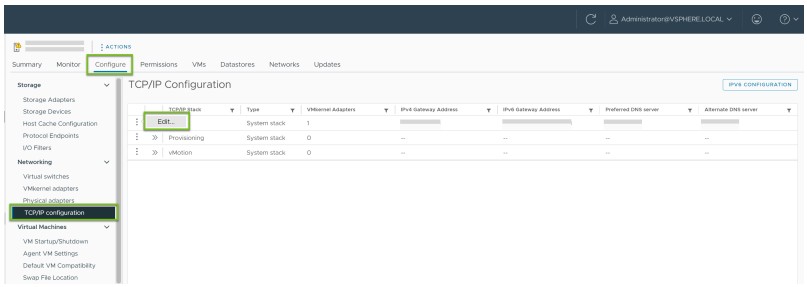
Activate or Deactivate IPv6 Support on a Host

The IPv6 support in vSphere lets hosts work in an IPv6 network that has a large address space, enhanced multicasting, simplified routing, and so on.

In ESX 6.0 and later releases, IPv6 is enabled by default.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **TCP/IP Configuration**.
3. Click **Edit**.

   Figure 65: TCP/IP Configuration



4. Use the toggle button to activate or deactivate IPv6 support.
5. Click **OK**.
6. Reboot the host to apply the changes in the IPv6 support.

Configure the IPv6 settings of VMkernel adapters on the host, for example, of the management network. See Set Up IPv6 on an ESX Host.

Set Up IPv6 on an ESX Host

To connect an ESX host over IPv6 to the management network, vSphere vMotion, shared storage, vSphere Fault Tolerance, and so on, edit the IPv6 settings of the VMkernel adapters on the host.

Verify that IPv6 is enabled on the ESX host. See Activate or Deactivate IPv6 Support on a Host .

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and select **VMkernel adapters**.
3. Select the VMkernel adapter on the target distributed or standard switch and click **Edit**.
4. In the Edit Settings dialog box, click **IPv6 settings**.
5. Configure the address assignment of the VMkernel adapter.

IPv6 Address Option	Description
Obtain IPv6 address automatically through DHCP	Receive an IPv6 address for the VMkernel adapter from a DHCPv6 server.
Obtain IPv6 address automatically through Router Advertisement	Receive an IPv6 address for the VMkernel adapter from a router through Router Advertisement.

IPv6 Address Option	Description
Static IPv6 addresses	Set one or more addresses. For each address entry, enter the IPv6 address of the adapter, subnet prefix length and IPv6 address of the default gateway.

You can select several assignment options according the configuration of your network.

6. Click **OK** to apply the changes on the VMkernel adapter.

Set Up IPv6 on vCenter

Use the vSphere Client to configure vCenter for communication with ESX hosts in an IPv6 network.

1. On the vSphere Client main page, click **Administration** > **Deployment** > **System Configuration**.
2. Under **System Configuration**, select a node from the list.
3. Click **Login**.
4. Use the vSphere Client to log in as administrator@*`your_domain_name`* to the vCenter instance.
   The address is of the type http://*`appliance-IP-address-or-FQDN`*/ui.
5. Under Common, select **Networking** and click **Edit**.
6. Expand the network interface name to edit the IP address settings.
7. Edit the IPv6 address settings.

Option	Description
Enable or Disable IPv6 settings	Enables or disables the IPv6 address based on the toggle switch option.
Obtain IPv6 settings automatically through DHCP	Assigns IPv6 addresses to the appliance automatically from the network by using DHCP.
Obtain IPv6 settings automatically through router advertisement	Assigns IPv6 addresses to the appliance automatically from the network by using router advertisement.
Use static IPv6 addresses	Uses static IPv6 addresses that you set up manually. 1. Click the checkbox. 2. Enter the IPv6 address and the subnet prefix length. 3. Click **Add** to enter additional IPv6 addresses. 4. Click **Save**. **Note:** For static IPv4 or IPv6 addresses, DNS server must be set manually.

You can configure the appliance to obtain the IPv6 settings automatically through both DHCP and router advertisement. You can assign a static IPv6 address at the same time.

Note:
Second and third-party solutions must re-register when there is a change in the IPv4 and IPv6 IP addresses.

8. Optional: To remove IPv6 addresses that are assigned automatically through Router Advertisement, click **Remove Addresses** and delete the addresses.

   You might want to delete certain IPv6 addresses that vCenter obtained through Router Advertisement to stop the communication on these addresses and to enforce the configured static addresses.

Connect the ESX hosts to vCenter over IPv6 by using their FQDNs.

Monitor Network Connection and Traffic

Monitor network connection and packets that pass through the ports of a vSphere Standard Switch or a vSphere Distributed Switch to analyze the traffic between virtual machines and hosts.

How to Capture Network Packets

Learn how to use the PacketCapture utility to diagnose networking problems such as slow connection, lost packets, and connectivity problems

PacketCapture is a lightweight tcpdump utility that captures and stores only the minimum amount of data that is needed to diagnose the network problem. PacketCapture is integrated in the rhttpproxy service of ESX and vCenter. You start and stop PacketCapture by editing the rhttpproxy service XML configuration file.

Note:

For ESX 8.0 and earlier, the process for enabling network packet captures was by editing the rhttpproxy service XML configuration file. Starting ESX 8.0 Update 1 and later, reverse proxy configuration has been moved from XML files to the configstore database. To enable, Capture Network Packets with configstorecli, see Broadcom knowledge base 312007.

1. Start capturing packets.
   a) Open an SSH connection and log in to the ESX host or vCenter.
   b) Open the `config.xml` file for editing.

vSphere Component	File Location
ESX	`/etc/vmware/rhttpproxy/config.xml`
vCenter	`/etc/vmware-rhttpproxy/config.xml`

   c) Make the following changes.

```
<config>
  <packetCapture>
    <enabled>true</enabled>
```

   d) Optional: Configure PacketCapture options.

Option and Default Value	Description
<validity>72</validity>	On startup delete all `pcap` and `pcap.gz` files that were last modified before the specified period of hours and are not part of the current process.
<directory>/directory_path</directory>	The directory in which `pcap` and `pcap.gz` files are stored. The directory must exist and be accessible.
<maxDataInPcapFile>52428800</maxDataInPcapFile>	The amount of captured data in bytes that each `pcap` and `pcap.gz` file can store before rolling over to the next file. The minimum size is 5 MB on vCenter and 2.5MB on ESX. **Note:** Storing 50 MB of captured data in a `pcap` file requires a `pcap` file of about 67.5 MB.
<maxPcapFilesCount>5</maxPcapFilesCount>	The number of `pcap` or `pcap.gz` files to rotate. The minimum number is 2.

   e) Save and close the `config.xml` file.
   f) Reload the `config.xml` file by running the following command.

```
kill -SIGHUP `pidof rhttpproxy`
```

2. Stop capturing packets.
   a) Open an SSH connection and log in to the ESX host or vCenter.
   b) Open the `config.xml` file for editing.
   c) Make the following changes.

```
<config>
  <packetCapture>
    <enabled>false</enabled>
```

   d) Save and close the `config.xml` file.
   e) Reload the `config.xml` file by running the following command.

```
kill -SIGHUP `pidof rhttpproxy`
```

3. Collect the captured data.

   The `pcap` or `pcap.gz` files are stored in the following default directories.

vSphere Component	File Location
ESX	`/var/run/log`
vCenter	`/var/log/vmware/rhttpproxy`

Copy the `pcap` and `pcap.gz` files to a system that runs a network analyzer tool, such as Wireshark, and examine the packet details.

Before you analyze the `pcap` and `pcap.gz` captured from an ESX host, use the TraceWrangler utility to fix the frame size metadata. For more information, see Broadcom knowledge base article 312012.

Capture and Trace Network Packets by Using the pktcap-uw Utility

Learn how to monitor the traffic that flows through physical network adapters, VMkernel adapters, and virtual machines adapters, and analyze packet information by using the graphical user interface of network analysis tools such as Wireshark.

In vSphere you can monitor packets on a host by using the `pktcap-uw` console utility. You can use the utility without additional installation on an ESX host. `pktcap-uw` provides many points in the host network stack at which you can monitor traffic.

For detailed analysis of captured packets, you can save packet content from the `pktcap-uw` utility to files in PCAP or PCAPNG format and open them in Wireshark. You can also troubleshoot dropped packets and trace a packet's path in the network stack.

> **Note:**
> The `pktcap-uw` utility is not fully supported for backward compatibility across vSphere releases. The options of the utility might change in the future.

pktcap-uw Command Syntax for Capturing Packets

Use the `pktcap-uw` utility to inspect the contents of packets while they traverse the network stack on an ESX host.

pktcap-uw Syntax for Capturing Packets

The `pktcap-uw` command has the following syntax for capturing packets at a certain place in the network stack:

```
pktcap-uw
switch_port_arguments
capture_point_options
filter_options
output_control_options
```

Note:
Certain options of the `pktcap-uw` utility are designed for VMware internal use only and you should use them only under the supervision of VMware Technical Support. These options are not described in the *vSphere Networking* guide.

Table 156: pktcap-uw Arguments for Capturing Packets

Argument Group	Argument	Description
switch_port_arguments	`--uplink vmnic`*X*	Capture packets that are related to a physical adapter. You can combine the --uplink and --capture options for monitoring packets at a certain place in the path between the physical adapter and the virtual switch. See Capture Packets That Arrive at a Physical Adapter.
	`--vmk vmk`*X*	Capture packets that are related to a VMKernel adapter. You can combine the vmk and --capture options for monitoring packets at a certain place in the path between the VMkernel adapter and the virtual switch. See How to Capture Packets for a VMkernel Adapter.
	`--switchport {`*vmxnet3_port_ID* `|` *vmkernel_adapter_port_ID*`}`	Capture packets that are related to a VMXNET3 virtual machine adapter or to a VMkernel adapter that is connected to a particular virtual switch port. You can view the ID of the port in the network panel of the `esxtop` utility. You can combine the switchport and capture options for monitoring packets at a certain place in the path between the VMXNET3 adapter or VMkernel adapter and the virtual switch. See How to Capture Packets for a VMXNET3 Virtual Machine Adapter.
	`--lifID` *lif_ID*	Capture packets that are related to the logical interface of a distributed router. See the *VMware NSX* documentation.
capture_point_options	`--capture` *capture_point*	Capture packets at a particular place in the network stack. For example, you can monitor packets right after they arrive from a physical adapter.

Argument Group	Argument	Description
	--dir {0|1|2}	Capture packets according to the direction of the flow with regard to the virtual switch. 0 stands for incoming traffic, 1 for outgoing traffic, and 2 for bidirectional traffic. By default, the `pktcap-uw` utility captures ingress traffic. Use the --dir option together with the --uplink, --vmk, or --switchport option.
	--stage {0|1}	Capture the packet closer to its source or to its destination. Use this option to examine how a package changes while it traverses the points in the stack. 0 stands for traffic closer to source and 1 for traffic closer to destination. Use the --stage option together with the --uplink, --vmk , --switchport, or --dvfilter option.
	`--dvfilter` *`filter_name`* `--capture PreDVFilter|PostDVFilter`	Capture packets before or after a vSphere Network Appliance (DVFilter) intercepts them. See How to Capture Packets at DVFilter Level.
	-A | --availpoints	View all capture points that the `pktcap-uw` utility supports.
	For details about the capture points of the `pktcap-uw` utility, see Capture Points of the pktcap-uw Utility.	
`filter_options`	Filter captured packets according to source or destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port. See pktcap-uw Options for Filtering Packets.	
`output_control_options`	Save the contents of a packet to a file, capture only a number of packets, and capture a number of bytes at the beginning of packets, and so on. See pktcap-uw Options for Output Control.	

The vertical bars | represent alternative values, and the curly brackets {} used with vertical bars specify a list of choices for an argument or option.

pktcap-uw Command Syntax for Tracing Packets

Use the `pktcap-uw` utility to view the path of a packet in the network stack on an ESX host for latency analysis.

pktcap-uw Syntax for Tracing Packets

The command of the `pktcap-uw` utility has the following syntax for tracing packets in the network stack:

```
pktcap-uw --trace filter_optionsoutput_control_options
```

Options to the pktcap-uw Utility for Tracing Packets

The `pktcap-uw` utility supports the following options when you use it to trace packets:

Table 157: pktcap-uw Options for Tracing Packets

Argument	Description
filter_options	Filter traced packets according to source or destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port. See pktcap-uw Options for Filtering Packets.
output_control_options	Save the contents of a packet to a file and trace only a number of packets. See pktcap-uw Options for Output Control.

pktcap-uw Options for Output Control

Use the options for output control of the `pktcap-uw` utility to save packet contents to a file, capture up to a certain number of bytes from each packet, and limit the number of captured packets.

pktcap-uw Options for Output Control

The options of the `pktcap-uw` utility for output control are valid when you capture and trace packets. For information about the command syntax of the `pktcap-uw` utility, see pktcap-uw Command Syntax for Capturing Packets and pktcap-uw Command Syntax for Tracing Packets.

Table 158: Options for Output Control That Are Supported by the pktcap-uw Utility

Option	Description
`{-o | --outfile}` *pcap_file*	Save captured or traced packets to a file in packet capture (PCAP) format. Use this option to examine packets in a visual analyzer tool such as Wireshark.
-P | --ng	Save packet content in the PCAPNG file format. Use this option together with the -o or --outfile option.
`--console`	Print packet details and content to the console output. By default, the `pktcap-uw` utility shows packet information in the console output.
`{-c | --count}` *number_of_packets*	Capture the first *number_of_packets* packets.
`{-s | --snaplen}` *snapshot_length*	Capture only the first *snapshot_length* bytes from each packet. If traffic on the host is intensive, use this option to reduce the load on the CPU and storage. To limit the size of captured contents, set a value greater than 24. To capture the complete packet, set this option to 0.
`-h`	View help about the `pktcap-uw` utility.

The vertical bars | represent alternative values, and the curly brackets {} used with vertical bars specify a list of choices for an argument or option.

pktcap-uw Options for Filtering Packets

Narrow the range of packets that you monitor by using the `pktcap-uw` utility to apply filtering options for source and destination address, VLAN, VXLAN, and next level protocol consuming the packet payload.

Filter Options

The filter options for `pktcap-uw` are valid when you capture and trace packets. For information about the command syntax of the `pktcap-uw` utility, see pktcap-uw Command Syntax for Capturing Packets and pktcap-uw Command Syntax for Tracing Packets.

Table 159: Filter Options of the pktcap-uw Utility

Option	Description
`--srcmac` *`mac_address`*	Capture or trace packets that have a specific source MAC address. Use colons to separate the octets in it.
`--dstmac` *`mac_address`*	Capture or trace packets that have a specific destination MAC address. Use colons to separate the octets in it.
`--mac` *`mac_address`*	Capture or trace packets that have a specific source or destination MAC address. Use colons to separate the octets in it.
`--ethtype 0x`*`Ethertype`*	Capture or trace packets at Layer 2 according to the next level protocol that consumes the packet payload. *`EtherType`* corresponds to the EtherType field in Ethernet frames. It represents the type of next level protocol that consumes the payload of the frame. For example, to monitor traffic for the Link Layer Discovery Protocol (LLDP) protocol, type `--ethtype 0x88CC`.
`--vlan` *`VLAN_ID`*	Capture or trace packets that belong to a VLAN.
`--srcip` *`IP_addess|IP_address/subnet_range`*	Capture or trace packets that have a specific source IPv4 address or subnet.
`--dstip` *`IP_addess|IP_address/subnet_range`*	Capture or trace packets that have a specific destination IPv4 address or subnet.
`--ip` *`IP_addess`*	Capture or trace packets that have a specific source or destination IPv4 address.
`--proto 0x`*`IP_protocol_number`*	Capture or trace packets at Layer 3 according to the next level protocol that consumes the payload. For example, to monitor traffic for the UDP protocol, type `--proto 0x11`.
`--srcport` *`source_port`*	Capture or trace packets according to their source TCP port.
`--dstport` *`destination_port`*	Capture or trace packets according to their destination TCP port.
`--tcpport` *`TCP_port`*	Capture or trace packets according to their source or destination TCP port.
`--vxlan` *`VXLAN_ID`*	Capture or trace packets that belong to a VXLAN.
`--rcf` *`pcap_filter_expression`*	Capture or trace packets using the rich common filter expression. For example, to capture all ingress and egress packets whose IP content length is larger than 1000 bytes, use the filter expression `--rcf "ip[2:2]>1000"`. To select a specific source host address and port number, use the filter expression `--rcf "src host 12.0.0.1 and port 5000"`. This example filters traffic for the host address 12.0.0.1 using port 5000. To learn more about filtering network traffic with the `--rcf` option, refer to the documentation for pcap filter expressions using command-line packet analyzers such as `tcpdump`. See pcap-filter - packet filter syntax. **Note:** When using the `--rcf` option, obey the following limitations. • Do not filter VLAN packets using the `--rcf` option. To trace VLAN or VXLAN use the `pktcap-uw--vlan` or `--vxlan` options. • Do not filter an IP broadcast address. • Do not use `--rcf` on ENS ports.

Option	Description
`--rcf-tcp-data` *tcp_packet_data_filter*	Capture or trace TCP data packets using the rich common filter expression. For example, to capture all HTTP/1.0 response packets with 200 OK, use the filter expression `--rcf-tcp-data "HTTP/1.0 200 OK"`. To filter on HTTP GET requests returning an index.html file, use the filter expression `--rcf-tcp-data "GET / index.html"`.

The vertical bars | represent alternative values.

Capturing Packets by Using the pktcap-uw Utility

Capture packets through the `pktcap-uw` utility in the path between a virtual switch and the physical adapters, VMkernel adapters and virtual machine adapters to troubleshoot data transfer in the network stack on an ESX host.

Capture Packets That Arrive at a Physical Adapter

Monitor host traffic related to the external network by capturing packets at certain points in the path between a vSphere Standard Switch or vSphere Distributed Switch and a physical adapter.

You can specify a certain capture point in the data path between a virtual switch and a physical adapter, or determine a capture point by traffic direction with regard to the switch and proximity to the packet source or destination. For information about supported capture points, see Capture Points of the pktcap-uw Utility.

1. Optional: Find the name of the physical adapter that you want to monitor in the host adapter list.
   - In the vSphere Client, on the **Configure** tab for the host, expand **Networking** and select **Physical adapters**.
   - In the ESX Shell to the host, to view a list of the physical adapters and examine their state, run the following ESXCLI command:
     ```
     esxcli network nic list
     ```

   Each physical adapter is represented as `vmnic`*X*. *X* is the number that ESX assigned to the physical adapter port.
2. In the ESX Shell to the host, run the `pktcap-uw` command with the `--uplink vmnic`*X* argument and with options to monitor packets at a particular point, filter captured packets and save the result to a file.

   ```
   pktcap-uw
   --uplink vmnicX [--capturecapture_point|--dir 0|1]  [filter_options] [--outfilepcap_file_path [--ng]] [--
   countnumber_of_packets]
   ```

   where the square brackets [] enclose the options of the `pktcap-uw--uplink vmnic`*X* command and the vertical bars | represent alternative values.

   If you run the `pktcap-uw--uplink vmnic`*X* command without options, you obtain the content of packets that are incoming to the standard or distributed switch in the console output at the point where they are switched.

   a) Use the --capture option to check packets at another capture point or the --dir option at another traffic direction.

pktcap-uw Command Option	Goal
`--capture UplinkSnd`	Monitor packets immediately before they enter the physical adapter device.
`--capture UplinkRcv`	Monitor packets immediately after they are received in the network stack from the physical adapter.
`--dir 1`	Monitor packets that leave the virtual switch.
`--dir 0`	Monitor packets that enter the virtual switch.

b) Use a *filter_options* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

c) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

- To save packets to a `.pcap` file, use the --outfile option.
- To save packets to a `.pcapng` file, use the --ng and --outfile options.

You can open the file in a network analyzer tool such as Wireshark.

By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

d) Use the --count option to monitor only a number of packets.

3. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

Capture Packets That Are Received at vmnic0 from an IP Address 192.168.25.113

To capture the first 60 packets from a source system that is assigned the IP address 192.168.25.113 at vmnic0 and save them to a file called `vmnic0_rcv_srcip.pcap`, run the following `pktcap-uw` command:

```
pktcap-uw --uplink vmnic0 --capture UplinkRcv --srcip 192.168.25.113 --outfile
 vmnic0_rcv_srcip.pcap --count 60
```

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

How to Capture Packets for a VMXNET3 Virtual Machine Adapter

Monitor traffic that flows between a virtual switch and a VMXNET3 virtual machine adapter by using the `pktcap-uw` utility.

Verify that the virtual machine adapter is of type VMXNET3.

You can specify a certain capture point in the data path between a virtual switch and a virtual machine adapter. You can also determine a capture point by traffic direction with regard to the switch and proximity to the packet source or destination. For information about supported capture points, see Capture Points of the pktcap-uw Utility.

1. On the host, learn the port ID of the virtual machine adapter by using the `esxtop` utility.
   a) In the ESX Shell to the host, to start the utility, run `esxtop`.
   b) To switch to the network panel of the utility, press n.
   c) In the USED-BY column, locate the virtual machine adapter, and write down the PORT-ID value for it.
      The USED-BY field contains the name of the virtual machine and the port to which the virtual machine adapter is connected.
   d) Press Q to exit `esxtop`.
2. In the ESX Shell, run `pktcap-uw--switchport`*port_ID*.
   port_ID is the ID that the `esxtop` utility displays for the virtual machine adapter in the PORT-ID column.
3. In the ESX Shell, run the `pktcap-uw` command with the `--switchport` *port_ID* argument and with options to monitor packets at a particular point, filter captured packets and save the result to a file.

```
            pktcap-uw
--switchport
port_ID [--capturecapture_point|--dir 0|1 --stage 0|1]  [filter_options] [--outfile pcap_file_path [--ng]]
 [--countnumber_of_packets]
```

where the square brackets [] enclose the options of the `pktcap-uw--switchport`*port_ID* command and the vertical bars | represent alternative values.

If you run the `pktcap-uw--switchport`*port_ID* command without options, you obtain the content of packets that are incoming to the standard or distributed switch in the console output at the point when they are switched.

a) To check packets at another capture point or direction in the path between the guest operating system and the virtual switch, use the --capture option or combine the values of the --dir and --stage options.

pktcap-uw Command Options	Goal
`--capture VnicTx`	Monitor packets when they pass from the virtual machine to the switch.
`--capture VnicRx`	Monitor packets when they arrive to the virtual machine.
--dir 1 --stage 0	Monitor packets immediately after they leave the virtual switch.
--dir 1	Monitor packets immediately before they enter the virtual machine.
--dir 0 --stage 1	Monitor packets immediately after they enter the virtual switch.

b) Use a *`filter_options`* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

c) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

- To save packets to a `.pcap` file, use the --outfile option.
- To save packets to a `.pcapng` file, use the --ng and --outfile options.

You can open the file in a network analyzer tool such as Wireshark.

By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

d) Use the --count option to monitor only a number of packets.

4. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

Capture Packets That Are Received at a Virtual Machine from an IP Address 192.168.25.113

To capture the first 60 packets from a source that is assigned the IP address 192.168.25.113 when they arrive at a virtual machine adapter with port ID 33554481 and save them to a file called `vmxnet3_rcv_srcip.pcap`, run the following `pktcap-uw` command:

```
pktcap-uw --switchport 33554481 --capture VnicRx --srcip 192.168.25.113 --outfile
  vmxnet3_rcv_srcip.pcap --count 60
```

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

How to Capture Packets for a VMkernel Adapter

Monitor packets that are exchanged between a VMkernel adapter and a virtual switch by using the `pktcap-uw` utility.

You can capture packets at a certain capture point in the flow between a virtual switch and a VMkernel adapter. You can also determine a capture point by traffic direction with regard to the switch and proximity to the packet source or destination. For information about supported capture points, see Capture Points of the pktcap-uw Utility.

1. Optional: Find the name of the VMkernel adapter that you want to monitor in the VMkernel adapter list.

- In the , expand **Networking** on the **Configure** tab for the host and select **VMkernel adapters**.
- In the ESX Shell to the host, to view a list of the physical adapters, run the following console command:

```
esxcli network ip interface list
```

Each VMkernel adapter is represented as `vmk`*`X`*, where *`X`* is the sequence number that ESX assigned to the adapter.

2. In the ESX Shell to the host, run the `pktcap-uw` command with the `--vmk vmk`*`X`* argument and with options to monitor packets at a particular point, filter captured packets and save the result to a file.

```
pktcap-uw --vmk vmkX [--capturecapture_point|--dir 0|1 --stage 0|1]  [filter_options] [--
outfilepcap_file_path [--ng]] [--countnumber_of_packets]
```

where the square brackets [] enclose the options of the `pktcap-uw--vmk vmkX` command and the vertical bars | represent alternative values.

You can replace the `--vmk vmkX` option with `--switchport`*`vmkernel_adapter_port_ID`*, where *`vmkernel_adapter_port_ID`* is the PORT-ID value that the network panel of the `esxtop` utility displays for the adapter.

If you run the `pktcap-uw--vmk vmkX` command without options, you obtain the content of packets that are leaving the VMkernel adapter.

a) To check transmitted or received packets at a specific place and direction, use the --capture option, or combine the values of the --dir and --stage options.

pktcap-uw Command Options	Goal
--dir 1 --stage 0	Monitor packets immediately after they leave the virtual switch.
--dir 1	Monitor packets immediately before they enter the VMkernel adapter.
--dir 0 --stage 1	Monitor packets immediately before they enter the virtual switch.

b) Use a *`filter_options`* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

c) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

- To save packets to a `.pcap` file, use the --outfile option.
- To save packets to a `.pcapng` file, use the --ng and --outfile options.

You can open the file in a network analyzer tool such as Wireshark.

By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

d) Use the --count option to monitor only a number of packets.

3. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

How to Capture Dropped Packets

Troubleshoot lost connectivity by capturing dropped packets through the `pktcap-uw` utility.

A packet might be dropped at a point in the network stream for many reasons, for example, a firewall rule, filtering in an IOChain and DVfilter, VLAN mismatch, physical adapter malfunction, checksum failure, and so on. You can use the `pktcap-uw` utility to examine where packets are dropped and the reason for the drop.

1. In the ESX Shell to the host, run the `pktcap-uw --capture Drop` command with options to monitor packets at a particular point, filter captured packets and save the result to a file.

```
pktcap-uw --capture Drop [filter_options] [--outfile pcap_file_path [--ng]] [--count number_of_packets]
```

where the square brackets [] enclose the options of the `pktcap-uw --capture Drop` command and the vertical bars | represent alternative values.

a) Use a *`filter_options`* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

b) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

•

You can open the file in a network analyzer tool such as Wireshark.

By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

Note:
You can see the reason and the place where a packet is dropped only when you capture packets to the console output. The `pktcap-uw` utility saves only the content of packets to a `.pcap` or `.pcapng` file.

c) Use the --count option to monitor only a number of packets.

2. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

Besides the contents of dropped packets, the output of the `pktcap-uw` utility displays the reason for the drop and the function in the network stack that handled the packet last.

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

How to Capture Packets at DVFilter Level

Examine how packets change when they pass through a vSphere Network Appliance (DVFilter).

DVFilters are agents that reside in the stream between a virtual machine adapter and a virtual switch. They intercept packets to protect virtual machines from security attacks and unwanted traffic.

1. Optional: To find the name of the DVFilter that you want to monitor, in the ESX Shell, run the `summarize-dvfilter` command.

   The output of the command contains the fast-path and slow-path agents of the DVFilters that are deployed on the host.

2. Run the `pktcap-uw` utility with the `--dvfilter`*`dvfilter_name`* argument and with options to monitor packets at a particular point, filter captured packets and save the result to a file.

```
            pktcap-uw
--dvFilter
dvfilter_name --capture PreDVFilter|PostDVFilter [filter_options] [--outfilepcap_file_path [--ng]] [--
countnumber_of_packets]
```

   where the square brackets [] enclose optional items of the `pktcap-uw--dvFilter vmnic`*`X`* command and the vertical bars | represent alternative values.

   a) Use the --capture option to monitor packets before or after the DVFilter intercepts them.

pktcap-uw Command Option	Goal
`--capture PreDVFilter`	Capture packets before they enter the DVFilter.
`--capture PostDVFilter`	Capture packets after they leave the DVFilter.

   b) Use a *`filter_options`* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
   For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

   c) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

   - To save packets to a `.pcap` file, use the --outfile option.
   - To save packets to a `.pcapng` file, use the --ng and --outfile options.

You can open the file in a network analyzer tool such as Wireshark.

By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

d) Use the --count option to monitor only a number of packets.

3. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

How to use the Capture Points of the pktcap-uw Utility

Learn how to use the capture points of the `pktcap-uw` utility to monitor packets when a function handles them at a specific place in the network stack on a host.

Overview of Capture Points

A capture point in the `pktcap-uw` utility represents a place in the path between a virtual switch on one side and a physical adapter, VMkernel adapter or a virtual machine adapter on the other.

You can use certain capture points in combination with an adapter option. For example, you use the UplinkRcv point when you capture uplink traffic. You can address other points standalone. For example, use the Drop point to inspect all dropped packets.

> **Note:**
> Certain capture points of the `pktcap-uw` utility are designed for VMware internal use only and you should use them only under the supervision of VMware Technical Support. These capture points are not described in the *vSphere Networking* guide.

Option for Using Capture Points in the pktcap-uw Utility

To examine a packet state or content at a capture point, add the --capture *`capture_point`* option to the `pktcap-uw` utility.

Auto-Selecting a Capture Point

For traffic that is related to a physical, VMkernel or VMXNET3 adapter, by combining the --dir and --stage options you can auto-select and switch between capture points to examine how a packet changes before and after a point.

Capture Points of the pktcap-uw Utility

The `pktcap-uw` utility supports capture points that can be used only when you monitor uplink, VMkernel or virtual machine traffic, and capture points that represent special places in the stack that are not related to the adapter type.

Capture Points That Are Relevant to Physical Adapter Traffic

The `pktcap-uw --uplink vmnic`*`X`* command supports capture points for functions that handle traffic at a specific place and direction in the path between the physical adapter and the virtual switch.

Capture Point	Description
UplinkRcv	The function that receives packets from the physical adapter.
UplinkSnd	The function that sends packets to the physical adapter.
PortInput	The function that passes a list of packets from UplinkRcv to a port on the virtual switch.
PortOutput	The function that passes a list of packets from a port on the virtual switch to the UplinkSnd point.

Capture Points That Are Relevant to Virtual Machine Traffic

The `pktcap-uw --switchport vmxnet3_port_ID` command supports capture points for functions that handle traffic packets at a specific place and direction in the path between a VMXNET3 adapter and a virtual switch.

Capture Point	Description
VnicRx	The function in the virtual machine NIC backend that receives packets from the virtual switch.
VnicTx	The function in the virtual machine NIC backend that sends packets from the virtual machine to the virtual switch.
PortOutput	The function that passes a list of packets from a port on the virtual switch to Vmxnet3Rx.
PortInput	The function that passes a list of packets from Vmxnet3Tx to a port on the virtual switch. Default capture point for traffic related to a VMXNET3 adapter.

Capture Points That Are Relevant to VMkernel Adapter Traffic

The `pktcap-uw --vmk vmkX` and `pktcap-uw` --switchport *`vmkernel_adapter_port_ID`* commands support capture points that represent functions at a specific place and direction in the path between a VMkernel adapter and a virtual switch.

Capture Point	Description
PortOutput	The function that passes a list of packets from a port on the virtual switch to the VMkernel adapter.
PortInput	The function that passes a list of packets from the VMkernel adapter to a port on the virtual switch. Default capture point for traffic related to a VMkernel adapter.

Capture Points That Are Relevant to Distributed Virtual Filters

The `pktcap-uw --dvfilter divfilter_name` command requires a capture point that indicates whether to capture packets when they enter the DVFilter or when they leave it.

Capture Point	Description
PreDVFilter	The point before a DVFilter intercepts a packet.
PostDVFilter	The point after a DVFilter intercepts a packet.

Standalone Capture Points

Certain capture points are mapped directly to the network stack rather than to a physical, VMkernel or VMXNET3 adapter.

Capture Point	Description
Drop	Captures dropped packets and shows the place where drops occur.
TcpipDispatch	Capture packets at the function that dispatches traffic to the TCP/IP stack of the VMkernel from the virtual switch, and the reverse.
PktFree	Capture packets right before they are released.
VdrRxLeaf	Capture packets at the receive leaf I/O chain of a dynamic router in VMware NSX. Use this capture point together with the --lifID option.

Capture Point	Description
VdrRxTerminal	Capture packets at the receive terminal I/O chain of a dynamic router in VMware NSX. Use this capture point together with the --lifID option.
VdrTxLeaf	Capture packets at the transmit leaf I/O chain of a dynamic router in VMware NSX. Use this capture point together with the --lifID option.
VdrTxTerminal	Capture packets at the transmit terminal I/O chain of a dynamic router in VMware NSX. Use this capture point together with the --lifID option.

For information about dynamic routers, see the *VMware NSX* documentation.

How to List the Capture Points of the pktcap-uw Utility

View all capture points of the `pktcap-uw` utility to find the name of capture point for monitoring traffic at a certain place in the network stack on the ESX host.

For information about the capture points of the `pktcap-uw` utility, see Capture Points of the pktcap-uw Utility.

In the ESX Shell to the host, run the `pktcap-uw -A` command to view all capture points that the `pktcap-uw` utility supports.

How to Trace Packets using the pktcap-uw Utility

Use the `pktcap-uw` utility to trace the path that packets traverse in the network stack for latency analysis and for locating the point where a packet is corrupted or dropped.

The `pktcap-uw` utility shows the path of packets together with timestamps that note the time when a packet is handled by a networking function on ESX. The utility reports the path of a packet immediately before it is released from the stack.

To view the full path information for a packet, you must print the result from the `pktcap-uw` utility in the console output or save it to a PCAPNG file.

1. In the ESX Shell to the host, run the `pktcap-uw`**--trace** command with options to filter traced packets, save the result to a file and limit the number of traced packets.

   ```
   pktcap-uw --trace [filter_options] [--outfilepcap_file_path [--ng]] [--
   countnumber_of_packets]
   ```

   where the square brackets [] enclose optional items of the `pktcap-uw --trace` command and the vertical bars | represent alternative values.

   a) Use a *`filter_options`* to filter packets according to source and destination address, VLAN ID, VXLAN ID, Layer 3 protocol, and TCP port.
   For example, to monitor packets from a source system that has IP address 192.168.25.113, use the `--srcip 192.168.25.113` filter option.

   b) Use options to save the contents of each packet or the contents of a limited number of packets to a `.pcap` or `.pcapng` file.

   •

   You can open the file in a network analyzer tool such as Wireshark.

   By default, the `pktcap-uw` utility saves the packet files to the root folder of the ESX file system.

   Note:
   A `.pcap` file contains only the contents of traced packets. To collect packet paths besides packet content, save the output to a `.pcapng` file.

c) Use the --count option to monitor only a number of packets.

2. If you have not limited the number of packets by using the --count option, press Ctrl+C to stop capturing or tracing packets.

If the contents of the packet are saved to a file, copy the file from the ESX host to the system that runs a graphical analyzer tool, such as Wireshark, and open it in the tool to examine the packet details.

Configure the NetFlow Settings of a vSphere Distributed Switch

Analyze virtual machine IP traffic that flows through a vSphere Distributed Switch by sending reports to a NetFlow collector.

vSphere Distributed Switch supports IPFIX (NetFlow version 10).

Note:

To configure IPFIX on a vSphere Distributed Switch backed by ESX on DPU, you must create vmknic on ops TCP/IP stack. Else, the flow information is not exported to collector.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. From the **Actions** menu, select **Settings** > **Edit Netflow**.
3. Type the **Collector IP address** and **Collector port** of the NetFlow collector.
   You can contact the NetFlow collector by IPv4 or IPv6 address.
4. Set an **Observation Domain ID** that identifies the information related to the switch.
5. To see the information from the distributed switch in the NetFlow collector under a single network device instead of under a separate device for each host on the switch, type an IPv4 address in the **Switch IP address** text box.
6. Optional: In the **Active flow export timeout** and **Idle flow export timeout** text boxes, set the time, in seconds, to wait before sending information after the flow is initiated.
7. Optional: To change the portion of data that the switch collects, configure **Sampling Rate**.
   The sampling rate represents the number of packets that NetFlow drops after every collected packet. A sampling rate of *x* instructs NetFlow to drop packets in a *collected packets*:*dropped packets* ratio 1:*x*. If the rate is 0, NetFlow samples every packet, that is, collect one packet and drop none. If the rate is 1, NetFlow samples a packet and drops the next one, and so on.
8. Optional: To collect data on network activity between virtual machines on the same host, enable **Process internal flows only**.
   Collect internal flows only if NetFlow is enabled on the physical network device to avoid sending duplicate information from the distributed switch and the physical network device.
9. Click **OK**.

Enable NetFlow reporting for traffic from virtual machines connected to a distributed port group or a port. See Manage NetFlow Monitoring on a Distributed Port Group or Distributed Port.

What is Port Mirroring

Port mirroring allows you to mirror a distributed port's traffic to other distributed ports or specific physical switch ports.

Port mirroring is used on a switch to send a copy of packets seen on one switch port (or an entire VLAN) to a monitoring connection on another switch port. Port mirroring is used to analyze and debug data or diagnose errors on a network.

What is Port Mirroring Interoperability

There are some interoperability issues to consider when using vSphere port mirroring with other features of vSphere.

vMotion

vMotion functions differently depending on which vSphere port mirroring session type you select. During vMotion, a mirroring path could be temporarily invalid, but it is restored when vMotion completes.

Table 160: vMotion Interoperability with port mirroring

Port mirroring session type	Source and destination	Interoperable with vMotion	Functionality
Distributed Port Mirroring	Non-uplink distributed port source and destination	Yes	Port mirroring between distributed ports can only be local. If the source and destination are on different hosts due to vMotion, mirroring between them will not work. However, if the source and destination move to the same host, port mirroring works.
Remote Mirroring Source	Non-uplink distributed port source	Yes	When a source distributed port is moved from host A to host B, the original mirroring path from the source port to A's uplink is removed on A, and a new mirroring path from the source port to B's uplink is created on B. Which uplink is used is determined by the uplink name specified in session.
	Uplink port destinations	No	Uplinks can not be moved by vMotion.
Remote Mirroring Destination	VLAN source	No	
	Non-uplink distributed port destination	Yes	When a destination distributed port is moved from host A to host B, all original mirroring paths from source VLANs to the destination port are moved from A to B.
Encapsulated Remote Mirroring (L3) Source	Non-uplink distributed port source	Yes	When a source distributed port is moved from host A to host B, all original mirroring paths from the source port to destination IPs are moved from A to B.
	IP destination	No	
Distributed Port Mirroring (legacy)	IP source	No	
	Non-uplink distributed port destination	No	When a destination distributed port is moved from host A to host B, all original mirroring paths from source IPs to the destination port are invalid because the port mirroring session source still sees the destination on A.

TSO and LRO

TCP Segmentation Offload (TSO) and large receive offload (LRO) might cause the number of mirroring packets to not equal to the number of mirrored packets.

When TSO is enabled on a vNIC, the vNIC might send a large packet to a distributed switch. When LRO is enabled on a vNIC, small packets sent to it might be merged into a large packet.

Source	Destination	Description
TSO	LRO	Packets from the source vNIC might be large packets, and whether they are split is determined by whether their sizes are larger than the destination vNIC LRO limitation.
TSO	Any destination	Packets from the source vNIC might be large packets, and they are split to standard packets at the destination vNIC.
Any source	LRO	Packets from the source vNIC are standard packets, and they might be merged into larger packets at the destination vNIC.

Create a Port Mirroring Session

Create a port mirroring session by using the vSphere Client to mirror vSphere Distributed Switch traffic to ports, uplinks, and remote IP addresses.

Verify that the vSphere Distributed Switch is version 5.0.0 and later.

Types of Port Mirroring Session

To begin a port mirroring session, you must specify the type of port mirroring session.

1. Browse to a distributed switch in the vSphere Client navigator.
2. Click the **Configure** tab and expand **Settings**.
3. Select the **Port mirroring** option and click **New**.
4. Select the session type for the port mirroring session.

   Note:

   When the session type for the port mirroring session is set to **Distributed Port Mirroring,Remote Mirror Source,**or **Remote Mirror Destination,**the switching VLAN tag will not be in the mirrored packets.

   Port Mirroring is not recommended for monitoring because when used for longer durations performance is impacted.

Option	Description
Distributed Port Mirroring	Mirror packets from a number of distributed ports to other distributed ports on the same host. If the source and the destination are on different hosts, this session type does not function.
Remote Mirroring Source	Mirror packets from a number of distributed ports to specific uplink ports on the corresponding host.
Remote Mirroring Destination	Mirror packets from a number of VLANs to distributed ports.
Encapsulated Remote Mirroring (L3) Source	Mirror packets from a number of distributed ports to the IP addresses of a remote agent. The virtual machine’s traffic is mirrored to a physical or virtual destination through an IP tunnel.

5. Click **Next**.

Port Mirroring Name and Session Details

To continue creating a port mirroring session, specify the name, description, and session details for the new port mirroring session.

1. Set the session properties. Different options are available for configuration depending on which session type you selected.

Option	Description
Name	You can enter a unique name for the port mirroring session, or accept the automatically generated session name.
Status	Use the drop down menu to enable or disable the session.
Session type	Displays the type of session you selected.
Encapsulation Type	Select **GRE**, **ERSPAN TWO**, or **ERSPAN THREE**. **Note:** This option is enabled when session type is set to **Encapsulated Remote Mirroring (L3) Source**.
Session ID	Specify an ERSPAN ID if encapsulation type is set to **ERSPAN TWO** or **ERSPAN THREE**. **Note:** This option is enabled when session type is set to **Encapsulated Remote Mirroring (L3) Source**.
Encapsulation VLAN ID	VLAN ID that encapsulates all frames at the destination ports. **Note:** If the original frames have a VLAN and Preserve original VLAN is not selected, the encapsulation VLAN replaces the original VLAN. This option is enabled when session type is set to **Remote Mirroring Source**.
Preserve original VLAN	Select Preserve original VLAN to keep the original VLAN in an inner tag so mirrored frames are double encapsulated. This option is enabled when session type is set to **Remote Mirroring Source**.
Normal I/O on destination ports	Use the drop-down menu to allow or disallow normal I/O on destination ports. This property is only available for uplink and distributed port destinations. If you disallow this option, mirrored traffic is allowed on outgoing destination ports, but incoming traffic is not allowed.
TCP/IP Stack	Use the drop-down menu to select the type of TCP/IP stack. • Default: The default TCP/IP stack. • Mirror: Using mirror stack instead of default TCP/IP netstack can separate mirror traffic from management traffic. Without mirror stack, mirror traffic is bound to default TCP/IP stack. The management traffic also uses the default TCP/IP stack. If the mirror traffic is large, then it affects the management traffic. If you want to separate the mirror traffic from the default TCP/IP stack, then you can have a dedicated mirror netstack on the ESX . You can enable this dedicated netstack while configuring Encapsulated Remote Mirroring sessions.

Option	Description
	Note: If you want to configure ERSPAN on a vSphere Distributed backed by ESX on DPU, create a vmknic on mirror TCP/IP stack. **Note:** This option is enabled when session type is set to **Encapsulated Remote Mirroring (L3) Source**.
Mirrored packet length (Bytes)	Use the check box to enable mirrored packet length in bytes. This puts a limit on the size of mirrored frames. If this option is selected, all mirrored frames are truncated to the specified length.
Description	You have the option to enter a description of the port mirroring session configuration.

2. Click **Next**.

Port Mirroring Sources

To continue creating a port mirroring session, select sources and traffic direction for the new port mirroring session.

You can create a port mirroring session without setting the source and destinations. When a source and destination are not set, a port mirroring session is created without the mirroring path. This allows you to create a port mirroring session with the correct properties set. Once the properties are set, you can edit the port mirroring session to add the source and destination information.

Note:
Consider the following limitations when selecting port mirroring sources.

- A source mirror port cannot be used in more than one mirror session.
- A port cannot be used as a mirror source and mirror destination in the same or different mirror sessions at the same time.

1. Select the source of the traffic to be mirrored and the traffic direction.

   Depending on the type of port mirroring session you selected, different options are available for configuration.

Option	Description
Add existing ports from a list	Click **Select distributed ports**. A dialog box displays a list of existing ports. Select the check box next to the distributed port and click **OK**. You can choose more than one distributed port.
Add existing ports by port number	Click **Add distributed ports**, enter the port number and click **OK.**
Set the traffic direction	After adding ports, select the port in the list and click the ingress, egress, or ingress/egress button. Your choice appears in the Traffic Direction column.
Specify the source VLAN	If you selected a Remote Mirroring Destination sessions type, you must specify the source VLAN. Click **Add** to add a VLAN ID. Edit the ID by using the up and down arrows, or by clicking in the field and entering the VLAN ID manually.

2. Click **Next**.

Port Mirroring Destinations

To complete the creation of a port mirroring session, select ports or uplinks as destinations for the port mirroring session.

You can create a port mirroring session without setting the source and destinations. When a source and destination are not set, a port mirroring session is created without the mirroring path. This allows you to create a port mirroring session

with the correct properties set. Once the properties are set, you can edit the port mirroring session to add the source and destination information.

Port mirroring is checked against the VLAN forwarding policy. If the VLAN of the original frames is not equal to or trunked by the destination port, the frames are not mirrored.

1. Select the destination for the port mirroring session.

   Depending on which type of session you chose, different options are available.

Option	Description
Select a destination-distributed port	Click **Select distributed ports** to select ports from a list, or click **Add distributed ports** to add ports by port number. You can add more than one distributed port.
Select an uplink	Select an available uplink from the list and click **Add** to add the uplink to the port mirroring session. You can select more than one uplink.
Select ports or uplinks	Click **Select distributed ports** to select ports from a list, or click **Add distributed ports** to add ports by port number. You can add more than one distributed port. Click **Add uplinks** to add uplinks as the destination. Select uplinks from the list and click **OK**.
Specify IP address	Click **Add**. A new list entry is created. Select the entry and either click **Edit** to enter the IP address, or click directly in the IP Address field and type the IP address. A warning appears if the IP address is invalid.

2. Click **Next**.
3. Review the information that you entered for the port mirroring session on the **Ready to complete** page.
4. Optional: Use the **Back** button to edit the information.
5. Click **Finish**.

The new port mirroring session appears in the Port Mirroring section of the **Settings** tab.

View Port Mirroring Session Details

View port mirroring session details, including status, sources, and destinations.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and click **Port mirroring**.
3. Select a port mirroring session from the list to display more detailed information at the bottom of the screen. Use the tabs to review configuration details.
4. Optional: Click **New** to add a port mirroring session.
5. Optional: Click **Edit** to edit the details for the selected port mirroring session.
6. Optional: Click **Remove** to delete the selected port mirroring session.

Edit Port Mirroring Session Details, Sources, and Destinations

Edit the details of a port mirroring session, including name, description, status, sources, and destinations.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Configure** tab, expand **Settings** and click **Port mirroring**.
3. Select a port mirroring session from the list and click **Edit**.
4. On the **Properties** page, edit the session properties.
   Depending on the type of port mirroring session being edited, different options are available for configuration.

Option	Description
Name	You can enter a unique name for the port mirroring session, or accept the automatically generated session name.
Status	Use the drop-down menu to enable or disable the session.
Normal I/O on destination ports	Use the drop-down menu to allow or disallow normal I/O on destination ports. This property is only available for uplink and distributed port destinations. If you do not select this option, mirrored traffic will be allowed out on destination ports, but no traffic will be allowed in.
Mirrored packet length (Bytes)	Use the check box to enable mirrored packet length in bytes. This puts a limit on the size of mirrored frames. If this option is selected, all mirrored frames are truncated to the specified length.
Description	You have the option to enter a description of the port mirroring session configuration.

5. On the **Sources** page, edit sources for the port mirroring session.
   Depending on the type of port mirroring session being edited, different options are available for configuration.

Option	Description
Add existing ports from a list	Click the **Select distributed ports to add to this port mirroring session.** button. A dialog opens with a list of existing ports. Select the check box next to the distributed port and click **OK**. You can choose more than one distributed port.
Set the traffic direction	After adding ports, select the port in the list and click the ingress, egress, or ingress/egress button. Your choice is displayed in the Traffic Direction column.

6. In the **Destinations** section, edit the destinations for the port mirroring session.
   Depending on the type of port mirroring session being edited, different options are available for configuration.

Option	Description
Select a destination-distributed port	Click the **Select distributed ports to add to this port mirroring session.** button to select ports from a list. You can add more than one distributed port.

7. Click **OK**.

vSphere Distributed Switch Health Check

The health check support helps you identify and troubleshoot configuration errors in a vSphere Distributed Switch.

Use vSphere Distributed Switch health check to examine certain settings on the distributed and physical switches to identify common errors in your environment's networking configuration. The default interval between two health checks is 1 minute.

Important:

Use health check to troubleshoot network problems, and then deactivate it after you identify and resolve the problem. After you deactivate the vSphere Distributed Switch health check, the generated MAC addresses age

out of your physical network environment according to your network policy. For more information, see Knowledge Base article 315308.

Configuration Error	Health Check	Required Configuration on the Distributed Switch
The VLAN trunk ranges configured on the distributed switch do not match the trunk ranges on the physical switch.	Checks whether the VLAN settings on the distributed switch match the trunk port configuration on the connected physical switch ports.	At least two active physical NICs
The MTU settings on the physical network adapters, distributed switch, and physical switch ports do not match.	Checks whether the physical access switch port MTU jumbo frame setting based on per VLAN matches the vSphere distributed switch MTU setting.	At least two active physical NICs
The teaming policy that is configured on the port groups does not match the policy on the physical switch port-channel.	Checks whether the connected access ports of the physical switch that participate in an EtherChannel are paired with distributed ports whose teaming policy is set to IP hash.	At least two active physical NICs and two hosts

Health check is limited only to the access switch port to which the distributed switch uplink connects.

Manage vSphere Distributed Switch Health Check

Health check monitors for changes in vSphere Distributed Switch configurations. You must enable vSphere Distributed Switch health check to perform checks on distributed switch configurations.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. Select the **Configure** tab and expand Setting.
3. Select **Health Check** and click the **Edit** button.
4. Use the drop-down menus to enable or disable the health check options.

Option	Description
VLAN and MTU	Reports the status of distributed uplink ports and VLAN ranges.
Teaming and failover	Checks for any configuration mismatch between the ESX host and the physical switch that is used in the teaming policy.
Advanced NIC Passthrough	The status of Advanced NIC Passthrough. • Up: Indicates that the Advanced NIC Passthrough is functioning normally. • Down: Indicates that the Advanced NIC Passthrough has failed or is not working as expected. • In Progress: Indicates that the Advanced NIC Passthrough status is being determined or is in a transitional state. • Unknown: Indicates that the status of Advanced NIC Passthrough cannot be determined.

5. Click **OK**.

When you change the configuration of a vSphere Distributed Switch, you can view information about the change in the **Monitor** tab in the vSphere Client. See View vSphere Distributed Switch Health Status.

View vSphere Distributed Switch Health Status

Once you have enabled health check on a vSphere Distributed Switch, you can view the network health status of the hosts that are connected in the vSphere Client .

Verify that health check for VLAN, MTU, teaming policy, and for Advanced NIC Passthrough is enabled on the vSphere Distributed Switch. See Manage vSphere Distributed Switch Health Check.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. On the **Monitor** tab, click **Health**.
3. In the **Host member health status** section, examine the overall, VLAN, MTU, Advanced NIC Passthrough, and teaming health of the hosts connected to the switch.

Switch Discovery Protocol

Switch discovery protocols help vSphere administrators to determine which port of the physical switch is connected to a vSphere standard switch or vSphere distributed switch.

vSphere 5.0 and later supports Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP). CDP is available for vSphere standard switches and vSphere distributed switches connected to Cisco physical switches. LLDP is available for vSphere distributed switches version 5.0.0 and later.

When CDP or LLDP is enabled for a particular vSphere distributed switch or vSphere standard switch, you can view properties of the peer physical switch such as device ID, software version, and time out from the vSphere Client.

Enable Cisco Discovery Protocol on a vSphere Distributed Switch

Cisco Discovery Protocol (CDP) enables vSphere administrators to determine which port of a physical Cisco switch connects to a vSphere Standard Switch or vSphere Distributed Switch. When CDP is enabled for a vSphere Distributed Switch, you can view the properties of the Cisco switch such as device ID, software version, and time out.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. From the **Actions** menu, select **Settings** > **Edit Settings**.
3. In the Edit Settings dialog box, click **Advanced**.
4. In the Discovery Protocol section, select **Cisco Discovery Protocol** from the **Type** drop-down menu.
5. From the **Operation** drop-down menu, select the operational mode of the ESX hosts connected to the switch.

Option	Description
Listen	ESX detects and displays information about the associated Cisco switch port, but information about the vSphere Distributed Switch is not available to the Cisco switch administrator.
Advertise	ESX makes information about the vSphere Distributed Switch available to the Cisco switch administrator, but does not detect and display information about the Cisco switch.
Both	ESX detects and displays information about the associated Cisco switch and makes information about the vSphere Distributed Switch available to the Cisco switch administrator.

6. Click **OK**.

Enable Link Layer Discovery Protocol on a vSphere Distributed Switch

With Link Layer Discovery Protocol (LLDP), vSphere administrators can determine which physical switch port connects to a given vSphere Distributed Switch. When LLDP is enabled for a particular distributed switch, you can view properties of the physical switch (such as chassis ID, system name and description, and device capabilities) from the .

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. From the **Actions** menu, select **Settings** > **Edit Settings**.
3. In the Edit Settings dialog box, click **Advanced**.
4. In the Discovery Protocol section, select **Link Layer Discovery Protocol** from the **Type** drop-down menu.
5. From the **Operation** drop-down menu, select the operational mode of the ESX hosts connected to the switch.

Operation	Description
Listen	ESX detects and displays information about the associated physical switch port, but information about the vSphere Distributed Switch is not available to the switch administrator.
Advertise	ESX makes information about the vSphere Distributed Switch available to the switch administrator, but does not detect and display information about the physical switch.
Both	ESX detects and displays information about the associated physical switch and makes information about the vSphere Distributed Switch available to the switch administrator.

6. Click **OK**.

View Switch Information

When Cisco Discovery Protocol (CDP) or Link Layer Discovery Protocol (LLDP) is enabled on the distributed switch and the hosts connected to the switch are in Listen or Both operational mode, you can view physical switch information from the vSphere Client.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **Networking** and click **Physical adapters**.
3. Select a physical adapter from the list to view its detailed information.

According to the enabled switch discovery protocol, the properties of the switch appear under the **CDP** or **LLDP** tab. If the information is available in the network, under Peer device capability you can examine the system capabilities of the switch.

Configure Protocol Profiles for Virtual Machine Networking

A network protocol profile contains a pool of IPv4 and IPv6 addresses that vCenter assigns to vApps or to virtual machines with vApp functionality that are connected to port groups associated with the profile.

Network protocol profiles also contain settings for the IP subnet, DNS, and HTTP proxy server.

To configure the networking settings of virtual machines by using from network protocol profiles, perform the following operations:

- Create network profiles at the level of a data center or a vSphere distributed switch.
- Associate a protocol profile with the port group of a vApp virtual machine.
- Enable the transient or static IP allocation policy from the settings of the vApp or from the vApp options of a virtual machine.

> **Note:**
> If you move a vApp or a virtual machine that retrieves its network settings from a protocol profile to another data center, to power it on you must assign a protocol profile to the connected port group on the destination data center.

Add a Network Protocol Profile

A network protocol profile contains a pool of IPv4 and IPv6 addresses that vCenter assigns to vApps or to virtual machines with vApp functionality that are connected to port groups associated with the profile.

You can configure network protocol profile ranges for IPv4, IPv6, or both. vCenter uses these ranges to dynamically allocate IP addresses to the virtual machines within a vApp, when the vApp uses transient IP allocation policy.

Network protocol profiles also contain settings for the IP subnet, the DNS, and HTTP proxy servers.

> **Note:**
> If you move a vApp or a virtual machine that retrieves its network settings from a protocol profile to another data center, to power on the vApp or virtual machine you must assign a protocol profile to the connected port group on the destination data center.

1. Navigate to a data center that is associated with a vApp.
2. On the **Configure** tab, select **More** > **Network Protocol Profiles**.
   Existing network protocol profiles are listed.
3. Click the **Add** button.
   The **Add Network Protocol Profile** wizard opens.
4. On the **Name and network** page, enter the name of the network protocol profile and select the networks that use this profile. Click **Next**.
   A network can be associated with one network protocol profile at a time.
5. On the **IPv4** page, configure the relevant IPv4 settings.
   a) In the **Subnet** and the **Gateway** text boxes, enter the IP subnet and gateway.
   b) To indicate that the DHCP server is available on the network, select the **DHCP Present** radio button.
   c) In the **DNS server addresses** text box, enter the DNS server information.
   d) To specify an IP pool range, enable the **IP Pool** option.
   e) If you enable IP pools, enter a comma-separated list of host address ranges in the **IP pool range** text box.
      A range consists of an IP address, a pound sign (#), and a number indicating the length of the range.

      For example, `10.20.60.4#10, 10.20.61.0#2` indicates that the IPv4 addresses can range from 10.20.60.4 to 10.20.60.13 and 10.20.61.0 to 10.20.61.1.

      The gateway and the ranges must be within the subnet. The ranges that you enter in the **IP pool range** text box cannot include the gateway address.
   f) Click **Next**.
6. On the **IPv6** page, configure the relevant IPv6 settings.
   a) In the **Subnet** and the **Gateway** text boxes, enter the IP subnet and gateway.
   b) Select the **DHCP Present** radio button to indicate that the DHCP server is available on this network.
   c) In the **DNS server addresses**, enter the DNS server information.
   d) Enable the **IP Pool** option to specify an IP pool range.
   e) If you enable IP pools, enter a comma-separated list of host address ranges in the **IP pool range** text box.
      A range consists of an IP address, a pound sign (#), and a number indicating the length of the range.

      For example, assume that you specify the following IP pool range: `fe80:0:0:0:2bff:fe59:5a:2b#10, fe80:0:0:0:2bff:fe59:5f:b1#2`. Then the addresses are in this range:

      fe80:0:0:0:2bff:fe59:5a:2b - fe80:0:0:0:2bff:fe59:5a:34

      and

      fe80:0:0:0:2bff:fe59:5f:b1 - fe80:0:0:0:2bff:fe59:5f:b2 .

      The gateway and the ranges must be within the subnet. The ranges that you enter in the **IP pool range** text box cannot include the gateway address.

f) Click **Next**.

7. On the **Other network configurations** page, specify more network configurations.
   a) Enter the DNS domain.
   b) Enter the host prefix.
   c) Enter the DNS search path.
      The search paths are specified as a list of DNS domains separated by commas, semi-colons, or spaces.
   d) Enter the server name and port number for the HTTP proxy server.
      The server name must include a colon and a port number. For example, `web-proxy:3912` is a valid proxy server.
   e) Click **Next**.
8. On the **Name and Network Assignment** page, review the settings and click **Finish**.

Select the Network Protocol Profile Name and Network

Name the network protocol profile and select the network that should use it.

1. Type the name of the network protocol profile.
2. Select the networks that use this network protocol profile.
   A network can be associated with one network protocol profile at a time.
3. Click **Next**.

Specify Network Protocol Profile IPv4 Configuration

A network protocol profile contains a pool of IPv4 and IPv6 addresses for use by vApps. When you create a network protocol profile, you set up its IPv4 configuration.

You can configure network protocol profile ranges for IPv4, IPv6, or both. vCenter uses these ranges to dynamically allocate IP addresses to virtual machines when a vApp is set up to use transient IP allocation.

1. Enter the **IP Subnet** and **Gateway** in their respective fields.
2. Select **DHCP Present** to indicate that the DHCP server is available on this network.
3. Enter the DNS server information.
   Specify the servers by IP addresses separated by a comma, semicolon, or space.
4. Select the **Enable IP Pool** check box to specify an IP pool range.
5. If you enable IP Pools, enter a comma-separated list of host address ranges in the **IP pool range** field.
   A range consists of an IP address, a pound sign (#), and a number indicating the length of the range.

   The gateway and the ranges must be within the subnet. The ranges that you enter in the **IP pool range** field cannot include the gateway address.

   For example, `10.20.60.4#10, 10.20.61.0#2` indicates that the IPv4 addresses can range from 10.20.60.4 to 10.20.60.13 and 10.20.61.0 to 10.20.61.1.
6. Click **Next**.

Specify Network Protocol Profile IPv6 Configuration

A network protocol profile contains a pool of IPv4 and IPv6 addresses for use by vApps. When you create a network protocol profile, you set up its IPv6 configuration.

You can configure network protocol profile ranges for IPv4, IPv6, or both.vCenter uses these ranges to dynamically allocate IP addresses to virtual machines when a vApp is set up to use transient IP allocation.

1. Enter the **IP Subnet** and **Gateway** in their respective fields.
2. Select **DHCP Present** to indicate that the DHCP server is available on this network.
3. Enter the DNS server information.
   Specify the servers by IP addresses separated by a comma, semicolon, or space.
4. Select the **Enable IP Pool** check box to specify an IP pool range.
5. If you enable IP Pools, enter a comma-separated list of host address ranges in the **IP pool range** field.
   A range consists of an IP address, a pound sign (#), and a number indicating the length of the range. For example, assume that you specify the following IP pool range:

   fe80:0:0:0:2bff:fe59:5a:2b#10,fe80:0:0:0:2bff:fe59:5f:b1#2

   Then the addresses are in this range:

   fe80:0:0:0:2bff:fe59:5a:2b - fe80:0:0:0:2bff:fe59:5a:34

   and

   fe80:0:0:0:2bff:fe59:5f:b1 - fe80:0:0:0:2bff:fe59:5f:b2

   The gateway and the ranges must be within the subnet. The ranges that you enter in the **IP pool range** field cannot include the gateway address.
6. Click **Next**.

Specify Network Protocol Profile DNS and Other Configuration

When you create a network protocol profile, you can specify the DNS domain, DNS search path, a host prefix, and HTTP proxy.

1. Enter the DNS domain.
2. Enter the host prefix.
3. Enter the DNS search path.
   The search paths are specified as a list of DNS domains separated by commas, semi-colons, or spaces.
4. Enter the server name and port number for the proxy server.
   The server name can optionally include a colon and a port number.
   For example, `web-proxy:3912` is a valid proxy server.
5. Click **Next**.

Complete the Network Protocol Profile Creation

Review the settings and click **Finish** to complete adding the network protocol profile.

Associate a Port Group with a Network Protocol Profile

To apply the range of IP addresses from a network protocol profile to a virtual machine that is a part of a vApp or has vApp functionality enabled, associate the profile with a port group that controls the networking of the virtual machine.

1. Navigate to a data center that is associated with a vApp.
2. On the **Configure** tab, select **More > Network Protocol Profiles**.
   Existing network protocol profiles are listed.
3. Select a network profile from thr list and click **Assign**.
   The **Assign Networks** dialog box opens.
4. Select a port group or a network to assign to the network profile.
   - On the **Distributed portgroups** tab, you see a list of the distributed port groups.

- On the **Networks** tab, you see a list of the port groups of standard switches.

You can select multiple port groups before you close the dialog box.

5. Click **Save**.

The port groups that you selected are now associated with the network protocol profile.

To apply to a virtual machine or vApp the range of IP addresses that the network protocol profile contains, configure the virtual machine or vApp to use the network protocol profile. For more information, see Use a Network Protocol Profile to Allocate IP Addresses to a Virtual Machine or vApp .

What is Multicast Filtering

In vSphere 6.0 and later, vSphere Distributed Switch supports basic and snooping models for filtering of multicast packets that are related to individual multicast groups. Choose a model according to the number of multicast groups to which the virtual machines on the switch subscribe.

Multicast FilterinBasic Multicast FilteringIn basic multicast filtering mog Modes

In addition to the default basic mode for filtering multicast traffic, vSphere Distributed Switch 6.0.0 and later releases support multicast snooping that forwards multicast traffic in a more precise way based on the Internet Group Management Protocol (IGMP) and Multicast Listener Discovery (MLD) messages from virtual machines.

de, a vSphere Standard Switch or vSphere Distributed Switch forwards multicast traffic for virtual machines according to the destination MAC address of the multicast group. When joining a multicast group, the guest operating system pushes the multicast MAC address of the group down to the network through the switch. The switch saves the mapping between the port and the destination multicast MAC address in a local forwarding table.

The switch does not interpret the IGMP messages that a virtual machine sends to join or leave a group. The switch sends them directly to the local multicast router, which then interprets them to join the virtual machine to or remove it from the group.

The basic mode has the following restrictions:

- A virtual machine might receive packets from groups that it is not subscribed for because the switch forwards packets according to the destination MAC address of a multicast group, which can be potentially mapped up to 32 IP multicast groups.
- A virtual machine that is subscribed for traffic from more than 32 multicast MAC addresses receives packets that it is not subscribed for because of a limitation in the forwarding model.
- The switch does not filter packets according to source address as defined in IGMP version 3.

Multicast Snooping

In multicast snooping mode, a vSphere Distributed Switch provides IGMP and MLD snooping according to RFC 4541. The switch dispatches multicast traffic more precisely by using IP addresses. This mode supports IGMPv1, IGMPv2, and IGMPv3 for IPv4 multicast group addresses, and MLDv1 and MLDv2 for IPv6 multicast group addresses.

The switch dynamically detects the membership of a virtual machine. When a virtual machine sends a packet which contains IGMP or MLD membership information through a switch port, the switch creates a record about the destination IP address of the group, and in the case of IGMPv3, about a source IP address that the virtual machine prefers to receive traffic from. If a virtual machine does not renew its membership to a group within a certain period of time, the switch removes the entry for the group from the lookup records.

In multicast snooping mode of a distributed switch, a virtual machine can receive multicast traffic on a single switch port from up to 512 groups and 10 sources.

> **Note:**
> In vSphere 6.7, the default multicast filtering mode is Basic. In vSphere 7.0, the default multicast filtering mode is IGMP/MLD snooping. When the DVS is upgraded to 7.0, the default multicast filtering mode will change from basic to IGMP/MLD snooping.

Enable Multicast Snooping on a vSphere Distributed Switch

Use multicast snooping on a vSphere Distributed Switch to forward traffic in a precise manner according to Internet Group Management Protocol (IGMP) or Multicast Listener Discovery (MLD) membership information that virtual machines send to subscribe for multicast traffic.

Verify that vSphere Distributed Switch is version 6.0.0 and later.

Use multicast snooping if virtualized workloads on the switch subscribe to more than 32 multicast groups or must receive traffic from specific source nodes. For information about the multicast filtering modes of vSphere Distributed Switch, see Multicast FilterinBasic Multicast FilteringIn basic multicast filtering mog Modes.

1. On the vSphere Client Home page, click **Networking** and navigate to the distributed switch.
2. From the **Actions** menu, select **Settings** > **Edit Settings**.
3. In the dialog box that displays the settings of the switch, click **Advanced**.
4. From the **Multicast filtering mode** drop-down menu, select **IGMP/MLD snooping**, and click **OK**.

Multicast snooping becomes active on hosts running ESX 6.0 and later.

Edit the Query Time Interval for Multicast Snooping

When IGMP or MLD multicast snooping is enabled on a vSphere Distributed Switch, the switch sends general queries about the membership of virtual machines in case a snooping querier is not configured on the physical switch. On ESX hosts that are attached to the distributed switch, you can edit the time interval in which the switch sends general queries.

The default time interval for sending snooping queries is 125 seconds.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System** and select **Advanced System Settings**.
3. Click **Edit**.
4. Locate the `Net.IGMPQueryInterval` system setting and enter a new value in seconds for the setting.
5. Click **OK**

Edit the Number of Source IP Addresses for IGMP and MLD

When you enable IGMP or MLD multicast snooping on a vSphere Distributed Switch, you can edit the maximum number of IP sources from which the members of a multicast group receive packets.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand **System** and select **Advanced System Settings**.
3. Click **Edit**.
4. Locate the `Net.IGMPV3MaxSrcIPNum` or `Net.MLDV2MaxSrcIPNum` system settingand enter a new value between 1 and 32 for the setting.
5. Click **OK**.

What is Stateless Network Deployment

Stateless is a mode of execution for ESX hosts with no local storage that formerly would save configuration or state. Configurations are abstracted into a host profile, which is a template that applies to a class of machines. Stateless allows easy replacement, removal, and addition of failed hardware, and improves the ease of scaling a hardware deployment.

Every stateless ESX boot is like a first boot. The ESX host boots with networking connectivity to vCenter through the built-in standard switch. If the host profile specifies distributed switch membership, vCenter joins the ESX host to VMware distributed switches.

When planning the network setup for stateless ESX hosts, you should keep the configuration as generic as possible and avoid host-specific items. Currently the design has no hooks to reconfigure physical switches when deploying a new host. Any such requirement would need special handling.

To set up stateless deployment, one ESX host must be installed in the standard fashion. Then find and record the following network-related information to save in the host profile:

- vSphere standard switch instances and settings (port groups, uplinks, MTU, and so forth)
- Distributed switch instances
- Selection rules for uplinks and uplink port or port groups
- vNIC information:
  - Address information (IPv4 or IPv6, static or DHCP, gateway)
  - Port groups and distributed port groups assigned to the physical network adapter (`vmknic`)
  - If there are distributed switches, record VLAN, physical NICs bound to the `vmknic`, and if `Etherchannel` is configured

The recorded information is used as a template for the host profile. Once the host profile virtual switch information has been extracted and placed in the host profile, you have the opportunity to change any of the information. Modifications are offered for both standard and distributed switches in these sections: uplink selection policy, based on either vmnic name or device number, and auto discovery based on VLAN ID. The (possibly modified) information is stored by the stateless boot infrastructure and applied to a stateless ESX host on its next boot. During network initialization, a generic network plug-in interprets the recorded host profile setting and does the following:

- Loads appropriate physical NIC drivers.
- Creates all standard switch instances, along with port groups. It selects uplinks based on policy. If the policy is based on the VLAN ID, there is a probing process to gather relevant information.
- For VMkernel network adapters connected to the standard switch, it creates VMkernel network adapters and connects them to port groups.
- For each VMkernel network adapter connected to a distributed switch, it creates a temporary standard switch (as needed) with uplinks bound to the VMkernel network adapter. It creates a temporary port group with VLAN and teaming policies based on recorded information. Specifically, IP-hash is used if Etherchannel was used in the distributed switch.
- Configures all VMkernel network adapter settings (assigns address, gateway, MTU, and so forth).

Basic connectivity is functioning, and the networking setup is complete if there is no distributed switch present.

If there is a distributed switch present, the system stays in maintenance mode until distributed switch remediation is complete. No virtual machines are started at this time. Because distributed switches requires vCenter, the boot process continues until vCenter connectivity is established, and vCenter notices that the host should be part of a distributed switch. It issues a distributed switch host join, creating a distributed switch proxy standard switch on the host, selects appropriate uplinks, and migrates the vmknic from the standard switch to the distributed switch. When this operation is complete, it deletes the temporary standard switch and port groups.

At the end of the remediation process, the ESX host is taken out of maintenance mode, and HA or DRS can start virtual machines on the host.

In the absence of a host profile, a temporary standard switch is created with “default networking” logic, which creates a management network switch (with no VLAN tag) whose uplink corresponds to the PXE booting vNIC. A vmknic is created on the management network port group with the same MAC address as the PXE booting vNIC. This logic was previously used for PXE booting. If there is a host profile, but the networking host profile is deactivated or fatally incomplete, vCenter falls back to default networking so that the ESX host can be managed remotely. This triggers a compliance failure, so vCenter then initiates recovery actions.

vSphere Networking Best Practices

Consider these best practices when you configure your network.

- To ensure a stable connection between vCenter, ESX, and other products and services, do not set connection limits and timeouts between the products. Setting limits and timeouts can affect the packet flow and cause services interruption.

- Isolate from one another the networks for host management, vSphere vMotion, vSphere FT, and so on, to improve security and performance.
- Dedicate a separate physical NIC to a group of virtual machines, or use Network I/O Control and traffic shaping to guarantee bandwidth to the virtual machines. This separation also enables distributing a portion of the total networking workload across multiple CPUs. The isolated virtual machines can then better handle application traffic, for example, from a vSphere Client.
- To physically separate network services and to dedicate a particular set of NICs to a specific network service, create a vSphere Standard Switch or vSphere Distributed Switch for each service. If this is not possible, separate network services on a single switch by attaching them to port groups with different VLAN IDs. In either case, verify with your network administrator that the networks or VLANs you choose are isolated from the rest of your environment and that no routers connect them.
- Keep the vSphere vMotion connection on a separate network. When migration with vMotion occurs, the contents of the guest operating system's memory is transmitted over the network. You can do this either by using VLANs to segment a single physical network or by using separate physical networks (the latter is preferable).

  For migration across IP subnets and for using separate pools of buffer and sockets, place traffic for vMotion on the vMotion TCP/IP stack, and traffic for migration of powered-off virtual machines and cloning on the Provisioning TCP/IP stack. See VMkernel Networking Layer.
- Multihoming in VMkernel networking is not supported. For more information, see Broadcom knowledge base article 318546.
- You can add and remove network adapters from a standard or distributed switch without affecting the virtual machines or the network service that is running behind that switch. If you remove all the running hardware, the virtual machines can still communicate among themselves. If you leave one network adapter intact, all the virtual machines can still connect with the physical network.
- To protect your most sensitive virtual machines, deploy firewalls in virtual machines that route between virtual networks with uplinks to physical networks and pure virtual networks with no uplinks.
- For best performance, use VMXNET 3 virtual machine NICs.
- Physical network adapters connected to the same vSphere Standard Switch or vSphere Distributed Switch should also be connected to the same physical network.
- Configure the same MTU on all VMkernel network adapters in a vSphere Distributed Switch. If several VMkernel network adapters, configured with different MTUs, are connected to vSphere distributed switches, you might experience network connectivity problems.

Troubleshoot vSphere Networking

The troubleshooting topics about networking in vSphere provide solutions to potential problems that you might encounter with the connectivity of ESX hosts, vCenter and virtual machines.

For general information about troubleshooting your implementation of vSphere and how to identify the symptoms of the problem, determine which of the components are affected, and test possible solutions, see the Guidelines for Troubleshooting Your Implementation of vSpheresection.

Troubleshoot MAC Address Allocation

In vSphere, certain restrictions on the range of MAC addresses that can be assigned to virtual machines might cause lost of connectivity or inability to power on workloads.

Duplicate MAC Addresses of Virtual Machines on the Same Network

You encounter loss of packets and connectivity because virtual machines have duplicate MAC addresses generated by vCenter.

The MAC addresses of virtual machines on the same broadcast domain or IP subnet are in conflict, or vCenter generates a duplicate MAC address for a newly created virtual machine.

A virtual machine powers on and functions properly, but shares a MAC address with another virtual machine. This situation might cause packet loss and other problems.

Virtual machines might have duplicate MAC addresses due to several reasons.

- Two vCenter instances with identical IDs generate overlapping MAC addresses for virtual machine network adapters. Each vCenter instance has an ID between 0 and 63 that is randomly generated at installation time, but can be reconfigured after installation. vCenter uses the instance ID to generate MAC addresses for the network adapters of the machine.
- A virtual machine has been transferred in power-off state from one vCenter instance to another in the same network, for example, by using shared storage, and a new virtual machine network adapter on the first vCenter receives the freed MAC address.
- Change the MAC address of a virtual machine network adapter manually.

  If you have an existing virtual machine with a conflicting MAC address, you must provide a unique MAC address in the **Virtual Hardware** settings.

  - Power off the virtual machine, configure the adapter to use a manual MAC address, and type the new address.
  - If you cannot power the virtual machine off for configuration, re-create the network adapter that is in conflict with enabled manual MAC address assignment and type the new address. In the guest operating system, set the same static IP address to the re-added adapter as before.

  For information about configuring the network adapters of virtual machines, see the *vSphere Networking* and *vSphere Virtual Machine Administration* documentation.
- If the vCenter instance generates the MAC addresses of virtual machines according to the default allocation, VMware OUI, change the vCenter instance ID or use another allocation method to resolve conflicts.

  Note:
  Changing the vCenter instance ID or switching to a different allocation scheme does not resolve MAC address conflicts in existing virtual machines. Only virtual machines created or network adapters added after the change receive addresses according to the new scheme.

  For information about MAC address allocation schemes and setup, see the *vSphere Networking* documentation.

Solution	Description
Change the vCenter ID	You can keep using the VMware OUI allocation scheme if your deployment contains a small number of vCenter instances. According to this scheme, a MAC address has the following format: `00:50:56:XX:YY:ZZ` where `00:50:56` represents the VMware OUI, *XX* is calculated as (80 + vCenter ID), and *YY:ZZ* is a random number. To change the vCenter ID, configure the **vCenter unique ID** option in the **Runtime Settings** section from the **General** settings of the vCenter instance and restart it. The VMware OUI allocation works with up to 64 vCenter instances and is suitable for small scale deployments.
Switch to prefix-based allocation	You can use a custom OUI. For example, for a `02:12:34` locally administered address range, MAC addresses have the form `02:12:34:XX:YY:ZZ`. You can use the fourth octet *XX* to distribute the OUI address space between the vCenter instances. This structure results in 255 address clusters, each cluster managed by a vCenter instance, and in about 65000 MAC addresses per vCenter. For example, `02:12:34:01:YY:ZZ` for vCenter A, `02:12:34:02:YY:ZZ` for vCenter B, and so on. Prefix-based allocation is suitable for deployments of a larger scale. For globally unique MAC addresses, the OUI must be registered in IEEE.

a) Configure MAC address allocation.
b) Apply the new MAC address allocation scheme to an existing virtual machine in its **Virtual Hardware** settings.
   - Power off a virtual machine, configure the adapter to use a manual MAC address, revert to automatic MAC address allocation, and power on the virtual machine.
   - If the virtual machine is in production and you cannot power it off for configuration, after you change the vCenter ID or the address allocation scheme, re-create the network adapter in conflict with enabled automatic MAC address assignment. In the guest operating system, set the same static IP address to the re-added adapter as before.

- Enforce MAC address regeneration when transferring a virtual machine between vCenter instances by using the virtual machine files from a datastore.
  a) Power off a virtual machine, remove it from the inventory, and in its configuration file (`.vmx`), set the `ethernetX.addressType` parameter to `generated`.
     `X` next to `ethernet` stands for the sequence number of the virtual NIC in the virtual machine.
  b) Import the virtual machine from one vCenter system to another by registering the virtual machine from a datastore in the target vCenter.
     The virtual machine files can reside in a datastore that is shared between the two vCenter instances or can be uploaded to a datastore that is accessible only from the target vCenter system.
     For information about registering a virtual machine from a datastore, see *vSphere Virtual Machine Administration*.
  c) Power on the virtual machines for the first time.
     While the virtual machine is starting up, an information icon appears on the virtual machine in the vSphere Client.
  d) Right-click the virtual machine and select **Guest OS** > **Answer Question**.
  e) Select the **I Copied It** option.

  The target vCenter re-generates the MAC address of the virtual machine. The new MAC address starts with the VMware OUI `00:0c:29` and is based on the BIOS UUID of the virtual machine. The BIOS UUID of the virtual machine is calculated from the BIOS UUID of the host.
- If the vCenter and hosts are version 6.0 and later and the vCenter instances are connected in Enhanced Linked Mode, migrate virtual machines by using vMotion across vCenter systems.
  When a virtual machine is migrated across vCenter systems, the source vCenter adds the MAC address of the virtual machine to a denylist and does not assign them to other virtual machines.

Attempt to Power On a Virtual Machine Fails Due to a MAC Address Conflict

After you set a certain static MAC address to a virtual machine adapter, you cannot power on the virtual machine.

In the vSphere Web Client, after you assign a MAC address within the range `00:50:56:40:`*`YY`*`:`*`ZZ`* – `00:50:56:7F:`*`YY`*`:`*`ZZ`* to a virtual machine, attempts to power the virtual machine on fail with a status message that the MAC address is in conflict.

```
00:50:56:XX:YY:ZZ is not a valid static Ethernet address. It
conflicts with VMware reserved MACs for other usage.
```

You attempt to assign a MAC address which starts with the VMware OUI `00:50:56` and is within the address range allocated for host VMkernel adapters on the vCenter system.

If you want to preserve the VMware OUI prefix, set a static MAC address within the range `00:50:56:00:00:00` – `00:50:56:3F:FF:FF`. Otherwise, set an arbitrary MAC address whose prefix is different from the VMware OUI one. For information about the ranges available for static MAC addresses that have the VMware OUI prefix, see the *vSphere Networking* documentation.

Unable to Remove a Host from a vSphere Distributed Switch

Under certain conditions, you might be unable to remove a host from the vSphere distributed switch.

- Attempts to remove a host from a vSphere distributed switch fail, and you receive a notification that resources are still in use. The notification that you receive might look like the following:
  ```
  The resource '16' is in use.
  ```

```
vDS DSwitch port 16 is still on host 10.23.112.2 connected to MyVM nic=4000 type=vmVnic
```

- Attempts to remove a host proxy switch that still exists on the host from a previous networking configuration fail. For example, you moved the host to a different data center or vCenter system, or upgraded the ESX and vCenter software, and created new networking configuration. When trying to remove the host proxy switch, the operation fails because resources on the proxy switch are still in use.

You cannot remove the host from the distributed switch or delete the host proxy switch because of the following reasons.

- There are VMkernel adapters on the switch that are in use.
- There are virtual machine network adapters connected to the switch.

Problem	Solution
Cannot remove a host from a distributed switch	1. In the vSphere Client, navigate to the distributed switch. 2. On the **Configure** tab, select **More** > **Ports**. 3. Locate all ports that are still in use and check which VMkernel or virtual machine network adapters on the host are still attached to the ports. 4. Migrate or delete the VMkernel and virtual machine network adapters that are still connected to the switch. 5. Use the **Add and Manage Hosts** wizard in the vSphere Client to remove the host from the switch. After the host is removed, the host proxy switch is deleted automatically.
Cannot remove a host proxy switch	1. In the vSphere Client, navigate to the host. 2. Delete or migrate any VMkernel or virtual machine network adapters that are still connected to the host proxy switch. 3. Delete the host proxy switch from the Networking view on the host.

Hosts on a vSphere Distributed Switch Lose Connectivity to vCenter

Hosts on a vSphere Distributed Switch cannot connect to vCenter after a port group configuration.

After you change the networking configuration of a port group on a vSphere Distributed Switch that contains the VMkernel adapters for the management network, the hosts on the switch lose connectivity to vCenter. In the vSphere Client the status of the hosts is nonresponsive.

On a vSphere Distributed Switch in the vCenter that has networking rollback deactivated, the port group containing the VMkernel adapters for the management network is misconfigured in vCenter and the invalid configuration is propagated to the hosts on the switch.

> **Note:**
> In vSphere networking rollback is enabled by default. However, you can activate or deactivate rollbacks at the vCenter level. For more information, see the *vSphere Networking* documentation.

1. From the Direct Console User Interface (DCUI) to an affected host, use the **Restore vDS** option from the **Network Restore Options** menu to configure the uplinks and the ID of the VLAN for the management network.

   The DCUI creates a local ephemeral port and applies the VLAN and uplink configuration to the port. The DCUI changes the VMkernel adapter for the management network to use the new host local port to restore connectivity to vCenter.

   After the host re-connects to vCenter, the vSphere Client displays a warning that some hosts on the switch have different networking configuration from the configuration stored in vSphere distributed switch.

2. In the vSphere Client, configure the distributed port group for the management network with correct settings.

Situation	Solution
You have altered the port group configuration only once	You can roll the configuration of the port group back one step. Right-click the port group, click **Restore Configuration**, and select **Restore to previous configuration**.

Situation	Solution
You have backed up a valid configuration of the port group	You can restore the configuration of the port group by using the backup file. Right-click the port group, click **Restore Configuration**, and select **Restore configuration from a file**. You can also restore the configuration for the entire switch, including the port group, from a backup file for the switch.
You have performed more than one configuration step and you do not have a backup file	You must provide valid settings for the port group manually.

For information about networking rollback, recovery, and restore, see the *vSphere Networking* documentation.

3. Migrate the VMkernel adapter for the management network from the host local ephemeral port to a distributed port on the switch by using the **Add and Manage Hosts** wizard.

   Unlike distributed ports, the ephemeral local port of the VMKernel has a non-numeric ID.

   For information about handling VMkernel adapters through the **Add and Manage Hosts** wizard, see the *vSphere Networking* documentation.

4. Apply the configuration of the distributed port group and VMkernel adapter from vCenter to the host.
   - Push the correct configuration of the distributed port group and VMkernel adapter from vCenter to the host.
     a. In the vSphere Client, navigate to the host.
     b. On the **Configure** tab, click **Networking**.
     c. From the **Virtual switches** list, select the distributed switch and click **Rectify the state of the selected distributed switch on the host**.
   - Wait until vCenter applies the settings within the next 24 hours.

Alarm for Loss of Network Redundancy on a Host

An alarm reports a loss of uplink redundancy on a vSphere standard or a distributed switch for a host.

No redundant physical NICs for a host are connected to a particular standard or a distributed switch, and the following alarm appears:

```
Host name or IP Network uplink redundancy lost
```

Only one physical NIC on the host is connected to a certain standard or a distributed switch. The redundant physical NICs are either down or are not assigned to the switch.

For example, assume that a host in your environment has physical NICs *vmnic0* and *vmnic1* connected to *vSwitch0*, and the physical NIC *vmnic1* goes offline, leaving only *vmnic0* connected to *vSwitch0*. As a result, the uplink redundancy for *vSwitch0* is lost on the host.

Check which switch has lost uplink redundancy on the host. Connect at least one more physical NIC on the host to this switch and reset the alarm to green. You can use the vSphere Client or the ESX Shell.

If a physical NIC is down, try to bring it back up by using the ESX Shell on the host.

For information about using the networking commands in the ESX Shell, see *ESXCLI Reference*. For information about configuring networking on a host in the vSphere Client, see *vSphere Networking*.

Virtual Machines Lose Connectivity After Changing the Uplink Failover Order of a Distributed Port Group

Changes in the failover NIC order on a distributed port group cause the virtual machines associated with the group to disconnect from the external network.

After you rearrange the uplinks in the failover groups for a distributed port group in vCenter, for example, by using the vSphere Client, some virtual machines in the port group can no longer access the external network.

After changing the failover order, many reasons might cause virtual machines to lose connectivity to the external network.

- The host that runs the virtual machines does not have physical NICs associated with the uplinks that are set to active or standby. All uplinks that are associated with physical NICs from the host for the port group are moved to unused.
- A Link Aggregation Group (LAG) that has no physical NICs from the host is set as the only active uplink according to the requirements for using LACP in vSphere.
- If the virtual machine traffic is separated in VLANs, the host physical adapters for the active uplinks might be connected to trunk ports on the physical switch that do not handle traffic from these VLANs.
- If the port group is configured with IP hash load balancing policy, an active uplink adapter is connected to a physical switch port that might not be in an EtherChannel.

You can examine the connectivity of the virtual machines in the port group to associated host uplinks and uplink adapters from the central topology diagram of the distributed switch or from the proxy switch diagram for the host.

- Restore the failover order with the uplink that is associated with a single physical NIC on the host back to active.
- Create a port group with identical settings, make it use the valid uplink number for the host, and migrate the virtual machine networking to the port group.
- Move the NIC to an uplink that participates in the active failover group.

  You can use the vSphere Client to move the host physical NIC to another uplink.

  - Use the **Add and Manage Hosts** wizard on the distributed switch.
    a. Navigate to the distributed switch in the vSphere Client.
    b. From the **Actions** menu select **Add and Manage Hosts**.
    c. On the **Select task** page, select the **Manage host networking** option and select the host.
    d. To assign the NIC of the host to an active uplink, navigate to the **Manage physical network adapters** page and associate the NIC to the switch uplink.
  - Move the NIC at the level of the host.
    a. Navigate to the host in the vSphere Client, and on the **Configure** tab, expand the **Networking** menu.
    b. Select **Virtual Switches** and select the distributed proxy switch.
    c. Click **Manage the physical network adapters connected to the selected switch**, and move the NIC to the active uplink

Unable to Add a Physical Adapter to a vDS That Has Network I/O Control

You might be unable to add a physical adapter with low speed, for example, 1 Gbps, to a vSphere Distributed Switch that has vSphere Network I/O Control version 3 configured.

You try to add a physical adapter with low speed, for example, 1 Gbps, to a vSphere Distributed Switch that is connected to physical adapters with high speed, for example, 10 Gbps. Network I/O Control version 3 is enabled on the switch and bandwidth reservations exist for one or more system traffic types, such as vSphere management traffic, vSphere vMotion traffic, vSphere NFS traffic, and so on. The task for adding the physical adapter fails with a status message that a parameter is incorrect.

```
A specified parameter was not correct: spec.host[].backing.pnicSpec[]
```

Network I/O Control aligns the bandwidth that is available for reservation to the 10-Gbps speed of the individual physical adapters that are already connected to the distributed switch. After you reserve a part of this bandwidth, adding a physical adapter whose speed is less than 10 Gbps might not meet the potential needs of a system traffic type.

For information about Network I/O Control version 3, see the *vSphere Networking* documentation.

1. In the vSphere Client, navigate to the host.
2. On the **Configure** tab, expand the **System** group of settings.
3. Select **Advanced System Settings** and click **Edit**.
4. Type the physical adapters that you want to use outside the scope of Network I/O Control as a comma-separated list for the `Net.IOControlPnicOptOut` parameter.
   For example: `vmnic2,vmnic3`

5. Click **OK** to apply the changes.
6. In the vSphere Client, add the physical adapter to the distributed switch.

Troubleshoot SR-IOV Enabled Workloads

Under certain conditions, you might experience connectivity or power-on problems with virtual machines that use SR-IOV to send data to physical network adapters.

SR-IOV Enabled Workload Cannot Communicate After You Change Its MAC Address

After you change the MAC address in the guest operating system of an SR-IOV enabled virtual machine, the virtual machine loses connectivity.

When you connect the network adapter of a virtual machine to an SR-IOV virtual function (VF), you create a passthrough network adapter for the virtual machine. After the (VF) driver in the guest operating system modifies the MAC address for the passthrough network adapter, the guest operating system shows that the change is successful but the VM network adapter loses connectivity. Although the guest operating system shows that the new MAC address is enabled, a log message in the `/var/log/vmkernel.log` file indicates that the operation has failed.

```
Requested mac address change to new MAC address on port VM NIC port number, disallowed by
 vswitch policy.
```

where

- *`new MAC address`* is the MAC address in the guest operation system.
- *`VM NIC port number`* is the port number of the VM network adapter in hexadecimal format.

The default security policy on the port group to which the passthrough network adapter is connected does not allow changes in the MAC address in the guest operating system. As a result, the networking interface in the guest operating system cannot acquire an IP address and loses connectivity.

In the guest operating system, reset the interface to cause the passthrough network adapter to regain its valid MAC address. If the interface is configured to use DHCP for address assignment, the interface acquires an IP address automatically.
For example, on a Linux virtual machine run the `ifconfig` console command.

```
ifconfig ethX down
ifconfig ethX up
```

where *`X`* in `ethX` represents the sequence number of the virtual machine network adapter in the guest operating system.

Virtual Machine That Uses an SR-IOV Virtual Function Fails to Power On Because the Host Is Out of Interrupt Vectors

On an ESX host, one or more virtual machines that use SR-IOV virtual functions (VFs) for networking are powered off.

On an ESX host, one or more virtual machines that use SR-IOV virtual functions (VFs) for networking fail to power on if the total number of assigned virtual functions is close to the maximum number of VFs specified in the *vSphere Configuration Maximums* guide.

The virtual machine log file `vmware.log` contains the following message about the VF:

```
PCIPassthruChangeIntrSettings: vf_name failed to register interrupt (error code 195887110)
```

The VMkernel log file `vmkernel.log` contains the following messages about the VF assigned to the virtual machine:

```
VMKPCIPassthru: 2565: BDF = vf_name intrType = 4 numVectors: 3
WARNING: IntrVector: 233: Out of interrupt vectors
```

The number of allocatable interrupt vectors scales up with the number of physical CPUs on an ESX host. An ESX host that has 32 CPUs can provide a total of 4096 interrupt vectors. When the host boots, devices on the host such as storage controllers, physical network adapters, and USB controllers consume a subset of the 4096 vectors. If these devices require more than 1024 vectors, the maximum number of potentially supported VFs is reduced.

When a virtual machine powers on and the guest operating system VF driver starts, interrupt vectors are consumed. If the required number of interrupt vectors is not available, the guest operating system shuts down unexpectedly without any error messages.

No rule presently exists to determine the number of interrupt vectors consumed or available on a host. This number depends on the hardware configuration of the host.

- To be able to power on the virtual machines, reduce the total number of VFs assigned to virtual machines on the host. For example, change the SR-IOV network adapter of a virtual machine to an adapter that is connected to a vSphere Standard Switch or vSphere Distributed Switch.

A Virtual Machine that Runs a VPN Client Causes Denial of Service for Virtual Machines on the Host or Across a vSphere HA Cluster

A virtual machine sending Bridge Protocol Data Unit (BPDU) frames, for example, a VPN client, causes some virtual machines connected to the same port group to lose connectivity. The transmission of BPDU frames might also break the connection of the host or of the parent vSphere HA cluster.

A virtual machine that is expected to send BPDU frames causes the traffic to the external network of the virtual machines in the same port group to be blocked.

If the virtual machine runs on a host that is a part of a vSphere HA cluster, and the host becomes network-isolated under certain conditions, you observe Denial of Service (DoS) on the hosts in the cluster.

As a best practice, a physical switch port that is connected to an ESX host has the Port Fast and BPDU guard enabled to enforce the boundary of the Spanning Tree Protocol (STP). A standard or distributed switch does not support STP, and it does not send any BPDU frames to the switch port. However, if any BPDU frame from a compromised virtual machine arrives at a physical switch port facing an ESX host , the BPDU guard feature deactivates the port to stop the frames from affecting the Spanning Tree Topology of the network.

In certain cases a virtual machine is expected to send BPDU frames, for example, when deploying VPN that is connected through a Windows bridge device or through a bridge function. If the physical switch port paired with the physical adapter that handles the traffic from this virtual machine has the BPDU guard on, the port is error-disabled, and the virtual machines and VMkernel adapters using the host physical adapter cannot communicate with the external network anymore.

If the teaming and failover policy of the port group contains more active uplinks, the BPDU traffic is moved to the adapter for the next active uplink. The new physical switch port becomes deactivated, and more workloads become unable to exchange packets with the network. Eventually, almost all entities on the ESX host might become unreachable.

If the virtual machine runs on a host that is a part of a vSphere HA cluster, and the host becomes network-isolated because most of the physical switch ports connected to it are deactivated, the active primary host in the cluster moves the BPDU sender virtual machine to another host. The virtual machine starts disabling the physical switch ports connected to the new host. The migration across the vSphere HA cluster eventually leads to accumulated DoS across the entire cluster.

- If the VPN software must continue its work on the virtual machine, allow the traffic out of the virtual machine and configure the physical switch port individually to pass the BPDU frames.

Network Device	Configuration
Distributed or standard switch	Set the Forged Transmit security property on the port group to **Accept** to allow BPDU frames to leave the host and reach the physical switch port. You can isolate the settings and the physical adapter for the VPN traffic by placing the virtual machine in a separate port group and assigning the physical adapter to the group. **CAUTION:** Setting the Forged Transmit security property to **Accept** to enable a host to send BPDU frames carries a security risk because a compromised virtual machine can perform spoofing attacks.
Physical switch	• Keep the Port Fast enabled. • Enable the BPDU filter on the individual port. When a BPDU frame arrives at the port, it is filtered out. **Note:** Do not enable the BPDU filter globally. If the BPDU filter is enabled globally, the Port Fast mode becomes deactivated and all physical switch ports perform the full set of STP functions.

- To deploy a bridge device between two virtual machine NICs connected to the same Layer 2 network, allow the BPDU traffic out of the virtual machines and deactivate Port Fast and BPDU loop prevention features.

Network Device	Configuration
Distributed or standard switch	Set the Forged Transmit property of the security policy on the port groups to **Accept** to allow BPDU frames to leave the host and reach the physical switch port. You can isolate the settings and one or more physical adapters for the bridge traffic by placing the virtual machine in a separate port group and assigning the physical adapters to the group. **CAUTION:** Setting the Forged Transmit security property to **Accept** to enable bridge deployment carries a security risk because a compromised virtual machine can perform spoofing attacks.
Physical switch	• Deactivate Port Fast on the ports to the virtual bridge device to run STP on them. • Deactivate BPDU guard and filter on the ports facing the bridge device.

- Protect the environment from DoS attacks in any case by activating the BPDU filter on the ESX host or on the physical switch.
  - On a host that does not have the Guest BPDU filter implemented enable the BPDU filter on the physical switch port to the virtual bridge device.

Network Device	Configuration
Distributed or standard switch	Set the Forged Transmit property of the security policy on the port group to **Reject**.
Physical switch	• Keep the Port Fast configuration. • Enable the BPDU filter on the individual physical switch port. When a BPDU frame arrives at the physical port, it is filtered out. **Note:** Do not enable the BPDU filter globally. If the BPDU filter is enabled globally, the Port Fast mode is deactivated and all physical switch ports perform the full set of STP functions.

Low Throughput for UDP Workloads on Windows Virtual Machines

When a Windows virtual machine in vSphere transmits large UDP packets, the throughput is lower than expected or is oscillating even when other traffic is negligible.

When a Windows virtual machine transmits UDP packets larger than 1024 bytes, you experience lower than expected or oscillating throughput even when other traffic is negligible. In case of a video streaming server, video playback pauses.

For every UDP packet larger than 1024 bytes, the Windows network stack waits for a transmit completion interrupt before sending the next packet. vSphere does not provide a transparent workaround of the situation.

- Increase the threshold in bytes at which Windows changes its behavior for UDP packets by modifying the registry of the Windows guest OS.
  a) Locate the `HKLM\System\CurrentControlSet\Services\Afd\Parameters` registry key.
  b) Add a value with the name `FastSendDatagramThreshold` of type DWORD equal to 1500.
- Modify the coalescing settings of the virtual machine NIC.

  If the Windows virtual machine has a VMXNET3 vNIC adapter, configure one of the following parameters in the `.vmx` file of the virtual machine. Use the vSphere Client, or directly modify the `.vmx` file.

Action	Parameter	Value
Increase the interrupt rate of the virtual machine to a higher rate than expected packet rate. For example, if the expected packet rate is 15000 interrupts per second, set the interrupt rate to 16000 interrupts per second. Set the `ethernetX.coalescingScheme` parameter to `rbc` and the `ethernetX.coalescingParams` parameter to `16000`. The default interrupt rate is 4000 interrupts per second.	`ethernetX.coalescingScheme` `ethernetX.coalescingParams`	rbc 16000
Deactivate coalescing for low throughput or latency-sensitive workloads.	`ethernetX.coalescingScheme`	disabled
Revert to the coalescing algorithm from earlier ESX releases. **Note:** The ability to revert to the earlier algorithm will not be available in later vSphere releases.	`ethernetX.coalescingScheme`	calibrate

  X next to `ethernet` stands for the sequence number of the vNIC in the virtual machine.

  For more information about configuring parameters in the `.vmx` file, see the *vSphere Virtual Machine Administration*documentation.
- Modify ESX host coalescing settings.

  This approach affects all virtual machines and all virtual machine NICs on the host.

  You can edit the advanced system settings list for the host in the vSphere Client, or by using a vCLI console command on the host from the ESX Shell.

Action	Parameter in the vSphere Client	Parameter for the `esxcli system settings sdvanced set` Command	Value
Set a default interrupt rate higher than the expected packet rate. For example, set the interrupt rate to 16000 if 15000 interrupts are expected per second.	`Net.CoalesceScheme` `Net.CoalesceParams`	`/Net/CoalesceScheme` `/Net/CoalesceParams`	rbc 16000
Deactivate coalescing for low throughput or latency-sensitive workloads.	`Net.CoalesceDefaultOn`	`/Net/CoalesceDefaultOn`	0

Action	Parameter in the vSphere Client	Parameter for the `esxcli system settings sdvanced set` Command	Value
Revert to the coalescing scheme from earlier ESX releases. **Note:** The ability to revert to the earlier algorithm will not be available in later vSphere releases.	`Net.CoalesceScheme`	`/Net/CoalesceScheme`	calibrate

For information about configuring a host from the vSphere Client, see the *vCenter and Host Management* documentation. For information about setting host properties by using a vCLI command, refer to the *ESXCLI Reference* documentation.

VMs on the Same Distributed Port Group are Unable to Communicate

Under certain conditions, the virtual machines that are on the same distributed port group but on different hosts cannot communicate with each other.

Virtual machines that reside on different hosts and on the same port group are unable to communicate. Pings from one virtual machine to another have no effect. You cannot migrate the virtual machines between the hosts by using vMotion.

- There are no physical NICs on some of the hosts assigned to active or standby uplinks in the teaming and failover order of the distributed port group.
- The physical NICs on the hosts that are assigned to the active or standby uplinks reside in different VLANs on the physical switch. The physical NICs in different VLANs cannot see each other and thus cannot communicate with each other.
- In the topology of the distributed switch, check which host does not have physical NICs assigned to an active or standby uplink on the distributed port group. Assign at least one physical NIC on that host to an active uplink on the port group.
- In the topology of the distributed switch, check the VLAN IDs of the physical NICs that are assigned to the active uplinks on the distributed port group. On all hosts, assign physical NICs that are from the same VLAN to an active uplink on the distributed port group.
- To verify that there is no problem at the physical layer, migrate the virtual machines to the same host and check the communication between them. Verify that inbound and outbound ICMP traffic is enabled in the guest OS. By default ICMP traffic is deactivated in Windows Server 2008 and Windows Server 2012.

Attempt to Power On a Migrated vApp Fails Because the Associated Protocol Profile Is Missing

You cannot power on a vApp or virtual machine that you transferred to a data center or a vCenter system because a network protocol profile is missing.

After you cold migrate a vApp or a virtual machine to another data center or vCenter system, an attempt to power it on fails. An error message states that a property cannot be initialized or allocated because the network of the vApp or virtual machine does not have an associated network protocol profile.

```
Cannot initialize property 'property'. Network 'port group' has no associated network
 protocol profile.
```

```
Cannot allocate IP address for property 'property'. Network 'port group' has no associated
 network protocol profile.
```

By using the OVF environment, the vApp or virtual machine retrieves network settings from a network protocol profile that is associated with the port group of the vApp or virtual machine.

vCenter creates such a network protocol profile for you when you install the OVF of a vApp and associates the profile with the port group that you specify during the installation.

The mapping between the protocol profile and port group is valid only in the scope of a data center. When you move the vApp, the protocol profile is not transferred to the target data center because of the following reasons:

- The network settings of the protocol profile might not be valid in the network environment of the target data center.
- A port group that has the same name and is associated with another protocol profile might already exist in the target data center, and vApps and virtual machines might be connected to this group. Replacing the protocol profiles for the port group might affect the connectivity of these vApp and virtual machines.

- Create a network protocol profile on the target data center or vCenter system with the required network settings and associate the protocol profile with the port group to which the vApp or virtual machine is connected. For example, this approach is suitable if the vApp or virtual machine is a vCenter extension that uses the vCenter Extension vService.

  For information about providing network settings to a vApp or virtual machine from a network protocol profile, see the *vSphere Networking* documentation.
- Use the vSphere Client to export the OVF file of the vApp or virtual machine from the source data center or vCenter system and deploy it on the target data center or vCenter system.

  When you use the vSphere Client to deploy the OVF file, the target vCenter system creates the network protocol profile for the vApp.

  For information about managing OVF files in the vSphere Client, see the *vSphere Virtual Machine Administration* documentation.

Networking Configuration Operation Is Rolled Back and a Host Is Disconnected from vCenter

When you attempt to add or configure networking on a vSphere Distributed Switch on a host, the operation is rolled back and the host is disconnected from vCenter.

An attempt to perform a networking configuration operation on a vSphere Distributed Switch on a host, such as creating a virtual machine adapter or a port group, causes the host to disconnect from vCenter and results in the error message `Transaction has rolled back on the host`.

Under stressful conditions on a host, that is, if many concurrent networking operations compete for limited resources, the time to perform some of the operations might exceed the default timeout for rollback of network configuration operations on the distributed switch. As a result, these operations are rolled back.

For example, such a condition might come up when you create a VMkernel adapter on a host that has a very high number of switch ports or virtual adapters, all of which consume system resources on the host.

The default timeout to roll an operation back is 30 seconds.

- Use the vSphere Client to increase the timeout for rollback on vCenter.

  If you encounter the same problem again, increase the rollback timeout with 60 seconds incrementally until the operation has enough time to succeed.

  a) On the **Configure** tab of a vCenter instance, expand **Settings**.
  b) Select **Advanced Settings** and click **Edit settings**.
  c) If the property is not present, add the `config.vpxd.network.rollbackTimeout` parameter to the settings.
  d) Type a new value, in seconds, for the `config.vpxd.network.rollbackTimeout` parameter
  e) Click **Save**.
  f) Restart the vCenter system to apply the changes.
- Increase the timeout for rollback by editing the `vpxd.cfg` configuration file.

  If you encounter the same problem again, increase the rollback timeout with 60 seconds incrementally until the operation has enough time to succeed.

  a) Log in to the vCenter shell by using an SSH client and navigate to `/etc/vmware-vpx`.
  b) Open the `vpxd.cfg` file for editing.
  c) Under the `<network>` section, increase the timeout, in the `<rollbackTimeout>` element.

```
<config>
  <vpxd>
    <network>
      <rollbackTimeout>60</rollbackTimeout>
```

```
        </network>
      </vpxd>
    </config>
```

d) Save and close the file.

e) Restart the vCenter system to apply the changes.

vSphere Storage

vSphere Storage describes virtualized and software-defined storage technologies that VMware ESX™ and VMware vCenter® offer, and explains how to configure and use these technologies.

Intended Audience

This information is for experienced system administrators who are familiar with the virtual machine and storage virtualization technologies, data center operations, and SAN storage concepts.

Introduction to Storage in vSphere Environment

vSphere supports various storage options and functionalities in traditional and software-defined storage environments. A high-level overview of vSphere storage elements and aspects helps you plan a proper storage strategy for your virtual data center.

Traditional Storage Virtualization Models in vSphere Environment

Generally, storage virtualization refers to a logical abstraction of physical storage resources and capacities from virtual machines and their applications. ESX provides host-level storage virtualization.

In a vSphere environment, a traditional model is built around the following storage technologies using ESX and vCenter virtualization functionalities.

Local and Networked Storage	vSphere storage management begins with capacity that your storage administrator preallocates on local host or network attached storage systems. See What Types of Physical Storage Does ESX Support.
Storage Area Networks	A storage area network (SAN) is a specialized high-speed network that connects computer systems to high-performance storage systems, such as Fibre Channel, iSCSI, or NVMe over Fabrics. See Using ESX with a SAN.
Fibre Channel	Fibre Channel (FC) is a storage protocol that the SAN uses to transfer data traffic from ESX host servers to shared storage. The protocol packages SCSI commands into FC frames. To connect to the FC SAN, your host uses Fibre Channel host bus adapters (HBAs) or Fibre Channel over Ethernet (FCoE) adapters. FCoE adapters support high-performance SAN traffic over ethernet cables and switches. See Using ESX with Fibre Channel SAN.
Internet SCSI	Internet iSCSI (iSCSI) is a SAN transport that can use Ethernet connections between computer systems, or ESX hosts, and high-performance storage systems. To connect to the storage systems, your hosts use hardware iSCSI adapters or software iSCSI initiators with standard network adapters. See Using ESX with iSCSI SAN.
NVMe over Fabrics (NVMe-oF)	A storage protocol that enables communication between ESX hosts and NVMe storage devices over a network fabric. vSphere supports NVMe over Fibre Channel (NVMe/FC) and NVMe over TCP (NVMe/TCP), as well as NVMe over RDMA using RoCE. See .
NVMe over TCP (NVMe/TCP)	A transport protocol that enables ESX hosts to access NVMe storage devices over standard TCP/IP networks. NVMe/TCP uses Ethernet networking infrastructure and does not require RDMA-capable network adapters. See .
NVMe over Fibre Channel (NVMe/FC)	A transport protocol that enables ESX hosts to access NVMe storage devices over a Fibre Channel fabric. NVMe/FC uses Fibre Channel infrastructure to transport NVMe commands between hosts and storage arrays. See .
NVMe over RDMA (NVMe/RDMA)	A transport protocol that enables ESX hosts to access NVMe storage devices using Remote Direct Memory Access (RDMA). vSphere supports RDMA over Converged Ethernet (RoCE), which enables low-latency communication between hosts and NVMe storage devices.

See .

Storage Device or LUN

In the ESX context, the terms storage device and LUN are used interchangeably. Typically, both terms mean a storage volume that is presented to the host from a block storage system and is available for formatting.

See Target and Device Representations and Managing ESX Storage Devices.

Virtual Disks

A virtual machine on an ESX host uses a virtual disk to store its operating system, application files, and other data associated with its activities. Virtual disks are large physical files, or sets of files, that can be copied, moved, archived, and backed up as any other files. You can configure virtual machines with multiple virtual disks.

To access virtual disks, a virtual machine uses virtual NVMe or SCSI controllers. These virtual controllers include BusLogic Parallel, LSI Logic Parallel, LSI Logic SAS, VMware Paravirtual, NVMe, and others.

Each virtual disk resides on a datastore that is deployed on physical storage. From the standpoint of the virtual machine, each virtual disk appears as if it were a SCSI or NVMe drive connected to a SCSI or NVMe controller. Whether the physical storage is accessed through storage or network adapters on the host is typically transparent to the VM guest operating system and applications.

VMware vSphere® VMFS

The datastores that you deploy on block storage devices use the native vSphere Virtual Machine File System (VMFS) format. It is a special high-performance file system format that is optimized for storing virtual machines.

See vSphere VMFS Datastore Concepts and Operations.

NFS

The NFS client built into ESX uses the Network File System (NFS) protocol version 3 and 4.1over TCP/IP to access an NFS volume that is located on a NAS server. The ESX host can mount the volume and use it as an NFS datastore.

See NFS Datastore Concepts and Operations in vSphere Environment.

Raw Device Mapping

In addition to virtual disks, vSphere offers a mechanism called raw device mapping (RDM). RDM is useful when a guest operating system inside a virtual machine requires direct access to a storage device. For information about RDMs, see Raw Device Mapping in vSphere.

Software-Defined Storage Models in vSphere Environment

In addition to abstracting underlying storage capacities from vSphere VMs, as traditional storage models do, software-defined storage abstracts storage capabilities.

With the software-defined storage model, a virtual machine becomes a unit of storage provisioning and can be managed through a flexible policy-based mechanism. The model involves the following vSphere technologies.

VMware vSphere® Virtual Volumes™ (vVols)

The Virtual Volumes functionality changes the storage management paradigm from managing space inside datastores to managing abstract storage objects handled by storage arrays. With Virtual Volumes, an individual virtual machine, not the datastore, becomes a unit of storage management. And storage hardware gains complete control over virtual disk content, layout, and management.

See Working with vSphere Virtual Volumes.

> **Note:**
>
> Starting with VMware Cloud Foundation 9.0 and VMware vSphere Foundation 9.0, the Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release of VCF and VMware vSphere Foundation. Support for Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VCF and VMware vSphere Foundation 5.x, and other supported versions until end-of-support of the respective release. For more information, see Broadcom Knowledge Base article 401070.

VMware vSAN

vSAN is a distributed layer of software that runs natively as a part of the hypervisor. vSAN aggregates local or direct-attached capacity devices of an ESX host cluster and creates a single storage pool shared across all hosts in the vSAN cluster.

See the *Administering VMware vSAN* documentation.

Storage Policy Based Management

Storage Policy Based Management (SPBM) is a framework that provides a single control panel across various data services and storage solutions, including vSAN. Using storage policies, the framework aligns the application demands of your virtual machines with the capabilities provided by storage entities.

See Storage Policy Based Management in vSphere.

I/O Filters	I/O filters are software components that can be installed on ESX hosts and can offer additional data services to virtual machines. Depending on implementation, the services might include replication, encryption, caching, and so on. See Filtering Virtual Machine I/O in vSphere.

vSphere Storage APIs

Storage APIs is a family of APIs used by third-party hardware, software, and storage providers to develop components that enhance several vSphere features and solutions.

This Storage publication describes several Storage APIs that contribute to your storage environment. For information about other APIs from this family, including vSphere APIs - Data Protection, see the VMware website.

vSphere APIs for Storage Awareness

Also known as VASA, these APIs, either supplied by third-party vendors or offered by VMware, enable communications between vCenter and underlying storage. Through VASA, storage entities can inform vCenter about their configurations, capabilities, and storage health and events. In return, VASA can deliver VM storage requirements from vCenter to a storage entity and ensure that the storage layer meets the requirements.

VASA becomes essential when you work with Virtual Volumes, vSAN, vSphere APIs for I/O Filtering (VAIO), and storage VM policies. See Using Storage Providers in vSphere.

vSphere APIs for Array Integration

These APIs, also known as VAAI, include the following components:

- Hardware Acceleration APIs. Help arrays to integrate with vSphere, so that vSphere can offload certain storage operations to the array. This integration significantly reduces CPU overhead on the host. See Storage Hardware Acceleration in vSphere.
- Array Thin Provisioning APIs. Help to monitor space use on thin-provisioned storage arrays to prevent out-of-space conditions, and to perform space reclamation. See ESX and Array Thin Provisioning.

vSphere APIs for Multipathing

Known as the Pluggable Storage Architecture (PSA), these APIs allow storage partners to create and deliver multipathing and load-balancing plug-ins that are optimized for each array. Plug-ins communicate with storage arrays and determine the best path selection strategy to increase I/O performance and reliability from the ESX host to the storage array. For more information, see Using Pluggable Storage Architecture and Path Management with ESX.

Getting Started with a Traditional Storage Model in vSphere Environment

Setting up your ESX storage in traditional environments, includes configuring your storage systems and devices, enabling storage adapters, and creating datastores.

What Types of Physical Storage Does ESX Support

In traditional storage environments, the ESX storage management process starts with storage space that your storage administrator preallocates on different storage systems. ESX supports local storage and networked storage.

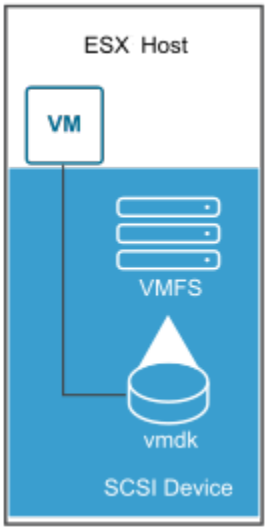
Local Storage

Local storage can be internal hard disks located inside your ESX host. It can also include external storage systems located outside and connected to the host directly through protocols such as SAS or SATA.

Local storage does not require a storage network to communicate with your host. You need a cable connected to the storage unit and, when required, a compatible storage controller or HBA in your host.

The following illustration depicts a virtual machine using local SCSI storage.

Figure 66: Local Storage



In this example of a local storage topology, the ESX host uses a single connection to a storage device. On that device, you can create a VMFS datastore, which you use to store virtual machine disk files.

Although this storage configuration is possible, it is not a best practice. Using single connections between storage devices and hosts creates single points of failure (SPOF) that can cause interruptions when a connection becomes unreliable or fails. However, because most of local storage devices do not support multiple connections, you cannot use multiple paths to access local storage.

ESX supports various local storage devices, including SCSI, IDE, SATA, USB, SAS, flash, and NVMe devices.

> **Note:**
> You cannot use IDE/ATA or USB drives to store virtual machines.

Local storage does not support sharing across multiple hosts. Only one host has access to a datastore on a local storage device. As a result, although you can use local storage to create VMs, you cannot use VMware features that require shared storage, such as HA and vMotion.

However, if you use a cluster of hosts that have just local storage devices, you can implement vSAN. vSAN transforms local storage resources into software-defined shared storage. With vSAN, you can use features that require shared storage. For details, see the *Administering VMware vSAN* documentation.

Networked Storage

Networked storage consists of external storage systems that your ESX host uses to store virtual machine files remotely. Typically, the host accesses these systems over a high-speed storage network.

Networked storage devices are shared. Datastores on networked storage devices can be accessed by multiple hosts concurrently. ESX supports multiple networked storage technologies.

In addition to traditional networked storage that this topic covers, VMware supports virtualized shared storage, such as vSAN. vSAN transforms internal storage resources of your ESX hosts into shared storage that provides such capabilities as High Availability and vMotion for virtual machines. For details, see the *Administering VMware vSAN* documentation.

> **Note:**
> The same LUN cannot be presented to an ESX host or multiple hosts through different storage protocols. To access the LUN, hosts must always use a single protocol, for example, either Fibre Channel only or iSCSI only.

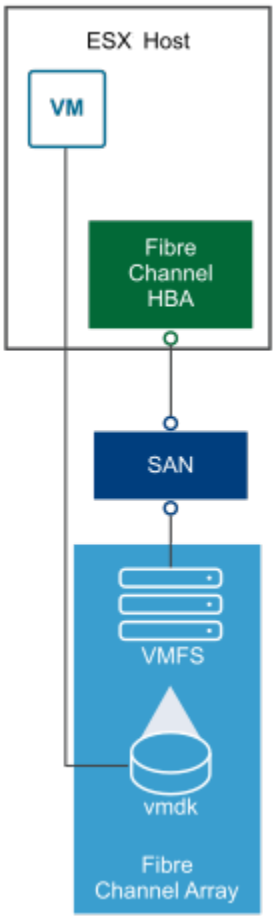
Fibre Channel (FC)

Stores virtual machine files remotely on an FC storage area network (SAN). FC SAN is a specialized high-speed network that connects your hosts to high-performance storage devices. The network uses Fibre Channel protocol to transport SCSI or NVMe traffic from virtual machines to the FC SAN devices.

To connect to the FC SAN, your host should be equipped with Fibre Channel host bus adapters (HBAs). Unless you use Fibre Channel direct connect storage, you need Fibre Channel switches to route storage traffic. If your host contains FCoE (Fibre Channel over Ethernet) adapters, you can connect to your shared Fibre Channel devices by using an Ethernet network.

Fibre Channel Storage depicts virtual machines using Fibre Channel storage.

Figure 67: Fibre Channel Storage



In this configuration, a host connects to a SAN fabric, which consists of Fibre Channel switches and storage arrays, using a Fibre Channel adapter. LUNs from a storage array become available to the host. You can access the LUNs and create datastores for your storage needs. The datastores use the VMFS format.

For specific information on setting up the Fibre Channel SAN, see Using ESX with Fibre Channel SAN.

Internet SCSI (iSCSI)

Stores virtual machine files on remote iSCSI storage devices. iSCSI packages SCSI storage traffic into the TCP/IP protocol, so that it can travel through standard TCP/IP networks instead of the specialized FC network. With an iSCSI connection, your host serves as the initiator that communicates with a target, located in remote iSCSI storage systems.

ESX offers the following types of iSCSI connections:

Hardware iSCSI	Your ESX host connects to storage through a third-party network adapter capable of offloading iSCSI and network processing. Modern server-grade network adapters, including Converged Network Adapters (CNAs), feature built-in hardware iSCSI offload capabilities and support booting from SAN. Hardware iSCSI adapters can be dependent or independent.
Software iSCSI	Your host uses a software-based iSCSI initiator in the VMkernel to connect to storage. With this type of iSCSI connection, your host needs only a standard network adapter for network connectivity.

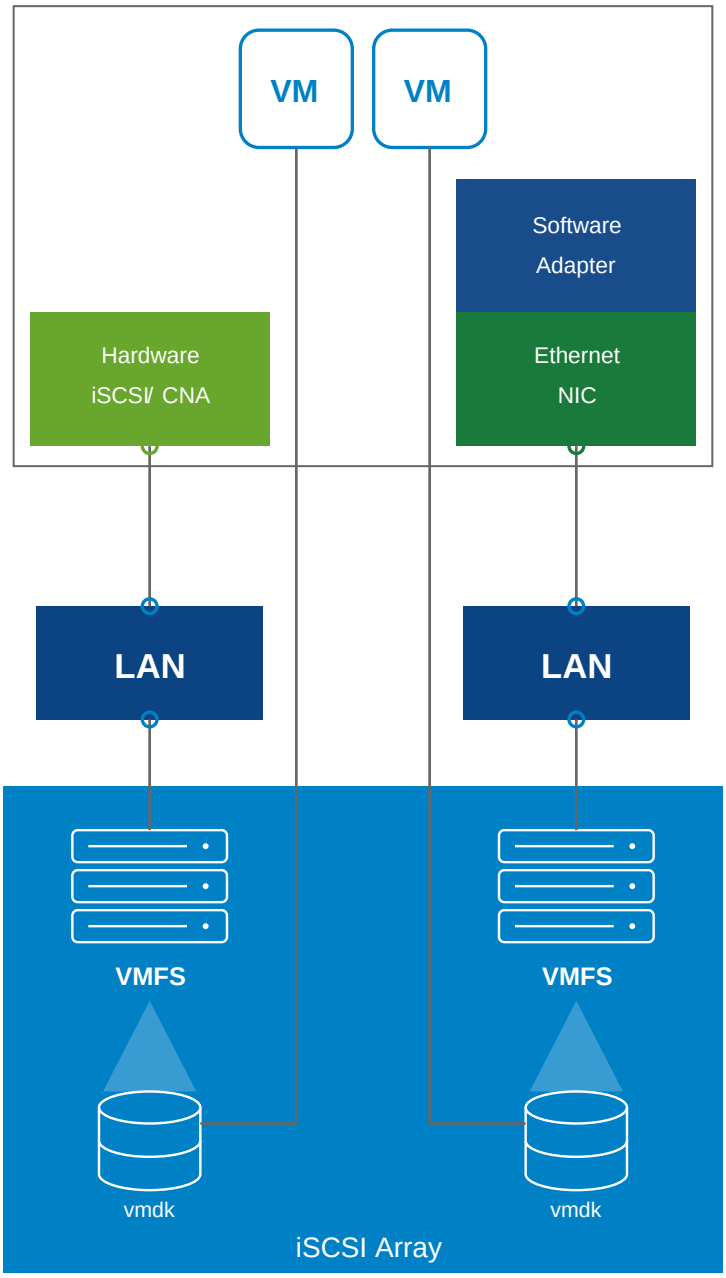
You must configure iSCSI initiators for the host to access and display iSCSI storage devices.

iSCSI Storage depicts different types of iSCSI initiators.

Note:

The graphic is for illustration only. The setup should not be used in real environment.

Figure 68: iSCSI Storage



In the left example, the host uses the hardware iSCSI adapter to connect to the iSCSI storage system.

In the right example, the host uses a software iSCSI adapter and an Ethernet NIC to connect to the iSCSI storage.

iSCSI storage devices from the storage system become available to the host. You can access the storage devices and create VMFS datastores for your storage needs.

For specific information on setting up the iSCSI SAN, see Using ESX with iSCSI SAN.

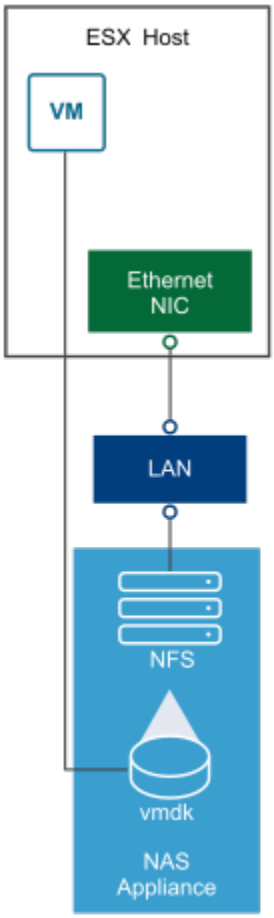
Network-attached Storage (NAS)

Stores virtual machine files on remote file servers accessed over a standard TCP/IP network. The NFS client built into ESX uses Network File System (NFS) protocol version 3 and 4.1 to communicate with the NAS/NFS servers. For network connectivity, the host requires a standard network adapter.

You can mount an NFS volume directly on the ESX host. You then use the NFS datastore to store and manage virtual machines in the same way that you use the VMFS datastores.

NFS Storage depicts a virtual machine using the NFS datastore to store its files. In this configuration, the host connects to the NAS server, which stores the virtual disk files, through a regular network adapter.

Figure 69: NFS Storage



For specific information on setting up NFS storage, see NFS Datastore Concepts and Operations in vSphere Environment.

Shared Serial Attached SCSI (SAS)

Stores virtual machines on direct-attached SAS storage systems that offer shared access to multiple hosts. This type of access permits multiple hosts to access the same VMFS datastore on a LUN.

NVMe over Fabrics Storage

VMware NVMe over Fabrics (NVMe-oF) provides a distance connectivity between a host and a target storage device on a shared storage array. VMware supports different technologies, including NVMe over RDMA (with RoCE v2 technology), NVMe over Fibre Channel, and NVMe over TCP/IP. For more information, see About VMware NVMe Storage.

Comparing Types of Storage

Whether certain vSphere functionality is supported might depend on the storage technology that you use.

The following table compares networked storage technologies that ESX supports.

Table 161: Networked Storage that ESX Supports

Technology	Protocols	Transfers	Interface
Fibre Channel	FC/SCSI, FC/ NVMe	Block access of data/LUN	FC HBA
Fibre Channel over Ethernet	FCoE/SCSI	Block access of data/LUN	Converged Network Adapter (hardware FCoE)
iSCSI	IP/SCSI	Block access of data/LUN	• Converged Network Adapter (hardware iSCSI) • Network adapter (software iSCSI)
NAS	IP/NFS	File (no direct LUN access)	Network adapter

The following table compares the vSphere features that different types of storage support.

Table 162: vSphere Features Supported by Storage

Storage Type	Boot VM	vMotion	Datastore	RDM	VM Cluster	vSphere HA and DRS	Storage APIs - Data Protection
Local Storage	Yes	No	VMFS	No	Yes	No	Yes
Fibre Channel	Yes	Yes	VMFS	Yes	Yes	Yes	Yes
iSCSI	Yes	Yes	VMFS	Yes	Yes	Yes	Yes
NAS over NFS	Yes	Yes	NFS 3 and NFS 4.1	No	No	Yes	Yes

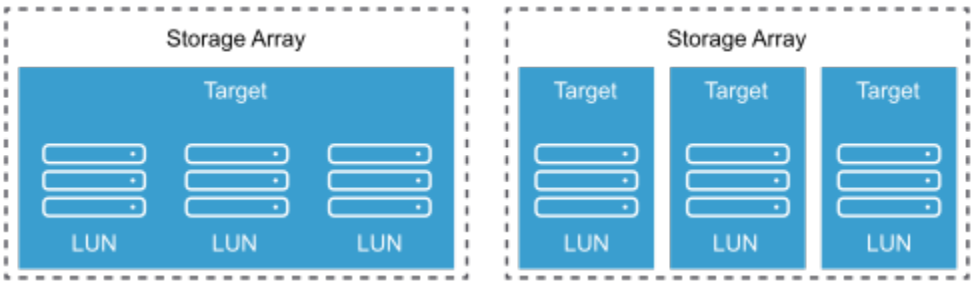
Note:
Local storage supports a cluster of virtual machines on a single host (also known as a cluster in a box). A shared virtual disk is required. For more information about this configuration, see the *vSphere Resource Management* documentation.

Target and Device Representations

In the ESX context, the term target identifies a single storage unit that the host can access. The terms storage device and LUN describe a logical volume that represents storage space on a target. In the ESX context, both terms also mean a storage volume that is presented to the host from a storage target and is available for formatting. Storage device and LUN are often used interchangeably.

Different storage vendors present the storage systems to ESX hosts in different ways. Some vendors present a single target with multiple storage devices or LUNs on it, while others present multiple targets with one LUN each.

Figure 70: Target and LUN Representations



In this illustration, three LUNs are available in each configuration. In one case, the host connects to one target, but that target has three LUNs that can be used. Each LUN represents an individual storage volume. In the other example, the host detects three different targets, each having one LUN.

Targets that are accessed through the network have unique names that are provided by the storage systems. The iSCSI targets use iSCSI names. Fibre Channel targets use World Wide Names (WWNs).

> **Note:**
> ESX does not support accessing the same LUN through different transport protocols, such as iSCSI and Fibre Channel.

A device, or LUN, is identified by its UUID name. If a LUN is shared by multiple hosts, it must be presented to all hosts with the same UUID.

How Virtual Machines Access Storage

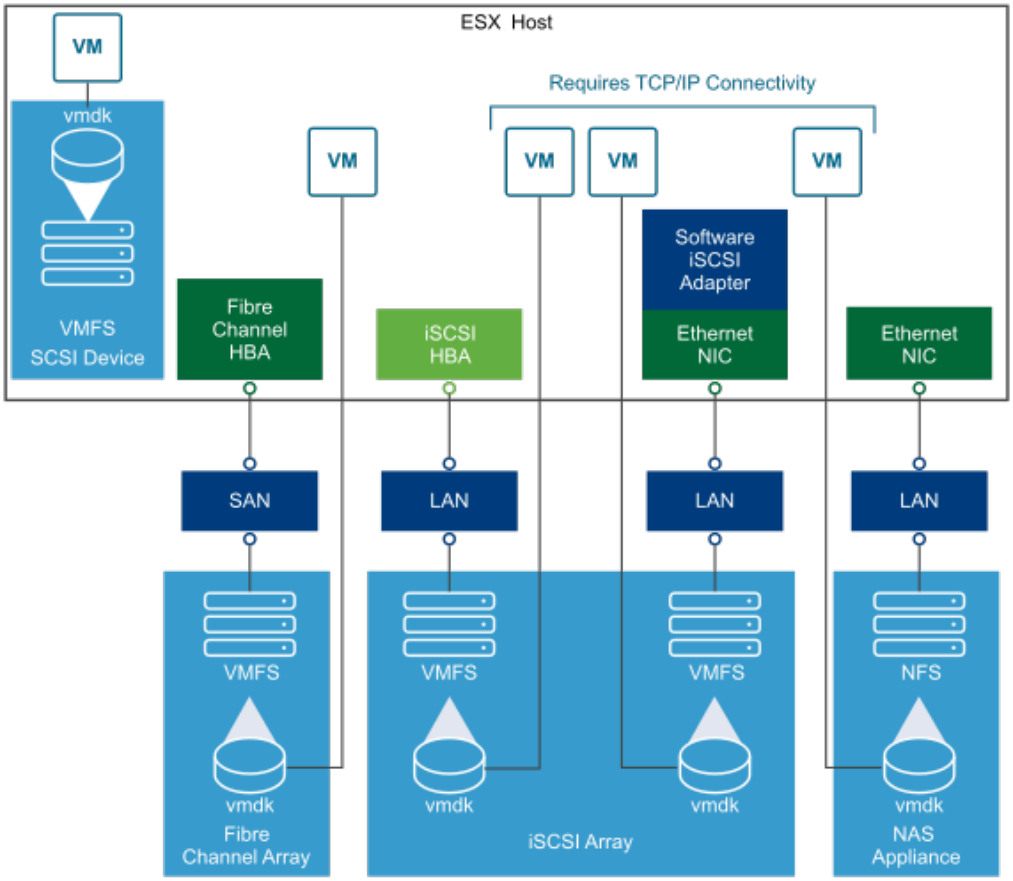
When a virtual machine communicates with its virtual disk stored on a datastore, it issues SCSI or NVMe commands. Because datastores can exist on various types of physical storage, these commands are encapsulated into other forms, depending on the protocol that the ESX host uses to connect to a storage device.

Regardless of the type of storage device your host uses, the virtual disk always appears to the virtual machine as a mounted SCSI or NVMe device. The virtual disk hides a physical storage layer from the virtual machine's operating system. This allows you to run operating systems that are not certified for specific storage equipment, such as SAN, inside the virtual machine.

The following graphic depicts five virtual machines using different types of storage to illustrate the differences between each type.

> **Note:**
>
> The graphic is for illustration only. The setup should not be used in real environment.

Figure 71: Virtual machines accessing different types of storage



Note:
This diagram is for conceptual purposes only. It is not a recommended configuration.

View Storage Adapters Available on an ESX Host

Storage adapters provide connectivity for your ESX host to a specific storage unit or network. ESX supports different classes of adapters, including SCSI, NVMe, iSCSI, RAID, Fibre Channel, Fibre Channel over Ethernet (FCoE), and Ethernet. ESX accesses the adapters directly through device drivers in the VMkernel.

You must enable certain adapters, for example software iSCSI, before you can view their information. To configure adapters, see Configuring iSCSI and iSER Adapters and Storage with ESX.

Depending on the type of storage you use, you might need to enable and configure a storage adapter on your host. You can display details for the available storage adapters and review their information.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**.
4. Use the icons to perform storage adapter tasks.

   Availability of specific icons depends on the storage configuration.

Icon	Description
Add Software Adapter	Add a storage adapter. Supports iSCSI, NVMe over RDMA, or NVMe over TCP adapters.
Refresh	Refresh information about storage adapters, topology, and file systems on the host.
Rescan Storage	Rescan all storage adapters on the host to discover newly added storage devices or VMFS datastores.
Rescan Adapter	Rescan the selected adapter to discover newly added storage devices.

5. To view details for a specific adapter, select the adapter from the list.
6. Use tabs under Adapter Details to access additional information and modify properties for the selected adapter.

   Depending on the type of your adapter, available tabs change. The following tabs are available for all types of adapters.

Tab	Description
Properties	Review general adapter properties that typically include a name and model of the adapter and unique identifiers formed according to specific storage standards. For some types of adapters, you can use this tab to configure additional properties, for example, authentication.
Devices	View storage devices the adapter can access. Use the tab to perform basic device management tasks. See Display Storage Devices for an Adapter.
Paths	List and manage all paths the adapter uses to access storage devices.

Characteristics of ESX Datastores

Datastores are logical containers, analogous to file systems, that hide the specifics of each storage device and provide a uniform model for storing virtual machine files. You can display all datastores available to your ESX hosts and analyze their properties.

Datastores are added to vCenter in the following ways:

- You can create a VMFS datastore or an NFS version 3 or 4.1 datastore using the New Datastore wizard. A vSAN datastore is automatically created when you enable vSAN.
- When you add an ESX host to vCenter, all datastores on the host are added to vCenter.

The following table describes datastore details that you can see when you review datastores through the vSphere Client. Certain characteristic might not be available or applicable to all types of datastores.

Important:

Starting with VMware Cloud Foundation 9.0 and vSphere Foundation 9.0, the vSphere Virtual Volumes capability, also known as vVols, is deprecated and will be removed in a future release. Support for vSphere Virtual Volumes will continue for critical bug fixes only for versions of vSphere 8.x, VCF and vSphere Foundation 5.x, and other supported versions until end-of-support for the respective release. For more information, see Broadcom Knowledge Base article 401070.

Table 163: Datastore Information

Datastore Information	Applicable Datastore Type	Description
Name	VMFS NFS vSAN vVol	Editable name that you assign to a datastore. For information on renaming a datastore, see Change Datastore Name.
Type	VMFS NFS vSAN vVol	File system that the datastore uses. For information about VMFS and NFS datastores and how to manage them, see Working with Datastores in vSphere Storage Environment. For information about vSAN datastores, see the *Administering VMware vSAN* documentation. For information about Virtual Volumes, see Working with vSphere Virtual Volumes.
Device Backing	VMFS NFS vSAN	Information about underlying storage, such as a storage device on which the datastore is deployed (VMFS), server and folder (NFS), or disk groups (vSAN).
Protocol Endpoints	vVol	Information about corresponding protocol endpoints. See Static Protocol Endpoints.
Extents	VMFS	Individual extents that the datastore spans and their capacity.
Drive Type	VMFS	Type of the underlying storage device, such as a flash drive or a regular HDD drive. For details, see Using Flash Devices with ESX Hosts.
Capacity	VMFS NFS vSAN vVol	Includes total capacity, provisioned space, and free space.
Mount Point	VMFS NFS vSAN vVol	A path to the datastore in the `/vmfs/volumes/` directory of the host.
Capability Sets	VMFS **Note:** A multi-extent VMFS datastore assumes capabilities of only one of its extents. NFS vSAN vVol	Information about storage data services that the underlying storage entity provides. You cannot modify them.
Storage I/O Control	VMFS NFS	Information on whether cluster-wide storage I/O prioritization is enabled. See the *vSphere Resource Management* documentation.
Hardware Acceleration	VMFS NFS vSAN vVol	Information on whether the underlying storage entity supports hardware acceleration. The status can be Supported, Not Supported, or Unknown. For details, see Storage Hardware Acceleration in vSphere. **Note:** NFS 4.1 does not support Hardware Acceleration.
Tags	VMFS NFS vSAN vVol	Datastore capabilities that you define and associate with datastores in a form of tags. For information, see Assign Tags to Datastores.

Datastore Information	Applicable Datastore Type	Description
Connectivity with Hosts	VMFS NFS vVol	Hosts where the datastore is mounted.
Multipathing	VMFS vVol	Path selection policy the host uses to access storage. For more information, see Understanding Multipathing and Failover in the ESX Environment.
Default Storage Policy	vSAN	When you provision a virtual machine on a datastore in vSphere, you must assign to the virtual machine a compatible VM storage policy. If you do not configure and explicitly assign the storage policy to the virtual machine, the system uses a default storage policy. See Default Storage Policies in vSphere.

Display Datastore Information

Access the Datastores view with the vSphere Client navigator.

Use the Datastores view to list all datastores available in the vSphere infrastructure inventory, analyze the information, and modify properties.

1. Navigate to any inventory object that is a valid parent object of a datastore, such as a host, a cluster, or a data center, and click the **Datastores** tab.

   Datastores that are available in the inventory appear in the center panel.

   You can check the status of the datastore.

Normal	The datastore is healthy and without problems.
Warning	The datastore experiences a potential problem that might require investigation. For example, the datastore is nearing capacity.
Critical	The datastore experiences a serious problem that requires immediate attention. For example, data corruption or connectivity issues.
Unknown	vCenter cannot determine the status due to datastore connectivity issues or other problems.

2. Use the options from a datastore right-click menu to perform basic tasks for a selected datastore.

   Availability of specific options depends on the type of the datastore and its configuration.

Option	Description
New Virtual Machine or **Register VM**	Create a new VM or register an existing virtual machine in the inventory. See the *vSphere Virtual Machine Administration* documentation.
Increase Datastore Capacity	Increase the capacity of the VMFS datastore or add an extent. See Increase Capacity of vSphere VMFS Datastore.
Browse Files	Navigate to the datastore file browser. See Navigate to the Datastore Browser.
Rename	Change the datastore name.See Change Datastore Name.
Mount Datastore	Mount the datastore to certain hosts.See Mount Datastores.
Unmount Datastore	Unmount the datastore from certain hosts. See Unmount Datastores.
Maintenance Mode	Use datastore maintenance mode. See the *vSphere Resource Management* documentation.
Configure Storage I/O Control (VMFS)	EnableStorageI/OControlforthe VMFS datastore. See the *vSphere Resource Management* documentation.
Edit Space Reclamation (VMFS)	Change space reclamation settings for the VMFS datastore. See Configure Fixed Reclamation for the Existing VMFS6 Datastore.
Add Permissions	Assign or manage user and group permissions for the datastore. See the *vSphere Security* documentation.

Option	Description
Alarms	Create alarm definitions to monitor your datastore environment. Alarms are notifications that are activated in response to an event, a set of conditions, or the state of the datastore. For more information, see the *vSphere Monitoring and Performance*.
Delete Datastore (VMFS)	Remove the VMFS datastore. See Remove VMFS Datastores.
Tags & Custom Attributes	Use tags to encode information about the datastore. See Assign Tags to Datastores.

3. To view specific datastore details, click a selected datastore.
4. Use tabs to access additional information and modify datastore properties.

Tab	Description
Summary	View statistics and configuration for the selected datastore.
Monitor	View alarms, performance data, resource allocation, events, and other status information for the datastore.
Configure	View and modify datastore properties. Menu items that you can see depend on the datastore type.
Permissions	Assign or edit permissions for the selected datastore.
Files	Navigate to the datastore file browser.
Hosts	View hosts where the datastore is mounted.
VMs	View virtual machines that reside on the datastore.

Using ESX with a SAN

Using ESX with a SAN improves flexibility, efficiency, and reliability. Using ESX with a SAN also supports centralized management, failover, and load balancing technologies.

The following are benefits of using ESX with a SAN:

- You can store data securely and configure multiple paths to your storage, eliminating a single point of failure.
- An ESX host can access storage devices presented from multiple storage arrays, including arrays from different vendors.
- Using a SAN with ESX systems extends failure resistance to the server. When you use SAN storage, all applications can instantly be restarted on another host after the failure of the original host.
- You can perform live migration of virtual machines using VMware vMotion.
- Use VMware High Availability (HA) in conjunction with a SAN to restart virtual machines in their last known state on a different server if their host fails.
- Use VMware Fault Tolerance (FT) to replicate protected virtual machines on two different hosts. Virtual machines continue to function without interruption on the secondary host if the primary one fails.
- Use VMware Distributed Resource Scheduler (DRS) to migrate virtual machines from one host to another for load balancing. Because storage is on a shared SAN array, applications continue running seamlessly.
- If you use VMware DRS clusters, put an ESX host into maintenance mode to have the system migrate all running virtual machines to other ESX hosts. You can then perform upgrades or other maintenance operations on the original host.

The portability and encapsulation of VMware virtual machines complements the shared nature of this storage. When virtual machines are located on SAN-based storage, you can quickly shut down a virtual machine on one server and power it up on another server, or suspend it on one server and resume operation on another server on the same network. This ability allows you to migrate computing resources while maintaining consistent shared access.

ESX and SAN Use Cases

When used with a SAN, ESX can benefit from multiple vSphere features, including Storage vMotion, Distributed Resource Scheduler (DRS), High Availability, and so on.

Using ESX with a SAN is effective for the following tasks:

Storage consolidation and simplification of storage layout	If you are working with multiple hosts, and each host is running multiple virtual machines, local storage might be insufficient. SAN storage can offer increased capacity, performance, and resiliency.
Host Maintenance with zero downtime	When performing ESX host or infrastructure maintenance, use vMotion to migrate virtual machines to other host. If shared storage is on the SAN, you can perform maintenance without interruptions to the users of the virtual machines. Virtual machine working processes continue throughout a migration.
Load balancing	Workloads can be spread across multiple physical paths and HBAs multipathing can prevent a single HBA or path from becoming a bottleneck.
High Availability and Redundancy	If a Host Bus Adapter (HBA), Fibre Channel switch, cable, or storage processor (SP) fails, traffic automatically reroutes to an alternate active path. This prevents application downtime or storage disconnects.
Simplified array migrations and storage upgrades	When you purchase new storage systems, use Storage vMotion to perform live migrations of virtual machines from existing storage to their new destinations. You can perform the migrations without interruptions of the virtual machines.

Specifics of Using SAN Storage with ESX

Using a SAN with an ESX host differs from a traditional SAN use in several ways.

- You cannot use SAN administration tools to access operating systems of virtual machines that reside on the storage. With traditional tools, you can monitor only the VMware ESX operating system. You use the vSphere Client to monitor virtual machines.
- The HBA visible to the SAN administration tools is part of the ESX system, not part of the virtual machine.
- Typically, your ESX system performs multipathing for you.

Making LUN Decisions

You must plan how to set up storage for your ESX systems before you format LUNs with VMFS datastores.

When you make your LUN decision, the following considerations apply:

- Each LUN must have the correct RAID level and storage characteristic for the applications running in virtual machines that use the LUN.
- Each LUN must contain only one VMFS datastore.
- If multiple virtual machines access the same VMFS, use disk shares to prioritize virtual machines.

You might want fewer, larger LUNs for the following reasons:

- More flexibility to create virtual machines without asking the storage administrator for more space.
- More flexibility for resizing virtual disks, doing snapshots, and so on.
- Fewer VMFS datastores to manage.
- Array and path limits**.** Array-side or host configuration limits, such as a maximum of 256 LUNs or paths per host, might require consolidating storage into fewer, larger LUNs to stay within supported thresholds.

You might want more, smaller LUNs for the following reasons:

- Less wasted storage space.
- Different applications might need different RAID characteristics.
- More flexibility, as the multipathing policy and disk shares are set per LUN.
- Use of Microsoft Cluster Service requires that each cluster disk resource is in its own LUN.
- Better performance because there is less contention for a single volume.

When the storage characterization for a virtual machine is unavailable, it might not be easy to determine the number and size of LUNs to provision. You can experiment using either a predictive or adaptive scheme.

Use the Predictive Scheme to Make LUN Decisions

Experiment using the predictive scheme.

1. Provision several LUNs with different storage characteristics.
2. Create a VMFS datastore on each LUN, labeling each datastore according to its characteristics.
3. Create virtual disks to contain the data for virtual machine applications in the VMFS datastores created on LUNs with the appropriate RAID level for the applications' requirements.
4. Use disk shares to distinguish high-priority from low-priority virtual machines.

   Note:
   Disk shares are relevant only within a given host. The shares assigned to virtual machines on one host have no effect on virtual machines on other hosts.
5. Run the applications to determine whether virtual machine performance is acceptable.

Use the Adaptive Scheme to Make LUN Decisions

You can experiment using the adaptive scheme.

1. Provision a large LUN (RAID 1+0 or RAID 5), with write caching enabled.
2. Create a VMFS on that LUN.
3. Create four or five virtual disks on the VMFS.
4. Run the applications to determine whether disk performance is acceptable.

If performance is acceptable, you can place additional virtual disks on the VMFS. If performance is not acceptable, create a new, large LUN, possibly with a different RAID level, and repeat the process. Use migration so that you do not lose virtual machines data when you recreate the LUN.

Selecting Virtual Machine Locations

When you are working on optimizing performance for your virtual machines, storage location is an important factor. Depending on your storage needs, you might select storage with high performance and high availability, or storage with lower performance.

Storage can be divided into different tiers depending on several factors:

- High Tier. Offers high performance and high availability. Might offer built-in snapshots to facilitate backups and point-in-time (PiT) restorations. Supports replication, full storage processor redundancy, and SAS drives. Uses high-cost spindles.
- Mid Tier. Offers mid-range performance, lower availability, some storage processor redundancy, and SCSI or SAS drives. Might offer snapshots. Uses medium-cost spindles.
- Lower Tier. Offers low performance, little internal storage redundancy. Uses low-end SCSI drives or SATA.

Not all VMs must be on the highest-performance and most-available storage throughout their entire life cycle.

When you decide where to place a virtual machine, the following considerations apply:

- Criticality of the VM
- Performance and availability requirements
- PiT restoration requirements
- Backup and replication requirements

A virtual machine might change tiers throughout its life cycle because of changes in criticality or changes in technology. Criticality is relative and might change for various reasons, including changes in the organization, operational processes, regulatory requirements, disaster planning, and so on.

Third-Party Management Applications

You can use third-party management applications with your ESX host.

Most SAN hardware is packaged with storage management software. In many cases, this software is a Web application that can be used with any Web browser connected to your network. In other cases, this software typically runs on the storage system or on a single server, independent of the servers that use the SAN for storage.

Use this third-party management software for the following tasks:

- Storage array management, including LUN creation, array cache management, LUN mapping, and LUN security.
- Setting up replication, check points, snapshots, or mirroring.

If you run the SAN management software on a virtual machine, you gain the benefits of a virtual machine, including failover with vMotion and VMware HA. Because of the additional level of indirection, however, the management software might not see the SAN. In this case, you can use an RDM.

Note:
Whether a virtual machine can run management software successfully depends on the particular storage system.

SAN Storage Backup Considerations

Having a proper backup strategy is one of the most important aspects of SAN management. In the SAN environment, backups have two goals. The first goal is to archive online data to offline media. This process is repeated periodically for all online data on a time schedule. The second goal is to provide access to offline data for recovery from a problem. For example, database recovery often requires a retrieval of archived log files that are not currently online.

Scheduling a backup depends on several factors:

- Identification of critical applications that require more frequent backup cycles within a given period.
- Recovery point and recovery time goals. Consider how precise your recovery point must be, and how long you are willing to wait for it.
- The rate of change (RoC) associated with the data. For example, if you are using synchronous/asynchronous replication, the RoC affects the amount of bandwidth required between the primary and secondary storage devices.
- Overall impact on a SAN environment, storage performance, and other applications.
- Identification of peak traffic periods on the SAN. Backups scheduled during those peak periods can slow the applications and the backup process.
- Time to schedule all backups within the data center.
- Time it takes to back up an individual application.
- Resource availability for archiving data, such as offline media access.

Include a recovery-time objective (RTO) for each application when you design your backup strategy. That is, consider the time and resources necessary to perform a backup. For example, if a scheduled backup stores so much data that recovery requires a considerable amount of time, examine the scheduled backup. Perform the backup more frequently, so that less data is backed up at a time and the recovery time decreases.

If an application requires recovery within a certain time frame, the backup process must provide a time schedule and specific data processing to meet the requirement. Fast recovery can require the use of recovery volumes that reside on online storage. This process helps to minimize or eliminate the need to access slow offline media for missing data components.

Using Third-Party Backup Packages

You can use third-party backup solutions to protect system, application, and user data in your virtual machines.

The vStorage APIs - Data Protection that VMware offers can work with third-party products. When using the APIs, third-party software can perform backups without loading ESX hosts with the processing of backup tasks.

The third-party products using the vStorage APIs - Data Protection can perform the following backup tasks:

- Perform a full, differential, and incremental image backup and restore of virtual machines.
- Perform a file-level backup of virtual machines that use supported Windows and Linux operating systems.
- Ensure data consistency by using Microsoft Volume Shadow Copy Services (VSS) for virtual machines that run supported Microsoft Windows operating systems.

Because vSphere Storage APIs - Data Protection (VADP) use the snapshot capabilities of VMFS, backups do not require that you stop virtual machines. These backups are generally nondisruptive, and can be performed at any time.

Note:

While VADP enables image-level backups using vSphere snapshots, VM snapshots are not database-aware backups and do not guarantee transaction consistency for high-I/O database workloads. For backing up database virtual machines, use native database backup utilities or certified application-aware backup solutions. For more information, see Broadcom KB 426571.

Using ESX with Fibre Channel SAN

ESX supports Fibre Channel (FC), a storage protocol that the SAN uses to transfer data traffic from hosts to shared storage. This section provides introductory information about how to use ESX with Fibre Channel SAN. For more information, check your vendor documentation.

Fibre Channel SAN Concepts

If you are a vSphere administrator planning to set up hosts to work with SANs, you must have a working knowledge of SAN concepts. You can find information about SANs in print and on the Internet. Because this industry changes constantly, check these resources frequently.

Storage Area Network (SAN)	A storage area network (SAN) is a specialized high-speed network that connects host servers to high-performance storage subsystems. The SAN components include host bus adapters (HBAs) in the host servers, switches that help route storage traffic, cables, storage processors (SPs), and storage disk arrays.
SAN Fabric	A SAN topology with at least one switch present on the network forms a SAN fabric.
Fibre Channel (FC) Protocol	To transfer traffic from host servers to shared storage, the SAN uses the Fibre Channel (FC) protocol that packages SCSI or NVMe commands into Fibre Channel frames.
Zoning	To restrict server access to storage arrays not allocated to that server, the SAN uses zoning. Typically, zones are created for each group of servers that access a shared group of storage devices and LUNs. Zones define which HBAs can connect to which SPs. Devices outside a zone are not visible to the devices inside the zone. Zoning has the following effects: • Reduces the number of targets and LUNs presented to a host. • Controls and isolates paths in a fabric. • Can prevent non- ESX systems from accessing a particular storage system, and from possibly destroying VMFS data. • Can be used to separate different environments, for example, a test from a production environment. With ESX hosts, use a single-initiator zoning or a single-initiator-single-target zoning. The latter is a preferred zoning practice. Using the more restrictive zoning prevents problems and misconfigurations that can occur on the SAN. For detailed instructions and best zoning practices, contact storage array or switch vendors.
LUN Masking	LUN masking is a process that makes a LUN available to some hosts and unavailable to other hosts.
Multipathing	When transferring data between the host server and storage, the SAN uses a technique known as multipathing. Multipathing allows you to have more than one physical path from the ESX host to a LUN on a storage system.
Path Failover	Generally, a single path from a host to a LUN consists of an HBA, switch ports, connecting cables, and the storage controller port. If any component of the path fails, the host selects another available path for I/O. The process of detecting a failed path and switching to another is called path failover.

Ports in Fibre Channel SAN

In the context of this document, a port is the connection from a device into the SAN. Each node in the SAN, such as a host, a storage device, or a fabric component has one or more ports that connect it to the SAN. Ports are identified in a number of ways.

WWPN (World Wide Port Name)	A globally unique identifier for a port that allows certain applications to access the port. The FC switches discover the WWPN of a device or host and assign a port address to the device.
Port_ID (or port address)	Within a SAN, each port has a unique port ID that serves as the FC address for the port. This unique ID enables routing of data through the SAN to that port. The FC switches assign the port ID when the device logs in to the fabric. The port ID is valid only while the device is logged on.

When N-Port ID Virtualization (NPIV) is used, a single FC HBA port (N-port) can register with the fabric by using several WWPNs. This method allows an N-port to claim multiple fabric addresses, each of which appears as a unique entity. When ESX hosts use a SAN, these multiple, unique identifiers allow the assignment of WWNs to individual virtual machines as part of their configuration.

Fibre Channel Storage Array Types

ESX supports different storage systems and arrays. They generally fall into these categories.

Active-active storage system	Supports access to the LUNs simultaneously through all the storage ports that are available without significant performance degradation. All the paths are active, unless a path fails.
Active-passive storage system	A system in which one storage processor is actively providing access to a given LUN. The other processors act as a backup for the LUN and can be actively providing access to other LUN I/O. I/O can be successfully sent only to an active port for a given LUN. If access through the active storage port fails, one of the passive storage processors can be activated by the servers accessing it.
Asymmetrical storage system	Supports Asymmetric Logical Unit Access (ALUA). ALUA-compliant storage systems provide different levels of access per port. With ALUA, the host can determine the states of target ports and prioritize paths. The host uses some of the active paths as primary, and uses others as secondary.

How Virtual Machines Access Data on a Fibre Channel SAN

ESX stores a virtual machine's disk files within a VMFS datastore that resides on a SAN storage device. When virtual machine guest operating systems send SCSI or NVMe commands to their virtual disks, the SCSI or NVMe virtualization layer translates these commands to VMFS file operations.

When a virtual machine interacts with its virtual disk stored on a SAN, the following process takes place:

1. When the guest operating system in a virtual machine reads or writes to a SCSI or NVMe disk, it sends SCSI or NVMe commands to the virtual disk.
2. Device drivers in the virtual machine’s operating system communicate with the virtual SCSI or NVMe controllers.
3. The virtual SCSI or NVMe controller forwards the command to the VMkernel.
4. The VMkernel performs the following tasks.
   a. Locates the appropriate virtual disk file in the VMFS volume.
   b. Maps the requests for the blocks on the virtual disk to blocks on the appropriate physical device.
   c. Sends the modified I/O request from the device driver in the VMkernel to the physical HBA.
5. The physical HBA performs the following tasks.
   a. Packages the I/O request according to the rules of the FC protocol.
   b. Transmits the request to the SAN.
6. Depending on a port the HBA uses to connect to the fabric, one of the SAN switches receives the request. The switch routes the request to the appropriate storage device.

Setting Up ESX Fibre Channel Environment

This topic provides an overview of setup steps that you need to follow when configuring your Fibre Channel SAN environment to work with ESX. In preparation for configuring your Fibre Channel SAN and setting up the ESX system, review the requirements and recommendations.

ESX Fibre Channel SAN Requirements

- Make sure that ESX systems support the SAN storage hardware and firmware combinations you use. For an up-to-date list, see the VMware by Broadcom Compatibility Guide.
- Configure your system to have only one VMFS volume per LUN.
- Unless you are using diskless servers, do not set up the diagnostic partition on a SAN LUN.

If you use diskless servers that boot from a SAN, a shared diagnostic partition is appropriate.

- Use RDMs to access raw disks. For information, see Raw Device Mapping in vSphere.
- Make sure that the storage device driver specifies a large enough queue. You can set the queue depth for the physical HBA during a system setup.
- VMtools increases the default disk timeout SCSI `TimeoutValue` parameter to 180 seconds. On virtual machines without VMtools, increase the value of the SCSI `TimeoutValue` parameter. With this parameter set up, the VMs can better tolerate delayed I/O resulting from a path failover. For information, see Understanding Multipathing and Failover in the ESX Environment.

ESX Fibre Channel SAN Restrictions

When you useESX with a SAN, certain restrictions apply.

- ESX does not support FC connected tape devices.
- You cannot use multipathing software inside a virtual machine to perform I/O load balancing to a single physical LUN. However, when your Microsoft Windows virtual machine uses dynamic disks, this restriction does not apply. For information about configuring dynamic disks, see Set Up Dynamic Disk Mirroring for Virtual Machines on vSphere VMFS.

Setting LUN Allocations

This topic provides general information about how to allocate LUNs when your ESX works with SAN.

When you set LUN allocations, be aware of the following points:

Storage provisioning	To ensure that the ESX system recognizes the LUNs at startup time, provision all LUNs to the appropriate HBAs before you connect the SAN to the ESX system. Provision all LUNs to all ESX HBAs at the same time. HBA failover works only if all HBAs see the same LUNs.
vMotion and VMware DRS	When you use vMotion or DRS, make sure that the LUNs for the virtual machines are provisioned to all ESX hosts. This action provides the most ability to move virtual machines.
Active-active compared to active-passive arrays	When you use vMotion or DRS with an active-passive SAN storage device, make sure that all ESX systems have consistent paths to all storage processors. Not doing so can cause path thrashing when a vMotion migration occurs. For active-passive storage arrays not listed in Storage/SAN Compatibility, VMware does not support storage port failover. In those cases, you must connect the server to the active port on the storage array. This configuration ensures that the LUNs are presented to the ESX host.

Setting Fibre Channel HBAs

Typically, FC HBAs that you use on your ESX host work correctly with the default configuration settings.

You should follow the configuration guidelines provided by your storage array vendor. During FC HBA setup, consider the following issues.

- Do not mix FC HBAs from different vendors in a single host. Having different models of the same HBA is supported, but a single LUN cannot be accessed through two different HBA types, only through the same type.
- Ensure that the firmware level on each HBA is the same.
- Set the timeout value for detecting a failover. To ensure optimal performance, do not change the default value.
- ESX supports 32 Gbps end-to-end Fibre Channel connectivity.

Configuring Fibre Channel over Ethernet

To access Fibre Channel storage, an ESX host can use the Fibre Channel over Ethernet (FCoE) protocol. The FCoE protocol encapsulates Fibre Channel frames into Ethernet frames. As a result, your host does not need special Fibre Channel links to connect to Fibre Channel storage. The host can use 10 Gbit lossless Ethernet to deliver Fibre Channel traffic.

Fibre Channel over Ethernet Adapters

VMware supports hardware FCoE adapters, a specialized offloaded Converged Network Adapters (CNAs) that contain network and Fibre Channel functionalities on the same card.

When such adapter is installed, your host detects and can use both CNA components. In the vSphere Client, the networking component appears as a standard network adapter (vmnic) and the Fibre Channel component as a FCoE adapter (vmhba).

For information on adapters that can be used with VMware FCoE, see the VMware by Broadcom Compatibility Guide.

Setup Workflow

This topic provides an overview of installation and setup steps that you need to follow when configuring your Fibre Channel SAN environment to work with ESX.

Follow these steps to configure your ESX SAN environment.

1. Design your SAN if it is not already configured. Most existing SANs require only minor modification to work with ESX.
2. Check that all SAN components meet requirements.
3. Perform any necessary storage array modification.
   Most vendors have vendor-specific documentation for setting up a SAN to work with VMware ESX.
4. Set up the HBAs for the hosts you have connected to the SAN.
5. Install ESX on the hosts.
6. Create virtual machines and install guest operating systems.
7. (Optional) Set up your system for VMware HA failover or for using Microsoft Clustering Services.
8. Upgrade or modify your environment as needed.

Booting ESX from Fibre Channel SAN

When you set up your ESX host to boot from a SAN, your host's boot image is stored on one or more LUNs in the SAN storage system. When the host starts, it boots from the LUN on the SAN rather than from its local disk.

ESX supports booting through a Fibre Channel host bus adapter (HBA) or a Fibre Channel over Ethernet (FCoE) converged network adapter (CNA).

Boot from SAN Benefits

Boot from SAN can provide numerous benefits to your ESX environment. However, in certain cases, boot from SAN is not compatible with your hosts. Before you set up your system for boot from SAN, decide whether it is appropriate for your environment.

> **CAUTION:**
> When you use boot from SAN with multiple ESX hosts, each host must have its own boot LUN. If you configure multiple hosts to share the boot LUN, ESX image corruption might occur.

If you use boot from SAN, the benefits for your environment include the following:

- Cheaper servers. Servers can be more dense and run cooler without internal storage.
- Easier server replacement. You can replace servers and have the new server point to the old boot location.
- Less wasted space. Servers without local disks often take up less space.
- Easier backup processes. You can back up the system boot images in the SAN as part of the overall SAN backup procedures. Also, you can use advanced array features such as snapshots on the boot image.
- Improved management. Creating and managing the operating system image is easier and more efficient.
- Better reliability. You can access the boot disk through multiple paths, which protects the disk from being a single point of failure.

Requirements and Considerations when Booting from Fibre Channel SAN

Your ESX boot configuration must meet specific requirements.

Table 164: Boot from SAN Requirements

Requirement	Description
ESX system requirements	Follow vendor recommendations for the server booting from a SAN.
Adapter requirements	Configure the adapter, so it can access the boot LUN. See your vendor documentation.
Access control	• Each host must have access to its own boot LUN only, not the boot LUNs of other hosts. Use storage system software to make sure that the host accesses only the designated LUNs. • Multiple servers can share a diagnostic partition. You can use array-specific LUN masking to achieve this configuration.
Multipathing support	Multipathing to a boot LUN on active-passive arrays is not supported because the BIOS does not support multipathing and is unable to activate a standby path.
SAN considerations	If the array is not certified for a direct connect topology, the SAN connections must be through a switched topology. If the array is certified for the direct connect topology, the SAN connections can be made directly to the array. Boot from SAN is supported for both switched topology and direct connect topology.
LUN and HLU considerations	When configuring Boot from SAN, consider how the host handles Host LUN (HLU) assignments. Consult your storage vendor documentation to determine specific host limitations and recommended HLU assignments. • Some hosts require the HLU number to be 0, or it might be considered a best practice. • Certain hosts can only detect LUNs in the range of 0–255. If the array LUN number is 2106, the host would not be able to see it. In such cases, an HLU is used to remap the LUN to a number within the host's supported range. For example, array LUN 2106 can appear as HLU 1 on the host.
Hardware-specific considerations	If you are running an IBM eServer BladeCenter and use boot from SAN, you must deactivate IDE drives on the blades.

Getting Ready to Boot ESX from SAN

When you prepare your ESX host to boot from a SAN, you perform several tasks.

This workflow describes the generic boot-from-SAN enablement process on the rack-mounted servers. For information on enabling the boot from SAN option on blade servers or other types of hosts, refer to partner documentation.

For more information, see VMware by Broadcom Compatibility Guide.

Configure SAN Components and Storage System

Before you set up your ESX host to boot from a SAN LUN, configure SAN components and a storage system.

Because configuring the SAN components is vendor-specific, refer to the product documentation for each item.

1. Connect network cable, referring to any cabling guide that applies to your setup.
   Check the switch wiring, if there is any.

2. Configure the storage array.
   a) From the SAN storage array, make the ESX host visible to the SAN. This process is often called creating an object.
   b) From the SAN storage array, set up the host to have the WWPNs of the host's adapters as port names or node names.
   c) Create and assign LUNs.
   d) Record the IP addresses of the switches and storage arrays.
   e) Record the WWPN for each SP.

   CAUTION:
   If you use a scripted installation process to install ESX in boot from SAN mode, take special steps to avoid unintended data loss.

Configure Storage Adapter to Boot from SAN

When you set up your host to boot from SAN, you enable the boot adapter in the host BIOS. You then configure the boot adapter to initiate a primitive connection to the target boot LUN.

Determine the WWPN for the storage adapter.

Configure the storage adapter to boot from SAN.

Because configuring boot adapters is vendor-specific, consult your vendor documentation.

Set Up Your System to Boot from Installation Media

When setting up your host to boot from SAN, you first boot the host from the VMware installation media. To boot from the installation media, change the system boot sequence in the BIOS setup.

Because changing the boot sequence in the BIOS is vendor-specific, refer to vendor documentation for instructions. The following procedure explains how to change the boot sequence on an IBM host.

1. Power on your system and enter the system BIOS Configuration/Setup Utility.
2. Select **Startup Options** and press Enter.
3. Select **Startup Sequence Options** and press Enter.
4. Change the **First Startup Device** to **[CD-ROM]**.

You can now install ESX.

Best Practices for Fibre Channel Storage with ESX

When you use ESX with Fibre Channel SAN, follow recommendations to avoid performance problems.

The vSphere Client offers extensive facilities for collecting performance information. The information is graphically displayed and frequently updated.

You can also use the `resxtop` or `esxtop` command-line utilities. The utilities provide a detailed look at how ESX uses resources. For more information, see the *vSphere Resource Management* documentation.

Check with your storage representative if your storage system supports Storage API - Array Integration hardware acceleration features. If it does, refer to your vendor documentation to enable hardware acceleration support on the storage system side. For more information, see Storage Hardware Acceleration in vSphere.

Preventing Fibre Channel SAN Problems

When you use ESX with a Fibre Channel SAN, follow specific guidelines to avoid SAN problems.

To prevent problems with your SAN configuration, observe these tips:

- Place only one VMFS datastore on each LUN.

- Do not change the path policy the system sets for you unless you understand the implications of making such a change.
- Document everything. Include information about zoning, access control, storage, switch, server and FC HBA configuration, software and firmware versions, and storage cable plan.
- Plan for failure:
  - Make several copies of your topology maps. For each element, consider what happens to your SAN if the element fails.
  - Verify different links, switches, HBAs, and other elements to ensure that you did not miss a critical failure point in your design.
- Ensure that the Fibre Channel HBAs are installed in the correct slots in the host, based on slot and bus speed. Balance PCI bus load among the available buses in the server.
- Become familiar with the various monitor points in your storage network, at all visibility points, including host's performance charts, FC switch statistics, and storage performance statistics.
- Be cautious when changing IDs of the LUNs that have VMFS datastores being used by your ESX host. If you change the ID, the datastore becomes inactive and its virtual machines fail. Resignature the datastore to make it active again. See vSphere VMFS Datastore Copies and Datastore Resignaturing.

  After you change the ID of the LUN, rescan the storage to reset the ID on your host. For information on using the rescan, see Rescan Operations for ESX Storage.

Deactivate Automatic ESX Host Registration

Certain storage arrays require that ESX hosts register with the arrays. ESX performs automatic host registration by sending the host's name and IP address to the array. If you prefer to perform manual registration using storage management software, deactivate the ESX auto-registration feature.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. Under Advanced System Settings, select the **Disk.EnableNaviReg** parameter and click the **Edit** icon.
5. Change the value to 0.

This operation deactivates the automatic host registration that is activated by default.

Optimizing Fibre Channel SAN Storage Performance

Several factors contribute to optimizing a typical SAN environment.

If the environment is properly configured, the SAN fabric components (particularly the SAN switches) are only minor contributors because of their low latencies relative to servers and storage arrays. Make sure that the paths through the switch fabric are not saturated, that is, that the switch fabric is running at the highest throughput.

Storage Array Performance

Storage array performance is one of the major factors contributing to the performance of the entire SAN environment.

If you encounter any problems with storage array performance, consult your storage array vendor documentation for any relevant information.

To improve the array performance in the vSphere environment, follow these general guidelines:

- When assigning LUNs, remember that several hosts might access the LUN, and that several virtual machines can run on each host. One LUN used by a host can service I/O from many different applications running on different operating systems. Because of this diverse workload, the RAID group containing the ESX LUNs typically does not include LUNs used by other servers that are not running ESX.

- Make sure that the read/write caching is available.
- SAN storage arrays require continual redesign and tuning to ensure that I/O is load-balanced across all storage array paths. To meet this requirement, distribute the paths to the LUNs among all the SPs to provide optimal load-balancing. Close monitoring indicates when it is necessary to rebalance the LUN distribution.

  Tuning statically balanced storage arrays is a matter of monitoring the specific performance statistics, such as I/O operations per second, blocks per second, and response time. Distributing the LUN workload to spread the workload across all the SPs is also important.

  Note:
  Dynamic load-balancing is not currently supported with ESX.

Server Performance with Fibre Channel

You must consider several factors to ensure optimal server performance.

Each server application must have access to its designated storage with the following conditions:

- High I/O rate (number of I/O operations per second)
- High throughput (megabytes per second)
- Minimal latency (response times)

Because each application has different requirements, you can meet these goals by selecting an appropriate RAID group on the storage array.

To achieve performance goals, follow these guidelines:

- Place each LUN on a RAID group that provides the necessary performance levels. Monitor the activities and resource use of other LUNs in the assigned RAID group. A high-performance RAID group that has too many applications doing I/O to it might not meet performance goals required by an application running on the ESX host.
- Ensure that each host has enough HBAs to increase throughput for the applications on the host for the peak period. I/O spread across multiple HBAs provides faster throughput and less latency for each application.
- To provide redundancy for a potential HBA failure, make sure that the host is connected to a dual redundant fabric.
- When allocating LUNs or RAID groups for ESX systems, remember that multiple operating systems use and share that resource. The LUN performance required by the ESX host might be much higher than when you use regular physical machines. For example, if you expect to run four I/O intensive applications, allocate four times the performance capacity for the ESX LUNs.
- When you use multiple ESX systems in with vCenter, the performance requirements for the storage subsystem increase correspondingly.
- The number of outstanding I/Os needed by applications running on the ESX system must match the number of I/Os the HBA and storage array can handle.

Using ESX with iSCSI SAN

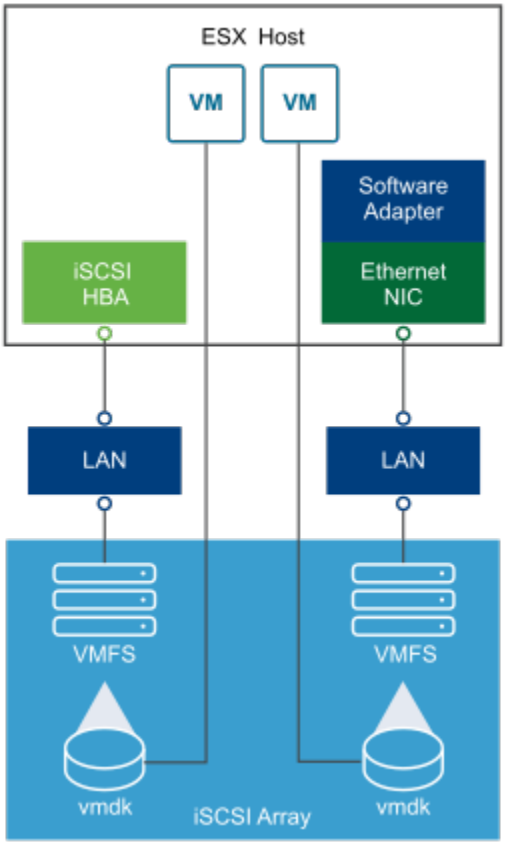
ESX can connect to external SAN storage using the Internet SCSI (iSCSI) protocol. In addition to traditional iSCSI, ESX also supports iSCSI Extensions for RDMA (iSER).

With the iSER protocol, the host can use the same iSCSI framework, but replaces the TCP/IP transport with the Remote Direct Memory Access (RDMA) transport.

About iSCSI SAN

iSCSI SAN uses Ethernet connections between hosts and high-performance storage subsystems.

On the host side, the iSCSI SAN components include iSCSI host bus adapters (HBAs) or Network Interface Cards (NICs). The iSCSI network also includes switches and routers that transport the storage traffic, cables, storage processors (SPs), and storage disk systems.



The iSCSI SAN uses a client-server architecture.

The client, called iSCSI initiator, operates on your ESX host. It initiates iSCSI sessions by issuing SCSI commands and transmitting them, encapsulated into the iSCSI protocol, to an iSCSI server. The server is known as an iSCSI target. Typically, the iSCSI target represents a physical storage system on the network.

The target can also be a virtual iSCSI SAN, for example, an iSCSI target emulator running in a virtual machine. The iSCSI target responds to the initiator's commands by transmitting required iSCSI data.

iSCSI Multipathing

When transferring data between the host server and storage, the SAN uses a technique known as multipathing. With multipathing, your ESX host can have more than one physical path to a LUN on a storage system.

Generally, a single path from a host to a LUN consists of an iSCSI adapter or NIC, switch ports, connecting cables, and the storage controller port. If any component of the path fails, the host selects another available path for I/O. The process of detecting a failed path and switching to another is called path failover.

For more information on multipathing, see Understanding Multipathing and Failover in the ESX Environment.

Nodes and Ports in the iSCSI SAN

A single discoverable entity on the iSCSI SAN, such as an initiator or a target, represents an iSCSI node.

Each node has a node name. ESX uses several methods to identify a node.

IP Address	Each iSCSI node can have an IP address associated with it so that routing and switching equipment on your network can establish the connection between the host and storage. This address is like the IP address that you assign to your computer to get access to your company's network or the Internet.
iSCSI Name	A worldwide unique name for identifying the node. iSCSI uses the iSCSI Qualified Name (IQN) and Extended Unique Identifier (EUI).

	By default, ESX generates unique iSCSI names for your iSCSI initiators, for example, `iqn.1998-01.com.vmware:iscsitestox-68158ef2`. Usually, you do not have to change the default value, but if you do, make sure that the new iSCSI name you enter is worldwide unique.
iSCSI Alias	A more manageable name for an iSCSI device or port used instead of the iSCSI name. iSCSI aliases are not unique and are intended to be a friendly name to associate with a port.

Each node has one or more ports that connect it to the SAN. iSCSI ports are end-points of an iSCSI session.

iSCSI Naming Conventions

iSCSI uses a special unique name to identify an iSCSI node, either target or initiator.

iSCSI names are formatted in two different ways. The most common is the IQN format.

For more details on iSCSI naming requirements and string profiles, see RFC 3721 and RFC 3722 on the IETF website.

iSCSI Qualified Name Format

The iSCSI Qualified Name (IQN) format takes the form `iqn.yyyy-mm.naming-authority:unique name`, where:

- *`yyyy-mm`* is the year and month when the naming authority was established.
- *`naming-authority`* is the reverse syntax of the Internet domain name of the naming authority. For example, the `iscsi.vmware.com` naming authority can have the iSCSI qualified name form of `iqn.1998-01.com.vmware.iscsi`. The name indicates that the `vmware.com` domain name was registered in January of 1998, and `iscsi` is a subdomain, maintained by `vmware.com`.
- *`unique name`* is any name you want to use, for example, the name of your host. The naming authority must make sure that any names assigned following the colon are unique, such as:
  - `iqn.1998-01.com.vmware.iscsi:name1`
  - `iqn.1998-01.com.vmware.iscsi:name2`
  - `iqn.1998-01.com.vmware.iscsi:name999`

Enterprise Unique Identifier Format

The Enterprise Unique Identifier (EUI) format takes the form `eui.16_hex_digits`.

For example, `eui.0123456789ABCDEF`.

The 16-hexadecimal digits are text representations of a 64-bit number of an IEEE EUI (extended unique identifier) format. The top 24 bits are a company ID that IEEE registers with a particular company. The remaining 40 bits are assigned by the entity holding that company ID and must be unique.

iSCSI Initiators

To access iSCSI targets, your ESX host uses iSCSI initiators.

The initiator is a software or hardware installed on your ESX host. The iSCSI initiator originates communication between your host and an external iSCSI storage system and sends data to the storage system.

In the ESX environment, iSCSI adapters configured on your host play the role of initiators. ESX supports several types of iSCSI adapters.

For information on configuring and using iSCSI adapters, see Configuring iSCSI and iSER Adapters and Storage with ESX.

Software iSCSI Adapter

A software iSCSI adapter is a VMware code built into the VMkernel. Using the software iSCSI adapter, your host can connect to the iSCSI storage device through standard network adapters. The software iSCSI adapter handles iSCSI processing while communicating with the network adapter. With the software iSCSI adapter, you can use iSCSI technology without purchasing specialized hardware.

Hardware iSCSI Adapter

A hardware iSCSI adapter is a third-party adapter that offloads iSCSI and network processing from your host. Hardware iSCSI adapters are divided into categories.

- `Dependent Hardware iSCSI Adapter`. Depends on VMware networking, and iSCSI configuration and management interfaces provided by VMware.
  This type of adapter can be a card that presents a standard network adapter and iSCSI offload functionality for the same port. The iSCSI offload functionality depends on the host's network configuration to obtain the IP, MAC, and other parameters used for iSCSI sessions. An example of a dependent adapter is the iSCSI licensed Broadcom 5709 NIC.

- `Independent Hardware iSCSI Adapter`. Implements its own networking and iSCSI configuration and management interfaces.

  Typically, an independent hardware iSCSI adapter is a card that either presents only iSCSI offload functionality or iSCSI offload functionality and standard NIC functionality. The iSCSI offload functionality has independent configuration management that assigns the IP, MAC, and other parameters used for the iSCSI sessions. An example of an independent adapter is the QLogic QLA4052 adapter.

Hardware iSCSI adapters might need to be licensed. Otherwise, they might not appear in the client. Contact your vendor for licensing information.

Using iSER Protocol with ESX

In addition to traditional iSCSI, ESX supports the iSCSI Extensions for RDMA (iSER) protocol. When the iSER protocol is enabled, the iSCSI framework on the ESX host can use the Remote Direct Memory Access (RDMA) transport instead of TCP/IP.

The traditional iSCSI protocol carries SCSI commands over a TCP/IP network between an iSCSI initiator on a host and an iSCSI target on a storage device. The iSCSI protocol encapsulates the commands and assembles that data in packets for the TCP/IP layer. When the data arrives, the iSCSI protocol disassembles the TCP/IP packets, so that the SCSI commands can be differentiated and delivered to the storage device.

iSER differs from traditional iSCSI as it replaces the TCP/IP data transfer model with the Remote Direct Memory Access (RDMA) transport. Using the direct data placement technology of the RDMA, the iSER protocol can transfer data directly between the memory buffers of the ESX host and storage devices. This method eliminates unnecessary TCP/IP processing and data coping, and can also reduce latency and the CPU load on the storage device.

In the iSER environment, iSCSI works exactly as before, but uses an underlying RDMA fabric interface instead of the TCP/IP-based interface.

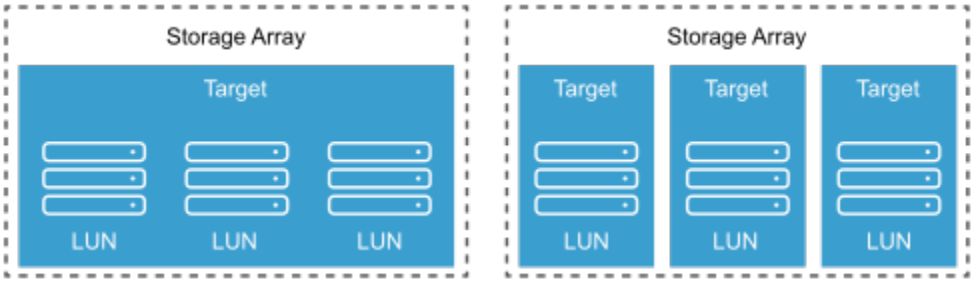
Because the iSER protocol preserves the compatibility with iSCSI infrastructure, the process of enabling iSER on the ESX host is similar to the iSCSI process. See Configure iSER with ESX.

Establishing iSCSI Connections

In the ESX context, the term target identifies a single storage unit that your host can access. The terms storage device and LUN describe a logical volume that represents storage space on a target. Typically, the terms device and LUN, in the ESX context, mean a SCSI volume presented to your host from a storage target and available for formatting.

Different iSCSI storage vendors present storage to hosts in different ways. Some vendors present multiple LUNs on a single target. Others present multiple targets with one LUN each.

Figure 72: Target Compared to LUN Representations



In these examples, three LUNs are available in each of these configurations. In the first case, the host detects one target but that target has three LUNs that can be used. Each of the LUNs represents individual storage volume. In the second case, the host detects three different targets, each having one LUN.

Host-based iSCSI initiators establish connections to each target. Storage systems with a single target containing multiple LUNs have traffic to all the LUNs on a single connection. With a system that has three targets with one LUN each, the host uses separate connections to the three LUNs.

This information is useful when you are trying to aggregate storage traffic on multiple connections from the host with multiple iSCSI adapters. You can set the traffic for one target to a particular adapter, and use a different adapter for the traffic to another target.

iSCSI Storage System Types

The types of ESX storage that your host supports include active-active, active-passive, and ALUA-compliant.

Active-active storage system	Supports access to the LUNs simultaneously through all the storage ports that are available without significant performance degradation. All the paths are always active, unless a path fails.
Active-passive storage system	A system in which one storage processor is actively providing access to a given LUN. The other processors act as a backup for the LUN and can be actively providing access to other LUN I/O. I/O can be successfully sent only to an active port for a given LUN. If access through the active storage port fails, one of the passive storage processors can be activated by the servers accessing it.
Asymmetrical storage system	Supports Asymmetric Logical Unit Access (ALUA). ALUA-compliant storage systems provide different levels of access per port. With ALUA, hosts can determine the states of target ports and prioritize paths. The host uses some of the active paths as primary and others as secondary.
Virtual port storage system	Supports access to all available LUNs through a single virtual port. Virtual port storage systems are active-active storage devices, but hide their multiple connections though a single port. ESX multipathing does not make multiple connections from a specific port to the storage by default. Some storage vendors supply session managers to establish and manage multiple connections to their storage. These storage systems handle port failovers and connection balancing transparently. This capability is often called transparent failover.

Discovery, Authentication, and Access Control

You can use several mechanisms to discover your storage and to limit access to it.

You must configure your host and the iSCSI storage system to support your storage access control policy.

Discovery	A discovery session is part of the iSCSI protocol. It returns the set of targets you can access on an iSCSI storage system. The two types of discovery available on ESX are dynamic and static. Dynamic discovery obtains a list of accessible targets from the iSCSI storage system. Static discovery can access only a particular target by target name and address. For more information, see Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host.
Authentication	iSCSI storage systems authenticate an initiator by a name and key pair. ESX supports the CHAP authentication protocol. To use CHAP authentication, the ESX host and the iSCSI storage system must have CHAP enabled and have common credentials. For information on enabling CHAP, see Configuring CHAP Parameters for iSCSI or iSER Storage Adapters on ESX Host.
Access Control	Access control is a policy set up on the iSCSI storage system. Most implementations support one or more of three types of access control: • By initiator name • By IP address • By the CHAP protocol Only initiators that meet all rules can access the iSCSI volume. Using only CHAP for access control can slow down rescans because the ESX host can discover all targets, but then fails at the authentication step. iSCSI rescans work faster if the host discovers only the targets it can authenticate.

How Virtual Machines Access Data on an iSCSI SAN

ESX stores a virtual machine's disk files within a VMFS datastore that resides on a SAN storage device. When virtual machine guest operating systems send SCSI commands to their virtual disks, the SCSI virtualization layer translates these commands to VMFS file operations.

When a virtual machine interacts with its virtual disk stored on a SAN, the following process takes place:

1. When the guest operating system in a virtual machine reads or writes to SCSI disk, it sends SCSI commands to the virtual disk.
2. Device drivers in the virtual machine’s operating system communicate with the virtual SCSI controllers.

3. The virtual SCSI controller forwards the commands to the VMkernel.
4. The VMkernel performs the following tasks.
   a. Locates an appropriate virtual disk file in the VMFS volume.
   b. Maps the requests for the blocks on the virtual disk to blocks on the appropriate physical device.
   c. Sends the modified I/O request from the device driver in the VMkernel to the iSCSI initiator, hardware or software.
5. If the iSCSI initiator is a hardware iSCSI adapter, independent or dependent, the adapter performs the following tasks.
   a. Encapsulates I/O requests into iSCSI Protocol Data Units (PDUs).
   b. Encapsulates iSCSI PDUs into TCP/IP packets.
   c. Sends IP packets over Ethernet to the iSCSI storage system.
6. If the iSCSI initiator is a software iSCSI adapter, the following takes place.
   a. The iSCSI initiator encapsulates I/O requests into iSCSI PDUs.
   b. The initiator sends iSCSI PDUs through TCP/IP connections.
   c. The VMkernel TCP/IP stack relays TCP/IP packets to a physical NIC.
   d. The physical NIC sends IP packets over Ethernet to the iSCSI storage system.
7. Ethernet switches and routers on the network carry the request to the appropriate storage device.

Error Correction

To protect the integrity of iSCSI headers and data, the iSCSI protocol offers error correction methods known as header digests and data digests.

Both methods are supported by the ESX host, and you can activate them. These methods are used to check the header and SCSI data transferred between iSCSI initiators and targets, in both directions.

Header and data digests check the noncryptographic data integrity beyond the integrity checks that other networking layers, such as TCP and Ethernet, provide. The digests check the entire communication path, including all elements that can change the network-level traffic, such as routers, switches, and proxies.

The existence and type of the digests are negotiated when an iSCSI connection is established. When the initiator and target agree on a digest configuration, this digest must be used for all traffic between them.

Enabling header and data digests requires additional processing for both the initiator and the target, and can affect throughput and CPU use performance.

> **Note:**
> Systems that use the Intel Nehalem processors offload the iSCSI digest calculations, as a result, reducing the impact on performance.

For information on enabling header and data digests, see Configuring Advanced Parameters for iSCSI on ESX Host.

Booting from iSCSI SAN with ESX

When you set up your ESX host to boot from a SAN, your host's boot image is stored on one or more LUNs in the SAN storage system. When the ESX host starts, it boots from the LUN on the SAN rather than from its local disk.

You can use boot from the SAN if you do not want to handle maintenance of local storage or have diskless hardware configurations, such as blade systems.

ESX supports different methods of booting from the iSCSI SAN.

Table 165: Boot from iSCSI SAN support

Independent Hardware iSCSI	Software iSCSI
Configure the iSCSI HBA to boot from the SAN. For information on configuring the HBA, see Configure Independent Hardware iSCSI Adapter for SAN Boot.	Use the software iSCSI adapter and a network adapter that supports the iSCSI Boot Firmware Table (iBFT) format. For information, see *VMware ESX Installation and Setup*.

General Recommendations for Boot from iSCSI SAN

If you plan to set up and use an iSCSI LUN as the boot device for your host, follow certain general guidelines.

The following guidelines apply to booting from the independent hardware iSCSI and iBFT.

- Review any vendor recommendations for the hardware you use in your boot configuration.
- For installation prerequisites and requirements, review *vSphere Installation and Setup*.
- Use static IP addresses to reduce the chances of DHCP conflicts.
- Use different LUNs for VMFS datastores and boot partitions.
- Configure proper ACLs on your storage system.
  - The boot LUN must be visible only to the host that uses the LUN. No other host on the SAN is permitted to see that boot LUN.
  - If a LUN is used for a VMFS datastore, multiple hosts can share the LUN.
- Configure a diagnostic partition.
  - With the independent hardware iSCSI only, you can place the diagnostic partition on the boot LUN. If you configure the diagnostic partition in the boot LUN, this LUN cannot be shared across multiple hosts. If a separate LUN is used for the diagnostic partition, multiple hosts can share the LUN.
  - If you boot from SAN using iBFT, you cannot set up a diagnostic partition on a SAN LUN. To collect your host's diagnostic information, use the vSphere ESX Dump Collector on a remote server. For information about the ESX Dump Collector, see *vCenter Deployment and Setup* and *vSphere Networking*.

Prepare the iSCSI SAN

Before you configure your host to boot from an iSCSI LUN, prepare and configure your storage area network.

> **CAUTION:**
> If you use scripted installation to install ESX when booting from a SAN, you must take special steps to avoid unintended data loss.

1. Connect network cables, referring to any cabling guide that applies to your setup.
2. Ensure IP connectivity between your storage system and server.

   Verify configuration of any routers or switches on your storage network. Storage systems must be able to ping the iSCSI adapters in your hosts.
3. Configure the storage system.
   a) Create a volume (or LUN) on the storage system for your host to boot from.
   b) Configure the storage system so that your host has access to the assigned LUN.

      This step might involve updating ACLs with the IP addresses, iSCSI names, and the CHAP authentication parameter you use on your host. On some storage systems, in addition to providing access information for the ESX host, you must also explicitly associate the assigned LUN with the host.
   c) Ensure that the LUN is presented to the host correctly.
   d) Ensure that no other system has access to the configured LUN.
   e) Record the iSCSI name and IP addresses of the targets assigned to the host.

      You must have this information to configure your iSCSI adapters.

Configure Independent Hardware iSCSI Adapter for SAN Boot

If your ESX host uses an independent hardware iSCSI adapter, such as QLogic HBA, you can configure the adapter to boot from the SAN.

This procedure discusses how to enable the QLogic iSCSI HBA to boot from the SAN. For more information and more up-to-date details about QLogic adapter configuration settings, see the QLogic website.

1. Start the installation media and reboot the host.
2. Use the BIOS to set the host to boot from the installation media first.
3. During server POST, press Crtl+q to enter the QLogic iSCSI HBA configuration menu.
4. Select the I/O port to configure.
   By default, the Adapter Boot mode is set to Disable.
5. Configure the HBA.
   a) From the **Fast!UTIL Options** menu, select **Configuration Settings** > **Host Adapter Settings**.
   b) Optional: Configure the following settings for your host adapter: initiator IP address, subnet mask, gateway, initiator iSCSI name, and CHAP.
6. Configure iSCSI settings.
   See Configure iSCSI Boot Settings.
7. Save your changes and restart the system.

Configure iSCSI Boot Settings

Configure iSCSI boot parameters, so that your ESX host can boot from an iSCSI LUN.

1. From the **Fast!UTIL Options** menu, select **Configuration Settings** > **iSCSI Boot Settings**.
2. Before you can set SendTargets, set Adapter Boot mode to **Manual**.
3. Select **Primary Boot Device Settings**.
   a) Enter the discovery **Target IP** and **Target Port**.
   b) Configure the **Boot LUN** and **iSCSI Name** parameters.
      - If only one iSCSI target and one LUN are available at the target address, leave **Boot LUN** and **iSCSI Name** blank.
        After your host reaches the target storage system, these text boxes are populated with appropriate information.
      - If more than one iSCSI target and LUN are available, supply values for **Boot LUN** and **iSCSI Name**.
   c) Save changes.
4. From the **iSCSI Boot Settings** menu, select the primary boot device.
   An auto rescan of the HBA finds new target LUNs.
5. Select the iSCSI target.
   If more than one LUN exists within the target, you can select a specific LUN ID by pressing **Enter** after you locate the iSCSI device.
6. Return to the **Primary Boot Device Setting** menu. After the rescan, **Boot LUN** and **iSCSI Name** are populated. Change the value of **Boot LUN** to the appropriate LUN ID.

Best Practices for iSCSI Storage with ESX

When you use ESX with iSCSI SAN, follow recommendations that VMware offers to avoid problems.

Check with your storage representative if your storage system supports Storage API - Array Integration hardware acceleration features. If it does, refer to your vendor documentation to enable hardware acceleration support on the storage system side. For more information, see Storage Hardware Acceleration in vSphere.

Preventing iSCSI SAN Problems

When using ESX with a SAN, you must follow specific guidelines to avoid SAN problems.

Observe the following tips:

- Place only one VMFS datastore on each LUN.

- Do not change the path policy the system sets for you unless you understand the implications of making such a change.
- Document everything. Include information about configuration, access control, storage, switch, server and iSCSI HBA configuration, software and firmware versions, and storage cable plan.
- Plan for failure:
  - Make several copies of your topology maps. For each element, consider what happens to your SAN if the element fails.
  - Cross off different links, switches, HBAs, and other elements to ensure that you did not miss a critical failure point in your design.
- Ensure that the iSCSI HBAs are installed in the correct slots in the ESX host, based on slot and bus speed. Balance PCI bus load among the available buses in the server.
- Become familiar with the various monitor points in your storage network, at all visibility points, including ESX performance charts, Ethernet switch statistics, and storage performance statistics.
- Change LUN IDs only when VMFS datastores deployed on the LUNs have no running virtual machines. If you change the ID, virtual machines running on the VMFS datastore might fail.

  After you change the ID of the LUN, you must rescan your storage to reset the ID on your host. For information on using the rescan, see Rescan Operations for ESX Storage.
- If you change the default iSCSI name of your iSCSI adapter, make sure that the name you enter is worldwide unique and properly formatted. To avoid storage access problems, never assign the same iSCSI name to different adapters, even on different hosts.

Optimizing iSCSI SAN Storage Performance

Several factors contribute to optimizing a typical SAN environment.

If the network environment is properly configured, the iSCSI components provide adequate throughput and low enough latency for iSCSI initiators and targets. If the network is congested and links, switches or routers are saturated, iSCSI performance suffers and might not be adequate for ESX environments.

Storage System Performance

Storage system performance is one of the major factors contributing to the performance of the entire iSCSI environment.

If issues occur with storage system performance, consult your storage system vendor’s documentation for any relevant information.

When you assign LUNs, remember that you can access each shared LUN through a number of hosts, and that a number of virtual machines can run on each host. One LUN used by the ESX host can service I/O from many different applications running on different operating systems. Because of this diverse workload, the RAID group that contains the ESX LUNs should not include LUNs that other hosts use that are not running ESX for I/O intensive applications.

Enable read caching and write caching.

Load balancing is the process of spreading server I/O requests across all available SPs and their associated host server paths. The goal is to optimize performance in terms of throughput (I/O per second, megabytes per second, or response times).

SAN storage systems require continual redesign and tuning to ensure that I/O is load balanced across all storage system paths. To meet this requirement, distribute the paths to the LUNs among all the SPs to provide optimal load balancing. Close monitoring indicates when it is necessary to manually rebalance the LUN distribution.

Tuning statically balanced storage systems is a matter of monitoring the specific performance statistics (such as I/O operations per second, blocks per second, and response time) and distributing the LUN workload to spread the workload across all the SPs.

Server Performance with iSCSI

To ensure optimal ESX host performance, consider several factors.

Each server application must have access to its designated storage with the following conditions:

- High I/O rate (number of I/O operations per second)
- High throughput (megabytes per second)
- Minimal latency (response times)

Because each application has different requirements, you can meet these goals by selecting an appropriate RAID group on the storage system.

To achieve performance goals, follow these guidelines:

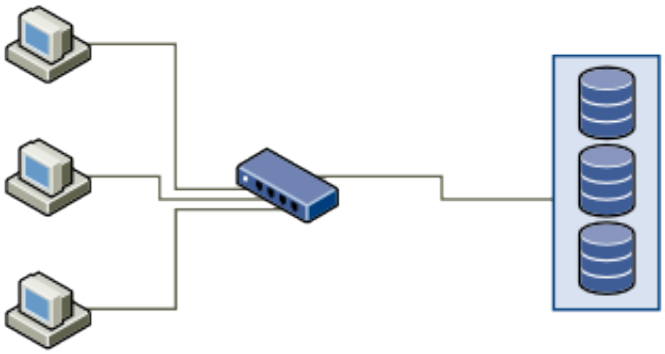
- Place each LUN on a RAID group that provides the necessary performance levels. Monitor the activities and resource use of other LUNS in the assigned RAID group. A high-performance RAID group that has too many applications doing I/O to it might not meet performance goals required by an application running on the ESX host.
- To achieve maximum throughput for all the applications on the host during the peak period, install enough network adapters or iSCSI hardware adapters. I/O spread across multiple ports provides faster throughput and less latency for each application.
- To provide redundancy for software iSCSI, make sure that the initiator is connected to all network adapters used for iSCSI connectivity.
- When allocating LUNs or RAID groups for ESX systems, remember that multiple operating systems use and share that resource. The LUN performance required by the ESX host might be much higher than when you use regular physical machines. For example, if you expect to run four I/O intensive applications, allocate four times the performance capacity for the ESX LUNs.
- When you use multiple ESX systems with vCenter, the storage performance requirements increase.
- The number of outstanding I/Os needed by applications running on an ESX system must match the number of I/Os the SAN can handle.

Network Performance

A typical SAN consists of a collection of computers connected to a collection of storage systems through a network of switches. Several computers often access the same storage.

The following graphic shows several computer systems connected to a storage system through an Ethernet switch. In this configuration, each system is connected through a single Ethernet link to the switch. The switch is connected to the storage system through a single Ethernet link.

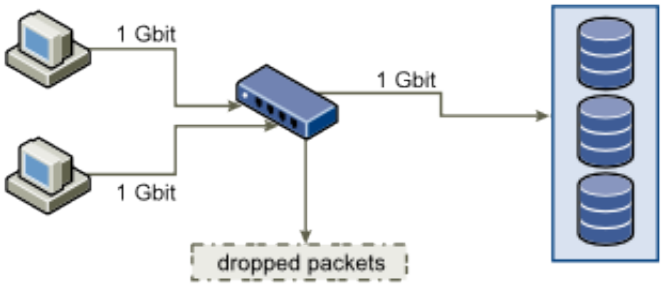
Figure 73: Single Ethernet Link Connection to Storage



When systems read data from storage, the storage responds with sending enough data to fill the link between the storage systems and the Ethernet switch. It is unlikely that any single system or virtual machine gets full use of the network speed. However, this situation can be expected when many systems share one storage device.

When writing data to storage, multiple systems or virtual machines might attempt to fill their links. As a result, the switch between the systems and the storage system might drop network packets. The data drop might occur because the switch has more traffic to send to the storage system than a single link can carry. The amount of data the switch can transmit is limited by the speed of the link between it and the storage system.

Figure 74: Dropped Packets



Recovering from dropped network packets results in large performance degradation. In addition to time spent determining that data was dropped, the retransmission uses network bandwidth that can otherwise be used for current transactions.

iSCSI traffic is carried on the network by the Transmission Control Protocol (TCP). TCP is a reliable transmission protocol that ensures that dropped packets are retried and eventually reach their destination. TCP is designed to recover from dropped packets and retransmits them quickly and seamlessly. However, when the switch discards packets with any regularity, network throughput suffers. The network becomes congested with requests to resend data and with the resent packets. Less data is transferred than in a network without congestion.

Most Ethernet switches can buffer, or store, data. This technique gives every device attempting to send data an equal chance to get to the destination. The ability to buffer some transmissions, combined with many systems limiting the number of outstanding commands, reduces transmissions to small bursts. The bursts from several systems can be sent to a storage system in turn.

If the transactions are large and multiple servers are sending data through a single switch port, an ability to buffer can be exceeded. In this case, the switch drops the data it cannot send, and the storage system must request a retransmission of the dropped packet. For example, if an Ethernet switch can buffer 32 KB, but the server sends 256 KB to the storage device, some of the data is dropped.

Most managed switches provide information on dropped packets, similar to the following:

```
*: interface is up
IHQ: pkts in input hold queue     IQD: pkts dropped from input queue
OHQ: pkts in output hold queue    OQD: pkts dropped from output queue
RXBS: rx rate (bits/sec)          RXPS: rx rate (pkts/sec)
TXBS: tx rate (bits/sec)          TXPS: tx rate (pkts/sec)
TRTL: throttle count
```

Table 166: Sample Switch Information

Interface	IHQ	IQD	OHQ	OQD	RXBS	RXPS	TXBS	TXPS	TRTL
* GigabitEthernet0/1	3	9922	0	0	476303000	62273	477840000	63677	0

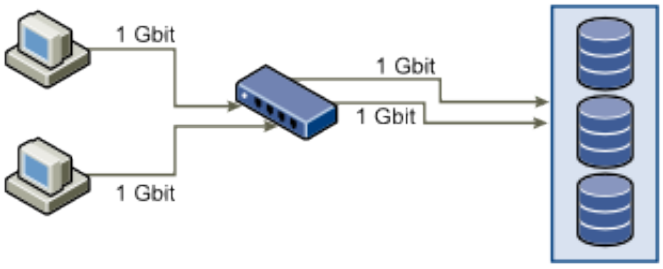
In this example from a Cisco switch, the bandwidth used is 476303000 bits/second, which is less than half of wire speed. The port is buffering incoming packets, but has dropped several packets. The final line of this interface summary indicates that this port has already dropped almost 10,000 inbound packets in the IQD column.

Configuration changes to avoid this problem involve making sure several input Ethernet links are not funneled into one output link, resulting in an oversubscribed link. When several links transmitting near capacity are switched to a smaller number of links, oversubscription becomes possible.

Generally, applications or systems that write much data to storage must avoid sharing Ethernet links to a storage device. These types of applications perform best with multiple connections to storage devices.

Multiple Connections from Switch to Storage shows multiple connections from the switch to the storage.

Figure 75: Multiple Connections from Switch to Storage



Using VLANs or VPNs does not provide a suitable solution to the problem of link oversubscription in shared configurations. VLANs and other virtual partitioning of a network provide a way of logically designing a network. However, they do not change the physical capabilities of links and trunks between switches. When storage traffic and other network traffic share physical connections, oversubscription and lost packets might become possible. The same is true of VLANs that share interswitch trunks. Performance design for a SAN must consider the physical limitations of the network, not logical allocations.

Checking Ethernet Switch Statistics

Many Ethernet switches provide different methods for monitoring switch health.

Switches that have ports operating near maximum throughput much of the time do not provide optimum performance. If you have ports in your iSCSI SAN running near the maximum, reduce the load. If the port is connected to an ESX system or iSCSI storage, you can reduce the load by using manual load balancing.

If the port is connected between multiple switches or routers, consider installing additional links between these components to handle more load. Ethernet switches also commonly provide information about transmission errors, queued packets, and dropped Ethernet packets. If the switch regularly reports any of these conditions on ports being used for iSCSI traffic, performance of the iSCSI SAN will be poor.

Configuring iSCSI and iSER Adapters and Storage with ESX

Before ESX can work with iSCSI SAN, you must set up your iSCSI environment. To work properly, your environment must follow specific recommendations. In addition, several restrictions exist when you use ESX with iSCSI SAN.

iSCSI Storage Recommendations

- Verify that your ESX host supports the iSCSI SAN storage hardware and firmware. For an up-to-date list, see the VMware by Broadcom Compatibility Guide.
- To ensure that the host recognizes LUNs at startup time, configure all iSCSI storage targets so that your host can access them and use them. Configure your host so that it can discover all available iSCSI targets.
- Unless you are using diskless servers, set up a diagnostic partition on local storage. If you have diskless servers that boot from iSCSI SAN, see General Recommendations for Boot from iSCSI SAN for information about diagnostic partitions with iSCSI.
- Set the SCSI controller driver in the guest operating system to a large enough queue.
- VMtools increases the default disk timeout SCSI `TimeoutValue` parameter to 180 seconds. On virtual machines without VMtools, increase the value of the SCSI `TimeoutValue` parameter. With this parameter set up, the VMs can better tolerate delayed I/O resulting from a path failover. For information, see Path Failover and Virtual Machines.
- Configure your environment to have only one VMFS datastore for each LUN.

iSCSI Storage Restrictions

- ESX does not support iSCSI-connected tape devices.
- You cannot use virtual-machine multipathing software to perform I/O load balancing to a single physical LUN.
- ESX does not support multipathing when you combine independent hardware adapters with either software or dependent hardware adapters.

Setting Up iSCSI ESX Environment

The process of preparing your iSCSI environment involves the following steps.

Step	Details
1. Set up iSCSI storage	For information, see your storage vendor documentation. In addition, follow recommendations in Best Practices for iSCSI Storage with ESX.
2. Configure iSCSI/iSER adapters	Use an appropriate workflow to configure your adapter: • Set Up Independent Hardware iSCSI Adapters with ESX • Configure Dependent Hardware iSCSI Adapters with ESX • Configure the Software iSCSI Adapter with ESX • Configure iSER with ESX
3. Create a datastore on iSCSI storage	Creating vSphere Datastores

iSCSI Networking

For certain types of iSCSI adapters, you must configure VMkernel networking.

You can verify the network configuration by using the `vmkping` utility.

The independent hardware iSCSI adapter does not require VMkernel networking. You can configure network parameters, such as an IP address, subnet mask, and default gateway on the independent hardware iSCSI adapter.

All types of iSCSI adapters support IPv4 and IPv6 protocols.

iSCSI Adapter (vmhba)	Description	VMkernel Networking	Adapter Network Settings
Independent Hardware iSCSI Adapter	Third-party adapter that offloads the iSCSI and network processing and management from your host.	Not required.	For information, see Edit Network Settings for Hardware iSCSI.
Software iSCSI Adapter	Uses standard NICs to connect your host to a remote iSCSI target on the IP network.	Required. For information, see Setting Up Network for iSCSI and iSER with ESX.	N/A
Dependent Hardware iSCSI Adapter	Third-party adapter that depends on VMware networking and iSCSI configuration and management interfaces.	Required For information, see Setting Up Network for iSCSI and iSER with ESX.	N/A
VMware iSER Adapter	Uses an RDMA-capable network adapter to connect your host to a remote iSCSI target.	Required For information, see Setting Up Network for iSCSI and iSER with ESX.	N/A

Discovery Methods

For all types of iSCSI adapters, you must set the dynamic discovery address or static discovery address. In addition, you must provide a target name of the storage system. For software iSCSI and dependent hardware iSCSI, the address is pingable using `vmkping`.

See Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host.

CHAP Authentication

Enable the CHAP parameter on the initiator and the storage system side. After authentication is enabled, it applies to all targets that are not yet discovered. It does not apply to targets that are already discovered.

See Configuring CHAP Parameters for iSCSI or iSER Storage Adapters on ESX Host.

Set Up Independent Hardware iSCSI Adapters with ESX

An independent hardware iSCSI adapter is a specialized third-party adapter capable of accessing iSCSI storage over TCP/IP. This iSCSI adapter handles all iSCSI and network processing and management for your ESX system.

- Verify whether the adapter must be licensed.
- Install the adapter on your ESX host.

For information about licensing, installation, and firmware updates, see vendor documentation.

The process of setting up the independent hardware iSCSI adapter includes these steps.

Step	Description
View Independent Hardware iSCSI Adapters	View an independent hardware iSCSI adapter and verify that it is correctly installed and ready for configuration.
Modify General Properties for iSCSI or iSER Adapters on ESX Hosts	If needed, change the default iSCSI name and alias assigned to your iSCSI adapters. For the independent hardware iSCSI adapters, you can also change the default IP settings.
Edit Network Settings for Hardware iSCSI	Change default network settings so that the adapter is configured properly for the iSCSI SAN.
Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host	Set up dynamic discovery. With dynamic discovery, each time the initiator contacts a specified iSCSI storage system, it sends the SendTargets request to the system. The iSCSI system responds by supplying a list of available targets to the initiator. In addition to the dynamic discovery method, you can use static discovery and manually enter information for the targets.
Set Up CHAP for iSCSI or iSER Storage Adapter	If your iSCSI environment uses the Challenge Handshake Authentication Protocol (CHAP), configure it for your adapter.
Enable Jumbo Frames for Independent Hardware iSCSI	If your iSCSI environment supports Jumbo Frames, enable them for the adapter.

View Independent Hardware iSCSI Adapters

On the ESX host, view an independent hardware iSCSI adapter and verify that it is correctly installed and ready for configuration.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

After you install an independent hardware iSCSI adapter on the host, it appears on the list of storage adapters available for configuration. You can view its properties.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**.
   If installed, the hardware iSCSI adapter appears on the list of storage adapters.
4. Select the adapter to view.
   The default details for the adapter appear.

Adapter Information	Description
Model	Model of the adapter.
iSCSI Name	Unique name formed according to iSCSI standards that identifies the iSCSI adapter. You can edit the iSCSI name.

Adapter Information	Description
iSCSI Alias	A friendly name used instead of the iSCSI name. You can edit the iSCSI alias.
IP Address	Address assigned to the iSCSI HBA.
Targets	Number of targets accessed through the adapter.
Devices	All storage devices or LUNs the adapter can access.
Paths	All paths the adapter uses to access storage devices.

Edit Network Settings for Hardware iSCSI

After you install an independent hardware iSCSI adapter on an ESX host, you might need to change its default network settings so that the adapter is configured properly for the iSCSI SAN.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
4. Click the **Network Settings** tab and click **Edit**.
5. In the IPv4 settings section, deactivate IPv4 or select the method to obtain IP addresses.

   Note:
   The automatic DHCP option and static option are mutually exclusive.

Option	Description
No IPv4 settings	Deactivate IPv4.
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings.
Use static IPv4 settings	Enter the IPv4 IP address, subnet mask, and default gateway for the iSCSI adapter.

6. In the IPv6 settings section, deactivate IPv6 or select an appropriate option for obtaining IPv6 addresses.

   Note:
   Automatic options and the static option are mutually exclusive.

Option	Description
No IPv6 settings	Deactivate IPv6.
Enable IPv6	Select an option for obtaining IPv6 addresses.
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses.
Obtain IPv6 addresses automatically through Router Advertisement	Use router advertisement to obtain IPv6 addresses.
Override Link-local address for IPv6	Override the link-local IP address by configuring a static IP address.
Static IPv6 addresses	1. Click **Add** to add a new IPv6 address. 2. Enter the IPv6 address and subnet prefix length, and click **OK**.

7. In the DNS settings section, provide IP addresses for a preferred DNS server and an alternate DNS server.
   You must provide both values.

Configure Dependent Hardware iSCSI Adapters with ESX

A dependent hardware iSCSI adapter in your ESX system is a third-party adapter that depends on VMware networking, and iSCSI configuration and management interfaces provided by VMware.

When installed on a host, it presents its two components, a standard network adapter and an iSCSI engine, to the same port. The iSCSI engine appears on the list of storage adapters as an iSCSI adapter (vmhba).

The iSCSI adapter is enabled by default. To make it functional, you must connect it, through a virtual VMkernel adapter (vmk), to a physical network adapter (vmnic) associated with it. You can then configure the iSCSI adapter.

After you configure the dependent hardware iSCSI adapter, the discovery and authentication data is passed through the network connection. The iSCSI traffic goes through the iSCSI engine, bypassing the network.

The entire setup and configuration process for the dependent hardware iSCSI adapters involves several steps.

Step	Description
View Dependent Hardware iSCSI Adapters	View a dependent hardware iSCSI adapter to verify that it is correctly loaded.
Modify General Properties for iSCSI or iSER Adapters on ESX Hosts	If needed, change the default iSCSI name and alias assigned to your adapter.
Determine Association Between iSCSI and Network Adapters	You must create network connections to bind dependent iSCSI and physical network adapters. To create the connections correctly, determine the name of the physical NIC with which the dependent hardware iSCSI adapter is associated.
Configure Port Binding for iSCSI or iSER on ESX	Configure connections for the traffic between the iSCSI component and the physical network adapters. The process of configuring these connections is called port binding.
Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host	Set up dynamic discovery. With dynamic discovery, each time the initiator contacts a specified iSCSI storage system, it sends the SendTargets request to the system. The iSCSI system responds by supplying a list of available targets to the initiator. In addition to the dynamic discovery method, you can use static discovery and manually enter information for the targets.
Set Up CHAP for iSCSI or iSER Storage Adapter	If your iSCSI environment uses the Challenge Handshake Authentication Protocol (CHAP), configure it for your adapter.
Set Up CHAP for Target	You can also configure different CHAP credentials for each discovery address or static target.
Enable Jumbo Frames for Networking	If your iSCSI environment supports Jumbo Frames, enable them for the adapter.

Dependent Hardware iSCSI Considerations

When you use dependent hardware iSCSI adapters withESX, certain considerations apply.

- When you use any dependent hardware iSCSI adapter, performance reporting for a NIC associated with the adapter might show little or no activity, even when iSCSI traffic is heavy. This behavior occurs because the iSCSI traffic bypasses the regular networking stack.
- If you use a third-party virtual switch deactivate automatic pinning. Use manual pinning instead, making sure to connect a VMkernel adapter (vmk) to an appropriate physical NIC (vmnic). For information, refer to your virtual switch vendor documentation.
- The Broadcom iSCSI adapter performs data reassembly in hardware, which has a limited buffer space. When you use the Broadcom iSCSI adapter in a congested network or under heavy load, activate flow control to avoid performance degradation.

  Flow control manages the rate of data transmission between two nodes to prevent a fast sender from overrunning a slow receiver. For best results, activate flow control at the end points of the I/O path, at the hosts and iSCSI storage systems.

  To activate flow control for the host, use the `esxcli system module parameters` command.
- Dependent hardware adapters support IPv4 and IPv6.

View Dependent Hardware iSCSI Adapters

On an ESX host, view a dependent hardware iSCSI adapter to verify that it is correctly loaded.

If installed, the dependent hardware iSCSI adapter (vmhba#) appears on the list of storage adapters under such category as, for example, Broadcom iSCSI Adapter. If the dependent hardware adapter does not appear on the list of storage adapters, check whether it needs to be licensed. See your vendor documentation.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**.
4. Select the adapter (vmhba#) to view.

   The default details for the adapter appear, including the iSCSI name, iSCSI alias, and the status.

Although the dependent iSCSI adapter is enabled by default, to make it functional, you must set up networking for the iSCSI traffic and bind the adapter to the appropriate VMkernel iSCSI port. You then configure discovery addresses and CHAP parameters.

Determine Association Between iSCSI and Network Adapters

On an ESX host, network connections bind dependent iSCSI and physical network adapters. To create the connections correctly, you must determine the name of the physical NIC with which the dependent hardware iSCSI adapter is associated.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**.
4. Select the iSCSI adapter (vmhba#) and click the **Network Port Binding** tab under adapter details.
5. Click **Add**.

   The network adapter (vmnic#) that corresponds to the dependent iSCSI adapter is listed in the Physical Network Adapter column.

If the VMkernel Adapter column is empty, create a VMkernel adapter (vmk#) for the physical network adapter (vmnic#) and then bind them to the associated dependent hardware iSCSI. See Setting Up Network for iSCSI and iSER with ESX.

Configure the Software iSCSI Adapter with ESX

With the software-based iSCSI implementation, you can use standard NICs to connect your host to a remote iSCSI target on the IP network. The software iSCSI adapter that is built into ESX facilitates this connection by communicating with the physical NICs through the network stack.

When you use the software iSCSI adapters, consider the following:

- Designate a separate network adapter for iSCSI. Do not use iSCSI on 100 Mbps or slower adapters.
- Avoid hard coding the name of the software adapter, vmhbaXX, in the scripts. It is possible for the name to change from one ESX release to another. The change might cause failures of your existing scripts if they use the hardcoded old name. The name change does not affect the behavior of the iSCSI software adapter.

The process of configuring the software iSCSI adapter involves several steps.

Step	Description
Add or Remove the Software iSCSI Adapter	Activate your software iSCSI adapter so that your host can use it to access iSCSI storage.
Modify General Properties for iSCSI or iSER Adapters on ESX Hosts	If needed, change the default iSCSI name and alias assigned to your adapter.

Step	Description
Configure Port Binding for iSCSI or iSER on ESX	Configure connections for the traffic between the iSCSI component and the physical network adapters. The process of configuring these connections is called port binding.
Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host	Set up dynamic discovery. With dynamic discovery, each time the initiator contacts a specified iSCSI storage system, it sends the SendTargets request to the system. The iSCSI system responds by supplying a list of available targets to the initiator. In addition to the dynamic discovery method, you can use static discovery and manually enter information for the targets.
Set Up CHAP for iSCSI or iSER Storage Adapter	If your iSCSI environment uses the Challenge Handshake Authentication Protocol (CHAP), configure it for your adapter.
Set Up CHAP for Target	You can also configure different CHAP credentials for each discovery address or static target.
Enable Jumbo Frames for Networking	If your iSCSI environment supports Jumbo Frames, enable them for the adapter.

Add or Remove the Software iSCSI Adapter

You must activate your software iSCSI adapter so that your ESX host can use it to access iSCSI storage. If you do not need the software iSCSI adapter after activation, you can remove it.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

If you plan to remove the software iSCSI adapter, perform these steps:

- Migrate all VMs from the iSCSI datastore.
- Unmount the datastore. See Unmount Datastores.
- Detach the iSCSI device and perform the adapter rescan. See Performing Planned Storage Device Removal.

Note:
If you boot from iSCSI using the software iSCSI adapter, the adapter is added and the network configuration is created at the first boot. If you remove the adapter, it is added again each time you boot the host.

You can activate only one software iSCSI adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Add or remove the adapter.

Option	Description
Add the software iSCSI adapter	1. Under **Storage**, click **Storage Adapters**, and click **Add Software Adapter**. 2. From the drop-down menu, select **Software iSCSI Adapter** and confirm that you want to add the adapter. The software iSCSI adapter (vmhba#) is added and appears on the list of storage adapters. After adding the adapter, the host assigns the default iSCSI name to it. You can now complete the adapter configuration.
Remove the software iSCSI adapter	1. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to remove. 2. Click the **Properties** tab. 3. From the **Adapter Status Actions** menu, select **Disable** and confirm that you want to deactivate the adapter. The adapter will be marked for removal. The adapter will be removed from the host on next host reboot. 4. Reboot the host. After the reboot, the adapter no longer appears on the list of storage adapters. All virtual machines and other data on the storage devices associated with this adapter become inaccessible. You can later activate the adapter.

Configure iSER with ESX

In addition to traditional iSCSI, ESX supports the iSCSI Extensions for RDMA (iSER) protocol. When the iSER protocol is enabled, the iSCSI framework on the ESX host can use the Remote Direct Memory Access (RDMA) transport instead of TCP/IP. You can configure iSER on your ESX host.

For more information about the iSER protocol, see Using iSER Protocol with ESX.

The entire setup and configuration process for VMware iSER involves several steps.

Step	Description
Install and View an RDMA Capable Network Adapter	To configure iSER with ESX, you must first install an RDMA capable network adapter, for example, Mellanox Technologies MT27700 Family ConnectX-4. After you install this type of adapter, the vSphere Client displays its two components, an RDMA adapter and a physical network adapter `vmnic#`.
Enable the VMware iSER Adapter	To be able to use the RDMA capable adapter for iSCSI, use the `esxcli` to enable the VMware iSER storage component. The component appears in the vSphere Client as a `vmhba#` storage adapter under the VMware iSCSI over RDMA (iSER) Adapter category.
Modify General Properties for iSCSI or iSER Adapters on ESX Hosts	If needed, change the default name and alias assigned to the iSER storage adapter `vmhba#`.
Configure Port Binding for iSCSI or iSER on ESX	You must create network connections to bind the iSER storage adapter `vmhba#` and the RDMA capable network adapter `vmnic#`. The process of configuring these connections is called port binding. **Note:** iSER does not support NIC teaming. When configuring port binding, use only one RDMA adapter per vSwitch.
Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host	Set up the dynamic or static discovery for your iSER storage adapter `vmhba#`. With the dynamic discovery, each time the initiator contacts a specified iSER storage system, it sends the SendTargets request to the system. The iSER system responds by supplying a list of available targets to the initiator. With the static discovery, you manually enter information for the targets.
Set Up CHAP for iSCSI or iSER Storage Adapter	If your environment uses the Challenge Handshake Authentication Protocol (CHAP), configure it for your iSER storage adapter `vmhba#`.
Set Up CHAP for Target	You can also configure different CHAP credentials for each discovery address or static target.
Enable Jumbo Frames for Networking	If your environment supports Jumbo Frames, enable them for the iSER storage adapter `vmhba#`.

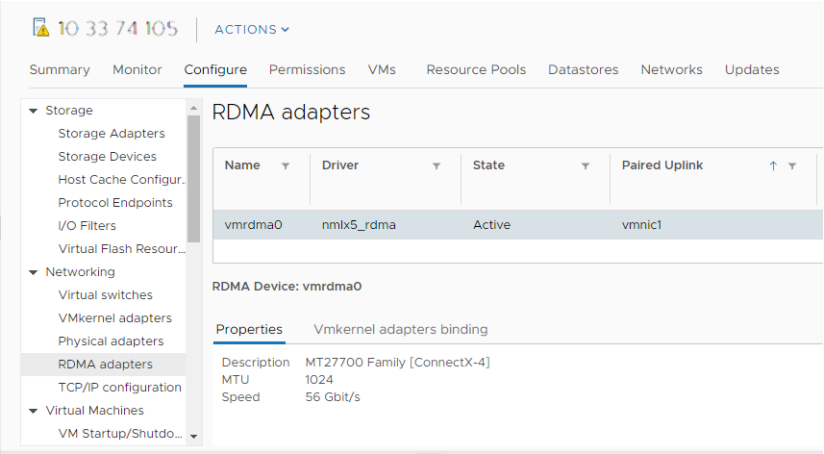
Install and View an RDMA Capable Network Adapter

ESX supports RDMA capable network adapters, for example, Mellanox Technologies MT27700 Family ConnectX-4. After you install such adapter on your host, the vSphere Client displays its two components, an RDMA adapter and a physical network adapter.

You can use the vSphere Client to view the RDMA adapter and its corresponding network adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Under **Networking**, click **RDMA adapters**.

   In this example, the RDMA adapter appears on the list as `vmrdma0`. The **Paired Uplink** column displays the network component as the `vmnic1` physical network adapter.



3. To verify the description of the adapter, select the RDMA adapter from the list, and click the **Properties** tab.

You can use the `vmnic#` network component of the adapter for such storage configurations as iSER or NVMe over RDMA. For the iSER configuration steps, see Configure iSER with ESX. For information about NVMe over RDMA, see Configuring NVMe over RDMA (RoCE v2) on ESX.

Enable the VMware iSER Adapter

To be able to use the RDMA capable adapter for iSCSI, use the `esxcli` to enable the VMware iSER storage component. After you enable the component, it appears in the vSphere Client as a `vmhba#` storage adapter under the VMware iSCSI over RDMA (iSER) Adapter category.

- Make sure that your iSCSI storage supports the iSER protocol.
- Install the RDMA capable adapter on your ESX host. For information, see Install and View an RDMA Capable Network Adapter.
- For RDMA capable adapters that support RDMA over Converged Ethernet (RoCE), determine the RoCE version that the adapter uses.
- Use the RDMA capable switch.
- Enable flow control on the ESX host. To enable flow control for the host, use the `esxcli system module parameters` command.
- Make sure to configure RDMA switch ports to create lossless connections between the iSER initiator and target.

1. Use ESXCLI to enable the VMware iSER storage adapter and set its RoCE version.
   a) Enable the iSER storage adapter.

      ```
      esxcli rdma iser add
      ```

   b) Verify that the iSER adapter has been added.

      ```
      esxcli iscsi adapter list
      ```

      The output is similar to the following.

      ```
      Adapter Driver State UID Description
      ------- ------ ------- ------------- -----------------------------------
      vmhba64 iser unbound iscsi.vmhba64 VMware iSCSI over RDMA (iSER) Adapter
      ```

   c) Specify the RoCE version that iSER uses to connect to the target.

      Use the RoCE version of the RDMA capable adapter. The command you enter is similar to the following:

      ```
      esxcli rdma iser params set -a vmhba64 -r 1
      ```

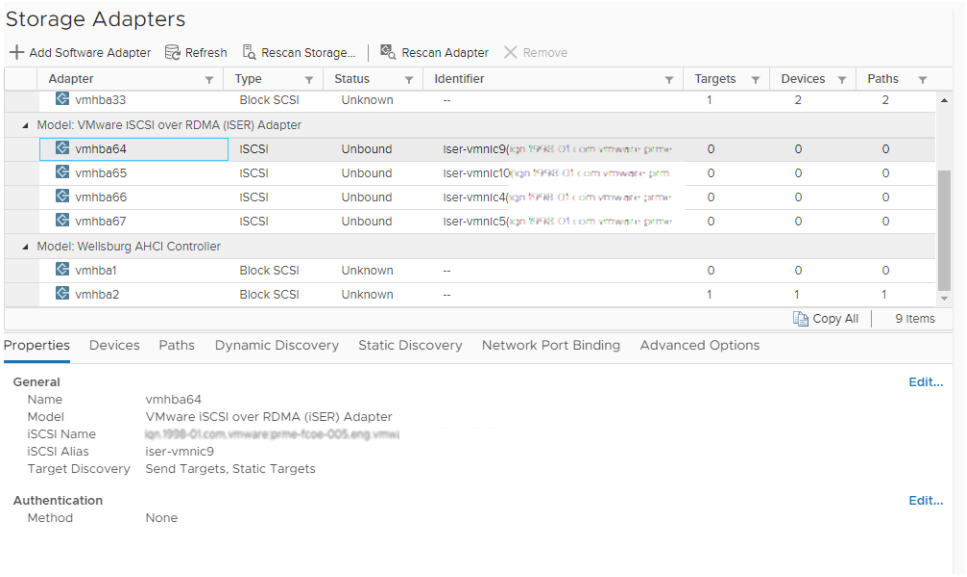
When the command completes, a message similar to the following appears in the VMkernel log.

```
vmkernel.0:2020-02-18T18:26:15.949Z cpu6:2100717 opID=45abe37e)iser: iser_set_roce: Setting roce type:
 1 for vmhba: vmhba64
vmkernel.0:2020-02-18T18:26:15.949Z cpu6:2100717 opID=45abe37e)iser: iser_set_roce: Setting rdma port:
 3260 for vmhba: vmhba64
```

If you do not specify the RoCE version, the host defaults to the highest RoCE version the RDMA capable adapter supports.

2. Use the vSphere Client to display the iSER adapter.
   a) In the vSphere Client, navigate to the ESX host.
   b) Click the **Configure** tab.
   c) Under **Storage**, click **Storage Adapters**, and review the list of adapters.

   If you enabled the adapter, it appears as a storage `vmhba#` adapter on the list under the VMware iSCSI over RDMA (iSER) Adapter category.



3. Select the iSER storage `vmhba#` to review its properties or perform the following tasks.

Option	Description
Configure port binding for the iSER storage adapter	You must create network connections to bind the iSER storage adapter `vmhba#` and the RDMA capable network adapter `vmnic#`. The process of configuring these connections is called port binding. For general information about port binding, see Setting Up Network for iSCSI and iSER with ESX. To configure port binding for iSER, see Configure Port Binding for iSCSI or iSER on ESX.
Set up dynamic or static discovery for the iSER storage adapter	See Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host.
Configure the Challenge Handshake Authentication Protocol (CHAP) for the iSER storage adapter	See Set Up CHAP for iSCSI or iSER Storage Adapter.

Modify General Properties for iSCSI or iSER Adapters on ESX Hosts

You can change default name and alias assigned to your iSCSI or iSER storage adapters by the ESX host. For the independent hardware iSCSI adapters, you can also change the default IP settings.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

Important:
When you modify any default properties for your adapters, make sure to use correct formats for their names and IP addresses.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
4. Click the **Properties** tab, and click **Edit** in the General panel.
5. Optional: Modify the following general properties.

Option	Description
iSCSI Name	Unique name formed according to iSCSI standards that identifies the iSCSI adapter. If you change the name, make sure that the name you enter is worldwide unique and properly formatted. Otherwise, certain storage devices might not recognize the iSCSI adapter.
iSCSI Alias	A friendly name you use instead of the iSCSI name.

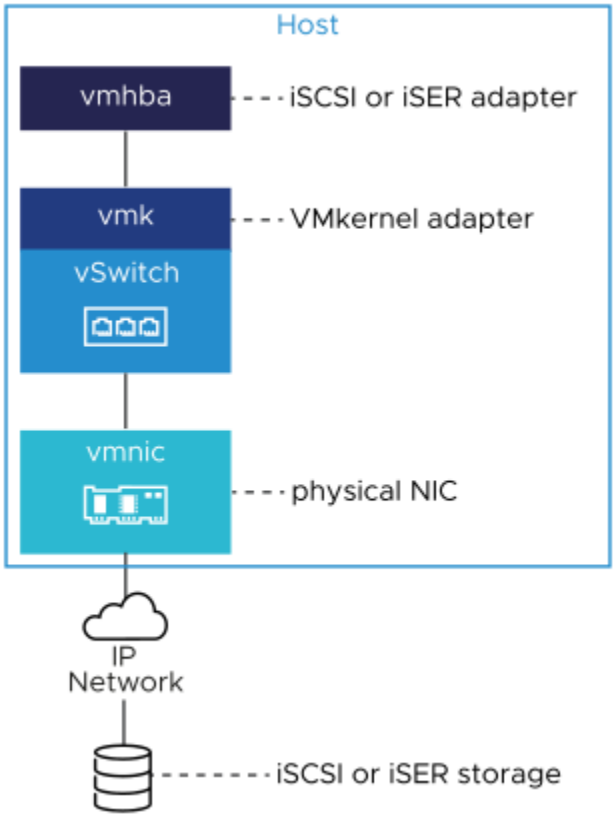
If you change the iSCSI name, it is used for new iSCSI sessions. For existing sessions, the new settings are not used until you log out and log in again.
For other configuration steps you can perform for the iSCSI or iSER storage adapters, see the following topics:

- Set Up Independent Hardware iSCSI Adapters with ESX
- Configure Dependent Hardware iSCSI Adapters with ESX
- Configure the Software iSCSI Adapter with ESX
- Configure iSER with ESX

Setting Up Network for iSCSI and iSER with ESX

Certain types of iSCSI adapters depend on the VMkernel networking. These adapters include the software or dependent hardware iSCSI adapters, and the VMware iSCSI over RDMA (iSER) adapter. If your ESX environment includes any of these adapters, you must configure connections for the traffic between the iSCSI or iSER component and the physical network adapters.

Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter. You use 1:1 mapping between each virtual and physical network adapter. You then associate the VMkernel adapter with an appropriate iSCSI or iSER adapter. This process is called port binding.



Follow these rules when configuring the port binding:

- You can connect the software iSCSI adapter with any physical NICs available on your host.
- The dependent iSCSI adapters must be connected only to their own physical NICs.
- You must connect the iSER adapter only to the RDMA-capable network adapter.

For specific considerations on when and how to use network connections with software iSCSI, see Broadcom knowledge base article 317719.

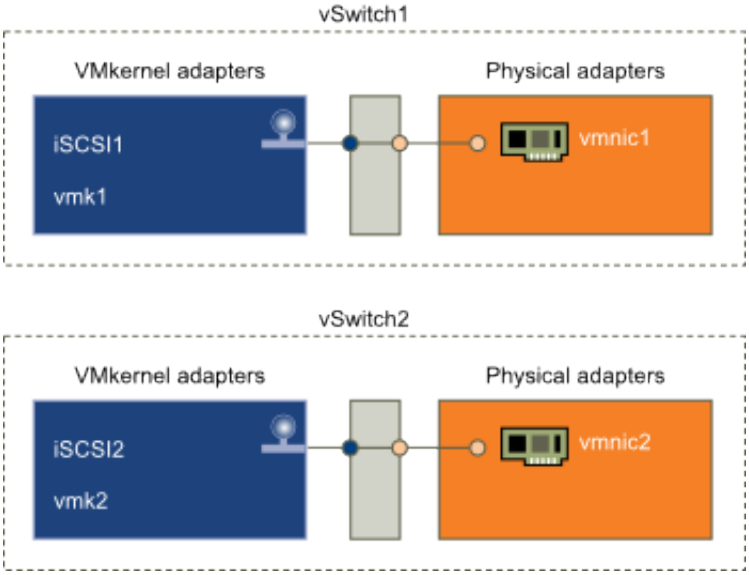
Multiple Network Adapters in iSCSI or iSER Configuration

If your host has more than one physical network adapter for iSCSI or iSER, you can use the adapters for multipathing.

You can use multiple physical adapters in a single or multiple switch configurations.

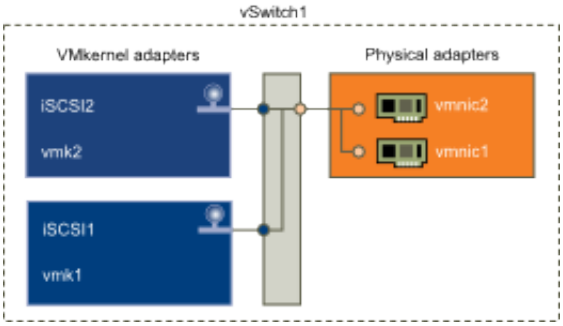
In the multiple switch configuration, you designate a separate vSphere switch for each virtual-to-physical adapter pair.

Figure 76: 1:1 Adapter Mapping on Separate vSphere Standard Switches



An alternative is to add all NICs and VMkernel adapters to the single vSphere switch. The number of VMkernel adapters must correspond to the number of physical adapters on the vSphere Standard switch. The single switch configuration is not appropriate for iSER because iSER does not support NIC teaming.

Figure 77: 1:1 Adapter Mapping on a Single vSphere Standard Switch



For that type of configuration, you must override the default network setup and make sure that each VMkernel adapter maps to only one corresponding active physical adapter, as the table indicates.

VMkernel Adapter (vmk#)	Physical Network Adapter (vmnic#)
vmk1 (iSCSI1)	**Active Adapters** vmnic1 **Unused Adapters** vmnic2
vmk2 (iSCSI2)	**Active Adapters** vmnic2 **Unused Adapters** vmnic1

You can also use distributed switches. For more information about vSphere distributed switches and how to change the default network policy, see the *vSphere Networking* documentation.

The following considerations apply when you use multiple physical adapters:

- Physical network adapters must be on the same subnet as the storage system they connect to.

- (Applies only to iSCSI and not to iSER) If you use separate vSphere switches, you must connect them to different IP subnets. Otherwise, VMkernel adapters might experience connectivity problems and the host fails to discover the LUNs.
- The single switch configuration is not appropriate for iSER because iSER does not support NIC teaming.

Do not use port binding when any of the following conditions exist:

- Array target iSCSI ports are in a different broadcast domain and IP subnet.
- VMkernel adapters used for iSCSI connectivity exist in different broadcast domains, IP subnets, or use different virtual switches.

Note:
In iSER configurations, the VMkernel adapters used for iSER connectivity cannot be used for converged traffic. The VMkernel adapters that you created to enable connectivity between the ESX host with iSER and the iSER target must be used only for iSER traffic.

Best Practices for Configuring Networking with Software iSCSI

When you configure networking with software iSCSI, consider several best practices.

Software iSCSI Port Binding

You can bind the software iSCSI initiator on the ESX host to a single or multiple VMkernel ports, so that iSCSI traffic flows only through the bound ports. Unbound ports are not used for iSCSI traffic.

When port binding is configured, the iSCSI initiator creates iSCSI sessions from all bound ports to all configured target portals.

See the following examples.

VMkernel Ports	Target Portals	iSCSI Sessions
2 bound VMkernel ports	2 target portals	4 sessions (2 x 2)
4 bound VMkernel ports	1 target portal	4 sessions (4 x 1)
2 bound VMkernel ports	4 target portals	8 sessions (2 x 4)

Note:
Make sure that all target portals are reachable from all VMkernel ports when port binding is used. Otherwise, iSCSI sessions might fail to create. As a result, the rescan operation might take longer than expected.

No Port Binding

If you do not use port binding, the ESX networking layer selects the best VMkernel port based on its routing table. The host uses the port to create an iSCSI session with the target portal. Without the port binding, only one session per each target portal is created.

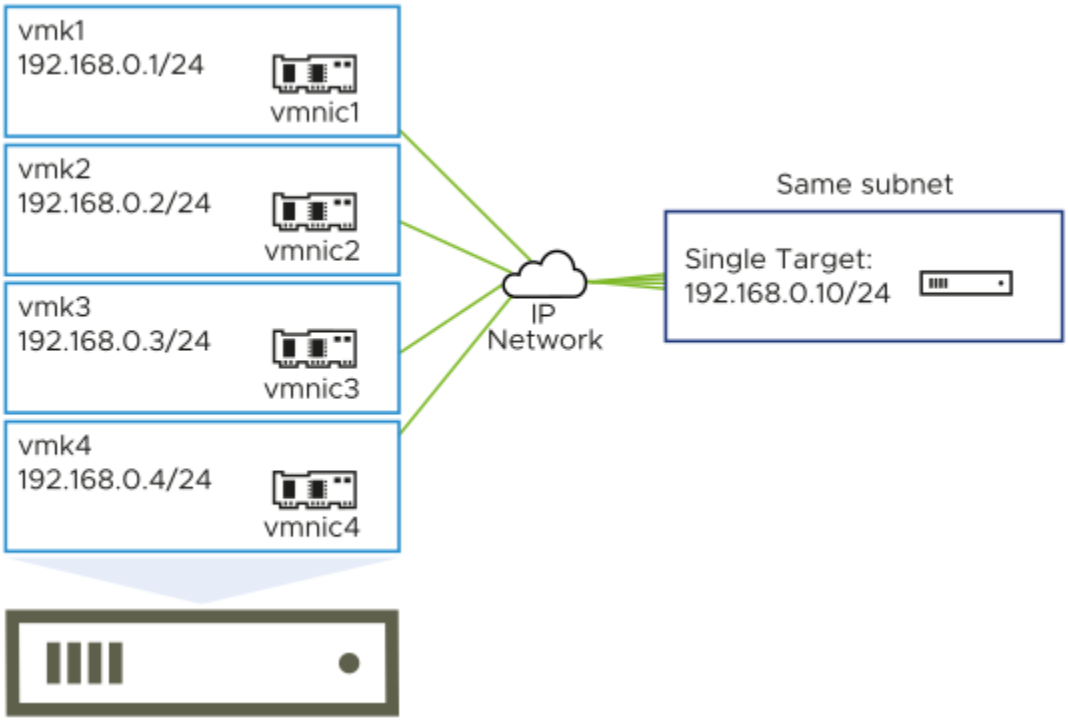
See the following examples.

VMkernel Ports	Target Portals	iSCSI Sessions
2 unbound VMkernel ports	2 target portals	2 sessions
4 unbound VMkernel ports	1 target portal	1 session
2 unbound VMkernel ports	4 target portals	4 sessions

Software iSCSI Multipathing

Example 1. Multiple paths for an iSCSI target with a single network portal

If your target has only one network portal, you can create multiple paths to the target by adding multiple VMkernel ports on your ESX host and binding them to the iSCSI initiator.

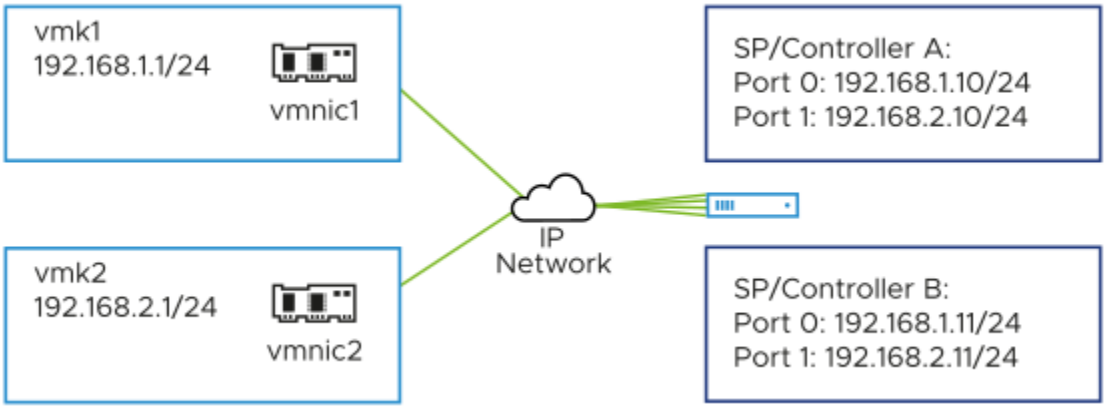


In this example, all initiator ports and the target portal are configured in the same subnet. The target is reachable through all bound ports. You have four VMkernel ports and one target portal, so total of four paths are created.

Without the port binding, only one path is created.

Example 2. Multiple paths with VMkernel ports in different subnets

You can create multiple paths by configuring multiple ports and target portals on different IP subnets. By keeping initiator and target ports in different subnets, you can force ESX to create paths through specific ports. In this configuration, you do not use port binding because port binding requires that all initiator and target ports are on the same subnet.



ESX selects vmk1 when connecting to Port 0 of Controller A and Controller B because all three ports are on the same subnet. Similarly, vmk2 is selected when connecting to Port 1 of Controller A and B. You can use NIC teaming in this configuration.

Total of four paths are created.

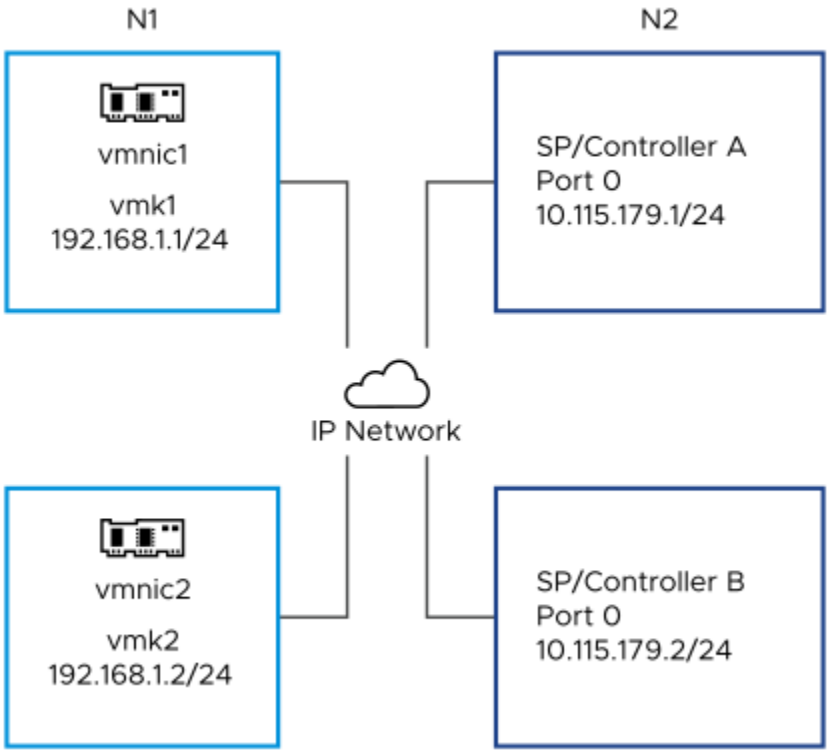
Paths	Description
Path 1	vmk1 and Port0 of Controller A
Path 2	vmk1 and Port0 of Controller B
Path 3	vmk2 and Port1 of Controller A
Path 4	vmk2 and Port1 of Controller B

Routing with Software iSCSI

You can use the `esxcli` command to add static routes for your iSCSI traffic. After you configure static routes, initiator and target ports in different subnets can communicate with each other.

Example 1. Using static routes with port binding

In this example, you keep all bound VMkernel ports in one subnet (N1) and configure all target portals in another subnet (N2). You can then add a static route for the target subnet (N2).

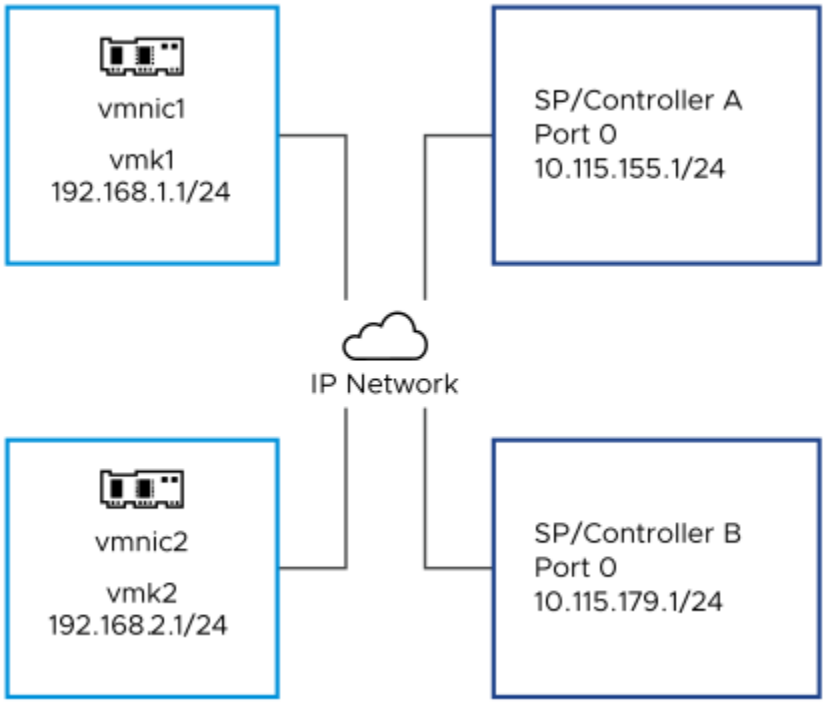


Use the following command:

```
# esxcli network ip route ipv4 add -gateway 192.168.1.253 -network 10.115.179.0/24
```

Example 2. Using static routes to create multiple paths

In this configuration, you use static routing when using different subnets. You cannot use the port binding with this configuration.



You configure vmk1 and vmk2 in separate subnets, 192.168.1.0 and 192.168.2.0. Your target portals are also in separate subnets, 10.115.155.0 and 10.155.179.0.

You can add the static route for 10.115.155.0 from vmk1. Make sure that the gateway is reachable from vmk1.

```
# esxcli network ip route ipv4 add -gateway 192.168.1.253 -network 10.115.155.0/24
```

You then add static route for 10.115.179.0 from vmk2. Make sure that the gateway is reachable from vmk2.

```
# esxcli network ip route ipv4 add -gateway 192.168.2.253 -network 10.115.179.0/24
```

When connecting with Port 0 of Controller A, vmk1 is used.

When connecting with Port 0 of Controller B, vmk2 is used.

Example 3. Routing with a separate gateway per vmkernel port

You can configure a separate gateway per VMkernel port. If you use DHCP to obtain IP configuration for a VMkernel port, gateway information can also be obtained using DHCP.

To see gateway information per VMkernel port, use the following command:

```
# esxcli network ip interface ipv4 address list
 Name  IPv4 Address    IPv4 Netmask   IPv4 Broadcast  Address Type  Gateway         DHCP DNS
 ----  -------------  -------------  --------------  ------------  --------------  --------
 vmk0  10.115.155.122  255.255.252.0  10.115.155.255  DHCP          10.115.155.253      true
 vmk1  10.115.179.209  255.255.252.0  10.115.179.255  DHCP          10.115.179.253      true
 vmk2  10.115.179.146  255.255.252.0  10.115.179.255  DHCP          10.115.179.253      true
```

With separate gateways per VMkernel port, you use port binding to reach targets in different subnets.

Managing iSCSI Network

Special considerations apply to network adapters, both physical and VMkernel, that are associated with an iSCSI adapter.

After you create network connections for iSCSI, an iSCSI initiator is enabled in the vSphere Client. The indicator shows that a particular virtual or physical network adapter is iSCSI-bound. To avoid disruptions in iSCSI traffic, follow these guidelines and considerations when managing iSCSI-bound virtual and physical network adapters:

- Verify that the VMkernel network adapters are assigned addresses on the same subnet as the iSCSI storage portal they connect to.
- iSCSI adapters using VMkernel adapters cannot connect to iSCSI ports on different subnets, even if the iSCSI adapters discover those ports.
- When using separate vSphere switches to connect physical network adapters and VMkernel adapters, make sure that the vSphere switches connect to different IP subnets.
- If VMkernel adapters are on the same subnet, they must connect to a single vSwitch.
- If you migrate VMkernel adapters to a different vSphere switch, move associated physical adapters.
- Do not make configuration changes to iSCSI-bound VMkernel adapters or physical network adapters.
- Do not make changes that might break association of VMkernel adapters and physical network adapters. You can break the association if you remove one of the adapters or the vSphere switch that connects them. Or if you change the 1:1 network policy for their connection.

iSCSI Network Troubleshooting

A warning sign indicates non-compliant port group policy for an iSCSI-bound VMkernel adapter.

The VMkernel adapter's port group policy is considered non-compliant in the following cases:

- The VMkernel adapter is not connected to an active physical network adapter.
- The VMkernel adapter is connected to more than one physical network adapter.
- The VMkernel adapter is connected to one or more standby physical adapters.
- The active physical adapter is changed.

Set up the correct network policy for the iSCSI-bound VMkernel adapter. See Configure Port Binding for iSCSI or iSER on ESX.

Configure Port Binding for iSCSI or iSER on ESX

The port binding creates connections for the traffic between certain types of iSCSI and iSER adapters and the physical network adapters on your ESX system.

The following types of adapters require the port binding:

- Software iSCSI adapter
- Dependent hardware iSCSI adapter
- VMware iSCSI over RDMA (iSER) adapter

The following tasks discuss the network configuration with a vSphere Standard switch and a single physical network adapter. If you have multiple network adapters, see Multiple Network Adapters in iSCSI or iSER Configuration.

> **Note:**
> iSER does not support NIC teaming. When configuring port binding for iSER, use only one RDMA-enabled physical adapter (vmnic#) and one VMkernel adapter (vmk#) per vSwitch.

You can also use the VMware vSphere®Distributed Switch™ and VMware NSX®Virtual Switch™ in the port binding configuration. For information about NSX virtual switches, see the *VMware NSX Data Center for vSphere* documentation.

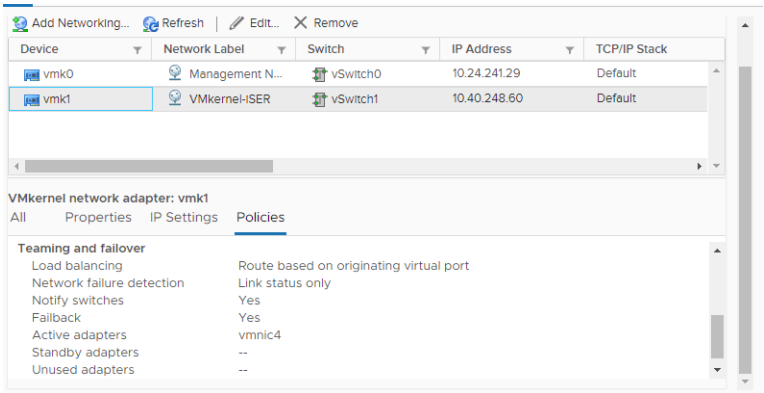
If you use a vSphere distributed switch with multiple uplink ports, for port binding, create a separate distributed port group per each physical NIC. Then set the team policy so that each distributed port group has only one active uplink port. For detailed information on distributed switches, see the *vSphere Networking* documentation.

Create a Single VMkernel Adapter for iSCSI or iSER

Connect the VMkernel, which runs services for iSCSI storage, to a physical network adapter on your ESX host. You then use the created VMkernel adapter in the port binding configuration with the iSCSI or iSER adapters.

- If you are creating a VMkernel adapter for dependent hardware iSCSI, you must use the physical network adapter (vmnic#) that corresponds to the iSCSI component. See Determine Association Between iSCSI and Network Adapters.
- With the iSER adapter, make sure to use an appropriate RDMA-capable vmnic#. See Install and View an RDMA Capable Network Adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Select **Add Networking** from the right-click menu.
3. Select **VMkernel Network Adapter**, and click **Next**.
4. Select **New standard switch** to create a vSphere Standard switch.
5. Click the **Add adapters** icon, and select an appropriate network adapter (vmnic#) to use for iSCSI.
   Make sure to assign the adapter to Active Adapters.
6. Enter a network label.
   A network label is a friendly name that identifies the VMkernel adapter that you are creating, for example, iSCSI or iSER.
7. Specify the IP settings.
8. Review the information and click **Finish**.
   You created the virtual VMkernel adapter (vmk#) for a physical network adapter (vmnic#) on your host.
9. Verify your configuration.
   a) Under **Networking**, select **VMkernel Adapters**, and select the VMkernel adapter (vmk#) from the list.
   b) Click the **Policies** tab, and verify that the corresponding physical network adapter (vmnic#) appears as an active adapter under **Teaming and failover**.



If your host has one physical network adapter for iSCSI traffic, bind the VMkernel adapter that you created to the iSCSI or iSER vmhba adapter.

If you have multiple network adapters, you can create additional VMkernel adapters and then perform iSCSI binding. The number of virtual adapters must correspond to the number of physical adapters on the host. For information, see Multiple Network Adapters in iSCSI or iSER Configuration.

Bind iSCSI or iSER Adapters to VMkernel Adapters

On the ESX host, bind an iSCSI or iSER adapter with a VMkernel adapter.

Create a virtual VMkernel adapter for each physical network adapter on your host. If you use multiple VMkernel adapters, set up the correct network policy.
Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

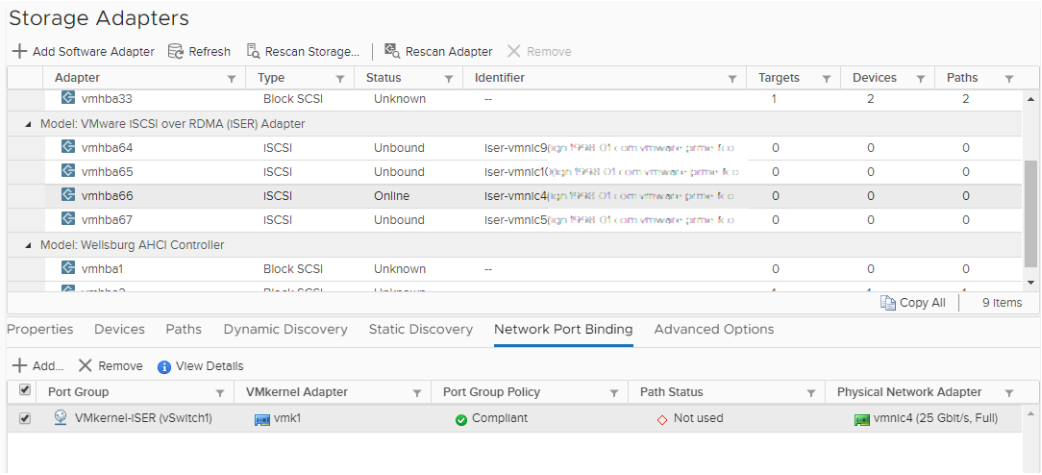
1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the appropriate iSCSI or iSER adapter (vmhba#) from the list.
4. Click the **Network Port Binding** tab and click the **Add** icon.
5. Select a VMkernel adapter to bind with the iSCSI or iSER adapter.

   Note:
   Make sure that the network policy for the VMkernel adapter is compliant with the binding requirements.

You can bind the software iSCSI adapter to one or more VMkernel adapters. For a dependent hardware iSCSI adapter or the iSER adapter, only one VMkernel adapter associated with the correct physical NIC is available.

6. Click **OK**.

   The network connection appears on the list of network port bindings for the iSCSI or iSER adapter.



Review Port Binding Details on the ESX Host

Review networking details of the VMkernel adapter that is bound to the iSCSI or iSER vmhba adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the appropriate iSCSI or iSER adapter from the list.
4. Click the **Network Port Binding** tab and select the VMkernel adapter from the list.
5. Click the **View Details** icon.
6. Review the VMkernel adapter and physical adapter information by switching between available tabs.

For other configuration steps you can perform for the iSCSI or iSER storage adapters, see the following topics:

- Configure Dependent Hardware iSCSI Adapters with ESX
- Configure the Software iSCSI Adapter with ESX
- Configure iSER with ESX

Using Jumbo Frames with iSCSI and iSER with ESX

ESX supports the use of Jumbo Frames with iSCSI and iSER protocols. Jumbo Frames are Ethernet frames with the size that exceeds 1500 Bytes. The maximum transmission unit (MTU) parameter is typically used to measure the size of Jumbo Frames.

When you use Jumbo Frames for iSCSI traffic, the following considerations apply:

- All network components must support Jumbo Frames.
- Check with your vendors to ensure your physical NICs and iSCSI adapters support Jumbo Frames.
- To set up and verify physical network switches for Jumbo Frames, consult your vendor documentation.

The following table explains the level of support that ESX provides to Jumbo Frames.

Table 167: Support of Jumbo Frames

Type of iSCSI Adapters	Jumbo Frames Support
Software iSCSI	Supported
Dependent Hardware iSCSI	Supported. Check with vendor.
Independent Hardware iSCSI	Supported. Check with vendor.
VMware iSER	Supported. Check with vendor.

Enable Jumbo Frames for Networking

You can enable Jumbo Frames for ESX storage adapters that use VMkernel networking for their traffic. These adapters include software iSCSI adapters, dependent hardware iSCSI adapters, and VMware iSER adapters.

To enable Jumbo Frames, change the default value of the maximum transmission units (MTU) parameter. You change the MTU parameter on the vSphere switch that you use for iSCSI traffic. For more information, see the *vSphere Networking* documentation.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Networking**, click **Virtual switches**, and select the vSphere switch that you want to modify from the list.
4. Click the **Edit settings** icon.
5. On the Properties page, change the MTU parameter.

   This step sets the MTU for all physical NICs on that standard switch. Set the MTU value to the largest MTU size among all NICs connected to the standard switch. ESX supports the MTU size of up to 9000 Bytes.

Enable Jumbo Frames for Independent Hardware iSCSI

To enable Jumbo Frames for independent hardware iSCSI adapters on your ESX host, change the default value of the maximum transmission units (MTU) parameter.

Use the Advanced Options settings to change the MTU parameter for the iSCSI HBA.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the independent hardware iSCSI adapter from the list of adapters.
4. Click the **Advanced Options** tab and click **Edit**.
5. Change the value of the MTU parameter.

   ESX supports the MTU size up to 9000 Bytes.

Configure Dynamic or Static Discovery for iSCSI and iSER on ESX Host

You need to set up target discovery addresses, so that the iSCSI or iSER storage adapter can determine which storage resource on the network is available for access on your ESX host.

Required privilege:**Host** > **Configuration** > **Storage Partition Configuration**

The ESX system supports these discovery methods:

Dynamic Discovery Also known as SendTargets discovery. Each time the initiator contacts a specified iSCSI server, the initiator sends the SendTargets request to the server. The server responds by supplying a list of available targets to the initiator. The names and IP addresses of these targets appear on the **Static Discovery** tab. If you remove a static target added by dynamic discovery, the target might be returned to the list the next time a rescan happens, the storage adapter is reset, or the host is rebooted.

Note:
With software and dependent hardware iSCSI, ESX filters target addresses based on the IP family of the iSCSI server address specified. If the address is IPv4, IPv6 addresses that might come in the SendTargets response from the iSCSI server are filtered out. When DNS names are used to specify an iSCSI server, or when the SendTargets response from the iSCSI server has DNS names, ESX relies on the IP family of the first resolved entry from DNS lookup.

Static Discovery In addition to the dynamic discovery method, you can use static discovery and manually enter information for the targets. The iSCSI or iSER adapter uses a list of targets that you provide to contact and communicate with the iSCSI servers.

When you set up static or dynamic discovery, you can only add new iSCSI targets. You cannot change any parameters of an existing target. To make changes, remove the existing target and add a new one.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
4. Configure the discovery method.

Discovery Method	Description
Dynamic Discovery	1. Click **Dynamic Discovery** and click **Add**. 2. Enter the IP address or DNS name of the storage system and click **OK**. 3. Rescan the iSCSI adapter. After establishing the SendTargets session with the iSCSI system, your host populates the Static Discovery list with all newly discovered targets. **Note:** A dynamically discovered target remains on the list even after it is removed from the array side.
Static Discovery	1. Click **Static Discovery** and click **Add**. 2. Enter the target's information and click **OK** 3. Rescan the iSCSI adapter.

For other configuration steps you can perform for the iSCSI or iSER storage adapters, see the following topics:

- Set Up Independent Hardware iSCSI Adapters with ESX
- Configure Dependent Hardware iSCSI Adapters with ESX
- Configure the Software iSCSI Adapter with ESX
- Configure iSER with ESX

Remove Dynamic or Static iSCSI Targets

Remove iSCSI servers connected to your ESX host.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the iSCSI adapter to modify from the list.
4. Switch between **Dynamic Discovery** and **Static Discovery**.
5. Select an iSCSI server to remove and click **Remove**.
6. Rescan the iSCSI adapter.

   If you are removing the static target that was dynamically discovered, you need to remove it from the storage system before performing the rescan. Otherwise, your host will automatically discover and add the target to the list of static targets when you rescan the adapter.

Configuring CHAP Parameters for iSCSI or iSER Storage Adapters on ESX Host

The IP networks that the iSCSI technology uses to connect your ESX host to remote targets do not protect the data they transport. As a result, you must ensure security of the connection. One of the protocols that iSCSI implements is the Challenge Handshake Authentication Protocol (CHAP). The CHAP protocol verifies the legitimacy of ESX initiators that access targets on the network.

CHAP uses a three-way handshake algorithm to verify the identity of your host and, if applicable, of the iSCSI target when the host and target establish a connection. The verification is based on a predefined private value, or CHAP secret, that the initiator and target share.

ESX supports CHAP authentication at the adapter level. In this case, all targets receive the same CHAP name and secret from the iSCSI initiator. For software and dependent hardware iSCSI adapters, and for iSER adapters, ESX also supports per-target CHAP authentication, which allows you to configure different credentials for each target to achieve greater level of security.

Selecting CHAP Authentication Method

ESX supports unidirectional CHAP for all types of iSCSI and iSER initiators, and bidirectional CHAP for software and dependent hardware iSCSI, and for iSER.

Before configuring CHAP, check whether CHAP is activated at the iSCSI storage system. Also, obtain information about the CHAP authentication method the system supports. If CHAP is activated, configure it for your initiators, making sure that the CHAP authentication credentials match the credentials on the iSCSI storage.

ESX supports the following CHAP authentication methods:

Unidirectional CHAP	In unidirectional CHAP authentication, the target authenticates the initiator, but the initiator does not authenticate the target.
Bidirectional CHAP	The bidirectional CHAP authentication adds an extra level of security. With this method, the initiator can also authenticate the target. VMware supports this method for software and dependent hardware iSCSI adapters, and for iSER adapters.

For software and dependent hardware iSCSI adapters, and for iSER adapters, you can set unidirectional CHAP and bidirectional CHAP for each adapter or at the target level. Independent hardware iSCSI supports CHAP only at the adapter level.

When you set the CHAP parameters, specify a security level for CHAP.

> **Note:**
> When you specify the CHAP security level, how the storage array responds depends on the array's CHAP implementation and is vendor-specific. For information on CHAP authentication behavior in different initiator and target configurations, consult the array documentation.

Table 168: CHAP Security Level

CHAP Security Level	Description	Supported Storage Adapters
None	The host does not use CHAP authentication. If authentication is activated, use this option to deactivate it.	Independent hardware iSCSI Software iSCSI Dependent hardware iSCSI iSER
Use unidirectional CHAP if required by target	The host prefers a non-CHAP connection, but can use a CHAP connection if required by the target.	Software iSCSI Dependent hardware iSCSI iSER
Use unidirectional CHAP unless prohibited by target	The host prefers CHAP, but can use non-CHAP connections if the target does not support CHAP.	Independent hardware iSCSI Software iSCSI Dependent hardware iSCSI iSER

CHAP Security Level	Description	Supported Storage Adapters
Use unidirectional CHAP	The host requires successful CHAP authentication. The connection fails if CHAP negotiation fails.	Independent hardware iSCSI Software iSCSI Dependent hardware iSCSI iSER
Use bidirectional CHAP	The host and the target support bidirectional CHAP.	Software iSCSI Dependent hardware iSCSI iSER

Set Up CHAP for iSCSI or iSER Storage Adapter

When you set up the CHAP name and secret at the iSCSI or iSER adapter level, all targets receive the same parameters from the adapter. By default, all discovery addresses or static targets inherit CHAP parameters that you set up at the adapter level.

- Before setting up CHAP parameters for software or dependent hardware iSCSI, determine whether to configure unidirectional or bidirectional CHAP. Independent hardware iSCSI adapters do not support bidirectional CHAP.
- Verify CHAP parameters configured on the storage side. Parameters that you configure must match the ones on the storage side.
- Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

The CHAP name cannot exceed 511 alphanumeric characters and the CHAP secret cannot exceed 255 alphanumeric characters. Some adapters, for example, the QLogic adapter, might have lower limits, 255 for the CHAP name and 100 for the CHAP secret.

1. Navigate to the iSCSI or iSER storage adapter.
   a) In the vSphere Client, navigate to the ESX host.
   b) Click the **Configure** tab.
   c) Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
2. Click the **Properties** tab and click **Edit** in the **Authentication** panel.
3. Specify an authentication method.
   - **None**
   - **Use unidirectional CHAP if required by target**
   - **Use unidirectional CHAP unless prohibited by target**
   - **Use unidirectional CHAP**
   - **Use bidirectional CHAP**. To configure bidirectional CHAP, you must select this option.
4. Specify the outgoing CHAP name.

   Make sure that the name you specify matches the name configured on the storage side.
   - To set the CHAP name to the iSCSI adapter name, select **Use initiator name**.
   - To set the CHAP name to anything other than the iSCSI initiator name, deselect **Use initiator name** and enter a name in the **Name** text box.
5. Enter an outgoing CHAP secret to be used as part of authentication. Use the same secret that you enter on the storage side.
6. If configuring bidirectional CHAP, specify incoming CHAP credentials.

   Make sure to use different secrets for the outgoing and incoming CHAP.
7. Click **OK**.
8. Rescan the iSCSI adapter.

If you change the CHAP parameters, they are used for new iSCSI sessions. For existing sessions, new settings are not used until you log out and log in again.

For other configuration steps you can perform for the iSCSI or iSER storage adapters, see the following topics:

- Set Up Independent Hardware iSCSI Adapters with ESX
- Configure Dependent Hardware iSCSI Adapters with ESX
- Configure the Software iSCSI Adapter with ESX
- Configure iSER with ESX

Set Up CHAP for Target

If you use software and dependent hardware iSCSI adapters, or an iSER storage adapter, you can configure different CHAP credentials for each discovery address or static target.

- Before setting up CHAP parameters, determine whether to configure unidirectional or bidirectional CHAP.
- Verify CHAP parameters configured on the storage side. Parameters that you configure must match the ones on the storage side.
- Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

The CHAP name cannot exceed 511 and the CHAP secret 255 alphanumeric characters.

1. Navigate to the iSCSI or iSER storage adapter.
   a) In the vSphere Client, navigate to the ESX host.
   b) Click the **Configure** tab.
   c) Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
2. Click either **Dynamic Discovery** or **Static Discovery**.
3. From the list of available targets, select a target to configure and click **Authentication**.
4. Deselect **Inherit settings from parent** and specify authentication method.
   - **None**
   - **Use unidirectional CHAP if required by target**
   - **Use unidirectional CHAP unless prohibited by target**
   - **Use unidirectional CHAP**
   - **Use bidirectional CHAP**. To configure bidirectional CHAP, you must select this option.
5. Specify the outgoing CHAP name.

   Make sure that the name you specify matches the name configured on the storage side.
   - To set the CHAP name to the iSCSI adapter name, select **Use initiator name**.
   - To set the CHAP name to anything other than the iSCSI initiator name, deselect **Use initiator name** and enter a name in the **Name** text box.
6. Enter an outgoing CHAP secret to be used as part of authentication. Use the same secret that you enter on the storage side.
7. If configuring bidirectional CHAP, specify incoming CHAP credentials.

   Make sure to use different secrets for the outgoing and incoming CHAP.
8. Click **OK**.
9. Rescan the storage adapter.

If you change the CHAP parameters, they are used for new iSCSI sessions. For existing sessions, new settings are not used until you log out and log in again.

Configuring Advanced Parameters for iSCSI on ESX Host

You might need to configure additional parameters for iSCSI initiators on your ESX host. For example, some iSCSI storage systems require ARP (Address Resolution Protocol) redirection to move iSCSI traffic dynamically from one port to another. In this case, you must activate the ARP redirection on your host.

The following table lists advanced iSCSI parameters that you can configure using the vSphere Client. In addition, you can use ESXCLI commands to configure some of the advanced parameters. For information, see the *Getting Started with ESXCLI* documentation. See ESXCLI.

Depending on the type of your adapters, certain parameters might not be available.

> **Important:**
> Do not change the advanced iSCSI settings unless VMware support or Storage Vendors direct you to change them.

Table 169: Additional Parameters for iSCSI Initiators

Advanced Parameter	Description
Header Digest	Increases data integrity. When the header digest parameter is enabled, the system performs a checksum over each header part of the iSCSI Protocol Data Unit (PDU). The system verifies the data using the CRC32C algorithm.
Data Digest	Increases data integrity. When the data digest parameter is enabled, the system performs a checksum over each PDU data part. The system verifies the data using the CRC32C algorithm. **Note:** Systems that use the Intel Nehalem processors offload the iSCSI digest calculations for software iSCSI. This offload helps to reduce the impact on performance.
ErrorRecoveryLevel	iSCSI Error Recovery Level (ERL) value that the iSCSI initiator on the host negotiates during a login.
LoginRetryMax	Maximum number of times the ESX iSCSI initiator attempts to log into a target before ending the attempts.
MaxOutstandingR2T	Defines the R2T (Ready to Transfer) PDUs that can be in transition before an acknowledged PDU is received.
FirstBurstLength	Specifies the maximum amount of unsolicited data an iSCSI initiator can send to the target during the execution of a single SCSI command, in bytes.
MaxBurstLength	Maximum SCSI data payload in a Data-In or a solicited Data-Out iSCSI sequence, in bytes.
MaxRecvDataSegLength	Maximum data segment length, in bytes, that can be received in an iSCSI PDU.
MaxCommands	Maximum SCSI commands that can be queued on the iSCSI adapter.
DefaultTimeToWait	Minimum time in seconds to wait before attempting a logout or an active task reassignment after an unexpected connection termination or reset.
DefautTimeToRetain	Maximum time in seconds, during which reassigning the active task is still possible after a connection termination or reset.
LoginTimeout	Time in seconds the initiator waits for the login response to finish.
LogoutTimeout	Time in seconds the initiator waits to get a response for Logout request PDU.
RecoveryTimeout	Specifies the amount of time, in seconds, that can lapse while a session recovery is performed. If the timeout exceeds its limit, the iSCSI initiator ends the session.
No-Op Interval	Specifies the time interval, in seconds, between NOP-Out requests sent from your iSCSI initiator to an iSCSI target. The NOP-Out requests serve as the ping mechanism to verify that a connection between the iSCSI initiator and the iSCSI target is active.
No-Op Timeout	Specifies the amount of time, in seconds, that can lapse before your host receives a NOP-In message. The iSCSI target sends the message in response to the NOP-Out request. When the no-op timeout limit is exceeded, the initiator ends the current session and starts a new one.
ARP Redirect	With this parameter enabled, storage systems can move iSCSI traffic dynamically from one port to another. Storage systems that perform array-based failovers require the ARP parameter.
Delayed ACK	With this parameter enabled, storage systems can delay an acknowledgment of received data packets.

Configure Advanced Parameters for iSCSI on ESX Host

The advanced iSCSI settings control such parameters as header and data digest, ARP redirection, delayed ACK, and so on.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

> **CAUTION:**
> Do not make any changes to the advanced settings unless you are working with the VMware support team or have thorough information about the values to provide for the settings.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
4. Configure advanced parameters.

Option	Description
At the adapter level	Click the **Advanced Options** tab and click **Edit**.
At the target level	1. Click either **Dynamic Discovery** or **Static Discovery**. 2. From the list of available targets, select a target to configure and click **Advanced**.

5. Enter any required values for the advanced parameters you want to modify.

Managing iSCSI Session on ESX Host

To communicate with each other, iSCSI initiators on the ESX host and targets establish iSCSI sessions. You can review and manage iSCSI sessions using ESXCLI.

By default, software iSCSI and dependent hardware iSCSI initiators start one iSCSI session between each initiator port and each target port. If your iSCSI initiator or target has more than one port, your host can have multiple sessions established. The default number of sessions for each target equals the number of ports on the iSCSI adapter times the number of target ports.

Using ESXCLI, you can display all current sessions to analyze and debug them. To create more paths to storage systems, you can increase the default number of sessions by duplicating existing sessions between the iSCSI adapter and target ports.

You can also establish a session to a specific target port. This capability is useful if your host connects to a single-port storage system that presents only one target port to your initiator. The system then redirects additional sessions to a different target port. Establishing a new session between your iSCSI initiator and another target port creates an additional path to the storage system.

The following considerations apply to iSCSI session management:

- Some storage systems do not support multiple sessions from the same initiator name or endpoint. Attempts to create multiple sessions to such targets can result in an unpredictable behavior of your iSCSI environment.
- Storage vendors can provide automatic session managers. Using the automatic session manages to add or delete sessions, does not guarantee lasting results and can interfere with the storage performance.

Review iSCSI Sessions

Use the ESXCLI command to display iSCSI sessions between an iSCSI adapter and a storage system.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To list iSCSI sessions, run the following command:

```
esxcli iscsi session list
```

The command takes these options:

Option	Description
**`-A|--adapter=`*`str`*	The iSCSI adapter name, for example, vmhba34.
**`-s|--isid=`*`str`*	The iSCSI session identifier.

Option	Description
-n|--name=*str*	The iSCSI target name, for example, iqn.X.

Add iSCSI Sessions

Use ESXCLI to add an iSCSI session for a target you specify or to duplicate an existing session. By duplicating sessions, you increase the default number of sessions and create additional paths to storage systems.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To add or duplicate an iSCSI session, run the following command:

```
esxcli iscsi session add
```

The command takes these options:

Option	Description
-A|--adapter=*str*	The iSCSI adapter name, for example, vmhba34. This option is required.
-s|--isid=*str*	The ISID of a session to duplicate. You can find it by listing all sessions.
-n|--name=*str*	The iSCSI target name, for example, iqn.X.

Rescan the iSCSI adapter.

Remove iSCSI Sessions

Use the ESXCLI command to remove an iSCSI session between an iSCSI adapter and a target.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To remove a session, run the following command:

```
esxcli iscsi session remove
```

The command takes these options:

Option	Description
-A|--adapter=*str*	The iSCSI adapter name, for example, vmhba34. This option is required.
-s|--isid=*str*	The ISID of a session to remove. You can find it by listing all session.
-n|--name=*str*	The iSCSI target name, for example, iqn.X.

Rescan the iSCSI adapter.

Using iSNS with ESX iSCSI

Starting with vSphere 9.1, ESX supports the Internet Storage Name Service (iSNS) protocol. iSNS is a lightweight discovery protocol that simplifies iSCSI storage configuration across IP networks. The protocol enables automated discovery, centralized management, and zoning control for iSCSI environments. Unlike manual iSCSI SendTargets discovery, iSNS uses a central server that ESX periodically queries to update the list of available iSCSI targets.

The iSNS protocol simplifies management of iSCSI targets by providing a consolidated configuration point for entire storage network.

Devices register with an iSNS server instead of requiring a per-host configuration. ESX periodically queries the iSNS server to update the iSCSI target list, so new targets are discovered automatically without manual rescans.

iSNS also provides zoning-like functionality through Discovery Domains, which limit discovery to authorized groups of initiators and targets. This helps prevent accidental or unauthorized access.

The iSNS protocol is defined in RFC 4171 and supports both iSCSI and iFCP. For more information, see Internet Storage Name Service (iSNS) RFC 4171 documentation.

ESX supports the iSNS client only with iSCSI. To be able to use iSNS with the ESX 9.1 release, you must activate the iSNS client support for iSCSI adapters. See Activate or Deactivate iSNS Client Support on ESX.

Note:

Contact your switch vendor to confirm iSNS server support. For details on storage partner iSNS compatibility, see the VMware by Broadcom Compatibility Guide.

iSNS Concepts

When using iSNS with ESX iSCSI, be familiar with the following concepts.

- **iSNS Protocol (iSNSP) –** iSNSP is a flexible and lightweight protocol that specifies how the iSNS clients and servers communicate.
- **iSNS Server** – Centralized service for initiator and target registration. It responds to iSNS protocol queries and requests from clients.
- **iSNS Client** – iSNS clients initiate transactions with iSNS servers using the iSNSP. In ESX, the client integrates with the ESX iSCSI initiator. It can register device attribute information and download information about other registered clients in a common Discovery Domain.
- **Discovery Domains** – Provide zoning-like control of initiators and targets.
- **Periodic iSNS Discovery** – ESX periodically queries the iSNS server to update the iSCSI target list automatically, so new targets are discovered without manual rescans.

Activate or Deactivate iSNS Client Support on ESX

To use iSNS with ESX 9.1, you must enable the iSNS client support for iSCSI adapters. You can later disable the iSNS client support.

Enable iSNS Client Support on ESX

To enable the iSNS client support for iSCSI adapters, set the advanced `IscsiIsnsSupport` configuration option that is disabled by default.

1. Set the `IscsiIsnsSupport` option.
   ```
   $ esxcli system settings advanced set -o /ISCSI/IscsiIsnsSupport -i 1
   ```
2. Restart the iSCSI daemon.
   ```
   $ /usr/lib/vmware/iscsid/vmkiscsid restart
   ```

Disable iSNS Client Support on ESX

If you want to disable the iSNS client support for iSCSI adapters, follow these steps.

1. Remove iSCSI sessions running on targets discovered through iSNS.
   ```
   $ esxcli iscsi session remove ...
   ```
2. Remove all iSNS discovery addresses.
   ```
   $ esxcli iscsi adapter discovery isns remove ...
   ```
3. Disable the advanced `IscsiIsnsSupport` option.
   ```
   $ esxcli system settings advanced set -o /ISCSI/IscsiIsnsSupport -i 0
   ```
4. Restart `vmkiscsid`.
   ```
   $ /usr/lib/vmware/iscsid/vmkiscsid restart
   ```

Manage iSNS with ESX iSCSI

Use the `esxcli` commands to manage iSNS with ESX iSCSI.

List iSNS Address

The command displays an IP address and discovery methods. The discovery method is static.

```
# esxcli iscsi adapter discovery isns list
```

Sample output:

```
Adapter   discoveryMethod    Host
-------   ---------------    -----------------
vmhba33   static             10.112.93.20:3205
```

Add an iSNS Address with the Static Discovery Method

```
# esxcli iscsi adapter discovery isns add -A adapter -a ip/dns -m static
```

Remove an iSNS Address with the Static Discovery Method

```
# esxcli iscsi adapter discovery isns remove -A adapter mode ip/dns -m static
```

Get or Set iSNS Parameters

```
# esxcli iscsi adapter isns param get/set
# esxcli iscsi adapter discovery isns param get/set
```

Managing ESX Storage Devices

Manage local and networked storage device that your ESX host has access to.

For information about storage devices that ESX supports, see What Types of Physical Storage Does ESX Support .

Viewing Storage Devices Available to ESX Hosts

When your ESX host connects to block-based storage systems, LUNs or storage devices that support ESX become available to the host.

After the devices get registered with your host, you can display all available local and networked devices and review their information. If you use third-party multipathing plug-ins, the storage devices available through the plug-ins also appear on the list.

> **Note:**
> If an array supports implicit asymmetric logical unit access (ALUA) and has only standby paths, the registration of the device fails. The device can register with the host after the target activates a standby path and the host detects it as active. The advanced system `/Disk/FailDiskRegistration` parameter controls this behavior of the host.

For each storage adapter, you can display a separate list of storage devices available for this adapter.

Generally, when you review storage devices, you see the following information.

Storage Device Information	Description
Name	Also called Display Name. It is a name that the ESX host assigns to the device based on the storage type and manufacturer. Generally, you can change this name to a name of your choice. See ESX Storage Device Names and Identifiers.
Identifier	A universally unique identifier that is intrinsic to the device. See ESX Storage Device Names and Identifiers.
Operational State	Indicates whether the device is attached or detached. See Detach Storage Devices.
LUN	Logical Unit Number (LUN) within the SCSI target. The LUN number is provided by the storage system. If a target has only one LUN, the LUN number is always zero (0). See Target and Device Representations .
Type	Type of device, for example, disk or CD-ROM.
Drive Type	Information about whether the device is a flash drive or a regular HDD drive. For information about flash drives and NVMe devices, see Using Flash Devices with ESX Hosts.
Transport	Transportation protocol your host uses to access the device. The protocol depends on the type of storage being used. See What Types of Physical Storage Does ESX Support.
Capacity	Total capacity of the storage device.

Storage Device Information	Description
Owner	The plug-in, such as the NMP or a third-party plug-in, that the host uses to manage paths to the storage device. See Using Pluggable Storage Architecture and Path Management with ESX.
Hardware Acceleration	Information about whether the storage device assists the host with virtual machine management operations. The status can be Supported, Not Supported, or Unknown. See Storage Hardware Acceleration in vSphere.
Sector Format	Indicates whether the device uses a traditional, 512n, or advanced sector format, such as 512e or 4Kn. See Sector Formats for Storage Devices.
Location	A path to the storage device in the `/vmfs/devices/` directory.
Partition Format	A partition scheme used by the storage device. It can be of a master boot record (MBR) or GUID partition table (GPT) format. The GPT devices can support datastores greater than 2 TB. See Sector Formats for Storage Devices.
Partitions	Primary and logical partitions, including a VMFS datastore, if configured. See Working with Datastores in vSphere Storage Environment.
Multipathing Policies	Path Selection Policy and Storage Array Type Policy the host uses to manage paths to storage. See Understanding Multipathing and Failover in the ESX Environment.
Paths	Paths used to access storage and their status. See Deactivate Storage Paths.

Display Storage Devices for an ESX Host

Display all storage devices available to a ESX host. If you use any third-party multipathing plug-ins, the storage devices available through the plug-ins also appear on the list.

The Storage Devices view allows you to list the storage devices of the host, analyze their information, and modify properties.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.

   All storage devices available to the host are listed in the Storage Devices table.
4. To view details for a specific device, select the device from the list.
5. Use the icons to perform basic storage management tasks.

   Availability of specific icons depends on the device type and configuration.
6. Use the following tabs to access additional information and modify properties for the selected device.

Tab	Description
Properties	View device properties and characteristics. View and modify multipathing policies for the device. See Viewing and Managing Storage Paths on ESX Hosts.
Paths	Display paths available for the device. Deactivate or enable a selected path. See Deactivate Storage Paths.
Partition Details	Displays information about partitions and their formats.

Display Storage Devices for an Adapter

Display a list of storage devices accessible through a specific storage adapter on an ESX host.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**.

   All storage adapters installed on the host are listed in the Storage Adapters table.

4. Select the adapter from the list and click the **Devices** tab.
   Storage devices that the host can access through the adapter are displayed.
5. Use the icons to perform basic storage management tasks.

   Availability of specific icons depends on the device type and configuration.

Icon	Description
Refresh	Refresh information about storage adapters, topology, and file systems.
Detach Storage Devices	Detach the selected device from the host.
Attach Storage Devices	Attach the selected device to the host.
Rename Storage Devices	Change the display name of the selected device.
Use the Locator LED on ESX Storage Devices	Turn on the locator LED for the selected devices.
Use the Locator LED on ESX Storage Devices	Turn off the locator LED for the selected devices.
Marking Storage Devices on ESX Hosts	Mark the selected devices as flash disks.
Marking Storage Devices on ESX Hosts	Mark the selected devices as HDD disks.
Marking Storage Devices on ESX Hosts	Mark the selected devices as local for the host.
Marking Storage Devices on ESX Hosts	Mark the selected devices as remote for the host.
Erase ESX Storage Devices	Erase partitions on the selected devices.
Change Perennial Reservation Settings on ESX Hosts	Mark the selected device as perennially reserved.
Change Perennial Reservation Settings on ESX Hosts	Clear the perennial reservation from the selected device.

Sector Formats for Storage Devices

ESX supports storage devices with traditional and advanced sector formats. In storage, a sector is a subdivision of a track on a storage disk or device. Each sector stores a fixed amount of data.

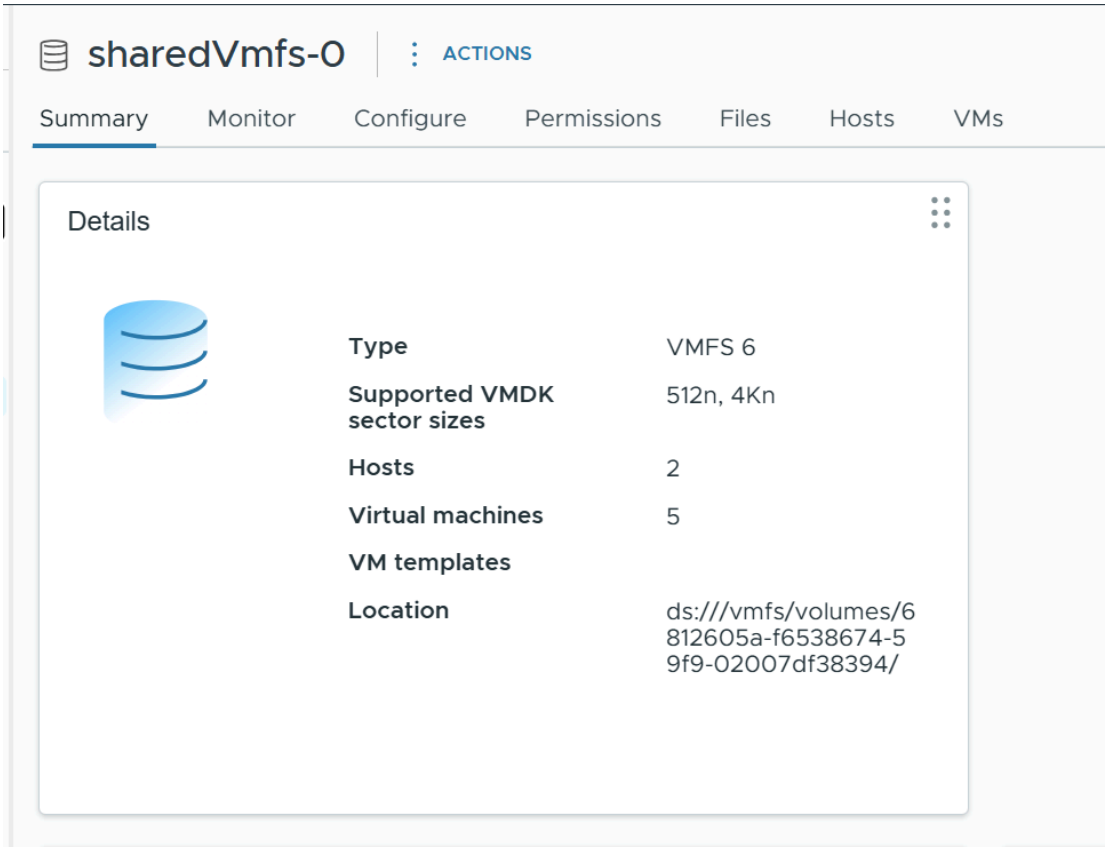
This table introduces different storage device formats that ESX supports.

Storage Device Format	Logical Sector Size	Physical Sector Size
512n	512	512
512e	512	4096
4Kn	4096	4096

Note:

4Kn SCSI local HDD is the device that ESX always presents as 512e. Read-modify-write (RMW) is done in the storage infrastructure layer.

ESX can detect different device formats. After you create a datastore, the **Details** pane in the datastore **Summary** shows the type of sector size that the datastore and its backing device support.



512-Byte Native Format

ESX supports traditional 512n storage devices that use a native 512-bytes sector size.

512-Byte Emulation Format

Due to the increasing demand for larger capacities, the storage industry has introduced advanced formats, such as 512-byte emulation, or 512e. 512e is the advanced format in which the physical sector size is 4096 bytes, but the logical sector emulates the 512-bytes sector size. Storage devices that use the 512e format can support legacy applications and guest operating systems. These devices serve as an intermediate step to 4Kn sector drives.

4Kn Format

ESX supports the 4Kn sector technology. In the 4Kn devices, both physical and logical sectors are 4096 bytes (4 KiB) in length. The device does not have an emulation layer, but exposes its 4Kn physical sector size directly to ESX.

ESX detects and registers the 4Kn devices.

Starting with vSphere 9.0, VMs can understand the backend device as 4Kn and send a 4K I/O through the 4Kn virtual disk to the device.

When you use 4Kn devices, the following considerations apply:

- ESX supports 4Kn local devices for use with VMFS and vSAN datastores.
- ESX can boot only from a 4Kn device with UEFI.
- You can use the 4Kn device to configure a coredump partition and coredump file.
- Only VMs with hardware version 21 (vSphee 9.0) and later can support 4Kn virtual disk.
- Once the VM disk is created, the sector size of the disk cannot be changed.

Determine Device Format

To determine whether the device uses the 512n, 512e, or 4Kn format, run the following command.

```
esxcli storage core device capacity list
```

The following sample output shows the format type.

```
Device                Physical Blocksize  Logical Blocksize  Logical Block Count         Size
 Format Type
--------------------  ------------------  -----------------  -------------------  -----------
 -----------
naa.5000xxxxxxxxx36f                 512                512           2344225968  1144641 MiB  512n
naa.5000xxxxxxxxx030                4096                512           3516328368  1716957 MiB  4Kn
 SWE
naa.5000xxxxxxxxx8df                 512                512           2344225968  1144641 MiB  512n
naa.5000xxxxxxxxx4f4                4096                512           3516328368  1716957 MiB  4Kn
 SWE
```

Supported Sector Formats for Virtual Disks

The following table shows which types of storage devices support different VMDK sector formats and lists compatible datastores.

Datastore Type	Backing Device Type	Supported Virtual Disk Format
VMFS Local	512n or 512e HDD/SSD/NVMe	512n
VMFS Local	4Kn SCSI Local HDD	512n only
VMFS Local	4Kn NVMe/SSD	4Kn
VMFS Remote	512n or 512e	512n and 4Kn
NFS	512n and 4Kn	512n and 4Kn
vSAN	512e, 512n, 4Kn SCSI/NVMe	512n and 4Kn

Sector Format Compatibility

The following considerations and limitations apply.

- 4Kn SCSI Local HDD Limitation: ESX presents the 4Kn SCSI local HDDs as 512e, and these devices only supports the 512n VMDK format.
- 4Kn Virtual Disks: These disks are supported on local storage only when the backing device is a 4Kn NVMe or SSD and the VM is using hardware version 21 (vSphere 9.0) or later.
- Software Emulation: For 4Kn SCSI devices, ESX performs an emulation process at the storage infrastructure layer to handle the 512e presentation.

ESX Storage Device Names and Identifiers

In the ESX environment, each storage device is identified by several names.

Device Identifiers

Depending on the type of storage, the ESX host uses different algorithms and conventions to generate an identifier for each storage device.

Storage-provided identifiers

The ESX host queries a target storage device for the device name. From the returned metadata, the host extracts or generates a unique identifier for the device. The identifier is based on specific storage standards, is unique and persistent across all hosts, and has one of the following formats:

- naa.*xxx*
- eui.*xxx*
- t10.*xxx*

Path-based identifier

When the device does not provide an identifier, the host generates an mpx.*path* name, where *path* represents the first path to the device, for example, `mpx.vmhba1:C0:T1:L3`. This identifier can be used in the same way as the storage-provided identifiers.

The mpx.*path* identifier is created for local devices on the assumption that their path names are unique. However, this identifier is not unique or persistent, and can change after every system restart.

Typically, the path to the device has the following format:

vmhba*Adapter*:C*Channel*:T*Target*:L*LUN*

- vmhba*Adapter* is the name of the storage adapter. The name refers to the physical adapter on the host, not to the SCSI controller used by the virtual machines.
- C*Channel* is the storage channel number.
  Software iSCSI adapters and dependent hardware adapters use the channel number to show multiple paths to the same target.
- T*Target* is the target number. Target numbering is determined by the host and might change when the mappings of targets visible to the host change. Targets that are shared by different hosts might not have the same target number.
- L*LUN* is the LUN number that shows the position of the LUN within the target. The LUN number is provided by the storage system. If a target has only one LUN, the LUN number is always zero (0).

For example, `vmhba1:C0:T3:L1` represents LUN1 on target 3 accessed through the storage adapter vmhba1 and channel 0.

Legacy identifier

In addition to the device-provided identifiers or mpx.*path* identifiers, ESX generates an alternative legacy name for each device. The identifier has the following format:

vml.*number*

The legacy identifier includes a series of digits that are unique to the device. The identifier can be derived in part from the metadata obtained through the SCSI INQUIRY command. For nonlocal devices that do not provide SCSI INQUIRY identifiers, the vml.*number* identifier is used as the only available unique identifier.

Displaying Device Names

You can use the `esxcli storage core device list` command to display all device names. The output is similar to the following example:

```
# esxcli storage core device list
naa.XXX
   Display Name: DGC Fibre Channel Disk(naa.XXX)
   ...
   Other UIDs: vml.000XXX
mpx.vmhba1:C0:T0:L0
   Display Name: Local VMware Disk (mpx.vmhba1:C0:T0:L0)
   ...
   Other UIDs: vml.0000000000XYZ
```

NVMe Devices with NGUID Device Identifiers

For NVMe devices, ESX generates device identifiers based on the information it retrieves from the devices. Generally, the NVMe devices support identifiers in EUI64 or NGUID formats, or use both formats. NGUID is a Namespace Globally Unique Identifier that uses the EUI64 16-byte designator format.

For the devices that support only the NGUID format, the host creates two identifiers: eui.*xxx (NGUID)* as primary and t10.*xxx_controller_serial_number* as alternative primary.

ID Formats Supported by Device		Device Identifier Generated by Host
EUI64 ID Format	**NGUID ID Format**	**ESX 8.0 and later**
yes	yes	t10.*xxx_EUI64*
yes	no	t10.*xxx_EUI64*

ID Formats Supported by Device		Device Identifier Generated by Host
EUI64 ID Format	**NGUID ID Format**	**ESX 8.0 and later**
no	yes	eui.*xxx (NGUID)* as primary ID t10.*xxx_controller_serial_number* as alternative primary ID

Verify Mapping Between Primary and Alternative Device Identifiers

Use the `esxcli storage core device uidmap list` command to verify the device identifiers. The output is similar to the following:

```
  esxcli storage core device uidmap list
eui.0000xyz.....
   Primary UID: eui.0000xyz.....
   Alternative Primary UIDs: t10.0000abc....
   Legacy UID: vml.0000000000766d68....
   Alternative Legacy UIDs: vml.000000000080906....
```

Rename Storage Devices

The ESX host assigns a display name to a storage device based on the storage type and manufacturer. You can change the display name of the device.

You cannot rename certain types of local devices.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. Select the device to rename and click **Rename**.
5. Change the device name to a friendly name.

Rescan Operations for ESX Storage

When you perform storage management tasks or make changes in the SAN configuration, you might need to rescan storage available to your ESX host.

When you perform VMFS datastore management operations, such as creating a VMFS datastore or RDM, adding an extent, and increasing or deleting a VMFS datastore, your host or vCenter automatically rescans and updates your storage. You can deactivate the automatic rescan feature by turning off the Host Rescan Filter. See Manage Filters for LUNs and vSphere VMFS Datastores.

In certain cases, you must perform a manual rescan. You can rescan all storage available to your host or to all hosts in a folder, cluster, and data center.

If the changes you make are isolated to storage connected through a specific adapter, perform a rescan for this adapter.

Perform the manual rescan each time you make one of the following changes.

- Zone a new disk array on a SAN.
- Create new LUNs on a SAN.
- Change the path masking on a host.
- Reconnect a cable.
- Change CHAP settings (iSCSI only).
- Add or remove discovery or static addresses (iSCSI only).
- Add a single host to the vCenter after you have edited or removed from the vCenter a datastore shared by the vCenter hosts and the single host.

Important:
If you rescan when a path is unavailable, the host removes the path from the list of paths to the device. The path reappears on the list as soon as it becomes available and starts working again.

Perform Storage Rescan

When you make changes in your SAN configuration, you might need to rescan storage. You can rescan all storage available to your ESX host, cluster, or data center. If the changes you make are isolated to storage accessed through a specific host, perform the rescan for only this host.

1. In the vSphere Client object navigator, browse to a host, a cluster, a data center, or a folder that contains hosts.
2. From the right-click menu, select **Storage** > **Rescan Storage**.
3. Specify extent of rescan.

Option	Description
Scan for New Storage Devices	Rescan all adapters to discover new storage devices. If new devices are discovered, they appear in the device list.
Scan for New VMFS Volumes	Rescan all storage devices to discover new datastores that have been added since the last scan. Any new datastores appear in the datastore list.

Perform Adapter Rescan

When you make changes in your SAN configuration and these changes are isolated to storage accessed through a specific adapter on ESX host, perform rescan for only this adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter to rescan from the list.
4. Click the **Rescan Adapter** icon.

Change the Number of Scanned Storage Devices

The range of scanned LUN IDs for an ESX host can be from 0 to 16,383. ESX ignores LUN IDs greater than 16,383. The configurable `Disk.MaxLUN` parameter controls the range of scanned LUN ID range. The parameter has a default value of 1024.

The `Disk.MaxLUN` parameter also determines how many LUNs the SCSI scan code attempts to discover using individual INQUIRY commands if the SCSI target does not support direct discovery using REPORT_LUNS.

You can modify the `Disk.MaxLUN` parameter depending on your needs. For example, if your environment has a smaller number of storage devices with LUN IDs from 1 through 100, set the value to 101. As a result, you can improve device discovery speed on targets that do not support REPORT_LUNS. Lowering the value can shorten the rescan time and boot time. However, the time to rescan storage devices might also depend on other factors, including the type of the storage system and the load on the storage system.

In other cases, you might need to increase the value if your environment uses LUN IDs that are greater than 1023.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. In the Advanced System Settings table, select **Disk.MaxLUN** and click the **Edit** icon.
5. Change the existing value to the value of your choice, and click **OK**.
   The value you enter specifies the LUN ID that is after the last one you want to discover.
   For example, to discover LUN IDs from 1 through 100, set **Disk.MaxLUN** to 101.

Identifying ESX Storage Connectivity Problems

When your ESX connects to a storage device, it might experiences a connectivity problem. Storage connectivity problems might be caused by various reasons. Although ESX cannot always detect the reason for a device or its paths being unavailable, the host can determine whether the problem is permanent or temporary. In other words, the host can differentiate between a permanent device loss (PDL) state of the device and a transient all paths down (APD) state of storage.

Permanent Device Loss (PDL)	A condition that occurs when a storage device permanently fails or is administratively removed or excluded. It is not expected to become available. When the device becomes permanently unavailable, ESX receives appropriate sense codes or a login rejection from storage arrays, and is able to recognize that the device is permanently lost.
All Paths Down (APD)	A condition that occurs when a storage device becomes inaccessible to the host and no paths to the device are available. ESX treats this as a transient condition because typically the problems with the device are temporary and the device is expected to become available again.

Connectivity Problems and vSphere High Availability

When the device enters the PDL or APD state, vSphere High Availability (HA) can detect connectivity problems and provide automated recovery for affected virtual machines on the ESX host. vSphere HA uses VM Component Protection (VMCP) to protect virtual machines running on the host in the vSphere HA cluster against accessibility failures. For more information about VMCP and how to configure responses for datastores and virtual machines when the APD or PDL condition occurs, see the *vSphere Availability* documentation.

Detecting PDL Conditions

A storage device is considered to be in the permanent device loss (PDL) state when it becomes permanently unavailable to your ESX host.

Typically, the PDL condition occurs when a device is unintentionally removed, or its unique ID changes, or when the device experiences an unrecoverable hardware error.

When the storage array determines that the device is permanently unavailable, it sends SCSI error sense codes or NVMe error codes to the ESX host. After receiving these errors, your host recognizes the device as failed and registers the state of the device as PDL. For the device to be considered permanently lost, the sense codes must be received on all its paths.

After registering the PDL state of the device, the host stops attempts to reestablish connectivity or to send commands to the device.

The vSphere Client displays the following information for the device:

- The operational state of the device changes to `Lost Communication`.
- All paths are shown as `Dead`.
- Datastores on the device are not available.

If no open connections to the device exist, or after the last connection closes, the host removes the PDL device and all paths to the device. You can deactivate the automatic removal of paths by setting the advanced host parameter `Disk.AutoremoveOnPDL` to 0.

If the device returns from the PDL condition, the host can discover it, but treats it as a new device. Data consistency for virtual machines on the recovered device is not guaranteed.

> **Note:**
> When a device fails without sending appropriate SCSI sense codes or NVMe error codes or an iSCSI login rejection, the host cannot detect PDL conditions. In this case, the host continues to treat the device connectivity problems as APD even when the device fails permanently.

Permanent Device Loss and SCSI Sense Codes

The following VMkernel log example of a SCSI sense code indicates that the device is in the PDL state.

```
H:0x0 D:0x2 P:0x0 Valid sense data: 0x5 0x25 0x0 or Logical Unit Not Supported
```

Permanent Device Loss and NVMe Error Codes

The following VMkernel log example of an NVMe error code indicates that the device is in the PDL state.

```
H:0x0 D:0xb P:0x0 or H:0x0 D:0x11a P:0x0
```

Permanent Device Loss and iSCSI

On iSCSI arrays with a single LUN per target, PDL is detected through iSCSI login failure. An iSCSI storage array rejects your host's attempts to start an iSCSI session with a reason `Target Unavailable`. As with the sense codes, this response must be received on all paths for the device to be considered permanently lost.

Permanent Device Loss and Virtual Machines

After registering the PDL state of the device, the host closes all I/O from virtual machines. vSphere HA can detect PDL and restart failed virtual machines.

Performing Planned Storage Device Removal

When a storage device is malfunctioning, you can avoid permanent device loss (PDL) conditions or all paths down (APD) conditions. Perform a planned removal and reconnection of a storage device.

Planned device removal is an intentional disconnection of a storage device. You might also plan to remove a device for such reasons as upgrading your hardware or reconfiguring your storage devices. When you perform an orderly removal and reconnection of a storage device, you complete a number of tasks.

Task	Description
Migrate virtual machines from the device you plan to detach.	*vCenter and Host Management*
Unmount the datastore deployed on the device.	See Unmount Datastores.
Detach the storage device.	See Detach Storage Devices.
For an iSCSI device with a single LUN per target, delete the static target entry from each iSCSI HBA that has a path to the storage device.	See Remove Dynamic or Static iSCSI Targets.
Perform any necessary reconfiguration of the storage device by using the array console.	See your vendor documentation.
Reattach the storage device.	See Attach Storage Devices.
Mount the datastore and restart the virtual machines.	See Mount Datastores.

Detach Storage Devices

Safely detach a storage device from your ESX host.

- The device does not contain any datastores.
- No virtual machines use the device as an RDM disk.
- The device does not contain a diagnostic partition or a scratch partition.

You might need to detach the device to make it inaccessible to your host, when, for example, you perform a hardware upgrade on the storage side.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. Select the device to detach and click the **Detach** icon.

The device becomes inaccessible. The operational state of the device changes to Unmounted.

If multiple hosts share the device, detach the device from each host.

Attach Storage Devices

Reattach a storage device that you previously detached from the ESX host.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. Select the detached storage device and click the **Attach** icon.

The device becomes accessible.

Recovering from PDL Conditions

An unplanned permanent device loss (PDL) condition occurs when a storage device becomes permanently unavailable without being properly detached from the ESX host.

The following items in the vSphere Client indicate that the device is in the PDL state:

- The datastore deployed on the device is unavailable.
- Operational state of the device changes to Lost Communication.
- All paths are shown as Dead.
- A warning about the device being permanently inaccessible appears in the VMkernel log file.

To recover from the unplanned PDL condition and remove the unavailable device from the host, perform the following tasks.

Task	Description
Power off and unregister all virtual machines that are running on the datastore affected by the PDL condition.	See *vSphere Virtual Machine Administration*.
Unmount the datastore.	See Unmount Datastores.
Rescan all ESX hosts that had access to the device. **Note:** If the rescan is not successful and the host continues to list the device, some pending I/O or active references to the device might still exist. Check for any items that might still have an active reference to the device or datastore. The items include virtual machines, templates, ISO images, raw device mappings, and so on.	See Perform Storage Rescan.

Handling Transient APD Conditions

A storage device is considered to be in the all paths down (APD) state when it becomes unavailable to your ESX host for an unspecified time period.

The reasons for an APD state can be, for example, a failed switch or a disconnected storage cable.

In contrast with the permanent device loss (PDL) state, the host treats the APD state as transient and expects the device to be available again.

The host continues to retry issued commands in an attempt to reestablish connectivity with the device. If the host's commands fail the retries for a prolonged period, the host might be at risk of having performance problems. Potentially, the host and its virtual machines might become unresponsive.

To avoid these problems, your host uses a default APD handling feature. When a device enters the APD state, the host turns on a timer. With the timer on, the host continues to retry non-virtual machine commands for a limited time period only.

By default, the APD timeout is set to 140 seconds. This value is typically longer than most devices require to recover from a connection loss. If the device becomes available within this time, the host and its virtual machine continue to run without experiencing any problems.

If the device does not recover and the timeout ends, the host stops its attempts at retries and stops any non-virtual machine I/O. Virtual machine I/O continues retrying. The vSphere Client displays the following information for the device with the expired APD timeout:

- The operational state of the device changes to `Dead or Error`.
- All paths are shown as `Dead`.
- Datastores on the device are dimmed.

Even though the device and datastores are unavailable, virtual machines remain responsive. You can power off the virtual machines or migrate them to a different datastore or host.

If later the device paths become operational, the host can resume I/O to the device and end the special APD treatment.

Deactivate Storage APD Handling

The storage all paths down (APD) handling on your ESX host is activated by default. When this functionality is activated and a storage device enters the APD state, the host continues to retry nonvirtual machine I/O commands only for a limited time period. After the time period expires, the host stops its retry attempts and terminates any nonvirtual machine I/O. You can deactivate the APD handling feature on your host.

If you deactivate the APD handling, the host will indefinitely continue to retry issued commands in an attempt to reconnect to the APD device. This behavior might cause virtual machines on the host to exceed their internal I/O timeout and become unresponsive or fail. The host might become disconnected from vCenter.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. In the Advanced System Settings table, select the **Misc.APDHandlingEnable** parameter and click the `Edit` icon.
5. Change the value to 0.

If you deactivate the APD handling, you can reactivate it and set its value to 1 when a device enters the APD state. The internal APD handling feature turns on immediately and the timer starts with the current timeout value for each device in APD.

Change Timeout Limits for Storage APD

The timeout parameter controls how many seconds the ESX host must retry the I/O commands to a storage device in an all paths down (APD) state. You can change the default timeout value.

The timeout period begins immediately after the device enters the APD state. After the timeout ends, the host marks the APD device as unreachable. The host stops its attempts to retry any I/O that is not coming from virtual machines. The host continues to retry virtual machine I/O.
By default, the timeout parameter on your host is set to 140 seconds. You can increase the value of the timeout if, for example, storage devices connected to your ESX host take longer than 140 seconds to recover from a connection loss.

> **Note:**
> If you change the timeout parameter after the device becomes unavailable, the change does not take effect for that particular APD incident.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. In the Advanced System Settings table, select the **Misc.APDTimeout** parameter and click the `Edit` icon.
5. Change the default value.
   You can enter a value between 20 and 99999 seconds.

Verify the Connection Status of a Storage Device on ESX Host

Use the `esxcli` command to verify the connection status of a particular storage device.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

1. Run the `esxcli storage core device list -d=`*`device_ID`* command.
2. Review the connection status in the `Status:` area.
   - `on` - Device is connected.
   - `dead` - Device has entered the APD state. The APD timer starts.
   - `dead timeout` - The APD timeout has expired.
   - `not connected` - Device is in the PDL state.

Use the Locator LED on ESX Storage Devices

Use the locator LED to identify specific storage devices connected to your ESX host, so that you can locate them among other devices. You can turn the locator LED on or off.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. From the list of storage devices, select one or more disks and set the locator LED indicator.
   - Click the **Turn On LED** icon.
   - Click the **Turn Off LED** icon.

Erase ESX Storage Devices

Certain functionalities, such as vSAN or virtual flash resource, require that your ESXhost uses clean storage devices. You can erase an HHD or flash device and remove all existing data.

- Make sure that the host is in the connected state.
- Verify that the devices you plan to erase are not in use.
- Required privilege: **Host.Config.Storage**

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. Select one or more devices and click the **Erase Partitions** icon.
5. Verify that the partition information you are erasing is not critical.
6. Click **OK** to confirm your change.

Change Perennial Reservation Settings on ESX Hosts

You can adjust the perennial reservation setting on ESX storage devices that are used as physical Raw Device Mappings (RDMs) in Windows Server Failover Clustering (WSFC) configurations.

Before marking a device as perennially reserved, verify that the device does not host a standard VMFS datastore. Storage devices backing VMFS datastores with Clustered VMDK support enabled are an exception and require a perennial reservation.

WSFC cluster nodes that are spread over several hosts require physical RDMs. The RDMs are shared among all hosts where cluster nodes run. The host with the active node holds persistent SCSI-3 reservations on all shared RDM devices. When the active node is running and devices are locked, no other host can write to the devices. If another participating host boots while the active node is holding the lock on the devices, the boot might take unusually long time because the host unsuccessfully attempts to contact the locked devices. The same issue might also affect rescan operations.

To prevent this problem, activate perennial reservation for all devices on the ESX hosts where secondary WSFC nodes with RDMs reside. This setting informs the ESX host about the permanent SCSI reservation on the devices, so that the host can skip the devices during the boot or storage rescan process.

If you later re-purpose the marked devices as standard VMFS datastores, remove the reservation to avoid unpredictable datastore behavior.

For information about WSFC clusters, see the *Setup for Windows Server Failover Clustering* documentation.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. From the list of storage devices, select the device and click one of the following icons.

Option	Description
Mark as Perennially Reserved	Mark the selected device as perennially reserved. **Note:** Repeat the procedure for each RDM device that is participating in the WSFC cluster.
Unmark as Perennially Reserved	Clear perennial reservation for the device that was previously marked.

The configuration is permanently stored with the ESX host and persists across restarts.

You can also use the `esxcli` command to mark the devices participating in the WSFC cluster.

1. Mark the devices as perennially reserved.
   ```
   esxcli storage core device setconfig -d naa.id --perennially-reserved=true
   ```
2. Verify that the device is perennially reserved.
   ```
   esxcli storage core device list -d naa.id
   ```
   In the output of the `esxcli` command, search for the entry `Is Perennially Reserved: true`.
3. To remove the perennially reserved flag, run the following command.
   ```
   esxcli storage core device setconfig -d naa.id --perennially-reserved=false
   ```

Using Flash Devices with ESX Hosts

In addition to regular storage hard disk drives (HDDs), ESX supports flash storage devices.

Unlike regular HDDs that are electromechanical devices containing moving parts, flash devices use semiconductors as their storage medium and have no moving parts. Typically, the flash devices are resilient and provide faster access to data.

To detect flash devices, ESX uses an inquiry mechanism based on T10 standards. Check with your vendor whether your storage array supports the ESX mechanism of flash device detection.

After the host detects the flash devices, you can use them for several tasks and functionalities.

If you use NVMe storage, enable the high-performance plug-in (HPP) to improve your storage performance. See VMware High Performance Plug-In and Path Selection Schemes.

For specifics about using NVMe storage with ESX, see About VMware NVMe Storage.

Table 170: Using Flash Devices with ESX

Functionality	Description
vSAN	vSAN requires flash devices. For more information, see the *Administering VMware vSAN* documentation.
VMFS Datastores	Create VMFS datastores on flash devices. Use the datastores for the following purposes: • Store virtual machines. Certain guest operating systems can identify virtual disks stored on these datastores as flash virtual disks. • Allocate datastore space for the ESX host swap cache. See Configure Host Cache with VMFS Datastore in vSphere Environment.
Virtual Flash Resource (VFFS)	If required by your vendor, set up a virtual flash resource and use it for I/O caching filters. See Filtering Virtual Machine I/O in vSphere.

Flash Devices and Virtual Machines

Guest operating systems can identify virtual disks that reside on flash-based datastores as flash virtual disks.

Guest operating systems can use standard inquiry commands such as SCSI VPD Page (B1h) for SCSI devices and ATA IDENTIFY DEVICE (Word 217) for IDE devices.

For linked clones, native snapshots, and delta-disks, the inquiry commands report the virtual flash status of the base disk.

Operating systems can detect that a virtual disk is a flash disk under the following conditions:

- Detection of flash virtual disks is supported on VMs with virtual hardware version 8 or later.
- Devices backing a shared VMFS datastore must be marked as flash on all hosts.
- If the VMFS datastore includes several device extents, all underlying physical extents must be flash-based.

Best Practices for Flash Devices on ESX Hosts

Follow these best practices when you use flash devices in vSphere environment.

- Use flash devices approved by the VMware by Broadcom Compatibility Guide.
- Make sure to use the latest firmware with flash devices. Frequently check with your storage vendors for any updates.
- Carefully monitor how intensively you use the flash device and calculate its estimated lifetime. The lifetime expectancy depends on how actively you continue to use the flash device. See Estimate Lifetime of Flash Devices.
- If you use NVMe devices for storage, enable the high-performance plug-in (HPP) to improve your storage performance. For specifics of using the NVMe devices, see VMware High Performance Plug-In and Path Selection Schemes

Monitor Flash Devices

You can monitor certain critical flash device parameters, including `Media Wearout Indicator`, `Temperature`, and `Reallocated Sector Count`, from an ESX host.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Use the `esxcli` command to monitor flash devices.

Display the flash device statistics by running the following command:

`esxcli storage core device smart get -d=`*`flash device_ID`*

Estimate Lifetime of Flash Devices

When working with flash devices, monitor how actively you use them and calculate their estimated lifetime.

Note the number of days passed since the last reboot of your ESX host. For example, ten days.

Typically, storage vendors provide reliable lifetime estimates for a flash device under ideal conditions. For example, a vendor might guarantee a lifetime of 5 years under the condition of 20 GB writes per day. However, the more realistic life

expectancy of the device depends on how many writes per day your ESX host actually generates. Follow these steps to calculate the lifetime of the flash device.

1. Obtain the total number of blocks written to the flash device since the last reboot.

   Run the `esxcli storage core device stats get -d=device_ID` command. For example:

   ```
   ~ # esxcli storage core device stats get -d t10.xxxxxxxxxxxxxxx
    Device: t10.xxxxxxxxxxxxxxx
    Successful Commands: xxxxxxx
    Blocks Read: xxxxxxxx
    Blocks Written: 629145600
    Read Operations: xxxxxxxx
   ```

   The Blocks Written item in the output shows the number of blocks written to the device since the last reboot. In this example, the value is 629,145,600. After each reboot, it resets to 0.

2. Calculate the total number of writes and convert to GB.

   One block is 512 bytes. To calculate the total number of writes, multiply the Blocks Written value by 512, and convert the resulting value to GB.

   In this example, the total number of writes since the last reboot is approximately 322 GB.

3. Estimate the average number of writes per day in GB.

   Divide the total number of writes by the number of days since the last reboot.

   If the last reboot was ten days ago, you get 32 GB of writes per day. You can average this number over the time period.

4. Estimate lifetime of your device by using the following formula:

   vendor provided number of writes per day times *vendor provided life span* divided by *actual average number of writes per day*

   For example, if your vendor guarantees a lifetime of 5 years under the condition of 20 GB writes per day, and the actual number of writes per day is 30 GB, the life span of your flash device will be approximately 3.3 years.

Keeping Flash Disks VMFS-Free

If you use the auto-partitioning boot option when installing or auto-deploying ESX, the auto-partitioning option creates a VMFS datastore on your host's local storage. In certain cases, you need to keep your local storage flash disks unformatted.

By default, auto-partitioning deploys VMFS file systems on any unused local storage disks on your host, including flash disks.

However, a flash disk formatted with VMFS becomes unavailable for such features as virtual flash and vSAN. Both features require an unformatted flash disk and neither can share the disk with any other file system.

To ensure that auto-partitioning does not format the flash disk with VMFS, use the following boot options when you install ESX or boot the ESX host for the first time:

- `autoPartition=TRUE`
- `skipPartitioningSsds=TRUE`

If you use Auto Deploy, set these parameters on a reference ESX host.

1. In the vSphere Client, navigate to the host to use as a reference host and click the **Configure** tab.
2. Click **System** to open the system options, and click **Advanced System Settings**.
3. Set the following items.

Parameter	Value
VMkernel.Boot.autoPartition	True
VMkernel.Boot.skipPartitioningSsds	True

4. Reboot the host.

If flash disks that you plan to use with the virtual flash resource and vSAN already have VMFS datastores, remove the datastores.

Marking Storage Devices on ESX Hosts

When you configure vSAN or set up a virtual flash resource, your storage environment must include local flash devices. However, ESX might not recognize certain storage devices as flash devices when their vendors do not support automatic flash device detection. In other cases, certain devices might not be detected as local, and ESX marks them as remote. When devices are not recognized as the local flash devices, they are excluded from the list of devices offered for vSAN or virtual flash resource. Marking these devices as local flash makes them available for vSAN and virtual flash resource.

Mark Storage Devices as Flash

If ESX does not recognize its devices as flash, mark them as flash devices.

Verify that the device is not in use.

ESX does not recognize certain devices as flash when their vendors do not support automatic flash disk detection. The Drive Type column for the devices shows HDD as their type.

> **CAUTION:**
> Marking the HDD devices as flash might deteriorate the performance of datastores and services that use them. Mark the devices only if you are certain that they are flash devices.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. From the list of storage devices, select one or several HDD devices and click the **Mark as Flash Disk** ( ) icon.
5. Click **Yes** to save your changes.

The type of the devices changes to flash.
If the flash device that you mark is shared among multiple hosts, make sure that you mark the device from all hosts that share the device.

Mark Storage Devices as Local

ESX enables you to mark devices as local. This action is useful in cases when ESX is unable to determine whether certain devices are local.

- Make sure that the device is not shared.
- Power off virtual machines that reside on the device and unmount an associated datastore.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. From the list of storage devices, select one or several remote devices and click the **Mark as Local** icon.
5. Click **Yes** to save your changes.

About Virtual Flash Resource in vSphere Environment

You can aggregate local flash devices on an ESX host into a single virtualized caching layer called virtual flash resource. When you set up the virtual flash resource, you create a new file system, Virtual Flash File System (VFFS). VFFS is a derivative of VMFS, which is optimized for flash devices and is used to group the physical flash devices into a single caching resource pool. As a non-persistent resource, it cannot be used to store virtual machines.

After you set up the virtual flash resource, you can use it for I/O caching filters. See Filtering Virtual Machine I/O in vSphere.

Considerations for Virtual Flash Resource

When you configure a virtual flash resource, several considerations apply.

- You can have only one virtual flash resource on a single ESX host. The virtual flash resource is managed at the host's level.
- You cannot use the virtual flash resource to store virtual machines. Virtual flash resource is a caching layer only.
- You can use only local flash devices for the virtual flash resource.
- You can create the virtual flash resource from mixed flash devices. All device types are treated equally and no distinction is made between SAS, SATA, or PCI express connectivity. When creating the resource from mixed flash devices, make sure to group similar performing devices together to maximize performance.
- You cannot use the same flash devices for the virtual flash resource and vSAN. Each requires its own exclusive and dedicated flash device.

Set Up Virtual Flash Resource

You can set up a virtual flash resource or add capacity to existing virtual flash resource.

To set up a virtual flash resource, you use local flash devices connected to your host or host cluster. To increase the capacity of your virtual flash resource, you can add more devices, up to the maximum number indicated in the *Configuration Maximums* documentation. An individual flash device must be exclusively allocated to the virtual flash resource. No other vSphere functionality, such as vSAN or VMFS, can share the device with the virtual flash resource.

1. In the vSphere Client, navigate to the host.
2. Click the **Configure** tab.
3. Under **Virtual Flash**, click **Virtual Flash Resource Management**.
4. Click one of the following options.

Option	Description
Add Capacity	If you are creating the virtual flash resource on an individual host.
Add Capacity on Cluster	If you are creating the virtual flash resource on a cluster.

5. From the list of available entities, select one or more to use for the virtual flash resource and click **OK**.

   If your flash devices do not appear on the list, see Marking Storage Devices on ESX Hosts.

Option	Description
Local VMware Disk	Select any combination of unclaimed flash devices. ESX creates the VFFS volume on one of the devices and then extends the volume on the rest of the devices. The system configures the virtual flash resource on the entire VFFS volume. If a VFFS volume exists on your host, you cannot select any unclaimed devices without first selecting the existing VFFS volume.
volume ID **- Configure using the existing VFFS volume extents**	If you previously created a VFFS volume on one of the host's flash devices using the `vmkfstools` command, the volume also appears on the list of eligible entities. You can select just this volume for the virtual flash resource. Or combine it with the unclaimed devices. ESX uses the existing VFFS volume to extend it over other devices.

The virtual flash resource is created. The Device Backing area lists all devices that you use for the virtual flash resource. Use the virtual flash resource for I/O caching filters developed through vSphere APIs for I/O Filtering. See Using Cache I/O Filters with Flash Storage Devices on ESX Hosts.

You can increase the capacity by adding more flash devices to the virtual flash resource.

Remove Virtual Flash Resource

You might need to remove a virtual flash resource deployed on local flash devices connected to the ESX host. Removing the virtual flash resource frees the devices for other services.

- Verify that the virtual flash resource is not used for I/O filters.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Virtual Flash**, click **Virtual Flash Resource Management** and click **Remove All**.

After you remove the virtual flash resource and erase the flash device, the device is available for other operations.

Set Alarm for Virtual Flash Use

Set an alarm to indicate when the use of a virtual flash resource on your ESX host exceeds specified threshold.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. Select the setting to change and click the **Edit** button.

Parameter	Description
VFLASH.ResourceUsageThreshold	The system triggers the `Host vFlash resource usage` alarm when a virtual flash resource use exceeds the threshold. The default threshold is 80%. You can change the threshold to an appropriate value. The alarm is cleared when the virtual flash resource use drops below the threshold.

5. Click **OK**.

Configure Host Cache with VMFS Datastore in vSphere Environment

Enable your ESX host to swap to the host cache. You can also change the amount of space allocated for the host cache.

Create a VMFS datastore using flash devices as backing. See Create a vSphere VMFS Datastore.

> **Note:**
>
> This functionality will be deprecated in the future vSphere releases.

Your ESX hosts can use a portion of a flash-backed storage entity as a swap cache shared by all virtual machines.

The host-level cache is made up of files on a low-latency disk that ESX uses as a write-back cache for virtual machine swap files. All virtual machines running on the host share the cache. Host-level swapping of virtual machine pages makes the best use of potentially limited flash device space.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Host Cache Configuration**.
4. Select the flash datastore in the list and click the **Edit** icon.
5. Allocate appropriate space for host cache.
6. Click **OK**.

About VMware NVMe Storage

Non-Volatile Memory (NVM) storage devices that use persistent memory have become popular in data centers. To connect to local and networked NVM devices, ESX uses the NVM Express (NVMe) protocol, a standardized protocol designed specifically for high-performance multi-queue communication with NVM devices.

VMware NVMe Concepts

Before you begin working with NVMe storage in the ESX environment, learn about basic NVMe concepts.

NVM Express (NVMe) NVMe is a method for connecting and transferring data between a host and a target storage system. NVMe is designed for use with faster storage media equipped with non-volatile memory, such as flash devices. This type of storage can achieve low latency, low CPU usage, and high performance, and generally serves as an alternative to SCSI storage.

NVMe Transports The NVMe storage can be directly attached to a host using a PCIe interface or indirectly through different fabric transports. VMware NVMe over Fabrics (NVMe-oF) provides a distance connectivity between a host and a target storage device on a shared storage array.

The following types of transports for NVMe currently exist.

NVMe Transport	ESX Support
NVMe over PCIe	Local storage.
NVMe over RDMA	Shared NVMe-oF storage. With the RoCE v2 technology.
NVMe over Fibre Channel	Shared NVMe-oF storage.
NVMe over TCP	Shared NVMe-oF storage.

For specific configuration requirements for each transport, see Requirements for VMware NVMe Storage.

NVMe Namespaces In the NVMe storage array, a namespace is a storage volume backed by some quantity of non-volatile memory. In the context of ESX, the namespace is analogous to a storage device, or LUN. After your ESX host discovers the NVMe namespace, a flash device that represents the namespace appears on the list of storage devices in the vSphere Client. You can use the device to create a datastore and store virtual machines.

Targets can present namespaces, equivalent to LUNs in SCSI, to a host in active/active or asymmetric access modes. ESX can discover and use namespaces presented in either way. ESX supports *NVMe end to end*, and *without any NVMe to SCSI* emulation.

NVMe Controllers A controller is associated with one or several NVMe namespaces and provides an access path between the ESX host and the namespaces in the storage array. To access the controller, the host can use two mechanisms, controller discovery and controller connection. For information, see Add Controllers for NVMe over Fabrics.

Controller Discovery With this mechanism, the ESX host first contacts a discovery controller. The discovery controller returns a list of available controllers. After you select a controller for your host to access, all namespaces associated with this controller become available to your host.

Controller Connection Your ESX host connects to the controller that you specify. All namespaces associated with this controller become available to your host.

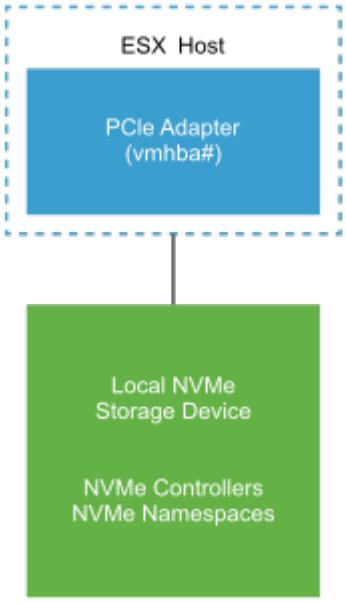
NVMe Subsystem Generally, an NVMe subsystem is a storage array that might include several NVMe controllers, several namespaces, a non-volatile memory storage medium, and an interface between the controller and non-volatile memory storage medium. The subsystem is identified by a subsystem NVMe Qualified Name (NQN).

VMware High-Performance Plug-in (HPP) By default, the ESX host uses the HPP to claim the NVMe-oF targets. When selecting physical paths for I/O requests, the HPP applies an appropriate Path Selection Scheme (PSS). For information about the HPP, see VMware High Performance Plug-In and Path Selection Schemes. To change the default path selection mechanism, see Change the Path Selection Policy.

Basic VMware NVMe Architecture and Components

ESX supports local NVMe over PCIe storage and shared NVMe-oF storage, such as NVMe over Fibre Channel, NVMe over RDMA (RoCE v2), and NVMe over TCP.

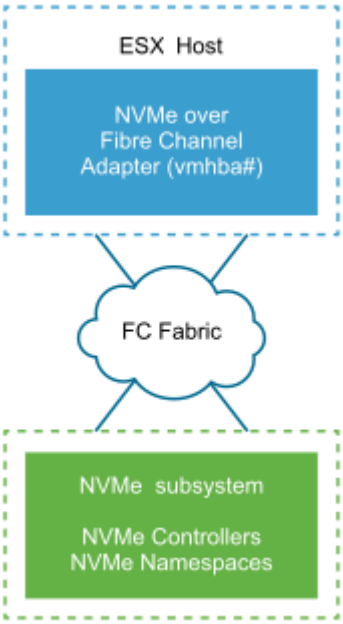
VMware NVMe over PCIe In this configuration, your ESX host uses a PCIe storage adapter to access one or more local NVMe storage devices. After you install the adapter on the host, the host discovers available NVMe devices, and they appear in the list of storage devices in the vSphere Client.



VMware NVMe over Fibre Channel

This technology maps NVMe onto the Fibre Channel protocol to enable the transfer of data and commands between a host and a target storage device. This transport can use existing Fibre Channel infrastructure upgraded to support NVMe.

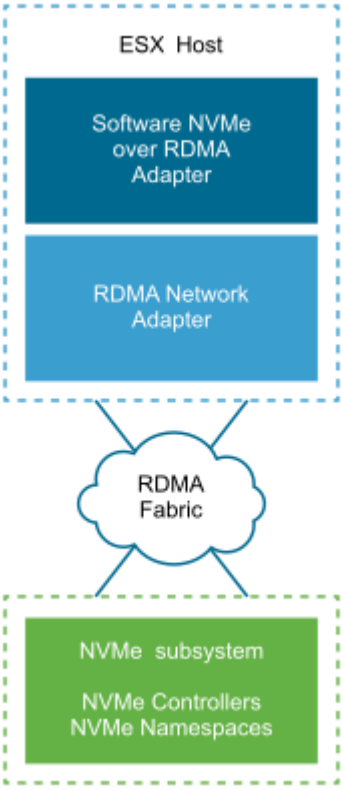
To access the NVMe over Fibre Channel storage, install a Fibre Channel storage adapter that supports NVMe on your ESX host. You do not need to configure the adapter. It automatically connects to an appropriate NVMe subsystem and discovers all shared NVMe storage devices that it can reach. You can later reconfigure the adapter and disconnect its controllers or connect other controllers that were not available during the host boot. For more information, see Add Controllers for NVMe over Fabrics.



NVMe over RDMA (RoCE v2)

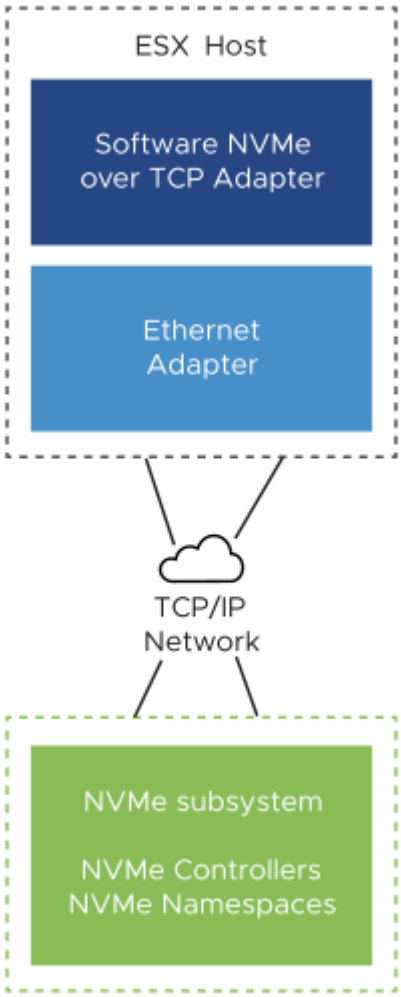
This technology uses a remote direct memory access (RDMA) transport between two systems on the network. The transport enables data exchange in the main memory bypassing the operating system or the processor of either system. ESX supports RDMA over Converged Ethernet v2 (RoCE v2) technology, which enables a remote direct memory access over an Ethernet network.

To access storage, the ESX host uses an RDMA network adapter installed on your host and a software NVMe over RDMA storage adapter. You must configure both adapters to use them for storage discovery. For more information, see Configuring NVMe over RDMA (RoCE v2) on ESX.



NVMe over TCP

This technology uses Ethernet connections between two systems. To access storage, the ESX host uses a network adapter installed on your host and a software NVMe over TCP storage adapter. You must configure both adapters to use them for storage discovery. For more information, see Configuring NVMe over TCP on ESX.



NVMe over Fabrics Authentication

NVMe storage security relies on two fundamental functions. Authentication, which verifies the identify of the NVMe entity, and a secure channel to guarantee the integrity of transferred data. Starting with VCF 9.0, ESX supports in-band authentication using the DH-HMAC CHAP protocol with NVMe-oF storage.

The VCF 9.0 release supports unidirectional authentication that is initiated by the target. If a remote NVMe target, such as a storage array or controller, requests authentication, ESX uses the DH-HMAC CHAP protocol to authenticate with the target.

If authentication is successful, ESX can establish and maintain a connection with the NVMe target. Otherwise, ESX cannot establish the connection and NVMe namespaces remain undiscoverable.

Key Considerations

- The functionality is enabled only if a remote target requests ESX to authenticate. NVMe controllers must support authentication.
- The host must authenticate on every queue of each NVMe controller.
- Because the authentication is unidirectional, the ESX host does not request authentication from the target.
- Authentication occurs only once during the administrator and I/O queue connection set-up phase before these queues can be used for I/O traffic. This transaction is not exposed to VMs.

Perform a Key Interchange for NVMe over Fabrics Authentication

To be able to authenticate the ESX host, the NVMe entity must know the key that identifies the host. When configuring the NVMe controller, you can communicate this key to the controller.

1. Use the `esxcli nvme auth get` command to get the host key.

   You can see the output similar to the following:

   ```
   Host Secret: DHHC-1:03:MNKVsxxxxxx
   ```

2. Use the `esxcli nvme info get` command to get the NQN of the host.

   ```
   Host NQN: nqn.2014-08.com.vmware.xxxxxx
   ```

3. Communicate the key to the target controller.

   In the target controller UI, enter the ESX NQN and the host key.

   The controller UI is vendor-specific. For information, see the vendor documentation.

   Note:

   ESX supports only the sha_512 algorithm and 3072_bit DH group. Make sure to select the appropriate algorithm and DH group.

   After you enter the host's information, the authentication transaction begins and the host is able to connect to subsystems.

4. Verify the host's connection to the NVMe controller.
   a) Use the `esxcli nvme controller list` command to verify that the host can establish connection with the NVMe controller.

      The output shows available subsystems and established I/O queues.
   b) Use the `esxcli nvme namespace list` to list namespaces available from the controller.

Requirements for VMware NVMe Storage

Learn how to prepare your ESX environment for NVMe storage. Configuration requirements might change depending on the type of NVMe transport you use. If you use NVMe over RDMA (RoCE v2), in addition to general requirements, you must also configure lossless Ethernet network.

Requirements for NVMe over PCIe

Your ESX storage environment must include the following components:

- Local NVMe storage devices.
- Compatible ESX host.
- Hardware NVMe over PCIe adapter. After you install the adapter, your ESX host detects it and displays in the vSphere Client as a storage adapter (vmhba) with the protocol indicated as PCIe. You do not need to configure the adapter.

Requirements for NVMe over RDMA (RoCE v2)

- NVMe storage array with NVMe over RDMA (RoCE v2) transport support.
- Compatible ESX host.
- Ethernet switches supporting a lossless network.
- Network adapter that supports RDMA over Converged Ethernet (RoCE v2). To configure the adapter, see Configuring NVMe over RDMA (RoCE v2) on ESX.
- Software NVMe over RDMA adapter. This software component must be enabled on your ESX host and connected to an appropriate network RDMA adapter. For information, see Add Software NVMe over RDMA or NVMe over TCP Adapters.
- NVMe controller. You must add a controller after you configure the software NVMe over RDMA adapter. See Add Controllers for NVMe over Fabrics.
- Lossless Ethernet. See Configuring Lossless Ethernet for NVMe over RDMA.

Requirements for NVMe over Fibre Channel

- Fibre Channel storage array that supports NVMe. For information, see Using ESX with Fibre Channel SAN.

- Compatible ESX host.
- Hardware NVMe adapter. Typically, it is a Fibre Channel HBA that supports NVMe. When you install the adapter, your ESX host detects it and displays in the vSphere Client as a standard Fibre Channel adapter (vmhba) with the storage protocol indicated as NVMe. You do not need to configure the hardware NVMe adapter to use it.
- NVMe controller. You do not need to configure the controller. After you install the required hardware NVMe adapter, it automatically connects to all targets and controllers that are reachable at the moment. You can later disconnect the controllers or connect other controllers that were not available during the host boot. See Add Controllers for NVMe over Fabrics.

Requirements for NVMe over TCP

- NVMe storage array with NVMe over TCP transport support.
- Compatible ESX host.
- An Ethernet adapter. To configure the adapter, see Configuring NVMe over TCP on ESX.
- Software NVMe over TCP adapter. This software component must be enabled on your ESX host and connected to an appropriate network adapter. For more information, see Add Software NVMe over RDMA or NVMe over TCP Adapters.
- NVMe controller. You must add a controller after you configure the software NVMe over TCP adapter. See Add Controllers for NVMe over Fabrics.

VMware NVMe over Fabrics Shared Storage Support

In the ESX environment, the NVMe storage devices appear similar to SCSI storage devices, and can be used as shared storage. Follow these rules when using the NVMe-oF storage.

- Do not mix transport types to access the same namespace.
- Make sure that active paths are presented to the host. The namespaces cannot be registered until the active path is discovered.

The following table compares the level of support that SCSI and NVMe storage provide in the ESX environment.

Table 171: SCSI and NVMe Storage Comparison

Shared Storage Functionality	SCSI over Fabric Storage	NVMe over Fabric Storage
RDM	Supported	Not supported
Core dump	Supported	Not supported
SCSI-2 reservations	Supported	Not supported
Clustered VMDK	Supported	Supported
Shared VMDK with multi-writer flag	Supported	Supported Starting with vSphere 7.0 Update 1. For more information, see the Knowledge Base article 313527.
Virtual Volumes	Supported	Supported In vSphere 8.0 and later. For more information, see NVMe and Virtual Volumes in vSphere.
Hardware acceleration with VAAI plug-ins	Supported	Not supported
Default MPP	NMP	HPP (NVMe-oF targets cannot be claimed by NMP)

The following table lists the features specific to NVMe that ESX supports.

Table 172: NVMe Specific Features

Feature	Description
Dispersed NVMe Namespaces	ESX 9.1 introduces support for dispersed NVMe namespaces. This feature allows a single namespace to be accessed by controllers across multiple NVM subsystems in a stretched cluster environment. With this enhancement, storage partners can support advanced workflows such as synchronous copy of data and remote replication. In these workflows, data can be mirrored or migrated between different physical storage units while maintaining a single logical namespace.

Configuring Lossless Ethernet for NVMe over RDMA

NVMe over RDMA in ESX requires lossless Ethernet network.

To establish lossless networks, you can select one of the available QoS settings.

Enable Global Pause Flow Control

In this network configuration, ensure global pause flow control is enabled on the network switch ports. Also, ensure that RDMA capable NICs in the host auto-negotiate to the correct flow control automatically.

To check flow control, run the following command:

```
#esxcli network nic get -n vmnicX
   Pause RX: true
   Pause TX: true
```

If the above command options are not set to true, run the following command.

```
#esxcli network nic pauseParams set -r true -t true -n vmnicX
```

Enable Priority Flow Control

For RoCE traffic to be lossless, you must configure the PFC priority value to 3 in the physical switch and hosts. You can configure the PFC in the ESX host in two ways:

- Automatic Configuration. Apply DCB PFC configuration automatically on the host RNIC, if the RNIC driver supports DCB and DCBx.
  You can verify the current DCB settings by running the following command:
  ```
  #esxcli network nic dcb status get -n vmnicX
  ```
- Manual configuration. In some cases, the RNIC drivers provide a method to manually configure the DCB PFC using driver specific parameters. To use this method, see vendor specific driver documentation. For example, in Mellanox ConnectX-4/5 driver, you can set the PFC priority value to three by running the following command and then rebooting the host.
  ```
  #esxcli system module parameters set -m nmlx5_core -p "pfctx=0x08 pfcrx=0x08"
  ```

Enable DSCP Based PFC

DSCP based PFC is another way to configure lossless network. In physical switches and hosts, you must set the DSCP value to 26. To use this option, see vendor specific driver documentation. For example, in Mellanox ConnectX-4/5 driver, you can set the DSCP tag value to 26 by running the following commands.

1. Enable PFC and DSCP trust mode.
   ```
   #esxcli system module parameters set -m nmlx5_core -p "pfctx=0x08 pfcrx=0x08 trust_state=2"
   ```
2. Set DSCP value to 26.
   ```
   #esxcli system module parameters set -m nmlx5_rdma -p "dscp_force=26"
   ```
3. Verify parameters to check to confirm if settings are correct and set.

```
esxcli system module parameters list -m nmlx5_core | grep 'trust_state\|pfcrx\|pfctx'
```

4. Reboot the host.

Configuring NVMe over RDMA (RoCE v2) on ESX

The adapter configuration process on the ESXhost involves setting up VMkernel binding for an RDMA network adapter, and then adding a software NVMe over RDMA adapter. You can then add an NVMe controller.

The entire configuration process includes these actions.

Action	Description
View RDMA Network Adapters	On your ESX host, install a network adapter that supports RDMA (RoCE v2). For example, Mellanox Technologies MT27700 Family ConnectX-4. After you install the network adapter, use the vSphere Client to review the RDMA adapter and a physical network adapter.
Configure VMkernel Binding for the RDMA Adapter	Port binding for NVMe over RDMA involves creating a switch and connecting the physical network adapter and the VMkernel adapter to the switch. Through this connection, the RDMA adapter becomes bound to the VMkernel adapter. In the configuration, you can use a vSphere standard switch or a vSphere distributed switch.
Add Software NVMe over RDMA or NVMe over TCP Adapters	Use the vSphere Client to activate the software storage adapters for NVMe over RDMA.
Add Controllers for NVMe over Fabrics	Use the vSphere Client to add an NVMe controller. After you add the controller, the NVMe namespaces associated with the controller become available to your ESX host. The NVMe storage devices that represent the namespaces in the ESX environment appear on the storage devices list.

The following video walks you through the steps of configuring NVMe over RDMA adapters.

View RDMA Network Adapters

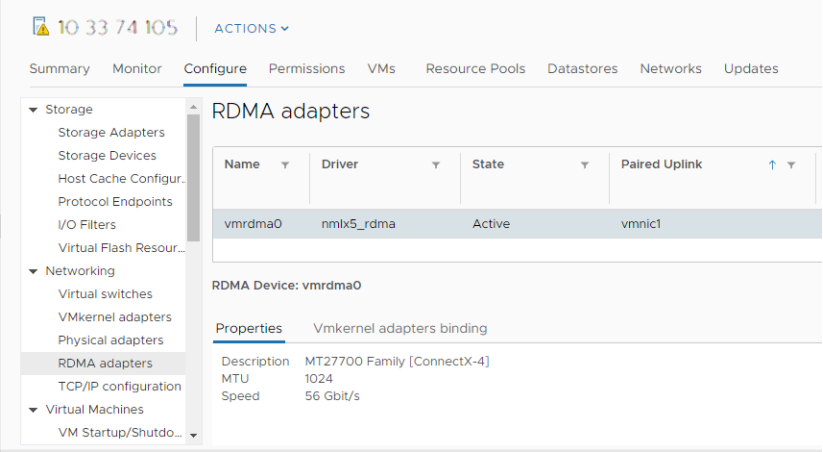
After you install a network adapter that supports RDMA (RoCE v2) on your ESX host, use the vSphere Client to review the RDMA adapter and a physical network adapter.

1. On your ESX host, install an adapter that supports RDMA (RoCE v2), for example, Mellanox Technologies MT27700 Family ConnectX-4.

   The host discovers the adapter and the vSphere Client displays its two components, an RDMA adapter and a physical network adapter.

2. In the vSphere Client, verify that the RDMA adapter is discovered by your host.
   a) Navigate to the host.
   b) Click the **Configure** tab.
   c) Under **Networking**, click **RDMA adapters**.

      In this example, the RDMA adapter appears on the list as `vmrdma0`. The **Paired Uplink** column displays the network component as the `vmnic1` physical network adapter.

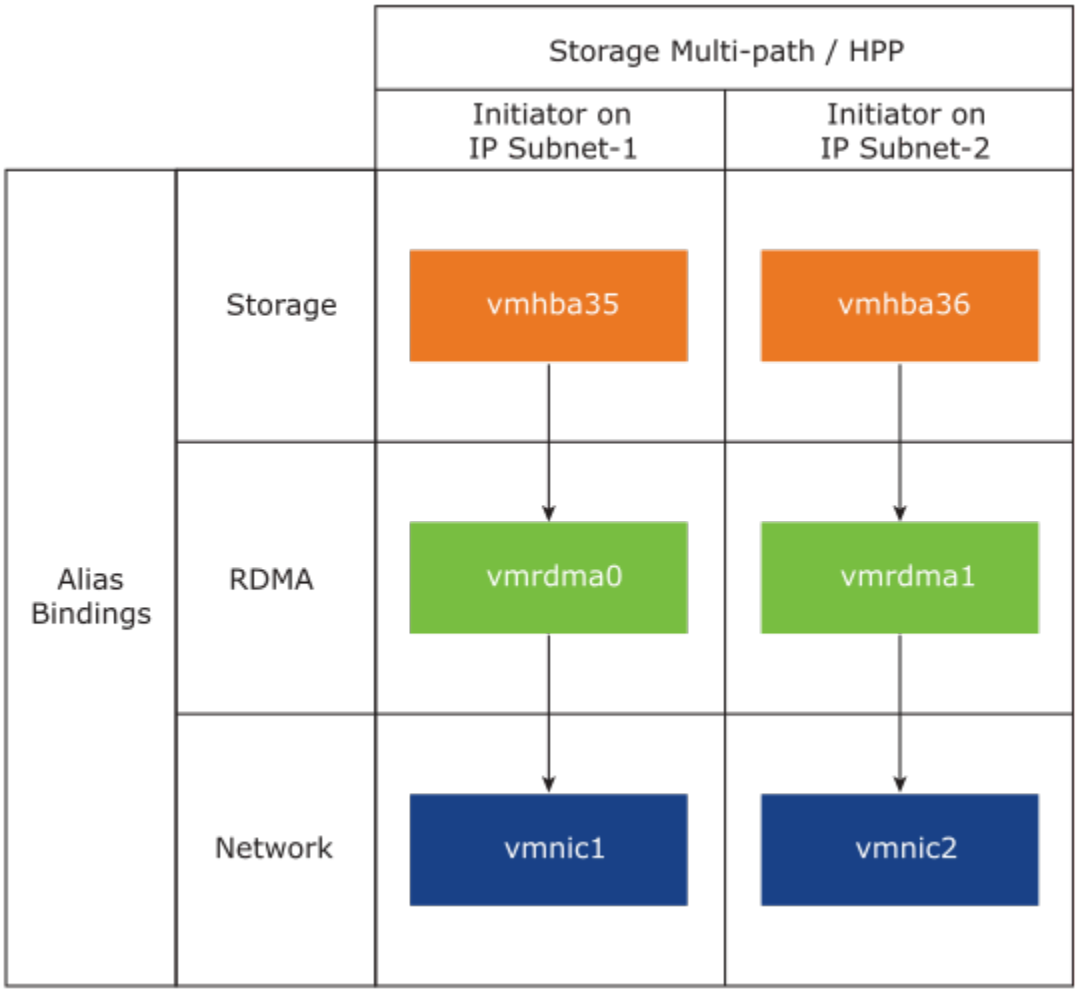


d) To verify the description of the adapter, select the RDMA adapter from the list, and click the **Properties** tab.

Configure VMkernel Binding for the RDMA Adapter

Port binding for NVMe over RDMA involves creating a switch and connecting the physical network adapter and the VMkernel adapter to the switch. Through this connection, the RDMA adapter becomes bound to the VMkernel adapter. In the configuration, you can use a vSphere standard switch or a vSphere distributed switch.

The following diagram displays the port binding for the NVMe over RDMA adapter.

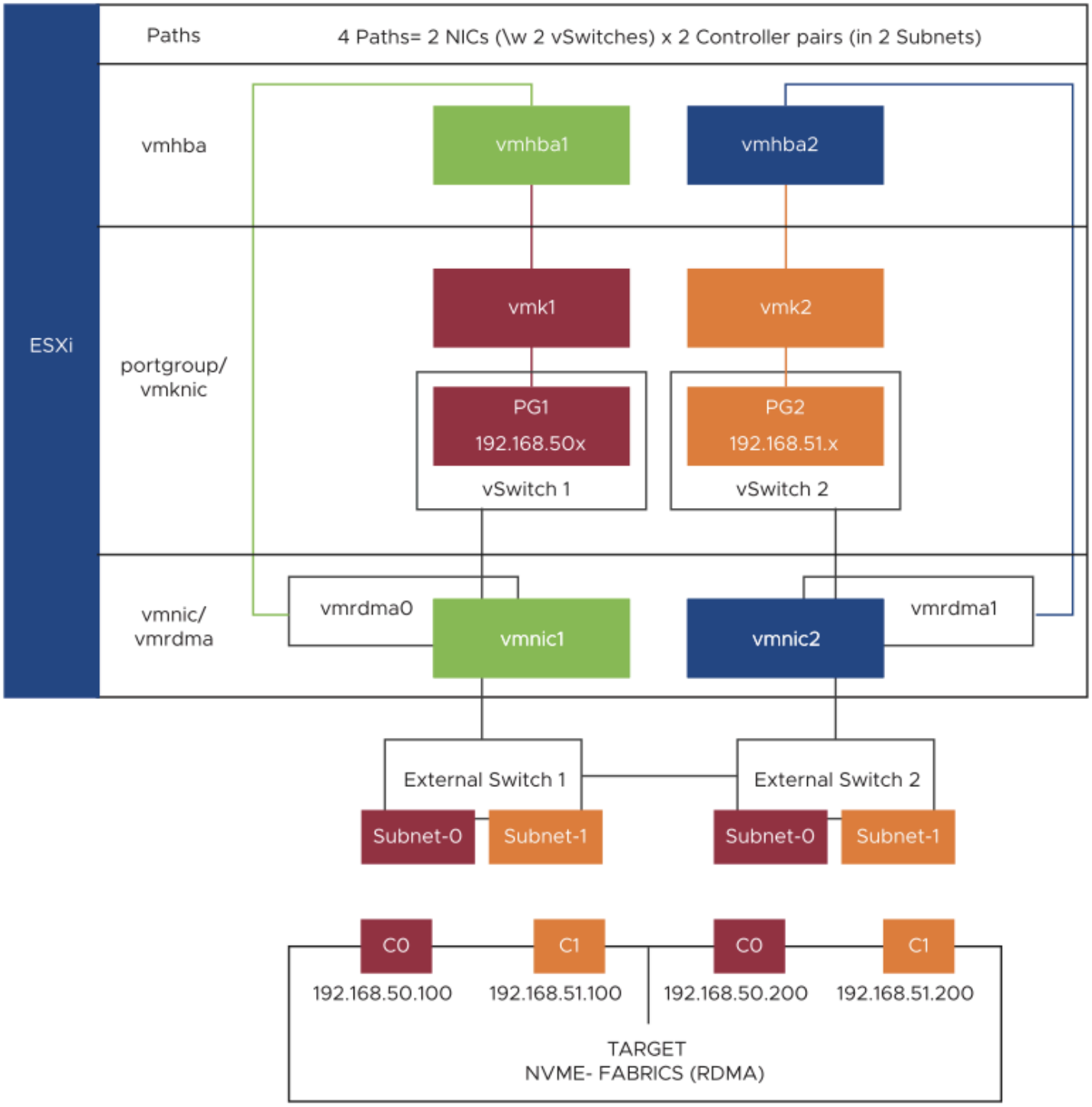


For more information about creating switches, see *Create a vSphere Standard Switch* or *Create a vSphere Distributed Switch* in the *vSphere Networking* documentation.

Example of Network Topology with NVMe over RDMA

In this example, two vSphere standard switches and two uplinks (RDMA capable NICs) provide high availability. They connect to two controller pairs in two subnets.

HA with Multiple vSwitches and Multiple Uplinks (RNICs)



Configure VMkernel Binding with a vSphere Standard Switch

You can configure VMkernel port binding for the RDMA adapter using a vSphere standard switch and one uplink per switch. Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter. You use 1:1 mapping between each virtual and physical network adapter.

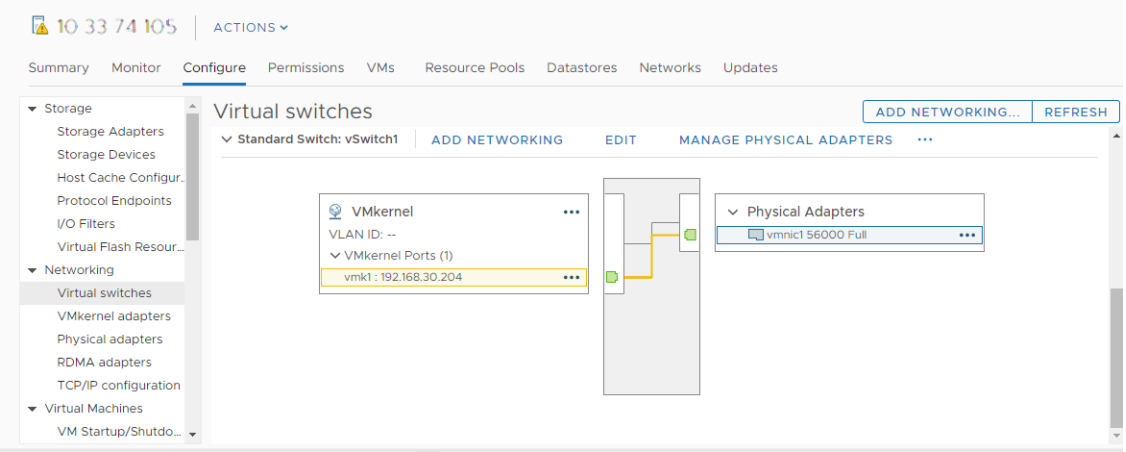
1. Create a vSphere standard switch with a VMkernel adapter and the network component.
   a) In the vSphere Client, select your host and click the **Networks** tab.
   b) Click **Actions** > **Add Networking**.
   c) Select **VMkernel Network Adapter** and click **NEXT**.
   d) Select **New standard switch** and click **NEXT**.
   e) Under **Assigned adapters**, click **+**.

      The list of available physical adapters is displayed.
   f) Select the required physical adapter `vmnic`, and click **OK**.

      Note:
      Ensure to select the physical network adapter that corresponds to the RDMA adapter. To see the association between the RDMA adapter `vmrdma`, and the physical network adapter `vmnic`, see View RDMA Network Adapters.
   g) Under **VMkernel port settings**, enter the required values.

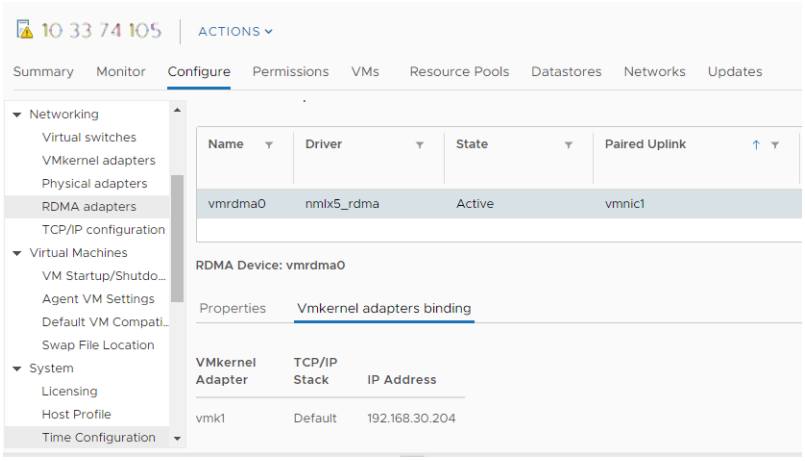
      If you are using VLAN for the storage path, enter the VLAN ID.
   h) In the **IP settings** list, enter the VMkernel IPv4 settings.
   i) Under Available services, select **NVMe over RDMA**.
2. Verify that your switch is correctly configured.
   a) On the **Configure** tab, select **Virtual switches** under **Networking**.
   b) Expand the switch and verify its configuration.

      The illustration shows that the physical network adapter and the VMkernel adapter are connected to the vSphere standard switch. Through this connection, the RDMA adapter is bound to the VMkernel adapter.



3. Verify the configuration of the VMkernel binding for the RDMA adapter.
   a) Under **Networking** list, click **RDMA adapters**, and select the RDMA adapter from the list.
   b) Click the **VMkernel adapters binding** tab and verify that the associated VMkernel adapter appears on the page.

      In this example, the `vmrdma0` RDMA adapter is paired to the `vmnic1` network adapter and is connected to the `vmk1` VMkernel adapter.



Configure VMkernel Binding with a vSphere Standard Switch and NIC Teaming

You can configure VMkernel port binding for the RDMA adapter using a vSphere standard switch with the NIC teaming configuration. You can use NIC teaming to achieve network redundancy. You can configure two or more network adapters (NICs) as a team for high availability and load balancing.

1. Create a vSphere standard switch with a VMkernel adapter and the network component with the NIC teaming configuration.
   a) In the vSphere Client, select your host and click **Networks** tab.
   b) Click **Actions** > **Add Networking**.
   c) Select **VMkernel Network Adapter** and click **NEXT**.
   d) Select **New standard switch** and click **NEXT**.
   e) Under **Assigned adapters**, click **+**.
      A list of available physical adapters is displayed.
   f) Select the required physical adapter `vmnic`, and add it under **Active adapters**.
   g) Select another physical adapter `vmnic`, and add it under **Unused adapters**.
   h) Under **VMkernel port settings**, enter the required values.
      If you are using VLAN for the storage path, enter the VLAN ID.
   i) In the **IP settings** list, specify VMkernel IPv4 settings.
   j) Under Available services, select **NVMe over RDMA**.

   Repeat step 1 to configure an existing standard switch.
2. Configure your switch for NIC teaming configuration.
   a) Click the **Configure** tab, and select **Virtual switches** under **Networking**.
   b) Select the appropriate VMkernel adapter.
   c) From the right-click menu, click **Edit Settings**.
   d) Select **Teaming and Failover**.
   e) Under **Active adapters**, move the required physical adapter `vmnic`.
   f) Under **Standby adapters** > **Failover order**, move the other physcial adapters.
   g) Set appropriate load balancing and other properties.
   h) Repeat the steps to configure additional VMkernel adapters.
3. Repeat steps 1 and 2 to add and configure additional set of teamed `rnic`s. To verify if the adapter is configured, click the **Configure** tab and select **VMkernel adapters**

Configure VMkernel Binding with a vSphere Distributed Switch

You can configure VMkernel port binding for the RDMA adapter using a vSphere distributed switch and one uplink per switch. Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter. You use 1:1 mapping between each virtual and physical network adapter.

1. Create a vSphere distributed switch with a VMkernel adapter and the network component.
   a) In the vSphere Client, select **Datacenter**, and click the **Networks** tab.
   b) Click **Actions**, and select **Distributed Switch** > **New Distributed Switch**.
   c) Select a name for the switch.
      Ensure that the location of the data center is present within your host, and click **Next**.
   d) Select a compatible ESX version, and click **Next**.
   e) Enter the required number of uplinks, and click **Finish**.
2. Add one or more hosts to your distributed virtual switch.
   a) In the vSphere Client, select **Datacenter**, and click **Distributed Switches**.
      A list of available DSwitches appears.
   b) Right-click the DSwitch, and select **Add and Manage Hosts** from the menu.
   c) Select **Add hosts**, and click **Next**.
   d) Select your host, and click **Next**.
   e) Select **Assign uplink**.
   f) Enter the relevant uplink to assign the `vmnic`.
   g) Assign a VMkernel adapter, and click **Next**.
   h) In the vSphere Client, select the DSwitch, and click the **Ports** tab.
      You can view the uplinks created for your switch here.
3. Create distributed port groups for the NVMe over RDMA storage path.
   a) In the vSphere Client, select the required DSwitch.
   b) Click **Actions** and select **Distributed Port Group** > **New Distributed Port Group**.
   c) Under **Configure Settings**, enter the general properties of the port group.
      If you have configured a specific VLAN, add it in the VLAN ID.

      Note:
      Network connectivity issues might occur if you do not configure VLAN properly.
4. Configure the VMkernel adapters.
   a) In the vSphere Client, expand the **DSwitch** list, and select the distributed port group.
   b) Click **Actions** > **Add VMkernel Adapters**.
   c) In the **Select Member Hosts** dialog box, select your host and click **OK**.
   d) In the **Configure VMkernel Adapter** dialog box, ensure that the MTU matches to the Switch MTU.
   e) Under **Available services**, select **NVMe over RDMA** for appropriate tagging.
   f) Click **Finish**.
   g) Repeat step b and step c to add multiple RDMA capable NICs.
5. Set NIC teaming policies for the distributed port groups.
   a) In the **Distributed Port Group**, click **Actions** > **Edit Settings**.
   b) Click **Teaming and Failover**, and verify the active uplinks.
   c) Assign one uplink as **Active** for the port group, and the other uplink as **Unused**.
      Repeat step c for each of the port groups created.

After you complete the configuration, click **Configure**, and verify whether the physical adapter tab on your host lists the DVSwitch for the NICs selected.

Add Software NVMe over RDMA or NVMe over TCP Adapters

ESX supports NVMe over RDMA and NVMe over TCP software adapters. Use thevSphere Client to add the software storage adapters for NVMe over RDMA or NVMe over TCP.

- On yourESX host, install an adapter that supports the following types of storage.
  – NVMe over RDMA adapter. For example, Mellanox Technologies MT27700 Family ConnectX-4.
  – NVMe over TCP adapter. For example, i40en.

- Configure the VMkernel binding for your adapters.
  - For NVMe over RDMA, see Configure VMkernel Binding for the RDMA Adapter.
  - For NVMe over TCP, see Configure VMkernel Binding for the NVMe over TCP Adapter.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and click the **Add Software Adapter** icon.
4. Select the adapter type as required.
   - **NVMe over RDMA adapter**
   - **NVMe over TCP adapter**
5. Depending on your selection in Step 4, select an appropriate RDMA adapter or TCP network adapter (`vmnic`) from the drop-down menu.

   Note:
   If you get an error message that prevents you from creating the software adapter, make sure that the VMkernel binding for the adapter is configured correctly. For more information, see Configure VMkernel Binding for the RDMA Adapter and Configure VMkernel Binding for the NVMe over TCP Adapter.

The software NVMe over RDMA and NVMe over TCP adapters appear in the list as `vmhba` storage adapters. You can remove the adapters if you need to free the underlying RDMA and TCP network adapter for other purposes. See Remove Software NVMe Adapters from the ESX Host.

Add Controllers for NVMe over Fabrics

Use the vSphere Client to add an NVMe controller. After you add the controller, the NVMe namespaces associated with the controller become available to your ESX host. The NVMe storage devices that represent the namespaces in the ESX environment appear on the storage devices list.

- Make sure that your ESX host has appropriate adapters for your type of storage. See Requirements for VMware NVMe Storage.
- If you use NVMe over RDMA (RoCE v2) storage, perform these tasks:
  - Configure VMkernel Binding for the RDMA Adapter
  - Add Software NVMe over RDMA or NVMe over TCP Adapters
- If you use NVMe over TCP storage, perform these tasks:
  - Configure VMkernel Binding for the NVMe over TCP Adapter
  - Add Software NVMe over RDMA or NVMe over TCP Adapters

  Note:
  With NVMe over Fibre Channel, after you install the required adapter, it automatically connects to all targets that are reachable at the moment. You can later reconfigure the adapter and disconnect its controllers or connect other controllers that were not available during the host boot.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (vmhba#) to configure.
4. Click the **Controllers** tab, and click **Add Controller**.
5. On the **Add controller** dialog box, select one of the following discovery methods.

Option	Description
Automatically	The automatic option indicates that your host can discover controllers automatically and accept connection to any available controller. 1. Specify the following parameters for to discover controllers. – For NVMe over RDMA (RoCE v2), the IP address and transport port number.

Option	Description
	– For NVMe over TCP, the IP address, transport port number, and the digest parameter. 2. Click **Discover Controllers**. 3. From the list of controllers, select the controller to use.
Manually	With the manual method, you manually enter controller details. The host requests a connection to a specific controller using the parameters you specify: • Subsystem NQN • Target port identification. – For NVMe over RDMA (RoCE v2), the IP address and transport port number (optional). – For NVMe over TCP, the IP address, transport port number (optional), and the digest parameter (optional). – For NVMe over Fibre Channel, the WorldWideNodeName and WorldWidePortName. • Admin queue size. An optional parameter that specifies the size of the admin queue of the controller. A default value is 16. • Keepalive timeout. An optional parameter to specify in seconds the keep alive timeout between the adapter and the controller. A default timeout value is 60 seconds. **Note:** IO Queue Size and IO Queue Number are optional parameters that can be set only through `esxcli`.

The controller appears on the list of controllers. Your host can now discover the NVMe namespaces that are associated with the controller. The NVMe storage devices that represent the namespaces in the ESX environment appear on the storage devices list in the vSphere Client.

Configuring NVMe over TCP on ESX

The adapter configuration process on the ESX host involves setting up VMkernel binding for a TCP network adapter, and then adding a software adapter for NVMe over TCP. After that, you can add an NVMe controller.

The entire configuration process includes these actions.

Action	Description
On your ESX host, install an adapter that supports NVMe over TCP technology	For example, i40en.
Configure VMkernel Binding for the NVMe over TCP Adapter	VMkernel binding for NVMe over TCP involves creating a virtual switch and connecting the physical network adapter and the VMkernel adapter to the virtual switch. Through this connection, the TCP adapter becomes bound to the VMkernel adapter. In the configuration, you can use a vSphere standard switch or a vSphere distributed switch.
Add Software NVMe over RDMA or NVMe over TCP Adapters	Use the vSphere Client to enable the software storage adapters for NVMe over TCP.
Add Controllers for NVMe over Fabrics	Use the vSphere Client to add an NVMe controller. After you add the controller, the NVMe namespaces associated with the controller become available to your ESX host. The NVMe storage devices that represent the namespaces in the ESX environment appear on the storage devices list.

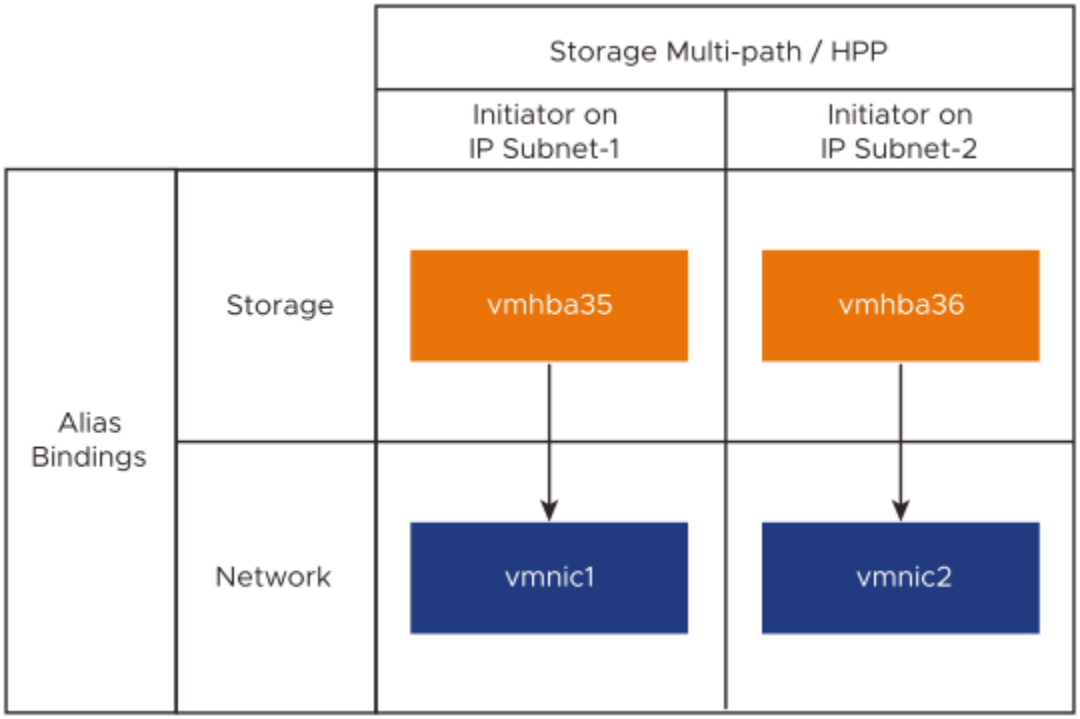
Configure VMkernel Binding for the NVMe over TCP Adapter

Port binding for NVMe over TCP involves creating a virtual switch and connecting the physical network adapter and the VMkernel adapter to the virtual switch. Through this connection, the TCP adapter becomes bound to the VMkernel adapter. In the configuration, you can use a vSphere standard switch or a vSphere distributed switch.

Note:

The ESX NVMe over TCP implementation does not support Link Aggregation Control Protocol (LACP), port-channel, or EtherChannel configurations on physical switches. NVMe over TCP relies on VMware Storage Multipathing to provide high availability, failover, and load balancing. Do not configure LACP or other unsupported configurations for VMkernel adapters used for NVMe over TCP.

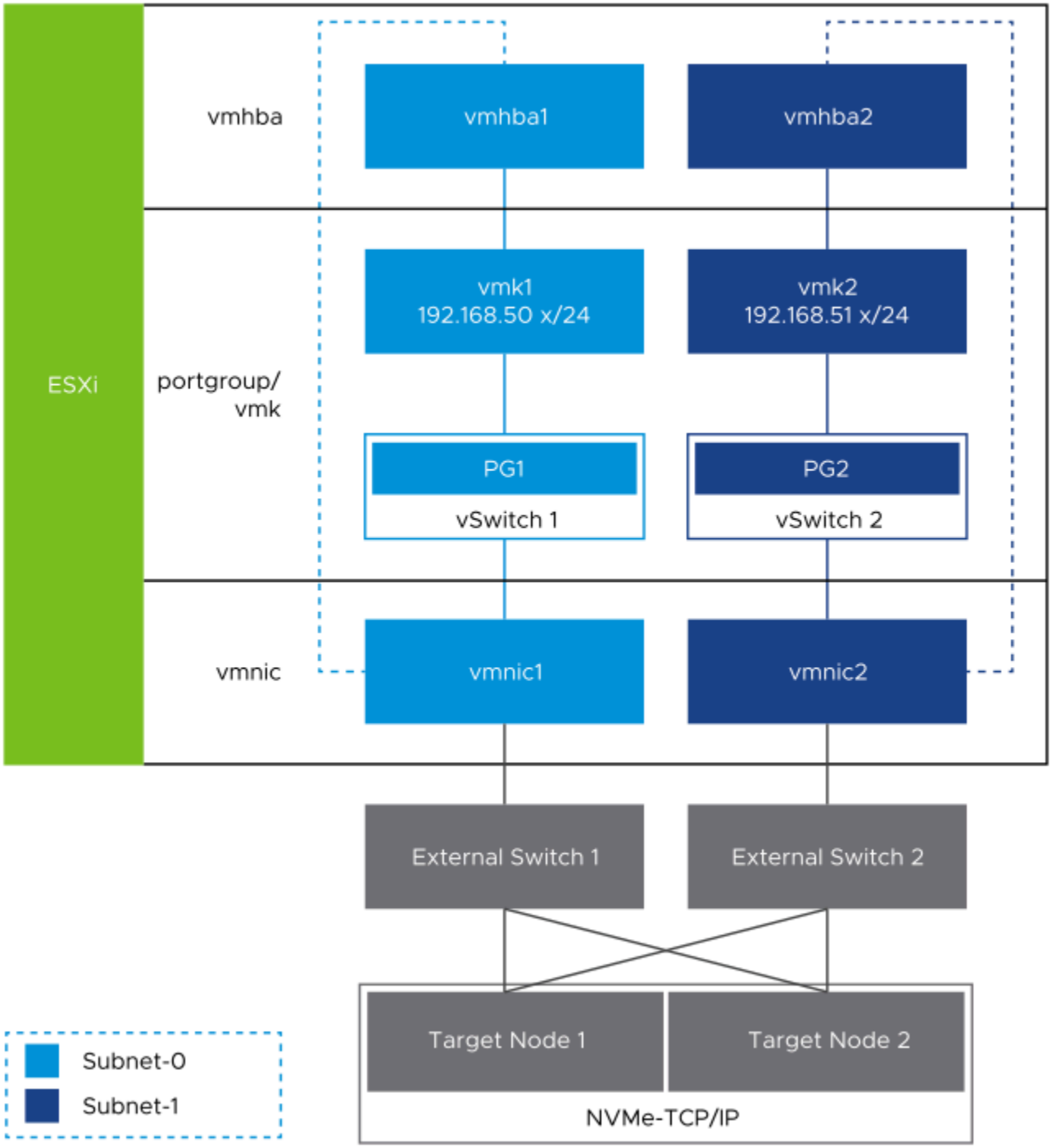
The following diagram displays the port binding for the NVMe over TCP adapter.



For more information about creating switches, see *Create a vSphere Standard Switch* or *Create a vSphere Distributed Switch* in the *vSphere Networking* documentation.

Example of Network Topology with NVMe over TCP

In this example, two vSphere standard switches and two network adapters (vmnic) on the host provide high availability. They connect to two external switches.



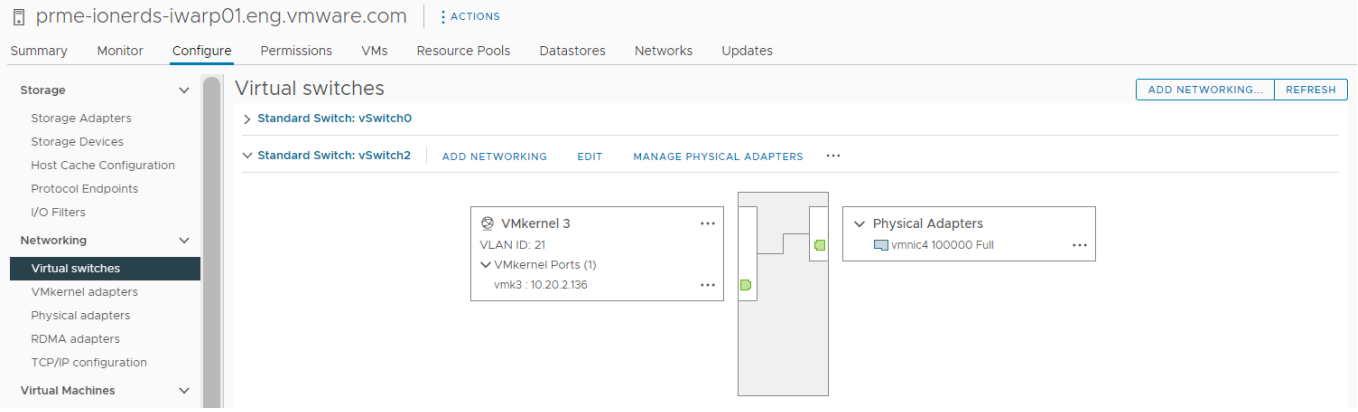
Configure VMkernel Binding for the TCP Adapter with a vSphere Standard Switch

You can configure VMkernel binding for the TCP adapter using a vSphere standard switch and one uplink per switch. Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter. You use 1:1 mapping between each virtual and physical network adapter.

1. Create a vSphere standard switch with a VMkernel adapter and the network component.
   a) In the vSphere Client, select your host and click the **Networks** tab.
   b) Click **Actions** > **Add Networking**.
   c) Select **VMkernel Network Adapter** and click **NEXT**.
   d) Select **New standard switch** and click **NEXT**.
   e) Under **Assigned adapters**, click **+**.
      The list of available physical adapters is displayed.
   f) Select the required physical adapter `vmnic`, and click **OK**.

      Note:
      Make sure to select the physical network adapter that corresponds to the TCP/IP adapter.
   g) Under **VMkernel port settings**, enter the required values.
      If you are using VLAN for the storage path, enter the VLAN ID.
   h) In the **IP settings** list, enter the VMkernel IPv4 settings.
   i) Under **Available services**, select **NVMe over TCP** for appropriate tagging.
2. Verify that your switch is correctly configured.
   a) On the **Configure** tab, select **Virtual switches** under **Networking**.
   b) Expand the switch and verify its configuration.

      The illustration shows that the physical network adapter and the VMkernel adapter are connected to the vSphere standard switch. Through this connection, the TCP adapter is bound to the VMkernel adapter.



3. Set NIC teaming policies for vSphere standard switch.

   Note:

   The NVMe over TCP adapter does not support NIC teaming, failover policies, load-balancing policies, or such configurations as LACP, port-channel, or EtherChannel. ESX NVMe over TCP relies on VMware Storage Multipathing to provide high availability, failover, and load balancing. You must use a strict 1:1 mapping between each VMkernel adapter and a single physical network adapter.

   a) Click the **Configure** tab, and select **Virtual switches** under **Networking**.
   b) Select the appropriate VMkernel adapter.
   c) From the right-click menu, click **Edit Settings**.
   d) Select **Teaming and Failover**.
   e) Under **Active adapters**, move the required physical adapter `vmnic`.
   f) Under **Standby adapters** > **Failover order**, move the other physical adapters.
   g) Set appropriate load balancing and other properties.
   h) Repeat the steps to configure additional VMkernel adapters.

   To verify if the adapter is configured, click the **Configure** tab and select **VMkernel adapters**.

Configure VMkernel Binding for the TCP Adapter with a vSphere Distributed Switch

You can configure VMkernel port binding for the TCP adapter using a vSphere distributed switch and one uplink per switch. Configuring the network connection involves creating a virtual VMkernel adapter for each physical network adapter. You use 1:1 mapping between each virtual and physical network adapter.

1. Create a vSphere distributed switch with a VMkernel adapter and the network component.
   a) In the vSphere Client, select **Datacenter**, and click the **Networks** tab.
   b) Click **Actions**, and select **Distributed Switch** > **New Distributed Switch**.
   c) Select a name for the switch.
      Ensure that the location of the data center is present within your host, and click **Next**.
   d) Select a compatible ESX version, and click **Next**.
   e) Enter the required number of uplinks, and click **Finish**.
2. Add one or more hosts to your distributed virtual switch.
   a) In the vSphere Client, select **Datacenter**, and click **Distributed Switches**.
      A list of available DSwitches appears.
   b) Right-click the DSwitch, and select **Add and Manage Hosts** from the menu.
   c) Select **Add hosts**, and click **Next**.
   d) Select your host, and click **Next**.
   e) Select **Assign uplink**.
   f) Enter the relevant uplink to assign the `vmnic`.
   g) Assign a VMkernel adapter, and click **Next**.
   h) In the vSphere Client, select the DSwitch, and click the **Ports** tab.
      You can view the uplinks created for your switch here.
3. Create distributed port groups for the NVMe over TCP storage path.
   a) In the vSphere Client, select the required DSwitch.
   b) Click **Actions** and select **Distributed Port Group** > **New Distributed Port Group**.
   c) Under **Configure Settings**, enter the general properties of the port group.
      If you have configured a specific VLAN, add it in the VLAN ID.

      Note:
      Network connectivity issues might occur if you do not configure VLAN properly.
4. Configure the VMkernel adapters.
   a) In the vSphere Client, expand the **DSwitch** list, and select the distributed port group.
   b) Click **Actions** > **Add VMkernel Adapters**.
   c) In the **Select Member Hosts** dialog box, select your host and click **OK**.
   d) In the **Configure VMkernel Adapter** dialog box, ensure that the MTU matches to the Switch MTU.
   e) Click **Finish**.
   f) Repeat step b and step c to add multiple TCP capable NICs.
5. Set NIC teaming policies for the distributed port groups.

   Note:

   The NVMe over TCP adapter does not support NIC teaming, failover policies, load-balancing policies, or such configurations as LACP, port-channel, or EtherChannel. Each distributed port group used for NVMe over TCP must have exactly one active uplink and all other uplinks set to Unused.

   a) In the **Distributed Port Group**, click **Actions** > **Edit Settings**.
   b) Click **Teaming and Failover**, and verify the active uplinks.
   c) Assign one uplink as **Active** for the port group, and the other uplink as **Unused**.
      Repeat step c for each of the port groups created.

After you complete the configuration, click **Configure**, and verify whether the physical adapter tab on your host lists the DVSwitch for the NICs selected.

Remove Software NVMe Adapters from the ESX Host

Use the vSphere Client to remove software NVMe over RDMA or NVMe over TCP adapters from the ESX host. You can remove the adapter if you need to free the underlying RDMA adapter or ethernet adapter for other purposes.

You cannot remove the NVMe over PCIe and NVMe over Fibre Channel adapters.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Adapters**, and select the adapter (`vmhba#`) to remove.
4. Remove the NVMe controller connected to the adapter.
   a) Click the **Controllers** tab.
   b) Select the controller and click **Remove**.
      The NVMe controller is disconnected and disappears from the list.
5. Click the **Remove** icon (Remove the host's storage adapter) to remove the NVMe over RDMA adapter or the NVMe over TCP adapter.

Booting ESX from NVMe Storage

To be able to boot ESX from NVMe-oF storage, you must install ESX on an NVMe namespace that is presented to ESX through NVMe over Fabrics transports, TCP, and Fibre Channel.

Get the Server SMBIOS UUID

The ESX installer uses SMBIOS UUID based Host NQN to connect to the NVMe over Fabrics target. To install ESX on an NVMe over Fabrics device, you must obtain and add the SMBIOS UUID based Host NQN of the host server to the allowed host list of the NVMe subsystem on the storage array in advance.

> **Note:**
>
> Only when ESX is installed on an NVMe over Fabrics device, the Host NQN based on SMBIOS UUID will continue to be used for connecting to the targets.

Use one of the following options to obtain the SMBIOS UUID.

- If ESX has been installed on the server, you can get the UUID using the CLI:

  ```
  smbiosDump | grep UUID
  ```
- If ESX has not been installed, you can get the UUID using the server management interface.

  The method to get the server SMBIOS UUID through the server management interface is vendor specific. For information, see the server vendor documentations.

The format of the Host NQN based on the UUID is `nqn.2014-08.org.nvmexpress:uuid:<UUID>`.

Typically, the UUID you get in the output looks similar to the following string: `UUID: 4c4c4544-0039-4610-8053-b2c04f424d32`.

In this example, the Host NQN based on this UUID will be `nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0039-4610-8053-b2c04f424d32`.

Booting ESX from NVMe over TCP

When you set up your ESX host to boot from NVMe over TCP SAN, your host's boot image is stored on a remote NVMe over TCP namespace in the SAN storage system. When the ESX host starts, it boots from the remote namespace on the SAN rather than from its local disk.

You can use boot from the SAN if you do not want to handle maintenance of local storage or have diskless hardware configurations, such as blade systems. ESX supports booting from the NVMe over TCP SAN using a network adapter.

Boot from SAN Benefits

As with boot from other types of storage, boot from NVMe over TCP SAN can provide numerous benefits to your ESX environment. However, in certain cases, boot from SAN is not compatible with your hosts. Before you set up your system for boot from SAN, decide whether it is appropriate for your environment.

For general information about boot from SAN benefits, see Boot from SAN Benefits.

Requirements and Considerations when Booting from NVMe over TCP SAN

Your ESX boot configuration must meet specific requirements.

Requirement	Description
ESX system requirements	• The server BIOS must support boot from NVMe over TCP SAN. Follow any vendor recommendations for the server. • For installation prerequisites and requirements, review *VMware ESX Installation and Setup*. • The server must support SMBIOS UUID based Host NQN for NVMe over TCP boot.
Access control	Configure proper ACLs on your storage system. • The boot namespace must be visible only to the host that uses it. No other host on the SAN is permitted to see that namespace.
Diagnostic partition	With NVMe over TCP boot, configure the diagnostic partition on a remote server using the ESX Dump Collector. See *vCenter Deployment and Setup* and *vSphere Networking*. Alternately, you can configure the diagnostic partition on a local device. See Configuring a diagnostic coredump partition on an ESX host. The diagnostic partition stores core dumps that are used for debugging and by the Broadcom technical support.
Multipathing support	Multipathing is not supported at boot time as the host BIOS does not support multipathing. After you install ESX on the remote namespace, you can configure additional paths to the boot device for high availability. See Configuring NVMe over TCP on ESX.
Additional considerations	• If the array is not certified for a direct connect topology, the SAN connections must be through a switched topology. If the array is certified for the direct connect topology, the SAN connections can be made directly to the array. Boot from SAN is supported for both switched topology and direct connect topology. • Use different namespaces for VMFS datastores and boot partitions. • Boot namespaces must have high availability as ESX is sensitive to boot device failures. • Use static IP addresses to reduce the chances of DHCP conflicts.

Configure SAN Components and Storage System for NVMe over TCP Boot

Before you configure your ESX host to boot from an NVMe over TCP SAN namespace, prepare and configure your storage area network (SAN).

Obtain and note down the SMBIOS UUID based Host NQN string from the ESX host BIOS. See Get the Server SMBIOS UUID.

1. Connect network cables, referring to any cabling guide that applies to your setup.
2. Ensure IP connectivity between your storage system and the host.

   Verify the configuration of any routers or switches on your storage network. Host network adapters must be able to ping the target storage system.
3. Configure the storage system.
   a) Create an NVMe over TCP volume, or namespace, on the storage system for your host to boot from.
   b) Use the SMBIOS UUID based Host NQN string from Step 1 and configure the storage system so that your host has access to the assigned volume.

      For information, see vendor documentation.

      This step might involve adding host initiator information to the storage system and mapping volumes to hosts.
   c) Make sure that the volume is presented to the host correctly.
   d) Make sure that no other system has access to the configured volume.
   e) Record the NVMe-oF Subsystem NQN, IP address, and port information of the target assigned to the host.

      If configuring the host to connect to a discovery controller on boot, record the discovery controller IP.

      You must have this information to configure the host BIOS for NVMe over TCP boot.

      Note:

      Even if discovery controllers are used for configuring NVMe over TCP boot in the server BIOS, ESX connects directly to the target subsystem to avoid any delays performing discovery for faster booting.

Configure Server BIOS and Install ESX on an NVMe over TCP Namespace

Because configuring the boot sequence in the BIOS is vendor-specific, before configuring your ESX host, review vendor documentation for information related to NVMe over TCP boot.

- Make sure that you have available the target information recorded in the previous procedure.
- Make sure you have the required host network configuration information, such as DHCP vs Static, IP address, subnet mask, gateway IP, VLAN ID, and so on.

1. Use the ESX installation media and reboot the host.
2. Enter the BIOS Configuration/Setup Utility and configure the server BIOS to set the host to boot from the installation media first.
3. Configure NVMe over TCP boot from SAN in the server BIOS.

   Typically, it involves the following steps.
   a) Enable NVMe over TCP boot in the BIOS.
   b) Set BIOS to use SMBIOS UUID based Host NQN string for NVMe over TCP connections.
   c) Configure NVMe-oF subsystem settings using the target and host network parameters noted previously.
4. Reboot the host so that it begins the ESX installation from the media.
5. Proceed with the installation.

   Note:

   The ESX network configuration is only set up once during the ESX installation based on information from the NVMe over TCP boot settings in the BIOS.

Booting ESX from NVMe over Fibre Channel

Starting from version 9.0, ESX supports boot from NVMe over Fibre Channel protocol.

Requirements

- Fibre Channel storage array that supports NVMe. For information, see Using ESX with Fibre Channel SAN.

- Hardware NVMe adapter. Typically, it is a Fibre Channel HBA that supports NVMe.
- A compatible server for ESX. The server should support SMBIOS UUID.

Configure Storage System and HBA for NVMe over Fibre Channel Boot

Before you configure your ESX host to boot from an NVMe over FC SAN namespace, prepare and configure your storage area network (SAN). You must also configure the Fibre Channel HBA on the host.

Obtain and note down the SMBIOS UUID based Host NQN string from the ESX host BIOS. See Get the Server SMBIOS UUID.

1. Configure the NVMe subsystem and namespace on the storage array, and add the Host NQN to the list of allowed hosts on the NVMe subsystem.

   See the array vendor documentation for specific steps.

2. Configure the Fibre Channel HBA on your host to enable NVMe over FC boot.

   For information, see vendor specific documentation.

   The following steps use Marvell FC HBA and Dell server as an example.

   a) Enter the host's BIOS.

      System Setup > **Device Settings** > **Fibre Channel in Slot 2 xxxxx** > **HBA Configuration Page**

   b) Enable **NVMe Boot from SAN** and configure SMBIOS UUID based Host NQN.

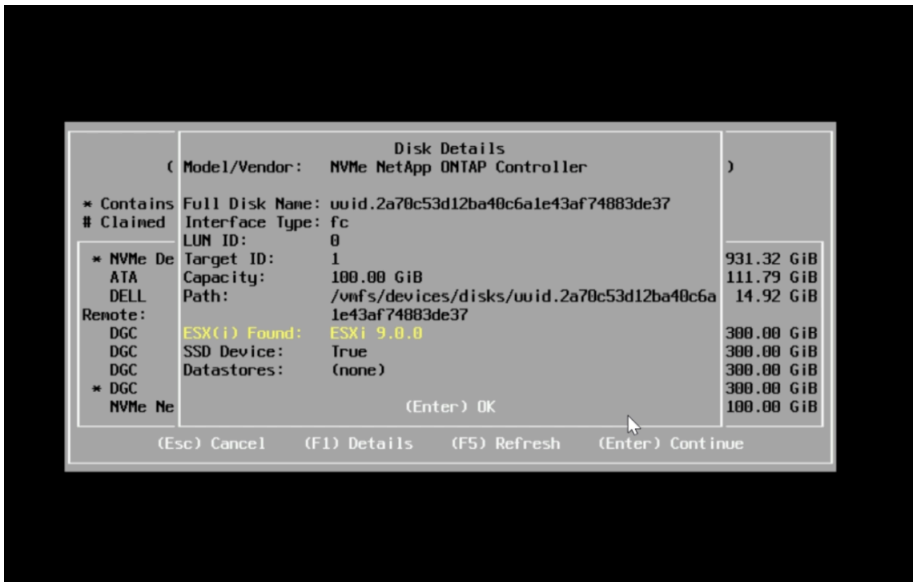
      The Host NQN and the host ID are generated based on the SMBIOS UUID by default. If the default value is not as expected, fill in the correct Host NQN manually.

   c) Enable **Connect to First NVMe Target**.

      You must fill in the WWNs of the storage array, the NVMe Qualified Name for the target, and the namespace ID.

Install ESX on an NVMe over Fibre Channel Namespace

1. Attach the ESX installer media to the server, and reboot or power on the server.
2. Follow the prompts to perform the installation.
3. On the **Select a Disk to Install or Upgrade** screen, select the same remote NVMe over FC device that you set in the FC HBA BIOS.



Working with Datastores in vSphere Storage Environment

In the vSphere environment, datastores are logical containers, analogous to file systems, that hide specifics of physical storage and provide a uniform model for storing virtual machine files. You can also use datastores for storing ISO images, virtual machine templates, and floppy images. vCenter and ESX support VMFS, NFS, vSAN, and Virtual Volumes datastores.

vCenter and ESX support the following types of datastores.

Table 173: Types of Datastores

Datastore Type	Description
VMFS (version 5 and 6)	Datastores that you deploy on block storage devices use the vSphere Virtual Machine File System (VMFS) format. VMFS is a special high-performance file system format that is optimized for storing virtual machines. See vSphere VMFS Datastore Concepts and Operations.
NFS (version 3 and 4.1)	An NFS client built into ESX uses the Network File System (NFS) protocol over TCP/IP to access a designated NFS volume. The volume is located on a NAS server. The ESX host mounts the volume as an NFS datastore, and uses it for storage needs. ESX supports versions 3 and 4.1 of the NFS protocol. See NFS Datastore Concepts and Operations in vSphere Environment
vSAN	vSAN aggregates all local capacity devices available on the hosts into a single datastore shared by all hosts in the vSAN cluster. See the *Administering VMware vSAN* documentation.
vVol	Virtual Volumes datastore represents a storage container in vCenter and vSphere Client. See Working with vSphere Virtual Volumes.

Depending on your storage type, some of the following tasks are available for the datastores.

- Create datastores. You can use the vSphere Client to create certain types of datastores.
- Perform administrative operations on the datastores. Several operations, such as renaming a datastore, are available for all types of datastores. Others apply to specific types of datastores.
- Organize the datastores. For example, you can group them into folders according to business practices. After you group the datastores, you can assign the same permissions and alarms on the datastores in the group at one time.
- Add the datastores to datastore clusters. A datastore cluster is a collection of datastores with shared resources and a shared management interface. When you create the datastore cluster, you can use Storage DRS to manage storage resources. For information about datastore clusters, see the *vSphere Resource Management* documentation.

Creating vSphere Datastores

You use the New Datastore wizard to create your datastores in the vSphere environment. Depending on the type of your storage and storage needs, you can create a VMFS, NFS, or Virtual Volumes datastore.

A vSAN datastore is automatically created when you enable vSAN. For information, see the *Administering VMware vSAN* documentation.

See the following topics to create datastores.

VMFS	NFS	Virtual Volumes
Create a vSphere VMFS Datastore You can also use the New Datastore wizard to manage VMFS datastore copies. See vSphere VMFS Datastore Copies and Datastore Resignaturing.	Creating NFS Datastores with the vSphere Client	Create a Virtual Volumes Datastore in vSphere Environment For information about deprecation of Virtual Volumes, see the Broadcom KB 401070.

To learn more about the vSphere datastores, see the following:

- vSphere VMFS Datastore Concepts and Operations
- NFS Datastore Concepts and Operations in vSphere Environment
- Working with vSphere Virtual Volumes

vSphere VMFS Datastore Concepts and Operations

To store virtual disks, ESX uses datastores. The datastores are logical containers that hide specifics of physical storage from virtual machines and provide a uniform model for storing the virtual machine files. The datastores that you deploy on block storage devices use the native vSphere Virtual Machine File System (VMFS) format. It is a special high-performance file system format that is optimized for storing virtual machines.

Use the vSphere Client to set up the VMFS datastore in advance on the block-based storage device that your ESX host discovers.

After creating a datastore, you can perform basic administration operations, such as renaming, unmounting, or removing the datastore. You can also perform more advanced tasks, for example, expand the datastore, manage duplicate datastores, or collect datastore diagnostic information.

The workflow that you use when managing the VMFS datastores generally includes the following actions.

Action	Description
1. Create a vSphere VMFS Datastore	Set up the VMFS datastore on the block-based storage device that your ESX host discovers.
2. Perform Basic Administration Operations for Datastores	Perform basic administration operations, such as renaming, unmounting, or removing the datastore.
3. Perform Advanced VMFS Datastore Tasks	Perform more advanced tasks, for example, expand the datastore, manage duplicate datastores, or collect datastore diagnostic information.

Action	Description
4. Browse the Datastore	Use the datastore file browser in your vSphere environment to manage contents of your datastores. You can upload files or folders, download files from the datastore, or move or copy folders or files to a new location. You can also use the datastore browser to inflate the thin virtual disk.

Versions of VMFS Datastores

Several versions of the VMFS file system have been released since its introduction. Currently, ESX supports VMFS5 and VMFS6.

For all supported VMFS version, ESX offers complete read and write support. On the supported VMFS datastores, you can create and power on virtual machines.

Table 174: Host Access to VMFS Versions

VMFS	ESX
VMFS6	Read and write
VMFS5	Read and write

The following table compares major characteristics of VMFS5 and VMFS6. For additional information, see *Configuration Maximums* .

Table 175: Comparing VMFS5 and VMFS6

Features and Functionalities	VMFS5	VMFS6
Access for ESX hosts version 6.5 and later	Yes	Yes
Access for ESX hosts version 6.0 and earlier	Yes	No
Datastores per host	512	512
512n storage devices	Yes	Yes (default)
512e storage devices	Yes. Not supported on local 512e devices.	Yes (default)
4Kn storage devices	No	Yes
Automatic space reclamation	No	Yes
Manual space reclamation through the esxcli command. See Manually Reclaim Accumulated Storage Space on VMFS5.	Yes	Yes
Space reclamation from guest OS	Limited	Yes
GPT storage device partitioning	Yes	Yes
MBR storage device partitioning	Yes For a VMFS5 datastore that has been previously upgraded from VMFS3.	No
Storage devices greater than 2 TB for each VMFS extent	Yes	Yes
Support for virtual machines with large capacity virtual disks, or disks greater than 2 TB	Yes	Yes
Support of small files of 1 KB	Yes	Yes
Default use of ATS-only locking mechanisms on storage devices that support ATS. See vSphere VMFS Locking Mechanisms.	Yes	Yes

Features and Functionalities	VMFS5	VMFS6
Block size	Standard 1 MB	Standard 1 MB
Default snapshots	SEsparse	SEsparse
Virtual disk emulation type	512n	512n, 4Kn
vMotion	Yes	Yes
Storage vMotion across different datastore types	Yes	Yes
High Availability and Fault Tolerance	Yes	Yes
DRS and Storage DRS	Yes	Yes
RDM	Yes	Yes

When you work with VMFS datastores, consider the following:

- Datastore Extents. A spanned VMFS datastore must use only homogeneous storage devices, either 512n, 512e, or 4Kn. The spanned datastore cannot extend over devices of different formats.
- Block Size. The block size on a VMFS datastore defines the maximum file size and the amount of space a file occupies. VMFS5 and VMFS6 datastores support the block size of 1 MB.
- Storage vMotion. Storage vMotion supports migration across VMFS, vSAN, and Virtual Volumes datastores. vCenter performs compatibility checks to validate Storage vMotion across different types of datastores.
- Storage DRS. VMFS5 and VMFS6 can coexist in the same datastore cluster. However, all datastores in the cluster must use homogeneous storage devices. Do not mix devices of different formats within the same datastore cluster.
- Device Partition Formats. Any new VMFS5 or VMFS6 datastore uses GUID partition table (GPT) to format the storage device. The GPT format enables you to create datastores larger than 2 TB. If your VMFS5 datastore has been previously upgraded from VMFS3, it continues to use the master boot record (MBR) partition format, which is characteristic for VMFS3. Conversion to GPT happens only after you expand the datastore to a size larger than 2 TB.

VMFS Datastores as Repositories

ESX can format SCSI-based storage devices as VMFS datastores. VMFS datastores primarily serve as repositories for virtual machines.

> **Note:**
> Always have only one VMFS datastore for each LUN.

You can store multiple virtual machines on the same VMFS datastore. Each virtual machine, encapsulated in a set of files, occupies a separate single directory. For the operating system inside the virtual machine, VMFS preserves the internal file system semantics, which ensures correct application behavior and data integrity for applications running in virtual machines.

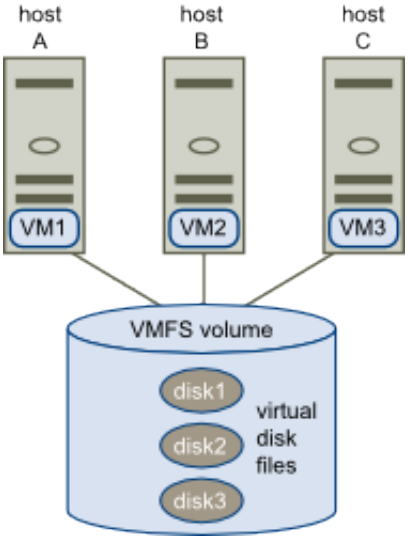
When you run multiple virtual machines, VMFS provides specific locking mechanisms for the virtual machine files. As a result, the virtual machines can operate safely in a SAN environment where multiple ESX hosts share the same VMFS datastore.

In addition to the virtual machines, the VMFS datastores can store other files, such as the virtual machine templates and ISO images.

Sharing a VMFS Datastore Across Hosts

As a cluster file system, VMFS lets multiple ESX hosts access the same VMFS datastore concurrently.

Figure 78: Sharing a VMFS Datastore Across Hosts



For information on the maximum number of hosts that can connect to a single VMFS datastore, see the *Configuration Maximums* document.

To ensure that multiple hosts do not access the same virtual machine at the same time, VMFS provides on-disk locking. For more details about the ATS-only and ATS+SCSI locking mechanisms that VMFS uses, see vSphere VMFS Locking Mechanisms. You can also learn how to display information about locking mechanisms, how to change from ATS+SCSI to ATS-only, or how to downgrade to ATS+SCSI.

Sharing the VMFS volume across multiple hosts offers several advantages, for example, the following:

- You can use VMware Distributed Resource Scheduling (DRS) and VMware High Availability (HA).
  You can distribute virtual machines across different physical servers. That means you run a mix of virtual machines on each server, so that not all experience high demand in the same area at the same time. If a server fails, you can restart virtual machines on another physical server. If the failure occurs, the on-disk lock for each virtual machine is released. For more information about VMware DRS, see the *vSphere Resource Management* documentation. For information about VMware HA, see the *vSphere Availability* documentation.
- You can use vMotion to migrate running virtual machines from one physical server to another. For information about migrating virtual machines, see the *vCenter and Host Management* documentation.

To create a shared datastore, mount the datastore on those ESX hosts that require the datastore access. See Mount Datastores.

VMFS Metadata Updates

A VMFS datastore holds virtual machine files, directories, symbolic links, RDM descriptor files, and so on. The datastore also maintains a consistent view of all the mapping information for these objects. This mapping information is called metadata.

Metadata is updated each time you perform datastore or virtual machine management operations. Examples of operations requiring metadata updates include the following:

- Creating, growing, or locking a virtual machine file
- Changing attributes of a file
- Powering a virtual machine on or off
- Creating or deleting a VMFS datastore
- Expanding a VMFS datastore
- Creating a template
- Deploying a virtual machine from a template
- Migrating a virtual machine with vMotion

When metadata changes are made in a shared storage environment, VMFS uses special locking mechanisms to protect its data and prevent multiple hosts from concurrently writing to the metadata.

Snapshot Formats on VMFS

When you take a snapshot, the virtual disk's state is preserved, preventing the guest OS from writing directly to it. Instead, a delta disk, also called a child disk, is created. This delta disk stores all changes made after the snapshot and represents the difference between the current VM disk state and the state at the time of the snapshot.

On a VMFS datastore, delta disks use a sparse format. They start empty and grow as data changes.

- Delta disks use copy-on-write mechanism when data is only stored in the delta disk when it is modified.
- Delta disks function as a redo log. It starts empty when the snapshot is taken and expands as the VM writes new data.
- Delta disk is stored as a file on the VMFS datastore.
- When a snapshot is created, the base VMDK is replaced by the new sparse delta disk, which tracks all changes.
- The specific sparse format used for delta disks depends on your datastore type and vSphere version.

Snapshot Formats	VMFS5 (vSphere 8.x and earlier)	VMFS5 (vSphere 9.0)	VMFS6
SEsparse	For virtual disks larger than 2 TB.	Default format for all virtual disks.	Default format for all virtual disks.
VMFSsparse	For virtual disks smaller than 2 TB.	Existing VMFSsparse snapshots continue to work.	N/A

SEsparse

SEsparse is a default format for all delta disks on the VMFS6 datastores and, starting from vSphere 9.0, on all VMFS5 datastores. On legacy VMFS5, SEsparse is used for virtual disks of the size 2 TB and larger.

- It uses 4 KB blocks and is optimized for thin-provisioning.
- It is space efficient and supports the space reclamation technique. With space reclamation, blocks that the guest OS deletes are marked. The system sends commands to the SEsparse layer in the hypervisor to unmap those blocks. The unmapping helps to reclaim space allocated by SEsparse once the guest operating system has deleted that data. For more information about space reclamation, see Storage Space Reclamation in vSphere.

VMFSsparse

VMFS5 on vSphere 8.x and earlier uses the VMFSsparse format for virtual disks smaller than 2 TB.

Snapshot Migration

You can migrate VMs with snapshots across different datastores. The following considerations apply:

- If you migrate a VM with the VMFSsparse snapshot to VMFS6, the snapshot format changes to SEsparse.
- You cannot mix VMFSsparse redo-logs with SEsparse redo-logs in the same hierarchy.

Upgrading VMFS Datastores

You cannot upgrade a VMFS5 datastore to VMFS6. If you have a VMFS5 datastore in your environment, create a VMFS6 datastore and migrate virtual machines from the VMFS5 datastore to VMFS6.

Create a vSphere VMFS Datastore

VMFS datastores serve as repositories for virtual machines. You can set up VMFS datastores on any SCSI or NVMe based storage devices that the host discovers, including Fibre Channel, iSCSI, and local storage devices.

- Install and configure any adapters that your storage requires.
  - For Fibre Channel, see Setting Up ESX Fibre Channel Environment.
  - For iSCSI, see Configuring iSCSI and iSER Adapters and Storage with ESX.
  - For NVMe, see About VMware NVMe Storage.
- To discover newly added storage devices, perform a rescan. See Rescan Operations for ESX Storage.
- Verify that storage devices you are planning to use for your datastores are available. See Viewing Storage Devices Available to ESX Hosts.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. From the right-click menu, select **Storage** > **New Datastore**.
3. Select VMFS as the datastore type.
4. Enter the datastore name and if necessary, select the placement location for the datastore.
   The system enforces an 80 character limit for the datastore name.
5. Select the device to use for your datastore.

   Important:
   The device you select must not have any values displayed in the Snapshot Volume column. If a value is present, the device contains a copy of an existing VMFS datastore. For information on managing datastore copies, see vSphere VMFS Datastore Copies and Datastore Resignaturing.
6. Specify the datastore version.

Option	Description
VMFS6	Default format on all hosts that support VMFS6. The ESX hosts of version 6.0 or earlier cannot recognize the VMFS6 datastore.
VMFS5	VMFS5 datastore supports access by the ESX hosts of version 6.7 or earlier.

7. Define configuration details for the datastore.

   Note:
   The required minimum size for a VMFS6 datastore is 2 GB.

   a) Specify partition configuration.

Option	Description
Use all available partitions	Dedicates the entire disk to a single VMFS datastore. If you select this option, all file systems and data currently stored on this device are destroyed.
Use free space	Deploys a VMFS datastore in the remaining free space of the disk.

   b) If the space allocated for the datastore is excessive for your purposes, adjust the capacity values in the Datastore Size field.
   By default, the entire free space on the storage device is allocated.
   c) For VMFS6, specify the block size and define space reclamation parameters. See Space Reclamation on vSphere VMFS Datastores.
8. In the Ready to Complete page, review the datastore configuration information and click **Finish**.

The datastore on the SCSI or NVMe based storage device is created. It is available to all hosts that have access to the device.
After you create the VMFS datastore, you can perform the following tasks:

- Change the capacity of the datastore. See Increase Capacity of vSphere VMFS Datastore.
- Edit space reclamation settings. See Configure Fixed Reclamation for the Existing VMFS6 Datastore.
- Enable shared vmdk support. See Configure Settings for Clustered Virtual Disks on the vSphere VMFS6 Datastore.

Basic Administrative Operations for vSphere Datastores

After creating datastores, you can perform several administrative operations on the datastores, such as renaming, unmounting, or removing the datastore. Certain operations, for example, renaming, are available for all types of datastores. Others, such as deleting the datastore, apply only to VMFS datastores.

Change Datastore Name

Use the vSphere Client to change the name of an existing datastore. You can rename the datastore that has virtual machines running on it without any negative impact.

> **Note:**
> If the host is managed by vCenter, you cannot rename the datastore by directly accessing the host from the VMware Host Client. You must rename the datastore from vCenter.

1. In the vSphere Client, navigate to the datastore.
2. Right-click the datastore to rename, and select **Rename**.
3. Enter a new datastore name.

   The system enforces an 80 character limit for the datastore name.

The new name appears on all hosts that have access to the datastore.

Decommissioning a Datastore

You place a datastore in maintenance mode when you need to take it out of use to service it. You can then unmount or remove the datastore.

When you initiate a maintenance mode or a removal of a datastore that does not have Storage DRS enabled, the system checks for the presence of the following items on the datastore:

- VMs that have virtual disks.
- Starting with vSphere 9.0, Cloud Native Storage volumes backed by First Class Disks (FCDs).

For the maintenance mode or datastore removal operations to proceed, you must migrate the VMs and CNS-managed FCDs residing on the datastore.

The maintenance mode task remains in a pending state until all migration tasks are complete. Datastore maintenance mode is successful only when all datastore resources have been migrated away from the source datastore.

When a datastore is entering maintenance mode or has successfully entered maintenance mode, it does not accept any incoming VM or other requests. Requests are accepted only when the datastore exits maintenance mode.

To avoid data loss when decommissioning your datastores, follow these guidelines:

- Always enter maintenance mode before unmounting or removing a VMFS or NFS datastore.
- Do not unmount a datastore if VMs and CNS volumes backed by FCDs are still present.
- Keep at least one connected host out of maintenance mode so vSphere can check for CNS volumes. The datastore removal operation fails if all connected hosts are in maintenance mode.

The maintenance mode behavior differs when your datastore is Storage DRS enabled. For more information, see Using Storage DRS Maintenance Mode.

For information about CNS volumes, see Getting Started with Cloud Native Storage in vSphere.

Place a Datastore in Maintenance Mode

If you need to take a datastore out of service, you can place it in maintenance mode.

1. Browse to the datastore in the vSphere Client.
2. Right-click the datastore and select **Maintenance Mode** > **Enter Maintenance Mode**.
3. Optional: Indicate if you want to use vMotion to migrate VMs.
4. Click **Continue** to initiate migration using Storage vMotion.

   After you place the datastore in maintenance mode, perform additional manual migration tasks.

Unmount Datastores

When you unmount a datastore, it remains intact, but can no longer be seen from the hosts that you specify. The datastore continues to appear on other hosts, where it remains mounted.

When appropriate, before unmounting datastores, make sure that the following prerequisites are met:

- No virtual machines or CNS volumes backed by FCDs reside on the datastore.
- Storage DRS does not manage the datastore.
- Storage I/O Control is deactivated for this datastore.

Do not perform any configuration operations that might result in I/O to the datastore while the unmounting is in progress.

> **Note:**
> Make sure that the datastore is not used by vSphere HA Heartbeating. vSphere HA Heartbeating does not prevent you from unmounting the datastore. However, if the datastore is used for heartbeating, unmounting it might cause the host to fail and restart any active virtual machine.

1. In the vSphere Client, navigate to the datastore.
2. Right-click the datastore and select **Unmount Datastore**.
3. If the datastore is shared, select the hosts from which to unmount the datastore.
4. Confirm that you want to unmount the datastore.

After you unmount a VMFS datastore from all hosts, the datastore is marked as inactive. If you unmount an NFS or a Virtual Volumes datastore from all hosts, the datastore disappears from the inventory. You can mount the unmounted VMFS datastore. To mount the NFS or Virtual Volumes datastore that has been removed from the inventory, use the New Datastore wizard.

If you unmounted the VMFS datastore as a part of a storage removal procedure, you can now detach the storage device that is backing the datastore. See Detach Storage Devices.

Remove VMFS Datastores

You can delete any type of VMFS datastore, including copies that you have mounted without resignaturing. When you delete a datastore, it is destroyed and disappears from all hosts that have access to the datastore.

- Remove or migrate all virtual machines and CNS volumes backed by FCDs from the datastore.
- Unmount the datastore from all hosts.
- Disable Storage DRS for the datastore. For more information, see Activate Storage DRS.
- Disable Storage I/O Control for the datastore. For more information, see Enable Storage I/O Control.
- Make sure that the datastore is not used for vSphere HA heartbeating. For more information, see Configure Heartbeat Datastores.

> **Note:**
> The delete operation for the datastore permanently deletes all files associated with virtual machines on the datastore. Although you can delete the datastore without unmounting, it is preferable that you unmount the datastore first.

1. In the vSphere Client, navigate to the datastore.
2. Right-click the datastore to remove, and select **Delete Datastore**.
3. Confirm that you want to remove the datastore.

Mount Datastores

You can mount a datastore you previously unmounted. You can also mount a datastore on additional hosts, so that it becomes a shared datastore.

A VMFS datastore that has been unmounted from all hosts remains in inventory, but is marked as inaccessible. See Unmount Datastores.

You can use this task to mount the VMFS datastore to a specified host or multiple hosts.

If you have unmounted an NFS or a Virtual Volumes datastore from all hosts, the datastore disappears from the inventory. To mount the NFS or Virtual Volumes datastore that has been removed from the inventory, use the New Datastore wizard.

A datastore of any type that is unmounted from some hosts while being mounted on others, is shown as active in the inventory.

1. In the vSphere Client, navigate to the datastore.
2. Right-click the datastore to mount and select one of the following options:
   - **Mount Datastore**
   - **Mount Datastore on Additional Hosts**

     You can also select the VMkernel binding option for the required datastore. For more information, see step g under Configure VMkernel Binding for NFS 3.

   Whether you see one or another option depends on the type of datastore you use.
3. Select the hosts that should access the datastore and click **OK**.
4. To list all hosts that share the datastore, navigate to the datastore, and click the **Hosts** tab.

Managing vSphere VMFS Datastores

After you create a VMFS datastore in your vSphere environment, you can perform such tasks as expanding the datastore, managing duplicate datastores, or collecting datastore diagnostic information.

For general information about VMFS datastores, see vSphere VMFS Datastore Concepts and Operations.

To create a VMFS datastore and perform basic operations, see these topics:

- Create a vSphere VMFS Datastore
- Basic Administrative Operations for vSphere Datastores

Renaming a VMFS Datastore

Renaming a VMFS datastore triggers a background metadata refresh that generates a spike in read I/O on connected ESX hosts. Renaming in bulk using automation can overload your storage network and degrade performance.

Renaming a VMFS datastore triggers a background metadata refresh on a per-host and per-datastore basis. This refresh generates a spike in read I/O as the ESX hosts process `SlowRefresh` operations.

Because a datastore's metadata footprint scales with its capacity, larger volumes require significantly more read processing. Consequently, renaming datastores in bulk using automation, such as PowerCLI or scripts, can trigger a concurrent I/O storm.

This influx of read requests can saturate storage array controllers, which can cause high latency across the SAN and degrade overall array performance.

vSphere VMFS Locking Mechanisms

In a shared storage environment, when multiple ESX hosts access the same VMFS datastore, locking mechanisms are used. The locking mechanisms prevent multiple hosts from concurrently writing to the metadata and ensure that no data corruption occurs. VMFS supports ATS-only and ATS+SCSI locking. Learn how to display information about locking mechanisms, how to change from ATS+SCSI to ATS-only, or how to downgrade to ATS+SCSI.

To learn more about VMFS as a cluster file system, see vSphere VMFS Datastore Concepts and Operations.

Depending on its configuration and the type of underlying storage, a VMFS datastore can use different types of locking mechanisms. It can exclusively use the atomic test and set locking mechanism (ATS-only), or use a combination of ATS and SCSI reservations (ATS+SCSI).

What is ATS-Only Mechanism?

For storage devices that support T10 standard-based VAAI specifications, VMFS provides ATS locking, also called hardware assisted locking. The ATS algorithm supports discrete locking per disk sector. All newly formatted VMFS5 and VMFS6 datastores use the ATS-only mechanism if the underlying storage supports it, and never use SCSI reservations.

When you create a multi-extent datastore where ATS is used, vCenter filters out non-ATS devices. This filtering allows you to use only those devices that support the ATS primitive.

In certain cases, you might need to change the default locking mechanism for a VMFS5 or VMFS6 datastore. For information, see Change Locking Mechanism to ATS+SCSI.

> **Note:**
>
> If you are running a VMware vSAN environment or have ATS-only VMFS volumes, do not deactivate ATS. Deactivating ATS might result in an outage as no locking mechanism is available. For more information, see a VMware at Broadcom Knowledge Base article 319797.

What is ATS+SCSI Mechanism?

A VMFS datastore that supports the ATS+SCSI mechanism is configured to use ATS and attempts to use it when possible. If ATS fails, the VMFS datastore reverts to SCSI reservations. In contrast with the ATS locking, the SCSI reservations lock an entire storage device while an operation that requires metadata protection is performed. After the operation completes, VMFS releases the reservation, and other operations can continue.

Datastores that use the ATS+SCSI mechanism include VMFS5 datastores that were upgraded from VMFS3. In addition, new VMFS5 or VMFS6 datastores on storage devices that do not support ATS use the ATS+SCSI mechanism.

If the VMFS datastore reverts to SCSI reservations, you might notice performance degradation caused by excessive SCSI reservations.

Display VMFS Locking Information

Use the `esxcli` command to obtain information about the locking mechanism that a VMFS datastore uses.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To display information related to VMFS locking mechanisms, run the following command:

```
esxcli storage vmfs lockmode list
```

The table lists items that the output of the command might include.

Table 176: VMFS Locking Information

Fields	Values	Descriptions
Locking Modes		Indicates the locking configuration of the datastore.
	`ATS`	The datastore is configured to use the ATS-only locking mode.
	`ATS+SCSI`	The datastore is configured to use the ATS mode. If ATS fails or is not supported, the datastore can revert to SCSI.
	`ATS upgrade pending`	The datastore is in the process of an online upgrade to the ATS-only mode.
	`ATS downgrade pending`	The datastore is in the process of an online downgrade to the ATS+SCSI mode.

Fields	Values	Descriptions
ATS Compatible		Indicates whether the datastore can be or cannot be configured for the ATS-only mode.
ATS Upgrade Modes		Indicates the type of upgrade that the datastore supports.
	None	The datastore is not ATS-only compatible.
	Online	The datastore can be used during its upgrade to the ATS-only mode.
	Offline	The datastore cannot be used during its upgrade to the ATS-only mode.
ATS Incompatibility Reason		If the datastore is not compatible with ATS-only, the item indicates the reason for the incompatibility.

Upgrading Legacy VMFS Datastores to ATS-Only Locking Mechanism

If your VMFS datastore uses the ATS+SCSI locking mechanism, you can change to ATS-only locking.

Typically, VMFS5 datastores that were previously upgraded from VMFS3 continue using the ATS+SCSI locking mechanism. If the datastores are deployed on ATS-enabled hardware, they are eligible for an upgrade to ATS-only locking. Depending on your vSphere environment, you can use one of the following upgrade modes:

- The online upgrade to the ATS-only mechanism is available for most single-extent VMFS5 datastores. While you perform the online upgrade on one of the hosts, other hosts can continue using the datastore.
- The offline upgrade to ATS-only must be used for VMFS5 datastores that span multiple physical extents. Datastores composed of multiple extents are not eligible for the online upgrade. These datastores require that no hosts actively use the datastores at the time of the upgrade request.

Prepare for an Upgrade to ATS-Only Locking

You must perform several steps to prepare your environment for an online or offline upgrade to ATS-only locking.

1. Upgrade all hosts that access the VMFS5 datastore to the newest version of vSphere.
2. Determine whether the datastore is eligible for an upgrade of its current locking mechanism by running the `esxcli storage vmfs lockmode list` command.

   The following sample output indicates that the datastore is eligible for an upgrade. It also shows the current locking mechanism and the upgrade mode available for the datastore.

```
Locking Mode  ATS Compatible  ATS Upgrade Modes
------------  --------------  -----------------
ATS+SCSI           true       Online or Offline
```

3. Depending on the upgrade mode available for the datastore, perform one of the following actions:

Upgrade Mode	Action
Online	Verify that all hosts have consistent storage connectivity to the VMFS datastore.
Offline	Verify that no hosts are actively using the datastore.

Upgrade Locking Mechanism to the ATS-Only Type

If a VMFS datastore is ATS-only compatible, you can upgrade its locking mechanism from ATS+SCSI to ATS-only.

If you plan to complete the upgrade of the locking mechanism by putting the datastore into maintenance mode, deactivate Storage DRS. This prerequisite applies only to an online upgrade.

Most datastores that do not span multiple extents are eligible for an online upgrade. While you perform the online upgrade on one of the ESX hosts, other hosts can continue using the datastore. The online upgrade completes only after all hosts have closed the datastore.

1. Perform an upgrade of the locking mechanism by running the following command:

   `esxcli storage vmfs lockmode set -a|--ats -l|--volume-label=VMFS label -u|--volume-uuid=VMFS UUID`.

2. For an online upgrade, perform additional steps.
   a) Close the datastore on all hosts that have access to the datastore, so that the hosts can recognize the change.

      You can use one of the following methods:

      - Unmount and mount the datastore.
      - Put the datastore into maintenance mode and exit maintenance mode.

   b) Verify that the Locking Mode status for the datastore changed to ATS-only by running:

      `esxcli storage vmfs lockmode list`

   c) If the Locking Mode displays any other status, for example ATS UPGRADE PENDING, check which host has not yet processed the upgrade by running:

      `esxcli storage vmfs host list`

Change Locking Mechanism to ATS+SCSI

When you create a VMFS5 datastore on a device that supports the atomic test and set (ATS) locking, the datastore uses the ATS-only locking mechanism. In certain circumstances, you might need to downgrade the ATS-only locking to ATS +SCSI.

You might need to switch to the ATS+SCSI locking mechanism when, for example, your storage device is downgraded. Or when firmware updates fail and the device no longer supports ATS.
The downgrade process is similar to the ATS-only upgrade. As with the upgrade, depending on your storage configuration, you can perform the downgrade in online or offline mode.

> **Note:**
> If you are running a VMware vSAN environment or have ATS-only VMFS volumes, do not deactivate ATS. Deactivating ATS might result in an outage as no locking mechanism is available. For more information, see a VMware at Broadcom Knowledge Base article 319797.

1. Change the locking mechanism to ATS+SCSI by running the following command:

   `esxcli storage vmfs lockmode set -s|--scsi -l|--volume-label= VMFS label -u|--volume-uuid= VMFS UUID`.

2. For an online mode, close the datastore on all hosts that have access to the datastore, so that the hosts can recognize the change.

vSphere VMFS Datastore Copies and Datastore Resignaturing

When a storage device contains a VMFS datastore copy, ESX detects the copy. You can mount the datastore copy with the original datastore UUID or change the UUID. The process of changing the UUID is called datastore resignaturing.

Each VMFS datastore created on a storage device has a unique signature, also called UUID, that is stored in the file system superblock. When the storage device is replicated or its snapshot is taken on the array side, the resulting device copy is identical, byte-for-byte, with the original device. For example, if the original storage device contains a VMFS datastore with UUIDX, the copy appears to contain a datastore copy with the same UUIDX.

In addition to LUN snapshots and replications, certain device operations, such as LUN ID changes, might produce a copy of the original datastore.

After the ESX host detects the VMFS datastore copy, you can mount the datastore copy with its original UUID or change the UUID using the datastore resignaturing.

Whether you select resignaturing or mounting without resignaturing depends on how the LUNs are masked in the storage environment. If your hosts can see both copies of the LUN, then resignaturing is the optimal method.

Keeping Existing Datastore Signature

If you do not need to resignature a VMFS datastore copy, you can mount it without changing its signature.

You can keep the signature if, for example, you maintain synchronized copies of virtual machines at a secondary site as part of a disaster recovery plan. In the event of a disaster at the primary site, you mount the datastore copy and power on the virtual machines at the secondary site.

Resignaturing a VMFS Datastore Copy

Use datastore resignaturing if you want to retain the data stored on the VMFS datastore copy.

When resignaturing a VMFS copy, ESX assigns a new signature (UUID) to the copy, and mounts the copy as a datastore distinct from the original. All references to the original signature in virtual machine configuration files are updated.

When you perform datastore resignaturing, consider the following points:

- Datastore resignaturing is irreversible.
- After resignaturing, the storage device replica that contained the VMFS copy is no longer treated as a replica.
- A spanned datastore can be resignatured only if all its extents are online.
- The resignaturing process is fault tolerant. If the process is interrupted, you can resume it later.
- You can mount the new VMFS datastore without a risk of its UUID conflicting with UUIDs of any other datastore from the hierarchy of device snapshots.

Resignature a VMFS Datastore Copy

Use datastore resignaturing if you want to retain the data stored on the VMFS datastore copy.

- Unmount the datastore copy.
- Perform a storage rescan on your host.

When resignaturing a VMFS copy, ESX assigns a new signature (UUID) to the copy, and mounts the copy as a datastore distinct from the original. All references to the original signature from virtual machine configuration files are updated.

When you perform datastore resignaturing, consider the following points:

- Datastore resignaturing is irreversible.
- After resignaturing, the storage device replica that contained the VMFS copy is no longer treated as a replica.
- A spanned datastore can be resignatured only if all its extents are online.
- The resignaturing process is fault tolerant. If the process is interrupted, you can resume it later.
- You can mount the new VMFS datastore without a risk of its UUID conflicting with UUIDs of any other datastore from the hierarchy of device snapshots.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. Click the **New Datastore** icon.
3. Enter the datastore name and if necessary, select the placement location for the datastore.
4. Select VMFS as the datastore type.
5. From the list of storage devices, select the device that has a specific value displayed in the Snapshot Volume column.

   The value present in the Snapshot Volume column indicates that the device is a copy that contains a copy of an existing VMFS datastore.
6. Under **Mount Options**, select **Assign a New Signature** and click **Next**.
7. Review the datastore configuration information and click **Finish**.

Increase Capacity of vSphere VMFS Datastore

You can increase the capacity of a VMFS datastore on the ESX host. Additional capacity might be required when you add virtual machines to the datastore, or when the virtual machines running on the datastore require more space.

You can increase the datastore capacity if the host storage meets one of the following conditions:

- The backing device for the existing datastore has enough free space.

- You added new storage devices to the host.

Before you increase the capacity of a shared VMFS datastore, make sure you meet the following requirements to prevent data corruption or other problems:

- Storage-Side Provisioning
  For datastore expansions, you must first increase the underlying volume or LUN size on the backing block storage array, such as iSCSI or Fibre Channel SAN.
- Mandatory Multi-Host Rescan (Before Expansion)
  In environments where multiple ESX hosts share the same datastore, you must perform a storage device rescan on all sharing hosts before executing the VMFS expansion. If you fail to perform a storage rescan across all sharing hosts before expanding the VMFS volume, problems might occur that can cause unexpected virtual machine outages, data loss, or active workload downtime.

If a shared datastore has powered on virtual machines and becomes 100% full, you can increase the datastore capacity. You can perform this action only from the host where the powered on virtual machines are registered.
Depending on your storage configuration, you can use one of the following methods to increase the datastore capacity. You do not need to power off virtual machines when using either method of increasing the datastore capacity.

Expand an Existing Datastore	Increase the size of an expandable datastore. The datastore is considered expandable when the backing storage device has free space immediately after the datastore extent.
Add an Extent	Increase the capacity of an existing VMFS datastore by adding new storage devices to the datastore. The datastore can span over multiple storage devices, yet appear as a single volume. The spanned VMFS datastore can use any or all its extents at any time. It does not need to fill up a particular extent before using the next one. **Note:** Datastores that support only the hardware assisted locking, also called the atomic test and set (ATS) mechanism, cannot span over non-ATS devices. For more information, see vSphere VMFS Locking Mechanisms.

1. In the vSphere Client, navigate to the datastore.
2. Select **Increase Datastore Capacity** from the datastore right-click menu.
3. Select a device from the list of storage devices.
   Your selection depends on whether an expandable storage device is available.

Option	Description
To expand an existing datastore extent	Select the device for which the Expandable column reads YES.
To add an extent	Select the device for which the Expandable column reads NO.

4. Review the **Partition Layout** to see the available configurations.
5. Select a configuration option from the bottom panel.
   Depending on the current layout of the disk and on your previous selections, the menu items you see might vary.

Menu Item	Description
Use free space to expand the datastore	Expands an existing extent to a required capacity.
Use free space	Deploys an extent in the remaining free space of the disk. This menu item is available only when you are adding an extent.
Use all available partitions	Dedicates the entire disk to a single extent. This menu item is available only when you are adding an extent and when the disk you are formatting is not blank. The disk is reformatted, and the datastores and any data that it contains are erased.

6. Set the capacity for the extent.
   The minimum extent size is 1.3 GB. By default, the entire free space on the storage device is available.

7. Click **Next**.
8. Review the proposed layout and the new configuration of your datastore, and click **Finish**.

   Note:
   It is recommended to perform a cluster level rescan at vCenter after you perform any form of expansion done at the host level. If you do not perform a cluster level rescan from vCenter, it may not detect any change. This causes a disparity in the current space/usage for the datastore.

   If the device where you increased the capacity is exposed to multiple hosts in multiple clusters, then the cluster level rescan must be triggered on all the clusters where the device is visible.

Configure Settings for Clustered Virtual Disks on the vSphere VMFS6 Datastore

If you plan to use a virtual disk in Windows Server Failover Clustering (WSFC) configurations, your VMFS6 datastore must support clustered virtual disks. Use the vSphere Client to configure the clustered disk support.

Follow these guidelines when you use a datastore for clustered virtual disks:

- The storage array supports the ATS, Write Exclusive – All Registrant (WEAR) SCSI-3 reservation type. For this type of configurations, ESX supports only Fibre Channel arrays.
- Storage devices that back the datastore are claimed by the NMP. ESX does not support third-party plug-ins (MPPs) in the clustered virtual disk configurations.
- The datastore is of VMFS6 type. You cannot expand or span the datastore.
- The VMFS6 datastore is deployed on the storage device with the attribute `Clustered VMDK Supported` set to Yes.
- Virtual disks you use for clustering are in the Thick Provision Eager Zeroed format.

For information on the use of clustered virtual disks in VM clusters, see the *Setup for Windows Server Failover Clustering* documentation.

1. In the vSphere Client, navigate to the datastore.
2. Click the **Configure** tab, and click **General**.
3. Under **Datastore Capabilities**, click one of the following options next to the **Clustered VMDK** item.

Option	Description
Enable	To activate support for clustered virtual disks on the datastore. After you activate the support, you can place the clustered virtual disks on this VMFS datastore.
Disable	To deactivate the support. Before deactivating, make sure to power off all virtual machines with the clustered virtual disks.

4. Confirm your configuration.

Manage Filters for LUNs and vSphere VMFS Datastores

When you perform VMFS datastore management operations, vCenter uses default storage protection filters. The filters help you to avoid storage corruption by retrieving only the storage devices that can be used for a particular operation. Unsuitable devices are not displayed for selection. You can turn off the filters to view all devices.

Before you change the device filters, consult with the VMware support team.

These filters are available by default.

Filter Name	Description
config.vpxd.filter.vmfsFilter (VMFS Filter)	Filters out storage devices, or LUNs, that are already used by a VMFS datastore on any host managed by vCenter. The LUNs do not show up as candidates to be formatted with another VMFS datastore or to be used as an RDM.
config.vpxd.filter.rdmFilter (RDM Filter)	Filters out LUNs that are already referenced by an RDM on any host managed by vCenter. The LUNs do not show up as candidates to be formatted with VMFS or to be used by a different RDM. For your virtual machines to access the same LUN, the virtual machines must share the same RDM mapping file. For information about this type of configuration, see the *vSphere Resource Management* documentation.
config.vpxd.filter.sameHostsAndTransportsFilter (Same Hosts and Transports Filter)	Filters out LUNs ineligible for use as VMFS datastore extents because of host or storage type incompatibility. Prevents you from adding the following LUNs as extents: • LUNs not exposed to all hosts that share the original VMFS datastore. • LUNs that use a storage type different from the one the original VMFS datastore uses. For example, you cannot add a Fibre Channel extent to a VMFS datastore on a local storage device.
config.vpxd.filter.hostRescanFilter (Host Rescan Filter)	Automatically rescans and updates VMFS datastores after you perform datastore management operations. The filter helps provide a consistent view of all VMFS datastores on all hosts managed by vCenter. **Note:** If you present a new LUN to a host or a cluster, the hosts automatically perform a rescan no matter whether you have the Host Rescan Filter on or off.

1. Browse to the vCenter instance.
2. Click the **Configure** tab.
3. Under **Settings**, click **Advanced Settings**, and click **EDIT SETTINGS**.
4. Specify the filter to turn off.
   In the **Name** and **Value** text boxes at the bottom of the page, enter appropriate information.

Name	Value
config.vpxd.filter.vmfsFilter	False
config.vpxd.filter.rdmFilter	False
config.vpxd.filter.sameHostsAndTransportsFilter	False
config.vpxd.filter.hostRescanFilter	False **Note:** If you turn off this filter, your hosts continue to perform a rescan each time you present a new LUN to a host or a cluster.

5. Click **ADD**, and click **SAVE** to save your changes.
   You are not required to restart the vCenter system.

Set Up Dynamic Disk Mirroring for Virtual Machines on vSphere VMFS

Typically, in the ESX environment, you cannot use LUN manager software on virtual machines to mirror virtual disks. However, if your Microsoft Windows virtual machines support dynamic disks, you can mirror virtual disks across two SAN LUNs. Mirroring helps you to protect the virtual machines from an unplanned storage device loss.

- Use a Windows virtual machine that supports dynamic disks.
- Required privilege: **Virtual machine** > **Configuration** > **Settings**

1. Create a virtual machine with two virtual disks.
   Place the disks on different datastores.
2. Log in to your virtual machine and configure the disks as dynamic mirrored disks.
   See Microsoft documentation.
3. After the disks synchronize, power off the virtual machine.
4. Change virtual machine settings to enable the dynamic disk mirroring.
   a) Right-click the virtual machine and select **Edit Settings**.
   b) Click the **VM Options** tab and expand the **Advanced** menu.
   c) Click **Edit Configuration** next to Configuration Parameters.
   d) Click **Add Configuration Params** and add the following parameters:
      Replace `#` with the boot device SCSI bus controller ID number. By default, it is set to 0.

Name	Value
`scsi#.returnNoConnectDuringAPD`	True
`scsi#.returnBusyOnNoConnectStatus`	False

   e) For ESX version 6.7 or later, include an additional parameter for each virtual disk participating in the software RAID-1 configuration.
      The parameter prevents guest OS I/O failures when a storage device fails.
      Replace `#` with the boot device SCSI bus controller ID number. By default, it is set to 0.

Name	Value
`scsi#:1.passthruTransientErrors`	True
`scsi#:2.passthruTransientErrors`	True

   f) Click **OK**.

Setting Up ESX Core Dump on a VMFS Datastore

During a host failure, ESX must be able to save diagnostic information to a preconfigured location for diagnostic and technical support purposes.

Typically, a partition to collect diagnostic information, also called a core dump, is created on a local storage device during ESX installation. You can also configure an ESX Dump Collector and keep core dumps on a network server for use during debugging. For information on setting up the ESX Dump Collector, see Configure an ESX Dump Collector in the *VMware ESX Installation and Setup* documentation.

Another option is to use a file on a VMFS datastore to collect the diagnostic information.

Set Up a File as Core Dump Location

If the size of your available core dump partition is insufficient, you can configure ESX to use a file on a VMFS datastore for diagnostic information.

- Make sure that the amount of available space in the datastore for a core dump file is twice as large as the recommended file size. For more information, see the Knowledge Base article 318862.

- Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

  Note:
  VMFS datastores on software iSCSI do not support core dump files.

1. Create a VMFS datastore core dump file by running the following command:

   ```
   esxcli system coredump file add
   ```

   The command takes the following options, but they are not required and can be omitted:

Option	Description
--datastore | -d *datastore_UUID or datastore_name*	Specify the datastore for the dump file. If not provided, the system selects a datastore of sufficient size.
--file | -f *file_name*	Specify the filename of the dump file. If not provided, the system creates a unique name for the file.
--size |-s *file_size_MB*	Set the size in MB of the dump file. If not provided, the system creates a file of the size appropriate for the memory installed in the host.

2. Verify that the file has been created:

   ```
   esxcli system coredump file list
   ```

   You can see the output similar to the following:

   ```
   Path                                          Active Configured Size
   --------------------------------------------- ------ ---------- ---------
   /vmfs/volumes/52b021c3-.../vmkdump/test.dumpfile false  false      104857600
   ```

3. Activate the core dump file for the host:

   ```
   esxcli system coredump file set
   ```

   The command takes the following options:

Option	Description
--enable |-e	Activate or deactivate the dump file. This option cannot be specified when unconfiguring the dump file.
--path | -p	The path of the core dump file to use. The file must be pre-allocated.
--smart | -s	This flag can be used only with **--enable | -e=true**. It causes the file to be selected using the smart selection algorithm. For example, `esxcli system coredump file set --smart --enable true`
--unconfigure | -u	Unconfigure the current VMFS dump file.

4. Verify that the core dump file is active and configured:

   ```
   esxcli system coredump file list
   ```

   The output similar to the following indicates that the core dump file is active and configured:

   ```
   Path                                          Active Configured Size
   --------------------------------------------- ------ ---------- ---------
   /vmfs/volumes/52b021c3-.../vmkdump/test.dumpfile True   True       104857600
   ```

For information about other commands you can use to manage the core dump files, see the *ESXCLI Reference* documentation.

Deactivate and Delete a Core Dump File

Deactivate a configured core dump file and, if needed, remove it from the VMFS datastore.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

You can temporarily deactivate the core dump file. If you do not plan to use the deactivated file, you can remove it from the VMFS datastore. To remove the file that has not been deactivated, you can use the `esxcli system coredump file remove` command with the `--force | -F` parameter.

1. List the core dump files:
   ```
   esxcli system coredump file list
   ```
2. Deactivate the core dump file by running the following command:
   ```
   esxcli system coredump file set --unconfigure | -u
   ```
3. Remove the file from the VMFS datastore:
   ```
   esxcli system coredump file remove --file | -f file_name
   ```
   The command takes the following options:

Option	Description
**`--file	-f`**	Enter the name of the dump file to be removed. If you do not enter the name, the command removes the default configured core dump file.
**`--force	-F`**	Deactivate and unconfigure the dump file being removed. This option is required if the file has not been previously deactivated and is active.

The core dump file is removed from the VMFS datastore.

Checking Metadata Consistency with vSphere On-disk Metadata Analyzer

Use the vSphere On-disk Metadata Analyzer (VOMA) to check metadata consistency and identify and fix incidents of metadata corruption on the VMFS datastores or logical volumes that back the VMFS datastores.

You can check metadata consistency when you experience problems with a VMFS datastore or a virtual flash resource. For example, perform a metadata check if one of the following occurs:

- You experience storage outages.
- After you rebuild RAID or perform a disk replacement.
- You see metadata errors in the `vmkernel.log` file similar to the following:
  ```
  cpu11:268057)WARNING: HBX: 599: Volume 50fd60a3-3aae1ae2-3347-0017a4770402 ("<Datastore_name>") may be
   damaged on disk. Corrupt heartbeat detected at offset 3305472: [HB state 0 offset 6052837899185946624 gen
   15439450 stampUS 5 $
  ```
- You are unable to access files on a VMFS.
- You see corruption being reported for a datastore in events tabs of vCenter.

To check metadata consistency, run VOMA from the CLI of an ESX host. VOMA can be used to check and fix minor inconsistency issues for a VMFS datastore or logical volumes that back the VMFS datastore.

VOMA can check and fix the following items.

Table 177: VOMA Functions

VOMA Functions	Description
Metadata check and fix	Examples of metadata check and fix include, but are not limited to, the following: • Validation of VMFS volume header for basic metadata consistency. • Checking consistency of VMFS resource files (system file). • Checking the pathname and connectivity of all files.
Directory validation	VOMA can detect and correct the following errors: • Directory hash block corruption. • Alloc map corruption. • Link blocks corruptions. • Directory entry block corruptions. Based on the nature of the corruption, VOMA can either fix only the corrupted entries or fully reconstruct the hash block, alloc map blocks and link blocks.
Lost and found files	During a filesystem check, VOMA can find files that are not referenced anywhere in the filesystem. These orphaned files are valid and complete, but do not have a name or directory entry on the system. If VOMA encounters orphaned files during scanning, it creates a directory named lost+found at the root of the volume to store the orphaned files. The names of the files use the `File`*`sequence-number`* format.

Command options that the VOMA tool takes include the following.

Table 178: VOMA Command Options

Command Option	Description	
`-m|--module`	The modules to run include the following:	
	`vmfs`	If you do not specify the name of the module, this option is used by default. You can check the VMFS file systems and the file systems that back virtual flash resources. If you specify this module, minimal checks are performed for LVM as well.
	`lvm`	Check logical volumes that back the VMFS datastores.
	`ptbl`	Check and validate VMFS partitions, such as MBR or GPT. If no partition exists, determine whether partitions should exist.
`-f|--func`	Functions to be performed include the following:	
	`query`	List functions supported by module.
	`check`	Check for errors.
	`fix`	Check and fix errors for PTBL module.
	`advfix`	Check and fix for VMFS and LVM modules.
	`dump`	Collect metadata dump.
`-A|--skipAffinityChk`	Skip affinity related check/fix for VMFS6.	
`-d|--device`	Indicate the device or disk to inspect. Make sure to provide the absolute path to the device partition backing the VMFS datastore. If the datastore spans multiple devices, provide the UUID of the head extent. For example, `voma -m vmfs -f check -d /vmfs/devices/disks/naa.xxxx:x` If you use the `-x|--extractDump` command, enter multiple device paths, with a partition qualifier, separated with a comma. The number of device paths you enter equals the number of spanned devices.	

Command Option	Description
-b|--blockSize	Indicate the disk block size.
-s|--logfile	Specify the path to log file to output the results.
-x|--extractDump	Extract the collected dump using VOMA.
-D|--dumpfile	Indicate the dump file to save the collected metadata dump.
-v|--version	Display the version of VOMA.
-h|--help	Display the help message for the VOMA command.
-Z| --file	Indicate that you run VOMA on extracted device files.
-p |--patchfiledir	Indicate the file to store the corrupted VMFS/LVM metadata.
-r | --restorefile	Restore previous state using patch file generated by advfix mode.
-l | --live	Perform dump operation on an active datastore.

Use the following as examples:

```
voma -m vmfs -f check -d /vmfs/devices/disks/naa.xxxx:x
```

```
voma -m vmfs -f dump -d /vmfs/devices/disks/naa.xxxx:x -D dumpfilename
```

```
voma -r patchfilename -d /vmfs/devices/disks/naa.xxxx:x
```

Use VOMA to Check Metadata Consistency

The task demonstrates how to use VOMA to check VMFS metadata consistency. VOMA can be used to check and fix minor inconsistency issues for a VMFS datastore or a virtual flash resource. Run VOMA from the CLI of an ESX host.

Power off any virtual machines that are running or migrate them to a different datastore.

1. Obtain the name and partition number of the device that backs the VMFS datastore that you want to check.

   ```
   #esxcli storage vmfs extent list
   ```

   The Device Name and Partition columns in the output identify the device. For example:

   ```
   Volume Name  .....  Device Name                              Partition
   1TB_VMFS6    .....  naa.xxxx                                    3
   ```

2. Check for VMFS errors.

   Provide the absolute path to the device partition that backs the VMFS datastore, and provide a partition number with the device name. For example:

   ```
   # voma -m vmfs -f check -d /vmfs/devices/disks/naa.xxxx:x
   ```

   The output lists possible errors. For example, the following output indicates that the heartbeat address is invalid.

   ```
   XXXXXXXXXXXXXXXXXXXXXXX
   Phase 2: Checking VMFS heartbeat region
    ON-DISK ERROR: Invalid HB address
   Phase 3: Checking all file descriptors.
   Phase 4: Checking pathname and connectivity.
   Phase 5: Checking resource reference counts.

   Total Errors Found:          1
   ```

Configuring Pointer Block Cache for vSphere VMFS Datastores

Pointer blocks, also called indirection blocks, are file system resources that contain addresses to VMFS file blocks. When you open a `vmdk` file on an ESX host, pointer blocks related to that file are stored in the pointer block cache. The size of the pointer block cache is a configurable parameter.

The pointer block cache is a host-wide cache that is independent from VMFS. The cache is shared across all datastores that are accessed from the same ESX host.

The size of the pointer block cache is controlled by `/VMFS3/MinAddressableSpaceTB` and `/VMFS3/MaxAddressableSpaceTB`. You can configure the minimum and maximum sizes on each ESX host.

Parameter	Description
`/VMFS3/MinAddressableSpaceTB`	The minimum value is minimum amount of memory that the system guarantees to the pointer block cache. For example, 1 TB of open file space requires approximately 4 MB of memory. Default value is 10 TB.
`/VMFS3/MaxAddressableSpaceTB`	The parameter defines the maximum limit of pointer blocks that can be cached in memory. Default value is 32 TB. Maximum value is 128 TB. Typically, the default value of the `/VMFS3/MaxAddressableSpaceTB` parameter is adequate. However, as the size of the open vmdk files increases, the number of pointer blocks related to those files also increases. If the increase causes any performance degradation, you can adjust the parameter to its maximum value to provide more space for the pointer block cache. Base the maximum size of the pointer block cache on the working set, or the active pointer blocks required.
`Pointer Block Eviction`	The `/VMFS3/MaxAddressableSpaceTB` parameter also controls the growth of the pointer block cache. When the size of the pointer block cache approaches the configured maximum size, a pointer block eviction process starts. The mechanism leaves active pointer blocks, but removes non-active or less active blocks from the cache, so that space can be reused.

To change the values for the pointer block cache, use the **Advanced System Settings** dialog box of the vSphere Client or the `esxcli system settings advanced set -o` command.

You can use the `esxcli storage vmfs pbcache` command to obtain information about the size of the pointer block cache and other statistics. This information assists you in adjusting minimum and maximum sizes of the pointer block cache, so that you can get maximum performance.

Obtain Information for VMFS Pointer Block Cache

You can get information about VMFS pointer block cache use. This information helps you understand how much space the pointer block cache consumes. You can also determine whether you must adjust the minimum and maximum sizes of the pointer block cache.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To obtain or reset the pointer block cache statistics, use the following command:

```
esxcli storage vmfs pbcache
```

Option	Description
`get`	Get VMFS pointer block cache statistics.
`reset`	Reset the VMFS pointer block cache statistics.

Getting Statistics for Pointer Block Cache

```
#esxcli storage vmfs pbcache get
   Cache Capacity Miss Ratio: 0 %
   Cache Size: 0 MiB
   Cache Size Max: 132 MiB
   Cache Usage: 0 %
   Cache Working Set: 0 TiB
   Cache Working Set Max: 32 TiB
   Vmfs Heap Overhead: 0 KiB
   Vmfs Heap Size: 23 MiB
   Vmfs Heap Size Max: 256 MiB
```

Change the Size of the Pointer Block Cache

You can adjust the minimum and maximum sizes of the pointer block cache.

> **CAUTION:**
> Changing advanced options is considered unsupported. Typically, the default settings produce the optimum result. Change the advanced options only when you get specific instructions from VMware technical support or a knowledge base article.

1. Browse to the host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. In Advanced System Settings, select the appropriate item.

Option	Description
`VMFS3.MinAddressableSpaceTB`	Minimum size of all open files that VMFS cache guarantees to support.
`VMFS3.MaxAddressableSpaceTB`	Maximum size of all open files that VMFS cache supports before eviction starts.

5. Click the **Edit** button and change the value.
6. Click **OK**.

Use the esxcli Command to Change the Pointer Block Cache

You can also use the `esxcli system settings advanced set -o` to modify the size of the pointer block cache. The following example describes how to set the size to its maximum value of 128 TB.

1. To change the value of `/VMFS3/MaxAddressableSpaceTB` to 128 TB, enter the following command:
   ```
   # esxcli system settings advanced set -i 128 -o /VMFS3/MaxAddressableSpaceTB
   ```
2. To confirm that the value is set correctly, enter this command:
   ```
   # esxcli system settings advanced list -o /VMFS3/MaxAddressableSpaceTB
   ```

NFS Datastore Concepts and Operations in vSphere Environment

An NFS client built into ESX uses the Network File System (NFS) protocol over TCP/IP to access a designated NFS volume that is located on a NAS server. The ESX host can mount the volume and use it for its storage needs. vSphere supports versions 3 and 4.1 of the NFS protocol.

Typically, the NFS volume or directory is created by a storage administrator and is exported from the NFS server. You do not need to format the NFS volume with a local file system, such as VMFS. Instead, you mount the volume directly on the ESX hosts and use it to store and boot virtual machines in the same way that you use the VMFS datastores.

In addition to storing virtual disks on NFS datastores, you can use NFS as a central repository for ISO images, virtual machine templates, and so on. If you use the datastore for the ISO images, you can connect the CD-ROM device of the virtual machine to an ISO file on the datastore. You then can install a guest operating system from the ISO file.

After you set up NFS storage environment, see the following topics to create and manage NFS datastores:

- Creating NFS Datastores with the vSphere Client
- Basic Administrative Operations for vSphere Datastores

NFS Protocols and ESX

ESX supports NFS protocols version 3 and 4.1. To support both versions, ESX uses two different NFS clients.

Comparing Versions of NFS Clients

The following table lists capabilities that the NFS version 3 and 4.1 support.

Characteristics	NFS version 3	NFS version 4.1
Security mechanisms	AUTH_SYS	AUTH_SYS and Kerberos (krb5, krb5i, krb5p)
Encryption algorithms with Kerberos	N/A	AES256-CTS-HMAC-SHA1-96 and AES128-CTS-HMAC-SHA1-96
Multipathing	Not supported	Supported through the session trunking
Locking mechanisms	Propriety client-side locking	Server-side locking
Hardware acceleration	Supported	Supported
Thick virtual disks	Supported	Supported
IPv6	Supported	Supported for AUTH_SYS and Kerberos
ISO images presented as CD-ROMs to virtual machines	Supported	Supported
Virtual machine snapshots	Supported	Supported
Virtual machines with virtual disks greater than 2 TB	Supported	Supported

NFS Protocols and vSphere Solutions

The following table lists major vSphere solutions that NFS versions support.

vSphere Features	NFS version 3	NFS version 4.1
vMotion and Storage vMotion	Yes	Yes
High Availability (HA)	Yes	Yes
Fault Tolerance (FT)	Yes	Yes
Distributed Resource Scheduler (DRS)	Yes	Yes
Host Profiles	Yes	Yes
Storage DRS	Yes	Initial placement.
Storage I/O Control	Yes	No
Site Recovery Manager	Yes	Site Recovery Manager does not support NFS 4.1 datastores for array-based replication and Virtual Volumes replication. You can use Site Recovery Manager with NFS v 4.1 datastores for vSphere Replication.
Virtual Volumes	Yes	Yes
vSphere Replication	Yes	Yes
VMware Cloud Foundation Operations (formerly Aria Operations)	Yes	Yes

NFS 4.1 and Fault Tolerance

Virtual machines on NFS 4.1 support the Fault Tolerance mechanism. The mechanism can accommodate symmetric multiprocessor (SMP) virtual machines with up to four vCPUs.

NFS Upgrades

When you upgrade ESX from a version earlier than 6.5, existing NFS 4.1 datastores automatically begin supporting functionalities that were not available in the previous ESX release. These functionalities include Virtual Volumes, hardware acceleration, and so on.

ESX does not support automatic datastore conversions from NFS version 3 to NFS 4.1.

If you want to upgrade your NFS 3 datastore, the following options are available:

- Create the NFS 4.1 datastore, and then use Storage vMotion to migrate virtual machines from the old datastore to the new one.
- Use conversion methods provided by your NFS storage server. For more information, contact your storage vendor.
- Unmount the NFS 3 datastore, and then mount as NFS 4.1 datastore.

  CAUTION:
  If you use this option, make sure to unmount the datastore from all hosts that have access to the datastore. The datastore can never be mounted by using both protocols at the same time.

Guidelines and Requirements for NFS Storage with ESX

When you use NFS storage with ESX, follow specific guidelines related to NFS server configuration, networking, NFS datastores, and so on.

NFS Server Configuration

When you configure NFS servers to work with ESX, follow recommendation of your storage vendor. In addition to these general recommendations, use specific guidelines that apply to NFS in vSphere environment.

The guidelines include the following items.

- Make sure that the NAS servers you use are listed in the *VMware HCL*. Use the correct version for the server firmware.
- Ensure that the NFS volume is exported using NFS over TCP.
- Make sure that the NAS server exports a particular share as either NFS 3 or NFS 4.1. The NAS server must not provide both protocol versions for the same share. The NAS server must enforce this policy because ESX does not prevent mounting the same share through different NFS versions.
- NFS 3 and non-Kerberos (AUTH_SYS) NFS 4.1 do not support the delegate user functionality that enables access to NFS volumes using nonroot credentials. If you use NFS 3 or non-Kerberos NFS 4.1, ensure that each host has root access to the volume. Different storage vendors have different methods of enabling this functionality, but typically the NAS servers use the `no_root_squash` option. If the NAS server does not grant root access, you can still mount the NFS datastore on the host. However, you cannot create any virtual machines on the datastore.
- If the underlying NFS volume is read-only, make sure that the volume is exported as a read-only share by the NFS server. Or mount the volume as a read-only datastore on the ESX host. Otherwise, the host considers the datastore to be read-write and might not open the files.

NFS Networking

An ESX host uses TCP/IP network connection to access a remote NAS server. Certain guidelines and best practices exist for configuring the networking when you use NFS storage.

For more information, see the *vSphere Networking* documentation.

- For network connectivity, use a standard network adapter in your ESX host.
- ESX supports Layer 2 and Layer 3 Network switches. If you use Layer 3 switches, ESX hosts and NFS storage arrays must be on different subnets and the network switch must handle the routing information.
- Configure a VMkernel port group for NFS storage. You can create the VMkernel port group for IP storage on an existing virtual switch (vSwitch) or on a new vSwitch. The vSwitch can be a vSphere Standard Switch (VSS) or a vSphere Distributed Switch (VDS).
- If you use multiple ports for NFS traffic, make sure that you correctly configure your virtual switches and physical switches.

- NFS 3 and NFS 4.1 support IPv6.
- You can configure NFS storage with multiple connections by using the `nconnect` option. For NFS 4.1, you can create multiple connections per session. For NFS 3, you can mount the datastore with multiple connections. You can set a maximum of four connections per NFS datastore by default. However, you can increase it up to eight by using the advanced NFS option. Ensure that the total number of connections across all mounted NFS datastores does not exceed 256. See Configure Multiple TCP Connections for NFS with the esxcli Command.
- You can isolate NFS traffic to specific VMkernel adapters. Without binding, if the VMkernel adapter that ESX uses for NFS traffic fails, the network infrastructure redirects the traffic to an alternative route. As a result, the NFS traffic might unintentionally flow through a random VMkernel adapter. VMkernel port binding for the NFS datastore allows you to bind an NFS volume to a specific VMkernel adapter to connect to an NFS server. See Configuring VMkernel Binding for NFS Datastores.

NFS File Locking

File locking mechanisms are used to restrict access to data stored on a server to only one user or process at a time. The locking mechanisms of the two NFS versions are not compatible. NFS 3 uses proprietary locking and NFS 4.1 uses native protocol specified locking.

NFS 3 locking on ESX does not use the Network Lock Manager (NLM) protocol. Instead, VMware provides its own locking protocol. NFS 3 locks are implemented by creating lock files on the NFS server. Lock files are named `.lck-`*`file_id`*`..`

NFS 4.1 uses share reservations as a locking mechanism.

Because NFS 3 and NFS 4.1 clients do not use the same locking protocol, you cannot use different NFS versions to mount the same datastore on multiple hosts. Accessing the same virtual disks from two incompatible clients might result in incorrect behavior and cause data corruption.

NFS Security

With NFS 3 and NFS 4.1, ESX supports the AUTH_SYS security. In addition, for NFS 4.1, the Kerberos security mechanism is supported.

NFS 3 supports the AUTH_SYS security mechanism. With this mechanism, storage traffic is transmitted in an unencrypted format across the LAN. Because of this limited security, use NFS storage on trusted networks only and isolate the traffic on separate physical switches. You can also use a private VLAN.

NFS 4.1 supports the Kerberos authentication protocol to secure communications with the NFS server. Nonroot users can access files when Kerberos is used. For more information, see Using Kerberos for NFS 4.1 with ESX.

In addition to Kerberos, NFS 4.1 supports traditional non-Kerberos mounts with the AUTH_SYS security. In this case, use root access guidelines for NFS version 3.

> **Note:**
> You cannot use two security mechanisms, AUTH_SYS and Kerberos, for the same NFS 4.1 datastore shared by multiple hosts.

NFS Multipathing

NFS 4.1 supports multipathing as per protocol specifications. For NFS 3 multipathing is not applicable.

NFS 3 uses one TCP connection for I/O. As a result, ESX supports I/O on only one IP address or hostname for the NFS server, and does not support multiple paths. Depending on your network infrastructure and configuration, you can use the network stack to configure multiple connections to the storage targets. In this case, you must have multiple datastores, each datastore using separate network connections between the host and the storage.

NFS 4.1 provides multipathing for servers that support the session trunking. When the trunking is available, you can use multiple IP addresses to access a single NFS volume. Client ID trunking is not supported.

NFS and Hardware Acceleration

Virtual disks created on NFS datastores are thin-provisioned by default. To be able to create thick-provisioned virtual disks, you must use hardware acceleration that supports the Reserve Space operation.

NFS 3 and NFS 4.1 support hardware acceleration that allows your host to integrate with NAS devices and use several hardware operations that NAS storage provides. For more information, see vSphere Hardware Acceleration on NAS Devices.

NFS Datastores

When you create an NFS datastore, make sure to follow specific guidelines.

The NFS datastore guidelines and best practices include the following items. To create an NFS datastores, see Create an NFS 4.1 Datastore.

- You cannot use different NFS versions to mount the same datastore on different hosts. NFS 3 and NFS 4.1 clients are not compatible and do not use the same locking protocol. As a result, accessing the same virtual disks from two incompatible clients might result in incorrect behavior and cause data corruption.
- NFS 3 and NFS 4.1 datastores can coexist on the same host.
- ESX cannot automatically upgrade NFS version 3 to version 4.1, but you can use other conversion methods. For information, see NFS Upgrades.
- When you mount the same NFS 3 volume on different hosts, make sure that the server and folder names are identical across the hosts. If the names do not match, the hosts see the same NFS version 3 volume as two different datastores. This error might result in a failure of such features as vMotion. An example of such discrepancy is entering `filer` as the server name on one host and `filer.domain.com` on the other. This guideline does not apply to NFS version 4.1.
- If you use non-ASCII characters to name datastores and virtual machines, make sure that the underlying NFS server offers internationalization support. If the server does not support international characters, use only ASCII characters, or unpredictable failures might occur.
- If you mount an NFS datastore using the DNS name and the NFS server IP address changes, you can refresh the DNS connection by running the `esxcli storage` commands. See Refreshing NFS Server IP Addresses for DNS-Mounted Datastores.

Using Layer 3 Routed Connections to Access NFS Storage with ESX

When you use Layer 3 (L3) routed connections to access NFS storage, consider certain requirements and restrictions.

Ensure that your environment meets the following requirements:

- Use Cisco's Hot Standby Router Protocol (HSRP) in IP Router. If you are using a non-Cisco router, use Virtual Router Redundancy Protocol (VRRP) instead.
- To prioritize NFS L3 traffic on networks with limited bandwidths, or on networks that experience congestion, use Quality of Service (QoS). See your router documentation for details.
- Follow Routed NFS L3 recommendations offered by storage vendor. Contact your storage vendor for details.
- Deactivate Network I/O Resource Management (NetIORM).
- If you are planning to use systems with top-of-rack switches or switch-dependent I/O device partitioning, contact your system vendor for compatibility and support.

In an L3 environment, the following restrictions apply:

- The environment does not support VMware Site Recovery Manager.
- The environment supports only the NFS protocol. Do not use other storage protocols such as FCoE over the same physical network.
- The NFS traffic in this environment does not support IPv6.
- The NFS traffic in this environment can be routed only over a LAN. Other environments such as WAN are not supported.

Firewall Configurations for NFS Storage with ESX

Learn about firewall rule set files, `nfsClient` and `nfs41client`, that ESX creates when you mount NFS datatores version 3 or 4.1

For general information about firewall configurations, see *Configuring the ESX Firewall* in the *vSphere Security* documentation.

NFS Client Firewall Behavior

The NFS Client firewall rule set behaves differently than other ESX firewall rule sets. ESX configures NFS Client settings when you mount or unmount an NFS datastore. The behavior differs for different versions of NFS.

When you add, mount, or unmount an NFS datastore, the resulting behavior depends on the version of NFS.

NFS v3 Firewall Behavior

When you add or mount an NFS v3 datastore, ESX checks the state of the NFS Client (`nfsClient`) firewall rule set.

- If the `nfsClient` rule set is deactivated, ESX activates the rule set and deactivates the Allow All IP Addresses policy by setting the `allowedAll` flag to `FALSE`. The IP address of the NFS server is added to the allowed list of outgoing IP addresses.
- If the `nfsClient` rule set is activated, the state of the rule set and the allowed IP address policy are not changed. The IP address of the NFS server is added to the allowed list of outgoing IP addresses.

> **Note:**
> If you manually activate the `nfsClient` rule set or manually set the Allow All IP Addresses policy, either before or after you add an NFS v3 datastore to the system, your settings are overridden when the last NFS v3 datastore is unmounted. The `nfsClient` rule set is deactivated when all NFS v3 datastores are unmounted.

When you remove or unmount an NFS v3 datastore, ESX performs one of the following actions.

- If none of the remaining NFS v3 datastores are mounted from the server of the datastore being unmounted, ESX removes the server's IP address from the list of outgoing IP addresses.
- If no mounted NFS v3 datastores remain after the unmount operation, ESX deactivates the `nfsClient` firewall rule set.

NFS v4.1 Firewall Behavior

When you mount the first NFS v4.1 datastore, ESX activates the `nfs41client` rule set and sets its `allowedAll` flag to TRUE. This action opens port 2049 for all IP addresses. Unmounting an NFS v4.1 datastore does not affect the firewall state. That is, the first NFS v4.1 mount opens port 2049 and that port remains activated unless you close it explicitly.

Verify Firewall Ports for NFS Clients

To enable access to NFS storage, ESX automatically opens firewall ports for the NFS clients when you mount an NFS datastore. For troubleshooting reasons, you might need to verify that the ports are open.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Firewall**, and click **Edit**.
4. Scroll down to an appropriate version of NFS to make sure that the port is open.

Set Up NFS Storage Environment with ESX

You must perform several configuration steps before you mount an NFS datastore in vSphere.

- Familiarize yourself with the guidelines in Guidelines and Requirements for NFS Storage with ESX.
- For details on configuring NFS storage, consult your storage vendor documentation.
- If you use Kerberos, make sure that AES256-CTS-HMAC-SHA1-96 or AES128-CTS-HMAC-SHA1-96 are enabled on the NAS server.

1. On the NFS server, configure an NFS volume and export it to be mounted on the ESX hosts.
   a) Note the IP address or the DNS name of the NFS server and the full path, or folder name, for the NFS share.
      For NFS 4.1, you can collect multiple IP addresses or DNS names to use the multipathing support that the NFS 4.1 datastore provides.
   b) If you plan to use Kerberos authentication with NFS 4.1, specify the Kerberos credentials to be used by ESX for authentication.
2. On each ESX host, configure a VMkernel Network port for NFS traffic.
   For more information, see the *vSphere Networking* documentation.
3. If you plan to use Kerberos authentication with the NFS 4.1 datastore, configure the ESX hosts for Kerberos authentication.
   See Configure ESX Hosts for Kerberos Authentication.

You can now create an NFS datastore on the ESX hosts. See Create an NFS 4.1 Datastore.

Using Kerberos for NFS 4.1 with ESX

With NFS version 4.1, ESX supports the Kerberos authentication mechanism. The ESX implementation of Kerberos for NFS 4.1 provides krb5, krb5i, and krb5p security models that offer different levels of security.

The RPCSEC_GSS Kerberos mechanism is an authentication service. It allows an NFS 4.1 client installed on ESX to prove its identity to an NFS server before mounting an NFS share. The Kerberos security uses cryptography to work across an insecure network connection.

The ESX implementation of Kerberos for NFS 4.1 provides the following security models that offer different levels of security.

Kerberos for authentication only (krb5)	Supports identity verification.
Kerberos for authentication and data integrity (krb5i)	In addition to identity verification, provides data integrity services. These services help to protect the NFS traffic from tampering by checking data packets for any potential modifications.
Kerberos for authentication, data integrity and privacy (krb5p)	In addition to identity verification and data integrity, provides end-to-end encryption for NFS. With krb5p, the system encrypts the NFS packets before sending them on the wire. Replies are decrypted before the system processes the results.

ESX supports the following Kerberos cryptographic algorithms that prevent unauthorized users from gaining access to NFS traffic. They use the key lengths of 256 bits and 128 bits. AES-256 provides higher security than AES-128.

- AES256-CTS-HMAC-SHA1-96
- AES128-CTS-HMAC-SHA1-96

During initial negotiation, the highest form of security algorithm supported by the NFS 4.1 client on ESX and NAS server is selected.

Before setting up your NFS 4.1 datastores, make sure that AES256-CTS-HMAC-SHA1-96 or AES128-CTS-HMAC-SHA1-96 are enabled on the NAS server.

When you use Kerberos authentication, the following considerations apply:

- ESX uses Kerberos with the Active Directory domain.
- As a vSphere administrator, you specify Active Directory credentials to provide access to NFS 4.1 Kerberos datastores for an NFS user. A single set of credentials is used to access all Kerberos datastores mounted on that host.
- When multiple ESX hosts share the NFS 4.1 datastore, you must use the same Active Directory credentials for all hosts that access the shared datastore. To automate the assignment process, set the user in host profiles and apply the profile to all ESX hosts.
- You cannot use two security mechanisms, AUTH_SYS and Kerberos, for the same NFS 4.1 datastore shared by multiple hosts.

Configure ESX Hosts for Kerberos Authentication

If you use NFS 4.1 with Kerberos, you must perform several tasks to set up your hosts for Kerberos authentication.

- Make sure that Microsoft Active Directory (AD) and NFS servers are configured to use Kerberos.
- Enable AES256-CTS-HMAC-SHA1-96 or AES128-CTS-HMAC-SHA1-96 encryption modes on AD. The NFS 4.1 client does not support the DES-CBC-MD5 encryption mode.
- Make sure that the NFS server exports are configured to grant full access to the Kerberos user.

When multiple ESX hosts share the NFS 4.1 datastore, you must use the same Active Directory credentials for all hosts that access the shared datastore. You can automate the assignment process by setting the user in host profiles and applying the profile to all ESX hosts.

Configure DNS for NFS 4.1 with Kerberos

When you use NFS 4.1 with Kerberos, you must change the DNS settings on ESX hosts. The settings must point to the DNS server that is configured to hand out DNS records for the Kerberos Key Distribution Center (KDC). For example, use the Active Directory server address if AD is used as a DNS server.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Networking**, click **TCP/IP configuration**.
4. Select **Default** and click the **Edit** icon.
5. Manually enter the DNS settings.

Option	Description
Domain	*AD Domain Name*
Preferred DNS server	*AD Server IP*
Search domains	*AD Domain Name*

Configure Network Time Protocol for NFS 4.1 with Kerberos

If you use NFS 4.1 with Kerberos, ESX hosts, the NFS server, and the Active Domain server need to be time synchronized. Typically, in the setup the Active Domain server is used as the Network Time Protocol (NTP) server.

The following task describes how to synchronize the ESX host with the NTP server.

The best practice is to use the Active Domain server as the NTP server.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, select **Time Configuration**.
4. Click **Edit** and set up the NTP server.
   a) Select **Use Network Time Protocol (Enable NTP client)**.
   b) To synchronize with the NTP server, enter its IP addresses.
   c) Select **Start NTP Service**.
   d) Set the NTP Service Startup Policy.
5. Click **OK**.

   The host synchronizes with the NTP server.

Enable Kerberos Authentication in Active Directory

If you use NFS 4.1 storage with Kerberos, you must add each ESX host to an Active Directory domain and enable Kerberos authentication. Kerberos integrates with Active Directory to enable single sign-on and provides an extra layer of security when used across an insecure network connection.

Set up an AD domain and a domain administrator account with the rights to add hosts to the domain.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Authentication Services**.
4. Add the ESX host to an Active Directory domain.
   a) In the Authentication Services pane, click **Join Domain**.
   b) Supply the domain settings, and click **OK**.

   The directory services type changes to Active Directory.
5. Configure or edit credentials for an NFS Kerberos user.
   a) In the NFS Kerberos Credentials pane, click **Edit**.
   b) Enter a user name and password.

      Files stored in all Kerberos datastores are accessed using these credentials.

   The state for NFS Kerberos credentials changes to Enabled.

After you configure your host for Kerberos, you can create an NFS 4.1 datastore with Kerberos enabled.

For information about creating an NFS user account to be used by the NFS 4.1 for Kerberos authentication, see VMware PowerCLI on the Broadcom Developer Portal page at https://developer.broadcom.com/powercli/latest/vmware.vimautomation.storage/commands/new-nfsuser.

Configuring VMkernel Binding for NFS Datastores

VMkernel port binding for the NFS datastore on an ESX host allows you to bind an NFS volume to a specific VMkernel adapter to connect to an NFS server. With NFS datastores, you can isolate NFS traffic to a specific VMkernel adapter. To route the traffic to this adapter, connect the datastore to the adapter.

> **Note:**
> Without binding, if the VMkernel adapter that ESX uses for NFS traffic fails, the network infrastructure redirects the traffic to an alternative route. As a result, the NFS traffic might unintentionally flow through a random VMkernel adapter.

Configure VMkernel Binding for NFS 3

For NFS 3 datastores, you can use the vSphere Client to connect the datastore to specific VMkernel adapters.

- Set up the NFS storage environment.
- Create and configure a VMkernel adapter on default TCP/IP stack.

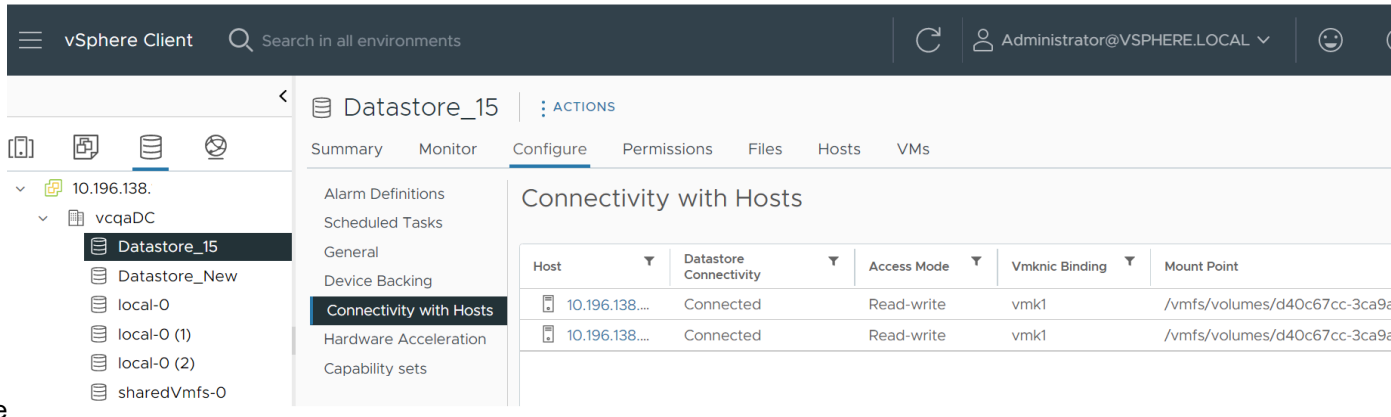
1. Bind an NFS 3 datastore to a VMkernel adapter.
   a) In the vSphere Client navigator, browse to a host, a cluster, or a data center.
   b) From the right-click menu, select **Storage** > **New Datastore**.
   c) Select **NFS** as the datastore type, and select **NFS 3**.
   d) Enter the datastore parameters. For more information, see Create an NFS 4.1 Datastore.
   e) Select **Bind to vmknic**.
   f) If you are creating the datastore at the data center or cluster level, select hosts that mount the datastore.
   g) For the datastore at the data center or cluster level, select one of the following options.
      - Select **Vmknics on all hosts**. With this option, selectable VMkernel adapters are available across all hosts where you mount your datastore.
      - Select **Vmknic per host**. With this option, you can specify a VMkernel adapter per host.

      Note:
      You can perform only one selection at a time.

h) Review the configuration options you made and click **Finish**.

2. Review the hosts connected to the NFS 3 datastore.
   a) In the vSphere Client, select the NFS 3 datastore with VMkernel adapter binding.
   b) Click the **Configure** tab, and click **Connectivity with Hosts**.

   You can view all the hosts connected to the NFS 3 datastore. The **Vmknic Binding** column shows the VMkernel adapters connected to the



datastore

Note:
If you delete the VMkernel adapter associated with the NFS 3 datastore, the datastore becomes inaccessible triggering a warning. The warning displays the text in the **Events** screen similar to the following.

```
vmknic vmkX removed, NFSv3 datastore datastore_name configured with the vmknic will be inaccessible.
```

Configure VMkernel Binding for NFS 4.1

To connect an NFS 4.1 datastore to a specific VMkernel adapter, use the `esxcli storage nfs41` command. You can specify the VMkernel adapters either while mounting the NFS 4.1 datastore or, if the NFS datastore is already mounted with a specific VMkernel adapter, you can modify the existing NFS volume configuration to specify the desired VMkernel adapters.

- Set up the NFS storage environment.
- Create and configure a VMkernel adapter on default TCP/IP stack.

1. To specify VMkernel adapters while mounting a new NFS v4.1 volume on the ESX host, use the following command:

   ```
   [root@:~] esxcli storage nfs41 add
   ```

   Example: `[root@:~]esxcli storage nfs41 add -I 10.187.114.32:vmk0 -I 10.187.214.27:vmk1 -s /store1 -v store1`

2. To set the VMkernel adapters on an existing NFS 4.1 datastore, use the following command:

   ```
   [root@:~] esxcli storage nfs41 param set
   ```

   Example: `[root@:~] esxcli storage nfs41 param set -I 10.187.114.32:vmk2 -v store1`

3. (Optional) Verify whether the specified VMkernel adapters are set:

   ```
   [root@:~] esxcli storage nfs41 list
   ```

   The output should be similar to the following:

```
Volume Name  Host(s)                      Share    Vmknics     Accessible  Mounted  Connections  Read-Only
  Security   isPE  Hardware Acceleration
-----------  -------------                -------  -------     ----------  -------  -----------  ---------
  --------  -----  ---------------------
store1       10.187.114.32,10.187.214.27   /store1   vmk2,vmk1        true     true            1
               false              AUTH_SYS  false     Not Supported
```

Creating NFS Datastores with the vSphere Client

You can use the vSphere Client or the `esxcli` command to configure the NFS datastore depending on the NFS version and the configuration you want to set.

Create an NFS 3 Datastore

You can use the **New Datastore** wizard to mount an NFS 3 volume.

- Set up NFS storage environment. See Set Up NFS Storage Environment with ESX.
- If you plan to isolate the NFS traffic to a specific VMkernel adapter that you connect to the datastore, configure the VMkernel binding. For more information, see Configuring VMkernel Binding for NFS Datastores.

  Note:

  If you plan to configure an existing datastore on new hosts in the datacenter, select the **Mount Datastore to Additional Hosts** option from the datastore right-click menu instead. See Mount Datastores.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. From the right-click menu, select **Storage** > **New Datastore**.
3. Select NFS as the datastore type and specify NFS 3 as the protocol.

   Important:

   If multiple hosts access the same datastore, you must use the same protocol on all hosts.
4. Specify datastore name and configuration parameters, and click **Next**.

Option	Description
Name	The datastore name. The system enforces an 80 character limit for the datastore name.
Folder	The mount point folder name
Server	The server name or IP address. You can use IPv6 or IPv4 formats.
Vmknic Binding	Select **Bind to vmknic**, to connect the NFS 3 datastore to a specific VMkernel adapter. This vSphere Client option is available only for the NFS 3 datastores. To configuring VMkernel binding for NFS 4.1 datastores, use the `esxcli` command. See Configure VMkernel Binding for NFS 4.1.
nConnect	Select this option to enable multiple TCP connections for datastore volumes.
Number of TCP Connections	If you enabled multiple connections, specify the number of TCP connections to the NFS server. You can set a maximum of four connections per NFS datastore by default. If you want to increase the number, use the advanced NFS option. See Configure Multiple TCP Connections for NFS with the esxcli Command.
Access Mode	Select **Mount NFS read only** if the volume is exported as read-only by the NFS server.

5. If you create a datastore at the data center or cluster level, on the **Hosts accessibility** page, select hosts that mount the datastore from the list of compatible hosts.
6. If you enabled VMkernel binding, select the VMkernel adapter to connect to the datastore.

   For information, see Configure VMkernel Binding for NFS 3.

7. Review the configuration options and click **Finish**.

Create an NFS 4.1 Datastore

Use the **New Datastore** wizard to mount an NFS 4.1 volume.

- Set up an NFS storage environment. See Set Up NFS Storage Environment with ESX.
- If you plan to use Kerberos authentication with the NFS 4.1 datastore, make sure to configure the ESX hosts for Kerberos authentication. See Using Kerberos for NFS 4.1 with ESX.
- The vSphere Client option to configure VMkernel binding is not available for NFS 4.1 datastores. If you plan to isolate the NFS traffic to a specific VMkernel adapter that you connect to the datastore, configure the VMkernel binding using the `esxcli` command. For more information, see Configuring VMkernel Binding for NFS Datastores.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. From the right-click menu, select **Storage** > **New Datastore**.
3. Select NFS as the datastore type and specify NFS 4.1 as the protocol.

   Important:
   If multiple hosts access the same datastore, you must use the same protocol on all hosts.

4. Specify datastore name and configuration parameters, and click **Next**.

Option	Description
Name	The datastore name. The system enforces an 80 character limit for the datastore name.
Folder	The mount point folder name
Server	The server name or IP address. You can use IPv6 or IPv4 formats. With NFS 4.1, you can add multiple IP addresses or server names if the NFS server supports trunking. The ESX host uses these values to achieve multipathing to the NFS server mount point.
nConnect	Select this option to enable multiple TCP connections for datastore volumes.
Number of TCP Connections	If you enabled multiple connections, specify the number of TCP connections to the NFS server. You can set a maximum of four connections per NFS datastore by default. If you want to increase the number, use the advanced NFS option. See Configure Multiple TCP Connections for NFS with the esxcli Command.
Access Mode	Select **Mount NFS read only** if the volume is exported as read only by the NFS server.

5. To use Kerberos security with NFS 4.1 datastore, enable Kerberos and select an appropriate Kerberos model.

Option	Description
Use Kerberos for authentication only (krb5)	Supports identity verification
Use Kerberos for authentication and data integrity (krb5i)	In addition to identity verification, provides data integrity services. These services help to protect the NFS traffic from tampering by checking data packets for any potential modifications.
Use Kerberos for authentication, data integrity, and privacy (krb5p)	In addition to identity verification and data integrity, provides end-to-end encryption for NFS. With krb5p, the system encrypts the NFS packets before sending them on the wire. Replies are decrypted before the system processes the results.

   If you do not enable Kerberos, the datastore uses the default AUTH_SYS security.

6. If you create a datastore at the data center or cluster level, select hosts that mount the datastore.
7. Review the configuration options and click **Finish**.

Creating and Configuring NFS Datastores with the esxcli Command

Use the `esxcli` command to configure specific settings on NFS.

Command	Description
NFS 4.1: `esxcli storage nfs41 add` NFS 3: `esxcli storage nfs add`	Adds a new NFS or NAS volume and mounts it with the given volume name on the ESX host.
NFS 4.1: `esxcli storage nfs41 param set` NFS 3: `esxcli storage nfs param set`	Sets the NFS volume parameters.

The commands accept the following options:

Option	Description
`-H|--hosts=<str>`	NFS 4.1: The hostname or IP address of the server for the NFS 4.1 volume you want to add. You can add multiple NFS volumes separated with a comma. NFS 3: The hostname or IP address of the NAS volume you want to add and mount on the system.
`-I|--host-vmknic=<str>`	NFS 4.1: The hostname or IP address of the server for the NFS 4.1 volume you want to add followed by a colon(:) and the VMknic name. You can specify multiple names. Example: `--host-vmknic=<host1>:<vmknic1> --host-vmknic=<ip2>:<vmknic2>` NFS 3: The hostname or IP address of the NAS volume you want to add. You can specify only one name. Example: `<ip1>:<vmknic1>`
`-p|--ispe`	If set, this flag will set PE as the mount point.
`-r|--readonly`	If set, this flag will make the mount point be read-only.
`-a|--sec=<str>`	Security setting. Acceptable values are: AUTH_SYS, SEC_KRB5, SEC_KRB5I, or SEC_KRB5P. This option is applicable only for NFS 4.1 and later.
`-s|--share=<str>`	The share name to use for the NFS or NAS mount point on the remote system. This option is required.
`-v|--volume-name=<str>`	The volume name to use for the NFS or NAS mount. The volume name must be unique and should not conflict with existing NAS, NFS, VMFS, or other volume names. This option is required.
`-c|--connections=<long>`	The number of RPC connections to be established for the NFS datastore. A higher number of connections results in better performance on datastores with higher I/O load.
`-q|--maxqueuedepth=<long>`	The maximum queue depth for the NFS volume. This option can be used only with the `param set` command.

Configure Multiple TCP Connections for NFS with the esxcli Command

You can configure NFS storage with multiple connections using the `esxcli` command.

Set up the NFS storage environment.

When you configure multiple TCP connections, the following considerations apply:

- You can set a maximum of four connections per NFS datastore by default. However, you can increase the number to eight by using the advanced NFS option:
  - `esxcfg-advcfg -s 8 /NFS41/MaxNConnectConns`
  - `esxcfg-advcfg -s 8 /NFS/MaxConnectionsPerDatastore`
- Ensure that the total number of connections across all mounted NFS datastores does not exceed 256.
- For NFS 4.1, setting up multiple connections does not impact multipathing. Multiple connections and multiple paths can co-exist.
- When the NFS datastore is mounted in a cluster, all the hosts in the cluster must use the same number of connections. Make sure to configure all hosts in the cluster with same `<number_of_connections>` value.

Note:
The commands in the following procedure are examples for NFS 4.1 datastores. For NFS 3, replace `nfs41` with `nfs`.

1. To specify the number of connections while mounting the NFS datastore, use the following command:

   ```
   esxcli storage nfs41 add -H <host> -v <volume-label> -s <remote_share> -c <number_of_connections>
   ```

   When the NFS 4.1 datastore is mounted with multiple paths, multiple connections are created for each of the path:

   ```
   esxcli storage nfs41 add -H <IP1,IP2> -v <volume-label> -s <remote_share> -c
   <number_of_connections>
   ```

2. To increase or decrease the number of connections on an existing NFS datastore, use the following command:

   ```
   esxcli storage nfs41 param set -v <volume-label> -c <number_of_connections>
   ```

Refreshing NFS Server IP Addresses for DNS-Mounted Datastores

Starting with vSphere 9.1, you can update the NFS connection when the NFS server IP address changes in DNS.

When you mount an NFS datastore and use the server name instead of the IP address, ESX resolves the name of an NFS server during mount. It then uses the resolved IP address for all communication. If the DNS record changes, ESX continues to use the old IP and the datastore enters an All Paths Down (APD) state.

You can use the `esxcli storage` commands to re-resolve the hostname and update the server IP for a specific volume or for all volumes that use the same hostname.

This operation applies only to NFS datastores mounted using DNS names.

Use the following commands to query DNS and update the server IP of the volumes. Use the `force` option when volumes are not in APD.

Action	Command
Re-resolve the IP address of a specific volume.	When the volume is in APD: `esxcli storage nfs/nfs41 param set -r <remotehostname> -v <volumename>`
	When the volume is not in APD: `esxcli storage nfs/nfs41 param set -r <remotehostname> -v <volumename> -f`
Re-resolve the IP address of all volumes mounted at the specified server.	When the volumes are in APD: `esxcli storage nfs/nfs41 param set -r <remotehostname>`
	When the volumes are not in APD: `esxcli storage nfs/nfs41 param set -r <remotehostname> -f`

When you use this command, keep in mind the following:

- For NFS 4.1, the volume parameter `-v` is ignored. DNS resolution happens for all NFS 4.1 volumes mounted with the same DNS.
- The command has no effect if the DNS entries do not change.
- Remote host name `-r` has precedence over all other options.

- Refresh operation might fail for NFS volumes that are in Mounted=False state. To resolve this issue, retry the operation after some time.

Collecting Statistical Information for NFS Storage on ESX Host

You can use the `nfsStats` tool in your ESX host to display statistical information about NFS calls and Remote Procedure Calls (RPC). The command displays statistical information for NFS 3 and NFS 4.1 mounts on the ESX host.

Generally, the `nfsStats` tool performs the following tasks.

- Gathers NFS statistics to investigate problems when you deploy a new configuration, such as a new NFS server or network, in the NFS environment.
- Provides statistics about the success and failure of NFS operations.
- Publishes latency statistics about the success and failure of NFS operations.
- Troubleshoots NFS performance issues.

The command syntax is `nfsStats`*`options`*.

The following command options are available.

Table 179: nfsStats commands

Command Option	Description
No option	Obtain both NFS statistics and RPC statistics for all NFS datastores.
-3	Display only NFS 3 statistics.
-4	Display only NFS 4.1 statistics.
-n	Display only NFS 3 and NFS 4.1 statistics.
-r	Display RPC statistics.
-i*`interval`*	Display NFS and RPC statistics at an interval equal to the specified value in seconds. For example, if the value you enter is 10, the statistics is refreshed every 10 seconds.
-j	Display statistics in json format.
-e	Display error statistics for NFS 3, NFS 4.1, and RPC protocols.
-d, --datastore-name*`DSNAME1, DSNAME2, ...`*	Display both NFS and RPC statistics for the specified NFS datastores. Use this option in conjunction with the type of the NFS datastore, for example, -3 or -4.

Using Datastore Browser in vSphere Environment

Use the datastore file browser in your vSphere environment to manage the contents of your datastores. You can upload files or folders, download files from the datastore, or move or copy folders or files to a new location. You can also use the datastore browser to inflate the thin virtual disk.

Navigate to the Datastore Browser

Navigate to your datastore to browse folders and files that are stored on the datastore and perform administrative tasks on your folders and files.

1. Open the datastore browser.
   a) Display the datastore in the inventory.
   b) Right-click the datastore and select **Browse Files**.
2. Explore the contents of the datastore by navigating to existing folders and files.
3. Perform administrative tasks by using the icons and options.

Icons and Options	Descriptions
Upload Files	Upload a file to the datastore.
Upload Folder (Available only in the vSphere Client)	Upload a folder to the datastore.
Download	Download from the datastore.
New Folder	Create a folder on the datastore.
Copy to	Copy selected folders or files to a new location, either on the same datastore or on a different datastore.
Move to	Move selected folders or files to a new location, either on the same datastore or on a different datastore.
Rename to	Rename selected files.
Delete	Delete selected folders or files.
Inflate	Convert a selected thin virtual disk to thick. This option applies only to thin-provisioned disks.

Upload Files or Folders to Datastores

Use the datastore file browser to upload files to datastores on the ESX host. If you use the vSphere Client, you can also upload folders.

Required privilege:**Datastore** > **Browse Datastore** and**Host** > **System Management**

In addition to their traditional use as storage for virtual machines files, datastores can serve to store data or files related to virtual machines. For example, you can upload ISO images of operating systems from a local computer to a datastore on the host. You then use these images to install guest operating systems on the new virtual machines.

> **Note:**
> You cannot upload files directly to the Virtual Volumes datastores. You must first create a folder on the Virtual Volumes datastore, and then upload the files into the folder. The created folders in Virtual Volumes datastores for block storage have a limited storage capacity space of 4GB. The Virtual Volumes datastore supports direct uploads of folders.

1. Open the datastore browser.
   a) Display the datastore in the inventory.
   b) Right-click the datastore and select **Browse Files**.
2. Optional: Create a folder to store the file or folder.
3. Upload the file or folder.

Option	Description
Upload a file	1. Select the target folder and click **Upload Files**. 2. Locate the item to upload on the local computer and click **Open**.
Upload a folder (available only in the vSphere Client)	1. Select the datastore or the target folder and click **Upload Folders**. 2. Locate the item to upload on the local computer and click **Ok**.

4. Refresh the datastore file browser to see the uploaded files or folders on the list.

Download Files from Datastores

Use the datastore file browser to download files from the datastore available on your ESX host to your local computer.

- Required privilege: **Datastore** > **Browse Datastore**
- Privileges when a VM is encrypted: **Cryptographic operations** > **Decrypt** and **Cryptographic operations** > **Direct Access**

1. Open the datastore browser.
   a) Display the datastore in the inventory.
   b) Right-click the datastore and select **Browse Files**.
2. Navigate to the file to download and click **Download**.
3. Follow the prompts to save the file to your local computer.

Rename Datastore Files

Use the datastore browser to rename files.

Required privilege: **Datastore** > **Browse Datastore**

1. Open the datastore browser.
   a) Display the datastore in the inventory.
   b) Right-click the datastore and select **Browse Files**.
2. Browse to a file you want to rename.
3. Select the file and click **Rename to**.
4. Specify the new name and click **OK**.

Move or Copy Datastore Folders or Files

Use the datastore browser to move or copy folders or files to a new location, either on the same datastore or on a different datastore.

Required privilege: **Datastore** > **Browse Datastore**

> **Note:**
> Virtual disk files are moved or copied without format conversion. If you move a virtual disk to a datastore that belongs to a host different from the source host, you might need to convert the virtual disk. Otherwise, you might not be able to use the disk.

You cannot copy VM files across vCenter instances.

1. Open the datastore browser.
   a) Display the datastore in the inventory.
   b) Right-click the datastore and select **Browse Files**.
2. Browse to an object you want to move or copy, either a folder or a file.
3. Select the object and click **Move to** or **Copy to**.
4. Specify the destination location.
5. Optional: Select **Overwrite files and folders with matching names at the destination**.
6. Click **OK**.

Inflate Thin Virtual Disks

If you created a virtual disk in the thin format, you can change the format to thick.

- Make sure that the datastore where the virtual machine resides has enough space.
- Make sure that the virtual disk is thin.
- Remove snapshots.
- Power off your virtual machine.

You use the datastore browser to inflate the thin virtual disk.

Note:

VMs residing on vSAN do not support virtual disk inflation. See the KB 423722 for more details.

1. In the vSphere Client, navigate to the folder of the virtual disk you want to inflate.
   a) Navigate to the virtual machine.
   b) Click the **Datastores** tab.
      The datastore that stores the virtual machine files is listed.
   c) Right-click the datastore and select **Browse Files**.
      The datastore browser displays contents of the datastore.
2. Expand the virtual machine folder and browse to the virtual disk file that you want to convert.
   The file has the `.vmdk` extension and is marked with the virtual disk
   ( )
   icon.
3. Select the virtual disk file and click **Inflate**.

Note:
The option might not be available if the virtual disk is thick or when the virtual machine is running.

The inflated virtual disk occupies the entire datastore space originally provisioned to it.

Understanding Multipathing and Failover in the ESX Environment

To maintain a constant connection between a host and its storage, ESX supports multipathing. With multipathing, you can use more than one physical path that transfers data between the host and an external storage device.

If a failure of any element in the SAN network, such as an adapter, switch, or cable, occurs, ESX can switch to another viable physical path. This process of path switching to avoid failed components is known as path failover.

In addition to path failover, multipathing provides load balancing. Load balancing is the process of distributing I/O loads across multiple physical paths. Load balancing reduces or removes potential bottlenecks.

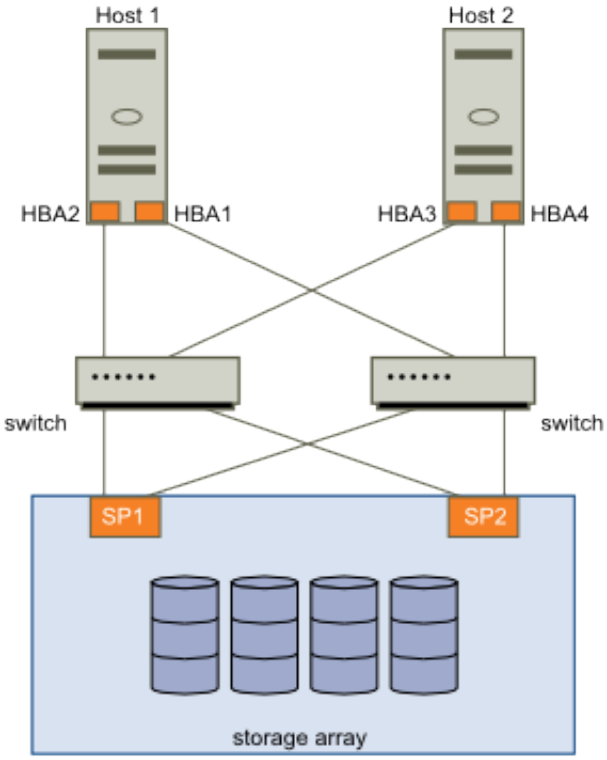
Note:
Virtual machine I/O might be delayed for up to 60 seconds while path failover takes place. With these delays, the SAN can stabilize its configuration after topology changes. In general, the I/O delays might be longer on active-passive arrays and shorter on active-active arrays.

Failover with Fibre Channel

To support multipathing, your host typically has two or more HBAs available. This configuration supplements the SAN multipathing configuration. Generally, the SAN multipathing provides one or more switches in the SAN fabric and one or more storage processors on the storage array device itself.

In the following illustration, multiple physical paths connect each server with the storage device. For example, if HBA1 or the link between HBA1 and the FC switch fails, HBA2 takes over and provides the connection. The process of one HBA taking over for another is called HBA failover.

Figure 79: Multipathing and Failover with Fibre Channel



Similarly, if SP1 fails or the links between SP1 and the switches breaks, SP2 takes over. SP2 provides the connection between the switch and the storage device. This process is called SP failover. VMware ESX supports both HBA and SP failovers.

Host-Based Failover with iSCSI

When setting up your ESX host for multipathing and failover, you can use multiple iSCSI HBAs or combine multiple NICs with the software iSCSI adapter.

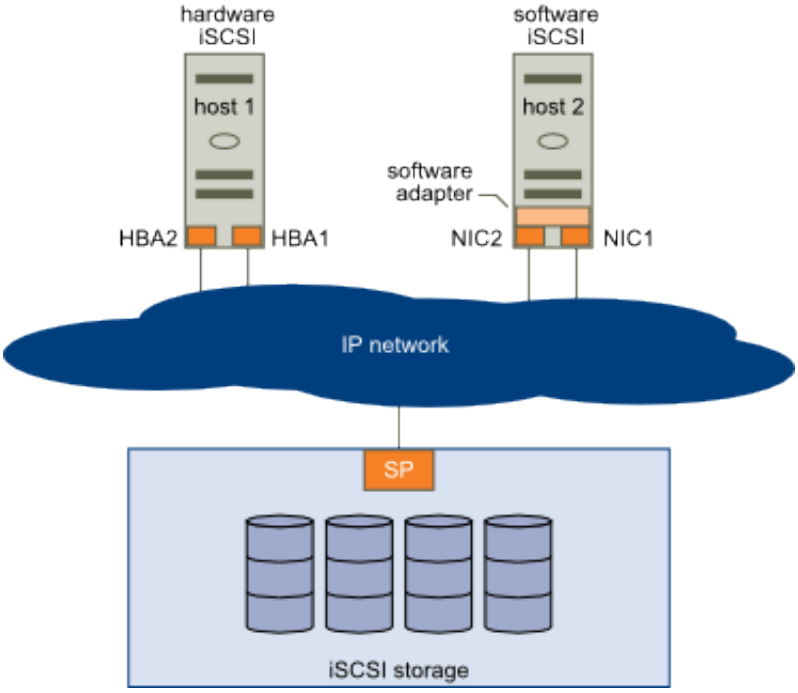
For information on different types of iSCSI adapters, see iSCSI Initiators.

When you use multipathing, specific considerations apply.

- ESX does not support multipathing when you combine an independent hardware adapter with software iSCSI or dependent iSCSI adapters in the same host.
- Multipathing between software and dependent adapters within the same host is supported.
- On different hosts, you can mix both dependent and independent adapters.

The following illustration shows multipathing setups possible with different types of iSCSI initiators.

Figure 80: Host-Based Path Failover



Hardware iSCSI and Failover	With hardware iSCSI, the host typically has two or more hardware iSCSI adapters. The host uses the adapters to reach the storage system through one or more switches. Alternatively, the setup might include one adapter and two storage processors, so that the adapter can use different paths to reach the storage system. On the illustration, Host1 has two hardware iSCSI adapters, HBA1 and HBA2, that provide two physical paths to the storage system. Multipathing plug-ins on your host, whether the VMkernel NMP or any third-party MPPs, have access to the paths by default. The plug-ins can monitor health of each physical path. If, for example, HBA1 or the link between HBA1 and the network fails, the multipathing plug-ins can switch the path over to HBA2.
Software iSCSI and Failover	With software iSCSI, as shown on Host 2 of the illustration, you can use multiple NICs that provide failover and load balancing capabilities for iSCSI connections. Multipathing plug-ins do not have direct access to physical NICs on your host. As a result, for this setup, you first must connect each physical NIC to a separate VMkernel port. You then associate all VMkernel ports with the software iSCSI initiator using a port binding technique. Each VMkernel port connected to a separate NIC becomes a different path that the iSCSI storage stack and its storage-aware multipathing plug-ins can use. For information about configuring multipathing for software iSCSI, see Setting Up Network for iSCSI and iSER with ESX.

Array-Based Failover with iSCSI

Some iSCSI storage systems manage path use of their ports automatically and transparently to ESX.

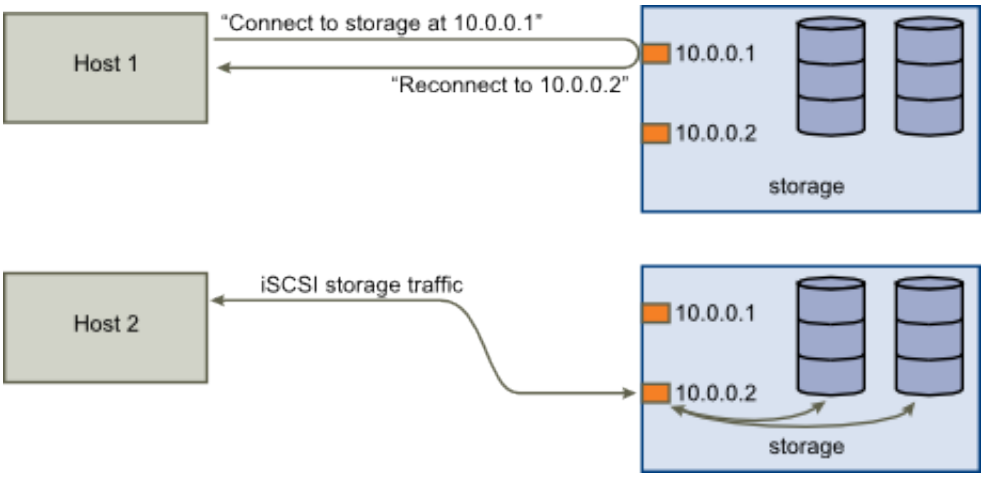
When using one of these storage systems, your host does not see multiple ports on the storage and cannot choose the storage port it connects to. These systems have a single virtual port address that your host uses to initially communicate. During this initial communication, the storage system can redirect the host to communicate with another port on the storage system. The iSCSI initiators in the host obey this reconnection request and connect with a different port on the system. The storage system uses this technique to spread the load across available ports.

If the ESX host loses connection to one of these ports, it automatically attempts to reconnect with the virtual port of the storage system, and should be redirected to an active, usable port. This reconnection and redirection happens quickly and generally does not disrupt running virtual machines. These storage systems can also request that iSCSI initiators reconnect to the system, to change which storage port they are connected to. This allows the most effective use of the multiple ports.

The Port Redirection illustration shows an example of port redirection. The host attempts to connect to the 10.0.0.1 virtual port. The storage system redirects this request to 10.0.0.2. The host connects with 10.0.0.2 and uses this port for I/O communication.

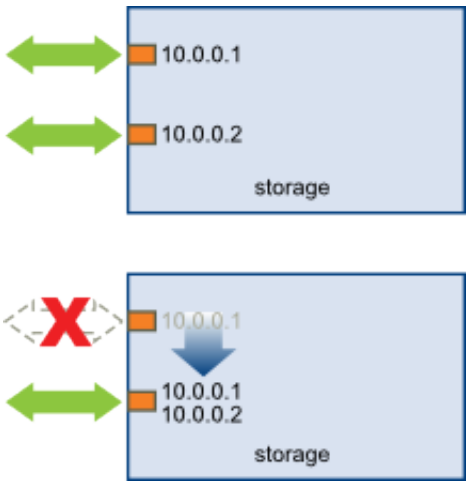
Note:
The storage system does not always redirect connections. The port at 10.0.0.1 could be used for traffic, also.

Figure 81: Port Redirection



If the port on the storage system that is acting as the virtual port becomes unavailable, the storage system reassigns the address of the virtual port to another port on the system. Port Reassignment shows an example of this type of port reassignment. In this case, the virtual port 10.0.0.1 becomes unavailable and the storage system reassigns the virtual port IP address to a different port. The second port responds to both addresses.

Figure 82: Port Reassignment



With this form of array-based failover, you can have multiple paths to the storage only if you use multiple ports on the ESX host. These paths are active-active. For additional information, see Managing iSCSI Session on ESX Host.

Path Failover and Virtual Machines

When a path fails, storage I/O might pause for 30-60 seconds until your host determines that the link is unavailable and performs the failover. If you attempt to display the host, its storage devices, or its adapters, the operation might appear to stall. Virtual machines with their disks installed on the SAN can appear unresponsive. After the failover, I/O resumes normally and the virtual machines continue to run.

A Windows virtual machine might interrupt the I/O and eventually fail when failovers take too long. To avoid the failure, set the disk timeout value for the Windows virtual machine to at least 60 seconds.

To avoid disruptions during a path failover, increase the standard disk timeout value on a Windows guest operating system.

This procedure explains how to change the timeout value by using the Windows registry.

Tip:
Before you begin: Back up the Windows registry.

1. Select **Start** > **Run**.
2. Type `regedit.exe`, and click **OK**.
3. In the left-panel hierarchy view, double-click **HKEY_LOCAL_MACHINE** > **System** > **CurrentControlSet** > **Services** > **Disk**.
4. Double-click **TimeOutValue**.
5. Set the value data to 0x3c (hexadecimal) or 60 (decimal) and click **OK**.
   After you make this change, Windows waits at least 60 seconds for delayed disk operations to finish before it generates errors.
6. Reboot guest OS for the change to take effect.

Using Pluggable Storage Architecture and Path Management with ESX

To manage multipathing, ESX uses a special VMkernel layer, Pluggable Storage Architecture (PSA). The PSA is an open and modular framework that coordinates various software modules responsible for multipathing operations, VMware NMP, VMware HPP, and third-party MPPs.

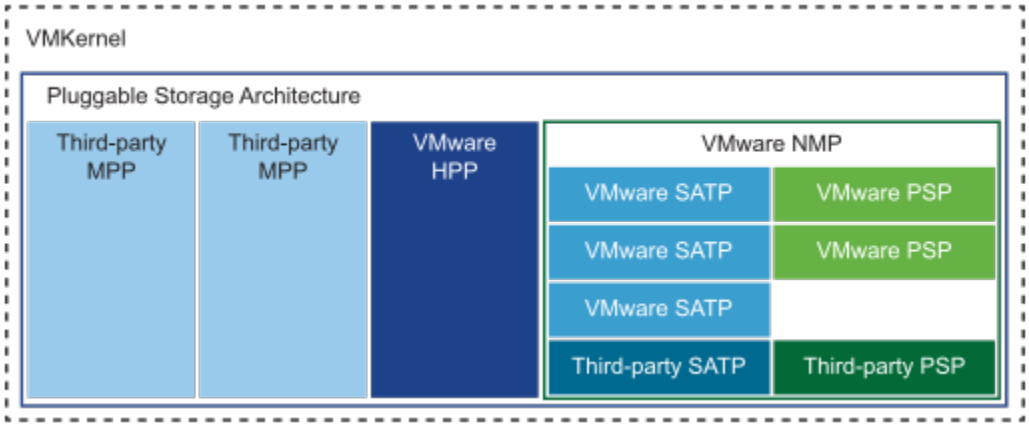
VMware provides generic native multipathing modules, called VMware NMP and VMware HPP. In addition, the PSA offers a collection of VMkernel APIs that third-party developers can use. The software developers can create their own load balancing and failover modules for a particular storage array. These third-party multipathing modules (MPPs) can be installed on the ESX host and run in addition to the VMware native modules, or as their replacement.

When coordinating the VMware native modules and any installed third-party MPPs, the PSA performs the following tasks:

- Loads and unloads multipathing plug-ins.
- Hides virtual machine specifics from a particular plug-in.
- Routes I/O requests for a specific logical device to the MPP managing that device.
- Handles I/O queueing to the logical devices.
- Implements logical device bandwidth sharing between virtual machines.
- Handles I/O queueing to the physical storage HBAs.
- Handles physical path discovery and removal.
- Provides logical device and physical path I/O statistics.

As the Pluggable Storage Architecture illustration shows, multiple third-party MPPs can run in parallel with the VMware NMP or HPP. When installed, the third-party MPPs can replace the behavior of the native modules. The MPPs can take control of the path failover and the load-balancing operations for the specified storage devices.

Figure 83: Pluggable Storage Architecture



Multipathing Concepts and Acronyms

Multipathing Concept	Acronym	Definition
Pluggable Storage Architecture	PSA	The PSA is an open and modular framework that coordinates various software modules responsible for multipathing operations. These modules include generic multipathing modules that VMware provides, NMP and HPP, and third-party MPPs.
Native Multipathing Plug-in	NMP	The NMP is the VMkernel multipathing module that ESX provides by default and is used for SCSI storage devices. The NMP associates physical paths with a specific storage device and provides a default path selection algorithm based on the array type. The NMP is extensible and manages additional submodules, called Path Selection Policies (PSPs) and Storage Array Type Policies (SATPs). PSPs and SATPs can be provided by VMware, or by a third party. For more information, see VMware Native Multipathing Plug-In.
Path Selection Plug-in	PSP	The PSPs are submodules of the VMware NMP. PSPs are responsible for selecting a physical path for SCSI I/O requests. For more information, see VMware Path Selection Plug-Ins and Policies.
Storage Array Type Plug-in	SATP	SATPs are responsible for array-specific operations for a SCSI storage array. The SATP can determine the state of a particular array-specific path, perform a path activation, and detect any path errors. For more information, see VMware SATPs.
Multipathing Plug-in	MPP (third-party)	The PSA offers a collection of VMkernel APIs that third parties can use to create their own multipathing plug-ins (MPPs). The modules provide specific load balancing and failover functionalities for a particular storage array. The MPPs can be installed on the ESX host. They can run in addition to the VMware native modules, or as their replacement.
Native High-Performance Plug-in	HPP	The HPP replaces the NMP for high-speed devices, such as NVMe. The HPP can improve the performance of ultra-fast flash devices that are installed locally on your ESX host, and is the default plug-in that claims NVMe-oF targets. For information, see VMware High Performance Plug-In and Path Selection Schemes.
Path Selection Scheme	PSS	To support multipathing, the HPP uses the Path Selection Schemes (PSS). A particular PSS is responsible for selecting physical paths for I/O requests and handles multipathing for NVMe storage devices. For information, see VMware High Performance Plug-In and Path Selection Schemes.
Claim Rules		The PSA uses claim rules to determine which plug-in owns the paths to a particular storage device. See Using Claim Rules to Control ESX Multipathing Modules.

VMware Native Multipathing Plug-In

By default, ESX provides an extensible multipathing module called Native Multipathing Plug-In (NMP).

Generally, the VMware NMP supports all storage arrays listed on the VMware storage HCL and provides a default path selection algorithm based on the array type. The NMP associates a set of physical paths with a specific storage device, or LUN.

For additional multipathing operations, the NMP uses submodules, called SATPs and PSPs. The NMP delegates to the SATP the specific details of handling path failover for the device. The PSP handles path selection for the device.

Typically, the NMP performs the following operations:

- Manages physical path claiming and unclaiming.
- Registers and de-registers logical devices.
- Associates physical paths with logical devices.
- Supports path failure detection and remediation.
- Processes I/O requests to logical devices:
  - Selects an optimal physical path for the request.
  - Performs actions necessary to handle path failures and I/O command retries.
- Supports management tasks, such as reset of logical devices.

ESX automatically installs an appropriate SATP for an array you use. You do not need to obtain or download any SATPs.

VMware NMP Flow of I/O

When a virtual machine issues an I/O request to a storage device managed by the NMP, the following process takes place.

1. The NMP calls the PSP assigned to this storage device.
2. The PSP selects an appropriate physical path on which to issue the I/O.
3. The NMP issues the I/O request on the path selected by the PSP.
4. If the I/O operation is successful, the NMP reports its completion.
5. If the I/O operation reports an error, the NMP calls the appropriate SATP.
6. The SATP interprets the I/O command errors and, when appropriate, activates the inactive paths.
7. The PSP is called to select a new path on which to issue the I/O.

Display Multipathing Modules

Use the `esxcli` command to list all multipathing modules loaded into the system. Multipathing modules manage physical paths that connect your host with storage. The modules include VMware native NMP and HPP, and any third-party MPPs.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To list multipathing modules, run the following command:

```
esxcli storage core plugin list --plugin-class=MP
```

This command typically shows the NMP and, if loaded, the HPP and the MASK_PATH module. If any third-party MPPs have been loaded, they are listed as well.

```
Plugin name  Plugin class
-----------  ------------
NMP          MP
```

For more information about the command, see the *ESXCLI Concepts and Examples* and *ESXCLI Reference* documentation.

Display NMP Storage Devices

Use the `esxcli` command to list all storage devices controlled by the VMware NMP and display SATP and PSP information associated with each device.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To list all storage devices, run the following command:

```
esxcli storage nmp device list
```

Use the `--device | -d=`*`device_ID`* parameter to filter the output of this command to show a single device.

Displaying NMP Storage Devices

```
# esxcli storage nmp device list
mpx.vmhba1:C0:T2:L0
```

```
   Device Display Name: Local VMware Disk (mpx.vmhba1:C0:T2:L0)
   Storage Array Type: VMW_SATP_LOCAL
   Storage Array Type Device Config: SATP VMW_SATP_LOCAL does not support device configuration.
   Path Selection Policy: VMW_PSP_FIXED
   Path Selection Policy Device Config: {preferred=vmhba1:C0:T2:L0;current=vmhba1:C0:T2:L0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba1:C0:T2:L0
   Is USB: false

......

eui.6238666462643332
   Device Display Name: SCST_BIO iSCSI Disk (eui.6238666462643332)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: {action_OnRetryErrors=off}
   Path Selection Policy: VMW_PSP_FIXED
   Path Selection Policy Device Config: {preferred=vmhba65:C0:T0:L0;current=vmhba65:C0:T0:L0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba65:C0:T0:L0
   Is USB: false
```

For more information about the command, see the *ESXCLI Concepts and Examples* and *ESXCLI Reference* documentation.

VMware Path Selection Plug-Ins and Policies

VMware Path Selection Plug-ins (PSPs) are responsible for selecting a physical path for I/O requests.

The plug-ins are submodules of the VMware NMP. The NMP assigns a default PSP for each logical device based on the device type. You can override the default PSP. For more information, see Change the Path Selection Policy.

Each PSP enables and enforces a corresponding path selection policy.

VMW_PSP_MRU - Most Recently Used (VMware)	The Most Recently Used (VMware) policy is enforced by VMW_PSP_MRU. It selects the first working path discovered at system boot time. When the path becomes unavailable, the host selects an alternative path. The host does not revert to the original path when that path becomes available. The Most Recently Used policy does not use the preferred path setting. This policy is default for most active-passive storage devices. The VMW_PSP_MRU supports path ranking. To set ranks to individual paths, use the `esxcli storage nmp psp generic pathconfig set` command. For details, see the *ESXCLI Reference* documentation.
VMW_PSP_FIXED - Fixed (VMware)	This Fixed (VMware) policy is implemented by VMW_PSP_FIXED. The policy uses the designated preferred path. If the preferred path is not assigned, the policy selects the first working path discovered at system boot time. If the preferred path becomes unavailable, the host selects an alternative available path. The host returns to the previously defined preferred path when it becomes available again. Fixed is the default policy for most active-active storage devices.
VMW_PSP_RR - Round Robin (VMware)	VMW_PSP_RR enables the Round Robin (VMware) policy. Round Robin is the default policy for many arrays. It uses an automatic path selection algorithm rotating through the configured paths. Both active-active and active-passive arrays use the policy to implement load balancing across paths for different LUNs. With active-passive arrays, the policy uses active paths. With active-active arrays, the policy uses available paths. The latency mechanism that can be activated for the policy makes it more adaptive. To achieve better load balancing results, the mechanism dynamically selects an optimal path by considering the following path characteristics: • I/O bandwidth • Path latency To change the default parameters for the adaptive latency Round Robin policy or to deactivate the latency mechanism, see the Change Default Parameters for Latency Round Robin. To set other configurable parameters for VMW_PSP_RR, use the `esxcli storage nmp psp roundrobin` command. For details, see the *ESXCLI Reference* documentation.

VMware SATPs

Storage Array Type Plug-ins (SATPs) are responsible for array-specific operations. The SATPs are submodules of the VMware NMP.

ESX offers an SATP for every type of array that VMware supports. ESX also provides default SATPs that support non-specific active-active, active-passive, ALUA, and local devices.

Each SATP accommodates special characteristics of a certain class of storage arrays. The SATP can perform the array-specific operations required to detect path state and to activate an inactive path. As a result, the NMP module itself can work with multiple storage arrays without having to be aware of the storage device specifics.

Generally, the NMP determines which SATP to use for a specific storage device and associates the SATP with the physical paths for that storage device. The SATP implements the tasks that include the following:

- Monitors the health of each physical path.
- Reports changes in the state of each physical path.
- Performs array-specific actions necessary for storage fail-over. For example, for active-passive devices, it can activate passive paths.

ESX includes several generic SATP modules for storage arrays.

VMW_SATP_LOCAL	SATP for local direct-attached devices. VMW_SATP_LOCAL provides multipathing support for the local devices, except the devices in 4K native format. To claim multiple paths to the local devices, you are no longer required to use other SATPs as you were in the earlier vSphere releases. VMW_SATP_LOCAL supports the VMW_PSP_MRU and VMW_PSP_FIXED path selection plug-ins, but does not support VMW_PSP_RR.
VMW_SATP_DEFAULT_AA	Generic SATP for active-active arrays.
VMW_SATP_DEFAULT_AP	Generic SATP for active-passive arrays.
VMW_SATP_ALUA	SATP for ALUA-compliant arrays.

For more information, see the VMware by Broadcom Compatibility Guide and the *ESXCLI Reference* documentation.

Display SATPs for the Host

Use the `esxcli` command to list VMware NMP SATPs loaded into the system. Display information about the SATPs.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To list VMware SATPs, run the following command:

```
esxcli storage nmp satp list
```

For each SATP, the output displays information that shows the type of storage array or system the SATP supports. The output also shows the default PSP for any LUNs that use this SATP. `Placeholder (plugin not loaded)` in the Description column indicates that the SATP is not loaded.

Displaying SATPs for the Host

```
# esxcli storage nmp satp list
Name                 Default PSP    Description
VMW_SATP_MSA         VMW_PSP_MRU    Placeholder (plugin not loaded)
VMW_SATP_ALUA        VMW_PSP_MRU    Placeholder (plugin not loaded)
VMW_SATP_DEFAULT_AP  VMW_PSP_MRU    Placeholder (plugin not loaded)
VMW_SATP_SVC         VMW_PSP_FIXED  Placeholder (plugin not loaded)
VMW_SATP_EQL         VMW_PSP_FIXED  Placeholder (plugin not loaded)
VMW_SATP_INV         VMW_PSP_FIXED  Placeholder (plugin not loaded)
VMW_SATP_EVA         VMW_PSP_FIXED  Placeholder (plugin not loaded)
VMW_SATP_ALUA_CX     VMW_PSP_RR     Placeholder (plugin not loaded)
VMW_SATP_SYMM        VMW_PSP_RR     Placeholder (plugin not loaded)
```

```
VMW_SATP_CX          VMW_PSP_MRU    Placeholder (plugin not loaded)
VMW_SATP_LSI         VMW_PSP_MRU    Placeholder (plugin not loaded)
VMW_SATP_DEFAULT_AA  VMW_PSP_FIXED  Supports non-specific active/active arrays
VMW_SATP_LOCAL       VMW_PSP_FIXED  Supports direct attached devices
```

For more information about the command, see the *ESXCLI Concepts and Examples* and *ESXCLI Reference* documentation.

VMware High Performance Plug-In and Path Selection Schemes

VMware provides the High-Performance Plug-in (HPP) to improve the performance of storage devices on your ESX host.

The HPP replaces the NMP for high-speed devices, such as NVMe. The HPP is the default plug-in that claims NVMe-oF targets. ESX supports end to end NVMe without emulation and SCSI to NVMe emulation. The HPP supports only active/active and implicit ALUA targets.

HPP Support	vSphere 8.0 Update 3 and Later
Storage devices	NVMe • Local - 512e/512n/4kn • Fabric - 512 SCSI • Local - 512e/512n/4kn • Fabric - 512e/512n, active/active and implicit ALUA targets only
Multipathing	Yes
Second-level plug-ins	No
SCSI-3 persistent reservations	No
4Kn SCSI HDD (with software emulation)	Yes

Path Selection Schemes

To support multipathing, the HPP uses the Path Selection Schemes (PSS) when selecting physical paths for I/O requests.

You can use the vSphere Client or the `esxcli` command to change the default path selection mechanism.

For information about configuring the path mechanisms in the vSphere Client, see Change the Path Selection Policy. To configure with the `esxcli` command, see ESX esxcli HPP Commands.

ESX supports the following path selection mechanisms.

FIXED

With this scheme, a designated preferred path is used for I/O requests. If the preferred path is not assigned, the host selects the first working path discovered at the boot time. If the preferred path becomes unavailable, the host selects an alternative available path. The host returns to the previously defined preferred path when it becomes available again.

When you configure **FIXED** as a path selection mechanism, select the preferred path.

LB-RR (Load Balance - Round Robin)

This is the default scheme for the devices claimed by HPP. After transferring a specified number of bytes or I/Os on a current path, the scheme selects the path using the round robin algorithm.

To configure the **LB-RR** path selection mechanism, specify the following properties:

- **IOPS** indicates the I/O count on the path to be used as criteria to switch a path for the device.
- **Bytes** indicates the byte count on the path to be used as criteria to switch a path for the device.

LB-IOPS (Load Balance - IOPs)

After transferring a specified number of I/Os on a current path, default is 1000, the system selects an optimal path that has the least number of outstanding I/Os.

When configuring this mechanism, specify the **IOPS** parameter to indicate the I/O count on the path to be used as criteria to switch a path for the device.

LB-BYTES (Load Balance - Bytes)

After transferring a specified number of bytes on a current path, default is 10 MB, the system selects an optimal path that has the least number of outstanding bytes.

	To configure this mechanism, use the **Bytes** parameter to indicate the byte count on the path to be used as criteria to switch a path for the device.
Load Balance - Latency (LB-Latency)	To achieve better load balancing results, the mechanism dynamically selects an optimal path by considering the following path characteristics: • **Latency evaluation time** parameter indicates at what time interval, in milliseconds, the latency of paths must be evaluated. • **Sampling I/Os per path** parameter controls how many sample I/Os must be issued on each path to calculate latency of the path.

HPP Best Practices

To achieve the fastest throughput from a high-speed storage device, follow these recommendations.

- Use the vSphere version that supports the HPP.
- Use the HPP for local NVMe and SCSI devices, and NVMe-oF devices.
- If you use NVMe over Fibre Channel devices, follow general recommendations for Fibre Channel storage. See Using ESX with Fibre Channel SAN.
- If you use NVMe-oF, do not mix transport types to access the same namespace.
- When using NVMe-oF namespaces, make sure that active paths are presented to the host. The namespaces cannot be registered until the active path is discovered.
- When you configure your VMs, you can use VMware Paravirtual controllers or add NVMe controllers. Both types have their advantages and disadvantages. To check which works best for your environment, see *SCSI, SATA, and NVMe Storage Controller Conditions, Limitations, and Compatibility* in the *vSphere Virtual Machine Administration* documentation.
- Set the latency sensitive threshold.
- If a single VM drives a significant share of the device's I/O workload, consider spreading the I/O across multiple virtual disks. Attach the disks to separate virtual controllers in the VM.

  Otherwise, I/O throughput might be limited due to saturation of the CPU core responsible for processing I/Os on a particular virtual storage controller.

For information about device identifiers for NVMe devices that support only the NGUID ID format, see NVMe Devices with NGUID Device Identifiers.

Enable the High-Performance Plug-In and the Path Selection Schemes

The high-performance plug-in (HPP) is the default plug-in that claims local NVMe and SCSI devices, and NVMe-oF targets. If necessary, you can replace it with NMP. In vSphere version 7.0 Update 1 and earlier, NMP remains the default plug-in for local NVMe and SCSI devices, but you can replace it with HPP.

Set up your vSphere NVMe storage environment. For more information, see About VMware NVMe Storage.

Use the `esxcli storage core claimrule add` command to enable the HPP or NMP on your ESX host.

For more information, see *Getting Started with ESXCLI* and *ESXCLI Reference*.

Examples in this topic demonstrate how to enable HPP and set up the path selection schemes (PSS).

Note:
Enabling the HPP is not supported on PXE booted ESX hosts.

1. Create an HPP claim rule by running the `esxcli storage core claimrule add` command.

   Use one of the following methods to add the claim rule.

Method	Description
Based on the NVMe controller model	`esxcli storage core claimrule add --type vendor --nvme-controller-model` For example, `esxcli storage core claimrule add --rule 429 --type vendor --nvme-controller-model "ABCD*" --plugin HPP`

Method	Description
Based on the PCI vendor ID and subvendor ID	`esxcli storage core claimrule add --type vendor --pci-vendor-id --pci-sub-vendor-id` For example, `esxcli storage core claimrule add --rule 429 --type vendor --pci-vendor-id 8086 --pci-sub-vendor-id 8086 --plugin HPP`.

2. Configure the PSS.

   Use one of the following methods.

Method	Description
Set the PSS based on the device ID	`esxcli storage hpp device set` For example, `esxcli storage hpp device set --device=`*`device`* `--pss=FIXED --path=preferred path`
Set the PSS based on the vendor/model	Use the `--config-string` option with the `esxcli storage core claimrule add` command. For example, `esxcli storage core claimrule add -r 914 -t vendor -V` *`vendor`* `-M` *`model`* `-P HPP --config-string "pss=LB-Latency,latency-eval-time=40000"`

3. Reboot your host for your changes to take effect.

Set Latency Sensitive Threshold

When you use the HPP for your storage devices, set the latency sensitive threshold for the device, so that I/O can avoid the I/O scheduler.

By default, ESX passes every I/O through the I/O scheduler. However, using the scheduler might create internal queuing, which is not efficient with the high-speed storage devices.

You can configure the latency sensitive threshold and enable the direct submission mechanism that helps I/O to bypass the scheduler. With this mechanism enabled, the I/O passes directly from PSA through the HPP to the device driver.

For the direct submission to work properly, the observed average I/O latency must be lower than the latency threshold you specify. If the I/O latency exceeds the latency threshold, the system stops the direct submission and temporarily reverts to using the I/O scheduler. The direct submission is resumed when the average I/O latency drops below the latency threshold again.

You can set the latency threshold for a family of devices claimed by HPP. Set the latency threshold using the vendor and model pair, the controller model, or PCIe vendor ID and sub vendor ID pair.

1. Set the latency sensitive threshold for the device by running the following command:

   `esxcli storage core device latencythreshold set -t` *`value in milliseconds`*

   Use one of the following options.

Option	Example
Vendor/model	Set the latency sensitive threshold parameter for all devices with the indicated vendor and model: `esxcli storage core device latencythreshold set -v 'vendor1' -m 'model1' -t 10`
NVMe controller model	Set the latency sensitive threshold for all NVMe devices with the indicated controller model: `esxcli storage core device latencythreshold set -c 'controller_model1' -t 10`

Option	Example
PCIe vendor/subvendor ID	Set the latency sensitive threshold for devices with 0x8086 as PCIe vendor ID and 0x8086 as PCIe sub vendor ID. `esxcli storage core device latencythreshold set -p '8086' -s '8086' -t 10`

2. Verify that the latency threshold is set:

```
esxcli storage core device latencythreshold list
```

```
Device                Latency Sensitive Threshold
--------------------  ---------------------------
naa.55cd2e404c1728aa               0 milliseconds
naa.500056b34036cdfd               0 milliseconds
naa.55cd2e404c172bd6              50 milliseconds
```

3. Monitor the status of the latency sensitive threshold. Check VMkernel logs for the following entries:
   - `Latency Sensitive Gatekeeper turned on for device` *`device`*`. Threshold of` *`XX`* `msec is larger than max completion time of` *`YYY`* `msec`
   - `Latency Sensitive Gatekeeper turned off for device` *`device`*`. Threshold of` *`XX`* `msec is exceeded by command completed in` *`YYY`* `msec`

ESX esxcli HPP Commands

You can use the ESX Shell to configure and monitor the high-performance plug-in.

See *Getting Started with ESXCLI* for an introduction, and *ESXCLI Reference* for details of the `esxcli` command use. See https://developer.broadcom.com/tools/esxcli/latest.

Command	Description	Options
`esxcli storage hpp path list`	List the paths currently claimed by the high-performance plug-in.	`-d|--device=`*`device`* Display information for a specific device. `-p|--path=`*`path`* Limit the output to a specific path.
`esxcli storage hpp device list`	List the devices currently controlled by the high-performance plug-in.	`-d|--device=`*`device`* Show a specific device.

Command	Description	Options
`esxcli storage hpp device set`	Configure settings for an HPP device.	`-B|--bytes=`*`long`* Maximum bytes on the path, after which the path is switched. `--cfg-file` Update the configuration file and runtime with the new setting. If the device is claimed by another PSS, ignore any errors when applying to runtime configuration. `-d|--device=`*`device`* The HPP device upon which to operate. Use any of the UIDs that the device reports. Required. `-I|--iops=`*`long`* Maximum IOPS on the path, after which the path is switched. `-T|--latency-eval-time=`*`long`* Control at what interval, in ms, the latency of paths must be evaluated. `-L|--mark-device-local=`*`bool`* Set HPP to treat the device as local or not. `-M|--mark-device-ssd=`*`bool`* Specify whether or not the HPP treats the device as an SSD. `-p|--path=`*`str`* The path to set as the preferred path for the device. `-P|--pss=`*`pss_name`* The path selection scheme to assign to the device. If you do not specify the value, the system selects the default. For the description of path selection schemes, see VMware High Performance Plug-In and Path Selection Schemes. Options include: • `FIXED` Use the `-p|--path=`*`str`* suboption to set the preferred path. • `LB-Bytes` Use the `-B|--bytes=`*`long`* suboption to specify the input. • `LB-IOPs` Use the `-I|--iops=`*`long`* suboption to specify the input. • `LB-Latency` Suboptions include: `-T|--latency-eval-time=`*`long`* `-S|--sampling-ios-per-path=`*`long`* • `LB-RR` Default Suboptions include: `-B|--bytes=`*`long`* `-I|--iops=`*`long`* `-S|--sampling-ios-per-path=`*`long`* Control how many sample I/Os must be issued on each path to calculate latency of the path. `-U|--use-ano=`*`bool`* Set the option to `true` to include non-optimized paths in the set of active paths used to issue I/Os on this device. Otherwise, set the option to `false`.

Command	Description	Options
`esxcli storage hpp device usermarkedssd list`	List the devices that were marked or unmarked as SSD by user.	`-d|--device=`*`device`* Limit the output to a specific device.

Viewing and Managing Storage Paths on ESX Hosts

When you start your ESX host or rescan your storage adapter, the host discovers all physical paths to storage devices available to the host. Based on a set of claim rules, the host determines which multipathing module, the NMP, HPP, or an MPP, owns the paths to a particular device.

The module that owns the device becomes responsible for managing the multipathing support for the device. By default, the host performs a periodic path evaluation every five minutes and assigns unclaimed paths to the appropriate module.

For the paths managed by the NMP module, a second set of claim rules is used. These rules assign an SATP and PSP modules to each storage device and determine which Storage Array Type Policy and Path Selection Policy to apply.

Use the vSphere Client to view the Storage Array Type Policy and Path Selection Policy assigned to a specific storage device. You can also check the status of all available paths for this storage device. If needed, you can change the default Path Selection Policy using the client.

To change the default multipathing module or SATP, modify claim rules using ESXCLI.

You can find some information about modifying claim rules in Using Claim Rules to Control ESX Multipathing Modules.

To learn more about multipathing modules, see Multipathing Concepts and Acronyms.

View Storage Device Paths

View which multipathing policies the host uses for a specific storage device and the status of all available paths for this storage device.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices**.
4. Select the storage device whose paths you want to view.
5. Click the **Properties** tab and review the module that owns the device, for example, NMP or HPP.

   Under Multipathing Policies, you can also see the Path Selection Policy and, if applicable, the Storage Array Type Policy assigned to the device.
6. Click the **Paths** tab to review all paths available for the storage device and the status of each path. The following path status information can appear:

Status	Description
Active (I/O)	Working path or multiple paths that currently transfer data.
Standby	Paths that are inactive. If the active path fails, they can become operational and start transferring I/O.
Disabled	Paths that are deactivated by the administrator.
Dead	Paths that are no longer available for processing I/O. A physical medium failure or array misconfiguration can cause this status.

   If you are using the **Fixed** path policy, you can see which path is the preferred path. The preferred path is marked with an asterisk (*) in the Preferred column.

View Datastore Paths

Review the paths that connect to storage devices backing your VMFS datastores.

1. In the vSphere Client, navigate to the datastore.
2. Click the **Configure** tab.
3. Click **Connectivity and Multipathing**.
4. Select a host to view multipathing details for its devices.
5. Under Multipathing Policies, review the module that owns the device, such as NMP. You can also see the Path Selection Policy and Storage Array Type Policy assigned to the device.

   For example, you might see the following:

Path Selection Policy	Preferred Path
Storage Array Type Policy	VMW_SATP_LOCAL
Owner Plugin	NMP

6. Under Paths, review the device paths and the status of each path. The following path status information can appear:

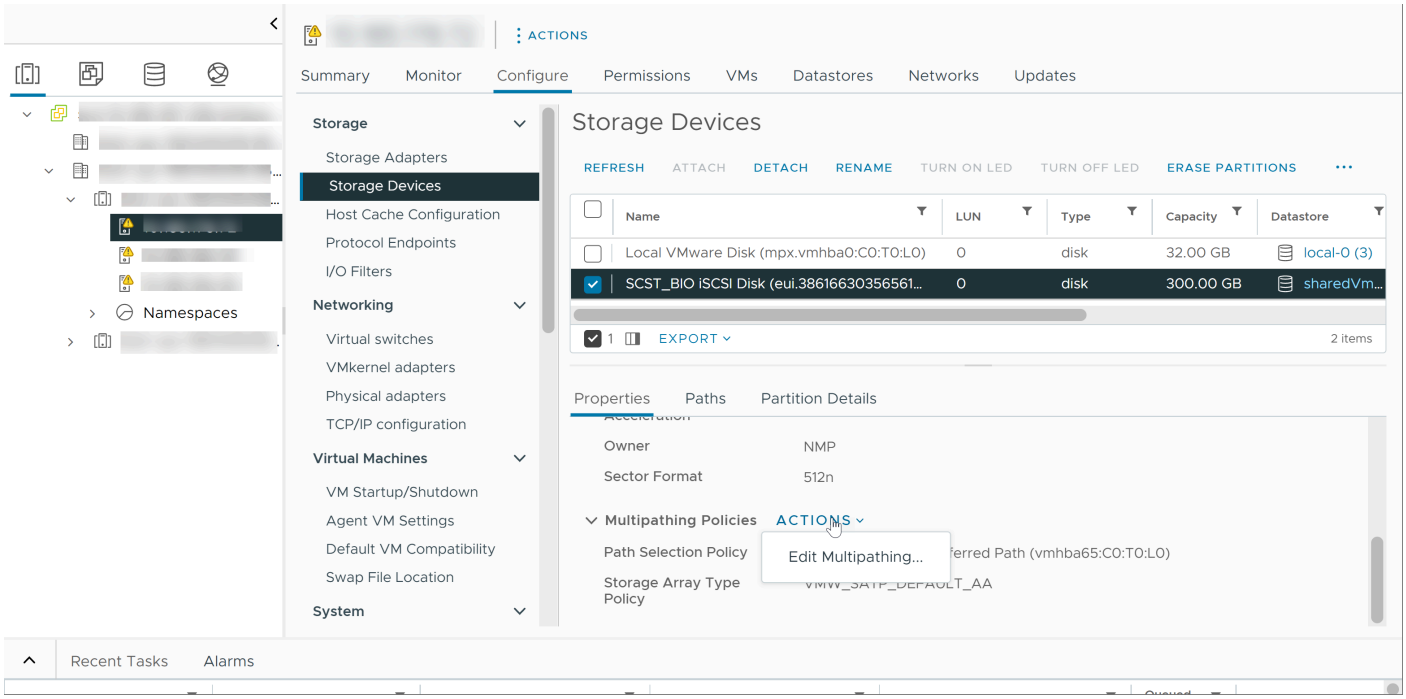
Status	Description
Active (I/O)	Working path or multiple paths that currently transfer data.
Standby	Paths that are inactive. If the active path fails, they can become operational and start transferring I/O.
Disabled	Paths that are deactivated by the administrator.
Dead	Paths that are no longer available for processing I/O. A physical medium failure or array misconfiguration can cause this status.

   If you are using the **Fixed** path policy, you can see which path is the preferred path. The preferred path is marked with an asterisk (*) in the Preferred column.

Change the Path Selection Policy

Generally, you do not need to change the default multipathing settings that your ESX host uses for a specific storage device. If you want to make any changes, you can use the **Edit Multipathing Policies** dialog box to modify the path selection policy. You can also use this dialog box to change multipathing for SCSI-based protocol endpoints.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Storage Devices** or **Protocol Endpoints**.
4. Select the item whose paths you want to change and click the **Properties** tab.
5. Under Multipathing Policies, select **Edit Multipathing** from the **Actions** menu.



6. Select a path policy and configure its settings. Your options change depending on the type of a storage device you use.
   - For information about path policies for SCSI devices, see VMware Path Selection Plug-Ins and Policies.
   - For information about path mechanisms for NVMe devices, see VMware High Performance Plug-In and Path Selection Schemes.
7. To save your settings and exit the dialog box, click **OK**.

Change Default Parameters for Latency Round Robin

On the ESX host, you can activate the latency mechanism for the Round Robin path selection policy. The mechanism considers I/O bandwidth and path latency to select an optimal path for I/O. When using the latency mechanism, the Round Robin policy can dynamically select the optimal path and achieve better load balancing results.

Set the path selection policy to Round Robin. See Change the Path Selection Policy.

When latency is activated, usage of paths for the I/Os depends on the workload. When a large number of paths is available, but the workload is not high enough, the system might not use some paths for I/Os.

You use the `esxcli` command to change the default parameters of the latency mechanism or deactivate the mechanism.

1. Configure the latency mechanism by using the following command.

   ```
   esxcli storage nmp psp roundrobin deviceconfig set --type=latency --device=device ID
   ```

   The command takes the following parameters:

Parameter	Description
-S|--num-sampling-cycles=*sampling value*	When `--type` is set to `latency`, this parameter controls how many I/Os to use to calculate the average latency of each path. The default value of this parameter is 16.

Parameter	Description
-T|--latency-eval-time=*time in ms*	When `--type` is set to `latency`, this parameter controls the frequency at which the latency of paths is updated. Default is 3 minutes.

2. Verify whether the latency Round Robin and its parameters are configured correctly.

   ```
   esxcli storage nmp psp roundrobin deviceconfig get --device=device ID
   ```

   or

   ```
   esxcli storage nmp device list --device=device ID
   ```

   The following sample output shows the path's configuration:

   ```
   Path Selection Policy: VMW_PSP_RR
     Path Selection Policy Device Config:
   {policy=latency,latencyEvalTime=180000,samplingCycles=16,curSamplingCycle=16,useANO=0;
   CurrentPath=vmhba1:C0:T0:L0: NumIOsPending=0,latency=0}
   ```

To deactivate the latency mechanism, in the Advanced System Settings for your host, change the `Misc.EnablePSPLatencyPolicy` parameter to 0.

Deactivate Storage Paths

You can temporarily deactivate paths for maintenance or other reasons.

You deactivate a path using the Paths panel. You have several ways to access the Paths panel, from a datastore, a storage device, an adapter, or a Virtual Volumes Protocol Endpoint view.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click one of the following items:
   - **Storage Adapters**
   - **Storage Devices**
   - **Protocol Endpoints**
4. In the right pane, select the item whose paths you want to deactivate, an adapter, storage device, or Protocol Endpoint, and click the **Paths** tab.
5. Select the path to deactivate and click **Disable**.

   The path's status changes to Disabled.

Using Claim Rules to Control ESX Multipathing Modules

Claim rules determine which multipathing module owns the paths to a particular storage device. They also define the type of multipathing support that the host provides to the device.

To learn more about multipathing modules, see Multipathing Concepts and Acronyms.

The claim rules are listed in the host's `/etc/vmware/esx.conf` file.

The rules fall into these categories:

- Core Claim Rules . These claim rules determine which multipathing module, the NMP, HPP, or a third-party MPP, claims the specific device.
- SATP Claim Rules. Depending on the device type, these rules assign a particular SATP submodule that provides vendor-specific multipathing management to the device.

You can use the `esxcli` commands to add or change the core and SATP claim rules. Typically, you add the claim rules to load a third-party MPP or to hide a LUN from your host. Changing claim rules might be necessary when default settings for a specific device are not sufficient.

For more information about commands available to manage PSA claim rules, see the *Getting Started with ESXCLI*.

For a list of storage arrays and corresponding SATPs and PSPs, see the Storage/SAN section of the *vSphere Compatibility Guide*.

Multipathing Considerations

Specific considerations apply when you manage storage multipathing plug-ins and claim rules.

The following considerations help you with multipathing:

- If no SATP is assigned to the device by the claim rules, the default SATP for iSCSI or FC devices is VMW_SATP_DEFAULT_AA. The default PSP is VMW_PSP_FIXED.
- When the system searches the SATP rules to locate a SATP for a given device, it searches the driver rules first. If there is no match, the vendor/model rules are searched, and finally the transport rules are searched. If no match occurs, NMP selects a default SATP for the device.
- If VMW_SATP_ALUA is assigned to a specific storage device, but the device is not ALUA-aware, no claim rule match occurs for this device. The device is claimed by the default SATP based on the device's transport type.
- The default PSP for all devices claimed by VMW_SATP_ALUA is VMW_PSP_MRU. The VMW_PSP_MRU selects an active/optimized path as reported by the VMW_SATP_ALUA, or an active/unoptimized path if there is no active/optimized path. This path is used until a better path is available (MRU). For example, if the VMW_PSP_MRU is currently using an active/unoptimized path and an active/optimized path becomes available, the VMW_PSP_MRU will switch the current path to the active/optimized one.
- While VMW_PSP_MRU is typically selected for ALUA arrays by default, certain ALUA storage arrays need to use VMW_PSP_FIXED. To check whether your storage array requires VMW_PSP_FIXED, see the VMware by Broadcom Compatibility Guide or contact your storage vendor. When using VMW_PSP_FIXED with ALUA arrays, unless you explicitly specify a preferred path, the ESX host selects the most optimal working path and designates it as the default preferred path. If the host selected path becomes unavailable, the host selects an alternative available path. However, if you explicitly designate the preferred path, it will remain preferred no matter what its status is.
- By default, the PSA claim rule 101 masks Dell array pseudo devices. Do not delete this rule, unless you want to unmask these devices.

List Multipathing Claim Rules for the Host

Use the `esxcli` command to list available multipathing claim rules.

Claim rules indicate whether the NMP, HPP, or a third-party MPP manages a given physical path. Each claim rule identifies a set of paths based on the following parameters:

- Vendor/model strings
- Transportation, such as SATA, IDE, Fibre Channel
- Adapter, target, or LUN location
- Device driver, for example, Mega-RAID

List the multipathing claim rules by running the `esxcli storage core claimrule list --claimrule-class=MP` command.

If you do not use the `claimrule-class` option, the MP rule class is implied.

Sample Output of the esxcli storage core claimrule list Command

```
Rule Class  Rule   Class    Type       Plugin     Matches
MP           10    runtime  vendor     HPP        vendor=NVMe model=*
MP           10    file     vendor     HPP        vendor=NVMe model=*
MP           50    runtime  transport  NMP        transport=usb
MP           51    runtime  transport  NMP        transport=sata
MP           52    runtime  transport  NMP        transport=ide
MP           53    runtime  transport  NMP        transport=block
MP           54    runtime  transport  NMP        transport=unknown
MP          101    runtime  vendor     MASK_PATH  vendor=DELL model=Universal Xport
MP          101    file     vendor     MASK_PATH  vendor=DELL model=Universal Xport
MP          200    runtime  vendor     MPP_1      vendor=NewVend model=*
```

```
MP          200     file     vendor     MPP_1      vendor=NewVend model=*
MP          201     runtime  location   MPP_2      adapter=vmhba41 channel=* target=* lun=*
MP          201     file     location   MPP_2      adapter=vmhba41 channel=* target=* lun=*
MP          202     runtime  driver     MPP_3      driver=megaraid
MP          202     file     driver     MPP_3      driver=megaraid
MP          65535   runtime  vendor     NMP        vendor=* model=*
```

This example indicates the following:

- The NMP claims all paths connected to storage devices that use the USB, SATA, IDE, and Block SCSI transportation.
- The rules for HPP, MPP_1, MPP_2, and MPP_3 have been added, so that the modules can claim specified devices. For example, the HPP claims all devices with vendor NVMe. All devices handled by the inbox nvme driver are claimed regardless of the actual vendor. The MPP_1 module claims all paths connected to any model of the NewVend storage array.
- You can use the MASK_PATH module to hide unused devices from your host. By default, the PSA claim rule 101 masks Dell array pseudo devices with a vendor string `DELL` and a model string `Universal Xport`.
- The Rule Class column in the output describes the category of a claim rule. It can be MP (multipathing plug-in), Filter, or VAAI.
- The Class column shows which rules are defined and which are loaded. The `file` parameter in the Class column indicates that the rule is defined. The `runtime` parameter indicates that the rule has been loaded into your system. For a user-defined claim rule to be active, two lines with the same rule number must exist, one line for the rule with the `file` parameter and another line with `runtime`. Several default system-defined claim rules have only one line with the Class of `runtime`. You cannot modify these rules.
- The default rule 65535 assigns all unclaimed paths to the NMP. Do not delete this rule.

Add Multipathing Claim Rules

Use the `esxcli` commands to add a multipathing PSA claim rule to the set of claim rules on the system. For the new claim rule to be active, you first define the rule and then load it into your system.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Examples when you add a PSA claim rule include the following:

- You load a new third-party MPP and must define the paths that this module claims.
- You must enable the native HPP.

> **Warning:**
> You cannot create rules where two different plug-ins claim paths to the same device. Your attempts to create these claim rules fail with a warning in `vmkernel.log`.

1. To define a new claim rule, use the following command:

   ```
   esxcli storage core claimrule add
   ```

   The command takes the following options:

Option	Description
-A|--adapter=<adapter>	Adapter of the paths to use. Valid only if `--type` is `location`.
-u|--autoassign	Adds a claim rule based on its characteristics. The rule number is not required.
-C|--channel=<channel>	Channel of the paths to use. Valid only if `--type` is `location`.
-c|--claimrule-class=<cl>	Claim rule class to use in this operation. You can specify `MP` (default), `Filter`, or `VAAI`. To configure hardware acceleration for a new array, add two claim rules, one for the VAAI filter and another for the VAAI plug-in. See Add Hardware Acceleration Claim Rules for detailed instructions.
-d|--device=<device_uid>	UID of the device. Valid only when `--type` is `device`.

Option	Description
-D|--driver=<driver>	Driver for the HBA of the paths to use. Valid only if `--type` is `driver`.
-f|--force	Force claim rules to ignore validity checks and install the rule anyway.
--force-reserved	Override protection of reserved rule ID ranges. Reserved claim rules are the rules with an ID below 100. You can use them to reassign local devices to specific plug-ins, for example, the NVMe device to HPP.
--if-unset=<str>	Run this command if this advanced user variable is not set to 1.
-i|--iqn=<iscsi_name>	iSCSI Qualified Name for the target. Valid only when `--type` is `target`.
-L|--lun=<lun_id>	LUN of the paths. Valid only if `--type` is `location`. LUN ID must not be higher than the value of the advanced configuration option /Disk/MaxLUN.
-M|--model=<model>	Model of the paths to use. Valid only if `--type` is `vendor`. Valid values are values of the Model string from the SCSI inquiry string. Run `vicfg-scsidevs <conn_options> -l` on each device to see model string values.
-P|--plugin=<plugin>	PSA plug-in to use. The values are `NMP`, `MASK_PATH`, or `HPP`. Third parties can also provide their own PSA plug-ins. Required.
-r|--rule=<rule_ID>	Rule ID to use. The rule ID indicates the order in which the claim rule is to be evaluated. User-defined claim rules are evaluated in numeric order starting with 101. You can run `esxcli storage core claimrule list` to determine which rule IDs are available.
-T|--target=<target>	Target of the paths to use. Valid only if `--type` is `location`.
-R|--transport=<transport>	Transport of the paths to use. Valid only if `--type` is `transport`. The following values are supported. • `block` — block storage • `fc` — Fibre Channel • `iscsivendor` — iSCSI • `iscsi` — not currently used • `ide` — IDE storage • `sas` — SAS storage • `sata` — SATA storage • `usb` — USB storage • `parallel` — parallel • `fcoe` — FCoE • `unknown`
-t|--type=<type>	Type of matching to use for the operation. Valid values are the following. Required. • `vendor` • `location` • `driver` • `transport` • `device` • `target`
-V|--vendor=<vendor>	Vendor of the paths to use. Valid only if `--type` is `vendor`. Valid values are values of the vendor string from the SCSI inquiry string. Run `vicfg-scsidevs <conn_options> -l` on each device to see vendor string values.
--wwnn=<wwnn>	World-Wide Node Number for the target.

Option	Description
--wwpn=<wwpn>	World-Wide Port Number for the target.
-a|--xcopy-use-array-values	Use the array reported values to construct the XCOPY command to be sent to the storage array. This applies to VAAI claim rules only.
-s|--xcopy-use-multi-segs	Use multiple segments when issuing an XCOPY request. Valid only if `--xcopy-use-array-values` is specified.
-m|--xcopy-max-transfer-size	Maximum data transfer size in MB when you use a transfer size different than array reported. Valid only if `--xcopy-use-array-values` is specified.
-k|--xcopy-max-transfer-size-kib	Maximum transfer size in KiB for the XCOPY commands when you use a transfer size different than array reported. Valid only if `--xcopy-use-array-values` is specified.

2. To load the new claim rule into your system, use the following command:

   ```
   esxcli storage core claimrule load
   ```

   This command loads all newly created multipathing claim rules from the `esx.conf` configuration file into the VMkernel. The command has no options.

3. To apply claim rules that are loaded, use the following command:

   ```
   esxcli storage core claimrule run
   ```

   The command takes the following options:

Option	Description
-A|--adapter=<adapter>	If `--type` is `location`, name of the HBA for the paths to run the claim rules on. To run claim rules on paths from all adapters, omit this option.
-C|--channel=<channel>	If `--type` is `location`, indicate the channel of the paths to use in this operation. To run claim rules on paths with any channel number, omit this option.
-c|--claimrule-class=<cl>	Claim rule class to use in this operation.
-d|--device=<device_uid>	UID of the device.
-L|--lun=<lun_id>	If `--type` is `location`, indicate the LUN of the paths to run claim rules on. To run claim rules on paths with any LUN, omit this option.
-p|--path=<path_uid>	If `--type` is `path`, this option indicates the unique path identifier (UID) or the runtime name of a path to run claim rules on.
-T|--target=<target>	If `--type` is `location`, indicate the target of the paths to run claim rules on. To run claim rules on paths with any target number, omit this option.
-t|--type=<location|path|all>	Type of claim to perform. By default, uses `all`, which means claim rules run without restriction to specific paths. Valid values are `location`, `path`, and `all`.
-w|--wait	You can use this option only if you also use `--type all`. If the option is included, the claim waits for paths to settle before running the claim operation. In that case, the system does not start the claiming process until it is likely that all paths on the system have appeared before starting the claim process. After the claiming process has started, the command does not return until device registration has completed. If you add or remove paths during the claiming or the discovery process, this option might not work correctly.

Defining Multipathing Claim Rules

In the following example, you add and load rule # 500. The rule claims all paths with the NewMod model string and the NewVend vendor string for the NMP plug-in.

```
# esxcli storage core claimrule add -r 500 -t vendor -V NewVend -M NewMod -P NMP
# esxcli storage core claimrule load
```

After you run the `esxcli storage core claimrule list` command, you can see the new claim rule appearing on the list.

The following output indicates that the claim rule 500 has been loaded into the system and is active.

```
Rule Class  Rule   Class    Type       Plugin     Matches
...         ...    ...      ...        ...        ...
MP          500    runtime  vendor     NMP        vendor=NewVend model=NewMod
MP          500    file     vendor     NMP        vendor=NewVend model=NewMod
```

Delete Multipathing Claim Rules

Use the `esxcli` commands to remove a multipathing PSA claim rule from the set of claim rules on the system.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

1. Delete a claim rule from the set of claim rules.

   ```
   esxcli storage core claimrule remove
   ```

   Note:
   By default, the PSA claim rule 101 masks Dell array pseudo devices. Do not delete this rule, unless you want to unmask these devices.

   The command takes the following options:

Option	Description
-c|--claimrule-class=<str>	Indicate the claim rule class (MP, Filter, VAAI).
-P|--plugin=<str>	Indicate the plug-in.
-r|--rule=<long>	Indicate the rule ID.

   This step removes the claim rule from the File class.

2. Remove the claim rule from the system.

   ```
   esxcli storage core claimrule load
   ```

   This step removes the claim rule from the Runtime class.

Mask Paths

You can prevent the host from accessing storage devices or LUNs or from using individual paths to a LUN. Use the `esxcli` commands to mask the paths. When you mask paths, you create claim rules that assign the MASK_PATH plug-in to the specified paths.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

1. Check what the next available rule ID is.

   ```
   esxcli storage core claimrule list
   ```

   The claim rules that you use to mask paths have rule IDs in the range from 101 through 200. If this command shows that rules 101 and 102 exist, you can specify 103 for the rule to add.

2. Assign the MASK_PATH plug-in to a path by creating a new claim rule for the plug-in.

   ```
   esxcli storage core claimrule add -P MASK_PATH
   ```

3. Load the MASK_PATH claim rule into your system.

   ```
   esxcli storage core claimrule load
   ```

4. Verify that the MASK_PATH claim rule was added correctly.

   ```
   esxcli storage core claimrule list
   ```

5. If a claim rule for the masked path exists, remove the rule.

   ```
   esxcli storage core claiming unclaim
   ```

6. Run the path claiming rules.
   ```
   esxcli storage core claimrule run
   ```

After you assign the MASK_PATH plug-in to a path, the path state becomes irrelevant and is no longer maintained by the host. As a result, commands that display the masked path's information might show the path state as dead.

Masking a LUN

In this example, you mask the LUN 20 on targets T1 and T2 accessed through storage adapters vmhba2 and vmhba3.

1. `#esxcli storage core claimrule list`
2. ```
   #esxcli storage core claimrule add -P MASK_PATH -r 109 -t location -A vmhba2 -C 0 -T 1 -L 20
   #esxcli storage core claimrule add -P MASK_PATH -r 110 -t location -A vmhba3 -C 0 -T 1 -L 20
   #esxcli storage core claimrule add -P MASK_PATH -r 111 -t location -A vmhba2 -C 0 -T 2 -L 20
   #esxcli storage core claimrule add -P MASK_PATH -r 112 -t location -A vmhba3 -C 0 -T 2 -L 20
   ```
3. `#esxcli storage core claimrule load`
4. `#esxcli storage core claimrule list`
5. ```
   # esxcli storage core claiming unclaim -t location -A vmhba2 -C 0 -T 1 -L 20
   # esxcli storage core claiming unclaim -t location -A vmhba2 -C 0 -T 2 -L 20
   # esxcli storage core claiming unclaim -t location -A vmhba3 -C 0 -T 1 -L 20
   # esxcli storage core claiming unclaim -t location -A vmhba3 -C 0 -T 2 -L 20
   ```
6. `#esxcli storage core claimrule run`

Unmask Paths

When you need the host to access the masked storage device, unmask the paths to the device.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

> **Note:**
> When you run an unclaim operation using a device property, for example, device ID or vendor, the paths claimed by the MASK_PATH plug-in are not unclaimed. The MASK_PATH plug-in does not track any device property of the paths that it claims.

1. Delete the MASK_PATH claim rule.
   ```
   esxcli storage core claimrule remove -r rule#
   ```
2. Verify that the claim rule was deleted correctly.
   ```
   esxcli storage core claimrule list
   ```
3. Reload the path claiming rules from the configuration file into the VMkernel.
   ```
   esxcli storage core claimrule load
   ```
4. Run the `esxcli storage core claiming unclaim` command for each path to the masked storage device.
   For example:
   ```
   esxcli storage core claiming unclaim -t location -A vmhba0 -C 0 -T 0 -L 149
   ```
5. Run the path claiming rules.
   ```
   esxcli storage core claimrule run
   ```

Your host can now access the previously masked storage device.

Define NMP SATP Rules

The NMP SATP claim rules define which SATP manages a storage device. Usually, you can use the default SATPs provided for storage devices. If default settings are not sufficient, use the `esxcli` commands to change the SATP for a specific device.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

You might need to create an SATP rule when you install a third-party SATP for a specific storage array.

1. To add a claim rule for a specific SATP, run the `esxcli storage nmp satp rule add` command. The command takes the following options.

Option	Description
-b|--boot	This rule is a system default rule added at boot time. Do not modify `esx.conf` or add to a host profile.
-c|--claim-option=*string*	Set the claim option string when adding a SATP claim rule.
-e|--description=*string*	Set the claim rule description when adding a SATP claim rule.
-d|--device=*string*	Set the device when adding SATP claim rules. Device rules are mutually exclusive with vendor/model and driver rules.
-D|--driver=*string*	Set the driver string when adding a SATP claim rule. Driver rules are mutually exclusive with vendor/model rules.
-f|--force	Force claim rules to ignore validity checks and install the rule anyway.
-h|--help	Show the help message.
-M|--model=*string*	Set the model string when adding SATP a claim rule. Vendor/ Model rules are mutually exclusive with driver rules.
-o|--option=*string*	Set the option string when adding a SATP claim rule.
-P|--psp=*string*	Set the default PSP for the SATP claim rule.
-O|--psp-option=*string*	Set the PSP options for the SATP claim rule.
-s|--satp=*string*	The SATP for which a new rule is added.
-R|--transport=*string*	Set the claim transport type string when adding a SATP claim rule.
-t|--type=*string*	Set the claim type when adding a SATP claim rule.
-V|--vendor=*string*	Set the vendor string when adding SATP claim rules. Vendor/ Model rules are mutually exclusive with driver rules.

   Note:
   When searching the SATP rules to locate a SATP for a given device, the NMP searches the driver rules first. If there is no match, the vendor/model rules are searched, and finally the transport rules. If there is still no match, NMP selects a default SATP for the device.

2. Reboot your host.

Defining an NMP SATP Rule

The following sample command assigns the VMW_SATP_INV plug-in to manage storage arrays with vendor string NewVend and model string NewMod.

```
# esxcli storage nmp satp rule add -V NewVend -M NewMod -s VMW_SATP_INV
```

When you run the `esxcli storage nmp satp list -s VMW_SATP_INV` command, you can see the new rule on the list of VMW_SATP_INV rules.

Scheduling Queues for Virtual Machine I/Os in vSphere Environment

By default, vSphere provides a mechanism that creates scheduling queues for every virtual machine file. Each file, for example, `.vmdk`, gets its own bandwidth controls. You can use the vSphere Client or the `esxcli` commands to turn off the mechanism.

This mechanism ensures that the I/O for a particular virtual machine file goes into its own separate queue and avoids interfering with I/Os from other files.

Edit Per File I/O Scheduling in the vSphere Client

The advanced `VMkernel.Boot.isPerFileSchedModelActive` parameter controls the per file I/O scheduling mechanism on VMFS and NFS 3 datastores. On the ESX host, the mechanism is turned on by default. You can turn off the mechanism using the **Advanced System Settings** dialog box.

If you turn off the per file I/O scheduling model, your host reverts to a legacy scheduling mechanism. The legacy scheduling maintains only one I/O queue for each virtual machine and storage device pair. All I/Os between the virtual machine and its virtual disks are moved into this queue. As a result, I/Os from different virtual disks might interfere with each other in sharing the bandwidth and affect each other's performance.

> **Note:**
> Do not deactivate per file scheduling if you have the HPP plug-in and the latency sensitive threshold parameter configured for high-speed local devices. Turning off per file scheduling might cause unpredictable behavior.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. Edit the value of the **VMkernel.Boot.isPerFileSchedModelActive** parameter.

Option	Description
True	Turn on the per file scheduling mechanism. By default, the parameter is activated.
False	Deactivate the per file scheduling mechanism.

5. Reboot the host for the changes to take effect.

Use esxcli Commands to Edit Per File I/O Scheduling

You can use the `esxcli` commands to change the I/O scheduling capability for VMFS, NFS 3, and NFS 4.1 datastores on your ESX host. The capability is turned on by default.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

To set or edit per file I/O scheduling, run the following commands:

Option	Description
`esxcli system settings kernel set -s isPerFileSchedModelActive -v FALSE`	Turn off per file I/O scheduling for VMFS and NFS 3.
`esxcli system settings kernel set -s isPerFileSchedModelActive -v TRUE`	Turn on per file I/O scheduling for VMFS and NFS 3.
`esxcli system module parameters list -m nfs41client`	Display the current status of NFS 4.1 file based scheduler.
`esxcli system module parameters set -m nfs41client -p fileBasedScheduler=0`	Turn off file based scheduler for NFS 4.1.
`esxcli system module parameters set -m nfs41client -p fileBasedScheduler=1`	Turn on file based scheduler for NFS 4.1.

Raw Device Mapping in vSphere

Raw device mapping (RDM) in vSphere provides a mechanism for a virtual machine to have direct access to a LUN on the physical storage subsystem.

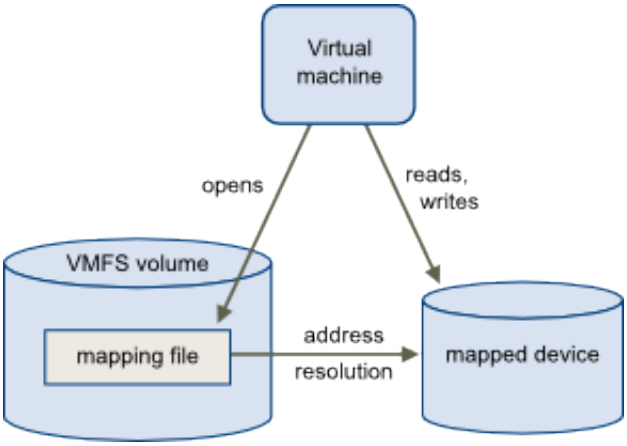
The following topics contain information about RDMs in vSphere and provide instructions on how to create and manage RDMs.

Introduction to Raw Device Mapping in vSphere

An RDM is a mapping file in a separate vSphere VMFS volume that acts as a proxy for a raw physical storage device. With RDM, a virtual machine can access and use the storage device directly. The RDM contains metadata for managing and redirecting disk access to the physical device.

The file gives you some of the advantages of direct access to a physical device, but keeps some advantages of a virtual disk in VMFS. As a result, it merges the VMFS manageability with the raw device access.

Figure 84: Raw Device Mapping



Typically, you use VMFS datastores for most virtual disk storage. On certain occasions, you might use raw LUNs or logical disks located in a SAN.

For example, you might use raw LUNs with RDMs in the following situations:

- When SAN snapshot or other layered applications run in the virtual machine. The RDM enables backup offloading systems by using features inherent to the SAN.
- In any MSCS clustering scenario that spans physical hosts, such as virtual-to-virtual clusters and physical-to-virtual clusters. In this case, cluster data and quorum disks are configured as RDMs rather than as virtual disks on a shared VMFS.

Think of an RDM as a symbolic link from a VMFS volume to a raw LUN. The mapping makes LUNs appear as files in a VMFS volume. The RDM, not the raw LUN, is referenced in the virtual machine configuration. The RDM contains a reference to the raw LUN.

Two compatibility modes are available for RDMs:

- In the virtual compatibility mode, the RDM acts like a virtual disk file. The RDM can use snapshots.
- In the physical compatibility mode, the RDM offers direct access to the SCSI device for those applications that require lower-level control.

Benefits of Raw Device Mapping

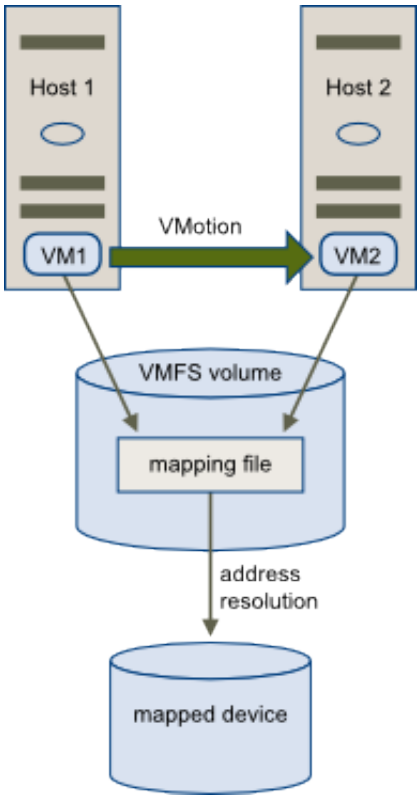
An RDM provides a number of benefits, but it should not be used in every situation. In general, virtual disk files are preferable to RDMs for manageability. However, when you need raw devices, you must use the RDM.

RDM offers several benefits.

User-Friendly Persistent Names	Provides a user-friendly name for a mapped device. When you use an RDM, you do not need to refer to the device by its device name. You refer to it by the name of the mapping file, for example: `/vmfs/volumes/myVolume/myVMDirectory/myRawDisk.vmdk`
Dynamic Name Resolution	Stores unique identification information for each mapped device. VMFS associates each RDM with its current SCSI device, regardless of changes in the physical configuration of the server because of adapter hardware changes, path changes, device relocation, and so on.
Distributed File Locking	Makes it possible to use VMFS distributed locking for raw SCSI devices. Distributed locking on an RDM makes it safe to use a shared raw LUN without losing data when two virtual machines on different servers try to access the same LUN.
File Permissions	Makes file permissions possible. The permissions of the mapping file are enforced at file-open time to protect the mapped volume.
File System Operations	Makes it possible to use file system utilities to work with a mapped volume, using the mapping file as a proxy. Most operations that are valid for an ordinary file can be applied to the mapping file and are redirected to operate on the mapped device.

Snapshots	Makes it possible to use virtual machine snapshots on a mapped volume. Snapshots are not available when the RDM is used in physical compatibility mode.
vMotion	Lets you migrate a virtual machine with vMotion. The mapping file acts as a proxy to allow vCenter to migrate the virtual machine by using the same mechanism that exists for migrating virtual disk files.

Figure 85: vMotion of a Virtual Machine Using Raw Device Mapping



SAN Management Agents	Makes it possible to run some SAN management agents inside a virtual machine. Similarly, any software that needs to access a device by using hardware-specific SCSI commands can be run in a virtual machine. This kind of software is called SCSI target-based software. When you use SAN management agents, select a physical compatibility mode for the RDM.

Vendors of storage management software must ensure that their software functions correctly in environments that include ESX.

Such software uses a physical compatibility mode for RDMs so that the software can access SCSI devices directly.

Various management products are best run centrally (not on the ESX machine), while others run well on the virtual machines. VMware at Broadcom does not certify these applications or provide a compatibility matrix. To find out whether a SAN management application is supported in an ESX environment, contact the SAN management software provider.

RDM Considerations and Limitations

Certain considerations and limitations exist when you use RDMs.

- The RDM is not available for direct-attached block devices or certain RAID devices. The RDM uses a SCSI serial number to identify the mapped device. Because block devices and some direct-attach RAID devices do not export serial numbers, they cannot be used with RDMs.
- If you are using the RDM in physical compatibility mode, you cannot use a snapshot with the disk. Physical compatibility mode allows the virtual machine to manage its own, storage-based, snapshot or mirroring operations. Virtual machine snapshots are available for RDMs with virtual compatibility mode.
- You cannot map to a disk partition. RDMs require the mapped device to be a whole LUN.
- If you use vMotion to migrate virtual machines with RDMs, make sure to maintain consistent LUN IDs for RDMs across all participating ESX hosts.

Characteristics of Raw Device Mapping in vSphere

An RDM is a special mapping file in a vSphere VMFS volume that manages metadata for its mapped device. The mapping file is presented to the management software as an ordinary disk file, available for the usual file-system operations. To the virtual machine, the storage virtualization layer presents the mapped device as a virtual SCSI device.

Key contents of the metadata in the mapping file include the location of the mapped device (name resolution), the locking state of the mapped device, permissions, and so on.

RDM Virtual and Physical Compatibility Modes

You can use RDMs in virtual compatibility or physical compatibility modes. Virtual mode specifies full virtualization of the mapped device. Physical mode specifies minimal SCSI virtualization of the mapped device, allowing the greatest flexibility for SAN management software.

In virtual mode, the VMkernel sends only READ and WRITE to the mapped device. The mapped device appears to the guest operating system exactly the same as a virtual disk file in a VMFS volume. The real hardware characteristics are hidden. If you are using a raw disk in virtual mode, you can realize the benefits of VMFS such as advanced file locking for data protection and snapshots for streamlining development processes. Virtual mode is also more portable across storage hardware than physical mode, presenting the same behavior as a virtual disk file.

In physical mode, the VMkernel passes all SCSI commands to the device, with one exception: the REPORT LUNs command is virtualized so that the VMkernel can isolate the LUN to the owning virtual machine. Otherwise, all physical characteristics of the underlying hardware are exposed. Physical mode is useful to run SAN management agents or other SCSI target-based software in the virtual machine. Physical mode also allows virtual-to-physical clustering for cost-effective high availability.

VMFS5 and VMFS6 support greater than 2 TB disk size for RDMs in virtual and physical modes.

Dynamic Name Resolution

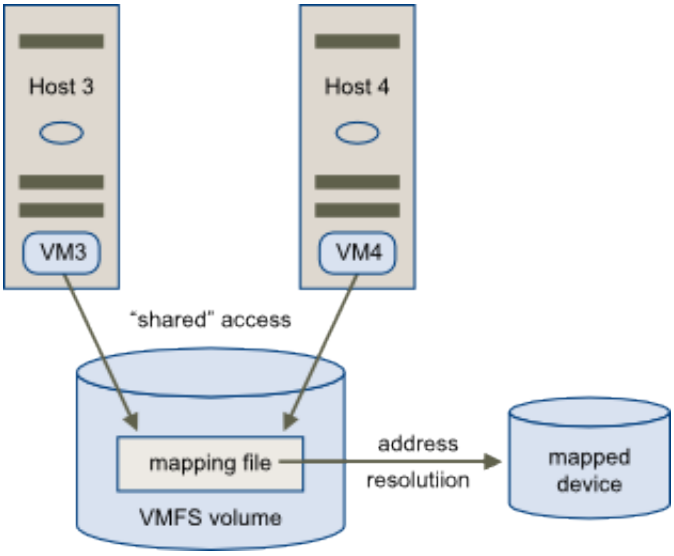
The RDM file supports dynamic name resolution when a path to a raw device changes.

VMFS uniquely identifies all mapped storage devices, and the identification is stored in its internal data structures. Any change in the path to a raw device, such as a Fibre Channel switch failure or the addition of a new HBA, can change the device name. Dynamic name resolution resolves these changes and automatically associates the original device with its new name.

Raw Device Mapping with Virtual Machine Clusters

Use an RDM with virtual machine clusters that require access to the same raw LUN for failover scenarios. The setup is similar to that of a virtual machine cluster that accesses the same virtual disk file, but an RDM replaces the virtual disk file.

Figure 86: Access from Clustered Virtual Machines



Comparing Available SCSI Device Access Modes

The ways of accessing a SCSI-based storage device include a virtual disk file on a VMFS datastore, virtual mode RDM, and physical mode RDM.

The following table provides a comparison of features available with the different modes.

Table 180: Features Available with Virtual Disks and Raw Device Mappings

ESX Features	Virtual Disk File	Virtual Mode RDM	Physical Mode RDM
SCSI Commands Passed Through	No	No	Yes `REPORT LUNs` is not passed through
vCenter Support	Yes	Yes	Yes
Snapshots	Yes	Yes	No
Distributed Locking	Yes	Yes	Yes
Clustering	Cluster-in-a-box only	Cluster-in-a-box cluster-across-boxes	Physical-to-virtual clustering cluster-across-boxes
SCSI Target-Based Software	No	No	Yes

Use virtual disk files for the cluster-in-a-box type of clustering. If you plan to reconfigure your cluster-in-a-box clusters as cluster-across-boxes clusters, use virtual mode RDMs for the cluster-in-a-box clusters.

Create vSphere Virtual Machines with RDMs

When you give your vSphere virtual machine direct access to a raw SAN LUN, you create an RDM disk that resides on a VMFS datastore and points to the LUN. You can create the RDM as an initial disk for a new virtual machine or add it to an existing virtual machine. When creating the RDM, you specify the LUN to be mapped and the datastore on which to put the RDM.

Although the RDM disk file has the same `.vmdk` extension as a regular virtual disk file, the RDM contains only mapping information. The actual virtual disk data is stored directly on the LUN.

This procedure assumes that you are creating a new virtual machine. For information, see the *vSphere Virtual Machine Administration* documentation.

1. Create a virtual machine.
   a) Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **New Virtual Machine**.
   b) Select **Create a new virtual machine** and click **Next**.
   c) Follow the steps required to create a virtual machine.
2. On the Customize Hardware page, click the **Virtual Hardware** tab.
3. Optional: To delete the default virtual hard disk that the system created for your virtual machine, move your cursor over the disk and click the **Remove** icon.
4. Add an RDM disk.
   a) Click **Add New Devices** and select **RDM Disk** from the list.
   b) From the list of LUNs, select a target raw LUN and click **OK**.

      The system creates an RDM disk that maps your virtual machine to the target LUN. The RDM disk is shown on the list of virtual devices as a new hard disk.
5. Configure the RDM disk.
   a) Click the **New Hard Disk** triangle to expand the properties for the RDM disk.
   b) Select a location for the RDM.

      You can place the RDM on the same datastore where your virtual machine configuration files reside, or select a different datastore.
   c) Select a compatibility mode.

Option	Description
Physical	Allows the guest operating system to access the hardware directly. Physical compatibility is useful if you are using SAN-aware applications on the virtual machine. However, a virtual machine with a physical compatibility RDM cannot be cloned, made into a template, or migrated if the migration involves copying the disk.
Virtual	Allows the RDM to behave as if it were a virtual disk, so you can use such features as taking snapshots, cloning, and so on. When you clone the disk or make a template out of it, the contents of the LUN are copied into a `.vmdk` virtual disk file. When you migrate a virtual compatibility mode RDM, you can migrate the mapping file or copy the contents of the LUN into a virtual disk.

   d) If you selected virtual compatibility mode, select a disk mode.

      Disk modes are not available for RDM disks using physical compatibility mode.

Option	Description
Dependent	Dependent disks are included in snapshots.
Independent - Persistent	Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode are written permanently to the disk.
Independent - Nonpersistent	Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset.

6. Complete your virtual machine configuration.

Deactivate SCSI INQUIRY Cache for Improved Performance in vSphere

Certain virtual machines with RDMs must obtain the SCSI INQUIRY information from the LUN instead of using SCSI INQUIRY data cached by ESX.

Certain guest operating systems or applications that run in the virtual machines with the RDMs display unpredictable behavior.

This behavior might be caused by cached SCSI INQUIRY data that interferes with specific guest operating systems and applications.

When the ESX host first connects to a target storage device, it issues the SCSI INQUIRY command to obtain basic identification data from the device. By default, ESX caches the received SCSI INQUIRY data (Standard, page 80, and page 83), and the data remains unchanged afterwards. Responses for subsequent SCSI INQUIRY commands are returned from the cache.

However, specific guest operating systems running in virtual machines with RDMs must query the LUN instead of using SCSI INQUIRY data cached by ESX. In these cases, you can configure the VM to ignore the SCSI INQUIRY cache.

Use one of the following methods.

Option	Description
Modify the .vmx file of the virtual machine with the RDM	Use this method for the VMs with hardware version 8 or later. 1. Add the following parameter to the file: `scsix:y.ignoreDeviceInquiryCache = "true"` where *x* is the SCSI controller number and *y* is the SCSI target number of the RDM. 2. Reboot the VM.
Use the `esxcli` command	Because you configure the setting at a host level, no VM hardware version limitations apply. `esxcli storage core device inquirycache set --device device id --ignore true` No VM reboot is required.

No matter which method you use to set the SCSI INQUIRY cache parameter to true, the VM starts contacting the LUN directly for the SCSI INQUIRY data.

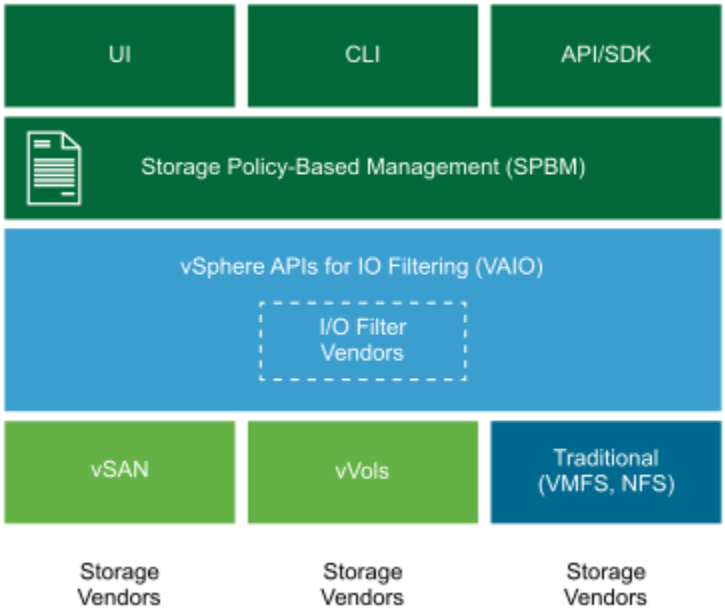
ignoreDeviceInquiryCache parameter in vmx	ignore inquirycache parameter in esxcli	Inquiry request serviced from
True	True	LUN
False (default if the parameter is not present)	True	LUN
True	False	LUN
False (default if the parameter is not present)	False	Cache

Storage Policy Based Management in vSphere

In vSphere, Storage Policy Based Management (SPBM) plays a major role by helping to align storage with application demands of your virtual machines. It provides a storage policy framework that serves as a single unified control panel across a broad range of data services and storage solutions.

As an abstraction layer, SPBM abstracts storage services delivered by Virtual Volumes, vSAN, I/O filters, or other storage entities.

Rather than integrating with each individual type of storage and data services, SPBM provides a universal framework for different types of storage entities.



SPBM offers the following mechanisms:

- Advertisement of storage capabilities and data services that storage arrays and other entities, such as I/O filters, offer.
- Bidirectional communications between ESX and vCenter on one side, and storage arrays and entities on the other.
- Virtual machine provisioning based on VM storage policies.

About Virtual Machine Storage Policies

Virtual machine storage policies are essential to virtual machine provisioning through SPBM. The policies control which type of storage is provided for the virtual machine and how the virtual machine is placed within storage. They also determine data services that the virtual machine can use.

vSphere offers default storage policies. See Default Storage Policies in vSphere.

In addition, you can define policies and assign them to the virtual machines. See Workflow for vSphere VM Storage Policies.

You use the VM Storage Policies interface to create a storage policy. When you define the policy, you specify various storage requirements for applications that run on the virtual machines. You can also use storage policies to request specific data services, such as caching or replication, for virtual disks.

You apply the storage policy when you create, clone, or migrate the virtual machine. After you apply the storage policy, the SPBM mechanism assists you with placing the virtual machine in a matching datastore. In certain storage environments, SPBM determines how the virtual machine storage objects are provisioned and allocated within the storage resource to guarantee the required level of service. The SPBM also enables requested data services for the virtual machine and helps you to monitor policy compliance.

Default Storage Policies in vSphere

When you provision a virtual machine on a datastore in vSphere, you must assign to the virtual machine a compatible VM storage policy. If you do not configure and explicitly assign the storage policy to the virtual machine, the system uses a default storage policy.

VMware-Provided Default Storage Policy	The generic default storage policy that ESX provides applies to all datastores and does not include rules specific to any storage type. In addition, ESX offers the default storage policies for object-based datastores, vSAN or Virtual Volumes. These policies guarantee the optimum placement for the virtual machine objects within the object-based storage. For information about the default storage policy for Virtual Volumes, see Virtual Volumes and VM Storage Policies.

	VMFS and NFS datastores do not have specific default policies and can use the generic default policy or a custom policy you define for them.
User-Defined Default Storage Policies	You can create a VM storage policy that is compatible with vSAN or Virtual Volumes. You can then designate this policy as the default for vSAN and Virtual Volumes datastores. The user-defined default policy replaces the default storage policy that vSphere provides. Each vSAN and Virtual Volumes datastore can have only one default policy at a time. However, you can create a single storage policy with multiple placement rule sets, so that it matches multiple vSAN and Virtual Volumes datastores. You can designate this policy as the default policy for all datastores. When the VM storage policy becomes the default policy for a datastore, you cannot delete the policy unless you disassociate it from the datastore.

Change the Default Storage Policy for a Datastore

For Virtual Volumes and vSAN datastores, VMware provides storage policies that are used as the default during the virtual machine provisioning. You can change the default storage policy for a selected Virtual Volumes or vSAN datastore.

Create a storage policy that is compatible with Virtual Volumes or vSAN. You can create a policy that matches both types of storage.

> **Note:**
> Do not designate a storage policy with replication rules as a default storage policy. Otherwise, the policy prevents you from selecting replication groups.

1. In the vSphere Client, navigate to the datastore.
2. Click the **Configure** tab, and click **General**.
3. In the Default Storage Policy pane, click **Edit**.
4. From the list of available storage policies, select a policy to designate as the default and click **OK**.

The selected storage policy becomes the default policy for the datastore. The system assigns this policy to any virtual machine objects that you provision on the datastore when no other policy is selected.

Workflow for vSphere VM Storage Policies

The entire process of creating and managing storage policies in vSphere typically includes several steps. Whether you must perform a specific step might depend on the type of storage or data services that your environment offers.

Step	Description
Populate the VM Storage Policies interface with appropriate data.	The VM Storage Policies interface is populated with information about datastores and data services that are available in your storage environment. This information is obtained from storage providers and datastore tags. • For entities represented by storage providers, verify that an appropriate provider is registered. Entities that use the storage provider include vSAN, Virtual Volumes, and I/O filters. Depending on the type of storage entity, some providers are self-registered. Other providers must be manually registered. See Use Storage Providers to Populate the VM Storage Policies Interface and Register Storage Providers for Virtual Volumes. • Tag datastores that are not represented by storage providers. You can also use tags to indicate a property that is not communicated through the storage provider, such as geographical location or administrative group. See Assign Tags to Datastores.
Create predefined storage policy components.	A storage policy component describes a single data service, such as replication, that must be provided for the virtual machine. You can define the component in advance and associate it with multiple VM storage policies. The components are reusable and interchangeable. See Create Storage Policy Components.
Create VM storage policies.	When you define storage policies for virtual machines, you specify storage requirements for applications that run on the virtual machines. See Creating and Managing vSphere VM Storage Policies.
Apply the VM storage policy to the virtual machine.	You can apply the storage policy when deploying the virtual machine or configuring its virtual disks. See Assign Storage Policies to Virtual Machines.
Check compliance for the VM storage policy.	Verify that the virtual machine uses the datastore that is compliant with the assigned storage policy. See Check Compliance for a VM Storage Policy.

To create and manage your storage policies, you use the VM Storage Policy interface of the vSphere Client.

Populating the VM Storage Policies Interface in vSphere

Before you start creating VM storage policies, you must populate the VM Storage Policy interface with information about storage entities and data services that are available in your vSphere storage environment.

This information is obtained from storage providers, also called VASA providers. Another source is datastore tags.

Storage Capabilities and Services

Certain datastores, for example, Virtual Volumes and vSAN, are represented by the storage providers. Through the storage providers, the datastores can advertise their capabilities in the VM Storage Policy interface. These datastore capabilities, data services, and other characteristics with ranges of values populate the VM Storage Policy interface.

You use these characteristics when you define datastore-based placement and service rules for your storage policy.

Data Services	I/O filters on your hosts are also represented by the storage providers. The storage provider delivers information about the data services of the filters to the VM Storage Policy interface. You use this information when defining the rules for host-based data services, also called common rules. Unlike the datastore-specific rules, these rules do not define storage placement and storage requirements for the virtual machine. Instead, they activate the requested I/O filter data services for the virtual machine.
Tags	Generally, VMFS and NFS datastores are not represented by a storage provider. They do not display their capabilities and data services in the VM Storage Polices interface. You can use tags to encode information about these datastores. For example, you can tag your VMFS datastores as VMFS-Gold and VMFS-Silver to represent different levels of service. For Virtual Volumes and vSAN datastores, you can use tags to encode information that is not advertised by the storage provider, such as geographical location (Palo Alto), or administrative group (Accounting). Similar to the storage capabilities and characteristics, all tags associated with the datastores appear in the VM Storage Policies interface. You can use the tags when you define the tag-based placement rules.

Use Storage Providers to Populate the VM Storage Policies Interface

For entities represented by storage (VASA) providers, verify that an appropriate provider is registered. After the storage providers are registered, the VM Storage Policies interface becomes populated with information about datastores and data services that the providers represent.

Register the storage providers that require manual registration. For more information, see the appropriate documentation:

- Administering VMware vSAN
- Working with vSphere Virtual Volumes
- Filtering Virtual Machine I/O in vSphere

Entities that use the storage provider include vSAN, Virtual Volumes, and I/O filters. Depending on the type of the entity, some providers are self-registered. Other providers, for example, the Virtual Volumes storage provider, must be manually registered. After the storage providers are registered, they deliver the following data to the VM Storage Policies interface:

- Storage capabilities and characteristics for such datastores as Virtual Volumes and vSAN.
- Data services the I/O filters provide.

1. Browse to the vCenter instance.
2. Click the **Configure** tab, and click **Storage Providers**.
3. In the Storage Providers list, view the storage providers registered with vCenter.

   The list shows general information including the name of the storage provider, its URL and status, storage entities that the provider represents, and so on.
4. To display more details, select a specific storage provider or its component from the list.

Assign Tags to Datastores

Use tags to encode information about a datastore. The tags are helpful when your datastore is not represented by a storage provider and does not advertise its services in the VM Storage Polices interface. You can also use the tags to indicate a property that is not communicated through a storage provider, such as a geographical location or administrative group.

Required privileges:

- **vSphere Tagging** > **Create vSphere Tag** on the root vCenter instance
- **vSphere Tagging** > **Create vSphere Tag Category** on the root vCenter instance
- **vSphere Tagging** > **Assign or Unassign vSphere Tag** on the root vCenter instance

You can apply a new tag that contains general storage information to a datastore. For more details about the tags, their categories, and how to manage the tags, see the *vCenter and Host Management* documentation.

1. In the vSphere Client, create a category for storage tags.
   a) From the Home menu, click **Tags & Custom Attributes**.
   b) Click the **Tags** tab and click **Categories**.
   c) Click the **Add Category** icon.
   d) Specify the category properties. See the following example.

Category Property	Example
Category Name	Storage Location
Description	Category for tags related to location of storage
Tags Per Object	**Many tags**
Associable Object Types	**Datastore** and **Datastore Cluster**

   e) Click **OK**.

2. Create a storage tag.
   a) On the **Tags** tab, click **Tags**.
   b) Click the **Add Tag** icon.
   c) Specify the properties for the tag. See the following example.

Tag Property	Example
Name	Texas
Description	Datastore located in Texas
Category	Storage Location

   d) Click **OK**.

3. Apply the tag to the datastore.
   a) Navigate to the datastore.
   b) Right-click the datastore, and select **Tags & Custom Attributes** > **Assign Tag**.
   c) From the list of tags, select an appropriate tag, for example, Texas in the Storage Location category, and click **Assign**.

The new tag is assigned to the datastore and appears on the datastore **Summary** tab in the **Tags** pane.

When creating a VM storage policy, you can reference the tag to include the tagged datastore in the list of compatible storage resources. See Create a VM Storage Policy for Tag-Based Placement.

Or you can exclude the tagged datastore from the VM storage policy. For example, your VM storage policy can include Virtual Volumes datastores located in Texas and California, but exclude datastores located in Nevada.

To learn more about how to use tags in VM storage policies, watch the following video.

About Rules and Rule Sets in vSphere Storage Policies

After the VM Storage Policies interface is populated with the appropriate data, you can start creating your vSphere storage policies. Creating a policy involves defining specific storage placement rules and rules to configure data services.

Rules	The rule is a basic element of the VM storage policy. Each individual rule is a statement that describes a single requirement for virtual machine storage and data services.
Rule Sets	Within a storage policy, individual rules are organized into collections of rules, or rule sets. Typically, the rule sets can be in one of the following categories: rules for host-based services and datastore-specific rules.
Datastore-Specific Rule Sets	Each rule set must include placement rules that describe requirements for virtual machine storage resources. All placement rules within a single rule set represent a single storage entity. These rules can be based on storage capabilities or tags.

	In addition, the datastore-specific rule set can include optional rules or storage policy components that describe data services to provide for the virtual machine. Generally, these rules request such services as caching, replication, other services provided by storage systems. To define the storage policy, one datastore-specific set is required. Additional rule sets are optional. A single policy can use multiple sets of rules to define alternative storage placement parameters, often from several storage providers.
Placement Rules: Capability-Based	Placement rules specify a particular storage requirement for the VM and enable SPBM to distinguish compatible datastores among all datastores in the inventory. These rules also describe how the virtual machine storage objects are allocated within the datastore to receive the required level of service. For example, the rules can list Virtual Volumes as a destination and define the maximum recovery point objective (RPO) for the Virtual Volumes objects. When you provision the virtual machine, these rules guide the decision that SPBM makes about the virtual machine placement. SPBM finds the Virtual Volumes datastores that can match the rules and satisfy the storage requirements of the virtual machine. See Create a VM Storage Policy for Virtual Volumes.
Placement Rules: Tag-Based	Tag-based rules reference datastore tags. These rules can define the VM placement, for example, request as a target all datastores with the VMFS-Gold tag. You can also use the tag-based rules to fine-tune your VM placement request further. For example, exclude datastores with the Palo Alto tag from the list of your Virtual Volumes datastores. See Create a VM Storage Policy for Tag-Based Placement.
Rules for Host-Based Services	This rule set activates data services provided by the host. The set for host-based services can include rules or storage policy components that describe particular data services, such as encryption or replication. Unlike datastore-specific rules, this set does not include placement rules. Rules for host-based services are generic for all types of storage and do not depend on the datastore. See Create a VM Storage Policy for Host-Based Data Services.

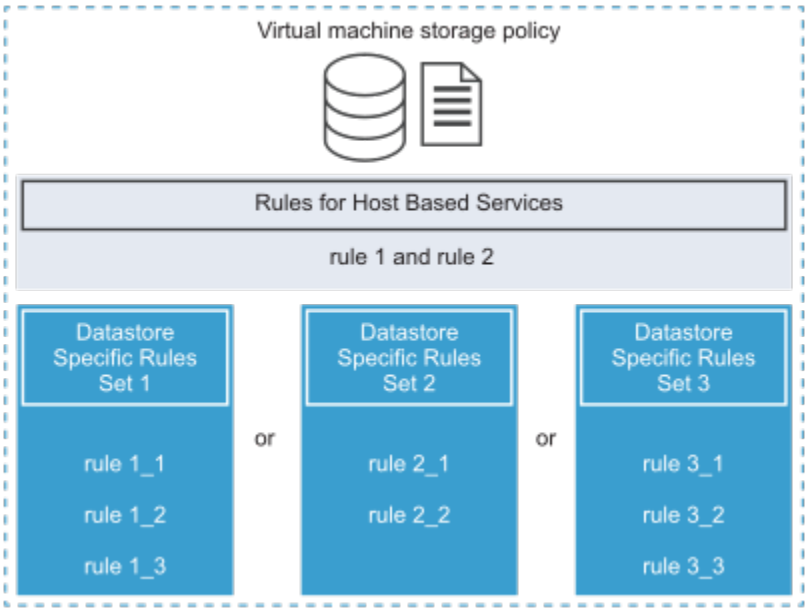
Table 181: Structure of a VM Storage Policy

Rules for Host-Based Services	Datastore-Specific Rule Sets
Rules or predefined storage policy components to activate data services installed on ESX hosts. For example, replication by I/O filters.	Capability-based or tag-based placement rules that describe requirements for virtual machine storage resources. For example, Virtual Volumes placement.
	Rules or predefined storage policy components that activate data services provided by storage. For example, caching by Virtual Volumes.

Relationships Between Rules and Rule Sets

The boolean operator `OR` defines the relationship between the datastore-specific rule sets within the policy. The `AND` operator defines the relationship between all rules within a single rule set. The policy can contain only a rule set for host-based services, or only a datastore-specific rule set, or both.

If the rule set for host-based services is not present, meeting all the rules of a single datastore-specific rule set is sufficient to satisfy the entire policy. If the rule set for host-based services is present, the policy matches the datastore that satisfies the host services rules and all rules in one of the datastore-specific sets.



Creating and Managing vSphere VM Storage Policies

To create and manage storage policies for your virtual machines in vSphere, you use the VM Storage Policies interface.

Create a VM Storage Policy for VMFS Datastore

Create a custom storage policy compatible with VMFS datastores. In the storage policy, you can specify disk or volume provisioning options.

- Be familiar with provisioning policies. See About Virtual Disk Provisioning Policies.
- Required privileges: **VM storage policies** > **Update** and **VM storage policies** > **View**.

1. Open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click **Create**.
2. Enter the policy name and description, and click **Next**.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy, for example VMFS Storage Policy.
K8s Compliant Name (optional)	Enter the name complaint with the RFC 1123 Kubernetes standard. If you do not enter this name, the system generates a unique valid Kubernetes compliant name. You cannot change this name later.
Description	Enter the description of the storage policy.

3. On the **Policy structure** page, enable rules for a target storage entity, VMFS storage.
   You can enable rules for several datastores. Multiple rule sets allow a single policy to define alternative storage placement parameters, often from several storage providers.

a) Under **Datastore specific rules**, select **Enable rules for "VMFS" storage**.
b) Under **Storage format**, select **Enable storage format** to be able to select a sector size.

4. On the **VMFS rules** page, define storage placement rules for the target VMFS datastore.
   a) Click the **Placement** tab.
   b) From the **Volume allocation** drop-down menu, select one of the following options.

Conserve space when possible	Use to create a virtual disk of volume in thin format. The option neither guarantees space reservation nor runtime performance. This option gives fast provisioning time but the space is allocated on demand and the blocks are also initialized on demand.
Fully initialized	Use to create a disk or volume in thick provision eager zeroed format. The option guarantees that the required space is reserved on the datastore and all the blocks of the disk are zeroed. This option gives the best runtime performance, but provisioning time is proportional to the size of the disk.
Reserve space	Use to create a disk or volume in thick provision lazy zeroed format. The option guarantees that the required space is reserved on the datastore. The blocks of the disk are not zeroed, so there are no guarantee on the runtime performance.

   c) To fine-tune your placement request further, click the **Tags** tab and add a tag-based rule.

      Tag-based rules can filter datastores by including or excluding specific placement criteria. For example, your VM storage policy can include VMFS datastores located in Texas and California, but exclude datastores located in Nevada.

5. If you enabled storage format, on the **Storage format** page, select the sector size for the virtual disk.

   Once the disk is created, you cannot change the sector size of the disk. For more information about datastore storage formats, see Sector Formats for Storage Devices.

Option	Description
512n	The sector size of the disk will be 512 bytes.
4Kn	The sector size of the disk will is 4096 bytes.

6. On the **Storage compatibility** page, review the list of compatible datastores.
7. On the **Review and finish** page, review the storage policy settings and click **Finish**.

   To change any settings, click **Back** to go to the relevant page.

Create a VM Storage Policy for Host-Based Data Services

To define the VM storage policy in the vSphere Client, use the **Create VM Storage Policy** wizard. In this task, you create rules for data services offered by ESX hosts. The VM storage policy that includes these rules activates specified data services for the virtual machine.

- For information about encrypting your virtual machines, see the *vSphere Security* documentation.
- For information about I/O filters, see Filtering Virtual Machine I/O in vSphere.
- For information about storage policy components, see About vSphere VM Storage Policy Components.
- Required privileges: **VM storage policies** > **Update** and **VM storage policies** > **View**.

Available data services include encryption, I/O control, caching, and so on. Certain data services, such as encryption, are provided by VMware. Others can be offered by third-party I/O filters that you install on your host.

The data services are usually generic for all types of storage and do not depend on a datastore. Adding datastore-specific rules to the storage policy is optional.

If you add datastore-specific rules, and both the I/O filters on the host and storage offer the same type of service, for example, encryption, your policy can request this service from both providers. As a result, the virtual machine data is

encrypted twice, by the I/O filter and your storage. However, replication provided by Virtual Volumes and replication provided by the I/O filter cannot coexist in the same storage policy.

1. Open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**
   c) Click **Create**.
2. Enter the policy name and description, and click **Next**.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy.
K8s Compliant Name (optional)	Enter the name complaint with the RFC 1123 Kubernetes standard. If you do not enter this name, the system generates a unique valid Kubernetes compliant name. You cannot change this name later.
Description	Enter the description of the storage policy.

3. On the **Policy structure** page under **Host based services**, click **Enable host based rules**.

   If needed, select **Enable storage format** to be able to select a sector size.
4. On the **Host based services** page, define rules and configure data services provided by your host.
   a) Click the tab for the data service category, for example, **Encryption**.
   b) Define custom rules for the data service category or use predefined components.

Option	Description
Disabled	This is a default setting.
Use storage policy component	Select a storage policy component from the drop-down menu. This option is available only if you have predefined components in your database.
Custom	Define custom rules for the data service category by specifying an appropriate provider and values for the rules.

Note:
You can activate several data services. If you use encryption with other data services, use the custom encryption option and set the **Allow I/O filters before encryption** parameter to **True**. The parameter ensures that other services, such as replication, can analyze clear text data before it is encrypted.

5. If you enabled storage format, on the **Storage format** page, select the sector size for the virtual disk.

   Once the disk is created, you cannot change the sector size of the disk. For more information about datastore storage formats, see Sector Formats for Storage Devices.

Option	Description
512n	The sector size of the disk will be 512 bytes.
4Kn	The sector size of the disk will is 4096 bytes.

6. On the **Storage compatibility** page, review the list of datastores that match this policy.

   To be compatible with the policy for host-based services, datastores must be connected to the host that provides these services. If you add datastore-specific rule sets to the policy, the compatible datastores must also satisfy storage requirements of the policy.
7. On the **Review and finish** page, review the storage policy settings and click **Finish**.

   To change any settings, click **Back** to go to the relevant page.

The new VM storage policy for host-based data services appears on the list.

Create a VM Storage Policy for Virtual Volumes

To define the VM storage policy in the vSphere Client, use the **Create VM Storage Policy** wizard. In this task, you create a custom storage policy compatible with Virtual Volumes. When you define the VM storage policy for Virtual Volumes, you create rules to configure storage and data services provided by the Virtual Volumes datastore. The rules are applied when the VM is placed on the Virtual Volumes datastore. The custom storage policy can replace the default No Requirements storage policy for Virtual Volumes.

- Verify that the Virtual Volumes storage provider is available and active. See Register Storage Providers for Virtual Volumes.
- Make sure that the VM Storage Policies interface is populated with information about storage entities and data services that are available in your storage environment. See Populating the VM Storage Policies Interface in vSphere.
- Define appropriate storage policy components. See Create Storage Policy Components.
- Required privileges: **VM storage policies** > **Update** and **VM storage policies** > **View**.

The procedure assumes that you are creating the VM storage policy for Virtual Volumes. For information about the vSAN storage policy, see the *Administering VMware vSAN* documentation.

1. Open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click **Create**.
2. Enter the policy name and description, and click **Next**.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy, for example Virtual Volumes Storage Policy.
K8s Compliant Name (optional)	Enter the name complaint with the RFC 1123 Kubernetes standard. If you do not enter this name, the system generates a unique valid Kubernetes compliant name. You cannot change this name later.
Description	Enter the description of the storage policy.

3. On the **Policy structure** page under Datastore specific rules, enable rules for a target storage entity, such as Virtual Volumes storage.
   You can enable rules for several datastores. Multiple rule sets allow a single policy to define alternative storage placement parameters, often from several storage providers.
4. On the `Virtual Volumes` rules page, define storage placement rules for the target Virtual Volumes datastore.
   a) Click the **Placement** tab and click **Add Rule**.
   b) From the Add Rule drop-down menu, select available capability and specify its value.
      For example, you can specify the number of read operations per second for the Virtual Volumes objects.
      You can include as many rules as you need for the selected storage entity. Verify that the values you provide are within the range of values that the Virtual Volumes datastore advertises.
   c) To fine-tune your placement request further, click the **Tags** tab and add a tag-based rule.
      Tag-based rules can filter datastores by including or excluding specific placement criteria. For example, your VM storage policy can include Virtual Volumes datastores located in Texas and California, but exclude datastores located in Nevada.
5. Optional: Define rules to configure datastore-specific services.
   The data services, such as encryption, caching, or replication, are offered by the storage. The VM storage policy that references data services, requests these services for the VM when the VM is placed to the Virtual Volumes datastore.

a) Click the tab for the data service category, for example, **Replication**.
b) Define custom rules for the data service category or use predefined components.

Option	Description
Disabled	This is a default setting.
Use storage policy component	Select a storage policy component from the drop-down menu. This option is available only if you have predefined components in your database.
Custom	Define custom rules for the data service category by specifying an appropriate provider and values for the rules.

6. On the **Storage compatibility** page, review the list of datastores that match this policy.
   If the policy includes several rule sets, the datastore must satisfy at least one rule set and all rules within this set.
7. On the **Review and finish** page, review the storage policy settings and click **Finish**.
   To change any settings, click **Back** to go to the relevant page.

The new VM storage policy compatible with Virtual Volumes appears on the list.

You can now associate this policy with a virtual machine, or designate the policy as default.

Create a VM Storage Policy for Tag-Based Placement

Tag-based rules reference the tags that you assign to the datastores and can filter the datastores to be used for placement of the VMs. To define tag-based placement in the vSphere Client, use the **Create VM Storage Policy** wizard.

- Make sure that the VM Storage Policies interface is populated with information about storage entities and data services that are available in your storage environment. See Populating the VM Storage Policies Interface in vSphere.
- Required privileges: **VM storage policies** > **Update** and **VM storage policies** > **View**.

Note:
At a datastore cluster level, you can use a tag-based storage policy only when all datastores in the cluster have consistent tags.

1. Open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click **Create**.
2. Enter the policy name and description, and click **Next**.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy.
K8s Compliant Name (optional)	Enter the name complaint with the RFC 1123 Kubernetes standard. If you do not enter this name, the system generates a unique valid Kubernetes compliant name. You cannot change this name later.
Description	Enter the description of the storage policy.

3. On the **Policy structure** page under Datastore specific rules, enable tag-based placement rules.
4. On the **Tag based placement** page, create the tag rules.
   a) Click **Add Tag Rule** and define tag-based placement criteria. Use the following as an example.

Option	Example
Tag category	Level of Service
Usage option	Use storage tagged with
Tags	Gold

   All datastores with the Gold tag become compatible as the storage placement target.
   b) Optional: Add more tag-based rules.
5. On the **Storage compatibility** page, review the list of datastores that match this policy.
6. On the **Review and finish** page, review the storage policy settings and click **Finish**.
   To change any settings, click **Back** to go to the relevant page.

The new VM storage policy compatible with tagged datastores appears on the list.

Edit or Clone a VM Storage Policy

If storage requirements for virtual machines and virtual disks change, you can modify the existing storage policy. You can also create a copy of the existing VM storage policy by cloning it. While cloning, you can optionally select to customize the original storage policy.

Required privilege: **StorageProfile.View**

1. In the vSphere Client, navigate to the storage policy.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
2. Select the storage policy, and click one of the following icons:
   - **Edit**
   - **Clone**
3. Optional: Modify the policy and click **OK**.
4. If editing the storage policy that is used by a virtual machine, reapply the policy to the virtual machine.

Option	Description
Manually later	If you select this option, the compliance status for all virtual disks and virtual machine home objects associated with the storage policy changes to Out of Date. To update configuration and compliance, manually reapply the storage policy to all associated entities. See Reapply Virtual Machine Storage Policy.
Now	Update virtual machine and compliance status immediately after editing the storage policy.

About vSphere VM Storage Policy Components

A VM storage policy in vSphere can include one or several reusable and interchangeable building blocks, called storage policy components. Each component describes a particular data service to be provided for the virtual machine. You can define the policy components in advance and associate them with multiple VM storage policies.

You cannot assign the predefined component directly to a virtual machine or virtual disk. Instead, you must add the component to the VM storage policy, and assign the policy to the virtual machine.

The component describes one type of service from one service provider. The services can vary depending on the providers that you use, but generally belong in one of the following categories.

- Compression

- Caching
- Encryption
- Replication

When you create the storage policy component, you define the rules for one specific type and grade of service.

The following example shows that virtual machines VM1 and VM2 have identical placement requirements, but must have different grades of replication services. You can create the storage policy components with different replication parameters and add these components to the related storage policies.

Table 182: Storage Policy Components

Virtual Machine	Placement Rules	Storage Policy Component
VM1 requires replication every 2 hours	Virtual Volumes Datastore	2-hour Replication
VM2 requires replication every 4 hours	Virtual Volumes Datastore	4-hour Replication

The provider of the service can be a storage system, an I/O filter, or another entity. If the component references an I/O filter, the component is added to the host-based rules of the storage policy. Components that reference entities other than the I/O filters, for example, a storage system, are added to the datastore-specific rule sets.

When you work with the components, follow these guidelines:

- Each component can include only one set of rules. All characteristics in this rule set belong to a single provider of the data services.
- If the component is referenced in the VM storage policy, you cannot delete the component. Before deleting the component, you must remove it from the storage policy or delete the storage policy.
- When you add components to the policy, you can use only one component from the same category, for example caching, per a set of rules.

Create Storage Policy Components

A storage policy component describes a single data service, such as replication, that must be provided for the virtual machine. You can define the component in advance and associate it with multiple VM storage policies. The components are reusable and interchangeable.

1. In the vSphere Client, open the **New Storage Policy Component** dialog box.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **Storage Policy Components**.
2. Click **Create Storage Policy Component**.
3. Select the vCenter instance.
4. Enter a name, for example, 4-hour Replication, and a description for the policy component.
   Make sure that the name does not conflict with the names of other components or storage policies.
5. Select the category of the service, for example, **Replication**.
6. Select the service provider.
7. Define rules for the selected category.
   For example, if you are configuring 4-hour replication, set the Recovery Point Objective (RPO) value to 4.

   For encryption based on I/O filters, set the **Allow I/O filters before encryption** parameter. Encryption provided by storage does not require this parameter.

Option	Description
False (default)	Does not allow the use of other I/O filters before the encryption filter.
True	Allows the use of other I/O filters before the encryption filter. Other filters, such as replication, can analyze clear text data before it is encrypted.

8. Click **OK**.

The new component appears in the list of storage policy components.

You can add the component to the VM storage policy. If the data service that the component references is provided by the I/O filters, you add the component to the host-based rules of the storage policy. Components that reference entities other than the I/O filters, for example, a storage system, are added to the datastore-specific rule sets.

Edit or Clone Storage Policy Components

You can modify the existing storage policy components. You can also create a copy of the existing component by cloning it.

1. In the vSphere Client, navigate to the storage policy component to edit or clone.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **Storage Policy Components**.
2. Select the component and click one of the following icons.

Option	Description
Edit Settings	When editing, you cannot change the category of the data service and the provider. For example, if the original component references replication provided by I/O filters, these settings must remain unchanged.
Clone	When cloning, you can customize any settings of the original component.

3. Modify appropriate values, and click **OK**.
4. If a VM storage policy that is assigned to a virtual machine references the policy component you edit, reapply the storage policy to the virtual machine.

Menu Item	Description
Manually later	If you select this option, the compliance status for all virtual disks and virtual machine home objects associated with the storage policy changes to Out of Date. To update configuration and compliance, manually reapply the storage policy to all associated entities. See Reapply Virtual Machine Storage Policy.
Now	Update virtual machine and compliance status immediately after editing the storage policy.

vSphere Storage Policies and Virtual Machines

After you define a VM storage policy in vSphere, you can apply it to a virtual machine. You apply the storage policy when provisioning the virtual machine or configuring its virtual disks. Depending on its type and configuration, the policy might serve different purposes. The policy can select the most appropriate datastore for the virtual machine and enforce the required level of service. Or it can enable specific data services for the virtual machine and its disks.

If you do not specify the storage policy, the system uses a default storage policy that is associated with the datastore. If your storage requirements for the applications on the virtual machine change, you can modify the storage policy that was originally applied to the virtual machine.

Assign Storage Policies to Virtual Machines

You can assign a VM storage policy in an initial deployment of a virtual machine or when you perform other virtual machine operations, such as cloning or migrating.

This topic describes how to assign the VM storage policy when you create a virtual machine. For information about other deployment methods that include cloning, deployment from a template, and so on, see Deploying Virtual Machines.

You can apply the same storage policy to the virtual machine configuration file and all its virtual disks. If storage requirements for your virtual disks and the configuration file are different, you can associate different storage policies with the VM configuration file and the selected virtual disks.

1. Start the virtual machine provisioning process and follow the appropriate steps.
2. Assign the same storage policy to all virtual machine files and disks.
   a) On the **Select storage** page, select a storage policy from the **VM Storage Policy** drop-down menu.

      Based on its configuration, the storage policy separates all datastores into compatible and incompatible. If the policy references data services offered by a specific storage entity, for example, Virtual Volumes, the compatible list includes datastores that represent only that type of storage.
   b) Select an appropriate datastore from the list of compatible datastores.

      The datastore becomes the destination storage resource for the virtual machine configuration file and all virtual disks.
   c) If you use the replication service with Virtual Volumes, specify the replication group.

      Replication groups indicate which VMs and virtual disks must be replicated together to a target site.

Option	Description
Preconfigured replication group	Replication groups that are configured in advance on the storage side. vCenter and ESX discover the replication groups, but do not manage their life cycle.
Automatic replication group	Virtual Volumes creates a replication group and assigns all VM objects to this group.

3. Change the VM storage policy for the virtual disk.

   Use this option if requirements for storage placement are different for virtual disks. You can also use this option to enable I/O filter services, such as caching and replication, for your virtual disks.
   a) On the **Customize hardware** page, expand the **New hard disk** pane.
   b) From the **VM storage policy** drop-down menu, select the storage policy to assign to the virtual disk.
   c) Optional: Change the storage location of the virtual disk.

      Use this option to store the virtual disk on a datastore other than the datastore where the VM configuration file resides.
4. Complete the virtual machine provisioning process.

After you create the virtual machine, the **Summary** tab displays the assigned storage policies and their compliance status. If storage placement requirements for the configuration file or the virtual disks change, you can later modify the virtual policy assignment.

Change Storage Policy Assignment for Virtual Machine Files and Disks

If your storage requirements for the applications on the virtual machine change, you can edit the storage policy that was originally applied to the virtual machine.

You can edit the storage policy for a powered-off or powered-on virtual machine.
When changing the VM storage policy assignment, you can apply the same storage policy to the virtual machine configuration file and all its virtual disks. You can also associate different storage policies with the VM configuration file and the virtual disks. You might apply different policies when, for example, storage requirements for your virtual disks and the configuration file are different.

1. In the vSphere Client, browse to the virtual machine.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click the storage policy you want to change, and click **VM Compliance**.

      You can see the list of virtual machines that use this storage policy.

d) Click the virtual machine whose policy you want to modify.

2. Click the **Configure** tab and click **Policies**.
3. Click **Edit VM Storage Policies**.
4. Specify the VM storage policy for your virtual machine.

Option	Actions
Apply the same storage policy to all virtual machine objects	Select the policy from the **VM storage policy** drop-down menu.
Apply different storage policies to the VM home object and virtual disks	1. Turn on the **Configure per disk** option. 2. Select the object, for example, VM home. 3. In the VM Storage Policy column, select the policy from the drop-down menu.

5. If you use Virtual Volumes policy with replication, configure the replication group.

   Replication groups indicate which VMs and virtual disks must be replicated together to a target site.

   All storage objects of a VM should belong to the same replication group. You cannot assign different replication groups to different storage objects of a VM.
6. Click **OK** to save the VM storage policy changes.

The storage policy is assigned to the virtual machine and its disks.

Check Compliance for a VM Storage Policy

You can check whether a virtual machine uses a datastore that is compatible with the storage requirements specified in the VM storage policy.

Verify that the virtual machine has a storage policy that is associated with it.

1. In the vSphere Client, navigate to the virtual machine.
2. Click the **Configure** tab and click **Policies**.
3. Click **Check VM Storage Policy Compliance**.

   The system verifies the compliance.
4. View the compliance status.

Compliance Status	Description
Compliant	The datastore that the virtual machine or virtual disk uses has the storage capabilities compatible with the policy requirements.
Noncompliant	The datastore that the virtual machine or virtual disk uses does not have the storage capabilities compatible with the policy requirements. You can migrate the virtual machine files and virtual disks to compliant datastores.
Out of Date	The status indicates that the policy has been edited, but the new requirements have not been communicated to the datastore where the virtual machine objects reside. To communicate the changes, reapply the policy to the objects that are out of date.
Not Applicable	This storage policy references datastore capabilities that are not supported by the datastore where virtual machine resides.

When you cannot bring the noncompliant datastore into compliance, migrate the files or virtual disks to a compatible datastore. See Find Compatible Storage Resource for Noncompliant Virtual Machine.

If the status is Out of Date, reapply the policy to the objects. See Reapply Virtual Machine Storage Policy.

Find Compatible Storage Resource for Noncompliant Virtual Machine

Determine which datastore is compatible with the storage policy that is associated with your virtual machine.

Occasionally, a storage policy that is assigned to a virtual machine can be in the noncompliant status. This status indicates that the virtual machine or its disks use datastores that are incompatible with the policy. You can migrate the virtual machine files and virtual disks to compatible datastores.

Use this task to determine which datastores satisfy the requirements of the policy.

1. Verify that the storage policy for the virtual machine is in the noncompliant state.
   a) In the vSphere Client, navigate to the virtual machine.
   b) Click the **Summary** tab.
      The VM Storage Policy Compliance panel on the VM Storage Policies pane shows the Noncompliant status.
2. Navigate to the noncompliant storage policy.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
3. Display the list of compatible datastores for the noncompliant storage policy.
   a) Click the storage policy.
   b) Click **Storage Compatibility**.
   The list of datastores that match the requirements of the policy appears.

You can migrate the virtual machine or its disks to one of the datastores in the list.

Reapply Virtual Machine Storage Policy

After you edit a storage policy that is already associated with a virtual machine object, you must reapply the policy. By reapplying the policy, you communicate new storage requirements to the datastore where the virtual machine object resides.

The compliance status for a virtual machine is Out of Date. The status indicates that the policy has been edited, but the new requirements have not been communicated to the datastore.

1. In the vSphere Client, navigate to the virtual machine.
2. Click the **Configure** tab and click **Policies**.
3. Verify that the compliance status is Out of Date.
4. Click **Reapply VM Storage Policy**.
5. Check the compliance status.

Compliance Status	Description
Compliant	The datastore that the virtual machine or virtual disk uses has the storage capabilities that the policy requires.
Noncompliant	The datastore supports specified storage requirements, but cannot currently satisfy the storage policy. For example, the status might become Noncompliant when physical resources of the datastore are unavailable. You can bring the datastore into compliance by making changes in the physical configuration of your host cluster. For example, by adding hosts or disks to the cluster. If additional resources satisfy the storage policy, the status changes to Compliant. When you cannot bring the noncompliant datastore into compliance, migrate the files or virtual disks to a compatible datastore. See Find Compatible Storage Resource for Noncompliant Virtual Machine.

Compliance Status	Description
Not Applicable	The storage policy references datastore capabilities that are not supported by the datastore.

Using Storage Providers in vSphere

A storage provider is a software component that is offered by VMware or developed by a third party through vSphere APIs for Storage Awareness (VASA). The storage provider can also be called VASA provider. The storage providers integrate with various storage entities that include external physical storage and storage abstractions, such as vSAN and Virtual Volumes. Storage providers can also support software solutions, for example, I/O filters.

Storage Provider Concepts

Generally, vCenter and ESX use the storage providers to obtain information about storage configuration, status, and storage data services offered in your environment. This information appears in the vSphere Client. The information helps you to make appropriate decisions about virtual machine placement, to set storage requirements, and to monitor your storage environment.

Persistence Storage Providers Storage providers that manage arrays and storage abstractions, are called persistence storage providers. Providers that support Virtual Volumes or vSAN belong to this category. In addition to storage, persistence providers can provide other data services, such as replication.

Data Service Providers Another category of providers is I/O filter storage providers, or data service providers. These providers offer data services that include host-based caching, compression, and encryption.

Both persistence storage and data service providers can belong to one of these categories.

Built-in Storage Providers Built-in storage providers are offered by VMware. Typically, they do not require registration. For example, the storage providers that support vSAN or I/O filters are build-in and become registered automatically.

Third-Party Storage Providers When a third party offers a storage provider, you typically must register the provider. An example of such a provider is the Virtual Volumes provider. You use the vSphere Client to register and manage each storage provider component.

For more information, Manage Storage Providers for vSphere Virtual Volumes.

Storage Providers and Data Representation

Information that the storage provider supplies can be divided into the following categories:

- Storage data services and capabilities. This type of information is essential for such functionalities as vSAN, Virtual Volumes, and I/O filters. The storage provider that represents these functionalities integrates with the Storage Policy Based Management (SPBM) mechanism. The storage provider collects information about data services that are offered by underlying storage entities or available I/O filters.

  You reference these data services when you define storage requirements for virtual machines and virtual disks in a storage policy. Depending on your environment, the SPBM mechanism ensures appropriate storage placement for a virtual machine or enables specific data services for virtual disks. For details, see Creating and Managing vSphere VM Storage Policies.
- Storage status. This category includes reporting about status of various storage entities. It also includes alarms and events for notifying about configuration changes.

  This type of information can help you troubleshoot storage connectivity and performance problems. It can also help you to correlate array-generated events and alarms to corresponding performance and load changes on the array.
- Storage DRS information for the distributed resource scheduling on block devices or file systems. This information helps to ensure that decisions made by Storage DRS are compatible with resource management decisions internal to the storage systems.

Storage Provider Requirements and Considerations

When you use third-party storage provider, vendors are responsible for supplying storage providers. The VASA program defines an architecture that integrates third-party storage providers into the vSphere environment, so that vCenter and ESX hosts can communicate with the storage providers.

To use storage providers, follow these requirements:

- Make sure that every storage provider you use is certified by VMware and properly deployed. For information about deploying the storage providers, contact your storage vendor.
- Make sure that the storage provider is compatible with the vCenter and ESX versions. See the VMware by Broadcom Compatibility Guide.
- Do not install the VASA provider on the same system as vCenter.
- If your environment contains older versions of storage providers, existing functionality continues to work. However, to use new features, upgrade your storage provider to a new version.

Managing Storage Providers

You might need to register certain types of storage providers to establish their connection to ESX. You can then review and manage the storage providers.

Register Storage Providers

Use the vSphere Client to register a separate storage provider for each host in a cluster.

Verify that the storage provider component is installed on the storage side and obtain its credentials from your storage administrator.

After registration, vCenter can detect and use the functionality of the latest VASA version.

> **Note:**
> Built-in storage providers do not require registration. For example, the storage providers that support vSAN or I/O filters are build-in. They become registered and appear on the list of storage providers automatically. For more information, see the *Administering VMware vSAN* documentation and Filtering Virtual Machine I/O in vSphere.

1. Navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.
3. Click the **Add** icon.
4. Enter connection information for the storage provider, including the name, URL, and credentials.
5. Click **OK**.

vCenter registers the storage provider and establishes a secure connection with it.

View Storage Provider Information

After you register a storage provider component with vCenter, the storage provider appears on the storage providers list. Certain storage providers are self-registered and automatically appear on the list after you set up the entity they represent, for example, vSAN or I/O filters.

Use the vSphere Client to view general storage provider information and details for each storage component.

1. Navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.
3. In the Storage Providers list, view the storage providers registered with vCenter.

   The list shows general information including the name of the storage provider, its URL and status, and version of VASA APIs.
4. To display additional details, click the icon next to the provider's name.

   > **Note:**
   > A single storage provider can support storage systems from multiple different vendors.

   You can display information about provider's certificates and storage entities it represents.

Re-authenticate Host VASA Client

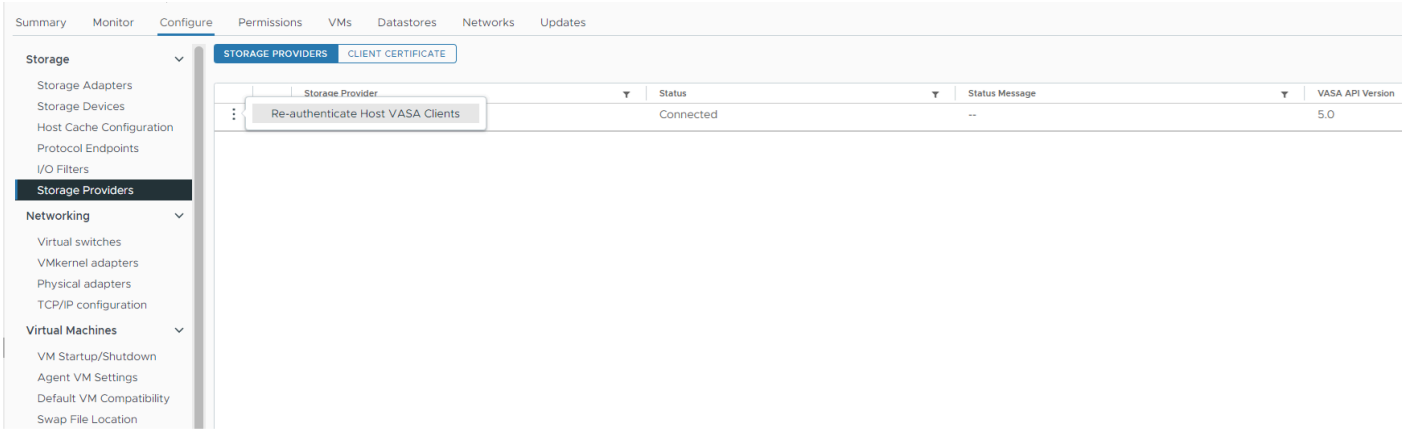
You can monitor the storage provider connection and authentication status on each ESX host. If there is an authentication issue in a specific host, you can re-authenticate the specific host against the storage provider.

Register storage providers. See Register Storage Providers for Virtual Volumes.

Use the vSphere Client to monitor the storage providers.

1. Navigate to the ESX host in vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.

   The **Storage Providers** tab lists the storage provider registered with the ESX host. You can view general information including the name of the storage provider, status, and version of the VASA API. The **Client Certificate** lists the VASA client certificate details.
3. If the status indicates an authentication error for a provider, click the row action (three dots) icon next to the provider's name and click **Re-authenticate Host VASA Clients**.



   The VASA client re-authorizes the ESX host with the storage provider.

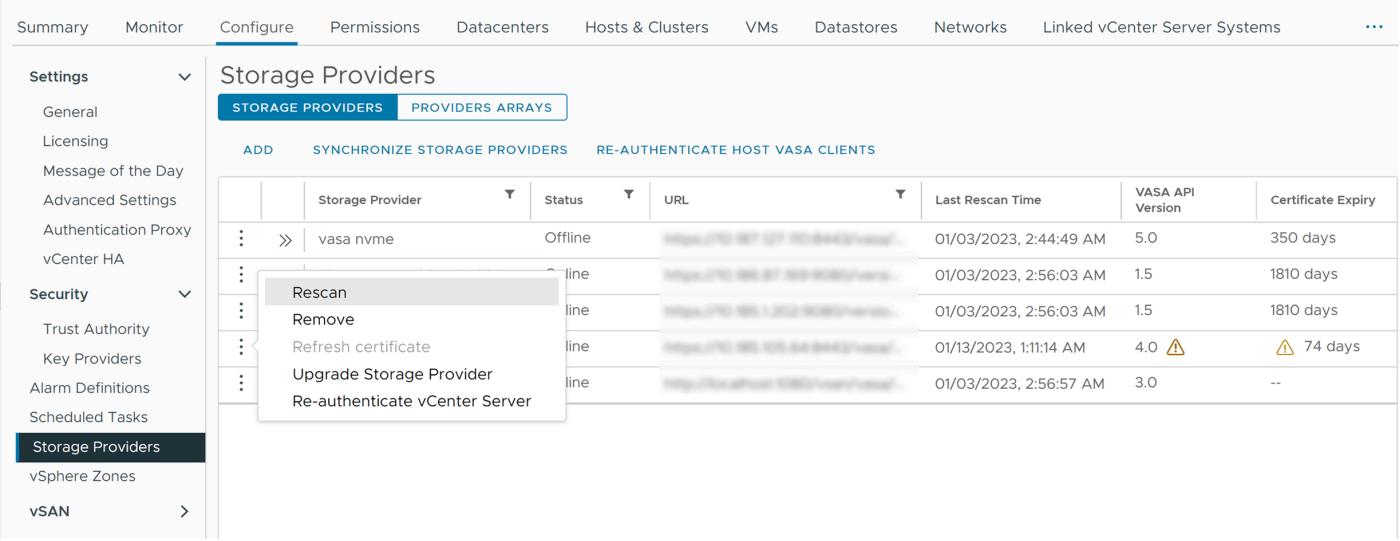
Manage Storage Providers

Use the vSphere Client to perform several management operations on the registered storage providers.

1. Navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.
3. Click the icon to perform one of the following global actions for all providers.

Icon	Description
Synchronize Storage Providers	Synchronize all storage providers with the current state of the environment.
Re-authenticate Host VASA Clients	Re-authenticate all hosts with Virtual Volumes datastores against their respective storage providers. This icon is available only for Virtual Volumes providers. **Note:** This operation is time consuming.

4. To manage a specific storage provider, navigate to the provider and click the **Row action** (three dots) icon next to the provider's name.



5. Select one of the following from the **Row actions** menu.

   Availability of the menu options depends on the type of the storage provider you use. For example, you cannot manually unregister or upgrade certain storage providers supplied by VMware, such as vSAN or I/O filter storage providers.

Option	Description
Rescan	Update storage data for the provider. vCenterperiodically updates storage data in its database. The updates are partial and reflect only those changes that storage providers communicate to vCenter at that moment. When needed, you can perform a full database synchronization for the selected storage provider.
Remove	Unregister storage providers that you do not use. After this operation, vCenter closes the connection and removes the storage provider from its configuration.
Refresh certificate	vCenter warns you when a certificate assigned to a storage provider is about to expire. You can refresh the certificate to continue using the provider. If you fail to refresh the certificate before it expires, vCenter discontinues using the provider.
Upgrade Storage Provider	When vCenter notifies you about a new available VASA version, accept this version to enable it in your vSphere environment. vCenter activates new capabilities exposed by the storage provider and communicates this information to all hosts in the inventory. When you perform this action, you must enter credentials of the storage provider. You cannot select specific VASA version for an upgrade if several versions are available. vCenterupgrades only to the latest version. **Note:** You have the flexibility to upgrade at your own pace. For more information, see vSphere Virtual Volumes Security.
Re-authenticate vCenter	Re-authenticate vCenter against the storage provider. This action re-authorizes vCenter with the storage provider if it has lost authentication and would need credentials. You must enter credentials of the storage provider.

Working with vSphere Virtual Volumes

vSphere Virtual Volumes, also known as vVols, virtualizes storage devices by abstracting physical hardware resources into logical pools of capacity. The Virtual Volumes functionality changes the storage management paradigm from managing space inside datastores to managing abstract storage objects handled by storage arrays. With Virtual Volumes, an individual virtual machine, not the datastore, becomes a unit of storage management, while storage hardware gains complete control over virtual disk content, layout, and management.

Important:

Starting with VMware Cloud Foundation 9.0 and VMware vSphere Foundation 9.0, the Virtual Volumes capability (vVols) is deprecated and will be removed in a future major release of both products. Support for Virtual Volumes continues for critical bug fixes only in vSphere 8.x, VCF and VVF 5.x, and other supported versions until the end of support for each respective release. For more information, see Broadcom knowledge base article 401070.

Virtual Volumes maps virtual disks and their derivatives, clones, snapshots, and replicas, directly to objects, called virtual volumes, on a storage system. This mapping allows vSphere to offload intensive storage operations such as snapshot, cloning, and replication to the storage system.

vSphere Virtual Volumes Concepts and Architecture

Learn about basic concepts and different components of the vSphere Virtual Volumes functionality.

Watch the following video to learn about Virtual Volumes concepts.

Virtual Volume Objects

Virtual volumes are encapsulations of virtual machine files, virtual disks, and their derivatives.

Virtual volumes are stored natively inside a storage system that is connected to your ESX hosts through Ethernet or SAN. They are exported as objects by a storage system and are managed entirely by hardware on the storage side. Typically, a unique GUID identifies a virtual volume. Virtual volumes are not preprovisioned, but created automatically when you perform virtual machine management operations. These operations include a VM creation, cloning, and snapshotting. ESX and vCenter associate one or more virtual volumes to a virtual machine.

Types of Virtual Volumes

The system creates the following types of virtual volumes for the core elements that make up the virtual machine:

Data-vVol	A data virtual volume that corresponds directly to each virtual disk `.vmdk` file. As virtual disk files on traditional datastores, virtual volumes are presented to virtual machines as SCSI or NVMe disks. Data-vVol can be either thick or thin-provisioned.
Config-vVol	A configuration virtual volume, or a home directory, represents a small directory that contains metadata files for a virtual machine. The files include a `.vmx` file, descriptor files for virtual disks, log files, and so forth. The configuration virtual volume is formatted with a file system. When ESX uses the SCSI or NVMe protocol to connect to storage, configuration virtual volumes are formatted with VMFS. With NFS protocol, configuration virtual volumes are presented as an NFS directory. Typically, it is thin-provisioned. Starting with vSphere 7.0 Update 2, partners can increase the config-vVol to above 4 GB. Work with your Virtual Volumes partner on implementing this if it is supported by your partner and applicable to your environment. vSphere 8.0 Update 2 and later, supports space reclamation for config-vVols that reside on Virtual Volumes datastores accessed through the SCSI or NVMe protocols. For more information, see Reclaim Space on the vSphere Virtual Volumes Datastores
Swap-vVol	Created when a VM is first powered on. It is a virtual volume to hold copies of VM memory pages that cannot be retained in memory. Its size is determined by the VM's memory size. It is thick-provisioned by default.
Snapshot-vVol	A virtual memory volume to hold the contents of virtual machine memory for a snapshot. Thick-provisioned. For more information, see vSphere Virtual Volumes Snapshots.
Other	A virtual volume for specific features. For example, a digest virtual volume is created for Content-Based Read Cache (CBRC).

Typically, a VM creates a minimum of three virtual volumes, data-vVol, config-vVol, and swap-vVol. The maximum depends on how many virtual disks and snapshots reside on the VM.

By using different virtual volumes for different VM components, you can apply and manipulate storage policies at the finest granularity level. For example, a virtual volume that contains a virtual disk can have a richer set of services than the virtual volume for the VM boot disk.

Disk Provisioning

The Virtual Volumes functionality supports a concept of thin and thick-provisioned virtual disks. However, from the I/O prospective, implementation and management of thin or thick provisioning by the arrays is transparent to the ESX host. ESX offloads to the storage arrays any functions related to thin provisioning.

You select the thin or thick type for your virtual disk at the VM creation time. If your disk is thin and resides on a Virtual Volumes datastore, you cannot change its type later by inflating the disk.

Shared Disks

You can place a shared disk on a Virtual Volumes storage that supports SCSI Persistent Reservations for Virtual Volumes. You can use this disk as a quorum disk and eliminate RDMs in the MSCS clusters. For more information, see the *vSphere Resource Management* documentation.

Virtual Volumes Storage Providers

A Virtual Volumes storage provider, also called a VASA provider, is a software component that acts as a storage awareness service for vSphere. The provider mediates out-of-band communication between vCenter and ESX hosts on one side and a storage system on the other.

The storage provider is implemented through VMware APIs for Storage Awareness (VASA) and is used to manage all aspects of Virtual Volumes storage.

The storage provider delivers information from the underlying storage container. The storage container capabilities appear in vCenter and the vSphere Client. Then, in turn, the storage provider communicates virtual machine storage requirements, which you can define in the form of a storage policy, to the storage layer. This integration process ensures that a virtual volume created in the storage layer meets the requirements outlined in the policy.

Typically, vendors are responsible for supplying storage providers that can integrate with vSphere and provide support to Virtual Volumes. Every storage provider must be certified by VMware and properly deployed. For information about deploying and upgrading the Virtual Volumes storage provider to a version compatible with current ESX release, contact your storage vendor.

After you deploy the storage provider, you must register it in vCenter. See Register Storage Providers for Virtual Volumes. To upgrade your storage providers or for other operations that you can perform, see Manage Storage Providers for vSphere Virtual Volumes.

Virtual Volumes Storage Containers

Unlike traditional block or file based storage, the Virtual Volumes functionality does not require preconfigured storage on a storage side. Instead, Virtual Volumes uses a storage container. It is a pool of raw storage capacity or an aggregation of storage capabilities that a storage system can provide to virtual volumes.

A storage container is a part of the logical storage fabric and is a logical unit of the underlying hardware. The storage container logically groups virtual volumes based on management and administrative needs. For example, the storage container can contain all virtual volumes created for a tenant in a multitenant deployment, or a department in an enterprise deployment. Each storage container serves as a virtual volume store and virtual volumes are allocated out of the storage container capacity.

Typically, a storage administrator on the storage side defines storage containers. The number of storage containers, their capacity, and their size depend on a vendor-specific implementation. At least one container for each storage system is required.

Note:
A single storage container cannot span different physical arrays.

After you register a storage provider associated with the storage system, vCenter discovers all configured storage containers along with their storage capability profiles, protocol endpoints, and other attributes. A single storage container can export multiple capability profiles. As a result, virtual machines with diverse needs and different storage policy settings can be a part of the same storage container.

Initially, all discovered storage containers are not connected to any specific host, and you cannot see them in the vSphere Client. To mount a storage container, you must map it to a Virtual Volumes datastore.

Static Protocol Endpoints

With SCSI or NFS transports, ESX hosts use a logical I/O proxy, called the protocol endpoint, to communicate with virtual volumes. ESX uses protocol endpoints to establish a data path on demand from virtual machines to their respective virtual volumes.

Note:
Information in this section applies only to static protocol endpoints that use the SCSI or NFS transports. For specifics about the NVMe protocol endpoints, see NVMe and Virtual Volumes in vSphere.

Each virtual volume is bound to a specific protocol endpoint. When a virtual machine on the host performs an I/O operation, the protocol endpoint directs the I/O to the appropriate virtual volume. Typically, a storage system requires just a few protocol endpoints. A single protocol endpoint can connect to hundreds or thousands of virtual volumes.

On the storage side, a storage administrator configures protocol endpoints, one or several per storage container. The protocol endpoints are a part of the physical storage fabric. The storage system exports the protocol endpoints with associated storage containers through the storage provider. After you map the storage container to a Virtual Volumes datastore, the ESX host discovers the protocol endpoints and they become visible in the vSphere Client. The protocol endpoints can also be discovered during a storage rescan. Multiple hosts can discover and mount the protocol endpoints.

In the vSphere Client, the list of available protocol endpoints looks similar to the host storage devices list. Different storage transports can be used to expose the protocol endpoints to ESX. When the SCSI-based transport is used, the protocol endpoint represents a proxy LUN defined by a T10-based LUN WWN. For the NFS protocol, the protocol endpoint is a mount point, such as an IP address and a share name. You can configure multipathing on the SCSI-based protocol endpoint, but not on the NFS-based protocol endpoint. No matter which protocol you use, the storage array can provide multiple protocol endpoints for availability purposes.

Protocol endpoints are managed per array. ESX and vCenter assume that all protocol endpoints reported for an array are associated with all containers on that array. For example, if an array has two containers and three protocol endpoints, ESX assumes that virtual volumes on both containers can be bound to all three protocol points.

For information about viewing the static protocol endpoints in the vSphere Client, see Review and Manage Static Protocol Endpoints.

Binding and Unbinding Virtual Volumes

At the time of creation, a virtual volume is a passive entity and is not immediately ready for I/O. To access the virtual volume, ESX or vCenter send a bind request.

The storage system replies with a protocol endpoint ID that becomes an access point to the virtual volume. The protocol endpoint accepts all I/O requests to the virtual volume. This binding exists until ESX sends an unbind request for the virtual volume.

For later bind requests on the same virtual volume, the storage system can return different protocol endpoint IDs.

When you use the NVMe protocol, the bind virtual volume response provides the NVMe subsytem NQN and the namespace ID (nsid) of the namespace virtual volume object. The ESX host uses this information and resolves this to the ANA group within the subsystem. Corresponding to this ANA group, a virtual protocol endpoint (vPE) is created if it does not exist. It is used to direct all I/O requests to Virtual Volumes.

When receiving concurrent bind requests to a virtual volume from multiple ESX hosts, the storage system can return the same or different endpoint bindings to each requesting ESX host. In other words, the storage system can bind different concurrent hosts to the same virtual volume through different endpoints.

The unbind operation removes the I/O access point for the virtual volume. The storage system might unbind the virtual volume from its protocol endpoint immediately, or after a delay, or take some other action. A bound virtual volume cannot be deleted until it is unbound.

Virtual Volumes Datastores

A Virtual Volumes datastore represents a storage container in vCenter and the vSphere Client.

After vCenter discovers storage containers exported by storage systems, you must mount them as Virtual Volumes datastores. The Virtual Volumes datastores are not formatted in a traditional way like, for example, VMFS datastores. You must still create them because all vSphere functionalities, including FT, HA, DRS, and so on, require the datastore construct to function properly.

You use the datastore creation wizard in the vSphere Client to map a storage container to a Virtual Volumes datastore. The Virtual Volumes datastore that you create corresponds directly to the specific storage container.

From a vSphere administrator prospective, the Virtual Volumes datastore is similar to any other datastore and is used to hold virtual machines. Like other datastores, the Virtual Volumes datastore can be browsed and lists virtual volumes by virtual machine name. Like traditional datastores, the Virtual Volumes datastore supports unmounting and mounting. However, such operations as upgrade and resize are not applicable to the Virtual Volumes datastore. The Virtual Volumes datastore capacity is configurable by the storage administrator outside of vSphere.

You can use the Virtual Volumes datastores with traditional VMFS and NFS datastores and with vSAN.

> **Note:**
> The size of a virtual volume must be a multiple of 1 MB, with a minimum size of 1 MB. As a result, all virtual disks that you provision on a Virtual Volumes datastore must be an even multiple of 1 MB. If the virtual disk you migrate to the Virtual Volumes datastore is not an even multiple of 1 MB, extend the disk to the nearest even multiple of 1 MB.

To create a Virtual Volumes datastore, see Create a Virtual Volumes Datastore in vSphere Environment.

Virtual Volumes and VM Storage Policies

A virtual machine that runs on a Virtual Volumes datastore requires a VM storage policy.

A VM storage policy is a set of rules that contains placement and quality-of-service requirements for a virtual machine. The policy enforces appropriate placement of the virtual machine within Virtual Volumes storage and guarantees that storage can satisfy virtual machine requirements.

You use the VM Storage Policies interface to create a Virtual Volumes storage policy. When you assign the new policy to the virtual machine, the policy enforces that the Virtual Volumes storage meets the requirements.

Virtual Volumes Default Storage Policy

For Virtual Volumes, VMware provides a default storage policy that contains no rules or storage requirements, called Virtual Volumes No Requirements Policy. This policy is applied to the VM objects when you do not specify another policy for the virtual machine on the Virtual Volumes datastore. With the No Requirements policy, storage arrays can determine the optimum placement for the VM objects.

The default No Requirements policy that VMware provides has the following characteristics:

- You cannot delete, edit, or clone this policy.
- The policy is compatible only with the Virtual Volumes datastores.
- You can create a VM storage policy for Virtual Volumes and designate it as the default.

Virtual Volumes and Storage Protocols

A Virtual Volumes storage system provides protocol endpoints that are discoverable on the physical storage fabric. ESX hosts use the protocol endpoints to connect to virtual volumes on the storage. Operation of the protocol endpoints depends on storage protocols that expose the endpoints to ESX hosts.

Virtual Volumes supports NFS version 3 and 4.1, iSCSI, Fibre Channel, FCoE, NVMe over Fibre Channel, and NVMe over TCP.

Irrespective of which storage protocol is used, protocol endpoints provide uniform access to both SAN and NAS storage. A virtual volume, like a file on other traditional datastore, is presented to a virtual machine as a SCSI or an NVMe disk.

> **Note:**
>
> A storage container is dedicated to SAN (SCSI or NVMe) or NAS and cannot be shared across those protocol types. An array can present one storage container with SCSI protocol endpoints and a different container with NFS protocol endpoints. The container cannot use a combination of SCSI, NVMe, and NFS storage access protocols.

Virtual Volumes and SCSI-Based Transports

On disk arrays, Virtual Volumes supports Fibre Channel, FCoE, and iSCSI protocols.

When the SCSI-based protocol is used, the protocol endpoint represents a proxy LUN defined by a T10-based LUN WWN.

As any block-based LUNs, the protocol endpoints are discovered using standard LUN discovery commands. The ESX host periodically rescans for new devices and asynchronously discovers block-based protocol endpoints. The protocol endpoint can be accessible by multiple paths. Traffic on these paths follows well-known path selection policies, as is typical for LUNs.

On SCSI-based disk arrays at VM creation time, ESX makes a virtual volume and formats it as VMFS. This small virtual volume stores all VM metadata files and is called the config-vVol. The config-vVol functions as a VM storage locator for vSphere.

Virtual Volumes on disk arrays supports the same set of SCSI commands as VMFS and use ATS as a locking mechanism.

CHAP Support for iSCSI Endpoints

Virtual Volumes supports Challenge Handshake Access Protocol (CHAP) with iSCSI targets. This support allows ESX hosts to share CHAP initiator credentials with Virtual Volumes storage providers, also called VASA providers, and for Virtual Volumes storage providers to raise system events notifying vCenter of changes to CHAP target credentials on the storage array.

Each ESX host can have multiple HBAs and each HBA can have properties configured on it. One of these properties is the authentication method that the HBA must use. Authentication is optional, but if implemented, it must be supported by both the initiator and target. CHAP is an authentication method that can be used both directions between initiator and target.

For more information about different CHAP authentication methods, see Selecting CHAP Authentication Method. To configure CHAP on your ESX host, see Configuring CHAP Parameters for iSCSI or iSER Storage Adapters on ESX Host.

Virtual Volumes and NFS Transports

With NAS storage, a protocol endpoint is an NFS share that the ESX host mounts using IP address or DNS name and a share name. Virtual Volumes supports NFS version 3 and 4.1 to access NAS storage. Both IPv4 and IPv6 formats are supported.

No matter which version you use, a storage array can provide multiple protocol endpoints for availability purposes.

In addition, NFS version 4.1 introduces trunking mechanisms that enable load balancing and multipathing.

Virtual Volumes on NAS devices supports the same NFS Remote Procedure Calls (RPCs) that ESX hosts use when connecting to NFS mount points.

On NAS devices, a config-vVol is a directory subtree that corresponds to a config-vVolID. The config-vVol must support directories and other operations that are necessary for NFS.

Virtual Volumes and NVMe

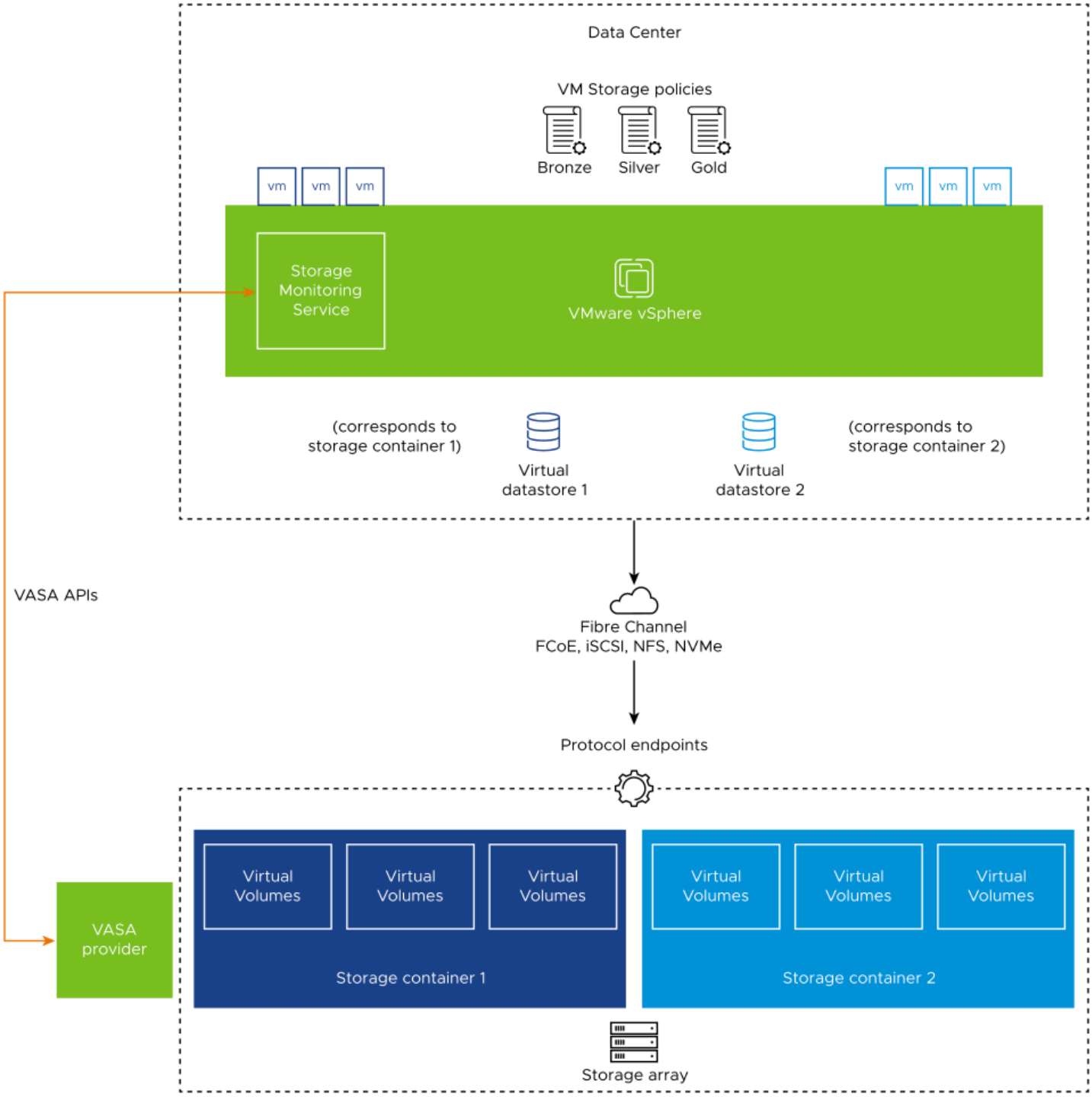
Virtual Volumes supports NVMe protocols, including NVMe over Fibre Channel and NVMe over TCP. A virtual volume object is mapped to a namespace within an NVMe subsystem. ANA groups within the NVMe subsystem are viewed as virtual protocol endpoints on the ESX host.

Virtual protocol endpoints are used for path state management as the ANA group state changes. The ESX host discovers the ANA groups dynamically, as and when required. This means that the virtual protocol endpoint is created only when the ESX host needs I/O access to a namespace virtual volume within the NVMe subsystem. Config-vVols on NVMe are similar to SCSI that are formatted with VMFS. They are also used to store the VM metadata files.

To configure NVMe with Virtual Volumes on your ESX host, see NVMe and Virtual Volumes in vSphere.

Virtual Volumes Architecture

An architectural diagram provides an overview of how all components of the Virtual Volumes functionality interact with each other.



Watch the video for information about Virtual Volumes architecture.

vSphere Virtual Volumes Security

Learn about security improvements that vSphere Virtual Volumes storage provider, also called VASA provider, version 5 offers in vSphere 8.0 Update 1 and later releases. You can also learn about the security model in vSphere 8.0 and earlier releases.

VASA 5 Support and Security with vSphere 8.0 Update 1 and Later

All storage providers of VASA 5 and later versions use a stricter authentication mechanism, which requires ESX to be authenticated in context of vCenter. VASA 5 improves security and offers a significantly modified management model that includes the following major changes:

- For each vCenter that registers with the array using VASA 5 or later, the VASA provider creates a dedicated web server instance, or a virtual host, which can be either a virtual web server instance or a completely separate instance. The VASA client in vCenter relies on certificate based authentication and authorization to access its dedicated virtual host created on the array. All VASA client certificates, including vCenter and ESX certificates, get registered with the virtual host. This creates strong isolation between different vCenter systems when the systems are authenticated. In addition, VASA providers can offer separate resource access and improved isolation to different vCenter systems.
- With VASA 5, the VASA client uses a dedicated certificate for VASA communications. Each vCenter provisions a certificate for the VASA provider, which is managed through a dedicated virtual host specific to vCenter. All ESX hosts that support VASA 5 use the dedicated virtual host created by their managing vCenter.
- vCenter provisions the VASA client certificate for each new ESX host 8.0 Update 1 or later and synchronizes the public key of the certificate with the VASA provider. In contrast with the previous security model that authenticated the CA issuer for the client certificate, the VASA provider now identifies and authorizes an individual client using the public key.
- To comply with VMware security requirements, vSphere does not trust self-signed certificates for TLS communications. The only exception is during a short period when the VASA provider gets registered and for backward compatibility. An array administrator can use a Custom CA certificate for the VASA provider to override the self-signed certificate at array for backward compatibility and bootstrapping.
- To avoid losing access to the VASA provider, follow these guidelines. For information, see Manage Storage Providers for vSphere Virtual Volumes.
  - Do not unregister and re-register your VASA provider to upgrade. Instead, use a proper upgrade mechanism for your VASA provider. When vCenter notifies you about a new available VASA version, make sure to accept this version within a reasonable time. To upgrade from the vSphere Client, use the **Upgrade Storage Provider** option.
  - Regularly refresh the VASA provider certificate. Make sure to refresh the certificate within a reasonable time after vCenter warns you that the certificate assigned to the VASA provider is about to expire. Use the vSphere Client **Refresh certificate** option.
  - When you renew the VMCA root certificate or vCenter client certificate, SMS might lose connection with the VASA provider. If the provider is offline, use the **Re-authenticate vCenter** option.
  - If a host loses authentication, you can remediate the authentication failure by using the **Re-authenticate Host VASA Clients** option.

Security Certificates with vSphere 8.0 and Earlier Releases

vSphere includes the VMware Certificate Authority (VMCA). By default, the VMCA creates all internal certificates used in vSphere environment. It generates certificates for newly added ESX hosts and storage VASA providers that manage or represent Virtual Volumes storage systems.

Communication with the VASA provider is protected by SSL certificates. These certificates can come from the VASA provider or from the VMCA.

- Certificates can be directly provided by the VASA provider for long-term use. They can be either self-generated and self-signed, or derived from an external Certificate Authority.
- Certificates can be generated by the VMCA for use by the VASA provider.

When a host or VASA provider is registered, VMCA follows these steps automatically, without involvement from the vSphere administrator.

1. When a VASA provider is first added to the vCenter storage management service (SMS), it produces a self-signed certificate.
2. After verifying the certificate, the SMS requests a Certificate Signing Request (CSR) from the VASA provider.
3. After receiving and validating the CSR, the SMS presents it to the VMCA on behalf of the VASA provider, requesting a CA signed certificate.

   The VMCA can be configured to function as a standalone CA, or as a subordinate to an enterprise CA. If you set up the VMCA as a subordinate CA, the VMCA signs the CSR with the full chain.

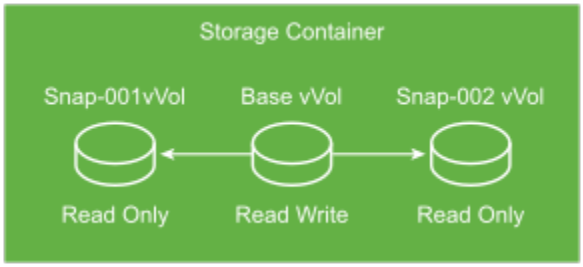
4. The signed certificate with the root certificates is passed to the VASA provider. The VASA provider can authenticate all future secure connections originating from the SMS on vCenter and on ESX hosts.

vSphere Virtual Volumes Snapshots

In vSphere, snapshots preserve the state and data of a virtual machine at the time you take the snapshot. Snapshots are useful when you must revert repeatedly to the same virtual machine state, but you do not want to create multiple virtual machines. Virtual Volumes snapshots serve many purposes. You can use them to create a quiesced copy for backup or archival purposes, or to create a test and rollback environment for applications. You can also use them to provision application images instantly.

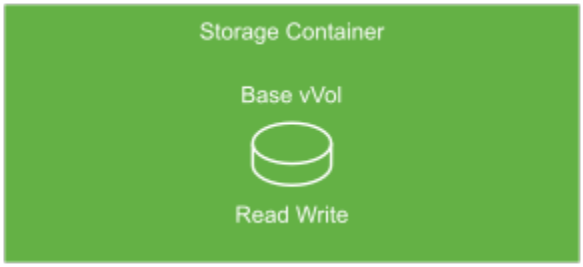
In Virtual Volumes environment, snapshots are managed by ESX and vCenter, but are performed by the storage array.

Each snapshot creates an extra virtual volume object, a snapshot virtual volume, that holds the contents of virtual machine memory. Original VM data is copied to this object, and it remains read-only, which prevents the guest operating system from writing to snapshot. You cannot resize the snapshot virtual volume. Typically, when you replicate the VM, its snapshot virtual volume is also replicated.



The base virtual volume remains active, or read-write. When another snapshot is created, it preserves the new state and data of the virtual machine at the time you take the snapshot.

When you delete snapshots, only the base virtual volume remains, while snapshot virtual volume objects are discarded. The base virtual volume represents the most current state of the virtual machine. Unlike snapshots on the traditional datastores, snapshot virtual volumes do not need to commit their contents to the base virtual volume.



For information about creating and managing snapshots, see the *vSphere Virtual Machine Administration* documentation.

vSphere Virtual Volumes and Shared Virtual Disks

Virtual Volumes support shared virtual disks that can be used with such clustering software as Windows Server Failover Clustering (WSFC). Starting with vSphere 8.0 Update 2, Virtual Volumes support the hot extension functionality for shared virtual disks on the Virtual Volumes datastore. With this functionality, you can increase the size of a shared disk without bringing the cluster down.

To learn more about the support Virtual Volumes provide to shared disks with WSFC, see the following information:

- VMware vSphere® Virtual Volumes Support for WSFC
- Virtual Volumes (vVols) now supports WSFC
- vVols and Microsoft WSFC

For information about setting up and extending the shared virtual disk on the Virtual Volumes datastore, see Add Hard Disks to the First Node of a vVols-Based WSFC Across Physical Hosts.

Before You Enable vSphere Virtual Volumes

To work with vSphere Virtual Volumes, ensure that your storage and vSphere environment are set up correctly.

Prepare Storage System for Virtual Volumes

To prepare your storage system environment for Virtual Volumes, follow these guidelines. For additional information, contact your storage vendor.

- The storage system or storage array that you use must support Virtual Volumes and integrate with the vSphere components through vSphere APIs for Storage Awareness (VASA). The storage array must support thin provisioning and snapshotting.
- The Virtual Volumes storage provider must be deployed.
- The following components must be configured on the storage side:
  - Protocol endpoints. Except NVMe.
  - Storage containers
  - Storage profiles
  - Replication configurations if you plan to use Virtual Volumes with replication. See Requirements for Replication with Virtual Volumes.
- If you use NVMe, create NVM subsystems and associate Virtual Volumes Host NQNs with relevant NVM subsystems. This configuration step depends on partner specific NVMe design. For more information, contact your array vendor.

Prepare vSphere Environment

- Ensure to follow appropriate setup guidelines for the type of storage you use, NVMe over Fibre Channel, NVMe over TCP, Fibre Channel, FCoE, iSCSI, or NFS. If necessary, install and configure storage adapters on your ESX hosts.
  - If you use iSCSI, activate the software iSCSI adapters on your ESX hosts. Configure Dynamic Discovery and enter the IP address of your Virtual Volumes storage system. See Configure the Software iSCSI Adapter with ESX.
  - If you use NVMe, follow the requirements for NVMe over Fibre Channel and NVMe over TCP. See Requirements for VMware NVMe Storage.
- Synchronize all components in the storage array with vCenter and all ESX hosts. Use Network Time Protocol (NTP) to do this synchronization.

For more information, contact your vendor and see the VMware by Broadcom Compatibility Guide.

Synchronize vSphere Storage Environment with a Network Time Server

If you use Virtual Volumes, configure Network Time Protocol (NTP) to make sure all ESX hosts on the vSphere network are synchronized.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, select **Time Configuration**.
4. Click **Edit** and set up the NTP server.
   a) Select **Use Network Time Protocol (Enable NTP client)**.
   b) Set the NTP Service Startup Policy.
   c) Enter the IP addresses of the NTP server to synchronize with.
   d) Click **Start** or **Restart** in the NTP Service Status section.
5. Click **OK**.

   The host synchronizes with the NTP server.

Configure vSphere Virtual Volumes

To configure Virtual Volumes in the vSphere environment, follow several steps.

- Be familiar with concepts and requirements related to Virtual Volumes in the vSphere environment. See vSphere Virtual Volumes Concepts and Architecture and NVMe and Virtual Volumes in vSphere.
- Follow guidelines in Before You Enable vSphere Virtual Volumes.

1. Register Storage Providers for Virtual Volumes.
2. Create a Virtual Volumes Datastore in vSphere Environment.

Register Storage Providers for Virtual Volumes

Your Virtual Volumes environment must include storage providers, also called VASA providers. Typically, third-party vendors develop storage providers through the VMware APIs for Storage Awareness (VASA). Storage providers facilitate communication between vSphere and the storage side. Use the vSphere Client to register the Virtual Volumes storage providers.

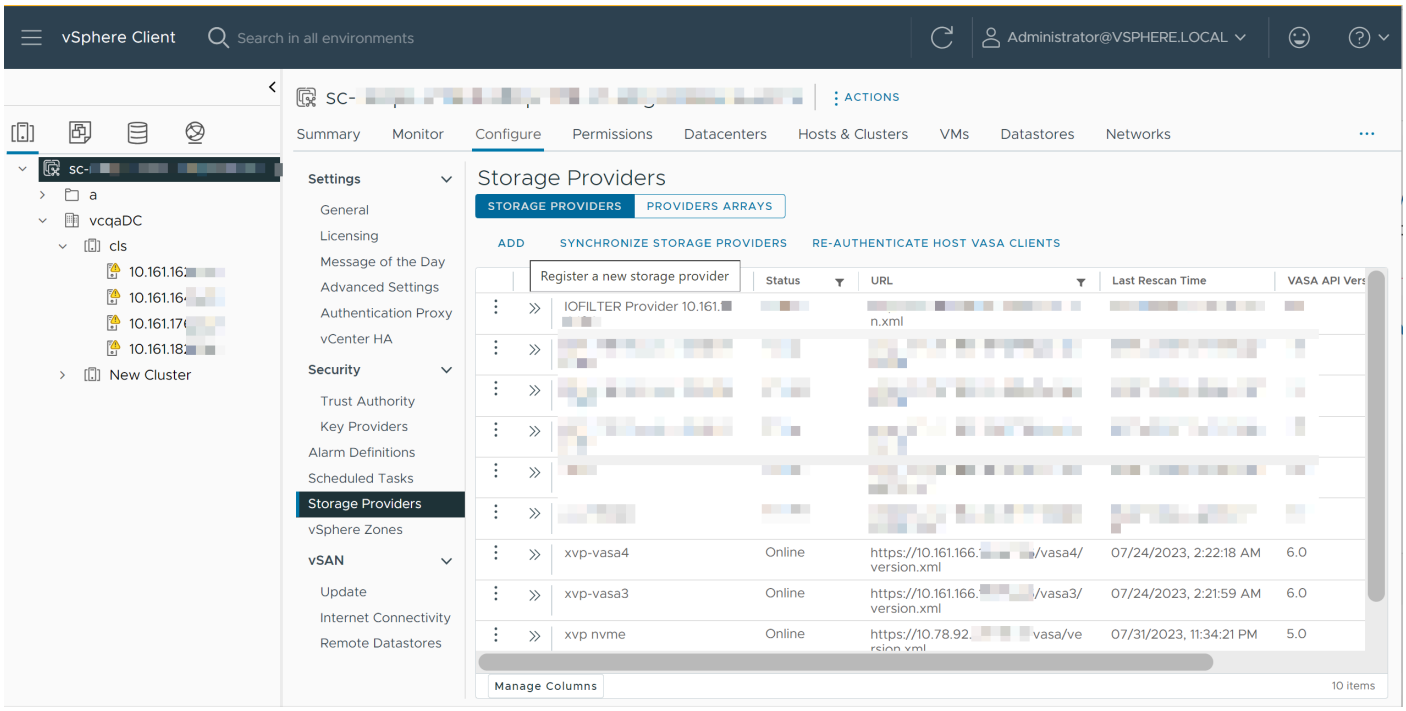
Verify that an appropriate version of the Virtual Volumes storage provider is installed on the storage side. Obtain credentials of the storage provider.

After registration, the Virtual Volumes provider communicates with vCenter. The provider reports characteristics of underlying storage and data services, such as replication, that the storage system provides. The characteristics appear in the VM Storage Policies interface and can be used to create a VM storage policy compatible with the Virtual Volumes datastore. After you apply this storage policy to a virtual machine, the policy is pushed to Virtual Volumes storage. The policy enforces optimal placement of the virtual machine within Virtual Volumes storage and guarantees that storage can satisfy virtual machine requirements. If your storage provides extra services, such as caching or replication, the policy enables these services for the virtual machine.

For more information about the storage providers, see Virtual Volumes Storage Providers.

1. Navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.
3. Click the **Add** icon.
4. Enter connection information for the storage provider, including the name, URL, and credentials.
5. To complete the registration, click **OK**.

   vCenter discovers and registers the Virtual Volumes storage provider.



6. Verify information available for your provider.
   Switch between **Storage Providers** and **Providers Arrays** to display information about your provider and the array it represents.

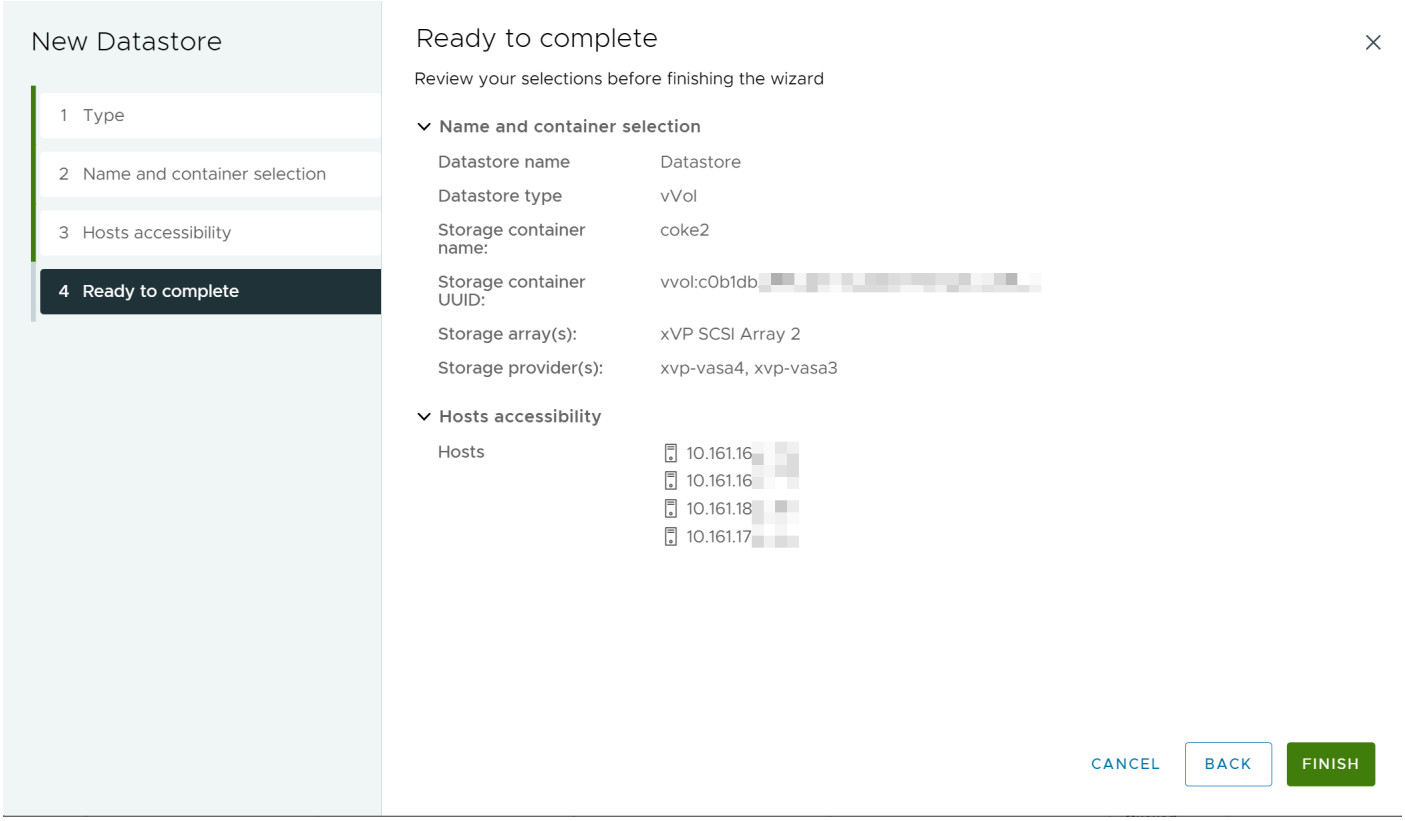
Create a Virtual Volumes Datastore in vSphere Environment

You use the **New Datastore** wizard to create a Virtual Volumes datastore.

For more information about the Virtual Volumes datastore, see Virtual Volumes Datastores.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. From the right-click menu, select **Storage** > **New Datastore**.
3. Select **vVol** as the datastore type.
4. Enter the datastore name and select a backing storage container from the list of storage containers.
   Make sure to use the name that does not duplicate another datastore name in your data center environment.

   If you mount the same Virtual Volumes datastore to several hosts, the name of the datastore must be consistent across all hosts.
5. Select the hosts that require access to the datastore.
6. Review the configuration options and click **Finish**.



After you create the Virtual Volumes datastore, you can perform the following operations:

- Rename the datastore, browse datastore files, unmount the datastore, and so on. See Basic Administrative Operations for vSphere Datastores.

  Note:
  You cannot add the Virtual Volumes datastore to a datastore cluster.
- Provision virtual machines on the Virtual Volumes datastore. See Provision VMs on vSphere Virtual Volumes Datastores and the *vSphere Virtual Machine Administration* documentation.
- Review and manage protocol endpoints. See Review and Manage Protocol Endpoints in vSphere.

- Manage or upgrade storage providers. See Manage Storage Providers for vSphere Virtual Volumes.

Provision VMs on vSphere Virtual Volumes Datastores

Follow this procedure to provision virtual machines on a vSphere Virtual Volumes datastore.

> **Note:**
> All virtual disks that you provision on a Virtual Volumes datastore must be an even multiple of 1 MB.

A virtual machine that runs on a Virtual Volumes datastore requires an appropriate VM storage policy.

After you provision the virtual machine, you can perform typical VM management tasks. For information, see the *vSphere Virtual Machine Administration* documentation.

1. Define a VM storage policy for Virtual Volumes.

   VMware provides a default No Requirements storage policy for Virtual Volumes. If you need, you can create a custom storage policy compatible with Virtual Volumes.

   See Create a VM Storage Policy for Virtual Volumes.

2. Assign the Virtual Volumes storage policy to the virtual machine.

   To guarantee that the Virtual Volumes datastore fulfills specific storage requirements when allocating a virtual machine, associate the Virtual Volumes storage policy with the virtual machine.

   See Assign Storage Policies to Virtual Machines.

3. Change default storage policy for a Virtual Volumes datastore.

   For virtual machines provisioned on Virtual Volumes datastores, VMware provides a default No Requirements policy. You cannot edit this policy, but you can designate a newly created policy as default.

   See Change the Default Storage Policy for a Datastore.

Review and Manage Protocol Endpoints in vSphere

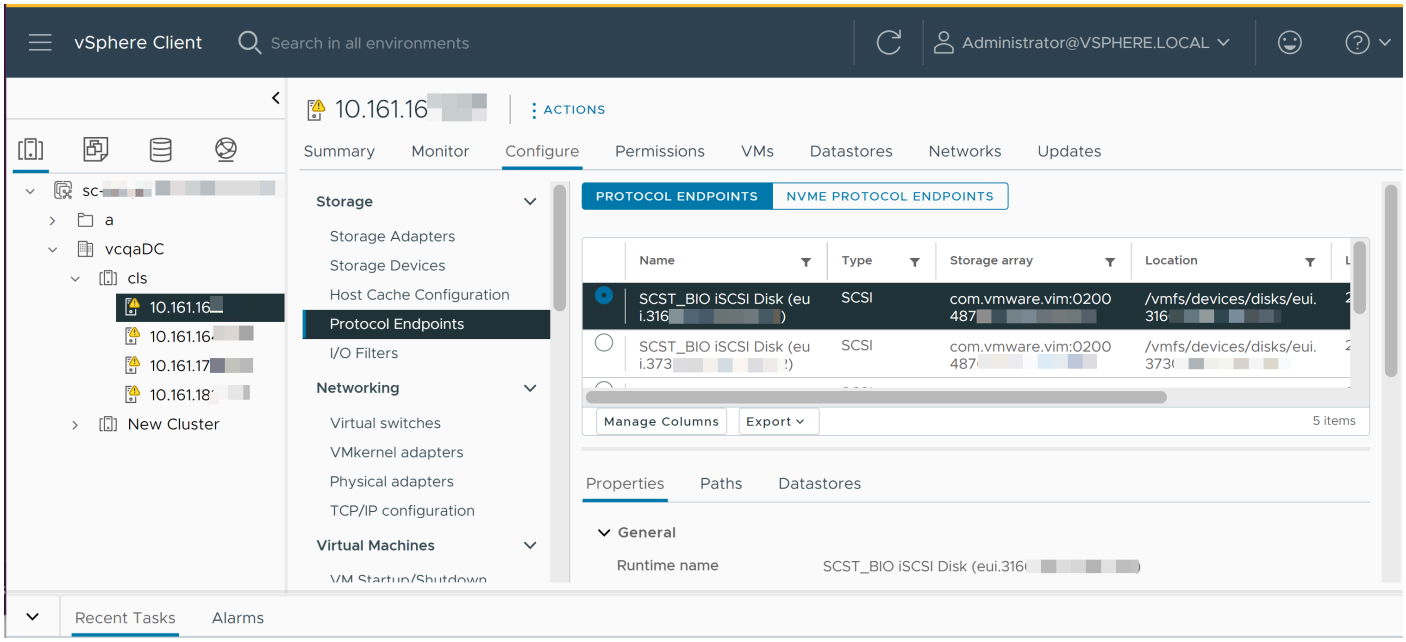
Learn how you can review and manage protocol endpoints that the ESX host uses to access Virtual Volumes on storage arrays. How the protocol endpoints are represented in the vSphere Client depends on whether the host uses the SCSI and NFS protocols or the NVMe protocol to access Virtual Volumes.

Review and Manage Static Protocol Endpoints

With the SCSI or NFS protocols, the protocol endpoints represent a proxy LUN or an NFS mount point. These protocol points are called static. After you map a storage container to a Virtual Volumes datastore, these protocol endpoints become visible in the vSphere Client. The list of available static protocol endpoints looks similar to the host storage devices list. You can review the properties of the protocol endpoints and modify specific settings.

For information about static protocol endpoints that use SCSI or NFS transports, see Static Protocol Endpoints.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Protocol Endpoints**.
4. Click the **Protocol Endpoints** tab.
5. To view details for a specific item, select this item from the list.



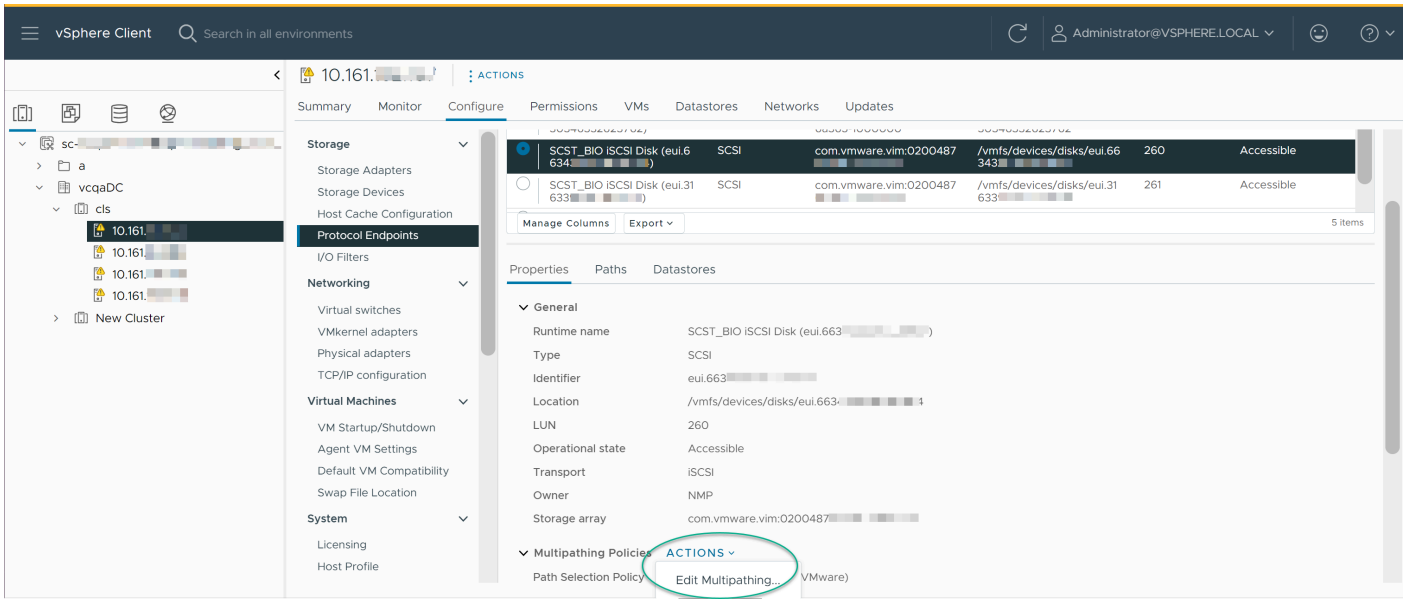
6. Use tabs to access additional information and modify properties for the selected protocol endpoint.

Tab	Description
Properties	View the item properties and characteristics. For SCSI (block) items, view and edit multipathing policies.
Paths (SCSI protocol endpoints only)	Display paths available for the protocol endpoint. Deactivate or activate a selected path. Change the Path Selection Policy.
Datastores	Display a corresponding Virtual Volumes datastore. Perform datastore management operations.

Change the Path Selection Policy for a SCSI Protocol Endpoint

If your ESX host uses SCSI-based transport to communicate with protocol endpoints representing a storage array, you can modify default multipathing policies assigned to protocol endpoints. Use the **Edit Multipathing Policies** dialog box to change a path selection policy.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Protocol Endpoints**.
4. Select the protocol endpoint whose paths you want to change and click the **Properties** tab.
5. Under Multipathing Policies, select **Edit Multipathing** from the **Actions** menu.



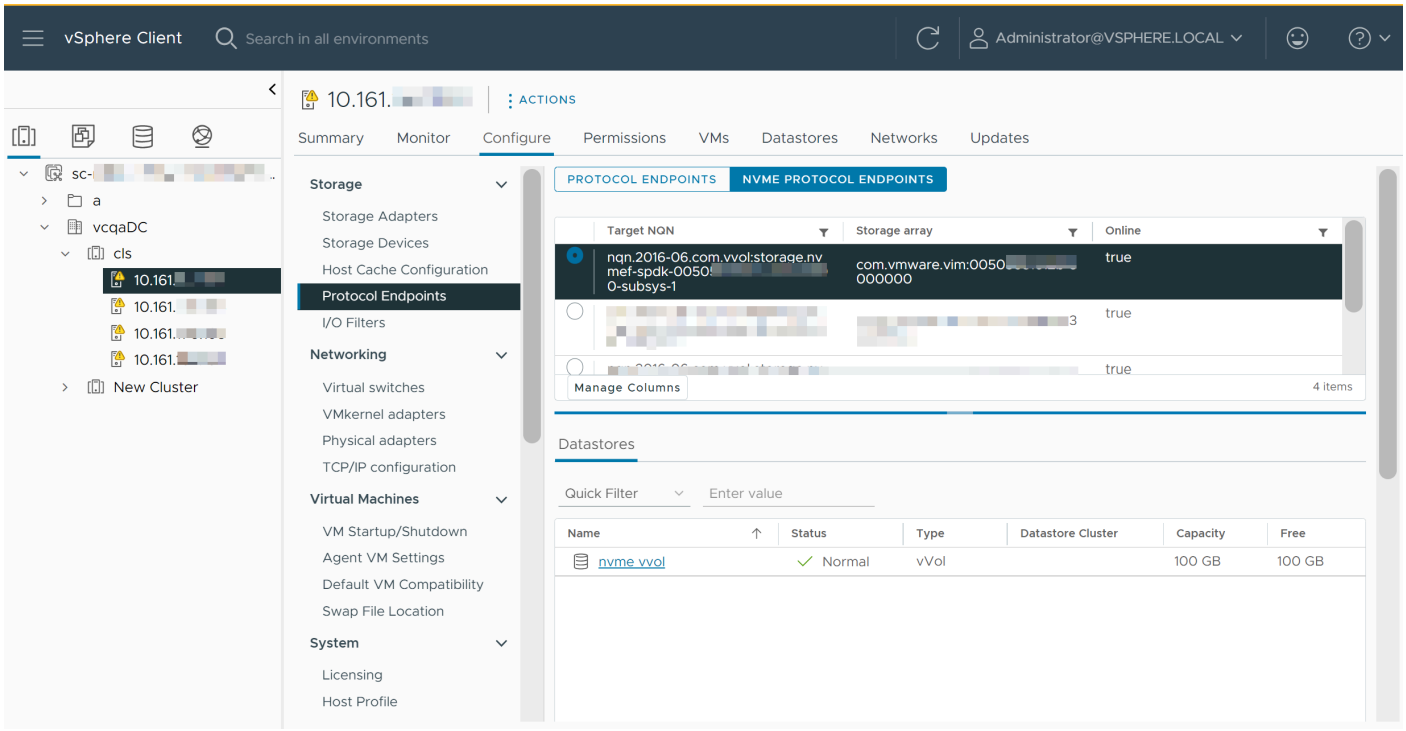
6. Select a path policy and configure its settings.
   For information about path policies for SCSI devices, see VMware Path Selection Plug-Ins and Policies.
7. To save your settings and exit the dialog box, click **OK**.

Review NVMe Protocol Endpoints

Unlike the static protocol endpoint, the NVMe protocol endpoint is virtual. This protocol endpoint is a logical representation of the ANA group where the virtual volumes reside. Until a VM is powered on, the NVMe protocol point doesn't exist. Once the VM is powered on, the NVMe protocol point is created, so that the host can access the virtual volumes in the ANA group. Use the vSphere Client to review NVMe virtual protocol endpoints.

For conceptual information that applies to Virtual Volumes with NVMe storage, see NVMe and Virtual Volumes in vSphere.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **Storage**, click **Protocol Endpoints**.
4. Click the **NVMe Protocol Endpoints** tab and review the list of available NVMe protocol endpoints and their information.
   You can see such details as target NVMe subsystems that are tracked for the NVMe Virtual Volumes datastore.



5. To display additional details, select a protocol endpoint from the list.
   You can display a Virtual Volumes datastore that corresponds to the specific protocol endpoint.

Manage Storage Providers for vSphere Virtual Volumes

After you register storage providers for Virtual Volumes, use the vSphere Client to manage the providers.

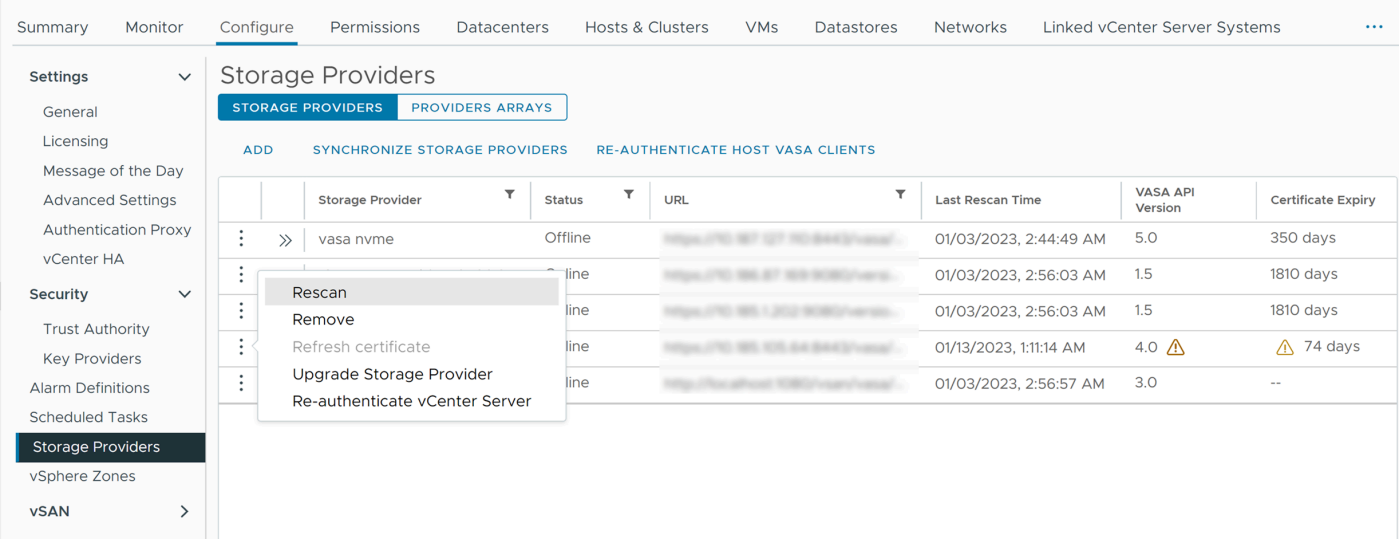
Register storage providers. See Register Storage Providers for Virtual Volumes.

For more information about the storage providers, see Virtual Volumes Storage Providers.

1. Navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.
3. Click the icon to perform one of the following global actions for all Virtual Volumes providers.

Icon	Description
Synchronize Storage Providers	Synchronize all storage providers with the current state of the environment.
Re-authenticate Host VASA Clients	Re-authenticate all hosts against their respective storage providers. Applies to ESX hosts version 8.0 Update 1 or later. The hosts must support VASA 5. **Note:** This operation is time consuming. To re-authenticate only specific hosts, see Re-authenticate Host VASA Client.

4. To manage a specific storage provider, navigate to the provider and click the **Row action** (three dots) icon next to the provider's name.



5. Select one of the following from the **Row actions** menu.

Option	Description
Rescan	Update storage data for the provider. vCenterperiodically updates storage data in its database. The updates are partial and reflect only those changes that storage providers communicate to vCenter at that moment. When needed, you can perform a full database synchronization for the selected storage provider.
Remove	Unregister storage providers that you do not use. After this operation, vCenter closes the connection and removes the storage provider from its configuration.
Refresh certificate	vCenter warns you when a certificate assigned to a storage provider is about to expire. You can refresh the certificate to continue using the provider. If you fail to refresh the certificate before it expires, vCenter discontinues using the provider.
Upgrade Storage Provider	When vCenter notifies you about a new available VASA version, accept this version to enable it in your vSphere environment. vCenter activates new capabilities exposed by the storage provider and communicates this information to all hosts in the inventory. When you perform this action, you must enter credentials of the storage provider. You cannot select specific VASA version for an upgrade if several versions are available. vCenterupgrades only to the latest version. **Note:** You have the flexibility to upgrade at your own pace. For more information, see vSphere Virtual Volumes Security.
Re-authenticate vCenter	Re-authenticate vCenter against the storage provider. This action re-authorizes vCenter with the storage provider if it has lost authentication and would need credentials. You must enter credentials of the storage provider.

Re-authenticate Host VASA Client

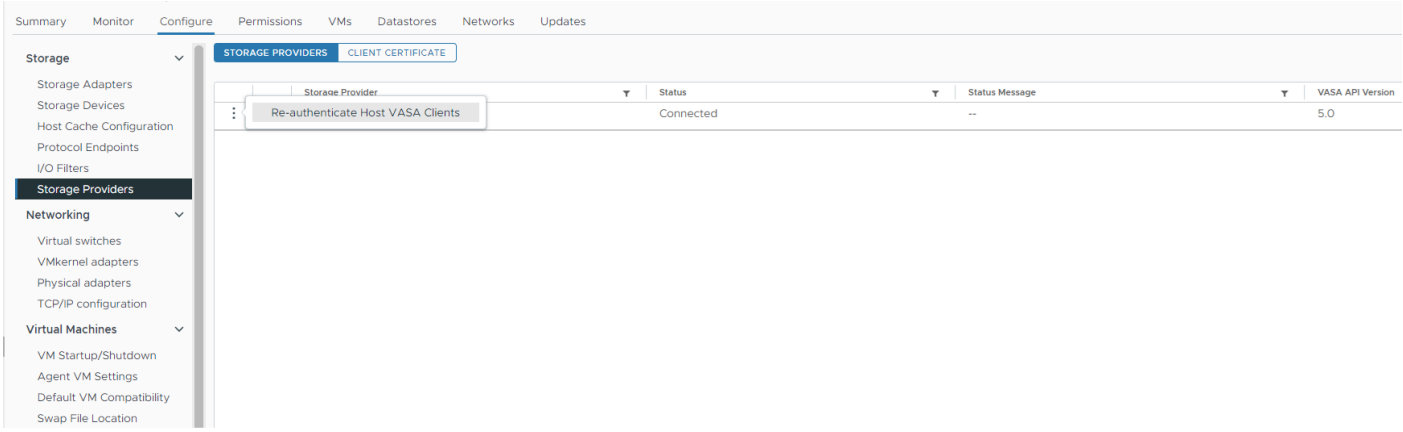
You can monitor the storage provider connection and authentication status on each ESX host. If there is an authentication issue in a specific host, you can re-authenticate the specific host against the storage provider.

Register storage providers. See Register Storage Providers for Virtual Volumes.

Use the vSphere Client to monitor the storage providers.

1. Navigate to the ESX host in vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.

   The **Storage Providers** tab lists the storage provider registered with the ESX host. You can view general information including the name of the storage provider, status, and version of the VASA API. The **Client Certificate** lists the VASA client certificate details.
3. If the status indicates an authentication error for a provider, click the row action (three dots) icon next to the provider's name and click **Re-authenticate Host VASA Clients**.



   The VASA client re-authorizes the ESX host with the storage provider.

NVMe and Virtual Volumes in vSphere

In vSphere, Virtual Volumes supports NVMe over Fibre Channel and NVMe over TCP protocols.

When you use this protocol, after you register the storage provider (VASA), the system automatically completes the underlying setup. You only create the datastore. The virtual protocol endpoints and connections are handled by the storage provider.

> **Note:**
> Starting with vSphere 8.0 U2, third-party MPPs support NVMe with Virtual Volumes. For more information about MPPs, see Multipathing Concepts and Acronyms.

Be familiar with the following concepts specific to the configuration of NVMe with Virtual Volumes.

Virtual Volume Namespace

Generally, a namespace can be considered an NVMe equivalent of a LUN. Each type of virtual volume, such as data, configuration, or swap virtual volume, creates a namespace with a 1:1 virtual volume to namespace ratio. The namespace limits depend on the array vendor.

Asymmetrical Namespace Access (ANA) Group

With SCSI or NFS Virtual Volumes implementation, the storage container logically groups the virtual volume objects. With NVMe, grouping of the virtual volumes depends on the array. Generally, at the array, the virtual volumes namespaces are grouped into an ANA group. The array determines the number of ANA groups, each having a unique ANAGRPID within the NVM subsystem. Virtual volume namespaces are allocated and active only on BIND request to the VASA provider. The BIND request also adds the namespace to the ANA group. A namespace remains allocated and active until the last host unbinds the virtual volume.

NVMe Virtual Protocol Endpoint

With SCSI or NFS protocols, protocol endpoints represent a proxy LUN or an NFS mount point. These protocol points are called static.

Unlike a static protocol point, the NVMe protocol endpoint is virtual. This protocol endpoint is a logical representation of the ANA group where the virtual volumes reside. Until a VM is powered on, the NVMe protocol point doesn't exist. Once the VM is powered on, the NVMe protocol point is created, so that the host can access the virtual volumes in the ANA group.

For information about viewing the NVMe virtual protocol endpoints in the vSphere Client, see Review NVMe Protocol Endpoints.

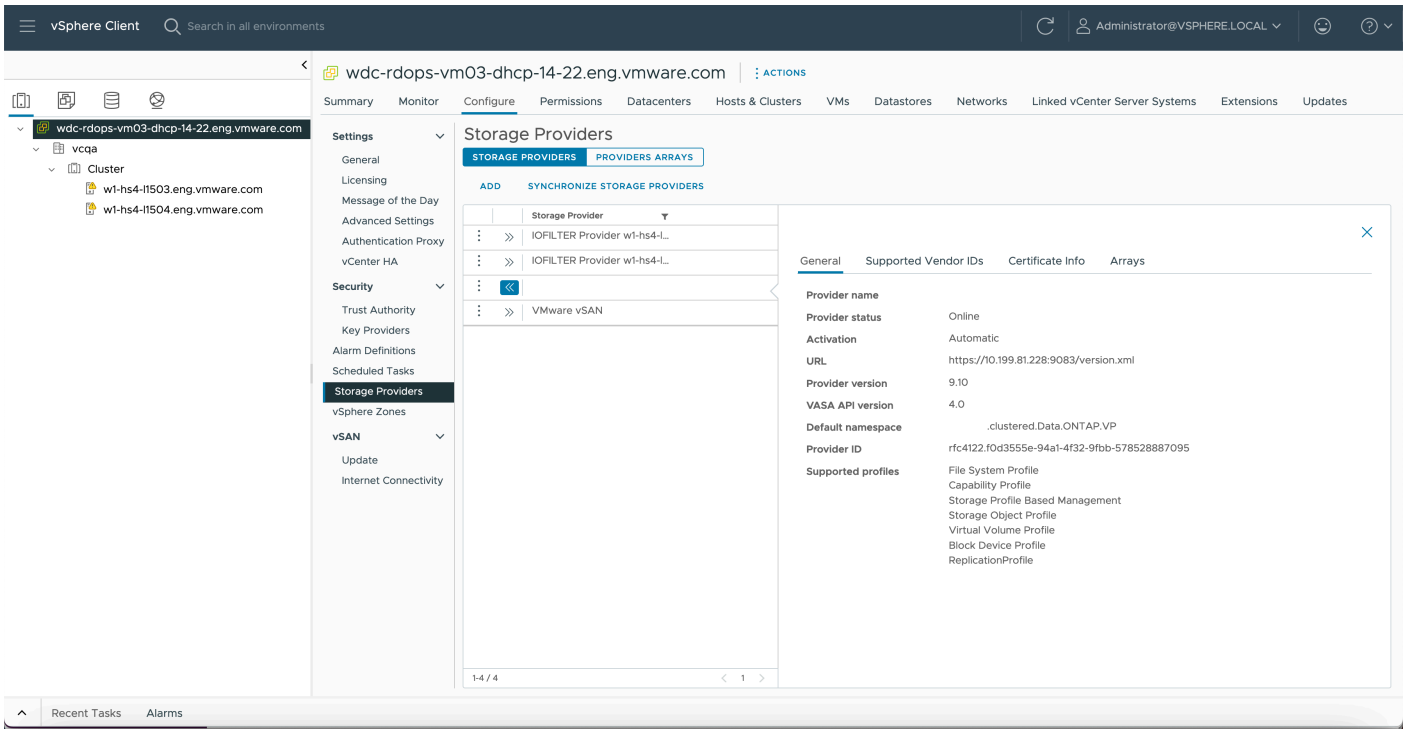
Configure NVMe with Virtual Volumes

Follow these steps to configure NVMe with Virtual Volumes on your ESX host.

- Make sure that your storage provider (VASA) supports NVMe with Virtual Volumes.
- Make sure that your storage array supports the NVMe with Virtual Volumes capability.
- Prepare your vSphere environment and a storage system. For more information, see Before You Enable vSphere Virtual Volumes and Requirements for VMware NVMe Storage.
- If you use NVMe over TCP, configure adapters for NVMe over TCP storage. See Configuring NVMe over TCP on ESX.

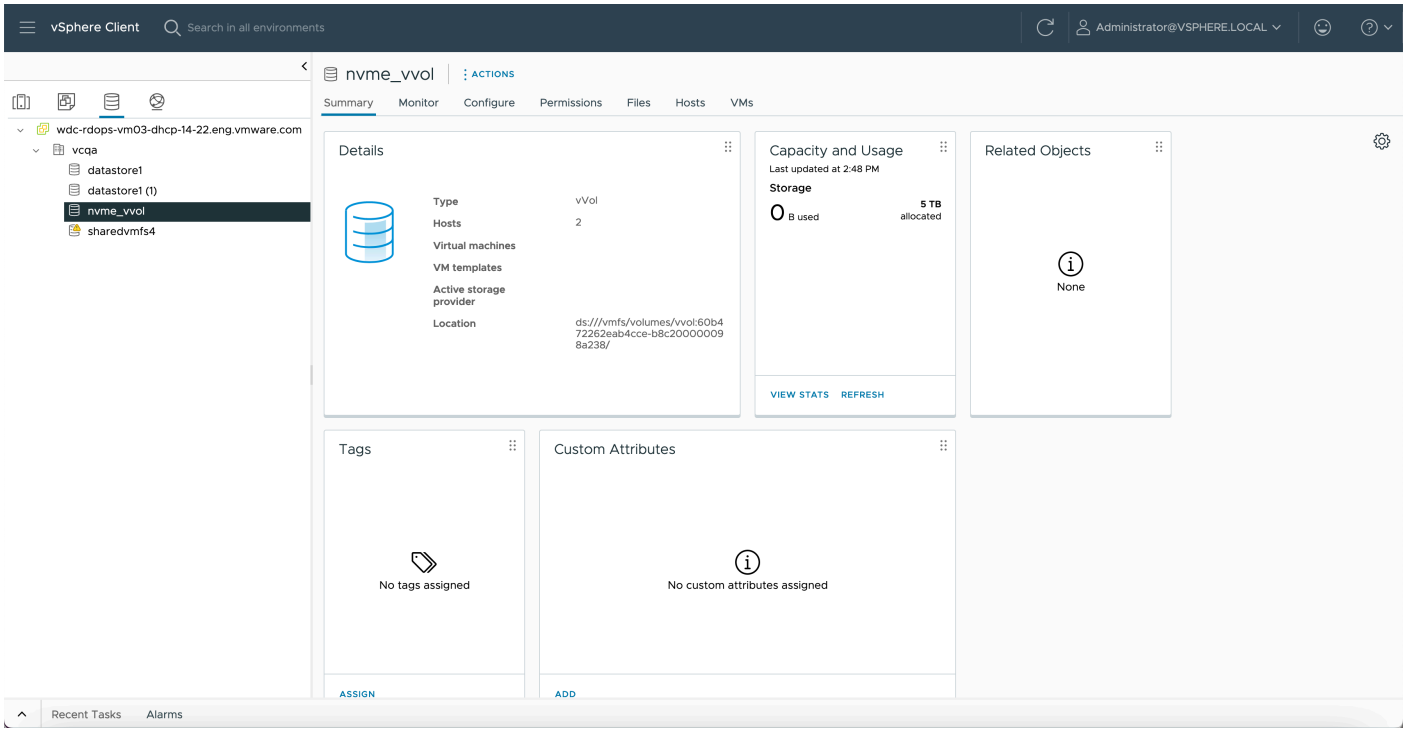
1. Register the storage provider.

   For information, see Register Storage Providers for Virtual Volumes.



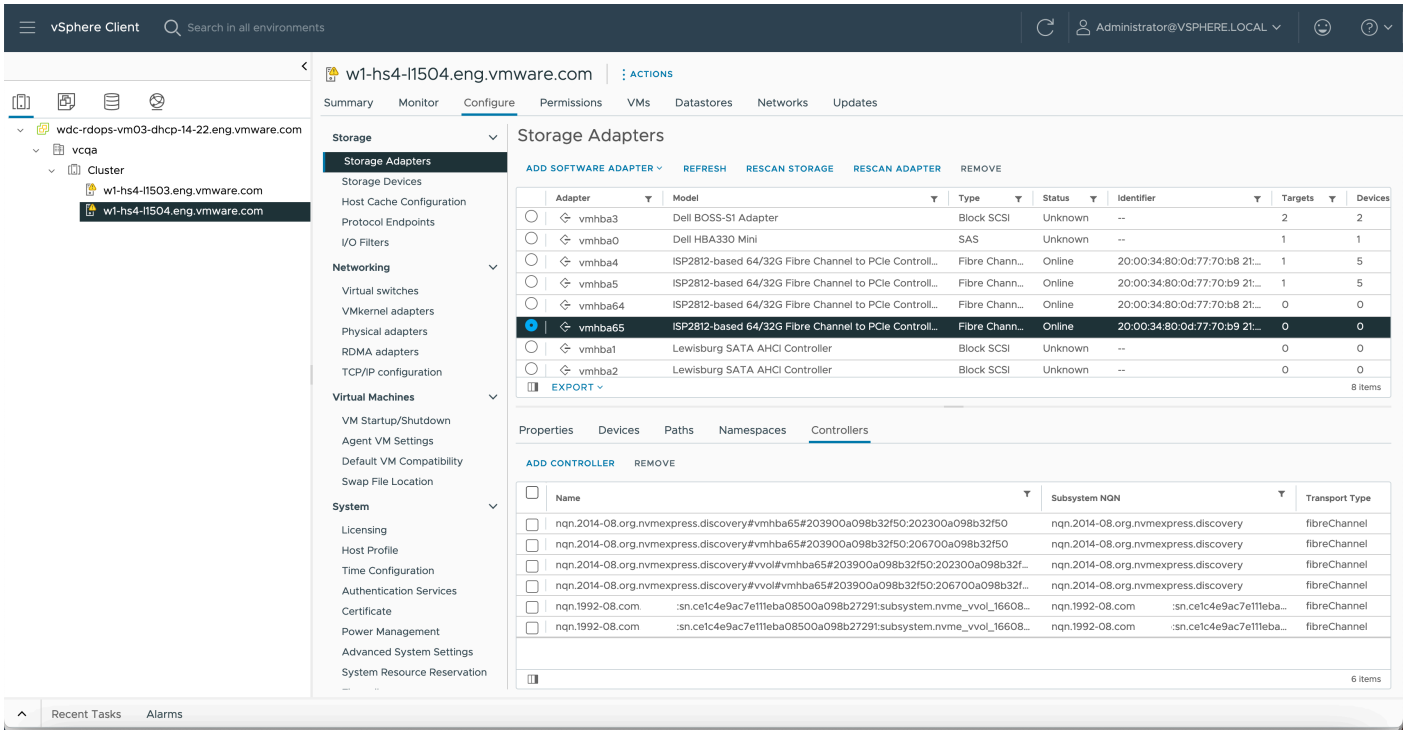
2. Create a Virtual Volumes datastore.

   When creating the datastore, select the storage containers that support NVMe capability. For information, see Create a Virtual Volumes Datastore in vSphere Environment.



3. If you use NVMe over TCP, add a controller.
   See Add Controllers for NVMe over Fabrics.

   Note:
   You do not need to configure the controller for NVMe over Fibre Channel. After you create the datastore, the system discovers controllers.



vSphere Virtual Volumes and Replication

vSphere Virtual Volumes support replication and disaster recovery. With the array-based replication, you can off-load replication of virtual machines to your storage array and use full replication capabilities of the array. You can replicate a single VM object, such as a virtual disk. You can also group several VM objects or virtual machines to replicate them as a single unit.

Array-based replication is policy driven. After you configure your Virtual Volumes storage for replication, information about replication capabilities and replication groups is delivered from the array by the storage provider. This information shows in the VM Storage Policy interface of vCenter.

You use the VM storage policy to describe replication requirements for your virtual machines. The parameters that you specify in the storage policy depend on how your array implements replication. For example, your VM storage policy might include such parameters as the replication schedule, replication frequency, or recovery point objective (RPO). The policy might also indicate the replication target, a secondary site where your virtual machines are replicated, or specify whether replicas must be deleted.

By assigning the replication policy during VM provisioning, you request replication services for your virtual machine. After that, the array takes over the management of all replication schedules and processes.

Requirements for Replication with Virtual Volumes

When you enable Virtual Volumes with replication, in addition to general Virtual Volumes requirements, your environment must satisfy several specific prerequisites.

For general Virtual Volumes requirements, see Before You Enable vSphere Virtual Volumes.

Storage Requirements

Implementation of Virtual Volumes replication depends on your array and might be different for storage vendors. Generally, the following requirements apply to all vendors.

- The storage arrays that you use to implement replication must be compatible with Virtual Volumes.
- The arrays must integrate with the version of the storage (VASA) provider compatible with Virtual Volumes replication.
- The storage arrays must be replication capable and configured to use vendor-provided replication mechanisms. Typical configurations usually involve one or two replication targets. Any required configurations, such as pairing of the replicated site and the target site, must also be performed on the storage side.
- When applicable, replication groups and fault domains for Virtual Volumes must be preconfigured on the storage side.

For more information, contact your vendor and see the VMware by Broadcom Compatibility Guide.

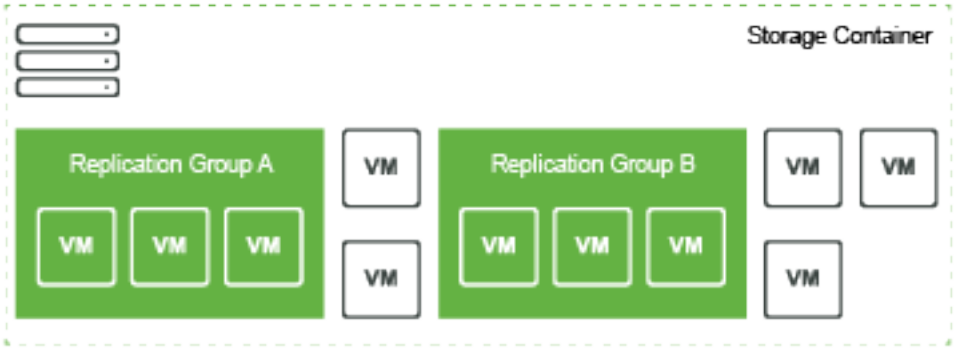
vSphere Requirements

- Use the vCenter and ESX versions that support Virtual Volumes storage replication.
- If you plan to migrate a virtual machine, make sure that target resources, such as the ESX hosts and Virtual Volumes datastores, support storage replication.

Virtual Volumes and Replication Groups

When your storage offers replication services, in addition to storage containers and protocol endpoints, your storage administrator can configure replication groups on the storage side.

vCenter and ESX can discover replication groups, but do not manage their life cycle. Replication groups, also called consistency groups, indicate which VMs and virtual disks must be replicated together to a target site. A single VM cannot span multiple replication groups.



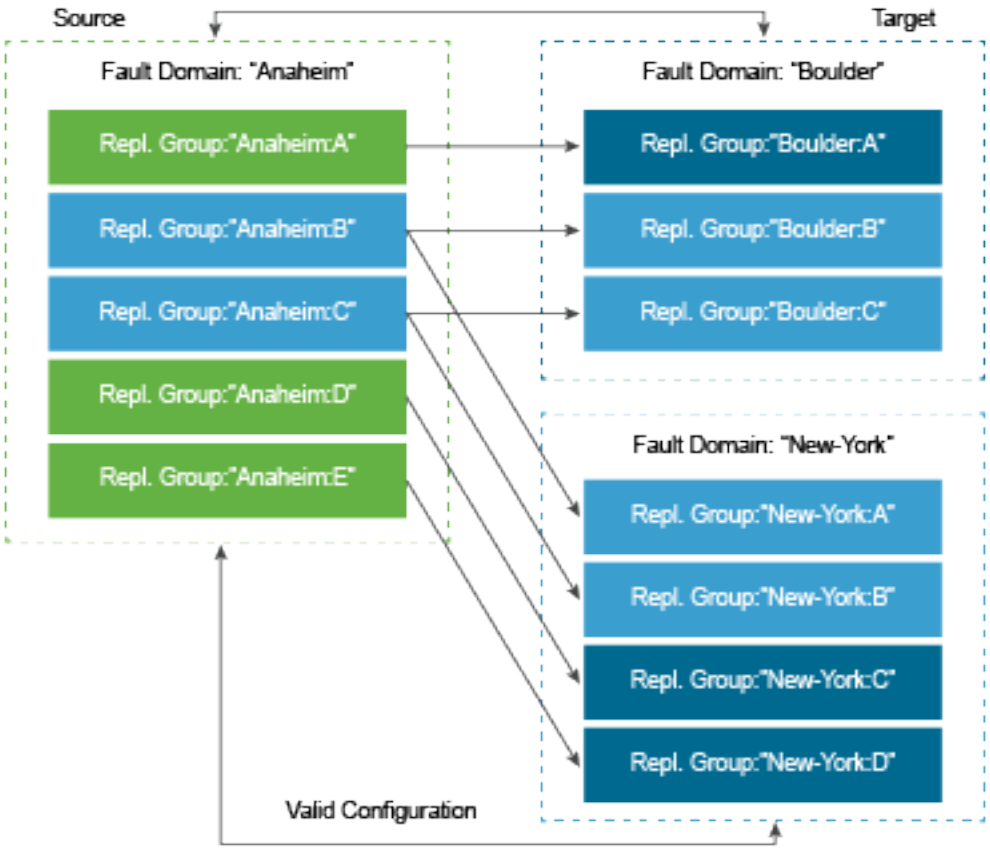
If no preconfigured groups are available, Virtual Volumes can use an automatic method. With the automatic method, Virtual Volumes creates a replication group on demand and associates this group with a Virtual Volumes object being provisioned. If you use the automatic replication group, all components of a virtual machine are assigned to the group. You cannot mix preconfigured and automatic replication groups for components of the same virtual machine.

Virtual Volumes and Fault Domains

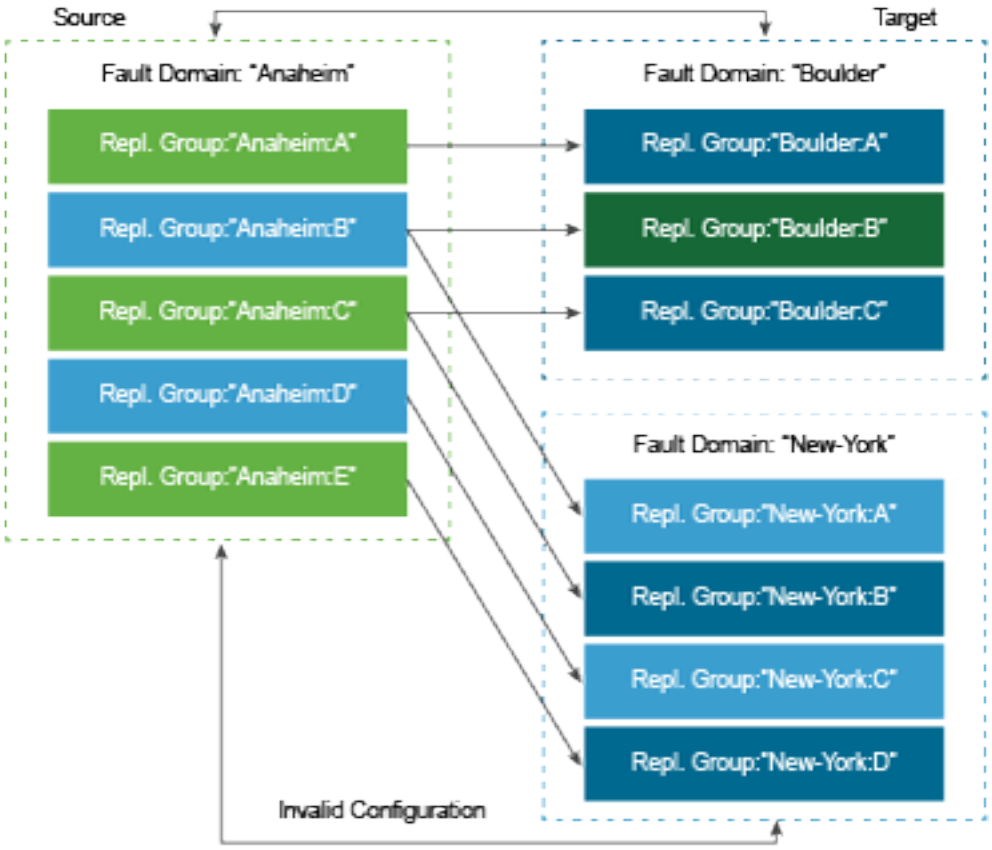
In the Virtual Volumes environment, fault domains define how specific replication groups must be combined when being replicated from a source to a target site.

Fault domains are configured and reported by the storage array, and are not exposed in the vSphere Client. The Storage Policy Based Management (SPBM) mechanism discovers fault domains and uses them for validation purposes during a virtual machine creation.

For example, provision two virtual machines, one associated with replication group Anaheim: B, the second associated with replication group Anaheim: C. SPBM validates the provisioning because each VM is in its own replication group with corresponding target fault domains.



Now provision two VMs, one associated with replication group Anaheim: B, the second associated with replication group Anaheim: D. This configuration is invalid. Both replication groups replicate to the New-York fault domain, however, only one replicates to the Boulder fault domain.



Virtual Volumes Replication Workflow

If information about replication capabilities of the Virtual Volumes storage array shows in vCenter, you can activate replication for your virtual machines.

The workflow to activate replication for your virtual machines includes steps typical for the virtual machine provisioning on Virtual Volumes storage.

1. Define the VM storage policy compatible with replication storage. The datastore-based rules of the policy must include the replication component. See Create a VM Storage Policy for Virtual Volumes.
   After you configure the storage policy that includes replication, vCenter discovers available replication groups.
2. Assign the replication policy to your virtual machine. If configured, select a compatible replication group, or use the automatic assignment. See Assign Storage Policies to Virtual Machines.

Replication Guidelines and Considerations

When you use replication with Virtual Volumes, specific considerations apply.

- You can apply the replication storage policy only to a configuration virtual volume and a data virtual volume. Other VM objects inherit the replication policy in the following way:
  - The memory virtual volume inherits the policy of the configuration virtual volume.
  - The digest virtual volume inherits the policy of the data virtual volume.
  - The swap virtual volume, which exists while a virtual machine is powered on, is excluded from replication.

- If you do not apply the replication policy to a VM disk, the disk is not replicated.
- The replication storage policy should not be used as a default storage policy for a datastore. Otherwise, the policy prevents you from selecting replication groups.
- Replication preserves snapshot history. If a snapshot was created and replicated, you can recover to the application consistent snapshot.
- You can replicate a linked clone. If a linked clone is replicated without its parent, it becomes a full clone.
- If a descriptor file belongs to a virtual disk of one VM, but resides in the VM home of another VM, both VMs must be in the same replication group. If the VMs are located in different replication groups, both of these replication groups must be failed over at the same time. Otherwise, the descriptor might become unavailable after the failover. As a result, the VM might fail to power on.
- In your Virtual Volumes with replication environment, you might periodically run a test failover workflow to ensure that the recovered workloads are functional after a failover.

  The resulting test VMs that are created during the test failover are fully functional and suitable for general administrative operations. However, certain considerations apply:
  - All VMs created during the test failover must be deleted before the test failover stops. The deletion ensures that any snapshots or snapshot-related virtual volumes that are part of the VM, such as the snapshot virtual volume, do not interfere with stopping of the test failover.
  - You can create full clones of the test VMs.
  - You can create fast clones only if the policy applied to the new VM contains the same replication group ID as the VM being cloned. Attempts to place the child VM outside of the replication group of the parent VM fail.

Best Practices for Working with vSphere Virtual Volumes

Observe the following recommendations when you use Virtual Volumes with ESX and vCenter.

Guidelines and Limitations when Using Virtual Volumes

For the best experience with Virtual Volumes functionality, you must follow specific guidelines.

Virtual Volumes supports the following capabilities, features, and VMware products:

- With Virtual Volumes, you can use advanced storage services that include replication, encryption, deduplication, and compression on individual virtual disks. Contact your storage vendor for information about services they support with Virtual Volumes.
- Virtual Volumes functionality supports backup software that uses vSphere APIs - Data Protection. Virtual volumes are modeled on virtual disks. Backup products that use vSphere APIs - Data Protection are as fully supported on virtual volumes as they are on VMDK files on a LUN. Snapshots that the backup software creates using vSphere APIs - Data Protection look as non-vVols snapshots to vSphere and the backup software.

  Note:
  Virtual Volumes do not support SAN transport mode. vSphere APIs - Data Protection automatically selects an alternative data transfer method.

  For more information about integration with the vSphere Storage APIs - Data Protection, consult your backup software vendor.
- Virtual Volumes supports vSphere features such as vSphere vMotion, Storage vMotion, snapshots, linked clones, and DRS.
- You can use clustering products, such as Oracle Real Application Clusters, with Virtual Volumes. To use these products, you activate the multiwrite setting for a virtual disk stored on the Virtual Volumes datastore.

For a list of features and products that Virtual Volumes functionality supports, see *VMware Product Interoperability Matrixes*.

Virtual Volumes Limitations

Improve your experience with Virtual Volumes by knowing the following limitations:

- Because the Virtual Volumes environment requires vCenter, you cannot use Virtual Volumes with a standalone host.

- Virtual Volumes functionality does not support RDMs.
- A Virtual Volumes storage container cannot span multiple physical arrays. Some vendors present multiple physical arrays as a single array. In such cases, you still technically use one logical array.
- Host profiles that contain Virtual Volumes datastores are vCenter specific. After you extract this type of host profile, you can attach it only to hosts and clusters managed by the same vCenter as the reference host. For more information, see *Configuration Maximums* document.

Virtual Volumes Limitations with NVMe

The following features are not supported for NVMe over Virtual Volumes:

- Replication and Site Recovery Manager operations.
- NVMe reservations.
- NVMe deallocate and unmap operations.
- Disk sharing across VMs.
- NVMe in-band migration.

Best Practices for Storage Container Provisioning

Follow these best practices when provisioning storage containers on the Virtual Volumes array side.

Creating Containers Based on Your Limits

Because storage containers apply logical limits when grouping virtual volumes, the container must match the boundaries that you want to apply.

Examples might include a container created for a tenant in a multitenant deployment, or a container for a department in an enterprise deployment.

- Organizations or departments, for example, Human Resources and Finance
- Groups or projects, for example, Team A and Red Team
- Customers

Putting All Storage Capabilities in a Single Container

Storage containers are individual datastores. A single storage container can export multiple storage capability profiles. As a result, virtual machines with diverse needs and different storage policy settings can be a part of the same storage container.

Changing storage profiles must be an array-side operation, not a storage migration to another container.

Avoiding Over-Provisioning Your Storage Containers

When you provision a storage container, the space limits that you apply as part of the container configuration are only logical limits. Do not provision the container larger than necessary for the anticipated use. If you later increase the size of the container, you do not need to reformat or repartition it.

Using Storage-Specific Management UI to Provision Protocol Endpoints

Every storage container needs protocol endpoints (PEs) that are accessible to ESX hosts.

When you use block storage, the PE represents a proxy LUN defined by a T10-based LUN WWN. For NFS storage, the PE is a mount point, such as an IP address or DNS name, and a share name. For NVMe storage, the PE is an ANA group and proxy namespaces are virtual.

Typically, configuration of PEs is array-specific. When you configure PEs, you might need to associate them with specific storage processors, or with certain hosts. To avoid errors when creating PEs, do not configure them manually. Instead, when possible, use storage-specific management tools.

No Assignment of IDs Above Disk.MaxLUN to Protocol Endpoint LUNs

By default, an ESX host can access LUN IDs that are within the range of 0 to 1023. If the ID of the protocol endpoint LUN that you configure is 1024 or greater, the host might ignore the PE.

If your environment uses LUN IDs that are greater than 1023, change the number of scanned LUNs through the `Disk.MaxLUN` parameter. See Change the Number of Scanned Storage Devices.

Best Practices for Virtual Volumes Performance

To ensure optimal Virtual Volumes performance results, follow these recommendations.

Using Different VM Storage Policies for Individual Virtual Volume Components

By default, all components of a virtual machine in the Virtual Volumes environment get a single VM storage policy. However, different components might have different performance characteristics, for example, a database virtual disk and a corresponding log virtual disk. Depending on performance requirements, you can assign different VM storage policies to individual virtual disks and to the VM home file, or config-vVol.

When you use the vSphere Client, you cannot change the VM storage policy assignment for swap-vVol, memory-vVol, or snapshot-vVol.

See Create a VM Storage Policy for Virtual Volumes.

Getting a Host Profile with Virtual Volumes

The best way to get a host profile with Virtual Volumes is to configure a reference host and extract its profile. If you manually edit an existing host profile in the vSphere Client and attach the edited profile to a new host, you might trigger compliance errors. Other unpredictable problems might occur.

Monitoring I/O Load on Individual Protocol Endpoint

- All virtual volume I/O goes through protocol endpoints (PEs). Arrays select protocol endpoints from several PEs that are accessible to an ESX host. Arrays can do load balancing and change the binding path that connects the virtual volume and the PE. See Binding and Unbinding Virtual Volumes.
- On block storage, ESX gives a large queue depth to I/O because of a potentially high number of virtual volumes. The `Scsi.ScsiVVolPESNRO` parameter controls the number of I/O that can be queued for PEs. You can configure the parameter on the Advanced System Settings page of the vSphere Client.

Monitoring Array Limitations

A single VM might occupy multiple virtual volumes. See Virtual Volume Objects.

Suppose that your VM has two virtual disks, and you take two snapshots with memory. Your VM might occupy up to 10 Virtual Volumes objects: a config-vVol, a swap-vVol, two data-vVols, four snapshot-vVols, and two memory snapshot-vVols.

Ensuring that Storage Provider Is Available

To access Virtual Volumes storage, your ESX host requires a storage provider (VASA provider). It is either built into the storage array firmware or deployed as an appliance in the form of a VM. If your VASA provider is a VM-based appliance, the storage provider is always available. To ensure that the storage provider is always available, follow these guidelines:

- Do not migrate a storage provider VM to Virtual Volumes storage.
- Back up your storage provider VM.
- When appropriate, use vSphere HA or Site Recovery Manager to protect the storage provider VM.

Using Stretched Storage Clustering with Virtual Volumes

Virtual Volumes support Stretched Storage Clustering (SSC) through SCSI.

SSC is a specific storage configuration that is commonly referred to as Metro Storage Clustering. A metro cluster is a host cluster that spans multiple sites located in different locations (usually two). The different sites that make up these types of clusters are almost always located in the same area if not city due to the latency requirements. The hosts may be in different data centers but still access the same shared storage. SSC configurations allow fully active and workload-balanced data centers to be used to their full potential and for fast recovery in the event of a host or even full site failure. These configurations are usually implemented in environments where disaster and downtime avoidance is a key requirement.

Virtual Volumes support the uniform host access SSC configuration where vSphere hosts from both sites are all connected to a storage node in the storage cluster across all sites.

Using the esxcli storage vvol Command to Troubleshoot vSphere Virtual Volumes

Learn about the command options that the `esxcli storage vvol` takes. The topics also provide solutions to problems that you might encounter when using vSphere Virtual Volumes.

Virtual Volumes and esxcli Commands

You can use the `esxcli storage vvol` commands to troubleshoot your Virtual Volumes environment.

The following command options are available.

Table 183: esxcli storage vvol commands

Namespace	Command Option	Description
`esxcli storage core device`	list	Identify protocol endpoints. The output entry `Is VVOL PE: true` indicates that the storage device is a protocol endpoint.
`esxcli storage vvol nvme info`	get	Get information for the Virtual Volumes NVMe host information.
`esxcli storage vvol storagearray`	list	List containers or protocol endpoints on the storage array.
`esxcli storage vvol storagecontainer`	list	List all available storage containers.
	abandonedvvol scan	Scan the specified storage container for abandoned Virtual Volumes.
`esxcli storage vvol daemon`	unbindall	Unbind all virtual volumes from all VASA providers known to the ESX host.
`esxcli storage vvol protocolendpoint`	list	List all protocol endpoints that your host can access.
	startforceunbind	Unbind all Virtual Volumes on the PE with zero open handle count.
`esxcli storage vvol stats`		Track performance statistics. For information, see Collecting Statistical Information for Virtual Volumes.
`esxcli storage vvol vasacontext`	get	Show the VASA context (VC UUID) associated with the host.
`esxcli storage vvol vasaprovider`	list	List all storage (VASA) providers associated with the host.
`esxcli storage vvol virtualvolume`	unmap	Reclaim the space by unmapping free blocks from VMFS config volumes. For information, see Reclaim Space on the vSphere Virtual Volumes Datastores.

Namespace	Command Option	Description
`esxcli storage vvol vmstats`	get	Get the Virtual Volumes information and statistics for a specific virtual machine.

Collecting Statistical Information for Virtual Volumes

You can use the `vvol stats` command in your ESX host to track performance statistics.

The following command options are available.

Command	Description	Options
`esxcli storage vvol stats get`	Get statistics for all VASA providers (default), or for specified namespace or entity in the given namespace.	`-e|--entity=`*`str`* Enter entity ID. `-n|--namespace=`*`str`* Enter node namespace expression. `-r|--raw` Use raw format output.
`esxcli storage vvol stats list`	List all the statistics nodes (default), or nodes under a specified namespace.	`-n|--namespace=`*`str`* Enter node namespace expression.
`esxcli storage vvol stats enable`	Activate statistics tracking for the complete namespace.	
`esxcli storage vvol stats disable`	Deactivate statistics tracking for the complete namespace.	
`esxcli storage vvol stats add`	Activate statistics tracking for a specific entity under a specific namespace.	`-e|--entity=`*`str`* Enter entity ID. `-n|--namespace=`*`str`* Enter node namespace expression.
`esxcli storage vvol stats remove`	Removes specific entity for statistics tracking under the specified namespace.	`-e|--entity=`*`str`* Enter entity ID. `-n|--namespace=`*`str`* Enter node namespace expression.
`esxcli storage vvol stats reset`	Reset the statistics counter for the specified statistics namespace or entity.	`-e|--entity=`*`str`* Enter entity ID. `-n|--namespace=`*`str`* Enter node namespace expression.

Virtual Volumes Datastore Is Inaccessible

After you create a Virtual Volumes datastore, it remains inaccessible.

The vSphere Client shows the datastore as inaccessible. You cannot use the datastore for virtual machine provisioning.

This problem might occur when you fail to configure protocol endpoints for the SCSI-based storage container or NVMe ANA groups mapped to the virtual datastore. Like traditional LUNs, SCSI protocol endpoints need to be configured so that an ESX host can detect them.

Before creating virtual datastores for SCSI-based containers, make sure to configure protocol endpoints on the storage side.

Failures When Migrating VMs or Deploying VM OVFs to Virtual Volumes Datastores

Your attempts to migrate a virtual machine or to deploy a VM OVF to Virtual Volumes datastores fail.

An OVF template or a VM being migrated from a nonvirtual datastore might include additional large files, such as ISO disk images, DVD images, and image files. If these additional files cause the configuration virtual volume to exceed its 4-GB limit, migration or deployment to a virtual datastore fails.

The configuration virtual volume, or config-vVol, contains various VM-related files. On traditional nonvirtual datastores, these files are stored in the VM home directory. Similar to the VM home directory, the config-vVol typically includes the VM configuration file, virtual disk and snapshot descriptor files, log files, lock files, and so on.

On virtual datastores, all other large-sized files, such as virtual disks, memory snapshots, swap, and digest, are stored as separate virtual volumes.

Config-vVols are created as 4-GB virtual volumes. Generic content of the config-vVol usually consumes only a fraction of this 4-GB allocation, so config-vVols are typically thin-provisioned to conserve backing space. Any additional large files, such as ISO disk images, DVD images, and image files, might cause the config-vVol to exceed its 4-GB limit. If such files are included in an OVF template, deployment of the VM OVF to Virtual Volumes storage fails. If these files are part of an existing VM, migration of that VM from a traditional datastore to Virtual Volumes storage also fails.

Note:
Starting with vSphere 7.0 Update 2, partners can increase the config-vVol to above 4 GB. Work with your Virtual Volumes partner on implementing this if it is supported by your partner and applicable to your environment.

- For VM migration. Before migrating a VM from a traditional datastore to a virtual datastore, remove excess content from the VM home directory to keep the config-vVol under the 4-GB limit.
- For OVF deployment. Because you cannot deploy an OVF template that contains excess files directly to a virtual datastore, first deploy the VM to a nonvirtual datastore. Remove any excess content from the VM home directory, and migrate the resulting VM to Virtual Volumes storage.

Filtering Virtual Machine I/O in vSphere

I/O filters are software components that can be installed on ESX hosts and can offer additional data services to virtual machines. The filters process I/O requests, which move between the guest operating system of a virtual machine and virtual disks.

The I/O filters can be offered by VMware or created by third parties through vSphere APIs for I/O Filtering (VAIO).

About I/O Filters

I/O filters can gain direct access to the virtual machine I/O path. You can enable the I/O filter for an individual virtual disk. The I/O filters are independent of the storage topology.

VMware offers certain categories of I/O filters. In addition, third-party vendors can create the I/O filters. Typically, they are distributed as packages that provide an installer to deploy the filter components on vCenter and ESX host clusters.

After the I/O filters are deployed, vCenter configures and registers an I/O filter storage provider, also called a VASA provider, for each host in the cluster. The storage providers communicate with vCenter and make data services offered by the I/O filter visible in the VM Storage Policies interface. You can reference these data services when defining common rules for a VM policy. After you associate virtual disks with this policy, the I/O filters are enabled on the virtual disks.

Datastore Support

I/O filters can support all datastore types including the following:

- VMFS
- NFS 3
- NFS 4.1
- vVol
- vSAN

System Requirements for I/O Filters

To be able to use I/O filters in your environment, you must follow specific requirements.

- Use the latest version of ESX and vCenter compatible with I/O filters. Older versions might not support I/O filters, or provide only partial support.
- Check for any additional requirements that individual partner solutions might have. In specific cases, your environment might need flash devices, extra physical memory, or network connectivity and bandwidth. For information, contact your vendor or your VMware representative.
- Web server to host partner packages for filter installation. The server must remain available after initial installation. When a new host joins the cluster, the server pushes appropriate I/O filter components to the host.

I/O Filter Guidelines and Best Practices

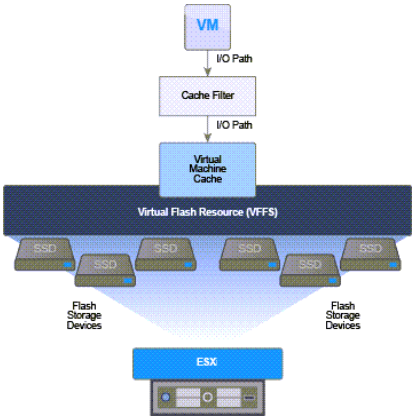
When you use I/O filters in your environment, follow specific guidelines and best practices.

- Because I/O filters are datastore-agnostic, all types of datastores, including VMFS, NFS, Virtual Volumes, and vSAN, are compatible with I/O filters.
- I/O filters support RDMs in virtual compatibility mode. No support is provided to RDMs in physical compatibility mode.
- You cannot change or assign the I/O filter policy while migrating or cloning a virtual machine. You can change the policy after you complete the migration or cloning.
- When you clone or migrate a virtual machine with I/O filter policy from one host to another, make sure that the destination host has a compatible filter installed. This requirement applies to migrations initiated by an administrator or by such functionalities as HA or DRS.
- When you convert a template to a virtual machine, and the template is configured with an I/O filter policy, the destination host must have the compatible I/O filter installed.
- If you use vCenter Site Recovery Manager to replicate virtual disks, the resulting disks on the recovery site do not have the I/O filter policies. You must create the I/O filter policies in the recovery site and reattach them to the replicated disks.
- If your virtual machine has a snapshot tree associated with it, you cannot add, change, or remove the I/O filter policy for the virtual machine.

Using Cache I/O Filters with Flash Storage Devices on ESX Hosts

A cache I/O filter can use a local flash device on your ESX host to cache virtual machine data.

If your caching I/O filter uses local flash devices, you need to configure a virtual flash resource, also known as VFFS volume. You configure the resource on your ESX host before activating the filter. While processing the virtual machine read I/Os, the filter creates a virtual machine cache and places it on the VFFS volume.



To set up a virtual flash resource, you use flash devices that are connected to your host. To increase the capacity of your virtual flash resource, you can add more flash drives. An individual flash drive must be exclusively allocated to a virtual flash resource and cannot be shared with any other vSphere service, such as vSAN or VMFS. See Set Up Virtual Flash Resource.

Deploying and Managing I/O Filters in vSphere

You can run the installer provided by your vendor to install, uninstall, or upgrade I/O filters in your vSphere environment.

When you work with I/O filters, the following considerations apply:

- vCenter uses ESX Agent Manager (EAM) to install and uninstall I/O filters. As an administrator, never invoke EAM APIs directly for EAM agencies that are created or used by vCenter. All operations related to I/O filters must go through VIM APIs. If you accidentally modify an EAM agency that was created by vCenter, you must revert the changes. If you accidentally destroy an EAM agency that is used by I/O filters, you must call `Vim.IoFilterManager#uninstallIoFilter` to uninstall the affected I/O filters. After uninstalling, perform a fresh reinstall.

- When a new host joins the cluster that has I/O filters, the filters installed on the cluster are deployed on the host. vCenter registers the I/O filter storage provider for the host. Any cluster changes become visible in the VM Storage Policies interface of the vSphere Client.
- When you move a host out of a cluster or remove it from vCenter, the I/O filters are uninstalled from the host. vCenter unregisters the I/O filter storage provider.
- If you use a stateless ESX host, it might lose its I/O filter VIBs during a reboot. vCenter checks the bundles installed on the host after it reboots, and pushes the I/O filter VIBs to the host if necessary.

Configure I/O Filters in the vSphere Environment

To set up data services that the I/O filters provide for your virtual machines, follow several steps.

- Create a cluster that includes at least one ESX host.
- For information about I/O filters provided by third parties, contact your vendor or your VMware representative.

Install I/O Filters in a Cluster

If you use I/O filters provided by third parties, install the I/O filters in an ESX host cluster.

- Required privileges: **Host.Configuration.Query patch**.
- Verify that the I/O filter solution is certified by VMware.

VMware partners create I/O filters through the vSphere APIs for I/O Filtering (VAIO) developer program.

The filter packages are distributed as solution bundle ZIP packages that can include I/O filter daemons, I/O filter libraries, CIM providers, and other associated components.

Typically, to deploy the filters, you run installers provided by vendors. Installation is performed at an ESX cluster level. You cannot install the filters on selected hosts directly.

Run the installer that the vendor provided.

The installer deploys the appropriate I/O filter extension on vCenter and the filter components on all hosts within a cluster.

A storage provider, also called a VASA provider, is automatically registered for every ESX host in the cluster. Successful auto-registration of the I/O filter storage providers triggers an event at the host level. If the storage providers fail to auto-register, the system raises alarms on the hosts.

View I/O Filters and Storage Providers

Use the vSphere Client to review I/O filters available in your environment and verify that the I/O filter providers appear as expected and are active.

When you install a third-party I/O filter, a storage provider, also called VASA provider, is automatically registered for every ESX host in the cluster. Successful auto-registration of the I/O filter storage providers triggers an event at the host level. If the storage providers fail to auto-register, the system raises alarms on the hosts.

1. Verify that the I/O filter storage providers appear as expected and are active.
   a) Navigate to vCenter.
   b) Click the **Configure** tab, and click **Storage Providers**.
   c) Review the storage providers for I/O filters.

   When the I/O filter providers are properly registered, capabilities and data services that the filters offer populate the VM Storage Policies interface.
2. Verify that the I/O filter components are listed on your cluster and ESX hosts.

Option	Actions
View I/O filters on a cluster	1. Navigate to the cluster. 2. Click the **Configure** tab. 3. Under **Configuration**, click **I/O Filters** to review the filters installed in the cluster.
View I/O filters on a host	1. Navigate to the host. 2. Click the **Configure** tab.

Option	Actions
	3. Under **Storage**, click **I/O Filters** to review the filters installed on the host.

Handling I/O Filter Installation Failures

Typically, all ESX hosts in a cluster have the same set of I/O filters installed. Occasionally, failures might happen during installation.

If an I/O filter installation fails on a host, the system generates events that report the failure. In addition, an alarm on the host shows the reason for the failure. Examples of failures include the following:

- The VIB URL is not accessible from the host.
- The VIB has an invalid format.
- The VIB requires the host to be in maintenance mode for an upgrade or uninstallation.
- The VIB requires the host to reboot after the installation or uninstallation.
- Attempts to put the host in maintenance mode fail because the virtual machine cannot be evacuated from the host.
- The VIB requires manual installation or uninstallation.

vCenter can resolve some failures. You might have to intervene for other failures. For example, you might need to edit the VIB URL, manually evacuate or power off virtual machines, or manually install or uninstall VIBs.

Install I/O Filters on a Single ESX Host

For troubleshooting purposes, you can download an ESX component of the I/O filter, packaged as a VIB file, and install it on the ESX host. Use the `esxcli` command to install the VIB file.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

1. Install the VIBs by running the following command:

   ```
   esxcli software vib install --depot path_to_VMware_vib_ZIP_file
   ```

   Options for the `install` command allow you to perform a dry run, specify a specific VIB, bypass acceptance-level verification, and so on. Do not bypass verification on production systems. See the *ESXCLI Reference* documentation.

2. Verify that the VIBs are installed on your ESX host.

   ```
   esxcli software vib list
   ```

Uninstall I/O Filters from a Cluster

You can uninstall I/O filters deployed in an ESX host cluster.

Required privileges: **Host.Config.Patch**.

1. Uninstall the I/O filter by running the installer that your vendor provides.

   During uninstallation, a third-party I/O filter installer automatically places the hosts into maintenance mode.

   If the uninstallation is successful, the filter and any related components are removed from the hosts.

2. Verify that the I/O filter components are properly uninstalled from your ESX hosts. Use one of the following methods:

   - Run the `esxcli software vib list` command.
   - View the I/O filters in the vSphere Client. See View I/O Filters and Storage Providers.

   The uninstalled filter no longer appears on the list.

Upgrade I/O Filters in a Cluster

After you upgrade your ESX hosts, use installers provided by I/O filter vendors to upgrade I/O filters deployed in the ESX host cluster.

- Required privileges: **Host.Config.Patch**.

- Upgrade your hosts to latest version of ESX. If you use vSphere Lifecycle Manager for the upgrade, see the *Managing Host and Cluster Lifecycle* documentation.

An upgrade consists of uninstalling the old filter components and replacing them with the new filter components. To determine whether an installation is an upgrade, vCenter checks the names and versions of existing filters. If the existing filter names match the names of the new filters but have different versions, the installation is considered an upgrade.

1. To upgrade the filter, run the vendor-provided installer.

   During the upgrade, a third party I/O filter installer automatically places the hosts into maintenance mode.

   The installer identifies any existing filter components and removes them before installing the new filter components.
2. Verify that the I/O filter components are properly upgraded in your ESX hosts. Use one of the following methods:
   - Run the `esxcli software vib list` command.
   - View the I/O filters in the vSphere Client. See View I/O Filters and Storage Providers.

After the upgrade, the system places the hosts back into operational mode.

Activate I/O Filter Data Services on Virtual Disks in vSphere

In vSphere, activating data services that I/O filters provide is a two-step process. You create a virtual machine policy based on data services that the I/O filters provide, and then attach this policy to a virtual machine.

For the cache I/O filters, configure the virtual flash resource on your ESX host before activating the filter. See Set Up Virtual Flash Resource.

For information about cache I/O filters, see Using Cache I/O Filters with Flash Storage Devices on ESX Hosts.

1. Define a VM policy based on I/O filter services.

   Make sure that the virtual machine policy lists data services provided by the I/O filters.

   See Create a VM Storage Policy for Host-Based Data Services.
2. Assign the I/O filter policy to a virtual machine.

   To activate data services that the I/O filter provides, associate the I/O filter policy with virtual disks. You can assign the policy when you provision the virtual machine.

   See Assign the I/O Filter Policy to Virtual Machines.

Assign the I/O Filter Policy to Virtual Machines

To activate data services that I/O filters provide, associate the I/O filter policy with virtual disks. You can assign the policy when you create or edit a virtual machine.

Verify that the I/O filter is installed on the ESX host where the virtual machine runs.

You can assign the I/O filter policy during an initial deployment of a virtual machine. This topic describes how to assign the policy when you create a new virtual machine. For information about other deployment methods, see the *vSphere Virtual Machine Administration* documentation.

> **Note:**
> You cannot change or assign the I/O filter policy when migrating or cloning a virtual machine.

1. Start the virtual machine provisioning process and follow the appropriate steps.
2. Assign the same storage policy to all virtual machine files and disks.
   a) On the **Select storage** page, select a storage policy from the **VM Storage Policy** drop-down menu.
   b) Select the datastore from the list of compatible datastores and click **Next**.

      The datastore becomes the destination storage resource for the virtual machine configuration file and all virtual disks. The policy also activates I/O filter services for the virtual disks.
3. Change the VM storage policy for the virtual disk.

   Use this option to enable I/O filters just for your virtual disks.

a) On the **Customize hardware** page, expand the **New hard disk** pane.
b) From the **VM storage policy** drop-down menu, select the storage policy to assign to the virtual disk.
c) Optional: Change the storage location of the virtual disk.

   Use this option to store the virtual disk on a datastore other than the datastore where the VM configuration file resides.

4. Complete the virtual machine provisioning process.

After you create the virtual machine, the **Summary** tab displays the assigned storage policies and their compliance status. You can later change the virtual policy assignment. See Change Storage Policy Assignment for Virtual Machine Files and Disks.

Migrating Virtual Machines with I/O Filters

When you migrate a virtual machine with I/O filters, specific considerations apply.

If you use Storage vMotion to migrate a virtual machine with I/O filters, a destination datastore must be connected to hosts with compatible I/O filters installed.

You might need to migrate a virtual machine with I/O filters across different types of datastores, for example, between VMFS and Virtual Volumes. If you do so, make sure that the VM storage policy includes rule sets for every type of datastore you are planning to use. For example, if you migrate your virtual machine between the VMFS and Virtual Volumes datastores, create a mixed VM storage policy that includes the following rules:

- Common Rules for the I/O filters
- Rule Set 1 for the VMFS datastore. Because Storage Policy Based Management does not offer an explicit VMFS policy, the rule set must include tag-based rules for the VMFS datastore.
- Rule Set 2 for the Virtual Volumes datastore

When Storage vMotion migrates the virtual machine, the correct rule set that corresponds to the target datastore is selected. The I/O filter rules remain unchanged.

If you do not specify rules for datastores and define only Common Rules for the I/O filters, the system applies default storage policies for the datastores.

Storage Hardware Acceleration in vSphere

The hardware acceleration functionality enables the ESX host to integrate with compliant storage systems. The host can offload certain virtual machine and storage management operations to the storage systems. With the storage hardware assistance, your host performs these operations faster and consumes less CPU, memory, and storage fabric bandwidth.

Block storage devices, Fibre Channel and iSCSI, and NAS devices support the hardware acceleration.

Hardware Acceleration Benefits

When the hardware acceleration functionality is supported, the host can get hardware assistance and perform the following tasks faster and more efficiently.

- Migrating virtual machines with Storage vMotion
- Deploying virtual machines from templates
- Cloning virtual machines or templates
- VMFS clustered locking and metadata operations for virtual machine files
- Provisioning thick virtual disks
- Creating fault-tolerant virtual machines
- Creating and cloning thick disks on NFS datastores

Hardware Acceleration Requirements

The hardware acceleration functionality works only if you use an appropriate host and storage array combination.

Table 184: Hardware Acceleration Storage Requirements

ESX	Block Storage Devices	NAS Devices
ESX	Support T10 SCSI standard, or block storage plug-ins for array integration (VAAI)	Support NAS plug-ins for array integration

Note:
If your SAN or NAS storage fabric uses an intermediate appliance in front of a storage system that supports hardware acceleration, the intermediate appliance must also support hardware acceleration and be properly certified. The intermediate appliance might be a storage virtualization appliance, I/O acceleration appliance, encryption appliance, and so on.

Hardware Acceleration Support Status

For each storage device and datastore, the vSphere Client display the hardware acceleration support status.

The status values are Unknown, Supported, and Not Supported. The initial value is Unknown.

For block devices, the status changes to Supported after the host successfully performs the offload operation. If the offload operation fails, the status changes to Not Supported. The status remains Unknown if the device provides partial hardware acceleration support.

With NAS, the status becomes Supported when the storage can perform at least one hardware offload operation.

When storage devices do not support or provide partial support for the host operations, your host reverts to its native methods to perform unsupported operations.

Hardware Acceleration Considerations

When you use the hardware acceleration functionality with ESX, certain considerations apply.

Several reasons might cause a hardware-accelerated operation to fail.

For any primitive that the array does not implement, the array returns an error. The error triggers the ESX host to attempt the operation using its native methods.

The VMFS data mover does not leverage hardware offloads and instead uses software data movement when one of the following occurs:

- The source and destination VMFS datastores have different block sizes.
- The source file type is RDM and the destination file type is non-RDM (regular file).
- The source VMDK type is eagerzeroedthick and the destination VMDK type is thin.
- The source or destination VMDK is in sparse or hosted format.
- The source virtual machine has a snapshot.
- The logical address and transfer length in the requested operation are not aligned to the minimum alignment required by the storage device. All datastores created with the vSphere Client are aligned automatically.
- The VMFS has multiple LUNs or extents, and they are on different arrays.

Hardware cloning between arrays, even within the same VMFS datastore, does not work.

vSphere Hardware Acceleration for Block Storage Devices

With hardware acceleration, your ESX host can integrate with block storage devices, Fibre Channel, or iSCSI, and use storage array operations that include full copy, block zeroing, and hardware assisted locking.

ESX hardware acceleration supports the following array operations:

- Full copy, also called clone blocks or copy offload. Storage arrays can make full copies of data within the array without having the host read and write the data. This operation reduces the time and network load when cloning virtual machines, provisioning from a template, or migrating with vMotion.

- Block zeroing, also called write same. Storage arrays can zero out a large number of blocks to provide newly allocated storage, free of previously written data. This operation reduces the time and network load when creating virtual machines and formatting virtual disks.
- Hardware assisted locking, also called atomic test and set (ATS). Supports discrete virtual machine locking without use of SCSI reservations. This operation allows disk locking per sector, instead of the entire LUN as with SCSI reservations.

Check with your vendor for the hardware acceleration support. Certain storage arrays require that you activate the support on the storage side.

On your host, the hardware acceleration is available by default. If your storage does not support the hardware acceleration, you can deactivate it.

In addition to hardware acceleration support, ESX includes support for array thin provisioning. For information, see ESX and Array Thin Provisioning.

Deactivate Hardware Acceleration for Block Storage Devices

On your host, the hardware acceleration for block storage devices is activated by default. You can use the vSphere Client advanced settings to deactivate the hardware acceleration operations.

As with any advanced settings, before you deactivate the hardware acceleration, consult with the VMware support team.

1. In the vSphere Client, navigate to the ESX host.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced System Settings**.
4. To deactivate the hardware acceleration, change the value for these options to 0:
   - VMFS3.HardwareAcceleratedLocking
   - DataMover.HardwareAcceleratedMove
   - DataMover.HardwareAcceleratedInit

Managing Hardware Acceleration on Block Storage Devices

To integrate with the block storage arrays, vSphere uses the ESX extensions called Storage APIs - Array Integration (VAAI). With this integration, vSphere can use the array hardware operations.

These extensions are implemented as the T10 SCSI commands. As a result, with the devices that support the T10 SCSI standard, your ESX host can communicate directly and does not require the VAAI plug-ins.

If the device does not support T10 SCSI or provides partial support, ESX reverts to using the VAAI plug-ins, installed on your host. The host can also use a combination of the T10 SCSI commands and plug-ins. The VAAI plug-ins are vendor-specific and can be either VMware or partner developed. To manage the VAAI capable device, your host attaches the VAAI filter and vendor-specific VAAI plug-in to the device.

For information about whether your storage requires VAAI plug-ins or supports hardware acceleration through T10 SCSI commands, see the VMware by Broadcom Compatibility Guide or contact your storage vendor.

You can use several `esxcli` commands to query storage devices for the hardware acceleration support information. For the devices that require the VAAI plug-ins, the claim rule commands are also available. For information about `esxcli` commands, see *Getting Started with ESXCLI*.

Display Hardware Acceleration Plug-Ins and Filter

To communicate with the devices that do not support the T10 SCSI standard, your host uses a single VAAI filter and a vendor-specific VAAI plug-in. Use the `esxcli` command to view the hardware acceleration filter and plug-ins currently loaded into your system.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Run the `esxcli storage core plugin list --plugin-class=value` command.

For *value*, enter one of the following parameters:

- Type `VAAI` to display plug-ins.

  The output of this command is similar to the following example:

  ```
  #esxcli storage core plugin list --plugin-class=VAAI
  Plugin name          Plugin class
  VMW_VAAIP_EQL        VAAI
  VMW_VAAIP_NETAPP     VAAI
  VMW_VAAIP_CX         VAAI
  ```

- Type `Filter` to display the Filter.

  The output of this command is similar to the following example:

  ```
  esxcli storage core plugin list --plugin-class=Filter
  Plugin name  Plugin class
  VAAI_FILTER  Filter
  ```

Verify Hardware Acceleration Support Status

Use the `esxcli` command to verify the hardware acceleration support status of a particular storage device.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Run the `esxcli storage core device list -d=device_ID` command.

The output shows the hardware acceleration, or VAAI, status that can be unknown, supported, or unsupported.

```
# esxcli storage core device list -d naa.XXXXXXXXXXXX4c
naa.XXXXXXXXXXX4c
 Display Name: XXXX Fibre Channel Disk(naa.XXXXXXXXXXX4c)
 Size: 20480
 Device Type: Direct-Access
 Multipath Plugin: NMP
 XXXXXXXXXXXXXXX
 Attached Filters: VAAI_FILTER
 VAAI Status: supported
 XXXXXXXXXXXXXXX
```

Verify Hardware Acceleration Support Details

Use the `esxcli` command to query whether the block storage device provides the hardware acceleration support.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Run the `esxcli storage core device vaai status get -d=device_ID` command.

If a VAAI plug-in manages the device, the output shows the name of the plug-in attached to the device. The output also shows the support status for each T10 SCSI based primitive, if available. Output appears in the following example:

```
# esxcli storage core device vaai status get -d naa.XXXXXXXXXXX4c
naa.XXXXXXXXXXX4c
 VAAI Plugin Name: VMW_VAAIP_SYMM
 ATS Status: supported
 Clone Status: supported
 Zero Status: supported
 Delete Status: unsupported
```

List Hardware Acceleration Claim Rules

Each block storage device managed by a VAAI plug-in needs two claim rules. One claim rule specifies the hardware acceleration filter and another specifies the hardware acceleration plug-in for the device. You can use the `esxcli` commands to list the hardware acceleration filter and plug-in claim rules.

1. To list the filter claim rules, run the `esxcli storage core claimrule list --claimrule-class=Filter` command.

   In this example, the filter claim rules specify devices that the VAAI_FILTER filter claims.

```
# esxcli storage core claimrule list --claimrule-class=Filter
Rule Class   Rule   Class   Type   Plugin      Matches                   XCOPY Use Array Reported Values
 XCOPY Use Multiple Segments  XCOPY Max Transfer Size KiB
Filter       65430  runtime vendor VAAI_FILTER vendor=EMC model=SYMMETRIX                          False
                       False                            0
Filter       65430  file    vendor VAAI_FILTER vendor=EMC model=SYMMETRIX                          False
                       False                            0
Filter       65431  runtime vendor VAAI_FILTER vendor=DGC model=*                                  False
                       False                            0
Filter       65431  file    vendor VAAI_FILTER vendor=DGC model=*                                  False
                       False                            0
```

2. To list the VAAI plug-in claim rules, run the `esxcli storage core claimrule list --claimrule-class=VAAI` command.

   In this example, the VAAI claim rules specify devices that the VAAI plug-in claims.

```
esxcli storage core claimrule list --claimrule-class=VAAI
Rule Class   Rule   Class   Type    Plugin         Matches                   XCOPY Use Array Reported
 Values  XCOPY Use Multiple Segments  XCOPY Max Transfer Size KiB
VAAI         65430  runtime vendor  VMW_VAAIP_SYMM vendor=EMC model=SYMMETRIX
 False                       False                            0
VAAI         65430  file    vendor  VMW_VAAIP_SYMM vendor=EMC model=SYMMETRIX
 False                       False                            0
VAAI         65431  runtime vendor  VMW_VAAIP_CX   vendor=DGC model=*
 False                       False                            0
VAAI         65431  file    vendor  VMW_VAAIP_CX   vendor=DGC model=*
 False                       False                            0
```

Add Hardware Acceleration Claim Rules

To configure the hardware acceleration for a new array, add two claim rules, one for the VAAI filter and another for the VAAI plug-in. For the new claim rules to be active, you first define the rules and then load them into your system.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

1. Define a new claim rule for the VAAI filter by running the `esxcli storage core claimrule add --claimrule-class=Filter --plugin=VAAI_FILTER` command.
2. Define a new claim rule for the VAAI plug-in by running the `esxcli storage core claimrule add --claimrule-class=VAAI` command.
3. Load both claim rules by running the following commands:

```
esxcli storage core claimrule load --claimrule-class=Filter

esxcli storage core claimrule load --claimrule-class=VAAI
```

4. Run the VAAI filter claim rule by using the `esxcli storage core claimrule run --claimrule-class=Filter` command.

   Note:
   Only the filter-class rules must be run. When the VAAI filter claims a device, it automatically finds the proper VAAI plug-in to attach.

Defining Hardware Acceleration Claim Rules

This example shows how to configure the hardware acceleration for IBM arrays using the VMW_VAAIP_T10 plug-in. Use the following sequence of commands. For information about the options that the command takes, see Add Multipathing Claim Rules.

```
# esxcli storage core claimrule add --claimrule-class=Filter --plugin=VAAI_FILTER
--type=vendor --vendor=IBM --autoassign
# esxcli storage core claimrule add --claimrule-class=VAAI --plugin=VMW_VAAIP_T10
--type=vendor --vendor=IBM --autoassign
# esxcli storage core claimrule load --claimrule-class=Filter
# esxcli storage core claimrule load --claimrule-class=VAAI
# esxcli storage core claimrule run --claimrule-class=Filter
```

Delete Hardware Acceleration Claim Rules

Use the `esxcli` command to delete existing hardware acceleration claim rules.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Run the following commands:

```
esxcli storage core claimrule remove -r claimrule_ID --claimrule-class=Filter
esxcli storage core claimrule remove -r claimrule_ID --claimrule-class=VAAI
```

Configure XCOPY Parameters

XCOPY is one of the VAAI primitives that is used for offloading tasks to the storage array. For example, you can use XCOPY to offload such operations as migration or cloning of virtual machines to the array instead of consuming vSphere resources to perform these tasks.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

You can use the XCOPY mechanism with all storage arrays that support the SCSI T10 based VMW_VAAIP_T10 plug-in developed by VMware. To enable the XCOPY mechanism, create a claim rule of the VAAI class.

Use the following command and enter the XCOPY options:

```
esxcli storage core claimrule add --claimrule-class=VAAI
```

For information about the options that the command takes, see Add Multipathing Claim Rules.

Option	Description
-a|--xcopy-use-array-values	Use array reported values for XCOPY commands.
-s|--xcopy-use-multi-segs	Use multiple segments for XCOPY commands. Valid only when `--xcopy-use-array-values` is specified.
-m|--xcopy-max-transfer-size	Maximum transfer size in MB for the XCOPY commands when you use a transfer size different than array reported. Valid only when `--xcopy-use-array-values` is specified.
-k|--xcopy-max-transfer-size-kib	Maximum transfer size in KiB for the XCOPY commands when you use a transfer size different than the array reported. Valid only if `--xcopy-use-array-values` is specified.
-e | --extended-xcopy-support	Use the extended XCOPY command to optimize the data copy between datastores that are in different storage arrays.

Configuring XCOPY

- `# esxcli storage core claimrule add -r 914 -t vendor -V XtremIO -M XtremApp -P VMW_VAAIP_T10 -c VAAI -a -s -k 64`
- `# esxcli storage core claimrule add -r 65430 -t vendor -V EMC -M SYMMETRIX -P VMW_VAAIP_SYMM -c VAAI -a -s -m 200`

Optimize Data Copying with Extended XCOPY Parameter

The extended XCOPY parameter optimizes the data copy between datastores across two different arrays. It allows you to migrate and clone workloads to offload XCOPY between two different arrays.

Extended XCOPY characteristics:

- Use a new extended XCOPY option in a claim rule.
  `esxcli storage core claimrule add -t vendor -V vendor -M model name -P VMW_VAAIP_T10 -c VAAI --autoassign -e`
  Example:
  `esxcli storage core claimrule add -t vendor -V DellEMC -M PowerStore -P VMW_VAAIP_T10 -c VAAI --autoassign -e`
- Extended XCOPY is supported only on T10 plug-in.

1. Use the following command to check the extended XCOPY status.
   `esxcli storage core device vaai status get -d device`
   Example:
   ```
   [root@tpp-h127:~] esxcli storage core device vaai status get -d naa.68ccf098007f554c61148eb1786e6749
   naa.68ccf098007f554c61148eb1786e6749
      VAAI Plugin Name: VMW_VAAIP_T10
      ATS Status: supported
      Clone Status: supported
      Zero Status: supported
      Delete Status: supported
      Ex Clone Status: supported"
   ```
2. Enable extended XCOPY for the supported storage array.
   a) Verify if the VAAI_FILTER plug-in device is added.
      `esxcli storage core claimrule list -c Filter`
   b) If the device is not added for VAAI_FILTER plug-in, use the following command.
      `esxcli storage core claimrule add -t vendor -V vendor -M model name -P VAAI_FILTER -c Filter --autoassign`
   c) Load the VAAI_FILTER claim rule.
      `esxcli storage core claimrule load -c Filter`
   d) Add the extended XCOPY claim rule with T10 plug-in.
      `esxcli storage core claimrule add -t vendor -V vendor -M model name -P VMW_VAAIP_T10 -c VAAI --autoassign -e`

      Note:

      - The claim rule in step d uses ESX default transfer size and single segment for XCOPY commands. To use array reported values of different transfer size and multiple segments, use optional parameters `-a`, `-s`, `-m|-k` with the above command.
      - Extended XCOPY setup works across different arrays.
      - Add a claim rule with optional parameters such as `-a`, `-s`, `-m|-k` for both source and destination devices. In this case, the minimum transfer size for source and destination arrays will be considered for data transfer.
        Source Device:
        `esxcli storage core claimrule add -t vendor -V vendor -M model name -P VMW_VAAIP_T10 -c VAAI --autoassign -a -s`
        Destination Device:
        `esxcli storage core claimrule add -t vendor -V vendor -M model name -P VMW_VAAIP_T10 -c VAAI --autoassign -a -s -e`
   e) Load the VAAI_T10 plug-in.
      `esxcli storage core claimrule load -c VAAI`

f) Run the newly added claim rule options.

```
esxcli storage core claimrule run --claimrule-class=Filter
```

vSphere Hardware Acceleration on NAS Devices

With the hardware acceleration, ESX hosts can integrate with NAS devices and use several hardware operations, such as Full File Clone and Fast File Clone that NAS storage provides. The hardware acceleration uses vSphere APIs for Array Integration (VAAI) to facilitate communications between the hosts and storage devices.

The VAAI NAS framework supports both versions of NFS storage, NFS 3 and NFS 4.1.

The VAAI NAS uses a set of storage primitives to offload storage operations from the host to the array. The following shows the supported NAS operations.

Storage Primitive	Support	Description
Full File Clone	NFS 3 and NFS 4.1	Supports an ability of NAS device to clone virtual disk files. This operation is similar to the VMFS block cloning, except that NAS devices clone entire files instead of file segments. Tasks that benefit from the full file clone operation include VM cloning, Storage vMotion, and deployment of VMs from templates. When the ESX host copies data with VAAI NAS, it does not need to read the data from the NAS and write back the data to the NAS. The host simply sends the copy command offloading it to the NAS. The copy process is done in the NAS, which reduces the load on the host.
Fast File Clone	NFS 3 and NFS 4.1	This operation, also called array-based or native snapshots, offloads creation of virtual machine snapshots and linked clones to the array.
Reserve Space	NFS 3 and NFS 4.1	Supports an ability of storage arrays to allocate space for a virtual disk file in the thick format. Typically, when you create a virtual disk on an NFS datastore, the NAS server determines the allocation policy. The default allocation policy on most NAS servers is thin and does not guarantee backing storage to the file. However, the reserve space operation can instruct the NAS device to use vendor-specific mechanisms to reserve space for a virtual disk. As a result, you can create thick virtual disks on the NFS datastore if the backing NAS server supports the reserve space operation.
Extended Statistics	NFS 3 and NFS 4.1	Supports visibility to space use on NAS devices. The operation enables you to query space utilization details for virtual disks on NFS datastores. The details include the size of a virtual disk and the space consumption of the virtual disk. This functionality is useful for thin provisioning.
Unmap	NFS 3 and NFS 4.1	The Unmap primitive supports the space reclamation for VMs that run on thin provisioned NFS datastores. It includes two primitives: • GET_UNMAP_GRANULARITY primitive gets called during a VM power-on operation. It fetches the Unmap related parameters from the storage backend. The parameters are expected to be filled by the storage vendor specific NAS plugin. For details, see the NAS VAAI Development Kit (NVDK) Guide. • UNMAP primitive gets called when the guest VM issues Unmap requests to free the space consumed by VMDK files. For example, file deletion can trigger Unmap if the guest OS filesystem supports the Discard option. The guest VM provides a region of a VMDK file, from which the backend storage can reclaim space. After receiving the guest OS Unmap request, the NFS storage layer in ESX passes the request to the NAS VAAI plugin to execute the Unmap request on the storage provider.

With NAS storage devices, the hardware acceleration integration is implemented through vendor-specific NAS plug-ins. These plug-ins are typically created by vendors and are distributed as vendor packages. No claim rules are required for the NAS plug-ins to function.

Several tools for installing and updating NAS plug-ins are available. They include the `esxcli` commands and vSphere Lifecycle Manager. For more information, see the *VMware ESX Upgrade* and *Managing Host and Cluster Lifecycle* documentation. For installation and update recommendations, see the VMware at Broadcom Knowledge Base article 332571.

Note:
NAS storage vendors might provide additional settings that can affect the performance and operation of VAAI. Follow the vendor's recommendations and configure the appropriate settings on both the NAS storage array and ESX. See your storage vendor documentation for more information.

Enable NAS Native Snapshots on Virtual Machines

If your deployment includes NAS arrays that support the vSphere APIs for Array Integration (VAAI), you can use the fast file clone technology, also called native NFS snapshots, to make virtual machine snapshots. With this technology, the NFS device copies the virtual machine without having the ESX host read and write the data. This operation might reduce the time and network load when VM snapshots are created.

- Verify that the NAS array supports the fast file clone operation with the VAAI NAS program.
- On your ESX host, install vendor-specific NAS plug-in that supports the fast file cloning with VAAI.
- Follow the recommendations of your NAS storage vendor to configure any required settings on both the NAS array and ESX. See your storage vendor documentation for more information.

By default, all newly created VMs support traditional ESX snapshot technology. To use the NFS native snapshot technology, enable it for the VM.

1. In the vSphere Client, right-click the virtual machine and select **Edit Settings**.
2. Click the **VM Options** tab and expand the **Advanced** menu.
3. Click **Edit Configuration** next to Configuration Parameters.
4. Configure the `snapshot.alwaysAllowNative` parameter.
   If the parameter exists, make sure that its value is set to True. If the parameter is not present, add it and set its value to True.

Name	Value
`snapshot.alwaysAllowNative`	True

Note:
VMs stored on NFS datastores and configured with `snapshot.alwaysAllowNative` might encounter the `Invalid Disk Chain` error. For more information, see Broadcom knowledge base article 313872.

Storage Provisioning and Space Reclamation in vSphere

vSphere supports two models of storage provisioning, thick provisioning and thin provisioning.

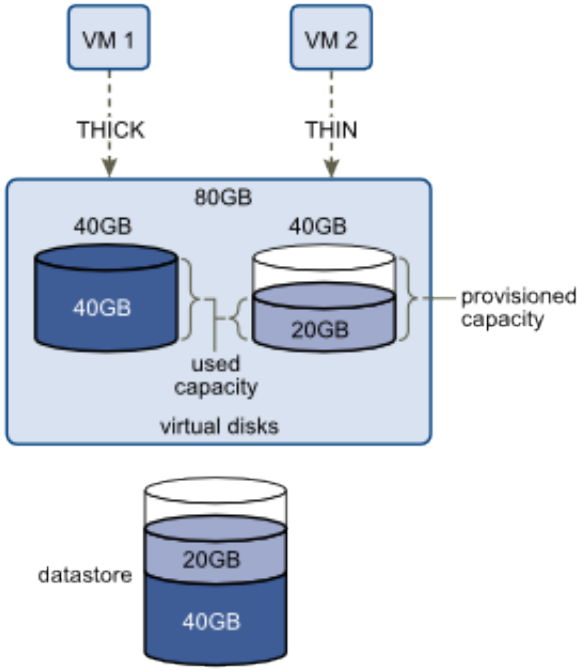
Thick provisioning	It is a traditional model of the storage provisioning. With the thick provisioning, large amount of storage space is provided in advance in anticipation of future storage needs. However, the space might remain unused causing underutilization of storage capacity.
Thin provisioning	This method contrast with thick provisioning and helps you eliminate storage underutilization problems by allocating storage space in a flexible on-demand manner. With ESX, you can use two models of thin provisioning, array-level and virtual disk-level. Thin provisioning allows you to report more virtual storage space than there is real physical capacity. This discrepancy can lead to storage over-subscription, also called over-provisioning. When you use thin provisioning, monitor actual storage usage to avoid conditions when you run out of physical storage space.

Virtual Disk Thin Provisioning with vSphere Storage

When you create a virtual machine, a certain amount of storage space on a datastore is provisioned to virtual disk files. ESX supports thick and thin provisioning for virtual disks.

By default, ESX offers a traditional storage provisioning method for virtual machines. With this method, you first estimate how much storage the virtual machine might need for its entire life cycle. You then provision a fixed amount of storage space to the VM virtual disk in advance, for example, 40 GB. The entire provisioned space is committed to the virtual disk. A virtual disk that immediately occupies the entire provisioned space is a thick disk.

ESX supports thin provisioning for virtual disks. With the disk-level thin provisioning feature, you can create virtual disks in a thin format. For a thin virtual disk, ESX provisions the entire space required for the disk's current and future activities, for example 40 GB. However, the thin disk uses only as much storage space as the disk needs for its initial operations. In this example, the thin-provisioned disk occupies only 20 GB of storage. If the disk requires more space, it can expand into its entire 40 GB of provisioned space.



About Virtual Disk Provisioning Policies

When you perform certain virtual machine management operations, you can specify a provisioning policy for the virtual disk file. The operations include creating a virtual disk, cloning a virtual machine to a template, or migrating a virtual machine.

NFS datastores with Hardware Acceleration and VMFS datastores support the following disk provisioning policies. On NFS datastores that do not support Hardware Acceleration, only a thin format is available.

You can use Storage vMotion or cross-host Storage vMotion to transform virtual disks from one format to another.

Thick Provision Lazy Zeroed	Creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the disk is created. Data remaining on the physical device is not erased during creation, but is zeroed out on demand later on first write from the virtual machine. Virtual machines do not read stale data from the physical device.
Thick Provision Eager Zeroed	A type of thick virtual disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the thick provision lazy zeroed format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take longer to create virtual disks in this format than to create other types of disks. Increasing the size of an Eager Zeroed Thick virtual disk causes a significant stun time for the virtual machine.
Thin Provision	Use this format to save storage space. For the thin disk, you provision as much datastore space as the disk would require based on the value that you enter for the virtual disk size. However, the thin disk starts small and at first, uses only as much datastore space as the disk needs for its initial operations. If the thin disk needs more space later, it can grow to its maximum capacity and occupy the entire datastore space provisioned to it. Thin provisioning is the fastest method to create a virtual disk because it creates a disk with just the header information. It does not allocate or zero out storage blocks. Storage blocks are allocated and zeroed out when they are first accessed. **Note:** If a virtual disk supports clustering solutions such as Fault Tolerance, do not make the disk thin.

Create Thin Provisioned Virtual Disks

To save storage space, you can create a virtual disk in a thin provisioned format. The thin provisioned virtual disk starts small and expands as more disk space is required. You can create thin disks only on the datastores that support disk-level thin provisioning.

This procedure assumes that you are creating a new virtual machine. For information, see Deploying Virtual Machines.

1. Create a virtual machine.
   a) Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **New Virtual Machine**.
   b) Select **Create a new virtual machine** and click **Next**.
   c) Follow the steps required to create a virtual machine.
2. Configure the thin virtual disk.
   a) On the Customize Hardware page, click the **Virtual Hardware** tab.
   b) Click the **New Hard disk** triangle to expand the hard disk options.
   c) Optional: Adjust the default disk size.

      With a thin virtual disk, the disk size value shows how much space is provisioned and guaranteed to the disk. At the beginning, the virtual disk might not use the entire provisioned space. The actual storage use value can be less than the size of the virtual disk.
   d) Select **Thin Provision** for Disk Provisioning.
3. Finish the virtual machine creation.

You created a virtual machine with a disk in the thin format.

If you created a virtual disk in the thin format, you can later inflate it to its full size.

View Virtual Machine Storage Resources

You can view how datastore storage space is allocated for your virtual machines.

1. Browse to the virtual machine.
2. Double-click the virtual machine and click the **Summary** tab.
3. Review the storage use information in the upper right area of the **Summary** tab.

Storage Usage shows how much datastore space is occupied by virtual machine files, including configuration and log files, snapshots, virtual disks, and so on. When the virtual machine is running, the used storage space also includes swap files.

For virtual machines with thin disks, the actual storage use value might be less than the size of the virtual disk.

Determine the Disk Format of a Virtual Machine

You can determine whether your virtual disk is in thick or thin format.

1. Right-click the virtual machine and select **Edit Settings**.
2. Click the **Virtual Hardware** tab.
3. Click the **Hard Disk** triangle to expand the hard disk options.

   The **Type** text box shows the format of your virtual disk.

If your virtual disk is in the thin format, you can inflate it to its full size.

Inflate Thin Virtual Disks

If you created a virtual disk in the thin format, you can change the format to thick.

- Make sure that the datastore where the virtual machine resides has enough space.

- Make sure that the virtual disk is thin.
- Remove snapshots.
- Power off your virtual machine.

You use the datastore browser to inflate the thin virtual disk.

Note:

VMs residing on vSAN do not support virtual disk inflation. See the KB 423722 for more details.

1. In the vSphere Client, navigate to the folder of the virtual disk you want to inflate.
   a) Navigate to the virtual machine.
   b) Click the **Datastores** tab.
      The datastore that stores the virtual machine files is listed.
   c) Right-click the datastore and select **Browse Files**.
      The datastore browser displays contents of the datastore.
2. Expand the virtual machine folder and browse to the virtual disk file that you want to convert.
   The file has the `.vmdk` extension and is marked with the virtual disk
   ( )
   icon.
3. Select the virtual disk file and click **Inflate**.

Note:
The option might not be available if the virtual disk is thick or when the virtual machine is running.

The inflated virtual disk occupies the entire datastore space originally provisioned to it.

Handling Datastore Over-Subscription

Because the provisioned space for thin disks can be greater than the committed space, a datastore over-subscription can occur, which results in the total provisioned space for the virtual machine disks on the datastore being greater than the actual capacity.

Over-subscription can be possible because usually not all virtual machines with thin disks need the entire provisioned datastore space simultaneously. However, if you want to avoid over-subscribing the datastore, you can set up an alarm that notifies you when the provisioned space reaches a certain threshold.

For information on setting alarms, see the *vCenter and Host Management* documentation.

If your virtual machines require more space, the datastore space is allocated on a first come first served basis. When the datastore runs out of space, you can add more physical storage and increase the datastore.

See Increase Capacity of vSphere VMFS Datastore.

ESX and Array Thin Provisioning

You can use thin-provisioned storage arrays with ESX.

The ESX host integrates with block-based storage and performs these tasks:

- The host can recognize underlying thin-provisioned LUNs and monitor their space use to avoid running out of physical space. The LUN space might change if, for example, your VMFS datastore expands or if you use Storage vMotion to migrate virtual machines to the thin-provisioned LUN. The host warns you about breaches in physical LUN space and about out-of-space conditions.
- The host can run the automatic T10 `unmap` command from VMFS6 and VM guest operating systems to reclaim unused space from the array. VMFS5 supports a manual space reclamation method.

Note:
ESX does not support activating or deactivating of thin provisioning on a storage device.

Requirements

To use the thin provisioning reporting and space reclamation features, follow these requirements:

- Use an appropriate ESX version.

Supported thin provisioning components	ESX 7.x and later
Thin provisioning	Yes
Unmap command originating from VMFS	Automatic for VMFS6
Unmap command originating from guest OS	Yes (VMFS6)

- Use storage systems that support T10-based vSphere Storage APIs - Array Integration (VAAI), including thin provisioning and space reclamation. For information, contact your storage provider and check the VMware by Broadcom Compatibility Guide.

Monitoring Space Use

The thin provision integration functionality helps you to monitor the use of space on thin-provisioned LUNs and to avoid running out of space.

The following sample flow demonstrates how the ESX host and the storage array interact to generate breach of space and out-of-space warnings for a thin-provisioned LUN. The same mechanism applies when you use Storage vMotion to migrate virtual machines to the thin-provisioned LUN.

1. Using storage-specific tools, your storage administrator provisions a thin LUN and sets a soft threshold limit that, when reached, triggers an alert. This step is vendor-specific.
2. Using the vSphere Client, you create a VMFS datastore on the thin-provisioned LUN. The datastore spans the entire logical size that the LUN reports.
3. As the space used by the datastore increases and reaches the set soft threshold, the following actions take place:
   a. The storage array reports the breach to your host.
   b. Your host triggers a warning alarm for the datastore.
      You can contact the storage administrator to request more physical space. Alternatively, you can use Storage vMotion to evacuate your virtual machines before the LUN runs out of capacity.
4. If no space is left to allocate to the thin-provisioned LUN, the following actions take place:
   a. The storage array reports out-of-space condition to your host.

      CAUTION:

      In certain cases, when a LUN becomes full, it might go offline or get unmapped from the host.
   b. The host pauses virtual machines and generates an out-of-space alarm.
      You can resolve the permanent out-of-space condition by requesting more physical space from the storage administrator.

Identify Thin-Provisioned Storage Devices

Use the `esxcli` command to verify whether a particular storage device is thin-provisioned.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

Run the `esxcli storage core device list -d=`*`device_ID`* command.

The following thin provisioning status indicates that the storage device is thin-provisioned.

```
# esxcli storage core device list -d naa.XXXXXXXXXXXX4c
naa.XXXXXXXXXXXX4c
  Display Name: XXXX Fibre Channel Disk(naa.XXXXXXXXXXXX4c)
  Size: 20480
  Device Type: Direct-Access
  Multipath Plugin: NMP
  ---------------------
```

```
Thin Provisioning Status: yes
---------------------
```

An unknown status indicates that a storage device is thick.

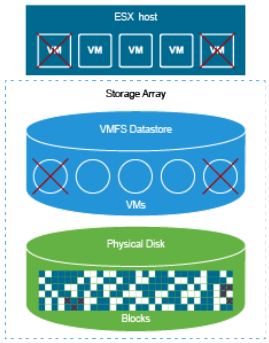
Note:
Some storage systems present all devices as thin-provisioned no matter whether the devices are thin or thick. Their thin provisioning status is always `yes`. For details, check with your storage vendor.

Storage Space Reclamation in vSphere

ESX supports the space reclamation command, also called SCSI unmap command or NVMe dataset management deallocate command, that originates from a VMFS datastore or a VM guest operating system. The command helps thin-provisioned storage arrays to reclaim unused space from the VMFS datastore and thin virtual disks on the datastore. The VMFS6 datastore can send the space reclamation command automatically. With the VMFS5 datastore, you can manually reclaim storage space.

For information about space reclamation with the Virtual Volumes datastores, see Reclaim Space on the vSphere Virtual Volumes Datastores.

You free storage space inside the VMFS datastore when you delete or migrate the VM, consolidate a snapshot, and so on. Inside the virtual machine, storage space is freed when you delete files on the thin virtual disk. These operations leave blocks of unused space on the storage array. However, when the array is not aware that the data was deleted from the blocks, the blocks remain allocated by the array until the datastore releases them. VMFS uses the SCSI unmap command to indicate to the array that the storage blocks contain deleted data, so that the array can unallocate these blocks.



The command can also originate directly from the guest operating system. Both VMFS5 and VMFS6 datastores can provide support to the unmap command that proceeds from the guest operating system. However, the level of support is limited on VMFS5.

Depending on the type of your VMFS datastore, you use different methods to configure space reclamation for the datastore and your virtual machines.

Watch the following video to learn more about how space reclamation works.

Space Reclamation on vSphere VMFS Datastores

Deleting or removing files from a VMFS datastore frees space within the file system. This free space is mapped to a storage device until the file system releases or unmaps it. ESX supports reclamation of free space, which is also called the unmap operation.

The operation helps the storage array to reclaim unused free space. Unmapped space can be then used for other storage allocation requests and needs.

Automatic Reclamation of Free Space on VMFS6 Datastore

On VMFS6 datastores, ESX supports the automatic asynchronous reclamation of free space. VMFS6 can run the unmap command to release free storage space in the background on thin-provisioned storage arrays that support unmap operations.

Automatic unmap processing has several advantages:

- Unmap requests are sent at a constant rate, which helps to avoid any instant load on the backing array.
- Freed regions are batched and unmapped together.
- I/O performance of other workloads is not impacted by the unmap command.

For VMFS6 datastores, you can configure the following space reclamation parameters.

Space reclamation granularity

Granularity defines the minimum size of a released space sector that underlying storage can reclaim. Storage cannot reclaim those sectors that are smaller in size than the specified granularity.

For VMFS6, reclamation granularity equals the block size. When you specify the block size as 1 MB, the granularity is also 1 MB. Storage sectors of the size smaller than 1 MB are not reclaimed.

Note:
Certain storage arrays recommend an optimal unmap granularity. ESX supports automatic unmap processing on arrays with the recommended unmap granularity of 1 MB or greater, for example, 16 MB. On the arrays with the optimal granularity of 1 MB and less, the unmap operation is supported if the granularity is a factor of 1 MB. For example, 1 MB is divisible by 512 bytes, 4 KB, 64 KB, and so on.

Space reclamation method

The method can be either priority or fixed. When the method you use is priority, you configure the priority rate. For the fixed method, you must indicate the reclamation rate in MB per second.

You can use the vSphere Client to set the priority method when you create the VMFS6 datastore. To enable the fixed method, you can later modify the space reclamation settings for the existing datastore.

Alternately, you can use the `esxcli storage vmfs reclaim config set` command to set space reclamation parameters.

Space reclamation priority rate

This parameter defines the rate at which the space reclamation operation is performed when you use the priority reclamation method. Typically, VMFS6 can send the unmap commands either in bursts or sporadically depending on the workload and configuration. For VMFS6, you can specify one of the following options.

Space Reclamation Priority	Description	Configuration
None	Deactivates the unmap operations for the datastore.	vSphere Client `esxcli` command
Low (default)	Sends the unmap command at a less frequent rate, 25–50 MB per second.	vSphere Client `esxcli` command
Medium	Sends the command at a rate twice faster than the low rate, 50–100 MB per second.	`esxcli` command
High	Sends the command at a rate three times faster than the low rate, over 100 MB per second.	`esxcli` command

Space reclamation hosts

This parameter defines the number of ESX hosts that can issue unmap commands to a VMFS6 datastore for automatic space reclamation. Use this parameter to limit the cumulative amount of space reclamation load issued to the target backing VMFS6 datastore.

After you enable space reclamation, the VMFS6 datastore can start releasing the blocks of unused space only when it has at least one open file. This condition can be fulfilled when, for example, you power on one of the VMs on the datastore.

Configure Priority Reclamation when Creating a VMFS6 Datastore

Space reclamation settings control how deleted or unmapped blocks are reclaimed on the devices that back the datastore. When you create a VMFS6 datastore in the vSphere Client, the only method for space reclamation you can specify is the priority method. You can also deactivate space reclamation when creating the datastore.

1. In the vSphere Client object navigator, browse to a host, a cluster, or a data center.
2. From the right-click menu, select **Storage** > **New Datastore**.
3. Follow the steps required to create a VMFS6 datastore.
4. On the **Partition configuration** page, specify the space reclamation parameters.

   The parameters define granularity and the priority rate at which space reclamation operations are performed. You can also use this page to deactivate space reclamation for the datastore.

Option	Description
Block size	The block size on a VMFS datastore defines the maximum file size and the amount of space the file occupies. VMFS6 supports the block size of 1 MB.
Space reclamation granularity	Specify granularity for the unmap operation. Unmap granularity equals the block size, which is 1 MB. Storage sectors of the size smaller than 1 MB are not reclaimed.
Space reclamation priority	Select one of the following options. • Low (default). Use the priority method for space reclamation. The unmap operation is performed at a low priority rate. • None. Select this option if you do not want to use the space reclamation operations for the datastore.

   Note:
   In the vSphere Client, the only available settings for the space reclamation priority are Low and None. To change the settings to Medium or High, use the `esxcli` command.
5. Finish the datastore creation process.

After you activate space reclamation, the VMFS6 datastore can start releasing the blocks of unused space only when it has at least one open file. This condition can be fulfilled when, for example, you power on one of the VMs on the datastore.

Configure Fixed Reclamation for the Existing VMFS6 Datastore

To activate the fixed reclamation method, modify the space reclamation settings for the datastore.

1. In the vSphere Client, navigate to the datastore.
2. Select **Edit Space Reclamation** from the right-click menu.
3. Specify the space reclamation setting.

Option	Description
Enable automatic space reclamation at fixed rate	Use the fixed method for space reclamation. Specify reclamation rate in MB per second. Minimum available value is 10 MB per second.
Disable automatic space reclamation	Deleted or unmapped blocks are not reclaimed. Select this option if you want to deactivate the space reclamation operations for the datastore.

4. Click **OK** to save the new settings.
5. Unmount and remount the datastore for the changes to take place.
   a) Unmount Datastores.
   b) Mount Datastores.
6. Repeat this procedure for all ESX hosts that access the datastore.

The modified value for the space reclamation priority appears on the **General** page for the datastore.

Use the ESXCLI Command to Change Space Reclamation on VMFS6

You can change the default space reclamation priority, granularity, and other parameters.

1. On the ESX host, use the following command to set space reclamation parameters.

   ```
   esxcli storage vmfs reclaim config set
   ```

   The command takes these options:

Option	Description
-b|--reclaim-bandwidth	Space reclamation fixed bandwidth in MB per second.
-g|--reclaim-granularity	Minimum granularity of automatic space reclamation in bytes.
-m|--reclaim-method	Method of automatic space reclamation. Supported options: • priority • fixed
-n|--reclaim-maxhosts	Maximum number of ESX hosts that can issue unmap commands to a VMFS6 datastore for automatic space reclamation.
-p|--reclaim-priority	Priority of automatic space reclamation. Supported options: • none • low • medium • high
-l|--volume-label	The label of the target VMFS volume.
-u|--volume-uuid	The uuid of the target VMFS volume.

   You can use the following examples.

   - Set the reclamation method to fixed and the rate to 100 MB per second.

     ```
     esxcli storage vmfs reclaim config set --volume-label datastore_name --reclaim-method fixed -b 100
     ```

   - Deactivate the VMFS automatic space reclamation.

     ```
     esxcli storage vmfs reclaim config set --volume-label datastore_name --reclaim-priority none
     ```

     Note:
     If you set only the `-n|--reclaim-maxhosts` parameter without any other parameters, you can skip Step 2. It takes around 30 minutes for the updated reclaim-maxhosts parameter value to take effect on all ESX hosts where the VMFS6 datastore mounted.

2. Unmount the VMFS6 datastore from all other ESX hosts where the datastore is mounted, and then remount it again.
   a) Unmount Datastores.
   b) Mount Datastores.

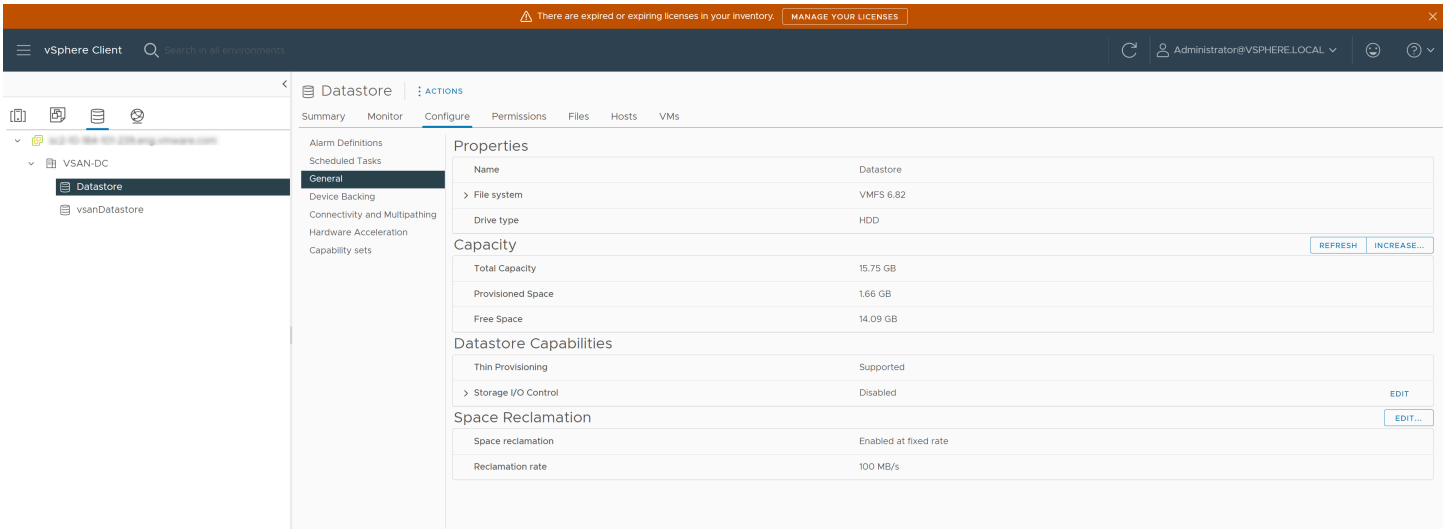
   This step ensures that all ESX hosts where the VMFS6 datastore was mounted update the reclamation settings for the datastore.

Verify Settings for Automatic Space Reclamation on VMFS6

After you configure or edit space reclamation parameters for a VMFS6 datastore, you can review your settings.

1. In the vSphere Client, navigate to the datastore.
2. Click the **Configure** tab.
3. Click **General**.
4. Under **Space Reclamation**, review the setting for the space reclamation priority.

   If you configured any values through the `esxcli` command, for example, Medium or High for the space reclamation priority, these values also appear in the vSphere Client.



Obtaining Parameters for VMFS6 Space Reclamation

You can also use the `esxcli storage vmfs reclaim config get -l=VMFS_label|-u=VMFS_uuid` command to obtain information for the space reclamation configuration.

```
# esxcli storage vmfs reclaim config get -l my_datastore
Reclaim Granularity: 1048576 Bytes
Reclaim Priority: low
Reclaim Method: priority
Reclaim Bandwidth: 26 MB/s
Reclaim Maximum Hosts: 128
```

Manually Reclaim Accumulated Storage Space on VMFS5

On VMFS datastores that do not support automatic space reclamation, such as VMFS5, you can use the `esxcli` command to reclaim unused storage space manually.

Install ESXCLI. See Getting Started with ESXCLI. For troubleshooting, run `esxcli` commands in the ESX Shell.

When you use the command, be aware that it might send many unmap requests at a time. This action can lock some of the resources during the operation. When you use the command, be aware that it might send many unmap requests at a time. This action can lock some of the resources during the operation.

1. To reclaim unused storage blocks on the thin-provisioned device, run the following command:

   ```
   esxcli storage vmfs unmap
   ```

   The command takes these options:

Option	Description
-l|--volume-label=*volume_label*	The label of the VMFS volume to unmap. A mandatory argument. If you specify this argument, do not use `-u|--volume-uuid=volume_uuid`.
-u|--volume-uuid=*volume_uuid*	The UUID of the VMFS volume to unmap. A mandatory argument. If you specify this argument, do not use `-l|--volume-label=volume_label`.
-n|--reclaim-unit=*number*	Number of VMFS blocks to unmap per iteration. An optional argument. If it is not specified, the command uses the default value of 200.

2. To verify whether the unmap process has finished, search for unmap in the `vmkernel.log` file.

Reclaim Space on the vSphere Virtual Volumes Datastores

When your ESX host uses the SCSI or NVMe protocol to connect to storage in the Virtual Volumes environment, the data and configuration virtual volumes are formatted with VMFS6. As a result, the space reclamation process available for VMFS can also be applied to both the data and configuration Virtual Volumes. When you reclaim space on the Virtual

Volumes datastore, you can use the `esxcli storage vvol` command to manually unmap free blocks from the virtual volumes.

- For information about virtual volumes, see Types of Virtual Volumes.
  For information about deprecation of Virtual Volumes, see the Broadcom KB 401070.
- For information about the space reclamation process, see Storage Space Reclamation in vSphere.

  Note:
  Starting with 8.0 Update 3:

  - NVMe backed data Virtual Volumes support space reclamation requests from guest operating systems.
  - Configuration Virtual Volumes support automatic space reclamation. To reclaim the space manually, follow the steps in the given procedure.

1. Obtain the storage container ID and meta vVol UUID for your configuration virtual volume formatted with VMFS.
   a) Navigate to the VM that contains virtual volumes.
      For example:
      ```
      [root@:~] cd /vmfs/volumes/vvol_datastore/VM_1
      ```
   b) From the path, extract the storage container ID and meta vVol UUID.
      ```
      [root@:/vmfs/volumes/vvol:4abe7eb4bf234ee0xxxxxxxxx/rfc4122.1ffee4bd-9c55-49b9-876d-xxxxxxxx
      ```
      In this example,

      Container ID is `vvol:4abe7eb4bf234ee0xxxxxxxxx`

      Meta vVol UUID is `rfc4122.1ffee4bd-9c55-49b9-876d-xxxxxxxx`
2. Use the following command to unmap free blocks from the configuration volume.
   ```
   esxcli storage vvol virtualvolume unmap
   ```
   The command takes the following options:

Option	Description
`-c|--container-id=<str>`	Storage container ID. The option is required.
`-n|--reclaim-unit=<long>`	Number of VMFS blocks to unmap per iteration.
`-u|--uuid=<str>`	Meta vVol UUID. The option is required.

3. Verify that you successfully unmapped the blocks by checking the `hostd` log file.
   ```
   grep "Total Unmapped blocks from vmfs" /var/run/log/hostd.log
   ```

   1. ```
      [root@:~]esxcli storage vvol virtualvolume unmap -c vvol:4abe7eb4bf234ee0xxxxxxxxx -u
       rfc4122.1ffee4bd-9c55-49b9-876d-xxxxxxxx
      ```
   2. ```
      [root@:~] grep "Total Unmapped blocks from vmfs" /var/run/log/hostd.log
      ```
      The output looks similar to the following:
      ```
      2023-06-01T10:27:45.836Z In(166) Hostd[1000342556]: [Originator@6876 sub=Libs
      opID=esxcli-5f-1861 sid=523fac9d user=root] Unmap: Done : Total Unmapped blocks from
      vmfs6 volume rfc4122.1ffee4bd-9c55-49b9-87 6d-xxxxxxxx : 0 (LFB Pass)
      2023-06-01T10:28:13.279Z In(166) Hostd[1000342556]: [Originator@6876 sub=Libs
      opID=esxcli-5f-1861 sid=523fac9d user=root] Unmap: Done : Total Unmapped blocks from
      vmfs6 volume rfc4122.1ffee4bd-9c55-49b9-87 6d-xxxxxxxx : 259421 (SFB Pass)
      ```

Space Reclamation Requests from Guest Operating Systems

ESX supports the unmap commands issued directly from a guest operating system to reclaim storage space. The level of support and requirements depend on the type of datastore where your virtual machine resides.

Inside a virtual machine, storage space is freed when, for example, you delete files on the thin virtual disk. The guest operating system notifies VMFS about freed space by sending the unmap command. The unmap command sent from the

guest operating system releases space within the VMFS datastore. The command then proceeds to the array, so that the array can reclaim the freed blocks of space.

Space Reclamation for VMFS6 Virtual Machines

VMFS6 generally supports automatic space reclamation requests that generate from the guest operating systems, and passes these requests to the array. Many guest operating systems can send the unmap command and do not require any additional configuration. The guest operating systems that do not support the automatic unmaps might require user intervention. For information about guest operating systems that support the automatic space reclamation on VMFS6, contact your vendor.

Generally, the guest operating systems send the unmap commands based on the unmap granularity they advertise. For details, see documentation provided with your guest operating system.

The following considerations apply when you use space reclamation with VMFS6:

- VMFS6 processes the unmap request from the guest OS only when the space to reclaim equals 1 MB or is a multiple of 1 MB. If the space is less than 1 MB or is not aligned to 1 MB, the unmap requests are not processed.
- For VMs with snapshots in the default SEsparse format, VMFS6 supports the automatic space reclamation only on ESX hosts version 6.7 or later.
  Space reclamation affects only the top snapshot and works when the VM is powered on.

Space Reclamation for VMFS5 Virtual Machines

Typically, the unmap command that generates from the guest operation system on VMFS5 cannot be passed directly to the array. You must run the `esxcli storage vmfs unmap` command to trigger unmaps for the array.

However, for a limited number of the guest operating systems, VMFS5 supports the automatic space reclamation requests.

To send the unmap requests from the guest operating system to the array, the virtual machine must meet the following prerequisites:

- The virtual disk must be thin-provisioned.
- Virtual machine hardware must be of version 11 or later.
- The advanced setting EnableBlockDelete must be set to 1.
- The guest operating system must be able to identify the virtual disk as thin.

Using vmkfstools in vSphere

`vmkfstools` is one of the ESX Shell commands for managing VMFS volumes, storage devices, and virtual disks. You can perform many storage operations using the `vmkfstools` command. For example, you can create and manage VMFS datastores on a physical partition, or manipulate virtual disk files, stored on VMFS or NFS datastores.

> **Note:**
> After you make a change using the `vmkfstools`, the vSphere Client might not be updated immediately. Use a refresh or rescan operation from the client.

For more information on the ESX Shell, see Getting Started with ESXCLI.

vmkfstools Command Syntax

Generally, you do not need to log in as the root user to run the `vmkfstools` commands. However, some commands, such as the file system commands, might require the root user login.

The `vmkfstools` command supports the following command syntax:

`vmkfstools` *`options target`*.

Target specifies a partition, device, or path to apply the command option to.

Table 185: `vmkfstools` Command Arguments

Argument	Description
options	One or more command-line options and associated arguments that you use to specify the activity for `vmkfstools` to perform. For example, selecting the disk format when creating a new virtual disk. After entering the option, specify a target on which to perform the operation. Target can indicate a partition, device, or path.
partition	Specifies disk partitions. This argument uses a *`disk_ID:P`* format, where *`disk_ID`* is the device ID returned by the storage array and *`P`* is an integer that represents the partition number. The partition digit must be greater than zero (0) and must correspond to a valid VMFS partition.
device	Specifies devices or logical volumes. This argument uses a path name in the ESX device file system. The path name begins with `/vmfs/devices`, which is the mount point of the device file system. Use the following formats when you specify different types of devices: • `/vmfs/devices/disks` for local or SAN-based disks. • `/vmfs/devices/lvm` for ESX logical volumes. • `/vmfs/devices/generic` for generic SCSI devices.
path	Specifies a VMFS file system or file. This argument is an absolute or relative path that names a directory symbolic link, a raw device mapping, or a file under `/vmfs`. • To specify a VMFS file system, use this format: `/vmfs/volumes/`*`file_system_UUID`* or `/vmfs/volumes/`*`file_system_label`* • To specify a file on a VMFS datastore, use this format: `/vmfs/volumes/`*`file_system_label`*`|`*`file_system_UUID`*`/[dir]/myDisk.vmdk` If the current working directory is the parent directory of `myDisk.vmdk`, do not enter the entire path.

The vmkfstools Command Options

The `vmkfstools` command has several options. Some of the options are suggested for advanced users only.

The long and single-letter forms of the options are equivalent. For example, the following commands are identical.

```
vmkfstools --createfs vmfs6 --blocksize 1m disk_ID:P
vmkfstools -C vmfs6 -b 1m disk_ID:P
```

-v Suboption

The `-v` suboption indicates the verbosity level of the command output.

The format for this suboption is as follows:

```
-v --verbose number
```

You specify the *`number`* value as an integer from 1 through 10.

You can specify the `-v` suboption with any `vmkfstools` option. If the output of the option is not suitable for use with the `-v` suboption, `vmkfstools` ignores `-v`.

> **Note:**
> Because you can include the `-v` suboption in any `vmkfstools` command line, `-v` is not included as a suboption in the option descriptions.

File System Options of vSphere vmkfstools Command

File system options allow you to create and manage VMFS datastores. These options do not apply to NFS. You can perform many of these tasks through the vSphere Client.

Listing Attributes of a VMFS Datastore

Use the `vmkfstools` command to list attributes of a VMFS datastore.

```
-P|--queryfs
      -h|--humanreadable
```

When you use this option on any file or directory that resides on a VMFS datastore, the option lists the attributes of the specified datastore. The listed attributes typically include the file system label, the number of extents for the datastore, the UUID, and a list of the devices where each extent resides.

> **Note:**
> If any device backing VMFS file system goes offline, the number of extents and available space change accordingly.

You can specify the `-h|--humanreadable` suboption with the `-P` option. If you do so, `vmkfstools` lists the capacity of the volume in a more readable form.

Example of Listing VMFS Attributes

```
~ vmkfstools -P -h /vmfs/volumes/my_vmfs
VMFS-5.81 (Raw Major Version: 14) file system spanning 1 partitions.
File system label (if any): my_vmfs
Mode: public
Capacity 99.8 GB, 97.5 GB available, file block size 1 MB, max supported file size 62.9 TB
UUID: 571fe2fb-ec4b8d6c-d375-XXXXXXXXXXXX
Partitions spanned (on "lvm"):
        eui.3863316131XXXXXX:1
Is Native Snapshot Capable: YES
```

Creating a VMFS Datastore or a Scratch Partition

Use the `vmkfstools` command to create a VMFS datastore or a scratch partition.

```
-C|--createfs [vmfs5|vmfs6|vfat]
```

This option creates the VMFS datastore on the specified SCSI or NVMe partition, such as *`disk_ID:P`*. The partition becomes the head partition of the datastore. For VMFS5 and VMFS6, the only available block size is 1 MB.

You can specify the following suboptions with the `-C` option.

- `-S|--setfsname` - Define the volume label of the VMFS datastore you are creating. Use this suboption only with the `-C` option. The label you specify can be up to 128 characters long and cannot contain any leading or trailing blank spaces.

  > **Note:**
  >
  > vCenter supports the 80 character limit for all its entities. If a datastore name exceeds this limit, the name gets shortened when you add this datastore to vCenter.

  After you define a volume label, you can use it whenever you specify the VMFS datastore for the `vmkfstools` command. The volume label appears in listings generated for the `ls -l` command and as a symbolic link to the VMFS volume under the `/vmfs/volumes` directory.
  To change the VMFS volume label, use the `ln -sf` command. Use the following as an example:

  ```
  ln -sf /vmfs/volumes/UUID /vmfs/volumes/datastore
  ```

  `datastore` is the new volume label to use for the *`UUID`* VMFS.

Note:

If your host is registered with vCenter, any changes you make to the VMFS volume label get overwritten by vCenter. This operation guarantees that the VMFS label is consistent across all vCenter hosts.

- `-Y|--unmapGranularity #[bBsSkKmMgGtT]` - This suboption applies to VMFS6 only. Define granularity for the unmap operation. The default granularity is 1 MB. As with the block size, enter the unit type.
- `-O|--unmapPriority <none|low|medium|high>` - This suboption applies to VMFS6 only. Define the priority for the unmap operation.

Example for Creating a VMFS File System

This example illustrates creating a VMFS6 datastore named my_vmfs on the `naa.ID:1` partition.

```
vmkfstools -C vmfs6 -S my_vmfs /vmfs/devices/disks/naa.ID:1
```

Adding an Extent to a VMFS Datastore

Use the `vmkfstools` command to add an extent to a VMFS datastore.

When you add an extent, you span the VMFS datastore from the head partition across the partition specified by *span_partition*.

```
-Z|--spanfs span_partitionhead_partition
```

You must specify the full path name for the head and span partitions, for example, `/vmfs/devices/disks/disk_ID:1`. Each time you use this option, you add an extent to the VMFS datastore, so that the datastore spans multiple partitions.

CAUTION:
When you run this option, you lose all data that previously existed on the SCSI or NVMe device you specified in *span_partition*.

Example for Extending a VMFS Datastore

In this example, you extend the existing head partition of the VMFS datastore over a new partition.

```
vmkfstools -Z /vmfs/devices/disks/naa.disk_ID_2:1 /vmfs/devices/disks/naa.disk_ID_1:1
```

The extended datastore spans two partitions, `naa.disk_ID_1:1` and `naa.disk_ID_2:1`. In this example, `naa.disk_ID_1:1` is the name of the head partition.

Expanding a VMFS Datastore

Instead of adding an extent to a VMFS datastore, you can increase the size of an existing datastore. Use the `vmkfstools -G` command.

You might increase the datastore size after the underlying storage had its capacity increased.

The command uses the following option:

```
-G|--growfs devicedevice
```

This option expands the VMFS datastore or its specific extent. For example,

```
vmkfstools --growfs /vmfs/devices/disks/disk_ID:1 /vmfs/devices/disks/disk_ID:1
```

Virtual Disk Options

Virtual disk options allow you to set up, migrate, and manage virtual disks stored on your datastores. You can also perform most of these tasks through the vSphere Client.

Supported Disk Formats

When you create or clone a virtual disk, you can use the `-d|--diskformat` suboption to specify the format for the disk.

Choose from the following formats:

- `zeroedthick` (default) – Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation, but is zeroed out on demand on first write from the virtual machine. The virtual machine does not read stale data from disk.
- `eagerzeroedthick` – Space required for the virtual disk is allocated at creation time. In contrast to `zeroedthick` format, the data remaining on the physical device is zeroed out during creation. It might take much longer to create disks in this format than to create other types of disks.
- `thin` – Thin-provisioned virtual disk. Unlike with the `thick` format, space required for the virtual disk is not allocated during creation, but is supplied, zeroed out, on demand.
- `rdm:`*`device`* – Virtual compatibility mode raw disk mapping.
- `rdmp:`*`device`* – Physical compatibility mode (pass-through) raw disk mapping.
- `2gbsparse` – A sparse disk with the maximum extent size of 2 GB. You can use disks in this format with hosted VMware products, such as VMware Fusion. However, you cannot power on the sparse disk on an ESX host unless you first re-import the disk with `vmkfstools` in a compatible format, such as `thick` or `thin`.

Disk Formats on NFS Datastores

The only disk formats you can use for NFS are `thin`, `thick`, `zeroedthick`, and `2gbsparse`.

`Thick`, `zeroedthick`, and `thin` formats usually behave the same because the NFS server and not the ESX host determines the allocation policy. The default allocation policy on most NFS servers is `thin`. However, on NFS servers that support Storage APIs - Array Integration, you can create virtual disks in `zeroedthick` format. The reserve space operation enables NFS servers to allocate and guarantee space.

For more information on array integration APIs, see Storage Hardware Acceleration in vSphere.

Creating a Virtual Disk

Use the `vmkfstools` command to create a virtual disk.

```
-c|--createvirtualdisk size[bB|sS|kK|mM|gG]
      -d|--diskformat [thin|zeroedthick|eagerzeroedthick]
      -W|--objecttype [file|vsan|vvol]
      --policyFile fileName
```

This option creates a virtual disk at the specified path on a datastore. Specify the size of the virtual disk. When you enter the value for *`size`*, you can indicate the unit type by adding a suffix of `k` (kilobytes), `m` (megabytes), or `g` (gigabytes). The unit type is not case-sensitive. `vmkfstools` interprets either `k` or `K` to mean kilobytes. If you do not specify a unit type, `vmkfstools` defaults to bytes.

You can specify the following suboptions with the `-c` option.

- `-d|--diskformat` specifies disk formats.
- `-W|--objecttype` specifies whether the virtual disk is a file on a VMFS or NFS datastore, or an object on a vSAN or Virtual Volumes datastore.
- `--policyFile` *`fileName`* specifies VM storage policy for the disk.

  Example for Creating a Virtual Disk

  This example shows how to create a two-gigabyte virtual disk file named `disk.vmdk`. You create the disk on the VMFS datastore named `myVMFS`. The disk file represents an empty virtual disk that virtual machines can access.

  ```
  vmkfstools -c 2048m /vmfs/volumes/myVMFS/disk.vmdk
  ```

Initializing a Virtual Disk

Use the `vmkfstools` command to initialize a virtual disk.

```
-w|--writezeros
```

This option cleans the virtual disk by writing zeros over all its data. Depending on the size of your virtual disk and the I/O bandwidth to the device hosting the virtual disk, completing this command might take a long time.

Note:
This command is not supported on NFS datastores.

CAUTION:
When you use this command, you lose any existing data on the virtual disk.

Inflating a Thin Virtual Disk

Use the `vmkfstools` command to inflate a thin virtual disk.

```
-j|--inflatedisk
```

This option converts a `thin` virtual disk to `eagerzeroedthick`, preserving all existing data. The option allocates and zeroes out any blocks that are not already allocated.

Converting a Zeroedthick Virtual Disk to an Eagerzeroedthick Disk

Use the `vmkfstools` command to convert any zeroedthick virtual disk to an eagerzeroedthick disk.

```
-k|--eagerzero
```

While performing the conversion, this option preserves any data on the virtual disk.

Follow this example:

```
vmkfstools --eagerzero /vmfs/volumes/myVMFS/VMName/disk.vmdk
```

Removing Zeroed Blocks

Use the `vmkfstools` command to remove zeroed blocks.

```
-K|--punchzero
```

This option deallocates all zeroed out blocks and leaves only those blocks that were allocated previously and contain valid data. The resulting virtual disk is in thin format.

Deleting a Virtual Disk

Use the vmkfstools command to delete a virtual disk file at the specified path on the VMFS volume.

Use the following option:

```
-U|--deletevirtualdisk
```

Renaming a Virtual Disk

Use the vmkfstools command to rename a virtual disk file at the specified path on the VMFS volume.

You must specify the original filename or file path *`oldName`* and the new filename or file path *`newName`*.

```
-E|--renamevirtualdisk oldNamenewName
```

Cloning or Converting a Virtual Disk or RDM

Use the vmkfstools command to create a copy of a virtual disk or raw disk you specify.

A non-root user cannot clone a virtual disk or an RDM. You must specify the original filename or file path *`oldName`* and the new filename or file path *`newName`*.

```
-i|--clonevirtualdisk oldNamenewName
     -d|--diskformat [thin|zeroedthick|eagerzeroedthick|rdm:device|rdmp:device|2gbsparse]
     -W|--objecttype [file|vsan|vvol]
     --policyFile fileName
     -N|--avoidnativeclone
```

Use the following suboptions to change corresponding parameters for the copy you create.

- `-d|--diskformat` specifies disk formats.
- `-W|--objecttype` specifies whether the virtual disk is a file on a VMFS or NFS datastore, or an object on a vSAN or Virtual Volumes datastore.
- `--policyFile` *`fileName`* specifies VM storage policy for the disk.

By default, ESX uses its native methods to perform the cloning operations. If your array supports the cloning technologies, you can off-load the operations to the array. To avoid the ESX native cloning, specify the `-N|--avoidnativeclone` option.

Example for Cloning or Converting a Virtual Disk

This example illustrates cloning the contents of a master virtual disk from the `templates` repository to a virtual disk file named `myOS.vmdk` on the `myVMFS` file system.

```
vmkfstools -i /vmfs/volumes/myVMFS/templates/gold-master.vmdk /vmfs/volumes/myVMFS/myOS.vmdk
```

You can configure a virtual machine to use this virtual disk by adding lines to the virtual machine configuration file, as in the following example:

```
scsi0:0.present = TRUE
scsi0:0.fileName = /vmfs/volumes/myVMFS/myOS.vmdk
```

If you want to convert the format of the disk, use the `-d|--diskformat` suboption.

This suboption is useful when you import virtual disks in a format not compatible with ESX, for example 2gbsparse format. After you convert the disk, you can attach this disk to a new virtual machine you create in ESX.

For example:

```
vmkfstools -i /vmfs/volumes/myVMFS/templates/gold-master.vmdk /vmfs/volumes/myVMFS/myOS.vmdk  -d
 thin
```

Extending a Virtual Disk

After you create a virtual machine, you can use the vmkfstools command to extend the size of a disk allocated to the virtual machine.

```
-X|--extendvirtualdisk newSize[bBsSkKmMgGtT]
```

Specify the `newSize` parameter adding an appropriate unit suffix. The unit type is not case-sensitive. `vmkfstools` interprets either `k` or `K` to mean kilobytes. If you do not specify the unit type, `vmkfstools` defaults to kilobytes.

The `newSize` parameter defines the entire new size, not just the increment you add to the disk.

For example, to extend a 4-g virtual disk by 1 g, enter: `vmkfstools -X 5g`*`disk name`*.

You can extend the virtual disk to the eagerzeroedthick format by using the `-d eagerzeroedthick` option.

When you use the `-X` option, the following considerations apply:

- Do not extend the base disk of a virtual machine that has snapshots associated with it. If you do, you can no longer commit the snapshot or revert the base disk to its original size.
- After you extend the disk, you might need to update the file system on the disk. As a result, the guest operating system recognizes the new size of the disk and can use it.

Upgrading Virtual Disks

This option converts the specified virtual disk file from ESX Server 2 formats to the ESX format.

Use this option to convert virtual disks of type LEGACYSPARSE, LEGACYPLAIN, LEGACYVMFS, LEGACYVMFS_SPARSE, and LEGACYVMFS_RDM.

```
-M|--migratevirtualdisk
```

Creating a Virtual Compatibility Mode Raw Device Mapping

Use the vmkfstools command to create a Raw Device Mapping (RDM) file on a VMFS volume and map a raw LUN to this file. After this mapping is established, you can access the LUN as you would a normal VMFS virtual disk. The file length of the mapping is the same as the size of the raw LUN it points to.

```
-r|--createrdm device
```

When specifying the *device* parameter, use the following format:

```
/vmfs/devices/disks/disk_ID:P
```

Example for Creating a Virtual Compatibility Mode RDM

In this example, you create an RDM file named *my_rdm.vmdk* and map the *disk_ID* raw disk to that file.

```
vmkfstools -r /vmfs/devices/disks/disk_ID my_rdm.vmdk
```

You can configure a virtual machine to use the `my_rdm.vmdk` mapping file by adding the following lines to the virtual machine configuration file:

```
scsi0:0.present = TRUE
scsi0:0.fileName = /vmfs/volumes/myVMFS/my_rdm.vmdk
```

Creating a Physical Compatibility Mode Raw Device Mapping

Use the vmkfstools command to map a pass-through raw device to a file on a VMFS volume. With the mapping, a virtual machine can bypass ESX SCSI command filtering when accessing its virtual disk. This type of mapping is useful when the virtual machine must send proprietary SCSI commands, for example, when SAN-aware software runs on the virtual machine.

```
-z|--createrdmpassthru deviceexample.vmdk
```

After you establish this type of mapping, you can use it to access the raw disk as you access any other VMFS virtual disk.

When specifying the *device* path, use the following format:

```
/vmfs/devices/disks/device_ID
```

For the .vmdk name, use this format. Make sure to create the datastore before using the command.

```
/vmfs/volumes/datastore_name/example.vmdk
```

For example,

```
vmkfstools -z /vmfs/devices/disks/naa.600a0000000000000... /vmfs/volumes/datastore1/mydisk.vmdk
```

Listing Attributes of an RDM

Use the vmkfstools command to list the attributes of a raw disk mapping. The attributes help you identify the storage device to which your RDM files maps.

```
-q|--queryrdm my_rdm.vmdk
```

This option prints the name of the raw disk RDM. The option also prints other identification information, like the disk ID, for the raw disk.

Example of Listing RDM Attributes

```
# vmkfstools -q /vmfs/volumes/VMFS/my_vm/my_rdm.vmdk

Disk /vmfs/volumes/VMFS/my_vm/my_rdm.vmdk is a Passthrough Raw Device Mapping

Maps to: vml.0200000000600507680190020770000000000000053231343520 20
```

Displaying Virtual Disk Geometry

Use the vmkfstools command to get information about the geometry of a virtual disk.

```
-g|--geometry
```

The output is in the form: `Geometry information C/H/S`, where `C` represents the number of cylinders, `H` represents the number of heads, and `S` represents the number of sectors.

> **Note:**
> When you import virtual disks from hosted VMware products to the ESX host, you might see a disk geometry mismatch error message. A disk geometry mismatch might also trigger problems when you load a guest operating system or run a newly created virtual machine.

Checking and Repairing Virtual Disks

Use the vmkfstools command to check or repair a virtual disk if it gets corrupted.

```
-x|--fix [check|repair]
```

For example,

```
vmkfstools -x check /vmfs/volumes/my_datastore/my_disk.vmdk
```

Checking Disk Chain for Consistency

Use the vmkfstools command to check the entire snapshot chain. You can determine if any of the links in the chain are corrupted or any invalid parent-child relationships exist.

```
-e|--chainConsistent
```

Storage Device Options of vSphere vmkfstools Command

You can use the device options of the vSphere vmkfstools command to perform administrative task for physical storage devices.

Managing SCSI Reservations of LUNs

Use the vmkfstools command to reserve a SCSI LUN for exclusive use by the ESX host. You can also release a reservation so that other hosts can access the LUN, and reset a reservation, forcing all reservations from the target to be released.

```
-L|--lock [reserve|release|lunreset|targetreset|busreset|readkeys|readresv] device
```

> **CAUTION:**
> Using the `-L` option can interrupt the operations of other servers on a SAN. Use the `-L` option only when troubleshooting clustering setups.

Unless advised by VMware, never use this option on a LUN hosting a VMFS volume.

You can specify the `-L` option in several ways:

- `-L reserve` – Reserves the specified LUN. After the reservation, only the server that reserved that LUN can access it. If other servers attempt to access that LUN, a reservation error appears.
- `-L release` – Releases the reservation on the specified LUN. Other servers can access the LUN again.
- `-L lunreset` – Resets the specified LUN by clearing any reservation on the LUN and making the LUN available to all servers again. The reset does not affect any of the other LUNs on the device. If another LUN on the device is reserved, it remains reserved.
- `-L targetreset` – Resets the entire target. The reset clears any reservations on all the LUNs associated with that target and makes the LUNs available to all servers again.
- `-L busreset` – Resets all accessible targets on the bus. The reset clears any reservations on all the LUNs accessible through the bus and makes them available to all servers again.
- `-L readkeys` – Reads the reservation keys registered with a LUN. Applies to SCSI-III persistent group reservation functionality.
- `-L readresv` – Reads the reservation state on a LUN. Applies to SCSI-III persistent group reservation functionality.

When entering the *`device`* parameter, use the following format:

```
/vmfs/devices/disks/disk_ID:P
```

Breaking Device Locks

Use the vmkfstools command to break the device lock on a particular partition.

```
-B|--breaklock device
```

When entering the *`device`* parameter, use the following format:

```
/vmfs/devices/disks/disk_ID:P
```

You can use this command when a host fails in the middle of a datastore operation, such as expand the datastore, add an extent, or resignature. When you run this command, make sure that no other host is holding the lock.

Getting Started with Cloud Native Storage in vSphere

Cloud Native Storage is a solution that provides comprehensive data management for stateful applications in vSphere. When you use Cloud Native Storage, you can create containerized stateful applications capable of surviving restarts and outages. Stateful containers leverage storage exposed by vSphere while using such primitives as standard volume, persistent volume, and dynamic provisioning.

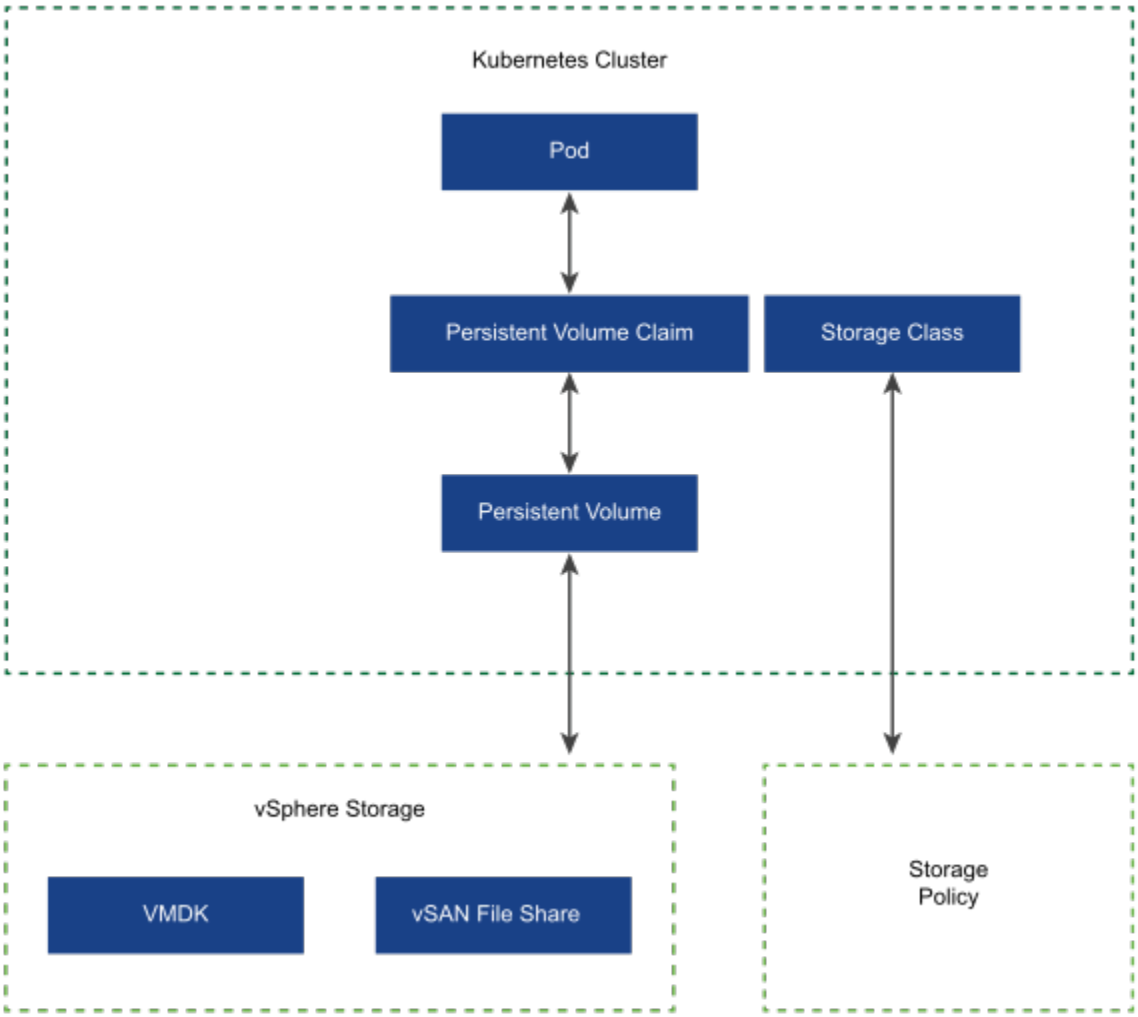
With Cloud Native Storage, you can create persistent container volumes independent of virtual machine and container life cycle. vSphere storage backs the volumes, and you can set a storage policy directly on the volumes. After you create the volumes, you can review them and their backing storage objects in the vSphere Client, and monitor their storage policy compliance.

vSphere Cloud Native Storage supports persistent volumes in the following Kubernetes distributions:

- Generic Kubernetes, also called vanilla, that you install from the official repositories. This *vSphere Storage* documentation covers only generic Kubernetes.
- vSphere Supervisor. For more information, see the vSphere Supervisor Platform documentation.

Cloud Native Storage Concepts and Terminology in vSphere

Be familiar with several concepts essential to the Cloud Native Storage environment in vSphere.



Kubernetes Cluster In the Cloud Native Storage environment, you can deploy a generic Kubernetes cluster in a cluster of virtual machines. On top of the Kubernetes cluster, you deploy your containerized applications. Applications can be stateful and stateless.

Note:

For information clusters you can run in vSphere Supervisor environment, see the vSphere Supervisor Platform documentation..

Pod A pod is a group of one or more containerized applications that share such resources as storage and network. Containers inside a pod are started, stopped, and replicated as a group.

Container Orchestrator Open-source platforms, such as Kubernetes, for deployment, scaling, and management of containerized applications across clusters of hosts. The platforms provide a container-centric infrastructure.

Stateful Application As containerized applications evolve from stateless to stateful, they require persistent storage. Unlike stateless applications that do not save data between sessions, stateful applications save data to persistent storage. The retained data is called the application's state. You can later retrieve the data and use it in the next session. Most applications are stateful. A database is as an example of a stateful application.

PersistentVolume Stateful applications use PersistentVolumes to store their data. A PersistentVolume is a Kubernetes volume capable of retaining its state and data. It is independent of a pod and can continue to exist even when the pod is deleted or reconfigured. In the vSphere environment, the PersistentVolume objects use vSphere virtual disks of the First Class Disk (FCD) type or vSAN file shares as their backing storage. First Class Disks are also referred to as Improved Virtual Disks (IVD) or managed virtual disks.

- Virtual disks support volumes that are mounted as ReadWriteOnce. These volumes can be used only by a single Pod in Kubernetes.
  You can use the vSphere encryption technology to protect FCD virtual disks that back persistent volumes. For more information, see Use Encryption with vSphere Cloud Native Storage.

- vSAN file shares support ReadWriteMany volumes that are mounted by many nodes. These volumes can be shared between multiple Pods or applications running across Kubernetes nodes or across Kubernetes clusters. For information about possible configurations with file shares, see Using vSAN File Service to Provision vSphere File Volumes.

StorageClass

Kubernetes uses a StorageClass to define different tiers of storage and to describe different types of requirements for storage backing the PersistentVolume. In the vSphere environment, a storage class can be linked to a storage policy. As a vSphere administrator, you create storage policies that describe different storage requirements. The VM storage policies can be used as a part of StorageClass definition for dynamic volume provisioning.

The following sample YAML file references the `Gold` storage policy that you created earlier using the vSphere Client. The resulting persistent volume VMDK is placed on a compatible datastore that satisfies the `Gold` storage policy requirements.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: gold-sc
  annotations:
    storageclass.kubernetes.io/is-default-class: "true"
provisioner: csi.vsphere.vmware.com
parameters:
  storagepolicyname: "Gold"
```

PersistentVolumeClaim

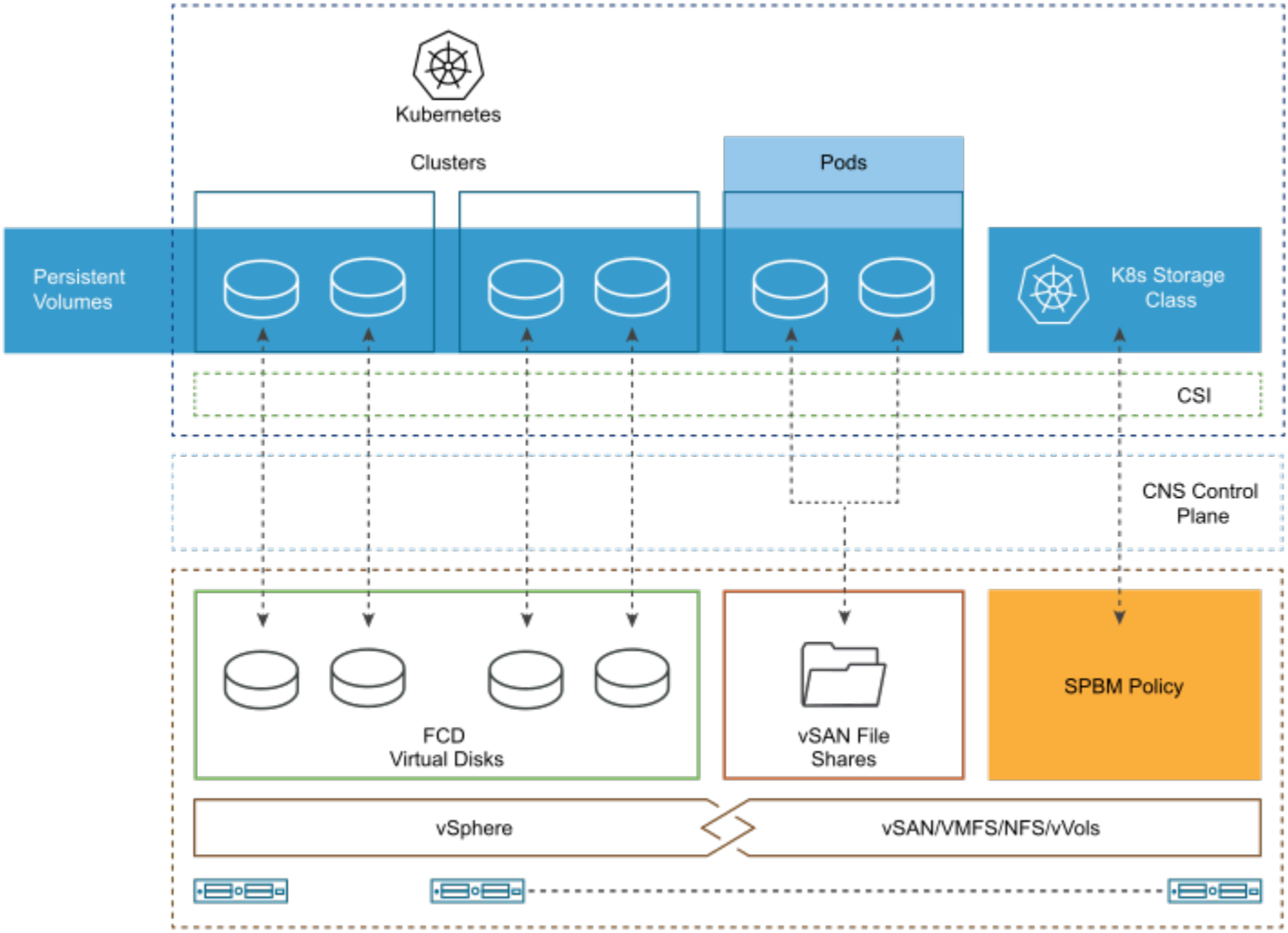
Typically, applications or pods can request persistent storage through a PersistentVolumeClaim. The PersistentVolumeClaim specifies the type and class of storage, the access mode, either ReadWriteOnce or ReadWriteMany, and other parameters for the PersistentVolume. The request can then dynamically provision the corresponding PersistentVolume object and the underlying virtual disk or vSAN file share in the vSphere environment.
Once the claim is created, the PersistentVolume is automatically bound to the claim. Pods use the claim to mount the PersistentVolume and access storage.
When you delete this claim, the corresponding PersistentVolume object and the underlying storage are deleted.

```
kind: PersistentVolumeClaim
metadata:
 name: persistent-VMDK
spec:
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 5Gi
 storageClassName: gold-sc
```

Cloud Native Storage Components

Cloud Native Storage uses several components to integrate with vSphere storage.

The following illustration shows how these components interact.



Kubernetes Cluster

In the Cloud Native Storage environment, a generic Kubernetes cluster is deployed on a cluster of virtual machines, or nodes, that run in vSphere. A Kubernetes user directly interacts with the cluster when deploying stateful applications on top of it.

Container Storage Interface (CSI) for vSphere

To consume underlying infrastructure resources, the cluster requires a CSI driver.

The vSphere CSI is an out-of-tree plug-in that exposes vSphere storage to containerized workloads on container orchestrators, such as Kubernetes. The plug-in enables vSAN and other types of vSphere storage.

The vSphere CSI communicates with the CNS control plane on vCenter for all storage provisioning operations. The vSphere CSI supports the following functionalities:

- Dynamic provisioning of container volumes.
- The vSphere First Class Disk functionality.
- Kubernetes zones.
- Conventional and raw mounts.
- Single vCenter, and multiple data centers and clusters.
- Provisioning from multiple datastores or datastore clusters.
- vSAN File Service

On Kubernetes, the CSI driver is used with the out-of-tree vSphere Cloud Provider Interface (CPI). The CSI driver is shipped as a container image and must be deployed by the cluster administrator. For information, see the *vSphere Container Storage Plug-in Deployment* section of the *Getting Started with VMware vSphere Container Storage Plug-in* documentation.

Cloud Native Storage Server Component

The CNS server component, or the CNS control plane, resides in vCenter. It is an extension of vCenter management that implements the provisioning and life cycle operations for the container volumes.

	When provisioning container volumes, it interacts with vCenter to create storage objects that back the volumes. The Storage Policy Based Management functionality guarantees a required level of service to the volumes. The CNS also performs query operations that allow you to manage and monitor container volumes and their backing storage objects through vCenter.
First Class Disk (FCD)	Also called Improved Virtual Disk (IVD) or managed virtual disk. It is a named virtual disk unassociated with a VM. These disks reside on a vSAN, VMFS, NFS, or vVols datastore and back ReadWriteOnce container volumes. The FCD technology allows to perform life cycle operations related to persistent volumes outside of the VM or pod life cycle. If the VM is a Kubernetes node that runs multiple container based applications and uses persistent volumes and virtual disks for many applications, CNS facilitates life cycle operations at the container and persistent volume granularity.
vSAN File Service	It is a vSAN layer that provides file shares. Currently, it supports NFSv3 and NFSv4.1 file shares. Cloud Native Storage uses vSAN file shares for persistent volumes of the ReadWriteMany type. A single ReadWriteMany volume can be mounted by multiple nodes. The volume can be shared between multiple pods or applications running across Kubernetes nodes or across Kubernetes clusters.
Storage Policy Based Management	Storage Policy Based Management is a vCenter service that supports provisioning of persistent volumes according to specified storage requirements. After provisioning, the service monitors compliance of the volume with the required policy characteristics.

Using vSAN File Service to Provision vSphere File Volumes

vSAN file service offers vSAN file shares that are consumed by persistent volumes of the ReadWriteMany (RWM) type. A single RWM volume can be mounted by multiple nodes. The volume can be shared between multiple pods or applications running across Kubernetes nodes or across Kubernetes clusters in vSphere.

When a Kubernetes pod request an RWM volume, Cloud Native Storage communicates with vSAN file service to create an NFS-based file share of the requested size and storage class. Cloud Native Storage then mounts the RWM volume into the Kubernetes worker node where the pod runs. If multiple nodes are requesting access to the RWM volume, Cloud Native Storage determines that the RWM volume already exists for that particular deployment and mounts the existing volume into the nodes.

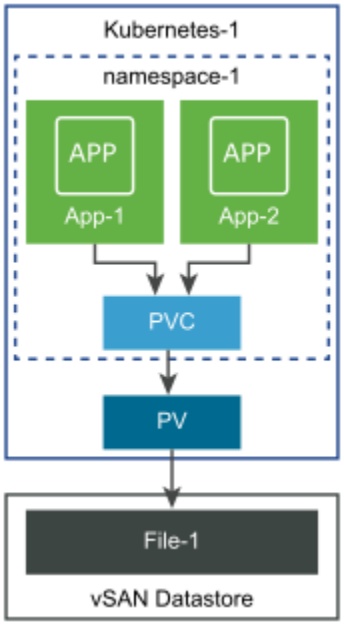
To be able to support RWM volumes, your environment must include the following items.

- vSphere 7.0 and later with vSAN
- vSAN file service enabled. For information, see *vSAN File Service* in the *Administering VMware vSAN* documentation.
- Kubernetes version 1.14 and later
- Compatible version of Container Storage Plug-in. For information, see Compatibility Matrices for vSphere Container Storage Plug-in.

You can use different configurations for file volumes.

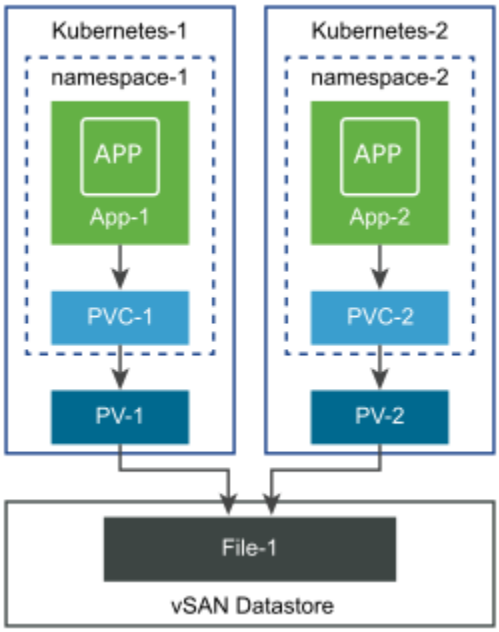
Single File Volume Shared Across Applications in the Same Namespace

In this example, a single file volume is used as shared storage across different applications in the same namespace. You use a single persistent volume claim to provision the file volume.



Single File Volume Shared Across Applications and Namespaces

This example uses a single file volume as shared storage across different applications and different namespaces. For each namespace, you create a separate persistent volume claim to provision the same file volume.



vSphere Cloud Native Storage Users

The types of users involved in the process of creating and monitoring Kubernetes volumes in the vSphere Cloud Native Storage environment generally fall into two categories, a Kubernetes user and a vSphere administrator. Both types of users have access to different tools and perform different tasks.

CNS Kubernetes User

The Kubernetes user might be a Kubernetes developer and an application owner, a Kubernetes administrator, or combine functions of both. The tasks that the Kubernetes user performs in the Cloud Native Storage environment include the following:

- Deploy and manage the vSphere CSI. For information, see Deploying the vSphere Container Storage Plug-in on a Native Kubernetes Cluster.
- Provision persistent volumes. For information about block volumes, see Provisioning Block Volumes. For information about file volumes, see Provisioning File Volumes.
- Perform life cycle operations for persistent volumes.
- Perform life cycle operations for storage classes.

CNS vSphere User

A CNS vSphere user, or a vSphere administrator, has access to the vSphere Client to perform the following tasks:

- Perform life cycle operations for the VM storage policies. For example, create a VM storage policy to be used for a Kubernetes storage class and communicate its name to the Kubernetes user. See Create a Storage Policy for Kubernetes in vSphere.
- Use the Cloud Native Storage section of the vSphere Client to monitor health and storage policy compliance of the container volumes across the Kubernetes clusters. See Monitor Container Volumes Across Kubernetes Clusters in vSphere.

Cloud Native Storage for vSphere Administrators

A vSphere administrator delivers storage resources to the Kubernetes team and creates VM storage policies that describe different storage requirements and classes of services. After the Kubernetes workloads with persistent storage are provisioned, the vSphere administrator can monitor the life cycle of the backing storage resources and their compliance to the requirements.

Requirements of vSphere Cloud Native Storage

Your Cloud Native Storage environment and virtual machines that participate in the Kubernetes cluster must meet several requirements.

Cloud Native Storage Requirements

- A compatible version of vSphere.
- A compatible version of Kubernetes.
- A Kubernetes cluster deployed on the virtual machines. For details about deploying the vSphere CSI plug-in and running the Kubernetes cluster on vSphere, see Deploying the vSphere Container Storage Plug-in on a Native Kubernetes Cluster.

Requirements for Kubernetes Cluster Virtual Machines

- Virtual machines with hardware version 15 or later. Install VMware Tools on each node virtual machine.
- Virtual machine hardware recommendations:
  - Set CPU and memory adequately based on workload requirements.
  - Use the VMware Paravirtual SCSI controller for the primary disk on the Node VM.
- All virtual machines must have access to a shared datastore, such as vSAN.
- Set the `disk.EnableUUID` parameter on each node VM. See Configure Kubernetes Cluster in vSphere Virtual Machines.
- To avoid errors and unpredictable behavior, do not take snapshots of CNS node VMs.

Requirements for CNS File Volume

- Use vSphere version 7.0 or later with a compatible Kubernetes version.
- Use a compatible version of vSphere Container Storage Plug-in.
- Enable and configure the vSAN file service. You must configure the necessary file service domains, IP pools, network, and so on. For information, see the *Administering VMware vSAN* documentation.
- Follow specific guidelines to configure network access from a guest OS in the Kubernetes node to a vSAN file share. See Configuring Network Access to vSAN File Share.

Configuring Network Access to vSAN File Share

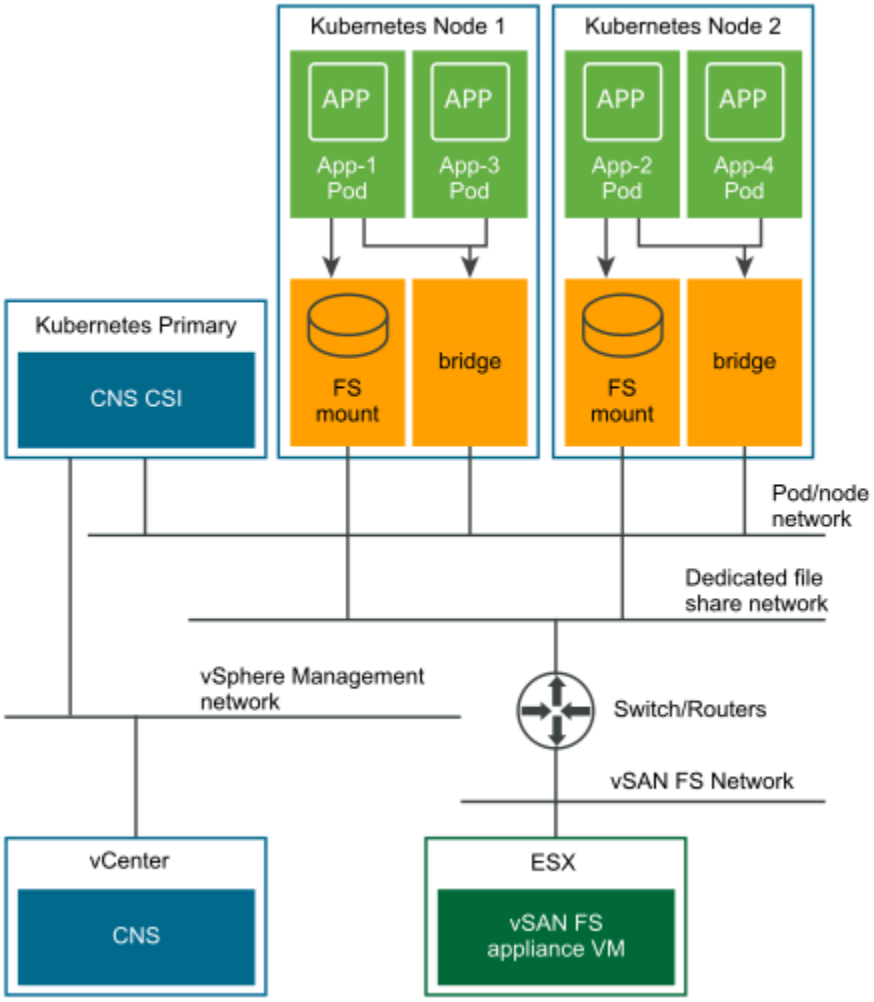
To be able to provision ReadWriteMany persistent volumes in your generic vSphere Kubernetes environment, configure necessary networks, switches, and routers from the Kubernetes nodes to the vSAN file service network.

Setting Up Network

When configuring the networks, follow these requirements:

- On every Kubernetes node, you can use a dedicated vNIC for the vSAN file share traffic. This option is required only if you want to use a secure data traffic path for your file volumes.
- If you use a dedicated vNIC, make sure that the traffic through the dedicated vNIC is routable to one or many vSAN file service networks.
- Make sure that only the guest OS on each Kubernetes node can directly access the vSAN file share through the file share IP address. The pods in the node cannot ping or access the vSAN file share by its IP address.
  CNS CSI driver ensures that only those pods that are configured to use the CNS file volume can access the vSAN file share by creating a mount point in the guest OS.
- Avoid creating an IP address clash between the node VMs and vSAN file shares.

The following illustration is an example of the CNS network configuration with the vSAN file share service.



In the illustration, the sample networking configuration follows these guidelines.

- The configuration uses separate networks for different items in the CNS environment.

Network	Description
vSphere management network	Typically, in a generic Kubernetes cluster, every node has access to this network.
Pod or node network	Kubernetes uses this network for the node to node or pod to pod communication.
Dedicated file share network	CNS file volume data traffic uses this network.
vSAN file share network	Network where the vSAN file share is enabled and where file shares are available.

- Every Kubernetes node has a dedicated vNIC for the file traffic. This vNIC is separate from the vNIC used for the node to node or pod to pod communication. This configuration is used only as an example, but is not mandatory.
- Only those applications that are configured to use the CNS file share have access to vSAN file shares through the mount point in the node guest OS. For example, in the illustration, the following takes place:
  - App-1 and App-2 pods are configured to use a file volume, and have access to the file share through the mount point created by the CSI driver.
  - App-3 and App-4 are not configured with a file volume and cannot access file shares.
- The vSAN file shares are deployed as containers in a vSAN file share appliance VM on the host. A Kubernetes deployer, which is a software or service that can configure, deploy, and manage Kubernetes clusters, configures necessary routers and switches, so that the guest OS in the Kubernetes node can access the vSAN file shares.

Security Limitations

Although the dedicated vNIC prevents an unauthorized pod from accessing the file shares directly, certain security limitations exist:

- The CNS file functionality assumes that anyone who has the CNS file volume ID is an authorized user of the volume. Any user that has the CNS file volume ID can access the data stored in the volume.
- CNS file volume supports only the AUTH_SYS authentication, which is a user ID-based authentication. To protect access to the data in the CNS file volume, you must use appropriate user IDs for the containers accessing the CNS file volume.
- An unbound ReadWriteMany persistent volume referring to a CNS file volume can be bound by a persistent volume claim created by any Kubernetes user under any namespace. Make sure that only authorized users have access to Kubernetes to avoid security issues.

Configuring the CSI Driver to Access vSAN File Service Clusters

Depending on the configuration, the CSI driver can provision file volumes on one or several vSAN clusters where the file service is enabled.

You can restrict access to only specific vSAN clusters where the file service is enabled. When deploying the Kubernetes cluster, configure the CSI driver with access to specific file service vSAN clusters. As a result, the CSI driver can provision the file volumes only on those vSAN clusters.

In the default configuration, the CSI driver uses any file service vSAN cluster available in vCenter for the file volume provisioning. The CSI driver does not verify which file service vSAN cluster is accessible while provisioning file volumes.

vSphere Cloud Native Storage Roles and Privileges

The CNS vSphere user must have specific privileges to perform operations related to Cloud Native Storage.

You can create several roles to assign sets of permissions on the objects that participate in the Cloud Native Storage environment.

> **Note:**
>
> These roles need to be created only for generic Kubernetes clusters. If you work in the vSphere Supervisor environment, use the Workload Storage Manager role for storage operations.

For more information about roles and permissions in vSphere, and how to create a role, see the *vSphere Security* documentation.

The following table lists the minimum privileges that are required for various CNS operations. Depending on your organization's security model, you can create these roles for different administrative personas. You can also consolidate the required privileges into a single custom role for a CNS service account.

Role Name	Privilege Name	Description	Required On
CNS-Datastore	Datastore > Low level file operations	Allows performing read, write, delete, and rename operations in the datastore browser.	Shared datastore where persistent volumes reside.
CNS-HOST-CONFIG-STORAGE	Host > Configuration > Storage partition configuration	Allows vSAN datastore management.	Required on a vSAN cluster with vSAN file service. Required for file volume only.
CNS-VM	Virtual machine > Change Configuration > Add existing disk	Allows adding an existing virtual disk to a virtual machine.	All cluster node VMs.
	Virtual Machine > Change Configuration > Add or remove device	Allows addition or removal of any non-disk device.	
CNS-SEARCH-AND-SPBM	CNS > Searchable	Allows storage administrator to see Cloud Native Storage UI.	Root vCenter.
	VM storage policies > View VM storage policies	Allows viewing of defined storage policies.	

Role Name	Privilege Name	Description	Required On
Read-only	Default role	Users with the Read Only role for an object are allowed to view the state of the object and details about the object. For example, users with this role can find the shared datastore accessible to all node VMs. For zone and topology-aware environments, all ancestors of node VMs, such as a host, cluster, and data center must have the Read-only role set for the vSphere user configured to use the CSI driver and CCM. This is required to allow reading tags and categories to prepare the nodes' topology.	All hosts where the nodes VMs reside Data center

Create a Storage Policy for Kubernetes in vSphere

The vSphere storage object that will back a Kubernetes containerized application should meet specific storage requirements. As a vSphere user, you create a VM storage policy based on the requirements given by the Kubernetes user.

- The Kubernetes user identifies the Kubernetes cluster where the stateful containerized application will be deployed.
- The Kubernetes user collects storage requirements for the containerized application and communicates them to the vSphere user.
- Required privileges:**VM storage policies** > **Update** and**VM storage policies** > **View**.

The storage policy will be associated with the virtual disk or vSAN file share that back the Kubernetes container.

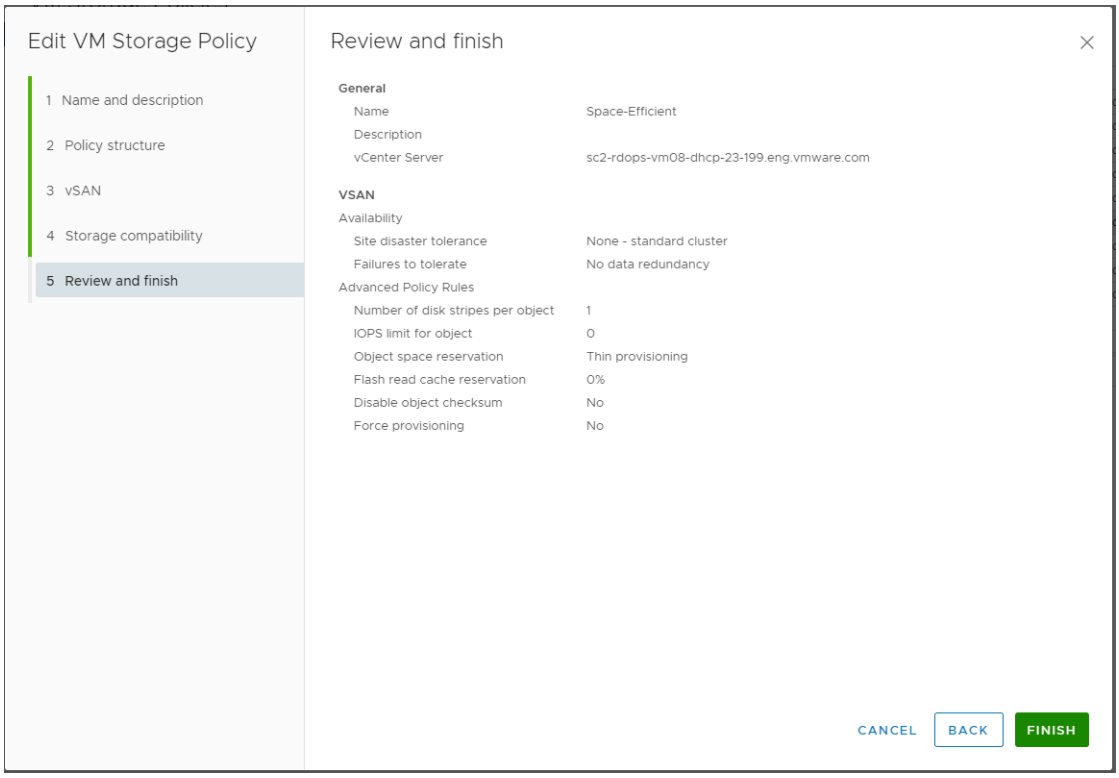
If you have multiple vCenter instances in your environment, create the VM storage policy on each instance. Use the same policy name across all instances.

If you want to create a storage policy for a datastore other than vSAN, see Creating and Managing vSphere VM Storage Policies. To use thick provisioning on a VMFS datastore, make sure to select an appropriate option in the **Volume allocation** drop-down menu when you define the policy.

1. In the vSphere Client, open the **Create VM Storage Policy** wizard.
   a) Click **Menu** > **Policies and Profiles**.
   b) Under **Policies and Profiles**, click **VM Storage Policies**.
   c) Click **Create**.
2. Enter the policy name and description, and click **Next**.

Option	Action
vCenter	Select the vCenter instance.
Name	Enter the name of the storage policy, for example `Space-Efficient`.
Description	Enter the description of the storage policy.

3. On the **Policy structure** page under Datastore-specific rules, select **Enable rules for vSAN storage** and click **Next**.
4. On the **vSAN** page, define the policy rule set and click **Next**.
   a) On the **Availability** tab, define the **Site disaster tolerance** and **Failures to tolerate**.
   b) On the **Advanced Policy Rules** tab, define advanced policy rules, such as number of disk stripes per object and flash read cache reservation.
5. On the **Storage compatibility** page, review the list of vSAN datastores that match this policy and click **Next**.
6. On the **Review and finish** page, review the policy settings, and click **Finish**.



You can now inform the Kubernetes user of the storage policy name. The VM storage policy you created will be used as a part of storage class definition for dynamic volume provisioning.

Configure Kubernetes Cluster in vSphere Virtual Machines

On each node VM, enable the `disk.EnableUUID` parameter, so that the vSphere VMs can successfully mount the virtual disks.

- Create several VMs for your Kubernetes cluster. For the VM requirements, see Requirements of vSphere Cloud Native Storage.
- Required privilege: **Virtual machine** > **Configuration** > **Settings**.

Note:
To avoid errors and unpredictable behavior, do not take snapshots of CNS node VMs.

Perform these steps for each of the VM nodes that participate in the cluster.

1. In the vSphere Client, right-click the virtual machine and select **Edit Settings**.
2. Click the **VM Options** tab and expand the **Advanced** menu.
3. Click **Edit Configuration** next to Configuration Parameters.
4. Configure the `disk.EnableUUID` parameter.

   If the parameter exists, make sure that its value is set to True. If the parameter is not present, add it and set its value to True.

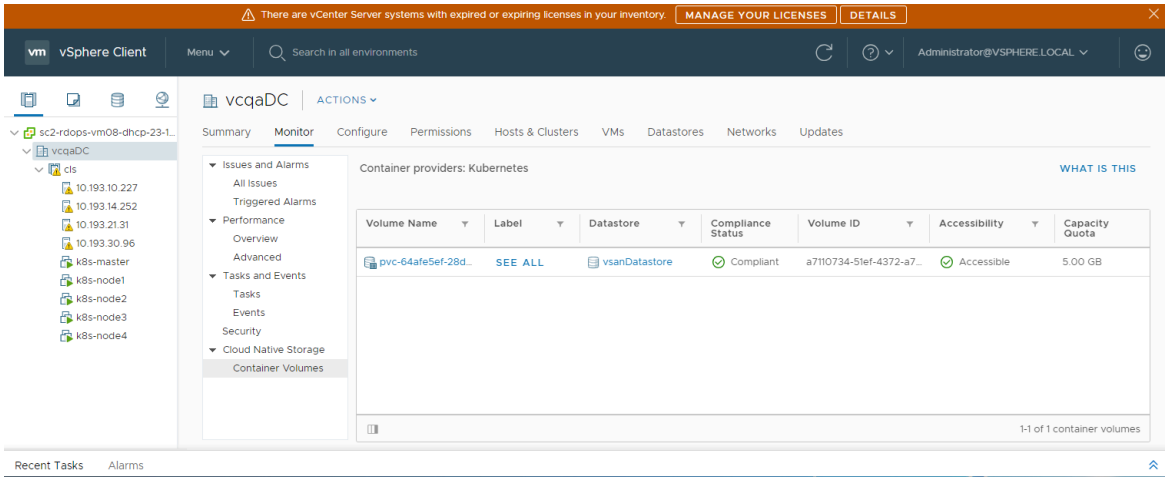
Name	Value
`disk.EnableUUID`	True

Monitor Container Volumes Across Kubernetes Clusters in vSphere

After a stateful application gets deployed in Kubernetes, the volumes and their backing vSphere storage objects become visible in the vSphere Client. You can display and monitor the volumes and troubleshoot any potential storage issues.

> **Note:**
> If you experience failures on the Kubernetes CNS server, the CNS objects in the vSphere Client might not display correctly until full synchronization takes place.

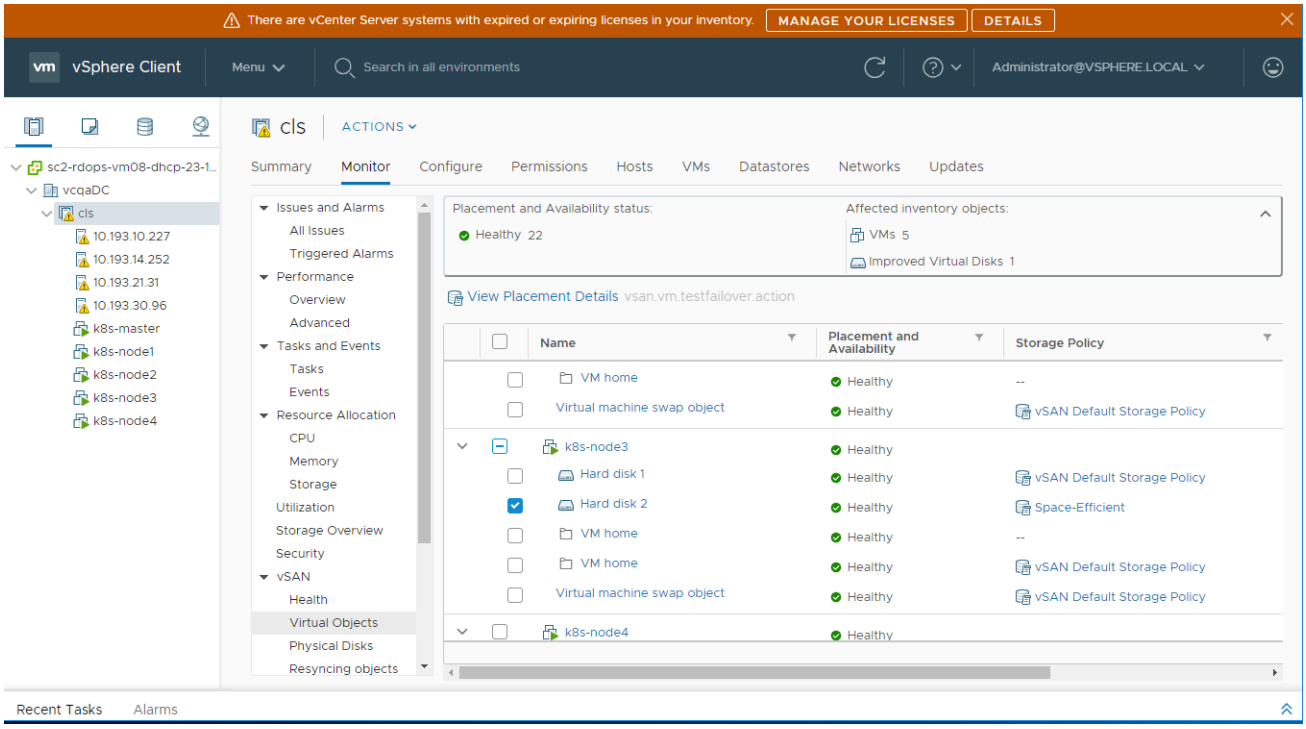
1. Navigate to the vCenter instance, a data center, or a datastore.
2. Click the **Monitor** tab and click **Container Volumes** under **Cloud Native Storage**.
3. Observe the container volumes available in your environment and monitor their storage policy compliance status.



4. Click the **SEE ALL** link in the Label column to view additional details.

   The details include the name of the PersistentVolumeClaim, StorageClass, and so on, and help you map the volume to the Kubernetes objects associated with it.
5. Click the link in the **Volume Name** column to review various components that back the volume and such details as placement, compliance, and storage policy.

> **Note:**
> The **Virtual Objects** screen is available only when the underlying datastore is vSAN.



Use Encryption with vSphere Cloud Native Storage

You can use the vSphere encryption technology to protect FCD virtual disks that back persistent volumes.

Using encryption in your vSphere environment requires some preparation, and includes setting up a trusted connection between vCenter and a key provider. vCenter can then retrieve keys from the key provider as needed. For information about components that participate in the vSphere encryption process, see vSphere Virtual Machine Encryption Components in the *vSphere Security* documentation.

1. Set up the key provider in your vSphere environment.
   For information, see Setting Up the Standard Key Provider.
2. Encrypt all node VMs on the Kubernetes cluster.
   Use the vSphere Client to perform this step.
   a) Navigate to a node VM.
   b) From the right-click menu, select **VM Policies** > **Edit VM Storage Policies**.
   c) From the **VM storage policy** drop-down menu, select **VM Encryption Policy** and click **OK**.
   To expedite the encryption process of the node VMs, you can encrypt only the VM home.
3. Create encrypted persistent volumes in the Kubernetes cluster with the vSphere CSI setup.
   a) Create a StorageClass that references the VM Encryption Storage Policy.
   Use the following YAML file as an example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
 name: encryption
provisioner: csi.vsphere.vmware.com
parameters:
 storagePolicyName: "VM Encryption Policy"
 datastore: vsanDatastore
```

b) Use the PersistentVolumeClaim to provision the persistent volume.

The PersistentVolumeClaim must include the name of the encryption storage class in the `storageClassName` field.

Migrating Container Volumes in vSphere

vSphere Cloud Native Storage supports migration of persistent container volumes between datastores.

Considerations and Limitations

When you perform the migration, keep in mind the following considerations:

- You can migrate only single block volumes.
- Make sure that original and target datastores are accessible to all nodes of the Kubernetes cluster. This applies to single zone deployments and to Kubernetes clusters deployed across multiple zones.
- If a volume is attached to a VM, the VM remains locked during volume relocation. If the volume you relocate is attached to a Kubernetes node VM, the system does not allow any other control operations on the VM. The operations include attaching or detaching other volumes to the VM, migrating the VM to a different datastore, changing any configuration of the VM, and so on.
- Cloud Native Storage does not support PMEM storage. Attempts to relocate a volume to a PMEM datastore fail.
- The target datastore must have sufficient free capacity to accommodate the migrated volume. If the free space of a datastore is less than the volume size, you cannot select the datastore for migration.
- The vSphere Client does not support automatic storage policy changes during the migration operation. However, the system doesn't prevent you from selecting a datastore with incompatible policy as a target datastore for migration.
- After the migration, you can locate the virtual disk that backs the volume in one of the following folders on a target datastore:
  - FCD directory on the datastore.
  - VM folder if the volume is attached to a VM.

For information about migrating volumes in the Supervisor environment, see Migrate Persistent Volumes.

For additional details and known issues, see the following Broadcom Knowledge Base articles:

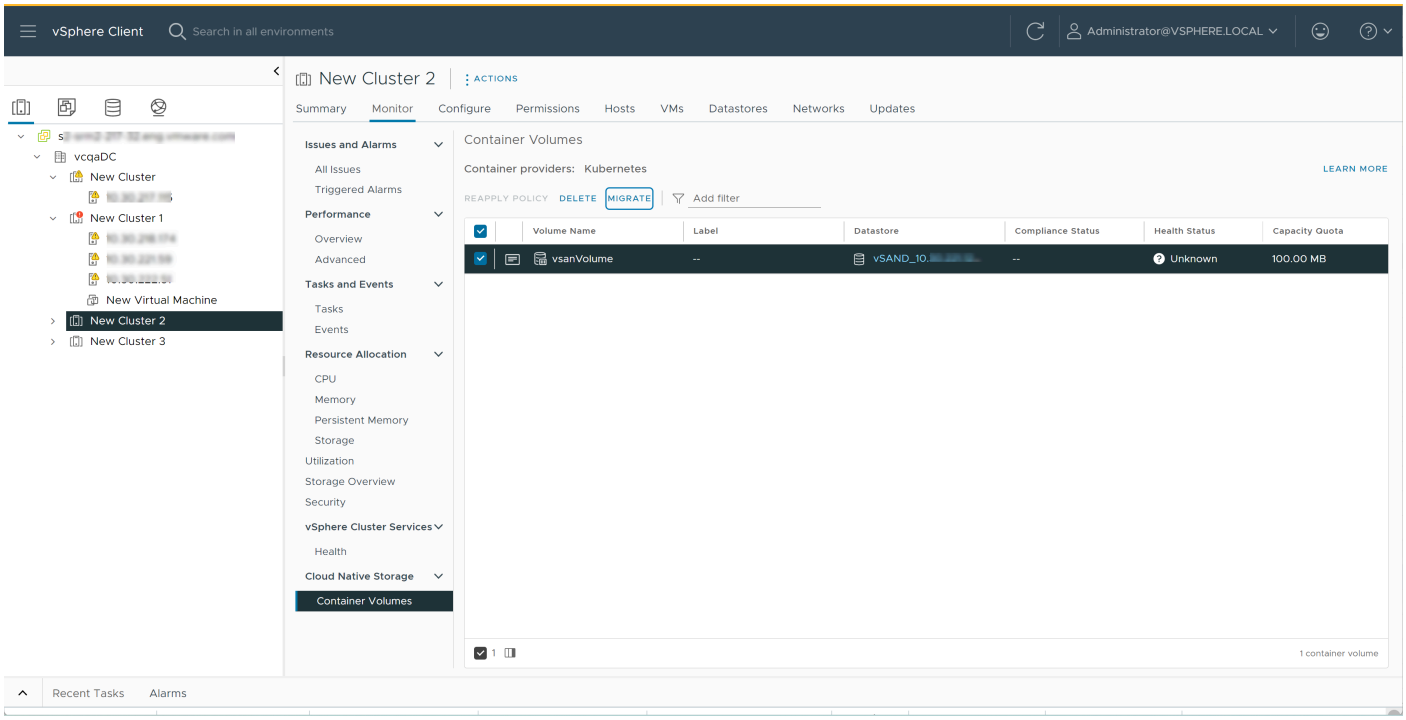
- 313416
- 313308
- 319402

Migrate Container Volumes

You can use the vSphere Client to migrate the persistent container volumes.

- Make sure that the target datastore has sufficient free capacity to accommodate the migrated volume.
- Make sure you have permissions for the target datastore.

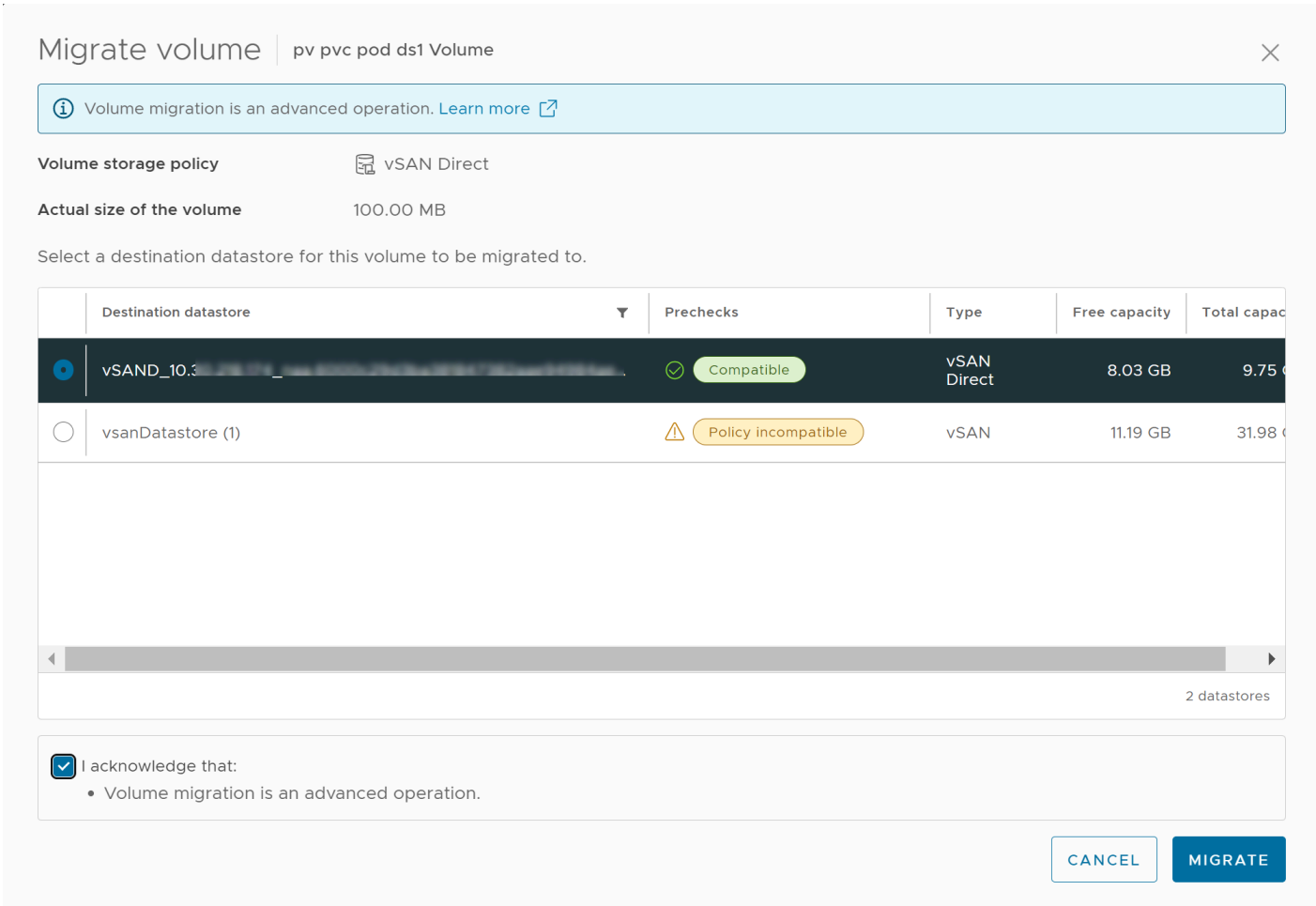
1. Navigate to the vCenter instance, a data center, a datastore, or a cluster.
2. Click the **Monitor** tab and click **Container Volumes** under **Cloud Native Storage**.
3. From the list of volumes, select a volume and click **Migrate**.



4. On the **Migrate volume** page, select the target datastore.

   The target datastore might display a warning. You can still proceed with the migration operation when the warnings include the following:

   - Policy incompatible. When the storage policy of the target datastore is not the same as the storage policy of the volume.
   - Inaccessible volume. When the volume is migrated from a datastore that is accessible from a set of hosts to a datastore that is not accessible from all hosts.



5. Select the **Acknowledge** check box and click **Migrate** to proceed with the migration operation.

vSphere Security

vSphere Security provides information about securing your vSphere® environment for VMware®vCenter® and VMware ESX.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

To help you protect your vSphere environment, this documentation describes available security features and the measures that you can take to safeguard your environment from attack.

Table 186: vSphere Security Highlights

Topics	Content Highlights
Permissions and User Management	• Permissions model (roles, groups, objects). • Creating custom roles. • Setting permissions. • Managing global permissions.
Host Security Features	• Lockdown mode and other security profile features. • Host smart card authentication. • vSphere Authentication Proxy. • UEFI Secure Boot. • Trusted Platform Module (TPM). • VMware® vSphere Trust Authority™. • Secure ESX Configuration and configuration sealing
Virtual Machine Encryption	• VMware vSphere® Native Key Provider™. • How does VM encryption work? • KMS setup. • Encrypting and decrypting VMs. • Troubleshooting and best practices.
Guest OS Security	• Virtual Trusted Platform Module (vTPM). • Virtualization Based Security (VBS).
Managing TLS Protocol Configuration	Changing TLS protocol configuration using a command-line utility.
Security Best Practices and Hardening	Best practices and advice from VMware security experts. • vCenter security • Host security • Virtual machine security • Networking security
vSphere Privileges	Complete listing of all vSphere privileges supported in this release.

Related Documentation

A companion document, *vSphere Authentication*, explains how you can use authentication services, for example, to manage authentication with vCenter Single Sign-On and to manage certificates in your vSphere environment.

In addition to these documents, VMware publishes the *vSphere Security Configuration Guide* (formerly known as the *Hardening Guide*) for each release of vSphere, accessible at https://core.vmware.com/security. The *vSphere Security Configuration Guide* contains guidelines on security settings that can or should be set by the customer, and security settings delivered by VMware that should be audited by the customer to ensure that they are still set to default.

What Happened to the Platform Services Controller

Beginning in vSphere 7.0, deploying a new vCenter or upgrading to vCenter 7.0 requires the use of the vCenter appliance, a preconfigured virtual machine optimized for running vCenter. The new vCenter contains all Platform Services Controller services, preserving the functionality and workflows, including authentication, certificate management, tags, and licensing. It is no longer necessary nor possible to deploy and use an external Platform Services Controller. All Platform Services Controller services are consolidated into vCenter, and deployment and administration are simplified.

As these services are now part of vCenter, they are no longer described as a part of Platform Services Controller. In vSphere 7.0, the *vSphere Authentication* publication replaces the *Platform Services Controller Administration* publication. The new publication contains complete information about authentication and certificate management. For information about upgrading or migrating from vSphere 6.5 and 6.7 deployments using an existing external Platform Services Controller to vSphere 7.0 using vCenter appliance, see the *vSphere Upgrade* documentation.

Intended Audience

This information is for experienced system administrators who are familiar with virtual machine technology and data center operations.

Certifications

VMware publishes a public list of VMware products that have completed Common Criteria certifications. To check if a particular VMware product version has been certified, see the Common Criteria Evaluation and Validation webpage at https://www.vmware.com/security/certifications/common-criteria.html.

Security in the vSphere Environment

The components of a vSphere environment are secured out of the box by several capabilities such as authentication and authorization, a firewall on each ESX host, and so on. You can modify the default setup in many ways. For example, you can set permissions on vCenter objects, open firewall ports, or change the default certificates. You can take security measures for different vSphere objects, for example, vCenter systems, ESX hosts, virtual machines, and network and storage objects.

A high-level overview of different areas of vSphere that require attention helps you plan your security strategy. You also benefit from other vSphere Security resources on the Broadcom website.

Securing the ESX Hypervisor

vSphere ESX is a lightweight, stateless architecture which minimizes the attack surface by eliminating a traditional OS console. You can further protect ESX hosts by using lockdown mode and other built-in features. For consistency, you can set up a reference host and can keep all hosts in sync with the host profile of the reference host. You can also protect your environment by performing scripted management, which ensures that changes apply to all hosts.

You can enhance protection of ESX hosts that are managed by vCenter with the following actions. Security considerations for standalone hosts are similar, though the management tasks might differ. See the *vSphere Single Host Management - VMware Host Client* documentation.

Limit ESX Access

By default, the ESX Shell and the SSH services are not running and only the root user can log in to the Direct Console User Interface (DCUI). If you decide to enable ESX Shell or SSH access, you can set timeouts to limit the risk of unauthorized access. Users who can access the ESX host must have permissions to manage the host. You set permissions on the host object from the vCenter system that manages the host.

See Using the ESX Shell.

Use Named Users and Least Privilege

By default, the root user can perform many tasks. Do not allow administrators to log in to the ESX host using the root user account. Instead, create named administrator users from vCenter and assign the Administrator role to those users. You can also assign a custom role to those users. See Create a vCenter Custom Role.

If you manage users directly on the host, role management options are limited. See the *vSphere Single Host Management - VMware Host Client* documentation.

Minimize the Number of Open ESX Firewall Ports

By default, firewall ports on your ESX host are opened only when you start a corresponding service. You can use the vSphere Client, ESXCLI, or PowerCLI commands to check and manage firewall port status.

See Configuring the ESX Firewall.

Automate ESX Host Management

Because it is often important that different hosts in the same data center are in sync, use scripted installation, vSphere Lifecycle Manager, or vSphere Auto Deploy (deprecated in 9.0) to provision hosts. You can manage the hosts using scripts. vSphere Configuration Profiles are an alternative to scripted management. You set up a cluster-level host profile that allows you to remediate hosts in the cluster that do not match the profile.

- See Use Scripts to Manage ESX Host Configuration Settingsfor more information on scripted host management.
- See What is vSphere Lifecycle Manager for more information on vSphere Lifecycle Manager.
- See What is vSphere Configuration Profiles for more information on vSphere Configuration Profiles.
- See Introduction to vSphere Auto Deploy for more information on Auto Deploy.

Configure ESX Lockdown Mode

In lockdown mode, ESX hosts can be accessed only through vCenter by default. You can select strict lockdown mode or normal lockdown mode. You can define Exception Users to allow direct access to service accounts such as backup agents.

See Configuring and Managing Lockdown Mode on ESX Hosts.

Check VIB Package Integrity

Each vSphere Installation Bundle (VIB) package has an associated acceptance level. You can add a VIB to an ESX host only if the VIB acceptance level is the same or better than the acceptance level of the host. You cannot add a CommunitySupported or PartnerSupported VIB to a host unless you explicitly change the acceptance level of the host.

See Manage the Acceptance Levels of ESX Hosts and vSphere Installation Bundles.

Manage ESX Certificates

The VMware Certificate Authority (VMCA) provisions each ESX host with a signed certificate that has VMCA as the root certificate authority by default. If your company policy requires it, you can replace the existing certificates with certificates that are signed by a third-party or an enterprise certificate authority.

See Managing Certificates for ESX Hosts.

Consider ESX Account Lockout

Account locking is supported for access through SSH and through the vSphere Web Services SDK. By default, a maximum of five failed attempts is allowed before the account is locked. The account is unlocked after 15 minutes by default.

> **Note:**
> The Direct Console Interface (DCUI) and the ESX Shell do not support account lockout.

See ESX Passwords and Account Lockout.

Securing vCenter Systems and Associated Services

Authentication through vCenter Single Sign-On, and authorization through the vCenter permissions model, protects your vCenter system and associated services. You can modify the default behavior, and you can take steps to limit access to your environment.

As you protect your vSphere environment, consider that all services that are associated with the vCenter instances must be protected. In some environments, you might protect several vCenter instances.

vCenter Uses Encrypted Communication

By default ("out of the box"), all data communication between the vCenter system and the other vSphere components is encrypted. In some cases, depending on how you configure your environment, some traffic might be unencrypted. For example, you can configure unencrypted SMTP for email alerts and unencrypted SNMP for monitoring. DNS traffic is also unencrypted. vCenter listens on port 80 (TCP) and port 443 (TCP). Port 443 (TCP) is the industry-standard HTTPS (secure HTTP) port and uses TLS encryption for protection. See vSphere TLS Configuration. Port 80 (TCP) is the industry-standard HTTP port and does not use encryption. The purpose of port 80 is to redirect requests from port 80 to port 443, where they are secure.

Harden vCenter Systems

The first step in protecting your vCenter environment is hardening each machine on which vCenter or an associated service runs. Similar considerations apply to a physical machine or a virtual machine. Always install the latest security patches for your operating system and follow industry standard best practices to protect the host machine.

Learn About the vSphere Certificate Model

By default, the VMware Certificate Authority (VMCA) provisions each ESX host and each machine in the environment with a certificate signed by VMCA. If your company policy requires it, you can change the default behavior. See the *vSphere Authentication* documentation for details.

For additional protection, explicitly remove expired or revoked certificates and failed installations.

Configure vCenter Single Sign-On

vCenter and associated services are protected by the vCenter Single Sign-On authentication framework. When you first install the software, you specify a password for the administrator of the vCenter Single Sign-On domain, administrator@vsphere.local by default. Only that domain is initially available as an identity source. You can add an external identity provider, such as Microsoft Active Directory Federation Services (AD FS), for federated authentication. You can add other identity sources, either Active Directory or LDAP, and set a default identity source. Users who can authenticate to one of those identity sources can view objects and perform tasks if they are authorized to do so. See the *vSphere Authentication* documentation for details.

> **Note:**
> You are encouraged to use federated authentication as vSphere moves towards token-based authentication. vCenter continues to have local accounts, for administrative access and error recovery.

Assign vCenter Roles to Named Users or Groups

For better logging, associate each permission that you give on an object with a named user or group and a predefined role or custom role. The vSphere permissions model allows great flexibility through multiple ways of authorizing users or groups. See Understanding Authorization in vSphere and Required vCenter Privileges for Common Tasks.

Restrict administrator privileges and the use of the administrator role. If possible, assign administrator privileges to specific named users rather than allowing users to log on as the Administrator user.

Set up Precision Time Protocol or Network Time Protocol

Set up Precision Time Protocol (PTP) or Network Time Protocol (NTP) for each node in your environment. VMware Certificate Authority requires an accurate time stamp and does not work correctly if the nodes are out of sync.

See Synchronizing Clocks on the vSphere Network.

Securing Virtual Machines

To secure your virtual machines, keep the guest operating systems patched and protect your virtual environment just as you protect your physical machine. Consider deactivating unnecessary functionality, minimize the use of the virtual machine console, and follow other best practices.

Protect the Guest Operating System

To protect your guest operating system, make sure that it uses the most recent patches and, if appropriate, anti-spyware and anti-malware applications. See the documentation from your guest operating system vendor and, potentially, other information available in books or on the Internet for that operating system.

Deactivate Unnecessary Virtual Machine Functionality

Check that unnecessary functionality is deactivated to minimize potential points of attack. Many of the features that are used infrequently are deactivated by default. Remove unnecessary hardware and deactivate certain features such as host-guest filesystem (HGFS) or copy and paste between the virtual machine and a remote console.

See Deactivate Unnecessary Functions Inside Virtual Machines.

Use Virtual Machine Templates and Scripted Management

Virtual machine templates enable you to set up the operating system so that it meets your requirements, and to create other virtual machines with the same settings.

If you want to change virtual machine settings after initial deployment, consider using PowerCLI scripts. For the most part, this documentation explains how to perform tasks using the vSphere Client. Consider using scripts or VCF Automation instead of the vSphere Client to keep your environment consistent. In large environments, you can group virtual machines into folders to optimize scripting.

For information on templates, see Use Templates to Deploy Virtual Machines and the *vSphere Virtual Machine Administration* documentation. For information on PowerCLI, see the VMware PowerCLI documentation.

Minimize Use of the Virtual Machine Console

The virtual machine console provides the same function for a virtual machine that a monitor on a physical server provides. Users with access to a virtual machine console have access to virtual machine power management and to removable device connectivity controls. As a result, virtual machine console access might allow a malicious attack on a virtual machine.

Consider UEFI Secure Boot for Virtual Machines

You can configure your virtual machines to use UEFI boot. If the operating system supports secure UEFI boot, you can select that option for your virtual machines for additional security. See Activate or Deactivate UEFI Secure Boot for a Virtual Machine.

Securing the Virtual Networking Layer

The virtual networking layer includes virtual network adapters, virtual switches, distributed virtual switches, and ports and port groups. ESX relies on the virtual networking layer to support communications between virtual machines and their users. In addition, ESX uses the virtual networking layer to communicate with iSCSI SANs, NAS storage, and so on.

vSphere includes the full array of features necessary for a secure networking infrastructure. You can secure each element of the infrastructure, such as virtual switches, distributed virtual switches, and virtual network adapters, separately. In addition, consider the following guidelines, discussed in more detail in Securing vSphere Networking.

Isolate Network Traffic

Isolation of network traffic is essential to a secure ESX environment. Different networks require different access and level of isolation. A management network isolates client traffic, command-line interface (CLI) or API traffic, and third-party

software traffic from normal traffic. Ensure that the management network is accessible only by system, network, and security administrators.

See ESX Networking Security Recommendations.

Use Firewalls to Secure Virtual Network Elements

You can open and close firewall ports and secure each element in the virtual network separately. For ESX hosts, firewall rules associate services with corresponding firewalls and can open and close the firewall according to the status of the service.

You can also open ports on vCenter instances explicitly.

For the list of all supported ports and protocols in VMware products, including vSphere and vSAN, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/. You can search ports by VMware product, create a customized list of ports, and print or save port lists.

Consider Network Security Policies

Network security policies provide protection of traffic against MAC address impersonation and unwanted port scanning. The security policy of a standard or distributed switch is implemented in Layer 2 (Data Link Layer) of the network protocol stack. The three elements of the security policy are promiscuous mode, MAC address changes, and forged transmits.

See the *vSphere Networking* documentation for instructions.

Secure Virtual Machine Networking

The methods that you use to secure virtual machine networking depend on several factors, including:

- The guest operating system that is installed
- Whether the virtual machines operate in a trusted environment

Virtual switches and distributed virtual switches provide significant protection when used with other common security practices, such as installing firewalls.

See Securing vSphere Networking.

Consider VLANs to Protect Your Environment

ESX supports IEEE 802.1q VLANs. VLANs let you segment a physical network. You can use VLANs to further protect the virtual machine network or storage configuration. When you use VLANs, two virtual machines on the same physical network cannot send packets to or receive packets from each other unless they are on the same VLAN. VLANs can protect your environment from broadcast storms by segmenting a single, large physical network into smaller, isolated logical broadcast domains. A VLAN can ensure that broadcast traffic such network loops are restricted to devices within the same VLAN, preventing a storm from overwhelming the entire network. VLANs may prevent unauthorized users from accessing restricted regions which can provide a strong security foundation for safeguarding valuable data.

See Securing Virtual Machines with VLANs.

Secure Connections to Virtualized Storage

A virtual machine stores operating system files, application files, and other data on a virtual disk. Each virtual disk appears to the virtual machine as a SCSI drive that is connected to a SCSI controller. A virtual machine is isolated from storage details and cannot access the information about the LUN where its virtual disk resides.

The Virtual Machine File System (VMFS) is a distributed file system and volume manager that presents virtual volumes to the ESX host. You are responsible for securing the connection to storage. For example, if you are using iSCSI storage, you can set up your environment to use Challenge Handshake Authentication Protocol (CHAP). If required by company policy, you can set up mutual CHAP. Use the vSphere Client or CLIs to set up CHAP.

See Storage Security Best Practices.

Evaluate the Use of Internet Protocol Security

ESX supports Internet Protocol Security (IPSec) over IPv6. You cannot use IPSec over IPv4.

See Using Internet Protocol Security on ESX Hosts.

Securing Passwords in Your vSphere Environment

Password restrictions, password expiration, and account lockout in your vSphere environment depend on the system that the user targets, who the user is, and how policies are set.

ESX password restrictions are determined by certain requirements. See ESX Passwords and Account Lockout.

vCenter Single Sign-On manages authentication for all users who log in to vCenter and other vCenter services. The password restrictions, password expiration, and account lockout depend on the domain of the user and on who the user is.

Password for the vCenter Single Sign-On Administrator

The password for the administrator@vsphere.local user, or the administrator@*mydomain* user if you selected a different domain during installation, does not expire and is not subject to the lockout policy. In all other regards, the password must follow the restrictions that are set in the vCenter Single Sign-On password policy. See the *vSphere Authentication* documentation for details.

If you forget the password for this user, search the VMware Knowledge Base system for information on resetting this password. The reset requires additional privileges such as root access to the vCenter system.

Passwords for Other Users of the vCenter Single Sign-On Domain

Passwords for other vsphere.local users, or users of the domain that you specified during installation, must follow the restrictions that are set by the vCenter Single Sign-On password policy and lockout policy. See the *vSphere Authentication* documentation for details. These passwords expire after 90 days by default. Administrators can change the expiration as part of the password policy.

If you forget your vsphere.local password, an administrator user can reset the password using the `dir-cli` command.

Passwords for Users from Other Identity Sources

Password restrictions, password expiration, and account lockout for all other users are determined by the domain (identity source) to which the user can authenticate.

vCenter Single Sign-On supports one default identity source. Users can log in to the corresponding domain with the vSphere Client with their user names. If users want to log in to a non-default domain, they can include the domain name, that is, specify *user*@*domain* or *domain**user*. The domain password parameters apply to each domain.

Passwords for vCenter Direct Console User Interface Users

The vCenter appliance is a preconfigured virtual machine that is optimized for running vCenter and the associated services.

When you deploy vCenter, you specify these passwords.

- Password for the root user.
- Password for the administrator of the vCenter Single Sign-On domain, administrator@vsphere.local by default.

You can change the root user password and perform other vCenter local user management tasks from the vCenter Management Interface. See the *vCenter Configuration* documentation.

vCenter and ESX Security Best Practices and Resources

If you follow best practices, your ESX hosts and vCenter systems can be as secure as or even more secure than an environment that does not include virtualization.

This manual includes best practices for the different components of your vSphere infrastructure. This manual is only one of the sources you must use to ensure a secure environment.

vSphere Security Resources

To learn more about specific aspects of vSphere security, use the following content in this manual.

Table 187: Security Best Practices

vSphere component	Resource
ESX host	Securing ESX Hosts
vCenter system	Securing vCenter Server Systems
Virtual machine	Virtual Machine Security Best Practices
vSphere Networking	vSphere Networking Security Best Practices

VMware Security Resources on the Web

VMware security resources, including security alerts and downloads, are available on the web.

Table 188: VMware Security Resources on the Web

Topic	Resource
Information about ESX and vCenter security and operations, including secure configuration and hypervisor security.	https://www.vmware.com/resources/resource-center?product_family[]=Cloud%20Infrastructure&term[]=security
Corporate security response policy	https://www.broadcom.com/support/vmware-services/security-response VMware is committed to helping you maintain a secure environment. Security issues are corrected in a timely manner. The VMware Security Response Policy states our commitment to resolve possible vulnerabilities in our products.
Third-party software support policy	https://compatibilityguide.broadcom.com/ VMware supports a variety of storage systems, software agents such as backup agents, system management agents, and plug-ins. You can find lists of agents, tools, and other software that support ESX in the ESX compatibility guides. The industry offers more products and configurations than VMware can test. If VMware does not list a product or configuration in a compatibility guide, Technical Support attempts to help you with any problems, but cannot guarantee that the product or configuration can be used. Always evaluate security risks for unsupported products or configurations carefully.
Compliance and security standards, and partner solutions and in-depth content about virtualization and compliance	https://github.com/vmware/vcf-security-and-compliance-guidelines
Security configuration guides (formerly known as hardening guides) for different versions of vSphere and other VMware products.	ESX Security Configuration Guide

vSphere Permissions and User Management Tasks

Authentication and authorization govern access to your vSphere environment. vCenter Single Sign-On supports authentication, which means it determines whether a user can log in to vSphere components at all. Each user must also be authorized to view or manipulate vSphere objects.

For an overview of assigning roles and permissions using the vSphere Client, watch the following video.

vCenter allows granular control over authorization with permissions and roles. When you assign a permission to an object in the vCenter object hierarchy, you specify which user or group has which privileges on that object. To specify the privileges, you use roles. A role is a predefined set of privileges. Privileges define rights to perform actions and read properties.

Initially, only the administrator user for the vCenter Single Sign-On domain is authorized to log in to the vCenter system. The default domain is vsphere.local and the default administrator is administrator@vsphere.local. You can change the default domain during installation of vSphere.

As an administrator user, you can:

1. Add an identity source in which users and groups are defined to vCenter Single Sign-On. See the *vSphere Authentication* documentation.
2. Give privileges to a user or group by selecting an object such as a virtual machine or a vCenter system and assigning a role on that object for the user or group.

Understanding Authorization in vSphere

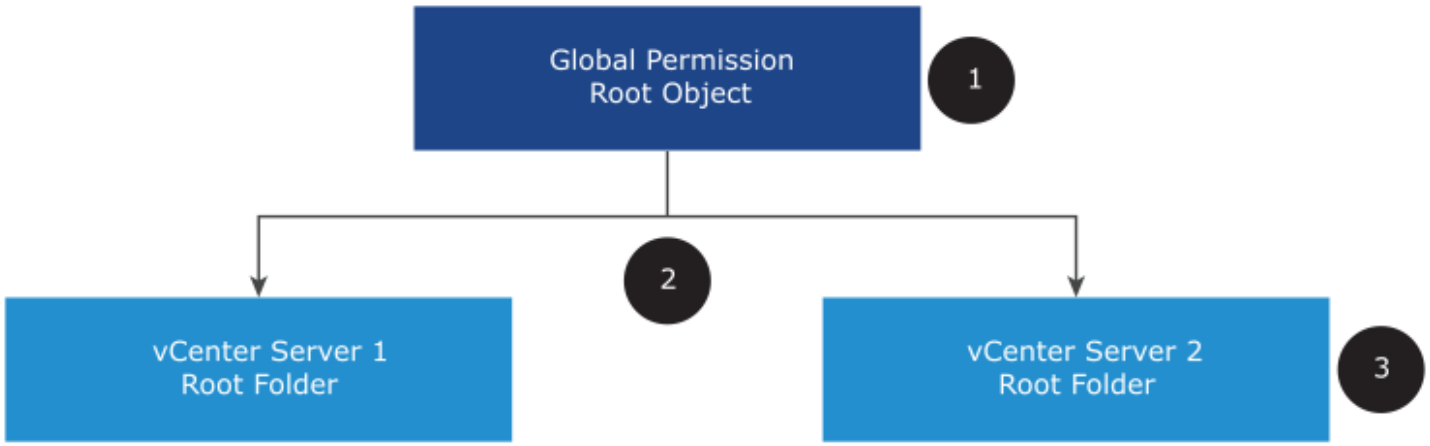
vSphere supports several models for determining whether a user is allowed to perform a task. Group membership in a vCenter Single Sign-On group decides what you are allowed to do. Your role on an object or your global permission determines whether you are allowed to perform other tasks.

How Permissions Work in vSphere

vSphere allows privileged users to give other users permissions to perform tasks. You can use global permissions, or you can use local vCenter permissions to authorize other users for individual vCenter instances.

The following figure illustrates how global and local permissions work.

Figure 87: Global Permissions and Local Permissions



In this figure:

1. You assign a global permission at the root object level with "Propagate to children" selected.
2. vCenter propagates the permissions to the vCenter 1 and vCenter 2 object hierarchies in the environment.
3. A local permission on the root folder on vCenter 2 overrides the global permission.

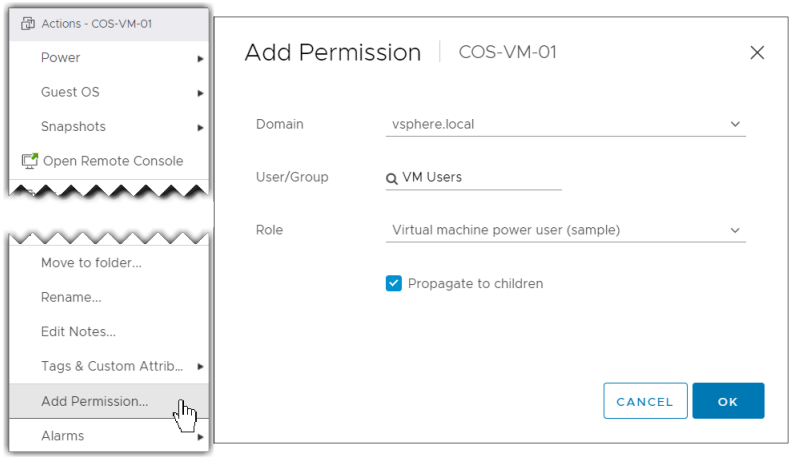
vCenter Permissions

The permission model for vCenter systems relies on assigning permissions to objects in the object hierarchy. Users get permissions in the following ways.

- From a specific permission for the user or from the groups that the user is a member of
- From a permission on the object or through the permission inheritance from a parent object

Each permission gives one user or group a set of privileges, that is, a role for a selected object. You can use the vSphere Client to add permissions. For example, you can right-click a virtual machine, select **Add Permission**, and complete the dialog box to assign a role to a group of users. That role gives those users the corresponding privileges on the virtual machine.

Figure 88: Adding Permissions to a Virtual Machine Using the vSphere Client



Global Permissions	Global permissions give a user or group privileges to view or manage all objects in each of the inventory hierarchies of the solutions in the deployment. That is, global permissions are applied to a global root object that spans solution inventory hierarchies. (Solutions include vCenter, VMware Cloud Foundation Operations orchestrator, and so on.) Global permissions also apply to global objects such as tags and content libraries. For example, consider a deployment that consists of two solutions, vCenter and VMware Cloud Foundation Operations orchestrator. You can use global permissions to assign a role to a group of users that has read-only privileges to all objects in both the vCenter and VMware Cloud Foundation Operations orchestrator object hierarchies. Global permissions are replicated across the vCenter Single Sign-On domain (vsphere.local by default). Global permissions do not provide authorization for services managed through the vCenter Single Sign-On domain groups. See Using vCenter Global Permissions. Global permissions are not replicated if your environment includes an on-premises vCenter and a vCenter in the cloud.
Group Membership in vCenter Single Sign-On Groups	For VMware Cloud on AWS, a Cloud Administrator group is predefined in vCenter Single Sign-On. When you use Hybrid Linked mode, you add this group to the linked domain. Members of a vCenter Single Sign-On domain group can perform certain tasks. For example, you can perform license management if you are a member of the LicenseService.Administrators group. See the *vSphere Authentication* documentation.
ESX Local Host Permissions	If you are managing a standalone ESX host that is not managed by a vCenter system, you can assign one of the predefined roles to users. See the *vSphere Single Host Management - VMware Host Client* documentation. For managed hosts, assign roles to the ESX host object in the vCenter inventory.

Understanding the Object-Level Permission Model

You authorize a user or group to perform tasks on vCenter objects by using permissions on the object. From a programmatic standpoint, when a user tries to perform an operation, an API method is executed. vCenter checks the permissions for that method to see if the user is authorized to perform the operation. For example, when a user tries to add a host, the `AddStandaloneHost_Task` method is invoked. This method requires that the role for the user has the Host.Inventory.AddStandaloneHost privilege. If the check does not find this privilege, the user is denied permission to add the host.

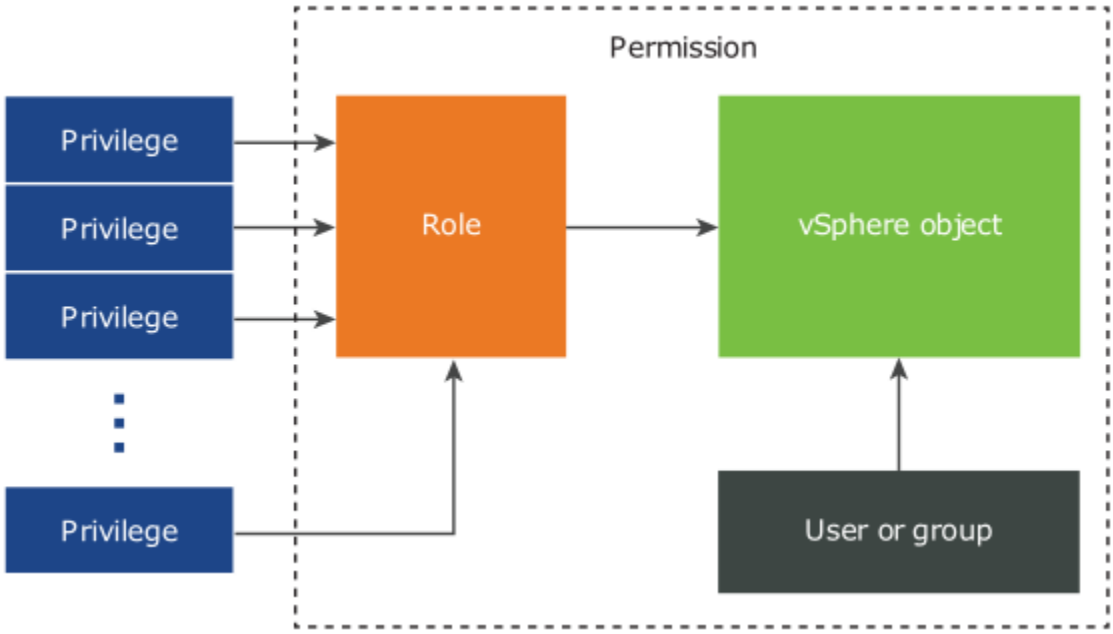
The following concepts are important.

Permissions	Each object in the vCenter object hierarchy has associated permissions. Each permission specifies for one group or user which privileges that group or user has on the object. Permissions can propagate to child objects.
Users and Groups	OnvCenter systems, you can assign privileges only to authenticated users or groups of authenticated users. Users are authenticated throughvCenter Single Sign-On. Users and groups must be defined in the identity source thatvCenter Single Sign-On uses to authenticate. Define users and groups using the tools in your identity source, for example, Active Directory.

Privileges	Privileges are fine-grained access controls. You can group those privileges into roles, which you can then map to users or groups.
Roles	Roles are sets of privileges. Roles allow you to assign permissions on an object based on a typical set of tasks that users perform. System roles, such as Administrator, are predefined onvCenter and cannot be changed.vCenter also provides some default sample roles, such as Resource Pool Administrator, that you can modify. You can create custom roles either from scratch or by cloning and modifying sample roles. See Create a vCenter Custom Role.

The following figure illustrates how a permission is constructed from privileges and roles, and assigned to a user or group for a vSphere object.

Figure 89: vSphere Permissions



To assign permissions to an object, you follow these steps:

1. Select the object to which you want to apply the permission in the vCenter object hierarchy.
   In VMware Cloud on AWS you cannot change permissions on objects that VMware manages for you, for example, the vCenter instance or ESX hosts.
2. Click the **Permissions** tab, and then click the **+** icon.
3. Select the group or user that should have privileges on the object.
4. Select individual privileges or a role, that is a set of privileges, that the group or user should have on the object.
   By default, Propagate to children is not selected. You must select the checkbox for the group or user to have the selected role on the selected object and its child objects.

vCenter offers sample roles, which combine frequently used privileges. To suit the access control needs of your environment, you can also create vCenter custom roles. You can create a role or clone an existing role.

Permissions must often be defined on both a source object and a destination object. For example, if you move a virtual machine, you need privileges on that virtual machine, but also privileges on the destination data center.

See the following information.

To find out about...	See...
Creating custom roles.	Create a vCenter Custom Role
All privileges and the objects to which you can apply the privileges	Defined Privileges
Sets of privileges that are required on different objects for different tasks.	Required vCenter Privileges for Common Tasks

The permissions model for standalone ESX hosts is simpler. See Assigning Privileges for ESX Hosts.

What Is vCenter User Validation

vCenter systems that use a directory service regularly validate users and groups against the user directory domain. Validation occurs at regular intervals specified in the vCenter settings. For example, assume that user Smith is assigned a role on several objects. If user Smith is removed from the domain, all permissions associated with that user are removed when the next validation occurs. If a new user Smith is added to the domain before the next validation occurs, the new user Smith replaces the old user Smith in permissions on any object.

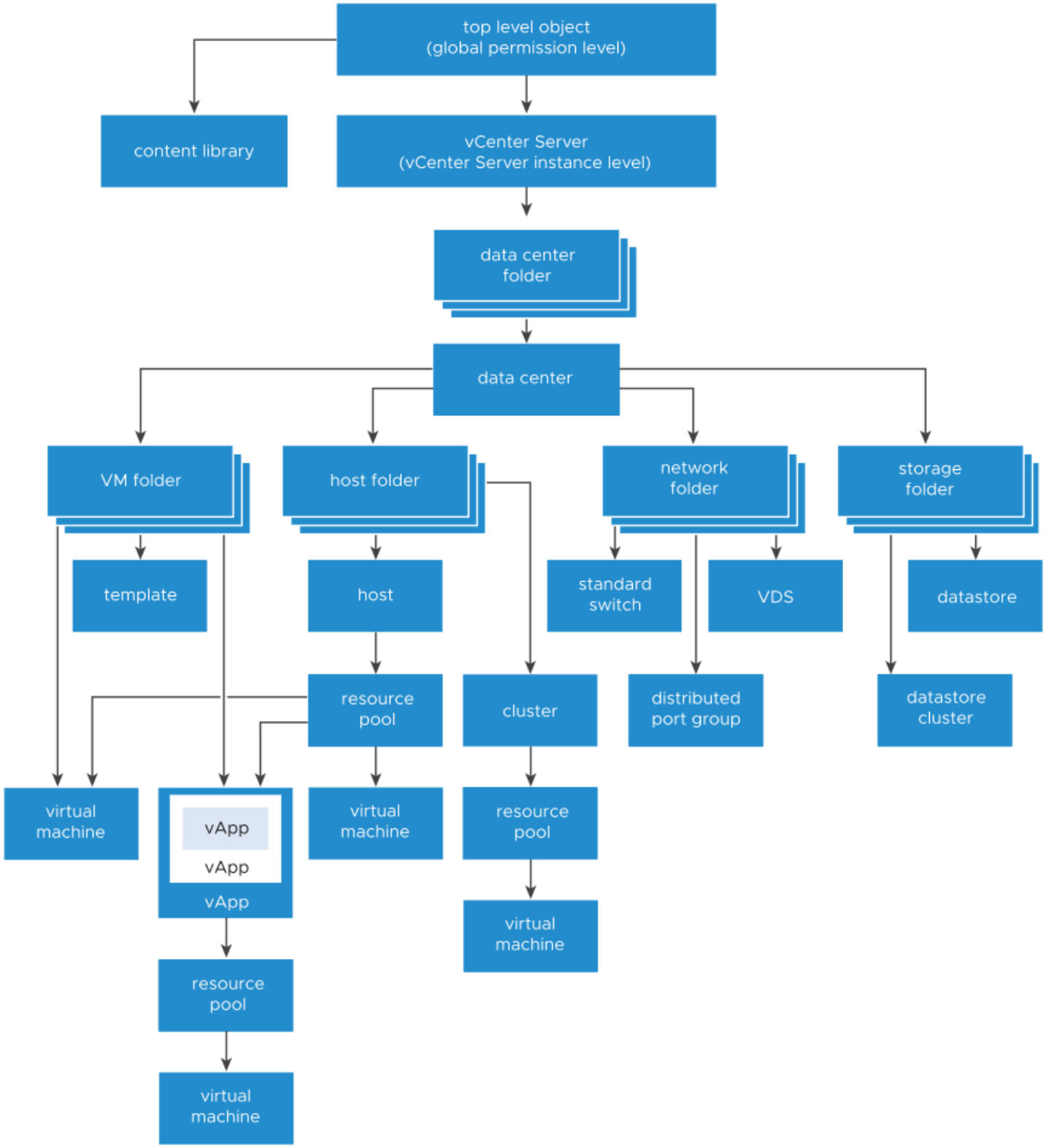
Hierarchical Inheritance of Permissions in vSphere

When you assign a permission to an object, you can choose whether the permission propagates down the object hierarchy. You set propagation for each permission. Propagation is not universally applied. Permissions defined for a child object always override the permissions that are propagated from parent objects.

The following figure illustrates the inventory hierarchy and the paths by which permissions can propagate.

> **Note:**
> Global permissions support assigning privileges across solutions from a global root object. See Using vCenter Global Permissions.

Figure 90: vSphere Inventory Hierarchy



About this figure:

- You cannot set direct permissions on the VM, host, network, and storage folders. That is, these folders act as containers, and as such are not visible to users.
- You cannot set permissions on standard switches.

Note:
To be able to set and propagate permissions to children on a vSphere Distributed Switch (VDS), the switch object must reside in a network folder created on the data center.

Most inventory objects inherit permissions from a single parent object in the hierarchy. For example, a datastore inherits permissions from either its parent datastore folder or parent data center. Virtual machines inherit permissions from both the parent virtual machine folder and the parent host, cluster, or resource pool simultaneously.

For example, you can set permissions for a distributed switch and its associated distributed port groups, by setting permissions on a parent object, such as a folder or data center. You must also select the option to propagate these permissions to child objects.

Permissions take several forms in the hierarchy.

Managed Entities

Managed entities refer to the following vSphere objects. Managed entities offer specific operations that vary depending on the entity type. Privileged users can define permissions on managed entities. See the vSphere API documentation for more information about vSphere objects, properties, and methods.

- Clusters
- Data centers
- Datastores
- Datastore clusters
- Folders
- Hosts
- Networks (except vSphere Distributed Switches)
- Distributed port groups
- Resource pools
- Templates
- Virtual machines
- vSphere vApps

Global Entities

You cannot modify permissions on entities that derive permissions from the root vCenter system.

- Custom fields
- Licenses
- Roles
- Statistics intervals
- Sessions

vCenter Management APIs for Authorization

Starting with 9.0, vCenter supports REST APIs for authorization in the vCenter instance.

vCenter authorization APIs completely manage and monitor the state of authorization configuration of the vCenter using a single endpoint. The new set of authorization APIs allows you to list and search privileges, and allows complete management of roles, global and local permissions including, listing, creating, deleting and updating them. The new set of REST APIs are available in addition to the existing authorization APIs supported by vCenter.

Privileges API

This API provides query operations for authorization privileges in the vCenter.

In vSphere, privileges are fine-grained access controls that can be grouped into roles and map them to users or groups. For example, privileges include the ability to create and delete virtual machines, change the configuration of a virtual machine, and power on and off a virtual machine.

The Privileges API provides methods to list the authorization privileges in the vCenter and to get details about a specific authorization privilege based on the privilege ID. For more information, see vCenter Autorization Privileges API.

Roles API

A role is a group of privileges brought together by a common purpose. The purpose of the role is to provide all required privileges in the system in order to administer tags.

The Roles API provides methods to list authorization roles, get details of a specific role, and create, update, and delete roles. For more information, see vCenter Authorization Roles API.

Permissions API

The permission is the combination between a specific object, an authorization role and a principal (user or group). The permission object grants a specific set of rights to a specific principal to invoke specific APIs. This is the primary mechanism used to control the access to our APIs.

The goal of this API is to allow granting both global and inventory permissions for any object with DynamicID that supports it.

To assign an inventory permission, you need to provide the DynamicID for the target inventory object. A DynamicID consists of two parts: `type` and `id`.

For inventory objects, the `type` field can either be `ManagedEntity`, which applies broadly to all inventory objects, or a specific object type. The supported object types are:

- `ClusterComputeResource`
- `ComputeResource`
- `Datacenter`
- `Datastore`
- `DistributedVirtualSwitch`
- `DistributedPortGroup`
- `Folder`
- `HostSystem`
- `Network`
- `OpaqueNetwork`
- `ResourcePool`
- `StoragePod`
- `VirtualApp`
- `VirtualMachine`

The `id` field should be set to the Managed Object ID (moID) of the inventory object.

To create a global permission, you also need to provide `type` and `id`. These values are the same for all global permission requests: `"type": "GlobalAcl"` and `"id": "GlobalAcl"`.

The Permissions API provides methods to list authorization permissions, get information about a specific authorization permission, and create, update, and delete permissions. For more information, see vCenter Authorization Permissions APIs.

How Do Multiple Permission Settings Work in vSphere

Objects can have multiple permissions, but only one permission for each user or group. For example, one permission might specify that GroupAdmin has the Administrator role on an object. Another permission might specify that the GroupVMAdmin has the Virtual Machine Administrator role on the same object. However, the GroupVMAdmin group cannot have another permission for the same GroupVMAdmin on this object.

A child object inherits the permissions of its parent if the parent's propagate property is set to true. A permission that is set directly on a child object overrides the permission in the parent object. See Example 2: Child Permissions Overriding Parent Permissions.

If multiple group roles are defined on the same object, and a user belongs to two or more of those groups, two situations are possible:

- No permission for the user is defined directly on the object. In that case, the user gets the union of the permissions that the groups have on the object.
- A permission for the user is defined directly on the object. In that case, the permissions for the user take precedence over all group permissions.

Example 1: Permission Inheritance from Multiple Groups

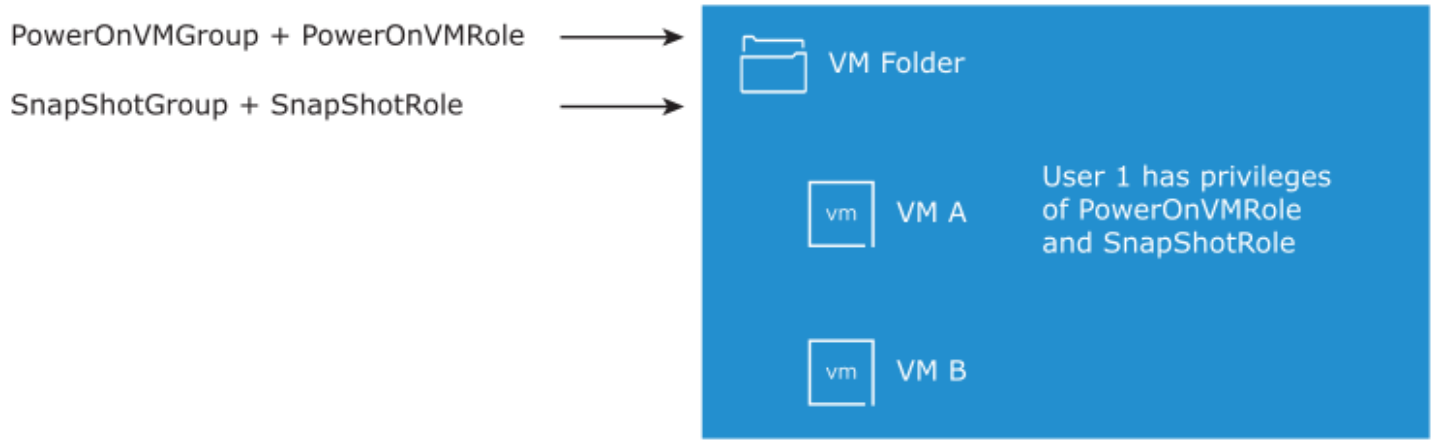
This example illustrates how an object can inherit multiple permissions from groups that are granted permission on a parent object.

In this example, two permissions are assigned on the same object for two different groups.

- PowerOnVMRole can power on virtual machines.
- SnapShotRole can take snapshots of virtual machines.
- PowerOnVMGroup is granted the PowerOnVMRole on VM Folder, with the permission set to propagate to child objects.
- SnapShotGroup is granted the SnapShotRole on VM Folder, with the permission set to propagate to child objects.
- User 1 is not assigned specific privileges.

User 1, who belongs to both the PowerOnVMGroup and the SnapShotGroup, logs in. User 1 can both power on and take snapshots of both VM A and VM B.

Figure 91: Example 1: Permission Inheritance from Multiple Groups



Example 2: Child Permissions Overriding Parent Permissions

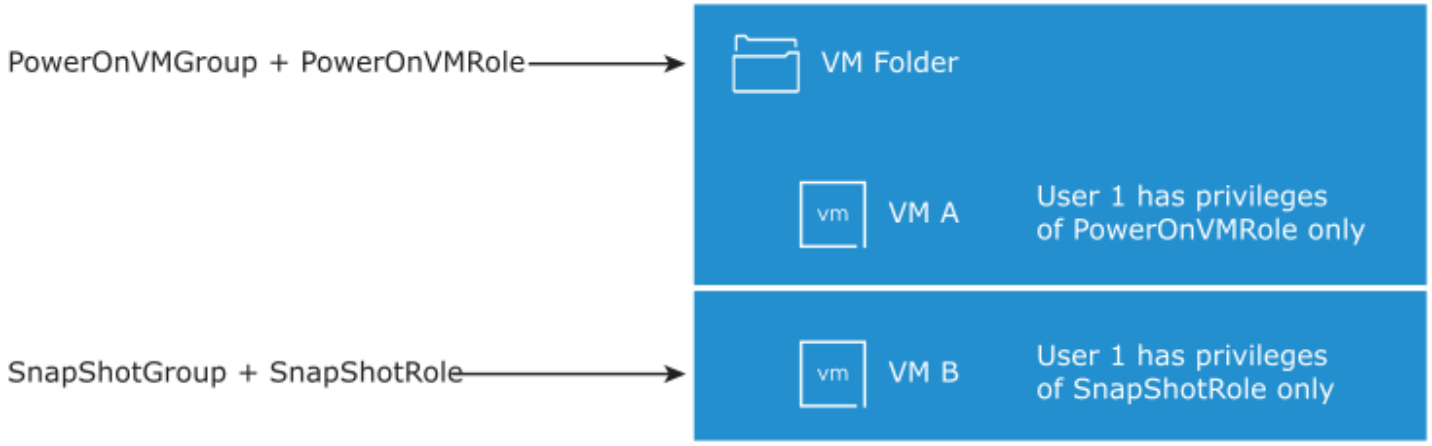
This example illustrates how permissions that are assigned on a child object can override permissions that are assigned on a parent object. You can use this overriding behavior to restrict user access to particular areas of the inventory.

In this example, permissions are defined on two different objects for two different groups.

- PowerOnVMRole can power on virtual machines.
- SnapShotRole can take snapshots of virtual machines.
- PowerOnVMGroup is granted the PowerOnVMRole on VM Folder, with the permission set to propagate to child objects.
- SnapShotGroup is granted the SnapShotRole on VM B.

User 1, who belongs to both the PowerOnVMGroup and the SnapShotGroup, logs in. Because the SnapShotRole is assigned at a lower point in the hierarchy than the PowerOnVMRole, it overrides PowerOnVMRole on VM B. User 1 can power on VM A, but not take snapshots. User 1 can take snapshots of VM B, but not power it on.

Figure 92: Example 2: Child Permissions Overriding Parent Permissions



Example 3: User Role Overriding Group Role

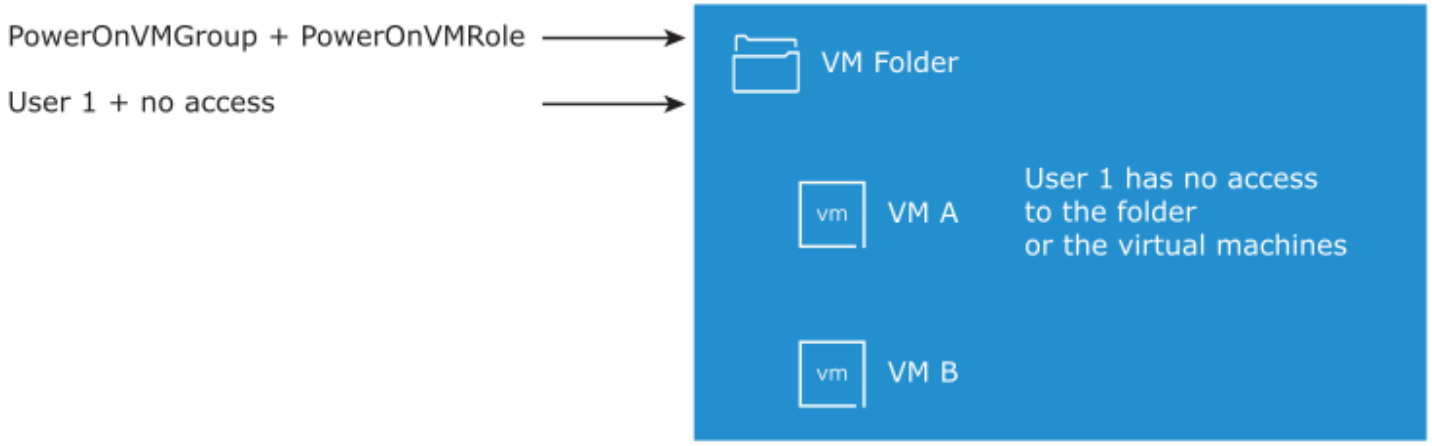
This example illustrates how the role assigned directly to an individual user overrides the privileges associated with a role assigned to a group.

In this example, permissions are defined on the same object. One permission associates a group with a role, the other permission associates an individual user with a role. The user is a member of the group.

- PowerOnVMRole can power on virtual machines.
- PowerOnVMGroup is granted the PowerOnVMRole on VM Folder.
- User 1 is granted the NoAccess role on VM Folder.

User 1, who belongs to PowerOnVMGroup, logs in. The NoAccess role granted to User 1 on VM Folder overrides the role assigned to the group. User 1 has no access to VM Folder or VMs A and B. VMs A and B are not visible in the hierarchy to User 1.

Figure 93: Example 3: User Permissions Overriding Group Permissions



Managing Permissions for vCenter Components

A permission is set on an object in the vCenter object hierarchy. Each permission associates the object with a group or user and the group's or user's access role. For example, you can select a virtual machine object, add one permission that gives the ReadOnly role to Group 1, and add a second permission that gives the Administrator role to User 2.

By assigning a different role to a group of users on different objects, you control the tasks that those users can perform in your vSphere environment. For example, to allow a group to configure memory for the host, select that host and add a permission that grants a role to that group that includes the**Host** > **Configuration** > **Memory Configuration** privilege.

For conceptual information about permissions, see the discussion in Understanding the Object-Level Permission Model.

You can assign permissions to objects at different levels of the hierarchy, for example, you can assign permissions to a host object or to a folder object that includes all host objects. See Hierarchical Inheritance of Permissions in vSphere. You can also assign propagating permissions to a global root object to apply the permissions to all object in all solutions. See Using vCenter Global Permissions.

Add a Permission to an Inventory Object

After you create users and groups and define roles, you must assign the users and groups and their roles to the relevant inventory objects. You can assign the same propagating permissions to multiple objects simultaneously by moving the objects into a folder and setting the permissions on the folder.

On the object whose permissions you want to modify, you must have a role that includes the **Permissions** > **Modify permission** privilege.

When you assign permissions, the user and the group names must match Active Directory precisely, including case. If you upgraded from earlier versions of vSphere, check for case inconsistencies if you experience problems with groups.

1. Browse to the object for which you want to assign permissions in the vSphere Client object navigator.
2. Click the **Permissions** tab.
3. Click **Add**.
4. Optional: If you have configured an external identity provider for federated authentication, the domain of that identity provider is available to select in the **Domain** drop-down menu.
5. If you select **VMware ID** from the **Domain** drop-down menu, enter the user or group name.

   Note:

   Enter the email address of the CSP account in the **Username** field. CSP accounts cannot be searched for in the VMwareID domain.
6. Select the user or group that will have the privileges defined by the selected role.
   a) From the **Domain** drop-down menu, select the domain for the user or group.
   b) Enter a name in the Search box.

      The system searches user names and group names.
   c) Select the user or group.
7. Select a role from the **Role** drop-down menu.
8. Optional: To propagate the permissions, select the **Propagate to children** check box.

   The role is applied to the selected object and propagates to the child objects.
9. Click **OK**.

Change or Remove Permissions on an Inventory Object

After a user or group and role pair is set for an inventory object, you can change the role paired with the user or group or change the setting of the **Propagate to children** check box. You can also remove the permission setting.

1. Browse to the object in the vSphere Client object navigator.
2. Click the **Permissions** tab.
3. Click a row to select a permission.

Task	Steps
Change permissions	1. Click **Edit**. 2. Select a role for the user or group from the **Role** drop-down menu. 3. Toggle the **Propagate to children** check box to change permission inheritance.

Task	Steps
	4. Click **OK**.
Remove permissions	1. Click **Delete**. 2. Click **Remove**.

Change vCenter User Validation Settings

vCenter periodically validates its user and group lists against the users and groups in the user directory. It then removes users or groups that no longer exist in the domain. You can deactivate validation or change the interval between validations. If you have domains with thousands of users or groups, or if searches take a long time to complete, consider adjusting the search settings.

These settings apply to vCenter Single Sign-On identity sources, and not an external identity source, such as Active Directory, that might be associated with vCenter .

Note:

This procedure applies only to vCenter user lists. You cannot search ESX user lists in the same way.

1. Browse to the vCenter system in the vSphere Client object navigator.
2. Select **Configure** and click **Settings** > **General**.
3. Click **Edit** and select **User directory**.
4. Change the values as needed and click **Save**.

Option	Description
User directory timeout	Timeout interval, in seconds, for searching this vCenter installation.
Query limit	Toggle on to set a maximum number of users and groups that vCenter displays.
Query limit size	Maximum number of users and groups from the selected domain that vCenter displays in the **Select Users or Groups** dialog box. If you enter 0 (zero), all users and groups appear.

Using vCenter Global Permissions

In vCenter, global permissions are applied to a global root object that spans VMware solutions. In an on-premises SDDC, global permissions might span both vCenter and VMware Cloud Foundation Operations orchestrator. But for any vSphere SDDC, global permissions apply to global objects such as tags and content libraries.

You can assign global permissions to users or groups, and decide on the role for each user or group. The role determines the set of privileges that the user or group has for all objects in the hierarchy. You can assign a predefined role or create custom roles. See Using vCenter Roles to Assign Privileges.

It is important to distinguish between vCenter permissions and global permissions.

Table 189: Differences Between vCenter Permissions and Global Permissions

Permission Type	Description
vCenter	vCenter permissions apply to specific objects in the inventory hierarchy, such as hosts, virtual machines, datastores, and so on. When you assign vCenter permissions, you specify that a user or group has a role (set of privileges) on the object.
Global	Global permissions give a user or group privileges to view or manage all objects in each of the inventory hierarchies in your deployment. Global permissions also apply to global objects such as tags and content libraries. See vCenter Permissions on Tag Objects. If you assign a global permission and do not select Propagate, the users or groups associated with this permission do not have access to the objects in the hierarchy. They only have access to some global functionality such as creating roles.

Add a Global Permission

You can use global permissions to give a user or group privileges for all objects in all inventory hierarchies in your deployment.

To perform this task, you must have **Permissions** > **Modify permission** privileges on the root object for all inventory hierarchies.

> **Important:**
> Use global permissions with care. Verify that you really want to assign permissions to all objects in all inventory hierarchies.

1. Log in to the vCenter by using the vSphere Client.
2. Select **Administration** and click **Global Permissions** in the Access Control area.
3. Click **Add**.
4. Optional: If you have configured an external identity provider for federated authentication, the domain of that identity provider is available to select in the **Domain** drop-down menu.
5. Select the user or group that will have the privileges defined by the selected role.
   a) From the **Domain** drop-down menu, select the domain for the user or group.
   b) Enter a name in the Search box.
      The system searches user names and group names.
   c) Select the user or group.
6. Select a role from the **Role** drop-down menu.
7. Decide whether to propagate the permissions by selecting the **Propagate to children** check box.
   If you assign a global permission and do not select **Propagate to children**, the users or groups associated with this permission do not have access to the objects in the hierarchy. They only have access to some global functionality such as creating roles.
8. Click **OK**.

vCenter Permissions on Tag Objects

In the vCenter object hierarchy, tag objects are not children of vCenter but are created at the vCenter top level. In environments with multiple vCenter instances, tag objects are shared across vCenter instances. Permissions for tag objects work differently than permissions for other objects in the vCenter object hierarchy.

Only Global Permissions or Permissions Assigned to the Tag Object Apply

If you grant permissions to a user on a vCenter inventory object, such as a virtual machine, that user can perform the tasks associated with the permission. However, the user cannot perform tag operations on the object.

For example, if you grant the **Assign vSphere Tag** privilege to user Dana on host TPA, that permission does not affect whether Dana can assign tags on host TPA. Dana must have the **Assign vSphere Tag** privilege at the top level, that is, a global permission, or must have the privilege for the tag object.

Table 190: How Global Permissions and Tag Object Permissions Affect What Users Can Do

Global Permission	Tag-Level Permission	vCenter Object-Level Permission	Effective Permission
No tagging privileges assigned.	Dana has **Assign or Unassign vSphere Tag** privileges for the tag.	Dana has **Delete vSphere Tag** privileges on ESX host TPA.	Dana has **Assign or Unassign vSphere Tag** privileges for the tag.
Dana has **Assign or Unassign vSphere Tag** privileges.	No privileges assigned for the tag.	Dana has **Delete vSphere Tag** privileges on ESX host TPA.	Dana has **Assign or Unassign vSphere Tag** global privileges. That includes privileges at the tag level.
No tagging privileges assigned.	No privileges assigned for the tag.	Dana has **Assign or Unassign vSphere Tag** privileges on ESX host TPA.	Dana does not have tagging privileges on any object, including host TPA.

Global Permissions Complement Tag Object Permissions

Global permissions, that is, permissions that are assigned on the top-level object, complement permissions on tag objects when the permissions on the tag objects are more restrictive. The vCenter permissions do not affect the tag objects.

For example, assume that you assign the **Delete vSphere Tag** privilege to user Robin at the top level by using global permissions. For the tag Production, you do not assign the **Delete vSphere Tag** privilege to Robin. In that case, Robin has the privilege for the tag Production because Robin has the global permission, which propagates from the top level. You cannot restrict privileges unless you modify the global permission.

Table 191: Global Permissions Complement Tag-Level Permissions

Global Permission	Tag-Level Permission	Effective Permission
Robin has **Delete vSphere Tag** privileges	Robin does not have **Delete vSphere Tag** privileges for the tag.	Robin has **Delete vSphere Tag** privileges.
No tagging privileges assigned	Robin does not have **Delete vSphere Tag** privileges assigned for the tag.	Robin does not have **Delete vSphere Tag** privileges

Tag-Level Permissions Can Extend Global Permissions

You can use tag-level permissions to extend global permissions. That means users can have both a global permission and a tag-level permission on a tag.

> **Note:**
> This behavior is different from how vCenter privileges are inherited. In vCenter, permissions defined for a child object always override the permissions that are propagated from parent objects.

Table 192: Global Permissions Extend Tag-Level Permissions

Global Permission	Tag-Level Permission	Effective Permission
Lee has **Assign or Unassign vSphere Tag** privilege.	Lee has **Delete vSphere Tag** privilege.	Lee has the **Assign vSphere Tag** privilege and the **Delete vSphere Tag** privilege for the tag.
No tagging privileges assigned.	Lee has **Delete vSphere Tag** privilege assigned for the tag.	Lee has the **Delete vSphere Tag** privilege for the tag.

Using vCenter Roles to Assign Privileges

In vCenter, a role is a predefined set of privileges that defines rights to perform actions and read properties. You create permissions by assigning a role to a user or group for an object. vCenter provides system roles and sample roles by default. You can also create custom roles.

Assigning Permissions in vCenter

When you assign permissions in vCenter, you pair a user or group with a role, and associate that pairing with an inventory object. For example, you can use the Virtual machine user sample role to allow a user to read and change virtual machine attributes.

A single user or group can have different roles for different objects in the inventory. For example, assume that you have two resource pools in your inventory, Pool A and Pool B. You can assign group Sales the Virtual machine user sample role on Pool A, and the Read-only role on Pool B. With these assignments, the users in group Sales can turn on virtual machines in Pool A, but can only view virtual machines in Pool B.

Users can schedule tasks only if they have a role that includes privileges to perform that task at the time the task is created.

What Are the Predefined vCenter Roles

vCenter provides predefined roles, as shown in the following table.

Table 193: Predefined vCenter Roles

Role Type	Role Names	Description
System	Administrator, Read-only, and No access.	System roles are permanent. You cannot delete system roles nor can you edit the privileges associated with these roles. The system roles are organized as a hierarchy. Each role inherits the privileges of the previous role. For example, the Administrator role inherits the privileges of the Read-only role. See the following section for more details on system roles.
Sample	vSphere provides a number of sample roles, for example, AutoUpdateUser, Resource pool administrator, and Virtual machine user.	vSphere provides sample roles for certain frequently performed combination of tasks. You can clone, modify, or remove these roles. **Note:** To avoid losing the predefined settings in a sample role, clone the role first and make modifications to the clone. You cannot reset the sample to its default settings.

To view the privileges associated with a role, navigate to the role in the vSphere Client (**Menu** > **Administration** > **Roles**) and click the **Privileges** tab.

To view all the vSphere privileges and descriptions, see Defined Privileges.

> **Note:**
> Changes to roles and privileges take effect immediately, even if the users involved are logged in. The exception is searches, where changes take effect after the user has logged out and logged back in.

vCenter System Roles

System roles cannot be altered or deleted.

Administrator Role	Users with the Administrator role for an object are allowed to view and perform all actions on the object. This role also includes all privileges of the Read Only role. If you have the Administrator role on an object, you can assign privileges to individual users and groups. If you are acting in the Administrator role in vCenter, you can assign privileges to users and groups in the default vCenter Single Sign-On identity source. See the *vSphere Authentication* documentation for supported identity services. By default, the administrator@vsphere.local user has the Administrator role on both vCenter Single Sign-On and vCenter after installation. That user can then associate other users with the Administrator role on vCenter. **Tip:** Best practice is to create a user at the root level and assign the Administrator role to that user. After creating a named user with Administrator privileges, you can remove the root user from any permissions or change its role to No Access.
Read Only Role	Users with the Read Only role for an object are allowed to view the state of the object and details about the object. For example, users with this role can view virtual machine, host, and resource pool attributes, but cannot view the remote console for a host. All actions through the menus and toolbars are disallowed.
No Access Role	Users with the No Access role for an object cannot view or change the object in any way. New users and groups are assigned this role by default. You can change the role on an object-by-object basis. The administrator of the vCenter Single Sign-On domain, administrator@vsphere.local by default, the root user, and vpxuser are assigned the Administrator role by default. Other users are assigned the No Access role by default.

Custom Roles in vCenter and ESX

You can create custom roles for vCenter and all objects that it manages, or for individual hosts.

vCenter Custom Roles (Recommended)	Create custom roles by using the role-editing facilities in the vSphere Client to create privilege sets that match your needs.
ESX Custom Roles	You can create custom roles for individual hosts by using a CLI or the VMware Host Client. See the *vSphere Single Host Management - VMware Host Client* documentation. Custom host roles are not accessible from vCenter. If you manage ESX hosts through vCenter, do not maintain custom roles in both the host and vCenter. Define roles at the vCenter level.

When you manage a host using vCenter, the permissions associated with that host are created through vCenter and stored on vCenter. If you connect directly to a host, only the roles that are created directly on the host are available.

> **Note:**
> When you add a custom role and do not assign any privileges to it, the role is created as a Read Only role with three system-defined privileges: **System** > **Anonymous**, **System** > **View**, and **System** > **Read**. These privileges are not visible in the vSphere Client but are used to read certain properties of some managed objects. All the predefined roles in vCenter contain these three system-defined privileges. See the *vSphere Web Services API* documentation for more information.

Create a vCenter Custom Role

To suit the access control needs of your environment, you can create vCenter custom roles. You can create a role or clone an existing role.

Verify that you have Administrator privileges on the vCenter system where you create the role.

You can create or edit a role on a vCenter system that is part of the same vCenter Single Sign-On domain as other vCenter systems. The VMware Directory Service (vmdir) propagates the role changes that you make to all other vCenter systems in the group. Assignments of roles to specific users and objects are not shared across vCenter systems.

1. Log in to the vCenter by using the vSphere Client.
2. Select **Administration** and click **Roles** in the **Access Control** area.
3. Create the role.

Option	Description
To create a role	1. Click **New**. 2. Enter a name for the new role. 3. Select and deselect privileges for the role. Scroll the privilege categories and select all privileges or a subset of privileges for that category. You can show all, selected, or unselected categories. You can also show all, selected, or unselected privileges. See Defined Privileges for more information. 4. Click **Create**.
To create the role by cloning	1. Select a role, and click **Clone**. 2. Enter a name for the role. 3. Click **OK**. **Note:** When creating a cloned role, you cannot change privileges. To change privileges, select the cloned role and click **Edit**.

You can now create permissions by selecting an object and assigning the role to a user or group for that object.

Using Privilege Recorder

In vSphere, privileges are fine-grained access controls that can be grouped into roles and map them to users or groups. Privilege recorder helps you identify the minimum set of privileges required to run a vCenter Server workflow.

To run a specific set of operations, it is very difficult to determine the minimal set of privileges that are required by the user. The privileges do not have one-one mapping with the specific workflow which usually consists of multiple calls to different APIs operating on the respective object. As a result, the user either has more access or too little access to the environment. With the aim to keep the environment secure, the privilege recorder feature helps you identify the minimum set of privileges required to run a vCenter Server workflow. It allows you to monitor and query the privileges that were checked while performing an operation. Privilege recorder is implemented using a REST API.

Note:
This feature is available as an API, and it supports only workflows run by a script. There is no UI support for Privilege Recorder.

Querying the ListAPI allows you to retrieve lists of privilege checks along with the corresponding sessions, users, managed objects, and operation IDs (opIDs). You can use the appropriate filters to obtain privileges for a particular workflow.

For example, assume that user A needs to create a VM. Creating a VM requires a certain set of privileges. User A must request for privileges from the system administrator. The system administrator can enable the privilege recorder and execute the create VM operation. While the privilege check is performed, the data for the privileges that were checked during the Create VM operation is stored. The data contains PrivilegeID, sessionID, OpID, and so on. In this example, this system admin will use the filters to get privileges for the create VM workflow. The system administrator can now create a role with minimum required privileges and assign it to the user.

Enable the Privilege Recorder

You enable the privilege recorder by using the vSphere Client to modify the vCenter configuration file, `vpxd.cfg`.

Verify that you have enough privileges to run your workflow. A user with the Administrator role is recommended.

1. Log in to the vCenter by using the vSphere Client.
2. Navigate to the vCenter instance.
3. Select **Configure > Advanced Settings**.
4. Click **Edit Settings**.
5. Add the settings.

   The following warning is displayed in the vSphere Client: `Adding or modifying configuration parameters is unsupported and can cause instability. Configuration parameters cannot be removed once they are added. Continue only if you know what you are doing.`

   Scroll down and in the **Name** field, enter the name of the setting, and in the **Value** field, enter the value for the specified setting.

Setting	Description
config.vpxd.privCheck.bufferSize	The count of privileges to be kept in memory. The default value is 0. If you do not change the default value, the privilege checks recorder does not record any data.
config.vpxd.privCheck.cleanupInterval	The interval on which privilege checks for unused sessions are cleaned up. The default value is 30 minutes.

6. Click **Add** and **Save**.

See Performing Privilege Checks Operations in *VMware Cloud Foundation Programming Guide*.

Export Roles and Permissions from the vSphere Client

You can export vCenter roles and permissions to a `.csv` file from the vSphere Client.

1. Browse to the object for which you want to export permissions in the vSphere Client object navigator.
2. Click the **Permissions** tab.
3. Optional: Filter the list by entering text in the **Quick Filter** field.
4. Export the list of permissions.
   - If you did not filter the permissions list, click **Export**.
   - If you filtered the permissions list, click **Export** and select **All Rows** to export all rows of the permissions list, or select **Matching Filters** to export only the rows that match the current filter.

The roles and permissions information is downloaded as a `.csv` file to your browser's default download location.

Best Practices for vCenter Roles and Permissions

Follow best practices for roles and permissions to maximize the security and manageability of your vCenter environment.

Follow these best practices when configuring roles and permissions in your vCenter environment:

- Where possible, assign a role to a group rather than individual users.
- Grant permissions only on the objects where they are needed, and assign privileges only to users or groups that must have them. Use the minimum number of permissions to make it easier to understand and manage the structure of your permissions.
- If you assign a restrictive role to a group, check that the group does not contain the Administrator user or other users with administrative privileges. Otherwise, you might unintentionally restrict privileges of administrators in the parts of the inventory hierarchy where you have assigned that group the restrictive role.

- Group objects into folders to make assigning permissions easier. For example, to grant the modify permission on one set of hosts and the view permission on another set of hosts, place each set of hosts in a folder.
- Use caution when adding a permission to the root vCenter objects. Users with privileges at the root level have access to global data on vCenter, such as roles, custom attributes, vCenter settings.
- Consider enabling propagation when you assign permissions to an object. Propagation ensures that new objects in the object hierarchy inherit permissions. For example, you can assign a permission to a virtual machine folder and enable propagation to ensure that the permission applies to all virtual machines in the folder, including any virtual machines that are moved into the folder in the future.
- Use the No Access role to mask specific areas of the hierarchy. The No Access role restricts access for the users or groups with that role. However, in the case of VMs and vAPPs, there are two permission propagation chains. Assigning a propagating permission with No Access role on one of the chains, does not imply that the respective vApp or VM would have no privileges propagated to it.
- Changes to licenses propagate to all linked vCenter systems in the same vCenter Single Sign-On domain.
- License propagation happens even if the user does not have privileges on all vCenter systems.

Required vCenter Privileges for Common Tasks

Many tasks require permissions on multiple objects in the vSphere inventory. If the user who attempts to perform the task only has privileges on one object, the task cannot complete successfully.

The following table lists common tasks that require more than one privilege. You can add permissions to inventory objects by pairing a user with one of the predefined roles or with multiple privileges. If you expect that you must assign a set of privileges multiple times, create custom roles. To learn more about required privileges for common tasks, see Using Privilege Recorder.

Refer to the *vSphere Web Services API Reference* documentation to learn how operations in the vSphere Client user interface map to API calls, and what privileges are required to perform operations. For example, the API documentation for the `AddHost_Task(addHost)` method specifies that the Host.Inventory.AddHostToCluster privilege is required to add a host to a cluster.

If the task that you want to perform is not in this table, the following rules explain where you must assign permissions to allow particular operations:

- Any operation that consumes storage space requires the **Datastore** > **Allocate Space** privilege on the target datastore, and the privilege to perform the operation itself. You must have these privileges, for example, when creating a virtual disk or taking a snapshot.
- Moving an object in the inventory hierarchy requires appropriate privileges on the object itself, the source parent object (such as a folder or cluster), and the destination parent object.
- Each host and cluster has its own implicit resource pool that contains all the resources of that host or cluster. Deploying a virtual machine directly to a host or cluster requires the **Resource** > **Assign Virtual Machine to Resource Pool** privilege.

Table 194: Required Privileges for Common Tasks

Task	Required Privileges	Applicable Role
Create a virtual machine	On the destination folder or data center: • **Virtual machine** > **Edit inventory** > **Create new** • **Virtual machine** > **Change Configuration** > **Add new disk** (if creating a new virtual disk) • **Virtual machine** > **Change Configuration** > **Add existing disk** (if using an existing virtual disk) • **Virtual machine** > **Configuration** > **Configure Raw device** (if using an RDM or SCSI pass-through device)	Administrator
	On the destination host, cluster, or resource pool: **Resource** > **Assign virtual machine to resource pool**	Resource pool administrator or Administrator

Task	Required Privileges	Applicable Role
	On the destination datastore or the folder that contains the datastore: **Datastore** > **Allocate space**	Datastore Consumer or Administrator
	On the network that the virtual machine will be assigned to: **Network** > **Assign network**	Network Consumer or Administrator
Power on a virtual machine	On the data center in which the virtual machine is deployed: **Virtual machine** > **Interaction** > **Power On**	Virtual Machine Power User or Administrator
	On the virtual machine or folder of virtual machines: **Virtual machine** > **Interaction** > **Power On**	
Deploy a virtual machine from a template	On the destination folder or data center: • **Virtual machine** > **Edit inventory** > **Create from existing** • **Virtual machine** > **Change Configuration** > **Add new disk**	Administrator
	On a template or folder of templates: **Virtual machine** > **Provisioning** > **Deploy template**	Administrator
	On the destination host, cluster or resource pool: • **Resource** > **Assign virtual machine to resource pool** • **vApp** > **Import**	Administrator
	On the destination datastore or folder of datastores: **Datastore** > **Allocate space**	Datastore Consumer or Administrator
	On the network that the virtual machine will be assigned to: **Network** > **Assign network**	Network Consumer or Administrator
Take a virtual machine snapshot	On the virtual machine or a folder of virtual machines: **Virtual machine** > **Snapshot management** > **Create snapshot**	Virtual Machine Power User or Administrator
Move a virtual machine into a resource pool	On the virtual machine or folder of virtual machines: • **Resource** > **Assign virtual machine to resource pool** • **Virtual machine** > **Edit inventory** > **Move**	Administrator
	On the destination resource pool: **Resource** > **Assign virtual machine to resource pool**	Administrator
Install a guest operating system on a virtual machine	On the virtual machine or folder of virtual machines: • **Virtual machine** > **Interaction** > **Answer question** • **Virtual machine** > **Interaction** > **Console interaction** • **Virtual machine** > **Interaction** > **Device connection** • **Virtual machine** > **Interaction** > **Power Off** • **Virtual machine** > **Interaction** > **Power On** • **Virtual machine** > **Interaction** > **Reset** • **Virtual machine** > **Interaction** > **Configure CD media** (if installing from a CD) • **Virtual machine** > **Interaction** > **Configure floppy media** (if installing from a floppy disk) • **Virtual machine** > **Interaction** > **VMware Tools install**	Virtual Machine Power User or Administrator
	On a datastore that contains the installation media ISO image: **Datastore** > **Browse datastore** (if installing from an ISO image on a datastore) On the datastore to which you upload the installation media ISO image: • **Datastore** > **Browse datastore** • **Datastore** > **Low level file operations** • **Host** > **Configuration** > **System Management**	Virtual Machine Power User or Administrator
Migrate a virtual machine with vMotion	On the virtual machine or folder of virtual machines: • **Resource** > **Migrate powered on virtual machine** • **Resource** > **Assign Virtual Machine to Resource Pool** (if destination is a different resource pool from the source)	Resource Pool Administrator or Administrator

Task	Required Privileges	Applicable Role
	On the destination host, cluster, or resource pool (if different from the source): **Resource** > **Assign virtual machine to resource pool**	Resource Pool Administrator or Administrator
Cold migrate (relocate) a virtual machine	On the virtual machine or folder of virtual machines: • **Resource** > **Migrate powered off virtual machine** • **Resource** > **Assign virtual machine to resource pool** (if destination is a different resource pool from the source)	Resource Pool Administrator or Administrator
	On the destination host, cluster, or resource pool (if different from the source): **Resource** > **Assign virtual machine to resource pool**	Resource Pool Administrator or Administrator
	On the destination datastore (if different from the source): **Datastore** > **Allocate space**	Datastore Consumer or Administrator
Migrate a virtual machine with Storage vMotion	On the virtual machine or folder of virtual machines: **Resource** > **Migrate powered on virtual machine**	Resource Pool Administrator or Administrator
	On the destination datastore: **Datastore** > **Allocate space**	Datastore Consumer or Administrator
Move a host into a cluster	On the host: **Host** > **Inventory** > **Add host to cluster**	Administrator
	On the destination cluster: • **Host** > **Inventory** > **Add host to cluster** • **Host** > **Inventory** > **Modify cluster**	Administrator
Add a single host to a data center by using the vSphere Client, or add a single host to a cluster by using PowerCLI or API (leveraging the addHost API)	On the host: **Host** > **Inventory** > **Add host to cluster**	Administrator
	On the cluster: • **Host** > **Inventory** > **Modify cluster** • **Host** > **Inventory** > **Add host to cluster**	Administrator
	On the data center: **Host** > **Inventory** > **Add standalone host**	Administrator
Add multiple hosts to a cluster	On the cluster: • **Host** > **Inventory** > **Modify cluster** • **Host** > **Inventory** > **Add host to cluster**	Administrator
	On the parent data center of the cluster (with propagate): • **Host** > **Inventory** > **Add standalone host** • **Host** > **Inventory** > **Move host** • **Host** > **Inventory** > **Modify cluster** • **Host** > **Configuration** > **Maintenance**	Administrator
Encrypt a virtual machine	Encryption tasks are possible only in environments that include vCenter. In addition, the ESX host must have encryption mode enabled for most encryption tasks. The user who performs the task must have the appropriate privileges. A set of **Cryptographic Operations** privileges allows fine-grained control. See Prerequisites and Required Privileges for Virtual Machine Encryption Tasks. Encryption tasks are possible only in environments that include vCenter. In addition, the ESX host must have encryption mode enabled for most encryption tasks. The user who performs the task must have the appropriate privileges. A set of **Cryptographic Operations** privileges allows fine-grained control. For more information, see the *vSphere Security* documentation.	Administrator
Protect a virtual machine (if using vSphere+ to protect the virtual machine)	On the data center in which the virtual machine is deployed: • **vSphere Tagging** > **Assign or Unassign vSphere Tag**	Administrator

Securing ESX Hosts

The ESX hypervisor architecture has many built-in security features such as CPU isolation, memory isolation, and device isolation. You can configure features such as lockdown mode and certificate replacement for enhanced security.

An ESX host is also protected with a firewall. You can open ports for incoming and outgoing traffic as needed, but should restrict access to services and ports. Using the ESX lockdown mode and limiting access to the ESX Shell can further contribute to a more secure environment. ESX hosts participate in the certificate infrastructure. Hosts are provisioned with certificates that are signed by the VMware Certificate Authority (VMCA) by default.

> **Note:**
>
> ESX is not built upon the Linux kernel or a commodity Linux distribution. It uses its own VMware specialized and proprietary kernel and software tools, delivered as a self-contained unit, and does not contain applications and components from Linux distributions.

Starting in vSphere 8.0 Update 1, ESX runs two reverse proxy services:

- VMware reverse proxy service, `rhttpproxy`
- Envoy

Envoy owns port 443, and all incoming ESX requests are routed through Envoy. Starting in vSphere 8.0 Update 1, rhttpproxy serves as a configuration management server for Envoy.

General ESX Security Recommendations

To secure an ESX host against unauthorized intrusion and misuse, VMware imposes constraints on several parameters, settings, and activities. To meet your configuration needs, you can loosen the constraints. If you do, make sure that you are working in a trusted environment and take other security measures.

What Are the ESX Built-In Security Features

ESX mitigates risks to your hosts as follows:

- The ESX Shell interface and the SSH interface are deactivated by default. Keep these interfaces deactivated unless you are performing troubleshooting or support activities. For day-to-day activities, use the vSphere Client, where activity is subject to role-based access control and modern access control methods.
- The ESX firewall is enabled by default, blocking all incoming/outgoing traffic except for core management services. You can explicitly open firewall ports that are associated with specific services.
- By default, all ports that are not required for management access to the host are closed. Open ports if you need additional services.
- ESX runs only services that are essential to managing its functions. The distribution is limited to the features required to run ESX.
- By default, weak ciphers are deactivated and communications from clients are secured by SSL. The exact algorithms used for securing the channel depend on the SSL handshake. Default certificates created on ESX use PKCS#1 SHA-256 with RSA encryption as the signature algorithm.
- An internal web service is used by ESX to support access by Web clients. The service has been modified to run only functions that a Web client requires for administration and monitoring. As a result, ESX is not vulnerable to web service security issues reported in broader use.
- VMware monitors all security alerts that can affect ESX security and issues a security patch if needed. To view a list of security alerts, go to https://blogs.vmware.com/security/2024/05/where-did-my-vmware-security-advisories-go.html.
- Insecure services such as FTP and Telnet are not installed, and the ports for these services are closed by default.
- To protect hosts from loading drivers and applications that are not cryptographically signed, use UEFI Secure boot. Enabling Secure Boot is done at the system BIOS. No additional configuration changes are required on the ESX host, for example, to disk partitions. See UEFI Secure Boot for ESX Hosts.
- If your ESX host has a TPM 2.0 chip, enable and configure the chip in the system BIOS. Working together with Secure Boot, TPM 2.0 provides enhanced security and trust assurance rooted in hardware. See Securing ESX Hosts with Trusted Platform Module.

- In ESX 8.0 and later, you can run the SSH process under a sandbox domain. The shell then has reduced privileges, and only permits access to a limited subset of commands. For more information, see the VMware knowledge base article at https://kb.vmware.com/s/article/87386.

Taking Further ESX Security Measures

Consider the following recommendations when evaluating host security and administration.

Limit access to ESX hosts	If you activate access to the Direct Console User Interface (DCUI), the ESX Shell, or SSH, enforce strict access security policies. The ESX Shell has privileged access to certain parts of the host. Provide only trusted users with ESX Shell login access.
Do not access managed ESX hosts directly	Use the vSphere Client to administer ESX hosts that are managed by a vCenter. Do not access managed hosts directly with the VMware Host Client, and do not change managed hosts from the DCUI. If you manage hosts with a scripting interface or API, do not target the host directly. Instead, target the vCenter system that manages the host and specify the host name.
Use DCUI only for troubleshooting	Access the host from the DCUI or the ESX Shell as the root user only for troubleshooting. To administer your ESX hosts, use the vSphere Client (or the VMware Host Client), or one of the VMware CLIs or APIs. See *ESXCLI Concepts and Examples*. If you use the ESX Shell or SSH, limit the accounts that have access and set timeouts.
Use only VMware sources to upgrade ESX components	The host runs several third-party packages to support management interfaces or tasks that you must perform. VMware only supports upgrades to these packages that come from a VMware source. If you use a download or patch from another source, you might compromise management interface security or functions. Check third-party vendor sites and the VMware knowledge base for security alerts.

ESX Advanced System Settings

Advanced system settings control aspects of ESX behavior, such as logging, system resources, and security.

The following table presents some of the important ESX advanced system settings for security. To view all the advanced system settings, consult either the vSphere Client (**Host** > **Configure** > **System** > **Advanced System Settings)** or the API for a given release.

Table 195: Partial List of Security Advanced System Settings

Advanced System Setting	Description	Default Value
Annotations.WelcomeMessage	Displays a welcome message in the Host Client prior to login, or in the DCUI on the default screen. In the DCUI, the welcome message replaces some text, such as the host IP address.	(Empty)
Config.Etc.issue	Displays a banner during an SSH login session.	(Empty)

Advanced System Setting	Description	Default Value
Config.Etc.motd	Displays the message of the day upon SSH login. **Note:** To insert new lines or returns into the issues and motd configurations, you can use both the vSphere API and CLI. For example, see https://williamlam.com/2021/03/adding-a-customized-notification-banner-in-the-vsphere-ui.html and https://williamlam.com/2015/02/easily-manage-esxi-vcsa-ssh-login-banner-motd-in-vsphere-6-0.html.	(Empty)
Config.HostAgent.vmacore.soap.sessionTimeout	Sets the idle time in minutes before the system automatically logs out a VIM API. A value of 0 (zero) deactivates the idle time. This setting applies only to new sessions.	30 (minutes)
Mem.MemEagerZero	Activates zeroing the user world and the guest memory pages in the VMkernel operating systems (including the VMM process) after a virtual machine exit. The default value (0) uses lazy zeroing. A value of 1 uses eager zeroing.	0 (deactivated)

Advanced System Setting	Description	Default Value
Security.AccountLockFailures	Sets the maximum number of failed login attempts before the system locks a user's account. For example, to lock the account on the fifth login failure, set this value to 4. A value of 0 (zero) deactivates account locking. For implementation reasons, some login mechanisms count unexpectedly: • VIM logins (including the VMware Host Client) and ESXCLI reflect the exact number of failed logins. • SSH connections count as a login attempt when displaying a password prompt, and undo that count on successful login. This behavior is normal for challenge and response communications. • CGI logins double-count login failures. **CAUTION:** Due to this problem, a user can be locked out faster than the number of failed logins when using the CGI interface.	5
Security.AccountUnlockTime	Sets the number of seconds that a user is locked out. Any login attempt within the specified lock timeout restarts the lock timeout.	900 (15 minutes)
Security.PasswordHistory	Sets the number of passwords to remember for each user. This setting prevents duplicate or similar passwords.	5
Security.PasswordMaxDays	Sets the maximum number of days between password changes.	99999

Advanced System Setting	Description	Default Value
Security.PasswordQualityControlPwquality	Configures the password quality requirements in the `pam_pwquality` configuration. The default setting requires 4 character classes and a minimum length of seven characters. If you are implementing the DoD Annex, the Security.PasswordQualityControlPwquality setting, with a PasswordMaxDays setting, satisfies the requirements: `minlen = 15 dcredit = -1 ucredit = -1 lcredit = -1 ocredit = -1 difok = 4 maxrepeat = 3` If this setting is set to `UNUSED`, the host uses the Security.PasswordQualityControl setting for password quality instead. Broadcom recommends using the older Security.PasswordQualityControl setting only if necessary for compatibility.	retry=3 dcredit=-1 ucredit=-1 lcredit=-1 ocredit=-1 minlen=7 difok=4 enforce_for_root
Security.PasswordQualityControl	Starting with ESX 9.1, it is recommended to use the Security.PasswordQualityControlPwquality variable to set password requirements rather than Security.PasswordQualityControl. Changes the required length and the character class requirement or allow pass phrases in the `Pam_passwdqc` configuration. You can use special characters in passwords. You can have password lengths of at least 15 characters. The default setting requires three character classes and a minimum length of seven characters. If implementing the DoD Annex, you can combine the `similar=deny` option plus a minimum password length to enforce a requirement that passwords are sufficiently different. The password history setting is only enforced for passwords changed through the VIM `LocalAccountManager.changePassword` API. To change the password requires that the user have administrator permission. The PasswordQualityControl setting, with a PasswordMaxDays setting, satisfies the requirements of the DoD Annex: `min=disabled,disabled,disabled,disabled,15 similar=deny`	retry=3 min=disabled,disabled,disabled,7,7

Advanced System Setting	Description	Default Value
UserVars.DcuiTimeOut	Sets the idle time in seconds before the system automatically logs out the DCUI. A value of 0 (zero) deactivates the timeout.	600 (10 minutes)
UserVars.ESXiShellInteractiveTimeOut	Sets the idle time in seconds before the system automatically logs out an interactive shell. This setting takes effect for new sessions only. A value of 0 (zero) deactivates the idle time. Applies to both the DCUI and the SSH shell.	0
UserVars.ESXiShellTimeOut	Sets the time in seconds a login shell waits for login. A value of 0 (zero) deactivates the timeout. Applies to both the DCUI and the SSH shell.	0
UserVars.HostClientSessionTimeout	Sets the idle time in seconds before the system automatically logs out the Host Client. A value of 0 (zero) deactivates the idle time.	900 (15 minutes)
UserVars.HostClientWelcomeMessage	Displays a welcome message in the Host Client upon login. The message is displayed following login as a “hint”.	(Empty)

Configure ESX Hosts with Host Profiles

Host profiles allow you to set up standard configurations for your ESX hosts and automate compliance to these configuration settings. Host profiles allow you to control many aspects of host configuration including memory, storage, networking, and so on.

Host Profiles provide an automated and centrally managed mechanism for host configuration and configuration compliance. Host Profiles can improve efficiency by reducing reliance upon repetitive, manual tasks. Host Profiles capture the configuration of a pre-configured and validated reference host, store the configuration as a managed object and use the catalog of parameters contained within to configure networking, storage, security, and other host-level parameters.

You can configure host profiles for a reference host from the vSphere Client and apply the host profile to all hosts that share the characteristics of the reference host. You can also use host profiles to monitor hosts for host configuration changes. See the *vSphere Host Profiles* documentation.

You can attach the host profile to a cluster to apply it to all hosts in the cluster.

1. Set up the reference host to specification and create a host profile.
2. Attach the profile to a host or cluster.
3. Apply the host profile of the reference host to other hosts or clusters.

Use Scripts to Manage ESX Host Configuration Settings

In environments with many ESX hosts, managing hosts with scripts is faster and less error prone than managing the hosts from the vSphere Client.

vSphere includes scripting languages for ESX host management. VMware PowerCLI is a Windows PowerShell interface to the vSphere API, and includes PowerShell cmdlets for administering vSphere components. ESXCLI includes a set of commands for managing ESX hosts and virtual machines. See https://developer.broadcom.com for reference information and programming tips. The vSphere Administrator documentation focuses on using the vSphere Client for management.

You can also use one of the scripting interfaces to the vSphere Automation SDK such as the vSphere Automation SDK for Python.

1. Create a custom role that has limited privileges.

   See Create a vCenter Custom Role.

   For example, consider creating a role that has a set of privileges for managing hosts but no privileges for managing virtual machines, storage, or networking. If the script you want to use only extracts information, you can create a role with read-only privileges for the host.

2. From the vSphere Client, create a service account and assign it the custom role.

   You can create multiple custom roles with different levels of access if you want access to certain hosts to be fairly limited.

3. Write scripts to perform parameter checking or modification, and run them.

   For example, you can check or set the shell interactive timeout of a host as follows:

Language	Commands
ESXCLI	`esxcli <conn_options> system settings advanced get /UserVars/ESXiShellTimeOut` `esxcli --formatter=csv --format-param=fields="Path,Int Value" system settings advanced list | grep /UserVars/ESXiShellTimeOut`

PowerCLI	`#List UserVars.ESXiShellInteractiveTimeOut for each host`
	`Get-VMHost | Select Name, @{N="UserVars.ESXiShellInteractiveTimeOut";E={$_ | Get-AdvancedSetting -Name UserVars.ESXiShellInteractiveTimeOut | Select -ExpandProperty Value}}`
	
	`# Set UserVars.ESXiShellTimeOut to 900 on all hosts`
	`Get-VMHost | Foreach { Get-AdvancedSetting -Entity $_ -Name UserVars.ESXiShellInteractiveTimeOut | Set-AdvancedSetting -Value 900 }`

4. In large environments, create roles with different access privileges and group hosts into folders according to the tasks that you want to perform. You can then run scripts over different folders from different service accounts.
5. Verify that the changes happened after you run the command.

ESX Passwords and Account Lockout

For ESX hosts, you must use a password with predefined requirements. You can change the required length and the character class requirement or allow pass phrases using the Security.PasswordQualityControlPwquality advanced system setting. You can also set the number of passwords to remember for each user using the Security.PasswordHistory advanced system setting.

> **Note:**
> The default requirements for ESX passwords can change from one release to the next. You can check and change the default password restrictions by using the Security.PasswordQualityControlPwquality advanced system setting.

ESX Passwords

ESX enforces password requirements for access from the Direct Console User Interface, the ESX Shell, SSH, or the VMware Host Client.

- By default, you must include a mix of at least one from each of the following four character classes: lowercase letters, uppercase letters, numbers, and special characters such as underscore or dash when you create a password.
- By default, password length is at least 7 characters and at most 128.
- Passwords must not contain the user name or parts of the user name.

> **Note:**
> An uppercase character that begins a password does not count toward the number of character classes used. A number that ends a password does not count toward the number of character classes used. A dictionary word used inside a password reduces the overall password strength.

Example ESX Passwords

The following password candidates illustrate potential passwords when the Security.PasswordQualityControlPwquality option is set as follows.

```
retry=3 dcredit=-1 ucredit=-1 lcredit=-1 ocredit=-1 minlen=7 difok=4 enforce_for_root
```

With this setting, a user is prompted up to three times (`retry=3`) for a new password that is not sufficiently strong or if the password was not entered correctly twice. Passwords with one, two, or three character classes are not allowed. Passwords from four-character classes require seven characters. New passwords must contain a minimum of four changes from the previous password to be accepted (difok=4). Restrictions are enforced even when the user changing the password is root (enforce_for_root). See the `pam_pwquality` man page for details on other options.

With these settings, the following password example password is allowed, because it contains seven characters from four character classes: `xQaT3#A`

Changing Default Password Restrictions

You can change the default restriction on passwords or pass phrases by using the Security.PasswordQualityControlPwquality advanced system setting for your ESX host. See the *vCenter and Host Management* documentation for information on changing ESX advanced system settings.

See the man page for `pam_pwquality` for details.

> **Note:**
> Not all possible combinations of password options have been tested. Perform testing after you change the default password settings.

The following example meets STIG requirements for ESX:

```
retry=3 dcredit=-1 ucredit=-1 lcredit=-1 ocredit=-1 minlen=15 maxrepeat=3 maxclassrepeat=4 difok=4
 enforce_for_root
```

If Security.PasswordQualityControlPwquality is set to `"UNUSED"`, the value defined in the legacy Security.PasswordQualityControl setting is used instead. Broadcom recommends against using Security.PasswordQualityControl unless it is necessary for compatibility reasons.

Password Restrictions and ESX Upgrade

ESX 9.0 and earlier used the Security.PasswordQualityControl advanced system setting to set restrictions on passwords or pass phrases. ESX 9.1 introduces the Security.PasswordQualityControlPwquality advanced system setting, which follows the `pam_pwquality` module format.

The behavior of these settings during an upgrade from an earlier ESX release to ESX 9.1 varies depending on whether the Security.PasswordQualityControl was modified from its default value:

- If Security.PasswordQualityControl setting was not modified from its default value, the new Security.PasswordQualityControlPwquality setting is enabled with its default value.
- If Security.PasswordQualityControl was modified from the default, Security.PasswordQualityControlPwquality is set to `"UNUSED"`, and the Security.PasswordQualityControl setting remains in effect. Review the password policy and set Security.PasswordQualityControlPwquality appropriately when possible.

ESX Account Lockout Behavior

Account locking is supported for access through SSH and through the vSphere Web Services SDK. The Direct Console Interface (DCUI) and the ESX Shell do not support account lockout. By default, a maximum of five failed attempts is allowed before the account is locked. The account is unlocked after 15 minutes by default.

Configuring Login Behavior

You can configure the login behavior for your ESX host with the following advanced system settings:

- Security.AccountLockFailures. Maximum number of failed login attempts before a user's account is locked. Zero deactivates account locking.
- Security.AccountUnlockTime. Number of seconds that a user is locked out.
- Security.PasswordHistory. Number of passwords to remember for each user. Starting in vSphere 8.0 Update 1, the default is five. Zero deactivates password history.

ESX Cryptographic Key Generation

ESX generates several asymmetric keys for normal operation. The Transport Layer Security (TLS) key secures communication with the ESX host using the TLS protocol. The SSH key secures communication with the ESX host using the SSH protocol.

Transport Layer Security Key

The Transport Layer Security (TLS) key secures communication with the host using the TLS protocol. Upon first boot, the ESX host generates the TLS key as a 2048-bit RSA key. Currently, ESX does not implement automatic generation of ECDSA keys for TLS. The TLS private key is not intended to be serviced by the administrator.

The TLS key resides at the following non-persistent location:

```
/etc/vmware/ssl/rui.key
```

The TLS public key (including intermediate certificate authorities) resides at the following non-persistent location as an X.509 v3 certificate :

```
/etc/vmware/ssl/rui.crt
```

When you use vCenter with your ESX hosts, vCenter generates a CSR automatically, signs it using the VMware Certificate Authority (VMCA), and generates the certificate. When you add an ESX host to vCenter, vCenter installs that resulting certificate on the ESX host.

The default TLS certificate is self-signed, with a subjectAltName field matching the host name at installation. You can install a different certificate, for example, to make use of a different subjectAltName or to include a particular Certificate Authority (CA) in the verification chain. See Replacing the Default ESX Certificate with a Custom Certificate.

SSH Key

The SSH key secures communication with the ESX host using the SSH protocol. Upon first boot, the system generates a nistp256 ECDSA key, and the SSH keys as 2048-bit RSA keys. The SSH server is deactivated by default. SSH access is intended primarily for troubleshooting purposes. The SSH keys are not intended to be serviced by the administrator. Logging in through SSH requires administrative privileges equivalent to full host control. To enable SSH access, see Activate Access to the ESX Shell Using the vSphere Client.

The SSH public keys reside at the following location:

```
/etc/ssh/ssh_host_rsa_key.pub
```

```
/etc/ssh/ssh_host_ecdsa_key.pub
```

The SSH private keys reside at the following location:

```
/etc/ssh/ssh_host_rsa_key
```

```
/etc/ssh/ssh_host_ecdsa_key
```

TLS Cryptographic Key Establishment

Configuration of TLS cryptographic key establishment is governed by choice of TLS cipher suites, which selects ECC-based key agreements using ephemeral Ecliptic Curve Diffie Hellman (ECDH) (as specified in NIST Special Publication 800-56A).

SSH Cryptographic Key Establishment

Configuration of SSH cryptographic key establishment is governed by the SSHD configuration. ESX provides a default configuration that permits ephemeral Diffie Hellman (DH) (as specified in NIST Special Publication 800-56A) key agreement, and ephemeral Ecliptic Curve Diffie Hellman (ECHD) (as specified in NIST Special Publication 800-56A). The SSHD configuration is not intended to be serviced by the administrator.

SSH Security in ESX

The ESX Shell interface and the SSH interface are deactivated by default. Keep these interfaces deactivated unless you are performing troubleshooting or support activities. For regular activities, use the vSphere Client, where activity is subject to role-based access control and modern access control methods.

SSH Configuration in ESX

The SSH configuration in ESX uses the following settings.

Version 1 SSH protocol deactivated	VMware does not support Version 1 SSH protocol and uses Version 2 protocol exclusively. Version 2 eliminates certain security problems present in Version 1 and provides you with a safe way to communicate with the management interface.
Improved cipher strength	SSH supports only 256-bit and 128-bit AES ciphers for your connections.

These settings are designed to provide solid protection for the data you transmit to the management interface through SSH. You cannot change these settings.

ESX SSH Keys

SSH keys can restrict, control, and secure access to an ESX host. An SSH key can allow a trusted user or script to log in to a host without entering a password.

You can use HTTPS PUT to copy the SSH key to the host.

Instead of generating the keys externally and uploading them, you can create the keys on the ESX host and download them. See the VMware knowledge base article at KB 313767 - Allowing SSH access to VMware vSphere ESXi/ESX hosts with public/private key authentication.

Enabling SSH and adding SSH keys to the host has inherent risks. Weigh the potential risk of exposing a user name and password against the risk of intrusion by a user who has a trusted key.

Upload an SSH Key Using HTTPS PUT

You can use authorized keys to log in to a host with SSH. You can upload authorized keys with HTTPS PUT.

Authorized keys allow you to authenticate remote access to a host. When users or scripts try to access a host with SSH, the key provides authentication without a password. With authorized keys you can automate authentication, which is useful when you write scripts to perform routine tasks.
You can upload the following types of SSH keys to a host using HTTPS PUT:

- Authorized keys file for root user
- DSA key
- DSA public key
- RSA key
- RSA public key

Important:
Do not modify the `/etc/ssh/sshd_config` file.

1. In your upload application, open the key file.
2. Publish the file to the following locations.

Type of key	Location
Authorized key files for the root user	`https://hostname_or_IP_address/host/ssh_root_authorized_keys` You must have full administrator privileges on the host to upload this file.
DSA keys	`https://hostname_or_IP_address/host/ssh_host_dsa_key`
DSA public keys	`https://hostname_or_IP_address/host/ssh_host_dsa_key_pub`
RSA keys	`https://hostname_or_IP_address/host/ssh_host_rsa_key`
RSA public keys	`https://hostname_or_IP_address/host/ssh_host_rsa_key_pub`

PCI and PCIe Devices and ESX

Using the VMware DirectPath I/O feature to pass through a PCI or PCIe device to a virtual machine results in a potential security vulnerability. The vulnerability can be triggered when buggy or malicious code, such as a device driver, is running in privileged mode in the guest OS. Industry-standard hardware and firmware do not currently have sufficient error containment support to protect ESX hosts from the vulnerability.

Use PCI or PCIe passthrough to a virtual machine only if a trusted entity owns and administers the virtual machine. You must be sure that this entity does not to attempt to crash or exploit the host from the virtual machine.

Your host might be compromised in one of the following ways.

- The guest OS might generate an unrecoverable PCI or PCIe error. Such an error does not corrupt data, but can crash the ESX host. Such errors might occur because of bugs or incompatibilities in the hardware devices that are being passed through. Other reasons for errors include problems with drivers in the guest OS.
- The guest OS might generate a Direct Memory Access (DMA) operation that causes an IOMMU page fault on the ESX host. This operation might be the result of a DMA operation that targets an address outside the virtual machine memory. On some machines, host firmware configures IOMMU faults to report a fatal error through a non-maskable interrupt (NMI). This fatal error causes the ESX host to crash. This problem might occur because of problems with the drivers in the guest OS.
- If the operating system on the ESX host is not using interrupt remapping, the guest OS might inject a spurious interrupt into the ESX host on any vector. ESX currently uses interrupt remapping on Intel platforms where it is available. Interrupt mapping is part of the Intel VT-d feature set. ESX does not use interrupt mapping on AMD platforms. A false interrupt can result in a crash of the ESX host. Other ways to exploit these false interrupts might exist in theory.

Deactivate the vSphere Managed Object Browser

The managed object browser (MOB) is a vSphere utility that provides a way to explore the VMkernel object model. However, attackers can use this interface to perform malicious configuration changes or actions because it is possible to change the host configuration by using the MOB. Use the MOB only for debugging, and ensure that it is deactivated in production systems.

The MOB is deactivated by default. However, for certain tasks, for example when extracting the old certificate from a system, you have to use the MOB. You can activate and deactivate the MOB as follows.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, click **Advanced System Settings**.
4. Check the value of **Config.HostAgent.plugins.solo.enableMob**, and click **Edit** to change it as appropriate.
   Do not use `vim-cmd` from the ESX Shell.

ESX Networking Security Recommendations

Isolation of network traffic is essential to a secure ESX environment. Different networks require a different access and level of isolation.

Your ESX host uses several networks. Use appropriate security measures for each network, and isolate traffic for specific applications and functions. For example, ensure that VMware vSphere® vMotion® traffic does not travel over networks where virtual machines are located. Isolation prevents snooping. Having separate networks is also recommended for performance reasons.

- vSphere infrastructure networks are used for features such as vSphere vMotion, VMware vSphere Fault Tolerance, VMware vSAN, and storage. Isolate these networks for their specific functions. It is often not necessary to route these networks outside a single physical server rack.
- A management network isolates client traffic, command-line interface (CLI) or API traffic, and third-party software traffic from other traffic. In general, the management network is accessible only by system, network, and security administrators. To secure access to the management network, use a bastion host or a virtual private network (VPN). Strictly control access within this network.
- Virtual machine traffic can flow over one or many networks. You can enhance the isolation of virtual machines by using virtual firewall solutions that set firewall rules at the virtual network controller. These settings travel with a virtual machine as it migrates from host to host within your vSphere environment.

Modifying ESX Web Proxy Settings

When you modify Web proxy settings, you have several encryption and user security guidelines to consider.

> **Note:**
> Restart the host process after making any changes to host directories or authentication mechanisms.

- Do not set up certificates that use a password or pass phrases. ESX does not support Web proxies that use passwords or pass phrases, also known as encrypted keys. If you set up a Web proxy that requires a password or pass phrase, ESX processes cannot start correctly.
- To support encryption for user names, passwords, and packets, SSL is activated by default for vSphere Web Services SDK connections. If you want to configure these connections so that they do not encrypt transmissions, deactivate SSL for your vSphere Web Services SDK connection by switching the connection from HTTPS to HTTP.
  Consider deactivating SSL only if you created a fully trusted environment for these clients, where firewalls are in place and transmissions to and from the host are fully isolated. Deactivating SSL can improve performance, because you avoid the overhead required to perform encryption.
- To protect against misuse of ESX services, most internal ESX services are accessible only through port 443, the port used for HTTPS transmission. Port 443 acts as a reverse proxy for ESX. You can see a list of services on ESX through an HTTP welcome page, but you cannot directly access the Storage Adapters services without proper authorization.
  You can change this configuration so that individual services are directly accessible through HTTP connections. Do not make this change unless you are using ESX in a fully trusted environment.
- When you upgrade your environment, the certificate remains in place.

vSphere Auto Deploy Security Considerations

When you use vSphere Auto Deploy, pay careful attention to networking security, boot image security, and potential password exposure through host profiles to protect your environment.

Note:

vSphere Auto Deploy is deprecated in the 9.0 release and will be removed in a future release.

Networking Security

Secure your network just as you secure the network for any other PXE-based deployment method. vSphere Auto Deploy transfers data over SSL to prevent casual interference and snooping. However, the authenticity of the client or of the Auto Deploy server is not checked during a PXE boot.

You can greatly reduce the security risk of Auto Deploy by completely isolating the network where Auto Deploy is used.

Boot Image and Host Profile Security

The boot image that the vSphere Auto Deploy server downloads to a machine can have the following components.

- The VIB packages that the image profile consists of are always included in the boot image.
- The host profile and host customization are included in the boot image if Auto Deploy rules are set up to provision the host with a host profile or host customization.
  - The administrator (root) password and user passwords that are included with host profile and host customization are hashed with SHA-512.
  - Any other passwords associated with profiles are in the clear. If you set up Active Directory by using host profiles, the passwords are not protected.

    Use the vSphere Authentication Proxy to avoid exposing the Active Directory passwords. If you set up Active Directory using host profiles, the passwords are not protected.
- The host's public and private SSL key and certificate are included in the boot image.

Control Access for CIM-Based Hardware Monitoring Tools

The Common Information Model (CIM) system provides an interface for remote applications to monitor hardware resources using a set of standard APIs. To ensure that the CIM interface is secure, provide only the minimum access necessary to these remote applications. If you provision a remote application with a root or Administrator account, and if the application is compromised, the virtual environment can be compromised.

CIM is an open standard that defines a framework for agent-less, standards-based monitoring of hardware resources for ESX hosts. This framework consists of a CIM object manager, often called a CIM broker, and a set of CIM providers.

CIM providers support management access to device drivers and underlying hardware. Hardware vendors, including server manufacturers and hardware device vendors, can write providers that monitor and manage their devices. VMware writes providers that monitor server hardware, ESX storage infrastructure, and virtualization-specific resources. These providers run inside the ESX host and are lightweight and focused on specific management tasks. The CIM broker takes information from all CIM providers and presents it to the outside world using standard APIs. The most common API is WS-MAN.

Do not provide root credentials to remote applications that access the CIM interface. Instead, create a less-privileged vSphere user account for these applications and use the VIM API ticket function to issue a sessionId (called a "ticket") to this less-privileged user account to authenticate to CIM. If the account has been granted permission to obtain CIM tickets, the VIM API can then supply the ticket to CIM. These tickets are then supplied as both the user ID and password to any CIM-XML API call. See the `AcquireCimServicesTicket()` method for more information.

The CIM service starts when you install a third-party CIM VIB, for example, when you run the `esxcli software vib install -n` *`VIBname`* command.

If you must activate the CIM service manually, run the following command:

```
esxcli system wbem set -e true
```

If necessary, you can deactivate wsman (WSManagement Service) so that only the CIM service is running:

```
esxcli system wbem set -W false
```

To confirm that wsman is deactivated, run the following command:

```
esxcli system wbem get
…
WSManagement PID: 0
WSManagement Service: false
```

For more information about ESXCLI commands, see *ESXCLI Documentation*. For more information about activating the CIM service, see the VMware knowledge base article at https://kb.vmware.com/s/article/1025757.

1. Create a non-root vSphere user account for CIM applications.

   See the topic on adding vCenter Single Sign-On users in *vSphere Authentication*. The required vSphere privilege for the user account is **Host.CIM** > **Interaction**.

2. Use the vSphere API SDK of your choice to authenticate the user account to vCenter. Then call `AcquireCimServicesTicket()` to return a ticket to authenticate with ESX as an administrator-level account using CIM-XML port 5989 or WS-Man port 433 APIs.

   See *vSphere Web Services API Reference* for more information.

3. Renew the ticket every two minutes as needed.

vSphere Distributed Services Engine Security Best Practices

To maximize the security of your ESX environment, follow best practices for vSphere Distributed Services Engine.

In vSphere 8.0 and later, vSphere Distributed Services Engine enables the offloading of infrastructure functions from the CPUs of a host or a server to data processing units (DPUs, also known as SmartNICs), thus freeing up CPU cycles to serve applications. For an introduction to vSphere Distributed Services Engine, see the *VMware ESX Installation and Setup* documentation. For more information about vSphere Distributed Services Engine, see the *Managing Host and Cluster Lifecycle* documentation.

In general, treat security aspects of vSphere Distributed Services Engine as you do when securing your ESXi environment.

- The ESXi Shell interface and the SSH interface to vSphere Distributed Services Engine are deactivated by default. Keep these interfaces deactivated unless you are performing troubleshooting or support activities.
- For day-to-day management activities of vSphere Distributed Services Engine, use the vSphere Client, where activity is subject to role-based access control and modern access control methods.

Controlling ESX Entropy

In ESX 8.0 and later, the ESX Entropy implementation supports the FIPS 140-3 and EAL4 certifications. Kernel boot options control which entropy sources to activate on an ESX host.

You must have root access on the ESX host.

In computing, the term “entropy” refers to random characters and data that are collected for use in cryptography, such as generating encryption keys to secure data transmitted over a network. Entropy is required by security for generating keys and communicating securely over the network. Entropy is often collected from a variety of sources on a system.

FIPS entropy handling is the default behavior if the following conditions are true.

1. The hardware supports RDSEED.
2. The disableHwrng VMkernel boot option isn’t present or is FALSE.
3. The entropySources VMkernel boot option isn’t present, is 0 (zero), or is 4.

> **Warning:**
> When you configure an ESX host with entropySources for external entropy only (that is, entropySources is set to 8), you must keep supplying the external entropy to the host using the entropy API. If the entropy becomes exhausted in the host, then the host becomes unresponsive. To recover from this situation, reboot the host. If the host is still unresponsive, you must reinstall ESX.

Starting with ESX 8.0 Update 1, you can configure external entropy sources in the kickstart file for scripted installation. You can configure ESX in a highly secure environment to consume entropy from external entropy sources, such as a

Hardware Security Module (HSM), and align with standards such as BSI Common criteria, EAL4, and NIST FIPS CMVP, by using the scripted installation method. For more information about configuring external entropy sources, see the *VMware ESX Installation and Setup* documentation.

You can configure the ESX Entropy subsystem using the following VMkernel boot options:

Table 196: ESX Entropy VMkernel Boot Options

VMkernel Boot Option	Option Type	Description	Default Value
disableHwrng (available prior to vSphere 8.0)	Boolean	Deactivates the RDRAND and the RDSEED entropy sources when set to TRUE (overrides "entropySources").	FALSE Activates hardware random number generator entropy sources if present.
entropySources (available starting in vSphere 8.0)	Integer, Bitmask	Specifies which entropy sources to activate. • 0 (default) Bitmask values: • 1=interrupts • 2=RDRAND • 4=RDSEED • 8=entropyd (EAL4 entropy handling is activated) Specifying entropySources=9 activates the interrupts and the userspace entropy sources, and deactivates the RDRAND and the RDSEED entropy sources.	0 (zero) If RDSEED is supported, the default is FIPS compliance. Otherwise the default is all entropy sources except entropyd.

Note:
Before making a change to use only RDRAND, RDSEED, or both entropy sources, check your vendor documentation to ensure that your ESX host supports those configurations. If your host does not support those configurations, vCenter notifies you with an alert, and the host falls back to using the interrupt and userspace entropy sources.

1. Use SSH or another remote console connection to start a session on the ESX host.
2. Log in as root.
3. Set the desired entropy VMkernel boot options.
   a) To deactivate the RDRAND and the RDSEED entropy sources for disableHwrng:

      ```
      esxcli system settings kernel set -s disableHwrng -v TRUE
      ```

   b) To set entropySources:

      ```
      esxcli system settings kernel set -s entropySources -v entropy_source_value
      ```

      See the preceding table for the values that you can set for entropySources.

Managing Certificates for ESX Hosts

The VMware Certificate Authority (VMCA) provisions each new ESX host with a signed certificate that has VMCA as the root certificate authority by default. Provisioning happens when you add a host to vCenter explicitly or as part of installation or upgrade of ESX.

You can view and manage ESX certificates from the vSphere Client and by using the `vim.CertificateManager` API in the vSphere Web Services SDK. You cannot view or manage ESX certificates by using certificate management CLIs that are available for managing vCenter certificates.

Starting in vSphere 8.0 Update 3, you can replace ESX certificates without placing the host into maintenance mode, and without having to restart the host or individual services.

Certificates and Certificate Modes

When ESX and vCenter communicate, they use TLS for almost all management traffic.

vCenter supports the following certificates and certificate modes for ESX hosts.

Table 197: Certificate Modes for ESX Hosts

Certificate Mode	Description
VMware Certificate Authority (default)	By default, the VMware Certificate Authority is used as the certificate authority (CA) for ESX host certificates. VMCA is the root CA by default, but it can be set up as the intermediary CA to another CA. In **vmca** mode, you can renew and refresh certificates from the vSphere Client. Also used if VMCA is a subordinate certificate.
Custom Certificate Authority	Use this mode if you want to use only custom certificates that are signed by a third-party or enterprise CA. In **custom** mode, you are responsible for managing the certificates. Starting in vSphere 8.0 Update 3, you can manage custom certificates from the vSphere Client. **Note:** Unless you change the certificate mode to Custom Certificate Authority (**custom**), VMCA might replace custom certificates, for example, when you select **Renew** in the vSphere Client.

To change the certificate mode to use a different type of certificate, see ESX Certificate Mode Switch Workflows and Change the ESX Certificate Mode.

ESX Certificate Expiration

You can view information about certificate expiration for certificates that are signed by VMCA or a third-party CA in the vSphere Client. You can view the information for all hosts that vCenter manages or for individual hosts. A yellow alarm is raised if the certificate is in the **Expiring Shortly** state (less than eight months). A red alarm is raised if the certificate is in the **Expiration Imminent** state (less than two months).

ESX Provisioning and Certificates

When you boot an ESX host from installation media, the host initially has an autogenerated certificate. When you add a host to the vCenter system, vCenter provisions the host with a certificate that is signed by VMCA as the root CA.

You can also use custom certificates that are signed by a third-party or an enterprise CA for ESX hosts.

ESX Provisioning and Certificates in Auto Deploy

Note:

vSphere Auto Deploy has been deprecated in the 9.0 release, and will be removed in a future release.

The process is similar for hosts that are provisioned with Auto Deploy. However, because those hosts do not store any state, the signed certificate is stored by the Auto Deploy server in its local certificate store. The certificate is reused during subsequent boots of the ESX hosts. An Auto Deploy server is part of any embedded deployment or vCenter system.

If VMCA is not available when an Auto Deploy host boots the first time, the host first attempts to connect. If the host cannot connect, it cycles through shutdown and reboot until VMCA becomes available and the host can be provisioned with a signed certificate.

You can make Auto Deploy a subordinate Certificate Authority of a third-party Certificate Authority. In this case, the generated certificates are signed with the Auto Deploy SSL key. See Make Auto Deploy a Subordinate Certificate Authority.

In ESX 8.0 and later, you can use custom certificates (certificates signed by a Certificate Authority) with Auto Deploy. When the host starts, Auto Deploy associates the custom certificate with either a MAC address or the BIOS UUID of the ESX host. See Use Custom Certificates with Auto Deploy.

Required Privileges for ESX Certificate Management

The **Certificates** > **Manage Certificates** privilege is required for users to manage your ESX host certificates.

ESX Host Name and IP Address Changes

vSphere 9.0 supports the renaming of ESX hosts, provided the new name exists in DNS and in the SAN of the certificate.

An ESX host name or IP address change might affect whether vCenter considers a host certificate valid. How you added the ESX host to vCenter affects whether manual intervention is necessary. Manual intervention means that you either reconnect the host, or you remove the host from vCenter and add it back.

Table 198: When Host Name or IP Address Changes Require Manual Intervention

ESX Host added to vCenter using...	ESX Host name changes	ESX IP address changes
Host name	vCenter connectivity problem. Manual intervention required.	No intervention required.
IP address	No intervention required.	vCenter connectivity problem. Manual intervention required.

ESX Host Upgrades and Certificates

If you upgrade an ESX host to ESX 9.0 or later, the upgrade process replaces the self-signed (thumbprint) certificates with VMCA-signed certificates. If the ESX host uses custom certificates, the upgrade process retains those certificates even if those certificates are expired or invalid.

The recommended upgrade workflow depends on the current certificates.

Host Provisioned with Thumbprint Certificates

If your host is currently using thumbprint certificates, it is automatically assigned VMCA certificates as part of the upgrade process.

> **Note:**
>
> You cannot provision legacy hosts with VMCA certificates. You must upgrade those hosts to ESX 9.0 or later.

Host Provisioned with Custom Certificates

If your host is provisioned with custom certificates, usually third-party CA-signed certificates, those certificates remain in place during upgrade. Change the certificate mode to **Custom** to ensure that the certificates are not replaced accidentally during a certificate refresh later.

> **Note:**
> If your environment is in VMCA mode, and you refresh the certificates from the vSphere Client, any existing certificates are replaced with certificates that are signed by VMCA.

Going forward, vCenter monitors the certificates and displays information, for example, about certificate expiration, in the vSphere Client.

Hosts Provisioned with Auto Deploy

Hosts that are being provisioned by Auto Deploy are always assigned new certificates when they are first booted with ESX 9.0 or later software. When you upgrade a host that is provisioned by Auto Deploy, the Auto Deploy server generates a certificate signing request (CSR) for the host and submits it to VMCA. VMCA stores the signed certificate for the host.

When the Auto Deploy server provisions the host, it retrieves the certificate from VMCA and includes it as part of the provisioning process.

You can use Auto Deploy with custom certificates.

Note:

In ESX 9.0, the Auto Deploy capability, where PXE boot infrastructure is used together with vSphere Host Profiles to provision and customize ESX hosts, is deprecated and will be removed in a future ESX release. For more information on the capability, see ESX Using vSphere Auto Deploy and Managing Stateless Hosts with an Image.

ESX Certificate Mode Switch Workflows

By default, VMware Certificate Authority (VMCA) provisions ESX with certificates. Use custom mode when replacing VMCA certificates with custom certificates. Use legacy thumbprint mode for debugging. If you do require a mode switch, review the potential impact before you start.

For an explanation of certificate modes, see Certificates and Certificate Modes.

Using Custom ESX Certificates

Note:
When switching from using VMCA certificates to custom certificates, make sure to allow time for organizational approval and fulfillment processes when generating certificates. Also, plan accordingly so that the current certificate does not expire during the switchover.

If your company policy requires that you use a different root CA than VMCA, you can switch the certificate mode in your environment after careful planning. The workflow is as follows.

1. Switch to **custom** mode. See Change the ESX Certificate Mode.
   Switching the mode enables the vSphere Client to activate the **Manage with External CA** drop-down, allowing you to generate the Certificate Signing Request.
2. Add the root certificate of the custom CA to VMware Endpoint Certificate Store (VECS).
3. Generate the Certificate Signing Request and obtain the certificates that you want to use.
   You might need to wait some time for the CSR to be returned.
4. Import the custom CA certificate to the vCenter host.
   Allow some time for the vCenter to distribute the custom CA certificate to the ESX hosts.

Switching from Custom CA Mode to VMCA Mode

If you are using custom CA mode and decide that using VMCA works better in your environment, you can perform the mode switch after careful planning. The workflow is as follows.

1. Remove all hosts from the vCenter system.
2. On the vCenter system, remove the root certificate of the third-party CA from VECS.
3. Switch to **vmca** mode. See Change the ESX Certificate Mode.
4. Add the hosts to the vCenter system.

Note:
Any other workflow for this mode switch might result in unpredictable behavior.

Retaining Thumbprint Mode Certificates During Upgrade

The switch from VMCA mode to thumbprint mode might be necessary if you encounter problems with the VMCA certificates. In thumbprint mode, the vCenter system checks only whether a certificate exists and is formatted correctly, and does not check whether the certificate is valid. See Change the ESX Certificate Mode for instructions.

Switching from Thumbprint Mode to VMCA Mode

If you use thumbprint mode and you want to start using VMCA-signed certificates, the switch requires some planning. The workflow is as follows.

1. Remove all ESX hosts from the vCenter system.
2. Switch to **vmca** mode. See Change the ESX Certificate Mode.
3. Add the ESX hosts to the vCenter system.

> **Note:**
> Any other workflow for this mode switch might result in unpredictable behavior.

Switching from Custom CA Mode to Thumbprint Mode

If you are encountering problems with your custom CA, consider switching to thumbprint mode temporarily. The switch works seamlessly if you follow the instructions in Change the ESX Certificate Mode. After the mode switch, the vCenter system checks only the format of the certificate and no longer checks the validity of the certificate itself.

Switching from Thumbprint Mode to Custom CA Mode

If you set your environment to thumbprint mode during troubleshooting, and you want to start using custom CA mode, you must first generate the required certificates. The workflow is as follows.

1. Remove all ESX hosts from the vCenter system.
2. Add the custom CA root certificate to the TRUSTED_ROOTS store on VECS on the vCenter system. See Update the vCenter TRUSTED_ROOTS Store (Custom Certificates).
3. For each ESX host:
   a. Deploy the custom CA certificate and key.
   b. Restart services on the host.
4. Switch to **custom** mode. See Change the ESX Certificate Mode.
5. Add the ESX hosts to the vCenter system.

ESX Certificate Default Settings

When a host is added to a vCenter system, vCenter sends a Certificate Signing Request (CSR) for the host to VMCA. Most of the default values are well suited for many situations, but company-specific information can be changed.

You can change many of the default settings using the vSphere Client. Consider changing the organization, and location information. See Change ESX Certificate Default Settings.

Table 199: ESX CSR Settings

Parameter	Default Value	Advanced Option
Key Size	2048	N.A.
Key Algorithm	RSA	N.A.
Certificate Signature Algorithm	sha256WithRSAEncryption	N.A.
Common Name	Name of the host if the host was added to vCenter by host name. IP address of the host if the host was added to vCenter by IP address.	N.A.
Country	US	vpxd.certmgmt.certs.cn.country
Email address	vmca@vmware.com	vpxd.certmgmt.certs.cn.email
Locality (City)	Palo Alto	vpxd.certmgmt.certs.cn.localityName
Organization Unit Name	VMware Engineering	vpxd.certmgmt.certs.cn.organizationalUnitName

Parameter	Default Value	Advanced Option
Organization Name	VMware	vpxd.certmgmt.certs.cn.organizationName
State or province	California	vpxd.certmgmt.certs.cn.state
Number of days the certificate is valid.	1825	vpxd.certmgmt.certs.daysValid
Hard threshold for the certificate expiration. vCenter raises a red alarm when this threshold is reached.	30 days	vpxd.certmgmt.certs.hardThreshold
Poll interval for vCenter certificate validity checks.	5 days	vpxd.certmgmt.certs.pollIntervalDays
Soft threshold for the certificate expiration. vCenter raises an event when this threshold is reached.	240 days	vpxd.certmgmt.certs.softThreshold
Mode that vCenter users to determine whether existing certificates are replaced. Change this mode to retain custom certificates during upgrade. See ESX Host Upgrades and Certificates.	vmca You can also specify thumbprint or custom. See Change the ESX Certificate Mode.	vpxd.certmgmt.mode

Change ESX Certificate Default Settings

When an ESX host is added to a vCenter system, vCenter sends a Certificate Signing Request (CSR) for the host to VMCA. You can change some of the default settings in the CSR using the vCenter Advanced Settings in the vSphere Client.

See the previous table for a list of default settings. Some of the defaults cannot be changed.

1. In the vSphere Client, select the vCenter system that manages the hosts.
2. Click **Configure**, and click **Advanced Settings**.
3. Click **Edit Settings**.
4. Click the **Filter** icon in the Name column, and in the Filter box, enter `vpxd.certmgmt` to display only certificate management parameters.
5. Change the value of the existing parameters to follow your company policy and click **Save**.

   The next time you add a host to vCenter, the new settings are used in the CSR that vCenter sends to VMCA and in the certificate that is assigned to the host.

Changes to certificate metadata only affect new certificates. If you want to change the certificates of hosts that are already managed by the vCenter system, you can disconnect and reconnect the hosts or renew the certificates.

View Certificate Expiration Information for ESX Hosts

For ESX hosts that are in VMCA mode or custom mode, you can view certificate details from the vSphere Client. The certificate information enables you to determine whether any of your certificates expire soon. You can also use this information to debug certificate problems.

You cannot view certificate status information for ESX hosts in thumbprint mode. You can view information for multiple ESX hosts or a single ESX host. The multiple-host view displays only the Certificate Valid To date information.

1. Log in to the vCenter by using the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Get the certificate information.

Single Host or Multiple Hosts	Steps
Single	1. Browse to the ESX host. 2. Click **Configure**. 3. Under **System**, click **Certificate**.
Multiple	1. Select **Hosts & Clusters** > **Hosts**. By default, the Hosts display does not include the certificate status. 2. To show or hide columns, click **Manage Columns**. 3. Select the **Certificate Valid To** check box, and scroll to the right if necessary to view the added column. The certificate information displays when the certificate expires. 4. (Optional) Deselect other columns to make it easier to see what you are interested in.

4. Review the certificate information.

   The following information is available only in the single-host view.

Field	Description
Subject	The subject used during certificate generation.
Issuer	The issuer of the certificate.
Valid From	Date on which the certificate was generated.
Valid To	Date on which the certificate expires.
Status	Status of the certificate, one of the following. **Good** Normal operation. **Expiring** Certificate expires soon. **Expiring shortly** Certificate is eight months or less away from expiration (Default). **Expiration imminent** Certificate is two months or less away from expiration (Default). **Expired** Certificate is not valid because it expired.

Note:
If a host is added to vCenter or reconnected after a disconnect, vCenter renews the certificate if the status is Expired, Expiring, Expiring shortly, or Expiration imminent. The status is Expiring if the certificate is valid for less than eight months, Expiring shortly if the certificate is valid for less than two months, and Expiration imminent if the certificate is valid for less than one month.

Renew the certificates that are about to expire. See Renew or Refresh ESX Certificates.

Renew or Refresh ESX Certificates

If you use the VMware Certificate Authority (VMCA) to assign certificates to your hosts, you can renew those certificates from the vSphere Client. If you use either VMCA certificates or custom certificates, you can refresh all certificates from the TRUSTED_ROOTS store associated with vCenter.

Verify the following:

- If using VMCA certificates, the certificate mode is set to **vmca**.
- If using custom certificates, the certificate mode is set to **custom**.
- The ESX hosts are connected to the vCenter system.
- There is proper time synchronization between the vCenter system and the ESX hosts.

- DNS resolution works between the vCenter system and the ESX hosts.
- The vCenter system's MACHINE_SSL_CERT and Trusted_Root certificates are valid and have not expired. See the VMware knowledge base article at https://kb.vmware.com/s/article/2111411.
- The ESX hosts are not in maintenance mode.

Note:
If you use custom certificates, and you need to renew them, re-import the certificates.

You can use the vSphere Client to renew your VMCA certificates when they are about to expire, or if you want to provision the host with a new certificate for other reasons. If you do not renew the VMCA certificate before it expires, disconnecting the host and reconnecting it causes vCenter to renew the certificate. The act of re-adding the host to vCenter reestablishes trust, and enables vCenter to unconditionally issue the renewed certificate.

By default, vCenter renews the VMCA certificate of a host with status Expired, Expiration imminent, or Expiring shortly, and each time the host is added to the inventory or reconnected.

You cannot renew an ESX certificate with an expiration date beyond that of the expiration date of the trusted root certificate. For example, even if the ESXvpxd.certmgmt.certs.daysValid advanced option is set to five years, and your trusted root certificate is set to expire in two years, the ESX certificate expiration date is limited to two years.

You can use the vSphere Client to push all certificates currently in the TRUSTED_ROOTS store in the vCenter VECS store to the ESX host. Use this capability if you need to refresh the trusted roots on an ESX host. This capability exists for both VMCA and custom certificates.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under **System**, click **Certificate**.
   You can view the details about the certificate of the selected host.
4. Select the appropriate option based on the type of certificate used.

Option	Description
Manage with VMCA > Renew	Retrieves a fresh signed certificate for the host from VMCA.
Manage with VMCA > Refresh CA Certificates **or** **Manage with External CA > Refresh CA Certificates**	Pushes all certificates in the TRUSTED_ROOTS store in the vCenter VECS store to the host.

Change the ESX Certificate Mode

Use VMware Certificate Authority (VMCA) to provision the certificates of the ESX hosts in your environment unless your corporate policy requires that you use custom certificates. To use custom certificates with a different root CA, edit the advanced vCenter setting, vpxd.certmgmt.mode. After the change, the hosts are no longer automatically provisioned with VMCA certificates when you refresh the certificates. You are responsible for the certificate management in your environment.

You can use the advanced vCenter settings to change to thumbprint mode or to custom CA mode. Use thumbprint mode only as a fallback option.

1. In the vSphere Client, select the vCenter system that manages the hosts.
2. Click **Configure**, and under Settings, click **Advanced Settings**.
3. Click **Edit Settings**.
4. Click the **Filter** icon in the Name column, and in the Filter box, enter `vpxd.certmgmt` to display only certificate management parameters.

Note:
The available options are **vmca**, **custom**, and **thumbprint**.

5. Change the value of vpxd.certmgmt.mode to **custom** if you intend to manage your own certificates, and to **thumbprint** if you temporarily want to use thumbprint mode, and click **Save**.

Replacing the Default ESX Certificate with a Custom Certificate

Your company's security policy might require that you replace the default ESX SSL certificate with a third-party certificate authority (CA) signed certificate on all your hosts.

By default, vSphere components use the VMCA-signed certificate and key that are created during installation. If you accidentally delete the VMCA-signed certificate, remove the host from its vCenter system, and add it back. When you add the host, vCenter requests a new certificate from VMCA and provisions the host with it.

You can replace VMCA-signed certificates with certificates from a trusted certificate authority, either a commercial CA or an organizational CA, if your company policy requires it.

You can replace the default certificates with custom certificates using the vSphere Client or the CLI.

> **Note:**
> You can also use the `vim.CertificateManager` and `vim.host.CertificateManager` managed objects in the vSphere Web Services SDK. See the vSphere Web Services SDK documentation.

Before you replace the certificate, you must update the TRUSTED_ROOTS store in VECS on the vCenter system that manages the host to ensure that the vCenter and the ESX host have a trust relationship.

> **Note:**
> If you are replacing SSL certificates on an ESX host that is part of a vSAN cluster, follow the steps that are in the VMware knowledge base article at https://kb.vmware.com/s/article/56441.

Requirements for ESX Certificate Signing Requests for Custom Certificates

Use a CSR with these characteristics:

- Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
- PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
- x509 version 3
- For root certificates, the CA extension must be set to true, and the cert sign must be in the list of requirements.
- SubjectAltName must contain DNS Name=<machine_FQDN>.
- CRT format
- Contains the following Key Usages: Digital Signature, Key Encipherment
- Start time of one day before the current time.
- CN (and SubjectAltName) set to the host name (or IP address) that the ESX host has in the vCenter inventory.

> **Note:**
> vSphere's FIPS certificate only validates RSA key sizes of 2048 and 3072. See Considerations When Using FIPS.

vSphere does not support the following certificates.

- Certificates with wildcards.
- The algorithms md2WithRSAEncryption, md5WithRSAEncryption, RSASSA-PSS, dsaWithSHA1, ecdsa_with_SHA1, and sha1WithRSAEncryption are not supported.

To generate the CSR using the vSphere Client, see Generate a Certificate Signing Request for a Custom Certificate Using the vSphere Client.

For information about generating the CSR using the CLI, see the VMware knowledge base article at https://kb.vmware.com/s/article/2113926.

Generate a Certificate Signing Request for a Custom Certificate Using the vSphere Client

Starting in vSphere 8.0 Update 3, you can use the vSphere Client to generate a Certificate Signing Request (CSR) for the ESX SSL certificate and to replace the certificate once it is ready.

Change the certificate mode to **custom**. See Change the ESX Certificate Mode. Switching the mode enables the vSphere Client to activate the **Manage with External CA** drop-down, allowing you to generate the Certificate Signing Request.

> **Warning:**
> Generating a Certificate Signing Request generates a new private key. Do not generate another Certificate Signing Request during the process of replacing certificates. If you do, the previously generated CSR and subsequent certificate will no longer be valid.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under **System**, click **Certificate**.
4. From the **Manage with External CA** drop-down, select either **Generate CSR Using IP** or **Generate CSR Using FQDN**.

   vCenter identifies the option previously used to generate the certificate on the ESX host.
5. Select either **Copy to Clipboard** or **Download**, depending on how you want to generate the certificate signing request.

You can now send the CSR to the certificate authority, or use the CSR to generate the certificate internally.

Replace the Default Certificate with a Custom Certificate Using the vSphere Client

Starting in vSphere 8.0 Update 3, you can replace the default VMCA-signed ESX certificates with custom certificates from the vSphere Client.

- Generate the certificate signing request and send it to the certificate authority. See Generate a Certificate Signing Request for a Custom Certificate Using the vSphere Client.
- When the certificate authority returns the certificate, store it on the ESX hosts.
- Make sure that the ESX certificate mode is set to **custom**. See Change the ESX Certificate Mode.
- Update the trusted root store. See Update the vCenter TRUSTED_ROOTS Store (Custom Certificates).

When importing the custom certificate, make sure that you:

- Add your entire CA certificate chain before proceeding with the replacement.
- Ensure that you provide the correct CA certificate for your environment. The import and replace process does not perform checking on the certificate that you use.
- Ensure that there are no SHA1 hashes in the certificate chain. SHA1 is not supported.
- Add the root CA to VECS before proceeding. If not, the host disconnects immediately after the certificate replacement.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under **System**, click **Certificate**.
4. From the **Manage with External CA** drop-down, select **Import and replace**.
5. Select the replace option.

Option	Description
Replace with external CA certificate where CSR generated by ESX (private key embedded)	Use this option if you generated the CSR on ESX, in which case, the private key is stored on ESX.
Replace with external CA certificate where CSR generated from a certificate authority (requires private key)	Use this option if you sent the CSR to a third-party certificate authority, and received back the certificate and private key.

6. Click **Next**.
7. Browse for the certificate, or certificate and private key.
8. Review the information, then click **Import and Replace**.

The custom certificate replaces the existing certificate.

Replace the Default Certificate and Key from theESX Shell

You can replace the default VMCA-signedESX certificates from theESX Shell.

- If you want to use third-party CA-signed certificates, generate the certificate request, send it to the certificate authority, and store the certificates on each ESX host.
- If necessary, enable the ESX Shell or enable SSH traffic from the vSphere Client.
- All file transfers and other communications occur over a secure HTTPS session. The user who is used to authenticate the session must have the privilege **Host** > **Config** > **AdvancedConfig** on the host.

> **Note:**
> Before you replace the certificates, update thevCenter Server TRUSTED_ROOTS store. See Update the vCenter TRUSTED_ROOTS Store (Custom Certificates).

1. Log in to theESX Shell, either directly from the DCUI or from an SSH client, as a user with administrator privileges.
2. In the directory `/etc/vmware/ssl`, rename the existing certificates using the following commands.

```
mv rui.crt orig.rui.crt
mv rui.key orig.rui.key
```

3. Copy the certificates that you want to use to `/etc/vmware/ssl`.
4. Rename the new certificate and key to `rui.crt` and `rui.key`.
5. Restart the host after you install the new certificate.

   Alternatively, you can put the host into maintenance mode, install the new certificate, use the Direct Console User Interface (DCUI) to restart the management agents, and set the host to exit maintenance mode.

Replace the Default Certificate Using HTTPS PUT

You can use third-party applications to upload certificates and key. Applications that support HTTPS PUT operations work with the HTTPS interface that is included withESX.

- If you want to use third-party CA-signed certificates, generate the certificate request, send it to the certificate authority, and store the certificates on each ESX host.
- If necessary, enable the ESX Shell or enable SSH traffic from the vSphere Client.
- All file transfers and other communications occur over a secure HTTPS session. The user who is used to authenticate the session must have the privilege **Host** > **Config** > **AdvancedConfig** on the host.

> **Note:**
> Before you replace the certificates, update the vCenter ServerTRUSTED_ROOTS store. See Update the vCenter TRUSTED_ROOTS Store (Custom Certificates).

1. Back up the existing certificates.
2. Set up basic access authentication, in which you supply a Base64 encoded username and password, separated with a single colon (:). For more information, see https://en.wikipedia.org/wiki/Basic_access_authentication.
3. In your upload application, process each file as follows:
   a) Open the file.
   b) Publish the file to one of these locations.

Option	Description
Certificates	`https://hostname/host/ssl_cert`
Keys	`https://hostname/host/ssl_key`

   The `/host/ssl_cert` and the `host/ssl_key` locations link to the certificate files in `/etc/vmware/ssl`.
4. Restart the host.

   Alternatively, you can put the host into maintenance mode, install the new certificate, use the Direct Console User Interface (DCUI) to restart the management agents, and set the host to exit maintenance mode.

Update the vCenter TRUSTED_ROOTS Store (Custom Certificates)

If you set up your ESX hosts to use custom certificates, you must update the `TRUSTED_ROOTS` store on the vCenter system that manages the hosts.

Replace the certificates on each host with custom certificates.

> **Note:**
> This step is not required if the vCenter system is also running with custom certificates issued by the same CA as those installed on the ESX hosts.

1. To update the vCenter Server TRUSTED_ROOTS store using vSphere Client, see Add a Trusted Root Certificate to the Certificate Store Using the vSphere Client .
2. To update the vCenter Server TRUSTED_ROOTS store using using command line interface, log in to the vCenter shell of the vCenter system that manages the ESX hosts.
3. To add the new certificates to the `TRUSTED_ROOTS` store, run `dir-cli`, for example:

   ```
   /usr/lib/vmware-vmafd/bin/dir-cli trustedcert publish --certpath_to_RootCA
   ```
4. When prompted, provide the Single Sign-On Administrator credentials.
5. If your custom certificates are issued by an intermediate CA, you must also add the intermediate CA to the `TRUSTED_ROOTS` store on the vCenter, for example:

   ```
   /usr/lib/vmware-vmafd/bin/dir-cli trustedcert publish --certpath_to_intermediateCA
   ```

Set certificate mode to Custom. If the certificate mode is VMCA, the default, and you perform a certificate refresh, your custom certificates are replaced with VMCA-signed certificates. See Change the ESX Certificate Mode.

Make Auto Deploy a Subordinate Certificate Authority

By default, the Auto Deploy server provisions each host with certificates that are signed by the VMware Certificate Authority (VMCA). You can set up the Auto Deploy server to provision all hosts with custom certificates that are not signed by VMCA. In that scenario, the Auto Deploy server becomes a subordinate certificate authority of your third-party certificate authority (CA).

- Request a certificate from your CA. The certificate must meet these requirements.
  - Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
  - PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
  - x509 version 3
  - For root certificates, the CA extension must be set to true, and the cert sign must be in the list of requirements.

  - SubjectAltName must contain DNS Name=<machine_FQDN>.
  - CRT format
  - Contains the following Key Usages: Digital Signature, Key Encipherment
  - Start time of one day before the current time.
  - CN (and SubjectAltName) set to the host name (or IP address) that the ESX host has in the vCenter inventory.

    Note:
    vSphere's FIPS certificate only validates RSA key sizes of 2048 and 3072. See Considerations When Using FIPS.
- Name the certificate and the key files `rbd-ca.crt` and `rbd-ca.key`.

1. Back up the default ESX certificates.

   The certificates are in the `/etc/vmware-rbd/ssl/` directory.
2. Stop the vSphere Authentication Proxy service.

Tool	Steps
vCenter Management Interface	1. In a Web browser, go to the vCenter Management Interface, https://*vcenter-IP-address-or-FQDN*:5480. 2. Log in as root. The default root password is the password that you set while deploying the vCenter. 3. Click **Services**, and click **VMware vSphere Authentication Proxy**. 4. Click **Stop**.
CLI	`service-control --stop vmcam`

3. On the system where the Auto Deploy service runs, replace `rbd-ca.crt` and `rbd-ca.key` in `/etc/vmware-rbd/ssl/` with your custom certificate and key files.
4. On the system where the Auto Deploy service runs, run the following command to update the TRUSTED_ROOTS store inside the VMware Endpoint Certificate Store (VECS) to use your new certificates.

   ```
   /usr/lib/vmware-vmafd/bin/dir-cli trustedcert publish --cert /etc/vmware-rbd/ssl/rbd-ca.crt
   /usr/lib/vmware-vmafd/bin/vecs-cli force-refresh
   ```
5. Create a `castore.pem` file that contains what is in the TRUSTED_ROOTS store and place the file in the `/etc/vmware-rbd/ssl/` directory.

   In custom mode, you are responsible for maintaining this file.
6. Change the ESX certificate mode for the vCenter system to `custom`.

   See Change the ESX Certificate Mode.
7. Restart the vCenter service and start the Auto Deploy service.

The next time you provision a host that is set up to use Auto Deploy, the Auto Deploy server generates a certificate. The Auto Deploy server uses the root certificate that you added to the TRUSTED_ROOTS store.

Use Custom Certificates with Auto Deploy

In vSphere 8.0 and later, you can set up the Auto Deploy server to provision ESX hosts with custom certificates that are signed by a third-party certificate authority (CA) or your own internal CA. By default, the Auto Deploy server provisions ESX hosts with certificates that are signed by the VMware Certificate Authority (VMCA).

Ensure that you have the following:

- Request a certificate from your certificate authority. The certificate must meet these requirements.
  - Key size: 2048 bits (minimum) to 8192 bits (maximum) (PEM encoded)
  - PEM format. VMware supports PKCS8 and PKCS1 (RSA keys). When keys are added to VECS, they are converted to PKCS8.
  - x509 version 3
  - CRT format

  - CA extension set to true
  - Key usage of Certificate Signing
  - Start time of one day before the current time

    Note:
    vSphere's FIPS certificate only validates RSA key sizes of 2048 and 3072. See Considerations When Using FIPS.
- ESX host MAC address or BIOS UUID. Evaluate which approach makes best sense for your environment. The BIOS UUID is more stable and less subject to change than the MAC address. If you change network adapters in an ESX host, the MAC address changes. However, the MAC address might be more familiar to work with, and easier to obtain than the BIOS UUID.
- At least PowerCLI version 12.6.0. For more information on Auto Deploy PowerCLI cmdlets, see the Auto Deploy PowerCLI Cmdlet Overview topic in the *VMware ESX Installation and Setup* documentation.

Ensure that you have the following privileges:

- Add custom certificate: **Autodeploy** > **Rule** > **Create**
- Get custom certificate information: **System** > **Read**

Note:

vSphere Auto Deploy is deprecated in the 9.0 release and will be removed in a future release.

In vSphere 8.0 and later, you can upload custom certificates to Auto Deploy that are signed by either a third-party CA or your own internal CA. Auto Deploy associates the custom certificate with either the MAC address or BIOS UUID of the ESX host. Each time an Auto Deploy host starts, Auto Deploy checks for a custom certificate. If Auto Deploy finds a custom certificate, it uses that certificate instead of generating one through the VMCA.

The high-level steps for this task include:

1. Generating the custom certificate request for either a third-party CA or for your own internal CA.
2. Obtaining the signed custom certificate (key and certificate) and storing it locally.
3. If you are using a third-party CA, and if not previously done, ensuring the root certificate of your CA is uploaded to the TRUSTED_ROOTS store on the vCenter.
4. Uploading the custom certificate to Auto Deploy and associating the certificate with either the MAC address or BIOS UUID of an ESX host.
5. Booting the ESX host.

When you assign a custom certificate to an ESX host, Auto Deploy pushes the certificate to the host on its next boot from Auto Deploy.

Be aware of the following considerations when using custom certificates and Auto Deploy.

- You must use the PowerCLI `Add-CustomCertificate`, `Remove-CustomCertificate`, and `List-CustomCertificate` cmdlets to manage custom certificates used with Auto Deploy. The capability to manage custom certificates is not available in the vSphere Client.
- To refresh a custom certificate used for Auto Deploy, you must run the `Add-CustomCertificate` cmdlet again.
- Be sure to examine your custom certificate for potential errors. Auto Deploy verifies only that the custom certificate complies with the X.509 certificate standards and that the expiration threshold of the certificate is set to at least 240 days. Auto Deploy does not perform any other certificate validation or checking. To change the certificate threshold, you can run the `Set-DeployOption -Key certificate-refresh-threshold` cmdlet.
- If you later remove a custom certificate from an ESX host using the `Remove-CustomCertificate` cmdlet, you must restart the host for the change to take effect.

For more information about custom certificates and Auto Deploy, see the *VMware ESX Installation and Setup* documentation.

1. Generate the certificate request.
   a) Using the requirements listed previously for the certificate request, create a configuration (.cfg) file.
   b) To generate a CSR file and a key file, run the `openssl req` command, passing in the configuration (.cfg) file.
      For example:
      ```
      openssl req -new -config custom_cert.cfg -days 4200 -sha256 -keyout rui.key -out rui.csr
      ```

In this command:

- `-new` generates a new certificate request.
- `-config custom_cert.cfg` specifies your custom .cfg file.
- `-days 4200` specifies 4200 days to certify the certificate for.
- `-sha256` specifies the message digest to sign the request with.
- `-keyout rui.key` specifies the file to write the newly created private key to.
- `-out rui.csr` specifies the output file to write to.

2. Either send the certificate request to your third-party CA, or, if you sign your own certificates, run the `openssl x509 -req` command to generate your custom certificate from your `rui.csr` file.

   For example:

   ```
   openssl x509 -req -in rui.csr -CA "/etc/vmware-rbd/ssl/rbd-ca.crt" -CAkey \
   "/etc/vmware-rbd/ssl/rbd-ca.key" -extfile \
   openssl.cfg -extensions x509 -CAserial "/etc/vmware-rbd/ssl/rbd-ca.srl" -days \
   4200 -sha256 -out signed_rui.crt
   ```

   In this command:

   - `-in rui.csr` specifies the input file.
   - `-CA "/etc/vmware-rbd/ssl/rbd-ca.crt"` specifies the directory to use for server certificate verification.
   - `-CAkey "/etc/vmware-rbd/ssl/rbd-ca.key"` sets the CA private key to sign a certificate with.
   - `-extfile openssl.cfg` specifies an additional, optional configuration file to read certificate extensions from.
   - `-extensions x509` specifies to use x509 certificate extensions.
   - `-CAserial "/etc/vmware-rbd/ssl/rbd-ca.srl"` uses the serial number in `rbd-ca.srl` to sign a certificate.
   - `-days 4200` specifies 4200 days to certify the certificate for.
   - `-sha256` specifies the message digest to sign the request with.
   - `-out signed_rui.crt` specifies the output file to write to.

3. Optional: If you haven't previously uploaded the certificate of your signing certificate authority to the TRUSTED_ROOTS store inside the VMware Endpoint Certificate Store (VECS), perform the following steps on the vCenter where the Auto Deploy service runs.
   a) Using a tool such as WinSCP, copy the certificate to the vCenter.
   b) Log in to the vCenter using SSH and run the following command.

      ```
      /usr/lib/vmware-vmafd/bin/dir-cli trustedcert publish --cert path_to_ca_certificate
      ```

4. Obtain either the ESX host MAC address or BIOS UUID.
5. Perform the following steps to add the custom certificate to Auto Deploy.
   a) To connect to the vCenter, run the `Connect-VIServer` cmdlet.

      ```
      Connect-VIServer -server VC_ip_address -User administrator_user -Password 'password'
      ```

   b) Optional: To view existing custom certificates, run the `Get-CustomCertificates` cmdlet.

      The first time you add custom certificates, you don't see any certificates returned by this cmdlet.

   c) To associate the custom certificate with the ESX host, run the `Add-CustomCertificate` cmdlet.

      ```
      Add-CustomCertificate -HostID [MAC_Address | BIOS_UUID] -Certificate "path_to_custom_cert" -Key
       "path_to_custom_cert_key"
      ```

      You can specify either the MAC address or the BIOS UUID of the host. Auto Deploy uploads the custom certificate to the host.

   d) To verify that the certificate was uploaded, run the `Get-CustomCertificates` cmdlet.

      You see output similar to the following:

      ```
      Name:     CustomHostCert-1
      CertificateId:      1
      HostId:             02:08:b0:8e:18:a2
      ExpirationTime: 1   2/28/2033 10:45:50 AM
      ```

```
TimeCreated:         9/29/2022 7:40:28 AM
LastModified:        9/29/2022 7:40:28 AM
AssociatedHostName:
```

The `AssociatedHostName` is blank for now. After you start the host, the output reflects the name of the ESX host associated with the custom certificate.

6. Start the ESX host.
7. To verify that the custom certificate is associated with the vCenter, run the `Get-CustomCertificates` cmdlet again.
   You see output resembling the following.

```
Name:     CustomHostCert-1
CertificateId:      1
HostId:             02:08:b0:8e:18:a2
ExpirationTime: 1   2/28/2033 10:45:50 AM
TimeCreated:        9/29/2022 7:40:28 AM
LastModified:       9/29/2022 7:40:28 AM
AssociatedHostName: host1.example.com
```

   Now the `AssociatedHostName` contains the name of the ESX host.

Restore ESX Certificate and Key Files When Certificate Replacement Fails

When replacing a certificate on an ESX host fails, the system creates certificate `.bak` files, which you can use to recover to the previous state.

The host certificate and key are located in `/etc/vmware/ssl/rui.crt` and `/etc/vmware/ssl/rui.key`. When you replace a host certificate by using either the vSphere Client or the vSphere Web Services SDK `vim.CertificateManager` managed object, and the replacement fails, the system creates `.bak` files for the previous key and certificate files.

When certificate replacement fails, you can restore previous certificates by copying over the `.bak` files to the current certificate and key files.

Customizing ESX Host Security

You can customize many of the essential security settings for your ESX host through the Firewall, Services, and Security Profile panels available in the vSphere Client. The Security Profile is especially useful for single host management. If you are managing multiple hosts, consider using one of the VMware CLIs or SDKs and automating the customization.

Configuring the ESX Firewall

ESX includes a firewall that is enabled by default. At installation time, the ESX firewall is configured to block incoming and outgoing traffic, except traffic for services that are enabled in the security profile of the host. You manage the firewall using the vSphere Client, the CLI, and the API.

As you open ports on the firewall, consider that unrestricted access to services running on an ESX host can expose a host to outside attacks and unauthorized access. Reduce the risk by configuring the ESX firewall to enable access only from authorized networks.

> **Note:**
> The firewall also allows Internet Control Message Protocol (ICMP) pings and communication with DHCP and DNS (UDP only) clients.

You can manage ESX firewall ports as follows:

- Use **Configure** > **Firewall** for each host in the vSphere Client. See Manage ESX Firewall Settings.
- Use ESXCLI commands from the command line or in scripts. See Using ESXCLI Firewall Commands to Configure ESX Behavior.
- Use a custom VIB if the port you want to open is not included in the security profile.
  To install the custom VIB, you have to change the acceptance level of the ESX host to CommunitySupported.

Note:
If you engage VMware Technical Support to investigate a problem on an ESX host with a CommunitySupported VIB installed, VMware Support can request you to uninstall this VIB. Such a request is a troubleshooting step to determine if that VIB is related to the problem being investigated.

The behavior of the NFS Client rule set (`nfsClient`) is different from other rule sets. When the NFS Client rule set is enabled, all outbound TCP ports are open for the destination hosts in the list of allowed IP addresses. See NFS Client Firewall Behavior for more information.

Manage ESX Firewall Settings

You can configure incoming and outgoing firewall connections for a service or a management agent from the vSphere Client or at the command line.

This task describes how to use the vSphere Client to configure ESX firewall settings. You can use the ESXi Shell or ESXCLI commands to configure ESXi at the command line to automate the firewall configuration. See Using ESXCLI Firewall Commands to Configure ESX Behavior for examples of using ESXCLI to manipulate firewalls and firewall rules.

Note:
If different services have overlapping port rules, enabling one service might implicitly activate other services. You can specify which IP addresses are allowed to access each service on the host to avoid this problem.

1. Log in to the vCenter by using the vSphere Client.
2. Browse to the host in the inventory.
3. Click **Configure**, then click **Firewall** under **System**.
   You can toggle between incoming and outgoing connections by clicking **Incoming** and **Outgoing**.
4. In the Firewall section, click **Edit**.
5. Select from one of the service groups, **Ungrouped**, **Secure Shell**, and **Simple Network Management Protocol**.
6. Select the rule sets to be activated, or deselect the rule sets to be deactivated.
7. For some services, you can also manage service details by navigating to **Configure** > **System** > **Services**.
   For more information about starting, stopping, and restarting services, see Activate or Deactivate an ESX Service.
8. For some services, you can explicitly specify IP addresses from which connections are allowed.
   See Add Allowed IP Addresses for an ESX Host.
9. Click **OK**.

Add Allowed IP Addresses for an ESX Host

By default, the firewall for each service allows access to all IP addresses. To restrict traffic, change each service to allow traffic only from your management subnet. You can also deselect some services if your environment does not use them.

To update the Allowed IP list for a service you can use the vSphere Client, ESXCLI, or PowerCLI. This task describes how to use the vSphere Client. See the topic titled "Manage the ESXi Firewall" in the *ESXCLI Concepts and Examples* documentation for instructions on using ESXCLI.

1. Log in to the vCenter by using the vSphere Client.
2. Browse to the ESX host.
3. Click **Configure**, then click **Firewall** under **System**.
   You can toggle between incoming and outgoing connections by clicking **Incoming** and **Outgoing**.

4. In the Firewall section, click **Edit**.
5. Select from one of the three service groups, **Ungrouped**, **Secure Shell**, and **Simple Network Management Protocol**.
6. To display the Allowed IP Addresses section, expand a service.
7. In the Allowed IP Addresses section, deselect **Allow connections from any IP address** and enter the IP addresses of networks that are allowed to connect to the host.
   Separate IP addresses with commas. You can use the following address formats:
   - 192.168.0.0/24
   - 192.168.1.2, 2001::1/64
   - fd3e:29a6:0a81:e478::/64
8. Ensure that the service itself is selected.
9. Click **OK**.
10. Verify your change in the **Allowed IP addresses** column for the service.

Incoming and Outgoing Firewall Ports for ESX Hosts

Open and close firewall ports for each service by using either the vSphere Client or the VMware Host Client.

ESX includes a firewall that is active by default. At installation time, the ESX firewall is configured to block incoming and outgoing traffic, except traffic for services that are enabled in the host security profile. For the list of supported ports and protocols in the ESX firewall, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/.

The VMware Ports and Protocols Tool lists port information for services that are installed by default. If you install other VIBs on your host, additional services and firewall ports might become available. The information is primarily for services that are visible in the vSphere Client but the VMware Ports and Protocols Tool includes some other ports as well.

NFS Client Firewall Behavior

The NFS Client firewall rule set behaves differently than other ESX firewall rule sets. ESX configures NFS Client settings when you mount or unmount an NFS datastore. The behavior differs for different versions of NFS.

When you add, mount, or unmount an NFS datastore, the resulting behavior depends on the version of NFS.

NFS v3 Firewall Behavior

When you add or mount an NFS v3 datastore, ESX checks the state of the NFS Client (`nfsClient`) firewall rule set.

- If the `nfsClient` rule set is deactivated, ESX activates the rule set and deactivates the Allow All IP Addresses policy by setting the `allowedAll` flag to `FALSE`. The IP address of the NFS server is added to the allowed list of outgoing IP addresses.
- If the `nfsClient` rule set is activated, the state of the rule set and the allowed IP address policy are not changed. The IP address of the NFS server is added to the allowed list of outgoing IP addresses.

  Note:
  If you manually activate the `nfsClient` rule set or manually set the Allow All IP Addresses policy, either before or after you add an NFS v3 datastore to the system, your settings are overridden when the last NFS v3 datastore is unmounted. The `nfsClient` rule set is deactivated when all NFS v3 datastores are unmounted.

When you remove or unmount an NFS v3 datastore, ESX performs one of the following actions.

- If none of the remaining NFS v3 datastores are mounted from the server of the datastore being unmounted, ESX removes the server's IP address from the list of outgoing IP addresses.
- If no mounted NFS v3 datastores remain after the unmount operation, ESX deactivates the `nfsClient` firewall rule set.

NFS v4.1 Firewall Behavior

When you mount the first NFS v4.1 datastore, ESX activates the `nfs41client` rule set and sets its `allowedAll` flag to TRUE. This action opens port 2049 for all IP addresses. Unmounting an NFS v4.1 datastore does not affect the firewall state. That is, the first NFS v4.1 mount opens port 2049 and that port remains activated unless you close it explicitly.

Using ESXCLI Firewall Commands to Configure ESX Behavior

If your environment includes multiple ESX hosts, automate firewall configuration by using ESXCLI commands or the vSphere Web Services SDK.

Firewall Command Reference

You can use the ESX Shell or ESXCLI commands to configure ESX at the command line to automate a firewall configuration. To manipulate firewalls and firewall rules, see *Getting Started with ESXCLI* for an introduction, and *ESXCLI Concepts and Examples* for examples of using ESXCLI.

In ESX 7.0 and later, access to the `service.xml` file, used to create custom firewall rules, is restricted. See VMware Knowledge Base article 2008226 for information about creating custom firewall rules using the `/etc/rc.local.d/local.sh` file.

Table 200: Firewall Commands

Command	Description
`esxcli network firewall get`	Return the status of the firewall and list the default actions.
`esxcli network firewall set --default-action`	Set to true to set the default action to pass. Set to false to set the default action to drop.
`esxcli network firewall set --enabled`	Activate or deactivate the ESX firewall.
`esxcli network firewall load`	Load the firewall module and the rule set configuration files.
`esxcli network firewall refresh`	Refresh the firewall configuration by reading the rule set files if the firewall module is loaded.
`esxcli network firewall unload`	Destroy filters and unload the firewall module.
`esxcli network firewall ruleset list`	List rule sets information.
`esxcli network firewall ruleset set --allowed-all`	Set to true to allow all access to all IPs. Set to false to use a list of allowed IP addresses.
`esxcli network firewall ruleset set --enabled --ruleset-id=<string>`	Set enabled to true to activate the specified ruleset. Set enabled to false to deactivate the specified ruleset.
`esxcli network firewall ruleset allowedip list`	List the allowed IP addresses of the specified rule set.
`esxcli network firewall ruleset allowedip add`	Allow access to the rule set from the specified IP address or range of IP addresses.
`esxcli network firewall ruleset allowedip remove`	Remove access to the rule set from the specified IP address or range of IP addresses.
`esxcli network firewall ruleset rule list`	List the rules of each ruleset in the firewall.

Activate or Deactivate an ESX Service

You can activate or deactivate ESX services from the vSphere Client.

Connect to vCenter with the vSphere Client.

An ESX host includes several services that are running by default. If your company policy allows it, you can deactivate services from the security profile, or activate services.

> **Note:**
> Activating services affects the security of your host. Do not activate a service unless strictly necessary.

After installation, certain services are running by default, while others are stopped. Sometimes, you must perform other steps before a service becomes available in the UI. For example, the NTP service is a way of getting accurate time information, but this service only works when required ports are opened in the firewall.

Available services depend on the VIBs that are installed on the ESX host. You cannot add services without installing a VIB. Some VMware products, for example, vSphere HA, install VIBs on hosts and make services and the corresponding firewall ports available.

In a default installation, you can modify the status of the following services from the vSphere Client.

Table 201: ESX Services in the Security Profile

Service	Default	Description
Direct Console UI	Running	The Direct Console User Interface (DCUI) service allows you to interact with an ESX host from the local console host using text-based menus.
ESX Shell	Stopped	The ESX Shell is available from the Direct Console User Interface and includes a set of fully supported commands and a set of commands for troubleshooting and remediation. You must activate access to the ESX Shell from the direct console of each system. You can activate access to the local ESX Shell or access to the ESX Shell with SSH.
SSH	Stopped	The SSH client service on the host that allows remote connections through the Secure Shell.
attestd	Stopped	vSphere Trust Authority Attestation Service.
dpd	Stopped	Data Protection daemon.
Load-Based Teaming Daemon	Running	Load-Based Teaming.
kmxd	Stopped	vSphere Trust Authority Key Provider Service.
Active Directory Service	Stopped	When you configure ESX for Active Directory, this service is started.
NTP Daemon	Stopped	Network Time Protocol daemon.
CIM Server	Running	Service that can be used by Common Information Model (CIM) applications.
slpd	Stopped	Service Location Protocol daemon.
SNMP Server	Stopped	SNMP daemon. See the *vSphere Monitoring and Performance* documentation for information on configuring SNMP v1, v2, and v3.
VDTC Service	Running	vSphere Distributed Tracing Collector service.
vltd	Stopped	VCDR LWD Transport daemon.
Syslog Server	Stopped	Syslog daemon. You can activate the syslog from the Advanced System Settings in the vSphere Client. See the *vCenter Deployment and Setup* documentation.
VMware vCenter Agent	Running	vCenter agent. Allows a vCenter to connect to an ESX host. Specifically, vpxa is the communication conduit to the host daemon, which in turn communicates with the ESX kernel.
X.Org Server	Stopped	X.Org Server. This optional feature is used internally for 3D graphics for virtual machines.

1. Browse to an ESX host in the inventory.
2. Click **Configure**, then click **Services** under **System**.
3. Select the service you want to change.
   a) Select **Restart**, **Start**, or **Stop** for a one-time change to the host's status.
   b) To change the status of the host across reboots, click **Edit Startup Policy** and select a policy.

   - **Start and stop with host**: The service starts shortly after the host starts, and closes shortly before the host shuts down. Much like **Start and stop with port usage**, this option means that the service regularly attempts to complete its tasks, such as contacting the specified NTP server. If the port was closed but is later opened, the client begins completing its tasks shortly thereafter.
   - **Start and stop manually**: The host preserves the user-determined service settings, regardless of whether ports are open or not. When a user starts the NTP service, that service is kept running if the host is powered on. If the service is started and the host is powered off, the service is stopped as part of the shutdown process. When the host is powered on, the service is started again, preserving the user-determined state.
   - **Start and stop with port usage**: The default setting for these services. If any port is open, the client attempts to contact the network resources for the service. If some ports are open, but the port for a particular service is closed, the attempt fails. If and when the applicable outgoing port is opened, the service begins completing its startup.

     Note:
     These settings apply only to service settings that are configured through the UI or to applications that are created with the vSphere Web Services SDK. Configurations made through other means, such as from the ESX Shell or with configuration files, are not affected by these settings.
4. Click **OK**.

Configuring and Managing Lockdown Mode on ESX Hosts

To increase the security of your ESX hosts, you can put them in lockdown mode. In lockdown mode, operations must be performed through vCenter by default.

You can select normal lockdown mode or strict lockdown mode, which offer different degrees of lockdown. You can also use the Exception User list. Exception users do not lose their privileges when the host enters lockdown mode. Use the Exception User list to add the accounts of third-party solutions and external applications that need to access the host directly when the host is in lockdown mode.

Lockdown Mode Behavior

In lockdown mode, some services are deactivated, and some services are accessible only to certain users.

Lockdown Mode Services Available for Different Users

When the host is running, available services depend on whether lockdown mode is activated, and on the type of lockdown mode.

- In strict and normal lockdown mode, privileged users can access the host through vCenter, from the vSphere Client, or by using the vSphere Web Services SDK.
- Direct Console Interface behavior differs for strict lockdown mode and normal lockdown mode.
  - In strict lockdown mode, the Direct Console User Interface (DCUI) service is deactivated.
  - In normal lockdown mode, accounts on the Exception User list can access the DCUI if they have administrator privileges. In addition, all users who are specified in the `DCUI.Access` advanced system setting can access the DCUI.
- If the ESXi Shell or SSH is activated and the host is placed in lockdown mode, accounts on the Exception Users list who have administrator privileges can use these services. For all other users, ESXi Shell or SSH access is deactivated. ESXi or SSH sessions for users who do not have administrator privileges are closed.

All access is logged for both strict and normal lockdown mode.

Table 202: Lockdown Mode Behavior

Service	Normal Mode	Normal Lockdown Mode	Strict Lockdown Mode
vSphere Web Services API	All users, based on permissions	vCenter (vpxuser) Exception users, based on permissions vCloud Director (vslauser, if available)	vCenter (vpxuser) Exception users, based on permissions vCloud Director (vslauser, if available)
CIM Providers	Users with administrator privileges on the host	vCenter (vpxuser) Exception users, based on permissions vCloud Director (vslauser, if available)	vCenter (vpxuser) Exception users, based on permissions vCloud Director (vslauser, if available)
Direct Console UI (DCUI)	Users with administrator privileges on the host, and users in the `DCUI.Access` advanced system setting	Users defined in the `DCUI.Access` advanced system setting Exception users with administrator privileges on the host	DCUI service is stopped.
ESXi Shell (if activated) and SSH (if activated)	Users with administrator privileges on the host	Users defined in the `DCUI.Access` advanced option Exception users with administrator privileges on the host	Users defined in the `DCUI.Access` advanced system setting Exception users with administrator privileges on the host

Lockdown Mode Behavior for Users Logged In to the ESX Shell When Lockdown Mode Is Activated

Users might log in to the ESX Shell or access the host through SSH before lockdown mode is activated. In that case, users who are on the list of Exception Users and who have administrator privileges on the host remain logged in. The session is closed for all other users. Termination applies to both normal and strict lockdown mode.

How Can You Deactivate Lockdown Mode

You can deactivate lockdown mode as follows.

From the vSphere Client	Users can deactivate both normal lockdown mode and strict lockdown mode from the vSphere Client. See Deactivate Lockdown Mode from the vSphere Client.
From the Direct Console User Interface	Users who can access the Direct Console User Interface on the ESX host can deactivate normal lockdown mode. In strict lockdown mode, the Direct Console Interface service is stopped. See Activate or Deactivate Normal Lockdown Mode from the Direct Console User Interface.

Activate Lockdown Mode from the vSphere Client

Select lockdown mode to require that all host configuration changes go through vCenter. vSphere supports normal lockdown mode and strict lockdown mode.

If you want to disallow all direct access to a host completely, you can select strict lockdown mode. Strict lockdown mode makes it impossible to access a host if the vCenter is unavailable and SSH and the ESXi Shell are deactivated. See Lockdown Mode Behavior.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Security Profile**.
4. In the Lockdown Mode panel, click **Edit**.
5. Click **Lockdown Mode** and select one of the lockdown mode options.

Option	Description
Normal	The host can be accessed through vCenter. Only users who are on the Exception Users list and have administrator privileges can log in to the Direct Console User Interface. If SSH or the ESXi Shell is activated, access might be possible.
Strict	The host can only be accessed through vCenter. If SSH or the ESXi Shell is activated, running sessions for accounts in the `DCUI.Access` advanced system setting and for Exception User accounts that have administrator privileges remain enabled. All other sessions are closed.

6. Click **OK**.

Deactivate Lockdown Mode from the vSphere Client

Deactivate lockdown mode to allow configuration changes from direct connections to the ESX host. Leaving lockdown mode activated results in a more secure environment.

Users can deactivate both normal lockdown mode and strict lockdown mode from the vSphere Client.

1. Browse to a host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Security Profile**.
4. In the Lockdown Mode panel, click **Edit**.
5. Click **Lockdown Mode** and select **Disabled** to deactivate lockdown mode.
6. Click **OK**.

The system exits lockdown mode, vCenter displays an alarm, and an entry is added to the audit log.

Activate or Deactivate Normal Lockdown Mode from the Direct Console User Interface

You can activate and deactivate normal lockdown mode from the Direct Console User Interface (DCUI). You can activate and deactivate strict lockdown mode only from the vSphere Client.

When the host is in normal lockdown mode, the following accounts can access the Direct Console User Interface:

- Accounts in the Exception Users list who have administrator privileges on the host. The Exception Users list is meant for service accounts such as a backup agent.
- Users defined in the `DCUI.Access` advanced option for the host. This option can be used to activate access in a catastrophic failure.

User permissions are preserved when you activate lockdown mode. User permissions are restored when you deactivate lockdown mode from the Direct Console Interface.

> **Note:**
> If you upgrade a host that is in lockdown mode to ESX version 6.0 without exiting lockdown mode, and if you exit lockdown mode after the upgrade, all permissions defined before the host entered lockdown mode are lost. The system assigns the administrator role to all users who are found in the `DCUI.Access` advanced option to guarantee that the host remains accessible.
>
> To retain permissions, deactivate lockdown mode for the host from the vSphere Client before the upgrade.

1. At the Direct Console User Interface of the host, press F2 and log in.
2. Scroll to the **Configure Lockdown Mode** setting and press Enter to toggle the current setting.
3. Press Esc until you return to the main menu of the Direct Console User Interface.

Specifying Accounts with Access Privileges in Lockdown Mode

You can specify service accounts that can access the ESX host directly by adding them to the Exception Users list. You can specify a single user who can access the ESX host in a catastrophic vCenter failure.

What Can Accounts Do When vSphere Is in Lockdown Mode

The vSphere version determines what different accounts can do by default when lockdown mode is activated, and how you can change the default behavior.

- In vSphere 5.0 and earlier, only the root user can log in to the Direct Console User Interface on an ESX host that is in lockdown mode.
- In vSphere 5.1 and later, you can add a user to the `DCUI.Access` advanced system setting for each host. The setting is meant for a catastrophic failure of vCenter. Companies usually lock the password for the user with this access into a safe. A user in the `DCUI.Access` list does not need to have full administrative privileges on the host.
- In vSphere 6.0 and later, the `DCUI.Access` advanced system setting is still supported. In addition, vSphere 6.0 and later supports an Exception User list, which is for service accounts that have to log in to the host directly. Accounts with administrator privileges that are on the Exception Users list can log in to the ESXi Shell. In addition, those users can log in to a host's DCUI in normal lockdown mode and can exit lockdown mode.

  You specify Exception Users from the vSphere Client.

  Note:
  Exception users are host local users or Active Directory users with privileges defined locally for the ESX host. Users that are members of an Active Directory group lose their permissions when the host is in lockdown mode.

Add Users to the DCUI.Access Advanced System Setting

If there is a catastrophic failure, the `DCUI.Access` advanced system setting allows you to exit lockdown mode when you cannot access the host from vCenter. You add users to the list by editing the Advanced Settings for the host from the vSphere Client.

Note:
Users in the DCUI.Access list can change lockdown mode settings regardless of their privileges. The ability to change lockdown modes can impact the security of your host. For service accounts that need direct access to the host, consider adding users to the Exception Users list instead. Exception users can only perform tasks for which they have privileges. See Specify Lockdown Mode Exception Users later in this topic.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, click **Advanced System Settings**, and click **Edit**.
4. Filter for DCUI.
5. In the **DCUI.Access** text box, enter the local ESX user names, separated by commas.

   Note:
   You cannot enter Active Directory users. Only local ESX users are supported.

   By default, the root user is included. Consider removing the root user from the DCUI.Access list, and specifying a named account for better auditability.
6. Click **OK**.

Specify Lockdown Mode Exception Users

You can add users to the Exception Users list from the vSphere Client. These users do not lose their permissions when the host enters lockdown mode.

Usually these users are accounts that represent third-party solutions and external applications that need to continue to function in lockdown mode. For example, it makes sense to add service accounts such as a backup agent to the Exception Users list.

> **Note:**
> The Exception Users list is meant for service accounts that perform very specific tasks, and not for administrators. Adding administrator users to the Exception Users list defeats the purpose of lockdown mode.

Exception users are host local users or Active Directory users with privileges defined locally for the ESX host. They are not members of an Active Directory group and are not vCenter users. These users are allowed to perform operations on the host based on their privileges. That means, for example, that a read-only user cannot deactivate lockdown mode on a host.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Security Profile**.
4. In the Lockdown Mode panel, click **Edit**.
5. Click **Exception Users** and click the **Add User** icon to add exception users.
6. Click **OK**.

Using vSphere Installation Bundles to Perform Secure Updates

Upgrading ESX with ESXCLI requires an understanding of vSphere Installation Bundles, image profiles, and software depots.

ESX consists of an image profile, which describes a set of vSphere Installation Bundles (VIBs) that contain the actual software. A VIB is a signed ramdisk representing a component of the system, roughly analogous to an RPM or DEB on a Linux system. An image profile is a collection of VIBs. A software depot is a collection of VIBs and image profiles. ESX patches and depots contain updated image profiles composed from a common set of VIBs.

You can install ESX updates on a standalone host using the `esxcli software` commands. For more information, see the *VMware ESX Upgrade* documentation.

> **Note:**
> Normally, in a vSphere 7.0 and later environment, you use VMware vSphere® vSphere Lifecycle Manager for lifecycle management of ESX hosts.

To list all installed VIBs and their current version, or the current image profile, you can use the following ESXCLI commands.

- `esxcli software vib list`
- `esxcli software profile get`

In general, the high-level steps to upgrade ESX securely are:

- Putting the ESX host in maintenance mode
- Running an `esxcli software profile update` command, which points to a URL or a ZIP file transferred to the host through SSH
- Restarting the ESX host

Because VMware cryptographically signs VIBs, secure transfer of VIBs or the entire depot is not necessary, and the update process verifies these signatures.

Manage the Acceptance Levels of ESX Hosts and vSphere Installation Bundles

The acceptance level of a vSphere Installation Bundle (VIB) depends on the amount of certification of that VIB. The acceptance level of the ESX host depends on the level of the lowest VIB. If you want to allow lower-level VIBs, you can change the acceptance level of the host. You can remove CommunitySupported VIBs to be able to change the host acceptance level.

VIBs are software packages that include a signature from VMware or a VMware partner. To protect the integrity of the ESX host, do not allow users to install unsigned (community-supported) VIBs. An unsigned VIB contains code that is

not certified by, accepted by, or supported by VMware or its partners. Community-supported VIBs do not have a digital signature.

The acceptance level of the ESX host must be the same or less restrictive than the acceptance level of any VIB you want to add to the host. For example, if the host acceptance level is VMwareAccepted, you cannot install VIBs at the PartnerSupported level. You can use ESXCLI commands to set an acceptance level for a host. To protect the security and integrity of your ESXi hosts, do not allow unsigned (CommunitySupported) VIBs to be installed on hosts in production systems.

The acceptance level for an ESX host is displayed in the **Security Profile** in the vSphere Client.

The following acceptance levels are supported.

VMwareCertified	The VMwareCertified acceptance level has the most stringent requirements. VIBs with this level go through thorough testing fully equivalent to VMware in-house Quality Assurance testing for the same technology. Today, only I/O Vendor Program (IOVP) program drivers are published at this level. VMware takes support calls for VIBs with this acceptance level.
VMwareAccepted	VIBs with this acceptance level go through verification testing, but the tests do not fully test every function of the software. The partner runs the tests and VMware verifies the result. Today, CIM providers and PSA plug-ins are among the VIBs published at this level. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.
PartnerSupported	VIBs with the PartnerSupported acceptance level are published by a partner that VMware trusts. The partner performs all testing. VMware does not verify the results. This level is used for a new or nonmainstream technology that partners want to enable for VMware systems. Today, driver VIB technologies such as Infiniband, ATAoE, and SSD are at this level with nonstandard hardware drivers. VMware directs customers with support calls for VIBs with this acceptance level to contact the partner's support organization.
CommunitySupported	The CommunitySupported acceptance level is for VIBs created by individuals or companies outside of VMware partner programs. VIBs at this level have not gone through any VMware-approved testing program and are not supported by VMware Technical Support or by a VMware partner.

1. Connect to each ESX host by using SSH.
2. Verify that the acceptance level is set to VMwareCertified, VMwareAccepted, or PartnerSupported by running the following command.

   ```
   esxcli software acceptance get
   ```

3. If the host acceptance level is CommunitySupported, determine whether any of the VIBs are at the CommunitySupported level by running the following commands.

   ```
   esxcli software vib list
   esxcli software vib get -n vibname
   ```

4. Remove any CommunitySupported VIBs by running the following command.

   ```
   esxcli software vib remove --vibname vib
   ```

5. Change the acceptance level of the host by using one of the following methods.

Option	Description
CLI command	`esxcli software acceptance set --level level` The `level` parameter is required and specifies the acceptance level to set. Should be one of `VMwareCertified`, `VMwareAccepted`, `PartnerSupported`, or `CommunitySupported`. See *ESXCLI Reference* for more information.
vSphere Client	1. Select a host in the inventory. 2. Click **Configure**. 3. Under System, select **Security Profile**. 4. Click **Edit** for Host Image Profile Acceptance Level and choose the acceptance level.

The new acceptance level is in effect.

Note:

ESX conducts integrity checks of VIBs governed by the Acceptance Level. You can use the `VMkernel.Boot.execInstalledOnly` setting to instruct ESX to only execute binaries that originate from a valid VIB installed on the host. Combined with Secure Boot, this setting ensures that every single process ever run on an ESX host is signed, allowed, and expected. By default, the `VMkernel.Boot.execInstalledOnly` setting is deactivated for partner compatibility in vSphere 7.0 and later. Activating this setting when possible improves security. For more information on configuring advanced options for ESX, see the VMware knowledge base article at https://kb.vmware.com/s/article/1038578.

Assigning Privileges for ESX Hosts

Usually, you give privileges to users by assigning permissions to ESX host objects that are managed by a vCenter system. If you are using a standalone ESX host, you can assign privileges directly.

Assigning Permissions to ESX Hosts That Are Managed by vCenter

If your ESX host is managed by a vCenter, perform management tasks through the vSphere Client.

You can select the ESX host object in the vCenter object hierarchy and assign the administrator role to a limited number of users. Those users can then perform direct management on the ESX host. See Using vCenter Roles to Assign Privileges.

Best practice is to create at least one named user account, assign it full administrative privileges on the host, and use this account instead of the root account. Set a highly complex password for the root account and limit the use of the root account. Do not remove the root account.

Assigning Permissions to Standalone ESX Hosts

You can add local users and define custom roles from the Management tab of the VMware Host Client. See the *vSphere Single Host Management - VMware Host Client* documentation.

For all versions of ESX, you can see the list of predefined users in the `/etc/passwd` file.

The following roles are predefined.

Read Only	Allows a user to view objects associated with the ESX host but not to make any changes to objects.
Administrator	Administrator role.
No Access	No access. This role is the default role. You can override the default role.

You can manage local users and groups and add local custom roles to an ESX host using a VMware Host Client connected directly to the ESX host. See the *vSphere Single Host Management - VMware Host Client* documentation.

In vSphere 6.0 and later, you can use ESXCLI account management commands for managing ESX local user accounts. You can use ESXCLI permission management commands for setting or removing permissions on both Active Directory accounts (users and groups) and on ESX local accounts (users only).

Note:
If you define a user for the ESX host by connecting to the host directly, and a user with the same name also exists in vCenter, those users are different. If you assign a role to the ESX user, the vCenter user is not assigned the same role.

Predefined ESX Users and Privileges

If your environment does not include a vCenter system, the following users are predefined.

root User	By default each ESX host has a single root user account with the Administrator role. That root user account can be used for local administration and to connect the host to vCenter. Assigning root user privileges can make it easier to break into an ESX host because the name is already known. Having a common root account also makes it harder to match actions to users.

For better auditing, create individual accounts with Administrator privileges. Set a highly complex password for the root account and limit the use of the root account, for example, for use when adding a host to vCenter. Do not remove the root account. For more information about assigning permissions to a user for an ESX host, see *vSphere Single Host Management - VMware Host Client* documentation.

Best practice is to ensure that any account with the Administrator role on an ESX host is assigned to a specific user with a named account. Use ESX Active Directory capabilities, which allow you to manage Active Directory credentials.

Important:
You can remove the access privileges for the root user. However, you must first create another permission at the root level that has a different user assigned to the Administrator role.

vpxuser User

vCenter uses vpxuser privileges when managing activities for the host.

The vCenter administrator can perform most of the same tasks on the host as the root user and also schedule tasks, work with templates, and so forth. However, the vCenter administrator cannot directly create, delete, or edit local users and groups for hosts. Only a user with Administrator privileges can perform these tasks directly on a host.

You cannot manage the vpxuser user using Active Directory.

CAUTION:
Do not change the vpxuser user in any way. Do not change its password. Do not change its permissions. If you do so, you might experience problems when working with hosts through vCenter.

dcui User

The dcui user runs on hosts and acts with Administrator rights. This user's primary purpose is to configure hosts for lockdown mode from the Direct Console User Interface (DCUI).

This user acts as an agent for the direct console and cannot be modified or used by interactive users.

Deactivating Shell Access for non-root ESX Users

In vSphere 8.0 and later, you can deactivate shell access for non-root ESXi users, such as the predefined vpxuser and dcui users. By deactivating shell access, you can enhance security by enforcing an "API only" stance for these users.

To deactivate shell access, you can use the `esxcli system account set --id`*`user`*`--shell-access false` command. The corresponding API is `LocalAccountManager.updateUser`. You can also use the VMware Host Client to change the Enable Shell Access flag of ESX local users.

Note:
When you deactivate the shell access for a user with administrative access, by virtue of being denied shell access, that user cannot grant shell access to other users, or change the passwords of users that have shell access. Other permissions, such as host profiles, will still allow users such as vpxuser and dcui to change passwords of other users.

When making changes of this kind, verify that they do not break existing third-party workflows.

Using Active Directory to Manage ESX Users

You can configure ESX to use a directory service such as Active Directory to manage users.

Creating local user accounts on each host presents challenges with having to synchronize account names and passwords across multiple hosts. Join ESX hosts to an Active Directory domain to eliminate the need to create and maintain local user accounts. Using Active Directory for user authentication simplifies the ESX host configuration and reduces the risk for configuration issues that could lead to unauthorized access.

When you use Active Directory, users supply their Active Directory credentials and the domain name of the Active Directory server when adding a host to a domain.

Configure an ESX Host to Use Active Directory

You can configure an ESX host to use a directory service such as Active Directory to manage users and groups.

- Verify that you have an Active Directory domain. See your directory server documentation.

- Verify that the host name of ESX is fully qualified with the domain name of the Active Directory forest.
  fully qualified domain name = *host_name.domain_name*

When you add an ESX host to Active Directory, the DOMAIN group **ESX Admins** is assigned full administrative access to the host if it exists. If you do not want to make full administrative access available, see VMware Knowledge Base article 1025569 for a workaround.

If a host is provisioned with Auto Deploy, Active Directory credentials cannot be stored on the hosts. You can use the vSphere Authentication Proxy to join the host to an Active Directory domain. Because a trust chain exists between the vSphere Authentication Proxy and the host, the Authentication Proxy can join the host to the Active Directory domain. See Using vSphere Authentication Proxy.

> **Note:**
> When you define user account settings in Active Directory, you can limit the computers that a user can log in to by the computer name. By default, no equivalent restrictions are set on a user account. If you set this limitation, LDAP Bind requests for the user account fail with the message `LDAP binding not successful`, even if the request is from a listed computer. You can avoid this problem by adding the netBIOS name for the Active Directory server to the list of computers that the user account can log in to.

1. Synchronize the time between ESX and the directory service system.

   See Synchronize ESX Clocks with a Network Time Server or the VMware Knowledge Base for information about how to synchronize ESX time with a Microsoft Domain Controller.
2. Ensure that the DNS servers that you configured for the host can resolve the host names for the Active Directory controllers.
   a) Browse to the host in the vSphere Client inventory.
   b) Click **Configure**.
   c) Under Networking, click **TCP/IP configuration**.
   d) Under TCP/IP Stack: Default, click **DNS** and verify that the host name and DNS server information for the host are correct.

Join the host to a directory service domain. See Add an ESX Host to a Directory Service Domain. For hosts that are provisioned with Auto Deploy, set up the vSphere Authentication Proxy. See Using vSphere Authentication Proxy. You can configure permissions so that users and groups from the joined Active Directory domain can access the vCenter components. For information about managing permissions, see Add a Permission to an Inventory Object .

Add an ESX Host to a Directory Service Domain

To have your ESX host use a directory service, you must join the host to the directory service domain.

You can enter the domain name in one of two ways:

- `name.tld` (for example, `domain.com`): The account is created under the default container.
- `name.tld/container/path` (for example, `domain.com/OU1/OU2`): The account is created under a particular organizational unit (OU).

To use the vSphere Authentication Proxy service, see Using vSphere Authentication Proxy.

1. Browse to a host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Authentication Services**.
4. Click **Join Domain**.
5. Enter a domain.

   Use the form `name.tld` or `name.tld/container/path`.

6. Enter the user name and password of a directory service user who has permissions to join the host to the domain, and click **OK**.
7. Optional: If you intend to use an authentication proxy, enter the proxy server IP address.
8. Click **OK** to close the Directory Services Configuration dialog box.

You can configure permissions so that users and groups from the joined Active Directory domain can access the vCenter components. For information about managing permissions, see Add a Permission to an Inventory Object .

View Directory Service Settings for an ESX Host

You can view the type of directory server, if any, that the ESX host uses to authenticate users and the directory server settings.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Authentication Services**.

   The Authentication Services page displays the directory service and domain settings.

You can configure permissions so that users and groups from the joined Active Directory domain can access the vCenter components. For information about managing permissions, see Add a Permission to an Inventory Object .

Securing the ESX LDAP Connection to Active Directory

ESX 9.1.1 provides several enhancements to improve the security of the Active Directory connection by enabling LDAPS.

When ESX is joined to an Active Directory domain, it uses an LDAP connection as part of the authentication process. This connection normally starts on the insecure port 389 and is then upgraded to an encrypted connection.

Starting from ESX 9.1.1, you can take additional steps to secure your Active Directory connection:

- ESX can now make the Active Directory connection on LDAPS port 636, improving security by ensuring that the connection is encrypted immediately. This requires the administrator to install the trusted root certificate for the AD server in the ESX server's certificate store.
- You can now enforce a minimum TLS level for the LDAPS connection. This is independent of TLS Profiles.
- You can set the Simple Authentication Security Layer (SASL) strength factor to require a higher level of encryption to increase security or to require a lower level in order to permit compability with older Active Directory servers.
- You can enable logging for LDAP or LDAPS connections, including the TLS handshake, allowing administrators to isolate problems without a packet capture.

The following configuration options can be used to enable LDAP and LDAPS connection logging:

Configuration Value	Description	Default	Configurable Options
`EnableConnectionLogging`	Logs LDAP/S connection handshakes and TLS negotiation to `hostd.log`. LDAP log entries are prefixed with `LDAP:` and can be easily found with `grep "LDAP\:"`.	Automatic (turns on only if system logging is verbose)	**1** (Always log), **0** (Never log) If no value is present, the setting is Automatic.
`ConnectOnTlsPort`	Connects to Active Directory on secure port 636 instead of insecure port 389	0 (FALSE)	**1** (Connect on port 636; requires AD root certificate)

Install an Active Directory Root Certificate on ESX

In order to enable use of LDAPS and automatically connecting on the secure LDAP port 636, you must install a root certificate on the ESX host.

This root certificate might be your Active Directory root certificate or another root certificate that your Active Directory server's LDAP certificate is generated from. This certificate must be updated when it expires, typically every 1 to 2 years.

> **Important:**
>
> If the root certificate expires, you will be unable to log in using Active Directory accounts. To avoid disruption, install the new certificate before the old one expires.

1. Export your enterprise root certificate to a CER file.
2. Login to your ESX host using your preferred SSH client.
3. Copy the CER file to a local file on the ESX host.
   ```
   scp myuser@source.mydomain.com:/fileshare/enterpriserootcert.cer /tmp/enterpriserootcert.cer
   ```
4. Backup the castore.pem file.
   ```
   cp /etc/vmware/ssl/castore.pem /etc/vmware/ssl/castore.pem.bak
   ```
5. Copy and append the root certificate to the castore.pem file.
   ```
   cat /tmp/enterpriserootcert.cer >> /etc/vmware/ssl/castore.pem
   ```
6. Restart the management services.
   ```
   /etc/init.d/hostd restart && /etc/init.d/vpxa restart
   ```

Enable Secure LDAPS Connection

You can configure ESX to make the Active Directory connection on LDAPs port 636, improving security by ensuring that the connection is encrypted immediately.

Ensure that the root certificate has been installed on the host. See Install an Active Directory Root Certificate on ESX.

> **Important:**
>
> If the root certificate expires, you will be unable to log in using Active Directory accounts. To avoid disruption, install the new certificate before the old one expires.

1. Login to your ESX host using your preferred SSH client.
2. Launch the likewise registry shell.
   ```
   /usr/lib/vmware/likewise/bin/lwregshell
   ```
   > **Note:**
   >
   > If you see the following error, ensure that the likewise service is running:
   >
   > ```
   > lwregshell (error = 40700 - LWREG_ERROR_NO_SUCH_KEY_OR_VALUE)
   > No such key or value
   > ```

   To check the status of the likewise service, type `/etc/init.d/lwsmd status`.

   To start the likewise service, type `/etc/init.d/lwsmd start`.
3. Enter the root of the machine's configuration registry.
   ```
   cd HKEY_THIS_MACHINE
   ```
4. If it does not already exist, create a new directory (or "key") specifically for managing the ESX LDAP settings.
   ```
   HKEY_THIS_MACHINE> add_key EsxLdapConfiguration
   ```
5. Navigate into the key you just created.
   ```
   HKEY_THIS_MACHINE> cd EsxLdapConfiguration
   ```

6. To enable Secure TLS Connections, add a registry value `ConnectOnTlsPort`.

   ```
   HKEY_THIS_MACHINE\EsxLdapConfiguration> add_value "ConnectOnTlsPort" REG_DWORD 1
   ```

7. List the registry values to confirm your changes.

   ```
   HKEY_THIS_MACHINE\EsxLdapConfiguration> list_values
   ```

8. Exit the configuration utility.

   ```
   exit
   ```

Wait 5 minutes for the new registry values to be read in.

Enable LDAP and LDAPS Logging on an ESX Host

You can enable logging for LDAP or LDAPS connections, including the TLS handshake, allowing administrators to isolate problems without a packet capture.

Enabling this logging can generate a large number of log entries. Enable this setting only for troubleshooting purposes or when the logging data is otherwise needed.

1. Login to your ESX host using your preferred SSH client.
2. Launch the likewise registry shell.

   ```
   /usr/lib/vmware/likewise/bin/lwregshell
   ```

   Note:

   If you see the following error, ensure that the likewise service is running:

   ```
   lwregshell (error = 40700 - LWREG_ERROR_NO_SUCH_KEY_OR_VALUE)
   No such key or value
   ```

   To check the status of the likewise service, type `/etc/init.d/lwsmd status`.

   To start the likewise service, type `/etc/init.d/lwsmd start`.

3. Enter the root of the machine's configuration registry.

   ```
   cd HKEY_THIS_MACHINE
   ```

4. Create a new directory (or "key") specifically for managing the ESX LDAP settings.

   ```
   HKEY_THIS_MACHINE> add_key EsxLdapConfiguration
   ```

5. Navigate into the key you just created.

   ```
   HKEY_THIS_MACHINE> cd EsxLdapConfiguration
   ```

6. Enable Connection Logging by adding the value, `EnableConnectionLogging`.

   ```
   HKEY_THIS_MACHINE\EsxLdapConfiguration> add_value "ConnectOnTlsPort" REG_DWORD 1
   ```

7. List the registry values to confirm your changes.

   ```
   HKEY_THIS_MACHINE\EsxLdapConfiguration> list_values
   ```

8. Exit the configuration utility.

   ```
   exit
   ```

Wait 5 minutes for the new registry values to be read in.

Additional LDAP and LDAPS Parameters Configurable in Likewise

Additional LDAP and LDAPS parameters can be configured in likewise in the same manner as described in Enable Secure LDAPS Connection and Enable LDAP and LDAPS Logging on an ESX Host.

Configuration Value	Description	Default	Configurable Options
`MinimumTlsVersion`	Sets the minimum TLS level required when connecting to Active Directory (if `ConnectOnTlsPort` is enabled).	12	10 (TLS 1.0), 11 (TLS 1.1), 12 (TLS 1.2), 13 (TLS 1.3)
`TrustAnyCertificate`	Disables LDAPS certificate trust checking. Should only be used for in-field diagnostics.	0 (FALSE)	1 (TRUE), 0 (FALSE)
`UseMinimumSecurityStrengthFactor`	Overrides the default minimum security strength factor with the value specified by `MinimumSecurityStrengthFactor`.	1 (TRUE)	1 (TRUE), 0 (FALSE)
`MinimumSecurityStrengthFactor`	Sets the minimum security strength factor (encryption bitness) for LDAP/S connections.	128	1-65535 (Special values: 0 = no protection, 1 = integrity/ replay protection, 56 = 56-bit encryption, 128 = AES-128 encryption, 256 = AES-256 encryption)
`NetworkConnectionTimeoutSeconds`	Specifies the maximum time in seconds to connect to the AD server (includes TLS negotiation if enabled).	5	1-65535

Using vSphere Authentication Proxy

You can add ESX hosts to an Active Directory domain by using vSphere Authentication Proxy instead of adding the hosts explicitly to the Active Directory domain.

Note:

vSphere Authentication Proxy has been deprecated and will be removed in a future release.

You only have to set up the host so it knows about the domain name of the Active Directory server and about the IP address of vSphere Authentication Proxy. When vSphere Authentication Proxy is enabled, it automatically adds hosts that are being provisioned with Auto Deploy to the Active Directory domain. You can also use vSphere Authentication Proxy with hosts that are not provisioned by using Auto Deploy.

See Required Ports for vCenter for information about TCP ports used by vSphere Authentication Proxy.

Auto Deploy

If you are provisioning hosts with Auto Deploy, you can set up a reference host that points to Authentication Proxy. You then set up a rule that applies the reference host's profile to any ESX host that is provisioned with Auto Deploy. vSphere Authentication Proxy stores the IP addresses of all hosts that Auto Deploy provisions using PXE in its access control list. When the host boots, it contacts vSphere Authentication Proxy, and vSphere Authentication Proxy joins those hosts, which are already in its access control list, to the Active Directory domain.

Even if you use vSphere Authentication Proxy in an environment that uses certificates that are provisioned by VMCA or third-party certificates, the process works seamlessly if you follow the instructions for using custom certificates with Auto Deploy.

See Make Auto Deploy a Subordinate Certificate Authority.

Other ESX Hosts

You can set up other hosts to use vSphere Authentication Proxy if you want to make it possible for the host to join the domain without using Active Directory credentials. That means you do not need to transmit Active Directory credentials to the host, and you do not save Active Directory credentials in the host profile.

In that case, you add the host's IP address to the vSphere Authentication Proxy access control list, and vSphere Authentication Proxy authorizes the host based on its IP address by default. You can enable client authentication to have vSphere Authentication Proxy check the host's certificate.

Note:
IPv6 is supported across vSphere. It is not supported on vSphere Supervisor.

Start the vSphere Authentication Proxy Service

The vSphere Authentication Proxy service is available on each vCenter system. By default, the service is not running. If you want to use vSphere Authentication Proxy in your environment, you can start the service from the vCenter Management Interface or from the command line.

Verify that you are using vCenter 6.5 or later. In earlier versions of vSphere, vSphere Authentication Proxy is installed separately. See the documentation for the earlier version of the product for instructions.

The vSphere Authentication Proxy service binds to an IPv4 address for communication with vCenter, and does not support IPv6. The vCenter instance can be on a host machine in an IPv4-only or IPv4/IPv6 mixed-mode network environment. However, when you specify the address of vSphere Authentication Proxy, you must specify an IPv4 address.

1. Start the VMware vSphere Authentication Proxy service.

Option	Description
vCenter Management Interface	1. In a Web browser, go to the vCenter Management Interface, https://*vcenter-IP-address-or-FQDN*:5480. 2. Log in as root. The default root password is the password that you set while deploying the vCenter. 3. Click **Services**, and click the **VMware vSphere Authentication Proxy** service. 4. Click **Start**. 5. After the service has started, click **Set Startup Type** and click **Automatic** to make the startup automatic.
CLI	`service-control --start vmcam`

2. Confirm that the service started successfully.

You can now set the vSphere Authentication Proxy domain. After that, vSphere Authentication Proxy handles all hosts that are provisioned with Auto Deploy, and you can explicitly add hosts to vSphere Authentication Proxy.

Add a Domain to vSphere Authentication Proxy Using the vSphere Client

You can add a domain to vSphere Authentication Proxy from the vSphere Client.

You can add a domain to vSphere Authentication Proxy only after you enable the proxy. After you add the domain, vSphere Authentication Proxy adds all hosts that you provision with Auto Deploy to that domain. For other hosts, you can also use vSphere Authentication Proxy if you do not want to give those hosts domain privileges.

1. Connect to a vCenter system with the vSphere Client.
2. Select the vCenter, and click **Configure**.
3. Click **Authentication Proxy** and click **Edit**.
4. Enter the name of the domain that vSphere Authentication Proxy will add hosts to, and the name and password of a user who has Active Directory privileges to add hosts to the domain.
5. Click **Save**.

Add a Domain to vSphere Authentication Proxy Using the camconfig Command

You can add a domain to vSphere Authentication using the `camconfig` command.

You can add a domain to vSphere Authentication Proxy only after you enable the proxy. After you add the domain, vSphere Authentication Proxy adds all hosts that you provision with Auto Deploy to that domain. For other hosts, you can also use vSphere Authentication Proxy if you do not want to give those hosts domain privileges.

1. Log in to the vCenter system as a user with administrator privileges.
2. Run the command to enable access to the Bash shell.

   ```
   shell
   ```

3. Go to the `/usr/lib/vmware-vmcam/bin/` directory where the **camconfig** script is located.
4. To add the domain and user Active Directory credentials to the Authentication Proxy configuration, run the following command.

   ```
   camconfig add-domain -d domain -u user
   ```

   You are prompted for a password.

   vSphere Authentication Proxy caches that user name and password. You can remove and recreate the user as needed. The domain must be reachable through DNS, but does not have to be a vCenter Single Sign-On identity source.

   vSphere Authentication Proxy uses the user name specified by *`user`* to create the accounts for ESX hosts in Active Directory. The user must have privileges to create accounts in the Active Directory domain to which you are adding the hosts. At the time of writing of this information, the Microsoft Knowledge Base article 932455 had background information for account creation privileges.
5. If you later want to remove the domain and user information from vSphere Authentication Proxy, run the following command.

   ```
   camconfig remove-domain -d domain
   ```

Use vSphere Authentication Proxy to Add a Host to a Domain

The Auto Deploy server adds all hosts that it provisions to vSphere Authentication Proxy, and vSphere Authentication Proxy adds those hosts to the domain. If you want to add other hosts to a domain using vSphere Authentication Proxy, you can add those hosts to vSphere Authentication Proxy explicitly. Afterwards, the vSphere Authentication Proxy server adds those hosts to the domain. As a result, user-supplied credentials no longer have to be transmitted to the vCenter system.

- If the ESX host is using a VMCA-signed certificate, verify that the host has been added to vCenter. Otherwise, the Authentication Proxy service cannot trust the ESX host.
- If the ESX host is using a root CA-signed certificate, verify that the appropriate root CA-signed certificate has been added to the vCenter system. See Managing Certificates for ESX Hosts.

You can enter the domain name in one of two ways:

- `name.tld` (for example, `domain.com`): The account is created under the default container.
- `name.tld/container/path` (for example, `domain.com/OU1/OU2`): The account is created under a particular organizational unit (OU).

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under **System**, select **Authentication Services**.
4. Click **Join Domain**.
5. Enter a domain.

   Use the form `name.tld`, for example `mydomain.com`, or `name.tld/container/path`, for example, `mydomain.com/organizational_unit1/organizational_unit2`.
6. Select **Using Proxy Server**.
7. Enter the IP address of the Authentication Proxy server, which is always the same as the IP address of the vCenter system.
8. Click **OK**.

Activate Client Authentication for vSphere Authentication Proxy

By default, vSphere Authentication Proxy adds any host if it has the IP address of that host in its access control list. For additional security, you can activate client authentication. If client authentication is activated, vSphere Authentication Proxy also checks the certificate of the host.

- Verify that the vCenter system trusts the host. By default, when you add a host tovCenter, the host is assigned a certificate that is signed by avCenter trusted root CA. vSphere Authentication Proxy trustsvCenter trusted root CA.
- If you plan on replacingESX certificates in your environment, perform the replacement before you activate vSphere Authentication Proxy. The certificates on the ESXi host must match that of the host's registration.

1. Log in to the vCenter system as a user with administrator privileges.
2. To activate access to the Bash shell, run the `shell` command.
3. Go to the `/usr/lib/vmware-vmcam/bin/` directory where the **camconfig** script is located.
4. To activate client authentication, run the following command.

   ```
   camconfig ssl-cliAuth -e
   ```

   Going forward, vSphere Authentication Proxy checks the certificate of each host that is added.
5. If you later want to deactivate client authentication again, run the following command.

   ```
   camconfig ssl-cliAuth -n
   ```

Import the vSphere Authentication Proxy Certificate to ESX Host

By default, ESX hosts require explicit verification of the vSphere Authentication Proxy certificate. If you are using vSphere Auto Deploy, the Auto Deploy service takes care of adding the certificate to hosts that it provisions. For other hosts, you must add the certificate explicitly.

- Upload the vSphere Authentication Proxy certificate to a datastore accessible to the ESX host. Using an SFTP application such WinSCP, you can download the certificate from the vCenter host at the following location.
  `/var/lib/vmware/vmcam/ssl/rui.crt`
- Verify that the UserVars.ActiveDirectoryVerifyCAMCertificateESX advanced setting is set to 1 (the default).

1. Select the ESX host and click **Configure**.
2. Under **System**, select **Authentication Services**.
3. Click **Import Certificate**.
4. Enter the certificate file path following the format `[datastore]/path/certname.crt`.
5. Enter the IP address of vSphere Authentication Proxy server.
6. Click **OK**.

Generate a New Certificate for vSphere Authentication Proxy

You can generate a new certificate that is provisioned by VMware Certificate Authority (VMCA), or a new certificate that includes VMCA as a subordinate certificate.

You must have root or Administrator privileges on the system on which vSphere Authentication Proxy is running.

See Set Up vSphere Authentication Proxy to Use Custom Certificates if you want to use a custom certificate that is signed by a third-party or enterprise CA.

1. Make a copy of `certool.cfg`.

   ```
   cp /usr/lib/vmware-vmca/share/config/certool.cfg /var/lib/vmware/vmcam/ssl/vmcam.cfg
   ```

2. Edit the copy with some information about your organization, as in the following example.

   ```
   Country = IE
   Name = vmcam
   Organization = VMware
   OrgUnit = vTSU
   State = Cork
   Locality = Cork
   Hostname = test-cam-1.test1.vmware.com
   ```

3. Generate the new private key in `/var/lib/vmware/vmcam/ssl/`.

   ```
   /usr/lib/vmware-vmca/bin/certool --genkey --privkey=/var/lib/vmware/vmcam/ssl/rui.key --pubkey=/tmp/
   vmcam.pub --server=localhost
   ```

   For *localhost*, supply the FQDN of the vCenter.

4. Generate the new certificate in `/var/lib/vmware/vmcam/ssl/` using the key and `vmcam.cfg` file that you created in Step 1 and Step 2.

   ```
   /usr/lib/vmware-vmca/bin/certool --server=localhost --gencert --privkey=/var/lib/vmware/vmcam/ssl/rui.key
    --cert=/var/lib/vmware/vmcam/ssl/rui.crt --config=/var/lib/vmware/vmcam/ssl/vmcam.cfg
   ```

   For *localhost*, supply the FQDN of the vCenter.

Set Up vSphere Authentication Proxy to Use Custom Certificates

Using custom certificates with vSphere Authentication Proxy consists of several steps. First you generate a CSR and send it to your CA for signing. Then you place the signed certificate and key file in a location that vSphere Authentication Proxy can access.

By default, vSphere Authentication Proxy generates a CSR during first boot and asks VMCA to sign that CSR. vSphere Authentication Proxy registers with vCenter Server using that certificate. You can use custom certificates in your environment, if you add those certificates to vCenter Server.

1. Generate a CSR for vSphere Authentication Proxy.
   a) Create a configuration file, `/var/lib/vmware/vmcam/ssl/vmcam.cfg`, as in the following example.

      ```
      [ req ]
      distinguished_name = req_distinguished_name
      encrypt_key = no
      prompt = no
      string_mask = nombstr
      ```

```
req_extensions = v3_req
[ v3_req ]
basicConstraints = CA:false
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
subjectAltName = DNS:vcenter1.example.com
[ req_distinguished_name ]
countryName = US
stateOrProvinceName = NY
localityName = New York
0.organizationName = Example Inc.
organizationalUnitName = IT Org
commonName = vcenter1.example.com
```

Note the following:

- subjectAltName: Use the format `DNS:`*`FQDN_of_vCenter_Appliance_to_use_the_CA-signed certificate`*.
- commonName: Use the same FQDN of the vCenter Appliance used in subjectAltName.

b) Run `openssl` to generate a CSR file and a key file, passing in the configuration file.

```
openssl req -new -nodes -out vmcam.csr -newkey rsa:2048 -keyout /var/lib/vmware/vmcam/ssl/rui.key -
config /var/lib/vmware/vmcam/ssl/vmcam.cfg
```

2. Back up the `rui.crt` certificate and `rui.key` files, which are stored in the following location.
   `/var/lib/vmware/vmcam/ssl/rui.crt`
3. Unregister vSphere Authentication Proxy.
   a) Go to the `/usr/lib/vmware-vmcam/bin` directory where the `camregister` script is located.
   b) Run the following command.

```
camregister --unregister -aVC_address -uuser
```

   `user` must be a vCenter Single Sign-On user that has administrator permissions on vCenter Server.
4. Stop the vSphere Authentication Proxy service.

Tool	Steps
vCenter Server Configuration Management Interface	1. In a Web browser, go to the vCenter Server Configuration Management Interface, https://*`vcenter-IP-address-or-FQDN`*:5480. 2. Log in as root. The default root password is the password that you set while deploying the vCenter Server. 3. Click **Services**, and click **VMware vSphere Authentication Proxy**. 4. Click **Stop**.
CLI	`service-control --stop vmcam`

5. Replace the existing `rui.crt` certificate and `rui.key` files with the files that you received from your CA.
6. Restart the vSphere Authentication Proxy service.
7. Reregister vSphere Authentication Proxy explicitly with vCenter Server by using the new certificate and key.

```
camregister --register -a VC_address -u user -cfull_path_to_rui.crt-kfull_path_to_rui.key
```

Using the ESX Shell

The ESX Shell provides essential maintenance commands and is deactivated by default on ESX hosts. You can activate local and remote access to the shell if necessary. To reduce the risk of unauthorized access, activate the ESX Shell for troubleshooting only.

The ESX Shell is independent of lockdown mode. Even if the host is running in lockdown mode, you can still log in to the ESX Shell if it is activated.

See *vSphere Security*.

The applicable services are as follows.

ESX Shell	Activate this service to access the ESX Shell locally.
SSH	Activate this service to access the ESX Shell remotely by using SSH.
Direct Console UI (DCUI)	When you activate this service while running in lockdown mode, you can log in locally to the direct console user interface as the root user and deactivate lockdown mode. You can then access the host using a direct connection to the VMware Host Client or by activating the ESX Shell.

The root user and users with the Administrator role can access the ESX Shell. Users who are in the Active Directory group ESX Admins are automatically assigned the Administrator role. By default, only the root user can run system commands (such as `vmware -v`) by using the ESX Shell.

> **Note:**
> Do not activate the ESX Shell unless you actually need access.

Set Idle Timeout for the ESX Shell Using the vSphere Client

If you enable the ESX Shell on a host, but forget to log out of the session, the idle session remains connected indefinitely. The open connection increases the potential for someone to gain privileged access to the host. Prevent this by setting a timeout for idle sessions.

The idle timeout is the amount of time that can elapse before a user is logged out of an idle interactive session. You can control the amount of time for both local and remote (SSH) sessions from the Direct Console Interface (DCUI) or from the vSphere Client.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Advanced System Settings**.
4. Click **Edit**, select `UserVars.ESXiShellInteractiveTimeOut`, and enter the timeout setting.
   A value of zero (0) deactivates the idle time.
5. Restart the ESXi Shell service and the SSH service for the timeout to take effect.
   a) Go to **System** > **Services**.
   b) One by one, select ESXi Shell and SSH, and click **Restart**.

If the session is idle, users are logged out after the timeout period elapses.

Set Availability Timeout for the ESX Shell Using the vSphere Client

The ESX Shell is deactivated by default. You can set an availability timeout for the ESX Shell to increase security when you activate the shell.

The availability timeout setting is the amount of time that can elapse before you must log in after the ESX Shell is activated. After the timeout period, the service is deactivated and users are not allowed to log in.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, select **Advanced System Settings**.
4. Click **Edit**, and select `UserVars.ESXiShellTimeOut`.
5. Enter the idle timeout setting.
6. Click **OK**.
7. Restart the ESXi Shell service and the SSH service for the timeout to take effect.
   a) Go to **System** > **Services**.
   b) One by one, select ESXi Shell and SSH, and click **Restart**.

If you are logged in when the timeout period elapses, your session will persist. However, after you log out or your session is terminated, users are not allowed to log in.

Set Availability Timeout or Idle Timeout for the ESX Shell Using the DCUI

The ESX Shell is deactivated by default. To increase security when you activate the shell, you can set an availability timeout, an idle timeout, or both.

Activate the ESX Shell. See Activate Access to the ESX Shell Using the DCUI.

The two types of timeout apply in different situations.

ESX Shell Idle Timeout	If a user activates the ESX Shell on a host, but forgets to log out of the session, the idle session remains connected indefinitely. The open connection can increase the potential for someone to gain privileged access to the host. You can prevent this situation by setting a timeout for idle sessions.
ESX Shell Availability Timeout	The availability timeout determines how much time can elapse before you log in after you initially activate the shell. If you wait longer, the service is deactivated and you cannot log in to the ESX Shell.

1. Log in to the ESX Shell.
2. From the Troubleshooting Mode Options menu, select **Modify ESXi Shell and SSH timeouts** and press Enter.
3. Enter the idle timeout (in seconds) or the availability timeout.
4. Press Enter and press Esc until you return to the main menu of the Direct Console User Interface.
5. Click **OK**.
6. Restart the ESXi Shell service and the SSH service for the timeout to take effect.
   a) In the vSphere Client, select the host, and go to **Configure** > **System** > **Services**.
   b) One by one, select ESXi Shell and SSH, and click **Restart**.

- If you set the idle timeout, users are logged out after the session is idle for the specified time.
- If you set the availability timeout, and you do not log in before that timeout elapses, logins become deactivated again.

Activate Access to the ESX Shell Using the vSphere Client

ESX Shell and SSH interfaces are deactivated by default. Keep these interfaces deactivated unless you are performing troubleshooting or support activities. For everyday activities, use the vSphere Client, where activity is subject to role-based access control and modern access control methods.

If you want to use an authorized SSH key, you can upload it. See ESX SSH Keys.

> **Note:**
> Access the host by using the vSphere Client, remote command-line tools (ESXCLI and PowerCLI), and published APIs. Do not activate remote access to the host using SSH unless special circumstances require it.

1. Browse to the host in the inventory.
2. Click **Configure**, then click **Services** under System.
3. Manage the ESXi, SSH, or Direct Console UI services.
   a) In the Services pane, select the service.
   b) Click **Edit Startup Policy** and select the startup policy **Start and stop manually**.
   c) To activate the service, click **Start**.

   When you select **Start and stop manually**, the service does not start when you reboot the host. If you want the service to start when you reboot the host, select **Start and stop with host**.

Set the availability and idle timeouts for the ESX Shell. See Set Availability Timeout for the ESX Shell Using the vSphere Client and Set Idle Timeout for the ESX Shell Using the vSphere Client.

Activate Access to the ESX Shell Using the DCUI

The Direct Console User Interface (DCUI) allows you to interact with the host locally using text-based menus. Evaluate whether the security requirements of your environment support activating the Direct Console User Interface.

You can use the Direct Console User Interface (DCUI) to activate local and remote access to the ESX Shell. You access the Direct Console User Interface from the physical console attached to the host. After the host reboots and loads ESX, press F2 to log in to the DCUI. Enter the credentials that you created when you installed ESX.

> **Note:**
> Changes made to the host using the Direct Console User Interface, the vSphere Client, ESXCLI, or other administrative tools are committed to permanent storage every hour or upon graceful shutdown. If the host fails before the changes are committed, they might be lost.

1. From the Direct Console User Interface, press F2 to access the System Customization menu.
2. Select **Troubleshooting Options** and press Enter.
3. From the Troubleshooting Mode Options menu, select a service to activate.
   - Enable ESX Shell
   - Enable SSH
4. Press Enter to activate the service.
5. Press Esc until you return to the main menu of the Direct Console User Interface.

Set the availability and idle timeouts for the ESX Shell. See Set Availability Timeout or Idle Timeout for the ESX Shell Using the DCUI.

Log in to the ESXi Shell for Troubleshooting

Perform ESX configuration tasks with the vSphere Client, ESXCLI, or VMware PowerCLI. Log in to the ESX Shell (formerly Tech Support Mode or TSM) for troubleshooting purposes only.

1. Log in to the ESX Shell using one of the following methods.
   - If you have direct access to the host, press Alt+F1 to open the login page on the machine's physical console.
   - If you are connecting to the host remotely, use SSH or another remote console connection to start a session on the host.
2. Enter a user name and password recognized by the host.

UEFI Secure Boot for ESX Hosts

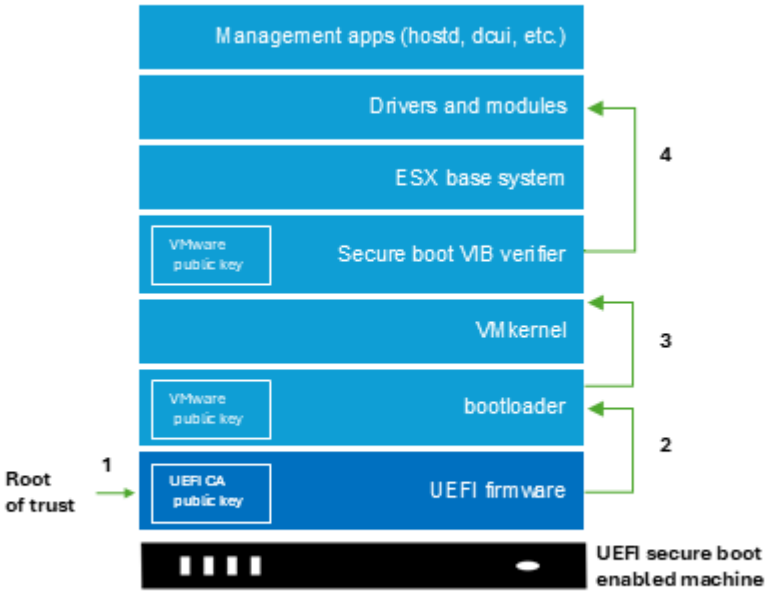
Secure boot is part of the UEFI firmware standard. The ESX bootloader consists of several UEFI apps and drivers. When secure boot is enabled in the hardware, a machine refuses to run any UEFI app or driver unless it is cryptographically signed by a key that the machine trusts. ESXsupports secure boot by having its bootloader components properly signed, and by having the bootloader begin the process of verifying that the rest of the ESX system is signed by a VMware key. ESX supports secure boot if it is enabled in the hardware.

How ESX Uses UEFI Secure Boot

ESX supports UEFI Secure Boot at each level of the boot stack.

> **Note:**
> Before you use UEFI Secure Boot on a host that was upgraded, check for compatibility by following the instructions in Run the Secure Boot Validation Script After ESX Upgrade.

Figure 94: UEFI Secure Boot



With secure boot in use, the boot sequence proceeds as follows.

1. One or more public key certificates that are stored in the UEFI firmware's "db" variable serve as the root of trust.
2. The UEFI firmware uses these keys to verify the signature of the ESX bootloader.
3. The ESX bootloader contains a VMware public key. The bootloader uses this key to verify the signature of the kernel and a small subset of the system that includes a secure boot VIB verifier.
4. The VIB verifier verifies every VIB package that is installed on the system.

> **Note:**
> When you install or upgrade to vSphere 7.0 Update 2 or later, and an ESX host has a TPM, the TPM seals the sensitive information by using a TPM policy based on PCR values for UEFI Secure Boot. This value is loaded during subsequent reboots if the policy is satisfied as true. To deactivate or activate UEFI Secure Boot in vSphere 7.0 Update 2 and later, see Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration.

Troubleshooting UEFI Secure Boot

If secure boot does not succeed at any level of the boot sequence, an error results.

The error message depends on the hardware vendor and on the level at which verification did not succeed.

- If you attempt to boot with a bootloader that is unsigned or has been tampered with, an error during the boot sequence results. The exact message depends on the hardware vendor. It might look like the following error, but might look different.

  ```
  UEFI0073: Unable to boot PXE Device...because of the Secure Boot policy
  ```

- If the kernel has been tampered with, an error like the following results.

```
Fatal error: 39 (Secure Boot Failed)
```

- If a package (VIB or driver) has been tampered with, a purple screen with the following message appears.

```
UEFI Secure Boot failed:
Failed to verify signatures of the following vibs (XX)
```

To resolve issues with secure boot, follow these steps.

1. Reboot the host with secure boot deactivated.
2. Run the secure boot verification script (see Run the Secure Boot Validation Script After ESX Upgrade).
3. Examine the information in the `/var/log/esxupdate.log` file.

Expiration of the Microsoft UEFI CA 2011 Certificate

During the secure boot process, public key certificates that are stored in the UEFI firmware's db variable are used to validate the signature of the ESX bootloader. The Microsoft Corporation UEFI CA 2011 certificate, the industry standard used to sign third party UEFI applications (including ESX bootloaders) expired on June 27, 2026. Microsoft has issued the Microsoft UEFI CA 2023 certificates (Microsoft UEFI CA 2023 and Microsoft Option ROM UEFI CA 2023) to replace the expiring Microsoft UEFI CA 2011 certificate.

After the Microsoft UEFI CA 2011 certificate expires, virtual machines will continue to boot. However, Microsoft cannot push db and dbx updates or update Key exchange keys until the expired certificate is replaced.

Virtual machines created on ESX 9.0 have the newer Microsoft UEFI CA 2023 certificates in place, unless they were deployed from templates that were created on older ESX versions and have an older platform key included. Virtual machines created on previous ESX might require remediation to update the platform key, depending on a number of factors:

- Virtual machine HW version: Virtual machines using HW version 13 must be upgraded to at least HW version 14 before remediation.
- Secure Boot enablement: Virtual machines without Secure Boot enabled do not require remediation.
- Guest OS: Guest OSes that do not expect to receive operating system updates requiring the platform key do not require remediation.

For non-TPM-enabled virtual machines, the platform key can be remediated using a Silent Platform Key update by rebooting the guest operating system. For certain Windows-based TPM-enabled virtual machines, the platform key can be remediated using a Capsule update that ensures that the key is updated without disrupting TPM-based applications.

For more information on identifying virtual machines that require remediation, and detailed instructions for updating the platform key, see Broadcom KB article 423893.

> **Important:**
>
> The expiration of this certificate does not introduce any new security vulnerabilities . The decision to revoke a UEFI Secure Boot certificate (removing it from the DB or adding it to the DBX) is a completely independent security decision, regardless of the certificate expiration. Specifically, the Microsoft Corporation UEFI CA 2011 is used to validate a wide range of pre-boot components beyond the OS bootloader. This includes, but is not limited to, firmware for hardware components such as RAID controllers, NICs, and HBAs, and independent UEFI applications and diagnostic tools developed by hardware vendors. The server vendors are responsible for these components, including their signature status. Consult your specific server vendor to understand the full impact of revoking the Microsoft Corporation UEFI CA 2011 in your environment before proceeding. Revoking this certificate without fully understanding its usage may result in an unbootable system.

Securing ESX Hosts with Trusted Platform Module

ESX hosts can use Trusted Platform Modules (TPM) chips, which are secure cryptoprocessors that enhance host security by providing a trust assurance rooted in hardware as opposed to software.

What Is a TPM

TPM is an industry-wide standard for secure cryptoprocessors. Today, TPM chips are found in most computers, from laptops, to desktops, to servers. vSphere supports TPM version 2.0.

A TPM 2.0 chip attests to an ESX identity of a host. Host attestation is the process of authenticating and attesting to the state of the software on a host at a given point in time. UEFI secure boot, which ensures that only signed software is loaded at boot time, is a requirement for successful attestation. The TPM 2.0 chip records and securely stores measurements of the software modules booted in the system, which vCenter remotely verifies.

The high-level steps of the remote attestation process are:

1. Establish the trustworthiness of the remote TPM and create an Attestation Key (AK) on it.
   When an ESX host is added to, rebooted from, or reconnected to vCenter, vCenter requests an AK from the host. Part of the AK creation process also involves the verification of the TPM hardware itself, to ensure that a known (and trusted) vendor has produced it.
2. Retrieve the Attestation Report from the host.
   vCenter requests that the host sends an Attestation Report, which contains a quote of Platform Configuration Registers (PCRs), signed by the TPM, and other signed host binary metadata. By checking that the information corresponds to a configuration it deems trusted, a vCenter identifies the platform on a previously untrusted host.
3. Verify the authenticity of the host.
   vCenter verifies the authenticity of the signed quote, infers the software versions, and determines the trustworthiness of said software versions. If vCenter determines the signed quote is invalid, remote attestation fails and the host is not trusted.

What Are the vSphere Requirements to Use a TPM

To use a TPM 2.0 chip, your vCenter environment must meet these requirements:

- ESX host with TPM 2.0 chip installed and enabled in UEFI
- UEFI Secure Boot enabled

Ensure that the TPM is configured in the BIOS of the ESX host to use the SHA-256 hashing algorithm and the TIS/FIFO (First-In, First-Out) interface and not CRB (Command Response Buffer). For information about setting these required BIOS options, refer to the vendor documentation.

Review the TPM 2.0 chips certified by VMware at the following location:

https://www.vmware.com/resources/compatibility/search.php

What Happens When You Boot a Host with a TPM

When you boot an ESX host with an installed TPM 2.0 chip, vCenter monitors the attestation status of the host. To view the hardware trust status, in the vSphere Client, select the vCenter, then the **Summary** tab under **Monitor**. The hardware trust status is one of the following:

- Green: Normal status, indicating full trust.
- Red: Attestation failed.

> **Note:**
> If you add a TPM 2.0 chip to an ESX host that vCenter already manages, you must first disconnect the host, then reconnect it. See the *vCenter and Host Management* documentation for information about disconnecting and reconnecting hosts.

View ESX Host Attestation Status

When added to an ESX host, a Trusted Platform Module 2.0 compatible chip attests the integrity of the platform. You can view the attestation status of the host in the vSphere Client. You can also view the Intel Trusted Execution Technology (TXT) status.

1. Connect to vCenter by using the vSphere Client.
2. Navigate to a data center and click the **Monitor** tab.
3. Click **Security**.
4. Review the host's status in the Attestation column and read the accompanying message in the **Message** column.

For a Failed or Warning attestation status, see Troubleshoot ESX Host Attestation Problems.

Troubleshoot ESX Host Attestation Problems

When you install a Trusted Platform Module (TPM) device on an ESX host, the host might fail to pass attestation. You can troubleshoot the potential causes of this problem.

1. View the ESX host alarm status and accompanying error message. See View ESX Host Attestation Status.
2. If the error message is `Host secure boot was disabled`, you must re-enable secure boot to resolve the problem.
3. If the attestation status of the host is failed, check the vCenter`vpxd.log` file for the following message:

   `No cached identity key, loading from DB`

   This message indicates that you are adding a TPM 2.0 chip to an ESX host that vCenter already manages. You must first disconnect the host, then reconnect it. See *vCenter and Host Management* documentation for information about disconnecting and reconnecting hosts.

   For more information about vCenter log files, including location and log rotation, see the VMware knowledge base article at https://kb.vmware.com/s/article/1021804.
4. For all other error messages, contact Customer Support.

ESX Log Files

Log files are an important component of troubleshooting attacks and obtaining information about breaches. All ESX hosts run a syslog service, which logs messages from the VMkernel and other system components to local files or to a remote host.

To increase the security of the host, take the following measures.

- Configure persistent logging to a datastore. By default, the logs on ESX hosts are stored in the in-memory file system. Therefore, they are lost when you reboot the host, and only 24 hours of log data is stored. When you enable persistent logging, you have a dedicated activity record for the host.
- Remote logging to a central host allows you to gather log files on a central host. From that host, you can monitor all hosts with a single tool, do aggregate analysis, and search log data. This approach facilitates monitoring and reveals information about coordinated attacks on multiple hosts.
- Configure the remote secure syslog on ESX hosts by using ESXCLI or PowerCLI, or by using an API client.
- Query the syslog configuration to make sure that the syslog server and port are valid.

Configure Syslog on ESX Hosts

You can use the vSphere Client, the VMware Host Client, or the `esxcli system syslog` command to configure the syslog service.

For information about using the `esxcli system syslog` command and other ESXCLI commands, see *Getting Started with ESXCLI*. For details how to open the ESX firewall for the port specified in each remote host specification, see the *vSphere Security* guide.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under **System**, click **Advanced System Settings**.
4. Click **Edit**.
5. Filter for `syslog`.
6. To set up logging globally and configure various advanced settings, see ESX Syslog Options.
7. Optional: To overwrite the default log size and log rotation for any of the logs:
   a) Click the name of the log that you want to customize.
   b) Enter the number of rotations and the log size you want.
8. Click **OK**.

Changes to the syslog options take effect.

> **Note:**
> Syslog parameter settings that you define by using the vSphere Client or VMware Host Client are effective immediately. However, most settings you define by using ESXCLI require an additional command to take effect. For more details, see ESX Syslog Options.

ESX Syslog Options

You can define the behavior of ESX syslog files and transmissions by using a set of syslog options.

Apart from the base settings, such as `Syslog.global.logHost`, starting from ESX 7.0 Update 1, a list of advanced options is available for customizations, and NIAP compliance.

> **Note:**
> Always configure persistent storage before you set any of the audit record parameters or the `Syslog.global.logDir` parameter.

> **Note:**
> All audit record settings, beginning with `Syslog.global.auditRecord`, take effect immediately. However, for other settings that you define by using ESXCLI, make sure to run the `esxcli system syslog reload` command to enable the changes.

Table 203: Legacy Syslog Options

Option	ESXCLI command	Description
`Syslog.global.logHost`	`esxcli system syslog config set --loghost=<str>`	Defines a comma-delimited list of remote hosts and specifications for message transmissions. If the `loghost=<str>` field is blank, no logs are forwarded. While no hard limit to the number of remote hosts to receive syslog messages exists, good practice is to keep the number of remote hosts to five or less. The format of a remote host specification is: `protocol://hostname|ipv4|'['ipv6']'[:port]`. The protocol must be one of TCP, UDP, or SSL. The value of a port can be any decimal number from 1 through 65535. If a port is not provided, SSL and TCP use 1514. UDP uses 514. For example: `ssl://hostName1:1514`.
`Syslog.global.defaultRotate`	`esxcli system syslog config set --default-rotate=<long>`	Maximum number of old log files to keep. You can set this number globally and for individual subloggers (see `Syslog.global.defaultSize`).
`Syslog.global.defaultSize`	`esxcli system syslog config set --default-size=<long>`	Default size of log files, in KiB. After a file reaches the default size, the syslog service creates a new file. You can set this number globally and for individual subloggers.
`Syslog.global.logDir`	`esxcli system syslog config set --logdir=<str>`	Directory where logs reside. The directory can be on mounted NFS or VMFS volumes. Only the `/scratch` directory on the local file system is persistent across reboots. Specify the directory as [*datastorename*] *path_to_file*, where the path is relative to the root of the volume backing the datastore. For example, the path `[storage1] /systemlogs` maps to the path `/vmfs/volumes/storage1/systemlogs`.
`Syslog.global.logDirUnique`	`esxcli system syslog config set --logdir-unique=<bool>`	Specifies the ESX host name to be concatenated to the value of `Syslog.global.logDir`. It is critical that you enable this setting when multiple ESX hosts log to a shared file system. Selecting this option creates a subdirectory with the name of the ESX host under the directory specified by **Syslog.global.LogDir**. A unique directory is useful if the same NFS directory is used by multiple ESX hosts.
`Syslog.global.certificate.checkSSLCerts`	`esxcli system syslog config set --check-ssl-certs=<bool>`	Enforces checking of SSL certificates when transmitting messages to remote hosts.

Table 204: Syslog Options Available Starting from ESX 7.0 Update 1 and later

Option	ESXCLI command	Description
`Syslog.global.auditRecord.storageCapacity`	`esxcli system auditrecords local set --size=<long>`	Specifies the capacity of the audit record storage directory located on the ESX host, in MiB. You cannot decrease the capacity of the audit record storage. You can increase the capacity before or after the audit record storage is enabled (see `Syslog.global.auditRecord.storageEnable`).
`Syslog.global.auditRecord.remoteEnable`	`esxcli system auditrecords remote enable`	Enables sending audit records to remote hosts. Remote hosts are specified by using the `Syslog.global.logHost` parameter.
`Syslog.global.auditRecord.storageDirectory`	`esxcli system auditrecords local set --directory=<dir>`	Creates an audit record storage directory and unless specified, sets `/scratch/auditLog` as the default location. You must not manually create an audit record storage directory and you cannot change the audit record storage directory while audit record storage is enabled (see `Syslog.global.auditRecord.storageEnable`).
`Syslog.global.auditRecord.storageEnable`	`esxcli system auditrecords local enable`	Enables the storage of audit records on an ESX host. If the audit record storage directory does not exist, it is created with the capacity specified by `Syslog.global.auditRecord.storageCapacity`.
`Syslog.global.certificate.checkCRL`	`esxcli system syslog config set --crl-check=<bool>`	Enables checking the revocation status of all the certificates in an SSL certificate chain. Enables verification of X.509 CRLs, which are not checked by default in compliance with industry conventions. A NIAP-validated configuration requires CRL checks. Due to implementation limitations, if CRL checks are enabled, then all certificates in a certificate chain must provide a CRL link. Do not enable the `crl-check` option for installations not related to certification, because of the difficulty in properly configuring an environment that uses CRL checks.
`Syslog.global.certificate.strictX509Compliance`	`esxcli system syslog config set --x509-strict=<bool>`	Enables strict compliance with X.509. Performs additional validity checks on CA root certificates during verification. These checks are generally not performed, as CA roots are inherently trusted, and might cause incompatibilities with existing, misconfigured CA roots. A NIAP-validated configuration requires even CA roots to pass validations. Do not enable the `x509-strict` option for installations not related to certification, because of the difficulty in properly configuring an environment that uses CRL checks.

Option	ESXCLI command	Description
Syslog.global.droppedMsgs.fileRotate	esxcli system syslog config set --drop-log-rotate=<long>	Specifies the number of old dropped message log files to keep.
Syslog.global.droppedMsgs.fileSize	esxcli system syslog config set --drop-log-size=<long>	Specifies the size of each dropped message log file before switching to a new one, in KiB.
Syslog.global.logCheckSSLCerts	esxcli system syslog config set --check-ssl-certs=<bool>	Enforces checking of SSL certificates when transmitting messages to remote hosts. **Note:** Deprecated. Use Syslog.global.certificate.checkSSLCerts in ESX 7.0 Update 1 and later.
Syslog.global.logFilters	esxcli system syslog config logfilter [add | remove | set] ...	Specifies one or more log filtering specifications. Each log filter must be separated by a double vertical bar "||". The format of a log filter is: numLogs | ident | logRegexp. numLogssets the maximum number of log entries for the specified log messages. After reaching this number, the specified log messages are filtered and ignored. ident specifies one or more system components to apply the filter to the log messages that these components generate. logRegexp specifies a case-sensitive phrase with Python regular expression syntax to filter the log messages by their content.
Syslog.global.logFiltersEnable		Enables the use of log filters.
Syslog.global.logLevel	esxcli system syslog config set --log-level=<str>	Specifies the log filtering level. You must change this parameter only when troubleshooting an issue with the syslog daemon. You can use the values debug for the most detailed level, info for the default detail level, warning for only warnings or errors, or error, only for errors.
Syslog.global.msgQueueDropMark	esxcli system syslog config --queue-drop-mark=<long>)	Specifies the percent of the message queue capacity at which messages are dropped.
Syslog.global.remoteHost.connectRetryDelay	esxcli system syslog config set --default-timeout=<long>	Specifies the delay before retrying to connect to a remote host after a connection attempt fails, in seconds.

Option	ESXCLI command	Description
Syslog.global.remoteHost.maxMsgLen	esxcli system syslog config set --remote-host-max-msg-len=<long>	Specifies the maximum length of a syslog transmission before truncation occurs, in bytes. The default maximum length for remote host messages is 1 KiB for TCP, SSL, and UDP protocols. You can increase the maximum message length to up to 16 KiB. However, raising this value above 1 KiB does not ensure that long transmissions arrive untruncated to a syslog collector. For example, when the syslog infrastructure that issues a message is external to ESX. For versions earlier than ESXi 8.0 Update 3i, this parameter applies only to TCP and SSL protocols. UDP maximum message transmission length is limited at 480 bytes for IPV4 and 1180 bytes for IPV6.
Syslog.global.vsanBacking	esxcli system syslog config set --vsan-backing=<bool>	Allows log files and the audit record storage directory to be placed on a vSAN cluster. However, enabling this parameter might cause the ESX host to become unresponsive.

ESX Log File Locations

ESX records host activity in log files, using a syslog facility.

Table 205: ESX Log File Locations

Component	Location	Purpose
Authentication	/var/log/auth.log	Contains all events related to authentication for the local system.
ESX host agent log	/var/log/hostd.log	Contains information about the agent that manages and configures the ESX host and its virtual machines.
Shell log	/var/log/shell.log	Contains a record of all commands typed into the ESX Shell and shell events (for example, when the shell was enabled).
System messages	/var/log/syslog.log	Contains all general log messages and can be used for troubleshooting. This information was formerly located in the messages log file.
vCenter agent log	/var/log/vpxa.log	Contains information about the agent that communicates with vCenter (if the host is managed by vCenter).
Virtual machines	The same directory as the affected virtual machine's configuration files, named vmware.log and vmware*.log. For example, /vmfs/volumes/datastore/virtual machine/vmware.log	Contains virtual machine power events, system failure information, tools status and activity, time sync, virtual hardware changes, vMotion migrations, machine clones, and so on.
VMkernel	/var/log/vmkernel.log	Records activities related to virtual machines and ESX.
VMkernel summary	/var/log/vmksummary.log	Used to determine uptime and availability statistics for ESX (comma separated).

Component	Location	Purpose
VMkernel warnings	`/var/log/vmkwarning.log`	Records activities related to virtual machines and ESX.
Quick Boot	`/var/log/loadESX.log`	Contains all events related to restarting an ESX host through Quick Boot.
Trusted infrastructure agent	`/var/run/log/kmxa.log`	Records activities related to the Client Service on the ESX Trusted Host.
Key Provider Service	`/var/run/log/kmxd.log`	Records activities related to the vSphere Trust Authority Key Provider Service.
Attestation Service	`/var/run/log/attestd.log`	Records activities related to the vSphere Trust Authority Attestation Service.
ESX Token Service	`/var/run/log/esxtokend.log`	Records activities related to the vSphere Trust Authority ESX Token Service.
ESX API Forwarder	`/var/run/log/esxapiadapter.log`	Records activities related to the vSphere Trust Authority API forwarder.

Securing Fault Tolerance Logging Traffic

VMware Fault Tolerance (FT) captures inputs and events that occur on a primary virtual machine and sends them to the secondary virtual machine, which is running on another host.

This logging traffic between the primary and secondary virtual machines is unencrypted and contains guest network and storage I/O data, as well as the memory contents of the guest operating system. This traffic might include sensitive data such as passwords in plain text. To avoid such data being divulged, ensure that this network is secured, especially to avoid man-in-the-middle attacks. For example, use a private network for FT logging traffic. You can also encrypt the FT logging traffic.

Activate Fault Tolerance Encryption

You can encrypt Fault Tolerance log traffic.

FT encryption requires SMP-FT. Encryption on Legacy FT (Record-Replay FT) is not supported.

vSphere Fault Tolerance performs frequent checks between a primary VM and secondary VM so that the secondary VM can quickly resume from the last successful checkpoint. The checkpoint contains the VM state that has been modified since the previous checkpoint. You can encrypt Fault Tolerance log traffic.

When you turn on Fault Tolerance, FT encryption is set to **Opportunistic** by default, which means it activates encryption only if both the primary and secondary host are capable of encryption. Follow this procedure if you need to change the FT encryption mode manually.

> **Note:**
> Fault Tolerance supports vSphere Virtual Machine Encryption with vSphere 7.0 Update 2 and later. In-guest and array-based encryption do not depend on or interfere with VM encryption. Having multiple encryption layers uses additional compute resources, which might impact virtual machine performance. The impact varies with hardware as well as the amount and type of I/O, but overall performance impact is negligible for most workloads. The effectiveness and compatibility of back-end storage features such as deduplication, compression, and replication might also be affected by VM encryption.

1. Select the VM and choose **Edit Settings**.
2. Under **VM Options** select the **Encrypted FT** drop-down menu.
3. Choose one of the following options:

Option	Description
Disabled	Do not turn on encrypted Fault Tolerance logging.
Opportunistic	Turn on encryption only if both sides are capable. A Fault Tolerance VM is allowed to move to an ESX host which does not support encrypted Fault Tolerance logging.

Option	Description
Required	Choose hosts for Fault Tolerance primary and secondary that both support encrypted FT logging.

Note:
While VM encryption is activated, FT encryption mode is set to **Required** by default and cannot be modified.

When FT encryption mode is set to **Required**:

- When you turn on FT, only FT encryption supported hosts are listed for the placement of FT secondary.
- FT failover can only happen on the FT encryption supported hosts.

4. Click **OK**.

Managing ESX Audit Records

Audit records conform to RFC 5424 and contain information about events pertaining to items such as the time, status, description, and user information logged for events that have occurred from actions on ESX hosts. Both local and remote audit record keeping are available. Audit record keeping is deactivated by default. You must manually activate both local and remote auditing modes.

Audit records, when enabled, are stored locally as a fixed-size buffer of recent additions. Once the audit records fill the buffer, new audit records overwrite the oldest records.

Audit records are stored in RFC 5424 format but are transmitted to remote hosts in compliance with the specified message formatting (RFC 3164 or RFC 5424 for ESX 8.0 and later). The audit records are part of the stream of syslog messages. You can configure the local storage and transmission of audit records independently of each other. When you activate both local storage and transmission, the audit records are stored and transmitted simultaneously.

During a loss of connection between the ESX host transmitting syslog data and a remote host, audit records are dropped if the available buffer space is exceeded. Upon reconnection, the system generates an audit message indicating potential message loss.

Configuring Audit Records

You use ESXCLI to configure the local audit record keeping. For more information, see *ESXCLI Concepts and Examples*.

Viewing Audit Records

You can view the audit records as follows.

- Local: Use the ESX`/bin/viewAudit` application.
- Remote: Configure a remote audit server using ESXCLI. For more information, see Enable the Transmission of Audit Records to a Remote Host with ESXCLI.

You can also use the FetchAuditRecords API (in the DiagnosticsManager managed object) to view audit records.

Securing the ESX Configuration

ESX configuration is protected by encryption.

What Is a Secure ESX Configuration

Many ESX services store secrets in their configuration files. These configurations persist in an ESX host's boot bank as an archived file. This archived configuration file is encrypted. As a result, attackers cannot read or alter this file directly, even if they have physical access to the ESX host's storage.

In addition to preventing an attacker from accessing secrets, a secure ESX configuration when used with a TPM can save virtual machine encryption keys across reboots. When the ESX host is configured with a TPM, the TPM is used to "seal" the configuration to the host, providing a strong security guarantee. As a result, encrypted workloads can continue to function when a key server is unavailable or unreachable. See vSphere Key Persistence on ESX Hosts.

You do not need to activate the ESX configuration encryption manually.

For tasks associated with a secure ESX configuration, see Manage a Secure ESX Configuration.

How Is Secure ESX Configuration Implemented

During the first boot after installing or upgrading the ESX host, the following occurs:

- If the ESX host has a TPM, and it is activated in the firmware, the archived configuration file is encrypted by an encryption key stored in the TPM. From this point on, the configuration of the host is sealed by the TPM.
- If the ESX host does not have a TPM, ESX uses a Key Derivation Function (KDF) to generate a secure configuration encryption key for the archived configuration file. The inputs to the KDF are stored on disk in the `encryption.info` file.

  Note:
  When an ESX host has an activated TPM device, you gain additional protection.

When the ESX host reboots after the first boot, the following occurs:

- If the ESX host has a TPM, the host must obtain the encryption key from the TPM for that specific host. If the TPM measurements satisfy the sealing policy that was used when creating the encryption key, then the host obtains the encryption key from the TPM.
- If the ESX host does not have a TPM, ESX reads information from the `encryption.info` file to unlock the secure configuration.

Secure ESX Configuration Requirements

- ESX 7.0 Update 2 or later
- TPM 2.0 for configuration encryption and ability to use a sealing policy

Secure ESX Configuration Recovery Key

A secure ESX configuration includes a recovery key. If you must recover the ESX secure configuration, you use a recovery key whose contents you enter as a command-line boot option. You can list the recovery key to create a recovery key backup. You can also rotate the recovery key as part of your security requirements.

Taking a backup of the recovery key is an important part of managing your secure ESX configuration. vCenter generates an alarm to remind you to back up the recovery key.

Secure ESX Configuration Recovery Key Alarm

Taking a backup of the recovery key is an important part of managing your secure ESX configuration. Whenever an ESX host in TPM mode is connected or reconnected to vCenter, vCenter generates an alarm to remind you to back up the recovery key. When you reset the alarm, it is not triggered again unless conditions change.

Best Practices for Secure ESX Configuration

Follow these best practices for the secure ESX recovery key:

- When you list a recovery key, it is temporarily displayed in an untrusted environment and is in memory. Remove traces of the key.
  - Rebooting the host removes the residual key in memory.
  - For enhanced protection, you can activate encryption mode on the host. See Activate Host Encryption Mode Explicitly.
- When you perform a recovery:
  - To eliminate any traces of the recovery key in an untrusted environment, reboot the host.
  - For enhanced security, rotate the recovery key to use a new key after having recovered the key one time.

What Are TPM Sealing Policies

A TPM can use Platform Configuration Register (PCR) measurements to implement policies that restrict unauthorized access to sensitive data. When you install or upgrade an ESX host with a TPM to vSphere 7.0 Update 2 and later, the

TPM seals the sensitive information by using a policy that incorporates the secure boot setting. This policy checks that if secure boot was activated when data was first sealed with the TPM, then secure boot must still be activated when attempting to unseal the data on a subsequent boot.

Secure boot is part of the UEFI firmware standard. With UEFI Secure Boot activated, a host refuses to load any UEFI driver or app unless the operating system bootloader has a valid digital signature.

You can choose to deactivate or activate UEFI Secure Boot enforcement. See Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration.

Note:
If you do not activate a TPM when you install or upgrade to vSphere 7.0 Update 2 or later, you can do so later with the following command.

```
esxcli system settings encryption set --mode=TPM
```

Once you have activated the TPM, you cannot undo the setting.

The `esxcli system settings encryption set` command fails on some TPMs even when the TPM is activated for the host.

- In vSphere 7.0 Update 2: TPMs from NationZ (NTZ), Infineon Technologies (IFX), and certain new models (like NPCT75x) from Nuvoton Technologies Corporation (NTC)
- In vSphere 7.0 Update 3: TPMs from NationZ (NTZ)

If an installation or upgrade of vSphere 7.0 Update 2 or later is unable to use the TPM during the first boot, the installation or upgrade continues, and the mode defaults to NONE (that is, `--mode=NONE`). The resulting behavior is as though the TPM is not activated.

The TPM can also enforce the setting for the execInstalledOnly boot option in the sealing policy. The execInstalledOnly enforcement is an advanced ESX boot option that guarantees that the VMkernel executes only binaries that have been properly packaged and signed as part of a VIB. The execInstalledOnly boot option has a dependency on the secure boot option. The secure boot enforcement must be activated before you can enforce the execInstalledOnly boot option in the sealing policy. See Activate or Deactivate the execInstalledOnly Enforcement for a Secure ESX Configuration.

Manage a Secure ESX Configuration

You can use ESXCLI commands to list the secure ESX configuration recovery key, rotate the recovery key, and change the TPM policies (for example, enforcing UEFI Secure Boot).

List the Contents of the Secure ESX Configuration Recovery Key

You can use ESXCLI to show the contents of the secure ESX configuration recovery key.

- Have access to the ESXCLI command set. You can run ESXCLI commands remotely, or run them in the ESX Shell.
- Required privilege for using ESXCLI standalone version or through PowerCLI: **Host** > **Config** > **Settings**

This task applies only to an ESX host that has a TPM. In general, you list the contents of the secure ESX configuration recovery key to create a backup, or as part of rotating recovery keys.

1. Run the following command on the ESX host.

   ```
   esxcli system settings encryption recovery list
   ```

2. Save the output in a secure, remote location as a backup, in case you must recover the secure configuration.

The recovery key ID and key are displayed.

List the Secure ESX Configuration Recovery Key

```
[root@host1] esxcli system settings encryption recovery list

Recovery ID                             Key
--------------------------------------  ---
{2DDD5424-7F3F-406A-8DA8-D62630F6C8BC}
 478269-039194-473926-430939-686855-231401-642208-184477-602511
```

```
-225586-551660-586542-338394-092578-687140-267425
```

Rotate the Secure ESX Configuration Recovery Key

You can use ESXCLI to rotate the secure ESX configuration recovery key.

- Have access to the ESXCLI command set. You can run ESXCLI commands remotely, or run them in the ESX Shell.
- Required privilege for using ESXCLI standalone version or through PowerCLI: **Host** > **Config** > **Settings**

This task applies only to an ESX host that has a TPM. You can rotate the ESX secure configuration recovery key as part of your security best practices.

1. List the recovery key.

   See List the Contents of the Secure ESX Configuration Recovery Key.

2. Run the following command.

   ```
   esxcli system settings encryption recovery rotate [-k keyID] -u uuid
   ```

   In this command, the optional *keyID* is the key ID in the VMkernel key cache and *uuid* is the Recovery ID (obtained from the `esxcli system settings encryption recovery list` command). If you do not supply the optional key ID, ESX replaces the old recovery key with a new recovery key that is randomly generated.

The recovery key is now set to the contents of the key referenced by key ID, if provided. Otherwise, ESX provides a new key ID.

Troubleshooting and Recovering the Secure ESX Configuration

You can troubleshoot and recover from boot problems that you might encounter with a secure ESX Configuration.

If you clear a TPM (that is, the seed values in the TPM are reset), if a TPM fails, or if you replace the motherboard or TPM device, or both, you must take steps to recover the ESX secure configuration. You must have the recovery key to recover the configuration. Until you recover the configuration, the ESX host cannot boot. See Recover the Secure ESX Configuration.

Although uncommon, it is possible that an ESX host might fail to restore or decrypt the secure configuration, preventing the host from booting. Possible situations include:

- Change to secure boot setting (or other policy)
- Actual tampering
- The recovery key is unavailable

To troubleshoot these conditions, see the VMware knowledge base article at https://kb.vmware.com/s/article/81446.

Recover the Secure ESX Configuration

If a TPM fails, or if you clear a TPM, you must recover the secure ESX Configuration. Until you recover the configuration, the ESX host cannot boot.

Get your recovery key. You should have previously listed and stored the recover key. See List the Contents of the Secure ESX Configuration Recovery Key.

Recovering the secure ESX configuration refers to the following situations:

- You cleared the TPM (that is, the seeds in the TPM were reset).
- The TPM failed.
- You replaced the motherboard or the TPM device, or both.

To troubleshoot other secure ESX configuration problems, see the VMware knowledge base article at https://kb.vmware.com/s/article/81446.

Perform the recovery manually. Do not perform the recovery as part of an installation or upgrade script.

1. Optional: If the TPM failed, move the disk (having the boot bank) to another host with a TPM.
2. Start the ESX host.
3. When the ESX installer window appears, press Shift+O to edit boot options.
4. To recover the configuration, at the command prompt, append the following boot option to any existing boot options.

   ```
   encryptionRecoveryKey=recovery_key
   ```

   The secure ESX configuration is recovered and the ESX host boots.
5. To persist the change, enter the following command:

   ```
   /sbin/auto-backup.sh
   ```

When you enter the recovery key, it is temporarily displayed in an untrusted environment and is in memory. Though not necessary, as a best practice, you can remove residual traces of the key in memory by rebooting the host. Or, you can rotate the key. See Rotate the Secure ESX Configuration Recovery Key.

Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration

You can choose to activate UEFI secure boot enforcement, or deactivate a previously activated UEFI secure boot enforcement. You can use ESXCLI to change the setting in the TPM on the ESX host.

- Have access to the ESXCLI command set. You can run ESXCLI commands remotely, or run them in the ESX Shell.
- Required privilege for using ESXCLI standalone version or through PowerCLI: **Host** > **Config** > **Settings**

This task applies only to ESX hosts that have a TPM. UEFI Secure boot is a firmware setting for ensuring that the software launched by the firmware is trusted. To learn more, see UEFI Secure Boot for ESX Hosts. The enablement of UEFI Secure boot can be enforced upon every boot by using the TPM.

Starting with ESX 9.0, UEFI secure boot enforcement can be activated or deactivated using vSphere Configuration Profiles. For more information on using vSphere Configuration Profiles, see What is vSphere Configuration Profiles.

1. List the current settings on the ESX host.

   ```
   esxcli system settings encryption get
      Mode: TPM
      Require Executables Only From Installed VIBs: false
      Require Secure Boot: true
   ```

   If secure boot enforcement is activated, Require Secure Boot displays true. If secure boot enforcement is deactivated, Require Secure Boot displays false.

   If Mode appears as NONE, you must activate the TPM in the firmware of the host and set the mode by running the following command:

   ```
   esxcli system settings encryption set --mode=TPM
   ```
2. Activate or deactivate the secure boot enforcement.

Option	Description

Activate	1. Shut down the host gracefully.
	For example, right-click the ESX host in the vSphere Client and select **Power** > **Shut Down**.
	2. Activate secure boot in the firmware of the host.
	See your specific vendor hardware documentation.
	3. Restart the host.
	4. Run the following ESXCLI command.
	`esxcli system settings encryption set --require-secure-boot=T`
	5. Verify the change.
	`esxcli system settings encryption get`
	`Mode: TPM`
	`Require Executables Only From Installed VIBs: false`
	`Require Secure Boot: true`

Option	Description
	Confirm that Required Secure Boot displays true. 6. To save the setting, run the following command. `/bin/backup.sh 0`

Deactivate	1. Run the following ESXCLI command.
	`esxcli system settings encryption set --require-secure-boot=F`
	2. Verify the change.
	`esxcli system settings encryption get`
	`Mode: TPM`
	`Require Executables Only From Installed VIBs: false`
	`Require Secure Boot: false`
	Confirm that Require Secure Boot displays false.
	3. To save the setting, run the following command.
	`/bin/backup.sh 0`
	You can choose to deactivate the secure boot in the firmware of the host, but at this point the dependency between the firmware setting and the TPM enforcement is no longer set.

The ESX host runs with secure boot enforcement activated or deactivated, depending on your choice.

Note:

If you do not activate a TPM when you install or upgrade vSphere, you can do so later with the following command.

```
esxcli system settings encryption set --mode=TPM
```

Once you have activated the TPM, you cannot undo the setting.

Activate or Deactivate the execInstalledOnly Enforcement for a Secure ESX Configuration

You can choose to activate execInstalledOnly enforcement, or deactivate a previously enabled execInstalledOnly enforcement. You must use ESXCLI to change the setting in the TPM on the ESX host. UEFI secure boot enforcement must be activated before you can activate the execInstalledOnly enforcement.

- To activate the execInstalledOnly enforcement, you must first activate the UEFI secure boot enforcement. The execInstalledOnly enforcement is built on top of the UEFI secure boot enforcement. See Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration.
- Have access to the ESXCLI command set. You can run ESXCLI commands remotely, or run them in the ESX Shell.
- Required privilege for using ESXCLI standalone version or through PowerCLI: **Host** > **Config** > **Settings**

This task applies only to ESX hosts that have a TPM. The execInstalledOnly vmkernel option, when set to TRUE, guarantees that the VMkernel executes only those binaries that have been packaged and signed as part of a VIB. The enablement of this vmkernel option can be enforced upon every boot by using the TPM.

Starting with ESX 9.0, execInstalledOnly enforcement can be activated or deactivated using vSphere Configuration Profiles. For more information on using vSphere Configuration Profiles, see What is vSphere Configuration Profiles.

1. List the current settings on the ESX host.

```
esxcli system settings encryption get
   Mode: TPM
   Require Executables Only From Installed VIBs: false
   Require Secure Boot: true
```

If execInstalledOnly enforcement is activated, Require Executables Only From Installed VIBs displays true. If execInstalledOnly enforcement is deactivated, Require Executables Only From Installed VIBs displays false. To activate the execInstalledOnly enforcement, the secure boot enforcement must be activated, and Require Secure Boot displays true in this case.

If Mode appears as NONE, you must enable the TPM in the firmware of the host and set the mode by running the following command:

```
esxcli system settings encryption set --mode=TPM
```

Also, if Require Secure Boot displays as False, see Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration to activate the enforcement.

2. Confirm that the execInstalledOnly boot option is enabled.

```
esxcli system settings kernel list -o execInstalledOnly
Name               Type  Configured  Runtime  Default  Description
-----------------  ----  ----------  -------  -------  -----------
execInstalledOnly  Bool  TRUE        TRUE     TRUE     Execute only those files that have been installed
 via a vib package and have not been modified.
```

3. Activate or deactivate the execInstalledOnly enforcement.

Option	Description

Activate	1. Verify that the secure boot option is activated.
	`esxcli system settings encryption get`
	`Mode: TPM`
	`Require Executables Only From Installed VIBs: false`
	`Require Secure Boot: true`
	Confirm that Require Secure Boot displays true. If not, see Activate or Deactivate the Secure Boot Enforcement for a Secure ESX Configuration.
	2. To configure the runtime value of the execInstalledOnly boot option to TRUE, run the following ESXCLI command.
	`esxcli system settings kernel set -s execInstalledOnly -v TRUE`
	3. Shut down the host gracefully.
	For example, right-click the ESX host in the vSphere Client and select **Power** > **Shut Down**.
	4. Restart the host.
	5. To set the execInstalledOnly enforcement, run the following ESXCLI command.
	`esxcli system settings encryption set --require-exec-installed-only=T`
	6. Verify the change.
	`esxcli system settings encryption get`
	`Mode: TPM`
	`Require Executables Only From Installed VIBs: true`
	`Require Secure Boot: true`
	Confirm that Require Executables Only From Installed VIBs displays true.
	7. To save the setting, run the following command.
	`/bin/backup.sh 0`

Deactivate	1. Run the following ESXCLI command.
	`esxcli system settings encryption set --require-exec-installed-only=F`
	2. Verify the change.
	`esxcli system settings encryption get`
	`Mode: TPM`
	`Require Executables Only From Installed VIBs: false`
	`Require Secure Boot: true`

Option	Description
	Confirm that Require Executables Only From Installed VIBs displays false. 3. To save the setting, run the following command. `/bin/backup.sh 0` The TPM no longer enforces the execInstalledOnly boot option.

The ESX host runs with execInstalledOnly enforcement activated or deactivated, depending on your choice.

Securing vCenter Server Systems

Securing vCenter Server includes ensuring security of the host where vCenter Server is running, following best practices for assigning privileges and roles, and verifying the integrity of the clients that connect to vCenter Server.

Best Practices for vCenter Access Control

Strictly control access to different vCenter components to increase security for the system.

The following guidelines help ensure security of your environment.

Use Named Accounts to Access vCenter

- Grant the Administrator role only to those administrators who are required to have it. You can create custom roles or use the No cryptography administrator role for administrators with more limited privileges. Do not apply this role to any group whose membership is not strictly controlled.
- Make sure that applications use unique service accounts when connecting to a vCenter system.

Monitor Privileges of vCenter Administrator Users

Not all administrator users must have the Administrator role. Instead, create a custom role with the appropriate set of privileges and assign it to other administrators.

Users with the vCenter Administrator role have privileges on all objects in the hierarchy. For example, by default the Administrator role allows users to interact with files and programs inside the guest operating system of a virtual machine. Assigning that role to too many users can lessen virtual machine data confidentiality, availability, or integrity. Create a role that gives the administrators the privileges they need, but remove some of the virtual machine management privileges. See also Using Privilege Recorder.

Minimize Access to the vCenter appliance

Do not allow users to log directly in to the vCenter appliance. Users who are logged in to the vCenter appliance can cause harm, either intentionally or unintentionally, by altering settings and modifying processes. Those users also have potential access to vCenter credentials, such as the SSL certificate. Allow only users who have legitimate tasks to perform to log in to the system and ensure that login events are audited.

Grant Minimal Privileges to Database Users

The database user requires only certain privileges specific to database access.

Some privileges are required only for installation and upgrade. You can remove these privileges from the database administrator after vCenter is installed or upgraded.

Restrict Datastore Browser Access

Assign the **Datastore** > **Browse datastore** privilege only to users or groups who really need that privilege. Users with the privilege can view, upload, or download files on datastores associated with the vSphere deployment through the Web browser or the vSphere Client.

Restrict Users from Running Commands in a Virtual Machine

By default, a user with the Administrator role can interact with the files and the programs of a guest operating system within a virtual machine. To reduce the risk of breaching guest confidentiality, availability, or integrity, create a custom nonguest access role without the **Virtual machine** > **Guest operations** privilege. See Restrict Users from Running Commands Within a Virtual Machine.

Consider Modifying the Password Policy for vpxuser

By default, vCenter changes the vpxuser password automatically every 30 days. Ensure that this setting meets company policy, or configure the vCenter password policy. See Set the vCenter Password Policy.

> **Note:**
> Make sure that password aging policy is not too short.

Check Privileges After Restarting vCenter

Check for privilege reassignment when you restart vCenter. If the user or group that has the Administrator role on the root folder cannot be validated during a restart, the role is removed from that user or group. In its place, vCenter grants the Administrator role to the vCenter Single Sign-On administrator, administrator@vsphere.local by default. This account can then act as the vCenter administrator.

Reestablish a named administrator account and assign the Administrator role to that account to avoid using the anonymous vCenter Single Sign-On administrator account (administrator@vsphere.local by default).

Use High Encryption Levels for Remote Desktop Protocol

On each Windows computer in the infrastructure, ensure that the Remote Desktop Protocol (RDP) Host Configuration settings are set to ensure the highest level of encryption appropriate for your environment.

Verify vSphere Client Certificates

Instruct users of the vSphere Client or other client applications to heed certificate verification warnings. Without certificate verification, the user might be the subject of a MiTM attack.

Set the vCenter Password Policy

By default, vCenter changes the vpxuser password automatically every 30 days. You can change that value from the vSphere Client.

1. Log in to the vCenter system by using the vSphere Client.
2. Select the vCenter system in the object hierarchy.
3. Click **Configure**.
4. Click **Advanced Settings** and click **Edit Settings**.
5. Click the **Filter** icon and enter `VimPasswordExpirationInDays`.
6. Set `VirtualCenter.VimPasswordExpirationInDays` to comply with your requirements.

Removing Expired or Revoked Certificates and Logs from Failed Installations

Leaving expired or revoked certificates or leaving vCenter installation logs for failed installation on your vCenter system can compromise your environment.

Removing expired or revoked certificates is required for the following reasons.

- If expired or revoked certificates are not removed from the vCenter system, the environment can be subject to a MiTM attack
- In certain cases, a log file that contains the database password in plain text is created on the system if vCenter installation fails. An attacker who breaks into the vCenter system, might gain access to this password and, at the same time, access to the vCenter database.

Limiting vCenter Network Connectivity

For improved security, avoid putting the vCenter system on any network other than a management network, and ensure that vSphere management traffic is on a restricted network. By limiting network connectivity, you limit certain types of attack.

vCenter requires access to a management network only. Avoid putting the vCenter system on other networks such as your production network or storage network, or on any network with access to the Internet. vCenter does not need access to the network where vMotion operates.

vCenter requires network connectivity to the following systems.

- All ESX hosts.
- The vCenter database.
- Other vCenter systems (if the vCenter systems are part of a common vCenter Single Sign-On domain for purposes of replicating tags, permissions, and so on).
- Systems that are authorized to run management clients. For example, the vSphere Client, a Windows system where you use the PowerCLI, or any other SDK-based client.
- Infrastructure services such as DNS, Active Directory, and PTP or NTP.
- Other systems that run components that are essential to functionality of the vCenter system.

Use the firewall on the vCenter to limit inbound access. Include IP-based access restrictions so that only necessary components can communicate with the vCenter system.

Evaluate the Use of Linux Clients with CLIs and SDKs

Communications between client components and a vCenter system or ESX hosts are protected by SSL-based encryption by default. Linux versions of these components do not perform certificate validation. Consider restricting the use of these clients.

To improve security, you can replace the VMCA-signed certificates on the vCenter system and on the ESX hosts with certificates that are signed by an enterprise or third-party CA. However, certain communications with Linux clients might still be vulnerable to machine-in-the-middle attacks. The following components are vulnerable when they run on the Linux operating system.

- ESXCLI commands
- vSphere SDK for Perl scripts
- Programs that are written using the vSphere Web Services SDK

You can relax the restriction against using Linux clients if you enforce proper controls.

- Restrict management network access to authorized systems only.
- Use firewalls to ensure that only authorized hosts are allowed to access vCenter.
- Use bastion hosts (jump-box systems) to ensure that the Linux clients are behind the "jump."

Examine vSphere Client Plug-Ins

vSphere Client extensions run at the same privilege level as the user who is logged in. A malicious extension can masquerade as a useful plug-in and perform harmful operations such as stealing credentials or changing the system configuration. To increase security, use an installation that includes only authorized extensions from trusted sources.

You must have privileges to access the vCenter Single Sign-On service. These privileges differ from vCenter privileges.

A vCenter installation includes an extensibility framework for the vSphere Client. You can use this framework to extend the client with menu selections or toolbar icons. The extensions can provide access to vCenter add-on components or external, Web-based functionality.

Using the extensibility framework results in a risk of introducing unintended capabilities. For example, if an administrator installs a plug-in in an instance of the vSphere Client, the plug-in can run arbitrary commands with the privilege level of that administrator.

To protect against a potential compromise of your vSphere Client, examine all installed plug-ins periodically and make sure that each plug-in comes from a trusted source.

1. Log in to the vSphere Client as administrator@vsphere.local or a user with vCenter Single Sign-On privileges.
2. From the Home page, select **Administration**, then select **Client Plug-Ins** under **Solutions**.
3. Examine the list of client plug-ins.

vCenter Security Best Practices

Follow all best practices for securing a vCenter system. Additional steps help you make your vCenter more secure.

Configure Precision Time Protocol or Network Time Protocol

Ensure that all systems use the same relative time source. This time source must be in sync with an agreed-upon time standard such as Coordinated Universal Time (UTC). Synchronized systems are essential for certificate validation. Precision Time Protocol (PTP) and Network Time Protocol (NTP) also make it easier to track an intruder in log files. Incorrect time settings make it difficult to inspect and correlate log files to detect attacks, and make auditing inaccurate. See Synchronize the Time in vCenter with an NTP Server.

Restrict vCenter Network Access

Restrict access to components that are required to communicate with the vCenter. Blocking access from unnecessary systems reduces the potential for attacks on the operating system.

For the list of all supported ports and protocols in VMware products, including vSphere and vSAN, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/. You can search ports by VMware product, create a customized list of ports, and print or save port lists.

Configure a Bastion Host

To help protect your assets, configure a bastion host (also called a jump box) to perform elevated administrative tasks. A bastion host is a special-purpose computer that hosts a minimal number of administrative applications. All other unnecessary services are removed. The host typically resides on the management network. A bastion host increases the protection of assets through restricting login to key individuals, requiring firewall rules to log in, and adding monitoring through auditing tools.

vCenter Password Requirements and Lockout Behavior

To manage your vSphere environment, you must be aware of thevCenter Single Sign-On password policy, ofvCenter passwords, and of lockout behavior.

This section discusses vCenter Single Sign-On passwords. See ESX Passwords and Account Lockout for a discussion of passwords ofESX local users.

vCenter Single Sign-On Administrator Password Requirements

The password for the administrator ofvCenter Single Sign-On, administrator@vsphere.local by default, is specified by thevCenter Single Sign-On password policy. By default, this password must meet the following requirements:

- At least eight characters
- At least one lowercase character
- At least one numeric character
- At least one special character

The password for this user cannot be more than 20 characters long. Non-ASCII characters are allowed. Administrators can change the default password policy. See the *vSphere Authentication* documentation.

vCenter Password Requirements

InvCenter, password requirements are dictated byvCenter Single Sign-On or by the configured identity source, which can be Active Directory or OpenLDAP.

vCenter Single Sign-On Lockout Behavior

Users are locked out after a preset number of consecutive failed attempts. By default, users are locked out after five consecutive failed attempts in three minutes and a locked account is unlocked automatically after five minutes. You can change these defaults using the vCenter Single Sign-On lockout policy. See the *vSphere Authentication* documentation.

ThevCenter Single Sign-On domain administrator, administrator@vsphere.local by default, is not affected by the lockout policy. The user is affected by the password policy.

vCenter Password Changes

If you know your password, you can change the password by using the `dir-cli password change` command. If you forget your password, avCenter Single Sign-On administrator can reset your password by using the `dir-cli password reset` command.

Search the VMware Knowledge Base for information on password expiration and related topics in different versions of vSphere.

Required Ports for vCenter

The vCenter appliance must be able to communicate with the vSphere Client. To enable migration and provisioning activities between managed ESX hosts, the source and destination ESX hosts must be able to receive data from each other through predetermined TCP and UDP ports.

vCenter is accessed through predetermined TCP and UDP ports. If you manage network components from outside a firewall, you might be required to configure the firewall to allow access on the appropriate ports. For the list of all supported ports and protocols in vSphere, see the VMware Ports and Protocols at https://ports.broadcom.com.

During deployment, if a port is in use or is blocked using a denylist, the vCenter installer displays an error message. You must use another port number to proceed with the deployment. There are internal ports that are used only for inter-process communication.

VMware uses designated ports for communication. Additionally, the managed hosts monitor designated ports for data from vCenter. If a built-in firewall exists between any of these elements, the installer opens the ports during the deployment or upgrade process. For custom firewalls, you must manually open the required ports. If you have a firewall between two managed hosts and you want to perform source or target activities, such as migration or cloning, you must configure a means for the managed hosts to receive data.

Using the File Integrity Monitoring Status API

The File Integrity Monitoring (FIM) Status API, introduced in VCF 9.1, allows administrators to programatically query the integrity status of vCenter instances.

The API provides critical security data, including file change detection status, scan timestamps, and cryptographic checksums of the monitored file systems. You can access it by using standard HTTPS requests.

API Specification

Endpoint Details

- **Method:** `GET`
- **URI:** `/fim/status`
- **Authentication:** Basic Authentication (Authorization: Basic `<base64_credentials>`)
- **Content-Type:** `application/json`

Permissions

Access to this API requires valid vCenter credentials and `System.Read` privilege on the vCenter root folder.

- **Unauthorized (403):** Returned if the authenticated user doesn't have the `System.Read` privilege required to read the status.
- **Unauthenticated (401):** Returned if credentials are missing or invalid.

Data Structures

Table 206: Response Body (JSON Object)

JSON Field	Type	Description
`files_changed`	boolean	`True` if monitored files have been altered; `False` otherwise.
`check_time`	long	Unix timestamp (seconds) of the last completed integrity scan.
`next_update`	long	Unix timestamp (seconds) when the next scan is scheduled.
`interval_seconds`	long	The configured frequency of integrity scans in seconds (for example, 3600 for 1 hour).
`checksums`	List	Array containing the cryptographic digest of the appliance state.

Table 207: Checksum Object

JSON Field	Type	Description
`hash_algorithm`	Enum	The hashing algorithm used. Currently supports SHA512.
`digest`	String	The hash string representing the file system state.

Example: cURL Command

You can use curl to query the endpoint directly from a terminal.

Bash syntax:

```
curl -k -u <username>:<password> -X GET https://<vc-fqdn>/fim/status
```

The `-u` flag handles the Basic Authentication encoding automatically.

The `-k` (insecure) flag is used to bypass self-signed certificate warnings in a lab environment. For production usage, ensure that you have valid certificates and omit this flag.

Bash example:

```
curl -k -u 'admin@vsphere.local:password' -X GET https://vcenter.local/fim/status
```

Example: HTTP Request (Raw)

The following example shows the raw HTTP request headers.

```
GET /fim/status HTTP/1.1
Host: <vc-fqdn>
Authorization: Basic <base64_auth>
Accept: application/json
```

Replace *<base64_auth>* with your base64-encoded credentials (for example, admin@vsphere.local:password).

A 200 OK response indicates a successful retrieval of the FIM status. The following is an example of a successful retrieval:

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 220

{
  "files_changed": false,
  "check_time": 1744731306,
  "next_update": 1744734906,
  "interval_seconds": 3600,
  "checksums": [
    {
      "hash_algorithm": "SHA512",
      "digest": "3rOQ1xC8/YcRXLTC8/1Gk0qXXz+yPPLj4RMhl57YF7cs+PwWU4yQoa6ILro/
rn1tm7cDAIBTHmBeYyPvG60n1Q=="
    }
  ]
}
```

FIM API Error Handling

Consumers of this API must be prepared to handle the following VMODL-mapped exceptions:

Status Code	Exception	Cause
`400`	`InvalidRequest`	The server cannot parse the request structure.
`401`	`Unauthenticated`	The user credentials that are provided are invalid or missing.
`403`	`Unauthorized`	The user lacks the necessary `System.Read` privilege to read FIM status.
`503`	`ServiceUnavailable`	The FIM service is not ready (for example, initializing or computing the first scan).
`500`	`Error`	An unexpected backend error occurred.

Troubleshooting the FIM API

Problem	Resolution
Service Unavailable (`503`)	If the API returns `503`, the FIM service might still be performing its initial scan after a reboot. Retry the request after a few minutes.
`files_changed` returns true, or the value of digest is not as expected for the vCenter build	Investigate the appliance immediately for unauthorized modifications or recent patch activity that might have altered system files. The scan results are available in the `aide-fim` application log messages in `/var/log/aide-fim/aide.log`.

Securing Virtual Machines

The guest operating system that runs in the virtual machine is subject to the same security risks as a physical system. Secure virtual machines like physical machines, and follow best practices discussed in this document and in the *Security Configuration Guide* (formerly known as the *Hardening Guide*).

The *Security Configuration Guide* is available at https://core.vmware.com/security.

Activate or Deactivate UEFI Secure Boot for a Virtual Machine

UEFI Secure Boot is a security standard that helps ensure that your PC boots using only software that is trusted by the PC manufacturer. For certain virtual machine hardware versions and operating systems, you can activate secure boot just as you can for a physical machine.

You can activate secure boot only if all prerequisites are met. If prerequisites are not met, the check box is not visible in the vSphere Client.

- Verify that the virtual machine operating system and firmware support UEFI boot.
  - EFI firmware
  - Virtual hardware version 14 or later.
  - Operating system that supports UEFI secure boot.

    Note:
    Some guest operating systems do not support changing from BIOS boot to UEFI boot without guest OS modifications. Consult your guest OS documentation before changing to UEFI boot. If you upgrade a virtual machine that already uses UEFI boot to an operating system that supports UEFI secure boot, you can activate Secure Boot for that virtual machine.
- Turn off the virtual machine. If the virtual machine is running, the check box is dimmed.

UEFI Secure Boot is a firmware-based security feature that ensures a virtual machine boots only with trusted, digitally signed software by enforcing a cryptographic chain of trust from firmware to OS loader. In VMware vSphere VMs, this is implemented in the virtual UEFI (vUEFI) firmware, which validates each boot component using secure boot databases stored in virtual NVRAM: the Platform Key (PK) establishes the root of trust, the Key Exchange Key (KEK) authorizes updates, the DB contains allowed signatures, and the DBX lists revoked ones. During boot, the firmware verifies the bootloader (e.g., Windows Boot Manager), which in turn verifies subsequent components, preventing execution of untrusted code.

For vSphere VMs, secure boot certificates are initialized by the host at the first VM power-on, and remain static for the life of the VM unless updated. The Microsoft Corporation KEK CA 2011 certificate and the Microsoft Corporation UEFI CA 2011 certificate used in the DB are due to expire in June 2026. The Microsoft Windows Production PCA 2011 certificate used in the DB is due to expire in October 2026. Therefore, VMs created with older versions of vSphere or earlier VM hardware versions require updates to use newer Microsoft certificates in order to retain full functionality when using secure boot. For details, see Broadcom knowledge base article 423893 .

VMware Tools version 10.1 or later is required for virtual machines that use UEFI secure boot. You can upgrade those virtual machines to a later version of VMware Tools when it becomes available.

For Linux virtual machines, VMware Host-Guest Filesystem is not supported in secure boot mode. Remove VMware Host-Guest Filesystem from VMware Tools before you activate secure boot.

Note:
If you turn on secure boot for a virtual machine, you can load only signed drivers into that virtual machine.

This task describes how to use the vSphere Client to activate and deactivate secure boot for a virtual machine. You can also write scripts to manage virtual machine settings. For example, you can automate changing the firmware from BIOS to EFI for virtual machines with the following PowerCLI code:

```
$vm = Get-VM TestVM

$spec = New-Object VMware.Vim.VirtualMachineConfigSpec
$spec.Firmware = [VMware.Vim.GuestOsDescriptorFirmwareType]::efi
$vm.ExtensionData.ReconfigVM($spec)
```

See *VMware PowerCLI User's Guide* for more information.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click the **VM Options** tab, and expand **Boot Options**.
4. Under **Boot Options**, ensure that firmware is set to **EFI**.
5. Select your task.
   - Select the **Secure Boot** check box to activate secure boot.
   - Deselect the **Secure Boot** check box to deactivate secure boot.

   Note:

   You cannot activate secure boot if the Virtualization-based security is activated.
6. Click **OK**.

When the virtual machine boots, only components with valid signatures are allowed. The boot process stops with an error if it encounters a component with a missing or invalid signature.

Virtual Machine Security Best Practices

Following virtual machine security best practices helps ensure the integrity of your vSphere deployment.

General Virtual Machine Protection

A virtual machine is, in most respects, the equivalent of a physical server. Employ the same security measures in virtual machines that you do for physical systems.

Follow these best practices to protect your virtual machines. For additional information, see the *vSphere Security Configuration Guide* at https://core.vmware.com/security-configuration-guide.

Patch Virtual Machines

Keep all security measures up-to-date, including applying appropriate patches. Keep track of updates for dormant virtual machines that are powered off, because it can be easy to overlook them. For example, ensure that anti-virus software, anti-spyware, intrusion detection, and other protections are enabled for virtual machines in your virtual infrastructure. Also, ensure that you have enough space for the virtual machine logs.

Scan Virtual Machines for Viruses

Because each virtual machine hosts a standard operating system, you must protect it from viruses by installing anti-virus software. Depending on how you use the virtual machine, you might also want to install a software firewall.

Stagger the schedule for virus scans, particularly in deployments with a large number of virtual machines. Performance of systems in your environment degrades significantly if you scan all virtual machines simultaneously. Because software firewalls and anti-virus software can be virtualization-intensive, balance the need for these two security measures against virtual machine performance, especially if you are confident that your virtual machines are in a fully trusted environment.

Deactivate Serial Ports on Virtual Machines

Serial ports are interfaces for connecting peripherals to the virtual machine. Administrators often use serial ports to provide a direct, low-level connection to the console of a server. A virtual serial port allows for the same access to a virtual machine. Because serial ports allow for low-level access, and do not have strong controls like logging or privileges, keep them deactivated on virtual machines.

Use Templates to Deploy Virtual Machines

When you manually install guest operating systems and applications on a virtual machine, you introduce a risk of misconfiguration. By using a template to capture a hardened base operating system image with no applications installed, you can ensure that all virtual machines are created with a known baseline level of security.

You can use templates that can contain a hardened, patched, and properly configured operating system to create other, application-specific templates, or you can use the application template to deploy virtual machines.

Provide templates for virtual machine creation that contain hardened, patched, and properly configured operating system deployments.

If possible, deploy applications in templates as well. Ensure that the applications do not depend on information specific to the virtual machine to be deployed.

For more information about templates, see the *vSphere Virtual Machine Administration* documentation.

Minimize Use of the Virtual Machine Console

The virtual machine console provides the same function for a virtual machine that a monitor provides on a physical server. Users with access to the virtual machine console have access to virtual machine power management and removable device connectivity controls. Console access might therefore allow a malicious attack on a virtual machine.

1. Use native remote management services, such as terminal services and SSH, to interact with virtual machines.

   Grant access to the virtual machine console only when necessary.

2. Limit the connections to the virtual machine console.

   For example, in a highly secure environment, limit the connection to one. In some environments, you can increase the limit if several concurrent connections are necessary to accomplish normal tasks.

   a) In the vSphere Client, power off the virtual machine.
   b) Right-click the virtual machine and select **Edit Settings**.
   c) Click the **VM Options** tab, and expand **VMware Remote Console Options**.
   d) Enter the maximum number of sessions, for example, `2`.
   e) Click **OK**.

Prevent Virtual Machines from Taking Over Resources

When one virtual machine consumes so much of the host resources that other virtual machines on the host cannot perform their intended functions, a Denial of Service (DoS) might occur. To prevent a virtual machine from causing a DoS, use host resource management features such as setting Shares and using resource pools.

By default, all virtual machines on an ESX host share resources equally. You can use Shares and resource pools to prevent a denial of service attack that causes one virtual machine to consume so much of the host's resources that other virtual machines on the same host cannot perform their intended functions.

Do not set limits or use resource pools until you fully understand the impact.

1. Provision each virtual machine with just enough resources (CPU and memory) to function properly.
2. Use Shares to guarantee resources to critical virtual machines.
3. Group virtual machines with similar requirements into resource pools.
4. In each resource pool, leave Shares set to the default to ensure that each virtual machine in the pool receives approximately the same resource priority.

   With this setting, a single virtual machine cannot use more than other virtual machines in the resource pool.

See the *vSphere Resource Management* documentation for information about shares and limits.

Deactivate Unnecessary Functions Inside Virtual Machines

Any service that runs in a virtual machine provides the potential for attack. By deactivating system components that are not necessary to support the application or service that is running on the system, you reduce the attack potential.

Virtual machines do not usually require as many services or functions as physical servers. When you virtualize a system, evaluate whether a particular service or function is necessary.

> **Note:**
> When possible, install guest operating systems using "minimal" or "core" installation modes to reduce the size, complexity, and attack surface of the guest operating system.

- Deactivate unused services in the operating system.

  For example, if the system runs a file server, turn off any Web services.
- Disconnect unused physical devices, such as CD/DVD drives, floppy drives, and USB adapters.
- Deactivate unused functionality, such as unused display features, or VMware Shared Folders, which enables sharing of host files to the virtual machine (Host Guest File System).
- Turn off screen savers.
- Do not run the X Window system on top of Linux, BSD, or Solaris guest operating systems unless it is necessary.

Remove Unnecessary Hardware Devices from Virtual Machines

Any activated or connected device in a virtual machines represents a potential attack channel. Users and processes with privileges on a virtual machine can connect or disconnect hardware devices, such as network adapters and CD-ROM drives. Attackers can use this capability to breach virtual machine security. Removing unnecessary hardware devices can help prevent attacks.

An attacker with access to a virtual machine can connect a disconnected hardware device and access sensitive information on media that is left in a hardware device. The attacker can potentially disconnect a network adapter to isolate the virtual machine from its network, resulting in a denial of service.

- Do not connect unauthorized devices to the virtual machine.
- Remove unneeded or unused hardware devices.
- Deactivate unnecessary virtual devices from within a virtual machine.
- Ensure that only required devices are connected to a virtual machine. Virtual machines rarely use serial or parallel ports. As a rule, CD/DVD drives are connected only temporarily during software installation.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and click **Edit Settings**.
3. In the **Virtual Hardware** tab, click the ellipsis icon and select **Remove device** to deactivate hardware devices that are not required.

   Include checks for the following devices:

   - Serial ports
   - Parallel ports
   - USB controllers
   - CD-ROM drives

   > **Note:**
   > You must use PowerCLI commands to manage floppy drive devices in vSphere 7.0 and later.

Deactivate Unused Display Features on Virtual Machines

Attackers can use an unused display feature as a vector for inserting malicious code into your environment. Deactivate features that are not in use in your environment.

Power off the virtual machine.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and click **Edit Settings**.
3. Select **Advanced Parameters**.
4. If appropriate, add or edit the following parameters.

Name	Description
svga.vgaonly	If you set this parameter to TRUE, advanced graphics functions no longer work. Do not set this parameter to TRUE with modern-day guest operating systems as they do not operate correctly. When `svga.vgaonly` is set to TRUE, only character-cell console mode is available. If you use this setting, `mks.enable3d` has no effect. **Note:** Apply this setting only to virtual machines that do not need a virtualized video card.
mks.enable3d	Set this parameter to FALSE on virtual machines that do not require 3D functionality.

5. Click **OK**.

Deactivate Copy and Paste Operations Between Guest Operating System and Remote Console

Copy and paste operations between the guest operating system and remote console are deactivated by default. For a secure environment, retain the default setting. If you require copy and paste operations, you must activate them using the vSphere Client.

Turn off the virtual machine.

The default values for these options are set to ensure a secure environment. However, you must set them to true explicitly for audit tools to be able to check that the setting is correct.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and click **Edit Settings**.
3. Select **Advanced Parameters**.
4. Add or edit the following parameters.

Name	Value
isolation.tools.copy.disable	`true`
isolation.tools.paste.disable	`true`
isolation.tools.setGUIOptions.enable	`false`

   These options override any settings made in the guest operating system's VMware Tools control panel.

5. Click **OK**.
6. Optional: If you made changes to the configuration parameters, restart the virtual machine.

Limiting Exposure of Sensitive Data Copied to the Virtual Machine Console Clipboard

Copy and paste operations are deactivated by default for hosts to prevent exposing sensitive data that has been copied to the clipboard.

When copy and paste is activated on a virtual machine running VMware Tools, you can copy and paste between the guest operating system and remote console. When the console window gains focus, processes running in the virtual machine and non-privileged users can access the virtual machine console clipboard. If a user copies sensitive information to the clipboard before using the console, the use might expose sensitive data to the virtual machine. To prevent this problem, copy and paste operations for the guest operating system are deactivated by default.

It is possible to activate copy and paste operations for virtual machines if necessary.

Restrict Users from Running Commands Within a Virtual Machine

By default, a user who has the vCenter Administrator role can interact with files and applications within the guest operating system of a virtual machine. To reduce the risk of breaching guest confidentiality, availability, or integrity, create a nonguest access role without the **Virtual machine** > **Guest operations** privilege. Assign that role to administrators who do not need virtual machine file access.

Verify that you have **Administrator** privileges on the vCenter system where you create the role.

For security, be as restrictive about allowing access to the virtual data center as you are to the physical data center. Apply a custom role that does not include the **Virtual machine** > **Guest operations** privilege to users who require administrator privileges, but who are not authorized to interact with guest operating system files and applications.

For example, a configuration might include a virtual machine on the infrastructure that has sensitive information on it.

If tasks such as vMotion migration require that data center administrators can access the virtual machine, deactivate some remote guest operating system operations to ensure that those administrators cannot access sensitive information.

1. Log in to the vSphere Client as a user who has **Administrator** privileges on the vCenter system where you want to create the role.
2. Select **Administration** and click **Roles**.
3. Click the Administrator role and click **Clone**.
4. Enter a role name and description and click **OK**.
   For example, enter `Administrator No Guest Access`.
5. Select the cloned role and click **Edit**.
6. Under the **Virtual machine** privilege, deselect Guests operations.
7. Click **Save**.

Select the vCenter system or the host and assign a permission that pairs the user or group that should have the new privileges to the newly created role. Remove those users from the Administrator role.

Prevent a Virtual Machine User or Process from Disconnecting Devices

Users and processes without root or administrator privileges within virtual machines can connect or disconnect devices, such as network adapters and CD-ROM drives, and can modify device settings. To increase virtual machine security, remove these devices.

Turn off the virtual machine.

You can prevent virtual machine users in the guest operating system, and processes running in the guest operating system, from making any changes to the devices by changing the virtual machine advanced settings.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and click **Edit Settings**.
3. Click the **Advanced Parameters** tab.
4. Verify the following parameter, or add it.

Name	Value
isolation.device.connectable.disable	true

   This setting does not affect a vSphere administrator's ability to connect or disconnect the devices attached to the virtual machine.
5. Click **OK**.

Prevent Guest Operating System Processes from Sending Configuration Messages to the Host

To ensure that the guest operating system does not modify configuration settings, you can prevent these processes from writing any name-value pairs to the configuration file.

Turn off the virtual machine.

1. Browse to the virtual machine in the vSphere Client inventory.
2. Right-click the virtual machine and click **Edit Settings**.
3. Select **Advanced Parameters**.
4. Verify the following parameter, or add it.

Name	Value
isolation.tools.setinfo.disable	true

5. Click **OK**.

Avoid Using Independent Nonpersistent Disks with Virtual Machines

When you use independent nonpersistent disks with virtual machines, successful attackers can remove any evidence that the machine was compromised by shutting down or rebooting the system. Without a persistent record of activity on a virtual machine, administrators might be unaware of an attack. Therefore, avoid using independent nonpersistent disks.

1. Ensure that virtual machine activity is logged remotely on a separate server, such as a syslog server or equivalent Windows-based event collector.
2. If remote logging of events and activity is not configured for the guest, ensure that scsiX:Y.mode is one of the following settings:
   - Not present
   - Not set to independent nonpersistent

When nonpersistent mode is not enabled, you cannot roll a virtual machine back to a known state when you reboot the system.

Securing Virtual Machines with Intel Software Guard Extensions

vSphere enables you to configure Virtual Intel® Software Guard Extensions (vSGX) for virtual machines. Using vSGX enables you to provide additional security to your workloads.

Some modern Intel CPUs implement a security extension called Intel® Software Guard Extensions (Intel® SGX). Intel SGX is a processor-specific technology for application developers who seek to protect select code and data from disclosure or modification. Intel SGX allows user-level code to define private regions of memory, called enclaves. The enclave contents are protected such that code running outside the enclave cannot access the enclave contents.

vSGX enables virtual machines to use Intel SGX technology if available on the hardware. To use vSGX, the ESX host must be installed on an SGX-capable CPU and SGX must be enabled in the BIOS of the ESX host. You can use the vSphere Client to enable SGX for a virtual machine.

In this release, you can use remote attestation for a vSGX-enabled virtual machine. Intel SGX remote attestation is a security mechanism that allows you to establish an authenticated and secure communication channel with a trusted remote entity. To use remote attestation for virtual machines using SGX enclaves, hosts with a single CPU socket do not require Intel registration. To enable the remote attestation on a virtual machine running on a host with multiple CPU sockets, you must first register the host with the Intel Registration Server. If an SGX-capable host with multiple CPU sockets is not registered with the Intel Registration Server, you can only power on vSGX-enabled virtual machines that do not require remote attestation.

See the *vCenter and Host Management* documentation for more information about registering a multi-socket ESX host with the Intel Registration Server.

Getting Started with vSGX

Virtual machines can use Intel SGX technology, if available on the hardware.

vSphere Requirements for vSGX

To use vSGX, your vSphere environment must meet these requirements:

- Virtual machine requirements:
  - EFI firmware
  - Hardware version 17 or later
  - To enable remote attestation, hardware version 20 or later
- Component requirements:
  - The ESX host must be installed on an SGX-capable CPU and SGX must be enabled in the BIOS of the ESX host.
  - To enable the remote attestation for the host, register the host with the Intel Registration Server. This way, the virtual machine running on the host can use the remote attestation. For more information on how to register a multi-socket ESX host, see the *vCenter and Host Management* documentation.
- Guest OS support:
  - Linux
  - Windows Server 2016 (64 bit) and later
  - Windows 10 (64 bit) and later

Supported Intel Hardware for vSGX

For supported Intel hardware for vSGX, consult the vSphere Compatibility Guide.

To search for compatible hardware, so the following:

1. Navigate to compatibility guide page at VMware by Broadcom Compatibility Guide/.
2. Under **Platform and Compute** section, select **CPU series**.
3. Enter `SGX` in the global search bar.
4. Click **Search**.

You might need to turn off hyperthreading on certain CPUs to enable SGX on the ESX host. For more information, see the KB article at https://knowledge.broadcom.com/external/article?legacyId=71367.

Unsupported VMware Features on vSGX

The following features are not supported in a virtual machine when vSGX is enabled:

- vMotion/DRS migration
- Virtual machine suspend and resume
- Virtual machine snapshots (Virtual machine snapshots are supported if you do not snapshot the virtual machine's memory.)
- Fault tolerance

> **Note:**
>
> These VMware features are not supported due to how the Intel SGX architecture functions. They are not the result of a VMware shortcoming.

Enable vSGX on a Virtual Machine

You can enable vSGX on a virtual machine at the same time that you create a virtual machine.

See vSphere Requirements for vSGX.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Right-click the object, select **New Virtual Machine**, and follow the prompts to create a virtual machine.
4. On the **Customize hardware** page, click the **VM Options** tab and expand **Confidential Compute**.
5. Select the **SGX** option under **Confidential VM mode**.
6. In the **Enclave page cache size (MB)** text box, enter the size of the cache size in MB.

   Note:
   The enclave page cache size must be multiple of 2 MB.
7. To prevent the virtual machine from powering on hosts that do not support SGX remote attestation, such as unregistered multi-socket SGX hosts, select the **Required** check box for **Remote attestation**.
8. From the **Launch control configuration** drop-down menu, select the appropriate mode.

Option	Action
Unlocked	This option enables the launch enclave configuration of the guest operating system.
Locked	This option allows you to configure the launch enclave. 1. Select the **Launch enclave public key hash** option. 2. To use one of the public keys configured on the host, select **Use from host** and from the drop-down menu, select a public key hash. 3. Enter a valid SHA256 hash (64) characters key.

9. Click **OK**.

Enable vSGX on an Existing Virtual Machine

You can enable vSGX on an existing virtual machine.

See vSphere Requirements for vSGX.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. In the **VM Options** tab, expand **Confidential Compute**.
4. To enable SGX, select the **SGX** option from the **Confidential VM mode** list.
5. In the **Enclave page cache size (MB)** text box, enter the size of the cache size in MB.

   Note:
   The enclave page cache size must be multiple of 2 MB.
6. To prevent the virtual machine from powering on hosts that do not support SGX remote attestation, such as unregistered multi-socket SGX hosts, select the **Required** check box for **Remote attestation**.
7. From the **Launch control configuration** drop-down menu, select the appropriate mode.

Option	Action
Unlocked	This option enables the launch enclave configuration of the guest operating system.
Locked	This option allows you to configure the launch enclave. 1. Select the **Launch enclave public key hash** option. 2. To use one of the public keys configured on the host, select **Use from host** and from the drop-down menu, select a public key hash.

Option	Action
	3. To enter the public key manually, select **Enter manually** and enter a valid SHA256 hash (64) characters key.

8. Click **OK**.

Remove vSGX from a Virtual Machine

You can remove vSGX from a virtual machine.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. In the **Edit Settings** dialog box, click on **VM Options** tab, under **Confidential Compute**, select the **None** option.
4. Click **OK**.
   Verify that the vSGX entry no longer appears in the virtual machine **Summary** tab in the **VM Hardware** pane.

Securing Virtual Machines with AMD Secure Encrypted Virtualization-Encrypted State

Secure Encrypted Virtualization-Encrypted State (SEV-ES) is a hardware feature enabled in recent AMD CPUs that keeps the guest operating system's memory and register state encrypted, protecting it against access from the hypervisor.

You can add SEV-ES to your virtual machines as an extra security enhancement. SEV-ES prevents CPU registers from leaking information in registers to components like the hypervisor. SEV-ES can also detect malicious modifications to a CPU register state.

vSphere and AMD Secure Encrypted Virtualization-Encrypted State

You can activate Secure Encrypted Virtualization-Encrypted State (SEV-ES) on supported AMD CPUs and guest operating systems.

Currently, SEV-ES supports only AMD EPYC 7xx2 CPUs (code named "Rome") and later CPUs, and only versions of Linux kernels that include specific support for SEV-ES.

SEV-ES Components and Architecture

The SEV-ES architecture consists of the following components.

- AMD CPU, specifically, the Platform Security Processor (PSP) that manages encryption keys and handles encryption.
- Enlightened operating system, that is, an operating system that uses guest-initiated calls to the hypervisor.
- Virtual Machine Monitor (VMM) and Virtual Machine Executable (VMX), to initialize an encrypted virtual machine state during virtual machine power-on, and also to handle calls from the guest operating system.
- VMkernel driver, to communicate unencrypted data between the hypervisor and the guest operating system.

Implementing and Managing SEV-ES on ESX

You must first activate SEV-ES in a system's BIOS configuration. See the documentation for your system for more information about accessing the BIOS configuration. After you have activated SEV-ES in the BIOS for your system, you can then add SEV-ES to a virtual machine.

You use either the vSphere Client (in vSphere 9.0 and later) or PowerCLI commands to activate and deactivate SEV-ES on virtual machines. You can create new virtual machines with SEV-ES, or activate SEV-ES on existing virtual machines. Privileges to manage virtual machines activated with SEV-ES are the same as for managing regular virtual machines.

Unsupported VMware Features on SEV-ES

The following features are not supported when SEV-ES is activated.

- System Management Mode

- vMotion
- Powered-on snapshots (however, no-memory snapshots are supported)
- Hot add or remove of CPU or memory
- Suspend/resume
- VMware Fault Tolerance
- Clones and instant clones
- Guest Integrity
- UEFI Secure Boot

Add AMD Secure Encrypted Virtualization-Encrypted State (SEV-ES) to a Virtual Machine Using the vSphere Client

You can use the vSphere Client to add SEV-ES to a virtual machine to provide enhanced security to the guest operating system.

- The system must be installed with an AMD EPYC 7xx2 (code named "Rome") or later CPU and supporting BIOS.
- SEV-ES must be enabled in the BIOS.
- The number of SEV-ES virtual machines per ESX host is controlled by the BIOS. When enabling SEV-ES in the BIOS, enter a value for the **Minimum SEV non-ES ASID** setting equal to the number of SEV-ES virtual machines plus one. For example, if you have 12 virtual machines that you want to run concurrently, enter `13`.

  Note:

  vSphere 9.0 supports 480 SEV-ES enabled virtual machines per ESX host.
- The ESX host running in your environment must be at ESX 9.0 or later.
- The vCenter must be at vSphere 9.0 or later.
- The guest operating system must support SEV-ES.
  Currently, only Linux kernels with specific support for SEV-ES are supported.
- The virtual machine must be at hardware version 18 or later.
- The virtual machine must have the **Reserve all guest memory** option enabled, otherwise power-on fails.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Right-click the object, select **New Virtual Machine**, and follow the prompts to create a virtual machine.

Option	Action
Select a creation type	Create a virtual machine.
Select a name and folder	Specify a name and target location.
Select a compute resource	Specify an object for which you have privileges to create virtual machines.
Select storage	In the VM storage policy, select the storage policy. Select a compatible datastore.
Select compatibility	Ensure that **ESX 7.0 and later** is selected.
Select a guest OS	Select Linux, and select a version of Linux with specific support for SEV-ES.
Customize hardware	Under **VM Options** > **Boot Options** > **Firmware**, ensure that EFI is selected. Under **VM Options** > **Confidential Compute** > **Confidential VM Mode**, select **AMD SEV-ES**.
Ready to complete	Review the information and click **Finish**.

The virtual machine is created with SEV-ES.

Add AMD Secure Encrypted Virtualization-Encrypted State (SEV-ES) to a Virtual Machine Using the Command Line

You can use the command line to add SEV-ES to a virtual machine to provide enhanced security to the guest operating system.

- The system must be installed with an AMD EPYC 7xx2 (code named "Rome") or later CPU and supporting BIOS.
- SEV-ES must be enabled in the BIOS.
- The number of SEV-ES virtual machines per ESX host is controlled by the BIOS. When enabling SEV-ES in the BIOS, enter a value for the **Minimum SEV non-ES ASID** setting equal to the number of SEV-ES virtual machines plus one. For example, if you have 12 virtual machines that you want to run concurrently, enter `13`.

  Note:

  vSphere 9.0 and later supports 480 SEV-ES enabled virtual machines per ESX host.
- The ESX host running in your environment must be at ESX 7.0 Update 1 or later.
- The guest operating system must support SEV-ES.
  Currently, only Linux kernels with specific support for SEV-ES are supported.
- The virtual machine must be at hardware version 18 or later.
- The virtual machine must have the **Reserve all guest memory** option enabled, otherwise power-on fails.
- PowerCLI 12.1.0 or later must be installed on a system with access to your environment.

1. In a PowerCLI session, run the `Connect-VIServer` cmdlet to connect as an administrator to the vCenter that manages the ESX host on which you want to add a virtual machine with SEV-ES.

   ```
   Connect-VIServer -server vCenter_Server_ip_address -User admin_user -Password 'password'
   ```

2. Create the virtual machine with the `New-VM` cmdlet, specifying `-SEVEnabled $true`.
   For example, first assign the host information to a variable, then create the virtual machine.

   ```
   $vmhost = Get-VMHost -Name 10.193.25.83
   New-VM -Name MyVM1 $vmhost -NumCPU 2 -MemoryMB 4 -DiskMB 4 -SEVEnabled $true
   ```

   If you must specify the virtual hardware version, run the `New-VM` cmdlet with the `-HardwareVersion vmx-18` parameter. For example:

   ```
   New-VM -Name MyVM1 $vmhost -NumCPU 2 -MemoryMB 4 -DiskMB 4 -SEVEnabled $true -HardwareVersion vmx-18
   ```

The virtual machine is created with SEV-ES.

Activate AMD Secure Encrypted Virtualization-Encrypted State (SEV-ES) on an Existing Virtual Machine Using the vSphere Client

You can use the vSphere Client to add SEV-ES to an existing virtual machine to provide enhanced security to the guest operating system.

- The system must be installed with an AMD EPYC 7xx2 (code named "Rome") or later CPU and supporting BIOS.
- SEV-ES must be activated in the BIOS.
- The number of SEV-ES virtual machines per ESX host is controlled by the BIOS. When activating SEV-ES in the BIOS, enter a value for the **Minimum SEV non-ES ASID** setting equal to the number of SEV-ES virtual machines plus one. For example, if you have 12 virtual machines that you want to run concurrently, enter `13`.

  Note:

  vSphere 9.0 supports 480 SEV-ES activated virtual machines per ESX host.
- The guest operating system must support SEV-ES.
  Currently, only Linux kernels with specific support for SEV-ES are supported.
- The virtual machine must be at hardware version 18 or later.
- The virtual machine must have the **Reserve all guest memory** option checked, otherwise power-on fails.
- Ensure that the virtual machine is powered off.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. Under **VM Options** > **Boot Options** > **Firmware**, ensure that EFI is selected.
4. In the **Edit Settings** dialog box, under **VM Options** > **Confidential Compute** > **Confidential VM Mode**, select **AMD SEV-ES**.
5. Click **OK**.

SEV-ES is added to the virtual machine.

Activate AMD Secure Encrypted Virtualization-Encrypted State (SEV-ES) on an Existing Virtual Machine Using the Command Line

You can use the command line to add SEV-ES to an existing virtual machine to provide enhanced security to the guest operating system.

- The system must be installed with an AMD EPYC 7xx2 (code named "Rome") or later CPU and supporting BIOS.
- SEV-ES must be activated in the BIOS.
- The number of SEV-ES virtual machines per ESX host is controlled by the BIOS. When activating SEV-ES in the BIOS, enter a value for the **Minimum SEV non-ES ASID** setting equal to the number of SEV-ES virtual machines plus one. For example, if you have 12 virtual machines that you want to run concurrently, enter `13`.

  Note:

  vSphere 9.0 supports 480 SEV-ES activated virtual machines per ESX host.
- The ESX host running in your environment must be ESX 7.0 Update 1 or later.
- The guest operating system must support SEV-ES.
  Currently, only Linux kernels with specific support for SEV-ES are supported.
- The virtual machine must be at hardware version 18 or later.
- The virtual machine must have the **Reserve all guest memory** option that is checked, otherwise power-on fails.
- PowerCLI 12.1.0 or later must be installed on a system with access to your environment.
- Ensure that the virtual machine is powered off.

1. In a PowerCLI session, run the `Connect-VIServer` cmdlet to connect as an administrator to the vCenter that manages the ESX host with the virtual machine to which you want to add SEV-ES.
   For example:
   ```
   Connect-VIServer -server vCenter_Server_ip_address -User admin_user -Password 'password'
   ```
2. Add SEV-ES to the virtual machine with the `Set-VM` cmdlet, specifying `-SEVEnabled $true`.
   For example:
   ```
   Set-VM -VM "MyVM2" -SEVEnabled $true
   ```
   If you must specify the virtual hardware version, run the `Set-VM` cmdlet with the `-HardwareVersion vmx-18` parameter. For example:
   ```
   Set-VM -VM "MyVM2" -HardwareVersion vmx-18
   Set-VM -VM "MyVM2" -SEVEnabled $true
   ```

SEV-ES is added to the virtual machine.

Deactivate AMD Secure Encrypted Virtualization-Encrypted State on a Virtual Machine Using the vSphere Client

You can use the vSphere Client to deactivate SEV-ES on a virtual machine.

- Ensure that the virtual machine is powered off.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. In the **Edit Settings** dialog box, **VM Options** > **Confidential Compute** > **Confidential VM Mode**, unselect **AMD SEV-ES**.
4. Click **OK**.

SEV-ES is deactivated on the virtual machine.

Deactivate AMD Secure Encrypted Virtualization-Encrypted State on a Virtual Machine Using the Command Line

You can use the command line to deactivate SEV-ES on a virtual machine.

- Ensure that the virtual machine is powered off.
- PowerCLI 12.1.0 or later must be installed on a system that has access to your environment.

1. In a PowerCLI session, run the `Connect-VIServer` cmdlet to connect as an administrator to the vCenter that manages the ESX host with the virtual machine from which you want to remove SEV-ES.

   For example:

   ```
   Connect-VIServer -server vCenter_Server_ip_address -User admin_user -Password 'password'
   ```

2. Deactivate SEV-ES on the virtual machine with the `Set-VM` cmdlet, specifying `-SEVEnabled $false`.

   For example, first assign the host information to a variable, then deactivate SEV-ES for the virtual machine.

   ```
   Set-VM -VM "MyVM2" -SEVEnabled $false
   ```

SEV-ES is deactivated on the virtual machine.

Securing Virtual Machines with Confidential Computing

Confidential Computing features, including AMD SEV-SNP and Intel TDX, are now part of the Advanced Cyber Compliance offering.

For more information, see Confidential Computing.

Virtual Machine Encryption

With vSphere Virtual Machine Encryption, you can encrypt your sensitive workloads in an even more secure way. Access to encryption keys can be made conditional to the ESX host being in a trusted state.

Before you can start with virtual machine encryption tasks, you must set up a key provider. The following key provider types are available.

Table 208: vSphere Key Providers

Key Provider	Description	For More Information
Standard key provider	Available in vSphere 6.5 and later, the standard key provider uses vCenter to request keys from an external key server. The key server generates and stores the keys, and passes them to vCenter for distribution.	See Configuring and Managing a Standard Key Provider.
VMware vSphere®Native Key Provider™	Available in vSphere 7.0 Update 2 and later, vSphere Native Key Provider is included in all vSphere editions and does not require an external key server.	See Configuring and Managing vSphere Native Key Provider.

Comparison of vSphere Key Providers

A high-level overview of the capabilities of the vSphere key providers requires your attention to help plan your encryption strategy.

In general, there is little difference in feature or product support between key provider daily operation. Though key providers look and behave similarly, you might have requirements and regulations to consider when choosing a key provider, as shown in the following table.

Table 209: Key Provider Considerations

Key Provider	External Key Server Required?	Quick Setup?	Works Only with vSphere?	Encryption Keys Stored Permanently on Host?	Rekey While Cloning?
Standard key provider	Yes	No	No	No	Yes
vSphere Native Key Provider	No	Yes	Yes	Yes	Yes

Note:
At host startup, vSphere Native Key Provider always writes the encryption key to the ESX hosts in the cluster. If you are concerned about the physical security of the cluster, consider using either a standard key provider, which requires that the key server be available for encrypted virtual machines to function.

Key Provider Encryption Features

The following encryption features are supported by each key provider type.

- Rekey using the same key provider or to another key provider
- Rotate keys
- Virtual Trusted Platform Module (vTPM)
- Disk encryption
- vSphere Virtual Machine Encryption
- Co-existence with other key providers
- Upgrade to a different key provider

Key Provider Support for vSphere Features

The following describes key provider support for some important vSphere features.

- Encrypted vSphere vMotion: Supported by all key provider types. The same key provider must be available on the destination host. See What Is Encrypted vSphere vMotion.
- vCenter File-based Backup and Restore: Standard key provider and vSphere Native Key Provider support vCenter file-based backup and restore.

Key Provider Support for VMware Products

The following table compares key provider support for some VMware products.

Table 210: Comparison of Support for VMware Products

Key Provider	vSAN Data-At-Rest Encryption	Site Recovery Manager	vSphere Replication
Standard key provider	Yes	Yes	Yes
vSphere Native Key Provider	Yes	Yes	Yes

Note:

Standard key provider and vSphere Native Key Provider support vSphere Virtual Machine Encryption on top of vSAN.

Required Hardware for Key Providers

The following table compares some minimum key provider hardware requirements.

Table 211: Comparison of Required Hardware for Key Providers

Key Provider	TPM on ESXi Host
Standard key provider	Not required
vSphere Native Key Provider	Not required vSphere Native Key Provider availability can optionally be restricted to hosts with a TPM.

Key Provider Naming

vSphere uses a key provider name to look up a key identifier. If two key providers have the same name, vSphere assumes that they are equivalent and have access to the same keys. Each logical key provider, regardless of its type (Standard, Trusted, and Native Key Provider), must have a unique name across all vCenter systems.

In a few instances, you configure the same key provider across multiple vCenter systems, such as:

- Migrating encrypted virtual machines between vCenter systems
- Setting up a vCenter as a disaster recovery site

How vSphere Virtual Machine Encryption Protects Your Environment

Regardless of which key provider you use, with vSphere Virtual Machine Encryption you can create encrypted virtual machines and encrypt existing virtual machines. Because all virtual machine files with sensitive information are encrypted, the virtual machine is protected. Only administrators with encryption privileges can perform encryption and decryption tasks.

Important:
ESX Shell users also have cryptographic operation privileges. For more information, see Prerequisites and Required Privileges for Virtual Machine Encryption Tasks.

What Storage Does vSphere Virtual Machine Encryption Support

vSphere Virtual Machine Encryption works with any supported storage type (NFS, iSCSI, Fibre Channel, direct-attached storage, and so on), including VMware vSAN. For more information about using encryption on a vSAN cluster, see the *Administering VMware vSAN* documentation.

vSphere Virtual Machine Encryption and vSAN use the same encryption libraries but they have different profiles. Virtual Machine Encryption is a per-VM encryption and vSAN is a datastore level encryption.

vSphere Encryption Keys and Key Providers

vSphere uses two levels of encryption in the form of a Key Encryption Key (KEK) and a Data Encryption Key (DEK). Briefly, an ESX host generates a DEK to encrypt virtual machines and disks. The KEK is provided by a key server, and encrypts (or "wraps") the DEK. The KEK encrypts the DEK using the AES256 algorithm and the DEK encrypts the VMDK using the XTS-AES-256 (512-bit key size) algorithm. Depending on the type of key provider, different methods are used to create and manage the DEK and KEK.

Standard key provider operates as follows.

1. The ESX host generates and uses internal keys to encrypt virtual machines and disks. These keys are used as DEKs.

2. vCenter requests keys from the key server (KMS). These keys are used as the KEKs. vCenter stores only the ID of each KEK, but not the key itself.
3. ESX uses the KEK to encrypt the internal keys, and stores the encrypted internal key on disk. ESX does not store the KEK on disk. If a host reboots, vCenter requests the KEK with the corresponding ID from the key server and makes it available to ESX. ESX can then decrypt the internal keys as needed.

Note:

The ESX hosts in vSphere clusters hold the KEK for encrypted virtual machines in host memory to enable availability features such as High Availability, vMotion, DRS, and so on. When a virtual machine is deleted or unregistered, the ESX hosts in the cluster delete the KEK from their memory. Thus, the ESX hosts can no longer use the KEK.

vSphere Native Key Provider operates as follows.

1. When you create the key provider, vCenter generates a primary key and pushes it to the ESX hosts in the cluster. (No external key server is involved.)
2. The ESX hosts generate a DEK on demand.
3. When you perform an encryption activity, data is encrypted with the DEK.
   Encrypted DEKs are stored alongside the encrypted data.
4. When you decrypt data, the primary key is used to decrypt the DEK, and then the data.

What Components Does vSphere Virtual Machine Encryption Encrypt

vSphere Virtual Machine Encryption supports encryption of virtual machine files, virtual disk files, and core dump files.

Virtual machine files

Most virtual machine files, in particular, guest data that is not stored in the VMDK file, are encrypted. This set of files includes but is not limited to the NVRAM, VSWP, and VMSN files. The key from the key provider unlocks an encrypted bundle in the VMX file that contains internal keys and other secrets. The key retrieval works as follows, depending on the key provider:

- Standard key provider: vCenter manages the keys from the key server and the ESXi hosts cannot directly access the key provider. The hosts wait for vCenter to push the keys.
- vSphere Native Key Provider: The ESX hosts directly access the key providers, and so fetch the requested keys from the vSphere Native Key Provider.

When you use the vSphere Client to create an encrypted virtual machine, you can encrypt and decrypt virtual disks separate from virtual machine files. All virtual disks are encrypted by default. For other encryption tasks, such as encrypting an existing virtual machine, you can encrypt and decrypt virtual disks separate from virtual machine files.

Note:
You cannot associate an encrypted virtual disk with a virtual machine that is not encrypted.

Virtual disk files

Data in an encrypted virtual disk (VMDK) file is never written in cleartext to storage or physical disk, and is never transmitted over the network in cleartext. The VMDK descriptor file is mostly cleartext, but contains a key ID for the KEK and the internal key (DEK) in the encrypted bundle.

You can use the vSphere Client or the vSphere API to perform a shallow recrypt operation with a new KEK, or use the vSphere API to perform a deep recrypt operation with a new internal key.

Core dumps

Core dumps on an ESX host that has encryption mode enabled are always encrypted. See vSphere Virtual Machine Encryption and Core Dumps. Core dumps on the vCenter system are not encrypted. Protect access to the vCenter system.

Virtual machine swap file

The virtual machine swap file is encrypted whenever you add a vTPM to a virtual machine. Environments low on RAM can experience encryption-related paging that could impact performance.

vTPMs

When you configure a vTPM, the virtual machine files are encrypted but not the disks. You can choose to add encryption explicitly for the virtual machine and its disks. For more information, see Securing Virtual Machines with Virtual Trusted Platform Module.

Note:
For information on some limitations concerning devices and features that vSphere Virtual Machine Encryption can interoperate with, see Virtual Machine Encryption Interoperability.

What Components Does vSphere Virtual Machine Encryption Not Encrypt

Some of the files that are associated with a virtual machine are not encrypted or partially encrypted.

Log files	Log files are not encrypted because they do not contain sensitive data.
Virtual machine configuration files	Most of the virtual machine configuration information, stored in the VMX and VMSD files, is not encrypted.
Virtual disk descriptor file	To support disk management without a key, most of the virtual disk descriptor file is not encrypted.

What Privileges Are Required to Perform Cryptographic Operations

Only users that are assigned the **Cryptographic Operations** privileges can perform cryptographic operations. The privilege set is fine grained. The default Administrator system role includes all **Cryptographic Operations** privileges. The No Cryptography Administrator role supports all Administrator privileges except for the **Cryptographic Operations** privileges.

In addition to using the **Cryptographer.*** privileges, vSphere Native Key Provider can use the **Cryptographer.ReadKeyServersInfo** privilege, which is specific to vSphere Native Key Providers.

See Cryptographic Operations Privileges for more information.

You can create additional custom roles, for example, to allow a group of users to encrypt virtual machines but to prevent them from decrypting virtual machines.

How Do You Perform Cryptographic Operations

The vSphere Client supports many of the cryptographic operations. For other tasks, you can use PowerCLI or the vSphere API.

Table 212: Interfaces for Performing Cryptographic Operations

Interface	Operations	Information
vSphere Client	Create encrypted virtual machine Encrypt and decrypt virtual machines Perform a shallow recrypt of a virtual machine (use a different KEK)	This book
PowerCLI	Create encrypted virtual machine Encrypt and decrypt virtual machines Configure vSphere Trust Authority	*VMware PowerCLI Cmdlets Reference*
vSphere Web Services SDK	Create encrypted virtual machine Encrypt and decrypt virtual machines Perform a deep recrypt of a virtual machine (use a different DEK) Perform a shallow recrypt of a virtual machine (use a different KEK)	*vSphere Web Services SDK Programming Guide* *vSphere Web Services API Reference*
`crypto-util`	Decrypt encrypted core dumps Check whether files are encrypted Perform other management tasks directly on the ESX host	Command-line help vSphere Virtual Machine Encryption and Core Dumps

How Do You Recrypt (Rekey) an Encrypted Virtual Machine

You can recrypt (also called rekey) a virtual machine with new keys, for example, in case a key expires or becomes compromised. The following rekeying options are available.

- A shallow recrypt, which replaces only the Key Encryption Key (KEK)
- A deep recrypt, which replaces both the Disk Encryption Key (DEK) and the KEK

A deep recrypt requires that the virtual machine is powered off and contains no snapshots. You can perform a shallow recrypt operation while the virtual machine is powered on, and if the virtual machine has snapshots present. Shallow recrypt of an encrypted virtual machine with snapshots is permitted only on a single snapshot branch (disk chain). Multiple snapshot branches are not supported. Also, shallow recrypt is not supported on a linked clone of a virtual machine or disk. If the shallow recrypt fails before updating all links in the chain with the new KEK, you can still access the encrypted virtual machine if you have the old and new KEKs. However, it is best to reissue the shallow recrypt operation before performing any snapshot operations.

You can perform a rekey of a virtual machine by using the vSphere Client, the CLI, or the API. See Rekey an Encrypted Virtual Machine Using the vSphere Client, Rekey an Encrypted Virtual Machine Using the CLI, and *vSphere Web Services SDK Programming Guide*.

vSphere Virtual Machine Encryption Components

Depending on which key provider you use, an external key server, the vCenter system, and your ESX hosts are potentially contributing to the encryption solution.

The following components comprise vSphere Virtual Machine Encryption:

- An external key server, also called a KMS (not required for vSphere Native Key Provider)
- vCenter
- ESX hosts

What Is the Role of a Key Server in vSphere Virtual Machine Encryption

The key server is a Key Management Interoperability Protocol (KMIP) management server that is associated with a key provider. A standard key provider requires a key server. vSphere Native Key Provider does not require a key server. The following table describes the differences in key provider and key server interaction.

Table 213: Key Providers and Key Server Interaction

Key Provider	Interaction with Key Server
Standard key provider	A standard key provider uses vCenter to request keys from a key server. The key server generates and stores the keys, and passes them to vCenter for distribution to the ESX hosts.
vSphere Native Key Provider	vSphere Native Key Provider does not require a key server. vCenter generates a primary key and pushes it to the ESX hosts. The ESX hosts then generate data encryption keys (even when not connected to vCenter). See vSphere Native Key Provider Overview.

You can use the vSphere Client or the vSphere API to add key provider instances to the vCenter system. If you use multiple key provider instances, all instances must be from the same vendor and must replicate keys.

If your environment uses different key server vendors in different environments, you can add a key provider for each key server and specify a default key provider. The first key provider that you add becomes the default key provider. You can explicitly specify the default later.

As a KMIP client, vCenter uses the Key Management Interoperability Protocol (KMIP) to make it easy to use the key server of your choice.

What Is the Role of vCenter in vSphere Virtual Machine Encryption

The following table describes the role of vCenter in the encryption process.

Table 214: Key Providers and vCenter

Key Provider	Role of vCenter	How Are Privileges Checked
Standard key provider	Only vCenter has the credentials for logging in to the key server. Your ESX hosts do not have those credentials. vCenter obtains keys from the key server and pushes them to the ESX hosts. vCenter does not store the key server keys, but keeps a list of key IDs.	vCenter checks the privileges of users who perform cryptographic operations.
vSphere Native Key Provider	The vCenter generates the keys.	vCenter checks the privileges of users who perform cryptographic operations.

You can use the vSphere Client to assign cryptographic operation privileges or to assign the **No cryptography administrator** custom role to groups of users. See Prerequisites and Required Privileges for Virtual Machine Encryption Tasks.

vCenter adds cryptography events to the list of events that you can view and export from the vSphere Client Event Console. Each event includes the user, time, key ID, and cryptographic operation.

The keys that come from the key server are used as key encryption keys (KEKs).

What Is the Role of ESX Hosts in vSphere Virtual Machine Encryption

ESX hosts are responsible for several aspects of the encryption workflow.

Table 215: Key Providers and ESX Hosts

Key Provider	ESX Host Aspects
Standard key provider	• vCenter pushes keys to an ESX host when the host needs a key. The host must have encryption mode enabled. • Ensuring that guest data for encrypted virtual machines is encrypted when stored on disk. • Ensuring that guest data for encrypted virtual machines is not sent over the network without encryption.
vSphere Native Key Provider	The ESX hosts fetch keys directly from the vSphere Native Key Provider.

The keys that the ESX host generates are called internal keys in this document. These keys typically act as data encryption keys (DEKs).

Encryption Process Flow

After you set up a key provider, users with the required privileges can create encrypted virtual machines and disks. Those users can also encrypt existing virtual machines and decrypt encrypted virtual machines, and add Virtual Trusted Platform Modules (vTPMs) to virtual machines.

Depending on the key provider type, the process flow can involve a key server, the vCenter, and the ESX host.

Standard Key Provider Encryption Process Flow

During the encryption process, different vSphere components interact as follows.

1. When the user performs an encryption task, for example, creating an encrypted virtual machine, vCenter requests a new key from the default key server. This key is used as the KEK.
2. vCenter stores the key ID and passes the key to the ESX host. If the ESX host is part of a cluster, vCenter sends the KEK to each host in the cluster.
   The key itself is not stored on the vCenter system. Only the key ID is known.
3. The ESX host generates internal keys (DEKs) for the virtual machine and its disks. It keeps the internal keys in memory only, and uses the KEKs to encrypt internal keys.
   Unencrypted internal keys are never stored on disk. Only encrypted data is stored. Because the KEKs come from the key server, the host continues to use the same KEKs.
4. The ESX host encrypts the virtual machine with the encrypted internal key.
   Any hosts that have the KEK and that can access the encrypted key file can perform operations on the encrypted virtual machine or disk.

vSphere Native Key Provider Encryption Process Flow

vSphere Native Key Provider is included in vSphere 7.0 Update 2 and later. When you configure a vSphere Native Key Provider, vCenter pushes a primary key to all ESX hosts in the cluster. Likewise, if you update or delete a vSphere Native Key Provider, the change is pushed to the hosts in the cluster. The difference is that vSphere Native Key Provider generates the keys and wraps them with the primary key, then hands them back to perform encryption.

Custom Attributes for Key Servers

The Key Management Interoperability Protocol (KMIP) supports adding custom attributes intended for vendor-specific purposes. Custom attributes enable you to more specifically identify keys stored in your key server. vCenter adds the following custom attributes for virtual machine keys and host keys.

Table 216: Virtual Machine Encryption Custom Attributes

Custom Attribute	Value
`x-Vendor`	VMware, Inc.
`x-Product`	VMware vSphere
`x-Product_Version`	vCenter version
`x-Component`	Virtual Machine
`x-Name`	Virtual machine name (gathered from ConfigInfo or ConfigSpec)
`x-Identifier`	Virtual machine's instanceUuid (gathered from ConfigInfo or ConfigSpec)

Table 217: Host Encryption Custom Attributes

Custom Attribute	Value
`x-Vendor`	VMware, Inc.
`x-Product`	VMware vSphere
`x-Product_Version`	vCenter version
`x-Component`	ESX Server
`x-Name`	Host name
`x-Identifier`	Host's hardware Uuid

vCenter adds the `x-Vendor`, `x-Product`, and `x-Product_Version` attributes when the key server creates a key. When the key is used to encrypt a virtual machine or host, vCenter sets the `x-Component`, `x-Identifier`, and `x-Name` attributes. You might be able to view these custom attributes in your key server user interface. Check with your key server vendor.

Both the host key and virtual machine key have the six custom attributes. `x-Vendor`, `x-Product`, and `x-Product_Version` might be the same for both keys. These attributes are set when the key is generated. Depending on if the key is for a virtual machine or a host, it might have `x-Component`, `x-Identifier`, and `x-Name` attributes appended.

Encryption Key Errors

When an error occurs sending keys from the key server to an ESX host, vCenter generates a message in the event log for the following events:

- Adding keys to the ESX host failed due to host connection or host support issues.
- Getting keys from the key server failed due to key missing in the key server.
- Getting keys from the key server failed due to the key server connection.

Decrypting Encrypted Virtual Machines

If you later want to decrypt an encrypted virtual machine, you change its storage policy. You can change the storage policy for the virtual machine and all disks. If you want to decrypt individual components, decrypt selected disks first, then decrypt the virtual machine by changing the storage policy for VM Home. Both keys are required for decryption of each component. See Decrypt an Encrypted Virtual Machine or Virtual Disk.

Virtual Disk Encryption

When you create an encrypted virtual machine from the vSphere Client, you can decide which disks to exclude from encryption. You can later add disks and set their encryption policies. You cannot add an encrypted disk to a virtual machine that is not encrypted, and you cannot encrypt a disk if the virtual machine is not encrypted.

Encryption for a virtual machine and its disks is controlled through storage policies. The storage policy for VM Home governs the virtual machine itself, and each virtual disk has an associated storage policy.

- Setting the storage policy of VM Home to an encryption policy encrypts only the virtual machine itself.
- Setting the storage policy of VM Home and all the disks to an encryption policy encrypts all components.

Consider the following use cases.

Table 218: Virtual Disk Encryption Use Cases

Use Case	Details
Create an encrypted virtual machine.	If you add disks while creating an encrypted virtual machine, the disks are encrypted by default. You can change the policy not to encrypt one or more of the disks. After virtual machine creation, you can explicitly change the storage policy for each disk. See Change the Encryption Policy for Virtual Disks.
Encrypt a virtual machine.	To encrypt an existing virtual machine, you change its storage policy. You can change the storage policy for the virtual machine and all virtual disks. To encrypt just the virtual machine, you can specify an encryption policy for VM Home and select a different storage policy, such as Datastore Default, for each virtual disk. See Create an Encrypted Virtual Machine.
Add an existing unencrypted disk to an encrypted virtual machine (encryption storage policy).	Fails with an error. You have to add the disk with the default storage policy, but can later change the storage policy. See Change the Encryption Policy for Virtual Disks.
Add an existing unencrypted disk to an encrypted virtual machine with a storage policy that does not include encryption, for example Datastore Default.	The disk uses the default storage policy. You can explicitly change the storage policy after adding the disk if you want an encrypted disk. See Change the Encryption Policy for Virtual Disks.
Add an encrypted disk to an encrypted virtual machine. VM Home storage policy is Encryption.	When you add the disk, it remains encrypted. The vSphere Client displays the size and other attributes, including encryption status.

Use Case	Details
Add an existing encrypted disk to an unencrypted virtual machine.	This use case is not supported. However, if you use the vSphere Client to encrypt the VM Home files, you can then reconfigure the unencrypted virtual machine with the encrypted disk.
Register an encrypted virtual machine.	If you remove an encrypted virtual machine from vCenter but do not delete it from disk, you can return it to the vCenter inventory by registering the VM's virtual machine configuration (.vmx) file. To register the encrypted VM, the user must have the **Cryptographic operations** > **Register VM** privilege. If the VM was encrypted using a standard key provider, when the encrypted VM is registered, vCenter pushes the required keys to the ESX host. If the user registering the VM does not have the **Cryptographic operations** > **Register VM** privilege, vCenter locks the VM upon registration, and the VM is not usable until it is unlocked. If the VM was encrypted using a vSphere Native Key Provider, when the encrypted VM is registered, vCenter no longer pushes keys to the ESX host. Instead, the keys are fetched from the host when the VM is registered. If the user registering the VM does not have the **Cryptographic operations** > **Register VM** privilege, vCenter does not permit the operation.

Virtual Machine Encryption Errors

If vCenter detects a critical error with virtual machine encryption, it creates an event. You can view these events to help troubleshoot and resolve encryption errors.

vCenter creates events for the following virtual machine encryption critical errors.

- Failure to generate a KEK.
- Insufficient disk space on the datastore to create an encrypted virtual machine.
- Insufficient user privilege to initiate encryption operation.
- The specified key is missing on the key provider and so the ESX host key is renewed with a new key.
- An error occurred on the key provider with the specified key and so the ESX host key is renewed with a new key.

Prerequisites and Required Privileges for Virtual Machine Encryption Tasks

Virtual machine encryption tasks are possible only in environments that include vCenter. Also, the ESX host must have encryption mode activated for most encryption tasks. The user who performs the task must have the appropriate privileges. A set of**Cryptographic Operations** privileges allows fine-grained control. If virtual machine encryption tasks require a change to the host encryption mode, additional privileges are required.

Using Cryptography Privileges and Roles

By default, the user with the vCenter Administrator role has all privileges, including cryptographic operations privileges. The**No cryptography administrator** role does not have the following privileges that are required for cryptographic operations.

> **Important:**
> ESX Shell users also have cryptographic operation privileges.

- Add**Cryptographic Operations** privileges.
- **Global** > **Diagnostics**
- **Host** > **Inventory** > **Add host to cluster**
- **Host** > **Inventory** > **Add standalone host**
- **Host** > **Local operations** > **Manage user groups**

You can assign the**No cryptography administrator** role to vCenter administrators that do not need**Cryptographic Operations** privileges.

To impose more limits on what users can do, you can clone the**No cryptography administrator** role and create a custom role with only some of the**Cryptographic Operations** privileges. For example, you can create a role that allows users to encrypt but not to decrypt virtual machines. See Using vCenter Roles to Assign Privileges.

What Is Host Encryption Mode

Host encryption mode determines if an ESX host is ready to accept cryptographic material for encrypting virtual machines and virtual disks. Before any cryptographic operations can occur on a host, encryption mode must be activated. Host encryption mode is often set automatically when it is required, but you can set it explicitly. You can check and explicitly set the current host encryption mode from the vSphere Client or by using the vSphere API.

When host encryption mode is activated, vCenter installs a host key on the host, which ensures that the host is cryptographically "safe." With the host key in place, other cryptographic operations can proceed, including vCenter obtaining keys from the key provider and pushing them to the ESXi hosts.

In "safe" mode, user worlds (that is, hostd) and encrypted virtual machines have their core dumps encrypted. Unencrypted virtual machines do not have their core dumps encrypted.

For more information about encrypted core dumps and how they are used by VMware Technical Support, see the VMware knowledge base article at https://kb.vmware.com/s/article/2147388.

For instructions, see Activate Host Encryption Mode Explicitly.

After host encryption mode is set, it cannot be deactivated easily. See Deactivate Host Encryption Mode Using the API.

Automatic changes occur when encryption operations attempt to set host encryption mode. For example, assume that you add an encrypted virtual machine to a standalone host. Host encryption mode is not set. If you have the required privileges on the host, encryption mode is automatically set.

Assume that a cluster has three ESX hosts, host A, B, and C. You create an encrypted virtual machine on host A. What happens depends on several factors.

- If hosts A, B, and C already have host encryption mode set, you need only**Cryptographic operations** > **Encrypt new** privileges to create the virtual machine.
- If hosts A and B are set for host encryption, and C is not, the system proceeds as follows.
  - Assume that you have both the**Cryptographic operations** > **Encrypt new** and the**Cryptographic operations** > **Register host** privileges on each host. In this case, the encryption process sets host encryption mode on host C, and pushes the key to each host in the cluster.

    For this case, you can also explicitly set host encryption mode on host C.
  - Assume that you have only**Cryptographic operations** > **Encrypt new** privileges on the virtual machine or virtual machine folder. In that case, virtual machine creation succeeds and the key becomes available on host A and host B. Host C remains deactivated for encryption and does not have the virtual machine key.
- If none of the hosts has host encryption mode set, and you have**Cryptographic operations** > **Register host** privileges on host A, then the virtual machine creation process sets host encryption mode on that host. Otherwise, for hosts B and C, an error results.
- You can also use the vSphere API to set the encryption mode of a cluster to "force enable." Force enable causes all hosts in the cluster to be cryptographically "safe," that is, vCenter has installed a host key on the host. See *vSphere Web Services SDK Programming Guide*.

Disk Space Requirements When Encrypting Virtual Machines

When you encrypt an existing virtual machine, you need at least twice the space that the virtual machine is currently using.

What Is Encrypted vSphere vMotion

Encrypted vSphere vMotion secures confidentiality, integrity, and authenticity of data that is transferred with vSphere vMotion. vSphere supports encrypted vMotion of unencrypted and encrypted virtual machines across vCenter instances.

vSphere vMotion always uses encryption when migrating encrypted virtual machines. For virtual machines that are not encrypted, you can select one of the encrypted vSphere vMotion options.

What Is Encrypted in Encrypted vSphere vMotion

For encrypted disks, the data is transmitted encrypted in all cases. For unencrypted disks, the following applies:

- If disk data is transferred within a host, that is without changing the host, you change only the datastore, the transfer is unencrypted.
- If disk data is transferred between hosts and encrypted vMotion is used, the transfer is encrypted. If encrypted vMotion is not used the transfer is unencrypted.

For virtual machines that are encrypted, migration with vSphere vMotion always uses encrypted vSphere vMotion. You cannot turn off encrypted vSphere vMotion for encrypted virtual machines.

Encrypted vSphere vMotion States for Unencrypted Virtual Machines

For virtual machines that are not encrypted, you can set encrypted vSphere vMotion to one of the following states. The default is Opportunistic.

Disabled	Do not use encrypted vSphere vMotion.
Opportunistic	Use encrypted vSphere vMotion if the source and the destination hosts support it. Only ESX hosts of version 6.5 and later support encrypted vSphere vMotion.
Required	Allow only encrypted vSphere vMotion. If the source or the destination host does not support encrypted vSphere vMotion, migration with vSphere vMotion is not allowed.

When you encrypt a virtual machine, the virtual machine keeps a record of the current encrypted vSphere vMotion setting. If you later deactivate encryption for the virtual machine, the encrypted vMotion setting remains at Required until you change the setting explicitly. You can change the settings using **Edit Settings**.

See the *vCenter and Host Management* documentation for information on activating and deactivating encrypted vSphere vMotion for virtual machines that are not encrypted.

> **Note:**
>
> Currently, you must use the vSphere APIs to migrate or clone encrypted virtual machines across vCenter instances. See *vSphere Web Services SDK Programming Guide* and *vSphere Web Services API Reference* documentation.

Migrating or Cloning Encrypted Virtual Machines Across vCenter Instances

vSphere vMotion supports migrating and cloning encrypted virtual machines across vCenter instances.

When migrating or cloning encrypted virtual machines across vCenter instances, the source and destination vCenter instances must be configured to share the key provider that was used to encrypt the virtual machine. In addition, the key provider name must be the same on both the source and destination vCenter instances, and must have the following characteristics:

- Standard key provider: The same key server (or key servers) must be in the key provider.
- vSphere Native Key Provider: Must have the same KDK.

> **Note:**
> You cannot clone or migrate an encrypted virtual machine using vSphere Native Key Provider to a standalone host, no matter that the source host resides in a cluster.

The destination vCenter ensures the destination ESX host has encryption mode set, ensuring the host is cryptographically "safe."

The following privileges are required when using vSphere vMotion to migrate or clone an encrypted virtual machine across vCenter instances.

- Migrating: **Cryptographic operations** > **Migrate** on the virtual machine
- Cloning: **Cryptographic operations** > **Clone** on the virtual machine

Also, the destination vCenter must have the **Cryptographic operations** > **EncryptNew** privilege. If the destination ESX host is not in "safe" mode, the **Cryptographic operations** > **RegisterHost** privilege must also be on the destination vCenter.

Certain tasks are not allowed when migrating virtual machines (non-encrypted or encrypted), either on the same vCenter or across vCenter instances.

- You cannot change the VM Storage Policy.
- You cannot perform a key change.

> **Note:**
> You can change the VM Storage Policy while cloning virtual machines.

vSphere Native Key Provider vMotion and Cross-vCenter vMotion

vSphere Native Key Provider supports vMotion and Encrypted vMotion across ESX hosts. Cross-vCenter vMotion is supported if vSphere Native Key Provider is configured on the destination host.

Encrypted vMotion and Intel QuickAssist Technology (QAT) hardware accelerator

The vMotion encryption and decryption tasks, which use CPU resources, are now offloaded to the hardware devices. Some processors, for example, 4th Gen Intel Xeon Scalable Processors and 5th Gen Intel Xeon Scalable Processors, have built-in QAT accelerators. Starting with vSphere 9.1, if your processor has QAT accelerator and the device is available for vMotion, by default the crypto operation for ecrypted vMotion is offloaded to the QAT accelerator. This reduces the CPU consumption for the vMotion task. If the QAT accelerator is needed for a virtual machine operation at the same time as an ecrypted vMotion task, the vMotion task does not interrupt the virtual machine operation.

The feature is activated by default during new deployment and after an upgrade to vSphere 9.1. You can see the following message in the vmkernel logs:

```
DAS Crypto AEAD session handle created successfully. Crypto workload will be offloaded.
```

Virtual Machine Encryption Best Practices

Follow virtual machine encryption best practices to avoid problems later, for example, when you generate a `vm-support` bundle.

Best Practices for Virtual Machine Encryption to Get Started With

To avoid problems when using virtual machine encryption, follow these general best practices.

- Do not encrypt any vCenter appliance virtual machines.
- If your ESX host fails, retrieve the support bundle as soon as possible. The host key must be available for generating a support bundle that uses a password, or for decrypting a core dump. If the host is rebooted, it is possible that the host key changes. If that happens, you can no longer generate a support bundle with a password or decrypt core dumps in the support bundle with the host key.
- Manage key provider names carefully. If the key provider name changes for a key server that is already in use, a VM that is encrypted with keys from that key server enters a locked state during power-on or register. In that case, remove the key server from the vCenter and add it with the key provider name that you used initially.
- Do not edit VMX files and VMDK descriptor files. These files contain the encryption bundle. It is possible that your changes make the virtual machine unrecoverable, and that the recovery problem cannot be fixed.
- The vSphere Virtual Machine Encryption process encrypts data on the host before writing the data to storage. The effectiveness of back-end storage features such as deduplication, compression, replication, and so on, might be affected when encrypting virtual machines in this manner.
- If you use multiple layers of encryption, for example, vSphere Virtual Machine Encryption and in-guest encryption (BitLocker, dm-crypt, and so on), overall virtual machine performance might be affected, because the encryption processes use additional CPU and memory resources.
- Ensure that replicated copies of virtual machines encrypted with vSphere Virtual Machine Encryption have access to the encryption keys at the recovery site. For standard key providers, this is handled as part of the design of the Key Management System, outside of vSphere. For vSphere Native Key Provider, ensure that a backup copy of the Native Key Provider key exists and is protected against loss. For more information, see Back up a vSphere Native Key Provider.
- Encryption is CPU intensive. AES-NI significantly improves encryption performance. Enable AES-NI in your BIOS.

Best Practices for Encrypted Core Dumps

Follow these best practices to avoid having problems when you want to examine a core dump to diagnose a problem.

- Establish a policy regarding core dumps. Core dumps are encrypted because they can contain sensitive information such as keys. If you decrypt a core dump, consider it sensitive information. ESX core dumps might contain keys for the ESX host and for the virtual machines on it. Consider changing the host key and recrypting encrypted virtual machines after you decrypt a core dump. You can perform both tasks by using the vSphere API.
  See vSphere Virtual Machine Encryption and Core Dumps for details.
- Always use a password when you collect a `vm-support` bundle. You can specify the password when you generate the support bundle from the vSphere Client or using the `vm-support` command.
  The password recrypts core dumps that use internal keys to use keys that are based on the password. You can later use the password to decrypt any encrypted core dumps that might be included in the support bundle. Unencrypted core dumps and logs are not affected by using the password option.
- The password that you specify during `vm-support` bundle creation is not persisted in vSphere components. You are responsible for keeping track of passwords for support bundles.
- Before you change the host key, generate a `vm-support` bundle with a password. You can later use the password to access any core dumps that might have been encrypted with the old host key.

Best Practices for Key Lifecycle Management

Implement best practices that guarantee key server availability and monitor keys on the key server.

- You are responsible for having policies in place that guarantee key server availability.
  If the key server is not available, virtual machine operations that require that vCenter request the key from the key server are not possible. That means running virtual machines continue to run, and you can power on, power off, and reconfigure those virtual machines. However, you cannot relocate the virtual machine to a host that does not have the key information.
  Most key server solutions include high availability features. You can use the vSphere Client or the API to specify a key provider and the associated key servers.

  Note:
  Starting in version 7.0 Update 2, encrypted virtual machines and virtual TPMs can continue to function even when the key server is temporarily offline or unavailable. The ESX hosts can persist the encryption keys to continue encryption and vTPM operations. See vSphere Key Persistence on ESX Hosts.
- You are responsible for keeping track of keys and for performing remediation if keys for existing virtual machines are not in the Active state.
  The KMIP standard defines the following states for keys.

  - Pre-Active
  - Active
  - Deactivated
  - Compromised
  - Destroyed
  - Destroyed Compromised

  vSphere Virtual Machine Encryption uses only Active keys for encryption. If a key is Pre-Active, vSphere Virtual Machine Encryption activates it. If the key state is Deactivated, Compromised, Destroyed, Destroyed Compromised, you cannot encrypt a virtual machine or disk with that key.
  For keys that are in other states, virtual machines using those keys continue to work. Whether a clone or migration operation succeeds depends on whether they key is already on the host.

  - If the key is on the destination host, the operation succeeds even if the key is not Active on the key server.
  - If the required virtual machine and virtual disk keys are not on the destination host, vCenter has to fetch the keys from the key server. If the key state is Deactivated, Compromised, Destroyed, or Destroyed Compromised, vCenter displays an error and the operation does not succeed.

  A clone or migration operation succeeds if the key is already on the host. The operation fails if vCenter has to pull the keys from the key server.

If a key is not Active, perform a rekey operation using the API. See the *vSphere Web Services SDK Programming Guide*.

- Develop key rotation policies so that keys are retired and rolled over after a specific time.
  - vSphere Native Key Provider: Change the `key_id` of a vSphere Native Key Provider.

Best Practices for Backup and Restore

Set up policies on backup and restore operations.

- Not all backup architectures are supported. See Virtual Machine Encryption Interoperability.
- Set up policies for restore operations. Because backup is always in cleartext, plan to encrypt virtual machines right after restore is finished. You can specify that the virtual machine is encrypted as part of the restore operation. If possible, encrypt virtual machine as part of the restore process to avoid exposing sensitive information. To change the encryption policy for any disks that are associated with the virtual machine, change the storage policy for the disk.
- Because the VM home files are encrypted, ensure that the encryption keys are available at the time of a restore.

Best Practices for Encryption Performance

- Encryption performance depends on the CPU and storage speed.
- Encrypting existing virtual machines is more time consuming than encrypting a virtual machine during creation. Encrypt a virtual machine when you create it if possible.

Best Practices for the Sample Storage Policy

Do not modify the bundled VM Encryption sample storage policy. Instead, clone the policy and edit the clone.

> **Note:**
> No automated way of returning VM Encryption Policy to its original settings exists.

See the *vSphere Storage* documentation for details customizing storage policies.

Best Practices for Removing Encryption Keys

To ensure that encryption keys are removed from a cluster, after deleting, unregistering, or moving the encrypted virtual machine to another vCenter, reboot the ESX hosts in the cluster.

Virtual Machine Encryption Caveats

Review Virtual Machine Encryption caveats to avoid problems later.

To understand which devices and features cannot be used with Virtual Machine Encryption, see Virtual Machine Encryption Interoperability.

Encrypted Virtual Machine Limitations

Consider the following caveats when you plan your virtual machine encryption strategy.

- When you clone an encrypted virtual machine or perform a Storage vMotion operation, you can attempt to change the disk format. Such conversions do not always succeed. For example, if you clone a virtual machine and attempt to change the disk format from lazy-zeroed thick format to thin format, the virtual machine disk keeps the lazy-zeroed thick format.
- When you detach a disk from a virtual machine, the storage policy information for the virtual disk is not retained.
  - If the virtual disk is encrypted, you must explicitly set the storage policy to VM Encryption Policy or to a storage policy that includes encryption.
  - If the virtual disk is not encrypted, you can change the storage policy when you add the disk to a virtual machine.

  See Virtual Disk Encryption for details.
- Decrypt core dumps before moving a virtual machine to a different cluster.

The vCenter does not store the key server keys but only tracks the key IDs. As a result, vCenter does not store the ESX host key persistently. However, in vSphere 7.0 Update 2 and later, encrypted devices can function even when access to a key server is disrupted. See vSphere Key Persistence on ESX Hosts.

Under certain circumstances, for example, when you move the ESX host to a different cluster and reboot the host, vCenter assigns a new host key to the host. You cannot decrypt any existing core dumps with the new host key.

- OVF Export is not supported for an encrypted virtual machine.
- Using the VMware Host Client to register an encrypted virtual machine is not supported.

Virtual Machine Locked State

If the virtual machine key or one or more of the virtual disk keys are missing, the virtual machine enters a locked state. In a locked state, you cannot perform virtual machine operations.

- When you encrypt both a virtual machine and its disks from the vSphere Client, the same key is used for both.
- When you perform the encryption using the API, you can use different encryption keys for the virtual machine and for disks. In that case, if you attempt to power on a virtual machine, and one of the disk keys is missing, the power on operation fails. If you remove the virtual disk, you can power on the virtual machine.

See Resolve Missing Encryption Key Issues for troubleshooting suggestions.

Virtual Machine Encryption Interoperability

vSphere Virtual Machine Encryption has some limitations regarding devices and features that it can interoperate with.

The following limitations and remarks refer to using vSphere Virtual Machine Encryption. For similar information about using vSAN encryption, see the *Administering VMware vSAN* documentation.

Limitations on Certain Encryption Tasks

Some restrictions apply when performing certain tasks on an encrypted virtual machine.

- For most virtual machine encrypted operations, you must power off the virtual machine. You can clone an encrypted virtual machine and you can perform a shallow recrypt while the virtual machine is powered on.

  Note:
  Virtual machines configured with IDE controllers must be powered off to perform a shallow rekey operation.
- You cannot perform a deep recrypt on a virtual machine with snapshots. You can perform a shallow recrypt on a virtual machine with snapshots.

Virtual Trusted Platform Module Devices and vSphere Virtual Machine Encryption

A virtual Trusted Platform Module (vTPM) is a software-based representation of a physical Trusted Platform Module 2.0 chip. You can add a vTPM to either a new or an existing virtual machine. To add a vTPM to a virtual machine, you must configure a key provider in your vSphere environment. When you configure a vTPM, the virtual machine “home” files are encrypted (memory swap, NVRAM files, and so on). The disk files, or VMDK files, are not automatically encrypted. You can choose to add encryption explicitly for the virtual machine disks.

CAUTION:
Cloning a virtual machine duplicates the entire virtual machine, including the virtual devices such as a vTPM. Information stored in the vTPM, including properties of the vTPM that software can use to determine a system’s identity, is also duplicated.

In vSphere 8.0 and later, when cloning a virtual machine that includes a vTPM, you can choose to start with a new, blank vTPM, which gets its own secrets and identity.

vSphere Virtual Machine Encryption and Suspended State and Snapshots

You can resume from a suspended state of an encrypted virtual machine, or revert to a memory snapshot of an encrypted machine. You can migrate an encrypted virtual machine with memory snapshot and suspended state between ESX hosts.

vSphere Virtual Machine Encryption and IPv6

You can use vSphere Virtual Machine Encryption with pure IPv6 mode or in mixed mode. You can configure the key server with IPv6 addresses. You can configure both the vCenter and the key server with only IPv6 addresses.

Limitations on Cloning in vSphere Virtual Machine Encryption

For all key provider types, cloning is supported conditionally. You can change encryption keys on clone. Certain cloning features do not work with vSphere Virtual Machine Encryption.

- Full clones are supported. The clone inherits the parent encryption state including keys. You can encrypt the full clone, re-encrypt the full clone to use new keys, or decrypt the full clone.
  Linked clones are supported and the clone inherits the parent encryption state including keys. You cannot decrypt the linked clone or re-encrypt a linked clone with different keys.

  Note:
  Verify that other applications support linked clones. For example, VMware Horizon® 7 supports both full clones and instant clones, but not linked clones.
- Instant clone is supported by all key provider types, but you cannot change encryption keys on clone.
- You can create a linked clone virtual machine from an encrypted virtual machine. The linked clone virtual machine contains the same keys. You can rekey the encrypted virtual machine “home” files of a linked clone, but you cannot rekey the disks.

Limitations with vSphere Native Key Provider

Certain operations are not supported with vSphere Native Key Provider.

- You cannot use vSphere Native Key Provider to encrypt virtual machines on a standalone host. The host must reside in a cluster to use vSphere Native Key Provider.
- You cannot move a host that contains virtual machines encrypted using vSphere Native Key Provider to a different cluster unless the target cluster contains the same vSphere Native Key Provider. (The encrypted virtual machines on the moved host become locked when the encryption keys are not present and the target cluster does not have the same vSphere Native Key Provider.)
- You cannot register a virtual machine encrypted by vSphere Native Key Provider to a legacy host because of the lack of support for vSphere Native Key Provider.
- You cannot register a virtual machine encrypted by vSphere Native Key Provider to a standalone host because of the requirements for the host to reside in a cluster.

Unsupported Disk Configurations with vSphere Virtual Machine Encryption

Certain types of virtual machine disk configurations are not supported with vSphere Virtual Machine Encryption.

- RDM (Raw Device Mapping). However, vSphere Virtual Volumes (vVols) are supported.
- Multi-writer or shared disks (MSCS, WSFC, or Oracle RAC). Encrypted virtual machine “home” files are supported for multi-writer disks. Encrypted virtual disks are not supported for multi-writer disks. If you attempt to select Multi-writer in the **Edit Settings** page of the virtual machine with encrypted virtual disks, the **OK** button is deactivated.

Miscellaneous Limitations in vSphere Virtual Machine Encryption

Other features that do not work with vSphere Virtual Machine Encryption include the following.

- vSphere ESX Dump Collector
- Content Library
  - Content libraries support two types of templates, the OVF Template type and the VM Template type. You cannot export an encrypted virtual machine to the OVF Template type. The OVF Tool does not support encrypted virtual machines. You can create encrypted VM templates using the VM Template type. In vSphere 8.0 and later, the `ovftool` command includes an option to add a vTPM placeholder to the OVF descriptor file. When deploying

a virtual machine from such a template, vCenter creates a vTPM with unique secrets on the destination virtual machine. See the *vSphere Virtual Machine Administration* documentation.

- Software for backing up encrypted virtual disks must use the VMware vSphere Storage API - Data Protection (VADP) to either back up the disks in hot-add mode or NBD mode with SSL enabled. However, not all backup solutions that use VADP for virtual disk backup are supported. Check with your backup vendor for details.
  - VADP SAN transport mode solutions are not supported for backing up encrypted virtual disks.
  - VADP Hot-Add solutions are supported for encrypted virtual disks. The backup software must support encryption of the proxy VM that is used as part of the hot-add backup workflow. The vendor must have the privilege **Cryptographic Operations** > **Encrypt Virtual Machine**.
  - Backup solutions using the NBD-SSL transport modes are supported for backing up encrypted virtual disks. The vendor application must have the privilege **Cryptographic Operations** > **Direct Access**.
- You cannot send output from an encrypted virtual machine to a serial port or parallel port. Even if the configuration appears to succeed, output is sent to a file.
- vSphere Virtual Machine Encryption is not supported in VMware Cloud on AWS. See the *Managing the VMware Cloud on AWS Data Center* documentation.

vSphere Key Persistence on ESX Hosts

In vSphere 7.0 Update 2 and later, encrypted virtual machines and virtual TPMs can continue to optionally function even when the key server is temporarily offline or unavailable. The ESX hosts can persist the encryption keys to continue encryption and vTPM operations.

In this vSphere release, encrypted devices can function even when access to a key server is disrupted. Encrypted vSAN clusters can also function even when access to a key provider is disrupted.

> **Note:**
> Key persistence is not necessary when using vSphere Native Key Provider. vSphere Native Key Provider is designed out-of-the-box to run without requiring access to a key server. See the following section, "Key Persistence and vSphere Native Key Provider."

How Does Key Persistence on ESX Hosts Work

When using a standard key provider, the ESX host relies on vCenter to manage the encryption keys. vSphere Native Key Provider handles keys differently. See the next section for more information.

Regardless of the type of key provider, the ESX host obtains the keys initially and retains them in its key cache. If the ESX host reboots, it loses its key cache. The ESX host then requests the keys again from the key server. When the ESX host tries to obtain keys and the key server is offline or unreachable, vTPMs and workload encryption cannot function. For edge-style deployments, in which a key server is typically not deployed on site, a loss of connectivity to a key server can cause unnecessary downtime for encrypted workloads.

In vSphere 7.0 Update 2 and later, encrypted workloads can continue to function even when the key server is offline or unreachable. If the ESX host has a TPM, the encryption keys are persisted in the TPM across reboots. So, even if an ESX host reboots, the host does not need to request encryption keys. Also, encryption and decryption operations can continue when the key server is unavailable, because the keys have persisted in the TPM. In essence, depending on the key provider, when the key server is unavailable, you can keep running encrypted workloads "key server free." Also, vTPMs can likewise continue to function even when the key server is unreachable.

Key Persistence and vSphere Native Key Provider

When using a vSphere Native Key Provider, vSphere generates encryption keys and no key server is required. The ESX hosts get a Key Derivation Key (KDK), which is used to derive other keys. After receiving the KDK and generating other keys, the ESX hosts do not need access to vCenter to do encryption operations. In essence, a vSphere Native Key Provider always runs "key server free."

The KDK persists on an ESX host by default even after reboot, and even when vCenter is not available after the host reboots.

You can activate key persistence with vSphere Native Key Provider, but it is normally unnecessary. The ESX hosts have complete access to vSphere Native Key Provider and so additional persistence of keys is redundant. One use case for

activating key persistence with vSphere Native Key Provider is when you also have a standard key provider (external KMIP server) configured.

How Do You Set Up Key Persistence

To activate or deactivate key persistence, see Activate and Deactivate Key Persistence on an ESX Host.

Configuring and Managing a Standard Key Provider

Using a standard key provider in your vSphere environment requires some preparation. After your environment is set up, you can create encrypted virtual machines and virtual disks and encrypt existing virtual machines and disks.

After your environment is set up for a standard key provider, you can use the vSphere Client to create encrypted virtual machines and virtual disks and encrypt existing virtual machines and disks. See Using Encryption in Your vSphere Environment.

You can perform additional tasks by using the API and by using the crypto-util CLI. See the *vSphere Web Services SDK Programming Guide* for the API documentation and the `crypto-util` command-line help for details about that tool.

What Is a Standard Key Provider

You can use a standard key provider to perform virtual machine encryption tasks.

In vSphere, a standard key provider gets encryption keys directly from a key server, and the vCenter distributes the keys to the required ESX hosts in a data center.

You can add separate standard key providers for different users and set the default standard key provider.

Standard Key Provider Requirements

- An external key server (KMS)

The key server must support the Key Management Interoperability Protocol (KMIP) 1.1 standard. See the *vSphere Compatibility Matrices* for details.

You can find information about VMware certified key server (KMS) vendors in the VMware Compatibility Guide under Platform and Compute. If you select Compatibility Guides, you can open the Key Management Server (KMS) compatibility documentation. This documentation is updated frequently.

Standard Key Provider Privileges

Standard key providers use the **Cryptographer.*** privileges. See Cryptographic Operations Privileges.

Wrapping Keys

vSphere 9.0 adds the ability to make use of wrapping keys with a standard key provider. Wrapping keys are useful to reduce key usage and prevent saturation of the KMS in situations where there is high key turnover, such as a VDI environment. This reduces the frequency with which administrators must manually delete or archive obsolete keys in order to free up capacity in the KMS.

To understand how wrapping keys work, it is useful to understand the following terms:

Raw Key	A key generated directly by the KMS. If the standard key provider is in raw key mode, raw keys are used directly for VM encryption.
Wrapping Key	A special kind of raw key that is used to encrypt other keys, rather than to directly encrypt VMs.
Wrapped Key	A randomly generated symmetric key that is encrypted by the wrapping key. A wrapped key can be transmitted securely or stored on disk without exposing secrets. The wrapping key is required to decrypt the wrapped key before using it to encrypt or decrypt a virtual machine.
Wrapped Key Mode	A standard key provider in wrapped key mode generates wrapping keys by default.
Raw Key Mode	A standard key provider in raw key mode generates only raw keys and uses them to directly encrypt VMs.

When in wrapped key mode, you can choose between two key types: autogenerated and custom.

Autogenerated keys are the default. When this option is selected, the KMS generates the wrapping key. In this mode, there is a rotation interval, after which the standard key provider replaces the wrapping key. The old wrapping key is stored in the vCenter database for tracking purposes. The rotation interval can be between 30 and 3650 days, and is 365 days by default.

When you select the custom key type, you specify the key ID. Use a wrapping key ID generated by KMIP server that is included in the standard key provider. When you use custom keys, you cannot configure a rotation interval. You can rotate the keys manually by entering a new wrapping key ID.

> **Note:**
>
> If you configure the same KMS server with a standard key provider in multiple vCenter instances, it is recommended that you use the same wrapping key ID.

After a wrapping key is rotated, either by automatic rotation or by manually entering a new wrapping key ID, existing encrypted VMs will continue to use the old keys. Re-key the VMs to re-encrypt them with the new keys.

Wrapped key mode is enabled by default. You can switch a standard key provider between wrapped key mode and raw key mode. If you enable wrapped key mode on a standard key provider that was previously using raw key mode, new VMs will be encrypted with wrapped keys, but existing VMs continue to use the original raw keys until they are re-keyed.

Setting Up the Standard Key Provider

Before you can start with virtual machine encryption tasks, you must set up the standard key provider.

Setting up a standard key provider includes adding the key provider and establishing trust with the key server. When you add a key provider, you are prompted to make it the default. You can explicitly change the default key provider. vCenter provisions keys from the default key provider.

> **Note:**
> What was previously called a Key Management Server cluster in vSphere 6.5 and 6.7 is now called a key provider.

Add a Standard Key Provider Using the vSphere Client

You can add a standard key provider to your vCenter system from the vSphere Client or by using the public API.

- Verify that the key server (KMS) is in the *VMware Compatibility Guide for Key Management Servers (KMS)* and is KMIP 1.1 compliant, and that it can be a symmetric key foundry and server.
- Verify that you have the required privileges: **Cryptographic operations** > **Manage key servers**.
- Ensure that the key server is highly available. Loss of connection to the key server, such as during a power outage or a disaster recovery event, renders encrypted virtual machines inaccessible.

  > **Note:**
  > In vSphere 7.0 Update 2 and later, encrypted virtual machines and virtual TPMs can continue to function even when the key server is temporarily offline or unavailable. See vSphere Key Persistence on ESX Hosts.
- Consider your infrastructure's dependencies on the key server carefully. Some KMS solutions are delivered as virtual appliances, making it possible to create a dependency loop or other availability problem with poor placement of the KMS appliance.

The vSphere Client enables you to add a standard key provider to your vCenter system, and establish trust between the key server and vCenter.

- You can add multiple key servers from the same vendor.
- If your environment supports solutions from different vendors, you can add multiple key providers.
- If your environment includes multiple key providers, and you delete the default key provider, you must set another default explicitly.
- You can configure the key server with IPv6 addresses.
  - Both the vCenter system and the key server can be configured with only IPv6 addresses.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Click **Add Standard Key Provider** and enter the key provider information.

Option	Value
Name	Name for the key provider. Each logical key provider, regardless of its type (Standard, Trusted, and Native Key Provider), must have a unique name across all vCenter systems. For more information, see Key Provider Naming.
KMS	Alias for the key server (KMS).
Address	IP address or FQDN of the key server.
Port	Port on which vCenter connects to the key server.
Use a single KMS wrapping key	Enables wrapped key mode. This allows you to use a single raw key to encrypt a number of wrapped keys, which can be used to encrypt or decrypt VMs. The use of wrapping keys helps conserve key capacity in the KMS for use cases where key turnover is high, such as VDI use cases. This option is on by default. For more information see Wrapping Keys.
Autogenerated	Optional wrapping key configuration. Select this option to use wrapping keys generated by the KMS. When you select this option, you can also configure a **Rotation interval**. This specifies the time interval at which the KMS will replace existing wrapping keys with newly generated keys, and may be between 30 and 3650 days. The default is 365 days.
Custom	Optional wrapping key configuration. Select this option to specify a custom key ID for the wrapping key. Enter the wrapping key ID in the **Wrapping Key ID** text box. Automatic rotation of keys is not available when you use a custom wrapping key. To rotate the key, enter a new wrapping key in the **Wrapping Key ID** text box.
Proxy server	Optional proxy server address for connecting to the key server.
Proxy port	Optional proxy port for connecting to the key server.
Username	Some key server vendors allow users to isolate encryption keys that are used by different users or groups by specifying a user name and password. Specify a user name only if your key server supports this functionality, and if you intend to use it.
Password	Some key server vendors allow users to isolate encryption keys that are used by different users or groups by specifying a user name and password. Specify a password only if your key server supports this functionality, and if you intend to use it.

   You can click **Add KMS** to add more key servers.
5. Click **Add Key Provider**.
6. Click **Trust**.
   vCenter adds the key provider and displays the status as Connected.

See Establish a Standard Key Provider Trusted Connection by Exchanging Certificates.

Establish a Standard Key Provider Trusted Connection by Exchanging Certificates

After you add the standard key provider to the vCenter system, you can establish a trusted connection. The exact process depends on the certificates that the key provider accepts, and on your company policy.

Add the standard key provider.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider.
   The KMS for the key provider is displayed.
4. Select the KMS.
5. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
6. Select the option appropriate for your server and follow the steps.

Option	See
vCenter Root CA certificate	Use the Root CA Certificate Option to Establish a Standard Key Provider Trusted Connection.
vCenter Certificate	Use the Certificate Option to Establish a Standard Key Provider Trusted Connection.
Upload certificate and private key	Use the Upload Certificate and Private Key Option to Establish a Standard Key Provider Trusted Connection.
New Certificate Signing Request	Use the New Certificate Signing Request Option to Establish a Standard Key Provider Trusted Connection.

Use the Root CA Certificate Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload your root CA certificate to the KMS. All certificates that are signed by your root CA are then trusted by this KMS.

The root CA certificate that vSphere Virtual Machine Encryption uses is a self-signed certificate that is stored in a separate store in the VMware Endpoint Certificate Store (VECS) on the vCenter system.

> **Note:**
> Generate a root CA certificate only if you want to replace existing certificates. If you do, other certificates that are signed by that root CA become invalid. You can generate a new root CA certificate as part of this workflow.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider with which you want to establish a trusted connection.
   The key server (KMS) for the key provider is displayed.
4. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
5. Select **vCenter Root CA Certificate** and click **Next**.
   The Download Root CA Certificate dialog box is populated with the root certificate that vCenter uses for encryption. This certificate is stored in VECS.
6. Copy the certificate to the clipboard or download the certificate as a file.
7. Follow the instructions from your KMS vendor to upload the certificate to their system.

> **Note:**
> Some KMS vendors require that the KMS vendor restarts the KMS to pick up the root certificate that you upload.

Finalize the certificate exchange. See Finish the Trust Setup for a Standard Key Provider.

Use the Certificate Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload the vCenter certificate to the KMS. After the upload, the KMS accepts traffic that comes from a system with that certificate.

vCenter generates a certificate to protect connections with the KMS. The certificate is stored in a separate key store in the VMware Endpoint Certificate Store (VECS) on the vCenter system.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider with which you want to establish a trusted connection.
   The key server (KMS) for the key provider is displayed.
4. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
5. Select **vCenter Certificate** and click **Next**.
   The Download Certificate dialog box is populated with the root certificate that vCenter uses for encryption. This certificate is stored in VECS.

   Note:
   Do not generate a new certificate unless you want to replace existing certificates.
6. Copy the certificate to the clipboard or download it as a file.
7. Follow the instructions from your KMS vendor to upload the certificate to the KMS.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Use the Upload Certificate and Private Key Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload the KMS server certificate and private key to the vCenter system.

- Request a certificate and private key from the KMS vendor. The files are X509 files in PEM format.

Some KMS vendors generate a certificate and private key for the connection and make them available to you. After you upload the files, the KMS trusts your vCenter instance.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider with which you want to establish a trusted connection.
   The key server (KMS) for the key provider is displayed.
4. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
5. Select **KMS certificate and private key** and click **Next**.
6. Paste the certificate that you received from the KMS vendor into the top text box or click **Upload a File** to upload the certificate file.
7. Paste the key file into the bottom text box or click **Upload a File** to upload the key file.
8. Click **Establish Trust**.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Use the New Certificate Signing Request Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that vCenter generate a Certificate Signing Request (CSR) and send that CSR to the KMS. The KMS signs the CSR and returns the signed certificate. You can upload the signed certificate to vCenter.

Using the **New Certificate Signing Request** option is a two-step process. First you generate the CSR and send it to the KMS vendor. Then you upload the signed certificate that you receive from the KMS vendor to vCenter.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider with which you want to establish a trusted connection.
   The key server (KMS) for the key provider is displayed.
4. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
5. Select **New Certificate Signing Request (CSR)** and click **Next**.
6. In the dialog box, copy the full certificate in the text box to the clipboard or download it as a file.
   Use the **Generate new CSR** button in the dialog box only if you explicitly want to generate a CSR.
7. Follow the instructions from your KMS vendor to submit the CSR.
8. When you receive the signed certificate from the KMS vendor, click **Key Providers** again, select the key provider, and from the **Establish Trust** drop-down menu, select **Upload Signed CSR Certificate**.
9. Paste the signed certificate into the bottom text box or click **Upload File** and upload the file, and click **Upload**.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Finish the Trust Setup for a Standard Key Provider

Unless the **Add Standard Key Provider** dialog box prompted you to trust the KMS, you must explicitly establish trust after certificate exchange is complete.

You can complete the trust setup, that is, make vCenter trust the KMS, either by trusting the KMS or by uploading a KMS certificate. You have two options:

- Trust the certificate explicitly by using the **Upload KMS certificate** option.
- Upload a KMS leaf certificate or the KMS CA certificate to vCenter by using the **Make vCenter Trust KMS** option.

> **Note:**
> If you upload the root CA certificate or the intermediate CA certificate, vCenter trusts all certificates that are signed by that CA. For strong security, upload a leaf certificate or an intermediate CA certificate that the KMS vendor controls.

1. Navigate to the vCenter.
2. Click **Configure** and select **Key Providers** under **Security**.
3. Select the key provider with which you want to establish a trusted connection.
   The key server (KMS) for the key provider is displayed.
4. Select the KMS.
5. Select one of the following options from the **Establish Trust** drop-down menu.

Option	Action
Make vCenter Trust KMS	In the dialog box that appears, click **Trust**.
Upload KMS certificate	1. In the dialog box that appears, either paste in the certificate, or click **Upload a file** and browse to the certificate file. 2. Click **Upload**.

Set Up Separate Key Providers for Different Users

You can set up your environment with different key providers for different users of the same KMS instance. Having multiple key providers is helpful, for example, if you want to grant different departments in your company access to different sets of encryption keys.

Set up the connection with the KMS.

You can use multiple key providers for the same KMS to separate keys. Having separate sets of keys is essential for use cases like different BUs or different customers.

Note:
Not all KMS vendors support multiple users.

1. Create two users with corresponding user names and passwords, for example C1 and C2, on the KMS.
2. Log in to vCenter and create the first key provider.
3. When prompted for a user name and password, give information that is unique to the first user.
4. Create a second key provider and add the same KMS, but use the second user name and password (C2).

The two key providers have independent connections to the KMS and use a different set of keys.

Delete a Standard Key Provider

You can use the vSphere Client to delete a standard key provider from vCenter.

Required privilege: **Cryptographic operations** > **Manage key servers**

Before you delete a standard key provider, rekey any encrypted virtual machines and datastores that were encrypted using that key provider to another key provider. See Rekey an Encrypted Virtual Machine Using the vSphere Client.

In addition, maintain a backup of the standard key provider in case you must rekey an encrypted virtual machine after deleting the key provider.

After you delete a standard key provider, virtual machines that have vTPMs or that are encrypted continue to run. If you reboot the ESX host, its encrypted virtual machines enter a locked state. After you unregister these virtual machines, they enter a locked state when you try to re-register them. The only way to unlock the virtual machines is to restore the previous standard key provider.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Select the standard key provider you want to delete.
5. Click **Delete**.
6. Read the warning message and slide the slider all the way to the right.
7. Click **Delete**.

The standard key provider is removed from the vCenter.

Configuring and Managing vSphere Native Key Provider

Using a VMware vSphere® Native Key Provider™ in your vSphere environment requires some preparation. After you configure vSphere Native Key Provider, you can create virtual Trusted Platform Modules (vTPMs) on your virtual machines.

After you set up your environment for vSphere Native Key Provider, you can use the vSphere Client and API to create vTPMs. You can also encrypt virtual machines and virtual disks, and encrypt existing virtual machines and disks.

vSphere Native Key Provider Overview

In vSphere 7.0 Update 2 and later, you can use the built-in vSphere Native Key Provider to enable encryption technologies, such as virtual TPMs (vTPM).

vSphere Native Key Provider is included in all vSphere editions and does not require an external key server (also called a Key Management Server (KMS) in the industry). You can also use vSphere Native Key Provider for vSphere Virtual Machine Encryption.

What Is vSphere Native Key Provider

With a standard key provide, you must configure an external key server. In a standard key provider setup, vCenter fetches the keys from the external key server and distributes them to the ESX hosts.

With vSphere Native Key Provider, you no longer need an external key server. vCenter generates a primary key, called the Key Derivation Key (KDK), and pushes it to all ESX hosts in the cluster. The ESX hosts then generate data encryption keys (even when not connected to vCenter) to enable security functionality such as vTPMs. vTPM functionality is included in all vSphere editions.

vSphere Native Key Provider:

- Enables the use of vTPMs, vSphere Virtual Machine Encryption, and vSAN Data at Rest Encryption, when you do not require or want an external key server.
- Works only with VMware infrastructure products.
- Does not provide external interoperability, KMIP support, hardware security modules, or other features that a traditional, third-party, external key server can offer for interoperability or regulatory compliance. If your organization requires this functionality for non-VMware products and components, install a traditional, third-party key server.
- Helps address the needs of organizations that either cannot use, or do not want not to use, an external key server.
- Improves data sanitization and system reuse practices by enabling earlier use of encryption technologies on media that is difficult to sanitize, such as flash and SSD.
- Provides a transition path between key providers. vSphere Native Key Provider is compatible with the VMware standard key provider.
- Works with multiple vCenter systems using an Enhanced Linked Mode configuration or a vCenter High Availability configuration.
- Can be used to enable vTPM in all editions of vSphere, and encrypt virtual machines with vSphere Virtual Machine Encryption. vSphere Virtual Machine Encryption works with vSphere Native Key Provider as it does with VMware standard key providers.
- Can be used to enable vSAN Data at Rest Encryption with the use of an appropriate vSAN license.
- Can use a Trusted Platform Module (TPM) 2.0 to increase security when one is installed in an ESX host. You can also configure vSphere Native Key Provider to be available only to hosts where a TPM 2.0 is installed. If you use a TPM, it must be TPM 2.0. vSphere Native Key Provider does not support TPM 1.2.

  Note:
  An ESX host does not require a TPM 2.0 to use a vSphere Native Key Provider. However, a TPM 2.0 does provide enhanced security.

As with all security solutions, consider the system design, implementation considerations, and tradeoffs of using Native Key Provider. For example, ESX key persistence avoids the dependency on a key server always being available. However, because key persistence stores the Native Key Provider cryptographic information on the clustered hosts, you still are at risk if malicious actors steal the ESX hosts themselves. Because environments differ, assess and implement your security controls in accordance with your organization’s regulatory and security needs, operational requirements, and tolerance for risk.

For more overview information about vSphere Native Key Provider, see https://core.vmware.com/native-key-provider.

vSphere Native Key Provider Requirements

To use vSphere Native Key Provider, you must:

- Ensure that both the vCenter system and ESX hosts are running vSphere 7.0 Update 2 or later.

- Configure the ESX hosts in a cluster.
- While not required, as a best practice, use ESX hosts that are as identical as possible, including TPMs. Cluster management and feature enablement is much easier when cluster hosts are identical.
- Configure the vCenter file-based backup and restore, and store the backups securely as they contain the Key Derivation Key. See the topic on vCenter backup and restore in the *vCenter Deployment and Setup* documentation.

To perform vSphere Virtual Machine Encryption or vSAN encryption using vSphere Native Key Provider, you must purchase the edition of those products containing the appropriate license.

vSphere Native Key Provider and Enhanced Linked Mode

Note:

In vSphere 9.0, Enhanced Linked Mode is deprecated and will be removed in a future release.

You can configure a single vSphere Native Key Provider that is shareable across vCenter systems configured in an Enhanced Linked Mode configuration. The high-level steps in this scenario are:

1. Creating the vSphere Native Key Provider on one of the vCenter systems
2. Backing up the Native Key Provider on the vCenter on which it was created
3. Exporting the Native Key Provider
4. Restoring the Native Key Provider to the other vCenter systems in the Enhanced Link Mode configuration (see Restore a vSphere Native Key Provider Using the vSphere Client)

vSphere Native Key Provider Privileges

As with standard key providers, vSphere Native Key Provider uses the **Cryptographer.*** privileges. In addition, vSphere Native Key Provider uses the **Cryptographer.ReadKeyServersInfo** privilege, which is specific to vSphere Native Key Providers, to list vSphere Native Key Providers. See Cryptographic Operations Privileges.

vSphere Native Key Provider Alarms

You must back up a vSphere Native Key Provider. When a vSphere Native Key Provider is not backed up, vCenter generates an alarm. When you back up the vSphere Native Key Provider for which an alarm was generated, vCenter resets the alarm. By default, vCenter checks for backed-up vSphere Native Key Providers once a day. You can change the checking interval by modifying the `vpxd.KMS.backupCheckInterval` option.

vSphere Native Key Provider Periodic Remediation Check

vCenter checks periodically that the vSphere Native Key Provider configuration on vCenter and ESX hosts matches. When a host state changes, for example, when you add a host to the cluster, the key provider configuration on the cluster drifts away from the configuration on the host. If the configuration (keyID) differs on the host, vCenter updates the configuration of the host automatically. No manual intervention is required.

By default, vCenter checks the configuration every five minutes. You can modify the interval by using the `vpxd.KMS.remediationInterval` option.

Using vSphere Native Key Provider with a Disaster Recovery Site

You can use vSphere Native Key Provider with a backup disaster recovery site. Importing the vSphere Native Key Provider backup from the primary vCenter to the vCenter backup at the disaster recovery site enables that cluster to decrypt and run your encrypted virtual machines.

Always test your DR solution. Never assume that your solution works without trying a recovery. Ensure that a copy of the vSphere Native Key Provider backup is also available to your DR site.

Unsupported Features in vSphere Native Key Provider

vSphere Native Key Provider does not support First Class Disk (FCD) encryption in vSphere 8.0 through vSphere 8.0 Update 2. Starting in vSphere 8.0 Update 3, vSphere Native Key Provider does support FCD encryption.

Migrating Virtual Machines Using vSphere Native Key Provider Across Non-Linked vCenter Systems

The high-level steps to migrate a virtual machine, either encrypted or enabled with a vTPM through vSphere Native Key Provider, from one non-linked vCenter system to another, include:

1. Restoring the vSphere Native Key Provider to the to-be-migrated-to vCenter system.
2. Migrating the virtual machine by using vMotion.

vSphere Native Key Provider Process Flows

Understanding vSphere Native Key Provider process flows is essential to learning how to configure and manage your vSphere Native Key Provider.

You can use the built-in vSphere Native Key Provider to power encryption-based virtual TPMs (vTPM). vSphere Native Key Provider is included in all vSphere editions and does not require an external key server (KMS).

Configuring vSphere Native Key Provider

Configuring vSphere Native Key Provider involves these basic operations:

1. A user with the appropriate administrative privileges uses the vSphere Client to create a vSphere Native Key Provider on a vCenter.
2. The vCenter then configures the vSphere Native Key Provider for all clusters of ESX hosts.
   In this step, vCenter pushes a primary key to all ESX hosts in the cluster. Likewise, if you update or delete a vSphere Native Key Provider, the change is pushed to the hosts in the cluster.
3. Users with the appropriate cryptographic privileges create vTPMs and encrypted virtual machines.
   See Using Encryption in Your vSphere Environment and Securing Virtual Machines with Virtual Trusted Platform Module.

vSphere Native Key Provider Encryption Process Flow

To understand how different components interact to perform an encryption task using vSphere Native Key Provider, see vSphere Native Key Provider Encryption Process Flow.

Configure a vSphere Native Key Provider

A key provider is required for performing encryption tasks. You can use the vSphere Client to configure a vSphere Native Key Provider on vCenter.

Required privilege: **Cryptographic operations** > **Manage key servers**

vSphere 7.0 Update 2 and later includes a key provider called vSphere Native Key Provider. vSphere Native Key Provider enables encryption-related functionality without requiring an external key server (KMS). Initially, vCenter is not configured with a vSphere Native Key Provider. You must manually configure a vSphere Native Key Provider.

An ESX host does not require a TPM 2.0 to use a vSphere Native Key Provider. However, a TPM 2.0 does provide enhanced security.

> **Note:**
> When you configure vSphere Native Key Provider, the key providers are available on all clusters for the vCenter on which you configure them. As a result, all hosts attached to the vCenter have access to all the vSphere Native Key Providers that you configure.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Click **Add** then click **Add Native Key Provider**.
5. Enter a name for the vSphere Native Key Provider.

   Each logical key provider, regardless of its type (Standard, Trusted, and Native Key Provider), must have a unique name across all vCenter systems.

   For more information, see Key Provider Naming.
6. If you want this vSphere Native Key Provider to be used only by hosts with a TPM 2.0, select the **Use key provider only with TPM protected ESXi hosts** check box.

   If enabled, the vSphere Native Key Provider is available only on hosts with a TPM 2.0.
7. Click **Add Key Provider**.

   Note:
   It takes about five minutes for all the clustered ESX hosts in a data center to get the key provider, and for the vCenter to update its cache. Because of the way the information is propagated, you might have to wait for a few minutes to use the key provider for key operations on some of the hosts.

The vSphere Native Key Provider is added and appears in the **Key Provider** pane. At this point, the vSphere Native Key Provider is not backed up. You must back up the vSphere Native Key Provider before you can use it.

See Back up a vSphere Native Key Provider.

Back up a vSphere Native Key Provider

In case you must restore the key provider configuration, backing up a vSphere Native Key Provider is required as part of a disaster recovery scenario. You can use the vSphere Client, PowerCLI, or API to back up the vSphere Native Key Provider.

Required privilege: **Cryptographic operations** > **Manage key servers**

Note:
In an Enhanced Link Mode configuration, you must perform the backup on the vCenter that the key provider belongs to.

vSphere Native Key Provider is backed up as part of the vCenter file-based backup. However, you must back up the vSphere Native Key Provider at least once before you can use it. When you create a vSphere Native Key Provider, it is not backed up.

A backup is necessary in case you must restore the configuration. To restore a vSphere Native Key Provider, see Restore a vSphere Native Key Provider Using the vSphere Client.

Keep the backup file in a secure location. You can password-protect the backup when you create it. The backup file is in PKCS#12 format.

vCenter creates an alarm if a vSphere Native Key Provider has not been backed up. You can acknowledge the alarm, but it reappears every 24 hours until you have backed up the vSphere Native Key Provider.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Select the vSphere Native Key Provider you want to back up.

   A status of "Not backed up" appears for key providers that you have not backed up.

5. Click **Back Up**.
6. To password-protect the backup, check the **Protect Native Key Provider data with password** box.
   a) Enter a password and save it in a secure location.
   b) Check the **I have saved the password in a secure place** box, indicating that you have saved the password to a secure place.
7. Click **Back Up Key Provider**.
   The backup file is in PKCS#12 format.
8. Save the backup file in a secure location.

The status of the vSphere Native Key Provider changes from Not Backed Up, to Warning, to Active. Warning indicates that the vCenter is still pushing the information to all the ESX hosts in the data center. Active means that the information has been pushed to all the hosts.

To add vTPMs to virtual machines, see Securing Virtual Machines with Virtual Trusted Platform Module. To encrypt virtual machines, see Using Encryption in Your vSphere Environment.

Recovering a vSphere Native Key Provider

You can recover the vSphere Native Key Provider either through the vSphere Client or from the vCenter appliance Backup.

When necessary, you can recover a vSphere Native Key Provider in the following ways.

1. If you do not need to rebuild your vCenter appliance, use the vSphere Client to restore the key provider. See Restore a vSphere Native Key Provider Using the vSphere Client.
2. If you must rebuild your vCenter appliance, you must restore the key provider from your vCenter appliance Backup. When you perform a vCenter appliance Backup, it saves the Native Key Provider. See https://blogs.vmware.com/vsphere/2018/05/vcenter-server-appliance-6-7-file-based-backup-and-restore-walkthroughs.html for information about restoring the vCenter appliance from backup.

Restore a vSphere Native Key Provider Using the vSphere Client

You can use the vSphere Client to restore the vSphere Native Key Provider.

- Required privilege: **Cryptographic operations** > **Manage key servers**
- The key provider backup file.
- The password for the key provider file, if you entered one when you backed up the key provider.

You can restore a vSphere Native Key Provider in case it was accidentally deleted or if you must perform a disaster recovery.

When you restore a vSphere Native Key Provider, you do not need to back up the key provider again. The initial backup suffices. Continue to maintain the backup file in a secure location.

> **Note:**
> You can also use this task to configure vSphere Native Key Provider for vCenter systems in an Enhanced Linked Mode configuration. After you create the vSphere Native Key Provider on one vCenter system in the Enhanced Linked Mode configuration, use the **Restore** function to import the encrypted key file to the other ELM-connected vCenter systems.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Select the vSphere Native Key Provider and click **Restore**.
5. Browse to the file location and select the backup encrypted key file.
   The file was saved in PKCS#12 format.

6. Optional: If the file is password protected, enter the password.
7. Click **Next**.
8. Optional: If you decided to use this key provider only with TPM-protected ESXi hosts, select the check box.
9. Click **Finish**.

The vSphere Native Key Provider is imported to the vCenter. To use the vSphere Native Key Provider for encryption tasks, ensure that you first select it in the **Key Provider** pane and click **Set as Default**.

Update a vSphere Native Key Provider

As part of your regular key rotation plans, you can use PowerCLI to update a vSphere Native Key Provider.

- Required privilege: **Cryptographic operations** > **Manage key servers**
- PowerCLI 12.3.0

If you have a policy for key rotation, you can update the vSphere Native Key Provider and rekey the virtual machines that you encrypted with that key provider. You must use PowerCLI to update the vSphere Native Key Provider. You can also rekey the encrypted virtual machines without updating the key provider. In this case, only the virtual machine keys are changed. To rekey a virtual machine, see Rekey an Encrypted Virtual Machine Using the vSphere Client.

1. In a PowerCLI session, run the `Connect-VIServer` cmdlet to connect as an administrator user to the vCenter where you configured the vSphere Native Key Provider that you want to update.

   ```
   Connect-VIServer -server VC_ip_address -User admin_user -Password 'password'
   ```

2. To get your vSphere Native Key Provider names, run the `Get-KeyProvider` cmdlet with the optional Type parameter.

   ```
   Get-KeyProvider -Type NativeKeyProvider
   ```

3. To update the key provider, run the `Set-KeyProvider` cmdlet, specifying your key provider name and GUID.
   You can generate a GUID to use by running the `New-Guid` cmdlet.

   ```
   Set-KeyProvider -KeyProvider KeyProvider_name -KeyId Guid
   ```

   A warning appears about backing up the configuration.

4. To back up the key provider, run the `Export-KeyProvider` cmdlet.

   ```
   Export-KeyProvider -KeyProvider KeyProvider_name -FilePath path_file_name
   ```

   You can also back up the key provider using the vSphere Client. See Back up a vSphere Native Key Provider.

When a key provider is updated, its status changes to Not Backed Up. After you back up the key provider, its status changes to Active.

Delete a vSphere Native Key Provider

You can use the vSphere Client to delete a vSphere Native Key Provider from vCenter.

Required privilege: **Cryptographic operations** > **Manage key servers**

Before you delete a vSphere Native Key Provider, rekey any encrypted virtual machines and datastores that were encrypted using that key provider to another key provider. See Rekey an Encrypted Virtual Machine Using the vSphere Client.

In addition, maintain a backup of the vSphere Native Key Provider in case you must rekey an encrypted virtual machine after deleting the key provider.

After you delete a vSphere Native Key Provider, virtual machines that have vTPMs or that are encrypted continue to run. If you reboot the ESX host, its encrypted virtual machines enter a locked state. After you unregister these virtual machines, they enter a locked state when you try to re-register them. The only way to unlock the virtual machines is to restore the previous vSphere Native Key Provider.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the vCenter instance.
3. Click **Configure**, and under **Security** click **Key Providers**.
4. Select the key provider you want to delete.
5. Click **Delete**.
6. Read the warning message and slide the slider all the way to the right.
7. Click **Delete**.

The vSphere Native Key Provider is removed from the vCenter.

Using Encryption in Your vSphere Environment

Whether you use a standard key provider or vSphere Native Key Provider, using encryption in your vSphere environment requires some preparation.

See the following information to set up your environment to use a key provider:

- Configuring and Managing a Standard Key Provider
- Configuring and Managing vSphere Native Key Provider

After your environment is set up, you can use the vSphere Client to create encrypted virtual machines and virtual disks and encrypt existing virtual machines and disks.

You can perform additional tasks by using the API and by using the `crypto-util` CLI. See the *vSphere Web Services SDK Programming Guide* for the API documentation and the `crypto-util` command-line help for details about that tool.

Create an Encryption Storage Policy

Before you can create encrypted virtual machines, you must create an encryption storage policy. You create the storage policy once, and assign it each time you encrypt a virtual machine or virtual disk.

- Set up the connection to a key provider.
  Although you can create a VM Encryption storage policy without the key provider connection in place, you cannot perform encryption tasks until trusted connection with the key provider is established.
- Required privileges: **Cryptographic operations** > **Manage encryption policies**.

If you want to use virtual machine encryption with other I/O filters, or to use the **Create VM Storage Policy** wizard in the vSphere Client, see the *vSphere Storage* documentation for details.

1. Log in to the vCenter by using the vSphere Client.
2. Select **Home**, click **Policies and Profiles**, then click **VM Storage Policies**.
3. Click **Create**.
4. Select the vCenter, enter a policy name, optionally enter a description, then click **Next**.
5. On the **Policy structure** page, check **Enable host based roles** then click **Next**.
6. On the **Host based services** page, select **Use storage policy component**, choose **Default encryption properties** from the drop-down menu, then click **Next**.
7. On the **Storage compatibility** page, leave **Compatible** selected, select a datastore, then click **Next**.
8. Review the information and click **Finish**.

The VM Encryption storage policy is added to the list, and is available for use when encrypting a virtual machine.

Activate Host Encryption Mode Explicitly

Host encryption mode must be set if you want to perform encryption tasks, such as creating an encrypted virtual machine, on an ESX host. In most cases, host encryption mode is activated automatically when you perform an encryption task.

Required privilege: **Cryptographic operations.Register host**

Sometimes, turning on encryption mode explicitly is necessary. See Prerequisites and Required Privileges for Virtual Machine Encryption Tasks.

1. Log in to the vCenter by using the vSphere Client.
2. Browse to the ESX host and click **Configure**.
3. Under System, click **Security Profile**.
4. Click **Edit** in the Host Encryption Mode panel.
5. Select **Enabled** and click **OK**.

Deactivate Host Encryption Mode Using the API

Host encryption mode is activated automatically when a user performs an encryption task, if the user has sufficient privilege. After host encryption mode is activated, all core dumps are encrypted to avoid the release of sensitive information to support personnel. If you no longer use virtual machine encryption with an ESX host, you can deactivate encryption mode.

After encryption mode is activated for an ESX host, you might need to deactivate it. For example, you might need to deactivate encryption mode to generate an ESX support bundle (using the `vm-support` command). Using the Host Encryption mode toggle (**Host** > **Configure** > **Security Profile** > **Edit Host Encryption Mode**) does not work when key material exists on the host.

You can use the API to deactivate host encryption mode by invoking the `CryptoManagerHostDisable` API method.

The crypto modes, or states, defined for an ESX host are:

- pendingIncapable: The host is crypto deactivated, that is, the host cannot perform vSphere Virtual Machine Encryption operations.
- incapable: The host is not safe for receiving sensitive material.
- prepared: The host is prepared for receiving sensitive material but does not have a host key set yet.
- safe: The host is crypto safe (activated), and has a host key set, that is, vSphere Virtual Machine Encryption operations are possible.

After you invoke `CryptoManagerHostDisable` on a host, the crypto state of the host changes as follows:

- If the original host crypto state is incapable or prepared, the host crypto state is changed to incapable.
- If the original host crypto state is safe, the host crypto state is changed to pendingIncapable.
- If the host crypto state is pendingIncapable, the host crypto state is still pendingIncapable.

This task shows how to deactivate host encryption mode by using the vCenter Managed Object Browser (MOB). For more information about using the API, see the *vSphere Web Services API* documentation at https://developer.vmware.com/apis/968/vsphere.

1. Log in to the vCenter as an administrator.
2. Unregister all encrypted virtual machines from the ESX host whose encryption mode you want to deactivate.
3. Access the MOB on the vCenter.

   `https://vcenter_server/mob`

   The MOB is disabled by default in vSphere 9.0. To enable it, see KB 401669.

4. Invoke the `CryptoManagerHostDisable` method on a host.
   a) Under content name, click **content**.
   b) Under rootFolder, click **group-D1 (Datacenters)**.
   c) Under childEntity, click the appropriate datacenter.
   d) Under hostFolder, click the appropriate host.
   e) Under childEntity, click the appropriate cluster.
   f) Under host, click the appropriate host.
   g) Under configManager, click **configManager**.
   h) Under cryptoManager, click **CryptoManagerHost-***number*.
   i) Click **CryptoManagerHostDisable**.
   The host crypto state is changed to either pendingIncapable or incapable, depending on its original crypto state.
5. Repeat step 4 for other hosts on which you want to deactivate encryption mode.
6. Reboot the hosts.

Once the host encryption mode is deactivated, you cannot perform encryption operations, such as adding encrypted virtual machines, unless you re-activate the host encryption mode.

Note:
After you reboot an ESX host on which you deactivated encryption mode, if the host crypto state was originally pendingIncapable, the host crypto state is still pendingIncapable. To re-activate host encryption mode, re-access the vCenter MOB and invoke the `ConfigureCryptoKey` API method. When re-activating host encryption mode, use the original host key ID if the host crypto state is pendingIncapable.

Create an Encrypted Virtual Machine

You an use the vSphere Client to create encrypted virtual machines.

- Configure a key provider and set it as the default.
- Create an encryption storage policy, or use the bundled sample, VM Encryption Policy.
- Ensure that the virtual machine is powered off.
- Verify that you have the required privileges:
  - **Cryptographic operations** > **Encrypt new**
  - If the host encryption mode is not Enabled, you also need**Cryptographic operations** > **Register host**.
  - **Profile-driven storage** > **Profile-driven storage update**
  - **Virtual machine** > **Change Configuration** > **Change Settings**

The vSphere Client filters by virtual machine encryption storage policies, easing creation of encrypted virtual machines.

Note:
Creating an encrypted virtual machine is faster and uses fewer storage resources than encrypting an existing virtual machine. If possible, encrypt virtual machines during the creation process.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Right-click the object and select **New Virtual Machine**.
4. Follow the prompts to create an encrypted virtual machine.

Option	Action
Select a creation type	Create a new virtual machine.
Select a name and folder	Specify a unique name and target location for the virtual machine.
Select a compute resource	Specify an object for which you have privileges to create encrypted virtual machines. See Prerequisites and Required Privileges for Virtual Machine Encryption Tasks.

Option	Action
Select storage	Select the **Encrypt this virtual machine** check box. Virtual machine storage policies that include encryption appear. Select a virtual machine storage policy (the bundled sample is VM Encryption Policy), and select a compatible datastore.
Select compatibility	Select the compatibility. You can migrate an encrypted virtual machine only to hosts with compatibility ESX 6.5 and later.
Select a guest OS	Select a guest OS that you plan to install on the virtual machine later.
Customize hardware	Customize the hardware, for example, by changing disk size or CPU. (Optional) Select the **VM Options** tab, and expand **Encryption**. Select which disks to exclude from encryption. When you deselect a disk, only the VM Home and any other selected disks are encrypted. Any New Hard disk that you add is encrypted. You can change the storage policy for individual hard disks later.
Ready to complete	Review the information and click **Finish**.

Clone an Encrypted Virtual Machine

A cloned, encrypted virtual machine is encrypted with the same keys unless you change them. To change keys, you can use the vSphere Client, the PowerCLI, or the API. If you use the PowerCLI or the API, you can clone the encrypted virtual machine and change keys in one step.

- A key provider must be configured and enabled.
- Create an encryption storage policy, or use the bundled sample, VM Encryption Policy.
- Required privileges (applies to all key providers):
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Encrypt**
  - **Cryptographic operations** > **Decrypt**
  - **Cryptographic operations** > **Recrypt**
  - If the host encryption mode is not enabled, you also must have **Cryptographic operations** > **Register host** privileges.

You can perform the following operations during clone.

- Create an encrypted virtual machine from an unencrypted virtual machine or template virtual machine.
- Create a decrypted virtual machine from an encrypted virtual machine or template virtual machine.
- Recrypt the destination virtual machine with different keys from that of source virtual machine.
- In vSphere 8.0 and later, selecting the **Replace** option for a virtual machine with a vTPM starts with a new, blank vTPM, which gets its own secrets and identity.

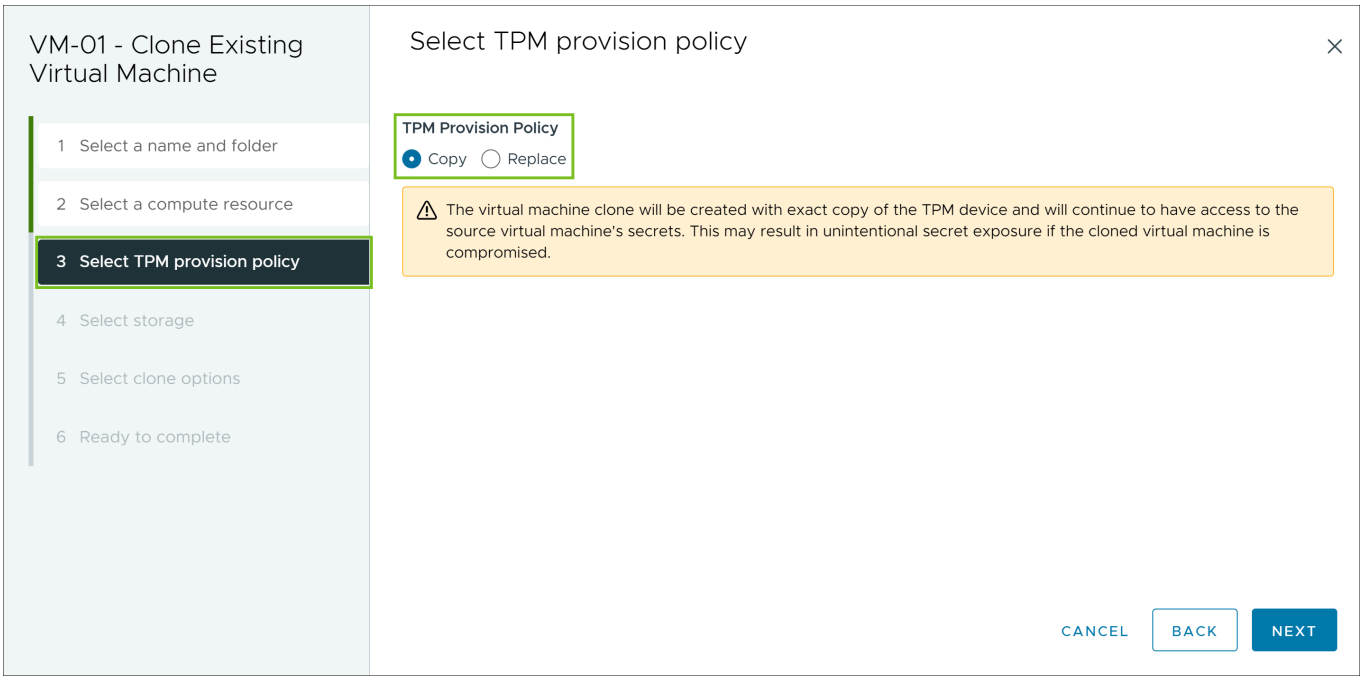
  Note:
  vSphere 8.0 and later includes the vpxd.clone.tpmProvisionPolicy advanced setting to make the default clone behavior for vTPMs to be "replace."

You can create an instant clone virtual machine from an encrypted virtual machine with the caution that the instant clone shares the same key with the source virtual machine. You cannot recrypt keys on either the source or the instant clone virtual machine.

To use the API to clone encrypted machines, see *vSphere Web Services SDK Programming Guide*.

1. Browse to the virtual machine in the vSphere Client inventory.
2. To create a clone of an encrypted machine, right-click the virtual machine, select **Clone** > **Clone to Virtual Machine**, and follow the prompts.
   a) On the **Select a name and folder** page, specify a name and the target location for the clone.
   b) On the **Select a compute resource** page, specify an object for which you have privileges.
   c) Optional: Change the keys for the cloned vTPM.

   Figure 95: Select TPM Provision Policy



   Cloning a virtual machine duplicates the entire virtual machine, including the vTPM and its secrets, which can be used to determine a system's identity. To change secrets on a vTPM, select **Replace** for **TPM Provision Policy**.

   Note:
   When you replace the secrets of a vTPM, all keys, including workload-related keys, are replaced. As a best practice, ensure that your workloads no longer use a vTPM before you replace the keys. Otherwise, the workloads in the cloned virtual machine might not function correctly.

   d) On the **Select storage** page, select a datastore. You can change the storage policy as part of the clone operation. For example, changing from using an encryption policy to a non-encryption policy decrypts the disks.
   e) On the **Select clone options** page, select clone options, as discussed in the *vSphere Virtual Machine Administration* documentation.
   f) On the **Ready to complete** page, review the information and click **Finish**.
3. Optional: Change the keys for the cloned virtual machine.

   By default, the cloned virtual machine is created with the same keys as its parent. Best practice is to change the keys of the cloned virtual machine to ensure that multiple virtual machines do not have the same keys.

   a) Decide upon a shallow or deep recrypt.

      To use a different DEK and KEK, perform a deep recrypt of the cloned virtual machine. To use a different KEK, perform a shallow recrypt of the cloned virtual machine. For a deep recrypt, you must power off the virtual machine. You can perform a shallow recrypt operation while the virtual machine is powered on, and if the virtual machine has snapshots present. Shallow recrypt of an encrypted virtual machine with snapshots is permitted only on a single snapshot branch (disk chain). Multiple snapshot branches are not supported. If the shallow recrypt fails before updating all links in the chain with the new KEK, you can still access the encrypted virtual machine if you have the old and new KEKs.

b) Perform a recrypt of the clone using the API. See *vSphere Web Services SDK Programming Guide*.

Encrypt an Existing Virtual Machine or Virtual Disk

You can encrypt an existing virtual machine or virtual disk by changing its storage policy. You can encrypt virtual disks only for encrypted virtual machines.

- Power off the virtual machine.
- Verify that you have the **Cryptographic operations** > **Manage encryption policies** privilege.

1. Right-click the virtual machine that you want to change and select **VM Policies** > **Edit VM Storage Policies**.

   You can set the storage policy for the virtual machine files, represented by VM home, and the storage policy for virtual disks.

2. Select the storage policy.

   - To encrypt the VM and its hard disks, select an encryption storage policy and click **OK**.
   - To encrypt the VM but not the virtual disks, toggle on **Configure per disk**, select the encryption storage policy for VM Home and other storage policies for the virtual disks, and click **OK**.

   You cannot encrypt the virtual disk of an unencrypted virtual machine. However, if you use the vSphere Client to encrypt the VM Home files, you can then reconfigure the unencrypted virtual machine with the encrypted disk.

3. If you prefer, you can encrypt the virtual machine, or both virtual machine and disks, from the **Edit Settings** menu in the vSphere Client.
   a) Right-click the virtual machine and select **Edit Settings**.
   b) Select the **VM Options** tab, and expand **Encryption**.
   c) Toggle on **Encrypt VM**.
   d) Choose an encryption policy. If you deselect all disks, only the VM home is encrypted.
   e) Click **OK**.

Decrypt an Encrypted Virtual Machine or Virtual Disk

You can decrypt a virtual machine, its disks, or both, by changing the storage policy.

- The virtual machine must be encrypted.
- The virtual machine must be powered off or in maintenance mode.
- Required privileges:
  - **Cryptographic operations** > **Decrypt**
  - **Cryptographic operations** > **Manage encryption policies**
  - **Virtual Machine** > **Change Configuration** > **Change settings**

This task describes how to decrypt an encrypted virtual machine using the vSphere Client.

All encrypted virtual machines require encrypted vMotion. During virtual machine decryption, the Encrypted vMotion setting remains. To change this setting so that Encrypted vMotion is no longer used, change the setting explicitly.

This task explains how to perform decryption using storage policies. For virtual disks, you can also perform decryption using the **Edit Settings** menu.

> **Note:**
> In the Virtual Machine Details pane, a vTPM-enabled virtual machine displays both a lock icon and an "Encrypted with *key_provider*" message. To remove a vTPM from a virtual machine, see Remove Virtual Trusted Platform Module from a Virtual Machine.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine that you want to change and select **VM Policies** > **Edit VM Storage Policies**.

   You can set the storage policy for the virtual machine files, represented by VM home, and the storage policy for virtual disks.

3. Select a storage policy.
   - To decrypt the VM and its hard disks, toggle off **Configure per disk**, select a storage policy from the drop-down menu, and click **OK**.
   - To decrypt a virtual disk but not the virtual machine, toggle on **Configure per disk**, select the encryption storage policy for VM Home and other storage policies for the virtual disks, and click **OK**.

   You cannot decrypt the virtual machine and leave the disk encrypted.
4. If you prefer, you can use the vSphere Client to decrypt the virtual machine and disks from the **Edit Settings** menu.
   a) Right-click the virtual machine and select **Edit Settings**.
   b) Select the **VM Options** tab and expand **Encryption**.
   c) To decrypt the VM and its hard disks, toggle off **Encrypt VM**.
   d) Click **OK**.
5. Optional: You can change the Encrypted vMotion setting.
   a) Right-click the virtual machine and click **Edit Settings**.
   b) Click **VM Options**, and expand **Encryption**.
   c) Set the **Encrypted vMotion** value.

Change the Encryption Policy for Virtual Disks

When you create an encrypted virtual machine from the vSphere Client, you can select which virtual disks that you add during virtual machine creation are encrypted. You can decrypt virtual disks that are encrypted by using the **Edit VM Storage Policies** option.

- You must have the **Cryptographic operations** > **Manage encryption policies** privilege.
- Ensure that the virtual machine is powered off.

> **Note:**
> An encrypted virtual machine can have virtual disks that are not encrypted. However, an unencrypted virtual machine cannot have encrypted virtual disks.

See Virtual Disk Encryption.

This task describes how to change the encryption policy using storage policies. You can also use the **Edit Settings** menu to make this change.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine and select **VM Policies** > **Edit VM Storage Policies** .
3. Change the storage policy.
   - To change the storage policy for the VM and its hard disks, select an encryption storage policy and click **OK**.
   - To encrypt the VM but not the virtual disks, toggle on **Configure per disk**, select the encryption storage policy for VM Home and other storage policies for the virtual disks, and click **OK**.

   You cannot encrypt the virtual disk of an unencrypted VM.
4. If you prefer, you can change the storage policy from the **Edit Settings** menu.
   a) Right-click the virtual machine and select **Edit Settings**.
   b) Select the **Virtual Hardware** tab, expand a hard disk, and select an encryption policy from the drop-down menu.
   c) Click **OK**.

Resolve Missing Encryption Key Issues

If the ESX host cannot get the key (KEK) from vCenter for an encrypted virtual machine or an encrypted virtual disk, the encrypted virtual machine becomes locked. After you make the keys available on the key server (KMS), you can unlock a locked encrypted virtual machine.

Under certain circumstances when using a standard key provider, the ESX host cannot get the key encryption key (KEK) for an encrypted virtual machine or an encrypted virtual disk from vCenter. In that case, you can still unregister or reload the virtual machine. However, you cannot perform other virtual machine operations such as powering on the virtual

machine. After taking the necessary steps to make the required keys available on the key server, you can unlock a locked encrypted virtual machine by using the vSphere Client.

If the virtual machine key is not available, a vCenter alarm notifies you and the state of the virtual machine displays as invalid. The virtual machine cannot power on. If the virtual machine key is available, but a key for an encrypted disk is not available, the virtual machine state does not display as invalid. However, the virtual machine cannot power on and the following error results:

```
The disk [/path/to/the/disk.vmdk] is encrypted and a required key was not found.
```

Note:
The following procedure illustrates the situations that can cause a virtual machine to become locked, the corresponding alarms and event logs that appear, and what to do in each case.

1. If the problem is the connection between the vCenter system and the key server, vCenter generates a virtual machine alarm. Also, an error message appears in the event log.

   Restore the connection to the key server. When the key server and keys become available, unlock the locked virtual machines. See Unlock Locked Virtual Machines. You can also reboot the host and re-register the virtual machine to unlock it after restoring the connection.

   Losing the connection to the key server does not automatically lock the virtual machine. The virtual machine only enters a locked state if the following conditions are met:

   - The key is not available on the ESX host.
   - vCenter cannot retrieve keys from the key server.

   After each ESX reboot, it is desirable, though not necessary, to have vCenter started first. vCenter requests the key with the corresponding ID from the key server and makes it available to ESX.

   Note:
   In vSphere 7.0 Update 2 and later, you can persist encryption keys across ESX reboots. See vSphere Key Persistence on ESX Hosts.

   If, after restoring connection to the key provider, the virtual machine remains locked, see Unlock Locked Virtual Machines.

2. If the connection is restored, register the virtual machine. If an error results, or if the operation succeeds but the virtual machine is in a locked state, verify that you have the **Cryptographic operations** > **RegisterVM** privilege for the vCenter system.

   This privilege is not required for powering on an encrypted virtual machine if the key is available. This privilege is required for registering the virtual machine if the key has to be retrieved.

3. If the key is no longer available on the key server, vCenter generates a virtual machine alarm. Also, an error message appears in the event log.

   Ask the key server administrator to restore the key. You might encounter an inactive key if you are powering on a virtual machine that had been removed from the inventory and that had not been registered for a long time. It also happens if you reboot the ESX host, and the key server is not available.

   a) Retrieve the key ID by using the Managed Object Browser (MOB) or the vSphere API.

      Retrieve the `keyId` from `VirtualMachine.config.keyId.keyId`.

   b) Ask the key server administrator to reactivate the key that is associated with that key ID.

   c) After restoring the key, see Unlock Locked Virtual Machines.

   If the key can be restored on the key server, vCenter retrieves it and pushes it to the ESX host the next time it is needed.

4. If the key server is accessible and the ESX host is powered on, but the vCenter system is unavailable, follow these steps to unlock virtual machines.

   a) Restore the vCenter system, or set up a different vCenter system, then establish trust with the key server.

      You must use the same key provider name, but the key server IP address can be different.

   b) Reregister all virtual machines that are locked.

      The new vCenter instance retrieves the keys from the key server and the virtual machines are unlocked.

5. If the keys are missing only on the ESX host, vCenter generates a virtual machine alarm and the following message appears in the event log:

   ```
   Virtual machine is locked because keys are missing on host.
   ```

   The vCenter system can retrieve the missing keys from the key provider. No manual recovery of keys is required. See Unlock Locked Virtual Machines.

Unlock Locked Virtual Machines

A vCenter alarm notifies you when an encrypted virtual machine is in a locked state. You can unlock a locked encrypted virtual machine by using the vSphere Client after taking the necessary steps to make the required keys available on the key server.

- Verify that you have the required privileges: **Cryptographic operations.RegisterVM**
- Other privileges might be required for optional tasks such as enabling host encryption.
- Before unlocking a locked virtual machine, troubleshoot the cause of the lock and attempt to fix the problem manually. See Resolve Missing Encryption Key Issues.

1. Connect to vCenter by using the vSphere Client.
2. Navigate to the virtual machine's **Summary** tab.

   When a virtual machine is locked, the Virtual Machine Locked alarm appears.
3. Decide if you want to either acknowledge the alarm, or reset the alarm to green but not unlock the virtual machine now.

   When you click either **Acknowledge** or **Reset to green**, the alarm goes away, but the virtual machine remains locked until you unlock it.
4. Navigate to the **Monitor** tab of the virtual machine and click **Events**.

   The **Events** pane displays information about why the virtual machine is locked.
5. Perform suggested troubleshooting before you unlock the virtual machine.
6. Navigate to the **Summary** tab of the virtual machine.

   A Virtual Machine Locked Alarm appears.
7. Select **Unlock VM** from the **Actions** drop-down menu on the right side.

Resolve ESX Host Encryption Mode Issues

Under certain circumstances, the ESX host's encryption mode can become deactivated.

An ESX host requires that host encryption mode is activated if it contains any encrypted virtual machines. If the host detects it is missing its host key, or if the key provider is unavailable, the host might fail to activate the encryption mode. vCenter generates an alarm when the host encryption mode cannot be activated.

1. If the problem is the connection between the vCenter system and the key provider, an alarm is generated and an error message appears in the event log.

   You must restore the connection to the key provider that contains the encryption keys in question.
2. If keys are missing, an alarm is generated and an error message appears in the event log.

   You must ensure that the keys are present in the key provider. Consult the documentation for your key management vendor for information about restoring from backup.

If, after restoring connection to the key provider, or manually recovering keys to the key provider, the host's encryption mode remains deactivated, re-activate the host encryption mode. See Re-Activate ESX Host Encryption Mode.

Re-Activate ESX Host Encryption Mode

In vSphere 6.7 and later, a vCenter alarm notifies you when an ESX host's encryption mode has become deactivated. You can re-activate the host encryption mode if it has become deactivated.

- Verify that you have the required privileges: **Cryptographic operations** > **Register host**
- Before re-activating encryption mode, troubleshoot the cause and attempt to fix the problem manually.

1. Connect to vCenter by using the vSphere Client.
2. Navigate to the **Summary** tab for the ESX host.
   When the encryption mode is deactivated, the Host Requires Encryption Mode Enabled alarm appears.
3. Decide if you want to either acknowledge the alarm, or reset the alarm to green but not re-activate the host encryption mode now.
   When you click either **Acknowledge** or **Reset to green**, the alarm goes away, but the encryption mode for the host remains deactivated until you re-activate it.
4. Navigate to the **Monitor** tab for the ESX host and click **Events**.
   More information is displayed about why encryption mode is deactivated. Perform suggested troubleshooting before you re-activate the encryption mode.
5. On the **Summary** tab, click **Enable Host Encryption Mode** to re-activate host encryption.
   A message appears, warning that encryption key data is transmitted to the host.
6. Click **Yes**.

Set Key Server Certificate Expiration Threshold

By default, vCenter notifies you 30 days before your key server (KMS) certificates expire. You can change this default value.

Key server certificates have an expiration date. When the threshold for the expiration date is reached, an alarm notifies you.

vCenter and key servers exchange two types of certificates: server and client. The VMware Endpoint Certificate Store (VECS) on the vCenter system stores the server certificates and one client certificate per key provider. Because there are two certificate types, there are two alarms for each certificate type (one for client, one for server).

1. Log in to a vCenter system by using the vSphere Client.
2. Select the vCenter system in the object hierarchy.
3. Click **Configure**.
4. Under **Settings**, click **Advanced Settings**, and click **Edit Settings**.
5. Click the **Filter** icon and enter `vpxd.kmscert.threshold`, or scroll to the configuration parameter itself.
6. Enter your value in days and click **Save**.

vSphere Virtual Machine Encryption and Core Dumps

If your environment uses vSphere Virtual Machine Encryption, and if an error occurs on the ESX host, the resulting core dump is encrypted to protect customer data. Core dumps that are included in the vm-support package are also encrypted.

> **Note:**
> Core dumps can contain sensitive information. Follow your organization's data security and privacy policy when handling core dumps.

Core Dumps on ESX Hosts

When an ESX host, a user world, or a virtual machine fails, a core dump is generated, and the host reboots. If the ESXi host has encryption mode enabled, the core dump is encrypted using a key that is in the ESXi key cache. (Depending on the key provider in use, the key comes from an external key server, the Key Provider Service, or vCenter). See How vSphere Virtual Machine Encryption Protects Your Environment for background information.

When an ESX host is cryptographically "safe," and a core dump is generated, an event is created. The event indicates that a core dump occurred along with the following information: world name, occurring times, keyID of the key used to encrypt the core dump, and core dump filename. You can view the event in the Events viewer under **Tasks and Events** for the vCenter.

The following table shows encryption keys used for each core dump type, by vSphere release.

Table 219: Core Dump Encryption Keys

Core Dump Type	Encryption Key (ESX 6.7 and Later)
ESXi Kernel	Host Key
User World (hostd)	Host Key
Encrypted Virtual Machine (VM)	Virtual Machine Key

What you can do after an ESX host reboot depends on several factors.

- In most cases, the key provider attempts to push the key to the ESX host after reboot. If the operation is successful, you can generate the vm-support package and you can decrypt or re-encrypt the core dump. See Decrypt or Re-Encrypt an Encrypted Core Dump.
- If vCenter cannot connect to the ESX host, you might be able to retrieve the key. See Resolve Missing Encryption Key Issues.
- If the host used a custom key, and that key differs from the key that vCenter pushes to the host, you cannot manipulate the core dump. Avoid using custom keys.

Core Dumps and vm-support Packages

When you contact VMware Technical Support because of a serious error, your support representative usually asks you to generate a vm-support package. The package includes log files and other information, including core dumps. If your support representatives cannot resolve the issues by looking at log files and other information, they might ask you to decrypt the core dumps and make relevant information available. To protect sensitive information such as keys, follow your organization's security and privacy policy. See Collect a vm-support Package for an ESX Host That Uses Encryption.

Core Dumps on vCenter Systems

A core dump on a vCenter system is not encrypted. vCenter already contains potentially sensitive information. At the minimum, ensure that the vCenter is protected. See Securing vCenter Server Systems. You might also consider turning off core dumps for the vCenter system. Other information in log files can help determine the problem.

Collect a vm-support Package for an ESX Host That Uses Encryption

If host encryption mode is enabled for the ESX host, any core dumps in the `vm-support` package are encrypted. You can collect the package from the vSphere Client, and you can specify a password if you expect to decrypt the core dump later.

Inform your support representative that host encryption mode is enabled for the ESX host. Your support representative might ask you to decrypt core dumps and extract relevant information.

> **Note:**
> Core dumps can contain sensitive information. Follow your organization's security and privacy policy to protect sensitive information such as host keys.

The `vm-support` package includes log files, core dump files, and more.

1. Log in to the vCenter system with the vSphere Client.
2. Click **Hosts & Clusters**, and right-click the ESX host.
3. Select **Export System Logs**.
4. In the dialog box, select **Password for encrypted core dumps**, and specify and confirm a password.
5. Leave the defaults for other options or make changes if requested by VMware Technical Support, and click **Export Logs**.

   If you have not configured your browser to ask where to save files before downloading, the download starts. If you have configured your browser to ask where to save files, specify a location for the file.

6. If your support representative asked you to decrypt the core dump in the `vm-support` package, log in to any ESX host and follow these steps.
   a) Log in to the ESX host and connect to the directory where the `vm-support` package is located.
      The filename follows the pattern esx.date_and_time.tgz.
   b) Make sure that the directory has enough space for the package, the uncompressed package, and the recompressed package, or move the package.
   c) Extract the package to the local directory.
      ```
      vm-support -x *.tgz .
      ```
      The resulting file hierarchy might contain core dump files for the ESX host, usually in `/var/core`, and might contain multiple core dump files for virtual machines.
   d) Decrypt each encrypted core dump file separately.
      ```
      crypto-util envelope extract --offset 4096 --keyfile vm-support-incident-key-file
      --password encryptedZdumpdecryptedZdump
      ```
      `vm-support-incident-key-file` is the incident key file that you find at the top level in the directory.

      `encryptedZdump` is the name of the encrypted core dump file.

      `decryptedZdump` is the name for the file that the command generates. Make the name similar to the *`encryptedZdump`* name.
   e) Provide the password that you specified when you created the `vm-support` package.
   f) Remove the encrypted core dumps, and compress the package again.
      ```
      vm-support --reconstruct
      ```
7. Remove any files that contain confidential information.

Decrypt or Re-Encrypt an Encrypted Core Dump

You can decrypt or re-encrypt an encrypted core dump on your ESX host by using the `crypto-util` CLI.

The key that was used to encrypt the core dump must be available on the ESX host that generated the core dump.

You can decrypt and examine the core dumps in the vm-support package yourself. Core dumps might contain sensitive information. Follow the security and privacy policy for your organization to protect sensitive information such as keys.

For details about re-encrypting a core dump and other features of `crypto-util`, see the command-line help.

Note:
`crypto-util` is for advanced users.

1. Log directly in to the ESX host on which the core dump happened.
   If the ESX host is in lockdown mode, or if SSH access is deactivated, you might have to activate access first.
2. Determine whether the core dump is encrypted.

Option	Description
Monitor core dump	`crypto-util envelope describe vmmcores.ve`
zdump file	`crypto-util envelope describe --offset 4096 zdumpFile`

3. Decrypt the core dump, depending on its type.

Option	Description
Monitor core dump	`crypto-util envelope extract vmmcores.ve vmmcores`
zdump file	`crypto-util envelope extract --offset 4096 zdumpEncryptedzdumpUnencrypted`

Activate and Deactivate Key Persistence on an ESX Host

You must activate key persistence on an ESX host. It is not activated by default.

Requirements to activate key persistence:

- ESX 7.0 Update 2 or later
- ESX host installed with TPM 2.0
- Have access to the ESXCLI command set. You can run ESXCLI commands remotely, or run them in the ESX Shell.

Note:
Key persistence is not necessary when using vSphere Native Key Provider. vSphere Native Key Provider is designed out-of-the-box to run without requiring access to a key server.

For additional security, the TPM can also use a sealing policy to prevent tampering during ESX host boot. See What Are TPM Sealing Policies.

For conceptual information about key persistence, see vSphere Key Persistence on ESX Hosts.

1. Start a session on the ESX host by using SSH or another remote console connection.
2. Log in as root.
3. Verify that the ESX host is in TPM mode.

   ```
   esxcli system settings encryption get
   ```

   If the Mode appears as NONE, you must enable the TPM in the firmware of the host, and set the mode by running the following command.

   ```
   esxcli system settings encryption set --mode=TPM
   ```

4. Activate or deactivate key persistence.
   a) To activate key persistence:

      ```
      esxcli system security keypersistence enable
      ```

   b) To deactivate persistence:

      ```
      esxcli system security keypersistence disable --remove-all-stored-keys
      ```

Rekey an Encrypted Virtual Machine Using the vSphere Client

You can use the vSphere Client to perform a shallow rekey of an encrypted virtual machine. You might perform a rekey of an encrypted virtual machine for business or compliance reasons.

Required privilege: **Cryptographic operations** > **Recrypt**

A shallow rekey (also called recrypt) replaces only the Key Encryption Key (KEK). You do not need to power off the encrypted virtual machine to perform a shallow rekey. If you need to replace both the Disk Encryption Key (DEK) and the KEK, you must perform a deep rekey.

Note:
Virtual machines configured with IDE controllers must be powered off to perform a shallow rekey operation.

For more conceptual information, see How Do You Recrypt (Rekey) an Encrypted Virtual Machine.

1. Log in to the vCenter system with the vSphere Client.
2. Browse the inventory list and select the encrypted virtual machine.
3. Right-click the encrypted virtual machine and select **VM Policies**.
4. Select **Re-encrypt**.
5. Click **Yes**.
   The encrypted virtual machine is rekeyed with the new KEK.

   Note:
   If the rekey fails, the events subsystem posts the following event:

   ```
   com.vmware.vc.vm.crypto.RekeyFail
   ```

Rekey an Encrypted Virtual Machine Using the CLI

You can use the CLI to perform a shallow rekey of an encrypted virtual machine. You might perform a rekey of an encrypted virtual machine for business or compliance reasons.

Required privilege: **Cryptographic operations** > **Recrypt**

Note:
Virtual machines configured with IDE controllers must be powered off to perform a shallow rekey operation.

A shallow key (also called recrypt) replaces only the Key Encryption Key (KEK). You do not need to power off the encrypted virtual machine to perform a shallow rekey. If you need to replace both the Disk Encryption Key (DEK) and the KEK, you must perform a deep rekey.

This task shows how to perform a shallow rekey on an encrypted virtual machine using the currently assigned key provider.

For more conceptual information, see How Do You Recrypt (Rekey) an Encrypted Virtual Machine.

1. In a PowerCLI session, run the `Connect-VIServer` cmdlet to connect as an administrator to the vCenter host.
2. Assign the current key provider to a variable.

   ```
   $kp = Get-KeyProvider keyprovider_name
   ```

3. Assign the encrypted virtual machine to a variable.

   ```
   $vm = Get-VM encrypted_vm_name
   ```

4. Check the security information for the encrypted virtual machine.

   ```
   Get-SecurityInfo -Entity $vm
   ```

   Note the EncryptionKeyId.
5. Perform the shallow rekey of the encrypted virtual machine.

   ```
   Set-VM -vm $vm -KeyProvider $kp
   ```

   Type `Y` to confirm the rekey.
6. To verify that the EncryptionKeyId is changed, check the security information for the encrypted virtual machine.

   ```
   Get-SecurityInfo -Entity $vm
   ```

Set the Default Key Provider Using the vSphere Client

You must set the default key provider if you do not make the first key provider the default, or if your environment uses multiple key providers and you remove the default one. You can use the vSphere Client to set the default key provider at the vCenter level.

As a best practice, verify that the Connection Status in the Key Providers tab shows Active and a green check mark.

1. Log in using the vSphere Client.
2. Navigate to the vCenter.
3. Click **Configure** and select **Key Providers** under **Security**.
4. Select the key provider.
5. Click **Set as Default**.
   A confirmation dialog box appears.
6. Click **Set as Default**.
   The key provider displays as the current default.

Set the Default Key Provider Using the Command Line

You must set the default key provider if you do not make the first key provider the default, or if your environment uses multiple key providers and you remove the default one. You can use PowerCLI to set the default key provider at the vCenter level, the cluster level, or the cluster folder level.

As a best practice, verify that the Connection Status in the Key Providers tab shows Active and a green check mark.

You must have a role that includes the **Cryptographic operations** > **Manage KMS** privilege.

1. Ensure that you are connected as an administrator to the vCenter where you created the key provider.

   ```
   Connect-VIServer -server VC_ip_address -User admin_user -Password 'password'
   ```

2. Obtain the key provider.

   ```
   Get-KeyProvider
   ```

   You can use the -Name*keyprovider* option to specify a single key provider.

3. Assign the `Get-KeyProvider` key provider information to a variable.
   For example, this command assigns the information to the variable `$kp`.

   ```
   $kp = Get-KeyProvider
   ```

   If you have multiple key providers, you can use `Select-Object` to select one of them.

   ```
   $kp = Get-KeyProvider | Select-Object -Index 0
   ```

4. Use one of the following PowerCLI commands.

Where to set the default	Command
vCenter level	`Set-KeyProvider -KeyProvider $kp -DefaultForSystem`
Cluster level	This example command sets the key provider for the cluster `CL-01`. `Add-EntityDefaultKeyProvider -KeyProvider $kp -Entity 'CL-01'`
Cluster Folder level	This example command sets the key provider for the cluster folder `Cluster-Folder-01`. `Add-EntityDefaultKeyProvider -KeyProvider $kp -Entity 'Cluster-Folder-01'`

Securing Virtual Machines with Virtual Trusted Platform Module

With the Virtual Trusted Platform Module (vTPM) feature, you can add a TPM 2.0 virtual cryptoprocessor to a virtual machine.

A vTPM is a software-based representation of a physical Trusted Platform Module 2.0 chip. A vTPM acts as any other virtual device. You can add a vTPM to a virtual machine in the same way you add virtual CPUs, memory, disk controllers, or network controllers. A vTPM does not require a hardware Trusted Platform Module chip.

What Is a Virtual Trusted Platform Module

A virtual Trusted Platform Module (vTPM) is a software-based representation of a physical Trusted Platform Module 2.0 chip. A vTPM acts as any other virtual device.

vTPMs provide hardware-based, security-related functions such as random number generation, attestation, key generation, and more. When added to a virtual machine, a vTPM enables the guest operating system to create and store keys that are private. These keys are not exposed to the guest operating system itself. Therefore, the virtual machine attack surface is reduced. Usually, compromising the guest operating system compromises its secrets, but enabling a vTPM greatly reduces this risk. These keys can be used only by the guest operating system for encryption or signing. With an attached vTPM, a client can remotely attest the identity of the virtual machine, and verify the software that it is running.

A vTPM does not require a physical Trusted Platform Module (TPM) 2.0 chip to be present on the ESX host. However, if you want to perform host attestation, an external entity, such as a TPM 2.0 physical chip, is required. For more information, see Securing ESX Hosts with Trusted Platform Module.

> **Note:**
> By default, no storage policy is associated with a virtual machine that has been enabled with a vTPM. Only the virtual machine files (VM Home) are encrypted. If you prefer, you can choose to add encryption explicitly for the virtual machine and its disks, but the virtual machine files would have already been encrypted.

How Do You Configure a vTPM for a Virtual Machine

From the perspective of the virtual machine, a vTPM is a virtual device. You can add a vTPM to either a new or an existing virtual machine. A vTPM depends on virtual machine encryption to secure vital TPM data, thus, it requires that you configure a key provider. When you configure a vTPM, the virtual machine files are encrypted but not the disks. You can choose to add encryption explicitly for the virtual machine and its disks.

When you back up a virtual machine enabled with a vTPM, the backup must include all virtual machine data, including the `*.nvram` file. If your backup does not include the `*.nvram` file, you cannot restore a virtual machine with a vTPM. Also, because the VM home files of a vTPM-enabled virtual machine are encrypted, ensure that the encryption keys are available at the time of a restore.

In vSphere 8.0 and later, when cloning a virtual machine with a vTPM, selecting the **Replace** option for a virtual machine with a vTPM starts with a new, blank vTPM, which gets its own secrets and identity. When you replace the secrets of a vTPM, all keys, including workload-related keys, are replaced. As a best practice, ensure that your workloads no longer use a vTPM before you replace the keys. Otherwise, the workloads in the cloned virtual machine might not function correctly.

vSphere Requirements for vTPMs

To use a vTPM, your vSphere environment must meet these requirements:

- Virtual machine requirements:
  - EFI firmware
  - Hardware version 14 and later
- Component requirements:
  - vCenter 6.7 and later for Windows virtual machines, vCenter 7.0 Update 2 and later for Linux virtual machines.
  - Virtual machine encryption (to encrypt the virtual machine home files).
  - Key provider configured for vCenter. See Comparison of vSphere Key Providers.
- Guest OS support:
  - Linux
  - Windows Server 2008 and later
  - Windows 7 and later

Differences Between a Hardware TPM and a Virtual TPM

You use a hardware Trusted Platform Module (TPM) to provide secure storage of credentials or keys. A vTPM performs the same functions as a TPM, but it performs cryptographic coprocessor capabilities in software. A vTPM uses the `.nvram` file, which is encrypted using virtual machine encryption, as its secure storage.

A hardware TPM includes a preloaded key called the Endorsement Key (EK). The EK has a private and public key. The EK provides the TPM with a unique identity. For a vTPM, this key is provided either by the VMware Certificate Authority (VMCA) or by a third-party Certificate Authority (CA). After the vTPM uses a key, it is typically not changed because doing so invalidates sensitive information stored in the vTPM. The vTPM does not contact the third-party CA at any time.

Create a Virtual Machine with a Virtual Trusted Platform Module

You can add a Virtual Trusted Platform Module (vTPM) when you create a virtual machine to provide enhanced security to the guest operating system. You must create a key provider before you can add a vTPM.

- Ensure that your vSphere environment is configured with a key provider. See the following for more information:
  - Configuring and Managing a Standard Key Provider
  - Configuring and Managing vSphere Native Key Provider
- The guest OS you use can be Windows Server 2008 and later, Windows 7 and later, or Linux.
- The ESX hosts running in your environment must be ESX 6.7 or later (Windows guest OS), or 7.0 Update 2 (Linux guest OS).
- The virtual machine must use EFI firmware.
- Verify that you have the required privileges:
  - **Virtual machine** > **Change Configuration** > **Add or remove device**

- **Cryptographic operations** > **Clone**
- **Cryptographic operations** > **Clone**
- **Cryptographic operations** > **Clone**
- **Cryptographic operations** > **Clone**
- **Cryptographic operations** > **Clone**

Note:
After creating a virtual machine with a vTPM, the **Cryptographic operations** > **Direct Access** privilege is required to open a console session.

The VMware virtual TPM is compatible with TPM 2.0 and creates a TPM-enabled virtual chip for use by the virtual machine and the guest OS it hosts.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Right-click the object, select **New Virtual Machine**, and follow the prompts to create a virtual machine.

Option	Action
Select a creation type	Create a new virtual machine.
Select a name and folder	Specify a name and target location.
Select a compute resource	Specify an object for which you have privileges to create a virtual machine. See Prerequisites and Required Privileges for Virtual Machine Encryption Tasks.
Select storage	Select a compatible datastore.
Select compatibility	You must select **ESX 6.7 and later** for Windows guest OS, or **ESX 7.0 Update 2ESX** and later for Linux guest OS.
Select a guest OS	Select Windows or Linux for use as the guest OS.
Customize hardware	Click **Add New Device** and select **Trusted Platform Module**. You can further customize the hardware, for example, by changing disk size or CPU.
Ready to complete	Review the information and click **Finish**.

The vTPM-enabled virtual machine appears in your inventory as specified.

Add Virtual Trusted Platform Module to an Existing Virtual Machine

You can add a Virtual Trusted Platform Module (vTPM) to an existing virtual machine to provide enhanced security to the guest operating system. You must create a key provider before you can add a vTPM.

- Ensure that your vSphere environment is configured for a key provider. See the following for more information:
  - Configuring and Managing a Standard Key Provider
  - Configuring and Managing vSphere Native Key Provider
- The guest OS you use can be Windows Server 2008 and later, Windows 7 and later, or Linux.
- Verify that the virtual machine is turned off.
- The ESX hosts running in your environment must be ESX 6.7 and later (Windows guest OS), or 7.0 Update 2 and later (Linux guest OS).
- The virtual machine must use EFI firmware.
- Verify that you have the required privileges:
  - **Virtual machine** > **Change Configuration** > **Add or remove device**
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Clone**
  - **Cryptographic operations** > **Clone**

Note:
After adding a vTPM to a virtual machine, the **Cryptographic operations** > **Direct Access** privilege is required to open a console session.

The VMware virtual TPM is compatible with TPM 2.0, and creates a TPM-enabled virtual chip for use by the virtual machine and the guest OS it hosts.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. In the **Edit Settings** dialog box, click **Add New Device** and select **Trusted Platform Module**.
4. Click **OK**.
   The **Virtual Machine Details** pane reflects that encryption has been applied to the virtual machine.

Remove Virtual Trusted Platform Module from a Virtual Machine

You can remove Virtual Trusted Platform Module (vTPM) security from a virtual machine.

- Ensure that the virtual machine is powered off.
- Verify that you have the required privileges: **Virtual machine** > **Change Configuration** > **Add or remove device** and **Cryptographic operations** > **Decrypt**

Removing a vTPM device causes all encrypted information on the virtual machine to become unrecoverable. Before removing a vTPM from a virtual machine, deactivate any applications in the Guest OS that use the vTPM device, such as BitLocker. Failure to do so can cause the virtual machine not to boot. Also, you cannot remove a vTPM from a virtual machine that contains snapshots.

1. Connect to vCenter by using the vSphere Client.
2. Right-click the virtual machine in the inventory that you want to modify and select **Edit Settings**.
3. In the **Virtual Hardware** tab, expand **Security Devices**.
4. Click the ellipses icon for Virtual TPM.
5. Click **Remove device**.
6. Click **Delete** to confirm you want to remove the vTPM.
   The vTPM device is marked for removal.
7. Click **OK**.

Identify Virtual Trusted Platform Module Enabled Virtual Machines

You can identify which of your virtual machines are enabled to use a Virtual Trusted Platform Module (vTPM).

You can generate a list of all virtual machines in your inventory showing virtual machine name, operating system, and vTPM status. You can also export this list to a CSV file for use in compliance audits.

1. Connect to vCenter by using the vSphere Client.
2. Select a vCenter instance, a host, or a cluster.
3. Click the **VMs** tab and click **Virtual Machines**.
4. To view all virtual machines on which a TPM is enabled, click **Manage Columns** and select **TPM**.
   The TPM column displays "Present" for virtual machines on which a TPM is enabled. Virtual machines without a TPM are listed as "Not present."
5. To export the contents of an inventory list view to a CSV file, click **Export**.

View Virtual Trusted Platform Module Device Certificates

Virtual Trusted Platform Module (vTPM) devices are pre-configured with default certificates, which you can review.

You must have a vTPM-enabled virtual machine in your environment.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. To identify vTPM-enabled virtual machines, click **VMs** and click **Virtual Machines**.
   If necessary, click **Manage Columns** and select **TPM** to display virtual machines with a TPM "Present."
4. Select the vTPM-enabled virtual machine whose certificate information you want to view.
5. Click the virtual machine's **Configure** tab.
6. Under **TPM**, select **Certificates**.
7. Select the certificate and view its information.
8. (Optional) To export the certificate information, click **Export**.
   The certificate is saved to disk.

You can replace the default certificate with a certificate issued by a third-party certificate authority (CA). See Export and Replace Virtual Trusted Platform Module Device Certificates.

Export and Replace Virtual Trusted Platform Module Device Certificates

You can replace the default certificate that comes with a Virtual Trusted Platform Module (vTPM) device.

You must have a vTPM-enabled virtual machine in your environment.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Select the vTPM-enabled virtual machine in the inventory whose certificate information you want to replace.
4. Click the **Configure** tab.
5. Under **TPM** select **Signing Requests**.
6. Select a certificate.
7. To export the certificate information, click **Export**.
   The certificate is saved to disk.
8. Get a certificate issued by a third-party certificate authority (CA) against the certificate signing request (CSR) you exported.
   You can use any CA that you might have in your IT environment.
9. When you have the new certificate, replace the existing certificate.
   a) Right-click the virtual machine in the inventory whose certificate you want to replace and select **Edit Settings**.
   b) In the **Edit Settings** dialog box, expand **Security Devices**, then expand **Virtual TPM**.
      The certificates appear.
   c) Click **Replace** for the certificate you want to replace.
      The **File Upload** dialog box appears.
   d) On your local machine, locate the new certificate and upload it.
      The new certificate replaces the default certificate that came with your vTPM device.
   e) The certificate information is updated in the Certificates list.

Securing Windows Guest Operating Systems with Virtualization-based Security

In vSphere 6.7 and later, you can enable Microsoft virtualization-based security (VBS) on supported Windows guest operating systems.

Microsoft VBS, a feature introduced in Windows 10 and Windows Server 2016 operating systems, uses hardware and software virtualization to enhance system security by creating an isolated, hypervisor-restricted, specialized subsystem.

VBS permits you to use the following Windows security features to harden your system and isolate key system and user secrets from being compromised:

- Credential Guard: Aims to isolate and harden key system and user secrets against compromise.
- Device Guard: Provides a set of features designed to work together to prevent and eliminate malware from running on a Windows system.
- Configurable Code Integrity: Ensures that only trusted code runs from the boot loader onwards.

See the topic on virtualization-based security in the Microsoft documentation for more information.

After you enable VBS for a virtual machine through vCenter, you enable VBS within the Windows guest operating system.

vSphere Virtualization-based Security Best Practices

Follow best practices for virtualization-based security (VBS) to maximize security and manageability of your Windows guest operating system environment.

Avoid problems by following these best practices.

VBS Hardware Requirements

Use the following hardware for VBS:

- Intel
  - Haswell CPU or later. For best performance, use the Skylake-EP CPU or later.
  - The Ivy Bridge CPU is acceptable.
  - The Sandy Bridge CPU might cause some slow performance.
- AMD
  - Zen 2 series CPUs (Rome) or later.
  - Older CPUs might cause some slow performance.

The mitigations for the Machine Check Exception on Page Size Change Intel CPU vulnerability can impact guest OS performance negatively when VBS is in use. For more information, see the VMware KB article at https://kb.vmware.com/kb/76050.

VBS and Windows Guest OS Compatibility

On Intel, VBS is supported for Windows 10 and Windows Server 2016 and later virtual machines, although Windows Server 2016 versions 1607 and 1703 require patches. Check the Microsoft documentation for ESX host hardware compatibility. Using Intel CPUs for VBS requires vSphere 6.7 or later and hardware version 14 or later.

On AMD, VBS is supported on Windows 10, version 1809, and Windows 2019 and later virtual machines. Using AMD CPUs for VBS requires vSphere 7.0 Update 2 or later and hardware version 19 or later.

Initially, Windows 10 required that you enable Hyper-V for VBS. Enabling Hyper-V is not required for Windows 10. The same applies to Windows Server 2016 and later. Consult the current Microsoft documentation and the *VMware vSphere Release Notes* for more information.

Unsupported VMware Features on VBS

The following features are not supported in a virtual machine when VBS is enabled:

- Fault tolerance
- PCI passthrough
- Hot add of CPU or memory

Installation and Upgrade Caveats with VBS

Before you configure VBS, understand the following installation and upgrade caveats:

- New virtual machines configured for Windows 10 and Windows Server 2016 and later on virtual hardware versions less than version 14 are created using Legacy BIOS by default. You must reinstall the guest operating system after changing the virtual machine's firmware type from Legacy BIOS to UEFI.
- If you plan to migrate your virtual machines from previous vSphere releases to vSphere 6.7 or later, and enable VBS on your virtual machines, use UEFI to avoid having to reinstall the operating system.

Activate Virtualization-based Security on a Virtual Machine

You can activate Microsoft virtualization-based security (VBS) for supported Windows guest operating systems at the same time you create a virtual machine.

See vSphere Virtualization-based Security Best Practices for acceptable CPUs.

Using Intel CPUs for VBS requires vSphere 6.7 or later. Create a virtual machine that uses hardware version 14 or later and one of the following supported guest operating systems:

- Windows 10 (64 bit) or later releases
- Windows Server 2016 (64 bit) or later releases

Using AMD CPUs for VBS requires vSphere 7.0 Update 2 or later. Create a virtual machine that uses hardware version 19 or later and one of the following supported guest operating systems:

- Windows 10 (64 bit), version 1809 or later releases
- Windows Server 2019 (64 bit) or later releases

Ensure that you install the latest patches for Windows 10, version 1809, and Windows Server 2019, before activating VBS.

For more information about activating VBS on virtual machines on AMD platforms, see the VMware KB article at https://kb.vmware.com/s/article/89880.

Configuring VBS is a process that involves first activating VBS in the virtual machine then activating VBS in the Windows guest OS.

1. Connect to vCenter by using the vSphere Client.
2. Select an object in the inventory that is a valid parent object of a virtual machine, for example, an ESX host or a cluster.
3. Right-click the object, select **New Virtual Machine**, and follow the prompts to create a virtual machine.

Option	Action
Select a creation type	Create a virtual machine.
Select a name and folder	Specify a name and target location.
Select a compute resource	Specify an object for which you have privileges to create virtual machines.
Select storage	In the VM storage policy, select the storage policy. Select a compatible datastore.
Select compatibility	Intel CPU: Ensure that **ESXi 6.7 and later** is selected. AMD CPU: Ensure that **ESXi 7.0 U2 and later** is selected.
Select a guest OS	Select the Windows guest operating system option that best corresponds to operating system release. Select the **Enable Windows Virtualization Based Security** check box.
Customize hardware	Customize the hardware, for example, by changing disk size or CPU.
Ready to complete	Review the information and click **Finish**.

The Virtual Machine Details tile under the **Summary** tab displays "Virtualization Based Security - Enable".

See Activate Virtualization-based Security on the Guest Operating System.

Activate Virtualization-based Security on an Existing Virtual Machine

You can activate Microsoft virtualization-based security (VBS) on existing virtual machines for supported Windows guest operating systems.

See vSphere Virtualization-based Security Best Practices for acceptable CPUs.

See the *vSphere Security* documentation for information about acceptable CPUs and VBS best practices.

Using Intel CPUs for VBS requires vSphere 6.7 or later. The virtual machine must have been created using hardware version 14 or later and one of the following supported guest operating systems:

- Windows 11 (64 bit) or later releases
- Windows Server 2016 (64 bit) or later releases

Using AMD CPUs for VBS requires vSphere 7.0 Update 2 or later. The virtual machine must have been created using hardware version 19 or later and one of the following supported guest operating systems:

- Windows 10 (64 bit), version 1809 or later releases
- Windows Server 2019 (64 bit) or later releases

Ensure that you install the latest patches for Windows 10, version 1809, and Windows Server 2019, before activating VBS.

For more information about activating VBS on virtual machines on AMD platforms, see the VMware KB article at https://knowledge.broadcom.com/external/article?legacyId=89880.

Configuring VBS is a process that involves first activating VBS in the virtual machine then activating VBS in the guest OS.

> **Note:**
> New virtual machines configured for Windows 10, Windows Server 2016, and Windows Server 2019 on hardware versions less than version 14 are created using Legacy BIOS by default. If you change the virtual machine's firmware type from Legacy BIOS to UEFI, you must reinstall the guest operating system.

1. In the vSphere Client, browse to the virtual machine.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click the **VM Options** tab.
4. Select the **Enable** check box for Virtualization Based Security.
5. Click **OK**.

The Virtual Machine Details tile under the **Summary** tab displays "Virtualization Based Security - Enable".

See Activate Virtualization-based Security on the Guest Operating System.

Activate Virtualization-based Security on the Guest Operating System

You can activate Microsoft virtualization-based security (VBS) for supported Windows guest operating systems.

- Ensure that virtualization-based security has been activated on the virtual machine.

You activate VBS from within the Windows Guest OS. Windows configures and enforces VBS through a Group Policy Object (GPO). The GPO gives you the ability to turn off and on the various services, such as Secure Boot, Device Guard, and Credential Guard, that VBS offers. Certain Windows versions also require you to perform the additional step of enabling the Hyper-V platform.

See Microsoft's documentation about deploying Device Guard to activate virtualization-based security for details.

1. In Microsoft Windows, edit the group policy to turn on VBS and choose other VBS-related security options.
2. Optional: For Microsoft Windows versions less than Redstone 4, in the Windows Features control panel, enable the Hyper-V platform.
3. Reboot the guest operating system.

Deactivate Virtualization-based Security

If you no longer use virtualization-based security (VBS) with a virtual machine, you can deactivate VBS. When you deactivate VBS for the virtual machine, the Windows VBS options remain unchanged but might induce performance issues. Before deactivating VBS on the virtual machine, deactivate VBS options within Windows.

Ensure that the virtual machine is powered off.

1. In the vSphere Client, browse to the virtual machine that is using VBS.
   See Identify VBS-Enabled Virtual Machines for help in locating virtual machines that use VBS.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click **VM Options**.
4. Deselect the **Enable** check box for Virtualization Based Security.
   A message reminds you to deactivate VBS in the guest OS.
5. Click **OK**.
6. Verify that the virtual machine's **Summary** tab no longer displays "Virtualization Based Security - Enable" in the Guest OS description.

Identify VBS-Enabled Virtual Machines

You can identify which of your virtual machines have VBS enabled, for reporting and compliance purposes.

1. Connect to vCenter by using the vSphere Client.
2. Select a vCenter instance, a data center, or a host in the inventory.
3. Click the **VMs** tab and click **Virtual Machines**.
4. To show the **VBS** column, click **Manage Columns** and select the **VBS** check box.
5. Scan for "Present" in the **VBS** column.

Securing vSphere Networking

Securing vSphere networking is an essential part of protecting your environment. You secure different vSphere components in different ways. See the *vSphere Networking* documentation for detailed information about networking in the vSphere environment.

Network security in the vSphere environment shares many characteristics of securing a physical network environment, but also includes some characteristics that apply only to virtual machines.

Using Firewalls

Add firewall protection to your virtual network by installing and configuring host-based firewalls on some or all its virtual machines.

For efficiency, you can set up private virtual machine Ethernet networks or virtual networks. With virtual networks, you install a host-based firewall on a virtual machine at the head of the virtual network. This firewall serves as a protective buffer between the physical network adapter and the remaining virtual machines in the virtual network.

Host-based firewalls can slow performance. Balance your security needs against performance goals before you install host-based firewalls on virtual machines elsewhere in the virtual network.

See Securing the Network with Firewalls.

Using Network Segmentation

Keep different virtual machine zones within a host on different network segments. If you isolate each virtual machine zone on its own network segment, you minimize the risk of data leakage from one zone to the next. Segmentation prevents various threats, including Address Resolution Protocol (ARP) spoofing. With ARP spoofing, an attacker manipulates the ARP table to remap MAC and IP addresses, and gains access to network traffic to and from a host. Attackers use ARP spoofing to generate man in the middle (MITM) attacks, perform denial of service (DoS) attacks, hijack the target system, and otherwise disrupt the virtual network.

Planning segmentation carefully lowers the chances of packet transmissions between virtual machine zones. Segmentation therefore prevents sniffing attacks that require sending network traffic to the victim. Also, an attacker cannot use a nonsecure service in one virtual machine zone to access other virtual machine zones in the host. You can implement segmentation by using one of two approaches.

- Use separate physical network adapters for virtual machine zones to ensure that the zones are isolated. Maintaining separate physical network adapters for virtual machine zones is probably the most secure method. After the initial segment creation. This approach is less prone to misconfiguration.
- Set up virtual local area networks (VLANs) to help safeguard your network. VLANs provide almost all the security benefits inherent in implementing physically separate networks without the hardware overhead. VLANs can save you the cost of deploying and maintaining additional devices, cabling, and so on. See Securing Virtual Machines with VLANs.

Preventing Unauthorized Access to Virtual Machines

Requirements for securing virtual machines are often the same as requirements for securing physical machines.

- If a virtual machine network is connected to a physical network, it can be subject to breaches like a network that consists of physical machines.
- Even if you do not connect a virtual machine to the physical network, the virtual machine can be attacked by other virtual machines.

Virtual machines are isolated from each other. One virtual machine cannot read or write another virtual machine's memory, access its data, use its applications, and so forth. However, within the network, any virtual machine or group of virtual machines can still be the target of unauthorized access from other virtual machines. Protect your virtual machines from such unauthorized access.

For additional information about protecting virtual machines, see the NIST document titled "Secure Virtual Network Configuration for Virtual Machine (VM) Protection" at:

https://csrc.nist.gov/pubs/sp/800/125/b/final

Securing the Network with Firewalls

Security administrators use firewalls to safeguard the network or selected components in the network from intrusion.

Firewalls control access to devices within their perimeter by closing all ports except for ports that the administrator explicitly or implicitly designates as authorized. The ports that administrators open allow traffic between devices on different sides of the firewall.

> **Important:**
>
> The ESX firewall does not allow per-network filtering of vMotion traffic. Therefore, you must install rules on your external firewall to ensure that no incoming connections can be made to the vMotion socket.

In a virtual machine environment, you can plan the layout for firewalls between components.

- Firewalls between physical machines such as vCenter systems and ESX hosts.
- Firewalls between one virtual machine and another, for example, between a virtual machine acting as an external Web server and a virtual machine connected to your company’s internal network.
- Firewalls between a physical machine and a virtual machine, such as when you place a firewall between a physical network adapter card and a virtual machine.

How you use firewalls in your ESX configuration is based on how you plan to use the network and how secure any given component has to be. For example, if you create a virtual network where each virtual machine is dedicated to running

a different benchmark test suite for the same department, the risk of unwanted access from one virtual machine to the next is minimal. Therefore, a configuration where firewalls are present between the virtual machines is not necessary. However, to prevent interruption of a test run from an outside host, you can configure a firewall at the entry point of the virtual network to protect the entire set of virtual machines.

For the list of all supported ports and protocols in VMware products, including vSphere and vSAN, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/. You can search ports by VMware product, create a customized list of ports, and print or save port lists.

Firewalls for Configurations with vCenter Server

If you access ESX hosts through vCenter, you typically protect vCenter using a firewall.

Firewalls must be present at entry points. A firewall might lie between the clients and vCenter or vCenter and the clients can both be behind a firewall.

For the list of all supported ports and protocols in VMware products, including vSphere and vSAN, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/. You can search ports by VMware product, create a customized list of ports, and print or save port lists.

Networks configured with vCenter can receive communications through the vSphere Client, other UI clients, or clients that use the vSphere API. During normal operation, vCenter listens for data from its managed hosts and clients on designated ports. vCenter also assumes that its managed hosts listen for data from vCenter on designated ports. If a firewall is present between any of these elements, you must ensure that the firewall has open ports to support data transfer.

You might also include firewalls at other access points in the network, depending on the network usage and on the level of security that clients require. Select the locations for your firewalls based on the security risks for your network configuration. The following firewall locations are commonly used.

- Between the vSphere Client or a third-party network-management client and vCenter.
- If your users access virtual machines through a Web browser, between the Web browser and the ESX host.
- If your users access virtual machines through the vSphere Client, between the vSphere Client and the ESX host. This connection is in addition to the connection between the vSphere Client and vCenter, and it requires a different port.
- Between vCenter and the ESX hosts.
- Between the ESX hosts in your network. Although traffic between hosts is usually considered trusted, you can add firewalls between them if you are concerned about security breaches from machine to machine.
  If you add firewalls between ESX hosts and plan to migrate virtual machines between them, open ports in any firewall that divides the source host from the target hosts.
- Between the ESX hosts and network storage such as NFS or iSCSI storage. These ports are not specific to VMware. Configure them according to the specifications for your network.

Connecting to vCenter Through a Firewall

Open TCP port 443 in the firewall to enable vCenter to receive data.

By default vCenter uses TCP port 443 to listen for data from its clients. If you have a firewall between vCenter and its clients, you must configure a connection through which vCenter can receive data from the clients. Firewall configuration depends on what is used at your site, ask your local firewall system administrator for information.

Connecting ESX Hosts Through Firewalls

If you have a firewall between your ESX hosts and vCenter, ensure that the managed hosts can receive data.

To configure a connection for receiving data, open ports for traffic from services such as vSphere High Availability, vMotion, and vSphere Fault Tolerance. See Configuring the ESX Firewall for a discussion of configuration files, vSphere Client access, and firewall commands. For a list of ports, see the VMware Ports and Protocols Tool™ at https://ports.broadcom.com/.

Firewalls for Configurations Without vCenter

If your environment does not include vCenter, clients can connect directly to the ESX network.

You can connect to a standalone ESX host in several ways.

- VMware Host Client
- ESXCLI interface
- vSphere Web Services SDK or vSphere Automation SDKs
- Third-party clients

The firewall requirements for standalone hosts are similar to requirements when a vCenter is present.

- Use a firewall to protect your ESX layer or, depending on your configuration, your clients, and the ESX layer. This firewall provides basic protection for your network.
- Licensing in this type of configuration is part of the ESX package that you install on each of the hosts. Because licensing is resident to ESX, a separate License Server with a firewall is not required.

You can configure firewall ports using ESXCLI or using the VMware Host Client. See *vSphere Single Host Management - VMware Host Client*.

Connecting to the Virtual Machine Console Through a Firewall

Certain ports must be open for user and administrator communication with the virtual machine console. Which ports must be open depends on the type of virtual machine console, and on whether you connect through vCenter with the vSphere Client or directly to the ESX host from the VMware Host Client.

For more information about ports, purpose, and classification (incoming, outgoing, or bidirectional), see the VMware Ports and Protocols Tool™ at https://ports.vmware.com.

Connecting to a Browser-Based Virtual Machine Console Through the vSphere Client

When you are connecting with the vSphere Client, you always connect to the vCenter system that manages the ESX host, and access the virtual machine console from there.

If you are using the vSphere Client and connecting to a browser-based virtual machine console, the following access must be possible:

- The firewall must allow vSphere Client to access vCenter on port 443.
- The firewall must allow vCenter to access the ESX host on port 902.

Connecting to a VMware Remote Console Through the vSphere Client

If you are using the vSphere Client and connecting to a VMware Remote Console (VMRC), the following access must be possible:

- The firewall must allow the vSphere Client to access vCenter on port 443.
- The firewall must allow the VMRC to access vCenter on port 443 and to access the ESX host on port 902 for VMRC versions before 11.0, and port 443 for VMRC version 11.0 and greater. For more information about VMRC version 11.0 and ESX port requirements, see the VMware knowledge base article at https://knowledge.broadcom.com/external/article?legacyId=76672.

Connecting to ESX Hosts Directly with the VMware Host Client

You can use the VMware Host Client virtual machine console if you connect directly to an ESX host.

> **Note:**
> Do not use the VMware Host Client to connect directly to hosts that are managed by a vCenter system. If you make changes to such hosts from the VMware Host Client, instability in your environment results.

The firewall must allow access to the ESX host on ports 443 and 902.

The VMware Host Client uses port 902 to provide a connection for guest operating system MKS activities on virtual machines. It is through this port that users interact with the guest operating systems and applications of the virtual machine. VMware does not support configuring a different port for this function.

Secure the Physical Switch on ESX Hosts

Secure the physical switch on each ESX host to prevent attackers from gaining access to the host and its virtual machines.

For best protection of your hosts, ensure that physical switch ports are configured with spanning tree deactivated and ensure that the non-negotiate option is configured for trunk links between external physical switches and virtual switches in Virtual Switch Tagging (VST) mode.

1. Log in to the physical switch and ensure that spanning tree protocol is deactivated or that Port Fast is configured for all physical switch ports that are connected to ESX hosts.
2. For virtual machines that perform bridging or routing, check periodically that the first upstream physical switch port is configured with BPDU Guard and Port Fast deactivated and with spanning tree protocol activated.

   To prevent the physical switch from potential Denial of Service (DoS) attacks, you can turn on the guest BPDU filter on the ESX hosts.
3. Log in to the physical switch and ensure that Dynamic Trunking Protocol (DTP) is not activated on the physical switch ports that are connected to the ESX hosts.
4. Routinely check physical switch ports to ensure that they are properly configured as trunk ports if connected to virtual switch VLAN trunking ports.

Securing Standard Switch Ports with Security Policies

The VMkernel port group or virtual machine port group on a standard switch has a configurable security policy. The security policy determines how strongly you enforce protection against impersonation and interception attacks on VMs.

Just like physical network adapters, virtual machine network adapters can impersonate another VM. Impersonation is a security risk.

- A VM can send frames that appear to be from a different machine so that it can receive network frames that are intended for that machine.
- A virtual machine network adapter can be configured so that it receives frames targeted for other machines

When you add a VMkernel port group or virtual machine port group to a standard switch, ESX configures a security policy for the ports in the group. You can use this security policy to ensure that the host prevents the guest operating systems of its VMs from impersonating other machines on the network. The guest operating system that might attempt impersonation does not detect that the impersonation was prevented.

The security policy determines how strongly you enforce protection against impersonation and interception attacks on VMs. To correctly use the settings in the security profile, see the Security Policy section in the *vSphere Networking* publication. This section explains:

- How VM network adapters control transmissions.
- How attacks are staged at this level.

Securing vSphere Standard Switches

You can secure standard switch traffic against Layer 2 attacks by restricting some of the MAC address modes of the VM network adapters.

Each VM network adapter has an initial MAC address and an effective MAC address.

Initial MAC Address

The initial MAC address is assigned when the adapter is created. Although the initial MAC address can be reconfigured from outside the guest operating system, it cannot be changed by the guest operating system.

Effective MAC Address

Each adapter has an effective MAC address that filters out incoming network traffic with a destination MAC address that is different from the effective MAC address. The guest operating system is responsible for setting the effective MAC address and typically matches the effective MAC address to the initial MAC address.

What Happens When You Create a Virtual Machine Network Adapter

Upon creating a virtual machine network adapter, the effective MAC address and initial MAC address are the same. The guest operating system can alter the effective MAC address to another value at any time. If an operating system changes the effective MAC address, its network adapter receives network traffic that is destined for the new MAC address.

When sending packets through a network adapter, the guest operating system typically places its own adapter effective MAC address in the source MAC address field of the Ethernet frames. It places the MAC address for the receiving network adapter in the destination MAC address field. The receiving adapter accepts packets only if the destination MAC address in the packet matches its own effective MAC address.

An operating system can send frames with an impersonated source MAC address. An operating system can therefore impersonate a network adapter that the receiving network authorizes, and stage malicious attacks on the devices in a network.

Using Security Policies to Protect Ports and Groups

Protect virtual traffic against impersonation and interception Layer 2 attacks by configuring a security policy on port groups or ports.

The security policy on distributed port groups and ports includes the following options:

- MAC address changes (see MAC Address Changes)
- Promiscuous mode (see Promiscuous Mode Operation)
- Forged transmits (see Forged Transmits)

You can view and change the default settings by selecting the virtual switch associated with the host from the vSphere Client. See *vSphere Networking* documentation.

MAC Address Changes

The security policy of a virtual switch includes a **MAC address changes** option. This option allows virtual machines to receive frames with a Mac Address that is different from the one configured in the VMX.

When the **Mac address changes** option is set to **Accept**, ESX accepts requests to change the effective MAC address of a virtual machine to a different address than the initial MAC address.

When the **Mac address changes** option is set to **Reject**, ESX does not honor requests to change the effective MAC address of a virtual machine to a different address than the initial MAC address. This setting protects the host against MAC impersonation. The port that the virtual machine adapter used to send the request is deactivated and the virtual machine adapter does not receive any more frames until the effective MAC address matches the initial MAC address. The guest operating system does not detect that the MAC address change request was not honored.

> **Note:**
> The iSCSI initiator relies on being able to get MAC address changes from certain types of storage. If you are using ESX iSCSI with iSCSI storage, set the **MAC address changes** option to **Accept**.

In some situations, you can have a legitimate need for more than one adapter to have the same MAC address on a network, for example, if you are using Microsoft Network Load Balancing in unicast mode. When Microsoft Network Load Balancing is used in the standard multicast mode, adapters do not share MAC addresses.

> **Note:**
> Starting in vSphere 7.0, the defaults for **Forged transmits** and **MAC address changes** have been changed to Reject instead of Accept. Contact your storage vendor to validate.

Forged Transmits

The **Forged transmits** option affects traffic that is transmitted from a virtual machine.

When the **Forged transmits** option is set to **Accept**, ESX does not compare source and effective MAC addresses.

To protect against MAC impersonation, you can set the **Forged transmits** option to **Reject**. If you do, the host compares the source MAC address being transmitted by the guest operating system with the effective MAC address for its virtual machine adapter to see if they match. If the addresses do not match, the ESX host drops the packet.

The guest operating system does not detect that its virtual machine adapter cannot send packets by using the impersonated MAC address. The ESX host intercepts any packets with impersonated addresses before they are delivered, and the guest operating system might assume that the packets are dropped.

> **Note:**
> Starting in vSphere 7.0, the defaults for **Forged transmits** and **MAC address changes** have been changed to Reject instead of Accept.

Promiscuous Mode Operation

Promiscuous mode eliminates any reception filtering that the virtual machine adapter performs so that the guest operating system receives all traffic observed on the wire. By default, the virtual machine adapter cannot operate in promiscuous mode.

Although promiscuous mode can be useful for tracking network activity, it is an insecure mode of operation, because any adapter in promiscuous mode has access to the packets even if some of the packets are received only by a particular network adapter. This means that an administrator or root user within a virtual machine can potentially view traffic destined for other guest or host operating systems.

See the topic on configuring the security policy for a vSphere Standard Switch or Standard Port Group in the *vSphere Networking* documentation for information about configuring the virtual machine adapter for promiscuous mode.

> **Note:**
> In some situations, you might have a legitimate reason to configure a standard or a distributed virtual switch to operate in promiscuous mode, for example, if you are running network intrusion detection software or a packet sniffer.

Standard Switch Protection and VLANs

VMware standard switches provide safeguards against certain threats to VLAN security. Because of the way that standard switches are designed, they protect VLANs against a variety of attacks, many of which involve VLAN hopping.

Having this protection does not guarantee that your virtual machine configuration is invulnerable to other types of attacks. For example, standard switches do not protect the physical network against these attacks; they protect only the virtual network.

Standard switches and VLANs can protect against the following types of attacks.

Because new security threats develop over time, do not consider this an exhaustive list of attacks. Regularly check VMware security resources on the Web to learn about security, recent security alerts, and VMware security tactics.

MAC Flooding

MAC flooding floods a switch with packets that contain MAC addresses tagged as having come from different sources. Many switches use a content-addressable memory table to learn and store the source address for each packet. When the table is full, the switch can enter a fully open state in which every incoming packet is broadcast on all ports, letting the attacker see all of the traffic on the switch. This state might result in packet leakage across VLANs.

Although VMware standard switches store a MAC address table, they do not get the MAC addresses from observable traffic and are not vulnerable to this type of attack.

802.1q and ISL Tagging Attacks

802.1q and ISL tagging attacks force a switch to redirect frames from one VLAN to another by tricking the switch into acting as a trunk and broadcasting the traffic to other VLANs.

VMware standard switches do not perform the dynamic trunking required for this type of attack and, therefore, are not vulnerable.

Double-encapsulation Attacks

Double-encapsulation attacks occur when an attacker creates a double-encapsulated packet in which the VLAN identifier in the inner tag is different from the VLAN identifier in the outer tag. For backward compatibility, native VLANs strip the outer tag from transmitted packets unless configured to do otherwise. When a native VLAN switch strips the outer tag, only the inner tag is left, and that inner tag routes the packet to a different VLAN than the one identified in the now-missing outer tag.

VMware standard switches drop any double-encapsulated frames that a virtual machine attempts to send on a port configured for a specific VLAN. Therefore, they are not vulnerable to this type of attack.

Multicast Brute-force Attacks

Involve sending large numbers of multicast frames to a known VLAN almost simultaneously to overload the switch so that it mistakenly allows some of the frames to broadcast to other VLANs.

VMware standard switches do not allow frames to leave their correct broadcast domain (VLAN) and are not vulnerable to this type of attack.

Spanning-tree Attacks

Spanning-tree attacks target Spanning-Tree Protocol (STP), which is used to control bridging between parts of the LAN. The attacker sends Bridge Protocol Data Unit (BPDU) packets that attempt to change the network topology, establishing themselves as the root bridge. As the root bridge, the attacker can sniff the contents of transmitted frames.

VMware standard switches do not support STP and are not vulnerable to this type of attack.

Random Frame Attacks

Random frame attacks involve sending large numbers of packets in which the source and destination addresses stay the same, but in which fields are randomly changed in length, type, or content. The goal of this attack is to force packets to be mistakenly rerouted to a different VLAN.

VMware standard switches are not vulnerable to this type of attack.

Secure vSphere Distributed Switches and Distributed Port Groups

Administrators have several options for securing vSphere Distributed Switches in their vSphere environment.

The same rules apply for VLANs in a vSphere Distributed Switch as they do in a standard switch. For more information, see Standard Switch Protection and VLANs.

1. For distributed port groups with static binding, deactivate the Auto Expand feature.

   Auto Expand is activated by default.

   To deactivate Auto Expand, configure the `autoExpand` property under the distributed port group with the vSphere Web Services SDK or with a command-line interface. See the *vSphere Web Services SDK* documentation.

2. Ensure that all private VLAN IDs of any vSphere Distributed Switch are fully documented.
3. If you are using VLAN tagging on a dvPortgroup, VLAN IDs must correspond to the IDs on external VLAN-aware upstream switches. If VLAN IDs are not tracked correctly, mistaken reuse of IDs might allow unintended traffic. Similarly, wrong or missing VLAN IDs might lead to traffic not passing between physical and virtual machines.
4. Ensure that no unused ports exist on a virtual port group associated with a vSphere Distributed Switch.
5. Label all vSphere Distributed Switches.

   vSphere Distributed Switches associated with an ESX host require a text box for the name of the switch. This label serves as a functional descriptor for the switch, just like the host name associated with a physical switch. The label on the vSphere Distributed Switch indicates the function or the IP subnet of the switch. For example, you can label the switch as internal to indicate that it is only for internal networking on a virtual machine's private virtual switch. No traffic goes over physical network adapters.
6. Deactivate network health check for your vSphere Distributed Switches if you are not actively using it.

   Network health check is deactivated by default. Once activated, the health check packets contain information about the host, switch, and port that an attacker can potentially use. Use network health check only for troubleshooting, and turn it off when troubleshooting is finished.
7. Protect virtual traffic against impersonation and interception Layer 2 attacks by configuring a security policy on port groups or ports.

   The security policy on distributed port groups and ports includes the following options:

   - MAC address changes (see MAC Address Changes)
   - Promiscuous mode (see Promiscuous Mode Operation)
   - Forged transmits (see Forged Transmits)

   You can view and change the current settings by selecting **Manage Distributed Port Groups** from the right-button menu of the distributed switch and selecting **Security** in the wizard. See the *vSphere Networking* documentation.

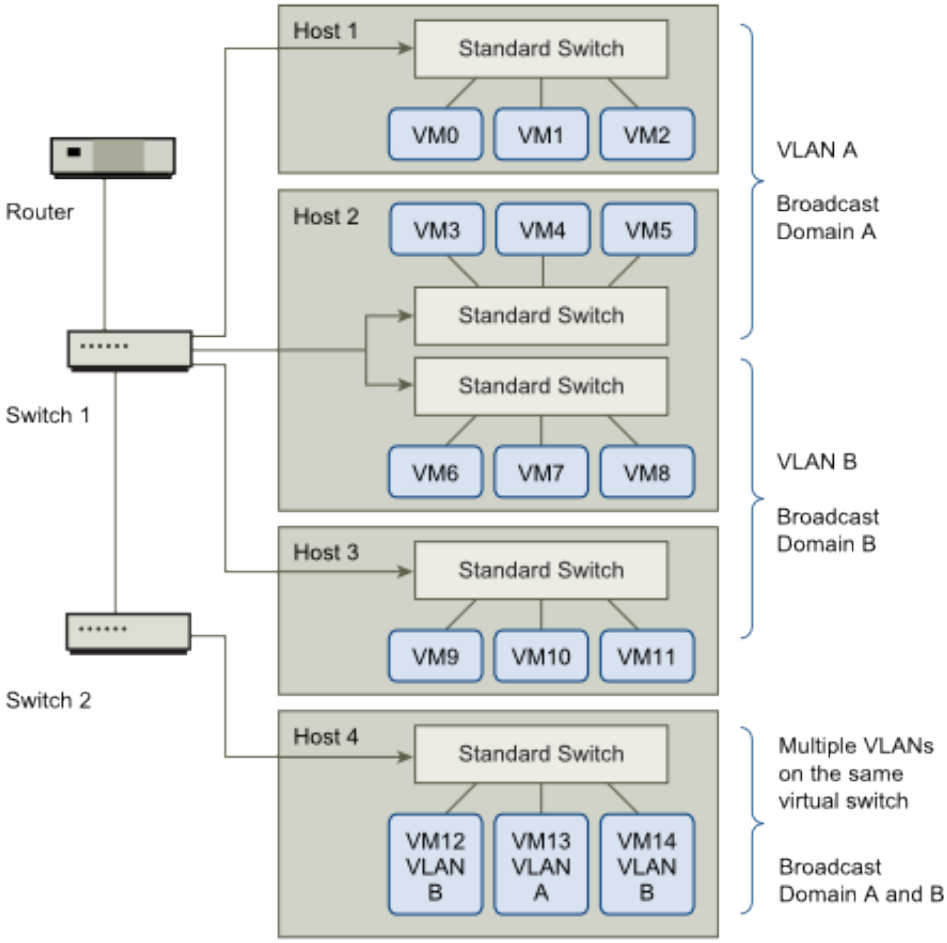
Securing Virtual Machines with VLANs

The network can be one of the most vulnerable parts of any system. Your virtual machine network requires as much protection as your physical network. Using VLANs can improve networking security in your environment.

VLANs are an IEEE standard networking scheme with specific tagging methods that allow routing of packets to only those ports that are part of the VLAN. When properly configured, VLANs provide a dependable means for you to protect a set of virtual machines from accidental or malicious intrusions.

VLANs let you segment a physical network so that two machines in the network are unable to transmit packets back and forth unless they are part of the same VLAN. For example, accounting records and transactions are among a company's most sensitive internal information. In a company whose sales, shipping, and accounting employees all use virtual machines in the same physical network, you might protect the virtual machines for the accounting department by setting up VLANs.

Figure 96: Sample VLAN Layout



In this configuration, all employees in the accounting department use virtual machines in VLAN A and the employees in sales use virtual machines in VLAN B.

The router forwards packets containing accounting data to the switches. These packets are tagged for distribution to VLAN A only. Therefore, the data is confined to Broadcast Domain A and cannot be routed to Broadcast Domain B unless the router is configured to do so.

This VLAN configuration prevents the sales force from intercepting packets destined for the accounting department. It also prevents the accounting department from receiving packets intended for the sales group. The virtual machines serviced by a single virtual switch can be in different VLANs.

Security Considerations for VLANs

The way you set up VLANs to secure parts of a network depends on factors such as the guest operating system and the way your network equipment is configured.

ESX features a complete IEEE 802.1q-compliant VLAN implementation. VMware cannot make specific recommendations on how to set up VLANs, but there are factors to consider when using a VLAN deployment as part of your security enforcement policy.

Secure VLANs

Administrators have several options for securing the VLANs in their vSphere environment.

1. Ensure that port groups are not configured to VLAN values that are reserved by upstream physical switches
   Do not set VLAN IDs to values reserved for the physical switch.

2. Ensure that port groups are not configured to VLAN 4095 unless you are using for Virtual Guest Tagging (VGT).

   Three types of VLAN tagging exist in vSphere:

   - External Switch Tagging (EST)
   - Virtual Switch Tagging (VST) - The virtual switch tags with the configured VLAN ID the traffic that is incoming to the attached virtual machines and removes the VLAN tag from the traffic that is leaving them. To set up VST mode, assign a VLAN ID between 1 and 4094.
   - Virtual Guest Tagging (VGT) - Virtual machines handle VLAN traffic. To activate VGT mode, set the VLAN ID to 4095. On a distributed switch, you can also allow virtual machine traffic based on its VLAN by using the **VLAN Trunking** option.

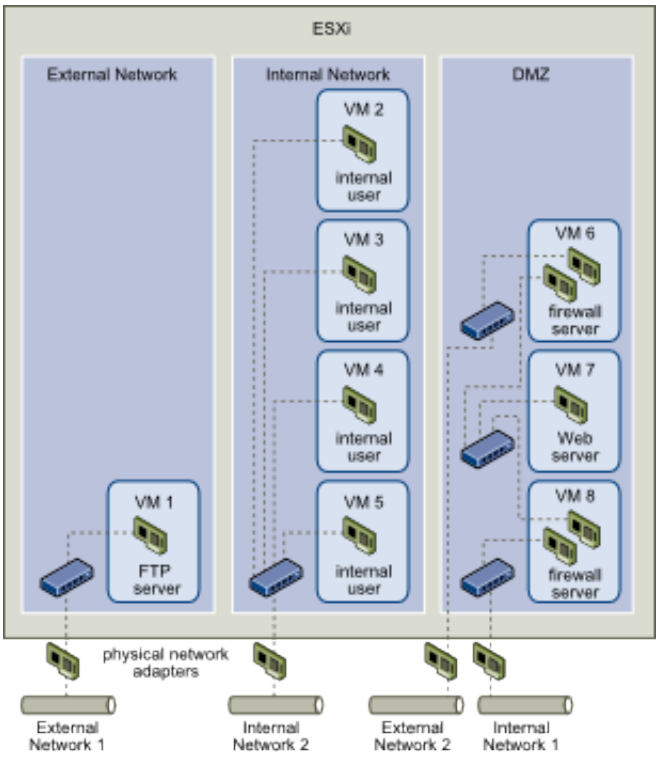
   On a standard switch you can configure VLAN networking mode at switch or port group level, and on a distributed switch at distributed port group or port level.

3. Ensure that all VLANs on each virtual switch are fully documented and that each virtual switch has all required VLANs and only required VLANs.

Creating Multiple Networks Within a Single ESX Host

The ESX system is designed so that you can connect some groups of virtual machines to the internal network, others to the external network, and still others to both—all on the same host. This capability is an outgrowth of basic virtual machine isolation coupled with a well-planned use of virtual networking features.

Figure 97: External Networks, Internal Networks, and a DMZ Configured on a Single ESX Host



In the figure, the system administrator configured a host into three distinct virtual machine zones: FTP server, internal virtual machines, and DMZ. Each zone serves a unique function.

FTP Server Zone

Virtual Machine 1 is configured with FTP software and acts as a holding area for data sent to and from outside resources such as forms and collateral localized by a vendor.

This virtual machine is associated with an external network only. It has its own virtual switch and physical network adapter that connect it to External Network 1. This network is dedicated to servers that the company uses to receive data from outside sources. For example, the company uses External Network 1 to receive FTP traffic from vendors and allow

vendors access to data stored on externally available servers though FTP. In addition to servicing Virtual Machine 1, External Network 1 services FTP servers configured on different ESX hosts throughout the site.

Because Virtual Machine 1 does not share a virtual switch or physical network adapter with any virtual machines in the host, the other resident virtual machines cannot transmit packets to or receive packets from the Virtual Machine 1 network. This restriction prevents sniffing attacks, which require sending network traffic to the victim. More importantly, an attacker cannot use the natural vulnerability of FTP to access any of the host's other virtual machines.

Internal Network Zone

Virtual Machines 2 through 5 are reserved for internal use. These virtual machines process and store company-private data such as medical records, legal settlements, and fraud investigations. As a result, the system administrators must ensure the highest level of protection for these virtual machines.

These virtual machines connect to Internal Network 2 through their own virtual switch and network adapter. Internal Network 2 is reserved for internal use by personnel such as claims processors, in-house lawyers, or adjustors.

Virtual Machines 2 through 5 can communicate with one another through the virtual switch and with internal virtual machines elsewhere on Internal Network 2 through the physical network adapter. They cannot communicate with externally facing machines. As with the FTP server, these virtual machines cannot send packets to or receive packets from the other virtual machines' networks. Similarly, the host's other virtual machines cannot send packets to or receive packets from Virtual Machines 2 through 5.

DMZ Zone

Virtual Machines 6 through 8 are configured as a DMZ that the marketing group uses to publish the company's external website.

This group of virtual machines is associated with External Network 2 and Internal Network 1. The company uses External Network 2 to support the Web servers that use the marketing and financial department to host the corporate website and other Web facilities that it hosts to outside users. Internal Network 1 is the conduit that the marketing department uses to publish its content to the corporate Web site, post downloads, and maintain services like user forums.

Because these networks are separate from External Network 1 and Internal Network 2, and the virtual machines have no shared points of contact (switches or adapters), there is no risk of attack to or from the FTP server or the internal virtual machine group.

Advantages of Using Virtual Machine Zones

By capitalizing on virtual machine isolation, correctly configuring virtual switches, and maintaining network separation, you can house all three virtual machine zones on the same ESX host and be confident that there will be no data or resource breaches.

The company enforces isolation among the virtual machine groups by using multiple internal and external networks and making sure that the virtual switches and physical network adapters for each group are separate from those of other groups.

Because none of the virtual switches straddle virtual machine zones, you succeed in eliminating the risk of packet leakage from one zone to another. A virtual switch, by design, cannot leak packets directly to another virtual switch. The only way for packets to travel from one virtual switch to another is under the following circumstances:

- The virtual switches are connected to the same physical LAN.
- The virtual switches connect to a common virtual machine, which might be used to transmit packets.

Neither of these conditions occur in the sample configuration. If you want to verify that no common virtual switch paths exist, you can check for possible shared points of contact by reviewing the network switch layout in the vSphere Client.

To safeguard the resources of virtual machines, configure a resource reservation and a limit for each virtual machine, which lowers the risk of DoS and DDoS attacks. You can further protect the ESX host and virtual machines by installing software firewalls at the front and back ends of the DMZ. Finally, ensure that the host is behind a physical firewall, and configure the networked storage resources so that each has its own virtual switch.

Using Internet Protocol Security on ESX Hosts

Internet Protocol Security (IPsec) secures IP communications coming from and arriving at a host. ESX hosts support IPsec using IPv6.

When you set up IPsec on an ESX host, you enable authentication and encryption of incoming and outgoing packets. When and how IP traffic is encrypted depends on how you set up the system's security associations and security policies.

A security association determines how the system encrypts traffic. When you create a security association, you specify the source and destination, encryption parameters, and a name for the security association.

A security policy determines when the system should encrypt traffic. The security policy includes source and destination information, the protocol and direction of traffic to be encrypted, the mode (transport or tunnel) and the security association to use.

List Available Security Associations on ESX Hosts

ESX can provide a list of all security associations available for use by security policies. The list includes both user created security associations and any security associations the VMkernel installed using Internet Key Exchange.

You can get a list of available security associations using the `esxcli` command.

At the command prompt, enter the command `esxcli network ip ipsec sa list`.

ESX displays a list of all available security associations.

Add an IPsec Security Association to an ESX Host

Add a security association to specify encryption parameters for associated IP traffic.

You can add a security association using the `esxcli` command.

At the command prompt, enter the command `esxcli network ip ipsec sa add` with one or more of the following options.

Option	Description
--sa-source= ***source address***	Required. Specify the source address.
--sa-destination= ***destination address***	Required. Specify the destination address.
--sa-mode= ***mode***	Required. Specify the mode, either `transport` or `tunnel`.
--sa-spi= ***security parameter index***	Required. Specify the security parameter index. The security parameter index identifies the security association to the host. It must be a hexadecimal with a 0x prefix. Each security association you create must have a unique combination of protocol and security parameter index.
--encryption-algorithm= ***encryption algorithm***	Required. Specify the encryption algorithm using one of the following parameters. • `3des-cbc` • `aes128-cbc` • `null` (provides no encryption)
--encryption-key= ***encryption key***	Required when you specify an encryption algorithm. Specify the encryption key. You can enter keys as ASCII text or as a hexadecimal with a 0x prefix.
--integrity-algorithm= ***authentication algorithm***	Required. Specify the authentication algorithm, either `hmac-sha1` or `hmac-sha2-256`.
--integrity-key= ***authentication key***	Required. Specify the authentication key. You can enter keys as ASCII text or as a hexadecimal with a 0x prefix.
--sa-name=*name*	Required. Provide a name for the security association.

New Security Association Command

The following example contains extra line breaks for readability.

```
esxcli network ip ipsec sa add
--sa-source 3ffe:501:ffff:0::a
--sa-destination 3ffe:501:ffff:0001:0000:0000:0000:0001
--sa-mode transport
--sa-spi 0x1000
--encryption-algorithm 3des-cbc
--encryption-key 0x697076367265616479 6c6f676f3364657363626f757432
--integrity-algorithm hmac-sha1
--integrity-key 0x69707636726561647 96c6f67736861316f757432
--sa-name sa1
```

Remove an IPsec Security Association from an ESX Host

You can remove a security association using the ESXCLI command.

Verify that the security association you want to use is not currently in use. If you try to remove a security association that is in use, the removal operation fails.

At the command prompt, enter the command `esxcli network ip ipsec sa remove --sa-name` *`security_association_name`*.

List Available IPsec Security Policies on an ESX Host

You can list available security policies using the ESXCLI command.

At the command prompt, enter the command `esxcli network ip ipsec sp list`.

The host displays a list of all available security policies.

Create an IPSec Security Policy on an ESX Host

Create a security policy to determine when to use the authentication and encryption parameters set in a security association. You can add a security policy using the ESXCLI command.

Before creating a security policy, add a security association with the appropriate authentication and encryption parameters as described in Add an IPsec Security Association to an ESX Host.

At the command prompt, enter the command `esxcli network ip ipsec sp add` with one or more of the following options.

Option	Description
`--sp-source=` ***`source address`***	Required. Specify the source IP address and prefix length.
`--sp-destination=` ***`destination address`***	Required. Specify the destination address and prefix length.
`--source-port=` ***`port`***	Required. Specify the source port. The source port must be a number between 0 and 65535.
`--destination-port=` ***`port`***	Required. Specify the destination port. The source port must be a number between 0 and 65535.
`--upper-layer-protocol=` ***`protocol`***	Specify the upper layer protocol using one of the following parameters. • `tcp` • `udp` • `icmp6` • `any`
`--flow-direction=` ***`direction`***	Specify the direction in which you want to monitor traffic using either `in` or `out`.

Option	Description
--action= ***action***	Specify the action to take when traffic with the specified parameters is encountered using one of the following parameters. • `none`: Take no action. • `discard`: Do not allow data in or out. • `ipsec`: Use the authentication and encryption information supplied in the security association to determine whether the data comes from a trusted source.
--sp-mode= ***mode***	Specify the mode, either `tunnel` or `transport`.
--sa-name=*security association name*	Required. Provide the name of the security association for the security policy to use.
--sp-name=*name*	Required. Provide a name for the security policy.

New Security Policy Command

The following example includes extra line breaks for readability.

```
esxcli network ip ipsec add
--sp-source=2001:db8:1::/64
--sp-destination=2002:db8:1::/64
--source-port=23
--destination-port=25
--upper-layer-protocol=tcp
--flow-direction=out
--action=ipsec
--sp-mode=transport
--sa-name=sa1
--sp-name=sp1
```

Remove an IPsec Security Policy from an ESX Host

You can remove a security policy from the ESX host using the ESXCLI command.

Verify that the security policy you want to use is not currently in use. If you try to remove a security policy that is in use, the removal operation fails.

At the command prompt, enter the command `esxcli network ip ipsec sp remove --sa-name` *`security policy name`*.

To remove all security policies, enter the command `esxcli network ip ipsec sp remove --remove-all`.

Ensure Proper SNMP Configuration on ESX Hosts

If SNMP is not properly configured, monitoring information can be sent to a malicious host. The malicious host can then use this information to plan an attack.

ESX includes an SNMP agent that can send notifications (traps and informs) and receive `GET`, `GETBULK`, and `GETNEXT` requests. SNMP is not activated by default. SNMP must be configured on each ESX host. You can use ESXCLI, PowerCLI, or the vSphere Web Services SDK for configuration.

See the *vSphere Monitoring and Performance* documentation for detailed information about configuring SNMP, including SNMP v3. SNMP v3 provides stronger security than SNMP v1 or SNMP v2c, including key authentication and encryption. See *ESXCLI Reference* for more information about the `esxcli system snmp` command options.

1. To determine whether SNMP is used, run the following command.

   ```
   esxcli system snmp get
   ```

2. To activate SNMP, run the following command.

   ```
   esxcli system snmp set --enable true
   ```

3. To deactivate SNMP, run the following command.

```
esxcli system snmp set --enable false
```

vSphere Networking Security Best Practices

Following networking security best practices helps ensure the integrity of your vSphere deployment.

General vSphere Networking Security Recommendations

Following general network security recommendations is the first step in securing your vSphere networking environment. You can then move on to special areas, such as securing the network with firewalls or using IPsec.

Recommendations to Secure a vSphere Networking Environment

- Spanning Tree Protocol (STP) detects and prevents loops from forming in the network topology. VMware virtual switches prevent loops in other ways, but do not support STP directly. When network topology changes occur, some time is required (30–50 seconds) while the network relearns the topology. During that time, no traffic is allowed to pass. To avoid these problems, network vendors have created features for switch ports to continue forwarding traffic. For more information, see the VMware knowledge base article at https://kb.vmware.com/s/article/1003804. Consult your network vendor documentation for the proper network and networking hardware configurations.
- Ensure that Netflow traffic for a Distributed Virtual Switch is only sent to authorized collector IP addresses. Netflow exports are not encrypted and can contain information about the virtual network. This information increases the potential for sensitive information to be viewed and captured in transit by attackers. If Netflow export is required, verify that all Netflow target IP addresses are correct.
- Ensure that only authorized administrators have access to virtual networking components by using the role-based access controls. For example, give virtual machine administrators only access to port groups in which their virtual machines reside. Give network administrators access to all virtual networking components but no access to virtual machines. Limiting access reduces the risk of misconfiguration, whether accidental or malicious, and enforces key security concepts of separation of duties and least privilege.
- Ensure that port groups are not configured to the value of the native VLAN. Physical switches are often configured with a native VLAN, and that native VLAN is often VLAN 1 by default. ESX does not have a native VLAN. Frames with VLAN specified in the port group have a tag, but frames with VLAN not specified in the port group are not tagged. This can cause a problem because virtual machines that are tagged with a 1 end up belonging to native VLAN of the physical switch.

  For example, frames on VLAN 1 from a Cisco physical switch are untagged because VLAN 1 is the native VLAN on that physical switch. However, frames from the ESX host that are specified as VLAN 1 are tagged with a 1. As a result, traffic from the ESX host that is destined for the native VLAN is not routed correctly because it is tagged with a 1 instead of being untagged. Traffic from the physical switch that is coming from the native VLAN is not visible because it is not tagged. If the ESX virtual switch port group uses the native VLAN ID, traffic from virtual machines on that port is not visible to the native VLAN on the switch because the switch is expecting untagged traffic.
- Ensure that port groups are not configured to VLAN values reserved by upstream physical switches. Physical switches reserve certain VLAN IDs for internal purposes and often disallow traffic configured to these values. For example, Cisco Catalyst switches typically reserve VLANs 1001–1024 and 4094. Using a reserved VLAN might result in a denial of service on the network.
- Ensure that port groups are not configured to VLAN 4095 except for Virtual Guest Tagging (VGT). Setting a port group to VLAN 4095 activates VGT mode. In this mode, the virtual switch passes all network frames to the virtual machine without modifying the VLAN tags, leaving it to the virtual machine to deal with them.
- Restrict port-level configuration overrides on a distributed virtual switch. Port-level configuration overrides are deactivated by default. When overrides are activated, you can use different security settings for a virtual machine than the port-group level settings. Certain virtual machines require unique configurations, but monitoring is essential. If overrides are not monitored, anyone who gains access to a virtual machine with a less secure distributed virtual switch configuration might attempt to exploit that access.
- Ensure that distributed virtual switch port mirror traffic is sent only to authorized collector ports or VLANs. A vSphere Distributed Switch can mirror traffic from one port to another to allow packet capture devices to collect specific traffic flows. Port mirroring sends a copy of all specified traffic in unencrypted format. This mirrored traffic contains the full data in the packets captured and can result in total compromise of that data if misdirected. If port mirroring is required, verify that all port mirror destination VLAN, port, and uplink IDs are correct.

Labeling vSphere Networking Components

Identifying the different components of your vSphere networking architecture is critical and helps ensure that no errors are introduced as your network expands.

Follow these best practices:

- Ensure that port groups are configured with a clear network label. These labels serve as a functional descriptor for the port group and help you identify each port group's function as the network becomes more complex.
- Ensure that each vSphere Distributed Switch has a clear network label that indicates the function or IP subnet of the switch. This label serves as a functional descriptor for the switch, just as physical switches require a host name. For example, you can label the switch as internal to show that it is for internal networking. You cannot change the label for a standard virtual switch.

Document and Check the vSphere VLAN Environment

Check your VLAN environment regularly to avoid addressing problems. Fully document the VLAN environment and ensure that VLAN IDs are used only once. Your documentation can help with troubleshooting and is essential when you want to expand the environment.

1. Ensure that all vSwitch and VLANS IDs are fully documented.

   If you are using VLAN tagging on a virtual switch, the IDs must correspond to the IDs on external VLAN-aware upstream switches. If VLAN IDs are not tracked completely, mistaken reuse of IDs might allow for traffic between the wrong physical and virtual machines. Similarly, if VLAN IDs are wrong or missing, traffic between physical and virtual machines might be blocked where you want traffic to pass.

2. Ensure that VLAN IDs for all distributed virtual port groups (dvPortgroup instances) are fully documented.

   If you are using VLAN tagging on a dvPortgroup the IDs must correspond to the IDs on external VLAN-aware upstream switches. If VLAN IDs are not tracked completely, mistaken reuse of IDs might allow for traffic between the wrong physical and virtual machines. Similarly, if VLAN IDs are wrong or missing, traffic between physical and virtual machines might be blocked where you want traffic to pass.

3. Ensure that private VLAN IDs for all distributed virtual switches are fully documented.

   Private VLANs (PVLANs) for distributed virtual switches require primary and secondary VLAN IDs. These IDs must correspond to the IDs on external PVLAN-aware upstream switches. If VLAN IDs are not tracked completely, mistaken reuse of IDs might allow for traffic between the wrong physical and virtual machines. Similarly, if PVLAN IDs are wrong or missing, traffic between physical and virtual machines might be blocked where you want traffic to pass.

4. Verify that VLAN trunk links are connected only to physical switch ports that function as trunk links.

   When connecting a virtual switch to a VLAN trunk port, you must properly configure both the virtual switch and the physical switch at the uplink port. If the physical switch is not properly configured, frames with the VLAN 802.1q header are forwarded to a switch that not expecting their arrival.

Adopting Network Isolation Practices in vSphere

Network isolation practices bolster network security in your vSphere environment.

Isolate the vSphere Management Network

The vSphere management network provides access to the vSphere management interface on each component. Services running on the management interface provide an opportunity for an attacker to gain privileged access to the systems. Remote attacks are likely to begin with gaining access to this network. If an attacker gains access to the management network, it provides the staging ground for further intrusion.

Strictly control access to management network by protecting it at the security level of the most secure VM running on an ESX host or cluster. No matter how the management network is restricted, administrators must have access to this network to configure the ESX hosts and vCenter system.

Place the vSphere management port group in a dedicated VLAN on a common standard switch. Production (VM) traffic can share the standard switch if the vSphere management port group's VLAN is not used by production VMs.

Check that the network segment is not routed, except to networks where other management-related entities are found. Routing a network segment might make sense for vSphere Replication. In particular, make sure that production VM traffic cannot be routed to this network.

Strictly control access to management functionality by using one of the following approaches.

- To access the management network in especially sensitive environments, configure a controlled gateway or other controlled method. For example, require that administrators connect to the management network through a VPN. Allow access to the management network only to trusted administrators.
- Configure bastion hosts that run management clients.

Isolate Storage Traffic

Ensure that IP-based storage traffic is isolated. IP-based storage includes iSCSI and NFS. Virtual machines might share virtual switches and VLANs with the IP-based storage configurations. This type of configuration might expose IP-based storage traffic to unauthorized virtual machine users.

IP-based storage frequently is not encrypted. Anyone with access to this network can view IP-based storage traffic. To restrict unauthorized users from viewing IP-based storage traffic, logically separate the IP-based storage network traffic from the production traffic. Configure the IP-based storage adapters on separate VLANs or network segments from the VMkernel management network to limit unauthorized users from viewing the traffic.

Isolate vMotion Traffic

vMotion migration information is transmitted in plain text. Anyone with access to the network over which this information flows can view it. Potential attackers can intercept vMotion traffic to obtain the memory contents of a VM. They might also stage an MITM attack in which the contents are modified during migration.

Separate vMotion traffic from production traffic on an isolated network. Set up the network to be nonroutable, that is, make sure that no layer-3 router is spanning this and other networks, to prevent outside access to the network.

Use a dedicated VLAN on a common standard switch for the vMotion port group. Production (VM) traffic can use the same standard switch if the vMotion port group's VLAN is not used by production virtual machines.

Isolate vSAN Traffic

When configuring your vSAN network, isolate vSAN traffic on its own Layer 2 network segment. You can perform this isolation by using dedicated switches or ports, or by using a VLAN.

Use Virtual Switches with the vSphere Network Appliance API Only If Required

Do not configure your host to send network information to a virtual machine unless you are using products that use the vSphere Network Appliance API (DvFilter). If the vSphere Network Appliance API is enabled, an attacker might attempt to connect a virtual machine to the filter. This connection might provide access to the network of other virtual machines on the host.

If you are using a product that uses this API, verify that the host is configured correctly. See the sections on DvFilter in the *Developing and Deploying vSphere Solutions, vServices, and ESX Agents* documentation. If your host is set up to use the API, make sure that the value of the `Net.DVFilterBindIpAddress` parameter matches the product that uses the API.

1. Browse to the host in the vSphere Client inventory.
2. Click **Configure**.
3. Under System, click **Advanced System Settings**.
4. Scroll down to `Net.DVFilterBindIpAddress` and verify that the parameter has an empty value.

   The order of parameters is not strictly alphabetical. Enter `DVFilter` in the Filter text box to display all related parameters.
5. Verify the setting.
   - If you are not using DvFilter settings, make sure that the value is blank.

- If you are using DvFilter settings, make sure that the value of the parameter is correct. The value must match the value that the product that uses the DvFilter is using.

Best Practices Involving Multiple vSphere Components

Some security best practices, such as setting up PTP or NTP in your environment, affect more than one vSphere component. Consider these recommendations when configuring your environment.

See Securing ESX Hosts and Securing Virtual Machines for related information.

Synchronizing Clocks on the vSphere Network

Verify that all components on the vSphere network have their clocks synchronized. If the clocks on the physical ESX servers in your vSphere network are not synchronized, SSL certificates and SAML tokens, which are time-sensitive, might not be recognized as valid in communications between network machines.

Unsynchronized clocks can result in authentication problems, which can cause the installation to fail or prevent the vCenter `vmware-vpxd` service from starting.

Time inconsistencies in vSphere can cause the first boot of a component in your environment to fail at different services depending on where in the environment time is not accurate and when the time is synchronized. Problems most commonly occur when the target ESX host for the destination vCenter is not synchronized with NTP or PTP. Similarly, issues can arise if the destination vCenter migrates to an ESX host set to a different time due to fully automated DRS.

vSphere supports three mechanisms for clock synchronization. To achieve optimal results, you must select the protocol that is best suited to your specific environment.

Table 220: Clock Synchronization Protocols on the vSphere Network

Feature	Network Time Protocol (NTP)	Software-Based Precision Time Protocol (PTP)	Hardware-Based Precision Time Protocol
Accuracy	Milliseconds level	10 to hundreds of microseconds	Sub-microsecond to nanoseconds
Timestamping	Software-based (OS layer)	Software-based (VMM layer)	Hardware-based (NIC/PHY layer)
Network Type	WAN and LAN	LAN, localized network	LAN, localized network
Deploy Cost	Low (runs on commodity hardware)	High (requires PTP-capable switches and grandmaster)	High (requires PTP-capable hardware)

NTP

NTP offers broad adoption and low hardware requirements. Kernel scheduling jitter, network queuing delay, and asymmetry between the send and receive paths limit NTP accuracy. On a local network, NTP achieves accuracy on the order of a few milliseconds. Over the Internet, NTP achieves accuracy on the order of tens of milliseconds.

NTP exchanges precision for ubiquity and simplicity. Choose NTP when millisecond-level accuracy is sufficient and special hardware is undesirable or unavailable.

Software-based PTP

Software-based PTP is a timestamping method that captures timestamps in the kernel networking stack at the moment the stack hands a packet to, or receives a packet from, the network driver. This approach avoids some of the scheduling jitter that affects pure userspace NTP daemons.

Software-based PTP achieves better accuracy than NTP, but its accuracy remains subject to variability from the following sources, all of which occur between the physical arrival of a packet and the point of timestamping:

- Interrupt handling
- Driver queuing
- Operating system scheduling

Software-based PTP occupies a middle ground between NTP and hardware-based PTP. By moving timestamping closer to the network stack, software-based PTP improves on NTP accuracy while still running on ordinary hardware. However, software-based PTP remains vulnerable to operating-system-induced jitter.

Hardware-based PTP

Hardware-based PTP is a timestamping method that captures packet timestamps directly in the network interface card (NIC), at or near the physical layer. Hardware-based PTP uses a dedicated PTP Hardware

Clock (PHC) that operates independently of the operating system clock and of any software scheduling delay. As a result, hardware-based PTP can achieve sub-microsecond, and even nanosecond-level, accuracy.

On an ESX host, the system passes the dedicated PTP-capable NIC through to a lightweight TRX VM that synchronizes the host clock. The ESX host and other guest virtual machines then benefit from the improved accuracy.

Hardware-based PTP delivers the highest accuracy of the three timestamping methods because the NIC timestamps packets independently of software scheduling. This accuracy comes at the cost of requiring PTP-aware network hardware end-to-end and, for best results, PTP-aware switches.

To avoid time synchronization issues, ensure that the following is correct before installing, migrating, or upgrading a vCenter instance.

- The target ESX host where the destination vCenter is to be deployed is synchronized to NTP or PTP.
- The ESX host running the source vCenter is synchronized to NTP or PTP.

To learn how to edit the time configuration for a host by using the vSphere Client, see Configuring the Time on Your ESX Host.

To learn how to configure guest virtual machines with hardware-based PTP clock synchronization, see How do I Add a Precision Clock Device to a Virtual Machine.

To synchronize ESX clocks with an NTP or a PTP server, you can use the VMware Host Client. For information about editing the time configuration of an ESX host, see topic Edit the Time Configuration of an ESX Host in the VMware Host Client in the *vSphere Single Host Management - VMware Host Client* documentation.

To learn how to change time synchronization settings for vCenter, see topic Configuring Time Synchronization Settings in vCenterin the *vCenter Configuration* documentation.

Storage Security Best Practices

Follow best practices for storage security, as outlined by your storage security provider. You can also take advantage of CHAP and mutual CHAP to secure iSCSI storage, mask and zone SAN resources, and configure Kerberos credentials for NFS 4.1.

See also the *Administering VMware vSAN* documentation.

Securing iSCSI Storage

The storage you configure for a host might include one or more storage area networks (SANs) that use iSCSI. When you configure iSCSI on a host, you can take measures to minimize security risks.

iSCSI supports accessing SCSI devices and exchanging data by using TCP/IP over a network port rather than through a direct connection to a SCSI device. An iSCSI transaction encapsulates blocks of raw SCSI data in iSCSI records and transmits the data to the requesting device or user.

iSCSI SANs support efficient use of the existing Ethernet infrastructure to provide hosts access to storage resources that they can dynamically share. iSCSI SANs are an economical storage solution for environments that rely on a common storage pool to serve many users. As with any networked system, your iSCSI SANs can be subject to security breaches.

Note:
The requirements and procedures for securing an iSCSI SAN are similar for hardware iSCSI adapters associated with hosts and for iSCSI configured directly through the host.

Securing iSCSI Devices

To secure iSCSI devices, require that the ESX host, or initiator, can authenticate to the iSCSI device, or target, whenever the host attempts to access data on the target LUN.

Authentication ensures that the initiator has the right to access a target. You grant this right when you configure authentication on the iSCSI device.

ESX does not support Secure Remote Protocol (SRP), or public-key authentication methods for iSCSI. You can use Kerberos only with NFS 4.1.

ESX supports both CHAP and Mutual CHAP authentication. The *vSphere Storage* documentation explains how to select the best authentication method for your iSCSI device and how to set up CHAP.

Ensure uniqueness of CHAP secrets. Set up a different mutual authentication secret for each host. If possible, set up a different secret for each client that to the ESX host. Unique secrets ensure that an attacker cannot create another arbitrary host and authenticate to the storage device even if one host is compromised. With a shared secret, compromise of one host might allow an attacker to authenticate to the storage device.

Protecting an iSCSI SAN

When you plan your iSCSI configuration, take measures to improve the overall security of the iSCSI SAN. Your iSCSI configuration is only as secure as your IP network, so by enforcing good security standards when you set up your network, you help safeguard your iSCSI storage.

The following are some specific suggestions for enforcing good security standards.

Protect Transmitted Data

A primary security risk in iSCSI SANs is that an attacker might sniff transmitted storage data.

Take additional measures to prevent attackers from easily seeing iSCSI data. Neither the hardware iSCSI adapter nor ESX iSCSI initiator encrypts the data that they transmit to and from the targets, making the data more vulnerable to sniffing attacks.

Allowing your virtual machines to share standard switches and VLANs with your iSCSI configuration potentially exposes iSCSI traffic to misuse by a virtual machine attacker. To help ensure that intruders cannot listen to iSCSI transmissions, make sure that none of your virtual machines can see the iSCSI storage network.

If you use a hardware iSCSI adapter, you can accomplish this by making sure that the iSCSI adapter and ESX physical network adapter are not inadvertently connected outside the host by virtue of sharing a switch or some other means. If you configure iSCSI directly through the ESX host, you can accomplish this by configuring iSCSI storage through a different standard switch than the one used by your virtual machines.

In addition to protecting the iSCSI SAN by giving it a dedicated standard switch, you can configure your iSCSI SAN on its own VLAN to improve performance and security. Placing your iSCSI configuration on a separate VLAN ensures that no devices other than the iSCSI adapter have visibility into transmissions within the iSCSI SAN. Also, network congestion from other sources cannot interfere with iSCSI traffic.

Secure iSCSI Ports

When you run iSCSI devices, ESX does not open any ports that listen for network connections. This measure reduces the chances that an intruder can break into ESX through spare ports and gain control over the host. Therefore, running iSCSI does not present any additional security risks at the ESX end of the connection.

Any iSCSI target device that you run must have one or more open TCP ports to listen for iSCSI connections. If any security vulnerabilities exist in the iSCSI device software, your data can be at risk through no fault of ESX. To lower this risk, install all security patches that your storage equipment manufacturer provides and limit the devices connected to the iSCSI network.

Masking and Zoning SAN Resources

You can use zoning and LUN masking to separate SAN activity and restrict access to storage devices.

You can protect access to storage in your vSphere environment by using zoning and LUN masking with your SAN resources. For example, you might manage zones defined for testing independently within the SAN so they do not interfere with activity in the production zones. Similarly, you might set up different zones for different departments.

When you set up zones, take into account any host groups that are set up on the SAN device.

Zoning and masking capabilities for each SAN switch and disk array and the tools for managing LUN masking are vendor specific.

See your SAN vendor's documentation and the *vSphere Storage* documentation.

Using Kerberos for NFS 4.1

With NFS version 4.1, ESX supports the Kerberos authentication mechanism.

The RPCSEC_GSS Kerberos mechanism is an authentication service. It allows an NFS 4.1 client installed on ESX to prove its identity to an NFS server before mounting an NFS share. The Kerberos security uses cryptography to work across an insecure network connection.

The ESX implementation of Kerberos for NFS 4.1 provides two security models, krb5 and krb5i, that offer different levels of security.

- Kerberos for authentication only (krb5) supports identity verification.
- Kerberos for authentication and data integrity (krb5i), in addition to identity verification, provides data integrity services. These services help to protect the NFS traffic from tampering by checking data packets for any potential modifications.

Kerberos supports cryptographic algorithms that prevent unauthorized users from gaining access to NFS traffic. The NFS 4.1 client on ESX attempts to use either the AES256-CTS-HMAC-SHA1-96 or AES128-CTS-HMAC-SHA1-96 algorithm to access a share on the NAS server. Before using your NFS 4.1 datastores, make sure that AES256-CTS-HMAC-SHA1-96 or AES128-CTS-HMAC-SHA1-96 are enabled on the NAS server.

The following table compares Kerberos security levels that ESX supports.

Table 221: Types of Kerberos Security

		ESX 6.5 and later
Kerberos for authentication only (krb5)	Integrity checksum for RPC header	Yes with AES
	Integrity checksum for RPC data	No
Kerberos for authentication and data integrity (krb5i)	Integrity checksum for RPC header	Yes with AES
	Integrity checksum for RPC data	Yes with AES

When you use Kerberos authentication, the following considerations apply:

- ESX uses Kerberos with the Active Directory domain.
- As a vSphere administrator, you specify Active Directory credentials to provide access to NFS 4.1 Kerberos datastores for an NFS user. A single set of credentials is used to access all Kerberos datastores mounted on that host.
- When multiple ESX hosts share the NFS 4.1 datastore, you must use the same Active Directory credentials for all hosts that access the shared datastore. To automate the assignment process, set the user in host profiles and apply the profile to all ESX hosts.
- You cannot use two security mechanisms, AUTH_SYS and Kerberos, for the same NFS 4.1 datastore shared by multiple hosts.

See the *vSphere Storage* documentation for step-by-step instructions.

Verify That Sending Host Performance Data to Guests Is Deactivated

vSphere includes virtual machine performance counters on Windows operating systems where VMware Tools is installed. Performance counters allow virtual machine owners to do accurate performance analysis within the guest operating system. By default, vSphere does not expose host information to the guest virtual machine.

By default, the capability to send host performance data to a virtual machine is deactivated. This default setting prevents a virtual machine from obtaining detailed information about the physical host. If a security breach of the virtual machine occurs, the setting does not make host data available to the attacker.

Note:
The following procedure illustrates the basic process. Consider using ESXCLI or VMware PowerCLI commands for performing this task on all hosts simultaneously.

1. On the ESX system that hosts the virtual machine, browse to the VMX file.
   Virtual machine configuration files are located in the `/vmfs/volumes/datastore` directory, where *`datastore`* is the name of the storage device where the virtual machine files are stored.
2. In the VMX file, verify that the following parameter is set.
   `tools.guestlib.enableHostInfo=FALSE`
3. Save and close the file.

You cannot retrieve performance information about the host from inside the guest virtual machine.

Setting Timeouts for the ESX Shell and the vSphere Client

To prevent intruders from using an idle session, set timeouts for the ESXi Shell and the vSphere Client.

ESX Shell Timeout

For the ESX Shell, you can set the following timeouts from the vSphere Client and from the Direct Console User Interface (DCUI).

Availability Timeout	The availability timeout setting is the amount of time that can elapse before you must log in after the ESX Shell is activated. After the timeout period, the service is deactivated and users are not allowed to log in.
Idle Timeout	The idle timeout is the amount of time that can elapse before the user is logged out of an idle interactive sessions. Changes to the idle timeout apply the next time a user logs in to the ESX Shell. Changes do not affect existing sessions.

Change the vSphere Client Timeout

You can configure the timeout intervals for vCenter operations. These intervals specify the amount of time after which the vSphere Client times out.

Prerequisites:

- Required privilege: **Global** > **Settings**

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **Advanced Settings**.
4. In **client.timeout.normal**, enter the timeout interval in seconds for normal operations.
   The value cannot be less than five (5).
5. In **client.timeout.long**, enter the timeout interval in minutes for long operations.
   The value cannot be less than five (5).
6. Click **Save**.
7. Restart the vCenter system for the changes to take effect.

vSphere TLS Configuration

Starting in 8.0 Update 3, vSphere supports TLS 1.3 and 1.2 through the use of TLS profiles. TLS profiles simplify the job of administering TLS parameters and also improve supportability.

vSphere 9.0 activates the default TLS profile, named COMPATIBLE, on ESX and vCenter hosts. The COMPATIBLE profile supports TLS 1.3, and some TLS 1.2 connections.

You can manage TLS profiles on ESX hosts either by using vSphere Configuration Profiles or `esxcli` commands. On vCenter hosts, you can manage TLS profiles by using the APIs. For example, you can use the Developer Center in the

vSphere Client. See *vSphere Automation SDKs Programming Guide* and *vSphere Automation REST API Programming Guide*.

vCenter and Envoy

vCenter runs two reverse proxy services:

- VMware reverse proxy service, `rhttpproxy`
- Envoy

Envoy is an open source edge and service proxy. Envoy owns port 443, and all incoming vCenter requests are routed through Envoy. The `rhttpproxy` serves as a configuration management server for Envoy. As a result, the TLS configuration is applied to `rhttpproxy`, which in turn sends the configuration to Envoy.

How vSphere Implements TLS Using TLS Profiles

vSphere 9.0 implements TLS 1.3 by grouping parameters, including protocol versions, groups (also called curves), and ciphers, into a single TLS profile. This TLS profile is applied system-wide. Using a single TLS profile eases administrative overhead of your hosts. You no longer need to manually configure individual TLS parameters, though that capability is still available if required. TLS profiles also significantly improve supportability. The grouping of parameters into TLS profiles simplifies the set of VMware verified TLS solutions from which to choose. On ESX, TLS profiles are integrated with vSphere Configuration Profiles.

The following ESX TLS profiles are provided:

- COMPATIBLE: The default profile. The exact mapping of the parameters in this profile can change from release to release but the profile is guaranteed to be compatible with all products and versions supported (currently N-2 versions). That is, an ESX host from release N using the COMPATIBLE profile can communicate with a host from release N-2.
- NIST_2024: A more restrictive profile that specifically supports the NIST 2024 standard. The exact mapping of parameters in this profile is guaranteed to satisfy the NIST 2024 standard across releases. This profile is guaranteed to be compatible only with current or newer releases, and not older releases.
- NIST_2024_TLS_13_ONLY: The same as NIST_2024, but with the additional restriction that only TLS 1.3 is allowed.
- MANUAL: Use this profile to create and test an ad hoc configuration in which you manually supply the TLS parameters. It is not guaranteed that a MANUAL profile functions error-free, because Broadcom does not test all possible combinations of parameters that are possible in the MANUAL profile. You must test a MANUAL profile, including across software upgrades. When you choose to use the MANUAL profile, the system behavior first defaults to the profile previously selected (COMPATIBLE, NIST_2024, or NIST_2024_TLS_1.3_ONLY), and remains as such until you make changes. You must use `esxcli` commands to manage the MANUAL TLS profile. See the help text that ships with `esxcli` for more information about changing the parameters in a MANUAL TLS profile.

When configuring the TLS profile to the desired state, you must reboot the ESX host or remediate the vLCM cluster in which the ESX host resides to apply changes.

The following tables show the details of TLS profiles for ESX and vCenter in vSphere 9.0. The Cipher List column shows the TLS ciphers for TLS 1.2 and below protocols. The Cipher Suites column shows the ciphers for the TLS 1.3 protocol.

Table 222: ESX TLS 1.3 Profiles

TLS Profile Name	TLS Protocol Versions	Cipher List	Cipher Suites	Curves	VMware Supported?
COMPATIBLE	TLS 1.3 and TLS 1.2	ECDHE+AESGCM:ECDHE+AES	TLS_AES_256_GCM_SHA384; TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes
NIST_2024	TLS 1.3 and TLS 1.2	ECDHE+AESGCM	TLS_AES_256_GCM_SHA384; TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes
NIST_2024_TLS_13_ONLY	TLS 1.3	N/A	TLS_AES_256_GCM_SHA384; TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes

TLS Profile Name	TLS Protocol Versions	Cipher List	Cipher Suites	Curves	VMware Supported?
MANUAL	Any	Any	Any	Any	Limited support

Notes:

- Supported settings (protocols, cipher list, cipher suites, and curves) represent at most what is supported.
- The NIST_2024 and NIST_2024_TLS_13_ONLY profiles apply to inbound connections only.

Starting in vSphere 9.0, all TLS profiles are FIPS compliant. The following ESX TLS 1.3 profiles are provided:

- COMPATIBLE: The default profile. The exact mapping of the parameters in this profile can change from release to release but the profile is guaranteed to be compatible with all products and versions supported (currently N-2 versions).
- NIST_2024: A more restrictive profile that specifically supports the NIST 2024 standard. The exact mapping of parameters in this profile is guaranteed to satisfy the NIST 2024 standard across releases. This profile is guaranteed to be compatible only with current or newer releases, and not older releases.
- NIST_2024_TLS_13_ONLY: The same as NIST 2024, but with the additional restriction that only TLS 1.3 is allowed. Note that there are some restrictions in using this profile:
  - vPostgres can't be fully configured to disable TLS 1.2.
  - Deploying hosts with autodeploy is not supported in this profile.

Table 223: vCenter TLS 1.3 Profiles

TLS Profile Name	TLS Protocol Versions	Cipher Suites	Curves	VMware Supported?
COMPATIBLE	TLS 1.3	TLS_AES_256_GCM_SHA384; TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes
	TLS 1.2	ECDHE-RSA-AES256-GCM-SHA384 ECDHE-RSA-AES128-GCM-SHA256 ECDHE-ECDSA-AES256-GCM-SHA384 ECDHE-ECDSA-AES128-GCM-SHA256 AES256-GCM-SHA384 AES128-GCM-SHA256 ECDHE-RSA-AES256-SHA ECDHE-RSA-AES128-SHA ECDHE-ECDSA-AES256-SHA ECDHE-ECDSA-AES128-SHA AES256-SHA AES128-SHA		
NIST_2024	TLS 1.3	TLS_AES_256_GCM_SHA384 TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes
	TLS 1.2	ECDHE-RSA-AES256-GCM-SHA384 ECDHE-RSA-AES128-GCM-SHA256 ECDHE-ECDSA-AES256-GCM-SHA384 ECDHE-ECDSA-AES128-GCM-SHA256		
NIST_2024_TLS_13_ONLY	TLS 1.3	TLS_AES_256_GCM_SHA384 TLS_AES_128_GCM_SHA256	prime256v1:secp384r1:secp521r1	Yes

Note:
You can switch back and forth between TLS profiles. For example, if you change from using the COMPATIBLE profile to using the NIST_2024 profile, you can switch back to using the COMPATIBLE profile. There is no limitation on switching back and forth between TLS profiles. For more information, see Managing vSphere TLS.

TLS and Inbound and Outbound Connections in ESX and vCenter

ESX 9.0 supports TLS 1.3 on both inbound (server) and outbound (client) connections. The ESX inbound (server) connections are of most concern, and where the more restrictive NIST_2024 profile applies.

For ESX, you can use the COMPATIBLE, NIST_2024, NIST_2024_TLS_13_ONLY, and MANUAL settings on inbound (server) connections. You can use the COMPATIBLE and MANUAL settings on outbound (client) connections.

vCenter TLS profiles apply their settings to both inbound and outbound connections.

Some vSphere services expose ports that accept TLS connections, while most services use the Reverse Proxy. All inbound connections accept TLS 1.2 and TLS 1.3 by default. Outbound connections support TLS 1.2 and TLS 1.3.

TLS and Lifecycle Management

Upgrading or migrating an ESX host or vCenter host to 9.0 enables the COMPATIBLE TLS profile by default. vSphere 9.0 supports TLS 1.3, and TLS 1.2 for bare minimum interoperability out of the box. In the future, upgrading to a later version of ESX or vCenter retains the current TLS profile in use as long as that profile has not been retired.

When upgrading to a new version, as a recommended best practice, first set the TLS profile to COMPATIBLE.

If you make local service-level edits before upgrading to vSphere 9.0, after the upgrade, the host is assigned the COMPATIBLE profile, which does not reflect those changes. To have the host reflect those changes, switch to using the MANUAL profile. See Change the TLS Profile of an ESX Host Using the vSphere Client or Change the TLS Profile of an ESX Host Using the CLI.

> **Warning:**
> The MANUAL TLS profile is not guaranteed to work error-free across upgrades. You must verify that an edited MANUAL TLS profile functions from one release to another, or switch to the COMPATIBLE TLS profile prior to upgrading.

Managing vSphere TLS

Starting in vSphere 8.0 Update 3, you can manage TLS profiles for ESX by using the vSphere Client, `esxcli` commands, or the APIs. For vCenter, you manage TLS profiles by using the APIs.

If you use vSphere Configuration Profiles, you can manage the TLS setting for ESX hosts at the vLCM cluster level. You can change the TLS setting for the cluster and remediate the cluster against this new configuration. For more information, see the chapter on managing vSphere Configuration Profiles in the *Managing Host and Cluster Lifecycle* documentation.

For standalone ESX hosts, and non-vLCM clusters, you must manage the TLS profile by using `esxcli` commands. See the *ESXCLI Concepts and Examples* documentation and the `esxcli` online help.

Currently, you can only manage vCenter TLS profiles by using the APIs. See *vSphere Automation SDKs Programming Guide* and *vSphere Automation REST API Programming Guide*.

View the TLS Profile of an ESX Host Using the vSphere Client

You can use the vSphere Client to view the TLS profile of an ESX host that is part of a vLCM cluster.

You have enabled vSphere Configuration Profiles and created a draft configuration for the cluster. See the *Managing Host and Cluster Lifecycle* documentation.

In vSphere Configuration Profiles, settings that are not explicitly configured use the default values from the appropriate profile. For TLS profiles, the default is COMPATIBLE.

To view the TLS profile of a stand-alone or non-vLCM cluster ESX host, see View the TLS Profile of an ESX Host Using the CLI.

1. In the vSphere Client, navigate to the vLCM cluster that you manage with a single image.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Settings** tab, click **system**.
4. Click **tls_client** or **tls_server** to view which TLS profile is defined in the current desired configuration document.

View the TLS Profile of an ESX Host Using the CLI

You can use the CLI to view the currently configured TLS profile of an ESX host.

Enable either SSH or the ESX Shell on the ESX host.

For standalone ESX hosts, and non-vLCM clusters, you must manage the TLS profile by using `esxcli` commands. For more information, see *ESXCLI Reference*. For ESX hosts in a vLCM cluster, you can use either vSphere Configuration Profiles or `esxcli` commands.

1. Connect to the ESX host.
   You can use SSH or the ESX Shell.
2. To view the currently configured TLS profile, run the following command.
   ```
   esxcli system tls [client | server] get
   ```
3. To view the parameters in the currently configured TLS profile, run the following command:
   ```
   esxcli system tls [client | server] get --show-profile-defaults
   ```

Change the TLS Profile of an ESX Host Using the vSphere Client

You can change the TLS profile of an ESX host. The default TLS profile is COMPATIBLE.

You have enabled vSphere Configuration Profiles and created a draft configuration for the cluster. See the *Managing Host and Cluster Lifecycle* documentation.

1. In the vSphere Client, navigate to a cluster that you manage with a single image.
2. On the **Configure** tab, click **Desired State** > **Configuration**.
3. On the **Settings** tab, click **system**.
4. Click either **tls_client** or **tls_server**.
   Depending on if the setting has been previously changed, click either **Configure Settings** or **Edit**.
5. Select the TLS profile from the drop-down.
6. Click **Save**.
7. Remediate the cluster against the draft configuration.
   a) To remediate the cluster against the draft configuration, on the **Draft** tab, click **Apply Changes**.
   b) Follow the steps in the **Remediate** wizard. For more information, see the *Managing Host and Cluster Lifecycle* documentation.

All the ESX hosts in the cluster are compliant with the desired configuration.

Change the TLS Profile of an ESX Host Using the CLI

You can change the TLS profile of an ESX host. The default TLS profile is COMPATIBLE.

Enable either SSH or the ESX Shell on the ESX host.

For standalone ESX hosts, and non-vLCM clusters, you must manage the TLS profile by using `esxcli` commands. For more information, see *ESXCLI Reference*. For ESX hosts in a vLCM cluster, you can use either vSphere Configuration Profiles or `esxcli` commands.

1. Connect to the ESX host.
   You can use either SSH or the ESX Shell.
2. Put the ESX host into maintenance mode.
3. To change the TLS profile, run the following command.
   ```
   esxcli system tls [client | server] set --profile [COMPATIBLE | NIST_2024 | MANUAL]
   ```
   Note:
   If you want to make changes to TLS parameters (either at the system-level or service-level), select the MANUAL profile.
4. Reboot the ESX host for the change to take effect.
5. After the ESX host reboots, take it out of maintenance mode.

Edit the Parameters in the MANUAL TLS Profile Using the CLI

You can edit the set of parameters in the MANUAL TLS profile. To change TLS parameters such as cipher list and cipher suite, you must first set the TLS profile to MANUAL.

Enable either SSH or the ESX Shell on the ESX host.

Change the TLS profile to MANUAL. See either Change the TLS Profile of an ESX Host Using the vSphere Client or Change the TLS Profile of an ESX Host Using the CLI.

Warning:

Broadcom does not test all possible combinations of TLS settings that can be configured in the MANUAL TLS profile. Before enabling custom settings using the MANUAL profile in your production environment, you should test these settings to ensure that your TLS servers and clients operate correctly, and that lifecycle operations such as upgrades can be completed successfully with these settings.

You must administer parameters in the MANUAL TLS profile by using `esxcli` commands. Administering the MANUAL TLS profile parameters is not integrated with vSphere Configuration Profiles.

You cannot set TLS parameters for individual vSphere services. The changes that you make by using the MANUAL TLS profile are applied at the system level.

1. Connect to the ESX host.
   You can use either SSH or the ESX Shell.
2. Put the ESX host into maintenance mode.
3. Ensure that the TLS profile is MANUAL.
   ```
   esxcli system tls [client | server] get
   ```
4. To change the parameters, run any of the following commands.
   ```
   esxcli system tls [client | server] set --cipher-list=str
   esxcli system tls [client | server] set --cipher-suite=str
   esxcli system tls [client | server] set --groups=str
   esxcli system tls [client | server] set --protocol-versions=str
   ```
   where *str* is a string in OpenSSL-style that is colon-, comma-, or space-delimited. For example: `--cipher-list=ECDHE+AESGCM:ECDHE+AES`

   For more information, run the following command:
   ```
   esxcli system tls [client | server] set --help
   ```
5. Reboot the ESX host for the change to take effect.
6. After the ESX host reboots, take it out of maintenance mode.

   The following example first sets the TLS profile to MANUAL then sets a more restrictive set of curves (groups). A reboot would be required to put the changes into effect.

```
[root@host1] esxcli system tls server get
   Profile: COMPATIBLE
   Cipher List: <profile default>
   Cipher Suite: <profile default>
   Groups: <profile default>
   Protocol Versions: <profile default>
   Reboot Required: false
[root@host1] esxcli system tls server set --profile MANUAL
[root@host1] esxcli system tls server get
   Profile: MANUAL
   Cipher List: ECDHE+AESGCM:ECDHE+AES
   Cipher Suite: TLS_AES_128_GCM_SHA256:TLS_AES_256_GCM_SHA384
   Groups: prime256v1:secp384r1:secp521r1
   Protocol Versions: tls1.2,tls1.3
   Reboot Required: true
[root@host1] esxcli system tls server set --groups=prime256v1:secp384r1
[root@host1] esxcli system tls server get
   Profile: MANUAL
   Cipher List: TLS_AES_128_CCM_SHA256
   Cipher Suite: TLS_AES_128_GCM_SHA256:TLS_AES_256_GCM_SHA384
   Groups: prime256v1:secp384r1
   Protocol Versions: tls1.2,tls1.3
   Reboot Required: true
```

Manage the TLS Profile of a vCenter Host

You use the APIs to view and change the TLS profile for a vCenter host.

You can use various ways to execute HTTP requests. This task shows how to use the Developer Center in the vSphere Client to manage TLS profiles. See *VMware vCenter Server Management Programming Guide* for more information about using APIs to manage the vCenter appliance.

Note:

The TLS profiles described here are the replacement for the deprecated TLS reconfigurator functionality.

1. Log in to the vCenter system with the vSphere Client.
2. From the Menu, select **Developer Center**.
3. Click **API Explorer**.
4. From the **Select API** drop-down, select **appliance**.

Note:

In previous versions, the APIs contained a `fipsEnforce` field which specified whether the associated profile fully supports FIPS or not. In vSphere 9.0 all profiles are FIPS compliant and this field has been deprecated. It is still present in the responses for the various GET APIs but is read-only and always set to true.

Table 224: vCenter TLS APIs

Option	API Category	Associated Action
Gets the list of all TLS profiles and their configuration.	tls/profiles/	GET
Gets the parameters of a specific TLS profile.	tls/profiles/{id}	GET
Gets the name of the current TLS profile configured globally.	tls/profiles/global/	GET

Option	API Category	Associated Action
Sets one of the standard profiles that you specify globally.	tls/profiles/global/	PUT **Note:** This action restarts the vCenter services.
Gets the parameters of the current TLS profile configured globally.	tls/manual-parameters/global	GET
Switch to manual parameters globally. Pass a specification to manually set parameters such as protocol versions, curves, fipsEnforced flag, and ciphers.	tls/manual-parameters/global	PUT
Gets the complete list of services integrated with TLS profiles.	tls/manual-parameters/services/	GET
Gets the current TLS parameters for the service specified by `{id}`. Replace `{id}` with the service name.	tls/manual-parameters/services/`{id}`	GET
Set manual TLS parameters for the service specified by `{id}`. Replace `{id}` with the service name.	tls/manual-parameters/services/`{id}`	PUT
Delete TLS parameters for the service specified by `{id}`. Replace `{id}` with the service name.	tls/manual-parameters/services/`{id}`	DELETE

5. Execute the desired command.

Defined Privileges

The following tables list the default privileges that, when selected for a role, can be paired with a user and assigned to an object.

vCenter Server allows fine-grained control over authorization with permissions and roles. When you assign a permission to an object in the vCenter Server object hierarchy, you specify which user or group has which privileges on that object. To specify the privileges, you use roles, which are sets of privileges. For more information, see:

- Managing Permissions for vCenter Components
- Using vCenter Roles to Assign Privileges

When setting permissions, verify all the object types are set with appropriate privileges for each particular action. Some operations require access permission at the root folder or parent folder in addition to access to the object being manipulated. Some operations require access or performance permission at a parent folder and a related object. See also Using Privilege Recorder.

vCenter extensions might define additional privileges not listed here. Refer to the documentation for the extension for more information on those privileges.

Alarms Privileges

Alarms privileges control the ability to create, modify, and respond to alarms on inventory objects.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 225: Alarms Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Acknowledge alarm	Allows suppression of all alarm actions on all triggered alarms.	Object on which an alarm is defined	Alarm.Acknowledge
Create alarm	Allows creation of a new alarm. When creating alarms with a custom action, privilege to perform the action is verified when the user creates the alarm.	Object on which an alarm is defined	Alarm.Create
Disable alarm action	Allows stopping an alarm action from occurring after an alarm has been triggered. The alarm itself is not deactivated.	Object on which an alarm is defined	Alarm.DisableActions
Disable or enable alarm on entity	Allows activating or deactivating a particular alarm on a particular target type.	Object on which the alarm can trigger	Alarm.ToggleEnableOnEntity
Modify alarm	Allows changing the properties of an alarm.	Object on which an alarm is defined	Alarm.Edit
Remove alarm	Allows deletion of an alarm.	Object on which an alarm is defined	Alarm.Delete
Set alarm status	Allows changing the status of the configured event alarm. The status can change to **Normal**, **Warning**, or **Alert**.	Object on which an alarm is defined	Alarm.SetStatus

Auto Deploy and Image Profile Privileges

Auto Deploy privileges control who can perform different tasks on Auto Deploy rules, and who can associate a host. Auto Deploy privileges also allow you to control who can create or edit an image profile.

The table describes privileges that determine who can manage Auto Deploy rules and rule sets and who can create and edit image profiles. For more information about Auto Deploy, see the *VMware ESX Installation and Setup* documentation.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 226: Auto Deploy Privileges

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
• **Host** – **AssociateMachine**	Allows users to associate a host with a machine.	vCenter	AutoDeploy.Host.AssociateMachine
• **Image Profile** – **Create** – **Edit**	**Create** allows creation of image profiles.**Edit** allows editing of image profiles.	vCenter	AutoDeploy.Profile.Create AutoDeploy.Profile.Edit
• **Rule** – **Create** – **Edit** – **Delete**	**Create** allows creation of Auto Deploy rules.**Edit** allows editing of Auto Deploy rules.**Delete** allows deletion of Auto Deploy rules.	vCenter	AutoDeploy.Rule.Create AutoDeploy.Rule.Edit AutoDeploy.Rule.Delete

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
• **RuleSet** – **Activate** – **Edit**	**Activate** allows activation of Auto Deploy rule sets.**Edit** allows editing of Auto Deploy rule sets.	vCenter	AutoDeploy.RuleSet.Activate AutoDeploy.RuleSet.Edit

Certificates Privileges

Certificates privileges control which users can manage ESX certificates.

This privilege determines who can perform certificate management forESX hosts. See Required Privileges for Certificate Management Operations in the *vSphere Authentication* documentation for information onvCenter certificate management.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 227: Host Certificates Privileges

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
Manage Certificates	Allows certificate management forESX hosts.	vCenter	Certificate.Manage

Certificate Authority Privileges

Certificate authority privileges control aspects of VMware Certificate Authority (VMCA) certificates.

Table 228: Certificate Authority Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create/Delete (Admins priv).	Allows full administrative-level access for managing vCenter certificates.	vCenter	CertificateAuthority.Administer
Create/Delete (below Admins priv).	Allows viewing the VMCA root certificate in the Certificate Management page in the vSphere Client.	vCenter	CertificateAuthority.Manage

Certificate Management Privileges

Certificate management privileges control which users can manage vCenter certificates.

Table 229: Certificate Management Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create/Delete (Admins priv).	Allows full administrative-level access to various internal APIs and functionality for vCenter certificate-related operations.	vCenter	CertificateManagement.Administer
Create/Delete (below Admins priv).	Allows reduced administrative access to various internal APIs and functionality. This privilege restricts certificate related operations so that the user cannot escalate non-administrator privileges. Allowed operations are: • Generating certificate signing requests • Creating and retrieving Trusted Root chains • Deleting Trusted Root chains created by a user with the privilege **Certificate Management** > **Create/Delete (below Admins priv).** • Retrieving Machine SSL certificates • Retrieving the signing certificate chains for validating tokens issued by vCenter	vCenter	CertificateManagement.Manage

Cns Privileges

Cloud Native Store (Cns) privileges control which users can access the Cloud Native Storage UI.

Table 230: Cns Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Searchable	Allows storage administrator to view the Cloud Native Storage UI.	Root vCenter	Cns.Searchable

Compute Policy Privileges

Compute policy privileges control the ability to manage compute policies.

Table 231: Compute Policy Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create and Delete Compute Policy	Allows creating and deleting compute policies.	Root vCenter	ComputePolicy.Manage

Content Library Privileges

Content Libraries provide simple and effective management for virtual machine templates and vApps. Content library privileges control who can view or manage different aspects of content libraries.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

> **Note:**
> Inheritance of permissions for content libraries works in the context of a single vCenter instance. However, content libraries are not direct children of avCenter system from an inventory perspective. The direct parent for content libraries is the global root. This relationship means that if you set a permission at a vCenter level and propagate it to the children objects, the permission applies to data centers, folders, clusters, hosts, virtual machines, and so on, but does not apply to the content libraries that you see and operate with in this vCenter instance. To assign a permission on a content library, an Administrator must grant the permission to the user as a global permission. Global permissions support assigning privileges across solutions from a global root object.

Table 232: Content Library Privileges

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
Add library item	Allows addition of items in a library.	Library	ContentLibrary.AddLibraryItem
Add root certificate to trust store	Allows addition of root certificates to the Trusted Root Certificates Store.	vCenter	ContentLibrary.AddCertToTrustStore
Check in a template	Allows checking in of templates.	Library	ContentLibrary.CheckInTemplate
Check out a template	Allows checking out of templates.	Library	ContentLibrary.CheckOutTemplate
Create a subscription for a published library	Allows creation of a library subscription.	Library	ContentLibrary.AddSubscription
Create local library	Allows creation of local libraries on the specifiedvCenter system.	vCenter	ContentLibrary.CreateLocalLibrary
Create or delete a Harbor registry	Allows creation or deletion of the VMware Tanzu Harbor Registry service.	vCenter for creation. Registry for deletion.	ContentLibrary.ManageRegistry
Create subscribed library	Allows creation of subscribed libraries.	vCenter	ContentLibrary.CreateSubscribedLibrary
Create, delete or purge a Harbor registry project	Allows creation, deletion, or purging of VMware Tanzu Harbor Registry projects.	Registry	ContentLibrary.ManageRegistryProject
Delete library item	Allows deletion of library items.	Library. Set this permission to propagate to all library items.	ContentLibrary.DeleteLibraryItem
Delete local library	Allows deletion of a local library.	Library	ContentLibrary.DeleteLocalLibrary

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
Delete root certificate from trust store	Allows deletion of root certificates from the Trusted Root Certificates Store.	vCenter	ContentLibrary.DeleteCertFromTrustStore
Delete subscribed library	Allows deletion of a subscribed library.	Library	ContentLibrary.DeleteSubscribedLibrary
Delete subscription of a published library	Allows deletion of a subscription to a library.	Library	ContentLibrary.DeleteSubscription
Download files	Allows download of files from the content library.	Library	ContentLibrary.DownloadSession
Evict library item	Allows eviction of items. The content of a subscribed library can be cached or not cached. If the content is cached, you can release a library item by evicting it if you have this privilege.	Library. Set this permission to propagate to all library items.	ContentLibrary.EvictLibraryItem
Evict subscribed library	Allows eviction of a subscribed library. The content of a subscribed library can be cached or not cached. If the content is cached, you can release a library by evicting it if you have this privilege.	Library	ContentLibrary.EvictSubscribedLibrary
Import Storage	Allows a user to import a library item if the source file URL starts with `ds://` or `file://`. This privilege is deactivated for content library administrator by default. Because an import from a storage URL implies import of content, activate this privilege only if necessary and if no security concern exists for the user who performs the import.	Library	ContentLibrary.ImportStorage
Manage Harbor registry resources on specified compute resource	Allows management of VMware Tanzu Harbor Registry resources.	Compute cluster	ContentLibrary.ManageClusterRegistryResource

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
Probe subscription information	This privilege allows solution users and APIs to probe a remote library's subscription info including URL, SSL certificate, and password. The resulting structure describes whether the subscription configuration is successful or whether there are problems such as SSL errors.	Library	ContentLibrary.ProbeSubscription
Publish a library item to its subscribers	Allows publication of library items to subscribers.	Library. Set this permission to propagate to all library items.	ContentLibrary.PublishLibraryItem
Publish a library to its subscribers	Allows publication of libraries to subscribers.	Library	ContentLibrary.PublishLibrary
Read storage	Allows reading of content library storage.	Library	ContentLibrary.ReadStorage
Sync library item	Allows synchronization of library items.	Library. Set this permission to propagate to all library items.	ContentLibrary.SyncLibraryItem
Sync subscribed library	Allows synchronization of subscribed libraries.	Library	ContentLibrary.SyncLibrary
Type introspection	Allows a solution user or API to introspect the type support plug-ins for the content library service.	Library	ContentLibrary.TypeIntrospection
Update configuration settings	Allows you to update the configuration settings. No vSphere Client user interface elements are associated with this privilege.	Library	ContentLibrary.UpdateConfiguration
Update files	Allows you to upload content into the content library. Also allows you to remove files from a library item.	Library	ContentLibrary.UpdateSession
Update library	Allows updates to the content library.	Library	ContentLibrary.UpdateLibrary
Update library item	Allows updates to library items.	Library. Set this permission to propagate to all library items.	ContentLibrary.UpdateLibraryItem
Update local library	Allows updates of local libraries.	Library	ContentLibrary.UpdateLocalLibrary
Update subscribed library	Allows you to update the properties of a subscribed library.	Library	ContentLibrary.UpdateSubscribedLibrary

Privilege Name in thevSphere Client	Description	Required On	Privilege Name in the API
Update subscription of a published library	Allows updates of subscription parameters. Users can update parameters such as the subscribed library'svCenter instance specification and placement of its virtual machine template items.	Library	ContentLibrary.UpdateSubscription
View configuration settings	Allows you to view the configuration settings. No vSphere Client user interface elements are associated with this privilege.	Library	ContentLibrary.GetConfiguration

Cryptographic Operations Privileges

Cryptographic operations privileges control who can perform which type of cryptographic operation on which type of object.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 233: Cryptographic Operations Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Direct Access	Allows users access to encrypted resources. Users can export virtual machines, have NFC access to virtual machines, and open a console session to an encrypted virtual machine.	Virtual machine, host, or datastore	Cryptographer.Access
Add disk	Allows users to add a disk to an encrypted virtual machine.	Virtual machine	Cryptographer.AddDisk
Clone	Allows users to clone an encrypted virtual machine.	Virtual machine	Cryptographer.Clone
Decrypt	Allows users to decrypt a virtual machine or disk.	Virtual machine	Cryptographer.Decrypt
Encrypt	Allows users to encrypt a virtual machine or a virtual machine disk.	Virtual machine	Cryptographer.Encrypt

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Encrypt new	Allows users to encrypt a virtual machine during virtual machine creation or a disk during disk creation.	Virtual machine folder	Cryptographer.EncryptNew
Manage encryption policies	Allows users to manage virtual machine storage policies with encryption IO filters. By default, virtual machines that use the Encryption storage policy do not use other storage policies.	vCenter root folder	Cryptographer.ManageEncryptionPolicy
Manage KMS	Allows users to manage the Key Management Server for the vCenter system. Management tasks include adding and removing KMS instances, and establishing a trust relationship with the KMS.	vCenter system	Cryptographer.ManageKeyServers
Manage keys	Allows users to perform key management operations. These operations are not supported from the vSphere Client but can be performed by using `crypto-util` or the API.	vCenter root folder	Cryptographer.ManageKeys
Migrate	Allows users to migrate an encrypted virtual machine to a different ESX host. Supports migration with or without vMotion and storage vMotion. Supports migration to a different vCenter instance.	Virtual machine	Cryptographer.Migrate

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Recrypt	Allows users to recrypt virtual machines or disks with a different key. This privilege is required for both deep and shallow recrypt operations.	Virtual machine	Cryptographer.Recrypt
Register VM	Allows users to register an encrypted virtual machine with an ESX host.	Virtual machine folder	Cryptographer.RegisterVM
Register host	Allows users to enable encryption on a host. You can enable encryption on a host explicitly, or the virtual machine creation process can enable it.	Host folder for standalone hosts, cluster for hosts in cluster	Cryptographer.RegisterHost
Read KMS information	Allows users to list vSphere Native Key Providers on the vCenter and on hosts. Also allows users to get vSphere Native Key Provider information.	vCenter or host	Cryptographer.ReadKeyServersInfo

dvPort Group Privileges

Distributed virtual port group privileges control the ability to create, delete, and modify distributed virtual port groups.

The table describes the privileges required to create and configure distributed virtual port groups.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 234: Distributed Virtual Port Group Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create	Allows creation of a distributed virtual port group.	Virtual port groups	DVPortgroup.Create
Delete	Allows deletion of distributed virtual port group. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Virtual port groups	DVPortgroup.Delete
Modify	Allows modification of a distributed virtual port group configuration.	Virtual port groups	DVPortgroup.Modify

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Policy operation	Allows setting the policy of a distributed virtual port group.	Virtual port groups	DVPortgroup.PolicyOp
Scope operation	Allows setting the scope of a distributed virtual port group.	Virtual port groups	DVPortgroup.ScopeOp

Distributed Switch Privileges

Distributed Switch privileges control the ability to perform tasks related to the management of Distributed Switch instances.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 235: vSphere Distributed Switch Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create	Allows creation of a distributed switch.	Data centers, Network folders	DVSwitch.Create
Delete	Allows removal of a distributed switch. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Distributed switches	DVSwitch.Delete
Host operation	Allows changing the host members of a distributed switch.	Distributed switches	DVSwitch.HostOp
Modify	Allows changing the configuration of a distributed switch.	Distributed switches	DVSwitch.Modify
Move	Allows moving a vSphere Distributed Switch to another folder.	Distributed switches	DVSwitch.Move
Network I/O control operation	Allow changing the resource settings for a vSphere Distributed Switch.	Distributed switches	DVSwitch.ResourceManagement
Policy operation	Allows changing the policy of a vSphere Distributed Switch.	Distributed switches	DVSwitch.PolicyOp
Port configuration operation	Allow changing the configuration of a port in a vSphere Distributed Switch.	Distributed switches	DVSwitch.PortConfig
Port setting operation	Allows changing the setting of a port in a vSphere Distributed Switch.	Distributed switches	DVSwitch.PortSetting
VSPAN operation	Allows changing the VSPAN configuration of a vSphere Distributed Switch.	Distributed switches	DVSwitch.Vspan

Datacenter Privileges

Datacenter privileges control the ability to create and edit data centers in the vSphere Client inventory.

All data center privileges are used in vCenter only. The **Create datacenter** privilege is defined on data center folders or the root object. All other data center privileges are pair with data centers, data center folders, or the root object.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 236: Datacenter Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create datacenter	Allows creation of new data center.	Data center folder or root object	Datacenter.Create
Move datacenter	Allows moving a data center. Privilege must be present at both the source and destination.	Data center, source and destination	Datacenter.Move
Network protocol profile configuration	Allows configuration of the network profile for a data center.	Data center	Datacenter.IpPoolConfig
Query IP pool allocation	Allows configuration of a pool of IP addresses.	Data center	Datacenter.IpPoolQueryAllocations
Reconfigure datacenter	Allows reconfiguration of a data center.	Data center	Datacenter.Reconfigure
Release IP allocation	Allows releasing the assigned IP allocation for a data center.	Data center	Datacenter.IpPoolReleaseIp
Remove datacenter	Allows removal of a data center. To have permission to perform this operation, you must have this privilege assigned to both the object and its parent object.	Data center plus parent object	Datacenter.Delete
Rename datacenter	Allows changing the name of a data center.	Data center	Datacenter.Rename
Update datacenter carbon info	Allows gathering metrics related to energy and carbon measurement.	Data center	Datacenter.UpdateCarbonInfo

Datastore Privileges

Datastore privileges control the ability to browse, manage, and allocate space on datastores.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 237: Datastore Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Allocate space	Allows allocating space on a datastore for a virtual machine, snapshot, clone, or virtual disk.	Data stores	Datastore.AllocateSpace
Browse datastore	Allows browsing files on a datastore.	Data stores	Datastore.Browse
Configure datastore IO management	Allows configuring Storage I/O Control.	Data stores	Datastore.ConfigIOManagement

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Configure datastore	Allows configuration of a datastore.	Data stores	Datastore.Config
Low level file operations	Allows performing read, write, delete, and rename operations in the datastore browser.	Data stores	Datastore.FileManagement
Move datastore	Allows moving a datastore between folders. Privileges must be present at both the source and destination.	Datastore, source and destination	Datastore.Move
Remove datastore	Allows removal of a datastore. This privilege is deprecated. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Data stores	Datastore.Delete
Remove file	Allows deletion of files in the datastore. This privilege is deprecated. Assign the **Low level file operations** privilege.	Data stores	Datastore.DeleteFile
Rename datastore	Allows renaming a datastore.	Data stores	Datastore.Rename
Update virtual machine files	Allows updating file paths to virtual machine files on a datastore after the datastore has been resignatured.	Data stores	Datastore.UpdateVirtualMachineFiles
Update virtual machine metadata	Allows updating virtual machine metadata associated with a datastore.	Data stores	Datastore.UpdateVirtualMachineMetadata

Datastore Cluster Privileges

Datastore cluster privileges control the configuration of datastore clusters for Storage DRS.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 238: Datastore Cluster Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Configure a datatstore cluster	Allows creation of and configuration of settings for datastore clusters for Storage DRS.	Datastore clusters	StoragePod.Config

ESX Agent Manager Privileges

ESX Agent Manager privileges control operations related to ESX Agent Manager and agent virtual machines. The ESX Agent Manager is a service that lets you install management virtual machines, which are tied to a host and not affected by VMware DRS or other services that migrate virtual machines.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 239: ESX Agent Manager

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Config	Allows deployment of an agent virtual machine on a host or cluster.	Virtual machines	EAM.Config
Modify	Allows modifications to an agent virtual machine such as powering off or deleting the virtual machine.	Virtual machines	EAM.Modify
View	Allows viewing of an agent virtual machine.	Virtual machines	EAM.View

Extension Privileges

Extension privileges control the ability to install and manage extensions.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 240: Extension Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Register extension	Allows registration of an extension (plug-in).	Root vCenter	Extension.Register
Unregister extension	Allows unregistering an extension (plug-in).	Root vCenter	Extension.Unregister

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Update extension	Allows updates to an extension (plug-in).	Root vCenter	Extension.Update

External Stats Provider Privileges

External stats provider privileges control the ability to notify vCenter Server of Proactive Distributed Resource Scheduler (DRS) statistics.

These privileges apply to an API that is VMware-internal only.

Folder Privileges

Folder privileges control the ability to create and manage folders.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 241: Folder Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create folder	Allows creation of a new folder.	Folders	Folder.Create
Delete folder	Allows deletion of a folder. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Folders	Folder.Delete
Move folder	Allows moving a folder. Privilege must be present at both the source and destination.	Folders	Folder.Move
Rename folder	Allows changing the name of a folder.	Folders	Folder.Rename

Global Privileges

Global privileges control global tasks related to tasks, scripts, and extensions.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 242: Global Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Act as vCenter	Allows preparation or initiation of a vMotion send operation or a vMotion receive operation.	Root vCenter	Global.VCServer
Cancel task	Allows cancelation of a running or queued task.	Inventory object related to the task	Global.CancelTask

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Capacity planning	Allows activating the use of capacity planning for planning consolidation of physical machines to virtual machines.	Root vCenter	Global.CapacityPlanning
Diagnostics	Allows retrieval of a list of diagnostic files, log header, binary files, or diagnostic bundle. To avoid potential security breaches, limit this privilege to the vCenter Server Administrator role.	Root vCenter	Global.Diagnostics
Disable methods	Allows servers for vCenter extensions to deactivate certain operations on objects managed by vCenter.	Root vCenter	Global.DisableMethods
Enable methods	Allows servers for vCenter extensions to activate certain operations on objects managed byvCenter.	Root vCenter	Global.EnableMethods
Global tag	Allows adding or removing global tags.	Root host or vCenter	Global.GlobalTag
Health	Allows viewing the health of vCenter components.	Root vCenter	Global.Health
Licenses	Allows viewing installed licenses and adding or removing licenses.	Root host or vCenter	Global.Licenses
Log event	Allows logging a user-defined event against a particular managed entity.	Any object	Global.LogEvent
Manage custom attributes	Allows adding, removing, or renaming custom field definitions.	Root vCenter	Global.ManageCustomFields
Proxy	Allows access to an internal interface for adding or removing endpoints to or from the proxy.	Root vCenter	Global.Proxy
Script action	Allows scheduling a scripted action along with an alarm.	Any object	Global.ScriptAction
Service managers	Allows use of the `resxtop` command in ESXCLI.	Root host or vCenter	Global.ServiceManagers
Set custom attribute	Allows viewing, creating, or removing custom attributes for a managed object.	Any object	Global.SetCustomField
Settings	Allows reading and modifying runtime vCenter configuration settings.	Root vCenter	Global.Settings
System tag	Allows adding or removing system tags.	Root vCenter	Global.SystemTag

Interact with the Guest Data Publisher Privileges

Interact with the Guest Data Publisher privileges control access to the published guest data on the host GDP service.

Table 243: Interact with the Guest Data Publisher Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Subscribe to the guest data publisher service on ESX hosts	Allows access to the published guest data on the host GDP service.	Hosts	GuestDataPublisher.GetData

Hybrid Linked Mode Privileges

Hybrid Linked Mode privileges control aspects of linking your cloud vCenter instance with an on-premises vCenter Single Sign-On domain. (Applies to VMware Cloud on AWS.)

Table 244: Hybrid Linked Mode Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create	Allows full administrative-level access for creating and deleting communities.	SDDC	HLM.Create
Manage	Allows creating trust for sources and accessing communities (read-level).	SDDC	HLM.Manage

Health Update Provider Privileges

Health update provider privileges control the ability for hardware vendors to notify vCenter Server of Proactive HA events.

These privileges apply to an API that is VMware-internal only.

Host CIM Privileges

Host CIM privileges control the use of CIM for host health monitoring.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 245: Host CIM Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **CIM** – **CIM Interaction**	Allows a client to obtain a ticket to use for CIM services.	Hosts	Host.Cim.CimInteraction

Host Configuration Privileges

Host configuration privileges control the ability to configure hosts.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 246: Host Configuration Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Configuration** – **Advanced Settings**	Allows setting advanced host configuration options.	Hosts	Host.Config.AdvancedConfig
• **Configuration** – **Authentication Store**	Allows configuring Active Directory authentication stores.	Hosts	Host.Config.AuthenticationStore

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Configuration** — **Change PciPassthru settings**	Allows changes to PciPassthru settings for a host.	Hosts	Host.Config.PciPassthru
• **Configuration** — **Change SNMP settings**	Allows changes to SNMP settings for a host.	Hosts	Host.Config.Snmp
• **Configuration** — **Change date and time settings**	Allows changes to date and time settings on the host.	Hosts	Host.Config.DateTime
• **Configuration** — **Change settings**	Allows setting of lockdown mode on ESXi hosts.	Hosts	Host.Config.Settings
• **Configuration** — **Connection**	Allows changes to the connection status of a host (connected or disconnected).	Hosts	Host.Config.Connection
• **Configuration** — **Firmware**	Allows updates to the ESXi host's firmware.	Hosts	Host.Config.Firmware
• **Configuration** — **GuestStore settings**	Allows changes to the GuestStore.	GuestStore repository	Host.Config.GuestStore
• **Configuration** — **Hyperthreading**	Allows activating and deactivting hyperthreading in a host CPU scheduler.	Hosts	Host.Config.HyperThreading
• **Configuration** — **Image configuration**	Allows changes to the image associated with a host.		Host.Config.Image
• **Configuration** — **Maintenance**	Allows putting the host in and out of maintenance mode and shutting down and restarting the host.	Hosts	Host.Config.Maintenance
• **Configuration** — **Memory configuration**	Allows modifications to the host configuration.	Hosts	Host.Config.Memory
• **Configuration** — **NVDIMM**	Allows reading and configuring Non-Volatile DIMMs.	Hosts	Host.Config.Nvdimm
• **Configuration** — **Network configuration**	Allows configuration of network, firewall, and vMotion network.	Hosts	Host.Config.Network
• **Configuration** — **Power**	Allows configuration of host power management settings.	Hosts	Host.Config.Power
• **Configuration** — **ProductLocker settings**	Allows configuration of the ESX productlocker folder.	Hosts	Host.Config.ProductLocker
• **Configuration** — **Quarantine**	Allows putting a host into Quarantine mode.	Hosts	Host.Config.Quarantine
• **Configuration** — **Query patch**	Allows querying for installable patches and installing patches on the host.	Hosts	Host.Config.Patch
• **Configuration** — **Security profile and firewall**	Allows configuration of Internet services, such as SSH, Telnet, SNMP, and of the host firewall.	Hosts	Host.Config.NetService

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Configuration** — **Storage partition configuration**	Allows VMFS datastore and diagnostic partition management. Users with this privilege can scan for new storage devices and manage iSCSI.	Hosts	Host.Config.Storage
• **Configuration** — **System Management**	Allows extensions to manipulate the file system on the host.	Hosts	Host.Config.SystemManagement
• **Configuration** — **System resources**	Allows updates to the configuration of the system resource hierarchy.	Hosts	Host.Config.Resources
• **Configuration** — **Virtual machine autostart configuration**	Allows changes to the auto-start and auto-stop order of virtual machines on a single host.	Hosts	Host.Config.AutoStart

Host Entropy Pool Privileges

Host Entropy Pool privileges control the ability to view and add ESX host entropy.

Table 247: Host Entropy Pool Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Entropy Pool** — **Read**	Allows reading the host entropy pool information.	Hosts	Host.Entropy.Read
• **Entropy Pool** — **Write**	Allows adding entropy to the host entropy pool.	Hosts	Host.Entropy.Write

Host Intel Software Guard Extensions Privileges

Host Intel Software Guard Extensions privileges control aspects of remote attestation on multi-socket ESX hosts.

Table 248: Host Intel Software Guard Extensions (SGX) Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Intel Software Guard Extensions (SGX)** — **Intel Software Guard Extensions (SGX) Register host**	Allows registering of hosts with Intel SGX registration service (for SGX workloads to be able to perform remote attestation of SGX when running on multi-socket SGX capable hosts).	Hosts	Host.Sgx.Register

Host Inventory Privileges

Host inventory privileges control adding hosts to the inventory, adding hosts to clusters, and moving hosts in the inventory.

The table describes the privileges required to add and move hosts and clusters in the inventory.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 249: Host Inventory Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Inventory** – **Add host to cluster**	Allows addition of a host to an existing cluster.	Clusters	Host.Inventory.AddHostToCluster
• **Inventory** – **Add standalone host**	Allows addition of a standalone host.	Host folders	Host.Inventory.AddStandaloneHost
• **Inventory** – **Create cluster**	Allows creation of a new cluster.	Host folders	Host.Inventory.CreateCluster
• **Inventory** – **Manage Cluster Lifecycle**	Allows managing the cluster.	Clusters	Host.Inventory.ManageClusterLifecyle
• **Inventory** – **Modify cluster**	Allows changing the properties of a cluster.	Clusters	Host.Inventory.EditCluster
• **Inventory** – **Move cluster or standalone host**	Allows moving a cluster or standalone host between folders. Privilege must be present at both the source and destination.	Clusters	Host.Inventory.MoveCluster
• **Inventory** – **Move host**	Allows moving a set of existing hosts into or out of a cluster. Privilege must be present at both the source and destination.	Clusters	Host.Inventory.MoveHost
• **Inventory** – **Remove cluster**	Allows deletion of a cluster or standalone host. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Clusters, Hosts	Host.Inventory.DeleteCluster
• **Inventory** – **Remove host**	Allows removal of a host. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Hosts plus parent object	Host.Inventory.RemoveHostFromCluster
• **Inventory** – **Rename cluster**	Allows renaming a cluster.	Clusters	Host.Inventory.RenameCluster

Host Local Operations Privileges

Host local operations privileges control actions performed when the VMware Host Client is connected directly to a host.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 250: Host Local Operations Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Local operations** – **Add host to vCenter**	Allows installation and removal of vCenter agents, such as vpxa and aam, on a host.	Root host	Host.Local.InstallAgent
• **Local operations** – **Create virtual machine**	Allows creation of a new virtual machine from scratch on a disk without registering it on the host.	Root host	Host.Local.CreateVM
• **Local operations** – **Delete virtual machine**	Allows deletion of a virtual machine on disk. Supported for registered and unregistered virtual machines.	Root host	Host.Local.DeleteVM
• **Local operations** – **Manage user groups**	Allows management of local accounts on a host.	Root host	Host.Local.ManageUserGroups
• **Local operations** – **Reconfigure virtual machine**	Allows reconfiguring a virtual machine.	Root host	Host.Local.ReconfigVM

Host Statistics Privileges

Hosts Statistics privileges control the ability to access statistical information from a data processing unit (DPU).

These privileges apply to an API that is VMware-internal only.

Host Trusted Platform Module Privileges

Host Trusted Platform Module privileges control operations related to managing Trusted Platform Module (TPM) chips.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 251: Host Trusted Platform Module Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Trusted Platform Module** – **Read** – **Unseal**	**Read** allows reading detailed information about the state of the TPM installed in the ESX host. **Unseal** allows requesting an ESX host to decrypt a challenge to prove its state.	Hosts	Host.Tpm.Read Host.Tpm.Unseal

Host vSphere Replication Privileges

Host vSphere replication privileges control the use of virtual machine replication by VMware vCenter Site Recovery Manager™ for a host.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 252: Host vSphere Replication Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **vSphere Replication** – **Manage Replication**	Allows management of virtual machine replication on this host.	Hosts	Host.Hbr.HbrManagement

Host Profile Privileges

Host Profile privileges control operations related to creating and modifying host profiles.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 253: Host Profile Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Clear	Allows clearing of profile related information.	Root vCenter	Profile.Clear
Create	Allows creation of a host profile.	Root vCenter	Profile.Create
Delete	Allows deletion of a host profile.	Root vCenter	Profile.Delete
Edit	Allows editing a host profile.	Root vCenter	Profile.Edit
Export	Allows exporting a host profile	Root vCenter	Profile.Export
View	Allows viewing a host profile.	Root vCenter	Profile.View

vCenter Profiles Privileges

vCenter Profiles privileges control aspects of listing profiles, and exporting and importing configurations from one vCenter to another.

Table 254: vCenter Profiles Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
vCenter Server Profiles Read Privileges	Allows listing and exporting of vCenter profiles.	vCenter	Infraprofile.Read
vCenter Server Profiles Write Privileges	Allows importing a profile to another vCenter and validating it.	vCenter	Infraprofile.Write

vSphere Namespaces Privileges

Namespaces privileges control who can create and manage VMware vSphere® with VMware Tanzu™ namespaces.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 255: Namespaces Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Allows disk decommission operations	Allows for decommissioning operations of data stores.	Data stores	Namespaces.ManageDisks
Backup Workloads component files	Allows for backing up the contents of the etcd cluster (used only in VMware Cloud on AWS).	Clusters	Namespaces.Backup
List accessible namespaces	Allows listing the accessible namespaces.	Clusters	Namespaces.ListAccess
Modify cluster-wide configuration	Allows modifying the cluster-wide configuration, and activating and deactivating cluster namespaces.	Clusters	Namespaces.Manage
Modify cluster-wide namespace self-service configuration	Allows modifying the namespace self-service configuration.	Clusters (for activating and deactivating) Templates (for modifying the configuration) vCenter (for creating a template)	Namespaces.SelfServiceManage
Modify namespace configuration	Allows modifying namespace configuration options such as resource allocation and user permissions.	Clusters	Namespaces.Configure
Toggle cluster capabilities	Allows manipulating the state of cluster capabilities (used internally only for VMware Cloud on AWS).	Clusters	Namespaces.ManageCapabilities
Upgrade clusters to newer versions	Allows initiation of the cluster upgrade.	Clusters	Namespaces.Upgrade

Network Privileges

Network privileges control tasks related to network management.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 256: Network Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Assign network	Allows assigning a network to a virtual machine.	Networks, Virtual Machines	Network.Assign
Configure	Allows configuring a network.	Networks, Virtual Machines	Network.Config

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Move network	Allows moving a network between folders. Privilege must be present at both the source and destination.	Networks	Network.Move
Remove	Allows removal of a network. This privilege is deprecated. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Networks	Network.Delete

NSX Privileges

NSX privileges control tasks related to NSX management.

Table 257: NSX Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Read NSX configuration	Allows reading of NSX objects.	NSX	Nsx.Read
Manage NSX configuration	Allows managing of NSX objects from a vSphere administrator's perspective.	NSX	Nsx.Manage
Modify NSX configuration	Allows managing of NSX objects from an enterprise administrator's perspective.	NSX	Nsx.ModifyAll

VMware Observability Privileges

VMware Observability privileges control the ability for an agent to access the observability APIs on vCenter Server.

These privileges apply to an API that is VMware-internal only.

OvfManager Privileges

OvfManager privileges control the ability to access vService Manager.

These privileges apply to an API that is VMware-internal only.

Interact with Partner Rest Daemons Privileges

Interact with Partner Rest Daemons privileges control access to read and write operations.

Table 258: Interact with Partner Rest Daemons Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Perform a GET operation from a Partner's Rest Daemon	Allows the partner-provisioned REST client to perform GET operations.	The partner's user that performs GET operations.	PartnerRestDaemon.Read
Perform a modifying operation to a Partner's Rest Daemon	Allows the partner-provisioned REST client to perform POST, PUT, and DELETE operations.	The partner's user that performs POST, PUT, or DELETE operations.	PartnerRestDaemon.Write

Performance Privileges

Performance privileges control modifying performance statistics settings.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 259: Performance Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Modify intervals	Allows creating, removing, and updating performance data collection intervals.	Root vCenter	Performance.ModifyIntervals

Plug-in Privileges

Plug-in privileges control the managing of vSphere Client plug-ins.

Table 260: Plug-in Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Manage Plug-ins	Allows for managing vSphere Client plug-ins.	vCenter	Plugin.Management

Replication as a Service Privileges

Replication as a Service privileges control access to various internal APIs and functionality related to vCenter Server linking.

These privileges apply to an API that is VMware-internal only.

Permissions Privileges

Permissions privileges control the assigning of roles and permissions.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 261: Permissions Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Modify permission	Allows defining one or more permission rules on an entity, or updating rules if rules are already present for the given user or group on the entity. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Any object plus parent object	Authorization.ModifyPermissions
Modify privilege	Allows modifying a privilege's group or description. No vSphere Client user interface elements are associated with this privilege.	Any object	Authorization.ModifyPrivileges
Modify role	Allows updating a role's name and the privileges that are associated with the role.	Any object	Authorization.ModifyRoles
Modify vTContainer	Allows creating, updating, and deleting vTContainer instances.	vTContainer objects	Authorization.ModifyVTContainers
Modify vTContainer mappings	Allows creating and deleting a vTContainer mapping.	vTContainer mapping objects	Authorization.ModifyVTContainerMappings
Reassign role permissions	Allows reassigning all permissions of a role to another role.	Any object	Authorization.ReassignRolePermissions

VM Storage Policies Privileges

VM storage privileges control operations related to storage profiles.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 262: VM Storage Privileges

Privilege Name	Description	Required On	Privilege Name in the API
Update VM storage policies	Allows changes to be made to storage profiles, such as creating and updating storage capabilities and virtual machine storage profiles.	Root vCenter	StorageProfile.Update
View VM storage policies	Allows using and viewing defined storage capabilities and storage profiles.	Root vCenter	StorageProfile.View

Resource Privileges

Resource privileges control the creation and management of resource pools, as well as the migration of virtual machines.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 263: Resource Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Apply recommendation	Allows accepting a suggestion by the server to perform a migration with vMotion.	Clusters	Resource.ApplyRecommendation
Assign vApp to resource pool	Allows assignment of a vApp to a resource pool.	Resource pools	Resource.AssignVAppToPool
Assign virtual machine to resource pool	Allows assignment of a virtual machine to a resource pool.	Resource pools	Resource.AssignVMToPool
Create resource pool	Allows creation of resource pools.	Resource pools, clusters	Resource.CreatePool
Migrate powered off virtual machine	Allows migration of a powered off virtual machine to a different resource pool or host.	Virtual machines	Resource.ColdMigrate
Migrate powered on virtual machine	Allows migration with vMotion of a powered on virtual machine to a different resource pool or host.	Virtual machines	Resource.HotMigrate
Modify resource pool	Allows changes to the allocations of a resource pool.	Resource pools	Resource.EditPool
Move resource pool	Allows moving a resource pool. Privilege must be present at both the source and destination.	Resource pools	Resource.MovePool
Query vMotion	Allows querying the general vMotion compatibility of a virtual machine with a set of hosts.	Root vCenter	Resource.QueryVMotion
Remove resource pool	Allows deletion of a resource pool. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Resource pools	Resource.DeletePool
Rename resource pool	Allows renaming of a resource pool.	Resource pools	Resource.RenamePool

Scheduled Task Privileges

Scheduled task privileges control creation, editing, and removal of scheduled tasks.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 264: Scheduled Task Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create tasks	Allows scheduling of a task. Required in addition to the privileges to perform the scheduled action at the time of scheduling.	Any object	ScheduledTask.Create
Modify task	Allows reconfiguration of the scheduled task properties.	Any object	ScheduledTask.Edit
Remove task	Allows removal of a scheduled task from the queue.	Any object	ScheduledTask.Delete
Run task	Allows running the scheduled task immediately. Creating and running a scheduled task also requires permission to perform the associated action.	Any object	ScheduledTask.Run

Sessions Privileges

Sessions privileges control the ability of extensions to open sessions on the vCenter system.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Note:
Assign Sessions privileges only to administrators or trusted users.

Table 265: Session Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Impersonate user	Allows impersonation of another user. This capability is used by extensions.	Root vCenter	Sessions.ImpersonateUser
Message	Allows setting of the global login message.	Root vCenter	Sessions.GlobalMessage
Validate session	Allows verification of session validity.	Root vCenter	Sessions.ValidateSession
View and stop sessions	Allows viewing sessions and forcing log out of one or more logged-on users.	Root vCenter	Sessions.TerminateSession
privilege.StorageProfile.ViewPermissions.label	Allows collection of sessions.	Root vCenter	Sessions.CollectPrivilegeChecks

Storage Views Privileges

Storage Views privileges control privileges for Storage Monitoring Service APIs.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 266: Storage Views Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Configure service	Allows privileged users to use all Storage Monitoring Service APIs. Use **Storage views** > **View** for privileges to read-only Storage Monitoring Service APIs.	Root vCenter	StorageViews.ConfigureService
View	Allows privileged users to use read-only Storage Monitoring Service APIs.	Root vCenter	StorageViews.View

Supervisor Services Privileges

Supervisor Services privileges control who can create and manage Supervisor Services on the vSphere with Tanzu environment.

Table 267: Supervisor Services Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Manage Supervisor Services	Allows for creating, updating, or deleting a Supervisor Service. Also allows for installing a Supervisor Service on a cluster, and creating or deleting a Supervisor Service version.	Clusters	SupervisorServices.Manage

Tasks Privileges

Tasks privileges control the ability of extensions to create and update tasks on the vCenter.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 268: Tasks Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create task	Allows an extension to create a user-defined task. No vSphere Client user interface elements are associated with this privilege.	Root vCenter	Task.Create
Update task	Allows an extension to update a user-defined task. No vSphere Client user interface elements are associated with this privilege.	Root vCenter	Task.Update

Tenant Management Privileges

Tenant Management privileges control aspects of defining and retrieving tenant management entities. (Applies to VMware Cloud on AWS.)

Table 269: Tenant Management Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Tenant provisioning operations	Allows defining a set of resources to use for tenant management.	Root folder and each entity currently marked as a service provider.	TenantManager.Update
Tenant query operations	Allows retrieving the list of tenant management resources.	Root folder and each entity currently marked as a service provider.	TenantManager.Query

Transfer Service Privileges

Transfer service privileges are VMware internal. Do not use these privileges.

VcTrusts/VcIdentity Privileges

VcTrusts/VcIdentity privileges control access to various internal APIs and functionality related to trust between vCenter systems.

Table 270: VcTrusts/VcIdentity Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create/Update/Delete (Admin privs)	Allows full administrative-level access to various internal APIs and functionality related to trust between vCenter systems.	N/A	Trust.Administer
Create/Update/Delete (below Admin privs)	Allows reduced administrative access to various internal APIs and functionality related to trust between vCenter systems. This privilege restricts creating/updating/deleting VcTrusts/VcIdentity so that the user cannot escalate non-administrator privileges.	N/A	Trust.Manage

vApp Privileges

vApp privileges control operations related to deploying and configuring a vApp.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 271: vApp Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Add virtual machine	Allows adding a virtual machine to a vApp.	vApps	VApp.AssignVM
Assign resource pool	Allows assigning a resource pool to a vApp.	vApps	VApp.AssignResourcePool
Assign vApp	Allows assigning a vApp to another vApp	vApps	VApp.AssignVApp
Clone	Allows cloning of a vApp.	vApps	VApp.Clone
Create	Allows creation of a vApp.	vApps	VApp.Create
Delete	Allows deletion a vApp. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	vApps	VApp.Delete
Export	Allows export of a vApp from vSphere.	vApps	VApp.Export
Import	Allows import of a vApp into vSphere.	vApps	VApp.Import
Move	Allows moving a vApp to a new inventory location.	vApps	VApp.Move
Power Off	Allows power off operations on a vApp.	vApps	VApp.PowerOff
Power On	Allows power on operations on a vApp.	vApps	VApp.PowerOn
Pull from URL	Allows listing of remote source file descriptors.	vApps	VApp.PullFromUrls
Rename	Allows renaming a vApp.	vApps	VApp.Rename
Suspend	Allows suspension of a vApp.	vApps	VApp.Suspend
Unregister	Allows unregistering a vApp. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	vApps	VApp.Unregister
View OVF Environment	Allows viewing the OVF environment of a powered-on virtual machine within a vApp.	vApps	VApp.ExtractOvfEnvironment
vApp application configuration	Allows modification of a vApp's internal structure, such as product information and properties.	vApps	VApp.ApplicationConfig
vApp instance configuration	Allows modification of a vApp's instance configuration, such as policies.	vApps	VApp.InstanceConfig
vApp managedBy configuration	Allows an extension or solution to mark a vApp as being managed by that extension or solution. No vSphere Client user interface elements are associated with this privilege.	vApps	VApp.ManagedByConfig

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
vApp resource configuration	Allows modification of a vApp's resource configuration. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	vApps	VApp.ResourceConfig

VcIdentityProviders Privileges

VcIdentityProviders privileges control access to the VcIdentityProviders API.

Table 272: VcIdentityProviders Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create	Allows create-only access to the VcIdentityProviders API (vCenter identity providers).	N/A	VcIdentityProviders.Create
Manage	Allows administrative-level write access (create, read, update, delete) to the VcIdentityProviders API (vCenter identity providers).	N/A	VcIdentityProviders.Manage
Read	Allows read access to the VcIdentityProviders API (vCenter identity providers).	N/A	VcIdentityProviders.Read

VMware vSphere Lifecycle Manager Configuration Privileges

VMware vSphere Lifecycle Manager configuration privileges control the ability to configure the vSphere Lifecycle Manager service.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

> **Note:**
> Assign privileges that authorize users to invoke VMware vSphere Lifecycle Manager APIs that accept URLs only to administrators or trusted users.

Table 273: VMware vSphere Lifecycle Manager Configuration Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Configure** – **Configure Service**	Allows configuring the vSphere Lifecycle Manager service and the scheduled patch download task.	Root vCenter	VcIntegrity.General.com.vmware.vcIntegrity.Configure

VMware vSphere Lifecycle Manager Manager Desired Configuration Management Privileges

VMware vSphere Lifecycle Manager Desired Configuration Management privileges control the ability to manage the vSphere Lifecycle Manager configuration.

Table 274: VMware vSphere Lifecycle Manager Desired Configuration Management Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Desired Configuration Management Privileges** – **Export desired cluster configuration** – **Modify desired cluster configuration** – **Read-only access to desired configuration management platform** – **Remediate cluster to the desired configuration**	**Export desired cluster configuration** allows exporting the configuration or the configuration schema. **Modify desired cluster configuration** allows importing a configuration or extracting the configuration from a reference host. **Read-only access to desired configuration management platform** allows checking compliance, running the remediation pre-check, viewing compliance, and viewing pre-check results. **Remediate cluster to the desired configuration** allows remediating a cluster and transitioning to vSphere Configuration Profiles.	Root vCenter	VcIntegrity.ClusterConfiguration.Export VcIntegrity.ClusterConfiguration.Modify VcIntegrity.ClusterConfiguration.View VcIntegrity.ClusterConfiguration.Remediate

VMware vSphere Lifecycle Manager ESX Health Perspectives Privileges

VMware vSphere Lifecycle Manager ESX health perspective privileges control the ability to check the health of ESX hosts and clusters.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 275: VMware vSphere Lifecycle Manager ESX Health Perspectives Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **ESXi Health Perspectives** – **Read** – **Write**	**Read** allows querying the health of ESX hosts and clusters. **Write** is currently not used.	Hosts Clusters	VcIntegrity.lifecycleHealth.Read VcIntegrity.lifecycleHealth.Write

VMware vSphere Lifecycle Manager Depots Privileges

VMware vSphere Lifecycle Manager Depots Privileges control the ability to manage depots.

Table 276: VMware vSphere Lifecycle Manager Depots Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: Depots Privileges** – **Delete**	Allows deleting a vSphere Lifecycle Manager depot.	Root vCenter	VcIntegrity.lifecycleDepots.Delete

VMware vSphere Lifecycle Manager General Privileges

VMware vSphere Lifecycle Manager General privileges control the ability to read and write Lifecyle Manager resources.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 277: VMware vSphere Lifecycle Manager General Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: General Privileges** – **Read** – **Write**	**Read** allows reading of the vSphere Lifecycle Manager resources. This privilege is required to get task information. **Write** allows writing of the vSphere Lifecycle Manager resources. This privilege is required to cancel a vSphere Lifecycle Manager task.	Root vCenter	VcIntegrity.lifecycleGeneral.Read VcIntegrity.lifecycleGeneral.Write

VMware vSphere Lifecycle Manager Hardware Compatibility Privileges

VMware vSphere Lifecycle Manager Hardware Compatibility privileges control the ability to discover and resolve potential hardware compatibility issues.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 278: VMware vSphere Lifecycle Manager Hardware Compatibility Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: Hardware Compatibility Privileges** – **Access Hardware Compatibility** – **Write**	**Access Hardware Compatibility** and **Write** allow accessing the hardware compatibility data and resolving potential hardware compatibility issues.	Hosts	VcIntegrity.HardwareCompatibility.Read VcIntegrity.HardwareCompatibility.Write

VMware vSphere Lifecycle Manager Image Privileges

VMware vSphere Lifecycle Manager Image privileges control the ability to manage images.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Note:
Assign privileges that authorize users to invoke VMware vSphere Lifecycle Manager APIs that accept URLs only to administrators or trusted users.

Table 279: VMware vSphere Lifecycle Manager Image Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: Image Privileges** – **Read** – **Write**	**Read** allows reading of vSphere Lifecycle Manager images. This privilege is required to: • List all the drafts for a cluster • Get more information on a draft • Perform a scan on a draft • Validate a draft • Retrieve the contents of a draft • Compute the effective component list • Get the contents of the current desired state document • Start a scan on a cluster • Get the compliance result • Get a recommendation • Export the current desired state as a depot, JSON file, or ISO **Write** allows managing of vSphere Lifecycle Manager images. This privilege is required to: • Create, delete, or commit a draft • Import the desired state • Generate recommendations • Set or delete different portions of a draft	Root vCenter	VcIntegrity.lifecycleSettings.Read VcIntegrity.lifecycleSettings.Write

VMware vSphere Lifecycle Manager Image Remediation Privileges

VMware vSphere Lifecycle Manager Image privileges control the ability to remediate images.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 280: VMware vSphere Lifecycle Manager Image Remediation Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: Image Remediation Privileges** – **Read** – **Write**	**Read** allows performing the remediation pre-check. **Write** allows performing the remediation.	Clusters	VcIntegrity.lifecycleSoftwareRemediation.Read VcIntegrity.lifecycleSoftwareRemediation.Write

VMware vSphere Lifecycle Manager Settings Privileges

VMware vSphere Lifecycle Manager Settings privileges control the ability to manage depots and remediation policies.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

> **Note:**
> Assign privileges that authorize users to invoke VMware vSphere Lifecycle Manager APIs that accept URLs only to administrators or trusted users.

Table 281: VMware vSphere Lifecycle Manager Settings Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Lifecycle Manager: Settings Privileges** – **Read** – **Write**	**Read** allows reading of vSphere Lifecycle Manager depots and remediation policies. **Write** allows writing of vSphere Lifecycle Manager depots and remediation policies.	Root vCenter	VcIntegrity.lifecycleSoftwareSpecification.Read VcIntegrity.lifecycleSoftwareSpecification.Write

VMware vSphere Lifecycle Manager Manage Baseline Privileges

VMware vSphere Lifecycle Manager Manage Baseline privileges control the ability to manage baselines and baseline groups.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 282: VMware vSphere Lifecycle Manager Manage Baseline Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Manage Baseline** – **Attach Baseline** – **Manage Baseline**	**Attach Baseline** allows attaching baselines and baseline groups to objects in the vSphere inventory. **Manage Baseline** allows creating, editing, or deleting baselines and baseline groups.	Root vCenter	VcIntegrity.Baseline.com.vmware.vcIntegrity.AssignBaselines VcIntegrity.Baseline.com.vmware.vcIntegrity.ManageBaselines

VMware vSphere Lifecycle Manager Manage Patches and Upgrades Privileges

VMware vSphere Lifecycle Manager Manage Patches and Upgrades privileges control the ability to view, scan, and remediate applicable patches, extensions, or upgrades.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 283: VMware vSphere Lifecycle Manager Manage Patches and Upgrades Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Manage Patches and Upgrades** – **Remediate to Apply Patches, Extensions, and Upgrades** – **Scan for Applicable Patches, Extensions, and Upgrades** – **Stage Patches and Extensions** – **View Compliance Status**	**Remediate to Apply Patches, Extensions, and Upgrades** allows remediation of virtual machines and hosts to apply patches, extensions, or upgrades when you are using baselines. In addition, this privilege allows viewing the compliance status. **Scan for Applicable Patches, Extensions, and Upgrades** allows scanning virtual machines and hosts to search for applicable patches, extensions, or upgrades when you are using baselines. **Stage Patches and Extensions** allows staging patches or extensions to ESX hosts when you are using baselines. In addition, this privilege allows viewing the compliance status of ESX hosts. **View Compliance Status** allows viewing the baseline compliance information for an object in the vSphere inventory.	Root vCenter	VcIntegrity.Updates.com.vmware.vcIntegrity.Remediate VcIntegrity.Updates.com.vmware.vcIntegrity.Scan VcIntegrity.Updates.com.vmware.vcIntegrity.Stage VcIntegrity.Updates.com.vmware.vcIntegrity.ViewStatus

VMware vSphere Lifecycle Manager Upload File Privileges

VMware vSphere Lifecycle Manager Upload File privileges control the ability to import updates to the vSphere Lifecycle Manager depot.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

> **Note:**
> Assign privileges that authorize users to invoke VMware vSphere Lifecycle Manager APIs that accept URLs only to administrators or trusted users.

Table 284: VMware vSphere Lifecycle Manager Upload File Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Upload file** – **Upload upgrade images and offline bundles**	Allows uploading upgrade ISO and offline patch bundles.	Root vCenter	VcLifecycle.Upgrade

Virtual Machine Change Configuration Privileges

Virtual Machine Change Configuration privileges control the ability to configure virtual machine options and devices.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 285: Virtual Machine Change Configuration Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Change Configuration** – **Acquire disk lease**	Allows disk lease operations for a virtual machine.	Virtual machines	VirtualMachine.Config.DiskLease
• **Change Configuration** – **Add existing disk**	Allows adding an existing virtual disk to a virtual machine.	Virtual machines	VirtualMachine.Config.AddExistingDisk
• **Change Configuration** – **Add new disk**	Allows creation of a new virtual disk to add to a virtual machine.	Virtual machines	VirtualMachine.Config.AddNewDisk
• **Change Configuration** – **Add or remove device**	Allows addition or removal of any non-disk device.	Virtual machines	VirtualMachine.Config.AddRemoveDevice
• **Change Configuration** – **Advanced configuration**	Allows addition or modification of advanced parameters in the virtual machine's configuration file.	Virtual machines	VirtualMachine.Config.AdvancedConfig
• **Change Configuration** – **Change CPU count**	Allows changing the number of virtual CPUs.	Virtual machines	VirtualMachine.Config.CPUCount
• **Change Configuration** – **Change Memory**	Allows changing the amount of memory allocated to the virtual machine.	Virtual machines	VirtualMachine.Config.Memory
• **Change Configuration** – **Change Settings**	Allows changing general virtual machine settings.	Virtual machines	VirtualMachine.Config.Settings
• **Change Configuration** – **Change Swapfile placement**	Allows changing the swapfile placement policy for a virtual machine.	Virtual machines	VirtualMachine.Config.SwapPlacement
• **Change Configuration** – **Change resource**	Allows changing the resource configuration of a set of virtual machine nodes in a given resource pool.	Virtual machines	VirtualMachine.Config.Resource
• **Change Configuration** – **Configure Host USB device**	Allows attaching a host-based USB device to a virtual machine.	Virtual machines	VirtualMachine.Config.HostUSBDevice

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Change Configuration** — **Configure Raw device**	Allows adding or removing a raw disk mapping or SCSI pass through device. Setting this parameter overrides any other privilege for modifying raw devices, including connection states.	Virtual machines	VirtualMachine.Config.RawDevice
• **Change Configuration** — **Configure managedBy**	Allows an extension or solution to mark a virtual machine as being managed by that extension or solution.	Virtual machines	VirtualMachine.Config.ManagedBy
• **Change Configuration** — **Display connection settings**	Allows configuration of virtual machine remote console options.	Virtual machines	VirtualMachine.Config.MksControl
• **Change Configuration** — **Extend virtual disk**	Allows expansion of the size of a virtual disk.	Virtual machines	VirtualMachine.Config.DiskExtend
• **Change Configuration** — **Modify device settings**	Allows changing the properties of an existing device.	Virtual machines	VirtualMachine.Config.EditDevice
• **Change Configuration** — **Query Fault Tolerance compatibility**	Allows checking if a virtual machine is compatible for Fault Tolerance.	Virtual machines	VirtualMachine.Config.QueryFTCompatibility
• **Change Configuration** — **Query unowned files**	Allows querying of unowned files.	Virtual machines	VirtualMachine.Config.QueryUnownedFiles
• **Change Configuration** — **Reload from path**	Allows changing a virtual machine configuration path while preserving the identity of the virtual machine. Solutions such as VMware vCenter Site Recovery Manager use this operation to maintain virtual machine identity during failover and failback.	Virtual machines	VirtualMachine.Config.ReloadFromPath
• **Change Configuration** — **Remove disk**	Allows removal of a virtual disk device.	Virtual machines	VirtualMachine.Config.RemoveDisk
• **Change Configuration** — **Rename**	Allows renaming a virtual machine or modifying the associated notes of a virtual machine.	Virtual machines	VirtualMachine.Config.Rename
• **Change Configuration** — **Reset guest information**	Allows editing the guest operating system information for a virtual machine.	Virtual machines	VirtualMachine.Config.ResetGuestInfo

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Change Configuration** — **Set annotation**	Allows adding or editing a virtual machine annotation.	Virtual machines	VirtualMachine.Config.Annotation
• **Change Configuration** — **Toggle disk change tracking**	Allows activating or deactivating of change tracking for the virtual machine's disks.	Virtual machines	VirtualMachine.Config.ChangeTracking
• **Change Configuration** — **Toggle fork parent**	Allows activating or deactivating a vmfork parent.	Virtual machines	VirtualMachine.Config.ToggleForkParent
• **Change Configuration** — **Upgrade virtual machine compatibility**	Allows upgrade of the virtual machine's virtual machine compatibility version.	Virtual machines	VirtualMachine.Config.UpgradeVirtualHardware

Virtual Machine Guest Operations Privileges

Virtual Machine Guest Operations privileges control the ability to interact with files and applications inside a virtual machine's guest operating system with the API.

See the *vSphere Web Services API Reference* documentation for more information on these operations.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 286: Virtual Machine Guest Operations

Privilege Name in the vSphere Client	Description	Effective on Object	Privilege Name in the API
• **Guest operations** — **Guest operation alias modification**	Allows virtual machine guest operations that involve modifying the alias for the virtual machine.	Virtual machines	VirtualMachine.GuestOperations.ModifyAliases
• **Guest operations** — **Guest operation alias query**	Allows virtual machine guest operations that involve querying the alias for the virtual machine.	Virtual machines	VirtualMachine.GuestOperations.QueryAliases
• **Guest operations** — **Guest operation modifications**	Allows virtual machine guest operations that involve modifications to a guest operating system in a virtual machine, such as transferring a file to the virtual machine. No vSphere Client user interface elements are associated with this privilege.	Virtual machines	VirtualMachine.GuestOperations.Modify

Privilege Name in the vSphere Client	Description	Effective on Object	Privilege Name in the API
• **Guest operations** — **Guest operation program execution**	Allows virtual machine guest operations that involve running an application in the virtual machine. No vSphere Client user interface elements are associated with this privilege.	Virtual machines	VirtualMachine.GuestOperations.Execute
• **Guest operations** — **Guest Operation Queries**	Allows virtual machine guest operations that involve querying the guest operating system, such as listing files in the guest operating system. No vSphere Client user interface elements are associated with this privilege.	Virtual machines	VirtualMachine.GuestOperations.Query

Virtual Machine Interaction Privileges

Virtual Machine Interaction privileges control the ability to interact with a virtual machine console, configure media, perform power operations, and install VMware Tools.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 287: Virtual Machine Interaction

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Interaction** — **Answer question**	Allows resolution of issues with virtual machine state transitions or runtime errors.	Virtual machines	VirtualMachine.Interact.AnswerQuestion
• **Interaction** — **Backup operation on virtual machine**	Allows performance of backup operations on virtual machines.	Virtual machines	VirtualMachine.Interact.Backup
• **Interaction** — **Configure CD media**	Allows configuration of a virtual DVD or CD-ROM device.	Virtual machines	VirtualMachine.Interact.SetCDMedia
• **Interaction** — **Configure floppy media**	Allows configuration of a virtual floppy device.	Virtual machines	VirtualMachine.Interact.SetFloppyMedia
• **Interaction** — **Console interaction**	Allows interaction with the virtual machine's virtual mouse, keyboard, and screen.	Virtual machines	VirtualMachine.Interact.ConsoleInteract
• **Interaction** — **Create screenshot**	Allows creation of a virtual machine screen shot.	Virtual machines	VirtualMachine.Interact.CreateScreenshot
• **Interaction** — **Defragment all disks**	Allows defragment operations on all disks of the virtual machine.	Virtual machines	VirtualMachine.Interact.DefragmentAllDisks

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Interaction** — **Device connection**	Allows changing the connected state of a virtual machine's disconnectable virtual devices.	Virtual machines	VirtualMachine.Interact.DeviceConnection
• **Interaction** — **Drag and Drop**	Allows drag and drop of files between a virtual machine and a remote client.	Virtual machines	VirtualMachine.Interact.DnD
• **Interaction** — **Guest operating system management by VIX API**	Allows management of the virtual machine's operating system through the VIX API.	Virtual machines	VirtualMachine.Interact.GuestControl
• **Interaction** — **Inject USB HID scan codes**	Allows injection of USB HID scan codes.	Virtual machines	VirtualMachine.Interact.PutUsbScanCodes
• **Interaction** — **Pause or Unpause**	Allows pausing or unpausing of the virtual machine.	Virtual machines	VirtualMachine.Interact.Pause
• **Interaction** — **Perform wipe or shrink operations**	Allows performing wipe or shrink operations on the virtual machine.	Virtual machines	VirtualMachine.Interact.SESparseMaintenance
• **Interaction** — **Power Off**	Allows powering off a powered-on virtual machine. This operation powers down the guest operating system.	Virtual machines	VirtualMachine.Interact.PowerOff
• **Interaction** — **Power On**	Allows powering on a powered-off virtual machine, and resuming a suspended virtual machine.	Virtual machines	VirtualMachine.Interact.PowerOn
• **Interaction** — **Record session on Virtual Machine**	Allows recording a session on a virtual machine.	Virtual machines	VirtualMachine.Interact.Record
• **Interaction** — **Replay session on Virtual Machine**	Allows replaying of a recorded session on a virtual machine.	Virtual machines	VirtualMachine.Interact.Replay
• **Interaction** — **Reset**	Allows resetting of a virtual machine and reboots the guest operating system.	Virtual machines	VirtualMachine.Interact.Reset
• **Interaction** — **Resume Fault Tolerance**	Allows resuming of fault tolerance for a virtual machine.	Virtual machines	VirtualMachine.Interact.EnableSecondary
• **Interaction** — **Suspend**	Allows suspending a powered-on virtual machine. This operation puts the guest in standby mode.	Virtual machines	VirtualMachine.Interact.Suspend
• **Interaction** — **Suspend Fault Tolerance**	Allows suspension of fault tolerance for a virtual machine.	Virtual machines	VirtualMachine.Interact.DisableSecondary
• **Interaction** — **Suspend to memory**	Allows suspension of memory for a virtual machine.	Virtual machines	VirtualMachine.Interact.SuspendToMemory

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Interaction** — **Test failover**	Allows testing of Fault Tolerance failover by making the Secondary virtual machine the Primary virtual machine.	Virtual machines	VirtualMachine.Interact.MakePrimary
• **Interaction** — **Test restart Secondary VM**	Allows termination of a Secondary virtual machine for a virtual machine using Fault Tolerance.	Virtual machines	VirtualMachine.Interact.DisableSecondary
• **Interaction** — **Turn Off Fault Tolerance**	Allows turning off Fault Tolerance for a virtual machine.	Virtual machines	VirtualMachine.Interact.TurnOffFaultTolerance
• **Interaction** — **Turn On Fault Tolerance**	Allows turning on Fault Tolerance for a virtual machine.	Virtual machines	VirtualMachine.Interact.CreateSecondary
• **Interaction** — **VMware Tools install**	Allows mounting and unmounting the VMware Tools CD installer as a CD-ROM for the guest operating system.	Virtual machines	VirtualMachine.Interact.ToolsInstall

Virtual Machine Edit Inventory Privileges

Virtual Machine Edit Inventory privileges control adding, moving, and removing virtual machines.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 288: Virtual Machine Edit Inventory Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Edit inventory** — **Create from existing**	Allows creation of a virtual machine based on an existing virtual machine or template, by cloning or deploying from a template.	Clusters, Hosts, Virtual machine folders	VirtualMachine.Inventory.CreateFromExisting
• **Edit inventory** — **Create new**	Allows creation of a virtual machine and allocation of resources for its execution.	Clusters, Hosts, Virtual machine folders	VirtualMachine.Inventory.Create
• **Edit inventory** — **Move**	Allows relocating a virtual machine in the hierarchy. The privilege must be present at both the source and destination.	Virtual machines	VirtualMachine.Inventory.Move
• **Edit inventory** — **Register**	Allows adding an existing virtual machine to a vCenter Server or host inventory.	Clusters, Hosts, Virtual machine folders	VirtualMachine.Inventory.Register

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Edit inventory** — **Remove**	Allows deletion of a virtual machine. Deletion removes the virtual machine's underlying files from disk. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Virtual machines	VirtualMachine.Inventory.Delete
• **Edit inventory** — **Unregister**	Allows unregistering a virtual machine from a vCenter or host inventory. To have permission to perform this operation, a user or group must have this privilege assigned in both the object and its parent object.	Virtual machines	VirtualMachine.Inventory.Unregister

Virtual Machine Provisioning Privileges

Virtual Machine Provisioning privileges control activities related to deploying and customizing virtual machines.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 289: Virtual Machine Provisioning Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Provisioning** — **Allow disk access**	Allows opening a disk on a virtual machine for random read and write access. Used mostly for remote disk mounting.	Virtual machines	VirtualMachine.Provisioning.DiskRandomAccess
• **Provisioning** — **Allow file access**	Allows operations on files associated with a virtual machine, including vmx, disks, logs, and nvram.	Virtual machines	VirtualMachine.Provisioning.FileRandomAccess
• **Provisioning** — **Allow read-only disk access**	Allows opening a disk on a virtual machine for random read access. Used mostly for remote disk mounting.	Virtual machines	VirtualMachine.Provisioning.DiskRandomRead
• **Provisioning** — **Allow virtual machine download**	Allows read operations on files associated with a virtual machine, including vmx, disks, logs, and nvram.	Root host or vCenter	VirtualMachine.Provisioning.GetVmFiles

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Provisioning** — **Allow virtual machine files upload**	Allows write operations on files associated with a virtual machine, including vmx, disks, logs, and nvram.	Root host or vCenter	VirtualMachine.Provisioning.PutVmFiles
• **Provisioning** — **Clone template**	Allows cloning of a template.	Templates	VirtualMachine.Provisioning.CloneTemplate
• **Provisioning** — **Clone virtual machine**	Allows cloning of an existing virtual machine and allocation of resources.	Virtual machines	VirtualMachine.Provisioning.Clone
• **Provisioning** — **Create template from virtual machine**	Allows creation of a new template from a virtual machine.	Virtual machines	VirtualMachine.Provisioning.CreateTemplateFromVM
• **Provisioning** — **Customize guest**	Allows customization of a virtual machine's guest operating system without moving the virtual machine.	Virtual machines	VirtualMachine.Provisioning.Customize
• **Provisioning** — **Deploy template**	Allows deployment of a virtual machine from a template.	Templates	VirtualMachine.Provisioning.DeployTemplate
• **Provisioning** — **Mark as template**	Allows marking an existing powered off virtual machine as a template.	Virtual machines	VirtualMachine.Provisioning.MarkAsTemplate
• **Provisioning** — **Mark as virtual machine**	Allows marking an existing template as a virtual machine.	Templates	VirtualMachine.Provisioning.MarkAsVM
• **Provisioning** — **Modify customization specification**	Allows creation, modification, or deletion of customization specifications.	Root vCenter	VirtualMachine.Provisioning.ModifyCustSpecs
• **Provisioning** — **Promote disks**	Allows promote operations on a virtual machine's disks.	Virtual machines	VirtualMachine.Provisioning.PromoteDisks
• **Provisioning** — **Read customization specifications**	Allows reading a customization specification.	Virtual machines	VirtualMachine.Provisioning.ReadCustSpecs

Virtual Machine Service Configuration Privileges

Virtual machine service configuration privileges control who can perform monitoring and management tasks on the service configuration.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 290: Virtual Machine Service Configuration Privileges

Privilege Name in the vSphere Client	Description	Privilege Name in the API
• **Service configuration** – **Allow notifications**	Allows generating and consuming notification about service status.	VirtualMachine.Namespace.Event
• **Service configuration** – **Allow polling of global event notifications**	Allows querying whether any notifications are present.	VirtualMachine.Namespace.EventNotify
• **Service configuration** – **Manage service configurations**	Allows creating, modifying, and deleting virtual machine services.	VirtualMachine.Namespace.Management
• **Service configuration** – **Modify service configuration**	Allows modification of existing virtual machine service configuration.	VirtualMachine.Namespace.ModifyContent
• **Service configuration** – **Query service configurations**	Allows retrieval of list of virtual machine services.	VirtualMachine.Namespace.Query
• **Service configuration** – **Read service configuration**	Allows retrieval of existing virtual machine service configuration.	VirtualMachine.Namespace.ReadContent

Virtual Machine Snapshot Management Privileges

Virtual machine snapshot management privileges control the ability to take, delete, rename, and restore snapshots.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 291: Virtual Machine Snapshot Management Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Snapshot management** – **Create snapshot**	Allows creation of a snapshot from the virtual machine's current state.	Virtual machines	VirtualMachine.State.CreateSnapshot
• **Snapshot management** – **Remove Snapshot**	Allows removal of a snapshot from the snapshot history.	Virtual machines	VirtualMachine.State.RemoveSnapshot
• **Snapshot management** – **Rename Snapshot**	Allows renaming a snapshot with a new name, a new description, or both.	Virtual machines	VirtualMachine.State.RenameSnapshot
• **Snapshot management** – **Revert to snapshot**	Allows setting the virtual machine to the state it was in at a given snapshot.	Virtual machines	VirtualMachine.State.RevertToSnapshot

Virtual Machine vSphere Replication Privileges

Virtual Machine vSphere replication privileges control the use of replication by VMware vCenter Site Recovery Manager™ for virtual machines.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 292: Virtual Machine vSphere Replication Privileges

Privilege Name in the	Description	Required On	Privilege Name in the API
• **vSphere Replication** – **Configure Replication**	Allows configuration of replication for the virtual machine.	Virtual machines	VirtualMachine.Hbr.ConfigureReplication
• **vSphere Replication** – **Manage Replication**	Allows triggering of full sync, online sync or offline sync on a replication.	Virtual machines	VirtualMachine.Hbr.ReplicaManagement
• **vSphere Replication** – **Monitor Replication**	Allows monitoring of replication.	Virtual machines	VirtualMachine.Hbr.MonitorReplication

Virtual Machine Classes Privileges

Virtual Machine Classes privileges control who can add and remove virtual machine classes on a Kubernetes namespace.

Table 293: Virtual Machine Classes Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Manage Virtual Machine Classes	Allows managing virtual machine classes on Kubernetes namespaces on a Supervisor Cluster.	Clusters	VirtualMachineClasses.Manage

vSAN Privileges

vSAN privileges control who can perform shallow rekey operations and update client information.

Table 294: vSAN Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
• **Cluster** – **ShallowRekey**	Allows performing a shallow rekey for a cluster.	Clusters	Vsan.Cluster.ShallowRekey
• **Xvc** – **UpdateClientInfo**	Used internally.	Service user	Vsan.Xvc.UpdateClientInfo

vSAN Stats Privileges

vSphere Stats privileges control the ability to access vSAN metrics.

Table 295: vSAN Stats Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Access service discovery endpoint of vSAN stats	Allows accessing the service discovery endpoint `https://`*`vCenterServer-IP`*`/vsan/metrics/serviceDiscovery`.	Service account role.	vSANStats.Access

vSphere Zones Privileges

vSphere Zones privileges control who can create and manage vSphere Zones on vSphere with Tanzu.

Table 296: vSphere Zones Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Attach and Detach vSphere objects for vSphere Zones	Allows for associating objects with a vSphere Zone.	Clusters	Zone.ObjectAttachable
Create, Update and Delete vSphere Zones and their associations	Allows for creating and deleting a vSphere Zone.	Clusters	Zone.Manage

vService Privileges

vService privileges control the ability to create, configure, and update vService dependencies for virtual machines and vApps.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 297: vService Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Create dependency	Allows creation of a vService dependency for a virtual machine or vApp.	vApps and virtual machines	vService.CreateDependency
Destroy dependency	Allows removal of a vService dependency for a virtual machine or vApp.	vApps and virtual machines	vService.DestroyDependency
Reconfigure dependency configuration	Allows reconfiguration of a dependency to update the provider or binding.	vApps and virtual machines	vService.ReconfigureDependency
Update dependency	Allows updates of a dependency to configure the name or description.	vApps and virtual machines	vService.UpdateDependency

vSphere Tagging Privileges

vSphere Tagging privileges control the ability to create and delete tags and tag categories, and assign and remove tags on vCenter inventory objects.

You can set this privilege at different levels in the hierarchy. For example, if you set a privilege at the folder level, you can propagate the privilege to one or more objects within the folder. The object listed in the Required On column must have the privilege set, either directly or inherited.

Table 298: vSphere Tagging Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Assign or Unassign vSphere Tag	Allows assignment or unassignment of a tag for an object in the vCenter inventory.	Any object	InventoryService.Tagging.AttachTag
Assign or Unassign vSphere Tag on Object	Allows objects to have tags assigned or unassigned. Use this privilege to limit which objects users are able to assign or unassign tags to.	Any object	InventoryService.Tagging.ObjectAttachable
Create vSphere Tag	Allows creation of a tag.	Any object	InventoryService.Tagging.CreateTag
Create vSphere Tag Category	Allows creation of a tag category.	Any object	InventoryService.Tagging.CreateCategory
Delete vSphere Tag	Allows deletion of a tag.	Any object	InventoryService.Tagging.DeleteTag
Delete vSphere Tag Category	Allows deletion of a tag category.	Any object	InventoryService.Tagging.DeleteCategory
Edit vSphere Tag	Allows editing of a tag.	Any object	InventoryService.Tagging.EditTag
Edit vSphere Tag Category	Allows editing of a tag category.	Any object	InventoryService.Tagging.EditCategory
Modify UsedBy Field for Category	Allows changing the UsedBy field for a tag category.	Any object	InventoryService.Tagging.ModifyUsedByForCategory
Modify UsedBy Field for Tag	Allows changing the UsedBy field for a tag.	Any object	InventoryService.Tagging.ModifyUsedByForTag

vSphere Client Privileges

vSphere Client privileges control offline access to vCenter.

These privileges apply to VMware Cloud only.

vSphere Data Protection Privileges

vSphere Data Protection privileges control the ability to manage the VMware vSphere® Data Protection™ backup and recovery solution.

Table 299: vSphere Data Protection Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Protection	Allows performing data protection operations such as creating and managing backups.	vCenter	vSphereDataProtection.Protection
Recovery	Allows performing data protection operations such as restoring backups.	vCenter	vSphereDataProtection.Recovery

vSphere Stats Privileges

vSphere Stats privileges control the ability to access vStats state and stats data on objects such as virtual machines and hosts.

Table 300: vSphere Stats Privileges

Privilege Name in the vSphere Client	Description	Required On	Privilege Name in the API
Collect Stats Data	Allows creating and updating acquisition specs in vStats.	The object for which stats data is being collected.	vStats.CollectAny
Modify Stats Configuration	Allows managing the vStats service configuration settings.	vCenter	vStats.Settings
Query Stats Data	Allows enumerating stats providers, and metrics and counters the providers expose, for which providers can collect stats data.	The object for which stats data is being queried.	vStats.QueryAny

vSphere Hardening and Compliance

Organizations expect to keep their data secure by reducing the risk of data theft, cyberattack, or unauthorized access. Organizations also must often comply with one or more regulations from government standards to private standards, such as the National Institute of Standards and Technology (NIST) and Defense Information Systems Agency Security Technical Implementation Guides (DISA STIG). Ensuring that your vSphere environment is in compliance with such standards involves understanding a broader set of considerations including people, processes, and technology.

For detailed guidance on hardening your vSphere system, see the VCF 9.0 Security Configuration & Hardening Guide.

Security Versus Compliance in the vSphere Environment

The terms security and compliance are often used interchangeably. However, they are unique and distinct concepts.

Security, often thought of as information security, is commonly defined as a set of technical, physical, and administrative controls that you implement to provide confidentiality, integrity, and availability. For example, you secure a host by locking down which accounts can log into it, and by what means (SSH, direct console, and so on). Compliance, by contrast, is a set of requirements necessary to meet the minimum controls established by different regulatory frameworks that provide limited guidance on any specific type of technology, vendor, or configuration. For example, the Payment Card Industry (PCI) has established security guidelines to help organizations proactively protect customer account data.

Security reduces the risk of data theft, cyberattack, or unauthorized access, while compliance is the proof that a security control is in place, typically within a defined time line. Security is primarily outlined in the design decisions and highlighted within the technology configurations. Compliance is focused on mapping the correlation between security controls and specific requirements. A compliance mapping provides a centralized view to list out many of the required security controls. Those controls are further detailed by including compliance citations for each respective security control as dictated by a domain such as NIST, PCI, FedRAMP, HIPAA, and so forth.

Effective cybersecurity and compliance programs are built on three pillars: people, process, and technology. A general misconception is that technology alone can solve all your cybersecurity needs. Technology does play a large and important role in the development and execution of an information security program. However, technology without process and procedures, and awareness and training, creates a vulnerability within your organization.

When defining your security and compliance strategies, keep the following in mind:

- People need general awareness and training, whereas IT staff need specific training.
- Process defines how activities, roles, and documentation within an organization are used to mitigate risk. Processes are only effective if people follow them correctly.
- Technology can be used to prevent or reduce the impact of cybersecurity risk to your organization. Which technology to use depends on the risk acceptance level within an organization.

Glossary of Compliance Terms

Compliance introduces specific terms and definitions that are important to understand.

Table 301: Compliance Terms

Term	Definition
CJIS	Criminal Justice Information Services. In the context of compliance, the CJIS produces a Security Policy for how local, state, and federal criminal justice and law enforcement agencies must take security precautions to protect sensitive information such as fingerprints and criminal backgrounds.
DISA STIG	Defense Information Systems Agency Security Technical Implementation Guide. The Defense Information Systems Agency (DISA) is the entity responsible for maintaining the security posture of the Department of Defense (DoD) IT infrastructure. DISA accomplishes this task by developing and using Security Technical Implementation Guides, or "STIGs."
FedRAMP	Federal Risk and Authorization Management Program. FedRAMP is a government-wide program that provides a standardized approach to security assessment, authorization, and continuous monitoring for cloud products and services.
HIPAA	Health Insurance Portability and Accountability Act. Passed by Congress in 1996, HIPAA does the following: • Gives millions of American workers and their families the ability to transfer and continue health insurance coverage for when they change or lose jobs • Reduces health care fraud and abuse • Mandates industry-wide standards for health care information on electronic billing and other processes • Requires the protection and confidential handling of protected health information The latter bullet is of most importance to *vSphere Security* documentation.
NCCoE	National Cybersecurity Center of Excellence. NCCoE is a U.S government organization that produces and publicly shares solutions to cybersecurity problems that U.S. businesses encounter. The center forms a team of people from cybersecurity technology companies, other federal agencies, and academia to address each problem.
NIST	National Institute of Standards and Technology. Founded in 1901, NIST is a non-regulatory federal agency within the U.S. Department of Commerce. The mission of NIST is to advocate for U.S. innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that increase economic security and improve our quality of life.
PAG	Product Applicability Guide. A document that provides general guidance for organizations that are considering a company's solutions to help them address compliance requirements.
PCI DSS	Payment Card Industry Data Security Standard. A set of security standards designed to ensure that all companies that accept, process, store, or transmit credit card information maintain a secure environment.

Term	Definition
VVD/VCF Compliance Solutions	VMware Validated Design/VMware Cloud Foundation. The VMware Validated Designs provide comprehensive and extensively tested blueprints to build and operate a Software-Defined Data Center. VVD/VCF compliance solutions enable customers to meet compliance requirements for multiple government and industry regulations.

About the National Institute of Standards and Technology

The National Institute of Standards and Technology (NIST) is a non-regulatory government agency that develops technology, metrics, standards, and guidelines. Compliance with NIST standards and guidelines has become a top priority in many industries today.

The National Institute of Standards and Technology (NIST) was founded in 1901 and is now part of the U.S. Department of Commerce. NIST is one of the nation's oldest physical science laboratories. Today, NIST measurements support the smallest of technologies to the largest and most complex of human-made creations, from nanoscale devices, up to earthquake-resistant skyscrapers and global communication networks.

The Federal Information Security Management Act (FISMA) is a United States federal law passed in 2002 that made it a requirement for federal agencies to develop, document, and implement an information security and protection program. NIST plays an important role in the FISMA implementation by producing key security standards and guidelines (for example, FIPS 199, FIPS 200, and SP 800 series).

Government and private organizations use NIST 800-53 to secure information systems. Cybersecurity and privacy controls are essential to protect organizational operations (including mission, functions, image, and reputation), organizational assets, and individuals from a diverse set of threats. Some of these threats include hostile cyber-attacks, natural disasters, structural failures, and human errors. VMware has enlisted a third-party audit partner to evaluate VMware products and solutions against the NIST 800-53 catalog of controls. For more information, visit the NIST webpage at https://www.nist.gov/cyberframework.

About DISA STIGs

The Defense Information Systems Agency (DISA) develops and publishes Security Technical Implementation Guides, or "STIGs." DISA STIGs provide technical guidance for hardening systems and reducing threats.

The Defense Information Systems Agency (DISA) is the U.S. Department of Defense (DoD) combat support agency responsible for maintaining the security posture of the DOD Information Network (DODIN). One of the ways DISA accomplishes this task is by developing, disseminating, and mandating the implementation of Security Technical Implementation Guides, or STIGs. In brief, STIGs are portable, standards-based guides for hardening systems. STIGs are mandatory for U.S. DoD IT systems and, as such, provide a vetted, secure baseline for non-DoD entities to measure their security posture.

Vendors such as VMware submit suggested security hardening guidance to DISA for evaluation, based on DISA protocols and feedback. Once that process is complete, the official STIG is published on the DISA organization's web site at https://public.cyber.mil/stigs/. VMware provides security baselines and hardening guidance for vSphere as part of the *vSphere Security Configuration Guide*. See https://brcm.tech/vcf-scg.

About NERC CIP

The North American Electric Reliability Corporation (NERC) is a nonprofit international regulatory authority tasked with ensuring the reliability and security of North America's bulk power system, including the United States, Canada, and part of Baja California in Mexico.

NERC develops, enforces, and monitors reliability standards, and educates, trains, and certifies industry personnel. The Critical Infrastructure Protection (CIP) standards, a key component of NERC's regulations, are designed to secure essential cyber assets of the bulk electric system. These standards mandate specific security management controls, system security management, information protection, and vulnerability assessments for utilities and other entities within the bulk power system.

About VMware Security Development Lifecycle

The VMware Security Development Lifecycle (SDL) program identifies and mitigates security risk during the development phase of VMware software products. VMware also operates the VMware Security Response Center (VSRC) to conduct the analysis and remediation of software security issues in VMware products.

The SDL is the software development methodology that the VMware Security Engineering, Communication, and Response (vSECR) group, and VMware product development groups, use to help identify and mitigate security issues.

The VMware Security Response Center (VSRC) works with customers and the security research community to achieve the goals of addressing security issues and providing customers with actionable security information in a timely manner. For more information about the VSRC, see the webpage at https://www.vmware.com/security/vsrc.html.

Audit Logging in vSphere

Audit logging of network traffic, compliance alerts, firewall activity, operating system changes, and provisioning activities is considered a best practice for maintaining the security of any IT environment. In addition, logging is a specific requirement of many regulations and standards.

One of the first steps to take for ensuring that you are aware of changes to your infrastructure is to audit your environment. By default, vSphere includes tools that enable you to view and track changes. For example, you can use the Tasks and Events tab in thevSphere Client on any object in your vSphere hierarchy to see what changes have occurred. You can also use the PowerCLI to retrieve events and tasks. In addition, VMware Cloud Foundation Operations for logs offers audit logging to support collection and retention of important system events. Finally, many third-party tools are available that providevCenter auditing.

Log files can also help determine who or what is accessing a host, a virtual machine, and so on. For more information, see ESX Log File Locations.

Single Sign-On Audit Events

Single Sign-On (SSO) audit events are records of user or system actions for accessing the SSO services.

vCenter Server 6.7 Update 2 and later improves VMware vCenter Single Sign-On auditing by adding events for the following operations:

- User management
- Login
- Group creation
- Identity source
- Policy updates

The supported identity sources are vsphere.local and Active Directory over LDAP.

When a user logs in to vCenter through Single Sign-On, or makes changes that affect SSO, the following audit events are written to the SSO audit log file:

- **Login and Logout Attempts:** Events for all the successful and failed login and logout operations.
- **Privilege Change:** Event for change in a user role or permissions.
- **Account Change:** Event for change in the user account information, for example, user name, password, or any additional account information.
- **Security Change:** Event for change in a security configuration, parameter, or policy.
- **Account Enabled or Disabled:** Event for when an account is activated or deactivated.
- **Identity Source:** Event for adding, deleting, or editing an identity source.

In the vSphere Client, event data is displayed in the **Monitor** tab. See the *vSphere Monitoring and Performance* documentation.

SSO audit event data includes the following details:

- Timestamp of when the event occurred.

- User who performed the action.
- Description of the event.
- Severity of the event.
- IP address of client used to connect to vCenter, if available.

SSO Audit Event Log Overview

The vSphere Single-Sign On process writes audit events to the `audit_events.log` file in the `/var/log/audit/sso-events/` directory.

> **CAUTION**:
> Never manually edit the `audit_events.log` file, as doing so might cause the audit logging to fail.

Keep the following in mind when working with the `audit_events.log` file:

- The log file is archived once it reaches 50 MB.
- A maximum of 10 archive files is kept. If the limit is reached, the oldest file is purged when a new archive is created.
- The archive files are named `audit_events-<index>.log.gz`, where the index is a numeral from 1 to 10. The first archive created is index 1, and is increased with each subsequent archive.
- The oldest events are in archive index 1. The highest indexed file is the latest archive.

Understanding Security and Compliance Next Steps

Conducting a security assessment is the first step in understanding any vulnerabilities in your infrastructure. A security assessment is part of a security audit, which looks at both systems and practices, including security compliance.

A security assessment generally refers to scanning your organization's physical infrastructure (firewalls, networks, hardware, and so on) to identify vulnerabilities and flaws. A security assessment is not the same as a security audit. A security audit includes not only a review of physical infrastructure but other areas such as policy and standard operating procedures, including security compliance. After you have the audit, you can decide on the steps to remedy the problems within the system.

You might ask these general questions when preparing to conduct a security audit:

1. Is our organization mandated to adhere to a compliance regulation? If so which one(s)?
2. What is our audit interval?
3. What is our internal self-assessment interval?
4. Do we have access to previous audit results and have we viewed them?
5. Do we use a third-party audit firm to help us prepare for an audit? If so, what is their level of comfort with virtualization?
6. Do we run vulnerability scans against the systems and applications? When and how often?
7. What are our internal cybersecurity policies?
8. Is your audit logging configured according to your needs? See Audit Logging in vSphere.

In the absence of specific guidance or direction on where to begin, you can jumpstart securing your vSphere environment by:

- Keeping your environment up-to-date with the latest software and firmware patches
- Maintaining good password management and hygiene for all accounts
- Reviewing vendor-approved security recommendations
- Referring to the VMware Security Configuration Guides (see vSphere Security Configuration Guide)
- Using readily available and proven guidance from policy frameworks such as NIST, ISO, and so forth
- Following guidance from regulatory compliance frameworks such as PCI, DISA, and FedRAMP

vCenter and FIPS

You can enable FIPS-validated cryptography on the vCenter appliance.

FIPS 140-3 is a U.S. and Canadian government standard that specifies security requirements for cryptographic modules. vSphere uses FIPS-validated cryptographic modules to match those specified by the FIPS 140-2 standard. The goal of vSphere FIPS support is to ease the compliance and security activities in various regulated environments.

ESX and vCenter use FIPS-validated cryptography to protect management interfaces and the VMware Certificate Authority (VMCA).

vSphere 7.0 Update 2 and later includes additional FIPS-validated cryptography to the vCenter appliance.

> **Note:**
> vSphere favors compatibility over FIPS, so some components have considerations to be aware of. See Considerations When Using FIPS.

FIPS Modules Used in ESX

A cryptographic module is a set of hardware, software, or firmware that implements security functions. ESX uses several FIPS 140-2 validated cryptographic modules.

The following table shows the set of FIPS 140-3 validated cryptographic modules in use by ESX.

Table 302: FIPS Modules

Cryptographic Module	Version	Algorithms (CAVP)	Certificate Number
VMkernel Cryptographic Module	2.0	AES-CBC, AES-CBC-CS3, AES-CTR, AES-ECB, AES-GCM, AES-XTS Testing Revision 2.0, Counter DRBG, HMAC-SHA-1, HMAC-SHA2-256, HMAC-SHA2-512, SHA-1, SHA2-256, SHA2-512 (A2792)	Certificate in progress
OpenSSL FIPS Object Module	3.0	AES, CKG, CVL, DRBG, DSA, ECDSA, HMAC, KAS-RSA-SSC, KAS-SSC, KBKDF, KDA, KMAC, KTS, KTS-RSA, PBKDF, RSA, SHA-3, SHS, Triple-DES (A1938)	Certificate #4282
VMware OpenSSL FIPS Object Module	2.0.20-vmw	AES, CKG, /drbg, DSA, ECDSA, HMAC, KAS-SSC, RSA, SHS, Triple-DES (C470)	Certificate #3857
VMware's ESXboot Cryptographic Module	1.0	HMAC-SHA2-224, RSA SigVer (FIPS186-4), SHA2-224, SHA2-256, SHA2-384, SHA2-512	Certificate #4442
VMware's Boring Crypto Module	6.0	AES, CVL, DRBG, ECDSA, HMAC, KAS, KAS-SSC, KTS, RSA, SHS, Triple-DES (A4970)	Certificate #4694

Activate and Deactivate FIPS on the vCenter appliance

You can activate or deactivate FIPS-validated cryptography on the vCenter appliance by using HTTP requests. FIPS-validated cryptography is deactivated by default.

You can use various ways to execute HTTP requests. This task shows how to use the Developer Center in the vSphere Client to activate and deactivate the FIPS-validated cryptography on the vCenter appliance. See *VMware vCenter Server Management Programming Guide* for more information about using APIs to work with the vCenter appliance.

1. Log in to the vCenter system with the vSphere Client.
2. From the Menu, select **Developer Center**.
3. Click **API Explorer**.
4. From the **Select API** drop-down menu, select **appliance**.
5. Scroll down through the categories and expand **system/security/global_fips**.
6. Expand **GET** and click **Execute** under **Try it out**.
   You can view the current setting under **Response**.
7. Change the setting.
   a) To activate FIPS, expand **PUT**, enter the following in the `request_body`, and click **Execute**.

```
{
"enabled":true
}
```

   b) To deactivate FIPS, expand **PUT**, enter the following in the `request_body`, and click **Execute**.

```
{
"enabled":false
}
```

The vCenter appliance reboots after you activate or deactivate the FIPS-validated cryptography.

Considerations When Using FIPS

When activating FIPS on vCenter appliance, some components present functional constraints currently.

You should see no differences after activating FIPS on vCenter, however there are some considerations to be aware of.

Table 303: FIPS Considerations

Product or Component	Consideration	Workaround
vSphere Single Sign-On	When you activate FIPS, vCenter supports only cryptographic modules for federated authentication.	Use federated authentication. See the *vSphere Authentication* documentation for details.
Non-VMware and partner vSphere Client UI plug-ins	These plug-ins might not work with FIPS enabled.	Upgrade plug-ins to use conformant encryption libraries. See the topic titled "Preparing Local Plug-ins for FIPS Compliance" in the *vSphere Client SDK* documentation.
Certificates	Certificates with key sizes greater than 3072 bits have not been tested.	Generate certificates with keys using 2048 or 3072 bit sizes.

vSphere Resource Management

vSphere Resource Management describes resource management for VMware®ESX and vCenter® environments.

This documentation focuses on the following topics.

- Resource allocation and resource management concepts
- Virtual machine attributes and admission control
- Resource pools and how to manage them
- Clusters, vSphere® Distributed Resource Scheduler (DRS), vSphere Distributed Power Management (DPM), and how to work with them
- Datastore clusters, Storage DRS, Storage I/O Control, and how to work with them
- Advanced resource management options
- Performance considerations

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

This information is for system administrators who want to understand how the system manages resources and how they can customize the default behavior. It's also essential for anyone who wants to understand and use resource pools, clusters, DRS, datastore clusters, Storage DRS, Storage I/O Control, or vSphere DPM.

This documentation assumes you have a working knowledge of VMware ESX and of vCenter.

> **Note:**
> In this document, "Memory" can refer to physical RAM or Persistent Memory.

Configure Your Resource Allocation Settings in vSphere

When available resource capacity does not meet the demands of the resource consumers (and virtualization overhead), administrators might need to customize the amount of resources that are allocated to virtual machines or resource pools in vSphere.

Use the resource allocation settings to determine the amount of CPU, memory, and storage resources provided for a virtual machine. Administrators have several options for allocating resources.

- Reserve the physical resources of the ESX host or cluster.
- Set an upper bound on the resources that can be allocated to a virtual machine.
- Guarantee that a particular virtual machine is always allocated a higher percentage of the physical resources than other virtual machines.

> **Note:**
> In this chapter, "Memory" refers to physical RAM.

Resource Allocation Limit

Limit specifies an upper bound for CPU or memory that can be allocated to a virtual machine.

A server can allocate more than the reservation to a virtual machine, but can never allocate more than the limit, even if there are unused resources on the system. The limit is expressed in concrete units (megahertz or megabytes).

CPU and memory limits default to unlimited. When the memory limit is unlimited, the amount of memory that is configured for the virtual machine when it was created becomes its effective limit.

Usually, it is not necessary to specify a limit. There are benefits and drawbacks:

- Benefits — Assigning a limit is useful if you start with a small number of virtual machines and you want to manage user expectations. Performance deteriorates as you add more virtual machines. You can simulate having fewer resources available by specifying a limit.
- Drawbacks — You might waste idle resources if you specify a limit. The system does not allow virtual machines to use more resources than the limit, even when the system is underutilized and idle resources are available. Specify the limit only if you have good reasons for doing so.

Resource Allocation Settings Suggestions

Select resource allocation settings (reservation, limit and shares) that are appropriate for your ESX environment.

The following guidelines can help you achieve better performance for your virtual machines.

- Use **Reservation** to specify the minimum acceptable amount of CPU or memory, not the amount you want to have available. The amount of concrete resources represented by a reservation does not change when you change the environment, such as by adding or removing virtual machines. The ESX host assigns additional resources as available based on the limit for your virtual machine, the number of shares and estimated demand.
- When specifying the reservations for virtual machines, do not commit all resources (plan to leave at least 10% unreserved). As you move closer to fully reserving all capacity in the system, it becomes increasingly difficult to make changes to reservations and to the resource pool hierarchy without violating admission control. In a DRS-enabled cluster, reservations that fully commit the capacity of the cluster or of individual ESX hosts in the cluster can prevent DRS from migrating virtual machines between ESX hosts.
- If you expect frequent changes to the total available resources, use **Shares** to allocate resources fairly across virtual machines. If you use **Shares**, and you upgrade the ESX host, for example, each virtual machine stays at the same priority (keeps the same number of shares) even though each share represents a larger amount of memory ot CPU.

Edit Settings in vSphere

Use the Edit Settings dialog box to change allocations for memory and CPU resources.

1. Browse to the virtual machine in the vSphere Client.
2. Right-click and select **Edit Settings**.
3. Edit the CPU Resources.

Option	Description
Shares	CPU shares for this resource pool with respect to the parent's total. Sibling resource pools share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values respectively in a 1:2:4 ratio. Select **Custom** to give each virtual machine a specific number of shares, which expresses a proportional weight.
Reservation	Guaranteed CPU allocation for this resource pool.
Limit	Upper limit for this resource pool's CPU allocation. Select **Unlimited** to specify no upper limit.

4. Edit the Memory Resources.

Option	Description
Shares	Memory shares for this resource pool with respect to the parent's total. Sibling resource pools share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values respectively in a 1:2:4 ratio. Select **Custom** to give each virtual machine a specific number of shares, which expresses a proportional weight.
Reservation	Guaranteed memory allocation for this resource pool.
Limit	Upper limit for this resource pool's memory allocation. Select **Unlimited** to specify no upper limit.

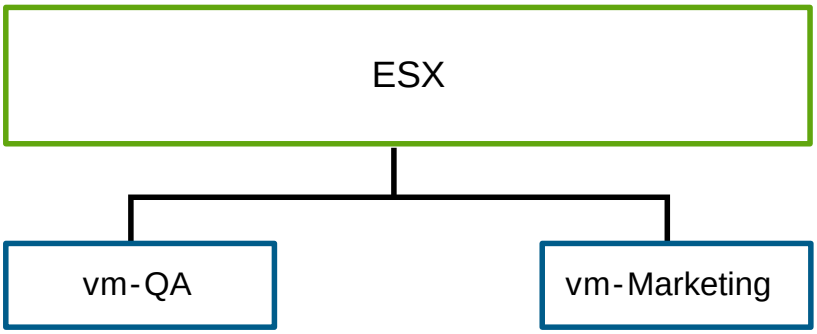
5. Click **OK**.

Changing Resource Allocation Settings—Example

The following example illustrates how you can change resource allocation settings to improve virtual machine performance.

Assume that on an ESX host, you have created two new virtual machines — one each for your QA (VM-QA) and Marketing (VM-Marketing) departments.

Figure 98: Single ESX Host with Two Virtual Machines



In the following example, assume that VM-QA is memory-intensive, and accordingly, you want to change the resource allocation settings for the two virtual machines to:

- Specify that, when system memory is overcommitted, VM-QA can use twice as much CPU and memory resources as the Marketing virtual machine. Set the CPU shares and memory shares for VM-QA to **High** and for VM-Marketing set them to **Normal**.
- Ensure that the Marketing virtual machine has a certain amount of guaranteed CPU resources. You can do so using a reservation setting.

1. Browse to the virtual machines in the vSphere Client.
2. Right-click **VM-QA**, the virtual machine for which you want to change shares, and select **Edit Settings**.
3. Under **Virtual Hardware**, expand CPU and select **High** from the **Shares** drop-down menu.
4. Under **Virtual Hardware**, expand Memory and select **High** from the **Shares** drop-down menu.
5. Click **OK**.
6. Right-click the marketing virtual machine (**VM-Marketing**) and select **Edit Settings**.
7. Under **Virtual Hardware**, expand CPU and change the **Reservation** value to the desired number.
8. Click **OK**.

Admission Control in vSphere

When you power on a virtual machine, the system checks the amount of CPU and memory resources that have not yet been reserved. Based on the available unreserved resources, the system determines whether it can guarantee the reservation for which the virtual machine is configured (if any). This process is called admission control.

If enough unreserved CPU and memory are available, or if there is no reservation, the virtual machine is powered on. Otherwise, an `Insufficient Resources` warning appears.

> **Note:**
> In addition to the user-specified memory reservation, for each virtual machine there is also an amount of overhead memory. This extra memory commitment is included in the admission control calculation.

When the vSphere DPM feature is enabled, ESX hosts might be placed in standby mode (that is, powered off) to reduce power consumption. The unreserved resources provided by these ESX hosts are considered available for admission control. If a virtual machine cannot be powered on without these resources, a recommendation to power on sufficient standby ESX hosts is made. For more information, see Managing Power Resources.

CPU Virtualization in vSphere

CPU virtualization emphasizes performance and runs directly on the processor whenever possible. The underlying physical resources are used whenever possible and the virtualization layer runs instructions only as needed to make virtual machines operate as if they were running directly on a physical machine.

CPU virtualization is not the same thing as emulation. ESX does not use emulation to run virtual CPUs. With emulation, all operations are run in software by an emulator. A software emulator allows programs to run on a computer system other than the one for which they were originally written. The emulator does this by emulating, or reproducing, the original computer's behavior by accepting the same data or inputs and achieving the same results. Emulation provides portability and runs software designed for one platform across several platforms.

When CPU resources are overcommitted, the ESX host time-slices the physical processors across all virtual machines so each virtual machine runs as if it has its specified number of virtual processors. When an ESX host runs multiple virtual machines, it allocates to each virtual machine a share of the physical resources. With the default resource allocation settings, all virtual machines associated with the same ESX host receive an equal share of CPU per virtual CPU. This means that a single-processor virtual machine is assigned only half of the resources of a dual-processor virtual machine.

Software-Based CPU Virtualization

With software-based CPU virtualization, the guest application code runs directly on the processor, while the guest privileged code is translated and the translated code runs on the processor.

The translated code is slightly larger and usually runs more slowly than the native version. As a result, guest applications, which have a small privileged code component, run with speeds very close to native. Applications with a significant privileged code component, such as system calls, traps, or page table updates can run slower in the virtualized environment.

Hardware-Assisted CPU Virtualization

Certain processors provide hardware assistance for CPU virtualization.

When using this assistance, the guest can use a separate mode of execution called guest mode. The guest code, whether application code or privileged code, runs in the guest mode. On certain events, the processor exits out of guest mode and enters root mode. The hypervisor executes in the root mode, determines the reason for the exit, takes any required actions, and restarts the guest in guest mode.

When you use hardware assistance for virtualization, there is no need to translate the code. As a result, system calls or trap-intensive workloads run very close to native speed. Some workloads, such as those involving updates to page tables, lead to a large number of exits from guest mode to root mode. Depending on the number of such exits and total time spent in exits, hardware-assisted CPU virtualization can speed up execution significantly.

Virtualization and Processor-Specific Behavior

Although VMware software virtualizes the CPU, the virtual machine detects the specific model of the processor on which it is running.

Processor models might differ in the CPU features they offer, and applications running in the virtual machine can use these features. Therefore, it is not possible to use vMotion® to migrate virtual machines between systems running on processors with different feature sets. You can avoid this restriction, in some cases, by using Enhanced vMotion Compatibility (EVC) with processors that support this feature. See the *vCenter and Host Management* documentation for more information.

Performance Implications of CPU Virtualization

CPU virtualization adds varying amounts of overhead depending on the workload and the type of virtualization used.

An application is CPU-bound if it spends most of its time executing instructions rather than waiting for external events such as user interaction, device input, or data retrieval. For such applications, the CPU virtualization overhead includes the additional instructions that must be executed. This overhead takes CPU processing time that the application itself can use. CPU virtualization overhead usually translates into a reduction in overall performance.

For applications that are not CPU-bound, CPU virtualization likely translates into an increase in CPU use. If spare CPU capacity is available to absorb the overhead, it can still deliver comparable performance in terms of overall throughput.

ESX supports up to 960 virtual processors (CPUs) for each virtual machine.

> **Note:**
> Deploy single-threaded applications on uniprocessor virtual machines, instead of on SMP virtual machines that have multiple CPUs, for the best performance and resource use.
>
> Single-threaded applications can take advantage only of a single CPU. Deploying such applications in dual-processor virtual machines does not speed up the application. Instead, it causes the second virtual CPU to use physical resources that other virtual machines could otherwise use.

Administering CPU Resources in vSphere

You can configure virtual machines with one or more virtual processors, each with its own set of registers and control structures.

When a virtual machine is scheduled, its virtual processors are scheduled to run on physical processors. The VMkernel Resource Manager schedules the virtual CPUs on physical CPUs, thereby managing the virtual machine's access to physical CPU resources.

View Processor Information

You can access information about current CPU configuration in the vSphere Client.

1. Browse to the ESX host in the vSphere Client.
2. Under **Hardware**, expand **CPU** to view the information about the number and type of physical processors and the number of logical processors.

> **Note:**
> In hyperthreaded systems, each hardware thread is a logical processor. For example, a dual-core processor with hyperthreading activated has two cores and four logical processors.

Specifying CPU Configuration

You can specify CPU configuration to improve resource management. However, if you do not customize CPU configuration, the ESX host uses defaults that work well in most situations.

You can specify CPU configuration in the following ways:

- Use the attributes and special features available through the vSphere Client. The vSphere Client allows you to connect to the ESX host or a vCenter instance.
- Use advanced settings under certain circumstances.
- Use the vSphere SDK for scripted CPU allocation.
- Use hyperthreading.

Multicore Processors

Multicore processors provide many advantages for a ESX host performing multitasking of virtual machines.

> **Note:**
> In this topic, "Memory" can refer to physical RAM or Persistent Memory.

Intel and AMD have developed processors which combine two or more processor cores into a single integrated circuit (often called a package or socket). VMware uses the term socket to describe a single package which can have one or more processor cores with one or more logical processors in each core.

A dual-core processor, for example, provides almost double the performance of a single-core processor, by allowing two virtual CPUs to run at the same time. Cores within the same processor are typically configured with a shared last-level cache used by all cores, potentially reducing the need to access slower main memory. A shared memory bus that

connects a physical processor to main memory can limit performance of its logical processors when the virtual machines running on them are running memory-intensive workloads which compete for the same memory bus resources.

Each logical processor of each processor core is used independently by the ESX CPU scheduler to run virtual machines, providing capabilities similar to SMP systems. For example, a two-way virtual machine can have its virtual processors running on logical processors that belong to the same core, or on logical processors on different physical cores.

The ESX CPU scheduler can detect the processor topology and the relationships between processor cores and the logical processors on them. It uses this information to schedule virtual machines and optimize performance.

The ESX CPU scheduler can interpret processor topology, including the relationship between sockets, cores, and logical processors. The scheduler uses topology information to optimize the placement of virtual CPUs onto different sockets. This optimization can maximize overall cache usage, and to improve cache affinity by minimizing virtual CPU migrations.

Hyperthreading with vSphere

Hyperthreading technology allows a single physical processor core to behave like two logical processors. The processor can run two independent applications at the same time. To avoid confusion between logical and physical processors, Intel refers to a physical processor as a socket, and the discussion in this chapter uses that terminology as well.

Intel Corporation developed hyperthreading technology to enhance the performance of its Pentium IV and Xeon processor lines. Hyperthreading technology allows a single processor core to execute two independent threads simultaneously.

While hyperthreading does not double the performance of a system, it can increase performance by better utilizing idle resources leading to greater throughput for certain important workload types. An application running on one logical processor of a busy core can expect slightly more than half of the throughput that it obtains while running alone on a non-hyperthreaded processor. Hyperthreading performance improvements are highly application-dependent, and some applications might see performance degradation with hyperthreading because many processor resources (such as the cache) are shared between logical processors.

> **Note:**
> On processors with Intel Hyper-Threading technology, each core can have two logical processors which share most of the core's resources, such as memory caches and functional units. Such logical processors are usually called threads.

Some processors do not support hyperthreading and as a result have only one thread per core. For such processors, the number of cores also matches the number of logical processors.

Hyperthreading and ESX Hosts

An ESX host that is enabled for hyperthreading should behave similarly to an ESX host without hyperthreading. You might need to consider certain factors if you enable hyperthreading, however.

ESX hosts manage processor time intelligently to guarantee that load is spread smoothly across processor cores in the system. Logical processors on the same core have consecutive CPU numbers, so that CPUs 0 and 1 are on the first core together, CPUs 2 and 3 are on the second core, and so on. Virtual machines are preferentially scheduled on two different cores rather than on two logical processors on the same core.

If there is no work for a logical processor, it is put into a halted state, which frees its execution resources and allows the virtual machine running on the other logical processor on the same core to use the full execution resources of the core. The VMware scheduler properly accounts for this halt time, and charges a virtual machine running with the full resources of a core more than a virtual machine running on a half core. This approach to processor management ensures that the server does not violate any of the standard ESX resource allocation rules.

Consider your resource management needs before you enable CPU affinity on ESX hosts using hyperthreading. For example, if you bind a high priority virtual machine to CPU 0 and another high priority virtual machine to CPU 1, the two virtual machines have to share the same physical core. In this case, it can be impossible to meet the resource demands of these virtual machines. Ensure that any custom affinity settings make sense for a hyperthreaded system.

Enable Hyperthreading

To enable hyperthreading, you must first enable it in your system's BIOS settings and then turn it on in the vSphere Client. Hyperthreading is enabled by default.

Consult your system documentation to determine whether your CPU supports hyperthreading.

1. Ensure that your system supports hyperthreading technology.
2. Enable hyperthreading in the system BIOS.
   Some manufacturers label this option **Logical Processor**, while others call it **Enable Hyperthreading**.
3. Ensure that hyperthreading is enabled for the ESX host.
   a) Browse to the ESX host in the vSphere Client.
   b) Click **Configure**.
   c) Under **System**, click **Advanced System Settings** and select **VMkernel.Boot.hyperthreading**.

      You must restart the ESX host for the setting to take effect. Hyperthreading is enabled if the value is `true`.
4. Under **Hardware**, click **Processors** to view the number of Logical processors.

Hyperthreading is enabled.

Using CPU Affinity with vSphere

By specifying a CPU affinity setting for each virtual machine, you can restrict the assignment of virtual machines to a subset of the available processors in multiprocessor systems. By using this feature, you can assign each virtual machine to processors in the specified affinity set.

CPU affinity specifies virtual machine-to-processor placement constraints and is different from the relationship created by a VM-VM or VM-Host affinity rule, which specifies virtual machine-to-virtual machine ESX host placement constraints.

In this context, the term CPU refers to a logical processor on a hyperthreaded system and refers to a core on a non-hyperthreaded system.

The CPU affinity setting for a virtual machine applies to all of the virtual CPUs associated with the virtual machine and to all other threads (also known as worlds) associated with the virtual machine. Such virtual machine threads perform processing required for emulating mouse, keyboard, screen, CD-ROM, and miscellaneous legacy devices.

In some cases, such as display-intensive workloads, significant communication might occur between the virtual CPUs and these other virtual machine threads. Performance might degrade if the virtual machine's affinity setting prevents these additional threads from being scheduled concurrently with the virtual machine's virtual CPUs. Examples of this include a uniprocessor virtual machine with affinity to a single CPU or a two-way SMP virtual machine with affinity to only two CPUs.

For the best performance, when you use manual affinity settings, VMware recommends that you include at least one additional physical CPU in the affinity setting to allow at least one of the virtual machine's threads to be scheduled at the same time as its virtual CPUs. Examples of this include a uniprocessor virtual machine with affinity to at least two CPUs or a two-way SMP virtual machine with affinity to at least three CPUs.

Assign a Virtual Machine to a Specific Processor

Using CPU affinity, you can assign a virtual machine to a specific processor. This allows you to restrict the assignment of virtual machines to a specific available processor in multiprocessor systems.

1. Create a virtual machine in the vSphere Client. For more information on how to create a virtual machine, see Create a Virtual Machine with the New Virtual Machine Wizard.
2. On the **Customize hardware** page, expand **CPU**.
3. In the **Scheduling Affinity** field, enter a comma-separated list of hyphenated processor ranges.
   Use '-' for ranges and ',' to separate values.
   For example, "0, 2, 4-7" would indicate processors 0, 2, 4, 5, 6 and 7.

4. Click **OK**.

Potential Issues with CPU Affinity

Before you use CPU affinity, you must consider certain issues.

Potential issues with CPU affinity include:

- For multiprocessor systems, ESX systems perform automatic load balancing. Avoid manual specification of virtual machine affinity to improve the scheduler's ability to balance load across processors.
- Affinity can interfere with the ESX host's ability to meet the reservation and shares specified for a virtual machine.
- Because CPU admission control does not consider affinity, a virtual machine with manual affinity settings might not always receive its full reservation.
  Virtual machines that do not have manual affinity settings are not adversely affected by virtual machines with manual affinity settings.
- When you move a virtual machine from one ESX host to another, affinity might no longer apply because the new ESX host might have a different number of processors.
- The NUMA scheduler might not be able to manage a virtual machine that is already assigned to certain processors using affinity. For more information, see Using NUMA Systems with ESX.
- Affinity can affect the ability of the ESX host to schedule virtual machines on multicore or hyperthreaded processors to take full advantage of resources shared on such processors.

Host Power Management Policies in ESX

You can apply several power management features in ESX that the host hardware provides to adjust the balance between performance and power. You can control how ESX uses these features by selecting a power management policy.

Selecting a high-performance policy provides more absolute performance but at lower efficiency and performance per watt. Low-power policies provide less absolute performance but at higher efficiency.

You can select a policy for the ESX host that you manage by using the VMware Host Client. If you do not select a policy, ESX uses Balanced by default.

Table 304: CPU Power Management Policies

Power Management Policy	Description
High Performance	Do not use any power management features.
Balanced (Default)	Reduce energy consumption with minimal performance compromise
Low Power	Reduce energy consumption at the risk of lower performance
Custom	User-defined power management policy. Advanced configuration becomes available.
With Intel Hardware Controlled P-states (Intel HWP) enabled, these policies provide hints to the hardware to deliver similar energy and performance trade-offs.	

Performance State (P-state) operates where a busy CPU executes at a specific frequency and voltage. Based on workload, this P-state can be varied while the CPU is running to save power while minimizing the impact on performance. This helps maximize performance per watt and is an important feature available on modern processors.

Legacy ACPI P-states provide a mechanism for software management of P-states, whereby ESX can periodically request a specific P-state based on the current system utilization.

Newer processors support hardware management of P-states, whereby ESX provides hints to the hardware about the desired operating range and performance bias, and the hardware makes load-based decisions about what the current P-state should be at any given time without software intervention.

Intel Hardware Controlled P-states (HWP) is an implementation of the ACPI Collaborative Processor Performance Control (CPPC) standard and is supported in ESX 9.0 and later on Intel Sapphire Rapids and newer platforms. This document refers to HWP or CPPC as Intel HWP for this feature on Intel processors.

These two types of P-state management are mutually exclusive, so ESX only supports one of these at a time, and changing between the two types requires a reboot. You can determine what type of P-states are managed by ESX from the Technology string in the Power Management section of the vCenter system.

If one of the P-state management types is supported, the string contains either **ACPI P-states** or **ACPI CPPC**.

Important:

Intel HWP, if available, is enabled and configured by default on supported platforms in ESX 9.0 and later. For more information on how to configure HWP for use by ESX, refer to your BIOS documentation.

When a CPU is idle, ESX can apply deep halt states, also known as C-states. The deeper the C-state, the less power the CPU uses, but it also takes longer for the CPU to start running again. When a CPU becomes idle, ESX applies an algorithm to predict the idle state duration and chooses an appropriate C-state to enter. In power management policies that do not use deep C-states, ESX uses only the shallowest halt state for idle CPUs, C1.

Select a CPU Power Management Policy

You set the CPU power management policy for a ESX host using the vSphere Client.

Verify that the BIOS settings on the ESX host system allow the operating system to control power management (for example, **OS Controlled**). If the ESX host hardware does not allow the operating system to manage power, only the Not Supported policy is available. (On some systems, only the High Performance policy is available.)

1. Browse to the ESX host in the vSphere Client.
2. Click **Configure**.
3. Under Hardware, select Overview and click **Edit Power Policy**.
4. Select a power management policy for the ESX host and click **OK**.

   The policy selection is saved in the ESX host configuration and can be used again at boot time. You can change it at any time, and it does not require a server reboot.

Configure Custom Policy Parameters for the ESX Host Power Management

When you use the Custom policy for the ESX host power management, ESX bases its power management policy on the values of several advanced configuration parameters.

Select **Custom** for the power management policy, as described in Select a CPU Power Management Policy.

1. Browse to the ESX host in the vSphere Client.
2. Click **Configure**.
3. Under **System**, select **Advanced System Settings**.
4. In the right pane, you can edit the power management parameters that affect the Custom policy.

   Power management parameters that affect the Custom policy have descriptions that begin with **In Custom policy**. All other power parameters affect all power management policies.
5. Select the parameter and click the **Edit** button.

   Note:
   The default values of power management parameters match the Balanced policy.

Parameter	Description
Power.UsePStates	Use ACPI P-states or CPPC to save power when the processor is busy. When deactivated, the ESX configures to use the highest available frequency.
Power.MaxCpuLoad	Use P-states to save power on a CPU only when the CPU is busy for less than the given percentage of real time. This parameter is not used when CPPC is enabled.
Power.MinFreqPct	Do not use any P-states slower than the given percentage of full CPU speed.

Parameter	Description
Power.MaxFreqPct	Do not use P-states that are faster than the given percentage of full CPU speed. On Intel processors, setting this to a value less than 100% will turn off turbo on all processors. **Note:** This limit does not apply to CPUs running low latency workloads, as they are run at full speed. For more information on how to adjust the latency sensitivity of a virtual machine, see Latency Sensitivity.
Power.TimerHz	Controls how many times per second ESX reevaluates which P-state each CPU should be in. When CPPC is enabled, it is only used for debugging.
Power.UseCStates	Use deep ACPI C-states (C2 or below) when the processor is idle.
Power.CStateMaxLatency	Do not use C-states whose latency is greater than this value.
Power.CStateResidencyCoef	When a CPU becomes idle, choose the deepest C-state whose latency multiplied by this value is less than the ESX host prediction of how long the CPU will remain idle. Larger values make ESX more conservative about using deep C-states, while smaller values are more aggressive.
Power.CStatePredictionCoef	A parameter in the ESX algorithm for predicting how long a CPU that becomes idle will remain idle. Changing this value is not recommended.
Power.PerfBias	Performance Energy Bias Hint (Intel-only). Sets an MSR on Intel processors to an Intel-recommended value. Intel recommends 0 for high performance, 6 for balanced, and 15 for low power. Other values are undefined.
Power.PerfBiasEnable	Use Performance Energy Bias Hint.

6. Click **OK**.

Memory Virtualization with vSphere

Before you manage memory resources, you should understand how they are being virtualized and used by ESX.

The VMkernel manages all physical RAM on the ESX host. The VMkernel dedicates part of this managed physical RAM for its own use. The rest is available for use by virtual machines.

The virtual and physical memory space is divided into blocks called pages. When physical memory is full, the data for virtual pages that are not present in physical memory are stored on disk. Depending on processor architecture, pages are typically 4 KB or 2 MB. See Advanced Memory Attributes.

Virtual Machine Memory

Each virtual machine consumes memory based on its configured size, plus additional overhead memory for virtualization.

The configured size is the amount of memory that is presented to the guest operating system. This is different from the amount of physical RAM that is allocated to the virtual machine. The latter depends on the resource settings (shares, reservation, limit) and the level of memory pressure on the ESX host.

For example, consider a virtual machine with a configured size of 1GB. When the guest operating system boots, it detects that it is running on a dedicated machine with 1GB of physical memory. In some cases, the virtual machine might be allocated the full 1GB. In other cases, it might receive a smaller allocation. Regardless of the actual allocation, the guest operating system continues to behave as though it is running on a dedicated machine with 1GB of physical memory.

Shares Specify the relative priority for a virtual machine if more than the reservation is available.

Reservation Is a guaranteed lower bound on the amount of physical RAM that the ESX host reserves for the virtual machine, even when memory is overcommitted. Set the reservation to a level that ensures the virtual machine has sufficient memory to run efficiently, without excessive paging.

After a virtual machine consumes all of the memory within its reservation, it is allowed to retain that amount of memory and this memory is not reclaimed, even if the virtual machine becomes idle. Some guest operating systems (for example, Linux) might not access all of the configured memory immediately after booting. Until the virtual machines consumes all of the memory within its reservation, VMkernel can

	allocate any unused portion of its reservation to other virtual machines. However, after the guest's workload increases and the virtual machine consumes its full reservation, it is allowed to keep this memory.
Limit	Is an upper bound on the amount of physical RAM that the ESX host can allocate to the virtual machine. The virtual machine's memory allocation is also implicitly limited by its configured size.

Memory Overcommitment

For each running virtual machine, the system reserves physical RAM for the virtual machine's reservation (if any) and for its virtualization overhead.

The total configured memory sizes of all virtual machines may exceed the amount of available physical memory on the ESX host. However, it doesn't necessarily mean memory is overcommitted. Memory is overcommitted when the combined working memory footprint of all virtual machines exceed that of the ESX host memory sizes.

Because of the memory management techniques the ESX host uses, your virtual machines can use more virtual RAM than there is physical RAM available on the ESX host. For example, you can have an ESX host with 2 GB of memory and run four virtual machines with 1 GB of memory each. In that case, the memory is overcommitted. For instance, if all four virtual machines are idle, the combined consumed memory may be well below 2 GB. However, if all 4 GB virtual machines are actively consuming memory, then their memory footprint may exceed 2 GB and the ESX host will become overcommitted.

Overcommitment makes sense because, typically, some virtual machines are lightly loaded while others are more heavily loaded, and relative activity levels vary over time.

To improve memory utilization, the ESX host transfers memory from idle virtual machines to virtual machines that need more memory. Use the Reservation or Shares parameter to preferentially allocate memory to important virtual machines. This memory remains available to other virtual machines if it is not in use. ESX implements various mechanisms such as ballooning, memory sharing, memory compression and swapping to provide reasonable performance even if the ESX host is not heavily memory overcommitted.

An ESX host can run out of memory if virtual machines consume all reservable memory in a memory-overcommitted environment. Although the powered-on virtual machines are not affected, a new virtual machine might fail to power on due to a lack of memory.

> **Note:**
> All virtual machine memory overhead is also considered reserved.

In addition, memory compression is enabled by default on ESX hosts to improve virtual machine performance when memory is overcommitted, as described in Memory Compression.

Memory Sharing

Memory sharing is a proprietary ESX technique that can help achieve greater memory density on an ESX host.

Memory sharing relies on the observation that several virtual machines might be running instances of the same guest operating system. These virtual machines might have the same applications or components loaded, or contain common data. In such cases, an ESX host uses a proprietary Transparent Page Sharing (TPS) technique to eliminate redundant copies of memory pages. With memory sharing, a workload running on a virtual machine often consumes less memory than it might when running on physical machines. As a result, higher levels of overcommitment can be supported efficiently. The amount of memory saved by memory sharing depends on whether the workload consists of nearly identical machines which might free up more memory. A more diverse workload might result in a lower percentage of memory savings.

> **Note:**
>
> Due to security concerns, inter-virtual machine transparent page sharing is deactivated by default and page sharing is being restricted to intra-virtual machine memory sharing. Page sharing does not occur across virtual machines and only occurs inside a virtual machine. See Sharing Memory Across Virtual Machines for more information.

Memory Virtualization

Because of the extra level of memory mapping introduced by virtualization, ESX can effectively manage memory across all virtual machines.

Some of the physical memory of a virtual machine might be mapped to shared pages or to pages that are unmapped, or swapped out.

A host performs virtual memory management without the knowledge of the guest operating system and without interfering with the guest operating system’s own memory management subsystem.

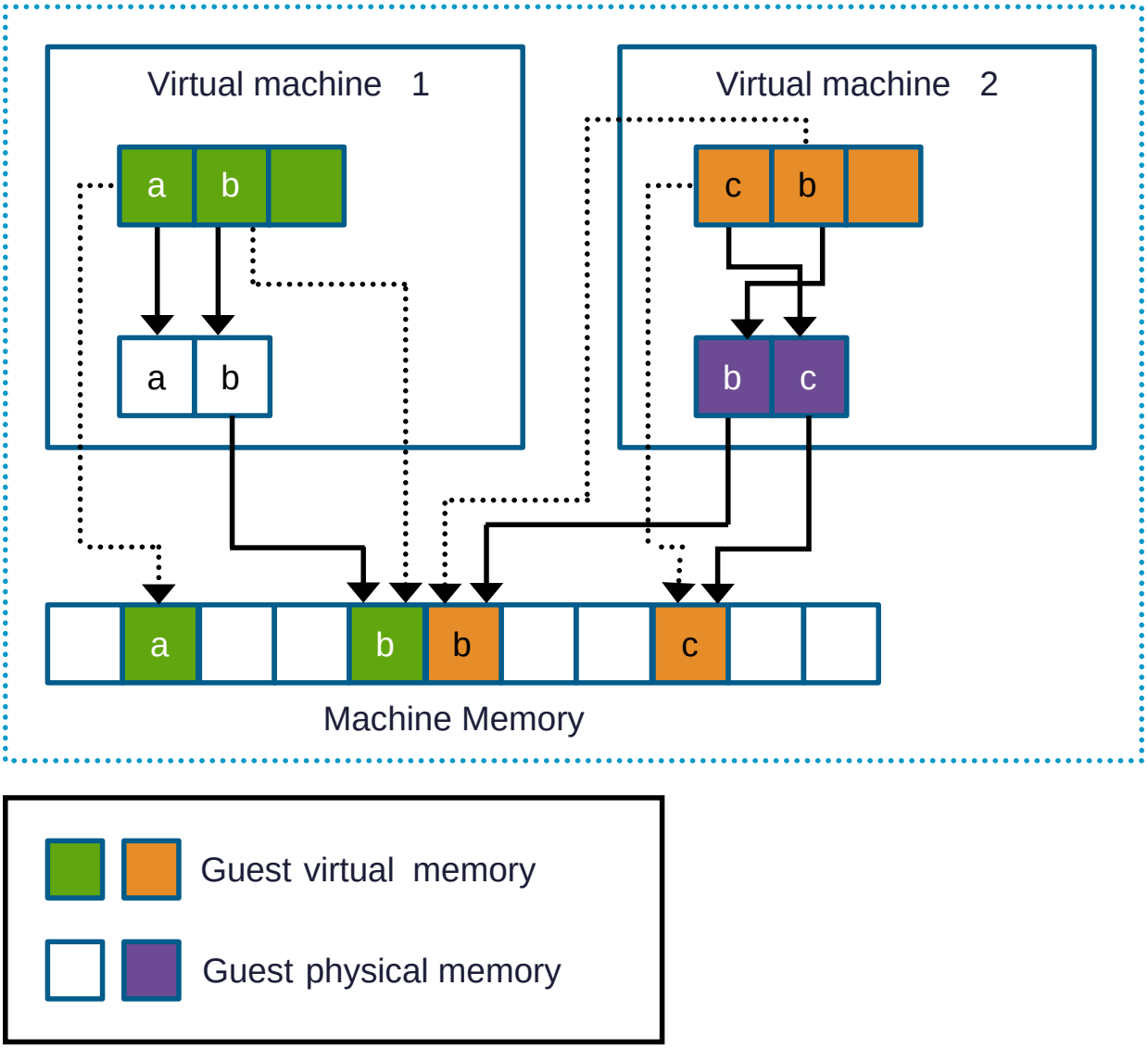
The Virtual Machine Manager (VMM) for each virtual machine maintains a mapping from the guest operating system's physical memory pages to the physical memory pages on the underlying machine. (VMware refers to the underlying ESX host physical pages as “machine” pages and the guest operating system’s physical pages as “physical” pages.)

Each virtual machine sees a contiguous, zero-based, addressable physical memory space. The underlying machine memory on the server used by each virtual machine is not necessarily contiguous.

The guest virtual to guest physical addresses are managed by the guest operating system. The hypervisor is only responsible for translating the guest physical addresses to machine addresses. Hardware-assisted memory virtualization utilizes the hardware facility to generate the combined mappings with the guest's page tables and the nested page tables maintained by the hypervisor.

The diagram illustrates the ESX implementation of memory virtualization.

Figure 99: ESX Memory Mapping



- The boxes represent pages, and the arrows show the different memory mappings.
- The arrows from guest virtual memory to guest physical memory show the mapping maintained by the page tables in the guest operating system. (The mapping from virtual memory to linear memory for x86-architecture processors is not shown).
- The arrows from guest physical memory to machine memory show the mapping maintained by the VMM.
- The dashed arrows show the mapping from guest virtual memory to machine memory in the shadow page tables also maintained by the VMM. The underlying processor running the virtual machine uses the shadow page table mappings.

Hardware-Assisted Memory Virtualization

Some CPUs, such as AMD SVM-V and the Intel Xeon 5500 series, provide hardware support for memory virtualization by using two layers of page tables.

> **Note:**
>
> In this topic, "Memory" can refer to physical RAM or persistent memory.

The first layer of page tables stores guest virtual-to-physical translations, while the second layer of page tables stores guest physical-to-machine translation. The TLB (translation look-aside buffer) is a cache of translations maintained by the processor's memory management unit (MMU) hardware. A TLB miss is a miss in this cache and the hardware needs to go to memory (possibly many times) to find the required translation. For a TLB miss to a certain guest virtual address, the

hardware looks at both page tables to translate guest virtual address to machine address. The first layer of page tables is maintained by the guest operating system. The VMM only maintains the second layer of page tables.

Performance Considerations

When you use hardware assistance, you eliminate the overhead for software memory virtualization. In particular, hardware assistance eliminates the overhead required to keep shadow page tables in synchronization with guest page tables. However, the TLB miss latency when using hardware assistance is significantly higher. By default the hypervisor uses large pages in hardware assisted modes to reduce the cost of TLB misses. As a result, whether or not a workload benefits by using hardware assistance primarily depends on the overhead the memory virtualization causes when using software memory virtualization. If a workload involves a small amount of page table activity (such as process creation, mapping the memory, or context switches), software virtualization does not cause significant overhead. Conversely, workloads with a large amount of page table activity are likely to benefit from hardware assistance.

By default the hypervisor uses large pages in hardware assisted modes to reduce the cost of TLB misses. The best performance is achieved by using large pages in both guest virtual to guest physical and guest physical to machine address translations.

The option `LPage.LPageAlwaysTryForNPT` can change the policy for using large pages in guest physical to machine address translations. For more information, see Advanced Memory Attributes.

Support for Large Page Sizes

ESX provides limited support for large page sizes.

x86 architecture allows system software to use 4KB, 2MB and 1GB pages. We refer to 4KB pages as small pages while 2MB and 1GB pages are referred to as large pages. Large pages relieve translation lookaside buffer (TLB) pressure and reduce the cost of page table walks, which results in improved workload performance.

In virtualized environments, large pages can be used by the hypervisor and the guest operating system independently. While the biggest performance impact is achieved if large pages are used by the guest and the hypervisor, in most cases a performance impact can be observed even if large pages are used only at the hypervisor level.

ESX hypervisor uses 2MB pages for backing guest vRAM by default. vSphere ESX provides a limited support for backing guest vRAM with 1GB pages. For more information, see *Backing Guest vRAM with 1GB Pages*.

Administer Your Memory Resources with vSphere

Using the vSphere Client you can view information about and make changes to memory allocation settings. To administer your memory resources effectively, you must also be familiar with memory overhead, idle memory tax, and how ESX hosts reclaim memory.

When administering memory resources, you can specify memory allocation. If you do not customize memory allocation, the ESX host uses defaults that work well in most situations.

You can specify memory allocation in several ways.

- Use the attributes and special features available through the vSphere Client. The vSphere Client allows you to connect to the ESX host or vCenter system.
- Use advanced settings.
- Use the vSphere SDK for scripted memory allocation.

> **Note:**
> In this chapter, "Memory" can refer to physical RAM or Persistent Memory.

Understanding Memory Overhead

Virtualization of memory resources has some associated overhead.

ESX virtual machines can incur two kinds of memory overhead.

- The additional time to access memory within a virtual machine.

- The extra space needed by the ESX host for its own code and data structures, beyond the memory allocated to each virtual machine.

ESX memory virtualization adds little time overhead to memory accesses. Because the processor's paging hardware uses page tables (shadow page tables for software-based approach or two level page tables for hardware-assisted approach) directly, most memory accesses in the virtual machine can execute without address translation overhead.

The memory space overhead has two components.

- A fixed, system-wide overhead for the VMkernel.
- Additional overhead for each virtual machine.

Overhead memory includes space reserved for the virtual machine frame buffer and various virtualization data structures, such as shadow page tables. Overhead memory depends on the number of virtual CPUs and the configured memory for the guest operating system.

Overhead Memory on Virtual Machines

Virtual machines require a certain amount of available overhead memory to power on. You should be aware of the amount of this overhead.

The amount of overhead memory needed for a virtual machine depends on a large number of factors, including the number of vCPUs and memory size, the number and types of devices, the execution mode that the monitor is using and the hardware version of the virtual machine. The version of vSphere you are using can also effect the amount of memory needed. VMX automatically calculates the amount of overhead memory needed for a virtual machine.

In order to find out how much overhead memory is needed for your specific configuration, first power on the virtual machine in question. Look in the vmware.log file. When the virtual machine powers on, the amount of overhead memory it needs is printed to the log. Search within the log for `VMMEM` to see the initial and precise amount of overhead memory reserved for the virtual machine.

How ESX Hosts Allocate Memory

An ESX host allocates the memory specified by the `Limit` parameter to each virtual machine, unless memory is overcommitted. ESX never allocates more memory to a virtual machine than its specified physical memory size.

For example, a 1GB virtual machine might have the default limit (unlimited) or a user-specified limit (for example 2GB). In both cases, the ESX host never allocates more than 1GB, the physical memory size that was specified for it.

When memory is overcommitted, each virtual machine is allocated an amount of memory somewhere between what is specified by **Reservation** and what is specified by **Limit**. The amount of memory granted to a virtual machine above its reservation usually varies with the current memory load.

An ESX host determines allocations for each virtual machine based on the number of shares allocated to it and an estimate of its recent working set size.

- Shares — ESX hosts use a modified proportional-share memory allocation policy. Memory shares entitle a virtual machine to a fraction of available physical memory.
- Working set size — ESX hosts estimate the working set for a virtual machine by monitoring memory activity over successive periods of virtual machine execution time. Estimates are smoothed over several time periods using techniques that respond rapidly to increases in working set size and more slowly to decreases in working set size.
  This approach ensures that a virtual machine from which idle memory is reclaimed can ramp up quickly to its full share-based allocation when it starts using its memory more actively.
  Memory activity is monitored to estimate the working set sizes for a default period of 60 seconds. To modify this default , adjust the `Mem.SamplePeriod` advanced setting. See Set Advanced Host Attributes.

Memory Tax for Idle Virtual Machines

If a virtual machine is not actively using all of its currently allocated memory, ESX charges more for idle memory than for memory that is in use. This is done to help prevent virtual machines from hoarding idle memory.

The idle memory tax is applied in a progressive fashion. The effective tax rate increases as the ratio of idle memory to active memory for the virtual machine rises. (In earlier versions of ESX that did not support hierarchical resource pools, all idle memory for a virtual machine was taxed equally.)

You can modify the idle memory tax rate with the `Mem.IdleTax` option. Use this option, together with the `Mem.SamplePeriod` advanced attribute, to control how the system determines target memory allocations for virtual machines. See Set Advanced Host Attributes.

Note:
In most cases, changes to `Mem.IdleTax` are not necessary nor appropriate.

VMX Swap Files

Virtual machine executable (VMX) swap files allow the ESX host to greatly reduce the amount of overhead memory reserved for the VMX process.

Note:

VMX swap files are not related to the swap to ESX host swap cache feature or to the regular ESX host-level swap files.

ESX reserves memory per virtual machine for a variety of purposes. Memory for the needs of certain components, such as the virtual machine monitor (VMM) and virtual devices, is fully reserved when a virtual machine is powered on. However, some of the overhead memory that is reserved for the VMX process can be swapped. The VMX swap feature reduces the VMX memory reservation significantly (for example, from about 50 MB or more per virtual machine to about 10 MB per virtual machine). This allows the remaining memory to be swapped out when the ESX host memory is overcommitted, reducing overhead memory reservation for each virtual machine.

The ESX host creates VMX swap files automatically, provided there is sufficient free disk space at the time a virtual machine is powered on.

Memory Reclamation

ESX hosts can reclaim memory from virtual machines.

An ESX host allocates the amount of memory specified by a reservation directly to a virtual machine. Anything beyond the reservation is allocated using the ESX host physical resources or, when physical resources are not available, handled using special techniques such as ballooning or swapping. The ESX hosts can use two techniques for dynamically expanding or contracting the amount of memory allocated to virtual machines.

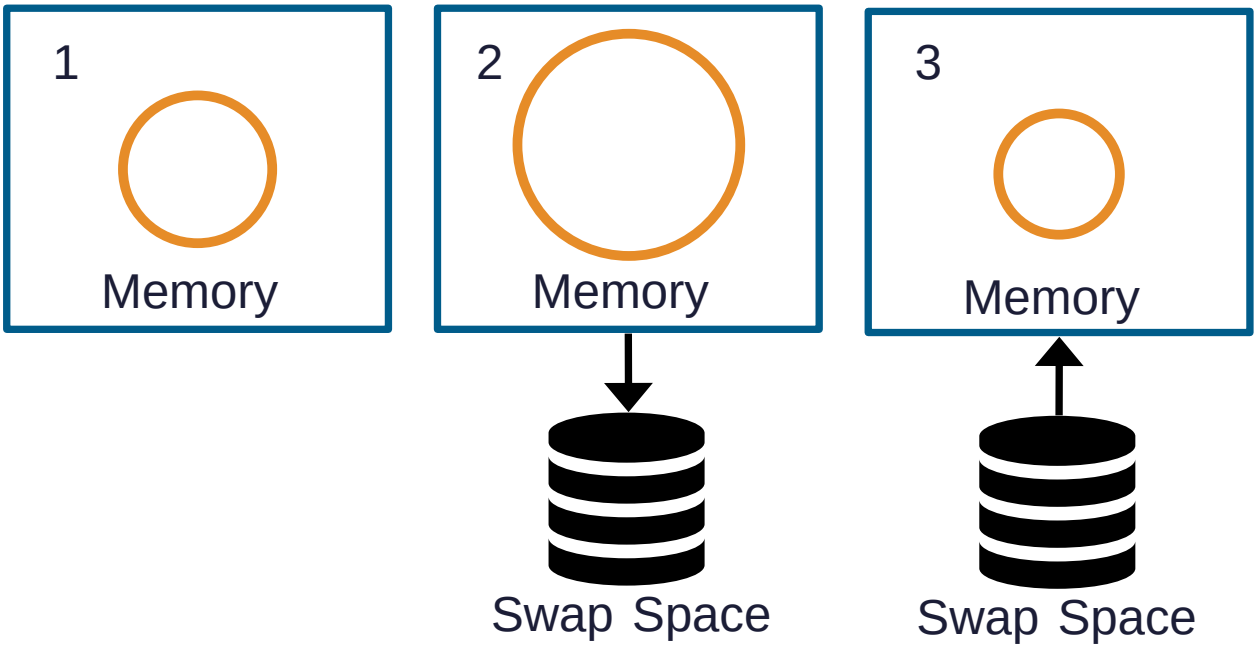
- ESX systems use a memory balloon driver (`vmmemctl`), loaded into the guest operating system running in a virtual machine. See Memory Balloon Driver.
- ESX system swaps out a page from a virtual machine to a server swap file without any involvement by the guest operating system. Each virtual machine has its own swap file.

Memory Balloon Driver

The memory balloon driver (`vmmemctl`) collaborates with the server to reclaim pages that are considered least valuable by the guest operating system.

The driver uses a proprietary ballooning technique that provides predictable performance that closely matches the behavior of a native system under similar memory constraints. This technique increases or decreases memory pressure on the guest operating system, causing the guest to use its own native memory management algorithms. When memory is tight, the guest operating system determines which pages to reclaim and, if necessary, swaps them to its own virtual disk.

Figure 100: Memory Ballooning in the Guest Operating System



Note:
You must configure the guest operating system with sufficient swap space. Some guest operating systems have additional limitations.

If necessary, you can limit the amount of memory `vmmemctl` reclaims by setting the **sched.mem.maxmemctl** parameter for a specific virtual machine. This option specifies the maximum amount of memory that can be reclaimed from a virtual machine in megabytes (MB). See Set Advanced Virtual Machine Attributes.

Sharing Memory Across Virtual Machines

Many ESX workloads present opportunities for sharing memory across virtual machines (as well as within a single virtual machine).

ESX memory sharing runs as a background activity that scans for sharing opportunities over time. The amount of memory saved varies over time. For a fairly constant workload, the amount generally increases slowly until all sharing opportunities are exploited.

To determine the effectiveness of memory sharing for a given workload, try running the workload, and use `resxtop` or `esxtop` to observe the actual savings. Find the information in the `PSHARE` field of the interactive mode in the Memory page.

Use the `Mem.ShareScanTime` and `Mem.ShareScanGHz` advanced settings to control the rate at which the system scans memory to identify opportunities for sharing memory.

You can also configure sharing for individual virtual machines by setting the `sched.mem.pshare.enable` option.

Due to security concerns, inter-virtual machine transparent page sharing is deactivated by default and page sharing is being restricted to intra-virtual machine memory sharing. This means page sharing does not occur across virtual machines and only occurs inside of a virtual machine. The concept of salting has been introduced to help address concerns system administrators may have over the security implications of transparent page sharing. Salting can be used to allow more granular management of the virtual machines participating in transparent page sharing than was previously possible. With the new salting settings, virtual machines can share pages only if the salt value and contents of the pages are identical. A new host config option `Mem.ShareForceSalting` can be configured to activate or deactivate salting.

See Advanced Attributes in vSphere for information on how to set advanced options.

Memory Compression

ESX provides a memory compression cache to improve virtual machine performance when you use memory overcommitment. Memory compression is activated by default. When an ESX host memory becomes overcommitted, ESX compresses virtual pages and stores them in memory.

Because accessing compressed memory is faster than accessing memory that is swapped to disk, memory compression in ESX allows you to overcommit memory without significantly hindering performance. When a virtual page needs to be swapped, ESX first attempts to compress the page. Pages that can be compressed to 2 KB or smaller are stored in the virtual machine's compression cache, increasing the capacity of the ESX host.

You can set the maximum size for the compression cache and deactivate memory compression using the Advanced Settings dialog box in the vSphere Client.

Activate or Deactivate the Memory Compression Cache

Memory compression is activated by default. You can use Advanced System Settings in the vSphere Client to activate or deactivate memory compression for an ESX host.

1. Browse to the ESX host in the vSphere Client.
2. Click **Configure**.
3. Under **System**, select **Advanced System Settings**.
4. Locate Mem.MemZipEnable and click the **Edit** button.
5. Enter 1 to activate or enter 0 to deactivate the memory compression cache.
6. Click **OK**.

Set the Maximum Size of the Memory Compression Cache

You can set the maximum size of the memory compression cache for the ESX host virtual machines.

You set the size of the compression cache as a percentage of the memory size of the virtual machine. For example, if you enter 20 and a virtual machine's memory size is 1000 MB, ESX can use up to 200MB of the ESX host memory to store the compressed pages of the virtual machine.

If you do not set the size of the compression cache, ESX uses the default value of 10 percent.

1. Browse to the ESX host in the vSphere Client.
2. Click **Configure**.
3. Under **System**, select **Advanced System Settings**.
4. Locate Mem.MemZipMaxPct and click the **Edit** button.
   The value of this attribute determines the maximum size of the compression cache for the virtual machine.
5. Enter the maximum size for the compression cache.
   The value is a percentage of the size of the virtual machine and must be between 5 and 100 percent.
6. Click **OK**.

Measuring and Differentiating Types of Memory Usage

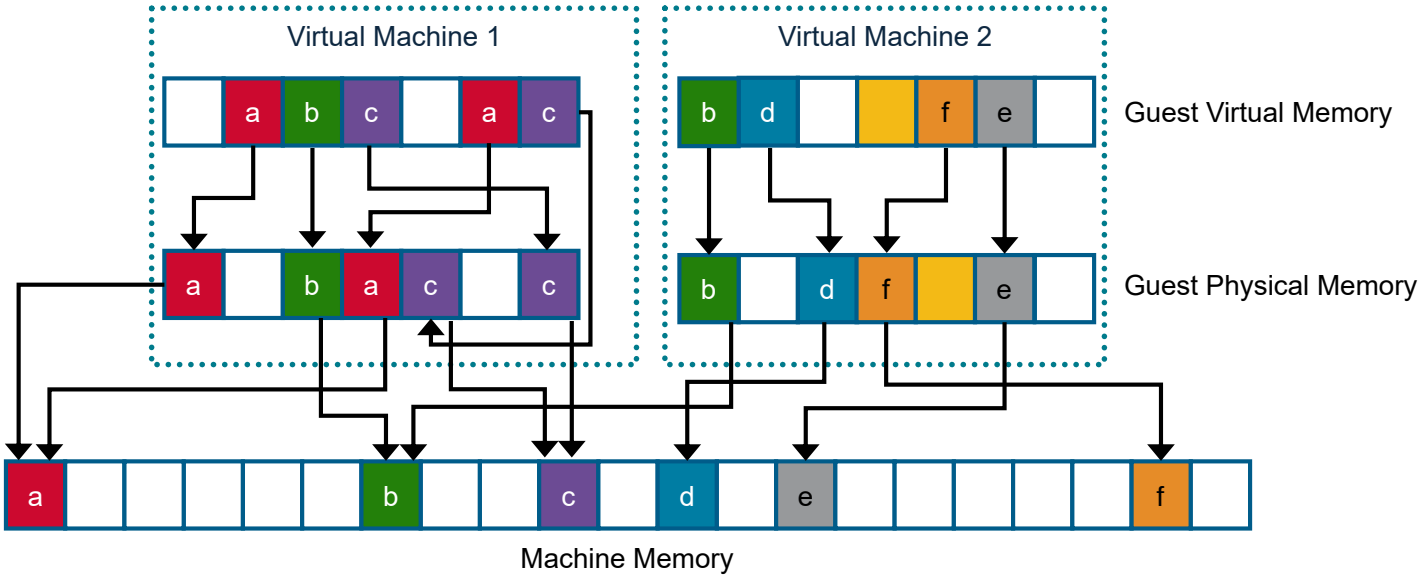
The **Performance** tab of the vSphere Client displays several metrics that can be used to analyze memory usage.

Some of these memory metrics measure guest physical memory while other metrics measure machine memory. For instance, two types of memory usage that you can examine using performance metrics are guest physical memory and machine memory. You measure guest physical memory using the Memory Granted metric (for a virtual machine) or Memory Shared (for an ESX host). To measure machine memory, however, use Memory Consumed (for a virtual machine) or Memory Shared Common (for an ESX host). Understanding the conceptual difference between these types of memory usage is important for knowing what these metrics are measuring and how to interpret them.

The VMkernel maps guest physical memory to machine memory, but they are not always mapped one-to-one. Multiple regions of guest physical memory might be mapped to the same region of machine memory (when memory sharing) or specific regions of guest physical memory might not be mapped to machine memory (when the VMkernel swaps out or balloons guest physical memory). In these situations, calculations of guest physical memory usage and machine memory usage for an individual virtual machine or an ESX host differ.

Consider the example in the following figure, which shows two virtual machines running on an ESX host. Each block represents 4 KB of memory and each color/letter represents a different set of data on a block.

Figure 101: Memory Usage Example



The performance metrics for the virtual machines can be determined as follows:

- To determine Memory Granted (the amount of guest physical memory that is mapped to machine memory) for virtual machine 1, count the number of blocks in virtual machine 1's guest physical memory that have arrows to machine memory and multiply by 4 KB. Since there are five blocks with arrows, Memory Granted is 20 KB.
- Memory Consumed is the amount of machine memory allocated to the virtual machine, accounting for savings from shared memory. First, count the number of blocks in machine memory that have arrows from virtual machine 1's guest physical memory. There are three such blocks, but one block is shared with virtual machine 2. So count two full blocks plus half of the third and multiply by 4 KB for a total of 10 KB Memory Consumed.

The important difference between these two metrics is that Memory Granted counts the number of blocks with arrows at the guest physical memory level and Memory Consumed counts the number of blocks with arrows at the machine memory level. The number of blocks differs between the two levels due to memory sharing and so Memory Granted and Memory Consumed differ. Memory is being saved through sharing or other reclamation techniques.

A similar result is obtained when determining Memory Shared and Memory Shared Common for the ESX host.

- Memory Shared for the ESX host is the sum of each virtual machine's Memory Shared. Calculate shared memory by looking at each virtual machine's guest physical memory and counting the number of blocks that have arrows to machine memory blocks that themselves have more than one arrow pointing at them. There are six such blocks in the example, so Memory Shared for the ESX host is 24 KB.
- Memory Shared Common is the amount of machine memory shared by virtual machines. To determine common memory, look at the machine memory and count the number of blocks that have more than one arrow pointing at them. There are three such blocks, so Memory Shared Common is 12 KB.

Memory Shared is concerned with guest physical memory and looks at the origin of the arrows. Memory Shared Common, however, deals with machine memory and looks at the destination of the arrows.

The memory metrics that measure guest physical memory and machine memory might appear contradictory. In fact, they are measuring different aspects of a virtual machine's memory usage. By understanding the differences between these metrics, you can better use them to diagnose performance issues.

Memory Reliability

Memory reliability, also known as error isolation, allows ESX to stop using parts of memory when it determines that a failure might occur, as well as when a failure did occur.

When enough corrected errors are reported at a particular address, ESX stops using this address to prevent the corrected error from becoming an uncorrected error.

Memory reliability provides better VMkernel reliability despite corrected and uncorrected errors in RAM. It also enables the system to avoid using memory pages that might contain errors.

Correcting an Error Isolation Notification

With memory reliability, VMkernel stops using pages that receive an error isolation notification.

The user receives an event in the vSphere Client when VMkernel recovers from an uncorrectable memory error, when VMkernel retires a significant percentage of system memory due to a large number of correctable errors, or if there are a large number of pages that are unable to retire.

1. Vacate the ESX host.
2. Migrate the virtual machines.
3. Run memory-related hardware tests.

System Swap

System swap is a memory reclamation process that can take advantage of unused memory resources across an entire system.

System swap allows the system to reclaim memory from memory consumers that are not virtual machines. When system swap is enabled you have a tradeoff between the impact of reclaiming the memory from another process and the ability to assign the memory to a virtual machine that can use it. The amount of space required for the system swap is 1GB.

Memory is reclaimed by taking data out of memory and writing it to background storage. Accessing the data from background storage is slower than accessing data from memory, so it is important to carefully select where to store the swapped data.

ESX determines automatically where the system swap should be stored, this is the **Preferred swap file location**. This decision can be aided by selecting a certain set of options. The system selects the best possible enabled option. If none of the options are feasible then system swap is not activated.

The available options are:

- Datastore - Allow the use of the datastore specified. Please note that a vSAN datastore or a VMware vSphere® Virtual VolumesTM datastore cannot be specified for system swap files.
- Host Swap Cache - Allow the use of part of the ESX host swap cache.
- Preferred swap file location - Allow the use of the preferred swap file location configured for the ESX host.

Configure System Swap

You can customize the options that determine the system swap location.

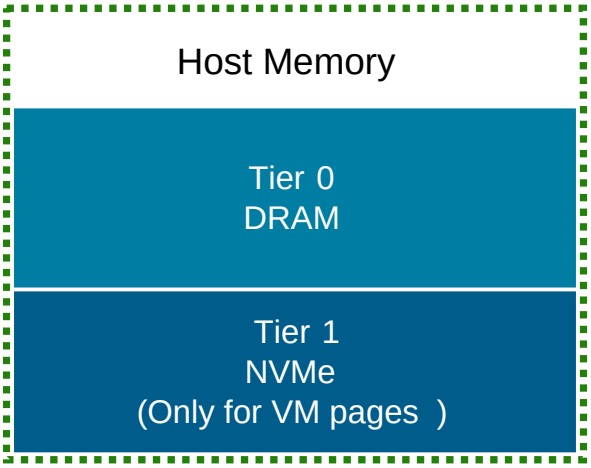
Select the **Enabled** check box in the **Edit System Swap Settings** dialog box.

1. Browse to the ESX host in the vSphere Client.
2. Click **Configure**.
3. Under **System**, select **System Swap**.
4. Click **Edit**.
5. Select the check boxes for each option that you want to enable.
6. If you select the **datastore** option, select a datastore from the drop-down menu.
7. Click **OK**.

Memory Tiering over NVMe

Memory Tiering over NVMe (Memory Tiering) allows you to add memory capacity to an ESX host by using NVMe devices that are installed locally on the ESX host as tiered memory. Memory Tiering optimizes performance by intelligently allocating VM memory to either NVMe devices or faster dynamic random access memory (DRAM) in the ESX host.

Figure 102: Memory Tiering Architecture with DRAM and NVMe Tiers



Memory Tiering combines DRAM (Tier 0) with the slower NVMe memory (Tier 1) to provide contiguous memory. Memory pages from NVMe are used only for VM memory allocations on an ESX host.

With the Memory Tiering feature, you can increase the workload capacity with a minimal impact on performance. This feature allows you to add more VMs and workloads to the ESX host, thus better using the available CPU resources.

By default, the Memory Tiering feature is deactivated.

You can activate the Memory Tiering feature and configure the NVMe device on the ESX host to serve as tiered memory. Memory Tiering integrates seamlessly with DRS, which continuously monitors the ESX hosts within a cluster and places VMs to load-balance memory usage across the cluster.

vSphere Requirements for Memory Tiering

You can configure Memory Tiering either at a cluster or at an ESX host level.

To use the Memory Tiering feature, your environment must meet the following requirements.

Component Requirements:

- vCenter 9.1 and later
- ESX 9.1 and later

NVMe Devices Requirements for Memory Tiering

- The NVMe devices that are used for Memory Tiering must be of a drive similar to vSAN cache (Mixed Use with 3 DWPD). An already configured vSAN cache or data disk cannot be used for Memory Tiering. However, you can use an existing vSAN data disk for Memory Tiering by removing it from the vSAN disk group.

For more information about the supported NVMe devices, see the Broadcom Compatibility Guide, and in the **Search Programs** field, enter `vSAN SSD` and from the **Storage & Availability** card, click **vSAN SSD**. On the vSAN SSD page, apply these additional filters.

Search Criteria	Value
Device Type	NVMe
Endurance Class	Class D >= 7300 TBW
Performance Class	Class F: 100,000-349,999 Writes Per Second or Class G: 350,000+ Writes Per Second
Endurance - Drive Writes Per Day (DWPD)	3 or higher

- NVMe devices cannot be over fabric or Ethernet. They must be installed locally on the ESX host. To determine the best NVMe slot to use on the ESX host for the best bandwidth and least contention with other PCIe-based devices, see the user guide or owner's manual for your server.
- The system automatically generates a dedicated partition for Memory Tiering that is based on the configuration. If no specific size is provided, the system uses the full available capacity of the drive. By default, the system limits NVMe tiering capacity to the DRAM size at a 1:1 ratio, up to a maximum of 4 TB. If the NVMe device capacity is smaller than the DRAM size, the system uses the smaller capacity. To ensure optimal performance, dedicate the NVMe device solely to Memory Tiering.

  Note:

  You must delete all existing partitions on the device to ensure a clean configuration for Memory Tiering. For more information on how to delete the partitions, see Broadcom knowledge base article 323144.
- To ensure device resiliency if there is a single device failure, consider using either RAID or software mirroring. For more information, see Memory Tiering Configuration.
- If the user interface signals a device failure, replace the drive by following the instructions in the "Modify Configuration After Activating Memory Tiering" section at Memory Tiering Configuration.
- Hot-plug and prepare to remove are not supported for an NVMe device being used for Memory Tiering. To remove or replace the NVMe device, you must deactivate Memory Tiering on the ESX host.
- Make sure to delete all partitions before using the NVMe device for Memory Tiering.
- Consider implementing either hardware RAID or software mirroring. For more information on configuring these redundancy options, see the "Using RAID with Memory Tiering" section.

Host Requirements for Memory Tiering

- Make sure that the ESX host is placed in maintenance mode before you activate, deactivate, or reconfigure Memory Tiering.
- The ESX host must have a compatible NVMe device.
- The ESX host cannot have Intel Optane Persistent Memory configured. For more information on how to deactivate Intel Optane Persistent Memory, see the user guide or owner's manual for your server.

  Important:

  The Optane Persistent Memory is not supported, and you must remove it from the ESX host.
- If an ESX host uses Memory Tiering, the suspend VMs to memory operation is not supported. Other operations, such as suspend to disk and snapshots, are supported.
- The ESX host cannot have NVDIMM-N Persistent Memory installed. To deactivate or remove NVMDIMM-N Persistent Memory, see the user guide or owner's manual for your server.
- Quick Boot is not supported for ESX hosts with activated Memory Tiering.

  Important:

  After you meet the ESX host requirements, exit the Maintenance Mode.

Capacity Planning Considerations

Before you configure Memory Tiering, consider the available capacity, including Active Memory and VM Profiles, to ensure optimal resource allocation.

The key considerations when you plan virtual machines and workloads include:

- Determine the amount of DRAM needed for virtual machines that require all their memory to be in DRAM.
- Determine the DRAM and NVMe needed based on VM Memory Resource Reservations.
- Memory pages are intelligently and proactively placed between the primary DRAM and the secondary NVMe tiers. The performance difference for workloads with Memory Tiering is negligible.
- Optimal performance requires sufficient memory allocation. The average active memory of the virtual machine must fit within the DRAM allocated to it at any given time. Ensure that active memory usage on the ESX host has not exceeded 50% of the DRAM capacity at any point.

If you experience issues with Memory Tiering during capacity planning, see Memory Tiering Troubleshooting Information.

Virtual Machine Profiles and Performance with Memory Tiering

The virtual machine profiles (VM profiles) and performance considerations are important for planning the memory capacity so that you can fully use the Memory Tiering feature.

Virtual Machine Profiles

Two VM profiles are available based on the resource configuration of the virtual machines and the intended workload: virtual machines that can use only Tier 0 memory (DRAM) and virtual machines that can have memory pages on NVMe.

Virtual machines that can only use Tier 0 memory

The following virtual machines require all their memory to be in Tier 0 (DRAM) because their attributes depend on having memory in DRAM only. Memory Tiering automatically ensures these virtual machines only use DRAM.

> **Note:**
>
> VMs with more than 512 GB of memory and 64 vCPUs on hosts configured for Memory Tiering may experience performance degradation. To prevent unexpected performance issues, disable Memory Tiering for these large VMs and configure them with full memory reservations.

The following VM types are now supported to run on a physical ESX 9.1 host with activated Memory Tiering, but are still excluded from the Memory Tiering process.

- 1 GB Page VM
- Confidential VM
- Fault Tolerant VM
- Large VM
- Low Latency VM
- Nested Virtualization VM
- PCI Passthrough VM
- Prealloc VM
- vQAT VM

> **Note:**
>
> With Nested VMs, Virtualization-Based Security (VBS) VMs are the only exception. By default, Memory Tiering is deactivated for a Nested Virtualization VM, but you can activate it for VBS by setting the `vmx` option `sched.mem.enableNestedTiering` to `TRUE`. The underlying ESX (base) can also be configured with Memory Tiering. You must understand and test the VM workloads before deploying on these configurations. Follow the same guidelines as with normal Memory Tiering deployment, as described in the performance white paper section Using and monitoring Memory Tiering.

Memory Tiering Configuration Parameters

The ESX host uses the DRAM:NVMe ratio to balance the memory size and performance.

Host Memory Size

The amount of NVMe used is the smaller of the two values - the DRAM:NVMe ratio or the NVMe Device Partition size.

The following table displays examples of the DRAM:NVMe ratio and the amount of NVMe that is used.

The system may only use a portion of the available NVMe capacity. However, the entire NVMe device must be dedicated to Memory Tiering and cannot be shared with other services, such as boot devices or vSAN.

DRAM Size	DRAM:NVMe Ratio	Percentage of DRAM	NVMe Device Partition Size	NVMe Used
256 GB	1:1 (default)	100%	512 GB	256 GB
256 GB	1:1	100%	256 GB	256 GB
256 GB	2:1	50%	512 GB	128 GB
256 GB	2:1	50%	128 GB	128 GB
1 TB	4:1	25%	4 TB	256 GB
2 TB	4:1	25%	2 TB	512 GB

To avoid latency issues, the total amount of active virtual machine memory must fit into the total DRAM available to virtual machines on the ESX host. To minimize any impact on performance, the active memory should not exceed 50% of the DRAM size.

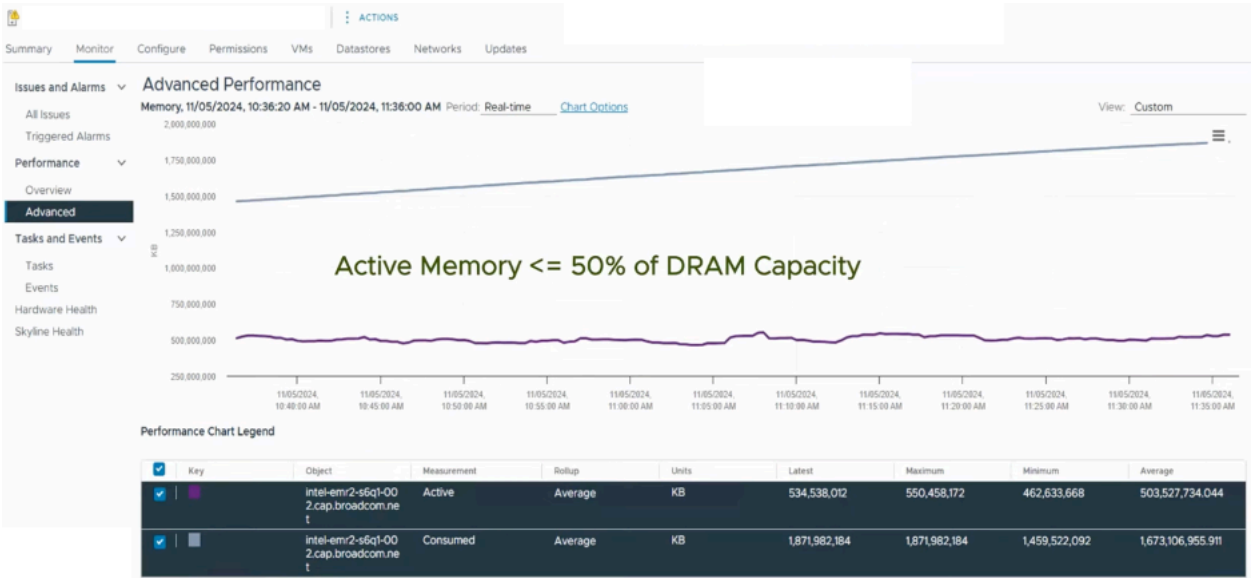
Assessing Customer Workloads for Memory Tiering

For more information about Memory Tiering performance monitoring, see the performance white paper.

Active Memory on the ESX Host

Memory Tiering is appropriate for environments where memory consumption is high (more than 50% allocated across all VMs), but active memory usage, which is memory actively used by workloads at any given time, remains low (less than 50% of total DRAM).

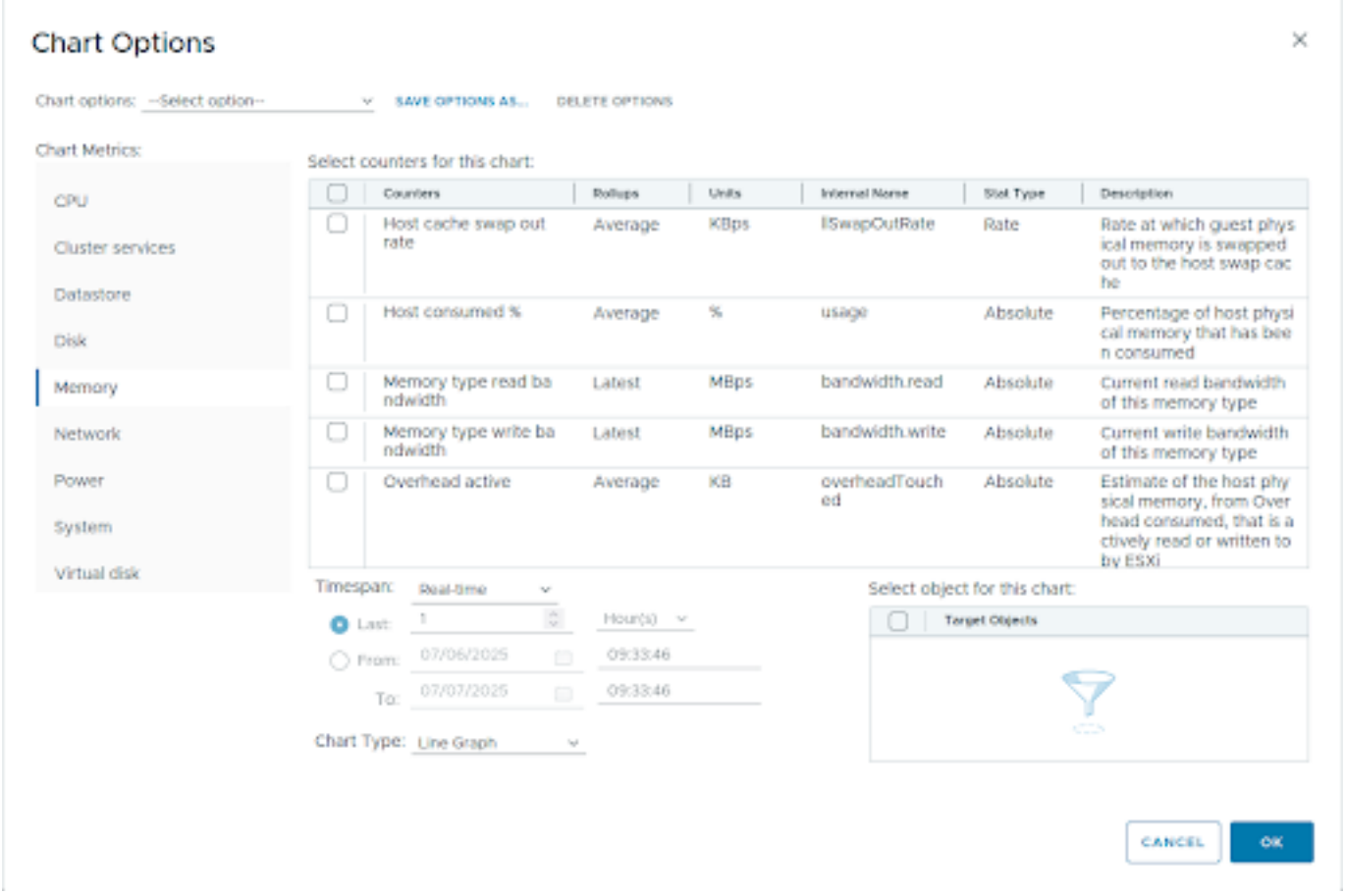
The screenshot below illustrates how Active Memory and DRAM Capacity can be monitored using vCenter.



Tier Bandwidth and Latency Metrics

The following vCenter screenshots display where to locate various performance metrics.

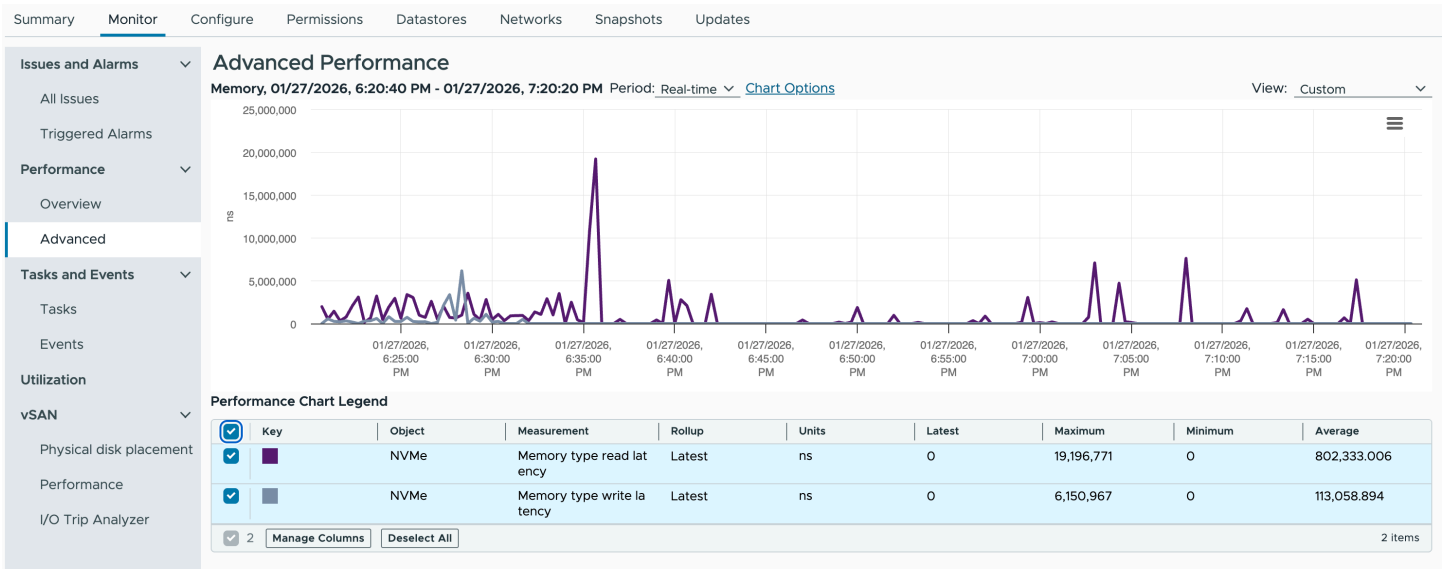
A screenshot that demonstrates how to select bandwidth and latency statistics in the Advanced Memory Performance chart. You can monitor these metrics at the per-VM level.



The screenshot below is from the vCenter (Memory View), captured after selecting **Memory Read Bandwidth** and **Memory Write Bandwidth** from the **Chart options**.

- **Write** represents cold pages being moved into the NVMe tier.
- **Read** represents pages being fetched from the NVMe tier when they become active, but are not present in DRAM.

If NVMe read bandwidth exceeds 200 MB/s, begin monitoring device latencies. Actual latency values vary depending on the class of the NVMe device that you use.

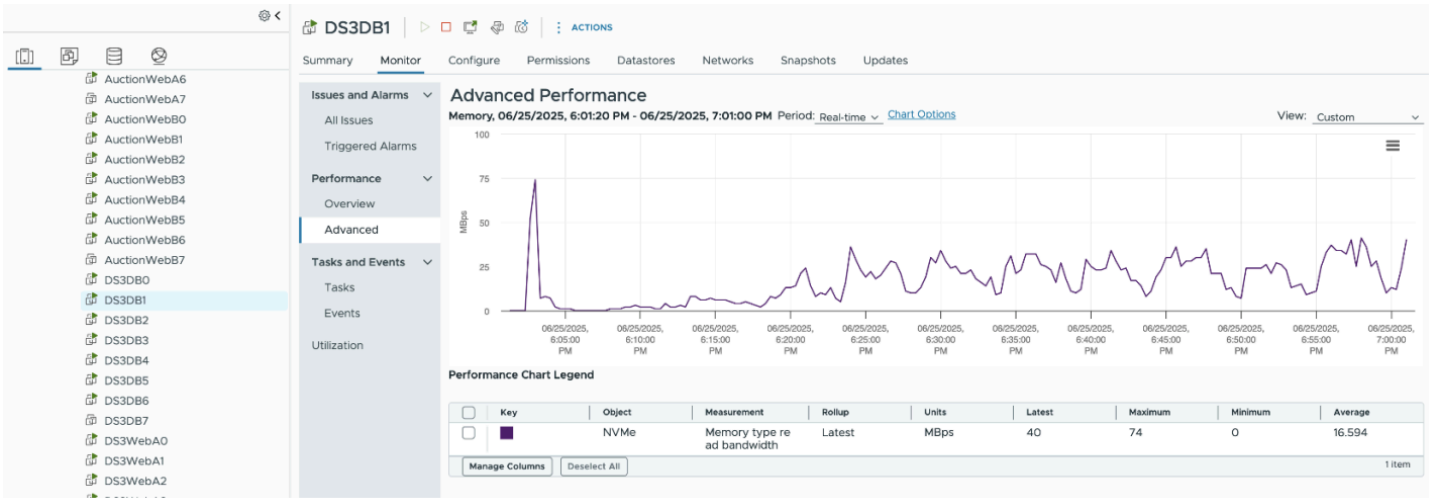


Read latencies remain below 200 microseconds, even at approximately 400 MB/sec of read bandwidth. This indicates strong performance from the NVMe device in use. However, with other NVMe devices, we have observed latencies closer to 300 microseconds under similar read bandwidth conditions.

Monitoring latency is critical because it has a direct impact on workload performance.

If you want to investigate the low performance for a specific VM, monitor the per-VM read bandwidth.

The following screenshot shows an example of a single VM.



Memory Tiering Configuration

You can perform Memory Tiering configuration using the vCenter Configuration Manager, ESXCLI commands, or PowerCLI scripts. You can activate or deactivate Memory Tiering at the cluster, ESX host, or deactivate Memory Tiering at the virtual machine level.

- Before you activate the Memory Tiering feature for an ESX host or a cluster, make sure that the vSphere environment meets the requirements that are described in Memory Tiering over NVMe.
- Before you activate, deactivate, or reconfigure Memory Tiering by using ESXCLI or PowerCLI commands, make sure that the ESX host is in maintenance mode.

You can perform the following operations by using the vCenter Configuration Manager, ESXCLI, or PowerCLI scripts.

- Identifying available NVMe devices.

- Configuring Memory Tiering for each cluster.
- Configuring Memory Tiering for each host.
- Verify that Memory Tiering is configured correctly.

Activate Memory Tiering at the Cluster Level

With vSphere 9.1, you can configure Memory Tiering at the cluster level by using the Configuration Manager in the vSphere Client. The Configuration Manager displays the desired configuration for this cluster.

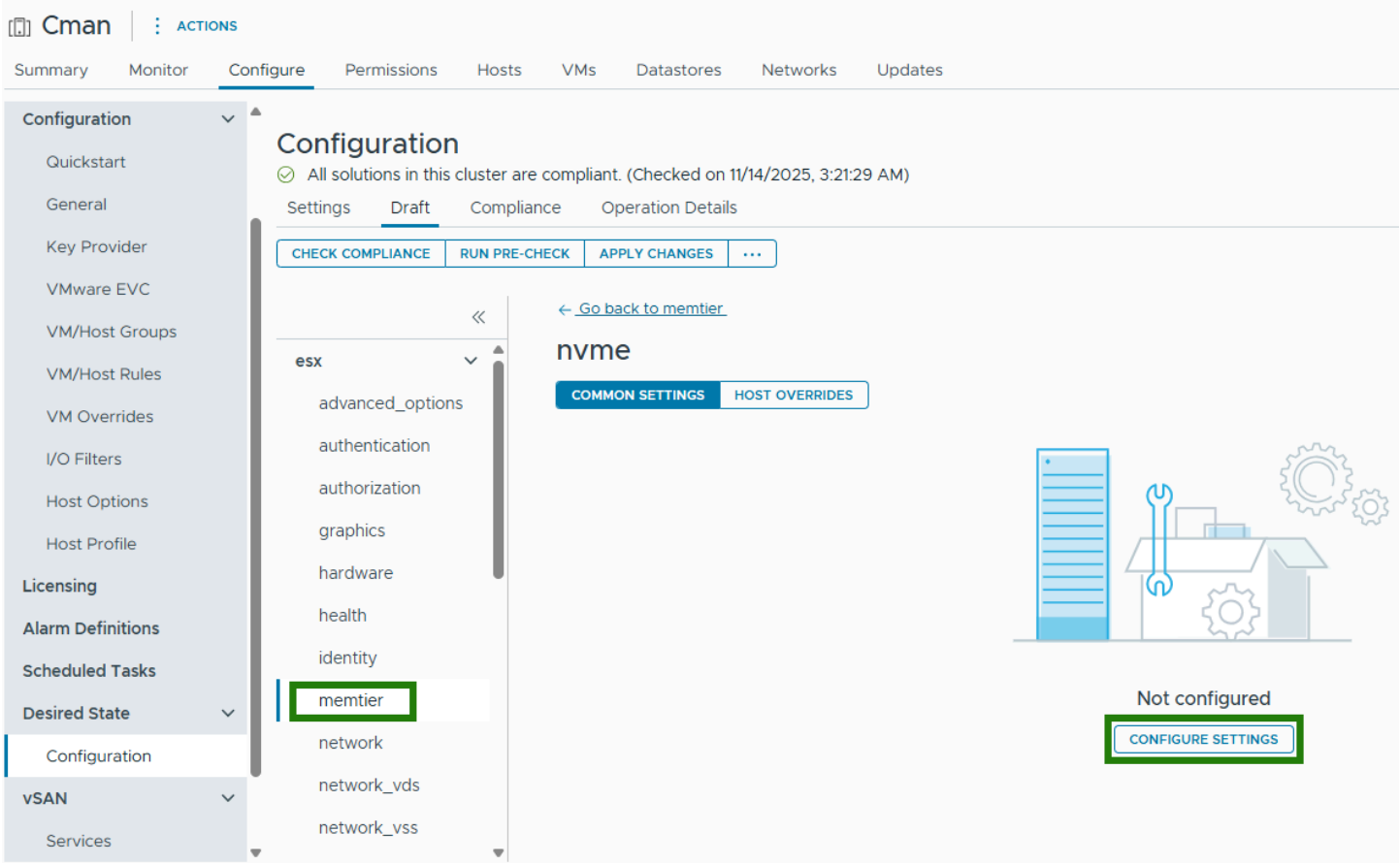
For more information about the configuration management of ESX hosts, see the vSphere Configuration Profiles documentation.

When activating Memory Tiering, the Configuration Manager automatically places the host in maintenance mode and applies all the configuration parameters that you set.

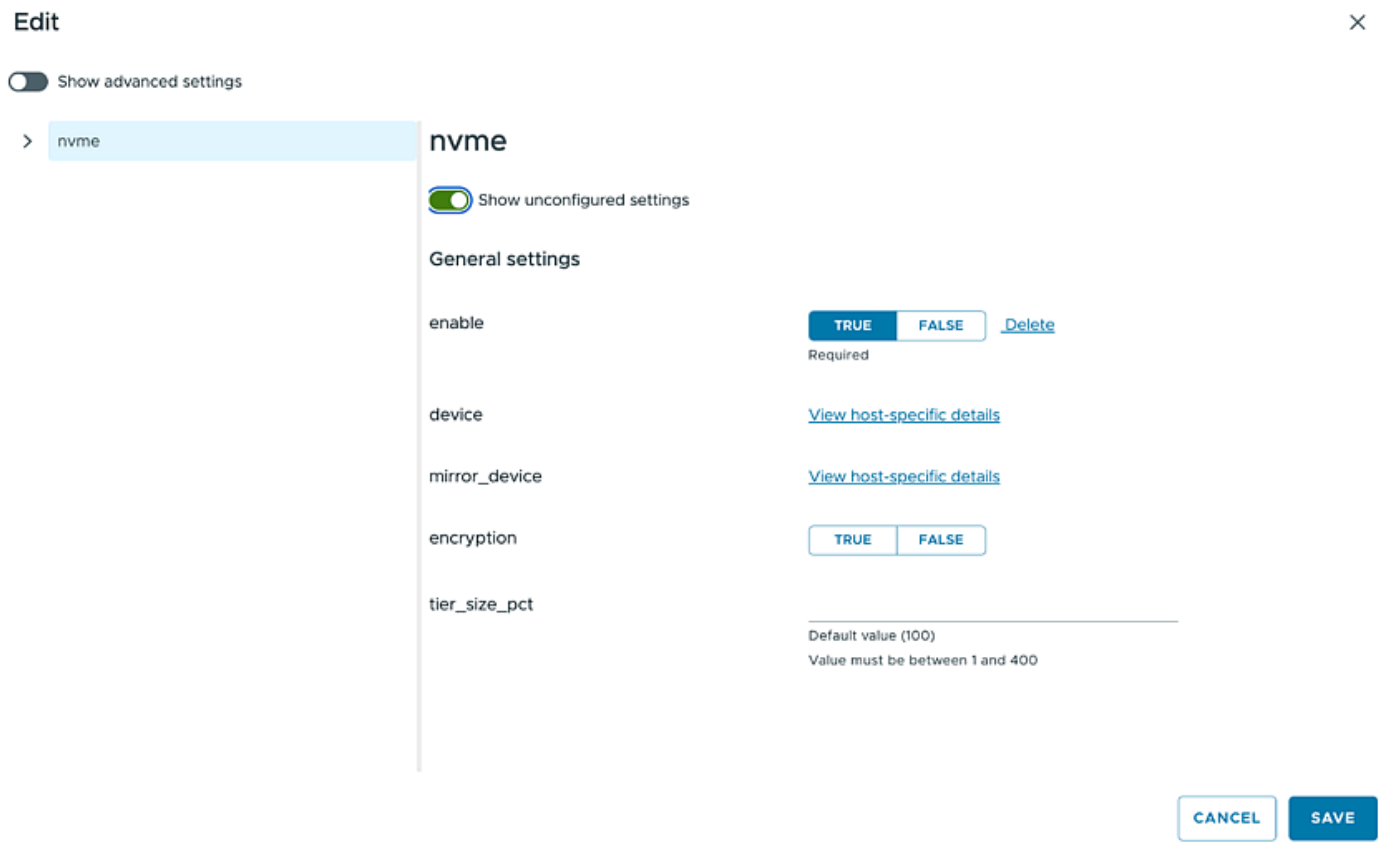
Memory Tiering is activated at the cluster level and applies to all hosts within the cluster, unless you configure specific host overrides.

When you activate Memory Tiering, you can configure a mirror device by setting the "device" and "mirror_device" values. The NVMe devices and mirror devices are specific to each host. If one NVMe device fails, the data remains available on the mirror device, allowing the VM to continue operating without interruption.

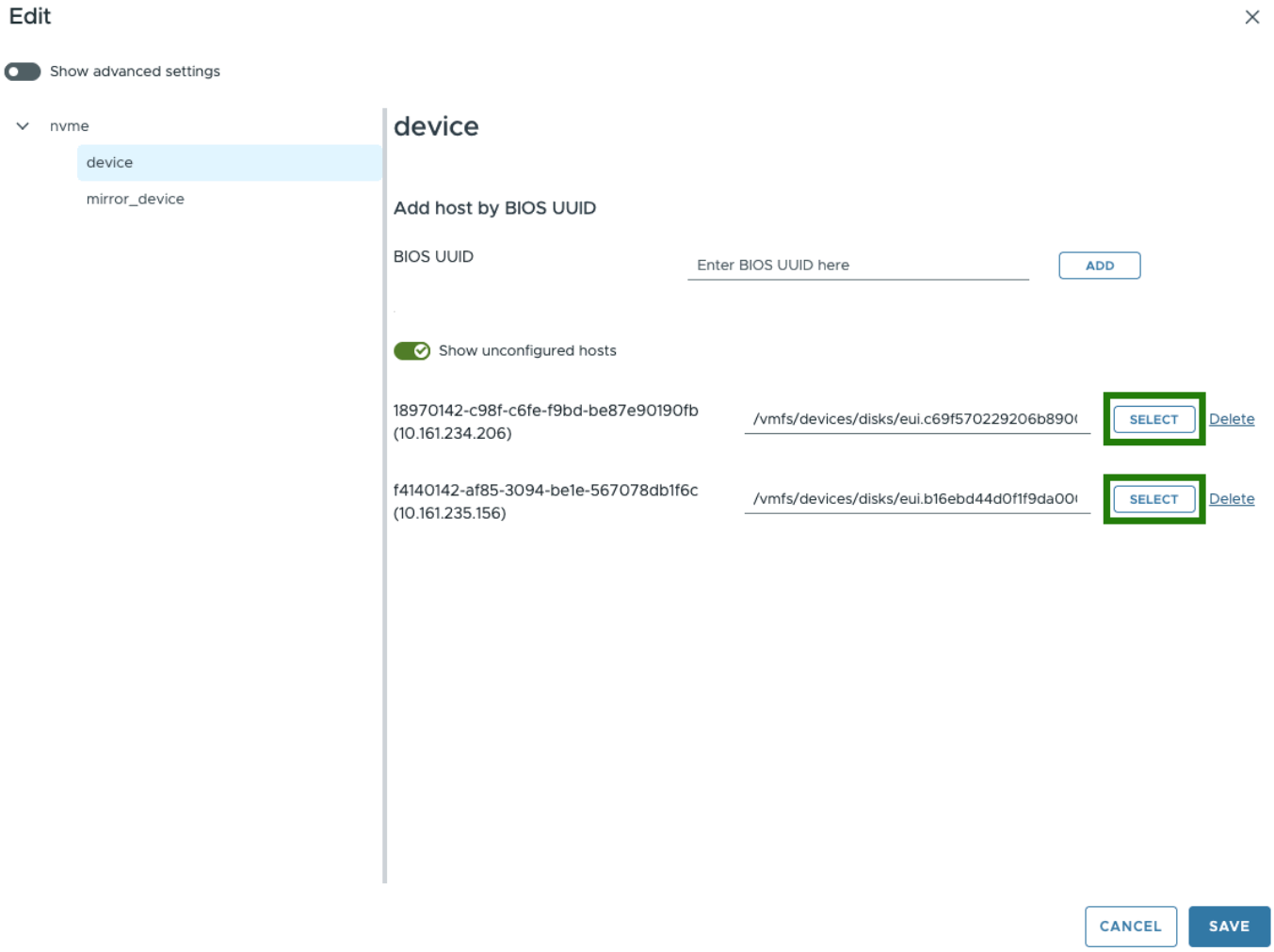
Perform the following steps to activate Memory Tiering at the cluster level.


Cman
ACTIONS
Summary
Monitor
Configure
Permissions
Hosts
VMs
Datastores
Networks
Updates
Configuration
Quickstart
General
Key Provider
VMware EVC
VM/Host Groups
VM/Host Rules
VM Overrides
I/O Filters
Host Options
Host Profile
Licensing
Alarm Definitions
Scheduled Tasks
Desired State
Configuration
vSAN
Services
Configuration
All solutions in this cluster are compliant. (Checked on 11/14/2025, 3:21:29 AM)
Settings
Draft
Compliance
Operation Details
CHECK COMPLIANCE
RUN PRE-CHECK
APPLY CHANGES
esx
advanced_options
authentication
authorization
graphics
hardware
health
identity
memtier
network
network_vds
network_vss
Go back to memtier
nvme
COMMON SETTINGS
HOST OVERRIDES
Not configured
CONFIGURE SETTINGS

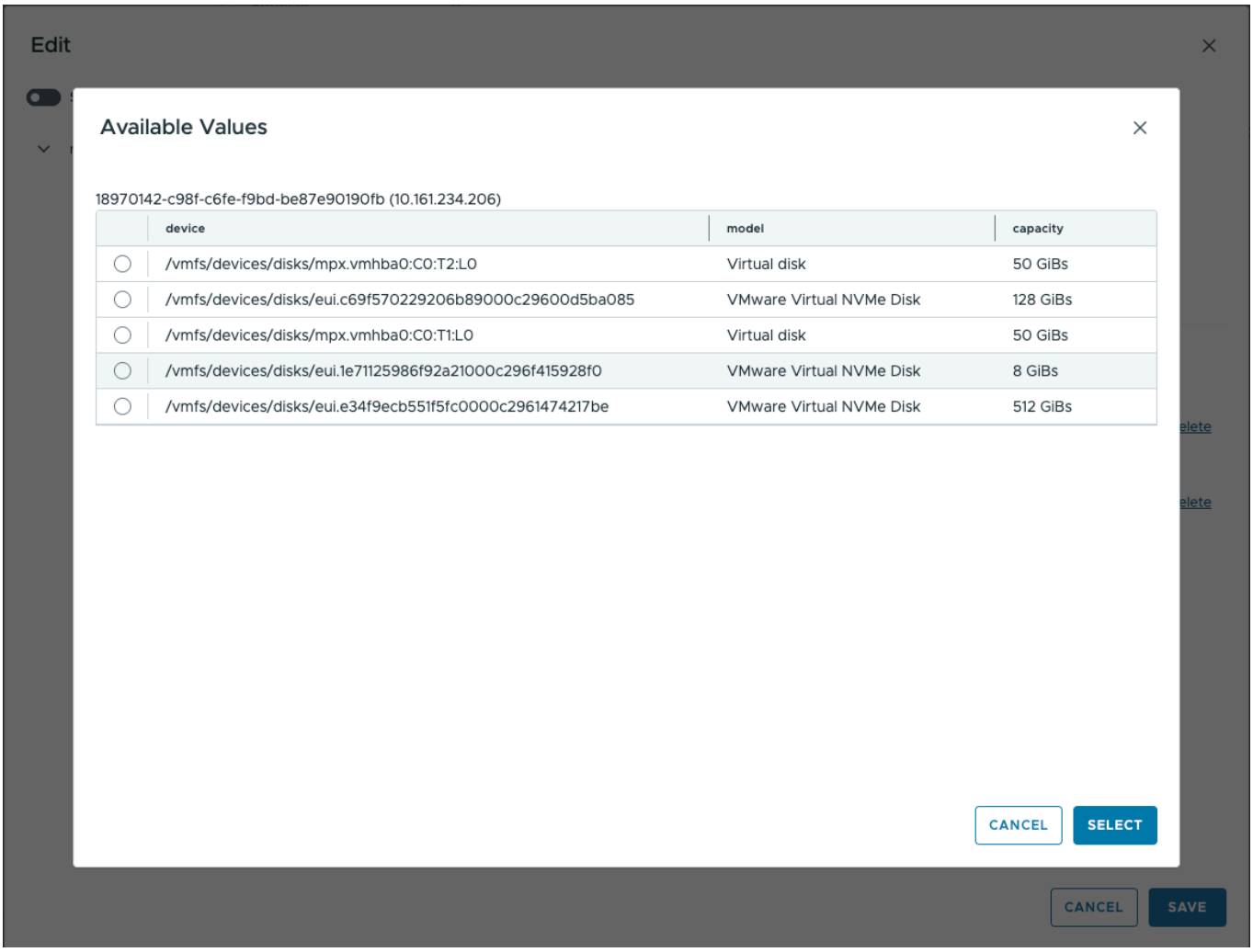
1. In the vSphere Client, select a cluster.
2. To access the Configuration Manager, click the **Configure** tab.
3. Navigate to **Desired State** > **Configuration**.
4. On the **Draft** tab, select **esx** > **memtier**.
5. Click the **nvme** link, and click the **Configure Settings** button.
6. Click the **Configure Settings** button.
7. From the **Edit** window, set the configuration parameters.



a) Required: To activate Memory Tiering, set the **enable** option to **True**, select at least one NVMe device, and optionally add a mirror device.
b) Required: To set a host-specific device, click the **View host-specific details** link of the **device** option, or click **device** from the left-hand panel.



c) From the **device** pane, add a host by BIOS UUID.
d) Select **Show unconfigured hosts** and click the **Select** button.
e) From the list of available devices, select a device and click **Select**.



Alternatively, you can select a host from the inventory, click **Configure**, navigate to **Storage** > **Storage Devices**, select a device, and under the **Properties** tab, copy the **Location** path. Paste the path in the device pane of the Configuration Manager (Required).

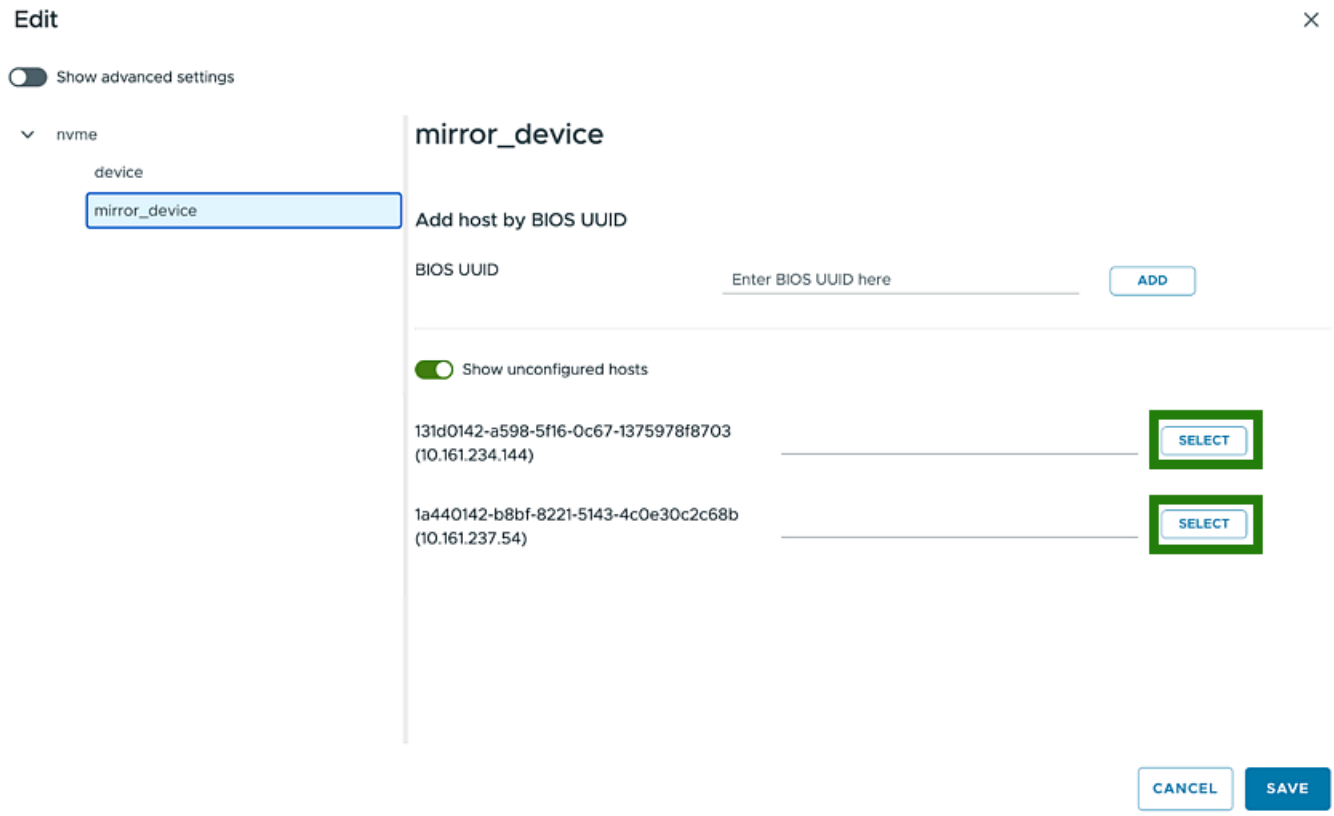
f) From the **device** pane, add a host by BIOS UUID.

g) Select **Show unconfigured hosts** and click the **Select** button.

h) From the list of available devices, select a device and click **Select**.

Alternatively, you can select a host from the inventory, click **Configure**, navigate to **Storage** > **Storage Devices**, select a device, and under the **Properties** tab, copy the **Location** path. Paste the path in the device pane of the Configuration Manager (Required).

i) To set a host-specific mirror device, click the **View host-specific details** link of the **mirror_device** option, or click **mirror_device** from the left-hand panel.



j) From the **mirror_device** pane, add a host by BIOS UUID.

k) Select **Show unconfigured hosts** and click the **Select** button.

l) From the list of available devices, select a device and click **Select**. If you configure both a primary and a mirror device, ensure that you select a different device for each device type.

m) To encrypt the stored data, set the **encryption** option to **True**. By default, **encryption** is set to **False**.

n) To define the size of the NVMe memory disk as a percentage of the main memory, in the **tier_size_pct** field, enter a value between 1 and 400.

Example: If the main memory is 16 GB and you set **tier_size_pct** to 50, the tier size is 8 GB, and the total available host memory is 24 GB.

Note:

For optimal performance, use the default value of 100, which configures a 1:1 ratio where the NVMe memory size equals the DRAM size.

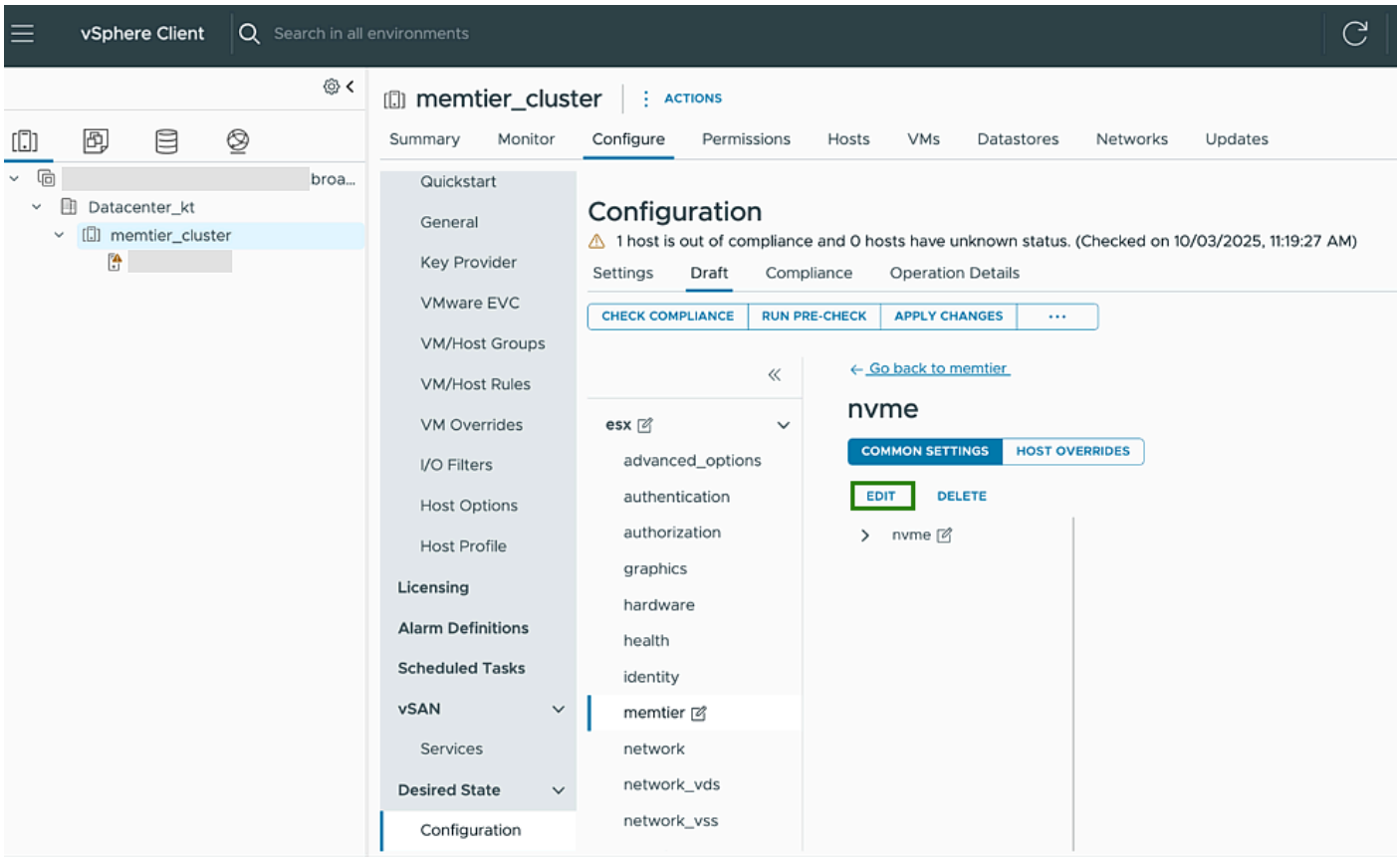
8. To apply the configuration, click **Save**.

Modify the Memory Tiering Configuration at the Cluster Level

You can modify the Memory Tiering configuration at the cluster level and can adjust your settings.

Perform the following steps to modify the Memory Tiering configuration at the cluster level.

1. In the vSphere Client, select a cluster.
2. To access the Configuration Manager, click the **Configure** tab.
3. Navigate to **Desired State** > **Configuration**.
4. On the **Draft** tab, select **esx** > **memtier**.
5. Click the **nvme** link and click the **Edit** button.

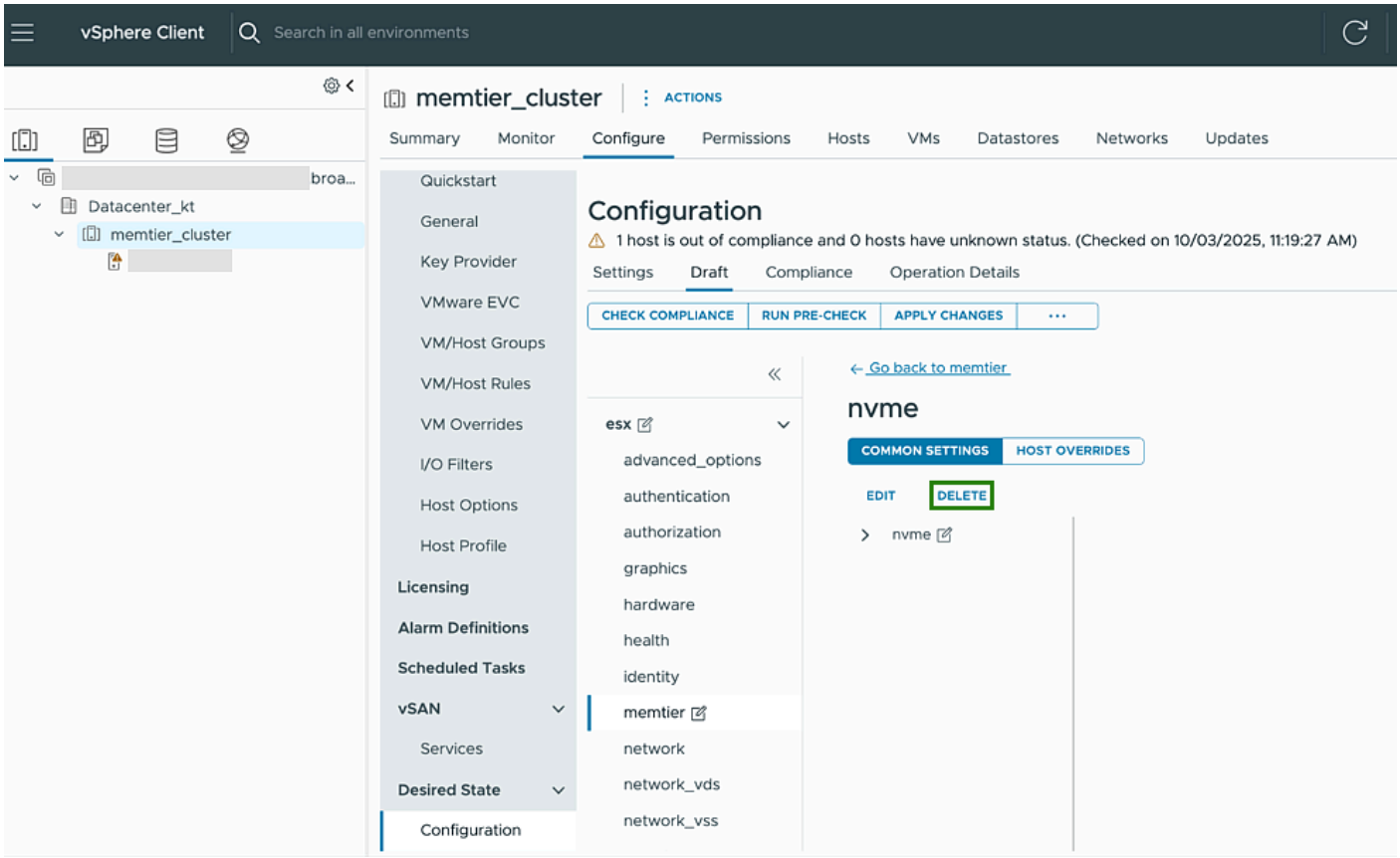


6. Make the necessary changes and click **Save**.

Delete the Memory Tiering Configuration at the Cluster Level

Perform the following steps to delete and deactivate the Memory Tiering configuration at the cluster level.

1. In the vSphere Client, select a cluster.
2. To access the Configuration Manager, click the **Configure** tab.
3. Navigate to **Desired State** > **Configuration**.
4. On the **Draft** tab, under **esx** click **memtier**.
5. Click the **nvme** link and select the NVMe configuration that you want to delete.
6. Click the **Delete** button to deactivate Memory Tiering at the cluster level.



Activate Memory Tiering at a Host Level with ESXCLI

- Configure the cluster to be managed by a single image and using vSphere Configuration Profiles. For more information, see Using vSphere Configuration Profiles to Manage Host Configuration at a Cluster Level.

- Place the host in maintenance mode.

Perform the following steps to activate the Memory Tiering feature at a host level.

1. To determine which device you want to use for Memory Tiering, run the following command.

   ```
   esxcli memtier device list
   ```

2. To activate Memory Tiering, run the following command.

   ```
   esxcli memtier enable -d <device_1>
   ```

3. Optional: Set other configuration parameters, such as activating Software Mirroring, setting tier size, and so on, during the activation process.
   a) To activate mirroring, run the following command.

      ```
      esxcli memtier enable -d <device_1,device_2>
      ```

   b) To activate the encryption and set the ratio to be 50%, run the following command.

      ```
      esxcli memtier enable -d <device_1> -c t -r 50
      ```

Modify the Configuration After Activating Memory Tiering

You can use ESXCLI commands to modify the configuration and manage the NVMe devices.

1. Replace an NVMe device.
   a) Place the host in maintenance mode.
   b) Run the following command where `set` sets the Memory Tiering configuration.

      ```
      esxcli memtier config set -d "new_device"
      ```

2. Change the DRAM: NVMe ratio.

How to Change the DRAM:NVMe Ratio

You can use the `"--tier-size-pct"` option to set the amount of NVMe to be used as Tiered Memory, expressed as a percentage of the total DRAM.

For example, setting the value to 25 configures 25% of the total DRAM as NVMe tiered memory, resulting in a DRAM:NVMe ratio of 4:1.

To change the tier ratio to 25%, run the command:

```
esxcli memtier config set --tier-size-pct 25
```

Note:

Make sure to place the host in Maintenance mode before you change the tier ratio.

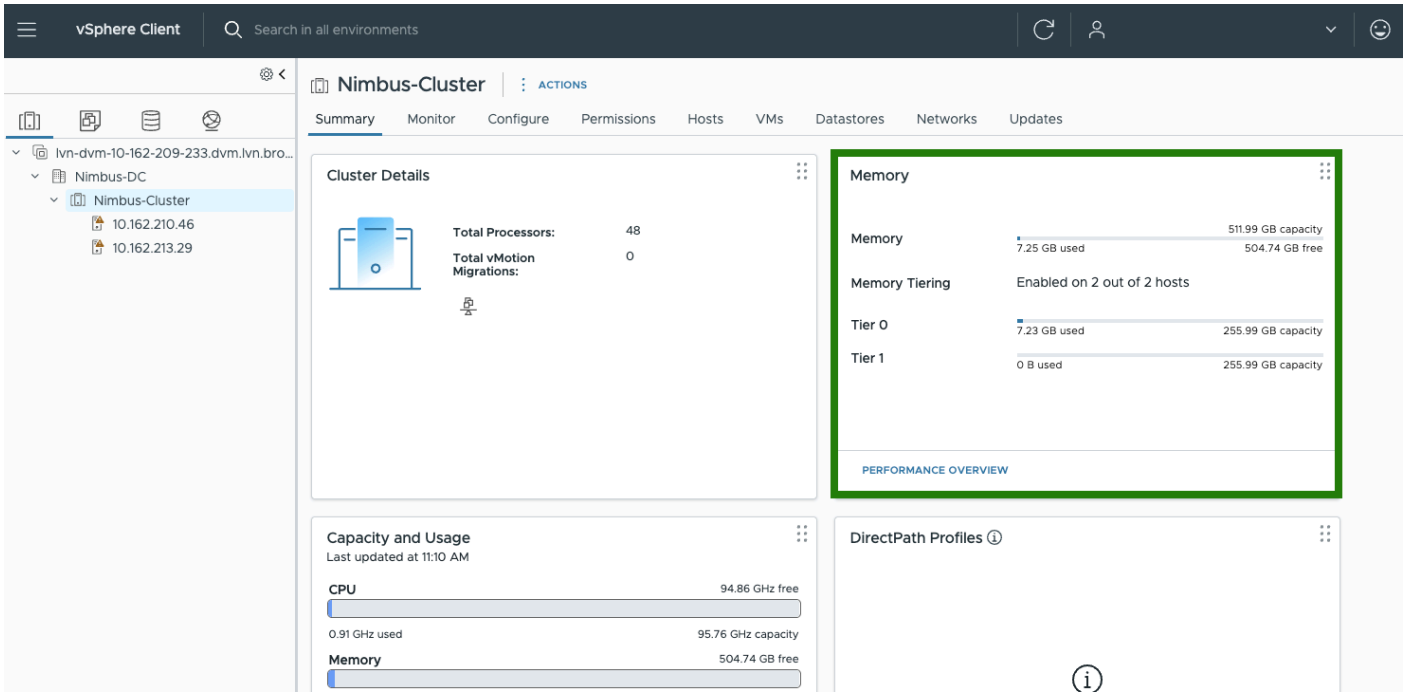
Alternatively, you can change the DRAM:NVMe ratio through the Configuration Manager.

For more information about the configuration management of ESX hosts, see the vSphere Configuration Profiles documentation.

How do I Check Whether a Cluster is Configured for Memory Tiering?

Perform the following steps to check whether a cluster is configured for Memory Tiering.

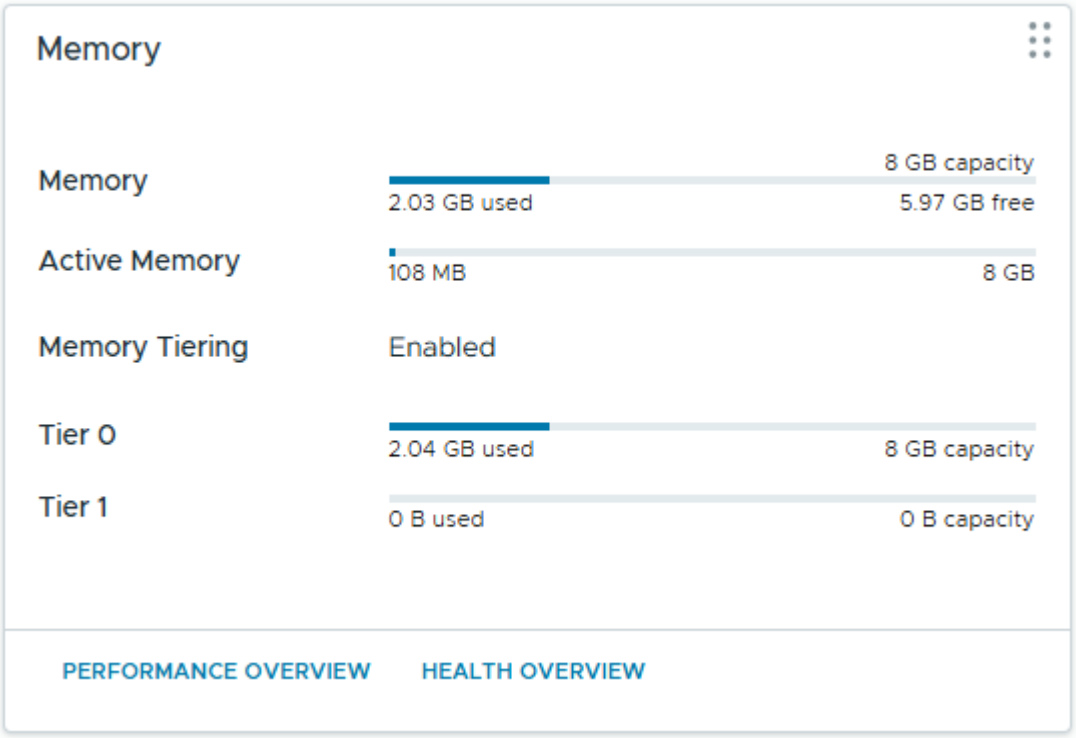
1. In the vSphere Client, navigate to and select a cluster, and click the **Summary** tab.
2. The **Memory** card provides information about the host configuration for Memory Tiering.



How do I Check Whether a Host is Configured for Memory Tiering?

You can check whether an ESX host is configured for Memory Tiering either through the vSphere Client or by ESXCLI commands.

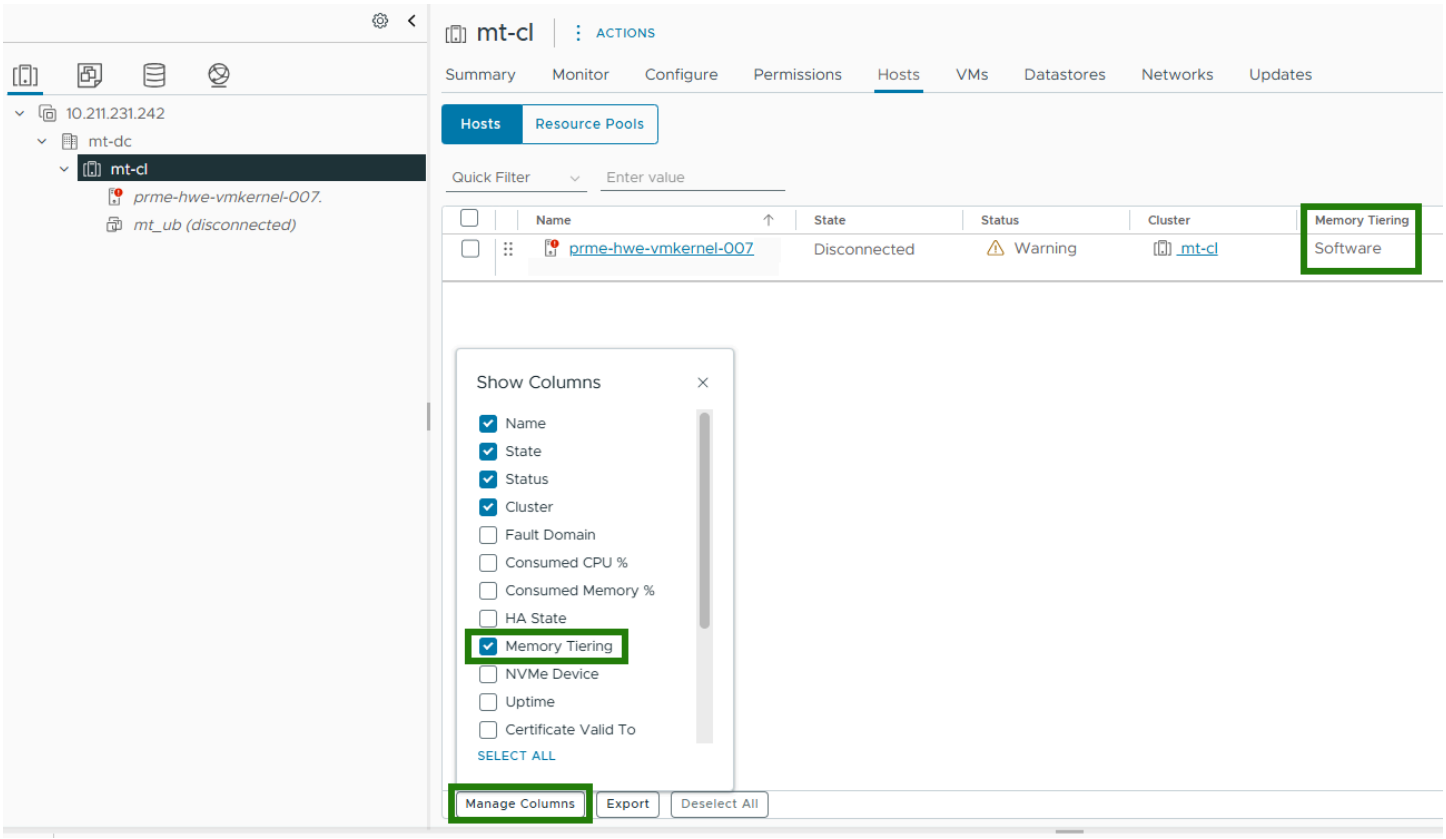
1. Check whether an ESX host is configured for Memory Tiering through the vSphere Client.
   a) In the vSphere Client, navigate to a host and click the **Summary** tab.
   b) The **Memory** card provides information about the host configuration for Memory Tiering.



2. Check whether an ESX host is configured for Memory Tiering by running the command `esxcli memtier status get`.
3. To check the current configuration state, run the command `esxcli memtier config get`.
   - If the **Enable** field is **False**, the ESX host is not configured for Memory Tiering.
   - If the **Enable** field is **True**, but the result of the `esxcli memtier status get` command returns**Status: Disabled**, an issue has occurred. Review the system logs or alarms to determine the reason why Memory Tiering failed to activate.

How do I Check Which Hosts Within a Cluster are Configured for Memory Tiering?

Perform the following steps to check which hosts are configured for Memory Tiering.



1. In the vSphere Client, navigate to a cluster and click the **Hosts** tab.
2. To display the **Memory Tiering** column if it does not show in the table:
   a) Click **Manage Columns** at the bottom part of the pane, and from the "Show Columns", select **Memory Tiering**.
3. The **Memory Tiering** column displays "Software" for all hosts that are configured for Memory Tiering.

   The number of hosts that are configured for Memory Tiering is also available in the **Memory** card under **Cluster** > **Summary**.

NVMe Device Encryption

VM memory pages that are stored on the NVMe device are not encrypted by default. If you require a higher level of data security, you can activate NVMe encryption that is configured at the host or VM level or encrypt data stored on the NVMe device. The host-level setting for encryption provides physical security guarantees, protecting data against device theft or unauthorized hardware access. In contrast, VM-level encryption offers more granular defense-in-depth, securing the environment against insider threats and intra-VM security.

The NVMe device encryption differs from the virtual machine encryption. For more information about the virtual machine encryption options, see Configure Virtual Machine Encryption Options.

Activate NVMe device encryption for a host

To activate NVMe device encryption for all VMs, place the host in Maintenance Mode and run the command

```
esxcli memtier config set --encryption true
```

Activate NVMe device encryption for a VM

To activate NVMe device encryption for a specific VM, power off the VM.

1. In the vSphere Client, right-click a VM and click **Edit Settings**.
2. Click the **Advanced Parameters** tab and set the `sched.mem.EncryptTierNVMe` option to `True`.
3. Power on the VM.

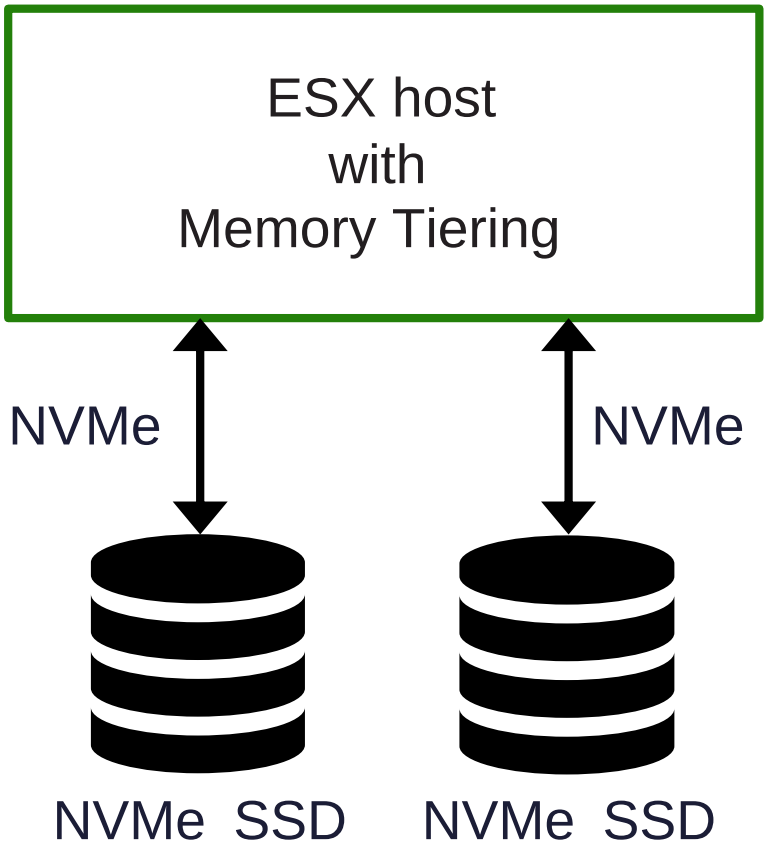
Software Mirroring (Software RAID)

With vSphere 9.1, use the Software Mirroring (software RAID) configuration instead of the hardware RAID configuration.

The Software Mirroring provides enhanced resiliency and performance, and provides solutions that ensure High Availability for NVMe devices on a host, such as:

- Redundancy support with two NVMe devices on a host ensures that a single device failure does not cause all VMs with pages on the failed device to fail.
- Simplified adoption of the Memory Tiering feature on existing servers that do not have Hardware RAID.

The following architecture diagram displays Software Mirroring with two NVMe devices.



When using Software Mirroring:

- Data is replicated to mirrored disks at the same partition and same block number.
- The size of mirroring disks must be the same or more than the first memory tier disk.

Activate Software Mirroring

- Place the host in maintenance mode.

To activate the Software Mirroring, perform the following steps.

1. Activate Software Mirroring during Memory Tiering activation.
   a. Run the command `esxcli memtier enable -d "device-1, device-2"`.
2. Activate Software Mirroring after Memory Tiering is activated.
   a. To reconfigure the Memory Tiering to include the mirror device, run the command `esxcli memtier config set -d "old-device, new-mirror-device"`.
3. Change the device after you activate Software Mirroring.
   a. Replace a device or a mirror device.
      a. Run the following commands.

Command	Description
`esxcli memtier config set --devices=<new-device>,<old-mirror-device-if-any>`	Replace the primary device.
`esxcli memtier config set --devices=<old-device>,<new-mirror-device>`	Replace the mirror device.

Deactivate Software Mirroring

To deactivate only the mirror device, perform the following steps.

1. Place the host in maintenance mode.
2. Run the command:

```
esxcli memtier config set -d "old-device-1"
```

Using Hardware RAID with Memory Tiering

Memory Tiering supports the use of NVMe devices behind a RAID controller.

To use RAID, perform the following steps.

1. To create a single logical device using multiple physical devices, configure the RAID controller. For more information, see Intel® Virtual RAID on CPU or one of the supported RAID controllers:
   - Tri-Mode Technology
   - 24G NVMe® Tri-mode RAID and HBA Storage Adapters
2. Configure specific NVMe devices to be used as tiered memory for each host. For more information, see the VMware PowerCLI and *ESXCLI Concepts and Examples* documentation.
3. Configure the cluster to be managed by a single image and using Config Manager Profiles. For more information, see Using vSphere Configuration Profiles to Manage Host Configuration at a Cluster Level.
4. Activate Memory Tiering at a cluster level. For more information, see Memory Tiering Configuration.
5. Create several VMs and ensure that their operations generate enough Active Memory to trigger the use of Tiered Memory.

Deactivate Memory Tiering for an ESX Host

Deactivating Memory Tiering on a host reduces the total memory amount available on the cluster. As a result, if an overcommitted cluster has insufficient memory, VMs might fail to power on.

Place the host in maintenance mode.

If you do not want a specific host to have Memory Tiering activated, perform one of the following options.

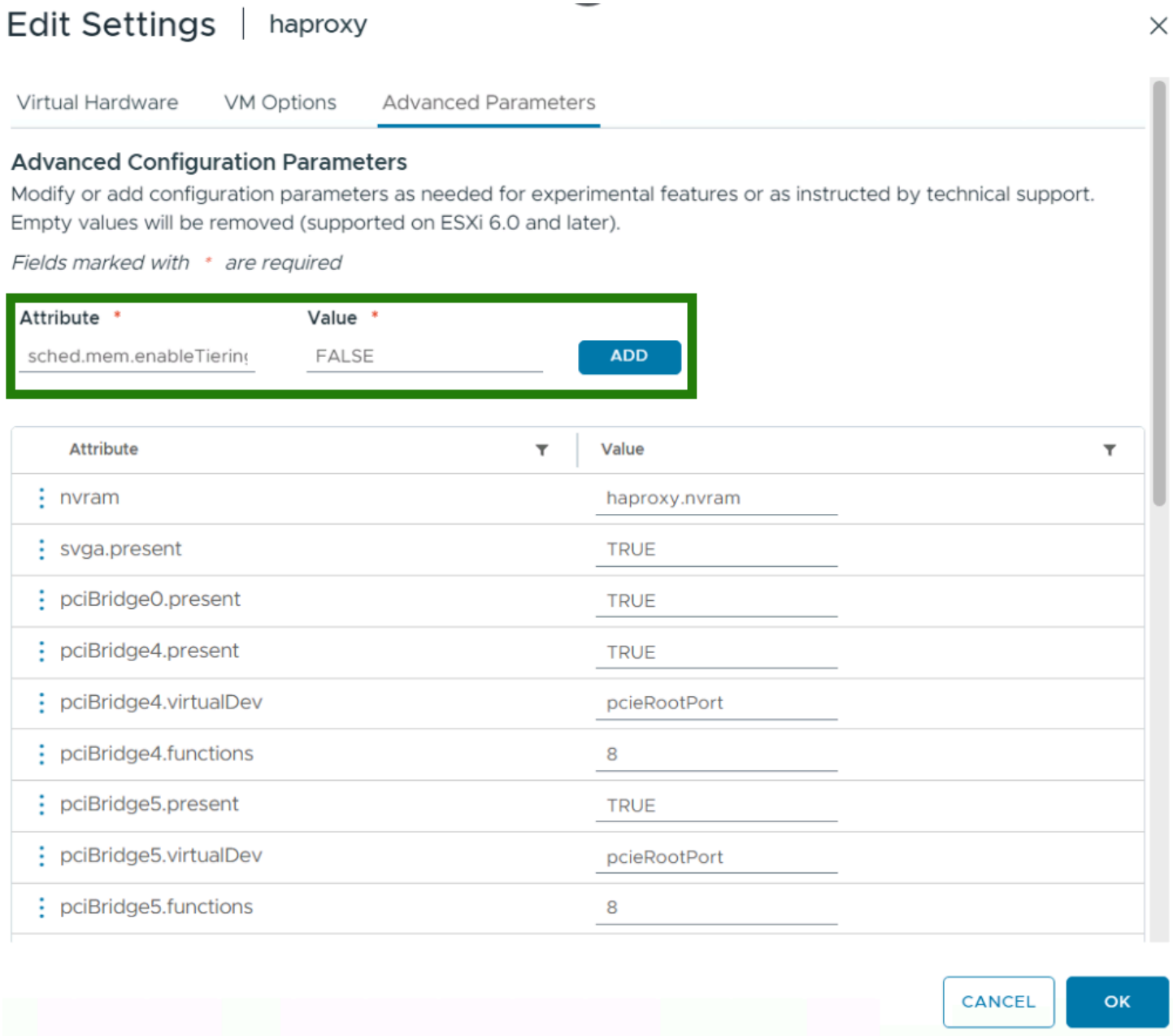
Option	Action
Deactivate Memory Tiering and retain the tier partition	Run the command `esxcli memtier disable`
Deactivate Memory Tiering and delete the tier partition	Run the command `esxcli memtier disable -d`

Deactivate Memory Tiering for a VM

Some VMs require Memory Tiering to be deactivated for them to meet the VM performance or latency requirements. The ESX host automatically deactivates Memory Tiering for all VMs described in the Virtual Machine Profiles and Performance with Memory Tiering.

Note:

Before you deactivate Memory Tiering for a specific VM, reserve the full amount of the VM memory to prevent potential memory reclamation issues.



To deactivate Memory Tiering for a specific VM, perform the following steps.

Note:

If you set `sched.mem.enableTiering` to be `False`, to prevent memory reclamation issues, fully reserve the VM memory.

1. Power off the VM.
2. Right-click the VM and select **Edit Settings**.
3. Click the **Advanced Parameters** tab.
4. Edit the **Advanced Configuration Parameters**.
   a) In the `Attribute` field, enter `sched.mem.enableTiering`.
   b) In the **Value** field, enter `FALSE`.
5. To fully reserve the VM memory, click the **Virtual Hardware** tab.
   a) Expand **Memory** and select the **Reserve all guest memory (All locked)** check box.
6. Power on the VM.

Delete the Memory Tiering Configuration at a Host Level

Place the host in maintenance mode.

When you no longer need the Memory Tiering feature, you can delete the Memory Tiering configuration at a host level.

To delete the Memory Tiering configuration and remove the tier partition, run the following command.

```
esxcli memtier config delete
```

Activate Memory Tiering with vSAN

You can configure Memory Tiering with vSAN by using one of the following approaches.

Approach 1

1. Activate vSAN on a cluster.
   a. In the vSphere Client, navigate to and select a cluster.
   b. Navigate to **Configure** > **vSAN** > **Services**.
   c. To start the **vSAN Enablement Wizard**, click **Configure**.
2. Complete the vSAN installation and verify Disk Assignments.
   a. To ensure that the disks you selected are claimed and part of vSAN, check **Disk Groups** or **Storage Devices**.
3. Deactivate Auto Disk Claim (vSAN Managed Disk Claim).
   a. In the **vSAN Configuration** step, select the **vSAN ESA (Express Storage Architecture)** if prompted.
   b. To ensure vSAN does not automatically claim any new eligible disks, deactivate the **vSAN Managed Disk Claim** option by switching off the toggle.
4. Plug a new NVMe disk into an available slot and activate Memory Tiering.
   a. Select the new disk to be used for Memory Tiering.

Approach 2

1. Activate vSAN on a cluster.
   a. In the vSphere Client, navigate to and select a cluster.
   b. Navigate to **Configure** > **vSAN** > **Services**.
   c. To start the **vSAN Enablement Wizard**, click **Configure**.
2. Deactivate Auto Disk Claim (vSAN Managed Disk Claim).
   a. In the **vSAN Configuration** step, select the **vSAN ESA (Express Storage Architecture)** if prompted.
   b. To ensure vSAN does not automatically claim any new eligible disks, deactivate the **vSAN Managed Disk Claim** option by switching off the toggle.
3. Do not claim the disks now.

a. Navigate through the pages of the wizard without claiming any disks.
b. Complete activating vSAN.

Note:

This prepares your cluster for disk configuration but leaves all disks unclaimed.

4. Reserve Disk for Memory Tiering.
   a. Identify the disk (typically a fast NVMe SSD) you want to reserve for Memory Tiering.
   b. Do not include this disk when you later claim disks manually.
   c. This disk must remain **unclaimed** and **compatible** for Memory Tiering use.
5. Manually claim storage disks.
   a. Navigate to **Cluster** > **Configure** > **vSAN** > **Disk Management**.
   b. Click **Claim Unused Disks for vSAN**.
   c. Select only the data disks you want to use for vSAN storage.

Note:

Do not select the reserved disk for Memory Tiering.

6. Verify disk assignments.
   a. To ensure that the disks you selected are claimed and part of vSAN, check **Disk Groups** or **Storage Devices**.

Note:

The reserved disk is still unclaimed.

7. Activate Memory Tiering.
   a. Select the reserved disk to use for Memory Tiering.

Approach 3

With this approach, you must remove a device from vSAN and must use the device for Memory Tiering.

- Verify that the cluster contains enough space to migrate the data from the storage pool device.

Each storage device of an ESX host claimed by vSAN from a storage pool. All storage devices that are claimed by vSAN contribute to capacity and performance.

1. In the vSphere Client, navigate to the cluster.
2. On the **Configure** tab, under vSAN click **Disk Management**.
3. Select the storage device pool, and click **Remove Disk**.
4. In the **Remove Disk** dialog box, select **Full data migration** to transfer all the data available on the host to other ESX hosts in the cluster.
5. Click **Go To Pre-Check** to find the impact on the cluster if the object is removed or placed in Maintenance mode.
6. Click **Remove** to remove the storage pool device.

Deployment in an Existing vSAN Environment

If you have a vSAN environment that does not contain NVMe drives available for Memory Tiering, you can evacuate drives and can remove them from vSAN groups.

For more information about deploying Memory Tiering in a vSAN-based environment, see NVMe Memory Tiering Design and Sizing on VMware Cloud Foundation 9 Part 4: vSAN Compatibility and Storage Considerations.

For more information about clearing an NVMe drive for Memory Tiering, see:

- vSAN OSA Device Removal UI
- vSAN ESA Device Removal UI
- vSAN Device Removal UI

Memory Tiering Considerations and Best Practices

Memory Tiering is a core platform feature that is integrated into the ESX kernel.

Workload Considerations

Optimal performance is achieved when the workload meets the active memory requirements. Good performance results are obtained with various different workloads, such as VDI, databases, and VMmark benchmarks. For more information about the performance analysis, see Memory Tiering Performance in VMware Cloud Foundation 9.0.

Note:

The active memory is always lower than consumed memory and represents the working set of currently running workloads.

Example of the ESX host memory usage

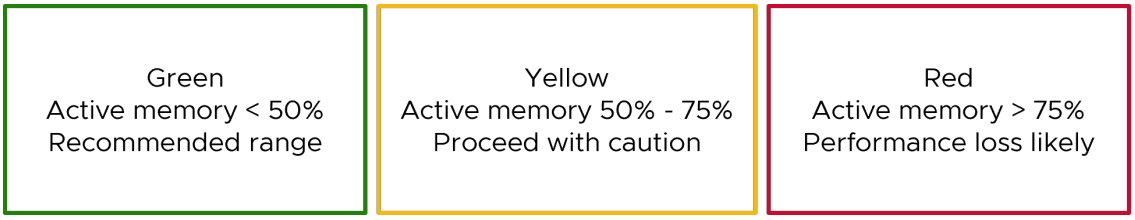
When an ESX host does not use Memory Tiering, a typical host memory usage may look like:

- Host Capacity: 1 TB
- DRAM Capacity: 1 TB
- Active Memory: 200 GB (20%)
- Consumed Memory: 800 GB (80%)

In this example, percentages refer to the total host capacity percentage.

Most workloads typically maintain an active memory range of 10% to 30%. To maximize the efficiency of Memory Tiering, identify hosts or clusters in your environment where current active memory usage is between 0% and 50% (pre–Memory Tiering) of the existing DRAM capacity.

When you activate suitable hosts and clusters for Memory Tiering, for optimal performance, the active memory of the workloads as a percentage of DRAM must adhere to the Memory Tiering Performance white paper.



For the example above, consider that the number of virtual machines on a host is doubled following the deployment of Memory Tiering. If both consumed and active memory also double, the new state of the host would be:

- Host Capacity: 2 TB
- DRAM Capacity: 1 TB
- NVMe Tier Capacity: 1 TB
- Active Memory: 400 GB (20%)
- Consumed Memory: 1600 GB (80%)

Note:

The new active memory of 400 GB remains well within 50% of DRAM capacity (1 TB DRAM). Therefore, the workloads running on the ESX host are expected to maintain optimal performance.

Memory Tiering over NVMe Mechanism

Cold pages start to get tiered to NVMe when the host memory consumption reaches around the 80% threshold of the DRAM capacity. This threshold is significant during initial power-on and load-up scenarios.

Note:

If a virtual machine is powered off, DRAM usage compared to NVMe can become lower until it stabilizes again.

Once this memory consumption threshold is reached, memory is allocated from Tier 1 (NVMe tier). The tiering algorithm, which is part of the memory management layer in ESX, thereafter monitors page activity across the entire memory,

including DRAM and NVMe capacity, and tries to move pages between Tier 0 (DRAM) and Tier 1. Page hotness and coldness monitoring, and cycling through the pages, are repeated at a frequency that is determined by the VM memory size. The entire VM memory is scanned in batches with short pauses, and once the scan is complete, the action to move the pages is taken at the end of the cycle. The scan rate at any point depends on the current state of the system, especially memory pressure. Page classification is performed based on both recency (how recently pages were accessed) and frequency (how often pages are accessed).

Note:

Except for VM memory pages, no other pages are subjected to Memory Tiering over NVMe; for example, kernel memory is never tiered. Software CPU overhead depends on the host memory, and you can see a small usage of about an eighth of a core for Memory Tiering for smaller-sized hosts.

Preparation, Sizing, and the DRAM:NVMe Ratio

Ensure that the host has dedicated NVMe devices that are available for Memory Tiering. To extend host memory capacity, use the default 1:1 DRAM:NVMe ratio. This ratio is suitable for a wide variety of workloads and customer environments, providing 2X the total memory capacity. For example, an environment with 1 TB DRAM and an additional 1 TB (allocated from a larger physical disk, such as a 1.6 TB NVMe drive) provides 2 TB of total host capacity.

You can change the Memory Tiering settings after evaluating your environment historical active memory usage. In general, NVMe sizes must be at least as large as the DRAM of the system.

Do not mix the same physical NVMe drive for NVMe tiering and vSAN, for example.

Difference from Swapping

Memory Tiering provides significant advantages, including:

- Memory Tiering uses sophisticated page aging techniques to classify hot and cold pages whereas swap code uses simplistic random selection of pages to swap out.
- Memory Tiering maintains consistent performance by using dedicated, direct-attached local NVMe devices. Swap often uses a shared device, such as NAS or SAN, that is used by multiple sub-systems including guest storage, that is VMDKs and swap can share a device.
- Memory Tiering algorithms are optimized to provide consistent performance in most workloads.
- Swap provides correctness under extreme conditions, such as migrating VMs with vMotion and other VM evacuation operations with 3X or 4X memory overcommitment. In contrast, Memory Tiering provides both correctness and high performance through thoughtful planning. You can achieve optimal results by choosing an appropriate DRAM:NVMe ratio, ensuring that the workloads active memory fits within the available DRAM capacity, and so on.

Memory Overcommit

Memory overcommit allows administrators to allocate additional memory than is physically present on the host. The memory overcommit relies on mechanisms, such as ballooning, transparent page sharing, and swap that manage over committed memory across multiple VMs. However, these mechanisms can lead to problems, especially under peak load periods or low memory conditions, resulting in performance issues.

With Memory Tiering, a large amount of additional capacity is available through a different tier and VMs can take advantage of this capacity. If some of the VM memory starts getting tiered to the NVMe tier more frequently, having memory overcommit might have a potential performance impact.

Note:

Do not use memory overcommit when Memory Tiering is activated.

vSphere vMotion and DRS

With Memory Tiering, DRS now includes new logic for VM placement and load balancing. There is almost no performance impact on a running VM during vMotion. However, vMotion will take longer (1.5X to 2X) to complete, since pages must be fetched from the NVMe tier first. Performing vMotion does not increase the host's DRAM capacity requirements, since

pages are streamed as they are selected and retrieved for migration to another host. If CPU utilization increases, DRS can also move VMs to a lightly loaded host and support this in a mixed environment of tiered and non-tiered hosts.

For more information about the vMotion performance with Memory Tiering, see Memory Tiering Performance in VMware Cloud Foundation 9.0.

Memory Tiering with Memory Reservations

Memory reservations (not pinned memory) can work with Memory Tiering. For example, if there are VMs on a host, and all these VMs have reserved their memory, and the combined consumed memory of these VMs exceeds the DRAM capacity, the memory reservations can then be split between DRAM and the NVMe tiers.

Server OEM Drive Selection

Selection of NVMe tiering drives for Dell Servers

When you select NVMe devices, make sure that you choose a **Mixed-Use Enterprise NVMe** drive from the Dell configuration. For example, for the Dell R750, such as "1.6 TB Enterprise NVMe Mixed Use AG Drive U.2 with carrier". These mixed read or write endurance classes (≥3 DWPD) are ideal for Memory Tiering and ensure optimal performance and longevity.

> **Note:**
>
> This may require selecting a different pair of NVMe drives for Memory Tiering than those used for vSAN in a ReadyNode configuration.

You can validate compatibility through the Broadcom Compatibility Guide under **vSAN SSD** using the following filters:

- **Supported Release**: ESX 9.1
- **Partner**: Dell
- **Device Type**: NVMe
- **Endurance Class**: D
- **Performance Class**: F or G
- **Drive Writes Per Day (DWPD)**: 3

You can also use the search link with the fields already populated.

For more information about the Memory Tiering for Dell Servers, see Enabling NVMe Memory Tiering on Dell PowerEdge servers using VMware vSphere 9.0.

Selection of NVMe Tiering Drives for HPE Servers

1. Open HPE ProLiant Compute SSD Selector Tool.
2. On the left side pane, under the **SSD Workload** drop-down menu, select **Mixed Use**.
3. Under the **Interface Type** drop-down menu, select **NVMe Mainstream Performance** and **NVMe High Performance**, and deselect the other options.
4. From the list of NVMe drives that appear, for more information about any NVMe drive entry you are interested in, click the **Excel** or **PDF** icon next to the drive name.

> **Note:**
>
> Make sure that "Endurance DWPD" is at least 3. NVMe Memory Tiering requires endurance >=3.

You can use the search link for VMware BCG for HPE branded drives with the fields already populated.

Selection of NVMe Tiering Drives for Lenovo Servers

You must select Mixed Use NVMe drives with endurance of 3 DWPD. For example, you must select the NVMe backplane and select a set of drives.

These drives must be Enterprise-class (or Advanced Data-Center class) drives.

NVMe Drive Examples

- ThinkSystem CD8P Mixed Use NVMe PCIe 5.0 SSDs

- ThinkSystem PM1745 Mixed Use NVMe PCIe 5.0 SSDs
- ThinkSystem P5620 Mixed Use NVMe PCIe 4.0 SSDs
- ThinkSystem 7450 Mixed Use NVMe PCIe 4.0 SSDs
- ThinkSystem Vendor Agnostic Mixed Use NVMe SSDs
- ThinkSystem PS1030 Mixed Use NVMe PCIe 5.0 x4 SSDs
- ThinkSystem Solidigm P5620 Mixed Use NVMe PCIe 4.0 SSDs
- ThinkSystem Intel P5600 Mainstream NVMe PCIe 4.0 x4 SSDs

You can use the search link for VMware BCG for Lenovo branded drives with the fields already populated.

For more information about Memory Tiering over NVMe for Lenovo, see Implementing Memory Tiering over NVMe using VMware ESX 9.0.

NVMe Failure Scenarios for Memory Tiering in vSphere

When an NVMe device used for Memory Tiering fails, the impact on virtual machines running on an ESX 9.1 and later host depends on whether those VMs have memory pages on the failed device. This topic describes the behavior of virtual machines during and after an NVMe device failure, and explains how vSphere HA, DRS, and maintenance mode, interact in an environment configured for Memory Tiering.

NVMe Device Failure Impact on Virtual Machines

When a single NVMe device that is used for Memory Tiering fails, the following behavior applies:

- Virtual machines that have memory pages on the NVMe device continue running if those pages are not accessed. If a page fault is raised against a page on the failed device and cannot be serviced, the virtual machine powers off.
- Virtual machines that use only DRAM are not affected by an NVMe device failure and continue running normally.

VM Restart Policies in HA-enabled Clusters

If a virtual machine powers off due to an NVMe device failure, you can configure vSphere HA VM restart policies to automatically restart that VM on another host in the cluster.

VM restart policies require only that:

- The virtual machine runs in a cluster with vSphere HA activated.
- The cluster has sufficient resources on other hosts to accommodate the restarted VMs.

A dedicated HA failover host is not required. Restart policies can recover VMs on any connected host in the cluster that is not in maintenance mode.

Deactivating Memory Tiering on an ESX Host

Memory tiering on an ESX host can be deactivated only when the host is in maintenance mode. When the host enters maintenance mode, vMotion evacuates all virtual machines on that host to other eligible hosts in the cluster.

After a virtual machine is migrated, the destination host immediately begins tiering the memory pages of the VM even if the previous host was not actively tiering memory for that VM.

> **Note:**
>
> In a future release, the source host will transfer memory heat information (page access frequency) to the destination host during vMotion. This data will help optimize the initial placement of frequently accessed pages immediately after migration.

When the host exits maintenance mode and rejoins the cluster, the virtual machine migration depends on the DRS automation level. If DRS determines that rebalancing the cluster improves resource utilization or performance, it migrates the VMs back to that host. When the destination host is configured for Memory Tiering, DRS also evaluates active memory utilization as part of the placement decision.

When an ESX host runs a mix of tiered and non-tiered VMs and Memory Tiering is deactivated, the host must enter maintenance mode. All VMs on the host, both tiered and non-tiered are evacuated by vMotion.

Triggering Maintenance Mode After an NVMe Failure

Placing a host in maintenance mode after an NVMe failure is a manual operation in vSphere 9.1. You can trigger maintenance mode in the following ways:

- Manually, from the vSphere Client or by using ESXCLI.
- Automatically through a hardware management plug-in, such as Dell OpenManage or HPE OneView, if the plugin is configured to detect hardware failures and place the host in maintenance mode automatically.

vSphere DRS and VM Recovery DRS Behavior in a Memory-tiered Cluster

DRS makes placement and load-balancing decisions that are based on both resource availability and virtual machine performance. If the performance of a VM decreases, DRS can migrate the VM with vMotion to a host that offers better performance. When a host that is configured for Memory Tiering is the target host, DRS also evaluates active memory utilization as part of the placement decision.

Deactivating Memory Tiering for a Specific Virtual Machine

You can deactivate memory tiering for individual VMs to ensure those VMs use only DRAM. You can also automate this process during VM power-on.

When you deactivate memory tiering for a VM, you must also configure a full memory reservation for that VM. Without a memory reservation, the VM may not have sufficient DRAM because other VMs on the same host continue to participate in memory tiering. The VM without a reservation can end up with its pages that are swapped to the host swap file, which performs worse than NVMe-tiered memory.

Effect on the DRAM:NVMe Ratio for Remaining Tiered Virutal Machines

Memory tiering is sized on a default 1:1 DRAM:NVMe ratio. When you apply full memory reservations to VMs and remove them from tiering, the effective ratio for the remaining tiered VMs changes.

For example, on a host with 1 TB of DRAM and 1 TB of NVMe, the default ratio is 1:1. If two VMs with 200 GB each are given full memory reservations and removed from tiering, 400 GB of DRAM is no longer available to the tiering pool. The effective ratio for the remaining VMs changes to approximately 0.6:1 (600 GB DRAM : 1 TB NVMe). This reduction in the DRAM cache available per tiered page can impact performance for the VMs that still use memory tiering.

Using Swap Files with vSphere

You can specify the location of your guest swap file, reserve swap space when memory is overcommitted, and delete a swap file.

ESX hosts use swapping to forcibly reclaim memory from a virtual machine when the `vmmemctl` driver is not available or it is not responsive.

- It was never installed.
- It is explicitly deactivated.
- It is not running (for example, while the guest operating system is booting).
- It is temporarily unable to reclaim memory quickly enough to satisfy current system demands.
- It is functioning properly, but the maximum balloon size is reached.

Standard demand-paging techniques swap pages back in when the virtual machine needs them.

Swap File Location

By default, the swap file is created in the same location as the virtual machine's configuration file, which may either be on a VMFS datastore, a vSAN datastore or a VMware vSphere® Virtual VolumesTM datastore. On a vSAN datastore or a vVols datastore, the swap file is created as a separate vSAN or vVols object.

The ESX host creates a swap file when a virtual machine is powered on. If this file cannot be created, the virtual machine cannot power on. Instead of accepting the default, you can also:

- Use per-virtual machine configuration options to change the datastore to another shared storage location.

- Use host-local swap, which allows you to specify a datastore stored locally on the host. This allows you to swap at a per-host level, saving space on the SAN. However, it can lead to a slight degradation in performance for vSphere vMotion because pages swapped to a local swap file on the source host must be transferred across the network to the destination host. Currently vSAN and vVols datastores cannot be specified for host-local swap.

Enable Host-Local Swap for a DRS Cluster

Host-local swap allows you to specify a datastore that is stored locally on the host as the swap file location. You can enable host-local swap for a DRS cluster.

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Configuration**, select **General** to view the swap file location and click **Edit** to change it.
4. Select the **Datastore specified by host** option and click **OK**.
5. Browse to one of the hosts in the cluster in the vSphere Client.
6. Click **Configure**.
7. Under Virtual Machines, select **Swap file location**.
8. Click Edit and select the local datastore to use and click **OK**.
9. Repeat 5 through 8 for each host in the cluster.

Host-local swap is now enabled for the DRS cluster.

Enable Host-Local Swap for a Standalone Host

Host-local swap allows you to specify a datastore that is stored locally on the host as the swap file location. You can enable host-local swap for a standalone host.

1. Browse to the host in the vSphere Client.
2. Click **Configure**.
3. Under **Virtual Machines**, select **Swap file location**.
4. Click **Edit** and select **Selected Datastore**.
5. Select a local datastore from the list and click **OK**.

Host-local swap is now enabled for the standalone host.

Swap Space and Memory Overcommitment

You must reserve swap space for any unreserved virtual machine memory (the difference between the reservation and the configured memory size) on per-virtual machine swap files.

This swap reservation is required to ensure that the ESX host is able to preserve virtual machine memory under any circumstances. In practice, only a small fraction of the host-level swap space might be used.

If you are overcommitting memory with ESX, to support the intra-guest swapping induced by ballooning, ensure that your guest operating systems also have sufficient swap space. This guest-level swap space must be greater than or equal to the difference between the virtual machine's configured memory size and its Reservation.

> **CAUTION:**
> If memory is overcommitted, and the guest operating system is configured with insufficient swap space, the guest operating system in the virtual machine can fail.

To prevent virtual machine failure, increase the size of the swap space in your virtual machines.

- Windows guest operating systems— Windows operating systems refer to their swap space as paging files. Some Windows operating systems try to increase the size of paging files automatically, if there is sufficient free disk space.

See your Microsoft Windows documentation or search the Windows help files for “paging files.” Follow the instructions for changing the size of the virtual memory paging file.

- Linux guest operating system — Linux operating systems refer to their swap space as swap files. For information on increasing swap files, see the following Linux man pages:
  - `mkswap` — Sets up a Linux swap area.
  - `swapon` — Enables devices and files for paging and swapping.

Guest operating systems with a lot of memory and small virtual disks (for example, a virtual machine with 8GB RAM and a 2GB virtual disk) are more susceptible to having insufficient swap space.

Note:
Do not store swap files on thin-provisioned LUNs. Running a virtual machine with a swap file that is stored on a thin-provisioned LUN can cause swap file growth failure, which can lead to termination of the virtual machine.

When you create a large swap file (for example, larger than 100GB), the amount of time it takes for the virtual machine to power on can increase significantly. To avoid this, set a high reservation for large virtual machines.

You can also place swap files on less costly storage using host-local swap files.

Configure Virtual Machine Swapfile Properties for the Host

Configure a swapfile location for the host to determine the default location for virtual machine swapfiles in the vSphere Client.

Required privilege:**Host machine** > **Configuration** > **Storage partition configuration**

By default, swapfiles for a virtual machine are located on a datastore in the folder that contains the other virtual machine files. However, you can configure your host to place virtual machine swapfiles on an alternative datastore.

You can use this option to place virtual machine swapfiles on lower-cost or higher-performance storage. You can also override this host-level setting for individual virtual machines.

Setting an alternative swapfile location might cause migrations with vMotion to complete more slowly. For best vMotion performance, store the virtual machine on a local datastore rather than in the same directory as the virtual machine swapfiles. If the virtual machine is stored on a local datastore, storing the swapfile with the other virtual machine files will not improve vMotion.

1. Browse to the host in the vSphere Client.
2. Click **Configure**.
3. Under **Virtual Machines**, click **Swap file location**.

   The selected swapfile location is displayed. If configuration of the swapfile location is not supported on the selected host, the tab indicates that the feature is not supported.

   If the host is part of a cluster, and the cluster settings specify that swapfiles are to be stored in the same directory as the virtual machine, you cannot edit the swapfile location from the host under **Configure**. To change the swapfile location for such a host, edit the cluster settings.
4. Click **Edit**.
5. Select where to store the swapfile.

Option	Description
Virtual machine directory	Stores the swapfile in the same directory as the virtual machine configuration file.
Use a specific datastore	Stores the swapfile in the location you specify. If the swapfile cannot be stored on the datastore that the host specifies, the swapfile is stored in the same folder as the virtual machine.

6. Optional: If you select **Use a specific datastore**, select a datastore from the list.
7. Click **OK**.

The virtual machine swapfile is stored in the location you selected.

Configure a Virtual Machine Swap File Location for a Cluster

By default, swap files for a virtual machine are on a datastore in the folder that contains the other virtual machine files. However, you can instead configure the hosts in your cluster to place virtual machine swap files on an alternative datastore of your choice.

Before you configure a virtual machine swap file location for a cluster, you must configure the virtual machine swap file locations for the hosts in the cluster as described in Configure Virtual Machine Swapfile Properties for the Host.

You can configure an alternative swap file location to place virtual machine swap files on either lower-cost or higher-performance storage, depending on your needs.

1. Browse to the cluster.
2. Click **Configure**.
3. Select **Configuration** > **General**.
4. Next to swap file location, click **Edit**.
5. Select where to store the swap file.

Option	Description
Virtual machine directory	Stores the swap file in the same directory as the virtual machine configuration file.
Datastore specified by host	Stores the swap file in the location specified in the host configuration. If the swap file cannot be stored on the datastore that the host specifies, the swap file is stored in the same folder as the virtual machine.

6. Click **OK**.

Delete Swap Files

If a host fails, and that host had running virtual machines that were using swap files, those swap files continue to exist and consume many gigabytes of disk space. You can delete the swap files to eliminate this problem.

1. Restart the virtual machine that was on the host that failed.
2. Stop the virtual machine.

The swap file for the virtual machine is deleted.

Using Persistent Memory with vSphere

Persistent Memory (PMem), also known as Non-Volatile Memory (NVM), is capable of maintaining data even after a power outage. PMem can be used by applications that are sensitive to downtime and require high performance.

> **CAUTION:**
>
> Persistent Memory and all the associated features are deprecated and will not be supported starting with VCF 9.0. The following sections are relevant to the previous versions of *vSphere*.

VMs can be configured to use PMem on a standalone host, or in a cluster. PMem is treated as a local datastore. Persistent memory significantly reduces storage latency. In ESX you can create VMs that are configured with PMem, and applications within these VMs can take advantage of this increased speed. Once a VM is initially powered on, PMem is reserved for it regardless of whether it is powered on or off. This PMem stays reserved until the VM is migrated or removed.

Persistent memory can be consumed by virtual machines in two different modes. Legacy guest OSes can still take advantage of virtual persistent memory disk feature.

- **Virtual Persistent Memory (vPMem)**
  Using vPMem, the memory is exposed to a guest OS as a virtual NVDIMM. This enables the guest OS to use PMem in byte addressable random mode.

  Note:

  You must use VM hardware version 14 and a guest OS that supports NVM technology.

  Note:

  You must use VM hardware version 19 when you configure vSphere HA for PMem VMs. For more information, see Configure vSphere HA for PMem VMs.
- **Virtual Persistent Memory Disk (vPMemDisk)**
  Using vPMemDisk, the memory can be accessed by the guest OS as a virtual SCSI device, but the virtual disk is stored in a PMem datastore.

When you create a VM with PMem, memory is reserved for it at the time of Hard disk creation. Admission control is also done at the time of Hard disk creation. For more information, see vSphere HA Admission Control PMem Reservation.

In a cluster, each VM has some capacity for PMem. The total amount of PMem must not be greater than the total amount available in the cluster. The consumption of PMem includes both powered on and powered off VMs. If a VM is configured to use PMem and you do not use DRS, then you must manually pick a host that has sufficient PMem to place the VM on.

NVDIMM and traditional storage

NVDIMM is accessed as memory. When you use traditional storage, software exists between applications and storage devices which can cause a delay in processing time. When you use PMem, the applications use the storage directly. This means that PMem performance is better than traditional storage. Storage is local to the host. However, since system software cannot track the changes, solutions such as backups do not currently work with PMem.

Solutions such as vSphere HA have limited scope if vPMem is used in a mode that is not write-through to a non-PMem datastore. When vSphere HA is activated for vPMem VMs with failover enabled, the VM can be failed over to different host. When this happens, the VM is using the PMem resources on the new host. To free up the resources on the old host, a garbage collector periodically identifies and frees up these resources for use by other VMs.

Name spaces

Name spaces for PMem are configured before ESX starts. Name spaces are similar to disks on the system. ESX reads name spaces and combines multiple name spaces into one logical volume by writing GPT headers. This is formatted automatically by default, if you have not previously configured it. If it has already been formatted, ESX attempts to mount the PMem.

GPT tables

If the data in PMem storage is corrupted it can cause ESX to fail. To avoid this, ESX checks for errors in the metadata during PMem mount time.

PMem regions

PMem regions are a continuous byte stream that represent a single vNVDimm or vPMemDisk. Each PMem volume belongs to a single host. This could be difficult to manage if an administrator has to manage each host in a cluster with a large number of hosts. However, you do not have to manage each individual datastore. Instead you can think of the entire PMem capacity in the cluster as one datastore.

VC and DRS automate initial placement of PMem datastores. Select a local PMem storage profile when the VM is created or when the device is added to the VM. The rest of the configuration is automated. One limitation is that ESX does not allow you to put the VM home on a PMem datastore. This is because it takes up valuable space to store VM log and stat files. These regions are used to represent the VM data and can be exposed as byte addressable nvDimms, or VpMem disks.

Migration

Since PMem is a local datastore, if you want to move a VM you must use storage vMotion. A VM with vPMem can only be migrated to an ESX host with PMem resource. A VM with vPMemDisk can be migrated to an ESX host without a PMem resource.

Error handling and NVDimm management

Host failures can result in a loss of availability on vPMem VMs which are not in write-through mode. In the case of catastrophic errors, you may lose all data and must take manual steps to reformat the PMem.

vSphere Persistent Memory with the vSphere Client

For a brief conceptual introduction to Persistent Memory, see:

Enhancements to Working with PMEM in the vSphere Client

For a brief overview of enhancements in the HTML5-based vSphere Client when working with PMem, see:

Migrating and Cloning VMs Using PMEM in the vSphere Client

For a brief overview of migrating and cloning virtual machines that use PMem, see:

Configure vSphere HA for PMem VMs

You can configure vSphere HA for PMem VMs in write-through mode, so that when a host fails VMs can be restored on another functioning host.

- You must select Hardware version 19.
- PMem VMs with vPMemDisks are not supported.

1. When creating a new VM in the **New Virtual Machine** wizard, select **Customize hardware**.
   a) Click **ADD NEW DEVICE** and select **Add NVDIMM** from the drop-down menu.
   b) Click the checkbox **Allow failover on another host for all NVDIMM devices**.
   c) Click **NEXT** and complete the **New Virtual Machine** wizard.

      On Host failure, NVDIMM PMem data cannot be recovered. By default, HA will not attempt to restart this virtual machine on another host. Allowing HA on host failure to failover the virtual machine, will restart the virtual machine on another host with a new, empty NVDIMM.
2. To enable HA on an existing VM, browse to the VM.
   a) Under **VM Hardware**, click **EDIT**.
   b) Select the NVDIMM.
   c) Click the checkbox **Allow failover on another host for all NVDIMM devices**.
   d) Click **OK**.

      On host failure, HA will restart this virtual machine on another host with new, empty NVDIMMs.

vSphere HA Admission Control PMem Reservation

Admission control is a policy that is used by vSphere HA to ensure failover capacity within a cluster.

Raising the number of potential host failures to tolerate will increase the availability restraints and capacity reserved. You can reserve a percentage of Persistent Memory for Host failover capacity. This is actual storage capacity that is blocked and must be considered for host power off.

Under **Edit Cluster Settings**, you can select **Admission Control** to specify the number of failures the host tolerates.

If you select a CPU/Memory reservation that is defined by:

- **Cluster resource percentage**, some amount of persistent memory capacity in the cluster is dedicated for failover purpose even if the virtual machines in the cluster are not using persistent memory currently. This percentage can either be specified through an override, or it is automatically calculated according to the **host failures to tolerate** setting. When PMem admission control is enabled, PMem capacity is reserved across the cluster even if there are VMs using PMem as disks.

- **Slot Policy (powered-on VMs)**, persistent memory admission control overrides the Slot Policy with the Cluster Resource Percentage policy, for persistent memory resources only. The percentage value is automatically calculated from the **host failures cluster tolerates** setting and cannot be overridden.
- **Dedicated failover hosts**, the persistent memory of the dedicated failover hosts is dedicated for failover purpose and you cannot provision virtual machines with persistent memory on these hosts.

  Note:
  After you select an admission control policy, you must also click the **Reserve Persistent Memory failover capacity** checkbox to enable PMem admission control.

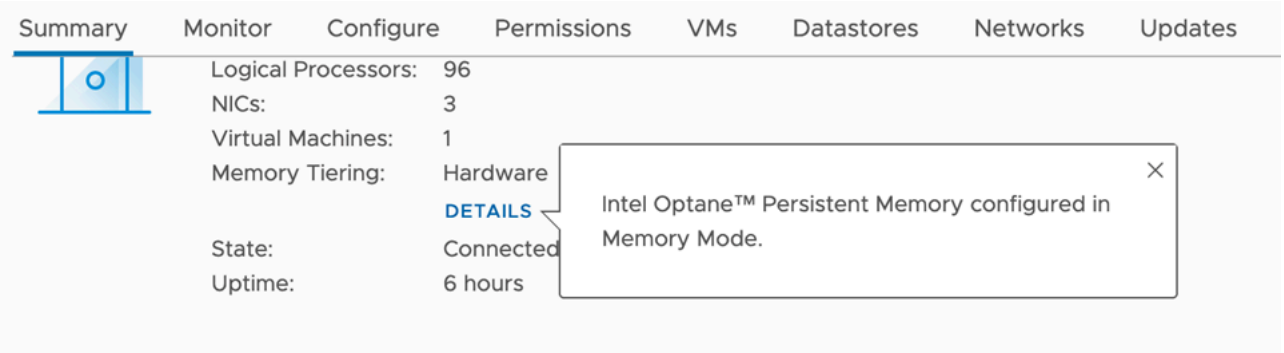
vSphere Memory Monitoring and Remediation

The vSphere Memory Monitoring and Remediation (vMMR) feature collects data and provides visibility of performance statistics so you can determine if your application workload is regressed due to Memory Mode. You can also use the vMMR feature to create custom alarms that trigger at desired threshold levels.

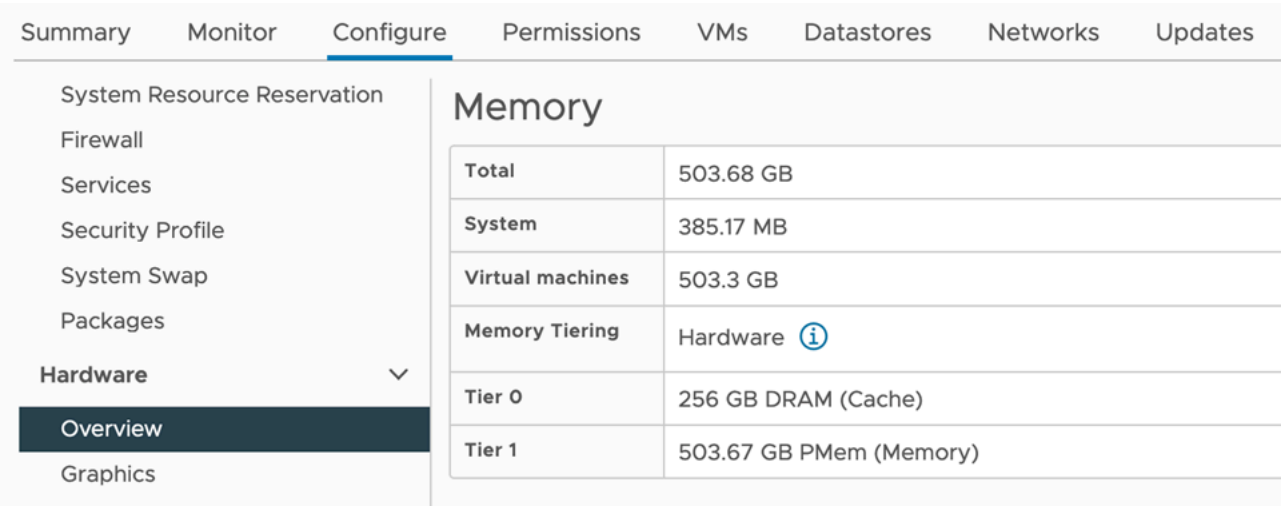
Intel Optane Persistent Memory can be configured in BIOS settings in App Direct or Memory Mode. In App Direct mode, persistent memory can be accessed as byte addressable, persistent memory along with DRAM. In Memory Mode, DRAM becomes the hardware cache and the larger PMem becomes volatile and appears as system memory.

Memory Mode is invisible and transparent to VMs. Once you configure the system in Memory Mode, the system appears as a traditional system with DRAM. A cluster can have a mix of hosts with different configurations. vSphere shows additional information indicating that the system is in Memory Mode. ESX programs provide performance counters that gather host-level and VM-level statistics. These performance statistics are used to create alarms and are also tracked in performance charts.

You can find out if the system is in Memory Mode in the host **Summary** tab, **Memory Tiering: Hardware with some additional details**.



You can also view the size of DRAM and PMEM under **Configure** > **Hardware** > **Overview** > **Memory**.



ESX gathers and exposes two kinds of memory statistics:

- Host Level Statistics: A memory sub-component measures DRAM and PMem performance by programming performance counters. The host level statistics are total, read/write bandwidth, read/write latency and miss rate for the different memory types (DRAM, PMem).
- VM Level Statistics: vSphere monitors performance counters to get data on the DRAM and PMEM read bandwidth of the VM.

Both Host and VM have a new Memory pane under Performance charts. This shows memory details like Memory Utilization and Memory Reclamation along with new statistics. On the ESX host level, you can monitor the Memory Bandwidth and Memory Miss Rate charts. On the VM level, you can view the PMem read bandwidth and DRAM read bandwidth.

From the **VMs** tab of an ESX host, you can view a list containing performance information about all virtual machines that reside on the host. To display information about the Memory Mode impact on a virtual machine, click the view columns ( ) icon and select the Active Memory, DRAM Read Bandwidth, and PMem Read Bandwidth metrics.

There are two preconfigured default alarms, one at the host level (Host Memory Mode High Active DRAM Usage) and another at the VM level (Virtual Machine High PMem Bandwidth Usage). If the alarm condition is met, an event is published to trigger the corresponding alarm. You can also create custom alarms that are based on performance metrics. vMMR alarms only work on hosts configured with Memory Mode.

When DRS is activated and fully automated in the cluster, if the active memory utilization of the host is above certain percentage of the size of the DRAM cache, DRS might move some VMs out of the host in order to balance the load.

For more information, see *vSphere Monitoring and Performance*.

> **Note:**
> vMMR is supported on Intel Broadwell, Skylake, Cascade Lake, and Ice Lake platforms. Host Level DRAM statistics are available on these platforms. Host and VM Level PMem statistics are only available in Cascade Lake and Ice Lake hosts configured in Memory Mode.

Configure Virtual Graphics on vSphere

You can edit graphics settings for supported graphics implementations.

vSphere supports multiple graphics implementations.

- VMware supports 3d graphics solutions from AMD, Intel and NVIDIA.
- NVIDIA GRID support.
- Allows single NVIDIA vib to support both vSGA and vGPU implementations.
- Provides vCenter GPU performance charts for Intel and NVIDIA.
- Enables graphics for Horizon View VDI desktops.
- You can create a VM with and up to 16 vGPU devices. HWv21 must be enabled on ESX and vCenter.

You can configure host graphics settings, and customize vGPU graphics settings on a per VM basis.

> **Note:**
> In this chapter, "Memory" refers to physical RAM.

View GPU Statistics

You can view detailed information for a host graphics card.

You can see GPU temperature, utilization, and memory usage.

> **Note:**
> These statistics are only displayed when the GPU driver is installed on the host.

1. In the vSphere Client, navigate to the host.
2. Click the **Monitor** tab and click **Performance**.
3. Click **Advanced** and select **GPU** from the drop-down menu.

Add an NVIDIA GRID vGPU to a Virtual Machine

If an ESX host has an NVIDIA GRID GPU graphics device, you can configure a virtual machine to use the NVIDIA GRID virtual GPU (vGPU) technology.

- Verify that an NVIDIA GRID GPU graphics device with an appropriate driver is installed on the host. For more information see How do I Add a PCI Device to a Virtual Machine.
- Verify that the virtual machine is compatible with ESX 6.0 and later.

NVIDIA GRID GPU graphics devices are designed to optimize complex graphics operations and enable them to run at high performance without overloading the CPU.

1. Right-click a virtual machine and select **Edit Settings**.
2. On the **Virtual Hardware** tab, select **Add New Device** and select **New PCI Device** from the drop-down menu.
3. Expand the **New PCI device**, and select the NVIDIA GRID vGPU passthrough device to which to connect your virtual machine.

   Note:
   Full memory reservation will be applied automatically, it's required for PCI device.
4. Select a GPU profile.

   A GPU profile represents the vGPU type.
5. Click **OK**.

The virtual machine can access the device.

Configuring Host Graphics

You can customize the graphics options on a per host basis.

- Power off the virtual machines.
- The NVIDIA vGPU Manager VIB must be installed on the host. Installing the VIB automatically configures the host for vendor shared passthrough graphics.

  Note:

  In vSphere 9.0 or later, the legacy Shared (vSGA) graphics mode is deprecated.

1. Select a host and select **Configure** > **Hardware** > **Graphics**.
2. Under **Host Graphics**, select **Edit**.
3. In the **Edit Host Graphics Settings** dialog box, select a shared passthrough GPU assignment policy.
   a) **Spread VMs across GPUs (best performance)**
   b) **Group VMs on GPU until full (GPU Consolidation)**
4. Click **OK**.
5. Restart the host for the settings to take effect.

Configuring Graphics Devices

You can edit graphics type for a video card.

Virtual machines must be powered off.

1. Under **Graphics Devices**, select a graphics card and click **Edit**.
   a) Select **Shared** for VMware shared virtual graphics.
   b) Select **Shared Direct** for Vendor shared passthrough graphics.
2. Click **OK**.

If you select a device, it shows which virtual machines are using that device if they are active.

After clicking **OK**, you must restart Xorg on the host.

Migrating VMs with vGPUs

You can migrate VM's that have vGPUs.

Starting with vSphere 8.0 Update 2, DRS can automatically migrate VMs with vGPUs. When the DRS Cluster Advanced Options are set and the Estimated VM Device's Stun Time for a VM is lower than the VM Device's vMotion Stun Time limit, DRS will automate VM migrations.

To enable this functionality, make sure the infrastructure meets the following requirements:

- Healthy vSphere Lifecycle Services. For more information, see Broadcom knowledge base article 316510.
- Configure the VM's vGPU devices through vCenter only.
- Healthy vMotion network

If these requirements are met, add the following DRS Cluster Advanced Options:

DRS Cluster Advanced Options	Value
`PassthroughDrsAutomation`	1
`LBMaxVmotionPerHost`	1

For vGPU VMs with Stun Times exceeding the "vMotion Stun Time Limit" (Default 100 seconds), a VI Admin can add the following DRS Cluster Advanced Option:

DRS Cluster Advanced Options	Value
`VmDevicesStunTimeTolerated`	<number of seconds, greater than any VM's Estimated Stun Time in the Cluster> (Default 100 seconds)

Or you can modify the "vMotion Stun Time Limit" in the **VM Configuration** > **Edit Settings** > **VM Options tab** > **Advanced section**.

For earlier releases:

For Maintenance Mode evacuations, see Broadcom knowledge base article 318908.

If VM placement issues arise, reduce DRS Automation to **Partially Automated**. For more information, see Edit Cluster Settings.

You can manually migrate VMs with vGPUs to another host. See "Using vMotion to Migrate vGPU Virtual Machines" in the *vCenter and Host Management* guide for more information.

Configuring vGPU Size

You can adjust the vGPU size in vCenter.

vGPU profiles on a single physical GPU can have different types or sizes. This means, for example, you can run vGPU VM profiles with 16 GB compute and 4 GB graphics simultaneously on a single device. This allows greater flexibility with vGPU workloads and better utilization of devices with mixed vGPU workloads.

You can activate or deactivate heterogeneous sizes on a per-device basis. Activating this feature reduces the maximum number of instances of some profiles. For example, the A16-1Q profile supports 24 instances when the feature is

deactivated, but it supports 16 instances when it is activated. You may prefer to deactivate the feature on some devices to maximize capacity for their workloads.

1. Browse to the ESX host.
2. You can adjust the vGPU size under: **Configure** > **Hardware** > **Graphics** > **Graphics Devices**.
3. Select **vGPU Mode** and choose either **SameSize** or **MixedSize**. The default value is **SameSize**.
   - **SameSize** indicates that the sizes are homogeneous.
   - **MixedSize** indicates that the sizes can be heterogeneous.

When editing a graphics device, the following message is displayed.

```
Settings will take effect after restarting the host or xorg service.
```

This is required when you edit the vGPU size.

Capacity Visualization and Management of DirectPath Profiles

DirectPath Profiles capture the settings of DirectPath passthrough devices, such as vGPU profiles (VMIOP), Enhanced DirectPath I/O, Dynamic DirectPath devices, and Virtual Device Groups, in the vSphere environment. DirectPath Profiles define specific parameters of virtual devices used by VMs, such as memory frame buffer size, sharing mode, and so on.

DirectPath Profiles allow you to define, save, and assign a user-friendly identifier to the configuration parameters of a DirectPath virtual device so that it can be referred across different VMs and workflows, saving time and improving efficiency.

Capacity Visualization of DirectPath Profiles

DirectPath Profiles allow you to monitor the utilization of DirectPath virtual devices. DirectPath Profiles provide enhanced visibility, and you can quickly assess the availability, capacity, and consumption of resources like vGPUs and other DirectPath virtual devices.

As an administrator, you can manage devices at cluster and host levels and gain information about the VMs that consume specific profiles. This makes it easier to troubleshoot, allocate, or release resources as needed.

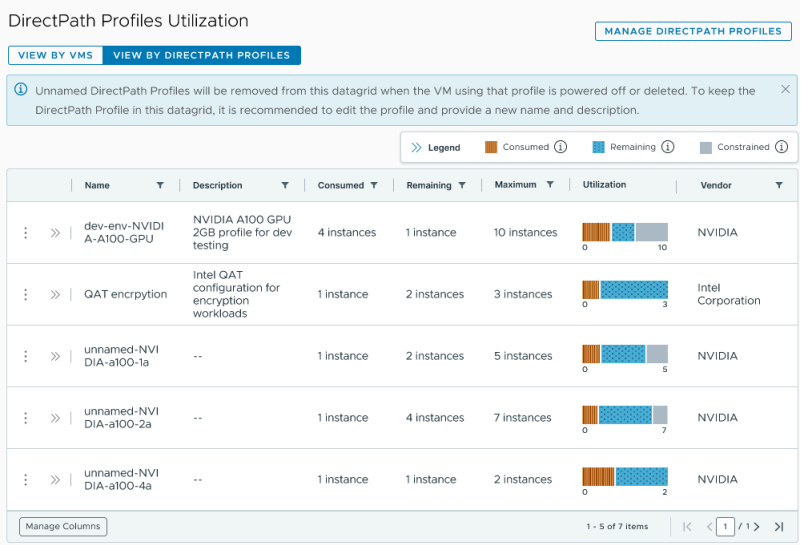
Where to Find DirectPath Profiles

You can view information about DirectPath Profiles on the **Summary** page of a cluster and a host, displayed in a separate **DirectPath Profiles** card, or the PCI Devices card on the **Summary** page of a virtual machine.

On the **Summary** page of a cluster or a host, you can see overview information, such as the number of consumed instances and the top five utilized profiles in the cluster. To view more details, click **View All**.

To view detailed information about the capacity and utilization, perform the following steps.

1. In the vSphere Client, navigate to and select a cluster or a host.
2. Click the **Monitor** tab and click **DirectPath Profiles Utilization**.
3. To view information about the DirectPath Profiles, click the **View By DirectPath Profiles** tab.



A list containing all DirectPath Profiles appears, and for each profile, you can monitor the following information related to capacity utilization. There are two types of DirectPath Profiles:

- DirectPath Profiles that users explicitly create through the workflow for creating DirectPath Profiles. A VM does not need to consume this DirectPath Profile.
- DirectPath Profiles that the system implicitly creates for DirectPath devices used by powered-on VMs. These DirectPath Profiles are prefixed with `unnamed-`.

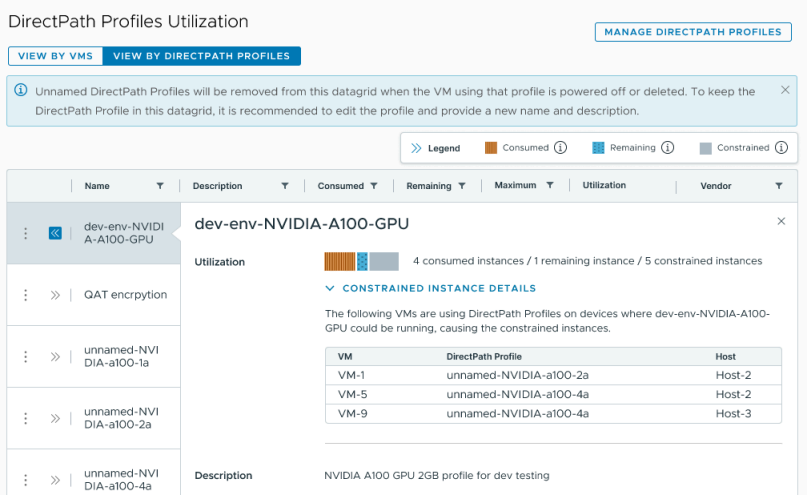
Parameter	Description
Consumed	Number of instances currently allocated to powered-on VMs within a target host or a cluster. This parameter does not account for devices allocated to VMs that are powering on or being migrated with vSphere vMotion into the target host or cluster.
Remaining	Number of instances that can still be allocated after accounting for the consumed capacity, assuming no additional VMs are created, and no existing VMs will compete for the device's resources.
Maximum	The total capacity that is available in a target host or a cluster if no VMs are running on the host or cluster. Note that the sum of **Consumed** and **Remaining** may not always equal **Maximum** due to placement constraints. Certain profiles, such as vGPU profiles, have specific placement rules defined by NVIDIA, which limit how they can be assigned to physical devices. As a result, not all capacity may be available for allocation. For more information, see the NVIDIA documentation.

Note:

If the DirectPath Profiles list contains unnamed DirectPath Profiles, they will be removed when you power on or delete the virtual machine that uses that profile.

If you encounter issues, for example, with VMs failing to power on due to DirectPath device configuration problems, you can review the DirectPath Profile utilization and available capacity. This information will help you identify any constraints or conflicts that might prevent a virtual machine from powering on.

Figure 103:



- To see which VMs use a specific DirectPath Profile, click the **View By VMs** tab.
  The list shows all VMs in the cluster, the hosts they reside on, and the DirectPath Profiles they consume.
- To create or edit a DirectPath Profile, click the **Manage DirectPath Profiles** button.

Manage DirectPath Profiles

Managing DirectPath Profiles allows you to capture specific configurations across different DirectPath passthrough devices without requiring manual configuration each time.

Create a DirectPath Profile

You can create, edit, and remove DirectPath Profiles and monitor their utilization across clusters, hosts, and VMs.

To create a DirectPath profile, perform the following steps.

1. In the *vSphere Client*, select **Menu** > **Policies and Profiles** > **DirectPath Profiles**.
2. From the DirectPath Profiles pane, click **Create**.
3. From the **vCenter and DirectPath Device** window, select the vCenter instance.
4. From the **Select DirectPath Device** list, select the DirectPath device, such as Enhanced DirectPath I/O or NVIDIA vGPu, and click **Next**.
5. In the **Name and Configuration** window, enter a unique profile name and description.
6. Configure the device-specific parameters for the selected device.
   a) If you select an NVDIA vGPU, the parameters will adjust according to your selected GPU profile.
   b) If you select an Enhanced DirectPath I/O device, set up the configuration parameters for the device depending on how it needs to be used.
7. Click **Finish**.

The DirectPath Profile you created appears in the list.

To monitor the detailed information about the profile configuration, click the angle icon » next to the DirectPath Profile name you want to view. You can review information about the profile name, description, access type, and the configurable parameters associated with the DirectPath passthrough device.

Edit and Delete a DirectPath Profile

You can modify the name and description of an existing DirectPath Profile.

To edit a DirectPath Profile, perform the following steps.

1. In the *vSphere Client*, select **Menu** > **Policies and Profiles** > **DirectPath Profiles**.
2. From the **DirectPath Profiles** window, select a profile and click **Edit**.
3. Update the name and the description, and click **OK**.

To delete a DirectPath Profile that is no longer needed, select the profile you want to remove from the list, click **Delete**, and confirm deletion to remove the DirectPath Profile from the environment.

If a powered-on virtual machine uses a device associated with this DirectPath profile, another DirectPath profile for the device, prefixed with `unnamed-`, will be automatically created in each cluster or host where the device is in use.

Manage Storage I/O Resources with vSphere

With VCF 9.0, vSphere Storage I/O Control Components are deprecated.

Virtual Machine Storage Policies

Virtual machine storage policies are essential to virtual machine provisioning. The policies control which type of storage is provided for the virtual machine, how the virtual machine is placed within the storage, and which data services are offered for the virtual machine.

vSphere includes default storage policies. However, you can define and assign new policies.

You use the VM Storage Policies interface to create a storage policy. When you define the policy, you specify various storage requirements for applications that run on virtual machines. You can also use storage policies to request specific data services, such as caching or replication, for virtual disks.

You apply the storage policy when you create, clone, or migrate the virtual machine. After you apply the storage policy, the Storage Policy Based Management (SPBM) mechanism places the virtual machine in a matching datastore and, in certain storage environments, determines how the virtual machine storage objects are provisioned and allocated within the storage resource to guarantee the required level of service. The SPBM also enables requested data services for the virtual machine. vCenter monitors policy compliance and sends an alert if the virtual machine is in breach of the assigned storage policy.

I/O Filters

I/O filters that are associated with virtual disks gain direct access to the virtual machine I/O path regardless of the underlying storage topology.

VMware offers certain categories of I/O filters. In addition, the I/O filters can be created by third-party vendors. Typically, they are distributed as packages that provide an installer to deploy filter components on vCenter and ESX host clusters.

When I/O filters are deployed on the ESX cluster, vCenter automatically configures and registers an I/O filter storage provider, also called a VASA provider, for each host in the cluster. The storage providers communicate with vCenter and make data services offered by the I/O filter visible in the VM Storage Policies interface. You can reference these data services when defining common rules for a VM policy. After you associate virtual disks with this policy, the I/O filters are enabled on the virtual disks.

For more information, see Filtering Virtual Machine I/O in vSphere.

Storage DRS Integration with Storage Profiles

Storage Policy Based Management (SPBM) allows you to specify the policy for a virtual machine which is enforced by Storage DRS. A datastore cluster can have set of datastores with different capability profiles. If the virtual machines have storage profiles associated with them, Storage DRS can enforce placement based on underlying datastore capabilities.

By default, Storage DRS will not enforce storage policies associated with a virtual machine. Please configure `EnforceStorageProfiles` option according to your requirements. The options are Default (0), Soft (1) or Hard (2).

As part of Storage DRS integration with storage profiles, the Storage DRS cluster level advanced option `EnforceStorageProfiles` is introduced. Advanced option `EnforceStorageProfiles` takes one of these integer values: 0,1 or 2. The default value is 0. When the option is set to 0, it indicates that there is no storage profile or policy

enforcement on the Storage DRS cluster. When the option is set to 1, it indicates that there is a storage profile or policy soft enforcement on the Storage DRS cluster. This is analogous with DRS soft rules. Storage DRS will comply with storage profile or policy in the optimum level. Storage DRS will violate the storage profile compliant if it is required to do so. Storage DRS affinity rules will have higher precedence over storage profiles only when storage profile enforcement is set to 1. When the option is set to 2, it indicates that there is a storage profile or policy hard enforcement on the Storage DRS cluster. This is analogous with DRS hard rules. Storage DRS will not violate the storage profile or policy compliant. Storage profiles will have higher precedence over affinity rules. Storage DRS will generate fault: `could not fix anti-affinity rule violation`.

1. Log in to the vSphere Client as an Administrator.
2. In the vSphere Client, select the Storage DRS cluster.
3. Click the **Configure** tab and navigate to **Services** > **Storage DRS**.
4. Under **vSphere Storage DRS is Turned ON**, click **Edit**.
5. From the **Edit Storage DRS Settings** window, click **Advanced options**.
6. Enter the following values under the Advanced options heading.
   a) In the **Option** field, enter `EnforceStorageProfiles`.
   b) In the **Value** field, enter the value of either 0, 1, or 2.
7. Click **Add**.
8. Click **OK**.

Managing Resource Pools with vSphere

A resource pool is a logical abstraction for flexible management of resources. Resource pools can be grouped into hierarchies and used to hierarchically partition available CPU and memory resources.

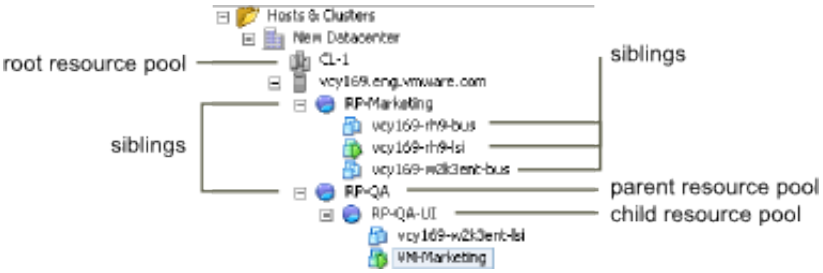
Each standalone host and each DRS cluster has an (invisible) root resource pool that groups the resources of that host or cluster. The root resource pool does not appear because the resources of the host (or cluster) and the root resource pool are always the same.

Users can create child resource pools of the root resource pool or of any user-created child resource pool. Each child resource pool owns some of the parent's resources and can, in turn, have a hierarchy of child resource pools to represent successively smaller units of computational capability.

A resource pool can contain child resource pools, virtual machines, or both. You can create a hierarchy of shared resources. The resource pools at a higher level are called parent resource pools. Resource pools and virtual machines that are at the same level are called siblings. The cluster itself represents the root resource pool. If you do not create child resource pools, only the root resource pools exist.

In the following example, RP-QA is the parent resource pool for RP-QA-UI. RP-Marketing and RP-QA are siblings. The three virtual machines immediately below RP-Marketing are also siblings.

Figure 104: Parents, Children, and Siblings in Resource Pool Hierarchy



For each resource pool, you specify a reservation, limit, shares, and whether the reservation should be expandable. The resource pool resources are then available to child resource pools and virtual machines.

Note:
In this chapter, "Memory" refers to physical RAM.

Why Use Resource Pools?

Resource pools allow you to delegate control over resources of a host (or a cluster), but the benefits are evident when you use resource pools to compartmentalize all resources in a cluster. Create multiple resource pools as direct children of the host or cluster and configure them. You can then delegate control over the resource pools to other individuals or organizations.

Using resource pools can result in the following benefits.

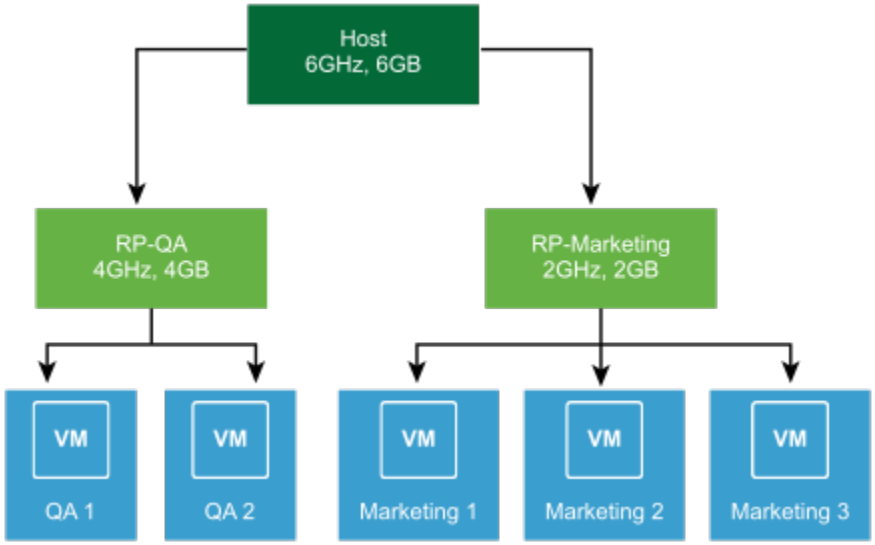
- Flexible hierarchical organization—Add, remove, or reorganize resource pools or change resource allocations as needed.
- Isolation between pools, sharing within pools—Top-level administrators can make a pool of resources available to a department-level administrator. Allocation changes that are internal to one departmental resource pool do not unfairly affect other unrelated resource pools.
- Access control and delegation—When a top-level administrator makes a resource pool available to a department-level administrator, that administrator can then perform all virtual machine creation and management within the boundaries of the resources to which the resource pool is entitled by the current shares, reservation, and limit settings. Delegation is usually done in conjunction with permissions settings.
- Separation of resources from hardware—If you are using clusters enabled for DRS, the resources of all hosts are always assigned to the cluster. That means administrators can perform resource management independently of the actual hosts that contribute to the resources. If you replace three 2 GB hosts with two 3 GB hosts, you do not need to make changes to your resource allocations.

  This separation allows administrators to think more about aggregate computing capacity and less about individual hosts.
- Management of sets of virtual machines running a multitier service— Group virtual machines for a multitier service in a resource pool. You do not need to set resources on each virtual machine. Instead, you can control the aggregate allocation of resources to the set of virtual machines by changing settings on their enclosing resource pool.

For example, assume a host has a number of virtual machines. The marketing department uses three of the virtual machines and the QA department uses two virtual machines. Because the QA department needs larger amounts of CPU and memory, the administrator creates one resource pool for each group. The administrator sets **CPU Shares** to **High** for the QA department pool and to **Normal** for the Marketing department pool so that the QA department users can run automated tests. The second resource pool with fewer CPU and memory resources is sufficient for the lighter load of the marketing staff. Whenever the QA department is not fully using its allocation, the marketing department can use the available resources.

The numbers in the following figure show the effective allocations to the resource pools.

Figure 105: Allocating Resources to Resource Pools



Create a Resource Pool

You can create a child resource pool of any ESX host, resource pool, or DRS cluster.

The vSphere Client is connected to the vCenter system.

> **Note:**
> If a host has been added to a cluster, you cannot create child resource pools of that host. If the cluster is enabled for DRS, you can create child resource pools of the cluster.

When you create a child resource pool, you are prompted for resource pool attribute information. The system uses admission control to make sure you cannot allocate resources that are not available. If you want your shares to scale dynamically when adding or removing VMs, you can select scalable shares.

> **Note:**
> Shares are scaled at the parent level. All descendant resource pools created from a parent with scalable shares have scalable shares by default.

1. In the vSphere Client, select a parent object for the resource pool (a host, another resource pool, or a DRS cluster).
2. Right-click the object and select **New Resource Pool**.
3. Type a name to identify the resource pool.
4. Select the checkbox if you want to enable scalable shares.
5. Specify how to allocate CPU and memory resources.

   The CPU resources for your resource pool are the guaranteed physical resources the host reserves for a resource pool. Normally, you accept the default and let the host handle resource allocation.

Option	Description
Shares	Specify shares for this resource pool with respect to the parent's total resources. Sibling resource pools share resources according to their relative share values bounded by the reservation and limit. • Select **Low**, **Normal**, or **High** to specify share values respectively in a 1:2:4 ratio. • Select **Custom** to give each virtual machine a specific number of shares, which expresses a proportional weight.
Reservation	Specify a guaranteed CPU or memory allocation for this resource pool. Defaults to 0. A nonzero reservation is subtracted from the unreserved resources of the parent (host or resource pool). The resources are considered reserved, regardless of whether virtual machines are associated with the resource pool.
Expandable Reservation	When the check box is selected (default), expandable reservations are considered during admission control. If you power on a virtual machine in this resource pool, and the combined reservations of the virtual machines are larger than the reservation of the resource pool, the resource pool can use resources from its parent or ancestors.
Limit	Specify the upper limit for this resource pool's CPU or memory allocation. You can usually accept the default (**Unlimited**). To specify a limit, deselect the **Unlimited** check box.

6. Click **OK**.

After you create a resource pool, you can add virtual machines to it. A virtual machine's shares are relative to other virtual machines (or resource pools) with the same parent resource pool.

Creating Resource Pools

Assume that you have a host that provides 6GHz of CPU and 3GB of memory that must be shared between your marketing and QA departments. You also want to share the resources unevenly, giving one department

(QA) a higher priority. This can be accomplished by creating a resource pool for each department and using the **Shares** attribute to prioritize the allocation of resources.

The example shows how to create a resource pool with the ESX host as the parent resource.

1. In the **New Resource Pool** dialog box, type a name for the QA department's resource pool (for example, RP-QA).
2. Specify **Shares** of **High** for the CPU and memory resources of RP-QA.
3. Create a second resource pool, RP-Marketing.
   Leave Shares at **Normal** for CPU and memory.
4. Click **OK**.

If there is resource contention, RP-QA receives 4GHz and 2GB of memory, and RP-Marketing 2GHz and 1GB. Otherwise, they can receive more than this allotment. Those resources are then available to the virtual machines in the respective resource pools.

Edit a Resource Pool

After you create the resource pool, you can edit its CPU and memory resource settings.

1. Browse to the resource pool in the vSphere Client.
2. Select **Edit Resource Settings** from the **Actions** drop-down menu.
3. Optional: You can change all attributes of the selected resource pool as described in Create a Resource Pool.
   - Select the checkbox if you want to enable sclalable shares.

     Note:
     Shares are scaled at the parent level. All descendant resource pools created from a parent with scalable shares have scalable shares by default.
   - Under **CPU**, select CPU resource settings.
   - Under **Memory**, select memory resource settings.
4. Click **OK** to save your changes.

Add a Virtual Machine to a Resource Pool

When you create a virtual machine, you can specify a resource pool location as part of the creation process. You can also add an existing virtual machine to a resource pool.

When you move a virtual machine to a new resource pool:

- The virtual machine's reservation and limit do not change.
- If the virtual machine's shares are high, medium, or low, %Shares adjusts to reflect the total number of shares in use in the new resource pool.
- If the virtual machine has custom shares assigned, the share value is maintained.

  Note:
  Because share allocations are relative to a resource pool, you might have to manually change a virtual machine's shares when you move it into a resource pool so that the virtual machine's shares are consistent with the relative values in the new resource pool. A warning appears if a virtual machine would receive a very large (or very small) percentage of total shares.
- Under **Monitor**, the information displayed in the **Resource Reservation** tab about the resource pool's reserved and unreserved CPU and memory resources changes to reflect the reservations associated with the virtual machine (if any).

  Note:
  If a virtual machine has been powered off or suspended, it can be moved but overall available resources (such as reserved and unreserved CPU and memory) for the resource pool are not affected.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and click **Migrate**.
   - You can move the virtual machine to another host.
   - You can move the virtual machine's storage to another datastore.
   - You can move the virtual machine to another host and move its storage to another datastore.
3. Select a resource pool in which to run the virtual machine.
4. Review your selections and click **Finish**.

If a virtual machine is powered on, and the destination resource pool does not have enough CPU or memory to guarantee the virtual machine’s reservation, the move fails because admission control does not allow it. An error dialog box displays available and requested resources, so you can consider whether an adjustment might resolve the issue.

Remove a Virtual Machine from a Resource Pool

You can remove a virtual machine from a resource pool either by moving the virtual machine to another resource pool or deleting it.

When you remove a virtual machine from a resource pool, the total number of shares associated with the resource pool decreases, so that each remaining share represents more resources. For example, assume you have a pool that is entitled to 6GHz, containing three virtual machines with shares set to **Normal**. Assuming the virtual machines are CPU-bound, each gets an equal allocation of 2GHz. If one of the virtual machines is moved to a different resource pool, the two remaining virtual machines each receive an equal allocation of 3GHz.

1. Browse to the resource pool in the vSphere Client.
2. Choose one of the following methods to remove the virtual machine from a resource pool.
   - Right-click the virtual machine and select **Move To...** to move the virtual machine to another resource pool.

     You do not need to power off the virtual machine before you move it.
   - Right-click the virtual machine and select **Delete from Disk**.

     You must power off the virtual machine before you can completely remove it.

Remove a Resource Pool

You can remove a resource pool from the inventory.

1. In the vSphere Client, right-click the resource pool and Select **Delete**.
   A confirmation dialog box appears.
2. Click **Yes** to remove the resource pool.

Resource Pool Admission Control

When you power on a virtual machine in a resource pool, or try to create a child resource pool, the system performs additional admission control to ensure the resource pool’s restrictions are not violated.

Before you power on a virtual machine or create a resource pool, ensure that sufficient resources are available using the **Resource Reservation** tab in the vSphere Client. The **Available Reservation** value for CPU and memory displays resources that are unreserved.

How available CPU and memory resources are computed and whether actions are performed depends on the **Reservation Type**.

Table 305: Reservation Types

Reservation Type	Description
Fixed	The system checks whether the selected resource pool has sufficient unreserved resources. If it does, the action can be performed. If it does not, a message appears and the action cannot be performed.
Expandable (default)	The system considers the resources available in the selected resource pool and its direct parent resource pool. If the parent resource pool also has the **Expandable Reservation** option selected, it can borrow resources from its parent resource pool. Borrowing resources occurs recursively from the ancestors of the current resource pool as long as the **Expandable Reservation** option is selected. Leaving this option selected offers more flexibility, but, at the same time provides less protection. A child resource pool owner might reserve more resources than you anticipate.

The system does not allow you to violate preconfigured **Reservation** or **Limit** settings. Each time you reconfigure a resource pool or power on a virtual machine, the system validates all parameters so all service-level guarantees can still be met.

Expandable Reservations Example 1

This example shows you how a resource pool with expandable reservations works.

Assume an administrator manages pool P, and defines two child resource pools, S1 and S2, for two different users (or groups).

The administrator knows that users want to power on virtual machines with reservations, but does not know how much each user will need to reserve. Making the reservations for S1 and S2 expandable allows the administrator to more flexibly share and inherit the common reservation for pool P.

Without expandable reservations, the administrator needs to explicitly allocate S1 and S2 a specific amount. Such specific allocations can be inflexible, especially in deep resource pool hierarchies and can complicate setting reservations in the resource pool hierarchy.

Expandable reservations cause a loss of strict isolation. S1 can start using all of P's reservation, so that no memory or CPU is directly available to S2.

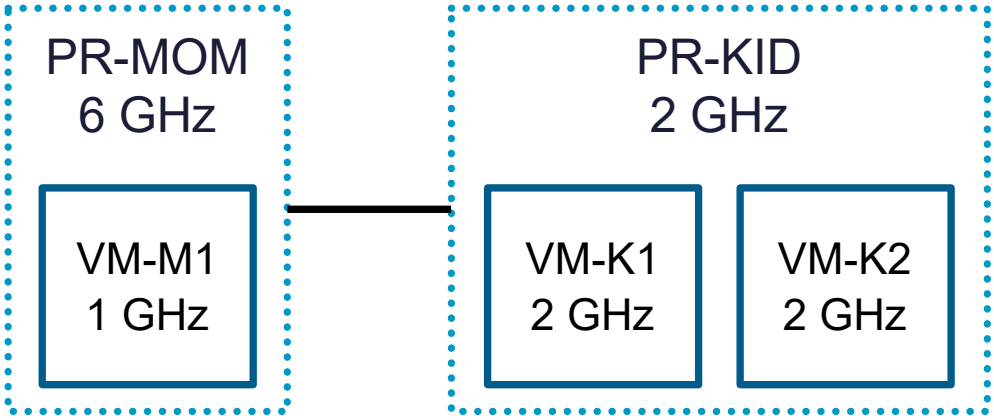
Expandable Reservations Example 2

This example shows how a resource pool with expandable reservations works.

Assume the following scenario, as shown in the figure.

- Parent pool RP-MOM has a reservation of 6GHz and one running virtual machine VM-M1 that reserves 1GHz.
- You create a child resource pool RP-KID with a reservation of 2GHz and with **Expandable Reservation** selected.
- You add two virtual machines, VM-K1 and VM-K2, with reservations of 2GHz each to the child resource pool and try to power them on.
- VM-K1 can reserve the resources directly from RP-KID (which has 2GHz).
- No local resources are available for VM-K2, so it borrows resources from the parent resource pool, RP-MOM. RP-MOM has 6GHz minus 1GHz (reserved by the virtual machine) minus 2GHz (reserved by RP-KID), which leaves 3GHz unreserved. With 3GHz available, you can power on the 2GHz virtual machine.

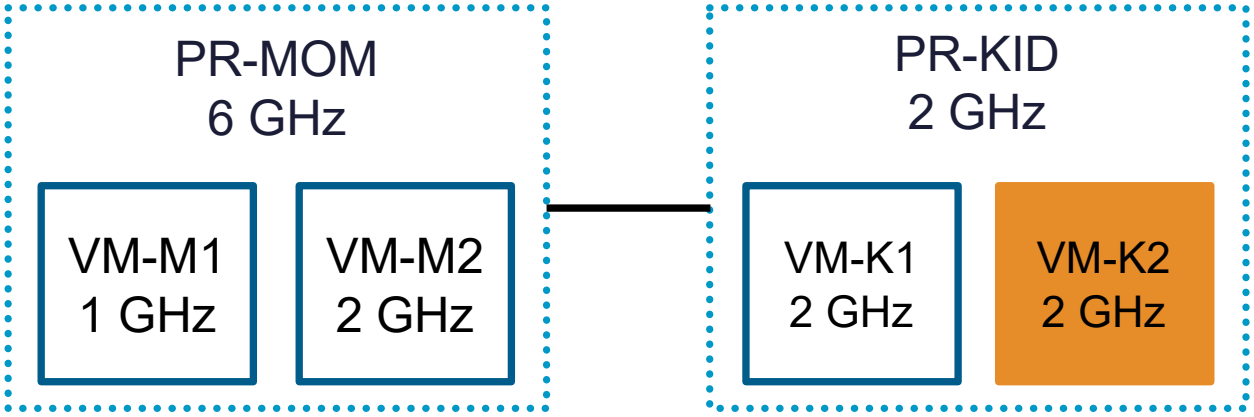
Figure 106: Admission Control with Expandable Resource Pools: Successful Power-On



Now, consider another scenario with VM-M1 and VM-M2.

- Power on two virtual machines in RP-MOM with a total reservation of 3GHz.
- You can still power on VM-K1 in RP-KID because 2GHz are available locally.
- When you try to power on VM-K2, RP-KID has no unreserved CPU capacity so it checks its parent. RP-MOM has only 1GHz of unreserved capacity available (5GHz of RP-MOM are already in use—3GHz reserved by the local virtual machines and 2GHz reserved by RP-KID). As a result, you cannot power on VM-K2, which requires a 2GHz reservation.

Figure 107: Admission Control with Expandable Resource Pools: Power-On Prevented



Create a vSphere DRS Cluster

A cluster is a collection of ESX hosts and associated virtual machines with shared resources and a shared management interface. Before you can obtain the benefits of cluster-level resource management you must create a cluster and activate DRS.

Depending on whether or not Enhanced vMotion Compatibility (EVC) is activated, vSphere DRS behaves differently when you use vSphere Fault Tolerance (vSphere FT) virtual machines in your cluster.

Table 306: vSphere DRS Behavior with vSphere FT Virtual Machines and EVC

EVC	DRS (Load Balancing)	DRS (Initial Placement)
Enabled	Enabled (Primary and Secondary VMs)	Enabled (Primary and Secondary VMs)
Disabled	Disabled (Primary and Secondary VMs)	Disabled (Primary VMs) Fully Automated (Secondary VMs)

Admission Control and Initial Placement

When you attempt to power on a single virtual machine or a group of virtual machines in a DRS-enabled cluster, vCenter performs admission control. It checks that there are enough resources in the cluster to support the virtual machine(s).

If the cluster does not have sufficient resources to power on a single virtual machine, or any of the virtual machines in a group power-on attempt, a message appears. Otherwise, for each virtual machine, DRS generates a recommendation of a host on which to run the virtual machine and takes one of the following actions

- Automatically executes the placement recommendation.
- Displays the placement recommendation, which the user can then choose to accept or override.

  Note:
  No initial placement recommendations are given for virtual machines on standalone hosts or in non-DRS clusters. When powered on, they are placed on the host where they currently reside.
- DRS considers network bandwidth. By calculating host network saturation, DRS is able to make better placement decisions. This can help avoid performance degradation of virtual machines with a more comprehensive understanding of the environment.

Single Virtual Machine Power On

In a DRS cluster, you can power on a single virtual machine and receive initial placement recommendations.

When you power on a single virtual machine, you have two types of initial placement recommendations:

- A single virtual machine is being powered on and no prerequisite steps are needed.
  The user is presented with a list of mutually exclusive initial placement recommendations for the virtual machine. You can select only one.
- A single virtual machine is being powered on, but prerequisite actions are required.
  These actions include powering on a host in standby mode or the migration of other virtual machines from one host to another. In this case, the recommendations provided have multiple lines, showing each of the prerequisite actions. The user can either accept this entire recommendation or cancel powering on the virtual machine.

Group Power-on

You can attempt to power on multiple virtual machines at the same time (group power-on).

Virtual machines selected for a group power-on attempt do not have to be in the same DRS cluster. They can be selected across clusters but must be within the same data center. It is also possible to include virtual machines located in non-DRS clusters or on standalone hosts. These virtual machines are powered on automatically and not included in any initial placement recommendation.

The initial placement recommendations for group power-on attempts are provided on a per-cluster basis. If all the placement-related actions for a group power-on attempt are in automatic mode, the virtual machines are powered on with no initial placement recommendation given. If placement-related actions for any of the virtual machines are in manual mode, the powering on of all the virtual machines (including the virtual machines that are in automatic mode) is manual. These actions are included in an initial placement recommendation.

For each DRS cluster that the virtual machines being powered on belong to, there is a single recommendation, which contains all the prerequisites (or no recommendation). All such cluster-specific recommendations are presented together under the **Power On Recommendations** tab.

When a nonautomatic group power-on attempt is made, and virtual machines not subject to an initial placement recommendation (that is, the virtual machines on standalone hosts or in non-DRS clusters) are included, vCenter attempts to power them on automatically. If these power-ons are successful, they are listed under the **Started Power-Ons** tab. Any virtual machines that fail to power-on are listed under the **Failed Power-Ons** tab.

Group Power-on

The user selects three virtual machines in the same data center for a group power-on attempt. The first two virtual machines (VM1 and VM2) are in the same DRS cluster (Cluster1), while the third virtual machine (VM3) is on a standalone host. VM1 is in automatic mode and VM2 is in manual mode. For this scenario, the user is

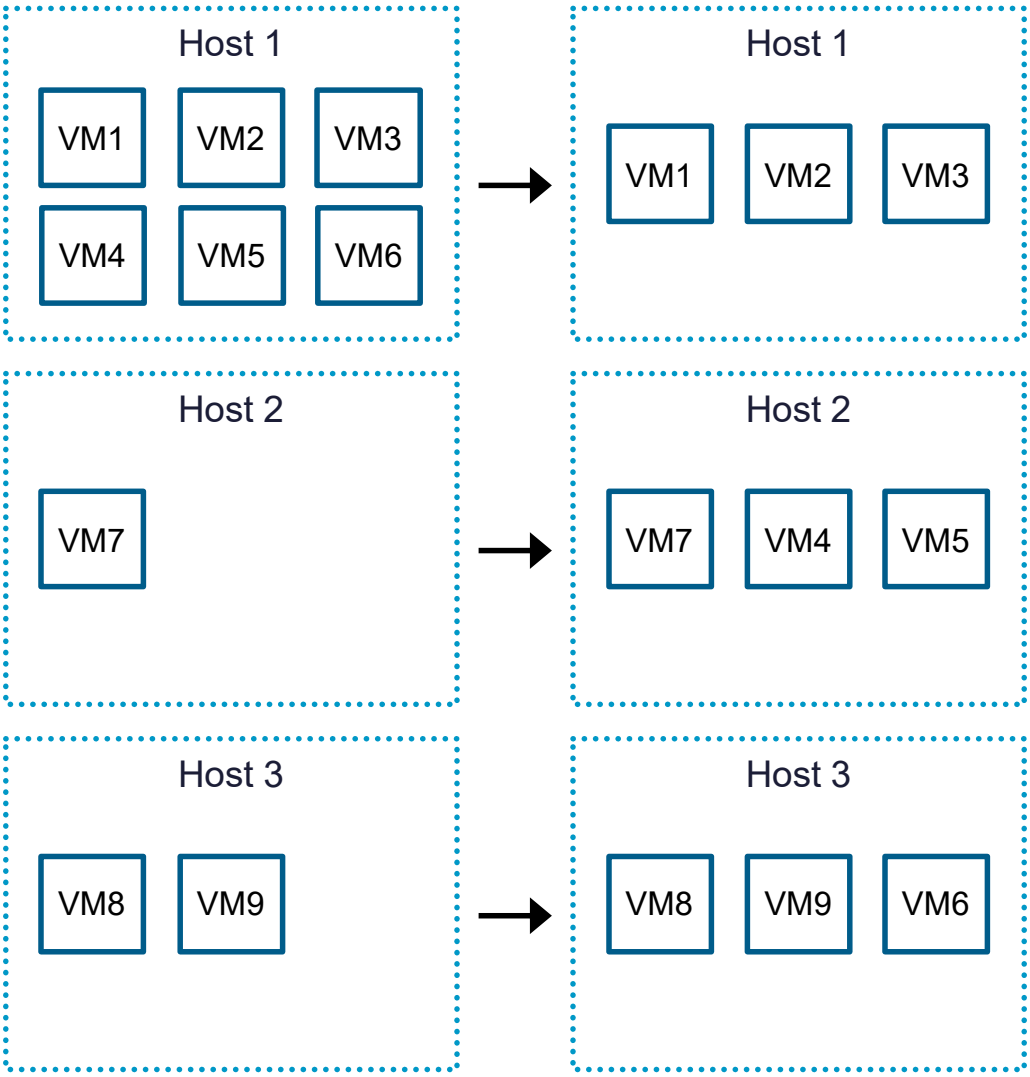
presented with an initial placement recommendation for Cluster1 (under the **Power On Recommendations** tab) which consists of actions for powering on VM1 and VM2. An attempt is made to power on VM3 automatically and, if successful, it is listed under the **Started Power-Ons** tab. If this attempt fails, it is listed under the **Failed Power-Ons** tab.

Virtual Machine Migration

Although DRS performs initial placements so that load is balanced across the cluster, changes in virtual machine load and resource availability can cause the cluster to become unbalanced. To correct such imbalances, DRS generates migration recommendations.

If DRS is enabled on the cluster, load can be distributed more uniformly to reduce the degree of this imbalance. For example, the three hosts on the left side of the following figure are unbalanced. Assume that Host 1, Host 2, and Host 3 have identical capacity, and all virtual machines have the same configuration and load (which includes reservation, if set). However, because Host 1 has six virtual machines, its resources might be overused while ample resources are available on Host 2 and Host 3. DRS migrates (or recommends the migration of) virtual machines from Host 1 to Host 2 and Host 3. On the right side of the diagram, the properly load balanced configuration of the hosts that results appears.

Figure 108: Load Balancing



When a cluster becomes unbalanced, DRS makes recommendations or migrates virtual machines, depending on the default automation level:

- If the cluster or any of the virtual machines involved are manual or partially automated, vCenter does not take automatic actions to balance resources. Instead, the Summary page indicates that migration recommendations are

available and the DRS Recommendations page displays recommendations for changes that make the most efficient use of resources across the cluster.

- If the cluster and virtual machines involved are all fully automated, vCenter migrates running virtual machines between hosts as needed to ensure efficient use of cluster resources.

  Note:
  Even in an automatic migration setup, users can explicitly migrate individual virtual machines, but vCenter might move those virtual machines to other hosts to optimize cluster resources.

By default, automation level is specified for the whole cluster. You can also specify a custom automation level for individual virtual machines.

DRS Migration Threshold

The DRS migration threshold allows you to specify which recommendations are generated and then applied (when the virtual machines involved in the recommendation are in fully automated mode) or shown (if in manual mode). This threshold is a measure of how aggressive DRS is in recommending migrations to improve VM happiness.

You can move the threshold slider to use one of five settings, ranging from Conservative to Aggressive. The higher the agressiveness setting, the more frequently DRS might recommend migrations to improve VM happiness. The Conservative setting generates only priority-one recommendations (mandatory recommendations).

After a recommendation receives a priority level, this level is compared to the migration threshold you set. If the priority level is less than or equal to the threshold setting, the recommendation is either applied (if the relevant virtual machines are in fully automated mode) or displayed to the user for confirmation (if in manual or partially automated mode.)

DRS Score

Each migration recommendation is computed using the VM happiness metric which measures execution efficiency. This metric is displayed as DRS Score in the cluster's Summary tab in the vSphere Client. DRS load balancing recommendations attempt to improve the DRS score of a VM. The Cluster DRS score is a weighted average of the VM DRS Scores of all the powered on VMs in the cluster. The Cluster DRS Score is shown in the gauge component. The color of the filled in section changes depending on the value to match the corresponding bar in the VM DRS Score histogram. The bars in the histogram show the percentage of VMs that have a DRS Score in that range. You can view the list with server-side sorting and filtering by selecting the Monitor tab of the cluster and selecting vSphere DRS, which shows a list of the VMs in the cluster sorted by their DRS score in ascending order.

Migration Recommendations

If you create a cluster with a default manual or partially automated mode, vCenter displays migration recommendations on the DRS Recommendations page.

The system supplies as many recommendations as necessary to enforce rules and balance the resources of the cluster. Each recommendation includes the virtual machine to be moved, current (source) host and destination host, and a reason for the recommendation. The reason can be one of the following:

- Balance average CPU loads or reservations.
- Balance average memory loads or reservations.
- Satisfy resource pool reservations.
- Satisfy an affinity rule.
- Host is entering maintenance mode or standby mode.

  Note:
  If you are using the vSphere Distributed Power Management (DPM) feature, in addition to migration recommendations, DRS provides host power state recommendations.

DRS Cluster Requirements

Hosts that are added to a DRS cluster must meet certain requirements to use cluster features successfully.

Shared Storage Requirements

A DRS cluster has certain shared storage requirements.

Ensure that the managed hosts use shared storage. Shared storage is typically on a SAN, but can also be implemented using NAS shared storage.

See the *vSphere Storage* documentation for information about other shared storage.

Shared VMFS Volume Requirements

A DRS cluster has certain shared VMFS volume requirements.

Configure all managed hosts to use shared VMFS volumes.

- Place the disks of all virtual machines on VMFS volumes that are accessible by source and destination hosts.
- Ensure the VMFS volume is sufficiently large to store all virtual disks for your virtual machines.
- Ensure all VMFS volumes on source and destination hosts use volume names, and all virtual machines use those volume names for specifying the virtual disks.

  Note:

  Virtual machine swap files must reside on a datastore accessible to both the source and destination hosts, like `.vmdk` virtual disk files. This requirement does not apply when all source and destination hosts run ESX 7.0 or later and use host-local swap. In that case, vMotion supports swap files on unshared storage. By default, swap files are stored with the virtual machine configuration file, but administrators can override this location by using advanced virtual machine configuration options.

Processor Compatibility Requirements

A DRS cluster has certain processor compatibility requirements.

To avoid limiting the capabilities of DRS, you should maximize the processor compatibility of source and destination hosts in the cluster.

vMotion transfers the running architectural state of a virtual machine between underlying ESX hosts. vMotion compatibility means that the processors of the destination host must be able to resume execution using the equivalent instructions where the processors of the source host were suspended. Processor clock speeds and cache sizes might vary, but processors must come from the same vendor class (Intel versus AMD) and the same processor family to be compatible for migration with vMotion.

Processor families are defined by the processor vendors. You can distinguish different processor versions within the same family by comparing the processors' model, stepping level, and extended features.

Sometimes, processor vendors have introduced significant architectural changes within the same processor family (such as 64-bit extensions and SSE3). VMware identifies these exceptions if it cannot guarantee successful migration with vMotion.

vCenter provides features that help ensure that virtual machines migrated with vMotion meet processor compatibility requirements. These features include:

- Enhanced vMotion Compatibility (EVC) – You can use EVC to help ensure vMotion compatibility for the hosts in a cluster. EVC ensures that all hosts in a cluster present the same CPU feature set to virtual machines, even if the actual CPUs on the hosts differ. This prevents migrations with vMotion from failing due to incompatible CPUs.
  Configure EVC from the Cluster Settings dialog box. The hosts in a cluster must meet certain requirements for the cluster to use EVC. For information about EVC and EVC requirements, see the *vCenter and Host Management* documentation.
- CPU compatibility masks – vCenter compares the CPU features available to a virtual machine with the CPU features of the destination host to determine whether to allow or disallow migrations with vMotion. By applying CPU compatibility masks to individual virtual machines, you can hide certain CPU features from the virtual machine and potentially prevent migrations with vMotion from failing due to incompatible CPUs.

vMotion Requirements for DRS Clusters

A DRS cluster has certain vMotion requirements.

To enable the use of DRS migration recommendations, the hosts in your cluster must be part of a vMotion network. If the hosts are not in the vMotion network, DRS can still make initial placement recommendations.

To be configured for vMotion, each host in the cluster must meet the following requirements:

- vMotion does not support raw disks or migration of applications clustered using Microsoft Cluster Service (MSCS).
- vMotion requires a private Gigabit Ethernet migration network between all of the vMotion enabled managed hosts. When vMotion is enabled on a managed host, configure a unique network identity object for the managed host and connect it to the private migration network.

Configuring DRS with Virtual Flash

DRS can manage virtual machines that have virtual flash reservations.

Virtual flash capacity appears as a statistic that is regularly reported from the host to the vSphere Client. Each time DRS runs, it uses the most recent capacity value reported.

You can configure one virtual flash resource per host. This means that during virtual machine power-on time, DRS does not need to select between different virtual flash resources on a given host.

DRS selects a host that has sufficient available virtual flash capacity to start the virtual machine. If DRS cannot satisfy the virtual flash reservation of a virtual machine, it cannot be powered-on. DRS treats a powered-on virtual machine with a virtual flash reservation as having a soft affinity with its current host. DRS will not recommend such a virtual machine for vMotion except for mandatory reasons, such as putting a host in maintenance mode, or to reduce the load on an over utilized host.

Create a Cluster

A cluster is a group of hosts. When a host is added to a cluster, the host's resources become part of the cluster's resources. The cluster manages the resources of all hosts within it.

- Verify that you have sufficient permissions to create a cluster object.
- Verify that a data center exists in the inventory.
- If you want to use vSAN, it must be enabled before you configure vSphere HA.

Clusters enable the vSphere High Availability (HA) and vSphere Distributed Resource Scheduler (DRS) solutions.

1. Browse to a data center in the vSphere Client.
2. Right-click the data center and select **New Cluster**.
3. Enter a name for the cluster.
4. Select DRS and vSphere HA cluster features.

Option	Description
To use DRS with this cluster	1. Select the DRS **Turn ON** check box. 2. Select an automation level and a migration threshold.
To use HA with this cluster	1. Select the vSphere HA **Turn ON** check box. 2. Select whether to enable host monitoring and admission control. 3. If admission control is enabled, specify a policy. 4. Select a VM Monitoring option. 5. Specify the virtual machine monitoring sensitivity.

5. Select an Enhanced vMotion Compatibility (EVC) setting.

   EVC ensures that all hosts in a cluster present the same CPU feature set to virtual machines, even if the actual CPUs on the hosts differ. This prevents migrations with vMotion from failing due to incompatible CPUs.

6. Click **OK**.

The cluster is added to the inventory.

Add hosts and resource pools to the cluster.

> **Note:**
> Under the **Cluster Summary** page, you can see **Cluster Services** which displays vSphere Cluster Services health status.

Edit Cluster Settings

When you add a host to a DRS cluster, the host's resources become part of the cluster's resources. In addition to this aggregation of resources, with a DRS cluster, you can support cluster-wide resource pools and enforce cluster-level resource allocation policies.

You can create a cluster without a special license, but you must have a license to enable a cluster for vSphere DRS or vSphere HA. vSphere

The following cluster-level resource management capabilities are also available.

Load Balancing	The distribution and usage of CPU and memory resources for all hosts and virtual machines in the cluster are continuously monitored. DRS compares these metrics to an ideal resource usage given the attributes of the cluster's resource pools and virtual machines, the current demand, and the imbalance target. DRS then provides recommendations or performs virtual machine migrations accordingly. See Virtual Machine Migration. When you power on a virtual machine in the cluster, DRS attempts to maintain proper load balancing by either placing the virtual machine on an appropriate host or making a recommendation. See Admission Control and Initial Placement.
Power management	When the vSphere Distributed Power Management (DPM) feature is enabled, DRS compares cluster and host-level capacity to the demands of the cluster's virtual machines, including recent historical demand. DRS then recommends you place hosts in standby, or places hosts in standby power mode when sufficient excess capacity is found. DRS powers-on hosts if capacity is needed. Depending on the resulting host power state recommendations, virtual machines might need to be migrated to and from the hosts as well. See Managing Power Resources.
Affinity Rules	You can control the placement of virtual machines on hosts within a cluster, by assigning affinity rules. See Using Affinity Rules with vSphere DRS.

1. Browse to a cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Under **vSphere DRS** click **Edit**.
4. Under **DRS Automation**, select a default automation level for DRS.

Automation Level	Action
Manual	• Initial placement: Recommended host is displayed. • Migration: Recommendation is displayed.
Partially Automated	• Initial placement: Automatic. • Migration: Recommendation is displayed.
Fully Automated	• Initial placement: Automatic. • Migration: Recommendation is run automatically.

5. Set the **Migration Threshold** for DRS.
6. Select the **Predictive DRS** check box. In addition to real-time metrics, DRS responds to forecasted metrics provided by vRealize Operations server. You must also configure **Predictive DRS** in a version of vRealize Operations that supports this feature.
7. Select **Virtual Machine Automation** check box to enable individual virtual machine automation levels.
   Override for individual virtual machines can be set from the VM Overrides page.

8. Under **Additional Options**, select a check box to enforce one of the default policies.

Option	Description
VM Distribution	For availability, distribute a more even number of virtual machines across hosts. This is secondary to DRS load balancing.
Memory Metric for Load Balancing	Load balance based on consumed memory of virtual machines rather than active memory. This setting is only recommended for clusters where host memory is not over-committed. **Note:** This setting is no longer supported and will not be displayed in vCenter 7.0.
CPU Over-Commitment	Control CPU over-commitment in the cluster.
Scalable Shares	Enable scalable shares for the resource pools on this cluster.

9. Under **Power Management**, select Automation Level.
10. If DPM is enabled, set the **DPM Threshold**.
11. Click **OK**.

Note:
Under the **Cluster Summary** page, you can see **Cluster Services** which displays vSphere Cluster Services health status.

You can view memory utilization for DRS in the vSphere Client. To find out more, see: Viewing Distributed Resource Scheduler Memory Utilization

Set a Custom Automation Level for a Virtual Machine

After you create a DRS cluster, you can customize the automation level for individual virtual machines to override the cluster's default automation level.

For example, you can select **Manual** for specific virtual machines in a cluster with full automation, or **Partially Automated** for specific virtual machines in a manual cluster.

If a virtual machine is set to **Disabled**, vCenter does not migrate that virtual machine or provide migration recommendations for it.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Under Services, select **vSphere DRS** and click **Edit**. Expand DRS Automation.
4. Select the **Enable individual virtual machine automation levels** check box.
5. To temporarily deactivate any individual virtual machine overrides, deselect the **Enable individual virtual machine automation levels** check box.

   Virtual machine settings are restored when the check box is selected again.
6. To temporarily suspend all vMotion activity in a cluster, put the cluster in manual mode and deselect the **Enable individual virtual machine automation levels** check box.
7. Select one or more virtual machines.
8. Click the **Automation Level** column and select an automation level from the drop-down menu.

Option	Description
Manual	Placement and migration recommendations are displayed, but do not run until you manually apply the recommendation.
Fully Automated	Placement and migration recommendations run automatically.
Partially Automated	Initial placement is performed automatically. Migration recommendations are displayed, but do not run.

Option	Description
Disabled	vCenter does not migrate the virtual machine or provide migration recommendations for it.

9. Click **OK**.

Note:

Other VMware products or features, such as vSphere vApp and vSphere Fault Tolerance, might override the automation levels of virtual machines in a DRS cluster. Refer to the product-specific documentation for details.

Deactivate vSphere DRS

You can turn off DRS for a cluster.

When DRS is deactivated:

- DRS affinity rules are not removed but are not applied until DRS is reactivated.
- Host and VM groups are not removed but are not applied until DRS is reactivated.
- Resource pools are permanently removed from the cluster. To avoid losing the resource pools, save a snapshot of the resource pool tree on your local machine. You can use the snapshot to restore the resource pool when you activate DRS.

Note:
Do not deactivate DRS if Workload Management is activated, this makes WCP services unrecoverable.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Under **vSphere DRS**, click **Edit**.
4. Deselect the **Turn On vSphere DRS** check box.
5. Click **OK** to turn off DRS.
6. Optional: Choose an option to save the resource pool.
   - Click **Yes** to save a resource pool tree snapshot on a local machine.
   - Click **No** to turn off DRS without saving a resource pool tree snapshot.

DRS is powered off.

Restore a Resource Pool Tree

You can restore a previously saved resource pool tree snapshot.

- vSphere DRS must be turned ON.
- You can restore a snapshot only on the same cluster that it was taken.
- No other resource pools are present in the cluster.
- Backup and restore must always be performed on the same version of vCenter and ESX.

1. Browse to the cluster in the vSphere Client.
2. Right-click on the cluster and select **Restore Resource Pool Tree**.
3. Click **Browse**, and locate the snapshot file on your local machine.
4. Click **Open**.
5. Click **OK** to restore the resource pool tree.

DRS Awareness of vSAN Stretched Cluster

DRS Awareness of vSAN Stretched Cluster is available on stretched clusters with DRS enabled. A vSAN stretched cluster has read locality, where the VM reads data from a local site. Fetching reads from a remote site can affect VM performance. With DRS Awareness of vSAN Stretched Cluster, DRS is now fully aware of VM read locality and will

place the VM on a site that can fully satisfy the read locality. This is automatic, there are no configurable options. DRS Awareness of vSAN Stretched Cluster works with existing affinity rules. It also works with VMware Cloud on AWS.

vSAN Stretched Cluster with vSphere HA and vSphere DRS provide resiliency by having two copies of data spread across two fault domains and a witness node in a third fault domain in case of failures. The two active fault domains provide replication of data so that both fault domains have a current copy of the data.

vSAN Stretched Cluster provides an automated method of moving workloads within the two fault domains. In case of full site failures, VMs are restarted on the secondary site by vSphere HA. This ensures that there is no downtime for critical production workloads. Once the primary site is back online, DRS immediately rebalances the VMs back to the primary site with soft affinity hosts. This process causes the VM to read and write from the secondary site while the VM data components are still rebuilding and might reduce VM performance.

In releases prior to vSphere 7.0 Update 2, we recommend that you change DRS from fully automated to partially automated mode, to avoid VMs migrating while resynchronization is in progress to the primary site. Set DRS back to fully automated only after the resynchronization is complete.

DRS Awareness of vSAN Stretched Cluster introduces a fully automated read locality solution for recovering from failures on a vSAN stretched cluster. The read locality information indicates the hosts the VM has full access to, and DRS uses this information when placing a VM on a host on vSAN Stretched Clusters. DRS prevents VMs from failing back to the primary site when vSAN resynchronization is still in progress during the site recovery phase. DRS automatically migrates a VM back to the primary affined site when its data components have achieved full read locality. This allows you to operate DRS in fully automatic mode in case of full site failures.

In case of partial site failures, if a VM loses read locality due to loss of data components greater than or equal to its Failures to Tolerate vSphere DRS will identify the VMs that consume a very high read bandwidth and try to rebalance them to the secondary site. This ensures that VMs with read-heavy workloads do not decrease during partial site failures. Once the primary site is back online and the data components have completed resynchronization, the VM is moved back to the site it is affined to.

DRS Placement of vGPUs

DRS distributes vGPU VMs across a cluster's hosts.

DRS will distribute vGPU VMs in a breadth-first manner across a cluster's hosts. Fractional vGPU profile allocation for a VM may be subject to homogeneous profile mutual exclusion rules.

- Manually migrate vGPU VMs to a desired host to open up unused Physical GPU capacity.
- Use the same vGPU profile configuration in all vGPU VMs in a Cluster.
- Enable Host "GPU Consolidation". For more information, see Configuring Host Graphics.
- If DRS Automation is active, consider putting the Cluster or VM into **Partially Automated** mode. For more information, see Edit Cluster Settings.

DRS Overhead Memory Management for VMs

In vSphere 8.0 Update 3 DRS has enhanced overhead memory management for VMs that are being reconfigured.

In VMware vSphere, overhead memory refers to the amount of memory that is used by ESX to manage a virtual machine (VM). This memory is necessary for ESX to perform its functions and is separate from the guest memory allocated to the VMs. The amount of overhead memory depends on several factors, including the number of virtual CPUs (vCPUs), the amount of memory allocated to the VM, and the VM's configuration and hardware version. More vCPUs and larger memory allocations lead to higher overhead memory consumption. In vSphere, DRS works with ESX memory management to ensure the VM has optimal overhead memory usage. DRS manages the overhead memory by setting up the VM overhead memory limit, and the ESX is allowed to consume the overhead memory within the limit.

Reconfiguring a VM in VMware vSphere can directly impact the overhead memory that ESX requires to manage a VM. When you change a VM's configuration, such as modifying the number of vCPUs, the amount of assigned RAM, or adding virtual hardware like network adapters or disk controllers, the overhead memory requirements can change. For example, reconfiguring a VM reservation from 250 GB to 0 GB requires about 25 MB of overhead memory usage. ESX allocates 25 MB of additional overhead memory to manage the page table mapping between virtual and physical pages. vSphere monitors and manages these changes. However, previous vSphere releases were not customized to accommodate

these overhead memory increases. If the new overhead memory increase exceeds the overhead limit, it can cause a reconfiguration failure.

In vSphere 8.0 Update 3, DRS proactively updates the VM's overhead memory limit before any reconfiguration. DRS checks various factors, including the VM's resource specifications, IO filters, and other elements impacting overhead memory. DRS ensures that the new overhead limit accommodates the expected increase in overhead memory due to the updated VM specifications after reconfiguration, which optimizes VM performance and stability.

DRS enhanced overhead memory management can help to prevent reconfiguration failures by intelligently managing overhead memory limits before a VM undergoes reconfiguration, which dramatically reduces the risk of reconfiguration failures. This proactive approach ensures a more reliable experience. By optimizing VM performance and stability, virtual environments can run efficiently and without interruption, particularly during critical reconfiguration processes. This enhancement seamlessly integrates into your existing vSphere environment, while increasing performance and reliability.

Use DRS Clusters to Manage Resources with vSphere

After you create a DRS cluster, you can customize it and use it to manage resources.

To customize your DRS cluster and the resources it contains you can configure affinity rules and you can add and remove hosts and virtual machines. When a cluster's settings and resources have been defined, you should ensure that it is and remains a valid cluster. You can also use a valid DRS cluster to manage power resources and interoperate with vSphere HA.

> **Note:**
> In this chapter, "Memory" can refer to physical RAM or persistent memory.

Adding Hosts to a Cluster

The procedure for adding hosts to a cluster is different for hosts managed by the same vCenter instance (managed hosts) than for hosts not managed by that server.

After a host has been added, the virtual machines deployed to the host become part of the cluster, and DRS can recommend migration of some virtual machines to other hosts in the cluster.

Add a Managed Host to a Cluster

When you add a standalone host already being managed by vCenter to a DRS cluster, the host's resources become associated with the cluster.

You can decide whether you want to associate existing virtual machines and resource pools with the cluster's root resource pool or graft the resource pool hierarchy.

> **Note:**
> If a host has no child resource pools or virtual machines, the host's resources are added to the cluster but no resource pool hierarchy with a top-level resource pool is created.

1. Browse to the host in the vSphere Client.
2. Right-click the host and select **Move To...**.
3. Select a cluster.
4. Click **OK** to apply the changes.
5. Select what to do with the host's virtual machines and resource pools.
   - **Put this host's virtual machines in the cluster's root resource pool**

     vCenter removes all existing resource pools of the host and the virtual machines in the host's hierarchy are all attached to the root. Because share allocations are relative to a resource pool, you might have to manually change a virtual machine's shares after selecting this option, which destroys the resource pool hierarchy.
   - **Create a resource pool for this host's virtual machines and resource pools**

vCenter creates a top-level resource pool that becomes a direct child of the cluster and adds all children of the host to that new resource pool. You can supply a name for that new top-level resource pool. The default is **Grafted from <host_name>**.

The host is added to the cluster.

Add an Unmanaged Host to a Cluster

You can add an unmanaged host to a cluster. Such a host is not currently managed by the same vCenter system as the cluster and it is not visible in the vSphere Client.

1. Browse to the cluster in the vSphere Client.
2. Right-click the cluster and select **Add Host**.
3. Enter the host name, user name, and password, and click **Next**.
4. View the summary information and click **Next**.
5. Assign an existing or a new license key and click **Next**.
6. Optional: You can enable lockdown mode to prevent remote users from logging directly into the host.

   If you do not enable lockdown mode, you can configure this option later by editing the Security Profile in the host settings.
7. Select what to do with the host's virtual machines and resource pools.
   - **Put this host's virtual machines in the cluster's root resource pool**

     vCenter removes all existing resource pools of the host and the virtual machines in the host's hierarchy are all attached to the root. Because share allocations are relative to a resource pool, you might have to manually change a virtual machine's shares after selecting this option, which destroys the resource pool hierarchy.
   - **Create a resource pool for this host's virtual machines and resource pools**

     vCenter creates a top-level resource pool that becomes a direct child of the cluster and adds all children of the host to that new resource pool. You can supply a name for that new top-level resource pool. The default is **Grafted from <host_name>**.
8. Review settings and click **Finish**.

The host is added to the cluster.

Adding Virtual Machines to a Cluster

You can add a virtual machine to a cluster in various ways.

- When you add a host to a cluster, all virtual machines on that host are added to the cluster.
- When a virtual machine is created, the **New Virtual Machine** wizard prompts you for the location to place the virtual machine. You can select a standalone host or a cluster and you can select any resource pool inside the host or cluster.
- You can migrate a virtual machine from a standalone host to a cluster or from a cluster to another cluster using the **Migrate Virtual Machine** wizard. To start this wizard, right-click the virtual machine name and select **Migrate**.

Move a Virtual Machine to a Cluster

You can move a virtual machine to a cluster.

1. Find the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click on the virtual machine and select **Move To...**.
3. Select a cluster.
4. Click **OK**.

Removing Virtual Machines from a Cluster

You can remove virtual machines from a cluster.

You can remove a virtual machine from a cluster in two ways.

- When you remove a host from a cluster, all of the powered-off virtual machines that you do not migrate to other hosts are removed as well. You can remove a host only if it is in maintenance mode or disconnected. If you remove a host from a DRS cluster, the cluster can become yellow because it is overcommitted.
- You can migrate a virtual machine from a cluster to a standalone host or from a cluster to another cluster using the **Migrate** wizard. To start this wizard, right-click the virtual machine name and select **Migrate**.

Move a Virtual Machine Out of a Cluster

You can move a virtual machine out of a cluster.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and select **Migrate**.
3. Select **Change datastore** and click **Next**.
4. Select a datastore and click **Next**.
5. Click **Finish**.

   If the virtual machine is a member of a DRS cluster rules group, vCenter displays a warning before it allows the migration to proceed. The warning indicates that dependent virtual machines are not migrated automatically. You have to acknowledge the warning before migration can proceed.

Removing a Host from a Cluster

When you remove a host from a DRS cluster, you affect resource pool hierarchies, virtual machines, and you might create invalid clusters. Consider the affected objects before you remove the host.

- Resource Pool Hierarchies – When you remove a host from a cluster, the host retains only the root resource pool, even if you used a DRS cluster and decided to graft the host resource pool when you added the host to the cluster. In that case, the hierarchy remains with the cluster. You can create a host-specific resource pool hierarchy.

  Note:
  Ensure that you remove the host from the cluster by first placing it in maintenance mode. If you instead disconnect the host before removing it from the cluster, the host retains the resource pool that reflects the cluster hierarchy.
- Virtual Machines – A host must be in maintenance mode before you can remove it from the cluster and for a host to enter maintenance mode all powered-on virtual machines must be migrated off that host. When you request that a host enter maintenance mode, you are also asked whether you want to migrate all the powered-off virtual machines on that host to other hosts in the cluster.
- Invalid Clusters – When you remove a host from a cluster, the resources available for the cluster decrease. If the cluster has enough resources to satisfy the reservations of all virtual machines and resource pools in the cluster, the cluster adjusts resource allocation to reflect the reduced amount of resources. If the cluster does not have enough resources to satisfy the reservations of all resource pools, but there are enough resources to satisfy the reservations for all virtual machines, an alarm is issued and the cluster is marked yellow. DRS continues to run.

Place a Host in Maintenance Mode

You place a host in maintenance mode when you must service it, for example, to install more memory. A host enters or leaves maintenance mode only as the result of a user request.

Virtual machines that are running on a host entering maintenance mode must be migrated to another host (either manually or automatically by DRS) or shut down. The host is in a state of **Entering Maintenance Mode** until all running virtual machines are powered down or migrated to different hosts. You cannot power on virtual machines or migrate virtual machines to a host entering maintenance mode.

When no more running virtual machines are on the host, the host's icon changes to include **under maintenance** and the host's Summary panel indicates the new state. While in maintenance mode, the host does not allow you to deploy or power on a virtual machine.

> **Note:**
> DRS does not recommend (or perform, in fully automated mode) any virtual machine migrations off of a host entering maintenance or standby mode if the vSphere HA failover level would be violated after the host enters the requested mode.

1. Browse to the host in the vSphere Client.
2. Right-click the host and select **Maintenance Mode** > **Enter Maintenance Mode**.
   - If the host is part of a partially automated or manual DRS cluster, browse to **Cluster** > **Monitor** > **DRS** > **Recommendations** and click **Apply Recommendations**.
   - If the host is part of an automated DRS cluster, virtual machines are migrated to different hosts when the host enters maintenance mode.
3. If applicable, click **Yes**.

The host is in maintenance mode until you select **Maintenance Mode** > **Exit Maintenance Mode**.

Place a Host in Workload Non-Disruptive Maintenance Mode

The workload non-disruptive maintenance mode ensures that placing a host into maintenance mode does not affect the running virtual machine workloads.

Workload non-disruptive maintenance mode ensures no workload interruptions during the VM evacuation process.

If a forceful evacuation might cause performance degradation, DRS will display the specific reason why the VM cannot be evacuated.

Perform the following steps.

1. Place the host into workload non-disruptive maintenance mode.
   a) Browse to the host in the vSphere Client.
   b) Right-click the host and select **Maintenance Mode** > **Enter Maintenance Mode**.

      The **Enter Maintenance Mode** dialog box displays.
   c) Select the **Non-Disruptive vMotion Evacuation** check box.
   d) Click the **Enter Maintenance Mode** button.
2. (Optional) If DRS anticipates that the forceful evacuation of VMs will result in performance degradation, DRS displays a notification through the DRS fault.
   a) Select a cluster.
   b) Click the **Monitor** tab.
   c) Under **vSphere DRS**, select **Faults**.

      The reason the VMs cannot be evacuated is displayed in the **DRS Faults** pane.

      To ensure that more compute resources are available, power off some VMs on the destination host.

DRS migrates VMs to other hosts in the cluster if those hosts can meet the compute resource demands of the VMs.

Remove a Host from a Cluster

You can remove hosts from a cluster.

1. Browse to the host in the vSphere Client.
2. Right-click the host and select **Maintenance Mode** > **Enter Maintenance Mode**.

   When the host is in maintenance mode, move it to a different inventory location, either the top-level data center or to a different cluster.
3. Right-click the host and select **Move To...**.
4. Select a new location for the host and click **OK**.

When you move the host, its resources are removed from the cluster. If you grafted the host's resource pool hierarchy onto the cluster, that hierarchy remains with the cluster.

After you remove a host from a cluster, you can perform the following tasks.

- Remove the host from vCenter.
- Run the host as a standalone host under vCenter.
- Move the host into another cluster.

Using Standby Mode

When a host machine is placed in standby mode, it is powered off.

Normally, hosts are placed in standby mode by the vSphere DPM feature to optimize power usage. You can also place a host in standby mode manually. However, DRS might undo (or recommend undoing) your change the next time it runs. To force a host to remain off, place it in maintenance mode and power it off.

Configuring VM Best Effort Restart Policy

This policy controls the evacuation behavior of a VM during maintenance mode.

When you enter maintenance mode, you must make sure that a host is ready for maintenance. This is often to upgrade the host. Whether you are performing an upgrade or not, you must evacuate all running VMs to put the host into maintenance mode. To do this you must determine how to deal with powered-on VMs. If you are performing an upgrade, you need to verify if the host can enter maintenance mode. If there are some VMs you cannot vMotion, you can power-off these VMs before you enter maintenance mode.

You can use compute policies to automate situations like this to evacuate VMs by powering-off the VMs when entering maintenance mode. Then you can tag all the desired VMs, and the system will power these VMs off when entering maintenance mode.

`BestEffortRestart` policy

During enter maintenance mode, `BestEffortRestart` policy attempts to shut down the VMs, if that fails the VMs are powered-off. Instead of waiting for the host to exit MM, the policy finds the best host for the VM while the original host is still in maintenance mode. `BestEffortRestart` creates tasks for VMs to be powered-on on the best hosts during enter maintenance mode so that these VMs are in powered-on state as soon as possible. If a VM cannot be powered-on for whatever reason, there is a remediation cycle that runs every 3 minutes. Exiting maintenance mode is no longer prerequisite for this VM to be running.

If you are moving a host out of a cluster and want VMs to remain in the cluster and not go away with the host you can use `BestEffortRestart`. In this case, you want VMs to remain in the cluster when the host is getting moved out of the cluster. So, during enter maintenance mode not only we have to power-off VMs but also find the best host for them excluding the current host and power them on there.

You can benefit from using `BestEffortRestart` for vGPU VMs and other passthrough devices. vGPU VMs have large memory framebuffers which are costly to migrate and may exceed the 100 second default vMotion timeout. These vMotions are likely to time out leaving vGPU VMs in an undesireable state. Instead, you can power-off vGPU VMs during

enter maintenance mode. Ideally pass-through VMs, including vGPU VMs, should be powered-on on a different host during enter maintenance mode so that they don't have to wait until exitMM completes.

DRS must be activated for this policy to work properly. If DRS is deactivated either at the host or VM level, a VM is powered-off during enter maintenance mode.

If `BestEffortRestart` policy is deleted, depending on the timing of when the policy is deleted, there could be multiple outcomes.

- If a policy is removed before the evacuation action for a VM, this VM is treated the same as any other VM. There is no connection to `BestEffortRestart` policy.
- If the policy is deleted after the evacuation action a VM can be powered-off. The VM will not be remediated because this VM is no longer associated with `BestEffortRestart` policy.
- If a tag associated with `BestEffortRestart` policy is deleted before the evacuation action for a VM, this VM is treated the same as any other VM. There is no connection to `BestEffortRestart` policy.

The VMs should be in the same state that they were before maintenance mode. A VM will eventually be powered-on if it was powered-on before maintenance mode.

When the `BestEffortRestart` policy is successful all `BestEffortRestart` policy VMs are powered-off and powered-on on the best hosts.

DRS Cluster Validity

The vSphere Client indicates whether a DRS cluster is valid, overcommitted (yellow), or invalid (red).

DRS clusters become overcommitted or invalid for several reasons.

- A cluster might become overcommitted if a host fails.
- A cluster becomes invalid if vCenter is unavailable and you power on virtual machines using the vSphere Client.
- A cluster becomes invalid if the user reduces the reservation on a parent resource pool while a virtual machine is in the process of failing over.
- If changes are made to hosts or virtual machines using the vSphere Client while vCenter is unavailable, those changes take effect. When vCenter becomes available again, you might find that clusters have turned red or yellow because cluster requirements are no longer met.

When considering cluster validity scenarios, you should understand these terms.

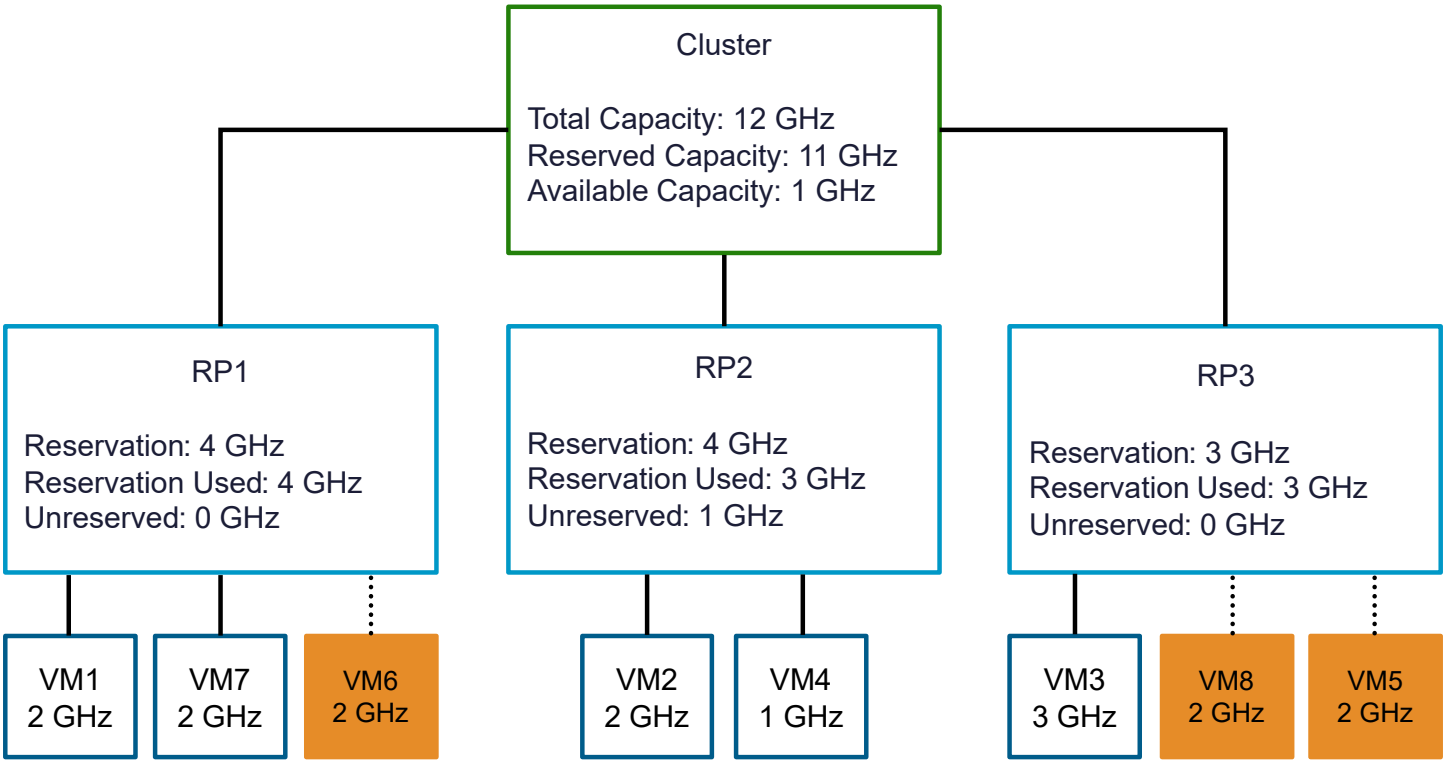
Reservation	A fixed, guaranteed allocation for the resource pool input by the user.
Reservation Used	The sum of the reservation or reservation used (whichever is larger) for each child resource pool, added recursively.
Unreserved	This nonnegative number differs according to resource pool type. • Nonexpandable resource pools: Reservation minus reservation used. • Expandable resource pools: (Reservation minus reservation used) plus any unreserved resources that can be borrowed from its ancestor resource pools.

Valid DRS Clusters

A valid cluster has enough resources to meet all reservations and to support all running virtual machines.

The following figure shows an example of a valid cluster with fixed resource pools and how its CPU and memory resources are computed.

Figure 109: Valid Cluster with Fixed Resource Pools

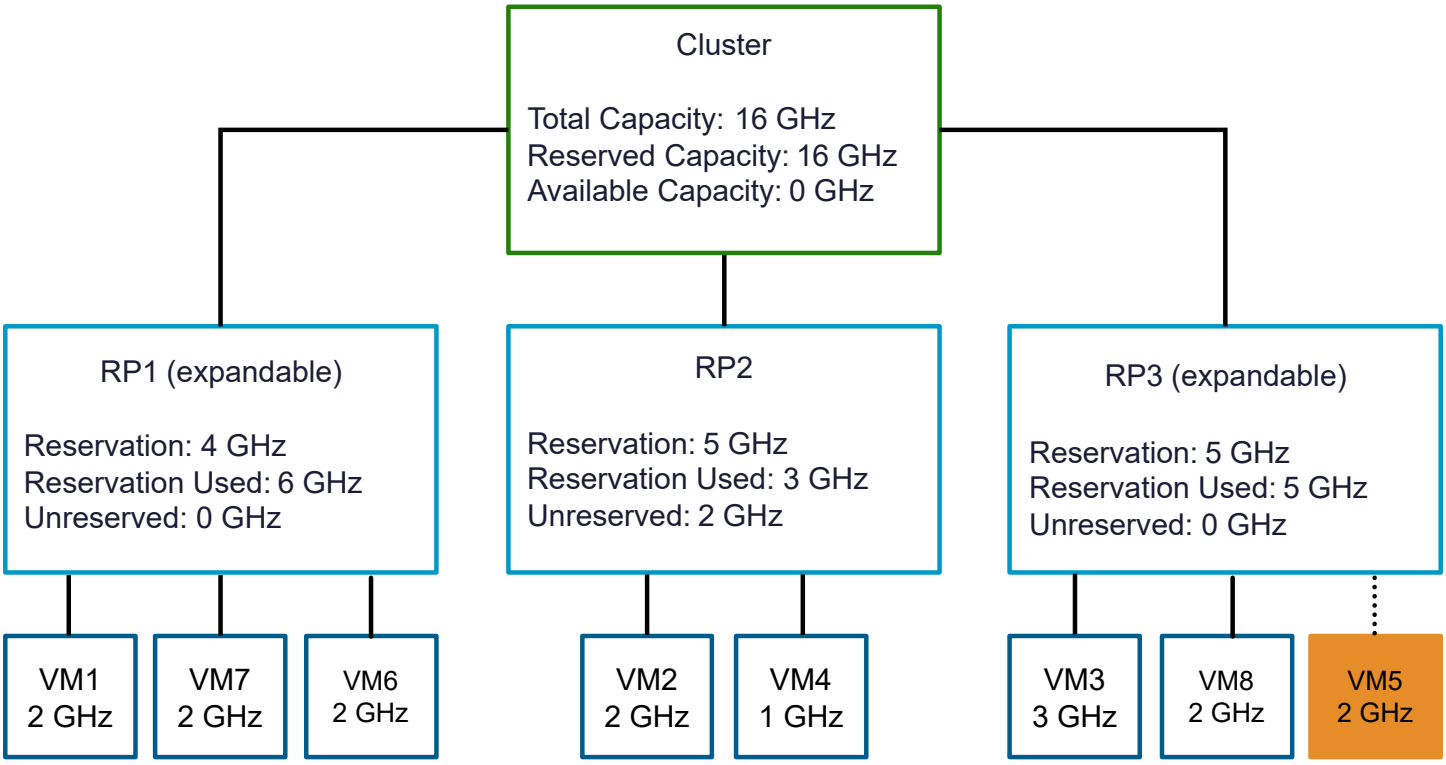


The cluster has the following characteristics:

- A cluster with total resources of 12 GHz.
- Three resource pools, each of type **Fixed** (**Expandable Reservation** is not selected).
- The total reservation of the three resource pools combined is 11 GHz (4+4+3 GHz). The total is shown in the **Reserved Capacity** field for the cluster.
- RP1 was created with a reservation of 4 GHz. Two virtual machines. (VM1 and VM7) of 2 GHz each are powered on (**Reservation Used**: 4 GHz). No resources are left for powering on additional virtual machines. VM6 is shown as not powered on. It consumes none of the reservation.
- RP2 was created with a reservation of 4 GHz. Two virtual machines of 1 GHz and 2 GHz are powered on (**Reservation Used**: 3 GHz). 1 GHz remains unreserved.
- RP3 was created with a reservation of 3 GHz. One virtual machine with 3 GHz is powered on. No resources for powering on additional virtual machines are available.

The following figure shows an example of a valid cluster with some resource pools (RP1 and RP3) using reservation type **Expandable**.

Figure 110: Valid Cluster with Expandable Resource Pools



A valid cluster can be configured as follows:

- A cluster with total resources of 16 GHz.
- RP1 and RP3 are of type **Expandable**, RP2 is of type Fixed.
- The total reservation used of the three resource pools combined is 16 GHz (6 GHz for RP1, 5 GHz for RP2, and 5 GHz for RP3). 16 GHz shows up as the **Reserved Capacity** for the cluster at top level.
- RP1 was created with a reservation of 4 GHz. Three virtual machines of 2 GHz each are powered on. Two of those virtual machines (for example, VM1 and VM7) can use RP1's reservations, the third virtual machine (VM6) can use reservations from the cluster's resource pool. (If the type of this resource pool were **Fixed**, you could not power on the additional virtual machine.)
- RP2 was created with a reservation of 5 GHz. Two virtual machines of 1 GHz and 2 GHz are powered on (**Reservation Used**: 3 GHz). 2 GHz remains unreserved.

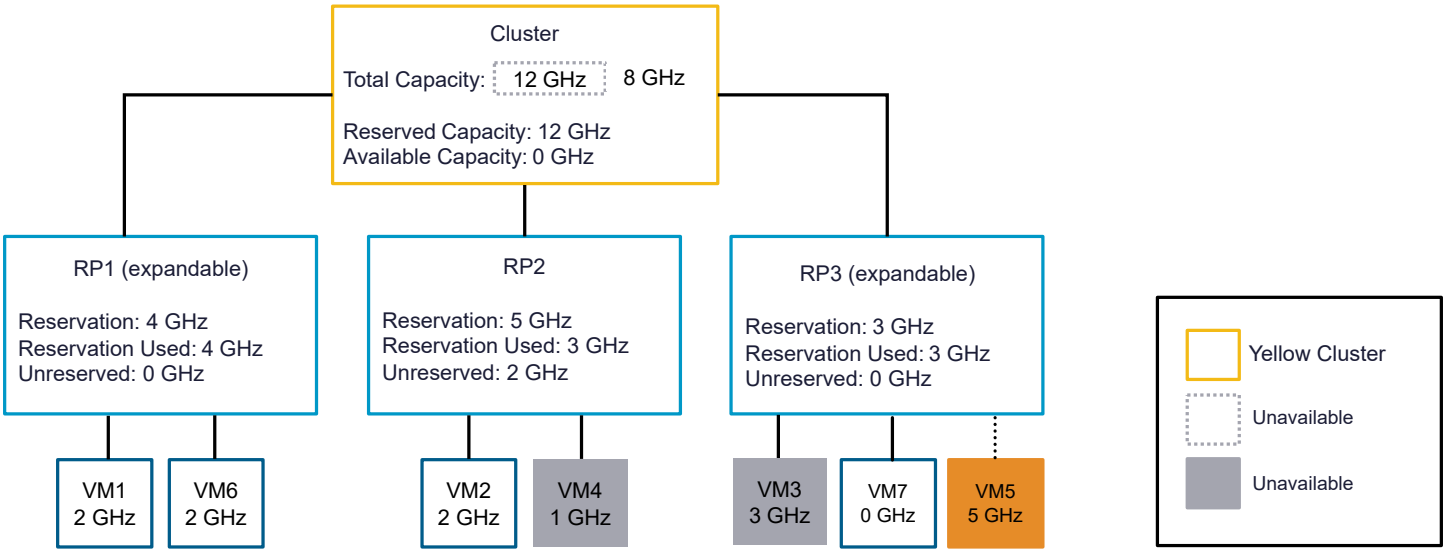
  RP3 was created with a reservation of 5 GHz. Two virtual machines of 3 GHz and 2 GHz are powered on. Even though this resource pool is of type **Expandable**, no additional 2 GHz virtual machine can be powered on because the parent's extra resources are already used by RP1.

Overcommitted DRS Clusters

A cluster becomes overcommitted (yellow) when its resource pool and virtual machine tree is internally consistent. However, the cluster does not have enough capacity to support all resources reserved by child resource pools.

Enough resources remain to support all running virtual machines because, when a host becomes unavailable, all its virtual machines become unavailable. A cluster typically turns yellow when cluster capacity is suddenly reduced, for example, when a host in the cluster becomes unavailable. Leave adequate additional cluster resources to avoid your cluster turning yellow.

Figure 111: Yellow Cluster



In this example:

- A cluster with total resources of 12 GHz coming from three hosts of 4 GHz each.
- Three resource pools reserve a total of 12 GHz.
- The total reservation used by the three resource pools combined is 12 GHz (4+5+3 GHz). This appears as the **Reserved Capacity** in the cluster.
- One of the 4 GHz hosts becomes unavailable, so total resources reduce to 8 GHz.
- At the same time, VM4 (1 GHz) and VM3 (3 GHz), which were running on the host that failed, are no longer running.
- The cluster is now running virtual machines that require a total of 6 GHz. The cluster still has 8 GHz available, which is sufficient to meet virtual machine requirements.
  The resource pool reservations of 12 GHz can no longer be met, so the cluster is marked as yellow.

Invalid DRS Clusters

A cluster enabled for DRS becomes invalid (red) when the tree is no longer internally consistent, that is, resource constraints are not observed.

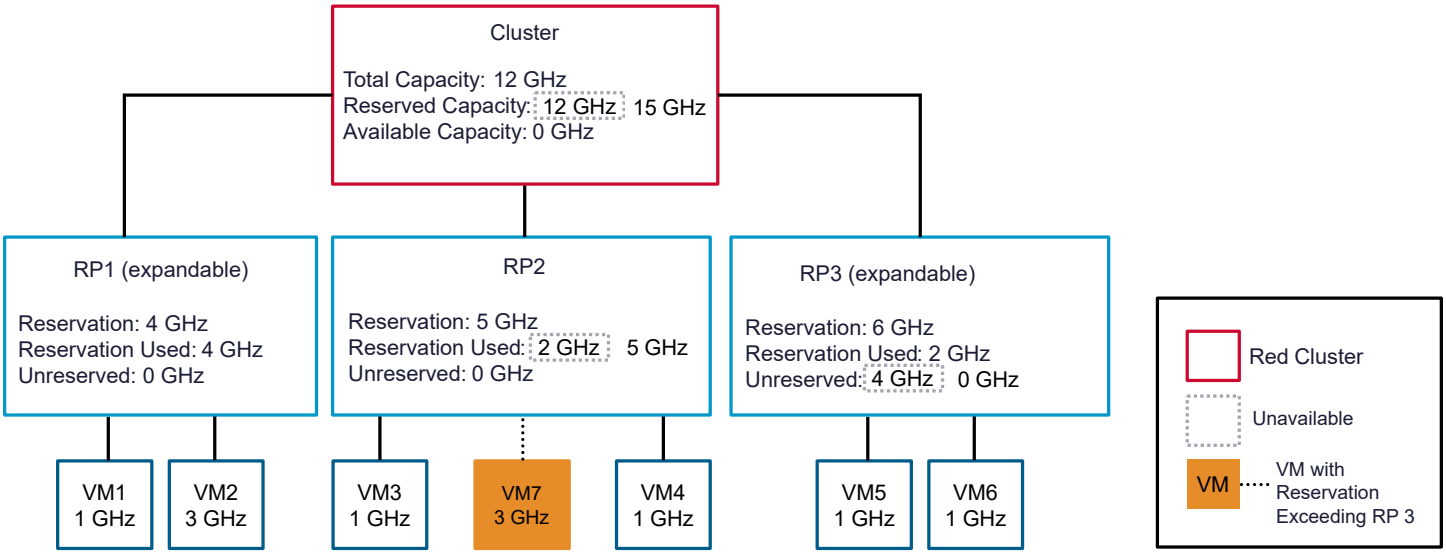
The total amount of resources in the cluster does not affect whether the cluster is red. A cluster can be red, even if enough resources exist at the root level, if there is an inconsistency at a child level.

You can resolve a red DRS cluster problem either by powering off one or more virtual machines, moving virtual machines to parts of the tree that have sufficient resources, or editing the resource pool settings in the red part. Adding resources typically helps only when you are in the yellow state.

A cluster can also turn red if you reconfigure a resource pool while a virtual machine is failing over. A virtual machine that is failing over is disconnected and does not count toward the reservation used by the parent resource pool. You might reduce the reservation of the parent resource pool before the failover completes. After the failover is complete, the virtual machine resources are again charged to the parent resource pool. If the pool's usage becomes larger than the new reservation, the cluster turns red.

If a user is able to start a virtual machine (in an unsupported way) with a reservation of 3 GHz under resource pool 2, the cluster would become red, as shown in the following figure.

Figure 112: Red Cluster



Managing Power Resources

The vSphere Distributed Power Management (DPM) feature allows a DRS cluster to reduce its power consumption by powering hosts on and off based on cluster resource utilization.

vSphere DPM monitors the cumulative demand of all virtual machines in the cluster for memory and CPU resources and compares this to the total available resource capacity of all hosts in the cluster. If sufficient excess capacity is found, vSphere DPM places one or more hosts in standby mode and powers them off after migrating their virtual machines to other hosts. Conversely, when capacity is deemed to be inadequate, DRS brings hosts out of standby mode (powers them on) and uses vMotion to migrate virtual machines to them. When making these calculations, vSphere DPM considers not only current demand, but it also honors any user-specified virtual machine resource reservations.

If you enable **Forecasted Metrics** when you create a DRS cluster, DPM will issue proposals in advance depending on the rolling forecast window you select.

> **Note:**
> ESX hosts cannot automatically be brought out of standby mode unless they are running in a cluster managed by vCenter.

vSphere DPM can use one of three power management protocols to bring a host out of standby mode: Intelligent Platform Management Interface (IPMI), Hewlett-Packard Integrated Lights-Out (iLO), or Wake-On-LAN (WOL). Each protocol requires its own hardware support and configuration. If a host does not support any of these protocols it cannot be put into standby mode by vSphere DPM. If a host supports multiple protocols, they are used in the following order: IPMI, iLO, WOL.

> **Note:**
> Do not disconnect a host in standby mode or move it out of the DRS cluster without first powering it on, otherwise vCenter is not able to power the host back on.

Configure IPMI or iLO Settings for vSphere DPM

IPMI is a hardware-level specification and Hewlett-Packard iLO is an embedded server management technology. Each of them describes and provides an interface for remotely monitoring and controlling computers.

Both IPMI and iLO require a hardware Baseboard Management Controller (BMC) to provide a gateway for accessing hardware control functions, and allow the interface to be accessed from a remote system using serial or LAN connections. The BMC is powered-on even when the host itself is powered-off. If properly enabled, the BMC can respond to remote power-on commands.

If you plan to use IPMI or iLO as a wake protocol, you must configure the BMC. BMC configuration steps vary according to model. See your vendor's documentation for more information. With IPMI, you must also ensure that the BMC LAN channel is configured to be always available and to allow operator-privileged commands. On some IPMI systems, when you enable "IPMI over LAN" you must configure this in the BIOS and specify a particular IPMI account.

vSphere DPM using only IPMI supports MD5- and plaintext-based authentication, but MD2-based authentication is not supported. vCenter uses MD5 if a host's BMC reports that it is supported and enabled for the Operator role. Otherwise, plaintext-based authentication is used if the BMC reports it is supported and enabled. If neither MD5 nor plaintext authentication is enabled, IPMI cannot be used with the host and vCenter attempts to use Wake-on-LAN.

You must perform the following procedure on each host.

1. Browse to the host in the vSphere Client.
2. Click the **Configure** tab.
3. Under **System**, click **Power Management**.
4. Click **Edit**.
5. Enter the following information.
   - User name and password for a BMC account. (The user name must have the ability to remotely power the host on.)
   - IP address of the NIC associated with the BMC, as distinct from the IP address of the host. The IP address should be static or a DHCP address with infinite lease.
   - MAC address of the NIC associated with the BMC.
6. Click **OK**.

Test Wake-on-LAN for vSphere DPM

The use of Wake-on-LAN (WOL) for the vSphere DPM feature is fully supported, if you configure and successfully test it according to the VMware guidelines. You must perform these steps before activating vSphere DPM for a cluster for the first time or on any host that is being added to a cluster that is using vSphere DPM.

Before testing WOL, ensure that your cluster meets the prerequisites.

- Your cluster must contain at least two hosts that are version ESX 7.0 or later.
- Each host's vMotion networking link must be working correctly. The vMotion network should also be a single IP subnet, not multiple subnets separated by routers.
- The vMotion NIC on each host must support WOL. To check for WOL support, first determine the name of the physical network adapter corresponding to the VMkernel port by selecting the host in the inventory panel of the vSphere Client, selecting the **Configuration** tab, and clicking **Networking**. After you have this information, click on **Network Adapters** and find the entry corresponding to the network adapter. The **Wake On LAN Supported** column for the relevant adapter should show Yes.
- To display the WOL-compatibility status for each NIC on a host, select the host in the inventory panel of the vSphere Client, select the **Configuration** tab, and click **Network Adapters**. The NIC must show Yes in the **Wake On LAN Supported** column.
- The switch port that each WOL-supporting vMotion NIC is plugged into should be set to auto negotiate the link speed, and not set to a fixed speed (for example, 1000 Mb/s). Many NICs support WOL only if they can switch to 100 Mb/s or less when the host is powered off.

After you verify these prerequisites, test each ESX host that is going to use WOL to support vSphere DPM. When you test these hosts, ensure that the vSphere DPM feature is deactivated for the cluster.

> **CAUTION:**
> Ensure that any host being added to a vSphere DPM cluster that uses WOL as a wake protocol is tested and deactivated from using power management if it fails the testing. If this is not done, vSphere DPM might power off hosts that it subsequently cannot power back up.

1. Browse to the host in the vSphere Client.
2. Right-click the host and select **Power** > **Enter Standby Mode**
   This action powers down the host.
3. Right-click the host and select **Power** > **Power On** to attempt to bring it out of standby mode.
4. Observe whether the host successfully powers on.
5. For any host that fails to exit standby mode successfully, perform the following steps.
   a) Select the host in the vSphere Client and select the **Configure** tab.
   b) Under **Hardware** > **Power Management**, click **Edit** to adjust the power management policy.
   After you do this, vSphere DPM does not consider that host a candidate for being powered off.

Activating vSphere DPM for a DRS Cluster

After you have performed the configuration or testing steps that are required by the wake protocol you are using on each host, you can enable vSphere DPM.

Configure the power management automation level, threshold, and host-level overrides. These settings are configured under **Power Management** in the cluster's Settings dialog box.

You can also create scheduled tasks to enable and deactivate DPM for a cluster using the **Schedule Task: Change Cluster Power Settings** wizard.

> **Note:**
> If a host in your DRS cluster has USB devices that are connected, deactivate DPM for that host. Otherwise, DPM might power off the host and sever the connection between the device and the virtual machine that was using it.

Automation Level

Whether the host power state and migration recommendations generated by vSphere DPM are run automatically or not depends on the power management automation level selected for the feature.

The automation level is configured under **Power Management** in the cluster's Settings dialog box.

> **Note:**
> The power management automation level is not the same as the DRS automation level.

Table 307: Power Management Automation Level

Option	Description
Off	The feature is deactivated and no recommendations are made.
Manual	Host power operation and related virtual machine migration recommendations are made, but not automatically run.
Automatic	Host power operations are automatically run if related virtual machine migrations can all be run automatically.

vSphere DPM Threshold

The power state (host power-on or power-off) recommendations generated by the vSphere DPM feature are assigned priorities that range from priority-one recommendations to priority-five recommendations.

These priority ratings are based on the amount of over- or under-utilization found in the DRS cluster and the improvement that is expected from the intended host power state change. A priority-one recommendation is mandatory, while a priority-five recommendation brings only slight improvement.

The threshold is configured under **Power Management** in the cluster's Settings dialog box. Each level you move the vSphere DPM Threshold slider to the right allows the inclusion of one more lower level of priority in the set of

recommendations that are executed automatically or appear as recommendations to be manually executed. At the Conservative setting, vSphere DPM only generates priority-one recommendations, the next level to the right only priority-two and higher, and so on, down to the Aggressive level which generates priority-five recommendations and higher (that is, all recommendations.)

> **Note:**
> The DRS threshold and the vSphere DPM threshold are essentially independent. You can differentiate the aggressiveness of the migration and host-power-state recommendations they respectively provide.

Host-Level Overrides

When you activate vSphere DPM in a DRS cluster, by default all hosts in the cluster inherit its vSphere DPM automation level.

You can override this default for an individual host by selecting the Host Options page of the cluster's Settings dialog box and clicking its **Power Management** setting. You can change this setting to the following options:

- Disabled
- Manual
- Automatic

> **Note:**
> Do not change a host's Power Management setting if it has been set to Disabled due to failed exit standby mode testing.

After activating and running vSphere DPM, you can verify that it is functioning properly by viewing each host's **Last Time Exited Standby** information displayed on the Host Options page in the cluster Settings dialog box and on the **Hosts** tab for each cluster. This field shows a timestamp and whether vCenter Succeeded or Failed the last time it attempted to bring the host out of standby mode. If no such attempt has been made, the field displays Never.

> **Note:**
>
> Times for the **Last Time Exited Standby** text box are derived from the vCenter event log. If this log is cleared, the times are reset to Never.

Monitoring vSphere DPM

You can use event-based alarms in vCenter to monitor vSphere DPM.

The most serious potential error you face when using vSphere DPM is the failure of a host to exit standby mode when its capacity is needed by the DRS cluster. You can monitor for instances when this error occurs by using the preconfigured **Exit Standby Error** alarm in vCenter. If vSphere DPM cannot bring a host out of standby mode (vCenter event `DrsExitStandbyModeFailedEvent`), you can configure this alarm to send an alert email to the administrator or to send notification using an SNMP trap. By default, this alarm is cleared after vCenter is able to successfully connect to that host.

To monitor vSphere DPM activity, you can also create alarms for the following vCenter events.

Table 308: vCenter Events

Event Type	Event Name
Entering Standby mode (about to power off host)	`DrsEnteringStandbyModeEvent`
Successfully entered Standby mode (host power off succeeded)	`DrsEnteredStandbyModeEvent`
Exiting Standby mode (about to power on the host)	`DrsExitingStandbyModeEvent`
Successfully exited Standby mode (power on succeeded)	`DrsExitedStandbyModeEvent`

For more information about creating and editing alarms, see the *vSphere Monitoring and Performance* documentation.

If you use monitoring software other than vCenter, and that software triggers alarms when physical hosts are powered off unexpectedly, you might have a situation where false alarms are generated when vSphere DPM places a host into standby mode. If you do not want to receive such alarms, work with your vendor to deploy a version of the monitoring software that is integrated with vCenter. You could also use vCenter itself as your monitoring solution, because starting with vSphere 4.x, it is inherently aware of vSphere DPM and does not trigger these false alarms.

VM Static Power

vSphere 8.0 Update 1 introduces a new power metric to estimate VM static power.

You can view this power metric in the **Advanced** pane of the **vCenter Performance Monitor** page in vCenter. This power metric is a semi-fixed value that is a function of the VM's configuration, specifically the number of vCPUs and memory size as well as the configuration of the host it is running on. VM static power estimates the minimal amount of required power based on the requirements of a physical machine with the same configuration, running on similar hardware as the ESX host. A VM may report different values of static power on two different ESX hosts, if one ESX host is more power efficient than the other. This is useful for dynamic workload placement strategies to optimize for overall power and may help estimate energy savings due to virtualization of workloads.

Using Affinity Rules with vSphere DRS

You can control the placement of virtual machines on hosts within a cluster by using affinity rules.

> **Note:**
> This topic describes using affinity rules with DRS. You can also use affinity rules without DRS.

You can create two types of rules.

- Used to specify affinity or anti-affinity between a group of virtual machines and a group of hosts. An affinity rule specifies that the members of a selected virtual machine DRS group can or must run on the members of a specific host DRS group. An anti-affinity rule specifies that the members of a selected virtual machine DRS group cannot run on the members of a specific host DRS group. For more information about creating and using this type of rule, see VM-VM Affinity Rules with vSphere DRS.
- Used to specify affinity or anti-affinity between individual virtual machines. A rule specifying affinity causes DRS to try to keep the specified virtual machines together on the same host, for example, for performance reasons. With an anti-affinity rule, DRS tries to keep the specified virtual machines apart, for example, so that when a problem occurs with one host, you do not lose both virtual machines. For more information about creating and using this type of rule, see VM-Host Affinity Rules with vSphere DRS.

When you add or edit an affinity rule, and the cluster's current state is in violation of the rule, the system continues to operate and tries to correct the violation. For manual and partially automated DRS clusters, migration recommendations based on rule fulfillment and load balancing are presented for approval. You are not required to fulfill the rules, but the corresponding recommendations remain until the rules are fulfilled.

To check whether any activated affinity rules are being violated and cannot be corrected by DRS, select the cluster's **DRS** tab and click **Faults**. Any rule currently being violated has a corresponding fault on this page. Read the fault to determine why DRS is not able to satisfy the particular rule. Rules violations also produce a log event.

> **Note:**
> VM-VM and VM-Host affinity rules are different from an individual host’s CPU affinity rules.

Interaction Between DRS Rules and Compute Policies

Starting with vSphere 9.1.1, DRS rules and compute policies are both available as mechanisms for expressing VM placement constraints. DRS rules are the legacy mechanism, and DRS compute policies are the newer mechanism. The two systems coexist in vSphere 9.1.

VM-VM Affinity Rules with vSphere DRS

A VM-VM affinity rule with DRS specifies whether selected individual virtual machines should run on the same host or be kept on separate hosts. This type of rule is used to create affinity or anti-affinity between individual virtual machines that you select.

Note:
This topic describes using affinity rules with DRS. You can also use affinity rules without DRS.

When an affinity rule is created, DRS tries to keep the specified virtual machines together on the same host. You might want to do this, for example, for performance reasons.

With an anti-affinity rule, DRS tries to keep the specified virtual machines apart. You could use such a rule if you want to guarantee that certain virtual machines are always on different physical hosts. In that case, if a problem occurs with one host, not all virtual machines would be placed at risk.

VM-Host Affinity Rules with vSphere DRS

A VM-Host affinity rule specifies whether or not the members of a selected virtual machine DRS group can run on the members of a specific host DRS group.

Note:
This topic describes using affinity rules for DRS. You can also use affinity rules without DRS.

Unlike a VM-VM affinity rule, which specifies affinity (or anti-affinity) between individual virtual machines, a VM-Host affinity rule specifies an affinity relationship between a group of virtual machines and a group of hosts. There are 'required' rules (designated by "must") and 'preferential' rules (designated by "should").

A VM-Host affinity rule includes the following components.

- One virtual machine DRS group.
- One host DRS group.
- A designation of whether the rule is a requirement ("must") or a preference ("should") and whether it is affinity ("run on") or anti-affinity ("not run on").

Because VM-Host affinity rules are cluster-based, the virtual machines and hosts that are included in a rule must all reside in the same cluster. If a virtual machine is removed from the cluster, it loses its DRS group affiliation, even if it is later returned to the cluster.

Create a VM-Host Group

A VM-Host affinity rule establishes an affinity (or anti-affinity) relationship between a virtual machine group with a host group. You must create both of these groups before you can create a rule that links them.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab.
3. Under **Configuration**, select **VM/Host Groups** and click **Add**.
4. In the **Create VM/Host Group** dialog box, type a name for the group.
5. Select **Host Group** from the **Type** drop-down menu and click **Add**.
6. Click the check box next to a host to add it. Continue this process until all desired hosts have been added.
7. Click **OK**.

Using this host group, you can create a VM-Host affinity rule that establishes an affinity (or anti-affinity) relationship with an appropriate virtual machine group.

Create a Virtual Machine Group

Affinity rules establish an affinity (or anti-affinity) relationship between groups. You must create groups before you can create a rule that links them.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab.
3. Under **Configuration**, select **VM/Host Groups** and click **Add**.
4. In the **Create VM/Host Group** dialog box, type a name for the group.
5. Select **VM Group** from the **Type** drop-down menu and click **Add**.
6. Click the check box next to a virtual machine to add it. Continue this process until all desired virtual machines have been added.
7. Click **OK**.

Create a VM-VM Affinity Rule

You can create VM-VM affinity rules to specify whether selected individual virtual machines should run on the same host or be kept on separate hosts.

> **Note:**
> If you use the vSphere HA Specify Failover Hosts admission control policy and designate multiple failover hosts, VM-VM affinity rules are not supported.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab.
3. Under **Configuration**, click **VM/Host Rules**.
4. Click **Add**.
5. In the **Create VM/Host Rule** dialog box, type a name for the rule.
6. From the **Type** drop-down menu, select either **Keep Virtual Machines Together** or **Separate Virtual Machines**.
7. Click **Add**.
8. Select at least two virtual machines to which the rule applies and click **OK**.
9. Click **OK**.

VM-VM Affinity Rule Conflicts

You can create and use multiple VM-VM affinity rules, however, this might lead to situations where the rules conflict with one another.

If the two VM-VM affinity rules are in conflict, you cannot activate both. For example, if one rule keeps two virtual machines together and another rule keeps the same two virtual machines apart, you cannot activate both rules. Select one of the rules to apply and deactivate or remove the conflicting rule.

When two VM-VM affinity rules conflict, the older one takes precedence and the newer rule is deactivated. DRS only tries to satisfy activated rules and deactivated rules are ignored. DRS gives a higher precedence to preventing violations of anti-affinity rules than violations of affinity rules.

Create a VM-Host Affinity Rule

You can create VM-Host affinity rules to specify whether the members of a selected virtual machine group can run on the members of a specific host group.

Create the virtual machine and host groups to which the VM-Host affinity rule applies.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab.
3. Under **Configuration**, click **VM/Host Rules**.
4. Click **Add**.
5. In the **Create VM/Host Rule** dialog box, type a name for the rule.
6. From the **Type** drop down menu, select **Virtual Machines to Hosts**.
7. Select the virtual machine group and the host group to which the rule applies.
8. Select a specification for the rule.
   - **Must run on hosts in group**. Virtual machines in VM Group 1 must run on hosts in Host Group A.
   - **Should run on hosts in group**. Virtual machines in VM Group 1 should, but are not required, to run on hosts in Host Group A.
   - **Must not run on hosts in group**. Virtual machines in VM Group 1 must never run on host in Host Group A.
   - **Should not run on hosts in group**. Virtual machines in VM Group 1 should not, but might, run on hosts in Host Group A.
9. Click **OK**.

Using VM-Host Affinity Rules

You use a VM-Host affinity rule to specify an affinity relationship between a group of virtual machines and a group of hosts. When using VM-Host affinity rules, you should be aware of when they could be most useful, how conflicts between rules are resolved, and the importance of caution when setting required affinity rules.

If you create more than one VM-Host affinity rule, the rules are not ranked, but are applied equally. Be aware that this has implications for how the rules interact. For example, a virtual machine that belongs to two groups, each of which belongs to a different required rule, can run only on hosts that belong to both of the host groups represented in the rules.

When you create a VM-Host affinity rule, its ability to function in relation to other rules is not checked. So it is possible for you to create a rule that conflicts with the other rules you are using. When two VM-Host affinity rules conflict, the older one takes precedence and the newer rule is deactivated. DRS only tries to satisfy activated rules and deactivated rules are ignored.

DRS, vSphere HA, and vSphere DPM never take any action that results in the violation of required affinity rules (those where the virtual machine group 'must run on' or 'must not run on' the host group). Accordingly, you should exercise caution when using this type of rule because of its potential to adversely affect the functioning of the cluster. If improperly used, required VM-Host affinity rules can fragment the cluster and inhibit the proper functioning of DRS, vSphere HA, and vSphere DPM.

A number of cluster functions are not performed if doing so would violate a required affinity rule.

- DRS does not evacuate virtual machines to place a host in maintenance mode.
- DRS does not place virtual machines for power-on or load-balance virtual machines.
- vSphere HA does not perform failovers.
- vSphere DPM does not optimize power management by placing hosts into standby mode.

To avoid these situations, exercise caution when creating more than one required affinity rule or consider using VM-Host affinity rules that are preferential only (those where the virtual machine group 'should run on' or 'should not run on' the host group). Ensure that the number of hosts in the cluster with which each virtual machine is affined is large enough that losing a host does not result in a lack of hosts on which the virtual machine can run. Preferential rules can be violated to allow the proper functioning of DRS, vSphere HA, and vSphere DPM.

> **Note:**
> You can create an event-based alarm that is triggered when a virtual machine violates a VM-Host affinity rule. Add a new alarm for the virtual machine and select **VM is violating VM-Host Affinity Rule** as the event trigger. For more information about creating and editing alarms, see the vSphere Monitoring and Performance documentation.

DRS Rules and Compute Policy Interactions in vSphere

In vSphere 9.1.1 and later, DRS rules and compute policies are two distinct mechanisms for expressing placement constraints on virtual machines. DRS rules specify which host a virtual machine must or must not reside on, or which virtual machines must or must not be on the same host. Compute policies use tags to express placement constraints that DRS enforces during placement and execution. Understanding how these mechanisms interact helps you avoid unexpected placement behavior.

Coexistence of DRS Rules and Compute Policies

For non-VM Service virtual machines, you can apply both DRS rules and compute policies to the same set of virtual machines. When both constraint types target the same virtual machines, the constraints can conflict with one another and produce unexpected placement behavior.

To prevent unexpected behavior, do not use both DRS rules and compute policies on the same set of non-VM Service virtual machines.

How DRS Evaluates Rules and Policies During Placement

When both DRS rules and compute policies apply to the same virtual machines, DRS evaluates both during placement. The outcome depends on the combination of constraint strengths. The following table describes the algorithm behavior for each combination.

Compute Policy Strictness	DRS Rule Type	Algorithm Behavior
Required (hard)	Hard (must rule)	Both constraints narrow the compatible host set. If the constraints conflict, the compatible host set becomes empty and placement fails with a `NoCompatibleHost` error.
Required (hard)	Soft (should rule)	DRS honors the soft rule only as a preference among policy-compliant hosts.
Preferred (firm)	Hard (must rule)	DRS honors the preferred policy only as a preference among rule-compliant hosts.
Preferred (firm)	Soft (should rule)	Both constraints act as preferences. The compute policy takes priority over the soft rule.

Note:

Compute policy strictness values of `Required` and `Preferred` correspond to hard and firm enforcement, respectively. DRS rule types of must and should correspond to hard and soft enforcement, respectively.

Compute Policy Strictness During Execution

In vSphere 9.1.1 and later, DRS enforces compute policy strictness during virtual machine execution, not only during initial placement. Compute policies use two enforcement levels:

- **Required (hard):** DRS does not violate the policy during any workflow. Hard enforcement applies during DRS load balancing, maintenance mode, mandatory moves, prerequisite moves, and vSphere HA failover.
- **Preferred (firm):** DRS treats the policy as a preference. DRS can violate the policy to satisfy a hard constraint or cluster operation.

Execution-time enforcement keeps compute policies active throughout the virtual machine lifecycle. When you delete a compute policy, the placement constraint that the policy represents is no longer enforced during DRS workflows.

Using Affinity Rules without vSphere DRS

You can use affinity rules without using vSphere DRS.

VM affinity rules allow you to control the placement of VMs within a vSphere cluster. VM-VM affinity means the VMs must be run on the same host within a cluster. Workloads are comprised of multiple VMs that have dependencies on each other. Using **Keep Virtual Machines Together** rules ensure that all selected VMs run on the same host.

VM-VM anti-affinity means the VMs must be run on different hosts within a cluster. You can use a **Separate Virtual Machines** rule when you have multiple resource intensive VMs to ensure that these VMs are always running on different hosts in the cluster to avoid resource contention.

VM-Host affinity rules control the VM to Host relationship and has 4 sub-rules. Use cases for VM-Host rules would include making sure workloads run (or do not run) on specific hosts in a mixed-hardware cluster. For licensing reasons, you may need to ensure that an application only runs on a specific host or specific subset of hosts.

The VM-Host rules leverage VM-Host groups. A VM group is a user-defined group of one or more VMs. A host group is a user-defined group of one or more hosts.

Rule	Description	Persistence
Must run on hosts in group	The VM group must be run on the defined host group.	A hard rule that cannot be violated.
Must not run on hosts in group	The VM group must not be run on the defined host group.	A hard rule that cannot be violated.
Should run on hosts in group	The VM group should be run on the defined host group.	A soft rule that can be violated.
Should not run on hosts in group	The VM group should not be run on the defined host group.	A soft rule that can be violated.

VM-Host groups and VM-Host rules can be created, deleted, and edited when vSphere DRS is not activated. The interface to manage VM-Host groups and VM-Host rules exists outside of the vSphere DRS configuration.

In the event of a vSphere HA failover of VMs, affinity rules are respected. This is true for rule types **Keep Virtual Machines Together** and **Separate Virtual Machines**, as well as hard VM-Host rule types. vSphere HA is unable to failover VMs if a hard rule would be violated. When using soft VM-Host rules, vSphere HA operations succeed even if the rule is violated. Soft rules allow for violation of their definitions.

For example, a 2-node cluster and a **Separate Virtual Machines** rule defined to place VM-1 on ESX-01 and VM-2 on ESX-02. ESX-01 experiences a failure and vSphere HA is initiated. vSphere HA is unable to power on VM-1 on ESX-02 as it would violate the rule to keep each virtual machine separate.

VM power-on operations do not allow affinity rules to be violated. This is true for rule types **Keep Virtual Machines Together** and **Separate Virtual Machines**, as well as hard VM-Host rules. When using soft VM-Host rules, power on operations succeed even if the rule is violated. Soft rules allow for violation of their definitions.

For example, if a rule is defined to **Keep Virtual Machines Together** on VM-1 and VM-2 and VM-1 is currently already powered on, an attempt to power on VM-2 will fail if it is registered to a different host. Since vSphere DRS is not activated, the user must manually migrate the VMs to the same host to be able to power on the VM.

When vSphere DRS is not activated and therefore not automatically migrating VMs, you can initiate migrations using vSphere vMotion on VMs with affinity rules defined. Both rule types **Keep Virtual Machines Together** and **Separate Virtual Machines** can be violated by migrating using vSphere vMotion. For example, if a rule is defined to **Separate Virtual Machines** on VM-1 and VM-2, an attempt to migrate the VMs to the same host will not be blocked. The operation will succeed even though it would violate the rule.

Any hard VM-Host rule cannot be violated by a user-initiated migration. For example, if a rule is defined that VM-1 and VM-2 must run on hosts ESX-01 and ESX-02, an attempt to migrate the VMs to hosts outside of the host group is blocked.

Soft VM-Host rules can be violated by a user-initiated migration. Soft rules allow for violation of their definitions. For example, if a rule is defined that VM-1 and VM-2 should not run on hosts ESX-03 and ESX-04, an attempt to migrate the VMs to either ESX-03 or ESX-04 is not blocked.

When vSphere DRS is not activated, there is no automatic remediation of VM placement. This includes violations because of manual migrations and violations that occur if new rules are defined that would violate current VM placement. Without vSphere DRS, you must correct VM placement to comply with rules.

Using Span Limit Policy

The Span Limit policy is a compute policy that allows you to establish the affinity relationship between a group of virtual machines and a specified number of hosts without explicitly identifying the hosts.

With the Span Limit policy, you can optimize the application licenses assigned to a fixed number of hosts, usually based on the number of cores.

Span Limit Policy Considerations and Limitations

Certain considerations and limitations exist when you use the Span Limit policy.

- To avoid violating the licensing instructions for the workloads, you must configure the Span Limit policy by setting the number of hosts (the group size) to be one less than the number of licenses.
  For example, if you have 5 licenses, you must configure the Span Limit policy with a group size of 4.
- VMs with the Span Limit policy (policy VMs) will be placed across no more than the group size number of hosts. This includes DRS operations like load-balancing, moving VMs in a cluster, and so on.

VM Operations

- If policy VMs are part of a cluster with DRS deactivated, the compliance status of such VMs with the Span Limit policy is "Not Applicable".
- If you power on a policy VM directly on the host, bypassing vCenter, the VM might be placed on a host that is not compliant with the Span Limit policy. The compliance status of such a VM is non-compliant, and the DRS will try to remediate the non-compliance.
- The cloning of a VM does not clone its tags to the cloned VM.
- If one of the hosts is in Maintenance mode, the VMs can spread across the group size plus one host.
  For example, you set the policy to have a group of 3 hosts and one of the policy hosts is in maintenance mode. While the host is entering maintenance, the policy VMs can spread across at most 4 hosts. Once the host successfully enters maintenance, the policy VMs will run on at most three hosts.

Maintenance Operations

- When using this policy, ensure you have sufficient spare capacity to perform maintenance operations.
- To ensure compliance with the policy, if a host selected by DRS for the policy is placed in maintenance mode, the policy VMs are moved to other hosts, including the additional replacement host selected by DRS for maintenance.
- If VMs cannot be moved to other hosts within the group, DRS will report a fault.

Host Operations

- You can add a host with VMs tagged with the policy’s VM tag. However, the move into the cluster fails if the host violates the rules of an existing group of hosts.
- A host is removed from the group only when explicitly removed from the cluster.

HA Failover Operations for Policy VMs

- DRS ensures that policy VMs are failed over to other hosts within the hosts group.
- If a host with policy VMs fails and HA cannot communicate with DRS, the policy VMs will be restarted on other compatible hosts in the HTP group.
- If a host with policy VMs fail and HA can communicate with DRS but there is not enough failover capacity, the policy VMs remain powered off on the failed host.

DRS Rules

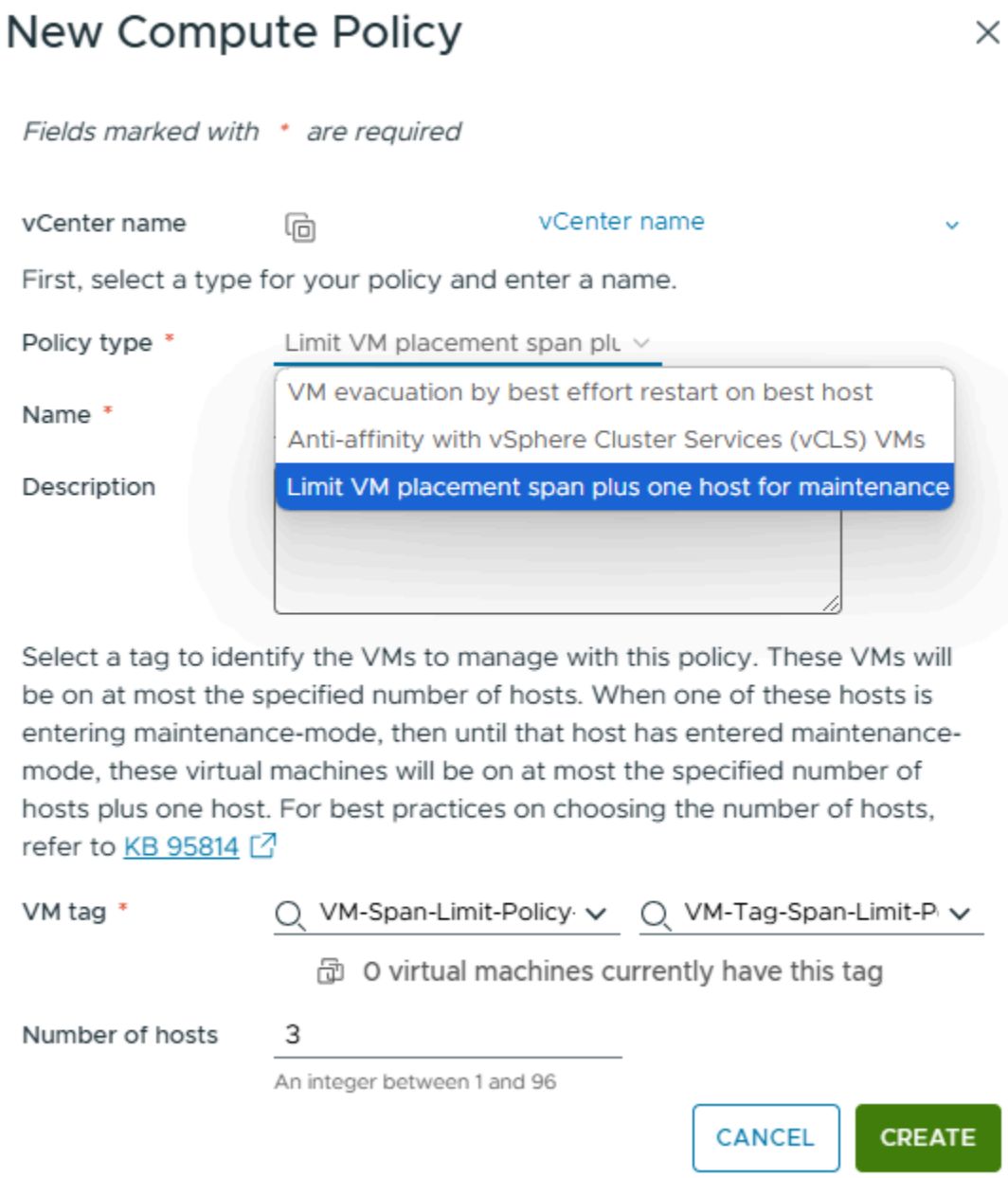
- This policy overrides the DRS soft rules, such as the VM-Host soft affinity or anti-affinity rules.
- If a policy VM is part of a VM-Host hard affinity rule, its compatible hosts are the ones that are both in the hard affinity rule and selected by DRS for this policy.

Cross vCenter vMotion

If a cross vCenter vMotion involves a tagged VM part of a span limit policy on the target vCenter, the vMotion will fail if the selected host on the target vCenter does not meet the policy requirements.

Create or Delete the Span Limit Policy

To create a Span Limit policy, perform the following steps.



1. In the vSphere Client, select **Menu** > **Policies and Profiles** > **Compute Policies**.
2. From the **Compute Policies** pane, click **Add** to open the **New Compute Policy** window.
3. From the **Policy type** drop-down menu, select `Limit VM placement span plus one host for maintenance`.
4. Enter a name and description for the policy.
5. From the **VM tag**, select the tags that workload VMs will have for **Category** and **Tag** and to which the policy applies.
6. In the **Number of hosts** field, enter the number of hosts. DRS ensures that any VM with this tag will run on a maximum of the given number of hosts.
7. Click **Create**.
8. (Optional) To delete the policy, navigate to **Policies and Profiles** > **Compute Policies** and delete the policy that appears as a card.

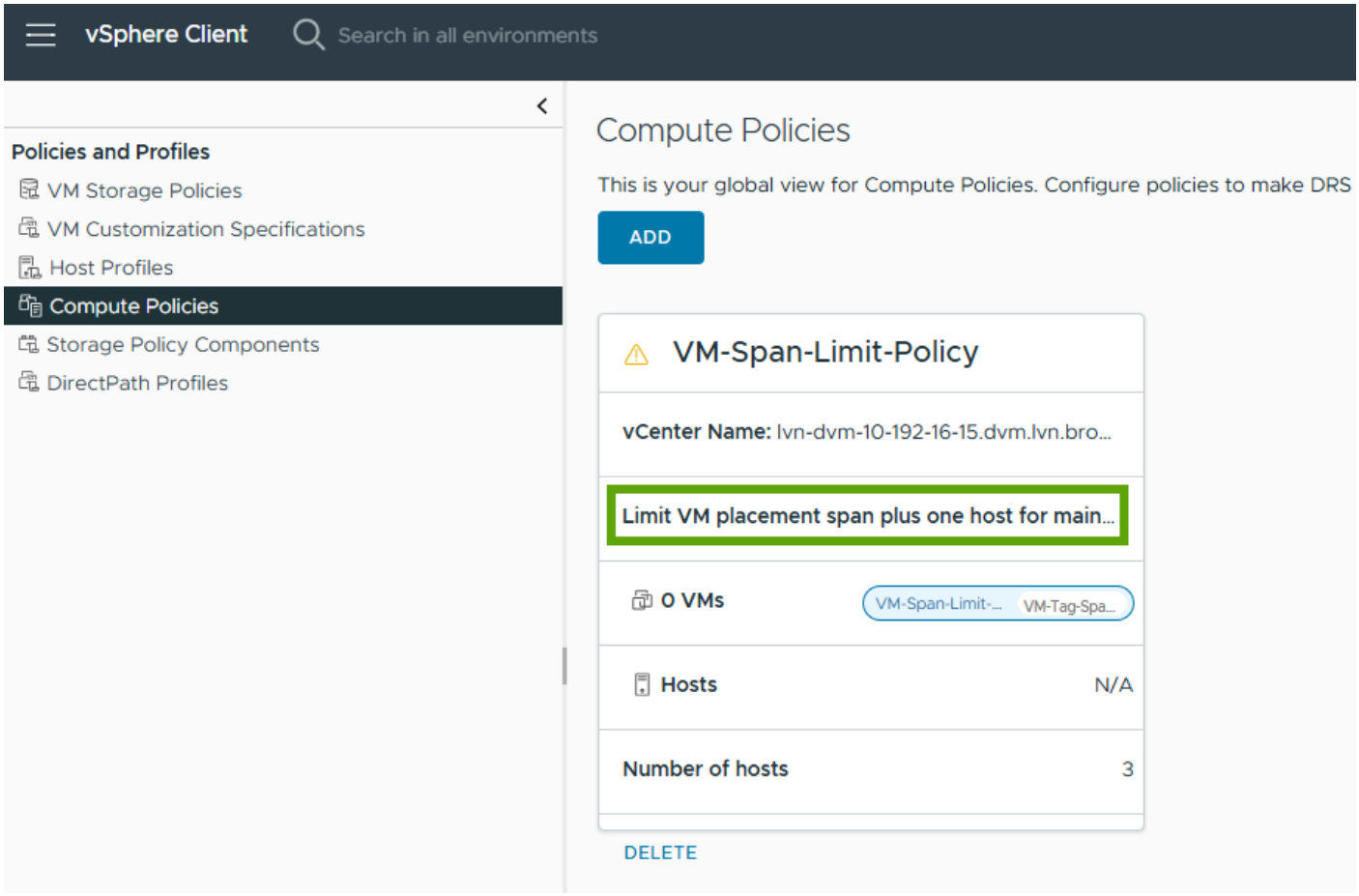
Once the policy is created, DRS in each cluster selects the designated number of hosts plus a replacement host if one of the hosts enters maintenance. Policy VMs within a cluster will only run on these selected hosts. If a chosen host enters maintenance, the VMs will be placed on the remaining selected hosts and the replacement host.

How the Span Limit Policy Works

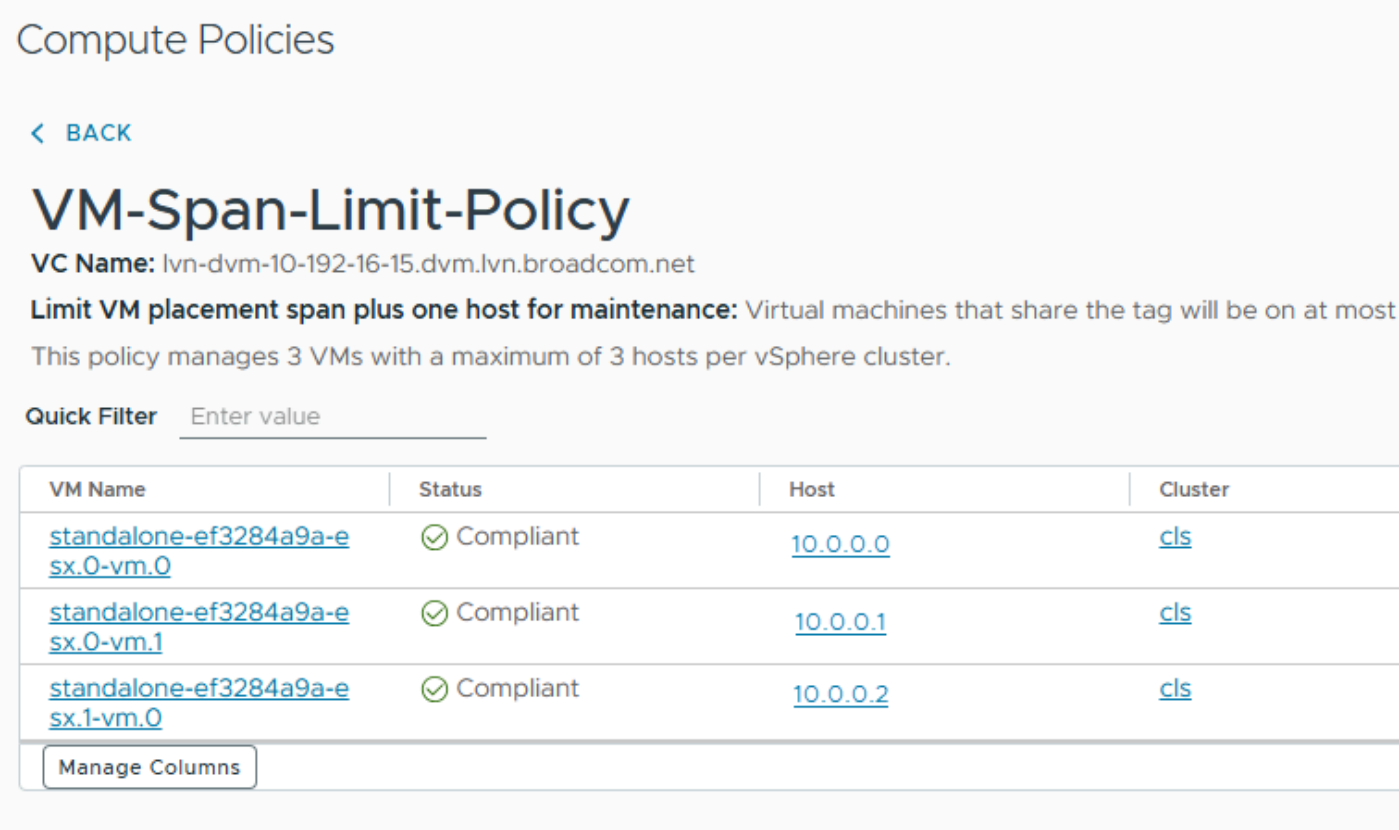
This section explains the operational behavior of the Span Limit policy when you migrate a VM with vMotion, place a Span Limit policy host in Maintenance Mode, or power on VMs directly on a host, bypassing the vCenter instance.

When you create the Span Limit policy to associate VMs in the inventory with the policy, perform the following steps.

1. From the vSphere Client inventory, select a cluster and click the VMs tab.
2. From the list of virtual machines, select all VMs you want to assign a VM tag to which you create the policy.
   a) Right-click the selected VMs and navigate to **Tags & Customs Attributes** > **Assign Tag**.
   a) From the **Assign Tag** window, click **Yes**.
   b) From the **Object - Assign Tag** window, select the tag name and click **Assign**.
3. To check the compliance of the VMs with the policy, power on a single VM, or the entire group of VMs, you assigned the tag and perform the following steps.
4. In the vSphere Client, select **Menu** > **Policies and Profiles** > **Compute Policies**.
5. From the **Compute Policies** pane, click **Limit VM placement span plus one host for maintenance** in the card.



The Host column displays the hosts selected for the Span Limit policy and the Status column shows `Compliant`.



The following section explains the behavior of the Span Limit policy when you perform particular operations.

Migrating a VM with vMotion to a different host

When you migrate a VM with vMotion to a host not selected for the Span Limit policy by DRS, the operation fails with a compatibility error.

Placing a Span Limit policy host into a Maintenance Mode

When you place a Span Limit policy host in Maintenance Mode, all the VMs on the host migrate to a replacement host selected by DRS for this policy.

> **Note:**
>
> While the evacuation of the host that is in maintenance is in progress, the policy VMs will be spread across one host more than the number of hosts configured for the policy.

From the **Compute Policy** pane, you can see the new host to which these VMs are moved.

Power on VMs directly on a policy host bypassing the vCenter instance

A mandatory remediation loop runs every 3 minutes to address non-compliance through such operations. When such violations are identified, DRS returns the VMs on the host they must run on.

Creating a Datastore Cluster in vSphere

A datastore cluster is a collection of datastores with shared resources and a shared management interface. Datastore clusters are to datastores what clusters are to hosts. When you create a datastore cluster, you can use vSphere Storage DRS to manage storage resources.

> **Note:**
> Datastore clusters are referred to as storage pods in the vSphere API.

When you add a datastore to a datastore cluster, the datastore's resources become part of the datastore cluster's resources. As with clusters of hosts, you use datastore clusters to aggregate storage resources, which enables you to support resource allocation policies at the datastore cluster level. The following resource management capabilities are also available per datastore cluster.

Space utilization load balancing	You can set a threshold for space use. When space use on a datastore exceeds the threshold, Storage DRS generates recommendations or performs Storage vMotion migrations to balance space use across the datastore cluster.
Anti-affinity rules	You can create anti-affinity rules for virtual machine disks. For example, the virtual disks of a certain virtual machine must be kept on different datastores. By default, all virtual disks for a virtual machine are placed on the same datastore.

Initial Placement and Ongoing Balancing

Storage DRS provides initial placement and ongoing balancing recommendations to datastores in a Storage DRS-enabled datastore cluster.

Initial placement occurs when Storage DRS selects a datastore within a datastore cluster on which to place a virtual machine disk. This happens when the virtual machine is being created or cloned, when a virtual machine disk is being migrated to another datastore cluster, or when you add a disk to an existing virtual machine.

Initial placement recommendations are made in accordance with space constraints and with respect to the goals of space. These goals aim to minimize the risk of over-provisioning one datastore and the performance impact on virtual machines.

Storage DRS is invoked at the configured frequency (by default, every eight hours) or when one or more datastores in a datastore cluster exceeds the user-configurable space utilization thresholds. When Storage DRS is invoked, it checks each datastore's space utilization.

Storage Migration Recommendations

vCenter displays migration recommendations on the Storage DRS Recommendations page for datastore clusters that have manual automation mode.

The system provides as many recommendations as necessary to enforce Storage DRS rules and to balance the space of the datastore cluster. Each recommendation includes the virtual machine name, the virtual disk name, the name of the datastore cluster, the source datastore, the destination datastore, and a reason for the recommendation.

- Balance datastore space use

Storage DRS makes mandatory recommendations for migration in the following situations:

- The datastore is out of space.
- Anti-affinity or affinity rules are being violated.
- The datastore is entering maintenance mode and must be evacuated.

In addition, optional recommendations are made when a datastore is close to running out of space or when space adjustments should be made.

Storage DRS considers moving virtual machines that are powered off or powered on for space balancing. Storage DRS includes powered-off virtual machines with snapshots in these considerations.

Create a Datastore Cluster

You can manage datastore cluster resources using Storage DRS.

1. Browse to data centers in the vSphere Client.
2. Right-click the data center and select **Storage**, **New Datastore Cluster**.
3. To complete the **New Datastore Cluster** wizard, follow the prompts.
4. Click **Finish**.

Activate Storage DRS

Storage DRS allows you to manage the aggregated resources of a datastore cluster. When Storage DRS is activated, it provides recommendations for virtual machine disk placement and migration to balance space across the datastores in the datastore cluster.

When you activate Storage DRS, you activate the following functions.

- Space load balancing among datastores within a datastore cluster.
- Initial placement for virtual disks based on space.

The Enable Storage DRS check box in the Datastore Cluster Settings dialog box activates or deactivates all of these components at once.

When you deactivate Storage DRS on a datastore cluster, Storage DRS settings are preserved. When you activate Storage DRS, the settings for the datastore cluster are restored to the point where Storage DRS was deactivated.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Select **Storage DRS** and click **Edit**.
4. Select **Turn ON vSphere DRS** and click **OK**.

Set the Automation Level for Datastore Clusters

The automation level for a datastore cluster specifies whether or not placement and migration recommendations from Storage DRS are applied automatically.

1. Browse to a datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Select **Storage DRS** and click **Edit**.
4. From the **Cluster automation level**, select the automation level.

   Fully Automated is the default automation level.

Option	Description
No Automation (Manual Mode)	Placement and migration recommendations are displayed, but do not run until you manually apply the recommendation.
Fully Automated	Placement and migration recommendations run automatically.

5. Click **OK**.

Setting the Aggressiveness Level for Storage DRS

The aggressiveness of Storage DRS is determined by specifying thresholds for space used.

Storage DRS collects resource usage information for the datastores in a datastore cluster. vCenter uses this information to generate recommendations for the placement of virtual disks on datastores.

When you set a low aggressiveness level for a datastore cluster, Storage DRS recommends Storage vMotion migrations only when absolutely necessary, for example, when space utilization or its imbalance is high. When you set a high aggressiveness level for a datastore cluster, Storage DRS recommends migrations whenever the datastore cluster can benefit from space.

In the vSphere Client, you can use the following thresholds to set the aggressiveness level for Storage DRS:

Space Utilization	Storage DRS generates recommendations or performs migrations when the percentage of space utilization on the datastore is greater than the threshold you set in the vSphere Client.

You can also set advanced options to further configure the aggressiveness level of Storage DRS.

Space utilization difference	This threshold ensures that there is some minimum difference between the space utilization of the source and the destination. For example, if the space used on datastore A is 82% and datastore B is 79%, the difference is 3. If the threshold is 5, Storage DRS will not make migration recommendations from datastore A to datastore B.

Set Storage DRS Runtime Rules

Set Storage DRS triggers and configure advanced options for the datastore cluster.

1. Browse to a datastore cluster in the vSphere Client.
2. Under **Services**, select **Storage DRS** and click **Edit**.
3. Click the **Runtime Settings** tab.
4. Optional: Set Storage DRS space thresholds.
   You set the aggressiveness level of Storage DRS by specifying thresholds for used space.
   - Use the Utilized Space slider to indicate the maximum percentage of consumed space allowed before Storage DRS is triggered. Storage DRS makes recommendations and performs migrations when space use on the datastores is higher than the threshold.
5. Optional: Configure advanced options.

Option	Description
Keep VMDKs together by default	Use the slide to specify if a virtual machine within the datastore cluster must have its virtual disk on the same datastore.
Check imbalances every	Specify how often Storage DRS should assess the space balancing.
Minimum space utilization difference	No recommendations until the utilization difference between source and destination is: Use the slider to specify the space utilization difference threshold. Utilization is usage * 100/ capacity. This threshold ensures that there is some minimum difference between the space utilization of the source and the destination. For example, if the space used on datastore A is 82% and datastore B is 79%, the difference is 3. If the threshold is 5, Storage DRS will not make migration recommendations from datastore A to datastore B.

6. Click **OK**.

Datastore Cluster Requirements

Datastores and hosts that are associated with a datastore cluster must meet certain requirements to use datastore cluster features successfully.

Follow these guidelines when you create a datastore cluster.

- Datastore clusters must contain similar or interchangeable datastores.
  A datastore cluster can contain a mix of datastores of different sizes, and can be from different arrays and vendors. However, the following types of datastores cannot coexist in a datastore cluster.
  - NFS and VMFS datastores cannot be combined in the same datastore cluster.
  - Replicated datastores cannot be combined with non-replicated datastores in the same Storage-DRS-enabled datastore cluster.
- All hosts attached to the datastores in a datastore cluster must be ESX 7.0 and later. If datastores in the datastore cluster are connected to ESX 7.x and earlier hosts, Storage DRS does not run.
- Datastores shared across multiple data centers cannot be included in a datastore cluster.

- As a best practice, do not include datastores that have hardware acceleration enabled in the same datastore cluster as datastores that do not have hardware acceleration enabled. Datastores in a datastore cluster must be homogeneous to guarantee hardware acceleration-supported behavior.

Adding and Removing Datastores from a Datastore Cluster

You add and remove datastores to and from an existing datastore cluster.

You can add to a datastore cluster any datastore that is mounted on a host in the vSphere Client inventory, with the following exceptions:

- All hosts attached to the datastore must be ESX 7.0 and later.
- The datastore cannot be in more than one data center in the same instance of the vSphere Client.

When you remove a datastore from a datastore cluster, the datastore remains in the vSphere Client inventory and is not unmounted from the host.

Use Datastore Clusters to Manage Storage Resources in vSphere

After you create a datastore cluster, you can customize it and use it to manage space utilization resources.

Using Storage DRS Maintenance Mode

You place a datastore in maintenance mode when you must take it out of use to service it. A datastore enters or leaves maintenance mode only as the result of a user request.

Maintenance mode is available to datastores within a Storage DRS-enabled datastore cluster. Standalone datastores cannot be placed in maintenance mode.

Virtual disks that are located on a datastore that is entering maintenance mode must be migrated to another datastore, either manually or using Storage DRS. When you attempt to put a datastore in maintenance mode, the **Placement Recommendations** tab displays a list of migration recommendations, datastores within the same datastore cluster where virtual disks can be migrated. On the **Faults** tab, vCenter displays a list of the disks that cannot be migrated and the reasons why. If Storage DRS affinity or anti-affinity rules prevent disks from being migrated, you can choose to enable the Ignore Affinity Rules for Maintenance option.

The datastore is in a state of Entering Maintenance Mode until all virtual disks have been migrated.

Place a Datastore in Maintenance Mode

If you must take a datastore out of service, you can place the datastore in Storage DRS maintenance mode.

Storage DRS is enabled on the datastore cluster that contains the datastore that is entering maintenance mode.

No CD-ROM image files are stored on the datastore.

There are at least two datastores in the datastore cluster.

1. Browse to the datastore in the vSphere Client.
2. Right-click the datastore and select **Maintenance Mode** > **Enter Maintenance Mode**.
   A list of recommendations appears for datastore maintenance mode migration.
3. Optional: On the Placement Recommendations tab, deselect any recommendations you do not want to apply.

   Note:
   The datastore cannot enter maintenance mode without evacuating all disks. If you deselect the recommendations, you must manually move the affected virtual machines.
4. If necessary, click **Apply Recommendations**.

   vCenter uses Storage vMotion to migrate the virtual disks from the source datastore to the destination datastore and the datastore enters maintenance mode.

The datastore icon might not be immediately updated to reflect the datastore's current state. To update the icon immediately, click **Refresh**.

Ignore Storage DRS Affinity Rules for Maintenance Mode

Storage DRS affinity or anti-affinity rules might prevent a datastore from entering maintenance mode. You can ignore these rules when you put a datastore in maintenance mode.

When you activate the Ignore Affinity Rules for Maintenance option for a datastore cluster, vCenter ignores Storage DRS affinity and anti-affinity rules that prevent a datastore from entering maintenance mode.

Storage DRS rules are ignored only for evacuation recommendations. vCenter does not violate the rules when making space and load balancing recommendations or initial placement recommendations.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Select **DRS** and click **Edit**.
4. Expand **Advanced Options** and click **Add**.
5. In the Option column, type `IgnoreAffinityRulesForMaintenance`.
6. In the Value column, type `1` to activate the option.
   Type `0` to deactivate the option.
7. Click **OK**.

The Ignore Affinity Rules for Maintenance Mode option is applied to the datastore cluster.

Applying Storage DRS Recommendations

Storage DRS collects resource usage information for all datastores in a datastore cluster. Storage DRS uses the information to generate recommendations for virtual machine disk placement on datastores in a datastore cluster.

Storage DRS recommendations appear on the **Storage DRS** tab in the vSphere Client datastore view. Recommendations also appear when you attempt to put a datastore into Storage DRS maintenance mode. When you apply Storage DRS recommendations, vCenter uses Storage vMotion to migrate virtual machine disks to other datastores in the datastore cluster to balance the resources.

You can apply a subset of the recommendations by selecting the Override Suggested DRS Recommendations check box and selecting each recommendation to apply.

Table 309: Storage DRS Recommendations

Label	Description
Priority	Priority level (1-5) of the recommendation. (Hidden by default.)
Recommendation	Action is being recommended by Storage DRS.
Reason	Why the action is needed.
Space Utilization % Before (source) and (destination)	Percentage of space used on the source and destination datastores before migration.
Space Utilization % After (source) and (destination)	Percentage of space used on the source and destination datastores after migration.

Refresh Storage DRS Recommendations

Storage DRS migration recommendations appear on the **Storage DRS** tab in the vSphere Client. You can refresh these recommendations by running Storage DRS.

At least one datastore cluster must exist in the vSphere Client inventory.

Enable Storage DRS for the datastore cluster. The **Storage DRS** tab appears only if Storage DRS is enabled.

1. In the vSphere Client datastore view, select the datastore cluster and click the **Storage DRS** tab.
2. Select the **Recommendations** view and click the **Run Storage DRS** link in the upper right corner.

The recommendations are updated. The Last Updated timestamp displays the time when Storage DRS recommendations were refreshed.

Change Storage DRS Automation Level for a Virtual Machine

You can override the datastore cluster-wide automation level for individual virtual machines. You can also override default virtual disk affinity rules.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Configuration**.
3. Under **VM Overrides**, select **Add**.
4. Select a virtual machine.
5. Click the Automation level drop-down menu, and select an automation level for the virtual machine.

Option	Description
Default (Manual)	Placement and migration recommendations are displayed, but do not run until you manually apply the recommendation.
Fully Automated	Placement and migration recommendations run automatically.
Disabled	vCenter does not migrate the virtual machine or provide migration recommendations for it.

6. Click the **Keep VMDKs together**, drop-down menu to override default VMDK affinity.
   See Override VMDK Affinity Rules.
7. Click **OK**.

Set Up Off-Hours Scheduling for Storage DRS

You can create a scheduled task to change Storage DRS settings for a datastore cluster so that migrations for fully automated datastore clusters are more likely to occur during off-peak hours.

Enable Storage DRS.

You can create a scheduled task to change the automation level and aggressiveness level for a datastore cluster. For example, you might configure Storage DRS to run less aggressively during peak hours, when performance is a priority, to minimize the occurrence of storage migrations. During non-peak hours, Storage DRS can run in a more aggressive mode and be invoked more frequently.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Services**.
3. Under **vSphere DRS** click the **Schedule DRS** button.
4. In the Edit Datastore Cluster dialog box, click **SDRS Scheduling**.
5. Expand **DRS Automation**.
   a) Select an automation level.
   b) Set the Migration threshold.

      Use the Migration slider to select the priority level of vCenter recommendations that adjust the cluster's load balance.
   c) Select whether to enable Virtual Machine Automation.

      Override for individual virtual machines can be set from the VM Overrides page.

6. Expand **Power Managment**.
   a) Select an automation level.
   b) Set the DPM threshold.

      Use the DPM slider to select the power recommendations that vCenter will apply.
7. Type a Task name.
8. Type a description of the Task you have created.
9. Under Configured Scheduler, click **Change** and select the time for the task to run and click **OK**.
10. Type an email address to send a notification email to when the task is complete.
11. Click **OK**.

The scheduled task runs at the specified time.

Storage DRS Anti-Affinity Rules

You can create Storage DRS anti-affinity rules to control which virtual disks should not be placed on the same datastore within a datastore cluster. By default, a virtual machine's virtual disks are kept together on the same datastore.

When you create an anti-affinity rule, it applies to the relevant virtual disks in the datastore cluster. Anti-affinity rules are enforced during initial placement and Storage DRS-recommendation migrations, but are not enforced when a migration is initiated by a user.

Note:
Anti-affinity rules do not apply to CD-ROM ISO image files that are stored on a datastore in a datastore cluster, nor do they apply to swapfiles that are stored in user-defined locations.

VM Anti-Affinity Rules	Specify which virtual machines should never be kept on the same datastore. See Create VM Anti-Affinity Rules.
VMDK Anti-Affinity Rules	Specify which virtual disks associated with a particular virtual machine must be kept on different datastores. See Create VMDK Anti-Affinity Rules.

If you move a virtual disk out of the datastore cluster, the affinity or anti-affinity rule no longer applies to that disk.

When you move virtual disk files into a datastore cluster that has existing affinity and anti-affinity rules, the following behavior applies:

- Datastore Cluster B has an intra-VM affinity rule. When you move a virtual disk out of Datastore Cluster A and into Datastore Cluster B, any rule that applied to the virtual disk for a given virtual machine in Datastore Cluster A no longer applies. The virtual disk is now subject to the intra-VM affinity rule in Datastore Cluster B.
- Datastore Cluster B has a VM anti-affinity rule. When you move a virtual disk out of Datastore Cluster A and into Datastore Cluster B, any rule that applied to the virtual disk for a given virtual machine in Datastore Cluster A no longer applies. The virtual disk is now subject to the VM anti-affinity rule in Datastore Cluster B.
- Datastore Cluster B has a VMDK anti-affinity rule. When you move a virtual disk out of Datastore Cluster A and into Datastore Cluster B, the VMDK anti-affinity rule does not apply to the virtual disk for a given virtual machine because the rule is limited to only specified virtual disks in Datastore Cluster B.

Note:
Storage DRS rules might prevent a datastore from entering maintenance mode. You can choose to ignore Storage DRS rules for maintenance mode by enabling the Ignore Affinity Rules for Maintenance option.

Create VM Anti-Affinity Rules

You can create an anti-affinity rule to indicate that all virtual disks of certain virtual machines must be kept on different datastores. The rule applies to individual datastore clusters.

Virtual machines that participate in a VM anti-affinity rule in a datastore cluster must be associated with an intra-VM affinity rule in the datastore cluster. The virtual machines must also comply with the intra-VM affinity rule.

If a virtual machine is subject to a VM anti-affinity rule, the following behavior applies:

- Storage DRS places the virtual machine's virtual disks according to the rule.
- Storage DRS migrates the virtual disks using vMotion according to the rule, even if the migration is for a mandatory reason such as putting a datastore in maintenance mode.
- If the virtual machine's virtual disk violates the rule, Storage DRS makes migration recommendations to correct the error or reports the violation as a fault if it cannot make a recommendation that will correct the error.

No VM anti-affinity rules are defined by default.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Configuration**.
3. Select **VM/Host Rules**.
4. Click **Add**.
5. Type a name for the rule.
6. From the Type menu, select **VM anti-affinity**.
7. Click **Add**.
8. Click **Select Virtual Machine**.
9. Select at least two virtual machines and click **OK**.
10. Click **OK** to save the rule.

Create VMDK Anti-Affinity Rules

You can create a VMDK anti-affinity rule for a virtual machine that indicates which of its virtual disks must be kept on different datastores.

VMDK anti-affinity rules apply to the virtual machine for which the rule is defined, not to all virtual machines. The rule is expressed as a list of virtual disks that are to be separated from one another.

If you attempt to set a VMDK anti-affinity rule and an intra-VM affinity rule for a virtual machine, vCenter rejects the most recently defined rule.

If a virtual machine is subject to a VMDK anti-affinity rule, the following behavior applies:

- Storage DRS places the virtual machine's virtual disks according to the rule.
- Storage DRS migrates the virtual disks using vMotion according to the rule, even if the migration is for a mandatory reason such as putting a datastore in maintenance mode.
- If the virtual machine's virtual disk violates the rule, Storage DRS makes migration recommendations to correct the error or reports the violation as a fault if it cannot make a recommendation that will correct the error.

No VMDK anti-affinity rules are defined by default.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Configuration**.
3. Select **VM/Host Rules**.
4. Click **Add**.
5. Type a name for the rule.
6. From the Type menu, select **VMDK anti-affinity**.
7. Click **Add**.
8. Click **Select Virtual Machine**.
9. Select a virtual machine and click **OK**.
10. Select at least two virtual disks to which the rule applies and click **OK**.
11. Click **OK** to save the rule.

Override VMDK Affinity Rules

VMDK affinity rules indicate that all virtual disks in a datastore cluster that are associated with a particular virtual machine are located on the same datastore in the datastore cluster. The rules apply to individual datastore clusters.

VMDK affinity rules are activated by default for all virtual machines that are in a datastore cluster. You can override the default setting for the datastore cluster or for individual virtual machines.

Virtual machines that are subject to VMDK affinity rules have the following behavior:

- Storage DRS places the virtual machine's virtual disks according to the rule.
- Storage DRS migrates the virtual disks using vMotion according to the rule, even if the migration is for a mandatory reason such as putting a datastore in maintenance mode.
- If the virtual machine's virtual disk violates the rule, Storage DRS makes migration recommendations to correct the error or reports the violation as a fault if it cannot make a recommendation that will correct the error.

When you add a datastore to a datastore cluster that is activated for Storage DRS, the VMDK affinity rule is deactivated for any virtual machine that has virtual disks on that datastore if it also has virtual disks on other datastores.

1. Browse to the datastore cluster in the vSphere Client.
2. Click the **Configure** tab and click **Configuration**.
3. Select **VM Overrides**.
4. Click **Add**.
5. Use the **+** button to select virtual machines.
6. Click the **Keep VMDKs together** drop-down menu and select **No**.
7. Click **OK**.

Clear Storage DRS Statistics

To diagnose problems with Storage DRS, you can clear Storage DRS statistics before you manually run Storage DRS.

Activate Storage DRS for the datastore cluster.

> **Important:**
> When you activate the option to clear Storage DRS statistics, statistics are cleared every time Storage DRS runs until you deactivate the option. Always deactivate the option after you diagnose the Storage DRS problem.

1. Activate the `ClearIoStatsOnSdrsRun` option.
   a) Browse to the datastore cluster in the vSphere Client.
   b) Click the **Configuration** tab and click **Services**.
   c) Select **vSphere DRS** and click **Edit**.
   d) Expand **Advanced Options** and click **Add**.
   e) In the Option column, type `ClearIoStatsOnSdrsRun`.
   f) In the corresponding Value text box, type `1`.
   g) Click **OK**.
2. Run Storage DRS on the datastore cluster.

   The current Storage DRS statistics for all datastores and virtual disks in all datastore clusters in the vSphere Client inventory are cleared, but no new statistics are collected.
3. Change the `ClearIoStatsOnSdrsRun` flag value to `0` to deactivate it.
4. Run Storage DRS again.

   Storage DRS runs normally. Allow several hours for the new setting to take effect.

Storage vMotion Compatibility with Datastore Clusters

A datastore cluster has certain vSphere Storage vMotion® requirements.

- The host must be running a version of ESX that supports Storage vMotion.
- The host must have write access to both the source datastore and the destination datastore.
- The host must have enough free memory resources to accommodate Storage vMotion.
- The destination datastore must have sufficient disk space.
- The destination datastore must not be in maintenance mode or entering maintenance mode.

Using NUMA Systems with ESX

ESX supports memory access optimization for Intel and AMD Opteron processors in server architectures that support NUMA (non-uniform memory access).

After you understand how ESX NUMA scheduling is performed and how the VMware NUMA algorithms work, you can specify NUMA controls to optimize the performance of your virtual machines.

What is NUMA?

NUMA systems are advanced server platforms with more than one system bus. They can harness large numbers of processors in a single system image with superior price to performance ratios.

A multi-gigahertz CPU needs to be supplied with a large amount of memory bandwidth to use its processing power effectively. Even a single CPU running a memory-intensive workload, such as a scientific computing application, can be constrained by memory bandwidth.

This problem is amplified on symmetric multiprocessing (SMP) systems, where many processors must compete for bandwidth on the same system bus. Some high-end systems often try to solve this problem by building a high-speed data bus. However, such a solution is expensive and limited in scalability.

NUMA is an alternative approach that links several small, cost-effective nodes using a high-performance connection. Each node contains processors and memory, much like a small SMP system. However, an advanced memory controller allows a node to use memory on all other nodes, creating a single system image. When a processor accesses memory that does not lie within its own node (remote memory), the data must be transferred over the NUMA connection, which is slower than accessing local memory. Memory access times are not uniform and depend on the location of the memory and the node from which it is accessed, as the technology's name implies.

Challenges for Operating Systems

Because a NUMA architecture provides a single system image, it can often run an operating system with no special optimizations.

The high latency of remote memory accesses can leave the processors under-utilized, constantly waiting for data to be transferred to the local node, and the NUMA connection can become a bottleneck for applications with high-memory bandwidth demands.

Furthermore, performance on such a system can be highly variable. It varies, for example, if an application has memory located locally on one benchmarking run, but a subsequent run happens to place all of that memory on a remote node. This phenomenon can make capacity planning difficult.

Some high-end UNIX systems provide support for NUMA optimizations in their compilers and programming libraries. This support requires software developers to tune and recompile their programs for optimal performance. Optimizations for one system are not guaranteed to work well on the next generation of the same system. Other systems have allowed an administrator to explicitly decide on the node on which an application should run. While this might be acceptable for certain applications that demand 100 percent of their memory to be local, it creates an administrative burden and can lead to imbalance between nodes when workloads change.

Ideally, the system software provides transparent NUMA support, so that applications can benefit immediately without modifications. The system should maximize the use of local memory and schedule programs intelligently without requiring constant administrator intervention. Finally, it must respond well to changing conditions without compromising fairness or performance.

How ESX NUMA Scheduling Works

ESX uses a sophisticated NUMA scheduler to dynamically balance processor load and memory locality or processor load balance.

1. Each virtual machine managed by the NUMA scheduler is assigned a home node. A home node is one of the system's NUMA nodes containing processors and local memory, as indicated by the System Resource Allocation Table (SRAT).
2. When memory is allocated to a virtual machine, the ESX host preferentially allocates it from the home node. The virtual CPUs of the virtual machine are constrained to run on the home node to maximize memory locality.
3. The NUMA scheduler can dynamically change a virtual machine's home node to respond to changes in system load. The scheduler might migrate a virtual machine to a new home node to reduce processor load imbalance. Because this might cause more of its memory to be remote, the scheduler might migrate the virtual machine's memory dynamically to its new home node to improve memory locality. The NUMA scheduler might also swap virtual machines between nodes when this improves overall memory locality.

Some virtual machines are not managed by the ESX NUMA scheduler. For example, if you manually set the processor or memory affinity for a virtual machine, the NUMA scheduler might not be able to manage this virtual machine. Virtual machines that are not managed by the NUMA scheduler still run correctly. However, they do not benefit from ESX NUMA optimizations.

The NUMA scheduling and memory placement policies in ESX can manage all virtual machines transparently, so that administrators do not need to address the complexity of balancing virtual machines between nodes explicitly.

The optimizations work seamlessly regardless of the type of guest operating system. ESX provides NUMA support even to virtual machines that do not support NUMA hardware, such as Windows NT 4.0. As a result, you can take advantage of new hardware even with legacy operating systems.

A virtual machine that has more virtual processors than the number of physical processor cores available on a single hardware node can be managed automatically. The NUMA scheduler accommodates such a virtual machine by having it span NUMA nodes. That is, it is split up as multiple NUMA clients, each of which is assigned to a node and then managed by the scheduler as a normal, non-spanning client. This can improve the performance of certain memory-intensive workloads with high locality. For information on configuring the behavior of this feature, see Advanced Virtual Machine Attributes.

ESX includes support for exposing virtual NUMA topology to guest operating systems. For more information about virtual NUMA control, see Using Virtual NUMA.

VMware NUMA Optimization Algorithms and Settings

This section describes the algorithms and settings used by ESX to maximize application performance while still maintaining resource guarantees.

Home Nodes and Initial Placement

When a virtual machine is powered on, ESX assigns it a home node. A virtual machine runs only on processors within its home node, and its newly allocated memory comes from the home node as well.

Unless a virtual machine's home node changes, it uses only local memory, avoiding the performance penalties associated with remote memory accesses to other NUMA nodes.

When a virtual machine is powered on, it is assigned an initial home node so that the overall CPU and memory load among NUMA nodes remains balanced. Because internode latencies in a large NUMA system can vary greatly, ESX determines these internode latencies at boot time and uses this information when initially placing virtual machines that are wider than a single NUMA node. These wide virtual machines are placed on NUMA nodes that are close to each other for the lowest memory access latencies.

Initial placement-only approaches are usually sufficient for systems that run only a single workload, such as a benchmarking configuration that remains unchanged as long as the system is running. However, this approach is unable to guarantee good performance and fairness for a datacenter-class system that supports changing workloads. Therefore, in addition to initial placement, ESX does dynamic migration of virtual CPUs and memory between NUMA nodes for improving CPU balance and increasing memory locality.

Dynamic Load Balancing and Page Migration

ESX combines the traditional initial placement approach with a dynamic rebalancing algorithm. Periodically (every two seconds by default), the system examines the loads of the various nodes and determines if it should rebalance the load by moving a virtual machine from one node to another.

This calculation takes into account the resource settings for virtual machines and resource pools to improve performance without violating fairness or resource entitlements.

The rebalancer selects an appropriate virtual machine and changes its home node to the least loaded node. When it can, the rebalancer moves a virtual machine that already has some memory located on the destination node. From that point on (unless it is moved again), the virtual machine allocates memory on its new home node and it runs only on processors within the new home node.

Rebalancing is an effective solution to maintain fairness and ensure that all nodes are fully used. The rebalancer might need to move a virtual machine to a node on which it has allocated little or no memory. In this case, the virtual machine incurs a performance penalty associated with a large number of remote memory accesses. ESX can eliminate this penalty by transparently migrating memory from the virtual machine's original node to its new home node:

1. The system selects a page (4 KB of contiguous memory) on the original node and copies its data to a page in the destination node.
2. The system uses the virtual machine monitor layer and the processor's memory management hardware to seamlessly remap the virtual machine's view of memory, so that it uses the page on the destination node for all further references, eliminating the penalty of remote memory access.

When a virtual machine moves to a new node, the ESX host immediately begins to migrate its memory in this fashion. It manages the rate to avoid overtaxing the system, particularly when the virtual machine has little remote memory remaining or when the destination node has little free memory available. The memory migration algorithm also ensures that the ESX host does not move memory needlessly if a virtual machine is moved to a new node for only a short period.

When initial placement, dynamic rebalancing, and intelligent memory migration work in conjunction, they ensure good memory performance on NUMA systems, even in the presence of changing workloads. When a major workload change occurs, for instance when new virtual machines are started, the system takes time to readjust, migrating virtual machines and memory to new locations. After a short period, typically seconds, or minutes, the system completes its readjustments and reaches a steady state.

Transparent Page Sharing Optimized for NUMA

Many ESX workloads present opportunities for sharing memory across virtual machines.

You might have multiple virtual machines running instances of the same guest operating system, have the same applications or components loaded, or contain common data. In such cases, ESX systems use a proprietary transparent

page-sharing technique to eliminate redundant copies of memory pages. With memory sharing, a workload running in virtual machines often consumes less memory than it might consume when running on physical machines. As a result, higher levels of overcommitment can be supported efficiently.

Transparent page sharing for ESX systems has also been optimized for use on NUMA systems. On NUMA systems, pages are shared per-node, so each NUMA node has its own local copy of heavily shared pages. When virtual machines use shared pages, they do not need access to remote memory.

Note:

This default behavior is the same in all previous versions of ESX.

Resource Management in NUMA Architectures

You can perform resource management with different types of NUMA architecture.

With the proliferation of highly multicore systems, NUMA architectures are becoming more popular as these architectures allow better performance scaling of memory intensive workloads. All modern Intel and AMD systems have NUMA support built into the processors. Additionally, there are traditional NUMA systems like the IBM Enterprise X-Architecture that extend Intel and AMD processors with NUMA behavior with specialized chipset support.

Typically, you can use BIOS settings to activate and deactivate NUMA behavior. For example, in AMD Opteron-based HP Proliant servers, NUMA can be deactivated by activating node interleaving in the BIOS. If NUMA is activated, the BIOS builds a system resource allocation table (SRAT) which ESX uses to generate the NUMA information used in optimizations. For scheduling fairness, NUMA optimizations are not activated for systems with too few cores per NUMA node or too few cores overall. You can modify the `numa.rebalancecorestotal` and `numa.rebalancecoresnode` options to change this behavior.

Using Virtual NUMA

vSphere includes support for exposing virtual NUMA topology to guest operating systems, which can improve performance by facilitating guest operating system and application NUMA optimizations.

Virtual NUMA topology is available to virtual machines and is activated by default when the number of virtual CPUs is greater than eight. You can also manually influence virtual NUMA topology using advanced configuration options.

The first time a virtual NUMA activated virtual machine is powered on, its virtual NUMA topology is based on the NUMA topology of the underlying physical host. Once a virtual machines virtual NUMA topology is initialized, it does not change unless the number of vCPUs in that virtual machine is changed.

The virtual NUMA topology does not consider the memory that is configured to a virtual machine. The virtual NUMA topology is not influenced by the number of virtual sockets and number of cores per socket for a virtual machine.

If the virtual NUMA topology must be overridden, see Virtual NUMA Controls.

Note:

Enabling CPU HotAdd deactivates virtual NUMA. For more information, see Broadcom knowledge base article 321931.

Virtual Topology in ESX

ESX 8.0 and later includes an enhanced virtual topology feature.

The virtual topology of a VM enables optimization within GOS for placement and load balancing. Selecting an accurate virtual topology that aligns with the underlying physical topology of the host where the VM is running is crucial for application performance.

ESX automatically selects optimal `coresPerSocket` for a VM and optimal virtual L3 size. It also includes a new virtual motherboard layout to expose NUMA for virtual devices and vNUMA topology when CPU hotplug is enabled.

Note:

Enhanced virtual topology is only available on ESX 8.0 and later. VMs must have hardware version 20 or later to use this feature.

1. To manually configure VM topology, first browse to the VM.
2. Select **VM Options**. Under **CPU Topology** you can adjust **Cores per socket** and **NUMA nodes**.
   In order to enable a hot-added CPU to a new NUMA node, add `numa.allowHotadd` under advanced configuration options. Then you can manually add a NUMA configuration.

   Note:
   By default, a VM with CPU hotplug enabled will enforce single NUMA node topology. Any hot-added CPUs go to the single NUMA node.
3. Under **Device Assignment** you can also assign a device to a virtual NUMA node, or leave it unassigned.

This newly configured topology is displayed as **Manual** in the existing VM topology section. Without manual configuration, this tab is displayed as **assigned at power on**.

Virtual NUMA Controls

For virtual machines with disproportionately large memory consumption, you can use advanced options to override the default virtual CPU settings.

You can add these advanced options to the virtual machine configuration file.

Table 310: Advanced Options for Virtual NUMA Controls

Option	Description	Default Value
`cpuid.coresPerSocket`	Determines the number of virtual cores per virtual CPU socket. This option does not affect the virtual NUMA topology unless numa.vcpu.followcorespersocket is configured. **Note:** ESX 8.0 automatically selects optimal `coresPerSocket` for a VM, the default value is displayed as 0.	1
`numa.vcpu.maxPerVirtualNode`	Determines the number of virtual NUMA nodes by splitting the total vCPU count evenly with this value as its divisor.	8
`numa.autosize.once`	When you create a virtual machine template with these settings, the settings remain the same every time you then power on the virtual machine with the default value TRUE. If the value is set to FALSE, the virtual NUMA topology is updated every time it is powered on. The virtual NUMA topology is reevaluated when the configured number of virtual CPUs on the virtual machine is modified at any time.	FALSE
`numa.vcpu.min`	The minimum number of virtual CPUs in a virtual machine that are required to generate a virtual NUMA topology. A virtual machine is always UMA when its size is smaller than numa.vcpu.min	9
`numa.vcpu.followcorespersocket`	If set to 1, reverts to the old behavior of virtual NUMA node sizing being tied to cpuid.coresPerSocket.	0
`numa.allowHotadd`	In order to activate the capability of hot-adding CPU to a new NUMA node, add `numa.allowHotadd` under advanced configuration options. Then you can manually add a NUMA configuration while CPU hotadd is activated.	FALSE

Option	Description	Default Value
numa.vcpu.coresPerNode	VMX parameter to configure size of virtual NUMA node, which is interpreted from the UI reconfiguration. This parameter is only effective for HWv20. Defaults to 0, which indicates ESX automatically selects vNUMA size. **Note:** If this option conflicts with numa.vcpu.maxPerVirtualNode, the VM cannot power on.	0
vcpu.hotadd	When this option is TRUE, virtual NUMA is deactivated. The VM always sees one virtual NUMA node when VM is capable of CPU hotplug.	
llc.multiLLCPerSocket	VM exposes truthful vLLC on AMD Epyc when this option is TRUE. The exposed vLLC size can be different than virtual socket size.	FALSE
llc.size.vcpu	Manually configured number of vcpus for vLLC on AMD Epyc. The value is ignored if it is incompatible with other settings of the VM.	
chipset.motherboardLayout	Type of virtual motherboard used by this VM. It can only have one of the following two values: acpi: New motherboard layout starting from HWv 20. i440bx: Legacy motherboard layout	
cpuid.coresPerSocket.cookie	This is a vmx entry generated by ESX to store the automatically generated coresPerSocket value. It is there to ensure vMotion consistency. Do not change or remove it manually.	

Specifying NUMA Controls

If you have applications that use a lot of memory or have a small number of virtual machines, you might want to optimize performance by specifying virtual machine CPU and memory placement explicitly.

Specifying controls is useful if a virtual machine runs a memory-intensive workload, such as an in-memory database or a scientific computing application with a large data set. You might also want to optimize NUMA placements manually if the system workload is known to be simple and unchanging. For example, an eight-processor system running eight virtual machines with similar workloads is easy to optimize explicitly.

Note:
In most situations, the ESX host's automatic NUMA optimizations result in good performance.

ESX provides three sets of controls for NUMA placement, so that administrators can control memory and processor placement of a virtual machine.

You can specify the following options.

NUMA Node Affinity	When you set this option, NUMA can schedule a virtual machine only on the nodes specified in the affinity.
CPU Affinity	When you set this option, a virtual machine uses only the processors specified in the affinity.
Memory Affinity	When you set this option, the server allocates memory only on the specified nodes.

A virtual machine is still managed by NUMA when you specify NUMA node affinity, but its virtual CPUs can be scheduled only on the nodes specified in the NUMA node affinity. Likewise, memory can be obtained only from the nodes specified in the NUMA node affinity. When you specify CPU or memory affinities, a virtual machine ceases to be managed by NUMA. NUMA management of these virtual machines is effective when you remove the CPU and memory affinity constraints.

Manual NUMA placement might interfere with ESX resource management algorithms, which distribute processor resources fairly across a system. For example, if you manually place 10 virtual machines with processor-intensive workloads on one node, and manually place only 2 virtual machines on another node, it is impossible for the system to give all 12 virtual machines equal shares of systems resources.

Associate Virtual Machines with Specific Processors

You might be able to improve the performance of the applications on a virtual machine by pinning its virtual CPUs to fixed processors. This allows you to prevent the virtual CPUs from migrating across NUMA nodes.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and click **Edit Settings**.
3. Select the **Virtual Hardware** tab, and expand **CPU**.
4. Under Scheduling Affinity, set the CPU affinity to the preferred processors.

   Note:
   You must manually select all processors in the NUMA node. CPU affinity is specified on a per-processor, not on a per-node, basis.

Associate Memory Allocations with Specific NUMA Nodes Using Memory Affinity

You can specify that all future memory allocations on a virtual machine use pages associated with specific NUMA nodes (also known as manual memory affinity).

Note:
Specify nodes to be used for future memory allocations only if you have also specified CPU affinity. If you make manual changes only to the memory affinity settings, automatic NUMA rebalancing does not work properly.

1. Browse to the virtual machine in the vSphere Client.
2. Click the **Configure** tab.
3. Click **Settings**, and click **VM Hardware**.
4. Click **Edit**.
5. Select the **Virtual Hardware** tab, and expand **Memory**.
6. Under NUMA Memory Affinity, set memory affinity.

   Binding a Virtual Machine to a Single NUMA Node

   The following example illustrates manually binding the last four physical CPUs to a single NUMA node for a two-way virtual machine on an eight-way server.

   The CPUs (for example, 4, 5, 6, and 7) are the physical CPU numbers.

   1. In the vSphere Client, right-click the virtual machine and select **Edit Settings**.
   2. Click the **Advanced Parameters** tab.
   3. In the vSphere Client, turn on CPU affinity for processors 4, 5, 6, and 7.

   Then, you want this virtual machine to run only on node 1.

   1. In the vSphere Client inventory panel, select the virtual machine and select **Edit Settings**.
   2. Click the **Advanced Parameters** tab.
   3. In the vSphere Client, set memory affinity for the NUMA node to 1.

   Completing these two tasks ensures that the virtual machine runs only on NUMA node 1 and, when possible, allocates memory from the same node.

Associate Virtual Machines with Specified NUMA Nodes

When you associate a NUMA node with a virtual machine to specify NUMA node affinity, you constrain the set of NUMA nodes on which ESX can schedule a virtual machine's virtual CPU and memory.

> **Note:**
> When you constrain NUMA node affinities, you might interfere with the ability of the ESX NUMA scheduler to rebalance virtual machines across NUMA nodes for fairness. Specify NUMA node affinity only after you consider the rebalancing issues.

1. Browse to the VM in the vSphere Client.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click the **Advanced Parameters** tab.
4. Optional: To add a parameter, enter an attribute name and value for the parameter and click **Add**.
5. Optional: To change a parameter, enter a new value in the **Value** text box for that parameter and click the check mark to confirm.
6.
   - To specify NUMA node for the virtual machine, in the Name column, enter `numa.nodeAffinity`.
   - To specify NUMA node for a specific Virtual NUMA node on the virtual machine, in the Name column, enter `sched.nodeX.affinity`, where X is the Virtual NUMA node number. For example, `sched.node0.affinity` specifies Virtual NUMA node 0 on the virtual machine.
7. In the Value column, enter the NUMA nodes where the virtual machine or the virtual NUMA node can be scheduled.
   Use a comma-separated list for multiple nodes. For example, enter `0,1` to constrain the virtual machine resource scheduling to NUMA nodes 0 and 1.
8. Click **OK**.

Advanced Attributes in vSphere

You can set advanced attributes for ESX hosts or individual virtual machines to help you customize resource management.

In most cases, adjusting the basic resource allocation settings (reservation, limit, shares) or accepting default settings results in appropriate resource allocation. However, you can use advanced attributes to customize resource management for a host or a specific virtual machine.

For more information about setting advanced attributes for ESX hosts, see the *vCenter and Host Management* documentation.

Set Advanced Virtual Machine Attributes

You can set advanced attributes for a virtual machine.

Power off the VM before setting advanced attributes.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and select **Edit Settings**.
3. Click the **Advanced Parameters** tab.
4. Optional: To add a parameter, enter an attribute name and value for the parameter and click **Add**.
5. Optional: To change a parameter, enter a new value in the **Value** text box for that parameter and click the check mark to confirm.
6. Click **OK**.

Advanced Virtual Machine Attributes

You can use the advanced virtual machine attributes to customize virtual machine configuration.

Table 311: Advanced Virtual Machine Attributes

Attribute	Description	Default
sched.mem.maxmemctl	Maximum amount of memory reclaimed from the selected virtual machine by ballooning, in megabytes (MB). If the ESX host needs to reclaim additional memory, it is forced to swap. Swapping is less desirable than ballooning.	-1 (Unlimited)
sched.mem.pshare.enable	Enables memory sharing for a selected virtual machine. This boolean value defaults to True. If you set it to False for a virtual machine, this turns off memory sharing.	True
sched.mem.pshare.salt	A salt value is a configurable VMX option for each virtual machine. If this option is not present in the virtual machine's VMX file, then the value of `vc.uuid vmx` option is taken as the default value. Since the `vc.uuid` is unique to each virtual machine, by default transparent page sharing happens only among the pages belonging to a particular virtual machine (intra-VM). If a group of virtual machines are considered trustworthy, it is possible to share pages among them by setting a common salt value for all of those virtual machines (inter-VM).	user configurable
sched.swap.persist	Specifies whether the virtual machine's swap files should persist or be deleted when the virtual machine is powered off. By default, the system creates the swap file for a virtual machine when the virtual machine is powered on, and deletes the swap file when the virtual machine is powered off.	False

Attribute	Description	Default
sched.swap.dir	Directory location of the virtual machine's swap file. Defaults to the virtual machine's working directory, that is, the directory that contains its configuration file. This directory must remain on a host that is accessible to the virtual machine. If you move the virtual machine (or any clones created from it), you might need to reset this attribute.	Equals `workingDir`

Advanced Virtual NUMA Attributes

You can use the advanced virtual NUMA attributes to customize virtual NUMA usage.

Table 312: Advanced NUMA Attributes

Attribute	Description	Default
cpuid.coresPerSocket	Determines the number of virtual cores per virtual CPU socket. If the value is greater than 1, also determines the size of virtual NUMA nodes if a virtual machine has a virtual NUMA topology. You can set this option if you know the exact virtual NUMA topology for each physical host.	1
numa.autosize	When you set this option, the virtual NUMA topology has the same number of virtual CPUs per virtual node as there are cores on each physical node.	FALSE
numa.autosize.once	When you create a virtual machine template with these settings, the settings are guaranteed to remain the same every time you subsequently power on the virtual machine. The virtual NUMA topology will be reevaluated if the configured number of virtual CPUs on the virtual machine is modified.	TRUE
numa.vcpu.maxPerVirtualNode	If `cpuid.coresPerSocket` is too restrictive as a power of two, you can set `numa.vcpu.maxPerVirtualNode` directly. In this case, do not set `cpuid.coresPerSocket`.	8
numa.vcpu.min	Minimum number of virtual CPUs in a virtual machine that are required in order to generate a virtual NUMA topology.	9
numa.vcpu.maxPerMachineNode	Maximum number of virtual CPUs that belong to the same virtual machine that can be scheduled on a NUMA node at the same time. Use this attribute to ensure maximum bandwidth, by forcing different NUMA clients on different NUMA nodes.	Number of cores per node on the physical host where a virtual machine is running.
numa.vcpu.maxPerClient	Maximum number of virtual CPUs in a NUMA client. A client is a group of virtual CPUs that are NUMA-managed as a single entity. By default, each virtual NUMA node is a NUMA client, but if a virtual NUMA node is larger than a physical NUMA node, a single virtual NUMA node can be backed by multiple NUMA clients.	Equals `numa.vcpu.maxPerMachineNode`
numa.nodeAffinity	Constrains the set of NUMA nodes on which a virtual machine's virtual CPU and memory can be scheduled. **Note:** When you constrain NUMA node affinities, you might interfere with the ability of the NUMA scheduler to rebalance virtual machines across NUMA nodes for fairness. Specify NUMA node affinity only after you consider the rebalancing issues.	

Attribute	Description	Default
numa.mem.interleave	Specifies whether the memory allocated to a virtual machine is statically interleaved across all the NUMA nodes on which its constituent NUMA clients are running and there is no virtual NUMA topology exposed.	True

Latency Sensitivity

You can adjust the latency sensitivity of a virtual machine to optimize the scheduling delay for latency sensitive applications.

ESX is optimized to deliver high throughput. You can optimize your virtual machine to meet the low latency requirement of latency sensitive applications. Examples of latency sensitive applications are VOIP or media player applications, or applications that require frequent access to the mouse or keyboard devices.

Adjust Latency Sensitivity

You can adjust the latency sensitivity of a virtual machine.

ESX requires full CPU and memory reservation to power on a VM with hardware version 14 when Latency Sensitivity is set to **High**. If you do not set these reservations, the VM fails to power on.

1. Browse to the virtual machine in the vSphere Client.
   a) To find a virtual machine, select a data center, folder, cluster, resource pool, or host.
   b) Click the **VMs** tab.
2. Right-click the virtual machine and click **Edit Settings**.
3. Click **VM Options** and click **Advanced**.
4. Select a setting from the **Latency Sensitivity** drop-down menu.
5. Click **OK**.

Virtual Hyperthreading support for VMs

Virtual Hyperthreading (vHT) is supported for VMs.

In vSphere 8.0 and later, vHT is supported for VMs. vHT is deactivated by default and can be activated under the latency sensitivity settings for each VM. vHT supports a maximum HT size of 2.

vHT is an extension of latency sensitivity high feature. Applications which benefit from hyperthreading awareness can see performance improvements with latency sensitivity high and vHT activated. The performance gain might come from sufficient resource reservation and that the VM has exclusive physical CPUs.

Without vHT activated on ESX, each virtual CPU (vCPU) is equivalent to a single non-hyperthreaded core available to the guest operating system. With vHT activated, each guest vCPU is treated as a single hyperthread of a virtual core (vCore).

Virtual hyperthreads of the same vCore occupy the same physical core. As a result, vCPUs of the VM can share the same core as opposed to using multiple cores on VMs with latency sensitivity high that have vHT deactivated.

ESX hosts and VM running older hardware versions cannot use this feature.

vHT Full CPU Reservation

You can use a formula to calculate the full CPU reservation for vHT.

For a low latency VM without vHT, each vCPU of the VM has exclusively affinity to a thread of the physical core. For hosts with Hyperthreading activated, the partner hyperthread has exclusive affinity to an idle world. Each vCPU of a low latency VM is allocated a dedicated physical core.

The CPU reservation for a low latency VM is computed as:

```
Low latency VM (without vHT)
CPU minimum reservation = numVcpus * cpuFrequency
```

However, when vHT is activated for a VM, each hypertwin of the physical core is shared between multiple vCPUs of the VM where each hypertwin has exclusive affinity to a vCPU of the VM. This means a core with numSMT physical hyperthreads is shared by numSMT virtual threads. In this case, the CPU reservation requirement would be computed as:

```
Low latency VM (with vHT)
CPU minimum reservation = (numVcpus / numSMT) * cpuFrequency
```

Table 313: Example of booting a low latency VM with 20 vCPUs on a host with CPU frequency as 2 GHz

	numSMT = 1(without vHT)	numSMT = 2(with vHT)
numVcpus	20	20
number of Physical cores	20	10 (each core shared by 2 vCPUs)
Required minimum CPU reservation	20 * 2.0 GHz = 40 GHz	(20/2) * 2.0 GHz = 20 GHz

Activate vHT for VMs

vHT is supported with ESX 8.0, but it is deactivated by default. You can activate vHT under the latency sensitivity settings for each VM.

When vHT is activated, CPU and Memory must be set to the full reservation. A warning is displayed if you set the reservation lower.

1. In vSphere Client, select a VM.
2. Select **Actions** and click **Edit Settings**.
3. Under Latency Sensitivity, click the drop-down menu and select **High with Hyperthreading**.
4. Click **OK**.

vHT is activated.

Reliable Memory

ESX supports reliable memory.

Some systems have reliable memory, which is a part of memory that is less likely to have hardware memory errors than other parts of the memory in the system. If the hardware exposes information about the different levels of reliability, ESX might be able to achieve higher system reliability.

View Reliable Memory

You can view whether or not the license permits reliable memory.

1. Browse to the host in the vSphere Client.
2. Click the **Configure** tab and click **System**.
3. Select **Licensing**.
4. Under **Licensed Features** verify Reliable Memory is displayed.

You can look up how much memory is considered reliable by using the ESXCLI `hardware memory get` command.

Backing Guest vRAM with 1 GB Pages

vSphere ESX provides a limited support for backing guest vRAM with 1 GB pages.

In order to use 1 GB pages for backing guest memory you must apply the option `sched.mem.lpage.enable1GPage = "TRUE"` for the VM. You can set this under Advanced options when you select **Edit Settings**. You can only enable 1 GB pages on a VM that is powered off.

A VM with 1 GB pages enabled must have full memory reservation. Otherwise, the VM will not be able to power on. All of the vRAM for VMs with 1 GB pages enabled is preallocated on power-on. Since these VMs have full memory reservation they are not affected by memory reclamation and their memory consumption stays at the maximum level for the entire lifetime of the VM.

1 GB page vRAM backing is opportunistic and 1 GB pages are allocated on a best effort basis. This includes cases where host CPUs do not have 1 GB capabilities. To maximize the chances of having guest vRAM backed with 1 GB pages, we recommended to start VMs requiring 1 GB pages on a freshly booted host because over time the host RAM is fragmented.

A VM with 1 GB pages enabled can be migrated to a different host. However, the 1 GB page size might not be allocated on the destination host in the same amount as it was on the source host. You might also see part of vRAM backed with a 1 GB page on the source host is no longer backed with a 1 GB page on the destination host.

The opportunistic nature of 1 GB pages extends to vSphere services such as HA and DRS that might not preserve 1 GB page vRAM backing. These services are not aware of the 1GB capabilities of the destination host and do not take 1GB memory backing into account while making placement decisions.

Fault Definitions

DRS faults indicate the reasons that prevent the generation of DRS actions (or the recommendation of those actions in manual mode).

The DRS faults are defined within this section.

> **Note:**
> In this chapter, "Memory" can refer to physical RAM or Persistent Memory.

Virtual Machine is Pinned

This fault occurs when DRS cannot move a virtual machine because DRS is deactivated on it. That is, the virtual machine is "pinned" on its registered host.

Virtual Machine not Compatible with any Host

This fault occurs when DRS cannot find a host that can run the virtual machine.

This might occur, for example, if no host can satisfy the virtual machine's CPU or memory resource needs or if no host currently has network or storage access needed by the virtual machine.

To address this problem, provide a host that can meet the virtual machine's requirements.

VM/VM DRS Rule Violated when Moving to another Host

This fault occurs when more than one virtual machines running on the same host and share affinity rules with each other cannot be moved to another host.

This might occur because not all the virtual machines can vMotion off the current host. For example, one of the virtual machines in the group is DRS-deactivated.

To prevent this, check for reasons why some virtual machines in the group cannot vMotion.

Host Incompatible with Virtual Machine

This fault occurs when DRS considers migrating a virtual machine to a host, but finds that the host is incompatible with the given virtual machine.

This might occur because the target host does not have access to the network or storage connection needed by the virtual machine. Another reason this fault occurs is if the target host has a CPU that differs sufficiently from the current host so that using vMotion amongst the hosts is not supported.

To avoid this, create clusters such that all hosts are configured consistently and vMotion is compatible amongst the hosts.

Another reason the host is incompatible with the virtual machine is that there is a required VM/Host DRS rule in place that instructs DRS to never place this virtual machine on this host.

Host Has Virtual Machine That Violates VM/VM DRS Rules

This fault occurs when the virtual machine, when powered on or moved by starting vMotion, might violate a VM/VM DRS rule.

The virtual machine can still be manually powered on or moved with vMotion, but vCenter cannot automatically do so.

Host has Insufficient Capacity for Virtual Machine

This fault occurs when the host does not have enough CPU or memory capacity for running the virtual machine.

Host in Incorrect State

This fault occurs when the host is entering maintenance or standby state when needed for DRS action to occur.

To address this fault, cancel the request for the host to enter standby or maintenance mode.

Host Has Insufficient Number of Physical CPUs for Virtual Machine

This fault occurs when the host hardware does not enough CPUs (hyperthreads) to support the number of virtual CPUs in the virtual machine.

Host has Insufficient Capacity for Each Virtual Machine CPU

This fault occurs when the host does not have enough CPU capacity for running the virtual machine.

The Virtual Machine Is in vMotion

This fault occurs when DRS cannot move a virtual machine because it is in vMotion.

No Active Host in Cluster

This fault occurs when the cluster in which the virtual machine is being moved does not contain any hosts that are connected and in a non-maintenance state.

This can occur, for example, when all the hosts are disconnected or in maintenance mode.

Insufficient Resources

This fault occurs when an attempted operation conflicts with a resource configuration policy.

This fault may occur, for example, if a power-on operation reserves more memory than is allocated to a resource pool.

Retry the operation after adjusting the resources to allow more memory.

Insufficient Resources to Satisfy Configured Failover Level for HA

This fault occurs when the HA configuration of CPU or memory resources reserved for failover is violated or is insufficient for the DRS operation.

This fault is reported when:

- The host is requested to enter maintenance or standby mode.
- The virtual machine violates failover when it attempts to power on.

No Compatible Hard Affinity Host

No host is available for the virtual machine that satisfies its mandatory VM/Host DRS affinity or anti-affinity rules.

No Compatible Soft Affinity Host

No host is available for the virtual machine that satisfied its preferred VM/Host DRS affinity or anti-affinity rules.

Soft Rule Violation Correction Disallowed

DRS migration threshold is set at mandatory-only.

This does not allow the generation of DRS actions to correct non-mandatory VM/Host DRS affinity rules.

Soft Rule Violation Correction Impact

Correcting the non-mandatory VM/Host DRS affinity rule does not occur because it impacts performance.

DRS Troubleshooting Information

This information describes vSphere® Distributed Resource Scheduler (DRS) problems for particular categories: cluster, host, and virtual machine problems.

> **Note:**
> In this chapter, "Memory" can refer to physical RAM or Persistent Memory.

Cluster Problems

Cluster problems can prevent DRS from performing optimally or from reporting faults.

Load Imbalance on Cluster

A cluster has a load imbalance of resources.

A cluster might become unbalanced because of uneven resource demands from virtual machines and unequal capacities of hosts.

The following are possible reasons why the cluster has a load imbalance:

- The migration threshold is too high.
  A higher threshold makes the cluster a more likely candidate for load imbalance.
- VM/VM or VM/Host DRS rules prevent virtual machines from being moved.
- DRS is deactivated for one or more virtual machines.
- A device is mounted to one or more virtual machines preventing DRS from moving the virtual machine in order to balance the load.
- Virtual machines are not compatible with the hosts to which DRS would move them. That is, at least one of the hosts in the cluster is incompatible for the virtual machines that would be migrated. For example, if host A's CPU is not vMotion-compatible with host B's CPU, then host A becomes incompatible for powered-on virtual machines running on host B.
- It would be more detrimental for the virtual machine's performance to move it than for it to run where it is currently located. This may occur when loads are unstable or the migration cost is high compared to the benefit gained from moving the virtual machine.
- vMotion is not activated or set up for the hosts in the cluster.

Address the problem that is causing the load imbalance.

Cluster is Yellow

The cluster is yellow due to a shortage of resources.

If the cluster does not have enough resources to satisfy the reservations of all resource pools and virtual machines, but does have enough resources to satisfy the reservations of all running virtual machines, DRS continues to run and the cluster is yellow.

A cluster can become yellow if the host resources are removed from the cluster (for example, if a host fails).

Add host resources to the cluster or reduce the resource pool reservations.

Cluster is Red Because of Inconsistent Resource Pool

A DRS cluster becomes red when it is invalid. It may become red because the resource pool tree is not internally consistent.

If the cluster resource pool tree is not internally consistent (for example, the sum of the children's reservations is greater than the parent pool's nonexpandable reservation), the cluster does not have enough resources to satisfy the reservations of all running virtual machines making the cluster red.

This can occur if vCenter is unavailable or if resource pool settings are changed while a virtual machine is in a failover state.

Revert the associated changes or otherwise revise the resource pool settings.

Cluster Is Red Because Failover Capacity Is Violated

A DRS cluster becomes red when it is invalid. It may become red because failover capacity is violated.

The cluster attempts to fail over virtual machines if there is host failure, but is not guaranteed to have enough resources available to fail over all virtual machines covered by the failover requirements.

If a cluster enabled for HA loses so many resources that it can no longer fulfill its failover requirements, a message appears and the cluster's status changes to red.

Review the list of configuration issues in the yellow box at the top of the cluster Summary page and address the issue that is causing the problem.

No Hosts are Powered Off When Total Cluster Load is Low

Hosts are not powered off when the total cluster load is low.

Hosts are not powered off when the total cluster load is low because extra capacity is needed for HA failover reservations.

Hosts might not be powered off for the following reasons:

- The `MinPoweredOn{Cpu|Memory}Capacity` advanced options settings need to be met.
- Virtual machines cannot be consolidated onto fewer hosts due to their resource reservations, VM/Host DRS rules, VM/VM DRS rules, not being DRS-activated, or not being compatible with the hosts having available capacity.
- Loads are unstable.
- DRS migration threshold is at the highest setting and only allows mandatory moves.
- vMotion is unable to run because it is not configured.
- DPM is deactivated on the hosts that might be powered off.
- Hosts are not compatible for virtual machines to be moved to another host.
- Host does not have Wake On LAN, IPMI, or iLO technology. Either one is required for DPM to enter a host in standby.

Address the issue that prevents hosts from being powered off when the total cluster load is low.

Hosts Are Powered-off When Total Cluster Load Is High

Hosts are powered off when total cluster load is high.

DRS determined that virtual machines can be run on fewer hosts without degrading the host or virtual machine performance. DRS is also constrained from moving the virtual machines running on the highly utilized hosts to the hosts scheduled for power-off.

The total cluster load is too high.

Reduce the cluster load.

DRS Seldom or Never Performs vMotion Migrations

DRS seldom or never performs vMotion migrations.

DRS does not perform vMotion migrations.

DRS never performs vMotion migrations when one or more of the following issues is present on the cluster.

- DRS is deactivated on the cluster.
- The hosts do not have shared storage.
- The hosts in the cluster do not contain a vMotion network.
- DRS is manual and no one has approved the migration.

DRS seldom performs vMotion when one or more of the following issues is present on the cluster:

- Loads are unstable, or vMotion takes a long time, or both. A move is not appropriate.
- DRS seldom or never migrates virtual machines.
- DRS migration threshold is set too high.

DRS moves virtual machines for the following reasons:

- Evacuation of host that a user requested enter maintenance or standby mode.
- VM/Host DRS rules or VM/VM DRS rules.
- Reservation violations.
- Load imbalance.
- Power management.

Address the issues that are causing DRS to avoid performing vMotion migrations.

Host Problems

Host problems might cause DRS to not perform as expected.

DRS Recommends Host Be Powered on to Increase Capacity When Total Cluster Load Is Low

The host must be powered on to help provide more capacity for the cluster or help hosts that are overloaded.

DRS recommends that the host is powered on to increase capacity when the total cluster load is low.

The recommendation might be made because:

- The cluster is a DRS-HA cluster. Additional powered-on hosts are needed to provide more failover capability.
- Some hosts are overloaded and virtual machines on currently powered-on hosts can be moved to hosts in standby mode to balance the load.
- The capacity is necessary to meet the `MinPoweredOn{Cpu|Memory}Capacity` advanced options.

Power on the host.

Total Cluster Load Is High

The total cluster load is high.

When the total cluster load is high, DRS does not power-on the host.

The following are possible reasons why DRS does not power-on the host:

- VM/VM DRS rules or VM/Host DRS rules prevent the virtual machine from being moved to this host.
- Virtual machines are pinned to their current hosts, so DRS cannot move these virtual machines to hosts in standby mode to balance the load.
- DRS or DPM is in manual mode and the recommendations were not applied.
- No virtual machines on highly used hosts are moved to that host.
- DPM is deactivated on the host because of a user setting or host previously failing to exit standby.

Address that issue that prevents DRS from powering on the host.

Total Cluster Load Is Low

The total cluster load is low.

When the total cluster load is low, DRS does not power off the host.

The following are possible reasons why DRS does not power off the host:

- Distributed Power Management (DPM) detected better candidates to power off.
- vSphere HA needs extra capacity for failover.
- The load is not low enough to trigger the host to power off.
- DPM projects that the load will increase.
- DPM is not enabled for the host.
- DPM threshold is set too high.
- While DPM is enabled for the host, no suitable power-on mechanism is present for the host.
- DRS cannot evacuate the host.
- The DRS migration threshold is at the highest setting and only performs mandatory moves.

Address the issue that is preventing DRS from powering off the host.

DRS Does Not Evacuate a Host Requested to Enter Maintenance or Standby Mode

DRS does not evacuate a host requested to enter maintenance mode or standby mode.

When you attempt to put a host into maintenance or standby mode, DRS does not evacuate the host as expected.

vSphere HA is activated and evacuating this host might violate HA failover capacity.

There is no solution. If appropriate, deactivate vSphere HA before you attempt to put the host into maintenance mode or standby mode.

DRS Does Not Move Any Virtual Machines onto a Host

DRS does not move any virtual machines onto a host.

DRS does not recommend migration of virtual machine to a host that has been added to a DRS-activated cluster.

After a host has been added to a DRS-activated cluster, the virtual machines deployed to the host become part of the cluster. DRS can recommend migration of some virtual machines to this host just added to the cluster. If that does not occur, there may be problems with vMotion, host compatibility, or affinity rules. The following are possible reasons:

- vMotion is not configured or activated on this host.
- Virtual machines on other hosts are not compatible with this host.
- The host does not have sufficient resources for any virtual machine.
- Moving any virtual machines to this host would violate a VM/VM DRS rule or VM/Host DRS rule.
- This host is reserved for HA failover capacity.
- A device is mounted to the virtual machine.
- The vMotion threshold is too high.
- DRS is deactivated for the virtual machines, hence the virtual machine could not be moved onto the destination host.

Address the issue that prevents DRS from moving virtual machines onto a host.

DRS Does Not Move Any Virtual Machines from a Host

DRS does not move any virtual machines from a host.

Virtual machines are not moved from this host.

This may be because of problems with vMotion, DRS, or host compatibility. The following are the possible reasons:

- vMotion is not configured or activated on this host.
- DRS is deactivated for the virtual machines on this host.
- Virtual machines on this host are not compatible with any other hosts.

- No other hosts have sufficient resources for any virtual machines on this host.
- Moving any virtual machines from this host would violate a VM/VM DRS rule or VM/Host DRS rule.
- DRS is deactivated for one or more virtual machines on the host.
- A device is mounted to the virtual machine.

Address the issues that are preventing DRS from moving virtual machines from the host.

VM migrations initiated by DRS due to incompatibility between the VMs and their host

In a DRS activated cluster, you may observe large amounts of VM migrations initiated by DRS targeting only some hosts in the cluster.

DRS scans hosts in the cluster every minute for possible recommendations as a part of its load balancing workflow. The outcome relies on the results of compatibility checks of powered-on VMs against the hosts in the cluster. If there are compatibility constraints that narrow a VM's compatible set of hosts to only certain hosts in the cluster, then DRS will try to satisfy these constraints by migrating that VM to one of the compatible hosts.

The constraints typically come from two sources, static user configuration and runtime state change in the cluster. This article focuses on the runtime state change that could be unexpected to the users.

The runtime state change that can impact the compatibility between powered-on VMs and hosts can be in one of the following areas.

vSphere High Availability

In a HA activated cluster, all hosts are expected to have a healthy HA status. If a host does not have healthy HA status at some time, it will generate a compatibility failure during VM compat-check. Some examples of such situations are: HA agent unreachable, HA agent isolated, HA agent partitioned. For more information, see, Troubleshooting vSphere HA Host States.

These state changes typically accompany the following event in vCenter.

"vSphere HA agent on a host has an error"

Storage Accessibility

If a VM cannot access its configuration file (VMX file), virtual disks (VMDK) or swap file from its current host, it will fail the compat-check leading to the current host being incompatible. If a different host in the cluster still has accessibility to these files DRS could try to migrate the VM to that host. The outcome of such migration varies depending on the accessibility of VM’s VMX file from its current host. If the VM only loses accessibility to its VMDK but not the VMX file, the migration could succeed. If the VM loses accessibility to its VMX file, the migration can fail.

Network Accessibility

For environments with NSX-T, the NSX component status could go down on some hosts, or all hosts at different times. In versions of vCenter prior to 7.0 Update 2, this could lead to compatibility check failures for VMs and the affected hosts.

Resolution

1. If a user is planning to perform an operation that could potentially lead to the runtime state change discussed above, temporarily setting DRS to manual mode can avoid undesired migrations. Alternatively, a user can also temporarily set the DRS advanced option, VmsPerLBIteration, to 0. It asks DRS to scan no VM during its load balance workflow, hence no migration recommendation.
2. With vCenter 7.0 Update 1, DRS introduced an advanced option to tolerate a powered-on VM's incompatibility with its current host for a predefined time period, CompatCheckTransientFailureTimeSeconds. Users can configure this option to avoid undesired migrations due to transient incompatibility.

   Note:
   The default value of this option is 600 (10 minutes), which means DRS will not move a VM out just because of incompatibility with its current host if the incompatibility lasts for less than 10 mins. The maximum value of this option is 3600 (60 minutes).

With vCenter 7.0 Update 3 and 8.0 Update 1, this option can also be set to -1, which disallows DRS to move VMs out due to its incompatibility with its current host.

With vCenter 8.0 Update 3, the default value of this option is set to -1.

To set the DRS advanced option perform the following steps from the vSphere client:

1. Right-click the DRS cluster and click `Settings > vSphere DRS > Edit > Advanced Options > Add`
2. In the Option column enter the option name.
3. Click in the Value column to enter the desired value, then click OK for this setting to take effect.

Virtual Machine Problems

Virtual machine problems might cause DRS to not perform as expected.

Insufficient CPU or Memory Resources

The virtual machine does not receive enough CPU or memory resources.

In some cases, the virtual machine's demand is greater than its resource entitlement. When this occurs, the virtual machine doesn't receive enough CPU or memory resources.

The following sections describe the factors that influence the entitlement for a virtual machine.

Cluster is Yellow or Red

If the cluster is yellow or red, the capacity is insufficient to meet the resource reservations configured for all virtual machines and resource pools in the cluster. The particular virtual machine might be one that is not receiving its reservation. Check the status of the cluster (red or yellow) and resolve the situation.

Resource Limit is Too Restrictive

The virtual machine, its parent resource pool, or its resource pool ancestors might have a configured resource limit that is too restrictive. Check whether demand is equal to or greater than any configured limits.

Cluster is Overloaded

The cluster on which the virtual machine is running might have insufficient resources. Also, the virtual machine's share value is such that other virtual machines are granted proportionally more of the resources. To determine the demand is larger than the capacity, check the cluster statistics.

Host is Overloaded

To determine if the host's resources are oversubscribed, check the host statistics. If they are oversubscribed, consider why DRS is not moving any of the virtual machines now running on the host to other hosts. This condition might exist for the following reasons:

- The VM/VM DRS rules and VM/Host DRS rules require the current virtual machine-to-host mapping. If such rules are configured in the cluster, consider deactivating one or more of them. Then run DRS and check whether the situation is corrected.
- DRS cannot move this virtual machine or enough of the other virtual machines to other hosts to free up capacity. DRS will not move a virtual machine for any of the following reasons:
  - DRS is deactivated for the virtual machine.
  - A host device is mounted to the virtual machine.
  - Either of its resource reservations is so large that the virtual machine cannot run on any other host in the cluster.
  - The virtual machine is not compatible with any other host in the cluster.

  Check whether any of these conditions exist for the virtual machine. If none exist, the conditions might exist for other virtual machines in the cluster. If this is the case, DRS cannot balance the cluster to address the virtual machine's demand.
- Decrease the DRS migration threshold setting and check whether the situation is resolved.
- Increase the virtual machine's reservation.

Address the problem that is causing the virtual machine to not receive enough CPU or memory resources.

VM/VM DRS Rule or VM/Host DRS Rule Violated

DRS rules specify which host a virtual machine must or must not reside on, or which virtual machines must be or must not be on the same host.

A VM/VM DRS rule or a VM/Host DRS rule is violated.

VM/VM DRS rules specify that selected virtual machines should be placed on the same host (affinity) or that virtual machines be placed on different hosts (anti-affinity). VM/Host DRS rules specify that selected virtual machines should be placed on specified hosts (affinity) or that selected virtual machines should not be placed on specified hosts (anti-affinity).

When a VM/VM DRS rule or VM/Host DRS rule is violated, it might be because DRS cannot move some or all of the virtual machines in the rule. The reservation of the virtual machine or other virtual machines in the affinity rule, or their parent resource pools, might prevent DRS from locating all virtual machines on the same host.

- Check the DRS faults panel for faults associated with affinity rules.
- Compute the sum of the reservations of all the virtual machines in the affinity rule. If that value is greater than the available capacity on any host, the rule cannot be satisfied.
- Compute the sum of the reservations of their parent resource pools. If that value is greater than the available capacity of any host, the rule cannot be satisfied if the resources are obtained from a single host.

Virtual Machine Power On Operation Fails

If an error message appears, stating that the virtual machine fails to power on, consider the following reasons.

The virtual machine might fail to power on because of insufficient resources or because there are no compatible hosts for the virtual machine.

If the cluster does not have sufficient resources to power on a single virtual machine or any of the virtual machines in a group power on attempt, check the resources required by the virtual machine against those available in the cluster or its parent resource pool. If necessary, reduce the reservations of the virtual machine to be powered on, reduce the reservations of its sibling virtual machines, or increase the resources available in the cluster or its parent resource pool.

If the virtual machine cannot run on a tiered host due to its attributes, add a non-tiered host to the cluster or move it to a cluster with compatible hosts that support the virtual machine.

DRS Does Not Move the Virtual Machine

DRS does not move the virtual machine when it is initially powered on despite insufficient resources on the host.

When you power on a virtual machine, DRS does not migrate it as expected when there are not enough resources on the host where the virtual machine is registered.

The following are possible reasons why DRS does not move the virtual machine.

- DRS is deactivated on the virtual machine.
- The virtual machine has a device mounted.
- The virtual machine is not compatible with any other hosts.
- No other hosts have a sufficient number of physical CPUs or capacity for each CPU for the virtual machine.
- No other hosts have sufficient CPU or memory resources to satisfy the reservations and required memory of this virtual machine.
- Moving the virtual machine will violate an affinity or anti-affinity rule.
- The DRS automation level of the virtual machine is manual and the user does not approve the migration recommendation.
- DRS will not move fault tolerance-activated virtual machines.
- Check if a VM Override definition exists for the VM which may impede an automatic migration.

Address the issue that prevents DRS from moving the virtual machine.

Memory Tiering Troubleshooting Information

If you experience issues when using Memory Tiering, consider one of the issues described in this section.

To validate the Memory Tiering requirements and ESX host configurations, see Capacity Planning Considerations.

Troubleshooting might require performing some initial configuration manually. When you troubleshoot, consider whether:

- The feature is enabled.
- The host displays the correct amount of DRAM and NVMe as tiered memory.
- The feature is operating in a limited way.
- The host fails with errors or warnings.
- The host actively uses NVMe.

- The feature was previously working.
- The feature is configured correctly.

If the Memory Tiering feature is configured correctly, the vSphere Client displays the following information.

1. On the **Hosts** pane, Memory Tiering appears as **Software**.
2. On the **Configure** > **Hardware Overview**, under **Memory**

Memory Value	Description
Tier 0	Displays the amount of DRAM in the system.
Tier 1	Displays the amount of NVMe storage that is used as tiered memory.
Total	Displays the total memory for the host as a sum of Tier 0 and Tier 1.

Total Memory Capacity Contains Only DRAM

If Total Memory Capacity contains only DRAM and no additional NVMe capacity, Memory Tiering is not configured correctly. The possible reasons may include:

NVMe device

- The NVMe device is not configured correctly.

  Solution
  1. Deactivate Memory Tiering for the host.
  2. Delete all partitions on the device.
  3. Create the Tier Partition.
  4. Activate Memory Tiering for the host.

- The NVMe device is not on the list of recommended device types.

  Solution
  1. Deactivate Memory Tiering for the host.
  2. Replace the NVMe device with one of the recommended device types listed.
  3. Create the Tier Partition.
  4. Activate Memory Tiering for the host.

- The tier partition on the NVMe device is corrupted.

  Solution
  1. Deactivate Memory Tiering for the host.
  2. Delete the Tier Partition.
  3. Create the Tier Partition.
  4. Activate Memory Tiering for the host.

- The NVMe device is a Self Encrypting Drive and is locked.

  Solution For more information on how to unlock the NVMe device, see the NVMe device's manual.

ESX hosts

- The NVMe tier is not part of the system memory.

Solution Verify that the Memory Tiering is activated and the NVMe device is configured. For more information, see Memory Tiering Configuration.

Intel Optane PMem DIMMs

- Verify that Intel Optane PMem DIMMs are not installed.

  Solution Remove the Intel Optane PMem DIMMs or use BIOS to deactivate the DIMM slots.

Dell NVDIMM-N DIMMs

- Verify that Dell NVDIMM-N DIMMs are not installed.

  Solution Remove the Dell NVDIMM-N DIMMs or use BIOS to deactivate the DIMM slots.

NVMe Driver Configuration Fails with an Error Message

When you configure an NVMe device for Memory Tiering, the NVMe driver configuration might fail with error messages.

- If you run the ESXCLI command `esxcli system tierdevice create -d <device>` to create a new tier partition while a partition that includes tier partition already exists, the NVMe driver configuration might fail with the following error message `Selected device already has an existing partition. Aborting partition creation. Please ensure the device name is correct and clean up any existing partitions.`
- If you run the `esxcli system tierdevice delete -d <device>` command to delete a non-tier partition, the NVMe driver configuration might fail with the following error message `The device specified does not have a tier partition. Aborting partition cleanup.`

Solution To delete non-tiered partitions, use the ESXCLI partedUtil utility. For more information, see Broadcom knowledge base article 323144.

CAUTION:

Deleting non-tiered partitions may delete existing data on the disk.

System Memory Total Displays Less NVMe Tiered Memory

The total system memory includes both DRAM and NVMe based on the configured DRAM:NVMe ratio and partition size.

If the total system memory shows less NVMe tiered memory, consider the following factors that can influence the NVMe tiered memory size.

The tier partition size on the NVMe device is not configured accurately

Solution Verify that the tier partition size is a maximum of 4 TB.

The amount of tiered NVMe is limited by the DRAM:NVMe ratio

Configuring a tier partition with a larger size may result in only the actual amount of tiered memory that adheres to the DRAM:NVMe ratio.

Example:

A host has 256 GB DRAM and a 4 TB NVMe device. The DRAM:NVMe ratio is 1:1. As a result, 256 GB of the NVMe tier partition is used as tiered memory, matching the amount of the available DRAM.

Solution Verify that the NVMe device does not have partitions other than a tier partition.

If the NVMe device has other partitions, delete all partitions, recreate the tier partition, and reboot the host.

The Reservable Memory is Lower than Expected

Each VM requires additional overhead memory, which reduces the amount available for reservation. Smaller VMs have proportionally higher overhead. For example, a 256 GB VM has an overhead of 4.5% of the VM size, 5.5% for a 64 GB VM, and 6% for a 16 GB VM.

Note:

These are representative values, and the exact overhead value can change depending on the VM configuration.

When calculating the maximum size of a VM that can be powered on with full reservation, you must consider this memory overhead. This overhead requirement applies to both tiered and non-tiered hosts.

The ESX hypervisor also requires some overhead memory to function. A memory-tiered host has higher overhead than a non-memory-tiered host. For example, a 1 TB non-tiered host can consume about 2% of memory for its overhead, whereas a tiered host with 512 GB DRAM and 512 GB NVMe can take around 7.96% of total memory for its overhead.

The overheads required by the hypervisor and VMs can reduce the available reservable memory in a host.

Tiered Memory Capacity and Storage Capacity Do Not Match

If Tiered Memory Capacity and Storage Capacity do not match on different vSphere Client screens, verify the following.

When you navigate to the **Configure** > **Storage Devices** pane for each host, the NVMe devices configured for Memory Tiering are marked with **Consumed for Memory Tiering**.

You can view the size of the NVMe device. You cannot share the NVMe device with any existing, non-memory tiering functions.

The Tier 1 capacity available as tiered memory may be smaller than the actual size of the tier partition. This configuration depends on the DRAM:NVMe ratio. For more information, see Memory Tiering Configuration Parameters.

NVMe Device Health Monitoring

NVMe device health is essential for maintaining system performance and reliability. ESX provides SMART technology for disk health monitoring.

You can monitor an NVMe device's remaining lifespan to determine when it needs to be replaced. Another key metric to monitor is the NVMe wear rate, which indicates how much the device has been used compared to its expected lifespan.

You can run the following ESXCLI commands.

To monitor the NVMe device health:

- `esxcli nvme device list`
- `esxcli nvme device log smart get -A <adapter>`

To monitor the storage health:

- `esxcli storage core device list`
- `esxcli storage core device smart get -d <device>`

vSphere Availability

vSphere Availability describes solutions that provide business continuity, including how to establish vSphere® High Availability (HA) and vSphere Fault Tolerance.

Intended Audience

This information is for anyone who wants to provide business continuity through the vSphere HA and Fault Tolerance solutions. The information in this book is for experienced Windows or Linux system administrators who are familiar with virtual machine technology and data center operations.

You Can Minimize Downtime with vSphere

Downtime, whether planned or unplanned, brings considerable costs. However, solutions that ensure higher levels of availability have traditionally been costly, hard to implement, and difficult to manage.

VMware software makes it simpler and less expensive to provide higher levels of availability for important applications. With vSphere, you can increase the baseline level of availability provided for all applications and provide higher levels of availability more easily and cost effectively. With vSphere, you can:

- Provide high availability independent of hardware, operating system, and applications.
- Reduce the planned downtime for common maintenance operations.
- Provide automatic recovery in cases of failure.

vSphere makes it possible to reduce planned downtime, prevent unplanned downtime, and recover rapidly from outages.

Reduce Planned Downtime with vSphere

Planned downtime typically accounts for over 80% of data center downtime. Hardware maintenance, server migration, and firmware updates all require downtime for physical servers. To minimize the impact of this downtime, organizations are forced to delay maintenance until inconvenient and difficult-to-schedule downtime windows.

vSphere makes it possible for organizations to dramatically reduce planned downtime. Because workloads in a vSphere environment can be dynamically moved to different physical servers without downtime or service interruption, server maintenance can be performed without requiring application and service downtime. With vSphere, organizations can:

- Eliminate downtime for common maintenance operations.
- Eliminate planned maintenance windows.
- Perform maintenance at any time without disrupting users and services.

The vSphere vMotion® and Storage vMotion functionality in vSphere makes it possible for organizations to reduce planned downtime because workloads in a VMware environment can be dynamically moved to different physical servers or to different underlying storage without service interruption. Administrators can perform faster and completely transparent maintenance operations, without being forced to schedule inconvenient maintenance windows.

Prevent Unplanned Downtime with vSphere

While an ESX host provides a robust platform for running applications, an organization must also protect itself from unplanned downtime caused from hardware or application failures. vSphere builds important capabilities into data center infrastructure that can help you prevent unplanned downtime.

These vSphere capabilities are part of virtual infrastructure and are transparent to the operating system and applications running in virtual machines. These features can be configured and utilized by all the virtual machines on a physical system, reducing the cost and complexity of providing higher availability. Key availability capabilities are built into vSphere:

- Shared storage. Eliminate single points of failure by storing virtual machine files on shared storage, such as Fibre Channel or iSCSI SAN, or NAS. The use of SAN mirroring and replication features can be used to keep updated copies of virtual disk at disaster recovery sites.

- Network interface teaming. Provide tolerance of individual network card failures.
- Storage multipathing. Tolerate storage path failures.

In addition to these capabilities, the vSphere HA and Fault Tolerance features can minimize or eliminate unplanned downtime by providing rapid recovery from outages and continuous availability, respectively.

vSphere HA Provides Rapid Recovery from Outages

vSphere HA leverages multiple ESX hosts configured as a cluster to provide rapid recovery from outages and cost-effective high availability for applications running in virtual machines.

vSphere HA protects application availability in the following ways:

- It protects against a server failure by restarting the virtual machines on other ESX hosts within the cluster.
- It protects against application failure by continuously monitoring a virtual machine and resetting it in the event that a failure is detected.
- It protects against datastore accessibility failures by restarting affected virtual machines on other ESX hosts which still have access to their datastores.
- It protects virtual machines against network isolation by restarting them if their ESX host becomes isolated on the management or vSAN network. This protection is provided even if the network has become partitioned.

Unlike other clustering solutions, vSphere HA provides the infrastructure to protect all workloads with the infrastructure:

- You do not need to install special software within the application or virtual machine. All workloads are protected by vSphere HA. After vSphere HA is configured, no actions are required to protect new virtual machines. They are automatically protected.
- You can combine vSphere HA with vSphere Distributed Resource Scheduler (DRS) to protect against failures and to provide load balancing across the hosts within a cluster.

vSphere HA has several advantages over traditional failover solutions:

Minimal setup	After a vSphere HA cluster is set up, all virtual machines in the cluster get failover support without additional configuration.
Reduced hardware cost and setup	The virtual machine acts as a portable container for the applications and it can be moved among hosts. Administrators avoid duplicate configurations on multiple machines. When you use vSphere HA, you must have sufficient resources to fail over the number of hosts you want to protect with vSphere HA. However, the VMware vCenter® instance automatically manages resources and configures clusters.
Increased application availability	Any application running inside a virtual machine has access to increased availability. Because the virtual machine can recover from hardware failure, all applications that start at boot have increased availability without increased computing needs, even if the application is not itself a clustered application. By monitoring and responding to VMware Tools heartbeats and restarting nonresponsive virtual machines, it protects against guest operating system crashes.
vSphere DRS and vMotion integration	If an ESX host fails and virtual machines are restarted on other hosts, DRS can provide migration recommendations or migrate virtual machines for balanced resource allocation. If one or both of the source and destination hosts of a migration fail, vSphere HA can help recover from that failure.

vSphere Fault Tolerance Provides Continuous Availability

vSphere HA provides a base level of protection for your virtual machines by restarting virtual machines in the event of an ESX host failure. vSphere Fault Tolerance provides a higher level of availability, allowing users to protect any virtual machine from a host failure with no loss of data, transactions, or connections.

Fault Tolerance provides continuous availability by ensuring that the states of the Primary and Secondary VMs are identical at any point in the instruction execution of the virtual machine.

If either the ESX host running the Primary VM or the host running the Secondary VM fails, an immediate and transparent failover occurs. The functioning ESX host seamlessly becomes the Primary VM host without losing network connections or in-progress transactions. With transparent failover, there is no data loss and network connections are maintained. After a transparent failover occurs, a new Secondary VM is respawned and redundancy is re-established. The entire process is transparent and fully automated and occurs even if vCenter is unavailable.

Protecting vCenter with vCenter High Availability

vCenter High Availability (vCenter HA) protects not only against host and hardware failures but also against vCenter application failures. Using automated failover from active to passive, vCenter HA supports high availability with minimal downtime.

You configure vCenter HA from the vSphere Client. The configuration wizard provides these options.

Option	Description
Automatic	The automatic option clones the Active node to the Passive node and witness node, and configures the nodes for you. If your environment meets the following requirements, you can use this option. • The vCenter that becomes the Active node is managing its own ESX host and its own virtual machine. This configuration is sometimes called a self-managed vCenter.
Manual	The manual option offers more flexibility. You can use this option provided that your environment meets hardware and software requirements. If you select this option, you are responsible for cloning the Active node to the Passive node and the Witness node. You must also perform some networking configuration.

Creating and Using vSphere HA Clusters

vSphere HA clusters enable a collection of ESX hosts to work together so that, as a group, they provide higher levels of availability for virtual machines than each ESX host can provide individually. When you plan the creation and usage of a new vSphere HA cluster, the options you select affect the way that cluster responds to failures of hosts or virtual machines.

Before you create a vSphere HA cluster, you should know how vSphere HA identifies host failures and isolation and how it responds to these situations. You also should know how admission control works so that you can choose the policy that fits your failover needs. After you establish a cluster, you can customize its behavior with advanced options and optimize its performance by following recommended best practices.

Note:

You might get an error message when you try to use vSphere HA. For information about error messages related to vSphere HA, see Broadcom knowledge base article 342336.

How vSphere HA Works

vSphere HA provides high availability for virtual machines by pooling the virtual machines and the hosts they reside on into a cluster. Hosts in the cluster are monitored and in the event of a failure, the virtual machines on a failed host are restarted on alternate hosts.

When you create a vSphere HA cluster, a single host is automatically elected as the primary host. The primary host communicates with vCenter and monitors the state of all protected virtual machines and of the secondary hosts. Different types of host failures are possible, and the primary host must detect and appropriately deal with the failure. The primary host must distinguish between a failed host and one that is in a network partition or that has become network isolated. The primary host uses network and datastore heartbeating to determine the type of failure.

Primary and Secondary Hosts

When you add a host to a vSphere HA cluster, an agent is uploaded to the host and configured to communicate with other agents in the cluster. Each host in the cluster functions as a primary host or a secondary host.

When vSphere HA is enabled for a cluster, all active hosts (that are not in standby, maintenance mode or not disconnected) participate in an election to choose the cluster's primary host. The host that mounts the greatest number of datastores has an advantage in the election. Only one primary host typically exists per cluster and all other hosts are secondary hosts. If the primary host fails, is shut down or put in standby mode, or is removed from the cluster a new election is held.

The primary host in a cluster has several responsibilities:

- Monitoring the state of secondary hosts. If a secondary host fails or becomes unreachable, the primary host identifies which virtual machines must be restarted.
- Monitoring the power state of all protected virtual machines. If one virtual machine fails, the primary host ensures that it is restarted. Using a local placement engine, the primary host also determines where the restart takes place.
- Managing the lists of cluster hosts and protected virtual machines.
- Acting as the vCenter management interface to the cluster and reporting the cluster health state.

The secondary hosts primarily contribute to the cluster by running virtual machines locally, monitoring their runtime states, and reporting state updates to the primary host. A primary host can also run and monitor virtual machines. Both secondary hosts and primary hosts implement the VM and Application Monitoring features.

One of the functions performed by the primary host is to orchestrate restarts of protected virtual machines. A virtual machine is protected by a primary host after vCenter observes that the virtual machine's power state has changed from powered off to powered on in response to a user action. The primary host persists the list of protected virtual machines in the cluster's datastores. A newly elected primary host uses this information to determine which virtual machines to protect.

> **Note:**
> If you disconnect a host from a cluster, the virtual machines registered to that host are unprotected by vSphere HA.

Host Failure Types

The primary host of a VMware vSphere® High Availability cluster is responsible for detecting the failure of secondary hosts. Depending on the type of failure detected, the virtual machines running on the hosts might need to be failed over.

In a vSphere HA cluster, three types of host failure are detected:

- Failure. A host stops functioning.
- Isolation. A host becomes network isolated.
- Partition. A host loses network connectivity with the primary host.

The primary host monitors the liveness of the secondary hosts in the cluster. This communication happens through the exchange of network heartbeats every second. When the primary host stops receiving these heartbeats from a secondary host, it checks for host liveness before declaring the host failed. The liveness check that the primary host performs is to determine whether the secondary host is exchanging heartbeats with one of the datastores. See Datastore Heartbeating . Also, the primary host checks whether the host responds to ICMP pings sent to its management IP addresses.

If a primary host cannot communicate directly with the agent on a secondary host, the secondary host does not respond to ICMP pings. If the agent is not issuing heartbeats, it is viewed as failed. The host's virtual machines are restarted on alternate hosts. If such a secondary host is exchanging heartbeats with a datastore, the primary host assumes that the secondary host is in a network partition or is network isolated. So, the primary host continues to monitor the host and its virtual machines. See Network Partitions .

Host network isolation occurs when a host is still running, but it can no longer observe traffic from vSphere HA agents on the management network. If a host stops observing this traffic, it attempts to ping the cluster isolation addresses. If this pinging also fails, the host declares that it is isolated from the network.

The primary host monitors the virtual machines that are running on an isolated host. If the primary host observes that the VMs power off, and the primary host is responsible for the VMs, it restarts them.

> **Note:**
> If you ensure that the network infrastructure is sufficiently redundant and that at least one network path is always available, host network isolation is less likely to occur.

Proactive HA Failures

A Proactive HA failure occurs when a host component fails, which results in a loss of redundancy or a noncatastrophic failure. However, the functional behavior of the VMs residing on the host is not yet affected. For example, if a power supply on the host fails, but other power supplies are available, that is a Proactive HA failure.

If a Proactive HA failure occurs, you can automate the remediation action taken in the vSphere Availability section of the vSphere Client. The VMs on the affected host can be evacuated to other hosts and the host is either placed in Quarantine mode or Maintenance mode.

> **Note:**
> Your cluster must use vSphere DRS for the Proactive HA failure monitoring to work.

Determining Responses to Host Issues

If a host fails and its virtual machines must be restarted, you can control the order in which the virtual machines are restarted with the VM restart priority setting. You can also configure how vSphere HA responds if hosts lose management network connectivity with other hosts by using the host isolation response setting. Other factors are also considered when vSphere HA restarts a virtual machine after a failure.

The following settings apply to all virtual machines in the cluster in the case of a host failure or isolation. You can also configure exceptions for specific virtual machines. See Customize an Individual Virtual Machine .

Host Isolation Response

Host isolation response determines what happens when a host in a vSphere HA cluster loses its management network connections, but continues to run. You can use the isolation response to have vSphere HA power off virtual machines that are running on an isolated host and restart them on a non-isolated host. Host isolation responses require that Host Monitoring Status is activated. If Host Monitoring Status is deactivated, host isolation responses are also suspended. A host determines that it is isolated when it is unable to communicate with the agents running on the other hosts, and it is unable to ping its isolation addresses. The host then executes its isolation response. The responses are Power off and restart VMs or Shutdown and restart VMs. You can customize this property for individual virtual machines.

> **Note:**
> If a virtual machine has a restart priority setting of Disabled, no host isolation response is made.

To use the Shutdown and restart VMs setting, you must install VMware Tools in the guest operating system of the virtual machine. Shutting down the virtual machine provides the advantage of preserving its state. Shutting down is better than powering off the virtual machine, which does not flush most recent changes to disk or commit transactions. Virtual machines that are in the process of shutting down take longer to fail over while the shutdown completes. Virtual Machines that have not shut down in 300 seconds, or the time specified in the advanced option das.isolationshutdowntimeout, are powered off.

After you create a vSphere HA cluster, you can override the default cluster settings for Restart Priority and Isolation Response for specific virtual machines. Such overrides are useful for virtual machines that are used for special tasks. For example, virtual machines that provide infrastructure services like DNS or DHCP might need to be powered on before other virtual machines in the cluster.

A virtual machine "split-brain" condition can occur when a host becomes isolated or partitioned from a primary host and the primary host cannot communicate with it using heartbeat datastores. In this situation, the primary host cannot determine that the host is alive and so declares it dead. The primary host then attempts to restart the virtual machines that are running on the isolated or partitioned host. This attempt succeeds if the virtual machines remain running on the isolated/partitioned host and that host lost access to the virtual machines' datastores when it became isolated or partitioned. A split-brain condition then exists because there are two instances of the virtual machine. However, only one instance is able to read or write the virtual machine's virtual disks. VM Component Protection can be used to prevent this split-brain condition. When you activate VMCP with the aggressive setting, it monitors the datastore accessibility of powered-on virtual machines, and shuts down those that lose access to their datastores.

To recover from this situation, ESX generates a question on the virtual machine that has lost the disk locks for when the host comes out of isolation and cannot reacquire the disk locks. vSphere HA automatically answers this question, allowing the virtual machine instance that has lost the disk locks to power off, leaving just the instance that has the disk locks.

Virtual Machine Dependencies

You can create dependencies between groups of virtual machines. To do so, you must first create the VM groups in the vSphere Client by going to the **Configure** tab for the cluster and selecting **VM/Host Groups**. Once the groups have been created, you can create restart dependency rules between the groups by browsing to**VM/Host Rules** and in the Type drop-down menu, select **Virtual Machines to Virtual Machines**. These rules can specify that certain VM groups cannot be restarted until other, specified VM groups have been Ready first.

Factors Considered for Virtual Machine Restarts

After a failure, the cluster's primary host attempts to restart affected virtual machines by identifying a host that can power them on. When choosing such a host, the primary host considers a number of factors.

File accessibility	Before a virtual machine can be started, its files must be accessible from one of the active cluster hosts that the primary can communicate with over the network
Virtual machine and host compatibility	If there are accessible hosts, the virtual machine must be compatible with at least one of them. The compatibility set for a virtual machine includes the effect of any required VM-Host affinity rules. For example, if a rule only permits a virtual machine to run on two hosts, it is considered for placement on those two hosts.
Resource reservations	Of the hosts that the virtual machine can run on, at least one must have sufficient unreserved capacity to meet the memory overhead of the virtual machine and any resource reservations. Four types of reservations are considered: CPU, Memory, vNIC, and Virtual flash. Also, sufficient network ports must be available to power on the virtual machine.
Host limits	In addition to resource reservations, a virtual machine can only be placed on a host if doing so does not violate the maximum number of allowed virtual machines or the number of in-use vCPUs.
Feature constraints	If the advanced option has been set that requires vSphere HA to enforce VM to VM anti-affinity rules, vSphere HA does not violate this rule. Also, vSphere HA does not violate any configured per host limits for fault tolerant virtual machines.

If no hosts satisfy the preceding considerations, the primary host issues an event stating that there are not enough resources for vSphere HA to start the VM and tries again when the cluster conditions have changed. For example, if the virtual machine is not accessible, the primary host tries again after a change in file accessibility.

VM and Application Monitoring

VM Monitoring restarts individual virtual machines if their VMware Tools heartbeats are not received within a set time. Similarly, Application Monitoring can restart a virtual machine if the heartbeats for an application it is running are not received. You can enable these features and configure the sensitivity with which vSphere HA monitors non-responsiveness.

When you enable VM Monitoring, the VM Monitoring service (using VMware Tools) evaluates whether each virtual machine in the cluster is running by checking for regular heartbeats and I/O activity from the VMware Tools process running inside the guest. If no heartbeats or I/O activity are received, this is most likely because the guest operating system has failed or VMware Tools is not being allocated any time to complete tasks. In such a case, the VM Monitoring service determines that the virtual machine has failed and the virtual machine is rebooted to restore service. For information on how to enable VM monitoring, see Activate VM Monitoring.

Occasionally, virtual machines or applications that are still functioning properly stop sending heartbeats. To avoid unnecessary resets, the VM Monitoring service also monitors a virtual machine's I/O activity. If no heartbeats are received within the failure interval, the I/O stats interval (a cluster-level attribute) is checked. The I/O stats interval determines if any disk or network activity has occurred for the virtual machine during the previous two minutes (120 seconds). If not, the virtual machine is reset. This default value (120 seconds) can be changed using the advanced option das.iostatsinterval.

To enable Application Monitoring, you must first obtain the appropriate SDK (or be using an application that supports VMware Application Monitoring) and use it to set up customized heartbeats for the applications you want to monitor. After you have done this, Application Monitoring works much the same way that VM Monitoring does. If the heartbeats for an application are not received for a specified time, its virtual machine is restarted.

You can configure the level of monitoring sensitivity. Highly sensitive monitoring results in a more rapid conclusion that a failure has occurred. While unlikely, highly sensitive monitoring might lead to falsely identifying failures when the virtual

machine or application in question is actually still working, but heartbeats have not been received due to factors such as resource constraints. Low sensitivity monitoring results in longer interruptions in service between actual failures and virtual machines being reset. Select an option that is an effective compromise for your needs.

You can also specify custom values for the monitoring sensitivity interval by selecting the **Custom** check box.

Table 314: VM Monitoring Settings

Preset Setting	Failure interval (seconds)	Minimum uptime (seconds)	Maximum per-VM resets	Maximum resets time window (hours)
High	30	120	3	1
Medium	60	240	3	24
Low	120	480	3	168

- **Failure interval**: If no VM heartbeat is received in the specified interval, vSphere HA restarts the VM.
- **Minimum uptime**: After a VM starts, vSphere HA waits until the specified interval is reached, before monitoring for VM heartbeats, storage and network traffic.
- **Maximum per-VM resets**: The maximum number of resets, that vSphere HA can restart a VM a within the specified interval for **Maximum resets time window**.
- **Maximum reset time window**: The maximum interval for the number of specified **Maximum per-VM resets**.

Note:

To set a custom value for the I/O stats interval, you can use the `das.iostatsInterval` advanced configuration option. For more information, see Broadcom knowledge base article 313261.

After failures are detected, vSphere HA resets virtual machines. The reset ensures that services remain available. To avoid resetting virtual machines repeatedly for nontransient errors, by default, virtual machines will be reset only three times during a certain configurable time interval. After virtual machines have been reset three times, vSphere HA makes no further attempts to reset the virtual machines after subsequent failures until after the specified time has elapsed. You can configure the number of resets using the **Maximum per-VM resets** custom setting.

Note:
The reset statistics are cleared when a virtual machine is powered off then back on, or when it is migrated using vMotion to another host. This causes the guest operating system to reboot, but this is not the same as a 'restart' in which the power state of the virtual machine is changed.

VM Component Protection

If VM Component Protection (VMCP) is activated, vSphere HA can detect datastore accessibility failures and provide automated recovery for affected virtual machines.

VMCP provides protection against datastore accessibility failures that can affect a virtual machine running on a host in a vSphere HA cluster. When a datastore accessibility failure occurs, the affected host can no longer access the storage path for a specific datastore. You can determine the response that vSphere HA will make to such a failure, ranging from the creation of event alarms to virtual machine restarts on other hosts.

Note:

When you use the VM Component Protection feature, your ESX hosts must be version 6.0 or higher.

Types of Failure

There are two types of datastore accessibility failure:

PDL	PDL (Permanent Device Loss) is an unrecoverable loss of accessibility that occurs when a storage device reports the datastore is no longer accessible by the host. This condition cannot be reverted without powering off virtual machines.
APD	APD (All Paths Down) represents a transient or unknown accessibility loss or any other unidentified delay in I/O processing. This type of accessibility issue is recoverable.

Configuring VMCP

VM Component Protection is configured in the vSphere Client. Go to the **Configure** tab and click **vSphere Availability** and **Edit**. Under **Failures and Responses** you can select **Datastore with PDL** or **Datastore with APD**. The storage protection levels you can choose and the virtual machine remediation actions available differ depending on the type of database accessibility failure.

PDL Failures	Under **Datastore with PDL**, you can select **Issue events** or **Power off and restart VMs**.
APD Failures	The response to APD events is more complex and accordingly the configuration is more fine-grained. You can select **Issue events**, **Power off and restart VMs--conservative restart policy**, or **Power off and restart VMs--aggressive restart policy**

Note:
If either the Host Monitoring or VM Restart Priority settings are deactivated, VMCP cannot perform virtual machine restarts. Storage health can still be monitored and events can be issued, however.

Network Partitions

When a management network failure occurs for a vSphere HA cluster, a subset of the cluster's hosts might be unable to communicate over the management network with the other hosts. Multiple partitions can occur in a cluster.

A partitioned cluster leads to degraded virtual machine protection and cluster management functionality. Correct the partitioned cluster as soon as possible.

- Virtual machine protection. vCenter allows a virtual machine to be powered on, but it can be protected only if it is running in the same partition as the primary host that is responsible for it. The primary host must be communicating with vCenter. A primary host is responsible for a virtual machine if it has exclusively locked a system-defined file on the datastore that contains the virtual machine's configuration file.
- Cluster management. vCenter can communicate with the primary host, but only a subset of the secondary hosts. As a result, changes in configuration that affect vSphere HA might not take effect until after the partition is resolved. This failure could result in one of the partitions operating under the old configuration, while another uses the new settings.

Datastore Heartbeating

When the primary host in a VMware vSphere® High Availability cluster cannot communicate with a secondary host over the management network, the primary host uses datastore heartbeating to determine whether the secondary host has failed, is in a network partition, or is network isolated. If the secondary host has stopped datastore heartbeating, it is considered to have failed and its virtual machines are restarted elsewhere.

VMware vCenter® selects a preferred set of datastores for heartbeating. This selection is made to maximize the number of hosts that have access to a heartbeating datastore and minimize the likelihood that the datastores are backed by the same LUN or NFS server.

You can use the advanced option das.heartbeatdsperhost to change the number of heartbeat datastores selected by vCenter for each host. The default is two and the maximum valid value is five.

vSphere HA creates a directory at the root of each datastore that is used for both datastore heartbeating and for persisting the set of protected virtual machines. The name of the directory is `.vSphere-HA`. Do not delete or modify the files stored in this directory, because this can have an impact on operations. Because more than one cluster might use a datastore, subdirectories for this directory are created for each cluster. Root owns these directories and files and only root can read and write to them. The disk space used by vSphere HA depends on several factors including which VMFS version is in use and the number of hosts that use the datastore for heartbeating. With vmfs5 and vmfs6, the maximum and typical usage is 3MB. vSphere HA use of the datastores adds negligible overhead and has no performance impact on other datastore operations.

vSphere HA limits the number of virtual machines that can have configuration files on a single datastore. See *Configuration Maximums* for updated limits. If you place more than this number of virtual machines on a datastore and power them on, vSphere HA protects virtual machines only up to the limit.

Note:
A vSAN datastore cannot be used for datastore heartbeating. Therefore, if no other shared storage is accessible to all hosts in the cluster, there can be no heartbeat datastores in use. However, if you have storage that is accessible by an alternate network path independent of the vSAN network, you can use it to set up a heartbeat datastore.

vSphere HA Security

vSphere HA is enhanced by several security features.

Select firewall ports opened	vSphere HA uses TCP and UDP port 8182 for agent-to-agent communication. The firewall ports open and close automatically to ensure they are open only when needed.
Configuration files that are protected using file system permissions	vSphere HA stores configuration information on the local storage or on ram disk if there is no local datastore. These files are protected using file system permissions and they are accessible only to the root user. Hosts without local storage are only supported if they are managed by Auto Deploy.
Detailed logging	The location where vSphere HA places log files depends on the version of the host. • For ESX hosts, vSphere HA writes to syslog only by default, so logs are placed where syslog is configured to put them. The log file names for vSphere HA are prepended with `fdm`, fault domain manager, which is a service of vSphere HA. • For legacy ESX hosts, vSphere HA writes to `/var/log/vmware/fdm` on local disk, as well as syslog if it is configured.
Secure vSphere HA logins	vSphere HA logs onto the vSphere HA agents using a user account, **vpxuser**, created by vCenter. This account is the same account used by vCenter to manage the host. vCenter creates a random password for this account and changes the password periodically. The time period is set by the vCenterVirtualCenter.VimPasswordExpirationInDays setting. Users with administrative privileges on the root folder of the host can log in to the agent.
Secure communication	All communication between vCenter and the vSphere HA agent is done over SSL. Agent-to-agent communication also uses SSL except for election messages, which occur over UDP. Election messages are verified over SSL so that a rogue agent can prevent only the host on which the agent is running from being elected as a primary host. In this case, a configuration issue for the cluster is issued so the user is aware of the problem.
Host SSL certificate verification required	vSphere HA requires that each host has a verified SSL certificate. Each host generates a self-signed certificate when it is booted for the first time. This certificate can then be regenerated or replaced with one issued by an authority. If the certificate is replaced, vSphere HA needs to be reconfigured on the host. If a host becomes disconnected from vCenter after its certificate is updated and the ESX or ESX host agent is restarted, then vSphere HA is automatically reconfigured when the host is reconnected to vCenter. If the disconnection does not occur because vCenter host SSL certificate verification is deactivated at the time, verify the new certificate and reconfigure vSphere HA on the host.

vSphere HA Admission Control

vSphere HA uses admission control to ensure that sufficient resources are reserved for virtual machine recovery when a host fails.

Admission control imposes constraints on resource usage. Any action that might violate these constraints is not permitted. Actions that might be disallowed include the following examples:

- Powering on a virtual machine
- Migrating a virtual machine
- Increasing the CPU or memory reservation of a virtual machine

The basis for vSphere HA admission control is how many host failures your cluster is allowed to tolerate and still guarantee failover. The host failover capacity can be set in three ways:

- Cluster resource percentage
- Slot policy
- Dedicated failover hosts

Note:

vSphere HA admission control can be deactivated. However, without vSphere HA admission control, there is no assurance that the expected number of virtual machines can be restarted after a failure. Do not permanently deactivate admission control.

Note:

Within a cluster, you must deactivate HA admission control temporarily to let vSphere vMotion proceed. This action prevents downtime for the machines on the hosts that you remediate. Disabling HA admission control before you remediate a two-node cluster causes the cluster to practically lose all its high availability guarantees. The reason is that when one of the two hosts enters maintenance mode, vCenter cannot failover virtual machines to that host and HA failovers are never successful.

Note:
To use vSphere HA admission control you must have at least three hosts in the cluster.

Regardless of the admission control option chosen, a VM resource reduction threshold also exists. You use this setting to specify the percentage of resource degradation to tolerate, but it is not available unless vSphere DRS is activated.

The resource reduction calculation is checked for both CPU and memory. It considers a virtual machine's reserved memory and memory overload to decide whether to permit it to power on, migrate, or have reservation changes. The actual memory consumed by the virtual machine is not considered in the calculation because the memory reservation does not always correlate with the actual memory usage of the virtual machine. If the actual usage is more than reserved memory, insufficient failover capacity is available, resulting in performance degradation on failover.

Setting a performance reduction threshold allows you to specify the occurrence of a configuration issue. For example:

- The default value is 100%, which produces no warnings.
- If you reduce the threshold to 0%, a warning is generated when cluster usage exceeds the available capacity.
- If you reduce the threshold to 20%, the performance reduction that can be tolerated is calculated as `performance reduction = current utilization * 20%`. When the current usage minus the performance reduction exceeds the available capacity, a configuration notice is issued.

Memory Tiering and HA Admission Control

In clusters with Memory Tiering activated, vSphere HA admission control tracks DRAM as an independent resource alongside total memory. In vSphere 9.1.1 and later, the cluster must retain sufficient DRAM capacity to guarantee that virtual machines can restart after a host failure, preventing failover operations from silently failing due to insufficient DRAM.

How Memory Tiering Works in a vSphere HA Cluster

DRAM is the primary memory resource that determines whether a virtual machine can power on. Its availability directly controls whether virtual machines can run and how well they perform.

To extend memory capacity beyond available DRAM, vSphere introduced the Memory Tiering feature. Memory Tiering allows ESX hosts to use Non-Volatile Memory Express (NVMe) devices as a secondary memory tier. This approach enables higher virtual machine density per host by expanding the total addressable memory beyond physical DRAM limits.

In a cluster that is configured for Memory Tiering, the ESX host memory consists of two tiers:

- Tier 0 (DRAM) that is the primary, high-performance, low-latency memory tier. DRAM is required for virtual machine overhead memory and for powering on virtual machines, regardless of their tier configuration.
- Tier 1 (NVMe) that is the secondary, high-capacity memory tier. NVMe extends the total memory available to virtual machines but cannot substitute for DRAM when a virtual machine powers on or restarts.

When you activate Memory Tiering, each virtual machine in the cluster is classified as one of the following types:

- Tiered VM - a virtual machine whose memory spans both Tier 0 (DRAM) and Tier 1 (NVMe). Overhead memory always resides in DRAM. The configured memory reservation is served from NVMe.

- Non-tiered VM - a virtual machine whose entire memory (both reservation and overhead) resides in DRAM. No NVMe memory is used.

Each virtual machine type has a different DRAM requirement at restart time.

Why Tier-Unaware Admission can Cause Silent Failures During a Failover

In vSphere 9.1 and earlier, vSphere HA admission control treated all host memory (DRAM and NVMe) as a single uniform pool. When calculating failover capacity, the cluster summed the total memory across hosts without distinguishing how much of that memory was DRAM and creating a reliability gap in the clusters configured for Memory Tiering.

Consider the following failure scenario:

1. You configure a cluster with a failover level of one host failure.
2. Admission control reports a green status, indicating that sufficient failover capacity is reserved.
3. A host fails. vSphere HA attempts to restart the affected virtual machines on surviving hosts.
4. The restart fails not because the total memory is insufficient, but because the surviving hosts lack enough DRAM to power on the virtual machines. The reserved capacity was backed by NVMe, which the virtual machines cannot use to restart.

From the perspective of an administrator, HA is configured correctly, admission control shows a green status, and failover is seemingly guaranteed. In reality, the cluster admitted more non-tiered vitrtual machines than the surviving hosts could restart, because admission control counted NVMe memory toward a reservation that strictly required DRAM.

This sequence results in a silent failure: no alert is triggered, the admission control status remains green, and the capacity gap is only discovered when an actual host failure occurs.

Tier-Aware Admission Control in vSphere 9.1.1 and Later

Starting in vSphere 9.1.1, vSphere HA admission control tracks DRAM as an independent resource alongside total memory and CPU. Before admitting a virtual machine, the cluster performs two independent memory checks.

- Total memory check - the cluster must retain sufficient total memory capacity (DRAM and NVMe) beyond the configured failover reservation.
- DRAM check: the cluster must also retain sufficient DRAM capacity to cover the DRAM requirement of the new virtual machine while preserving the configured DRAM failover reservation.

If either check fails, a virtual machine is not admitted, even if the other check passes. This prevents the cluster from being over-admitted on DRAM when total memory appears adequate.

The DRAM failover reservation is calculated as follows:

- Auto mode (Cluster Resources Percentage policy): vSphere HA sums the DRAM capacities of the `n` largest DRAM hosts in the cluster, where `n` equals the configured failover level. This ensures that the simultaneous failure of any `n` hosts leaves the cluster with enough DRAM to restart all affected virtual machines.
- Manual mode (Cluster Resources Percentage policy): The memory percentage that you specify is applied independently to the total cluster DRAM capacity to compute the DRAM failover reservation. This DRAM reservation is counted within the total memory reservation — it is not added on top of it.

The existing admission control status indicators (green, yellow, and red) are extended to reflect DRAM capacity shortfalls. No new alert types are introduced. If a DRAM shortfall causes a violation, the cluster reports the condition using the existing failover capacity violation alerts.

Supported Admission Control Policies for Tier-Aware Enforcement

Tier-aware DRAM enforcement applies to the following admission control policies in vSphere 9.1.1:

Admission Control Policy	Tier-Aware DRAM Enforcement	Description
Cluster Resource Percentage	Supported auto mode and manual mode	DRAM failover capacity is calculated and enforced independently alongside total memory.
Dedicated Failover Hosts	Supported	The designated failover DRAM capacity of an ESX host is tracked and reported to DRS.
Slot Policy (Host Failures Cluster Tolerates)	Not supported	The cluster continues to treat memory as a single pool. DRAM is not tracked independently.

Important

If the cluster uses Memory Tiering, do not use the Slot Policy for admission control. The Slot Policy calculates the slot size from the total memory reservation plus overhead, without distinguishing DRAM from NVMe. Use the Cluster Resource Percentage policy or the Dedicated Failover Hosts policy to ensure that DRAM availability is enforced as part of the admission control.

Cluster Resources Percentage Admission Control

You can configure vSphere HA to perform admission control by reserving a specific percentage of cluster CPU and memory resources for recovery from host failures. Starting in vSphere 9.1.1, clusters with configured Memory Tiering also reserve a percentage of DRAM capacity independently, ensuring that sufficient DRAM is available to restart virtual machines after a host failure.

With the cluster resources percentage option, vSphere HA enforces admission control as follows:

1. Calculates the total resource requirements for all powered-on virtual machines in the cluster.
2. Calculates the total host resources available for virtual machines.
3. Calculates the Current CPU Failover Capacity and Current Memory Failover Capacity for the cluster. In clusters with Memory Tiering activated, vSphere HA also calculates the Current DRAM Failover Capacity.
4. Determines whether the Current CPU Failover Capacity, Current Memory Failover Capacity, or Current DRAM Failover Capacity (for clusters with activated Memory Tiering) is less than the corresponding Configured Failover Capacity.

   If any check fails, admission control disallows the operation.

vSphere HA uses the actual reservations of the virtual machines. If a virtual machine does not have reservations, meaning that the reservation is 0, vSphere HA applies a default of 0 MB memory and 32 MHz CPU.

> **Note:**
>
> The cluster resources percentage option also ensures that at least two vSphere HA activated hosts exist in the cluster, excluding hosts entering maintenance mode. If only one vSphere HA-enabled host remains, vSphere HA does not allow the operation, even if sufficient resources are available. vSphere HA cannot perform a failover with a single host in the cluster.

Computing the Current Available Capacity

The total resource requirements for the powered-on virtual machines consist of two components: CPU and memory. vSphere HA calculates each value as follows:

- The CPU component: vSphere HA sums the CPU reservations of the powered-on virtual machines. If you have not specified a CPU reservation for a virtual machine, vSphere HA assigns a default value of 32 MHz. You can change this default by using the das.vmcpuminmhz advanced option.

- The memory component: vSphere HA sums the memory reservation plus memory overhead of each powered-on virtual machine. If you have not specified a memory reservation for a virtual machine, vSphere HA uses 0 MB as the default.

vSphere HA calculates the total host resources available for virtual machines by adding the CPU and memory resources of the host. These amounts come from the root resource pool of the host, not the total physical resources of the host. vSphere HA considers only hosts that are connected, not in maintenance mode, and have no vSphere HA errors.

vSphere HA computes the Current CPU Failover Capacity by subtracting the total CPU resource requirements from the total host CPU resources and dividing the result by the total host CPU resources. vSphere HA calculates the Current Memory Failover Capacity in the same way.

Tier-Aware Failover Capacity Calculations in Clusters with Memory Tiering Activated

With vSphere 9.1.1, when Memory Tiering is active in the cluster, vSphere HA computes two independent memory failover capacities instead of one. This ensures that the cluster reserves not only enough total memory, but also enough DRAM to restart virtual machines after a host failure.

Auto Mode

In auto mode, vSphere HA derives the failover reservation from the configured failover level. For a failover level of `n`, vSphere HA computes the total memory failover capacity and the DRAM failover capacity where:

- The total memory failover capacity is the sum of the total memory (both DRAM and NVMe) across the `n` largest hosts, ranked by total memory.
- The DRAM failover capacity is the sum of DRAM capacity across the `n` largest hosts, ranked by DRAM capacity.

The DRAM reservation is counted within the total memory reservation, not added on top of it.

A host with high NVMe capacity, but relatively low DRAM capacity may rank in the top `n` hosts for total memory, but not for DRAM. As a result, the set of top `n` hosts for total memory and the set of top `n` hosts for DRAM may differ. vSphere HA accounts for this automatically - you do not need to manually configure which hosts are included in each calculation.

The Current Memory Failover Capacity calculation is extended to separately report the Current DRAM Failover Capacity. Both values appear in the admission control status.

Manual Mode

In manual mode, you specify a memory percentage instead of having vSphere HA derive it from the failover level. vSphere HA applies the specified percentage independently to both total cluster memory and total cluster DRAM capacity to derive the two failover reservations.

For example, if you configure a memory percentage of 25% in a cluster with two hosts, where each host has 20 MB DRAM and 30 MB NVMe, vSphere HA computes the following:

Resource	Total Cluster Capacity	Failover Reservation (25%)
Total memory (DRAM + NVMe)	100 MB	25 MB
DRAM	40 MB	10 MB

The DRAM failover reservation of 10 MB is counted within the total memory failover reservation of 25 MB - it is not added on top of it.

VM Admission Checks in Clusters with Memory Tiering Activated

A virtual machine is admitted only if two independent checks pass:

1. Total memory check: the available total memory capacity must exceed the total memory failover reservation.
2. DRAM check: the available DRAM capacity must exceed the DRAM failover reservation.

If either check fails, vSphere HA generates an Insufficient Resource event and blocks the power-on operation. This prevents DRAM over-admission even when the aggregate memory appears adequate.

Virtual machines with no memory reservation use 0 MB as the default. The DRAM admission check considers the DRAM requirement of the virtual machine based on whether the virtual machine is configured for Memory Tiering or not.

Virtual Machine Type	DRAM Requirement at Admission
VM that is not configured for Memory Tiering	Full memory reservation plus overhead - all in DRAM.
Tiered VM	Overhead memory only - the configured reservation is served from NVMe. If NVMe capacity is insufficient, the reservation falls back to DRAM.

Failover Violation Checks in Clusters with Memory Tiering Activated

The FailoverResource violation check runs every 5 minutes to verify that actual cluster utilization does not exceed the available capacity. Starting in vSphere 9.1.1, vSphere HA extends this check to include a DRAM capacity check alongside the existing total memory check.

If actual DRAM utilization causes the available DRAM to drop below the configured failover reservation threshold, vSphere HA triggers a configuration issue on the cluster. No new alert types are introduced, as the existing admission control alerts cover both total memory and DRAM violations.

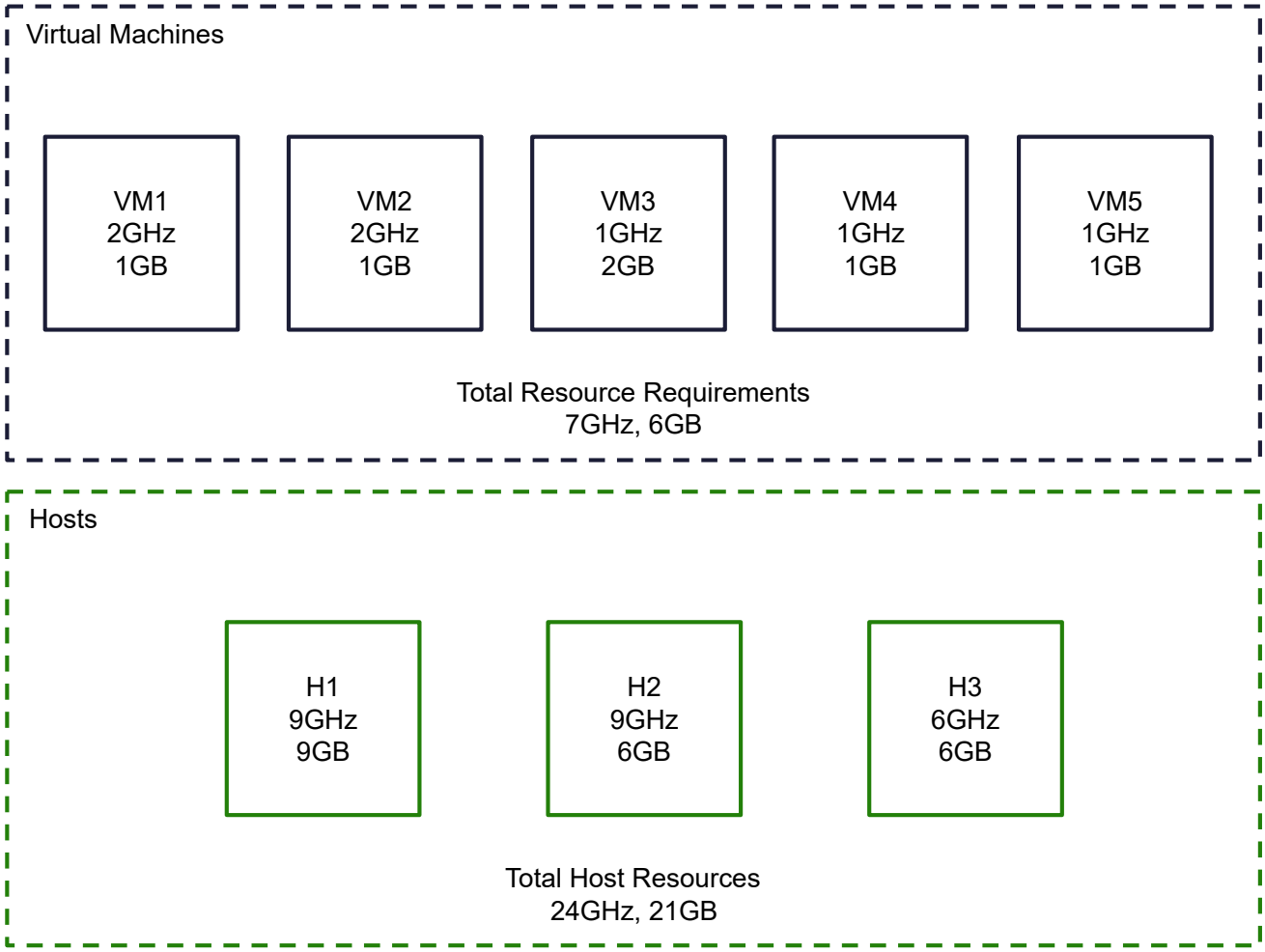
Admission Control Example Without Memory Tiering

The following example illustrates how vSphere HA calculates failover capacity in a cluster that does not use Memory Tiering. The example uses manual mode, in which you specify the failover capacity percentage directly.

Cluster Configuration

The cluster consists of:

- Three hosts, each with a different amount of available CPU and memory resources.
  - Host 1 (H1): 9 GHz CPU and 9 GB memory
  - Host 2 (H2): 9 GHz CPU and 6 GB memory
  - Host 3 (H3): 6 GHz CPU and 6 GB memory
- Five powered-on virtual machines run in the cluster with differing CPU and memory requirements.
  - VM1: 2 GHz CPU and 1 GB memory
  - VM2: 2 GHz CPU and 1 GB memory
  - VM3: 1 GHz CPU and 2 GB memory
  - VM4: 1 GHz CPU and 1 GB memory
  - VM5: 1 GHz CPU and 1 GB memory
- The Configured Failover Capacity for CPU and memory is set to 25%.



The total resource requirements for the powered-on virtual machines are 7 GHz CPU and 6 GB memory. The total host resources available for virtual machines are 24 GHz CPU and 21 GB memory. Based on this:

- The Current CPU Available Capacity is 70%. Calculated as (24 GHz - 7 GHz)/24 GHz.
- The Current Memory Available Capacity is 71%. Calculated as (21 GB - 6 GB)/21 GB.

Because the Configured Failover Capacity is set to 25%, 45% of the cluster's total CPU resources and 46% of the cluster's memory remain available to power on additional virtual machines.

Admission Control Example with Memory Tiering

The following example illustrates how vSphere HA calculates tier-aware failover capacity in a cluster with Memory Tiering activated. The example uses manual mode, in which you specify the failover capacity percentage directly.

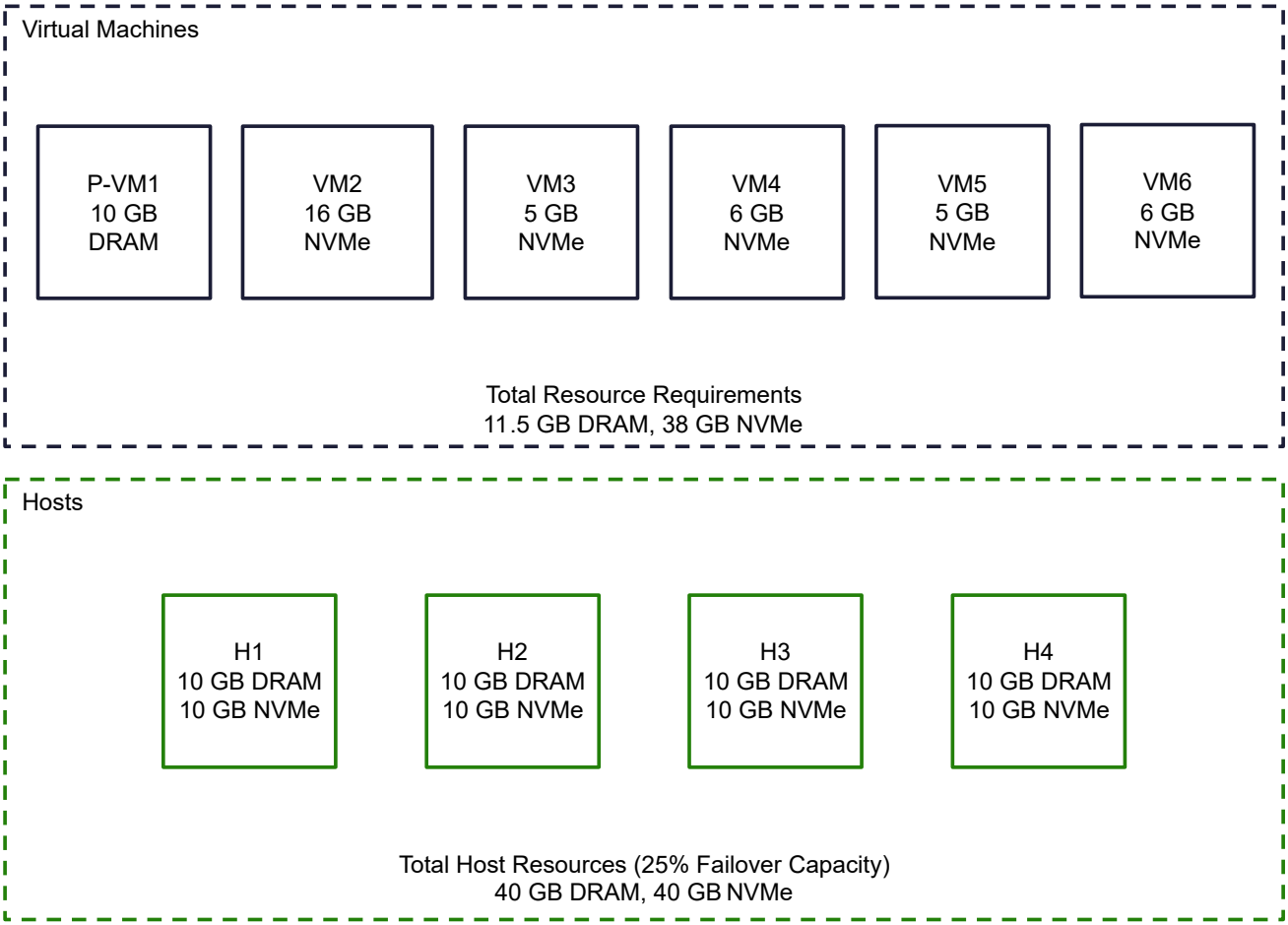
Cluster Configuration

The cluster consists of:

- Four hosts. Each host has 10 GB DRAM and 10 GB NVMe, for a total of 20 GB memory per host:
  - Host 1 (H1): 10 GB DRAM, 10 GB NVMe
  - Host 2 (H2): 10 GB DRAM, 10 GB NVMe
  - Host 3 (H3): 10 GB DRAM, 10 GB NVMe
  - Host 4 (H4): 10 GB DRAM, 10 GB NVMe
- Six powered-on virtual machines that are run in the cluster with the following reservations and tier classifications.
  - P-VM1 (pre-allocated, non-tiered VM): 10 GB reservation, all in DRAM

  - VM2 (tiered VM): 16 GB reservation, served from NVMe; 0.5 GB overhead in DRAM
  - VM3 (tiered VM): 5 GB reservation, served from NVMe; 0.25 GB overhead in DRAM
  - VM4 (tiered VM): 6 GB reservation, served from NVMe; 0.25 GB overhead in DRAM
  - VM5 (tiered VM): 5 GB reservation, served from NVMe; 0.25 GB overhead in DRAM
  - VM6 (tiered VM): 6 GB reservation, served from NVMe; 0.25 GB overhead in DRAM
- The Configured Failover Capacity for CPU and memory is set to 25% (manual mode).

Figure 113: Admission Control Example with Percentage of Cluster Resources Reserved Policy



The total memory that is used by all six virtual machines is 48 GB. The DRAM that is used by all six virtual machines is 11.5 GB (10 GB for P-VM1 full reservation plus 1.5 GB total overhead for VM2 through VM6).

The total host memory available for virtual machines is 40 GB of DRAM and 40 GB of NVMe (80 GB total).

The failover reservations at 25% are calculated as follows:

- Total memory failover reservation: 25% of 80 GB = 20 GB
- DRAM failover reservation: 25% of 40 GB = 10 GB

The DRAM reservation is counted within the total memory reservation - it is not added on top of it.

The current failover capacities are calculated as follows:

- Current Total Memory Failover Capacity: (80 GB − 49.5 GB) / 80 GB = 38.1%
- Current DRAM Failover Capacity: (40 GB − 11.5 GB) / 40 GB = 71.3%

Both current failover capacities exceed the configured 25%, so admission control allows the operation.

Available Capacity for Additional Virtual Machines

Because the configured failover capacity is 25%, the cluster retains the following capacity to power on additional virtual machines:

- Available total memory: 13.1% of 80 GB, or approximately 10.5 GB, remains available beyond the failover reservation.
- Available DRAM: 46.3% of 40 GB, or approximately 18.5 GB, remains available beyond the DRAM failover reservation.

A new virtual machine is admitted only if both checks pass: the available total memory capacity exceeds the total memory failover reservation, and the available DRAM capacity exceeds the DRAM failover reservation. If either check fails, vSphere HA raises an Insufficient Resource event and blocks the power-on. This prevents the cluster from being over-admitted on DRAM when total memory appears adequate.

How Tier-Aware Admission Control Prevents Silent Failures

If Host 1 fails, P-VM1 must restart on a surviving host and requires 10 GB of DRAM. Without tier-aware admission control, the cluster would appear to have 32 GB of available memory—more than enough to restart P-VM1. However, the surviving hosts might not have 10 GB of available DRAM, because their DRAM is already consumed by the overhead of other tiered virtual machines. With tier-aware admission control, vSphere HA verifies that sufficient DRAM capacity exists before admitting new virtual machines, preventing this silent failure.

Slot Policy Admission Control

With the slot policy option, vSphere HA admission control ensures that a specified number of hosts can fail and sufficient resources remain in the cluster to fail over all the virtual machines from those hosts.

Using the slot policy, vSphere HA performs admission control in the following way:

1. Calculates the slot size.
   A slot is a logical representation of memory and CPU resources. By default, it is sized to satisfy the requirements for any powered-on virtual machine in the cluster.
2. Determines how many slots each host in the cluster can hold.
3. Determines the Current Failover Capacity of the cluster.
   This is the number of hosts that can fail and still leave enough slots to satisfy all of the powered-on virtual machines.
4. Determines whether the Current Failover Capacity is less than the Configured Failover Capacity (provided by the user).
   If it is, admission control disallows the operation.

> **Note:**
> You can set a specific slot size for both CPU and memory in the admission control section of the vSphere HA settings in the vSphere Client.

Slot Size Calculation

Slot size is comprised of two components, CPU and memory.

- vSphere HA calculates the CPU component by obtaining the CPU reservation of each powered-on virtual machine and selecting the largest value. If you have not specified a CPU reservation for a virtual machine, it is assigned a default value of 32 MHz. You can change this value by using the das.vmcpuminmhz advanced option.)
- vSphere HA calculates the memory component by obtaining the memory reservation, plus memory overhead, of each powered-on virtual machine and selecting the largest value. There is no default value for the memory reservation.

If your cluster contains any virtual machines that have larger reservations than the others, they distort slot size calculation. To avoid this, you can specify an upper bound for the CPU or memory component of the slot size by using the das.slotcpuinmhz or das.slotmeminmb advanced options, respectively. See vSphere HA Advanced Options.

You can also determine the risk of resource fragmentation in your cluster by viewing the number of virtual machines that require multiple slots. This can be calculated in the admission control section of the vSphere HA settings in the vSphere Client. Virtual machines might require multiple slots if you have specified a fixed slot size or a maximum slot size using advanced options.

Using Slots to Compute the Current Failover Capacity

After the slot size is calculated, vSphere HA determines each host's CPU and memory resources that are available for virtual machines. These amounts are those contained in the host's root resource pool, not the total physical resources of the host. The resource data for a host that is used by vSphere HA can be found on the host's **Summary** tab on the vSphere Client. If all hosts in your cluster are the same, this data can be obtained by dividing the cluster-level figures by the number of hosts. Resources being used for virtualization purposes are not included. Only hosts that are connected, not in maintenance mode, and that have no vSphere HA errors are considered.

The maximum number of slots that each host can support is then determined. To do this, the host's CPU resource amount is divided by the CPU component of the slot size and the result is rounded down. The same calculation is made for the host's memory resource amount. These two numbers are compared and the smaller number is the number of slots that the host can support.

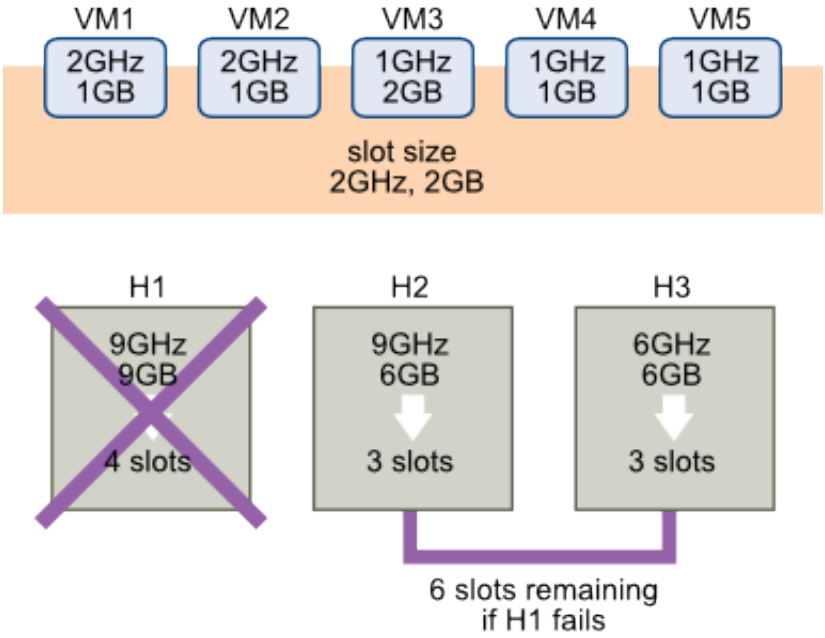
The Current Failover Capacity is computed by determining how many hosts (starting from the largest) can fail and still leave enough slots to satisfy the requirements of all powered-on virtual machines.

Admission Control Using Slot Policy

The way that slot size is calculated and used with this admission control policy is shown in an example. Make the following assumptions about a cluster:

- The cluster is comprised of three hosts, each with a different amount of available CPU and memory resources. The first host (H1) has 9 GHz of available CPU resources and 9 GB of available memory, while Host 2 (H2) has 9 GHz and 6 GB and Host 3 (H3) has 6 GHz and 6 GB.
- There are five powered-on virtual machines in the cluster with differing CPU and memory requirements. VM1 needs 2 GHz of CPU resources and 1 GB of memory, while VM2 needs 2 GHz and 1 GB, VM3 needs 1 GHz and 2 GB, VM4 needs 1 GHz and 1 GB, and VM5 needs 1 GHz and 1 GB.
- The Host Failures Cluster Tolerates is set to one.

Figure 114: Admission Control Example with Host Failures Cluster Tolerates Policy



1. Slot size is calculated by comparing both the CPU and memory requirements of the virtual machines and selecting the largest.
   The largest CPU requirement (shared by VM1 and VM2) is 2GHz, while the largest memory requirement (for VM3) is 2GB. Based on this, the slot size is 2GHz CPU and 2GB memory.
2. The maximum number of slots that each host can support is determined.
   H1 can support four slots. H2 can support three slots (which is the smaller of 9GHz/2GHz and 6GB/2GB) and H3 can also support three slots.
3. Current Failover Capacity is computed.

The largest host is H1 and if it fails, six slots remain in the cluster, which is sufficient for all five of the powered-on virtual machines. If both H1 and H2 fail, only three slots remain, which is insufficient. Therefore, you must configure the Failover Capacity as one host. A two-host failover capacity is not supported because the remaining resources are insufficient for the active virtual machines.

The cluster still has one available slot (the six slots on H2 and H3 minus the five used slots) to power on one new VM that fits the slot size.

vSphere HA Dedicated Failover Hosts Admission Control

You can configure vSphere HA to designate specific hosts as dedicated failover hosts. When a host fails, vSphere HA attempts to restart its virtual machines on the designated failover hosts. Starting in vSphere 9.1.1, clusters with Memory Tiering activated require the failover host to have sufficient DRAM capacity - not just total memory - to guarantee reliable restart.

With dedicated failover hosts admission control, you designate specific hosts as failover hosts. When a host fails, vSphere HA attempts to restart its virtual machines on any of the specified failover hosts. If restarting the virtual machines is not possible, for example, the failover hosts have failed or have insufficient resources, vSphere HA attempts to restart those virtual machines on other hosts in the cluster.

To ensure that spare capacity is available on a failover host, vSphere HA prevents you from powering on virtual machines or using vMotion to migrate virtual machines to a failover host. DRS does not use a failover host for load balancing.

Note:
If you use dedicated failover hosts admission control and designate multiple failover hosts, DRS does not attempt to enforce VM-VM affinity rules for virtual machines that are running on failover hosts.

Choosing a Dedicated Failover Host in a Cluster with Memory Tiering

Starting in vSphere 9.1.1, if your cluster is configured for Memory Tiering, you must consider DRAM capacity when you select a dedicated failover host. A failover host with sufficient total memory, but insufficient DRAM may fail to restart all virtual machines from a failing host, because virtual machines require DRAM to power on regardless of their configuration.

To guarantee reliable failover in a cluster with Memory Tiering, the dedicated failover host must meet the following capacity requirements.

- The failover host total memory capacity (DRAM + NVMe) must be at least equal to the largest total memory capacity among all active hosts in the cluster. This ensures that the failover host can accommodate all virtual machines from any failing host.
- The failover host DRAM capacity must be at least equal to the largest DRAM capacity among all active hosts in the cluster. This ensures that the failover host has enough DRAM to power on all virtual machines from any failing host.

Important:

If the dedicated failover host has less DRAM than the host with the largest DRAM capacity in the cluster, vSphere HA may fail to restart all virtual machines from that host. This condition silently undermines the failover guarantee, even though the total memory appears sufficient. The admission control status does not automatically warn about this condition. Verify the DRAM capacity manually when selecting a dedicated failover host for a memory-tiered cluster.

You must select a dedicated failover host with DRAM capacity at least equal to the largest DRAM capacity among all active hosts in the cluster. If the cluster violates the configured failover policy, including insufficient total memory, DRAM, or CPU capacity, vSphere HA raises an `InsufficientFailoverLevelEvent` as a configuration issue on the cluster.

How vSphere HA Tracks DRAM Capacity on the Failover Host

Starting in vSphere 9.1.1, vSphere HA tracks the unreserved DRAM capacity of the dedicated failover host and reports it to DRS whenever DRS queries for failover host resources. This ensures that DRS placement and load-balancing decisions account for DRAM as an independent resource, alongside CPU and total memory.

By tracking DRAM separately, vSphere HA ensures that DRS does not inadvertently consume the DRAM capacity that HA needs for virtual machine restart. This prevents a scenario in which DRS load-balancing actions fill the failover host's DRAM with virtual machine overhead, leaving insufficient DRAM for failover restart.

Spare VM Reservation for Dedicated Failover Hosts

The spare VM reservation for the Dedicated Failover Hosts policy includes unreserved DRAM capacity alongside CPU and total memory. DRS uses this figure to exclude the DRAM capacity from load-balancing decisions, ensuring that DRS actions do not inadvertently consume the DRAM that HA needs for restart.

When DRS queries vSphere HA for the spare VM reservation, vSphere HA returns the following resources for each failover host:

- CPU - unreserved CPU capacity of the failover host.
- Total memory - unreserved total memory capacity (DRAM + NVMe) of the failover host.
- DRAM - unreserved DRAM capacity of the failover host.

DRS excludes all three resources from its load-balancing calculations, ensuring that the failover host retains sufficient CPU, total memory, and DRAM capacity to restart virtual machines after a host failure.

vSphere HA Interoperability

vSphere HA can interoperate with many other features, such as DRS and vSAN.

Before configuring vSphere HA, you should be aware of the limitations of its interoperability with these other features or products.

Using vSphere HA with vSAN

You can use vSAN as the shared storage for a vSphere HA cluster. If activated, vSAN aggregates the specified local storage disks available on the hosts into a single datastore shared by all hosts.

To use vSphere HA with vSAN, you must be aware of certain considerations and limitations for the interoperability of these two features.

For information about vSAN, see *Administering VMware vSAN*.

> **Note:**
> You can use vSphere HA with vSAN stretched clusters.

ESX Host Requirements

You can use vSAN with a vSphere HA cluster only if the following conditions are met:

- All the cluster's ESX hosts must be version 5.5 or later.
- The cluster must have a minimum of three ESX hosts.

Networking Differences

vSAN has its own network. If vSAN and vSphere HA are activated for the same cluster, the HA interagent traffic flows over this storage network rather than the management network. vSphere HA uses the management network only if vSAN is deactivated. vCenter chooses the appropriate network if vSphere HA is configured on a host.

> **Note:**
> You can activate vSAN only if vSphere HA is deactivated.

If you change the vSAN network configuration, the vSphere HA agents do not automatically pick up the new network settings. To change the vSAN network, you must take the following steps in the vSphere Client:

1. Deactivate Host Monitoring for the vSphere HA cluster.
2. Make the vSAN network changes.
3. Right-click all hosts in the cluster and select **Reconfigure for vSphere HA**.

4. Re-activate Host Monitoring for the vSphere HA cluster.

The Table 315: vSphere HA Networking Differences table shows the differences in vSphere HA networking whether or not vSAN is used.

Table 315: vSphere HA Networking Differences

	vSAN Activated	vSAN Deactivated
Network used by vSphere HA	vSAN storage network	Management network
Heartbeat datastores	Any datastore mounted to > 1 host, but not vSAN datastores	Any datastore mounted to > 1 host
Host declared isolated	Isolation addresses not pingable and vSAN storage network inaccessible	Isolation addresses not pingable and management network inaccessible

Capacity Reservation Settings

When you reserve capacity for your vSphere HA cluster with an admission control policy, you must coordinate this setting with the corresponding vSAN setting that ensures data accessibility on failures. Specifically, the Number of Failures Tolerated setting in the vSAN rule set must not be lower than the capacity that the vSphere HA admission control setting reserved.

For example, if the vSAN rule set allows for only two failures, the vSphere HA admission control policy must reserve capacity that is equivalent to only one or two host failures. If you are using the Percentage of Cluster Resources Reserved policy for a cluster that has eight hosts, you must not reserve more than 25% of the cluster resources. In the same cluster, with the Host Failures Cluster Tolerates policy, the setting must not be higher than two hosts. If vSphere HA reserves less capacity, failover activity might be unpredictable. Reserving too much capacity overly constrains the powering on of virtual machines and intercluster vSphere vMotion migrations.

Using vSphere HA and DRS Together

Using vSphere HA with Distributed Resource Scheduler (DRS) combines automatic failover with load balancing. Starting in vSphere 9.1.1, clusters with memory tiering activated extend this integration to include DRAM as a third tracked resource, ensuring that DRS placement and load-balancing decisions do not consume the DRAM capacity that HA needs for virtual machine restart.

Using vSphere HA with Distributed Resource Scheduler (DRS) combines automatic failover with load balancing. This combination can result in a more balanced cluster after vSphere HA moves virtual machines to different hosts.

When vSphere HA performs failover and restarts virtual machines on different hosts, its first priority is the immediate availability of all virtual machines. After vSphere HA restarts the virtual machines, the hosts on which they were powered on might be heavily loaded, while other hosts are comparatively lightly loaded. vSphere HA uses the virtual machine's CPU and memory reservation and overhead memory to determine whether a host has enough spare capacity to accommodate the virtual machine. In clusters with memory tiering enabled, vSphere HA also uses the virtual machine's DRAM requirement to make this determination.

Admission Control and Maintenance Mode

In a cluster using DRS and vSphere HA with admission control turned on, virtual machines might not be evacuated from hosts entering maintenance mode. This behavior occurs because of the resources reserved for restarting virtual machines in the event of a failure. You must manually migrate the virtual machines with vMotion.

Scenarios Where vSphare HA Cannot Fail over Virtual Machines

In some scenarios, vSphere HA might not be able to fail over virtual machines because of resource constraints. This can occur for several reasons.

- HA admission control is deactivated and Distributed Power Management (DPM) is activated. This can result in DPM consolidating virtual machines onto fewer hosts and placing the empty hosts in standby mode leaving insufficient powered-on capacity to perform a failover.

- VM-Host affinity (required) rules might limit the hosts on which certain virtual machines can be placed.
- There might be sufficient aggregate resources but these can be fragmented across multiple hosts so that they can not be used by virtual machines for failover.

In such cases, vSphere HA can use DRS to try to adjust the cluster (for example, by bringing hosts out of standby mode or migrating virtual machines to defragment the cluster resources) so that HA can perform the failovers.

If DPM is in manual mode, you must confirm host power-on recommendations. Similarly, if DRS is in manual mode, you might need to confirm migration recommendations.

If you are using VM-Host affinity rules that are required, be aware that these rules cannot be violated. vSphere HA does not perform a failover if doing so would violate such a rule.

For more information about DRS, see the *vSphere Resource Management* documentation.

DRAM-Aware DRS Integration in Clusters with Memory Tiering

Starting with vSphere 9.1.1, clusters with Memory Tiering activated extend DRS and vSphere HA integration to include DRAM as a third tracked resource, alongside CPU and total memory. This ensures that DRS placement and load-balancing decisions account for DRAM reservations, preventing DRS actions from inadvertently consuming the DRAM capacity that vSphere HA needs for virtual machine restarts.

The DRAM-aware integration affects the following three areas of DRS and vSphere HA interaction.

DRAM-Aware Unreserved Capacity

In clusters using Memory Tiering, DRS receives DRAM-aware unreserved capacity from vSphere HA in addition to the existing CPU and total memory figures. This allows DRS to account for DRAM reservations when making placement and load-balancing decisions, ensuring that DRS moves do not inadvertently consume the DRAM capacity that HA needs for virtual machine restart.

Unreserved Capacity Rule

When vSphere HA runs admission control to generate the unreserved capacity rule that is used during host maintenance mode entry, vMotion, and load balancing, vSphere HA communicates the minimum DRAM capacity required to be kept free across the cluster to DRS alongside the existing CPU and total memory minimums. This ensures that DRS respects the DRAM failover reservation when generating placement recommendations and load-balancing migrations.

Spare VM Reservation

The internal vSphare HA call that DRS invokes to determine how much capacity is reserved for HA (`GetSpareVMReservation`) is extended to return DRAM spare capacity as an independent output. This extension applies to both the Cluster Resources Percentage policy and the Dedicated Failover Hosts policy. DRS uses the DRAM spare capacity figure to exclude the DRAM capacity from its load-balancing calculations, ensuring that DRS actions do not consume the DRAM that HA needs for restart.

The following table summarizes the three resources that DRS now receives from DAS in memory-tiered clusters.

Resource	What RDS Receives from HA	How DRS Uses it
CPU	Unreserved CPU capacity and CPU failover reservation	Excludes the CPU capacity that is reserved for HA from load-balancing decisions.
Total memory	Unreserved total memory capacity (DRAM + NVMe) and total memory failover reservation	Excludes total memory capacity for HA from load-balancing decisions.
DRAM	Unreserved DRAM capacity and DRAM failover reservation	Excludes the DRAM capacity that is reserved for HA from load-balancing decisions.

Note:

vSphere DRS is a critical feature of vSphere which is required to maintain the health of the workloads running inside vSphere Cluster.

vSphere HA and DRS Affinity Rules

If you create a DRS affinity rule for your cluster, you can specify how vSphere HA applies that rule during a virtual machine failover.

The two types of rules for which you can specify vSphere HA failover behavior are the following:

- VM anti-affinity rules force specified virtual machines to remain apart during failover actions.
- VM-Host affinity rules place specified virtual machines on a particular host or a member of a defined group of hosts during failover actions.

When you edit a DRS affinity rule, you must use vSphere HA advanced options to enforce the desired failover behavior for vSphere HA.

- **HA must respect VM anti-affinity rules during failover** -- When the advanced option for VM anti-affinity rules is set, vSphere HA does not fail over a virtual machine if doing so violates a rule. Instead, vSphere HA issues an event reporting there are insufficient resources to perform the failover.
- **HA should respect VM to Host affinity rules during failover** --vSphere HA attempts to place VMs with this rule on the specified hosts if at all possible.

For more information, see vSphere HA Advanced Options.

> **Note:**
> vSphere HA can restart a VM in a DRS-deactivated cluster, overriding a VM-Host affinity rules mapping if the host failure happens soon (by default, within 5 minutes) after setting the rule.

Other vSphere HA Interoperability Issues

To use vSphere HA, you must be aware of the following additional interoperability issues.

VM Component Protection

VM Component Protection (VMCP) has the following interoperability issues and limitations:

- VMCP does not detect or respond to accessibility issues for files located on vSAN datastores. If a virtual machine's configuration and VMDK files are located only on vSAN datastores, they are not protected by VMCP.
- VMCP does not detect or respond to accessibility issues for files located on Virtual Volume datastores. If a virtual machine's configuration and VMDK files are located only on Virtual Volume datastores, they are not protected by VMCP.
- VMCP does not protect against inaccessible Raw Device Mapping (RDM)s.

> **Note:**
>
> vSphere Virtual Volumes (vVols) framework is deprecated in VMware vSphere 9.0.

IPv6

vSphere HA can be used with IPv6 network configurations, which are fully supported if the following considerations are observed:

- The cluster contains only ESX 6.0 or later hosts.
- The management network for all hosts in the cluster must be configured with the same IP version, either IPv6 or IPv4. vSphere HA clusters cannot contain both types of networking configuration.
- The network isolation addresses used by vSphere HA must match the IP version used by the cluster for its management network.
- IPv6 cannot be used in vSphere HA clusters that also utilize vSAN.

In addition to the previous restrictions, the following types of IPv6 address types are not supported for use with the vSphere HA isolation address or management network: link-local, ORCHID, and link-local with zone indices. Also, the loopback address type cannot be used for the management network.

Note:
To upgrade an existing IPv4 deployment to IPv6, you must first deactivate vSphere HA.

Creating a vSphere HA Cluster

vSphere HA operates in the context of a cluster of ESX (or legacy ESX) hosts. You must create a cluster, populate it with hosts, and configure vSphere HA settings before failover protection can be established.

When you create a vSphere HA cluster, you must configure a number of settings that determine how the feature works. Before you do this, identify your cluster's nodes. These nodes are the ESX hosts that will provide the resources to support virtual machines and that vSphere HA will use for failover protection. You should then determine how those nodes are to be connected to one another and to the shared storage where your virtual machine data resides. After that networking architecture is in place, you can add the hosts to the cluster and finish configuring vSphere HA.

You can activate and configure vSphere HA before you add host nodes to the cluster. However, until the hosts are added, your cluster is not fully operational and some of the cluster settings are unavailable. For example, the Specify a Failover Host admission control policy is unavailable until there is a host that can be designated as the failover host.

Note:
The Virtual Machine Startup and Shutdown (automatic startup) feature is deactivated for all virtual machines residing on hosts that are in (or moved into) a vSphere HA cluster. Automatic startup is not supported when used with vSphere HA.

vSphere HA Checklist

The vSphere HA checklist contains requirements that you must be aware of before creating and using a vSphere HA cluster.

Review this list before you set up a vSphere HA cluster. For more information, follow the appropriate cross reference.

- A cluster must contain at least two hosts.
- All hosts must be configured with static IP addresses. If you are using DHCP, you must ensure that the address for each host persists across reboots.
- All hosts must have at least one management network in common. The best practice is to have at least two management networks in common. You should use the VMkernel network with the **Management traffic** checkbox enabled. The networks must be accessible to each other and vCenter and the hosts must be accessible to each other on the management networks. For more information, see Best Practices for Networking.
- To ensure that any virtual machine can run on any host in the cluster, all hosts must have access to the same virtual machine networks and datastores. Similarly, virtual machines must be located on shared, not local, storage otherwise they cannot be failed over in the case of a host failure.

  Note:
  vSphere HA uses datastore heartbeating to distinguish between partitioned, isolated, and failed hosts. So if some datastores are more reliable in your environment, configure vSphere HA to give preference to them.
- For VM Monitoring to work, VMware tools must be installed. For more information, see VM and Application Monitoring.
- vSphere HA supports both IPv4 and IPv6. For more information, see Other vSphere HA Interoperability Issues for considerations when using IPv6.
- For VM Component Protection to work, hosts must have the All Paths Down (APD) Timeout feature enabled.
- To use VM Component Protection, clusters must contain ESX 7.0 hosts or later.
- Only vSphere HA clusters that contain ESX 7.0 or later hosts can be used to enable VMCP. Clusters that contain hosts from an earlier release cannot enable VMCP, and such hosts cannot be added to a VMCP-enabled cluster.
- If your cluster uses Virtual Volume datastores, when vSphere HA is enabled a configuration Virtual Volume is created on each datastore by vCenter. In these containers, vSphere HA stores the files it uses to protect virtual machines. vSphere HA does not function correctly if you delete these containers. Only one container is created per Virtual Volume datastore.

  Note:

  vSphere Virtual Volumes (vVols) framework is deprecated in VMware vSphere 9.0.

Create a vSphere HA Cluster in the vSphere Client

To enable your cluster for vSphere HA, you must first create an empty cluster. After you plan the resources and networking architecture of your cluster, use the vSphere Client to add hosts to the cluster and specify the cluster's vSphere HA settings.

- Verify that all virtual machines and their configuration files reside on shared storage.
- Verify that the hosts are configured to access the shared storage so that you can power on the virtual machines by using different hosts in the cluster.
- Verify that hosts are configured to have access to the virtual machine network.
- Verify that you are using redundant management network connections for vSphere HA. For information about setting up network redundancy, see Best Practices for Networking.
- Verify that you have configured hosts with at least two datastores to provide redundancy for vSphere HA datastore heartbeating.
- Connect vSphere Client to vCenter by using an account with cluster administrator permissions.

A vSphere HA-enabled cluster is a prerequisite for vSphere Fault Tolerance.

1. In the vSphere Client, browse to the data center where you want the cluster to reside, and click **New Cluster**.
2. Complete the **New Cluster** wizard.

   Do not turn on vSphere HA (or DRS).
3. Click **OK** to close the wizard and create an empty cluster.
4. Based on your plan for the resources and networking architecture of the cluster, use the vSphere Client to add hosts to the cluster.
5. Browse to the cluster and enable vSphere HA.
   a) Click the **Configure** tab.
   b) Select **vSphere Availability** and click **Edit**.
   c) Select **vSphere HA**.
6. Under **Failures and Responses** select **Enable Host Monitoring**.

   With Host Monitoring enabled, hosts in the cluster can exchange network heartbeats and vSphere HA can take action when it detects failures. Host Monitoring is required for the vSphere Fault Tolerance recovery process to work properly.
7. Select a setting for **VM Monitoring**.

   Select **VM Monitoring Only** to restart individual virtual machines if their heartbeats are not received within a set time. You can also select **VM and Application Monitoring** to enable application monitoring.
8. Click **OK**.

You have a vSphere HA cluster, populated with hosts.

Configure the appropriate vSphere HA settings for your cluster.

- Failures and responses
- Admission Control
- Heartbeat Datastores
- Advanced Options

See Configuring vSphere Availability Settings.

Configuring vSphere Availability Settings

When you create a vSphere HA cluster or configure an existing cluster, you must configure settings that determine how the feature works.

In the vSphere Client, you can configure following the vSphere HA settings:

Failures and responses	Provide settings here for host failure responses, host isolation, VM monitoring, and VM Component Protection.

Admission Control	Activate or deactivate admission control for the vSphere HA cluster and choose a policy for how it is enforced.
Heartbeat Datastores	Specify preferences for the datastores that vSphere HA uses for datastore heartbeating.
Advanced Options	Customize vSphere HA behavior by setting advanced options.

Configuring Responses to Failures

The **Failure and Responses** pane of the vSphere HA settings allows you to configure how your cluster should function when problems are encountered.

In this part of the vSphere Client, you can determine the specific responses the vSphere HA cluster has for host failures and isolation. You can also configure VM Component Protection (VMCP) actions when Permanent Device Loss (PDL) and All Paths Down (APD) situations occur and you can enable VM monitoring.

The following tasks are available:

Respond to Host Failure

You can set specific responses to host failures that occur in your vSphere HA cluster.

This page is editable only if you have activated vSphere HA.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Failures and Responses** and then expand **Host Failure Response**.
5. Select from the following configuration options.

Option	Description
Failure Response	If you select **Disabled**, this setting turns off host monitoring and VMs are not restarted when host failures occur. If **Restart VMs** is selected, VMs are failed over based on their restart priority when a host fails.
Default VM Restart Priority	The restart priority determines the order in which virtual machines are restarted when the host fails. Higher priority virtual machines are started first. If multiple hosts fail, all virtual machines are migrated from the first host in order of priority, then all virtual machines from the second host in order of priority, and so on.
VM Restart Priority Condition	A specific condition must be selected as well as a delay after that condition has been met, before vSphere HA is allowed to continue to the next VM restart priority.

6. Click **OK**.

Your settings for the host failure response take effect.

Respond to Host Isolation

You can set specific responses to host isolation that occurs in your vSphere HA cluster.

This page is editable only if you have activated vSphere HA.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Failures and Responses** and expand **Response for Host Isolation**.
5. To configure the host isolation response, select **Disabled**, **Shut down and restart VMs**, or **Power off and restart VMs**.
6. Click **OK**.

Your setting for the host isolation response takes effect.

Configure VMCP Responses

Configure the response that VM Component Protection (VMCP) makes when a datastore encounters a PDL or APD failure.

This page is editable only if you have activated vSphere HA.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Failures and Responses**, and expand either **Datastore with PDL** or **Datastore with APD** .
5. If you clicked **Datastore with PDL**, you can set the VMCP failure response for this type of issue, either **Disabled**, **Issue Events**, or **Power off and restart VMs**.
6. If you clicked **Datastore with APD**, you can set the VMCP failure response for this type of issue, either **Disabled**, **Issue Events**, **Power off and restart VMs--Conservative restart policy**, or **Power off and restart VMs--Aggressive restart policy**. You can also set **Response recovery**, which is the number of minutes that VMCP waits before taking action.
7. Click **OK**.

Your settings for the VMCP failure response take effect.

Activate VM Monitoring

You can turn on VM and Application Monitoring and also set the monitoring sensitivity for your vSphere HA cluster.

This page is editable only if you have activated vSphere HA.

> **Note:**
> vSphere HA might failover a non-healthy VM on a healthy host if VM monitoring is deactivated. The host could be selected because of a DRS recommendation or based on which host has the lowest available resources.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Failures and Responses** and expand **VM Monitoring**.
5. Select **VM Monitoring** and **Application Monitoring**.
   These settings turn on VMware Tools heartbeats and application heartbeats, respectively.
6. To set the heartbeat monitoring sensitivity, move the slider between **Low** and **High** or select **Custom** to provide custom settings.
7. Click **OK**.

Your monitoring settings take effect.

Configure Proactive HA

You can configure how Proactive HA responds when a provider has notified its health degradation to vCenter, indicating a partial failure of that host.

This page is editable only if you have enabled vSphere DRS.

1. In the vSphere Client, browse to the Proactive HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Select **Turn on Proactive HA**.
5. Click **Proactive HA Failures and Responses**.
6. Select from the following configuration options.

Option	Description
Automation Level	Determine whether host quarantine or maintenance mode and VM migrations are recommendations or automatic. • **Manual**. vCenter suggests migration recommendations for virtual machines. • **Automated**. Virtual machines are migrated to healthy hosts and degraded hosts are entered into quarantine or maintenance mode depending on the configured Proactive HA automation level.
Remediation	Determine what happens to partially degraded hosts. • **Quarantine mode for all failures**. Balances performance and availability, by avoiding the usage of partially degraded hosts provided that virtual machine performance is unaffected. • **Quarantine mode for moderate and Maintenance mode for severe failure (Mixed)**. Balances performance and availability, by avoiding the usage of moderately degraded hosts provided that virtual machine performance is unaffected. Ensures that virtual machines do not run on severely failed hosts. • **Maintenance mode for all failures**. Ensures that virtual machines do not run on partially failed hosts. `Host.Config.Quarantine` and `Host.Config.Maintenance` privileges are required to put hosts in Quarantine mode and Maintenance mode, respectively.

   To enable Proactive HA providers for this cluster, select the check boxes. Providers appear when their corresponding vSphere Client plugin has been installed and the providers monitor every host in the cluster. To view or edit the failure conditions supported by the provider, click the edit link.

7. Click **OK**.

Configure vSphere HA Admission Control

After you create a cluster, you can configure vSphere HA admission control to specify whether virtual machines can power on if they violate availability constraints. Starting in vSphere 9.1.1, clusters with Memory Tiering activated automatically enforce DRAM failover reservations for the Cluster Resource Percentage and Dedicated Failover Hosts policies.

- You must activate vSphere HA on the cluster before you can configure admission control.
- You must create the cluster before configuring admission control.

The Admission Control page appears only if you activated vSphere HA.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Admission Control** to display the configuration options.
5. Select a number for the **Host failures cluster tolerates**. This value specifies the maximum number of host failures the cluster can tolerate while guaranteeing a successful failover.
6. Select an option for **Define host failover capacity by**.

Option	Description
Cluster resource percentage	Specify a percentage of the cluster's CPU and memory resources to reserve as spare capacity to support failovers. **Note:** When a cluster includes hosts configured for Memory Tiering, vSphere HA automatically applies the specified memory percentage to the cluster's DRAM capacity, computing and enforcing a separate DRAM failover reservation. No additional manual configuration is required.
Slot Policy (powered-on VMs)	Select a slot size policy that covers all powered-on VMs or is a fixed size. You can also calculate how many VMs require multiple slots. **Note:** In a memory-tiered cluster, ensure that the designated failover host has a DRAM capacity at least equal to the largest DRAM capacity among all active hosts. For more information, see vSphere HA Dedicated Failover Hosts Admission Control.
Dedicated failover hosts	Select hosts to use for failover actions. Failovers can still occur on other hosts in the cluster if a default failover host does not have enough resources.
Disabled	Select this option to deactivate admission control and allow virtual machines power on that violate availability constraints.

7. Set the percentage for the **Performance degradation VMs tolerate**.

   This setting determines what percentage of performance degradation the VMs in the cluster are allowed to tolerate during a failure.
8. Click **OK**.

vSphere HA admission control is configured for the cluster. The cluster reserves CPU, memory, and in clusters that are configured for Memory Tiering - DRAM resources according to the policy you selected. If you selected the Cluster Resources Percentage policy, vSphere HA enforces both a total memory failover reservation and a DRAM failover reservation. If you selected the Dedicated Failover Hosts policy, vSphere HA tracks the unreserved DRAM capacity of the designated failover host and reports it to DRS.

Configure Heartbeat Datastores

vSphere HA uses datastore heartbeating to distinguish between hosts that have failed and hosts that reside on a network partition. With datastore heartbeating, vSphere HA can monitor hosts when a management network partition occurs and continue to respond to failures.

You can specify the datastores that you want to be used for datastore heartbeating.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Heartbeat Datastores** to display the configuration options for datastore heartbeating.
5. To instruct vSphere HA about how to select the datastores and how to treat your preferences, select from the following options.

Table 316:

Datastore Heartbeating Options
Automatically select datastores accessible from the host
Use datastores only from the specified list
Use datastores from the specified list and complement automatically if needed

6. In the Available heartbeat datastores pane, select the datastores that you want to use for heartbeating.
   The listed datastores are shared by more than one host in the vSphere HA cluster. When a datastore is selected, the lower pane displays all the hosts in the vSphere HA cluster that can access it.
7. Click **OK**.

Set Advanced Options

To customize vSphere HA behavior, set advanced vSphere HA options.

Verify that you have cluster administrator privileges.

> **Note:**
> Because these options affect the functioning of vSphere HA, change them with caution.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Select **vSphere Availability** and click **Edit**.
4. Click **Advanced Options**.
5. Click **Add** and type the name of the advanced option in the text box.
   You can set the value of the option in the text box in the Value column.
6. Repeat step 5 for each new option that you want to add and click **OK**.

The cluster uses the options that you added or modified.

Once you have set an advanced vSphere HA option, it persists until you do one the following:

- Using the vSphere Client, reset its value to the default value.
- Manually edit or delete the option from the fdm.cfg file on all hosts in the cluster.

vSphere HA Advanced Options

You can set advanced options that affect the behavior of your vSphere HA cluster. These options control isolation detection, slot size bounds, admission control defaults, VM monitoring sensitivity, and failover tolerance parameters. Starting in vSphere 9.1.1, the existing memory tolerance parameter also applies to DRAM failover capacity calculations in clusters configured for Memory Tiering.

Note:

With vSphere 9.1.1, DRAM failover capacity calculation uses the same tolerance parameters as total memory. The vpdx.das.memToleranceInMB advanced option controls the tolerance threshold for both total memory and DRAM failover violation checks.

If you do not specify a value, vSphere HA uses a default of 5% of the desired resource. No separate or additional advanced options are introduced for DRAM-specific tolerance configuration. For more information about how tolerance parameters apply to tier-aware failover violation checks, see Cluster Resources Percentage Admission Control.

Table 317: vSphere HA advanced options

Option	Description
das.isolationaddress[...]	Sets the address to ping to determine if a host is isolated from the network. vSphere HA pings this address only when heartbeats are not received from any other host in the cluster. If not specified, the default gateway of the management network is used. This default gateway has to be a reliable address that is available, so that the host can determine if it is isolated from the network. You can specify multiple isolation addresses (up to 10) for the cluster: das.isolationAddressX, where X = 0-9. Typically you should specify one per management network. Specifying too many addresses makes isolation detection take too long.
das.usedefaultisolationaddress	By default, vSphere HA uses the default gateway of the console network as an isolation address. This option specifies whether or not this default is used (true|false).
das.isolationshutdowntimeout	The period of time the system waits for a virtual machine to shut down before powering it off. This only applies if the host's isolation response is Shut down VM. Default value is 300 seconds.
das.slotmeminmb	Defines the maximum bound on the memory slot size. If this option is used, the slot size is the smaller of this value or the maximum memory reservation plus memory overhead of any powered-on virtual machine in the cluster.
das.slotcpuinmhz	Defines the maximum bound on the CPU slot size. If this option is used, the slot size is the smaller of this value or the maximum CPU reservation of any powered-on virtual machine in the cluster.
das.vmmemoryminmb	Defines the default memory resource value assigned to a virtual machine if its memory reservation is not specified or zero. This is used for the Host Failures Cluster Tolerates admission control policy. If no value is specified, the default is 0 MB.
das.vmcpuminmhz	Defines the default CPU resource value assigned to a virtual machine if its CPU reservation is not specified or zero. This is used for the Host Failures Cluster Tolerates admission control policy. If no value is specified, the default is 32MHz.
das.iostatsinterval	Changes the default I/O stats interval for VM Monitoring sensitivity. The default is 120 (seconds). Can be set to any value greater than, or equal to 0. Setting to 0 deactivates the check. **Note:** Values of less than 50 are not recommended since smaller values can result in vSphere HA unexpectedly resetting a virtual machine.

Option	Description
das.ignoreinsufficienthbdatastore	Deactivates configuration issues created if the host does not have sufficient heartbeat datastores for vSphere HA. Default value is false.
das.heartbeatdsperhost	Changes the number of heartbeat datastores required. Valid values can range from 2-5 and the default is 2.
das.config.fdm.isolationPolicyDelaySec	The number of seconds system waits before executing the isolation policy once it is determined that a host is isolated. The minimum value is 30. If set to a value less than 30, the delay will be 30 seconds.
das.respectvmvmantiaffinityrules	Determines if vSphere HA enforces VM-VM anti-affinity rules. The default value is "true" and rules are enforced even if vSphere DRS is not activated. In this case, vSphere HA does not fail over a virtual machine if doing so violates a rule, but it issues an event reporting there are insufficient resources to perform the failover. This option can also be set to "false", whereby the rules are not enforced. See *vSphere Resource Management* for more information on anti-affinity rules.
das.maxresets	The maximum number of reset attempts made by VMCP. If a reset operation on a virtual machine affected by an APD situation fails, VMCP retries the reset this many times before giving up
das.maxterminates	The maximum number of retries made by VMCP for virtual machine termination.
das.terminateretryintervalsec	If VMCP fails to terminate a virtual machine, this is the number of seconds the system waits before it retries a terminate attempt
das.config.fdm.reportfailoverfailevent	When set to 1, activates generation of a detailed per-VM event when an attempt by vSphere HA to restart a virtual machine is unsuccessful. Default value is 0. In versions earlier than vSphere 6.0, this event is generated by default.
vpxd.das.completemetadataupdateintervalsec	The period of time (seconds) after a VM-Host affinity rule is set during which vSphere HA can restart a VM in a DRS-deactivated cluster, overriding the rule. Default value is 300 seconds.
das.config.fdm.memReservationMB	By default vSphere HA agents run with a configured memory limit of 250 MB. A host might not allow this reservation if it runs out of reservable capacity. You can use this advanced option to lower the memory limit to avoid this issue. Only integers greater than 100, which is the minimum value, can be specified. Conversely, to prevent problems during primary agent elections in a large cluster (containing 6,000 to 8,000 VMs) you should raise this limit to 325 MB. **Note:** Once this limit is changed, for all hosts in the cluster you must run the Reconfigure HA task. Also, when a new host is added to the cluster or an existing host is rebooted, this task should be performed on those hosts in order to update this memory setting.
das.reregisterrestartdisabledvms	When vSphere HA is deactivated on a specific VM this option ensures that the VM is registered on another host after a failure. This allows you to power-on that VM without needing to re-register it manually. **Note:** When this option is used, vSphere HA does not power on the VM, but only registers it.

Option	Description
das.respectvmhostsoftaffinityrules	Determines if vSphere HA restarts a respective VM on a host that belongs to the same VM-Host group. If no such host is available or if the value of this option is set to “false”, vSphere HA restarts the VM on any available host in the cluster. In vSphere 6.5 or later, the default value is "true". This value might not be visibly defined in the advanced HA options of the cluster. If you want to deactivate the option, you must manually set this option as "false" in the advanced HA options for the cluster.

Note:
If you change the value of any of the following advanced options, you must deactivate and then re-activate vSphere HA before your changes take effect.

- das.isolationaddress[...]
- das.usedefaultisolationaddress
- das.isolationshutdowntimeout

Tolerance Parameters for Failover Violation Checks

In vSphere 9.1.1, the existing tolerance parameters that govern failover violation checks for total memory and CPU also apply to DRAM capacity in clusters that are configured for Memory Tiering.

The following table describes the tolerance parameters.

Option	Description
vpdx.das.memToleranceInMB	Defines the tolerance threshold for memory failover violation checks. vSphere HA applies this threshold to both total memory and DRAM capacity calculations in memory-tiered clusters. If you do not specify a value, vSphere HA uses a default of 5% of the desired resource.
vpxd.das.cpuToleranceInMhz	Defines the tolerance threshold for CPU failover violation checks. If you do not specify a value, vSphere HA uses a default of 5% of the desired resource.

Customize an Individual Virtual Machine

Each virtual machine in a vSphere HA cluster is assigned the cluster default settings for VM Restart Priority, Host Isolation Response, VM Component Protection, and VM Monitoring. You can specify specific behavior for each virtual machine by changing these defaults. If the virtual machine leaves the cluster, these settings are lost.

1. In the vSphere Client, browse to the vSphere HA cluster.
2. Click the **Configure** tab.
3. Under Configuration, select **VM Overrides** and click **Add**.
4. Use the **+** button to select virtual machines to which to apply the overrides.
5. Click **OK**.
6. Optional: You can change other settings, such as the **Automation level**, **VM restart priority**, **Response for Host Isolation**, VMCP settings,**VM Monitoring**, or **VM monitoring sensitivity** settings.

   Note:
   You can view the cluster defaults for these settings by first expanding **Relevant Cluster Settings** and then expanding **vSphere HA**.

7. Click **OK**.

The virtual machine's behavior now differs from the cluster defaults for each setting that you changed.

Best Practices for vSphere High Availability Clusters

To ensure optimal vSphere HA cluster performance, you must follow certain best practices. This section highlights key best practices for networking, interoperability, cluster monitoring, and starting in vSphere 9.1.1 - Memory Tiering and admission control policy selection.

You can also refer to the *vSphere High Availability Deployment Best Practices* publication for further discussion.

Best Practices for Memory Tiering and Admission Control

Starting in vSphere 9.1.1, you can observe the following best practices for clusters that use memory tiering with vSphere HA admission control.

- In clusters configured with Memory Tiering, verify that the DRAM capacity of any designated failover host equals or exceeds the largest DRAM capacity of any active host in the cluster. A shortfall in DRAM (even when total memory appears sufficient) can cause virtual machines to fail to restart after a host failure.
- Do not use the Slot Policy admission control policy in clusters with Memory Tiering. Use Cluster Resource Percentage or Dedicated Failover Hosts to ensure DRAM-aware failover capacity enforcement.

Best Practices for Networking

Observe the following best practices for the configuration of host NICs and network topology for vSphere HA. Best Practices include recommendations for your ESX hosts, and for cabling, switches, routers, and firewalls.

Network Configuration and Maintenance

The following network maintenance suggestions can help you avoid the accidental detection of failed hosts and network isolation because of dropped vSphere HA heartbeats.

- When changing the networks that your clustered ESX hosts are on, suspend the Host Monitoring feature. Changing your network hardware or networking settings can interrupt the heartbeats that vSphere HA uses to detect host failures, which might result in unwanted attempts to fail over virtual machines.
- When you change the networking configuration on the ESX hosts themselves, for example, adding port groups, or removing vSwitches, suspend Host Monitoring. After you have made the networking configuration changes, you must reconfigure vSphere HA on all hosts in the cluster, which causes the network information to be reinspected. Then re-enable Host Monitoring.

  Note:

  Because networking is a vital component of vSphere HA, inform the vSphere HA administrator before you perform network maintenance.

Networks Used for vSphere HA Communications

To identify which network operations might disrupt the functioning of vSphere HA, you must know which management networks are being used for heartbeating and other vSphere HA communications.

- On ESX hosts in the cluster, vSphere HA communications, by default, travel over VMkernel networks. With an ESX host, if you want to use a network other than the one vCenter uses to communicate with the host for vSphere HA, you must explicitly enable the **Management traffic** check box.

To keep vSphere HA agent traffic on the networks you have specified, configure hosts so vmkNICs used by vSphere HA do not share subnets with vmkNICs used for other purposes. vSphere HA agents send packets using any pNIC that is associated with a given subnet when there is also at least one vmkNIC configured for vSphere HA management traffic. Therefore, to ensure network flow separation, the vmkNICs used by vSphere HA and by other features must be on different subnets.

Network Isolation Addresses

A network isolation address is an IP address that is pinged to determine whether a host is isolated from the network. vSphere HA pings this address only when a host has stopped receiving heartbeats from all other hosts in the cluster. If a host can ping its network isolation address, the host is not network isolated, and the other hosts in the cluster have either failed or are network partitioned. However, if the host cannot ping its isolation address, the host has likely become isolated from the network and no failover action is taken.

By default, the network isolation address is the default gateway for the host. Only one default gateway is specified, regardless of how many management networks have been defined. Use the das.isolationaddress[...] advanced option to add isolation addresses for additional networks. See vSphere HA Advanced Options.

Network Path Redundancy

Network path redundancy between cluster nodes is important for vSphere HA reliability. A single management network is a single point of failure and can result in failovers although only the network has failed. If you have only one management network, any failure between the host and the cluster can cause an unnecessary or false failover activity if heartbeat datastore connectivity is not retained during the networking failure. Possible failures include NIC failures, network cable failures, network cable removal, and switch resets. Consider these possible sources of failure between hosts and try to minimize them, typically by providing network redundancy.

The first way you can implement network redundancy is at the NIC level with NIC teaming. Using a team of two NICs connected to separate physical switches improves the reliability of a management network. Because servers connected through two NICs (and through separate switches) have two independent paths for sending and receiving heartbeats, the cluster is more resilient. To configure a NIC team for the management network, configure the vNICs in vSwitch configuration for Active or Standby configuration. The recommended parameter settings for the vNICs are:

- Default load balancing = route based on originating port ID
- Failback = No

After you add a NIC to a host in your vSphere HA cluster, you must reconfigure vSphere HA on that host.

In most implementations, NIC teaming provides sufficient heartbeat redundancy. As an alternative, you can create a second management network connection attached to a separate virtual switch. Redundant management networking allows the reliable detection of failures and prevents isolation or partition conditions from occurring, because heartbeats can be sent over multiple networks. The original management network connection is used for network and management purposes. When you create the second management network connection, vSphere HA sends heartbeats over both management network connections. If one path fails, vSphere HA still sends and receives heartbeats over the other path.

> **Note:**
> Configure the fewest possible number of hardware segments between the servers in a cluster. The goal is to limit single points of failure. Also, routes with too many hops can cause networking packet delays for heartbeats, and increase the possible points of failure.

Using IPv6 Network Configurations

Only one IPv6 address can be assigned to a given network interface used by your vSphere HA cluster. Assigning multiple IP addresses increases the number of heartbeat messages sent by the cluster's primary host with no corresponding benefit.

Best Practices for Interoperability

Observe the following best practices for allowing interoperability between vSphere HA and other features.

vSphere HA and Storage vMotion Interoperability in a Mixed Cluster

vSphere HA might respond to a host failure by restarting a virtual machine on a host with an ESX version different from the one on which the virtual machine was running before the failure. A problem can occur if, at the time of failure, the virtual machine was involved in a Storage vMotion action on an ESX 5.x host, and vSphere HA restarts the virtual machine on a host with a version earlier than ESX 5.0. While the virtual machine might power-on, any subsequent attempts at snapshot operations might corrupt the vdisk state and leave the virtual machine unusable.

Using Auto Deploy with vSphere HA

You can use vSphere HA and Auto Deploy together to improve the availability of your virtual machines. Auto Deploy provisions hosts when they power-on and you can also configure it to install the vSphere HA agent on hosts during the boot process. See the Auto Deploy documentation included in vSphere Installation and Setup for details.

Upgrading Hosts in a Cluster Using vSAN

If you are upgrading the ESX hosts in your vSphere HA cluster to version 5.5 or later, and you also plan to use vSAN, follow this process.

1. Upgrade all of the hosts.
2. Deactivate vSphere HA.
3. Activate vSAN.
4. Re-activate vSphere HA.

Best Practices for Cluster Monitoring

Observe the following best practices for monitoring the status and validity of your vSphere HA cluster.

Setting Alarms to Monitor Cluster Changes

When vSphere HA or Fault Tolerance take action to maintain availability, for example, a virtual machine failover, you can be notified about such changes. Configure alarms in vCenter to be triggered when these actions occur, and have alerts, such as emails, sent to a specified set of administrators.

Several default vSphere HA alarms are available.

- Insufficient failover resources (a cluster alarm)
- Cannot find primary (a cluster alarm)
- Failover in progress (a cluster alarm)
- Host HA status (a host alarm)
- VM monitoring error (a virtual machine alarm)
- VM monitoring action (a virtual machine alarm)
- Failover failed (a virtual machine alarm)

> **Note:**
> The default alarms include the feature name, vSphere HA.

Monitor and Analyze High Availability Events

As an administrator, you can efficiently monitor and track vSphere HA events that are related to host failures and failover operations. You can monitor HA events in a single data grid in the user interface, accelerating the root cause analysis.

The vSphere HA events are listed in chronological order, ensuring precise tracking of the event sequence. The events show their logical relationships, allowing you to quickly understand the sequence of actions that led to a state change. You can track information about the failed host, the VMs that attempted failover, the VMs that succeeded failover, and those VMs with failed failover attempts.

For each event, you can trace the chain of events and you can view individual events that occur independently.

Selecting an event displays the following information:

- Date and time of event
- Event type
- User
- Host
- Detailed description
- Related events

To view the HA events interface, perform the following steps.

1. In the vSphere Client, navigate to and select a cluster.
2. Click the **Monitor** tab.
3. Navigate to **vSphere HA** > **HA Events**.
   The HA events appear in the **HA Events** pane.
4. To display the entire chain of events, expand a specific event from the list.
5. (Optional) Export all events or filter the information that you want to export.
   a. To export all events, from the **Events** pane, select **Export** > **All events**.
      All events are exported in a comma-separated values (CSV) file.
   b. To customize more precisely the information you want to export, select **Export** > **Advanced export**, and click **Export**.
      You can filter the data based on various criteria, such as category, type, time range, the number of events to export.

Change in behavior for HA VIBs

In vSphere 7.0 or later, it is possible the HA VIBs might be removed in some cases when HA is activated on a Lifecycle Manager (vLCM) cluster. In previous releases, vCenter would not attempt to remove HA VIBs from ESX hosts.

This situation can occur only on the Lifecycle managed cluster with vSphere HA activated. When a vLCM **Remediate** operation occurs (either as a user-initiated operation or an API invocation) after vSphere HA is deactivated on the cluster, the vSphere HA VIBs might be removed as a consequence.

Note:

This change in behavior is harmless because vCenter pushes the required vSphere HA VIBs when HA is activated again.

Providing Fault Tolerance for Virtual Machines

You can use vSphere Fault Tolerance for your virtual machines to ensure continuity with higher levels of availability and data protection.

Fault Tolerance is built on the ESX host platform, and it provides availability by having identical virtual machines run on separate hosts.

To obtain the optimal results from Fault Tolerance you must be familiar with how it works, how to enable it for your cluster, virtual machines and the best practices for its usage.

How Fault Tolerance Works

You can use vSphere Fault Tolerance (FT) for most mission-critical virtual machines. FT provides continuous availability for such a virtual machine by creating and maintaining another VM that is identical and continuously available to replace it if a failover situation occurs.

The protected virtual machine is called the Primary VM. The duplicate virtual machine, the Secondary VM, is created and runs on another host. The primary VM is continuously replicated to the secondary VM so that the secondary VM can take over at any point, thereby providing Fault Tolerant protection.

The Primary and Secondary VMs continuously monitor the status of one another to ensure that Fault Tolerance is maintained. A transparent failover occurs if the host running the Primary VM fails, or encounters an uncorrectable hardware error in the memory of the Primary VM, in which case the Secondary VM is immediately activated to replace the Primary VM. A new Secondary VM is started and Fault Tolerance redundancy is reestablished automatically. If the host running the Secondary VM fails, it is also immediately replaced. In either case, users experience no interruption in service and no loss of data.

A fault tolerant virtual machine and its secondary copy are not allowed to run on the same host. This restriction ensures that a host failure cannot result in the loss of both VMs.

Note:
You can also use VM-Host affinity rules to dictate which hosts designated virtual machines can run on. If you use these rules, be aware that for any Primary VM that is affected by such a rule, its associated Secondary VM is also affected by that rule. For more information about affinity rules, see the vSphere Resource Management documentation.

Fault Tolerance avoids "split-brain" situations, which can lead to two active copies of a virtual machine after recovery from a failure. Atomic file locking on shared storage is used to coordinate failover so that only one side continues running as the Primary VM and a new Secondary VM is respawned automatically.

vSphere Fault Tolerance can accommodate symmetric multiprocessor (SMP) virtual machines with up to 8 vCPUs.

Fault Tolerance Use Cases

Several typical situations can benefit from the use of vSphere Fault Tolerance.

Fault Tolerance provides a higher level of business continuity than vSphere HA. When a Secondary VM is called upon to replace its Primary VM counterpart, the Secondary VM immediately takes over the Primary VM's role with the entire state of the virtual machine preserved. Applications are already running, and data stored in memory does not need to be reentered or reloaded. Failover provided by vSphere HA restarts the virtual machines affected by a failure.

This higher level of continuity and the added protection of state information and data informs the scenarios when you might want to deploy Fault Tolerance.

- Applications which must always be available, especially applications that have long-lasting client connections that users want to maintain during hardware failure.
- Custom applications that have no other way of doing clustering.
- Cases where high availability might be provided through custom clustering solutions, which are too complicated to configure and maintain.

Another key use case for protecting a virtual machine with Fault Tolerance can be described as On-Demand Fault Tolerance. In this case, a virtual machine is adequately protected with vSphere HA during normal operation. During certain critical periods, you might want to enhance the protection of the virtual machine. For example, you might be running a quarter-end report which, if interrupted, might delay the availability of critical information. With vSphere Fault Tolerance, you can protect this virtual machine before running this report and then turn off or suspend Fault Tolerance after the report has been produced. You can use On-Demand Fault Tolerance to protect the virtual machine during a critical time period and return the resources to normal during non-critical operation.

Fault Tolerance Requirements and Limits

Before using vSphere Fault Tolerance (FT), consider the high-level requirements and limits that apply to this feature.

Requirements

The following CPU and networking requirements apply to FT.

CPUs that are used in host machines for fault tolerant VMs must be compatible with vSphere vMotion. Also, CPUs that support Hardware MMU virtualization (Intel EPT or AMD RVI) are required. The following CPUs are supported.

- Intel Sandy Bridge or later. Avoton is not supported.
- AMD Bulldozer or later.

Use a 10-Gbit logging network for FT and verify that the network is low latency. A dedicated FT network is highly recommended.

Note:
Fault Tolerance is currently unsupported to be enabled on any VM utilizing an NSX T created portgroup (vlan or overlay segment). Fault Tolerance is also unsupported for NSX T Managers and Edge nodes.

Limits

In a cluster configured to use Fault Tolerance, two limits are enforced independently.

das.maxftvmsperhost	The maximum number of fault tolerant VMs allowed on a host in the cluster. The default value is 4. There is no FT VMs per host maximum, you can use larger numbers if the workload performs well in FT VMs. You can deactivate checking by setting the value to `0`.
das.maxftvcpusperhost	The maximum number of vCPUs aggregated across all fault tolerant VMs on a host. The default value is 8. There is no FT vCPU per host maximum, you can use larger numbers if the workload performs well. You can deactivate checking by setting the value to `0`.

Fault Tolerance Interoperability

Before configuring vSphere Fault Tolerance, you must be aware of the features and products Fault Tolerance cannot interoperate with.

vSphere Features Not Supported with Fault Tolerance

When configuring your cluster, you should be aware that not all vSphere features can interoperate with Fault Tolerance.

Starting with vSphere 9.1.1, a cross-reference to Memory Tiering and HA admission control is included for clusters that use both Fault Tolerance and Memory Tiering.

The following vSphere features are not supported for fault tolerant virtual machines.

Feature	Description
Snapshots	You must remove or commit snapshots before you can activate Fault Tolerance on a virtual machine. You cannot take snapshots of virtual machines on which Fault Tolerance is activated. Disk-only snapshots created for vStorage APIs - Data Protection (VADP) backups are supported with Fault Tolerance. However, legacy FT does not support VADP.
Storage vMotion	You cannot invoke Storage vMotion for virtual machines with Fault Tolerance turned on. To migrate the storage, temporarily turn off Fault Tolerance, perform the Storage vMotion action, and then turn Fault Tolerance back on.
Linked clones	You cannot use Fault Tolerance on a virtual machine that is a linked clone, nor can you create a linked clone from a Fault Tolerance-activated virtual machine.
Virtual Volume datastores	Virtual Volume datastores are not supported with Fault Tolerance.
Storage-based policy management	Storage-based policy management is not supported with Fault Tolerance.
I/O filters	I/O filters are not supported with Fault Tolerance.
VBS-activated virtual machines	Virtual machines with Virtualization-Based Security (VBS) activated are not supported with Fault Tolerance.
VM Namespace DB and VM DataSets	VM Namespace DB and VM DataSets are not supported with Fault Tolerance.
Memory Tiering	You cannot turn on Fault Tolerance for powered-on virtual machines on hosts configured for Memory Tiering. For more information, see Memory Tiering and vSphere HA Admission Control.

- Snapshots. Snapshots must be removed or committed before Fault Tolerance can be enabled on a virtual machine. In addition, it is not possible to take snapshots of virtual machines on which Fault Tolerance is enabled.

  Note:
  Disk-only snapshots created for vStorage APIs - Data Protection (VADP) backups are supported with Fault Tolerance. However, legacy FT does not support VADP.

- Storage vMotion. You cannot invoke Storage vMotion for virtual machines with Fault Tolerance turned on. To migrate the storage, you should temporarily turn off Fault Tolerance, and perform the storage vMotion action. When this is complete, you can turn Fault Tolerance back on.
- Linked clones. You cannot use Fault Tolerance on a virtual machine that is a linked clone, nor can you create a linked clone from an FT-enabled virtual machine.
- Virtual Volume datastores.
- Storage-based policy management.
- I/O filters.
- VBS enabled VMs.
- VM Namespace DB and VM DataSets.

Features and Devices Incompatible with Fault Tolerance

Not all third party devices, features, or products can interoperate with Fault Tolerance.

For a virtual machine to be compatible with Fault Tolerance, the Virtual Machine must not use the following features or devices.

Table 318: Features and Devices Incompatible with Fault Tolerance and Corrective Actions

Incompatible Feature or Device	Corrective Action
Memory Tiering	Turning on Fault Tolerance on powered VMs is not supported on hosts configured for Memory Tiering.
Physical Raw Disk mapping (RDM).	With legacy FT, you can reconfigure virtual machines with physical RDM-backed virtual devices to use virtual RDMs instead.
CD-ROM or floppy virtual devices backed by a physical or remote device.	Remove the CD-ROM or floppy virtual device or reconfigure the backing with an ISO installed on shared storage.
USB and sound devices.	Remove these devices from the virtual machine.
N_Port ID Virtualization (NPIV).	Deactivate the NPIV configuration of the virtual machine.
NIC passthrough.	This feature is not supported by Fault Tolerance, so it must be turned off.
Hot-plugging devices.	The hot plug feature is automatically deactivated for fault tolerant virtual machines. To hot plug devices (either adding or removing), you must momentarily turn off Fault Tolerance, perform the hot plug, and then turn on Fault Tolerance. **Note:** When using Fault Tolerance, changing the settings of a virtual network card while a virtual machine is running is a hot-plug operation, since it requires "unplugging" the network card and then "plugging" it in again. For example, with a virtual network card for a running virtual machine, if you change the network that the virtual NIC is connected to, FT must be turned off first.
Serial or parallel ports	Remove these devices from the virtual machine.
Video devices that have 3D activated.	Fault Tolerance does not support video devices that have 3D activated.
Virtual Machine Communication Interface (VMCI)	Not supported by Fault Tolerance.
2 TB+ VMDK	Fault Tolerance is not supported with a 2 TB+ VMDK.

Using Fault Tolerance with DRS

You can use vSphere Fault Tolerance with vSphere Distributed Resource Scheduler (DRS).

FT VMs do not require EVC to support DRS. You can use FT with DRS on vSphere 6.5 and 6.0 hosts that are managed by a vSphere 6.7 or higher VC.

Note:
vSphere DRS is a critical feature of vSphere which is required to maintain the health of the workloads running inside vSphere Cluster. Starting with vSphere 7.0 Update 1, DRS depends on the availability of vCLS VMs. See *vSphere Cluster Services (vCLS)* in *vSphere Resource Management* for more information.

Preparing Your Cluster and Hosts for Fault Tolerance

To enable vSphere Fault Tolerance for your cluster, you must meet the feature's prerequisites and you must perform certain configuration steps on your hosts. After those steps are accomplished and your cluster has been created, you can also check that your configuration complies with the requirements for enabling Fault Tolerance.

The tasks you should complete before attempting to set up Fault Tolerance for your cluster include the following:

- Ensure that your cluster, hosts, and virtual machines meet the requirements outlined in the Fault Tolerance Checklist.
- Configure networking for each host.
- Create the vSphere HA cluster, add hosts, and check compliance.

After your cluster and hosts are prepared for Fault Tolerance, you are ready to turn on Fault Tolerance for your virtual machines. See Turn On Fault Tolerance.

Fault Tolerance Checklist

The following checklist contains cluster, host, and virtual machine requirements that you need to be aware of before using vSphere Fault Tolerance.

Review this list before setting up Fault Tolerance.

Note:

The failover of fault tolerant virtual machines is independent of vCenter, but you must use vCenter to set up your Fault Tolerance clusters.

Cluster Requirements for Fault Tolerance

You must meet the following cluster requirements before you use Fault Tolerance.

- Fault Tolerance logging and VMotion networking configured. For more information, see Configure Networking for Host Machines.
- vSphere HA cluster created and enabled. For more information, see Creating a vSphere HA Cluster. vSphere HA must be enabled before you can power on fault tolerant virtual machines or add a host to a cluster that already supports fault tolerant virtual machines.

Host Requirements for Fault Tolerance

You must meet the following host requirements before you use Fault Tolerance.

- Hosts must use supported processors.
- Hosts must be certified for Fault Tolerance. For more information, see Broadcom Compatibility Guide and select **Search by Fault Tolerant Compatible Sets** to determine if your hosts are certified.
- The configuration for each host must have Hardware Virtualization (HV) enabled in the BIOS.

Note:
VMware recommends that the hosts you use to support FT VMs have their BIOS power management settings turned to "Maximum performance" or "OS-managed performance".

To confirm the compatibility of the hosts in the cluster to support Fault Tolerance, you can also run profile compliance checks as described in Create Cluster and Check Compliance.

Virtual Machine Requirements for Fault Tolerance

You must meet the following virtual machine requirements before you use Fault Tolerance.

- No unsupported devices attached to the virtual machine. For more information, see Fault Tolerance Interoperability.
- Incompatible features must not be running with the fault tolerant virtual machines. For more information, see Fault Tolerance Interoperability.
- Virtual machine files (except for the VMDK files) must be stored on shared storage. Acceptable shared storage solutions include Fibre Channel, (hardware and software) iSCSI, vSAN, NFS, and NAS.

Other Configuration Recommendations

You should also observe the following guidelines when configuring Fault Tolerance.

- If you are using NFS to access shared storage, use dedicated NAS hardware with at least a 1Gbit NIC to obtain the network performance required for Fault Tolerance to work properly.
- The memory reservation of a fault tolerant virtual machine is set to the VM's memory size when Fault Tolerance is turned on. Ensure that a resource pool containing fault tolerant VMs has memory resources above the memory size of the virtual machines. Without this excess in the resource pool, there might not be any memory available to use as overhead memory.
- To ensure redundancy and maximum Fault Tolerance protection, you should have a minimum of three hosts in the cluster. In a failover situation, this provides a host that can accommodate the new Secondary VM that is created.

Configure Networking for Host Machines

On each host that you want to add to a vSphere HA cluster, you must configure two different networking switches (vMotion and FT logging) so that the host can support vSphere Fault Tolerance.

Multiple gigabit Network Interface Cards (NICs) are required. For each host supporting Fault Tolerance, a minimum of two physical NICs is recommended. For example, you need one dedicated to Fault Tolerance logging and one dedicated to vMotion. Use three or more NICs to ensure availability. See Fault Tolerance Requirements and Limits.

To set up Fault Tolerance for a host, you must complete this procedure for each port group option (vMotion and FT logging) to ensure that sufficient bandwidth is available for Fault Tolerance logging. Select one option, finish this procedure, and repeat the procedure a second time, selecting the other port group option.

1. In the vSphere Client, browse to the host.
2. Click the **Configure** tab and click **Networking**.
3. Select **VMkernel adapters**.
4. Click the **Add Networking** icon.
5. Provide appropriate information for your connection type.
6. Click **Finish**.

After you create both a vMotion and Fault Tolerance logging virtual switch, you can create other virtual switches, as needed. Add the host to the cluster and complete any steps needed to turn on Fault Tolerance.

> **Note:**
> If you configure networking to support FT but subsequently suspend the Fault Tolerance logging port, pairs of fault tolerant virtual machines that are powered on remain powered on. If a failover situation occurs, when the Primary VM is replaced by its Secondary VM a new Secondary VM is not started, causing the new Primary VM to run in a Not Protected state.

Create Cluster and Check Compliance

vSphere Fault Tolerance is used in the context of a vSphere HA cluster. After you configure networking on each host, create the vSphere HA cluster and add the hosts to it. You can check to see whether the cluster is configured correctly and complies with the requirements for the enablement of Fault Tolerance.

1. In the vSphere Client, browse to the cluster.
2. Click the **Monitor** tab and click **Profile Compliance**.
3. Click **Check Compliance Now** to run the compliance tests.

The results of the compliance test appear and the compliance or noncompliance of each host is shown.

Using Fault Tolerance

After you have taken all of the required steps for activating vSphere Fault Tolerance for your cluster, you can use the feature by turning it on for individual virtual machines.

Before Fault Tolerance can be turned on, validation checks are performed on a virtual machine.

After these checks are passed and you turn on vSphere Fault Tolerance for a virtual machine, new options are added to the Fault Tolerance section of its context menu. These include turning off or deactivating Fault Tolerance, migrating the Secondary VM, testing failover, and testing restart of the Secondary VM.

Validation Checks for Turning On Fault Tolerance

If the option to turn on Fault Tolerance is available, this task still must be validated and can fail if certain requirements are not met.

Several validation checks are performed on a virtual machine before Fault Tolerance can be turned on.

- SSL certificate checking must be activated in the vCenter settings.
- The host must be in a vSphere HA cluster or a mixed vSphere HA and DRS cluster.
- The host must have ESX 6.x or greater installed.
- The virtual machine must not have snapshots.
- The virtual machine must not be a template.
- The virtual machine must not have vSphere HA deactivated.
- The virtual machine must not have a video device with 3D activated.

Checks for Powered-On Virtual Machines

Several additional validation checks are performed for powered-on virtual machines (or those that are in the process of being powered on).

- The BIOS of the hosts where the fault tolerant virtual machines reside must have Hardware Virtualization (HV) activated.
- The host that supports the Primary VM must have a processor that supports Fault Tolerance.
- Your hardware should be certified as compatible with Fault Tolerance. To confirm that it is, use the Broadcom Compatibility Guide and select **Search by Fault Tolerant Compatible Sets**.
- The configuration of the virtual machine must be valid for use with Fault Tolerance (for example, it must not contain any unsupported devices).

Secondary VM Placement

When your effort to turn on Fault Tolerance for a virtual machine passes the validation checks, the Secondary VM is created. The placement and immediate status of the Secondary VM depends upon whether the Primary VM was powered-on or powered-off when you turned on Fault Tolerance.

If the Primary VM is powered on:

- The entire state of the Primary VM is copied and the Secondary VM is created, placed on a separate compatible host, and powered on if it passes admission control.
- The Fault Tolerance Status displayed for the virtual machine is **Protected**.

If the Primary VM is powered off:

- The Secondary VM is immediately created and registered to a host in the cluster (it might be re-registered to a more appropriate host when it is powered on.)

- The Secondary VM is not powered on until after the Primary VM is powered on.
- The Fault Tolerance Status displayed for the virtual machine is **Not Protected, VM not Running**.
- When you attempt to power on the Primary VM after Fault Tolerance has been turned on, the additional validation checks listed above are performed.

  After these checks are passed, the Primary and Secondary VMs are powered on and placed on separate, compatible hosts. The virtual machine's Fault Tolerance Status is tagged as **Protected**.

Turn On Fault Tolerance

You can turn on vSphere Fault Tolerance through the vSphere Client.

The option to turn on Fault Tolerance is unavailable (dimmed) if any of these conditions apply:

- The virtual machine resides on a host that does not have a license for the feature.
- The virtual machine resides on a host that is in maintenance mode or standby mode.
- The virtual machine is disconnected or orphaned (its .vmx file cannot be accessed).
- The user does not have permission to turn the feature on.

When Fault Tolerance is turned on, vCenter resets the virtual machine's memory limit and sets the memory reservation to the memory size of the virtual machine. While Fault Tolerance remains turned on, you cannot change the memory reservation, size, limit, number of vCPUs, or shares. You also cannot add or remove disks for the VM. When Fault Tolerance is turned off, any parameters that were changed are not reverted to their original values.

Connect vSphere Client to vCenter using an account with cluster administrator permissions.

1. In the vSphere Client, browse to the virtual machine for which you want to turn on Fault Tolerance.
2. Right-click the virtual machine and select **Fault Tolerance** > **Turn On Fault Tolerance**.
3. Click **Yes**.
4. Select a datastore on which to place the Secondary VM configuration files. Then click **Next**.
5. Select a host on which to place the Secondary VM. Then click **Next**.
6. Review your selections and then click **Finish**.

The specified virtual machine is designated as a Primary VM, and a Secondary VM is established on another host. The Primary VM is now fault tolerant.

> **Note:**
> The VM datastores and memory are replicated during the FT Turn On process. This can take several minutes depending on the size of the replicated data. The VM state does not appear as protected until replication is complete.

Turn Off Fault Tolerance

Turning off vSphere Fault Tolerance deletes the secondary virtual machine, its configuration, and all history.

Use the **Turn Off Fault Tolerance** option if you do not plan to reenable the feature. Otherwise, use the **Suspend Fault Tolerance** option.

> **Note:**
> If the Secondary VM resides on a host that is in maintenance mode, disconnected, or not responding, you cannot use the **Turn Off Fault Tolerance** option. In this case, you should suspend and resume Fault Tolerance instead.

1. In the vSphere Client, browse to the virtual machine for which you want to turn off Fault Tolerance.
2. Right-click the virtual machine and select **Fault Tolerance** > **Turn Off Fault Tolerance**.
3. Click **Yes**.

Fault Tolerance is turned off for the selected virtual machine. The history and the secondary virtual machine for the selected virtual machine are deleted.

Note:
Fault Tolerance cannot be turned off when the secondary VM is in the process of being started. Since this involves syncing up the primary VM's full state to the secondary VM, this process may take longer than expected.

Suspend Fault Tolerance

Suspending vSphere Fault Tolerance for a virtual machine suspends its Fault Tolerance protection, but preserves the Secondary VM, its configuration, and all history. Use this option to resume Fault Tolerance protection in the future.

1. In the vSphere Client, browse to the virtual machine for which you want to suspend Fault Tolerance.
2. Right-click the virtual machine and select **Fault Tolerance** > **Suspend Fault Tolerance**.
3. Click **Yes**.

Fault Tolerance is suspended for the selected virtual machine. Any history and the Secondary VM for the selected virtual machine are preserved and will be used if the feature is resumed.

After you suspend Fault Tolerance, to resume the feature select **Resume Fault Tolerance**.

Migrate Secondary Virtual Machines

After vSphere Fault Tolerance is turned on for a Primary VM, you can migrate its associated Secondary VM.

1. In the vSphere Client, browse to the Primary VM for which you want to migrate its Secondary VM.
2. Right-click the virtual machine and select **Fault Tolerance** > **Migrate Secondary**.
3. Complete the options in the Migrate dialog box and confirm the changes that you made.
4. Click **Finish** to apply the changes.

The Secondary VM associated with the selected fault tolerant virtual machine is migrated to the specified host.

Test Fault Tolerance Protection

You can induce a failover situation for a selected Primary VM to test your Fault Tolerance protection.

This option is unavailable (dimmed) if the virtual machine is powered off.

1. In the vSphere Client, browse to the Primary VM for which you want to test failover.
2. Right-click the virtual machine and select **Fault Tolerance** > **Test Failover**.
3. View details about the failover in the Task Console.

This task induces failure of the Primary VM to ensure that the Secondary VM replaces it. A new Secondary VM is also started placing the Primary VM back in a Protected state.

Test Restart Secondary Virtual Machines

You can induce the failure of a Secondary VM to test the Fault Tolerance protection provided for a selected Primary VM.

This option is unavailable (dimmed) if the virtual machine is powered off.

1. In the vSphere Client, browse to the Primary VM for which you want to conduct the test.
2. Right-click the virtual machine and select **Fault Tolerance** > **Test Restart Secondary**.
3. View details about the test in the Task Console.

This task results in the termination of the Secondary VM that provided Fault Tolerance protection for the selected Primary VM. A new Secondary VM is started, placing the Primary VM back in a Protected state.

Upgrade Hosts Used for Fault Tolerance

Use the following procedure to upgrade hosts used for Fault Tolerance.

Verify that you have cluster administrator privileges.

Verify that you have sets of four or more ESX hosts that are hosting fault tolerant virtual machines that are powered on. If the virtual machines are powered off, the Primary and Secondary VMs can be relocated to hosts with different builds.

> **Note:**
> This upgrade procedure is for a minimum four-node cluster. The same instructions can be followed for a smaller cluster, though the unprotected interval will be slightly longer.

1. Using vMotion, migrate the fault tolerant virtual machines off of two hosts.
2. Upgrade the two evacuated hosts to the same ESX build.
3. Suspend Fault Tolerance on the Primary VM.
4. Using vMotion, move the Primary VM for which Fault Tolerance has been suspended to one of the upgraded hosts.
5. Resume Fault Tolerance on the Primary VM that was moved.
6. Repeat 1 to 5 for as many fault tolerant virtual machine pairs as can be accommodated on the upgraded hosts.
7. Using vMotion, redistribute the fault tolerant virtual machines.

All ESX hosts in a cluster are upgraded.

Activate Fault Tolerance Encryption

You can encrypt Fault Tolerance log traffic.

FT encryption requires SMP-FT. Encryption on Legacy FT (Record-Replay FT) is not supported.

vSphere Fault Tolerance performs frequent checks between a primary VM and secondary VM so that the secondary VM can quickly resume from the last successful checkpoint. The checkpoint contains the VM state that has been modified since the previous checkpoint. You can encrypt Fault Tolerance log traffic.

When you turn on Fault Tolerance, FT encryption is set to **Opportunistic** by default, which means it activates encryption only if both the primary and secondary host are capable of encryption. Follow this procedure if you need to change the FT encryption mode manually.

> **Note:**
> Fault Tolerance supports vSphere Virtual Machine Encryption with vSphere 7.0 Update 2 and later. In-guest and array-based encryption do not depend on or interfere with VM encryption. Having multiple encryption layers uses additional compute resources, which might impact virtual machine performance. The impact varies with hardware as well as the amount and type of I/O, but overall performance impact is negligible for most workloads. The effectiveness and compatibility of back-end storage features such as deduplication, compression, and replication might also be affected by VM encryption.

1. Select the VM and choose **Edit Settings**.
2. Under **VM Options** select the **Encrypted FT** drop-down menu.
3. Choose one of the following options:

Option	Description
Disabled	Do not turn on encrypted Fault Tolerance logging.
Opportunistic	Turn on encryption only if both sides are capable. A Fault Tolerance VM is allowed to move to an ESX host which does not support encrypted Fault Tolerance logging.
Required	Choose hosts for Fault Tolerance primary and secondary that both support encrypted FT logging.

> **Note:**
> While VM encryption is activated, FT encryption mode is set to **Required** by default and cannot be modified.
>
> When FT encryption mode is set to **Required**:

- When you turn on FT, only FT encryption supported hosts are listed for the placement of FT secondary.
- FT failover can only happen on the FT encryption supported hosts.

4. Click **OK**.

Best Practices for Fault Tolerance

To ensure optimal Fault Tolerance results, you should follow certain best practices.

The following recommendations for host and networking configuration can help improve the stability and performance of your cluster.

Host Configuration

Hosts running the Primary and Secondary VMs should operate at approximately the same processor frequencies, otherwise the Secondary VM might be restarted more frequently. Platform power management features that do not adjust based on workload (for example, power capping and enforced low frequency modes to save power) can cause processor frequencies to vary greatly. If Secondary VMs are being restarted on a regular basis, deactivate all power management modes on the hosts running fault tolerant virtual machines or ensure that all hosts are running in the same power management modes.

Host Networking Configuration

The following guidelines allow you to configure your host's networking to support Fault Tolerance with different combinations of traffic types (for example, NFS) and numbers of physical NICs.

- Distribute each NIC team over two physical switches ensuring L2 domain continuity for each VLAN between the two physical switches.
- Use deterministic teaming policies to ensure particular traffic types have an affinity to a particular NIC (active/standby) or set of NICs (for example, originating virtual port-id).
- Where active/standby policies are used, pair traffic types to minimize impact in a failover situation where both traffic types will share a vmnic.
- Where active/standby policies are used, configure all the active adapters for a particular traffic type (for example, FT Logging) to the same physical switch. This minimizes the number of network hops and lessens the possibility of oversubscribing the switch to switch links.

  Note:
  FT logging traffic between Primary and Secondary VMs is unencrypted and contains guest network and storage I/O data, as well as the memory contents of the guest operating system. This traffic can include sensitive data such as passwords in plaintext. To avoid such data being divulged, ensure that this network is secured, especially to avoid 'man-in-the-middle' attacks. For example, you could use a private network for FT logging traffic.

Homogeneous Clusters

vSphere Fault Tolerance can function in clusters with nonuniform hosts, but it works best in clusters with compatible nodes. When constructing your cluster, all hosts should have the following configuration:

- Common access to datastores used by the virtual machines.
- The same virtual machine network configuration.
- The same BIOS settings (power management and hyperthreading) for all hosts.

Run **Check Compliance** to identify incompatibilities and to correct them.

Performance

To increase the bandwidth available for the logging traffic between Primary and Secondary VMs use a 10Gbit NIC, and activate the use of jumbo frames.

You can select multiple NICs for the FT logging network. By selecting multiple NICs, you can take advantage of the bandwidth from multiple NICs even if all of the NICs are not dedicated to running FT.

Store ISOs on Shared Storage for Continuous Access

Store ISOs that are accessed by virtual machines with Fault Tolerance activated on shared storage that is accessible to both instances of the fault tolerant virtual machine. If you use this configuration, the CD-ROM in the virtual machine continues operating normally, even when a failover occurs.

Avoid Network Partitions

A network partition occurs when a vSphere HA cluster has a management network failure that isolates some of the hosts from vCenter and from one another. See Network Partitions . When a partition occurs, Fault Tolerance protection might be degraded.

In a partitioned vSphere HA cluster using Fault Tolerance, the Primary VM (or its Secondary VM) could end up in a partition managed by a primary host that is not responsible for the virtual machine. When a failover is needed, a Secondary VM is restarted only if the Primary VM was in a partition managed by the primary host responsible for it.

To ensure that your management network is less likely to have a failure that leads to a network partition, follow the recommendations in Best Practices for Networking.

Using vSAN Datastores

vSphere Fault Tolerance can use vSAN datastores, but you must observe the following restrictions:

- A mix of vSAN and other types of datastores is not supported for both Primary VMs and Secondary VMs.

To increase performance and reliability when using FT with vSAN, the following conditions are also recommended.

- vSAN and FT should use separate networks.
- Keep Primary and Secondary VMs in separate vSAN fault domains.

Activate Metro Cluster Fault Tolerance

In vSphere 8.0 U3 you can activate Metro Cluster Fault Tolerance in the Fault Tolerance wizard.

In the fault tolerance wizard, you can select a checkbox labeled **Enable Metro Cluster Fault Tolerance** to activate FT Metro Cluster functionality and a drop-down list for choosing a HostGroup as the preferred location of FT VM. By default, the checkbox is unchecked and the drop-down list is deactivated, indicating that FT Metor Cluster is deactivated for the VM (`ConfigInfo.metroFtEnabed is 'FALSE'`).

When the checkbox is checked, the drop-down list will be activated for choosing a HostGroup. The wizard will prevent proceeding to the next step if no HostGroup is selected for the VM. To ensure the legality of the selected HostGroup, the wizard will call the function `queryFaultToleranceCompatibleHosts` and retrieve the result through the returned messages.

Before checking the checkbox **Enable Metro Cluster Fault Tolerance**, the drop-down list of HostGroup is deactivated. A signpost button will be added next to the label `Host Group`, the signpost content is: `Based on the selected Host Group, FT can divide the hosts in the cluster into two groups and ensure to place FT primary and FT secondary in different groups.`

After checking the checkbox **Enable Metro Cluster Fault Tolerance**, the drop-down list is activated in the wizard for the HostGroup selection. FT handles the validation check for the FT Metro Cluster flag and HostGroup, which is triggered by clicking the **NEXT** button. The wizard sets **FaultToleranceConfigSpec.metroFtEnabled** to **TRUE** and **FaultToleranceConfigSpec.preferredLocation** to the selected HostGroup. Then, the wizard fetchs the compatible host list.

After checking the checkbox **Enable Metro Cluster Fault Tolerance**, but not selecting any HostGroup, the wizard can't be moved forward with **NEXT** button with the error message: `Please assign a host group for Fault Tolerance VM before enabling Metro Cluster.` FT also checks the HostGroup for FT Metro Cluster activated VM and the task might fail if the HostGroup is not configured.

You can remove the configured host group while FT VM is running. FT Metro Cluster is deactivated in such case. However, the host group name is still displayed on the FT information card, since FT Metro Cluster can be re-activated after configuring the host group again.

If you delete the HostGroup for a running FT VM, the FT Metro Cluster is deactivated. In case of a failover, `Metro Cluster status Not Applicable Host Group is missing.` is displayed on the FT information card. However, if you add the HostGroup back, the FT Metro Cluster will be activated again.

If you delete the HostGroup for a powered-off FT VM, the Fault Tolerance Secondary VM cannot be powered-on.

Troubleshooting Fault Tolerant Virtual Machines

To maintain a high level of performance and stability for your fault tolerant virtual machines and also to minimize failover rates, you should be aware of certain troubleshooting issues.

The troubleshooting topics discussed focus on problems that you might encounter when using the vSphere Fault Tolerance feature on your virtual machines. The topics also describe how to resolve problems.

To troubleshoot Fault Tolerance, see Broadcom knowledge base article 342336. This article contains a list of error messages that you might encounter when you attempt to use the feature and, where applicable, advise on how to resolve each error.

Hardware Virtualization Not Enabled

You must enable Hardware Virtualization (HV) before you use vSphere Fault Tolerance.

When you attempt to power on a virtual machine with Fault Tolerance enabled, an error message might appear if you did not enable HV.

This error is often the result of HV not being available on the ESX server on which you are attempting to power on the virtual machine. HV might not be available either because it is not supported by the ESX server hardware or because HV is not enabled in the BIOS.

If the ESX server hardware supports HV, but HV is not currently enabled, enable HV in the BIOS on that server. The process for enabling HV varies among BIOSes. See the documentation for your hosts' BIOSes for details on how to enable HV.

If the ESX server hardware does not support HV, switch to hardware that uses processors that support Fault Tolerance.

Compatible Hosts Not Available for Secondary VM

If you power on a virtual machine with Fault Tolerance enabled and no compatible hosts are available for its Secondary VM, you might receive an error message.

You might encounter the following error message:

```
Secondary VM could not be powered on as there are no compatible hosts that can accommodate
 it.
```

This can occur for a variety of reasons including that there are no other hosts in the cluster, there are no other hosts with HV enabled, Hardware MMU Virtualization is not supported by host CPUs, data stores are inaccessible, there is no available capacity, or hosts are in maintenance mode.

If there are insufficient hosts, add more hosts to the cluster. If there are hosts in the cluster, ensure they support HV and that HV is enabled. The process for enabling HV varies among BIOSes. See the documentation for your hosts' BIOSes for details on how to enable HV. Check that hosts have sufficient capacity and that they are not in maintenance mode.

Secondary VM on Overcommitted Host Degrades Performance of Primary VM

If a Primary VM appears to be executing slowly, even though its host is lightly loaded and retains idle CPU time, check the host where the Secondary VM is running to see if it is heavily loaded.

When a Secondary VM resides on a host that is heavily loaded, the Secondary VM can affect the performance of the Primary VM.

A Secondary VM running on a host that is overcommitted (for example, with its CPU resources) might not get the same amount of resources as the Primary VM. When this occurs, the Primary VM must slow down to allow the Secondary VM to keep up, effectively reducing its execution speed to the slower speed of the Secondary VM.

If the Secondary VM is on an overcommitted host, you can move the VM to another location without resource contention problems. Or more specifically, do the following:

- For FT networking contention, use vMotion technology to move the Secondary VM to a host with fewer FT VMs contending on the FT network. Verify that the quality of the storage access to the VM is not asymmetric.
- For storage contention problems, turn FT off and on again. When you recreate the Secondary VM, change its datastore to a location with less resource contention and better performance potential.
- To resolve a CPU resources problem, set an explicit CPU reservation for the Primary VM at an MHz value sufficient to run its workload at the desired performance level. This reservation is applied to both the Primary and Secondary VMs, ensuring that both VMs can execute at a specified rate. For guidance in setting this reservation, view the performance graphs of the virtual machine (before Fault Tolerance was enabled) to see how many CPU resources it used under normal conditions.

Increased Network Latency Observed in FT Virtual Machines

If your FT network is not optimally configured, you might experience latency problems with the FT VMs.

FT VMs might see a variable increase in packet latency (on the order of milliseconds). Applications that demand very low network packet latency or jitter (for example, certain real-time applications) might see a degradation in performance.

Some increase in network latency is expected overhead for Fault Tolerance, but certain factors can add to this latency. For example, if the FT network is on a particularly high latency link, this latency is passed on to the applications. Also, if the FT network has insufficient bandwidth (fewer than 10 Gbps), greater latency might occur.

Verify that the FT network has sufficient bandwidth (10 Gbps or more) and uses a low latency link between the Primary VM and Secondary VM. These precautions do not eliminate network latency, but minimize its potential impact.

Some Hosts Are Overloaded with FT Virtual Machines

You might encounter performance problems if your cluster's hosts have an imbalanced distribution of FT VMs.

Some hosts in the cluster might become overloaded with FT VMs, while other hosts might have unused resources.

vSphere DRS does not load balance FT VMs (unless they are using legacy FT). This limitation might result in a cluster where hosts are unevenly distributed with FT VMs.

Manually rebalance the FT VMs across the cluster by using vSphere vMotion. Generally, the fewer FT VMs that are on a host, the better they perform, due to reduced contention for FT network bandwidth and CPU resources.

Losing Access to FT Metadata Datastore

Access to the Fault Tolerance metadata datastore is essential for the proper functioning of an FT VM. Loss of this access can cause a variety of problems.

These problems include the following:

- FT can terminate unexpectedly.
- If both the Primary VM and Secondary VM cannot access the metadata datastore, the VMs might fail unexpectedly. Typically, an unrelated failure that terminates FT must also occur when access to the FT metadata datastore is lost by both VMs. vSphere HA then tries to restart the Primary VM on a host with access to the metadata datastore.
- The VM might stop being recognized as an FT VM by vCenter. This failed recognition can allow unsupported operations such as taking snapshots to be performed on the VM and cause problematic behavior.

Lack of access to the Fault Tolerance metadata datastore can lead to the undesirable outcomes in the previous list.

When planning your FT deployment, place the metadata datastore on highly available storage. While FT is running, if you see that the access to the metadata datastore is lost on either the Primary VM or the Secondary VM, promptly address the storage problem before loss of access causes one of the previous problems. If a VM stops being recognized as an FT VM by vCenter, do not perform unsupported operations on the VM. Restore access to the metadata datastore. After access is restored for the FT VMs and the refresh period has ended, the VMs are recognizable.

Turning On vSphere FT for Powered-On VM Fails

If you try to turn on vSphere Fault Tolerance for a powered-on VM, this operation can fail.

When you select **Turn On Fault Tolerance** for a powered-on VM, the operation fails and you see an `Unknown error` message.

This operation can fail if the host that the VM is running on has insufficient memory resources to provide fault tolerant protection. vSphere Fault Tolerance automatically tries to allocate a full memory reservation on the host for the VM. Overhead memory is required for fault tolerant VMs and can sometimes expand to 1 to 2 GB. If the powered-on VM is running on a host that has insufficient memory resources to accommodate the full reservation plus the overhead memory, trying to turn on Fault Tolerance fails. Subsequently, the `Unknown error` message is returned.

Choose from these solutions:

- Free up memory resources on the host to accommodate the VM's memory reservation and the added overhead.
- Move the VM to a host with ample free memory resources and try again.

FT Virtual Machines not Placed or Evacuated by vSphere DRS

FT virtual machines in a cluster that is activated with vSphere DRS do not function correctly if Enhanced vMotion Compatibility (EVC) is currently deactivated.

Because EVC is a prerequisite for using DRS with FT VMs, DRS does not place or evacuate them if EVC has been deactivated (even if it is later reactivated).

When EVC is deactivated on a DRS cluster, a VM override that deactivates DRS on an FT VM might be added. Even if EVC is later reactivated, this override is not canceled.

If DRS does not place or evacuate FT VMs in the cluster, check the VMs for a VM override that is deactivating DRS. If you find one, remove the override that is deactivating DRS.

> **Note:**
> For more information on how to edit or delete VM overrides, see *vSphere Resource Management*.

Fault Tolerant Virtual Machine Failovers

A Primary or Secondary VM can fail over even though its ESX host has not crashed. In such cases, virtual machine execution is not interrupted, but redundancy is temporarily lost. To avoid this type of failover, be aware of some of the situations when it can occur and take steps to avoid them.

Partial Hardware Failure Related to Storage

This problem can arise when access to storage is slow or down for one of the hosts. When this occurs there are many storage errors listed in the VMkernel log. To resolve this problem you must address your storage-related problems.

Partial Hardware Failure Related to Network

If the logging NIC is not functioning or connections to other hosts through that NIC are down, this can trigger a fault tolerant virtual machine to be failed over so that redundancy can be reestablished. To avoid this problem, dedicate a separate NIC each for vMotion and FT logging traffic and perform vMotion migrations only when the virtual machines are less active.

Insufficient Bandwidth on the Logging NIC Network

This can happen because of too many fault tolerant virtual machines being on a host. To resolve this problem, more broadly distribute pairs of fault tolerant virtual machines across different hosts.

Use a10-Gbit logging network for FT and verify that the network is low latency.

vMotion Failures Due to Virtual Machine Activity Level

If the vMotion migration of a fault tolerant virtual machine fails, the virtual machine might need to be failed over. Usually, this occurs when the virtual machine is too active for the migration to be completed with only minimal disruption to the activity. To avoid this problem, perform vMotion migrations only when the virtual machines are less active.

Too Much Activity on VMFS Volume Can Lead to Virtual Machine Failovers

When a number of file system locking operations, virtual machine power ons, power offs, or vMotion migrations occur on a single VMFS volume, this can trigger fault tolerant virtual machines to be failed over. A symptom that this might be occurring is receiving many warnings about SCSI reservations in the VMkernel log. To resolve this problem, reduce the number of file system operations or ensure that the fault tolerant virtual machine is on a VMFS volume that does not have an abundance of other virtual machines that are regularly being powered on, powered off, or migrated using vMotion.

Lack of File System Space Prevents Secondary VM Startup

Check whether or not your /(root) or /vmfs/*datasource* file systems have available space. These file systems can become full for many reasons, and a lack of space might prevent you from being able to start a new Secondary VM.

vCenter High Availability

vCenter High Availability (vCenter HA) protects vCenter against host and hardware failures. The active-passive architecture of the solution can also help you reduce downtime significantly when you patch vCenter.

After some network configuration, you create a three-node cluster that contains Active, Passive, and Witness nodes. Different configuration paths are available. What you select depends on your existing configuration.

Plan the vCenter HA Deployment

Before you can configure vCenter HA, you have to consider several factors. A deployment with components that use different versions of vSphere requires different considerations than a deployment that includes only vSphere 8.0 components. Resource and software requirements and the networking setup must also be considered carefully.

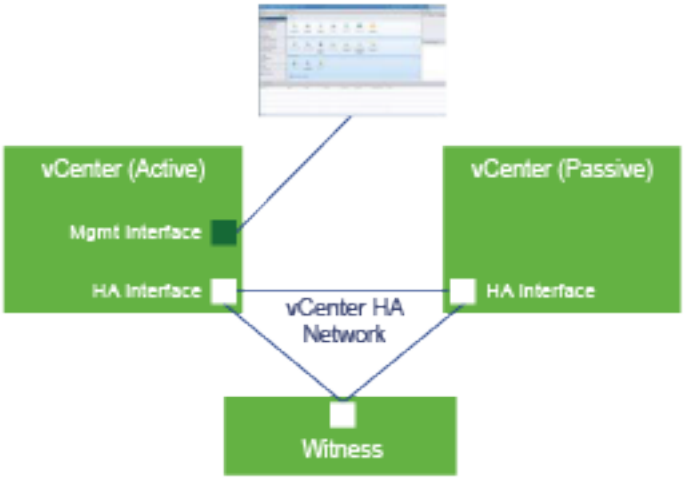
vCenter HA Architecture Overview

A vCenter HA cluster consists of three vCenter instances. The first instance, initially used as the Active node, is cloned twice to a Passive node and to a Witness node. Together, the three nodes provide an active-passive failover solution.

Deploying each of the nodes on a different ESX instance protects against hardware failure. Adding the three ESX hosts to a DRS cluster can further protect your environment.

When vCenter HA configuration is complete, only the Active node has an active management interface (public IP). The three nodes communicate over a private network called vCenter HA network that is set up as part of the configuration. The Active node is continuously replicating data to the Passive node.

Figure 115: vCenter Three-Node Cluster



All three nodes are necessary for the functioning of this feature. Compare the node responsibilities.

Table 319: vCenter HA Nodes

Node	Description
Active	• Runs the active vCenter instance • Uses a public IP address for the management interface • Uses the vCenter HA network for replication of data to the Passive node. • Uses the vCenter HA network to communicate with the Witness node.
Passive	• Is initially a clone of the Active node • Constantly receives updates from and synchronizes state with the Active node over the vCentervCenter HA network • Automatically takes over the role of the Active node if a failure occurs
Witness	• Is a lightweight clone of the Active node • Provides a quorum to protect against a split-brain situations

vCenter HA Hardware and Software Requirements

Before you set up vCenter HA, ensure that you have sufficient memory, CPU, and datastore resources, and ensure that you are using versions of vCenter and ESX that support vCenter HA.

Your environment must meet the following requirements.

Table 320: vCenter HA Requirements

Component	Requirements
ESX	• ESX 6.0 or later is required. • A minimum of three ESX hosts is strongly recommended. Each vCenter HA node can then run on a different host for better protection.
Management vCenter (if used)	Your environment can include a management vCenter system, or you can set up your vCenter to manage the ESX host on which it runs (self-managed vCenter) • vCenter 6.0 or later is required.

Component	Requirements
vCenter	• vCenter 6.5 or later is required. • Deployment size Small (4 CPU and 16GB RAM) or bigger is required to meet the RTO. Do not use Tiny in production environments. • vCenter HA is supported and tested with VMFS, NFS, and vSAN datastores. • Ensure you have enough disk space to collect and store support bundles for all three nodes on the Active node. See Collecting Support Bundles for a vCenter HA Node.
Network connectivity	• vCenter HA network latency between Active, Passive, and Witness nodes must be less than 10 ms. • The vCenter HA network must be on a different subnet than the management network.

vCenter HA Configuration Workflow

To configure the passive and witness nodes in your vCenter HA environment, you can use the **Set Up vCenter HA** wizard. With the manual option, you must manually clone the active node and create the passive and witness nodes.

> **Note:**
> For automatic and manual configurations, verify that your environment is satisfying the hardware and software requirements. See vCenter HA Hardware and Software Requirements.

Automatic Configuration

You can use the **Set Up vCenter HA** wizard to automatically create the passive and witness nodes as part of vCenter HA configuration.

1. You deploy the first vCenter, which will become the active node.
2. You add a second network (port group) for vCenter HA traffic on each ESX host.
3. You start the vCenter HA configuration and supplies the IP addresses, the target ESX host or cluster, and the datastore for each clone.
4. The system clones the active node and creates a passive node with precisely the same settings, including the same host name.
5. The system clones the active node again and creates a more light-weight witness node.
6. The system sets up the vCenter HA network on which the three nodes communicate, for example, by exchanging heartbeats and other information.

Manual Configuration

If you want more control over your deployment, you can perform a manual vCenter HA configuration, where you clone the active node and create the passive and witness nodes of your configuration. If you select this option and remove the vCenter HA configuration later, you are responsible for deleting the nodes that you created.

1. You deploy the first vCenter, which becomes the active node.
2. You add a second network (port group) for vCenter HA traffic on each ESX host.
3. You must add a second network adapter (NIC) to the active node if the credentials of the active management vCenter are unknown.
4. You log in to the vCenter (active node) with the vSphere Client.
5. You start the vCenter HA configuration, select the checkbox to manually configure and supply IP address and subnet information for the passive and witness nodes. Optionally, you can override the failover management IP addresses.
6. After the vCenter HA deployment task finishes, you log in to the management vCenter and create two clones of the vCenter (active node).
7. The system sets up the vCenter HA network on which the three nodes exchange heartbeats and replication information.
8. The vCenter is protected by vCenter HA.

For more information, see Configure vCenter HA With the vSphere Client.

Configure the Network

Regardless of the deployment option and inventory hierarchy that you select, you have to set up your network before you can start configuration. To set the foundation for the vCenter HA network, you add a port group to each ESX host.

- The vCenter that later becomes the Active node, is deployed.
- You can access and have privileges to modify that vCenter and the ESX host on which it runs.
- During network setup, you need static IP addresses for the management network. The management and cluster network addresses must be IPv4 or IPv6. They cannot be mixed mode IP addresses.

After configuration is complete, the vCenter HA cluster has two networks, the management network on the first virtual NIC and the vCenter HA network on the second virtual NIC.

Management network
The management network serves client requests (public IP). The management network IP addresses must be static.

vCenter HA network
The vCenter HA network connects the Active, Passive, and Witness nodes and replicates the server state. It also monitors heartbeats.

- The vCenter HA network IP addresses for the Active, Passive, and Witness nodes must be static.
- The vCenter HA network must be on a different subnet than the management network. The three nodes can be on the same subnet or on different subnets.
- Network latency between the Active, Passive, and Witness nodes must be less than 10 milliseconds.
- You must not add a default gateway entry for the cluster network.

Note:

vCenter HA setup is not supported when using a Wide Area Network (WAN) for the networking configuration. VCHA requires a dedicated, high-fidelity, and low-latency network. For more information, see kB 80551.

1. Log in to the management vCenter and find the ESX host on which the Active node is running.
2. Add a port group to the ESX host.

   This port group can be on an existing virtual switch or, for improved network isolation, you can create a new virtual switch. It must be different from the management network.
3. If your environment includes the recommended three ESX hosts, add the port group to each of the hosts.

Configure vCenter HA With the vSphere Client

When you use the vSphere Client, the **Set Up vCenter HA** wizard creates and configures a second network adapter on the vCenter, clones the Active node, and configures the vCenter HA network.

- Deploy vCenter that you want to use as the initial Active node.
  - The vCenter must have a static IP address.
  - SSH must be activated on the vCenter.
- Verify that your environment meets the following requirements.
  - The vCenter that will become the Active node is managing its own ESX host and its own virtual machine. This configuration is sometimes called a self-managed vCenter.
- Set up the infrastructure for the vCenter HA network. For more information, see Configure the Network.
- Determine which static IP addresses to use for the two vCenter nodes that will become the Passive node and Witness node.

Note:

In order to use an NSX segment on the Active node, you must create NIC2/eth1 by using **Edit VM Settings** to add the second NIC with the NSX segment. You do not need to specify any resources for Passive or Witness nodes, because the Clone must be created using **Clone VM** after adding necessary Guest customization specifications for Passive and Witness containing NIC1/eth0 and NIC2/eth1 with IP addresses. When you configure VCHA IP addresses for eth1 in vCenter, the eth1 on the Active Node is automatically filled in.

1. Log in to the Active node with the vSphere Client.
2. Select the vCenter object in the inventory and select the **Configure** tab.
3. Under setting, select **vCenter HA**.
4. Click the **Set Up vCenter HA** button to start the setup wizard.
   - If the vCenter is self-managed, the **Resource settings** page is displayed. Proceed to step 7.
   - If your vCenter is managed by another vCenter in the same SSO domain, proceed to step 7.
   - If your vCenter is managed by another vCenter in a different SSO domain, input the location and credential details of that management vCenter.
5. Click **Management vCenter credentials**. Specify the Management vCenter FQDN or IP address, Single Sign-On user name and password and click **Next**.

   If you do not have the Single Sign-On administrator credentials, select the second bullet and click **Next**.
6. You may see a **Certificate warning** displayed. Review the SHA1 thumbprint and select **Yes** to continue.
7. In the **Resource settings** section, first select the vCenter HA network for the active node from the drop-down menu.

   Note:
   The network selector is no longer visible once NIC2/eth1 is created.
8. Click on the checkbox if you want to automatically create clones for Passive and Witness nodes.

   Note:
   If you do not select the checkbox, you must manually create clones for Passive and Witness nodes after you click **Finish**.
9. For the Passive node, click **Edit**.
   a) Specify a unique name and target location.
   b) Select the destination compute resource for the operation.
   c) Select the datastore in which to store the configuration and disk files.
   d) Select virtual machine Management (NIC 0) and vCenter HA (NIC 1) networks.

      If there are issues with your selections, errors or compatibility warnings are displayed.
   e) Review your selections and click **Finish**.
10. For the Witness node, click **Edit**.
    a) Specify a unique name and target location.
    b) Select the destination compute resource for the operation.
    c) Select the datastore in which to store the configuration and disk files.
    d) Select vCenter HA (NIC 1) network.

       If there are issues with your selections, errors or compatibility warnings are displayed.
    e) Review your selections and click **Finish**.
11. Click **Next**.
12. In the **IP settings** section, select the IP version from the drop-down menu.
13. Enter the IPv4 address (NIC 1) and Subnet mask or prefix length information for the Active, Passive and Witness nodes.

    You can Edit management network settings for the Passive Node. Customizing these settings are optional. By default, the management network settings of the Active node are applied.
14. Click **Finish**.

The Passive and Witness nodes are created. When **Set Up vCenter HA** is complete, vCenter has high availability protection. After vCenter HA is activated, you can click **Edit** to enter Maintenance Mode, Enable or Disable vCenter HA. There are separate buttons to remove vCenter HA or initiate vCenter HA failover.

See Manage the vCenter HA Configuration for a list of cluster management tasks.

For a brief overview of enhancements in the vSphere Client when working with vCentervCenter HA, see:

Manage the vCenter HA Configuration

After you configure your vCenter HA cluster, you can perform management tasks. These tasks include certificate replacement, replacement of SSH keys, and SNMP setup. You can also edit the cluster configuration to deactivate or activate vCenter HA, enter maintenance mode, and remove the cluster configuration.

Set Up SNMP Traps

You can set up Simple Network Management Protocol (SNMP) traps to receive SNMP notifications for your vCenter HA cluster.

The traps default to SNMP version 1.

Set up SNMP traps for the Active node and the Passive node. You tell the agent where to send related traps, by adding a target entry to the snmpd configuration.

1. Log in to the Active node by using the Virtual Machine Console or SSH.
2. Run the `vicfg-snmp` command, for example:

   ```
   vicfg-snmp -t 10.160.1.1@1166/public
   ```

   In this example, `10.160.1.1` is the client listening address, `1166` is the client listening port, and `public` is the community string.
3. Activate the SNMP agent (snmpd) by running the following command.

   ```
   vicfg-snmp -e
   ```

You might also find these commands useful.

- To view the complete help for the command, run `vicfg-snmp -h`.
- To deactivate the SNMP agent, run `vicfg-snmp -D`.
- To show the SNMP agent's configuration, run `vicfg-snmp -s`.
- To reset the configuration to the default, run `vicfg-snmp -r`.

Set Up Your Environment to Use Custom Certificates

The machine SSL certificate on each node is used for cluster management communication and for encryption of replication traffic. If you want to use custom certificates, you have to remove the vCenter HA configuration, delete the Passive and Witness nodes, provision the Active node with the custom certificate, and reconfigure the cluster.

If possible, replace certificates in the vCenter that will become the Active node before you clone the node.

1. Edit the cluster configuration and select **Remove**.
2. Delete the Passive node and the Witness node.
3. On the Active node, which is now a standalone vCenter, replace the machine SSL Certificate with a custom certificate.
4. Reconfigure the cluster.

Manage vCenter HA SSH Keys

vCenter HA uses SSH keys for password-less authentication between the Active, Passive, and Witness nodes. The authentication is used for heartbeat exchange and file and data replication. To replace the SSH keys in the nodes of a vCentervCenter HA cluster, you deactivate the cluster, generate new SSH keys on the Active node, transfer the keys to the passive node, and activate the cluster.

1. Edit the cluster and change the mode to **Disabled**.
2. Log in to the Active node by using the Virtual Machine Console or SSH.
3. Activate the bash shell.

   ```
   bash
   ```

4. Optionally, back up the contents of the `/home/vcha/.ssh` folder to the Active node.
5. To stop VCHA service on the Active node, run the following command.

   ```
   vmon-cli --stop vcha
   ```

6. To generate new SSH keys on the Active node, run the following command.

   ```
   /usr/sbin/vcha-reset-ssh-keys
   ```

7. Copy the SSH keys to the Passive node and Witness node.

   ```
   scp /home/vcha/.ssh/*
   ```

8. To start the VCHA service on the Active node, run the following command.

   ```
   vmon-cli --start vcha
   ```

9. Edit the cluster configuration and set the vCenter HA cluster to **Enabled**.

The VCHA state changes to **Healthy**.

Initiate a vCenter HA Failover

You can manually initiate a failover and have the Passive node become the Active node.

A vCenter HA cluster supports two types of failover.

Automatic failover	The Passive node attempts to take over the active role in case of an Active node failure.
Manual failover	The user can force a Passive node to take over the active role by using the Initiate Failover action.

Initiate a manual failover for troubleshooting and testing.

1. Log in to the Active node vCenter with the vSphere Client and click **Configure** for the vCenter where you need to initiate failover.
2. Under **Settings** select **vCenter HA** and click **Initiate Failover**.
3. Click **Yes** to start the failover.

   A dialog offers you the option to force a failover without synchronization. In most cases, performing synchronization first is best.
4. After the failover, you can verify that the Passive node has the role of the Active node in the vSphere Client.

Edit the vCenter HA Cluster Configuration

When you edit the vCenter HA cluster configuration, you can deactivate or activate the cluster, place the cluster in maintenance mode, or remove the cluster.

Verify that the vCenter HA cluster is deployed and contains the Active, Passive, and Witness nodes.

The operating mode of a vCenter controls the failover capabilities and state replication in a vCenter HA cluster.

A vCenter HA cluster can operate in one of the following modes.

Table 321: vCenter HA Cluster Modes of Operation

Mode	Automatic Failover	Manual Failover	Replication	
Enabled	Yes	Yes	Yes	This default mode of operation protects the vCenter from hardware and software failures by performing automatic failover.
Maintenance	No	Yes	Yes	Used for some maintenance tasks. For other tasks, you have to deactivate vCenter HA.

Mode	Automatic Failover	Manual Failover	Replication	
Disabled	No	No	No	If the Passive or Witness nodes are lost or recovering from a failure, a vCenter HA configuration can be deactivated. The Active node continues as a standalone vCenter.

Note:
If the cluster is operating in either Maintenance or Disabled mode, an Active node can continue serving client requests even if the Passive and Witness nodes are lost or unreachable.

1. Log in to the Active node vCenter with the vSphere Client and click **Configure**.
2. Under **Settings** select **vCenter HA** and click **Edit**.
3. Select one of the options.

Option	Result
Enable vCenter HA	Activates replication between the Active and Passive nodes. If the cluster is in a healthy state, your Active node is protected by automatic failover from the Passive node.
Maintenance Mode	In maintenance mode, replication still occurs between the Active and Passive nodes. However, automatic failover is deactivated.
Disable vCenter HA	Deactivates replication and failover. Keeps the configuration of the cluster. You can later activate vCenter HA again.
Remove vCenter HA cluster	Removes the cluster. Replication and failover no longer are provided. The Active node continues to operate as a standalone vCenter. For more information, see Remove a vCenter HA Configuration.

4. Click OK.

Perform Backup and Restore Operations

For additional security, you can back up the Active node in the vCenter HA cluster. You can then restore the node in case of catastrophic failure.

Verify the interoperability of vCenter HA and the backup and restore solution. One solution is vCenter file-based restore.

Note:
Remove the cluster configuration before you restore the Active node. Results are unpredictable if you restore the Active node and the Passive node is still running or other cluster configuration is still in place.

1. Back up the Active node.
   Do not back up the Passive node and Witness node.
2. Before you restore the cluster, power off and delete all vCenter HA nodes.
3. Restore the Active node.
   The Active node is restored as a standalone vCenter.

Remove a vCenter HA Configuration

You can remove a vCenter HA configuration from the vSphere Client.

1. Log in to the Active node vCenter and click **Configure**.
2. Under **Settings** select **vCenter HA** and click **Remove VCHA**.
   - The vCenter HA cluster's configuration is removed from the Active, Passive, and Witness nodes.
   - You can choose to delete the Passive and Witness nodes.
   - The Active node continues to run as a standalone vCenter.

- You cannot reuse the Passive and Witness nodes in a new vCenter HA configuration.
- If you performed a manual configuration, or if the Passive and Witness nodes are not discoverable, you must delete these nodes explicitly.
- Even if the second virtual NIC was added by the configuration process, the removal process does not remove the virtual NIC.

Reboot All vCenter HA Nodes

If you have to shut down and reboot all nodes in the cluster, you must follow a specific shutdown order to prevent the Passive node from assuming the role of Active node.

1. Shut down the nodes in this order.
   - Passive node
   - Active node
   - Witness node
2. Restart each node.

   You can restart nodes in any order.
3. Verify that all nodes join the cluster successfully, and that the previous Active node resumes that role.

Change the Server Size

When you deploy a vCenter, you select an environment. For vCenter HA, Small, Medium, Large, and X-Large are supported for production environments. If you need more space and want to change the environment, you have to delete the Passive node virtual machine before you change the configuration.

1. Log in to the Active node with the vSphere Client, edit the cluster configuration, and select **Disable**.
2. Delete the Passive node virtual machine.
3. Change the vCenter configuration for the Active node, for example, from a Small environment to a Medium environment.
4. Reconfigure vCenter HA.

Collecting Support Bundles for a vCenter HA Node

Collecting a support bundle from all the nodes in a vCenter HA cluster helps with troubleshooting.

When you collect a support bundle from the Active node in a vCenter HA cluster, the system proceeds as follows.

- Collects support bundle information from the Active node itself.
- Collects support bundles from Passive and Witness nodes and places them in the `commands` directory on the Active node support bundle.

  Note:
  The collection of support bundles from the Passive and Witness nodes is a best effort and happens if the nodes are reachable.

Troubleshoot Your vCenter HA Environment

In case of problems, you can troubleshoot your environment. The task you must perform depends on the failure symptoms.

vCenter HA Clone Operation Fails During Deployment

If the vCenter HA configuration process does not create the clones successfully, you have to resolve that cloning error.

Clone operation fails.

Note:
Cloning a Passive or Witness VM for a VCHA deployment to the same NFS 3.1 datastore as the source Active node VM fails. You must use NFS4 or clone the Passive and Witness VMs to a datastore different from the Active VM.

Look for the clone exception. It might indicate one of the following problems.

- You have a DRS-enabled cluster, but do not have three hosts.
- The host or database connection is lost.
- Not enough disk space.
- Other **Clone Virtual Machine** errors

1. Resolve the error that caused the problem.
2. Remove the cluster and start configuration again.

Redeploy the Passive or Witness node

If the passive or witness node fails and vCenter HA cluster was configured using the automatic cloning method, you can redeploy it in the **vCenter HA Settings** page.

1. Log in to the Active node with the vSphere Client.
2. Select the vCenter object in the inventory and select the **Configure** tab.
3. Select **vCenter HA** under **Settings**.
4. Click on the **REDEPLOY** button next to the node to start the Redeploy wizard.
5. - If your vCenter is managed by another vCenter in the same SSO domain, proceed to step 6.
   - If your vCenter is managed by another vCenter in a different SSO domain, input the location and credential details of that management vCenter. Enter the **Management vCenter FQDN or IP address** and **Single Sign-On** credentials.
6. Specify a unique name and target location.
7. Select the destination compute resource for the operation.
8. Select the datastore in which to store the configuration and disk files.
9. Configure the virtual machine networks.
   - If you are redeploying the Passive node, select virtual machine Management (NIC 0) and vCenter HA (NIC 1) networks.
   - If you are redeploying the Witness node, select vCenter HA (NIC 1) network.

   If there are issues with your selections, errors or compatibility warnings are displayed.
10. Review your selections and click **Finish** to redeploy the node.

vCenter HA Deployment Fails with an Error

Deployment failures can be caused by configuration issues, especially problems with the networking setup.

You start a vCenter HA cluster configuration, and it fails with an error. The error might show the cause of the problem, for example, you might see an SSH Connection Failed message.

If deployment fails, take steps to resolve the network issues.

1. Verify that the Passive and Witness nodes can be reached from the Active node.
2. Verify that routing between the nodes is set up correctly.
3. Check network latency.

Troubleshooting a Degraded vCenter HA Cluster

For a vCenter HA cluster to be healthy, each of the Active, Passive, and Witness nodes must be fully operational and be reachable over the vCenter HA cluster network. If any of the nodes fails, the cluster is considered to be in a degraded state.

If the cluster is in a degraded state, failover cannot occur. For information about failure scenarios while the cluster is in a degraded state, see Resolving Failover Failures.

The cluster can be in a degraded state for a number of reasons.

One of the nodes fails	• If the Active node fails, a failover of the Active node to the Passive node occurs automatically. After the failover, the Passive node becomes the Active node. At this point, the cluster is in a degraded state because the original Active node is unavailable. After the failed node is repaired or comes online, it becomes the new Passive node and the cluster returns to a healthy state after the Active and Passive nodes synchronize. • If the Passive node fails, the Active node continues to function, but no failover is possible and the cluster is in a degraded state. If the Passive node is repaired or comes online, it automatically rejoins the cluster and the cluster state is healthy after the Active and Passive nodes synchronize. • If the Witness node fails, the Active node continues to function and replication between Active and Passive node continues, but no failover can occur. If the Witness node is repaired or comes online, it automatically rejoins the cluster and the cluster state is healthy.
Database replication fails	If replication fails between the Active and Passive nodes, the cluster is considered degraded. The Active node continues to synchronize with the Passive node. If it succeeds, the cluster returns to a healthy state. This state can result from network bandwidth problems or other resource shortages.
Configuration file replication issues	If configuration files are not properly replicated between the Active and Passive nodes, the cluster is in a degraded state. The Active node continues to attempt synchronization with the Passive node. This state can result from network bandwidth problems or other resource shortages.

How you recover depends on the cause of the degraded cluster state. If the cluster is in a degraded state, events, alarms, and SNMP traps show errors.

If one of the nodes is down, check for hardware failure or network isolation. Check whether the failed node is powered on.

In case of replication failures, check if the vCenter HA network has sufficient bandwidth and ensure network latency is 10 ms or less.

Recovering from Isolated vCenter HA Nodes

If all nodes in a vCenter HA cluster cannot communicate with each other, the Active node stops serving client requests.

Node isolation is a network connectivity problem.

1. Attempt to resolve the connectivity problem. If you can restore connectivity, isolated nodes rejoin the cluster automatically and the Active node starts serving client requests.
2. If you cannot resolve the connectivity problem, you have to log in to Active node's console directly.
   a) Power off and delete the Passive node and the Witness node virtual machines.
   b) Log in to the Active node by using SSH or through the Virtual Machine Console.
   c) To enable the Bash shell, enter `shell` at the `appliancesh` prompt.
   d) Run the following command to remove the vCenter HA configuration.

      ```
      vcha-destroy -f
      ```

   e) Reboot the Active node.
      The Active node is now a standalone vCenter.

f) Perform vCenter HA cluster configuration again.

Resolving Failover Failures

When a Passive node does not become the Active node during a failover, you can force the Passive node to become the Active node.

The Passive node fails while trying to assume the role of the Active node.

A vCenter HA failover might not succeed for these reasons.

- The Witness node becomes unavailable while the Passive node is trying to assume the role of the Active node.
- An server state synchronization issue between the nodes exists.

You recover from this issue as follows.

1. If the Active node recovers from the failure, it becomes the Active node again.
2. If the Witness node recovers from the failure, follow these steps.
   a. Log in to the Passive node through the Virtual Machine Console.
   b. To enable the Bash shell, enter `shell` at the `appliancesh` prompt.
   c. Run the following command.
      ```
      vcha-reset-primary
      ```
   d. Reboot the Passive node.
3. If both Active node and Witness node cannot recover, you can force the Passive node to become a standalone vCenter.
   a. Delete the Active node and Witness node virtual machines.
   b. Log in to the Passive node through the Virtual Machine Console.
   c. To enable the Bash shell, enter `shell` at the `appliancesh` prompt.
   d. Run the following command.
      ```
      vcha-destroy
      ```
   e. Reboot the Passive node.

VMware vCenter HA Alarms and Events

If a vCenter HA cluster is in a degraded state, alarms and events show errors.

Table 322: The following events will raise VCHA health alarm in vpxd:

Event Name	Event Description	Event Type	Category
vCenter HA cluster state is currently healthy	vCenter HA cluster state is currently healthy	com.vmware.vcha.cluster.state.healthy	info
vCenter HA cluster state is currently degraded	vCenter HA cluster state is currently degraded	com.vmware.vcha.cluster.state.degraded	warning
vCenter HA cluster state is currently isolated	vCenter HA cluster state is currently isolated	com.vmware.vcha.cluster.state.isolated	error
vCenter HA cluster is destroyed	vCenter HA cluster is destroyed	com.vmware.vcha.cluster.state.destroyed	info

Table 323: The following events will raise PSC HA health alarm in vpxd:

Event Name	Event Description	Event Type	Category
PSC HA state is currently healthy	PSC HA state is currently healthy	com.vmware.vcha.psc.ha.health.healthy	info
PSC HA state is currently degraded	PSC HA state is currently degraded	com.vmware.vcha.psc.ha.health.degraded	info

Event Name	Event Description	Event Type	Category
PSC HA is not monitored after vCenter HA cluster is destroyed	PSC HA state is not being monitored	com.vmware.vcha.psc.ha.health.unknown	info

Table 324: Cluster Status Related Events

Event Name	Event Description	Event Type	Category
Node {nodeName} joined back to the cluster	One node joined back to the cluster	com.vmware.vcha.node.joined	info
Node {nodeName} left the cluster	One node left the cluster	com.vmware.vcha.node.left	warning
Failover succeeded	Failover succeeded	com.vmware.vcha.failover.succeeded	info
Failover cannot proceed when cluster is in disabled mode	Failover cannot proceed when cluster is in disabled mode	com.vmware.vcha.failover.failed.disabledmode	warning
Failover cannot proceed when cluster does not have all three nodes connected	Failover cannot proceed when cluster does not have all three nodes connected	com.vmware.vcha.failover.failed.nodemissing	warning
Failover cannot proceed when vPostgres on Passive node is not ready to takeover	Failover cannot proceed when Passive node is not ready to takeover	com.vmware.vcha.failover.failed.passivenotready	warning
vCenter HA cluster mode changed to {clusterMode}	vCenter HA cluster mode changed	com.vmware.vcha.cluster.mode.changed	info

Table 325: Database replication-related events

vent Name	Event Description	Event Type	Category
Database replication mode changed to {newState}	Database replication state changed: sync, async or no replication	com.vmware.vcha.DB.replication.state.changed	info

Table 326: File replication-related events

Event Name	Event Description	Event Type	Category
Appliance {fileProviderType} is {state}	Appliance File replication state changed	com.vmware.vcha.file.replication.state.changed	info

Patching a vCenter High Availability Environment

You can patch a vCenter which is in a vCenter High Availability cluster by using the `software-packages` utility available in the vCenter shell.

For more information, see *Patch a vCenter High Availability Environment* in *vSphere Upgrade*.

Reduced Downtime Upgrade for vCenter HA

With VMware vSphere 9.0, Reduced Downtime Upgrade (RDU) is integrated with automatic and manual vCenter HA deployment. RDU ensures an upgrade process with minimum downtime, while preserving your vCenter HA deployment and configuration before, during, and after the upgrade process.

Reduced Downtime Upgrade (RDU) is migration-based upgrade process for vCenter appliance (VCSA), designed to minimize downtime during the upgrade process by migrating critical configurations, such as the VCSA configuration, vCenter Database (VCDB), and network configuration, from the source VCSA to the target VCSA prior to shutting down

the source appliance. The switchover to the new VCSA occurs only after all necessary data has been transferred, ensuring minimal service disruption. For more information about RDU and detailed upgrade tasks, see Reduced Downtime Upgrade in the *vCenter Upgrade* guide.

Note:

In a vCenter HA cluster, RDU is supported only with IPv4 node addresses. IPv6 addresses are not supported.

During the staging phase of the upgrade, vCenter HA is undeployed, which results in temporary period where VCHA protection is not available. After the RDU switchover is completed, VCHA is automatically redeployed on the target nodes, restoring the protection to your vCenter environment without additional intervention.

RDU supports both automatic and manual vCenter HA deployments on self-managed and non-self-managed vCenter instances. This flexibility allows for seamless integration regardless of the management model in place. For more information about VCHA deployment types, see Protecting vCenter with vCenter High Availability.

- For upgrades of automatic vCenter HA deployments in self-managed vCenter instances, no credentials are required during the process. When upgrading non-self-managed vCenter instances, you must provide the credentials of the managing vCenter. After the upgrade, the passive and witness nodes are automatically cloned and redeployed.
- For upgrades of manual vCenter HA deployments, during the process, you can provide the credentials of the vCenter instances that manage the passive and witness nodes. After a successful upgrade, the passive and witness nodes are cloned and redeployed to the vCenter instances, for which you initially provided the credentials. Alternatively, if you do not provide the vCenter credentials, the source vCenter instance is upgraded, but vCenter HA is not configured.

In the event of an upgrade failure or rollback, RDU ensures that vCenter HA is restored to the original functional state as before the upgrade, which guarantees the continued reliability of your vCenter environment.

vSphere Monitoring and Performance

VMware provides several tools to help you monitor your virtual environment and to locate the source of potential issues and current problems.

Performance Charts	Allow you to see performance data on a variety of system resources including CPU, Memory, Storage, and so on.
Performance Monitoring Command-line Utilities	Allow you to access detailed information on system performance through the command line.
Host Health	Allows you to quickly identify which hosts are healthy and which are experiencing problems.
Events, Alerts, and Alarms	Allow you to configure alerts and alarms and to specify the actions the system should take when they are triggered.
System Log Files	System logs contain additional information about activities in your vSphere environment.

Intended Audience

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

The content in this section is intended for vSphere administrators who perform the following tasks:

- Monitor the health and performance of physical hardware backings for the virtual environment.
- Monitor the health and performance of virtual devices in the virtual environment.
- Troubleshoot problems in the system.
- Configure alarms.
- Configure SNMP messages.
- Use vCenter Events for forensic analysis and auditing of the user actions in the virtual environment.

Virtual machine administrators also might find the section on Monitoring Guest Operating System Performance helpful.

Monitoring Inventory Objects with Performance Charts

The vSphere statistics subsystem collects data on the resource usage of inventory objects. Data on a wide range of metrics is collected at frequent intervals, processed, and archived in the vCenter database. You can access statistical information through command-line monitoring utilities or by viewing performance charts in the vSphere Client.

Counters and Metric Groups

vCenter systems and hosts use data counters to query for statistics. A data counter is a unit of information relevant to a given inventory object or device. Each counter collects data for a different statistic in a metric group. For example, the disk metric group includes separate data counters to collect data for disk read rate, disk write rate, and disk usage. Statistics for each counter are rolled up after a specified collection interval. Each data counter consists of several attributes that are used to determine the statistical value collected.

For a complete list and description of performance metrics, see the *vSphere API Reference*.

> **Note:**
>
> Counters might not contain data from hosts of version earlier than the version when the counter is introduced.

Collection Levels and Collection Intervals

Collection levels determine the number of counters for which data is gathered during each collection interval. Collection intervals determine the time period during which statistics are aggregated, calculated, rolled up, and archived in the vCenter database. Together, the collection interval and collection level determine how much statistical data is collected and stored in your vCenter database.

Data Availability

Real-time data appears in the performance charts only for hosts and virtual machines that are powered on. Historical data appears for all supported inventory objects, but might be unavailable during certain circumstances.

Performance Chart Types

Performance metrics are displayed in different types of charts, depending on the metric type and object.

Table 327: Performance Chart Types

Chart Type	Description
Line graph	Displays metrics for a single inventory object. The data for each performance counter is plotted on a separate line in the chart. For example, a network chart for a host can contain two lines: one showing the number of packets received, and one showing the number of packets transmitted.
Bar chart	Displays storage metrics for datastores in a selected data center. Each datastore is represented as a bar in the chart. Each bar displays metrics based on the file type: virtual disks, snapshots, swap files, and other files. You can view space utilization bar charts for datastore clusters and data centers.
Pie chart	Displays space utilization information for a single object, based on the file types, or virtual machines. For example, a pie chart for a datastore can display the amount of storage space occupied by the virtual machines taking up the largest space. You can view space utilization pie charts for datastores and virtual machines.
Stacked graph	Displays metrics for the child objects that have the highest statistical values. The metrics for the host itself are displayed in separate line charts. Stacked charts are useful in comparing the resource allocation and usage across multiple hosts or virtual machines. By default, the 10 child objects with the highest data counter values are displayed.

Data Counters

Each data counter includes several attributes that are used to determine the statistical value collected. See the *vSphere API Reference* documentation for a complete list and description of supported counters.

Table 328: Data Counter Attributes

Attribute	Description
Units	Gives information about the units of measure, the standard in which the statistic quantity is measured. • Kilobytes (KB) – 1024 bytes **Note:** Technically, 1 Kilobytes (KB) = 1000 bytes and 1 Kibibyte (KiB) = 1024 bytes. However, based on the context, kilobyte is also used for 1024 bytes in the computer science literature interchangeably. • Kilobytes per second (KBps) – 1024 bytes per second • Kilobits (kb) – 1000 bits • Kilobits per second (kbps) – 1000 bits per second • Megabytes (MB) • Megabytes per second (MBps) • Megabits (Mb), megabits per second (Mbps) • Megahertz (MHz) • Microseconds (µs) • Milliseconds (ms) • Number (#) • Percent (%) • Seconds (s) • Watts (watt) • Joules (joule) • Terabytes (teraBytes) • Temperature in celsius (celsius)
Description	Text description of the data counter.
Stat Type	Gives information about the statistics type, the measurement used during the statistics interval. Related to the unit of measurement. • Rate – Value over the current statistics interval. For example: CPU Usage: Amount of CPU actively used by the host, resource pool, or virtual machine in the cluster. Counter: usagemhz Stats Type: Rate Unit: MegaHertz (MHz) • Delta – Change from previous statistics interval. For example: CPU System Time for vCPU (%): Amount of time spent on system processes on each virtual CPU in the virtual machine. **Note:** This is the host view of the CPU usage, not the guest operating system view. Counter: system Stats Type: Delta Unit: Percentage (%) • Absolute – Absolute value (independent of the statistics interval). For example: Memory: Amount of host machine memory used by all powered on virtual machines in the cluster. A cluster's consumed memory consists of virtual machine consumed memory and overhead memory. It does not include host-specific overhead memory, such as memory used by the service console or VMkernel. Counter: consumed Stats Type: Absolute Unit: MegaBytes (MB)

Attribute	Description
Rollups	Calculation method used during the statistics interval to aggregate data. Determines the type of statistical values that are returned for the counter. • Average – Data collected during the interval is aggregated and averaged. For example: Virtual Disk Read Requests: Number of virtual disk read commands completed on each virtual disk on the virtual machine. The aggregate number of all virtual disk read commands is also displayed in the chart. Counter: numberRead Stats Type: Absolute Unit: Number Rollup Type: Average • Minimum – The minimum value is rolled up. • Maximum – The maximum value is rolled up. The Minimum and Maximum values are collected and displayed only in statistics level 4. Minimum and maximum rollup types are used to capture peaks in data during the interval. For real-time data, the value is the current minimum or current maximum. For historical data, the value is the maximum or minimum of the aggregated values. For example, the following information for the CPU usage chart shows that the average is collected at statistics level 1. The minimum and maximum values are collected at statistics level 4. – Counter: usage – Unit: Percentage (%) – Rollup Type: Average (Minimum/Maximum) – Collection Level: 1 (4) • Summation – Data collected is summed. The measurement displayed in the chart represents the sum of data collected during the interval. For example: Network Packets Received: Number of network packets received across the top ten physical NIC instances on the host. The chart also displays the aggregated value for all NICs. Counter: packetRx Stats Type: Absolute Unit: Number Rollup Type: Summation • Latest – Data collected during the interval is a set value. The value displayed in the performance charts represents the current value. For example: Space in GB (Allocated): Total amount of logical datastore space provisioned by an administrator for the virtual machine. It is the storage size up to which the virtual machine files on datastores can grow. This includes log files, VMX files, and other miscellaneous files. Allocated space is not always in use. Counter: provisioned Stats Type: Absolute Unit: Gigabytes (GB) Rollup Type: Latest **Note:** Minimum and maximum rollups are available only when statistics level 4 is configured for the vCenter instance.
Statistics level	The collection level determines the number of data counters used to collect statistics data. Collection levels are also referred as statistics levels. These collection levels vary from 1 to 4, with Level 4 having the most counters. • Level 1 is the least detailed statistics level and only includes the most critical statistics, such as aggregate CPU, memory, and network usage. • Level 2 introduces a number of additional statistics • Level 3 incorporates per instance statistics, for example, CPU usage of a host on a per-CPU basis. • Level 4 is the most detailed and is inclusive of all the other levels For more information about collection levels, see Statistics Levels. **Note:** Be careful when you set a higher collection level, as the process requires a significant increase of resource usage.

Metric Groups

The performance data collection subsystem for vSphere collects performance data on various inventory items and their devices. Data counters define individual performance metrics. Performance metrics are organized into logical groups based on the object or object device. Statistics for one or more metrics can be displayed in a chart.

Table 329: Metric Groups

Metric group	Description
Cluster Services	Performance statistics for clusters configured by using vSphere Distributed Resource Scheduler, vSphere High Availability, or both.
CPU	CPU utilization per host, virtual machine, resource pool, or compute resource.
Datastore	Statistics for datastore utilization. **Note:** NFS statistics are collected under datastore statistics.
Disk	Disk utilization per host, virtual machine, or datastore. Disk metrics include I/O performance, such as latency and read/write speeds, and utilization metrics for storage as a finite resource.
GPU	GPU utilization per cluster, or host. GPU metrics include GPU utilization in percentage of the total, memory usage, temperature of the GPU devices.
GPU Power Usage	GPU power consumption per virtual machine. **Important:** This metric group shows only for virtual machines that have NVIDIA GPUs.
Memory	Memory utilization per host, virtual machine, resource pool, or compute resource. The value obtained is one of the following: • For virtual machines, memory refers to the guest physical memory. Guest physical memory is the amount of physical memory presented as a virtual-hardware component to the virtual machine, at creation time, and made available when the virtual machine is running. • For hosts, memory refers to the machine memory. Machine memory is the RAM that is installed on the hardware that comprises the host.
Network	Network utilization for both physical and virtual network interface controllers (NICs) and other network devices. The virtual switches that support connectivity among all components, such as hosts, virtual machines, VMkernel.
Power	Energy usage statistics per host.
VM power	Power consumption for CPU, memory, virtual GPU, and all other virtual machine devices. You can include those data counters in the avanced VM power performance charts that you create for your virtual machines.
Storage Adapter	Data traffic statistics per host bus adapter (HBA).
Storage Path	Data traffic statistics per path.
System	Overall system availability, such as the system heartbeat and uptime. These counters are available directly from hosts and from vCenter.
Virtual Disk	Disk utilization and disk performance metrics for virtual machines.
Virtual Flash	Virtual flash counters.
Virtual Machine Operations	Virtual machine power and provisioning operations in a cluster or data center.
vSphere Replication	Statistics for the virtual machine replication performed by vSphere Replication.

Data Collection Intervals

Collection intervals determine the duration for which statistics are aggregated, calculated, rolled up, and archived. Together, the collection interval and collection level determine how much statistical data is gathered and stored in your vCenter database.

Table 330: Collection Intervals

Collection Interval/ Archive Length	Collection Frequency	Default Behavior
1 Day	5 Minutes	Real-time (20s) statistics are rolled up to create one data point every 5 minutes. The result is 12 data points every hour and 288 data points every day. After 30 minutes, the six data points collected are aggregated and rolled up as a data point for the 1-Week time range. You can change the interval duration and archive length of the 1-Day collection interval by configuring the statistics settings.
1 Week	30 Minutes	1-Day statistics are rolled up to create one data point every 30 minutes. The result is 48 data points every day and 336 data points every week. Every 2 hours, the 12 data points collected are aggregated and rolled up as a data point for the 1-Month time range. You cannot change the default settings of the 1-Week collection interval.
1 Month	2 Hours	1-Week statistics are rolled up to create one data point every 2 hours. The result is 12 data points every day and 360 data points every month (assuming a 30-day month). After 24 hours, the 12 data points collected are aggregated and rolled up as a data point for the 1-Year time range. You cannot change the default settings of the 1-Month collection interval.
1 Year	1 Day	1-Month statistics are rolled up to create one data point every day. The result is 365 data points each year. You can change the archive length of the 1-Year collection interval by configuring the statistics settings.

Note:
If you change the duration of data collection intervals, you might need to allocate more storage resources.

Statistics Levels

Each collection interval has a default statistics level that determines the amount of data gathered and which counters are available for display in the charts.

Table 331: Statistics Levels

Level	Metrics	Best Practice
Level 1	• Cluster Services (VMware Distributed Resource Scheduler) – all metrics • CPU – cpuentitlement, totalmhz, usage (average), usagemhz • Disk – capacity, maxTotalLatency, provisioned, unshared, usage (average), used • Memory – consumed, mementitlement, overhead, swapinRate, swapoutRate, swapused, totalmb, usage (average), vmmemctl (balloon), total bandwidth (DRAM, PMem, NVMe) • Network – usage (average), IPv6 • System – heartbeat, uptime • Virtual Machine Operations – numChangeDS, numChangeHost, numChangeHostDS • Virtual machine power - power capacity usage, GPU(watts)	Use for long-term performance monitoring when device statistics are not required. Level 1 is the default Collection Level for all Collection Intervals.
Level 2	• Level 1 metrics • CPU – idle, reservedCapacity • Disk – All metrics, excluding numberRead and numberWrite. • Memory – All metrics, excluding memUsed, maximum and minimum rollup values, read or write latency (DRAM, PMem, NVMe). • Virtual Machine Operations – All metrics	Use for long-term performance monitoring when device statistics are not required but you want to monitor more than the basic statistics.
Level 3	• Level 1 and Level 2 metrics • Metrics for all counters, excluding minimum and maximum rollup values. • Device metrics	Use for short-term performance monitoring after encountering problems or when device statistics are required.
Level 4	All metrics supported by the vCenter, including minimum and maximum rollup values.	Use for short-term performance monitoring after encountering problems or when device statistics are required.

Note:

When the statistics levels, level 3 or level 4 are used beyond the default value, it may cause one particular process, vpxd, to sustain memory growth, if it cannot save the statistics information to the database as quickly as required. If the usage limit of these statistics levels is not monitored closely, it may cause vpxd to grow out of memory and eventually crash.

So, if you decide to elevate any of these levels, you must monitor the size of the vpxd process to make sure that it is not growing boundlessly after the change.

View Performance Charts

The vCenter statistics settings, the type of object selected, and the features that are enabled on the selected object determine the amount of information displayed in charts. Charts are organized into views. You can select a view to see related data together on one screen. You can also specify the data collection period. The duration extends from the selected time range to the present time.

Overview charts display multiple data sets in one panel to evaluate different resource statistics, display thumbnail charts for child objects. It also displays charts for a parent and a child object. You can export overview charts in the PNG, JPEG, SVG, or CSV formats.

Advanced charts display more information than overview charts and are configurable. You can print or export data from advanced charts in the PNG, JPEG, SVG, or CSV formats. See Working with Advanced and Custom Charts .

1. Select a valid inventory object in the vSphere Client.

   Overview and advanced performance charts are available for datacenter, cluster, host, resource pool, vApp, and virtual machine objects. Overview charts are also available for datastores and datastore clusters. Performance charts are not available for network objects.

2. Click the **Monitor** tab, and expand **Performance**.
3. Select the type of chart to view.

   Available views depend on the type of object. For views that might contain many charts in a large environment, the vSphere Client displays the charts distributed on multiple pages. You can use the arrow buttons to navigate between pages.

   - Select **Overview** to view overview performance charts for the selected object.
   - Select **Advanced** to view advanced performance charts for the selected object.
4. From the **Period** drop-down menu, select a predefined or custom time range.

Overview Performance Charts Options Available Under the View Menu

The performance chart options that you can access under the **View** menu vary depending on the type of inventory object you select.

For example, the **Virtual Machines** view is available when you view host performance charts only if there are virtual machines on the selected host. Likewise, the **Fault Tolerance** view for virtual machine performance charts is available only when that feature is enabled for the selected virtual machine.

Table 332: Performance Chart Views by Inventory Object

Object	View List Items
Data center	• **Space utilization** - space utilization charts for datastores in the data center, including space by file type and storage space used by each datastore in the data center. • **Clusters** - thumbnail CPU and memory charts for each cluster, and stacked charts for total CPU and memory usage in the data center. This view is the default.
Datastore and datastore cluster	• **Space** - space utilization charts for the datastore: – space utilization by file type – space utilization by virtual machine – space usage • **Performance** - performance charts for the datastore or datastore cluster and for virtual machine disks on the resource. **Note:** The **Performance** view for datastores is only available when all hosts that are connected to the datastores are ESX/ESXi 4.1 or later. The Performance view for datastore clusters is only available when the Storage DRS is enabled.
Cluster	• **Overview** - CPU and memory charts for the cluster. • **Resource Pools & Virtual Machines** - thumbnail charts for resource pools and virtual machines, and stacked charts for total CPU and memory usage in the cluster. • **Hosts** - thumbnail charts for each host in the cluster, and stacked charts for total CPU, memory, disk usage, and network usage.
Host	• **Overview** - CPU, memory, disk, GPU compute, GPU memory and network charts for the host. • **Virtual Machines** - thumbnail charts for virtual machines, and stacked charts for total CPU usage and total memory usage on the host. • **Disks** - read rates, write rates, and disk usage for the host. • **Network Adapters** - network bandwidth on a host. • **GPUs** - thumbnail charts for GPUs compute usage, memory allocation, and temperature.
Resource Pool and vApps	• **Overview** - CPU and memory charts for the resource pool. • **Resource Pools & Virtual Machines** - thumbnail charts for resource pools, and virtual machines and stacked charts for CPU and memory usage in the resource pool or vApp.

Object	View List Items
Virtual Machine	• **Storage** - space utilization charts for the virtual machine: space by file type, space by datastore, and total gigabytes. • **Fault Tolerance** - CPU and memory charts that display comparative metrics for the fault-tolerant primary and secondary virtual machines. • **Overview** - CPU, memory, network, host (thumbnail charts), and disk usage charts for the virtual machine.

Overview Performance Charts

The overview and advanced performance charts display the metrics for an object in the inventory. Use these charts to monitor and troubleshoot performance problems.

The metrics provided in performance charts are a subset of those collected for hosts and the vCenter instance. For a complete list of all metrics collected by hosts and the vCenter instance, see the *vSphere API Reference* documentation.

Clusters

The cluster charts contain information about CPU, disk, memory, and network usage for clusters. The help topic for each chart contains information about the data counters displayed in that chart. The statistics level set for vCenter determines the available counters.

CPU (MHz)

The CPU (MHz) chart displays CPU usage for the cluster.

Cluster Counters

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Advanced** and open the **Chart Options** dialog box.

Table 333: Data Counters

Chart Label	Description
Usage	Sum of the average CPU usage values, in Megahertz, of all virtual machines in the cluster. • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Total	Total amount of CPU resources available in the cluster. The maximum value is equal to the number of cores multiplied by the frequency of the processors. For example, a cluster has two hosts, each of which has four CPUs that are 3 GHz each, and one virtual machine that has two virtual CPUs. VM totalmhz = 2 vCPUs * 3000 MHz = 6000 MHz Host totalmhz = 4 CPUs * 3000 MHz = 12000 MHz Cluster totalmhz = 2 x 4 * 3000 MHz = 24000 MHz • Counter: totalmhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Summation • Collection Level: 1

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of cluster resources. However, if the value is constantly high, the CPU demanded is likely greater than the CPU capacity available. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the hosts in the cluster.

If performance is impacted, consider taking the following actions.

Table 334: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
3	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. This might help avoid hot spots in the cluster.
4	Migrate one or more virtual machines to a new host.
5	Upgrade the physical CPUs or cores on each host in the cluster if necessary.
6	Enable CPU-saving features, such as TCP Segmentation Offload.
7	Replace software I/O with the dedicated hardware, such as iSCSI HBAs or TCP Segmentation Offload NICs.

CPU Usage

The cluster CPU Usage charts monitor the CPU utilization of the hosts, resource pools, and virtual machines in the cluster. This chart displays the 10 child objects in the cluster with the most CPU usage.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Overview** and select an inventory object from the **View** drop-down menu.

Table 335: Data Counters

Chart Label	Description
<host>, <resource pool>, or <virtual machine>	Amount of CPU actively used by the host, resource pool, or virtual machine in the cluster. • Counter: usagemhz • Stats Type: Rate • Unit: MegaHertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of cluster resources. However, if the value is constantly high, the CPU demanded is likely greater than the CPU capacity available. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the hosts in the cluster.

If performance is impacted, consider taking the following actions.

Table 336: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
3	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. This might help avoid hot spots in the cluster.
4	Migrate one or more virtual machines to a new host.
5	Upgrade the physical CPUs or cores on each host in the cluster if necessary.
6	Enable CPU-saving features, such as TCP Segmentation Offload.
7	Replace software I/O with the dedicated hardware, such as iSCSI HBAs or TCP Segmentation Offload NICs.

Disk (KBps)

The Disk (KBps) chart displays the disk I/O of the 10 hosts in the cluster with the most disk usage.

This chart is located on the **Monitor** tab. Select **Performance** > **Overview** and select **Hosts** from the **View** drop-down menu.

Table 337: Data Counters

Chart Label	Description
host_name	Average data I/O rate across all hosts in the cluster. • Counter: usage • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.

- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 338: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Memory (MB)

The Memory (MB) chart displays consumed memory for the cluster. The chart appears only at collection level 1.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 339: Data Counters

Chart Label	Description
Consumed	Amount of host machine memory used by all powered on virtual machines in the cluster. A cluster's consumed memory consists of virtual machine consumed memory and overhead memory. It does not include host-specific overhead memory, such as memory used by the service console or VMkernel. • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Total	Total amount of machine memory of all hosts in the cluster that is available for virtual machine memory (physical memory for use by the Guest OS) and virtual machine overhead memory. Memory Total = Aggregate host machine memory - (VMkernel memory + Service Console memory + other service memory) **Note:** The totalmb data counter is the same as the effectivemem data counter, which is supported only for backward compatibility. • Counter: totalmb • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Memory usage is not necessarily an indicator of performance problems. Memory can be high if a host is swapping or ballooning, which can result in virtual machine guest swapping. In such cases, check for other problems, such as CPU over-commitment or storage latencies.

If you have constantly high memory usage in a cluster, resource pool, or vApp, consider taking the following actions.

Table 340: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Verify that the balloon driver is enabled. The balloon driver is installed with VMware Tools and is critical to performance. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	If the balloon value is high, check the resource shares, reservations, and limits for the virtual machines and resource pools on the hosts. Verify that the host's settings are adequate and not lower than those set for the virtual machine. If free memory is available on the hosts and the virtual machines are experiencing high swap or balloon memory, the virtual machine (or resource pool, if it belongs to one) has reached its resource limit. Check the maximum resource limit set on that host.
4	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
5	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. It might help avoid hot spots in the cluster.
6	Add more physical memory to one or more hosts.

Memory (MB)

The Memory (MB) chart displays memory data counters for clusters. The chart appears at all collection levels except level 1.

Description

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

> **Note:**
>
> These data counter definitions are for hosts. At the cluster level, the values are collected and totaled. The counter values in the chart represent the aggregate amounts of the host data. The counters that appear in the chart depend on the statistics level set for your vCenter.

Table 341: Data Counters

Chart Label	Description
Active	Sum of the active guest physical memory of all powered on virtual machines on the host, plus memory used by basic VMkernel applications. Active memory is estimated by the VMkernel. • Counter: active • Stats Type: Absolute • Unit: kilobytes (kb) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Balloon	Sum of the guest physical memory reclaimed by the balloon driver for all powered on virtual machines on the host. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Consumed	Amount of machine memory used on the host. Consumed memory includes virtual machine memory, service console memory, and VMkernel memory. consumed memory = total host memory - free host memory • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Granted	Sum of the guest physical memory granted for all powered on virtual machines. Granted memory is mapped to the host's machine memory. Granted memory for a host includes the shared memory of each virtual machine on the host. • Counter: granted • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)

Chart Label	Description
Swap Used	Sum of the memory swapped by all powered on virtual machines on the host. • Counter: swapused • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Tier memory consumed	Amount of machine memory consumed by the host per tier. • Counter: tier.consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Latest • Collection Level: 2
Total	Aggregate total memory available to the cluster. • Counter: totalmb • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

To ensure best performance, the host memory must be large enough to accommodate the active memory of the virtual machines. The active memory can be smaller than the virtual machine memory size. It allows you to over-provision memory, but still ensures that the virtual machine active memory is smaller than the host memory.

Transient high-usage values usually do not cause performance degradation. For example, memory usage can be high when several virtual machines are started at the same time or when a spike occurs in virtual machine workload. However, a consistently high memory usage value (94% or greater) indicates that the host is probably lacking the memory required to meet the demand. If the active memory size is the same as the granted memory size, the demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the memory usage value is high, and the host has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot handle the demand for memory. It leads to memory reclamation, which might degrade performance.

If the host has enough free memory, check the resource shares, reservation, and limit settings of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machines.

If the host has little free memory available, or if you notice a degradation in performance, consider taking the following actions.

Table 342: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.

#	Resolution
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory Consumed

The Memory Consumed chart displays memory usage for the 10 child objects in the cluster with the most consumed memory.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Overview** and select an object from the **View** drop-down menu.

Table 343: Data Counters

Chart Label	Description
Resource Pools and Virtual Machines, or *Hosts*	Amount of machine memory used by all resource pools and virtual machines in the cluster or by all hosts in the cluster, depending on the cluster view. Consumed memory includes virtual machine memory, service console memory, and VMkernel memory. consumed memory = total host memory - free host memory • Counter: consumed • Stats Type: Absolute • Unit: MegaBytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Memory usage is not necessarily an indicator of performance problems. Memory can be high if a host is swapping or ballooning, which can result in virtual machine guest swapping. In such cases, check for other problems, such as CPU over-commitment or storage latencies.

If you have constantly high memory usage in a cluster, resource pool, or vApp, consider taking the following actions.

Table 344: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Verify that the balloon driver is enabled. The balloon driver is installed with VMware Tools and is critical to performance. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	If the balloon value is high, check the resource shares, reservations, and limits for the virtual machines and resource pools on the hosts. Verify that the host's settings are adequate and not lower than those set for the virtual machine. If free memory is available on the hosts and the virtual machines are experiencing high swap or balloon memory, the virtual machine (or resource pool, if it belongs to one) has reached its resource limit. Check the maximum resource limit set on that host.
4	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
5	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. It might help avoid hot spots in the cluster.

#	Resolution
6	Add more physical memory to one or more hosts.

Network (Mbps)

The Network (Mbps) chart displays network speed for the 10 hosts in the cluster with the most network usage.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Overview** and select **Hosts**from the **View** drop-down menu.

Table 345: Data Counters

Chart Label	Description
<host>	Average rate at which data is transmitted and received across all NIC instances on the host. • Counter: usage • Stats Type: Rate • Unit: Megabits per second (Mbps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 346: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.

#	Resolution
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Data centers

The data center charts contain information about CPU, disk, memory, and storage usage for data centers. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for vCenter.

CPU (MHz)

The CPU (MHz) chart displays CPU usage for the 10 clusters in the data center with the most CPU usage.

This chart is located on the **Monitor** tab for a data center. Select **Performance** > **Overview** and select **Clusters** from the **View** drop-down menu.

Table 347: Data Counters

Chart Label	Description
<cluster>	Amount of CPU currently in use by the cluster. The active CPU usage is approximately equal to the ratio of the used CPU cycles to the available CPU cycles. The maximum possible value is the frequency of the processors multiplied by the number of cores. For example, a two-way SMP virtual machine using 4000MHz on a host that has four 2GHz processors is using 50% of the CPU (4000 ÷ 4 × 2000) = 0.5). • Counter: usagemhz • Stats Type: Rate • Unit: MegaHertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of cluster resources. However, if the value is constantly high, the CPU demanded is likely greater than the CPU capacity available. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the hosts in the cluster.

If performance is impacted, consider taking the following actions.

Table 348: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
3	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. This might help avoid hot spots in the cluster.
4	Migrate one or more virtual machines to a new host.
5	Upgrade the physical CPUs or cores on each host in the cluster if necessary.
6	Enable CPU-saving features, such as TCP Segmentation Offload.
7	Replace software I/O with the dedicated hardware, such as iSCSI HBAs or TCP Segmentation Offload NICs.

Memory (MB)

The Memory (MB) chart displays the average amount of consumed memory for the 10 clusters in the data center with the most consumed memory.

This chart is located on the **Monitor** tab for a data center. Select **Performance** > **Overview** and select **Clusters** from the **View** drop-down menu.

Table 349: Data Counters

Chart Label	Description
<cluster>	Amount of host machine memory used by all powered on virtual machines in the cluster. • Counter: consumed • Stats Type: Absolute • Unit: MegaBytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A cluster's consumed memory consists of virtual machine consumed memory and overhead memory. It does not include host-specific overhead memory, such as memory used by the service console or VMkernel.

If you experience problems with cluster memory usage, use the thumbnail cluster charts to examine memory usage for each cluster and increase memory resources if needed.

If the cluster is a DRS cluster, check the aggressiveness threshold. If the value is low, increase the threshold. Increasing the threshold might help avoid hot spots in the cluster.

Space Utilization

The **Space Utilization By File Type** bar chart displays datastore space usage for virtual disks, swap files, snapshot files, and other virtual machine files.

Note:
This chart does not show historical statistics. It only shows the most recently available data, which may be up to 30 minutes late, depending on when the last statistics rollup occurred. In addition, statistics are not collected across all datastores at one time. They are collected asynchronously.

The **Space Utilization By File Type** chart is located on the **Monitor** tab for a data center. Select **Performance** > **Overview** and select **Space Utilization** from the **View** drop-down menu.

Datastore Counters

Table 350: Data Counters

File Type	Description
Virtual Disks	Amount of disk space used by virtual disk files. Virtual disk files store the contents of the virtual machine's hard disk drive. It includes information that you write to a virtual machine's hard disk, such as the operating system, program files, and data files. The files have the extension `.vmdk` and appear as a physical disk drive to a guest operating system. **Note:** Delta disks, which also have an extension `.vmdk`, are not included in this file type. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Swap Files	Amount of disk space used by swap files. Swap files back up the virtual machine's physical memory. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Snapshots	Amount of disk space used by virtual machine snapshot files. Snapshot files store information about virtual machine snapshots. They include snapshot state files and delta disk files. A snapshot state file stores the running state of the virtual machine at the time of the snapshot. It has the extension `.vmsn`. A delta disk file stores the updates made by the virtual machine to the virtual disks after a snapshot is taken. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Other VM Files	Amount of disk space used by all other virtual machine files, such as configuration files and log files. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Other	Amount of disk space used by all other non-virtual machine files, such as documentation files and backup files.
Free Space	Amount of disk space not currently in use.
Total Space	Amount of disk space available to the datastore. It defines the datastore capacity. The chart displays the information for datastores but not for data centers. total space = virtual disk space + swap file space + snapshot space + other VM file space + other space + free space

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Datastores and Datastore Clusters

The datastore charts contain information about disk usage for datastores or the datastores that are part of a cluster. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for vCenter.

Space

The Space chart displays space usage data counters for datastores.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Space** from the **View** drop-down menu.

Table 351: Data Counters

Chart Label	Description
Allocated	Amount of physical space provisioned by an administrator for the datastore. It is the storage size up to which files on the datastore can grow. Allocated space is not always in use. • Counter: provisioned • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1
Used	Amount of physical datastore space in use. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1
Capacity	Maximum capacity of the datastore. capacity = virtual machine file space + non-virtual machine file space + free space **Note:** Storage data is collected and updated in the overview charts every 30 minutes. Therefore, if you refresh the datastore, the capacity value might only be updated in the datastore **Summary** tab, and not in the overview charts. • Counter: capacity • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Utilization By File Type

The **Space Utilization by File Type** chart displays space used by virtual disks, swap files, snapshot files, and other virtual machine files on the datastore or the datastore cluster.

> **Note:**
> This chart does not show historical statistics. It only shows the most recently available data, which may be up to 30 minutes late, depending on when the last statistics rollup occurred. In addition, statistics are not collected across all datastores at one time. They are collected asynchronously.

The **Space Utilization by File Type** chart on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Space Utilization** from the **View** drop-down menu.

Datastore Counters

Table 352: Data Counters

File Type	Description
Virtual Disks	Amount of disk space used by virtual disk files. Virtual disk files store the contents of the virtual machine's hard disk drive. It includes information that you write to a virtual machine's hard disk, such as the operating system, program files, and data files. The files have the extension `.vmdk` and appear as a physical disk drive to a guest operating system. **Note:** Delta disks, which also have an extension `.vmdk`, are not included in this file type. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Swap Files	Amount of disk space used by swap files. Swap files back up the virtual machine's physical memory. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Snapshots	Amount of disk space used by virtual machine snapshot files. Snapshot files store information about virtual machine snapshots. They include snapshot state files and delta disk files. A snapshot state file stores the running state of the virtual machine at the time of the snapshot. It has the extension `.vmsn`. A delta disk file stores the updates made by the virtual machine to the virtual disks after a snapshot is taken. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)

File Type	Description
Other VM Files	Amount of disk space used by all other virtual machine files, such as configuration files and log files. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Other	Amount of disk space used by all other non-virtual machine files, such as documentation files and backup files.
Free Space	Amount of disk space not currently in use.
Total Space	Amount of disk space available to the datastore. It defines the datastore capacity. The chart displays the information for datastores but not for data centers. total space = virtual disk space + swap file space + snapshot space + other VM file space + other space + free space

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space used by Datastore in GB

The Space used by Datastore in GB chart displays the 10 datastores in the data center with the most used disk space.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 353: Data Counters

Chart Label	Description
<datastore>	Amount of used storage space on the 10 datastores with the most used space. • Counter: used • Stats Type: Absolute • Unit: GigaBytes (GB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Utilization by Virtual Machines

The Space Utilization by Virtual Machines chart displays the amount of space used by the five virtual machines with the most space used on the datastore or the datastores in the cluster.

> **Note:**
> This chart does not show historical statistics. It only shows the most recently available data, which may be up to 30 minutes late, depending on when the last statistics rollup occurred. In addition, statistics are not collected across all datastores at one time. They are collected asynchronously.

The **Space Utilization by File Virtual Machines** chart on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Space Utilization** from the **View** drop-down menu.

Table 354: Data Counters

Chart Label	Description
virtual_machine	Amount of datastore space used by the five virtual machines with the most used datastore space. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Allocated by Datastore in GB

The Space allocated by Datastore in GB displays the top 10 datastores, virtual machines in the datastore-cluster with most provisioned space.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 355: Data Counters

Chart Label	Description
<datastore>	Amount of provisioned storage space on the top 10 datastores with the most provisioned space. • Counter: provisioned • Stats Type: Absolute • Unit: KiloBytes (KB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Capacity by Datastore in GB

The Space capacity by Datastore in GB displays the top 10 configured size of the datastores in the datastore cluster.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 356: Data Counters

Chart Label	Description
<datastore>	Configured size of the datastores in the datastore cluster. • Counter: capacity • Stats Type: Absolute • Unit: KiloBytes (KB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Storage I/O Control Normalized Latency

This chart displays the normalized latency in microseconds on the datastore. Storage I/O Control monitors latency to detect congestion on the datastore. This metric computes a weighted response time across all hosts and VMs accessing the datastore. I/O count is used as the weight for the response time. It captures the device level latency and does not include any queuing inside the hypervisor storage stack or inside the VM. It is adjusted for the I/O size. High latencies that are the result of large I/Os are discounted so as not to make the datastore seem slower than it really is. Data for all virtual machines is combined. This chart displays zero values when Storage I/O Control is deactivated.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 357: Data Counters

Chart Label	Description
Storage I/O Control Normalized Latency	Storage I/O Control monitors latency to detect congestion on the datastore. • Counter: sizeNormalizedDatastoreLatency • Stats Type: Absolute • Unit: Microseconds • Rollup Type: Average • Collection Level: 3

Storage I/O Control Aggregate IOPs

This chart displays the number of I/O operations per one second on the datastore, aggregated across all hosts, and virtual machines accessing this datastore. The chart displays zero values when Storage I/O Control is deactivated.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 358: Data Counters

Chart Label	Description
Storage I/O Control Aggregate IOPs	Number of I/O operations per second on the datastore, aggregated across all hosts, and virtual machines accessing the datastore. • Counter: datastoreIops • Stats Type: Absolute • Unit: Number • Rollup Type: Average • Collection Level: 3

Storage I/O Control Activity

This chart displays the percentage of time during which Storage I/O Control actively controlled latency of the datastore.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 359: Data Counters

Chart Label	Description
Storage I/O Control Activity	This is the percentage of time during which the Storage I/O Control actively controlled the I/O latency for the datastore. • Counter: siocActiveTimePercentage • Stats Type: Absolute • Unit: Percent • Rollup Type: Average • Collection Level: 3

Average Device Latency per Host

This chart displays the average amount of latency on a host device. The chart displays the 10 hosts with the highest device latency.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Hosts** from the **View** drop-down menu.

Table 360: Data Counters

Chart Label	Description
Average Device Latency per Host	Measures the amount of time, in milliseconds, to complete a SCSI command issued from the physical device. • Counter: deviceLatency • Stats Type: Absolute • Unit: Milliseconds (ms) • Rollup Type: Average • Collection Level: 3

Maximum Queue Depth per Host

This chart displays maximum queue depth that hosts are currently maintaining for the datastore. When Storage I/O is enabled, queue depth can change over time when congestion is detected at the array.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Hosts** from the **View** drop-down menu.

Table 361: Data Counters

Chart Label	Description
Max Queue Depth per Host	Maximum queue depth. Queue depth is the number of commands the SCSI driver queues to the HBA. • Counter: maxQueueDepth • Stats Type: Absolute • Unit: Number • Rollup Type: Average • Collection Level: 3

Read IOPs per Host

This chart displays the per-host disk read rates for a datastore. The chart displays information about the ten hosts with the highest values.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Hosts** from the **View** drop-down menu.

Table 362: Data Counters

Chart Label	Description
Read IOPs per Host	Number of disk read commands completed on each disk on the host, per second. Read rate = blocks read per second × block size • Counter: numberReadAveraged • Stats Type: Rate • Unit: Number • Rollup Type: Average • Collection Level: 3

Write IOPs Per Host

This chart displays the per-host disk write rates for a datastore. The chart displays information about the 10 hosts with the highest values.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Hosts** from the **View** drop-down menu.

Table 363: Data Counters

Chart Label	Description
Write IOPs per Host	Number of disk write commands completed on each disk on the host, per second. Write rate = blocks written per second × block size • Counter: numberWriteAveraged • Stats Type: Rate • Unit: Number • Rollup Type: Average • Collection Level: 3

Average Read Latency per Virtual Machine Disk

This chart displays the top ten virtual machine disks with the highest average read latency in milliseconds. Data is not displayed when the virtual machine is powered off.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 364: Data Counters

Chart Label	Description
Average Read Latency per Virtual Machine Disk	Latency measures the time used to process a SCSI command issued by the guest OS to the virtual machine. The kernel latency is the time VMkernel takes to process an I/O request. The device latency is the time it takes the hardware to handle the request. Total latency = kernelLatency + deviceLatency. • Counter: totalReadLatency • Stats Type: Absolute • Unit: Milliseconds (ms) • Rollup Type: Average • Collection Level: 3

Average Write Latency per Virtual Machine Disk

This chart displays the top ten virtual machine disks with the highest average write latency in milliseconds. Data is not displayed when the virtual machine is powered off.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 365: Data Counters

Chart Label	Description
Average Write Latency per Virtual Machine Disk	Latency measures the time used to process a SCSI command issued by the guest OS to the virtual machine. The kernel latency is the time VMkernel takes to process an I/O request. The device latency is the time it takes the hardware to handle the request. Total latency = kernelLatency + deviceLatency. • Counter: totalWriteLatency • Stats Type: Absolute • Unit: Milliseconds (ms) • Rollup Type: Average • Collection Level: 3

Read IOPs per Virtual Machine Disk

This chart displays the top ten virtual machines with the highest number of read operations. Data is not displayed when the virtual machine is powered off.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 366: Data Counters

Chart Label	Description
Read IOPs per Virtual Machine Disk	Number of disk read commands completed on each virtual machine disk, per second. Read rate = blocks read per second × block size • Counter: numberReadAveraged • Stats Type: Rate • Unit: Number • Rollup Type: Average • Collection Level: 3

Write IOPs Per Virtual Machine Disk

This chart displays the 10 virtual machines with the highest number of write operations. Data is not displayed when the virtual machine is powered off.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 367: Data Counters

Chart Label	Description
Write IOPs per Virtual Machine Disk	Number of disk write commands completed on each virtual machine disk on the host. Write rate = blocks read per second × block size • Counter: numberWriteAveraged • Stats Type: Rate • Unit: Number • Rollup Type: Average • Collection Level: 3

Virtual Machine Observed Latency per Datastore

This chart displays the average datastore latency as observed by the virtual machines.

This chart is located on the **Monitor** tab for a datastore or datastore cluster when you select **Performance** > **Overview**.

Table 368: Data Counters

Chart Label	Description
VM observed latency report per Datastore	This is the average datastore latency as observed by the virtual machines in the datastore cluster. • Counter: datastoreVMObservedLatency • Stats Type: Absolute • Unit: Microseconds • Rollup Type: Latest • Collection Level: 3

Hosts

The hosts charts contain information about CPU, disk, memory, network, GPU, and storage usage for hosts. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for vCenter.

CPU usage in %

The CPU (%) chart displays CPU usage for the host.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **CPU usage in %** from the **View** drop-down menu.

Table 369: Data Counters

Chart Label	Description
Usage	Actively used CPU, as a percentage of the total available CPU, for each physical CPU on the host. Active CPU is approximately equal to the ratio of the used CPU to the available CPU. Available CPU = # of physical CPUs × clock rate. 100% represents all CPUs on the host. For example, if a four-CPU host is running a virtual machine with two CPUs, and the usage is 50%, the host is using two CPUs completely. • Counter: usage • Stats Type: Rate • Unit: Percentage (%) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of the host resources. However, if the value is constantly high, the host is probably lacking the CPU required to meet the demand. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the host.

If performance is impacted, consider taking the following actions.

Table 370: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.
3	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
4	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
5	Upgrade the physical CPUs or cores on the host if necessary.
6	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

CPU Usage in MHz

The CPU (MHz) chart displays CPU usage for the host.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter.

Table 371: Data Counters

Chart Label	Description
Usage	The sum, in megahertz, of the actively used CPU of all powered on virtual machines on a host. The maximum possible value is the frequency of the processors multiplied by the number of processors. For example, if you have a host with four 2GHz CPUs running a virtual machine that is using 4000MHz, the host is using two CPUs completely. 4000 ÷ (4 × 2000) = 0.50 • Counter: usagemhz • Stats Type: Rate • Unit: MegaHertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of the host resources. However, if the value is constantly high, the host is probably lacking the CPU required to meet the demand. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the host.

If performance is impacted, consider taking the following actions.

Table 372: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.
3	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
4	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
5	Upgrade the physical CPUs or cores on the host if necessary.
6	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

CPU (DPU)

The CPU (DPU) performance chart displays CPU usage of DPU.

This chart is available in the **DPU** pane of the **View** dropdown menu for a host instance at **Monitor** > **Performance** > **Overview** tab.

Table 373: Data Counters

Chart Label	Description
Usage (%)	Get the CPU usage of a DPU. • Counter: usage • Stats Type: Rate • Unit: % • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Usage (Mhz)	Get the sum of actively used CPUs of all powered on virtual machines of a DPU. • Counter: usage • Stats Type: Rate • Unit: Mhz • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

CPU Usage

The CPU Usage chart displays CPU usage of the 10 virtual machines on the host with the most CPU usage.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 374: Counters

Name	Description
virtual_machine	Amount of CPU actively being used by each virtual machine on the host. 100% represents all CPUs. For example, if a virtual machine has one virtual CPU that is running on a host with four CPUs and the CPU usage is 100%, the virtual machine is using one CPU resource. virtual CPU usage = usagemhz ÷ (number of virtual CPUs × core frequency) **Note:** The host's view of the CPU usage, not the guest operating system view. • Counter: usage • Stats Type: Rate • Unit: Percentage (%). Precision is to 1/100%. A value between 0 and 100. • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage or CPU ready indicates that you are making the best use of the virtual machine resources. However, if the CPU usage value for a virtual machine is above 90% and the CPU ready value is above 20%, performance is being impacted.

If performance is impacted, consider taking the following actions.

Table 375: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.

#	Resolution
3	Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked line chart on the host's **Virtual Machine** view shows the CPU usage for virtual machines on the host.
4	Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.
5	Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time doesn't decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
6	Increase the amount of memory allocated to the virtual machine. This decreases disk and or network activity for applications that cache. This might lower disk I/O and reduce the need for the host to virtualize the hardware. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
7	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
8	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
9	Upgrade the physical CPUs or cores on the host if necessary.
10	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

Disk (KBps)

The Disk (KBps) chart displays disk I/O of the host.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter.

Table 376: Data Counters

Chart Label	Description
Usage	Average data I/O rate across all LUNs on the host. • Counter: usage • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.

- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 377: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk Rate (KBps)

The Disk Rate chart displays disk read and write rates for LUNs on a host, including average rates.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu.

Table 378: Data Counters

Chart Label	Description
Read	Number of disk read commands completed on each disk on the host, per second. The aggregate number of all disk read commands is also displayed in the chart. Read rate = blocksRead per second × blockSize • Counter: read • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average • Collection Level: 3
Write	Number of disk write commands completed on each disk on the host, per second. The aggregate number of all disk write commands is also displayed in the chart. Write rate = blocksWritten per second × blockSize • Counter: write • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average • Collection Level: 3

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 379: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.

#	Resolution
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk Requests (Number)

The Disk Requests chart displays disk usage for the host.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu.

Table 380: Data Counters

Chart Label	Description
Read Requests	Number of disk read commands completed on each LUN on the host. The aggregate number of all disk read commands is also displayed in the chart. • Counter: numberRead • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3
Write Requests	Number of disk write commands completed on each LUN on the host. The aggregate number of all disk write commands is also displayed in the chart. • Counter: numberWrite • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 381: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk (Number)

The Disk (Number) chart displays maximum queue depth for the top ten LUNs on a host.

This chart is located on the **Monitor** tab for the host.

Table 382: Data Counters

Chart Label	Description
Maximum Queue Depth	Maximum queue depth. Queue depth is the number of commands the SCSI driver queues to the HBA. • Counter: maxQueueDepth • Stats Type: Absolute • Unit: Number • Rollup Type: Average • Collection Level: 1

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 383: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.

#	Resolution
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk (ms)

The Disk (ms) chart displays the amount of time taken to process commands on a host.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Oveview** from the **View** drop-down menu.

Table 384: Data Counters

Chart Label	Description
Highest Disk Latency	Highest latency value of all disks used by the host. Latency measures the time used to process a SCSI command issued by the guest OS to the virtual machine. The kernel latency is the time VMkernel takes to process an I/O request. The device latency is the time it takes the hardware to handle the request. Total latency = kernelLatency + deviceLatency. • Counter: maxTotalLatency • Stats Type: Absolute • Unit: Milliseconds (ms) • Rollup Type: Latest (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 385: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk (KBps)

The Disk (KBps) chart displays disk usage for the 10 virtual machines on the host with the most disk usage.

This chart is located on the **Monitor** tab for a datastore or datastore cluster. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 386: Data Counters

Chart Label	Description
virtual_machine	Sum of the data read from the virtual machine. • Counter: usage • Stats Type: Rate • Unit: KiloBytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 387: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

GPU Compute

The GPU Compute chart displays GPU usage for the host.

This chart is located on the **Monitor** tab for the host.

Note:
The GPU charts are visible only if the host supports GPUs.

Table 388: Data Counters

Chart Labels	Description
Usage	The utilization of a GPU in percentages. • Counter: Utilization • Stats Type: Absolute • Unit: Percent (%) • Rollup Type: Average • Collection Level: 4

Chart Analysis

GPU Compute Usage (%) is vendor and architecture specific. High utilization can suggest that the GPU is used effectively, or it can indicate potential bottlenecks during peak workloads.

If a virtual machine is experiencing a Quality of Service (QoS) issue, please refer to the GPU vendor QoS documentation. If you use NVIDIA virtual GPU (vGPU) technology, see the NVIDIA Virtual GPU Software for VMware vSphere documentation for more information.

GPU Memory

The GPU Memory chart displays the amount of GPU memory used for the host in kilobytes and in percentage of the total available.

This chart is located on the **Monitor** tab for the host.

Note:
The GPUs view is visible only if the host supports GPUs.

Table 389: Data Counters

Chart Labels	Description
Memory Usage in KB for *device_id*	The amount of GPU memory used in kilobytes. • Counter: Memory used • Stats Type: Absolute • Unit: Kilobytes (KB) • Rollup Type: Average • Collection Level: 4
Memory Usage for *device_id*	The amount of GPU memory used in percentage of the total available. • Counter: Memory used • Stats Type: Absolute • Unit: Percent (%) • Rollup Type: Average • Collection Level: 4

Chart Analysis

GPU Memory Usage (in Kilobytes) might not represent the active workload in the virtual machine, but the memory reserved for a virtual GPU.

Monitoring memory usage is essential to prevent out-of-memory errors, which can lead to application crashes or reduced performance.

If a virtual machine is experiencing a Quality of Service (QoS) issue, refer to the GPU vendor QoS documentation. If you use NVIDIA virtual GPU (vGPU) technology, see the NVIDIA Virtual GPU Software for VMware vSphere documentation for more information.

GPU Compute Usage

The GPU Compute Usage chart displays GPU compute usage for the top 10 graphics devices on the host with the most GPU compute usage.

This chart is located on the **Monitor** tab for the host.

> **Note:**
> The GPUs view is visible only if the host supports GPUs.

Table 390: Data Counters

Chart Labels	Description
Usage for *device_id*	The utilization of a GPU in percentages. • Counter: Utilization • Stats Type: Absolute • Unit: Percent (%) • Rollup Type: Average • Collection Level: 4

Chart Analysis

GPU Compute Usage (%) is vendor and architecture specific. High utilization may suggest that the GPU is used effectively, or it may indicate potential bottlenecks during peak workloads.

If a virtual machine is experiencing a Quality of Service (QoS) issue, please refer to the GPU vendor QoS documentation. If you use NVIDIA virtual GPU (vGPU) technology, see the NVIDIA Virtual GPU Software for VMware vSphere documentation for more information.

GPU Memory Allocation

The GPU Memory Allocation chart displays GPU memory allocation for the top 10 graphics devices on the host with the most GPU memory usage.

This chart is located on the **Monitor** tab for the host.

> **Note:**
> The GPUs view is visible only if the host supports GPUs.

Table 391: Data Counters

Chart Labels	Description
Usage for *device_id*	The amount of GPU memory used in percentages of the total available. • Counter: Memory usage • Stats Type: Absolute • Unit: Percent (%) • Rollup Type: Average • Collection Level: 4

Chart Analysis

GPU memory usage tracks how much of the GPU onboard memory (VRAM) is in use.

GPU details measurements can show usage trends related to virtual machine power on failures. Device and driver specific placement constraints of vGPU on physical GPU devices, including fragmentation of Device Memory, exhaustion of Device Memory, or exhaustion of Assignable Processor Engines could contribute to power on failures.

If GPU memory usage is almost at capacity and a virtual machine power on failure occurs with a Virtual GPU that would exceed capacity, increase the number of compatible GPU hosts, and migrate one or more virtual machines to the new host to free up GPU Memory resources for the new VM. For further guidance see Knowledge Base article 312021.

If a virtual machine is experiencing a Quality of Service (QoS) issue, please refer to the GPU vendor QoS documentation. If you use NVIDIA virtual GPU (vGPU) technology, see the NVIDIA Virtual GPU Software for VMware vSphere documentation for more information.

GPU Temperature

The GPU Temperature chart displays GPU temperature in degrees Celsius for the top 10 graphics devices on the host with the highest GPU temperature.

This chart is located on the **Monitor** tab for the host.

> **Note:**
> The GPUs view is visible only if the host supports GPUs.

Table 392: Data Counters

Chart Labels	Description
GPU temperature for *device_id*	The temperature of a GPU in degrees Celsius. • Counter: Temperature • Stats Type: Absolute • Unit: Celsius (°C) • Rollup Type: Average • Collection Level: 4

Chart Analysis

Monitoring GPU temperature helps prevent overheating, which can result in throttling and reduced performance. Overheating can also lead to hardware damage.

If a virtual machine is experiencing a Quality of Service (QoS) issue, please refer to the GPU vendor QoS documentation. If you use NVIDIA virtual GPU (vGPU) technology, see the NVIDIA Virtual GPU Software for VMware vSphere documentation for more information.

Memory (%)

The Memory (%) chart displays host memory usage.

Chart Analysis

To ensure best performance, the host memory must be large enough to accommodate the active memory of the virtual machines. The active memory can be smaller than the virtual machine memory size. It allows you to over-provision memory, but still ensures that the virtual machine active memory is smaller than the host memory.

Transient high-usage values usually do not cause performance degradation. For example, memory usage can be high when several virtual machines are started at the same time or when a spike occurs in virtual machine workload. However, a consistently high memory usage value (94% or greater) indicates that the host is probably lacking the memory required to meet the demand. If the active memory size is the same as the granted memory size, the demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the memory usage value is high, and the host has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot handle the demand for memory. It leads to memory reclamation, which might degrade performance.

If the host has enough free memory, check the resource shares, reservation, and limit settings of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machines.

If the host has little free memory available, or if you notice a degradation in performance, consider taking the following actions.

Table 393: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (Balloon)

The Memory (Balloon) chart displays balloon memory on a host.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Advanced**, open the **Chart Options** dialog box and select the Ballooned memory counter.

Table 394: Data Counters

Chart Label	Description
Balloon	Sum of the guest physical memory reclaimed by the balloon driver for all powered on virtual machines on the host. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

To ensure best performance, the host memory must be large enough to accommodate the active memory of the virtual machines. The active memory can be smaller than the virtual machine memory size. It allows you to over-provision memory, but still ensures that the virtual machine active memory is smaller than the host memory.

Transient high-usage values usually do not cause performance degradation. For example, memory usage can be high when several virtual machines are started at the same time or when a spike occurs in virtual machine workload. However, a consistently high memory usage value (94% or greater) indicates that the host is probably lacking the memory required to meet the demand. If the active memory size is the same as the granted memory size, the demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the memory usage value is high, and the host has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot handle the demand for memory. It leads to memory reclamation, which might degrade performance.

If the host has enough free memory, check the resource shares, reservation, and limit settings of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machines.

If the host has little free memory available, or if you notice a degradation in performance, consider taking the following actions.

Table 395: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (DPU)

The Memory (DPU) performace chart displays memory used on DPU.

This chart is available in the **DPU** pane of the **View** dropdown menu of vSphere Client for a host instance at **Monitor** > **Performance** > **Overview** tab.

Table 396: Data Counters

Chart Label	Description
consumed (%)	Amount of memory consumed by a DPU. • Counter: consumed • Stats Type: Rate • Unit: % • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Memory Rate (MBps)

The Memory (MBps) chart displays the swap in and swap out rates for a host.

This chart is located on the **Monitor** tab for a cluster. . Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 397: Data Counters

Chart Label	Description
Swap in rate	Average rate at which memory is swapped in from the host swap file. • Counter: swapinRate • Stats Type: Rate • Unit: MegaBytes per second (MBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Swap out rate	Average rate at which memory is swapped out to the host swap file. • Counter: swapoutRate • Stats Type: Rate • Unit: MegaBytes per second (MBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Tier in rate	Average rate at which memory is brought in from the NVMe tier.
Tier out rate	Average rate at which memory is moved to the NVMe tier.

Chart Analysis

Host memory must be large enough to accommodate virtual machine workload. Transient high-usage values usually do not cause performance degradation. For example, memory usage can be high when several virtual machines are started at the same time or when there is a spike in virtual machine workload.

However, a consistently high memory usage value (94% or greater) indicates the host does not have the memory resources required to meet the demand. If the memory balloon and swap values are not high, performance is probably not affected. If the memory usage value is high, and the host has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host requires more memory resources.

If the host is not lacking memory resources, check the resource shares, reservation, and limit settings of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machines.

If the host is lacking memory resources or you notice a degredation in performance, consider taking the following actions.

Table 398: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of a virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (MB)

The Memory (MB) chart displays memory data counters for hosts.

This chart is located on the **Monitor** tab for a host. . Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

> **Note:**
> Guest physical memory refers to the virtual hardware memory presented to a virtual machine for its guest operating system. Machine memory is the actual physical RAM in the host.

Not all counters are collected at collection level 1.

Table 399: Data Counters

Chart Label	Description
Active	Sum of the active guest physical memory of all powered on virtual machines on the host, plus memory used by basic VMKernel applications. Active memory is estimated by the VMkernel and is based on the current workload of the host. • Counter: active • Stats Type: Absolute • Unit: kilobytes (kb) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Balloon	Sum of the guest physical memory reclaimed by the balloon driver for all powered on virtual machines on the host. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Label	Description
Balloon Target	Sum of the balloon target memory of all powered on virtual machines on the host. If the balloon target value is greater than the balloon value, the VMkernel inflates the balloon, causing more virtual machine memory to be reclaimed. If the balloon target value is less than the balloon value, the VMkernel deflates the balloon, which allows the virtual machine to consume additional memory if needed. Virtual machines initiate memory reallocation. Therefore, it is possible to have a balloon target value of 0 and a balloon value greater than 0. • Counter: vmmemctltarget • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Consumed	Amount of machine memory used on the host. Consumed memory includes virtual machine memory, service console memory, and VMkernel memory. consumed memory = total host memory - free host memory • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Granted	Sum of the guest physical memory granted for all powered on virtual machines. Granted memory is mapped to the host's machine memory. Granted memory for a host includes the shared memory of each virtual machine on the host. • Counter: granted • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Shared Common	Amount of machine memory shared by all powered on virtual machines. Shared common memory consists of the entire pool of memory from which sharing is possible, including the amount of physical RAM required by the guest memory. memory shared - memory shared common = amount of memory saved on the host from sharing • Counter: sharedcommon • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Swap Used	Sum of the memory swapped by all powered on virtual machines on the host. • Counter: swapused • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Tier memory consumed	Amount of machine memory consumed by the host per tier. • Counter: tier.consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Latest • Collection Level: 2

Chart Label	Description
Total	Aggregate total memory available to the cluster. • Counter: totalmb • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

To ensure best performance, the host memory must be large enough to accommodate the active memory of the virtual machines. The active memory can be smaller than the virtual machine memory size. It allows you to over-provision memory, but still ensures that the virtual machine active memory is smaller than the host memory.

Transient high-usage values usually do not cause performance degradation. For example, memory usage can be high when several virtual machines are started at the same time or when a spike occurs in virtual machine workload. However, a consistently high memory usage value (94% or greater) indicates that the host is probably lacking the memory required to meet the demand. If the active memory size is the same as the granted memory size, the demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the memory usage value is high, and the host has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot handle the demand for memory. It leads to memory reclamation, which might degrade performance.

If the host has enough free memory, check the resource shares, reservation, and limit settings of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machines.

If the host has little free memory available, or if you notice a degradation in performance, consider taking the following actions.

Table 400: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory Usage

The Memory Usage chart displays memory usage for the 10 virtual machines on the host with the most memory usage.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Virtual Machine Counters

Note:
Guest physical memory refers to the virtual hardware memory presented to a virtual machine for its guest operating system.

Table 401: Data Counters

Chart Label	Description
Usage	Amount of guest physical memory currently in use on the virtual machine. • Counter: usage • Stats Type: Absolute • Unit: Percentage (%) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 402: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory Bandwidth (MBps)

The **Memory Bandwidth** chart displays DRAM and NVMe bandwidth graphs for the host.

To access this chart, navigate the ESX host in the inventory.

1. Click the **Monitor** tab.
2. Under **Performance**, click **Overview**.
3. From the **View** drop-down menu, select **Memory**.

The **Memory** pane displays information about memory bandwidth in addition to the memory utilization and memory reclamation information. Memory miss rate is displayed, as well, but only in Memory mode.

Note:

PMem bandwidth is only available on supported hosts configured in memory mode.

If the host does not have vMMR or is not in memory mode, the **Memory** option is not listed in the **View** drop-down menu.

Note:

The NVMe bandwidth graph is only available on hosts with configured Memory Tiering over NVMe (Memory Tiering). For more information about Memory Tiering, see Memory Tiering over NVMe.

Table 403: Data Counters

Chart Label	Description
Estimated DRAM Bandwidth	Total read and write bandwidth of the DRAM memory type. • Counter: bandwidth.total • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 1 (4)
Estimated PMem Bandwidth	Total read and write bandwidth of the PMem memory type. • Counter: bandwidth.total • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 1 (4)
Estimated NVMe Bandwidth	Total read and write bandwidth of the NVMe memory type. • Counter: bandwidth.total • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 1 (4)
DRAM (Tier 0) read bandwith	• Counter: bandwidth.read • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: latest • Collection Level: 2
DRAM (Tier 0) write bandwith	• Counter: bandwidth.write • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: latest • Collection Level: 2

Chart Label	Description
NVMe (Tier 1) read bandwidth	• Counter: bandwidth.read • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: latest • Collection Level: 2
NVMe (Tier 1) write bandwidth	• Counter: bandwidth.write • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: latest • Collection Level: 2

Memory Latency (ns)

The **Memory Latency** chart displays DRAM and NVMe latency graphs for the host.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Memory** from the **View** drop-down menu.

Table 404: Data Counters

Chart Label	Description
DRAM read latency	• Counter: latency.read • Stats Type: Absolute • Unit: nanosecond • Rollup Type: latest • Collection Level: 3
DRAM write latency	• Counter: latency.write • Stats Type: Absolute • Unit: nanosecon • Rollup Type: latest • Collection Level: 3
NVMe read latency	• Counter: latency.read • Stats Type: Absolute • Unit: nanosecond • Rollup Type: latest • Collection Level: 3
NVMe write latency	• Counter: latency.write • Stats Type: Absolute • Unit: nanosecon • Rollup Type: latest • Collection Level: 3

Memory Miss Rate (%)

The Memory Miss Rate performance chart displays DRAM Miss Rate graph for the host.

This chart is available in the **Memory** pane of the **View** dropdown menu of vSphere Client for a host instance at **Performance** > **Overview** tab. The Memory pane gives information about Memory Miss Rate (DRAM) in addition to the memory consumption, memory reclamation, and memory bandwidth information.

> **Note:**
> Miss rate is only supported on supported hosts in Memory Mode. If the host is not supported by vMMR, then the **Memory** option is not listed in the **View** dropdown menu.

Table 405: Data Counters

Chart Label	Description
Estimated DRAM Miss Rate	Get the current miss rate of DRAM memory type. • Counter: missrate • Stats Type: Absolute • Unit: % • Rollup Type: Latest • Collection Level: 2 (4)

Network (Mbps)

The Network (Mbps) chart displays network usage for the host.

This chart is located on the **Monitor** tab for a cluster. Select **Performance** > **Advanced** and select **Network** from the **View** drop-down menu.

Table 406: Host Counters

Chart Label	Description
Usage	Average rate at which data is transmitted and received across all NIC instances connected to the host. • Counter: usage • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 407: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Network Rate (Mbps)

The Network Rate chart displays network bandwidth on a host.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Network Adapters** from the **View** drop-down menu.

Table 408: Data Counters

Chart Label	Description
Data Receive Rate	Rate at which data is received across the top ten physical NIC instances on the host. This represents the bandwidth of the network. The chart also displays the aggregated data receive rate of all physical NICs. • Counter: received • Stats Type: Rate • Unit: Kilobytes per second (Kbps) • Rollup Type: Average • Collection Level: 3 (4)
Data Transmit Rate	Rate at which data is transmitted across the top ten physical NIC instances on the host. This represents the bandwidth of the network. The chart also displays the aggregated data transmit rate of all physical NICs. • Counter: transmitted • Stats Type: Rate • Unit: Kilobytes per second (Kbps) • Rollup Type: Average • Collection Level: 3 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 409: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Network Packets (Number)

The Network Packets chart displays the network bandwidth on a host.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Network Adapters** from the **View** drop-down menu.

Table 410: Data Counters

Chart Label	Description
Packets Received	Number of network packets received across the top ten physical NIC instances on the host. The chart also displays the aggregated value for all NICs. • Counter: packetRx • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3
Packets Transmitted	Number of network packets transmitted across the top ten physical NIC instances on the host. The chart also displays the aggregated value for all NICs. • Counter: packetTx • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 411: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.

#	Resolution
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Network (Mbps)

The Network (Mbps) chart displays network usage for the 10 virtual machines on the host with the most network usage.

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Overview** and select **Virtual Machines** from the **View** drop-down menu.

Table 412: Data Counters

Chart Label	Description
<virtual machine>	Sum of the data transmitted and received across all virtual NIC instances connected to the virtual machine. • Counter: usage • Stats Type: Rate • Unit: Megabits per second (Mbps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 413: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Resource Pools

The resource pool charts contain information about CPU and memory usage for resource pools. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for vCenter.

CPU Usage in MHz

The CPU (MHz) chart displays CPU usage in the resource pool or vApp.

This chart is located on the **Monitor** tab for a resource pool. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Counters

Table 414: Data Counters

Chart Label	Description
Usage	CPU usage is the sum of the average CPU usage values of the virtual machines in the resource pool or vApp. CPU usage = number of cores * CPU frequency • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of the resources available. However, if the value is constantly high, the CPU demanded is likely greater than the CPU capacity available. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines in the resource pool. Generally, if the CPU usage

value for a virtual machine is above 90% and the CPU ready value for a virtual machine is above 20%, performance is impacted.

If performance is impacted, consider taking the following actions.

Table 415: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Deploy single-threaded applications on uniprocessor virtual machines instead of SMP virtual machines.
3	Migrate one or more virtual machines to a new host.
4	Upgrade the physical CPUs or cores on each host if necessary.
5	Enable CPU-saving features, such as TCP Segmentation Offload.
6	Replace software I/O with dedicated hardware, such as iSCSI HBAs or TCP Segmentation Offload NICs.

CPU Usage

The CPU Usage chart displays CPU usage of virtual machines in the resource pool or vApp. The chart displays the top 10 virtual machines with the highest CPU usage.

This chart is located on the **Monitor** tab for a resource pool. Select **Performance** > **Overview** and select **Resource Pools and Virtual Machines** from the **View** drop-down menu.

Table 416: Data Counters

Chart Label	Description
virtual_machine	Amount of CPU actively used by virtual machines. • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage or CPU ready indicates that you are making the best use of the virtual machine resources. However, if the CPU usage value for a virtual machine is above 90% and the CPU ready value is above 20%, performance is being impacted.

If performance is impacted, consider taking the following actions.

Table 417: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.
3	Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked line chart on the host's **Virtual Machine** view shows the CPU usage for virtual machines on the host.
4	Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.

#	Resolution
5	Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time doesn't decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
6	Increase the amount of memory allocated to the virtual machine. This decreases disk and or network activity for applications that cache. This might lower disk I/O and reduce the need for the host to virtualize the hardware. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
7	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
8	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
9	Upgrade the physical CPUs or cores on the host if necessary.
10	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

Memory (MB)

The Memory (MB) chart displays memory usage in the resource pool or vApp.

This chart is located on the **Monitor** tab for a resource pool. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 418: Data Counters

Chart Label	Description
resource_pool or *vApp*	Sum of the active memory used by all virtual machines in the resource pool or vApp. Active memory is determined by the VMkernel and includes overhead memory. memory usage = active memory / configured virtual machine memory size • Counter: used • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average • Collection Level: 1

Chart Analysis

Memory usage is not necessarily an indicator of performance problems. Memory can be high if a host is swapping or ballooning, which can result in virtual machine guest swapping. In such cases, check for other problems, such as CPU over-commitment or storage latencies.

If you have constantly high memory usage in a cluster, resource pool, or vApp, consider taking the following actions.

Table 419: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Verify that the balloon driver is enabled. The balloon driver is installed with VMware Tools and is critical to performance. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.

#	Resolution
3	If the balloon value is high, check the resource shares, reservations, and limits for the virtual machines and resource pools on the hosts. Verify that the host's settings are adequate and not lower than those set for the virtual machine. If free memory is available on the hosts and the virtual machines are experiencing high swap or balloon memory, the virtual machine (or resource pool, if it belongs to one) has reached its resource limit. Check the maximum resource limit set on that host.
4	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
5	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. It might help avoid hot spots in the cluster.
6	Add more physical memory to one or more hosts.

Memory Consumed

The Memory Consumed chart displays the memory performance of all virtual machines in the resource pool or vApp.

This chart is located on the **Monitor** tab for a resource pool. Select **Performance** > **Overview** and select **Resource Pools and Virtual Machines** from the **View** drop-down menu.

For resource pools and virtual machines in a resource pool or vApp, this chart is located in the **Resource Pools & Virtual Machines** view of the resource pool or vApp **Performance** tab.

Table 420: Data Counters

Chart Label	Description
virtual_machine	Amount of host memory used by the virtual machine for its guest operating system's physical memory. Memory overhead is not included in consumed memory. consumed memory = memory granted - memory saved from page sharing For example, if a virtual machine has 100 MB of memory that is shared equally with three other virtual machines, its portion of the shared memory is 25 MB (100 MB ÷ 4 VMs). This amount is counted in the memory consumed data counter. • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 421: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (MB)

The Memory (MB) chart displays memory data counters for resource pools or vApps.

Description

This chart is located on the **Monitor** tab for a host. Select **Performance** > **Advanced**, open the **Chart Options** dialog box, and select a counter to view.

> **Note:**
>
> These data counter definitions are for virtual machines. At the resource pool level, the values are collected and totaled. The counter values in the chart represent the aggregate amounts of the virtual machine data. The counters that appear in the chart depend on the collection level set for your vCenter.

Table 422: Data Counters

Chart Label	Description
Active	Sum of the active guest physical memory of all powered on virtual machines in the resource pool. • Counter: active • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Balloon	Sum of the guest physical memory reclaimed by the balloon driver for all powered on virtual machines in the resource pool. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Label	Description
Consumed	Amount of the physical memory consumed by the virtual machine for the guest memory. Consumed memory does not include the overhead memory. It includes the shared memory and memory that might be reserved, but not actually used. consumed memory = memory granted – memory saved due to memory sharing • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Granted	Sum of the guest physical memory granted for all powered on virtual machines. Granted memory is mapped to the host's machine memory. • Counter: granted • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Shared	Amount of the guest physical memory shared with other virtual machines in the resource pool.
Swapped	Sum of the memory swapped by all powered on virtual machines in the resource pool. • Counter: swapused • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 423: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.

#	Resolution
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

vApps

The vApp charts contain information about CPU and memory usage for vApps. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for vCenter.

CPU (MHz)

The CPU (MHz) chart displays CPU usage in the vApp or resource pool.

This chart is located on the **Monitor** tab for a vApp. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Counters

Table 424: Data Counters

Chart Label	Description
Usage	CPU usage is the sum of the average CPU usage values of the virtual machines in the resource pool or vApp. CPU usage = number of cores * CPU frequency • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage indicates that you are making the best use of the resources available. However, if the value is constantly high, the CPU demanded is likely greater than the CPU capacity available. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines in the resource pool. Generally, if the CPU usage value for a virtual machine is above 90% and the CPU ready value for a virtual machine is above 20%, performance is impacted.

If performance is impacted, consider taking the following actions.

Table 425: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Deploy single-threaded applications on uniprocessor virtual machines instead of SMP virtual machines.
3	Migrate one or more virtual machines to a new host.
4	Upgrade the physical CPUs or cores on each host if necessary.
5	Enable CPU-saving features, such as TCP Segmentation Offload.
6	Replace software I/O with dedicated hardware, such as iSCSI HBAs or TCP Segmentation Offload NICs.

CPU Usage

The CPU Usage chart displays CPU usage of each virtual machine in the vApp or resource pool.

This chart is located on the **Monitor** tab for a vApp. Select **Performance** > **Overview** and select **Resource Pools and Virtual Machines** from the **View** drop-down menu.

Table 426: Data Counters

Chart Label	Description
virtual_machine	Amount of CPU actively used by virtual machines. • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage or CPU ready indicates that you are making the best use of the virtual machine resources. However, if the CPU usage value for a virtual machine is above 90% and the CPU ready value is above 20%, performance is being impacted.

If performance is impacted, consider taking the following actions.

Table 427: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.
3	Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked line chart on the host's **Virtual Machine** view shows the CPU usage for virtual machines on the host.
4	Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.
5	Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time doesn't decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
6	Increase the amount of memory allocated to the virtual machine. This decreases disk and or network activity for applications that cache. This might lower disk I/O and reduce the need for the host to virtualize the hardware. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
7	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
8	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
9	Upgrade the physical CPUs or cores on the host if necessary.
10	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

Memory (MB)

The Memory (MB) chart displays memory usage in the vApp or resource pool.

This chart is located on the **Monitor** tab for a vApp. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 428: Data Counters

Chart Label	Description
resource_pool or *vApp*	Sum of the active memory used by all virtual machines in the resource pool or vApp. Active memory is determined by the VMkernel and includes overhead memory. memory usage = active memory / configured virtual machine memory size • Counter: used • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average • Collection Level: 1

Chart Analysis

Memory usage is not necessarily an indicator of performance problems. Memory can be high if a host is swapping or ballooning, which can result in virtual machine guest swapping. In such cases, check for other problems, such as CPU over-commitment or storage latencies.

If you have constantly high memory usage in a cluster, resource pool, or vApp, consider taking the following actions.

Table 429: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	Verify that the balloon driver is enabled. The balloon driver is installed with VMware Tools and is critical to performance. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, it does not impact virtual machine performance.
3	If the balloon value is high, check the resource shares, reservations, and limits for the virtual machines and resource pools on the hosts. Verify that the host's settings are adequate and not lower than those set for the virtual machine. If free memory is available on the hosts and the virtual machines are experiencing high swap or balloon memory, the virtual machine (or resource pool, if it belongs to one) has reached its resource limit. Check the maximum resource limit set on that host.
4	If the cluster is not a DRS cluster, enable DRS. To enable DRS, perform the following tasks: 1. Select the cluster, and click the **Configure** tab. 2. Under **Services**, click **vSphere DRS**. 3. click **Edit**. An Edit Cluster Settings dialog box opens. 4. Click **Turn ON vSphere DRS**, and click **OK**.
5	If the cluster is a DRS cluster: • Increase the number of hosts, and migrate one or more virtual machines to the new host. • Check the aggressiveness threshold. If the value is low, increase the threshold. It might help avoid hot spots in the cluster.
6	Add more physical memory to one or more hosts.

Memory Consumed

The Memory Consumed chart displays the memory performance of the top ten virtual machines in the vApp or resource pool.

This chart is located on the **Monitor** tab for a vApp. Select **Performance** > **Overview** and select **Resource Pools and Virtual Machines** from the **View** drop-down menu.

For resource pools and virtual machines in a resource pool or vApp, this chart is located in the **Resource Pools & Virtual Machines** view of the resource pool or vApp **Performance** tab.

Table 430: Data Counters

Chart Label	Description
virtual_machine	Amount of host memory used by the virtual machine for its guest operating system's physical memory. Memory overhead is not included in consumed memory. consumed memory = memory granted - memory saved from page sharing For example, if a virtual machine has 100 MB of memory that is shared equally with three other virtual machines, its portion of the shared memory is 25 MB (100 MB ÷ 4 VMs). This amount is counted in the memory consumed data counter. • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 431: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Virtual Machines

The virtual machine charts contain information about CPU, disk, memory, network, storage, and fault tolerance for virtual machines. The help topic for each chart contains information about the data counters displayed in that chart. The counters available are determined by the collection level set for *vCenter*.

CPU (%)

The CPU (%) chart displays virtual machine CPU usage and ready values.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 432: Data Counters

Chart Label	Description
Usage	Amount of actively used virtual CPU as a percentage of total available CPU. CPU usage is the average CPU utilization over all available virtual CPUs in the virtual machine. For example, if a virtual machine with one virtual CPU is running on a host that has four physical CPUs and the CPU usage is 100%, the virtual machine is using one physical CPU completely. virtual CPU usage = usagemhz ÷ (number of virtual CPUs × core frequency) **Note:** This is the host's view of the CPU usage, not the guest operating system view. • Counter: usage • Stats Type: Rate • Unit: Percentage (%). Precision is to 1/100%. A value between 0 and 100. • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Ready	Percentage of time that the virtual machine was ready, but could not get scheduled to run on the physical CPU. CPU ready time is dependent on the number of virtual machines on the host and their CPU loads. At collection level 1, the average CPU ready time of all virtual CPUs on the virtual machine is displayed. At collection level 3, the average CPU ready time of each virtual CPU is also displayed. • Counter: ready • Stats Type: Rate • Unit: Percentage (%) • Rollup Type: Summation • Collection Level: 1

Chart Analysis

A short spike in CPU usage or CPU ready indicates that you are making the best use of the virtual machine resources. However, if the CPU usage value for a virtual machine is above 90% and the CPU ready value is above 20%, performance is being impacted.

If performance is impacted, consider taking the following actions.

Table 433: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.
3	Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked line chart on the host's **Virtual Machine** view shows the CPU usage for virtual machines on the host.

#	Resolution
4	Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.
5	Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time doesn't decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
6	Increase the amount of memory allocated to the virtual machine. This decreases disk and or network activity for applications that cache. This might lower disk I/O and reduce the need for the host to virtualize the hardware. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
7	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
8	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
9	Upgrade the physical CPUs or cores on the host if necessary.
10	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

CPU Usage in MHz

The CPU Usage (MHz) chart displays virtual machine CPU usage.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter.

Table 434: Data Counters

Chart Label	Description
Usage	Amount of actively used virtual CPU. **Note:** The host's view of the CPU usage, not the guest operating system view. • Counter: usagemhz • Stats Type: rate • Unit: MegaHertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A short spike in CPU usage or CPU ready indicates that you are making the best use of the virtual machine resources. However, if the CPU usage value for a virtual machine is above 90% and the CPU ready value is above 20%, performance is being impacted.

If performance is impacted, consider taking the following actions.

Table 435: CPU Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on every virtual machine on the host.
2	Set the CPU reservations for all high-priority virtual machines to guarantee that they receive the CPU cycles required.

#	Resolution
3	Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked line chart on the host's **Virtual Machine** view shows the CPU usage for virtual machines on the host.
4	Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.
5	Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time doesn't decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
6	Increase the amount of memory allocated to the virtual machine. This decreases disk and or network activity for applications that cache. This might lower disk I/O and reduce the need for the host to virtualize the hardware. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
7	Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the hypervisor's maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
8	If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
9	Upgrade the physical CPUs or cores on the host if necessary.
10	Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

Disk (KBps)

The Disk (KBps) chart displays disk usage for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 436: Data Counters

Chart Label	Description
Usage	Average data I/O rate across all virtual disks on the virtual machine. • Counter: usage • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.

- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 437: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk Rate (KBps)

The Disk Rate chart displays disk usage for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu.

Table 438: Data Counters

Chart Label	Description
Read	Number of disk read commands completed on each virtual disk on the virtual machine, per second. The aggregate number of all disk read commands per second is also displayed in the chart. Read rate = blocksRead per second × blockSize • Counter: read • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average • Collection Level: 2
Write	Number of disk write commands completed on each virtual disk on the virtual machine, per second. The aggregate number of all disk write commands per second is also displayed in the chart. Write rate = blocksWritten per second × blockSize • Counter: write • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average • Collection Level: 2

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 439: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.

#	Resolution
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Disk Requests (Number)

The Disk Requests chart displays disk usage for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu.

Table 440: Data Counters

Chart Label	Description
Read Requests	Number of disk read commands completed on each virtual disk on the virtual machine. The aggregate number of all disk read commands is also displayed in the chart. • Counter: numberRead • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3
Write Requests	Number of disk write commands completed on each virtual disk on the virtual machine. The aggregate number of all disk write commands is also displayed in the chart. • Counter: numberWrite • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3

Chart Analysis

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read or write requests, check whether any such applications were running then.

The best ways to determine if your vSphere environment is experiencing disk problems is to monitor the disk latency data counters. You can use the advanced performance charts to view these statistics.

- The kernelLatency data counter measures the average amount of time, in milliseconds, that the VMkernel spends processing each SCSI command. For best performance, the value must be 0 -1 milliseconds. If the value is greater than 4 ms, the virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- The deviceLatency data counter measures the average amount of time, in milliseconds, to complete a SCSI command from the physical device. Depending on your hardware, a number greater than 15 ms indicates probable problems with the storage array. Move the active VMDK to a volume with more spindles or add disks to the LUN.
- The queueLatency data counter measures the average amount of time taken per SCSI command in the VMkernel queue. This value must always be zero. If not, the workload is too high and the array cannot process the data fast enough.

If the disk latency values are high, or if you notice other problems with disk I/O performance, consider taking the following actions.

Table 441: Disk I/O Performance Enhancement Advice

#	Resolution
1	Increase the virtual machine memory. It allows more operating system caching, which reduces I/O activity. Note: It might require you to increase the host memory. Increasing memory might reduce the need to store data because databases can utilize the system memory to cache data and avoid disk access. To verify that virtual machines have adequate memory, check swap statistics in the guest operating system. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
2	Defragment the file systems on all guests.
3	Deactivate antivirus on-demand scans on the VMDK and VMEM files.
4	Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. To increase throughput, consider array-side improvements.
5	Use Storage vMotion to migrate I/O-intensive virtual machines across multiple hosts.
6	Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
7	Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter. For more information, see *vSphere Storage*.
8	For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. It alleviates disk spindle contention during periods of high use.
9	On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
10	If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
11	For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
12	Use the most current host hardware.

Virtual Disk Requests (Number)

The Virtual Disk Requests chart displays virtual disk usage for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu. It is available at collection (display) levels 3 and 4.

Table 442: Data Counters

Chart Label	Description
Read Requests	Number of virtual disk read commands completed on each virtual disk on the virtual machine. The aggregate number of all virtual disk read commands is also displayed in the chart. • Counter: numberRead • Stats Type: Absolute • Unit: Number • Rollup Type: Average • Collection Level: 2
Write Requests	Number of virtual disk write commands completed on each virtual disk on the virtual machine. The aggregate number of all virtual disk write commands is also displayed in the chart. • Counter: numberWrite • Stats Type: Absolute • Unit: Number • Rollup Type: Average • Collection Level: 2

Virtual Disk Rate (KBps)

The Virtual Disk Rate chart displays virtual disk usage rate for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Disks** from the **View** drop-down menu. It is available only at collection levels 3 and 4.

Table 443: Data Counters

Chart Label	Description
Read Requests	Number of virtual disk read commands completed on each virtual disk on the virtual machine. The aggregate number of all virtual disk read commands per second is also displayed in the chart. Read rate = blocksRead per second × blockSize • Counter: read • Stats Type: Rate • Unit: KiloBytes per second (KBps) • Rollup Type: Average • Collection Level: 3
Write Requests	Number of virtual disk write commands completed on each virtual disk on the virtual machine per second. The aggregate number of all virtual disk write commands per second is also displayed in the chart. Write rate = blocksWritten per second × blockSize • Counter: write • Stats Type: Rate • Unit: KiloBytes per second (KBps) • Rollup Type: Average • Collection Level: 3

Memory (%)

The Memory (%) chart monitors virtual machine memory usage.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Virtual Machine Counters

Note:
Guest physical memory refers to the virtual hardware memory presented to a virtual machine for its guest operating system.

Table 444: Data Counters

Chart Label	Description
Usage	Amount of guest physical memory currently in use on the virtual machine. • Counter: usage • Stats Type: Absolute • Unit: Percentage (%) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 445: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (KB)

The Memory (MB) chart displays virtual machine balloon memory.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 446: Data Counters

Chart Label	Description
Balloon	Amount of guest physical memory reclaimed from the virtual machine by the balloon driver. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 447: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (KBps)

The Memory (MBps) chart displays virtual machine memory swap rates.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter.

Table 448: Data Counters

Chart Label	Description
swapinRate	Average rate at which memory is swapped into the virtual machine. • Counter: swapinRate • Stats Type: Rate • Unit: MegaBytes per second (MBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
swapoutRate	Average rate at which memory is swapped out of the virtual machine. • Counter: swapoutRate • Stats Type: Rate • Unit: MegaBytes per second (MBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If there is sufficient swap space, a high balloon value is not a performance issue. However, if the swapin and swapout values for the host are large, the host is probably lacking the memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. The host might require more memory resources. If it does not, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If memory usage is high or you notice degradation in performance, consider taking the following actions.

Table 449: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of a virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory (MB)

The Memory (MB) chart displays memory data counters for virtual machines.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter. The chart appears only at collection levels 2, 3, and 4.

In the following descriptions, the guest physical memory refers to the virtual-hardware memory presented to a virtual machine for its guest operating system. Machine memory is actual physical RAM in the host. Note that not all counters are collected at collection level 1.

Table 450: Data Counters

Chart Label	Description
Active	Amount of guest physical memory in use by the virtual machine. Active memory is estimated by VMkernel statistical sampling and represents the actual amount of memory the virtual machine needs. The value is based on the current workload of the virtual machine. • Counter: active • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Balloon	Amount of guest physical memory reclaimed from the virtual machine by the balloon driver. • Counter: vmmemctl • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Balloon Target	Desired amount of virtual machine balloon memory. Balloon target memory is estimated by the VMkernel. If the balloon target amount is greater than the balloon amount, the VMkernel inflates the balloon amount, which reclaims more virtual machine memory. If the balloon target amount is less than the balloon amount, the VMkernel deflates the balloon, which allows the virtual machine to reallocate memory when needed. • Counter: vmmemctltarget • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)
Consumed	Amount of guest physical memory consumed by the virtual machine for guest memory. Consumed memory does not include overhead memory. It includes shared memory and memory that might be reserved, but not actually used. consumed memory = memory granted - memory saved due to memory sharing • Counter: consumed • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)
Shared	Amount of guest physical memory available for sharing. Memory sharing occurs through transparent page sharing. • Counter: shared • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)

Chart Label	Description
Swapped	The amount of guest physical memory swapped out to the disk by the VMkernel. This data counter measures VMkernel swapping and not to guest OS swapping. swapped = swapout – swapin **Note:** In some cases, vMotion can skew these values and cause a virtual machine to arrive on a host with some memory already swapped out. As a result, the swapped value can be greater than the swapout – swapin value. • Counter: swapped • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4)

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 451: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory Bandwidth (MBps)

The **Memory Bandwidth** chart for a virtual machine displays graphs for the estimated DRAM read and write bandwidth, PMem read and write bandwidth, and NVMe read and write bandwidth.

To access this chart, navigate to a virtual machine in the inventory.

1. Click the **Monitor** tab.
2. Under **Performance**, click **Overview**.
3. From the **View** drop-down menu, select **Memory**.

The **Memory** pane displays information about memory bandwidth, memory utilization, and memory reclamation.

Note:

PMem bandwidth is only available on supported hosts configured in memory mode.

If the host does not vMMR or is not in memory mode, the **Memory** option is not listed in the **View** drop-down menu.

Note:

The NVMe bandwidth graph is only available on hosts with configured Memory Tiering over NVMe (Memory Tiering). For more information about Memory Tiering, see Memory Tiering over NVMe.

Table 452: Data Counters

Chart Label	Description
DRAM Read Bandwidth	Current read bandwidth of DRAM memory type. • Counter: bandwidth.read • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 2 (4)
PMem Read Bandwidth	Current read bandwidth of PMem memory type. • Counter: bandwidth.read • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 2 (4)
Read Bandwidth NVMe	Current read bandwidth of NVMe memory type. • Counter: bandwidth.read • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 2 (4)
Write Bandwidth NVMe	Current write bandwidth of NVMe memory type. • Counter: bandwidth.write • Stats Type: Absolute • Unit: MegaBytes per second (MBps) • Rollup Type: Latest • Collection Level: 2 (4)

Network (Mbps)

The Network (Mbps) chart displays network bandwidth for the virtual machine.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Overview** from the **View** drop-down menu.

Table 453: Virtual Machine Counters

Chart Label	Description
Usage	Average rate at which data is transmitted and received across all virtual NIC instances connected to the virtual machine. • Counter: usage • Stats Type: Rate • Unit: Kilobytes per second (KBps) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 1 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 454: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.

#	Resolution
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Network Rate (Mbps)

The Network Rate chart displays network usage for virtual machines.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter. The chart appears only at collection levels 3 and 4.

Table 455: Data Counters

Chart Label	Description
Data Receive Rate	Rate at which data is received across each virtual NIC instance on the virtual machine. • Counter: received • Stats Type: Rate • Unit: Kilobytes per second (Kbps) • Rollup Type: Average • Collection Level: 2 (4)
Data Transmit Rate	Rate at which data is transmitted across each virtual NIC instance on the virtual machine. • Counter: transmitted • Stats Type: Rate • Unit: Kilobytes per second (Kbps) • Rollup Type: Average • Collection Level: 2 (4)

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

> **Note:**
> In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 456: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.

#	Resolution
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Network Packets (Number)

The Network Packets monitors network bandwidth for virtual machines.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Advanced** and open the **Chart Options** dialog box and select the respective counter. The chart appears only at collection levels 3 and 4.

Table 457: Data Counters

Chart Label	Description
Packets Transmitted	Number of network packets transmitted across the top ten virtual NIC instances on the virtual machine. The chart also displays the aggregated value for each NIC. • Counter: packetTx • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3
Packets Received	Number of network packets received across the top ten virtual NIC instances on the virtual machine. The chart also displays the aggregated value for each NIC. • Counter: packetRx • Stats Type: Absolute • Unit: Number • Rollup Type: Summation • Collection Level: 3

Chart Analysis

Network performance depends on the application workload and network configuration. Dropped network packets indicate a bottleneck in the network. To determine whether packets are being dropped, use `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values.

If packets are being dropped, adjust the virtual machine shares. If packets are not being dropped, check the size of the network packets and the data receive and transfer rates. In general, the larger the network packets, the faster the network speed. When the packet size is large, fewer packets are transferred, which reduces the amount of CPU required

to process the data. When network packets are small, more packets are transferred but the network speed is slower because more CPU is required to process the data.

Note:
In some instances, large packets might result in a high network latency. To check the network latency, use the VMware AppSpeed performance monitoring application or a third-party application.

If packets are not being dropped and the data receive rate is slow, the host is probably lacking the CPU resources required to handle the load. Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different vSwitches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.

If you experience network-related performance problems, also consider taking the following actions.

Table 458: Networking Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine.
2	If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
3	If virtual machines running on the same host communicate with each other, connect them to the same vSwitch to avoid transferring packets over the physical network.
4	Assign each physical NIC to a port group and a vSwitch.
5	Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
6	Ensure that the physical NIC capacity is large enough to handle the network traffic on that vSwitch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10 Gbps). Alternatively, consider moving some virtual machines to a vSwitch with a lighter load or to a new vSwitch.
7	If packets are being dropped at the vSwitch port, increase the virtual network driver ring buffers where applicable.
8	Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that NICs with 1 Gbps are not reset to 100 Mbps because they are connected to an older switch.
9	Verify that all NICs are running in full duplex mode. Hardware connectivity problems might result in a NIC resetting itself to a lower speed or half duplex mode.
10	Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TSO-Jumbo Frames are enabled where possible.

Space

The Space chart displays space utilization data counters for virtual machines.

This chart is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Space** from the **View** drop-down menu.

Table 459: Data Counters

Chart Label	Description
Allocated	Total amount of logical datastore space provisioned by an administrator for the virtual machine. It is the storage size up to which the virtual machine files on datastores can grow. This includes log files, VMX files, and other miscellaneous files. Allocated space is not always in use. • Counter: provisioned • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1
Used	Amount of physical datastore space in use by the virtual machine files. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1
Not Shared	Amount of datastore space that belongs only to this virtual machine and is not shared with other virtual machines. Only unshared space is guaranteed to be reclaimed for the virtual machine if, for example, it is moved to a different datastore and then back again. The value is an aggregate of all unshared space for the virtual machine, across all datastores. • Counter: unshared • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Utilization by Datastores

The **Space Utilization by Datastores** chart displays the amount of space used by a virtual machine on different datastores in the data center.

> **Note:**
> This chart does not show historical statistics. It only shows the most recently available data, which may be up to 30 minutes late, depending on when the last statistics rollup occurred. In addition, statistics are not collected across all datastores at one time. They are collected asynchronously.

The Space **Utilization by Datastores** is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Space Utilization** from the **View** drop-down menu.

Table 460: Data Counters

Chart Label	Description
datastore_name	Amount of disk space in the datastore currently in use by the virtual machine. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Space Utilization By File Type

The **Space Utilization by File Type** chart displays the datastore usage by virtual machine files.

> **Note:**
> This chart does not show historical statistics. It only shows the most recently available data, which may be up to 30 minutes late, depending on when the last statistics rollup occurred. In addition, statistics are not collected across all datastores at one time. They are collected asynchronously.

The Space **Utilization by File Type** is located on the **Monitor** tab for a virtual machine. Select **Performance** > **Overview** and select **Space Utilization** from the **View** drop-down menu.

Datastore counters

Table 461: Data Counters

File Type	Description
Virtual Disks	Amount of disk space used by virtual disk files. Virtual disk files store the contents of the virtual machine's hard disk drive, including information that you write to a virtual machine's hard disk - the operating system, program files, and data files. The files have the extension `.vmdk` and appear as a physical disk drive to a guest operating system. **Note:** Delta disks, which also have an extension `.vmdk`, are not included in this file type. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Swap Files	Amount of disk space used by swap files. Swap files back up the virtual machine's physical memory. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Snapshots	Amount of disk space used by virtual machine snapshot files. Snapshot files store information about virtual machine snapshots. They include snapshot state files and delta disk files. A snapshot state file stores the running state of the virtual machine at the time of the snapshot. It has the extension `.vmsn`. A delta disk file stores the updates made by the virtual machine to the virtual disks after a snapshot is taken. • Counter: used • Stats Type: Absolute • Unit: GigaBytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Other VM Files	Amount of disk space used by all other virtual machine files, such as configuration files and log files. • Counter: used • Stats Type: Absolute • Unit: Gigabytes (GB) • Rollup Type: Latest • Collection Level: 1 (4)
Total Space	Amount of disk space used by the virtual machine. total space = virtual disk space + swap file space + snapshot space + other VM file space

Chart Analysis

The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks. If possible, you can provision more space to the datastore, or you can add disks to the datastore or use shared datastores.

If snapshot files are consuming high datastore space, consider consolidating them to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface. For information about consolidating the data center, see the vSphere documentation.

Fault Tolerance Performance Counters

The fault tolerance charts contain information about CPU and memory for fault tolerant virtual machines.

Note:

The performance charts and help topics for fault tolerance are available only when you have vSphere Fault Tolerance enabled. If you select a link for a secondary virtual machine in the thumbnail section of the Resource Pools and Virtual Machines view of the cluster **Monitor** tab, the navigation in the inventory updates to the primary virtual machine. This occurs because secondary machines are not displayed in the inventory.

CPU (MHz)

The CPU (MHz) chart displays virtual CPU usage for fault tolerant virtual machines.

This chart is located in the **Fault Tolerance** view of the virtual machine **Peformance** tab. It is available only at collection levels 3 and 4.

Table 462: Data Counters

Name	Description
Usage	The average amount of virtual CPU, per CPU instance, in use on the primary and secondary fault tolerant virtual machines. • Counter: usagemhz • Stats Type: Rate • Unit: Megahertz (MHz) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 3 (4)

Chart Analysis

A large discrepancy in CPU usage between the primary and secondary virtual machines might indicate performance problems. The CPU ready, system, and wait times of each virtual machine should be synchronized. A large discrepancy in these values might indicate performance problems. Consider taking the following actions.

Table 463: CPU Performance Enhancement Advice

#	Resolution
1	Verify that the primary and secondary hosts are in the same CPU model family and have similar CPU configurations. For best results, use CPUs with the same stepping level.
2	Verify that the CPU resource reservations set for both virtual machines are consistent within the cluster. VMware HA plans for a worst-case scenario by considering all powered on virtual machines in a cluster and finding the maximum memory and CPU reservations.
3	Verify that the network and datastore connections for both virtual machines are similar.
4	Turn off power management (also known as power-capping) in the BIOS. If power management is enabled, the secondary host might enter lower performance, power-saving modes. Such modes can leave the secondary virtual machine with insufficient CPU resources, potentially making it impossible for the secondary to complete all tasks completed on a primary in a timely fashion.
5	Turn off hyperthreading in the BIOS. If hyperthreading is enabled and the secondary virtual machine is sharing a CPU with another demanding virtual machine, the secondary virtual machine might run too slowly to complete all tasks completed on the primary in a timely fashion.

CPU System Time for vCPU (%)

The CPU System Time chart displays virtual CPU usage for fault tolerant virtual machines.

This chart is located in the **Fault Tolerance** view of the Virtual Machine **Peformance** tab. It is available only at collection levels 3 and 4.

Table 464: Data Counters

Chart Label	Description
System	Amount of time spent on system processes on each virtual CPU in the virtual machine. **Note:** This is the host view of the CPU usage, not the guest operating system view. • Counter: system • Stats Type: Delta • Unit: Percentage (%) • Rollup Type: Summation • Collection Level: 3

Chart Analysis

A large discrepancy in CPU usage between the primary and secondary virtual machines might indicate performance problems. The CPU ready, system, and wait times of each virtual machine should be synchronized. A large discrepancy in these values might indicate performance problems. Consider taking the following actions.

Table 465: CPU Performance Enhancement Advice

#	Resolution
1	Verify that the primary and secondary hosts are in the same CPU model family and have similar CPU configurations. For best results, use CPUs with the same stepping level.
2	Verify that the CPU resource reservations set for both virtual machines are consistent within the cluster. VMware HA plans for a worst-case scenario by considering all powered on virtual machines in a cluster and finding the maximum memory and CPU reservations.
3	Verify that the network and datastore connections for both virtual machines are similar.
4	Turn off power management (also known as power-capping) in the BIOS. If power management is enabled, the secondary host might enter lower performance, power-saving modes. Such modes can leave the secondary virtual machine with insufficient CPU resources, potentially making it impossible for the secondary to complete all tasks completed on a primary in a timely fashion.
5	Turn off hyperthreading in the BIOS. If hyperthreading is enabled and the secondary virtual machine is sharing a CPU with another demanding virtual machine, the secondary virtual machine might run too slowly to complete all tasks completed on the primary in a timely fashion.

CPU Used Time for vCPU (%)

The CPU Used Time chart displays virtual CPU usage for fault tolerant virtual machines.

This chart is located in the **Fault Tolerance** view of the Virtual Machine **Peformance** tab. The chart is available only for collection levels 3 and 4.

Table 466: Data Counters

Chart Label	Description
used	Amount of used virtual CPU as a percentage of total available CPU on the primary and secondary virtual machines. A high value indicates excessive use of CPU resources. • Counter: used • Stats Type: Delta • Unit: Percentage (%) • Rollup Type: Summation • Collection Level: 1

Chart Analysis

A large discrepancy in CPU usage between the primary and secondary virtual machines might indicate performance problems. The CPU ready, system, and wait times of each virtual machine should be synchronized. A large discrepancy in these values might indicate performance problems. Consider taking the following actions.

Table 467: CPU Performance Enhancement Advice

#	Resolution
1	Verify that the primary and secondary hosts are in the same CPU model family and have similar CPU configurations. For best results, use CPUs with the same stepping level.
2	Verify that the CPU resource reservations set for both virtual machines are consistent within the cluster. VMware HA plans for a worst-case scenario by considering all powered on virtual machines in a cluster and finding the maximum memory and CPU reservations.
3	Verify that the network and datastore connections for both virtual machines are similar.
4	Turn off power management (also known as power-capping) in the BIOS. If power management is enabled, the secondary host might enter lower performance, power-saving modes. Such modes can leave the secondary virtual machine with insufficient CPU resources, potentially making it impossible for the secondary to complete all tasks completed on a primary in a timely fashion.
5	Turn off hyperthreading in the BIOS. If hyperthreading is enabled and the secondary virtual machine is sharing a CPU with another demanding virtual machine, the secondary virtual machine might run too slowly to complete all tasks completed on the primary in a timely fashion.

Memory Active (MB)

The Memory Active chart displays active memory usage for fault tolerant virtual machines.

This chart is located in the **Fault Tolerance** view of the Virtual Machine **Peformance** tab. It is not available at collection level 1.

Table 468: Data Counters

Chart Label	Description
Active	Amount of guest physical memory in use by the fault tolerant virtual machine. Active memory is estimated by VMkernel statistical sampling and represents the actual amount of memory the virtual machine needs. Additional, unused memory may be swapped out or ballooned with no performance impact. • Counter: active • Stats Type: Absolute • Unit: Megabytes (MB) • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4) Make sure that the primary and secondary virtual machines have enough memory. If the secondary system is not provisioned well, it might slow down performance of the primary virtual machine or fail.

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If sufficient swap space is available, a high balloon value does not cause performance problems. However, if the swapin and swapout values for the host are large, the host is probably lacking the amount of memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. A free memory value of 6% or less indicates that the host cannot meet the memory requirements. This leads to memory reclamation, which might degrade performance. If the active memory size is the same as the granted memory size, demand for memory is greater than the memory resources available. If the active memory is consistently low, the memory size might be too large.

If the host has enough free memory, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If little free memory is available, or if you notice degradation in performance, consider taking the following actions.

Table 469: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Memory Swapout (MB)

The Memory Swapout chart displays the swapout memory usage for fault tolerant virtual machines.

This chart is located in the **Fault Tolerance** view of the Virtual Machine **Peformance** tab. It is not available at collection level 1.

Table 470: Data Counters

Chart Label	Description
Swapout	Amount of machine memory written to the VMkernel swap file. • Counter: swapout • Stats Type: Absolute • Unit: MegaBytes • Rollup Type: Average (Minimum/Maximum) • Collection Level: 2 (4) Make sure that the primary and secondary virtual machines have enough memory and that the swapout value is not high. If the secondary system is not provisioned well, it might slow down performance of the primary virtual machine or fail.

Chart Analysis

A virtual machine's memory size must be slightly larger than the average guest memory usage. This enables the host to accommodate workload spikes without swapping memory among guests. Increasing the virtual machine memory size results in more overhead memory usage.

If there is sufficient swap space, a high balloon value is not a performance issue. However, if the swapin and swapout values for the host are large, the host is probably lacking the memory required to meet the demand.

If a virtual machine has high ballooning or swapping, check the amount of free physical memory on the host. The host might require more memory resources. If it does not, check the resource shares, reservation, and limit of the virtual machines and resource pools on the host. Verify that the host settings are adequate and not lower than those set for the virtual machine.

If memory usage is high or you notice degradation in performance, consider taking the following actions.

Table 471: Memory Performance Enhancement Advice

#	Resolution
1	Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
2	Verify that the balloon driver is enabled. The VMkernel regularly reclaims unused virtual machine memory by ballooning and swapping. Generally, this does not impact virtual machine performance.
3	Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
4	If the memory reservation of a virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
5	Migrate one or more virtual machines to a host in a DRS cluster.
6	Add physical memory to the host.

Working with Advanced Charts

Use advanced charts, or create your own custom charts, to see more performance data. Advanced charts can be useful when you are aware of a problem but need more statistical data to pinpoint the source of the issue.

Advanced charts include the following features:

- More information. Hover over a data point in a chart and details about that specific data point are displayed.
- Customizable charts. Change chart settings. To create your own charts, save custom settings.
- Export to spreadsheet.
- Save to image file or spreadsheet.

View Advanced Performance Charts

Advanced charts support data counters that are not supported in other performance charts.

For example, you can use the Power view to display the `vm.power` counter and monitor virtual machine power information for hosts.

1. Navigate to an inventory object in the vSphere Client.
2. Click the **Monitor** tab, and click **Performance** > **Advanced**.
3. Optional: Select an option from the **View** drop-down menu.

   The amount of historical data displayed in a chart depends on the collection interval and statistics level set for the vCenter instance.
4. Optional: To view the selected performance chart in a pop-up window, click the **Chart Context Menu** icon ( ), and click **Popup Chart**.

   The chart opens in a new browser window enabling you to navigate within the vSphere Client while keeping the performance chart open in a separate window.

Change Advanced Performance Chart Settings

You can customize an advanced performance chart by specifying the objects to monitor, the data counters to include, the data collection interval, and chart type. You can customize preconfigured chart views and create chart views.

1. Select an inventory object in the vSphere Client.
2. Click the **Monitor** tab, and click **Performance** > **Advanced**.
3. Click **Chart Options**.

   The **Chart Options** dialog box appears.
4. From the **Chart Metrics** list, select a metric group for the chart.
5. From the table, select the data counters to display in the chart.
6. From the **Timespan** drop-down menu, select the data collection interval for the metric group.

   If you want to specify a custom time range, you must select **Custom interval** in the **Timespan** menu.

   - Select **Last** and set the number of hours, days, weeks, or months to monitor the object.
   - Select **From** and specify the beginning and end dates and times of the data collection period.

   You can also customize the time range options by customizing the statistics collection interval setting.
7. From the **Chart Type** drop-down menu, select the chart type.

   When selecting the stacked graph option, consider the following.

   - You can select only one item from the list of measurements.
   - Per-virtual-machine stacked graphs are available only for hosts.
   - To display information about the counter's function and whether the selected metric can be stacked for per-virtual-machine graphs, click a counter description name.
8. In Target Objects, select the inventory objects to display in the chart.
9. Click **OK**.

Create a Custom Advanced Chart

You can create your own charts by saving customized chart settings. New charts are added to the list under the **View** menu and appear only when charts for the selected object are displayed.

1. Select an inventory object in the vSphere Client.
2. Click the **Monitor** tab, and click **Performance** > **Advanced**.
3. Click Chart Options.

   The **Chart Options** dialog box appears.
4. Customize chart settings by selecting the chart metrics, counters, objects, timespan and chart type that you want to save as a view.
5. Click **Save options as**.
6. Enter a name for your custom chart options and click **OK**.
7. Click **OK** to close the **Chart Options** dialog box.

The chart settings are saved and an entry for your chart is added to the list under the **View** menu in the **Advanced Performance** pane.

Delete a Custom Advanced Chart View

You can delete custom performance charts when you no longer need them.

1. Select an inventory object in the vSphere Client.
2. Click the **Monitor** tab, and click **Performance** > **Advanced**.
3. Click Chart Options.

   The **Chart Options** dialog box appears.
4. From the **Chart options** drop-down menu, select a custom chart and click **Delete Options**.
5. Click **OK** to confirm deletion.

   The chart is deleted, and it is removed from the **View** menu.
6. Click **OK** to close the **Chart Options** dialog box.

Save Chart Data to a File

You can save data from the advanced performance charts to a file in various graphic formats or in comma-separated values (CSV) format.

1. In the vSphere Client, select an inventory object.
2. Click the **Monitor** tab, and click **Performance** > **Advanced**.
3. Click the **Chart Context Menu** icon

   ( ).
4. Select a file type to export the data to.

The file is saved to your local system in the file format that you specified.

Troubleshoot and Enhance Performance

This section presents tips for identifying and solving performance issues.

The suggestions in this section are not meant to be a comprehensive guide to diagnosing and troubleshooting problems in the virtual environment. It is meant to provide information about some common problems that can be solved without contacting VMware Technical Support.

Solutions for Consistently High CPU Usage

Temporary spikes in CPU usage are not necessarily a concern, but consistently high CPU usage might indicate a problem. You can use the CPU performance charts to monitor CPU usage for hosts, clusters, resource pools, virtual machines, and vApps.

- Host CPU usage constantly is high. A high CPU usage value can lead to increased ready time and processor queuing of the virtual machines on the host.
- Virtual machine CPU usage is above 90% and the CPU ready value is above 20%. Application performance is impacted.
- The host probably is lacking the CPU resources required to meet the demand.
- There might be too many virtual CPUs relative to the number of physical processor cores.
- There might be an IO storage or networking operation that places the CPU in a wait state.
- The Guest OS generates too much load for the CPU.
- Verify that VMware Tools is installed on every virtual machine on the host.
- Compare the CPU usage value of a virtual machine with the CPU usage of other virtual machines on the host or in the resource pool. The stacked bar chart on the host's **Virtual Machine** view shows the CPU usage for all virtual machines on the host.
- Determine whether the high ready time for the virtual machine resulted from its CPU usage time reaching the CPU limit setting. If so, increase the CPU limit on the virtual machine.
- Increase the CPU shares to give the virtual machine more opportunities to run. The total ready time on the host might remain at the same level if the host system is constrained by CPU. If the host ready time does not decrease, set the CPU reservations for high-priority virtual machines to guarantee that they receive the required CPU cycles.
- Increase the amount of memory allocated to the virtual machine. This can potentially decrease disk and/or network activity for applications that cache. This might lower disk I/O and/or network traffic, which could in turn reduce CPU utilization. Virtual machines with smaller resource allocations generally accumulate more CPU ready time.
- Reduce the number of virtual CPUs on a virtual machine to only the number required to execute the workload. For example, a single-threaded application on a four-way virtual machine only benefits from a single vCPU. But the ESX maintenance of the three idle vCPUs takes CPU cycles that could be used for other work.
- If the host is not already in a DRS cluster, add it to one. If the host is in a DRS cluster, increase the number of hosts and migrate one or more virtual machines onto the new host.
- Upgrade the physical CPUs on the host if necessary.
- Use the newest version of hypervisor software, and enable CPU-saving features such as TCP Segmentation Offload, large memory pages, and jumbo frames.

Solutions for Memory Performance Problems

Host machine memory is the hardware backing for guest virtual memory and guest physical memory. For best performance, the host machine memory should be at least slightly larger than the combined active memory of the virtual machines on the host. A virtual machine's memory size should be slightly larger than the average guest memory usage. Increasing the virtual machine memory size results in more overhead memory usage.

- Memory usage is constantly high (94% or greater) or constantly low (24% or less).
- Free memory consistently is 6% or less and swapping frequently occurs.
- The host probably is lacking the memory resources required to meet the combined active memory size of all the running VMs.
- Host machine memory resources are not enough to meet the demand, which leads to memory reclamation (like swapping) and degraded performance.

- Verify that VMware Tools is installed on each virtual machine. The balloon driver is installed with VMware Tools and is critical to performance.
- Verify that the balloon driver is enabled. The VMkernel regularly attempts to reclaim the unused virtual machine memory by ballooning and, if necessary, swapping. Generally, this does not impact virtual machine performance.
- Reduce the memory space on the virtual machine, and correct the cache size if it is too large. This frees up memory for other virtual machines.
- If the memory reservation of the virtual machine is set to a value much higher than its active memory, decrease the reservation setting so that the VMkernel can reclaim the idle memory for other virtual machines on the host.
- Migrate one or more virtual machines to a host in a DRS cluster.
- Add physical memory to the host.

Solutions for Storage Performance Problems

Datastores represent storage locations for virtual machine files. A storage location can be a VMFS volume, a directory on Network Attached Storage, or a local file system path. Datastores are platform-independent and host-independent.

- Snapshot files are consuming a lot of datastore space.
- The datastore is at full capacity when the used space is equal to the capacity. Allocated space can be larger than datastore capacity, for example, when you have snapshots and thin-provisioned disks.
- Consider consolidating snapshots to the virtual disk when they are no longer needed. Consolidating the snapshots deletes the redo log files and removes the snapshots from the vSphere Client user interface.
- You can provision more space to the datastore if possible, or you can add disks to the datastore or use shared datastores.

Solutions for Disk Performance Problems

Use the disk charts to monitor average disk loads and to determine trends in disk usage. For example, you might notice a performance degradation with applications that frequently read from and write to the hard disk. If you see a spike in the number of disk read/write requests, check if any such applications were running then.

- The value for the kernelLatency data counter is greater than 4 ms.
- The value for the deviceLatency data counter is greater than 15 ms indicates that there are probably problems with the storage array.
- The queueLatency data counter measures above zero.
- Spikes in latency.
- Unusual increases in read/write requests.
- The virtual machines on the host are trying to send more throughput to the storage system than the configuration supports.
- The storage array probably is experiencing internal problems.
- The workload is too high and the array cannot process the data fast enough.

- The virtual machines on the host are trying to send more throughput to the storage system than the configuration supports. Check the CPU usage, and increase the queue depth.
- Move the active VMDK to a volume with more spindles or add disks to the LUN.
- Increase the virtual machine memory. Doing so should allow for more operating system caching, which can reduce I/O activity. Note: It may require you to increase the host memory. Increasing memory might reduce the need to store data because some workloads can utilize system memory to cache data and avoid disk access.
- Check swap statistics in the guest operating system to verify that virtual machines have adequate memory. Increase the guest memory, but not to an extent that leads to excessive host memory swapping. Install VMware Tools so that memory ballooning can occur.
- Defragment the file systems on all guests.
- If appropriate for your environment, deactivate antivirus on-demand scans on the VMDK and VMEM files.
- Use the vendor's array tools to determine the array performance statistics. When too many servers simultaneously access common elements on an array, the disks might have trouble keeping up. Consider array-side improvements to increase throughput.
- Use Storage vMotion to distribute I/O-intensive virtual machines across multiple hosts.
- Balance the disk load across all physical resources available. Spread heavily used storage across LUNs that are accessed by different adapters. Use separate queues for each adapter to improve disk efficiency.
- Configure the HBAs and RAID controllers for optimal use. Verify that the queue depths and cache settings on the RAID controllers are adequate. If not, increase the number of outstanding disk requests for the virtual machine by adjusting the `Disk.SchedNumReqOutstanding` parameter.
- For resource-intensive virtual machines, separate the virtual machine's physical disk drive from the drive with the system page file. This alleviates disk spindle contention during periods of high use.
- On systems with sizable RAM, deactivate memory trimming by adding the line `MemTrimRate=0` to the virtual machine's VMX file.
- If the combined disk I/O is higher than a single HBA capacity, use multipathing or multiple links.
- For ESX hosts, create virtual disks as preallocated. When you create a virtual disk for a guest operating system, select **Allocate all disk space now**. The performance degradation associated with reassigning additional disk space does not occur, and the disk is less likely to become fragmented.
- Use the most current hypervisor software.

Solutions for Poor Network Performance

Network performance is dependent on application workload and network configuration. Dropped network packets indicate a bottleneck in the network. Slow network performance can be a sign of load-balancing problems.

Network problems can manifest in many ways:

- Packets are being dropped.
- Network latency is high.
- Data transfer rate is low.

Network problems can have several causes:

- Virtual machine network resource shares are too few.
- Network packet size is too large, which results in high network latency. Use the VMware AppSpeed performance monitoring application or a third-party application to check network latency.
- Network packet size is too small, which increases the demand for the CPU resources needed for processing each packet. Host CPU, or possibly virtual machine CPU, resources are not enough to handle the load.

- Determine whether packets are being dropped by using `esxtop` or the advanced performance charts to examine the `droppedTx` and `droppedRx` network counter values. Verify that VMware Tools is installed on each virtual machine.
- Check the number of virtual machines assigned to each physical NIC. If necessary, perform load balancing by moving virtual machines to different virtual switches or by adding more NICs to the host. You can also move virtual machines to another host or increase the host CPU or virtual machine CPU.
- If possible, use vmxnet3 NIC drivers, which are available with VMware Tools. They are optimized for high performance.
- If virtual machines running on the same host communicate with each other, connect them to the same virtual switch to avoid the cost of transferring packets over the physical network.
- Assign each physical NIC to a port group and a virtual switch.
- Use separate physical NICs to handle the different traffic streams, such as network packets generated by virtual machines, iSCSI protocols, vMotion tasks.
- Ensure that the physical NIC capacity is large enough to handle the network traffic on that virtual switch. If the capacity is not enough, consider using a high-bandwidth physical NIC (10Gbps) or moving some virtual machines to a virtual switch with a lighter load or to a new virtual switch.
- If packets are being dropped at the virtual switch port, increase the virtual network driver ring buffers where applicable.
- Verify that the reported speed and duplex settings for the physical NIC match the hardware expectations and that the hardware is configured to run at its maximum capability. For example, verify that 1Gbps NICs are not reset to 100Mbps because they are connected to an older switch.
- Verify that all NICs are running in full duplex mode. Hardware connectivity issues might result in a NIC resetting itself to a lower speed or half duplex mode.
- Use vNICs that are TCP Segmentation Offload (TSO)-capable, and verify that TCP Segmentation Offload-Jumbo Frames are enabled where possible.

Empty Performance Charts

No graphics or data are displayed in performance charts.

When data is missing for a performance chart, the chart appears empty and you see the message `No data available`.

The causes for missing data in performance charts that are described herein are based on the assumption that the default roll-up configuration for the vCenter system has not changed. The causes include but are not limited to the following scenarios:

- Metrics introduced in ESX 5.0 are not available for hosts running earlier versions.
- Data is deleted when you remove or add objects to vCenter.
- Performance charts data for inventory objects that were moved to a new site by protection and recover is deleted from the old site and not copied to the new site.
- Performance charts data is deleted when you use VMware vMotion across vCenter instances.
- Real-time statistics are not available for disconnected hosts or powered off virtual machines.
- Real-time statistics are collected on hosts and aggregated every 5 minutes. After six data points are collected for approximately 30 minutes, they are rolled up to the vCenter database to create the 1-Day statistic. 1-Day statistics might not be available for 30 minutes after the current time, depending on when the sample period began.
- The 1-Day statistics are rolled up to create one data point every 30 minutes. If a delay occurs in the roll-up operation, the 1-Week statistics might not be available for 1 hour after the current time. It takes 30 minutes for the 1-Week collection interval, plus 30 minutes for the 1-Day collection interval.
- The 1-Week statistics are rolled up to create one data point every two hours. If a delay occurs in the roll-up operations, the 1-Month statistics might not be available for 3 hours. It takes 2 hours for the 1-Month collection interval, plus 1 hour for the 1-Week collection interval.
- The 1-Month statistics are rolled up to create one data point every day. If a delay occurs in the roll-up operations, the statistics might not be available for 1-day and 3 hours. It takes one day for the past year collection interval, plus 3 hours for the past month collection interval. During this time, the charts are empty.

- There is no solution available.

Solutions for Performance Issues With Memory Mode

Intel provides an Intel Optane Persistent Memory Mode (PMem), in which the hardware hides the DRAM as cache and exposes PMem as the system memory. Although PMem is cheaper than DRAM, it has higher access latency which could lead to performance degradation issues.

Performance degradation issues while using PMem in Memory Mode:

- If the active memory is higher than some percentage of the available DRAM memory, the VM performance could degrade as memory accesses may have to go to PMem.
- Any two random VMs can have higher level of page collision due to hardware implementation resulting in degraded VM performance even if the available DRAM memory is fully utilized.

vSphere performs real-time monitoring using vSphere Memory Monitoring and Remediation (vMMR). vMMR collects both Host level and VM level memory statistics like DRAM/PMem bandwidth, latency, miss rate which provides additional insights. This is helpful in analyzing if the host is experiencing issues due to running in Memory Mode and if there is a need to re-distribute workload. If the analysis indicates that some workloads are having performance degradation due to running on systems configured in Memory mode, the VMs can be migrated from the current host to other hosts to balance the load.

- Two preconfigured default alarms have been added based on the newly collected statistics. One at the host level (Host Memory Mode High Active DRAM Usage) and another at the VM level (Virtual Machine High PMem Bandwidth Usage). If the alarm condition is met, an event will be published to trigger the corresponding alarm. If alarm is triggered, then it is indicative that there may be some issue with Memory Mode on this system. You can further analyze, if it is a real problem by using Performance charts.
- You can also create custom alarms based on new performance metrics at cluster/host or VM level. For example, you can create an alarm if the PMem bandwidth is observed to be higher than some value. vMMR alarms only work on systems with Memory Mode. For more information on how to create a custom alarm, see the Create or Edit Alarms section.
- If the host is experiencing a performance issue, then it can be narrowed down to CPU, Memory, Disk, or Network issue by looking at the existing performance charts.

  In vSphere Client, a new **Memory** pane is added under the **Monitor**tab for both Host and VM. The host level performance chart displays read/write bandwidth, missrate for the different memory types (DRAM, PMem). The VM Level performance chart displays DRAM and PMem read bandwidth of the VM. These performance charts help customers to analyze the statistics and determine if their application workload is regressed due to Memory Mode. For example, if significant higher PMem bandwidth is observed, it is indicative of issues originating due to Memory Mode and can be investigated further.
- You can also plot custom based performance charts at host and VM level by using **Advanced** option and plotting some of Memory Mode related metrices.
- From the VMs tab of an ESX host, you can view a list containing performance information about all virtual machines that reside on the host. To display information about the Memory Mode impact on a virtual machine, click the view columns (

) icon and select the newly added Active Memory, DRAM Read Bandwidth, and PMem Read Bandwidth metrices. This is helpful in identifying the most impacted VMs.

For more information on vMMR, see the "vSphere Memory Monitoring and Remediation" section in Using Persistent Memory with vSphere.

Monitoring Guest Operating System Performance

This section describes how to install and view VMware-specific performance data for virtual machines that run Microsoft Windows operating systems. VMware provides performance counters that enable you to view data on many aspects of guest operating system performance for the Microsoft Windows Perfmon utility.

Some virtualization processes dynamically allocate available resources depending on the status, or utilization rates, of virtual machines in the environment. It can make obtaining accurate information about the resource utilization (CPU

utilization, in particular) of individual virtual machines, or applications running within virtual machines, difficult. VMware now provides virtual machine-specific performance counter libraries for the Windows Perfmon utility. It enables application administrators to access accurate virtual machine resource utilization statistics from within the Windows Perfmon utility.

You can take advantage of virtualized CPU performance counters to use performance tuning tools inside the guest operating system. See the *vSphere Virtual Machine Administration* documentation.

View Performance Statistics for Windows Guest Operating Systems

You can display VMware-specific statistics in the Microsoft Windows Perfmon utility.

Verify that a virtual machine with a Microsoft Windows operating system and VMware Tools is installed.

1. Open a console to the virtual machine and log in.
2. Select **Start > Run**.
3. Enter **Perfmon** and press **Enter**.
4. In the Performance dialog box, click **Add** .
5. In the Add Counters dialog box, select **Use local computer counters**.
6. Select a virtual machine performance object.
   Virtual machine performance object names begin with **VM**.
7. Select the counters that you want to display for that object.
8. If the performance object has multiple instances, select the instances you want to display.
9. Click **Add**.
   The Performance dialog box displays data for the selected performance object.
10. Click **Close** to close the Add Counter dialog box and return to the Performance dialog box.

Monitoring Host Health Status

You can use the vSphere Client to monitor the state of host hardware components, such as CPU processors, memory, fans, and other components.

The host health monitoring tool allows you to monitor the health of a variety of host hardware components, including:

- CPU processors
- Memory
- Fans
- Temperature
- Voltage
- Power
- Network
- Battery
- Storage
- Cable/Interconnect
- Software components
- Watchdog
- PCI devices
- Other

The host health monitoring tool presents data gathered using Systems Management Architecture for Server Hardware (SMASH) profiles. The information displayed depends on the sensors available on your server hardware. SMASH is an industry standard specification providing protocols for managing a variety of systems in the data center. For more information, see the SMASH documentation.

You can monitor host health status by connecting the vSphere Client to a vCenter system. You can also set alarms to trigger when the host health status changes.

Note:
The interpretation of hardware monitoring information is specific for each hardware vendor. Your hardware vendor can help you understand the results of the host hardware components monitoring.

View the Hardware Health Status of a Host

You can monitor the health status of host hardware in the vSphere Client

1. Select a host in the vSphere Client .
2. Click the **Monitor** tab, and click **Hardware Health**.
3. Select the type of information to view.

Option	Description
Sensors	Displays all sensors arranged in a tree view. If the status is blank, the health monitoring service cannot determine the status of the component.
DPU Sensors	Displays the storage sensors.
Alerts and Warnings	Displays alerts and warnings.
System Event Log	Displays the system event log.

Monitoring Events, Alarms, and Automated Actions

vSphere includes a user-configurable events and alarms subsystem. This subsystem tracks events happening in the vSphere environment and stores the data in log files and the vCenter database. This subsystem also enables you to specify the conditions under which alarms are triggered. Alarms can change state from mild warnings to more serious alerts as system conditions change, and can trigger automated alarm actions. This functionality is useful when you want to be informed, or take immediate action when certain events or conditions occur for a specific inventory object or group of objects.

Events

Events are records of user actions or system actions that occur on objects in the vCenter inventory or on a host. Actions that might be recorded as events include, but are not limited to, the following examples:

- A license key expires
- A virtual machine is powered on
- A user logs in to a virtual machine
- A host connection is lost

Event data includes details about the event, such as who generated it, when it occurred, and what type of event it is.

The types of events are:

Table 472: Event Types

Event Type	Description
Error	Indicates that a fatal problem has occurred in the system and terminates the process or operation.
Warning	Indicates that there is a potential risk to the system which needs to be fixed. This event does not terminate the process or operation.
Information	Describes that the user or system operation is completed successfully.
Audit	Provides important audit log data which is crucial for the security framework. The audit log data includes information about what is the action, who did it, when it occurred, and the IP address of the user. You can learn more about this in the *vSphere Security* guide.

Alarms

Alarms are notifications that are activated in response to an event, a set of conditions, or the state of an inventory object. An alarm definition consists of the following elements in the vSphere Client:

- Name and description - Provides an identifying label and description.
- Targets - Defines the type of object that is monitored.
- Alarm Rules - Defines the event, condition, or state that triggers the alarm and defines the notification severity. It also defines operations that occur in response to triggered alarms.
- Last modified - The last modified date and time of the defined alarm.

Alarms have the following severity levels:

- Normal – green
- Warning – yellow
- Alert – red

Alarm definitions are associated with the object selected in the inventory. An alarm monitors the type of inventory objects specified in its definition.

For example, you might want to monitor the CPU usage of all virtual machines in a specific host cluster. You can select the cluster in the inventory, and add a virtual machine alarm to it. When enabled, that alarm monitors all virtual machines running in the cluster and triggers when any one of them meets the criteria defined in the alarm. To monitor a specific virtual machine in the cluster, but not others, select that virtual machine in the inventory and add an alarm to it. To apply the same alarms to a group of objects, place those objects in a folder and define the alarm on the folder.

> **Note:**
>
> You can activate, deactivate, and modify alarms only from the object in which the alarm is defined. For example, if you defined an alarm in a cluster to monitor virtual machines, you can only modify that alarm through the cluster. You cannot change the alarm at the individual virtual machine level. However, you can disable or re-enable an alarm on a specific leaf inventory entity. For example, if a virtual machine related alarm is enabled on a vCenter level it can be disabled on a specific virtual machine and thus the alarm status for the specific entity will not be calculated and alarm state will be reset to gray.

Alarm Actions

Alarm actions are operations that occur in response to the trigger. For example, you can have an email notification sent to one or more administrators when an alarm is triggered.

> **Note:**
> Default alarms are not preconfigured with actions. You must manually set what action occurs when the triggering event, condition, or state occurs.

Live Refresh of Recent Tasks and Alarms

Live refresh for recent tasks and alarms that result from operations that other users perform in your environment is enabled by default. Now, all recent tasks and alarms are visible to all of the users with permissions to view them.

View Events for an Inventory Object

You can view events associated with a single object or view all vSphere events. The events list for a selected inventory object includes events associated with child objects. vSphere keeps information about tasks and events for a specified time period. You can configure this time period. By default, the period is 30 days.

1. To view the events associated with an inventory object, select the object in the vSphere Client.
2. Click the **Monitor** tab, and click **Events**.
3. Click the expand row icon in each row to view the details for the corresponding event.

   You can expand multiple rows simultaneously.

4. Optional: Click the **Open in new tab** button at the top right corner of the datagrid to open the events view in a new tab.
   When you open the new tab from **Entity** > **Monitor** > **Events** menu, the name of the entity is displayed next to the title in the new tab. Click **REFRESH** at the top right of the datagrid in the new tab to refresh the events in the current page.
5. Click the **Filter** option to filter the events based on event category, type of user, and a selected time range.
6. Optional: Select an event and click **Copy to clipboard** option to copy the event to clipboard.
7. Optional: To view the event that triggered an alarm, click the **Alarms** tab in the bottom panel adjacent to the `Recent Tasks` tab.
   a) Select an alarm and navigate to the **Triggering Event**column.

      The **Triggering Event**column on the **Alarms** tab displays a hyperlink text in the following form: `"object-name: alarm-name"`. Alarms that are not triggered by a particular event, you see plain text in the **Triggering Event**column.
   b) Click on the hyperlink text to navigate to the **Events** view for the object of the alarm.

      In the **Events** view for the object of the alarm, you see detailed information about the specific event that triggered the alarm.
   c) To return to the default event-list view for the object of the alarm, click the **Back to all events** button.

Export Events in vSphere Client

You can export selected or all events into a `.csv` file.

1. Select an inventory object in the vSphere Client.
2. Click the **Monitor** tab, and click **Events**.
3. Click **Export** to export all or selected events into a `.csv` file.
   - Select **All events** option from the dropdown menu to export all the events.
   - Select **Only selected** option from the dropdown menu to export only the selected events from the event list.
   - Select **Advanced export** option from the dropdown menu to export events based on customized settings, like event category, user type, time range and so on.

View System Event Log

vSphere records events in the vCenter database. System log entries include information, such as who generated the event, when the event was created, and the type of event.

- Required privilege: **Global** > **Diagnostics**

1. Select a host in the vSphere Client navigator.
2. Click **Monitor** tab, and click **Hardware Health**.
3. Click **System event log**.

Export System Event Log Data

You can export all or part of the system event log data stored in the vCenter database.

Required Role: **Read-only**

1. Select a host in the vSphere Client.
2. Click the **Monitor** tab, and click **Hardware Health**.
3. Click the **System event log** option.
4. Click **Export**.

Consolidating Identical Events

The event burst filter monitors the incoming stream of events for identical events over a short period of time. To optimize the storage size of events, events that occur repeatedly are consolidated into a single event before storing them in the database or the remote syslog server.

Event bursts can be seen in various scenarios, which include,

- An existing hardware failure.
- An automated solution which logs in and logs out of vCenter often.

The events burst filter is enabled by default. It consolidates all types of events except those events on the allow list. An event burst is defined as more than one identical event per second. Two events are identical if,

- The events are of the same type.
- The events are on the same inventory object.
- The events are issued by the same user.

> **Note:**
> The remaining event-specific data is not used to determine if two events are identical.

An event burst is detected after seeing 30 identical events for less than 30 seconds. These 30 events are stored in the vCenter database or the remote syslog server. Consolidation of the bursting events starts from the 31st event. The bursting events do not go into the database, but the consolidated event gets stored in the database.

In cases of bursting events, the burst filter affects only the events that go to the database and remote syslog stream. Alarms triggered by bursting events and `EventHistoryCollector` objects are not affected.

- `com.vmware.vc.EventBurstStartedEvent` - the start of an event burst.
- `com.vmware.vc.EventBurstEndedEvent` - the end of an event burst.
- `com.vmware.vc.AllEventBurstsEndedEvent` - the end of all the events bursts.
- `com.vmware.vc.EventBurstCompressedEvent` - the consolidated event after the event burst.

Each of this event contains,

- `eventType` - the event type of the bursting event.
- `objectId` - the entity of the bursting event.
- `userName` - the user name of the bursting event.

Also the compressed event contains,

- `count` - the number of identical events since the start of an event burst. The count starts from the 31st event.
- `burstStartTime` - Time of the 31st event of an event burst.

> **Note:**
> The timestamps of the burst filter events are not related to the bursting events.

Configure Events Burst Filter

You can configure the basic and advanced settings for the events burst filter in the advanced vCenter settings.

Burst filter can be in the following configurations.

- **Enabled**: If `compressToDb` or `compressToSyslog` are enabled, then the burst filter detects bursts, posts events for them, and consolidates events before storing them in the database or sending them to the remote syslog server.
- **Disabled**: The burst filter does not detect bursts, does not post events for them, and does not consolidate events before storing them in the database or sending them to the remote syslog server. If both `compressToDb` and

`compressToSyslog` are disabled, then the burst filter detects bursts and posts events for them but does not consolidate events.

1. In the vSphere Client, navigate to the vCenter Server instance.
2. Select the **Configure** tab.
3. Click **Settings** > **Advanced Settings**.
4. Click **Edit**.
5. Click on the filter text box present in the **Name** column of the table header. Type `vpxd.event`, and press Enter.
6. For the basic setting, follow the steps.
   a) Enable or disable the vpxd.event.burstFilter.compressToDb option.

      This option enables you to compress the bursting of events in the database. The default value for the setting is enabled.
   b) Enable or disable the vpxd.event.burstFilter.compressToSyslog option.

      This option enables you to compress the bursting of events in the syslog server. The default value for the setting is disabled.
   c) Click **Save** to close the **Edit Advanced vCenter Server Settings**.
7. The burst settings are not visible with their default values in **Advanced Settings**. However, a key-value can be entered for the configuration to take effect.

   Note:
   Adding or modifying advanced settings might result in an unstable environment. Configuration parameters cannot be removed once they are added.

   a) Configure a key-value to enable the config.vpxd.event.burstFilter.enabled option.

      This parameter enables burst detection of the VC database and syslog. On disabling burst filter, bursts are not detected and the events are stored in the database or remote syslog server by default without compression. The default value is Enabled.
   b) Configure a key-value to enable the config.vpxd.event.burstFilter.rateEvents option.

      You can configure the number of events after which a burst is detected. If the configured limit is reached, the event sequence is considered a burst and is compressed while storing it in the VC database. The default value is 30. If the default is set, then the first 30 events are not compressed. They are used to detect a burst. If a burst is detected, then a single consolidated event with their count replaces the subsequent events after the first 30 events.

      Note:
      It does not compress the real-time stream of events like alarms and `EventManager` property collector updates.
   c) Configure a key-value to enable the config.vpxd.event.burstFilter.rateSeconds option.

      This parameter allows you to configure the number of seconds the data is stored for an event from the last time the event occurred. Higher values load the burst filter cache, and the bursts are detected even if the incoming flow of events is not even. Lower values exert lesser load on burst filter cache, and bursts may not be detected if there is a temporary halt of incoming events. The default value is 30 seconds.
   d) Configure a key-value to enable the config.vpxd.event.burstFilter.cacheSize option.

      The number of unique events tracked by the burst filter. When the cache is full, the burst filter stops monitoring the incoming new unique events and passes them through the VC database and syslog. The default value is 128000.

      Note:
      Setting a lower value than default reduces the memory footprint but it also reduces the capacity to detect bursts for a large inventory.
   e) Configure a key-value to enable the config.vpxd.event.burstFilter.whitelist option.

      You can configure the burst filter to compress all types of events except the events that are on the allow list. This parameter allows you to separate event types. It can avoid in monitoring few event types by the burst filter. The default value is `<whiteList>vim.event.UserLoginSessionEvent;vim.event.UserLogoutSessionEvent</whiteList>`

f) Configure a key-value to enable the config.vpxd.event.burstFilter.compressRatio option.

When the burst filter starts compressing an event, it posts a compressed event when the burst ends or on every X event. If a burst occurs for days, it posts some events instead of muting them all until the burst ends. The default value is 3600. The events are logged every 3600 events.

g) Click **Save** to close the **Edot Advanced vCenter Server Settings**.

Note:

Restart the `vmware-vpxd` service for all changes to take effect. For more information how to restart services in the vSphere Client, see *vCenter and Host Management* documentation.

Streaming Events to a Remote Syslog Server

After you enable remote streaming, vCenter starts sending newly generated events to a remote syslog server as syslog protocol messages over the network in real time. The events are not sent as log files. vCenter streams each event individually as it occurs, and only events generated after streaming is enabled are transmitted.

All syslog messages begin with a specific prefix. You can distinguish the vCenter events from other syslog messages by their `Event` prefix.

The syslog protocol limits the length of syslog messages to 1024 characters. Messages that are longer than 1024 characters split into multiple syslog messages.

In the syslog server, events have the following format:

```
<syslog-prefix> : Event [eventId] [partInfo] [createdTime] [eventType] [severity] [user]
 [target] [chainId] [desc]
```

Item	Description
syslog-prefix	Displays the syslog prefix. The <*syslog-prefix*> is determined by the remote syslog server configuration.
eventId	Displays the unique ID of the event message. The default value is Event.
partInfo	Displays whether the message is split into parts.
createdTime	Displays the time when the event was generated.
eventType	Displays the event type.
severity	Displays whether the event is a piece information, a warning, or an error.
user	Displays the name of the user who generated the event.
target	Displays the object the event refers to.
chainId	Displays information about the parent or the group ID.
desc	Displays the description of the event.

Split of Long Event Message into Multiple Syslog Messages

Events that are longer than 1024 characters split into multiple syslog messages in the following manner:

```
<syslog-prefix> : Event [eventId][1-X][payload-part-1]<syslog-prefix> : Event [eventId][2-
X][payload-part-2]
...
<syslog-prefix> : Event [eventId][X-X][payload-part-X]
```

The *X* stands for the number of the event message parts.

Forwarding vCenter Log Files to Remote Syslog Server

Remote event streaming is distinct from forwarding vCenter log files to a remote syslog server. Event streaming sends individual vCenter events as they occur, while log file forwarding sends the contents of vCenter log files. In addition to streaming syslog messages to a remot syslog server, you can also forward the log files to a remote syslog server to conduct an analysis of your logs. For more information, see Forward vCenter Log Files to Remote Syslog Server.

Activate the Streaming of Events to a Remote Syslog Server

You can activate the option to have events written to the vCenter streaming facility. Streaming events is supported only for vCenter. The streaming of events to a remote syslog server is deactivated by default. You can enable and configure the streaming of vCenter events to a remote syslog server from the vCenter Management Interface.

After you activate remote streaming, vCenter sends newly generated events to a remote syslog server — a host on your network that runs syslog receiver software. The remote syslog server must be configured to receive syslog messages before you activate event streaming. For information about configuring the remote destination host, see Forward vCenter Log Files to Remote Syslog Server.

1. In the vSphere Client, navigate to the vCenter Server instance.
2. Select the **Configure** tab.
3. Expand **Settings** option, and select **Advanced Settings**.
4. Click **EDIT SETTINGS**.
5. Click on the filter text box present in the **Name** column of the table header. Type `vpxd.event`, and press Enter.
6. Activate or deactivate the vpxd.event.syslog.enabled option.
   By default, this option is deactivated.
7. Click **SAVE**.

Retention of Events in the vCenter Database

You can configure vCenter to retain events in the database for a limited period of time. Discarding events periodically ensures optimal performance of the database.

In new installations of vCenter, the event clean up option is enabled by default and the default number of days to retain event messages in the database is 30. You can change this value to the number of days that you want to retain the event messages in the database.

If you are upgrading or migrating from vCenter9.0 or earlier, and you had the event cleanup option enabled, your setting to retain events is preserved after the upgrade or migration to vCenter 6.5.

After the retention period ends, the events are deleted from the database. However, there might be latency in the deletion of the events that are older than the configured retention setting.

Configure Database Settings

You can configure the maximum number of database connections that can occur simultaneously. To limit the growth of the vCenter database and save storage space, you can configure the database to discard information about tasks or events periodically.

> **Note:**
> Do not use the database retention options, if you want to keep a complete history of tasks and events for your vCenter.

1. In the vSphere Client, navigate to the vCenter instance.
2. Select the **Configure** tab.
3. Under **Settings**, select **General**.
4. Click **Edit**.
5. In **Edit vCenter general settings** window, click **Database**.
6. In the**Maximum connections** field, enter the required number of connections.

> **Note:**
> Do not change this value unless one of these issues exists in your system.

- If your vCenter system performs many operations frequently and performance is critical, increase the number of connections.
- If the database is shared and connections to the database are costly, decrease the number of connections.

7. Enable **Task cleanup** option for vCenter to delete the retained tasks periodically.
8. Optional: In **Task retention (days)** field, enter a value (in days).
   Information about tasks that are performed on this vCenter system is discarded after the specified number of days.
9. Enable the **Event cleanup** option for vCenter to clean up the retained events periodically.
10. Optional: In the **Event retention (days)** field, enter a value (in days).
    Information about events for this vCenter system is discarded after the specified number of days.

    Note:
    Monitor vCenter database consumption and disk partition in the vCenter Appliance Management Interface.

    Warning:

    Increasing the events retention to more than 30 days results in significant increase of vCenter database size and can shutdown the vCenter. Ensure that you increase the vCenter database accordingly.
11. Click **Save**.
12. Restart the vCenter to apply changes manually.

View Triggered Alarms

Triggered alarms for a selected inventory object are visible in several locations throughout the vSphere Client.

1. In the vSphere Client, select an inventory object.
2. On the **Monitor** tab, click **Issues and Alarms** > **Triggered Alarms**.

Create or Edit Alarms

To monitor your environment, you can create alarm definitions in the vSphere Client . You can access alarm definitions from the **Configure** tab.

Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**

1. Select an inventory object and click the **Configure** tab.
2. Click **Alarm Definitions**.
3. Select your task.
   - Click **Add** to add an alarm.
   - Select an alarm and click **Enable** to activate an alarm.
   - Select an alarm and click **Disable** to deactivate an alarm.
   - Select an alarm and click **Delete** to delete an alarm.
   - Select an alarm and cick **Edit** to edit an alarm.

Define an Alarm in the vSphere Client

In the vSphere Client, you define alarms in the **Alarm Definitions** wizard. You access the **Alarm Definitions** wizard from the **Configure** tab for a selected object

- Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**.
- Start the **Alarm Definitions** wizard. See Create or Edit Alarms.

1. Specify the alarm name, description, and target. See Specify Alarm Name, Description, and Target.
2. Specify one or more alarm rules. See Specify Alarm Rules.
3. Specify alarm reset rules. See Specify Alarm Reset Rules.
4. Review your selections and enable the alarm. See Review and Enable Alarm.

Specify Alarm Name, Description, and Target

Settings of an alarm definition include alarm name, description, and target. \

- Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**
- Start the **Alarm Definitions** wizard. See Create or Edit Alarms.

1. Type a name and description for the alarm.
2. Select the type of inventory object that this alarm monitors from the **Target type** drop-down menu

   Depending on the type of target that you choose to monitor, the summary that follows the **Target**, change.
3. Click **Next**.

   Note:
   Depending on the type of activity that you choose to monitor, the options on the Alarm Rule page, change.

Specify Alarm Rules

You can select and configure the events, states, or conditions that trigger the alarm from the **Alarm Rule** page in the **New Alarm Definition** wizard.

- Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**
- Start the **Alarm Definitions** wizard. See Create or Edit Alarms.

An alarm definition must contain at least one trigger before you can save it.

1. Select a trigger from the drop-down menu.

   The combined event triggers are displayed. You can set the rule for a single event only. You must create multiple rules for multiple events.
2. Click **Add Argument** to select an argument from the drop-down menu.

   It supports **ALL** based expression, option to select **ANY** is not available. You must create a separate alarm definition for each trigger. The **OR** operator is not supported in the vSphere Client. However, you can combine more than one condition trigger with **AND** operator.
3. Select an operator from the drop-down menu.
4. Select an option from the drop-down menu to set the threshold for triggering an alarm
5. Select severity of the alarm from the drop-down menu.

   You can set the condition to either **Show as Warning** or **Show as Critical**, but not for both. You must create a separate alarm definition for warning and critical status. **Keep the target's current state** option can be enabled only for an event based alarm condition. For example, if **Cannot power off the VM** and **Keep the target's current state** is selected for an alarm rule, then it cannot be combined with any other alarm rule with different severity like **Warning** or **Critical**. These alarms will not have reset rules as the alarm rule does not change the state of the target object.

6. Send email notifications
   a) To send email notifications when alarms are triggered, enable **Send email notifications**.
   b) In the **Email to**, enter recipient addresses. Use commas to separate multiple addresses.
7. To send traps when alarms are triggered on a vCenterinstance, enable **Send SNMP traps**.
8. Run scripts
   a) To run scripts when alarms are triggered, enable **Run script**.
   b) In **Run this script** column, enter the full path to the script.

      ```
      /var/myscripts/myAlarmActionScript
      ```

      You can use environment variables to define more complex scripts and attach them to multiple alarms or inventory objects so that the alarm action occurs when the alarm triggers.

Table 473: Alarm Environment Variables

Variable Name	Variable Description	Supported Alarm Type
VMWARE_ALARM_NAME	Name of the triggered alarm.	Condition, State, Event
VMWARE_ALARM_ID	MOID of the triggered alarm.	Condition, State, Event
VMWARE_ALARM_TARGET_NAME	Name of the entity on which the alarm triggered.	Condition, State, Event
VMWARE_ALARM_TARGET_ID	MOID of the entity on which the alarm triggered.	Condition, State, Event
VMWARE_ALARM_OLDSTATUS	Old status of the alarm.	Condition, State, Event
VMWARE_ALARM_NEWSTATUS	New status of the alarm.	Condition, State, Event
VMWARE_ALARM_TRIGGERINGSUMMARY	Multiline summary of the alarm.	Condition, State, Event
VMWARE_ALARM_DECLARINGSUMMARY	Single-line declaration of the alarm expression.	Condition, State, Event
VMWARE_ALARM_ALARMVALUE	Value that triggered the alarm.	Condition, State
VMWARE_ALARM_EVENTDESCRIPTION	Description text of the alarm status change event.	Condition, State
VMWARE_ALARM_EVENTDESCRIPTION	Description of the event that triggered the alarm.	Event
VMWARE_ALARM_EVENT_USERNAME	User name associated with the event.	Event
VMWARE_ALARM_EVENT_DATACENTER	Name of the datacenter in which the event occurred.	Event
VMWARE_ALARM_EVENT_COMPUTERESOURCE	Name of the cluster or resource pool in which the event occurred.	Event
VMWARE_ALARM_EVENT_HOST	Name of the host on which the event occurred.	Event
VMWARE_ALARM_EVENT_VM	Name of the virtual machine on which the event occurred.	Event
VMWARE_ALARM_EVENT_NETWORK	Name of the network on which the event occurred.	Event
VMWARE_ALARM_EVENT_DATASTORE	Name of the datastore on which the event occurred.	Event
VMWARE_ALARM_EVENT_DVS	Name of the vNetwork Distributed Switch on which the event occurred.	Event

If your script does not make use of the alarm environment variables, include any necessary parameters in the configuration field. Enclose parameters in curly brackets. For example:

```
/var/myscripts/myAlarmActionScript {alarmName} {targetName}
```

Check that the script can be executed:

```
chmod +x /var/myscripts/myAlarmActionScript
```

Starting with vCenter 8.0, the owner of the script must have vpxuser privileges :

```
chown vpxd /var/myscripts/myAlarmActionScript
```

For more information about scripts which require root user privileges, see Knowledge Base article 313285.

9. (Optional) Configure alarm transitions and frequency.
10. Select an advanced action from the drop-down menu.

    You can define the advanced actions for virtual machine and hosts. These advanced actions are applicable only for virtual machines and hosts. There are different sets of advanced actions based on target types of virtual machines and hosts.

    You can add multiple advanced actions for an alarm.
11. (Optional) Configure frequency for advanced actions.
12. Click **Add Another Rule** to add an additional rule for an alarm.
13. Click **Duplicate Rule** to create an identical rule for an alarm.
14. Click **Remove Rule** to remove an existing rule set for an alarm.

Click **Next** to set the Reset Rule.

Specify Alarm Reset Rules

You can select and configure the events, states, or conditions that trigger the alarm from the **Reset Rule** page in the **New Alarm Definition** wizard.

- Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**
- Start the **Alarm Definitions** wizard. See Create or Edit Alarms.

You can set the Alarm Reset Rule

1. Enable the **Reset the alarm to green** option.
2. Select a trigger from the drop-down menu.

   The combined event triggers are displayed. You can set the rule for a single event only. You must create multiple rules for multiple events.
3. Click **Add Argument** to select an argument from the drop-down menu.

   It supports **ALL** based expression, option to select **ANY** is not available. You must create a separate alarm definition for each trigger. The **OR** operator is not supported in the vSphere Client. However, you can combine more than one condition trigger with **AND** operator.

4. Select an operator from the drop-down menu.
5. Send email notifications
   a) To send email notifications when alarms are triggered, enable **Send email notifications**.
   b) In the **Email to**, enter recipient addresses. Use commas to separate multiple addresses.
6. To send traps when alarms are triggered on a vCenter instance, enable **Send SNMP traps**.
7. Run scripts. To run scripts when alarms are triggered, enable **Run script**.
8. In **Run this script** column, type script or command information:

For this type of command...	Enter this...
EXE executable files	Full pathname of the command. For example, to run the `cmd.exe` command in the `C:\tools` directory, type: `c:\tools\cmd.exe`.
BAT batch file	Full pathname of the command as an argument to the `c:\windows\system32\cmd.exe` command. For example, to run the `cmd.bat` command in the `C:\tools` directory, type: `c:\windows\system32\cmd.exe /c c:\tools\cmd.bat`. **Note:** The command and its parameters must be formatted into one string.

   If your script does not make use of the alarm environment variables, include any necessary parameters in the configuration field. Enclose parameters in curly brackets. For example:

```
c:\tools\cmd.exe {alarmName} {targetName}
c:\windows\system32\cmd.exe /c c:\tools\cmd.bat {alarmName} {targetName}
```

   The script can run on any platform. You must provide the path to the script and argument keys. For example:

```
/var/myscripts/myAlarmActionScript {alarmName} {targetName}
```

9. (Optional) Configure alarm transitions and frequency.
10. Select an advanced action from the **Add Advanced Actions** drop-down menu.

    You can add multiple advanced actions for an alarm reset rule. You can define the advanced actions for virtual machines and hosts. These advanced actions are applicable only for virtual machines and hosts. There are different sets of advanced actions based on target types of virtual machines and hosts.

    You can add multiple advanced actions for an alarm.
11. (Optional) Configure frequency for advanced actions.
12. Click **Add Another Rule** to add an additional reset rule for an alarm.
13. Click **Duplicate Rule** to create an identical reset rule for an alarm.
14. Click **Remove Rule** to remove an existing reset rule set for an alarm.

    Click **Next** to review the alarm definition.

Review and Enable Alarm

You can review and enable the alarm in the vSphere Client

- Required Privilege: **Alarms** > **Create alarm** or **Alarms** > **Modify alarm**
- Start the **Alarm Definitions** wizard. See Create or Edit Alarms.

After setting the alarm rule, review the alarm before enabling the alarm.

1. Review the **Alarm Name**, **Description**, **Targets**, and **Alarm Rule**.
2. (Optional) Configure alarm transitions and frequency.
3. Select **Enable this alarm** to enable the alarm.

The alarm is enabled.

Disable Alarms on Specific Objects

vSphere includes a feature to disable alarms on specific leaf object in the inventory, even if those are defined on one of its parents. This feature is helpful, if a specific alarm is not applicable for a given object.

Required Privilege: Alarms.Disable or enable alarm on entity

You can access alarms on the **Configure** tab for a selected inventory object.

> **Note:**
>
> The **Enable** and **Disable** buttons are automatically enabled or disabled.

1. Select an inventory object, click the **Configure** tab, and click **Alarm Definitions**.
2. Select an alarm.
3. Click **Disable**.

Acknowledge Triggered Alarms

After you acknowledge an alarm in the vSphere Client, its alarm actions are discontinued. Alarms are not cleared or reset when they are acknowledged.

Required privilege: **Alarm** > **Alarm Acknowledge**

Acknowledging an alarm lets other users know that you are taking ownership of the issue. For example, a host has an alarm set to monitor CPU usage. It sends an email to an administrator when the alarm is triggered. The host CPU usage spikes, triggering the alarm which sends an email to the host's administrator. The administrator acknowledges the triggered alarm to let other administrators know the problem is being addressed, and to prevent the alarm from sending more email messages. The alarm, however, is still visible in the system.

1. In the vSphere Client, click the **Alarms** tab in the bottom pane.
2. Right-click an alarm and select **Acknowledge**.

   You can also acknowledge the alarm from the Monitor tab for an object.

   1. Select an inventory object in the object navigator.
   2. Click the **Monitor** tab.
   3. Click **Issues and Alarms**, and click **Triggered Alarms**.
   4. Select an alarm and select **Acknowledge**.

Reset Triggered Event Alarms

An alarm triggered by an event might not reset to a normal state if vCenter does not retrieve the event that identifies the normal condition. In such cases, reset the alarm manually in the vSphere Client to return it to a normal state.

Required privilege: **Alarm** > **Set Alarm Status**

1. In the vSphere Client, click the **Alarms** tab in the bottom pane.
2. Right-click an alarm and **Reset to green**.

   You can also reset triggered alarms from the **Monitor** tab of an inventory object.

   1. Select an inventory object.
   2. Click the **Monitor** tab.
   3. Click **Issuesand Alarms**, and click **Triggered Alarms**.
   4. Select the alarms you want to reset.

Use Shift+left-click or Ctrl+left-click to select multiple alarms is supported in the vSphere Client.

5. Right-click an alarm and select **Reset to Green**.

Preconfigured vSphere Alarms

vCenter provides a list of default alarms, which monitor the operations of vSphere inventory objects. You must only set up actions for these alarms.

Some alarms are stateless. vCenter does not keep data on stateless alarms, does not compute, or display their status. Stateless alarms cannot be acknowledged or reset. Stateless alarms are indicated by an asterisk next to their name.

Table 474: Default vSphere Alarms

Alarm Name	Description
Host connection and power state	Monitors the power state of the host and whether the host is reachable.
Host CPU usage	Monitors host CPU usage.
Host memory usage	Monitors host memory usage.
Virtual machine CPU usage	Monitors virtual machine CPU usage.
Virtual machine memory usage	Monitors virtual machine memory usage.
Datastore usage on disk	Monitors datastore disk usage. **Note:** This alarm controls the Status value for datastores in vSphere Client. If you deactivate this alarm, the datastore status is displayed as Unknown.
Virtual machine CPU ready	Monitors virtual machine CPU ready time.
Virtual machine total disk latency	Monitors virtual machine total disk latency.
Virtual machine disk commands canceled	Monitors the number of virtual machine disk commands that are canceled.
Virtual machine disk reset	Monitors the number of virtual machine bus resets.
License inventory monitoring	Monitors the license inventory for compliance.
License user threshold monitoring	Monitors whether a user-defined license threshold is exceeded.
License capacity monitoring	Monitors whether a license capacity is exceeded.
The host license edition is not compatible with the vCenter license edition	Monitors the compatibility of the vCenter and host license editions.
Host flash capacity exceeds the licensed limit for vSAN	Monitors whether the flash disk capacity on the host exceeds the limit of the vSAN license.
Expired vSAN license	Monitors the expiry of the vSAN license and the end of the evaluation period.
Errors occurred on the disk(s) of a vSAN host	Default alarm that monitors whether the host disks in the vSAN cluster have errors.
Timed out starting Secondary VM *	Monitors whether starting a secondary virtual machine has timed out.
No compatible host for Secondary VM	Monitors the availability of compatible hosts on which a secondary virtual machine can be created and run.
Virtual machine Fault Tolerance state changed	Monitors changes in the Fault Tolerance state of a virtual machine.
Virtual Machine Fault Tolerance vLockStep interval Status Changed	Monitors changes in the Fault Tolerance Secondary vLockStep interval.
Host processor status	Monitors the host processors.

Alarm Name	Description
Host memory status	Monitors host memory usage.
Host hardware fan status	Monitors host fans.
Host hardware voltage	Monitors host hardware voltage.
Host hardware temperature status	Monitors the temperature status of the host hardware.
Host hardware power status	Monitors the host power status.
Host hardware system board status	Monitors the status of host system boards.
Host battery status	Monitors the battery status of hosts.
Status of other host hardware objects	Monitors other host hardware objects.
Host storage status	Monitors host connectivity to storage devices.
Host IPMI System Event Log status	Monitors the capacity of the IPMI system event log.
Host Baseboard Management Controller status	Monitors the status of the Baseboard Management Controller.
Host error *	Monitors host error and warning events.
Virtual machine error *	Monitors virtual machine error and warning events.
Host connection failure *	Monitors host connection failures.
Unmanaged workload detected on SIOC-enabled datastore	Monitors the unmanaged I/O workload on a SIOC-enabled datastore.
Thin-provisioned volume capacity threshold exceeded	Monitors whether the thin provisioning threshold on the storage array exceeds for volumes backing the datastore.
Datastore capability alarm	Monitors the change in the capability status for volumes backing the datastore.
VASA provider disconnected	Monitors the changes in the connection state of VASA providers.
VASA Provider certificate expiration alarm	Monitors whether VASA provider certificates are getting close to their expiry date.
VM storage compliance alarm	Monitors the virtual disk compliance with the object-based storage.
Datastore compliance alarm	Monitors whether the virtual disk on the datastore complies with the object-based storage.
Refreshing CA certificates and CRLs for a VASA provider failed	Monitors whether the refreshing of CA certificates and CRLs for some of the VASA providers has failed.
Insufficient vSphere HA failover resources	Monitors the sufficiency of failover cluster resources required for vSphere High Availability.
vSphere HA failover in progress	Monitors the failover progress of vSphere High Availability.
Cannot find vSphere HA primary agent	Monitors whether vCenter is able to connect to a vSphere High Availability primary agent.
vSphere HA host status	Monitors the host health status reported by vSphere High Availability.
vSphere HA virtual machine failover failed	Monitors whether a failover operation that uses vSphere High Availability failed.
vSphere HA virtual machine monitoring action	Monitors whether vSphere High Availability has restarted a virtual machine.
vSphere HA virtual machine monitoring error	Monitors whether vSphere High Availability failed to reset a virtual machine.
vSphere HA VM Component Protection could not power off a virtual machine	Monitors whether vSphere High Availability VM Component Protection cannot power off a virtual machine with an inaccessible datastore.

Alarm Name	Description
License error *	Monitors license errors.
Health status changed *	Monitors changes to service and extension health status.
Virtual machine component protection restart error	Monitors whether the vSphere HA VM Component Protection fails to restart a virtual machine.
Storage DRS recommendation	Monitors Storage DRS recommendations.
Storage DRS is not supported on a host	Monitors and alerts when Storage DRS is not supported on a host.
Datastore cluster is out of space	Monitors whether a datastore cluster runs out of disk space.
Datastore is in multiple datacenters	Monitors whether a datastore in a datastore cluster is visible in more than one data center.
vSphere Distributed Switch VLAN trunked status	Monitors changes in vSphere Distributed Switch VLAN trunked status.
vSphere Distributed Switch MTU matched status	Monitors changes in vSphere Distributed Switch MTU matched status.
vSphere Distributed Switch MTU supported status	Monitors changes in vSphere Distributed Switch MTU supported status.
vSphere Distributed Switch teaming matched status	Monitors changes in vSphere Distributed Switch teaming matched status.
Virtual Machine network adapter reservation status	Monitors changes in the reservation status of a virtual machine network adapter.
Virtual machine Consolidation Needed status	Monitors changes in the virtual machine Consolidation Needed status.
Host virtual flash resource status	Monitors the Flash Read Cache resource status on the host.
Host virtual flash resource usage	Monitors the Flash Read Cache resource usage on the host.
Registration/unregistration of a VASA vendor provider on a vSAN host fails	Default alarm that monitors whether the registration or unregistration of a VASA vendor provider on a vSAN host fails.
Registration/unregistration of third-party IO filter storage providers fails on a host	Default alarm that monitors whether vCenter fails to register or unregister third-party IO filter storage providers on a host.
Service Control Agent Health Alarm	Monitors the health status of the VMware Service Control Agent.
Identity Health Alarm	Monitors the health status of the Identity Management Service.
vSphere Client Health Alarm	Monitors the health status of the vSphere Client.
ESX Agent Manager Health Alarm	Monitors the health status of the ESX Agent Manager.
Message Bus Config Health Alarm	Monitors the health status of the Message Bus Configuration Service.
Cis License Health Alarm	Monitors the health status of the License Service.
vCenter Health Alarm	Monitors the health status of vCenter.
Database Health Alarm	Monitors the database health status. When database space reaches 80%, vCenter displays a warning event. When database space reaches 95%, vCenter displays an error event and shuts down. You can clean up the database, or increase database storage capacity, and start vCenter. The alarm is triggered only for PostgreSQL and Microsoft SQL Server database health issues, and does not work with Oracle databases.
Data Service Health Alarm	Monitors the health status of the Data Service.
RBD Health Alarm	Monitors the health status of the vSphere Auto Deploy Waiter.

Alarm Name	Description
vService Manager Health Alarm	Monitors the health status of the vService Manager.
Performance Charts Service Health Alarm	Monitors the health status of the Performance Charts Service.
Content Library Service Health Alarm	Monitors the health status of the VMware Content Library Service.
VMware vSphere ESX Dump Collector Health Alarm	Monitors the health status of the VMware vSphere ESX Dump Collector Service.
VMware vAPI Endpoint Service Health Alarm	Monitors the health status of the VMware vAPI Endpoint Service.
VMware vSphere Profile-Driven Storage Service Health Alarm	Monitors the health status of the VMware vSphere Profile-Driven Storage Service.
VMware vFabric Postgres Service Health Alarm	Monitors the health status of the VMware vFabric Postgres Service.
ESX Host Certificates Update Failure Status	Monitors whether the update of the ESX host certificates failed.
ESX Host Certificate Status	Monitors the certificate status of an ESX host.
ESX Host Certificate Verification Failure Status	Monitors whether the verification of an ESX host certificate failed.
vSphere vCenter Host Certificate Management Mode	Monitors changes in the certificate management mode of vCenter.
Root Certificate Status	Monitors whether a root certificate is getting close to its expiration date.
GPU ECC Uncorrected Memory Alarm	Monitors the GPU ECC uncorrected memory status.
GPU ECC Corrected Memory Alarm	Monitors the GPU ECC corrected memory status.
GPU Thermal Condition Alarm	Monitors the GPU Thermal condition status.
Network connectivity lost	Monitors the network connectivity on a virtual switch.
Network uplink redundancy lost	Monitors network uplink redundancy on a virtual switch.
Network uplink redundancy degraded *	Monitors network uplink redundancy degradation on a virtual switch.
VMKernel NIC not configured correctly *	Monitors incorrectly configured VMkernel NICs.
Cannot connect to storage *	Monitors host connectivity to a storage device.
Migration error *	Monitors whether a virtual machine cannot be migrated or relocated, or is orphaned.
Exit standby error	Monitors whether a host cannot exit standby mode.

View Alarm Definitions for Preconfigured vSphere Alarms

You can view the definitions of all preconfigured vSphere alarms and all alarms that you configured for each inventory object.

When a preconfigured alarm is triggered and you want to see what rules triggered the alarm, you can view the alarm definition.

1. Select an object from the inventory, and click the **Configure** tab.
2. In the left navigation pane, click **Alarm Definitions**.
   A list with all configured alarms for the selected object appears.
3. Optional: Use the filters above each column to quickly find the alarm definition.
4. Click the arrow to the left of the alarm name.

You see all information related to the configuration of the alarm, including the description of the alarm, the target object, and the alarm rules that trigger the alarm.

Monitoring vCenter Capacity by Using the Configuration Maximums Checker

Starting with vSphere 9.1, you can check at any point the configuration capacity of your vCenter instance by using the Configuration Maximums checker integrated into the vSphere Client. You no longer need to spend time performing manual verification of the allowed configuration maximums and inspecting your current infrastructure against the defined limits. Also, the Configuration Maximums checker provides you with a means to determine if a target vCenter deployment size is suitable for your environment before you perform a vCenter upgrade.

By using the Configuration Maximums checker that is integrated with vCenter, you can view the limits for certain inventory objects as well as your current value for those inventory objects. You can easily spot the components that exceed the threshold limits for vCenter and also those that exceed the maximum prescribed by Broadcom.

For optimal performance and to prevent errors, it is essential that you stay within the configuration limits. Although you can go beyond the prescribed limits, crossing the configuration maximum limits leads to serious issues, such as services crashing, upgrades failing, and so on.

For detailed information about the prescribed limits for each component, visit the Configuration Maximums tools at https://configmax.broadcom.com/home.

Components That the Configuration Maximums Checker Supports

Your virtual environment has various configurable components, such as virtual machines, hosts, data centers, host profiles, resource pools, vApps, and so on. Each of those components has a limit to create or configure. In vSphere 9.1, the Configuration Maximums checker supports the following vCenter components:

- Powered-on virtual machines
- Registered virtual machines
- Linked vCenter instances
- Host profiles
- Attached host profiles
- Datastore clusters

Consequences of Crossing the Prescribed Configuration Maximums

Going beyond the configuration maximums that your vCenter instance deployment size supports has the following serious consequences:

- Performance issues
  Exceeding the maximum values for CPU, memory, or storage might lead to reduced virtual machine performance and slower application response times, increased latency, and decreased throughput.
- Increased downtime
  Overutilization might increase the number of host crashes or freezes and virtual machine failures, and it might also cause storage or network congestion and prolonged recovery times.
- Reduced resource availability
  Exceeding the allowed maximums might prevent you from successfully scaling applications or services, adding new virtual machines or hosts to the inventory, migrating workloads, and consequently meeting your business demands.
- Security risks
  Overconfiguring hosts or virtual machines might increase the attack surface, increasing the chances for malware and ransomware, unauthorized access, and data breaches.
- Support and maintenance issues
  Exceeding maximums might lead to increased troubleshooting complexity, longer resolution times, higher support costs, and the possibility for misconfigured or incompatible components.
- Compatibility issues
  Overconfiguring might lead to compatibility issues with VMware tools and agents, third-party applications and plug-ins, and hardware components and devices.

Most Commonly Exceeded Configuration Maximums

- Memory size
  The maximum memory size for a virtual machine depends on the physical memory of the ESX host and virtual machine compatibility settings. Virtual machines with memory sizes greater than 6128 GB require EFI firmware, and virtual machines whose memory excedes 24560 GB (for BIOS firmware) might cause performance issues.
- Max vCPUs Limit Exceeded
  This error occurs when the number of running virutal CPUs on a host exceeds the recommended limit, which is dynamically calculated based on the resources available in your environemnt.
- Max Hosts per vCenter Exceeded
  This error occurs when the number of hosts connected to a vCenter instance exceeds the recommended limit.
- Max Powered On Virtual Machines Exceeded
  This error occurs when the number of powered-on virtual machines on a host exceeds the recommended limit.
- Max Registered Virtual Machines Exceeded
  This error occurs when the number of registered virtual machines on a host exceeds the recommended limit.
- Max Linked vCenter instances Exceeded
  This error occurs when the number of linked vCenter instances exceeds the recommended limit.
- Max Hosts in Linked vCenter Exceeded
  This error occurs when the number of hosts in linked vCenter instances exceeds the recommended limit.
- Max Powered On Virtual Machines in Linked vCenter Exceeded
  This error occurs when the number of powered-on virtual machines in linked vCenter instances exceeds the recommended limit.
- Max Registered Virtual Machines in Linked vCenter Exceeded
  This error occurs when the number of registered virtual machines in linked vCenter instances exceeds the recommended limit.

Important:

For detailed information about the prescribed limits for each component, visit the Configuration Maximums tools at https://configmax.broadcom.com/home.

View the Configuration Maximums for Your vCenter Instance

You can view the configuration maximum limits corresponding to your vCenter deployment size to ensure that you are well within the prescribed limits and to avoid failure. Staying within the prescribed limits ensures optimal performance.

- Ensure that the size of the vCenter deployment is medium or large. You cannot view the Configuration Maximums checker for small vCenter deployments.
- Ensure that you have **Administrator** privileges.

In the **Configuration Maximums** pane, the **Usage** value for a component is color-coded.

- Green usage bar indicates that you are have not reached 70% of the allowed configuration maximum value for the respective component.
- Yellow usage bar indicates that you have reached and exceeded 70% of the allowed configuration maximum value for the respective component.
- Red usage bar indicates that you are beyond the defined configuration maximum threshold for the respective object.

The information in the Configuration Maximums checker is not updated automatically. To update the table, you need to refresh vSphere Client by clicking the refresh icon.

1. In the vSphere Client, select a vCenter instance.
2. On the **Monitor** tab, click **Configuration Maximums**.

   In certain cases, you might be unable to see the Configuration Maximums table. Usually, this issue is caused by a server error, and it must resolve itself when you retry viewing the Configuration Maximums table.

   When the backend cannot fetch the current usage value for a component, you see an error for that component in the **Current Value** column, and `N/A` in the **Usage** column.

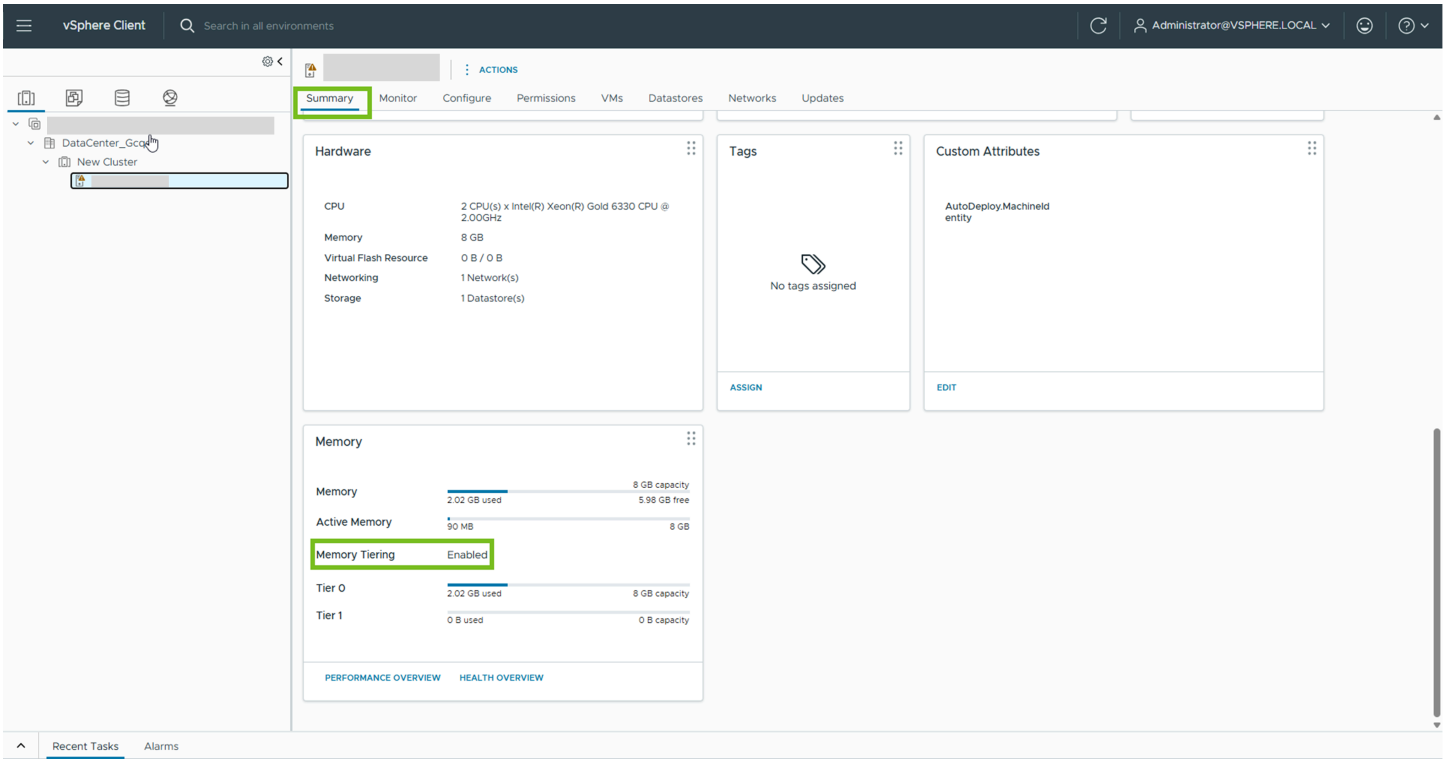
In the **Configuration Maximums** pane, you see a table that shows the prescribed configuration maximum for each supported component as well as its current value in your environment.

View Memory Tiering Device Information

For each ESX host that has memory tiering enabled, you can view a list of the devices used for memory tiering. For each device, you can view important information, such as capacity, tier size, consumed tiered memory, and health.

Ensure that memory tiering is enabled on the ESX host. For more information, see Memory Tiering Configuration

You can check if a host has memory tiering enabled in the **Memory** card on the **Summary** tab for a host.



On the **Summary** tab for a cluster, you can view information about the number of hosts in the cluster that have memory tiering enabled.

1. In the vSphere Client, select an ESX host.
2. On the **Monitor** tab, click **Memory Tiering**.

   On the **Memory Tiering** pane, you can see information about device health and can take actions, if the health status is `Warning` or `Error`.

   `Healthy` This status indicates that there are no issues with the device.

   `Warning` This status indicates that the device is unreliable due to various issues, such as:

   - The device spare capacity is below the threshold
   - Device temperature is above the threshold
   - Disk backup from memory to non-volatile memory failed
   - The available spare space in the device is below 30%

   When the device health status is `Warning`, you must replace the device as soon as possible.

   `Error` This status indicates that the device is unreliable due to various issues, such as:

   - The available spare space in the device is below 10%
   - The disk is read-only and you cannot use it for write operations

   When the device health status is `Error`, you must replace the device immediately.

On the **Memory Tiering** pane, you see a device list. You can use the expand arrows to view more detailed information for each device on the list.

Monitoring Solutions with the vCenter Solutions Manager

A solution is an extension of the vCenter that adds new functions to the vCenter instance. You can monitor the detailed information of an inventory and the health status of the solutions.

VMware products that integrate with vCenter are also considered solutions. For example, vSphere ESX Agent Manager is a VMware solution that lets you manage host agents that add new capabilities to ESX and ESX hosts.

You can also install a solution to add functionality from third-party technologies to the standard functions of vCenter. Solutions are typically delivered as OVF packages. You can install and deploy solutions from the vSphere Client. You can integrate the solutions into the vCenter Solutions Manager, which provides a list of all installed solutions.

If a virtual machine or vApp runs a solution, a custom icon represents it in the inventory of the vSphere Client. Each solution registers a unique icon to show that the solution manages the virtual machine or vApp. The icons show the power states (powered on, paused, or powered off). Solutions display more than one type of icon if they manage more than one type of virtual machine or vApp.

When you power on or power off a virtual machine or vApp, you are notified that you are operating with an object that is managed by the vCenter Solutions Manager. When you attempt another operation on a virtual machine or a vApp that is managed by a solution, an informational warning message appears.

For more information about developing and deploying solutions, vServices, and ESX agents, see https://techdocs.broadcom.com/bin/gethidpage?ux-context-string=api-sdk_8&appid=vcf-9-1&language=&format=rendered.

View Solutions

vCenter Solutions Manager helps to deploy, monitor, and interact with solutions that are installed in a vCenter instance.

The vCenter Solutions Manager displays information about the solution, such as the solution name, the vendor name, and the version of the product. The vCenter Solutions Manager also displays information about the health of a solution.

1. Navigate to the vCenter Solutions Manager.
   a) In the vSphere Client, select **Menu** > **Administration**.
   b) Expand **Solutions** and click **vCenter Server Extensions**.
2. Click a solution from the list.

   For instance, vService Manager or vSphere ESX Agent Manager.
3. Navigate through the tabs to view information about the solution.
   - **Summary**: You can view details about the solution, such as the product name, a short description, and links to the product and vendor websites. You can also view the solution configuration and the solution UI.
     Select the vCenter link to view the **Summary** page of the Virtual Machines or vApp.
   - **Monitor**: You can view the tasks and events related to the solution.
   - **VMs**: You can view a list of all the Virutal Machines and vApps that belong to the solution.

Monitoring the Health of Services and Nodes

You can monitor the health status of services and nodes to determine whether problems exist in your environment.

The vSphere Client provides an overview of all services and nodes across the management stack of the vCenter system. A list of default services is available for each vCenter instance.

View the Health Status of Nodes

In the vSphere Client, you can view the health status of vCenter nodes.

vCenter instances and machines that run vCenter services are considered nodes. Graphical badges represent the health status of nodes.

1. Use the vSphere Client to log in as administrator@*your_domain_name* to the vCenter instance.
   The address is of the type http://*appliance-IP-address-or-FQDN*/ui.
2. From the vSphere Client menu, select **Administration**.
3. Select **Deployment** > **System Configuration**.
4. Select a node to view its health status.

Table 475: Health States

Badge Icon	Description
	Good. The health of the object is normal.
	Warning. The object is experiencing some problems.
	Critical. The object is either not functioning properly or will stop functioning soon.
	Unknown. No data is available for this object.

View the Health Status of Services

You can view the health status of various vCenter services using the vCenter Management Interface

- Verify that the vCenter is successfully deployed and running.
- If you are using Internet Explorer, verify that TLS 1.0, TLS 1.1, and TLS 1.2 are enabled in the security settings.

  Note:
  The login session expires if you leave the vCenter Management Interface idle for 10 minutes.

1. In a Web browser, go to the vCenter Management Interface, https://*appliance-IP-address-or-FQDN*:5480.
2. Log in as root.
   The default root password is the password that you set while deploying vCenter.
3. In the vCenter Management Interface, click **Services**.
4. You can view the health of all the listed services in the **Health** column.

Performance Monitoring Utilities: resxtop and esxtop

The `resxtop` and `esxtop` command-line utilities provide a detailed look at how ESX uses resources in real time. You can start either utility in one of three modes: interactive (default), batch, or replay.

The fundamental difference between `resxtop` and `esxtop` is that you can use `resxtop` remotely, whereas you can start `esxtop` only through the ESX Shell of a local ESX host.

Using the esxtop Utility

You can run the `esxtop` utility using the ESX Shell to communicate with the management interface of the ESX host. You must have root user privileges.

esxtop Utility

To use the `esxtop` utility, type the following command with desired options:

```
esxtop [-h] [-v] [-b] [-l] [-s] [-a] [-c config file] [-R vm-support_dir_path]  [-d delay] [-n iterations] [-
export-entity entity-file] [-import-entity entity-file]
```

Command-Line Help Option	Description
-h	Prints this help menu.
-v	Prints version.
-b	Enables batch mode.
-l	Locks the esxtop objects to those available in the first snapshot.
-s	Enables secure mode.
-a	Displays all statistics.
-c	Sets the esxtop configuration file, which by default is .esxtop60rc.
-R	Enables replay mode.
-d	Sets the delay between updates in seconds.
-n	Runs esxtop for only n iterations. Use -n infinity to run esxtop for infinite period of time.
-u	Suppresses server-wide physical CPU statistics.

For example:

```
[root@localhost:~] esxtop
```

The above command will open the interactive screen of esxtop, by default.

```
[root@localhost:~] esxtop -b -a -d 10 -n 3 > /vmfs/volumes/localhost/test.csv
```

The above command runs the batch mode of esxtop with all counters, updates delay of 10 secs for 3 iterations. The output is written to the test.csv file which can be opened with other tools like Windows Perfmon.

> **Important:**
>
> The list of command-line options might not be complete. You can view all command-line options in the help menu by using the -h option.

esxtop Configuration

The esxtop utility reads its default configuration from .esxtop50rc on the ESX system. This configuration file consists of nine lines.

The first eight lines contain lowercase and uppercase letters to specify which fields appear in which order on the CPU, memory, storage adapter, storage device, virtual machine storage, network, interrupt, and CPU power panels. The letters correspond to the letters in the Fields or Order panels for the respective esxtop panel.

The ninth line contains information on the other options. Most important, if you saved a configuration in secure mode, you do not get an insecure esxtop without removing the s from the seventh line of your .esxtop50rc file. A number specifies the delay time between updates. As in interactive mode, typing c, m, d, u, v, n, I, or p determines the panel with which esxtop starts.

> **Note:**
> Do not edit the .esxtop50rc file. Instead, select the fields and the order in a running esxtop process, make changes, and save this file using the W interactive command.

Using the resxtop Utility

resxtop is a command-line utility or tool that runs on linux and provides a detailed look at how ESX uses resources in real time.

Before you can use any resxtop commands, you must download and install resxtop to your system.

Note:
`resxtop` is supported only on Linux.

After it is set up, start `resxtop` from the command line. For remote connections, you can connect to a host either directly or through vCenter.

To launch `resxtop` and connect to a remote server, enter this command

```
resxtop --server <hostname> --username <user>
```

The command-line options listed in the following table are the same as for `esxtop` (except for the `R` option) with additional connection options.

Note:
`resxtop` does not use all the options shared by other ESXCLI commands.

Table 476: resxtop Command-Line Options

Option	Description
`[server]`	Name of the remote host to connect to (required). If connecting directly to the ESX host, use the name of that host. If your connection to the ESX host is indirect (that is, through vCenter), use the name of the vCenter system for this option.
`[vihost]`	If you connect indirectly (through vCenter), this option should contain the name of the ESX host you connect to. If you connect directly to the host, this option is not used. Note that the host name needs to be the same as what appears in the vSphere Client.
`[portnumber]`	Port number to connect to on the remote server. The default port is 443, and unless this is changed on the server, this option is not needed.
`[username]`	User name to be authenticated when connecting to the remote host. The remote server prompts you for a password.

You can also use `resxtop` on a local ESX host by omitting the `server` option on the command line. The command defaults to localhost.

Installing resxtop on Ubuntu Linux

Installing RESXTOP on Linux systems is a simple process the involves extracting the files from the archive, running and install script, and updating the path.

1. Download `resxtop` from https://developer.broadcom.com/tools/vmware-vsphere-resxtop/latest.
2. Unpack the archive using the linux `tar` command: `tar -xvf resxtop-<version>-<build_number>-lin64.tgz.`
3. Execute the install script by running the linux command "`./install.sh`".
4. Update the path by running the linux command "`export LD_LIBRARY_PATH=/usr/lib/vmware/resxtop`".

Using esxtop or resxtop in Interactive Mode

By default, `resxtop` and `esxtop` run in interactive mode. Interactive mode displays statistics in different panels.

A help menu is available for each panel.

Interactive Mode Command-Line Options

You can use various command-line options with `esxtop` and `resxtop` in interactive mode.

Table 477: Interactive Mode Command-Line Options

Option	Description
`h`	Prints help for `resxtop` (or `esxtop`) command-line options.
`v`	Prints `resxtop` (or `esxtop`) version number.

Option	Description
s	Calls resxtop (or esxtop) in secure mode. In secure mode, the -d command, which specifies delay between updates, is disabled.
d	Specifies the delay between updates. The default is five seconds. The minimum is two seconds. Change this with the interactive command s. If you specify a delay of less than two seconds, the delay is set to two seconds.
n	Number of iterations. Updates the display n times and exits. Default value is 10000.
server	The name of the remote server host to connect to (required for resxtop only).
vihost	If you connect indirectly (through vCenter), this option should contain the name of the ESX host you connect to. If you connect directly to the ESX host, this option is not used. Note that the host name needs to be the same as what is displayed in the vSphere Client.
portnumber	The port number to connect to on the remote server. The default port is 443, and unless this is changed on the server, this option is not needed. (resxtop only)
username	The user name to be authenticated when connecting to the remote host. The remote server prompts you for a password, as well (resxtop only).
a	Show all statistics. This option overrides configuration file setups and shows all statistics. The configuration file can be the default ~/.esxtop50rc configuration file or a user-defined configuration file.
c*filename*	Load a user-defined configuration file. If the -c option is not used, the default configuration filename is ~/.esxtop50rc. Create your own configuration file, specifying a different filename, using the W single-key interactive command.

Common Statistics Description

Several statistics appear on the different panels while resxtop (or esxtop) is running in interactive mode. These statistics are common across all four panels.

The Uptime line, found at the top of each of the four resxtop (or esxtop) panels, displays the current time, time since last reboot, number of currently running worlds and load averages. A world is an ESX VMkernel schedulable entity, similar to a process or thread in other operating systems.

Below that the load averages over the past one, five, and fifteen minutes appear. Load averages consider both running and ready-to-run worlds. A load average of 1.00 means that there is full utilization of all physical CPUs. A load average of 2.00 means that the ESX system might need twice as many physical CPUs as are currently available. Similarly, a load average of 0.50 means that the physical CPUs on the ESX system are half utilized.

Statistics Columns and Order Pages

You can define the order of fields displayed in interactive mode.

If you press f, F, o, or O, the system displays a page that specifies the field order on the top line and short descriptions of the field contents. If the letter in the field string corresponding to a field is uppercase, the field is displayed. An asterisk in front of the field description indicates whether a field is displayed.

The order of the fields corresponds to the order of the letters in the string.

From the Field Select panel, you can:

- Toggle the display of a field by pressing the corresponding letter.
- Move a field to the left by pressing the corresponding uppercase letter.
- Move a field to the right by pressing the corresponding lowercase letter.

Interactive Mode Single-Key Commands

When running in interactive mode, resxtop (or esxtop) recognizes several single-key commands.

All interactive mode panels recognize the commands listed in the following table. The command to specify the delay between updates is disabled if the `s` option is given on the command line. All sorting interactive commands sort in descending order.

Table 478: Interactive Mode Single-Key Commands

Key	Description
h or ?	Displays a help menu for the current panel, giving a brief summary of commands, and the status of secure mode.
space	Immediately updates the current panel.
^L	Erases and redraws the current panel.
f or F	Displays a panel for adding or removing statistics columns (text boxes) to or from the current panel.
o or O	Displays a panel for changing the order of statistics columns on the current panel.
#	Prompts you for the number of statistics rows to display. Any value greater than 0 overrides automatic determination of the number of rows to show, which is based on window size measurement. If you change this number in one `resxtop` (or `esxtop`) panel, the change affects all four panels.
s	Prompts you for the delay between updates, in seconds. Fractional values are recognized down to microseconds. The default value is five seconds. The minimum value is two seconds. This command is not available in secure mode.
W	Write the current setup to an esxtop (or resxtop) configuration file. This is the recommended way to write a configuration file. The default filename is the one specified by -c option, or `~/.esxtop50rc` if the -c option is not used. You can also specify a different filename on the prompt generated by this W command.
q	Quit the interactive mode.
c	Switch to the CPU resource utilization panel.
p	Switch to the CPU Power utilization panel.
m	Switch to the memory resource utilization panel.
d	Switch to the storage (disk) adapter resource utilization panel.
u	Switch to storage (disk) device resource utilization screen.
v	Switch to storage (disk) virtual machine resource utilization screen.
n	Switch to the network resource utilization panel.
i	Switch to the interrupt panel.
k	Kill a world.
e	Expand/Rollup CPU statistics.
V	View only VM instances.
L	Change the length of the `Name` field.
l	Limit display to a single group based on GID.
Gg	Limit display to groups (G) or worlds (g) that match the regular expression.
Aa	Limit display to groups (A) or worlds (a) that do not match the regular expression.
x	Switch to the vSAN device panel.
z	Switch to the RDMA device panel.

Important:

The list of single-key commands might not be complete. You can see an up-to-date list of the single-key commands in the help menu by using h or ?.

CPU Panel

The CPU panel displays server-wide statistics as well as statistics for the individual world, resource pool, and virtual machine CPU utilization.

Resource pools, virtual machines that are running, or other worlds are at times called groups. For worlds belonging to a virtual machine, statistics for the virtual machine that is running are displayed. All other worlds are logically aggregated into the resource pools that contain them.

Table 479: CPU Panel Statistics

Line	Description
PCPU USED(%)	A PCPU refers to a physical hardware execution context. It can be a physical CPU core if the hyperthreading is unavailable or deactivated, or a logical CPU (LCPU or SMT thread) if the hyperthreading is activated. PCPU USED(%) displays the following percentages: • percentage of CPU usage per PCPU • percentage of CPU usage averaged over all PCPUs CPU Usage (%USED) is the percentage of PCPU nominal frequency that was used since the last screen update. It equals the total sum of %USED for Worlds that ran on this PCPU. **Note:** If a PCPU is running at frequency that is higher than its nominal (rated) frequency, then PCPU USED(%) can be greater than 100%. If a PCPU and its partner are busy when hyperthreading is activated, each PCPU account for half of the CPU usage.
PCPU UTIL(%)	A PCPU refers to a physical hardware execution context. It can be a physical CPU core if the hyperthreading is unavailable or deactivated, or a logical CPU (LCPU or SMT thread) if the hyperthreading is activated. PCPU UTIL(%) represents the percentage of real time that the PCPU was not idle (raw PCPU utilization). It displays the percentage CPU utilization per PCPU, and the percentage CPU utilization averaged over all PCPUs. **Note:** PCPU UTIL(%) might differ from PCPU USED(%) due to power management technologies or hyperthreading.
ID	Resource pool ID or virtual machine ID of the resource pool or virtual machine of the world that is running. Alternatively, the world ID of the world that is running.
GID	Resource pool ID of the resource pool or virtual machine of the world that is running.
NAME	Name of the resource pool or virtual machine of the world that is running, or name of the world that is running.
NWLD	Number of members in the resource pool or virtual machine of the world that is running. If a Group is expanded using the interactive command `e`, then NWLD for all the resulting worlds is 1.
%STATE TIMES	Set of CPU statistics made up of the following percentages. For a world, the percentages are a percentage of one physical CPU core.
%USED	Percentage of physical CPU core cycles used by the resource pool, virtual machine, or world. %USED might depend on the frequency with which the CPU core is running. When running with lower CPU core frequency, %USED can be smaller than %RUN. On CPUs which support the turbo mode, CPU frequency can also be higher than the nominal (rated) frequency, and %USED can be larger than %RUN. %USED = %RUN + %SYS - %OVRLP
%SYS	Percentage of time spent in the ESX VMkernel on behalf of the resource pool, virtual machine, or world to process interrupts and to perform other system activities. This time is part of the time used to calculate %USED. %USED = %RUN + %SYS - %OVRLP
%WAIT	Percentage of time the resource pool, virtual machine, or world spent in the blocked or busy wait state. This percentage includes the percentage of time the resource pool, virtual machine, or world was idle. 100% = %RUN + %RDY + %CSTP + %WAIT
%VMWAIT	The total percentage of time the Resource Pool/World spent in a blocked state waiting for events.

Line	Description
%IDLE	Percentage of time the resource pool, virtual machine, or world was idle. Subtract this percentage from %WAIT to see the percentage of time the resource pool, virtual machine, or world was waiting for some event. The difference, %WAIT- %IDLE, of the VCPU worlds can be used to estimate guest I/O wait time. To find the VCPU worlds, use the single-key command `e` to expand a virtual machine and search for the world NAME starting with "vcpu". (The VCPU worlds might wait for other events in addition to I/O events, so this measurement is only an estimate.)
%RDY	Percentage of time the resource pool, virtual machine, or world was ready to run, but was not provided CPU resources on which to execute. 100% = %RUN + %RDY + %CSTP + %WAIT
%MLMTD (max limited)	Percentage of time the ESX VMkernel deliberately did not run the resource pool, virtual machine, or world because doing so would violate the resource pool, virtual machine, or world's limit setting. Because the resource pool, virtual machine, or world is ready to run when it is prevented from running in this way, the %MLMTD (max limited) time is included in %RDY time.
%SWPWT	Percentage of time a resource pool or world spends waiting for the ESX VMkernel to swap memory. The %SWPWT (swap wait) time is included in the %WAIT time.
EVENT COUNTS/s	Set of CPU statistics made up of per second event rates. These statistics are for VMware internal use only.
CPU ALLOC	Set of CPU statistics made up of the following CPU allocation configuration parameters.
AMIN	Resource pool, virtual machine, or world attribute Reservation.
AMAX	Resource pool, virtual machine, or world attribute Limit. A value of -1 means unlimited.
ASHRS	Resource pool, virtual machine, or world attribute Shares.
SUMMARY STATS	Set of CPU statistics made up of the following CPU configuration parameters and statistics. These statistics apply only to worlds and not to virtual machines or resource pools.
AFFINITY BIT MASK	Bit mask showing the current scheduling affinity for the world.
HTSHARING	Current hyperthreading configuration.
CPU	The physical or logical processor on which the world was running when `resxtop` (or `esxtop`) obtained this information.
HTQ	Indicates whether the world is quarantined or not. N means no and Y means yes.
TIMER/s	Timer rate for this world.
%OVRLP	Percentage of system time spent during scheduling of a resource pool, virtual machine, or world on behalf of a different resource pool, virtual machine, or world while the resource pool, virtual machine, or world was scheduled. This time is not included in %SYS. For example, if virtual machine A is being scheduled and a network packet for virtual machine B is processed by the ESX VMkernel, the time spent appears as %OVRLP for virtual machine A and %SYS for virtual machine B. %USED = %RUN + %SYS - %OVRLP
%RUN	Percentage of total time scheduled. This time does not account for hyperthreading and system time. On a hyperthreading activated server, the %RUN can be twice as large as %USED. %USED = %RUN + %SYS - %OVRLP 100% = %RUN + %RDY + %CSTP + %WAIT
%CSTP	Percentage of time a resource pool spends in a ready, co-deschedule state. **Note:** You might see this statistic displayed, but it is intended for VMware use only. 100% = %RUN + %RDY + %CSTP + %WAIT
POWER	Current CPU power consumption for a resource pool (in Watts).
%LAT_C	Percentage of time the resource pool or world was ready to run but was not scheduled to run because of CPU resource contention.
%LAT_M	Percentage of time the resource pool or world was ready to run but was not scheduled to run because of memory resource contention.
%DMD	CPU demand in percentage. It represents the average active CPU load in the past minute.

Line	Description
CORE UTIL(%)	Percentage of CPU cycles per core when at least one of the PCPUs in this core is unhalted, and its average over all cores. This statistic only appears when hyperthreading is activated. In batch mode, the corresponding CORE UTIL(%) statistic is displayed for each PCPU. For example, PCPU 0 and PCPU 1 have the same the CORE UTIL(%) number, and that is the number for core 0.

You can change the display using single-key commands.

Table 480: CPU Panel Single-Key Commands

Command	Description
e	Toggles whether CPU statistics are displayed expanded or unexpanded. The expanded display includes CPU resource utilization statistics broken down by individual worlds belonging to a resource pool or virtual machine. All percentages for the individual worlds are percentage of a single physical CPU. Consider these examples: • If the %Used by a resource pool is 30% on a two-way server, the resource pool is utilizing 30 percent of one physical core. • If the %Used by a world belonging to a resource pool is 30 percent on a two-way server, that world is utilizing 30% of one physical core.
U	Sorts resource pools, virtual machines, and worlds by the resource pool's or virtual machine's %Used column. This is the default sort order.
R	Sorts resource pools, virtual machines, and worlds by the resource pool's or virtual machine's %RDY column.
N	Sorts resource pools, virtual machines, and worlds by the GID column.
V	Displays virtual machine instances only.
L	Changes the displayed length of the NAME column.

CPU Power Panel

The CPU Power panel displays CPU Power utilization statistics.

On the CPU Power panel, statistics are arranged per physical CPU. A physical CPU is a physical hardware execution context. It is the physical CPU core when hyper-threading is unavailable or deactivated, or a logical CPU (LCPU or SMT thread) when hyper-threading is activated.

Table 481: CPU Power Panel Statistics

Line	Description
Power Usage	Current total power usage (in Watts).
Power Cap	Total power cap (in Watts).
PSTATE MHZ	Clock frequency per state. **Note:** When ACPI Collaborative Processor Performance Control (CPPC) is enabled, `Hardware Controlled` is displayed. For more information about P-state management types, see Host Power Management Policies in ESX.
%USED	Percentage of physical CPU nominal frequency used since the last screen update. It is the same as PCPU USED(%) shown in the CPU Screen.
%UTIL	Raw physical CPU utilization is the percentage of time that physical CPU was not idle. It is the same as PCPU UTIL(%) shown in the CPU Screen.

Line	Description
%Cx	Percentage of time the physical CPU spent in C-State 'x'.
%Px	Percentage of time the physical CPU spent in P-State 'x'. On systems with Processor Clocking Control, P-states are not directly visible to ESX . The `esxtop` shows the percentage of time spent at full speed under the heading 'P0' and the percentage of time spent at any lower speed under 'P1'. **Note:** When ACPI CPPC is enabled, the hardware controls the frequency and the individual P-states do not exist. In this case, %Px values are not displayed. The %A/MPERF value can be used to determine the current frequency.
%Tx	Percentage of time the physical CPU spent in T-State 'x'.
%A/MPERF	aperf and mperf are two hardware registers used to keep track of the actual frequency and nominal frequency of the processor. Displays the real-time aperf to mperf ratio in the last esxtop update period. %A/MPERF *nominal frequency of the processor = current frequency of the processor

Memory Panel

The Memory panel displays server-wide and group memory utilization statistics. As on the CPU panel, groups correspond to resource pools, running virtual machines, or other worlds that are consuming memory.

The first line, found at the top of the Memory panel displays the current time, time since last reboot, number of currently running worlds, and memory overcommitment averages. The memory overcommitment averages over the past one, five, and fifteen minutes appear. Memory overcommitment of 1.00 means a memory overcommitment of 100 percent.

Table 482: Memory Panel Statistics

Field	Description
PMEM (MB)	Displays the machine memory statistics for the server. All numbers are in megabytes. **total** Total amount of the machine memory in the server. **vmk** Amount of the machine memory being used by the ESX VMkernel. **other** Amount of the machine memory being used by everything other than the ESX VMkernel. **free** Amount of the machine memory that is free.
VMKMEM (MB)	Displays the machine memory statistics for the ESX VMkernel. All numbers are in megabytes. **managed** Total amount of the machine memory managed by the ESX VMkernel. **min free** Minimum amount of the machine memory that the ESX VMkernel aims to keep free. **rsvd** Total amount of the machine memory currently reserved by resource pools. **ursvd** Total amount of the machine memory currently unreserved. **state** Current machine memory availability state. Possible values are high, soft, hard and low. High means that the machine memory is not under any pressure and low means that it is.
NUMA (MB)	Displays the ESX NUMA statistics. This line appears only if the ESX host is running on a NUMA server. All numbers are in megabytes. For each NUMA node in the server, two statistics are displayed: • The total amount of machine memory in the NUMA node that is managed by ESX. • The amount of machine memory in the node that is currently free (in parentheses). Shared memory for the ESX host might be larger than the total amount of memory if memory is over-committed.

Field	Description
PSHARE (MB)	Displays the ESX page-sharing statistics. All numbers are in megabytes. **shared** Amount of the physical memory that is being shared. **common** Amount of the machine memory that is common across worlds. **saving** Amount of the machine memory that is saved because of page sharing. shared = common + saving
SWAP (MB)	Displays the ESX swap usage statistics. All numbers are in megabytes. **curr** Current swap usage. **rclmtgt** Where the ESX system expects the reclaimed memory to be. Memory can be reclaimed by swapping or compression. **r/s** Rate at which the memory is swapped in by the ESX system from disk. **w/s** Rate at which the memory is swapped to disk by the ESX system.
ZIP (MB)	Displays the ESX memory compression statistics. All numbers are in megabytes. **zipped** Total compressed physical memory. **saved** Saved memory by compression.
MEMCTL (MB)	Displays the memory balloon statistics. All numbers are in megabytes. **curr** Total amount of the physical memory reclaimed using the `vmmemctl` module. **target** Total amount of physical memory the ESX host attempts to reclaim using the `vmmemctl` module. **max** Maximum amount of the physical memory the ESX host can reclaim using the `vmmemctl` module.
AMIN	Memory reservation for this resource pool or virtual machine.
AMAX	Memory limit for this resource pool or virtual machine. A value of -1 means Unlimited.
ASHRS	Memory shares for this resource pool or virtual machine.
NHN	Current home node for the resource pool or virtual machine. This statistic is applicable only on NUMA systems. If the virtual machine has no home node, a dash (-) appears.
NRMEM (MB)	Current amount of remote memory allocated to the virtual machine or resource pool. This statistic is applicable only on NUMA systems.
N% L	Current percentage of memory allocated to the virtual machine or resource pool that is local.
MEMSZ (MB)	Amount of physical memory allocated to a resource pool or virtual machine. The values are the same for the VMM and VMX groups. MEMSZ = GRANT + MCTLSZ + SWCUR + "never touched"
GRANT (MB)	Amount of guest physical memory mapped to a resource pool or virtual machine. The consumed host machine memory is equal to GRANT - SHRDSVD. The values are the same for the VMM and VMX groups.
CNSM	Amount of the memory currently consumed by the virtual machine. The memory currently consumed by the virtual machine is equal to the amount of memory that the VM guest operating system currently uses, excluding the amount of memory saved for sharing if memory sharing is enabled on the VM, excluding the amount of memory saved if some of the VM memory is compressed. For more information on memory sharing and memory compression, see the *vSphere Resource Management* documentation.
SZTGT (MB)	Amount of machine memory the ESX VMkernel wants to allocate to a resource pool or virtual machine. The values are the same for the VMM and VMX groups.
TCHD (MB)	Working set estimate for the resource pool or virtual machine. The values are the same for the VMM and VMX groups.
%ACTV	Percentage of guest physical memory that is being referenced by the guest. This is an instantaneous value.

Field	Description
%ACTVS	Percentage of guest physical memory that is being referenced by the guest. This is a slow moving average.
%ACTVF	Percentage of guest physical memory that is being referenced by the guest. This is a fast moving average.
%ACTVN	Percentage of guest physical memory that is being referenced by the guest. This is an estimation. (You might see this statistic displayed, but it is intended for VMware use only.)
MCTL?	Memory balloon driver is installed or not. **N** means no, **Y** means yes.
MCTLSZ (MB)	Amount of physical memory reclaimed from the resource pool by way of ballooning.
MCTLTGT (MB)	Amount of physical memory the ESX system attempts to reclaim from the resource pool or virtual machine by way of ballooning.
MCTLMAX (MB)	Maximum amount of physical memory the ESX system can reclaim from the resource pool or virtual machine by way of ballooning. This maximum depends on the guest operating system type.
SWCUR (MB)	Current swap usage by this resource pool or virtual machine.
SWTGT (MB)	Target where the ESX host expects the swap usage by the resource pool or virtual machine to be.
SWR/s (MB)	Rate at which the ESX host swaps in memory from disk for the resource pool or virtual machine.
SWW/s (MB)	Rate at which the ESX host swaps resource pool or virtual machine memory to disk.
LLSWR/s (MB)	Rate at which memory is read from the host cache. The reads and writes are attributed to the VMM group only, LLSWAP statistics are not displayed for the VM group.
LLSWW/s (MB)	Rate at which memory is written to the host cache from various sources. The reads and writes are attributed to the VMM group only, LLSWAP statistics are not displayed for the VM group.
CPTRD (MB)	Amount of data read from checkpoint file.
CPTTGT (MB)	Size of checkpoint file.
ZERO (MB)	Resource pool or virtual machine physical pages that are zeroed.
SHRD (MB)	Resource pool or virtual machine physical pages that are shared.
SHRDSVD (MB)	Machine pages that are saved because of resource pool or virtual machine shared pages.
OVHD (MB)	Current space overhead for resource pool.
OVHDMAX (MB)	Maximum space overhead that might be incurred by resource pool or virtual machine.
OVHDUW (MB)	Current space overhead for a user world. (You might see this statistic displayed, but it is intended for VMware use only.)
GST_NDx (MB)	Guest memory allocated for a resource pool on NUMA node x. This statistic is applicable on NUMA systems only.
OVD_NDx (MB)	VMM overhead memory allocated for a resource pool on NUMA node x. This statistic is applicable on NUMA systems only.
TCHD_W (MB)	Write working set estimate for resource pool.
CACHESZ (MB)	Compression memory cache size.
CACHEUSD (MB)	Used compression memory cache.
ZIP/s (MB/s)	Compressed memory per second.
UNZIP/s (MB/s)	Decompressed memory per second.

Table 483: Memory Panel Interactive Commands

Command	Description
M	Sort resource pools or virtual machines by MEMSZ column. This is the default sort order.
B	Sort resource pools or virtual machines by Group Memctl column.
N	Sort resource pools or virtual machines by GID column.
V	Display virtual machine instances only.
L	Changes the displayed length of the NAME column.

Storage Adapter Panel

Statistics in the Storage Adapter panel are aggregated per storage adapter by default. Statistics can also be viewed per storage path.

Table 484: Storage Adapter Panel Statistics

Column	Description
ADAPTR	Name of the storage adapter.
PATH	Storage path name. This name is only visible if the corresponding adapter is expanded. See interactive command `e` in *Storage Adapter Panel Interactive Commands.*
NPTH	Number of paths.
AQLEN	Current queue depth of the storage adapter.
CMDS/s	Number of commands issued per second.
READS/s	Number of read commands issued per second.
WRITES/s	Number of write commands issued per second.
MBREAD/s	Megabytes read per second.
MBWRTN/s	Megabytes written per second.
RESV/s	Number of SCSI reservations per second.
CONS/s	Number of SCSI reservation conflicts per second.
DAVG/cmd	Average device latency per command, in milliseconds.
KAVG/cmd	Average ESX VMkernel latency per command, in milliseconds.
GAVG/cmd	Average virtual machine operating system latency per command, in milliseconds.
QAVG/cmd	Average queue latency per command, in milliseconds.
DAVG/rd	Average device read latency per read operation, in milliseconds.
KAVG/rd	Average ESX VMkernel read latency per read operation, in milliseconds.
GAVG/rd	Average guest operating system read latency per read operation, in milliseconds.
QAVG/rd	Average queue latency per read operation, in milliseconds.
DAVG/wr	Average device write latency per write operation, in milliseconds.
KAVG/wr	Average ESX VMkernel write latency per write operation, in milliseconds.
GAVG/wr	Average guest operating system write latency per write operation, in milliseconds.
QAVG/wr	Average queue latency per write operation, in milliseconds.
FCMDS/s	Number of failed commands issued per second.
FREAD/s	Number of failed read commands issued per second.
FWRITE/s	Number of failed write commands issued per second.

Column	Description
FMBRD/s	Megabytes of failed read operations per second.
FMBWR/s	Megabytes of failed write operations per second.
FRESV/s	Number of failed SCSI reservations per second.
ABRTS/s	Number of commands cancelled per second.
RESETS/s	Number of commands reset per second.
PAECMD/s	The number of PAE (Physical Address Extension) commands per second.
PAECP/s	The number of PAE copies per second.
SPLTCMD/s	The number of split commands per second.
SPLTCP/s	The number of split copies per second.

The following table displays the interactive commands you can use with the storage adapter panel.

Table 485: Storage Adapter Panel Interactive Commands

Command	Description
`e`	Toggles whether storage adapter statistics appear expanded or unexpanded. Allows you to view storage resource utilization statistics broken down by individual paths belonging to an expanded storage adapter. You are prompted for the adapter name.
`r`	Sorts by READS/s column.
`w`	Sorts by WRITES/s column.
`R`	Sorts by MBREAD/s read column.
`T`	Sorts by MBWRTN/s written column.
`N`	Sorts first by ADAPTR column, then by PATH column. This is the default sort order.

Storage Device Panel

The storage device panel displays server-wide storage utilization statistics.

By default, the information is grouped per storage device. You can also group the statistics per path, per world, or per partition.

Table 486: Storage Device Panel Statistics

Column	Description
DEVICE	Name of the storage device.
PATH	Path name. This name is visible only if the corresponding device is expanded to paths. See the interactive command `p` in *Storage Device Panel Interactive Commands.*
WORLD	World ID. This ID is visible only if the corresponding device is expanded to worlds. See the interactive command `e` in *Storage Device Panel Interactive Commands.* The world statistics are per world per device.
PARTITION	Partition ID. This ID is visible only if the corresponding device is expanded to partitions. See interactive command `t` in *Storage Device Panel Interactive Commands*.
NPH	Number of paths.
NWD	Number of worlds.
NPN	Number of partitions.
SHARES	Number of shares. This statistic is applicable only to worlds.
BLKSZ	Block size in bytes.

Column	Description
NUMBLKS	Number of blocks of the device.
DQLEN	Current device queue depth of the storage device.
WQLEN	World queue depth. This is the maximum number of ESX VMkernel active commands that the world is allowed to have. This is a per device maximum for the world. It is valid only if the corresponding device is expanded to worlds.
ACTV	Number of commands in the ESX VMkernel that are currently active. This statistic applies to only worlds and devices.
QUED	Number of commands in the ESX VMkernel that are currently queued. This statistic applies to only worlds and devices.
%USD	Percentage of the queue depth used by ESX VMkernel active commands. This statistic applies to only worlds and devices.
LOAD	Ratio of ESX VMkernel active commands plus ESX VMkernel queued commands to queue depth. This statistic applies to only worlds and devices.
CMDS/s	Number of commands issued per second.
READS/s	Number of read commands issued per second.
WRITES/s	Number of write commands issued per second.
MBREAD/s	Megabytes read per second.
MBWRTN/s	Megabytes written per second.
DAVG/cmd	Average device latency per command in milliseconds.
KAVG/cmd	Average ESX VMkernel latency per command in milliseconds.
GAVG/cmd	Average guest operating system latency per command in milliseconds.
QAVG/cmd	Average queue latency per command in milliseconds.
DAVG/rd	Average device read latency per read operation in milliseconds.
KAVG/rd	Average ESX VMkernel read latency per read operation in milliseconds.
GAVG/rd	Average guest operating system read latency per read operation in milliseconds.
QAVG/rd	Average queue read latency per read operation in milliseconds.
DAVG/wr	Average device write latency per write operation in milliseconds.
KAVG/wr	Average ESX VMkernel write latency per write operation in milliseconds.
GAVG/wr	Average guest operating system write latency per write operation in milliseconds.
QAVG/wr	Average queue write latency per write operation in milliseconds.
ABRTS/s	Number of commands cancelled per second.
RESETS/s	Number of commands reset per second.
PAECMD/s	Number of PAE commands per second. This statistic applies to only paths.
PAECP/s	Number of PAE copies per second. This statistic applies to only paths.
SPLTCMD/s	Number of split commands per second. This statistic applies to only paths.
SPLTCP/s	Number of split copies per second. This statistic applies to only paths.

The following table displays the interactive commands you can use with the storage device panel.

Table 487: Storage Device Panel Interactive Commands

Command	Description
e	Expand or roll up storage world statistics. This command allows you to view storage resource utilization statistics separated by individual worlds belonging to an expanded storage device. You are prompted for the device name. The statistics are per world per device.
P	Expand or roll up storage path statistics. This command allows you to view storage resource utilization statistics separated by individual paths belonging to an expanded storage device. You are prompted for the device name.
t	Expand or roll up storage partition statistics. This command allows you to view storage resource utilization statistics separated by individual partitions belonging to an expanded storage device. You are prompted for the device name.
r	Sort by READS/s column.
w	Sort by WRITES/s column.
R	Sort by MBREAD/s column.
T	Sort by MBWRTN column.
N	Sort first by DEVICE column, then by PATH, WORLD, and PARTITION column. This is the default sort order.
L	Changes the displayed length of the DEVICE column.

Virtual Machine Storage Panel

This panel displays virtual machine-centric storage statistics.

By default, statistics are aggregated on a per-resource-pool basis. One virtual machine has one corresponding resource pool, so the panel displays statistics on a per-virtual-machine basis. You can also view the statistics on per-VSCSI-device basis.

Table 488: Virtual Machine Storage Panel Statistics

Column	Description
ID	Resource pool ID or VSCSI ID of VSCSI device.
GID	Resource pool ID.
VMNAME	Name of the resource pool.
VSCSINAME	Name of the VSCSI device.
NDK	Number of VSCSI devices
CMDS/s	Number of commands issued per second.
READS/s	Number of read commands issued per second.
WRITES/s	Number of write commands issued per second.
MBREAD/s	Megabytes read per second.
MBWRTN/s	Megabytes written per second.
LAT/rd	Average latency (in milliseconds) per read.
LAT/wr	Average latency (in milliseconds) per write.

The following table lists the interactive commands you can use with the virtual machine storage panel.

Table 489: Virtual Machine Storage Panel Interactive Commands

Command	Description
e	Expand or roll up storage VSCSI statistics. Allows you to view storage resource utilization statistics broken down by individual VSCSI devices belonging to a group. You are prompted to enter the group ID. The statistics are per VSCSI device.
r	Sort by READS/s column.
w	Sort by WRITES/s column.
R	Sort by MBREAD/s column.
T	Sort by MBWRTN/s column.
N	Sort first by VMNAME column, and then by VSCSINAME column. It is the default sort order.

vSAN Storage Panel

This panel displays statistics about vSAN storage.

By default, statistics are aggregated on a per-resource-pool basis. One virtual machine has one corresponding resource pool, so the panel displays statistics on a per-virtual-machine basis. You can also view the statistics on per-VSCSI-device basis.

Table 490: vSAN Storage Panel Statistics

Column	Description
ROLE	vSAN role: Client/Owner/CompMgr
READS/s	IOPS (read)
MBREAD/s	Bandwidth MB/s (read)
AVGLAT	Average latency
SDLAT	Standard deviation
WRITES/s	IOPS (write)
MBWRITE/s	Bandwidth MB/s (write)
AVGLAT	Average latency
SDLAT	Standard deviation
UNMAP/s	IOPS (unmap operations)
GBUNMAP/s	Bandwidth GB/s (unmaps)
AVGLAT	Average latency
SDLAT	Standard deviation

RDMA Device Panel

This panel displays storage statistics about RDMA devices

Table 491: RDMA Device Panel Statistics

Column	Description
NAME	Device name
TEAM-PNIC	Device team NIC
PKTTX/s	Packet transmits/sec
MbTX/s	Bandwidth MB/s of transmits

Column	Description
PKTRX/s	Packet receives/sec
MbRX/s	Bandwidth MB/s of receives
QP	Number of queue pairs allocated
MR	Number of memory regions allocated

Table 492: Virtual Machine Storage Panel Interactive Commands

Command	Description
S	Sort by the State column.
N	Use the default sort order.
T	Sort by the MbTX/s column.
R	Sort by the MbRX/s column.
t	Sort by the PKTTX/s column.
e	Sort by the PKTRX/s column.

Network Panel

The Network panel displays server-wide network utilization statistics.

Statistics are arranged by port for each virtual network device configured. For physical network adapter statistics, see the row in the table that corresponds to the port to which the physical network adapter is connected. For statistics on a virtual network adapter configured in a particular virtual machine, see the row corresponding to the port to which the virtual network adapter is connected.

Table 493: Network Panel Statistics

Column	Description
PORT-ID	Virtual network device port ID.
UPLINK	Y means that the corresponding port is an uplink. N means it is not.
UP	Y means that the corresponding link is up. N means it is not.
SPEED	Link speed in Megabits per second.
FDUPLX	Y means the corresponding link is operating at full duplex. N means it is not.
USED-BY	Virtual network device port user.
DTYP	Virtual network device type. H means HUB and S means switch.
DNAME	Virtual network device name.
PKTTX/s	Number of packets transmitted per second.
PKTRX/s	Number of packets received per second.
MbTX/s	MegaBits transmitted per second.
MbRX/s	MegaBits received per second.
%DRPTX	Percentage of transmit packets dropped.
%DRPRX	Percentage of receive packets dropped.
TEAM-PNIC	Name of the physical NIC used for the team uplink.
PKTTXMUL/s	Number of multicast packets transmitted per second.
PKTRXMUL/s	Number of multicast packets received per second.
PKTTXBRD/s	Number of broadcast packets transmitted per second.

Column	Description
PKTRXBRD/s	Number of broadcast packets received per second.

The following table displays the interactive commands you can use with the network panel.

Table 494: Network Panel Interactive Commands

Command	Description
`T`	Sorts by Mb Tx column.
`R`	Sorts by Mb Rx column.
`t`	Sorts by Packets Tx column.
`r`	Sorts by Packets Rx column.
`N`	Sorts by PORT-ID column. This is the default sort order.
`L`	Changes the displayed length of the DNAME column.

Interrupt Panel

The interrupt panel displays information about the use of interrupt vectors.

Table 495: Interrupt Panel Statistics

Column	Description
VECTOR	Interrupt vector ID.
COUNT/s	Total number of interrupts per second. This value is cumulative of the count for every CPU.
COUNT_x	Interrupts per second on CPU x.
TIME/int	Average processing time per interrupt (in microseconds).
TIME_x	Average processing time per interrupt on CPU x (in microseconds).
DEVICES	Devices that use the interrupt vector. If the interrupt vector is not enabled for the device, its name is enclosed in angle brackets (< and >).

Using Batch Mode

Batch mode allows you to collect and save resource utilization statistics in a file.

After you prepare for batch mode, you can use `esxtop` or `resxtop` in this mode.

Prepare for Batch Mode

To run in batch mode, you must first prepare for batch mode.

1. Run `resxtop` (or `esxtop`) in interactive mode.
2. In each of the panels, select the columns you want.
3. Save this configuration to a file (by default `~/.esxtop50rc`) using the `W` interactive command.

You can now use `resxtop` (or `esxtop`) in batch mode.

Use esxtop or resxtop in Batch Mode

After you have prepared for batch mode, you can use `esxtop` or `resxtop` in this mode.

1. Start `resxtop` (or `esxtop`) to redirect the output to a file.
   For example:
   ```
   esxtop -b > my_file.csv
   ```
   The filename must have a `.csv` extension. The utility does not enforce this, but the post-processing tools require it.
2. Process statistics collected in batch mode using tools such as Microsoft Excel and Perfmon.

In batch mode, `resxtop` (or `esxtop`) does not accept interactive commands. In batch mode, the utility runs until it produces the number of iterations requested (see command-line option `n`, below, for more details), or until you end the process by pressing Ctrl+c.

Batch Mode Command-Line Options

You can use batch mode with command-line options.

Table 496: Command-Line Options in Batch Mode

Option	Description
`a`	Show all statistics. This option overrides configuration file setups and shows all statistics. The configuration file can be the default `~/.esxtop50rc` configuration file or a user-defined configuration file.
`b`	Runs `resxtop` (or `esxtop`) in batch mode.
`c` *filename*	Load a user-defined configuration file. If the `-c` option is not used, the default configuration filename is ~/.esxtop41rc. Create your own configuration file, specifying a different filename, using the `W` single-key interactive command.
`d`	Specifies the delay between statistics snapshots. The minimum supported interval is 10 seconds. A shorter interval might lead to issues. If a delay of less than two seconds is specified, the delay is set to two seconds.
`n`	Number of iterations. `resxtop` (or `esxtop`) collects and saves statistics this number of times, and then exits.
`server`	The name of the remote server host to connect to (required, `resxtop` only).
`vihost`	If you connect indirectly (through vCenter), this option should contain the name of the ESX host you connect to. If you connect directly to the ESX host, this option is not used. **Note:** The host name needs to be the same as what appears in the vSphere Client.
`portnumber`	The port number to connect to on the remote server. The default port is 443, and unless this is changed on the server, this option is not needed. (`resxtop` only)
`username`	The user name to be authenticated when connecting to the remote host. You are prompted by the remote server for a password, as well (`resxtop` only).

Using Replay Mode

In replay mode, `esxtop` replays resource utilization statistics collected using `vm-support`.

After you prepare for replay mode, you can use esxtop in this mode. See the `vm-support` man page.

In replay mode, `esxtop` accepts the same set of interactive commands as in interactive mode and runs until no more snapshots are collected by `vm-support` to be read or until the requested number of iterations are completed.

Prepare for Replay Mode

To run in replay mode, you must prepare for replay mode.

1. Run `vm-support` in snapshot mode in the ESX Shell.

   Use the following command.

```
vm-support -p -i interval -d duration
```

2. Unzip and untar the resulting tar file so that `esxtop` can use it in replay mode.

You can now use `esxtop` in replay mode.

Use esxtop in Replay Mode

You can use `esxtop` in replay mode.

Replay mode can be run to produce output in the same style as batch mode (see the command-line option `b`, below).

Note:
Batch output from `esxtop` cannot be played back by `resxtop`.

Snapshots collected by vm-supported can be replayed by `esxtop`. However, vm-support output generated by ESX can be replayed only by `esxtop` running on the same version of ESX.

To activate replay mode, enter the following at the command-line prompt.

```
esxtop -R vm-support_dir_path
```

Replay Mode Command-Line Options

You can use replay mode with command-line options.

The following table lists the command-line options available for `esxtop` replay mode.

Table 497: Command-Line Options in Replay Mode

Option	Description
`R`	Path to the vm-support collected snapshot's directory.
`a`	Show all statistics. This option overrides configuration file setups and shows all statistics. The configuration file can be the default `~/.esxtop50rc` configuration file or a user-defined configuration file.
`b`	Runs `esxtop` in Batch mode.
`c` *filename*	Load a user-defined configuration file. If the `-c` option is not used, the default configuration filename is ~/.esxtop50rc. Create your own configuration file and specify a different filename using the `W` single-key interactive command.
`d`	Specifies the delay between panel updates. The default is five seconds. The minimum is two seconds. If a delay of less than two seconds is specified, the delay is set to two seconds.
`n`	Number of iterations `esxtop` updates the display this number of times and then exits.

Using the vimtop Plug-In to Monitor the Resource Use of Services

You can use the `vimtop` utility plug-in to monitor vSphere services that run in the vCenter.

`vimtop` is a tool similar to `esxtop`, which runs in the environment of the vCenter. By using the text-based interface of `vimtop` in the appliance shell, you can view overall information about the vCenter, and a list of vSphere services and their resource use.

Monitor Services By Using vimtop in Interactive Mode

You can use the `vimtop` plug-in to monitor services in real time.

The default view of the `vimtop` interactive mode consists of the overview tables and the main table. You can use single-key commands in interactive mode to switch the view from processes to disks or network.

1. From an SSH client application, log in to the vCenter shell.
2. Run the `vimtop` command to access the plug-in in interactive mode.

Interactive Mode Command-Line Options

You can use various command-line options when you run the `vimtop` command to enter the plug-in interactive mode.

Table 498: Interactive Mode Command-Line Options

Option	Description
-h	Prints help for the `vimtop` command-line options.
-v	Prints the `vimtop` version number.
-c *filename*	Loads a user-defined `vimtop` configuration file. If the -c option is not used, the default configuration file is `/root/vimtop/vimtop.xml`. You can create your own configuration file, specifying a different filename and path by using the `W` single-key interactive command.
-n *number*	Sets the number of performed iterations before the `vimtop` exits interactive mode. `vimtop` updates the display *number* number of times and exits. The default value is 10000.
-p / -d*seconds*	Sets the update period in seconds.

Interactive Mode Single-Key Commands for vimtop

When running in interactive mode, `vimtop` recognizes several single-key commands.

All interactive mode panels recognize the commands listed in the following table.

Table 499: Interactive Mode Single-Key Commands

Key Names	Description
h	Show a help menu for the current panel, giving a brief summary of commands, and the status of secure mode.
i	Show or hide the top line view of the overview panel of the `vimtop` plug-in.
t	Show or hide the Tasks section, which displays information in the overview panel about the tasks currently running on the vCenter instance.
m	Show or hide the Memory section in the overview panel.
f	Show or hide the CPU section which displays information in the overview panel about all available CPUs.
g	Show or hide the CPUs section which displays information in the overview panel about the top 4 physical CPUs.
spacebar	Immediately refreshes the current pane.
p	Pause the displayed information about the services resource use in the current panels.
r	Refresh the displayed information about the services resource use in the current panels.
s	Set refresh period.
q	Exit the interactive mode of the `vimtop` plug-in.
k	Displays the Disks view of the main panel.
o	Switch the main panel to Network view.
Esc	Clear selection or return to the Processes view of the main panel.
Enter	Select a service to view additional details.
n	Show or hide names of the headers in the main panel.

Key Names	Description
u	Show or hide the measurement units in the headers in the main panel.
left, right arrows	Select columns.
up, down arrows	Select rows.
<,>	Move a selected column.
Delete	Remove selected column.
c	Add a column to the current view of the main panel. Use spacebar to add or remove columns from the displayed list.
a	Sort the selected column in ascending order.
d	Sort the selected column in descending order.
z	Clear the sort order for all columns.
l	Set width for the selected column.
x	Return the column widths to their default values.
+	Expand selected item.
-	Collapse selected item.
w	Write the current setup to a `vimtop` configuration file. The default file name is the one specified by -c option, or `/root/vimtop/vimtop.xml` if the -c option is not used. You can also specify a different file name on the prompt generated by the `w` command.

Monitoring Networked Devices with SNMP and vSphere

Simple Network Management Protocol (SNMP) is commonly used by management programs to monitor a variety of networked devices.

vSphere systems run SNMP agents, which can provide information to a management program in at least one of the following ways:

- In response to a `GET`, `GETBULK`, or `GETNEXT` operation, which is a specific request for information from the management system.
- By sending a notification which is an alert sent by the SNMP agent to notify the management system of a particular event or condition.

Management Information Base (MIB) files define the information that can be provided by managed devices. The MIB files define managed objects, described by object identifiers (OIDs) and variables arranged in a hierarchy.

vCenter and ESX have SNMP agents. The agent provided with each product has different capabilities.

Using SNMP Traps with vCenter

The SNMP agent included with vCenter can be used to send traps when vCenter starts and when an alarm is triggered on vCenter. The vCenter SNMP agent functions only as a trap emitter and does not support other SNMP operations, such as receiving `GET`, `GETBULK`, and `GETNEXT` requests.

vCenter can send SNMPv1 traps to other management applications. You must configure your management server to interpret the SNMP traps sent by vCenter.

To use the vCenter SNMP traps, configure the SNMP settings on vCenter and your management client software to accept the traps from vCenter.

The traps sent by vCenter are defined in `VMWARE-VC-EVENT-MIB.mib`.

Configure SNMP Settings for vCenter

If you plan to use SNMP with vCenter, you must use the vSphere Client to configure the SNMP settings.

- Verify that the vSphere Client is connected to a vCenter instance.

- Verify that you have the domain name or IP address of the SNMP receiver, the port number of the receiver, and the community string.

1. In the vSphere Client, navigate to a vCenter instance.
2. Click the **Configure** tab.
3. Under **Settings**, click **General**.
4. On the vCenter Settings central pane, click **Edit**.
   The **Edit vCenter general settings** wizard opens.
5. Click **SNMP receivers** to edit their settings.
6. Enter the following information for the primary receiver of the SNMP traps.

Option	Description
Enable receiver	Activate the toggle to enable the SNMP receiver.
Primary Receiver URL	Enter the domain name or IP address of the receiver of SNMP traps.
Receiver port	Enter the port number of the receiver to which the SNMP agent sends traps. If the port value is empty, vCenter uses port 162 by default.
Community string	Enter the community string that is used for authentication.

7. Optional: Activate the toggle for receiver 2, receiver 3, or reciver 4 and configure their settings.
8. Click **Save**.

The vCenter system is now ready to send traps to the management system you have specified.
Configure your SNMP management software to receive and interpret data from the vCenter SNMP agent. See Configure SNMP Management Client Software.

Configure SNMP for ESX

ESX includes an SNMP agent that can send notifications (traps and informs) and receive `GET`, `GETBULK`, and `GETNEXT` requests.

The SNMP agent adds support for version 3 of the SNMP protocol, offering increased security and improved functionality, including the ability to send informs. You can use `esxcli` commands to enable and configure the SNMP agent. You configure the agent differently depending on whether you want to use SNMP v1/v2c or SNMP v3.

As an alternative to configuring SNMP manually using `esxcli` commands, you can use host profiles to configure SNMP for an ESX host. See the *vSphere Host Profiles* documentation for more information.

Configure the SNMP Agent for Polling

If you configure the ESX SNMP agent for polling, it can listen for and respond to requests from SNMP management client systems, such as `GET`, `GETNEXT` and `GETBULK` requests.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

By default, the embedded SNMP agent listens on UDP port 161 for polling requests from management systems. You can use the `esxcli system snmp set` command with the --port option to configure an alternative port. To avoid conflicting with other services, use a UDP port that is not defined in `/etc/services`.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. Run the `esxcli system snmp set` command with the --port option to configure the port.
   For example, run the following command:
   ```
   esxcli system snmp set --port port
   ```

Here, *port* is the port the SNMP agent uses to listen for polling requests.

Note:
The port you specify must not be already in use by other services. Use IP addresses from the dynamic range, port 49152 and up.

2. Optional: If the ESX SNMP agent is not enabled, run the following command:
   ```
   esxcli system snmp set --enable true
   ```

Configure ESX for SNMPv1 and SNMPv2c

When you configure the ESX SNMP agent for SNMPv1 and SNMPv2c, the agent supports sending notifications (traps and informs) and receiving `GET`, `GETBULK`, and `GETNEXT` requests.

In SNMPv1 and SNMPv2c, authentication is performed by using community strings. Community strings are namespaces which contain one or more managed objects. This form of authentication does not secure the communication between the SNMP agent and the management system. To secure the SNMP communication in your environment, use SNMPv3.

Configure SNMP Communities

To enable the ESX SNMP agent to send and receive SNMP v1 and v2c messages, you must configure at least one community for the agent.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

An SNMP community defines a group of devices and management systems. Only devices and management systems that are members of the same community can exchange SNMP messages. A device or management system can be a member of multiple communities.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

Run the `esxcli system snmp set` command with the --communities option to configure an SNMP community.
For example, to configure public, East, and West network operation centers communities, run the following command:
```
esxcli system snmp set --communities public,eastnoc,westnoc
```

Each time you specify a community with this command, the settings you specify overwrite the previous configuration. To specify multiple communities, separate the community names with a comma.

Configure the SNMP Agent to Send SNMP v1 or v2c Notifications

You can use the ESX SNMP agent to send virtual machine and environmental notifications to management systems.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

To send SNMP v1/v2c notifications with the SNMP agent, you must configure the target (receiver) unicast address, community, and an optional port. If you do not specify a port, the SNMP agent sends traps to UDP port 162 on the target management system by default.

To configure SNMP v3 traps, see Configure SNMP v3 Targets.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. Run the `esxcli system snmp set` command with the --targets option:
   ```
   esxcli system snmp set --targets target_address@port/community
   ```
   Here, *target_address* is the address of the target system, *port* is the port number to send the notifications to, and *community* is the community name.

   Each time you specify a target with this command, the settings you specify overwrite all previously specified settings. To specify multiple targets, separate them with a comma.

For example, run the following command for configuring the targets 192.0.2.1@163/westnoc and 2001:db8::1@163/eastnoc:

```
esxcli system snmp set --targets 192.0.2.1@163/westnoc,2001:db8::1@163/eastnoc
```

2. Optional: If the ESX SNMP agent is not enabled, run the following command:

```
esxcli system snmp set --enable true
```

3. Optional: Send a test trap to verify that the agent is configured correctly by running the `esxcli system snmp test` command.

   The agent sends a `warmStart` trap to the configured target.

Configure ESX for SNMP v3

When you configure the ESX SNMP agent for SNMPv3, the agent supports sending notifications (traps and informs) and receiving `GET`, `GETBULK`, and `GETNEXT` requests. SNMPv3 also provides stronger security than SNMPv1 or SNMPv2c, including key authentication and encryption.

Inform is a notification that the sender resends up to three times or until the receiver acknowledges the notification.

Configure the SNMP Engine ID

Every SNMP v3 agent has an engine ID which serves as a unique identifier for the agent. The engine ID is used with a hashing function to generate keys for authentication and encryption of SNMP v3 messages.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

If you do not specify an engine ID, when you enable the SNMP agent, an engine ID is automatically generated.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

Run the `esxcli system snmp set` command with the --engineid option to configure the SNMP engine ID.
For example, run the following command:

```
esxcli system snmp set --engineid id
```

Here, *`id`* is the engine ID and it must be a hexadecimal string between 5 and 32 characters long.

Configure SNMP Authentication and Privacy Protocols

SNMPv3 optionally supports authentication and privacy protocols.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

Authentication is used to ensure the identity of users. Privacy allows for encryption of SNMP v3 messages to ensure confidentiality of data. These protocols provide a higher level of security than is available in SNMPv1 and SNMPv2c, which use community strings for security.

Both authentication and privacy are optional. However, you must enable authentication to enable privacy.

The SNMPv3 authentication and privacy protocols are licensed vSphere features and might not be available in some vSphere editions.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. Optional: Run the `esxcli system snmp set` command with the --authentication option to configure authentication.
   For example, run the following command:

```
esxcli system snmp set --authentication protocol
```

   Here, *`protocol`* must be either `none` (for no authentication) or `SHA1`.

2. Optional: Run the `esxcli system snmp set` command with the --privacy option to configure privacy.
   For example, run the following command:
   ```
   esxcli system snmp set --privacy protocol
   ```
   Here, *protocol* must be either `none` (for no privacy) or `AES128`.

Configure SNMP Users

You can configure up to 5 users who can access SNMP v3 information. User names must be no more than 32 characters long.

- Verify that you have configured the authentication and privacy protocols before configuring users.
- Verify that you know the authentication and privacy passwords for each user you plan to configure. Passwords must be at least 7 characters long. Store these passwords in files on the host system.
- Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

While configuring a user, you generate authentication and privacy hash values based on the user's authentication and privacy passwords and the SNMP agent's engine ID. If you change the engine ID, the authentication protocol, or the privacy protocol after configuring users, the users are no longer valid and must be reconfigured.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. If you are using authentication or privacy, get the authentication and privacy hash values for the user by running the `esxcli system snmp hash` command with the --auth-hash and --priv-hash flags.
   For example, run the following command:
   ```
   esxcli system snmp hash --auth-hash secret1 --priv-hash secret2
   ```
   Here, *secret1* is the path to the file containing the user's authentication password and *secret2* is the path to the file containing the user's privacy password.

   Alternatively, you can pass the --raw-secret flag and specify the passwords directly on the command line.

   For example, you can run the following command:
   ```
   esxcli system snmp hash --auth-hash authsecret --priv-hash privsecret --raw-secret
   ```
   The produced output might be the following:
   ```
   Authhash: 08248c6eb8b333e75a29ca0af06b224faa7d22d6
   Privhash: 232ba5cbe8c55b8f979455d3c9ca8b48812adb97
   ```
   The authentication and privacy hash values are displayed.
2. Configure the user by running the `esxcli system snmp set` command with the --user flag.
   For example, you can run the following command:
   ```
   esxcli system snmp set --users userid/authhash/privhash/security
   ```
   The command accepts the following parameters:

Parameter	Description
userid	The user name.
authhash	The authentication hash value.
privhash	The privacy hash value.
security	The level of security enabled for that user, which can be *auth* (for authentication only), *priv* (for authentication and privacy), or *none* (for no authentication or privacy).

   For example, run the following command to configure user1 for access with authentication and privacy:
   ```
   esxcli system snmp set --users user1/08248c6eb8b333e75a29ca0af06b224faa7d22d6/
   232ba5cbe8c55b8f979455d3c9ca8b48812adb97/priv
   ```
   You must run the following command to configure user2 for access with no authentication or privacy:

```
esxcli system snmp set --users user2/-/-/none
```

3. Optional: Test the user configuration by running the following command:
   ```
   esxcli system snmp test --user username --auth-hash secret1 --priv-hash secret2
   ```
   If the configuration is correct, this command returns the following message: "User *username* validated correctly using engine id and security level: *protocols*". Here, *protocols* indicates the security protocols configured.

Configure SNMP v3 Targets

Configure SNMP v3 targets to allow the ESX SNMP agent to send SNMP v3 traps and informs.

- Ensure that the users who access the traps or informs are configured as SNMP users for both the ESX SNMP agent and the target management system.
- If you are configuring informs, you need the engine ID for the SNMP agent on the remote system that receives the inform message.
- Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

SNMP v3 allows for sending both traps and informs. An inform message is a type of a message that the sender resends a maximum of three times. The sender waits for 5 seconds between each attempt, unless the message is acknowledged by the receiver.

You can configure a maximum of three SNMP v3 targets, in addition to a maximum of three SNMP v1/v2c targets.

To configure a target, you must specify a hostname or IP address of the system that receives the traps or informs, a user name, a security level, and whether to send traps or informs. The security level can be either `none` (for no security), `auth` (for authentication only), or `priv` (for authentication and privacy).

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. Optional: If you are configuring informs, configure the remote users by running the `esxcli system snmp set` command with the --remote-users option.
   For example, run the following command:
   ```
   esxcli system snmp set --remote-users userid/auth-protocol/auth-hash/priv-protocol/priv-hash/engine-id
   ```
   The command accepts the following parameters:

Parameter	Description
userid	The user name.
auth-protocol	The authentication protocol, `none` (for no authentication) or `SHA1`.
auth-hash	The authentication hash or − if the authentication protocol is `none`.
priv-protocol	The privacy protocol, `AES128`, or `none`.
priv-hash	The privacy hash, or − if the privacy protocol is `none`.
engine-id	The engine ID of the SNMP agent on the remote system that receives the inform message.

2. Run the `esxcli system snmp set` command with the --v3targets option.
   For example, run the following command:
   ```
   esxcli system snmp set --v3targets hostname@port/userid/secLevel/message-type
   ```
   The parameters of the command are as follows.

Parameter	Description
hostname	The host name or IP address of the management system that receives the traps or informs.
port	The port on the management system that receives the traps or informs. If you do not specify a port, the default port, 162, is used.

Parameter	Description
userid	The user name.
secLevel	The level of authentication and privacy you have configured. Use `auth` if you have configured authentication only, `priv` if you have configured both authentication and privacy, and `none` if you have configured neither.
message-type	The type of the messages received by the management system. Use `trap` or `inform`.

3. Optional: If the ESX SNMP agent is not enabled, run the following command:
   ```
   esxcli system snmp set --enable true
   ```
4. Optional: Send a test notification to verify that the agent is configured correctly by running the `esxcli system snmp test` command.

   The agent sends a `warmStart` notification to the configured target.

Configure the Source of Hardware Events Received by the SNMP Agent

You can configure the ESX SNMP agent to receive hardware events from IPMI sensors.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

IPMI sensors are used for hardware monitoring in ESX/ESXi 4.x and earlier.

If you run ESXCLI commands through ESXCLI, you must supply connection options that specify the target host and login credentials. If you use ESXCLI commands directly on a host using the ESX Shell, you can use the commands as given without specifying connection options. For more information on connection options see *ESXCLI Concepts and Examples*.

1. Run the `esxcli system snmp set --hwsrc source` command to configure the source for hardware events.

   Here, *source* is `sensors`, for hardware event received from IPMI sensors. Default configuration is none, for no hardware events source set.
2. Optional: If the ESX SNMP agent is not enabled, run the following command:
   ```
   esxcli system snmp set --enable true
   ```

Configure the SNMP Agent to Filter Notifications

You can configure the ESX SNMP agent to filter out notifications if you don't want your SNMP management software to receive those notifications.

Configure the ESX SNMP agent by using the ESXCLI commands. See *Getting Started with ESXCLI* for more information on how to use ESXCLI.

1. Run the `esxcli system snmp set` command to filter notifications:
   ```
   esxcli system snmp set --notraps oid_list
   ```
   Here, *oid_list* is a list of OIDs for the notifications to filter, separated by commas. This list replaces any OIDs that were previously specified using this command.

   For example, to filter out `coldStart` (OID `1.3.6.1.4.1.6876.4.1.1.0`) and `warmStart` (OID `1.3.6.1.4.1.6876.4.1.1.1`) traps, run the following command:
   ```
   esxcli system snmp set --notraps 1.3.6.1.4.1.6876.4.1.1.0,1.3.6.1.4.1.6876.4.1.1.1
   ```
2. Optional: If the ESX SNMP agent is not enabled, run the following command:
   ```
   esxcli system snmp set --enable true
   ```

The traps identified by the specified OIDs are filtered out of the output of the SNMP agent, and are not sent to SNMP management software.

To clear all notification filters, run the `esxcli system snmp set --notraps reset` command.

Configure SNMP Management Client Software

After you have configured a vCenter instance or an ESX host to send traps, configure your management client software to receive and interpret those traps.

Download the VMware MIB files, see Broadcom knowledge base article at https://knowledge.broadcom.com/external/article/313538.

To configure your management client software, specify the communities for the managed device, configure the port settings, and load the VMware MIB files. See the documentation for your management system for specific instructions for these steps.

1. In your management software, specify the vCenter instance or ESX host as an SNMP-based managed device.
2. If you are using SNMPv1 or SNMPv2c, set up appropriate community names in the management software.

   These names must correspond to the communities set for the SNMP agent on the vCenter instance or ESX host.
3. If you are using SNMPv3, configure users and authentication and privacy protocols to match those configured on the ESX host.
4. If you configured the SNMP agent to send traps to a port on the management system other than the default UDP port 162, configure the management client software to listen on the port you configured.
5. Load the VMware MIBs into the management software so you can view the symbolic names for vCenter or the host variables.

   To prevent lookup errors, load these MIB files in the following order before loading other MIB files:

   1. `VMWARE-ROOT-MIB.mib`
   2. `VMWARE-TC-MIB.mib`
   3. `VMWARE-PRODUCTS-MIB.mib`

The management software can now receive and interpret traps from vCenter or ESX hosts.

SNMP Diagnostics

You can use SNMP tools to diagnose configuration problems.

- Run the `esxcli system snmp test` command from the ESXCLI set to prompt the SNMP agent to send a test `warmStart` trap.
- Run the `esxcli system snmp get` command to display the current configuration of the SNMP agent.
- The `SNMPv2-MIB.mib` file provides several counters to aid in debugging SNMP problems. See SNMPv2 Diagnostic Counters.
- The `VMWARE-ESX-AGENTCAP-MIB.mib` file defines the capabilities of the VMware SNMP agents by product version. Use this file to determine if the SNMP functionality that you want to use is supported.

Monitor Guest Operating Systems with SNMP

You can use SNMP to monitor guest operating systems or applications running in virtual machines.

The virtual machine uses its own virtual hardware devices. Do not install agents in the virtual machines that are intended to monitor physical hardware.

- Install the SNMP agents you normally would use for that purpose in the guest operating systems.

VMware MIB Files

VMware MIB files contain information provided by ESX hosts and vCenter to SNMP management software.

Management Information Base (MIB) files define the information that can be provided by managed devices. The MIB files define managed objects, described by object identifiers (OIDs) and variables arranged in a hierarchy. SMI structure of management information (RFC 2578) is the syntax used to write MIB files for specific product and function. These MIB files are versioned independently of product and can be used to identify event types and event data related information.

For information about how you can download these MIB files refer to the Broadcom knowledge base article at https://knowledge.broadcom.com/external/article/313538.

The **VMware MIB Files** table lists the MIB files provided by VMware and describes the information provided by each file.

Table 500: VMware MIB Files

MIB File	Description
VMWARE-ROOT-MIB.mib	Contains VMware's enterprise OID and top-level OID assignments.
VMWARE-ESX-AGENTCAP-MIB.mib	Defines the capabilities of the VMware agents by product versions. This file is optional and might not be supported by all management systems.
VMWARE-CIMOM-MIB.mib	Defines variables and trap types used to report on the state of the CIM Object Management subsystem.
VMWARE-ENV-MIB.mib	Defines variables and trap types used to report on the state of physical hardware components of the host computer. Enables conversion of CIM indications to SNMP traps.
VMWARE-OBSOLETE-MIB.mib	For use with versions of ESX/ESXi prior to 4.0. Defines OIDs that have been made obsolete to maintain backward compatibility with earlier versions of ESX. Includes variables formerly defined in the files VMWARE-TRAPS-MIB.mib and VMWARE-VMKERNEL-MIB.mib.
VMWARE-PRODUCTS-MIB.mib	Defines OIDs to uniquely identify each SNMP agent on each VMware platform by name, version, and build platform.
VMWARE-RESOURCES-MIB.mib	Defines variables used to report information on resource usage of the VMkernel, including physical memory, CPU, and disk utilization.
VMWARE-SYSTEM-MIB.mib	The VMWARE-SYSTEM-MIB.mib file is obsolete. Use the SNMPv2-MIB to obtain information from sysDescr.0 and sysObjec ID.0.
VMWARE-TC-MIB.mib	Defines common textual conventions used by VMware MIB files.
VMWARE-VC-EVENT-MIB.mib	Defines traps sent by vCenter. Load this file if you use vCenter to send traps.
VMWARE-VMINFO-MIB.mib	Defines variables for reporting information about virtual machines, including virtual machine traps.

The Other MIB Files table lists MIB files included in the VMware MIB files package that are not created by VMware. These can be used with the VMware MIB files to provide additional information.

Table 501: Other MIB Files

MIB File	Description
ENTITY-MIB.mib	Allows the description of relationships between physical entities and logical entities managed by the same SNMP agent. See RFC 4133 for more information.
HOST-RESOURCES-MIB.mib	Defines objects that are useful for managing host computers.
HOST-RESOURCES-TYPES.mib	Defines storage, device, and filesystem types for use with HOST-RESOURCES-MIB.mib.
IEEE8021-BRIDGE-MIB	Defines objects for managing devices that support IEEE 802.1D.
IEEE8023-LAG-MIB	Defines objects for managing devices that support IEEE 802.3ad link aggregation.
IEEE8021-Q-BRIDGE-MIB	Defines objects for managing Virtual Bridged Local Area Networks.
IF-MIB.mib	Defines attributes related to physical NICs on the host system.
IP-MIB.mib	Defines objects for managing implementations of the Internet Protocol (IP) in an IP version-independent manner.
IP-FORWARD-MIB.mib	Defines objects for managing IP forwarding.
LLDP-V2-MIB.mib	Defines objects for managing devices using Linked Layer Discovery Protocol (LLDP).
SNMPv2-CONF.mib	Defines conformance groups for MIBs.

MIB File	Description
SNMPv2-MIB.mib	Defines the SNMP version 2 MIB objects.
SNMPv2-SMI.mib	Defines the Structure of Management Information for SNMP version 2.
SNMPv2-TC.mib	Defines textual conventions for SNMP version 2.
TCP-MIB.mib	Defines objects for managing devices using the TCP protocol.
UDP-MIB.mib	Defines objects for managing devices using the UDP protocol.

SNMPv2 Diagnostic Counters

The SNMPv2-MIB.mib file provides a number of counters to aid in debugging SNMP problems.

Table 502: Diagnostic Counters from SNMPv2-MIB lists some of these diagnostic counters.

Table 502: Diagnostic Counters from SNMPv2-MIB

Variable	ID Mapping	Description
snmpInPkts	snmp 1	The total number of messages delivered to the SNMP entity from the transport service.
snmpInBadVersions	snmp 3	The total number of SNMP messages that were delivered to the SNMP entity and were for an unsupported SNMP version.
snmpInBadCommunityNames	snmp 4	The total number of community-based SNMP messages delivered to the SNMP entity that used an invalid SNMP community name.
snmpInBadCommunityUses	snmp 5	The total number of community-based SNMP messages delivered to the SNMP entity that represented an SNMP operation that was not allowed for the community named in the message.
snmpInASNParseErrs	snmp 6	The total number of ASN.1 or BER errors encountered by the SNMP entity when decoding received SNMP messages.
snmpEnableAuthenTraps	snmp 30	Indicates whether the SNMP entity is permitted to generate authenticationFailure traps. The value of this object overrides any configuration information. It therefore provides a means of disabling all authenticationFailure traps.
snmpSilentDrops	snmp 31	The total number of Confirmed Class PDUs delivered to the SNMP entity that were silently dropped because the size of a reply containing an alternate Response Class PDU with an empty variable-bindings field was greater than either a local constraint or the maximum message size associated with the originator of the request.
snmpProxyDrops	snmp 32	The total number of Confirmed Class PDUs delivered to the SNMP entity that were silently dropped because the transmission of the message to a proxy target failed in a manner other than a time-out such that no Response Class PDU could be returned.

System Log Files

In addition to lists of events and alarms, vSphere components generate assorted logs. These logs contain additional information about activities in your vSphere environment. Broadcom technical support might request several files to help resolve any issues you might have with the product.

Log files are an important component of troubleshooting attacks and obtaining information about breaches. All ESX hosts run a syslog service, which logs messages from the VMkernel and other system components to local files or to a remote host.

For information how to configure the syslog service on ESX hosts, see Configure Syslog on ESX Hosts.

For information about Configure Log Filtering on ESX Hosts

Table 503: Type and Location of System Log Files

Log File Type	Log File Location
ESX System Logs	`/var/run/log`
vSphere Client Logs	`/var/log/vmware/vsphere-ui/logs` **Note:** The main vSphere Client log file is `vsphere_client_virgo.log`.

ESX Log File Locations

ESX records host activity in log files, using a syslog facility.

Table 504: ESX Log File Locations

Component	Location	Purpose
Authentication	`/var/log/auth.log`	Contains all events related to authentication for the local system.
ESX host agent log	`/var/log/hostd.log`	Contains information about the agent that manages and configures the ESX host and its virtual machines.
Shell log	`/var/log/shell.log`	Contains a record of all commands typed into the ESX Shell and shell events (for example, when the shell was enabled).
System messages	`/var/log/syslog.log`	Contains all general log messages and can be used for troubleshooting. This information was formerly located in the messages log file.
vCenter agent log	`/var/log/vpxa.log`	Contains information about the agent that communicates with vCenter (if the host is managed by vCenter).
Virtual machines	The same directory as the affected virtual machine's configuration files, named `vmware.log` and `vmware*.log`. For example, `/vmfs/volumes/datastore/virtual machine/vmware.log`	Contains virtual machine power events, system failure information, tools status and activity, time sync, virtual hardware changes, vMotion migrations, machine clones, and so on.
VMkernel	`/var/log/vmkernel.log`	Records activities related to virtual machines and ESX.
VMkernel summary	`/var/log/vmksummary.log`	Used to determine uptime and availability statistics for ESX (comma separated).
VMkernel warnings	`/var/log/vmkwarning.log`	Records activities related to virtual machines and ESX.
Quick Boot	`/var/log/loadESX.log`	Contains all events related to restarting an ESX host through Quick Boot.

Component	Location	Purpose
Trusted infrastructure agent	`/var/run/log/kmxa.log`	Records activities related to the Client Service on the ESX Trusted Host.
Key Provider Service	`/var/run/log/kmxd.log`	Records activities related to the vSphere Trust Authority Key Provider Service.
Attestation Service	`/var/run/log/attestd.log`	Records activities related to the vSphere Trust Authority Attestation Service.
ESX Token Service	`/var/run/log/esxtokend.log`	Records activities related to the vSphere Trust Authority ESX Token Service.
ESX API Forwarder	`/var/run/log/esxapiadapter.log`	Records activities related to the vSphere Trust Authority API forwarder.

View System Log Files on an ESX Host

You can use the Direct Console User Interface (DCUI) to view system logs on an ESX host. These logs provide information about system operational events.

1. From the ESX Host Client, click **Manage** > **Services**. Confirm that the DCUI service is running.
2. Open an SSH client (such as PuTTY) and connect to the ESX host.
3. Log in with the root account.
4. Enter `DCUI` to start the Direct Console User Interface.
5. Click inside the console window and press F2 to customize the system.
6. Type the password for the root user and press Enter.
7. Use the arrow keys to select **View System Logs**.
8. Press a corresponding number key to view a log.
   vCenter agent (vpxa) logs appear if the host is managed by vCenter.
9. Press Enter or the spacebar to scroll through the messages.
10. Optional: Perform a regular expression search.
    a) Press the slash key (/).
    b) Type the text to find.
    c) Press Enter
       The found text is highlighted on the screen.
11. Press q to return to the direct console.

Collecting Log Files

Broadcom technical support might request several files to help resolve technical issues. The following sections describe script processes for generating and collecting some of these files.

Set Verbose Logging

You can set verbose logging for vpxd logs.

1. Select a vCenter instance.
2. Click **Configure** tab.
3. Click **General** and in the **vCenter Settings** pane, click **Edit**.
4. In the **Edit vCenter general settings** window, Click **Logging settings** and select **verbose** from the **Log level** drop-down menu.
5. Click **Save**.

Collect vSphere Log Files

You can collect vSphere log files in to a single location.

- Verify that you have the **System.View** privilege.
- Verify tat you have **SystemConfiguration.Adminsitartors** group membership.

You can download the log bundle from vSphere Client connected to a vCenter system.

1. In vSphere Client menu, select **Administration** > **Deployment** > **System Configuration**.
2. Select a vCenter node and click **Export Support Bundle**.
3. Select the support bundle type.
   - Full support bundle: Contains information on all services.
   - Minimal support bundle: Contains basic product and configuration information.
   - Custom support bundle: Contains customized information. Select any of the manifest properties to download the corresponding log files.
4. Click **Export**.
5. Optional: You can also download the vCenter log bundle using another method:
   1. Right-click a vCenter instance and click **Export System Logs**.
   2. On the **Select hosts** page, select the hosts to include in the export bundle.
   3. Check **Include vCenter and vSphere UI Client logs** to optionally include vCenter and vSphere UI logs in the bundle.
   4. Click **Next**.
   5. On the **Select logs** page, select specific system logs to export.
   6. Click **Finish**.

      Note:
      The log bundle is generated as a `.zip` file. By default, the vpxd logs within the bundle are compressed as `.tgz` files. You must use `gunzip` to uncompress these files.

Collect ESX Log Files

You can collect and package all relevant ESX system and configuration information, a well as ESX log files. This information can be used to analyze the problems.

- Run the following script on the ESX Shell: `/usr/bin/vm-support`

  The resulting file has the following format: `esx-`*`date-unique-xnumber`*`.tgz`

Turn Off Compression for vpxd Log Files

By default, vCenter `vpxd` log files are rolled up and compressed into `.gz` files. You can turn off this setting to leave the `vpxd` logs uncompressed.

1. Log in to the vCenter using the vSphere Client.
2. Select **Administration** > **vCenter Settings**.
3. Select **Advanced Settings**.
4. In the **Key** text box, type `log.compressOnRoll`.
5. In the **Value** text box, type `false`.
6. Click **Add**, and click **OK**.

ESX VMkernel Files

If the VMkernel fails, an error message appears and then the virtual machine reboots. If you specified a VMware core dump partition when you configured your virtual machine, the VMkernel also generates a core dump and error log.

More serious problems in the VMkernel can freeze the machine without an error message or core dump.

Export System Log Files

When the vSphere Client is connected to vCenter, you can select hosts from which to download system log files.

To save diagnostic data for ESX hosts and vCenter, the vSphere Clientmust be connected to the vCenter system.

Required privileges:

- To view diagnostic data: **Read-Only User**
- To manage diagnostic data: **Global** > **Diagnostics**

1. In the inventory, right click a vCenter instance.
   Export System Logs option is available for Hosts and Virtual Machines.
2. Click **Export System Logs**.
3. If you are connected to vCenter, select the object for which you want to export data.
   Selecting an object selects all of its child objects.
4. If you are connected to vCenter, select the **Include vCenter and vSphere UI Client logs** checkbox to download vCenter and vSphere Client log files and host log files, and click **Next**.
5. If the selected host supports manifest driven exports of system log files, select the system log files to collect. Select the specific system log files to download.
   If the host does not support manifest exports of log files, all system log files are exported.
6. Select **Gather performance data** to include performance data information in the log files.
   You can update the duration and interval time you want the data collected.
7. Optional: Select **Password for encrypted core dumps** to apply a password for encrypted core dumps to the support package.

   You can make that password available to your support representative in a secure channel.

   If only some of the host in your environment use encryption, some of the files in the package are encrypted.
8. Click **Finish**.
9. Specify the location to which to save the log files.

   The host or vCenter generates a `.zip` file containing the log files.

   The **Recent Tasks** panel shows the Generate diagnostic bundles task in progress.

   The Downloading Log Bundles dialog box appears when the Generating Diagnostic Bundle task is finished. The download status of each bundle appears in the dialog box.

Some network errors can cause download failures. When you select an individual download in the dialog box, the error message for that operation appears under the name and location of the log bundle file.

10. Click **Save**.
11. Verify the information in the Summary and click **Finish** to download the log files.

Diagnostic bundles containing log files for the specified objects are downloaded to the location specified.

Upload the Logs Package to a VMware Service Request.

Upload the Logs Package to a VMware Service Request

If you already have a Service Request ID with VMware, you can use the vSphere Client to upload the system log bundles directly to your service request.

Request a Service Request ID from VMware Technical Support.

1. In the vSphere Client, navigate to **Administration**.
2. Click **Support**, and click **Upload File to Service Request**.

   You see a link to a KB article that gives detailed information about how to upload your files to cases by using the Broadcom Support Portal.
3. Follow the steps in the KB article.

Manage the Amount of Virtual Machine Log Data

Virtual machines can write support and troubleshooting information into a virtual machine log file stored on a VMFS volume.

The default settings for virtual machines are appropriate for most situations.

If your environment relies heavily on using vMotion, or if the defaults do not seem suitable for other reasons, you can modify the logging settings for virtual machine guest operating systems.

A new log file is ALWAYS created during virtual machine power-on, resume, migration with vMotion, and so on.

You can further adjust the amount of log data generated by a virtual machine by making adjustments to two parameters:

- `vmx.log.keepOld`: The number of log files retained per virtual machine is controlled by the `vmx.log.keepOld` property.

  Note:
  VMware recommends saving a minimum of 10 log files. If you need log files for a longer time span, you can set `vmx.log.keepOld` to 20 or more.
- `vmx.log.rotateSize`: The maximum size of each log file, in bytes, is controlled by `vmx.log.rotateSize` property.
  To ensure that a new log file is created more frequently, you can limit the maximum size of the log files with this configuration parameter. Each time an entry is written to the log, the size of the log is checked. If `vmx.log.rotateSize` is set to a non default value, and the size is over the limit, the next entry is written to a new log file. If the maximum number of log files exists, the oldest log file is deleted.

  Note:
  If this parameter is used, VMware recommends setting it to a minimum of 2000000 bytes.

The total amount of storage space consumed by all the log files is the product of these two parameters.

These settings may be set on an individual virtual machine basis or for all virtual machines on an ESX host.

The following procedure discusses limiting the amount of virtual machine log data for all virtual machines on an ESX host:

- To place limits on the amount of virtual machine log data, edit `/etc/vmware/config`
- If the `vmx.log.keepOld` and `vmx.log.rotateSize` properties are not defined in the file, add them.

For example, to retain ten log files, each one no larger than 2000000 bytes, add the following to `/etc/vmware/config`:

```
vmx.log.keepOld = "10"
vmx.log.rotateSize = "2000000"
```

The maximum amount of storage space consumed by each virtual machine is 10 * 2000000 bytes.

Note:

You can use a PowerCLI script to change these properties for all the virtual machines on a host.

Handle Virtual Machine Log Throttling

You can deal with the log throttling of a virtual machine.

By default, virtual machine logs monitor the logging rate. If the logging rate exceeds an established rate, logging is throttled (that is, suspended for a time before resuming). This prevents problematic logging from consuming all of the available log file storage space.

Virtual machine log throttle is identified by the presence of `<< file Throttled>>` in virtual machine log files.

Note:
If throttling occurs, you may be asked to temporarily disable log throttling for debugging purposes. When the debugging has completed, it is very important is re-enable throttling.

The virtual machine log throttling rate is controlled by the `vmx.log.throttleBytesPerSec` property.

This setting may be set on an individual virtual machine basis or for all virtual machines on an ESX host.

To disable virtual machine throttling for all virtual machines on an ESX host, take the following actions:

- Edit the `/etc/vmware/config` file.
- If the `vmx.log.throttleBytesPerSec` property is not defined in the file, add it.
  For example:
  ```
  vmx.log.throttleBytesPerSec = "-1"
  ```

Note:
You can use a PowerCLI script to change these properties for all the virtual machines on a host.

vSphere Single Host Management - VMware Host Client

vSphere Single Host Management - VMware Host Client provides information about managing single hosts with the VMware Host Client.

The VMware Host Client can be used to conduct emergency management when vCenter is unavailable. You can use the VMware Host Client to perform administrative tasks, basic troubleshooting tasks, and advanced administrative tasks.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we have updated this guide to remove instances of non-inclusive language.

Intended Audience

This information is intended for anyone who wants to use the VMware Host Client to manage single ESX hosts. The information is written for experienced Windows or Linux system administrators who are familiar with virtual machine technology and data center operations.

VMware Host Client Release Notes

Release notes include product enhancements and notices, bug fixes, and resolved issues.

VMware Host Client 9.1 Release Notes

VMware Host Client 9.1 Release Notes

This document contains the following sections

- Introduction
- What's New
- What's in the Release Notes
  - VMware Host Client Overview
  - Previous Releases
- Product Support Notices
- Internationalization
- Known Issues
- Known Issues from Previous Releases

Introduction

VMware Host Client 9.1 | 12 MAY 2026 | Build 25370933 Check for additions and updates to these release notes.

What's New

- The VMware Host Client 9.1 features a new, modernized user interface aligned with the vSphere Client design. This update provides full feature parity with previous versions, allowing consistent management of compute, storage, and networking at the single ESX host level.
- The VMware Host Client 9.1 provides the following improvements.
  - Virtual machine operations: create, review, and register VMs; perform VM power operations; and modify VM settings.
  - ESX host management: manage host parameters, power operations, datastores, and networking.
  - Enhanced grid navigation: introduced virtual scrolling within managed grids to provide a faster and more responsive browsing experience for large data sets.
  - Roles and Permissions: added the ability to export user roles and assigned permissions to a CSV file.

What's in the Release Notes

The release notes cover the following topics:

VMware Host Client Overview

The VMware Host Client is an HTML5-based client that is used to connect to and manage single ESX hosts. It can be used to perform administrative tasks to manage host resources such as virtual machines, networking, and storage. The VMware Host Client can also be helpful for troubleshooting individual virtual machines or hosts if vCenter and the vSphere Client are unavailable.

For more information, see the vSphere 9.1 Documentation.

Previous Releases

- VMware Host Client 9.0
- VMware Host Client 2.18.0
- VMware Host Client 2.14.0
- VMware Host Client 2.12.0
- VMware Host Client 2.5.0
- VMware Host Client 1.37.0
- VMware Host Client 1.34.8
- VMware Host Client 1.34.4
- VMware Host Client 1.34.0
- VMware Host Client 1.33.3
- VMware Host Client 1.30.0
- VMware Host Client 1.25.0
- VMware Host Client 1.21
- VMware Host Client 1.8
- VMware Host Client 1.4
- VMware Host Client 1.2
- VMware Host Client 1.0

Product Support Notices

Starting with VMware Host Client 9.1, the **Client Build Number** is no longer displayed in the **About** tab.

Internationalization

vSphere Single Host Management - VMware Host Client 9.1 is available in the following languages:

- English
- French
- Japanese
- Spanish

Known Issues

Setting the NTP service startup policy to "Start and stop with port usage" fails with an error

In VMware Host Client 9.1, when you navigate to **Host > Configure > System > Services** and set the NTP service startup policy "Start and stop with port usage", the following error message displays: "ServerFaultCode: A specified parameter was not correct: policy".

This issue occurs because the **ntpClient firewall ruleset is now system-managed**. The system automatically activates or deactivates the required firewall rules whenever you start or stop the NTP service. As a result, the "Start and stop with port usage" policy is obsolete for this service and can no longer be set explicitly in the user interface.

Note:

There is no functional impact when upgrading from version 9.0 to 9.1. If the host previously had this policy activated, the NTP service continues to function as expected after the upgrade. The error only appears if you attempt to manually reapply or modify the policy to "Start and stop with port usage" within the version 9.1 user interface.

Workaround: Start or stop the NTP service. The system automatically manages the required firewall rules and startup behavior, so you do not need to manually set this startup policy. For more information about starting and stoping the NTP service, see Edit the Time Configuration of an ESX Host in the VMware Host Client.

Known Issues from Previous Releases

The Current reading parameter of the Fan Redundancy health sensor displays Redundancy lost

Although the system fan health sensor is in green status, the **Current reading** parameter displays **Redundancy lost**. When you run the ESXCLI command `esxcli hardware ipmi sdr list`, the **Computed reading** returns a **Fully Redundant** value. As both **Current reading** and **Computed reading** parameters are the same, the **Redundancy lost** value is false.

Perform the following steps to reproduce this issue:

1. Install an ESXi 8.0 guest operating system.
2. Log in to the VMware Host Client.
3. Click **Monitor** in the VMware Host Client inventory.
4. On the **Hardware tab**, click **System sensors**.
5. Search for Fan sensors.
   Result: **Current reading** column for **System Board 1 Fan Redundancy 0** displays **Redundancy lost**.

Workaround: None

When you create a VM through the vSphere Client and the VMware Host Client, the maximum number of cores per socket might be different for the same guest OS

When you create a virtual machine for the same guest operating system through the vSphere Client and the VMware Host Client, the CPU Cores per Socket drop-down menu might display different numbers.

Workaround: Use the vSphere Client because the VMware Host Client is in maintenance mode.

You might not change the Load balancing setting of NIC teaming

The Load balancing setting of NIC teaming might not change when you select the **Inherit from vSwitch** option.

Perform the following steps to reproduce the issue:

1. Open Google Chrome and log in to the VMware Host Client.
2. Click **Networking** in the VMware Host Client inventory and click **Management Network**.
3. Click **Edit settings**.
   Result: The **Edit port group - Management Network** window appears.
4. Expand **NIC teaming**.
   a. From the **Load balancing** drop-down menu, select **Inherit from vSwitch** and click **Save**.
5. Complete steps 1-4.
6. The **Load balancing** option remains unchanged.

Workaround:

Override the **Policy** with the same settings as vSwitch, without using **Inherit from vSwitch**.

Do not edit **configstore**.

You cannot customize the firewall rule configuration with the option 'Only allow connections from the following networks' on ESX hosts

Starting with vSphere 8.0 Update 2, you cannot customize the firewall rule configuration with the option **Only allow connections from the following networks** on ESX hosts.

Perform the following steps.

1. Navigate to **Networking > Firewall rules**, and select **DHCP client**.
2. Click **Edit settings**.
3. Select **Only allow connections from the following networks** and enter an IP address.

When you click **OK**, the operation fails with an error such as `Operation failed, diagnostics report: Invalid operation requested: Can not change allowed ip list this ruleset, it is owned by system service`.

This is expected behavior.

Workaround: None

The vendor's name of some local storage devices displays incorrect or incomplete

In the VMware Host Client inventory, when you navigate to **Storage > Devices**, in the **Vendor** column of the table, you might see part of the device model name instead of the vendor's name.

Workaround: None

The names of future Windows OS releases appear with placeholders or code names

Because the official names of the future Windows OS releases Windows 12 and Windows 2025 are still unknown when you create a virtual machine, you might see placeholders or code names on the **Select a name and guest OS** page of the **New virtual machine** wizard.

Workaround: None

What is VMware Host Client

The VMware Host Client is an HTML5-based client that is used to connect to and manage a single ESX host.

You can use the VMware Host Client to:

- Perform administrative and basic troubleshooting tasks, and advanced administrative tasks on your target ESX host.
- Conduct an emergency management when vCenter is not available.

It is important to know that the VMware Host Client is different from the vSphere Client. You use the vSphere Client to connect to vCenter and manage multiple ESX hosts, whereas you use the VMware Host Client to manage a single ESX host.

VMware Host Client functions include, but are not limited to the following operations:

- Basic virtualization operations, such as deploying and configuring virtual machines of various complexity
- Creating and managing networking and datastores
- Advanced tuning of host level options to improve performance

VMware Host Client System Requirements

Make sure that your browser supports the VMware Host Client.

VMware Host Client supports the following guest operating systems and web browser versions.

Supported Browsers	Mac OS	Windows 32-bit and 64-bit	Linux
Google Chrome	120+	120+	120+
Mozilla Firefox	115+	115+	115+
Microsoft Edge	120+	120+	N/A

Use the VMware Host Client

You can use the VMware Host Client to conduct emergency management when vCenter is temporarily unavailable.

Start the VMware Host Client and Log In

You can use the VMware Host Client to manage single ESX hosts and perform various administrative and troubleshooting tasks on your virtual machines.

To log in to the ESX host, perform the following steps.

1. In a web browser, enter the target host name or IP address using the form `https://host-name/ui` or `https://host-IP-address/ui`.

   A login screen appears.
2. Enter your user name and your password.
3. Click **Log In** to continue.
4. Review the VMware Customer Experience Improvement Program (CEIP) page and choose whether you want to join the program.

   To learn about the program and how to configure it at any time, see Leave and Rejoin the Customer Experience Improvement Program in the VMware Host Client.
5. Click **OK**.

Log Out of the VMware Host Client

When you no longer must view or manage your target ESX host, log out of the VMware Host Client.

> **Note:**
> Closing a VMware Host Client session does not stop the host.

To log out of the ESX host, click the user name in the top-right corner of the VMware Host Client window and select **Log out** from the drop-down menu.

You are now logged out of the VMware Host Client. Your target ESX host continues to run all its normal activities.

Manage the Language and Keyboard Layout Settings of the VMware Host Client

You can change the language preferences of the VMware Host Client and determine the keyboard layout settings for the Web Console.

Select the language for the VMware Host Client user interface and specify the keyboard layout setting. By default, the display language of the VMware Host Client and the keyboard layout is the default language of the browser you use.

1. Click the user name in the top-right corner of the VMware Host Client window and select **Settings** > **My Preferences**.
2. To change the default browser language, click the **Language** tab and select **Specify a language**.
3. From the drop-down menu, select the language and click **Save**.

Configure the VMware Host Client Theme for the User Interface

To enhance your experience while performing your tasks, you can customize the look and feel of the VMware Host Client

After you customize the user interface, the VMware Host Client saves your individual preferences locally in your browser. By default, the VMware Host Client uses a white background theme. You can change to a dark mode theme.

Click the user name in the top-right corner of the VMware Host Client window and select **Settings** > **Switch Theme**.

The theme changes to a dark mode.

Leave and Rejoin the Customer Experience Improvement Program in the VMware Host Client

You can participate in the Customer Experience Improvement Program (CEIP) to provide anonymous feedback or information to VMware for quality, reliability, and functionality improvements of VMware products and services.

You can choose to leave the Customer Experience Improvement Program (CEIP), or rejoin the CEIP at any time.

For more information about the data collected through CEIP, see VMware Products in CEIP Program.

1. To leave and rejoin the CEIP, click the user name in the top-right corner of the VMware Host Client window.
2. Select **Settings** > **Send usage statistics**.
3. From the **Customer Experience Improvement Program** window, select or deselect **Join the Broadcom's Customers Experience Program** and click **OK**.

Configure the VMware Host Client Timeout Value

By default, the VMware Host Client session closes after 15 minutes of idle time and requires the user to log in again to resume using the VMware Host Client.

During login, the ESX host reads the `UserVars.HostClientSessionTimeout` advanced settings parameter that controls the idle time before the system automatically logs you out.

You can change the timeout value or deactivate the timeout.

1. Click the user name in the top-right corner of the VMware Host Client window and select **Settings** > **Application Timeout**.
2. Select the option you prefer.
3. To deactivate the timeout, select **Off**.

Host Management with the VMware Host Client

With the VMware Host Client, you can manage single ESX hosts during vCenter upgrades or when vCenter stops responding or becomes unavailable.

The VMware Host Client has a crucial set of troubleshooting functions, which allow you to perform tasks on the ESX host that you are logged in to if vCenter is unavailable. These functions include but are not limited to configuring advanced host settings, managing certificates, using the ESX Shell, enabling Lockdown mode, and so on.

Manage System Settings in the VMware Host Client

With the VMware Host Client, you can manage advanced host settings, configure start and stop policies for host services, and manage time and date configuration for the host.

Manage Advanced Settings in the VMware Host Client

You can change the settings of a host by using the VMware Host Client.

> **CAUTION:**
> Changing advanced options is considered unsupported unless VMware technical support or a Broadcom knowledge base article instructs you to do so. In all other cases, changing these options is considered unsupported. Usually, the default settings produce the optimum result.

1. In the VMware Host Client inventory, click **Host**.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the appropriate item and update its value.
6. Click **OK** to apply the changes.

Create an Initial Welcome Message for the Direct Console User Interface in the VMware Host Client

By using the VMware Host Client, you can create a welcome message that appears on the initial screen of the Direct Console User Interface (DCUI).

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the `Annotations.WelcomeMessage` key and in the Value text box, enter the welcome message.

   To set the default message, leave the **Value** text box blank.
6. Click **OK**.

Configure the SOAP Session Timeout in the VMware Host Client

In the VMware Host Client, you can configure the SOAP session timeout. The default value is 30 minutes.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the `Config.HostAgent.vmacore.soap.sessionTimeout` key and in the **Value** text box, enter the timeout setting in minutes.

   The default value is (30). A value of zero (0) deactivates the timeout.
6. Click **OK**.

Configure the Passwords and Account Lockout Policy in the VMware Host Client

For ESX hosts, you must use a password with predefined requirements. You can change the required password length, character class requirements, or allow passphrases, all using the Security.PasswordQualityControl advanced option. You can also set the number of passwords to remember for each user using the Security.PasswordHistory advanced option. This setting prevents duplicate or similar passwords. The Security.PasswordMaxDays advanced option allows you to set up the maximum number of days between password changes.

Note:
Always perform additional testing after you change the default password settings.

If you attempt to log in with incorrect credentials, the account lockout policy specifies when and for how long the system locks your account.

ESX Passwords

ESX enforces password requirements for access.

- By default, when you create a password, you must include a mix of characters from any three of the following four character classes: lowercase letters, uppercase letters, numbers, and special characters such as underscore or dash.
- By default, the password must contain a length of at least 7 characters and a maximum of 40 characters.
- Passwords must not contain a dictionary word or part of a dictionary word.
- Passwords must not contain the user name or parts of the user name.

Note:

An uppercase character that begins a password does not count toward the number of character classes used. A number that ends a password does not count toward the number of character classes used.

Example of ESX Passwords

The following password candidates illustrate potential passwords if the option is set as follows:

```
retry=3 min=disabled,disabled,disabled,7,7
```

With this setting, a user is prompted up to three times (retry=3) for a new password that is not sufficiently strong or if the password was not entered correctly twice. Passwords with one or two character classes and password phrases are not allowed, because the first three items are deactivated. Passwords from three and four character classes require 7 characters.

The following password candidates meet the password requirements:

- xQaTEhb!: Contains eight characters from three character classes.
- xQaT3#A: Contains seven characters from four character classes.

The following password candidates do not meet the password requirements:

- Xqat3hi: Begins with an uppercase character, reducing the effective number of character classes to two. The minimum number of required character classes is three.
- xQaTEh2: Ends with a number, reducing the effective number of character classes to two. The minimum number of required character classes is three.

Password Quality Control

You can control the quality of passwords by using the Security.PasswordQualityControl advanced option.

Security.PasswordQualityControl consists of several settings that follow the pattern:

```
retry=N min=N0,N1,N2,N3,N4 max=N passphrase=N similar=permit|deny
```

Password Quality Control Settings	Description	Default
`retry=N`	The number of times the user must provide a new password if the password is incorrect or not sufficiently strong.	`retry=3`
`min=N0,N1,N2,N3,N4`	Character class and the passphrase minimum length requirement. • `N0` is minimum length of passwords from a single character class. • `N1` is minimum length of passwords from two character classes. • `N2` is minimum length for a passphrase. • `N3` is minimum length for three character classes. • `N4` is minimum length for four character classes. You can use disabled to disallow a password with the specified number of character classes.	`min=disabled,disabled,disabled,7,7`
`max=N`	The maximum allowed password length.	`max=40`
`passphrase=N`	The number of words required for a passphrase. To make sure that the `passphrase` is recognized, do not set `N2` from the `min` setting to disabled.	`passphrase=3`

Password Quality Control Settings	Description	Default
`similar=permit|deny`	Indicates whether a password is allowed to be similar to the old password. To use this setting, make sure that you set the Security.PasswordHistory option to a non-zero value. Starting with vSphere 8.0 Update 1, the default value is 5.	`similar=deny`

ESX Passphrase

Instead of a password, you can use a passphrase. Passphrases are deactivated by default. You can change the default setting by using the Security.PasswordQualityControl advanced option.

For example, you can change the option to the following.

```
retry=3 min=disabled,disabled,16,7,7
```

This example allows passphrases of at least 16 characters. The passphrase must consist of at least 3 words, separated by spaces.

Example Password History and Rotation Policy

To remember history of 6 passwords, set the Security.PasswordHistory option to 6.

To enforce a 90-day password rotation policy, set the Security.PasswordMaxDays option to 90.

ESX Account Lockout Policy

Users are locked out after a preset number of consecutive failed attempts. By default, users are locked out after 5 consecutive failed attempts in 3 minutes and a locked account is unlocked automatically after 15 minutes by default. You can change the maximum allowed failed attempts and the period of time in which the user account is locked out by using the Security.AccountLockFailures and Security.AccountUnlockTime advanced options.

To configure the administrator passwords and account lockout behavior, perform the following steps.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.

Option	Action
Configure the required password length, character class requirement, or allow passphrases	1. In the **Edit Advanced System Settings** window, find the `Security.PasswordQualityControl` key. 2. In the **Value** text box, enter the new value.
Configure the number of passwords to remember for each user	1. In the **Edit Advanced System Settings** window, find the `Security.PasswordHistory` key. 2. In the **Value** text box, enter the new value. **Note:** Zero (0) deactivates password history.
Configure the maximum number of days between password changes	1. In the **Edit Advanced System Settings** window, find the `Security.PasswordMaxDays` key. 2. In the **Value** text box , enter the new value.
Configure the number of failed login attempts allowed before lockout	1. In the **Edit Advanced System Settings** window, find the `Security.AccountLockFailures` key. 2. In the **Value** text box, enter the new value. **Note:** Zero (0) deactivates account locking.
Configure the period of time in which the user's account is locked out	1. In the **Edit Advanced System Settings** window, find the `Security.AccountUnlockTime` key.

Option	Action
	2. In the **Value** text box, enter the new value.

5. Click **OK**.

Configure Syslog in the VMware Host Client

To configure the syslog service, you can use the VMware Host Client.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.

Option	Description
Syslog.global.LogHost	Remote host to which syslog messages are forwarded and the port on which the remote host receives syslog messages. You can include the protocol and the port, for example, `protocol://hostName1:port` where `protocol` can be udp, tcp, or ssl. You can use only port 514 for UDP. The ssl protocol uses TLS 1.2. For example: `ssl://hostName1:1514`. The value of `port` can be any decimal number between 1 and 65535. While no hard limit to the number of remote hosts to receive syslog messages exists, it is recommended to keep the number of remote hosts to five or less.
Syslog.global.logCheckSSLCerts	Enforce checking of the SSL certificates when you log in to a remote host.
Syslog.global.defaultRotate	Maximum number of archives to keep. You can set this number globally and for individual subloggers.
Syslog.global.defaultSize	Default size of the log, in KB, before the system rotates logs. You can set this number globally and for individual subloggers.
Syslog.global.LogDir	Directory where logs are stored. The directory can be on mounted NFS or VMFS volumes. Only the `/scratch` directory on the local file system is persistent across reboots. Specify the directory as [*datastorename*] *path_to_file*, where the path is relative to the root of the volume backing the datastore. For example, the path `[storage1] /systemlogs` maps to the path `/vmfs/volumes/storage1/systemlogs`.
Syslog.global.logDirUnique	Selecting this option creates a subdirectory with the name of the ESX host under the directory specified by **Syslog.global.LogDir**. A unique directory is useful if the same NFS directory is used by multiple ESX hosts.

5. In the **Edit Advanced System Settings** window, find the appropriate item and update its value.
6. Click **OK**.

Configure Advanced TLS/SSL Key Options

You can configure the security protocols and cryptographic algorithms that are used to encrypt communications with the ESX host.

The Transport Layer Security (TLS) key secures communication with the host using the TLS protocol. Upon first boot, the ESX host generates the TLS key as a 2048-bit RSA key. Currently, ESX does not implement automatic generation of ECDSA keys for TLS. The TLS private key is not intended to be serviced by the administrator.

The SSH key secures communication with the ESX host using the SSH protocol. Upon first boot, the system generates the SSH key as a 2048-bit RSA key. The SSH server is deactivated by default. SSH access is intended primarily for troubleshooting purposes. The SSH key is not intended to be serviced by the administrator. Logging in through SSH requires administrative privileges equivalent to full host control. To enable SSH access, see Activate the Secure Shell (SSH) in the VMware Host Client.

You can configure the following ESX host security key settings.

Note:
The UserVars.ESXiVPsAllowedCiphers security key setting only affects I/O filters.

Key	Default	Description
UserVars.ESXiVPsAllowedCiphers	!aNULL:kECDH+AESGCM:ECDH+AESGCM:RSA+AESGCM:kECDH+AES:ECDH+AES:RSA+AES	The default cipher control string.
Config.HostAgent.ssl.keyStore.allowAny	False	You can add any certificate to the ESX CA trust store.
Config.HostAgent.ssl.keyStore.allowSelfSigned	False	You can add non-CA self-signed certificates to the ESX CA trust store, that is, certificates that do not have the CA bit set.
Config.HostAgent.ssl.keyStore.discardLeaf	True	Discards leaf certificates added to the ESX CA trust store.

To configure the ESX security key settings, perform the following steps.

1. In the VMware Host Client inventory, click **Host**.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the appropriate item and update its value.
6. In the **Value** field, enter the new value and click **OK**.

Configure Userworld Memory Zeroing

With VMware Host Client, you can use the advanced option Mem.MemEagerZero to determine how pages are zeroed out for virtual machines and user space applications.

To zero all pages when they allocated to virtual machines and user space applications, set Mem.MemEagerZero to one (1). If the memory is not reused, this setting prevents exposing the information from a virtual machine or user space applications to other clients while preserving the previous content in memory.

When you set Mem.MemEagerZero to 1, pages are zeroed when a user space application exits. For virtual machines, such pages are zeroed if:

- The virtual machine is powered off.
- The virtual machine pages are migrated.
- The ESX host reclaims virtual machines memory.

Note:

For information about setting the advanced virtual machine options, see the *vSphere Resource Management* documentation.

To configure the userworld memory zeroing, perform the following steps.

1. In the VMware Host Client inventory, click **Host**.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the **Mem.MemEagerZero** key.
6. In the **Value** text box, enter the new value and click **OK**.

   The default value is zero (0).

Edit the Time Configuration of an ESX Host in the VMware Host Client

By using the VMware Host Client, you can configure the time settings of a host manually or can synchronize the time and date of the host with an NTP or a PTP server. NTP provides millisecond timing accuracy and PTP maintains microsecond timing accuracy.

The NTP service on the host periodically takes the time and date from the NTP server. You can use the **Start**, **Stop**, or **Restart** buttons to change the status of the NTP service on the host at any time regardless of the selected startup policy for the NTP service.

PTP provisions precise time synchronization for the virtual machines within a network. To change the PTP service on the host at any time, you can use the **Start**, **Stop**, or **Restart** buttons. Starting or stopping the PTP service automatically activates or deactivates PTP. To apply the change when you activate or deactivate PTP manually, start or stop the PTP service.

For more information about the services, see Manage Services in the VMware Host Client.

Note:

The NTP and PTP services cannot run simultaneously. Activating one service automatically deactivates the other.

1. In the VMware Host Client inventory, click **Host**.
2. Click the **Configure** tab.
3. Under **System**, click **Time Configuration**.
4. Set the time and date for the host.

Option	Action
Manually configure the date and time	1. Click **Edit**. The **Edit Manual Time Configuration** window appears. 2. Set the date and time for the host manually. 3. Click **OK**.
Network Time Protocol (Enable NTP client)	1. Click **Edit**. The **Edit Network Time Protocol** dialog box appears. 2. Select the **Enable** check box. 3. In the **NTP Servers** text box, enter the IP addresses or host names, comma-separated, of the NTP servers that you want to use. 4. From the **NTP Service Startup Policy** drop-down menu, select an option for starting and stopping the NTP service on the host. a. **Start and stop with host**. Starts and stops the NTP service when the host powers on and shuts down. b. **Start and stop manually**. Activates manual starting and stopping of the NTP service. If you select the **Start and stop manually** policy, the status of the NTP service changes only when you use the UI controls. 5. Click **OK**.

Option	Action
	Note: Activating or deactivating the NTP protocol is sufficient; you do not need to manually toggle the service start-up policy separately.
Use Precision Time Protocol (Enable PTP client)	1. Click **Edit**. 2. Select the **Enable** check box. 3. From the **Network interface** drop-down menu, select a network interface. The IPv4 and Subnet mask appear. 4. Click **OK**.

Manage Hardware for an ESX Host by Using the VMware Host Client

When you log in to an ESX host by using the VMware Host Client, you can manage PCI devices and can configure power management settings.

For more information about the host power management policies, see Host Power Management Policies in ESX.

Change Power Management Policies in the VMware Host Client

Change the power management policies of the host that you are managing to control the energy consumption of your host.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Hardware**, click **Power Management**.
4. Click the **Edit Power Policy** button.

   The available power management policies are displayed.
5. Select the policy that you want to apply and click **OK**.

Manage Services in the VMware Host Client

In the VMware Host Client, you can start, stop, and restart services that are running on the host that you are logged in to, and you can configure the host service policy.

You can restart services when you change host configurations or in case of suspected functional or performance issues.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Services**.
4. From the **Services** list, select a service.
5. Select an operation.
   - **Restart**
   - **Start**
   - **Stop**
6. To configure the service policy, click the **Edit Startup Policy** button.
7. In the **Edit Startup Policy** window, select an option for the service.
   - **Start and stop with host**
   - **Start and stop manually**
   - **Start and stop with port usage**

8. Click **OK**.

Managing Security and Users for an ESX Host by Using the VMware Host Client

The ESX hypervisor architecture has many built-in security features that you can configure to improve security.

By using the VMware Host Client, you can configure features, such as active directory, and you can also manage certificates.

Manage Host Authentication by Using the VMware Host Client

When you log in to an ESX host by using the VMware Host Client, you can check whether active directory and smart card authentication are enabled, and you can also join the host to a directory service domain.

You can configure ESX to use a directory service such as Active Directory to manage users. For more information, see *vSphere Security* documentation.

Join an ESX Host to a Directory Service Domain by Using the VMware Host Client

To use a directory service for your host, you must join the host to the directory service domain.

You can enter the domain name in one of two ways:

- `name.tld` (for example, `domain.com`): The account is created under the default container.
- `name.tld/container/path` (for example, `domain.com/OU1/OU2`): The account is created under a particular organizational unit (OU).

To use the vSphere Authentication Proxy service, see *vSphere Security*.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Authentication**.
4. Click **Join Domain**.
5. Enter a domain name.

   Use the form `name.tld` or `name.tld/container/path`.
6. Enter the user name and password of a directory service user account that has permissions to join the host to the domain.
7. If you intend to use an authentication proxy, enter the proxy server IP address.
8. Click **OK**.

Manage Host Certificates by Using the VMware Host Client

When you log in to an ESX host by using the VMware Host Client, you can view the certificate details of your host, such as the issuer and the validity period, and you can also import new certificates.

View Certificate Details for an ESX Host in the VMware Host Client

You can use the certificate information for debugging.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Certificate**.
   You can view the following certificate details.

Field	Description
Subject	The subject used during certificate generation.
Issuer	The issuer of the certificate.
Valid from	Date on which the certificate is generated.
Valid to	Date on which the certificate expires.

Import a New Certificate for an ESX Host in the VMware Host Client

You can import a certificate from a trusted certificate authority when you are logged in to an ESX host with the VMware Host Client.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Certificate**.
4. To import a new certificate, click the **Import New Certificate** button, paste the generated signed certificate in PEM format, and click **Import**.
5. To generate a certificate signing request (CSR), perform the following steps.

Option	Description
Generate CSR using IP	• Click **Generate CRS Using IP**, click the **Copy to Clipboard** button, and click **Close**. • To generate the signed certificate, pass the certificate signing request to the CA (CA). • In the **Certificate** text box, paste the generated signed certificate in PEM format and click **Import**.
Generate CSR using FQDN	• Click **Generate CSR Using FQDN**, click the **Copy to Clipboard** button, and click **Close**. • To generate the signed certificate, pass the certificate signing request to the certificate authority. • In the **Certificate** text box, paste the generated signed certificate in PEM format and click **Import**.

   You do not have to import the certificate immediately. To make sure that you can use the signed certificate, do not restart the host between generating the certificate signing request and importing the certificate.

   The certificate signing request is then passed to the certificate authority to generate the official certificate.

   An FQDN request has the fully qualified host name of the host in the resulting common name field of the certificate. The IP signing request has the current IP address of the host in the common name field.

Manage Users with the VMware Host Client

Manage users to control who is authorized to log in to ESX.

Users and roles control who has access to the ESX host components and what actions each user can perform.

The ESX user management has the following caveats .

- The users created when you connect directly to an ESX host are not the same as the vCenter users. When the host is managed by vCenter, vCenter ignores users created directly on the host.

- You cannot create ESX users by using the vSphere Client. You must log in to the host directly with the VMware Host Client to create ESX users .
- ESX 5.1 and later does not support local groups. However, Active Directory groups are supported.

To prevent anonymous users, such as root, from accessing the host with the Direct Console User Interface (DCUI) or ESX Shell, remove the user's administrator privileges on the root folder of the host. This applies to both local users and Active Directory users and groups.

Add an ESX User in the VMware Host Client

Adding a user to the users' table updates the internal user list that the host maintains.

For information about password requirements, see Configure the Passwords and Account Lockout Policy in the VMware Host Client or the *vSphere Security* documentation.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Users**.

   You cannot create ESX users with the vSphere Client. To create ESX users, you must directly log in to the host with the VMware Host Client.
4. Click **Add a User**.
5. Enter a user name, and a password.

   Note:
   Do not create a user who is named `ALL`. Privileges associated with the name `ALL` might not be available to all users in some situations. For example, if a user named `ALL` has Administrator privileges, a user with the **ReadOnly** privileges might be able to log in to the host remotely. This is not the intended behavior.

   - Do not include any spaces in the user name.
   - Do not include any non-ASCII characters in the user name.
   - Create a password that meets the required length and complexity. The host checks for password compliance using the default authentication plug-in, `pam_passwdqc.so`. If the password is not compliant, an error message indicates password requirements.
6. To activate the local access to the ESX Shell, select the **Enable Shell Access** check box.
7. Click **Add**.

Update an ESX User in the VMware Host Client

You can change the description and password for an ESX user in the VMware Host Client.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Users**.
4. Select a user from the list and click **Edit User**.
5. Update the user details and click **Save**.

Remove a Local ESX User from a Host in the VMware Host Client

You can remove a local ESX user from the host.

If you remove a user from the host, they lose permissions to all objects on the host and cannot log in again.

Note:
Users who are logged in and are removed from the domain keep their host permissions until you restart the host.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Users**.
4. Select the user that you want to remove from the list, click **Remove User**, and click **Yes**.

   You cannot remove the root user.

Manage ESX Roles in the VMware Host Client

ESX grants access to objects only to users who are assigned permissions for the object. When you assign a user permission for the object, you do so by pairing the user with a role.

A role is a predefined set of privileges. For more information about privileges, see the *vSphere Security* documentation.

ESX hosts provide a number of default roles, and you cannot change the privileges associated with these roles. Each subsequent default role includes the privileges of the previous role. For example, the Administrator role inherits the privileges of the Read Only role. Roles that you create do not inherit privileges from any of the default roles.

You can create custom roles by using the role-editing functions in the VMware Host Client to create privilege sets that match your user needs. Also, the roles you create directly on a host are not accessible in vCenter. You can work with these roles only if you log in to the host directly from the VMware Host Client.

> **Note:**
> When you add a custom role and do not assign any privileges to it, the role is created as a read-only role with the **System.Anonymous**, **System.View**, and **System.Read** system-defined privilege.

If you manage an ESX host through vCenter, maintaining custom roles in the host and vCenter can result in confusion and misuse. In this type of configuration, maintain custom roles only in vCenter.

You can create host roles and set permissions through a direct connection to the ESX host with the VMware Host Client.

Add a Role in the VMware Host Client

You can create roles to suit the access control needs of your environment.

Verify that you are logged in as a user with Administrator privileges, such as root or vpxuser.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Roles**.
4. Click **Add Role**.
5. Enter a name for the new role.
6. Select privileges for the role.
   a) Scroll through the privilege categories and select all privileges or a subset of privileges for that category. You can show all, selected, or unselected privileges.
7. Click **Create**.

Update a Role in the VMware Host Client

When you edit a role, you can change the privileges selected for that role. When complete, these privileges are applied to any user or group that is assigned the edited role.

Verify that you are logged in as a user with Administrator privileges, such as root or vpxuser.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Roles**.
4. Select a role from the list and click **Edit Role**.
5. Update the role details and click **Save**.

Remove a Role in the VMware Host Client

When you remove a role that is not assigned to any users or groups, the definition is removed from the list of roles. When you remove a role that is assigned to a user or group, you can remove assignments or replace them with an assignment to another role.

Verify that you are logged in as a user with Administrator privileges, such as root or vpxuser.

> **CAUTION:**
> You must understand how users will be affected before removing or replacing all assignments. Users who have not been granted any permissions cannot log in.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Roles**.
4. Select the name of the role that you want to remove from the list.
5. Click **Remove Role**, select **Remove only if unused**, and click **Yes**.

Managing Hosts in vCenter

To monitor all hosts in your vSphere environment from one place and to simplify host configuration, connect the hosts to a vCenter instance.

For information about the configuration management of ESX hosts, see the *vSphere Networking* documentation, the *vSphere Storage* documentation, and the *vSphere Security* documentation.

Update Your VMware Host Client Environment to the Latest Version

To evaluate whether you are using the latest version of the VMware Host Client, check what VIBs are installed in your environment and examine the VIBs version information. You can update your VMware Host Client environment by entering a URL or a datastore path to either a VIB or the `metadata.zip` file in an ESX offline bundle package.

If you provide a VIB file, an existing VIB that is installed on your VMware Host Client environment is updated to the new VIB.

If you provide an offline bundle, you update the entire ESX host to the version described by the `metadata.zip` file in the bundle. Make sure that the entire offline bundle is available through the URL or is uploaded to the datastore.

To update your environment to the latest version, perform the following tasks:

Task	Steps
Upload a VIB to a datastore	1. Click **Storage** in the VMware Host Client environment. 2. Click the **Datastores** tab. 3. Click on a datastore from the list of available datastores. 4. Click the **Files** tab and click **Upload**. 5. To store the VIB, select a directory and click **Upload**. 6. Browse to and double-click the file.
Upload an offline bundle to a datastore	1. Download the ESX offline bundle package. 2. Upload the ESX offline bundle package to the ESX host. a. Click **Storage** in the VMware Host Client environment. b. Click the **Datastores** tab. c. Click on a datastore from the list of available datastores.

Task	Steps
	d. Click the **Files** tab and click **Upload**. You can also upload the offline bundle package by using SCP or WinSCP. 3. Extract the contents of the offline bundle on the ESX host. For example, log in to the host by using SSH. 4. Navigate to the directory where you uploaded the offline bundle. 5. Extract the contents by using the `unzip` command.
Update your environment	1. Click **Host** in the VMware Host Client inventory. 2. Click the **Configure** tab. 3. Under **Security & Users**, click **Packages**. 4. Click **Install Update** and enter the URL or the datastore path to a VIB or a `metadata.zip` file in an offline bundle. 5. (Optional) If you intend to use an authentication proxy, activate **Use proxy server**. 6. Click **Update**. **CAUTION:** If you update an ESX host managed by vSphere Lifecycle Manager, the host might become non-compliant.

Unable to Connect from the VMware Host Client to an ESX Host After Upgrading to a Newer Version of ESX

After you upgrade your ESX host to a newer version, your browser console might display an error message when you attempt to access the ESX host by using the VMware Host Client, and your connection might fail.

After you upgrade your ESX host to a newer version, attempting to navigate to `https://host-fqdn/ui` or `https://host-IP-address/ui` might result in the following error:

```
503 Service Unavailable (Failed to connect to endpoint:
 [N7Vmacore4Http16LocalServiceSpecE:0xffa014e8] _serverNamespace = /ui _isRedirect = false
 _port = 8308)
```

A change to `/etc/vmware/rhttpproxy/endpoints.conf` remains after an upgrade and causes the `/ui` endpoint to override the VMware Host Client.

When the `/ticket` is missing from the `endpoint.conf` file on your 6.0 or later ESX host, your in-browser virtual machine console displays a `Failed to connect` error message, but the VMware Remote Console continues to work.

1. Log in to your ESX host either by using SSH or ESX Shell.
   If you use SSH, you must first enable SSH. You can enable SSH by using DCUI.
2. Back up the `endpoints.conf` file.
   ```
   cp /etc/vmware/rhttpproxy/endpoints.conf /tmp
   ```
3. Open the `/etc/vmware/rhttpproxy/endpoints.conf` file in an editor and remove the following line.
   ```
   /ui local 8308 redirect  allow
   ```
4. Restart the **rhttpproxy** configuration management server.
   ```
   /etc/init.d/rhttpproxy restart
   ```
5. Access the VMware Host Client using the specified full name of the host in the secure URL with `https://host-fqdn/ui` or a valid numeric IP address `https://1.2.3.4/ui`.

Reboot or Shut Down an ESX Host in the VMware Host Client

You can power off or restart any ESX host by using the VMware Host Client. Powering off a managed host disconnects it from vCenter, but does not remove it from the inventory.

To be able to reboot or shut down a host, you need these privileges.

- **Host** > **Configuration** > **Maintenance**
- **Global** > **Log event**

Always perform the following tasks before you reboot or shut down a host:

- Power off all virtual machines on the host.
- Place the host in maintenance mode.

1. From the **Actions** menu, select **Shut down** or **Reboot**.

   Note:
   If the host is not in maintenance mode, shutting down or rebooting it does not stop the virtual machines that are running on this host safely and unsaved data may be lost. If the host is part of a vSAN cluster, you might lose access to the vSAN data on the host.

2. Enter a reason for the operation and click **OK**.

Using the ESX Shell in the VMware Host Client

The ESX Shell provides essential maintenance commands for an ESX host and is deactivated by default. You can activate the ESX Shell from the VMware Host Client for troubleshooting, but activate it only when you need access.

The ESX Shell provides essential maintenance commands and is deactivated by default on ESX hosts. When you manage a host by using a direct connection to the VMware Host Client, you can activate local and remote access to the shell if necessary. To reduce the risk of unauthorized access, activate the ESX Shell for troubleshooting only.

The ESX Shell is independent of lockdown mode. Even if the host is running in lockdown mode, you can still log in to the ESX Shell if it is activated.

The applicable services are as follows.

Service	Description
ESX Shell	Activate this service to access the ESX Shell locally.
SSH	Activate this service to access the ESX Shell remotely by using SSH.
Direct Console UI (DCUI)	When you activate this service while running in lockdown mode, you can log in locally to the direct console user interface as the root user and deactivate lockdown mode. You can then access the host by using a direct connection to the VMware Host Client or by activating the ESX Shell.

The root user and users with the Administrator role can access the ESX Shell. Users who are in the Active Directory group are automatically assigned the Administrator role. By default, only the root user can run system commands (such as `vmware -v`) by using the ESX Shell.

Activate the Secure Shell (SSH) in the VMware Host Client

Activate the Secure Shell (SSH) to access the ESX Shell remotely by using SSH.

1. To activate or deactivate the Secure Shell (SSH), click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Services**.
4. Select **SSH** from the list and click **Start**.

Create a Timeout for ESX Shell Availability in the VMware Host Client

The ESX Shell is deactivated by default. To increase security when you enable the shell, you can set an availability timeout for the ESX Shell.

The availability timeout defines how long both local and remote shell logins are allowed before the ability to log in through the shell is deactivated. When the availability timeout expires, any existing shell sessions remains, but new shell sessions are not allowed.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the `UserVars.ESXiShellTimeOut` key and in the **Value** text box, enter the timeout setting.

   A value of zero (0) deactivates the timeout.
6. Click **OK**.

   You must restart the SSH service and the ESX Shell service for the timeout to take effect.

Create a Timeout for Idle ESX Shell Sessions in the VMware Host Client

If you enable the ESX Shell on a host, but forget to log out of the session, the idle session remains connected indefinitely. The open connection increases the potential for someone to gain privileged access to the ESX host. Prevent this by setting a timeout for idle sessions.

The idle timeout is the amount of time that can elapse before you are logged out of an idle interactive session.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Advanced Settings**.
4. Click the **Edit** button.
5. In the **Edit Advanced System Settings** window, find the `UserVars.ESXiShellInteractiveTimeOut` key and in the **Value** text box, enter the timeout setting.

   A value of zero (0) deactivates the timeout.
6. Click **OK**.

   The timeout takes effect only for newly logged in sessions.

If the session is idle, users are logged out after the timeout period elapses.

Place an ESX Host in Maintenance Mode in the VMware Host Client

You place an ESX host in maintenance mode when you need to upgrade it, for example, to install more memory. A host enters or leaves maintenance mode only as the result of a user request.

Before you place a host in maintenance mode, power off all virtual machines that are running on that host or migrate them to another host, either manually or automatically by DRS.

The host is in a state of **Entering Maintenance Mode** until all running virtual machines are powered off or migrated to different hosts. You cannot power off virtual machines or migrate virtual machines to a host that is entering or in maintenance mode.

To place a host in maintenance mode, all virtual machines that are running on the host must be powered off or migrated to different hosts. If you attempt to place a host that has running virtual machines on it in maintenance mode, DRS must power off or migrate the running virtual machines for the task to complete. If a time out occurs before the virtual machines are powered off or migrated, an error message appears.

When all virtual machines on the host are inactive, the host's icon displays **under maintenance** and the host's Summary panel indicates the new state. While in maintenance mode, the host does not allow you to deploy or power on a virtual machine.

1. Click **Host** in the VMware Host Client inventory.
2. From the **Actions** drop-down menu, select **Enter Maintenance Mode**.
3. Click **Yes**.

The host is in maintenance mode until you select **Exit Maintenance Mode**.

Manage Permissions in the VMware Host Client

In the VMware Host Client, permissions refer to the access roles assigned to users for various objects, such as virtual machines or ESX hosts. The roles allow users to perform various tasks on the assigned objects.

For example, to configure memory for the host, a user must be granted a role that includes the **Host** > **Configuration** > **Memory Configuration** privilege. By assigning different roles to users for different objects, you can control the tasks that users can perform by using the VMware Host Client.

When connecting directly to a host with the VMware Host Client, the root and vpxuser user accounts have the same access rights as any user assigned the Administrator role on all objects.

All other users initially have no permissions on any object, which means the users cannot view or perform tasks on these objects. A user with Administrator privileges must assign permissions to these users to allow them to perform tasks.

Many tasks require permissions on more than one object. The following rules can help you determine which roles to assign to users to allow particular tasks:

- Any task that consumes hard disk space, such as creating a virtual disk or taking a snapshot, requires the **Datastore** > **Allocate Space** privilege on the target datastore and the privilege to perform the operation itself.
- Each host and cluster has its own implicit resource pool that contains all the resources of that host or cluster. Deploying a virtual machine directly to a host or cluster requires the **Resource** > **Assign Virtual Machine to Resource Pool** privilege.

The list of privileges is the same for both ESX and vCenter.

You can create roles and set permissions through a direct connection to the ESX host.

Permission Validation

vCenter and ESX hosts that use Active Directory regularly validate users and groups against the Windows Active Directory domain. Validation occurs whenever the host system starts and at regular intervals specified in the vCenter settings.

For example, if user Smith was assigned permissions and in the domain the user's name was changed to Smith2, the host concludes that Smith no longer exists and removes permissions for that user when the next validation occurs.

Similarly, if user Smith is removed from the domain, all permissions are removed when the next validation occurs. If a new user Smith, is added to the domain before the next validation occurs, the new user Smith receives all the permissions the old user Smith was assigned.

Assign Permissions to a User for an ESX Host in the VMware Host Client

To perform particular activities on an ESX host, a user must have permissions that are associated with a particular role. In the VMware Host Client, you can assign roles to users and give the users the permissions necessary to perform various tasks on the host.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Permissions** tab and click **Add**.

   The **Add Permission** window opens.

3. From the **Domain** drop-down menu, select the domain that contains the user or group you want to add to.
4. In the **User/Group** field, enter the user or group name.
5. From the **Role** drop-down menu, select the desired permission role.
6. If you want the permission to apply to child objects as well, select the **Propagate to children** check box.

   If you set a permission at a vCenter level and propagate it to the child objects, the permission applies to data centers, folders, clusters, hosts, virtual machines, and other objects in the vCenter instance.
7. Click **OK**.

Export Roles and Permissions from the VMware Host Client

You can export a list of user roles and assigned permissions from the VMware Host Client in a .CSV file format.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Permissions** tab.
3. Select a user or group from the list.
4. Select the **Export** drop-down menu at the bottom, and select to export only the selected row from the list or all rows.

   The .CSV file download starts automatically.

Remove Permissions for a User in the VMware Host Client

Removing a permission for a user does not remove the user from the list of users available. It also does not remove the role from the list of available items. It removes the user and role pair from the selected inventory object.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Permissions** tab.
3. Select a user or group from the list and click **Delete**.

   The **Delete Permission** window opens.
4. To delete the selected permission, click **Remove**.

Assign User Permissions for a Virtual Machine in the VMware Host Client

Assign a role to a user to grant them permissions to perform specific tasks on a virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, click a virtual machine.
3. Click the **Permissions** tab and click **Add**.
   The **Add Permission** window appears.
4. From the **Domain** drop-down menu, select the domain that contains the user or group you want to add to.
5. In the **User/Group** field, enter the user or group name.
6. From the **Role** drop-down menu, select the desired permission role.
7. If you want the permission to apply to child objects as well, select the **Propagate to children** check box.

   If you set a permission at a vCenter level and propagate it to the child objects, the permission applies to data centers, folders, clusters, hosts, virtual machines, and other objects in the vCenter instance.

8. Click **OK**.

Remove Permissions for a Virtual Machine in the VMware Host Client

To prevent a user from performing tasks on a particular virtual machine, remove the user permissions for that virtual machine.

Removing a permission for a user does not remove the user from the list of users available. It also does not remove the role from the list of available items. It removes the user and role pair from the selected inventory object.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, click a virtual machine.
3. Click the **Permissions** tab.
4. Select a user or group from the list and click **Delete**.

   The **Delete Permission** window opens.
5. To delete the selected permission, click **Remove**.

Lockdown Mode in the VMware Host Client

To increase the security of your ESX hosts, you can put them in lockdown mode. In lockdown mode, operations must be performed through vCenter by default.

Normal Lockdown Mode and Strict Lockdown Mode

You can select normal lockdown mode or strict lockdown mode.

Normal Lockdown Mode	In normal lockdown mode, the DCUI service remains active. If the connection to the vCenter system is lost, and access through the vSphere Client is unavailable, privileged accounts can log in to the ESX host's Direct Console Interface and exit lockdown mode. Only the following accounts can access the Direct Console User Interface: • Accounts in the Exception User list for lockdown mode who have administrative privileges on the host. The Exception Users list is meant for service accounts that perform specific tasks. Adding ESX administrators to this list defeats the purpose of lockdown mode. • Users defined in the DCUI.Access advanced option for the host. This option is for emergency access to the Direct Console Interface in case the connection to vCenter is lost. These users do not require administrative privileges on the host.
Strict Lockdown Mode	In strict lockdown mode, the DCUI service is stopped. If the connection to vCenter is lost and the vSphere Client is no longer available, the ESX host becomes unavailable, unless the ESX Shell and SSH services are enabled and Exception Users are defined. If you cannot restore the connection to the vCenter system, you must reinstall the host.

Lockdown Mode and the ESX Shell and SSH Services

Strict lockdown mode stops the DCUI service. However, the ESX Shell and SSH services are independent of lockdown mode. For lockdown mode to be an effective security measure, ensure that ESX Shell and SSH services are also deactivated. These services are deactivated by default.

When a host is in lockdown mode, users on the Exception Users list can access the host from the ESX Shell and through SSH if they have the Administrator role on the host. This access is possible even in strict lockdown mode. Leaving the ESX Shell service and the SSH service deactivated is the most secure option.

> **Note:**
> The Exception Users list is meant for service accounts that perform specific tasks such as host backups, and not for administrators. Adding administrator users to the Exception Users list defeats the purpose of lockdown mode.

Put an ESX Host in Lockdown Mode by Using the VMware Host Client

You can use the VMware Host Client to enter normal and strict lockdown mode.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Security & Users**, click **Lockdown Mode**.
4. Click the **Edit** button.
5. In the **Lockdown mode** window, select the host lockdown mode.
6. Click **OK**.

Manage CPU Resources by Using the VMware Host Client

When you connect to an ESX host with the VMware Host Client, you have access to a limited number of resource management settings.

View Processor and CPU Information by Using the VMware Host Client

In the VMware Host Client, you can access information about the current processor type and CPU configuration of the ESX host that you are logged in to.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Summary** tab.
3. In the **Host Details** card, you can see details about the number of logical processors.
4. In the **Hardware** card, you can see the CPU model.

Assign a Virtual Machine to a Specific Processor in the VMware Host Client

By using CPU affinity, you can assign a virtual machine to a specific processor. This way, you can assign a virtual machine only to a particular available processor in multiprocessor systems.

Power off the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and click **Edit Settings**.
3. Expand **CPU**.
4. Under **Scheduling Affinity**, select physical processor affinity for the virtual machine.
   Use a hyphen to indicate ranges and a comma to separate values.
   For example, `0, 2, 4-7` would indicate processors 0, 2, 4, 5, 6, and 7.
5. Click **OK** to apply your changes.

Monitor an ESX Host in the VMware Host Client

When you connect to a host using the VMware Host Client, you can monitor the host health status, and view performance charts, events, tasks, system logs, and notifications.

View Charts in the VMware Host Client

When you are logged in to the VMware Host Client, you can view information about the resource usage of the ESX host that you are managing in line chart form.

To reduce memory consumption, the VMware Host Client only contains statistics for the last hour.

1. Click **Host** in the VMware Host Client inventory.
2. Click **Monitor** and click **Performance**.
3. Optional: To view the host usage for the last hour, select an option from the drop-down menu.
   - To view the percentage of CPU that the host used during the last hour, select **CPU**.

- To view the percentage of memory that the host consumed during the last hour, select **Memory**.
- To view the disk usage that the host consumed during the last hour, select **Disk**.
- To view the percentage of the network that the host consumed during the last hour, select **Network**.

Monitor Hardware Health Status in the VMware Host Client

When you are logged in to the VMware Host Client, you can monitor the health status of the ESX host hardware.

Note:
Hardware health status is only available when the underlying hardware supports it.

1. Click **Host** in the VMware Host Client inventory.
2. Click **Monitor** and under **Hardware**, click **System sensors**.
3. Select the type of information to view.
4. Optional: Use the filter controls above the list to filter the list.
5. Optional: Click a column heading to sort the list.
6. Optional: Click **Manage Columns** at the bottom-left corner of the **System sensors** pane.

   The **Show Columns** pop-up appears.
7. To show a specific parameter in the **System sensors** pane, select the respective check box from the list.

View Events in the VMware Host Client

Events are records of user actions or system actions that occur on an ESX host. When you are logged in the VMware Host Client, you can view all events that are associated with the host that you are managing.

Required privilege: **Read-only**.

1. Click **Host** in the VMware Host Client inventory.
2. Click **Monitor** and under **Tasks and Events**, click **Events**.
   a) Optional: To see event details, click the icon (») next to the event.

      Note:

      To ensure a responsive experience, the **Events** grid uses virtual scrolling. This feature allows rendering only the rows visible on your screen and dynamically loads new data as you scroll.
   b) Optional: To filter the list by severity, select an option from the **Select severity** drop-down menu above the list.

View Tasks in the VMware Host Client

When you are logged in to the VMware Host Client, you can view tasks that are related to the ESX host. You can view information about task initiator, task state, task result, task description, and so on.

1. Click **Host** in the VMware Host Client inventory.
2. Click **Monitor** and under **Tasks and Events**, click **Tasks**.
   a) Optional: To see event details, click the icon (») next to the task.
   b) Optional: To filter the list by severity, select an option from the **Select severity** drop-down menu above the list.

Virtual Machine Management with the VMware Host Client

You can configure virtual machines to perform the same tasks as physical computers. Unlike physical computers, virtual machines support special features.

You can use the VMware Host Client to create, register, and manage virtual machines, and to conduct daily administrative and troubleshooting tasks.

Create a Virtual Machine in the VMware Host Client

Virtual machines are the key component of a virtual infrastructure. When you create a virtual machine, you can add it to the host inventory, associate it with a particular datastore, and select an operating system and virtual hardware options.

Verify that you have the **VirtualMachine** > **Inventory** > **Create** privileges.

Depending on the properties of the virtual machine you want to create, you might need the following extra privileges:

- **VirtualMachine** > **Config** > **AddExistingDisk** if including a virtual disk device that refers to an existing virtual disk file (not RDM).
- **VirtualMachine** > **Config** > **AddNewDisk** if including a virtual disk device that creates a new virtual disk file (not RDM).
- **VirtualMachine** > **Config** > **RawDevice** if including a raw device mapping (RDM) or SCSI passthrough device.
- **VirtualMachine** > **Config** > **HostUSBDevice** if including a VirtualUSB device backed by a host USB device.
- **VirtualMachine** > **Config** > **AdvancedConfig** if setting values in `ConfigSpec.extraConfig`.
- **VirtualMachine** > **Config** > **SwapPlacement** if setting swapPlacement.
- **Datastore** > **AllocateSpace** required on all datastores where the virtual machine and its virtual disks are created.
- **Network** > **Assign** required on the network which is assigned to the new virtual machine that is being created.

After you power on the virtual machine, it consumes resources dynamically as the workload increases, or returns resources dynamically as the workload decreases.

Every virtual machine has virtual devices that provide the same function as physical hardware. A virtual machine gets CPU and memory, access to storage, and network connectivity from the host it runs on.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click **New Virtual Machine**.

   The **New Virtual Machine** wizard opens.
3. On the **Select a name and compatibility** page, provide a unique name for the virtual machine and select the virtual machine compatibility.
   a) In the **Virtual machine name** text box, enter a name for the virtual machine.
   b) From the **Compatible with** drop-down menu, select the virtual machine compatibility.
4. Click **Next**.
5. On the **Select a guest OS** page, select the guest operating system and version.
   a) From the **Guest OS Family** drop-down menu, select the guest operating system.
   b) From the **Guest OS Version** drop-down menu, select the guest operating system version.
   c) To enable VBS on the virtual machine, select the **Enable Windows Virtualization Based Security** check box.

      Note:

      The **Enable Windows Virtualization Based Security** option only appears for the latest Windows OS versions, for example, Windows 10 and Windows Server 2016, and if the virtual machine compatibility is ESX 6.7 and later.

      When you enable this option, hardware virtualization, IOMMU, EFI, and secure boot become available to the guest operating system. You must also enable **Virtualization Based Security** within the guest operating system of this virtual machine.
6. Click **Next**.
7. On the **Select storage** page, select a datastore from the list where to store the virtual machine files and click **Next**.
8. On the **Customize hardware** page, configure the virtual machine hardware and options.

   You can leave the defaults and can configure the virtual machine hardware and options later.

   For information about virtual machine options and virtual disk configuration, including instructions for adding different types of devices, see *vSphere Virtual Machine Administration*.

a) Optional: To view and configure device settings, expand any device.

Option	Description
CPU	The CPU or processor is the portion of a computer system that carries out the instructions of a computer program and is the primary element carrying out the computer's functions. CPUs contain cores. The number of virtual CPUs that are available to a virtual machine depends on the number of licensed CPUs on the host, and the number of CPUs supported by the guest operating system. To use the VMware multicore virtual CPUs feature, you must comply with the requirements of the guest operating system EULA.
Memory	You can add, change, or configure virtual machine memory resources or options to enhance virtual machine performance. You can set most of the memory parameters during virtual machine creation or after the guest operating system is installed. The memory resource settings for a virtual machine determine how much of the host memory is allocated to the virtual machine. The virtual hardware memory size determines how much memory is available to applications that run in the virtual machine.
Hard disk	You can add large-capacity virtual disks to virtual machines and can add more space to existing disks, even when the virtual machine is running. You can set most of the virtual disk parameters during virtual machine creation or after you install the guest operating system.
SCSI controller	Storage controllers appear to a virtual machine as different types of SCSI controllers, including BusLogic Parallel, LSI Logic Parallel, LSI Logic SAS, and VMware Paravirtual SCSI. You can set the type of SCSI bus sharing for a virtual machine and indicate whether the SCSI bus is shared. Depending on the type of sharing, virtual machines can access the same virtual disk simultaneously on the same server or on any server. You can change the SCSI controller configuration for a virtual machine on an ESX host only.
SATA controller	If a virtual machine has multiple hard disks or CD/DVD-ROM devices, you can add up to three additional SATA controllers to assign the devices to. When you spread the devices among several controllers, you can improve performance and avoid data traffic congestion. You can also add additional controllers if you exceed the thirty-device limit for a single controller. You can boot virtual machines from SATA controllers and use them for large-capacity virtual hard disks.
Network adapter	When you configure a virtual machine, you can add network adapters (NICs) and specify the adapter type. The type of network adapters that are available depend on the following factors: • The virtual machine compatibility, which depends on the host that created or most recently updated it. • Whether the virtual machine compatibility has been updated to the latest version for the current host. • The guest operating system.
CD/DVD drive	You can configure DVD or CD devices to connect to client devices, host devices, or Datastore ISO files.
Video Card	You can select the default settings or can specify custom settings. You can specify the number of displays, the total video memory, and can enable 3D support for guest operating systems on which VMware supports 3D.
PCI device	You can configure PCI devices on an ESXhost to make them available for passthrough. You can also change the hardware label to restrict the virtual machine placement to specific hardware instances.
Dynamic PCI device	PCI passthrough devices are automatically grouped by their vendor and model name. You can configure the desired devices by the vendor and model name instead of selecting a physical PCI device by the hardware address. You can add all available devices with the same hardware label or with a blank hardware label to a virtual machine. When you power on a virtual machine, specific physical PCI passthrough devices with matching vendor and model names attach to the virtual machine.
Security devices	You can configure Virtual Intel® Software Guard Extensions (vSGX) for virtual machines and can provide additional security to your workloads. You can activate or deactivate vSGX when you create a virtual machine or when you edit an existing virtual machine.

b) Optional: To remove a device, click the vertical ellipsis icon (⋮) next to the device and click **Remove device**. This option appears only for virtual hardware that you can remove safely.

9. Optional: To use the available virtual machine options and to customize the settings and behavior of your virtual machine, click the **VM Options** tab and make the necessary changes.
10. Optional: To modify or add various configuration parameters, click the **Advanced Parameters** tab, and make the necessary changes.
11. On the **Ready to complete** page, review the details and click **Finish**.

Use Consoles in the VMware Host Client

You can access a virtual machine through a browser console or through VMware Remote Console (VMRC) in the VMware Host Client and perform different tasks on the virtual machine.

Using Browser Console

Note:
The browser console is not supported for any version of ESX 7.0 and earlier. To access the browser console, you must use VMRC.

You can use a browser console to gain access to the guest operating system without installing additional software. For other console functionalities, such as attaching local hardware, install VMware Remote Console.

Note:
Currently browser consoles support only US, Japanese and German keyboard layouts. You must select the desired keyboard layout before opening the console.

Using VMware Remote Console

VMware Remote Console provides access to virtual machines on remote hosts and performs console and device operations, such as configuring operating system settings and monitoring the virtual machine console for *VMware vSphere*. You can perform a variety of tasks on the virtual machine, such as restarting and shutting down the virtual machine guest operating system, resuming and suspending the virtual machine, configuring VMware Tools updates, configuring and managing the virtual machine and different devices, and so on. VMRC can also modify virtual machine settings such as RAM, CPU cores, and disks. VMware Workstation™ and VMware Fusion™ work as VMRC clients, so you do not need to download and install VMRC if you have any of the three installed on your system.

For a full set of console features, download and install VMRC.

Install the VMware Remote Console Application in the VMware Host Client

The VMware Remote Console (VMRC) is a standalone console application that enables you to connect to client devices and launch virtual machine consoles on remote hosts.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list.
3. Click **Download VMRC** and follow the steps described in Broadcom knowledge base article 368995.

Note:

You must have a profile at the Broadcom Support Portal to download the VMRC installer.

Launch the VMware Remote Console for a Virtual Machine in the VMware Host Client

You can access virtual machines in the VMware Host Client by using VMware Remote Console. You can launch one or more consoles to access several remote virtual machines at the same time.

- Verify that VMware Remote Console is installed on your local system.
- Power on the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Select a virtual machine from the list.
3. Click the **Summary** tab and from the **Guest OS** card, click **Launch Remote Console**.

   Note:

   You can also access the VMware Remote Console from the **Actions** drop-down menu.

   The VMware Remote Console opens as a standalone application for the selected virtual machine.

Open a Virtual Machine Console in the VMware Host Client

With the VMware Host Client, you can access the desktop of a virtual machine by launching a console to the virtual machine. From the console, you can perform tasks in the virtual machine, such as configuring operating system settings, running applications, monitoring performance, and so on.

Power on the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Select a virtual machine from the list.
3. From the **Actions** menu, select **Open Console in a new tab** to open the web console, or **Open Remote Console**.

Manage a Guest Operating System in the VMware Host Client

With the VMware Host Client, you can manage the guest operating system of a virtual machine. You can install and upgrade VMware Tools, shut down, reboot, and change the configured guest operating system.

Shut Down or Restart a Guest Operating System by Using the VMware Host Client

To shut down and restart the guest operating system on a virtual machine, you must install VMware Tools on that virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Select a virtual machine, and select your task.
   - To shut down a virtual machine, right-click the virtual machine and select **Guest OS** > **Shut Down Guest OS**.
   - To restart a virtual machine, right-click the virtual machine and select **Guest OS** > **Restart Guest OS**.

Change the Guest Operating System in the VMware Host Client

When you change the guest operating system type in the virtual machine settings, you change the setting for the guest operating system in the configuration file of the virtual machine. To change the guest operating system itself, you must install the new operating system in the virtual machine.

Power off the virtual machine.

When you set the guest operating system type for a new virtual machine, vCenter applies configuration defaults that are based on the type of guest operating system. Changing the guest operating system type setting affects the available ranges and recommendations of the virtual machine settings.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **VM Options** tab and expand **General Options**.
4. Select a guest operating system family and version.

   For a virtual machine with ESX 6.7 and later compatibility, if you select a Windows OS version that supports VBS, the VBS row appears on the **VM Options** tab.

   If you choose a Windows OS version that supports VBS, and if the virtual machine compatibility is ESX 6.7 and later, the VBS row appears on the **VM Options** tab.

5. Optional: Click the **Enable Virtualization Based Security** to enable VBS.

   Important:
   Enabling VBS requires that you use EFI to boot the virtual machine. Changing the firmware might make the guest OS unbootable.

6. Click **OK** to apply your changes.

The virtual machine configuration parameters for the guest operating system are changed. You can now install the guest operating system.

Introduction to VMware Tools

VMware Tools is a set of services and modules that enable several features in VMware products for better management of guests operating systems and seamless user interactions with them.

VMware Tools has the ability to:

- Pass messages from the host operating system to the guest operating system.
- Customize guest operating systems as a part of the vCenter and other VMware products.
- Run scripts that help automate guest operating system operations. The scripts run when the power state of the virtual machine changes.
- Synchronize the time in the guest operating system with the time on the host operating system

VMware Tools Lifecycle Management provides a simplified and scalable approach for the installation and upgrade of VMware Tools. It includes a number of feature enhancements, driver-related enhancements, and support for new guest operating systems.

You must run the latest version of VMware Tools or use open-vm-tools distributed with the Linux OS distribution. Although a guest operating system can run without VMware Tools, you must always run the latest version of VMware Tools in your guest operating systems to access the latest features and updates.

You can configure your virtual machine to automatically check and apply VMware Tools upgrades each time you power on your virtual machines.

For information about enabling automatic upgrade of VMware Tools on your virtual machines, see the *vSphere Virtual Machine Administration* documentation.

Installing VMware Tools

Although you can use guest operating systems without VMware Tools, many VMware features are not available unless you install VMware Tools. VMware Tools enhances the performance of the guest operating system of your virtual machines.

Installing VMware Tools is part of the process of creating new virtual machines. Upgrade VMware Tools when updates become available. For information about creating virtual machines, see the *VMware Tools Administration* documentation.

The installers for VMware Tools are ISO image files. An ISO image file looks like a CD-ROM to your guest operating system. Each type of guest operating system, including Windows, Linux, Solaris, FreeBSD, and NetWare, has an ISO image file. When you install or upgrade VMware Tools, the first virtual CD-ROM disk drive of the virtual machine temporarily connects to the VMware Tools ISO file of your guest operating system.

For information about installing or upgrading VMware Tools in Windows virtual machines, Linux virtual machines, Mac OS X virtual machines, Solaris virtual machines, NetWare virtual machines, or FreeBSD virtual machines, see the *VMware Tools Administration* documentation.

Install VMware Tools from the VMware Host Client

VMware Tools is a suite of utilities that you install in the operating system of a virtual machine. VMware Tools enhances the performance and management of the virtual machine.

Power on the virtual machines.

You can install VMware Tools in one or more virtual machines by using the VMware Host Client.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list.
3. Select **Guest OS** > **Install VMware Tools**.
4. In the **Install VMware Tools** dialog box, click the **Mount** button.

Upgrading VMware Tools

You can upgrade VMware Tools manually, or you can configure virtual machines to check for newer versions of VMware Tools and install them.

The guest operating system checks the version of VMware Tools when you power on a virtual machine. The status bar of your virtual machine displays a message when a new version is available.

For vSphere virtual machines, when the installed version of VMware Tools is outdated, the status bar displays the message:

```
A newer version of Tools is available for this VM
```

In Windows virtual machines, you can set VMware Tools to notify you when an upgrade is available. If this notification option is enabled, the VMware Tools icon in the Windows taskbar includes a yellow caution icon when a VMware Tools upgrade is available.

For Windows and Linux guest operating systems, you can configure the virtual machine to automatically upgrade VMware Tools. Although the version check is performed when you power on the virtual machine, if Windows guest operating systems, the automatic upgrade occurs when you power off or restart the virtual machine. The status bar displays the message `Installing VMware Tools ...` when an upgrade is in progress. The procedure is mentioned below.

> **Note:**
>
> Upgrading VMware Tools on Windows guest operation systems automatically installs the SVGA graphics drivers. The SVGA graphics driver allows the sleep mode available in guest OS power settings to adjust the sleep options. For example, you can use the sleep mode setting **Change when the computer sleeps** to configure your guest OS to automatically go to sleep mode after a certain time or prevent your guest OS from automatically switching to sleep mode after being idle for some time.

For vSphere virtual machines, you can use one of the following processes to upgrade multiple virtual machines at the same time.

You can use one of the following processes to upgrade multiple virtual machines at the same time.

- Log in to vCenter Server, select a host or cluster, and on the **Virtual Machines** tab specify the virtual machines on which to perform a VMware Tools upgrade.
- Use vSphere Lifecycle Manager to perform an orchestrated upgrade of virtual machines at the folder or data center level.

Some features in a particular release of a VMware product might depend on installing or upgrading to the version of VMware Tools included in that release. Upgrading to the latest version of VMware Tools is not always necessary, however, VMware highly recommends that you upgrade to the most updated version of the VMware Tools. Newer versions of VMware Tools are compatible with several ESX host versions. To avoid unnecessary upgrades, evaluate whether the added features and capabilities are necessary for your environment. See Hardware Features Available with Virtual Machine Compatibility Settings. However, VMware highly recommends installing and using the latest version of VMware Tools.

Some features in a particular release of a VMware product might depend on installing or upgrading to the version of VMware Tools included in that release. Upgrading to the latest version of VMware Tools is not always necessary. Newer versions of VMware Tools are compatible with several host versions. To avoid unnecessary upgrades, evaluate whether the added features and capabilities are necessary for your environment.

Table 505: Virtual Machine Compatibility Options

Compatibility	Description
ESX 9.1 and later	This virtual machine (hardware version 22) is compatible with ESX 9.1 and later.
ESX 9.0 and later	This virtual machine (hardware version 22) is compatible with ESX 9.0 and ESX 9.1 and later.
ESX 8.0 Update 2 and ESX 8.0 Update 3	This virtual machine (hardware version 21) is compatible with ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later.
ESX 8.0 and ESX 8.0 Update 1	This virtual machine (hardware version 20) is compatible with ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later..
ESX 7.0 Update 3 and later	This virtual machine (hardware version 19) is compatible with ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later..
ESX 7.0 Update 2 and later	This virtual machine (hardware version 19) is compatible with ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later.
ESX 7.0 Update 1 and later	This virtual machine (hardware version 18) is compatible with ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later.
ESX 7.0 and later	This virtual machine (hardware version 17) is compatible with ESX 7.0, ESX 7.0 Update 1, ESX 7.0 Update 2, ESX 7.0 Update 3, ESX 8.0, ESX 8.0 Update 1, ESX 8.0 Update 2, ESX 8.0 Update 3, ESX 9.0, and ESX 9.1 and later.

For more information, see the Broadcom Compatibility Guide at https://compatibilityguide.broadcom.com/.

Upgrade VMware Tools in the VMware Host Client

You can upgrade VMware Tools on a virtual machine by using the VMware Host Client.

Power on the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list.
3. Select **Guest OS** > **Upgrade VMware Tools**.

Unmount VMware Tools Installer

To disconnect the CD or DVD drive from the VMware Host Client, you can unmount the VMware Tools installer.

Power on the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list.
3. Select **Guest OS** > **Unmount VMware Tools Installer**.

Configuring a Virtual Machine in the VMware Host Client

You can add or configure most virtual machine properties during the virtual machine creation process or after you create the virtual machine and install the guest operating system.

You can configure three types of virtual machine properties.

Hardware	View the existing hardware configuration and add or remove hardware.
Options	View and configure various virtual machine properties, such as power management interaction between the guest operating system and the virtual machine, and VMware Tools settings.
Resources	Configure CPUs, CPU hyperthreading sources, memory, and disks.

Change the Name of a Virtual Machine in the VMware Host Client

You can change the name of a virtual machine after you finish the creation process. Changing the name does not change the name of any virtual machine files or the name of the directory in which the files reside.

Power off the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **VM Options** tab and expand **General Options**.
4. In the **VM Name** text box, enter a new name for the virtual machine.
5. Click **OK**.

View the Location of the Virtual Machine Configuration File in the VMware Host Client

You can view the location of the configuration files and the working files of a virtual machine by using the VMware Host Client.

Power off the virtual machine.

This information is useful when you configure backup systems.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **VM Options** tab and expand **General Options**.
4. Record the location of the configuration files and the working files.
5. Click **Cancel** to exit the screen.

Edit the Configuration File Parameters in the VMware Host Client

To fix specific problems with your system, VMware documentation or a VMware Technical Support representative might instruct you to change or add virtual machine configuration parameters.

Power off the virtual machine.

> **Important:**
> Changing or adding parameters when a system is functioning properly might lead to decreased system performance and instability.

The following conditions apply:

- To change a parameter, you must change the existing value for the keyword/value pair. For example, if the existing pair is keyword/value, and you change it to keyword/value2, the new keyword is value2.
- You cannot delete a configuration parameter entry.

> **CAUTION:**
> You must assign a value to the configuration parameter keywords. If you do not assign a value, the keyword might receive a value of 0 or false, which might result in a virtual machine that cannot power on.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **Advanced Parameters** tab.
4. Optional: To add a parameter, enter an attribute name and value for the parameter and click **Add**.
5. Optional: To change a parameter, type a new value in the **Value** text box for that parameter.
6. Click **OK** to save your changes.

Configure Autostart Options for a Virtual Machine in the VMware Host Client

You can configure autostart options for a virtual machine to set up the virtual machine to start before or after the other virtual machines on the host. If the host is part of a vSphere HA cluster, the automatic startup and shutdown of virtual machines is deactivated. Automatic startup is not supported with vSphere HA.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click a virtual machine from the list.
3. To configure any autostart option, select **Actions** > **Autostart**.

   The **Virtual Machine Startup and Shutdown** window displays.
4. Click the **Edit** button.
5. To set up the default timing and startup order for virtual machines, select an option from the **Edit VM Startup/Shutdown Configuration** window.

Option	Description
System influence	Select the **System influence** check box so that the host can automatically start and stop VMs with the system.
Startup delay	Select the time in seconds. After the ESX host powers on the first virtual machine, the host waits for the specified delay time and powers on the next virtual machine.
Shutdown delay	Select the time in seconds. Stop delay is the maximum time the ESX host waits for a shutdown command to complete. The order in which virtual machines are shut down is the reverse of their startup order. After the ESX host shuts down the first virtual machine within the time you specify, the host shuts down the next virtual machine.
Continue is VMware Tools is started	Associated with the "Shutdown delay" input. Allows the shutdown sequence to proceed once the ESX host detects that VMware Tools has successfully started in the guest operating system.
Shutdown action	From the **Shutdown action** drop-down menu, select a shutdown action that is applicable to the virtual machines on the host when the host shuts down. • **Guest shutdown** • **Power off** • **Suspend** • **System default** • **None**

6. Click **OK**.

Upgrade Virtual Machine Compatibility by Using the VMware Host Client

The virtual machine compatibility determines the virtual hardware available to the virtual machine, which corresponds to the physical hardware available on the host machine.

- Create a backup or snapshot of the virtual machines. See Manage Virtual Machines With Snapshots.
- Upgrade VMware Tools. On virtual machines that run Microsoft Windows, if you upgrade the compatibility before you upgrade VMware Tools, the virtual machine might lose its network settings.
- Verify that all `.vmdk` files are available to the ESX host on a VMFS3, VMFS5, or NFS datastore.
- Verify that the virtual machine is stored on VMFS3, VMFS5, or NFS datastores.
- Verify that the compatibility settings for the virtual machines are not set to the latest supported version.
- Determine the ESX versions that you want the virtual machines to be compatible with. See *vSphere Virtual Machine Administration*.

You can upgrade the compatibility level to make a virtual machine compatible with the latest version of ESX running on the host.

For information about virtual machine hardware versions and compatibility, see *vSphere Virtual Machine Administration*.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list and select **Upgrade VM Compatibility**.
3. In the **VM Compatibility Upgrade** dialog box, click **OK**.
4. In the **Configure VM Compatibility** dialog box, select the latest supported version and click **OK**.

Manage Virtual Machines in the VMware Host Client

After creating a virtual machine in the VMware Host Client, you can perform different management tasks on the virtual machine.

You can delete the virtual machine from the host, remove the virtual machine from a datastore, register it back on a datastore, and so on. You can also return the virtual machine to the host.

Access a Virtual Machine in the VMware Host Client

You can access the virtual machines on the host that you are logged into to configure the virtual machine hardware and options, to perform administrative tasks, and to perform basic troubleshooting tasks.

To access the virtual machines that are available on the host that you are logged into, click **Virtual Machines** in the VMware Host Client inventory.

You can now edit the virtual machine settings and perform different administrative and troubleshooting tasks on the virtual machines in the list.

Power States of a Virtual Machine in the VMware Host Client

The basic power operations for a virtual machine include powering on, powering off, suspending, and resetting.

- Verify that you have the **VirtualMachine** > **Interaction** > **PowerOn** privilege.
- Verify that you have the **VirtualMachine** > **Interaction** > **PowerOff** privilege.
- Verify that you have the **VirtualMachine** > **Interaction** > **Suspend** privilege.
- Verify that you have the **VirtualMachine** > **Interaction** > **Reset** privilege.

For information about how to change the virtual machine power states, see Configure the Virtual Machine Power States in the VMware Host Client.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine, select **Power**, and select a power operation.

Option	Description
Power On ( )	Powers on a virtual machine when the virtual machine is stopped.
Power off ( )	Powers off a virtual machine and shuts down the guest operating system. Powering off a virtual machine might cause loss of data.
Suspend ( )	Suspends a running virtual machine and leaves it connected to the network. When you resume a suspended virtual machine, the virtual machine continues operating at the same point the virtual machine was at when it was suspended.
Reset ( )	Shuts down and restarts the guest operating system without powering off the virtual machine.

Configure the Virtual Machine Power States in the VMware Host Client

Changing virtual machine power states is useful when you do maintenance on the host. You can use the system default settings for the virtual machine power controls, or you can configure the controls to interact with the guest operating system.

- Verify that you have privileges to perform the intended power operation on the virtual machine.
- To set optional power functions, install VMware Tools on the virtual machine.
- Power off the virtual machine before editing the VMware Tools options.

For example, you can configure the **Power off** control to power off the virtual machine or shut down the guest operating system.

You can modify many virtual machine configurations while the virtual machine is running, but you must change the virtual machine power state for some configurations.

You cannot configure a **Power on** ( ) action. This action powers on a virtual machine that is stopped, or starts a virtual machine and runs a script if the virtual machine is suspended and VMware Tools is installed and available. If VMware Tools is not installed, it starts the suspended virtual machine, but does not run a script.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **VM Options** tab, expand **VMware Tools**.
4. Select an option for the virtual machine **Shut Down Guest** ( ) control from the drop-down menu.

Option	Description
Power Off	Immediately stops the virtual machine. A Power off action shuts down the guest operating system or powers off the virtual machine. A message indicates that the guest operating system might not shut down properly. Use this power off option only when necessary.
Shut Down Guest	Uses VMware Tools to initiate an orderly system shut down of the virtual machine. Soft power operations are possible only if the tools are installed in the guest operating system.

5. Select an option for the **Suspend** ( ) control from the drop-down menu.

Option	Description
Suspend (Default)	Pauses all virtual machine activity. When VMware Tools is installed and available, a suspend action runs a script and suspends the virtual machine. If VMware Tools is not installed, a Suspend action suspends the virtual machine without running a script.
Suspend Guest	Puts the guest operating system on standby. This option stops all processes, but all virtual devices remain connected to the virtual machine.

6. Select an option for the **Reset** ( ) control from the drop-down menu.

Option	Description
Reset	Shuts down and restarts the guest operating system without powering off the virtual machine. If VMware Tools is not installed, a Reset action resets the virtual machine.
Restart Guest	Uses VMware Tools to initiate an orderly restart. Soft power operations are possible only if the tools are installed in the guest operating system.

7. Click **OK**.

Edit Virtual Machine Startup and Shutdown Settings in the VMware Host Client

You can configure virtual machines running on an ESX host to start up and shut down with the host or after a delay. You can also set the default timing and startup order for virtual machines. This way, the operating system has enough time to save data when the host enters maintenance mode or is being powered off for another reason.

The Virtual Machine Startup and Shutdown (automatic startup) setting is deactivated for all virtual machines on hosts in a vSphere HA cluster. Automatic startup is not supported with vSphere HA.

1. Click **Host** in the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **Virtual Machines**, click **VM Startup/Shutdown** and click **Edit**.

   The **Edit VM Startup/Shutdown Configuration** dialog box opens.
4. Select **Automatically start and stop the virtual machines with the system**.
5. In the **Default VM Settings** pane, configure the default startup and shutdown behavior for all virtual machines on the host.

Option	Description
Startup delay	After you start the ESX host, it starts powering on the virtual machines that are configured for automatic startup. After the ESX host powers on the first virtual machine, the host waits for the specified delay time and then powers on the next virtual machine. The virtual machines are powered on in the startup order that is specified under the **Default VM Settings** pane.
Shutdown delay	Shutdown delay is the maximum time the ESX host waits for a shutdown command to complete. When you power off the ESX host, the autostart manager initiates the automatic shutdown on the first virtual machine and waits within the specific delay time for the virtual machine to complete the power action. The power action can be Power Off, Guest Shutdown, or Suspended. The order in which virtual machines are shut down is the reverse of their startup order. After the ESX host shuts down the first

Option	Description
	virtual machine within the time that you specify, the host shuts down the next virtual machine. If a virtual machine does not shut down within the specified delay time, the host runs a power off command and then starts shutting down the next virtual machine. The ESX host shuts down only after all virtual machines are shut down.
Continue if VMware Tools is started	Shortens the startup delay of the virtual machine. If VMware Tools starts before the specified delay time passes, the ESX host powers on the next virtual machine without waiting for the delay time to pass.
Shutdown action	Select a shutdown action for the virtual machines on the host when the host shuts down. • **Guest Shutdown** • **Power Off** • **Suspend** • **None**

6. You can also configure the startup order and behavior for individual virtual machines. Use this option when you need the virtual machine's delay to differ from the default delay for all machines. The settings that you configure for individual virtual machines override their default settings.
   a) To change the startup order of virtual machines, click a virtual machine from the **Manual Startup** category and use the up arrow to move it up to the **Automatic** or **Automatic Ordered** categories. Use the up and down arrows to change the startup order for virtual machines in the **Automatic** and **Manual Startup** categories. During the shutdown, the virtual machines shut down in reverse order.
   b) To edit the startup and shutdown behavior of a virtual machine, select a virtual machine, use the up arrow to move it, and click the **Edit** button.

      The **Virtual Machine Startup/Shutdown setting** dialog box opens.
   a) In the **Startup Settings** pane, configure the startup behavior of the virtual machine. You can decide to use the default startup delay or you can specify a new one. If you select **Continue immediately if VMware Tools starts**, the ESX host powers on the next virtual machine without waiting for the delay to pass.
   b) In the **Shutdown Settings** pane, configure the shutdown behavior of the virtual machine. You can use the default shutdown delay or specify a new one and select the shutdown action.
7. Click **OK**.

Use Virtual Machine Column Configuration in the VMware Host Client

The virtual machines panel in the VMware Host Client allows you to configure the information that you want to display. You can show or hide different columns, such as status, used space, host name, host CPU, and so on.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click **Manage Columns** at the bottom-left corner of the **Virtual Machines** panel.

   The **Show Columns** pop-up appears.
3. Select the information that you want to display in the virtual machine panel.

Remove Virtual Machines from a Host in the VMware Host Client

You can remove a virtual machine from the inventory if you want to keep it on the datastore, but you no longer want the VMware Host Client inventory to display the virtual machine. For more information on how to return the virtual machine back to the inventory, see Register a Virtual Machine in the VMware Host Client.

Power off the virtual machine.

You can return the virtual machine back to the inventory,

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Remove from Inventory**.
3. To confirm that you want to remove the virtual machine from the inventory, click **OK**.

The host removes the virtual machine from the inventory and no longer tracks its condition. However the virtual machine files remain on the datastore.

Remove Virtual Machines from a Datastore in the VMware Host Client

To free space on the datastore, you can remove the virtual machines that you no longer need. Removing a virtual machine from the VMware Host Client inventory deletes all virtual machine files from the datastore, including the configuration file and virtual disk files. You can delete multiple virtual machines

- Power off the virtual machine.
- Verify that the virtual machine does not share the disk with another virtual machine. If two virtual machines share one disk, the disk files are not deleted.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Select one or more checkboxes next to the virtual machines that you want to delete from disk.
3. Right-click the selection and select **Delete from Disk**.
4. In the **Confirm Deletion** dialog box, click **OK**.

Register a Virtual Machine in the VMware Host Client

If you remove a virtual machine or template from a host but do not remove it from the host's datastore, you can return it to the host's inventory.

1. Click **Storage** in the VMware Host Client inventory.
2. Right-click a datastore in the list and click **Register VM**.

   The **Select File** dialog box opens.
3. Select the virtual machine that you want to register.
4. Select the `.vmx` check box and click **OK**.

Manage Virtual Machines With Snapshots in the VMware Host Client

Snapshots preserve the state and data of a virtual machine at the time that you take the snapshot. When you take a snapshot of a virtual machine, an image of the virtual machine in a given state is copied and stored.

When you take a snapshot, you capture the state of the virtual machine settings and the virtual disk. If you are taking a memory snapshot, you also capture the memory state of the virtual machine. These states are saved to files that reside with the base files of the virtual machines. For more information, see Snapshot Files and Limitations.

Take a Snapshot of a Virtual Machine in the VMware Host Client

You can take one or more snapshots of a virtual machine to capture the settings state, disk state, and memory state at the time you take the snapshot. When you take a snapshot, you can also quiesce the virtual machine files and exclude the virtual machine disks from snapshots. You can take a snapshot when a virtual machine is powered on, powered off, or suspended. To take a snapshot of a suspended virtual machine, wait until the suspend operation finishes before you take a snapshot.

- If you are taking a memory snapshot of a virtual machine that has multiple disks in different disk modes, verify that the virtual machine is powered off. For example, if you have a special purpose configuration that requires you to use an independent disk, you must power off the virtual machine before taking a snapshot.

- To capture the memory state of the virtual machine, verify that the virtual machine is powered on.
- To quiesce the virtual machine files, verify that the virtual machine is powered on and that VMware Tools is installed.
- Verify that you have the **Virtual machine** > **Snapshot management** > **Create snapshot** privilege on the virtual machine.

When you take a snapshot, other activity that is occurring in the virtual machine might affect the snapshot process when you revert to that snapshot. The best time to take a snapshot from a storage perspective is when you are not incurring a large I/O load. The best time to take a snapshot from a service perspective is when no applications in the virtual machine are communicating with other computers. The potential for problems is greatest if the virtual machine is communicating with another computer, especially in a production environment. For example, if you take a snapshot while the virtual machine is downloading a file from a server on the network, the virtual machine continues downloading the file and communicating its progress to the server. If you revert to the snapshot, communications between the virtual machine and the server are confused and the file transfer fails. Depending on the task that you are performing, you can create a memory snapshot or you can quiesce the file system in the virtual machine.

Memory Snapshots	The default selection for taking snapshots. When you capture the virtual machine memory state, the snapshot retains the live state of the virtual machine. Memory snapshots create a snapshot at a precise time, for example, to upgrade software that is still working. If you take a memory snapshot and the upgrade does not complete as expected, or the software does not meet your expectations, you can revert the virtual machine to its previous state. When you capture the memory state, the virtual machine files do not require quiescing. If you do not capture the memory state, the snapshot does not save the live state of the virtual machine and the disks are crash consistent unless you quiesce them. Snapshots that capture the memory state of a virtual machine take longer to complete. You might also see a momentary lapse in response over the network.
Quiesced Snapshots	When you quiesce a virtual machine, VMware Tools quiesces the file system of the virtual machine. A quiesce operation ensures that a snapshot disk represents a consistent state of the guest file systems. The quiesce operation pauses or alters the state of running processes on the virtual machine, especially processes that might modify information stored on the disk during a restore operation. Quiesced snapshots are appropriate for automated or periodic backups. For example, if you are unaware of the virtual machine activity, but want several recent backups to revert to, you can quiesce the files. If the virtual machine is powered off or VMware Tools is not available, the `Quiesce` parameter is not available. You cannot quiesce virtual machines that have large capacity disks. Application-consistent quiescing is not supported for virtual machines with IDE or SATA disks.

Important:
Do not use snapshots as your only backup solution or as a long-term backup solution.

Note:
If you take a snapshot of a Dynamic Disk (Microsoft-specific disk type), the snapshot technology preserves the quiesced state of the file system, but does not preserve the quiesced state of the application.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click a virtual machine from the list and select **Snapshots** > **Take snapshot**.
3. In the **Take snapshot** dialog box, enter a name for the snapshot.
4. Optional: Enter a description for the snapshot.
5. Optional: Select the **Include virtual machine's memory** check box to capture the memory of the virtual machine.
6. Optional: Deselect **Include virtual machine's memory** and select **Quiesce guest file system (requires VMware Tools installed)** check box to pause running processes on the guest operating system so that file system contents are in a known consistent state when you take the snapshot.

   Quiesce the virtual machine files only when the virtual machine is powered on and you do not want to capture the virtual machine's memory.

7. Click **Create**.

Revert to the Latest Snapshot in the VMware Host Client

To return a virtual machine to its original state, or to return to another snapshot in the snapshot hierarchy, you can restore a snapshot.

Verify that you have the **Virtual machine** > **Snapshot management** > **Revert to snapshot** privilege on the virtual machine.

When you restore a snapshot, you return the virtual machine memory, settings, and the state of the virtual machine disks to the state they were in at the time you took the snapshot. If you want the virtual machine to be suspended, powered on, or powered off when you start it, make sure that it is in the correct state when you take the snapshot.

You can restore snapshots in the following ways:

Revert to Latest Snapshot	Restores the parent snapshot, one level up in the hierarchy from the **You are Here** position. **Revert to Latest Snapshot** activates the parent snapshot of the current state of the virtual machine.
Revert To	Lets you restore any snapshot in the snapshot tree and makes that snapshot the parent snapshot of the current state of the virtual machine. Subsequent snapshots from this point create a new branch of the snapshot tree.

Restoring snapshots has the following effects:

- The current disk and memory states are discarded, and the virtual machine reverts to the disk and memory states of the parent snapshot.
- Existing snapshots are not removed. You can restore those snapshots at any time.
- If the snapshot includes the memory state, the virtual machine will be in the same power state as when you created the snapshot.

Table 506: Virtual Machine Power State After Restoring a Snapshot

Virtual Machine State When Parent Snapshot Is Taken	Virtual Machine State After Restoration
Powered on (includes memory)	Reverts to the parent snapshot, and the virtual machine is powered on and running.
Powered on (does not include memory)	Reverts to the parent snapshot and the virtual machine is powered off.
Powered off (does not include memory)	Reverts to the parent snapshot and the virtual machine is powered off.

Virtual machines running certain kinds of workloads can take several minutes to resume responsiveness after reverting from a snapshot.

> **Note:**
> vApp metadata for virtual machines in vApps does not follow the snapshot semantics for virtual machine configuration. vApp properties that are deleted, modified, or defined after a snapshot is taken remain intact (deleted, modified, or defined) after the virtual machine reverts to that snapshot or any previous snapshots.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click the virtual machine in the list and select **Snapshots** > **Revert to Latest Snapshot**.

> **Note:**
> The current state of the virtual machine will be lost unless you save it in a snapshot.

3. Click **OK** to revert the virtual machine to the most recent snapshot.

Delete a Snapshot in the VMware Host Client

You can use the Snapshot Manager to delete a single snapshot or all snapshots in a tree. Deleting a snapshot removes the snapshot from the Snapshot Manager. The snapshot files are consolidated and written to the parent snapshot disk and merge with the virtual machine base disk.

Deleting a snapshot does not change the virtual machine or other snapshots. Deleting a snapshot consolidates the changes between snapshots and previous disk states and writes all the data from the delta disk that contains the information about the deleted snapshot to the parent disk. When you delete the base parent snapshot, all changes merge with the base virtual machine disk.

To delete a snapshot, a large amount of information needs to be read and written to a disk. This process can reduce virtual machine performance until consolidation is complete. Consolidating snapshots removes redundant disks, which improves virtual machine performance and saves storage space. The time it takes to delete snapshots and consolidate the snapshot files depends on the amount of data that the guest operating system writes to the virtual disks after you take the last snapshot. The required time is proportional to the amount of data the virtual machine is writing during consolidation if the virtual machine is powered on.

Failure of disk consolidation can reduce the performance of virtual machines. You can check whether any virtual machines require separate consolidation operations by viewing a list. For information about locating and viewing the consolidation state of multiple virtual machines and running a separate consolidation operation, see *vSphere Virtual Machine Administration*.

Delete	Use the **Delete** option to remove a single parent or child snapshot from the snapshot tree. **Delete** writes disk changes that occur between the state of the snapshot and the previous disk state to the parent snapshot. **Note:** Deleting a single snapshot preserves the current state of the virtual machine and does not affect any other snapshot. You can also use the **Delete** option to remove a corrupt snapshot and its files from an abandoned branch of the snapshot tree without merging them with the parent snapshot.
Delete All	Use the **Delete All** option to delete all snapshots from the Snapshot Manager. **Delete all** consolidates and writes the changes that occur between snapshots and the previous delta disk states to the base parent disk and merges them with the base virtual machine disk.

To prevent snapshot files from merging with the parent snapshot if, for example, an update or installation fails, first use the **Restore** command to restore to a previous snapshot. This action invalidates the snapshot delta disks and deletes the memory file. You can then use the **Delete** option to remove the snapshot and any associated files.

Be careful not to accidentally delete a snapshot that you need. You cannot restore a deleted snapshot. For example, you might want to install several browsers, a, b, and c, and capture the virtual machine state after you install each browser. The first, or base snapshot, captures the virtual machine with browser a and the second snapshot captures browser b. If you restore the base snapshot that includes browser a and take a third snapshot to capture browser c and delete the snapshot that contains browser b, you cannot return to the virtual machine state that includes browser b.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click the virtual machine in the list and select **Snapshots** > **Manage Snapshots**.
3. Click the snapshot that you want to delete and click **Delete**.
4. Click **Delete** to confirm the deletion.

Why to Use Snapshot Manager in the VMware Host Client

You can review all snapshots for your virtual machines and use the Snapshot Manager to manage the snapshots.

After you take a snapshot, you can restore the virtual machine to its snapshot state at any time.

If you have a series of snapshots, you can use the Snapshot Manager to restore any parent or child snapshot. Subsequent child snapshots that you take from the restored snapshot create a branch in the snapshot tree. Use the Snapshot Manager to delete a snapshot from the tree.

Table 507: Snapshot Manager

Option	Description
Snapshot tree	Displays all snapshots for the virtual machine.
You are here ( ) icon	The **You are here** icon represents the current and active state of the virtual machine. The **Revert**, **Edit**, and **Delete** actions are deactivated for the **You are here** state.
Take Snapshot, **Revert**, **Edit**, **Delete**, **Delete All**	Snapshot options.
Details	Shows the snapshot name and description, and the date you created the snapshot. The Console shows the power state of the virtual machine when a snapshot was taken. The Name, Description, and Created text boxes are blank if you do not select a snapshot.

Monitor a Virtual Machine in the VMware Host Client

You can monitor various performance aspects and keep track of the actions that take place on virtual machines that you create in the VMware Host Client.

View Virtual Machine Performance Charts in the VMware Host Client

You can view line charts with information about the resource usage of virtual machines that you create in the VMware Host Client.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click a virtual machine from the list.
3. Click the **Monitor** tab and click **Performance**.
4. To view the virtual machine resource usage for the last hour, select an option from the drop-down menu.
   - To view the percentage of CPU that the virtual machine used during the last hour, select **CPU usage**.
   - To view the memory that the host consumed during the last hour, select **Memory usage**.

View Virtual Machine Events in the VMware Host Client

Events are records of the actions that a user performs on a virtual machine. When you create a virtual machine in the VMware Host Client, you can view the events associated with the virtual machine.

Required privilege: **Read only**.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click a virtual machine from the list.
3. Click the **Monitor** tab and under **Tasks and Events**, click **Events**.
   A list of all virtual machine events is displayed.
4. Optional: Click an event from the list to view event details.
5. Optional: Use the filter controls above the list to filter the list.
6. Optional: Click a column heading to sort the list.

View Virtual Machine Tasks in the VMware Host Client

When you create a virtual machine in the VMware Host Client, you can view all virtual machine tasks and information about the task target, initiator, queue time, start time, result, and time of completion.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click a virtual machine from the list.
3. Expand the virtual machine in the VMware Host Client inventory and click **Monitor**.
4. Click **Virtual Machines** in the VMware Host Client inventory.
5. Click a virtual machine from the list.
6. Click the **Monitor** tab and under **Tasks and Events**, click **Tasks**.
7. Optional: Click a task from the list to view task details.
8. Optional: Use the filter controls above the list to filter the list.
9. Optional: Click a column heading to sort the list.

Configure Virtual Machine Hardware in the VMware Host Client

You can configure most virtual machine hardware settings when you create a virtual machine or after you create the virtual machine and install the guest operating system.

Virtual CPU Limitations

The maximum number of virtual CPUs that you can assign to a virtual machine is 960. The number of virtual CPUs depends on the number of logical CPUs on the host, and the type of guest operating system that is installed on the virtual machine.

Be aware of the following limitations:

- A virtual machine cannot have more virtual CPUs than the number of logical cores on the host. The number of logical cores is equal to the number of physical cores if hyperthreading is deactivated or two times that number if hyperthreading is activated.
- If a running virtual machine has 128 virtual CPUs or less, you cannot use hot adding to further increase the number of virtual CPUs. To change the number of virtual CPUs beyond that limit, you must first power off the virtual machine. By contrast, if a running virtual machine already has more than 128 virtual CPUs, you can use hot adding to further increase the number of virtual CPUs to up to 960.
- The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.
- Not every guest operating system supports Virtual SMP, and guest operating systems that support this functionality might support fewer processors than are available on the host. For information about Virtual SMP support, see the Broadcom Compatibility Guide.

Configuring Multicore Virtual CPUs in the VMware Host Client

VMware multicore virtual CPU support lets you control the number of cores per virtual socket in a virtual machine. This capability lets operating systems with socket restrictions use more of the host CPU cores, which increases overall performance.

> **Important:**
> When you configure your virtual machine for multicore virtual CPU settings, you must ensure that your configuration complies with the requirements of the guest operating system EULA.

Using multicore virtual CPUs can be useful when you run operating systems or applications that can take advantage of only a limited number of CPU sockets.

You can configure a virtual machine to have up to 960 virtual CPUs. A virtual machine cannot have more virtual CPUs than the actual number of logical CPUs on the host. The number of logical CPUs means the number of physical processor cores or two times that number if hyperthreading is enabled. For example, if a host has 128 logical CPUs, you can configure the virtual machine for 128 virtual CPUs.

You configure how the virtual CPUs are assigned in terms of cores and cores per socket. Determine how many CPU cores you want in the virtual machine, then select the number of cores you want in each socket, depending on whether

you want a single-core CPU, dual-core CPU, tri-core CPU, and so on. Your selection determines the number of sockets that the virtual machine has.

The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.

For more information about multicore CPUs, see the *vSphere Resource Management* documentation.

Change the Number of Virtual CPUs in the VMware Host Client

A virtual machine with ESX 9.1 and later compatibility can have up to 960 virtual CPUs. You can change the number of virtual CPUs while your virtual machine is powered off. If virtual CPU hot add is enabled, you can increase the number of virtual CPUs while the virtual machine is running.

- If CPU hot add is not enabled, power off the virtual machine before adding virtual CPUs.
- To hot add multicore CPUs, verify that the virtual machine is compatible with ESX 7.0 and later.
- Verify that you have the following privileges:
  - **Virtual machine** > **Change Configuration** > **Change CPU Count**
  - **Virtual machine** > **Change Configuration** > **Change resource**
  - **Virtual machine** > **Change Configuration** > **Change**

Virtual CPU hot add is supported for virtual machines with multicore CPU support and ESX 6.5 and later compatibility. When the virtual machine is powered on and CPU hot add is enabled, you can hot add virtual CPUs to the running virtual machine. You can add only multiples of the number of cores per socket.

If a virtual machine has 128 virtual CPUs or less, you cannot use hot adding to further increase the number of virtual CPUs. To change the number of virtual CPUs beyond that limit, you must first power off the virtual machine. By contrast, if a virtual machine already has more than 128 virtual CPUs, you can use hot adding to further increase the number of virtual CPUs to up to 960.

The maximum number of virtual CPU sockets that a virtual machine can have is 128. If you want to configure a virtual machine with more than 128 virtual CPUs, you must use multicore virtual CPUs.

> **Important:**
>
> When you configure your virtual machine for multicore virtual CPU settings, you must ensure that your configuration complies with the requirements of the guest operating system EULA.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **CPU**.
4. From the **CPU** drop-down menu, select the number of cores and click **OK**.

Allocate CPU Resources in the VMware Host Client

To manage workload demands, you can change the amount of CPU resources allocated to a virtual machine by using the shares, reservations, and limits settings.

Power off the virtual machine.

A virtual machine has the following user-defined settings that affect its CPU resource allocation.

Limit	Places a limit on the consumption of CPU time for a virtual machine. This value is expressed in MHz or GHz.
Reservation	Specifies the guaranteed minimum allocation for a virtual machine. The reservation is expressed in MHz or GHz.
Shares	Each virtual machine is granted CPU shares. The more shares a virtual machine has, the more often it receives a time slice of a CPU when there is no CPU idle time. Shares represent a relative metric for allocating CPU capacity.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **CPU**, and allocate CPU capacity for the virtual machine.

Option	Description
Reservation	Guaranteed CPU allocation for this virtual machine.
Limit	Upper limit for this virtual machine's CPU allocation. Select **Unlimited** to specify no upper limit.
Shares	CPU shares for this virtual machine in relation to the parent's total. Sibling virtual machines share resources according to their relative share values bounded by the reservation and limit. Select **Low**, **Normal**, or **High**, which specify share values in a 1:2:4 ratio. Select **Custom** to assign each virtual machine a specific number of shares, which expresses a proportional weight.

4. Click **OK**.

Virtual Memory Configuration in the VMware Host Client

You can add, change, or configure virtual machine memory resources or options to improve virtual machine performance. You can set most of the memory parameters during virtual machine creation or after you install the guest operating system.

Some actions require that you power off the virtual machine before you change the settings.

How Memory Resources Work

The memory resource settings for a virtual machine determine how much of the ESX host memory is allocated to the virtual machine. The virtual hardware memory size determines how much memory is available to applications that run in the virtual machine. A virtual machine uses memory resources only up to its configured virtual hardware memory size. Hosts limit memory resource use to the maximum amount useful for the virtual machine, so you can accept the default Unlimited memory resource setting.

Change the Memory Configuration in the VMware Host Client

You can reconfigure the amount of memory allocated to a virtual machine to improve performance.

Verify that you have the following privileges on the virtual machine:

- **Virtual machine** > **Configuration** > **Change Memory**
- **Virtual machine** > **Change Configuration** > **Change resource**

The minimum memory size is 4 MB for virtual machines that use BIOS firmware. Virtual machines that use EFI firmware require at least 96 MB of RAM, or they cannot power on.

The maximum memory size for a virtual machine depends on the host's physical memory and the virtual machine's compatibility setting.

If the virtual machine memory exceeds the host memory size, swapping occurs, which can have a severe effect on virtual machine performance. The memory size must be a multiple of 4 MB. The maximum for best performance represents the threshold above which the host's physical memory is insufficient to run the virtual machine at full speed. This value fluctuates as conditions on the host change, for example, as virtual machines are powered on or off.

Table 508: Maximum Virtual Machine Memory

Introduced in Host Version	Virtual Machine Compatibility	Maximum Memory Size
ESX 9.1	ESX 9.1 and later	24560 GB
ESX 9.0	ESX 9.0 and later	24560 GB

Introduced in Host Version	Virtual Machine Compatibility	Maximum Memory Size
ESX 8.0 Update 3	ESX 8.0 Update 3 and later	24560 GB
ESX 8.0 Update 2	ESX 8.0 Update 2 and later	24560 GB
ESX 8.0 Update 1	ESX 8.0 Update 1 and later	24560 GB
ESX 8.0	ESX 8.0 and later	24560 GB

1. Click **Virtual Machines** in the inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Memory** drop-down menu.
   a) From the **Memory** drop-down menu, select the amount of RAM to assign to the virtual machine.
   b) Select whether the memory is specified in MB, GB, or TB.
4. Click **OK**.

Allocate Memory Resources to a Virtual Machine in the VMware Host Client

You can change the amount of memory resources allocated to a virtual machine by using the shares, reservations, and limits settings. The host determines the appropriate amount of physical RAM to allocate to virtual machines based on these settings. You can assign a high or low shares value to a virtual machine, depending on its load and status.

Power off the virtual machine.

The following user-defined settings affect the memory resource allocation of a virtual machine.

Limit	Places a limit on the consumption of memory for a virtual machine. This value is expressed in megabytes.
Reservation	Specifies the guaranteed minimum allocation for a virtual machine. The reservation is expressed in megabytes. If the reservation cannot be met, the virtual machine will not power on.
Shares	Each virtual machine is granted a number of memory shares. The more shares a virtual machine has, the greater share of host memory it receives. Shares represent a relative metric for allocating memory capacity. For more information about share values, see the *vSphere Resource Management* documentation.

You cannot assign a reservation to a virtual machine that is larger than the virtual machine configured memory. If you give a virtual machine a large reservation and reduce the virtual machine configured memory size, the reservation is reduced to match the new configured memory size.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **Memory**, and allocate the memory capacity for the virtual machine.

Option	Description
Reservation	Guaranteed memory allocation for this virtual machine.
Limit	Upper limit for this virtual machine's memory allocation. Select **Unlimited** to specify no upper limit.
Shares	The values **Low**, **Normal**, **High**, and **Custom** are compared to the sum of all shares of all virtual machines on the server.

4. Click **OK**.

Change Memory Hot Add Settings

Memory hot add lets you add memory resources for a virtual machine while that virtual machine is powered on.

- Power off the virtual machine.
- Verify that the virtual machine has a guest operating system that supports memory hot add functionality.
- Verify that the virtual machine compatibility is ESX 6.x and later.

- Verify that VMware Tools is installed.

Enabling memory hot add produces some memory overhead on the ESX host for the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **Memory**, and select the **Memory Hot Plug** check box to activate adding memory to the virtual machine while it is powered on.
4. Click **OK**.

You can now add memory to a virtual machine, even if the virtual machine is powered on.

Network Virtual Machine Configuration

ESX networking features enable communication between virtual machines on the same host, between virtual machines on different hosts, and between other virtual and physical machines.

The networking features also allow management of ESX hosts and enable communication between VMkernel services, such as NFS, iSCSI, or vSphere vMotion, and the physical network. When you configure networking for a virtual machine, you select or change an adapter type, a network connection, and whether to connect the network when the virtual machine powers on.

Change the Configuration of the Virtual Network Adapter in the VMware Host Client

You can configure the power-on connection setting, the MAC address, and the network connection of the virtual network adapter of a virtual machine.

Required privileges:

- **Virtual Machine** > **Configuration** > **Modify device settings** for editing the MAC address and network.
- **Virtual Machine** > **Interaction** > **Device connection** for changing **Connect** and **Connect at power on**.
- **Network** > **Assign network**

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **Virtual Hardware** tab and select the appropriate Network Adapter (NIC) from the hardware list.
4. Optional: To connect the virtual NIC when the virtual machine is powered on, select **Connect At Power On**.
5. Optional: Select the adapter type from the **Adapter Type** drop-down menu.
6. Select an option for MAC address configuration.

Option	Description
Automatic	vSphere assigns a MAC address automatically.
Manual	Enter the MAC address to use.

7. Click **OK**.

Add a Network Adapter to a Virtual Machine in the VMware Host Client

When you add a network adapter (NIC) to a virtual machine, you must select the adapter type, the network connection, and whether the device connects when the virtual machine is powered on.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click **Add New Device** and select **Network Adapter** from the drop-down menu.
4. Expand **New Network** and select the standard or distributed port group to connect to.
5. To connect the virtual NIC when the virtual machine is powered on, select **Connect At Power On**.
6. Select the adapter type from the **Adapter Type** drop-down menu.
7. Optional: To configure the virtual NIC to connect when the virtual machine is powered on, select **Connect At Power On**.
8. Select an option for MAC address configuration.
9. Click **OK**.

Virtual Disk Configuration in the VMware Host Client

You can add large-capacity virtual disks to virtual machines and add more space to existing disks, even when the virtual machine is running. You can set most of the virtual disk parameters during virtual machine creation or after you install the guest operating system.

Virtual Disk Storage Options

You can store virtual machine data in a new virtual disk, an existing virtual disk, or a mapped SAN LUN. A virtual disk appears as a single hard disk to the guest operating system. The virtual disk consists of one or more files on the host file system. You can copy or move virtual disks on the same host or between hosts.

Raw Device Mapping

When you map a LUN to a VMFS volume, the host creates a raw device mapping (RDM) file that points to the raw LUN. Encapsulating disk information in a file allows the host to lock the LUN so that only one virtual machine can write to it. This file has a .vmdk extension, but the file contains only disk information that describes the mapping to the LUN on the system. The actual data is stored on the LUN.

To deploy a virtual machine from a template, store its data in a virtual disk file. You can store virtual machine data on a LUN only for a virtual machine that you create directly, not from a template.

Change the Virtual Disk Configuration in the VMware Host Client

If you run out of disk space, you can increase the size of the disk. You can change the virtual device node and the persistence mode of the virtual disk configuration of a virtual machine.

Power off the virtual machine.

Verify that you have the following privileges:

- **Virtual machine** > **Configuration** > **Modify device settings** on the virtual machine.
- **Virtual machine** > **Configuration** > **Extend virtual disk** on the virtual machine.
- **Datastore** > **Allocate space** on the datastore.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand the hard disk to view or change the disk settings.

Option	Description
Maximum Size	Shows the maximum size of this hard disk on this virtual machine.
Type	Shows the storage type. You cannot change this setting for an existing hard disk. You choose the storage type of a hard disk when you add the hard disk to the virtual machine. For more

Option	Description
	information about storage types and available disk formats, see the *vSphere Storage* documentation.
Sector Size	The sector size of the disk. By default, the sector size is 512n (512 bytes). For more information about the datastore storage format and sector sizes, see the *vSphere Storage* documentation.
Sharing	Specifies sharing information.
Disk File	Lists disk files on the datastore.
Disk Mode	Disk mode determines how a virtual disk is affected by snapshots. You have the following options: • Dependent: Dependent mode is the default disk mode. When you take a snapshot of a virtual machine, dependent disks are included in snapshots. When you revert to the previous snapshot, all data is reverted to the point of taking a snapshot. • Independent - Persistent: Disks in persistent mode behave like conventional disks on your physical computer. All data written to a disk in persistent mode is written permanently to the disk, even if you revert a snapshot. When you power off or reset a virtual machine, the disk and all its snapshots are preserved. • Independent - Nonpersistent: Disks in nonpersistent mode behave like read-only disks. Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. With nonpersistent mode, you can restart the virtual machine with a virtual disk in the same state every time. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset a virtual machine, or you delete a snapshot.
Virtual Device Node	Displays the virtual device node.

4. Click **OK**.

Add a New Hard Disk to a Virtual Machine in the VMware Host Client

You can add a virtual hard disk to an existing virtual machine, or you can add a hard disk when you customize the virtual machine hardware during the virtual machine creation process. For example, you might need to provide additional disk space for an existing virtual machine with a heavy workload. During virtual machine creation, you might want to add a hard disk that is preconfigured as a boot disk.

- Verify that you are familiar with configuration options and caveats for adding virtual hard disks. See Virtual Disk Configuration.
- Before you add disks larger than 2 TB in size to a virtual machine, see *vSphere Virtual Machine Administration*.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add new disk** privilege on the destination folder or datastore.

Power off the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Optional: To delete an existing hard disk, click the vertical ellipsis icon (⋮) and click **Remove device**.
   The disk is removed from the virtual machine. If other virtual machines share the disk, the disk files are not deleted.

4. On the **Virtual Hardware** tab, click the **Add New Device** button.
5. From the drop-down menu, under **Disks, Drives and Storage**, select **Hard Disk**.
   The hard disk appears in the Virtual Hardware devices list.
6. Expand **New Hard disk**.
7. Optional: Enter a value for the hard disk size and select the units from the drop-down menu.
8. Select the datastore location where you want to store the virtual machine files.
9. Select the format for the virtual machine disk.

Option	Description
Thick Provision Lazy Zeroed	Create a virtual disk in a default thick format. Space required for the virtual disk is allocated during creation. Any data remaining on the physical device is not erased during creation, but is zeroed out on demand at a later time on first write from the virtual machine.
Thick Provision Eager Zeroed	Create a thick disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out during creation. It might take much longer to create disks in this format than to create other types of disks.
Thin Provision	Use the thin-provisioned format. At first, a thin-provisioned disk uses only as much datastore space as the disk initially requires. If the thin disk needs more space later, it can grow to the maximum capacity allocated to it.

10. From the **Sharing** drop-down menu, select a value for the shares to allocate to the virtual disk.
11. Accept the default or select a different virtual device node.
    In most cases, you can accept the default device node. For a hard disk, using a nondefault device node makes controlling the boot order or having different SCSI controller types easier. For example, you might want to boot from an LSI Logic controller and share a data disk with another virtual machine that is using a Buslogic controller with bus sharing turned on.
12. Optional: Select a disk mode.

Option	Description
Dependent	Dependent disks are included in snapshots.
Independent-Persistent	Disks in persistent mode behave like conventional physical computer disks. All data written to a disk in persistent mode are written permanently to the disk.
Independent-Nonpersistent	Changes to disks in nonpersistent mode are discarded when you power off or reset the virtual machine. The virtual disk returns to the same state every time you restart the virtual machine. Changes to the disk are written to and read from a redo log file that is deleted when you power off or reset.

13. From the **Virtual Device Note** drop-down, add a controller and assign the disk to a virtual device node on that controller.
14. Click **OK**.

Add an Existing Hard Disk to a Virtual Machine in the VMware Host Client

You can add an existing virtual hard disk to a virtual machine when you customize the virtual machine hardware during the virtual machine creation process or after the virtual machine is created. For example, you might want to add an existing hard disk that is preconfigured as a boot disk.

- Verify that you are familiar with controller and virtual device node behavior for different virtual hard disk configurations.

- Verify that you have the **Virtual machine** > **Change Configuration** > **Add existing disk** privilege on the destination folder or datastore.

Power off the virtual machine.

During virtual machine creation, a hard disk and a SCSI or SATA controller are added to the virtual machine by default, based on the guest operating system that you select. If this disk does not meet your needs, you can remove it and add an existing hard disk at the end of the creation process.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Optional: To delete an existing hard disk, click the vertical ellipsis icon
   (● ● ●)
   and click **Remove device**.

   The disk is removed from the virtual machine. If other virtual machines share the disk, the disk files are not deleted.
4. On the **Virtual Hardware** tab, click the **Add New Device** button.
5. From the drop-down menu, under **Disks, Drives and Storage**, select **Existing Hard Disk**.

   The **Select File** dialog box opens.
6. In the **Select File**, expand a datastore, select a virtual machine folder, and select the disk to add.
7. Click **OK**.
8. Optional: Expand the **New Hard disk** and make further customizations for the hard disk.
9. Click **OK** to add the existing hard disk.

Use Disk Shares to Prioritize Virtual Machines in the VMware Host Client

You can change the disk resources for a virtual machine. If multiple virtual machines access the same VMFS datastore and the same logical unit number (LUN), use disk shares to prioritize the level of access that virtual machines have to resources. Disk shares distinguish high-priority from low-priority virtual machines.

Power off the virtual machine.

You can allocate the I/O bandwidth of the host to the virtual hard disks of a virtual machine. You cannot pool disk I/O across a cluster.

The shares value represents the relative metric for controlling disk bandwidth to all virtual machines.

Disk shares are relevant only within a given host. The shares assigned to virtual machines on one host have no effect on virtual machines on other hosts.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand the hard disk to view the disk options.
4. In the **Sharing** drop-down menu, select a value for the shares to allocate to the virtual machine. For more information about the **multi-writer** option, see Broadcom knowledge base article 313527.
5. Click **OK**.

Virtual Machine Controller Configuration in the VMware Host Client

In the VMware Host Client, you can add various controllers to virtual machines, such as USB controllers, SCSI controllers, Paravirtual SCSI controllers, and SATA controllers. You can also change the SCSI Bus sharing configuration and the SCSI controller type.

Add SCSI Controllers in the VMware Host Client

You can add SCSI controllers to an existing virtual machine by adding hard disks on unused SCSI Bus numbers. You can set the type of SCSI bus sharing for a virtual machine and indicate whether to share the SCSI bus. Depending on the type of sharing, virtual machines can access the same virtual disk simultaneously on the same server or on any server.

Power off the virtual machine.

Adding a new hard disk on an unused SCSI Bus number creates a new SCSI controller.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and from the drop-down menu select **SCSI Controller**.

   The controller appears in the **Virtual Hardware** device list.
4. From the **Change Type** drop-down menu, select the controller type.

   Do not select a BusLogic Parallel controller for virtual machines with disks larger than 2 TB. This controller does not support large capacity hard disks.
5. From the **SCSI Bus Sharing** drop-down menu, select the type of sharing.

Option	Description
None	Virtual disks cannot be shared by other virtual machines.
Physical	Virtual disks can be shared by virtual machines on any ESX host.
Virtual	Virtual disks can be shared by virtual machines on the same ESX host.

6. Click **OK**.

You can now add a hard disk or other SCSI device to the virtual machine and assign it to the new SCSI controller.

Change the SCSI Controller Type in the VMware Host Client

You can attach virtual disks and RDMs to virtual machines by configuring virtual SCSI controller on the virtual machines.

Power off the virtual machine.

The choice of SCSI controller does not affect whether your virtual disk is an IDE or SCSI disk. The IDE adapter is always ATAPI. The default for your guest operating system is already selected. Older guest operating systems have BusLogic adapter as their default controller.

If you create an LSI Logic virtual machine and add a virtual disk that uses BusLogic adapters, the virtual machine boots from the BusLogic adapters disk. LSI Logic SAS is available only for virtual machines with hardware version 7 or later. Disks with snapshots might not experience performance gains when used on LSI Logic SAS, VMware Paravirtual, and LSI Logic Parallel adapters.

> **CAUTION:**
> Changing the SCSI controller type might result in a virtual machine boot failure.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **Virtual Hardware** tab and expand a SCSI controller.
4. Select a SCSI controller type from the **Change Type** drop-down menu.
5. Click **OK**.

About VMware Paravirtual SCSI Controllers

VMware Paravirtual SCSI controllers are high performance storage controllers that can result in greater throughput and lower CPU use. These controllers are best suited for high performance storage environments.

VMware Paravirtual SCSI controllers are available for virtual machines with ESX 6.x and later compatibility. Disks on such controllers might not experience optimal performance gains if they have snapshots or if memory on the ESX host is over committed. This behavior does not mitigate the overall performance gain of using VMware Paravirtual SCSI controllers as compared to other SCSI controller options.

For platform support for VMware Paravirtual SCSI controllers, see the Broadcom Compatibility Guide.

Add a Paravirtual SCSI Controller in the VMware Host Client

You can add a VMware Paravirtual SCSI high performance storage controller to provide greater throughput and lower CPU utilization.

- Verify that the virtual machine has a guest operating system with VMware Tools installed.
- Verify that the virtual machine has hardware version 7 or later.
- Familiarize yourself with VMware Paravirtual SCSI limitations. See *vSphere Virtual Machine Administration*.
- To access boot disk devices attached to a VMware Paravirtual SCSI controller, verify that the virtual machine has a Windows 2003 or Windows 2008 guest operating system.
- In some operating systems, before you change the controller type you must create a virtual machine with an LSI Logic controller and install VMware Tools.

Power off the virtual machine.

VMware Paravirtual SCSI controllers are best suited for environments, especially SAN environments, that run I/O-intensive applications.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. Click the **Virtual Hardware** tab and expand a SCSI controller.
4. From the **Change Type** drop-down menu, select **VMware Paravirtual**.
5. Click **OK**.

Add a SATA Controller to a Virtual Machine in the VMware Host Client

If a virtual machine has multiple hard disks or CD/DVD-ROM devices, you can add up to three additional SATA controllers to assign the devices to. When you assign the devices to different controllers, you can improve performance and avoid data traffic congestion. You can also add controllers if you need to exceed the thirty-device limit for a single controller.

- Verify that the virtual machine compatibility is ESX 6.5 and later.
- Verify that you are familiar with storage controller behavior and limitations. See *vSphere Virtual Machine Administration*.
- Verify that you have the **Virtual machine** > **Configuration** > **Add or remove device** privilege on the virtual machine.
- Power off the virtual machine.

You can boot virtual machines from SATA controllers and use them for large-capacity virtual hard disks.

Not all guest operating systems support AHCI SATA controllers. Typically, when you create virtual machines with ESX 6.5 and later compatibility and Mac OS X guest operating systems, a SATA controller is added by default for the virtual hard disk and CD/DVD-ROM devices. Most guest operating systems, including Windows Vista and later, have a default SATA controller for CD/DVD-ROM devices. For verification, see the Broadcom Compatibility Guide.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and from the drop-down menu select **SATA Controller**.
4. Click **OK**.

Add an NVMe Controller in the VMware Host Client

If a virtual machine has multiple hard disks, you can add up to four virtual NVMe controllers to which to assign the disks. Using a NVMe controller significantly reduces the software overhead for processing guest OS I/O, as compared to AHCI SATA or SCSI controllers.

- Verify that the virtual machine has a guest operating system that supports NVMe.
- Verify that the virtual machine compatibility is ESX 7.0 or later.
- Verify that you are familiar with storage controllers behavior and limitations. For more information, see the *Virtual Machine Administration* guide.
- Verify that you have the **Virtual machine** > **Change Configuration** > **Add new disk** privilege on the virtual machine.

NVMe controllers perform best with virtual disks on an all-flash disk array, local NVMe SSD, and PMem storage.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and from the drop-down menu select **NVMe Controller**.
4. If a new NVMe Controller is created, set **NVMe Bus Sharing** to **Physical**.
5. Click **OK**.

A new NVMe controller is added to the virtual machine.

You can add a hard disk to the virtual machine and assign it to the NVMe controller.

Other Virtual Machine Device Configuration in the VMware Host Client

In addition to configuring virtual machine CPU and Memory, and adding hard disks and virtual network adapters, you can also add and configure virtual hardware, such as DVD/CD-ROM drives and SCSI devices. You can also add a virtual Watchdog Timer (VWDT) device, Precision Clock device, and PCI devices.

Add a CD or DVD Drive to a Virtual Machine in the VMware Host Client

You can use a physical drive on a client or host, or you can use an ISO image to add a CD/DVD drive to a virtual machine.

Power off the virtual machine.

If you want to add a CD/DVD drive that is backed up by USB CD/DVD drive on the host, you must add the drive as a SCSI device. Hot adding or removing SCSI devices from an ESX host is not supported.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and select **CD/DVD Drive** from the drop-down menu.
4. From the **New CD/DVD drive** drop-down menu, select an option.

Option	Description
Client Device	1. Select **Client Device** as the location. 2. From the **Device Mode** drop-down menu, select **Emulate CD-ROM** or **Passthrough CD-ROM**.
Datastore ISO File	1. Select **Datastore ISO File** as the location. 2. Enter the path and filename for the image file, or click **Browse** to navigate to the file.

5. If you do not want the CD-ROM drive to connect when the virtual machine starts, deselect **Connect At Power On**.
6. Select the virtual device node that the drive uses in the virtual machine.
7. Click **OK**.

Add a USB Device to a Virtual Machine in the VMware Host Client

By using the VMware Host Client, you can add a USB controller to a virtual machine.

- Verify that a USB controller is present. See Add a USB Controller to a Virtual Machine.
- Add a physical USB device to the ESX host where the virtual machine is located by plugging the USB device into the host.

Note:
If the ESX host does not have available USB devices, you cannot add a USB device to the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and select **USB Controller** from the drop-down menu.
   The USB device appears in the hardware list of available hardware devices for the virtual machine.
4. From the **USB device** drop-down, select which USB device to add to the virtual machine.
5. Click **OK**.

Serial Port Configuration in the VMware Host Client

Parallel and serial ports are interfaces for connecting peripherals to the virtual machine. The virtual serial port can connect to a physical serial port or to a file on the host computer.

You can also use the virtual serial port to establish a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. You can add parallel and serial ports and change the serial port configuration.

Add a Serial Port to a Virtual Machine in the VMware Host Client

A virtual machine can use up to four virtual serial ports. You can connect the virtual serial port to a physical serial port or to a file on the host computer. You can also use a host-side-named pipe to set up a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. In addition, you can use a port or a Virtual Serial Port Concentrator (vSPC) URI to connect a serial port over the network.

- Familiarize yourself with the different media types that the port can access, vSPC connections, and any conditions that might apply. See *vSphere Virtual Machine Administration*.

- To connect a serial port over a network, add a Firewall rule set. See *vSphere Virtual Machine Administration*.
- Required privilege:**Virtual Machine** > **Configuration** > **Add or Remove Device**

Power off the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and select **Serial Port**.

   The Serial Port appears in the hardware list.
4. From the **New Serial port** drop-down menu, select the type of media port to access.

Option	Description
Use output file	Select this option to send output from the virtual serial port to a file on the host computer. To select an output file to connect to the serial port, click **Browse**.
Use named pipe	Select this option to set a direct connection between two virtual machines or a connection between a virtual machine and an application on the host computer. 1. Enter a name for the pipe in the **Pipe name** field. 2. Select the **Near End** and **Far End** of the pipe from the drop-down menus.
Use physical serial port	Select this option to have the virtual machine use a physical serial port on the host computer. Select the serial port from the drop-down menu.
Use network	To connect through a remote network, select **Use network**. 1. From the **Direction** drop-down menu, select **Server** or **Client**. a. To have the virtual machine monitor incoming connections from other hosts, select **Server**. b. To have the virtual machine initiate a connection to another host, select **Client**. 2. Enter a port URI. The URI is the remote end of the serial port to which the serial port of the virtual machine should connect. 3. If vSPC is used as an intermediate step to access all virtual machines through a single IP address, select **Use Virtual Serial Port Concentrator** and enter the vSPC URI location.

5. Optional: Select **Yield CPU on poll**.

   Select this option only for guest operating systems that use serial ports in polled mode. This option prevents the guest operating system from consuming excessive CPUs.
6. Optional: Deselect **Connect at power on** if you do not want the parallel port device to connect when the virtual machine powers on.
7. Click **OK**.

Establishing Serial Port Network Connections to a Client or Server Without Authentication Parameters

If you do not use vSPC and you configure your virtual machine with a serial port connected as a server with a `telnet://:12345` URI, you can connect to your virtual machine's serial port from your Linux or Windows operating system.

```
telnetyourESXiServerIPAddress12345
```

Similarly, if you run the Telnet Server on your Linux system on port 23 (`telnet://yourLinuxBox:23`), you configure the virtual machine as a client URI.

```
telnet://yourLinuxBox:23
```

The virtual machine initiates the connection to your Linux system on port 23.

Add a Virtual Watchdog Timer Device to a Virtual Machine in the VMware Host Client

To ensure self-reliance that is related to the system performance within a virtual machine, you can add a virtual Watchdog Timer (VWDT) device. You can add a virtual Watchdog Timer (VWDT) device to a virtual machine to prevent the virtual machine from a guest operating system failure for an extended period of time.

- Power off the virtual machine.
- Verify that you have the **Virtual machine** > **Configuration** > **Add or remove device** privilege on the virtual machine.
- Verify that the guest operating system of the virtual machine supports the VWDT device.
- Verify that the virtual hardware version is 17.

You can enable the VWDT to start either by the guest operating system, or by the BIOS or EFI firmware. If you chose the VWDT to start with the BIOS or EFI firmware, it starts before the guest operating system boots.The VWDT has an important role in guest-based clustering solutions where each virtual machine in the cluster can recover on its own if it fails.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and select **Watchdog Timer**.

   The Watchdog Timer device appears in the hardware list.
4. Optional: Select **Start with BIOS/EFI boot** to start the watchdog timer by the BIOS or EFI firmware.

   When you select this option, the VWDT device starts before the guest operating system. If the guest OS boot takes too long or does not support the watchdog timer, the device might constantly restart the virtual machine.
5. Click **OK**.

Add a Precision Clock Device to a Virtual Machine in the VMware Host Client

A precision clock is a virtual device that runs on a virtual machine and accesses the system time of a host. By adding a precision clock to a virtual machine, you ensure time synchronization and high precision timestamping.

- Power off the virtual machine.
- Verify that the virtual hardware version is 17.
- Verify that you have the **Virtual machine** > **Configuration** > **Add or remove device** privilege on the virtual machine.
- Verify that you have the **Virtual machine** > **Configuration** > **Modify device settings** privilege on the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button and select **Precision Clock**.

   The precision clock device appears in the hardware list.
4. Optional: Select the time synchronization protocol.
5. Click **OK**.

Add a PCI Device to a Virtual Machine in the VMware Host Client

DirectPath I/O allows the guest operating system of a virtual machine to access the physical PCI and PCIe devices that are connected to a host directly. By using this technology, you can connect each virtual machine to up to 16 physical PCI devices.

- Power off the virtual machine.
- Verify that you have the **Virtual machine** > **Configuration** > **Add or remove device** privilege on the virtual machine.
- Verify that the PCI devices are connected to the host and marked as available for passthrough.
- If you want to add a dynamic PCI device to a virtual machine, verify that the virtual hardware version is 17.

You can use Dynamic DirectPath I/O to assign multiple PCI passthrough devices to a virtual machine. You can identify the PCI passthrough devices by their vendor and model name.

Note:
Some virtual machine operations become unavailable when you add a PCI or PCIe passthrough device to the virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. On the **Virtual Hardware** tab, click the **Add New Device** button.
4. From the drop-down menu, under **Other Devices**, select **PCI Device**.

   The **Device Selection** window that contains a list of devices appears.
5. Select the PCI device to connect to the virtual machine and click **Select**.

   The PCI device appears in the Virtual Hardware device list.

Option	Description
DirectPath I/O **Dynamic DirectPath I/O** **Enhanced DirectPath I/O**	• Expand the **New PCI device** section and, if present, enter the configuration parameters for the PCI device. • To remove the PCI device, click the vertical ellipsis icon (⋮) and click **Remove device**. • To undo the operation, click **Undo**.
NVIDIA GRID vGPU	• Expand the **New PCI device** section. • From the **vGPU Profile** drop-down menu, select the NVIDIA GRID vGPU passthrough device to connect to the virtual machine. • To remove the NVIDIA GRID vGPU passthrough device, click the vertical ellipsis icon (⋮) and click **Remove device**. • To undo the operation, click **Undo**. **Note:** You can add only one NVIDIA GRID vGPU device in MIG mode to a virtual machine.
Group	• Expand the **New Vendor Device Group** section. • If you add a Vendor Device Group that has a NIC, add an appropriate network. • From the **Network** drop-down menu, click **Browse**, and select an appropriate network. • To remove the Vendor Device Group, click the vertical ellipsis icon (⋮) and click **Remove device**. (⋮) • To undo the operation, click **Undo**.

6. Click **OK**.
7. Power on the virtual machine.

   The connected PCI devices appear:

   - On the **Hardware** tab of the **Edit Settings** wizard.
   - On the **Summary** tab in the **PCI Devices** panel.

   You can clone the virtual machine with the PCI passthrough devices.

Securing Virtual Machines in the VMware Host Client

The guest operating system that runs in the virtual machine is vulnerable to the same security risks as any physical system.

To boost security in your virtual environment, you can add a virtual Trusted Platform Module (vTPM) to your ESX hosts. You can also enable virtualization-based security (VBS) for the virtual machines that run the latest Windows 10 and Windows Server 2016 operating systems. You can provide additional security to your workloads by using Virtual Intel® Software Guard Extensions (vSGX) for virtual machines.

Activate or Deactivate Virtualization-based Security on an Existing VM in the VMware Host Client

Virtualization-based security (VBS) uses the Microsoft Hyper-V-based virtualization technology to isolate core Windows OS services in a separate virtualized environment. Such isolation provides an additional level of protection, because it makes it impossible for the key services in your environment to be manipulated.

Configuring VBS is a process that involves first activating VBS in the virtual machine then activating VBS in the guest operating system.

> **Note:**
> New virtual machines configured for Windows 10, Windows Server 2016, and Windows Server 2019 on hardware versions less than version 14 are created using Legacy BIOS by default. If you change the firmware type of a virtual machine from Legacy BIOS to UEFI, you must reinstall the guest operating system.
>
> You can activate VBS on a virtual machine only if the TPM validation of the host is successful.

Using Intel CPUs for VBS requires vSphere 6.7 or later. The virtual machine must have been created using hardware version 14 or later and one of the following supported guest operating systems:

- Windows 10 (64 bit) or later releases
- Windows Server 2016 (64 bit) or later releases

Using AMD CPUs for VBS requires vSphere 7.0 Update 2 or later. The virtual machine must have been created using hardware version 19 or later and one of the following supported guest operating systems:

- Windows 10 (64 bit), version 1809 or later releases
- Windows Server 2019 (64 bit) or later releases

Ensure that you install the latest patches for Windows 10, version 1809, and Windows Server 2019, before enabling VBS.

For more information about activating VBS on virtual machines on AMD platforms, see Broadcom knowledge base article 313862.

You can change the level of security of a virtual machine by enabling or disabling Microsoft virtualization-based security (VBS) on existing virtual machines for supported Windows guest operating systems.

Activating VBS on a virtual machine automatically activates the virtual hardware that Windows requires for the VBS feature. By enabling VBS, a variant of Hyper-V starts in the virtual machine and Windows starts running inside the Hyper-V root partition.

VBS is available on the latest Windows OS versions, for example, Windows 10 and Windows Server 2016. To use VBS on a virtual machine, the virtual machine compatibility must be ESX 6.7 and later.

In the VMware Host Client, you can activate VBS during a virtual machine creation. Alternatively, you can activate or deactivate VBS for an existing virtual machine.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. From the list of virtual machines, right-click a virtual machine and select **Edit Settings**.
3. When you create and configure a virtual machine with the **Virtualization Based Security** option, on the **VM Options** tab, activate or deactivate VBS for the virtual machine.
   - To activate VBS for the virtual machine, select the **Enable Virtualization Based Security** check box.
   - To deactivate VBS for the virtual machine, deselect the **Enable Virtualization Based Security** check box.

When you activate VBS, several options are automatically selected and become dimmed in the wizard.

4. Click **OK**.

Managing Storage in the VMware Host Client

When you connect to an ESX host by using the VMware Host Client, you can perform different storage management tasks on the ESX host, including configuring adapters, creating datastores, and viewing storage device information.

Datastores in the VMware Host Client

Datastores are logical containers, similar to file systems, that contain specific information of each storage device and provide a uniform model for storing virtual machine files.

You can use datastores to store ISO images, virtual machine templates, and floppy images.

Depending on the type of storage you use, datastores can be of the following types:

- Virtual Machine File System (VMFS)
- Network File System (NFS)

Block storage devices, iSCSI, and NAS devices support the hardware acceleration.

The hardware acceleration functionality enables the ESX host to integrate with compliant storage systems. The host can offload certain virtual machine and storage management operations to the storage systems. With the storage hardware assistance, your host performs these operations faster and consumes less CPU, memory, and storage fabric bandwidth.

For more information, see Broadcom knowledge base article 318941.

View Datastore Information in the VMware Host Client

Use the VMware Host Client to display the datastores available to the hosts and analyze their properties.

1. Click **Storage** in the VMware Host Client inventory and click the **Datastores** tab.
2. To view the details for a specific datastore, click the datastore from the list.

Create a VMFS Datastore in the VMware Host Client

VMFS datastores serve as repositories for virtual machines. You can set up VMFS datastores on any SCSI-based storage devices that your host detects, including iSCSI, and local storage devices.

Install and configure any adapters that your storage requires. Rescan the adapters to discover newly added storage devices.

To create datastores in the VMware Host Client, you can use the **New Datastore** wizard.

1. Click **Storage** in the VMware Host Client inventory.
2. Click **New Datastore.**

   The **New Datastore** wizard opens.
3. On the **Type** page, select **VMFS** as the datastore type and click **Next**.
4. On the **Name and device selection** page, enter the datastore name and select the placement location for the datastore.

   The system enforces an 80 characters limit for the datastore name.
5. Click **Next**.
6. On the **VMFS version** page, specify the datastore version.

Option	Description
VMFS6	Default format on all hosts that support VMFS6. The ESX hosts of version 6.0 or earlier cannot recognize the VMFS6 datastore.

Option	Description
VMFS5	VMFS5 datastore supports access by the ESX hosts of version 6.7 or earlier.

7. On the **Partition configuration** page, define configuration details for the datastore.

   Note:

   The required minimum size for a VMFS6 datastore is 2 GB.

   a) From the **Partition Configuration** drop-down menu, select **Use all available partitions** to dedicate the entire disk to a single VMFS datastore. If you select this option, all file systems and data currently stored on this device are destroyed.
   b) If the space allocated for the datastore is excessive for your purposes, adjust the capacity values in the **Datastore Size** field.
   c) For VMFS6, specify the block size and define space reclamation parameters.
8. On the **Ready To Complete** page, review the configuration details and click **Finish**.

Mount an NFS Datastore in the VMware Host Client

With the VMware Host Client, you can create a Network File System (NFS) datastore to store virtual disks and to use as a central repository for ISO images, virtual machines, and so on.

Because NFS requires network connectivity to access data on remote servers, before configuring NFS, you must first configure VMkernel networking.

An NFS client that is built into ESX uses the Network File System (NFS) protocol over TCP/IP to access a designated NFS volume that is located on a NAS server. vSphere supports versions 3 and 4.1 of the NFS protocol.

The ESX host can mount an NFS volume and use it for its storage needs.

Typically, the NFS volume or directory is created by a storage administrator and is exported from the NFS server. You do not need to format the NFS volume with a local file system, such as VMFS. Instead, you mount the volume directly on the ESX hosts and use it to store and boot virtual machines in the same way that you use the VMFS datastores.

In addition to storing virtual disks on NFS datastores, you can use NFS as a central repository for ISO images, virtual machine templates, and so on. If you use the datastore for the ISO images, you can connect the CD-ROM device of the virtual machine to an ISO file on the datastore. You can then install a guest operating system from the ISO file.

When you use NFS storage, follow the specific guidelines that are related to NFS server configuration, networking, NFS datastores, and so on.

Use the **New Datastore** wizard to mount a Network File System (NFS) datastore in the VMware Host Client.

1. Click **Storage** in the VMware Host Client inventory.
2. Click **New Datastore.**

   The **New Datastore** wizard opens.
3. On the **Type** page, select **NFS** as the datastore type and click **Next**.
4. On the **NFS version** page, select the datastore version and click **Next**.
5. On the **Name and configuration** page, specify the datastore name and configuration parameters, and click **Next**.

Option	Description
Name	The datastore name. The system enforces an 80 characters limit for the datastore name.
Folder	The mount point folder name.
Server	The server name or IP address. You can use IPv6 or IPv4 formats.

Option	Description
Access Mode	Select **Mount NFS as read-only** if the volume is exported as read-only by the NFS server.

6. On the **Ready To Complete** page, review the configuration settings for the NFS datastore and click **Finish**.

Unmount a Datastore in the VMware Host Client

When you unmount a datastore in the VMware Host Client, it remains intact, but you can no longer view it in the inventory of the host that you manage. The datastore continues to appear on other hosts on which it is mounted.

> **Note:**
> Make sure that the datastore is not used by vSphere HA heartbeating. vSphere HA heartbeating does not prevent you from unmounting the datastore. However, if the datastore is used for heartbeating, unmounting it might cause the host to fail and restart all active virtual machines.

Before unmounting a datastore, also make sure that the following prerequisites are met:

- No virtual machines reside on the datastore.
- Storage DRS does not manage the datastore.
- Storage I/O Control is deactivated for this datastore.

Do not perform any configuration operations that might result in I/O to the datastore while the unmounting is in progress.

1. Click **Storage** in the VMware Host Client inventory.
2. Right-click the datastore to unmount it from the list and click **Unmount Datastore**.
3. Confirm that you want to unmount the datastore.

Unmounting or Removing Datastore Fails

When you try to unmount or remove a datastore, the operation fails.

The operation to unmount or remove a datastore fails if the datastore has any opened files. For these user operations, the vSphere HA agent closes all the files that it has opened, for example, heartbeat files. If the agent is not reachable by vCenter or the agent cannot flush out pending I/Os to close the files, a `The HA agent on host '{hostName}' failed to quiesce file activity on datastore '{dsName}` fault is triggered.

If the datastore to be unmounted or removed is used for heartbeating, vCenter excludes it from heartbeating and chooses a new one. However, the agent does not receive the updated heartbeat datastores if it is not reachable, that is, if the host is isolated or in a network partition. In such cases, heartbeat files are not closed and the user operation fails. The operation can also fail if the datastore is not accessible because of storage failures such as all paths down.

> **Note:**
> When you remove a VMFS datastore, the datastore is removed from all the hosts in inventory. So if there are any hosts in a vSphere HA cluster that are unreachable or that cannot access the datastore, the operation fails.

Ensure that the datastore is accessible and the affected hosts are reachable.

Using the Datastore File Browser in the VMware Host Client

Use the datastore file browser to manage the contents of your datastore. You can perform a number of tasks, including uploading files to the datastore, downloading datastore files to your system, moving and copying datastore folders or files, and creating new datastore directories.

Upload Files to a Datastore in the VMware Host Client

Use the datastore file browser to upload files to datastores on your host.

Required privilege: **Datastore** > **Browse Datastore**

> **Note:**
> Virtual Volumes do not support uploading files directly to the virtual datastores. You must first create a folder on the virtual datastore, and then upload the files into the folder.

In addition to their traditional use as a storage for virtual machine files, datastores can serve to store data or files related to virtual machines. For example, you can upload ISO images of operating systems from a local computer to a datastore on the host. You then use these images to install guest operating systems on the new virtual machines.

1. Click **Storage** in the VMware Host Client inventory.
2. Click the datastore that you want to store the file on.
3. Click the **Files** tab.
4. Select the target folder and click **Upload**.
5. Locate the item that you want to upload from your local computer and click **Open**.
   The file uploads to the datastore that you selected.
6. Optional: Refresh the datastore file browser to see the uploaded file on the list.

Download Files from a Datastore to Your System in the VMware Host Client

Use the datastore file browser to download files from the datastores available on the host that you are managing to your local system.

Required privilege: **Datastore** > **Browse Datastore**

1. Click **Storage** in the VMware Host Client inventory.
2. Click the datastore that you want to download the file from.
3. Click the **Files** tab.
4. Select the folder that contains the file that you want to download.
   The available files in the folder appear.
5. Select the file that you want to download.
6. Click **Download**.
7. On the new window, log in with the system credentials.
   The file is downloaded to your system.

Delete Files from a Datastore in the VMware Host Client

You can permanently remove files from any datastore if you no longer need them.

Required privilege: **Datastore** > **Browse Datastore**

1. Click **Storage** in the VMware Host Client inventory.
2. Click the datastore that you want to download the file from.
3. Click the **Files** tab.
4. Select the folder that contains the file that you want to download.
5. Select the file that you want to remove from the datastore and click **Delete**.
6. Click **OK**.

Move Datastore Folders or Files in the VMware Host Client

Use the datastore file browser to move files or folders to a new location, either on the same datastore or on a different datastore.

Required privilege: **Datastore** > **Browse Datastore**

> **Note:**
> Virtual disk files are moved and copied without format conversion. If you move a virtual disk to a datastore on a type of host that is different from the type of the source host, you might need to convert the virtual disks before you can use them.

1. Click **Storage** in the VMware Host Client inventory.
2. Click a datastore.
3. Click the **Files** tab.
4. Select the file or folder that you want to move to another location and click **Move To**.
5. Select your target destination and click **OK**.

Copy Datastore Folders or Files in the VMware Host Client

Use the datastore file browser to copy folders or files to a new location, either on the same datastore or on a different datastore.

Required privilege: **Datastore** > **Browse Datastore**

> **Note:**
> Virtual disk files are moved and copied without format conversion. If you move a virtual disk to a datastore on a type of host that is different from the type of the source host, you might need to convert the virtual disks.

1. Click **Storage** in the VMware Host Client inventory.
2. Click a datastore.
3. Click the **Files** tab.
4. Select the file or folder that you want to move to another location and click **Copy To**.
5. Select your target destination and click **OK**.

Create a New Datastore Directory in the VMware Host Client

You can create new datastore directories if you want to store files in a particular location.

Required privilege: **Datastore** > **Browse Datastore**

1. Click **Storage** in the VMware Host Client inventory.
2. Click a datastore.
3. Click the **Files** tab.
4. Click **New Foldeer**.
5. Enter a name for the new directory.
6. Click **OK**.

Rename a Datastore in the VMware Host Client

You can change the display name of a datastore in the VMware Host Client.

> **Note:**
> If the host is managed by vCenter, you cannot rename the datastore from the VMware Host Client. You can only perform the task from the vCenter instance that manages the host.

1. Click **Storage** in the VMware Host Client inventory and click **Datastores**.
2. Right-click a datastore in the list and select **Rename** from the drop-down menu.
3. Enter a new name for the datastore and click **OK** to apply your changes.

Delete a VMFS Datastore in the VMware Host Client

You can delete any type of VMFS datastore, including copies that you have mounted without resignaturing. When you delete a datastore, the datastore and all the files associated to the datastore are removed from the host.

Remove all virtual machines from the datastore.

> **Note:**
> The datastore delete operation permanently deletes all files associated with virtual machines on the datastore. Although you can delete the datastore without unmounting, it is preferable that you unmount the datastore first.

1. Click **Storage** in the VMware Host Client inventory and click **Datastores**.
2. Right-click the datastore from the list and select **Delete Datastore** from the drop-down menu.
3. Click **Confirm** to delete the datastore.

Storage Thin Provisioning in the VMware Host Client

With ESX, you can use two models of thin provisioning, array-level and virtual-disk level thin provisioning.

Thin provisioning is a method that optimizes storage utilization by allocating storage space in a flexible on-demand manner. Thin provisioning contrasts with the traditional model, called thick provisioning. With thick provisioning, a large amount of storage space is provided in advance in anticipation of future storage needs. However, the space might remain unused causing underutilization of storage capacity.

The VMware thin provisioning features help you eliminate storage underutilization problems at the datastore and storage array level.

Create Thin Provisioned Virtual Disks in the VMware Host Client

To save storage space, you can create thin provisioned virtual disks. The thin provisioned virtual disk starts small and grows as more disk space is required. You can create thin disks only on the datastores that support disk-level thin provisioning.

The following procedure assumes that you are creating a new virtual machine. For more information, see Create a Virtual Machine in the VMware Host Client.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click **New Virtual Machine**.
   The **New Virtual Machine** wizard opens.

3. On the **Select a name and compatibility** page, enter a name for the virtual machine and select the virtual machine compatibility.
4. Click **Next**.
5. On the **Select a guest OS** page, select the guest operating system and version, and click **Next**.
6. On the **Select storage** page, select a datastore from the list where to store the virtual machine files and click **Next**.
7. On the **Customize hardware** page, expand **New Hard disk**.
8. From the **Disk Provisioning** drop-down menu, select the **Thin Provision** option and click **Next**.
9. On the **Ready to complete** page of the **New Virtual Machine** wizard, review the configuration settings for the virtual machine and click **Finish** to save your settings.

View Virtual Machine Storage Resources in the VMware Host Client

You can view how datastore storage space is allocated for your virtual machines in the VMware Host Client.

Resource Consumption shows how much datastore space is occupied by virtual machine files, including configuration files, log files, snapshots, virtual disks, and so on. When the virtual machine is running, the used storage space also includes swap files.

For virtual machines with thin disks, the actual storage usage value might be less than the size of the virtual disk.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Click a virtual machine and click the **Summary** tab.
3. Review the **Usage** card of the virtual machine summary page.

Determine the Disk Format of a Virtual Machine in VMware Host Client

You can determine whether your virtual disk is provisioned in thick or thin format.

1. Click **Virtual Machines** in the VMware Host Client inventory.
2. Right-click the virtual machine from the list and click **Edit Settings**.
3. On the **Virtual Hardware** tab, expand **Hard disk**.
   The **Type** text box shows the format of your virtual disk.

Managing Storage Adapters in the VMware Host Client

When you connect to a host or to vCenter by using the VMware Host Client, you can perform various tasks on your storage adapters, such as configuring various iSCSI components.

When you enable iSCSI on the host that you are managing in your VMware Host Client environment, you can configure and add new network port bindings, static and dynamic targets, you can manage CHAP authentication, and configure various advanced settings on your host storage.

View Storage Adapters in the VMware Host Client

View the storage adapters that your host uses and related information.

1. Click **Storage** in the VMware Host Client inventory and click **Storage Adapters**.

   All storage adapters available to the host appear.
2. To view details for a specific adapter, select the adapter from the list.

Configuring Software iSCSI Adapters in the VMware Host Client

With the software-based iSCSI implementation, you can use standard NICs to connect your host to a remote iSCSI target on the IP network. The software iSCSI adapter that is built into ESX communicates with the physical NICs through the network stack.

Note:
Before you can use the software iSCSI adapter, you must set up networking, activate the adapter, and configure parameters such as CHAP.

The iSCSI adapter configuration workflow includes the following procedures:

- Enabling iSCSI on your host. See Add an iSCSI Adapter for an ESX Host in the VMware Host Client.
- Adding a port binding. See Add Port Binding in the VMware Host Client.
- Removing port binding. See Remove Port Binding in the VMware Host Client.

Add an iSCSI Adapter for an ESX Host in the VMware Host Client

Add an iSCSI adapter for your host in your VMware Host Client environment to configure storage adapters parameters, such as CHAP authentication, network port bindings, static and dynamic targets, and various advanced settings.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Click **Add iSCSI Adapter**.
3. Optional: To edit an adapter, select the adapter from the list.
   a) Click the **Properties** tab.
   b) From the **Actions** drop-down menu, select **Edit**.
   c) Configure the parameters that you want to change, and click **OK**.

Add Port Binding in the VMware Host Client

Use the VMware Host Client to bind an iSCSI adapter with a VMkernel adapter on your host.

- Create a virtual VMkernel adapter for each physical network adapter on your host. If you use multiple VMkernel adapters, set up the correct network policy.
- Required privilege:**Host** > **Configuration** > **Storage Partition Configuration**

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI adapter from the list and click the **Network Port Binding** tab.
3. In the **Network port bindings** section, click **Add**.
4. Select a VMkernel adapter to bind with the iSCSI adapter.

   Note:
   Make sure that the network policy for the VMkernel adapter is compliant with the binding requirements.

   You can bind the software iSCSI adapter to one or more VMkernel adapters. For a dependent hardware iSCSI adapter, only one VMkernel adapter associated with the correct physical NIC is available.
5. Click **OK**.

Remove Port Binding in the VMware Host Client

Edit the iSCSI configuration on your host to remove a port binding.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI adapter from the list and click the **Network Port Binding** tab.
3. In the **Network port bindings** section, select a **Port Group** and click **Remove**.
4. Click **OK**.

Configure Dynamic Discovery in the VMware Host Client

You must set up target discovery addresses, so that the iSCSI adapter can determine which storage resource on the network is available for access. The ESX host supports dynamic and static discovery methods. With Dynamic Discovery,

each time the initiator contacts a particular iSCSI storage system, the initiator sends the SendTargets request to the iSCSI system. The iSCSI system responds by supplying a list of available targets to the initiator.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

Dynamic Discovery

Also known as SendTargets discovery. Each time the initiator contacts a specified iSCSI server, the initiator sends the SendTargets request to the server. The server responds by supplying a list of available targets to the initiator. The names and IP addresses of these targets appear on the **Static Discovery** tab. If you remove a static target added by dynamic discovery, the target might be returned to the list the next time a rescan happens, the iSCSI adapter is reset, or the host is rebooted.

> **Note:**
> With software and dependent hardware iSCSI, ESX filters target addresses based on the IP family of the iSCSI server address specified. If the address is IPv4, IPv6 addresses that might come in the SendTargets response from the iSCSI server are filtered out. When DNS names are used to specify an iSCSI server, or when the SendTargets response from the iSCSI server has DNS names, ESX relies on the IP family of the first resolved entry from DNS lookup.

When you set up Dynamic Discovery, you can only add a new iSCSI system. You cannot change the IP address, DNS name, or port number of an existing iSCSI system. To modify the parameters, delete the existing system and add a new one.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI adapter from the list and click the **Dynamic Discovery** tab.
3. Click **Add**.
4. Enter the IP address and the port number.

   By default, all discovery addresses inherit the authentication settings that you set up at the adapter level.
5. Click **OK**.
6. Optional: To remove an iSCSI server, select the server from the list, click **Remove**, and click **OK**.

Configure Static Discovery in the VMware Host Client

With iSCSI initiators, you can use static discovery to manually enter information for the targets.

Required privileges: **Host** > **Configuration** > **Storage Partition Configuration**

When you set up Static Discovery, you can only add new iSCSI targets. You cannot change the IP address, DNS name, iSCSI target name, or port number of an existing target. To make changes, remove the existing target and add a new one.

In addition to the dynamic discovery method, you can use static discovery and manually enter information for the targets. The iSCSI adapter uses a list of targets that you provide to contact and communicate with the iSCSI servers.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI adapter from the list and click the **Static Discovery** tab.
3. Click **Add**.
4. Enter the IP address and the port number.

   By default, all discovery addresses inherit the authentication settings that you set up at the adapter level.

5. In the **iSCSI Target Name** enter the iSCSI target information.
6. Click **OK**.
7. To remove an iSCSI server, select the server from the list, click **Remove**, and click **OK**.

Edit Advanced Settings for iSCSI in the VMware Host Client

The advanced iSCSI settings control such parameters as header and data digest, ARP redirection, delayed ACK, and so on. Generally, you do not need to change these settings because your host works with the assigned predefined values.

Required privilege: **Host** > **Configuration** > **Storage Partition Configuration**

> **CAUTION:**
> Do not change the advanced iSCSI settings unless you are working with the VMware support team or otherwise have thorough information about the values to provide for the settings modification.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI adapter from the list and click either **Dynamic Discovery** or **Static Discovery**.
3. From the list of available targets, select a target.
4. Click **Advanced** to display the entire list of settings.
5. Edit the parameters that you want to change and click **OK**.

Configure CHAP Authentication for an iSCSI Adapter in the VMware Host Client

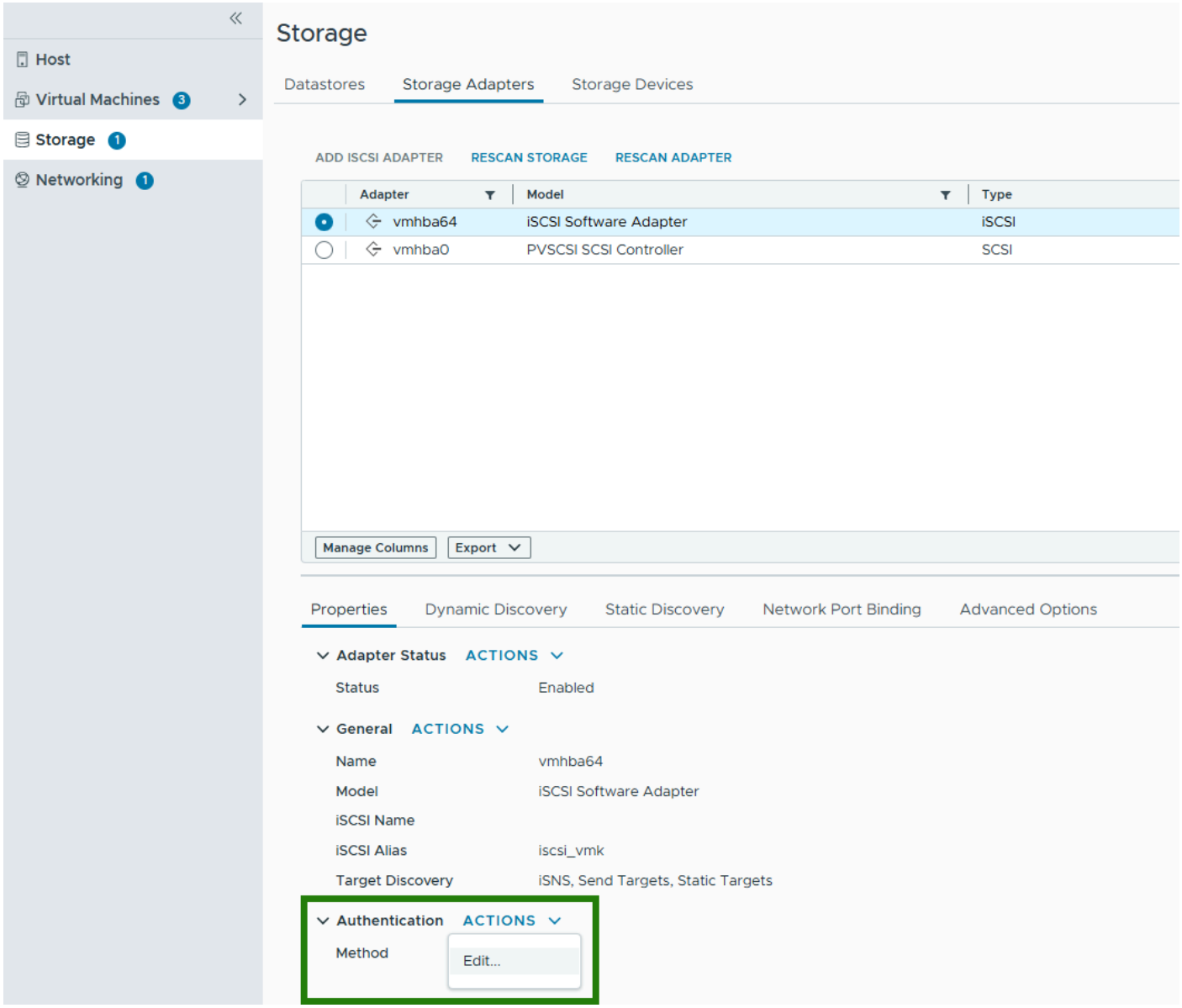
You can set up all targets to receive the same Challenge Handshake Authentication Protocol (CHAP) name and secret from the iSCSI initiator at the initiator level. By default, all discovery addresses or static targets inherit the CHAP parameters that you set up at the initiator level.

- Before you set up CHAP parameters for software or dependent hardware iSCSI, determine whether to configure one-way, also known as normal, or mutual CHAP. Independent hardware iSCSI adapters do not support mutual CHAP.
  - In one-way CHAP, the target authenticates the initiator.
  - In mutual CHAP, both the target and the initiator authenticate each other. Use different secrets for CHAP and mutual CHAP.

  When you configure CHAP parameters, verify that they match the parameters on the storage side.
- Required privileges: **Host** > **Configuration** > **Storage Partition Configuration**

The CHAP name cannot exceed 511 alphanumeric characters and the CHAP secret cannot exceed 255 alphanumeric characters. Some adapters, for example, the QLogic adapter, might have lower limits, 255 for the CHAP name and 100 for the CHAP secret.

1. Click **Storage** in the VMware Host Client inventory and click the **Storage Adapters** tab.
2. Select an iSCSI software adapter (`vmhba#`) from the list.
3. Click the **Properties** tab.
4. From the **Authentication** section, select the **Actions** drop-down menu, and select **Edit**.



5. The **Edit Authentication** window appears.
6. Select an authentication method.
   - None
   - Use undirectional CHAP if required by target
   - Use undirectional CHAP unless prohibited by target
   - Use undirectional CHAP
   - Use bidirectional CHAP.
7. Specify the outgoing CHAP name.

   Note:

   Make sure that the name you specify matches the name that is configured on the storage side.

   a) To set the CHAP name to the iSCSI adapter name, select the **Use initiator name** check box.
   b) To set the CHAP name to anything other than the iSCSI initiator name, deselect the **Use initiator name** check box and enter a name in the **Name** text box.
8. Enter an outgoing CHAP secret to be used as part of the authentication. Use the same secret that you enter on the storage side.
9. If you configure bidirectional CHAP, specify incoming CHAP credentials. Make sure to use different secrets for the outgoing and incoming CHAP.
10. Click **OK**.
11. Click **Rescan Adapter** to rescan the storage adapter.

If you change the authentication settings for an iSCSI adapter, you only use the updated credentials for new iSCSI sessions. Existing sessions persist until either the connection is lost due to some outside factor, such as force re-authentication, or you remove and add the adapter iSCSI targets.

Managing Storage Devices in the VMware Host Client

You can use the VMware Host Client to manage local and networked storage devices that the ESX host you are managing has access to.

View Storage Devices in the VMware Host Client

View all storage devices available to a host. If you use third-party multipathing plug-ins, the storage devices available through the plug-ins also appear on the list.

The Storage Devices view allows you to list the host storage devices, analyze their information, and modify properties.

1. Click **Storage** in the VMware Host Client inventory and click **Storage Devices**.

   All storage devices available to the host are listed.
2. To view details for a specific device, select the device from the list by clicking the double-angle icon

   ( )

   in front of the device name.

Clear a Storage Device Partition in the VMware Host Client

When you are logged in to an ESX host with the VMware Host Client, you can clear the partition table of a disk device that is accessible from the host.

Verify that the device is not in use by ESX as boot disk, VMFS datastore, or vSAN.

1. Click **Storage** in the VMware Host Client and click **Storage Devices**.
2. Select a device from the list.
3. Click the **Erase Partitions** button, and click **OK**.
   Clearing the partition table might cause data loss.

Performing Storage Refresh and Rescan Operations in the VMware Host Client

The refresh operation for datastores, storage devices, and storage adapters updates the lists and storage information that the VMware Host Client displays. It updates information such as the datastore capacity.

When you perform storage management tasks or change the SAN configuration, you must rescan your storage.

Perform Adapter Rescan in the VMware Host Client

When you change your SAN configuration and these changes are isolated to storage accessed through a specific adapter, perform rescan for only this adapter. When you rescan an adapter, you discover any new LUNs that are available on that adapter.

1. Click **Storage** in the VMware Host Client inventory and click **Storage Adapters**.
2. Click **Rescan Storage**.

Perform a Device Rescan in the VMware Host Client

When you rescan a device, you discover any new VMFS volumes that are available on the device.

1. Click **Storage** in the VMware Host Client inventory and click **Storage Devices**.
2. Click **Rescan Storage**.

Networking in the VMware Host Client

When you connect to an ESX host using the VMware Host Client, you can view and configure vSphere standard switches, port groups, physical NICs, VMkernel NICs, and TCP/IP stacks.

Manage Port Groups in the VMware Host Client

You can manage port group settings to configure traffic management, enhance networking security, and enhance performance.

By using the VMware Host Client, you can add and remove port groups. You can also examine port group information and edit port group settings, such as NIC teaming and traffic shaping.

View Port Group Information in the VMware Host Client

In the VMware Host Client, you can view information about port group configuration, network details, virtual switch topology, NIC teaming policy, offload policy, and security policy.

1. Click **Networking** in the VMware Host Client inventory and click **Port Groups**.
2. Click an item from the list of available port groups.
   Information about network details, virtual switch topology, NIC teaming policy, offload policy, and security policy is displayed.

Add a Virtual Machine Port Group for a new Standard Switch in the VMware Host Client

You can add a port group to a virtual switch in the VMware Host Client. Port groups provide networking for virtual machines.

1. Click **Networking** in the VMware Host Client inventory and click **Port Groups**.
2. Click **Add Networking**.
3. To create a new port group for virtual machine networking, select **Virtual Machine Port Group for a Standard Switch** as a connection type for which you want to use the new standard switch and click **Next**.
4. On the **Select target device** page, select **New standard switch**.
5. Enter a value for the MTU size and click **Next**.
6. On the **Connection settings** page, identify traffic through the ports of the group.
   a) Enter a **Network label** for the port group or accept the generated label.
   b) Set the VLAN ID to configure VLAN handling in the port group and click **Next**.

      The VLAN ID also reflects the VLAN tagging mode in the port group.

VLAN Tagging Mode	VLAN ID	Description
External Switch Tagging (EST)	0	The virtual switch does not pass traffic associated with a VLAN.
Virtual Switch Tagging (VST)	From 1 to 4094	The virtual switch tags traffic with the tag that you entered.
Virtual Guest Tagging (VGT)	4095	Virtual machines handle VLANs. The virtual switch permits traffic from any VLAN.

7. On the **Create a Standard Switch** page, assign free physical network adapters to the new switch.
   a) Select unclaimed adapters and move them to the **Active adapters**, **Standby adapters**, or **Unused adapters** of the switch.
8. Click **Next**.
9. On the **Ready to Complete** page, review the port group settings and click **Finish**.

Add a Virtual Machine Port Group for an Existing Standard Switch in the VMware Host Client

You can add a virtual machine port group for an existing standard switch in the VMware Host Client.

1. Click **Networking** in the VMware Host Client inventory and click **Port Groups**.
2. Click **Add Networking**.
3. To create port group for virtual machine networking, select **Virtual Machine Port Group for a Standard Switch** as a connection type for which you want to use the new standard switch and click **Next**.
4. On the **Select target device** page, select **Select an existing standards switch**.
5. Select a switch from the list and click **Next**.
6. On the **Connection settings** page, identify traffic through the ports of the group.
   a) Enter a **Network label** for the port group or accept the generated label.

      Note:

      Port group names must not contain a colon character (:).
   b) Set the VLAN ID to configure VLAN handling in the port group and click **Next**.

      The VLAN ID also reflects the VLAN tagging mode in the port group.

VLAN Tagging Mode	VLAN ID	Description
External Switch Tagging (EST)	0	The virtual switch does not pass traffic that is associated with a VLAN.
Virtual Switch Tagging (VST)	From 1 to 4094	The virtual switch tags traffic with the tag that you entered.

VLAN Tagging Mode	VLAN ID	Description
Virtual Guest Tagging (VGT)	4095	Virtual machines handle VLANs. The virtual switch permits traffic from any VLAN.

7. Click **Next**.
8. On the **Ready to Complete** page, review the port group settings and click **Finish**.

Edit Virtual Machine Port Group Settings in the VMware Host Client

To enhance networking security and improve networking performance in the VMware Host Client, you can edit various port group settings, such as the port group name, VLAN ID, and virtual switch. You can also configure security, NIC teaming, and traffic shaping components.

1. Click **Networking** in the VMware Host Client inventory and click **Port Groups**.
2. Click the vertical ellipsis icon
   (● ● ●)
   in front of the port group name and click **Edit settings**.
3. Optional: Click the **Properties** tab and enter a new network label.
4. Optional: Enter a new value for the VLAN ID.
   The VLAN ID reflects the VLAN tagging mode in the port group.

VLAN Tagging Mode	VLAN ID	Description
External Switch Tagging (EST)	0	The virtual switch does not pass traffic associated with a VLAN.
Virtual Switch Tagging (VST)	From 1 to 4094	The virtual switch tags traffic with the tag that you entered.
Virtual Guest Tagging (VGT)	4095	Virtual machines handle VLANs. The virtual switch permits traffic from any VLAN.

5. Optional: Click the **Security** tab and select whether to reject, accept, or inherit the Security policy exceptions from vSwitch.

Option	Description
Promiscuous Mode	• Select the **Override** check box. • **Reject**. Placing a guest adapter in promiscuous mode has no effect on which frames are received by the adapter. • **Accept**. Placing a guest adapter in promiscuous mode causes it to detect all frames passed on the vSphere distributed switch that are allowed under the VLAN policy for the port group that the adapter is connected to. • **Inherit from vSwitch**. Placing a guest adapter in promiscuous mode causes it to inherit the configuration from the associated virtual switch.
MAC Address Changes	• Select the **Override** check box. • **Reject**. If you set the **MAC Address Changes** to **Reject** and the guest operating system changes the MAC address of the adapter to anything other than what is in the `.vmx` configuration file, all inbound frames are dropped. If the guest operating system changes back the MAC address to match the MAC address in the `.vmx` configuration file, inbound frames are passed again. • **Accept**. Changing the MAC address from the guest operating system has the intended effect: frames to the new MAC address are received.

Option	Description
	• **Inherit from vSwitch**. If you set **MAC Address Changes** to **Inherit from vSwitch**, the MAC address changes to one of the associated virtual switches.
Forged Transmits	• Select the **Override** check box. • **Reject**. Any outbound frame with a source MAC address that is different from the one set on the adapter are dropped. • **Accept**. No filtering is performed and all outbound frames are passed. • **Inherit from vSwitch**. The outbound frame configuration is inherited from the associated virtual switch.

6. Optional: To configure traffic shaping, click **Traffic shaping**.
7. Optional: Select the **Override** check box and specify the following parameters.

Option	Description
Average Bandwidth	Establishes the number of bits per second to limit across a port, averaged over time—the allowed average load.
Peak Bandwidth	The maximum number of bits per second to limit across a port when it is sending/receiving a burst of traffic. This is the maximum bandwidth used by a port whenever it is using its burst bonus.
Burst Size	The maximum number of bytes to limit in a burst. If this parameter is set, a port might gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than specified by **Average Bandwidth**, it might be allowed to temporarily transmit data at a higher speed if a burst bonus is available. This parameter represents the maximum number of bytes that might be accumulated in the burst bonus and so transferred at a higher speed.

   Traffic shaping policy is applied to the traffic of each virtual network adapter attached to the virtual switch.
8. Click **Teaming and failover** to override the teaming and failover settings inherited from the standard switch.
9. Click **OK** to apply your changes.

Remove a Virtual Switch Port Group in the VMware Host Client

You can remove port groups from virtual switches in case you no longer need the associated labeled networks.

Verify that there are no VMkernel NICs and no powered-on virtual machines connected to the port group that you want to remove.

1. Click **Networking** in the VMware Host Client inventory and click **Port Groups**.
2. Click the vertical ellipsis icon
   ●
   ●
   (●)
   in front of the port group name and click **Remove**.
3. To remove the port group, click **Yes**.

Manage Virtual Switches in the VMware Host Client

In the VMware Host Client, you can configure various virtual switch settings, such as link discovery, NIC teaming, and traffic shaping.

View Virtual Switch Information in the VMware Host Client

In the VMware Host Client, you can view information about virtual switches, such as configuration, network details, virtual switch topology, and so on.

1. Click **Networking** in the VMware Host Client inventory and click the**Virtual Switches** tab.
2. Click on a switch from the list of available virtual switches.

   Information about virtual switch configuration, network details, and virtual switch topology is displayed.

Edit Virtual Switch Settings in the VMware Host Client

In the VMware Host Client, you can edit virtual switch settings, such as the virtual switch uplinks.

1. Click **Networking** in the VMware Host Client inventory and click **Virtual Switches**.
2. Select a virtual switch and click **Edit**.
3. Click the **Properties** tab, enter a new MTU value, and click **Next**.
4. Optional: Click the **Security** tab and select whether to reject, accept, or inherit the Security policy exceptions from vSwitch.

Option	Description
Promiscuous Mode	• **Reject**. Placing a guest adapter in promiscuous mode has no effect on which frames are received by the adapter. • **Accept**. Placing a guest adapter in promiscuous mode causes it to detect all frames passed on the vSphere distributed switch that are allowed under the VLAN policy for the port group that the adapter is connected to. • **Inherit from vSwitch**. Placing a guest adapter in promiscuous mode causes it to inherit the configuration from the associated virtual switch.
MAC Address Changes	• **Reject**. If you set the **MAC Address Changes** to **Reject** and the guest operating system changes the MAC address of the adapter to anything other than what is in the `.vmx` configuration file, all inbound frames are dropped. If the guest operating system changes back the MAC address to match the MAC address in the `.vmx` configuration file, inbound frames are passed again. • **Accept**. Changing the MAC address from the guest operating system has the intended effect: frames to the new MAC address are received. • **Inherit from vSwitch**. If you set **MAC Address Changes** to **Inherit from vSwitch**, the MAC address changes to one of the associated virtual switches.
Forged Transmits	• **Reject**. Any outbound frame with a source MAC address that is different from the one set on the adapter are dropped. • **Accept**. No filtering is performed and all outbound frames are passed. • **Inherit from vSwitch**. The outbound frame configuration is inherited from the associated virtual switch.

5. Optional: To configure traffic shaping, click **Traffic shaping**, and specify the following parameters.

Option	Description
Average Bandwidth	Establishes the number of bits per second to limit across a port, averaged over time—the allowed average load.
Peak Bandwidth	The maximum number of bits per second to limit across a port when it is sending/receiving a burst of traffic. This is the maximum bandwidth used by a port whenever it is using its burst bonus.

Option	Description
Burst Size	The maximum number of bytes to limit in a burst. If this parameter is set, a port might gain a burst bonus when it does not use all its allocated bandwidth. Whenever the port needs more bandwidth than specified by **Average Bandwidth**, it might be allowed to temporarily transmit data at a higher speed if a burst bonus is available. This parameter represents the maximum number of bytes that might be accumulated in the burst bonus and so transferred at a higher speed.

Traffic shaping policy is applied to the traffic of each virtual network adapter attached to the virtual switch.

6. Click **Teaming and failover** to override the teaming and failover settings inherited from the standard switch.
7. Click **OK** to apply your changes.

Assign Physical Adapters to a Virtual Switch

You can assign a physical adapter to a standard switch to provide connectivity to virtual machines and VMkernel adapters on the host.

1. Click **Networking** in the VMware Host Client inventory, and click the **Virtual Switches** tab.
2. Select a virtual switch that you want to add a physical adapter to.
3. Click **Manage Physical Adapters**.
4. To add one or more available physical network adapters to the switch, select unclaimed adapters and move them to the **Active adapters**, **Standby adapters**, or **Unused adapters** of the switch.

   Use the up and down arrows to change the position of an adapter in the failover groups.
5. Click **OK**.

Remove a Standard Virtual Switch in the VMware Host Client

You can remove the virtual standard switch if you no longer need it.

1. Click **Networking** in the VMware Host Client inventory and click the **Virtual Switches** tab.
2. Click **Remove**.
3. Click **Yes**.

Manage Physical Network Adapters in the VMware Host Client

Assign a physical adapter to a standard switch to provide connectivity to virtual machines.

View Physical Network Adapter Information in the VMware Host Client

In the VMware Host Client, you can view various information about physical network adapter (NIC) configuration and settings.

1. Click **Networking** in the VMware Host Client inventory and click the **Physical Adapters** tab.
2. To view details for a specific physical network adapter, select the adapter from the list by clicking the double-angle icon

   ( )

   in front of the adapter name.

Edit Physical NICs in the VMware Host Client

You can edit physical NIC speed by using the VMware Host Client.

1. Click **Networks** in the VMware Host Client inventory and click the **Physical Adapters** tab.
2. Click the vertical ellipsis icon
   (⋮)
   in front of the device name and click **Edit**.
3. From the **Configured speed, Duplex** drop-down menu, select a speed and click **OK**.

Manage VMkernel Network Adapters in the VMware Host Client

In the VMware Host Client, you can add and remove VMkernel network adapters (NICs), as well as view and modify the VMkernel NIC settings.

View VMkernel Network Adapter Information in the VMware Host Client

In the VMware Host Client, you can view information about VMkernel network adapters (NICs), such as TCP/IP configuration, network details, virtual switch topology, and so on.

1. Click **Networking** in the VMware Host Client inventory and click **VMkernel Adapters**.
2. To view details for a specific VMkernel NIC, select the NIC from the list by clicking the double-angle icon
   (»)
   in front of the device name.

Add a VMkernel Network Adapter for a New Standard Switch in the VMware Host Client

You can add a VMkernel network adapter (NIC) on a vSphere Standard Switch to provide network connectivity for hosts. The VMkernel NIC also handles the system traffic for vSphere vMotion, IP storage, Fault Tolerance, logging, vSAN, and so on.

1. Click **Networking** in the VMware Host Client inventory and click **VMKernel Adapters**.
2. Click **Add Networking**.
3. To create a VMKernel NIC, select **VMKernel Network Adapter** as a connection type for which you want to use the new standard switch and click **Next**.
4. On the **Select target device** page, select **New standard switch**.
5. Enter a value for the MTU size and click **Next**.
6. On the **Create a Standard Switch** page, assign physical adapters to the switch. To add one or more available physical network adapters to the switch, select unclaimed adapters and move them to the **Active adapters**, **Standby adapters**, or **Unused adapters** of the switch.

   Use the up and down arrows to change the position of an adapter in the failover groups.
7. Click **Next**.
8. On the **Port properties** page, specify the VMKernel settings and click **Next**.

Option	Description
Network label	Adding a VMkernel NIC also adds a port group. Specify a name for that port group.
VLAN ID	Enter a VLAN ID to determine the VLAN for the network traffic of the VMkernel adapter to use.
IP settings	Select IPv4, IPv6, or both.

Option	Description
	Note: The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	To get a better network connection, determine the MTU size.
TCP/IP stack	Select a TCP/IP stack. After you select a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select vMotion or the Provisioning TCP/IP stack, you can use only this stack to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you use the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated, and you cannot perform some operations. Such operations include traffic Provisioning, such as virtual machine cold migration, cloning, and snapshot migration.

9. Optional: Select the services to enable for the default TCP/IP stack on the host.

   vMotion enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. You cannot use vMotion to perform migrations to selected hosts if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or if no adapters use the vMotion TCP/IP stack.

10. On the **IPv4 settings** page, specify the VMkernel IPv4 settings and click **Next**.

Option	Description
Obtain IPv4 settings automatically	IP settings are obtained automatically. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack.

11. On the **Ready To Complete** page, review the port group settings and click **Finish**.

Add a VMkernel Network Adapter for an Existing Standard Switch in the VMware Host Client

You can add a VMkernel Network Adapter for an existing standard switch.

1. Click **Networking** in the VMware Host Client inventory and click **VMKernel Adapters**.
2. Click **Add Networking** at the top-right corner of the **Networks** pane.
3. To create a VMKernel NIC, select **VMKernel Network Adapter** as a connection type for which you want to use the new standard switch and click **Next**.
4. On the **Select target device** page, select **Select an exsisting standard switch**.
5. From the list of standard switches, select a switch and click **Next**.
6. On the **Port properties** page, specify the VMKernel settings and click **Next**.

Option	Description
Network label	Adding a VMkernel NIC also adds a port group. Specify a name for that port group.
VLAN ID	Enter a VLAN ID to determine the VLAN for the network traffic of the VMkernel adapter to use.
IP Version	Select IPv4, IPv6, or both.

Option	Description
	Note: The IPv6 option does not appear on hosts that do not have IPv6 enabled.
MTU	To get a better network connection, determine the MTU size.
TCP/IP stack	Select a TCP/IP stack. After you select a TCP/IP stack for the VMkernel adapter, you cannot change it later. If you select vMotion or the Provisioning TCP/IP stack, you can use only this stack to handle vMotion or Provisioning traffic on the host. All VMkernel adapters for vMotion on the default TCP/IP stack are deactivated for future vMotion sessions. If you use the Provisioning TCP/IP stack, VMkernel adapters on the default TCP/IP stack are deactivated, and you cannot perform some operations. Such operations include traffic Provisioning, such as virtual machine cold migration, cloning, and snapshot migration.

7. Optional: Select the services to enable for the default TCP/IP stack on the host.

   vMotion enables the VMkernel adapter to advertise itself to another host as the network connection where vMotion traffic is sent. You cannot use vMotion to perform migrations to selected hosts if the vMotion service is not enabled for any VMkernel adapter on the default TCP/IP stack, or if no adapters use the vMotion TCP/IP stack.

8. On the **IPv4 settings** page, specify the VMkernel IPv4 settings and click **Next**.

Option	Description
Obtain IPv4 settings automatically	IP settings are obtained automatically. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack.

9. On the **Ready To Complete** page, review the port group settings and click **Finish**.

Edit VMkernel Network Adapter Settings in the VMware Host Client

You might have to change the supported traffic type for a VMkernel network adapter, or the way IPv4 or IPv6 addresses are obtained.

1. Click **Networks** in the VMware Host Client inventory and click the **VMKernel Adapters** tab.
2. Click the vertical ellipsis icon
   (● ● ●)
   in front of the device name and click **Edit**.
3. In the **MTU** field, enter a new MTU size and select the services to activate for the default TCP/IP stack on the host.
4. On the **IPv4 settings** page, select an option to obtain the IPv4 address.

Option	Description
Option	Description
Obtain IPv4 settings automatically	Use DHCP to obtain IP settings. A DHCP server must be present on the network.
Use static IPv4 settings	Enter the IPv4 IP address and subnet mask for the VMkernel adapter. The VMkernel Default Gateway and DNS server addresses for IPv4 are obtained from the selected TCP/IP stack. Select the **Override default gateway for this adapter**

Option	Description
	check box and enter a gateway address, if you want to specify a different gateway for the VMkernel adapter. **Note:** Selected netstack should have the default gateway explicitly on each host.

5. On the **IPv6 settings** page, select the options to obtain the IPv6 address.

Option	Description
Obtain IPv6 addresses automatically through DHCP	Use DHCP to obtain IPv6 addresses. A DHCPv6 server must be present on the network.
Obtain IPv6 addresses automatically through Router Advertisement	Use a router advertisement to obtain IPv6 addresses. In 6.5 and later, router advertisement is enabled by default and supports the M and O flags in accordance with RFC 4861.
Static IPv6 addresses	• Click **Add** to add an IPv6 address. • Enter the IPv6 address and subnet prefix length, and click **OK**. • To change the VMkernel default gateway, click **Override default gateway for this adapter**. The VMkernel Default Gateway address for IPv6 is obtained from the selected TCP/IP stack.

6. Review the settings and click **OK**.

Remove a VMkernel Network Adapter in the VMware Host Client

In the VMware Host Client, you can remove a VMkernel network adapter if you no longer need it.

1. Click **Networking** in the VMware Host Client inventory and click the **VMkernel Adapters** tab.
2. Click the vertical ellipsis icon
   (● ● ●)
   in front of the device name and click **Remove**.
3. Click **OK** to remove the network adapter.

View TCP/IP Stack Configuration on a Host in the VMware Host Client

You can view the DNS and routing configuration of a TCP/IP stack on a host. You can also view the IPv4 and IPv6 routing tables, the congestion control algorithm, and the maximum number of allowed connections.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **TCP/IP Configuration**.
4. Click the double-angle icon
   (»)
   in front of the TCP/IP stack.

   The configuration setting of the stack you selected appear.

Change the Configuration of a TCP/IP Stack on a Host in the VMware Host Client

You can change the DNS and default gateway configuration of a TCP/IP stack on a host. You can also change the congestion control algorithm, the maximum number of connections, and the name of custom TCP/IP stacks.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **TCP/IP Configuration**.
4. Click the vertical ellipsis icon
   (● ● ●)
   in front of the **Default** and click **Edit**.

   The **Edit TCP/IP Stack Configuration** window appears.
5. Make the appropriate changes.

Page	Option
DNS configuration	Select a method of obtaining the DNS server. • Select **Obtain settings automatically from a VMkernel network adapter** and select a network adapter from the **VMKernel network adapter** drop-down menu. • Select **Enter settings manually** and edit the DNS configuration settings. – Edit the Host name. – Edit the Domain name. – Enter a preferred DNS server IP address. – Enter an alternate DNS server IP address. – Use the **Search domains** text box to specify DNS suffixes to use in DNS search when resolving unqualified domain names.
Routing	Edit the VMkernel and IPv6 gateway information. **Note:** Removing the default gateway might cause you to lose your connection to the host.
Name	Change the name of a custom TCP/IP stack.
Advanced	Edit the congestion control algorithm and the maximum number of connections.

6. Click **OK**.

Configure ESX Firewall in the VMware Host Client

ESX includes a firewall that is enabled by default. During installation, the ESX firewall is configured to block incoming and outgoing traffic, except traffic for services that are enabled in the host security profile.

As you open ports on the firewall, consider that unrestricted access to services running on an ESX host may expose a host to outside attacks and unauthorized access. Reduce the risk by configuring the ESX firewall to allow access only from authorized networks.

Note:
The firewall also allows Internet Control Message Protocol, or ICMP, pings, and communication with DHCP and DNS (UDP only) clients.

Manage ESX Firewall Settings by Using the VMware Host Client

When you are logged in to an ESX host with the VMware Host Client, you can configure incoming and outgoing firewall connections for a service or a management agent.

> **Note:**
> If different services have overlapping port rules, enabling one service might implicitly enable other services. You can specify which IP addresses are allowed to access each service on the host to avoid this problem.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Firewall**.

   The VMware Host Client displays a list of active incoming and outgoing connections with the corresponding firewall ports.
4. Select a firewall rule.
   a) To edit a firewall rule, click **Edit firewall rule**, make the necessary changes, and click **Save**.
   b) To activate or deactivate a firewall rule, click **Enable Firewall Rule** or **Disable Firewall Rule**.

Add Allowed IP Addresses for an ESX Host by Using the VMware Host Client

By default, the firewall for each service allows access to all IP addresses. To restrict traffic, configure each service to allow traffic only from your management subnet. You can also deselect some services if your environment does not use them.

1. Click **Host** the VMware Host Client inventory.
2. Click the **Configure** tab.
3. Under **System**, click **Firewall**.
4. Select a service from the list and click **Edit Firewall Rule**.
5. Click **Only allow connections from the following networks** and enter the IP addresses of networks that you want to connect to the host.
   Separate IP addresses with commas. You can use the following address formats:
   - 192.168.0.0/24
   - 192.168.1.2, 2001::1/64
   - fd3e:29a6:0a81:e478::/64
6. Click **Save**.

Setup for Windows Server Failover Clustering on VMware vSphere

Setup for Windows Server Failover Clustering describes the supported configurations for a WSFC with shared disk resources you can implement using virtual machines with Failover Clustering for Windows Server 2012 and above releases. You get step-by-step instructions for each configuration and a checklist of clustering requirements and recommendations.

Unless stated otherwise, the term Windows Server Failover Clustering (WSFC) applies to Failover Clustering with Windows Server 2012 and above releases.

Setup for Windows Server Failover Clustering covers ESX and VMware® vCenter®.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

This information is for system administrators who are familiar with VMware technology and Failover Clustering for Windows.

> **Note:**
> This is not a guide to using Failover Clustering for Windows. Use your Microsoft documentation for information about installation and configuration of Microsoft Cluster Service or Failover Clustering.
>
> **Note:**
>
> All instances of "WSFC" in this document and elsewhere is intended to refer to all Windows OS and Microsoft application clustering configurations which utilize one or more shared disks among the clustered nodes, namely:
>
> - The Microsoft Cluster Service (MSCS)
> - The Windows Server Failover Clustering (WSFC)
> - Microsoft SQL Server Always On Failover Cluster Instances (FCI)
>
> **Note:**
> Other WSFC-based solutions not accessing shared storage such as SQL Server Always On Availability Groups or Exchange Database Availability Group, require no special storage configurations on the vSphere side using VMFS or NFS. This guide should not be used for such configurations.

Getting Started with WSFC on VMware®vSphere®

VMware® vSphere® supports Windows Server Failover Clustering (WSFC) across virtual machines. Clustering virtual machines can reduce the hardware costs of traditional high-availability Windows clusters.

> **Note:**
> vSphere High Availability (vSphere HA) supports a Windows clustering solution. *vSphere Availability* describes vSphere HA functionality.

Clustering Configuration on VMware® vSphere®

Several applications use clustering, including stateless applications such as Web servers, and applications with built-in recovery features, such as database servers. You can set up WSFC clusters in several configurations, depending on your environment.

A typical clustering setup includes disks resources that are shared between nodes. A shared disk is required as a quorum disk. In a cluster of virtual machines across physical hosts, shared disks can be Raw Device Mappings (RDMs), Virtual Volumes (vVol) or VMFS VMDKs.

Note:
VMDKs in this guide refers to Shared disk files from VMFS datastore shared in physical compatibility mode (Attached to SCSI/NVMe controller with bus sharing mode set to **Physical**) for cluster across physical machines. This is also known as Clustered VMDKs.

If you are using RDMs or vVol as shared disks, they can be accessed using Fibre Channel (FC) SAN, FCoE, or iSCSI. In vSphere 8.0 Update 3 or later, vVol also supports shared disk from NVMe storage arrays (FC, TCP).

If you are using VMFS VMDKs as shared disks, only FC SAN connected storage devices are supported. FC SAN can be from SCSI and NVMe storage arrays. VMFS VMDKs are not supported by other SANs such as iSCSI or FCoE.

In vSphere 8.0 Update 3 or later, VMFS VMDKs also support storage from NVMe TCP arrays.

In ESX, clustered VMDKs on VMFS datastores are supported in configurations when the VMs hosting the nodes of a cluster are located on different ESX hosts, which are known as cluster-across-boxes (CAB) configuration.

Hosting Virtual Nodes of a WSFC on a Single ESX Host

A cluster of WSFC virtual machines on a single ESX host (also known as a cluster in a box or CIB) consists of clustered virtual machines on the same ESX host. The virtual machines are connected to the same storage, either local or remote. This configuration protects against failures at the operating system and application level, but it does not protect against hardware failures.

Note:

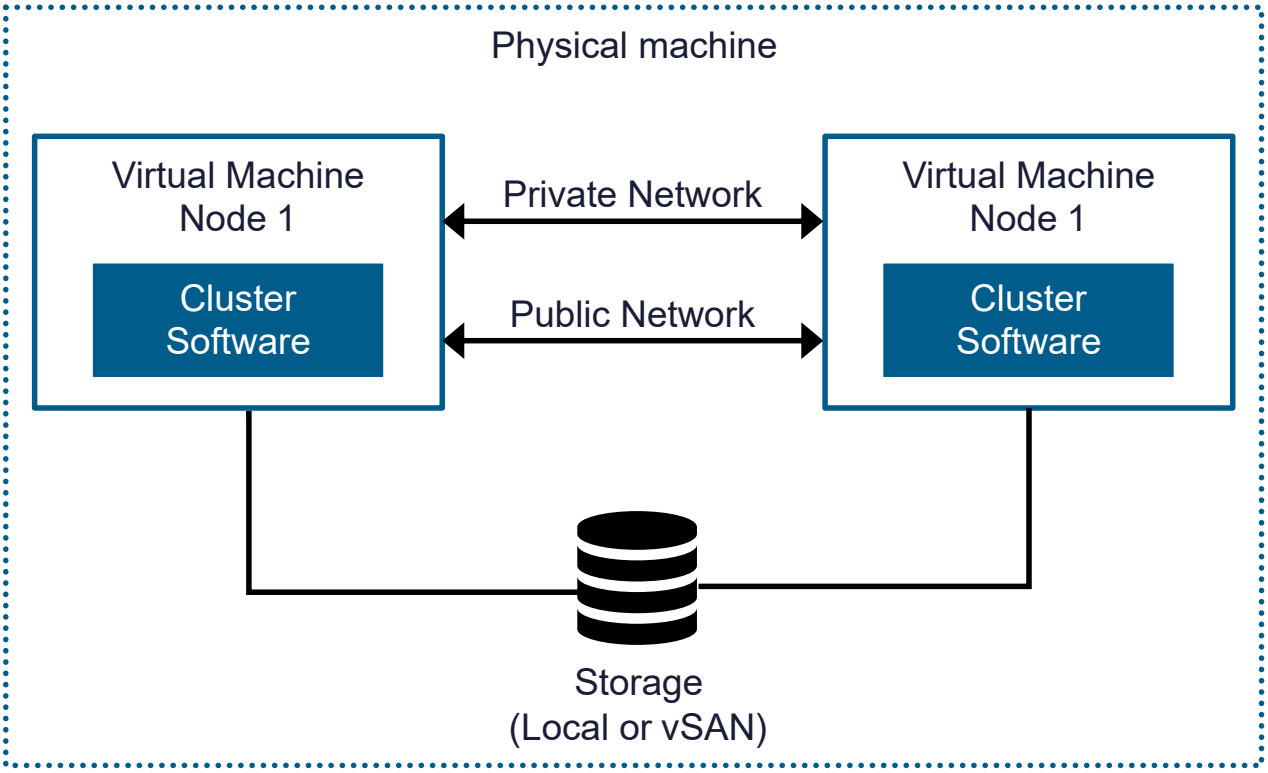
Do not host virtual nodes of a WSFC on the same ESX host for a production deployment.

Note:
Windows Server 2012 and later releases support up to five nodes (virtual machines).

The following figure shows a cluster in a box setup.

- Two virtual machines on the same ESX host run WSFC.
- The virtual machines share a private network connection for the private heartbeat and a public network connection.
- Each virtual machine is connected to shared storage, which can be local or on a vSAN.
- Each shared disk should be attached to SCSI controller with bus sharing mode set to **Virtual**.

Figure 116: Virtual Machines Clustered on a Single ESX Host



Clustering Virtual Machines Across Physical Hosts

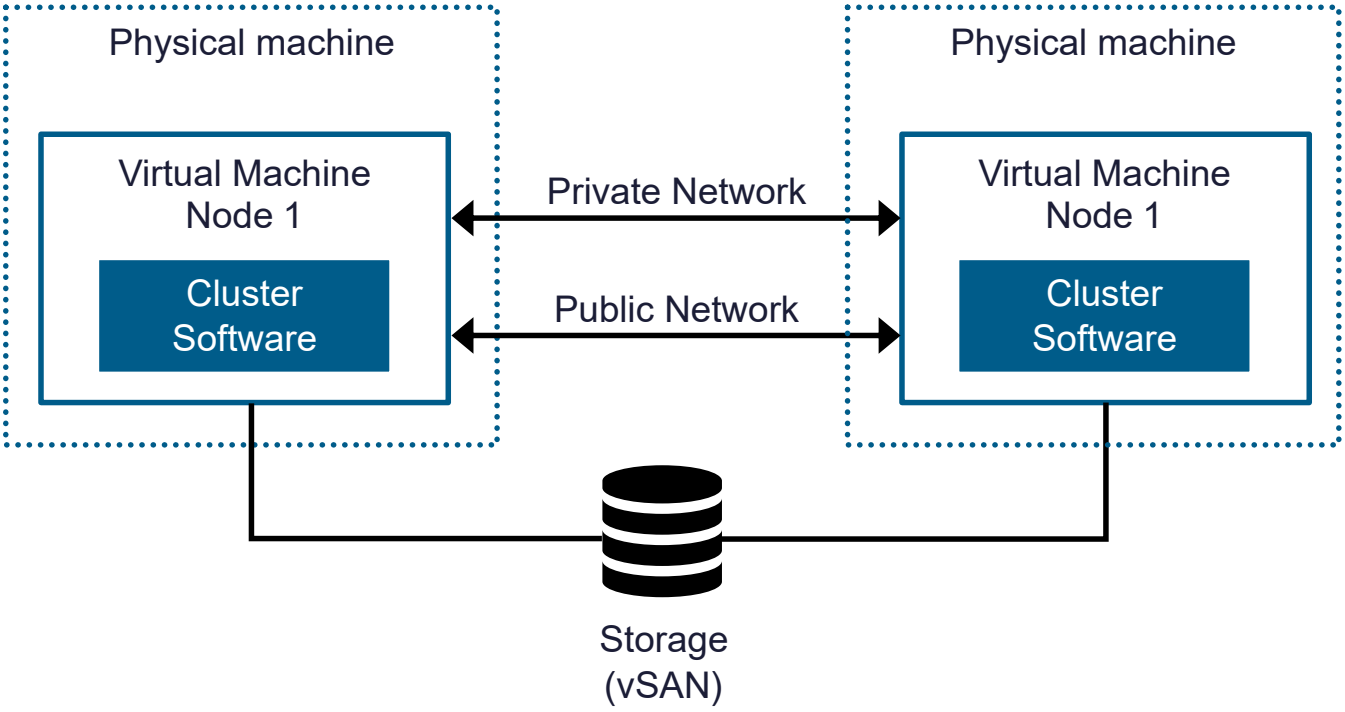
A cluster of virtual machines across physical ESX hosts (also known as a cluster across boxes or CAB) protects against software failures and hardware failures on the physical ESX hosts by placing the cluster nodes on separate ESX hosts. This configuration requires shared storage for the clustered disk resources.

The following figure shows a cluster-across-boxes setup.

- Two virtual machines on two different ESX hosts run WSFC.
- The virtual machines share a private network connection for the private heartbeat and a public network connection.
- Each virtual machine is connected to shared storage.
- An ESX host supports up to 16 WSFC clusters (multi-cluster) running on the same ESX host.

> **Note:**
> In a cluster of virtual machines across physical hosts, shared disks can be physical mode or vVol with vSphere 6.7 or later. You can use VMFS VMDKs from SCSI SAN with vSphere 7.0 or later and from NVMe SAN with vSphere 8.0 or later. If you are using RDMs or vVol as shared disks, they can be on Fibre Channel (FC) SAN, FCoE, or iSCSI. If you use VMFS VMDKs as shared disks, they can be on FC SAN. vSphere 8.0 Update 3 or later vVol also supports a shared disk from NVMe storage arrays (FC, TCP). vSphere 8.0 Update 3 or later VMFS VMDKs also supports storage from NVMe TCP array.

Figure 117: Virtual Machines Clustered Across ESX Hosts



Note:
Windows Server 2012 and above systems support up to five nodes (virtual machines). For more information about the supported guest operating systems, see Table 512: Other Clustering Requirements and Recommendations.

Note:
Private and Public network interconnect can share a single virtual NIC in a virtual machine.

You can expand the cluster-across-boxes model and place multiple virtual machines on multiple ESX hosts.

Clustering Physical Machines with Virtual Machines

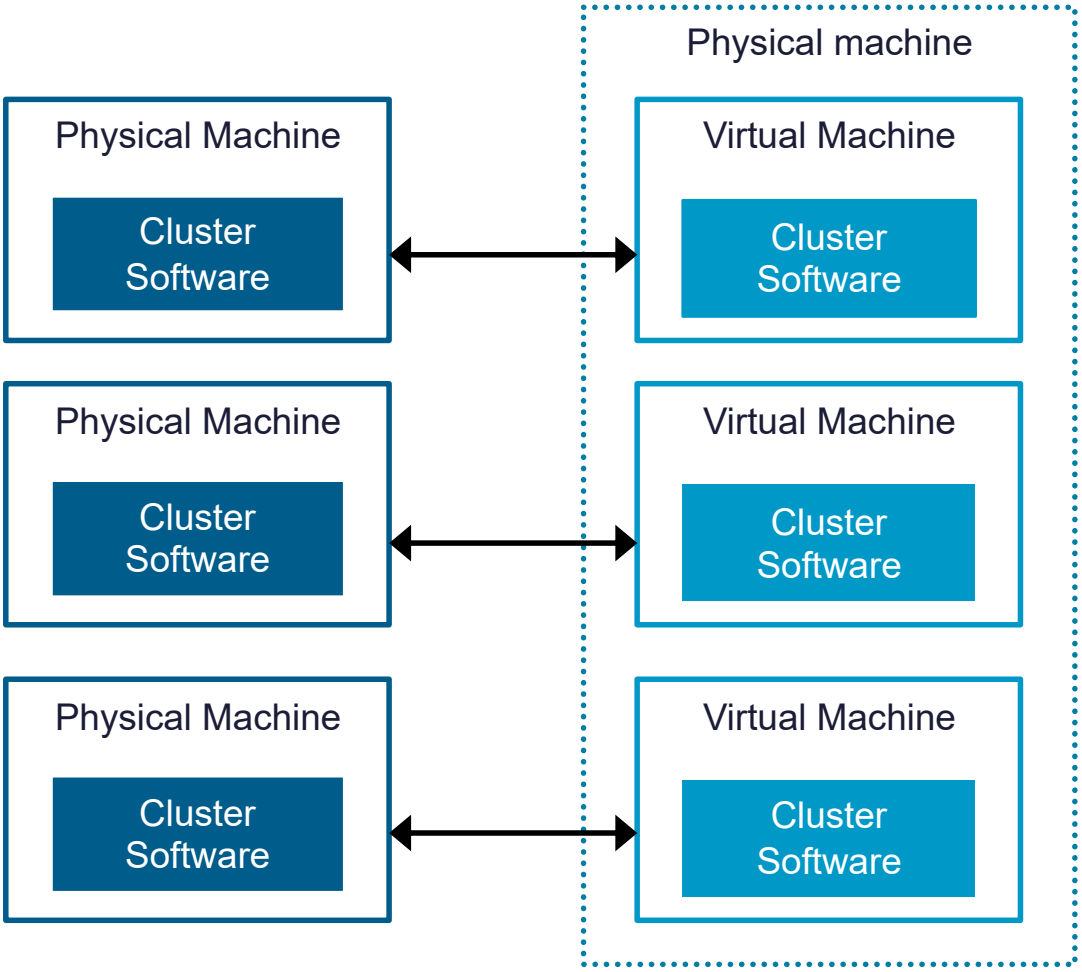
For a simple WSFC clustering solution with low hardware requirements with RDMs, you might choose to have one standby host.

Set up your system to have a virtual machine corresponding to each physical machine on the standby host, and create clusters, one each for each physical machine and its corresponding virtual machine. If there is hardware failure in one of the physical machines, the virtual machine on the standby host can take over for that physical host.

The following figure shows a standby host using three virtual machines on a single ESX host. Each virtual machine is running WSFC.

Note:
This solution is not applicable for vVol and clustered VMDKs.

Figure 118: Clustering Physical and Virtual Machines



Hardware and Software Requirements for WSFC on VMware vSphere

All vSphere WSFC configurations require certain hardware and software components.

The following table lists hardware and software requirements that apply to all vSphere WSFC configurations.

Table 509: Clustering Requirements

Component	Requirement
Virtual SCSI adapter	LSI Logic SAS for Windows Server 2012 or later. VMware Paravirtual for Windows Server 2012, 2012 R2. VMware Paravirtual for Windows Server 2016 or later. (recommended) NVMe Controller for Windows Server 2022 with OS Build 20348.1547 and later, virtual machine hardware version must be 21 or later.
Operating system	Windows Server 2012 or later releases. For more information about the supported guest operating systems, see Table 512: Other Clustering Requirements and Recommendations.
Virtual NIC	VMXNET3 is required. You must activate in-guest Windows Receive Side Scaling (RSS) on the virtual NIC. **Note:** Drivers, improvements and updates to the VMXNET3 virtual NIC are delivered through VMware Tools. Upgrade to the latest version of VMware Tools on any VM using the VMXNET3 virtual NIC.

Component	Requirement
I/O timeout	Set to 60 seconds or more. Modify `HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\Disk\TimeOutValue`. The system might reset this I/O timeout value if you re-create a cluster. You must reset the value in that case.
Disk format	pRDM (physical compatibility mode) vVol disks can be thin or thick format VMFS VMDKs with vSphere 7.0 `eagerzeroedthick` format
Disk and networking setup	Add networking before the disks.
Number of nodes	Windows Server 2012 or later: up to five-node clustering For more information about supported guest operating systems, see Table 512: Other Clustering Requirements and Recommendations.
NTP server	Synchronize Windows Active Directory domain controllers and WSFC nodes with a common NTP server, and deactivate host-based time synchronization. For more information, see Broadcom knowledge base article 326306.

The following table lists the requirements that apply to Clustered VMDK as a disk resource for a WSFC.

Table 510: Additional Requirements for Clustered VMDK WSFC

Component	Requirement
Windows Cluster Parameters	Set the Windows Cluster Parameter `QuorumArbitrationTimeMax` to 60.
Physical disk /Storage Array LUN	• The physical disk that virtual disks are stored on must support ATS SCSI • The physical disk must also support SCSI-3 Persistent Reservations, specifically Write Exclusive All Registrants (WEAR) type reservations
VMFS Datastore	• VMFS version must be VMFS6 • Fibre Channel (FC) only • The datastore must have attribute **ClusteredVMDK** set. For more information, see Activating Clustered VMDK Support.

What Shared Storage Configurations are Supported with WSFC?

Different WSFC cluster setups support different types of shared storage configurations. Some setups support more than one type. Select the recommended type of shared storage for the best results.

Differences between Physical compatibility RDMs and Virtual compatibility RDMs

An RDM (Raw Device Mapped) is a special mapping file in a VMFS volume that manages metadata for its mapped device. The mapping file is presented to the management software as an ordinary disk file, available for file-system operations. To the virtual machine, the storage virtualization layer presents the mapped device as a virtual SCSI device. There are two types of compatibility modes for RDMs.

- A Physical compatibility RDM (pRDM), specifies minimal SCSI virtualization of the mapped device.
  - Virtual Machine Snapshots are not available when the RDM is used in physical compatibility mode.
  - RDMs in physical compatibility mode are required to allow SCSI commands be directly passed to a LUN to satisfy the requirements of SCSI-3 Persistent Reservations used by WSFC.
- A Virtual compatibility RDM (vRDM), specifies full virtualization of the mapped device.
  - VMkernel sends only READ and WRITE to the mapped device. The mapped device appears to the guest operating system the same as a virtual disk file in a VMFS volume.
  - If you are using a raw disk in virtual mode, you can realize the benefits of VMFS, such as advanced file locking for data protection and snapshots for streamlining development processes.
  - Virtual mode is more portable across storage hardware than physical mode, presenting the same behavior as a virtual disk file.

– Supported for CIB configuration only.

Note:
Do not use cluster-in-a-box configuration for a production deployment.

For more information, see Broadcom knowledge base article 315481.

Table 511: Shared Storage Requirements

Storage Type	Clusters on One Physical Machine (Cluster in a Box)	Clusters Across Physical Machines (Cluster Across Boxes)	Clusters of Physical and Virtual Machines (Standby Host Clustering)
Clustered VMDK	No	Yes	No
VMDK in Virtual compatibility mode (SCSI controller sharing mode set to virtual)	Yes	No	No
pRDM (physical compatibility mode)	No	Yes (recommended)	Yes
vRDM (virtual compatibility mode)	Yes (not recommended)	No	No

Note:
The multi-writer feature must not be used for a clustered disk resource for WSFC.

In-guest storage options — such as iSCSI or SMB shares that are configured within guest operating systems running WSFC — are transparent to the ESX host. Any configuration that is supported by Microsoft requires no explicit support statement from Broadcom.

Round Robin Path Selection Policy (PSP_RR) for WSFC Disk Resources on ESX

ESX supports PSP_RR for WSFC.

- ESX supports PSP_RR for Windows Server 2012 and later.
- PSPs configured in mixed mode is supported. In a 2 node, WSFC cluster in CAB configuration one ESX host can be configured to use PSP_FIXED and the other ESX host can use PSP_RR.
- All hosts must be running ESX 6.7 or later.
- Mixed mode configurations of ESX 6.7 or later with previous ESX versions are not supported.
- If the number of paths to a storage device exceeds 5, WSFC cluster storage validation may fail if Round Robin path policy (PSP_RR) with iops =1 is used. In such cases, set the number of iops to 5 or more when the Round Robin path policy is used.

iSCSI Support for WSFC on ESX

ESX supports iSCSI storage and up to 5 node WSFC clusters.

- ESX supports iSCSI for Windows Server 2012 and later.
- Cluster-across-box (CAB) and cluster-in-a-box (CIB) are supported. A mixture of CAB and CIB is not supported.
- Clustered VMDKs on VMFS datastore in CAB are not supported over iSCSI.
- No qualification is needed for software iSCSI initiator in a guest operating system.
- N+1 cluster configuration comprising a cluster between "N" virtual machines on separate ESX hosts and one physical machine running Windows natively is supported.
- All hosts must be running ESX 7.0 or later.
- Cluster nodes registered to hosts with different ESX versions are not supported.
- Mixed mode of iSCSI config is supported.

FCoE Support for WSFC on ESX

ESX supports FCoE storage and up to 5 node WSFC clusters.

- ESX supports FCoE for Windows Server 2012 and later.
- Cluster-across-box (CAB) and cluster-in-a-box (CIB) are supported. A mixture of CAB and CIB is not supported.
- CAB configurations are supported with some cluster nodes on physical hosts. In a CAB configuration, a max of one virtual machine in a host can see a LUN.
- Clustered VMDK is not supported on a datastore that is connected over FCoE.
- N+1 cluster configuration, in which one ESX host has virtual machines which are secondary nodes and one primary node is a physical box are supported.
- Required DRS affinity (CIB) or Anti-affinity (CAB) rules for WSFC virtual machines.
- All hosts must be running FCoE initiators. Mixed cluster nodes running FC and FCoE are not supported.
- Mixed mode FCoE configuration is supported.

vMotion Support for WSFC on ESX

vSphere supports vMotion of a VM hosting a node of a WSFC.

Pre-requisites for vMotion support:

- vMotion is supported only for a cluster of virtual machines across physical hosts (CAB).
- Migrating more than 8 WSFC virtual machines at the same time for VMs with cluster shared resources might cause failover of cluster roles to other VMs.
- The vMotion network requires a 10 Gbps Ethernet link.
- The vMotion network must be a 10Gbps Ethernet link. A 1 Gbps Ethernet link is not supported for vMotion of WSFC virtual machines.
- vMotion is supported for Windows Server 2012 and later. Windows Server 2008 SP2 and earlier is not supported.
- The WSFC cluster heartbeat time-out must be modified at least to the values listed below:
  - `(get-cluster -name <cluster-name>).SameSubnetThreshold = 10`
  - `(get-cluster -name <cluster-name>).CrossSubnetThreshold = 20`
  - `(get-cluster -name <cluster-name>).RouteHistoryLength = 40`
- The WSFC virtual machine must be of hardware version 11 and later.

Modifying the WSFC heartbeat time-out:

WSFC nodes use the network to send heartbeat packets to other nodes of the cluster. If a node does not receive a response from another node for a specified time, the cluster removes the node from cluster membership. By default, a guest cluster node is considered down if it does not respond within 5 seconds. Other nodes that are members of the cluster take over any clustered roles that were running on the removed node.

During vMotion, a WSFC virtual machine may stall for a few seconds. If the stall exceeds the heartbeat timeout interval, the guest cluster treats the node as down and triggers an unnecessary failover. To prevent this, increase the heartbeat timeout interval to tolerate at least 10 missed heartbeats. The **SameSubnetThreshold** property controls this value. Its default is below 10 and must be updated.

From any node in the WSFC cluster, run the following command:

```
(get-cluster -name <cluster-name>).SameSubnetThreshold = 10
```

You can also adjust other properties to control the workload tolerance for failover. The delay setting controls how frequently heartbeats are sent between cluster nodes.

The default setting is 1 second and the maximum setting is 2 seconds. Set the **SameSubnetDelay** value to 1.

The threshold controls how many consecutive heartbeats can be missed before the node considers its partner to be unavailable and triggers the failover process. The default threshold is 5 heartbeats and the maximum is 120 heartbeats.

The combination of delay and threshold determines the total time clustered Windows nodes can lose communication before a failover is triggered. When the clustered nodes are in different subnets, they are called **CrossSubnetDelay** and **CrossSubnetThreshold**. Set the **CrossSubnetDelay** value to 2 and the **CrossSubnetThreshold** value to 20.

Note:
Recommended values for WSFC hearbeats settings are now defaults in Windows Server 2016 and above.

VMware vSphere® Virtual Volumes™ Support for WSFC

ESX supports the vVols framework for WSFC clusters of up to five nodes. The storage array must support SCSI-3 persistent reservations (PRs) at the individual LUN level. For NVMe, persistent reservations must be supported at the namespace level.

Note:

vSphere Virtual Volumes (vVols) framework is deprecated in VMware vSphere 9.0 and later.

- ESX supports vVols for Windows Server 2012 and later.
- Only Cluster-across-box (CAB) is supported.
- WSFC on vVols can work with any type of disk, "Thin" and "Thick"-provisioned disks.
- This feature enables customers to move away from using pRDM.
- WSFC on vVols supports vSphere HA, DRS, and vMotion.
- The underlying transport protocol can be FC, iSCSI, FCoE, or NVMe Fabric storage (FC, TCP).
- Virtual adapter NVMe support is added as part of vSphere 8.0 Update 3 along with PVSCSI.
- Cluster-in-a-box (CIB) and a mixture of CAB and CIB is not supported.
- N+1 cluster configuration, in which one ESX host has virtual machines which are secondary nodes and one primary node is a physical box is not supported.

Hot Extend a Shared vVol Disk

Hot extension of a shared vVol disk is supported with ESX 8.0 Update 2 or later. This allows you to increase the size of a shared disk without deactivating the cluster.

Extending the size of the shared vVol disk is a two-step process. First you must increase the size of shared vVol disk using vCenter. Then extend the partition size of the disk from the guest OS.

Best Practices for Hot Extending Shared vVol Disk:

- You can only extend one shared vVol disk at a time from a single VM. You cannot extend multiple shared vVol disks simultaneously.
- Do not reconfigure any other VM parameter while you are changing the size of a shared vVol disk. Any other change that you make in the VM configuration is not performed during this operation.
- All VMs sharing the disk must be accessible and registered with vCenter. VMs cannot be in a suspended state or in maintenance mode. VMs cannot be in APD or PDL state.
- Do not perform a shared disk extension when a cluster upgrade is in progress and there are hosts in the cluster that do not support this feature.
- Hot extend of a shared vVol disk is not supported for disks that have IO Filters that are attached.

1. Select any one VM that is sharing the vVol disk that you want to extend and click **Edit Settings**.
2. Select the disk that you want to extend and in the **Hard disk** field, enter the new size.
3. Click **OK**.

   All VM operations on all VMs that are sharing the vVol disk that is being extended are blocked until the extend operation is complete.

   After the operation is complete, each VM that is sharing the vVol disk shows the new disk size.

   Note:

   If the operation fails, vCenter displays an error message including the list of hosts where the disk extension failed. Run the extend operation again from one of the failed hosts.

After the size of the vVol disk is extended, you must also extend the disk partition size from the guest OS of the VM which owns the disk resource. Use the appropriate utility depending on your guest OS. Before you extend the size, verify all VMs that are sharing the disk display the new extended size. You can verify using the "disk mgmt tool" where the amount of increased size is displayed as unallocated. The Windows Fail over clustering tool shows the increased size of the shared vVol disk.

To extend the size of shared vVol disk for Windows Guest OS:

1. Open the Disk Management snap-in and right-click the data volume that you want to extend.
2. Select **Extend Volume** and follow the instructions in the **Extend Volume Wizard**.
3. After the Extend Volume operation is complete, the disk displays the increased size.

vSphere WSFC Setup Limitations

Before you set up WSFC in a virtual environment, review the list of functions that are not supported, and the requirements and recommendations that apply to your configuration.

The following environments and functions are not supported for WSFC setups with this release of vSphere:

- Using VMDKs on NFS datastore as a shared disk resource for WSFC.
- Mixed environments, such as configurations where one cluster node is running a different version of ESX than another cluster node.
- vSphere Fault Tolerance (FT).
- Migration with vMotion® of clustered virtual machines using vRDMs.
- N-Port ID Virtualization (NPIV).
- ESX hosts or clusters that have memory overcommitment are not suitable for deploying WSFC virtual machines. Memory overcommitment can cause virtual machines to stall for short durations. This can be disruptive as the WSFC clustering mechanism is time-sensitive and timing delays can cause the virtual machines to behave incorrectly.
- Suspend or resume of a VM node of a WSFC.
- Storage spaces are not supported with Failover clustering on Windows 2012 and later.
- VM configurations changes:
  - Hot adding memory.
  - Hot adding CPU.
  - Increasing the size of a shared disk is supported only for the disks created from vSphere Virtual Volumes storage with vSphere 8.0 U2 or later.
  - Pausing or resuming the virtual machine state. This I/O intensive operation is disruptive of the timing sensitive WSFC clustering software.
  - Hot adding storage controllers LSI Logic SAS, VMware Paravirtual or NVMe Controller.
  - Hot adding network adapters.
  - Any other hardware changes while cluster VMs are in powered on state except hot add disk or hot share disk.
  - Using snapshots.
  - Cloning a VM.
- Sharing disks between virtual machines without a clustering solution may lead to data corruptions.
- NVMe controller for WSFC is only supported for Windows Server 2022 with OS Build 20348.1547 and the shared disks should be created only from Clustered VMDK activated datastores (SCSI or NVMe). To use NVMe Controller, virtual machine hardware version must be 21 or later.

WSFC and Booting from a vSAN

You can boot a VM from a vSAN node of a WSFC.

Booting from a vSAN is complex. Problems that you encounter in physical environments extend to virtual environments. For general information about booting from a vSAN, see the *vSphere Storage* documentation.

Follow these guidelines when you place the boot disk of a virtual machine on a SAN-based VMFS volume:

- For more information about the best practices for boot-from-SAN that Microsoft publishes, see the Microsoft knowledge base article

Support for booting from a Storage Area Network (SAN).
- Test clustered configurations in different failover scenarios before you put them into production environments.

Cluster Virtual Machines Across Physical ESX Hosts

You can create a WSFC cluster that consists of two or more virtual machines on two or more ESX hosts. Use this method for production deployments.

A cluster across physical hosts requires specific hardware and software.
- Up to 5 nodes in a single WSFC is supported.
- Supported shared storage configuration, such as RDM in physical compatibility mode, vVol, clustered VMDK. For more information, see What Shared Storage Configurations are Supported with WSFC?.
- If the cluster validation wizard completes with the warning: `Validate Storage Spaces Persistent Reservation` you can safely ignore the warning. This check applies to the Microsoft Storage Spaces feature, which does not apply to VMware vSphere.
- The ESX host supports up to 16 WSFC clusters (multi-cluster) running on the same ESX host.
- With vSphere 8.0 Update 2 or later, you can use an NVMe controller in addition to the existing Paravirtual controller for WSFC with Clustered VMDK for Windows Server 2022 (OS Build 20348.1547) and later. To use an NVMe controller, the virtual machine hardware version must be 21 or later.

Clustered VMDK support for WSFC

vSphere supports VMDKs on clustered datastore as shared disk resources for a WSFC. Using VMDKs reduces the management overhead compared to pRDMs.

WSFC uses SCSI-3 persistent reservations (PR) commands - PR-IN and PR-OUT - to coordinate access to clustered disk resources. These commands are emulated at the vSCSI layer on the datastore. The datastore must be configured to host clustered VMDKs; such a datastore is called a clustered VMDK datastore.

WSFC with clustered VMDKs on Windows Server 2022 and later supports NVMe virtual adapters. With vSphere 8.0 Update 3 and later, VMFS VMDKs also support storage from NVMe-oF TCP arrays using PVSCSI or NVMe controllers.

Activating Clustered VMDK Support

You can activate clustered VMDK support when you create a VMFS6 datastore, or activate clustered VMDK on an existing VMFS6 datastore.

Before you activate clustered VMDK support, ensure all ESX hosts connected to the datastore use ESX 7.0 or later and are managed by vCenter 7.0 or later. All hosts connected to the datastore must be managed by the same vCenter while deactivating or activating the clustered VMDK flag on the datastore. Once the clustered VMDK flag is activated or deactivated, hosts can be managed by any vCenter with version 7.0 or later.

NVMe FC datastores are supported for activating clustered VMDK from vSphere 8.0 and later. WSFC with clustered VMDKs on Windows Server 2022 and later supports NVMe virtual adapters. vSphere 8.0 Update 3 and later support NVMe/TCP datastores.

1. Navigate to any inventory object that is a valid parent object of a datastore, such as a host, a cluster, or a data center, and click the **Datastores** tab.
2. Right-click the parent object and select **Create a New Datastore**.
3. On the **New Datastore** page, select a device which attribute **Clustered VMDK Supported** is set to `Yes`.
4. Create a VMFS6 datastore.
5. After the datastore is created, click the **Configure Datastore** tab.
6. Under **General**, go to the **Datastore Capabilities** section and click the `Enable` button for **Clustered VMDK**.
7. When activated, to deactivate the clustered VMDK, click the **Disable** button.

   Note:

   If there are WSFC VMs using clustered VMDKs on the datastore, you cannot deactivate the clustered VMDK.

Power off all virtual machines with shared disks on the datastore.

Configuration Maximums for Clustered VMDK Support

The following table provides the configuration limits for Clustered VMDK support.

Configuration	Maximum
Max number of VMs in a single WSFC cluster	5
Max number of WSFC clusters (multi-cluster) running on the same set of ESX hosts	16
Max number of clustered VMDKs per ESX host	192

Recommendations for Using Clustered VMDKs with WSFC

Follow these recommendations when using Clustered VMDKs with WSFC.

1. Do not present LUNs used for clustered VMDKs to ESX hosts of version earlier than 7.0. Doing so can cause slow boot times, an unresponsive hostd, and other issues. An ESX host of version earlier than ESX 7.0 cannot mount a clustered VMDK datastore. ESX hosts running WSFC VMs must have a physical persistent reservation (SCSI or NVMe) of type WEAR on the device. An ESX host must have ESX 8.0 or later to mount a clustered VMDK datastore when the backend LUN is created from NVMe FC SAN.

   WSFC with clustered VMDK and Windows Server 2022 or later supports NVMe virtual adapters.

   Clustered VMDKs also support storage from NVMe-oF TCP arrays using PVSCSI or NVMe controllers for Windows Server 2022 or later.
2. Before removing a virtual machine from a clustered VMDK datastore, ensure that all virtual machines hosting WSFC nodes are either migrated off or powered off properly. This ensures shared resources, such as heartbeat (HB) slots are released. If a VM fails or an All Paths Down (APD) condition occurs on the clustered VMDK datastore during power-off, always power the virtual machine on and then off before removing the VM from the cluster.
3. Virtual machines using non-shared disks on a clustered datastore continue to function normally and support operations such as snapshot and clone, but mixing disk types on a clustered datastore is not recommended.
4. Do not place clustered VMDKs for different WSFC clusters on the same shared datastore. Use a dedicated clustered datastore per WSFC cluster.
5. Set the virtual machine hardware version (virtual compatibility) to vSphere 7.0 or later when using the clustered VMDK feature.

Requirements for Using Clustered VMDKs with WSFC

You must follow these requirements when using Clustered VMDKs with WSFC.

1. You must set the Windows Cluster Parameter **QuorumArbitrationTimeMax** to 60.
2. The physical disk that virtual disks are stored on must support ATS SCSI commands.
3. The physical disk must support SCSI-3 Persistent Reservations, specifically Write Exclusive All Registrants (WEAR) type reservations.
4. VMDKs must be Eager Zeroed Thick Provisioned.
5. Storage devices can be claimed by VMware Native Multipathing Plugin (NMP) or by any other third-party MultiPathing Plugin (MPP). Check with vendors regarding the support for Clustered VMDK before using a vendor's third-party MultiPathing Plugin (MPP).

Limitations of Clustered VMDK support for WSFC

Clustered VMDK support is only possible with the following limitations:

1. Cluster in a Box (CIB) configuration is not supported.
2. Spanned VMFS datastores cannot store clustered VMDKs.
3. A physical LUN with a clustered datastore cannot have any additional VMFS datastores.

4. On-line Storage vMotion and snapshots are not supported.
5. Only disks with 512 sector size are supported.
6. VMFS5 and earlier are not supported for storing clustered VMDKs.
7. Only datastores connected through Fibre Channel (FC) and NVMe TCP are supported.
8. Clustered VMDK datastore cannot be used as an ESX installation disk.
9. Diagnostic partition is not supported on clustered disks.
10. SCSI-2 reservations are not supported on clustered VMDKs.
11. Hot expansion of a VMDK that is associated with a clustered VM is not supported.
12. Moving more than one node of WSFC to the same ESX host is not allowed and causes vMotion to fail. You must use anti-affinity DRS Rules to separate VMs and nodes of the cluster on different ESX hosts.
13. Cloning and Fault Tolerance are not supported.

Create the First Node for WSFC Clusters Across Physical Hosts

To create the first node, you create and configure a virtual machine and install a guest operating system on the virtual machine.

1. Open the vSphere Client and connect to the vCenter instance.

   Use the user name and password of the user with administrator permissions on the virtual machine.

2. In the vSphere Client, right-click the host and select **New Virtual Machine**.
3. Follow the prompts in the wizard to create the virtual machine.

Page	Action
Creation type	Select **Create a new virtual machine**.
Name and folder	Enter a name and select a location.
Compute resource	Select a cluster, host, vApp, or resource pool to run the virtual machine.
Storage	Select a datastore as the location for the virtual machine configuration file and the virtual machine disk (`.vmdk`) file.
Compatibility	The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.
Guest operating system	Select the guest operating system that you intend to install.
Customize hardware	Select virtual hardware, advanced virtual machine options, and SDRS rules, if necessary, or applicable.
Ready to Complete	Review your selections.

4. Click **Finish** to complete creating the virtual machine.

   Note:
   Do not add shared cluster disks now.

5. Install a Windows Server operating system on the virtual machine.

Create Additional Nodes for a WSFC Across Physical Hosts

To add nodes to a WSFC cluster of virtual machines across physical hosts, you can create a template from the first virtual machine and deploy additional virtual machines from it. Alternatively, follow the procedure that is described in the previous section to manually create additional virtual machines on another ESX host.

CAUTION:
If you clone a virtual machine with an RDM configuration, the cloning process converts the RDMs to virtual disks. Unmap all RDMs before cloning, and remap them after cloning is complete.

CAUTION:
If you clone a virtual machine with clustered VMDKs, unmap all clustered VMDKs before cloning and remap them after cloning is complete.

CAUTION:

Use Windows Sysprep before clone the virtual machine to a template.

1. In the vSphere Client, right-click the first virtual machine that you created and select **Clone** > **Clone to Template**.
2. Follow the prompts in the wizard to create the virtual machine template.

Page	Action
Name and folder	Enter a name (for example, Node2_Template) and select a location.
Compute Resource	Select the host or cluster where you run the virtual machine.
Disk Format	Select **Same format as source**.
Storage	Select a datastore as the location for the virtual machine configuration file and the `.vmdk` file.
Ready to Complete	Click **Finish** to create the virtual machine template.

3. Right-click the virtual machine template and select **Deploy VM from this Template**.
4. Follow the prompts in the wizard to deploy the virtual machine.

Page	Action
Name and folder	Enter a name (for example, Node2) and select a location.
Compute resource	Select the host or cluster where you run the virtual machine.
Disk Format	Select **Same format as source**.
Datastore	Select a datastore as the location for the virtual machine configuration file and the `.vmdk` file.
Clone options	Select **Customize the operating system**.

5. Select a new guest operating system from the list.
   a) Click the **Create a new specification** button to add a guest operating system. Proceed through the **New VM Guest Customization Spec** wizard.
   b) Click **Finish** to exit the wizard.
6. Click **Finish** to deploy the virtual machine.

Add Hard Disks to the First VM, first node of a WSFC, for Clusters Across Physical Hosts with pRDMs

In a WSFC cluster with shared disk, storage disks are shared between nodes.

Before you add hard disks to the first node, complete the following tasks:

- If you are using physical compatibility mode RDMs, ask your vSAN administrator for the location of unformatted vSAN LUNs. The hard disks that you create in this task must point to vSAN LUNs.
- If you use vVols for a shared disk, see *Add Hard Disks to the First Node for Clusters Across Physical Hosts with VMware vSphere® Virtual VolumesTM*.
- If you are use Clustered VMDKs on VMFS datastores, see *Add Hard Disks to the First Node for Cluster Across Physical Hosts with Clustered VMDKs on VMFS datastores*.

Note:
Use RDMs in physical compatibility mode.

1. In the vSphere Client, right-click the newly created virtual machine and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller**.
3. Select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Physical**, and click **OK**.

Note:
Although LSI SAS is supported, use VMware Paravirtual.

4. Right-click the newly created virtual machine in step 1 and select **Edit Settings**.
5. From the **New device** drop-down menu, select **RDM Disk**.
6. Select an unformatted LUN and click **OK**.
7. Expand **New Hard disk** and select a datastore for **Location**.
   This datastore must be on shared storage because you need a single shared RDM file for each shared disk.
8. Select **Physical** as the compatibility mode.
9. Select the newly created SCSI controller in step 2, for virtual device node (for example, select **SCSI (1:0)**).

   Note:
   You must select a new SCSI controller. You cannot use SCSI 0. SCSI controller 0 is reserved for boot disks and disks which should not be shared

10. Click **OK**.
    The wizard creates a new hard disk.

Add Hard Disks to the First Node of a vVols-Based WSFC Across Physical Hosts

In a WSFC cluster, storage disks are shared between nodes.

Before you add hard disks to the first node, make sure that:

- The virtual machines have hardware version 13 or later. For more information, see Virtual Volumes (vVols) now supports WSFC.
- vVols are prepared for deployment. For more information, see Working with vSphere Virtual Volumes in the *vSphere Storage* documentation.

Note:

The vSphere Virtual Volumes (vVols) framework is deprecated in vSphere 9.0 and later.

1. In the vSphere Client, right-click the newly created virtual machine and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller**.
3. In the new SCSI controller, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Physical**, and click **OK**.

   Note:

   Although LSI SAS is supported, use VMware Paravirtual.

4. Right-click the newly created virtual machine in step 1 and select **Edit Settings**
5. From the **New device** drop-down menu, select **New Hard Disk**.
6. Expand **New Hard disk** and select the disk size required.
7. Select a datastore for **Location**.
   This datastore must be on shared vVols.
8. Under Disk Provisioning, select either Thick or Thin Provision.
9. Expand the New Hard Disk.
10. Select the newly created SCSI controller in step 2, for virtual device node (for example, select **SCSI (1:0)**).

    Note:
    This must be a new SCSI controller. You cannot use SCSI 0.

11. Click **OK**.

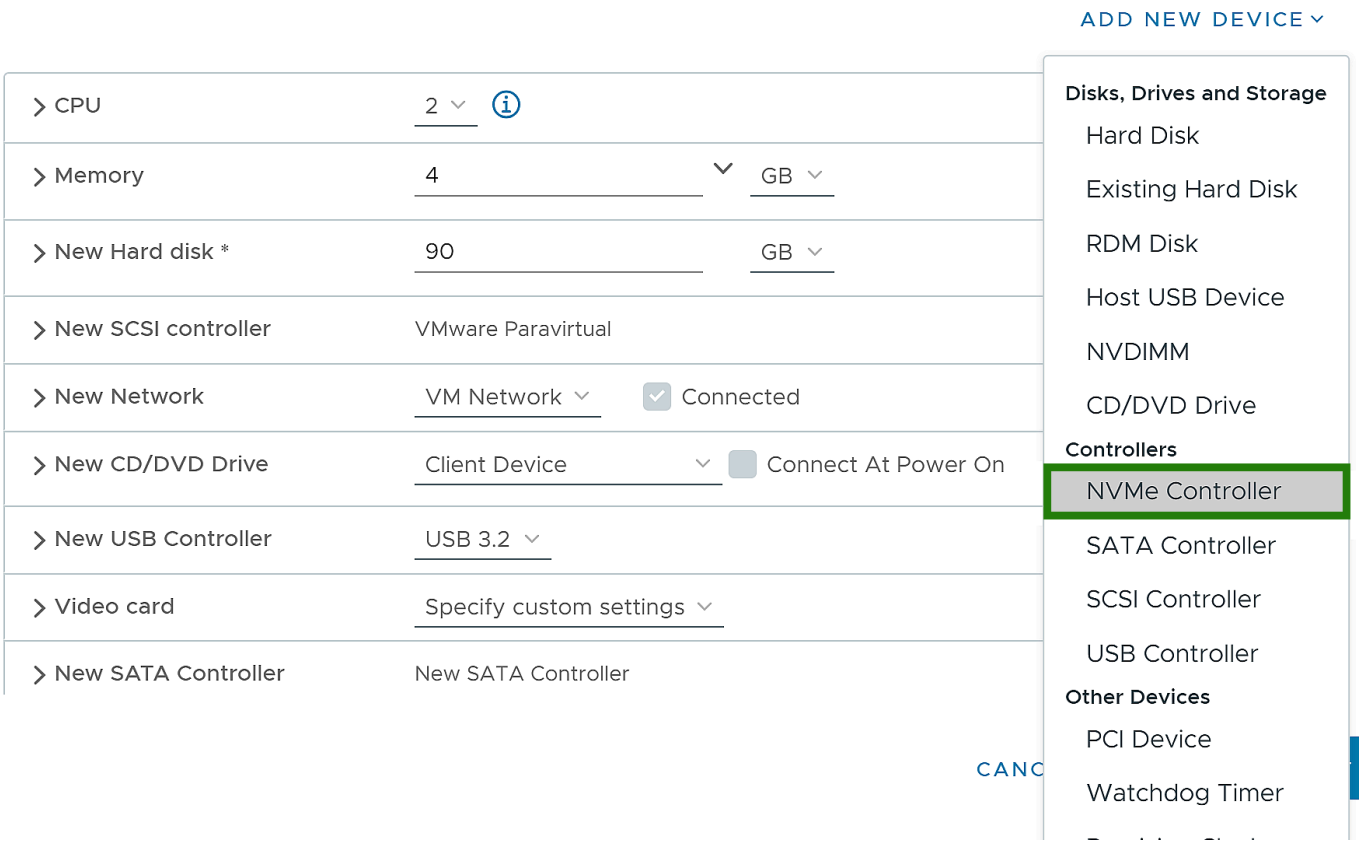
    The new hard disk is created.

Add Hard Disks to the First Node for a Cluster Across Physical Hosts with Clustered VMDKs on VMFS

You can add hard disks to the first node for a cluster across physical hosts with clustered VMDKs on VMFS datastores.

1. In the vSphere Client, right-click the newly created virtual machine, and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller** or **NVMe Controller**.
   - If a new SCSI controller is created, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Physical**, and click **OK**.

- If a new NVMe controller is created, set **NVME Bus Sharing** to **Physical**, and click



3. Right-click the newly created virtual machine in step 1, and select **Edit Settings**.
4. From the **New device** drop-down menu, select **New Hard Disk**.
5. Expand **New Hard disk** and select the disk size required.
6. Select a datastore for **Location**.
   This datastore must be the datastore with clustered VMDK flag activated.
7. Under Disk Provisioning, select **Thick Provision with Eager zeroed**.
8. Select the newly created SCSI controller or NVMe controller in step 2, for virtual device node (for example, select **SCSI (1:0)** or **NVMe (1:0)**).

   Note:
   This must be a new SCSI controller. You cannot use SCSI 0.
9. Click **OK**.

   The new hard disk is created.

Add Hard Disks to Additional WSFC Nodes for Clusters Across Physical Hosts

You can give additional nodes in a WSFC cluster access to shared disk resources already configured on the first node.

Before you begin, obtain the following information:

- The SCSI IDs for all virtual disks are shared.
- The datastore file path for each shared disk.

Note:
The procedure for clustered VMDK, pRDMs, and vVol is the same.

1. In the vSphere Client, right-click the newly created virtual machine, and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller** or **NVMe Controller**.
   a) If a new SCSI controller is created, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Physical**, and click **OK**.
   b) If a new NVMe controller is created, set **NVMe Bus Sharing** to **Physical**, and click **OK**.

   You must select the same SCSI controller or NVMe controller type.
3. Right-click the newly created virtual machine in step 1 and select **Edit Settings**.
4. From the **New device** drop-down menu, select **Existing Hard Disk**.
5. In **Disk File Path**, browse to the disk location you want to share for the first node.
6. Expand **New Hard disk**.
7. Select the same SCSI or NVMe ID that you selected for the first virtual machine's shared storage disks. For example, select **SCSI (1:0)** or **NVMe (1:0)**.

   Note:
   The disk SCSI or NVMe ID for this virtual machine's shared storage must match the corresponding SCSI or NVMe ID for the first virtual machine.
8. Click **OK**.

Use WSFC in a vSphere HA and vSphere DRS Environment

When you use WSFC in a vSphere High Availability (vSphere HA) or vSphere Distributed Resource Scheduler (DRS) environment, you must configure your ESX hosts and virtual machines to use certain settings. All ESX hosts that run WSFC virtual machines must be managed by a vCenter instance.

vSphere supports vMotion for WSFC virtual machine nodes.

Enable vSphere HA and vSphere DRS in ESX

ESX hosts that are running WSFC virtual machines can be part of a vSphere cluster with both vSphere HA and vSphere DRS activated. You can activate vSphere HA and vSphere DRS in the Cluster Settings dialog box.

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Services**, select **vSphere DRS** and click **Edit**.
4. Select the **Turn ON vSphere DRS** and **Turn on vSphere HA** check boxes.
5. Click **OK**.

Create VM-VM Affinity Rules for WSFC Virtual Machines

For WSFC virtual machines in a cluster, you must create VM-VM affinity or anti-affinity rules. VM-VM affinity rules specify which virtual machines should be kept together on the same ESX host (for example, a cluster of WSFC virtual machines on one physical ESX host). VM-VM anti-affinity rules specify which virtual machines should be kept apart on different physical ESX hosts (for example, a cluster of WSFC virtual machines across physical ESX hosts).

For a cluster of virtual machines on one physical ESX host, use affinity rules.

Note:

Do not use a cluster on one physical ESX host for production deployment.

For a cluster of virtual machines across physical hosts, use anti-affinity rules.

Note:

vSphere supports vMotion for WSFC virtual machine nodes across physical ESX hosts.

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Configuration**, click **VM/Host Rules**.
4. Click **Add**.
5. In the **Create VM/Host Rule** dialog box, enter a name for the rule.
6. From the **Type** drop-down menu, select a rule.
   - For a cluster of virtual machines on one physical ESX host, select **Keep Virtual Machines Together**.
   - For a cluster of virtual machines across physical ESX hosts, select **Separate Virtual Machines**.
7. Click **Add**.
8. Select the two virtual machines to which the rule applies and click **OK**.
9. Click **OK**.

Set DRS Automation Level for WSFC Virtual Machines

When you set the vSphere DRS automation level for the virtual machine to Partially Automated, vCenter performs initial placement of virtual machines when they are powered on and provides migration recommendations.

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Services**, select **vSphere DRS**, and click **Edit**.
4. Expand **DRS Automation**.
5. Under **Virtual Machine Automation**, select the **Enable individual virtual machine automation levels** check box, and click **OK**.
6. Under **Configuration**, select **VM Overrides**, and click **Add**.
7. Click the **+** button, select the WSFC virtual machines in the cluster, and click **OK**.
8. From the **Automation level** drop-down menu, select **Partially Automated**.
9. Click **OK**.

Note:

Set the DRS automation level for WSFC virtual machines to Partially Automated. Although there are no longer technical restrictions preventing the use of Fully Automated mode, using it requires you to adjust advanced WSFC settings to prevent unintended failovers during vMotion

Using vSphere DRS Groups and VM-Host Affinity Rules with WSFC Virtual Machines

You can set up two types of DRS groups: virtual machine DRS groups, which contain at least one virtual machine, and host DRS groups, which contain at least one ESX host. A VM-Host rule establishes a relationship between a virtual machine DRS group and a host DRS group.

You must use VM-Host affinity rules because vSphere HA does not enforce VM-VM affinity rules by default during failover. If the advanced configuration `das.respectVmVmAntiAffinityRules` is not activated, vSphere HA might separate clustered virtual machines that are meant to stay together, or place clustered virtual machines that are meant to stay apart on the same ESX host. VM-Host affinity rules, which vSphere HA does observe during failover placement, eliminate this risk and are therefore required for WSFC configurations.

Note:

VM-VM anti-affinity rule enforcement during failover can be activated through the `das.respectVmVmAntiAffinityRules` advanced configuration option in vSphere HA settings. However, even with this option activated, vSphere HA does not enforce VM-VM affinity (keep-together) rules during failover. Therefore, strict VM-Host affinity rules remain the supported mechanism for WSFC virtual machine placement.

For a cluster of virtual machines on one physical ESX host, all WSFC virtual machines must be in the same virtual machine DRS group, linked to the same host DRS group with the affinity rule "Must run on hosts in group."

For a cluster of virtual machines across physical ESX hosts, all WSFC virtual machines must be in the same virtual machine DRS group and all ESX hosts must be in the same host DRS group. The virtual machine and host DRS groups must be linked to the affinity rule "Must run on hosts in group."

CAUTION:

Limit the number of ESX hosts to two when you define host DRS group rules for a cluster of virtual machines on one physical ESX host. (This does not apply to clusters of virtual machines across physical ESX hosts.) Because vSphere HA does not enforce VM-VM affinity rules by default, virtual machines in the configuration could be spread across ESX hosts during a vSphere HA recovery from a host failure if more than two ESX hosts are included in a host DRS group rule.

Create a Virtual Machine DRS Group (WSFC)

Before you can create a VM-Host affinity rule, you must create the host DRS group and the virtual machine DRS group that the rule applies to.

For both, a cluster of virtual machines on one physical host (CIB) and a cluster of virtual machines across physical hosts (CAB), create one virtual machine DRS group that contains all WSFC virtual machines in a single WSFC cluster. For example, VMGroup_1 contains WSFC_VM_1, WSFC_VM_2 … WSFC_VM_5.

If your environment has multiple WSFC clusters, create a separate virtual machine DRS group for each WSFC cluster.

1. Browse to the cluster in the vSphere Client.
2. Click the **Configure** tab.
3. Under **Configuration**, select **VM/Host Groups**, and click **Add**.
4. In the **Create VM/Host Group** dialog box, enter a name for the group.
5. From the **Type** drop-down menu, select **VM Group**, and click **Add**.
6. Select the check box next to each virtual machine to add it to the group. Add all Continue this process until all desired virtual machines have been added.
   - For a cluster of virtual machines on one physical ESX host (CIB), add all WSFC virtual machines to one group.
   - For a cluster of virtual machines across physical ESX hosts (CAB), add all WSFC virtual machines to one group.
7. Click **OK**.

You created a virtual machine DRS group that contains all WSFC virtual machines in a single WSFC cluster. For both CIB and CAB configurations, all WSFC virtual machines must be in the same virtual machine DRS group so that a VM-Host affinity rule can link the group to a host DRS group. This ensures that vSphere HA respects the intended topology during a host failure.

Create a Host DRS Group (WSFC)

Before you can create a VM-Host affinity rule, you must create the host DRS group and the virtual machine DRS group that the rule applies to.

For a cluster of virtual machines on one physical ESX host, create one host DRS group that contains both the ESX hosts. For example, HostGroup_1 contains ESX_HOST_1, ESX_HOST_2.

For a cluster of virtual machines across physical ESX hosts, create one host DRS group that contains all the ESX hosts. For example, HostGroup_1 contains ESX_HOST_1 running WSFC_VM_1, ESX_HOST_2 running WSFC_VM_2 …... ESX_HOST_5 running WSFC_VM_5 and ESX_HOST_6 as a standby host.

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Configuration**, select **VM/Host Groups** and click **Add**.
4. In the **Create VM/Host Group** dialog box, type a name for the group (for example, `HostGroup_1`).
5. From the **Type** drop-down menu, select **Host Group**, and click **Add**.
6. Click the check box next to a host to add it. Continue this process until all desired hosts have been added.
7. Click **OK**.

Set up VM-Host Affinity Rules for DRS Groups (WSFC)

Create VM-Host affinity rules to specify whether the members of a selected virtual machine DRS group can run on the members of a specific host DRS group.

Create virtual machine DRS groups that contain one or more WSFC virtual machines as described in Create a Virtual Machine DRS Group (WSFC).

Create host DRS groups that contain one or more ESX hosts, as described in Create a Host DRS Group (WSFC).

1. Browse to the cluster in the vSphere Client.
2. Click **Configure**.
3. Under **Configuration**, select **VM/Host Rules**, and click **Add**.
4. In the **Create VM/Host Rule** dialog box, enter a name for the rule.
5. From the **Type** drop-down menu, select **Virtual Machines to Hosts**.
6. Select the virtual machine DRS group and the host DRS group to which the rule applies.
   For example, select VMGroup_1 and HostGroup_1.
7. Select **Must run on hosts in group**.
8. Click **OK**.

vSphere WSFC Setup Checklist

Before setting up a WSFC on vSphere, review the requirements checklists to configure your environment properly and determine if you need technical support.

Requirements for Clustered Disks

Each type of clustered disk has its own requirements, depending on whether it is in a single-host cluster or in a multi-host cluster.

Component	Single-Host Clustering	Multihost Clustering
Clustered VMDK	SCSI bus sharing mode set to virtual is not supported.	SCSI bus sharing or NVMe Bus Sharing mode set to physical is supported. vSphere 8.0 Update 2 or later supports NVMe with Windows Server 2022 or later.
VMDK (In virtual compatibility mode)	SCSI bus sharing mode set to virtual is supported. (Not Recommended)	SCSI bus sharing mode set to Physical is not supported.
vRDM	Device type must be set to virtual compatibility mode. SCSI bus sharing mode must be set to virtual mode. A single, shared RDM-mapping file for each clustered disk is required.	Not supported.
pRDM	Not supported.	RDM device type must be set to Physical compatibility mode during hard disk creation. SCSI bus sharing mode of vSCSI Controller hosting pRDM must be set to physical. A single, shared RDM-mapping file for each clustered disk is required.
vVols	Not Supported.	The bus sharing mode of the PVSCSI or NVMe controller must be set to physical.
All types	All clustered nodes must use the same target SCSI ID (on the virtual SCSI adapter) or same target NVMe ID (on the virtual NVMe adapter) for the same clustered disk. You must use a separate virtual SCSI or NVMe adapter for clustered disks. To enhance performance, you must use multiple virtual SCSI or NVMe adapters for clustered disks (up to supported maximum). In this setup, make sure that at least one controller is dedicated to a non-shared disk.	

Other Requirements and Recommendations

The following table lists the components in your environment that have requirements for options or settings.

Table 512: Other Clustering Requirements and Recommendations

Component	Requirement
Disk	Select Thick Provision with Eager zeroed during disk provisioning for VMDKs (for clustered VMDKs or VMDKs in virtual compatibility mode). For vVols select either thick or thin. Modify the .vmx file for the WSFC VMs to add these lines: • `scsiX.returnNoConnectDuringAPD = "TRUE"` • `scsiX.returnBusyOnNoConnectStatus = "FALSE"` • `nvmeX.returnNoConnectDuringAPD = "TRUE"` (if you use NVMe controller for boot device) • `nvmeX.returnBusyOnNoConnectStatus = "FALSE"` (if you use NVMe controller for boot device) • Where X is the boot device SCSI bus controller ID number. By default, X is set to 0. The only disks that you should not create with the `Thick Provision` option are RDM files (both physical and virtual compatibility mode). Disk I/O timeout is 60 seconds or more (`HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\Disk\TimeOutValue`).
Windows	The following Windows Server versions are supported: • Windows Server 2012 • Windows Server 2012 R2 • Windows Server 2016 • Windows Server 2019 with vSphere 6.7 Update 3 • Windows Server 2022 with vSphere 7.0 or later For Windows Server 2012 and later, you can use up to five cluster nodes. **Note:** If you recreate the cluster, this value might be reset to its default and must be reconfigured. You must configure the cluster service to restart automatically on failure - for the first, second, and subsequent times.
ESX	Do not overcommit memory. Set the **Memory Reservation** (minimum memory) option to the same as the amount of memory that is assigned to the virtual machine. ESX uses a different technique to determine if Raw Device Mapped (RDM) LUNs are used for WSFC cluster devices, by introducing a configuration flag to mark each device as "perennially reserved" that is participating in a WSFC cluster. For ESX hosts that have passive WSFC nodes with RDM LUNs, to mark the device as perennially reserved, use the ESXCLI command `esxcli storage core device setconfig -d <naa.id> --perennially-reserved=true`. For more information, see Broadcom knowledge base article 318851.
Multipathing	Contact your multipathing software vendor for information and support of non-VMware multipathing software in vSphere.

Required Information for Technical Support

Before you contact technical support, review the following table that contains information about the files and settings to collect. Technical support uses these files and settings to analyze issues with a WSFC setup.

Note:
Verify that your setup complies with the checklists in #unique_3225/unique_3225_Connect_42_AH975846 and Table 512: Other Clustering Requirements and Recommendations before you contact technical support.

Table 513: Information Required by Technical Support

File or Information	Description or Location
`vm-support tarball`	Contains the vmkernel log, virtual machine configuration files and logs, and so on.
Application and system event logs of all virtual machines with the problem	
Cluster log of all virtual machines with the problem	`%ClusterLog%`, which is set to `%SystemRoot%\cluster\cluster.log` or use `Get-ClusterLog Powershell CMdlet`
Disk I/O timeout	`HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\Disk\TimeOutValue`
vSphere Client display names and Windows NETBIOS names of the virtual machines experiencing the problem	
Date and time that the problem occurred	
SAN configuration of the ESX system	Details about LUNs, paths, and adapters.
(Optional) Memory dump of the virtual machine	Required if a guest virtual machine fails with a blue screen with an error message.

Cluster Virtual Machines on One Physical ESX Host

You can create a WSFC cluster with up to five nodes on a single ESX host.

Note:

For production deployments, do not set up a WSFC on VMs residing on a single ESX host.

Note:
Windows Server 2012 and later support up to five nodes (virtual machines). For more information about supported guest operating systems, see Table 512: Other Clustering Requirements and Recommendations.

Create the First Node of a WSFC on One Physical ESX Host

To create the first node, you create and configure a virtual machine and install a guest operating system on the virtual machine.

1. Open the vSphere Client and connect to the vCenter system.

   Use the user name and password of the user with administrator permissions on the virtual machine.
2. In the vSphere Client, right-click the host and select **New Virtual Machine**.
3. Follow the prompts in the wizard to create the virtual machine.

Page	Action
Creation type	Select **Create a new virtual machine**.
Name and folder	Enter a name and select a location.
Compute resource	Select a cluster, host, vApp, or resource pool to run this virtual machine.
Storage	Select a datastore as the location for the virtual machine configuration file and the virtual machine disk (`.vmdk`) file.
Compatibility	The host or the cluster supports more than one VMware virtual machine version. Select a virtual machine compatibility.
Guest operating system	Select the guest operating system that you intend to install.
Customize hardware	Select virtual hardware, advanced virtual machine options, and SDRS rules.
Ready to Complete	Review your selections.

4. Click **Finish** to complete creating the virtual machine.

   Note:
   Do not add shared cluster disks at this time.

5. Install a Windows Server operating system on the virtual machine.

Create Additional Nodes for Clusters on One Physical ESX Host

Create a template from the first virtual machine and deploy the second node from that template. You can have up to five nodes on Windows Server 2012 or later.

Note:
For a new template, use Windows Sysprep.

CAUTION:
If you clone a virtual machine with an RDM setup, the cloning process converts the RDMs to virtual disks. Unmap all RDMs before cloning, and remap them after cloning is complete.

1. In the vSphere Client, right-click the first virtual machine that you created and select **Clone** > **Clone to Template**.
2. Follow the prompts in the wizard to create the virtual machine template.

Page	Action
Name and folder	Enter a name (for example, Node2_Template) and select a location.
Compute Resource	Select the host or cluster where you will run the virtual machine.
Disk Format	Select **Same format as source**.
Storage	Select a datastore as the location for the virtual machine configuration file and the `.vmdk` file.
Ready to Complete	Click **Finish** to create the virtual machine template.

3. Right-click the virtual machine template and select **Deploy VM from this Template**.
4. Follow the prompts in the deployment wizard to deploy the virtual machine.

Page	Action
Name and folder	Enter a name (for example, Node2) and select a location.
Compute resource	Select the host or cluster where you will run the virtual machine.
Disk Format	Select **Same format as source**.
Datastore	Select a datastore as the location for the virtual machine configuration file and the `.vmdk` file.
Clone options	Select **Customize the operating system**.

5. Select a new guest operating system from the list.
   a) To add a guest operating system, click the **Create a new specification** button. Follow the prompts in the **New VM Guest Customization Spec** wizard.
   b) Click **Finish** to exit the wizard.
6. Click **Finish** to deploy the virtual machine.

Add Hard Disks to the First Node for Clusters on One Physical ESX Host

In a WSFC cluster, storage disks are shared between nodes. You set up a quorum disk and at least one shared storage disk.

1. In the vSphere Client, right-click the newly created virtual machine and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller**.
3. In the new SCSI controller, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Virtual**, and click **OK**.

   Note:

   Although LSI SAS is supported, use VMware Paravirtual.

4. Right-click the newly created virtual machine in step 1 and select **Edit Settings**
5. From the **New device** drop-down menu, select **Hard Disk**.
6. Expand **New Hard disk** and select the disk size required.
7. Select a datastore for **Location**.
8. Under Disk Provisioning, select **Thick Provision with Eager zeroed**.
9. Select the newly created SCSI controller in step 2, for virtual device node (for example, select **SCSI (1:0)**).

   Note:
   This must be a new SCSI controller. You cannot use SCSI 0.

10. Click **OK**.

    The new hard disk is created.

Add Hard Disks to Additional Nodes for Clusters on One Physical ESX Host

To allow shared access to clustered services and data, use **Add an Existing disk** on the second VM to point to the disks from the first VM that should be shared. Use the same SCSI IDs.

Before you begin, obtain the following information:

- The SCSI IDs for all virtual disks are shared.

1. In the vSphere Client, right-click the newly created virtual machine, and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller**.
3. In the new SCSI controller, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Virtual**, and click **OK**.

   Note:

   Although LSI SAS is supported, use VMware Paravirtual.

4. Right-click the newly created virtual machine in step 1 and select **Edit Settings**
5. From the **New device** drop-down menu, select **Existing Hard Disk**.
6. In **Disk File Path**, browse to the disk location specified for the first node you want to share.
7. Expand **New Hard disk**.
8. Select the same SCSI ID that you selected for the first virtual machine's shared storage disks. For example, select **SCSI (1:0)**.

   Note:
   The disk SCSI ID for this virtual machine's shared storage must match the corresponding SCSI ID for the first virtual machine.

9. Click **OK**.

Cluster Physical and Virtual Machines

You can create a WSFC cluster in which each physical server has a corresponding virtual machine. This type of configuration is known as a standby host cluster.

A standby host cluster has specific hardware and software requirements.

- The network configuration should be consistent between a physical and virtual node of a WSFC. Ensure that all required VLANs are available to ESX hosts, having a virtual node of a WSFC cluster.
- Use RDMs in physical compatibility mode. You cannot use virtual disks or RDMs in virtual compatibility mode for shared storage.
- Use the STORport Miniport driver for the Fibre Channel (FC) HBA (QLogic or Emulex) in the physical Windows machine.
- Do not run multipathing software in the physical or virtual machines.

- Use only a single physical path from the host to the storage arrays in standby host configurations.

Create the First Node for a Cluster of Physical and Virtual Machines

The first node in a standby host setup is a physical machine.

For more information about setting up a physical machine that participates in a WSFC cluster, see the official Microsoft Windows Server Failover Cluster documentation.

- Set up the physical machine using the settings listed in the table.

Component	Requirement
Windows Cluster Administrator application	Advanced minimum configuration if Windows 2012 is used.
Storage	Access to the same storage on a SAN as the ESX host on which the corresponding virtual machine will run.
Operating system	Installed on each physical machine.

Create the Second Node for a Cluster of Physical and Virtual Machines

To create the second node, you set up a virtual machine as a second node of a WSFC cluster, where the first node is a physical server.

Before you begin, make sure that the shared storage that is visible from the physical machine that you configured in Create the First Node for a Cluster of Physical and Virtual Machines is also visible from the ESX host.

1. Open the vSphere Client and connect to the vCenter system.

   Use the user name and password of the user who has administrator permissions on the virtual machine.
2. In the vSphere Client, right-click the host and select **New Virtual Machine**.
3. Follow the prompts in the wizard to create the virtual machine.

Page	Action
Creation type	Select **Create a new virtual machine**.
Name and folder	Enter a name and select a location.
Compute resource	Select a cluster, host, vApp, or resource pool to run this virtual machine.
Storage	Select a datastore as the location for the virtual machine configuration file and the virtual machine disk (`.vmdk`) file.
Compatibility	The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.
Guest operating system	Select the guest operating system that you intend to install.
Customize hardware	Select virtual hardware, advanced virtual machine options, and SDRS rules.
Ready to Complete	Review your selections.

4. Click **Finish** to complete creating the virtual machine.
5. Install a Windows Server operating system on the virtual machine.

Add Hard Disks to the Second Node for a Cluster of Physical and Virtual Machines

When you add hard disks to the second node, you set up the disks to point to the shared storage disks for the first node. The setup allows shared access to clustered disks.

1. In the vSphere Client, right-click the newly created virtual machine, and select **Edit Settings**.
2. From the **New device** drop-down menu, select **SCSI Controller**.
3. In the new SCSI controller, select **VMware Paravirtual**, set **SCSI Bus Sharing** to **Physical**, and click **OK**.

   Note:

   Although LSI SAS is supported, use VMware Paravirtual.

4. Select the newly created virtual machine in step 1, right-click, and select **Edit Settings**
5. From the **New device** drop-down menu, select **RDM Disk**.
6. Select an unformatted LUN and click **OK**.
7. Expand **New Hard disk**.
8. Select a datastore for **Location**.
   This datastore must be on shared storage because you need a single shared RDM file for each shared disk.
9. Select compatibility mode **Physical**.
10. Select the newly created SCSI controller in step 2, for virtual device node (for example, select **SCSI (1:0)**).

    Note:
    This must be a new SCSI controller. You cannot use SCSI 0.
11. Click **OK**.
    The wizard creates a new hard disk.

Install Microsoft Cluster Service

After you configure the first and second nodes, you must configure Microsoft Failover Clustering.

For more information about creating and configuring server clusters, see the Microsoft documentation.

In complex storage solutions, such as an FC switched fabric, a storage unit might have a different identity (target ID or raw disk ID) on each computer in the cluster. Although this is a valid storage configuration, it might cause a problem when you add a node to the cluster.

Create Additional Physical-Virtual Pairs

If you have additional physical machines, you can create additional clusters for each.

1. On the ESX host, configure an additional virtual machine for the physical machine.
2. Cluster the physical machine with the new virtual machine.

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